

Chapter 13

Multimedia

13.1 Multimedia Overview

13.1.1 Multimedia Overview

13.1.1.1 Display Interface

Display Interface consists of the following:

- Enhanced LCD Interface (eLCDIF)
 - Support 8-bit / 16-bit / 18-bit / 24-bit / 32-bit pixel depth
 - Support DOTCLK mode for MIPI-DPI interface
 - Support MPU mode for MIPI-DBI interface
 - Support resolution up to 1920x1080p60 and 1800x1200p60
 - Support one base layer and one graphics overlay with alpha blending
- Display Controller (DCSS) - provides a mechanism to display frame buffers in memory out to UltraHD or HDTVs. It has the capability to combine up to 3 layers of graphics or video overlay to the HDMI output. The key features of the display controller includes:
 - Supports up to 3 layers of graphics or video
 - Arbitrary offset
 - One plane can be graphics with 8 bit alpha support
 - Upscale 1920x1080p60 video or graphics to 3840x2160p60
 - Downscale 3840x2160p30 video to 1920x1080p30 or 1280x720p30
 - HDR support:
 - HDR10 with 2084 and 2020
 - Dolby Vision single and dual layer formats
 - HLG
 - HDMI 2.0a supporting one display:
 - Resolutions of: 640x480p60, 720x480p60, 1280x720p60, 1920x1080p60, 3840x2160p60, 4096x2160p60

- HDCP 2.2 and HDCP 1.4
- Pixel clock up to 596 MHz
- Audio support:
 - 32 channel audio output support
 - 1 S/PDIF audio ARC input support
- Output can also go to MIPI DSI output
- Frame Buffer Compression – Lossless compression of buffers
- HD Display Transmitter Controller (HDMI TX) - The HDMI firmware and HDCP keys are loaded by the boot core running ROM code. Once they are loaded, they need to be retained in the HDMI logic until next chip power-on reset. More information about the HDCP key handling can be found in the system boot chapter.
 - HDMI 2.0a complaint
 - HDCP 2.2 and HDCP 1.4 encryption
 - Support a maximum display resolution of 4096x2160@60Hz with a pixel clock frequency of 596MHz
- MIPI DSI Host Controller (MIPI_DSI) - one 4-lane MIPI DSI display with pixels from either the DCSS HDR Display Controller or the eLCDIF. The key features of the MIPI DSI (controller and PHY) include:
 - Complaint to MIPI-DSI standard v1.1
 - Support up to 4 data lanes
 - Support 80Mbps - 1.5Gbps data rate in high speed operation
 - Support 10Mbps data rate in low power operation
- MIPI CSI Host Controller (MIPI_CSI) - two 4-lane MIPI CSI2 camera input. The key features of the MIPI CSI2 (controller and PHY) includes:
 - Complaint to MIPI-CSI2 standard
 - Support up to 4 data lanes
 - Support 80Mbps - 1.5Gbps data rate in high speed operation
 - Support 10Mbps data rate in low power operation
 - Supports 4Kp30, 1080p60/30, 5MP at 15 fps, 720p60, VGA at 60 fps

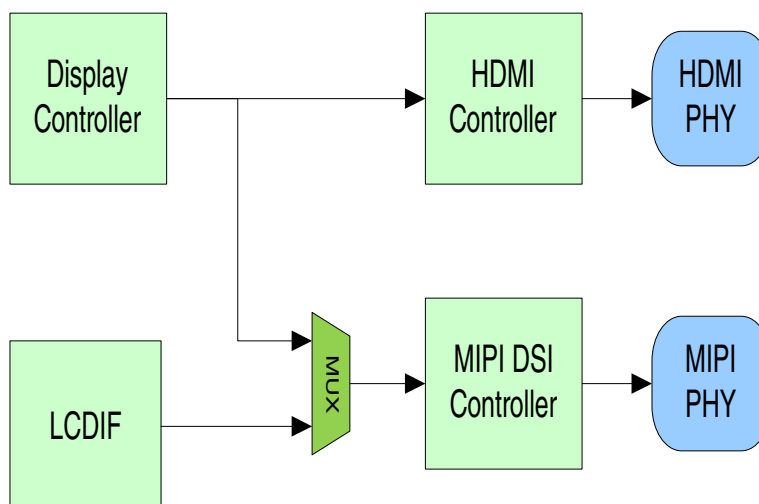


Figure 13-1. Display Interface

The following use cases are supported:

- Dual display: HDMI driven by Display Controller, with video playback support on HDMI; MIPI driven by LCDIF, without video playback support on MIPI
- Single display: HDMI driven by Display Controller, with video playback support on HDMI
- Single display: MIPI driven by Display Controller, with video playback support on MIPI
- Single display: MIPI driven by LCDIF, without video playback support on MIPI.

The i.MX 8M Dual/8M QuadLite/8M Quad do not support parallel display interface.

13.1.1.2 Graphics Processing Unit (GPU)

The chip utilizes a GPU to provide 2D/3D acceleration. Key features include:

- Run up to 800 MHz at 0.9V nominal voltage
- Support 267 million triangles/sec
- Support 1.6 Giga pixel/sec fill rate
- Support 25.6 GFLOPs 32-bit, 51.2 GFLOPs 16-bit
- Support OpenGL ES 1.1, 2.0, 3.0, 3.1
- Support OpenCL 1.1/1.2 FP
- TrustZone support using a local MMU to manage secure regions

For the best power and performance optimization, the GPU has following physical implementation:

- Independently powered with a dedicate power supply (VDD_GPU) to allow DVFS

- Support power down with internal power switch or its supply (VDD_GPU) shut off externally
- Dedicated PLL to allow most flexible clock frequency setting

13.1.1.3 Video Processing Unit (VPU)

The chip supports video decoding for various video format using the Hantro G2 + G1. Features include:

- 4Kp60 HEVC/H.265 Main, Main10 (Hantro G2) with best effort for 4Kp75
 - 4Kp75 is needed to support PiP (Picture in Picture) for live TV and dual layer Dolby Vision decoding one 4Kp60 stream and one 1080p60 stream
- 4Kp60 VP9 Profile 0, 2 (10 bit) (Hantro G2)
- 4Kp30 AVC/H.264 Baseline, Main, High decoder (Hantro G1)
- 1080p60 MPEG-2, MPEG-4p2, VC-1, VP8, RV9, AVS, MJPEG, H.263 decoder (Hantro G1)
- Lossless Frame Buffer Compression
- TrustZone support

For the best power and performance optimization, the VPU has following physical implementation:

- Independently powered with a dedicate power supply (VDD_VPU) to allow DVFS
- Support power down with internal power switch or its supply (VDD_VPU) shut off externally
- Dedicated PLL to allow most flexible clock frequency setting

13.1.1.4 Audio Interface

The device supports multiple audio interfaces as listed below:

Table 13-1. Audio Interface Summary

Interface	Function	RX Data Line	TX Data Line	Note
SAI-1	External audio	8	8	
SAI-2	External audio	1	1	
SAI-3	External audio	1	1	
SAI-4	HDMI-TX	0	4	Internal connection
SAI-5	External audio	4	4	
SAI-6	External audio	1	1	
SPDIF-1	External audio	1	1	
SPDIF-2	HDMI ARC	1	0	Internal connection

In addition to the general audio input/output functions, the audio interfaces support the following features:

- SAI-1 supports up to 16-channels TX (8 lanes) and 16-channels RX (8 lanes) at 384KHz/32-bit
- SAI-5 supports up to 8-channels TX (4 lanes) and 8-channels RX (4 lanes) at 384KHz/32-bit
- SAI-2/3/6 supports up to 2-channels TX (1 lanes) and 2-channels RX (1 lanes) at 384KHz/32-bit
- SAI-2/3/6 support up to 2-channels TX (1 lane) and 2-channels RX (1 lane) at 384KHz/32-bit
- SAI-1 supports glue-less switching between PCM and DSD operation for popular audio DACs
- SPDIF-1/2 supports raw capture mode that can save all incoming bits into an audio buffer

The SAI-1/2/3/5/6 and SPDIF-1 share GPIO pads on the chip through IOMUX. Common use cases supported by the audio interfaces are listed in the table below (many other configurations are possible). The number is the data lanes supported

Table 13-2. Audio Interface Use Cases

		UC 1	UC 2	UC 3	UC 4	UC 5	UC 6	UC 7	UC 8
SAI-1	TX	8 + DSD	8 + DSD	8 + DSD	8	8	8	4	4
	RX	8	8	8	8	8	8	4	4
SAI-2	TX	1		1		1	1		1
	RX	1		1		1	1		1
SAI-3	TX	1			1	1	1	1	1
	RX	1			1	1	1	1	1
SAI-5	TX		4		4	1		4	1
	RX		4	4	4	1	4	4	1
SAI-6	TX							1	1
	RX							1	1
SPDIF-1	TX	1	1	1	1	1	1	1	1
	RX	1	1	1	1	1	1	1	1

13.1.1.4.1 SAI Master Clock Inputs/Outputs

The MCLK pin on each SAI module can be configured as either an input or an output. When configured as an output, the SAI_{In}_CLK_ROOT from the CCM is routed to the pad output. When configured as an input, the external input to the pad is routed to SAI_{In}_MCLK, which can be used as master clock for SAI. Below is the diagram showing the both input/output options, by using SAI1 as the example.

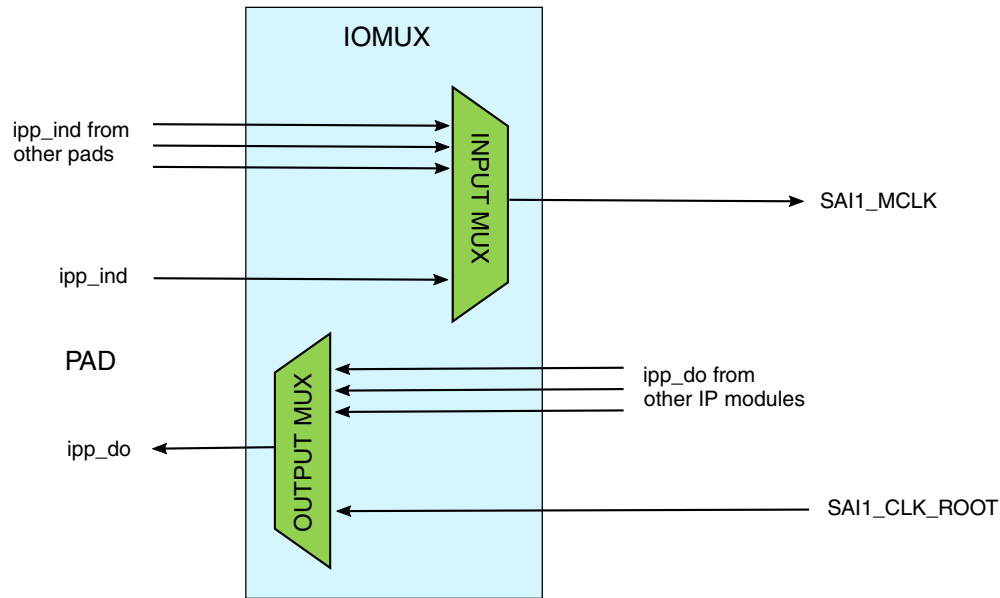


Figure 13-2. SAI MCLK Input and Output

Each SAI module supports up to 3 master clock inputs. The TX and RX sub-module inside each SAI can independently select one of the clock inputs as its master clock. This allows TX and RX of one SAI to run from different clock sources. The master clock inputs have the following options:

- SAI_n_MCLK[1] can be selected from SAI_n_CLK_ROOT from CCM or SAI_n_MCLK from IOMUX. This is the most straight-forward clock routing in which SAI_n only uses its own clock source from CCM or IO pad
- SAI_n_MCLK[2] can be selected from the following clock sources:
 - Any of the SAI_n_CLK_ROOT from CCM
 - Any of the SAI_n_MCLK from IOMUX
 - Other clock sources from SPIDF
- SAI_n_MCLK[3] has the same clock source options as SAI_n_MCLK[2]. This allows both TX and RX to have access to all the options without any dependency between each other

The clock options for master clock on SAI are shown in the diagram blow, by using SAI-1 as an example.

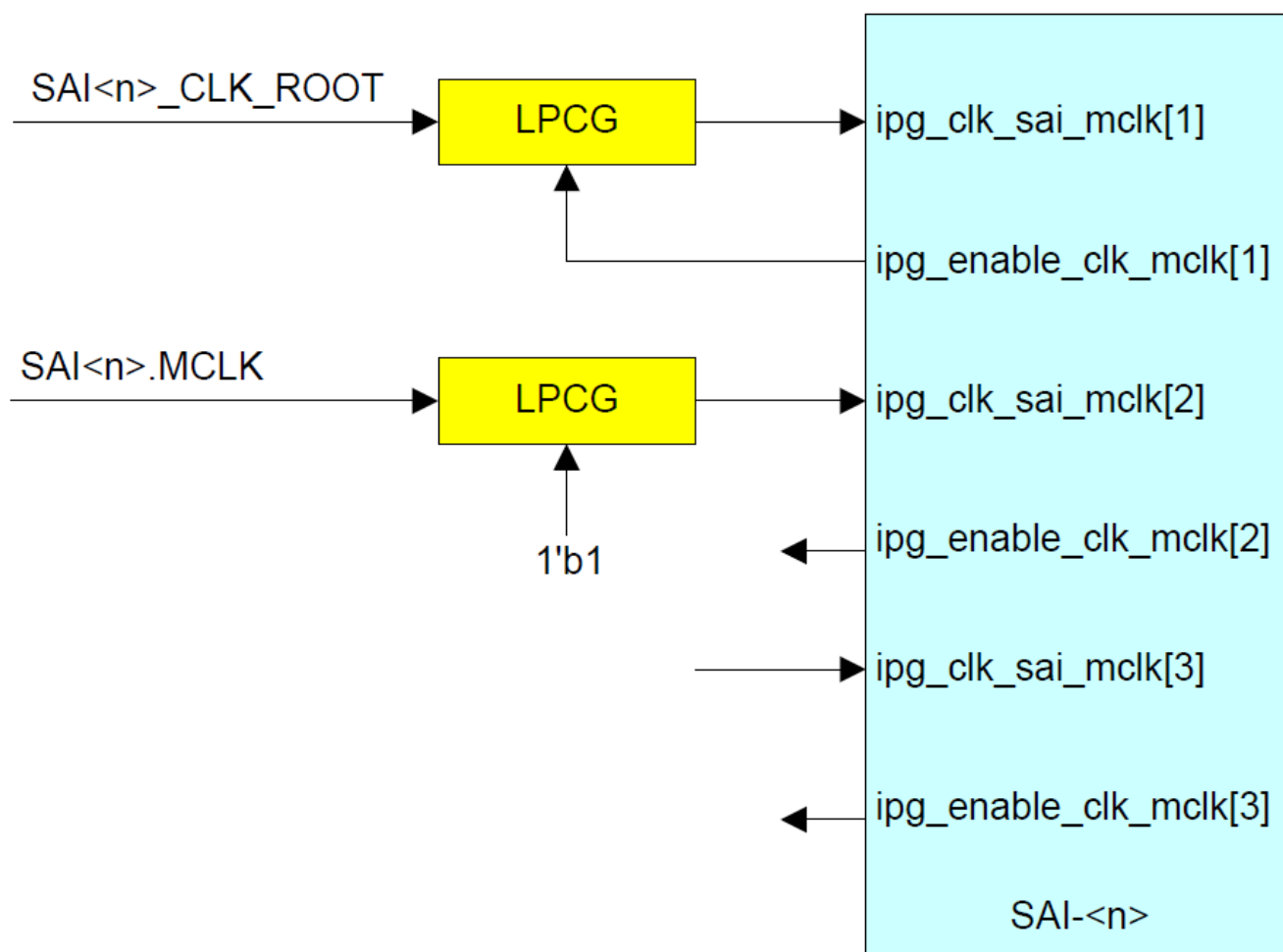


Figure 13-3. SAI Master Clock Options

The options on MCLK[1] are also available on MCLK[2] and MCLK[3]. This provides a similar SAI clock structure as i.MX6/i.MX7 processors.

The configuration of the MUX for master clock is controlled by IOMUXC_GPR registers. They should be configured before the SAI clock is enabled to avoid glitches on the clock.

13.1.1.4.2 SDMA Support

SDMA support is provided for SAI and SPDIF. In order to meet the audio data rate, two SDMA modules are used.

Because the SAI-2/3 and SPDIF-1/2 do not require high data throughput, they are assigned to SDMA-1, shared with other peripherals such as UART/SPI. SAI-1/4/5/6 need to support high sample rate and multi-channel audio, they are assigned to SDMA-2, which is a dedicated SDMA engine for audio. The SDMA-2 frequency is increased to 500/250 instead of 133/66 to make sure it has enough throughput.

Table 13-3. SDMA Assignment for Audio Interface

SDMA	Audio Interface	Note
SDMA-1 133MHz AHB, 66MHz IPG	SAI-2	1 TX + 1 RX
	SAI-3	1 TX + 1 RX
	SPDIF-1	TX + RX
	SPDIF-2	RX only, for HDMI ARC
SDMA-2 500MHz AHB, 250MHz IPG	SAI-1	8 TX + 8 RX
	SAI-4	4 TX only, for HDMI
	SAI-5	4 TX + 4 RX
	SAI-6	1 TX + 1 RX

13.2 Enhanced LCD Interface (eLCDIF)

13.2.1 Overview

The enhanced Liquid Crystal Display Interface (LCDIF) is a general purpose display controller used to drive a wide range of display devices varying in size and capability.

The LCDIF block supports the following:

- Displays with an asynchronous parallel MPU interface for command and data transfer to an integrated frame buffer.
- Displays that support moving pictures and require the RGB interface mode (DOTCLK interface).
- VSYNC mode for high-speed data transfers.
- Digital video encoders that accept ITU-R BT.656 format 4:2:2 YCbCr digital component video and convert it to analog TV signals.

The LCDIF provides fully programmable functionality to supported interfaces:

- Bus master interface to source frame buffer data for display refresh. This interface can also be used to drive data for "Smart" displays.
- PIO interface to manage data transfers between "Smart" displays and SoC.
- 8/16/18/24/32 bit LCD data bus support available.
- Programmable timing and parameters for MPU, VSYNC, and DOTCLK LCD interfaces to support a wide variety of displays.
- ITU-R BT.656 mode (called Digital Video Interface or DVI mode here) including progressive-to-interlace feature and RGB to YCbCr 4:2:2 color space conversion to support 525/60 and 625/50 operation.

13.2.2 Clocks

The following table describes the clock sources for LCDIF. Please see [Clock Controller Module \(CCM\)](#) for clock setting, configuration and gating information.

Table 13-4. LCDIF Clocks

Clock name	Clock Root	Description
apb_clk	MAIN_AXI_CLK_ROOT	AXI clock
pix_clk	LCDIF_PIXEL_CLK_ROOT	Pixel clock

13.2.3 Functional Description

[Bus Interface Mechanisms](#) through [Initializing the LCDIF](#), describe the internal pipeline for the LCDIF interfaces. Differences for each mode are then described in separate sections, as follows:

- [MPU Interface](#)
- [VSYNC Interface](#)
- [DOTCLK Interface](#)
- [ITU-R BT.656 Digital Video Interface \(DVI\)](#)

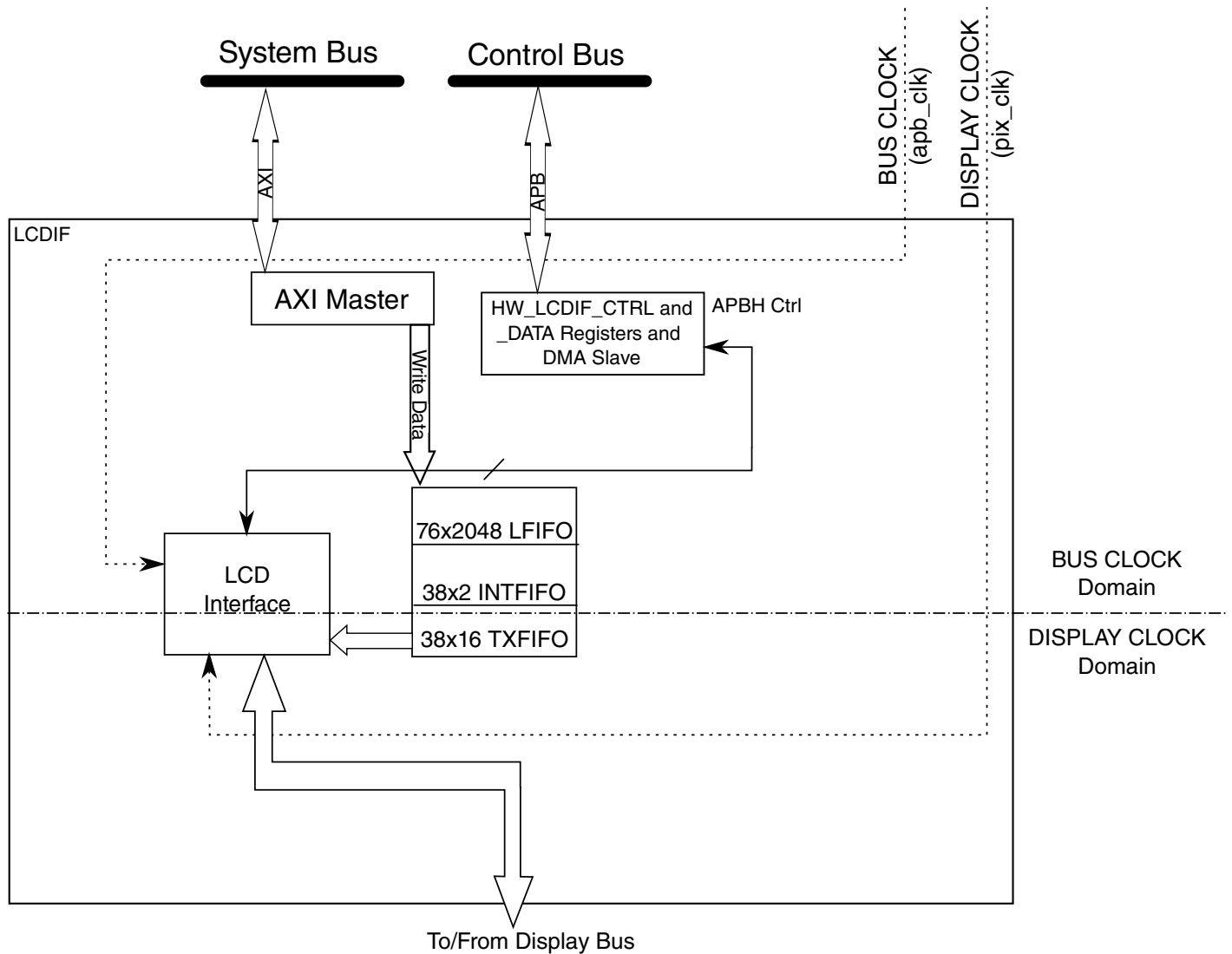


Figure 13-4. Top-Level Block Diagram of LCDIF subsystem

13.2.3.1 Bus Interface Mechanisms

The LCDIF module has memory-mapped control, data and status registers. It provides several interfaces to transfer data between the display and SoC.

The bus master interface is used to initiate the requests to transfer data from external memory to the display. It is completely autonomous, or no CPU intervention is required, to manage the cyclical nature of refreshing standard display types. Bus mastering can also be used for MPU mode data writes.

The PIO interface is used to interface to "Smart" displays to transfer frame buffer data and control information to/from the external display. The host CPU executes display drivers to manage the display solution. The following sections describe the system bus interface mechanisms.

13.2.3.1.1 Bus Master Operation in Write/Display Modes

The LCDIF block has a bus master interface that initiates requests for data to drive the display. The LCDIF_MASTER bit must be set to 1 to enable the bus master interface. Software should program all control registers required to transfer the frame sequence.

In the MPU and VSYNC mode, single frames are transferred. When a complete frame is transferred, LCDIF enters idle and clears the RUN bit in the CTRL register. For subsequent frame transmission, the LCDIF setup sequence should be repeated.

The DOTCLK and DVI mode is used to refresh the display at the desired refresh rate and resolution, and drive displays that don't integrate a display buffer memory. When the display is refreshed, the LCDIF will automatically update the LCDIF_CUR_BUF_ADDR register with the value in LCDIF_NEXT_BUF_ADDR at the end of current frame and start fetching the next frame from the new address. If the LCDIF_NEXT_BUF_ADDR register was not updated within a frame refresh cycle, LCDIF will keep transmitting the last frame until a new value is programmed into that register.

LCDIF also provides the capability of interlacing a progressive frame by fetching odd lines in the first field and then fetching even lines in the second field. This feature can be used in the DVI mode and can be turned on by setting the INTERLACE_FIELDS bit in the LCDIF_CTRL1 register.

13.2.3.1.2 System Bus Master Performance

The performance of the LCDIF block can be controlled by changing the burst length and the outstanding cycle issuing capability depending on the memory bandwidth requirements. Two fields in the LCDIF_CTRL2 register will throttle system memory requests. The LCDIF_CTRL2_OUTSTANDING_REQS field will control how many requests the LCDIF can have in flight on any given clock cycle. This should be programmed based on the expected system bus latency for returned read data. Also, the LCDIF_CTRL2_BURST_LEN_8 bit will set the number of 64 bit words requested for each LCDIF system bus request to either 8 or 16 QWORDS. Generally, 4 outstanding requests of length 16 will provide enough performance to drive any standard display resolution. These configuration bits are intended to change the access pattern of the LCDIF to optimize system bus throughput when other system masters will contend for system memory resources.

The LCDIF_THRES register can also be used to optimize bus throughput and power consumption.

The LCDIF_THRES_PANIC value can be used to raise the priority of requests initiated by the LCDIF to alter how the LCDIF requests are arbitrated by the system bus infrastructure. The panic output control signal is raised when the number of 32bpp pixel equivalents in the LFIFO is less than this programmed value. Since the LFIFO is arranged as a 256x64bit quadword FIFO, it contains two 32bpp pixels per quadword, or 512 32bpp pixels total. To set the panic output when 3/4s of the LFIFO is empty, set the LCDIF_THRES_PANIC value to $3/4 * 512$, or 128. The panic signal output is used to assess higher priority to LCDIF system requests to avoid LCDIF under run errors during periods of high system bandwidth utilization.

The features available with the LCDIF_THRES register require support from system clocking and dynamic priority control. Refer to the appropriate block documentation to assess the system support for these features.

13.2.3.2 Write Data Path

LCDIF supports raster based frame buffers and there is no support for tiled buffers.

There are several options to accommodate endianness of display buffers in memory before the data is processed for the external display. The LCDIF_CTRL[INPUT_DATA_SWIZZLE] field provides the following options for data word multiplexing:

```
00 (0): No swizzle (little-endian)
01 (1): Swap bytes 0 and 3, swap bytes 1 and 2 (big-endian)
10 (2): Swap half-words
11 (3): Swap bytes within each half-word
```

The LCDIF_CTRL[WORD_LENGTH] field indicates the input data/pixel format. LCDIF_TRANSFER_COUNT register denotes how much data is contained in each frame. The LCDIF_TRANSFER_COUNT[H_COUNT] field indicates the number of pixels per line and LCDIF_TRANSFER_COUNT[V_COUNT] indicates the total number of lines per frame. The LCDIF_CTRL1[BYTE_PACKING_FORMAT] field can be used to specify which bytes within the 32-bit word are going to be valid. For example, if the entire 32-bit word is valid, LCDIF_CTRL1[BYTE_PACKING_FORMAT] should be set to 0xF, if only lower 3 bytes of each word in the frame buffer are valid, then LCDIF_CTRL1[BYTE_PACKING_FORMAT] should be set to 0x7.

The LCDIF_CTRL[LCD_DATABUS_WIDTH] field suggests the width of the bus going to the display controller. There is an option to source all 32 bits of the input word and transfer it to the output display interface. If the LCDIF_CTRL[LCD_DATABUS_WIDTH] is not the same as

LCDIF_CTRL[WORD_LENGTH], LCDIF will perform RGB to RGB color space conversion. For example, if the input frame has fewer bits per pixel than the display, as in a 16 bpp input frame going to 24 bpp LCD, LCDIF will pad the MSBs of each color to the LSBs of the same color for each pixel. If the input frame has more bits per pixel than the display, for example, 24 bpp input frame going to 16 bpp LCD, LCDIF will drop the LSBs of each color channel to convert to the lower color depth. LCDIF also has the capability to support delta pixel displays by swizzling the R, G and B colors of each pixel in the odd and even lines of the frame separately by programming the LCDIF_CTRL2[ODD_LINE_PATTERN] and the LCDIF_CTRL2[EVEN_LINE_PATTERN] bit fields. This operation occurs after the RGB-to-RGB color space conversion operation.

LCDIF also supports RGB to YCbCr 4:2:2 color space conversion. This is useful in the DVI mode since the TV encoder requires input in YCbCr 4:2:2 format. The LCDIF_CSC* registers have complete programmability over the CSC coefficients and offsets. The values must be written into these registers in the signed two's complement format.

The following list shows how the different input/output combinations can be obtained:

- LCDIF_CTRL[WORD_LENGTH]=1 indicates that the input is 8-bit data. This is most likely going to be used for sending commands in MPU interface, or maybe a gray scale image. Any combination of LCDIF_CTRL1[BYTE_PACKING_FORMAT] is permissible.

Limitation: LCDIF_TRANSFER_COUNT[H_COUNT] must be a multiple of the sum of BYTE_PACKING_FORMAT [3], BYTE_PACKING_FORMAT [2], BYTE_PACKING_FORMAT [1] and BYTE_PACKING_FORMAT [0].
LCDIF_CTRL[LCD_DATABUS_WIDTH] must be 1, indicating an 8-bit data bus.

- LCDIF_CTRL[WORD_LENGTH]=0 implies the input frame buffer is RGB 16 bits per pixel. LCDIF_CTRL[DATA_FORMAT_16_BIT] field determines the pixels are RGB 555 or RGB 565.

Limitation: LCDIF_CTRL1[BYTE_PACKING_FORMAT] should be 0x3 or 0xC if there is only one pixel per word. If there are two pixels per word, it should be 0xF and LCDIF_TRANSFER_COUNT[H_COUNT] will be restricted to be a multiple of 2 pixels.

- LCDIF_CTRL[WORD_LENGTH]=2 indicates that input frame buffer is RGB 18 bits per pixel, that is, RGB 666. The valid RGB values can be left-aligned or right-aligned within a 32-bit word. The alignment of the valid 18 bits within a word is indicated by the LCDIF_CTRL[DATA_FORMAT_18_BIT] bit.

Limitation: LCDIF_CTRL1[BYTE_PACKING_FORMAT] can be 0x7, 0xE or 0xF. Packed pixels are not supported in this case.
LCDIF_TRANSFER_COUNT[H_COUNT] can be any number.

- LCDIF_CTRL[WORD_LENGTH]=3 indicates that the input frame-buffer is RGB 24 bits per pixel (RGB 888). If LCDIF_CTRL1[BYTE_PACKING_FORMAT] is 0x7, it indicates that there is only one pixel per 32-bit word and there is no restriction on LCDIF_TRANSFER_COUNT[H_COUNT].

Limitation: If LCDIF_CTRL1[BYTE_PACKING_FORMAT] is 0xF, it indicates that the pixels are packed, that is, there are 4 pixels in 3 words or 12 bytes and LCDIF_TRANSFER_COUNT[H_COUNT] must be a multiple of 4 pixels.

- LCDIF_CTRL1[YCBCR422_INPUT]=1 implies that the input frame is in YCbCr 4:2:2 format. LCDIF_CTRL1[BYTE_PACKING_FORMAT] must be 0xF.

Limitation: LCDIF_CTRL[LCD_DATABUS_WIDTH] must be 8-bit and LCDIF_TRANSFER_COUNT[H_COUNT] must be a multiple of 2 pixels.

LCDIF_CTRL2[ODD_LINE_PATTERN] and LCDIF_CTRL2[EVEN_LINE_PATTERN] must be 0 when any of LCDIF_CTRL[RGB_TO_YCBCR422_CSC] or LCDIF_CTRL1[INTERLACE_FIELDS] or LCDIF_CTRL[YCBCR422_INPUT] bits is 1.

After the RGB to RGB or RGB to YCbCr 4:2:2 color space conversions, there is one more opportunity to swizzle the data before sending it out to the display or the encoder. This can be done with the LCDIF_CTRL[CSC_DATA_SWIZZLE] field, and it provides the same options as the LCDIF_CTRL[INPUT_DATA_SWIZZLE] register.

Finally, there is an option to shift the output data before sending it out to the display. This is done based on the LCDIF_CTRL[SHIFT_DIR] and LCDIF_CTRL[SHIFT_NUM_BITS] fields.

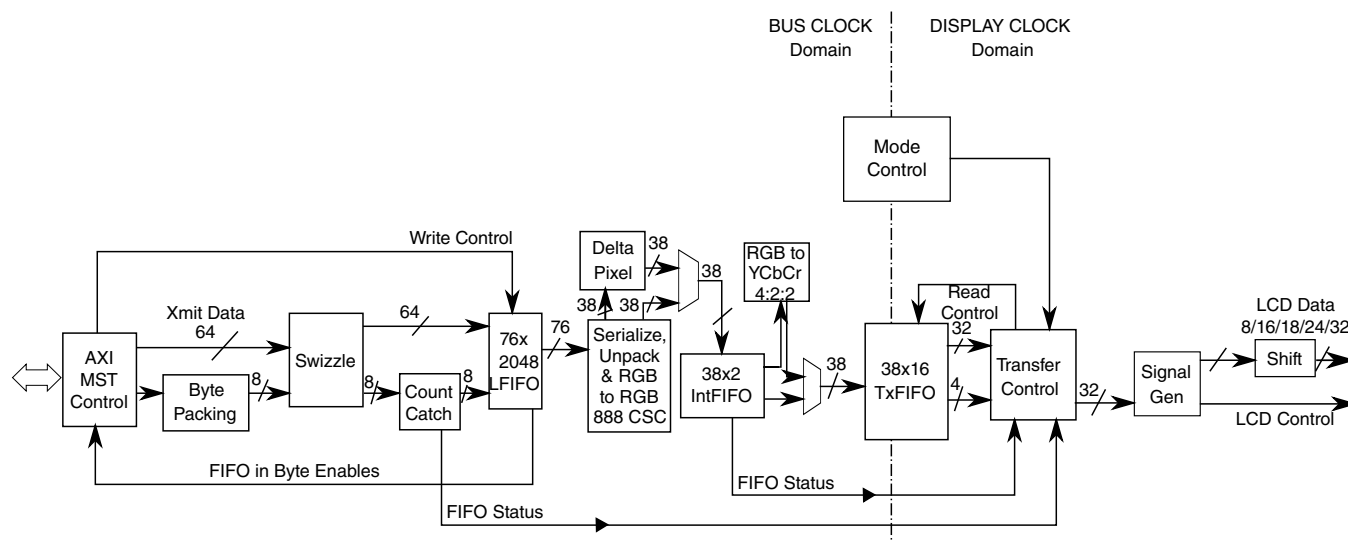


Figure 13-5. General Operations in Write Data Path

The examples in the following figures illustrate some different combinations of register programming for write mode. Assume that the data transferred over the system bus within a 32 bit word is organized as {A7-A0, B7-B0, C7-C0, D7-D0} in 8-bit mode and {A15-A0, B15-B0} in 16-bit mode.

In this example, all 32 bits of the input word are transferred out over an 8 bit display bus. Each byte within the 32 bit word is shifted to the right with zeros appended to bits D[7:6]. The input data bits [7:2] are shifted to the right by 2 bits and presented on the D[5:0].

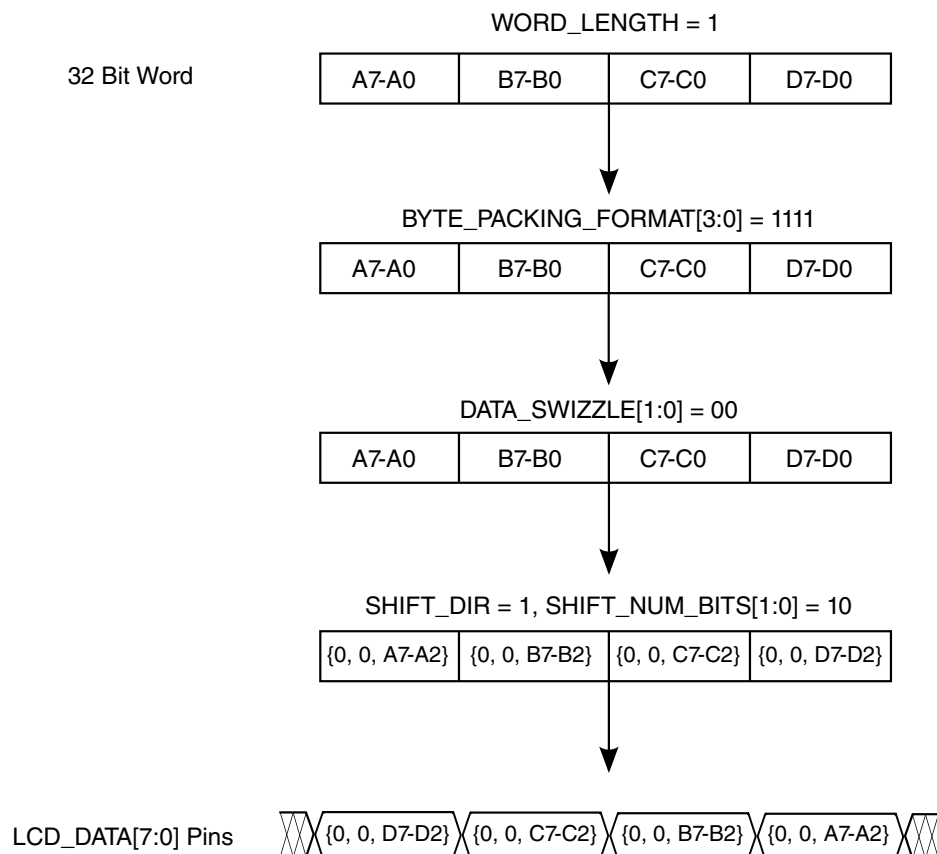


Figure 13-6. Register programming for write mode

In this 8 bit display interface example, one byte of the input word is deleted and not transferred over the external 8 bit display interface. This mode could be used to transfer 24bpp pixels over the 8 bit interface. In this case, the 4th unused byte is not transferred.

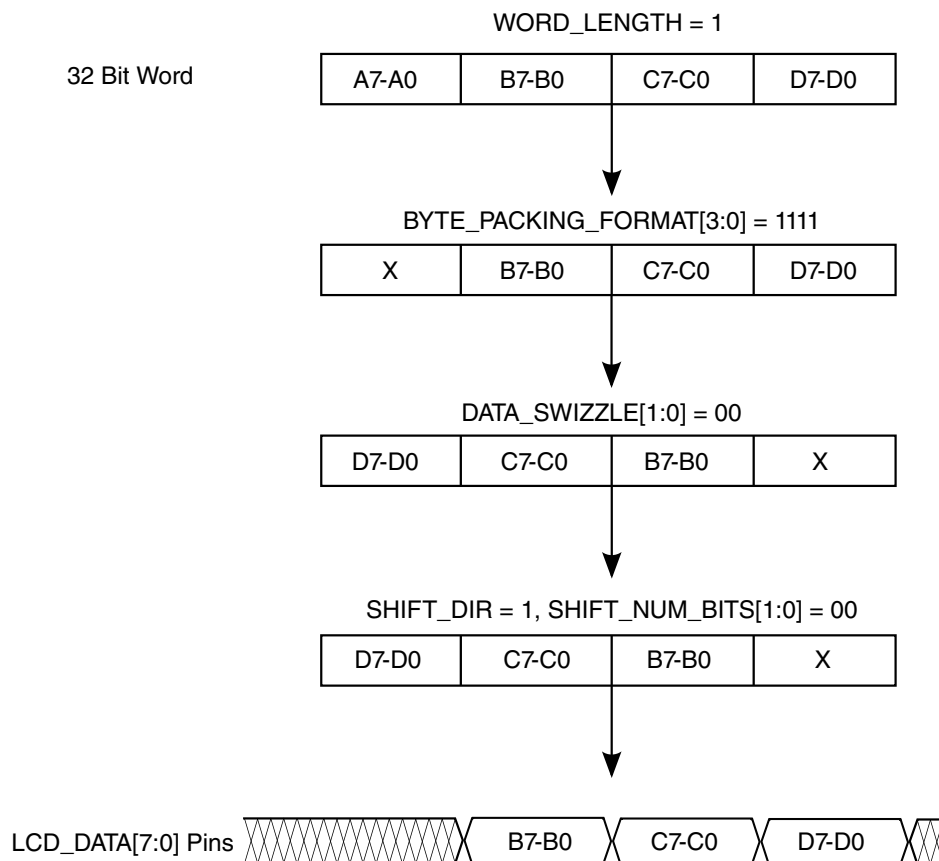


Figure 13-7. Register programming for write mode

The following example uses a 16 bit display interface. Each 16 bit half word is shifted to the right by two bits with zeros appended to the most significant two bits.

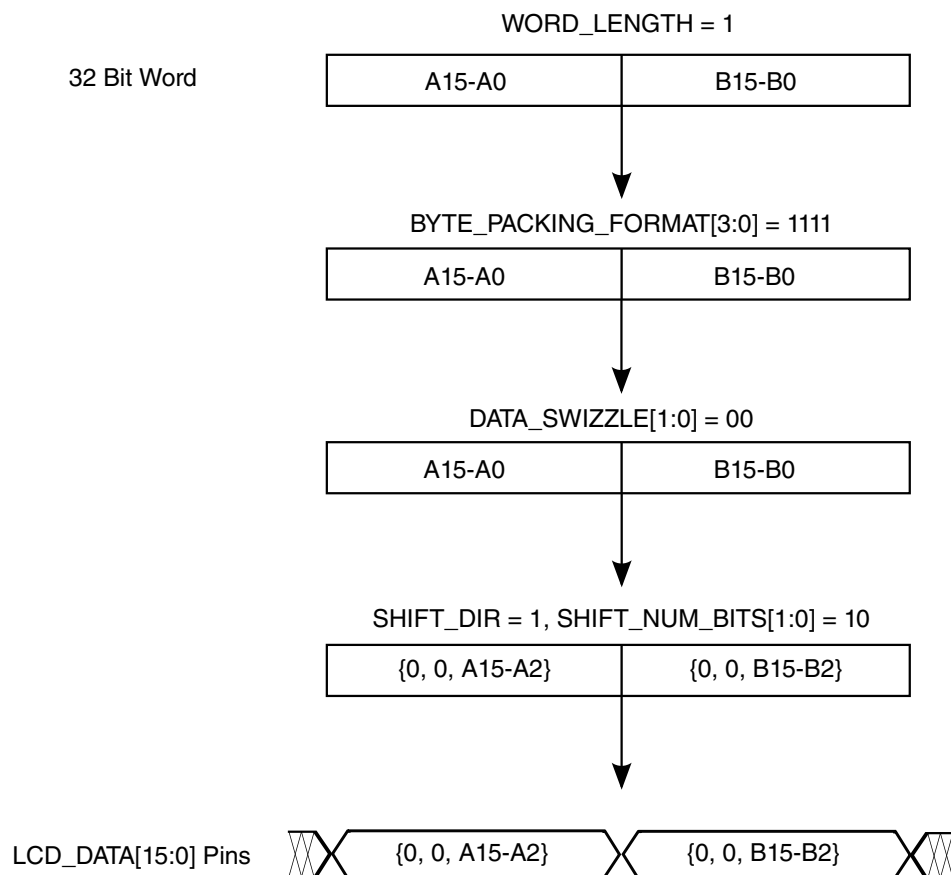


Figure 13-8. Register programming for write mode

This example indicates how an unpacked frame buffer can be sourced for display. Only a single 16 bit half word within the 32 bit word is transferred out via the 16 display bus.

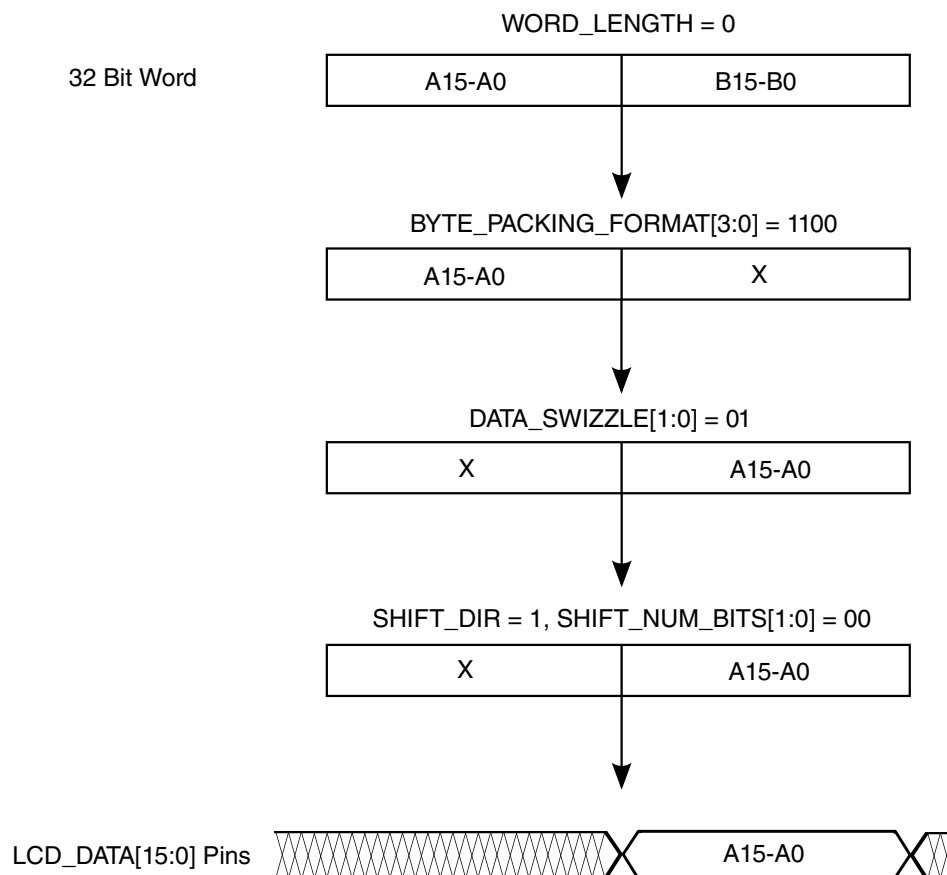


Figure 13-9. Register programming for write mode

13.2.3.3 Read Data Path

Figure 13-10 shows the MPU read data path in detail.

LCDIF can read from an external display that follows the 6800/8080 MPU protocol.

The display bus width is determined by the LCD_DATABUS_WIDTH bit field. The data sampled at every read strobe is called a subword and the number of subwords that can be packed in a 32-bit word is given by the READ_MODE_NUM_PACKED_SUBWORDS bit field. The INITIAL_DUMMY_READ bit field directs the LCDIF to skip the number of programmed subwords before starting to process read data. This feature is useful in the case of an LCD controller that returns the last written data the first time a read is issued, and then sends the correct data thereafter. SHIFT_DIR and SHIFT_NUM_BITS bit fields indicate whether the data needs to be shifted before getting stored in the internal registers. For example, a value of 2 in READ_MODE_NUM_PACKED_SUBWORDS if lcd databus width is 8 bits indicates two bytes should be packed in a 32-bit word, while if the lcd databus width is 16 bits, it indicates that two half words (or 4 bytes) should be packed.

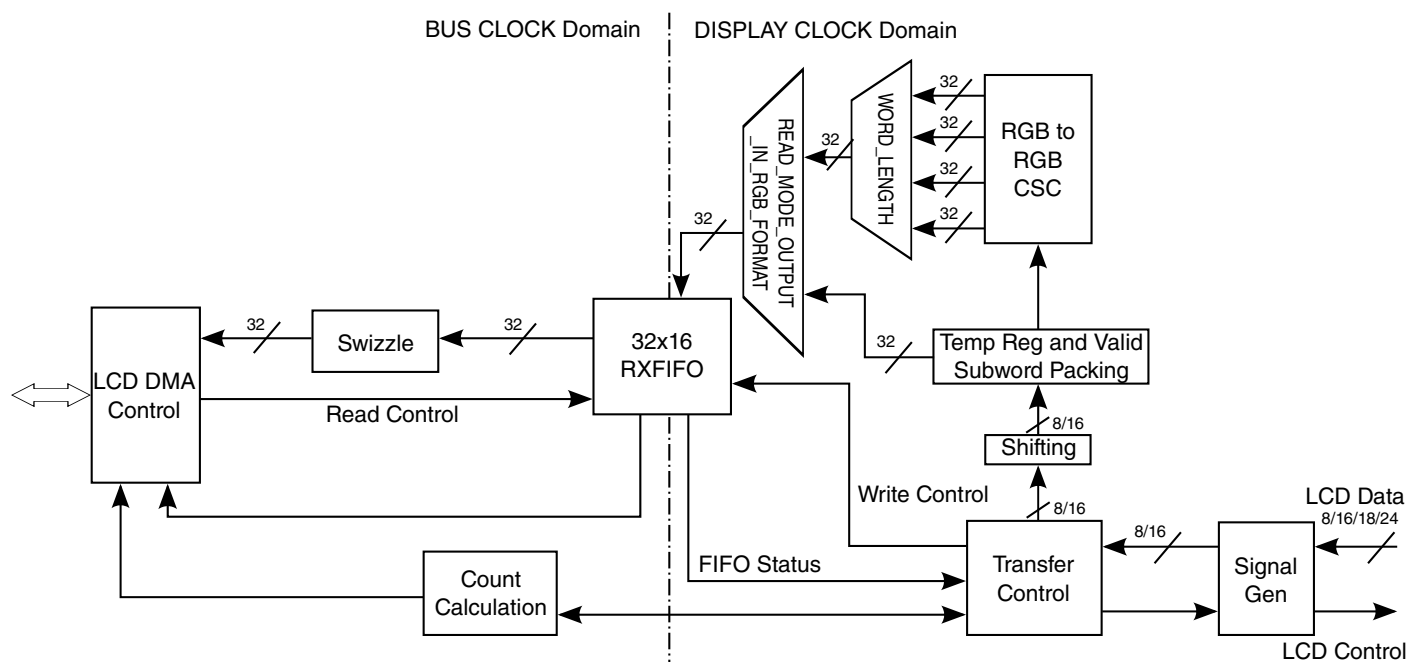


Figure 13-10. MPU Read Data Path

After the last subword within a word is reached, the block looks at the `READ_PACK_DIR` in the `HW_LCDIF_CTRL2` register. If this bit is set, the block will swizzle the data, but only within the valid bytes, unlike in the write mode, where swizzle occurs across all 4 bytes. If the `READ_MODE_OUTPUT_IN_RGB_FORMAT` bit is set, LCDIF will convert the data obtained from the `READ_PACK_DIR` operation into 24-bit unpacked RGB and then re-convert it into 16/18/24 bpp RGB depending on the `WORD_LENGTH` field. The `DATA_FORMAT_16/18/24_BIT` bit fields are also considered while converting to 24-bit unpacked RGB format. For example, if `DATA_FORMAT_18_BIT` is 1, the RGB666 data will be packed in the upper bits [31:4] of a 32-bit word, and that bit is 0, the data will be packed in the lower bits [17:0]. After all these operations, the data gets written into the RXFIFO.

The following figures show some examples of how data is handled in different MPU read modes.

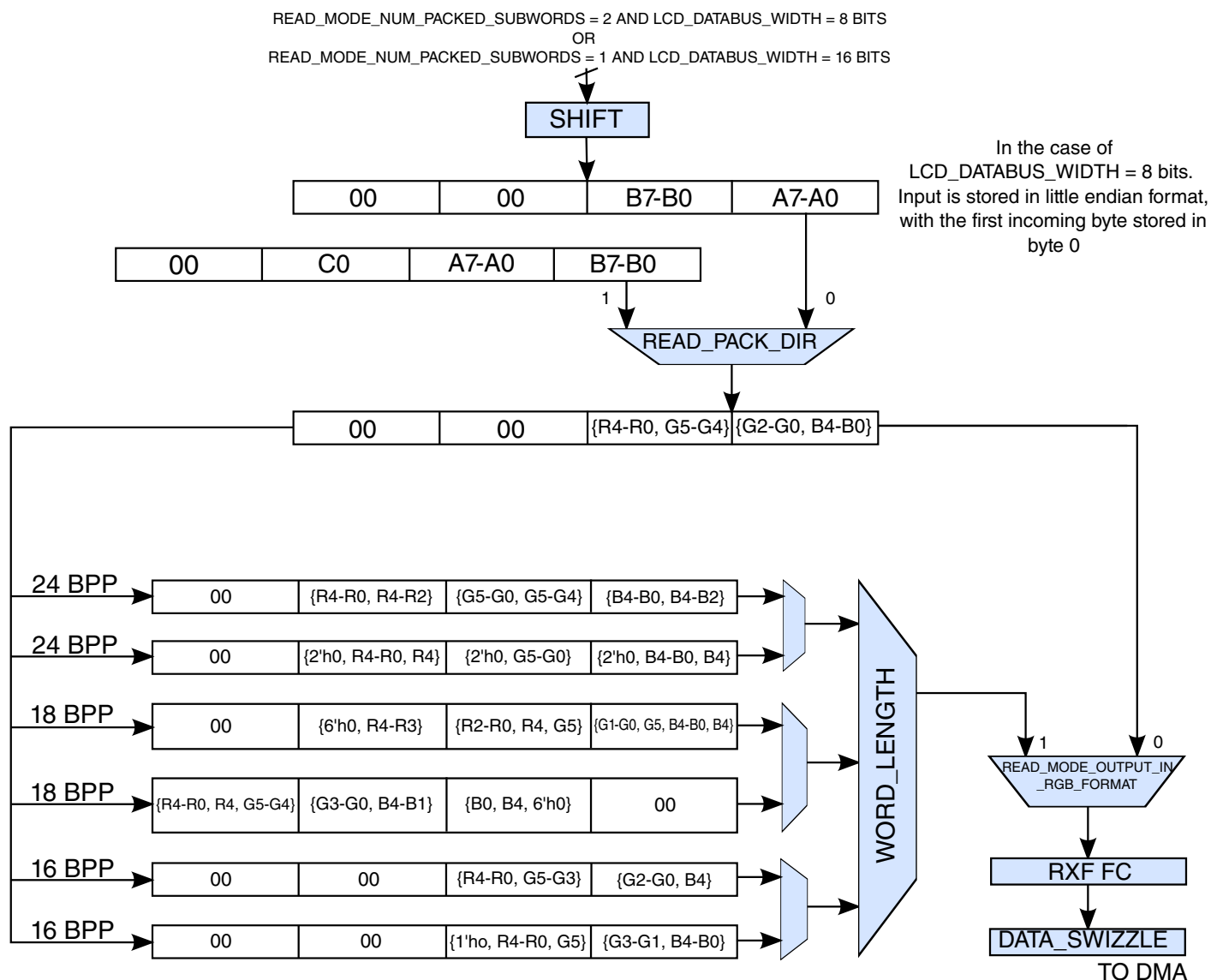


Figure 13-11. Data in MPU read mode

Enhanced LCD Interface (eLCDIF)

READ_MODE_NUM_PACKED_SUBWORDS = 3 AND LCD_DATABUS_WIDTH = 8 BITS
OR
READ_MODE_NUM_PACKED_SUBWORDS = 1 AND LCD_DATABUS_WIDTH = 24 BITS

8/24

SHIFT

00	C7-C0	B7-B0	A7-A0
----	-------	-------	-------

00	A7-A0	B7-B0	C7-C0
----	-------	-------	-------

1

0

READ_PACK_DIR

00	R7-R0	G7-G0	B7-B0
----	-------	-------	-------

In the case of
LCD_DATABUS_WIDTH = 8 bits.
Input is stored in little endian format,
with the first incoming byte stored in
byte 0

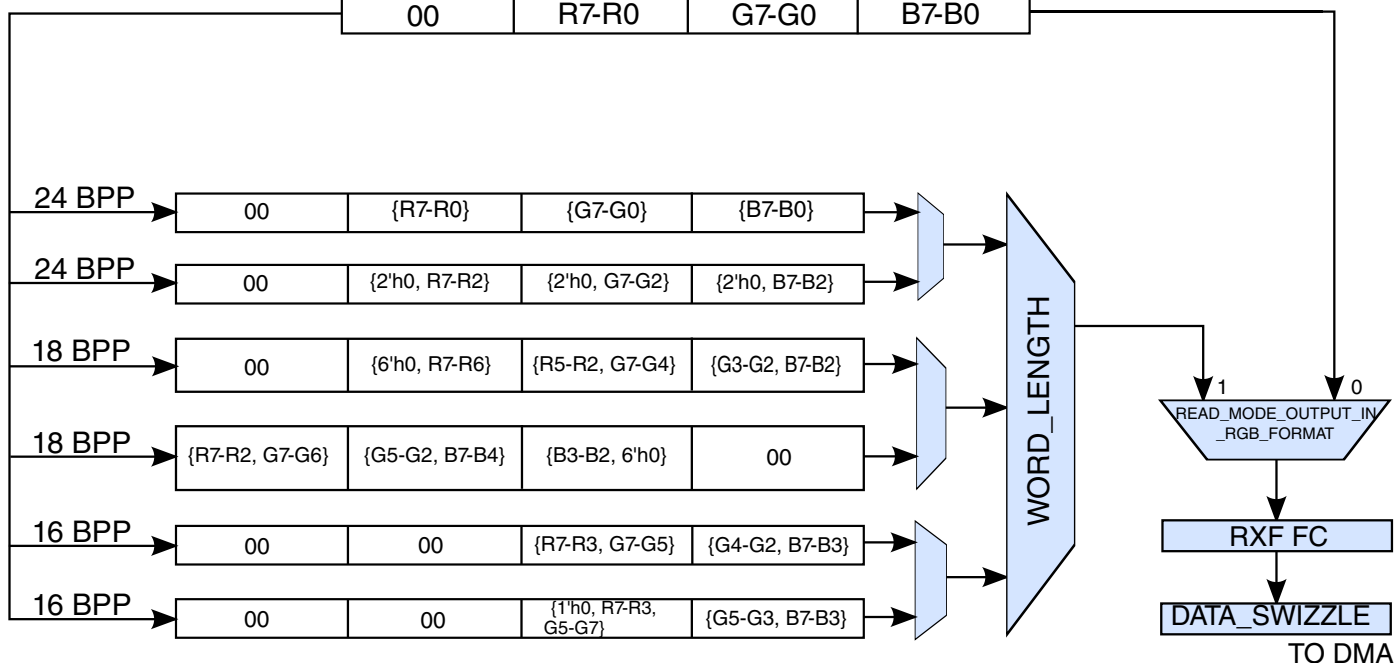


Figure 13-12. Data in MPU read mode

READ_MODE_NUM_PACKED_SUBWORDS = 1 AND LCD_DATABUS_WIDTH = 18 BITS

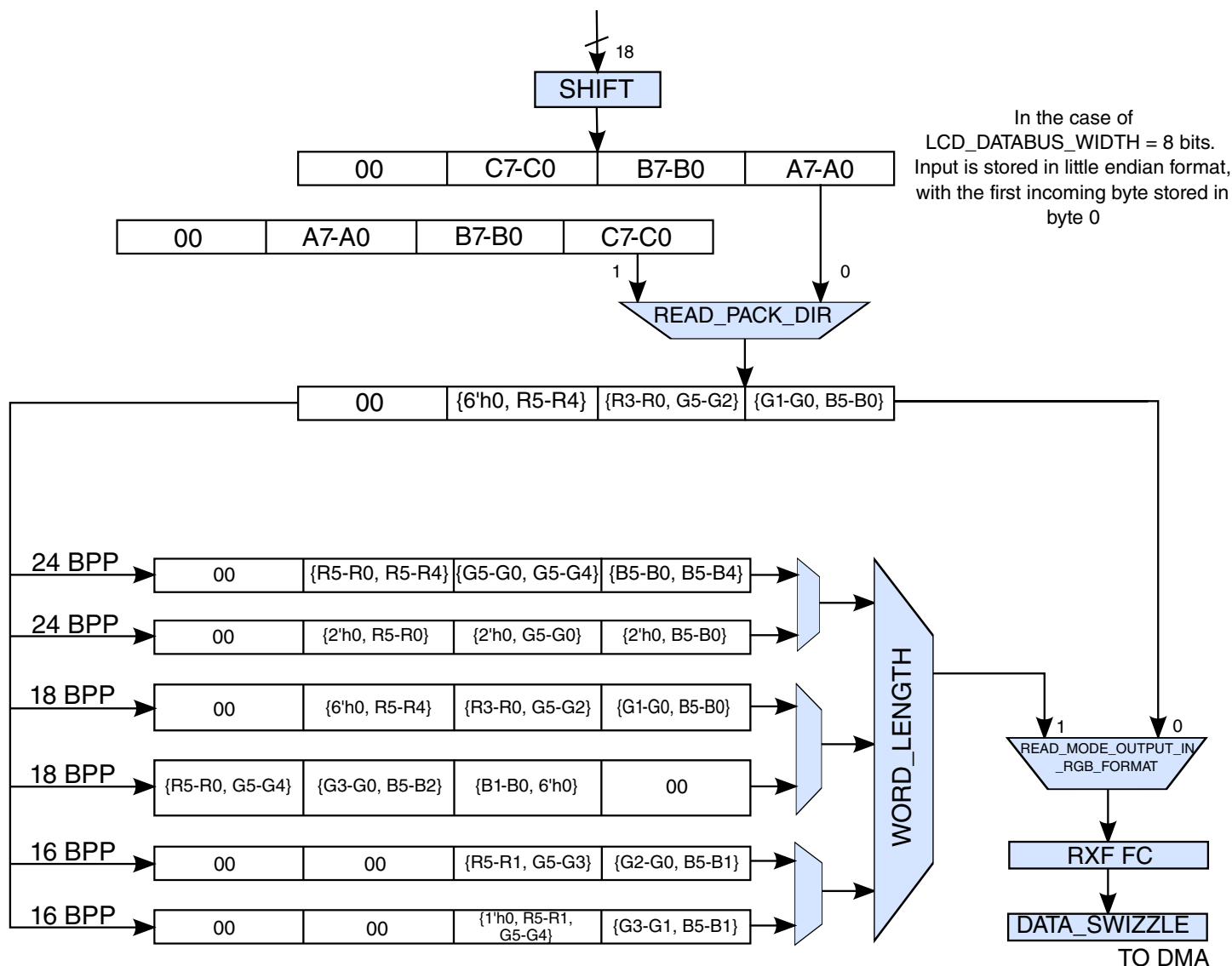


Figure 13-13. Data in MPU read mode

Restrictions:

READ_PACK_DIR should only be used if it is required to swizzle the subwords before doing RGB to RGB CSC, otherwise the DATA_SWIZZLE field should be used to swizzle across bytes.

READ_PACK_DIR must be 0 if LCD_DATABUS_WIDTH is 8 bits and READ_MODE_NUM_PACKED_SUBWORDS = 1

If READ_MODE_OUTPUT_IN_RGB_FORMAT bit is set, the following restrictions should be followed:

- If LCD_DATABUS_WIDTH = 8 bits, then
READ_MODE_NUM_PACKED_SUBWORDS <= 3.
- If LCD_DATABUS_WIDTH = 16/18/24 bits, then
READ_MODE_NUM_PACKED_SUBWORDS = 1.

13.2.3.4 LCDIF Interrupts

LCDIF supports a number of interrupts to aid controlling and status reporting of the block.

All the interrupts have individual mask bits for enabling or disabling each of them. They all get funneled through a single interrupt line connected to the interrupt collector (ICOLL).

The following list describes the different interrupts supported by LCDIF:

- Underflow interrupt is asserted when the clock domain crossing FIFO (TXFIFO) becomes empty but the block is in active display portion during that time. Software should take corrective action to make sure that this does not happen.
- In the bus master mode, the overflow interrupt will be asserted if the block has requested more data than its FIFOs could hold. In the read mode, it will be asserted if the RxFIFO becomes full and the block reads more data.
- VSYNC edge interrupt will be asserted every time a leading VSYNC edge occurs.
- Cur_frame_done interrupt occurs at the end of every frame in all modes except DVI. In DVI mode, if IRQ_ON_ALTERNATE_FIELDS bit is set, it will occur at the end of every frame, otherwise it will occur at the end of every field.

13.2.3.5 Initializing the LCDIF

This section describes write modes and MPU read mode.

13.2.3.5.1 Write Modes

The following initialization steps are common to all LCDIF write modes of operation before entering any particular mode.

Initialization steps:

1. Configure the external I/Os to correctly interface the external display, when required.
2. Start the DISPLAY CLOCK (pix_clk) clock and set the appropriate frequency by programming the registers in CCM.

3. Start the BUS CLOCK (apb_clk) and set the appropriate frequency by programming the registers in CCM.
4. Bring the LCDIF out of soft reset and disable the clock gate bit.
5. Reset the LCD controller by setting LCDIF_CTRL1[RESET] bit appropriately, being careful to observe the reset requirements of the controller. See [Behavior During Reset](#) for more information on Reset requirements.
6. Make sure LCDIF_CTRL[READ_WRITEB] bit is 0.
7. Set the transfer mode of operation to bus master. The LCDIF_CTRL[MASTER] bit determines the transfer mode selected. Bus master (LCDIF_CTRL[MASTER] =1), or PIO (LCDIF_CTRL[MASTER] =0) mode is the transfer mode to select.
8. Set the LCDIF_CTRL[INPUT_DATA_SWIZZLE] according to the endianness of the LCD controller. Also, set the LCDIF_CTRL[DATA_SHIFT_DIR] and LCDIF_CTRL[SHIFT_NUM_BITS] if it is required to shift the data left or right before it is output.
9. Set the LCDIF_CTRL[WORD_LENGTH] field appropriately: 0 = 16-bit input, 1 = 8-bit input, 2 = 18-bit input, 3 = 24/32-bit input. Also, select the correct 16/18/24 bit data format with the corresponding fields in LCDIF_CTRL register.
10. Set the LCDIF_CTRL1[BYTE_PACKING_FORMAT] field according to the input frame.
11. Set the LCDIF_CTRL[LCD_DATABUS_WIDTH] appropriately: 0 = 16-bit output, 1 = 8-bit output, 2 = 18-bit output, 3 = 24/32-bit output.
12. Enable the necessary IRQs.

13.2.3.5.2 MPU Read Mode

The following initialization steps should be done to enter the MPU read mode of operation:

Initialization steps:

1. Configure the external I/Os to correctly interface the external display.
2. Start the DISPLAY CLOCK (pix_clk) and set the appropriate frequency by programming the registers in CCM.
3. Start the BUS CLOCK (apb_clk) and set the appropriate frequency by programming the registers in CCM.
4. Bring the LCDIF out of soft reset and clock gate.
5. Reset the LCD controller by setting LCDIF_CTRL1_RESET bit appropriately, being careful to observe the reset requirements of the controller.
6. Set the READ_WRITEB bit in LCDIF_CTRL register to 1.
7. Set the LCDIF_MASTER bit in LCDIF_CTRL register to 0. Bus master mode is not supported for reading data from the display.

8. Also, set the DATA_SHIFT_DIR and SHIFT_NUM_BITS if it is required to shift the data left or right before it is output.
9. Indicate if the read data needs to color-space-converted and stored in a different RGB format by setting the READ_MODE_OUTPUT_IN_RGB_FORMAT field accordingly.
10. Set the WORD_LENGTH field appropriately: 0 = 16-bit input, 1 = 8-bit input, 2 = 18-bit input, 3 = 24-bit input if READ_MODE_OUTPUT_IN_RGB_FORMAT is required. Also, select the correct 16/18/24 bit data format with the corresponding fields in LCDIF_CTRL register.
11. Set the READ_MODE_NUM_PACKED_SUBWORDS field in LCDIF_CTRL2 according to the number of subwords per word required to be packed.
12. Set the READ_PACK_DIR to 1 if it is required to store the data in big-endian format.
13. Set the LCD_DATABUS_WIDTH appropriately: 0 = 16-bit output, 1 = 8-bit output, 2 = 18-bit output, 3 = 24-bit output.
14. Enable the necessary IRQs.

13.2.3.6 MPU Interface

The MPU interface is used to transfer data and commands between the SoC via the LCDIF and the external display at modest data rates.

Bus master or PIO transactions using the LCDIF_DATA register can be used for MPU mode write operations. For MPU mode read operations, only PIO can be used. LCDIF can support the 6800 as well as the 8080 MPU protocol. If DOTCLK_MODE, DVI_MODE and VSYNC_MODE bits in LCDIF_CTRL registers are 0, it implies that the block is in MPU interface mode of operation. The LCDIF MPU mode has four basic timing parameters: Setup and Hold for the Command/Data register selection (TCS, TCH) and Setup and Hold for the Data bus (TDS, TDH). These parameters are expressed in DISPLAY CLOCK (pix_clk) cycles. The LCD_WR signal is used as the write strobe while LCD_RS signal is typically used to switch between command and data modes.

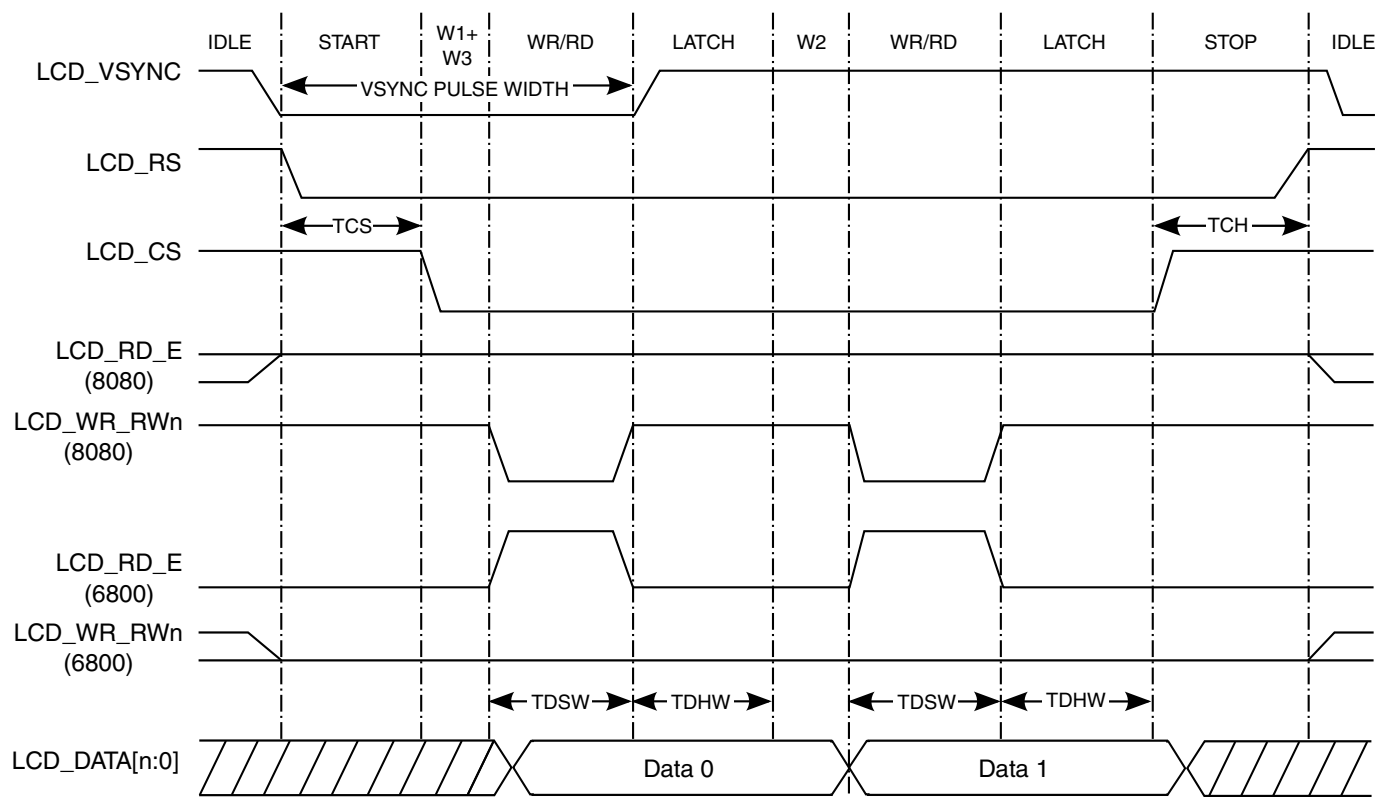


Figure 13-14. Timing in write mode of 6800 and 8080 protocols

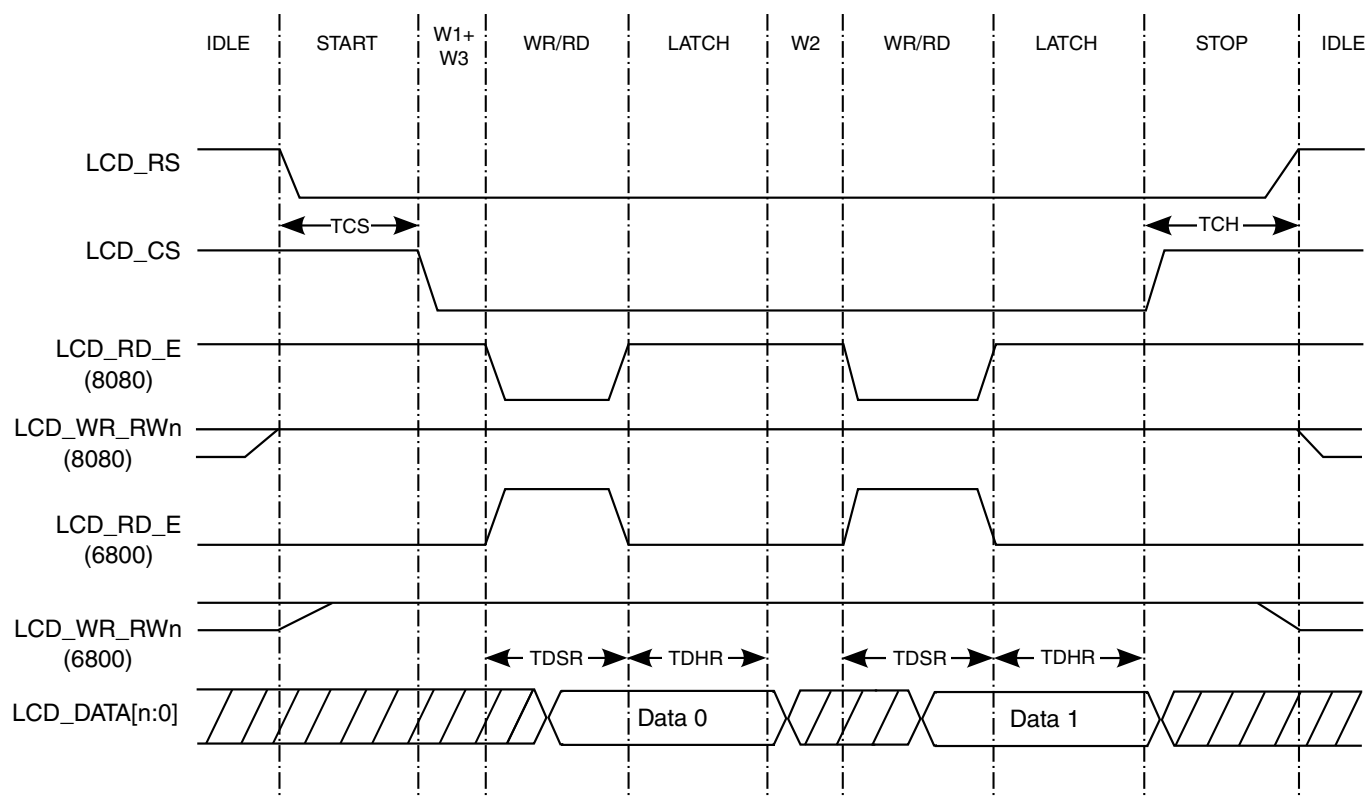


Figure 13-15. Read timing interface in 6800 and 8080 protocols

The LCDIF has flexible pin and strobe timings which enable it to optimally support a wide range of LCDs. The minimum cycle time is two DISPLAY CLOCK (pix_clk) cycles (TDS=TDH=1). For example, this results in a maximum LCD data rate of 12 MB/s when DISPLAY CLOCK (pix_clk) is 24 MHz. TDS and TDH are 8-bit values, so the minimum LCDIF period is 510 DISPLAY CLOCK (pix_clk) cycles (47 KHz with a 24 MHz DISPLAY CLOCK (pix_clk)). The timings are not automatically adjusted if the DISPLAY CLOCK (pix_clk) frequency changes, so it may be necessary to adjust the timings if DISPLAY CLOCK (pix_clk) changes.

In the MPU interface mode, the LCDIF_CTRL_BYPASS_COUNT bit must be 0. The RUN bit is cleared automatically once the LCDIF has received/transmitted all the data as per the LCDIF_TRANSFER_COUNT register and has completed the transfer to the panel. The current transfer can be cancelled/aborted if the RUN bit is manually made 0.

13.2.3.6.1 Code Example to Initialize the LCDIF in MPU Write Mode

```
// Note: Common initialization steps in Initializing the LCDIF must also be
// executed along with the following code
BF_CS1(LCDIF_CTRL, DATA_SELECT, 1); // 0 if sending command, 1 if sending data. Note that the
// idle state for LCD_RS signal is high, regardless of the
// programming of the DATA_SELECT register.
BF_CS1 (LCDIF_CTRL, MODE86, 8080_MODE);
BF_CS1 (LCDIF_CTRL, READ_WRITEB, 0);
BF_CS1 (LCDIF_CTRL, BYPASS_COUNT, 0); //Must be 0 in MPU mode
BF_CS1 (LCDIF_CTRL1, BUSY_ENABLE, 1); //Only if LCD controller implements a busy line
BF_CS4 (LCDIF_TIMING, CMD_HOLD, 2, CMD_SETUP, 2, DATA_HOLD, 2, DATA_SETUP, 2); //Values
based
// on DISPLAY CLOCK (pix_clk) frequency and timing requirements
of controller.
// Note that these register must be non-zero for correct
operation.
BF_CS2 (LCDIF_TRANSFER_COUNT, H_COUNT, 320, V_COUNT, 240); //For a 320 RGB x 240 display
BF_CS1 (LCDIF_CTRL, RUN, 1);
```

The LCDIF is now ready to receive data via bus master PIO write transactions using the LCDIF_DATA register. Note that when using the PIO write operations to the LCDIF_DATA register, the software will need to poll the FIFO STATUS bits to ensure that it does not overflow the LCDIF data buffers. When LCDIF is done transmitting H_COUNT x V_COUNT pixels, it will stop, turn off the RUN bit and assert the cur_frame_done interrupt.

13.2.3.7 VSYNC Interface

The VSYNC interface uses the same protocol as the MPU interface, with an additional signal VSYNC at the frame rate of the display, as shown in the figure given in MPU Interface section.

It is used in the moving picture display mode where data has to be written to the internal LCD buffer at a speed higher than the display rate and displayed in synchronization with the VSYNC signal. This mode is selected by setting the VSYNC_MODE bit in LCDIF_CTRL register. The VSYNC signal is programmable for period, polarity and direction. Many other programmable parameters are shared with the MPU interface. The VSYNC_OEB bit in LCDIF_VDCTRL0 register indicates whether the display controller will send the VSYNC signal, or whether it should be generated by LCDIF. The timing of the VSYNC signal is based on the DISPLAY CLOCK (pix_clk) (make sure VSYNC_PULSE_WIDTH_UNIT = VSYNC_PERIOD_UNIT = 0 and VSYNC_ONLY = 1) and it is determined by the VSYNC_PERIOD, VSYNC_PULSE_WIDTH and VSYNC_POL fields in LCDIF_VDCTRL0-4 registers. The SYNC_SIGNALS_ON bit in LCDIF_VDCTRL4 register must be set if the target requires the VSYNC signal to be generated by LCDIF. If the WAIT_FOR_VSYNC_EDGE bit in LCDIF_CTRL register is set, it indicates that the hardware should wait until it sees the leading VSYNC edge before starting the data transfer. The VERTICAL_WAIT_CNT indicates the number of DISPLAY CLOCK (pix_clk) cycles from the leading VSYNC edge after which data transfer will be started on the interface.

In the VSYNC interface mode, the LCDIF_CTRL_BYPASS_COUNT bit must be 0. The RUN bit is cleared automatically once the LCDIF has received/transmitted all the data as per the LCDIF_TRANSFER_COUNT register and has completed the transfer to the panel. The current transfer can be cancelled/aborted if the RUN bit is manually made 0.

13.2.3.7.1 Code Example to Initialize LCDIF in VSYNC Mode

```
// Note: Common initialization steps in Initializing the LCDIF must also be
// executed along with the following code
BF_CS1 (LCDIF_CTRL, DATA_SELECT, 1); // 0 if sending command, 1 if sending data. Note that
//the idle state for LCD_RS signal is high, regardless of the programming of the DATA_SELECT
//register.

BF_CS1 (LCDIF_CTRL, MODE86, 8080_MODE);
BF_CS1 (LCDIF_CTRL, BYPASS_COUNT, 0); //Must be 0 in MPU mode
BF_CS1 (LCDIF_CTRL1, BUSY_ENABLE, 0);
BF_CS4 (LCDIF_TIMING, CMD_HOLD, 2, CMD_SETUP, 2, DATA_HOLD, 2, DATA_SETUP, 2); //Values
//based on DISPLAY CLOCK (pix_clk) frequency and timing requirements of controller. Note
that these
//register must be non-zero for the MPU and VSYNC modes.
BF_CS2 (LCDIF_TRANSFER_COUNT, H_COUNT, 320, V_COUNT, 240); //For a 320 RGB x 240 display
//The following section indicates setting up the VSYNC signal timing when VSYNC is an output
BF_CS1 (LCDIF_VDCTRL0, VSYNC_OEB, 0); //Making VSYNC signal an output
BF_CS1 (LCDIF_VDCTRL4, VSYNC_ONLY, 1); //Only need to generate VSYNC signal
BF_CS1 (VDCTRL0, VSYNC_POL, 0); //Setting the polarity of VSYNC signal to be low during
//VSYNC_PULSE_WIDTH time
BF_CS2 (LCDIF_VDCTRL0, VSYNC_PERIOD_UNIT, 0, VSYNC_PULSE_WIDTH_UNIT, 0);
BF_CS2 (LCDIF_VDCTRL1, VSYNC_PERIOD, 400000, VSYNC_PULSE_WIDTH, 100); //Frame display rate in
//terms of number of DISPLAY CLOCKS (pix_clk).
BF_CS2 (LCDIF_VDCTRL2, HSYNC_PULSE_WIDTH, 0, HSYNC_PERIOD, 0);
BF_CS1 (LCDIF_VDCTRL3, VERTICAL_WAIT_CNT, 50);
BF_CS1 (LCDIF_VDCTRL4, SYNC_SIGNALS_ON, 1);
BF_CS2 (LCDIF_CTRL, VSYNC_MODE, 1, WAIT_FOR_VSYNC_EDGE, 1); //set WAIT_FOR_VSYNC_EDGE if
```

```
//software wishes to transfer the next frame after the VSYNC edge occurs.
BF_CS1 (LCDIF_CTRL, RUN, 1);
```

The LCDIF is now ready to receive data via bus master requests or PIO writes to the LCDIF_DATA register. When LCDIF is done transmitting $H_COUNT \times V_COUNT$ pixels, it will stop, turn off the RUN bit and assert the `cur_frame_done` interrupt.

13.2.3.8 DOTCLK Interface

The DOTCLK interface is another mode used in moving picture displays.

It includes the VSYNC, HSYNC, DOTCLK and (optional) ENABLE signals. The interface is popularly called the RGB interface if the ENABLE signal is present.

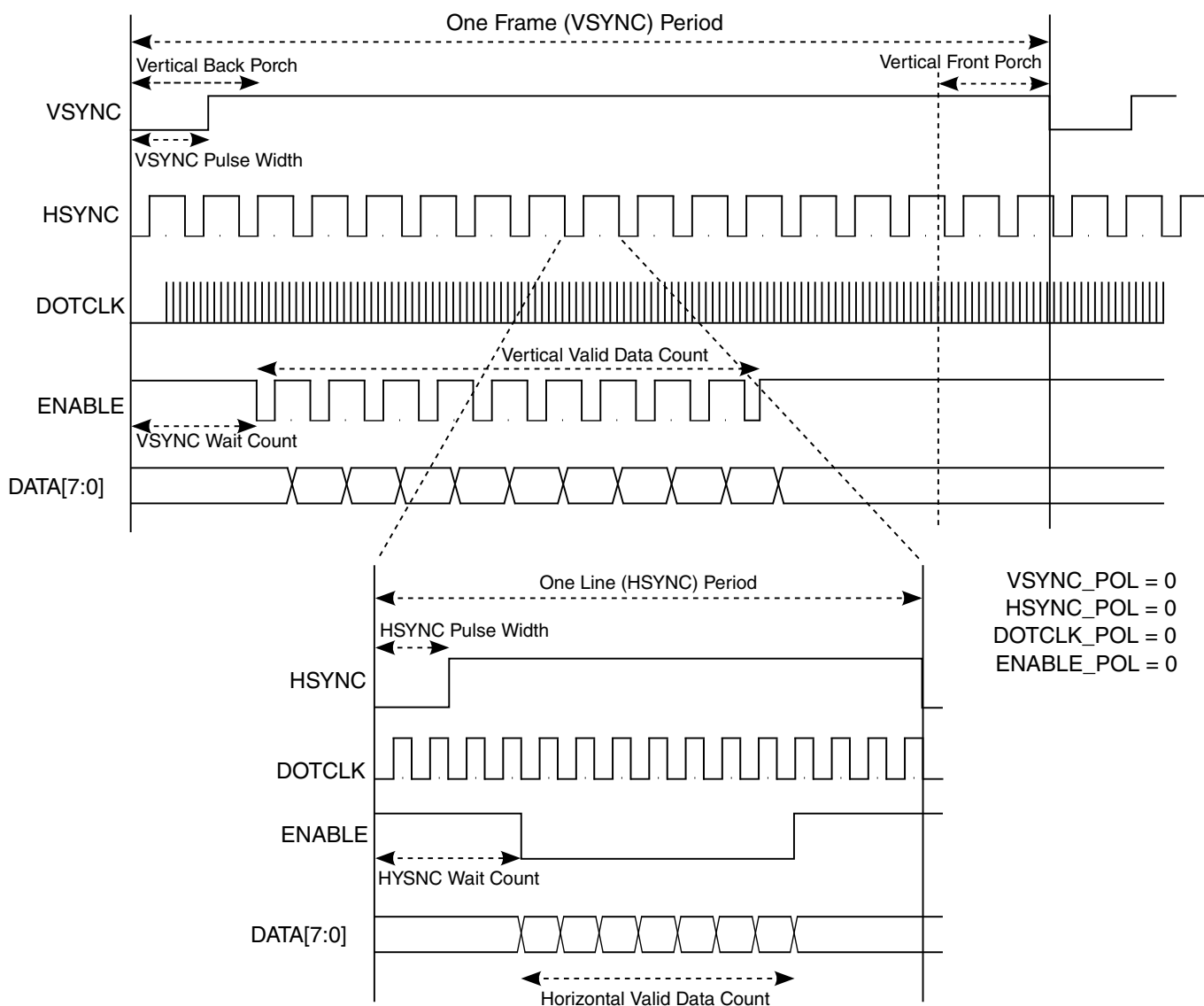


Figure 13-16. DOTCLK protocol with programmable parameters

The DOTCLK mode writes data at high speed to the LCD, and the display operation is synchronized with the VSYNC, HSYNC, ENABLE and DOTCLK signals. The polarities, periods and pulse-widths of the sync signals are programmable using the LCDIF_VDCTRL0-4 registers. The units for the VSYNC signal must be number of horizontal lines and can be selected using the VSYNC_PULSE_WIDTH_UNIT and VSYNC_PERIOD_UNIT bit fields. The VERTICAL_WAIT_CNT is by default given the same unit as the VSYNC_PERIOD. The DISPLAY CLOCK (pix_clk) frequency is managed by the CCM.

In DOTCLK mode, LCDIF_CTRL_BYPASS_COUNT bit must be set to 1. To end the current transfer, the software should make the DOTCLK_MODE bit 0, so that all data that is currently in the LCDIF LFIFO and TXFIFO is transmitted. Once that transfer is complete, the block will automatically clear the RUN bit and issue the cur_frame_done interrupt.

13.2.3.8.1 Code Example

The following code shows an example for programming a 320x240 display.

NOTE

Setting up the display must be done through the MPU mode or via SPI.

```
// Note: Common initialization steps in Initializing the LCDIF must also be
// executed along with the following code
BF_CS1 (LCDIF_CTRL, DOTCLK_MODE, 1);
BF_CS1 (LCDIF_CTRL, BYPASS_COUNT, 1); //Always for DOTCLK mode
BF_CS1 (LCDIF_VDCTRL0, VSYNC_OEB, 0); //Vsync is always an output in the DOTCLK mode
BF_CS4 (LCDIF_VDCTRL0, VSYNC_POL, 0, HSYNC_POL, 0, DOTCLK_POL, 0, ENABLE_POL, 0);
BF_CS1 (LCDIF_VDCTRL0, ENABLE_PRESENT, 1);
BF_CS2 (LCDIF_VDCTRL0, VSYNC_PERIOD_UNIT, 1, VSYNC_PULSE_WIDTH_UNIT, 1);
BF_CS1 (LCDIF_VDCTRL0, VSYNC_PULSE_WIDTH, 2);
BF_CS1 (LCDIF_VDCTRL1, VSYNC_PERIOD, 280);
BF_CS2 (LCDIF_VDCTRL2, HSYNC_PULSE_WIDTH, 10, HSYNC_PERIOD, 360); //Assuming
                                                                    // LCD_DATABUS_WIDTH is 24bit
BF_CS2 (LCDIF_VDCTRL3, VSYNC_ONLY, 0);
BF_CS2 (LCDIF_VDCTRL3, HORIZONTAL_WAIT_CNT, 20, VERTICAL_WAIT_CNT, 20);
BF_CS1 (LCDIF_VDCTRL4, DOTCLK_H_VALID_DATA_CNT, 320); //Note that DOTCLK_V_VALID_DATA_CNT is
                                                                    //implicitly assumed to be HW_LCDIF_TRANSFER_COUNT_V_COUNT
BF_CS1 (LCDIF_VDCTRL4, SYNC_SIGNALS_ON, 1);
BF_CS1 (LCDIF_CTRL, RUN, 1);
```

To stop the transfer completely, the ideal way is to make DOTCLK_MODE = 0. In that case, the block will transmit the contents in the FIFO and reset the RUN bit.

13.2.3.9 Alpha Blending Interface

The LCDIF have the capability to add an extra overlay on the normal display buffer, LCDIF can fetch data from two buffers and combine them before display, one buffer data can have the alpha value with the RGB pixels. With LCDIF_AS_CTRL[AS_ENABLE] is set, the LCDIF will start fetching alpha surface buffer data in bus master mode and combine it with another buffer.

The LCDIF_AS_CTRL[ALPHA_CTRL] bits determines how the alpha value is constructed for the alpha surface.

13.2.3.9.1 Alpha Blending/Color Key

Regardless of pixel input format, the PS and AS pixels are normalized to 32-bits, organized as one alpha and three data bytes. Alpha blending occurs in the RGB space, if blending is required, PS pixels should be converted to RGB space. If no alpha blending is required, then YUV pixels can bypass the alpha blending ALU without color space conversion.

13.2.3.9.2 Alpha Blend

The alpha value for an individual pixel represents a mathematical weighting factor applied to the AS pixel. An alpha value of 0x00 corresponds to a transparent pixel and a value of 0xFF corresponds to an opaque pixel.

The effective alpha value for an AS pixel is determined by the AS_CTRL[ALPHA] and AS_CTRL[ALPHA_CTRL] register fields. If AS_CTRL[ALPHA_CTRL] = ALPHA_OVERRIDE, the alpha value for the pixel is taken from the AS_CTRL[ALPHA]. This can be useful for applying a constant alpha to an entire image or for image formats that don't include an alpha value. If AS_CTRL[ALPHA_CTRL] = ALPHA_MULTIPLY, the pixel's alpha value will be multiplied by the pixel's ALPHA value in order to allow scaling of the pixel's alpha or to provide better control for pixel formats such as RGB1555, which only contains a single bit of alpha.

For each color channel, the equation used to blend two source pixels is defined below:

$G\alpha$ = PIO programmed global alpha (8-bit value).

$E\alpha$ = Embedded alpha associated with AS pixel.

$$\acute{\alpha} = G\alpha * E\alpha + 0x80$$

The result for the red channel as an example:

$$R[7:0] = (\acute{\alpha} * PS.r) + ((1 - \acute{\alpha}) * AS.r)$$

When α is 0xff, the PS pixel will not be blended with the AS pixel, but PS will be passed as the output pixel and will not be blended with AS. In this case, AS will be discarded. Likewise, if α is 0x00 for a given pixel, PS will be loaded as the output pixel.

AS_CTRL[ALPHA_INVERT] provides the option to invert the final alpha value. This essentially inverts the effect the alpha value has on the AS and PS blending operation.

13.2.3.9.3 Color Key

The color key function is provided to create transparent effects on the output pixel.

Color keying is applied on the input pixels after they are converted to 8-bits for each red, green, and blue color channels (color keys are not applied directly to 16-bit pixel formats but to their corresponding 24-bit representation). A color key range is programmable for both PS and AS pixels. If the PS 24-bit pixel is within the PS color key range, then AS is passed through the pixel pipeline. In this case, alpha blending does NOT occur.

Conversely, if PS is within the AS color key range, then PS is passed via the LCDIF data pipeline. If both PS and AS color key tests pass, then the back ground color register is passed onto following LCDIF processing components in the pipeline.

The condition for color keying to be satisfied is:

$$CK0.r.low \leq PS.r \leq CK0.r.high$$

$$CK0.g.low \leq PS.g \leq CK0.g.high$$

$$CK0.b.low \leq PS.b \leq CK0.b.high$$

For example, if the "red" 8-bit value for the PS pixel (or PS.r) is between the color key low and high values (CK0.r.l and CK0.r.h), the condition is true for the red color plane. When ALL three color planes meet this condition, then only the PS pixel is loaded into the output register.

To disable color keying, program the low color key register value to 0xff and the high value to 0x00. This will guarantee that the color key range test will never be true.

13.2.3.9.4 Color Key Processing (AS_CTRL)

The AS_CTRL register also contains an ENABLE_COLORKEY bit that can be used to enable or disable color key substitution for the AS.

When enabled, the pixel values are compared to the ASCOLORKEYLOW and ASCOLORKEYHIGH registers to determine if a match has occurred. When an AS pixel matches the color key range, the pixel from the AS image is considered transparent and the corresponding PS pixel is rendered. If both the PS and AS pixels match their corresponding color key ranges, the AS pixel is displayed unmodified.

AS color keys are handled in a manner similar to PS color keys. The same images used in the PS color key example could be used with the images swapped. In this case, matches on the AS image to the ASCOLORKEY register would display the PS pixels.

13.2.3.10 ITU-R BT.656 Digital Video Interface (DVI)

ITU-R BT.656 Digital Video Interface shown below transmits 4:2:2 YCbCr digital component video to a digital video encoder that can translate it into 525/60 or 625/50 analog TV signal.

Unique timing codes (timing reference signals) are embedded within the video stream to indicate the different timing events that would have been otherwise indicated by VSYNC, HSYNC and BLANK signals. The hardware supports 8-bit data transfers; the pins are shared with the lower 8 bits of LCD data bus. The LCD_RS pin is shared with the clock signal of the interface (called CCIRCLK here for uniqueness). CCIRCLK also can be obtained on the LCD_DOTCLK pin. The mode shares the write FIFO with the LCD interface and the associated pipeline. The programmable parameters in registers LCDIF_DVICTRL0-3 allow setting the total number of horizontal lines per frame, vertical and horizontal blanking interval, odd and even field start and end positions, and so on. In short, these parameters are provided to ensure that the hardware has enough flexibility to generate the right 525/60 or 625/50 data streams. Most of the initialization steps in [Initializing the LCDIF](#) such as data shifting, swizzle, and so on, are applicable to DVI mode also. The register descriptions in the programmable registers section at the end of this chapter include example code for programming the DVICTRL0-3 registers.

In DVI mode, LCDIF_CTRL_BYPASS_COUNT bit must be set to 1. To end the current transfer, the software should make the DVI_MODE bit the value 0, so that all data that is currently in the LCDIF LFIFO and TXFIFO is transmitted. Once that transfer is complete, the block will automatically clear the RUN bit and assert the cur_frame_done interrupt.

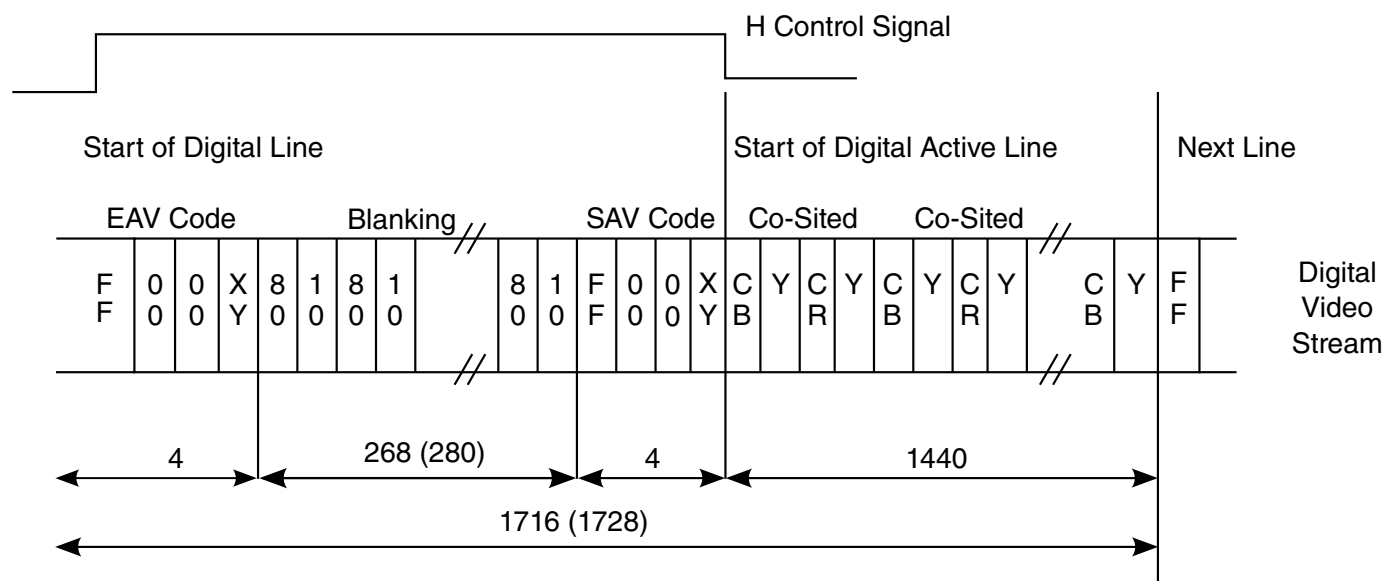


Figure 13-17. Digital Video Interface

13.2.4 Behavior During Reset

BUS CLOCK (apb_clk) and DISPLAY CLOCK (pix_clk) must be running before making any changes to SFTRST or CLKGATE bits.

A soft reset (SFTRST) can take multiple clock periods to complete, so do not set CLKGATE when setting SFTRST.

The reset process gates the clocks automatically.

13.2.5 LCDIF Memory Map/Register Definition

Some of the LCDIF registers (XXX_SET, XXX_CLR, and XXX_TOG) allow direct bit field masking and access.

- When writing 1 to XXX_SET bit fields, these registers allow setting the masked 1 bit fields, while keeping unchanged all bit fields which remain on 0 logic state.

- When writing 1 to XXX_CLR bit fields, these registers allow clearing the masked 1 bit fields, while keeping unchanged all other bit fields which remained on 0 logic state.
- When writing 1 to XXX_TOG bit fields, these registers allow inverting the logic state of all masked 1 bit fields, while they keep unchanged the remaining bit fields which were kept on 0 logic state.

LCDIF Hardware Register Format Summary

LCDIF memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
3032_0000	LCDIF General Control Register (LCDIF_CTRL)	32	R/W	C000_0000h	13.2.5.1/3901
3032_0004	LCDIF General Control Register (LCDIF_CTRL_SET)	32	R/W	C000_0000h	13.2.5.1/3901
3032_0008	LCDIF General Control Register (LCDIF_CTRL_CLR)	32	R/W	C000_0000h	13.2.5.1/3901
3032_000C	LCDIF General Control Register (LCDIF_CTRL_TOG)	32	R/W	C000_0000h	13.2.5.1/3901
3032_0010	LCDIF General Control1 Register (LCDIF_CTRL1)	32	R/W	000F_0000h	13.2.5.2/3904
3032_0014	LCDIF General Control1 Register (LCDIF_CTRL1_SET)	32	R/W	000F_0000h	13.2.5.2/3904
3032_0018	LCDIF General Control1 Register (LCDIF_CTRL1_CLR)	32	R/W	000F_0000h	13.2.5.2/3904
3032_001C	LCDIF General Control1 Register (LCDIF_CTRL1_TOG)	32	R/W	000F_0000h	13.2.5.2/3904
3032_0020	LCDIF General Control2 Register (LCDIF_CTRL2)	32	R/W	0020_0000h	13.2.5.3/3906
3032_0024	LCDIF General Control2 Register (LCDIF_CTRL2_SET)	32	R/W	0020_0000h	13.2.5.3/3906
3032_0028	LCDIF General Control2 Register (LCDIF_CTRL2_CLR)	32	R/W	0020_0000h	13.2.5.3/3906
3032_002C	LCDIF General Control2 Register (LCDIF_CTRL2_TOG)	32	R/W	0020_0000h	13.2.5.3/3906
3032_0030	LCDIF Horizontal and Vertical Valid Data Count Register (LCDIF_TRANSFER_COUNT)	32	R/W	0001_0000h	13.2.5.4/3909
3032_0040	LCD Interface Current Buffer Address Register (LCDIF_CUR_BUF)	32	R/W	0000_0000h	13.2.5.5/3909
3032_0050	LCD Interface Next Buffer Address Register (LCDIF_NEXT_BUF)	32	R/W	0000_0000h	13.2.5.6/3910
3032_0060	LCD Interface Timing Register (LCDIF_TIMING)	32	R/W	0000_0000h	13.2.5.7/3910

Table continues on the next page...

LCDIF memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
3032_0070	LCDIF VSYNC Mode and Dotclk Mode Control Register0 (LCDIF_VDCTRL0)	32	R/W	0000_0000h	13.2.5.8/3911
3032_0074	LCDIF VSYNC Mode and Dotclk Mode Control Register0 (LCDIF_VDCTRL0_SET)	32	R/W	0000_0000h	13.2.5.8/3911
3032_0078	LCDIF VSYNC Mode and Dotclk Mode Control Register0 (LCDIF_VDCTRL0_CLR)	32	R/W	0000_0000h	13.2.5.8/3911
3032_007C	LCDIF VSYNC Mode and Dotclk Mode Control Register0 (LCDIF_VDCTRL0_TOG)	32	R/W	0000_0000h	13.2.5.8/3911
3032_0080	LCDIF VSYNC Mode and Dotclk Mode Control Register1 (LCDIF_VDCTRL1)	32	R/W	0000_0000h	13.2.5.9/3912
3032_0090	LCDIF VSYNC Mode and Dotclk Mode Control Register2 (LCDIF_VDCTRL2)	32	R/W	0000_0000h	13.2.5.10/3913
3032_00A0	LCDIF VSYNC Mode and Dotclk Mode Control Register3 (LCDIF_VDCTRL3)	32	R/W	0000_0000h	13.2.5.11/3913
3032_00B0	LCDIF VSYNC Mode and Dotclk Mode Control Register4 (LCDIF_VDCTRL4)	32	R/W	0000_0000h	13.2.5.12/3914
3032_00C0	Digital Video Interface Control0 Register (LCDIF_DVICTRL0)	32	R/W	0000_0000h	13.2.5.13/3915
3032_00D0	Digital Video Interface Control1 Register (LCDIF_DVICTRL1)	32	R/W	0000_0000h	13.2.5.14/3916
3032_00E0	Digital Video Interface Control2 Register (LCDIF_DVICTRL2)	32	R/W	0000_0000h	13.2.5.15/3917
3032_00F0	Digital Video Interface Control3 Register (LCDIF_DVICTRL3)	32	R/W	0000_0000h	13.2.5.16/3918
3032_0100	Digital Video Interface Control4 Register (LCDIF_DVICTRL4)	32	R/W	0000_0000h	13.2.5.17/3919
3032_0110	RGB to YCbCr 4:2:2 CSC Coefficient0 Register (LCDIF_CSC_COEFF0)	32	R/W	0000_0000h	13.2.5.18/3920
3032_0120	RGB to YCbCr 4:2:2 CSC Coefficient1 Register (LCDIF_CSC_COEFF1)	32	R/W	0000_0000h	13.2.5.19/3921
3032_0130	RGB to YCbCr 4:2:2 CSC Coefficient2 Register (LCDIF_CSC_COEFF2)	32	R/W	0000_0000h	13.2.5.20/3921
3032_0140	RGB to YCbCr 4:2:2 CSC Coefficient3 Register (LCDIF_CSC_COEFF3)	32	R/W	0000_0000h	13.2.5.21/3922
3032_0150	RGB to YCbCr 4:2:2 CSC Coefficient4 Register (LCDIF_CSC_COEFF4)	32	R/W	0000_0000h	13.2.5.22/3923
3032_0160	RGB to YCbCr 4:2:2 CSC Offset Register (LCDIF_CSC_OFFSET)	32	R/W	0080_0010h	13.2.5.23/3924
3032_0170	RGB to YCbCr 4:2:2 CSC Limit Register (LCDIF_CSC_LIMIT)	32	R/W	00FF_00FFh	13.2.5.24/3924
3032_0180	LCD Interface Data Register (LCDIF_DATA)	32	R/W	0000_0000h	13.2.5.25/3925
3032_0190	Bus Master Error Status Register (LCDIF_BM_ERROR_STAT)	32	R/W	0000_0000h	13.2.5.26/3926

Table continues on the next page...

LCDIF memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
3032_01A0	CRC Status Register (LCDIF_CRC_STAT)	32	R/W	0000_0000h	13.2.5.27/3926
3032_01B0	LCD Interface Status Register (LCDIF_STAT)	32	R	9500_0000h	13.2.5.28/3927
3032_0200	LCDIF Threshold Register (LCDIF_THRES)	32	R/W	0100_000Fh	13.2.5.29/3929
3032_0210	LCDIF AS Buffer Control Register (LCDIF_AS_CTRL)	32	R/W	0000_0000h	13.2.5.30/3930
3032_0220	Alpha Surface Buffer Pointer (LCDIF_AS_BUF)	32	R/W	0000_0000h	13.2.5.31/3932
3032_0230	LCDIF_AS_NEXT_BUF	32	R/W	0000_0000h	13.2.5.32/3933
3032_0240	LCDIF Overlay Color Key Low (LCDIF_AS_CLRKEYLOW)	32	R/W	00FF_FFFFh	13.2.5.33/3933
3032_0250	LCDIF Overlay Color Key High (LCDIF_AS_CLRKEYHIGH)	32	R/W	0000_0000h	13.2.5.34/3934
3032_0260	LCD working insync mode with CSI for VSYNC delay (LCDIF_SYNC_DELAY)	32	R/W	0000_0000h	13.2.5.35/3934

13.2.5.1 LCDIF General Control Register (LCDIF_CTRLn)

The LCD Interface Control Register provides overall control of the LCDIF block. The LCDIF Control Register provides a variety of control functions to the programmer. These functions allow the interface to be very flexible to work with a variety of LCD controllers, and to minimize overhead and increase performance of LCD programming. The register has been organized such that switching between the different LCD modes can be done with minimum PIO writes.

Address: 3032_0000h base + 0h offset + (4d × i), where i=0d to 3d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SFTRST	CLKGATE	YCBCR422_INPUT	READ_WRITEB	WAIT_FOR_VSYNC_EDGE	DATA_SHIFT_DIR	SHIFT_NUM_BITS					DVI_MODE	BYPASS_COUNT	VSYNC_MODE	DOTCLK_MODE	DATA_SELECT
W																
Reset	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	INPUT_DATA_SWIZZLE		CSC_DATA_SWIZZLE		LCD_DATABUS_WIDTH		WORD_LENGTH		RGB_TO_YCBCR422_CSC	Reserved	MASTER	Reserved	DATA_FORMAT_16_BIT	DATA_FORMAT_18_BIT	DATA_FORMAT_24_BIT	RUN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_CTRLn field descriptions

Field	Description
31 SFTRST	This bit must be set to zero to enable normal operation of the LCDIF. When set to one, it forces a block level reset.
30 CLKGATE	This bit must be set to zero for normal operation. When set to one it gates off the clocks to the block.
29 YCBCR422_INPUT	Zero implies input data is in RGB color space. One implies input data is in YCbCr 4:2:2 format, such that YCbYCr are packed in a 32-bit word. It also means that there are 2 pixels in 4 bytes. If this bit is set, software should program the H_COUNT field in the TRANSFER_COUNT register to the total number of pixels that will have to be fetched by the LCDIF block per line and the BYTE_PACKING_FORMAT should be 0xF. The WORD_LENGTH does not matter in this case.

Table continues on the next page...

LCDIF_CTRLn field descriptions (continued)

Field	Description
28 READ_WRITEB	By default, LCDIF is in the write mode. Setting this bit to 1 will make the hardware go into 6800/8080 MPU read mode. The LCDIF_MASTER bit must be 0, since bus master mode can only be used for writing the display.
27 WAIT_FOR_VSYNC_EDGE	Setting this bit to 1 will make the hardware wait for the triggering VSYNC edge before starting write transfers to the LCD. Used only in the VSYNC mode of operation.
26 DATA_SHIFT_DIR	Use this bit to determine the direction of shift of transmit data. In the DVI mode, it works only on the active data, not on the timing codes and ancillary data. 0x0 TXDATA_SHIFT_LEFT — Data to be transmitted is shifted LEFT by SHIFT_NUM_BITS bits. 0x1 TXDATA_SHIFT_RIGHT — Data to be transmitted is shifted RIGHT by SHIFT_NUM_BITS bits.
25–21 SHIFT_NUM_BITS	The data to be transmitted is shifted left or right by this number of bits.
20 DVI_MODE	Set this bit to 1 to get into the ITU-R BT.656 digital video interface mode. Toggle this bit from 1 to 0 to make the hardware go out of DVI mode after completing all data transfer and after the RUN bit has been deasserted.
19 BYPASS_COUNT	When this bit is 0, it means that LCDIF will stop the block operation and turn off the RUN bit after the amount of data indicated by the LCDIF_TRANSFER_COUNT register has been transferred out. When this bit is set to 1, the block will continue normal operation indefinitely until it is told to stop. This bit must be 0 in MPU and VSYNC modes, and must be 1 in DOTCLK and DVI modes of operation.
18 VSYNC_MODE	Setting this bit to 1 will make the LCDIF hardware go into VSYNC mode. WAIT_FOR_VSYNC_EDGE can be used only if this bit is set. If VSYNC signal is required to be an output from the block, SYNC_SIGNALS_ON bit in LCDIF_VDCTRL4 register must be set.
17 DOTCLK_MODE	Set this bit to 1 to make the hardware go into the DOTCLK mode, i.e. VSYNC/HSYNC/DOTCLK/ENABLE interface mode. ENABLE is optional, selected by the ENABLE_PRESENT bit. Toggle this bit from 1 to 0 to make the hardware go out of DOTCLK mode after completing all data transfer and deasserting the RUN bit.
16 DATA_SELECT	Command Mode polarity bit. This bit should only be changed when RUN is 0. 0x0 CMD_MODE — Command Mode. LCD_RS signal is Low. 0x1 DATA_MODE — Data Mode. LCD_RS signal is High.
15–14 INPUT_DATA_SWIZZLE	This field specifies how to swap the bytes fetched by the bus master interface. The swizzle function is independent of the WORD_LENGTH bit. The supported swizzle configurations are: 0x0 NO_SWAP — No byte swapping.(Little endian) 0x0 LITTLE_ENDIAN — Little Endian byte ordering (same as NO_SWAP). 0x1 BIG_ENDIAN_SWAP — Big Endian swap (swap bytes 0,3 and 1,2). 0x1 SWAP_ALL_BYTES — Swizzle all bytes, swap bytes 0,3 and 1,2 (aka Big Endian). 0x2 HWD_SWAP — Swap half-words. 0x3 HWD_BYTE_SWAP — Swap bytes within each half-word.
13–12 CSC_DATA_SWIZZLE	This field specifies how to swap the bytes after the data has been converted into an internal representation of 24 bits per pixel and before it is transmitted over the LCD interface bus. The data is always transmitted with the least significant byte/hword (half word) first after the swizzle takes place. So, INPUT_DATA_SWIZZLE takes place first on the incoming data, and then CSC_DATA_SWIZZLE is applied. The swizzle function is independent of the WORD_LENGTH or the LCD_DATABUS_WIDTH fields. If RGB_TO_YCRCB422_CSC bit is set, the swizzle occurs on the Y, Cb, Cr values. The supported swizzle configurations are: 0x0 NO_SWAP — No byte swapping.(Little endian)

Table continues on the next page...

LCDIF_CTRLn field descriptions (continued)

Field	Description
	0x0 LITTLE_ENDIAN — Little Endian byte ordering (same as NO_SWAP). 0x1 BIG_ENDIAN_SWAP — Big Endian swap (swap bytes 0,3 and 1,2). 0x1 SWAP_ALL_BYTES — Swizzle all bytes, swap bytes 0,3 and 1,2 (aka Big Endian). 0x2 HWD_SWAP — Swap half-words. 0x3 HWD_BYTE_SWAP — Swap bytes within each half-word.
11–10 LCD_DATABUS_ WIDTH	LCD Data bus transfer width. 0x0 16_BIT — 16-bit data bus mode. 0x1 8_BIT — 8-bit data bus mode. 0x2 18_BIT — 18-bit data bus mode. 0x3 24_BIT — 24-bit data bus mode.
9–8 WORD_LENGTH	Input data format. 0x0 16_BIT — Input data is 16 bits per pixel. 0x1 8_BIT — Input data is 8 bits wide. 0x2 18_BIT — Input data is 18 bits per pixel. 0x3 24_BIT — Input data is 24 bits per pixel.
7 RGB_TO_ YCBCR422_CSC	Set this bit to 1 to enable conversion from RGB to YCbCr colorspace. See the LCDIF_CSC_ registers for further details.
6 -	This field is reserved. Reserved
5 MASTER	Set this bit to make the LCDIF act as a bus master.
4 RSRVD0	This field is reserved. Reserved bits. Write as 0.
3 DATA_ FORMAT_16_ BIT	When this bit is 1 and WORD_LENGTH = 0, it implies that the 16-bit data is in ARGB555 format. When this bit is 0 and WORD_LENGTH = 0, it implies that the 16-bit data is in RGB565 format. When WORD_LENGTH is not 0, this bit does not care.
2 DATA_ FORMAT_18_ BIT	Used only when WORD_LENGTH = 2, i.e. 18-bit. 0x0 LOWER_18_BITS_VALID — Data input to the block is in 18 bpp format, such that lower 18 bits contain RGB 666 and upper 14 bits do not contain any useful data. 0x1 UPPER_18_BITS_VALID — Data input to the block is in 18 bpp format, such that upper 18 bits contain RGB 666 and lower 14 bits do not contain any useful data.
1 DATA_ FORMAT_24_ BIT	Used only when WORD_LENGTH = 3, i.e. 24-bit. Note that this applies to both packed and unpacked 24-bit data. 0x0 ALL_24_BITS_VALID — Data input to the block is in 24 bpp format, such that all RGB 888 data is contained in 24 bits. 0x1 DROP_UPPER_2_BITS_PER_BYTE — Data input to the block is actually RGB 18 bpp, but there is 1 color per byte, hence the upper 2 bits in each byte do not contain any useful data, and should be dropped.
0 RUN	When this bit is set by software, the LCDIF will begin transferring data between the SoC and the display. This bit must remain set until the operation is complete.

13.2.5.2 LCDIF General Control1 Register (LCDIF_CTRL1n)

The LCDIF Control Register provides overall control of the LCDIF block.

The LCDIF Control1 Register provides additional programming to the LCDIF. It implements some bits which are unlikely to change often in a particular application. It also carries interrupt-related bits which are common across more than one mode of operation.

Address: 3032_0000h base + 10h offset + (4d × i), where i=0d to 3d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved	Reserved	Reserved			COMBINE_MPU_WR_STRB	BM_ERROR_IRQ_EN	BM_ERROR_IRQ	RECOVER_ON_UNDERFLOW	INTERLACE_FIELDS	START_INTERLACE_FROM_SECOND_FIELD	FIFO_CLEAR	IRQ_ON_ALTERNATE_FIELDS	BYTE_PACKING_FORMAT		
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	OVERFLOW_IRQ_EN	UNDERFLOW_IRQ_EN	CUR_FRAME_DONE_IRQ_EN	VSYNC_EDGE_IRQ_EN	OVERFLOW_IRQ	UNDERFLOW_IRQ	CUR_FRAME_DONE_IRQ	VSYNC_EDGE_IRQ	Reserved					BUSY_ENABLE	MODE86	RESET
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_CTRL1n field descriptions

Field	Description
31 -	This field is reserved. Reserved bits. Write as 0.
30 -	This field is reserved. Reserved bits. Write as 0.
29–28 -	This field is reserved. Reserved bits. Write as 0.
27 COMBINE_MPU_WR_STRB	If this bit is not set, the write strobe will be driven on LCD_WR_RWn pin in the 8080 mode and on the LCD_RD_E pin in the 6800 mode. If it is set, the write strobe of both the 6800 and 8080 modes will be driven only on the LCD_WR_RWn pin. Note that this does not work for read strobe.
26 BM_ERROR_IRQ_EN	This bit is set to enable bus master error interrupt in the LCDIF master mode.

Table continues on the next page...

LCDIF_CTRL1n field descriptions (continued)

Field	Description
25 BM_ERROR_IRQ	This bit is set to indicate that an interrupt is requested by the LCDIF block. This bit is cleared by software by writing a one to its SCT clear address. This bit will be set when the LCDIF is in master mode and an error response was returned by the slave. 0x0 NO_REQUEST — No Interrupt Request Pending. 0x1 REQUEST — Interrupt Request Pending.
24 RECOVER_ON_UNDERFLOW	Set this bit to enable the LCDIF block to recover in the next field/frame if there was an underflow in the current field/frame.
23 INTERLACE_FIELDS	Set this bit if it is required that the LCDIF block fetches odd lines in one field and even lines in the other field. It will work only in LCDIF_MASTER is set to 1.
22 START_INTERLACE_FROM_SECOND_FIELD	The default is to grab the odd lines first and then the even lines. Set this bit if it is required to grab the even lines first and then the odd lines. (Line numbers start from 1, so odd lines are 1,3,5,etc. and even lines are 2,4,6, etc.)
21 FIFO_CLEAR	Set this bit to clear all the data in the latency FIFO (LFIFO), TXFIFO and the RXFIFO.
20 IRQ_ON_ALTERNATE_FIELDS	If this bit is set, the LCDIF block will assert the cur_frame_done interrupt only on alternate fields, otherwise it will issue the interrupt on both odd and even field. This bit is mostly relevant if INTERLACE_FIELDS is set.
19–16 BYTE_PACKING_FORMAT	This bitfield is used to show which data bytes in a 32-bit word are valid. Default value 0xf indicates that all bytes are valid. For 8-bit transfers, any combination in this bitfield will mean valid data is present in the corresponding bytes. In the 16-bit mode, a 16-bit half-word is valid only if adjacent bits [1:0] or [3:2] or both are 1. A value of 0x0 will mean that none of the bytes are valid and should not be used. For example, set the bit field value to 0x7 if the display data is arranged in the 24-bit unpacked format (A-R-G-B where A value does not have to be transmitted). When input data is in YCbCr 4:2:2 format (YCBCR422_INPUT is 1), H_COUNT should be the number of pixels that should be fetched by the block and the BYTE_PACKING_FORMAT should be 0xF. (Note - YCBCR422_INPUT = 1 implies 2 pixels per 32 bits).
15 OVERFLOW_IRQ_EN	This bit is set to enable an overflow interrupt in the TXFIFO in the write mode.
14 UNDERFLOW_IRQ_EN	This bit is set to enable an underflow interrupt in the TXFIFO in the write mode.
13 CUR_FRAME_DONE_IRQ_EN	This bit is set to 1 enable an interrupt every time the hardware enters in the vertical blanking state.
12 VSYNC_EDGE_IRQ_EN	This bit is set to enable an interrupt every time the hardware encounters the leading VSYNC edge in the VSYNC and DOTCLK modes, or the beginning of every field in DVI mode.
11 OVERFLOW_IRQ	This bit is set to indicate that an interrupt is requested by the LCDIF block. This bit is cleared by software by writing a one to its SCT clear address. A latency FIFO (LFIFO) overflow in the write mode (MPU/VSYNC/DOTCLK/DVI mode) was detected, data samples have been lost. 0x0 NO_REQUEST — No Interrupt Request Pending. 0x1 REQUEST — Interrupt Request Pending.

Table continues on the next page...

LCDIF_CTRL1n field descriptions (continued)

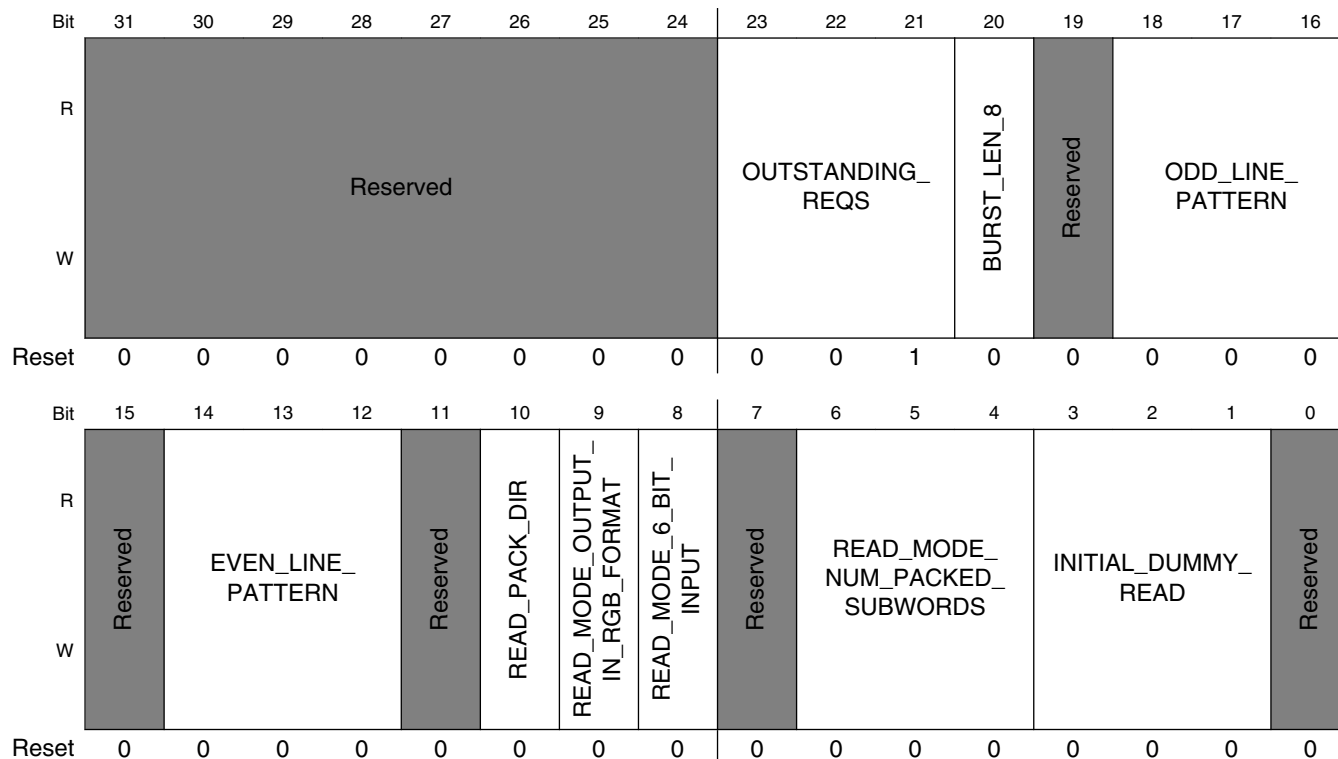
Field	Description
10 UNDERFLOW_ IRQ	This bit is set to indicate that an interrupt is requested by the LCDIF block. This bit is cleared by software by writing a one to its SCT clear address. A TXFIFO underflow in the write mode (MPU/VSYNC/DOTCLK/DVI mode) was detected. Could produce an error in the DOTCLK / DVI modes. 0x0 NO_REQUEST — No Interrupt Request Pending. 0x1 REQUEST — Interrupt Request Pending.
9 CUR_FRAME_ DONE_IRQ	This bit is set to indicate that an interrupt is requested by the LCDIF block. This bit is cleared by software by writing a one to its SCT clear address. It indicates that the hardware has completed transmitting the current frame and is in the vertical blanking period in the DOTCLK/DVI modes. In the MPU and VSYNC modes, this IRQ is asserted at the end of the data transfer indicated by LCDIF_TRANSFER_COUNT register. 0x0 NO_REQUEST — No Interrupt Request Pending. 0x1 REQUEST — Interrupt Request Pending.
8 VSYNC_EDGE_ IRQ	This bit is set to indicate that an interrupt is requested by the LCDIF block. This bit is cleared by software by writing a one to its SCT clear address. It is set whenever the leading VSYNC edge is detected in the VSYNC and DOTCLK modes. In the DVI mode, it is asserted every time the block enters a new field. 0x0 NO_REQUEST — No Interrupt Request Pending. 0x1 REQUEST — Interrupt Request Pending.
7–3 RSRVD0	This field is reserved. Reserved bits. Write as 0.
2 BUSY_ENABLE	This bit enables the use of the interface's busy signal input. This should be enabled for LCD controllers that implement a busy line (to stall the LCDIF from sending more data until ready). Otherwise this bit should be cleared. 0x0 BUSY_DISABLED — The busy signal from the LCD controller will be ignored. 0x1 BUSY_ENABLED — Enable the use of the busy signal from the LCD controller.
1 MODE86	This bit is used to select between the 8080 and 6800 series of microprocessor modes. This bit should only be changed when RUN is 0. 0x0 8080_MODE — Pins LCD_WR_RWn and LCD_RD_E function as active low WR and active low RD signals respectively. 0x1 6800_MODE — Pins LCD_WR_RWn and LCD_RD_E function as Read/Write and active high Enable signals respectively.
0 RESET	Reset bit for the external LCD controller. This bit can be changed at any time. It CANNOT be reset by SFTIRST. 0x0 LCDRESET_LOW — LCD_RESET output signal is low. 0x1 LCDRESET_HIGH — LCD_RESET output signal is high.

13.2.5.3 LCDIF General Control2 Register (LCDIF_CTRL2n)

The LCDIF Control Register provides overall control of the LCDIF block.

The LCDIF Control2 Register provides additional programming to the LCDIF. It implements some bits which are unlikely to change often in a particular application.

Address: 3032_0000h base + 20h offset + (4d × i), where i=0d to 3d



LCDIF_CTRL2n field descriptions

Field	Description
31–24 RSRVD5	This field is reserved. Reserved bits. Write as 0.
23–21 OUTSTANDING_ REQS	This bitfield indicates the maximum number of outstanding transactions that LCDIF should request when it is acting as a bus master. Default is 2 outstanding transactions. 0x0 REQ_1 — 0x1 REQ_2 — 0x2 REQ_4 — 0x3 REQ_8 — 0x4 REQ_16 —
20 BURST_LEN_8	By default, when the LCDIF is in the bus master mode, it will issue AXI bursts of length 16 (except when in packed 24 bpp mode, it will issue bursts of length 15). When this bit is set to 1, the block will issue bursts of length 8 (except when in packed 24 bpp mode, it will issue bursts of length 9). Note that this bitfield is only applicable when LCDIF_MASTER is set to 1.
19 RSRVD4	This field is reserved. Reserved bits. Write as 0.
18–16 ODD_LINE_ PATTERN	This field determines the order of the RGB components of each pixel in ODD lines (line numbers 1,3,5,...). This bitfield must be 0 in DVI mode. 0x0 RGB — 0x1 RBG — 0x2 GBR — 0x3 GRB —

Table continues on the next page...

LCDIF_CTRL2n field descriptions (continued)

Field	Description
	0x4 BRG — 0x5 BGR —
15 RSRVD3	This field is reserved. Reserved bits. Write as 0.
14–12 EVEN_LINE_ PATTERN	This field determines the order of the RGB components of each pixel in EVEN lines (line numbers 2,4,6,...). This bitfield must be 0 in DVI mode. 0x0 RGB — 0x1 RBG — 0x2 GBR — 0x3 GRB — 0x4 BRG — 0x5 BGR —
11 RSRVD2	This field is reserved. Reserved bits. Write as 0.
10 READ_PACK_DIR	The default value of 0 indicates data is stored in the little endian format. When LCD_DATABUS_WIDTH is 8-bit, this bit provides the option of rearranging the data byte-wise in the big endian format. For example, if READ_MODE_NUM_PACKED_SUBWORDS = 3 and the order of incoming data is 0x11, 0x22 and 0x33, then setting this bit to 1 will cause the data to be stored as 0x00112233 as opposed to the default 0x00332211. This operation occurs after the shifting operation done by SHIFT_NUM_BITS bitfield.
9 READ_MODE_ OUTPUT_IN_ RGB_FORMAT	Setting this bit will enable the LCDIF to convert the incoming data to the RGB format given by WORD_LENGTH bitfield. This feature is not available when WORD_LENGTH is set to 8 bits. LCDIF performs this operation of converting to RGB format after the endianness has been determined by the READ_PACK_DIR bitfield.
8 READ_MODE_6_ BIT_INPUT	Setting this bit to 1 indicates to LCDIF that even though LCD_DATABUS_WIDTH is set to 8 bits, the input data is actually only 6 bits wide and exists on D5-D0.
7 RSRVD1	This field is reserved. Reserved bits. Write as 0.
6–4 READ_MODE_ NUM_PACKED_ SUBWORDS	Indicates the number of valid 8/16/18/24-bit subwords that will be packed into the 32-bit word in read mode. The subword size (8, 16, 18 or 24 bits) is determined by the LCD_DATABUS_WIDTH field. The swizzle operation is performed after READ_MODE_NUM_PACKED_SUBWORDS number of subwords has been received and stored in little-endian format. For example, if LCD_DATABUS_WIDTH is set to 8-bit and data to be read back has to be stored in memory in 24-bit unpacked RGB format, set READ_MODE_NUM_PACKED_SUBWORDS to 0x3 so that each 32-bit word will contain only 3 valid bytes (RGB). Maximum value of READ_MODE_NUM_PACKED_SUBWORDS is 4 for 8-bit databus, 2 for 16-bit databus and 1 for 18/24-bit databus.
3–1 INITIAL_DUMMY_ READ	The value in this field determines the number of dummy 8/16/18/24-bit subwords that have to be read back from the LCD panel/controller. They will then not be stored in the read FIFO.
0 RSRVD0	This field is reserved. Reserved bits. Write as 0.

13.2.5.4 LCDIF Horizontal and Vertical Valid Data Count Register (LCDIF_TRANSFER_COUNT)

This register tells the LCDIF how much data will be sent for this frame, or transaction. The total number of words is a product of the V_COUNT and H_COUNT fields. The word size is specified by the WORD_LENGTH field.

This register gives the dimensions of the input frame. For normal operation, but V_COUNT and H_COUNT should be non-zero.

Address: 3032_0000h base + 30h offset = 3032_0030h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	V_COUNT																H_COUNT															
W	V_COUNT																H_COUNT															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_TRANSFER_COUNT field descriptions

Field	Description
31–16 V_COUNT	Number of horizontal lines per frame which contain valid data. In DOTCLK mode, V_COUNT should be the same as the number of active horizontal lines in a progressive frame. In DVI mode, V_COUNT should be the number of active horizontal lines per frame, and not per field.
H_COUNT	Total valid data (pixels) in each horizontal line. The data size is given by the WORD_LENGTH. When input data is in YCbCr 4:2:2 format (YCBCR422_INPUT is 1), H_COUNT should be the number of 32-bit words that should be fetched by the block and the BYTE_PACKING_FORMAT should be 0xF. In 24-bit packed format (WORD_LENGTH=0x3, BYTE_PACKING_FORMAT=0xF), the H_COUNT must be a multiple of 4 pixels. In 16-bit packed format (WORD_LENGTH=0x0, BYTE_PACKING_FORMAT=0xF), the H_COUNT must be a multiple of 2 pixels.

13.2.5.5 LCD Interface Current Buffer Address Register (LCDIF_CUR_BUF)

This register indicates the address of the current frame being transmitted by LCDIF.

When the LCDIF is behaving as a master, this address points to the address of the current frame of data being sent out via the LCDIF. When the current frame is done, the LCDIF block will assert the cur_frame_done interrupt for software to take action. The block will also copy the LCDIF_NEXT_BUF_ADDR into this bitfield so that the software can program the next frame address into the LCDIF_NEXT_BUF_ADDR bitfield. This address must always be double-word aligned.

Address: 3032_0000h base + 40h offset = 3032_0040h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
	ADDR																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_CUR_BUF field descriptions

Field	Description
ADDR	Address of the current frame being transmitted by LCDIF.

13.2.5.6 LCD Interface Next Buffer Address Register (LCDIF_NEXT_BUF)

This register indicates the address of next frame that will be transmitted by LCDIF.

When the LCDIF is behaving as a master, this address points to the address of the next frame of data that will be sent out via the LCDIF. It is up to the software to make sure that this register is programmed before the end of the current frame, otherwise it might result in old data going out the LCDIF. This address must always be double-word aligned.

Address: 3032_0000h base + 50h offset = 3032_0050h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
	ADDR																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_NEXT_BUF field descriptions

Field	Description
ADDR	Address of the next frame that will be transmitted by LCDIF.

13.2.5.7 LCD Interface Timing Register (LCDIF_TIMING)

The LCD interface timing register controls the various setup and hold times enforced by the LCD interface in the 6800/8080 MPU and VSYNC modes of operation.

The values used in this register are dependent on the particular LCD controller used, consult the users manual for the particular controller for required timings. Each field of the register must be non-zero, therefore the minimum value is: 0x01010101. NOTE: the timings are not automatically adjusted if the DISPLAY CLOCK (pix_clk) frequency changes—it may be necessary to adjust the timings if DISPLAY CLOCK (pix_clk) changes. NOTE: Each field in this register must be non-zero for the MPU and VSYNC modes to function. The settings in this register do not affect the DOTCLK and DVI modes.

Address: 3032_0000h base + 60h offset = 3032_0060h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CMD_HOLD								CMD_SETUP								DATA_HOLD								DATA_SETUP							
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_TIMING field descriptions

Field	Description
31–24 CMD_HOLD	Number of DISPLAY CLOCK (pix_clk) cycles that the LCD_RS signal is active after LCD_CS is deasserted.
23–16 CMD_SETUP	Number of DISPLAY CLOCK (pix_clk) cycles that the LCD_RS signal is active before LCD_CS is asserted.
15–8 DATA_HOLD	Data bus hold time in DISPLAY CLOCK (pix_clk) cycles. Also the time that the data strobe is deasserted in a cycle
DATA_SETUP	Data bus setup time in DISPLAY CLOCK (pix_clk) cycles. Also the time that the data strobe is asserted in a cycle.

13.2.5.8 LCDIF VSYNC Mode and Dotclk Mode Control Register0 (LCDIF_VDCTRL0n)

This register is used to control the VSYNC and DOTCLK modes of the LCDIF so as to work with different types of LCDs like moving picture displays and delta pixel displays.

This register gives general programmability to the VSYNC signal including polarity, direction, pulse width, etc.

Address: 3032_0000h base + 70h offset + (4d × i), where i=0d to 3d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								Reserved							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	VSYNC_PULSE_WIDTH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_VDCTRL0n field descriptions

Field	Description
31–30 RSRVD2	This field is reserved. Reserved bits. Write as 0.
29 VSYNC_OEB	0 means the VSYNC signal is an output, 1 means it is an input. Should be set to 0 in the DOTCLK mode. 0x0 VSYNC_OUTPUT — The VSYNC pin is in the output mode and the VSYNC signal has to be generated by the LCDIF block. 0x1 VSYNC_INPUT — The VSYNC pin is in the input mode and the LCD controller sends the VSYNC signal to the block.
28 ENABLE_PRESENT	Setting this bit to 1 will make the hardware generate the ENABLE signal in the DOTCLK mode, thereby making it the true RGB interface along with the remaining three signals VSYNC, HSYNC and DOTCLK.
27 VSYNC_POL	Default 0 active low during VSYNC_PULSE_WIDTH time and will be high during the rest of the VSYNC period. Set it to 1 to invert the polarity.
26 HSYNC_POL	Default 0 active low during HSYNC_PULSE_WIDTH time and will be high during the rest of the HSYNC period. Set it to 1 to invert the polarity.
25 DOTCLK_POL	Default is data launched at negative edge of DOTCLK and captured at positive edge. Set it to 1 to invert the polarity. Set it to 0 in DVI mode.
24 ENABLE_POL	Default 0 active low during valid data transfer on each horizontal line.
23–22 RSRVD1	This field is reserved. Reserved bits. Write as 0.
21 VSYNC_PERIOD_UNIT	Default 0 for counting VSYNC_PERIOD in terms of DISPLAY CLOCK (pix_clk) cycles. Set it to 1 to count in terms of complete horizontal lines. DISPLAY CLOCK (pix_clk) cycles should be used in the VSYNC mode, while horizontal line should be used in the DOTCLK mode.
20 VSYNC_PULSE_WIDTH_UNIT	Default 0 for counting VSYNC_PULSE_WIDTH in terms of DISPLAY CLOCK (pix_clk) cycles. Set it to 1 to count in terms of complete horizontal lines.
19 HALF_LINE	Setting this bit to 1 will make the total VSYNC period equal to the VSYNC_PERIOD field plus half the HORIZONTAL_PERIOD field (i.e. VSYNC_PERIOD field plus half horizontal line), otherwise it is just VSYNC_PERIOD. Should be only used in the DOTCLK mode, not in the VSYNC interface mode.
18 HALF_LINE_MODE	When this bit is 0, the first field (VSYNC period) will end in half a horizontal line and the second field will begin with half a horizontal line. When this bit is 1, all fields will end with half a horizontal line, and none will begin with half a horizontal line.
VSYNC_PULSE_WIDTH	Number of units for which VSYNC signal is active. For the DOTCLK mode, the unit is determined by the VSYNC_PULSE_WIDTH_UNIT. If the VSYNC_PULSE_WIDTH_UNIT is 0 for DOTCLK mode, VSYNC_PULSE_WIDTH must be less than HSYNC_PERIOD. For the VSYNC interface mode, it should be in terms of number of DISPLAY CLOCK (pix_clk) cycles only.

13.2.5.9 LCDIF VSYNC Mode and Dotclk Mode Control Register1 (LCDIF_VDCTRL1)

This register is used to control the VSYNC signal in the VSYNC and DOTCLK modes of the block.

This register determines the period and duty cycle of the VSYNC signal when it is generated in the block.

Address: 3032_0000h base + 80h offset = 3032_0080h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	VSYNC_PERIOD																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_VDCTRL1 field descriptions

Field	Description
VSYNC_PERIOD	Total number of units between two positive or two negative edges of the VSYNC signal. If HALF_LINE is set, it is implicitly calculated to be VSYNC_PERIOD plus half HSYNC_PERIOD.

13.2.5.10 LCDIF VSYNC Mode and Dotclk Mode Control Register2 (LCDIF_VDCTRL2)

This register is used to control the HSYNC signal in the DOTCLK mode of the block.

This register determines the period and duty cycle of the HSYNC signal when it is generated in the block.

Address: 3032_0000h base + 90h offset = 3032_0090h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	HSYNC_PULSE_WIDTH																HSYNC_PERIOD															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_VDCTRL2 field descriptions

Field	Description
31–18 HSYNC_PULSE_WIDTH	Number of DISPLAY CLOCK (pix_clk) cycles for which HSYNC signal is active.
HSYNC_PERIOD	Total number of DISPLAY CLOCK (pix_clk) cycles between two positive or two negative edges of the HSYNC signal.

13.2.5.11 LCDIF VSYNC Mode and Dotclk Mode Control Register3 (LCDIF_VDCTRL3)

This register is used to determine the vertical and horizontal wait counts.

This register determines the back porches of HSYNC and VSYNC signals when they are generated by the block.

Enhanced LCD Interface (eLCDIF)

Address: 3032_0000h base + A0h offset = 3032_00A0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved			MUX_SYNC- SIGNALS	VSYNC_ONLY	HORIZONTAL_WAIT_CNT										
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	VERTICAL_WAIT_CNT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_VDCTRL3 field descriptions

Field	Description
31–30 RSRVD0	This field is reserved. Reserved bits, write as 0.
29 MUX_SYNC- SIGNALS	When this bit is set, the LCDIF block will internally mux HSYNC with LCD_D14, DOTCLK with LCD_D13 and ENABLE with LCD_D12, otherwise these signals will go out on separate pins. This feature can be used to maintain backward compatible with 37xx.
28 VSYNC_ONLY	This bit must be set to 1 in the VSYNC mode of operation, and 0 in the DOTCLK mode of operation.
27–16 HORIZONTAL- WAIT_CNT	In the DOTCLK mode, wait for this number of clocks from falling edge (or rising if HSYNC_POL is 1) of HSYNC signal to account for horizontal back porch plus the number of DOTCLKs before the moving picture information begins.
VERTICAL- WAIT_CNT	In the VSYNC interface mode, wait for this number of DISPLAY CLOCK (pix_clk) cycles from the falling VSYNC edge (or rising if VSYNC_POL is 1) before starting LCD transactions and is applicable only if WAIT_FOR_VSYNC_EDGE is set. Minimum is CMD_SETUP+5. In the DOTCLK mode, it accounts for the vertical back porch lines plus the number of horizontal lines before the moving picture begins. The unit for this parameter is inherently the same as the VSYNC_PERIOD_UNIT.

13.2.5.12 LCDIF VSYNC Mode and Dotclk Mode Control Register4 (LCDIF_VDCTRL4)

This register is used to control the DOTCLK mode of the block.

This register determines the active data in each horizontal line in the DOTCLK mode. Note that the total number of active horizontal lines in the DOTCLK mode is the same as the V_COUNT bitfield in the LCDIF_TRANSFER_COUNT register.

Address: 3032_0000h base + B0h offset = 3032_00B0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DOTCLK_DLY_SEL			Reserved												
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DOTCLK_H_VALID_DATA_CNT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_VDCTRL4 field descriptions

Field	Description
31–29 DOTCLK_DLY_SEL	This bitfield selects the amount of time by which the DOTCLK signal should be delayed before coming out of the LCD_DOTCK pin. 0 = 2ns; 1=4ns;2=6ns;3=8ns. Remaining values are reserved.
28–19 RSRVD0	This field is reserved. Reserved bits, write as 0.
18 SYNC_SIGNALS_ON	Set this field to 1 if the LCD controller requires that the VSYNC or VSYNC/HSYNC/DOTCLK control signals should be active at least one frame before the data transfers actually start and remain active at least one frame after the data transfers end. The hardware does not count the number of frames automatically. Rather, the VSYNC edge interrupt can be monitored by software to count the number of frames that have occurred after this bit is set and then the RUN bit can be set to start the data transactions. This bit must always be set in the DOTCLK mode of operation, and it must be set in the VSYNC mode of operation when VSYNC signal is an output.
DOTCLK_H_VALID_DATA_CNT	Total number of DISPLAY CLOCK (pix_clk) cycles on each horizontal line that carry valid data in DOTCLK mode.

13.2.5.13 Digital Video Interface Control0 Register (LCDIF_DVCTRL0)

The Digital Video interface Control0 register provides the overall control of the Digital Video interface.

This register gives information about the horizontal active, horizontal blanking and total number of lines in the ITU-R BT.656 interface.

EXAMPLE

```
//525/60 video system
```

Enhanced LCD Interface (eLCDIF)

```
HW_LCDIF_DVICTRL0_H_ACTIVE_CNT_WR(0x5A0); //1440
HW_LCDIF_DVICTRL0_H_BLANKING_CNT_WR(0x106); //262
//625/50 video system
HW_LCDIF_DVICTRL0_H_ACTIVE_CNT_WR(0x5A0); //1440
HW_LCDIF_DVICTRL0_H_BLANKING_CNT_WR(0x112); //274
```

Address: 3032_0000h base + C0h offset = 3032_00C0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved				H_ACTIVE_CNT												Reserved				H_BLANKING_CNT											
W	Reserved				H_ACTIVE_CNT												Reserved				H_BLANKING_CNT											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_DVICTRL0 field descriptions

Field	Description
31–28 RSRVD1	This field is reserved. Reserved bits, write as 0.
27–16 H_ACTIVE_CNT	Number of active video samples to be transmitted. (Mostly will be 1440 for both PAL and NTSC). Must always be a multiple of 4.
15–12 RSRVD0	This field is reserved. Reserved bits, write as 0.
H_BLANKING_CNT	Number of blanking samples to be inserted between EAV and SAV during horizontal blanking interval.

13.2.5.14 Digital Video Interface Control1 Register (LCDIF_DVICTRL1)

The Digital Video interface Control1 register provides the overall control of the Digital Video interface.

This register contains information about the Field1 start and end, and the Field2 start in the ITU-R BT.656 interface.

EXAMPLE

```
//525/60 video system
HW_LCDIF_DVICTRL1_F1_START_LINE_WR(0x4); //4
HW_LCDIF_DVICTRL1_F1_END_LINE_WR(0x109); //265
HW_LCDIF_DVICTRL1_F2_START_LINE_WR(0x10A); //266
//625/50 video system
HW_LCDIF_DVICTRL1_F1_START_LINE_WR(0x1); //1
HW_LCDIF_DVICTRL1_F1_END_LINE_WR(0x138); //312
HW_LCDIF_DVICTRL1_F2_START_LINE_WR(0x139); //313
```

Address: 3032_0000h base + D0h offset = 3032_00D0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_DVICTRL1 field descriptions

Field	Description
31–30 RSRVD0	This field is reserved. Reserved bits, write as 0.
29–20 F1_START_LINE	Vertical line number from which Field 1 begins.
19–10 F1_END_LINE	Vertical line number at which Field1 ends.
F2_START_LINE	Vertical line number from which Field 2 begins.

13.2.5.15 Digital Video Interface Control2 Register (LCDIF_DVICTRL2)

The Digital Video interface Control2 register provides the overall control of the Digital Video interface.

This register contains information about the Field2 end, and the Vertical Blanking1 interval in the ITU-R BT.656 interface.

EXAMPLE

```
//525/60 video system
HW_LCDIF_DVICTRL2_F2_END_LINE_WR(0x3); //3
HW_LCDIF_DVICTRL2_V1_BLANK_START_LINE_WR(0x108); //264
HW_LCDIF_DVICTRL2_V1_BLANK_END_LINE_WR(0x11A); //282
//625/50 video system
HW_LCDIF_DVICTRL2_F2_END_LINE_WR(0x271); //625
HW_LCDIF_DVICTRL2_V1_BLANK_START_LINE_WR(0x137); //311
HW_LCDIF_DVICTRL2_V1_BLANK_END_LINE_WR(0x14F); //335
```

Address: 3032_0000h base + E0h offset = 3032_00E0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_DVICTRL2 field descriptions

Field	Description
31–30 RSRVD0	This field is reserved. Reserved bits, write as 0.
29–20 F2_END_LINE	Vertical line number at which Field 2 ends.
19–10 V1_BLANK_START_LINE	Vertical line number towards the end of Field1 where first Vertical Blanking interval starts.
V1_BLANK_END_LINE	Vertical line number in the beginning part of Field2 where first Vertical Blanking interval ends.

13.2.5.16 Digital Video Interface Control3 Register (LCDIF_DVICTRL3)

The Digital Video interface Control3 register provides the overall control of the Digital Video interface.

This register contains information about the Vertical Blanking2 interval in the ITU-R BT. 656 interface.

EXAMPLE

```
//525/60 video system
HW_LCDIF_DVICTRL3_V2_BLANK_START_LINE_WR(0x1); //1
HW_LCDIF_DVICTRL3_V2_BLANK_END_LINE_WR(0x13); //19
HW_LCDIF_DVICTRL0_V_LINES_CNT_WR(0x20D); //525
//625/50 video system
HW_LCDIF_DVICTRL3_V2_BLANK_START_LINE_WR(0x270); //624
HW_LCDIF_DVICTRL3_V2_BLANK_END_LINE_WR(0x16); //22
HW_LCDIF_DVICTRL0_V_LINES_CNT_WR(0x271); //625
```

Address: 3032_0000h base + F0h offset = 3032_00F0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
	Reserved		V2_BLANK_START_LINE										V2_BLANK_END_LINE			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																
	V2_BLANK_END_LINE							V_LINES_CNT								
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_DVICTRL3 field descriptions

Field	Description
31–30 RSRVD0	This field is reserved. Reserved bits, write as 0.

Table continues on the next page...

LCDIF_DVICTRL3 field descriptions (continued)

Field	Description
29–20 V2_BLANK_START_LINE	Vertical line number towards the end of Field2 where second Vertical Blanking interval starts.
19–10 V2_BLANK_END_LINE	Vertical line number in the beginning part of Field1 where second Vertical Blanking interval ends.
V_LINES_CNT	Total number of vertical lines per frame (generally 525 or 625)

13.2.5.17 Digital Video Interface Control4 Register (LCDIF_DVICTRL4)

The Digital Video interface Control4 register provides the overall control of the Digital Video interface.

This register is used to add side borders to the output if the input frame width is less than 720 pixels.

EXAMPLE

```

720 pixels per line. //If input frame has only 640 pixels per line, but output is supposed to have
HW_LCDIF_DVICTRL4_H_FILL_CNT_WR(0x50); //80
HW_LCDIF_DVICTRL4_Y_FILL_VALUE_WR(0x10); //16
HW_LCDIF_DVICTRL4_CB_FILL_VALUE_WR(0x80); //128
HW_LCDIF_DVICTRL4_CR_FILL_VALUE_WR(0x80); //128

```

Address: 3032_0000h base + 100h offset = 3032_0100h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Y_FILL_VALUE								CB_FILL_VALUE								CR_FILL_VALUE								H_FILL_CNT							
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_DVICTRL4 field descriptions

Field	Description
31–24 Y_FILL_VALUE	Value of Y component of filler data
23–16 CB_FILL_VALUE	Value of CB component of filler data
15–8 CR_FILL_VALUE	Value of CR component of filler data.
H_FILL_CNT	Number of active video samples that have to be filled with the filler data in the front and back portions of the active horizontal interval. Must be a multiple of 4. This field will have to be programmed if the input frame has less than 720 pixels per line.

13.2.5.18 RGB to YCbCr 4:2:2 CSC Coefficient0 Register (LCDIF_CSC_COEFF0)

LCDIF_CSC_COEFF0 register provides overall control over color space conversion from RGB to 4:2:2 YCbCr. The equations for the conversion are given by: $Y = C0 \cdot R + C1 \cdot G + C2 \cdot B + Y_offset$ $Cb = C3 \cdot R + C4 \cdot G + C5 \cdot B + CbCr_offset$ $Cr = C6 \cdot R + C7 \cdot G + C8 \cdot B + CbCr_offset$

This register carries programming information about RGB to YCbCr 4:2:2 CSC.

EXAMPLE

```
HW_LCDIF_CSC_COEFF0_C0_WR(0x41); //0.257x256=65
HW_LCDIF_CSC_COEFF0_CSC_SUBSAMPLE_FILTER_WR(0x3);
```

Address: 3032_0000h base + 110h offset = 3032_0110h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								C0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_CSC_COEFF0 field descriptions

Field	Description
31–26 RSRVD1	This field is reserved. Reserved bits, write as 0.
25–16 C0	Two's complement red multiplier coefficient for Y
15–2 RSRVD0	This field is reserved. Reserved bits, write as 0.
CSC_ SUBSAMPLE_ FILTER	This register describes the filtering and subsampling scheme to be performed on the chroma components in order to convert from YCbCr 4:4:4 to YCbCr 4:2:2 space. Note that the following descriptions apply individually to Cb and Cr. 0x0 SAMPLE_AND_HOLD — No filtering, simply keep every chroma value for samples numbered 2n and discard chroma values associated with all samples numbered 2n+1. 0x1 RSRVD — Reserved 0x2 INTERSTITIAL — Chroma samples numbered 2n and 2n+1 are averaged (weights 1/2, 1/2) and that chroma value replaces the two chroma values at 2n and 2n+1. This chroma now exists horizontally halfway between the two luma samples. 0x3 COSITED — Chroma samples numbered 2n-1, 2n, and 2n+1 are averaged (weights 1/4, 1/2, 1/4) and that chroma value exists at the same site as the luma sample numbered 2n and the chroma samples at 2n+1 are discarded.

13.2.5.19 RGB to YCbCr 4:2:2 CSC Coefficient1 Register (LCDIF_CSC_COEFF1)

LCDIF_CSC_COEFF1 register provides overall control over color space conversion from RGB to 4:2:2 YCbCr. The equations for the conversion are given by: $Y = C0 \cdot R + C1 \cdot G + C2 \cdot B + Y_offset$ $Cb = C3 \cdot R + C4 \cdot G + C5 \cdot B + CbCr_offset$ $Cr = C6 \cdot R + C7 \cdot G + C8 \cdot B + CbCr_offset$

This register carries programming information about RGB to YCbCr 4:2:2 CSC.

EXAMPLE

```
HW_LCDIF_CSC_COEFF1_C1_WR(0x81); //0.504x256=129
HW_LCDIF_CSC_COEFF1_C2_WR(0x19); //0.098x256=25
```

Address: 3032_0000h base + 120h offset = 3032_0120h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved						C2										Reserved						C1									
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_CSC_COEFF1 field descriptions

Field	Description
31–26 RSRVD1	This field is reserved. Reserved bits, write as 0.
25–16 C2	Two's complement blue multiplier coefficient for Y
15–10 RSRVD0	This field is reserved. Reserved bits, write as 0.
C1	Two's complement green multiplier coefficient for Y

13.2.5.20 RGB to YCbCr 4:2:2 CSC Coefficient2 Register (LCDIF_CSC_COEFF2)

LCDIF_CSC_COEFF2 register provides overall control over color space conversion from RGB to 4:2:2 YCbCr. The equations for the conversion are given by: $Y = C0 \cdot R + C1 \cdot G + C2 \cdot B + Y_offset$ $Cb = C3 \cdot R + C4 \cdot G + C5 \cdot B + CbCr_offset$ $Cr = C6 \cdot R + C7 \cdot G + C8 \cdot B + CbCr_offset$

This register carries programming information about RGB to YCbCr 4:2:2 CSC.

EXAMPLE

```
HW_LCDIF_CSC_COEFF2_C3_WR(0x3DB); //-0.148x256=-37
HW_LCDIF_CSC_COEFF2_C4_WR(0x3B6); //-0.291x256=-74
```

Address: 3032_0000h base + 130h offset = 3032_0130h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved						C4										Reserved						C3									
W	Reserved						C4										Reserved						C3									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_CSC_COEFF2 field descriptions

Field	Description
31–26 RSRVD1	This field is reserved. Reserved bits, write as 0.
25–16 C4	Two's complement green multiplier coefficient for Cb
15–10 RSRVD0	This field is reserved. Reserved bits, write as 0.
C3	Two's complement red multiplier coefficient for Cb

13.2.5.21 RGB to YCbCr 4:2:2 CSC Coefficient3 Register (LCDIF_CSC_COEFF3)

LCDIF_CSC_COEFF3 register provides overall control over color space conversion from RGB to 4:2:2 YCbCr. The equations for the conversion are given by: $Y = C0 \cdot R + C1 \cdot G + C2 \cdot B + Y_offset$ $Cb = C3 \cdot R + C4 \cdot G + C5 \cdot B + CbCr_offset$ $Cr = C6 \cdot R + C7 \cdot G + C8 \cdot B + CbCr_offset$

This register carries programming information about RGB to YCbCr 4:2:2 CSC.

EXAMPLE

```
HW_LCDIF_CSC_COEFF3_C5_WR(0x70); //0.439x256=112
HW_LCDIF_CSC_COEFF3_C6_WR(0x70); //0.439x256=112
```

Address: 3032_0000h base + 140h offset = 3032_0140h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved						C6										Reserved						C5									
W	Reserved						C6										Reserved						C5									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_CSC_COEFF3 field descriptions

Field	Description
31–26 RSRVD1	This field is reserved. Reserved bits, write as 0.

Table continues on the next page...

LCDIF_CSC_COEFF3 field descriptions (continued)

Field	Description
25–16 C6	Two's complement red multiplier coefficient for Cr
15–10 RSRVD0	This field is reserved. Reserved bits, write as 0.
C5	Two's complement blue multiplier coefficient for Cb

13.2.5.22 RGB to YCbCr 4:2:2 CSC Coefficient4 Register (LCDIF_CSC_COEFF4)

LCDIF_CSC_COEFF4 register provides overall control over color space conversion from RGB to 4:2:2 YCbCr. The equations for the conversion are given by: $Y = C0*R + C1*G + C2*B + Y_offset$ $Cb = C3*R + C4*G + C5*B + CbCr_offset$ $Cr = C6*R + C7*G + C8*B + CbCr_offset$

This register carries programming information about RGB to YCbCr 4:2:2 CSC.

EXAMPLE

```
HW_LCDIF_CSC_COEFF4_C7_WR(0x3A2); //-0.368x256=-94
HW_LCDIF_CSC_COEFF4_C8_WR(0x3EE); //-0.071x256=-18
```

Address: 3032_0000h base + 150h offset = 3032_0150h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								C8								Reserved								C7							
W	Reserved								C8								Reserved								C7							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_CSC_COEFF4 field descriptions

Field	Description
31–26 RSRVD1	This field is reserved. Reserved bits, write as 0.
25–16 C8	Two's complement blue multiplier coefficient for Cr
15–10 RSRVD0	This field is reserved. Reserved bits, write as 0.
C7	Two's complement green multiplier coefficient for Cr

13.2.5.23 RGB to YCbCr 4:2:2 CSC Offset Register (LCDIF_CSC_OFFSET)

LCDIF_CSC_ register provides overall control over color space conversion from RGB to 4:2:2 YCbCr. The equations for the conversion are given by: $Y = C0*R + C1*G + C2*B + Y_offset$ $Cb = C3*R + C4*G + C5*B + CbCr_offset$ $Cr = C6*R + C7*G + C8*B + CbCr_offset$

This register carries programming information about RGB to YCbCr 4:2:2 CSC.

Address: 3032_0000h base + 160h offset = 3032_0160h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								CBCR_OFFSET								Reserved								Y_OFFSET							
W	Reserved								CBCR_OFFSET								Reserved								Y_OFFSET							
Reset	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

LCDIF_CSC_OFFSET field descriptions

Field	Description
31–25 RSRVD1	This field is reserved. Reserved bits, write as 0.
24–16 CBCR_OFFSET	Two's complement offset for the Cb and Cr components
15–9 RSRVD0	This field is reserved. Reserved bits, write as 0.
Y_OFFSET	Two's complement offset for the Y component

13.2.5.24 RGB to YCbCr 4:2:2 CSC Limit Register (LCDIF_CSC_LIMIT)

LCDIF_CSC_CTRL0 register provides overall control over color space conversion from RGB to 4:2:2 YCbCr. The equations for the conversion are given by: $Y = C0*R + C1*G + C2*B + Y_offset$ $Cb = C3*R + C4*G + C5*B + CbCr_offset$ $Cr = C6*R + C7*G + C8*B + CbCr_offset$

This register carries programming information about RGB to YCbCr 4:2:2 CSC. Note that the values in this register are unsigned.

EXAMPLE

```
HW_LCDIF_CSC_LIMIT_CBCR_MIN_WR(0x10); //16
HW_LCDIF_CSC_LIMIT_CBCR_MAX_WR(0xF0); //240
HW_LCDIF_CSC_LIMIT_Y_MIN_WR(0x10); //16
HW_LCDIF_CSC_LIMIT_Y_MAX_WR(0xEB); //235
```

Address: 3032_0000h base + 170h offset = 3032_0170h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CBCR_MIN								CBCR_MAX								Y_MIN								Y_MAX							
W																																
Reset	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1

LCDIF_CSC_LIMIT field descriptions

Field	Description
31–24 CBCR_MIN	Lower limit of Cb and Cr after RGB to 4:2:2 YCbCr conversion
23–16 CBCR_MAX	Upper limit of Cb and Cr after RGB to 4:2:2 YCbCr conversion
15–8 Y_MIN	Lower limit of Y after RGB to 4:2:2 YCbCr conversion
Y_MAX	Upper limit of Y after RGB to 4:2:2 YCbCr conversion

13.2.5.25 LCD Interface Data Register (LCDIF_DATA)

This register is used to transfer data using the PIO interface mode of operation. In MPU mode, data written to this register will be transferred out to the display device. When receiving data from the display, data is read from this register using PIO operations. During write operations, data can be written to this register (from the processor's perspective) as bytes, half-words (16 bits), or words (32 bits) as desired.

This register holds the 32-bit word written by the Arm platform into LCDIF. This data then gets sent out by the block across the interface.

Address: 3032_0000h base + 180h offset = 3032_0180h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DATA_THREE								DATA_TWO								DATA_ONE								DATA_ZERO							
W	DATA_THREE								DATA_TWO								DATA_ONE								DATA_ZERO							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_DATA field descriptions

Field	Description
31–24 DATA_THREE	Byte 3 (most significant byte) of data written to LCDIF.
23–16 DATA_TWO	Byte 2 of data written to LCDIF.
15–8 DATA_ONE	Byte 1 of data written to LCDIF.
DATA_ZERO	Byte 0 (least significant byte) of data written to LCDIF.

13.2.5.26 Bus Master Error Status Register (LCDIF_BM_ERROR_STAT)

This register reflects the virtual address at which the AXI master received an error response from the slave.

When the BM_ERROR_IRQ is asserted, the address of the bus error is updated in the register.

Address: 3032_0000h base + 190h offset = 3032_0190h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ADDR																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_BM_ERROR_STAT field descriptions

Field	Description
ADDR	Virtual address at which bus master error occurred.

13.2.5.27 CRC Status Register (LCDIF_CRC_STAT)

This register reflects the CRC value of each frame sent out by LCDIF. The CRC is done on the final output bus, so the value will be dependent on the LCD_DATABUS_WIDTH bitfield even if the input data is the same.

This register will be updated when the CUR_FRAME_DONE_IRQ is asserted. In the case of DVI mode, the CRC is calculated for the entire frame, not separately for each field in the frame.

Address: 3032_0000h base + 1A0h offset = 3032_01A0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CRC_VALUE																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_CRC_STAT field descriptions

Field	Description
CRC_VALUE	Calculated CRC value.

13.2.5.28 LCD Interface Status Register (LCDIF_STAT)

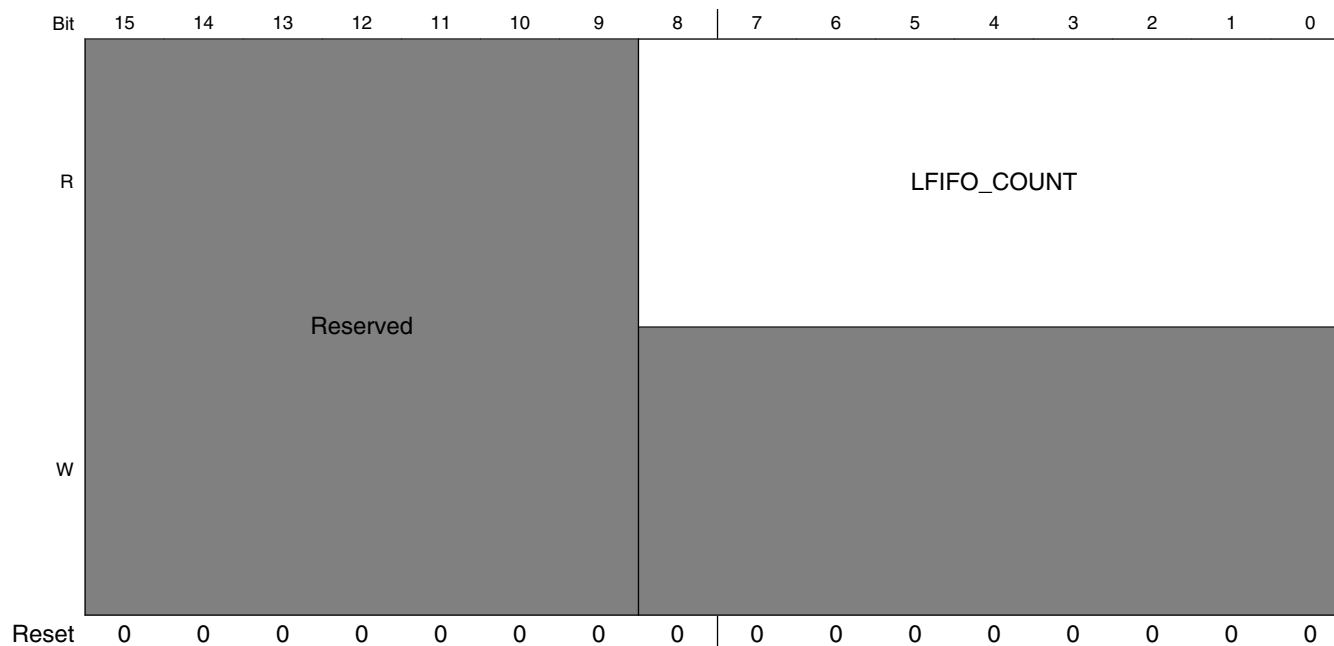
The LCD interface status register can be used to check the current status of the LCDIF block.

The LCD interface status register that contains read only views of some parameters or current state of the block.

Address: 3032_0000h base + 1B0h offset = 3032_01B0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	PRESENT	Reserved		LIFO_FULL	LIFO_EMPTY	TXFIFO_FULL	TXFIFO_EMPTY	BUSY	DVI_CURRENT_FIELD	Reserved						
W																
Reset	1	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0

Enhanced LCD Interface (eLCDIF)



LCDIF_STAT field descriptions

Field	Description
31 PRESENT	0: LCDIF not present on this product 1: LCDIF is present.
30 -	This field is reserved. Reserved.
29 LFIFO_FULL	Read only view of the signals that indicates LCD LFIFO is full.
28 LFIFO_EMPTY	Read only view of the signals that indicates LCD LFIFO is empty.
27 TXFIFO_FULL	Read only view of the signals that indicates LCD TXFIFO is full.
26 TXFIFO_EMPTY	Read only view of the signals that indicates LCD TXFIFO is empty.
25 BUSY	Read only view of the input busy signal from the external LCD controller.
24 DVI_CURRENT_FIELD	Read only view of the current field being transmitted. DVI_CURRENT_FIELD = 0 means field 1. DVI_CURRENT_FIELD = 1 means field 2.
23–9 RSRVD0	This field is reserved. Reserved bits. Write as 0.
LFIFO_COUNT	Read only view of the current count in Latency buffer (LFIFO).

13.2.5.29 LCDIF Threshold Register (LCDIF_THRES)

This register is used to activate control signals when the number of pixels reaches the programmed threshold. These control signals, in turn, can be used to manipulate access priority or dynamically change the input clock frequency to meet the required pixel throughput.

Memory request priority threshold register.

Address: 3032_0000h base + 200h offset = 3032_0200h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								FASTCLOCK								Reserved								PANIC							
W	Reserved								FASTCLOCK								Reserved								PANIC							
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

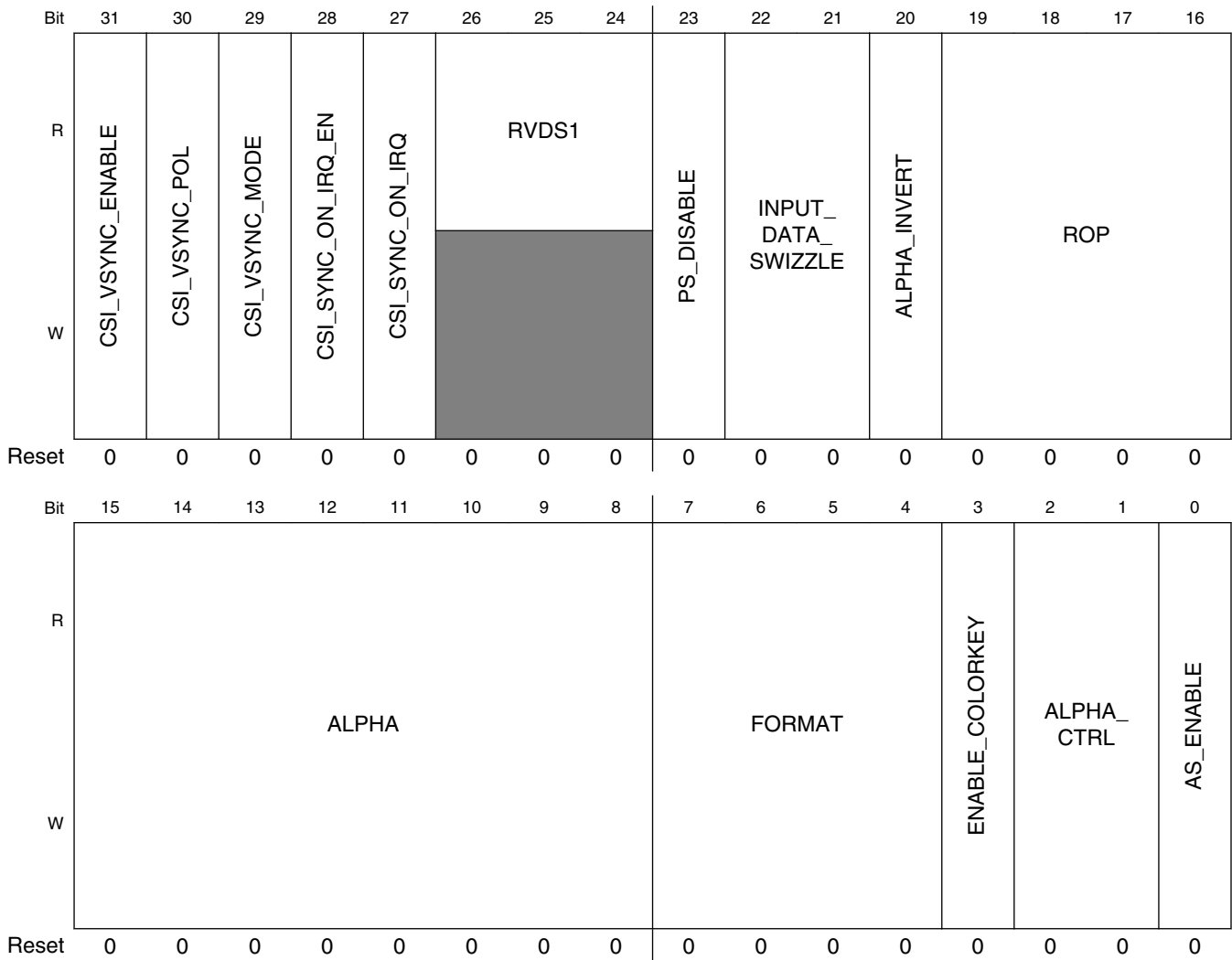
LCDIF_THRES field descriptions

Field	Description
31–25 RSRVD2	This field is reserved. Reserved bits. Write as 0.
24–16 FASTCLOCK	This value should be set to a value of pixels, from 0 to 511. When the number of pixels in the input pixel FIFO is LESS than this value, the fast clock control output will be raised. This signal can be used to reduce the system bus clock frequency to save power during horizontal or vertical blanking intervals. This value should also be programmed to a value that is greater than the "PANIC" threshold value. This will allow a faster clock to recover the number of pixels in the FIFO before a "panic" level is encountered.
15–9 RSRVD1	This field is reserved. Reserved bits. Write as 0.
PANIC	This value should be set to a value of pixels from 0 to 511. When the number of pixels in the input pixel FIFO is less than this value, the internal panic control output will be raised. This signal can be used to raise the access LCDIF's access priority.

13.2.5.30 LCDIF AS Buffer Control Register (LCDIF_AS_CTRL)

The Alpha Surface Parameter register provides additional controls for AS.

Address: 3032_0000h base + 210h offset = 3032_0210h



LCDIF_AS_CTRL field descriptions

Field	Description
31 CSI_VSYNC_ENABLE	When this bit is set by software, the LCDIF work as sync mode with CSI input.
30 CSI_VSYNC_POL	Default 0 active low during VSYNC_PULSE_WIDTH time and will be high during the rest of the VSYNC period. Set it to 1 to invert the polarity.

Table continues on the next page...

LCDIF_AS_CTRL field descriptions (continued)

Field	Description
29 CSI_VSYNC_MODE	this bit is set by software to decide which vsync generate mode. LCDIF vsync generate by internal counter when set to 0, LCDIF vsync delayed by each csi_vsync_in when set to 1; INT_SYNC_MODE = 0x0 LCDIF vsync generate by internal counter. EXT_SYNC_MODE = 0x1 LCDIF vsync delayed by each csi_vsync_in.
28 CSI_SYNC_ON_IRQ_EN	This bit is set to enable an interrupt when LCDIF lock with CSI vsync input.
27 CSI_SYNC_ON_IRQ	this bit is set by software to decide which vsync generate mode. LCDIF vsync generate by internal counter when set to 0, LCDIF vsync delayed by each csi_vsync_in when set to 1; INT_SYNC_MODE = 0x0 LCDIF vsync generate by internal counter. EXT_SYNC_MODE = 0x1 LCDIF vsync delayed by each csi_vsync_in.
26–24 RVDS1	Reserved, always set to zero.
23 PS_DISABLE	When this bit is set by software, the LCDIF will disable PS buffer data.
22–21 INPUT_DATA_SWIZZLE	This field specifies how to swap the bytes either in the HW_LCDIF_DATA register or those fetched by the AXI master part of LCDIF. The swizzle function is independent of the WORD_LENGTH bit. See the explanation of the HW_LCDIF_DATA below for names and definitions of data register fields. The supported swizzle configurations are: NO_SWAP = 0x0 No byte swapping.(Little endian) LITTLE_ENDIAN = 0x0 Little Endian byte ordering (same as NO_SWAP). BIG_ENDIAN_SWAP = 0x1 Big Endian swap (swap bytes 0, 3 and 1, 2). SWAP_ALL_BYTES = 0x1 Swizzle all bytes, swap bytes 0, 3 and 1, 2 (aka Big Endian). HWD_SWAP = 0x2 Swap half-words. HWD_BYTE_SWAP = 0x3 Swap bytes within each half-word.
20 ALPHA_INVERT	Setting this bit to logic 0 will not alter the alpha value. A logic 1 will invert the alpha value and apply (1-alpha) for image composition.
19–16 ROP	Indicates a raster operation to perform when enabled. Raster operations are enabled through the ALPHA_CTRL field. MASKAS = 0x0 AS AND PS MASKNOTAS = 0x1 nAS AND PS MASKASNOT = 0x2 AS AND nPS MERGEAS = 0x3 AS OR PS MERGENOTAS = 0x4 nAS OR PS MERGEASNOT = 0x5 AS OR nPS NOTCOPYAS = 0x6 nAS NOT = 0x7 nPS NOTMASKAS = 0x8 AS NAND PS NOTMERGEAS = 0x9 AS NOR PS XORAS = 0xA AS XOR PS NOTXORAS = 0xB AS XNOR PS

Table continues on the next page...

LCDIF_AS_CTRL field descriptions (continued)

Field	Description
15–8 ALPHA	Alpha modifier used when the ALPHA_MULTIPLY or ALPHA_OVERRIDE values are programmed in REG_AS_CTRL[ALPHA_CTRL]. The output alpha value will either be replaced (ALPHA_OVERRIDE) or scaled (ALPHA_MULTIPLY) when selected.
7–4 FORMAT	Indicates the input buffer format for AS. ARGB8888 = 0x0 32-bit pixels with alpha RGB888 = 0x4 32-bit pixels without alpha (unpacked 24-bit format) ARGB1555 = 0x8 16-bit pixels with alpha ARGB4444 = 0x9 16-bit pixels with alpha RGB555 = 0xC 16-bit pixels without alpha RGB444 = 0xD 16-bit pixels without alpha RGB565 = 0xE 16-bit pixels without alpha
3 ENABLE_ COLORKEY	Indicates that colorkey functionality is enabled for this alpha surface. Pixels found in the alpha surface colorkey range will be displayed as transparent (the PS pixel will be used).
2–1 ALPHA_CTRL	Determines how the alpha value is constructed for this alpha surface. Indicates that the value in the ALPHA field should be used instead of the alpha values present in the input pixels. Embedded = 0x0 Indicates that the AS pixel alpha value will be used to blend the AS with PS. The ALPHA field is ignored. Override = 0x1 Indicates that the value in the ALPHA field should be used instead of the alpha values present in the input pixels. Multiply = 0x2 Indicates that the value in the ALPHA field should be used to scale all pixel alpha values. Each pixel alpha is multiplied by the value in the ALPHA field. ROPs = 0x3 Enable ROPs. The ROP field indicates an operation to be performed on the alpha surface and PS pixels.
0 AS_ENABLE	When this bit is set by software, the LCDIF will start fetching AS buffer data in bus master mode and combine it with another buffer.

13.2.5.31 Alpha Surface Buffer Pointer (LCDIF_AS_BUF)

This register is used to indicate the base address of the AS buffer.

Address: 3032_0000h base + 220h offset = 3032_0220h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ADDR																															
W	ADDR																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

LCDIF_AS_BUF field descriptions

Field	Description
ADDR	Address pointer for the alpha surface 0 buffer.

13.2.5.32 LCDIF_AS_NEXT_BUF

When the LCDIF is behaving as a master, this address points to the address of the next frame of data that will be sent out via the LCDIF. It is upto the software to make sure that this register is programmed before the end of the current frame, otherwise it might result in old data going out the LCDIF. This address must always be double-word aligned.

Address: 3032_0000h base + 230h offset = 3032_0230h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_AS_NEXT_BUF field descriptions

Field	Description
ADDR	Address of the next frame that will be transmitted by LCDIF.

13.2.5.33 LCDIF Overlay Color Key Low (LCDIF_AS_CLRKEYLOW)

If a pixel in the current overlay image with a color that falls in the range from the ASCOLORKEYLOW to ASCOLORKEYHIGH range, it will use the PS pixel value for that location. Colorkey operations are higher priority than alpha or ROP operations.

Address: 3032_0000h base + 240h offset = 3032_0240h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
	RSVD1								PIXEL																							
Reset	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	

LCDIF_AS_CLRKEYLOW field descriptions

Field	Description
31–24 RSVD1	Reserved, always set to zero.
PIXEL	Low range of RGB color key applied to AS buffer

13.2.5.34 LCDIF Overlay Color Key High (LCDIF_AS_CLRKEYHIGH)

If a pixel in the current overlay image with a color that falls in the range from the ASCOLORKEYLOW to ASCOLORKEYHIGH range, it will use the PS pixel value for that location. Colorkey operations are higher priority than alpha or ROP operations.

Address: 3032_0000h base + 250h offset = 3032_0250h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RSVD1								PIXEL																							
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_AS_CLRKEYHIGH field descriptions

Field	Description
31–24 RSVD1	Reserved, always set to zero.
PIXEL	High range of RGB color key applied to AS buffer

13.2.5.35 LCD working insync mode with CSI for VSYNC delay (LCDIF_SYNC_DELAY)

The LCDIF DOTCLK mode VSYNC will delay from CSI_VSYNC as
(V_COUNT_DELAY * HSYNC_PERIOD + H_COUNT_DELAY) PIXCLK cycles

Address: 3032_0000h base + 260h offset = 3032_0260h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	V_COUNT_DELAY																H_COUNT_DELAY															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LCDIF_SYNC_DELAY field descriptions

Field	Description
31–16 V_COUNT_DELAY	LCDIF VSYNC delayed counter for CSI_VSYNC.
H_COUNT_DELAY	LCDIF VSYNC delayed counter for CSI_VSYNC.

13.3 Graphics Processing Unit (GPU)

13.3.1 Overview

The graphics processing unit (GPU) provides high performance, high quality graphics, low power consumption, and the smallest silicon footprint for its class. The dynamic power consumption is minimized by extensive use of multi-level hierarchical clock gating.

An optimized software stack, complete software development tools, and a growing application ecosystem are supported by a robust graphics pipeline designed for industry-standard APIs, and with full support for Android, Linux, and Windows embedded development platforms. The GPU supports the following graphics APIs:

- OpenGL ES 3.1/3.0
- OpenGL ES 2.0/1.1
- EGL 1.4
- DirectX 11 (9_3)
- OpenGL 3.1/3.0
- OpenGL 2.1/2.0
- OpenVG 1.1
- OpenCL 1.2 FP / 1.1 FP w/ Global memory

13.3.1.1 Block Diagram

A block diagram and a description of the functional blocks of the complete graphics pipeline is given below.

- Host Interface - Allows GCCORE to communicate with external memory and the CPU through AXI or AHB bus. In this block data crosses clock domain boundaries.
- Memory Controller - Internal memory management unit that controls the block-to-host memory request interface.
- Fetch Engine - Inserts high level primitives and commands into the graphics pipeline.
- Input Assembly - Assembles vertices and primitives for further processing.
- Compute Modules:
 - Fixed Function 3D Rendering Engine Units:
 - Setup/Raster - Converts triangles and lines into pixels. Computes slopes of color attributes and texture coordinates. Performs clipping.

- Texture Engine - Retrieves texture information from memory upon request by the vertex shader or fragment shader. Performs interpolation and filtering, and transfers the computed value to the fragment shader or the vertex shader. The texture unit can process two pixels or two vertices/cycle.
- Pixel Engine/Blit - The Pixel Engine does alpha blending and visible surface determination. The Blit Engine does resolve, clear, tiling and de-tiling as well as FSAA filtering. The Pixel Engine can process two pixels/cycle.
- Programmable Engine Units:
 - Shader Execution Units - SIMD processor that performs as vertex shader and fragment shader. When used as a vertex shader, it performs geometry transformations and lighting computations. When used as a fragment shader it applies texture data and computes color values for each pixel. The shaders also act as Compute Units for OpenCL. The GPU has 4 such vec4 shaders.
- Universal Storage Cache - Cache shared between the Fetch Engine and the Compute Modules. A portion of this cache can be locked to stay on-chip.

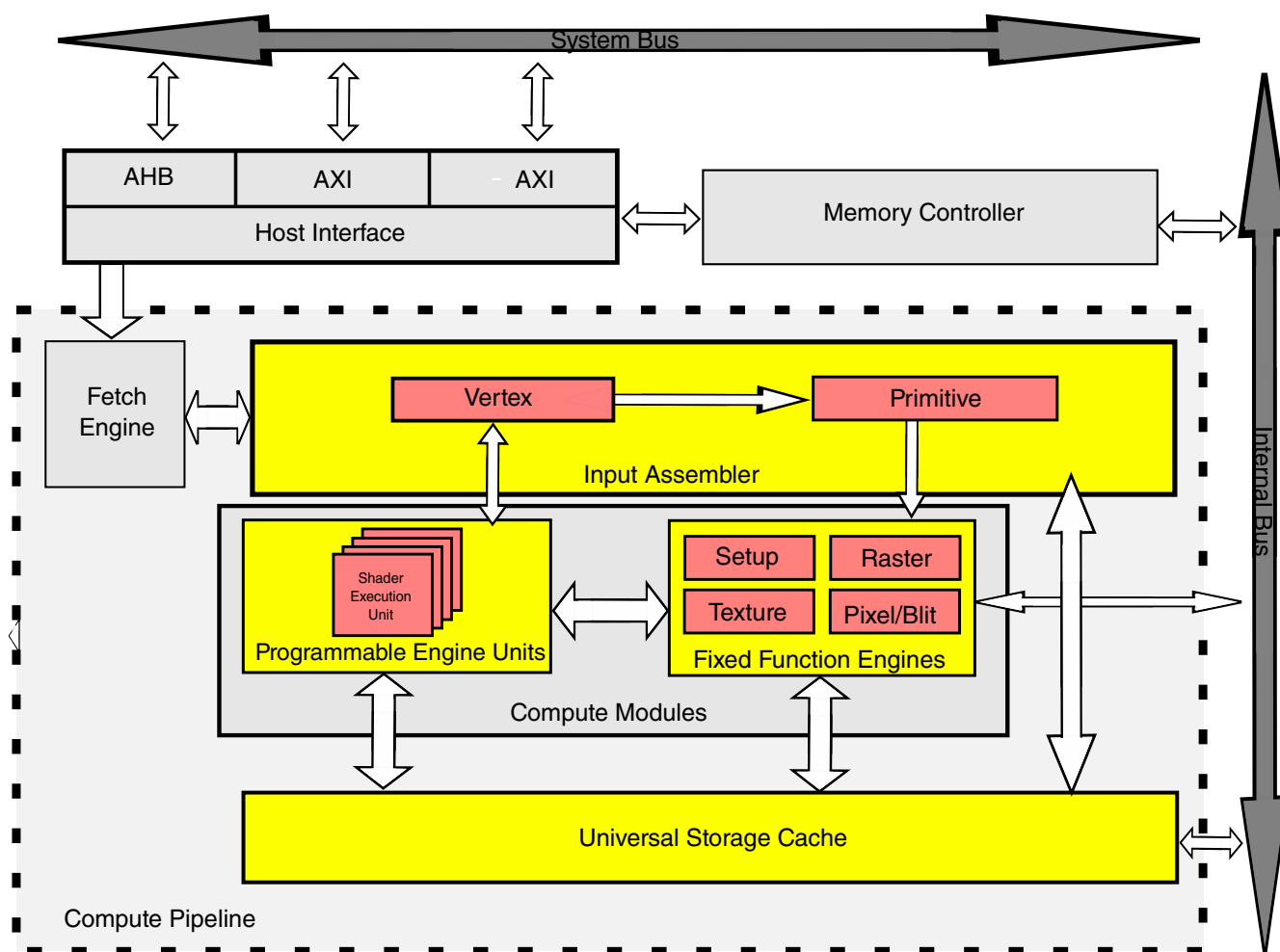


Figure 13-18. GPU Block Diagram

13.3.2 GPU Features

The features of the GPU 3D unit include:

- OpenGL ES 3.1 / 3.0 / 2.0 / 1.1 compliance, including extensions; OpenVG 1.1
- IEEE 32-bit floating-point pipeline
- Ultra-threaded, unified vertex and fragment (pixel) shaders
- Low bandwidth at both high and low data rates
- Low CPU loading
- Up to 16 programmable elements per vertex
- Dependent texture operation with high-performance
- Alpha blending
- Depth and stencil compare

- Point sampling, bi-linear sampling, tri-linear filtering, and cubic textures
- Resolve and fast clear
- 8k x 8k texture size and 8k x 8k rendering target
- 16 Vertex DMA streams
- Generates 2 texels and 2 pixels/cycle.
- Supports YUV formats for display output (YUV422, YUYV 4:2:2)
- 3D Composition engine
- MMU functionality supported
- Wide AA line support
- Primitive restart support
- Software programmable Big Endian support
- Performance Counters for DMA Profiling

The following table describes API support and select architectural features of the GPU:

Table 13-5. GPU Architecture Features

Feature	GPU Support
Primary API	OpenGL ES 3.1 / 3.0 / 2.0 / 1.1 OpenCL 1.1 / 1.2
Additional APIs	OpenVG 1.1 DirectX 11 (9_3) OpenGL 3.1 / 3.0 / 2.0 / 2.1
Other graphics support	EGL 1.4
Drivers	OpenGL ES 3.1 / 3.0 / 2.0 / 1.1 OpenVG 1.1 EGL 1.4 DirectX 11 (9_3) OpenGL 3.1 / 3.0 / 2.0 / 2.1 OpenCL 1.1 / 1.2
Operating systems	Windows Embedded Compact / Embedded CE 6 Embedded Linux Android 8.0/7.x / 6.0 / 5.1 / 4.x
Z (depth)	Early Z support included
Stencil	Early stencil support included
GLSL ES Shader languages	3.1 / 3.0 / 1.0
DirectX Shader model compatibility	SM3.0
Shader types and execution units	Four programmable Scalable Ultra-threaded Unified Shaders (SIMD4:transcendental,ctl-flow,tx-load) one instruction issue per shader per clock; IEEE 32-bit floating-point pipeline supports long shader instructions
FSAA anti-aliasing mechanisms	High quality MSAA 4x

13.3.2.1 GPU Host and Memory Interface Features

Table 13-6. GPU host and memory interface features

Feature	GPU Support
AHB interface	32-bit
AXI interface	2 128-bit AXI / ACE-Lite interfaces for external memory access
Virtual memory support	Yes
Code and data memory location restrictions	Unrestricted; arbitrary memory reads and writes
Physical address	32 bits
Resource locks with CPU	Semaphore lock

13.3.2.2 GPU Power Management Features

Table 13-7. Power management features

Feature	GPU Support
Low power CMOS technology compatible	Yes
Dynamic frequency scaling based on current load	Yes
Automatic clock gating of flip flops and rams	Yes
Global clock gating of unused macro blocks	Yes
Independent clock gating for the shader core	Yes
Software controlled effective clock frequency without changing the PLL	Yes

13.3.2.3 GPU Command Processor Features

Table 13-8. GPU command processor features

Feature	GPU Support
Counters	Variety of hardware counters for performance profiling

13.3.2.4 OpenCL Support

Support for OpenCL includes the following capabilities in a full memory configuration:

Table 13-9. OpenCL Support

OpenCL Support	GPU Support
Profile	Full Profile with Global memory
Compute Devices (GPGPU cores)	1
Compute Units per device	1
Processing Elements per compute unit (vec4 sh cores)	4
Preferred work-group/ thread group size	8
Max count global work-items each dim	64KB
Max count work-items each dim per work-group	1 KB
Instruction Cache Size	512

13.3.2.5 Texture processing

Table 13-10. Texture Processing Features

Feature	GPU Support					
Fixed-point input texture formats	A8, L8, I8, A8L8, ARGB4, XRGB4, ARGB8, XRGB8, ABGR8, XBGR8, R5G6B5, A1RGB5, X1RGB5, YUY2, UYVY, D16, D24X8, A8_OES, DXT1, DXT2, DXT3, DXT4, DXT5, HDR7E3, HDR6E4, HDR5E5, HDR6E5, RGBE8, RGBE8F, RGB9E5, RGB9E5F, ETC1, ETC2_RGB8, ETC2_RGB8A1, ETC2_RGB8A8, EAC_R11_UNSIGNED, EAC_RG11_UNSIGNED, EAC_RG11_SIGNED, R8G8, RF16, RF16GF16, RF16GF16BGF16AF16, RF32, RF32GF32, R10G10B10A2, EAC_R11_SIGNED, R8_SNORM, RG8_SNORM, RGBX8_SNORM, RGBA8_SNORM, YUV_ASSEMBLY, RI8, RI8GI8, RI8GI8BI8AI8, RI16, RI16GI16, RI16GI16BI16AI16, RI32, RI32GI32, RF11GF11BF10, RF16GF16BF16AF16, RI10GI10BI10AI2, R8G8B8G8, G8R8G8B8, RI8GI8BI8GI8, GI8RI8GI8BI8, RF11GF11BF10. All fixed-point formats are filtered. sRGB conversion and Swizzling supported for color channels.					
	Bits	Format	Alpha/X/s	R	B	G
	16	ARGB4444	4	4	4	4
	16	XRGB4444	4 don't care	4	4	4
	16	ARGB1555	1	5	5	5
	16	XRGB1555	1 don't care	5	5	5
	16	RGB565	0	5	6	5
	32	ARGB8888	8	8	8	8
	32	sARGB8888	8	8	8	8
	32	XRGB8888	8 don't care	8	8	8
	32	ABGR8888	8	8	8	8
	32	XBGR8888	8 don't care	8	8	8

Table continues on the next page...

Table 13-10. Texture Processing Features (continued)

Feature	GPU Support								
	Planes	Format	Mode	Y	U	V	UV	YUYV	UYVY
	1	YUY2(YUYV)	4:2:2					1	
	1	UYVY	4:2:2						1
Additional texture formats supported through Resolve conversion	Resolve converts planar to YUV 4:2:2 packed:								
	Planes	Format	Mode	Y	U	V	UV	YUYV	UYVY
	3	YV12	4:2:0	1	1	1			
	2	NV12	4:2:0	1			1		
Texture compression	4 bits and 8 bits per texel								
Compressed texture formats	DXT1, DXT2, DXT3, DXT4, DXT5, ETC1, ETC2, EAC11. All compressed formats are filtered.								
Texture Formats for Optimal Performance	Formats with 4 bits per texel: DXT1, ETC1, EXT2_RGB8A1, EAC_R11_UNSIGNED, EAC_R11_SIGNED								
Texture size maximum	8k x 8k								
Addressing modes	wrap, mirror, clamp								
Mipmap support	14 mipmap levels; programmable LOD biasing and replacement								
Shadow texture	Depth texture PCF filtering								
Texture coordinate fraction bits	5 bits								
Texture samplers	32 samplers, indexable								
Textures per fragment maximum	16 texture samplers								
Dependent texture operation	High performance; unlimited dependent texture reads								
Dependent tx per fragment max, relative sampling	No Limit								
Texture repeat max	256								
Texture types	2D, seamless cube map, 1D, projected, depth, bump map, displacement map, PCF, 3D, texture array								
Texture filters	Point sample, bi-linear, fast tri-linear (2 pixels/cycle), quad-linear, Anisotropic								
Texture component mapping: D3D, OGL ES options	Supports both D3D and OES options								
Texture size types	Power-of-2, Non-square, Non-power-of-2								
Texture swizzle	Yes								
OES3 half-float support	Yes								

13.3.2.6 Fragment Processing and Render Targets

Table 13-11. Fragment Processing and Render Target Features

Feature	GPU Support					
FSAA mechanisms	MSAA 4x, SSAA 4x					
Fragment color, alpha, Z, stencil precision	Bits	Format	Alpha	R	B	G
	16	ARGB4444	4	4	4	4
	16	XRGB4444	4 don't care	4	4	4
	16	ARGB1555	1	5	5	5
	16	XRGB1555	1 don't care	5	5	5
	16	RGB565	0	5	6	5
	16	RF16	0	16	0	0
	32	ARGB8888	8	8	8	8
	32	sRGB8888	8	8	8	8
	32	XRGB8888	8 don't care	8	8	8
	32	ABGR8888	8	8	8	8
	32	XBGR8888	8 don't care	8	8	8
	32	RGB10_a2ui	2	10	10	10
	32	RF16GF16	0	16	16	0
	32	RF32	0	32	0	0
	64	RF32GF32	0	32	32	0
	64	RF16GF16BF16AF16	16	16	16	16

13.3.2.7 Destination / Alpha Blending

Table 13-12. Destination / Alpha Blending Features

Feature	GPU Support					
	Bits	Format	Alpha	R	B	G
Destination / Resolve (3DBlit) color formats	16	ARGB4444	4	4	4	4
	16	ARGB1555	1	5	5	5
	16	RGB565	0	5	6	5
	32	ARGB8888	8	8	8	8
	32	sRGB8888	8	8	8	8
	32	ABGR8888	8	8	8	8
	32	RGB10_a2ui	2	10	10	10

13.3.3 Usage Mode

The GPU should be programmed through the NXP provided driver. NXP does not provide support for software that directly programs the GPU registers. APIs for programming the GPU through the software driver are described in separate driver documentation.

13.4 HD Display Transmitter Controller (HDMI TX)

13.4.1 Overview

The HD Display Transmitter Controller IP offers multi-protocol support of standards such as HDMI, DisplayPort, eDP, with one of these standards supported at a time.

the mentioned protocol options enable switching between the modes to be applied on a system level and performed by means of software configuration.

Protocols supported by the controller are:

- HDMI 1.4 Specification
- HDMI 2.0a Specification
- DisplayPort Specification, Version 1.3
- eDP Specification, Version, 1.4
- High-bandwidth Digital Content Protection system, Mapping HDCP to HDMI, Revision 2.2

- High-bandwidth Digital Content Protection system, Mapping HDCP to DisplayPort, Revision 2.2
- High-bandwidth Digital Content Protection system, Mapping HDCP to HDMI, Revision 1.4
- High-bandwidth Digital Content Protection system, Mapping HDCP to DisplayPort, Revision 1.3
- CEA-861-F
- IEC60958
- ARM AMBA 3 APB Specification, Version 1.0

13.4.1.1 Features

The HDMI_TX includes the following generic features:

- Compliant with HDCP2.2 (and back compatible with HDCP1.3/HDCP1.4)
- Variety of video resolutions and formats
 - Supports up to 4k2k at 60Hz resolution
 - 25-600Mhz pixel clk
 - Supports 8, 10, 12, and 16bpp
 - Supports RGB, YCbCr422 formats
- Variety of audio formats – PCM and compressed, over I2S interface
 - I2S PCM samples are converted to comply with IEC60958 sub-frames
 - Supporting I2S-TDM
- Enhanced testability using uCPU debugger (over JTAG) and Integrated Pattern Generator

HDMI 2.0a and DisplayPort 1.3 specific features:

- In HDMI mode, supports up to 600Mhz pixel clk
- In DisplayPort mode, supports up to HBR2

DisplayPort-specific features:

- 1Mbps AUX channel
- Configurable number of lanes (1, 2, or 4)
- SSC injection is supported by PHY
- Support for Single Stream-only (SST)
- 6bpp support

eDP-specific features:

- Supports backlights and multi-touch commands
- eDP DPCD registers

- Variable link rate R162/R216/R243/R270/R324/R432/R540
- Fast link training

13.4.2 Block diagram

This is a high-level block diagram illustrating the HD Display Transmitter Controller IP.

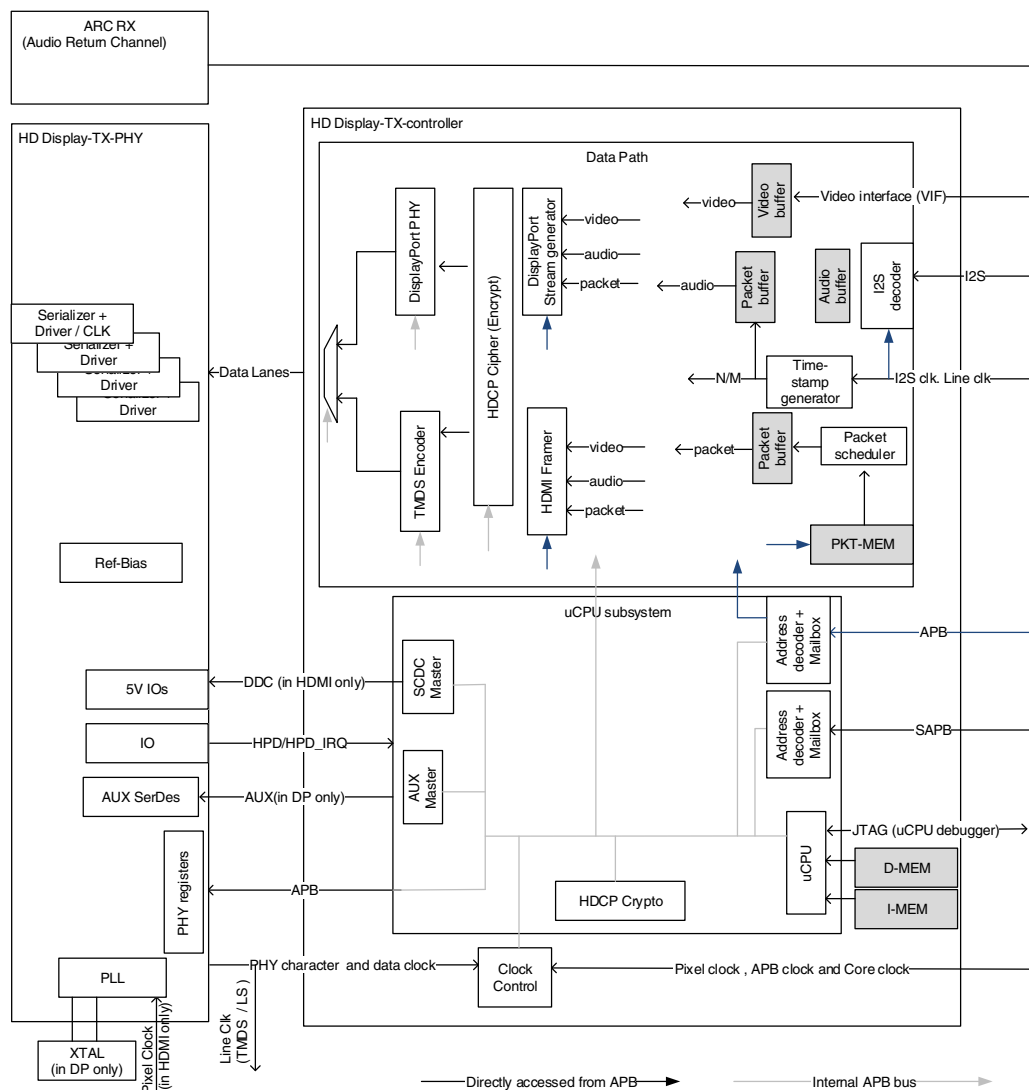


Figure 13-19. HDMI_TX block diagram

13.4.3 Video Packing

In HDMI, the Controller automatically detects the video timing from the video interface bus (VIF) and pack the video over the TMDS stream. The PHY PLL generates the character clock from the pixel clock configured by the HD Display Controller to a specific ratio (1:1, 1:1.25, 1:1.5, 1:2).

In DisplayPort, the Controller is configured by the external host with the video timing of the video interface bus (VIF) and packs the video over the main-stream. Video time stamp information (Nvid, Mvid) is generated by the Controller based on the ratio between the pixel clock and the ls clock.

The video data is transparently mapped to the video stream, where the Controller does not perform any color conversion.

13.4.4 Audio Packing

Audio info-frames and time-stamp information are generated by the Controller and encoded over the TMDS stream blanking period on HDMIsecondary stream in Displayport.

The Audio time-stamp information (NCTS in HDMI Maud and Naud in DisplayPort) is generated by the controller based on the ratio between the audio sample clock and the link clock (tmcls in HDMIls clock in DisplayPort).

A peripheral module converting I2S stream into AIF bus can easily interface with the Controller.

13.4.5 Info-Frame Packing

Info-frames are encoded over the TMDS stream blanking period on HDMI the secondary stream in DisplayPort.

Time sensitive info-frames (i.e. GC-packet, CTS in HDMI) are generated autonomously by the Controller. Other info-frames (i.e. audio and video meta-data) are written into the PKT-MEM by the external host (directly accessed over the APB bus), to be then read and encoded by the Controller over the TMDS stream blanking period (on HDMI) the secondary stream (in DisplayPort).

Checksum (in HDMI) is parity bytes (in DisplayPort) are generated by the Controller.

13.4.6 HDCP

HDCP 2.2 is supported by the Controller. A dedicated HW provides the cipher and the cryptographic acceleration, the HDCP authentication state-machine is implemented in the FW (running over the uCPU).

13.4.7 Clock Control

The clock control block integrates clock-gates and provides clock metering functionality.

13.4.8 Functional Description

13.4.8.1 Functional description

The following sections describe functional details of the HDMI_TX module.

13.4.8.1.1 Management And Control Interface

The HD Display TX controller has the following APB slave ports that are controlled by the host processor:

- APB Slave port (APB):
 - Used as the main control interface between the host processor and the HD Display TX controller.
 - Used in different methods during boot.
 - Used for directly loading the FW into the memories.
 - Used for directly accessing several HW modules and for communicating the internal uCPU over the mailbox channel.
- SAPB Slave port (SAPB):
 - Used for secure connection, that is, HDCP key loading between the host processor and the uCPU through mailbox and pre-defined commands.

The internal APB bus has an output APB master port which is used to manage the PHY.

Each Mailbox channel (APB or SAPB) generates an interrupt-level signal when its received FIFO is not empty or when its transmitted FIFO is full. The interrupt is cleared when the MAILBOX_INT_STATUS register is read. There are two sets of register, one for APB and one for SAPB.

13.4.8.1.2 Video Interface

The video interface (VIF) is compliant with CEA video timing and composed of 48-bit pixel data and the following control signals: data-enable, hsync, and vsync. The video timing of the video-interface bus (VIF) complies with the specification in CEA-861-F Annex L.

NOTE

Only a single video stream is supported

The video interface has the following control signals:

- VSYNC – Video Vertical Blanking indication signal as defined in the above standards.
- HSYNC - Video Horizontal Blanking indication signal as defined in the above standards.
- DATA ENABLE – Video data enable indication signal as defined in the above standards.
- DATA - 48 bits of data for supporting up to 16bpp.
- Clock – As defined for each format from 25MHz-600MHz.

13.4.8.1.2.1 Pixel Mapping

The following table shows the pixel mapping in different video formats.

Video Bus	RGB / YCbCr 4:4:4				4:2:0 / 4:2:2				Y Only
Bit Width	8	10	12	16	8	10	12	16 (4:2:0 only)	16/ 12/ 10/ 8
Channel 2	R/Cr				Cb/Cr				
[47:40]	Cr[15:8]	Cr[15:6]	Cr[15:8]	Cb[15:0]	C[15:8]	C[15:6]	C[15:4]	C[15:0]	
[39:38]									
[37:36]									
[35:32]									
Channel 1	G/Y				Y				
[31:24]	Y[15:8]	Y[15:6]	Y[15:4]	Y[15:0]	Y[15:8]	Y[15:6]	Y[15:4]	Y[15:0]	Y[15:0] Y[15:4] Y[15:6] Y[15:8]
[23:22]									
[21:20]									
[19:16]									
Channel 0	B/Cb								
[15:12]	Cb[15:8]	Cb[15:6]	Cb[15:4]	Cb[15:0]					
[11:8]									
[7:6]									
[5:4]									
[3:0]									

Figure 13-20. Pixel Mapping

NOTE

- In YCbCr4:2:2 or YCbCr4:2:0 the Cb and Cr are alternating as defined in timing diagram below.
- 6bpp in all formats and 16bpp in YCbCr422 format are supported only in DisplayPortoreDP.

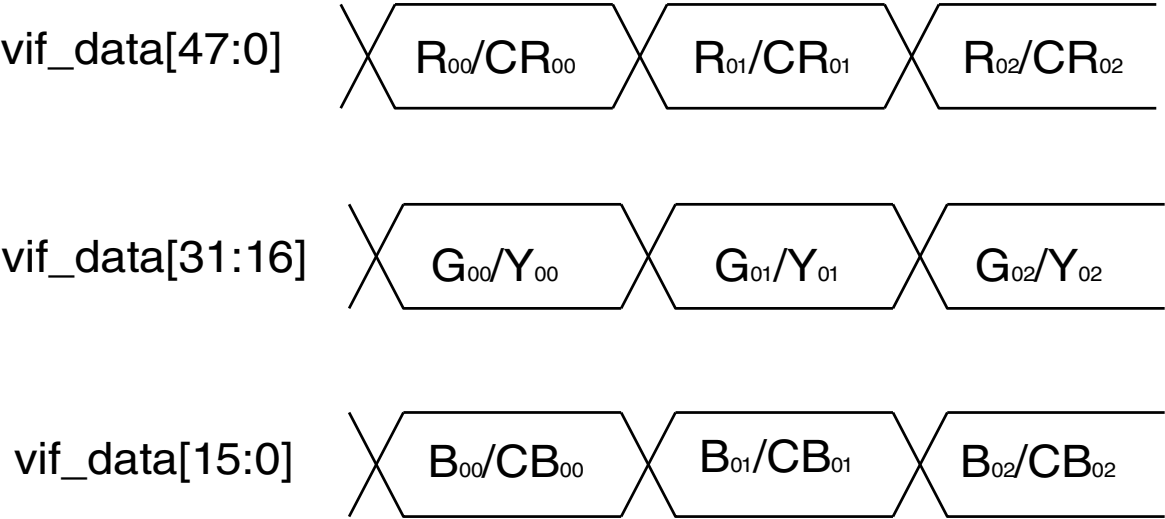


Figure 13-21. YCbCr 4:4:4 pixel mapping

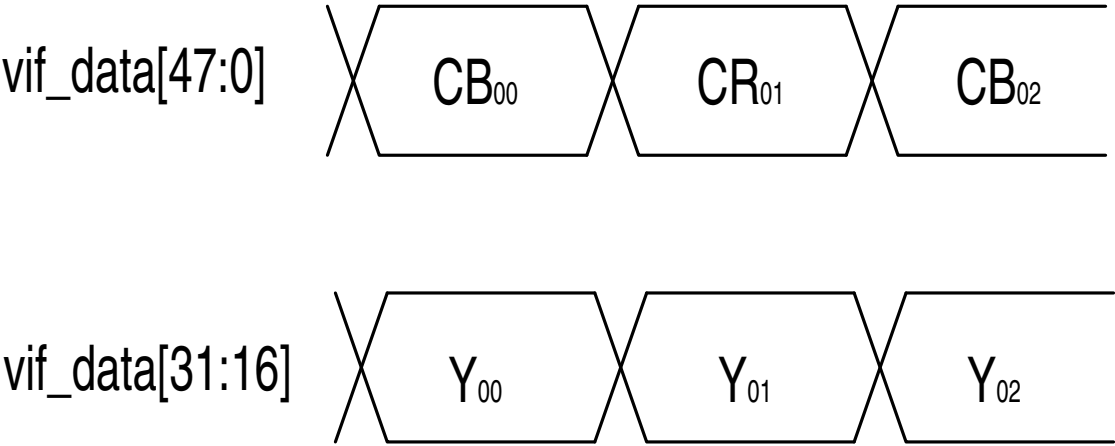


Figure 13-22. YCbCr 4:2:2 pixel mapping

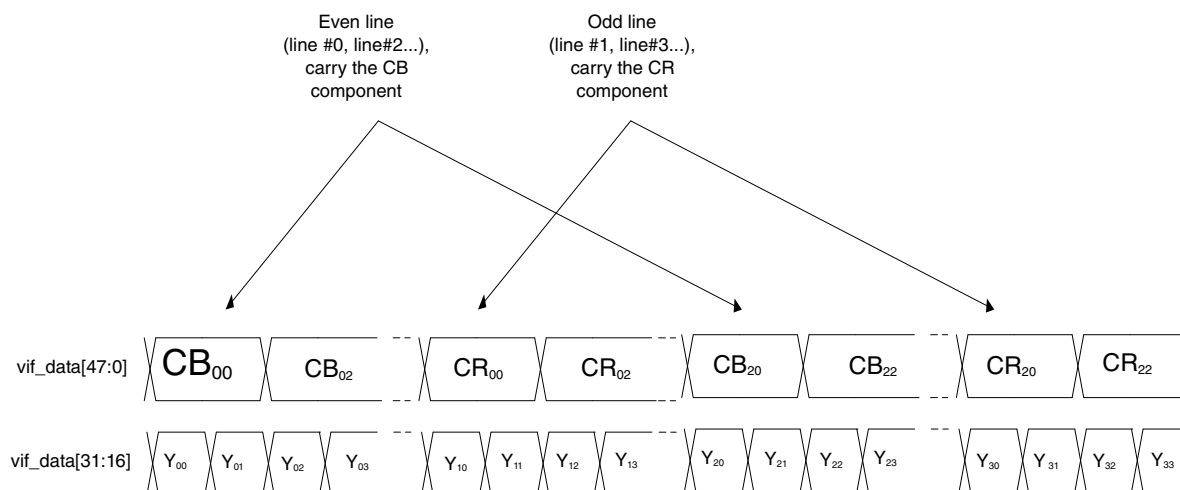


Figure 13-23. YCbCr 4:0:0 pixel mapping

13.4.8.1.2.2 Interlaced Video Timing

The HD Display TX controller expects the interlaced timing format to comply with the specification defined in CEA-861-F. The field information will be extracted from the input signal (hsync, vsync and de) and will be propagated towards the framer (i.e VB-ID field bit in DisplayPort).

13.4.8.1.2.3 3D Stereo Video Timing

The HD Display TX controller expects the stereo timing format to comply with one of the following formats:

- Frame-packing (in HD1.4)/Stacked-Top (in DP1.3).
- Top-Bottom (in HD1.4).
- Side-By-Side (in DP1.3 or HD1.4).

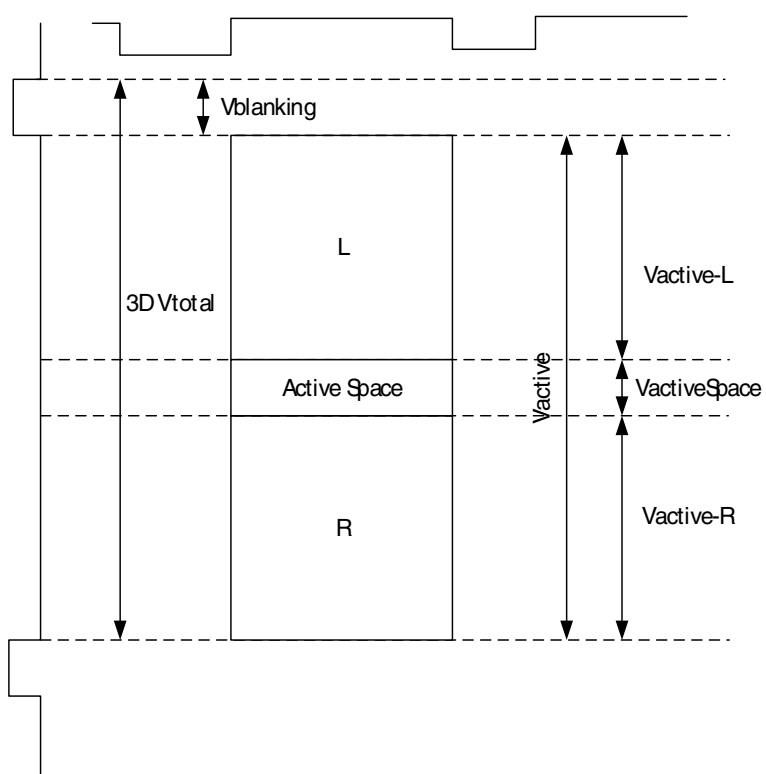


Figure 13-24. Frame-packing/Stacked-Top 3D Video Timing

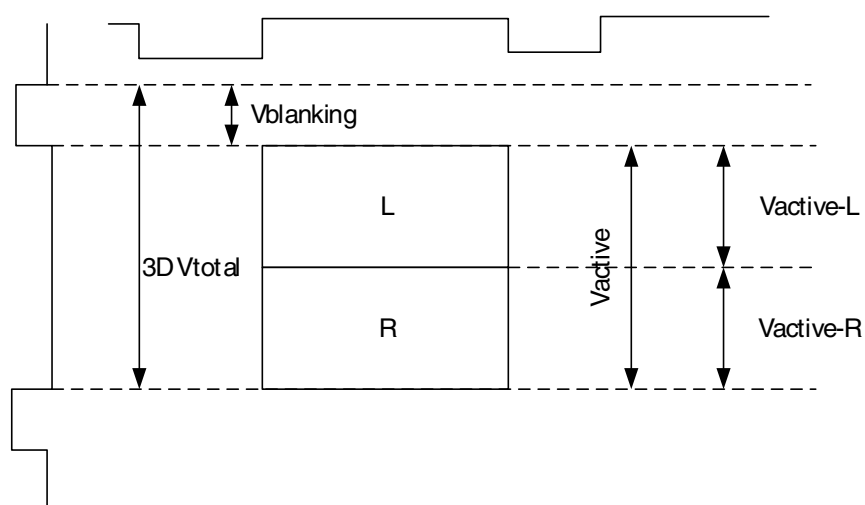


Figure 13-25. Top-Bottom 3D Video Timing

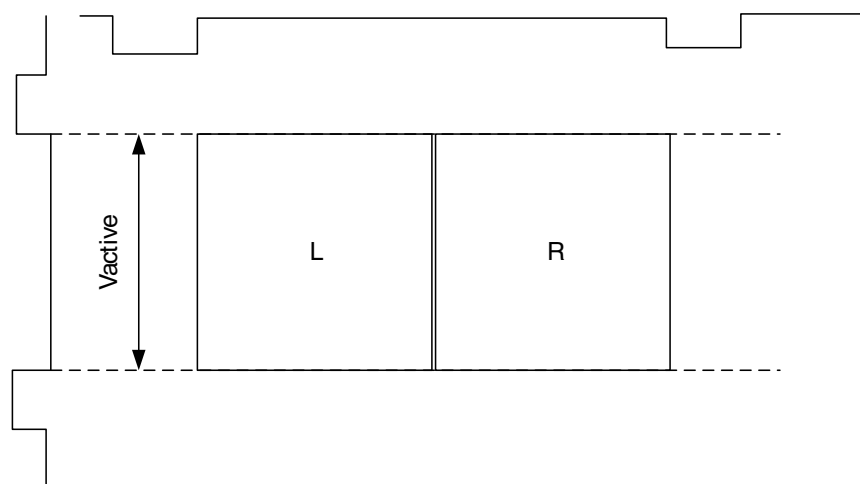


Figure 13-26. Side-By-Side 3D Video Timing

In the DisplayPort mode, the HD Display TX controller can convert the Frame-packing video timing into Field-sequential (in DP1.3). While converting into Field-sequential, the Vactive-space height should be equal to the Vertical-blank height.

13.4.8.1.2.4 Color Space Conversion Module

An add-on Color Space Conversion (CSC) module is optionally provided, it supports the video-timing defined in CEA-861-F.

The following conversion operations are supported:

- Color space conversion from RGB to YCbCr
- Color space conversion from YCbCr to RGB
- YCbCr up-sampling from 4:2:2 to 4:4:4
- YCbCr down-sampling from 4:4:4 to 4:2:2

The CSC module is configurable over its dedicated APB slave port.

13.4.8.1.3 Audio Interface

The Audio interface can be configured to support I2S.

When I2S is supported, the integrated I2S decoder extracts the PCM audio samples and converts them into 32-bit SPDIF sub-frame format compliant with IEC60958 to be carried over the data-stream. The Validity Flag (V), User Data (U), and the Channel Status (C) fields are configured by the external host. Parity bit (P) is calculated by the HD Display Transmitter Controller IP .

The I2S interface supports configurable slot time (16, 24, 32 bits), up to 8 channels, where each audio sample can occupy up to 24bits (right justified).

The interface is composed from three signals – I2S-data[3:0], I2S-bit-clk, and I2S-ws.

NOTE

Only a single audio stream is supported.

The sequence of operation is as follows:

1. Samples are decoded.
2. buffered (compensating for the video active period).
3. Mapped into an audio info-frame packet.

13.4.8.1.3.1 I2S Interface

The I2S interface could be configured to support multiple physical channels, i2s_data[n], (n=1, 2, 4), where channels could be configured with variable TDM time slots (m = 2 or 8). Time slot size is 32, 24 or 16 bits, supporting variable word length (28, 24, 20, and 16) configured as left or right justified.

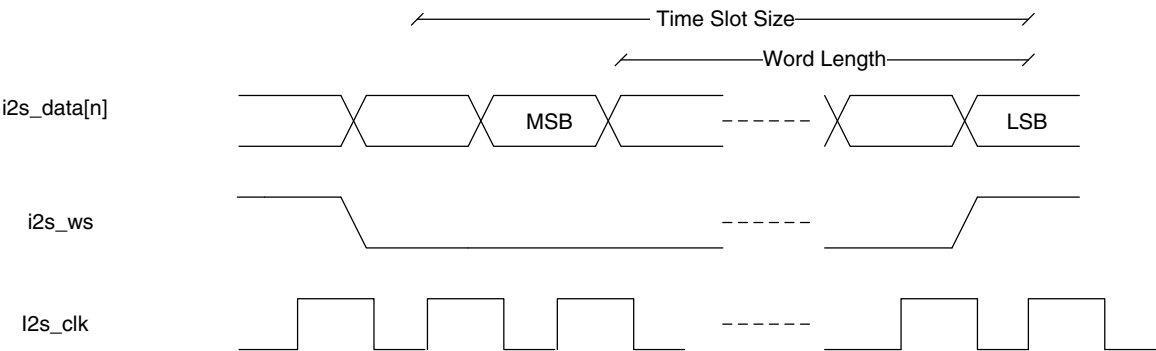


Figure 13-27. I2S bit Allocation (right justification)

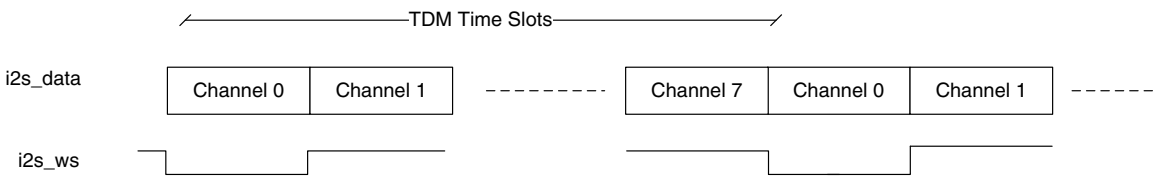


Figure 13-28. I2S TDM time allocation (M=8)

The I2S interface can carry audio samples in one of the following formats:

-
-

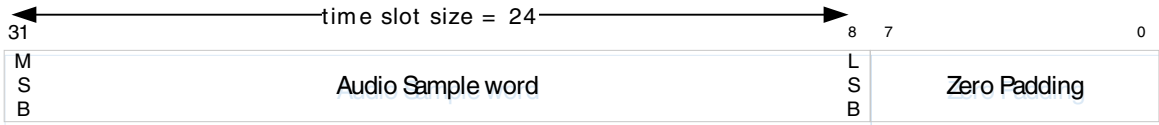


Figure 13-29. Audio L-PCM Sample Format

-

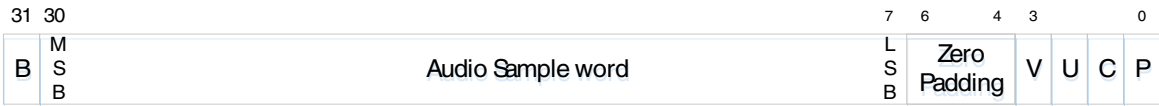


Figure 13-30. Audio SPDIF Sample Format

13.4.8.1.3.2 Audio Path

The following block diagram represents the audio path from I2S interface to creation of audio info-frame.

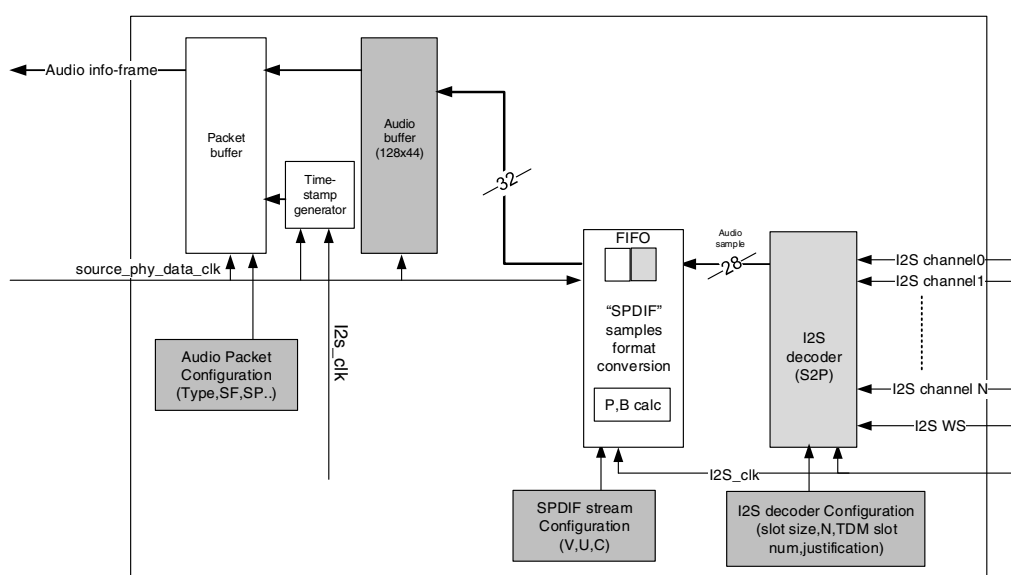


Figure 13-31. Audio Path

13.4.8.1.3.3 Audio Buffer Size

The Audio buffer is used to store audio samples during video active period, its default configuration is 128 samples (each sample occupy 44 bits).

You can configure its size to meet different worst case scenarios by using the following formula:

$$\text{Audio buffer size (in samples)} = \text{MAX}[H_{\text{active}} * \text{MaxFrameRate} * 2] / F_{\text{pixel}}$$

That is, if VIC10 (2880x480i) represents a worst-case scenario, its related video/audio parameters are (refer to Table 7-5 in HDMI1.4b):

$H_{\text{active}} = 2880$, $F_{\text{pixel}} = 54.05\text{MHz}$, Max Frame Rate = 768Khz (i.e 8 ch L-PCM @192Khz)

This results in 82 samples.

13.4.8.1.4 Info-frame Processing

Info-frames are encoded over the video stream vertical blanking period.

Time-sensitive info-frames, that is, GC-packet and CTS in HDMI are generated autonomously by the HD Display TX controller.

Other info-frames, that is, audio and video meta-data are written into the PKT-MEM by the host processor (directly accessed over the APB bus) only when changed, to be then read and encoded by the HD Display TX controller packet-scheduler over the stream once per frame.

Checksum (in HDMI) Parity bytes (in DisplayPort) are generated by the HD Display TX controller.

Up to 16 info-frames are supported. When writing info-frames into the PKT-MEM the host must update the packet allocation table.

13.4.8.1.5 PHY Interface

The Controller interfaces to the PHY over interfaces described in the following sections:

13.4.8.1.5.1 Data Lanes

In HDMI, three data lanes are supported.

In DisplayPort, 1, 2, or 4 data lanes are active – actual number of active lanes is configured during link-training.

Each data lane is of 20-bit width is composed of two characters and is synchronous to phy_data_clk.

13.4.8.1.5.2 APB Master Interface

The Controller has control over the PHY by means of an APB interface.

13.4.8.1.5.3 AUX Interface

The AUX master supports various protocols: native-AUX, I2C over AUX.

13.4.8.1.5.4 DDC Interface

DDC interface is used for reading the sink EDID content and to carry the HDCP and SCDC commands.

13.4.8.1.5.5 HPD/HDP_IRQ

The HPD signal is input to the Controller and provides the HPD functionality in HDMIHPD_IRQ functionality in (DisplayPort mode).

Glitch filtering is integrated in the Controller.

13.4.8.1.6 APB Slave Interface

The Controller interfaces to the SoC as a slave over the APB bus. Up to three APB slave ports are supported.

The primary APB slave interface used for the main external host to IP communication where mailbox is used for communicating with the embedded FW (control and status), packet memory is directly mapped to the APB bus and I-MEM / D-MEM are directly accessed during boot mode (for FW download) . An interrupt signal is driven by the Controller.

The secondary APB slave is used to carry designated commands which are part of the HDCP functionality, i.e. key loading and content encryption control.

13.4.9 Clocks And Resets

13.4.9.1 Clocks Overview

The section describes the clocks and the resets of the HD Display TX IP.

13.4.9.1.1 Overview

The clock domains in the IP are as shown:

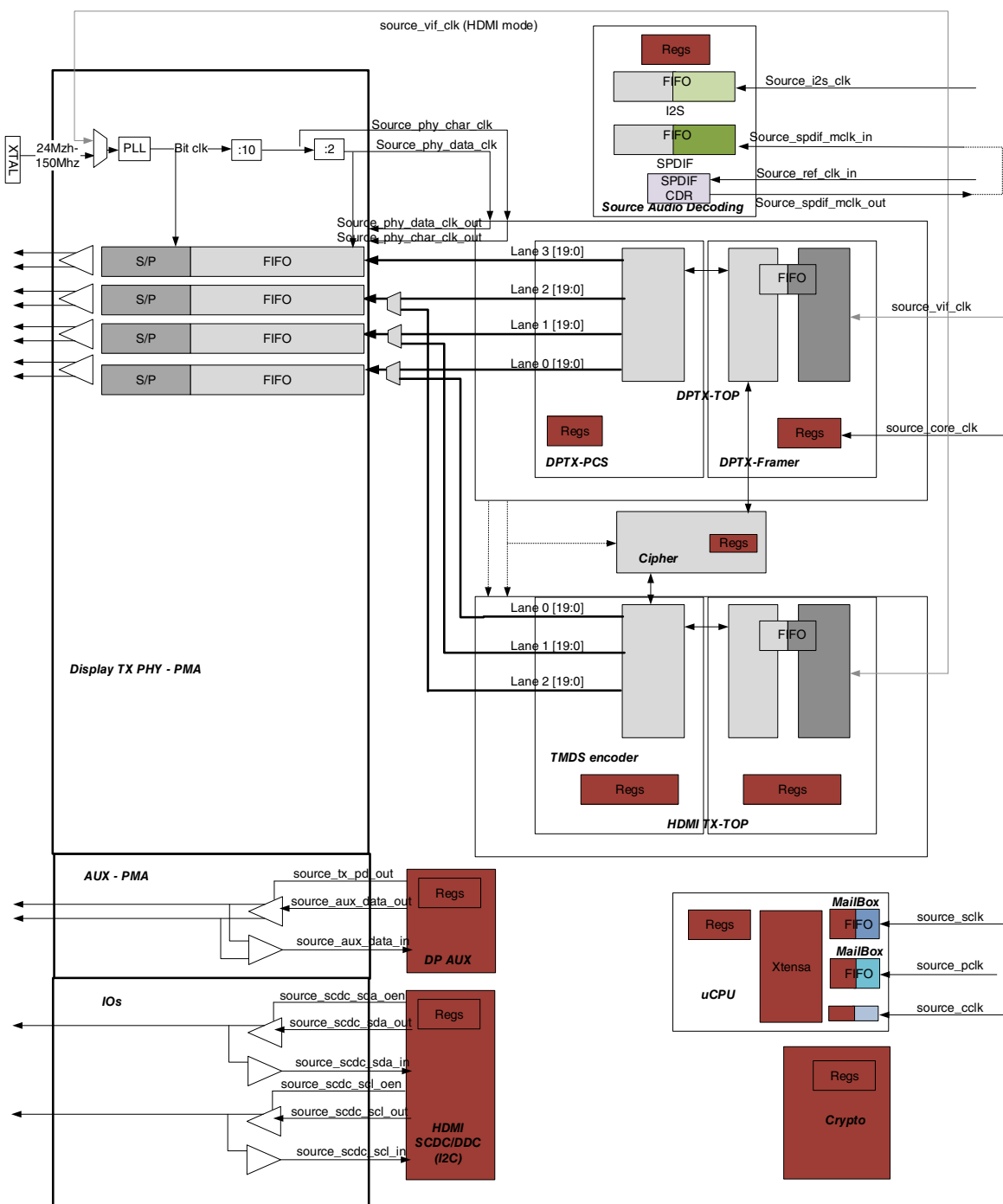


Figure 13-32. HD Display TX Clock Domains

NOTE

- Clock-gates are not represented in the clock scheme.
- Generated clocks are considered as the same clock domain.

The following table shows the list of all input clocks.

Table 13-13. Input Clock Signals

SignalName	Frequency	Source	Description
source_core_clk	100-200Mhz	Host	main HDisplay TX controller clock, used by the CPU and by the internal register bus
source_pclk	50-200MHz	Host	APB slave clock
source_sclk	50-200MHz	Host	SAPB slave clock
source_cclk	50-200MHz	Host	CAPB slave clock
xt_jtck	<10Mhz	JTAG debugger	Debugger JTAG clock
source_ref_clk_in	200-400Mhz	Host	SPDIF CDR reference clock. Must be >10 times the SPDIF BMC rate (i.e for 192Khz , $10 \times 192\text{Khz} \times 64 \times 2 = 245.76\text{Mhz}$)
source_spdif_mclk_in	<24.57Mhz	SPDIF CDR	SPDIF BMC decoder clock , need to be connected to source_spdif_mclk_out after scan clk mux insertion
source_i2s_clk	<50Mhz	Host	Audio I2S stream clock
source_vif_clk	24-600Mhz	Host	Video Pixel Clock
source_phy_char_clk	In DP: 162/270/540 /810MHZ In HDMI: <600Mhz	PHY	Lane clock (in DP:ls_clk) generated by the PHY and passed to use in the HD Display TX controller. In HDMI source_phy_char_clk is derived from source_vif_clk (ratio defined by HDMI standard) but its phase is independent , in DP independent clock
source_phy_data_clk	In DP: 81/135/270/405MHZ In HDMI: <300Mhz	PHY	Lane clock divided by two generated by the PHY and passed to use in the HD Display TX controller. source_phy_data_clk is derived from source_phy_char_clk but its phase is independent

The following table shows the clock signals that are internally generated in the CAR module.

Table 13-14. Generated Clock Signals

SignalName	Frequency	Source	Description
source_aux_clk_out	2MHz	source_core_clk	Not connected– for future use. Divided by configurable number according to the source_core_clk frequency
source_spdif_mclk_out	<24.57Mhz	source_ref_clk_in	Generated by SPDIF CDR, need to be fed-back (after scan clk muxing) to source_spdif_mclk_in
source_xt_pclk	100-200Mhz	source_core_clk	PHY apb clock

13.4.10 Memory Map and register definition

This section includes the HDMI_TX module memory map and detailed descriptions of all registers.

13.4.10.1 HDMI TX register descriptions

13.4.10.1.1 HDMI_TX memory map

Offset	Register	Width (In bits)	Access	Reset value
0h	APB Control Register (APB_CTRL)	32	RW	0000_0007h
4h	Internal CPU Interrupt Polarity Control Register (XT_INT_CTRL)	32	RW	0000_0000h
8h	Mailboxes full indication register (MAILBOX_FULL_ADDR)	32	RO	0000_0000h
Ch	Mailboxes Empty indication register (MAILBOX_EMPTY_ADDR)	32	RO	0000_0001h
10h	Write Data to Mailbox register (MAILBOX0_WR_DATA)	32	RW	0000_0000h
14h	Mailbox Read data register (MAILBOX0_RD_DATA)	32	RO	0000_0000h
18h	Software keep alive counter (KEEP_ALIVE)	32	RO	0000_0000h
1Ch	Software Version LSB (VER_L)	32	RO	0000_0000h
20h	Software Version MSB (VER_H)	32	RO	0000_0000h
24h	Software Lib version written by CPU (LSB) (VER_LIB_L_ADDR)	32	RO	0000_0000h
28h	Software Lib version written by CPU (MSB) (VER_LIB_H_ADDR)	32	RO	0000_0000h
2Ch	Software Debug Register H (SW_DEBUG_L)	32	RO	0000_0000h
30h	Software Debug Register L (SW_DEBUG_H)	32	RO	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
34h	Mailbox Interrupt mask register (MAILBOX_INT_MASK)	32	RW	0000_0000h
38h	Mailbox Interrupt Status register (MAILBOX_INT_STATUS)	32	RO	0000_0000h
3Ch	Core Clock frequency (SW_CLK_L)	32	RW	0000_0000h
40h	Core Clock frequency (SW_CLK_H)	32	RW	0000_0064h
44h	SW Event 0 register (SW_EVENTS0)	32	RO	0000_0000h
48h	SW Event 1 register (SW_EVENTS1)	32	RO	0000_0000h
4Ch	SW Event 2 register (SW_EVENTS2)	32	RO	0000_0000h
50h	SW Event 3 register (SW_EVENTS3)	32	RO	0000_0000h
60h	Internal CPU - On Chip Debug (OCD) Ctrl Register (XT_OCD_CTRL)	32	RW	0000_0003h
64h	Internal CPU - OCD R0 mode configuration (XT_OCD_CTRL_RO)	32	RO	0000_0000h
6Ch	APB Interrupt Mask Register (APB_INT_MASK)	32	RW	0000_0007h
70h	APB interrupt status register (APB_STATUS_MASK)	32	RO	0000_0000h
800h	HDMI shift pattern 3-0 (SHIFT_PATTERN_IN_3_0)	32	RW	0000_0000h
804h	HDMI shift pattern 4-7 (SHIFT_PATTERN_IN_4_7)	32	RW	0000_0000h
808h	HDMI shift pattern 9-8 with control bits (SHIFT_PATTERN_IN9_8)	32	RW	0000_0000h
80Ch	PRBS control (PRBS_CNTRL)	32	RW	0000_2222h
810h	PRBS error insertion (PRBS_ERR_INSERTION)	32	RW	0000_0000h
814h	Lane control register: swap, order, polarity (LANES_CONFIG)	32	RW	0060_001Bh
818h	PHY data select DP/HDMI and HDMI data source (PHY_DATA_SEL)	32	RW	0000_0000h
81Ch	Lane delay control (LANES_DEL_VAL)	32	RW	0000_6420h
900h	Register implemented only for configuration with HDMI. (SOURCE_H_DTX_CAR)	32	RW	0000_0000h
904h	DP TX clock and reset ctrl register (SOURCE_DPTX_CAR)	32	RW	0000_0000h
908h	Source PHY clock and reset ctrl register (SOURCE_PHY_CAR)	32	RW	0000_0000h
90Ch	Register implemented only for configuration with HDMI. (SOURCE_C_EC_CAR)	32	RW	0000_0000h
910h	CBUS clock and reset ctrl register. (SOURCE_CBUS_CAR)	32	RW	0000_0000h
918h	PKT clock and reset ctrl register (SOURCE_PKT_CAR)	32	RW	0000_0000h
91Ch	AIF clock and reset ctrl register (SOURCE_AIF_CAR)	32	RW	0000_0000h
920h	Cipher clock and reset ctrl register (SOURCE_CIPHER_CAR)	32	RW	0000_0000h
924h	Crypto clock and reset ctrl register (SOURCE_CRYPTOCAR)	32	RW	0000_0000h
A00h	Clock Meter control (CM_CTRL)	32	RW	0000_0000h
A04h	I2S clock control (CM_I2S_CTRL)	32	RW	0400_1000h
A08h	SPDIF clock control (CM_SPDIF_CTRL)	32	RW	0400_1000h
A0Ch	Video clock control (CM_VID_CTRL)	32	RW	0400_8000h
A10h	Lane control (CM_LANE_CTRL)	32	RW	0000_0000h
A14h	I2S clock stable, audio clock measured (I2S_NM_STABLE)	32	RO	0000_0000h
A18h	I2S clock stable, lane clock measured (I2S_NCTS_STABLE)	32	RO	0000_0000h
A1Ch	SPDIF clock stable, audio clock measured (SPDIF_NM_STABLE)	32	RO	0000_0000h
A20h	SPDIF clock stable, lane clock measured (SPDIF_NCTS_STABLE)	32	RO	0000_0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
A24h	Video clock stable (NMVID_MEAS_STABLE)	32	RO	0000_0000h
A28h	Video cycles measure (CM_VID_MEAS)	32	RW	0000_0000h
A2Ch	Audio cycles measure (CM_AUD_MEAS)	32	RW	0000_0000h
A40h	I2S clock measurment HDMI (I2S_MEAS)	32	RO	0000_0000h
A44h	I2S clock measurment DP (I2S_DP_MEAS)	32	RO	0000_0000h
A80h	SPDIF clock measurment DP (SPDIF_DP_MEAS)	32	RO	0000_0000h
A84h	SPDIF clock measurment HDMI (SPDIF_MEAS)	32	RO	0000_0000h
AC0h	Video clock measurment (NMVID_MEAS)	32	RO	0000_0000h
B00h	Video Input Interface Setting Register (BND_HSYNC2VSYNC)	32	RW	0000_2000h
B04h	Status of HSYNC to VSYNC Distance Counter 1 (HSYNC2VSYNC_F1_L1)	32	RO	0000_0000h
B08h	Status of HSYNC to VSYNC Distance Counter 2 (HSYNC2VSYNC_F2_L1)	32	RO	0000_0000h
B0Ch	Video Interface Status Register (HSYNC2VSYNC_STATUS)	32	RO	0000_0000h
B10h	Setting Polarity of HSYNC and VSYNC (HSYNC2VSYNC_POL_CTRL)	32	RW	0000_0000h
1000h	Video HSize configuration (SCHEDULER_H_SIZE)	32	RW	FFFF_FFFFh
1004h	Video VSize configuration (SCHEDULER_V_SIZE)	32	RW	FFFF_FFFFh
1008h	Video Blank (Keep Out) Size configuration (SCHEDULER_KEEP_OUT)	32	RW	08E3_F83Ah
100Ch	Video Blank (Front) Size configuration (HDTX_SIGNAL_FRONT_WIDTH)	32	RW	FFFF_FFFFh
1010h	Video Sync Size configuration (HDTX_SIGNAL_SYNC_WIDTH)	32	RW	FFFF_FFFFh
1014h	Video Back Size configuration (HDTX_SIGNAL_BACK_WIDTH)	32	RW	FFFF_FFFFh
1018h	Video Mode configuration (HDTX_CONTROLLER)	32	RW	000C_1F91h
101Ch	HDMI HDCP configuration (HDTX_HDCP)	32	RW	0000_2100h
1020h	HDMI HPD configuration (HDTX_HPD)	32	RW	0000_000Fh
1024h	HDMI CLOCK configuration - REG0 (HDTX_CLOCK_REG_0)	32	RW	0000_7C1Fh
1028h	HDMI CLOCK configuration - REG1 (HDTX_CLOCK_REG_1)	32	RW	0000_7C1Fh
102Ch	HDMI HPD status (HPD_PLUG_IN)	32	RO	0000_0000h
1030h	HDMI status (HDCP_IN)	32	RO	0000_0100h
1034h	HDMI GCP coding configuration (GCP_FORCE_COLOR_DEPTH_CODING)	32	RW	0000_0000h
1038h	HDMI SSCP positioning (SSCP_POSITIONING)	32	RW	0002_0005h
103Ch	HDMI HDCP window opportunity (HDCP_WIN_OF_OPP_POSITION)	32	RW	0000_3DFFh
1800h	HDMI I2C configuration (CTRL)	32	RW	0000_0000h
1804h	HDMI I2C status (STATUS)	32	RO	0000_0000h
1808h	HDMI I2C Address configuration (I2C_ADDR)	32	RW	0000_0000h
180Ch	HDMI I2C data register (I2C_DATA)	32	RW	0000_0000h
1810h	HDMI I2C interrupt status (INT_STATUS)	32	RO	0000_0000h
1814h	HDMI I2C transaction size (TRANS_SIZE)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
1818h	HDMI I2C Slave Monitor (SLAVE_MON)	32	RW	0000_0000h
181Ch	HDMI I2C Timeout (TIMEOUT)	32	RW	0000_001Fh
1820h	HDMI SCDC interrupt mask (INT_SCDC_MASK)	32	RO	0001_F2FFh
1824h	HDMI I2C interrupt enable (INT_ENABLE)	32	RW	0000_0000h
1828h	HDMI I2C interrupt disable (INT_DISABLE)	32	RW	0000_0000h
182Ch	HDMI I2C glitch filter configuration (GLITCH_FILTER_CTRL)	32	RW	0000_0000h
1830h	HDMI I2C RR configuration (RR_CTRL)	32	RW	0000_0000h
1834h	HDMI I2C RR timeout (RR_TIMEOUT)	32	RW	0000_208Dh
2000h	DPTX PHY control (DP_TX_PHY_CONFIG_REG)	32	RW	0000_0000h
2004h	DPTC PHY software reset (DP_TX_PHY_SW_RESET)	32	RW	0000_0000h
2008h	Scrambler seed (DP_TX_PHY_SCRAMBLER_SEED)	32	RW	0000_FFFFh
200Ch	Custom training value bytes 1-4 (DP_TX_PHY_TRAINING_01_04)	32	RW	0000_0000h
2010h	Custom training value bytes 5-8 (DP_TX_PHY_TRAINING_05_08)	32	RW	0000_0000h
2014h	Custom training value bytes 9-10 (DP_TX_PHY_TRAINING_09_10)	32	RW	0000_0000h
2018h	Custom CP2520 SR interval (DP_TX_PHY_SR_INTERVAL)	32	RW	0000_00FCh
2100h	HPD min timer for irq, define the minimum pclk cycles that the HPD pulse will be considered as IRQ (HPD_IRQ_DET_MIN_TIMER)	32	RW	0000_C350h
2104h	HPD max timer for irq, define the maximum pclk cycles that the HPD pulse will be considered as IRQ (HPD_IRQ_DET_MAX_TIMER)	32	RW	0001_86A0h
2108h	HPD min timer for HPD detect, define the minimum pclk cycles that the HPD is low (HPD_UNPLGED_DET_MIN_TIMER)	32	RW	0003_0D40h
210Ch	Timer for detecting HPD stable, count in pclk cycles (HPD_STABLE_TIMER)	32	RW	0001_86A0h
2110h	Timer for filtering small pulses on hpd input (HPD_FILTER_TIMER)	32	RW	0000_7A12h
2114h	HPD debouncer control (HPD_DBNC_TIMER)	32	RW	0000_0000h
211Ch	Mask of HPD interrupt and status (HPD_EVENT_MASK)	32	RW	0000_0000h
2120h	HPD interrupt and status (HPD_EVENT_DET)	32	R2C	0000_0004h
2200h	Global configuration of the framer module (DP_FRAMER_GLOBAL_CONFIG)	32	RW	0000_0023h
2204h	SW reset (DP_SW_RESET)	32	RW	0000_0000h
2208h	TU related configuration (DP_FRAMER_TU)	32	RW	0000_A002h
220Ch	Video pixel format configuration (DP_FRAMER_PXL_REPR)	32	RW	0000_0102h
2210h	Polarity and 3D configuration (DP_FRAMER_SP)	32	RW	0000_0000h
2214h	Audio packet configuration (AUDIO_PACK_CONTROL)	32	RW	0000_0000h
2218h	VC table values 0-3 (MST feature) (DP_VC_TABLE_0)	32	RW	0000_0000h
221Ch	VC table values 4-7 (MST feature) (DP_VC_TABLE_1)	32	RW	0000_0000h
2220h	VC table values 8-11 (MST feature) (DP_VC_TABLE_2)	32	RW	0000_0000h
2224h	VC table values 12-15 (MST feature) (DP_VC_TABLE_3)	32	RW	0000_0000h
2228h	VC table values 16-19 (MST feature) (DP_VC_TABLE_4)	32	RW	0000_0000h
222Ch	VC table values 20-23 (MST feature) (DP_VC_TABLE_5)	32	RW	0000_0000h
2230h	VC table values 24-27 (MST feature) (DP_VC_TABLE_6)	32	RW	0000_0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
2234h	VC table values 28-31 (MST feature) (DP_VC_TABLE_7)	32	RW	0000_0000h
2238h	VC table values 32-35 (MST feature) (DP_VC_TABLE_8)	32	RW	0000_0000h
223Ch	VC table values 36-39 (MST feature) (DP_VC_TABLE_9)	32	RW	0000_0000h
2240h	VC table values 40-43 (MST feature) (DP_VC_TABLE_10)	32	RW	0000_0000h
2244h	VC table values 44-47 (MST feature) (DP_VC_TABLE_11)	32	RW	0000_0000h
2248h	VC table values 48-51 (MST feature) (DP_VC_TABLE_12)	32	RW	0000_0000h
224Ch	VC table values 52-55 (MST feature) (DP_VC_TABLE_13)	32	RW	0000_0000h
2250h	VC table values 56-59 (MST feature) (DP_VC_TABLE_14)	32	RW	0000_0000h
2254h	Video FIFO latency threshold (LINE_THRESH)	32	RW	0000_0020h
2258h	Vertical blanking ID (DP_VB_ID)	32	RW	0000_0009h
225Ch	LVP insertion in MTP header enable (MST feature) (DP_MTPH_LVP_CONTROL)	32	RW	0000_0000h
2260h	Values to be inserted in the MTP header (MST feature) (DP_MTPH_SYMBOL_VALUES)	32	RW	0000_0000h
2264h	ECF insertion in MTP header enable (MST feature) (DP_MTPH_ECF_CONTROL)	32	RW	0000_0000h
2268h	Supporting configuration to switch from top/bottom on the input to field sequential on the output (DP_FIELDSEQ_3D)	32	RW	0000_0000h
226Ch	MTP header status (MST feature) (DP_MTPH_STATUS)	32	RO	0000_5555h
2270h	Interrupt sources of the framer module, active high. (DP_INTERRUPT_SOURCE)	32	RO	0000_0000h
2274h	Masks for the interrupt sources in the DP_INTERRUPT_SOURCE register, when set high, these bits disable the corresponding interrupts (DP_INTERRUPT_MASK)	32	RW	0000_0000h
2278h	MSA parameters to be used in the HW (DP_FRONT_BACK_PORCH)	32	RW	0010_003Ch
227Ch	Number of bytes per lane/chunk parameters (DP_BYTE_COUNT)	32	RW	0000_02A4h
2280h	MSA horizontal parameters first part (MSA_HORIZONTAL_0)	32	RW	0000_0000h
2284h	MSA horizontal parameters second part (MSA_HORIZONTAL_1)	32	RW	0000_0000h
2288h	MSA vertical parameters first part (MSA_VERTICAL_0)	32	RW	0000_0000h
228Ch	MSA vertical parameters second part (MSA_VERTICAL_1)	32	RW	0000_0000h
2290h	MISC0 and MISC1 values of the MSA, as per DPv1. (MSA_MISC)	32	RW	0002_0000h
2294h	MSA stream number (MST feature) (STREAM_CONFIG)	32	RW	0000_0000h
2298h	Status signals for the audio pack (AUDIO_PACK_STATUS)	32	RW	0012_0002h
229Ch	Status signals for the VIF module (VIF_STATUS)	32	RO	0000_5555h
22A0h	Status signals for the video stuff module, part 1 (PCK_STUFF_STATUS_0)	32	RO	0101_0101h
22A4h	Status signals for the video stuff module, part 2 (PCK_STUFF_STATUS_1)	32	RO	0000_0101h
22A8h	Status signals for the info pack module, as well as final VB-ID value (INFO_PACK_STATUS)	32	RO	0000_0002h
22ACh	Rate governor status, as well as additional video configuration (RATE_GOVERNOR_STATUS)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
22B0h	Video line parameters to be used in the RTL calculations (DP_HORIZONTAL)	32	RW	0000_0000h
22B4h	Video frame parameters to be used in the RTL calculations, part 1 (DP_VERTICAL_0)	32	RW	0000_0000h
22B8h	Video frame parameters to be used in the RTL calculations, part 2 (DP_VERTICAL_1)	32	RW	0000_0000h
22BCh	(DP_BLOCK_SDP)	32	RW	0FBE_0505h
2300h	DPTX lane enable (DPTX_LANE_EN)	32	RW	0000_0000h
2304h	DPTX enhanced mode (DPTX_ENHNCD)	32	RW	0000_0000h
2308h	DPTX Interrupt mask (DPTX_INT_MASK)	32	RW	0000_0003h
230Ch	DPTX interrupt status (DPTX_INT_STATUS)	32	RO	0000_0000h
2400h	HDCP DP status register (HDCP_DP_STATUS)	32	RO	0000_0000h
2404h	HDCP DP config register (HDCP_DP_CONFIG)	32	RW	0000_0009h
2408h	HDCP DP software reset register (HDCP_DP_SW_RST)	32	RW	0000_0000h
240Ch	HDCP DP FIFO status register (HDCP_DP_FIFO_STATUS)	32	RO	0000_0000h
2800h	Control bits for DP_AUX (DP_AUX_HOST_CONTROL)	32	RW	0000_0004h
2804h	Status of the DP_AUX interrupt sources (DP_AUX_INTERRUPT_SOURCE)	32	RO	0000_0000h
2808h	Mask vector of the DP_AUX interrupt sources (DP_AUX_INTERRUPT_MASK)	32	RW	0000_0000h
280Ch	Ordering and inversion of transmit/receive on Auxiliary Channel (DP_AUX_SWAP_INVERSION_CONTROL)	32	RW	0000_0000h
2810h	NACK transaction send (DP_AUX_SEND_NACK_TRANSACTION)	32	RW	0000_0000h
2814h	RX bits clear (DP_AUX_CLEAR_RX)	32	RW	0000_0000h
2818h	TX bits clear (DP_AUX_CLEAR_TX)	32	RW	0000_0000h
281Ch	Stop timer operation (DP_AUX_TIMER_STOP)	32	RW	0000_0000h
2820h	Clear timer operation (DP_AUX_TIMER_CLEAR)	32	RW	0000_0000h
2824h	Soft reset of the DP_AUX (DP_AUX_RESET_SW)	32	RW	0000_0000h
2828h	SYS_CLK and 2 MHz clock ratio (DP_AUX_DIVIDE_2M)	32	RW	0000_000Bh
282Ch	Pre charge field length (DP_AUX_TX_PRECHARGE_LENGTH)	32	RW	0000_0010h
2830h	Maximum legal receiving frequency (DP_AUX_FREQUENCY_1M_MAX)	32	RW	0000_0022h
2834h	Minimum legal receiving frequency (DP_AUX_FREQUENCY_1M_MIN)	32	RW	0000_0009h
2838h	Minimum received preamble length (DP_AUX_RX_PRE_MIN)	32	RW	0000_001Ah
283Ch	Maximum received preamble length (DP_AUX_RX_PRE_MAX)	32	RW	0000_0020h
2840h	DP_AUX_MAIN start value (DP_AUX_TIMER_PRESET)	32	RW	0000_1D4Ch
2844h	Transmit pattern (DP_AUX_NACK_FORMAT)	32	RW	0000_0020h
2848h	Mailbox write data (DP_AUX_TX_DATA)	32	RW	0000_0000h
284Ch	Mailbox read data (DP_AUX_RX_DATA)	32	RO	0000_0000h
2850h	AUX_TX status (DP_AUX_TX_STATUS)	32	RO	0000_0101h
2854h	AUX_RX status (DP_AUX_RX_STATUS)	32	RO	0101_0141h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
2858h	RX counter status (DP_AUX_RX_CYCLE_COUNTER)	32	RO	0000_0000h
285Ch	DP_AUX MAIN State Machines status (DP_AUX_MAIN_STATES)	32	RO	0000_0000h
2860h	DP_AUX MAIN timer status (DP_AUX_MAIN_TIMER)	32	RO	0000_0000h
2864h	Test mode configuration (DP_AUX_AFE_OUT)	32	RW	0000_0000h
3800h	CEC TX message header (TX_MSG_HEADER)	32	RW	0000_0000h
3804h	CEC TX message opcode (TX_MSG_OPCODE)	32	RW	0000_0000h
3808h	CEC TX message operand - byte 1 (TX_MSG_OP1)	32	RW	0000_0000h
380Ch	CEC TX message operand - byte 2 (TX_MSG_OP2)	32	RW	0000_0000h
3810h	CEC TX message operand - byte 3 (TX_MSG_OP3)	32	RW	0000_0000h
3814h	CEC TX message operand - byte 4 (TX_MSG_OP4)	32	RW	0000_0000h
3818h	CEC TX message operand - byte 5 (TX_MSG_OP5)	32	RW	0000_0000h
381Ch	CEC TX message operand - byte 6 (TX_MSG_OP6)	32	RW	0000_0000h
3820h	CEC TX message operand - byte 7 (TX_MSG_OP7)	32	RW	0000_0000h
3824h	CEC TX message operand - byte 8 (TX_MSG_OP8)	32	RW	0000_0000h
3828h	CEC TX message operand - byte 9 (TX_MSG_OP9)	32	RW	0000_0000h
382Ch	CEC TX message operand - byte 10 (TX_MSG_OP10)	32	RW	0000_0000h
3830h	CEC TX message operand - byte 11 (TX_MSG_OP11)	32	RW	0000_0000h
3834h	CEC TX message operand - byte 12 (TX_MSG_OP12)	32	RW	0000_0000h
3838h	CEC TX message operand - byte 13 (TX_MSG_OP13)	32	RW	0000_0000h
383Ch	CEC TX message operand - byte 14 (TX_MSG_OP14)	32	RW	0000_0000h
3840h	CEC TX message length (TX_MSG_LENGTH)	32	RW	0000_0000h
3844h	CEC TX message command (TX_MSG_CMD)	32	RW	0000_0000h
3848h	CEC TX buffer write enable (TX_WRITE_BUF)	32	RW	0000_0000h
384Ch	CEC TX buffer clear (TX_CLEAR_BUF)	32	RW	0000_0000h
3850h	CEC RX message command (RX_MSG_CMD)	32	RW	0000_0000h
3854h	CEC RX buffer clear (RX_CLEAR_BUF)	32	RW	0000_0000h
3858h	CEC Logical Address 0 (LOGICAL_ADDRESS_LA0)	32	RW	0000_000Fh
385Ch	CEC Logical Address 1 (LOGICAL_ADDRESS_LA1)	32	RW	0000_000Fh
3860h	CEC Logical Address 2 (LOGICAL_ADDRESS_LA2)	32	RW	0000_000Fh
3864h	CEC Logical Address 3 (LOGICAL_ADDRESS_LA3)	32	RW	0000_000Fh
3868h	CEC Logical Address 4 (LOGICAL_ADDRESS_LA4)	32	RW	0000_000Fh
386Ch	CEC Clock Divider - MSB part (CLK_DIV_MSB)	32	RW	0000_0003h
3870h	CEC Clock Divider - LSB part (CLK_DIV_LSB)	32	RW	0000_00E7h
387Ch	CDC message enable (cdc_msg)	32	RW	0000_0000h
3900h	CEC RX message header (RX_MSG_DATA1)	32	RO	0000_0000h
3904h	CEC RX message opcode (RX_MSG_DATA2)	32	RO	0000_0000h
3908h	CEC RX message operand 1 (RX_MSG_DATA3)	32	RO	0000_0000h
390Ch	CEC RX message operand 2 (RX_MSG_DATA4)	32	RO	0000_0000h
3910h	CEC RX message operand 3 (RX_MSG_DATA5)	32	RO	0000_0000h
3914h	CEC RX message operand 4 (RX_MSG_DATA6)	32	RO	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
3918h	CEC RX message operand 5 (RX_MSG_DATA7)	32	RO	0000_0000h
391Ch	CEC RX message operand 6 (RX_MSG_DATA8)	32	RO	0000_0000h
3920h	CEC RX message operand 7 (RX_MSG_DATA9)	32	RO	0000_0000h
3924h	CEC RX message operand 8 (RX_MSG_DATA10)	32	RO	0000_0000h
3928h	CEC RX message operand 9 (RX_MSG_DATA11)	32	RO	0000_0000h
392Ch	CEC RX message operand 10 (RX_MSG_DATA12)	32	RO	0000_0000h
3930h	CEC RX message operand 11 (RX_MSG_DATA13)	32	RO	0000_0000h
3934h	CEC RX message operand 12 (RX_MSG_DATA14)	32	RO	0000_0000h
3938h	CEC RX message operand 13 (RX_MSG_DATA15)	32	RO	0000_0000h
393Ch	CEC RX message operand 14 (RX_MSG_DATA16)	32	RO	0000_0000h
3940h	CEC RX message length (RX_MSG_LENGTH)	32	RO	0000_0000h
3944h	CEC RX status (RX_MSG_STATUS)	32	RO	0000_0000h
3948h	CEC RX status - number of received messages (NUM_OF_MSG_RX_BUF)	32	RO	0000_0000h
394Ch	CEC TX status (TX_MSG_STATUS)	32	RO	0000_0000h
3950h	CEC TX status - number of messages in TX buffer (NUMBER_OF_MSG_IN_TX_BUF)	32	RO	0000_0000h
3954h	CEC Free-Time counter (FREE_TIME_CNTR_CONFIG)	32	RW	0000_0022h
3958h	CEC Interrupt Mask (INT_CEC_MASK)	32	RW	0000_0003h
395Ch	CEC Interrupt status (INT_STAT)	32	RO	0000_0000h
3980h	CEC DB L timer (DB_L_TIMER)	32	RW	0000_0000h
3984h	CEC DB M timer (DB_M_TIMER)	32	RW	0000_0000h
3988h	CEC DB H timer (DB_H_TIMER)	32	RW	0000_0000h
4000h	Contains the revision of the internal HDCP 1. (CRYPTO_HDCP_REVISION)	32	RO	0000_0000h
4004h	Contains global configuration information for the HDCP Crypto module (HDCP_CRYPT_CONFIG)	32	RW	0000_0000h
4008h	Contains the status of the HDCP interrupt sources (CRYPTO_INTERRUPT_SOURCE)	32	RO	0000_0000h
400Ch	Contains the mask vector of the HDCP interrupt sources (CRYPTO_INTERRUPT_MASK)	32	RW	0000_0703h
4018h	Contains global configuration information for the HDCP 2. (CRYPTO22_CONFIG)	32	RW	0000_0000h
401Ch	Crypto 2. (CRYPTO22_STATUS)	32	RO	0000_0000h
403Ch	Holds 32-bit input data word of the SHA-256 module (SHA_256_DATA_IN)	32	RW	0000_0000h
4050h	Result of operation SHA-256 - 1' dw (SHA_256_DATA_OUT_0)	32	RO	0000_0000h
4054h	Result of operation SHA-256 - 2' dw (SHA_256_DATA_OUT_1)	32	RO	0000_0000h
4058h	Result of operation SHA-256 - 3' dw (SHA_256_DATA_OUT_2)	32	RO	0000_0000h
405Ch	Result of operation SHA-256 - 4' dw (SHA_256_DATA_OUT_3)	32	RO	0000_0000h
4060h	Result of operation SHA-256 - 5' dw (SHA_256_DATA_OUT_4)	32	RO	0000_0000h
4064h	Result of operation SHA-256 - 6' dw (SHA_256_DATA_OUT_5)	32	RO	0000_0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
4068h	Result of operation SHA-256 - 7' dw (SHA_256_DATA_OUT_6)	32	RO	0000_0000h
406Ch	Result of operation SHA-256 - 8' dw (SHA_256_DATA_OUT_7)	32	RO	0000_0000h
4070h	Input key word of the AES-32 module - 1' dw (AES_32_KEY_0)	32	RW	0000_0000h
4074h	Input key word of the AES-32 module - 2' dw (AES_32_KEY_1)	32	RW	0000_0000h
4078h	Input key word of the AES-32 module - 3' dw (AES_32_KEY_2)	32	RW	0000_0000h
407Ch	Input key word of the AES-32 module - 4' dw (AES_32_KEY_3)	32	RW	0000_0000h
4080h	Input data word to the AES-32 module (AES_32_DATA_IN)	32	RW	0000_0000h
4084h	AES-32 module - 128-bits output data word - 1' dw (AES_32_DATA_OUT_0)	32	RO	0000_0000h
4088h	AES-32 module - 128-bits output data word - 2' dw (AES_32_DATA_OUT_1)	32	RO	0000_0000h
408Ch	AES-32 module - 128-bits output data word - 3' dw (AES_32_DATA_OUT_2)	32	RO	0000_0000h
4090h	AES-32 module - 128-bits output data word - 4' dw (AES_32_DATA_OUT_3)	32	RO	0000_0000h
40A0h	Contains global configuration information for the HDCP 1. (CRYPTO14_CONFIG)	32	RW	0000_0000h
40A4h	Contains global status information for the HDCP 1. (CRYPTO14_STATUS)	32	RO	0000_0000h
40A8h	Contains 24-bit pseudo random data (CRYPTO14_PRNM_OUT)	32	RO	0000_0000h
40ACh	Contains the first word of the Km value (CRYPTO14_KM_0)	32	RO	0000_0000h
40B0h	Contains the most significant 3 bytes of the Km value (CRYPTO14_KM_1)	32	RO	0000_0000h
40B4h	First word of An value generated by hdcpRngCipher operation. (CRYPTO14_AN_0)	32	RW	0000_0000h
40B8h	Second word of An value generated by hdcpRngCipher operation (CRYPTO14_AN_1)	32	RW	0000_0000h
40BCh	First 32 bits of the KSV from the other HDCP device (CRYPTO14_YOUR_KSV_0)	32	RW	0000_0000h
40C0h	Last byte of the KSV from other HDCP device (CRYPTO14_YOUR_KSV_1)	32	RW	0000_0000h
40C4h	Mi value - 1' dw (CRYPTO14_MI_0)	32	RO	0000_0000h
40C8h	Mi value - 2' dw (CRYPTO14_MI_1)	32	RO	0000_0000h
40CCh	Ti value (CRYPTO14_TI_0)	32	RO	0000_0000h
40D0h	First 32 bits of the Ki frame key from this HDCP device (CRYPTO14_KI_0)	32	RO	0000_0000h
40D4h	Last 3 bytes of the Ki frame key from this HDCP device. (CRYPTO14_KI_1)	32	RO	0000_0000h
40D8h	This register defines number of iterations for SHA-1 calculations (CRYPTO14_BLOCKS_NUM)	32	RW	0000_0000h
40DCh	Key memory control register. (CRYPTO14_KEY_MEM_DATA_0)	32	RW	0000_0000h
40E0h	Key memory control register. (CRYPTO14_KEY_MEM_DATA_1)	32	RW	0000_0000h
40E4h	SHA1 message control register. (CRYPTO14_SHA1_MSG_DATA)	32	RW	0000_0000h
40E8h	SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_0)	32	RO	6745_2301h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
40ECh	SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_1)	32	RO	EFCD_AB89h
40F0h	SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_2)	32	RO	98BA_DCFEh
40F4h	SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_3)	32	RO	1032_5476h
40F8h	SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_4)	32	RO	C3D2_E1F0h
40FCh	TRNG control (TRNG_CTRL)	32	RW	0000_0000h
4100h	TRNG status (TRNG_DATA_RDY)	32	RO	0000_0000h
4104h	TRNG data register (TRNG_DATA)	32	RO	0000_0000h
3_0000h	Audio source control (AUDIO_SRC_CNTL)	32	RW	0000_0000h
3_0004h	Audio source configuration (AUDIO_SRC_CNFG)	32	RW	0000_0000h
3_0008h	Common channels configuration (COM_CH_STTS_BITS)	32	RW	0000_0000h
3_000Ch	Channels 0,1 configuration (STTS_BIT_CH01)	32	RW	0000_0000h
3_0010h	Channels 2,3 configuration (STTS_BIT_CH23)	32	RW	0000_0000h
3_0014h	Channels 4,5 configuration (STTS_BIT_CH45)	32	RW	0000_0000h
3_0018h	Channels 6,7 configuration (STTS_BIT_CH67)	32	RW	0000_0000h
3_001Ch	Channels 8,9 configuration (STTS_BIT_CH89)	32	RW	0000_0000h
3_0020h	Channels 10,11 configuration (STTS_BIT_CH1011)	32	RW	0000_0000h
3_0024h	Channels 12,13 configuration (STTS_BIT_CH1213)	32	RW	0000_0000h
3_0028h	Channels 14,15 configuration (STTS_BIT_CH1415)	32	RW	0000_0000h
3_002Ch	Channels 16,17 configuration (STTS_BIT_CH1617)	32	RW	0000_0000h
3_0030h	Channels 18,19 configuration (STTS_BIT_CH1819)	32	RW	0000_0000h
3_0034h	Channels 20,21 configuration (STTS_BIT_CH2021)	32	RW	0000_0000h
3_0038h	Channels 22,23 configuration (STTS_BIT_CH2223)	32	RW	0000_0000h
3_003Ch	Channels 24,25 configuration (STTS_BIT_CH2425)	32	RW	0000_0000h
3_0040h	Channels 26,27 configuration (STTS_BIT_CH2627)	32	RW	0000_0000h
3_0044h	Channels 28,29 configuration (STTS_BIT_CH2829)	32	RW	0000_0000h
3_0048h	Channels 30,31 configuration (STTS_BIT_CH3031)	32	RW	0000_0000h
3_004Ch	SPDIF control (SPDIF_CTRL_ADDR)	32	RW	0000_0000h
3_0050h	SPDIF channel 1 status [31:00] (SPDIF_CH1_CS_3100_ADDR)	32	RO	0000_0000h
3_0054h	SPDIF channel 1 status [63:32] (SPDIF_CH1_CS_6332_ADDR)	32	RO	0000_0000h
3_0058h	SPDIF channel 1 status [95:64] (SPDIF_CH1_CS_9564_ADDR)	32	RO	0000_0000h
3_005Ch	SPDIF channel 1 status [127:96] (SPDIF_CH1_CS_12796_ADDR)	32	RO	0000_0000h
3_0060h	SPDIF channel 1 status [159:128] (SPDIF_CH1_CS_159128_ADDR)	32	RO	0000_0000h
3_0064h	SPDIF channel 1 status [191:160] (SPDIF_CH1_CS_191160_ADDR)	32	RO	0000_0000h
3_0068h	SPDIF channel 2 status [31:00] (SPDIF_CH2_CS_3100_ADDR)	32	RO	0000_0000h
3_006Ch	SPDIF channel 2 status [63:32] (SPDIF_CH2_CS_6332_ADDR)	32	RO	0000_0000h
3_0070h	SPDIF channel 2 status [95:64] (SPDIF_CH2_CS_9564_ADDR)	32	RO	0000_0000h
3_0074h	SPDIF channel 2 status [127:96] (SPDIF_CH2_CS_12796_ADDR)	32	RO	0000_0000h
3_0078h	SPDIF channel 2 status [159:128] (SPDIF_CH2_CS_159128_ADDR)	32	RO	0000_0000h
3_007Ch	SPDIF channel 2 status [191:160] (SPDIF_CH2_CS_191160_ADDR)	32	RO	0000_0000h
3_0080h	Sample 2 Packets Control Register (SMPL2PKT_CNTL)	32	RW	0000_0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
3_0084h	Sample 2 Packets Config Register (SMPL2PKT_CNFG)	32	RW	0000_0800h
3_0088h	FIFO control register (FIFO_CNTL)	32	RW	0000_0000h
3_008Ch	FIFO Status register (FIFO_STTS)	32	RO	0000_0000h
3_0090h	SUB Packet Threshold register (SUB_PCKT_THRSH)	32	RW	0030_2010h
3_0800h	4 MSB of the packet memory address in which the data is written. (SOURCE_PIF_WR_ADDR)	32	RW	0000_0000h
3_0804h	Write request bit for the host write transaction. (SOURCE_PIF_WR_REQ)	32	RW	0000_0000h
3_0808h	4 MSB of the packet memory address from which the data is read. (SOURCE_PIF_RD_ADDR)	32	RW	0000_0000h
3_080Ch	Read request bit for the host read transaction. (SOURCE_PIF_RD_REQ)	32	RW	0000_0000h
3_0810h	The 32 bits of the data to be written to the packet memory. (SOURCE_PIF_DATA_WR)	32	RW	0000_0000h
3_0814h	The 32 bits of the data to be read from the packet memory. (SOURCE_PIF_DATA_RD)	32	RO	0000_0000h
3_0818h	Fifo1 flush (SOURCE_PIF_FIFO1_FLUSH)	32	RW	0000_0000h
3_081Ch	Fifo2 flush (SOURCE_PIF_FIFO2_FLUSH)	32	RW	0000_0000h
3_0820h	Status bits for the PIF module (SOURCE_PIF_STATUS)	32	RO	0000_0010h
3_0824h	Interrupt sources of the PIF module, active high. (SOURCE_PIF_INTERRUPT_SOURCE)	32	RO	0000_0000h
3_0828h	Masks for the interrupt sources in the SOURCE_PIF_INTERRUPT_SOURCE register, when set high, these bits disable the corresponding interrupts (SOURCE_PIF_INTERRUPT_MASK)	32	RW	0000_0000h
3_082Ch	Packet configuration to be stored in the allocation table (SOURCE_PIF_PKT_ALLOC_REG)	32	RW	0000_0000h
3_0830h	Enable bit for writing to the allocation table (SOURCE_PIF_PKT_ALLOC_WR_EN)	32	RW	0000_0000h
3_0834h	Software reset. (SOURCE_PIF_SW_RESET)	32	RW	0000_0000h
6_0000h	Contains the revision information of the HDCP Cipher module (HDCP_REVISION)	32	RO	0000_0000h
6_0004h	Contains the status of the HDCP Cipher interrupt sources (HDCP_INTERRUPT_SOURCE)	32	RO	0000_0000h
6_0008h	Contains the mask vector of the HDCP Cipher interrupt sources. (HDCP_INTERRUPT_MASK)	32	RW	0000_0000h
6_000Ch	Contains the configuration details for the Cipher core module (HDCP_CIPHER_CONFIG)	32	RW	0000_0000h
6_0010h	Holds the least significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_KEY_0)	32	RW	0000_0000h
6_0014h	Holds the second significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_KEY_1)	32	RW	0000_0000h
6_0018h	Holds the third significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_KEY_2)	32	RW	0000_0000h

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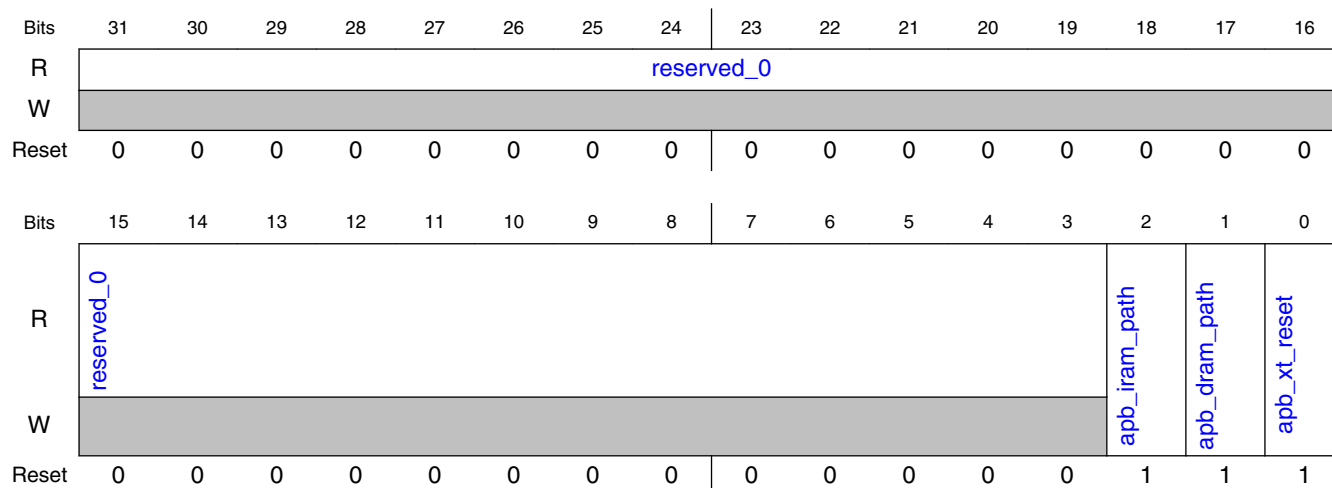
Offset	Register	Width (In bits)	Access	Reset value
6_001Ch	Holds the most significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_KEY_3)	32	RW	0000_0000h
6_0020h	Holds the least significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_RANDOM_0)	32	RW	0000_0000h
6_0024h	Holds the most significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_RANDOM_1)	32	RW	0000_0000h
6_0028h	Holds the first word of the Km value (CIPHER14_KM_0)	32	RW	0000_0000h
6_002Ch	Holds the most significant 3 bytes of the Km value (CIPHER14_KM_1)	32	RW	0000_0000h
6_0030h	Cipher 1. (CIPHER14_STATUS)	32	RO	0000_0000h
6_0034h	Cipher 1. (CIPHER14_RI_PJ_STATUS)	32	RO	0000_0000h
6_0038h	Cipher mode status register (CIPHER_MODE)	32	RO	0000_0000h
6_003Ch	Holds the First word of An value generated by hdcpRngCipher operation. (CIPHER14_AN_0)	32	RW	0000_0000h
6_0040h	Holds the second word of An value generated by hdcpRngCipher operation (CIPHER14_AN_1)	32	RW	0000_0000h
6_0044h	HDCCP2. (CIPHER22_AUTH)	32	RW	0000_0000h
6_0048h	This register contains Ri value in A5: Link Integrity Check state [HDCCP 1. (CIPHER14_R0_DP_STATUS)]	32	RO	0000_0000h
6_004Ch	Cipher for HDCCP1. (CIPHER14_BOOTSTRAP)	32	RW	0000_0000h

13.4.10.1.2 APB Control Register (APB_CTRL)

13.4.10.1.2.1 Offset

Register	Offset
APB_CTRL	0h

13.4.10.1.2.2 Diagram



13.4.10.1.2.3 Fields

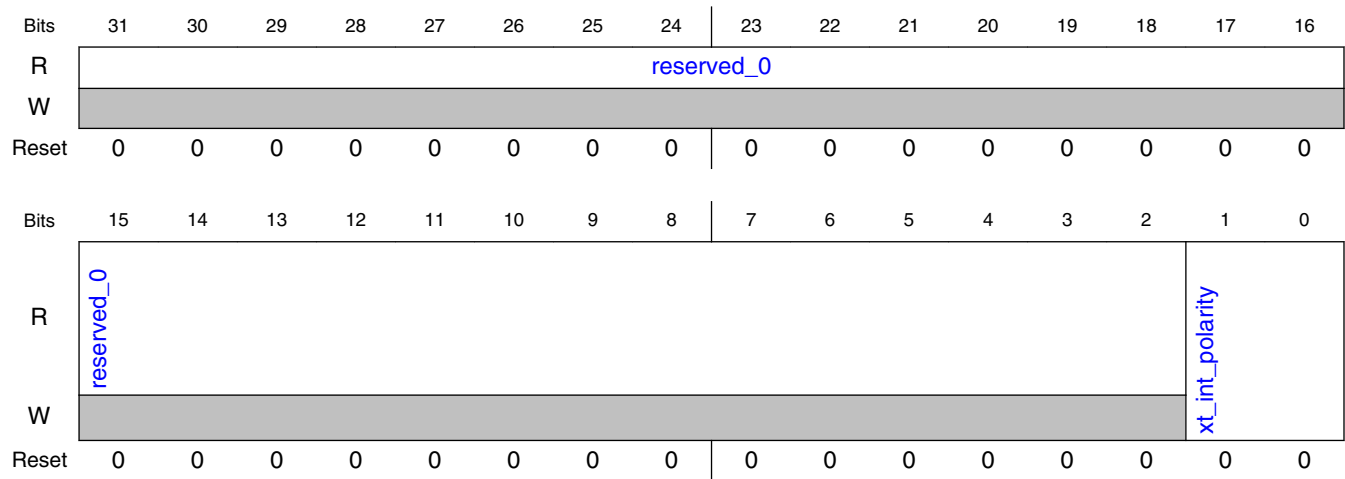
Field	Function
31-3 reserved_0	
2 apb_iram_path	When 1 then enable APB to R/W the IRAM
1 apb_dram_path	When 1 then enable APB to R/W the DRAM
0 apb_xt_reset	APB Control on the CPU reset. APB Control on the CPU reset. Active High

13.4.10.1.3 Internal CPU Interrupt Polarity Control Register (XT_INT_CTRL)

13.4.10.1.3.1 Offset

Register	Offset
XT_INT_CTRL	4h

13.4.10.1.3.2 Diagram



13.4.10.1.3.3 Fields

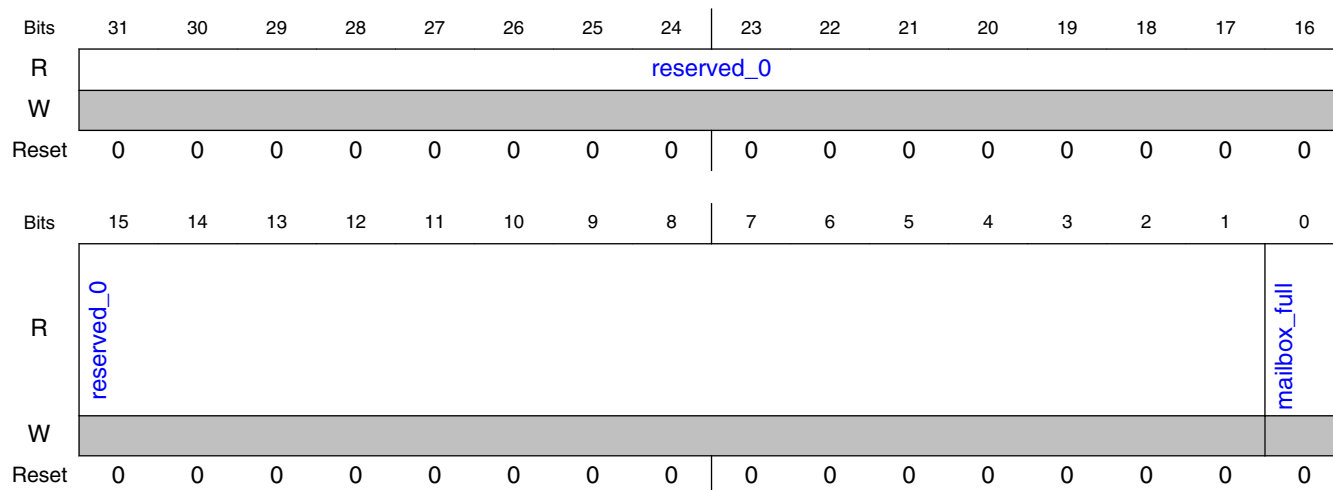
Field	Function
31-2 reserved_0	
1-0 xt_int_polarity	Each bit inverts appropriate interrupt signal provided do internal CPU interrupt input

13.4.10.1.4 Mailboxes full indication register (MAILBOX_FULL_ADDR)

13.4.10.1.4.1 Offset

Register	Offset
MAILBOX_FULL_ADDR	8h

13.4.10.1.4.2 Diagram



13.4.10.1.4.3 Fields

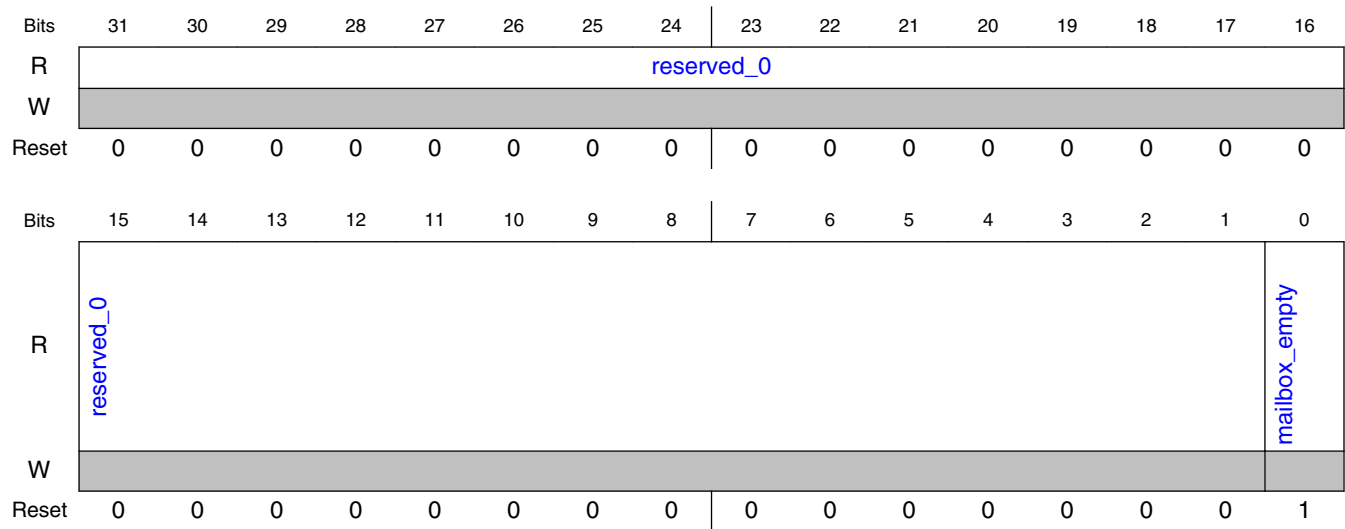
Field	Function
31-1 reserved_0	
0 mailbox_full	Mailboxes full indication

13.4.10.1.5 Mailboxes Empty indication register (MAILBOX_EMPTY_AD DR)

13.4.10.1.5.1 Offset

Register	Offset
MAILBOX_EMPTY_AD DR	Ch

13.4.10.1.5.2 Diagram



13.4.10.1.5.3 Fields

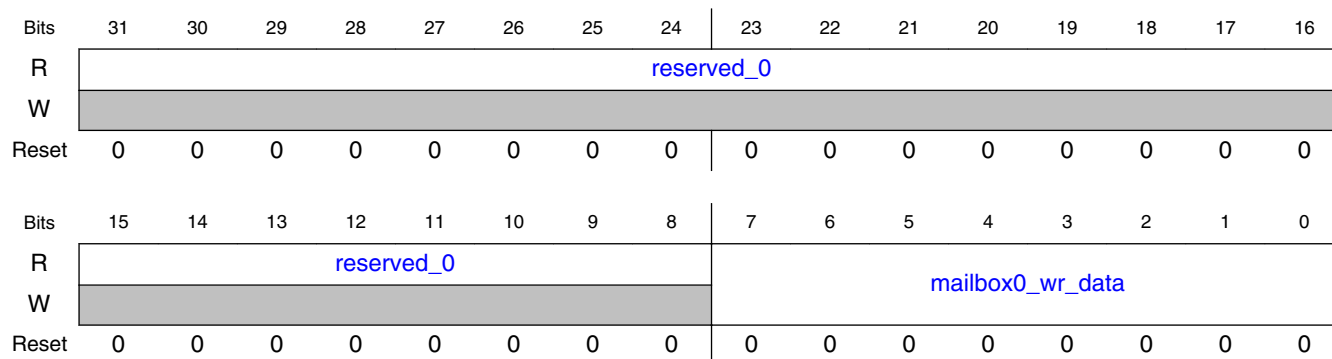
Field	Function
31-1 reserved_0	
0 mailbox_empty	Mailboxes Empty indication

13.4.10.1.6 Write Data to Mailbox register (MAILBOX0_WR_DATA)

13.4.10.1.6.1 Offset

Register	Offset
MAILBOX0_WR_DATA	10h

13.4.10.1.6.2 Diagram



13.4.10.1.6.3 Fields

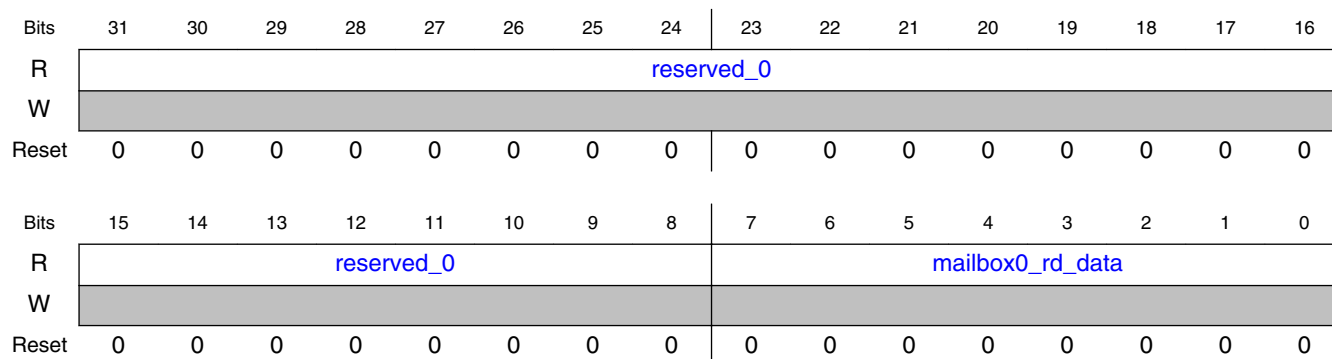
Field	Function
31-8 reserved_0	
7-0 mailbox0_wr_data	Write Data to Mailbox

13.4.10.1.7 Mailbox Read data register (MAILBOX0_RD_DATA)

13.4.10.1.7.1 Offset

Register	Offset
MAILBOX0_RD_DATA	14h

13.4.10.1.7.2 Diagram



13.4.10.1.7.3 Fields

Field	Function
31-8 reserved_0	
7-0 mailbox0_rd_data	Mailbox Read data

13.4.10.1.8 Software keep alive counter (KEEP_ALIVE)

13.4.10.1.8.1 Offset

Register	Offset
KEEP_ALIVE	18h

13.4.10.1.8.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								keep_alive_cnt							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.8.3 Fields

Field	Function
31-8 reserved_0	
7-0 keep_alive_cnt	Software keep alive counter. Software keep alive counter. Counter is initialized to 0x0 after reset and incremented by 0x1 with every FW kernel loop. It can be used to determine if internal CPU started running correctly.

13.4.10.1.9 Software Version LSB (VER_L)

13.4.10.1.9.1 Offset

Register	Offset
VER_L	1Ch

13.4.10.1.9.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								ver_lsb							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.9.3 Fields

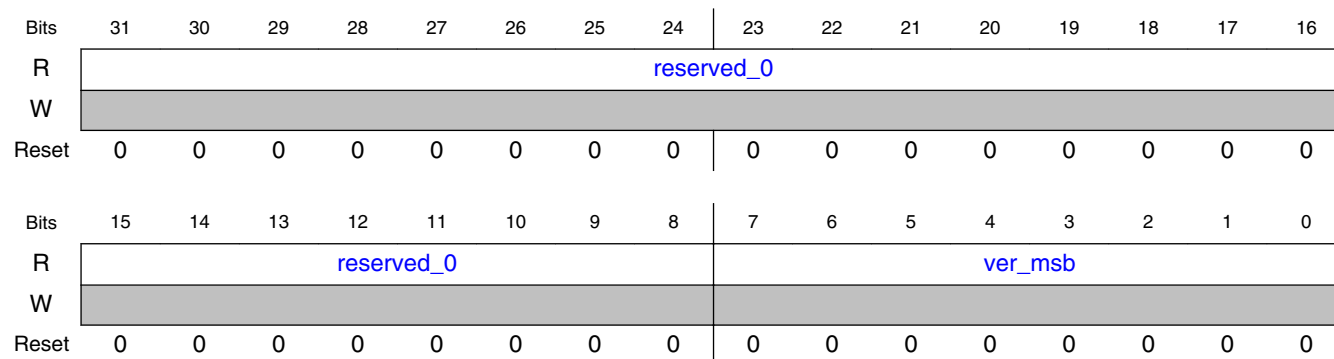
Field	Function
31-8 reserved_0	
7-0 ver_lsb	Software Version LSB. Software Version LSB. Loaded by FW

13.4.10.1.10 Software Version MSB (VER_H)

13.4.10.1.10.1 Offset

Register	Offset
VER_H	20h

13.4.10.1.10.2 Diagram



13.4.10.1.10.3 Fields

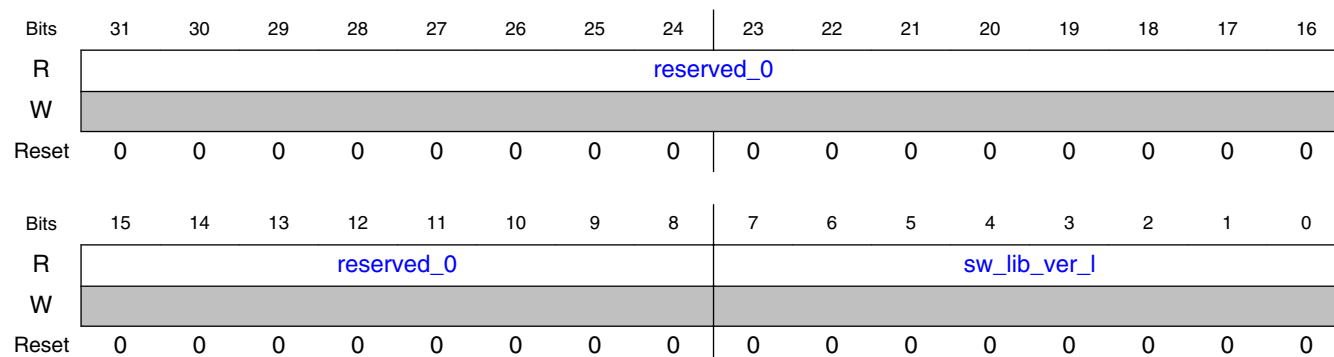
Field	Function
31-8 reserved_0	
7-0 ver_msb	Software Version MSB. Software Version MSB. Loaded by FW

13.4.10.1.11 Software Lib version written by CPU (LSB) (VER_LIB_L_ADDR)

13.4.10.1.11.1 Offset

Register	Offset
VER_LIB_L_ADDR	24h

13.4.10.1.11.2 Diagram



13.4.10.1.11.3 Fields

Field	Function
31-8 reserved_0	
7-0 sw_lib_ver_l	Software Lib version written by CPU (FW)

13.4.10.1.12 Software Lib version written by CPU (MSB) (VER_LIB_H_ADDR)

13.4.10.1.12.1 Offset

Register	Offset
VER_LIB_H_ADDR	28h

13.4.10.1.12.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								sw_lib_ver_h							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.12.3 Fields

Field	Function
31-8 reserved_0	
7-0 sw_lib_ver_h	Software Lib version written by CPU (FW)

13.4.10.1.13 Software Debug Register H (SW_DEBUG_L)

13.4.10.1.13.1 Offset

Register	Offset
SW_DEBUG_L	2Ch

13.4.10.1.13.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								sw_debug_7_0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.13.3 Fields

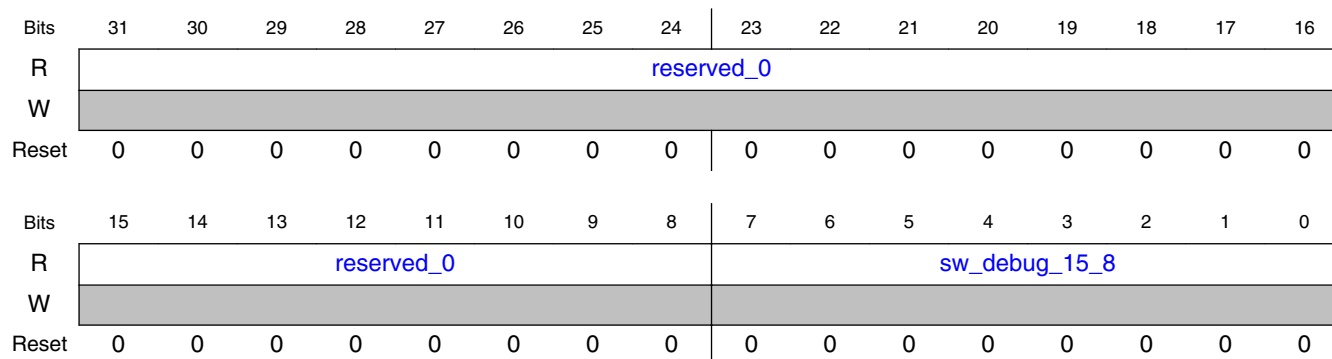
Field	Function
31-8 reserved_0	
7-0 sw_debug_7_0	For software debug purposes. For software debug purposes. Not used during normal operation.

13.4.10.1.14 Software Debug Register L (SW_DEBUG_H)

13.4.10.1.14.1 Offset

Register	Offset
SW_DEBUG_H	30h

13.4.10.1.14.2 Diagram



13.4.10.1.14.3 Fields

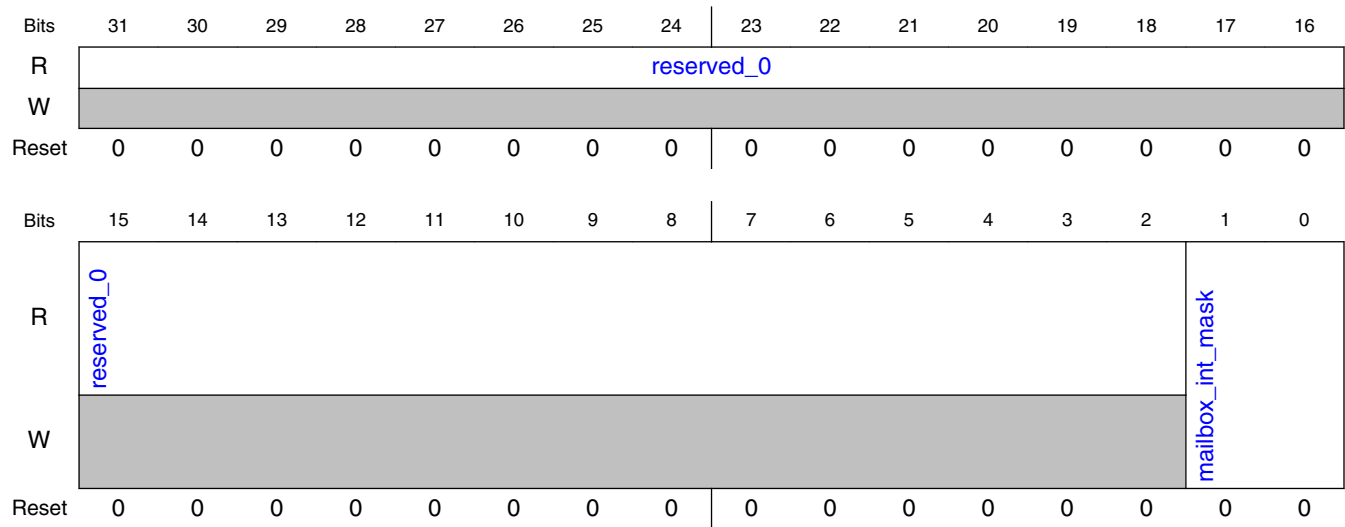
Field	Function
31-8 reserved_0	
7-0 sw_debug_15_8	For software debug purposes. For software debug purposes. Not used during normal operation.

13.4.10.1.15 Mailbox Interrupt mask register (MAILBOX_INT_MASK)

13.4.10.1.15.1 Offset

Register	Offset
MAILBOX_INT_MASK	34h

13.4.10.1.15.2 Diagram



13.4.10.1.15.3 Fields

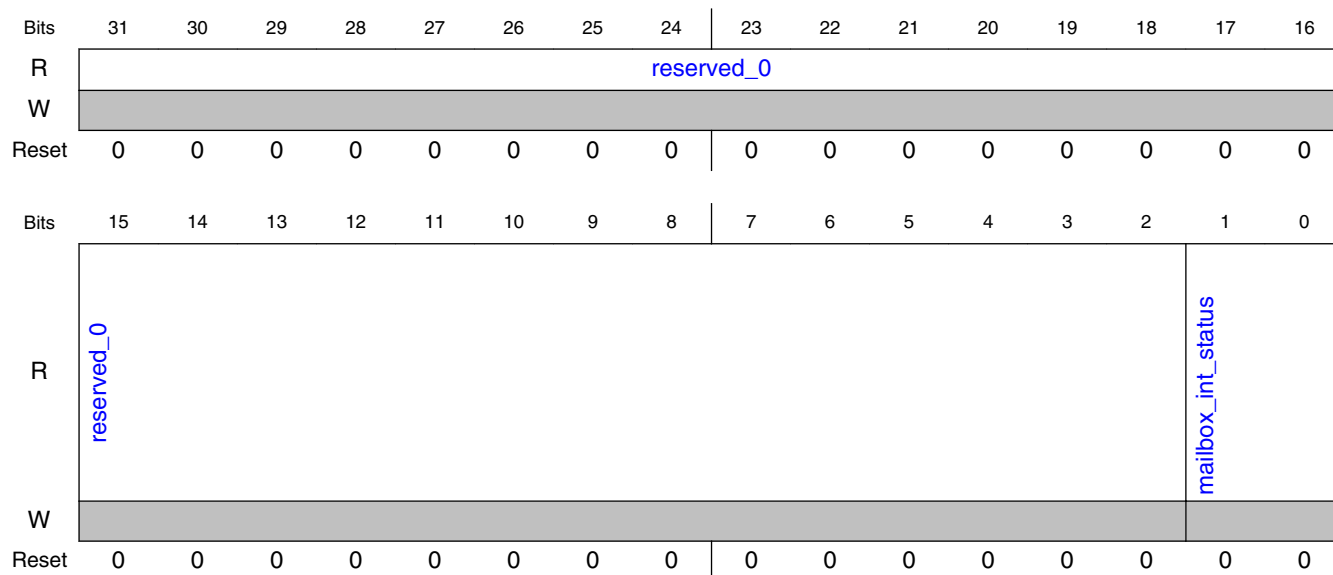
Field	Function
31-2 reserved_0	
1-0 mailbox_int_ma sk	Mailbox Interrupt mask-Bit[0] - Empty-Bit[1] - Full

13.4.10.1.16 Mailbox Interrupt Status register (MAILBOX_INT_STATUS)

13.4.10.1.16.1 Offset

Register	Offset
MAILBOX_INT_STATUS	38h

13.4.10.1.16.2 Diagram



13.4.10.1.16.3 Fields

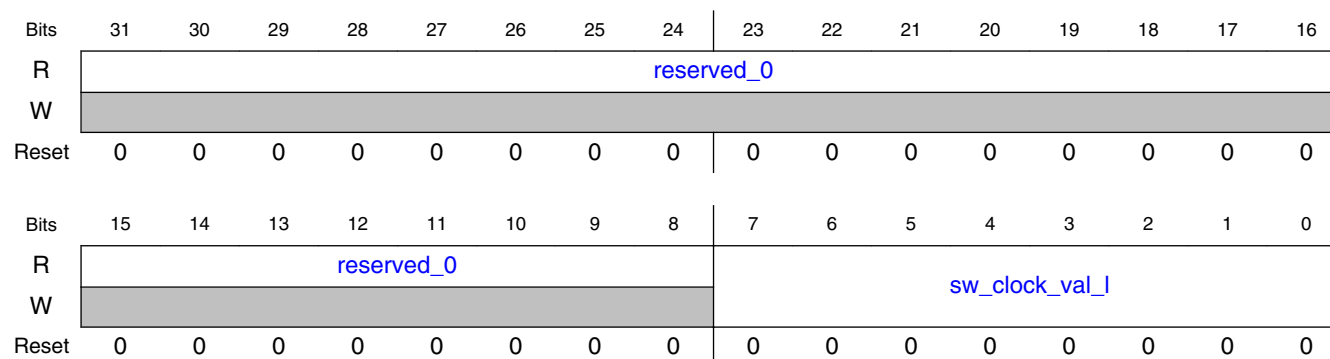
Field	Function
31-2 reserved_0	
1-0 mailbox_int_status	Mailbox Interrupt Status-Bit[0] - Empty-Bit[1] - Full-

13.4.10.1.17 Core Clock frequency (SW_CLK_L)

13.4.10.1.17.1 Offset

Register	Offset
SW_CLK_L	3Ch

13.4.10.1.17.2 Diagram



13.4.10.1.17.3 Fields

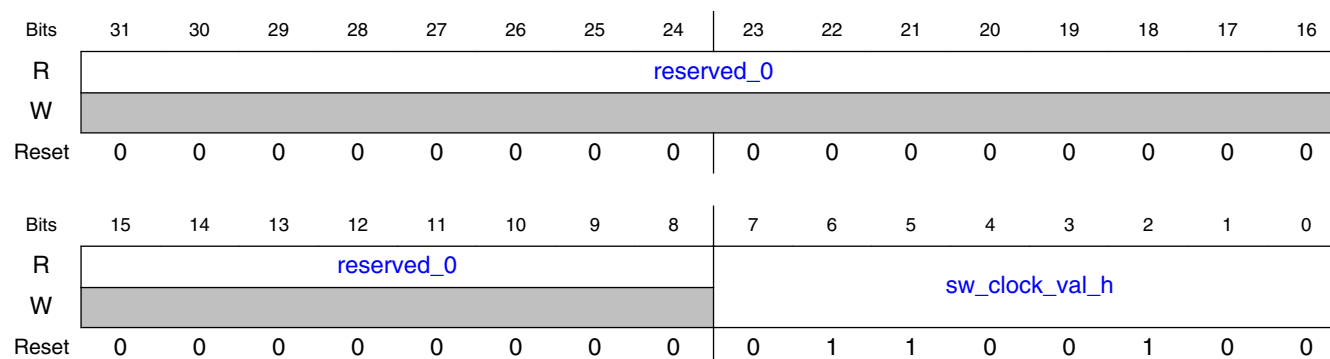
Field	Function
31-8 reserved_0	
7-0 sw_clock_val_l	Fractial of the clock decimal value. Fractial of the clock decimal value. Should be loaded by API to the value that reflects the frequency of clock provided to core clock input.

13.4.10.1.18 Core Clock frequency (SW_CLK_H)

13.4.10.1.18.1 Offset

Register	Offset
SW_CLK_H	40h

13.4.10.1.18.2 Diagram



13.4.10.1.18.3 Fields

Field	Function
31-8 reserved_0	
7-0 sw_clock_val_h	Clock frequency in decimal values. Clock frequency in decimal values. Should be loaded by API to the value that reflects the frequency of clock provided to core clock input.

13.4.10.1.19 SW Event 0 register (SW_EVENTS0)

13.4.10.1.19.1 Offset

Register	Offset
SW_EVENTS0	44h

13.4.10.1.19.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								sw_events7_0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.19.3 Fields

Field	Function
31-8 reserved_0	
7-0 sw_events7_0	When SW writes it updates just the extra event bit. When SW writes it updates just the extra event bit. When Host reads it is cleared

13.4.10.1.20 SW Event 1 register (SW_EVENTS1)

13.4.10.1.20.1 Offset

Register	Offset
SW_EVENTS1	48h

13.4.10.1.20.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								sw_events15_8							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.20.3 Fields

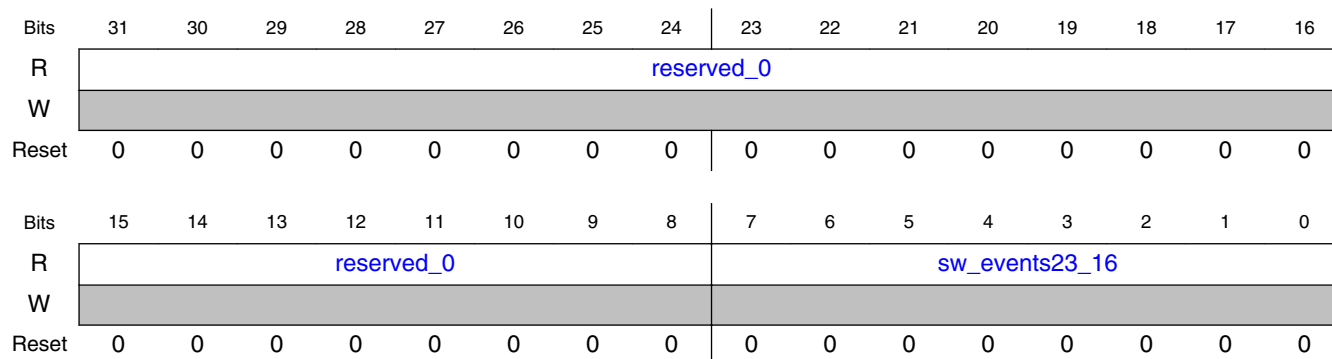
Field	Function
31-8 reserved_0	
7-0 sw_events15_8	When SW writes it updates just the extra event bit. When SW writes it updates just the extra event bit. When Host reads it is cleared

13.4.10.1.21 SW Event 2 register (SW_EVENTS2)

13.4.10.1.21.1 Offset

Register	Offset
SW_EVENTS2	4Ch

13.4.10.1.21.2 Diagram



13.4.10.1.21.3 Fields

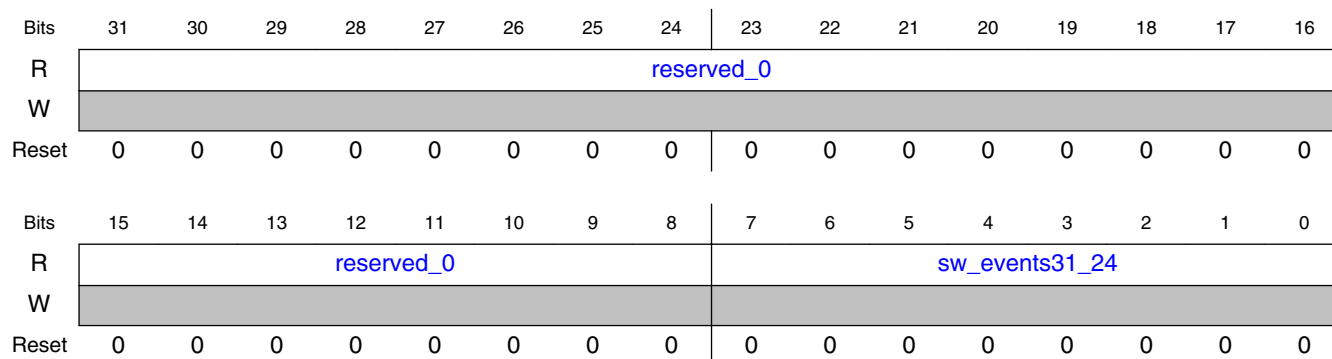
Field	Function
31-8 reserved_0	
7-0 sw_events23_16	When SW writes it updates just the extra event bit. When SW writes it updates just the extra event bit. When Host reads it is cleared

13.4.10.1.22 SW Event 3 register (SW_EVENTS3)

13.4.10.1.22.1 Offset

Register	Offset
SW_EVENTS3	50h

13.4.10.1.22.2 Diagram



13.4.10.1.22.3 Fields

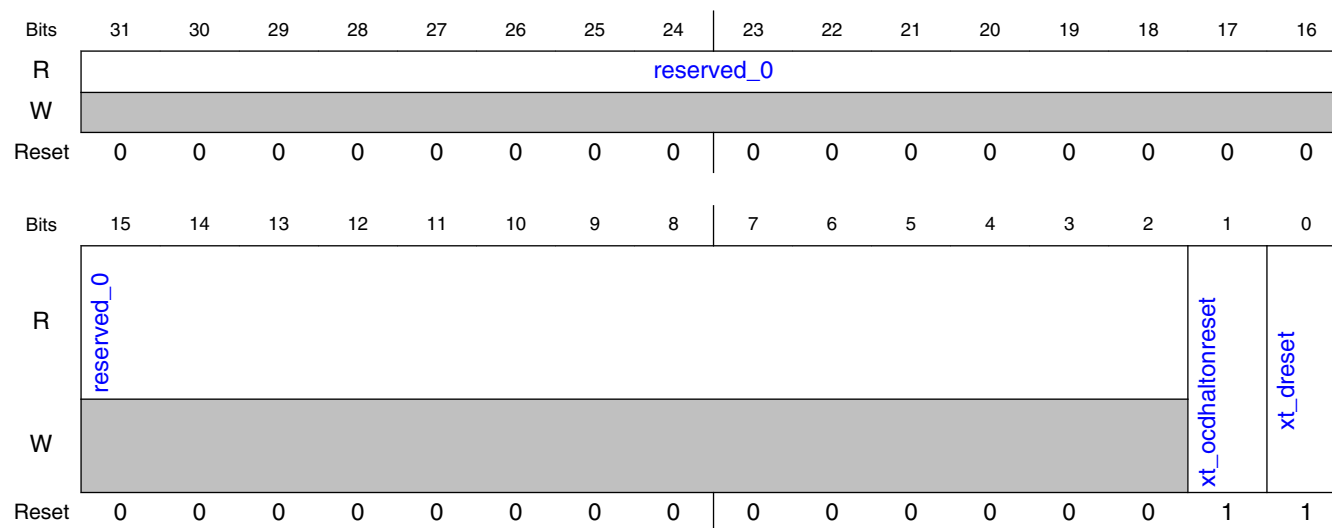
Field	Function
31-8 reserved_0	
7-0 sw_events31_2 4	When SW writes it updates just the extra event bit. When SW writes it updates just the extra event bit. When Host reads it is cleared

13.4.10.1.23 Internal CPU - On Chip Debug (OCD) Ctrl Register (XT_OCD_CTRL)

13.4.10.1.23.1 Offset

Register	Offset
XT_OCD_CTRL	60h

13.4.10.1.23.2 Diagram



13.4.10.1.23.3 Fields

Field	Function
31-2	

Table continues on the next page...

Clocks And Resets

Field	Function
reserved_0	
1 xt_ocdhaltonreset	Internal CPU - Halt On Reget configuration register
0 xt_dreset	Internal CPU - Dreset control register

13.4.10.1.24 Internal CPU - OCD R0 mode configuration (XT_OCD_CTRL_RO)

13.4.10.1.24.1 Offset

Register	Offset
XT_OCD_CTRL_RO	64h

13.4.10.1.24.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0															xt_xocdmode
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.24.3 Fields

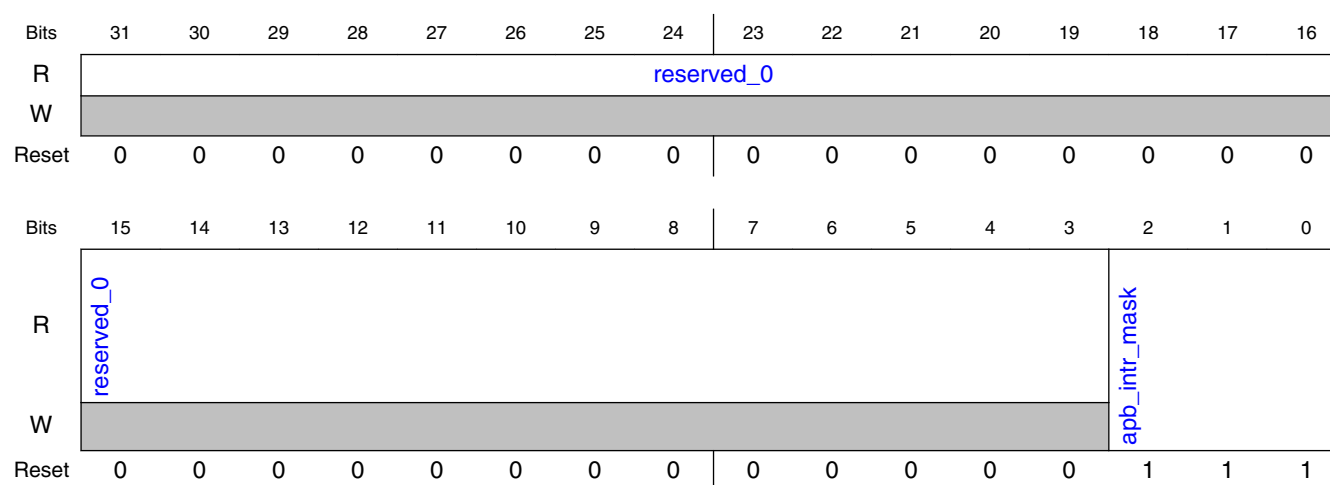
Field	Function
31-1 reserved_0	
0 xt_xocdmode	Internal CPU - OCD mode configuration

13.4.10.1.25 APB Interrupt Mask Register (APB_INT_MASK)

13.4.10.1.25.1 Offset

Register	Offset
APB_INT_MASK	6Ch

13.4.10.1.25.2 Diagram



13.4.10.1.25.3 Fields

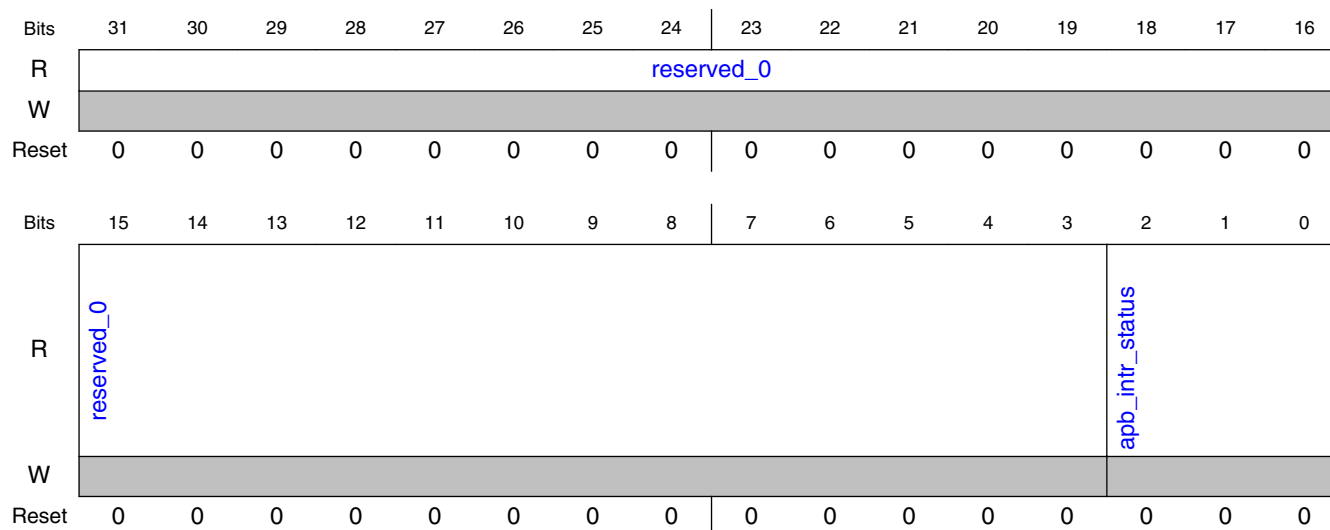
Field	Function
31-3 reserved_0	
2-0 apb_intr_mask	Mask the APB interrupt-Bit0 - Mailbox Interrupt-Bit1 - PIF Interrupt-Bit2 - CEC Interrupt

13.4.10.1.26 APB interrupt status register (APB_STATUS_MASK)

13.4.10.1.26.1 Offset

Register	Offset
APB_STATUS_MASK	70h

13.4.10.1.26.2 Diagram



13.4.10.1.26.3 Fields

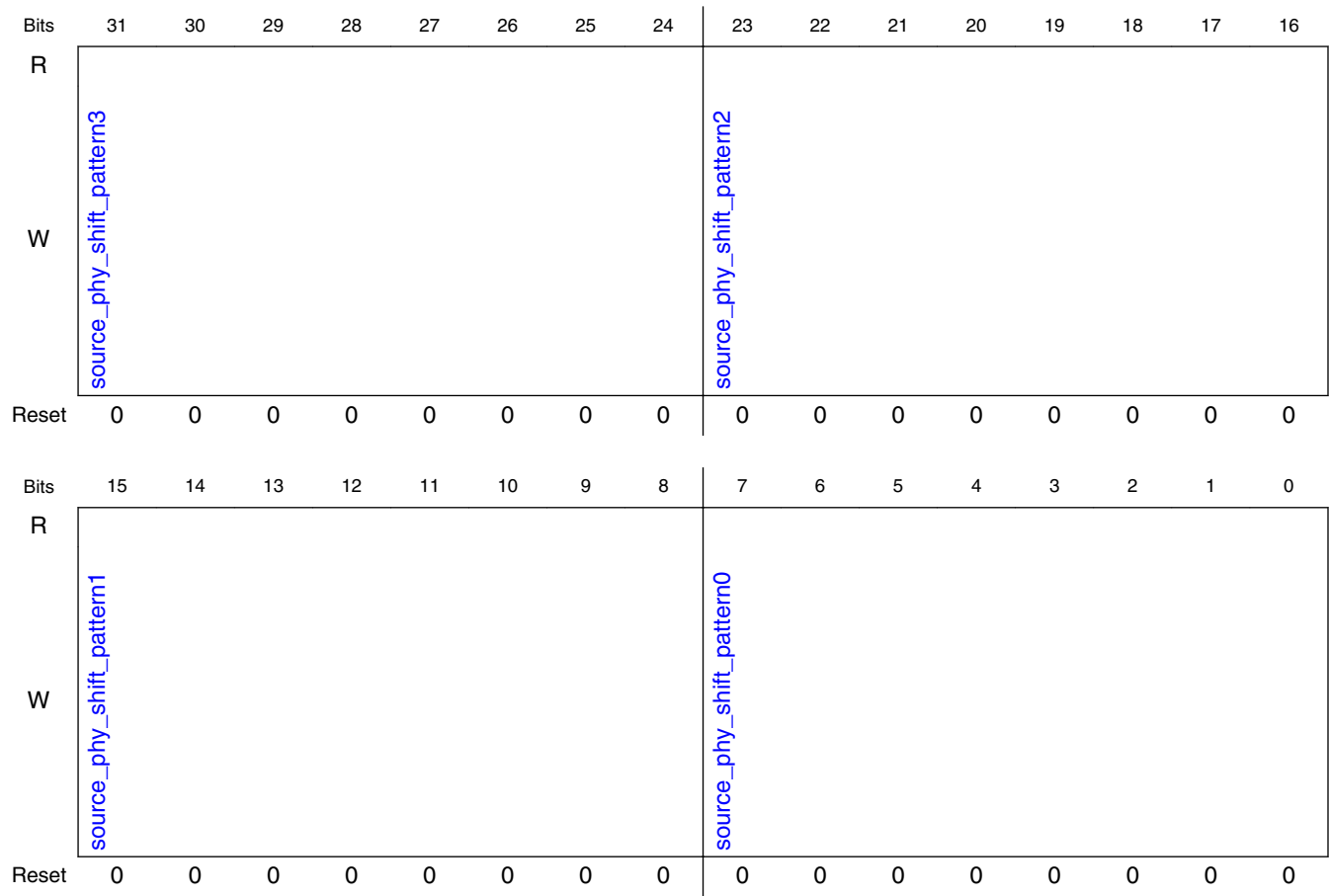
Field	Function
31-3 reserved_0	
2-0 apb_intr_status	APB interrupt STATUS-Bit0 - Mailbox Interrupt-Bit1 - PIF Interrupt-Bit2 - CEC Interrupt

13.4.10.1.27 HDMI shift pattern 3-0 (SHIFT_PATTERN_IN_3_0)

13.4.10.1.27.1 Offset

Register	Offset
SHIFT_PATTERN_IN_3_0	800h

13.4.10.1.27.2 Diagram



13.4.10.1.27.3 Fields

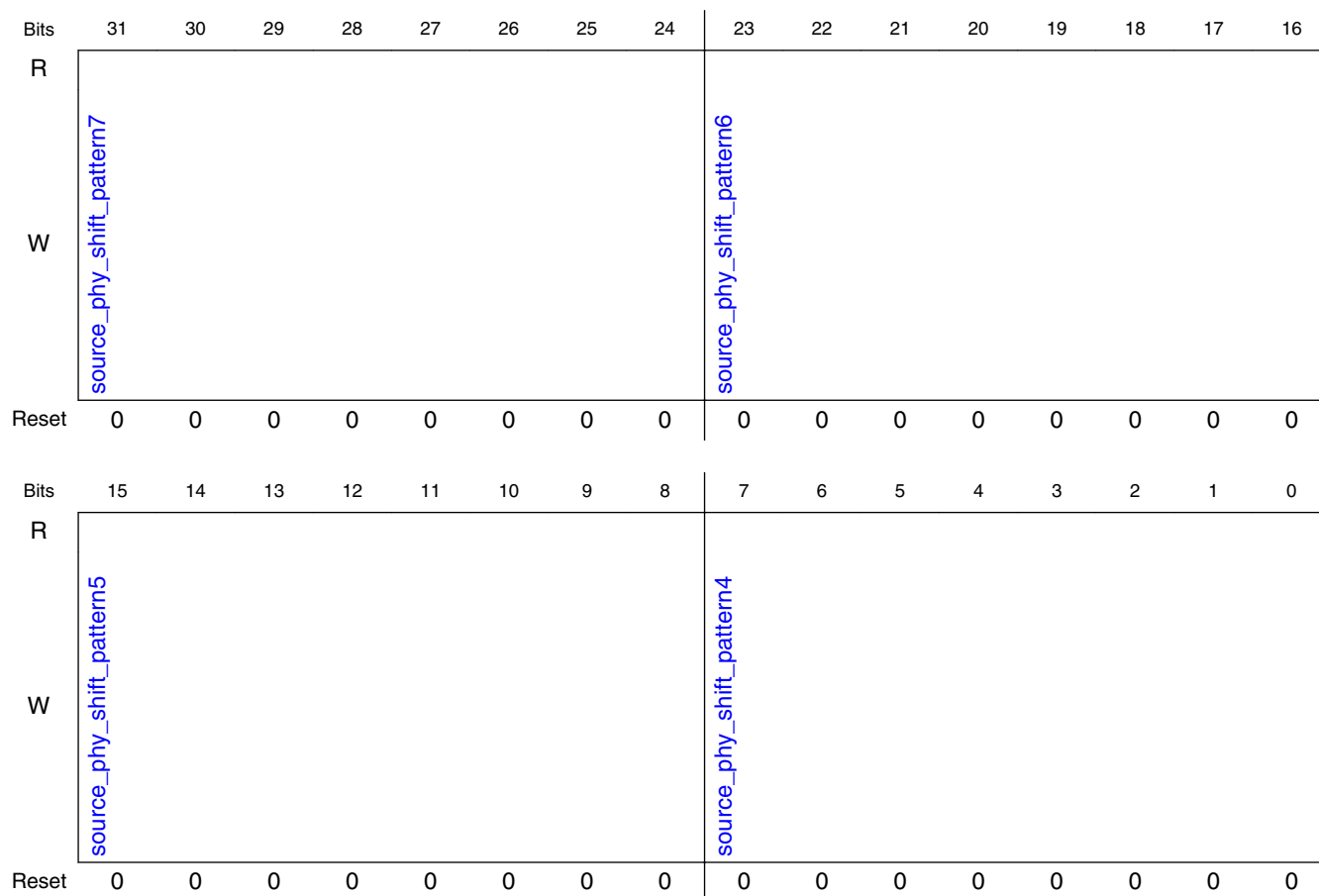
Field	Function
31-24 source_phy_shift_pattern3	Input to hdmi_pattern_shift
23-16 source_phy_shift_pattern2	Input to hdmi_pattern_shift
15-8 source_phy_shift_pattern1	Input to hdmi_pattern_shift
7-0 source_phy_shift_pattern0	Input to hdmi_pattern_shift

13.4.10.1.28 HDMI shift pattern 4-7 (SHIFT_PATTERN_IN_4_7)

13.4.10.1.28.1 Offset

Register	Offset
SHIFT_PATTERN_IN_4_7	804h

13.4.10.1.28.2 Diagram



13.4.10.1.28.3 Fields

Field	Function
31-24 source_phy_shift_pattern7	Input to hdmi_pattern_shift

Table continues on the next page...

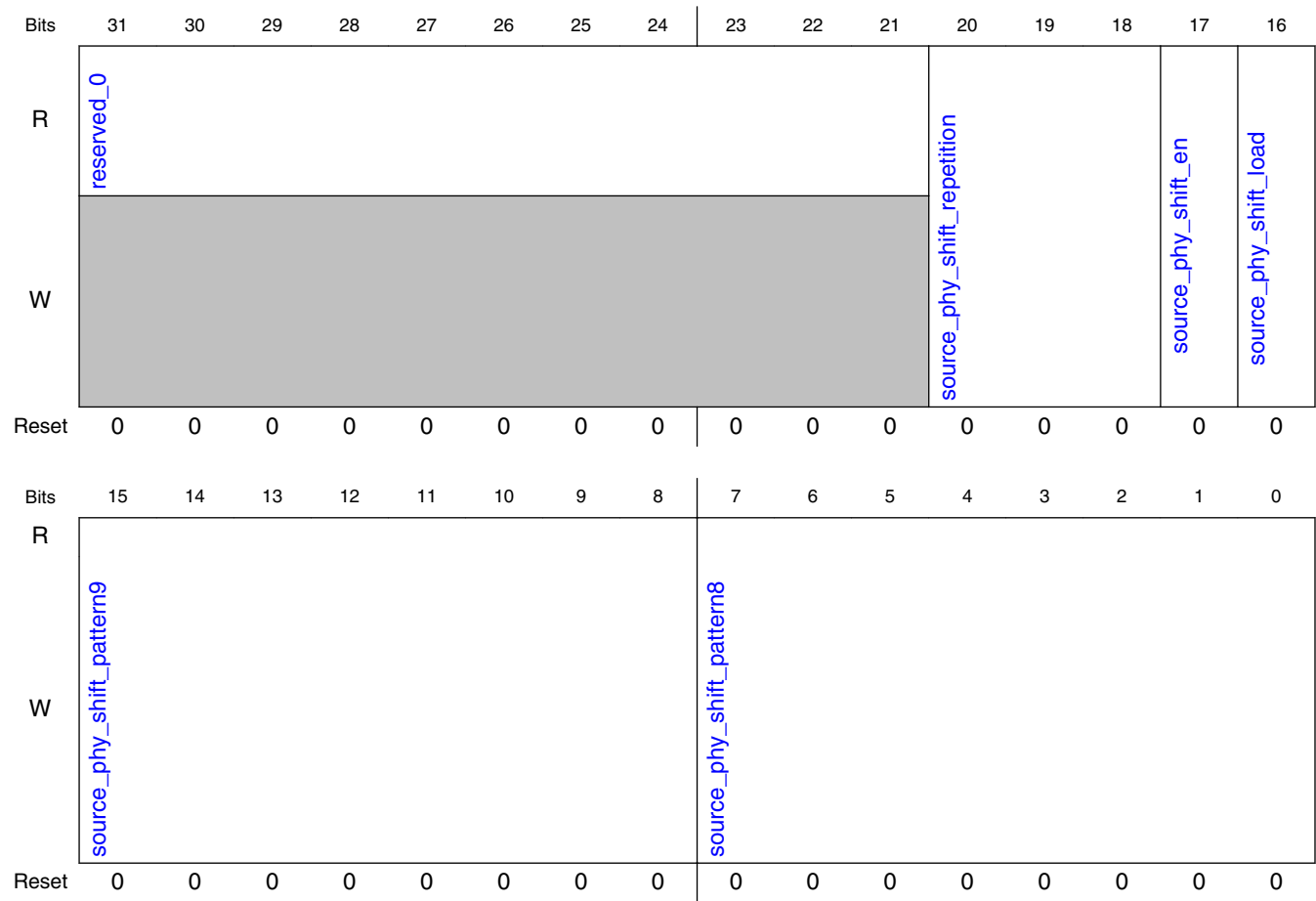
Field	Function
23-16 source_phy_shift t_pattern6	Input to hdmi_pattern_shift
15-8 source_phy_shift t_pattern5	Input to hdmi_pattern_shift
7-0 source_phy_shift t_pattern4	Input to hdmi_pattern_shift

13.4.10.1.29 HDMI shift pattern 9-8 with control bits (SHIFT_PATTERN_IN_9_8)

13.4.10.1.29.1 Offset

Register	Offset
SHIFT_PATTERN_IN9_8	808h

13.4.10.1.29.2 Diagram



13.4.10.1.29.3 Fields

Field	Function
31-21 reserved_0	
20-18 source_phy_shif t_repetition	Shift repetition Number
17 source_phy_shif t_en	When 1 enable the Shift pattern Mechanism
16 source_phy_shif t_load	When 1 load the 80 bits of data
15-8	Input to hdmi_pattern_shift

Table continues on the next page...

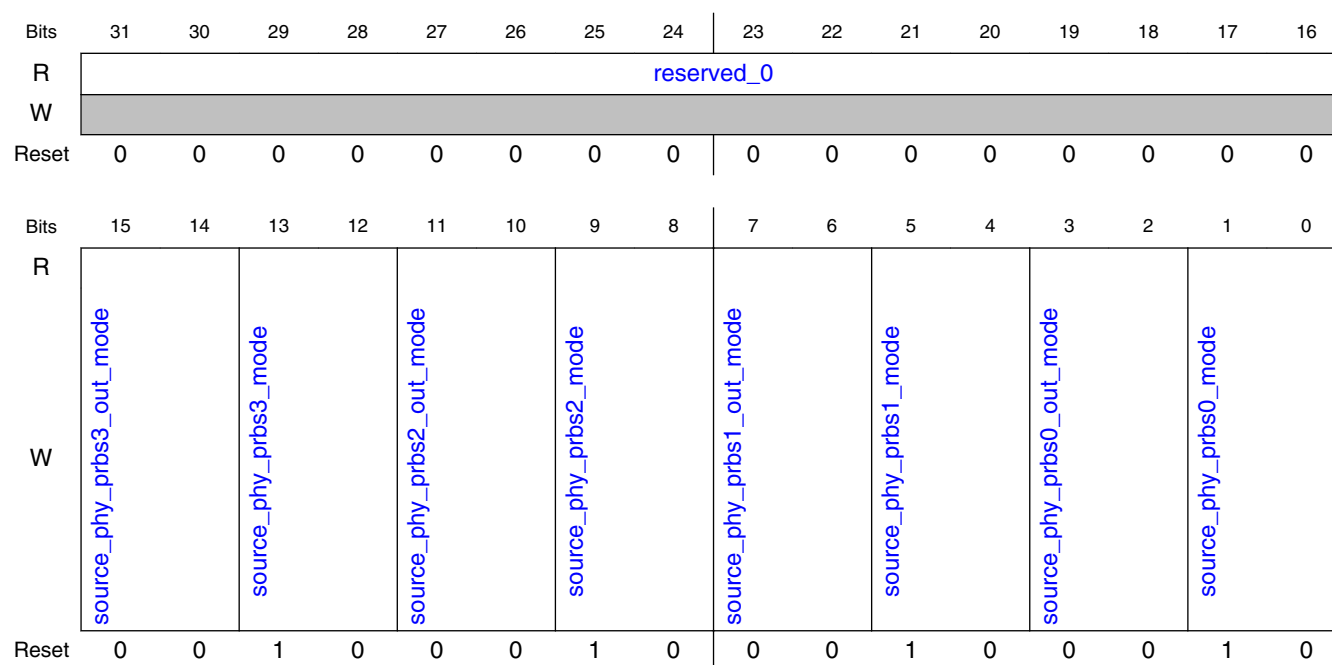
Field	Function
source_phy_shif t_pattern9	
7-0 source_phy_shif t_pattern8	Input to hdmi_pattern_shift

13.4.10.1.30 PRBS control (PRBS_CNTRL)

13.4.10.1.30.1 Offset

Register	Offset
PRBS_CNTRL	80Ch

13.4.10.1.30.2 Diagram



13.4.10.1.30.3 Fields

Field	Function
31-16 reserved_0	

Table continues on the next page...

Clocks And Resets

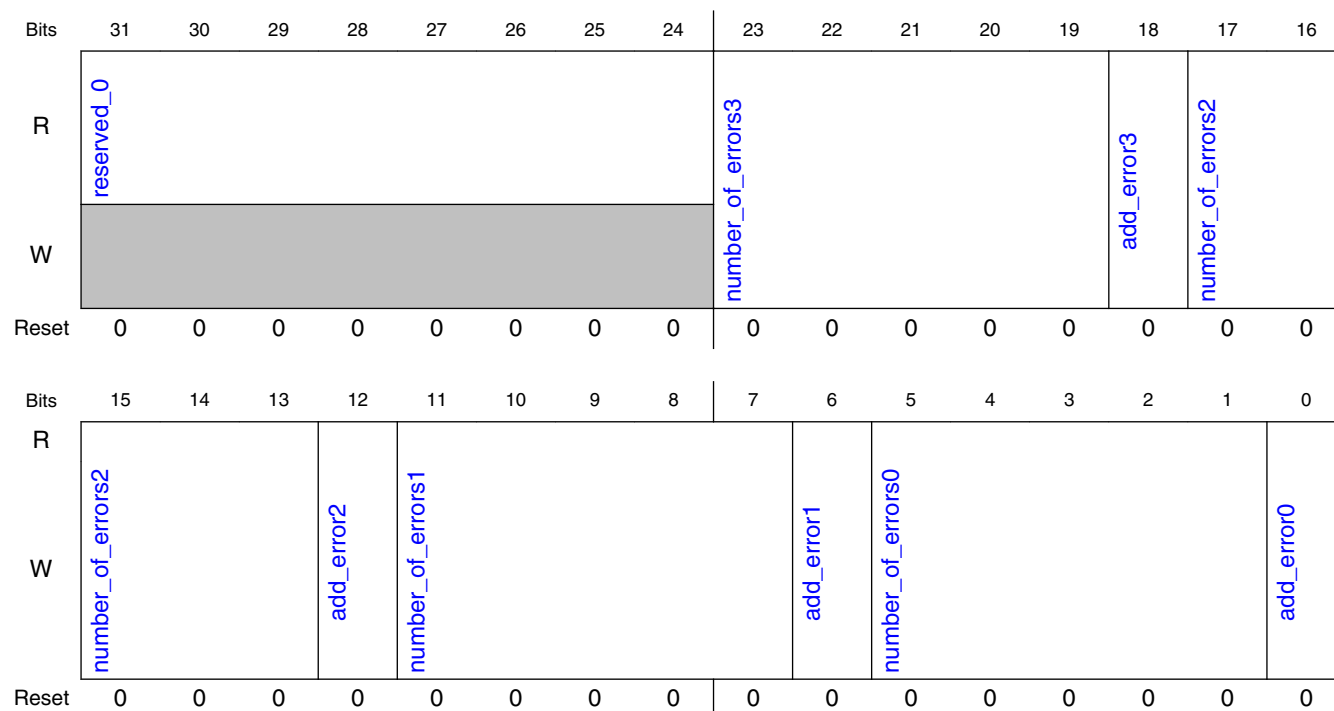
Field	Function
15-14 source_phy_prb s3_out_mode	00 = idle, output all zeros -01 = output 8 bits on pattern[7:0]10 = output 1 bit on pattern[9] and output inverted bit on pattern[8] -11 = output 10 bits on pattern[9:0]
13-12 source_phy_prb s3_mode	00 = PRBS11 -01 = PRBS15 -10 = PRBS7 -11 = PRBS31
11-10 source_phy_prb s2_out_mode	00 = idle, output all zeros01 = output 8 bits on pattern[7:0] -10 = output 1 bit on pattern[9] and output inverted bit on pattern[8] -11 = output 10 bits on pattern[9:0]
9-8 source_phy_prb s2_mode	00 = PRBS11 -01 = PRBS15 -10 = PRBS7 -11 = PRBS31
7-6 source_phy_prb s1_out_mode	00 = idle, output all zeros -01 = output 8 bits on pattern[7:0] -10 = output 1 bit on pattern[9] and output inverted bit on pattern[8] -11 = output 10 bits on pattern[9:0]
5-4 source_phy_prb s1_mode	00 = PRBS11 -01 = PRBS15 -10 = PRBS7 -11 = PRBS31
3-2 source_phy_prb s0_out_mode	00 = idle, output all zeros01 = output 8 bits on pattern[7:0]10 = output 1 bit on pattern[9] and output inverted bit on pattern[8]11 = output 10 bits on pattern[9:0]
1-0 source_phy_prb s0_mode	00 = PRBS1101 = PRBS15 -10 = PRBS7 -11 = PRBS31

13.4.10.1.31 PRBS error insertion (PRBS_ERR_INSERTION)

13.4.10.1.31.1 Offset

Register	Offset
PRBS_ERR_INSERTION	810h

13.4.10.1.31.2 Diagram



13.4.10.1.31.3 Fields

Field	Function
31-24 reserved_0	
23-19 number_of_errors3	The number of errors to be inserted when add_error is high. The number of errors to be inserted when add_error is high.
18 add_error3	When high the PRBS generator inserts the number of errors written in number_of_errors field. When high the PRBS generator inserts the number of errors written in number_of_errors field.
17-13 number_of_errors2	The number of errors to be inserted when add_error is high. The number of errors to be inserted when add_error is high.
12 add_error2	When high the PRBS generator inserts the number of errors written in number_of_errors field. When high the PRBS generator inserts the number of errors written in number_of_errors field.
11-7 number_of_errors1	The number of errors to be inserted when add_error is high. The number of errors to be inserted when add_error is high.
6 add_error1	When high the PRBS generator inserts the number of errors written in number_of_errors field
5-1	The number of errors to be inserted when add_error is high.

Table continues on the next page...

Clocks And Resets

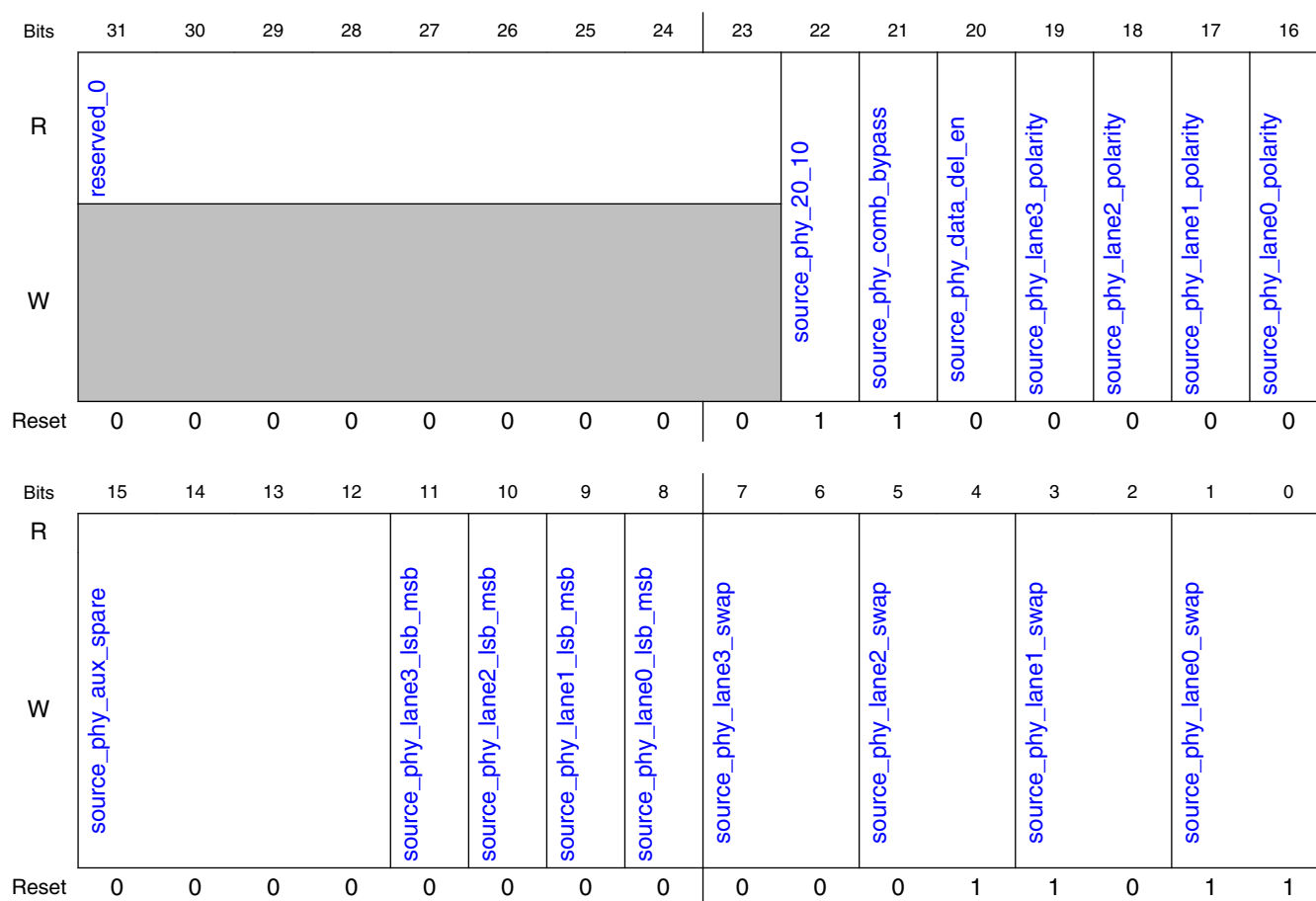
Field	Function
number_of_errors0	The number of errors to be inserted when add_error is high.
0 add_error0	When high the PRBS generator inserts the number of errors written in number_of_errors field

13.4.10.1.32 Lane control register: swap, order, polarity (LANES_CONFIG)

13.4.10.1.32.1 Offset

Register	Offset
LANES_CONFIG	814h

13.4.10.1.32.2 Diagram



13.4.10.1.32.3 Fields

Field	Function
31-23 reserved_0	
22 source_phy_20_10	1'b0: Data to PHY is 10 bit with char clock -1'd1: Data to PHY is 20 bit with data clock
21 source_phy_comb_bypass	Bypass swap invert and all combination
20 source_phy_data_del_en	enable configurable delay of lanes to be activated. enable configurable delay of lanes to be activated.if this bit is 0 the delay is only activated for DisplayPort mode with source_phy_data_sel=prbs or shift-mem
19 source_phy_lane3_polarity	Reverse polarity of data, lane3
18 source_phy_lane2_polarity	Reverse polarity of data, lane2
17 source_phy_lane1_polarity	Reverse polarity of data, lane1
16 source_phy_lane0_polarity	Reverse polarity of data, lane0
15-12 source_phy_aux_spare	Spare bits for aux **1. Spare bits for aux **1.1**
11 source_phy_lane3_lsb_msb	Reverse order of data, lane3
10 source_phy_lane2_lsb_msb	Reverse order of data, lane2
9 source_phy_lane1_lsb_msb	Reverse order of data, lane1
8 source_phy_lane0_lsb_msb	Reverse order of data, lane0
7-6 source_phy_lane3_swap	Swap control lane3
5-4	Swap control lane2

Table continues on the next page...

Clocks And Resets

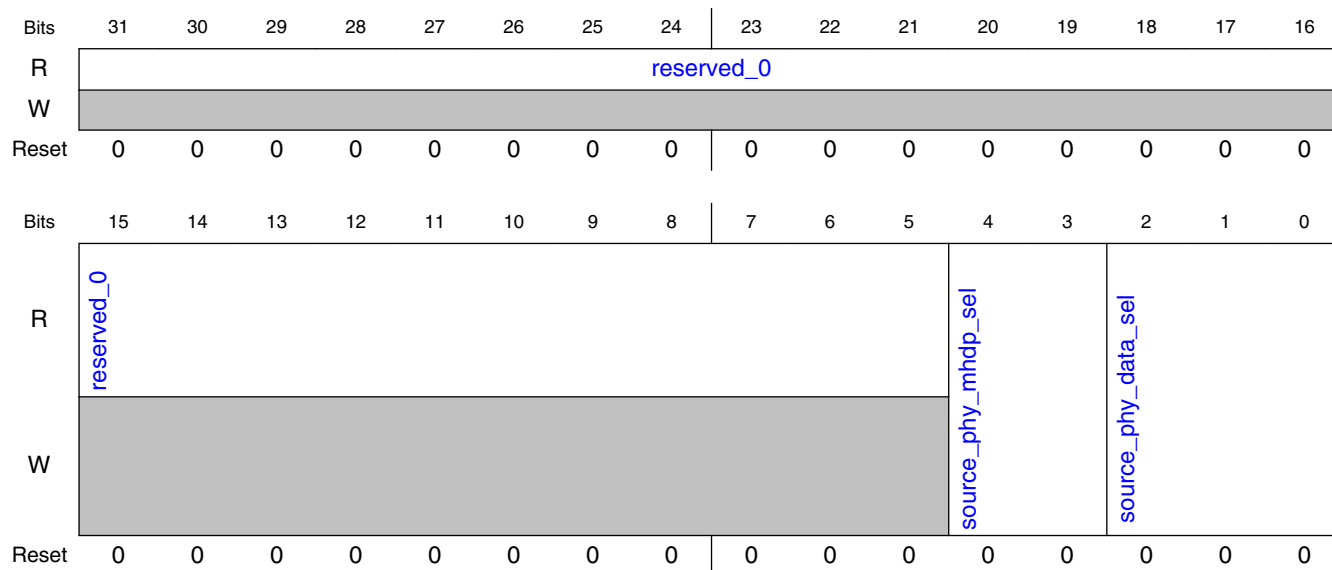
Field	Function
source_phy_lane2_swap	Swap control lane1
source_phy_lane1_swap	
source_phy_lane0_swap	Swap control lane0

13.4.10.1.33 PHY data select DP/HDMI and HDMI data source (PHY_DATA_SEL)

13.4.10.1.33.1 Offset

Register	Offset
PHY_DATA_SEL	818h

13.4.10.1.33.2 Diagram



13.4.10.1.33.3 Fields

Field	Function
31-5	

Table continues on the next page...

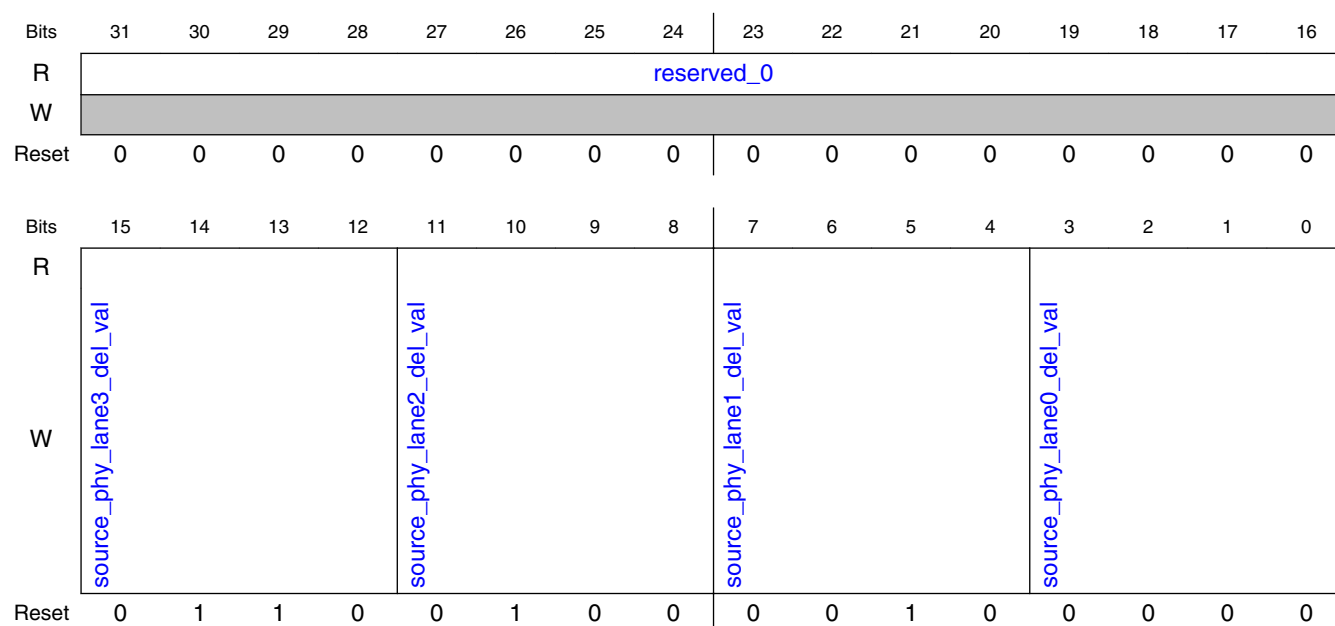
Field	Function
reserved_0	
4-3 source_phy_mh dp_sel	3'd0: tx_data = DP -3'd1: tx_data = HDMI -3'd2: tx_data = RSRV -3'd3: tx_data = RSRV
2-0 source_phy_dat a_sel	3'd0: tx_data = phy_dout -3'd1: tx_data = phy_dout_bypass -3'd2: tx_data = source_phy_prbs_pout -3'd3: tx_data = source_phy_shift_pout

13.4.10.1.34 Lane delay control (LANES_DEL_VAL)

13.4.10.1.34.1 Offset

Register	Offset
LANES_DEL_VAL	81Ch

13.4.10.1.34.2 Diagram



13.4.10.1.34.3 Fields

Field	Function
31-16	

Table continues on the next page...

Clocks And Resets

Field	Function
reserved_0	
15-12 source_phy_lane3_del_val	delay for lane 3 -this parameter can take values from 0 up to 8. delay for lane 3 -this parameter can take values from 0 up to 8. -All other values are reserved
11-8 source_phy_lane2_del_val	delay for lane 2 -this parameter can take values from 0 up to 8. delay for lane 2 -this parameter can take values from 0 up to 8. -All other values are reserved
7-4 source_phy_lane1_del_val	delay for lane 1 -this parameter can take values from 0 up to 8. delay for lane 1 -this parameter can take values from 0 up to 8. -All other values are reserved
3-0 source_phy_lane0_del_val	delay for lane 0 -this parameter can take values from 0 up to 8. delay for lane 0 -this parameter can take values from 0 up to 8. -All other values are reserved

13.4.10.1.35 Register implemented only for configuration with HDMI. (SOURCE_HDTX_CAR)

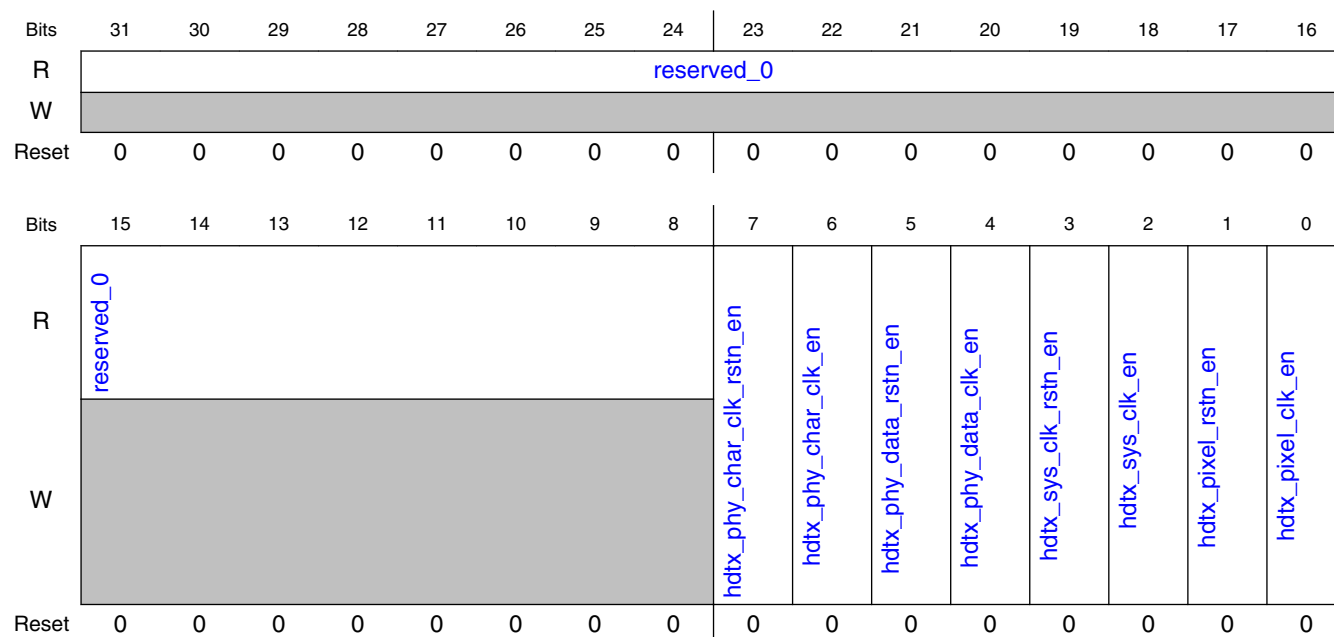
13.4.10.1.35.1 Offset

Register	Offset
SOURCE_HDTX_CAR	900h

13.4.10.1.35.2 Function

Register implemented only for configuration with HDMI. Each bit is responsible for clock or reset enable

13.4.10.1.35.3 Diagram



13.4.10.1.35.4 Fields

Field	Function
31-8 reserved_0	Reserved - ignored on write
7 hdtx_phy_char_clk_rstn_en	hdtx_phy_char_clk_rstn enable - active low (only when HDMI used)
6 hdtx_phy_char_clk_en	hdtx_phy_char_clk enable - active high (only when HDMI used)
5 hdtx_phy_data_rstn_en	hdtx_phy_data_rstn enable - active low (only when HDMI used)
4 hdtx_phy_data_clk_en	hdtx_phy_data_clk enable - active high (only when HDMI used)
3 hdtx_sys_clk_rstn_en	hdtx_sys_clk_rstn enable - active low (only when HDMI used)
2 hdtx_sys_clk_en	hdtx_sys_clk enable - active high (only when HDMI used)
1	hdtx_pixel_rstn enable - active low (only when HDMI used)

Table continues on the next page...

Clocks And Resets

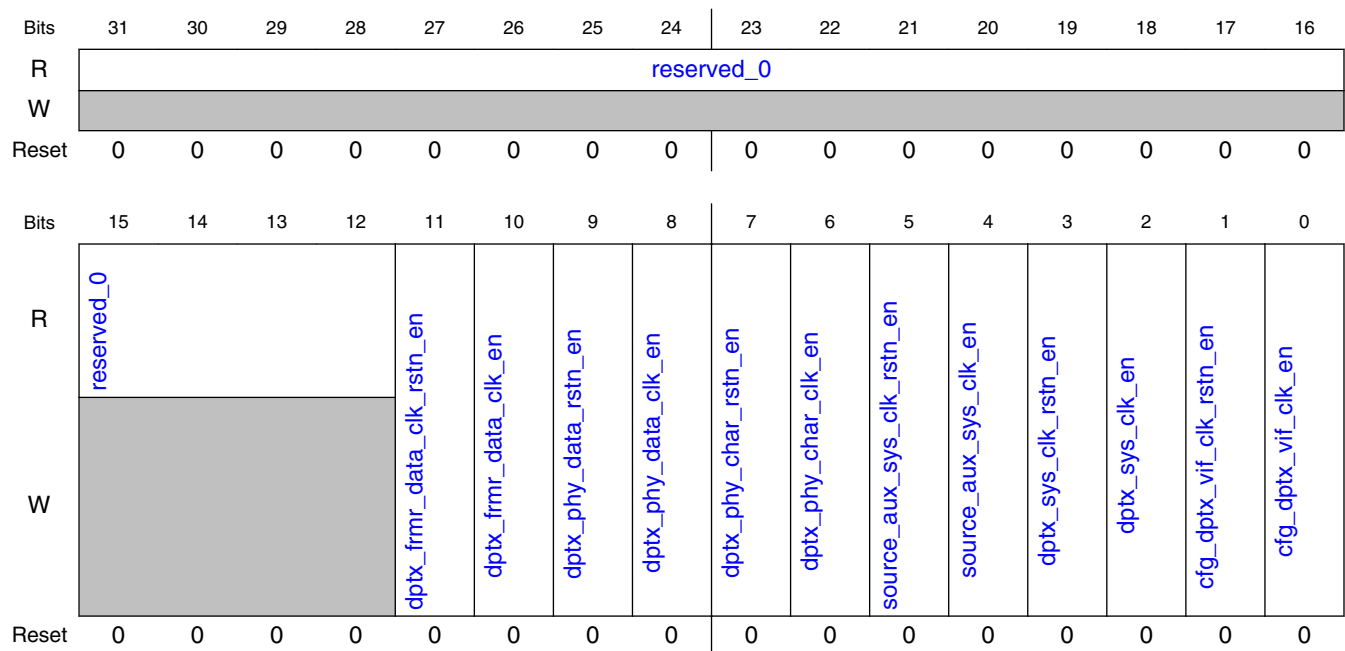
Field	Function
hdtx_pixel_rstn_en	
0	hdtx_pixel_clk enable - active high (only when HDMI used)
hdtx_pixel_clk_en	

13.4.10.1.36 DP TX clock and reset ctrl register (SOURCE_DPTX_CAR)

13.4.10.1.36.1 Offset

Register	Offset
SOURCE_DPTX_CAR	904h

13.4.10.1.36.2 Diagram



13.4.10.1.36.3 Fields

Field	Function
31-12 reserved_0	Reserved - ignored on write

Table continues on the next page...

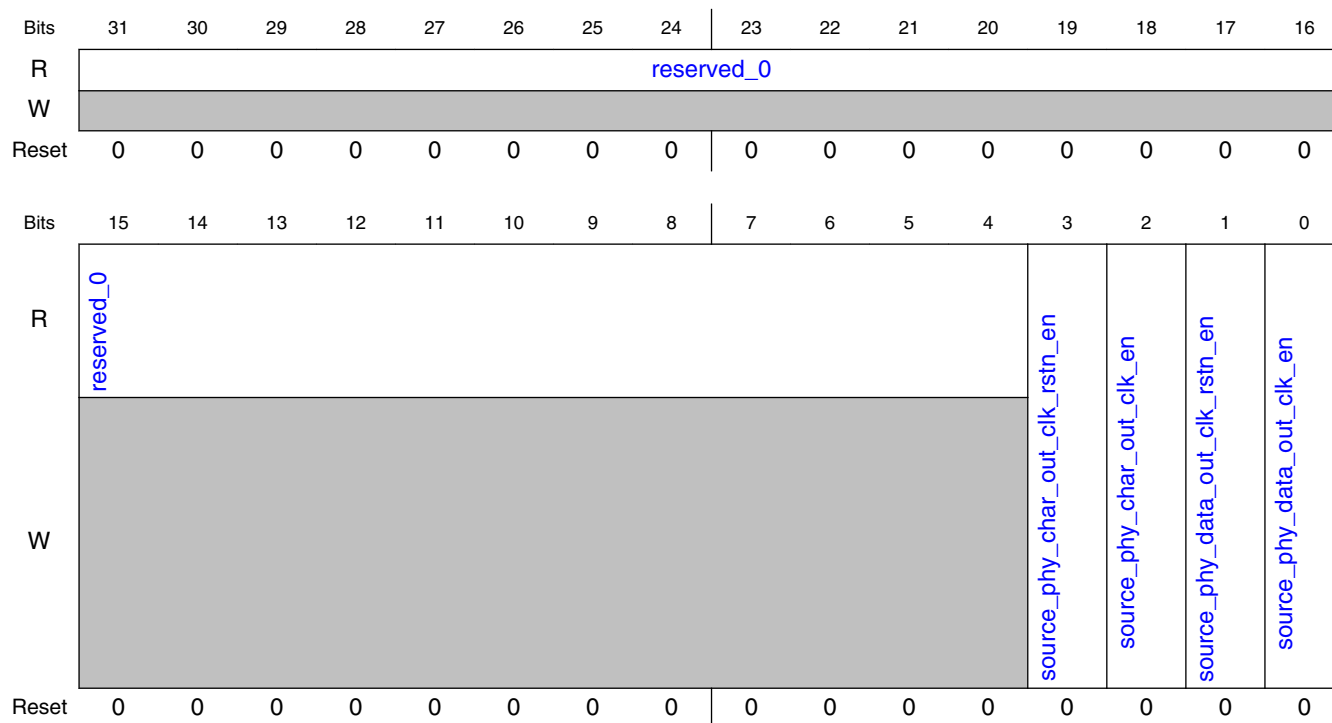
Field	Function
11 dptx_frmr_data_clk_rstn_en	dptx_frmr_data_clk_rstn enable - active low
10 dptx_frmr_data_clk_en	dptx_frmr_data_clk enable - active high
9 dptx_phy_data_rstn_en	dptx_phy_data_rstn enable - active low
8 dptx_phy_data_clk_en	dptx_phy_data_clk enable - active high
7 dptx_phy_char_rstn_en	dptx_phy_char_rstn enable - active low
6 dptx_phy_char_clk_en	dptx_phy_char_clk enable - active high
5 source_aux_sys_clk_rstn_en	source_aux_sys_clk_rstn enable - active low
4 source_aux_sys_clk_en	source_aux_sys_clk enable - active high
3 dptx_sys_clk_rstn_en	dptx_sys_clk_rstn enable - active low
2 dptx_sys_clk_en	dptx_sys_clk enable - active high
1 cfg_dptx_vif_clk_rstn_en	dptx_vif_clk_rstn enable - active low
0 cfg_dptx_vif_clk_en	dptx_vif_clk enable - active high

13.4.10.1.37 Source PHY clock and reset ctrl register (SOURCE_PHY_CAR)

13.4.10.1.37.1 Offset

Register	Offset
SOURCE_PHY_CAR	908h

13.4.10.1.37.2 Diagram



13.4.10.1.37.3 Fields

Field	Function
31-4 reserved_0	Reserved - ignored on write
3 source_phy_char_out_clk_rstn_en	source_phy_char_out_clk_rstn enable - active low
2 source_phy_char_out_clk_en	source_phy_char_out_clk enable - active high
1	source_phy_data_out_clk_rstn enable - active low

Table continues on the next page...

Field	Function
source_phy_data_out_clk_rstn_en	
0	source_phy_data_out_clk enable - active high
source_phy_data_out_clk_en	

13.4.10.1.38 Register implemented only for configuration with HDMI. (SOURCE_CEC_CAR)

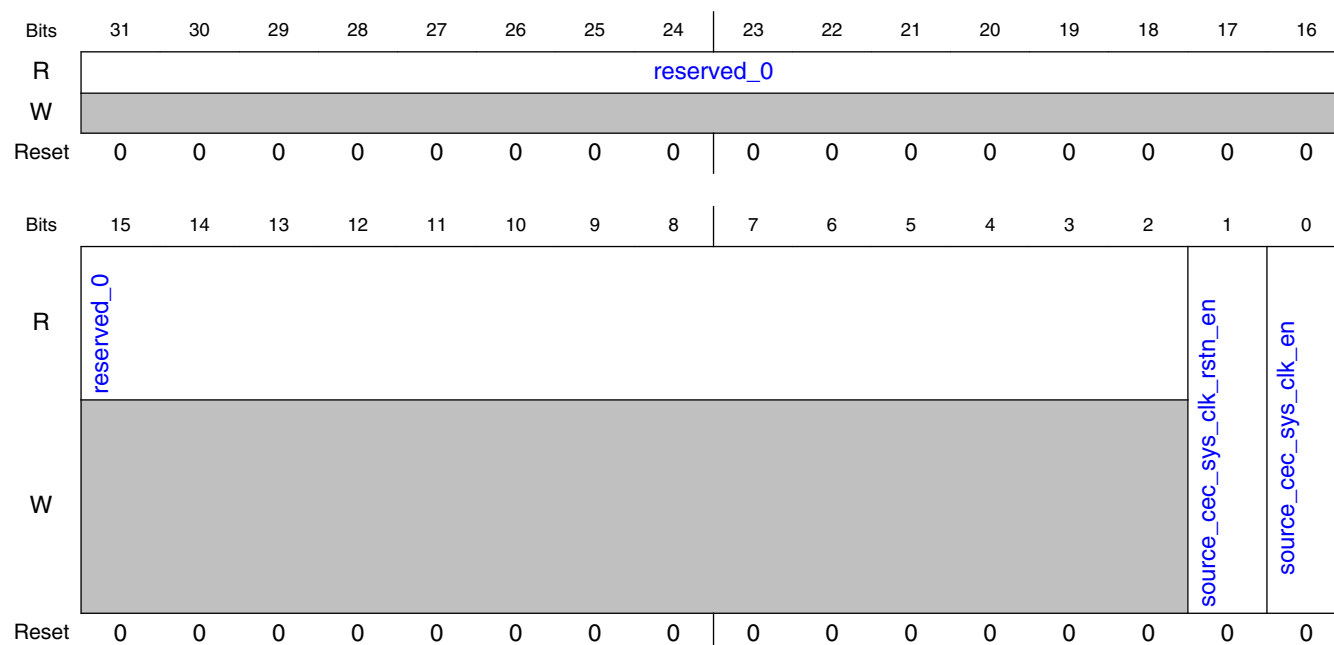
13.4.10.1.38.1 Offset

Register	Offset
SOURCE_CEC_CAR	90Ch

13.4.10.1.38.2 Function

Register implemented only for configuration with HDMI. Each bit is responsible for clock or reset enable

13.4.10.1.38.3 Diagram



13.4.10.1.38.4 Fields

Field	Function
31-2 reserved_0	Reserved - ignored on write (only when HDMI used)
1 source_cec_sys_clk_rstn_en	source_cec_sys_clk_rstn enable - active low (only when HDMI used)
0 source_cec_sys_clk_en	source_cec_sys_clk enable - active high (only when HDMI used)

13.4.10.1.39 CBUS clock and reset ctrl register. (SOURCE_CBUS_CAR)

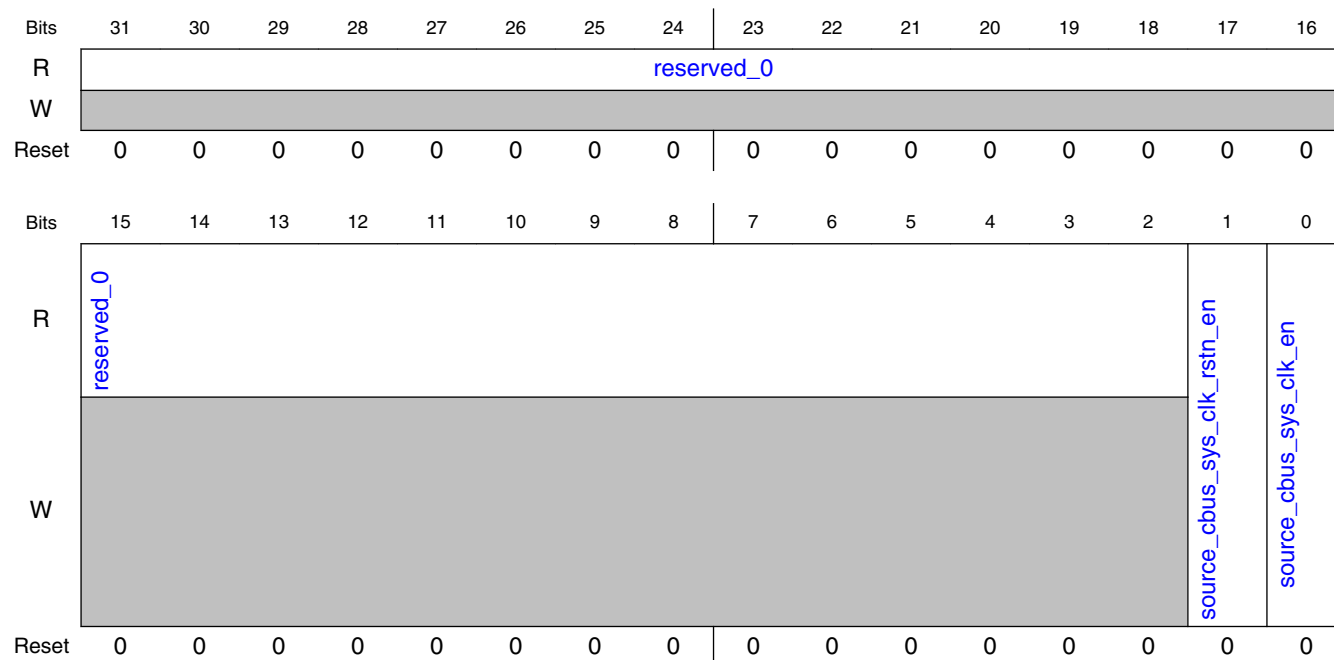
13.4.10.1.39.1 Offset

Register	Offset
SOURCE_CBUS_CAR	910h

13.4.10.1.39.2 Function

CBUS clock and reset ctrl register. Implemented only if CAPB is implemented

13.4.10.1.39.3 Diagram



13.4.10.1.39.4 Fields

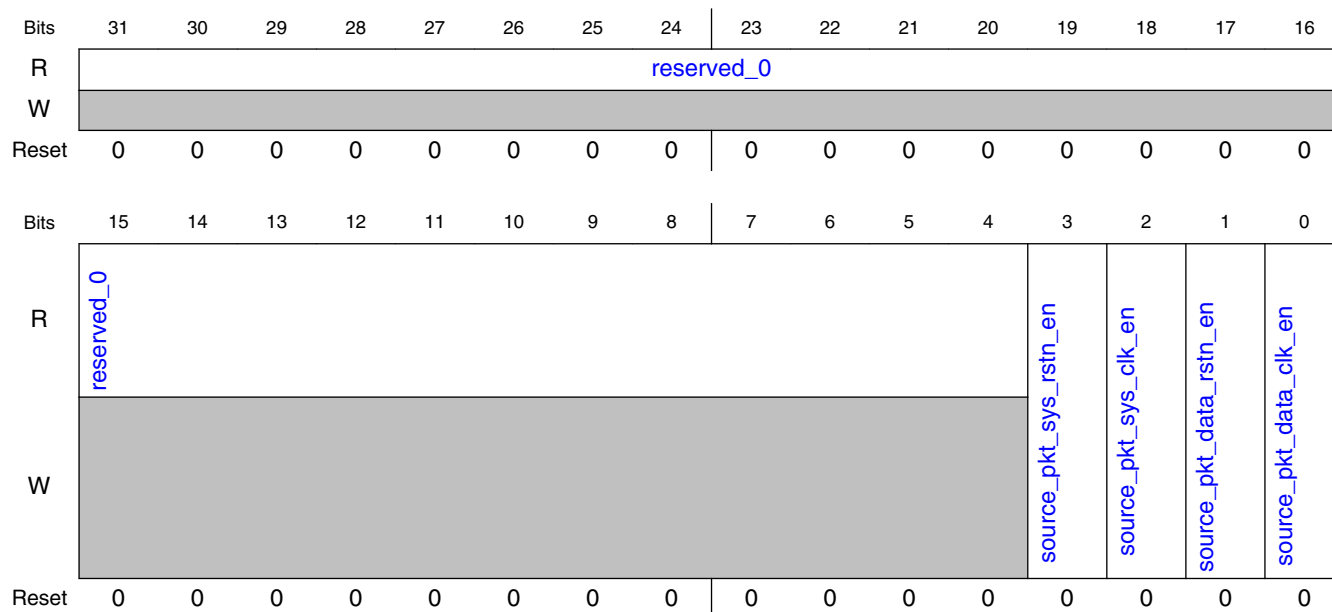
Field	Function
31-2 reserved_0	Reserved - ignored on write
1 source_cbus_sys_clk_rstn_en	source_cbus_sys_clk_rstn enable - active low
0 source_cbus_sys_clk_en	source_cbus_sys_clk enable - active high

13.4.10.1.40 PKT clock and reset ctrl register (SOURCE_PKT_CAR)

13.4.10.1.40.1 Offset

Register	Offset
SOURCE_PKT_CAR	918h

13.4.10.1.40.2 Diagram



13.4.10.1.40.3 Fields

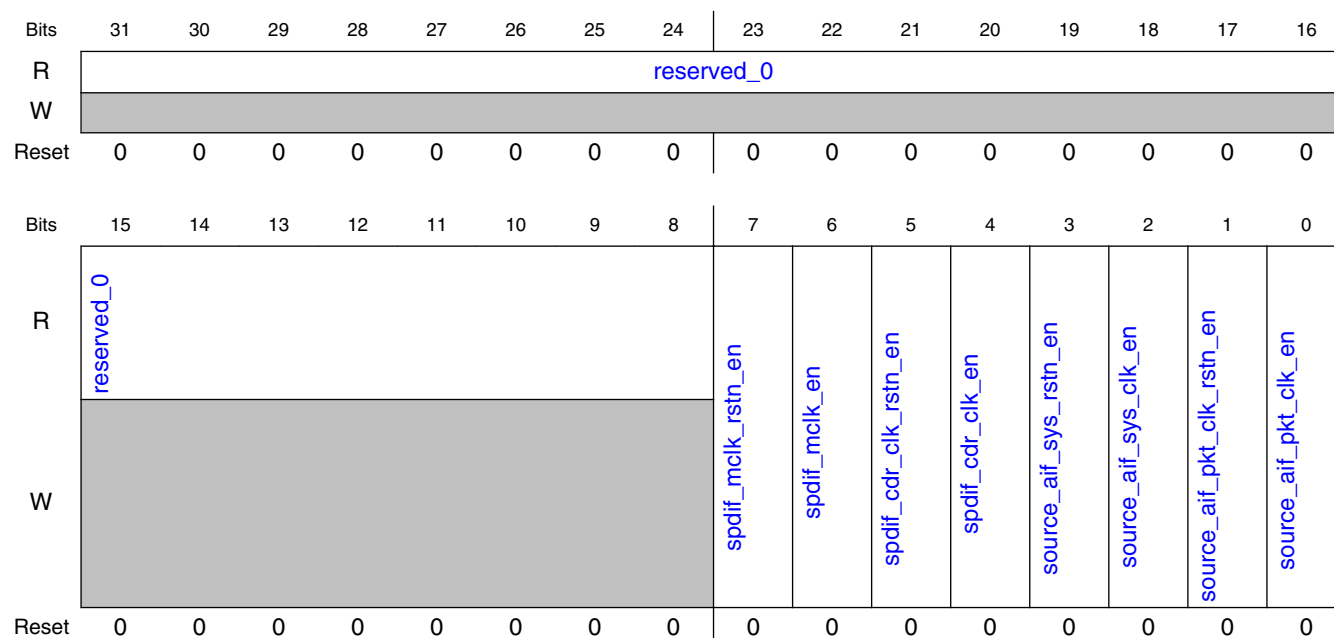
Field	Function
31-4 reserved_0	Reserved - ignored on write
3 source_pkt_sys_rstn_en	source_pkt_sys_rstn enable - active low
2 source_pkt_sys_clk_en	source_pkt_sys_clk enable - active high
1 source_pkt_data_rstn_en	source_pkt_data_rstn enable - active low
0 source_pkt_data_clk_en	source_pkt_data_clk enable - active high

13.4.10.1.41 AIF clock and reset ctrl register (SOURCE_AIF_CAR)

13.4.10.1.41.1 Offset

Register	Offset
SOURCE_AIF_CAR	91Ch

13.4.10.1.41.2 Diagram



13.4.10.1.41.3 Fields

Field	Function
31-8 reserved_0	Reserved - ignored on write
7 spdif_mclk_rstn_en	spdif_mclk enable - active low
6 spdif_mclk_en	spdif_mclk enable - active high
5 spdif_cdr_clk_rstn_en	spdif_cdr_clk enable - active low
4 spdif_cdr_clk_en	spdif_cdr_clk enable - active high

Table continues on the next page...

Clocks And Resets

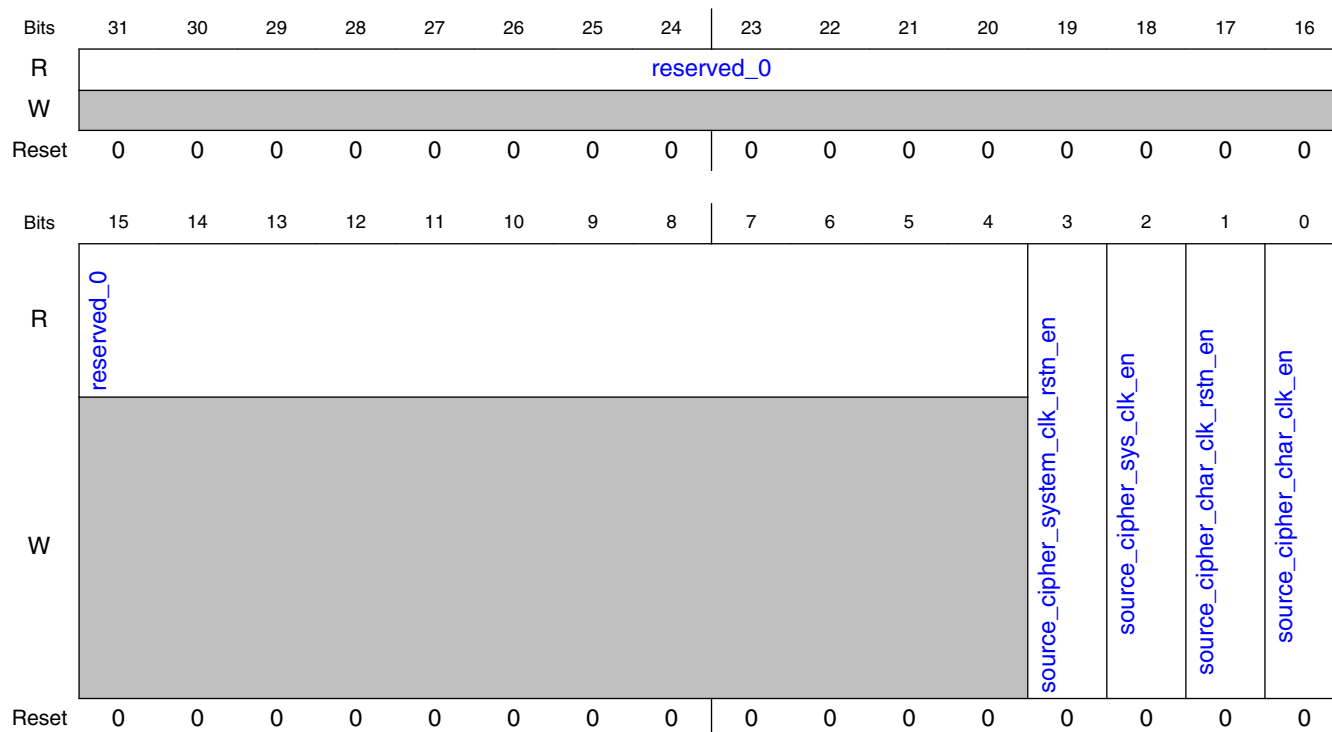
Field	Function
3 source_aif_sys_rstn_en	source_aif_sys_rstn enable - active low
2 source_aif_sys_clk_en	source_aif_sys_clk enable - active high
1 source_aif_pkt_clk_rstn_en	source_aif_pkt_clk_rstn enable - active low
0 source_aif_pkt_clk_en	source_aif_pkt_clk enable - active high

13.4.10.1.42 Cipher clock and reset ctrl register (SOURCE_CIPHER_CAR)

13.4.10.1.42.1 Offset

Register	Offset
SOURCE_CIPHER_CAR	920h

13.4.10.1.42.2 Diagram



13.4.10.1.42.3 Fields

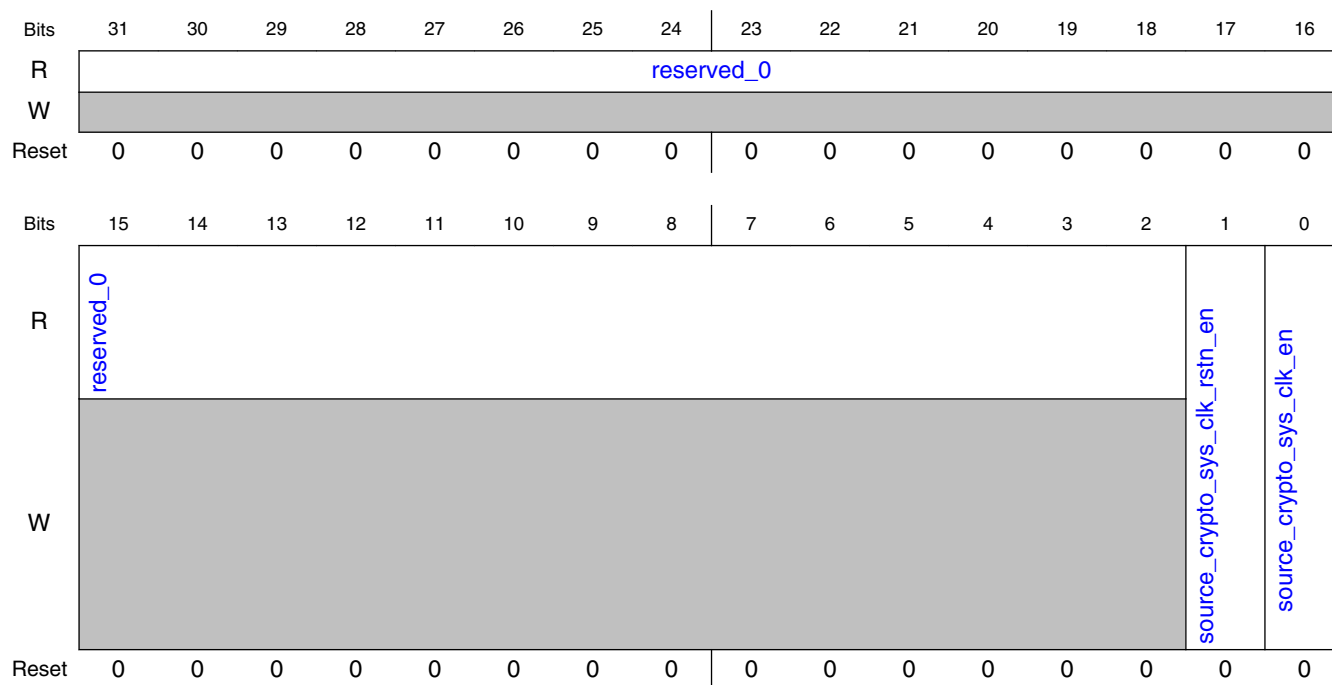
Field	Function
31-4 reserved_0	Reserved - ignored on write
3 source_cipher_system_clk_rstn_en	source_cipher_system_clk_rstn enable - active low (Only when HDCP used)
2 source_cipher_sys_clk_en	source_cipher_sys_clk enable - active high (Only when HDCP used)
1 source_cipher_char_clk_rstn_en	source_cipher_char_clk_rstn enable - active low (Only when HDCP used)
0 source_cipher_char_clk_en	source_cipher_char_clk enable - active high (Only when HDCP used)

13.4.10.1.43 Crypto clock and reset ctrl register (SOURCE_CRYPTO_CAR)

13.4.10.1.43.1 Offset

Register	Offset
SOURCE_CRYPTO_CAR	924h

13.4.10.1.43.2 Diagram



13.4.10.1.43.3 Fields

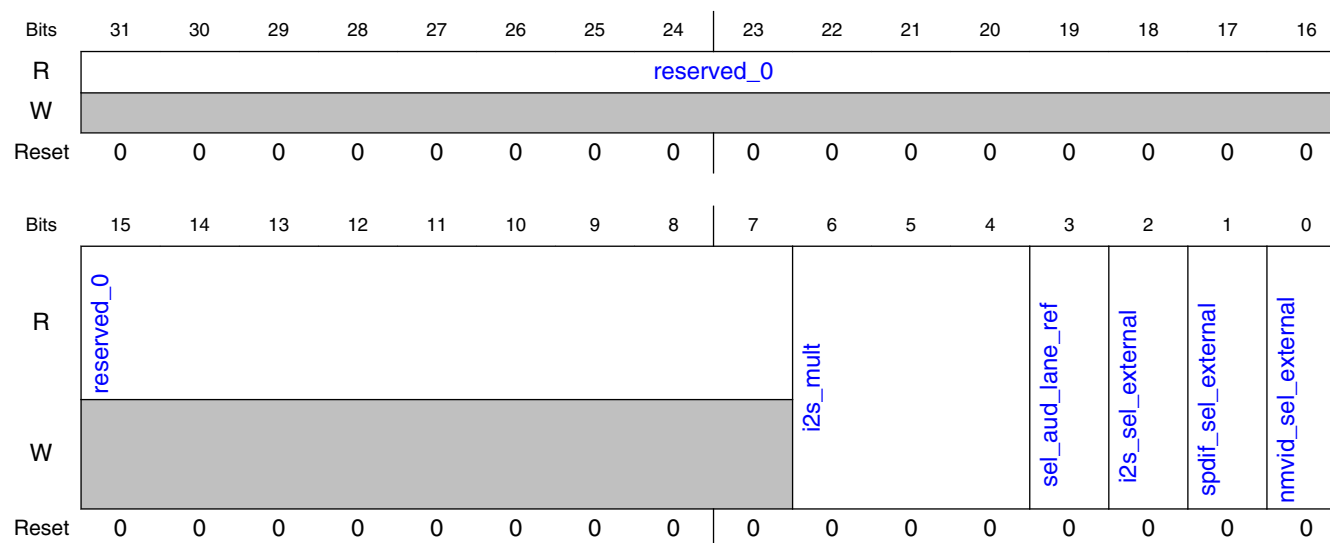
Field	Function
31-2 reserved_0	Reserved - ignored on write
1 source_crypto_sys_clk_rstn_en	source_crypto_sys_clk_rstn enable - active low (Only when HDCP used)
0 source_crypto_sys_clk_en	source_crypto_sys_clk enable - active high (Only when HDCP used)

13.4.10.1.44 Clock Meter control (CM_CTRL)

13.4.10.1.44.1 Offset

Register	Offset
CM_CTRL	A00h

13.4.10.1.44.2 Diagram



13.4.10.1.44.3 Fields

Field	Function
31-7 reserved_0	
6-4 i2s_mult	Select the division of N value for different I2S TDM configuration
3 sel_aud_lane_ref	When 1 Select Audio CLK as a reference (HDMI) -When 0 Select LANE CLK as a reference (DP) -
2 i2s_sel_external	When 1 Select external values of NMAUD (N/A) for I2S
1 spdif_sel_external	When 1 Select external values of NMAUD (N/A) for SPDIF

Table continues on the next page...

Clocks And Resets

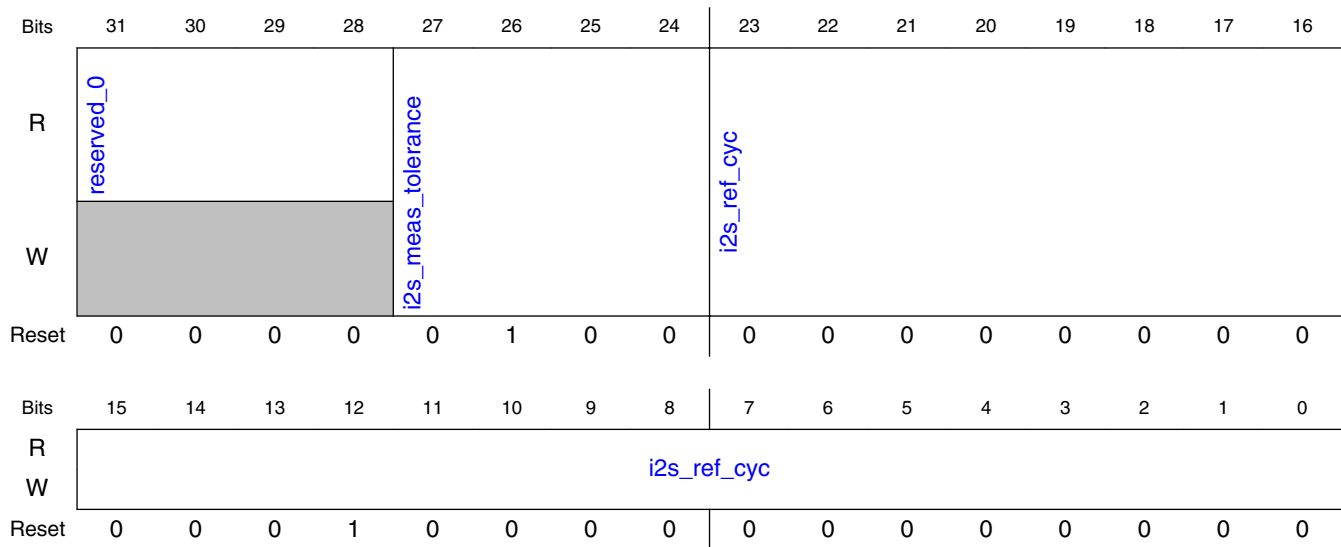
Field	Function
0 nmvid_sel_external	When 1 Select external values of NMVID (N/A)

13.4.10.1.45 I2S clock control (CM_I2S_CTRL)

13.4.10.1.45.1 Offset

Register	Offset
CM_I2S_CTRL	A04h

13.4.10.1.45.2 Diagram



13.4.10.1.45.3 Fields

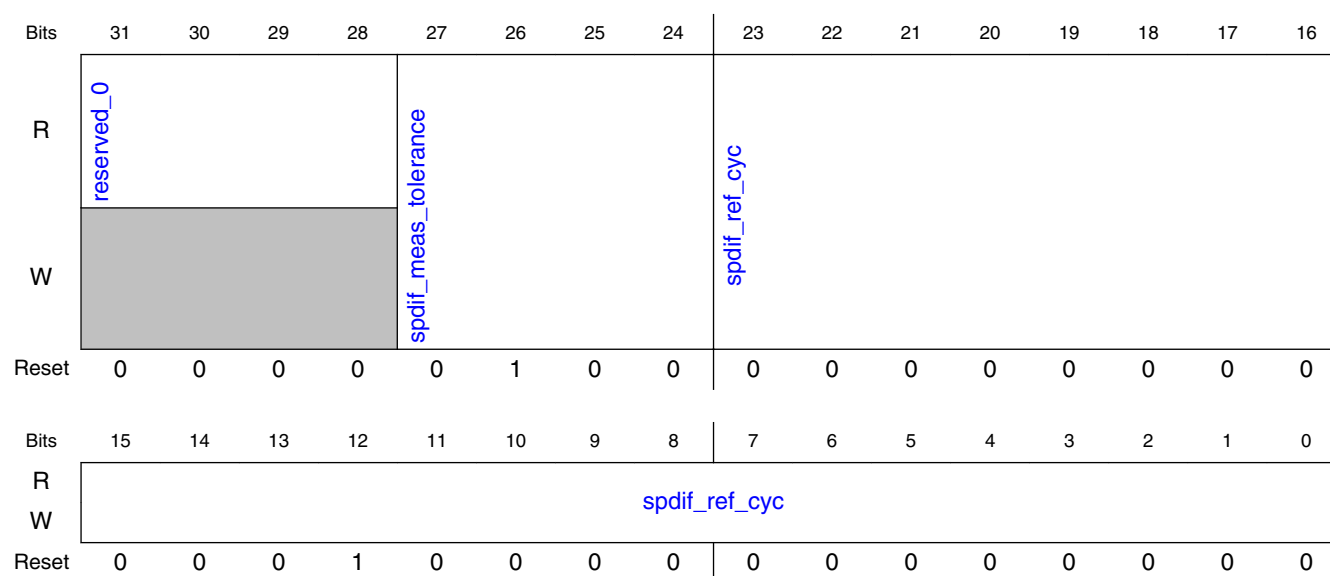
Field	Function
31-28 reserved_0	
27-24 i2s_meas_tolerance	Measurement tolerance of Audio clock to be stable in clocks
23-0 i2s_ref_cyc	Refernce cycles for I2S Audio meter

13.4.10.1.46 SPDIF clock control (CM_SPDIF_CTRL)

13.4.10.1.46.1 Offset

Register	Offset
CM_SPDIF_CTRL	A08h

13.4.10.1.46.2 Diagram



13.4.10.1.46.3 Fields

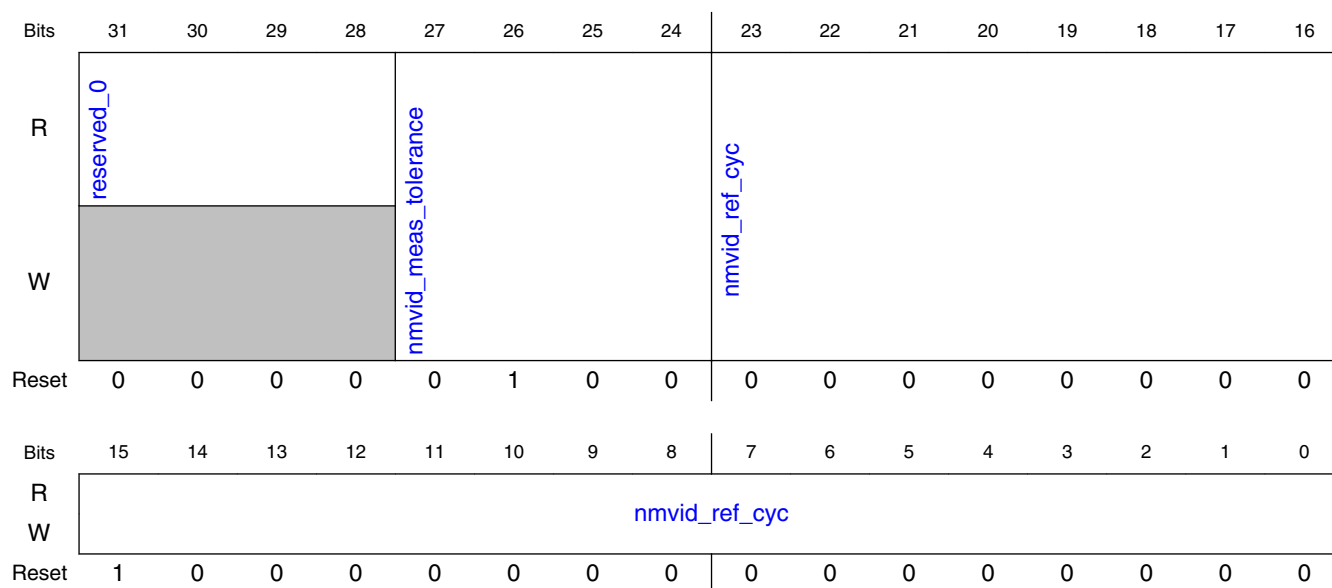
Field	Function
31-28 reserved_0	
27-24 spdif_meas_tolerance	SPDIF measurement tolerance to be stable in clocks
23-0 spdif_ref_cyc	Reference cycles of SPDIF measurement

13.4.10.1.47 Video clock control (CM_VID_CTRL)

13.4.10.1.47.1 Offset

Register	Offset
CM_VID_CTRL	A0Ch

13.4.10.1.47.2 Diagram



13.4.10.1.47.3 Fields

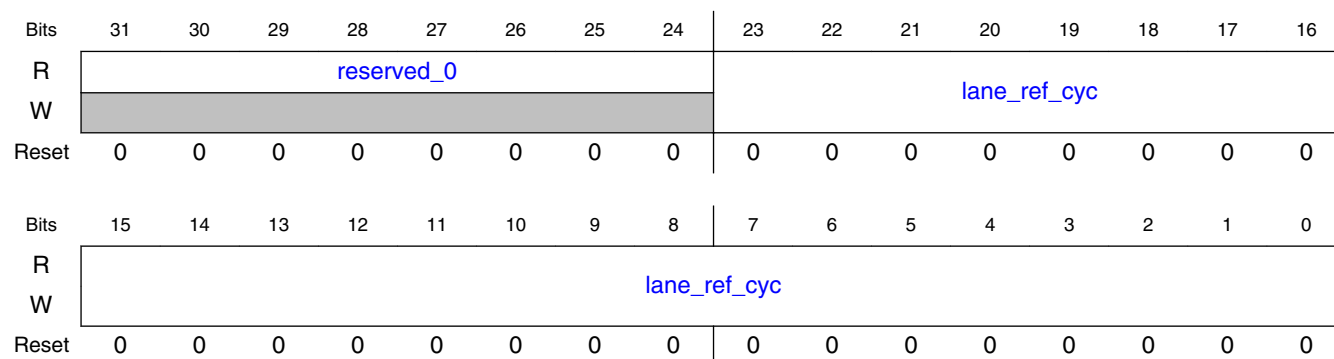
Field	Function
31-28 reserved_0	
27-24 nmvid_meas_tolerance	Video measurment tolerance in pixel clock cycles
23-0 nmvid_ref_cyc	Video refernce cycles

13.4.10.1.48 Lane control (CM_LANE_CTRL)

13.4.10.1.48.1 Offset

Register	Offset
CM_LANE_CTRL	A10h

13.4.10.1.48.2 Diagram



13.4.10.1.48.3 Fields

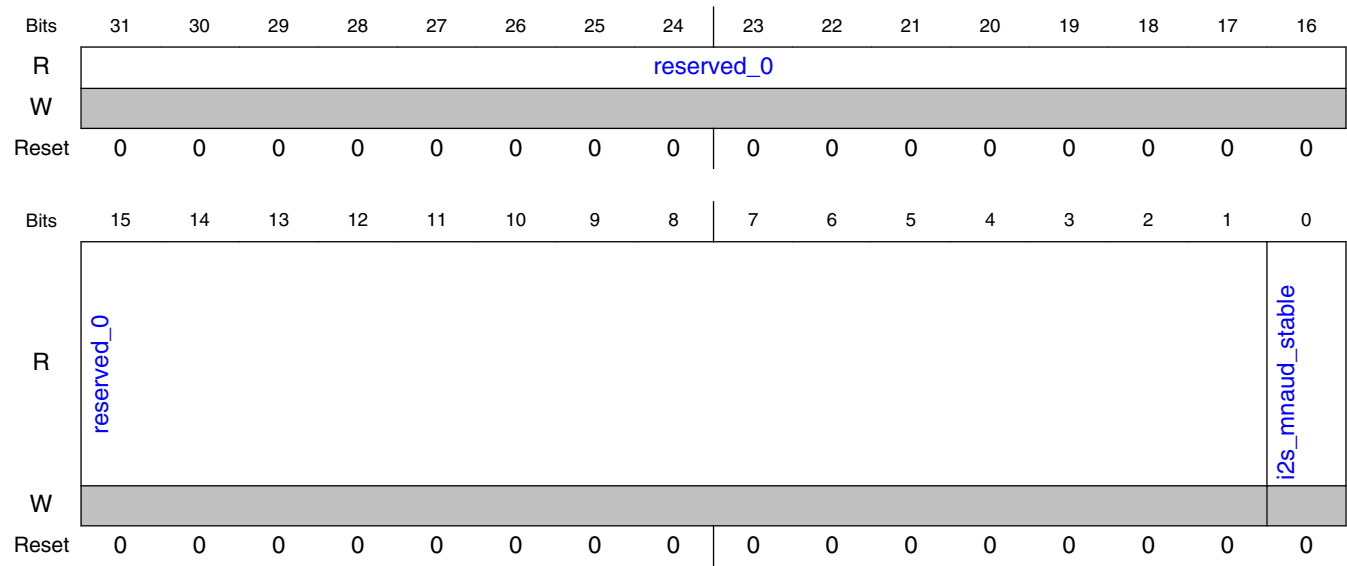
Field	Function
31-24 reserved_0	
23-0 lane_ref_cyc	Reference cycles when using lane clock as reference (DP)

13.4.10.1.49 I2S clock stable, audio clock measured (I2S_NM_STABLE)

13.4.10.1.49.1 Offset

Register	Offset
I2S_NM_STABLE	A14h

13.4.10.1.49.2 Diagram



13.4.10.1.49.3 Fields

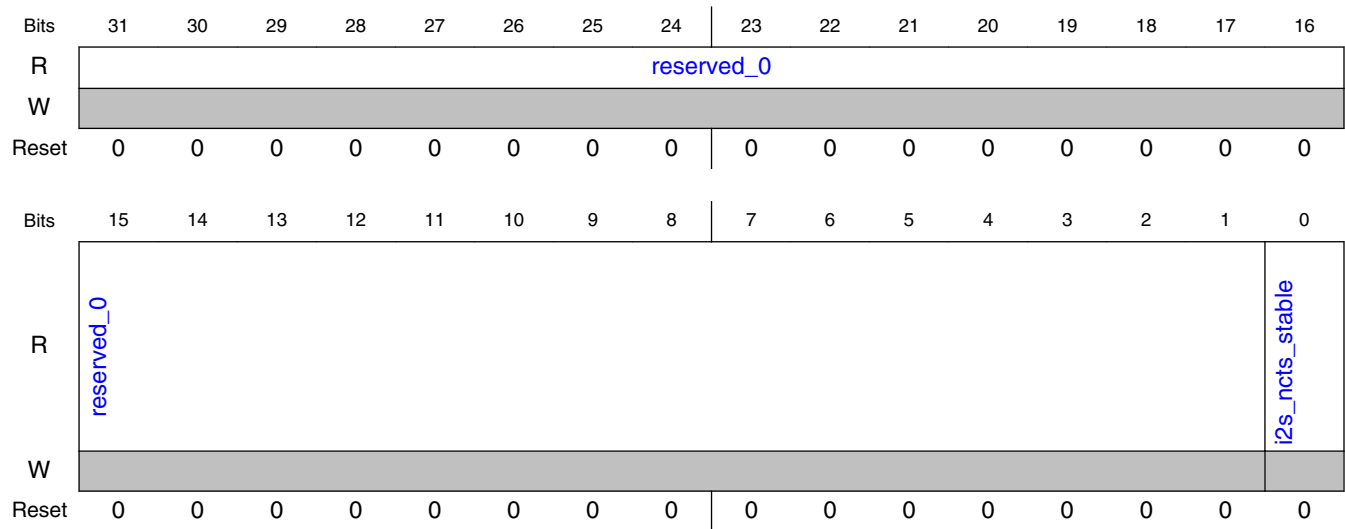
Field	Function
31-1 reserved_0	
0 i2s_mnaud_stable	I2S NMAUD Mesurment stable

13.4.10.1.50 I2S clock stable, lane clock measured (I2S_NCTS_STABLE)

13.4.10.1.50.1 Offset

Register	Offset
I2S_NCTS_STABLE	A18h

13.4.10.1.50.2 Diagram



13.4.10.1.50.3 Fields

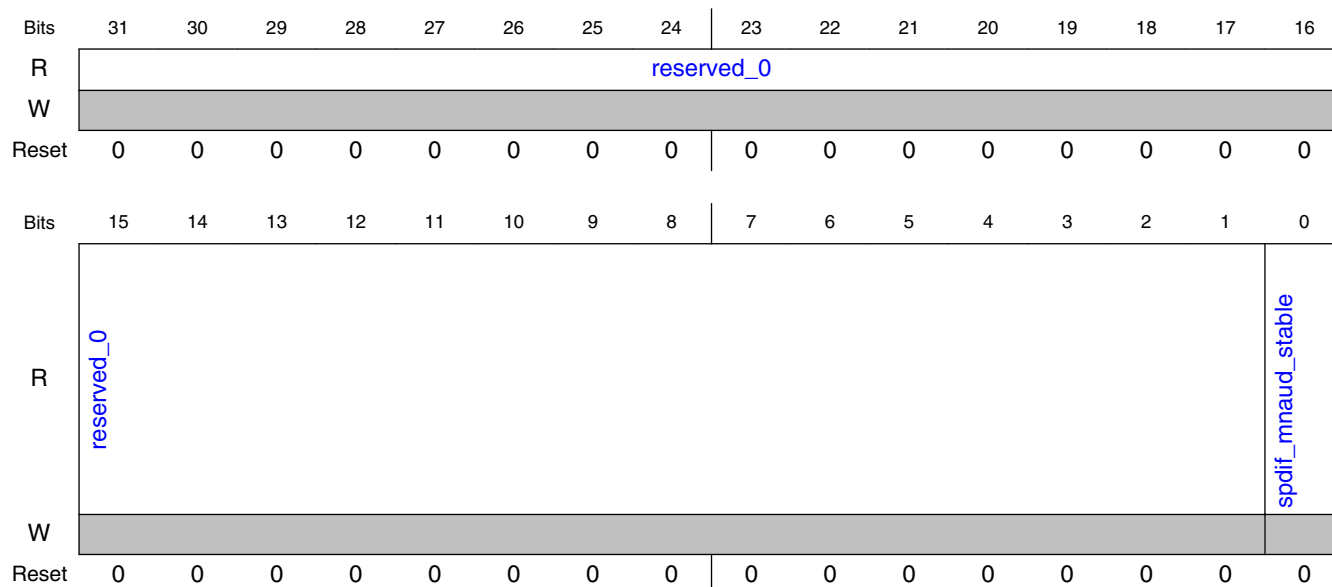
Field	Function
31-1 reserved_0	
0 i2s_ncts_stable	i2s CTS measurment stable

13.4.10.1.51 SPDIF clock stable, audio clock measured (SPDIF_NM_STABLE)

13.4.10.1.51.1 Offset

Register	Offset
SPDIF_NM_STABLE	A1Ch

13.4.10.1.51.2 Diagram



13.4.10.1.51.3 Fields

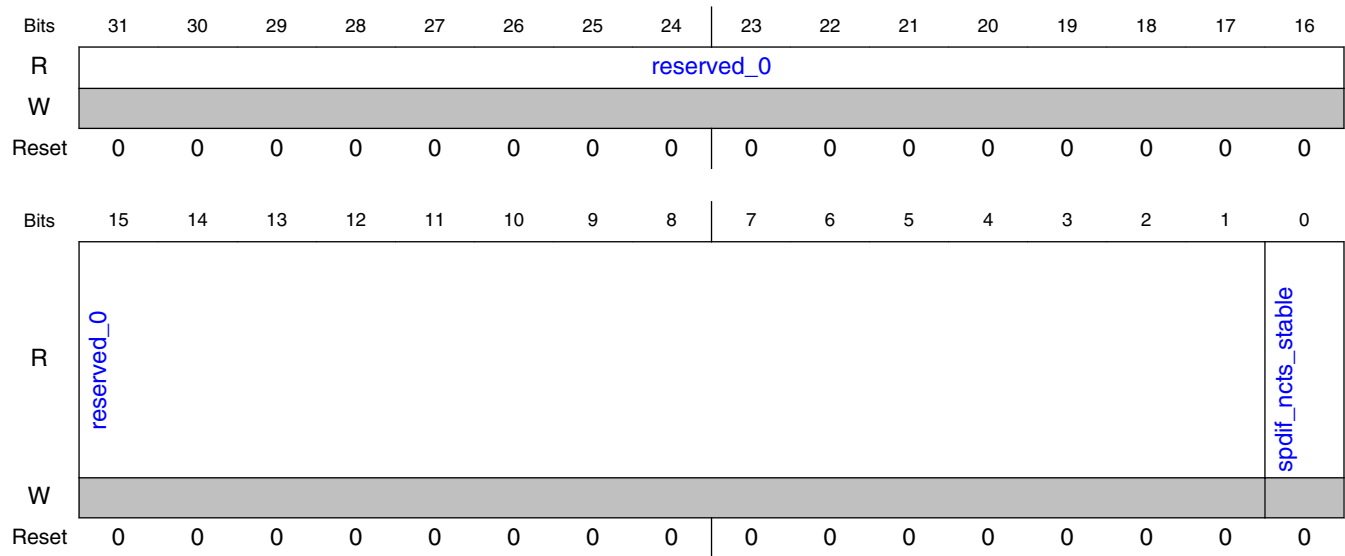
Field	Function
31-1 reserved_0	
0 spdif_mnaud_stable	SPDIF NMAUD measurment stable

13.4.10.1.52 SPDIF clock stable, lane clock measured (SPDIF_NCTS_STABLE)

13.4.10.1.52.1 Offset

Register	Offset
SPDIF_NCTS_STABLE	A20h

13.4.10.1.52.2 Diagram



13.4.10.1.52.3 Fields

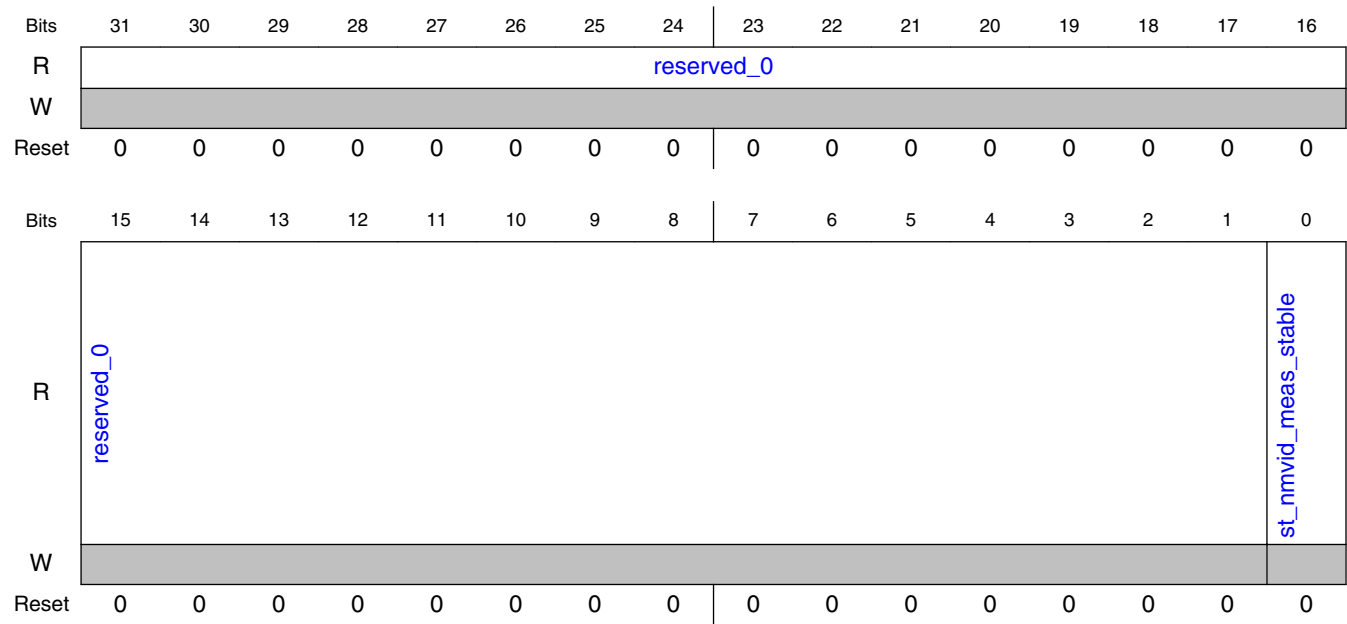
Field	Function
31-1 reserved_0	
0 spdif_ncts_stable	SPDIF CTS measurment stable

13.4.10.1.53 Video clock stable (NMVID_MEAS_STABLE)

13.4.10.1.53.1 Offset

Register	Offset
NMVID_MEAS_STABLE	A24h

13.4.10.1.53.2 Diagram



13.4.10.1.53.3 Fields

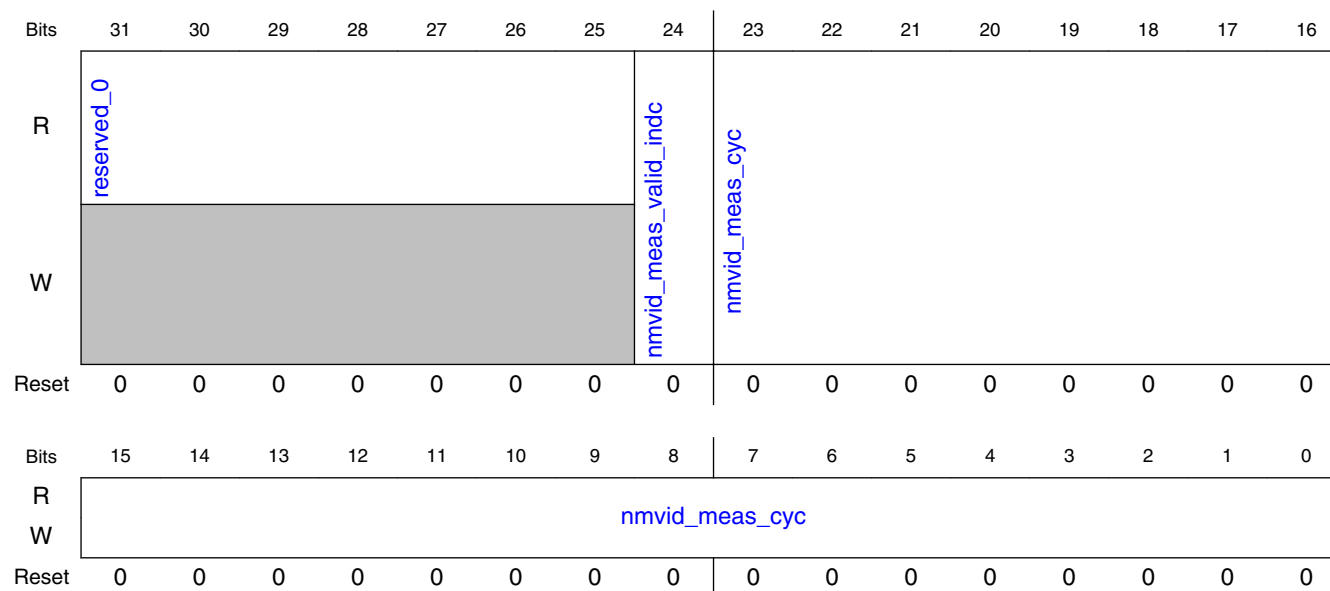
Field	Function
31-1 reserved_0	
0 st_nmvid_meas_stable	Pixel clock NMVID measurment stable

13.4.10.1.54 Video cycles measure (CM_VID_MEAS)

13.4.10.1.54.1 Offset

Register	Offset
CM_VID_MEAS	A28h

13.4.10.1.54.2 Diagram



13.4.10.1.54.3 Fields

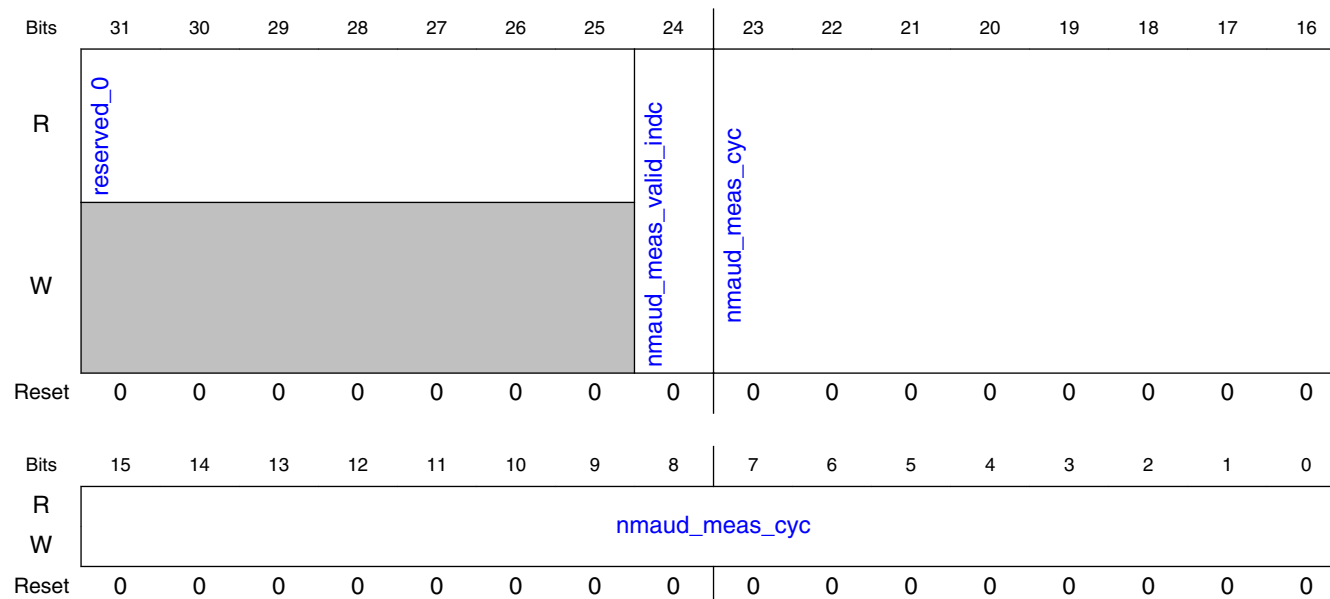
Field	Function
31-25 reserved_0	
24 nmvid_meas_valid_indc	When Toggle Valid pulse is generated to sample MNVID fix value
23-0 nmvid_meas_cyc	Fixed Value for NVID , The MVID is nmvid_ref_cyc[23:0]

13.4.10.1.55 Audio cycles measure (CM_AUD_MEAS)

13.4.10.1.55.1 Offset

Register	Offset
CM_AUD_MEAS	A2Ch

13.4.10.1.55.2 Diagram



13.4.10.1.55.3 Fields

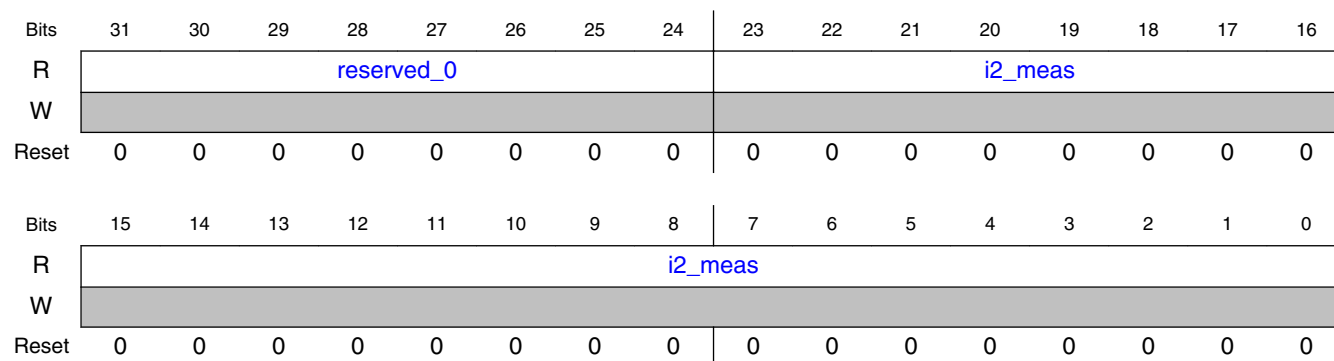
Field	Function
31-25 reserved_0	
24 nmaud_meas_v alid_indc	When Toggle Valid pulse is generated to sample MNAUD fix value
23-0 nmaud_meas_c yc	Fixed Value for NAUD, The MAUD is lane_ref_cyc[23:0]

13.4.10.1.56 I2S clock measurment HDMI (I2S_MEAS)

13.4.10.1.56.1 Offset

Register	Offset
I2S_MEAS	A40h

13.4.10.1.56.2 Diagram



13.4.10.1.56.3 Fields

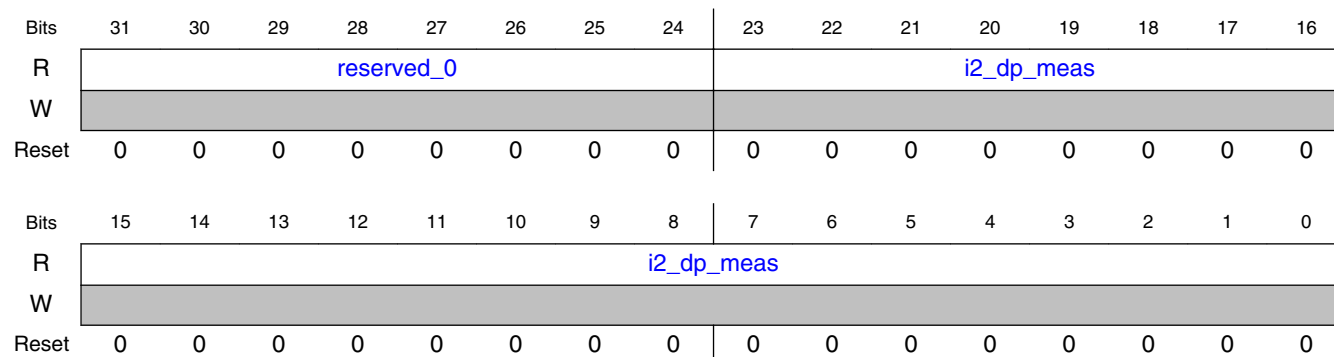
Field	Function
31-24 reserved_0	
23-0 i2_meas	I2S measurment value

13.4.10.1.57 I2S clock measurment DP (I2S_DP_MEAS)

13.4.10.1.57.1 Offset

Register	Offset
I2S_DP_MEAS	A44h

13.4.10.1.57.2 Diagram



13.4.10.1.57.3 Fields

Field	Function
31-24 reserved_0	
23-0 i2_dp_meas	I2S Clock Meter measurment value (in DP)

13.4.10.1.58 SPDIF clock measurment DP (SPDIF_DP_MEAS)

13.4.10.1.58.1 Offset

Register	Offset
SPDIF_DP_MEAS	A80h

13.4.10.1.58.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0								spdif_dp_meas							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	spdif_dp_meas															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.58.3 Fields

Field	Function
31-24 reserved_0	
23-0 spdif_dp_meas	SPDIF measurment value

13.4.10.1.59 SPDIF clock measurment HDMI (SPDIF_MEAS)

13.4.10.1.59.1 Offset

Register	Offset
SPDIF_MEAS	A84h

13.4.10.1.59.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0								spdif_meas							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	spdif_meas															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.59.3 Fields

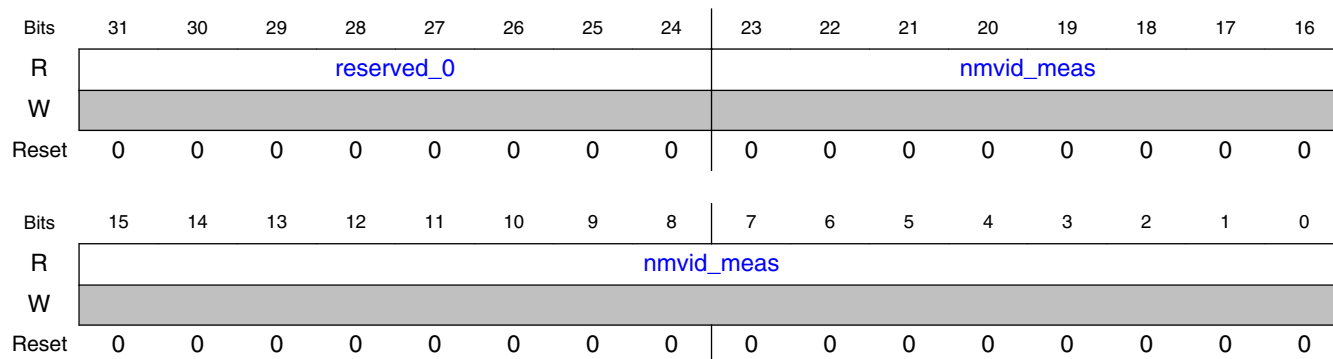
Field	Function
31-24 reserved_0	
23-0 spdif_meas	SPDIF Clock Meter measurment value (in DP)

13.4.10.1.60 Video clock measurment (NMVID_MEAS)

13.4.10.1.60.1 Offset

Register	Offset
NMVID_MEAS	AC0h

13.4.10.1.60.2 Diagram



13.4.10.1.60.3 Fields

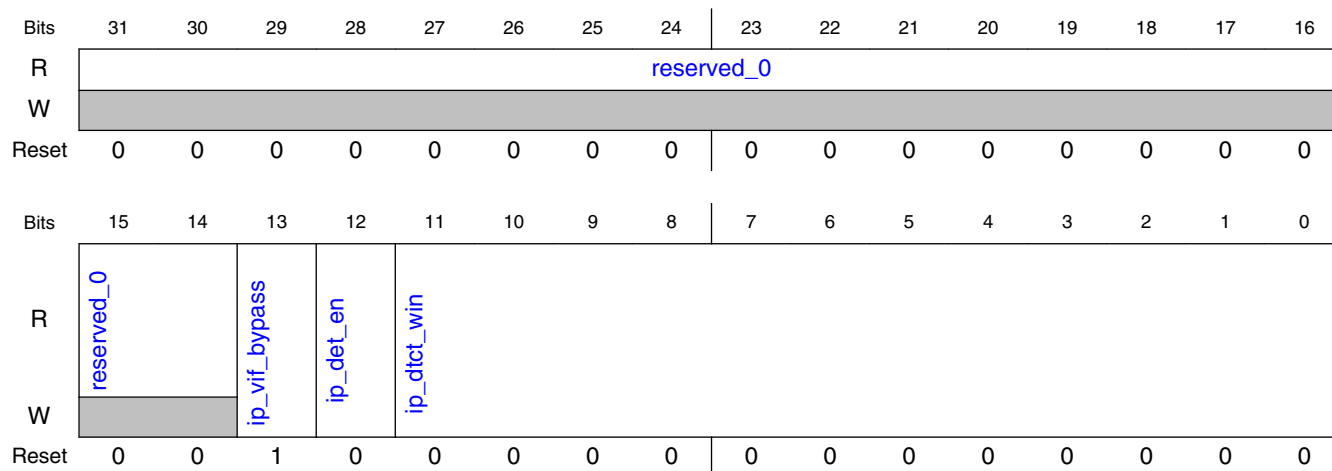
Field	Function
31-24 reserved_0	
23-0 nmvid_meas	Video clock measurment value

13.4.10.1.61 Video Input Interface Setting Register (BND_HSYNC2VSYNC)

13.4.10.1.61.1 Offset

Register	Offset
BND_HSYNC2VSYNC	B00h

13.4.10.1.61.2 Diagram



13.4.10.1.61.3 Fields

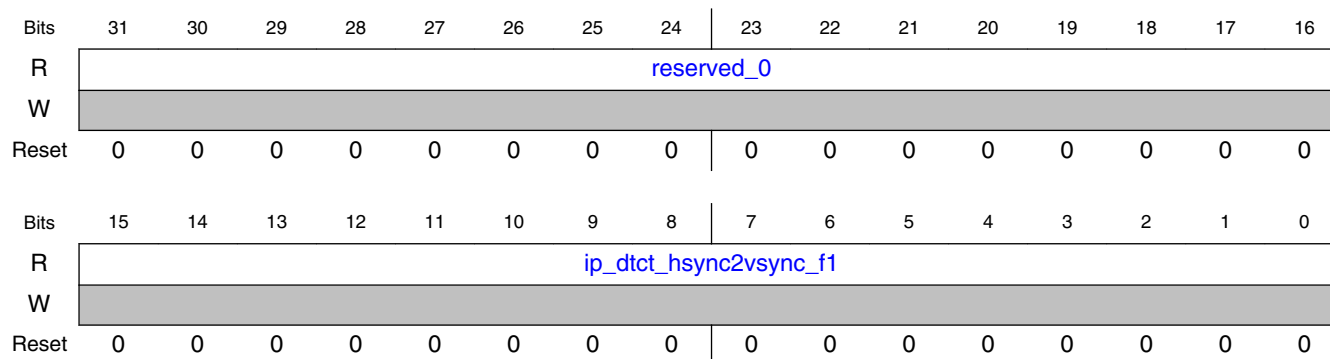
Field	Function
31-14 reserved_0	Reserved ignored on write. Reserved ignored on write.
13 ip_vif_bypass	Bypass video interface. Bypass video interface.
12 ip_det_en	Enable detection of Interlace formats after decided if the polarity is Automatic or Manual detection. Enable detection of Interlace formats after decided if the polarity is Automatic or Manual detection.
11-0 ip_dtct_win	Bound for HSYNC to VSYNC for all fields. Bound for HSYNC to VSYNC for all fields.

13.4.10.1.62 Status of HSYNC to VSYNC Distance Counter 1 (HSYNC2VSYNC_F1_L1)

13.4.10.1.62.1 Offset

Register	Offset
HSYNC2VSYNC_F1_L1	B04h

13.4.10.1.62.2 Diagram



13.4.10.1.62.3 Fields

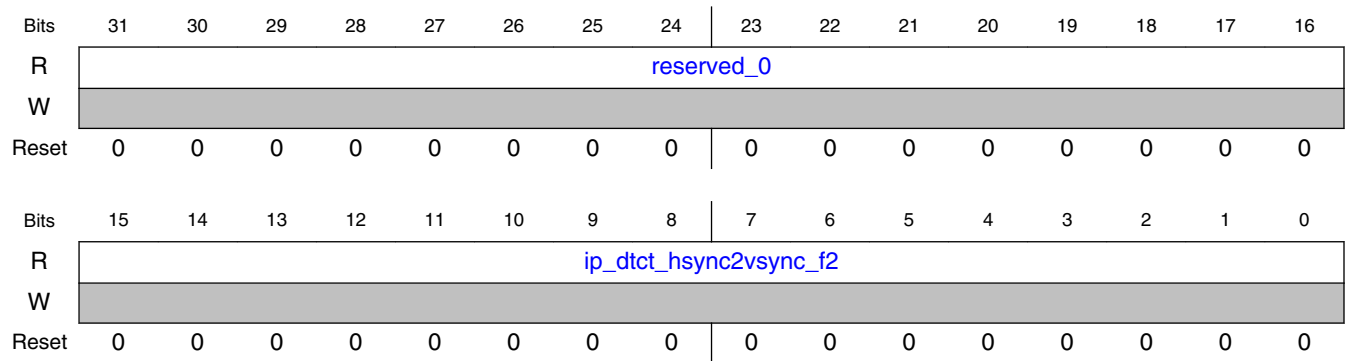
Field	Function
31-16 reserved_0	Reserved
15-0 ip_dtct_hsync2v sync_f1	Value of HSYNC to VSYNC field 1. Value of HSYNC to VSYNC field 1.

13.4.10.1.63 Status of HSYNC to VSYNC Distance Counter 2 (HSYNC2VSYNC_F2_L1)

13.4.10.1.63.1 Offset

Register	Offset
HSYNC2VSYNC_F2_L1	B08h

13.4.10.1.63.2 Diagram



13.4.10.1.63.3 Fields

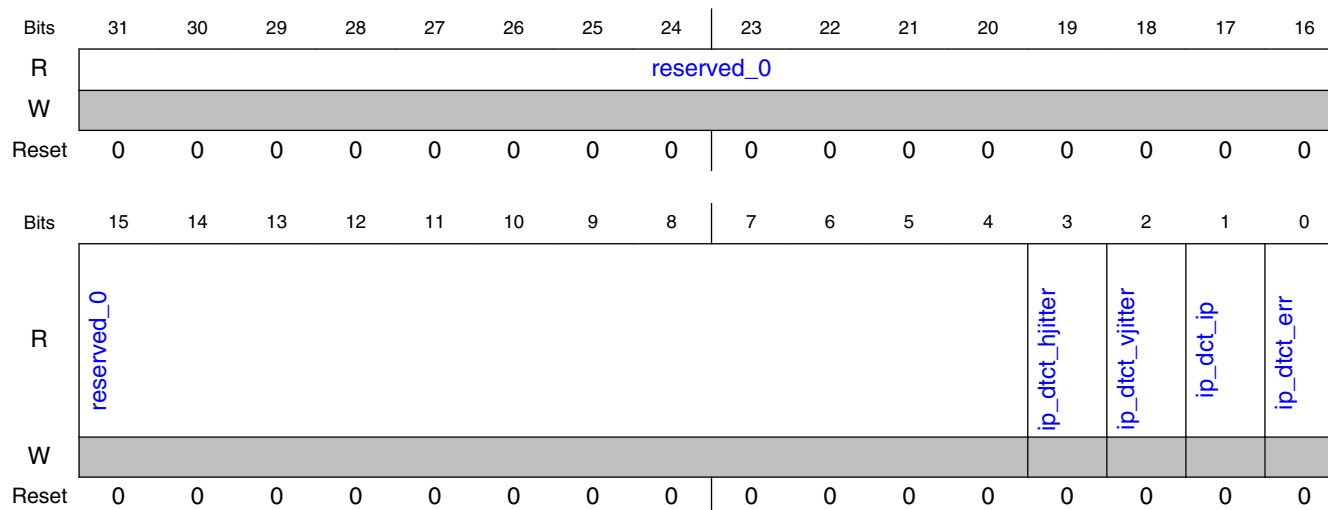
Field	Function
31-16 reserved_0	Reserved
15-0 ip_dtct_hsync2v sync_f2	Value of HSYNC to VSYNC field 2. Value of HSYNC to VSYNC field 2.

13.4.10.1.64 Video Interface Status Register (HSYNC2VSYNC_STATUS)

13.4.10.1.64.1 Offset

Register	Offset
HSYNC2VSYNC_STAT US	B0Ch

13.4.10.1.64.2 Diagram



13.4.10.1.64.3 Fields

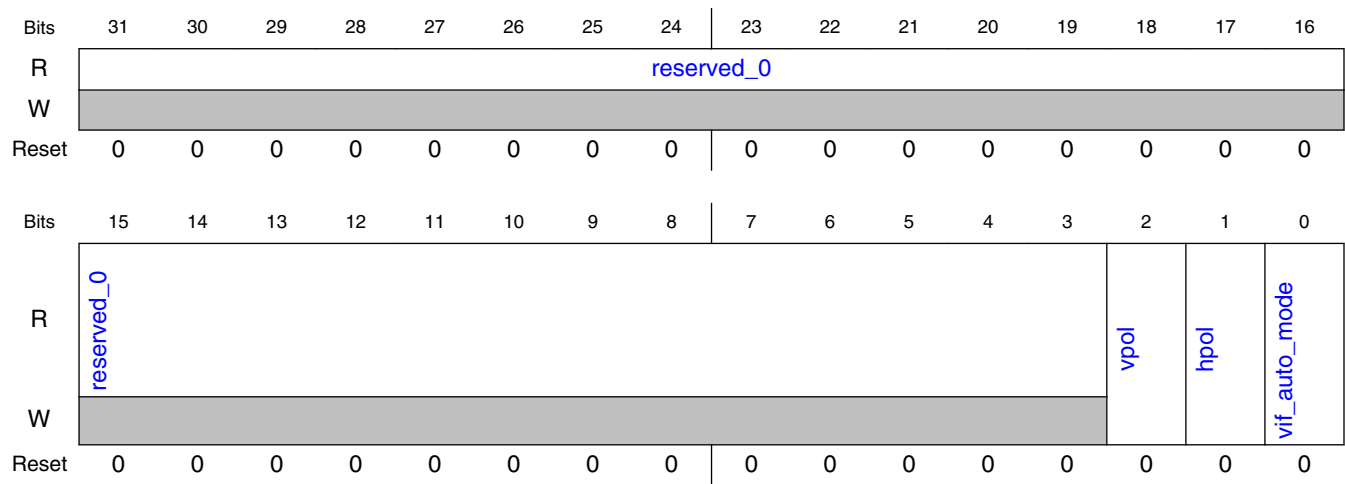
Field	Function
31-4 reserved_0	Reserved
3 ip_dtct_hjitter	Asserted when jitter is observed on httotal i. Asserted when jitter is observed on httotal i.e. HSYNC rising edge to next HSYNC rising edge delay count. Clear on Read.
2 ip_dtct_vjitter	Asserted when jitter is observed on vttotal i. Asserted when jitter is observed on vttotal i.e. VSYNC rising edge to next VSYNC rising edge delay count. Clear on Read.
1 ip_dct_ip	When asserted interlaced format is detected else progressive format. When asserted interlaced format is detected else progressive format.
0 ip_dtct_err	Asserted when HSYNC to VSYNC bound is violated. Asserted when HSYNC to VSYNC bound is violated. Clear on Read.

13.4.10.1.65 Setting Polarity of HSYNC and VSYNC (HSYNC2VSYNC_POL_CTRL)

13.4.10.1.65.1 Offset

Register	Offset
HSYNC2VSYNC_POL_CTRL	B10h

13.4.10.1.65.2 Diagram



13.4.10.1.65.3 Fields

Field	Function
31-3 reserved_0	Reserved ignored on write. Reserved ignored on write.
2 vpol	VSYNC polarity: 0-active HIGH 1-active LOW. VSYNC polarity: 0-active HIGH 1-active LOW.
1 hpol	HSYNC polarity: 0-active HIGH 1-active LOW. HSYNC polarity: 0-active HIGH 1-active LOW.
0 vif_auto_mode	Automatic or Manual configuration of the polarity: 0-vpol and hpol settings are used 1-automatic detection of polarity of input VSYNC and HSYNC

13.4.10.1.66 Video HSize configuration (SCHEDULER_H_SIZE)

13.4.10.1.66.1 Offset

Register	Offset
SCHEDULER_H_SIZE	1000h

13.4.10.1.66.2 Diagram



13.4.10.1.66.3 Fields

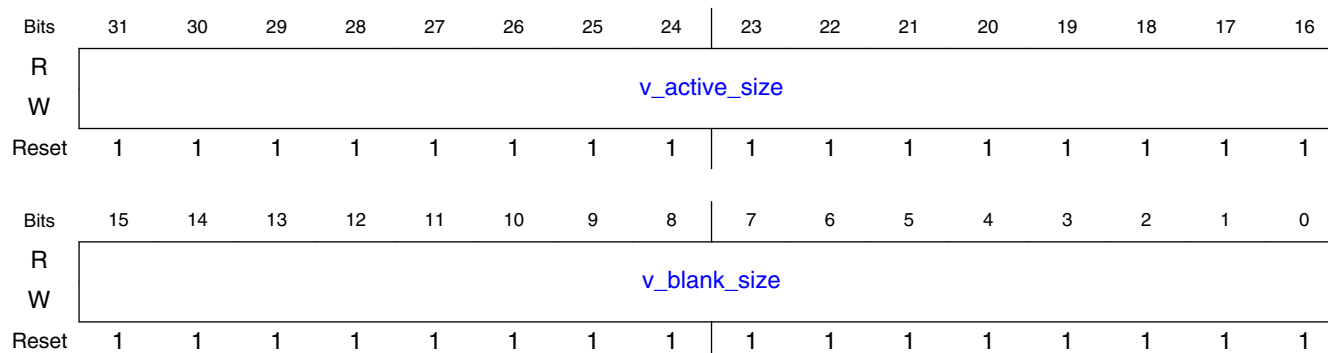
Field	Function
31-16 h_active_size	H-Active Size of video
15-0 h_blank_size	H-Blank Size of video

13.4.10.1.67 Video VSize configuration (SCHEDULER_V_SIZE)

13.4.10.1.67.1 Offset

Register	Offset
SCHEDULER_V_SIZE	1004h

13.4.10.1.67.2 Diagram



13.4.10.1.67.3 Fields

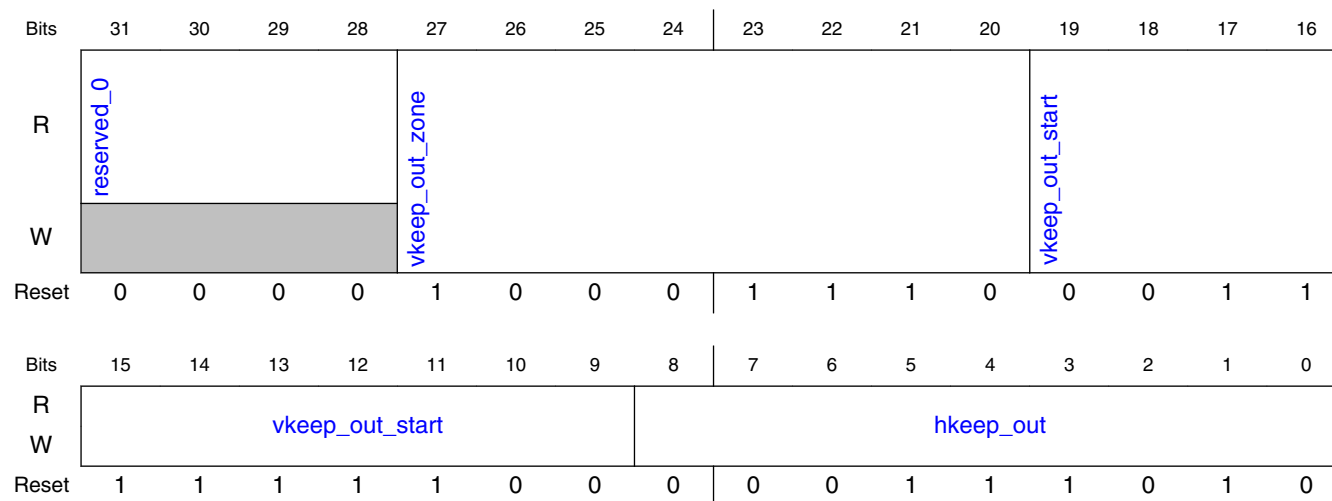
Field	Function
31-16 v_active_size	V-Active Size of video
15-0 v_blank_size	V-Blank Size of video

13.4.10.1.68 Video Blank (Keep Out) Size configuration (SCHEDULER_KEEP_OUT)

13.4.10.1.68.1 Offset

Register	Offset
SCHEDULER_KEEP_OUT	1008h

13.4.10.1.68.2 Diagram



13.4.10.1.68.3 Fields

Field	Function
31-28	

Table continues on the next page...

Clocks And Resets

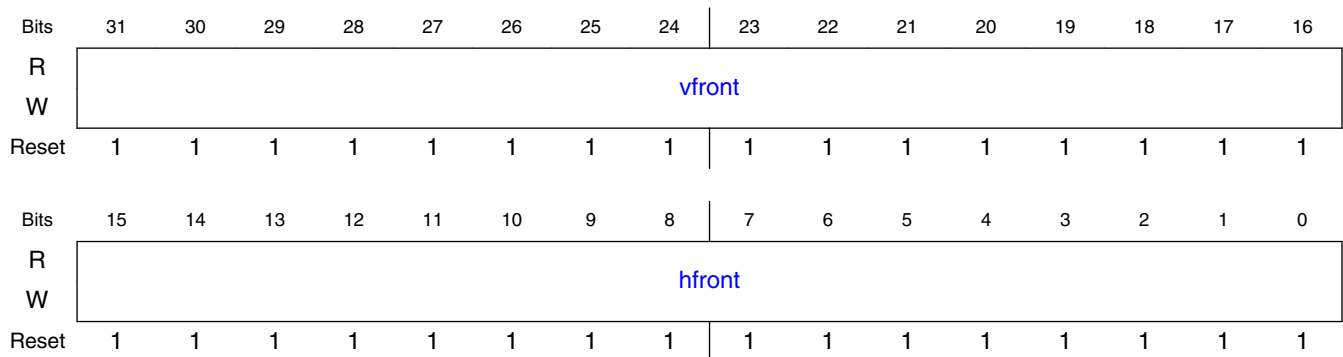
Field	Function
reserved_0	
27-20 vkeep_out_zone	V Blank Keep out zone
19-9 vkeep_out_start	V Blank Keep out buffer from rise of VSYNC
8-0 hkeep_out	H Blank Keep out size

13.4.10.1.69 Video Blank (Front) Size configuration (HDTX_SIGNAL_FRONT_WIDTH)

13.4.10.1.69.1 Offset

Register	Offset
HDTX_SIGNAL_FRONT_WIDTH	100Ch

13.4.10.1.69.2 Diagram



13.4.10.1.69.3 Fields

Field	Function
31-16 vfront	V Front Size. V Front Size.
15-0 hfront	H Front Size. H Front Size.

13.4.10.1.70 Video Sync Size configuration (HDTX_SIGNAL_SYNC_WIDTH)

13.4.10.1.70.1 Offset

Register	Offset
HDTX_SIGNAL_SYNC_WIDTH	1010h

13.4.10.1.70.2 Diagram



13.4.10.1.70.3 Fields

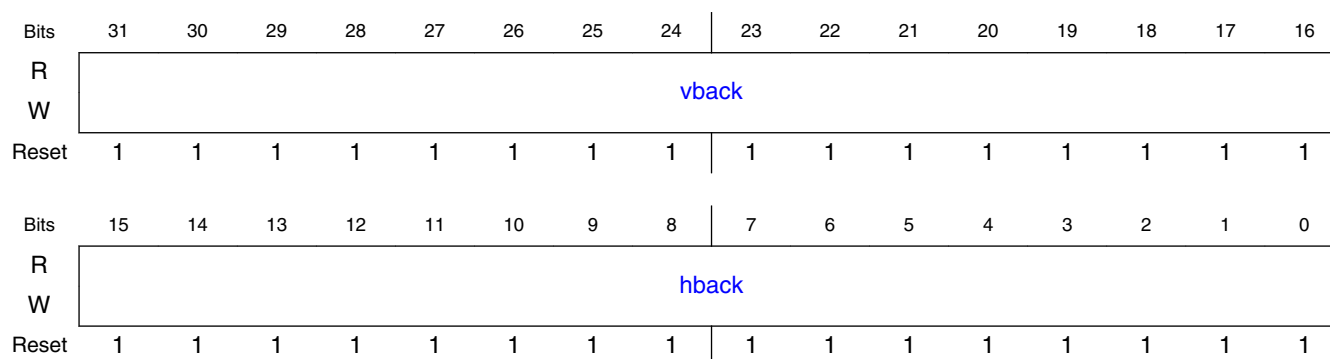
Field	Function
31-16	V Sync Size.
vsync	V Sync Size.
15-0	H Sync Size.
hsync	H Sync Size.

13.4.10.1.71 Video Back Size configuration (HDTX_SIGNAL_BACK_WIDTH)

13.4.10.1.71.1 Offset

Register	Offset
HDTX_SIGNAL_BACK_WIDTH	1014h

13.4.10.1.71.2 Diagram



13.4.10.1.71.3 Fields

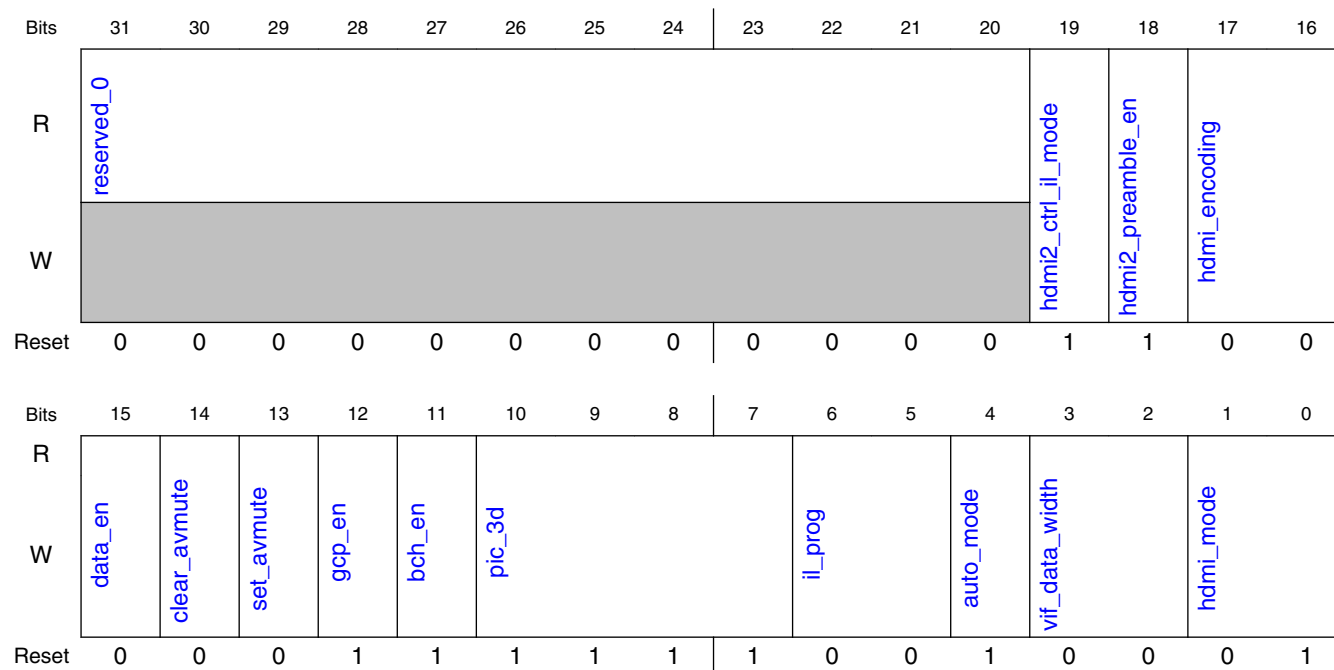
Field	Function
31-16	V Back Size.
vback	V Back Size.
15-0	H Back Size.
hback	H Back Size.

13.4.10.1.72 Video Mode configuration (HDTX_CONTROLLER)

13.4.10.1.72.1 Offset

Register	Offset
HDTX_CONTROLLER	1018h

13.4.10.1.72.2 Diagram



13.4.10.1.72.3 Fields

Field	Function
31-20 reserved_0	
19 hdmi2_ctrl_il_mode	HDMI2 Interlace mode: 1 - SSCP is generated after every vsync in frame 0 - SSCP is generated after first vsync of frame only
18 hdmi2_preamble_en	HDMI2 preamble enabled:
17-16 hdmi_encoding	HDMI pixel encoding
15 data_en	Data Transmission Enable:
14 clear_avmute	Clear_AVMute Value
13 set_avmute	Set_AVMute Value
12 gcp_en	Transmit GCP Enable:
11	BCH Error Correction Algorithm is enabled:

Table continues on the next page...

Clocks And Resets

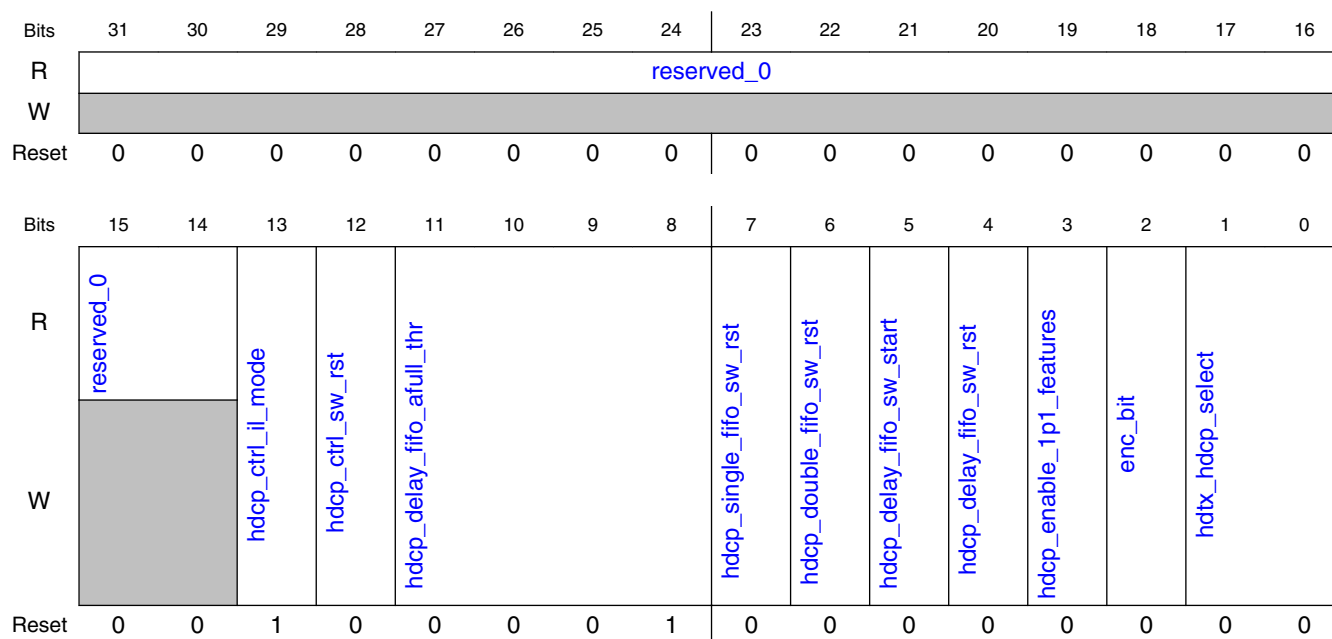
Field	Function
bch_en	
10-7 pic_3d	Picture in 3d mode:
6-5 il_prog	Interlaced or progressive video mode:
4 auto_mode	Frame sizing auto mode:
3-2 vif_data_width	Video I/F Data width - color depth feature:
1-0 hdmi_mode	HDMI mode operation

13.4.10.1.73 HDMi HDCP configuration (HDTX_HDCP)

13.4.10.1.73.1 Offset

Register	Offset
HDTX_HDCP	101Ch

13.4.10.1.73.2 Diagram



13.4.10.1.73.3 Fields

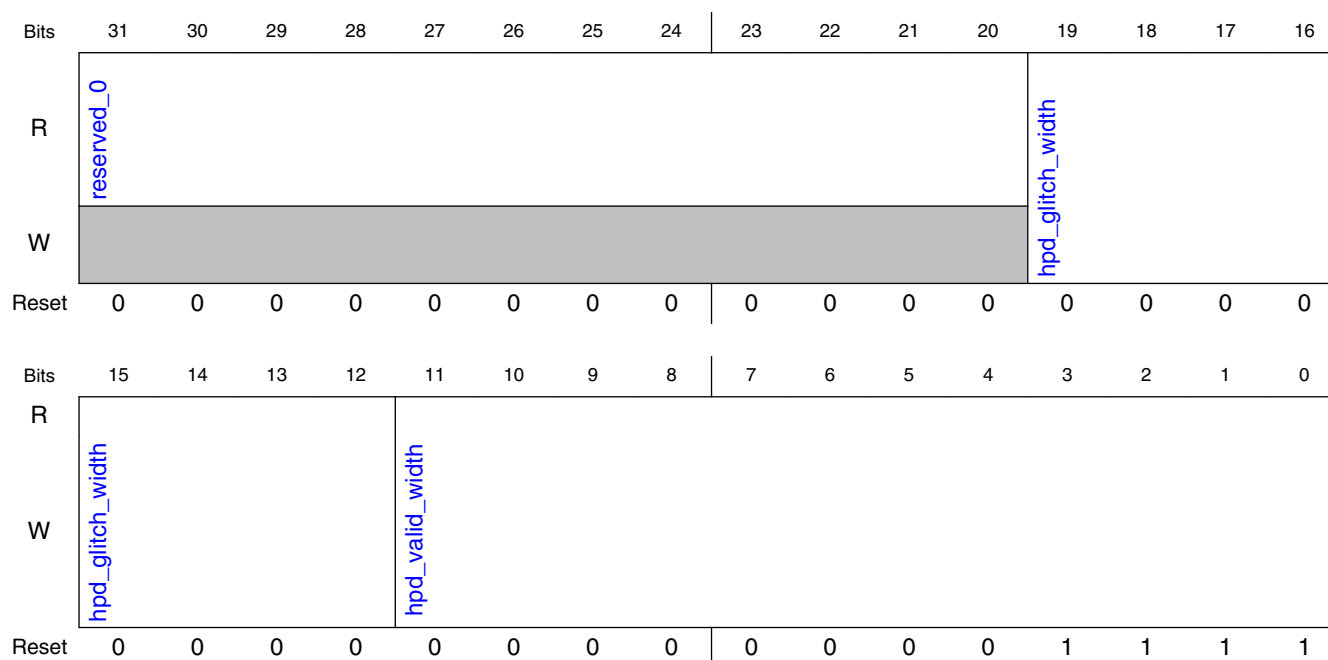
Field	Function
31-14 reserved_0	
13 hdcp_ctrl_il_mode	HDCP Interlace mode: 1 - encoding and vertical keep out on all vsync's in frame 0 - encoding and vertical keep out on first vsync of frame only
12 hdcp_ctrl_sw_reset	HDCP Control path SW reset
11-8 hdcp_delay_fifo_alfull_thr	hdcp delay fifo full threshold
7 hdcp_single_fifo_sw_rst	hdcp single fifo sw reset
6 hdcp_double_fifo_sw_rst	hdcp double fifo sw reset
5 hdcp_delay_fifo_sw_start	hdcp delay fifo sw start
4 hdcp_delay_fifo_sw_rst	hdcp delay fifo sw reset
3 hdcp_enable_1p1_features	hdcp enabling 1p1
2 enc_bit	hdcp encoding bit
1-0 hdtx_hdcp_select	HDCP Select

13.4.10.1.74 HDMI HPD configuration (HDTX_HPDP)

13.4.10.1.74.1 Offset

Register	Offset
HDTX_HPDP	1020h

13.4.10.1.74.2 Diagram



13.4.10.1.74.3 Fields

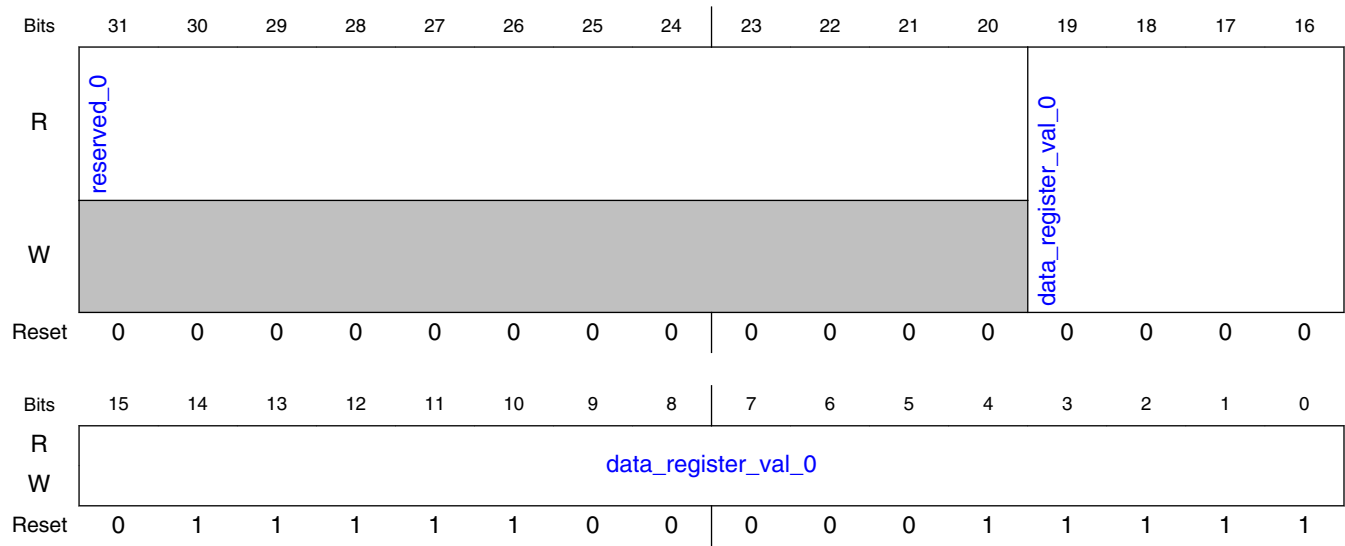
Field	Function
31-20 reserved_0	
19-12 hpd_glitch_width	hdp glitch valid
11-0 hpd_valid_width	hdp valid valid

13.4.10.1.75 HDMI CLOCK configuration - REG0 (HDTX_CLOCK_REG_0)

13.4.10.1.75.1 Offset

Register	Offset
HDTX_CLOCK_REG_0	1024h

13.4.10.1.75.2 Diagram



13.4.10.1.75.3 Fields

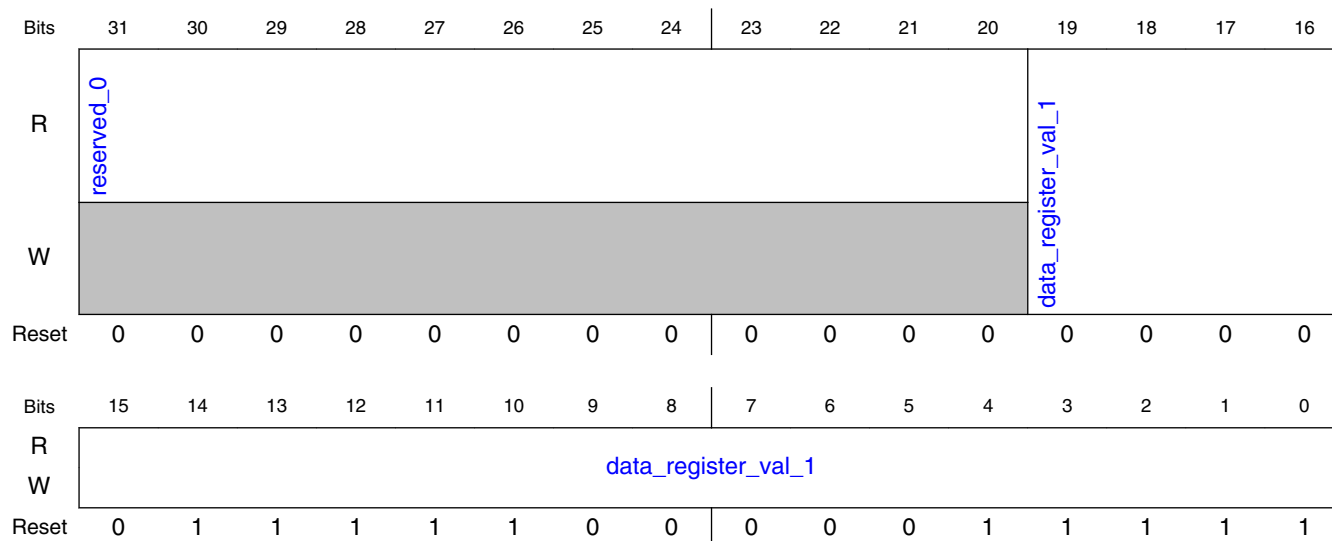
Field	Function
31-20 reserved_0	
19-0 data_register_val_0	data register 0 - 10 LSB is used for character clock

13.4.10.1.76 HDMI CLOCK configuration - REG1 (HDTX_CLOCK_REG_1)

13.4.10.1.76.1 Offset

Register	Offset
HDTX_CLOCK_REG_1	1028h

13.4.10.1.76.2 Diagram



13.4.10.1.76.3 Fields

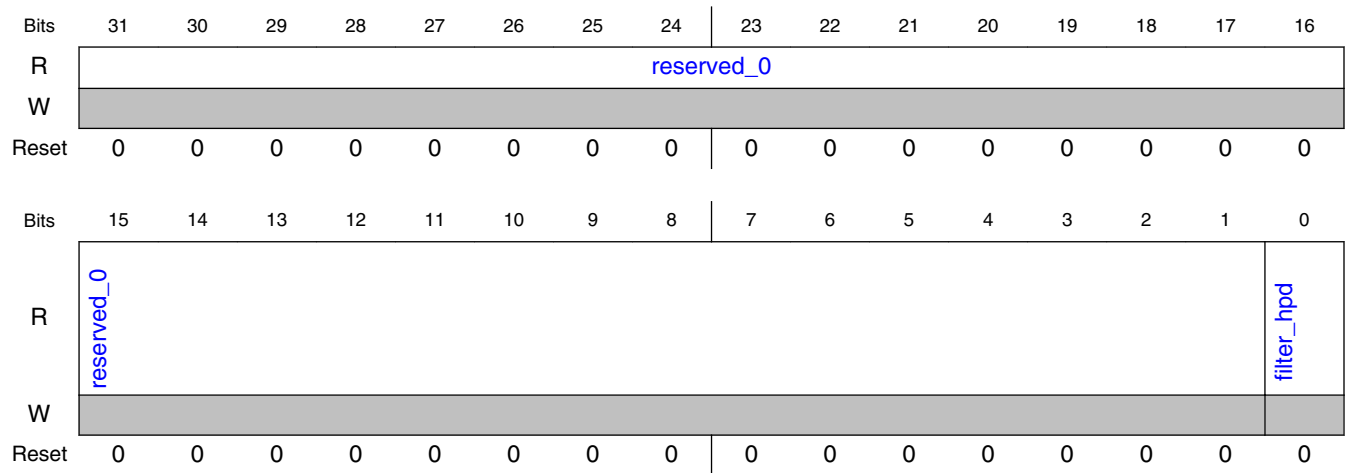
Field	Function
31-20 reserved_0	
19-0 data_register_val_1	data register 1

13.4.10.1.77 HDMI HPD status (HPD_PLUG_IN)

13.4.10.1.77.1 Offset

Register	Offset
HPD_PLUG_IN	102Ch

13.4.10.1.77.2 Diagram



13.4.10.1.77.3 Fields

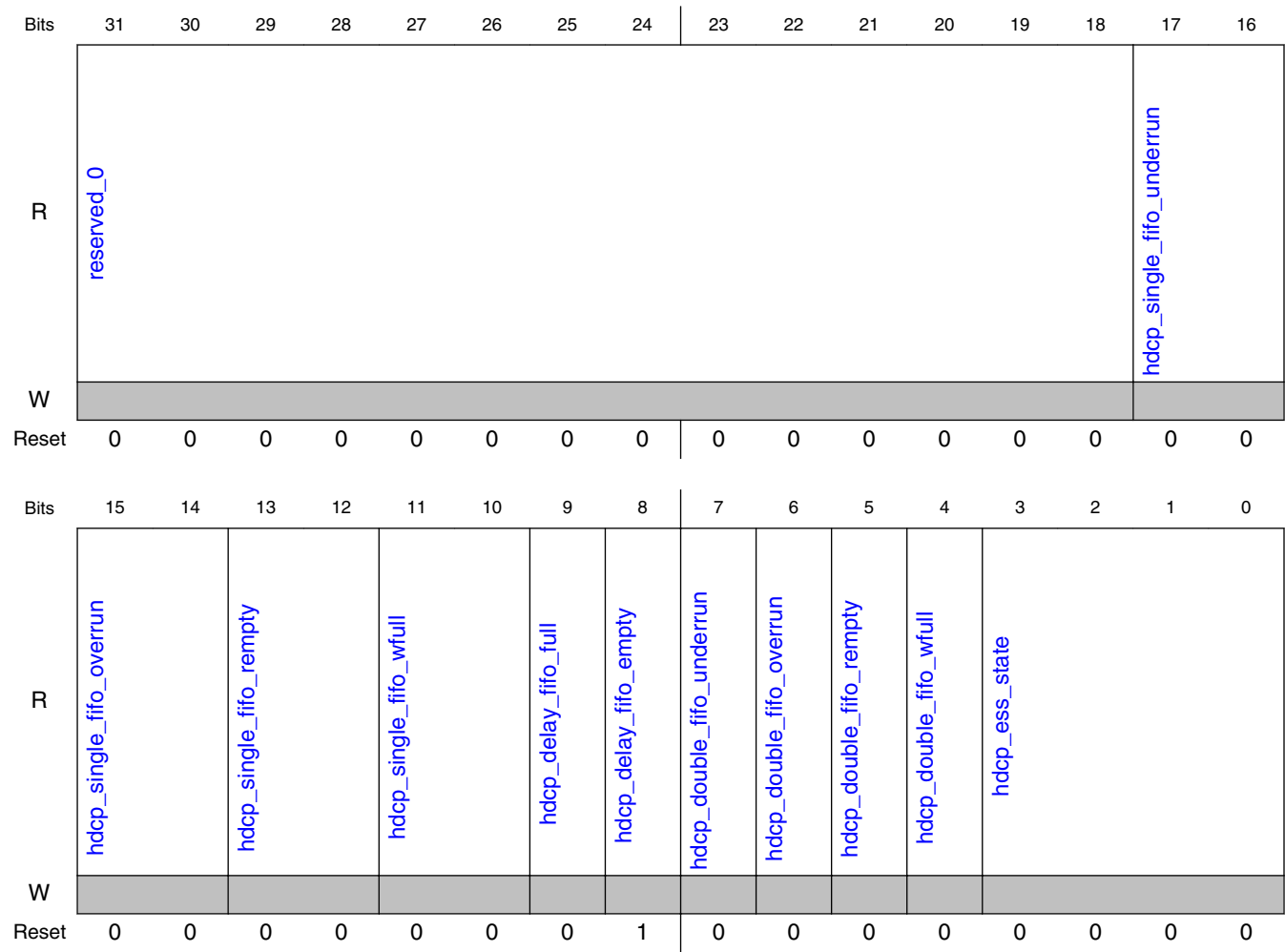
Field	Function
31-1 reserved_0	
0 filter_hpd	Filter HPD input

13.4.10.1.78 HDMI status (HDCP_IN)

13.4.10.1.78.1 Offset

Register	Offset
HDCP_IN	1030h

13.4.10.1.78.2 Diagram



13.4.10.1.78.3 Fields

Field	Function
31-18 reserved_0	
17-16 hdc_p_single_fifo_underrun	
15-14 hdc_p_single_fifo_overnun	
13-12 hdc_p_single_fifo_rempy	

Table continues on the next page...

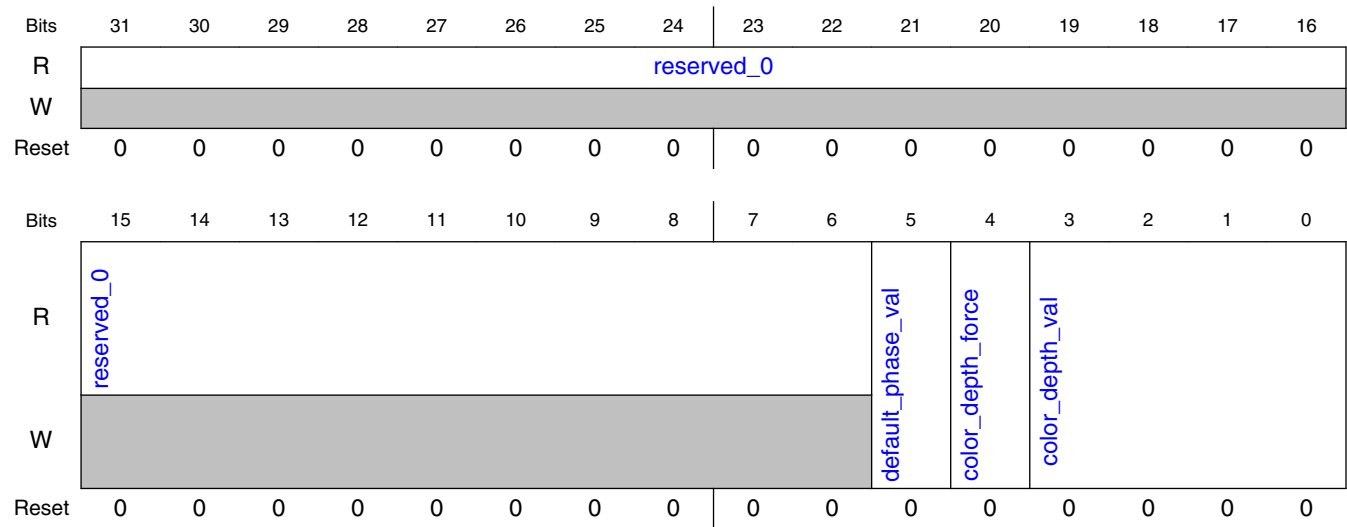
Field	Function
11-10 hdcp_single_fifo_wfull	
9 hdcp_delay_fifo_full	
8 hdcp_delay_fifo_empty	
7 hdcp_double_fifo_underrun	
6 hdcp_double_fifo_overrun	
5 hdcp_double_fifo_empty	
4 hdcp_double_fifo_wfull	
3-0 hdcp_ess_state	ESS State Machine

13.4.10.1.79 HDMI GCP coding configuration (GCP_FORCE_COLOR_DEPTH_CODING)

13.4.10.1.79.1 Offset

Register	Offset
GCP_FORCE_COLOR_DEPTH_CODING	1034h

13.4.10.1.79.2 Diagram



13.4.10.1.79.3 Fields

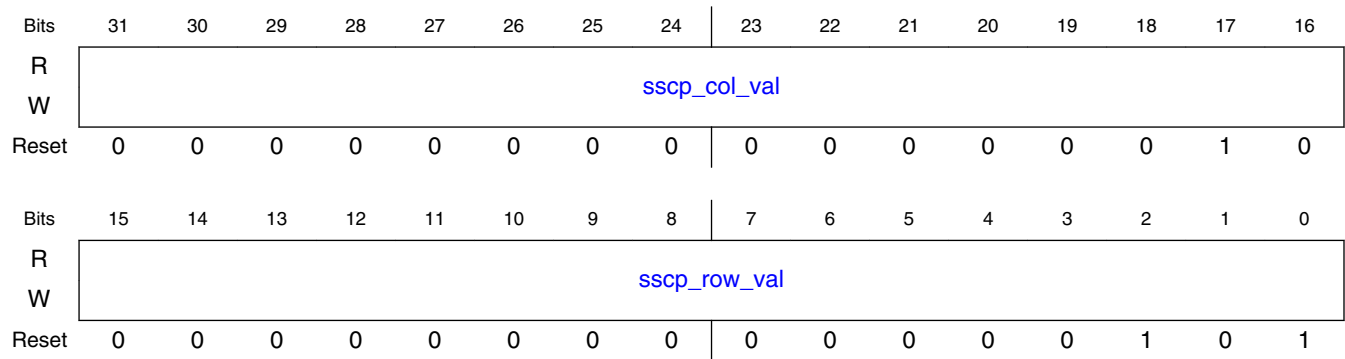
Field	Function
31-6 reserved_0	
5 default_phase_val	Default Phase Value Forced
4 color_depth_force	Enable Forcing Color Depth Value.
3-0 color_depth_val	Color Depth Value Forced

13.4.10.1.80 HDMI SSCP positioning (SSCP_POSITIONING)

13.4.10.1.80.1 Offset

Register	Offset
SSCP_POSITIONING	1038h

13.4.10.1.80.2 Diagram



13.4.10.1.80.3 Fields

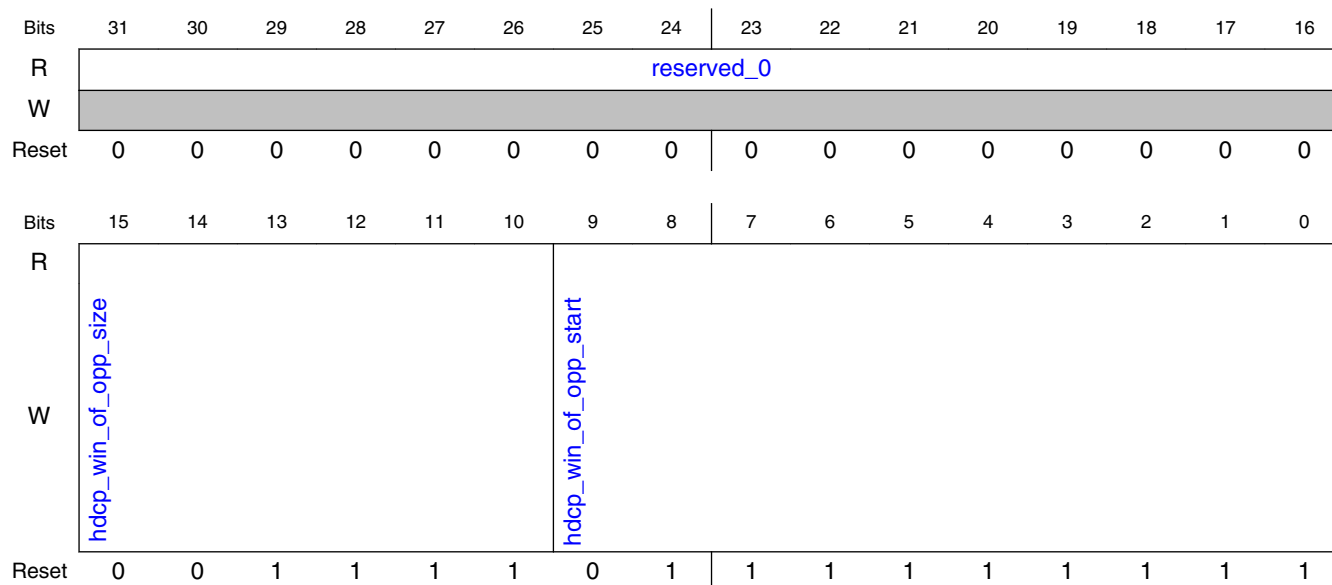
Field	Function
31-16 sscp_col_val	Column value of start of SSCP from end of Default
15-0 sscp_row_val	Row value of DE of the SSCP

13.4.10.1.81 HDMI HDCP window opportunity (HDCP_WIN_OF_OPP_POSITION)

13.4.10.1.81.1 Offset

Register	Offset
HDCP_WIN_OF_OPP_POSITION	103Ch

13.4.10.1.81.2 Diagram



13.4.10.1.81.3 Fields

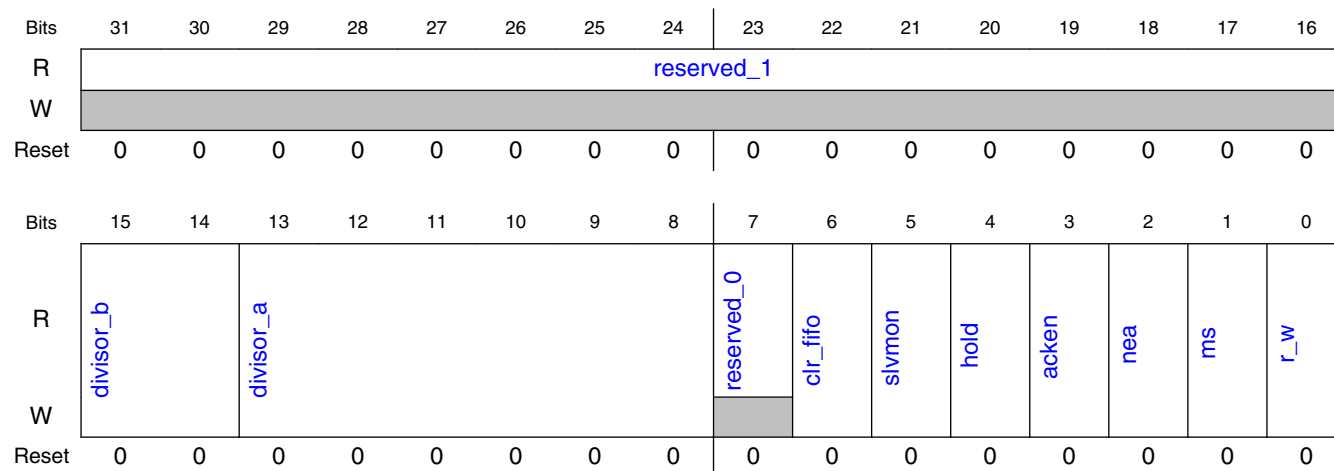
Field	Function
31-16 reserved_0	
15-10 hdcp_win_of_opp_size	Size of window of opportunity
9-0 hdcp_win_of_opp_start	Position of start of window of opportunity

13.4.10.1.82 HDMI I2C configuration (CTRL)

13.4.10.1.82.1 Offset

Register	Offset
CTRL	1800h

13.4.10.1.82.2 Diagram



13.4.10.1.82.3 Fields

Field	Function
31-16 reserved_1	
15-14 divisor_b	
13-8 divisor_a	
7 reserved_0	
6 clr_fifo	
5 slvmon	
4 hold	
3 acken	
2 nea	
1 ms	
0 r_w	

13.4.10.1.83 HDMI I2C status (STATUS)

13.4.10.1.83.1 Offset

Register	Offset
STATUS	1804h

13.4.10.1.83.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_3															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	rra	reserved_2						ba	rxovf	txdv	rxdv	reserved_1	rxrw	reserved_0		
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.83.3 Fields

Field	Function
31-16 reserved_3	
15 rra	
14-9 reserved_2	
8 ba	
7 rxovf	
6 txdv	
5	

Table continues on the next page...

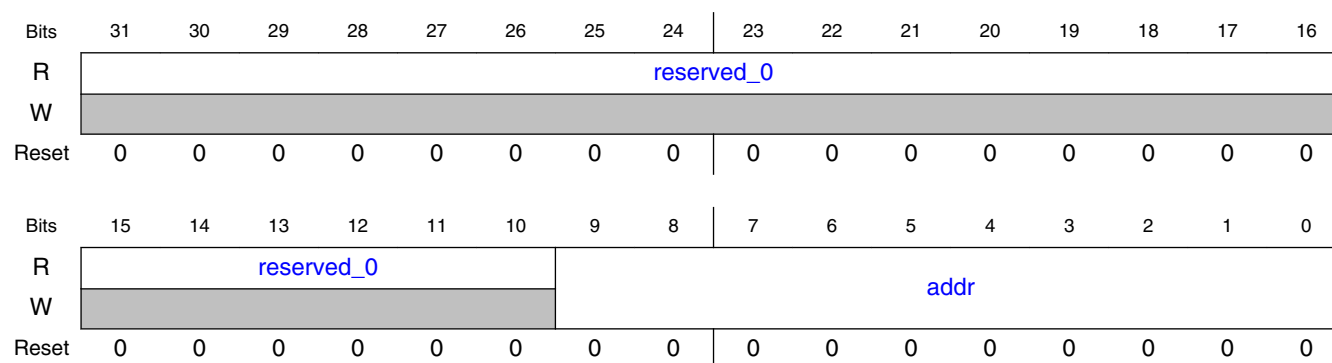
Field	Function
rxdv	
4 reserved_1	
3 rxrw	
2-0 reserved_0	

13.4.10.1.84 HDMI I2C Address configuration (I2C_ADDR)

13.4.10.1.84.1 Offset

Register	Offset
I2C_ADDR	1808h

13.4.10.1.84.2 Diagram



13.4.10.1.84.3 Fields

Field	Function
31-10 reserved_0	
9-0 addr	

13.4.10.1.85 HDMI I2C data register (I2C_DATA)

13.4.10.1.85.1 Offset

Register	Offset
I2C_DATA	180Ch

13.4.10.1.85.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								data							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.85.3 Fields

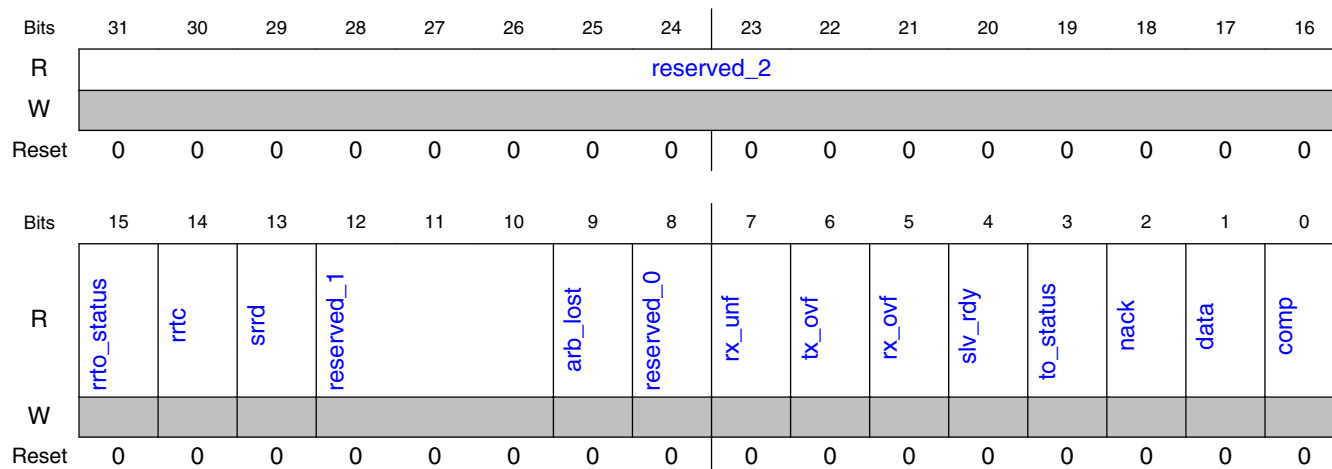
Field	Function
31-8 reserved_0	
7-0 data	

13.4.10.1.86 HDMI I2C interrupt status (INT_STATUS)

13.4.10.1.86.1 Offset

Register	Offset
INT_STATUS	1810h

13.4.10.1.86.2 Diagram



13.4.10.1.86.3 Fields

Field	Function
31-16 reserved_2	
15 rrto_status	
14 rrtc	
13 srrd	
12-10 reserved_1	
9 arb_lost	
8 reserved_0	
7 rx_unf	
6 tx_ovf	
5 rx_ovf	
4 slv_rdy	
3	

Table continues on the next page...

Clocks And Resets

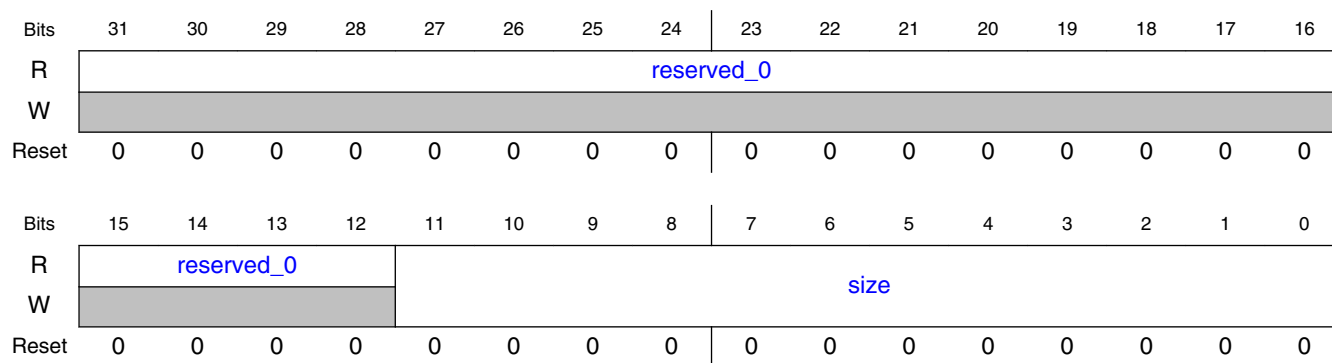
Field	Function
to_status	
2 nack	
1 data	
0 comp	

13.4.10.1.87 HDMI I2C transaction size (TRANS_SIZE)

13.4.10.1.87.1 Offset

Register	Offset
TRANS_SIZE	1814h

13.4.10.1.87.2 Diagram



13.4.10.1.87.3 Fields

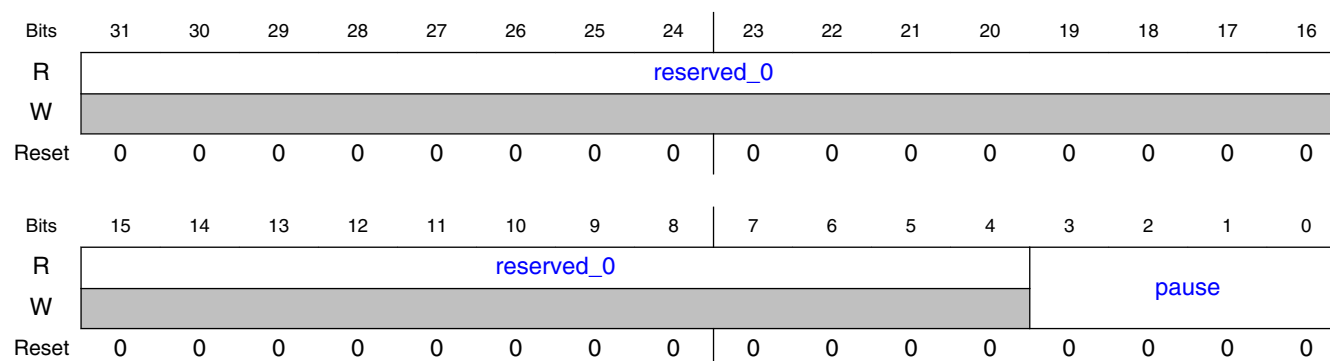
Field	Function
31-12 reserved_0	
11-0 size	

13.4.10.1.88 HDMI I2C Slave Monitor (SLAVE_MON)

13.4.10.1.88.1 Offset

Register	Offset
SLAVE_MON	1818h

13.4.10.1.88.2 Diagram



13.4.10.1.88.3 Fields

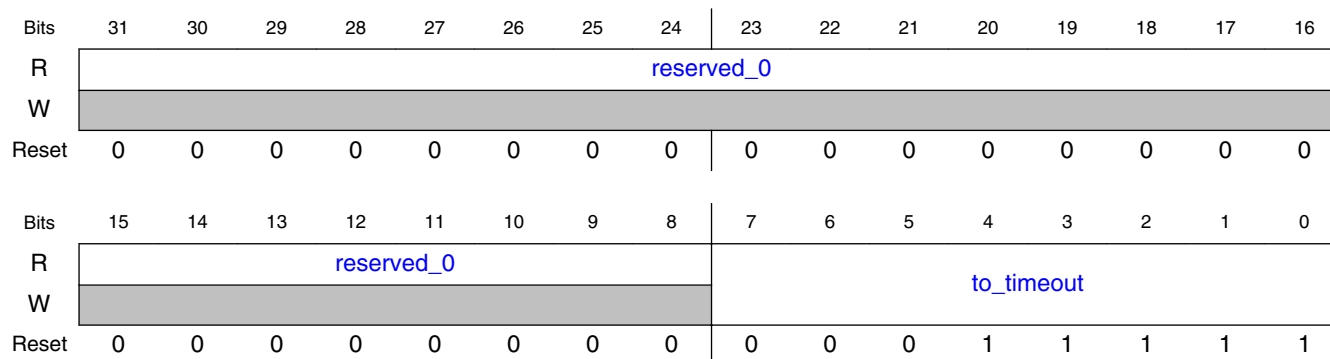
Field	Function
31-4 reserved_0	
3-0 pause	

13.4.10.1.89 HDMI I2C Timeout (TIMEOUT)

13.4.10.1.89.1 Offset

Register	Offset
TIMEOUT	181Ch

13.4.10.1.89.2 Diagram



13.4.10.1.89.3 Fields

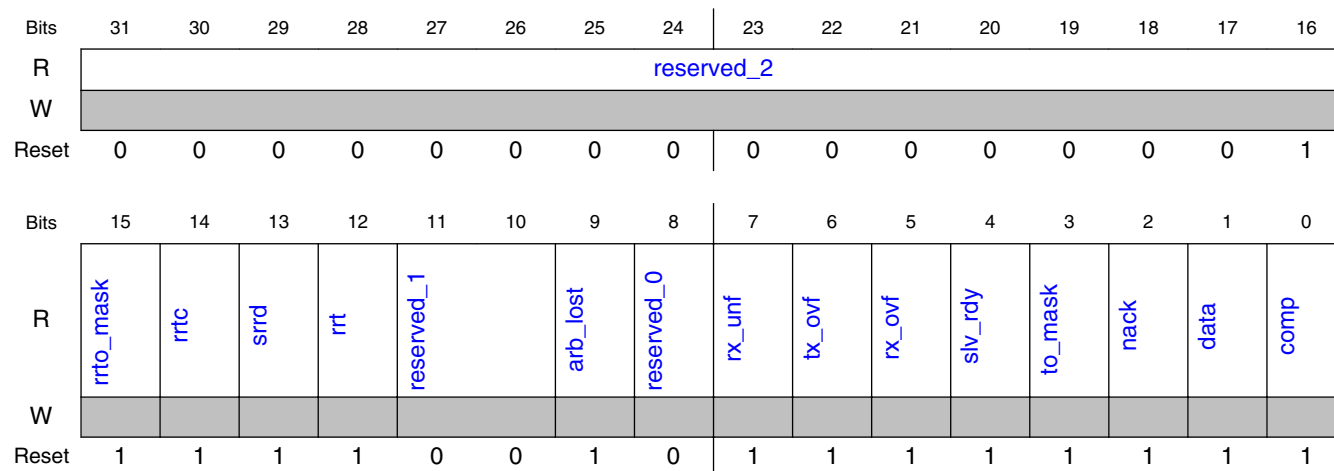
Field	Function
31-8 reserved_0	
7-0 to_timeout	

13.4.10.1.90 HDMI SCDC interrupt mask (INT_SCDC_MASK)

13.4.10.1.90.1 Offset

Register	Offset
INT_SCDC_MASK	1820h

13.4.10.1.90.2 Diagram



13.4.10.1.90.3 Fields

Field	Function
31-16 reserved_2	
15 rrto_mask	
14 rrtc	
13 srrd	
12 rrt	
11-10 reserved_1	
9 arb_lost	
8 reserved_0	
7 rx_unf	
6 tx_ovf	
5 rx_ovf	
4	

Table continues on the next page...

Clocks And Resets

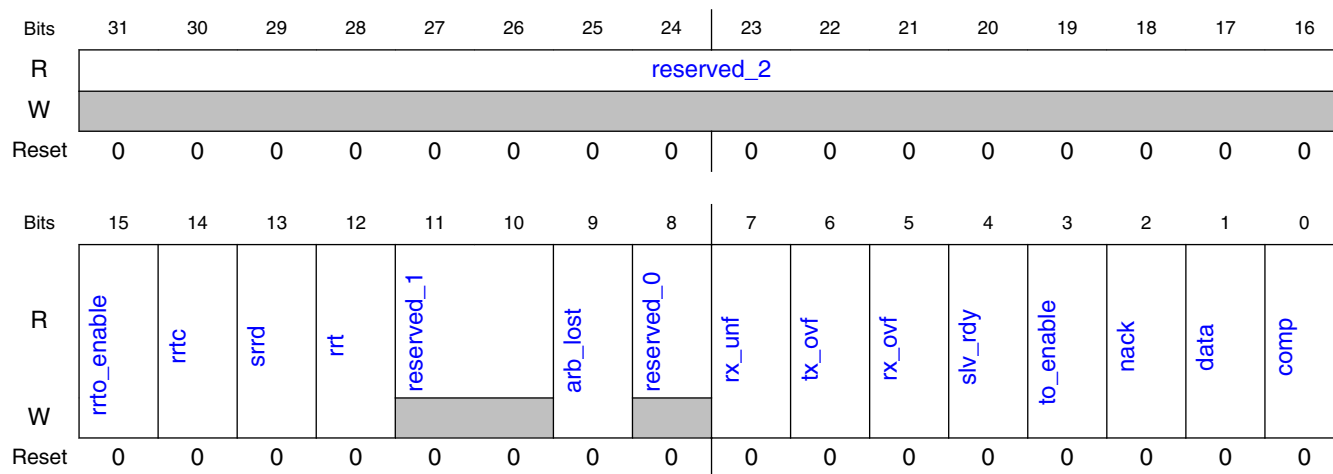
Field	Function
slv_rdy	
3 to_mask	
2 nack	
1 data	
0 comp	

13.4.10.1.91 HDMI I2C interrupt enable (INT_ENABLE)

13.4.10.1.91.1 Offset

Register	Offset
INT_ENABLE	1824h

13.4.10.1.91.2 Diagram



13.4.10.1.91.3 Fields

Field	Function
31-16	

Table continues on the next page...

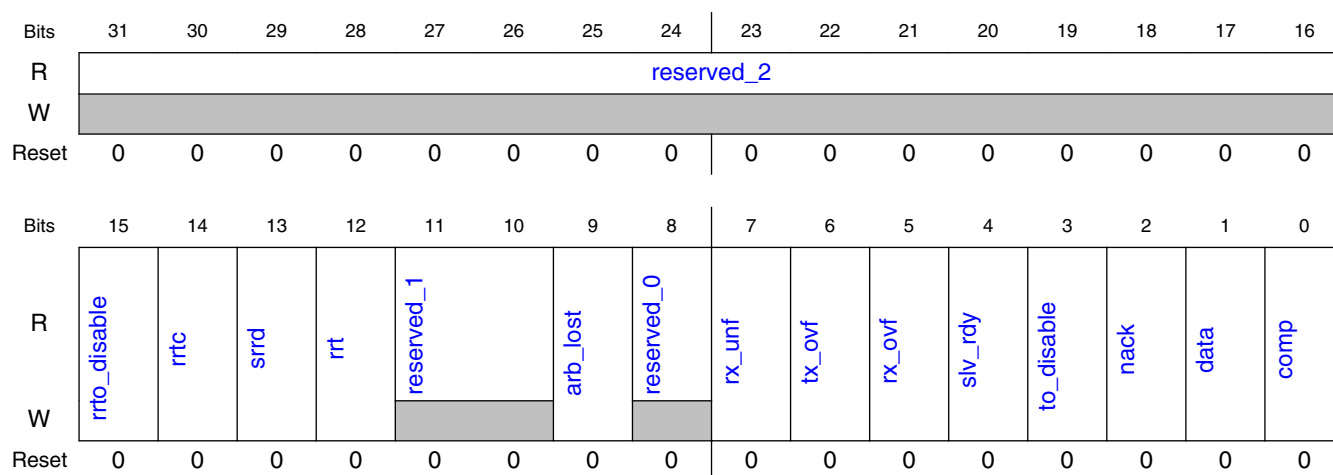
Field	Function
reserved_2	
15 rrto_enable	
14 rrtc	
13 srrd	
12 rrt	
11-10 reserved_1	
9 arb_lost	
8 reserved_0	
7 rx_unf	
6 tx_ovf	
5 rx_ovf	
4 slv_rdy	
3 to_enable	
2 nack	
1 data	
0 comp	

13.4.10.1.92 HDMI I2C interrupt disable (INT_DISABLE)

13.4.10.1.92.1 Offset

Register	Offset
INT_DISABLE	1828h

13.4.10.1.92.2 Diagram



13.4.10.1.92.3 Fields

Field	Function
31-16 reserved_2	
15 rrto_disable	
14 rrtc	
13 srrd	
12 rrt	
11-10 reserved_1	
9 arb_lost	
8 reserved_0	
7 rx_unf	
6 tx_ovf	
5 rx_ovf	

Table continues on the next page...

Field	Function
4 slv_rdy	
3 to_disable	
2 nack	
1 data	
0 comp	

13.4.10.1.93 HDMI I2C glitch filter configuration (GLITCH_FILTER_CTRL)

13.4.10.1.93.1 Offset

Register	Offset
GLITCH_FILTER_CTRL	182Ch

13.4.10.1.93.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								gf							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.93.3 Fields

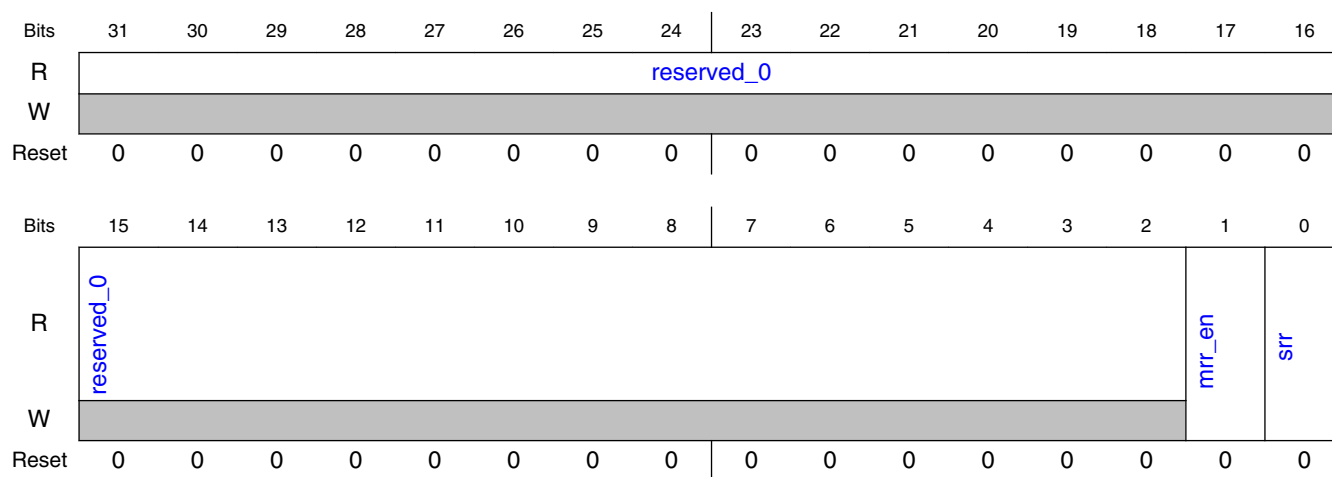
Field	Function
31-8 reserved_0	
7-0 gf	

13.4.10.1.94 HDMI I2C RR configuration (RR_CTRL)

13.4.10.1.94.1 Offset

Register	Offset
RR_CTRL	1830h

13.4.10.1.94.2 Diagram



13.4.10.1.94.3 Fields

Field	Function
31-2 reserved_0	
1 mrr_en	
0 srr	

13.4.10.1.95 HDMI I2C RR timeout (RR_TIMEOUT)

13.4.10.1.95.1 Offset

Register	Offset
RR_TIMEOUT	1834h

13.4.10.1.95.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	rrto_timeout															
W																
Reset	0	0	1	0	0	0	0	0	1	0	0	0	1	1	0	1

13.4.10.1.95.3 Fields

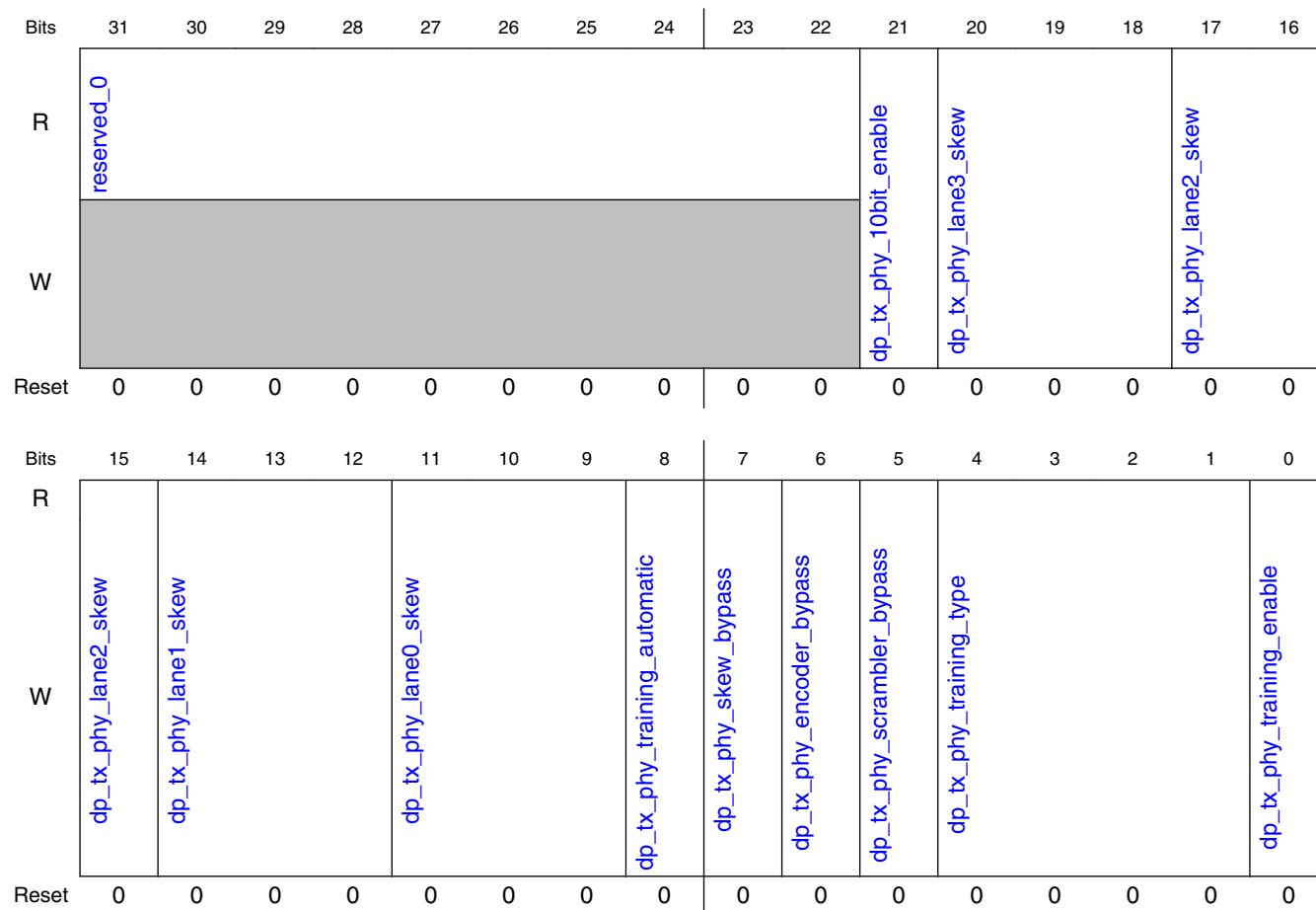
Field	Function
31-16 reserved_0	
15-0 rrto_timeout	

13.4.10.1.96 DPTX PHY control (DP_TX_PHY_CONFIG_REG)

13.4.10.1.96.1 Offset

Register	Offset
DP_TX_PHY_CONFIG_REG	2000h

13.4.10.1.96.2 Diagram



13.4.10.1.96.3 Fields

Field	Function
31-22 reserved_0	
21 dp_tx_phy_10bit_enable	Used to enable the 10-bit mode. Used to enable the 10-bit mode. Active high.
20-18 dp_tx_phy_lane3_skew	Specifies the programmable lane3 skew. Specifies the programmable lane3 skew.
17-15 dp_tx_phy_lane2_skew	Specifies the programmable lane2 skew. Specifies the programmable lane2 skew.
14-12	Specifies the programmable lane1 skew. Specifies the programmable lane1 skew.

Table continues on the next page...

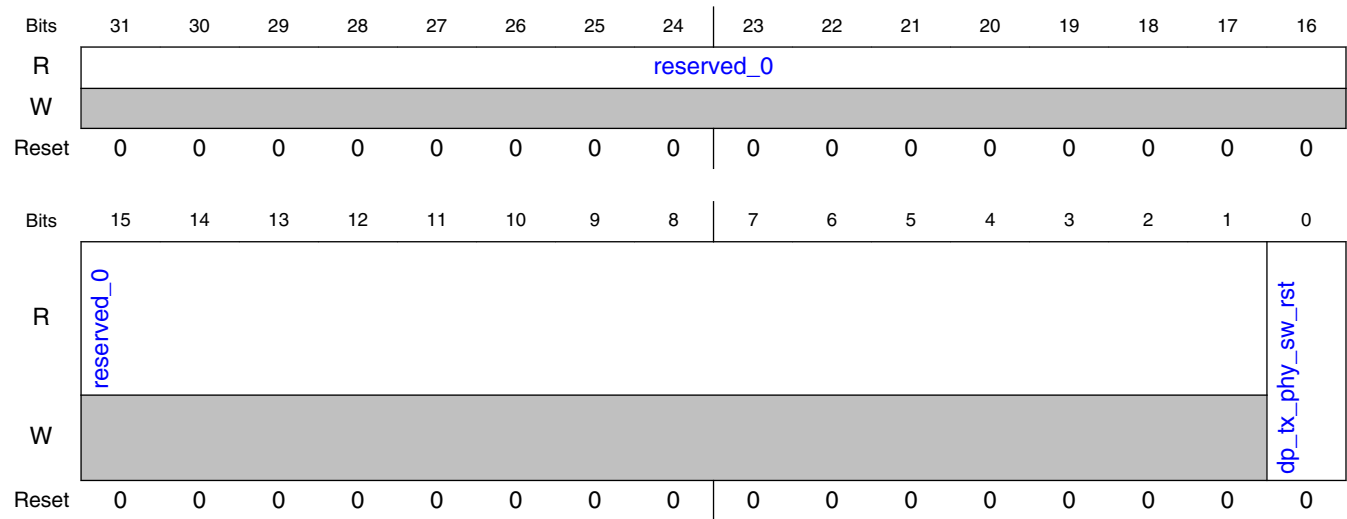
Field	Function
dp_tx_phy_lane1_skew	
11-9	Specifies the programmable lane0 skew.
dp_tx_phy_lane0_skew	Specifies the programmable lane0 skew.
8	Used to bypass 8/10 encoder and scrambler during link training.
dp_tx_phy_training_automatic	Used to bypass 8/10 encoder and scrambler during link training. Bit is sampled only when dp_tx_phy_training_enable bit is set.
7	Used to bypass the lane skew.
dp_tx_phy_skew_bypass	Used to bypass the lane skew. Active high.
6	Used to bypass the encoder.
dp_tx_phy_encoder_bypass	Used to bypass the encoder. Active high.
5	Used to bypass the scrambler.
dp_tx_phy_scrambler_bypass	Used to bypass the scrambler. Active high.
4-1	Specifies the training pattern type used as follows: 0000 PRBS7, 0001 TPS1, 0010 TPS2, 0011 TPS3, 0100 TPS4, 0101 custom 80-bit pattern, 0110 D10.
dp_tx_phy_training_type	Specifies the training pattern type used as follows: 0000 PRBS7, 0001 TPS1, 0010 TPS2, 0011 TPS3, 0100 TPS4, 0101 custom 80-bit pattern, 0110 D10.2 training pattern, 0111 Symbol Error Rate Measurement pattern, 1000 CP2520 pattern1, 1001 CP2520 pattern2, 1010 CP2520 pattern3, others reserved
0	Enables the training sequence (when set to 1).
dp_tx_phy_training_enable	Enables the training sequence (when set to 1).

13.4.10.1.97 DPTC PHY software reset (DP_TX_PHY_SW_RESET)

13.4.10.1.97.1 Offset

Register	Offset
DP_TX_PHY_SW_RESET	2004h

13.4.10.1.97.2 Diagram



13.4.10.1.97.3 Fields

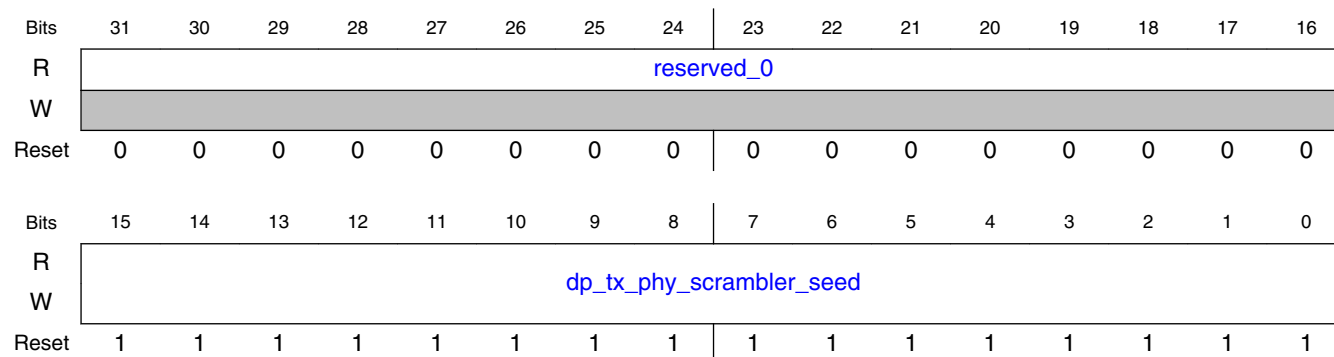
Field	Function
31-1 reserved_0	
0 dp_tx_phy_sw_rst	Software reset. Software reset. Active high.

13.4.10.1.98 Scrambler seed (DP_TX_PHY_SCRAMBLER_SEED)

13.4.10.1.98.1 Offset

Register	Offset
DP_TX_PHY_SCRAMBLER_SEED	2008h

13.4.10.1.98.2 Diagram



13.4.10.1.98.3 Fields

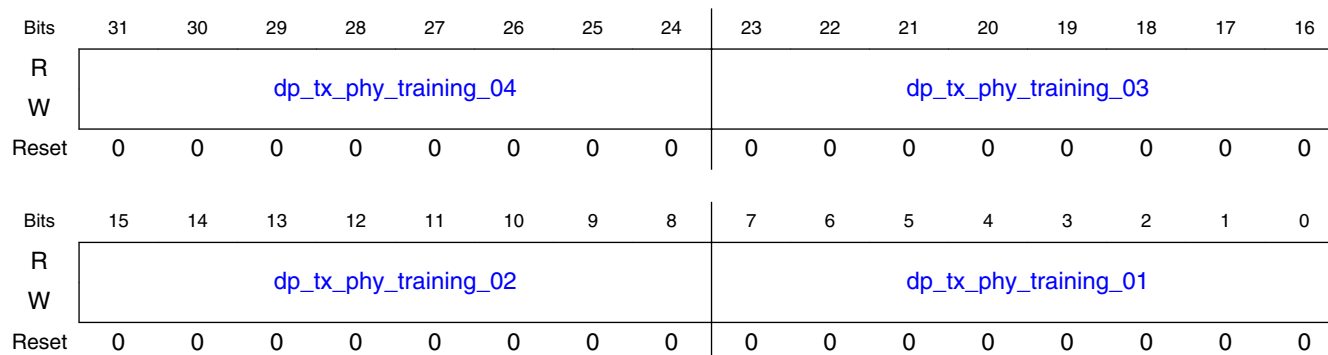
Field	Function
31-16 reserved_0	
15-0 dp_tx_phy_scrambler_seed	Scrambler seed range 0-0xFFFF

13.4.10.1.99 Custom training value bytes 1-4 (DP_TX_PHY_TRAINING_01_04)

13.4.10.1.99.1 Offset

Register	Offset
DP_TX_PHY_TRAINING_01_04	200Ch

13.4.10.1.99.2 Diagram



13.4.10.1.99.3 Fields

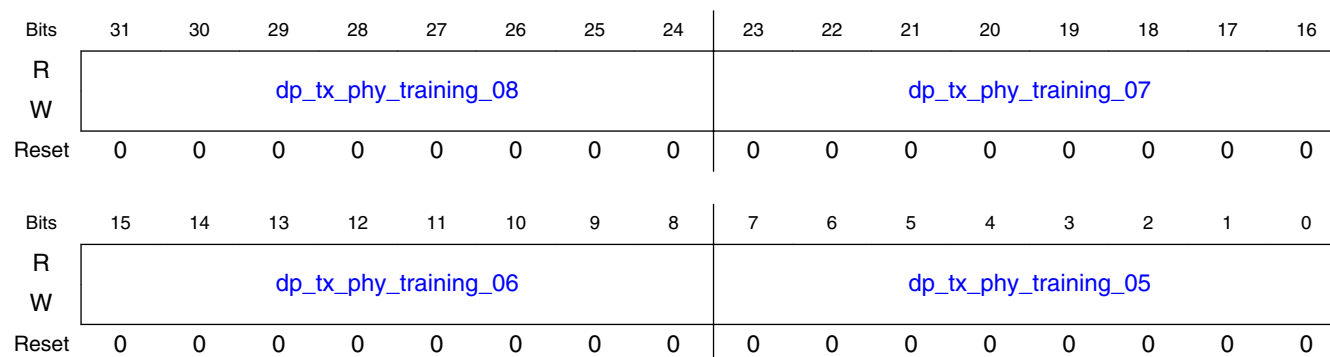
Field	Function
31-24 dp_tx_phy_training_04	Byte 4 of the 80-bit custom training data.
23-16 dp_tx_phy_training_03	Byte 3 of the 80-bit custom training data.
15-8 dp_tx_phy_training_02	Byte 2 of the 80-bit custom training data.
7-0 dp_tx_phy_training_01	Byte 1 of the 80-bit custom training data.

13.4.10.1.100 Custom training value bytes 5-8 (DP_TX_PHY_TRAINING_05_08)

13.4.10.1.100.1 Offset

Register	Offset
DP_TX_PHY_TRAINING_05_08	2010h

13.4.10.1.100.2 Diagram



13.4.10.1.100.3 Fields

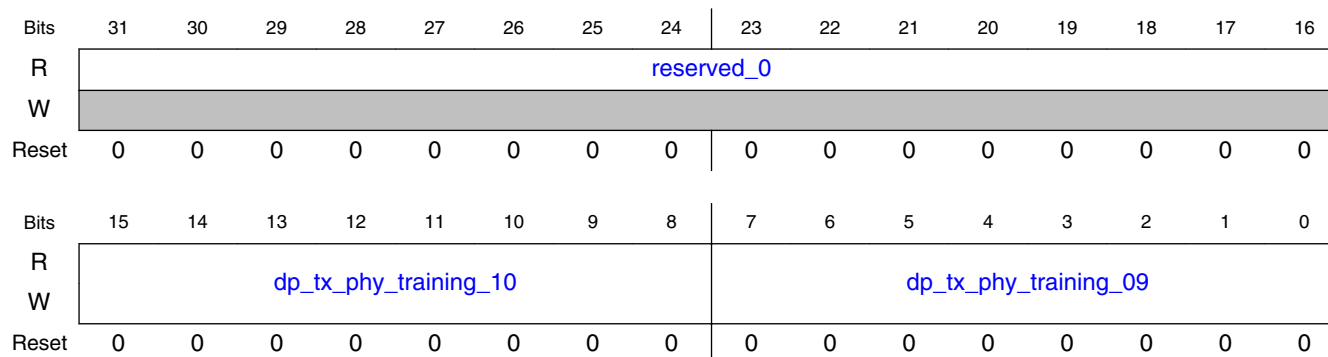
Field	Function
31-24 dp_tx_phy_training_08	Byte 8 of the 80-bit custom training data.
23-16 dp_tx_phy_training_07	Byte 7 of the 80-bit custom training data.
15-8 dp_tx_phy_training_06	Byte 6 of the 80-bit custom training data.
7-0 dp_tx_phy_training_05	Byte 5 of the 80-bit custom training data.

13.4.10.1.101 Custom training value bytes 9-10 (DP_TX_PHY_TRAINING_09_10)

13.4.10.1.101.1 Offset

Register	Offset
DP_TX_PHY_TRAINING_09_10	2014h

13.4.10.1.101.2 Diagram



13.4.10.1.101.3 Fields

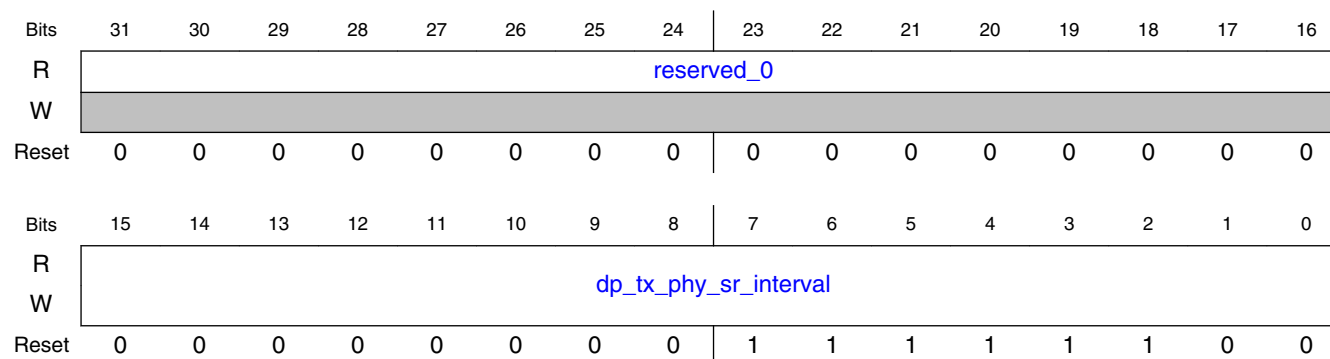
Field	Function
31-16 reserved_0	
15-8 dp_tx_phy_training_10	Byte 10 of the 80-bit custom training data.
7-0 dp_tx_phy_training_09	Byte 9 of the 80-bit custom training data.

13.4.10.1.102 Custom CP2520 SR interval (DP_TX_PHY_SR_INTERVAL)

13.4.10.1.102.1 Offset

Register	Offset
DP_TX_PHY_SR_INTERVAL	2018h

13.4.10.1.102.2 Diagram



13.4.10.1.102.3 Fields

Field	Function
31-16 reserved_0	
15-0 dp_tx_phy_sr_interval	CP2520 test pattern SR Interval definition

13.4.10.1.103 HPD min timer for irq, define the minimum pclk cycles that the HPD pulse will be considered as IRQ (HPD_IRQ_DET_MIN_TIMER)

13.4.10.1.103.1 Offset

Register	Offset
HPD_IRQ_DET_MIN_TIMER	2100h

13.4.10.1.103.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0								hpd_irq_det_min_timer							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	hpd_irq_det_min_timer															
W																
Reset	1	1	0	0	0	0	1	1	0	1	0	1	0	0	0	0

13.4.10.1.103.3 Fields

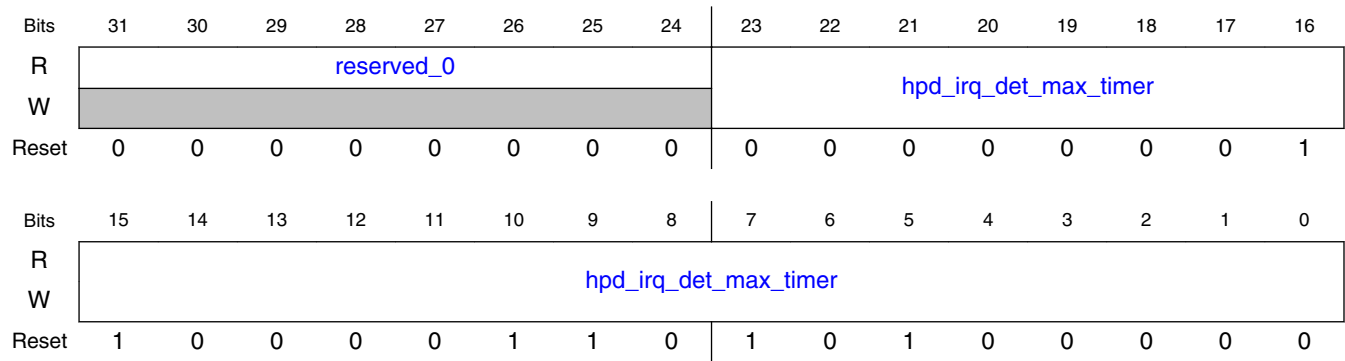
Field	Function
31-24 reserved_0	
23-0 hpd_irq_det_min_timer	hpd min timer for irq

13.4.10.1.104 HPD max timer for irq, define the maximum pclk cycles that the HPD pulse will be considered as IRQ (HPD_IRQ_DET_MAX_TIMER)

13.4.10.1.104.1 Offset

Register	Offset
HPD_IRQ_DET_MAX_TIMER	2104h

13.4.10.1.104.2 Diagram



13.4.10.1.104.3 Fields

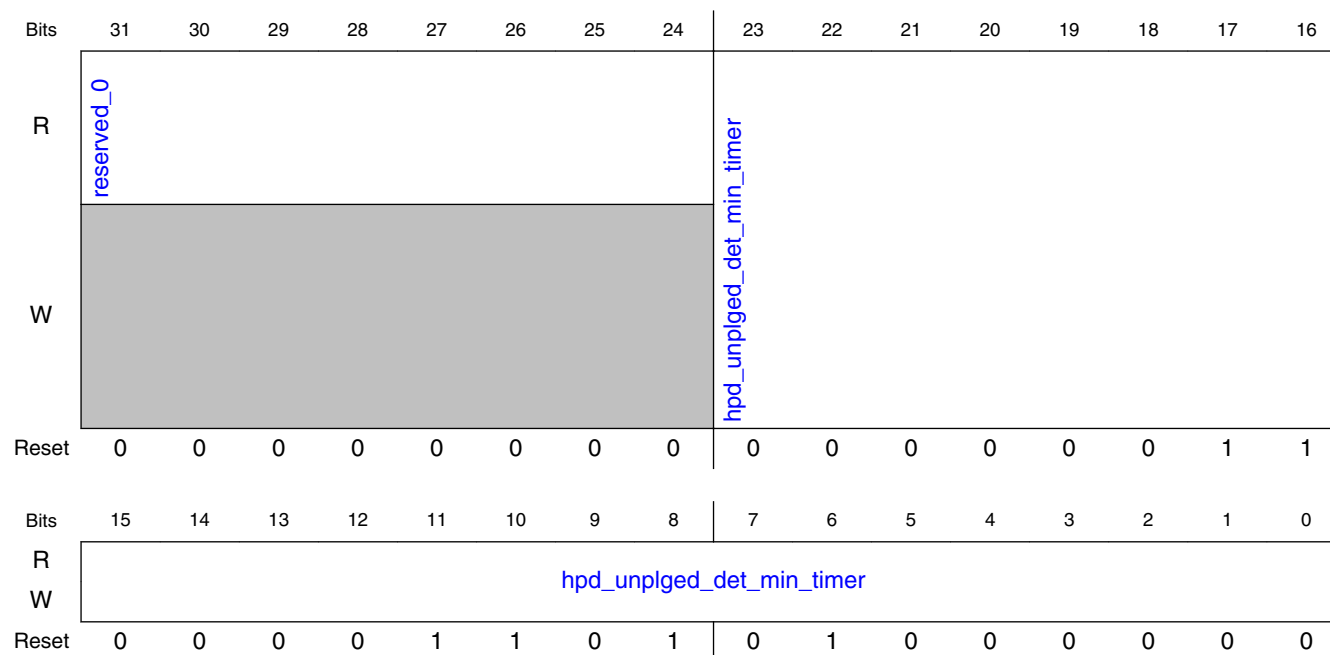
Field	Function
31-24 reserved_0	
23-0 hpd_irq_det_max_timer	hpd max timer

13.4.10.1.105 HPD min timer for HPD detect, define the minimum pclk cycles that the HPD is low (HPD_UNPLGED_DET_MIN_TIME R)

13.4.10.1.105.1 Offset

Register	Offset
HPD_UNPLGED_DET_MIN_TIMER	2108h

13.4.10.1.105.2 Diagram



13.4.10.1.105.3 Fields

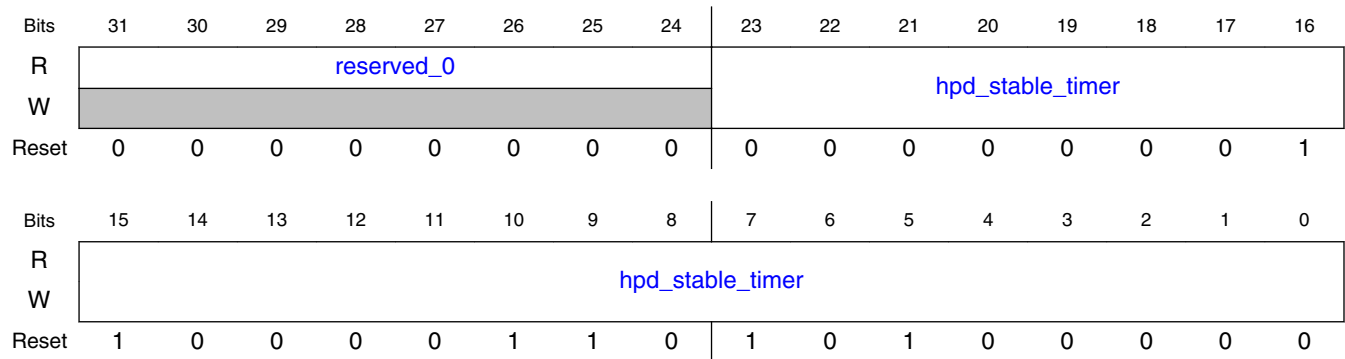
Field	Function
31-24 reserved_0	
23-0 hpd_unplged_det_min_timer	hpd unplugged timer

13.4.10.1.106 Timer for detecting HPD stable, count in pclk cycles (HPD_STABLE_TIMER)

13.4.10.1.106.1 Offset

Register	Offset
HPD_STABLE_TIMER	210Ch

13.4.10.1.106.2 Diagram



13.4.10.1.106.3 Fields

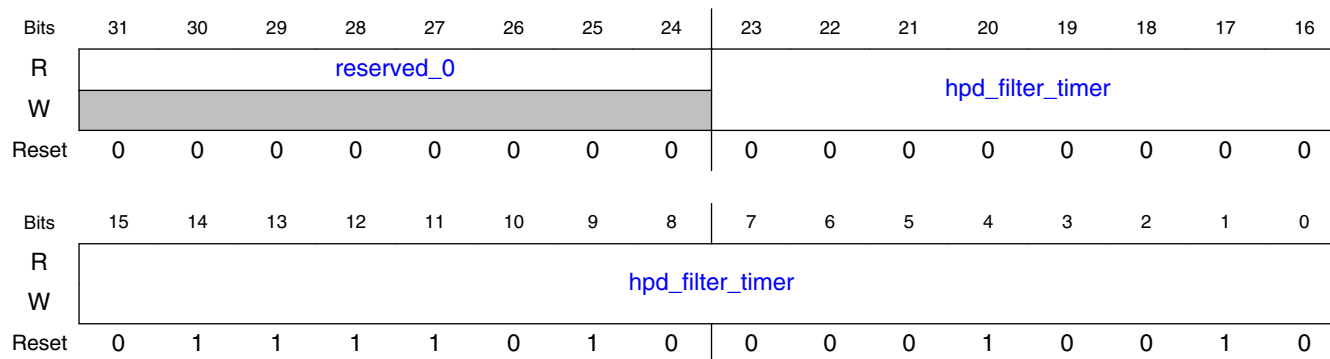
Field	Function
31-24 reserved_0	
23-0 hpd_stable_time r	hpd stable timer bits

13.4.10.1.107 Timer for filtering small pulses on hpd input (HPD_FILTER_TIMER)

13.4.10.1.107.1 Offset

Register	Offset
HPD_FILTER_TIMER	2110h

13.4.10.1.107.2 Diagram



13.4.10.1.107.3 Fields

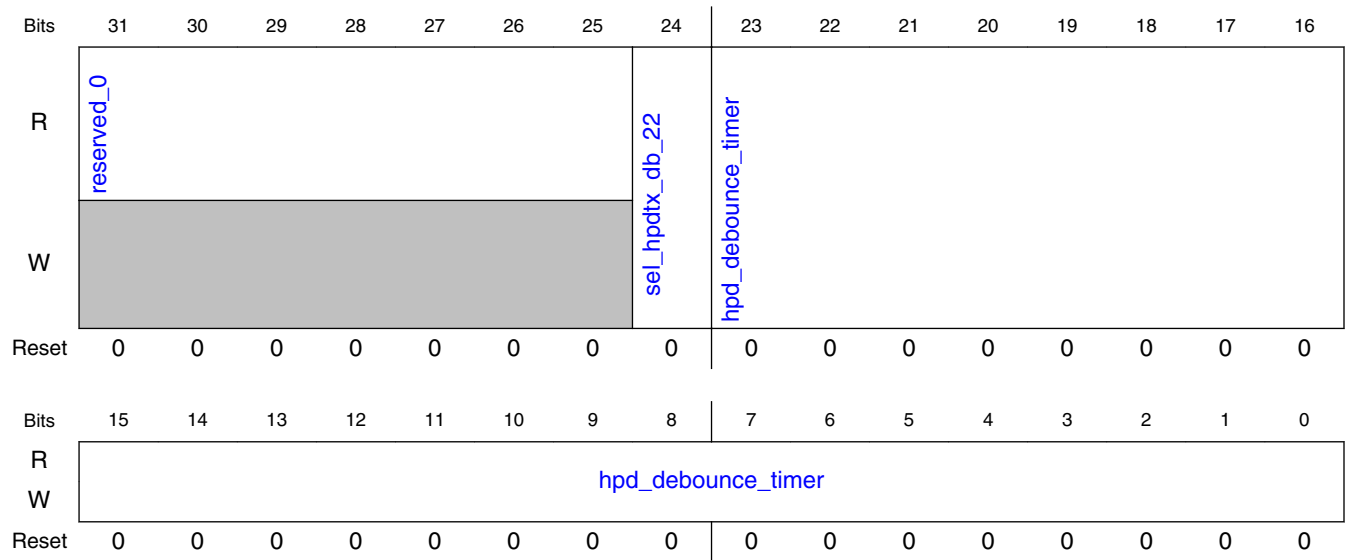
Field	Function
31-24 reserved_0	
23-0 hpd_filter_timer	hpd glitch filter timer bits

13.4.10.1.108 HPD debouncer control (HPD_DBNC_TIMER)

13.4.10.1.108.1 Offset

Register	Offset
HPD_DBNC_TIMER	2114h

13.4.10.1.108.2 Diagram



13.4.10.1.108.3 Fields

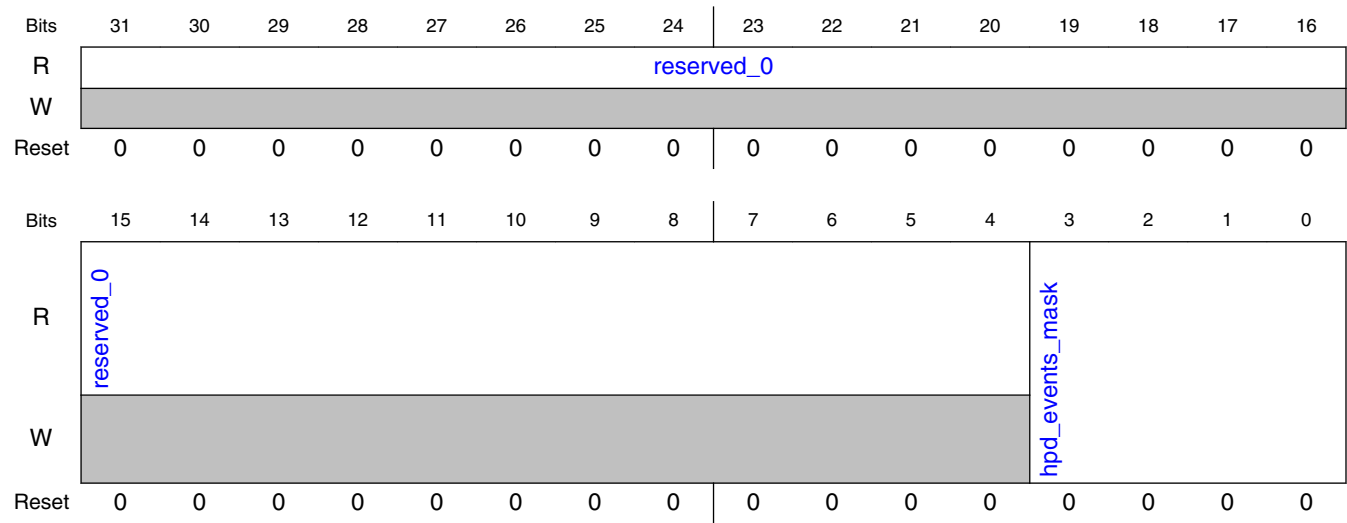
Field	Function
31-25 reserved_0	
24 sel_hpdtx_db_22	use debouncer
23-0 hpd_debounce_timer	hpd debounce timer bits

13.4.10.1.109 Mask of HPD interrupt and status (HPD_EVENT_MASK)

13.4.10.1.109.1 Offset

Register	Offset
HPD_EVENT_MASK	211Ch

13.4.10.1.109.2 Diagram



13.4.10.1.109.3 Fields

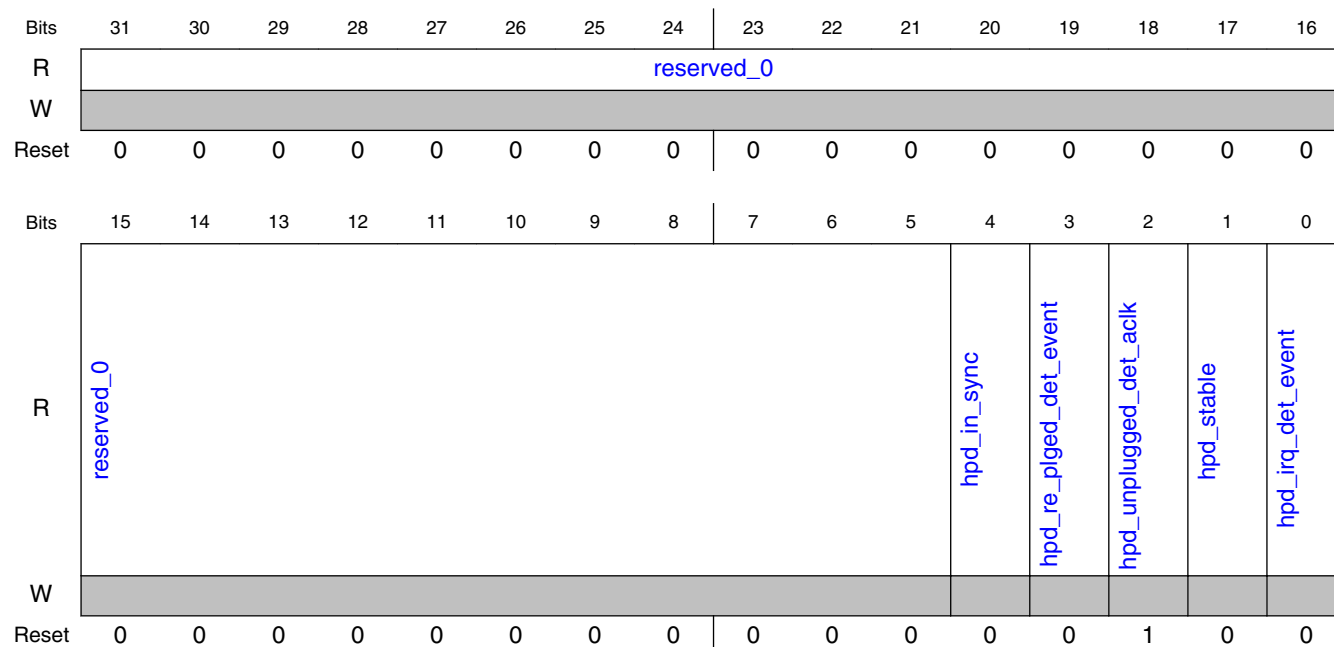
Field	Function
31-4 reserved_0	
3-0 hpd_events_ma sk	hpd mask events

13.4.10.1.110 HPD interrupt and status (HPD_EVENT_DET)

13.4.10.1.110.1 Offset

Register	Offset
HPD_EVENT_DET	2120h

13.4.10.1.110.2 Diagram



13.4.10.1.110.3 Fields

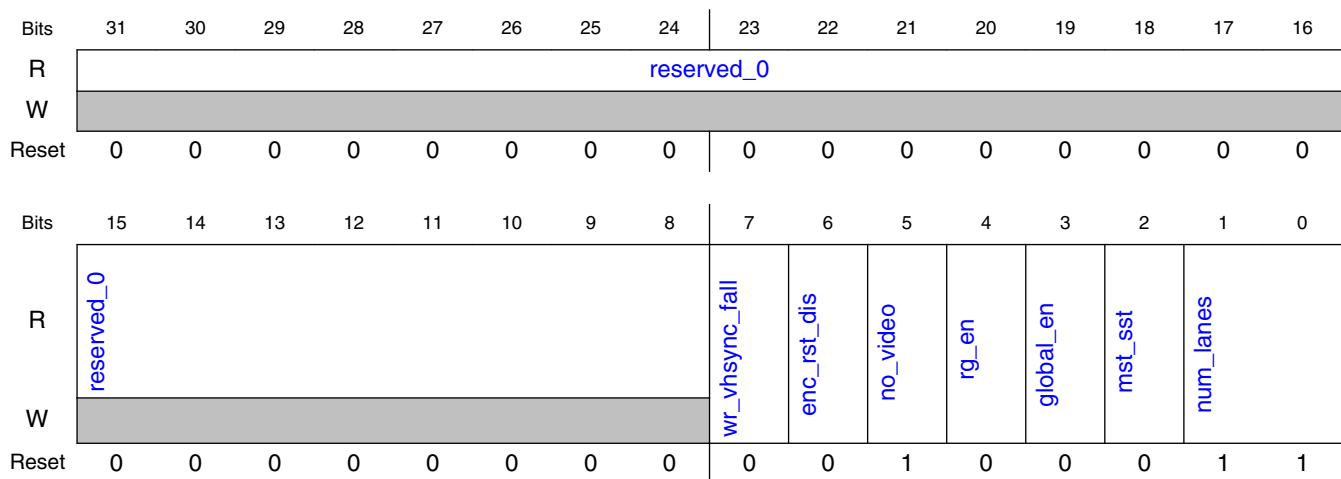
Field	Function
31-5 reserved_0	
4 hpd_in_sync	HPD in sync detected After a read operation, the field's value clears to 0.
3 hpd_re_plged_det_event	HPD Re-Plugged event detected. After a read operation, the field's value clears to 0.
2 hpd_unplugged_det_ack	HPD Un-Plugged event detected. After a read operation, the field's value clears to 0.
1 hpd_stable	HPD Stable indication After a read operation, the field's value clears to 0.
0 hpd_irq_det_event	Bit 0 - HPD irq event After a read operation, the field's value clears to 0.

13.4.10.1.111 Global configuration of the framer module (DP_FRAMER_GLOBAL_CONFIG)

13.4.10.1.111.1 Offset

Register	Offset
DP_FRAMER_GLOBAL_CONFIG	2200h

13.4.10.1.111.2 Diagram



13.4.10.1.111.3 Fields

Field	Function
31-8 reserved_0	
7 wr_vhsync_fall	When set to 1 change the write state machine to sync on falling edge of vsync
6 enc_rst_dis	Disable reset of phy by the framer (during no video mode), 1 - reset disabled, 0 - reset enabled
5 no_video	No_video mode configuration bit, active high
4 rg_en	Rate governor enable bit, active high
3 global_en	Global enable for complete Framer module, active high. Global enable for complete Framer module, active high. It is deasserted during configuration phase. Once configuration is finished, it is asserted.

Table continues on the next page...

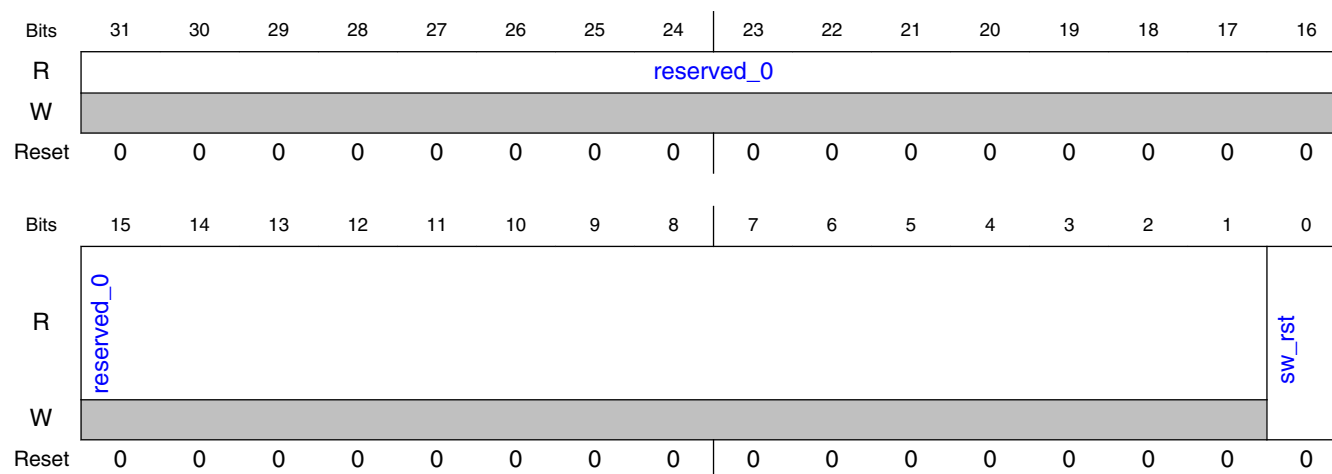
Field	Function
2 mst_sst	Mode select: 0 - SST mode 1 - MST mode
1-0 num_lanes	Number of lanes: 0h - One lane (Lane 0 only) 1h - Two lanes (Lanes 0 and 1 only) 2h - Reserved 3h - Four lanes (Lanes 0, 1, 2, and 3)

13.4.10.1.112 SW reset (DP_SW_RESET)

13.4.10.1.112.1 Offset

Register	Offset
DP_SW_RESET	2204h

13.4.10.1.112.2 Diagram



13.4.10.1.112.3 Fields

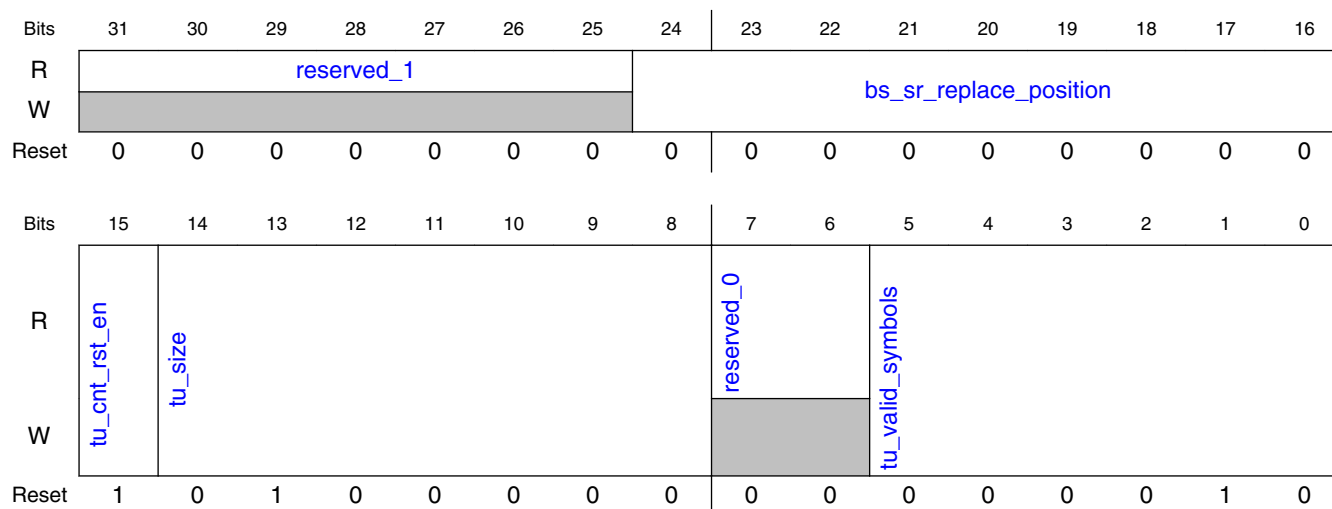
Field	Function
31-1 reserved_0	
0 sw_rst	SW reset for the entire framer

13.4.10.1.113 TU related configuration (DP_FRAMER_TU)

13.4.10.1.113.1 Offset

Register	Offset
DP_FRAMER_TU	2208h

13.4.10.1.113.2 Diagram



13.4.10.1.113.3 Fields

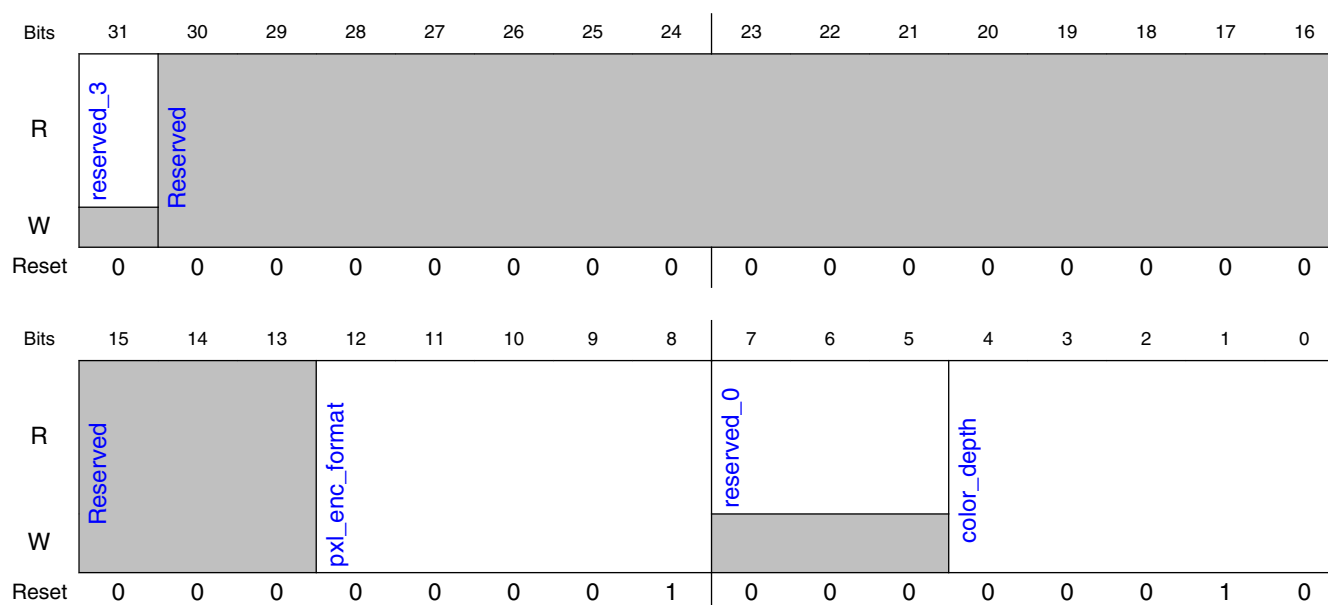
Field	Function
31-25 reserved_1	
24-16 bs_sr_replace_position	When set to non-zero value, the RTL BS counter will be reinitialized to this value.
15 tu_cnt_rst_en	When set high allows VIF logic to reset the TU_start counter in order to synchronize it to the VIF read fsm
14-8 tu_size	TU size (min=32, max=64)
7-6 reserved_0	
5-0 tu_valid_symbols	Number of valid symbols in TU, rounded to lower integer value (refer to equation in standard). Number of valid symbols in TU, rounded to lower integer value (refer to equation in standard). Always lower or equal to TU size

13.4.10.1.114 Video pixel format configuration (DP_FRAMER_PXL_REPR)

13.4.10.1.114.1 Offset

Register	Offset
DP_FRAMER_PXL_REPR	220Ch

13.4.10.1.114.2 Diagram



13.4.10.1.114.3 Fields

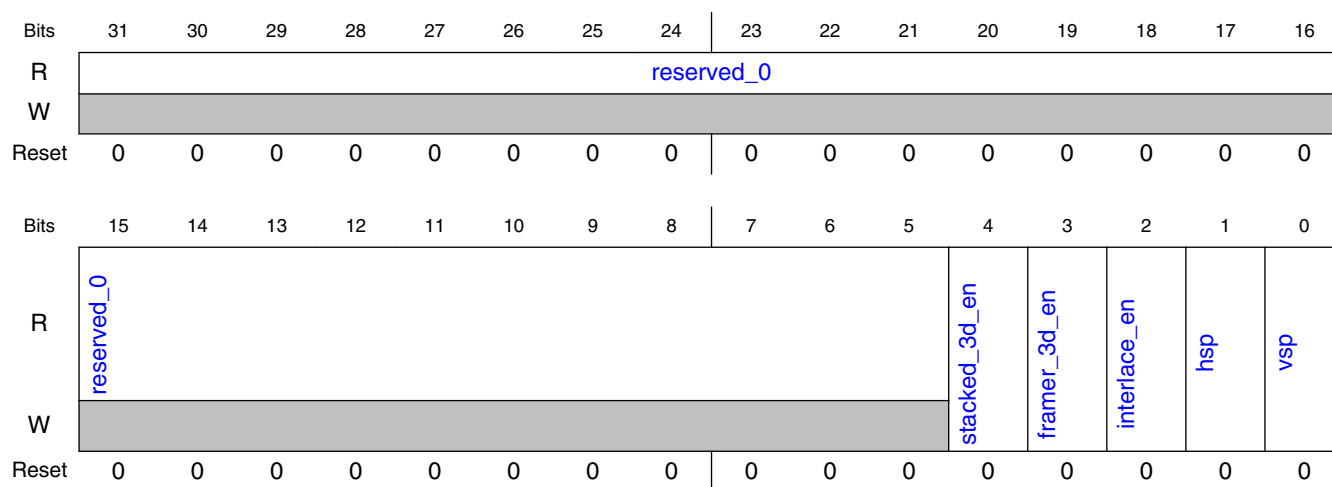
Field	Function
31 reserved_3	
30-13 —	Reserved
12-8 pxl_enc_format	Pixel encoding format: 1h - RGB2h - YCbCr 4:4:4, 4h - YCbCr 4:2:2, 8h - Y CbCr 4:2:0, 10h - Y only, All other values - RESERVED
7-5 reserved_0	
4-0 color_depth	Color depth: 1h - 6 bpc, 2h - 8 bpc, 4h - 10 bpc, 8h - 12 bpc, 10h - 16 bpc, All other values - RESERVED

13.4.10.1.115 Polarity and 3D configuration (DP_FRAMER_SP)

13.4.10.1.115.1 Offset

Register	Offset
DP_FRAMER_SP	2210h

13.4.10.1.115.2 Diagram



13.4.10.1.115.3 Fields

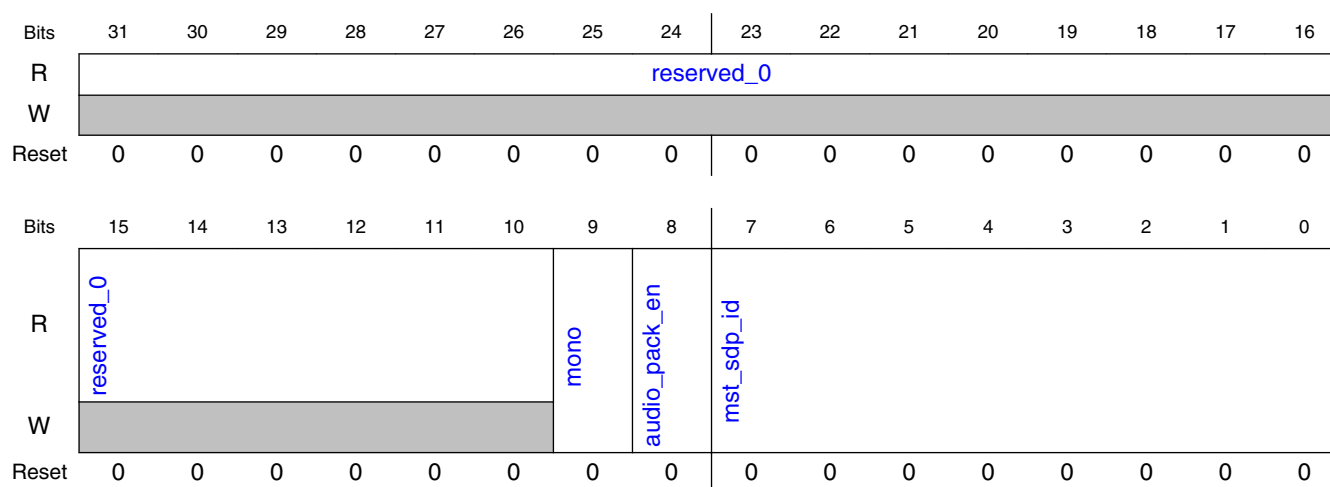
Field	Function
31-5 reserved_0	
4 stacked_3d_en	3D mode, 1 - stacked, 0 - field sequential
3 framer_3d_en	3D video enabled, active high
2 interlace_en	Interlaced video enabled, active high
1 hsp	HSYNC polarity: 0 - active high, 1 - active low
0 vsp	VSNC polarity: 0 - active high, 1 - active low

13.4.10.1.116 Audio packet configuration (AUDIO_PACK_CONTROL)

13.4.10.1.116.1 Offset

Register	Offset
AUDIO_PACK_CONTROL	2214h

13.4.10.1.116.2 Diagram



13.4.10.1.116.3 Fields

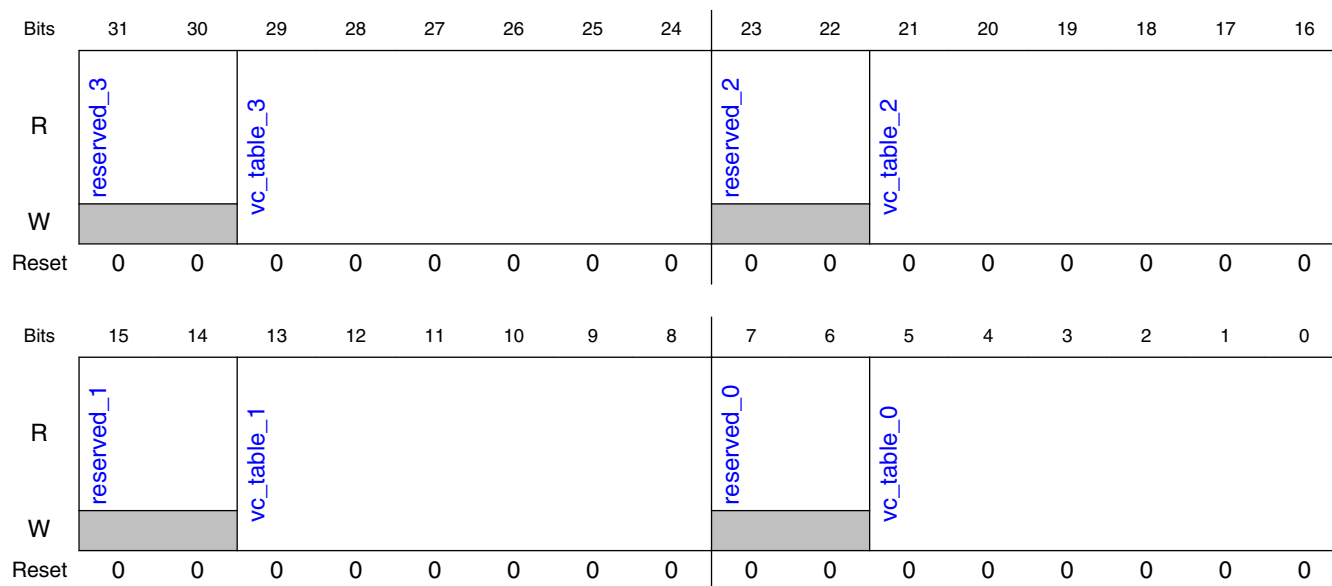
Field	Function
31-10 reserved_0	
9 mono	In case of 2-channel layout and one lane configuration SW decides whether it is a stereo or mono transfer. In case of 2-channel layout and one lane configuration SW decides whether it is a stereo or mono transfer. Relevant for SDP Header: ChannelCount field
8 audio_pack_en	Enables the Audio_Pack module, active high
7-0 mst_sdp_id	Secondary-Data Packet ID

13.4.10.1.117 VC table values 0-3 (MST feature) (DP_VC_TABLE_0)

13.4.10.1.117.1 Offset

Register	Offset
DP_VC_TABLE_0	2218h

13.4.10.1.117.2 Diagram



13.4.10.1.117.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_3	VC Table value for time slot 3
23-22 reserved_2	
21-16 vc_table_2	VC Table value for time slot 2
15-14 reserved_1	
13-8	VC Table value for time slot 1

Table continues on the next page...

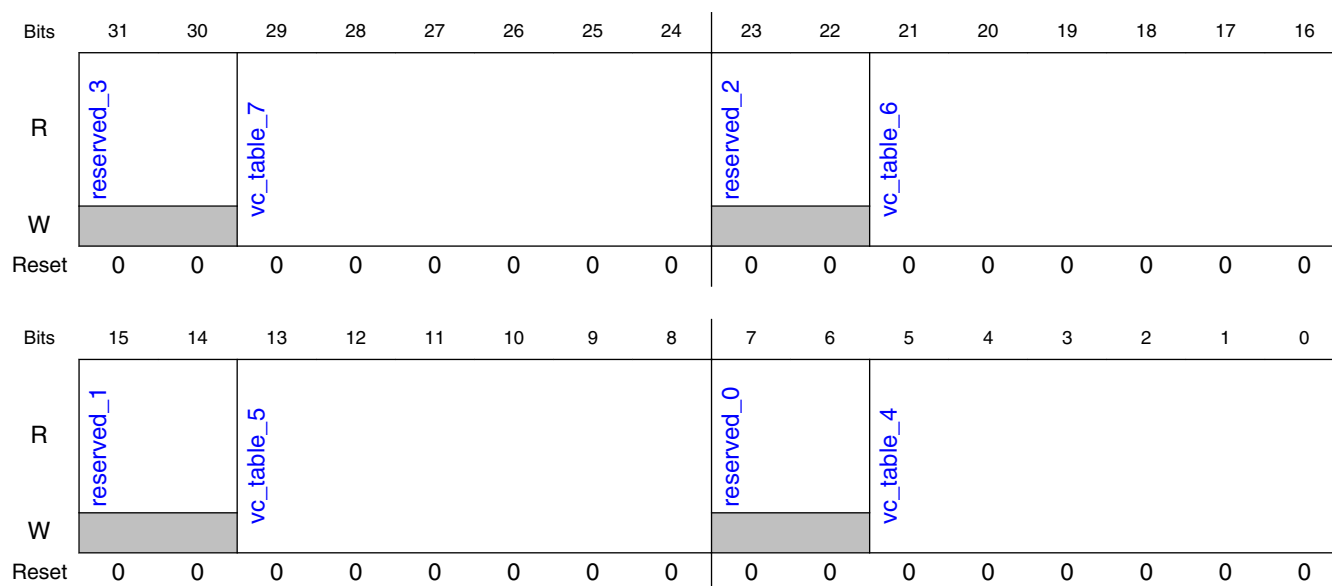
Field	Function
vc_table_1	
7-6 reserved_0	
5-0 vc_table_0	VC Table value for time slot 0

13.4.10.1.118 VC table values 4-7 (MST feature) (DP_VC_TABLE_1)

13.4.10.1.118.1 Offset

Register	Offset
DP_VC_TABLE_1	221Ch

13.4.10.1.118.2 Diagram



13.4.10.1.118.3 Fields

Field	Function
31-30 reserved_3	
29-24	VC Table value for time slot 7

Table continues on the next page...

Clocks And Resets

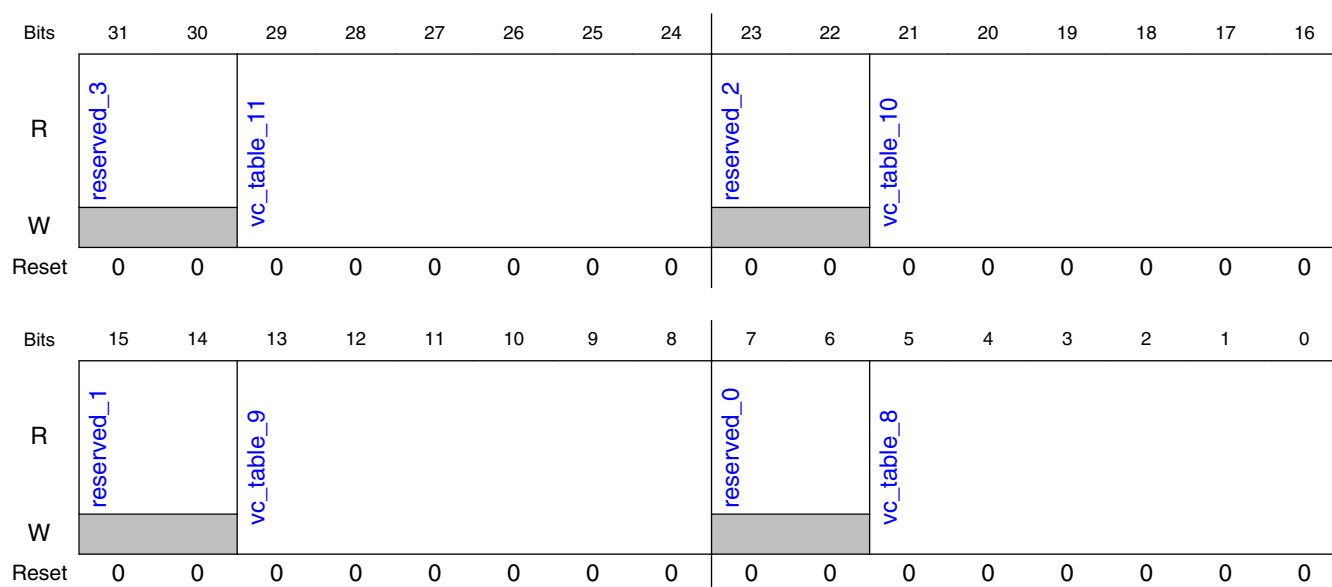
Field	Function
vc_table_7	
23-22 reserved_2	
21-16 vc_table_6	VC Table value for time slot 6
15-14 reserved_1	
13-8 vc_table_5	VC Table value for time slot 5
7-6 reserved_0	
5-0 vc_table_4	VC Table value for time slot 4

13.4.10.1.119 VC table values 8-11 (MST feature) (DP_VC_TABLE_2)

13.4.10.1.119.1 Offset

Register	Offset
DP_VC_TABLE_2	2220h

13.4.10.1.119.2 Diagram



13.4.10.1.119.3 Fields

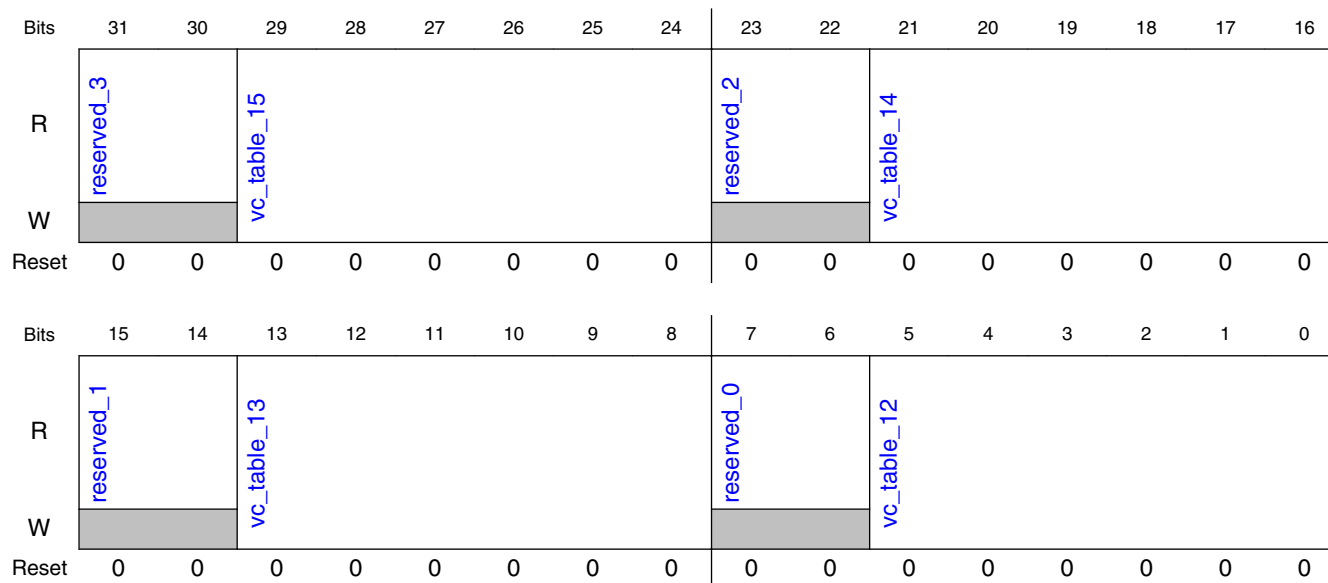
Field	Function
31-30 reserved_3	
29-24 vc_table_11	VC Table value for time slot 11
23-22 reserved_2	
21-16 vc_table_10	VC Table value for time slot 10
15-14 reserved_1	
13-8 vc_table_9	VC Table value for time slot 9
7-6 reserved_0	
5-0 vc_table_8	VC Table value for time slot 8

13.4.10.1.120 VC table values 12-15 (MST feature) (DP_VC_TABLE_3)

13.4.10.1.120.1 Offset

Register	Offset
DP_VC_TABLE_3	2224h

13.4.10.1.120.2 Diagram



13.4.10.1.120.3 Fields

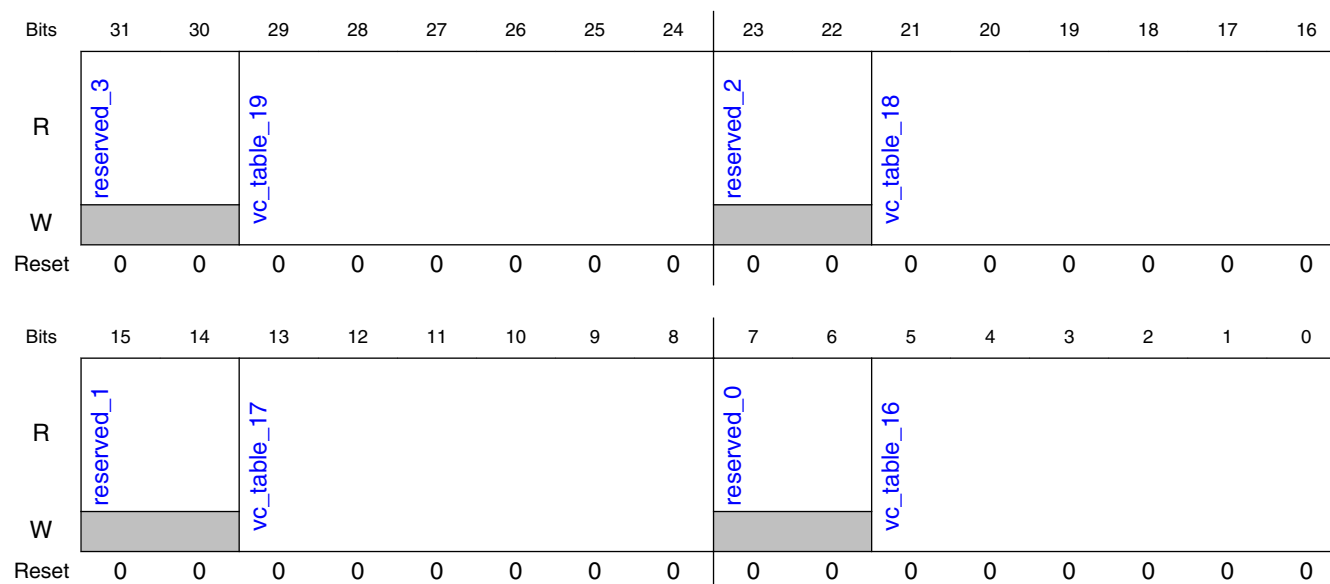
Field	Function
31-30 reserved_3	
29-24 vc_table_15	VC Table value for time slot 15
23-22 reserved_2	
21-16 vc_table_14	VC Table value for time slot 14
15-14 reserved_1	
13-8 vc_table_13	VC Table value for time slot 13
7-6 reserved_0	
5-0 vc_table_12	VC Table value for time slot 12

13.4.10.1.121 VC table values 16-19 (MST feature) (DP_VC_TABLE_4)

13.4.10.1.121.1 Offset

Register	Offset
DP_VC_TABLE_4	2228h

13.4.10.1.121.2 Diagram



13.4.10.1.121.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_19	VC Table value for time slot 19
23-22 reserved_2	
21-16 vc_table_18	VC Table value for time slot 18
15-14 reserved_1	
13-8 vc_table_17	VC Table value for time slot 17
7-6 reserved_0	

Table continues on the next page...

Clocks And Resets

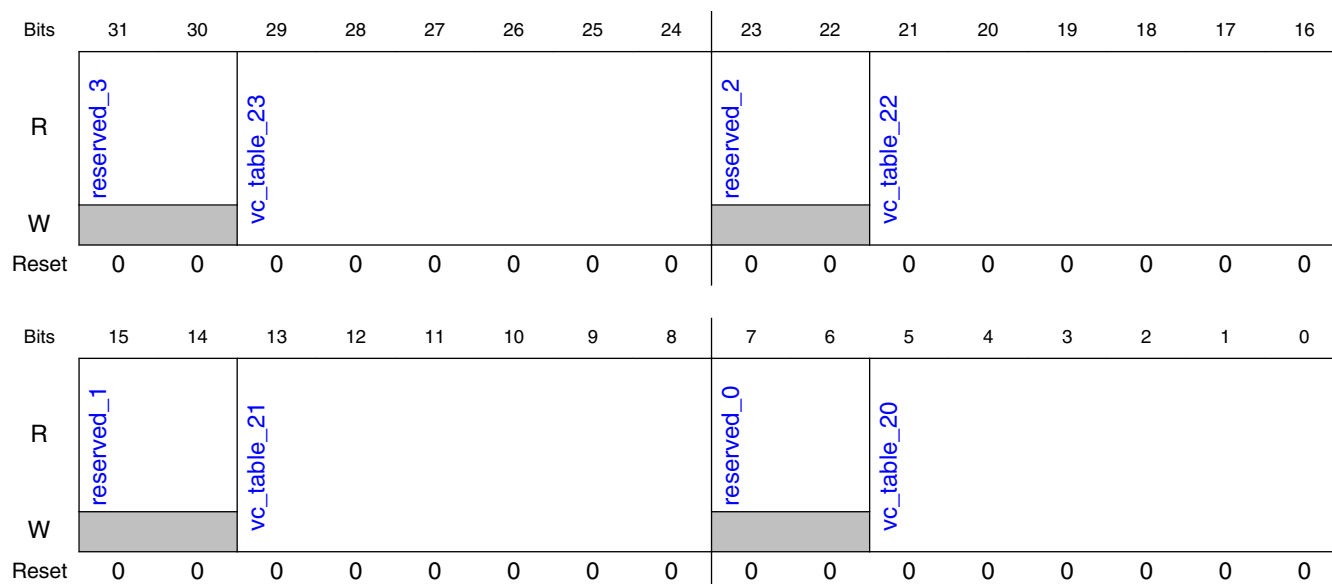
Field	Function
5-0 vc_table_16	VC Table value for time slot 16

13.4.10.1.122 VC table values 20-23 (MST feature) (DP_VC_TABLE_5)

13.4.10.1.122.1 Offset

Register	Offset
DP_VC_TABLE_5	222Ch

13.4.10.1.122.2 Diagram



13.4.10.1.122.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_23	VC Table value for time slot 23
23-22 reserved_2	

Table continues on the next page...

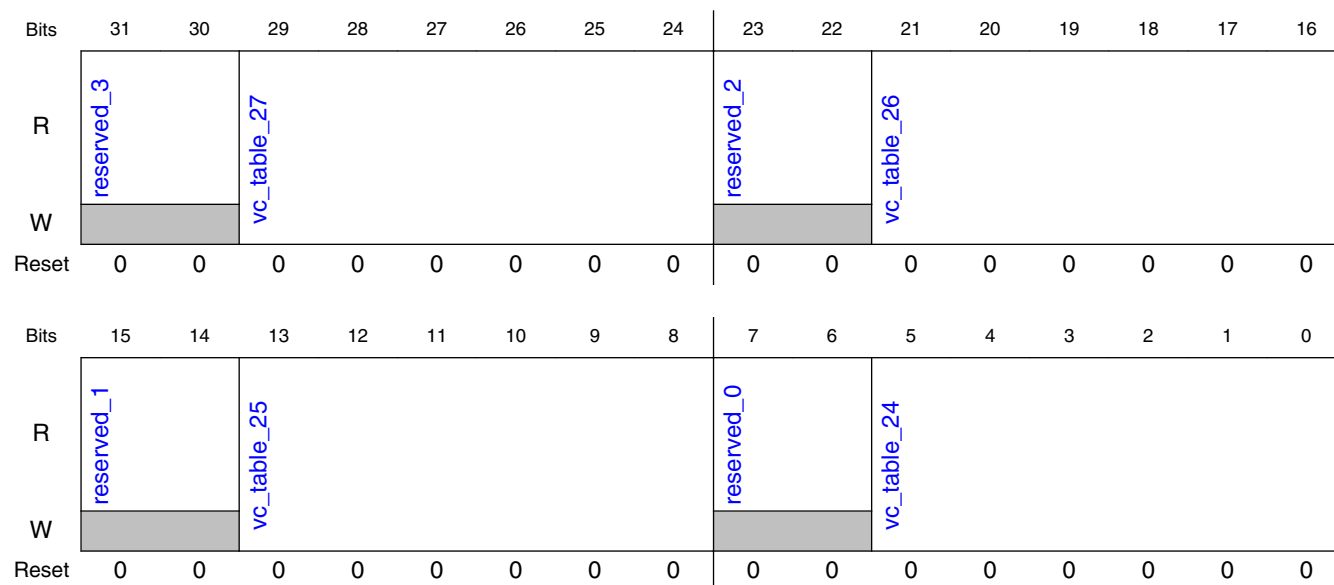
Field	Function
21-16 vc_table_22	VC Table value for time slot 22
15-14 reserved_1	
13-8 vc_table_21	VC Table value for time slot 21
7-6 reserved_0	
5-0 vc_table_20	VC Table value for time slot 20

13.4.10.1.123 VC table values 24-27 (MST feature) (DP_VC_TABLE_6)

13.4.10.1.123.1 Offset

Register	Offset
DP_VC_TABLE_6	2230h

13.4.10.1.123.2 Diagram



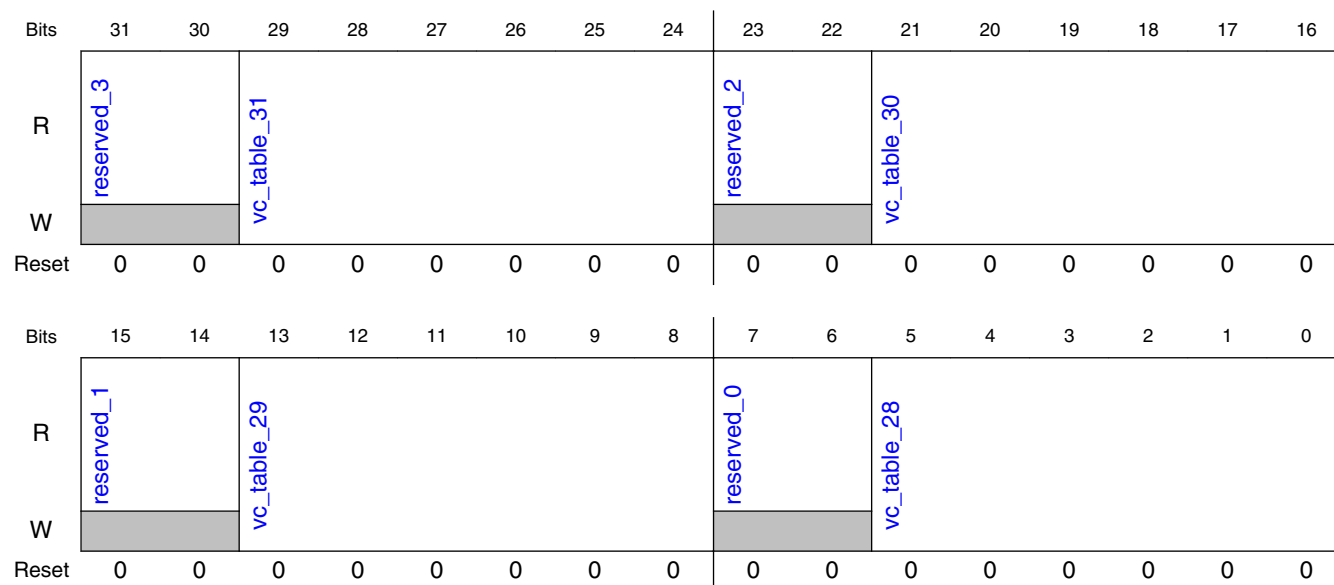
13.4.10.1.123.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_27	VC Table value for time slot 27
23-22 reserved_2	
21-16 vc_table_26	VC Table value for time slot 26
15-14 reserved_1	
13-8 vc_table_25	VC Table value for time slot 25
7-6 reserved_0	
5-0 vc_table_24	VC Table value for time slot 24

13.4.10.1.124 VC table values 28-31 (MST feature) (DP_VC_TABLE_7)**13.4.10.1.124.1 Offset**

Register	Offset
DP_VC_TABLE_7	2234h

13.4.10.1.124.2 Diagram



13.4.10.1.124.3 Fields

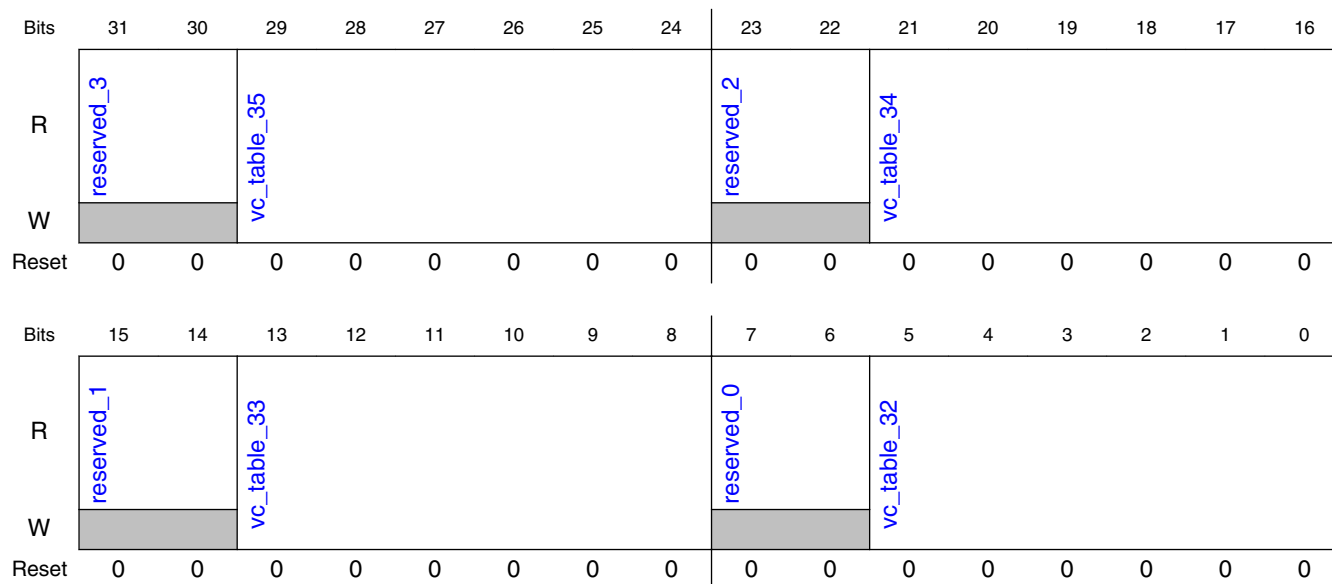
Field	Function
31-30 reserved_3	
29-24 vc_table_31	VC Table value for time slot 31
23-22 reserved_2	
21-16 vc_table_30	VC Table value for time slot 30
15-14 reserved_1	
13-8 vc_table_29	VC Table value for time slot 29
7-6 reserved_0	
5-0 vc_table_28	VC Table value for time slot 28

13.4.10.1.125 VC table values 32-35 (MST feature) (DP_VC_TABLE_8)

13.4.10.1.125.1 Offset

Register	Offset
DP_VC_TABLE_8	2238h

13.4.10.1.125.2 Diagram



13.4.10.1.125.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_35	VC Table value for time slot 35
23-22 reserved_2	
21-16 vc_table_34	VC Table value for time slot 34
15-14 reserved_1	
13-8 vc_table_33	VC Table value for time slot 33
7-6 reserved_0	

Table continues on the next page...

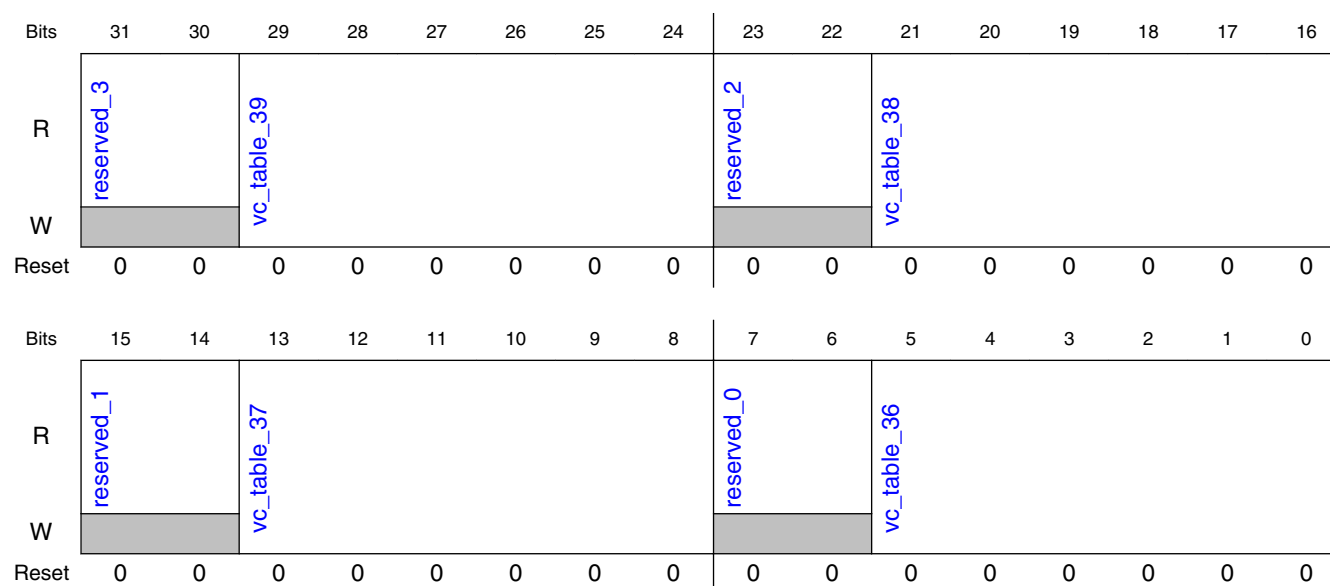
Field	Function
5-0 vc_table_32	VC Table value for time slot 32

13.4.10.1.126 VC table values 36-39 (MST feature) (DP_VC_TABLE_9)

13.4.10.1.126.1 Offset

Register	Offset
DP_VC_TABLE_9	223Ch

13.4.10.1.126.2 Diagram



13.4.10.1.126.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_39	VC Table value for time slot 39
23-22 reserved_2	

Table continues on the next page...

Clocks And Resets

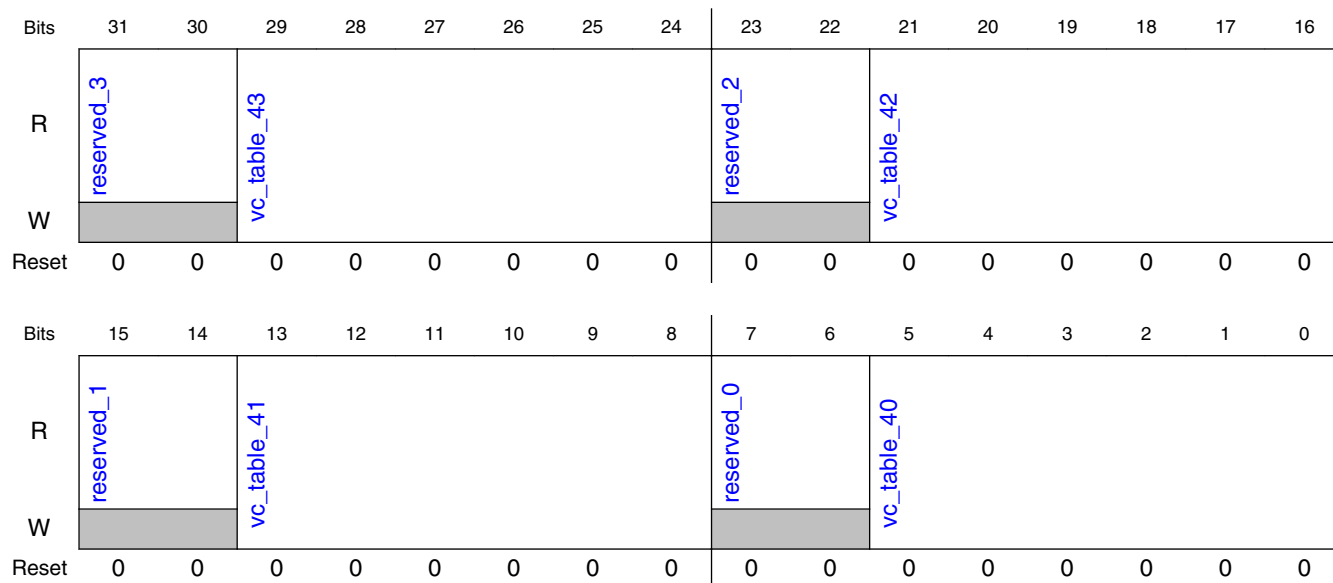
Field	Function
21-16 vc_table_38	VC Table value for time slot 38
15-14 reserved_1	
13-8 vc_table_37	VC Table value for time slot 37
7-6 reserved_0	
5-0 vc_table_36	VC Table value for time slot 36

13.4.10.1.127 VC table values 40-43 (MST feature) (DP_VC_TABLE_10)

13.4.10.1.127.1 Offset

Register	Offset
DP_VC_TABLE_10	2240h

13.4.10.1.127.2 Diagram



13.4.10.1.127.3 Fields

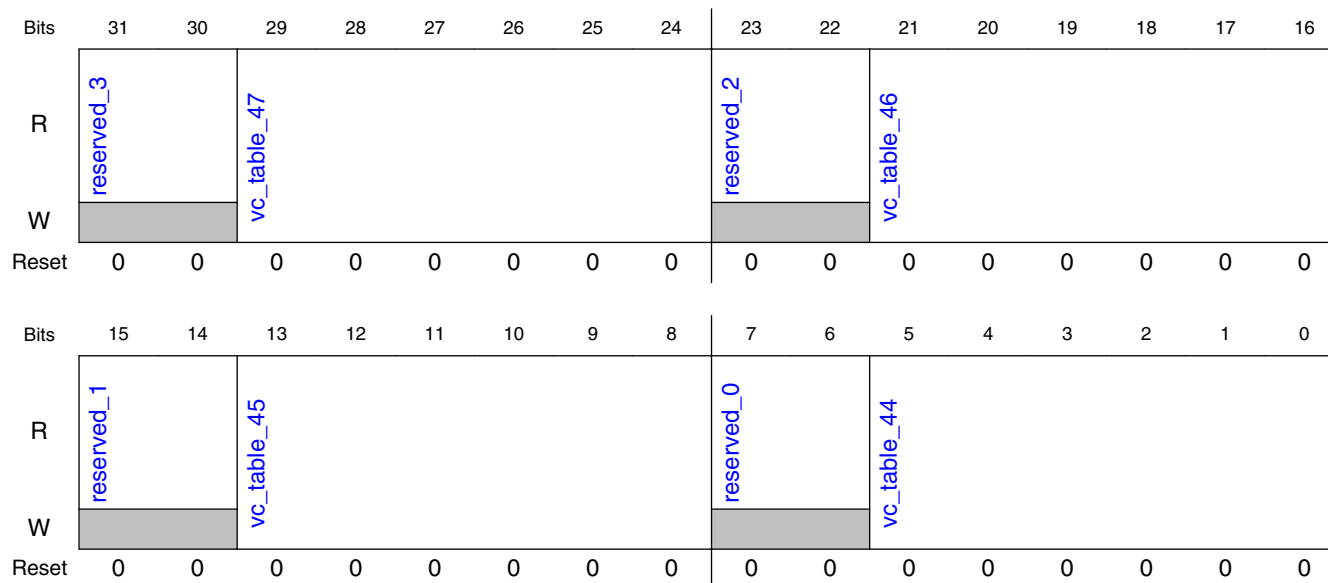
Field	Function
31-30 reserved_3	
29-24 vc_table_43	VC Table value for time slot 43
23-22 reserved_2	
21-16 vc_table_42	VC Table value for time slot 42
15-14 reserved_1	
13-8 vc_table_41	VC Table value for time slot 41
7-6 reserved_0	
5-0 vc_table_40	VC Table value for time slot 40

13.4.10.1.128 VC table values 44-47 (MST feature) (DP_VC_TABLE_11)

13.4.10.1.128.1 Offset

Register	Offset
DP_VC_TABLE_11	2244h

13.4.10.1.128.2 Diagram



13.4.10.1.128.3 Fields

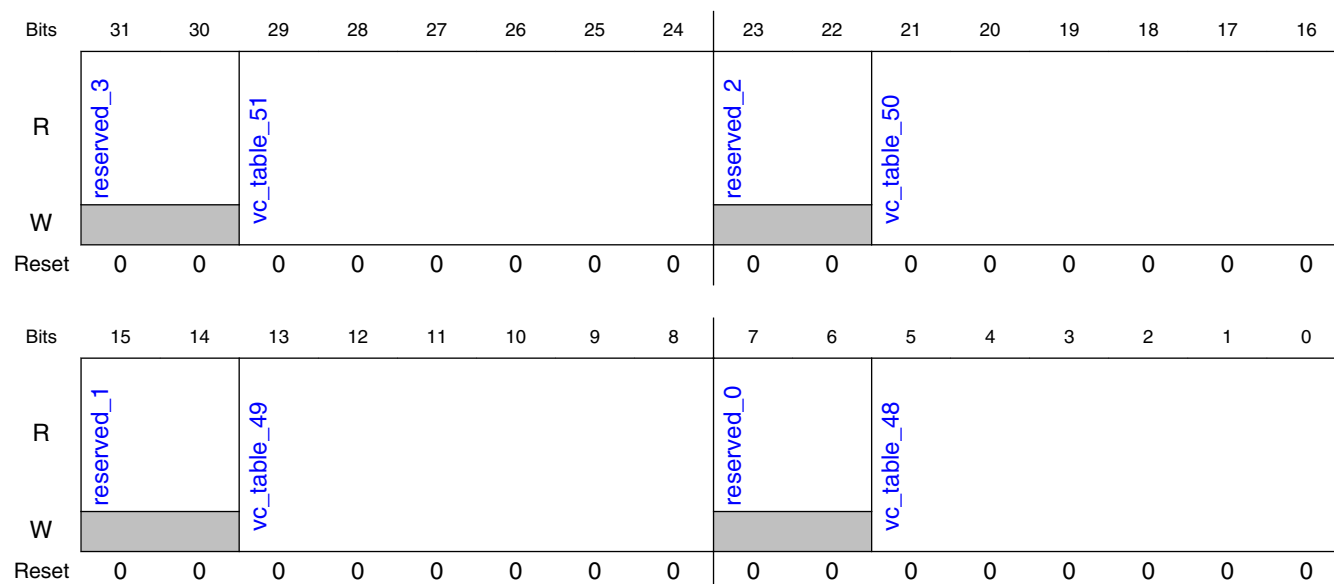
Field	Function
31-30 reserved_3	
29-24 vc_table_47	VC Table value for time slot 47
23-22 reserved_2	
21-16 vc_table_46	VC Table value for time slot 46
15-14 reserved_1	
13-8 vc_table_45	VC Table value for time slot 45
7-6 reserved_0	
5-0 vc_table_44	VC Table value for time slot 44

13.4.10.1.129 VC table values 48-51 (MST feature) (DP_VC_TABLE_12)

13.4.10.1.129.1 Offset

Register	Offset
DP_VC_TABLE_12	2248h

13.4.10.1.129.2 Diagram



13.4.10.1.129.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_51	VC Table value for time slot 51
23-22 reserved_2	
21-16 vc_table_50	VC Table value for time slot 50
15-14 reserved_1	
13-8 vc_table_49	VC Table value for time slot 49
7-6 reserved_0	

Table continues on the next page...

Clocks And Resets

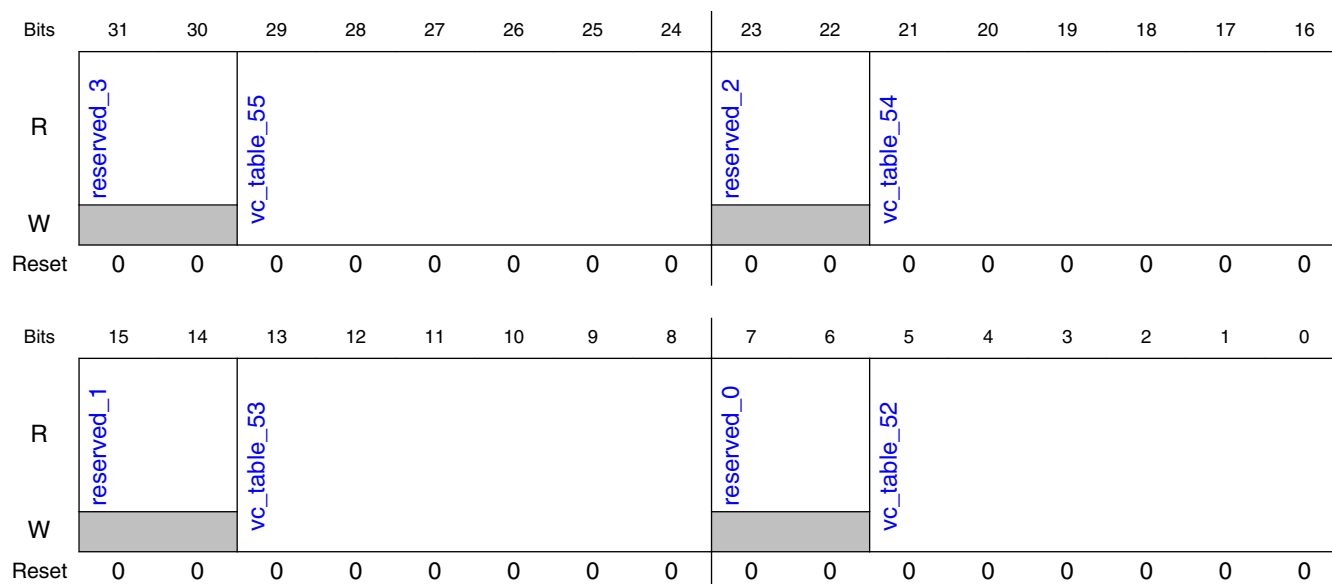
Field	Function
5-0 vc_table_48	VC Table value for time slot 48

13.4.10.1.130 VC table values 52-55 (MST feature) (DP_VC_TABLE_13)

13.4.10.1.130.1 Offset

Register	Offset
DP_VC_TABLE_13	224Ch

13.4.10.1.130.2 Diagram



13.4.10.1.130.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_55	VC Table value for time slot 55
23-22 reserved_2	

Table continues on the next page...

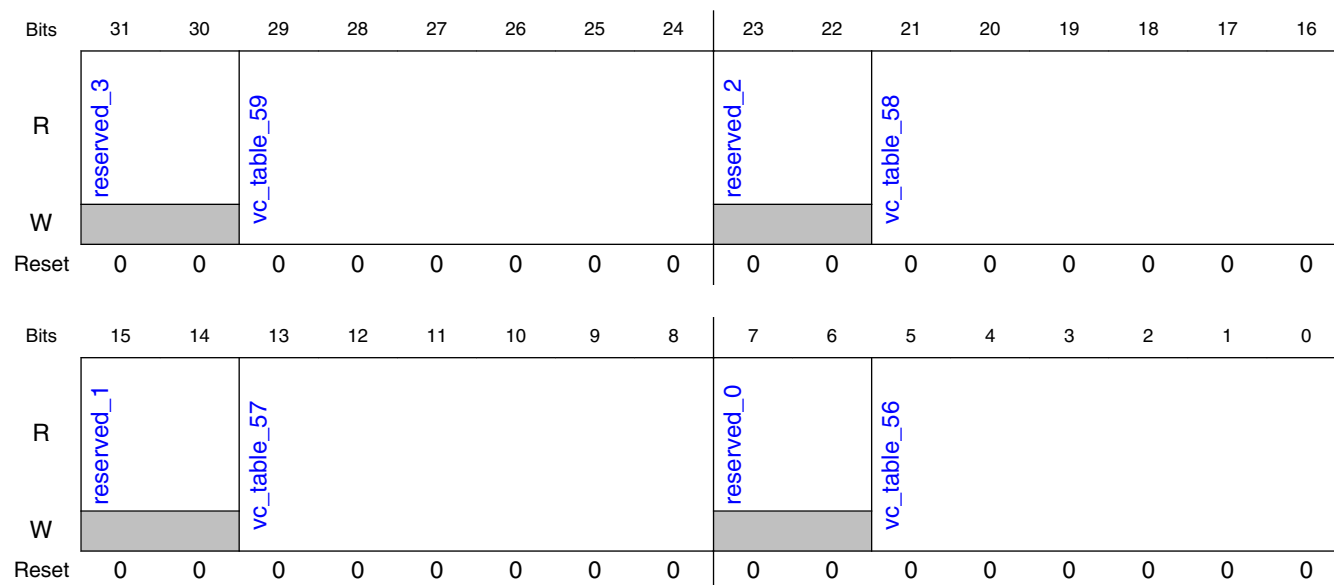
Field	Function
21-16 vc_table_54	VC Table value for time slot 54
15-14 reserved_1	
13-8 vc_table_53	VC Table value for time slot 53
7-6 reserved_0	
5-0 vc_table_52	VC Table value for time slot 52

13.4.10.1.131 VC table values 56-59 (MST feature) (DP_VC_TABLE_14)

13.4.10.1.131.1 Offset

Register	Offset
DP_VC_TABLE_14	2250h

13.4.10.1.131.2 Diagram



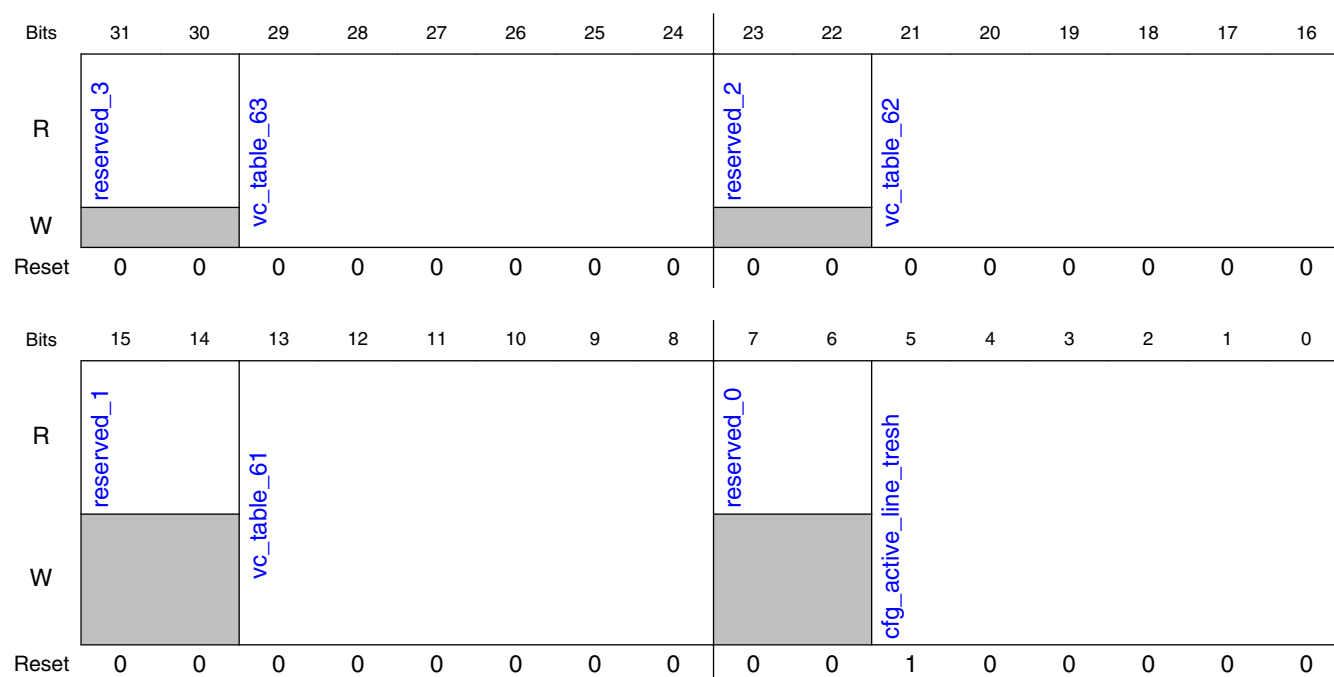
13.4.10.1.131.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_59	VC Table value for time slot 59
23-22 reserved_2	
21-16 vc_table_58	VC Table value for time slot 58
15-14 reserved_1	
13-8 vc_table_57	VC Table value for time slot 57
7-6 reserved_0	
5-0 vc_table_56	VC Table value for time slot 56

13.4.10.1.132 Video FIFO latency threshold (LINE_THRESH)**13.4.10.1.132.1 Offset**

Register	Offset
LINE_THRESH	2254h

13.4.10.1.132.2 Diagram



13.4.10.1.132.3 Fields

Field	Function
31-30 reserved_3	
29-24 vc_table_63	VC Table value for time slot 63
23-22 reserved_2	
21-16 vc_table_62	VC Table value for time slot 62
15-14 reserved_1	
13-8 vc_table_61	VC Table value for time slot 61
7-6 reserved_0	
5-0 cfg_active_line_tresh	Video Fifo Latency threshold. Defines the number of FIFO rows before reading starts.

13.4.10.1.133 Vertical blanking ID (DP_VB_ID)

13.4.10.1.133.1 Offset

Register	Offset
DP_VB_ID	2258h

13.4.10.1.133.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								vb_id							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1

13.4.10.1.133.3 Fields

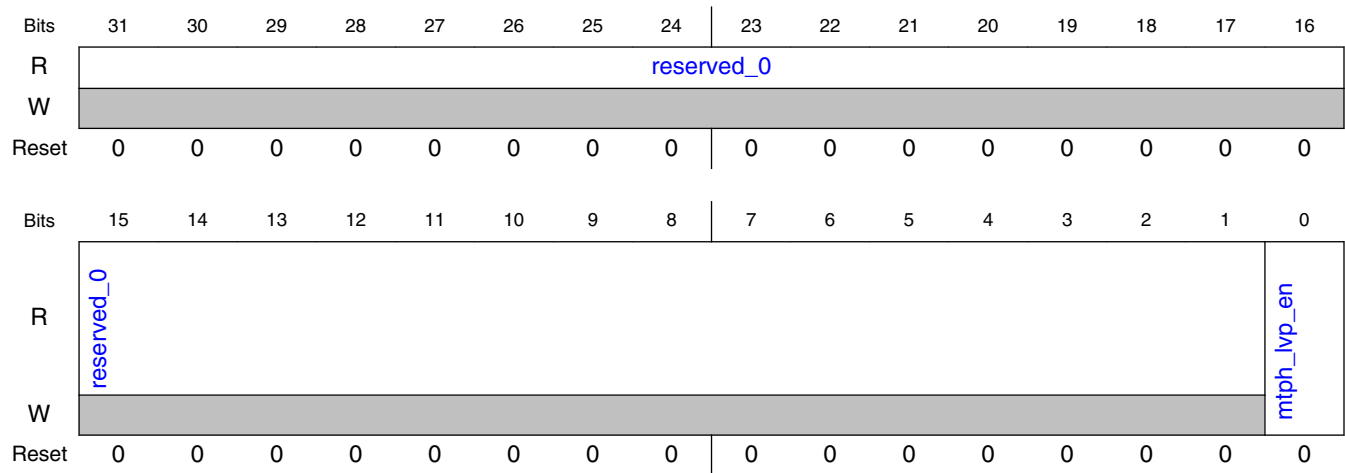
Field	Function
31-8 reserved_0	
7-0 vb_id	Vertical blanking ID

13.4.10.1.134 LVP insertion in MTP header enable (MST feature) (DP_MTPH_LVP_CONTROL)

13.4.10.1.134.1 Offset

Register	Offset
DP_MTPH_LVP_CONTROL	225Ch

13.4.10.1.134.2 Diagram



13.4.10.1.134.3 Fields

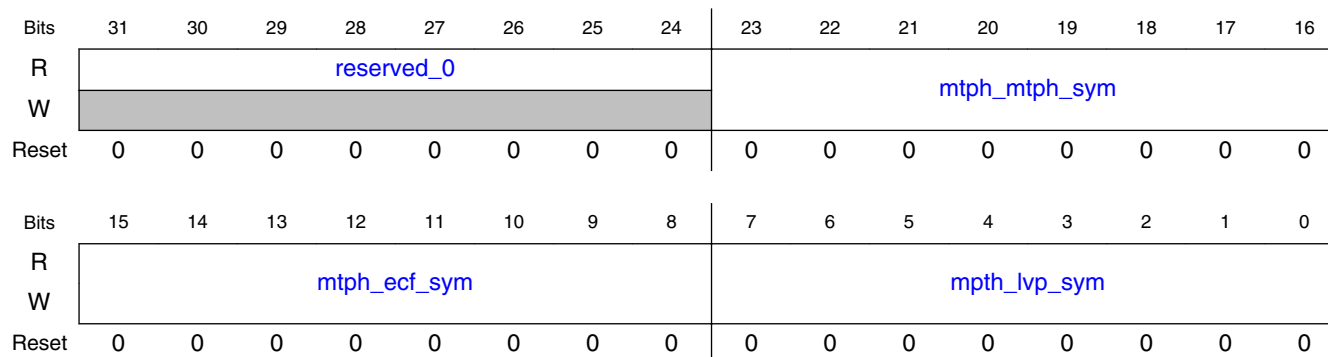
Field	Function
31-1 reserved_0	
0 mtp_h_lvp_en	This bit is used in MST mode. This bit is used in MST mode. When set high insertion of LVP symbols in MTP Header time slots is enabled.

13.4.10.1.135 Values to be inserted in the MTP header (MST feature) (DP_MTPH_SYMBOL_VALUES)

13.4.10.1.135.1 Offset

Register	Offset
DP_MTPH_SYMBOL_VALUES	2260h

13.4.10.1.135.2 Diagram



13.4.10.1.135.3 Fields

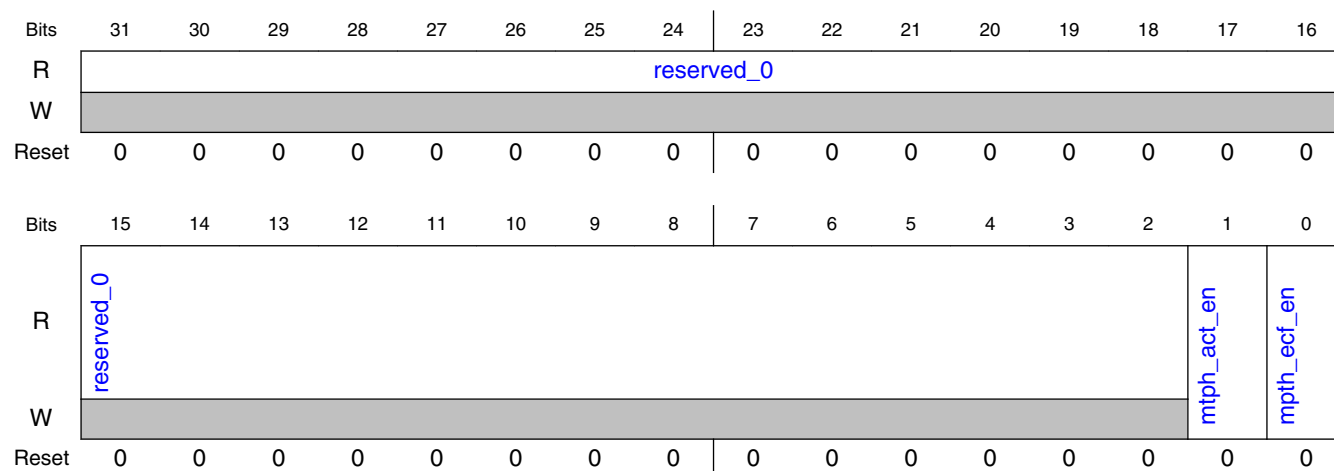
Field	Function
31-24 reserved_0	
23-16 mtph_mtph_sym	Value of this vector is inserted in MTP Header time slot when neither SR, ACT, LVP or ECF symbols are inserted in MTP Header time slot. Value of this vector is inserted in MTP Header time slot when neither SR, ACT, LVP or ECF symbols are inserted in MTP Header time slot.
15-8 mtph_ecf_sym	Value of this vector is inserted in MTP Header time slot when mtph_ecf_en signal is active
7-0 mpth_lvp_sym	Value of this vector is inserted in MTP Header time slot when mpth_lvp_en signal is active

13.4.10.1.136 ECF insertion in MTP header enable (MST feature) (DP_MTPH_ECF_CONTROL)

13.4.10.1.136.1 Offset

Register	Offset
DP_MTPH_ECF_CONTROL	2264h

13.4.10.1.136.2 Diagram



13.4.10.1.136.3 Fields

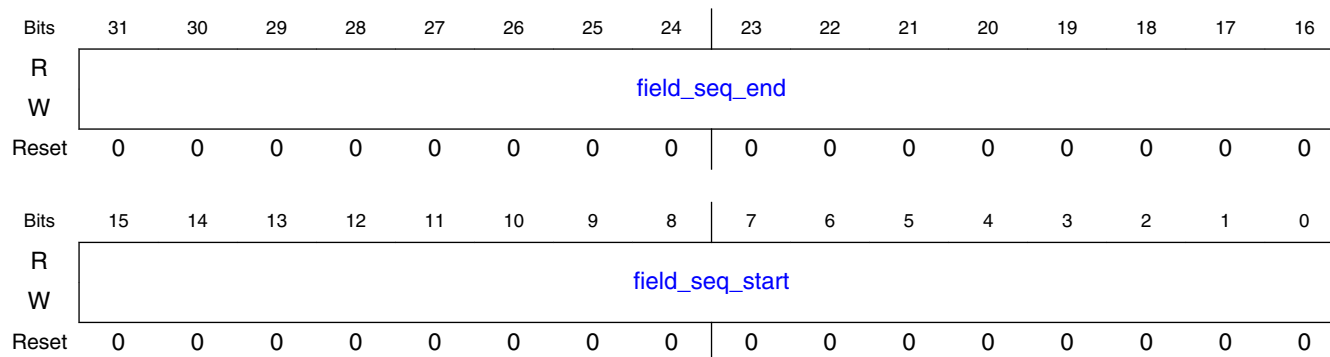
Field	Function
31-2 reserved_0	
1 mtph_act_en	MST feature
0 mtph_ecf_en	This bit is used in MST mode. This bit is used in MST mode. When set high insertion of ECF symbols in MTP Header time slots is enabled.

13.4.10.1.137 Supporting configuration to switch from top/bottom on the input to field sequential on the output (DP_FIELDSEQ_3D)

13.4.10.1.137.1 Offset

Register	Offset
DP_FIELDSEQ_3D	2268h

13.4.10.1.137.2 Diagram



13.4.10.1.137.3 Fields

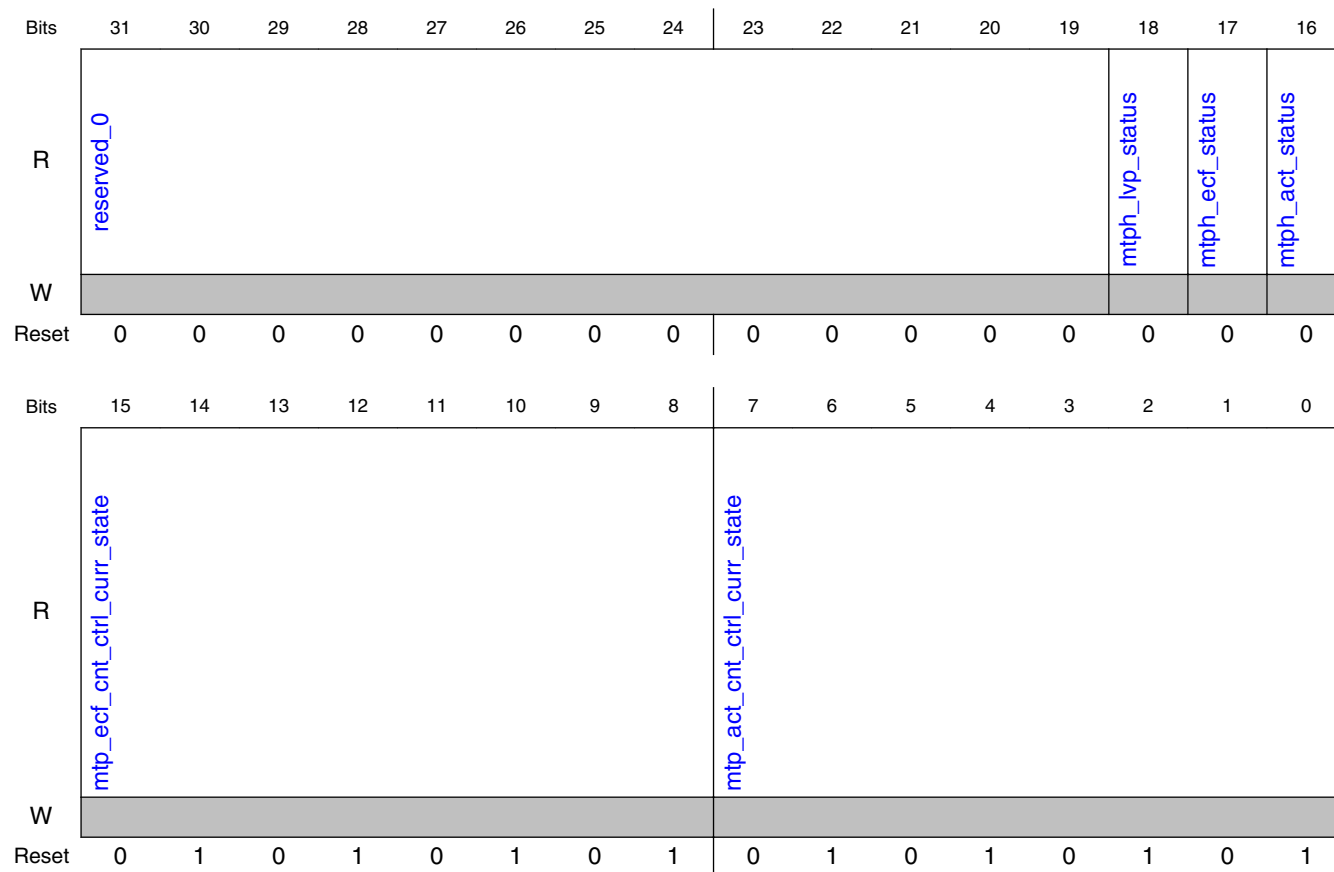
Field	Function
31-16 field_seq_end	Number of line in the frame where the Vblank part in the field sequential format ends
15-0 field_seq_start	Number of line in the frame where the Vblank part in the field sequential format starts

13.4.10.1.138 MTP header status (MST feature) (DP_MTPH_STATUS)

13.4.10.1.138.1 Offset

Register	Offset
DP_MTPH_STATUS	226Ch

13.4.10.1.138.2 Diagram



13.4.10.1.138.3 Fields

Field	Function
31-19 reserved_0	
18 mtp_h_lvp_status	Status of mtp_h_lvp
17 mtp_h_ecf_status	Status of mtp_h_ecf
16 mtp_h_act_status	Status of mtp_h_act
15-8 mtp_ecf_cnt_ctrl_curr_state	mtp_ecf fsm current state
7-0 mtp_act_cnt_ctrl_curr_state	mtp_act fsm current state

13.4.10.1.139 Interrupt sources of the framer module, active high. (DP_INTERRUPT_SOURCE)

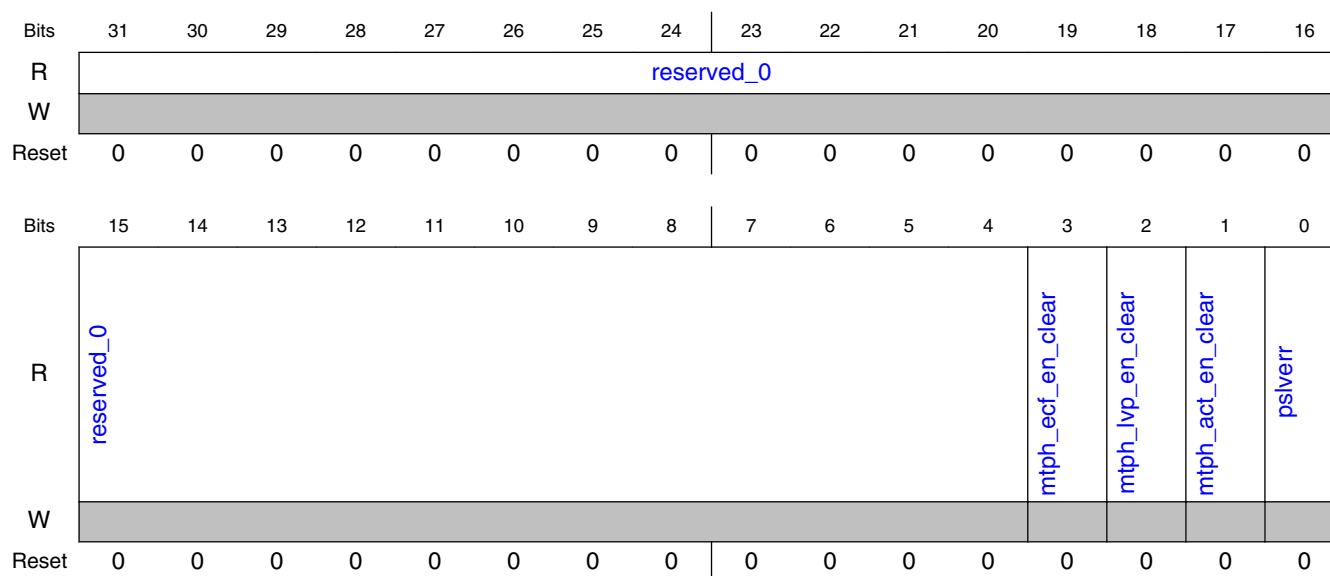
13.4.10.1.139.1 Offset

Register	Offset
DP_INTERRUPT_SOURCE	2270h

13.4.10.1.139.2 Function

Interrupt sources of the framer module, active high. Automatically cleared on read.

13.4.10.1.139.3 Diagram



13.4.10.1.139.4 Fields

Field	Function
31-4 reserved_0	
3 mtph_ecf_en_clear	Indication to clear mtph_ecf_en field in configuration register, active high

Table continues on the next page...

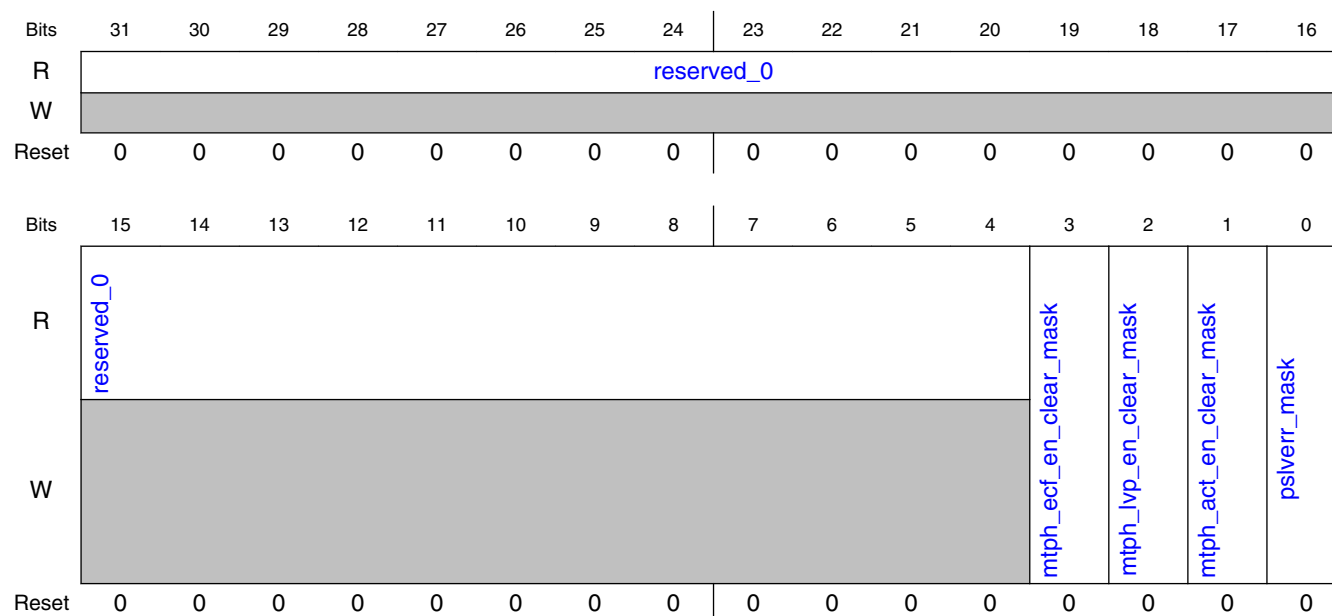
Field	Function
2 mtp_h_lvp_en_clear	Indication to clear mtp_h_lvp_en field in configuration register, active high
1 mtp_h_act_en_clear	Indication to clear mtp_h_act_en field in configuration register, active high
0 pslverr	APB slave error

13.4.10.1.140 Masks for the interrupt sources in the DP_INTERRUPT_SOURCE register, when set high, these bits disable the corresponding interrupts (DP_INTERRUPT_MASK)

13.4.10.1.140.1 Offset

Register	Offset
DP_INTERRUPT_MASK	2274h

13.4.10.1.140.2 Diagram



13.4.10.1.140.3 Fields

Field	Function
31-4 reserved_0	
3 mtph_ecf_en_clear_mask	Bit that masks mtph_ecf_en_clear interrupt
2 mtph_lvp_en_clear_mask	Bit that masks mtph_lvp_en_clear interrupt
1 mtph_act_en_clear_mask	Bit that masks mtph_act_en_clear interrupt
0 pslverr_mask	Bit that masks pslverr interrupt

13.4.10.1.141 MSA parameters to be used in the HW (DP_FRONT_BACK_PORCH)

13.4.10.1.141.1 Offset

Register	Offset
DP_FRONT_BACK_PORCH	2278h

13.4.10.1.141.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	front_porch															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	back_porch															
W																
Reset	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0

13.4.10.1.141.3 Fields

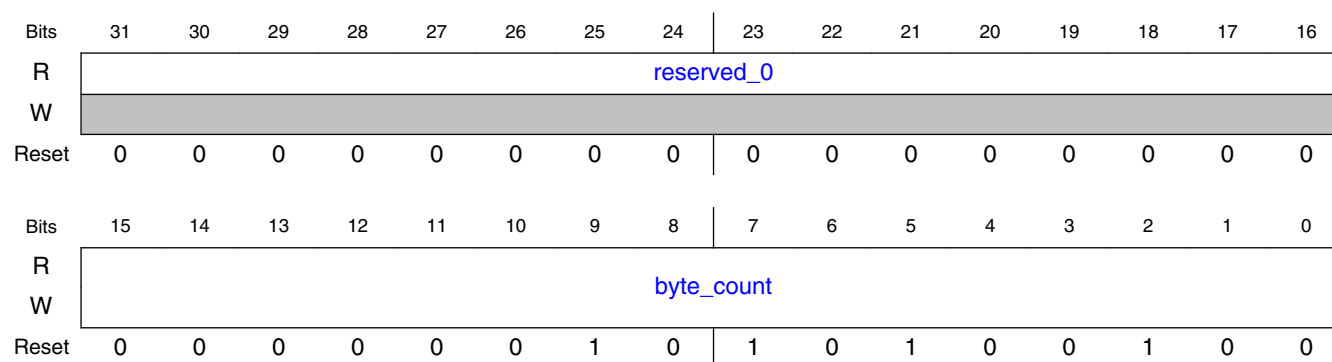
Field	Function
31-16 front_porch	Value of the front porch
15-0 back_porch	Value of the back porch

13.4.10.1.142 Number of bytes per lane/chunk parameters (DP_BYTE_COUNT)

13.4.10.1.142.1 Offset

Register	Offset
DP_BYTE_COUNT	227Ch

13.4.10.1.142.2 Diagram



13.4.10.1.142.3 Fields

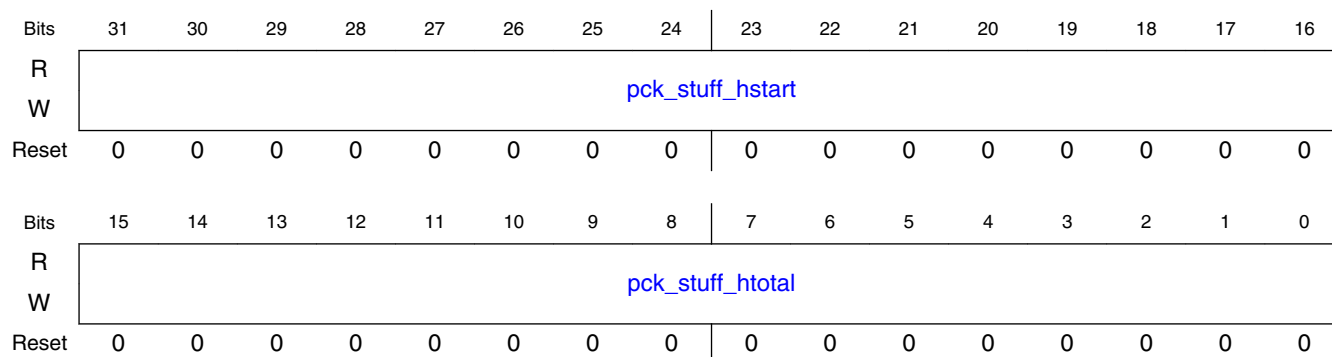
Field	Function
31-16 reserved_0	
15-0 byte_count	Total number of bytes in a line in case of non-DSC video. Total number of bytes in a line in case of non-DSC video. When DSC is enabled should be total number of bytes in a line *per lane*, including the additional EOC symbol(s).

13.4.10.1.143 MSA horizontal parameters first part (MSA_HORIZONTAL_0)

13.4.10.1.143.1 Offset

Register	Offset
MSA_HORIZONTAL_0	2280h

13.4.10.1.143.2 Diagram



13.4.10.1.143.3 Fields

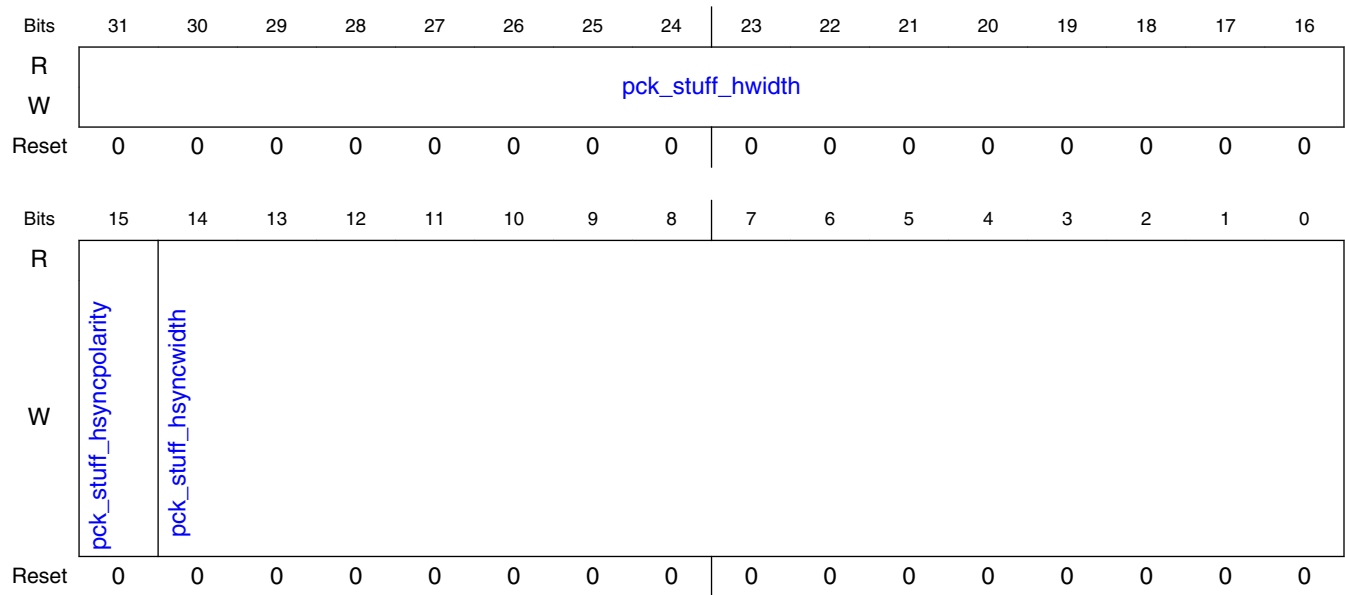
Field	Function
31-16 pck_stuff_hstart	Horizontal Active Start, index of pixel in a line where active region begins
15-0 pck_stuff_htotal	Horizontal Total, number of pixels per line

13.4.10.1.144 MSA horizontal parameters second part (MSA_HORIZONTAL_1)

13.4.10.1.144.1 Offset

Register	Offset
MSA_HORIZONTAL_1	2284h

13.4.10.1.144.2 Diagram



13.4.10.1.144.3 Fields

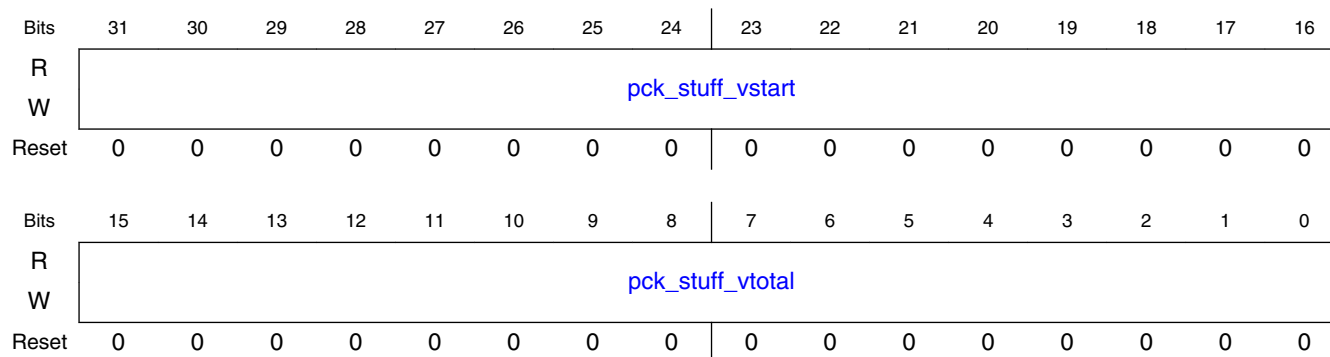
Field	Function
31-16 pck_stuff_hwid	Horizontal Active Video Width, number of active pixels per line
15 pck_stuff_hsync polarity	Horizontal Sync Polarity: 0 - active high, 1 - active low
14-0 pck_stuff_hsync width	Horizontal Sync Width, number of pixels Hsync is active

13.4.10.1.145 MSA vertical parameters first part (MSA_VERTICAL_0)

13.4.10.1.145.1 Offset

Register	Offset
MSA_VERTICAL_0	2288h

13.4.10.1.145.2 Diagram



13.4.10.1.145.3 Fields

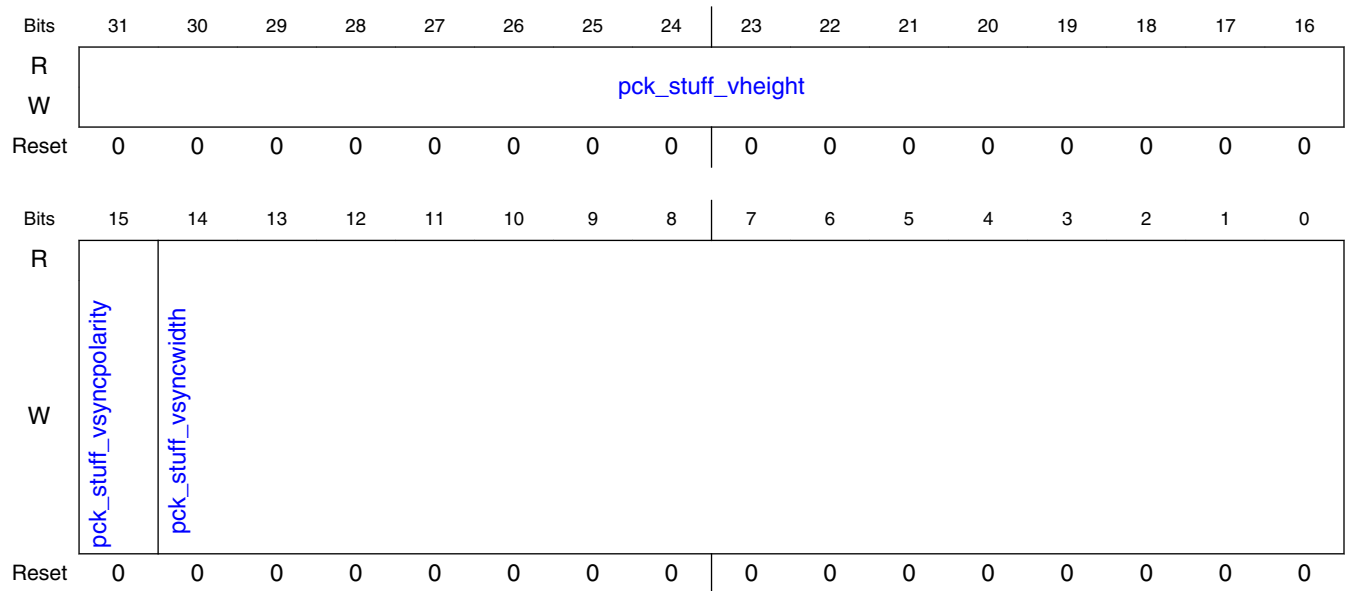
Field	Function
31-16 pck_stuff_vstart	Vertical Active Start, index of the first active line in a frame
15-0 pck_stuff_vtotal	Vertical Total, number of lines per frame

13.4.10.1.146 MSA vertical parameters second part (MSA_VERTICAL_1)

13.4.10.1.146.1 Offset

Register	Offset
MSA_VERTICAL_1	228Ch

13.4.10.1.146.2 Diagram



13.4.10.1.146.3 Fields

Field	Function
31-16 pck_stuff_vheight	Vertical Active Video Height, number of active lines in a frame
15 pck_stuff_vsyncpolarity	Vertical Sync Polarity, 0 - active high, 1- active low
14-0 pck_stuff_vsyncwidth	Vertical Sync Width, number of lines Vsync is active

13.4.10.1.147 MISC0 and MISC1 values of the MSA, as per DPv1. (MSA_MISC)

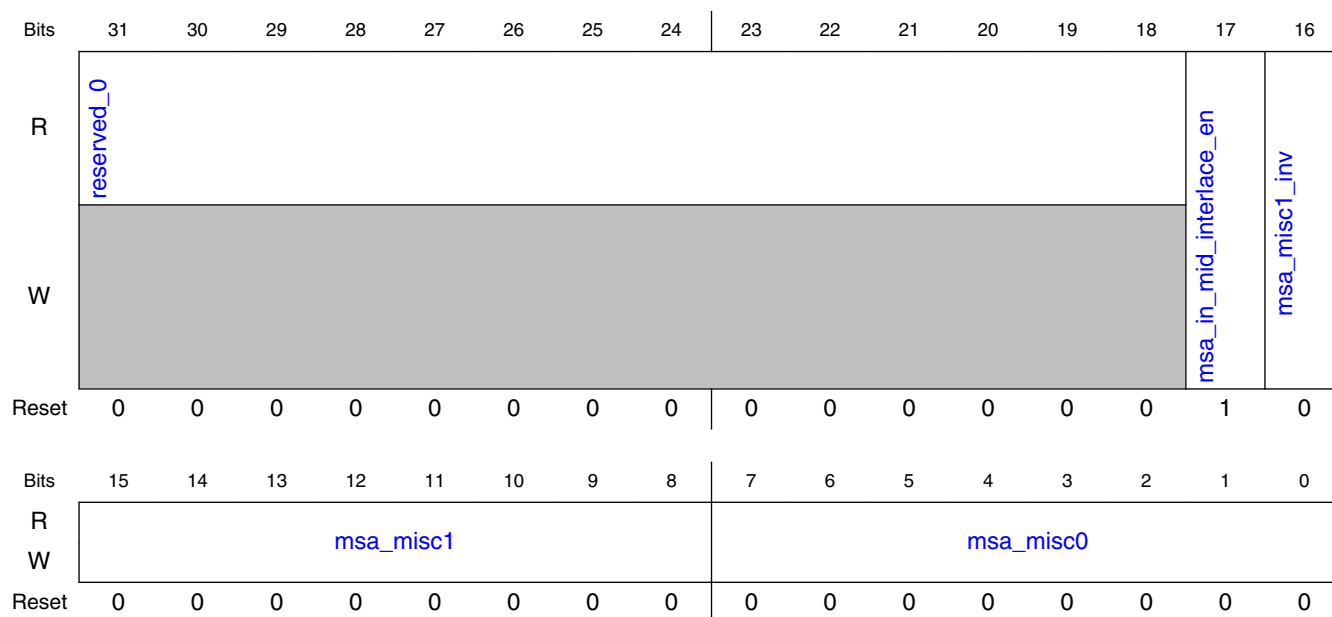
13.4.10.1.147.1 Offset

Register	Offset
MSA_MISC	2290h

13.4.10.1.147.2 Function

MISC0 and MISC1 values of the MSA, as per DPv1.4 standard

13.4.10.1.147.3 Diagram



13.4.10.1.147.4 Fields

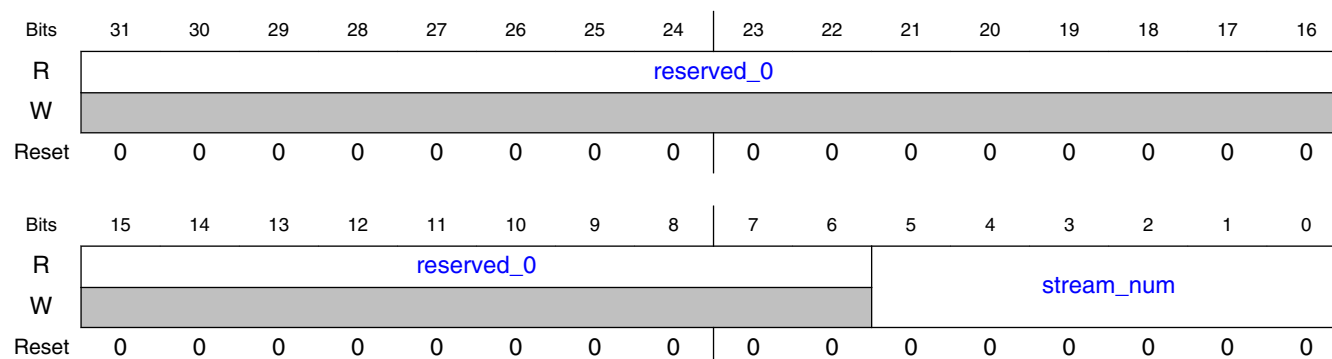
Field	Function
31-18 reserved_0	
17 msa_in_mid_interface_en	0 - enable transmission of MSA on each field , 1 - MSA is transmitted on Top only
16 msa_misc1_inv	L/R toggle for interlaced and field sequential video, 0 - left, 1 - right
15-8 msa_misc1	Miscellaneous1 value
7-0 msa_misc0	Miscellaneous0 value

13.4.10.1.148 MSA stream number (MST feature) (STREAM_CONFIG)

13.4.10.1.148.1 Offset

Register	Offset
STREAM_CONFIG	2294h

13.4.10.1.148.2 Diagram



13.4.10.1.148.3 Fields

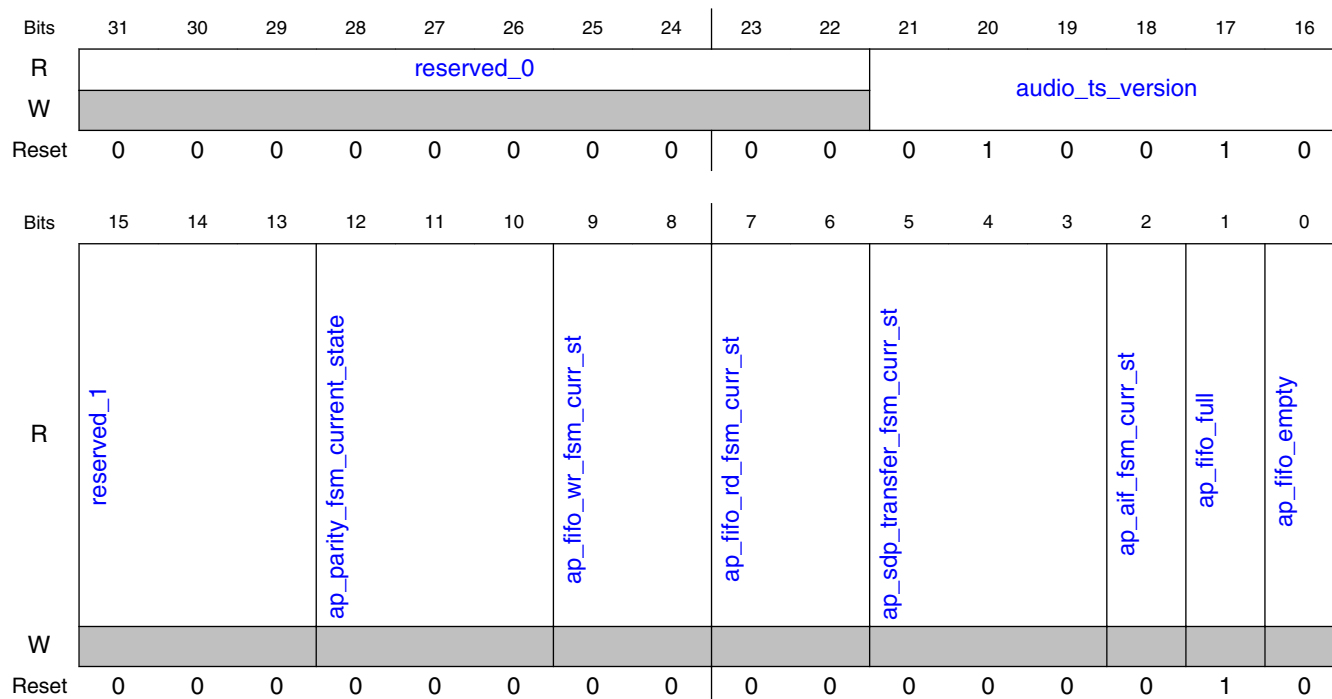
Field	Function
31-6 reserved_0	
5-0 stream_num	Number of current data stream. Number of current data stream.

13.4.10.1.149 Status signals for the audio pack (AUDIO_PACK_STATUS)

13.4.10.1.149.1 Offset

Register	Offset
AUDIO_PACK_STATUS	2298h

13.4.10.1.149.2 Diagram



13.4.10.1.149.3 Fields

Field	Function
31-22 reserved_0	
21-16 audio_ts_version	Audio timestamp version
15-13 reserved_1	
12-10 ap_parity_fsm_current_state	Audio pack parity calc fsm state
9-8 ap_fifo_wr_fsm_curr_st	Audio pack FIFO write fsm state
7-6 ap_fifo_rd_fsm_curr_st	Audio pack FIFO read fsm state
5-3 ap_sdp_transfer_fsm_curr_st	Audio pack sdp transfer fsm state

Table continues on the next page...

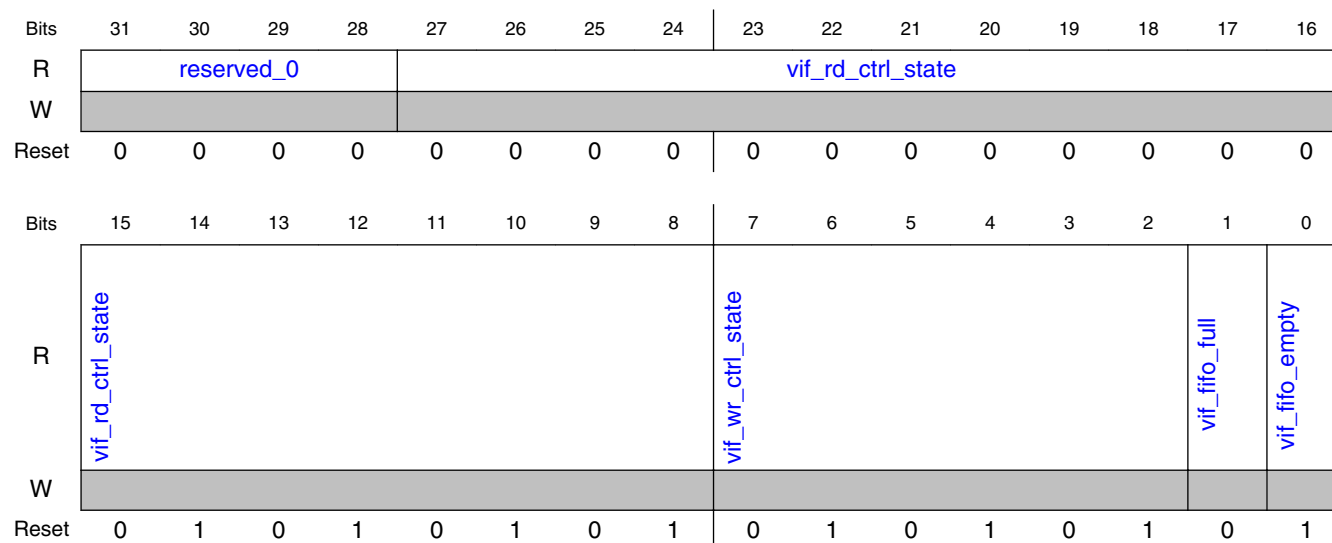
Field	Function
2 ap_aif_fsm_curr_st	Audio pack aif fsm state
1 ap_fifo_full	Audio Pack Sync FIFO full flag, active high
0 ap_fifo_empty	Audio Pack Sync FIFO empty flag, active high

13.4.10.1.150 Status signals for the VIF module (VIF_STATUS)

13.4.10.1.150.1 Offset

Register	Offset
VIF_STATUS	229Ch

13.4.10.1.150.2 Diagram



13.4.10.1.150.3 Fields

Field	Function
31-28 reserved_0	

Table continues on the next page...

Clocks And Resets

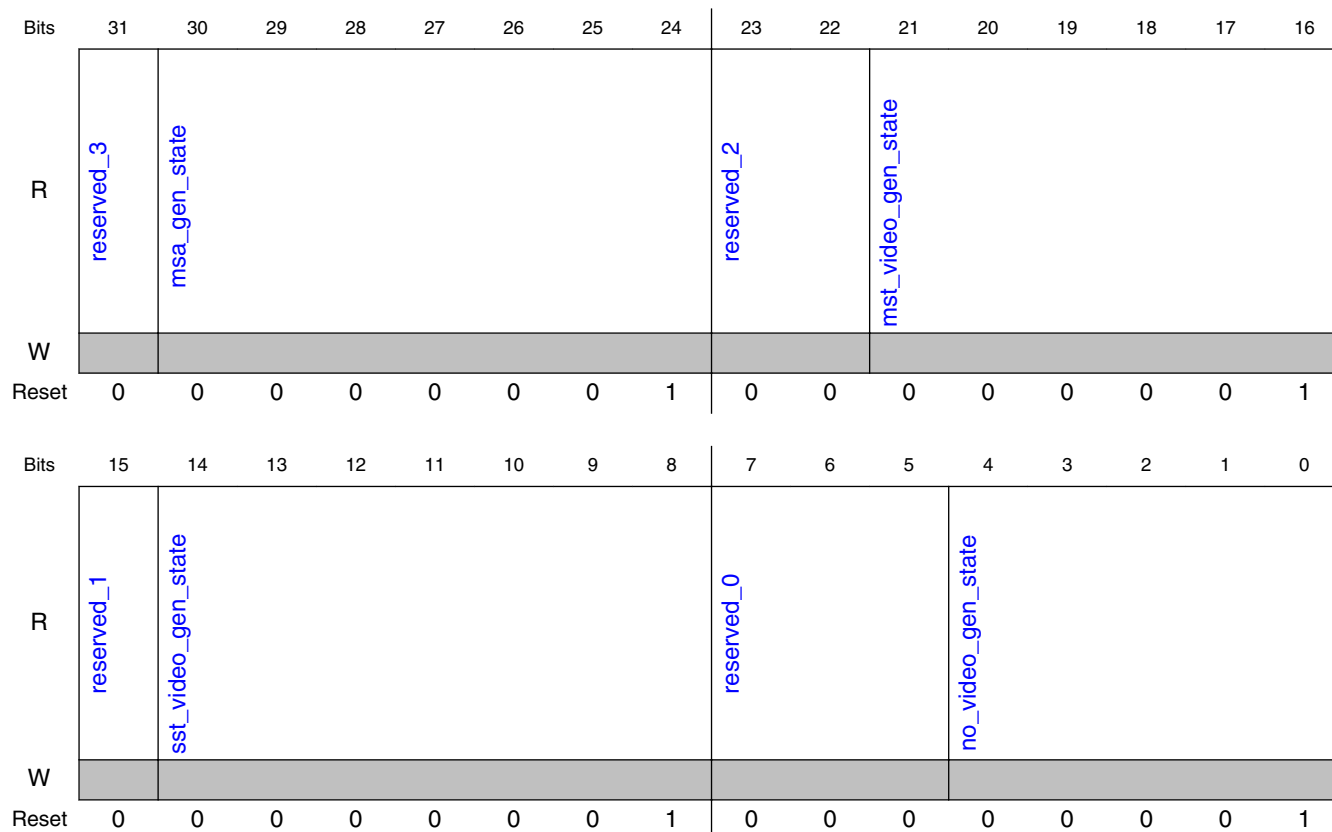
Field	Function
27-8 vif_rd_ctrl_state	VIF rd fsm current state
7-2 vif_wr_ctrl_state	VIF wr fsm current state
1 vif_fifo_full	VIF ASync FIFO full flag, active high
0 vif_fifo_empty	VIF ASync FIFO empty flag, active high

13.4.10.1.151 Status signals for the video stuff module, part 1 (PCK_STUFF_STATUS_0)

13.4.10.1.151.1 Offset

Register	Offset
PCK_STUFF_STATUS_0	22A0h

13.4.10.1.151.2 Diagram



13.4.10.1.151.3 Fields

Field	Function
31 reserved_3	
30-24 msa_gen_state	Secondary Data generator FSM status
23-22 reserved_2	
21-16 mst_video_gen_state	MST video generator FSM status. MST video generator FSM status.
15 reserved_1	
14-8 sst_video_gen_state	SST video generator FSM status. SST video generator FSM status.
7-5	

Table continues on the next page...

Clocks And Resets

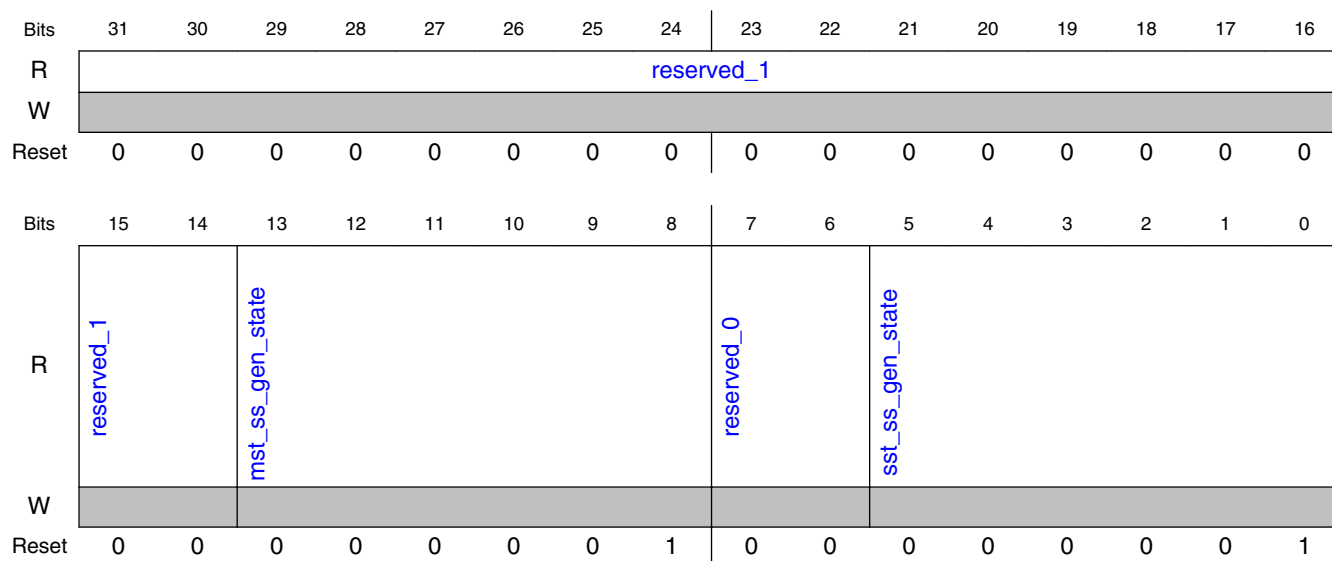
Field	Function
reserved_0	
4-0	No video generator FSM status.
no_video_gen_state	No video generator FSM status.

13.4.10.1.152 Status signals for the video stuff module, part 2 (PCK_STUFF_STATUS_1)

13.4.10.1.152.1 Offset

Register	Offset
PCK_STUFF_STATUS_1	22A4h

13.4.10.1.152.2 Diagram



13.4.10.1.152.3 Fields

Field	Function
31-14	
reserved_1	
13-8	MST SS generator FSM status

Table continues on the next page...

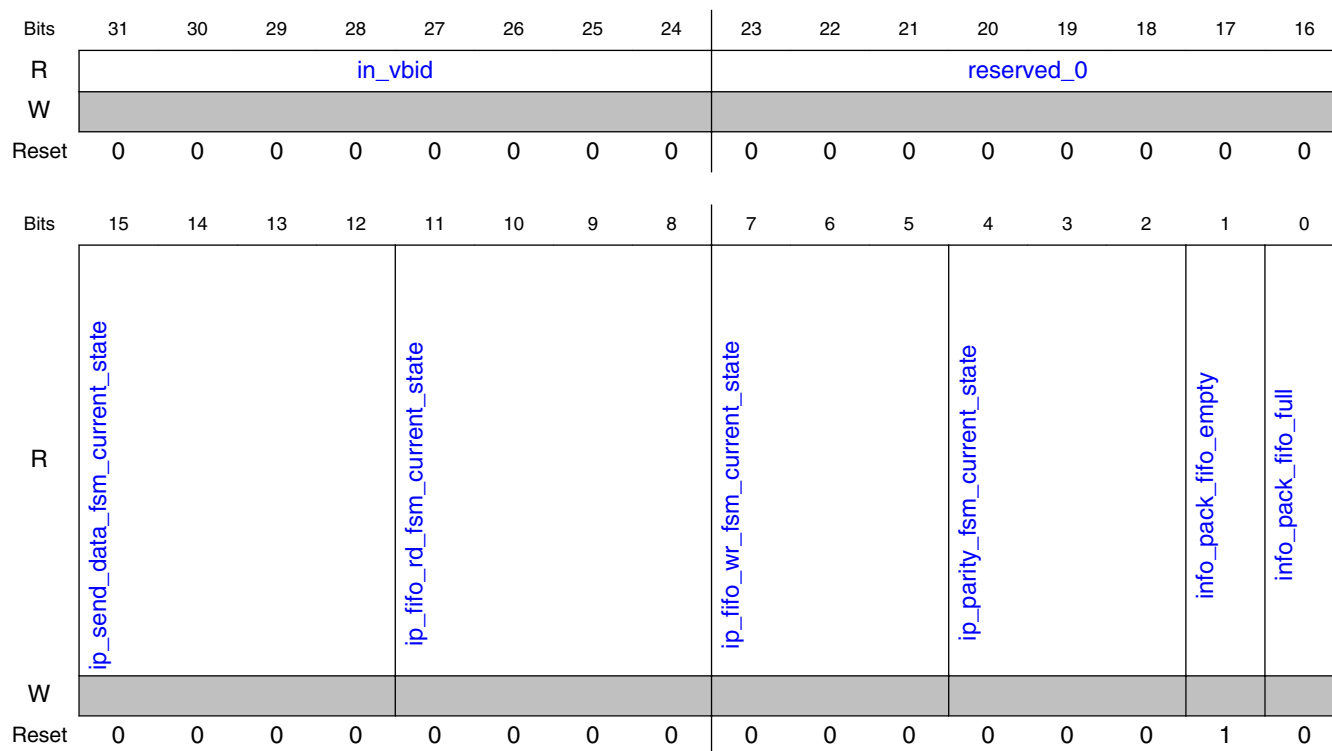
Field	Function
mst_ss_gen_state	
7-6 reserved_0	
5-0 sst_ss_gen_state	MSA generator FSM status

13.4.10.1.153 Status signals for the info pack module, as well as final VB-ID value (INFO_PACK_STATUS)

13.4.10.1.153.1 Offset

Register	Offset
INFO_PACK_STATUS	22A8h

13.4.10.1.153.2 Diagram



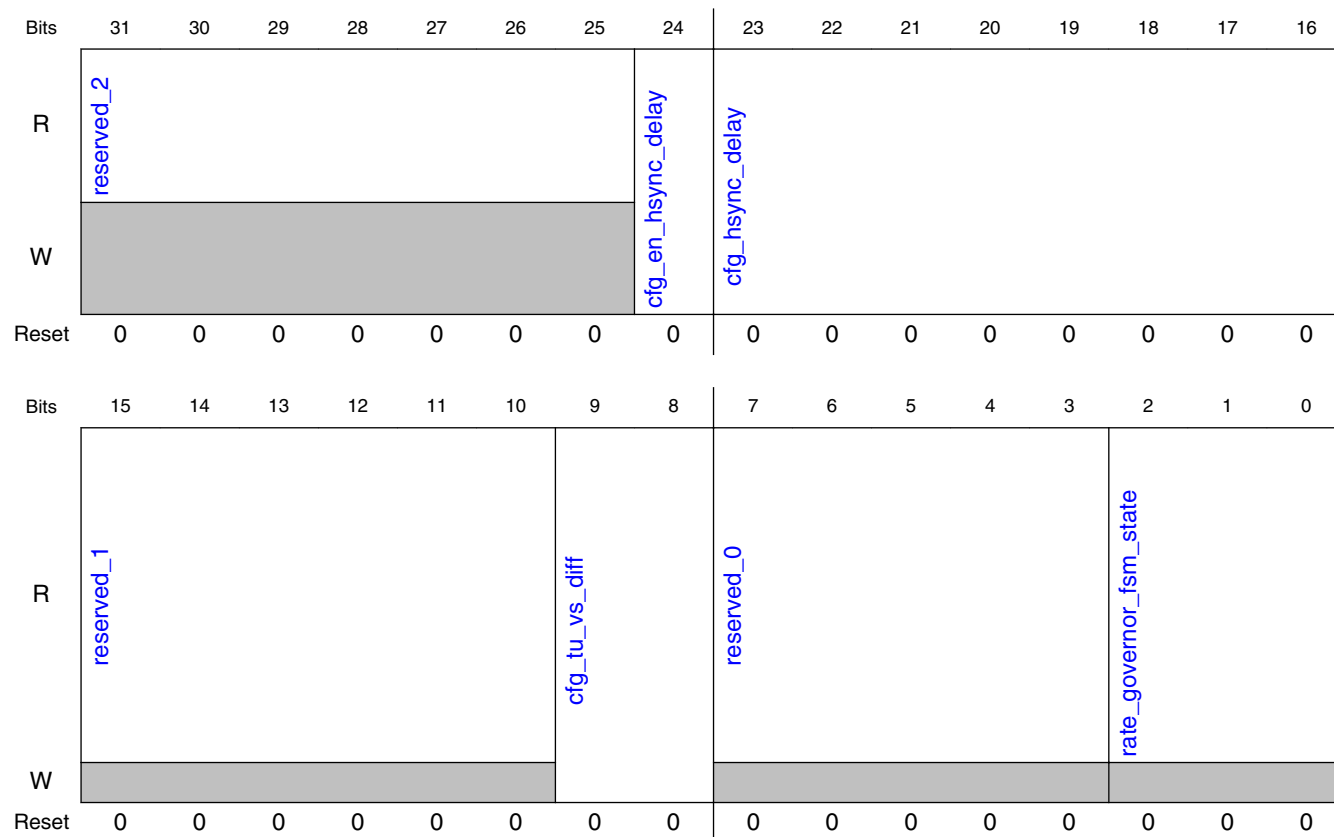
13.4.10.1.153.3 Fields

Field	Function
31-24 in_vbid	Value of the sent VB-ID (vb_id_final)
23-16 reserved_0	
15-12 ip_send_data_fsm_current_state	State of the send_data fsm
11-8 ip_fifo_rd_fsm_current_state	State of the fifo_rd fsm
7-5 ip_fifo_wr_fsm_current_state	State of the fifo_wr fsm
4-2 ip_parity_fsm_current_state	State of the parity fsm
1 info_pack_fifo_empty	Info_pack fifo empty flag, active high
0 info_pack_fifo_full	Info_pack fifo full flag, active high

13.4.10.1.154 Rate governor status, as well as additional video configuration (RATE_GOVERNOR_STATUS)**13.4.10.1.154.1 Offset**

Register	Offset
RATE_GOVERNOR_STATUS	22ACh

13.4.10.1.154.2 Diagram



13.4.10.1.154.3 Fields

Field	Function
31-25 reserved_2	
24 cfg_en_hsync_delay	Hsync Delay Enable. Hsync Delay Enable. Active high
23-16 cfg_hsync_delay	Hsync delay value (in pixel clocks). Hsync delay value (in pixel clocks). Actual delay is: (cfg_hsync_delay+2) pixel clocks.
15-10 reserved_1	
9-8 cfg_tu_vs_diff	TU-VS difference. TU-VS difference. b'00: TU-VS>3, b'01:NA, b'10:TU-VS=2, b'11:TU-VS=3
7-3 reserved_0	
2-0	Status of the governor FSM, relevant only for MST

Clocks And Resets

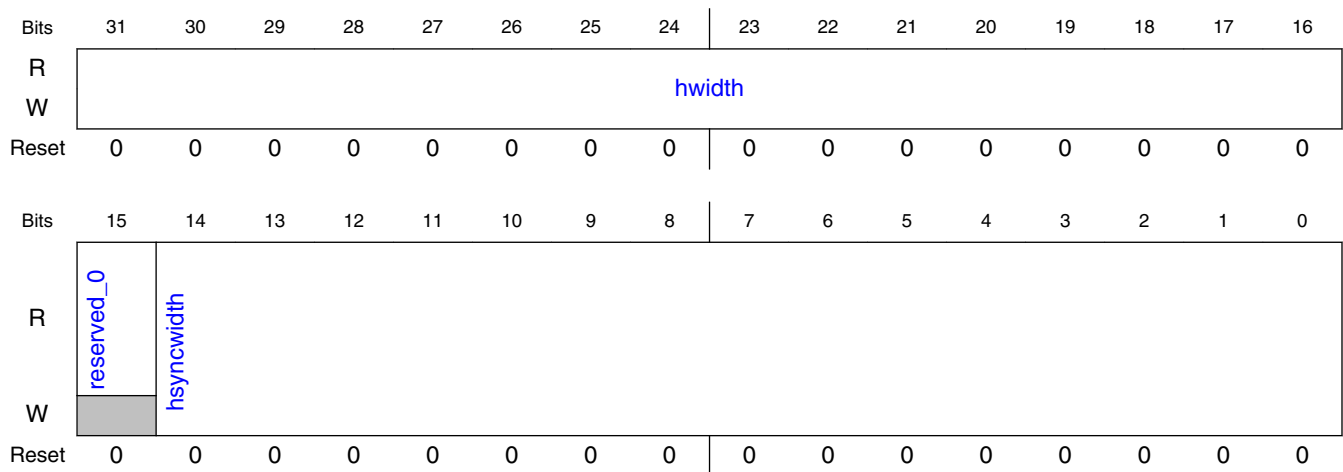
Field	Function
rate_governor_f sm_state	

13.4.10.1.155 Video line parameters to be used in the RTL calculations (DP_HORIZONTAL)

13.4.10.1.155.1 Offset

Register	Offset
DP_HORIZONTAL	22B0h

13.4.10.1.155.2 Diagram



13.4.10.1.155.3 Fields

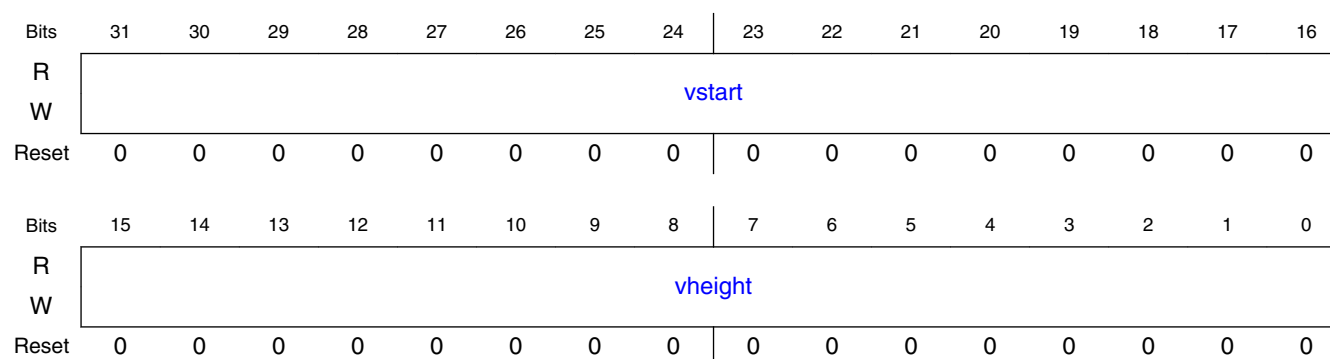
Field	Function
31-16 hwidth	Horizontal Active Video Width, number of active pixels per line
15 reserved_0	
14-0 hsyncwidth	Horizontal Sync Width, number of pixels Hsync is active

13.4.10.1.156 Video frame parameters to be used in the RTL calculations, part 1 (DP_VERTICAL_0)

13.4.10.1.156.1 Offset

Register	Offset
DP_VERTICAL_0	22B4h

13.4.10.1.156.2 Diagram



13.4.10.1.156.3 Fields

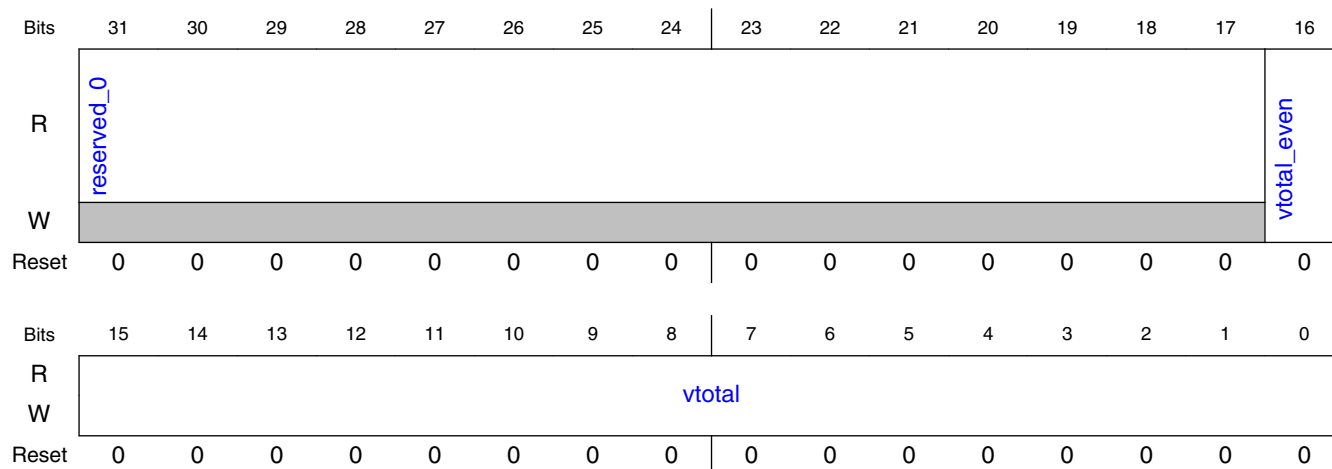
Field	Function
31-16 vstart	Vertical Active Start, index of the first active line in a frame
15-0 vheight	Vertical Active Video Height, number of active lines in a frame

13.4.10.1.157 Video frame parameters to be used in the RTL calculations, part 2 (DP_VERTICAL_1)

13.4.10.1.157.1 Offset

Register	Offset
DP_VERTICAL_1	22B8h

13.4.10.1.157.2 Diagram



13.4.10.1.157.3 Fields

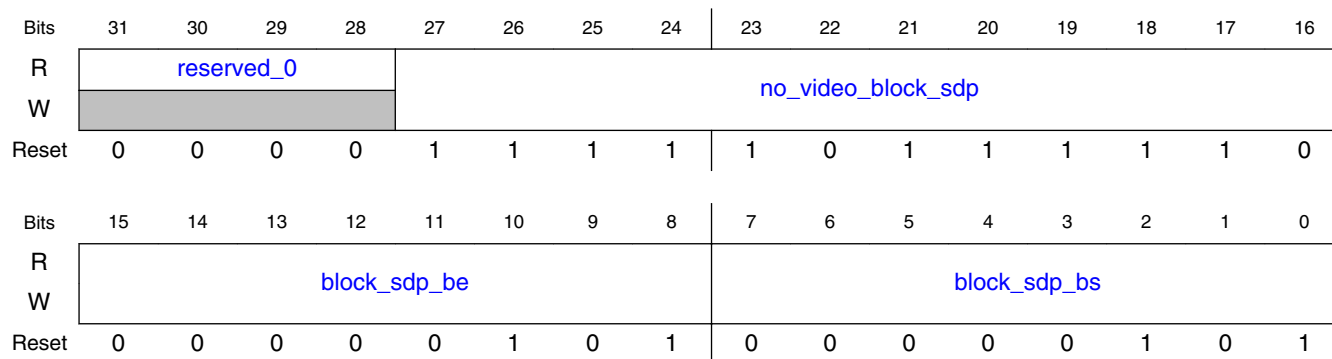
Field	Function
31-17 reserved_0	
16 vtotal_even	Indicate Vtotal is Even Number like MISC, active high
15-0 vtotal	Vertical Total, number of lines per frame

13.4.10.1.158 (DP_BLOCK_SDP)

13.4.10.1.158.1 Offset

Register	Offset
DP_BLOCK_SDP	22BCh

13.4.10.1.158.2 Diagram



13.4.10.1.158.3 Fields

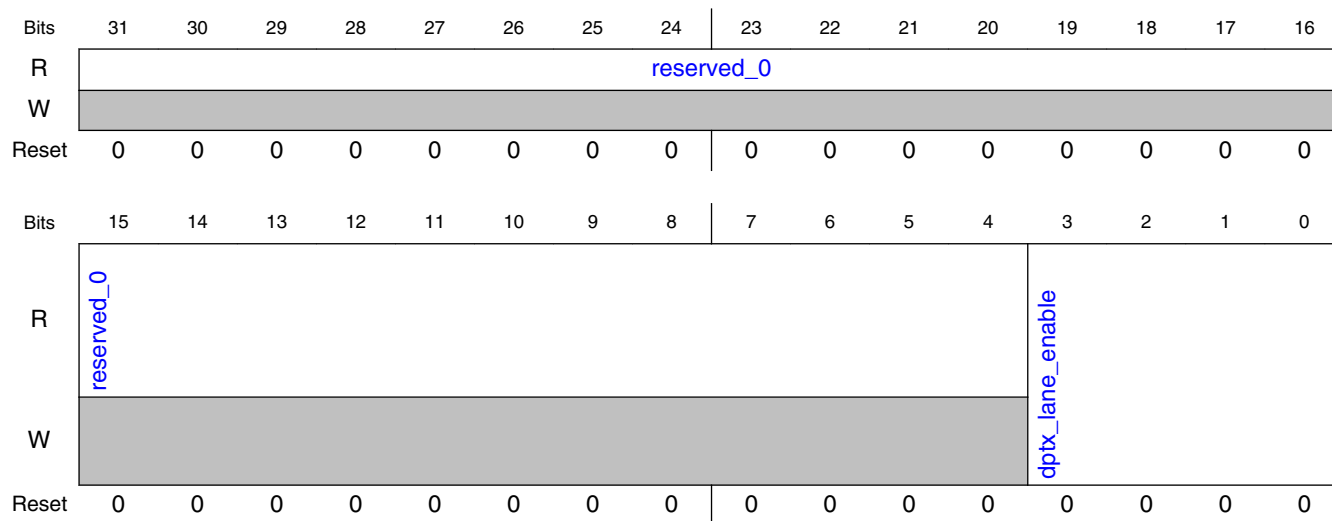
Field	Function
31-28 reserved_0	
27-16 no_video_block_sdp	
15-8 block_sdp_be	
7-0 block_sdp_bs	

13.4.10.1.159 DPTX lane enable (DPTX_LANE_EN)

13.4.10.1.159.1 Offset

Register	Offset
DPTX_LANE_EN	2300h

13.4.10.1.159.2 Diagram



13.4.10.1.159.3 Fields

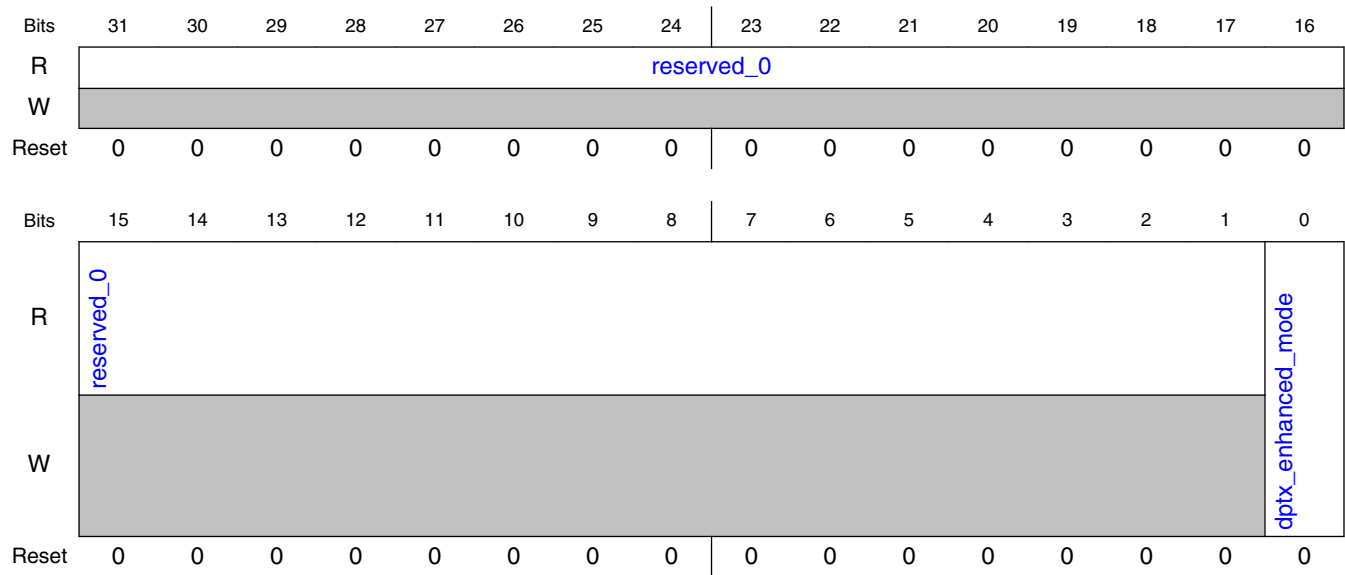
Field	Function
31-4 reserved_0	
3-0 dptx_lane_enabl e	DPTX lane enable each lane as a bit when 1 lane is enabled

13.4.10.1.160 DPTX enhanced mode (DPTX_ENHNCD)

13.4.10.1.160.1 Offset

Register	Offset
DPTX_ENHNCD	2304h

13.4.10.1.160.2 Diagram



13.4.10.1.160.3 Fields

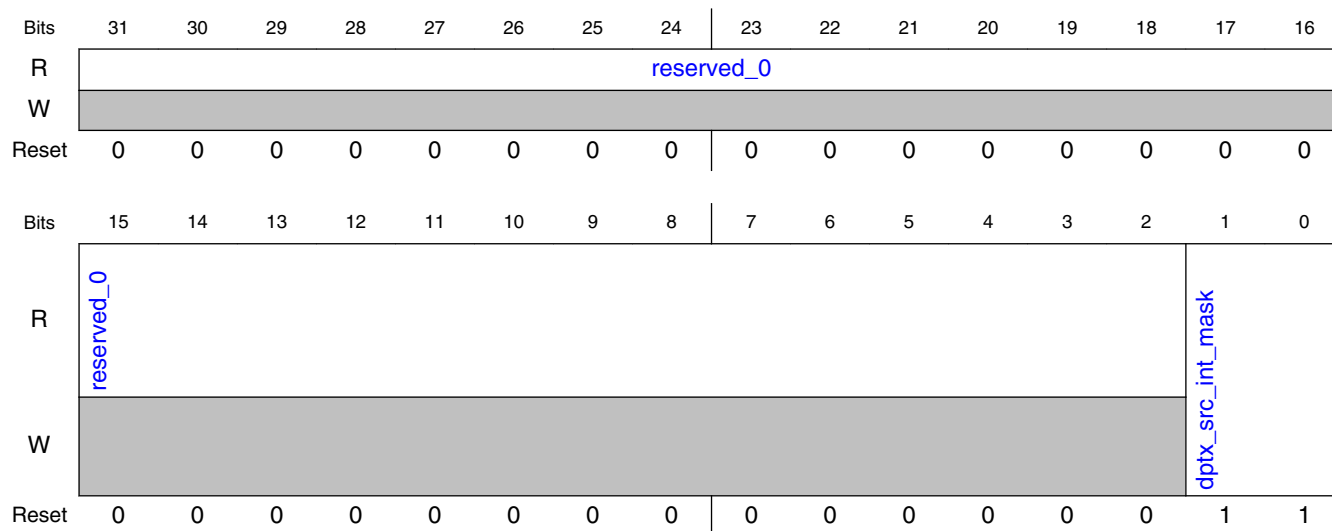
Field	Function
31-1 reserved_0	
0 dptx_enhanced_mode	DPTX is in enhanced mode when 1

13.4.10.1.161 DPTX Interrupt mask (DPTX_INT_MASK)

13.4.10.1.161.1 Offset

Register	Offset
DPTX_INT_MASK	2308h

13.4.10.1.161.2 Diagram



13.4.10.1.161.3 Fields

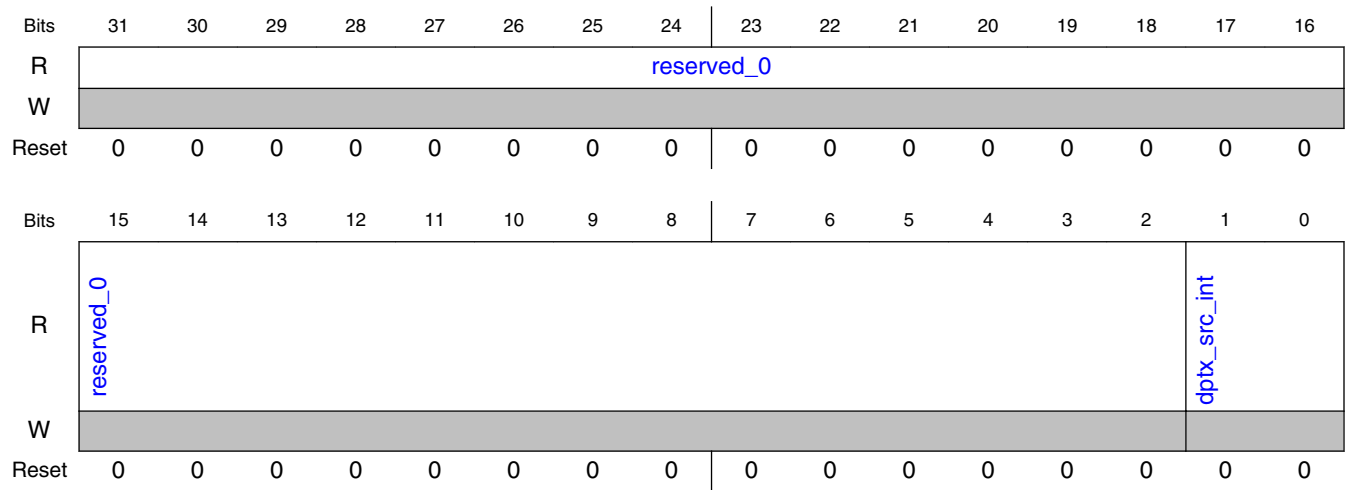
Field	Function
31-2 reserved_0	
1-0 dptx_src_int_ma sk	DPTX mask interrupts :-0-HPD interrupt Mask -1-Framer Interrupt Mask

13.4.10.1.162 DPTX interrupt status (DPTX_INT_STATUS)

13.4.10.1.162.1 Offset

Register	Offset
DPTX_INT_STATUS	230Ch

13.4.10.1.162.2 Diagram



13.4.10.1.162.3 Fields

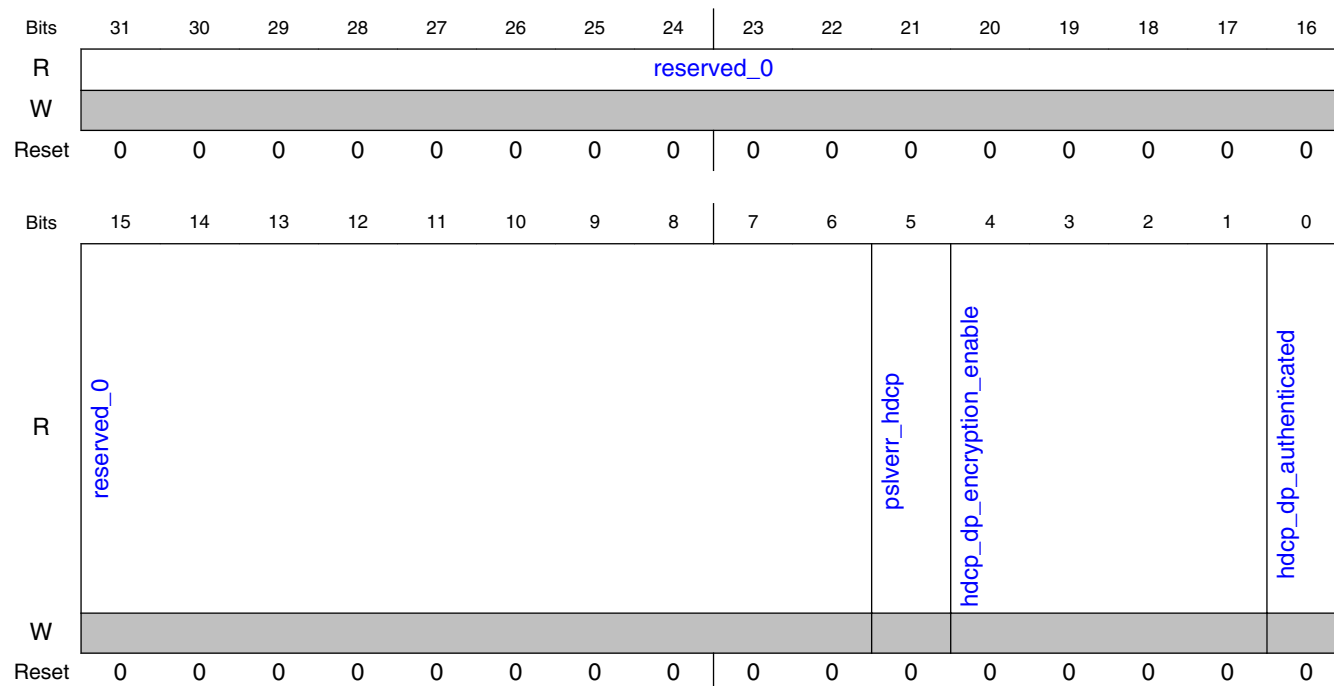
Field	Function
31-2 reserved_0	
1-0 dptx_src_int	HPD event Status-FRAMER event Status

13.4.10.1.163 HDCP DP status register (HDCP_DP_STATUS)

13.4.10.1.163.1 Offset

Register	Offset
HDCP_DP_STATUS	2400h

13.4.10.1.163.2 Diagram



13.4.10.1.163.3 Fields

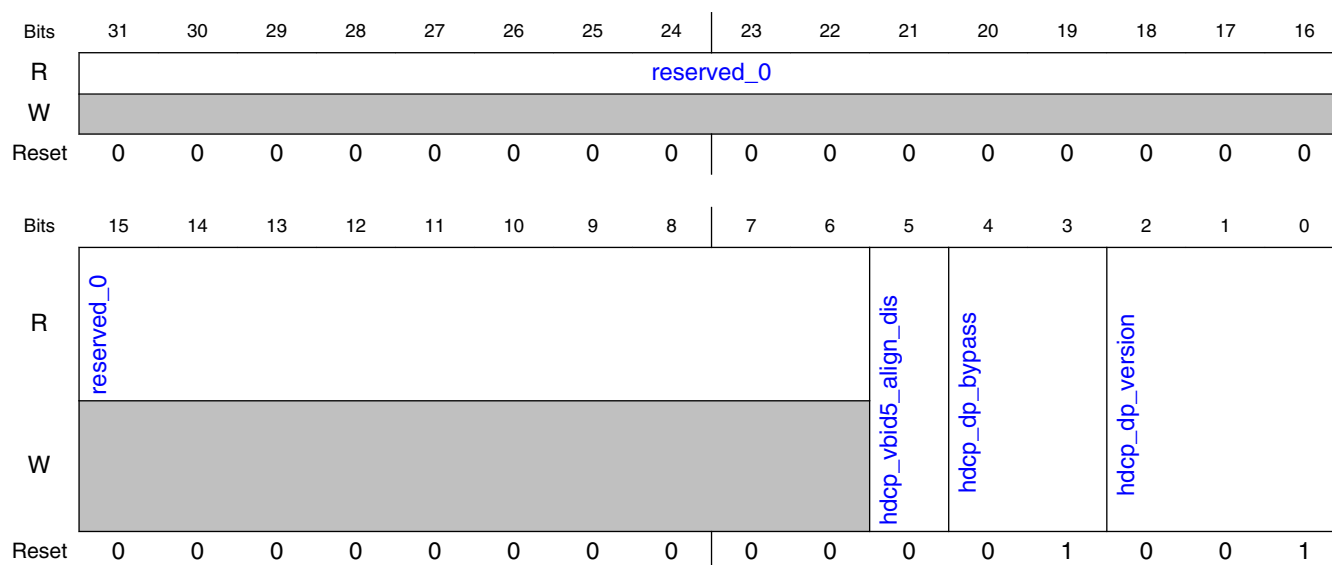
Field	Function
31-6 reserved_0	
5 pslverr_hdcp	APB slave error status from HDCP module. APB slave error status from HDCP module.
4-1 hdcp_dp_encryption_enable	Encryption enabled
0 hdcp_dp_authenticated	Authentication enabled HDCP 1. Authentication enabled HDCP 1.4

13.4.10.1.164 HDCP DP config register (HDCP_DP_CONFIG)

13.4.10.1.164.1 Offset

Register	Offset
HDCP_DP_CONFIG	2404h

13.4.10.1.164.2 Diagram



13.4.10.1.164.3 Fields

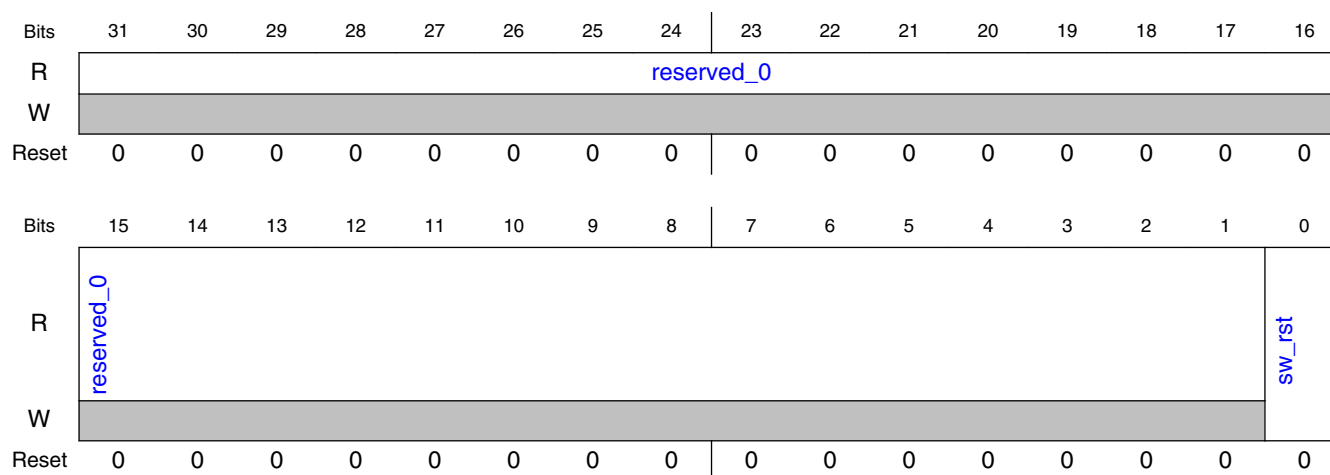
Field	Function
31-6 reserved_0	
5 hdcp_vbid5_align_dis	
4-3 hdcp_dp_bypass	HDCP DP bypass. HDCP DP bypass. 0x0-No bypass(HDCP enabled); 0x1-Bypass enabled; 0x2-Data from FIFO6 (debug, not intended for normal usage); 0x3-Data from FIFO4 (debug, not intended for normal usage).
2-0 hdcp_dp_version	HDCP version. HDCP version. 0x1-HDCP2.2; 0x2-HDCP1.4; Other-Reserved

13.4.10.1.165 HDCP DP software reset register (HDCP_DP_SW_RST)

13.4.10.1.165.1 Offset

Register	Offset
HDCP_DP_SW_RST	2408h

13.4.10.1.165.2 Diagram



13.4.10.1.165.3 Fields

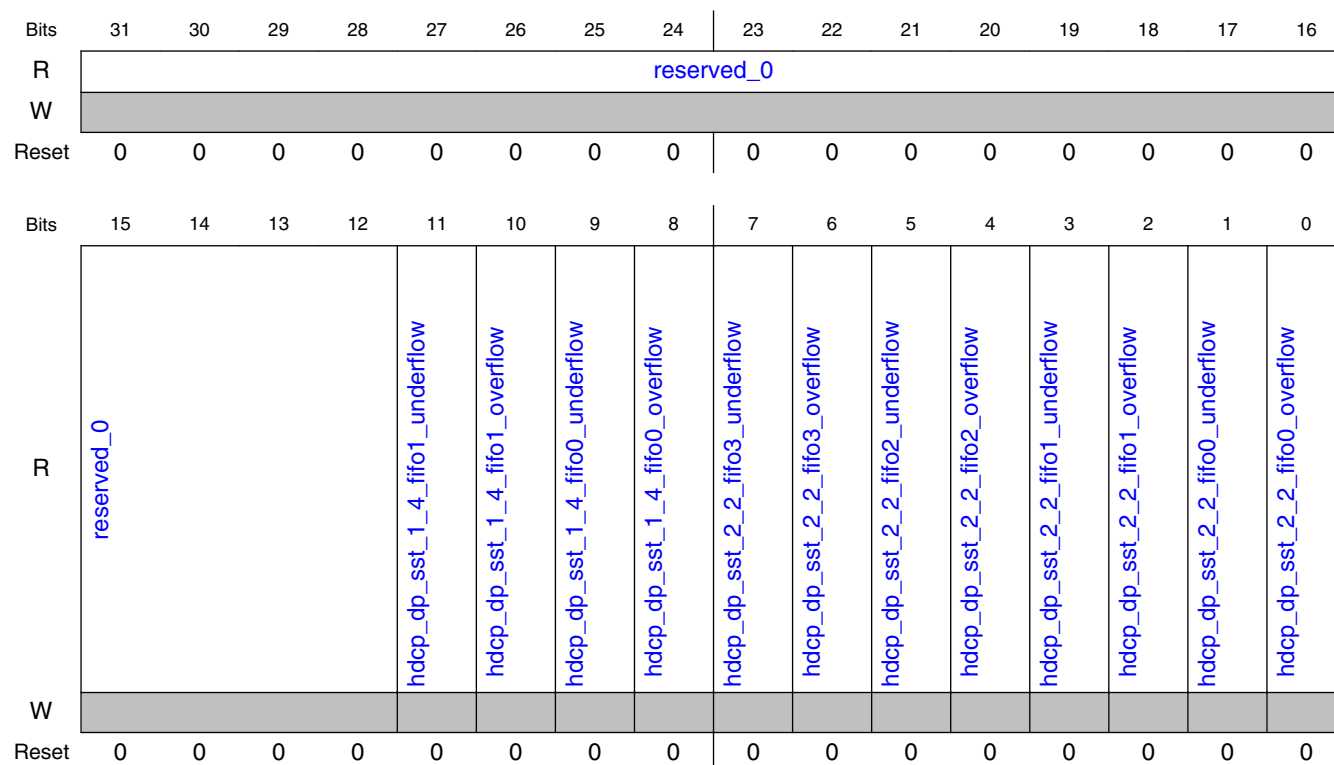
Field	Function
31-1 reserved_0	
0 sw_rst	Software reset

13.4.10.1.166 HDCP DP FIFO status register (HDCP_DP_FIFO_STATUS)

13.4.10.1.166.1 Offset

Register	Offset
HDCP_DP_FIFO_STATUS	240Ch

13.4.10.1.166.2 Diagram



13.4.10.1.166.3 Fields

Field	Function
31-12 reserved_0	
11 hdcp_dp_sst_1_4_fifo1_underflow	SST HDCP1. SST HDCP1.4 fifo1 underflow
10 hdcp_dp_sst_1_4_fifo1_overflow	SST HDCP1. SST HDCP1.4 fifo1 overflow
9 hdcp_dp_sst_1_4_fifo0_underflow	SST HDCP1. SST HDCP1.4 fifo0 underflow
8 hdcp_dp_sst_1_4_fifo0_overflow	SST HDCP1. SST HDCP1.4 fifo0 overflow
7	SST HDCP2. SST HDCP2.2 fifo3 underflow

Table continues on the next page...

Clocks And Resets

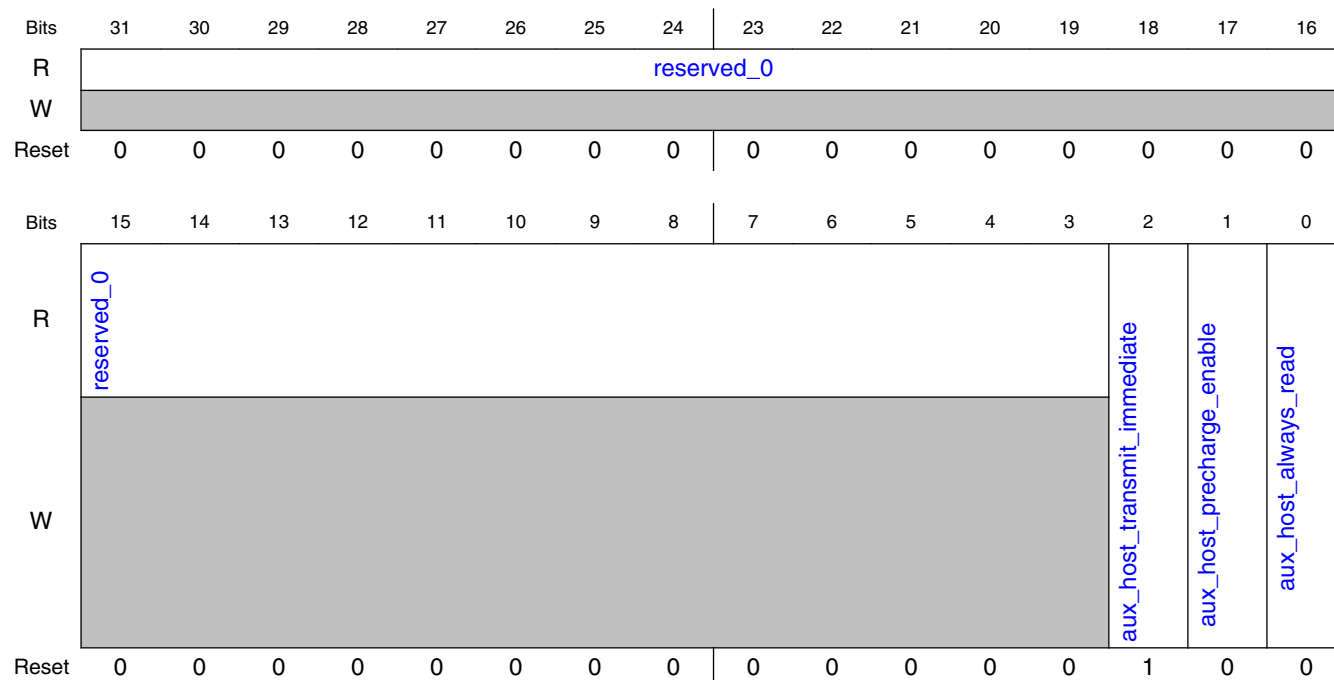
Field	Function
hdcp_dp_sst_2_2_fifo3_underflow	
6	SST HDCP2.
hdcp_dp_sst_2_2_fifo3_overflow	SST HDCP2.2 fifo3 overflow
5	SST HDCP2.
hdcp_dp_sst_2_2_fifo2_underflow	SST HDCP2.2 fifo2 underflow
4	SST HDCP2.
hdcp_dp_sst_2_2_fifo2_overflow	SST HDCP2.2 fifo2 overflow
3	SST HDCP2.
hdcp_dp_sst_2_2_fifo1_underflow	SST HDCP2.2 fifo1 underflow
2	SST HDCP2.
hdcp_dp_sst_2_2_fifo1_overflow	SST HDCP2.2 fifo1 overflow
1	SST HDCP2.
hdcp_dp_sst_2_2_fifo0_underflow	SST HDCP2.2 fifo0 underflow
0	SST HDCP2.
hdcp_dp_sst_2_2_fifo0_overflow	SST HDCP2.2 fifo0 overflow

13.4.10.1.167 Control bits for DP_AUX (DP_AUX_HOST_CONTROL)

13.4.10.1.167.1 Offset

Register	Offset
DP_AUX_HOST_CONTROL	2800h

13.4.10.1.167.2 Diagram



13.4.10.1.167.3 Fields

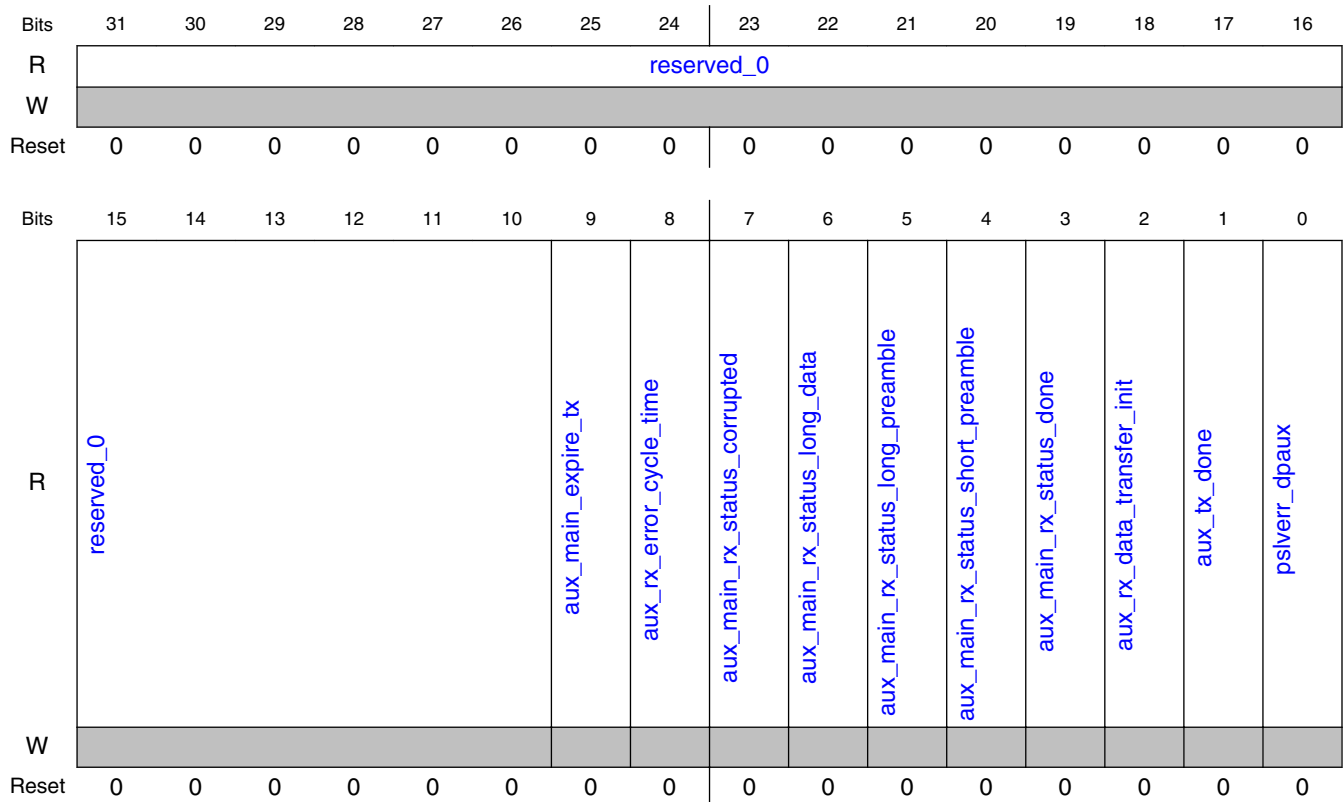
Field	Function
31-3 reserved_0	
2 aux_host_transmit_immediate	This bit is used only in DP_OUT mode. This bit is used only in DP_OUT mode. If SET, a transaction that comes from the adapter will be sent immediately without waiting for send_external_transaction pulse. If CLEAR, the MC controls the traffic to/from the adapter.
1 aux_host_precharge_enable	According to the current standard the tx precharge is done by sending 10 to 16 data_0 on the line before the SYNC. According to the current standard the tx precharge is done by sending 10 to 16 data_0 on the line before the SYNC. Old standard define the precharge by forcing the AFE to be in precharge mode before transmitting the SYNC.
0 aux_host_always_read	Normally, the aux_rx is disabled during transmit. Normally, the aux_rx is disabled during transmit. Setting this bit allow loopback operation and all transmit transactions will go to the receiver. Used for debug.

13.4.10.1.168 Status of the DP_AUX interrupt sources (DP_AUX_INTERRUPT_SOURCE)

13.4.10.1.168.1 Offset

Register	Offset
DP_AUX_INTERRUPT_SOURCE	2804h

13.4.10.1.168.2 Diagram



13.4.10.1.168.3 Fields

Field	Function
31-10 reserved_0	
9 aux_main_expire_tx	timer expire (external) in DP_OUT
8 aux_rx_error_cycle_time	Cycle time error. Cycle time error. Asserted if aux_rx_last_cycle is less then aux_host_1m_min or greater then aux_host_1m_max.
7	The received transaction corrupted during the data phase (bad STOP, or unaligned STOP).

Table continues on the next page...

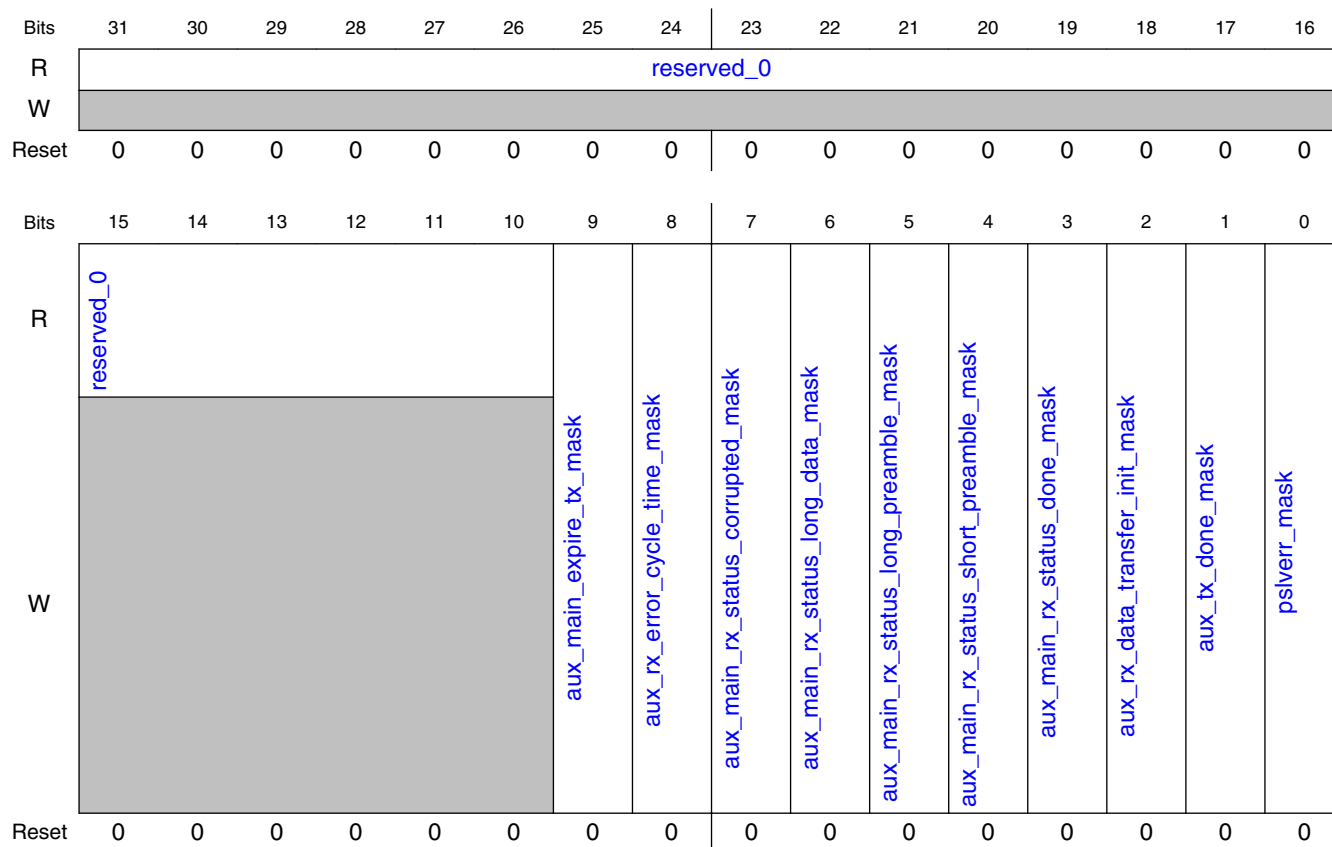
Field	Function
aux_main_rx_status_corrupted	The received transaction corrupted during the data phase (bad STOP, or unaligned STOP).
6 aux_main_rx_status_long_data	The received transaction had more than 20 data bytes.
5 aux_main_rx_status_long_preamble	The received transaction had preamble greater than the preamble_max.
4 aux_main_rx_status_short_preamble	The received transaction had preamble shorter than the preamble_max.
3 aux_main_rx_status_done	This module control the packet extraction and packet read from the memory.
2 aux_rx_data_transfer_init	Rx data transfer may be initiated.
1 aux_tx_done	Rx data transfer may be initiated. Falling edge of the aux_mailbox_empty.
0 pslverr_dpaux	Tx data transfer finished.
	Tx data transfer finished.
	APB slave error interrupt from DP AUX module.
	APB slave error interrupt from DP AUX module.

13.4.10.1.169 Mask vector of the DP_AUX interrupt sources (DP_AUX_INTERRUPT_MASK)

13.4.10.1.169.1 Offset

Register	Offset
DP_AUX_INTERRUPT_MASK	2808h

13.4.10.1.169.2 Diagram



13.4.10.1.169.3 Fields

Field	Function
31-10 reserved_0	
9 aux_main_expire_tx_mask	aux_main_expire_external
8 aux_rx_error_cycle_time_mask	aux_rx_error_cycle_time mask
7 aux_main_rx_status_corrupted_mask	aux_main_rx_status_corrupted_mask
6 aux_main_rx_status_long_data_mask	aux_main_rx_status_long_data_mask

Table continues on the next page...

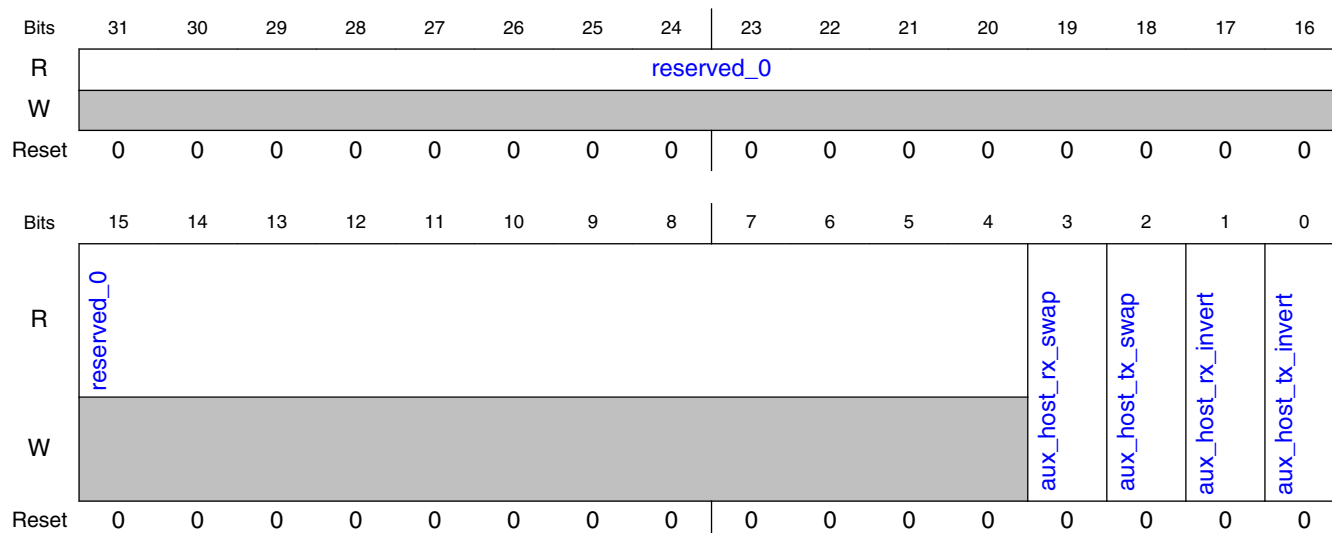
Field	Function
5 aux_main_rx_status_long_preamble_mask	aux_main_rx_status_long_preamble_mask
4 aux_main_rx_status_short_preamble_mask	aux_main_rx_status_short_preamble_mask
3 aux_main_rx_status_done_mask	aux_main_rx_status_done_mask
2 aux_rx_data_transfer_init_mask	rx_data_transfer_init mask
1 aux_tx_done_mask	aux_tx_done_mask
0 pslverr_mask	Mask for pslverr_dpaux interrupt

13.4.10.1.170 Ordering and inversion of transmit/receive on Auxiliary Channel (DP_AUX_SWAP_INVERSION_CONTROL)

13.4.10.1.170.1 Offset

Register	Offset
DP_AUX_SWAP_INVERSION_CONTROL	280Ch

13.4.10.1.170.2 Diagram



13.4.10.1.170.3 Fields

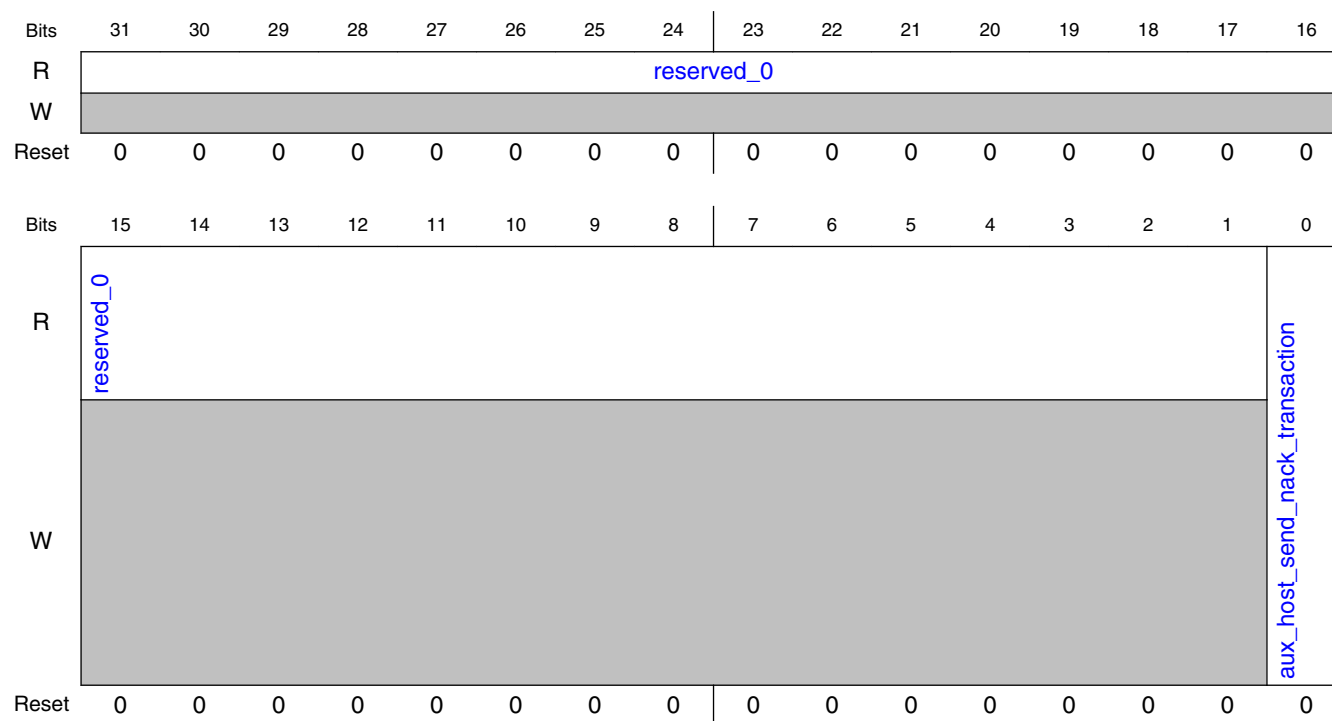
Field	Function
31-4 reserved_0	
3 aux_host_rx_swap	Shift right (LSB first) of the income data
2 aux_host_tx_swap	Shift right the output data (LSB first)
1 aux_host_rx_invert	Invert rx input and output data to AUXILIARY CHANNEL
0 aux_host_tx_invert	Invert tx input and output data to AUXILIARY CHANNEL

13.4.10.1.171 NACK transaction send (DP_AUX_SEND_NACK_TRANSACTION)

13.4.10.1.171.1 Offset

Register	Offset
DP_AUX_SEND_NACK_TRANSACTION	2810h

13.4.10.1.171.2 Diagram



13.4.10.1.171.3 Fields

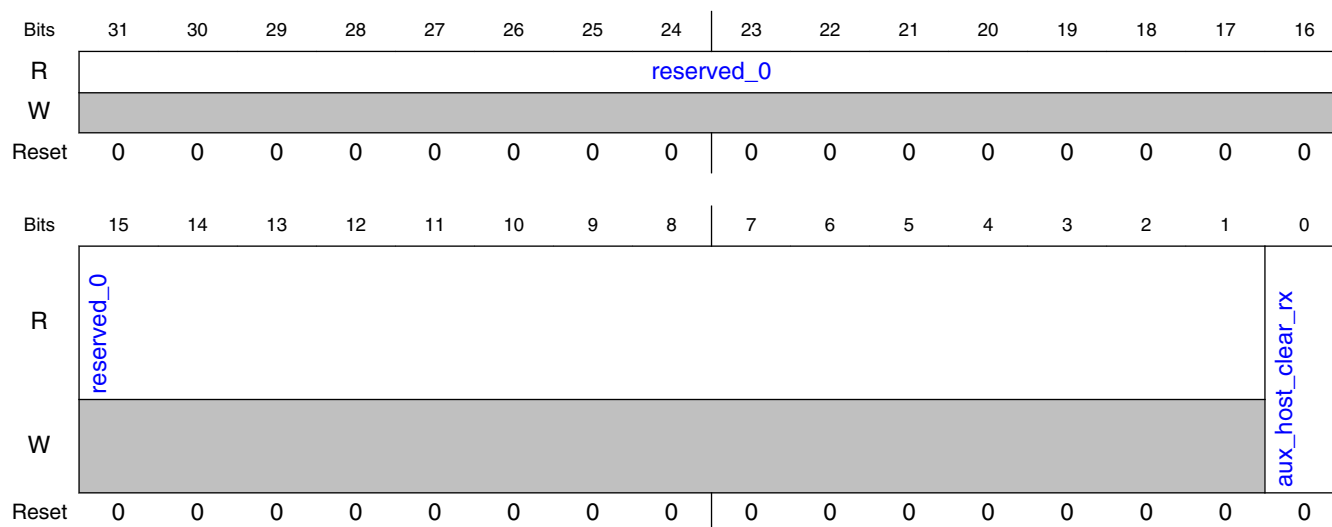
Field	Function
31-1 reserved_0	
0 aux_host_send_nack_transaction	Send nack transaction by AUX_TX. Send nack transaction by AUX_TX. This bit is automatically cleared when operation in completed.

13.4.10.1.172 RX bits clear (DP_AUX_CLEAR_RX)

13.4.10.1.172.1 Offset

Register	Offset
DP_AUX_CLEAR_RX	2814h

13.4.10.1.172.2 Diagram



13.4.10.1.172.3 Fields

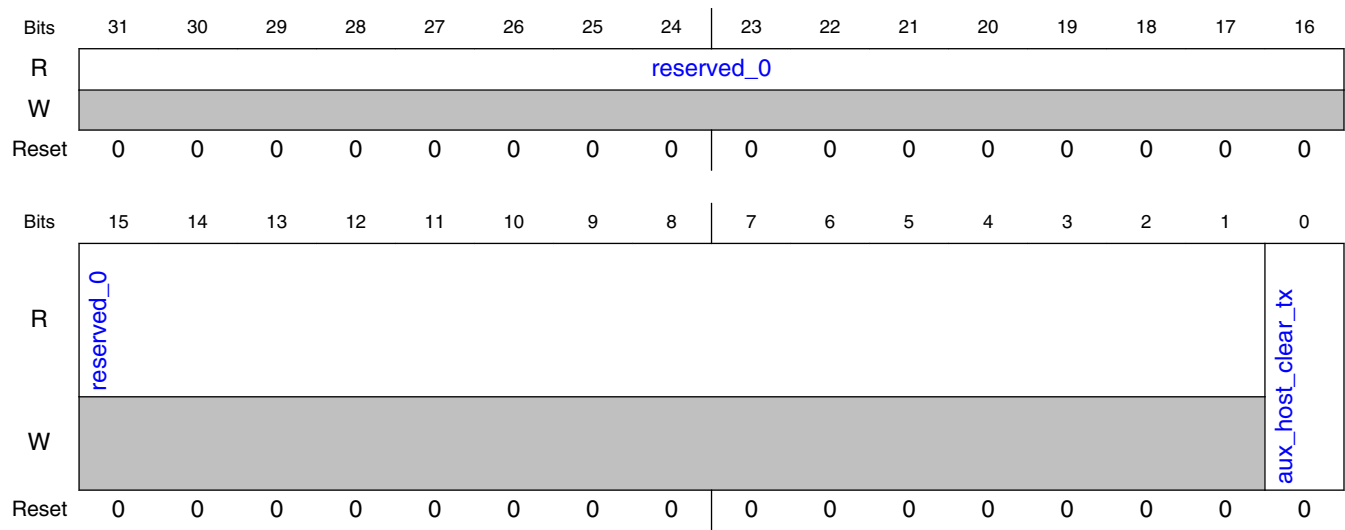
Field	Function
31-1 reserved_0	
0 aux_host_clear_rx	Clear all rx bits in register 64,65. Clear all rx bits in register 64,65. This command is an indication that the processing of last receive transaction was completed and the AUX_RX can start looking for new receive transaction. This bit is automatically cleared when operation is completed.

13.4.10.1.173 TX bits clear (DP_AUX_CLEAR_TX)

13.4.10.1.173.1 Offset

Register	Offset
DP_AUX_CLEAR_TX	2818h

13.4.10.1.173.2 Diagram



13.4.10.1.173.3 Fields

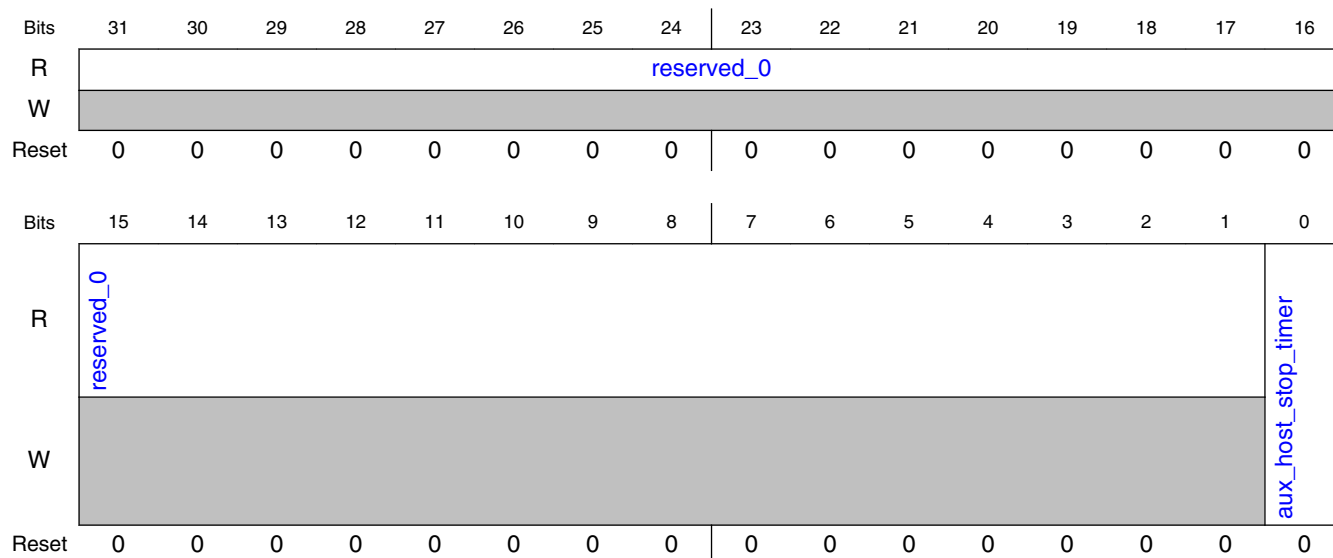
Field	Function
31-1 reserved_0	
0 aux_host_clear_tx	Clear all external bits in registers 64,67. Clear all external bits in registers 64,67.This command used in DP_IN mode. It is an indication that the processing of last external transaction was completed and the DP_AUX can start receive new external transaction from the adapter. This bit is automatically cleared when operation in completed.

13.4.10.1.174 Stop timer operation (DP_AUX_TIMER_STOP)

13.4.10.1.174.1 Offset

Register	Offset
DP_AUX_TIMER_STOP	281Ch

13.4.10.1.174.2 Diagram



13.4.10.1.174.3 Fields

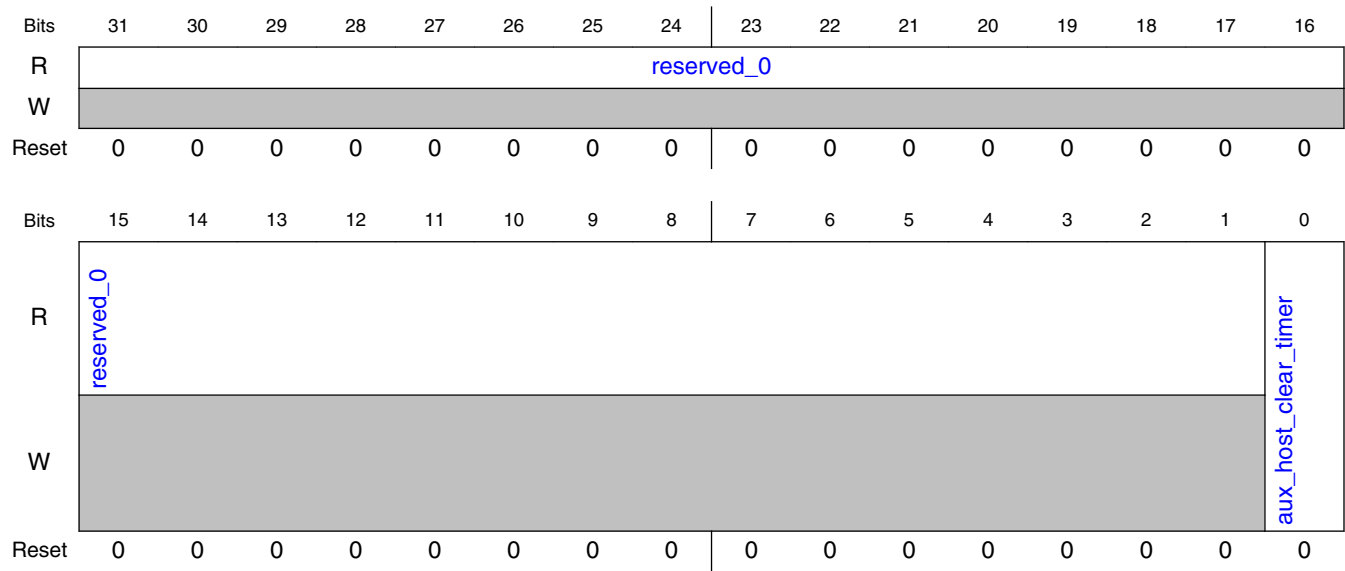
Field	Function
31-1 reserved_0	
0 aux_host_stop_timer	Stop timer operation

13.4.10.1.175 Clear timer operation (DP_AUX_TIMER_CLEAR)

13.4.10.1.175.1 Offset

Register	Offset
DP_AUX_TIMER_CLEAR	2820h

13.4.10.1.175.2 Diagram



13.4.10.1.175.3 Fields

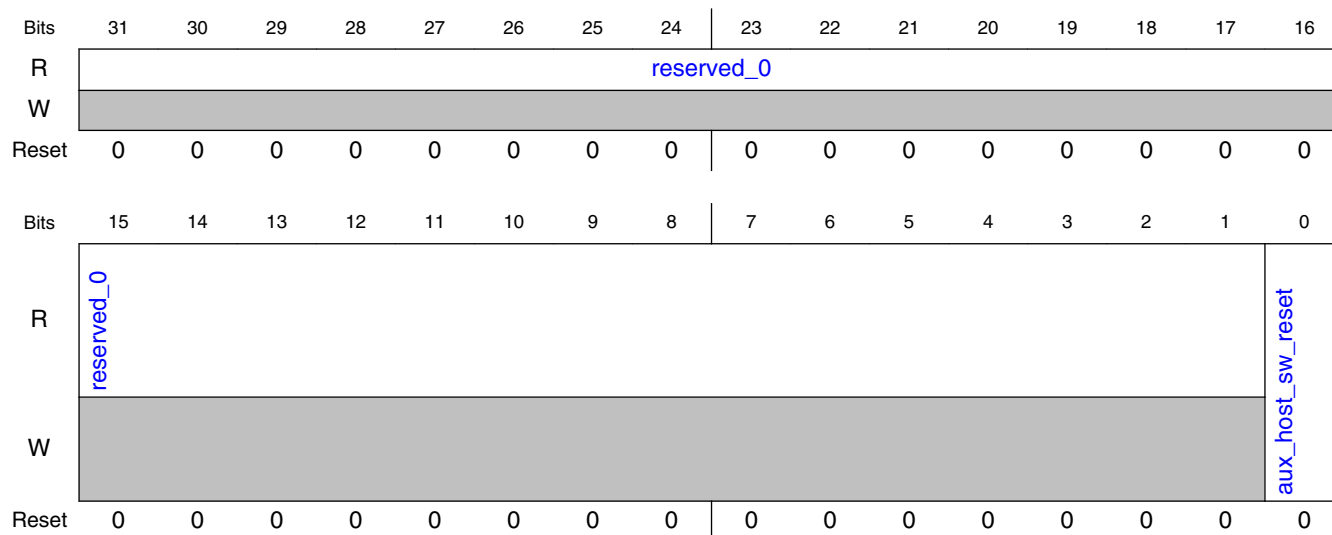
Field	Function
31-1 reserved_0	
0 aux_host_clear_timer	Stop timer operation. Stop timer operation. This bit is automatically cleared when operation in completed.

13.4.10.1.176 Soft reset of the DP_AUX (DP_AUX_RESET_SW)

13.4.10.1.176.1 Offset

Register	Offset
DP_AUX_RESET_SW	2824h

13.4.10.1.176.2 Diagram



13.4.10.1.176.3 Fields

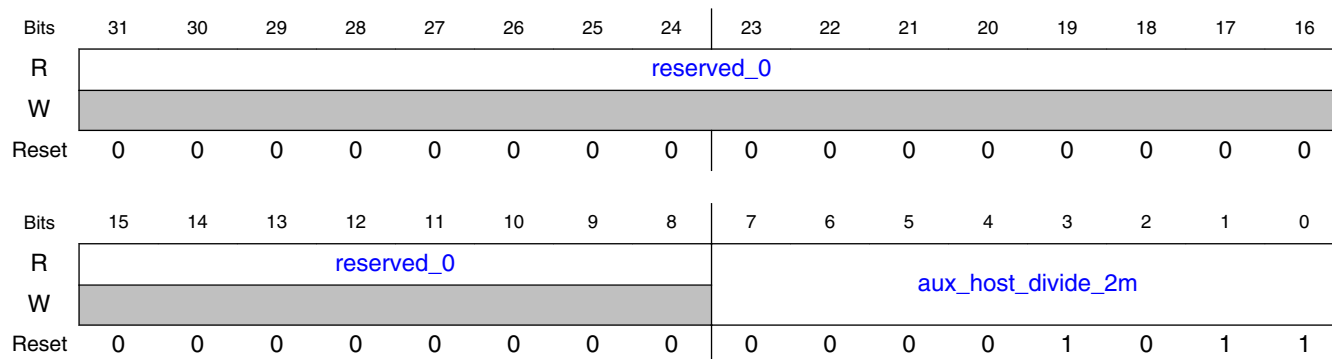
Field	Function
31-1 reserved_0	
0 aux_host_sw_reset	Reset all DP_AUX state machines and clear all the status bits. Reset all DP_AUX state machines and clear all the status bits. The registers value remains. (S/W reset). This bit is automatically cleared when operation in completed.

13.4.10.1.177 SYS_CLK and 2 MHz clock ratio (DP_AUX_DIVIDE_2M)

13.4.10.1.177.1 Offset

Register	Offset
DP_AUX_DIVIDE_2M	2828h

13.4.10.1.177.2 Diagram



13.4.10.1.177.3 Fields

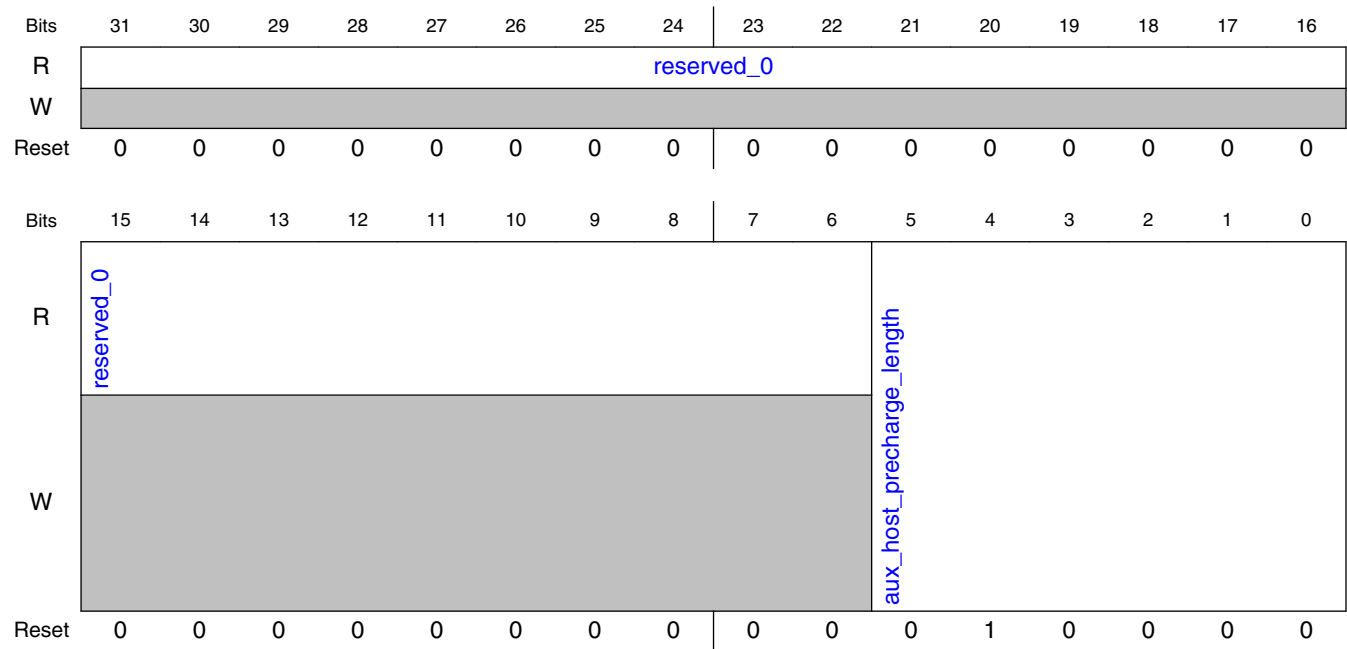
Field	Function
31-8 reserved_0	
7-0 aux_host_divide_2m	The ratio between sys_clk and 2MHz, [(sys_clk frequency/2MHz) - 1], for 25MHz sys_clk the value is 11. This register is used by AUX_TX for generating the AUX_TX clock.

13.4.10.1.178 Pre charge field length (DP_AUX_TX_PRECHARGE_LENGTH)

13.4.10.1.178.1 Offset

Register	Offset
DP_AUX_TX_PRECHARGE_LENGTH	282Ch

13.4.10.1.178.2 Diagram



13.4.10.1.178.3 Fields

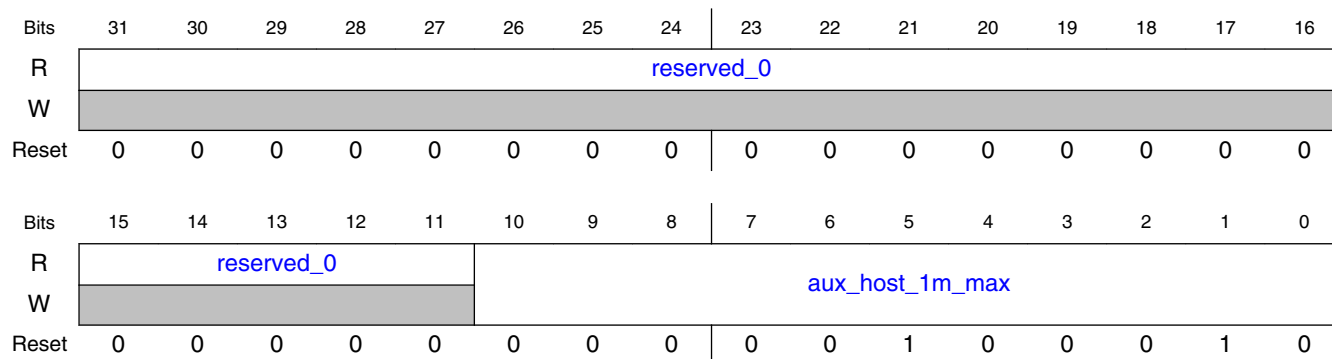
Field	Function
31-6 reserved_0	
5-0 aux_host_prech arge_length	Length of pre charge field, standard definition is 10 to 16 bits/clocks

13.4.10.1.179 Maximum legal receiving frequency (DP_AUX_FREQUENCY_1M_MAX)

13.4.10.1.179.1 Offset

Register	Offset
DP_AUX_FREQUENCY_1M_MAX	2830h

13.4.10.1.179.2 Diagram



13.4.10.1.179.3 Fields

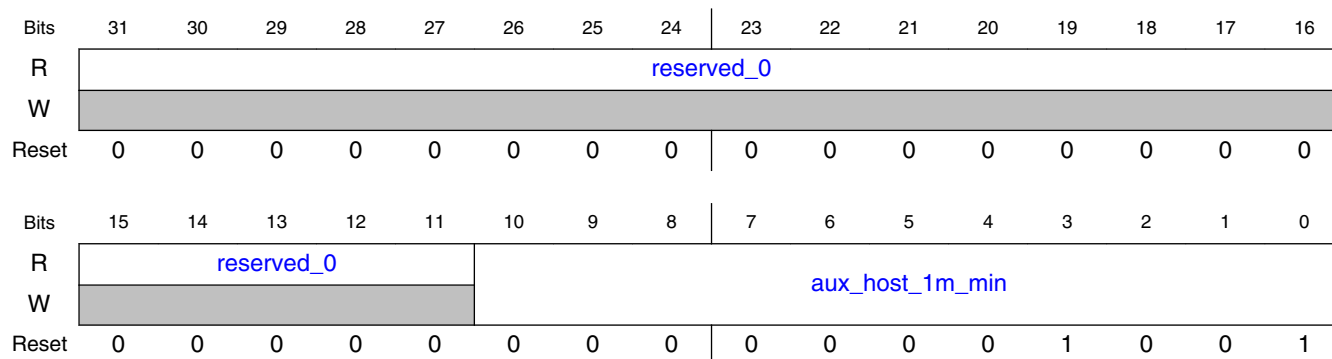
Field	Function
31-11 reserved_0	
10-0 aux_host_1m_max	The maximum legal frequency receiving from the line by the standard is 1. The maximum legal frequency receiving from the line by the standard is 1.25MHz.The calculation is:(1.25 MHz cycle time)/(sys_clk(-15%) cycle time) 800/46 =17

13.4.10.1.180 Minimum legal receiving frequency (DP_AUX_FREQUENCY_1M_MIN)

13.4.10.1.180.1 Offset

Register	Offset
DP_AUX_FREQUENCY_1M_MIN	2834h

13.4.10.1.180.2 Diagram



13.4.10.1.180.3 Fields

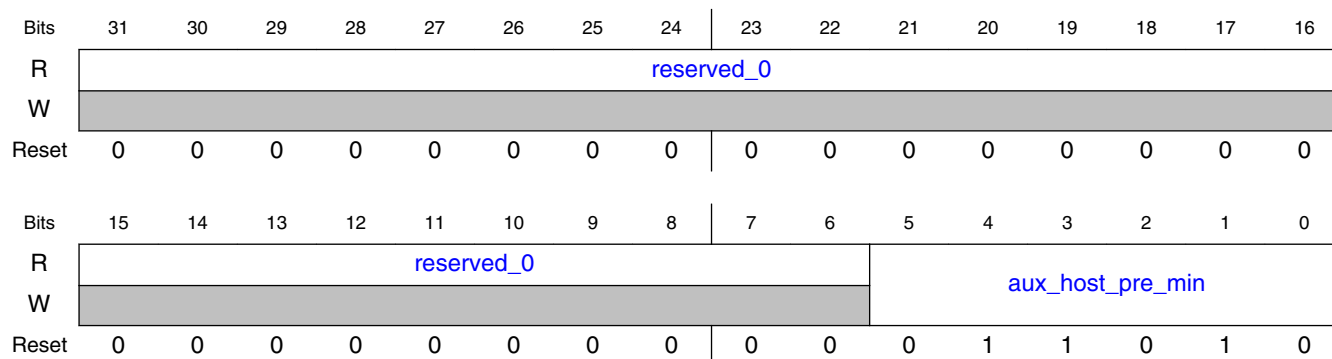
Field	Function
31-11 reserved_0	
10-0 aux_host_1m_min	The minimum legal frequency receiving from the line by the standard is 0. The minimum legal frequency receiving from the line by the standard is 0.83MHz.The calculation is:(0.83 MHz cycle time)/(sys_clk(+15%) cycle time) 1200/34 =35

13.4.10.1.181 Minimum received preamble length (DP_AUX_RX_PRE_MIN)

13.4.10.1.181.1 Offset

Register	Offset
DP_AUX_RX_PRE_MIN	2838h

13.4.10.1.181.2 Diagram



13.4.10.1.181.3 Fields

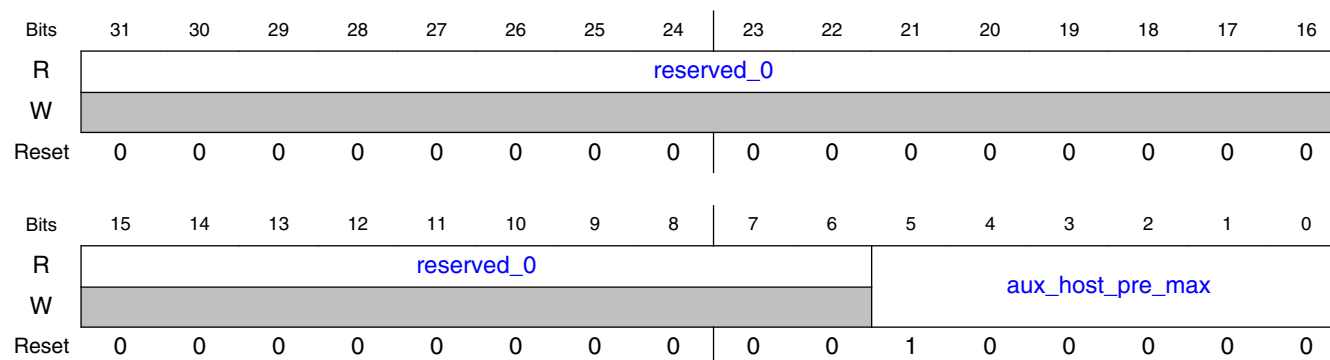
Field	Function
31-6 reserved_0	
5-0 aux_host_pre_max	Valid minimum length of preamble during receive. The standard defines pre_min=26 The value of this register should be greater then the average_number_of_cycles defined in reg 0 (2 , 4 or 8)

13.4.10.1.182 Maximum received preamble length (DP_AUX_RX_PRE_MAX)

13.4.10.1.182.1 Offset

Register	Offset
DP_AUX_RX_PRE_MAX	283Ch

13.4.10.1.182.2 Diagram



13.4.10.1.182.3 Fields

Field	Function
31-6 reserved_0	
5-0 aux_host_pre_max	Valid maximum length of preamble during receive. The standard defines pre_max = 32

13.4.10.1.183 DP_AUX_MAIN start value (DP_AUX_TIMER_PRESET)

13.4.10.1.183.1 Offset

Register	Offset
DP_AUX_TIMER_PRESET	2840h

13.4.10.1.183.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	aux_host_timer_preset															
W																
Reset	0	0	0	1	1	1	0	1	0	1	0	0	1	1	0	0

13.4.10.1.183.3 Fields

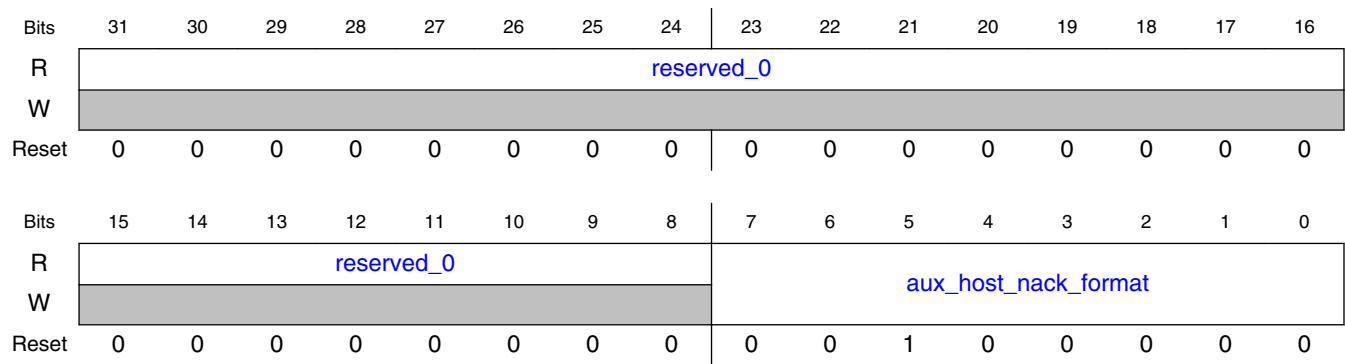
Field	Function
31-16 reserved_0	
15-0 aux_host_timer_preset	The preset value of the timer in DP_IN mode. The preset value of the timer in DP_IN mode. With sys_clk= 25MHz the Timer can measure up to ~2500 micro seconds. The defaults value is 300 microseconds (1D4c in Hex)

13.4.10.1.184 Transmit pattern (DP_AUX_NACK_FORMAT)

13.4.10.1.184.1 Offset

Register	Offset
DP_AUX_NACK_FORMAT	2844h

13.4.10.1.184.2 Diagram



13.4.10.1.184.3 Fields

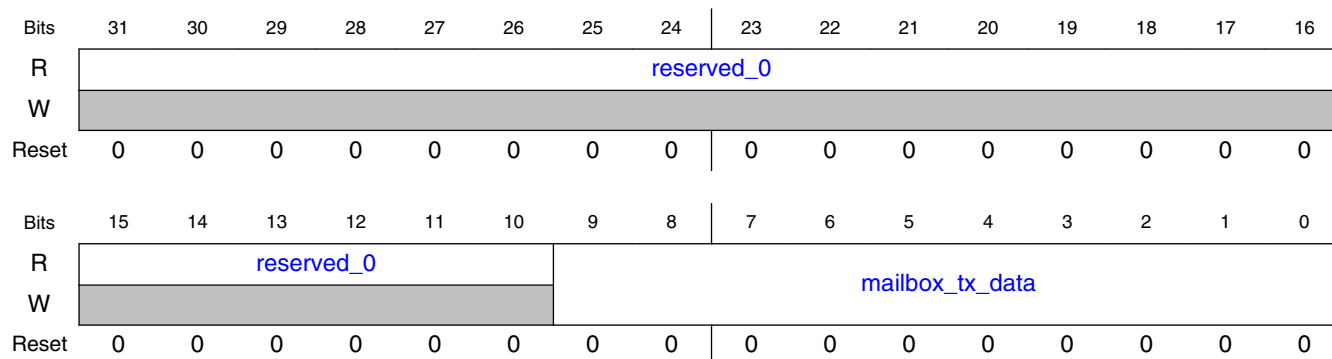
Field	Function
31-8 reserved_0	
7-0 aux_host_nack_ format	Nack or defer pattern for transmit (00100000 for defer, 00010000 for nack)

13.4.10.1.185 Mailbox write data (DP_AUX_TX_DATA)

13.4.10.1.185.1 Offset

Register	Offset
DP_AUX_TX_DATA	2848h

13.4.10.1.185.2 Diagram



13.4.10.1.185.3 Fields

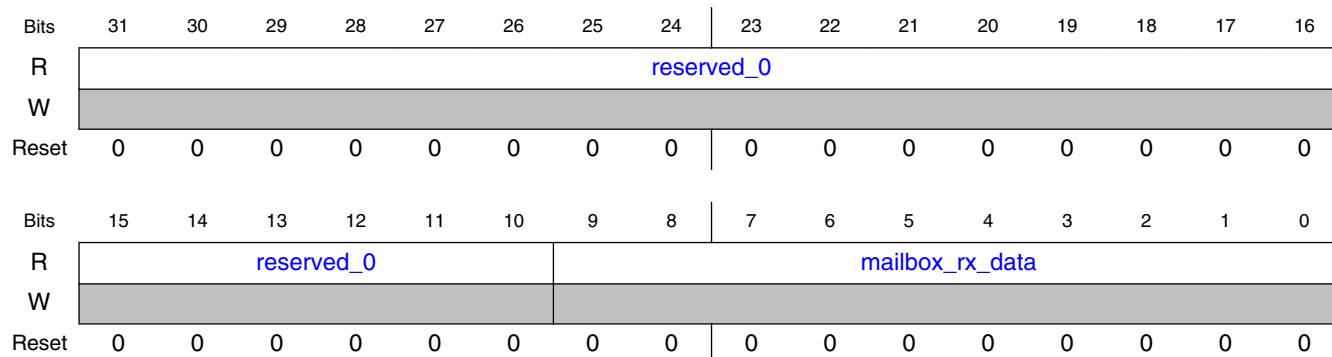
Field	Function
31-10 reserved_0	
9-0 mailbox_tx_data	TX data byte written to the mailbox. It is written 20 times and directly transferred into TX mailbox. First 8 bits are regular data. When the first data is transferred into mailbox mailbox_tx_data[8] (frame start) is set to 1. When the last data is transferred into mailbox, mailbox_tx_data[9] (frame_end) is set to 1.

13.4.10.1.186 Mailbox read data (DP_AUX_RX_DATA)

13.4.10.1.186.1 Offset

Register	Offset
DP_AUX_RX_DATA	284Ch

13.4.10.1.186.2 Diagram



13.4.10.1.186.3 Fields

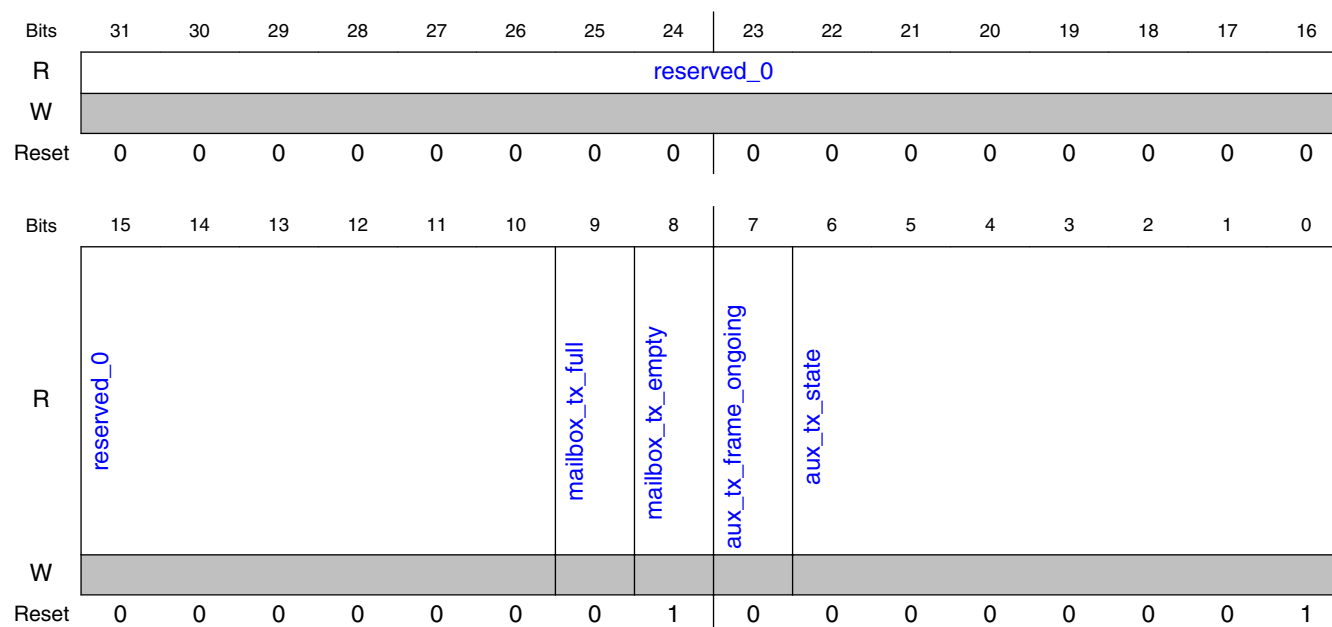
Field	Function
31-10 reserved_0	
9-0 mailbox_rx_data	Read data from the mailbox. Read data from the mailbox. Whenever read to this register occurs, aux_mailbox_read to RX Mailbox shall be asserted for one clock cycle.

13.4.10.1.187 AUX_TX status (DP_AUX_TX_STATUS)

13.4.10.1.187.1 Offset

Register	Offset
DP_AUX_TX_STATUS	2850h

13.4.10.1.187.2 Diagram



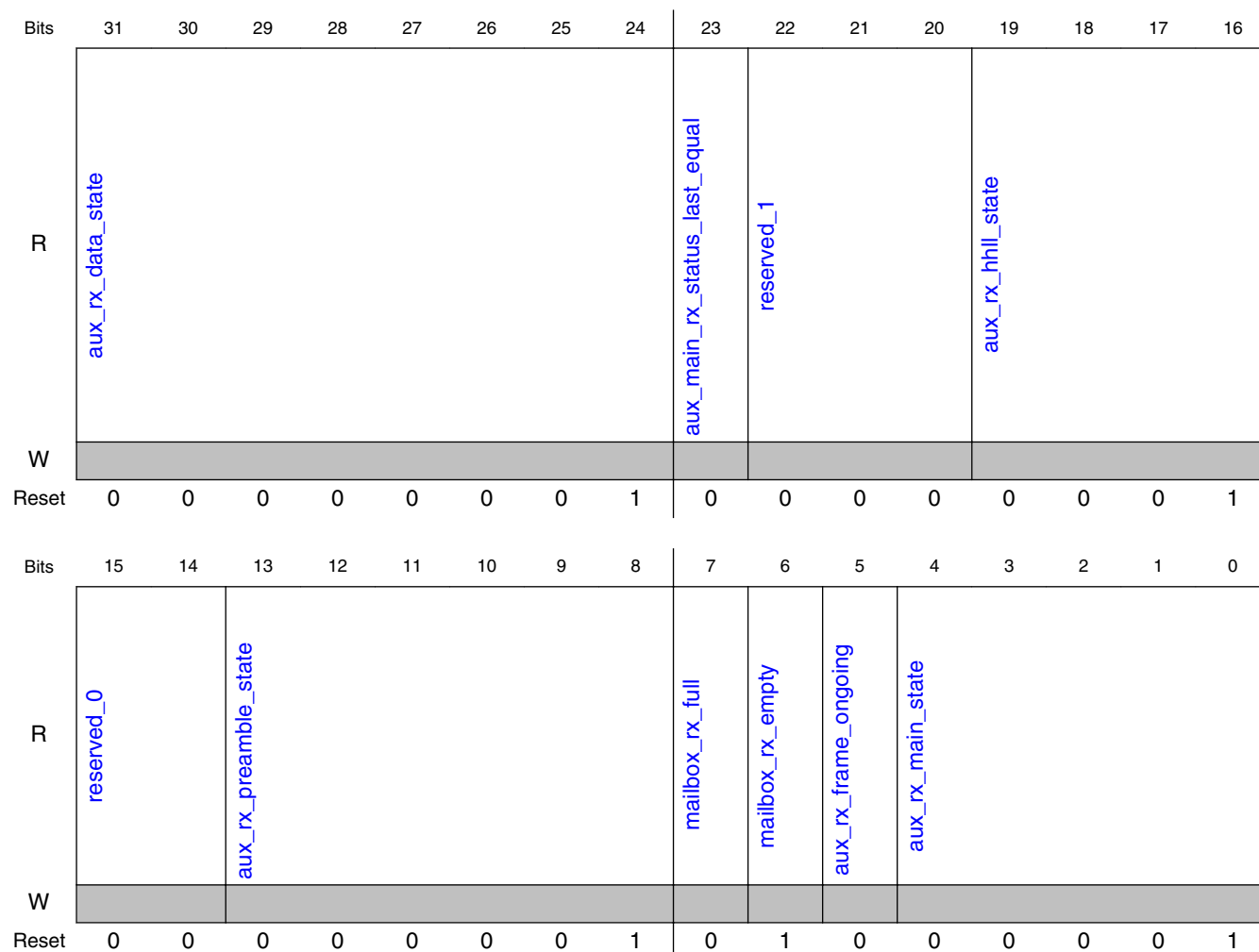
13.4.10.1.187.3 Fields

Field	Function
31-10 reserved_0	
9 mailbox_tx_full	Mailbox TX full flag
8 mailbox_tx_empty	Mailbox TX empty flag
7 aux_tx_frame_ongoing	Frame transmission status
6-0 aux_tx_state	Aux_tx state machine register

13.4.10.1.188 AUX_RX status (DP_AUX_RX_STATUS)**13.4.10.1.188.1 Offset**

Register	Offset
DP_AUX_RX_STATUS	2854h

13.4.10.1.188.2 Diagram



13.4.10.1.188.3 Fields

Field	Function
31-24 <code>aux_rx_data_state</code>	AUX_RX SM state
23 <code>aux_main_rx_status_last_equal</code>	The receive transaction is equal to the previous transaction. The receive transaction is equal to the previous transaction. (THIS IS NOT ERROR BIT !!)
22-20 <code>reserved_1</code>	
19-16 <code>aux_rx_hhll_state</code>	AUX_RX hhll state machine register

Table continues on the next page...

Clocks And Resets

Field	Function
15-14 reserved_0	
13-8 aux_rx_preamble_state	AUX_RX preamble state machine register
7 mailbox_rx_full	Mailbox RX full flag
6 mailbox_rx_empty	Mailbox RX empty flag
5 aux_rx_frame_ongoing	Frame reception status
4-0 aux_rx_main_state	AUX_RX main state machine register

13.4.10.1.189 RX counter status (DP_AUX_RX_CYCLE_COUNTER)

13.4.10.1.189.1 Offset

Register	Offset
DP_AUX_RX_CYCLE_COUNTER	2858h

13.4.10.1.189.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								aux_rx_cycle_counter							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.189.3 Fields

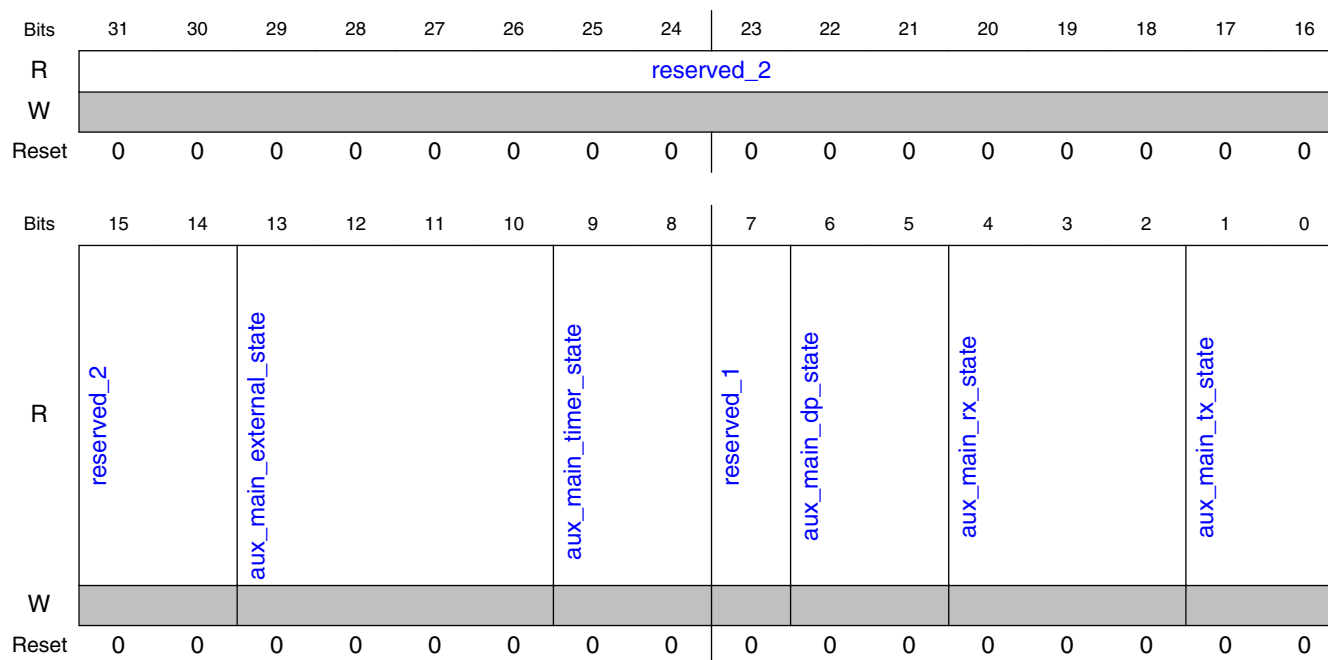
Field	Function
31-11 reserved_0	
10-0 aux_rx_cycle_counter	Count system clocks from last change in the auxiliary line input

13.4.10.1.190 DP_AUX MAIN State Machines status (DP_AUX_MAIN_STATES)

13.4.10.1.190.1 Offset

Register	Offset
DP_AUX_MAIN_STATES	285Ch

13.4.10.1.190.2 Diagram



13.4.10.1.190.3 Fields

Field	Function
31-14 reserved_2	
13-10 aux_main_external_state	AUX_MAIN external state machine register
9-8 aux_main_timer_state	AUX_MAIN timer state machine
7 reserved_1	
6-5 aux_main_dp_state	AUX_MAIN dp state machine
4-2 aux_main_rx_state	AUX_MAIN rx state machine
1-0 aux_main_tx_state	AUX_MAIN tx state machine

13.4.10.1.191 DP_AUX MAIN timer status (DP_AUX_MAIN_TIMER)

13.4.10.1.191.1 Offset

Register	Offset
DP_AUX_MAIN_TIMER	2860h

13.4.10.1.191.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	aux_main_timer															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.191.3 Fields

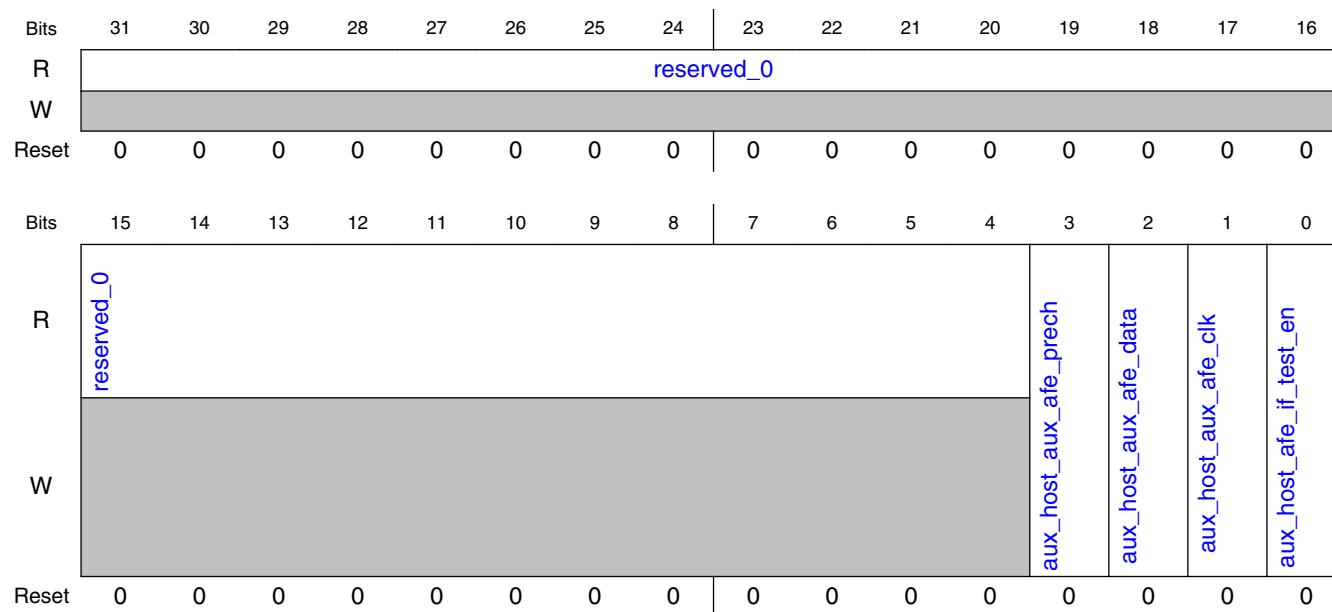
Field	Function
31-16 reserved_0	
15-0 aux_main_timer	DP_AUX MAIN timer status

13.4.10.1.192 Test mode configuration (DP_AUX_AFE_OUT)

13.4.10.1.192.1 Offset

Register	Offset
DP_AUX_AFE_OUT	2864h

13.4.10.1.192.2 Diagram



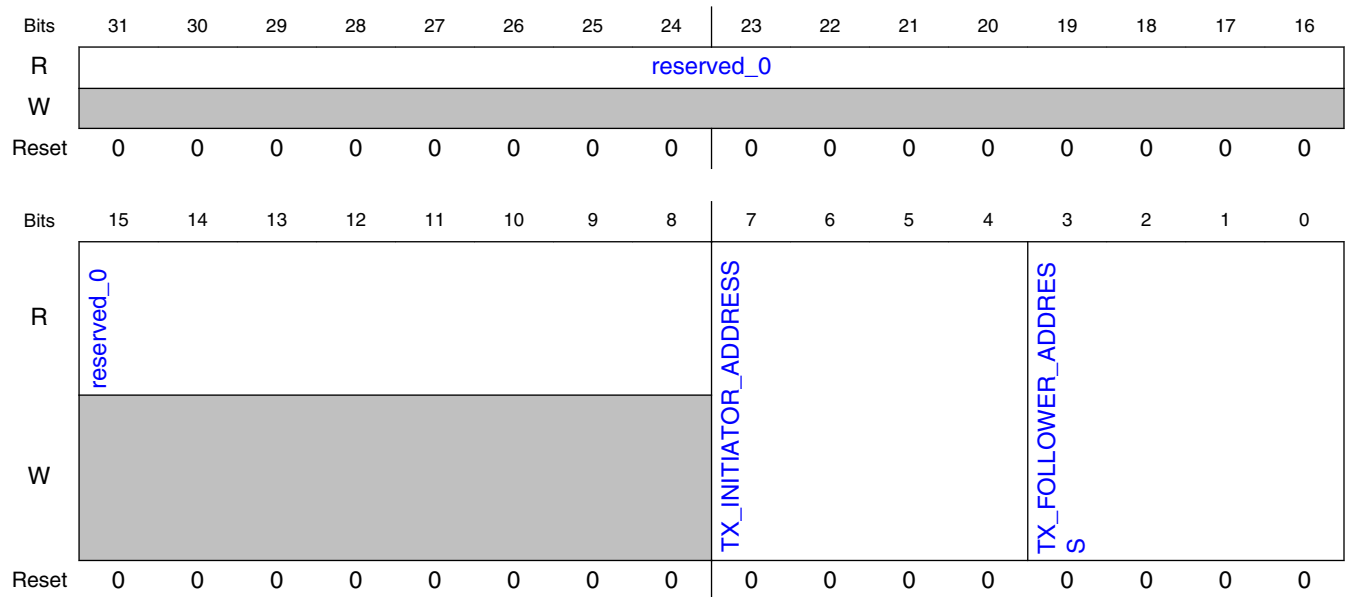
13.4.10.1.192.3 Fields

Field	Function
31-4 reserved_0	
3 aux_host_aux_afe_prech	Drive the aux_data_prech output to the AFE when aux_host_afe_if_test_en (bit 0) is set
2 aux_host_aux_afe_data	Drive the aux_data_out output to the AFE when aux_host_afe_if_test_en (bit 0) is set
1 aux_host_aux_afe_clk	Drive the aux_clk_out output to the AFE when aux_host_afe_if_test_en (bit 0) is set
0 aux_host_afe_if_test_en	TESTER mode enable. TESTER mode enable. Give the TESTER direct interface to the AFE_AUX.

13.4.10.1.193 CEC TX message header (TX_MSG_HEADER)**13.4.10.1.193.1 Offset**

Register	Offset
TX_MSG_HEADER	3800h

13.4.10.1.193.2 Diagram



13.4.10.1.193.3 Fields

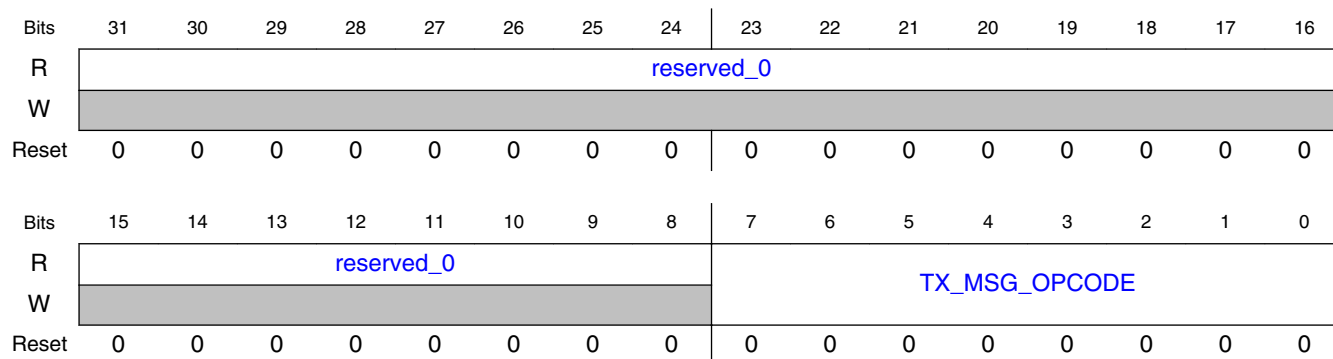
Field	Function
31-8 reserved_0	
7-4 TX_INITIATOR_ADDRESS	The 4 bit logical address of the initiator
3-0 TX_FOLLOWER_ADDRESS	The 4 bit logical address of the follower

13.4.10.1.194 CEC TX message opcode (TX_MSG_OPCODE)

13.4.10.1.194.1 Offset

Register	Offset
TX_MSG_OPCODE	3804h

13.4.10.1.194.2 Diagram



13.4.10.1.194.3 Fields

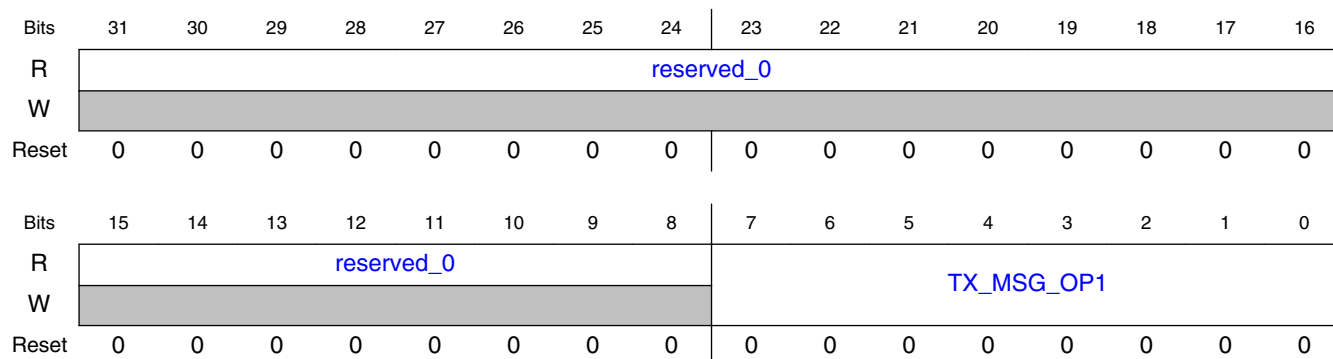
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OPCODE	The opcode of the current message

13.4.10.1.195 CEC TX message operand - byte 1 (TX_MSG_OP1)

13.4.10.1.195.1 Offset

Register	Offset
TX_MSG_OP1	3808h

13.4.10.1.195.2 Diagram



13.4.10.1.195.3 Fields

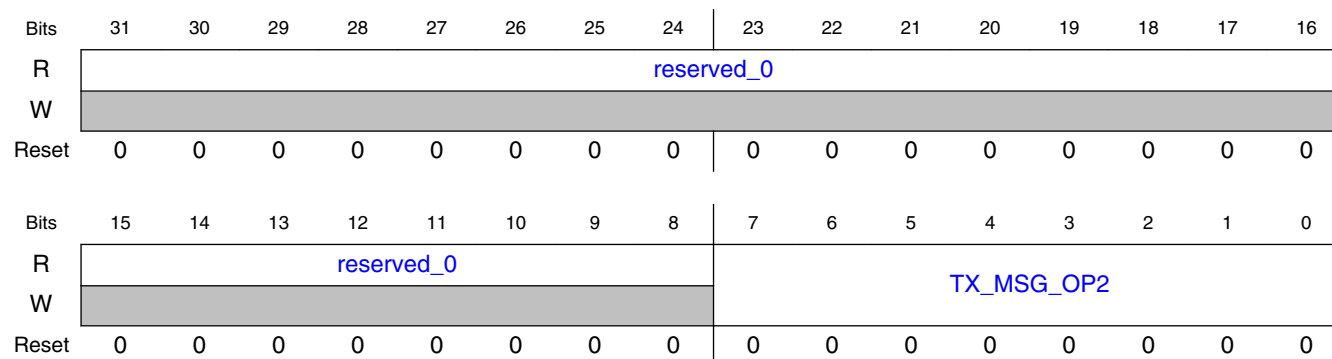
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP1	The first operand required by the current opcode. The first operand required by the current opcode.

13.4.10.1.196 CEC TX message operand - byte 2 (TX_MSG_OP2)

13.4.10.1.196.1 Offset

Register	Offset
TX_MSG_OP2	380Ch

13.4.10.1.196.2 Diagram



13.4.10.1.196.3 Fields

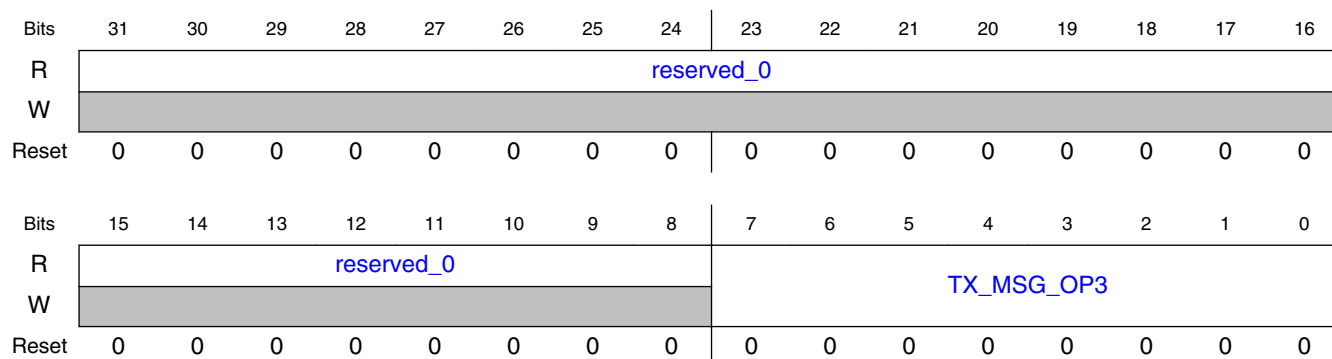
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP2	The second operand required by the current opcode. The second operand required by the current opcode.

13.4.10.1.197 CEC TX message operand - byte 3 (TX_MSG_OP3)

13.4.10.1.197.1 Offset

Register	Offset
TX_MSG_OP3	3810h

13.4.10.1.197.2 Diagram



13.4.10.1.197.3 Fields

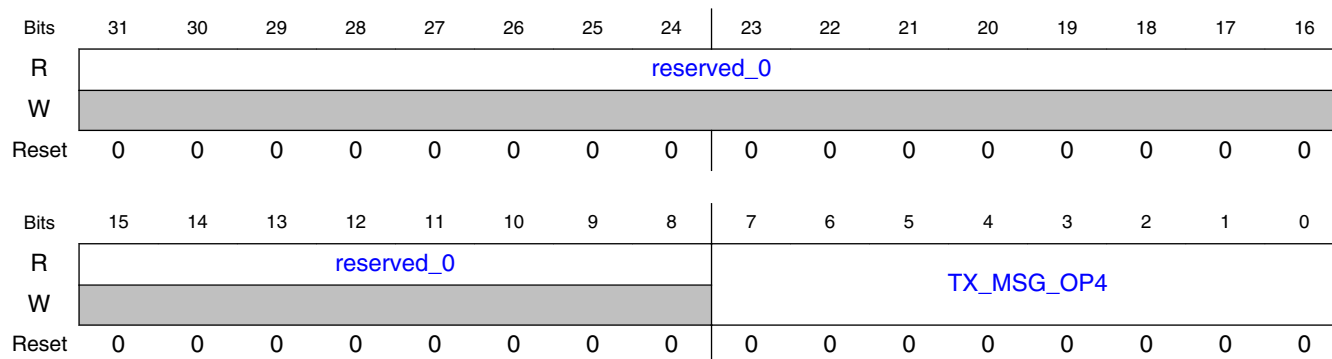
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP3	The third operand required by the current opcode.

13.4.10.1.198 CEC TX message operand - byte 4 (TX_MSG_OP4)

13.4.10.1.198.1 Offset

Register	Offset
TX_MSG_OP4	3814h

13.4.10.1.198.2 Diagram



13.4.10.1.198.3 Fields

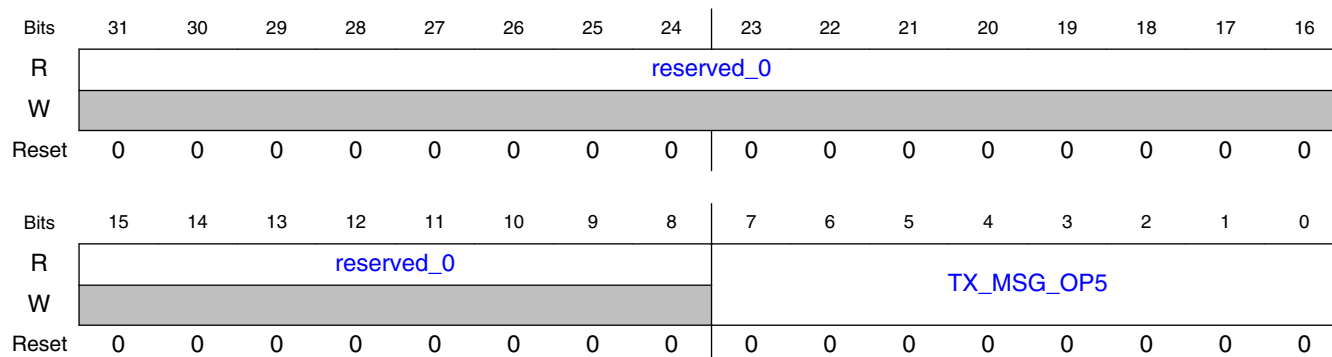
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP4	The fourth operand required by the current opcode.

13.4.10.1.199 CEC TX message operand - byte 5 (TX_MSG_OP5)

13.4.10.1.199.1 Offset

Register	Offset
TX_MSG_OP5	3818h

13.4.10.1.199.2 Diagram



13.4.10.1.199.3 Fields

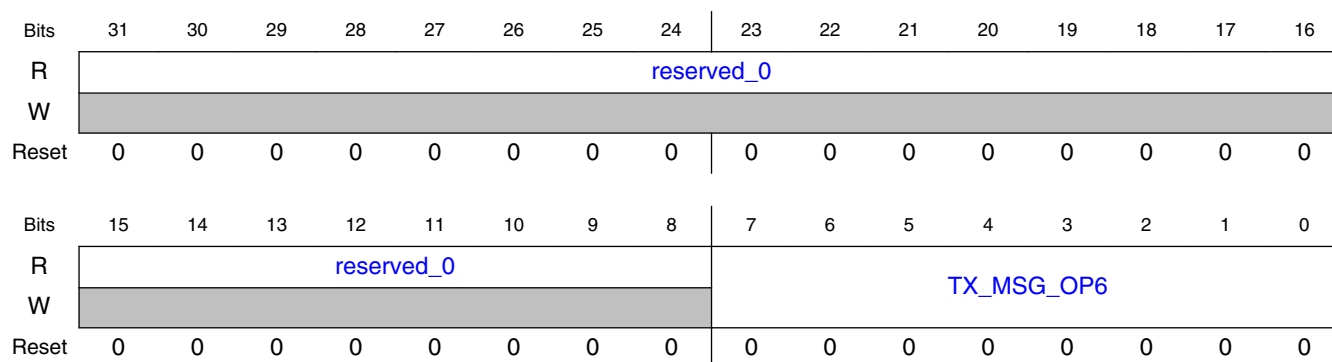
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP5	The fifth operand required by the current opcode. The fifth operand required by the current opcode.

13.4.10.1.200 CEC TX message operand - byte 6 (TX_MSG_OP6)

13.4.10.1.200.1 Offset

Register	Offset
TX_MSG_OP6	381Ch

13.4.10.1.200.2 Diagram



13.4.10.1.200.3 Fields

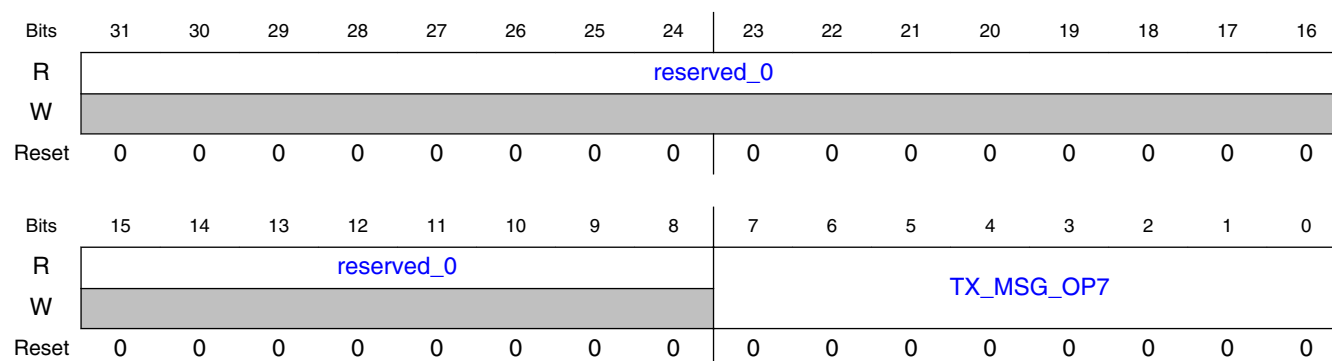
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP6	The sixth operand required by the current opcode. The sixth operand required by the current opcode.

13.4.10.1.201 CEC TX message operand - byte 7 (TX_MSG_OP7)

13.4.10.1.201.1 Offset

Register	Offset
TX_MSG_OP7	3820h

13.4.10.1.201.2 Diagram



13.4.10.1.201.3 Fields

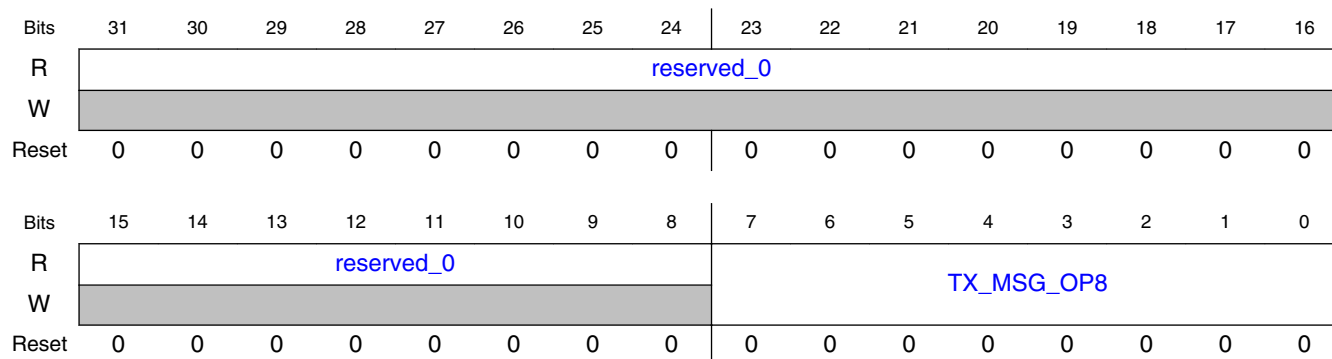
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP7	The seventh operand required by the current opcode.

13.4.10.1.202 CEC TX message operand - byte 8 (TX_MSG_OP8)

13.4.10.1.202.1 Offset

Register	Offset
TX_MSG_OP8	3824h

13.4.10.1.202.2 Diagram



13.4.10.1.202.3 Fields

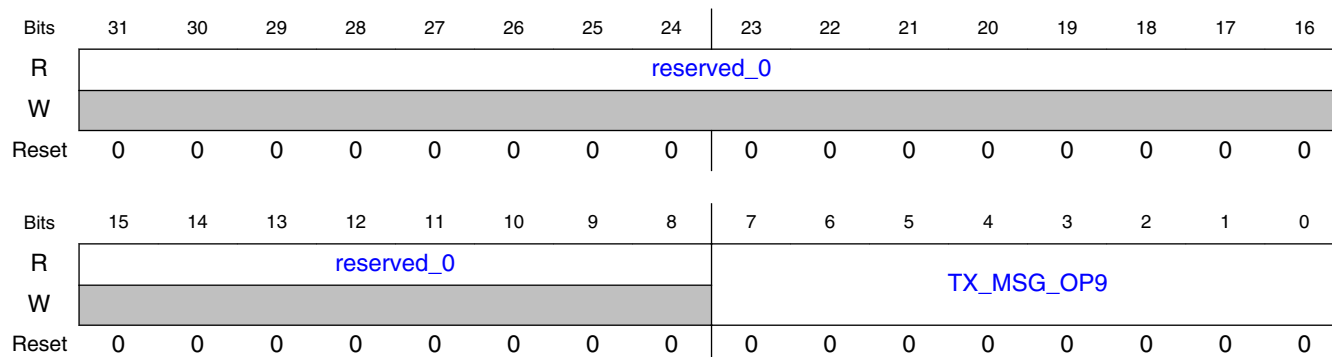
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP8	The eighth operand required by the current opcode.

13.4.10.1.203 CEC TX message operand - byte 9 (TX_MSG_OP9)

13.4.10.1.203.1 Offset

Register	Offset
TX_MSG_OP9	3828h

13.4.10.1.203.2 Diagram



13.4.10.1.203.3 Fields

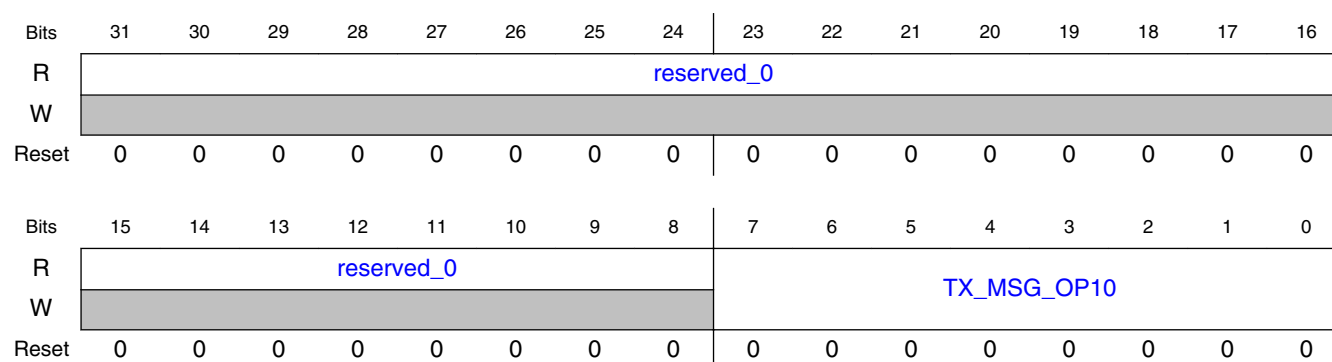
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP9	The ninth operand required by the current opcode. The ninth operand required by the current opcode.

13.4.10.1.204 CEC TX message operand - byte 10 (TX_MSG_OP10)

13.4.10.1.204.1 Offset

Register	Offset
TX_MSG_OP10	382Ch

13.4.10.1.204.2 Diagram



13.4.10.1.204.3 Fields

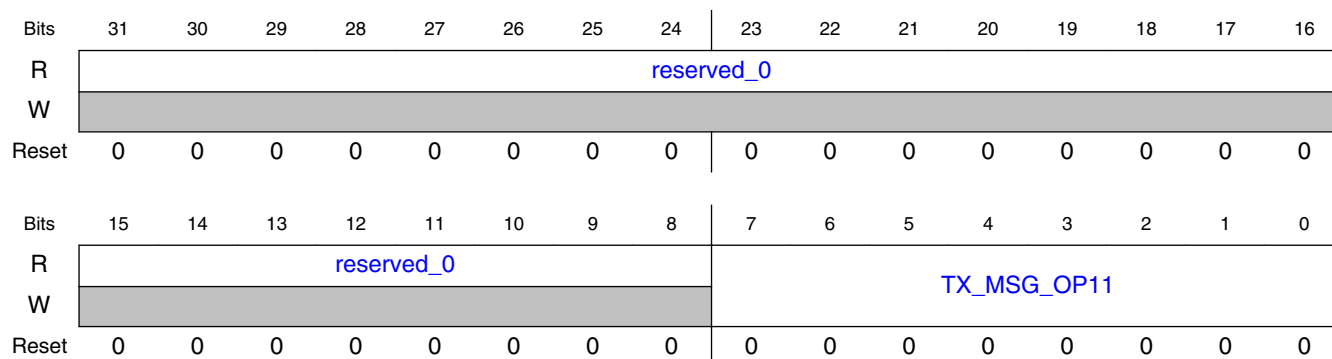
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP10	The tenth operand required by the current opcode. The tenth operand required by the current opcode.

13.4.10.1.205 CEC TX message operand - byte 11 (TX_MSG_OP11)

13.4.10.1.205.1 Offset

Register	Offset
TX_MSG_OP11	3830h

13.4.10.1.205.2 Diagram



13.4.10.1.205.3 Fields

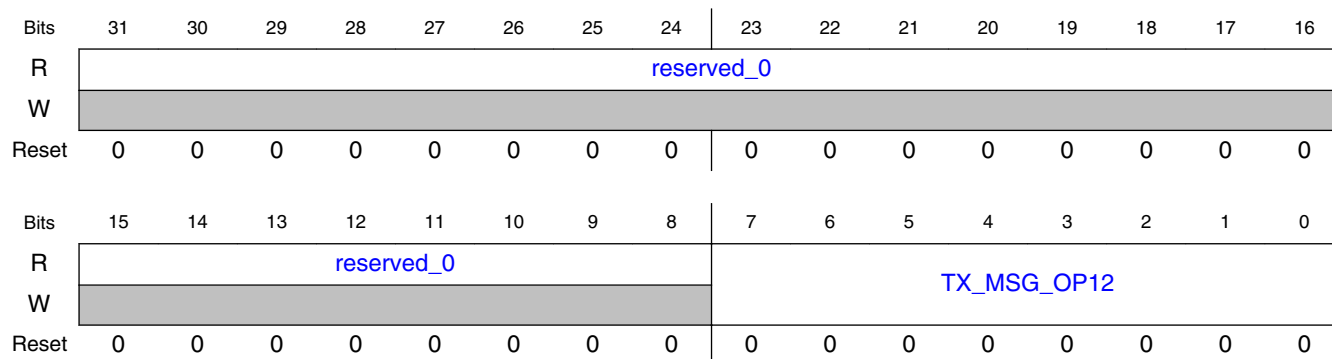
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP11	The eleventh operand required by the current opcode.

13.4.10.1.206 CEC TX message operand - byte 12 (TX_MSG_OP12)

13.4.10.1.206.1 Offset

Register	Offset
TX_MSG_OP12	3834h

13.4.10.1.206.2 Diagram



13.4.10.1.206.3 Fields

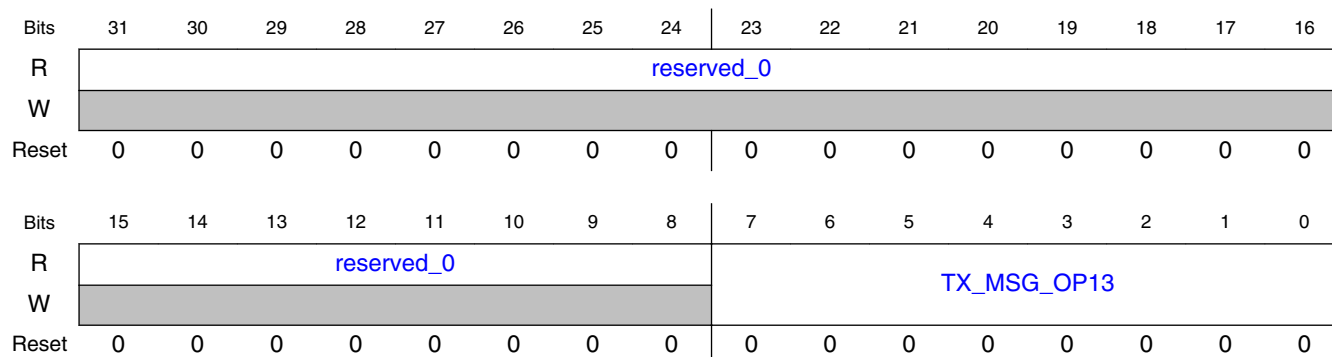
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP12	The twelvth operand required by the current opcode.

13.4.10.1.207 CEC TX message operand - byte 13 (TX_MSG_OP13)

13.4.10.1.207.1 Offset

Register	Offset
TX_MSG_OP13	3838h

13.4.10.1.207.2 Diagram



13.4.10.1.207.3 Fields

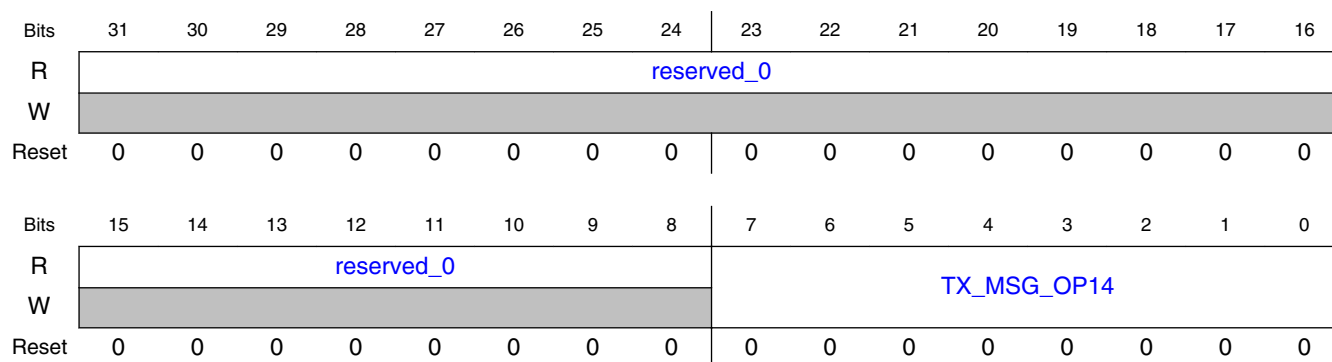
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP13	The thirteenth operand required by the current opcode.

13.4.10.1.208 CEC TX message operand - byte 14 (TX_MSG_OP14)

13.4.10.1.208.1 Offset

Register	Offset
TX_MSG_OP14	383Ch

13.4.10.1.208.2 Diagram



13.4.10.1.208.3 Fields

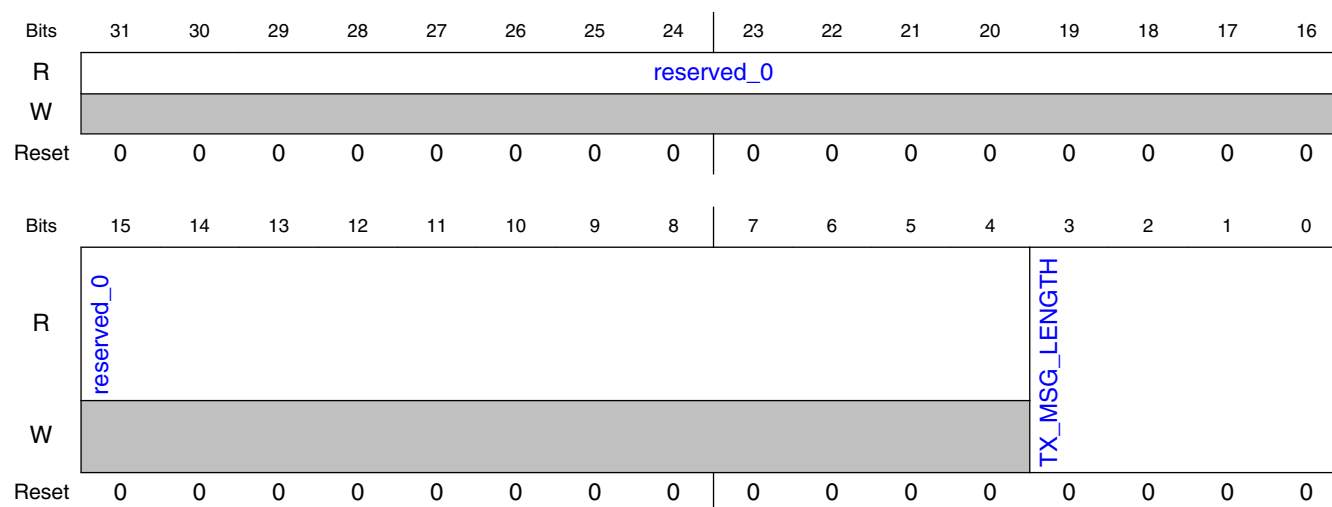
Field	Function
31-8 reserved_0	
7-0 TX_MSG_OP14	The fourteenth operand required by the current opcode.

13.4.10.1.209 CEC TX message length (TX_MSG_LENGTH)

13.4.10.1.209.1 Offset

Register	Offset
TX_MSG_LENGTH	3840h

13.4.10.1.209.2 Diagram



13.4.10.1.209.3 Fields

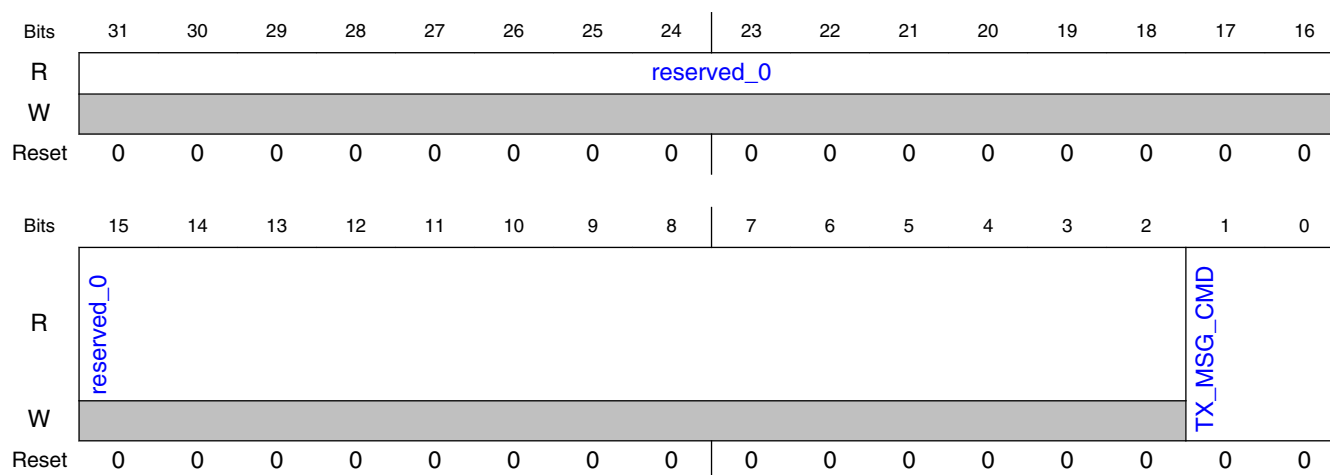
Field	Function
31-4 reserved_0	
3-0 TX_MSG_LENGTH	The number of bytes in the current message. The number of bytes in the current message. Zero means only header.

13.4.10.1.210 CEC TX message command (TX_MSG_CMD)

13.4.10.1.210.1 Offset

Register	Offset
TX_MSG_CMD	3844h

13.4.10.1.210.2 Diagram



13.4.10.1.210.3 Fields

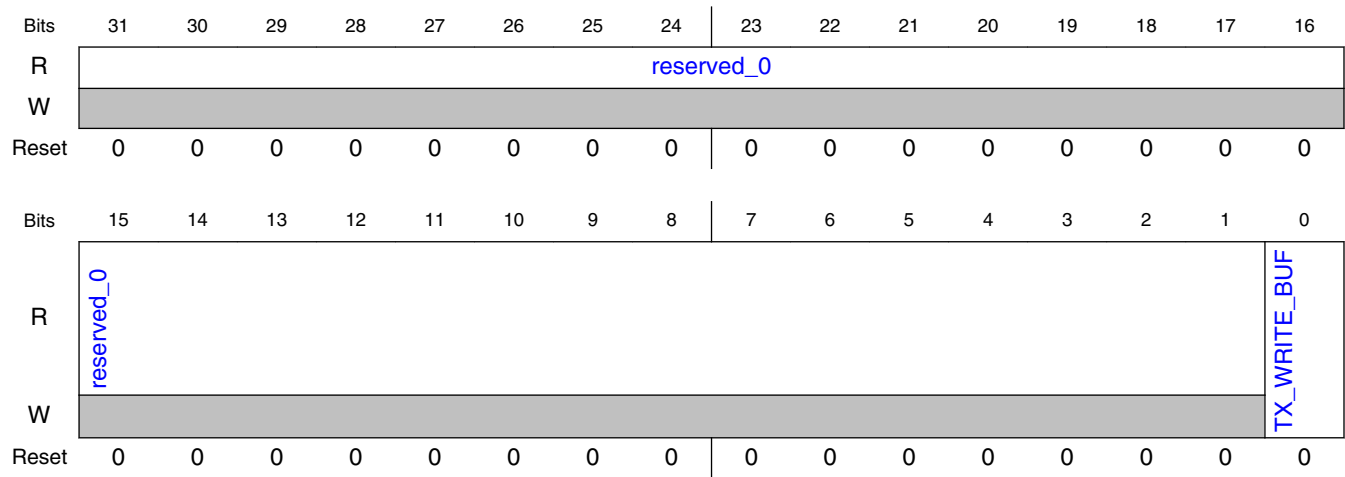
Field	Function
31-2 reserved_0	
1-0 TX_MSG_CMD	00: No transaction, 01: transmit earliest message in buffer, 10: abort transmitting earliest message, 11: overwrite earliest message and transmit next.

13.4.10.1.211 CEC TX buffer write enable (TX_WRITE_BUF)

13.4.10.1.211.1 Offset

Register	Offset
TX_WRITE_BUF	3848h

13.4.10.1.211.2 Diagram



13.4.10.1.211.3 Fields

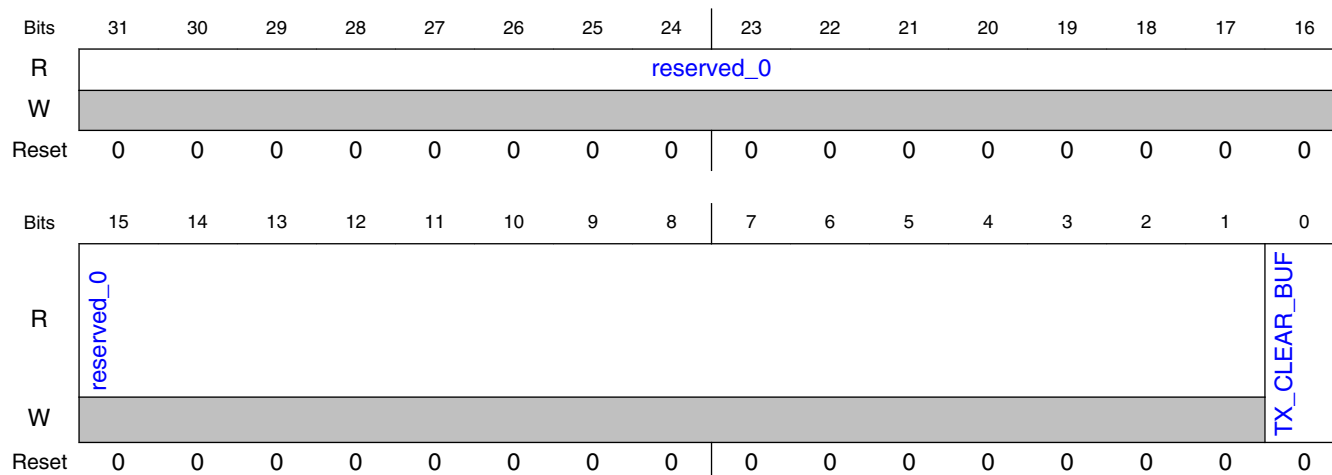
Field	Function
31-1 reserved_0	
0 TX_WRITE_BUF	Write enable (from host). Write enable (from host). -- Used only if BUFDEPTH > 1

13.4.10.1.212 CEC TX buffer clear (TX_CLEAR_BUF)

13.4.10.1.212.1 Offset

Register	Offset
TX_CLEAR_BUF	384Ch

13.4.10.1.212.2 Diagram



13.4.10.1.212.3 Fields

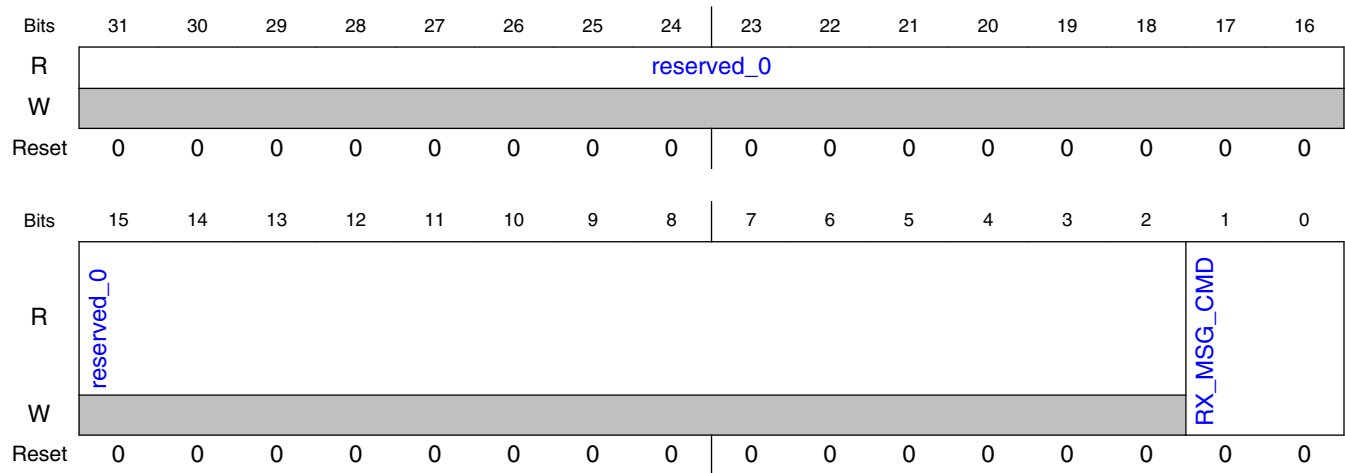
Field	Function
31-1 reserved_0	
0 TX_CLEAR_BUF	Clear all messages in Tx buf. Clear all messages in Tx buf. (from host). -- Used only if BUFDEPTH > 1.

13.4.10.1.213 CEC RX message command (RX_MSG_CMD)

13.4.10.1.213.1 Offset

Register	Offset
RX_MSG_CMD	3850h

13.4.10.1.213.2 Diagram



13.4.10.1.213.3 Fields

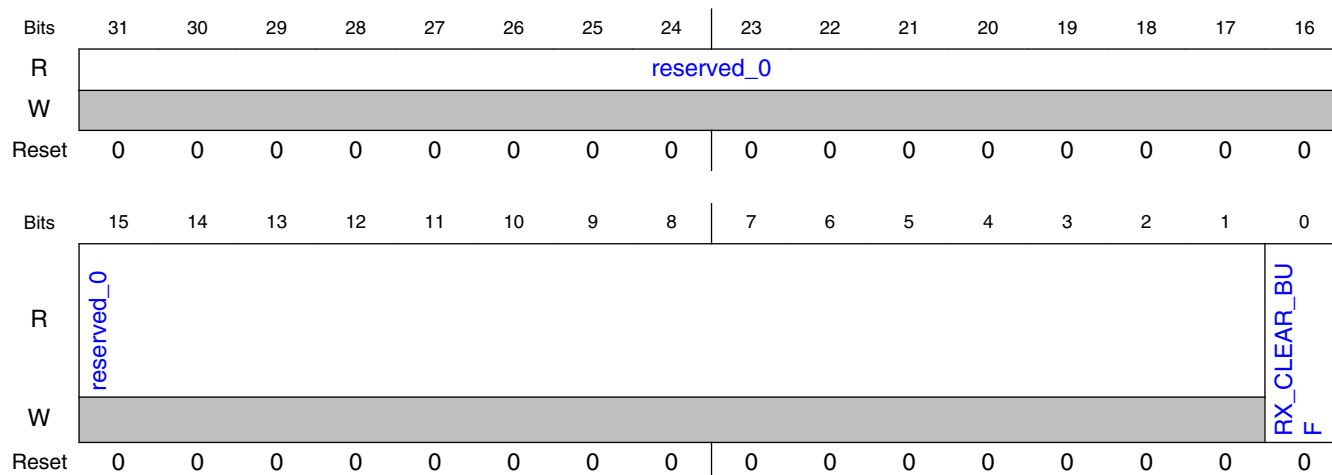
Field	Function
31-2 reserved_0	
1-0 RX_MSG_CMD	00: No transaction, 01: read earliest message in buffer, 10: disable receiving latest message, 11: clear earliest message and read next.

13.4.10.1.214 CEC RX buffer clear (RX_CLEAR_BUF)

13.4.10.1.214.1 Offset

Register	Offset
RX_CLEAR_BUF	3854h

13.4.10.1.214.2 Diagram



13.4.10.1.214.3 Fields

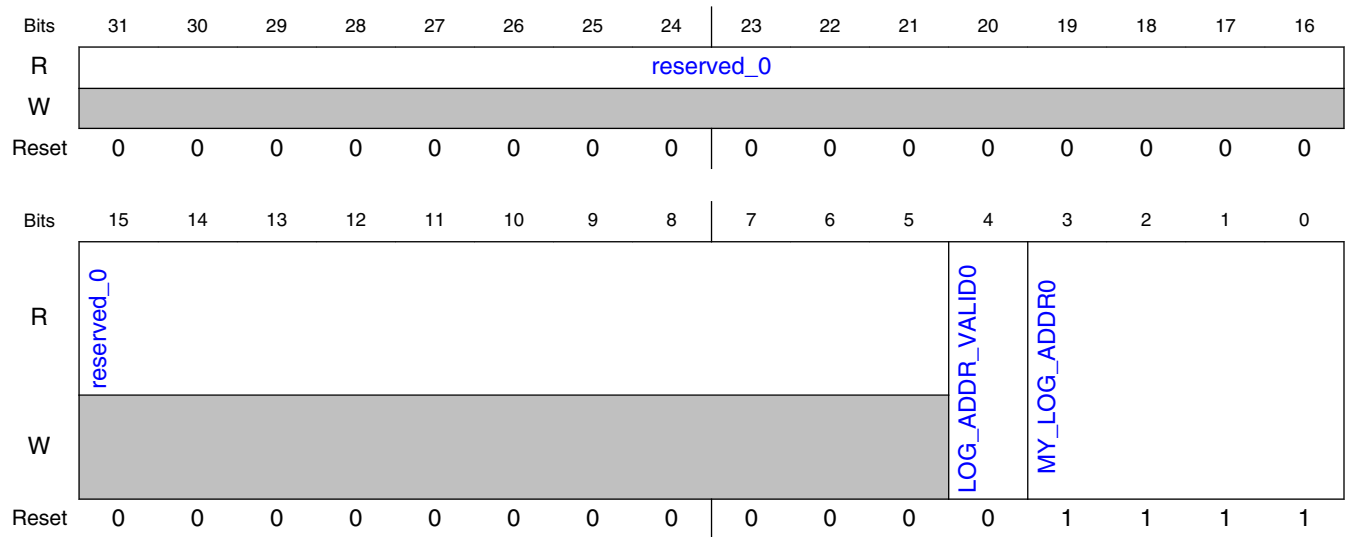
Field	Function
31-1 reserved_0	
0 RX_CLEAR_BU F	Clear all messages in Rx buf. Clear all messages in Rx buf.

13.4.10.1.215 CEC Logical Address 0 (LOGICAL_ADDRESS_LA0)

13.4.10.1.215.1 Offset

Register	Offset
LOGICAL_ADDRESS_ LA0	3858h

13.4.10.1.215.2 Diagram



13.4.10.1.215.3 Fields

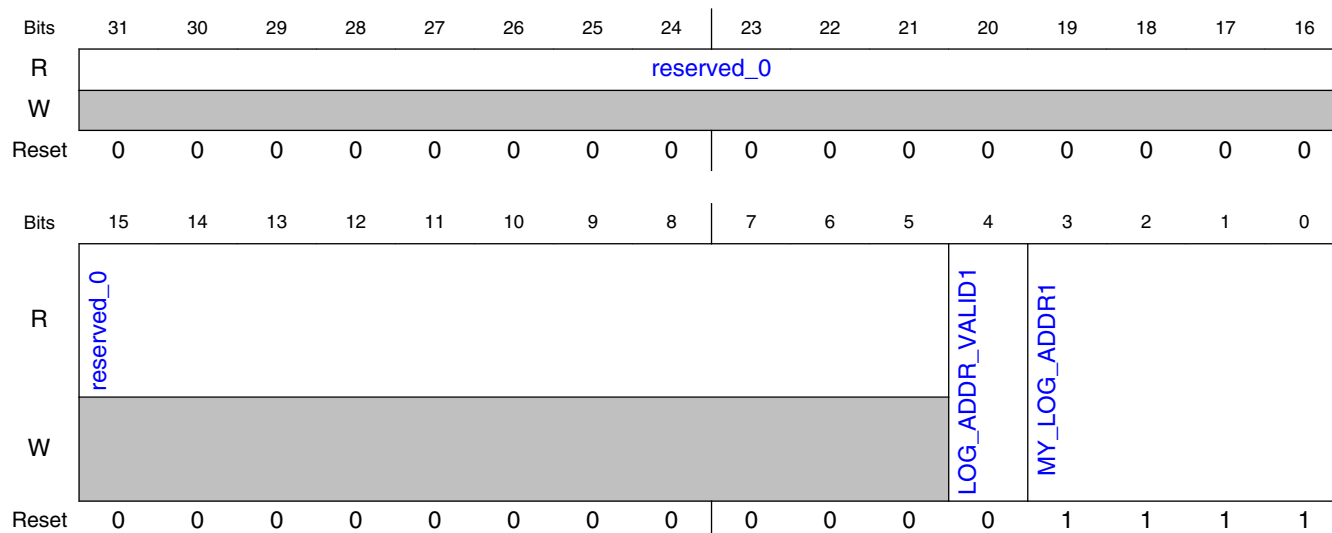
Field	Function
31-5 reserved_0	
4 LOG_ADDR_VALID0	Status of logical address ('1' valid, '0' not valid).
3-0 MY_LOG_ADDR0	The logical address for the first device.

13.4.10.1.216 CEC Logical Address 1 (LOGICAL_ADDRESS_LA1)

13.4.10.1.216.1 Offset

Register	Offset
LOGICAL_ADDRESS_LA1	385Ch

13.4.10.1.216.2 Diagram



13.4.10.1.216.3 Fields

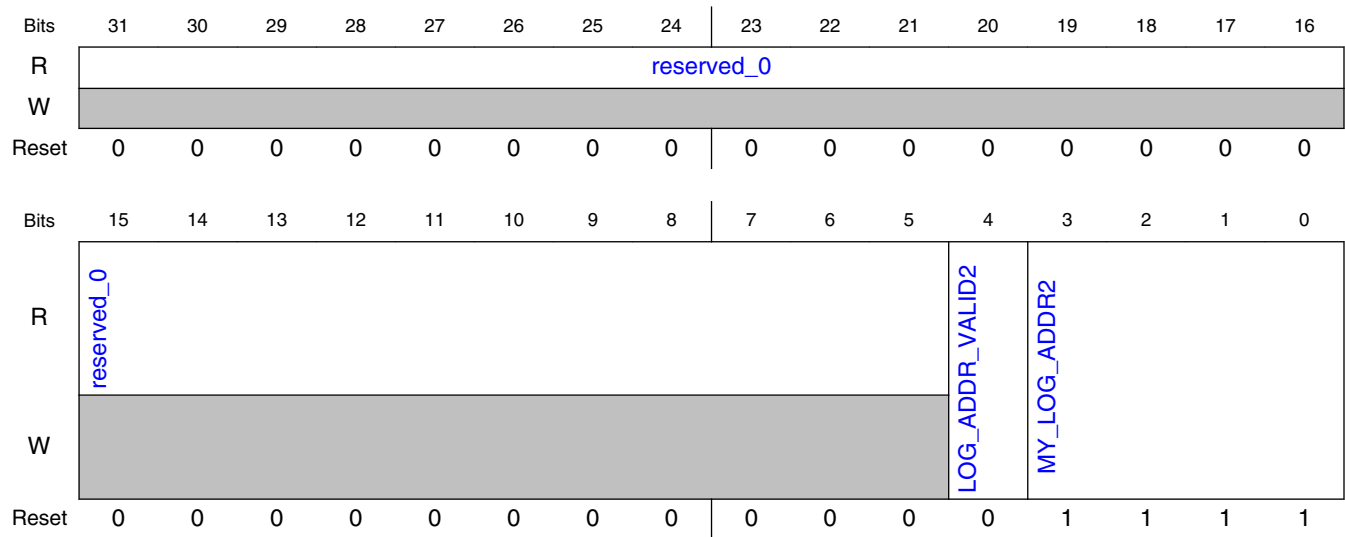
Field	Function
31-5 reserved_0	
4 LOG_ADDR_VALID1	Status of logical address ('1' valid, '0' not valid).
3-0 MY_LOG_ADDR1	The logical address for the second device.

13.4.10.1.217 CEC Logical Address 2 (LOGICAL_ADDRESS_LA2)

13.4.10.1.217.1 Offset

Register	Offset
LOGICAL_ADDRESS_LA2	3860h

13.4.10.1.217.2 Diagram



13.4.10.1.217.3 Fields

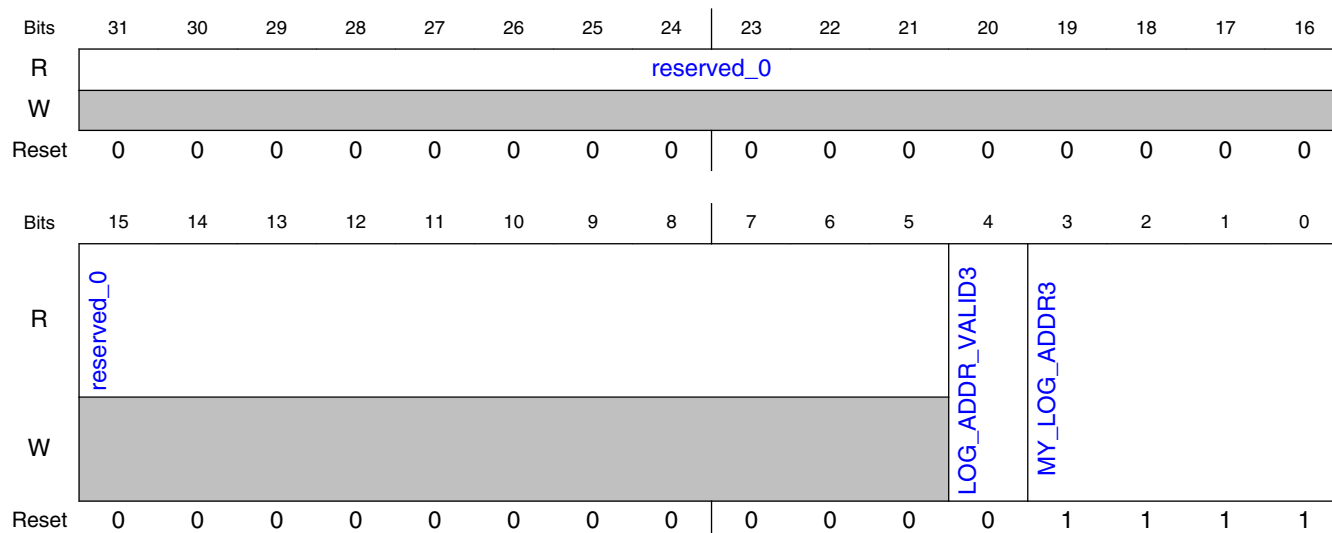
Field	Function
31-5 reserved_0	
4 LOG_ADDR_VALID2	Status of logical address ('1' valid, '0' not valid).
3-0 MY_LOG_ADDR2	The logical address for the third device.

13.4.10.1.218 CEC Logical Address 3 (LOGICAL_ADDRESS_LA3)

13.4.10.1.218.1 Offset

Register	Offset
LOGICAL_ADDRESS_LA3	3864h

13.4.10.1.218.2 Diagram



13.4.10.1.218.3 Fields

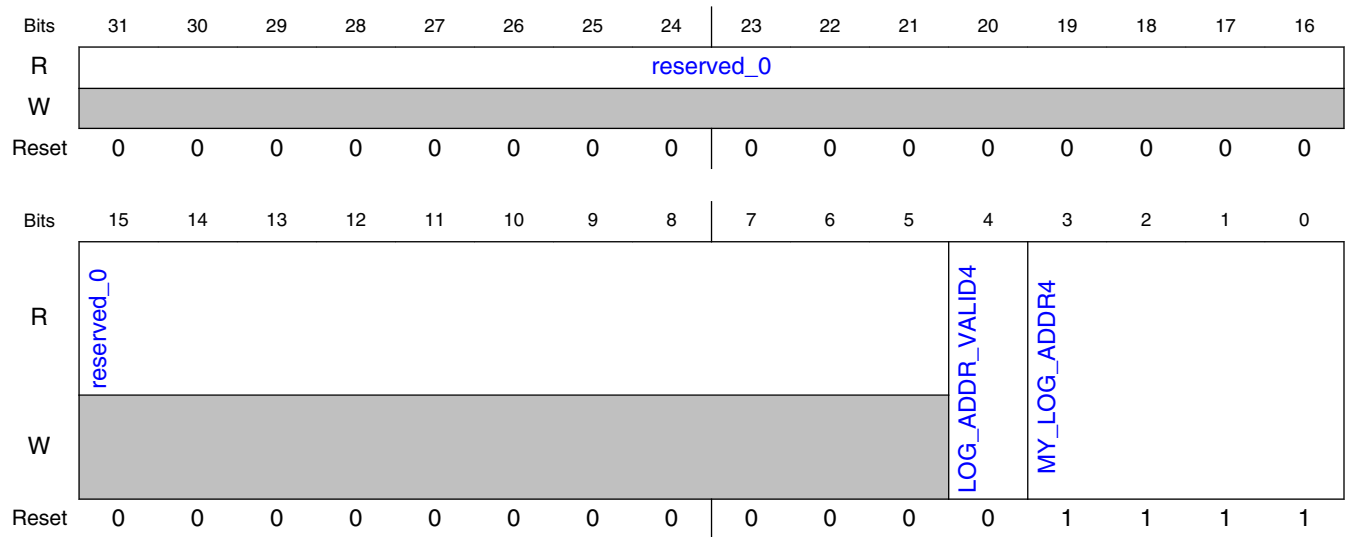
Field	Function
31-5 reserved_0	
4 LOG_ADDR_VALID3	Status of logical address ('1' valid, '0' not valid).
3-0 MY_LOG_ADDR3	The logical address for the fourth device.

13.4.10.1.219 CEC Logical Address 4 (LOGICAL_ADDRESS_LA4)

13.4.10.1.219.1 Offset

Register	Offset
LOGICAL_ADDRESS_LA4	3868h

13.4.10.1.219.2 Diagram



13.4.10.1.219.3 Fields

Field	Function
31-5 reserved_0	
4 LOG_ADDR_VA LID4	Status of logical address ('1' valid, '0' not valid).
3-0 MY_LOG_ADD R4	The logical address for the fifth device.

13.4.10.1.220 CEC Clock Divider - MSB part (CLK_DIV_MSB)

13.4.10.1.220.1 Offset

Register	Offset
CLK_DIV_MSB	386Ch

13.4.10.1.220.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								CLK_DIV_MSB							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

13.4.10.1.220.3 Fields

Field	Function
31-8 reserved_0	
7-0 CLK_DIV_MSB	Clock divider ratio. Clock divider ratio. It divides the system clock in order to have enable every 0.01ms. Default is #h3e7 for 100MHz clock.

13.4.10.1.221 CEC Clock Divider - LSB part (CLK_DIV_LSB)

13.4.10.1.221.1 Offset

Register	Offset
CLK_DIV_LSB	3870h

13.4.10.1.221.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								CLK_DIV_LSB							
W																
Reset	0	0	0	0	0	0	0	0	1	1	1	0	0	1	1	1

13.4.10.1.221.3 Fields

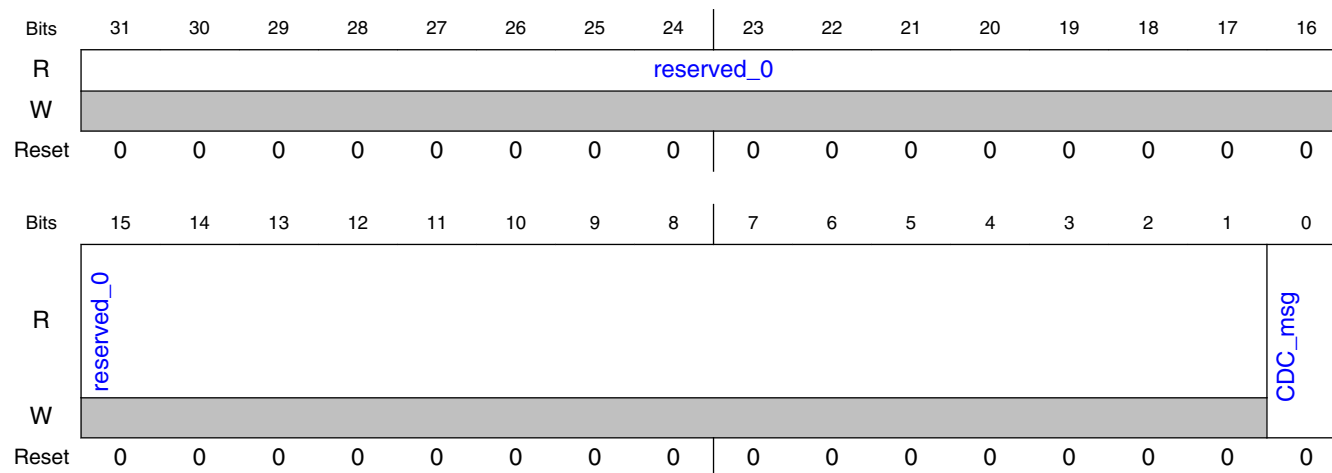
Field	Function
31-8 reserved_0	
7-0 CLK_DIV_LSB	

13.4.10.1.222 CDC mesage enable (cdc_msg)

13.4.10.1.222.1 Offset

Register	Offset
cdc_msg	387Ch

13.4.10.1.222.2 Diagram



13.4.10.1.222.3 Fields

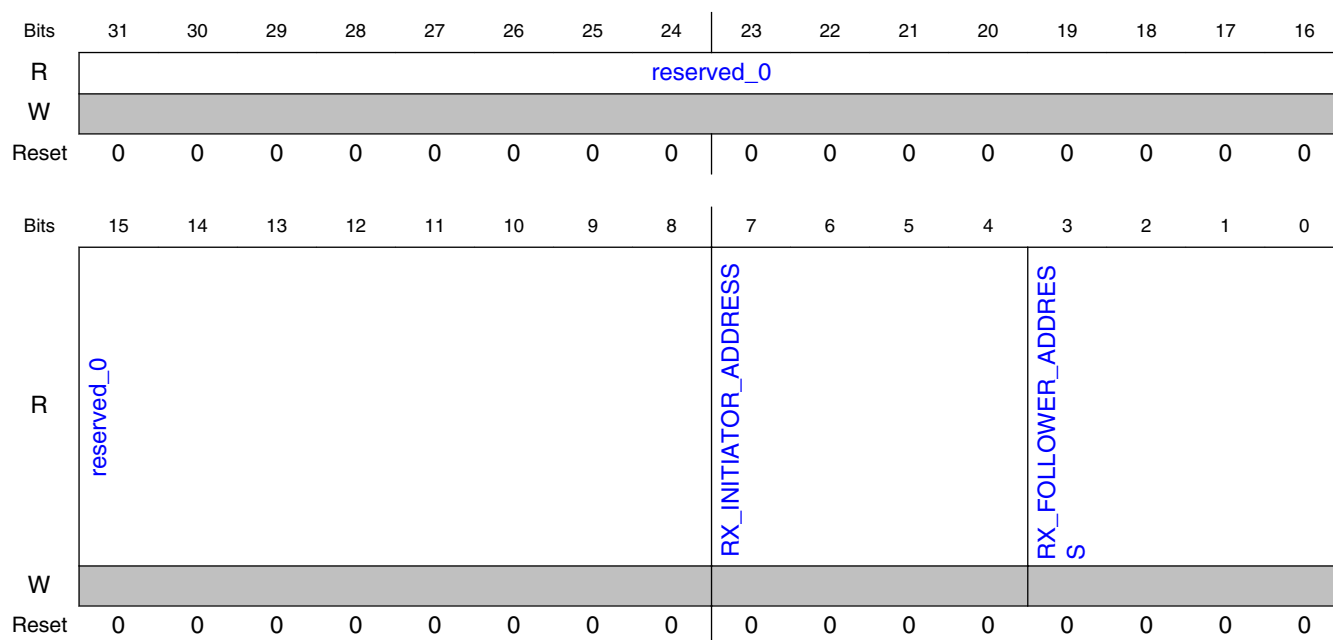
Field	Function
31-1 reserved_0	
0 CDC_msg	0: regular CEC message, 1: CDC message. 0: regular CEC message, 1: CDC message.

13.4.10.1.223 CEC RX message header (RX_MSG_DATA1)

13.4.10.1.223.1 Offset

Register	Offset
RX_MSG_DATA1	3900h

13.4.10.1.223.2 Diagram



13.4.10.1.223.3 Fields

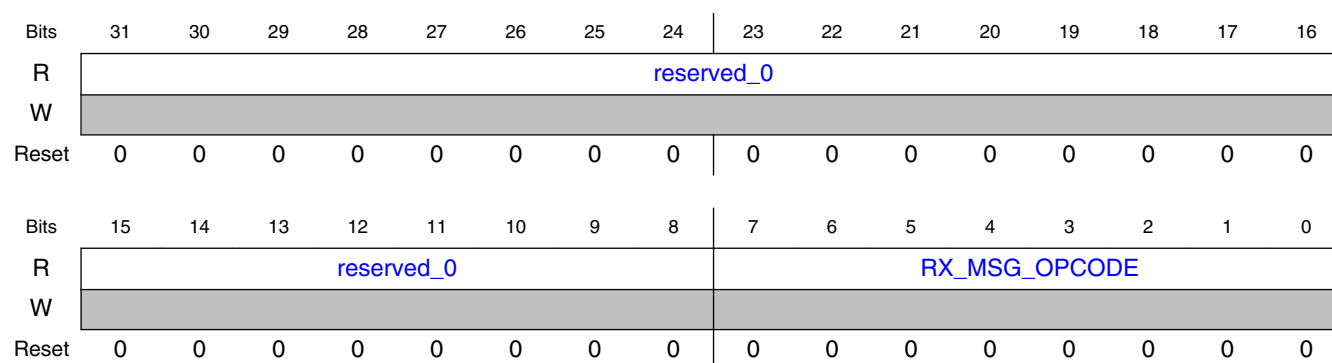
Field	Function
31-8 reserved_0	
7-4 RX_INITIATOR_ADDRESS	
3-0 RX_FOLLOWER_ADDRESS	Header of the received message. Header of the received message.

13.4.10.1.224 CEC RX message opcode (RX_MSG_DATA2)

13.4.10.1.224.1 Offset

Register	Offset
RX_MSG_DATA2	3904h

13.4.10.1.224.2 Diagram



13.4.10.1.224.3 Fields

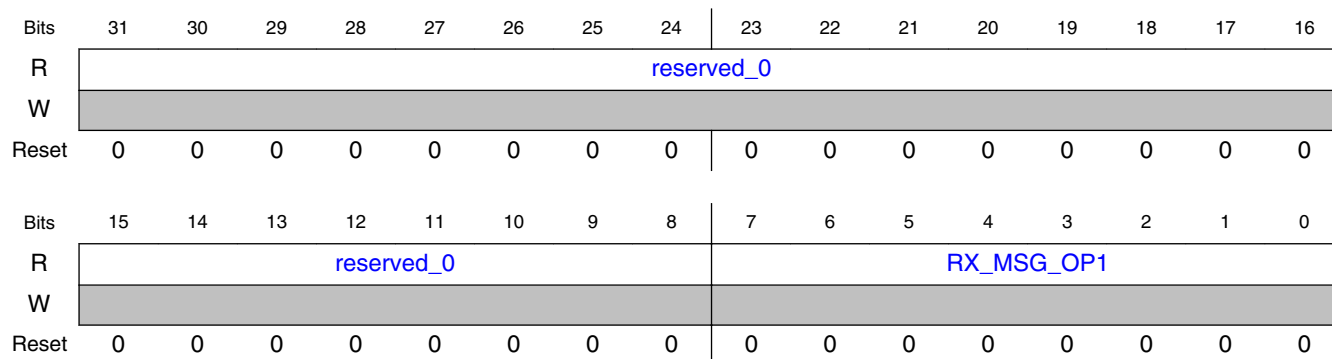
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OPCODE	Opcode of the received message.

13.4.10.1.225 CEC RX message operand 1 (RX_MSG_DATA3)

13.4.10.1.225.1 Offset

Register	Offset
RX_MSG_DATA3	3908h

13.4.10.1.225.2 Diagram



13.4.10.1.225.3 Fields

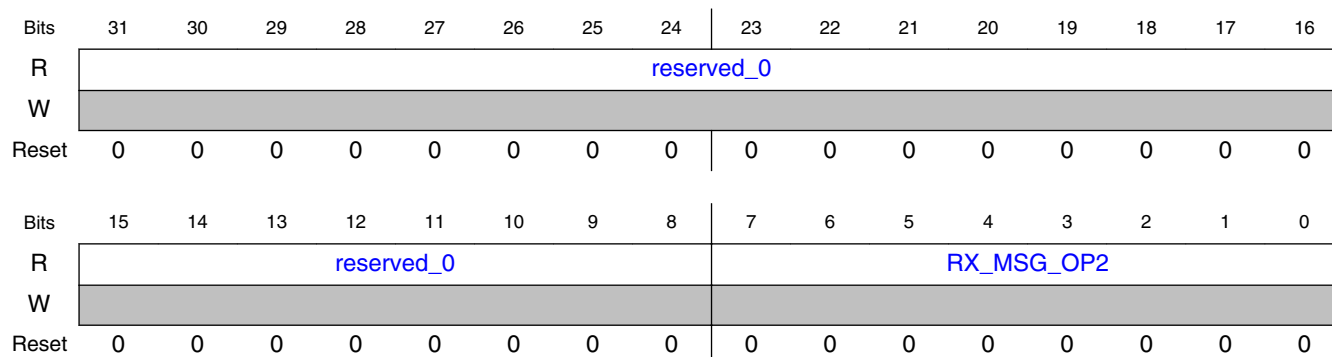
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP1	First operand for the opcode of the received message.

13.4.10.1.226 CEC RX message operand 2 (RX_MSG_DATA4)

13.4.10.1.226.1 Offset

Register	Offset
RX_MSG_DATA4	390Ch

13.4.10.1.226.2 Diagram



13.4.10.1.226.3 Fields

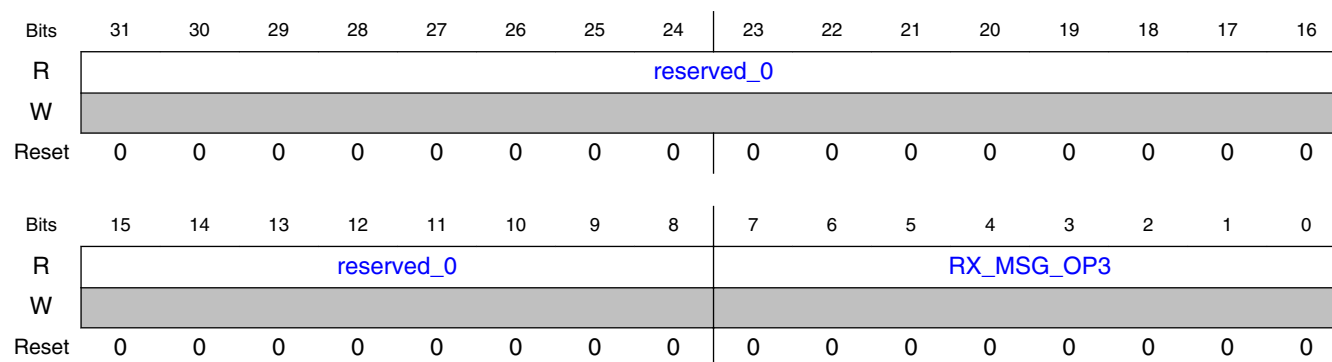
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP2	Second operand for the opcode of the received message.

13.4.10.1.227 CEC RX message operand 3 (RX_MSG_DATA5)

13.4.10.1.227.1 Offset

Register	Offset
RX_MSG_DATA5	3910h

13.4.10.1.227.2 Diagram



13.4.10.1.227.3 Fields

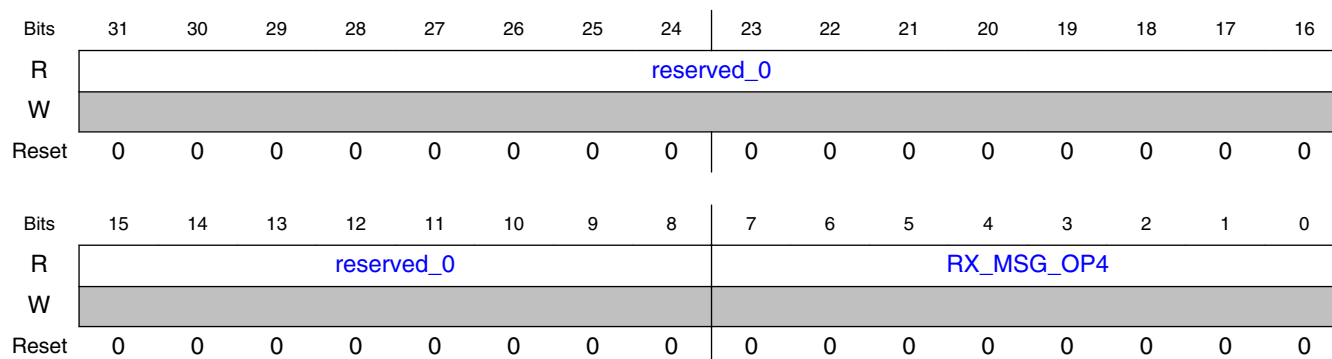
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP3	Third operand for the opcode of the received message.

13.4.10.1.228 CEC RX message operand 4 (RX_MSG_DATA6)

13.4.10.1.228.1 Offset

Register	Offset
RX_MSG_DATA6	3914h

13.4.10.1.228.2 Diagram



13.4.10.1.228.3 Fields

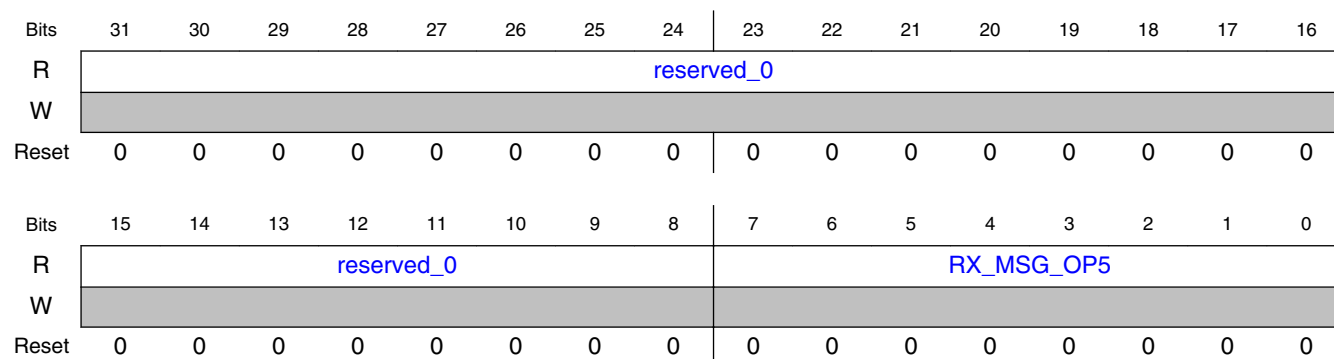
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP4	Fourth operand for the opcode of the received message.

13.4.10.1.229 CEC RX message operand 5 (RX_MSG_DATA7)

13.4.10.1.229.1 Offset

Register	Offset
RX_MSG_DATA7	3918h

13.4.10.1.229.2 Diagram



13.4.10.1.229.3 Fields

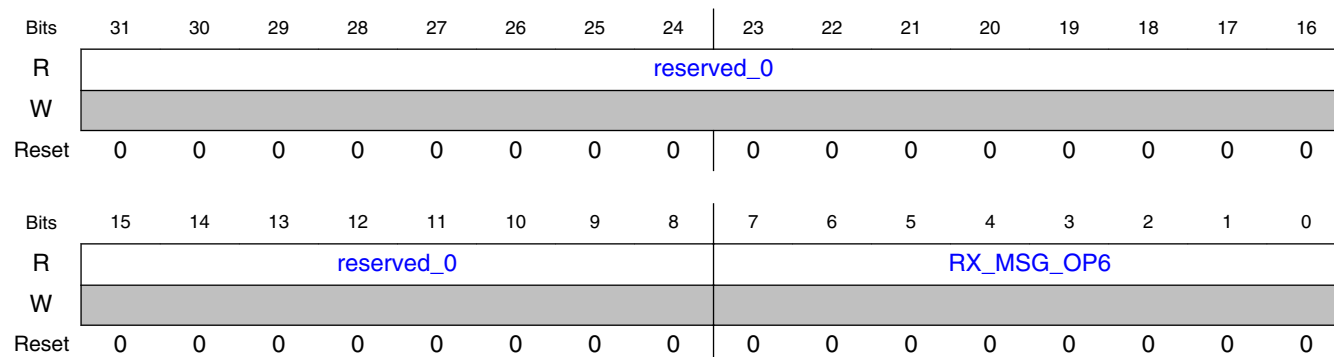
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP5	Fifth operand for the opcode of the received message.

13.4.10.1.230 CEC RX message operand 6 (RX_MSG_DATA8)

13.4.10.1.230.1 Offset

Register	Offset
RX_MSG_DATA8	391Ch

13.4.10.1.230.2 Diagram



13.4.10.1.230.3 Fields

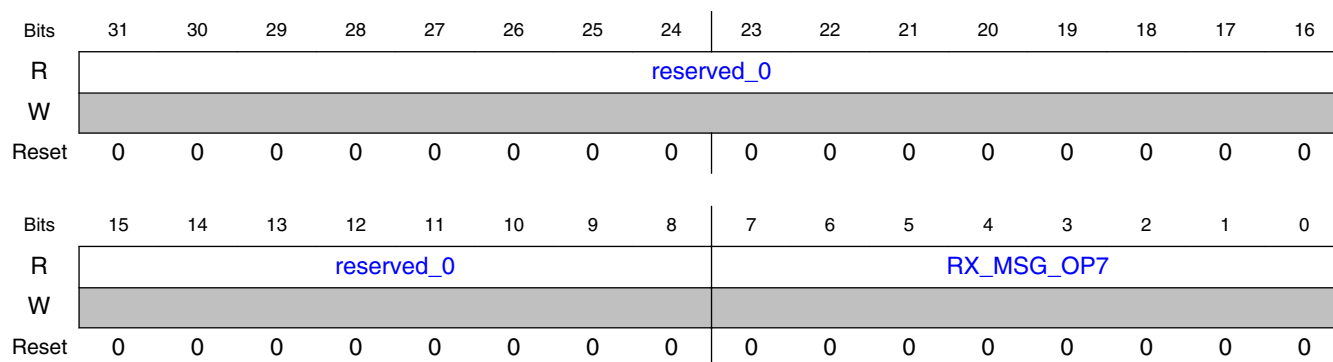
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP6	Sixth operand for the opcode of the received message.

13.4.10.1.231 CEC RX message operand 7 (RX_MSG_DATA9)

13.4.10.1.231.1 Offset

Register	Offset
RX_MSG_DATA9	3920h

13.4.10.1.231.2 Diagram



13.4.10.1.231.3 Fields

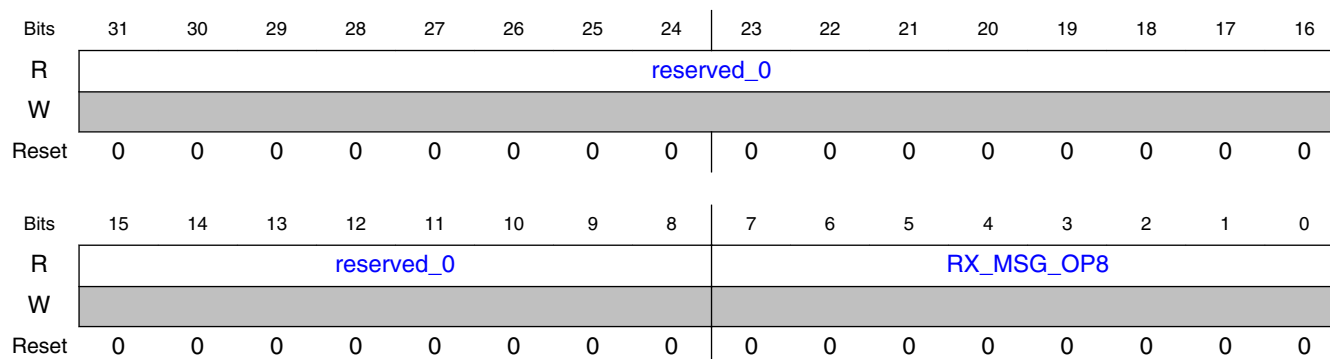
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP7	Seventh operand for the opcode of the received message.

13.4.10.1.232 CEC RX message operand 8 (RX_MSG_DATA10)

13.4.10.1.232.1 Offset

Register	Offset
RX_MSG_DATA10	3924h

13.4.10.1.232.2 Diagram



13.4.10.1.232.3 Fields

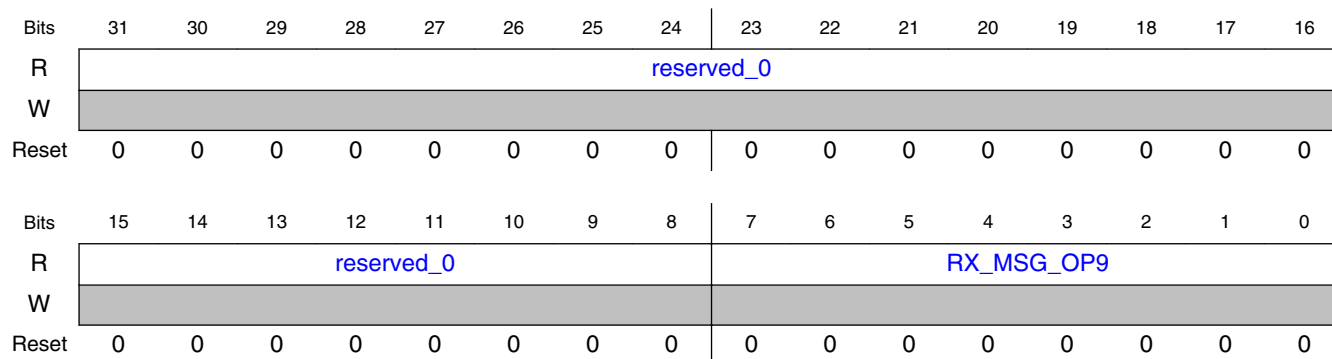
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP8	Eighth operand for the opcode of the received message.

13.4.10.1.233 CEC RX message operand 9 (RX_MSG_DATA11)

13.4.10.1.233.1 Offset

Register	Offset
RX_MSG_DATA11	3928h

13.4.10.1.233.2 Diagram



13.4.10.1.233.3 Fields

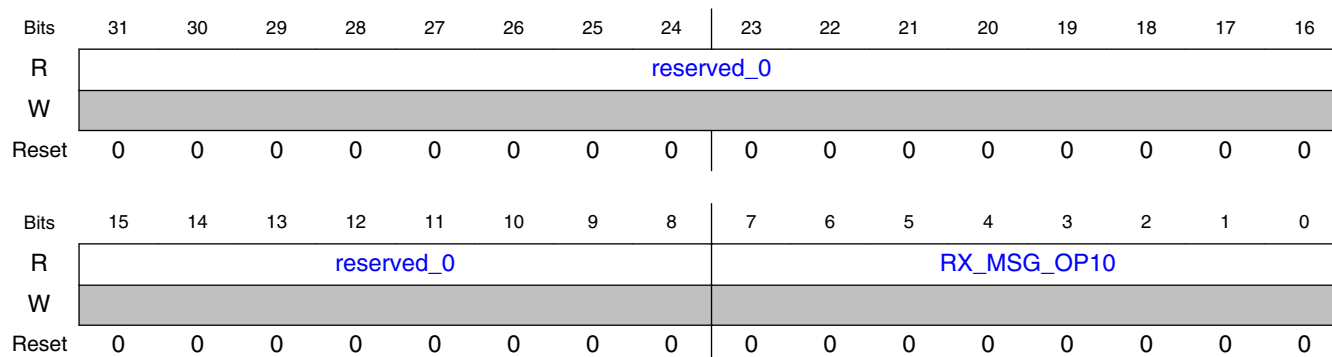
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP9	Ninth operand for the opcode of the received message.

13.4.10.1.234 CEC RX message operand 10 (RX_MSG_DATA12)

13.4.10.1.234.1 Offset

Register	Offset
RX_MSG_DATA12	392Ch

13.4.10.1.234.2 Diagram



13.4.10.1.234.3 Fields

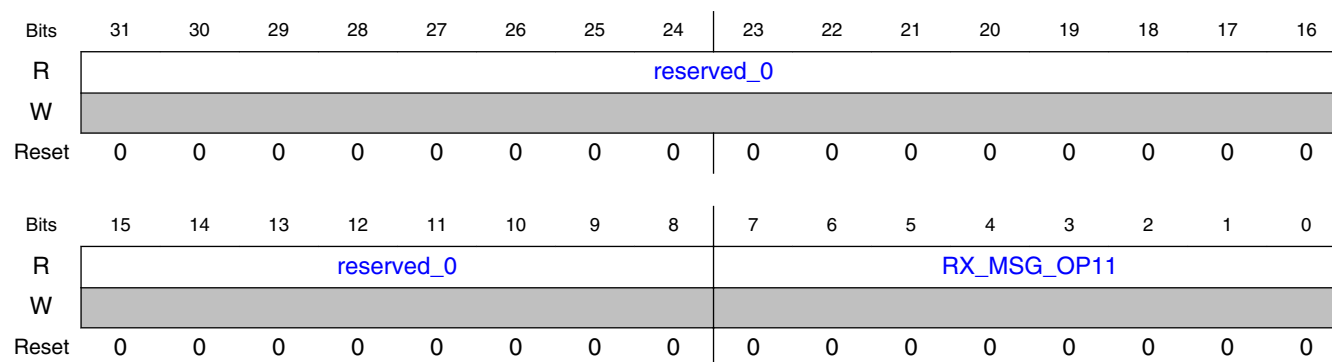
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP10	Tenth operand for the opcode of the received message.

13.4.10.1.235 CEC RX message operand 11 (RX_MSG_DATA13)

13.4.10.1.235.1 Offset

Register	Offset
RX_MSG_DATA13	3930h

13.4.10.1.235.2 Diagram



13.4.10.1.235.3 Fields

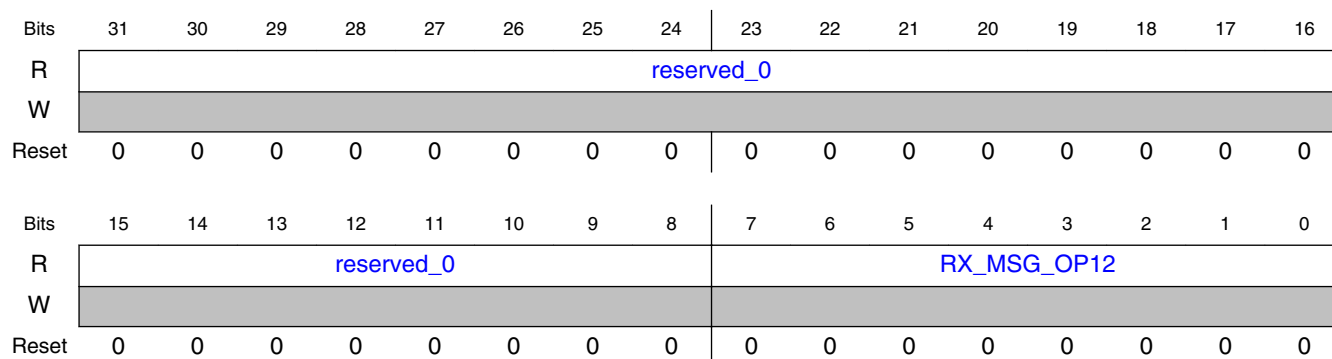
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP11	The eleventh operand for the opcode of the received message.

13.4.10.1.236 CEC RX message operand 12 (RX_MSG_DATA14)

13.4.10.1.236.1 Offset

Register	Offset
RX_MSG_DATA14	3934h

13.4.10.1.236.2 Diagram



13.4.10.1.236.3 Fields

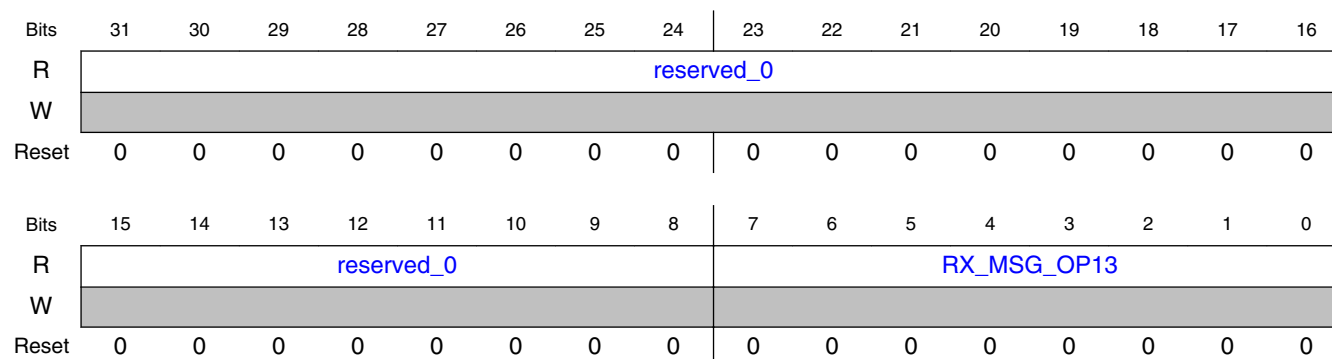
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP12	The twelfth operand for the opcode of the received message.

13.4.10.1.237 CEC RX message operand 13 (RX_MSG_DATA15)

13.4.10.1.237.1 Offset

Register	Offset
RX_MSG_DATA15	3938h

13.4.10.1.237.2 Diagram



13.4.10.1.237.3 Fields

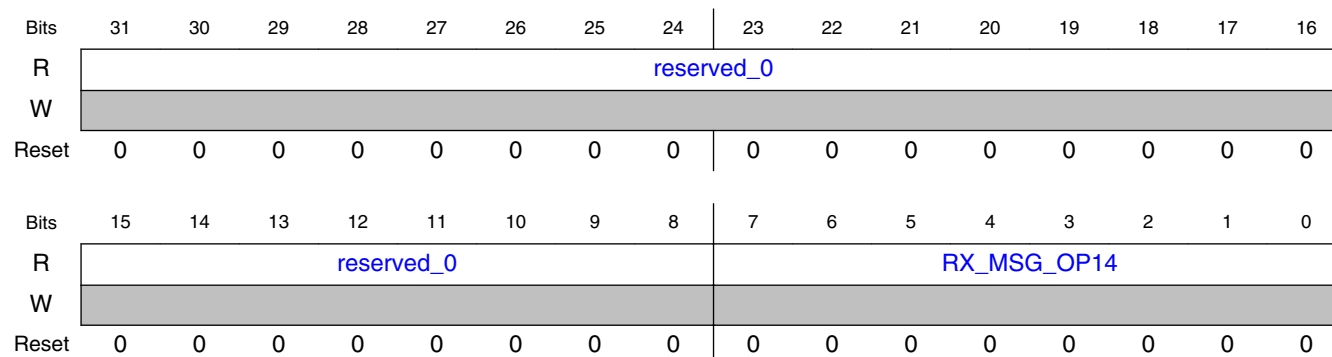
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP13	The thirteenth operand for the opcode of the received message.

13.4.10.1.238 CEC RX message operand 14 (RX_MSG_DATA16)

13.4.10.1.238.1 Offset

Register	Offset
RX_MSG_DATA16	393Ch

13.4.10.1.238.2 Diagram



13.4.10.1.238.3 Fields

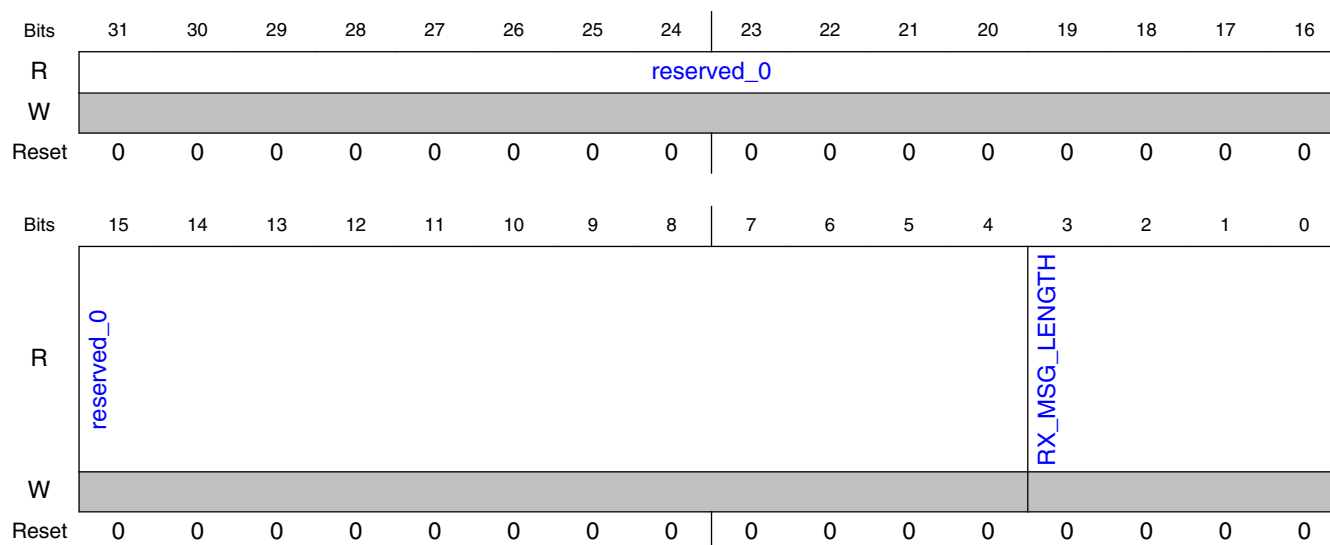
Field	Function
31-8 reserved_0	
7-0 RX_MSG_OP14	The fourteenth operand for the opcode of the received message.

13.4.10.1.239 CEC RX message length (RX_MSG_LENGTH)

13.4.10.1.239.1 Offset

Register	Offset
RX_MSG_LENGTH	3940h

13.4.10.1.239.2 Diagram



13.4.10.1.239.3 Fields

Field	Function
31-4 reserved_0	

Table continues on the next page...

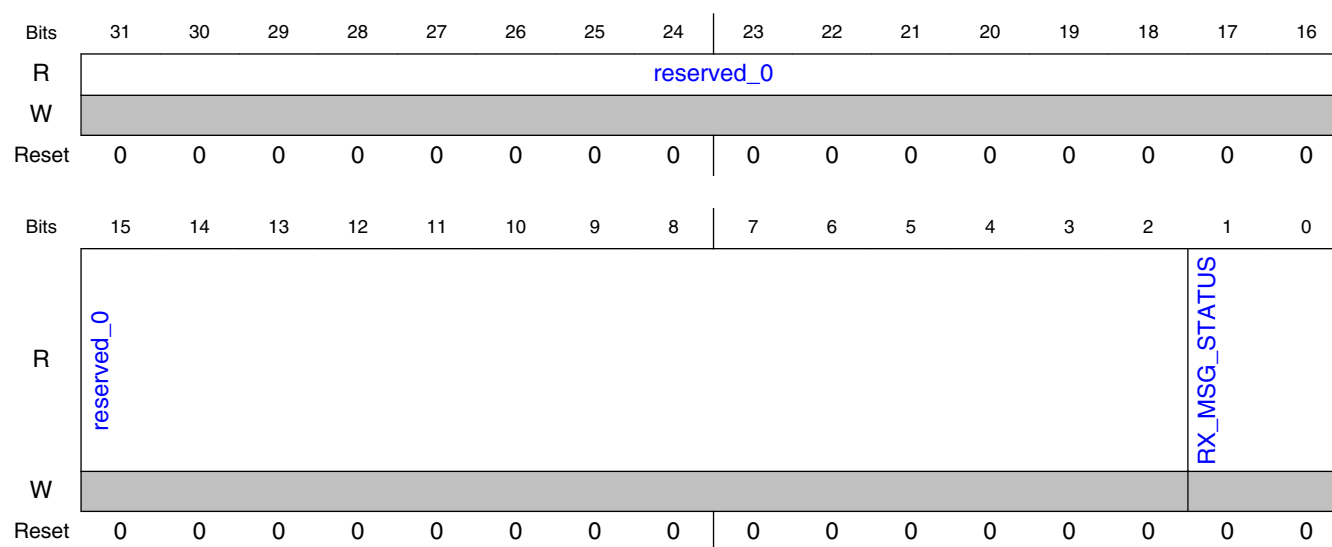
Field	Function
3-0	The number of bytes in the received message.
RX_MSG_LEN GTH	The number of bytes in the received message.

13.4.10.1.240 CEC RX status (RX_MSG_STATUS)

13.4.10.1.240.1 Offset

Register	Offset
RX_MSG_STATUS	3944h

13.4.10.1.240.2 Diagram



13.4.10.1.240.3 Fields

Field	Function
31-2 reserved_0	
1-0 RX_MSG_STAT US	00: no transaction, 01: busy, 10: message received successfully, 11: message received with error.

13.4.10.1.241 CEC RX status - number of received messages (NUM_OF_MSG_RX_BUF)

13.4.10.1.241.1 Offset

Register	Offset
NUM_OF_MSG_RX_BUF	3948h

13.4.10.1.241.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0												RX_NUM_MSG			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.241.3 Fields

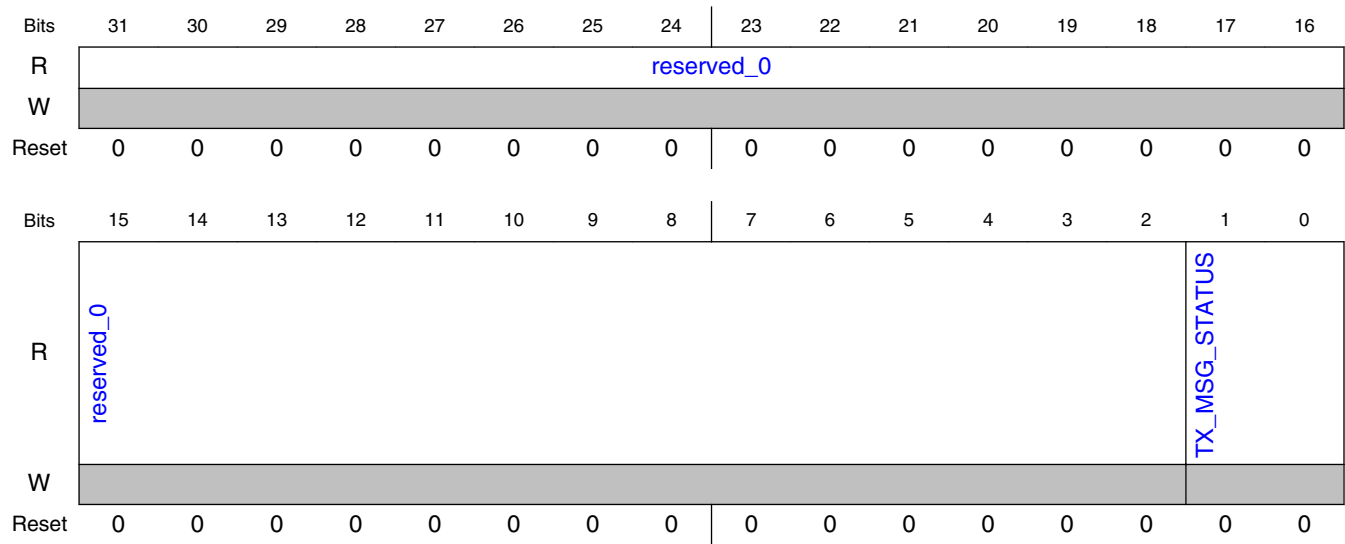
Field	Function
31-4 reserved_0	
3-0 RX_NUM_MSG	The number of received messages in Rx buffer waiting to be read.

13.4.10.1.242 CEC TX status (TX_MSG_STATUS)

13.4.10.1.242.1 Offset

Register	Offset
TX_MSG_STATUS	394Ch

13.4.10.1.242.2 Diagram



13.4.10.1.242.3 Fields

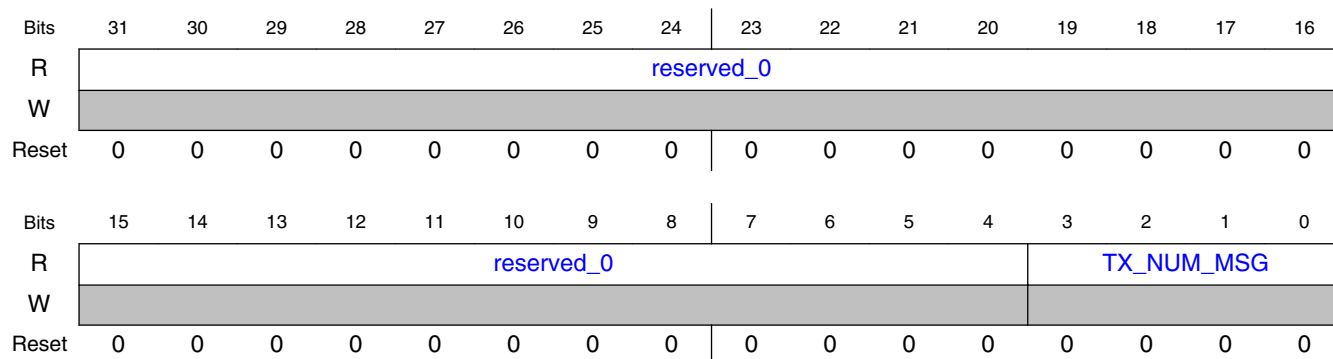
Field	Function
31-2 reserved_0	
1-0 TX_MSG_STAT US	00: no transaction, 01: busy, 10: message transmitted successfully, 11: message transmitted with error. 00: no transaction, 01: busy, 10: message transmitted successfully, 11: message transmitted with error.

13.4.10.1.243 CEC TX status - number of messages in TX buffer (NUMBER_OF_MSG_IN_TX_BUF)

13.4.10.1.243.1 Offset

Register	Offset
NUMBER_OF_MSG_IN_TX_BUF	3950h

13.4.10.1.243.2 Diagram



13.4.10.1.243.3 Fields

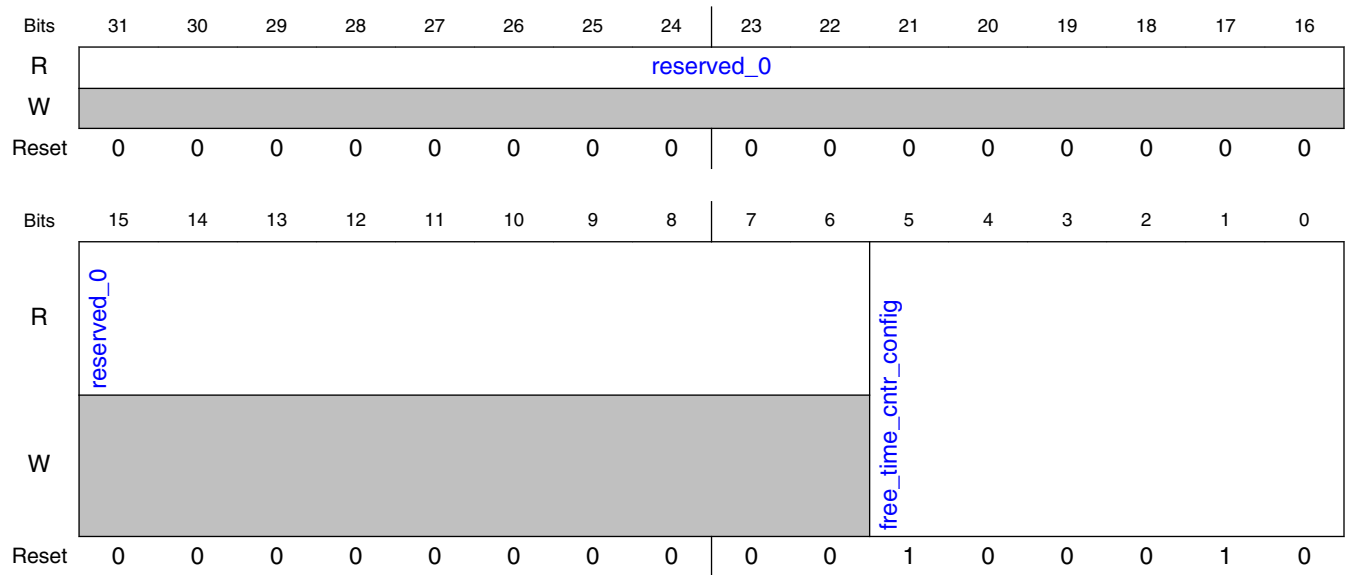
Field	Function
31-4 reserved_0	
3-0 TX_NUM_MSG	

13.4.10.1.244 CEC Free-Time counter (FREE_TIME_CNTR_CONFIG)

13.4.10.1.244.1 Offset

Register	Offset
FREE_TIME_CNTR_CONFIG	3954h

13.4.10.1.244.2 Diagram



13.4.10.1.244.3 Fields

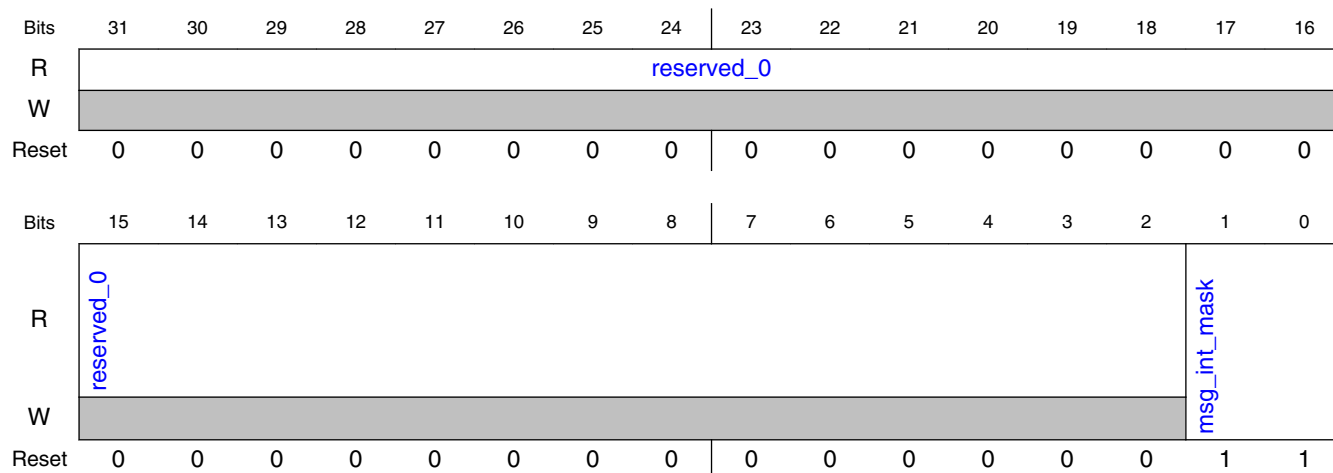
Field	Function
31-6 reserved_0	
5-0 free_time_cntr_c onfig	Parallel load for 6 MSB in free_time_cntr counter. Parallel load for 6 MSB in free_time_cntr counter.

13.4.10.1.245 CEC Interrupt Mask (INT_CEC_MASK)

13.4.10.1.245.1 Offset

Register	Offset
INT_CEC_MASK	3958h

13.4.10.1.245.2 Diagram



13.4.10.1.245.3 Fields

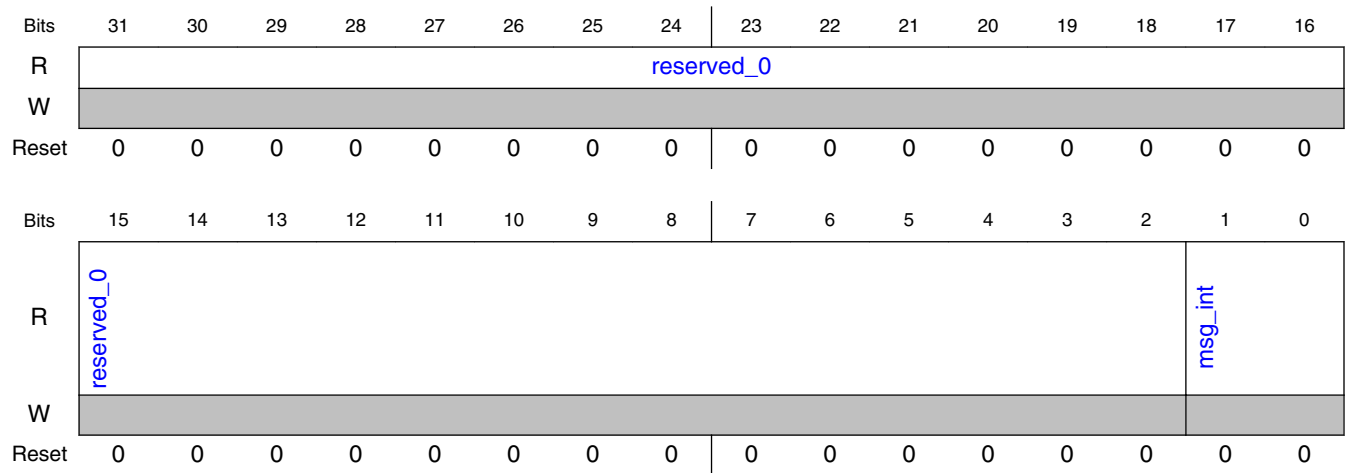
Field	Function
31-2 reserved_0	
1-0 msg_int_mask	Interrupt Mask. Interrupt Mask.

13.4.10.1.246 CEC Interrupt status (INT_STAT)

13.4.10.1.246.1 Offset

Register	Offset
INT_STAT	395Ch

13.4.10.1.246.2 Diagram



13.4.10.1.246.3 Fields

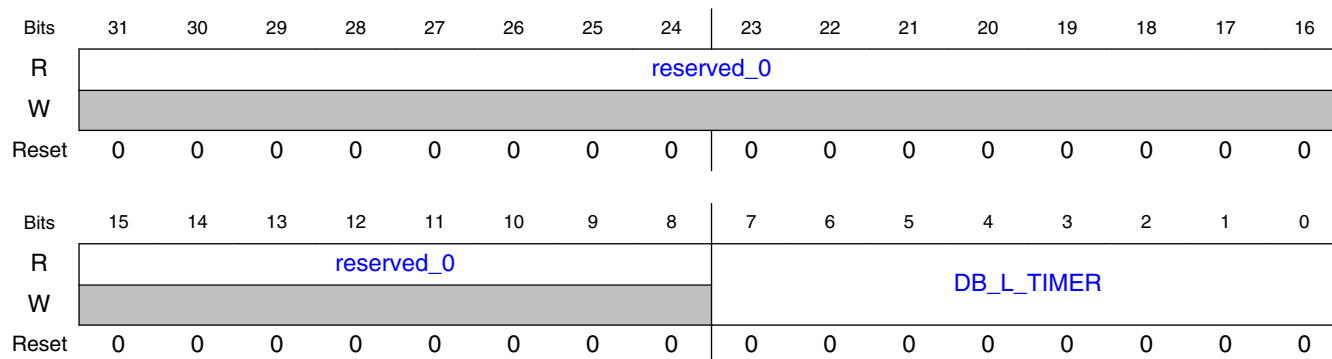
Field	Function
31-2 reserved_0	
1-0 msg_int	Interrupt Stat. Interrupt Stat.

13.4.10.1.247 CEC DB L timer (DB_L_TIMER)

13.4.10.1.247.1 Offset

Register	Offset
DB_L_TIMER	3980h

13.4.10.1.247.2 Diagram



13.4.10.1.247.3 Fields

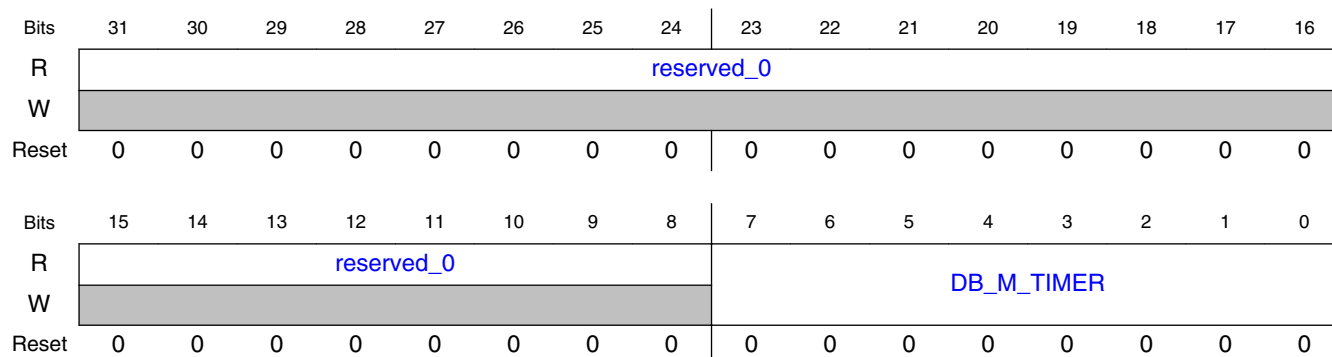
Field	Function
31-8 reserved_0	
7-0 DB_L_TIMER	

13.4.10.1.248 CEC DB M timer (DB_M_TIMER)

13.4.10.1.248.1 Offset

Register	Offset
DB_M_TIMER	3984h

13.4.10.1.248.2 Diagram



13.4.10.1.248.3 Fields

Field	Function
31-8 reserved_0	
7-0 DB_M_TIMER	

13.4.10.1.249 CEC DB H timer (DB_H_TIMER)

13.4.10.1.249.1 Offset

Register	Offset
DB_H_TIMER	3988h

13.4.10.1.249.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								DB_H_TIMER							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.249.3 Fields

Field	Function
31-8 reserved_0	
7-0 DB_H_TIMER	

13.4.10.1.250 Contains the revision of the internal HDCP 1. (CRYPTO_HDCP_REVISION)

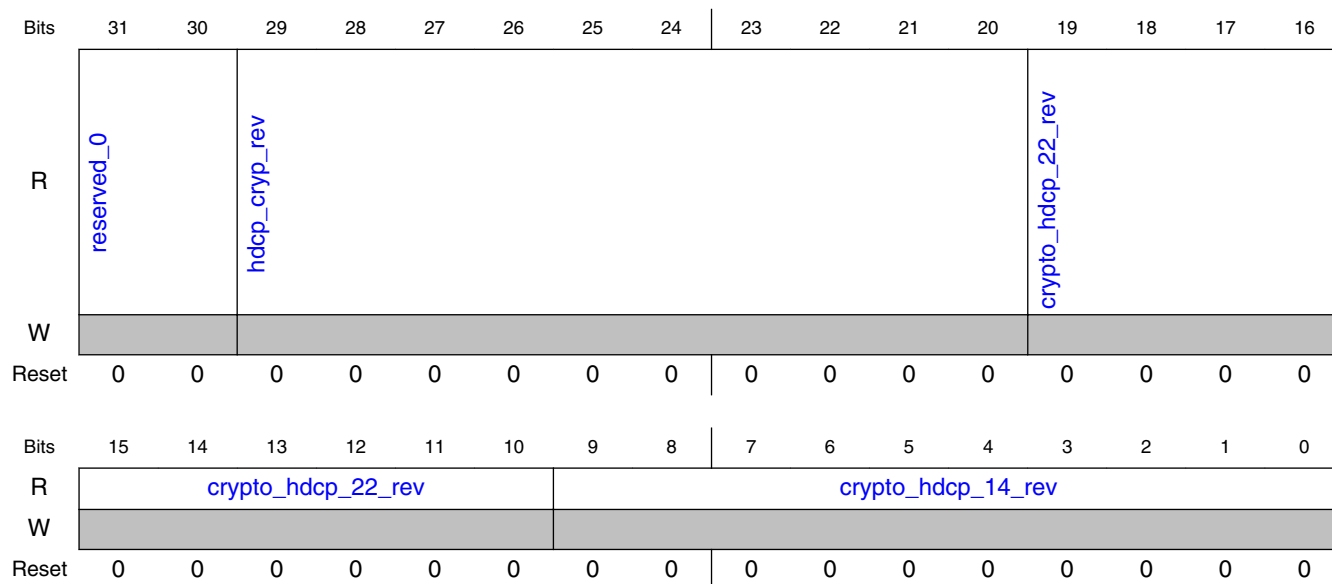
13.4.10.1.250.1 Offset

Register	Offset
CRYPTO_HDCP_REVISION	4000h

13.4.10.1.250.2 Function

Contains the revision of the internal HDCP 1.4 and 2.2 module

13.4.10.1.250.3 Diagram



13.4.10.1.250.4 Fields

Field	Function
31-30 reserved_0	
29-20 hdcp_cryp_rev	Revision of the HDCP Crypto block. Revision of the HDCP Crypto block.
19-10	Revision of the HDCP Crypto 2. Revision of the HDCP Crypto 2.2 block.

Table continues on the next page...

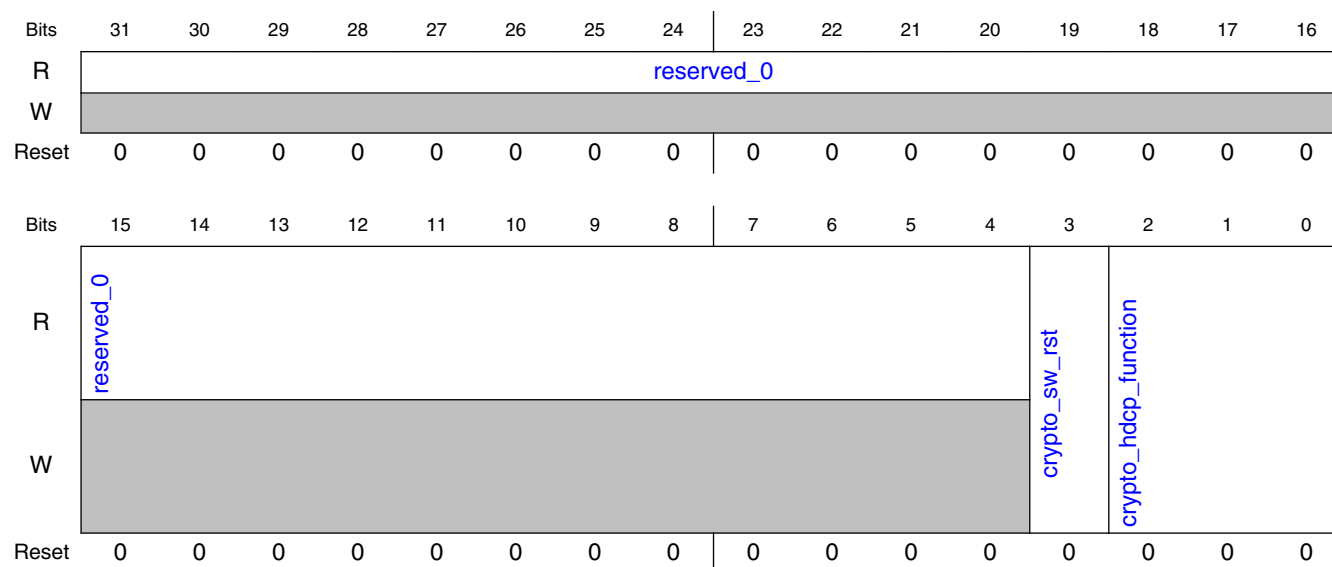
Field	Function
crypto_hdcv_22 _rev	
9-0	Revision of the HDCP Crypto 1.
crypto_hdcv_14 _rev	Revision of the HDCP Crypto 1.4 block.

13.4.10.1.251 Contains global configuration information for the HDCP Crypto module (HDCP_CRYPT0_CONFIG)

13.4.10.1.251.1 Offset

Register	Offset
HDCP_CRYPT0_CONFIG	4004h

13.4.10.1.251.2 Diagram



13.4.10.1.251.3 Fields

Field	Function
31-4 reserved_0	

Table continues on the next page...

Clocks And Resets

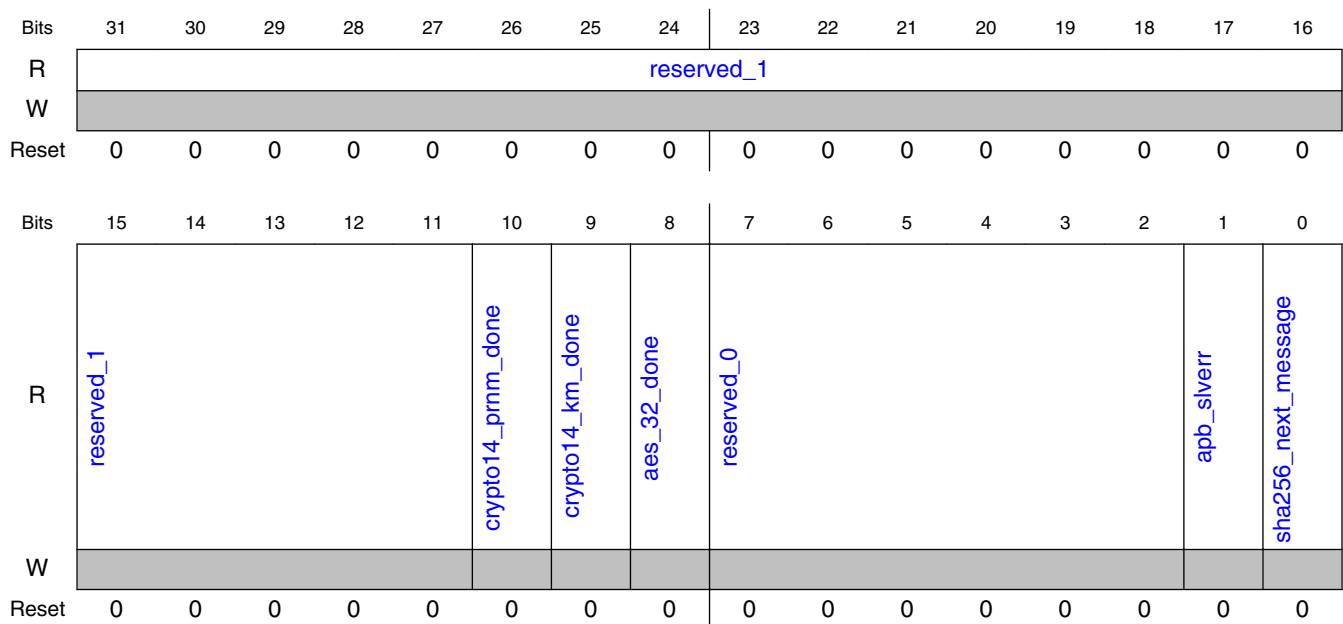
Field	Function
3 crypto_sw_rst	Software reset for the Crypto module.
2-0 crypto_hdcp_function	Enables a version of the Crypto function: 0: HDCP 1. Enables a version of the Crypto function: 0: HDCP 1.4 1: HDCP 2.2 Others: Reserved

13.4.10.1.252 Contains the status of the HDCP interrupt sources (CRYPTO_INTERRUPT_SOURCE)

13.4.10.1.252.1 Offset

Register	Offset
CRYPTO_INTERRUPT_SOURCE	4008h

13.4.10.1.252.2 Diagram



13.4.10.1.252.3 Fields

Field	Function
31-11	

Table continues on the next page...

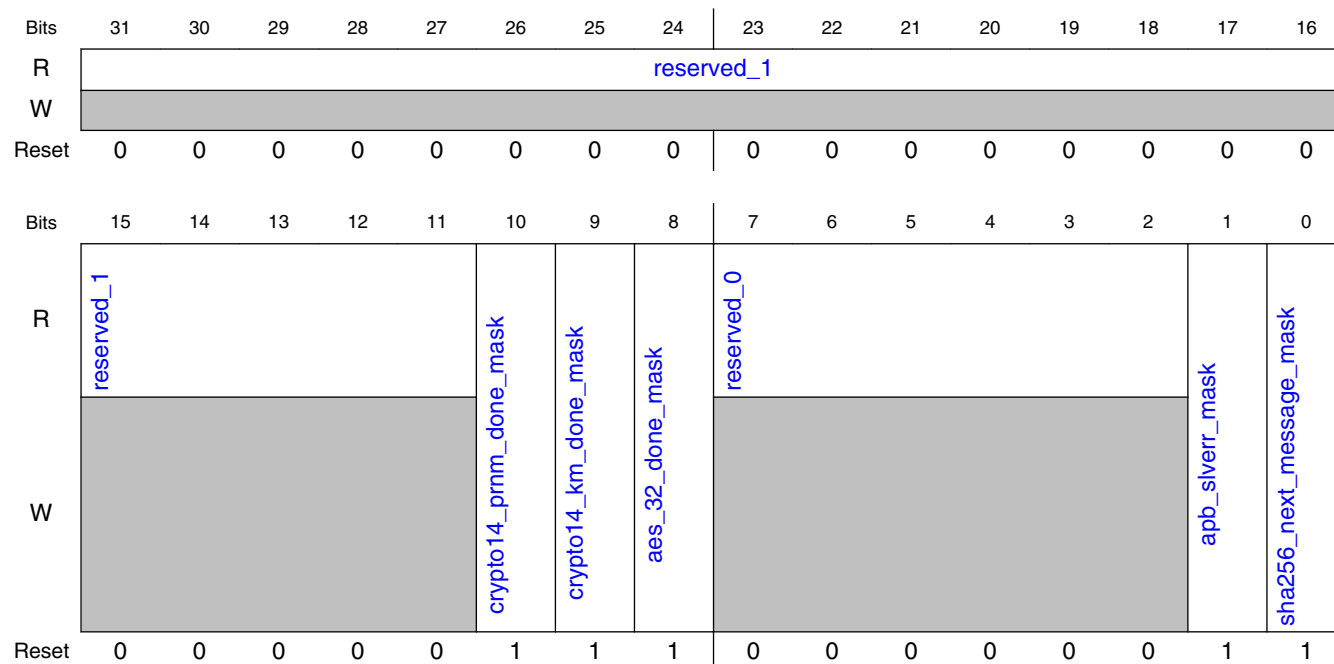
Field	Function
reserved_1	
10 crypto14_prnm_ done	LFSR and block output finished calculation. LFSR and block output finished calculation.
9 crypto14_km_do ne	Done reading/calculating Km. Done reading/calculating Km.
8 aes_32_done	Asserted when the rising edge of the AES-32 done output is detected. Asserted when the rising edge of the AES-32 done output is detected.
7-2 reserved_0	
1 apb_slvrr	APB slave error. APB slave error. Asserted when APB address is out of address range.
0 sha256_next_m essage	Asserted when the rising edge of the SHA256 done output is detected. Asserted when the rising edge of the SHA256 done output is detected.

13.4.10.1.253 CContains the mask vector of the HDCP interrupt sources (CRYPTO_INTERRUPT_MASK)

13.4.10.1.253.1 Offset

Register	Offset
CRYPTO_INTERRUPT_ MASK	400Ch

13.4.10.1.253.2 Diagram



13.4.10.1.253.3 Fields

Field	Function
31-11 reserved_1	
10 crypto14_prnm_done_mask	Set to 1 to mask the crypto14_prnm_done interrupt. Set to 1 to mask the crypto14_prnm_done interrupt.
9 crypto14_km_done_mask	Set to 1 to mask the crypto14_km_done interrupt. Set to 1 to mask the crypto14_km_done interrupt.
8 aes_32_done_mask	Set to 1 to mask the AES32_done interrupt. Set to 1 to mask the AES32_done interrupt.
7-2 reserved_0	
1 apb_slvrr_mask	Set to 1 for the apb_slvrr interrupt. Set to 1 for the apb_slvrr interrupt.
0 sha256_next_message_mask	Set to 1 to mask the SHA256_done interrupt. Set to 1 to mask the SHA256_done interrupt.

13.4.10.1.254 Contains global configuration information for the HDCP 2. (CRYPTO22_CONFIG)

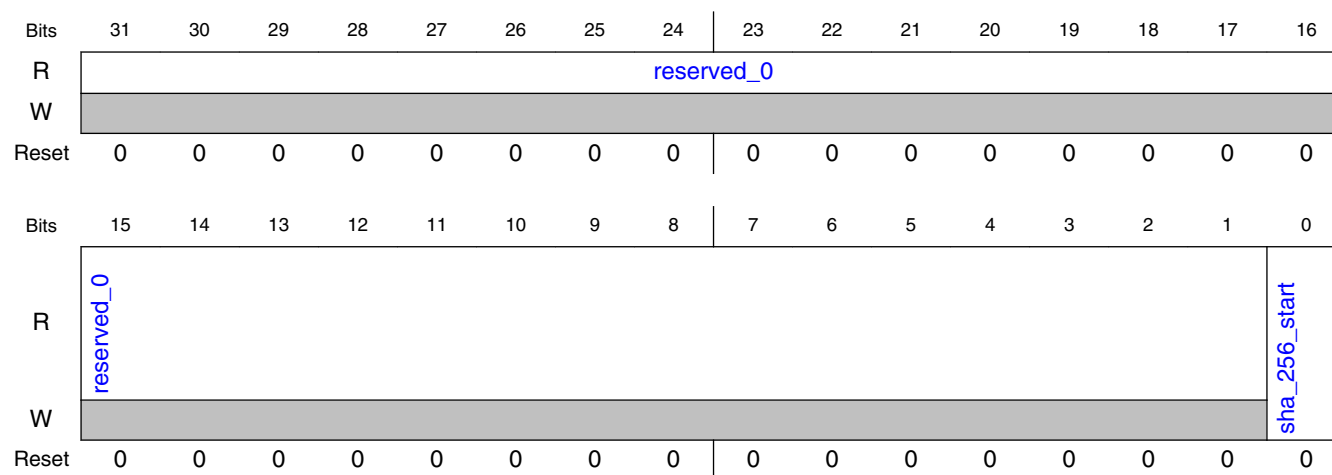
13.4.10.1.254.1 Offset

Register	Offset
CRYPTO22_CONFIG	4018h

13.4.10.1.254.2 Function

Contains global configuration information for the HDCP 2.2 Crypto module

13.4.10.1.254.3 Diagram



13.4.10.1.254.4 Fields

Field	Function
31-1 reserved_0	
0 sha_256_start	Set to 1 for Sha-256 start

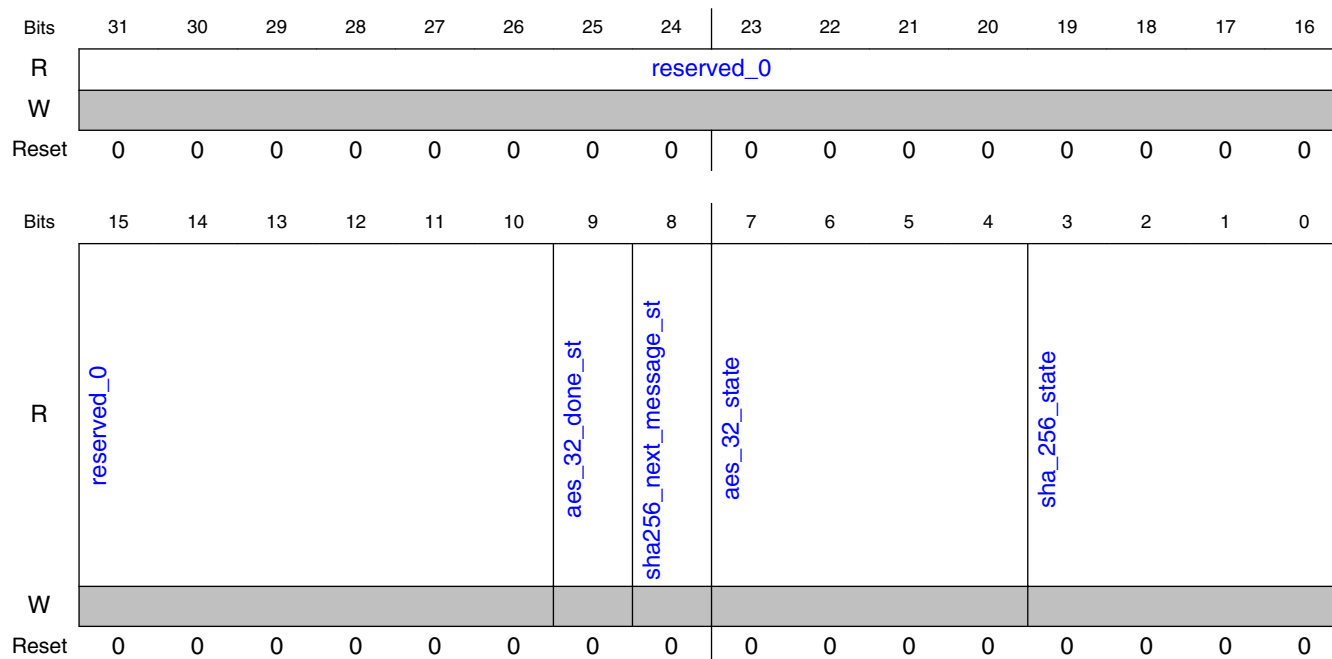
13.4.10.1.255 Crypto 2. (CRYPTO22_STATUS)

13.4.10.1.255.1 Offset

Register	Offset
CRYPTO22_STATUS	401Ch

13.4.10.1.255.2 Function

Crypto 2.2 global status register

13.4.10.1.255.3 Diagram**13.4.10.1.255.4 Fields**

Field	Function
31-10 reserved_0	
9 aes_32_done_st	Asserted when the rising edge of the AES-32 done output is detected.
8 sha256_next_message_st	Asserted when the SHA-256 module is ready to receive the next message
7-4	AES-32 current state

Table continues on the next page...

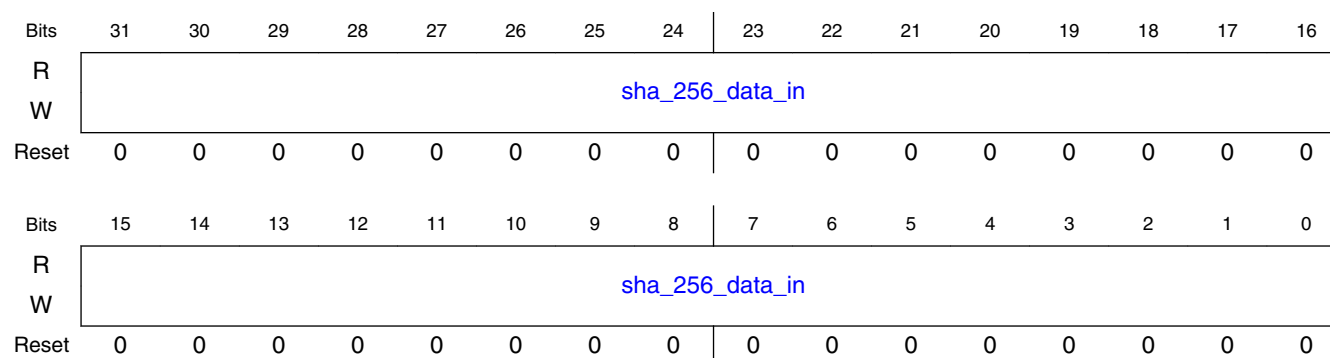
Field	Function
aes_32_state	
3-0 sha_256_state	SHA-256 current state

13.4.10.1.256 Holds 32-bit input data word of the SHA-256 module (SHA_256_DATA_IN)

13.4.10.1.256.1 Offset

Register	Offset
SHA_256_DATA_IN	403Ch

13.4.10.1.256.2 Diagram



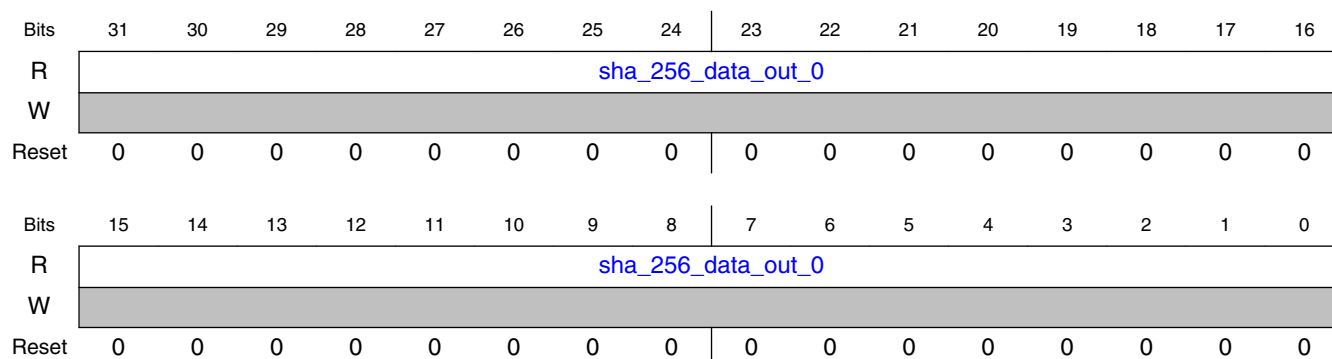
13.4.10.1.256.3 Fields

Field	Function
31-0	Holds the 32-bit input data word of the SHA-256 module.
sha_256_data_in	Holds the 32-bit input data word of the SHA-256 module.

13.4.10.1.257 Result of operation SHA-256 - 1' dw (SHA_256_DATA_OUT_0)

13.4.10.1.257.1 Offset

Register	Offset
SHA_256_DATA_OUT_0	4050h

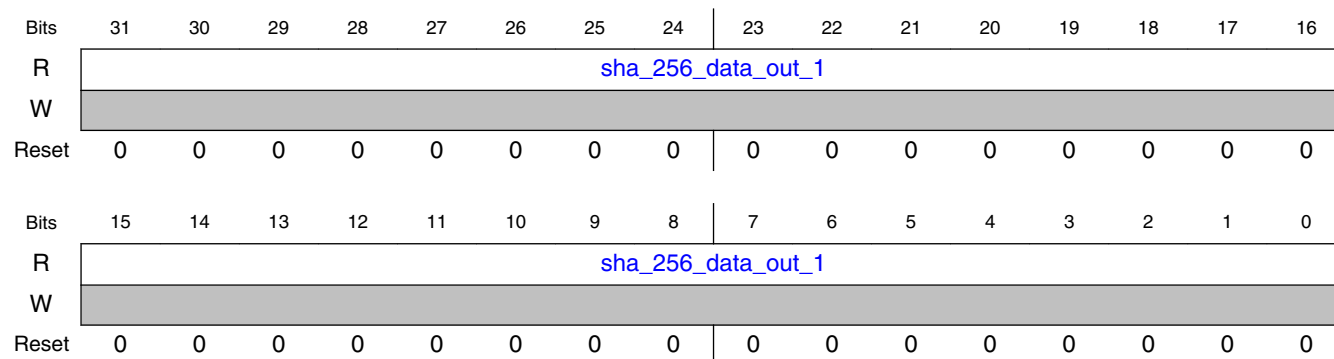
13.4.10.1.257.2 Diagram**13.4.10.1.257.3 Fields**

Field	Function
31-0	Holds the least significant 32-bits word of the 256-bits output data word of the SHA-256 module.
sha_256_data_out_0	Holds the least significant 32-bits word of the 256-bits output data word of the SHA-256 module.

13.4.10.1.258 Result of operation SHA-256 - 2' dw (SHA_256_DATA_OUT_1)**13.4.10.1.258.1 Offset**

Register	Offset
SHA_256_DATA_OUT_1	4054h

13.4.10.1.258.2 Diagram



13.4.10.1.258.3 Fields

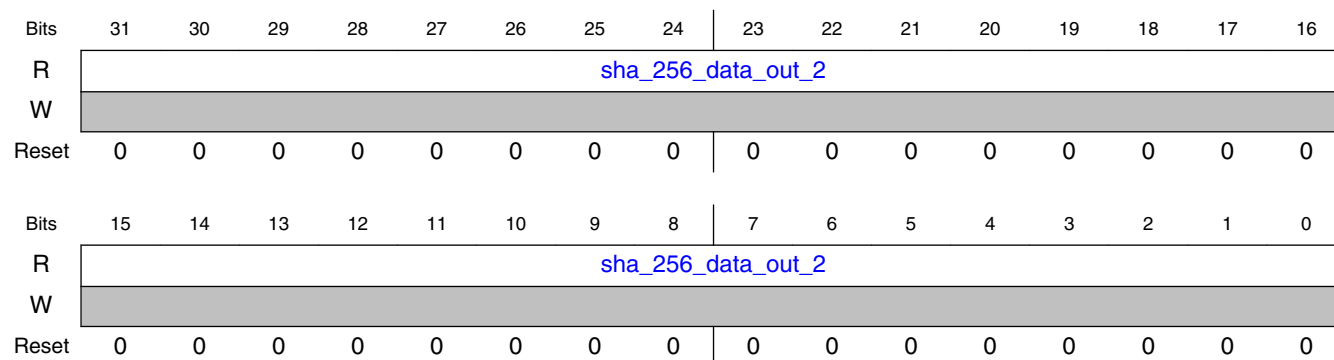
Field	Function
31-0	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.
sha_256_data_out_1	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.

13.4.10.1.259 Result of operation SHA-256 - 3' dw (SHA_256_DATA_OUT_2)

13.4.10.1.259.1 Offset

Register	Offset
SHA_256_DATA_OUT_2	4058h

13.4.10.1.259.2 Diagram



13.4.10.1.259.3 Fields

Field	Function
31-0 sha_256_data_out_2	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module

13.4.10.1.260 Result of operation SHA-256 - 4' dw (SHA_256_DATA_OUT_3)

13.4.10.1.260.1 Offset

Register	Offset
SHA_256_DATA_OUT_3	405Ch

13.4.10.1.260.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	sha_256_data_out_3															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	sha_256_data_out_3															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.260.3 Fields

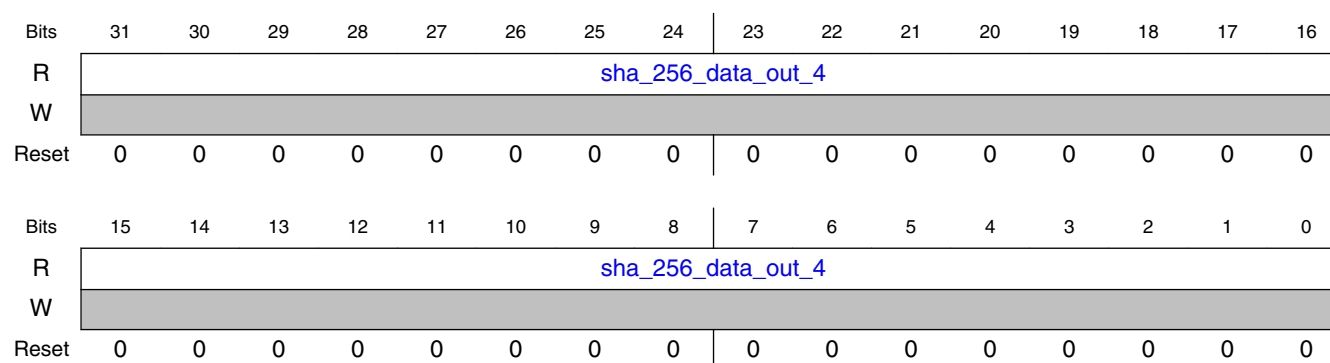
Field	Function
31-0 sha_256_data_out_3	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.

13.4.10.1.261 Result of operation SHA-256 - 5' dw (SHA_256_DATA_OUT_4)

13.4.10.1.261.1 Offset

Register	Offset
SHA_256_DATA_OUT_4	4060h

13.4.10.1.261.2 Diagram



13.4.10.1.261.3 Fields

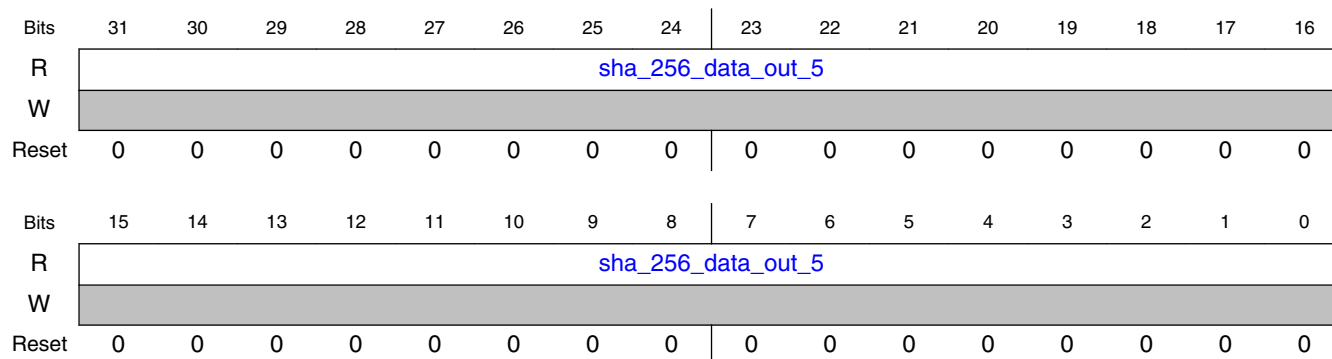
Field	Function
31-0	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.
sha_256_data_out_4	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.

13.4.10.1.262 Result of operation SHA-256 - 6' dw (SHA_256_DATA_OUT_5)

13.4.10.1.262.1 Offset

Register	Offset
SHA_256_DATA_OUT_5	4064h

13.4.10.1.262.2 Diagram



13.4.10.1.262.3 Fields

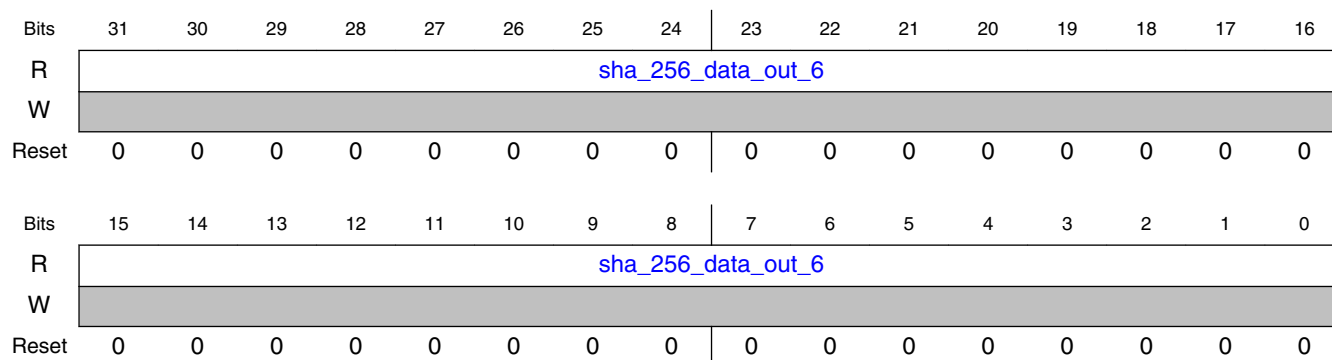
Field	Function
31-0	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.
sha_256_data_out_5	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.

13.4.10.1.263 Result of operation SHA-256 - 7' dw (SHA_256_DATA_OUT_6)

13.4.10.1.263.1 Offset

Register	Offset
SHA_256_DATA_OUT_6	4068h

13.4.10.1.263.2 Diagram



13.4.10.1.263.3 Fields

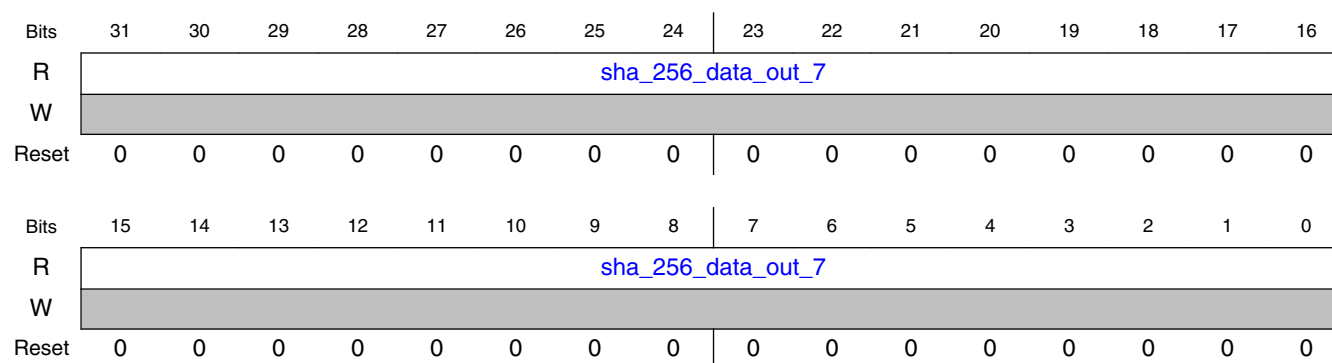
Field	Function
31-0	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.
sha_256_data_out_6	Holds the next significant 32-bits word of the 256-bits output data word of the SHA-256 module.

13.4.10.1.264 Result of operation SHA-256 - 8' dw (SHA_256_DATA_OUT_7)

13.4.10.1.264.1 Offset

Register	Offset
SHA_256_DATA_OUT_7	406Ch

13.4.10.1.264.2 Diagram



13.4.10.1.264.3 Fields

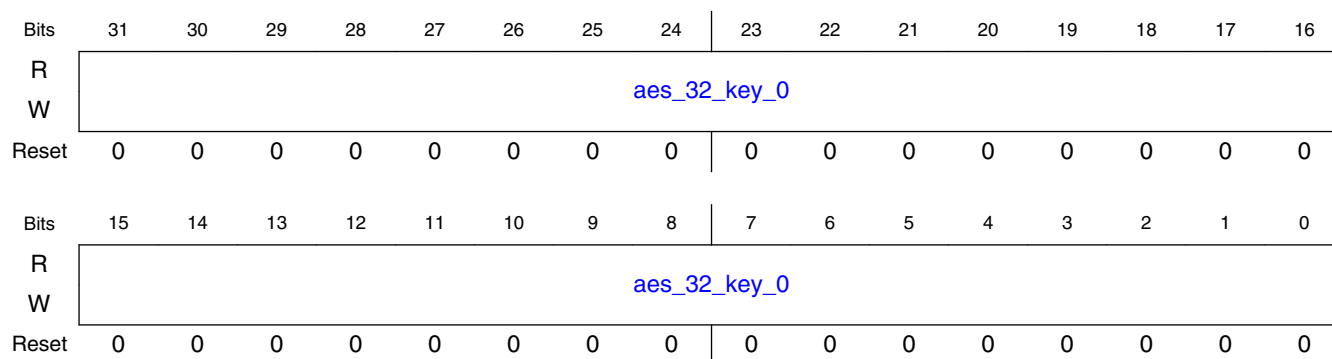
Field	Function
31-0	Holds the most significant 32-bits word of the 256-bits output data word of the SHA-256 module.
sha_256_data_out_7	Holds the most significant 32-bits word of the 256-bits output data word of the SHA-256 module.

13.4.10.1.265 Input key word of the AES-32 module - 1' dw (AES_32_KEY_0)

13.4.10.1.265.1 Offset

Register	Offset
AES_32_KEY_0	4070h

13.4.10.1.265.2 Diagram



13.4.10.1.265.3 Fields

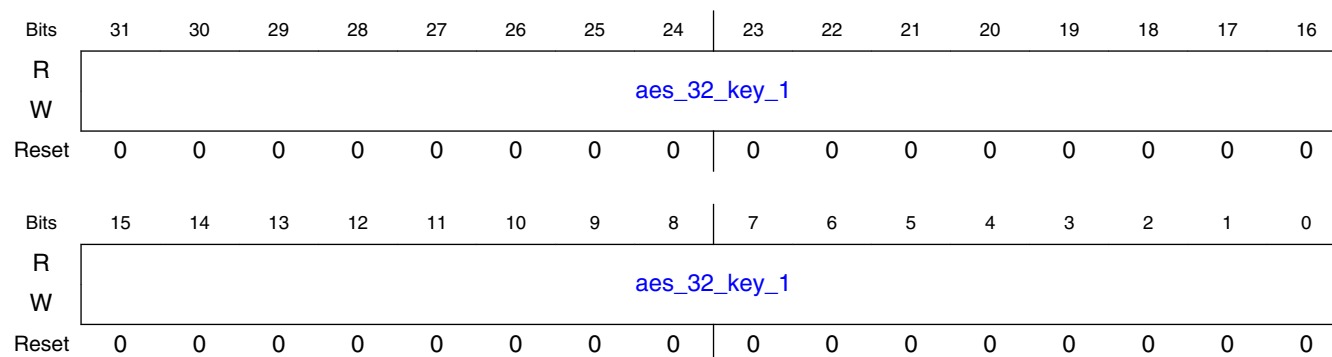
Field	Function
31-0	Holds the least significant 32-bits word of the 128-bits input key word of the AES-32 module.
aes_32_key_0	Holds the least significant 32-bits word of the 128-bits input key word of the AES-32 module.

13.4.10.1.266 Input key word of the AES-32 module - 2' dw (AES_32_KEY_1)

13.4.10.1.266.1 Offset

Register	Offset
AES_32_KEY_1	4074h

13.4.10.1.266.2 Diagram



13.4.10.1.266.3 Fields

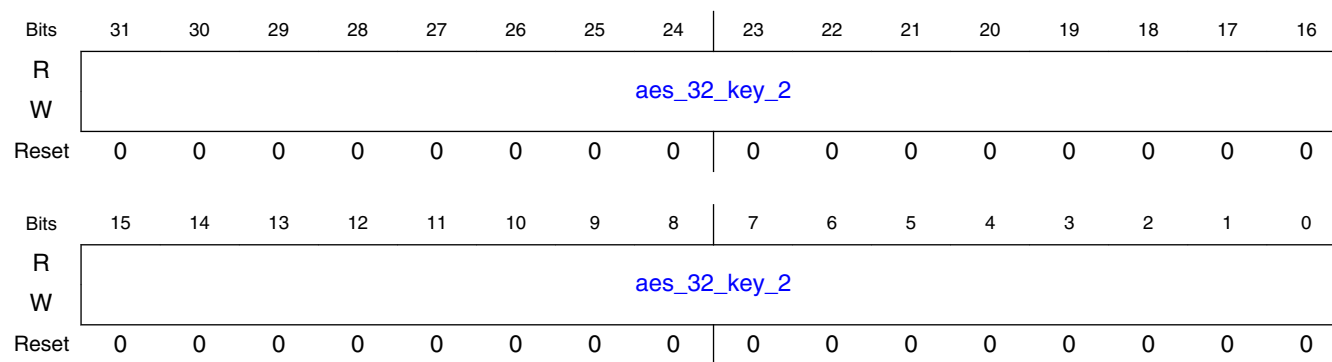
Field	Function
31-0	Holds the next significant 32-bits word of the 128-bits input key word of the AES-32 module.
aes_32_key_1	Holds the next significant 32-bits word of the 128-bits input key word of the AES-32 module.

13.4.10.1.267 Input key word of the AES-32 module - 3' dw (AES_32_KEY_2)

13.4.10.1.267.1 Offset

Register	Offset
AES_32_KEY_2	4078h

13.4.10.1.267.2 Diagram



13.4.10.1.267.3 Fields

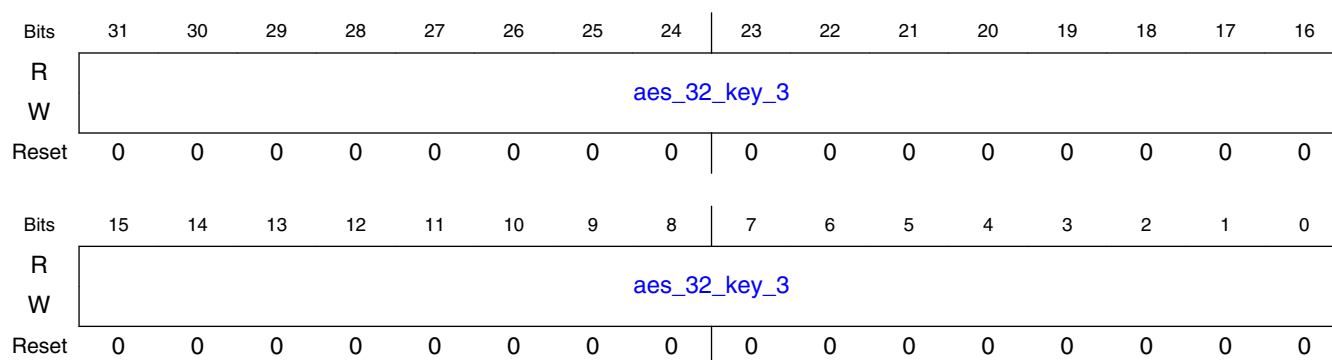
Field	Function
31-0	Holds the next significant 32-bits word of the 128-bits input key word of the AES-32 module.
aes_32_key_2	Holds the next significant 32-bits word of the 128-bits input key word of the AES-32 module.

13.4.10.1.268 Input key word of the AES-32 module - 4' dw (AES_32_KEY_3)

13.4.10.1.268.1 Offset

Register	Offset
AES_32_KEY_3	407Ch

13.4.10.1.268.2 Diagram



13.4.10.1.268.3 Fields

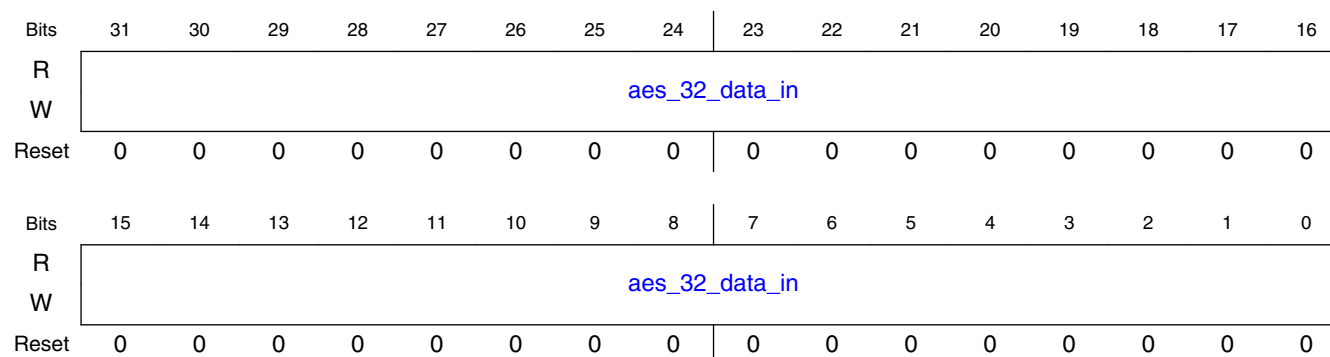
Field	Function
31-0	Holds the most significant 32-bits word of the 128-bits input key word of the AES-32 module.
aes_32_key_3	Holds the most significant 32-bits word of the 128-bits input key word of the AES-32 module.

13.4.10.1.269 Input data word to the AES-32 module (AES_32_DATA_IN)

13.4.10.1.269.1 Offset

Register	Offset
AES_32_DATA_IN	4080h

13.4.10.1.269.2 Diagram



13.4.10.1.269.3 Fields

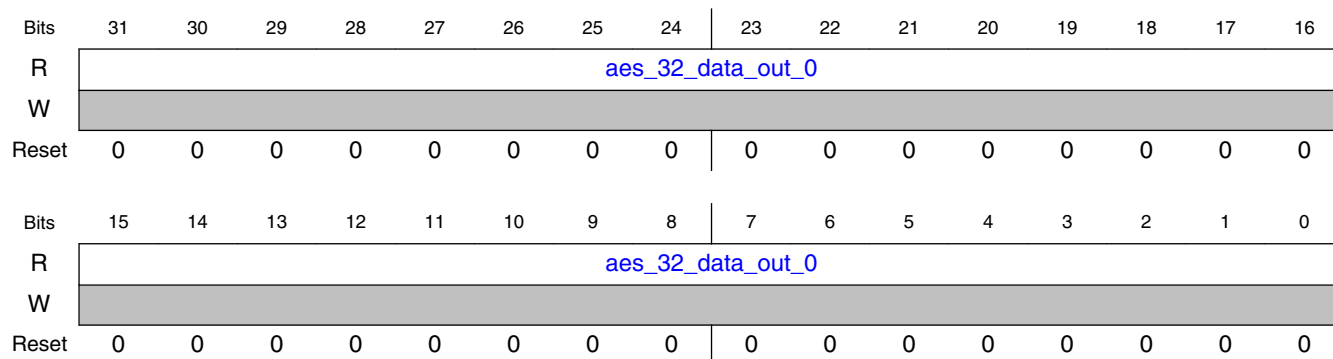
Field	Function
31-0	Holds the input data word to the AES-32 module.
aes_32_data_in	Holds the input data word to the AES-32 module.

13.4.10.1.270 AES-32 module - 128-bits output data word - 1' dw (AES_32_DATA_OUT_0)

13.4.10.1.270.1 Offset

Register	Offset
AES_32_DATA_OUT_0	4084h

13.4.10.1.270.2 Diagram



13.4.10.1.270.3 Fields

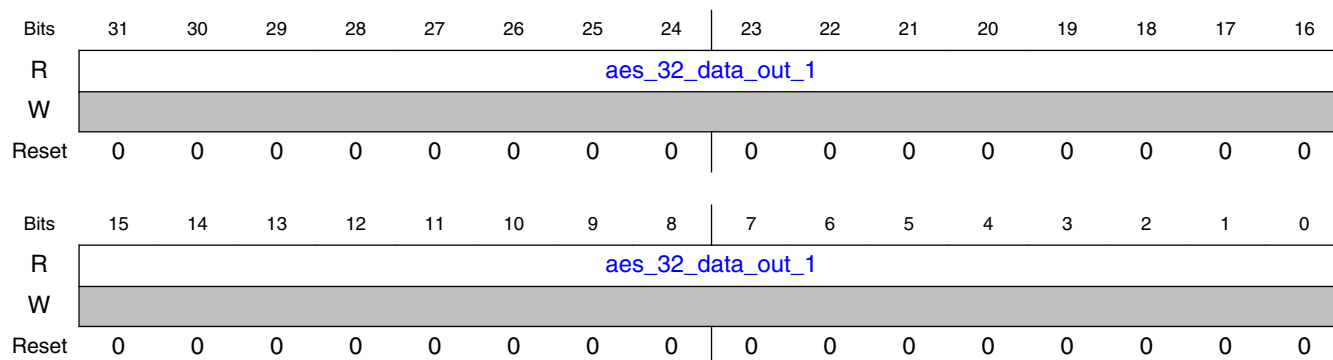
Field	Function
31-0	Holds the least significant 32-bits word of the 128-bits output data word of the AES-32 module.
aes_32_data_out_0	Holds the least significant 32-bits word of the 128-bits output data word of the AES-32 module.

13.4.10.1.271 AES-32 module - 128-bits output data word - 2' dw (AES_32_DATA_OUT_1)

13.4.10.1.271.1 Offset

Register	Offset
AES_32_DATA_OUT_1	4088h

13.4.10.1.271.2 Diagram



13.4.10.1.271.3 Fields

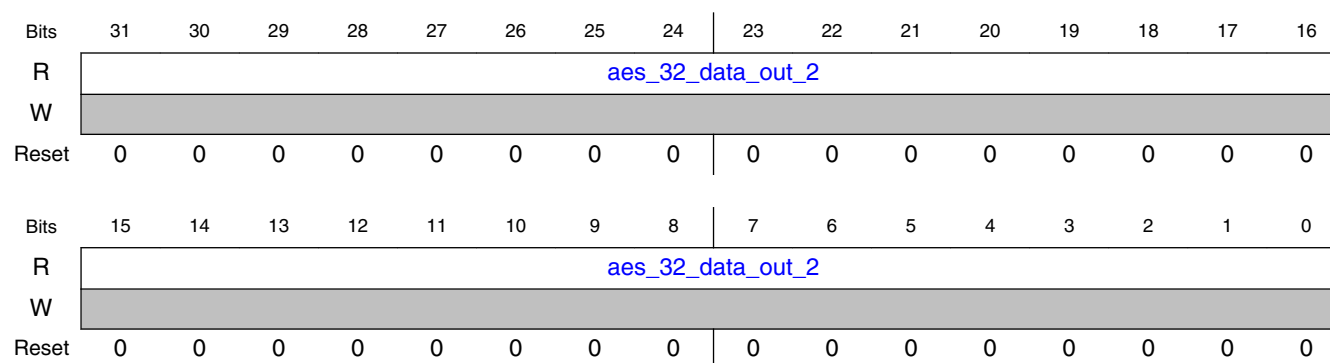
Field	Function
31-0	Holds the next significant 32-bits word of the 128-bits output data word of the AES-32 module.
aes_32_data_out_1	Holds the next significant 32-bits word of the 128-bits output data word of the AES-32 module.

13.4.10.1.272 AES-32 module - 128-bits output data word - 3' dw (AES_32_DATA_OUT_2)

13.4.10.1.272.1 Offset

Register	Offset
AES_32_DATA_OUT_2	408Ch

13.4.10.1.272.2 Diagram



13.4.10.1.272.3 Fields

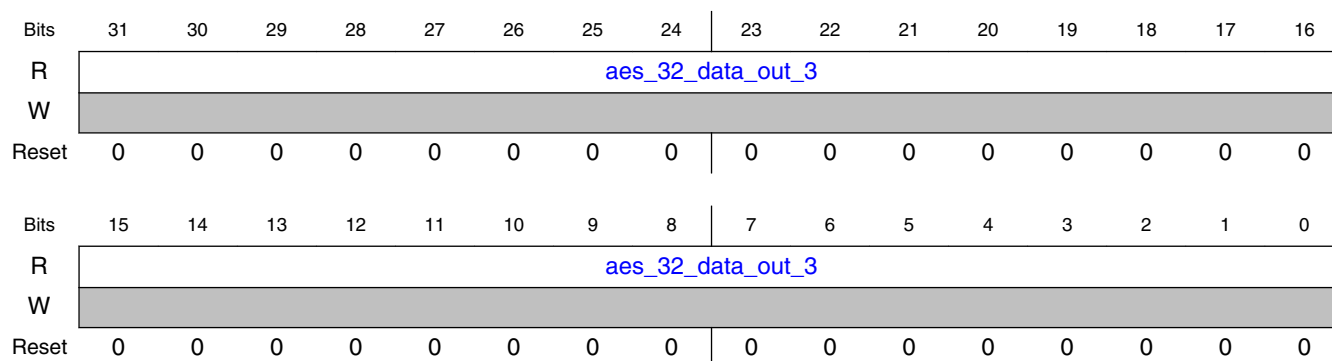
Field	Function
31-0	Holds the next significant 32-bits word of the 128-bits output data word of the AES-32 module.
aes_32_data_out_2	Holds the next significant 32-bits word of the 128-bits output data word of the AES-32 module.

13.4.10.1.273 AES-32 module - 128-bits output data word - 4' dw (AES_32_DATA_OUT_3)

13.4.10.1.273.1 Offset

Register	Offset
AES_32_DATA_OUT_3	4090h

13.4.10.1.273.2 Diagram



13.4.10.1.273.3 Fields

Field	Function
31-0	Holds the most significant 32-bits word of the 128-bits output data word of the AES-32 module.
aes_32_data_out_3	Holds the most significant 32-bits word of the 128-bits output data word of the AES-32 module.

13.4.10.1.274 Contains global configuration information for the HDCP 1. (CRYPTO14_CONFIG)

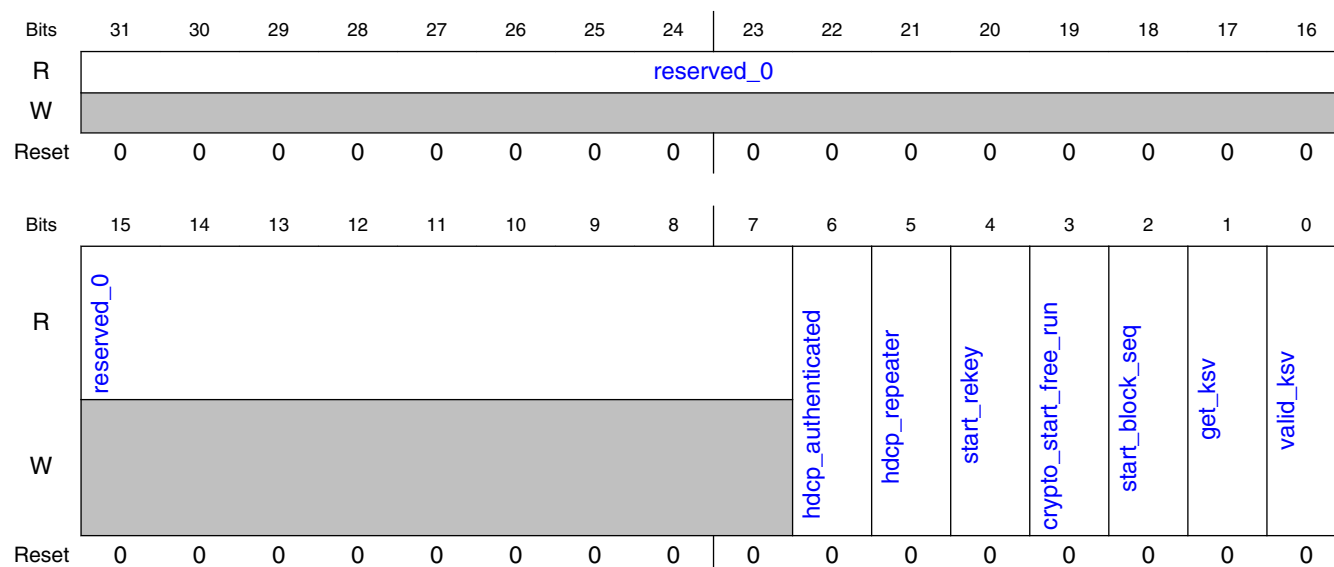
13.4.10.1.274.1 Offset

Register	Offset
CRYPTO14_CONFIG	40A0h

13.4.10.1.274.2 Function

Contains global configuration information for the HDCP 1.4 Crypto module

13.4.10.1.274.3 Diagram



13.4.10.1.274.4 Fields

Field	Function
31-7 reserved_0	
6 hdcp_authenticated	Authenticated finished.
5 hdcp_repeater	Repeater bit : 0: for the receiver, 1: for the repeater
4 start_rekey	Crypto 1. Crypto 1.4 command to start hdcpRekeyCipher
3 crypto_start_free_run	Crypto 1. Crypto 1.4 command to start free running enable for operation hdcpRngCipher
2 start_block_seq	Crypto 1. Crypto 1.4 command to start LFSR calculation
1 get_ksv	Read it's own KSV enable bit. Read it's own KSV enable bit.'0' reading not allowed'1' start reading.
0 valid_ksv	Enable for Km calculation. Enable for Km calculation. When high,start calculating Km. Indicates a good moment for ri_out sampling.

13.4.10.1.275 Contains global status information for the HDCP 1. (CRYPTO14_STATUS)

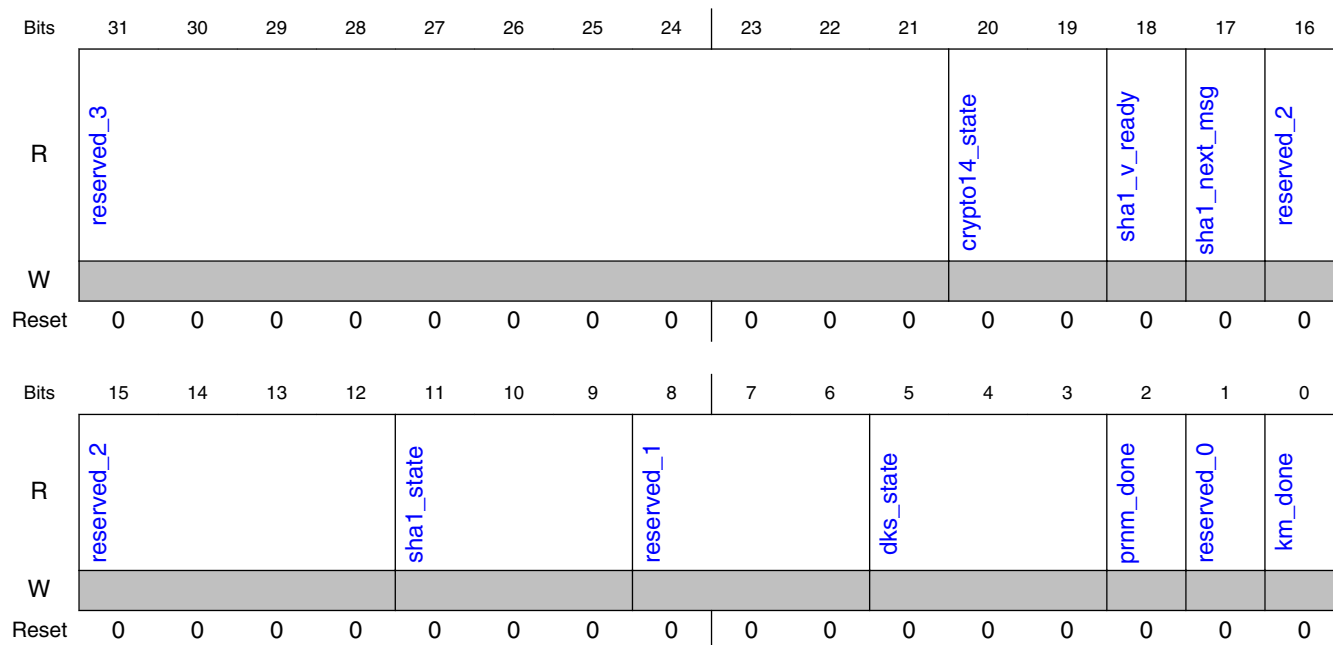
13.4.10.1.275.1 Offset

Register	Offset
CRYPTO14_STATUS	40A4h

13.4.10.1.275.2 Function

Contains global status information for the HDCP 1.4 Crypto module

13.4.10.1.275.3 Diagram



13.4.10.1.275.4 Fields

Field	Function
31-21 reserved_3	
20-19	Crypto operation SM state: Possible values: 00- HDCP_RNG_CIPHER; 01 - HDCP_BLOCK_CIPHER; 10 - HDCP_STREAM_CIPHER; 11 - HDCP_REKEY_CIPHER

Table continues on the next page...

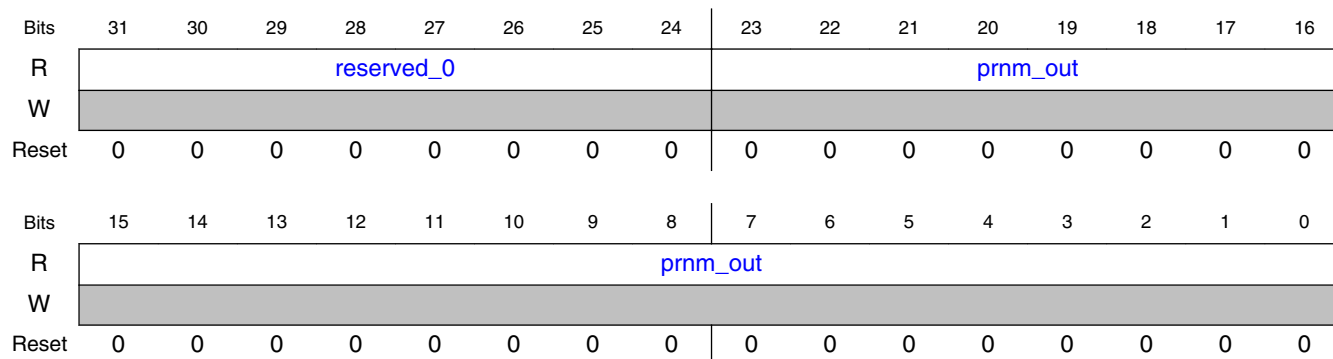
Field	Function
crypto14_state	
18 sha1_v_ready	Indication that V value from SHA-1 CRYPTO14_SHA1_V_VALUE_4 is ready.
17 sha1_next_msg	Request for the next message block. Request for the next message block. When set high, CRYPTO14_SHA1_MSG_DATA_0-15 registers shall be written.
16-12 reserved_2	
11-9 sha1_state	Current state for Crypto 1. Current state for Crypto 1.4 SHA-1 FSM. Used for debug purpose. Possible values: 000 - IDLE; 001 - PREPARE; 010 - CALCULATE; 011 - RESULT; 100 - BLOCK_WAIT
8-6 reserved_1	
5-3 dks_state	Crypto 1. Crypto 1.4 DKS current state. Used for debug purpose. Possible values: 000 - HDCP_IDLE_KSV; 010 - HDCP_IDLE; 100 - HDCP_PRECALC; 101 - HDCP_POSTCALC; 110 - HDCP_CALC; 111 - HDCP_READY
2 prnm_done	LFSR and block output finished calculation. LFSR and block output finished calculation.
1 reserved_0	
0 km_done	Done reading/calculating Km. Done reading/calculating Km. Used as interrupt event.

13.4.10.1.276 Contains 24-bit pseudo random data (CRYPTO14_PRNM_OUT)

13.4.10.1.276.1 Offset

Register	Offset
CRYPTO14_PRNM_OUT	40A8h

13.4.10.1.276.2 Diagram



13.4.10.1.276.3 Fields

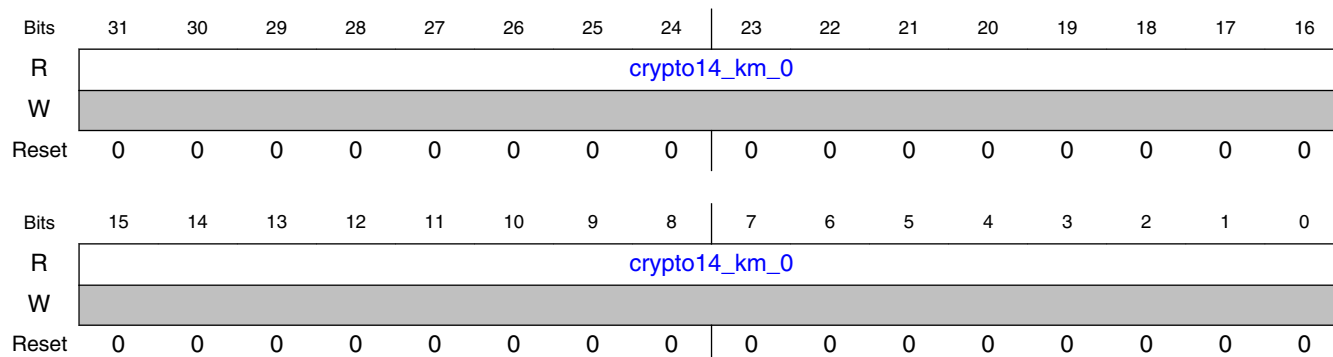
Field	Function
31-24 reserved_0	
23-0 prnm_out	24-bit pseudo-random data. 24-bit pseudo-random data.

13.4.10.1.277 Contains the first word of the Km value (CRYPTO14_KM_0)

13.4.10.1.277.1 Offset

Register	Offset
CRYPTO14_KM_0	40ACh

13.4.10.1.277.2 Diagram



13.4.10.1.277.3 Fields

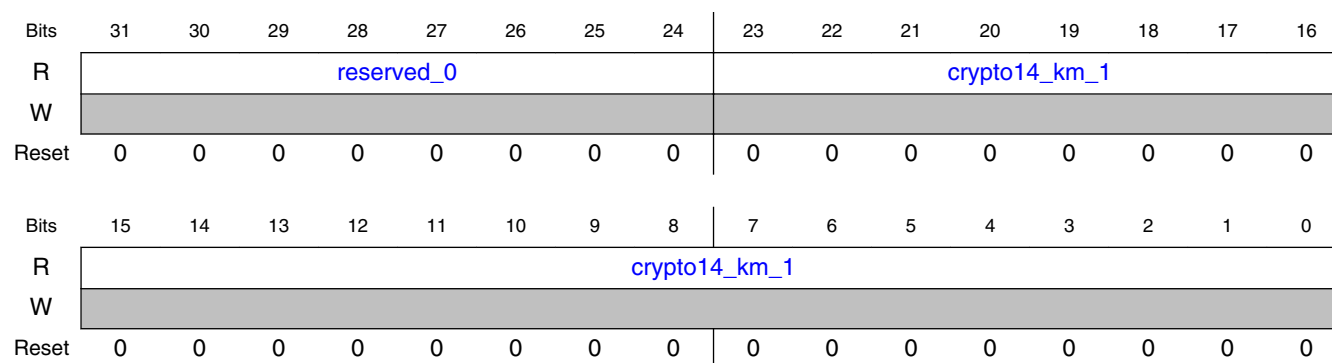
Field	Function
31-0	Holds the first word of the Km value.
crypto14_km_0	Holds the first word of the Km value.

13.4.10.1.278 Contains the most significant 3 bytes of the Km value (CRYPTO14_KM_1)

13.4.10.1.278.1 Offset

Register	Offset
CRYPTO14_KM_1	40B0h

13.4.10.1.278.2 Diagram



13.4.10.1.278.3 Fields

Field	Function
31-24 reserved_0	
23-0	Holds the most significant 3 bytes of the Km value.
crypto14_km_1	Holds the most significant 3 bytes of the Km value.

13.4.10.1.279 First word of An value generated by hdcpRngCipher operation. (CRYPTO14_AN_0)

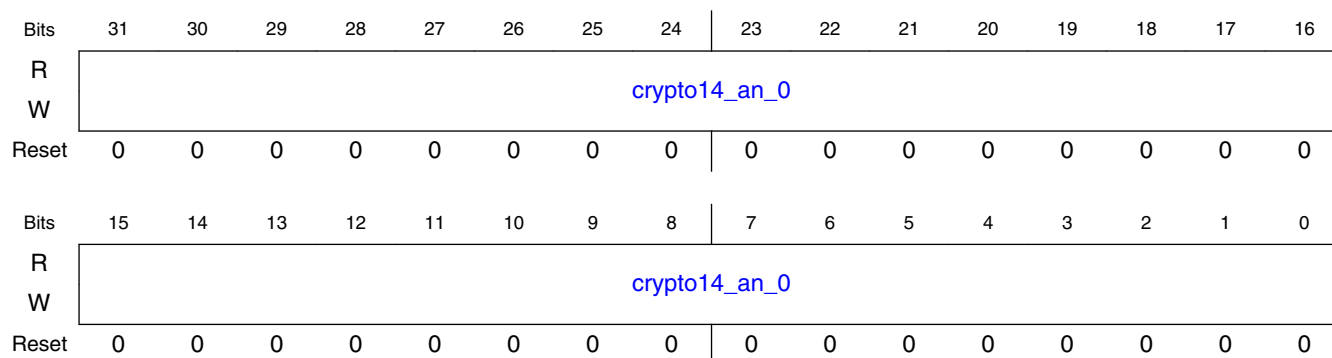
13.4.10.1.279.1 Offset

Register	Offset
CRYPTO14_AN_0	40B4h

13.4.10.1.279.2 Function

First word of An value generated by hdcpRngCipher operation.

13.4.10.1.279.3 Diagram



13.4.10.1.279.4 Fields

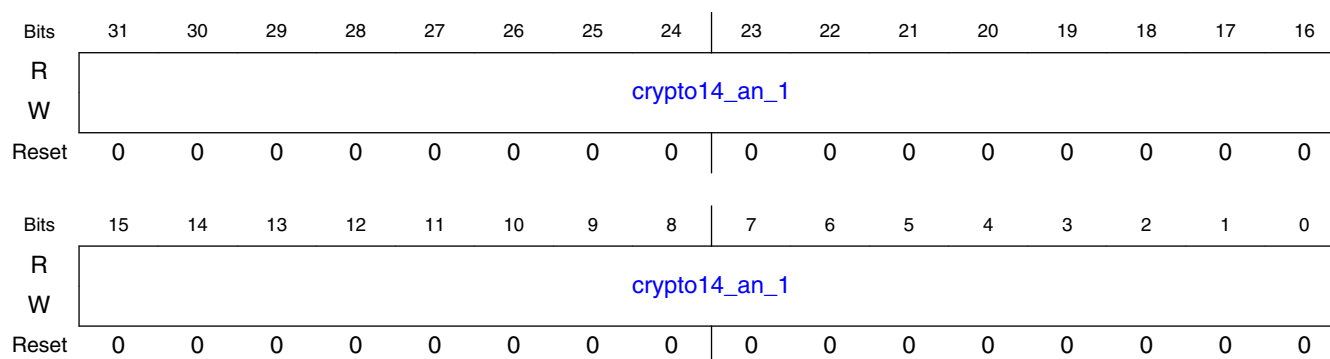
Field	Function
31-0	Holds the first 4 bytes of the An value.
crypto14_an_0	Holds the first 4 bytes of the An value.

13.4.10.1.280 Second word of An value generated by hdcpRngCipher operation (CRYPTO14_AN_1)

13.4.10.1.280.1 Offset

Register	Offset
CRYPTO14_AN_1	40B8h

13.4.10.1.280.2 Diagram



13.4.10.1.280.3 Fields

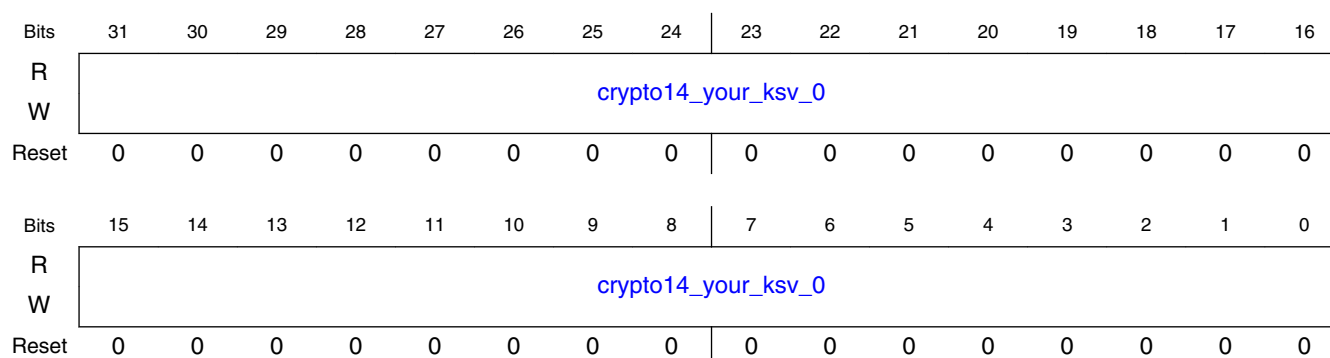
Field	Function
31-0	Holds the most significant 4 bytes of the An value.
crypto14_an_1	Holds the most significant 4 bytes of the An value.

13.4.10.1.281 First 32 bits of the KSV from the other HDCP device (CRYPTO14_YOUR_KSV_0)

13.4.10.1.281.1 Offset

Register	Offset
CRYPTO14_YOUR_KSV_0	40BCh

13.4.10.1.281.2 Diagram



13.4.10.1.281.3 Fields

Field	Function
31-0	Holds the first 32 bits of the KSV from the other HDCP device.
crypto14_your_ksv_0	Holds the first 32 bits of the KSV from the other HDCP device.

13.4.10.1.282 Last byte of the KSV from other HDCP device (CRYPTO14_YOUR_KSV_1)

13.4.10.1.282.1 Offset

Register	Offset
CRYPTO14_YOUR_KSV_1	40C0h

13.4.10.1.282.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0								crypto14_your_ksv_1							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.282.3 Fields

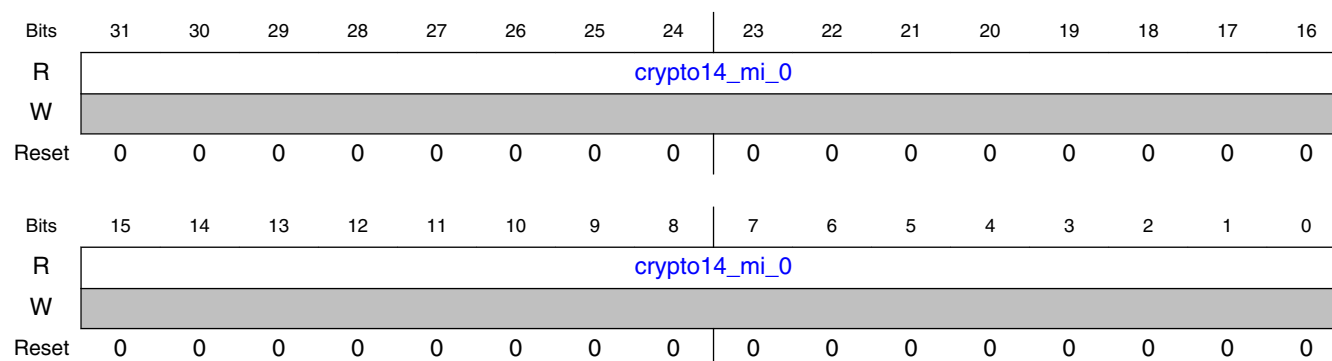
Field	Function
31-8 reserved_0	
7-0 crypto14_your_ksv_1	Holds the last byte of the KSV from other HDCP device

13.4.10.1.283 Mi value - 1' dw (CRYPTO14_MI_0)

13.4.10.1.283.1 Offset

Register	Offset
CRYPTO14_MI_0	40C4h

13.4.10.1.283.2 Diagram



13.4.10.1.283.3 Fields

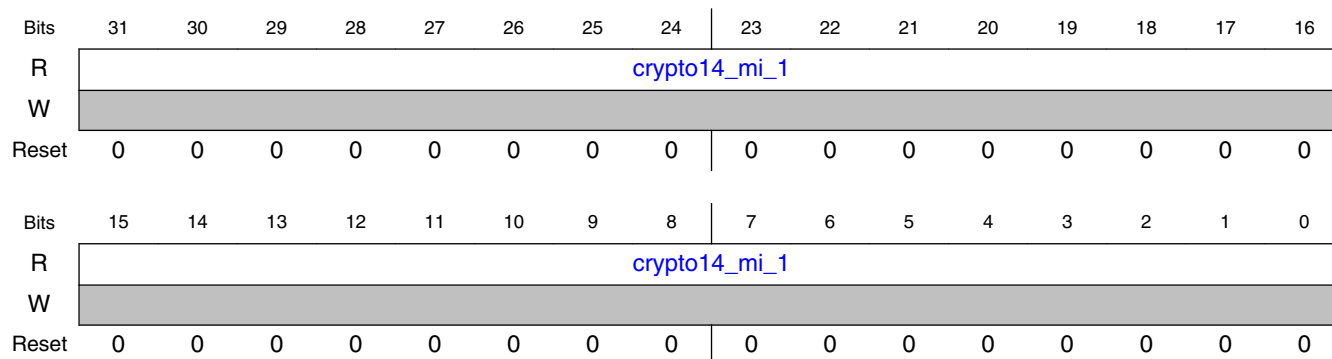
Field	Function
31-0 crypto14_mi_0	Mi value first 32 bits

13.4.10.1.284 Mi value - 2' dw (CRYPTO14_MI_1)

13.4.10.1.284.1 Offset

Register	Offset
CRYPTO14_MI_1	40C8h

13.4.10.1.284.2 Diagram



13.4.10.1.284.3 Fields

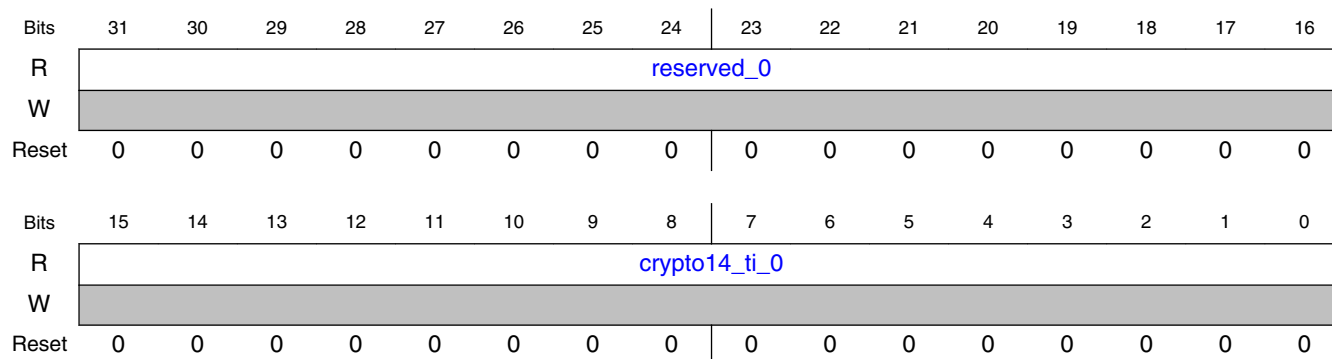
Field	Function
31-0 crypto14_mi_1	Mi value second 32 bits

13.4.10.1.285 Ti value (CRYPTO14_TI_0)

13.4.10.1.285.1 Offset

Register	Offset
CRYPTO14_TI_0	40CCh

13.4.10.1.285.2 Diagram



13.4.10.1.285.3 Fields

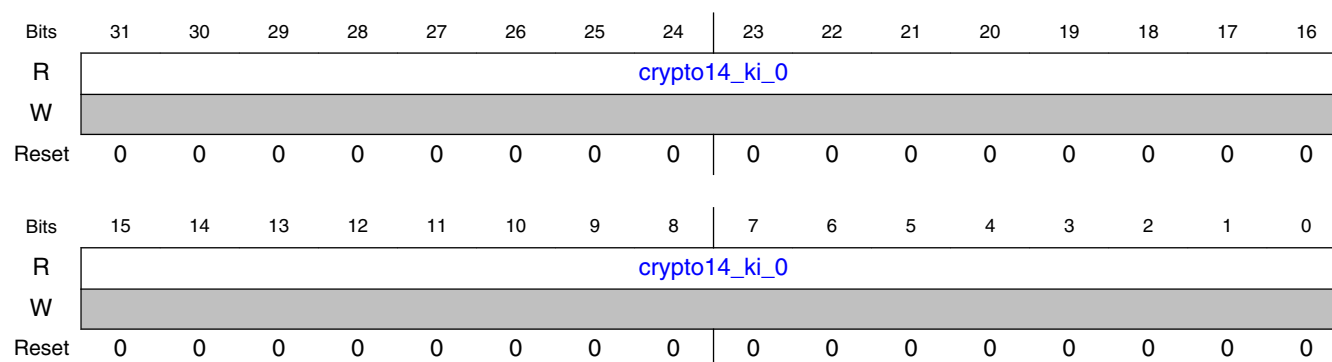
Field	Function
31-16 reserved_0	
15-0 crypto14_ti_0	Ti value

13.4.10.1.286 First 32 bits of the Ki frame key from this HDCP device (CRYPTO14_KI_0)

13.4.10.1.286.1 Offset

Register	Offset
CRYPTO14_KI_0	40D0h

13.4.10.1.286.2 Diagram



13.4.10.1.286.3 Fields

Field	Function
31-0 crypto14_ki_0	Holds the first 32 bits of the Ki frame key from this HDCP device.

13.4.10.1.287 Last 3 bytes of the Ki frame key from this HDCP device. (CRYPTO14_KI_1)

13.4.10.1.287.1 Offset

Register	Offset
CRYPTO14_KI_1	40D4h

13.4.10.1.287.2 Function

Last 3 bytes of the Ki frame key from this HDCP device.

13.4.10.1.287.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0								crypto14_ki_1							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	crypto14_ki_1															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.287.4 Fields

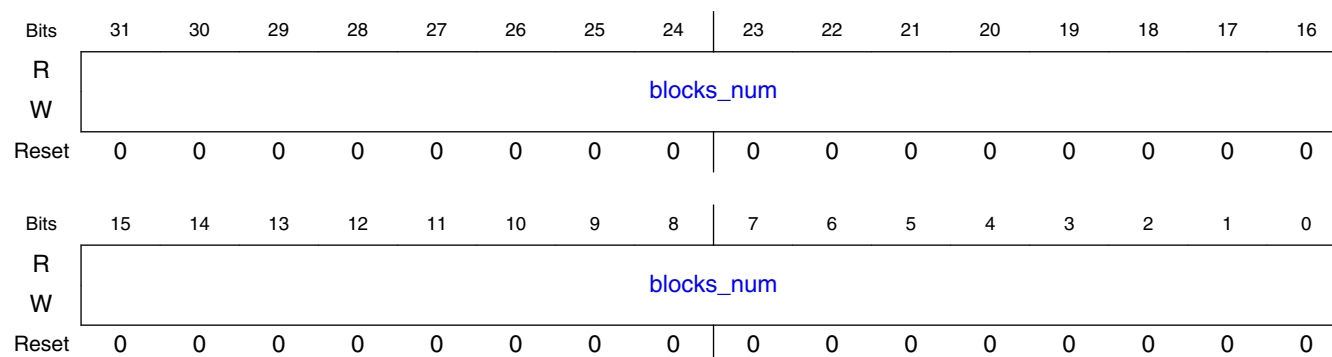
Field	Function
31-24 reserved_0	
23-0 crypto14_ki_1	Holds the last 3 bytes of the Ki frame key from this HDCP device. Holds the last 3 bytes of the Ki frame key from this HDCP device.

13.4.10.1.288 This register defines number of iterations for SHA-1 calculations (CRYPTO14_BLOCKS_NUM)

13.4.10.1.288.1 Offset

Register	Offset
CRYPTO14_BLOCKS_NUM	40D8h

13.4.10.1.288.2 Diagram



13.4.10.1.288.3 Fields

Field	Function
31-0	Number of iterations for SHA-1 calculation.
blocks_num	Number of iterations for SHA-1 calculation.

13.4.10.1.289 Key memory control register. (CRYPTO14_KEY_MEM_DATA_0)

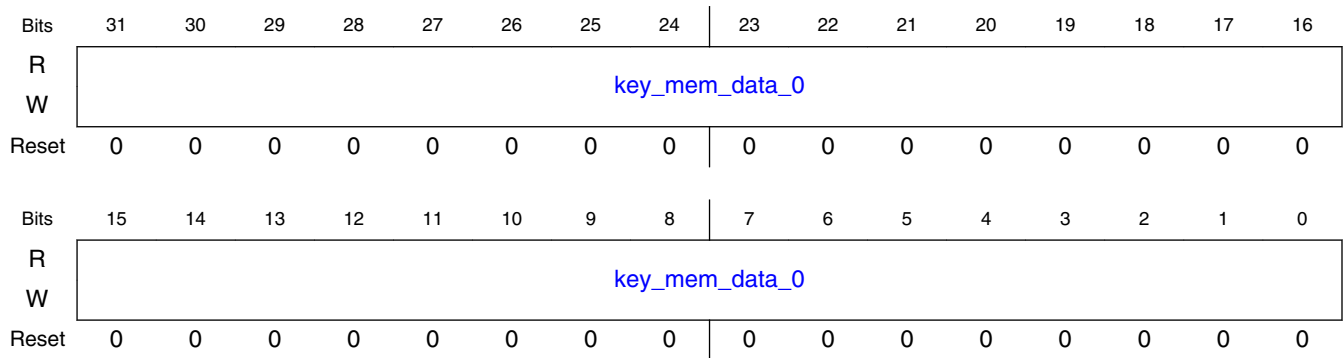
13.4.10.1.289.1 Offset

Register	Offset
CRYPTO14_KEY_MEM_DATA_0	40DCh

13.4.10.1.289.2 Function

Key memory control register. Writing data to this register transfers data to the key RAM. This is input for DKS block. First 32 bits

13.4.10.1.289.3 Diagram



13.4.10.1.289.4 Fields

Field	Function
31-0	Output data from keys RAM.
key_mem_data_0	Output data from keys RAM. Input for DKS block. First 32 bits.

13.4.10.1.290 Key memory control register. (CRYPTO14_KEY_MEM_DATA_1)

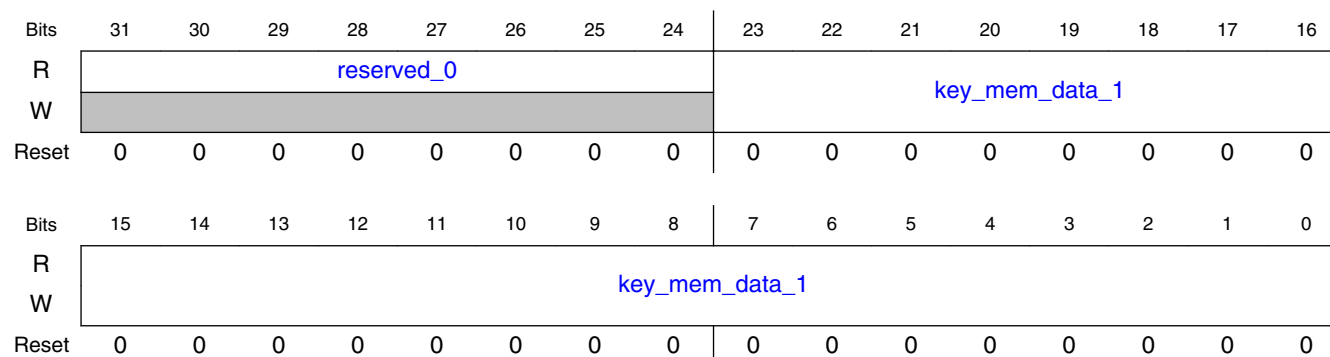
13.4.10.1.290.1 Offset

Register	Offset
CRYPTO14_KEY_MEM_DATA_1	40E0h

13.4.10.1.290.2 Function

Key memory control register. Writing data to this register transfers data to the key RAM. When data is written to this register, pulse key_mem_vld shall be also generated to the core_clk clock domain, as indication of new key_mem_data. This is input for DKS block. Last 3 bytes

13.4.10.1.290.3 Diagram



13.4.10.1.290.4 Fields

Field	Function
31-24 reserved_0	
23-0 key_mem_data_1	Output data from keys RAM. Output data from keys RAM. Input for DKS block. Last 3 bytes.

13.4.10.1.291 SHA1 message control register. (CRYPTO14_SHA1_MSG_DATA)

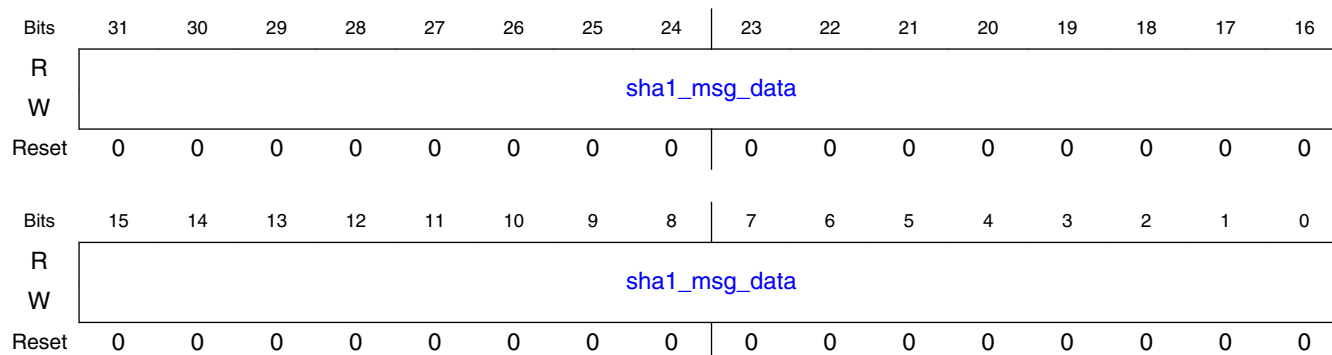
13.4.10.1.291.1 Offset

Register	Offset
CRYPTO14_SHA1_MSG_DATA	40E4h

13.4.10.1.291.2 Function

SHA1 message control register. 32-bit word for SHA-1 message. Input for SHA-1 block. Writing data to this register transfers data to the SHA-1 block. Write to this register shall be repeated 16 times.

13.4.10.1.291.3 Diagram



13.4.10.1.291.4 Fields

Field	Function
31-0	32-bit word for SHA-1 message.
sha1_msg_data	32-bit word for SHA-1 message. Input for SHA-1 block.

13.4.10.1.292 SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_0)

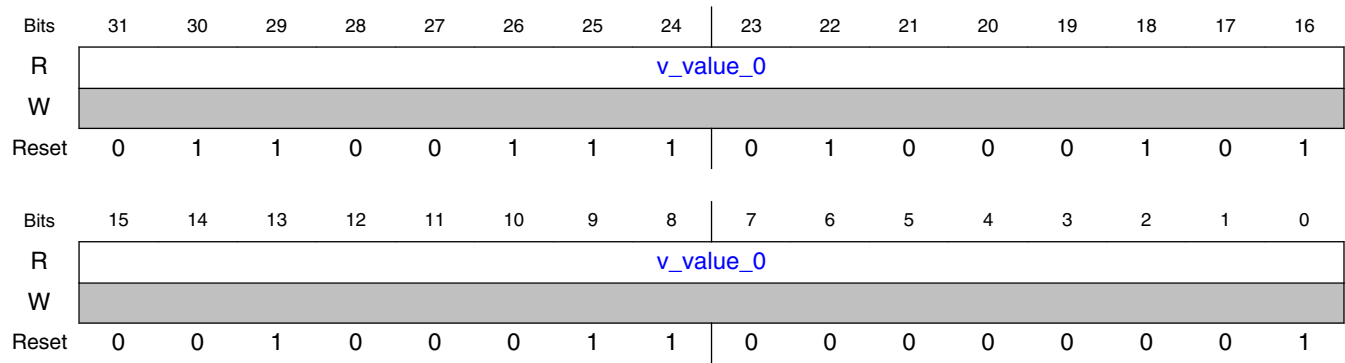
13.4.10.1.292.1 Offset

Register	Offset
CRYPTO14_SHA1_V_VALUE_0	40E8h

13.4.10.1.292.2 Function

SHA1 message status register. First 32-bit word for SHA-1 calculation. Output from SHA-1 block

13.4.10.1.292.3 Diagram



13.4.10.1.292.4 Fields

Field	Function
31-0	First 32-bit word for SHA-1 calculation value.
v_value_0	First 32-bit word for SHA-1 calculation value. Output from SHA-1 block.

13.4.10.1.293 SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_1)

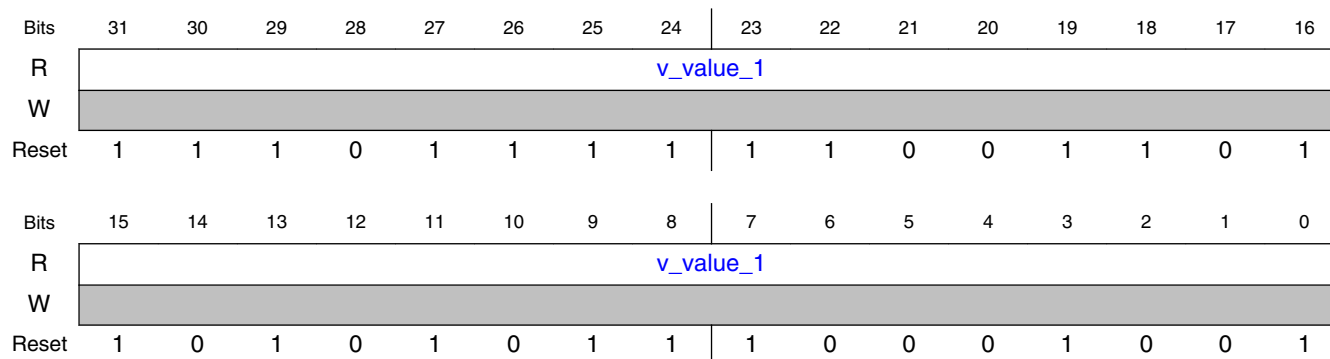
13.4.10.1.293.1 Offset

Register	Offset
CRYPTO14_SHA1_V_VALUE_1	40ECh

13.4.10.1.293.2 Function

SHA1 message status register. Second 32-bit word for SHA-1 calculation. Output from SHA-1 block

13.4.10.1.293.3 Diagram



13.4.10.1.293.4 Fields

Field	Function
31-0	Second 32-bit word for SHA-1 calculation value.
v_value_1	Second 32-bit word for SHA-1 calculation value. Output from SHA-1 block.

13.4.10.1.294 SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_2)

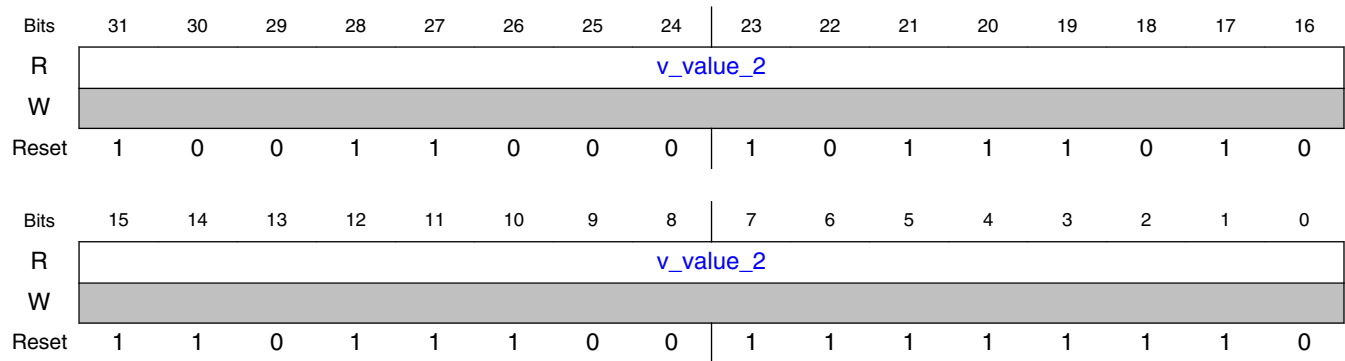
13.4.10.1.294.1 Offset

Register	Offset
CRYPTO14_SHA1_V_VALUE_2	40F0h

13.4.10.1.294.2 Function

SHA1 message status register. Third 32-bit word for SHA-1 calculation. Output from SHA-1 block

13.4.10.1.294.3 Diagram



13.4.10.1.294.4 Fields

Field	Function
31-0	Thir32-bit word for SHA-1 calculation value.
v_value_2	Thir32-bit word for SHA-1 calculation value. Output from SHA-1 block.

13.4.10.1.295 SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_3)

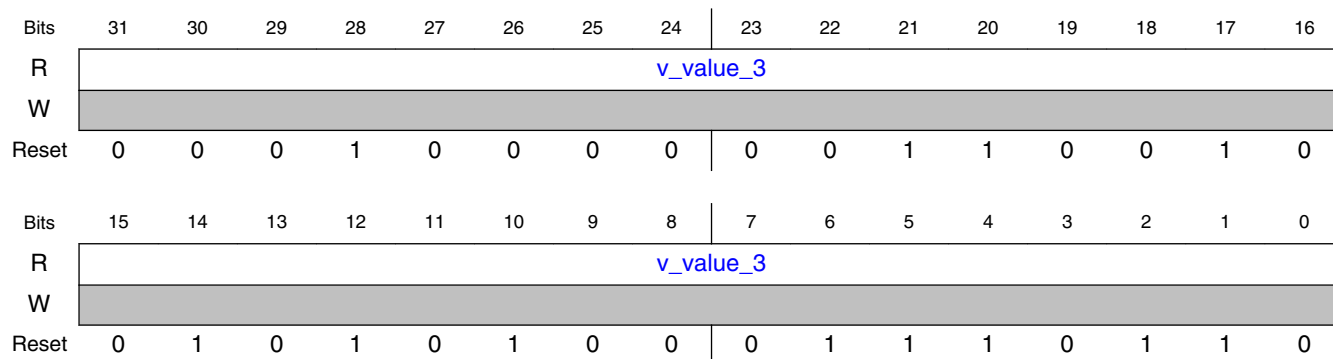
13.4.10.1.295.1 Offset

Register	Offset
CRYPTO14_SHA1_V_VALUE_3	40F4h

13.4.10.1.295.2 Function

SHA1 message status register. Thir32-bit word for SHA-1 calculation. Output from SHA-1 block

13.4.10.1.295.3 Diagram



13.4.10.1.295.4 Fields

Field	Function
31-0	4th 32-bit word for SHA-1 calculation value.
v_value_3	4th 32-bit word for SHA-1 calculation value. Output from SHA-1 block.

13.4.10.1.296 SHA1 message status register. (CRYPTO14_SHA1_V_VALUE_4)

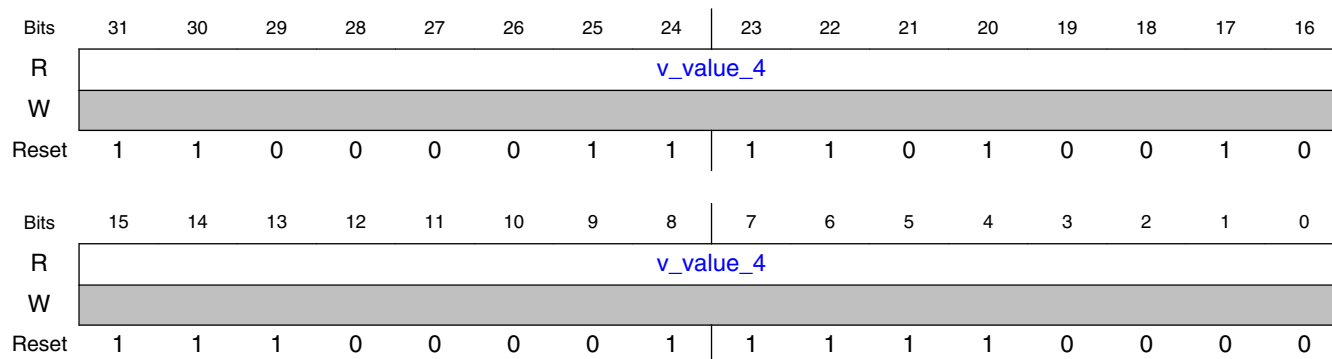
13.4.10.1.296.1 Offset

Register	Offset
CRYPTO14_SHA1_V_VALUE_4	40F8h

13.4.10.1.296.2 Function

SHA1 message status register. 5th 32-bit word for SHA-1 calculation. Output from SHA-1 block

13.4.10.1.296.3 Diagram



13.4.10.1.296.4 Fields

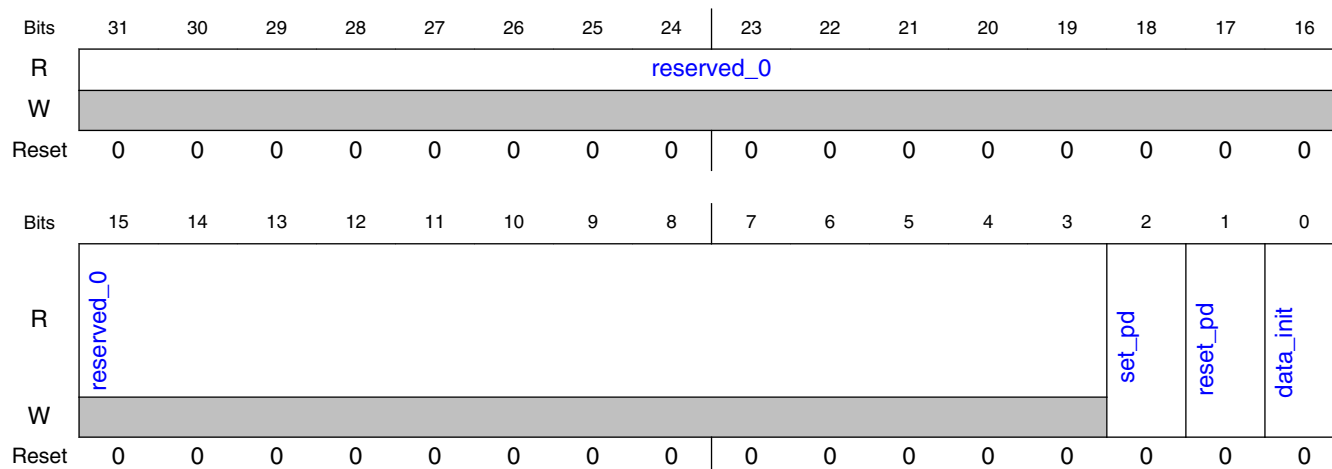
Field	Function
31-0	5th 32-bit word for SHA-1 calculation value.
v_value_4	5th 32-bit word for SHA-1 calculation value. Output from SHA-1 block.

13.4.10.1.297 TRNG control (TRNG_CTRL)

13.4.10.1.297.1 Offset

Register	Offset
TRNG_CTRL	40FCh

13.4.10.1.297.2 Diagram



13.4.10.1.297.3 Fields

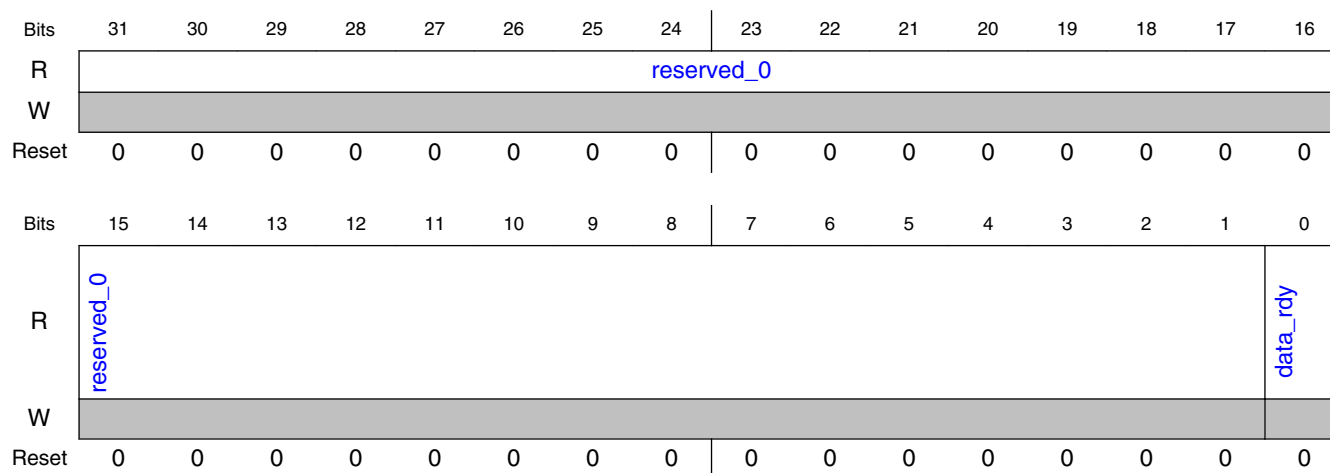
Field	Function
31-3 reserved_0	
2 set_pd	set pd
1 reset_pd	resetpd output
0 data_init	init trng read

13.4.10.1.298 TRNG status (TRNG_DATA_RDY)

13.4.10.1.298.1 Offset

Register	Offset
TRNG_DATA_RDY	4100h

13.4.10.1.298.2 Diagram



13.4.10.1.298.3 Fields

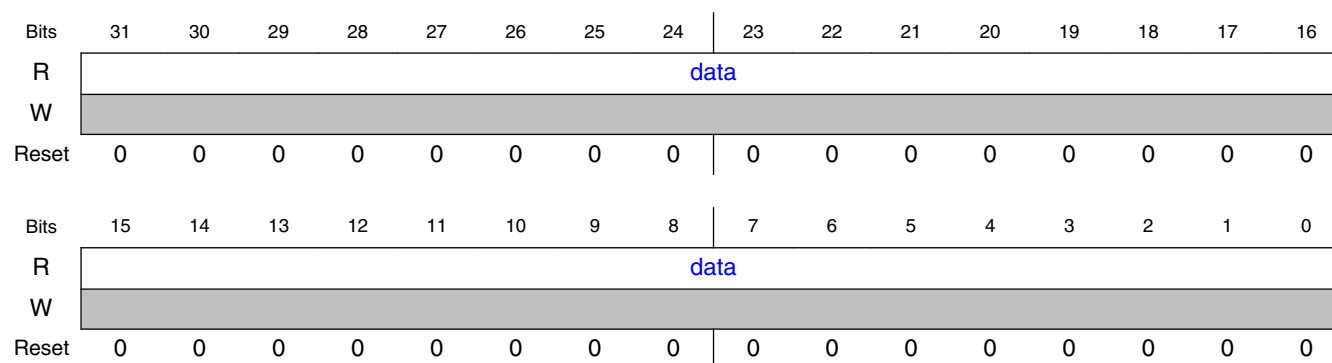
Field	Function
31-1 reserved_0	
0 data_rdy	

13.4.10.1.299 TRNG data register (TRNG_DATA)

13.4.10.1.299.1 Offset

Register	Offset
TRNG_DATA	4104h

13.4.10.1.299.2 Diagram



13.4.10.1.299.3 Fields

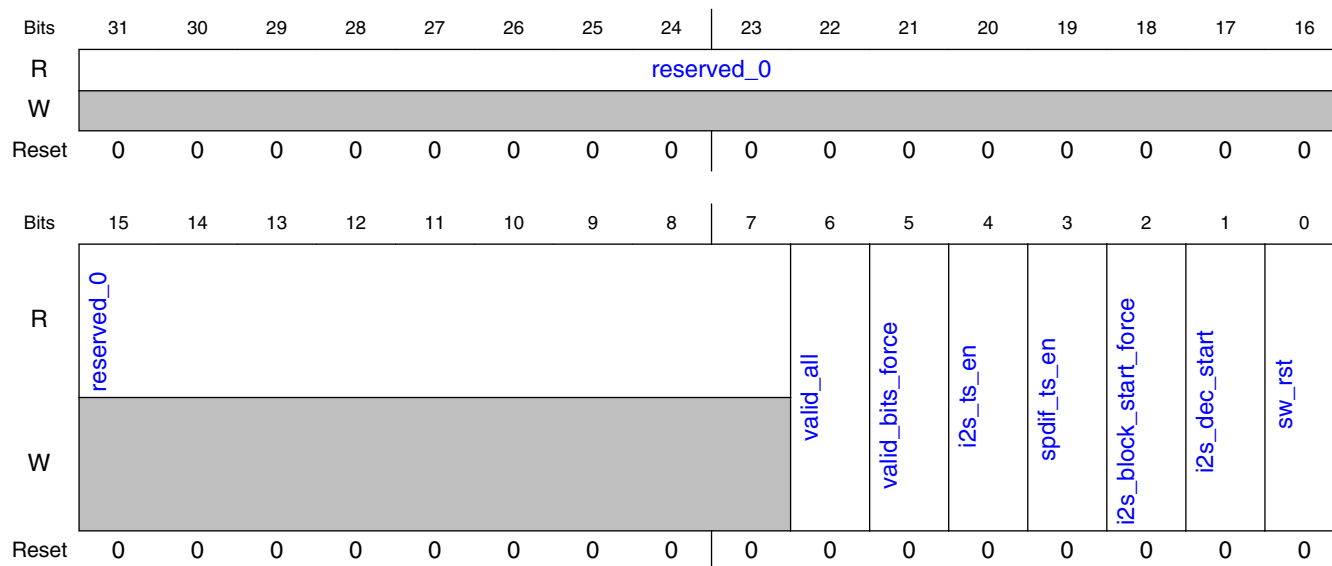
Field	Function
31-0 data	dat to read

13.4.10.1.300 Audio source control (AUDIO_SRC_CNTL)

13.4.10.1.300.1 Offset

Register	Offset
AUDIO_SRC_CNTL	3_0000h

13.4.10.1.300.2 Diagram



13.4.10.1.300.3 Fields

Field	Function
31-7 reserved_0	
6 valid_all	valid bit for all samples
5 valid_bits_force	Force valid bits of the channels
4 i2s_ts_en	Enble I2S Time Stamp when decoders are disabled
3 spdif_ts_en	Enble SPDIF Time Stamp when decoders are disabled
2 i2s_block_start_force	Force a Block Start in the audio stream
1 i2s_dec_start	When high Source Decoder starts. When high Source Decoder starts.

Table continues on the next page...

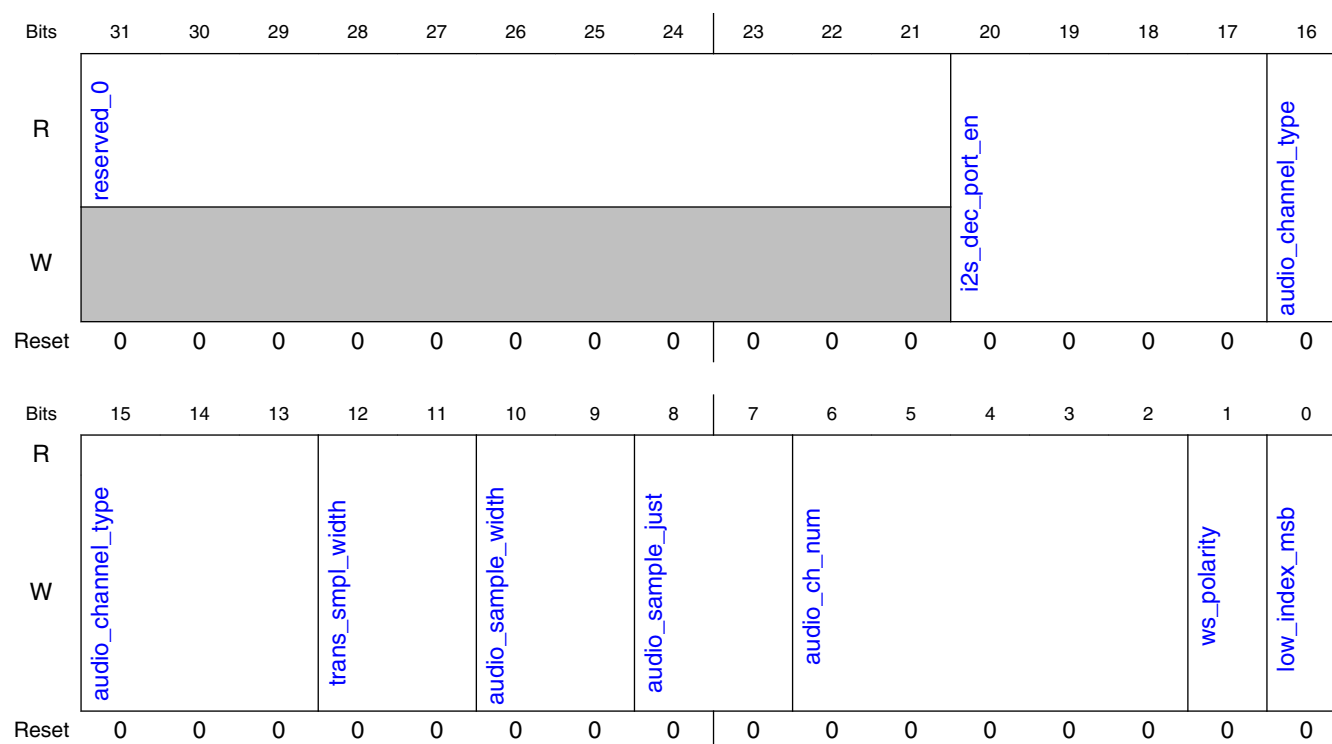
Field	Function
0	Software reset.
sw_rst	Software reset. Active high.

13.4.10.1.301 Audio source configuration (AUDIO_SRC_CNFG)

13.4.10.1.301.1 Offset

Register	Offset
AUDIO_SRC_CNFG	3_0004h

13.4.10.1.301.2 Diagram



13.4.10.1.301.3 Fields

Field	Function
31-21	
reserved_0	

Table continues on the next page...

Clocks And Resets

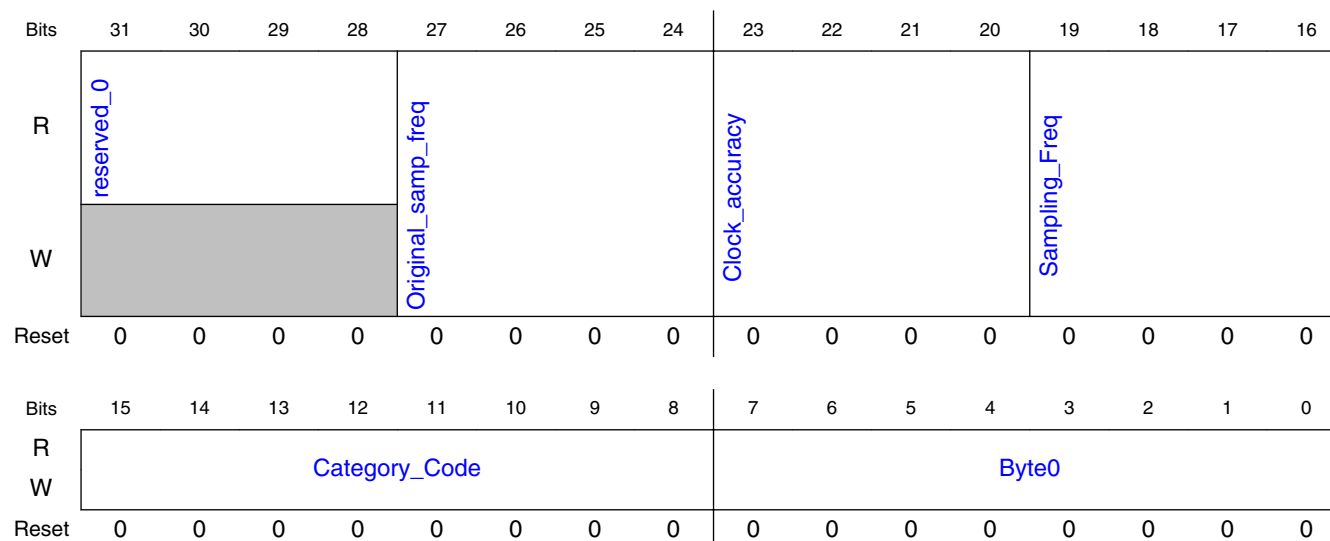
Field	Function
20-17 i2s_dec_port_en	Enables the I2S Decoder ports. Enables the I2S Decoder ports. Allowed values are:0001 - I2S port 0 is enabled.0011 - I2S ports 0,1 are enabled.1111 - I2S ports 0,1,2,3 are enabled. No other values are allowed.
16-13 audio_channel_type	Set the transmission type. Set the transmission type.
12-11 trans_smpl_width	Decoder Word Select width: 00-16 bit, 01-24 bit, 10-32 bit
10-9 audio_sample_width	Decoder sample width:00-16 bit, 01-24 bit, 10-32 bit
8-7 audio_sample_just	Data justification setting:00 left-justified, 01 right-justified
6-2 audio_ch_num	Number of channels to decode
1 ws_polarity	Word Select Polarity. Word Select Polarity. 0: No change, 1: Inverted.
0 low_index_msb	When low MSB is transmitted first. When low MSB is transmitted first. When high LSB is transmitted first.

13.4.10.1.302 Common channels configuration (COM_CH_STTS_BITS)

13.4.10.1.302.1 Offset

Register	Offset
COM_CH_STTS_BITS	3_0008h

13.4.10.1.302.2 Diagram



13.4.10.1.302.3 Fields

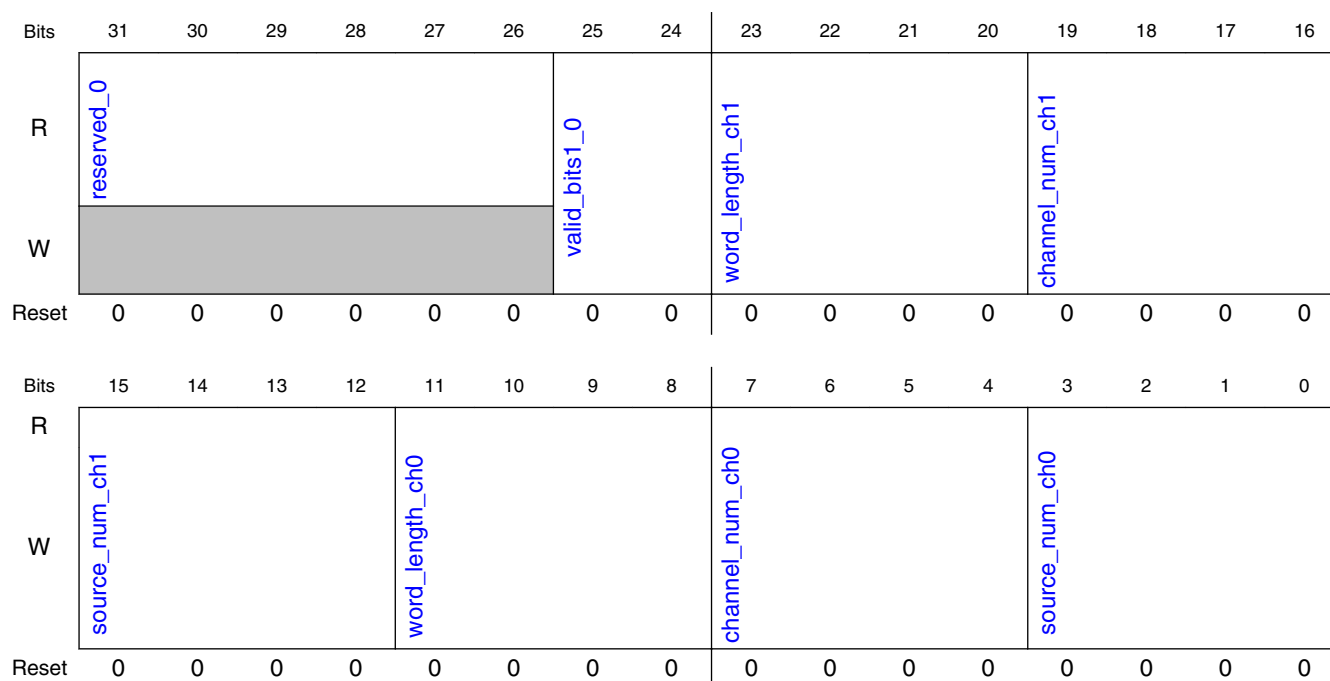
Field	Function
31-28 reserved_0	
27-24 Original_samp_freq	Original Sampling Freq. Original Sampling Freq. of transmitted channel. Same for all channels.
23-20 Clock_accuracy	Clock Accuracy of transmitted channel. Clock Accuracy of transmitted channel. Same for all channels.
19-16 Sampling_Freq	Sampling Frequency of transmitted channel. Sampling Frequency of transmitted channel. Same for all channels.
15-8 Category_Code	Category Code of transmitted channel. Category Code of transmitted channel. Same for all channels.
7-0 Byte0	Byte 0 of transmitted channel. Byte 0 of transmitted channel. Same for all channels.

13.4.10.1.303 Channels 0,1 configuration (STTS_BIT_CH01)

13.4.10.1.303.1 Offset

Register	Offset
STTS_BIT_CH01	3_000Ch

13.4.10.1.303.2 Diagram



13.4.10.1.303.3 Fields

Field	Function
31-26 reserved_0	
25-24 valid_bits1_0	Valid Bits for channel 1 and 0 if force is enabled
23-20 word_length_ch1	Channel 1 word length. Channel 1 word length.
19-16 channel_num_ch1	Channel 1 channel number. Channel 1 channel number.
15-12 source_num_ch1	Channel 1 Source number. Channel 1 Source number.
11-8 word_length_ch0	Channel 0 word length. Channel 0 word length.
7-4	Channel 0 channel number. Channel 0 channel number.

Table continues on the next page...

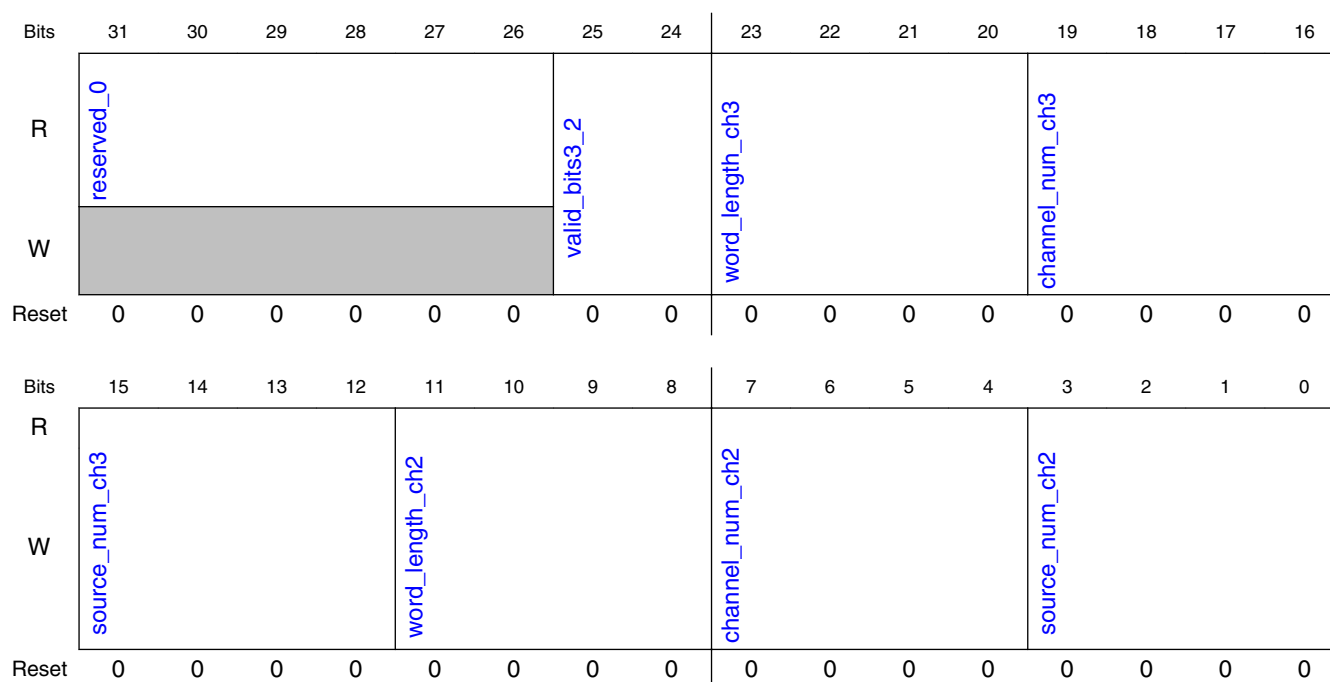
Field	Function
channel_num_ch0	
3-0	Channel 0 Source number.
source_num_ch0	Channel 0 Source number.

13.4.10.1.304 Channels 2,3 configuration (STTS_BIT_CH23)

13.4.10.1.304.1 Offset

Register	Offset
STTS_BIT_CH23	3_0010h

13.4.10.1.304.2 Diagram



13.4.10.1.304.3 Fields

Field	Function
31-26	

Table continues on the next page...

Clocks And Resets

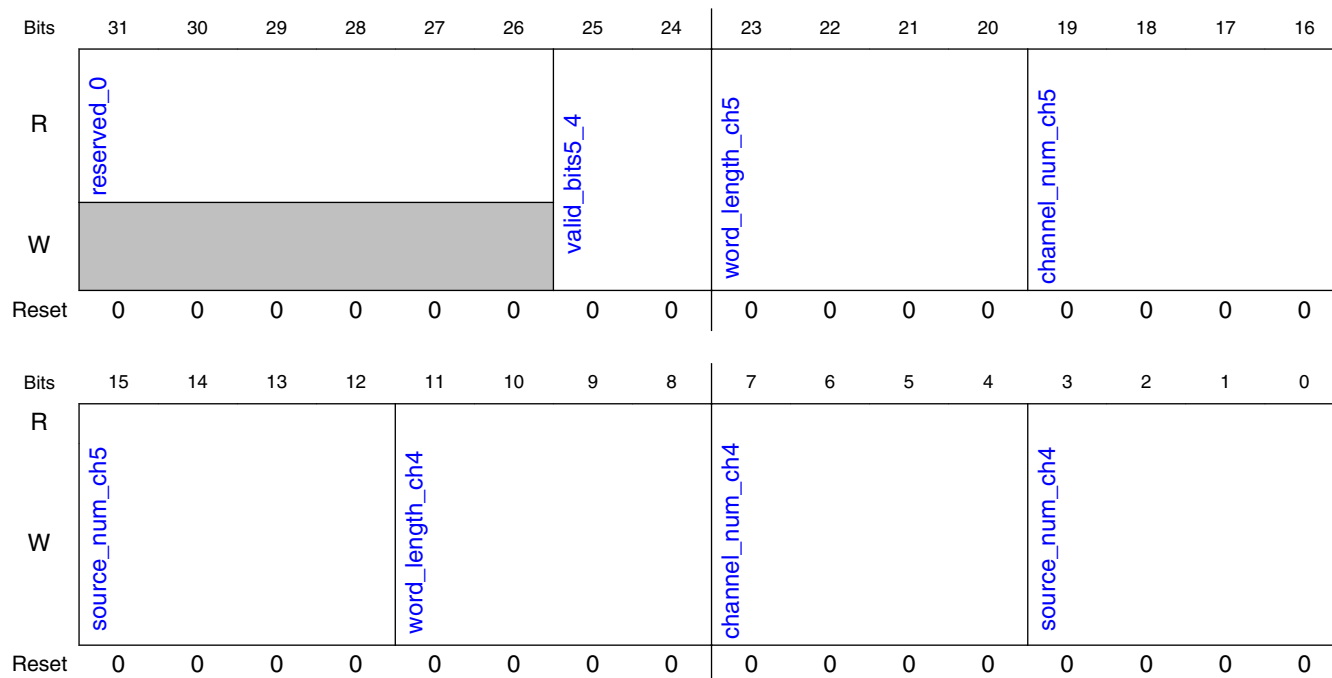
Field	Function
reserved_0	
25-24 valid_bits3_2	Valid Bits for channel 3 and 2 if force is enabled
23-20 word_length_ch 3	Channel 3 word length. Channel 3 word length.
19-16 channel_num_c h3	Channel 3 channel number. Channel 3 channel number.
15-12 source_num_ch 3	Channel 3 Source number. Channel 3 Source number.
11-8 word_length_ch 2	Channel 2 word length. Channel 2 word length.
7-4 channel_num_c h2	Channel 2 channel number. Channel 2 channel number.
3-0 source_num_ch 2	Channel 2 Source number. Channel 2 Source number.

13.4.10.1.305 Channels 4,5 configuration (STTS_BIT_CH45)

13.4.10.1.305.1 Offset

Register	Offset
STTS_BIT_CH45	3_0014h

13.4.10.1.305.2 Diagram



13.4.10.1.305.3 Fields

Field	Function
31-26 reserved_0	
25-24 valid_bits5_4	Valid Bits for channel 5 and 4 if force is enabled
23-20 word_length_ch5	Channel 5 word length. Channel 5 word length.
19-16 channel_num_ch5	Channel 5 channel number. Channel 5 channel number.
15-12 source_num_ch5	Channel 5 Source number. Channel 5 Source number.
11-8 word_length_ch4	Channel 4 word length. Channel 4 word length.
7-4 channel_num_ch4	Channel 4 channel number. Channel 4 channel number.

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Clocks And Resets

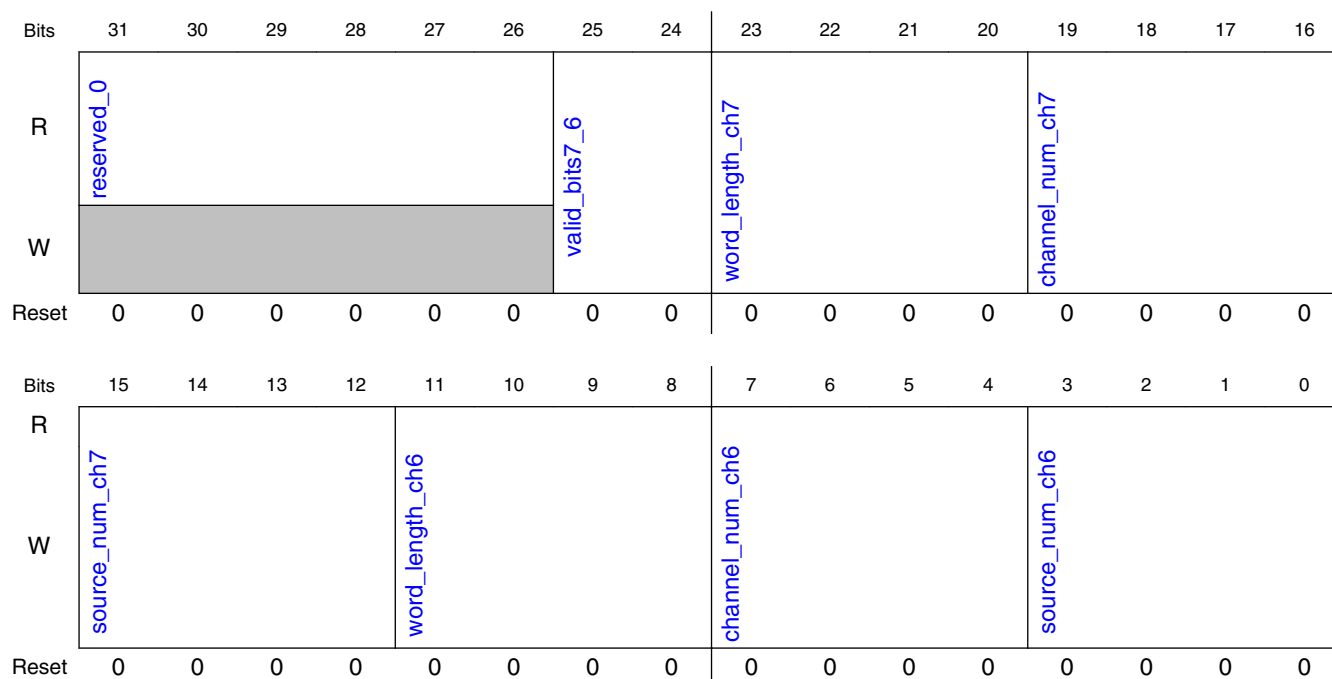
Field	Function
3-0	Channel 4 Source number.
source_num_ch4	Channel 4 Source number.

13.4.10.1.306 Channels 6,7 configuration (STTS_BIT_CH67)

13.4.10.1.306.1 Offset

Register	Offset
STTS_BIT_CH67	3_0018h

13.4.10.1.306.2 Diagram



13.4.10.1.306.3 Fields

Field	Function
31-26	
reserved_0	
25-24	Valid Bits for channel 7 and 6 if force is enabled

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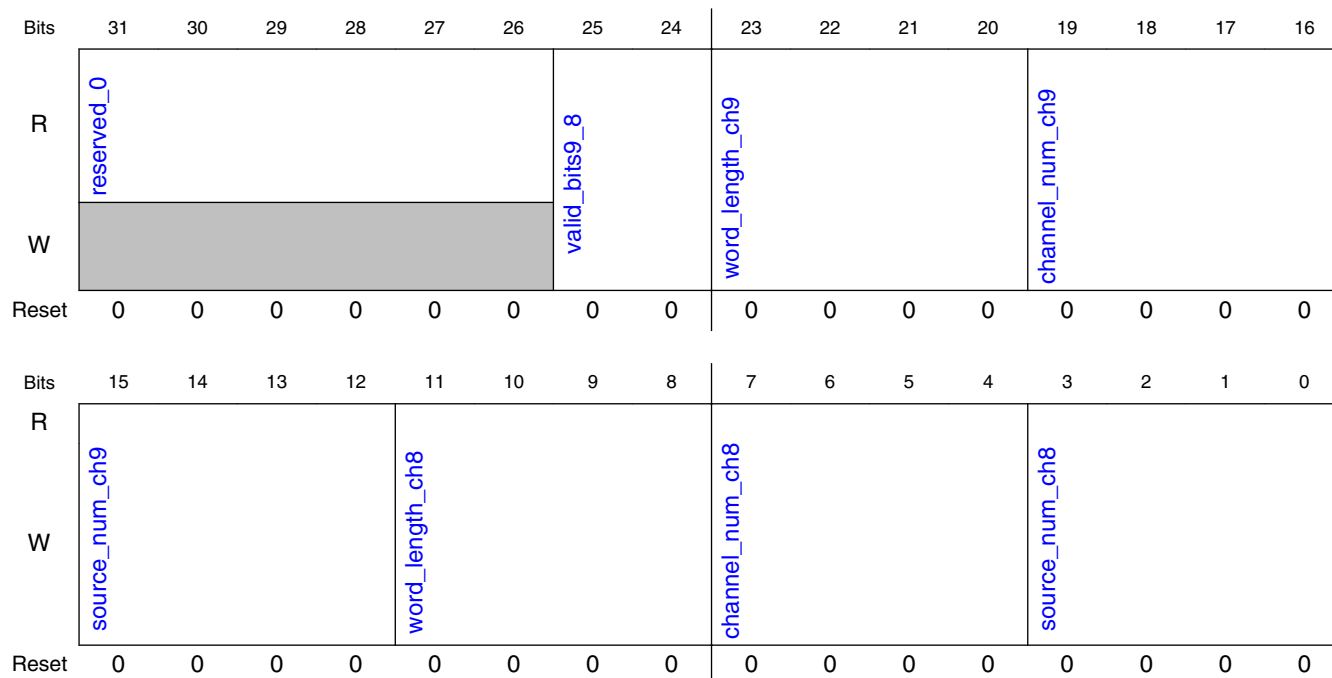
Field	Function
valid_bits7_6	
23-20 word_length_ch 7	Channel 7 word length. Channel 7 word length.
19-16 channel_num_c h7	Channel 7 channel number. Channel 7 channel number.
15-12 source_num_ch 7	Channel 7 Source number. Channel 7 Source number.
11-8 word_length_ch 6	Channel 6 word length. Channel 6 word length.
7-4 channel_num_c h6	Channel 6 channel number. Channel 6 channel number.
3-0 source_num_ch 6	Channel 6 Source number. Channel 6 Source number.

13.4.10.1.307 Channels 8,9 configuration (STTS_BIT_CH89)

13.4.10.1.307.1 Offset

Register	Offset
STTS_BIT_CH89	3_001Ch

13.4.10.1.307.2 Diagram



13.4.10.1.307.3 Fields

Field	Function
31-26 reserved_0	
25-24 valid_bits9_8	Valid Bits for channel 9 and 8 if force is enabled
23-20 word_length_ch9	Channel 9 word length. Channel 9 word length.
19-16 channel_num_ch9	Channel 9 channel number. Channel 9 channel number.
15-12 source_num_ch9	Channel 9 Source number. Channel 9 Source number.
11-8 word_length_ch8	Channel 8 word length. Channel 8 word length.
7-4 channel_num_ch8	Channel 8 channel number. Channel 8 channel number.

Table continues on the next page...

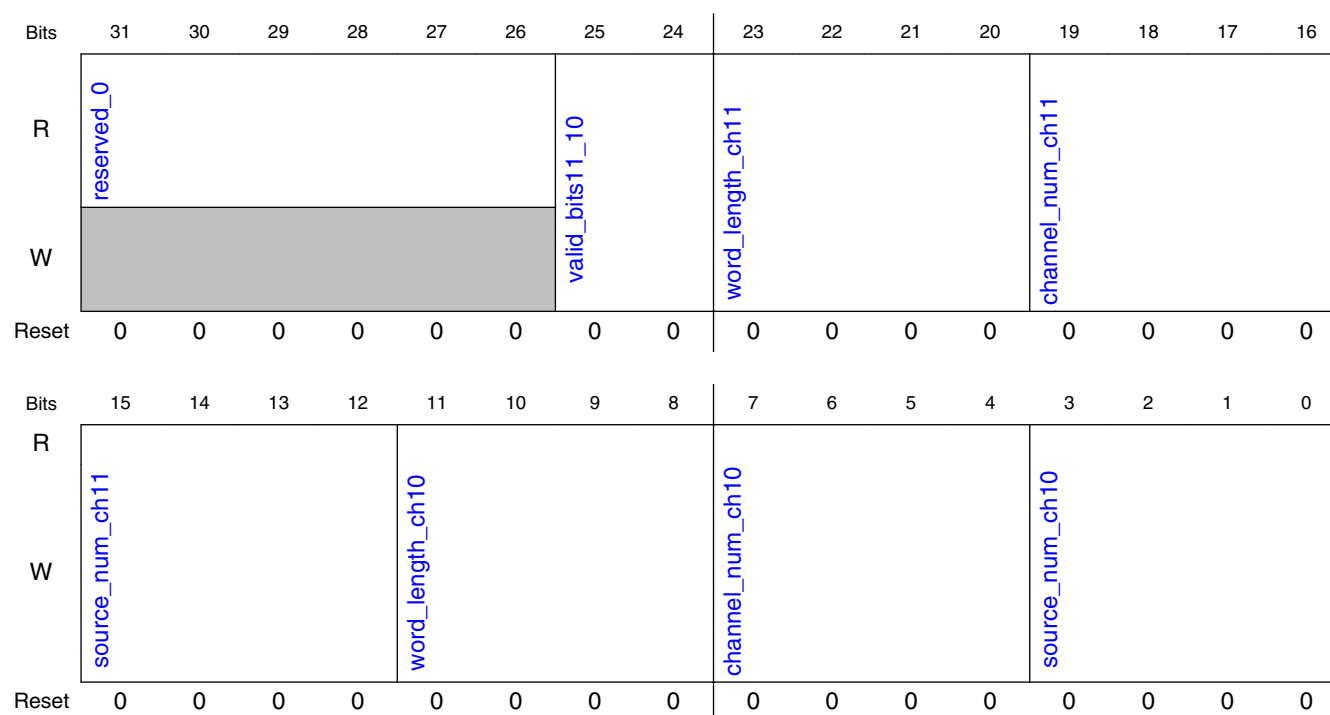
Field	Function
3-0 source_num_ch 8	Channel 8 Source number. Channel 8 Source number.

13.4.10.1.308 Channels 10,11 configuration (STTS_BIT_CH1011)

13.4.10.1.308.1 Offset

Register	Offset
STTS_BIT_CH1011	3_0020h

13.4.10.1.308.2 Diagram



13.4.10.1.308.3 Fields

Field	Function
31-26 reserved_0	

Table continues on the next page...

Clocks And Resets

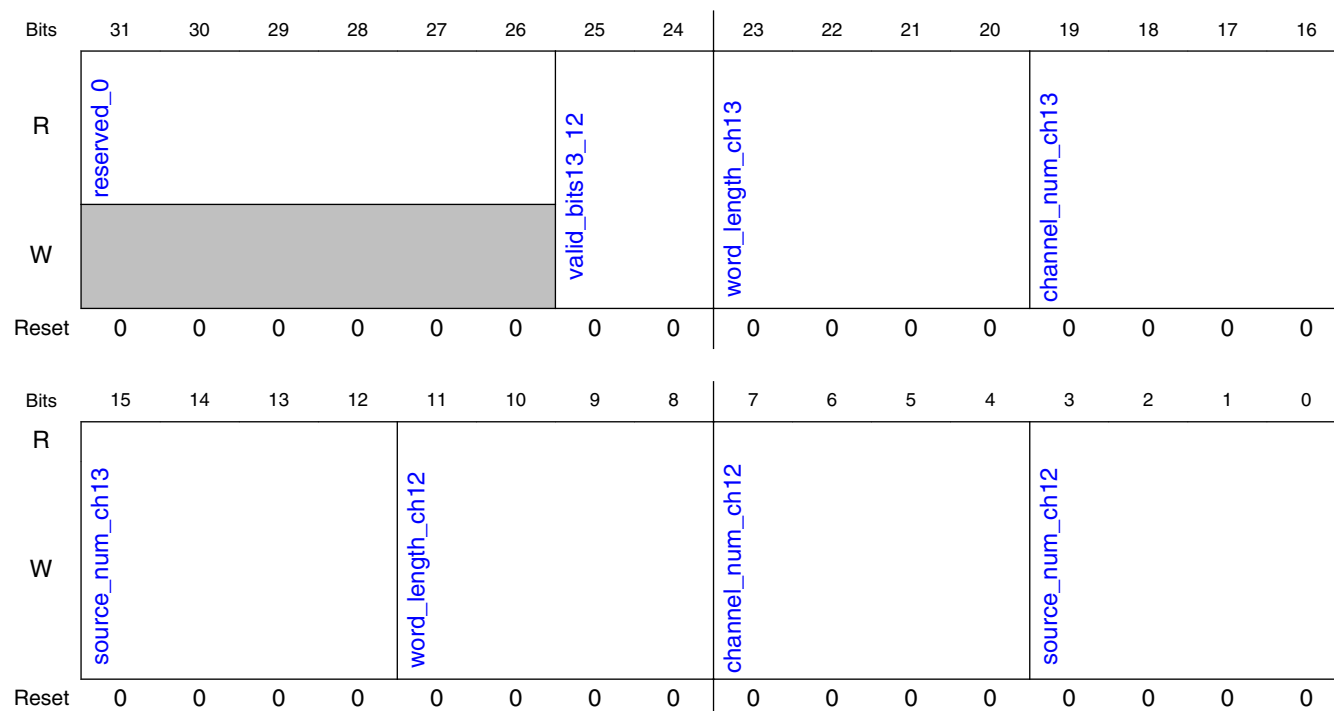
Field	Function
25-24 valid_bits11_10	Valid Bits for channel 11 and 10 if force is enabled
23-20 word_length_ch 11	Channel 11 word length. Channel 11 word length.
19-16 channel_num_c h11	Channel 11 channel number. Channel 11 channel number.
15-12 source_num_ch 11	Channel 11 Source number. Channel 11 Source number.
11-8 word_length_ch 10	Channel 10 word length. Channel 10 word length.
7-4 channel_num_c h10	Channel 10 channel number. Channel 10 channel number.
3-0 source_num_ch 10	Channel 10 Source number. Channel 10 Source number.

13.4.10.1.309 Channels 12,13 configuration (STTS_BIT_CH1213)

13.4.10.1.309.1 Offset

Register	Offset
STTS_BIT_CH1213	3_0024h

13.4.10.1.309.2 Diagram



13.4.10.1.309.3 Fields

Field	Function
31-26 reserved_0	
25-24 valid_bits13_12	Valid Bits for channel 13 and 12 if force is enabled
23-20 word_length_ch13	Channel 13 word length. Channel 13 word length.
19-16 channel_num_ch13	Channel 13 channel number. Channel 13 channel number.
15-12 source_num_ch13	Channel 13 Source number. Channel 13 Source number.
11-8 word_length_ch12	Channel 12 word length. Channel 12 word length.
7-4 channel_num_ch12	Channel 12 channel number. Channel 12 channel number.

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Clocks And Resets

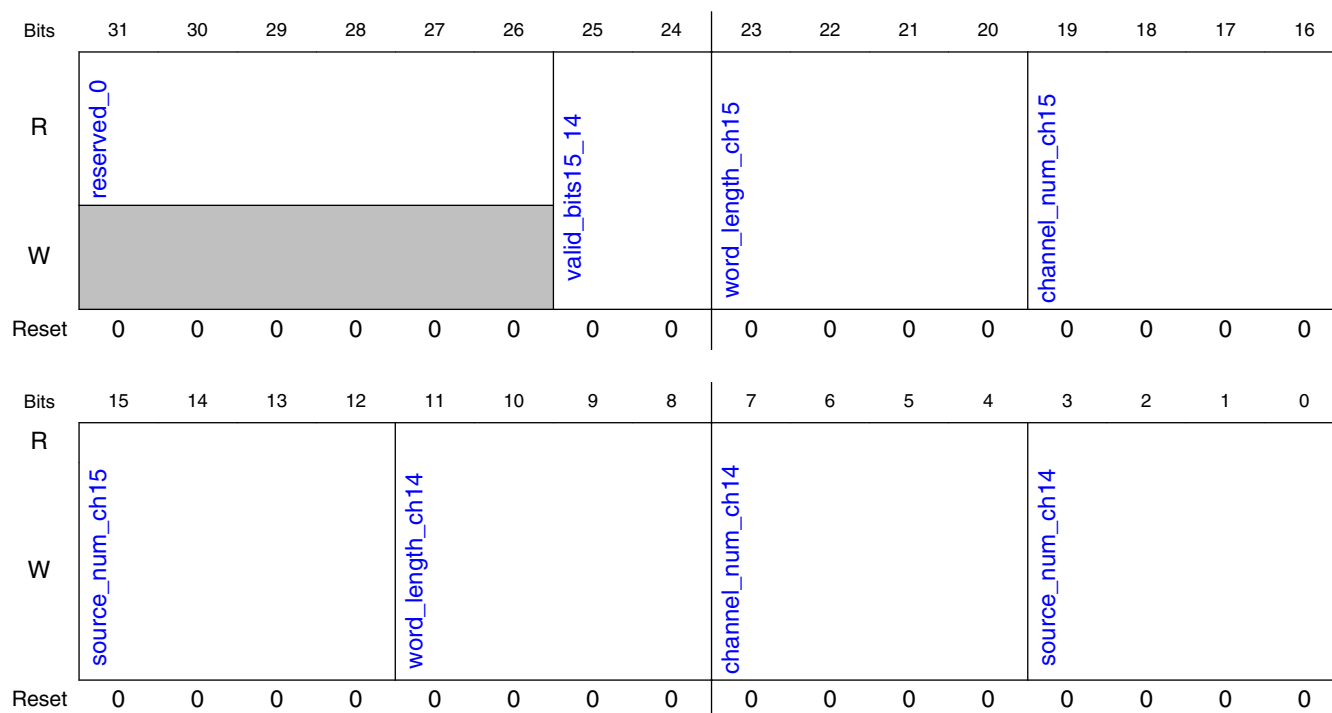
Field	Function
3-0	Channel 12 Source number.
source_num_ch12	Channel 12 Source number.

13.4.10.1.310 Channels 14,15 configuration (STTS_BIT_CH1415)

13.4.10.1.310.1 Offset

Register	Offset
STTS_BIT_CH1415	3_0028h

13.4.10.1.310.2 Diagram



13.4.10.1.310.3 Fields

Field	Function
31-26	
reserved_0	

Table continues on the next page...

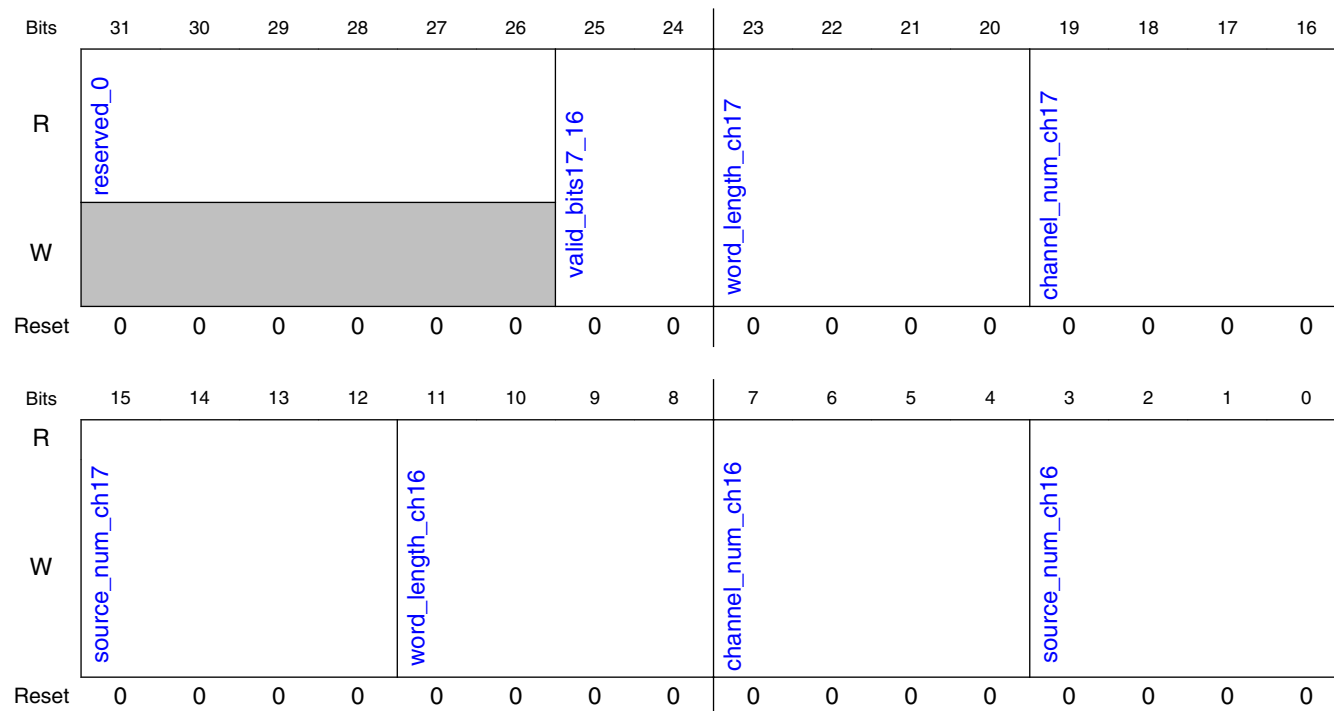
Field	Function
25-24 valid_bits15_14	Valid Bits for channel 15 and 14 if force is enabled
23-20 word_length_ch 15	Channel 15 word length. Channel 15 word length.
19-16 channel_num_c h15	Channel 15 channel number. Channel 15 channel number.
15-12 source_num_ch 15	Channel 15 Source number. Channel 15 Source number.
11-8 word_length_ch 14	Channel 14 word length. Channel 14 word length.
7-4 channel_num_c h14	Channel 14 channel number. Channel 14 channel number.
3-0 source_num_ch 14	Channel 14 Source number. Channel 14 Source number.

13.4.10.1.311 Channels 16,17 configuration (STTS_BIT_CH1617)

13.4.10.1.311.1 Offset

Register	Offset
STTS_BIT_CH1617	3_002Ch

13.4.10.1.311.2 Diagram



13.4.10.1.311.3 Fields

Field	Function
31-26 reserved_0	
25-24 valid_bits17_16	Valid Bits for channel 17 and 16 if force is enabled
23-20 word_length_ch17	Channel 17 word length. Channel 17 word length.
19-16 channel_num_ch17	Channel 17 channel number. Channel 17 channel number.
15-12 source_num_ch17	Channel 17 Source number. Channel 17 Source number.
11-8 word_length_ch16	Channel 16 word length. Channel 16 word length.
7-4 channel_num_ch16	Channel 16 channel number. Channel 16 channel number.

Table continues on the next page...

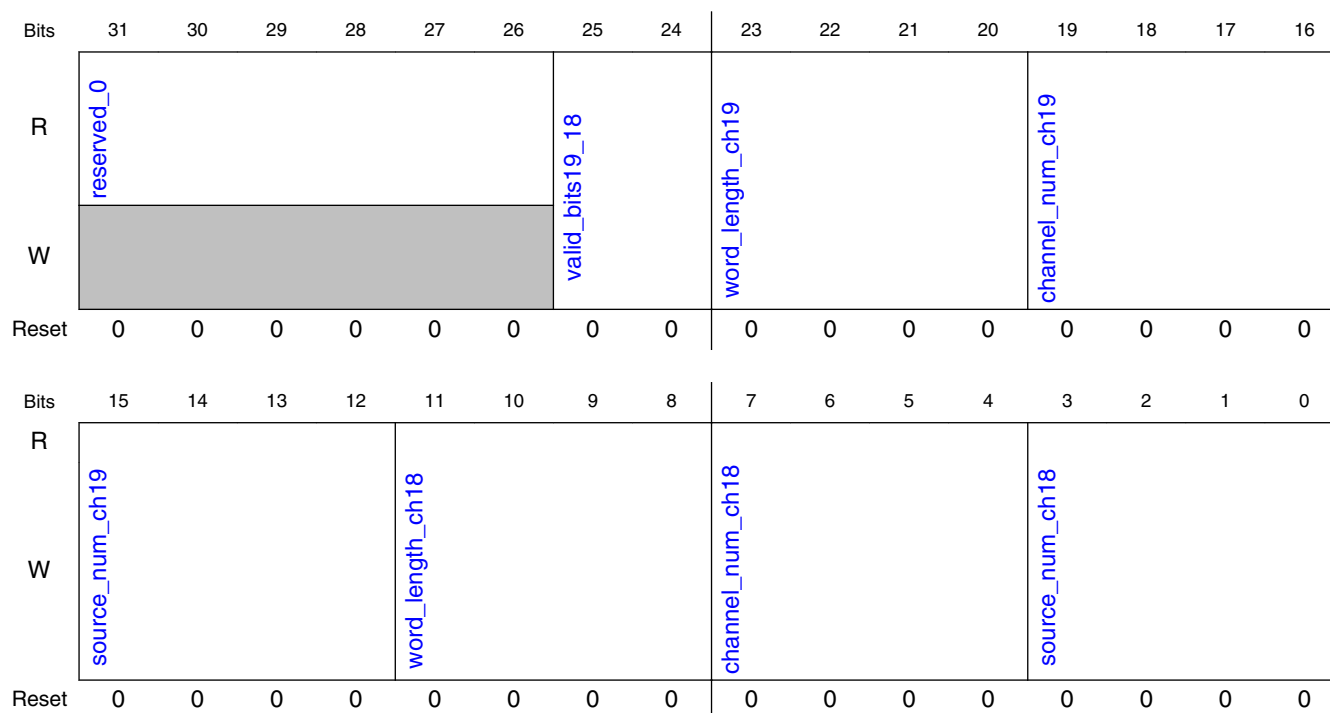
Field	Function
3-0 source_num_ch 16	Channel 16 Source number. Channel 16 Source number.

13.4.10.1.312 Channels 18,19 configuration (STTS_BIT_CH1819)

13.4.10.1.312.1 Offset

Register	Offset
STTS_BIT_CH1819	3_0030h

13.4.10.1.312.2 Diagram



13.4.10.1.312.3 Fields

Field	Function
31-26 reserved_0	

Table continues on the next page...

Clocks And Resets

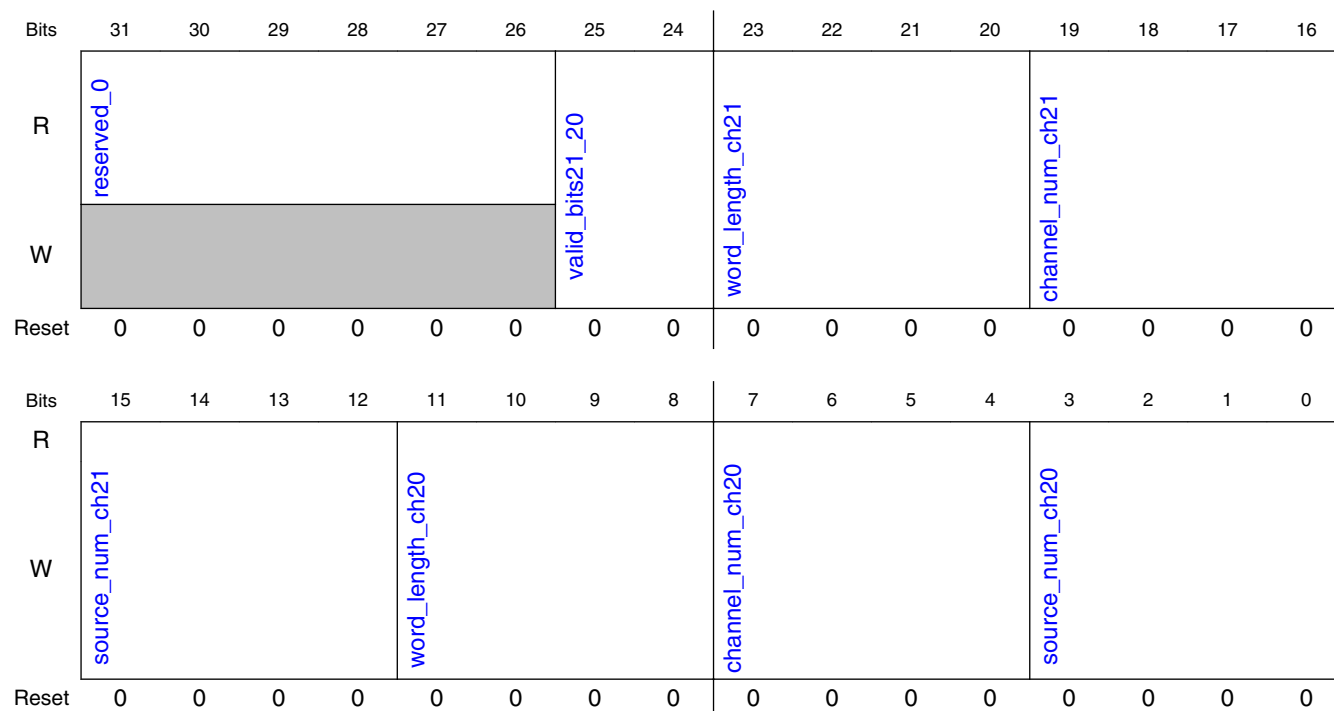
Field	Function
25-24 valid_bits19_18	Valid Bits for channel 19 and 18 if force is enabled
23-20 word_length_ch 19	Channel 19 word length. Channel 19 word length.
19-16 channel_num_c h19	Channel 19 channel number. Channel 19 channel number.
15-12 source_num_ch 19	Channel 19 Source number. Channel 19 Source number.
11-8 word_length_ch 18	Channel 18 word length. Channel 18 word length.
7-4 channel_num_c h18	Channel 18 channel number. Channel 18 channel number.
3-0 source_num_ch 18	Channel 18 Source number. Channel 18 Source number.

13.4.10.1.313 Channels 20,21 configuration (STTS_BIT_CH2021)

13.4.10.1.313.1 Offset

Register	Offset
STTS_BIT_CH2021	3_0034h

13.4.10.1.313.2 Diagram



13.4.10.1.313.3 Fields

Field	Function
31-26 reserved_0	
25-24 valid_bits21_20	Valid Bits for channel 21 and 20 if force is enabled
23-20 word_length_ch21	Channel 21 word length. Channel 21 word length.
19-16 channel_num_ch21	Channel 21 channel number. Channel 21 channel number.
15-12 source_num_ch21	Channel 21 Source number. Channel 21 Source number.
11-8 word_length_ch20	Channel 20 word length. Channel 20 word length.
7-4 channel_num_ch20	Channel 20 channel number. Channel 20 channel number.

Table continues on the next page...

Clocks And Resets

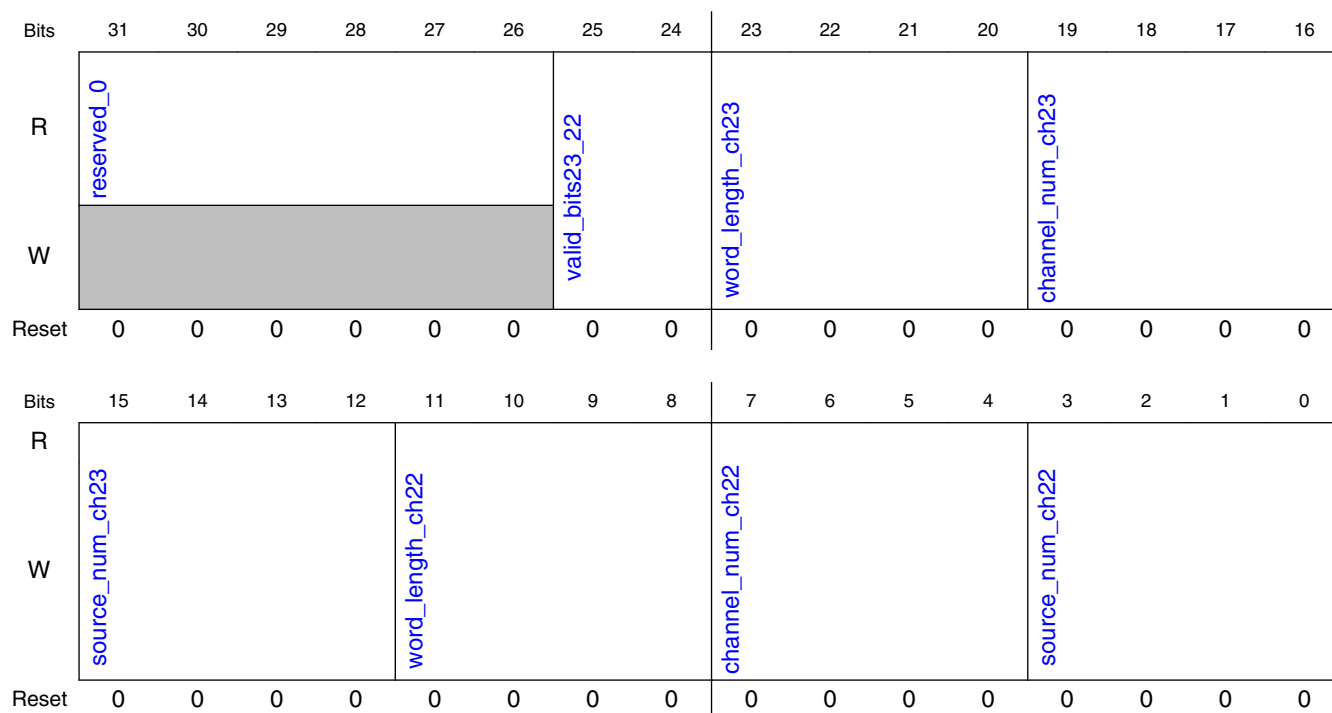
Field	Function
3-0	Channel 20 Source number.
source_num_ch20	Channel 20 Source number.

13.4.10.1.314 Channels 22,23 configuration (STTS_BIT_CH2223)

13.4.10.1.314.1 Offset

Register	Offset
STTS_BIT_CH2223	3_0038h

13.4.10.1.314.2 Diagram



13.4.10.1.314.3 Fields

Field	Function
31-26	
reserved_0	

Table continues on the next page...

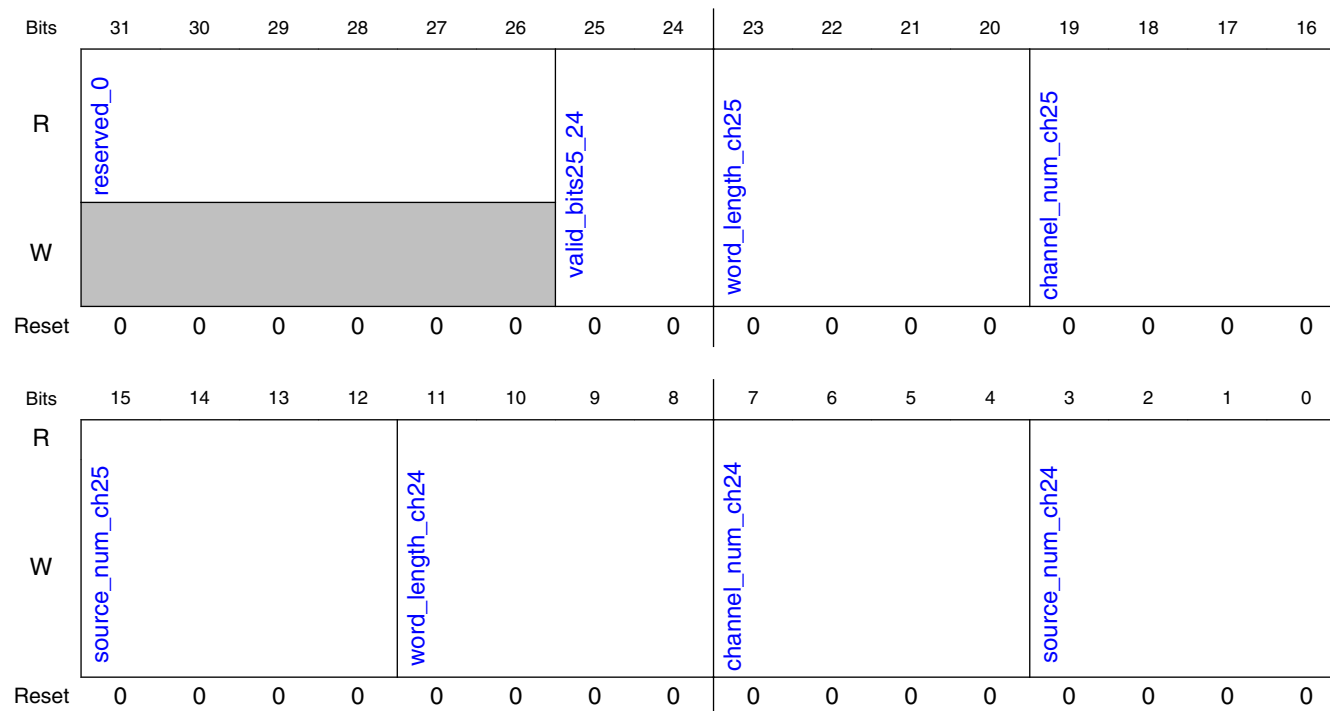
Field	Function
25-24 valid_bits23_22	Valid Bits for channel 23 and 22 if force is enabled
23-20 word_length_ch 23	Channel 23 word length. Channel 23 word length.
19-16 channel_num_c h23	Channel 23 channel number. Channel 23 channel number.
15-12 source_num_ch 23	Channel 23 Source number. Channel 23 Source number.
11-8 word_length_ch 22	Channel 22 word length. Channel 22 word length.
7-4 channel_num_c h22	Channel 22 channel number. Channel 22 channel number.
3-0 source_num_ch 22	Channel 22 Source number. Channel 22 Source number.

13.4.10.1.315 Channels 24,25 configuration (STTS_BIT_CH2425)

13.4.10.1.315.1 Offset

Register	Offset
STTS_BIT_CH2425	3_003Ch

13.4.10.1.315.2 Diagram



13.4.10.1.315.3 Fields

Field	Function
31-26 reserved_0	
25-24 valid_bits25_24	Valid Bits for channel 25 and 24 if force is enabled
23-20 word_length_ch25	Channel 25 word length. Channel 25 word length.
19-16 channel_num_ch25	Channel 25 channel number. Channel 25 channel number.
15-12 source_num_ch25	Channel 25 Source number. Channel 25 Source number.
11-8 word_length_ch24	Channel 24 word length. Channel 24 word length.
7-4 channel_num_ch24	Channel 24 channel number. Channel 24 channel number.

Table continues on the next page...

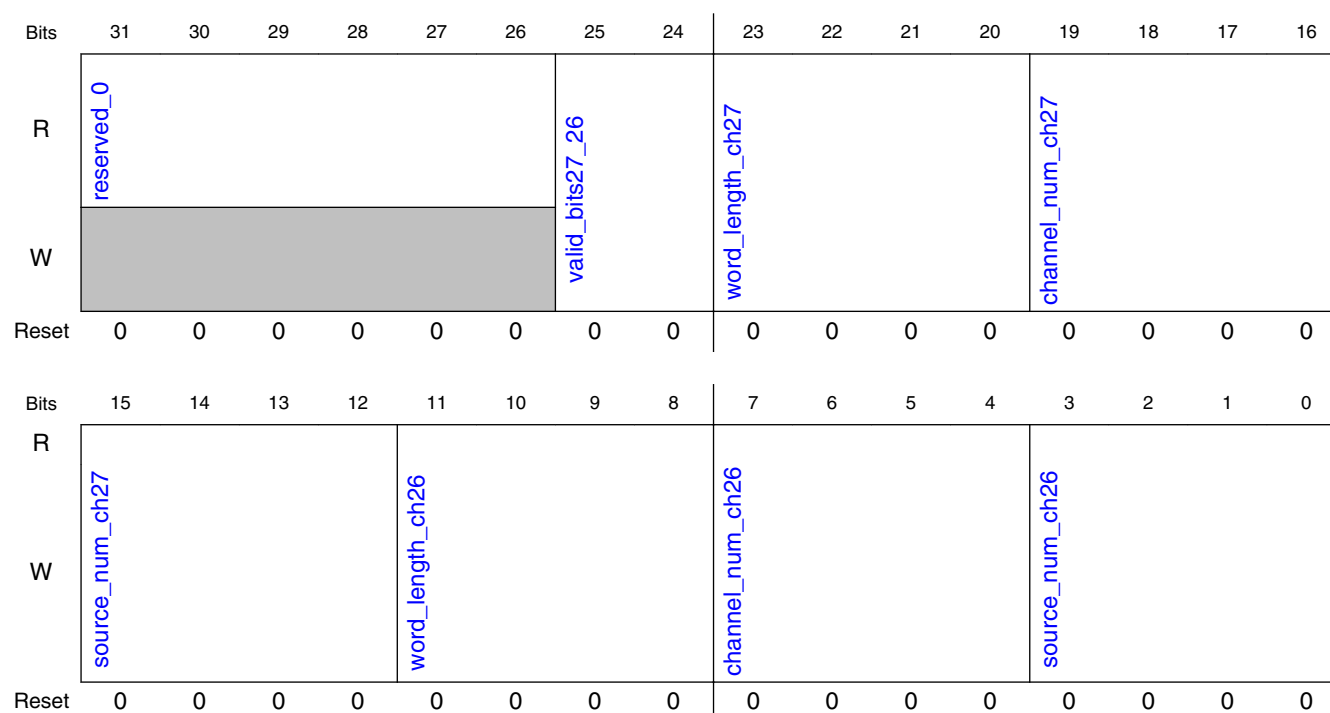
Field	Function
3-0 source_num_ch 24	Channel 24 Source number. Channel 24 Source number.

13.4.10.1.316 Channels 26,27 configuration (STTS_BIT_CH2627)

13.4.10.1.316.1 Offset

Register	Offset
STTS_BIT_CH2627	3_0040h

13.4.10.1.316.2 Diagram



13.4.10.1.316.3 Fields

Field	Function
31-26 reserved_0	

Table continues on the next page...

Clocks And Resets

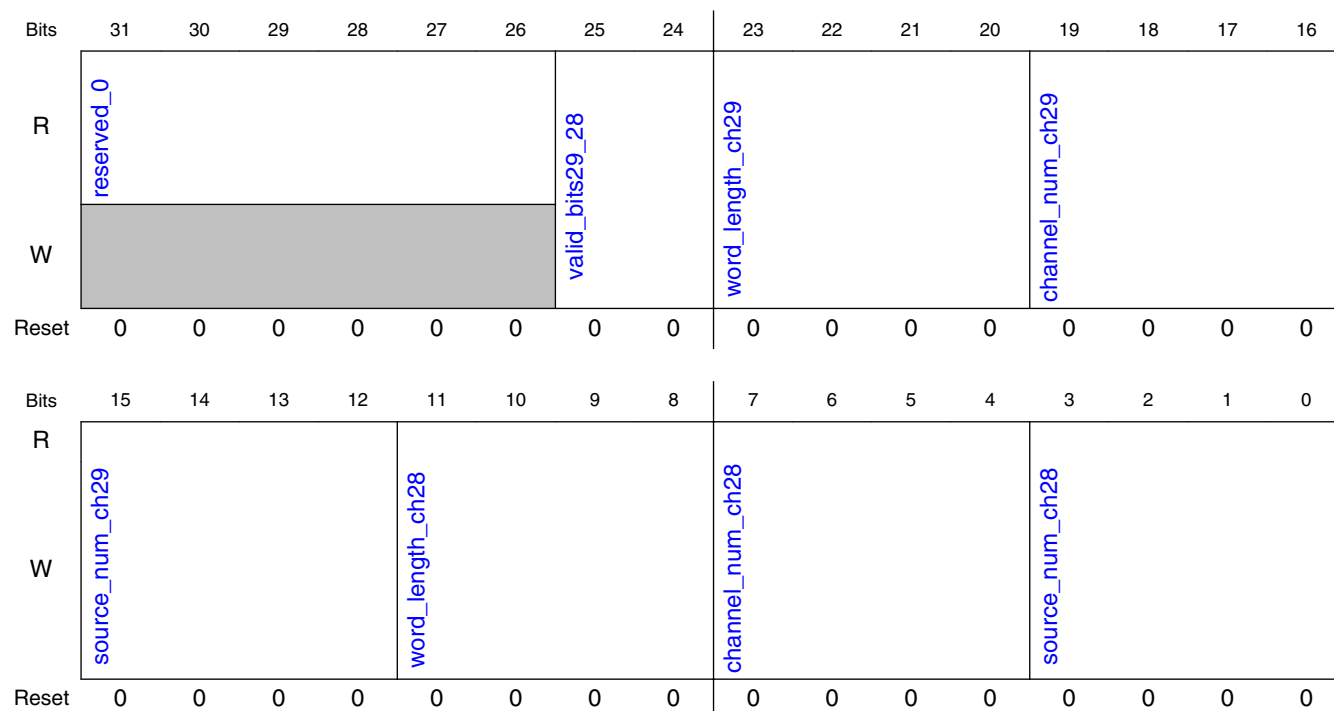
Field	Function
25-24 valid_bits27_26	Valid Bits for channel 27 and 26 if force is enabled
23-20 word_length_ch 27	Channel 27 word length. Channel 27 word length.
19-16 channel_num_ch h27	Channel 27 channel number. Channel 27 channel number.
15-12 source_num_ch 27	Channel 27 Source number. Channel 27 Source number.
11-8 word_length_ch 26	Channel 26 word length. Channel 26 word length.
7-4 channel_num_ch h26	Channel 26 channel number. Channel 26 channel number.
3-0 source_num_ch 26	Channel 26 Source number. Channel 26 Source number.

13.4.10.1.317 Channels 28,29 configuration (STTS_BIT_CH2829)

13.4.10.1.317.1 Offset

Register	Offset
STTS_BIT_CH2829	3_0044h

13.4.10.1.317.2 Diagram



13.4.10.1.317.3 Fields

Field	Function
31-26 reserved_0	
25-24 valid_bits29_28	Valid Bits for channel 29 and 28 if force is enabled
23-20 word_length_ch29	Channel 29 word length. Channel 29 word length.
19-16 channel_num_ch29	Channel 29 channel number. Channel 29 channel number.
15-12 source_num_ch29	Channel 29 Source number. Channel 29 Source number.
11-8 word_length_ch28	Channel 28 word length. Channel 28 word length.
7-4 channel_num_ch28	Channel 28 channel number. Channel 28 channel number.

Table continues on the next page...

Clocks And Resets

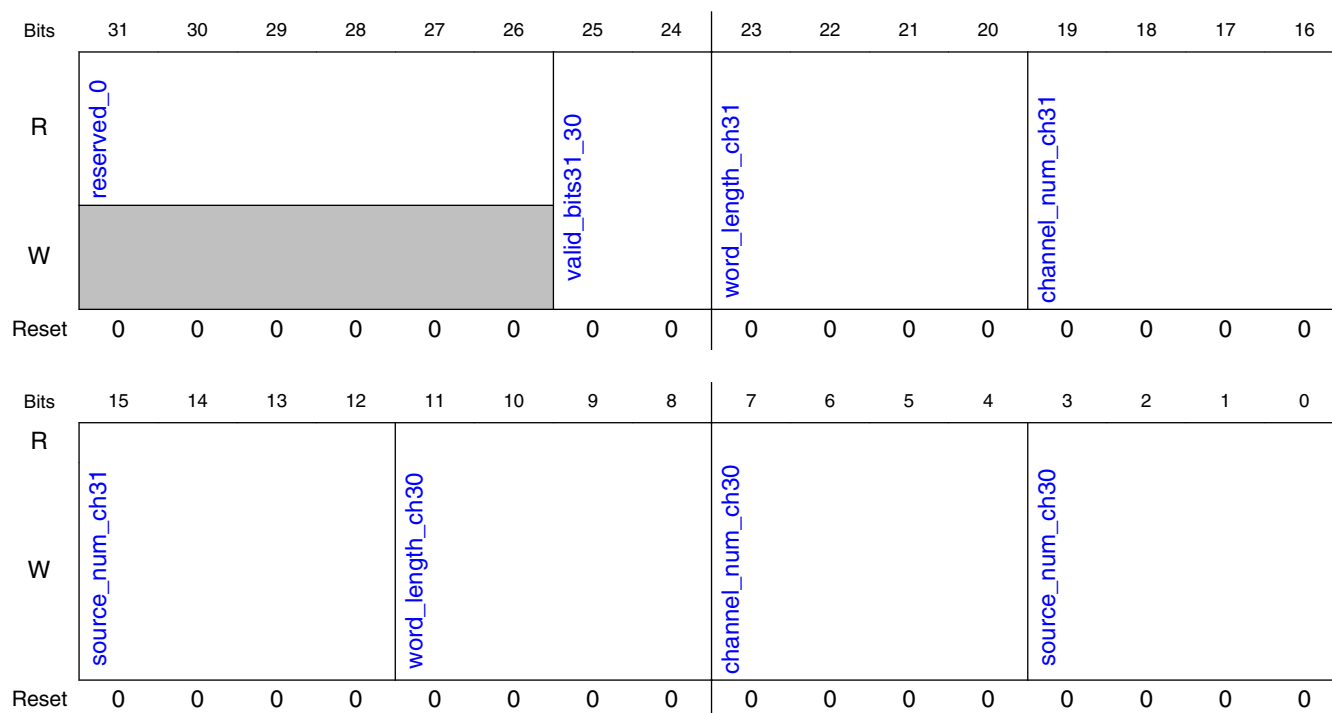
Field	Function
3-0	Channel 28 Source number.
source_num_ch 28	Channel 28 Source number.

13.4.10.1.318 Channels 30,31 configuration (STTS_BIT_CH3031)

13.4.10.1.318.1 Offset

Register	Offset
STTS_BIT_CH3031	3_0048h

13.4.10.1.318.2 Diagram



13.4.10.1.318.3 Fields

Field	Function
31-26	
reserved_0	

Table continues on the next page...

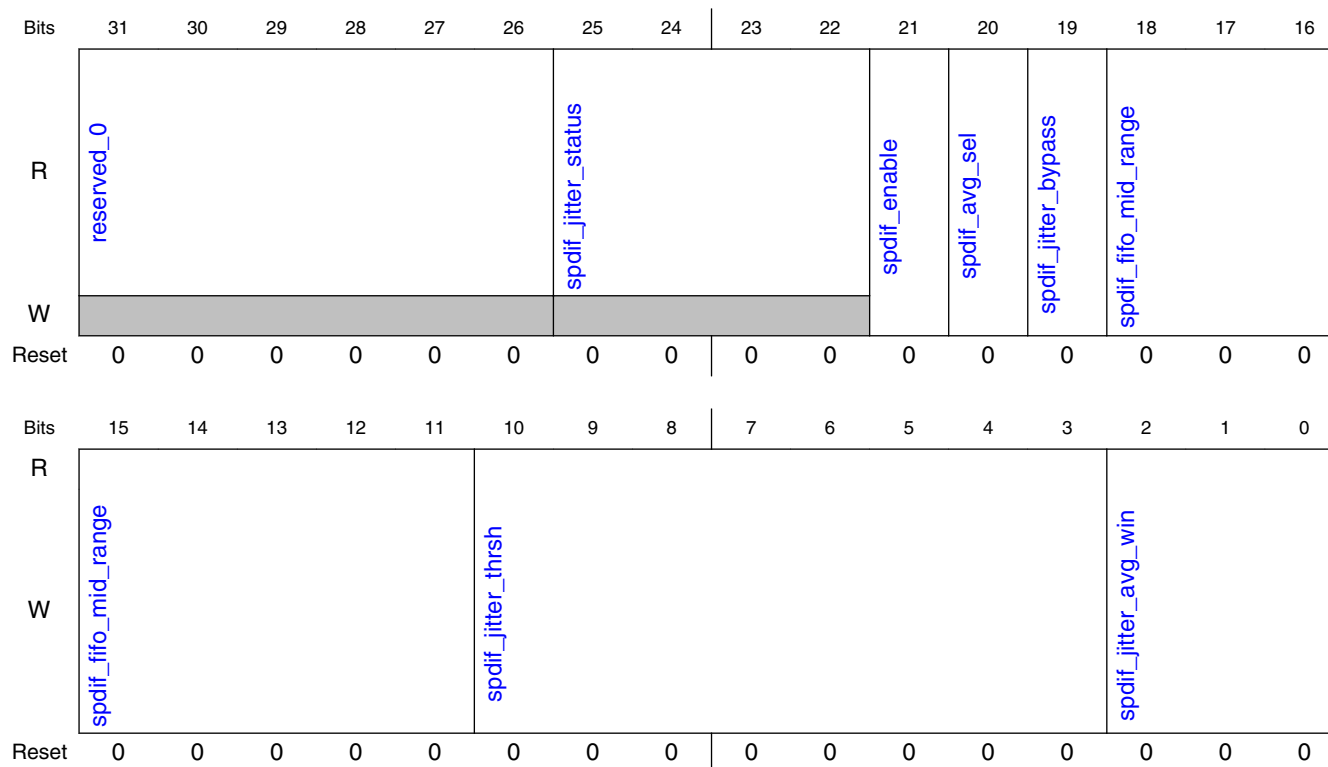
Field	Function
25-24 valid_bits31_30	Valid Bits for channel 31 and 30 if force is enabled
23-20 word_length_ch 31	Channel 31 word length. Channel 31 word length.
19-16 channel_num_c h31	Channel 31 channel number. Channel 31 channel number.
15-12 source_num_ch 31	Channel 31 Source number. Channel 31 Source number.
11-8 word_length_ch 30	Channel 30 word length. Channel 30 word length.
7-4 channel_num_c h30	Channel 30 channel number. Channel 30 channel number.
3-0 source_num_ch 30	Channel 30 Source number. Channel 30 Source number.

13.4.10.1.319 SPDIF control (SPDIF_CTRL_ADDR)

13.4.10.1.319.1 Offset

Register	Offset
SPDIF_CTRL_ADDR	3_004Ch

13.4.10.1.319.2 Diagram



13.4.10.1.319.3 Fields

Field	Function
31-26 reserved_0	
25-22 spdif_jitter_status	SPDIF Jitter Status
21 spdif_enable	SPDIF Enable
20 spdif_avg_sel	SPDIF average Select
19 spdif_jitter_bypass	SPDIF Jitter Bypass
18-11 spdif_fifo_mid_range	SPDIF fifo mid range
10-3 spdif_jitter_thrsh	SPDIF Jitter threshold

Table continues on the next page...

Field	Function
2-0 spdif_jitter_avg_win	Spdif Jitter AVG Window

13.4.10.1.320 SPDIF channel 1 status [31:00] (SPDIF_CH1_CS_3100_ADDR)

13.4.10.1.320.1 Offset

Register	Offset
SPDIF_CH1_CS_3100_ADDR	3_0050h

13.4.10.1.320.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	spdif_ch1_st_stts_bits3100															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	spdif_ch1_st_stts_bits3100															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

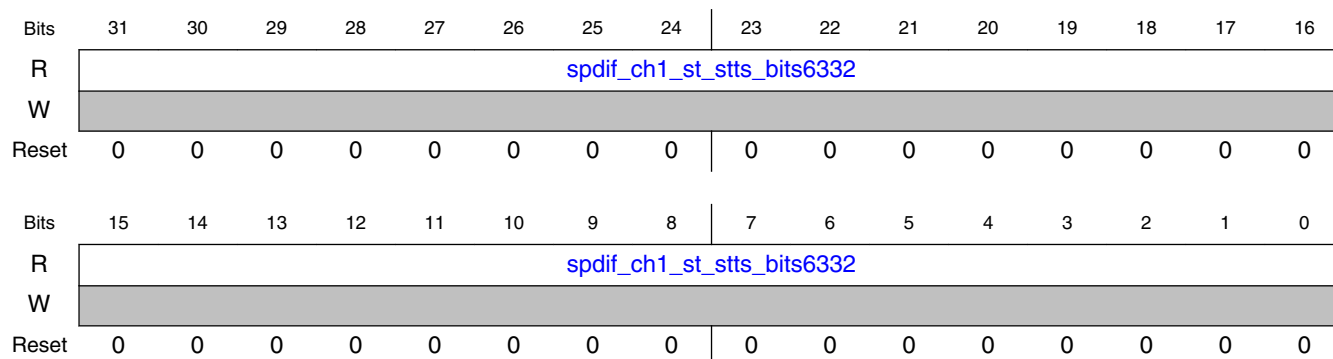
13.4.10.1.320.3 Fields

Field	Function
31-0 spdif_ch1_st_stts_bits3100	SPDIF Channel 1 Status bits[31:0]

13.4.10.1.321 SPDIF channel 1 status [63:32] (SPDIF_CH1_CS_6332_ADDR)

13.4.10.1.321.1 Offset

Register	Offset
SPDIF_CH1_CS_6332_ADDR	3_0054h

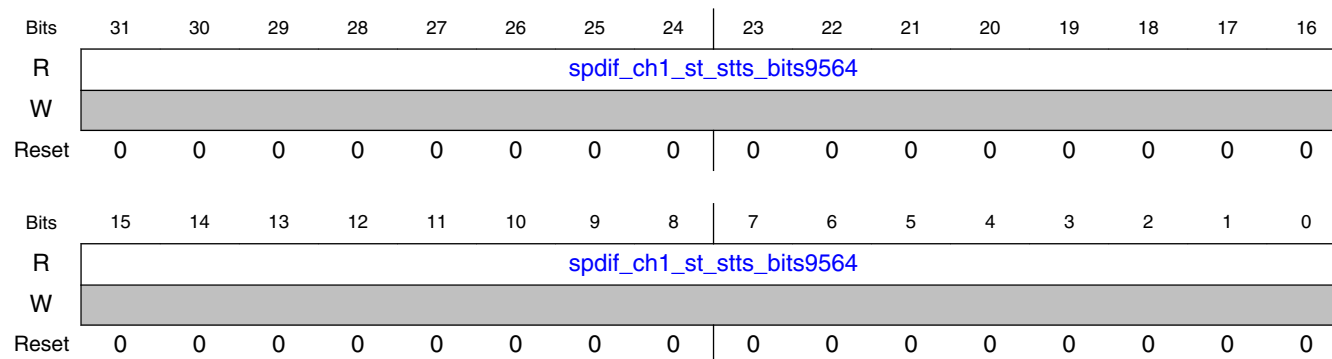
13.4.10.1.321.2 Diagram**13.4.10.1.321.3 Fields**

Field	Function
31-0 spdif_ch1_st_stts_bits6332	SPDIF Channel 1 Status bits[63:32]

13.4.10.1.322 SPDIF channel 1 status [95:64] (SPDIF_CH1_CS_9564_ADDR)**13.4.10.1.322.1 Offset**

Register	Offset
SPDIF_CH1_CS_9564_ADDR	3_0058h

13.4.10.1.322.2 Diagram



13.4.10.1.322.3 Fields

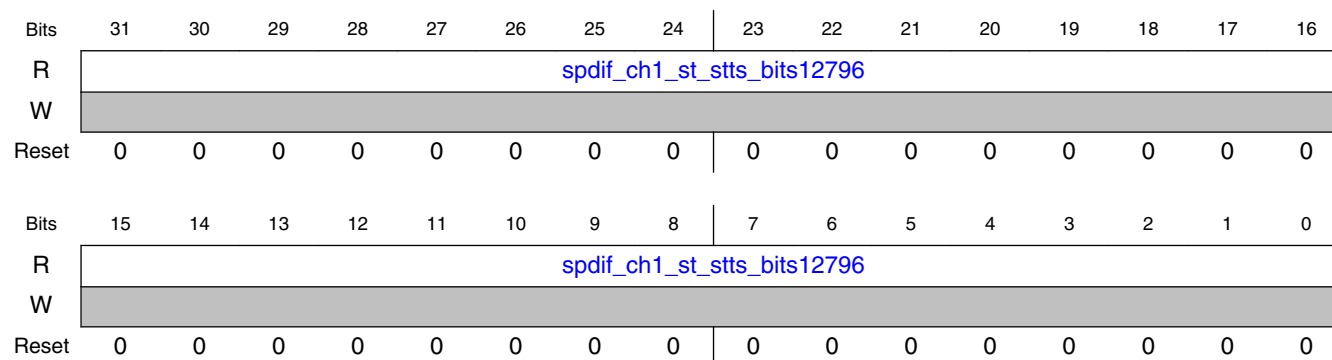
Field	Function
31-0 spdif_ch1_st_stts_bits9564	SPDIF Channel 1 Status bits[95:64]

13.4.10.1.323 SPDIF channel 1 status [127:96] (SPDIF_CH1_CS_12796_A DDR)

13.4.10.1.323.1 Offset

Register	Offset
SPDIF_CH1_CS_12796_ADDR	3_005Ch

13.4.10.1.323.2 Diagram



13.4.10.1.323.3 Fields

Field	Function
31-0 spdif_ch1_st_stts_bits12796	SPDIF Channel 1 Status bits[127:96]

13.4.10.1.324 SPDIF channel 1 status [159:128] (SPDIF_CH1_CS_159128_ADDR)

13.4.10.1.324.1 Offset

Register	Offset
SPDIF_CH1_CS_159128_ADDR	3_0060h

13.4.10.1.324.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	spdif_ch1_st_stts_bits159128															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	spdif_ch1_st_stts_bits159128															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.324.3 Fields

Field	Function
31-0 spdif_ch1_st_stts_bits159128	SPDIF Channel 1 Status bits[159:128]

13.4.10.1.325 SPDIF channel 1 status [191:160] (SPDIF_CH1_CS_191160_ADDR)

13.4.10.1.325.1 Offset

Register	Offset
SPDIF_CH1_CS_191160_ADDR	3_0064h

13.4.10.1.325.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	spdif_ch1_st_stts_bits191160															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	spdif_ch1_st_stts_bits191160															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.325.3 Fields

Field	Function
31-0 spdif_ch1_st_stts_bits191160	SPDIF Channel 1 Status bits[191:160]

13.4.10.1.326 SPDIF channel 2 status [31:00] (SPDIF_CH2_CS_3100_ADDR)

13.4.10.1.326.1 Offset

Register	Offset
SPDIF_CH2_CS_3100_ADDR	3_0068h

13.4.10.1.326.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	spdif_ch2_st_stts_bits3100															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	spdif_ch2_st_stts_bits3100															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.326.3 Fields

Field	Function
31-0 spdif_ch2_st_stts_bits3100	SPDIF Channel 2 Status bits[31:0]

13.4.10.1.327 SPDIF channel 2 status [63:32] (SPDIF_CH2_CS_6332_ADDR)

13.4.10.1.327.1 Offset

Register	Offset
SPDIF_CH2_CS_6332_ADDR	3_006Ch

13.4.10.1.327.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	spdif_ch2_st_stts_bits6332															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	spdif_ch2_st_stts_bits6332															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.327.3 Fields

Field	Function
31-0 spdif_ch2_st_stts_bits6332	SPDIF Channel 2 Status bits[63:32]

13.4.10.1.328 SPDIF channel 2 status [95:64] (SPDIF_CH2_CS_9564_ADDR)

13.4.10.1.328.1 Offset

Register	Offset
SPDIF_CH2_CS_9564_ADDR	3_0070h

13.4.10.1.328.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	spdif_ch2_st_stts_bits9564															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	spdif_ch2_st_stts_bits9564															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.328.3 Fields

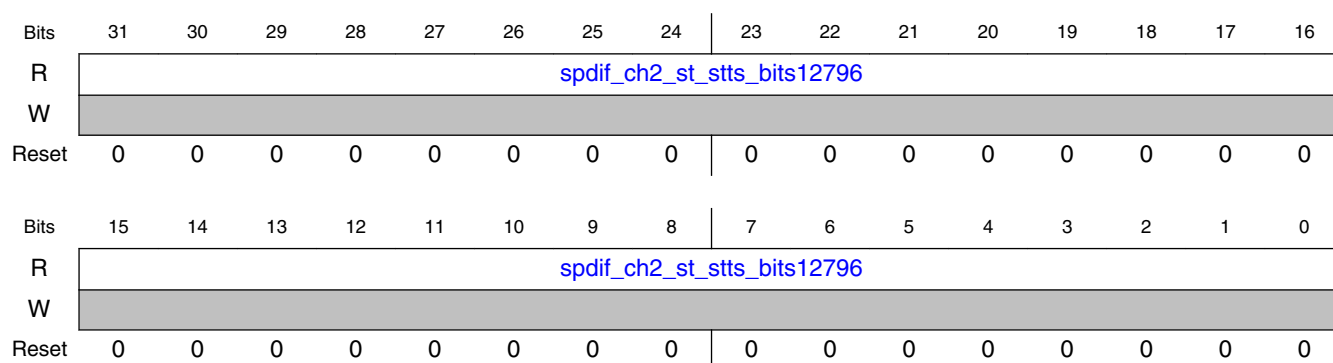
Field	Function
31-0 spdif_ch2_st_stts_bits9564	SPDIF Channel 2 Status bits[95:64]

13.4.10.1.329 SPDIF channel 2 status [127:96] (SPDIF_CH2_CS_12796_A DDR)

13.4.10.1.329.1 Offset

Register	Offset
SPDIF_CH2_CS_12796_ADDR	3_0074h

13.4.10.1.329.2 Diagram



13.4.10.1.329.3 Fields

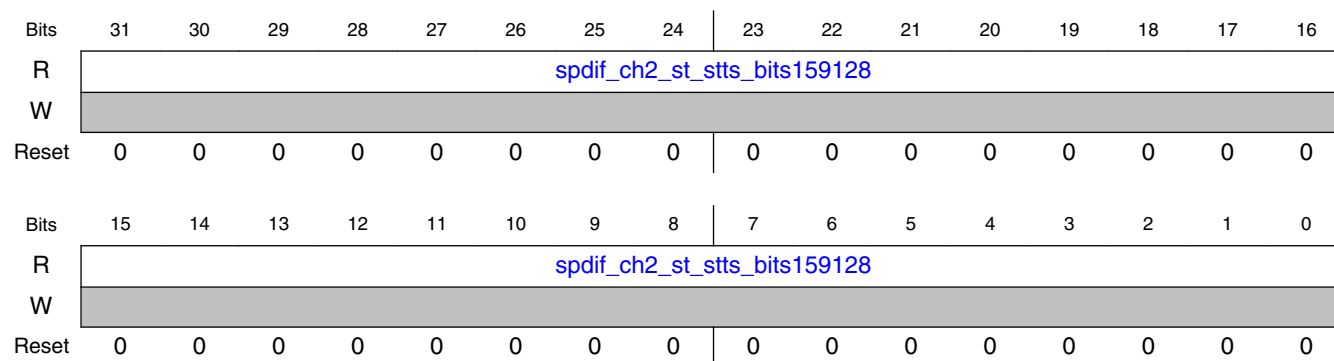
Field	Function
31-0 spdif_ch2_st_stts_bits12796	SPDIF Channel 2 Status bits[127:96]

13.4.10.1.330 SPDIF channel 2 status [159:128] (SPDIF_CH2_CS_159128_ADDR)

13.4.10.1.330.1 Offset

Register	Offset
SPDIF_CH2_CS_159128_ADDR	3_0078h

13.4.10.1.330.2 Diagram



13.4.10.1.330.3 Fields

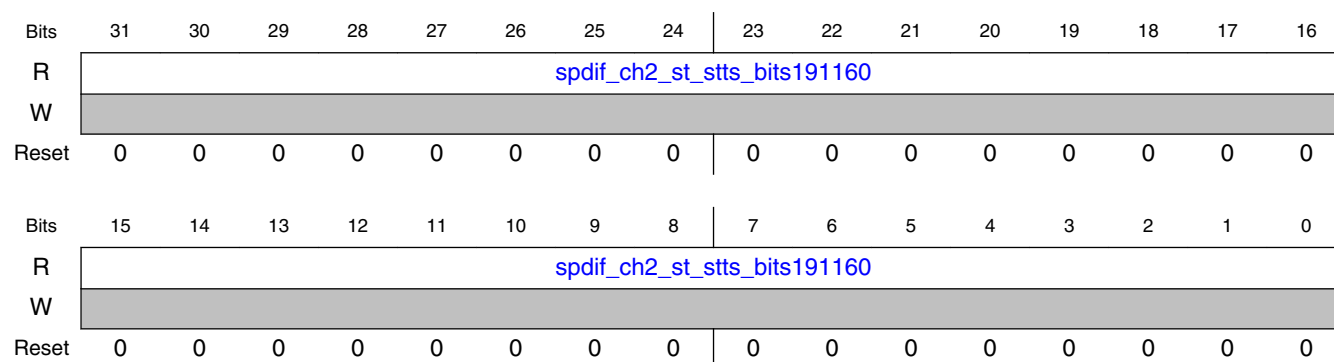
Field	Function
31-0 spdif_ch2_st_stts_bits159128	SPDIF Channel 2 Status bits[159:128]

13.4.10.1.331 SPDIF channel 2 status [191:160] (SPDIF_CH2_CS_191160_ADDR)

13.4.10.1.331.1 Offset

Register	Offset
SPDIF_CH2_CS_191160_ADDR	3_007Ch

13.4.10.1.331.2 Diagram



13.4.10.1.331.3 Fields

Field	Function
31-0 spdif_ch2_st_stt s_bits191160	SPDIF Channel 2 Status bits[191:160]

13.4.10.1.332 Sample 2 Packets Control Register (SMPL2PKT_CNTL)

13.4.10.1.332.1 Offset

Register	Offset
SMPL2PKT_CNTL	3_0080h

13.4.10.1.332.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	reserved_0													smpl2pkt_en		sw_rst
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.1.332.3 Fields

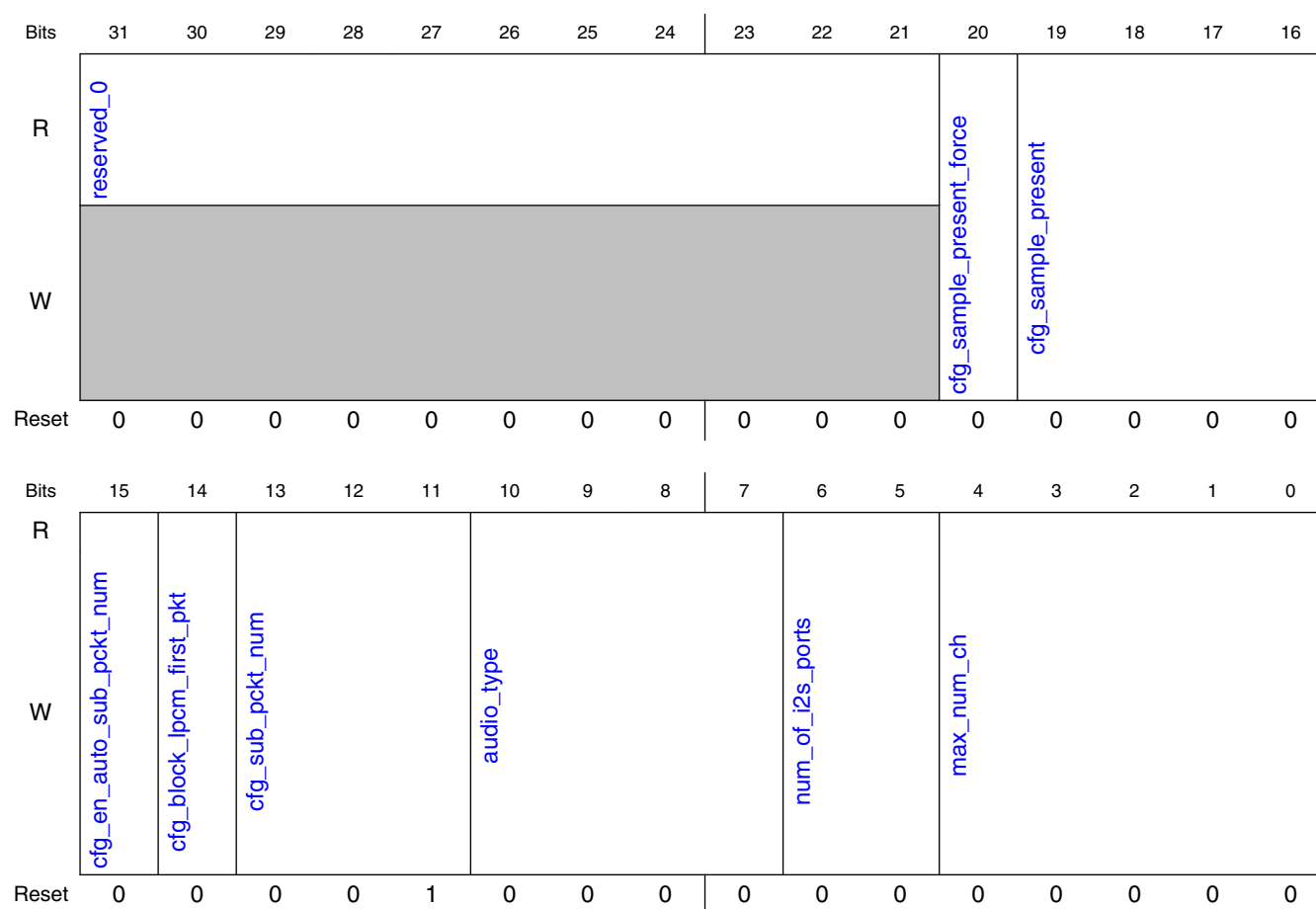
Field	Function
31-2 reserved_0	
1 smpl2pkt_en	When high Sample to Packets Block starts. When high Sample to Packets Block starts.
0 sw_rst	Software reset. Software reset. Active high.

13.4.10.1.333 Sample 2 Packets Config Register (SMPL2PKT_CNFG)

13.4.10.1.333.1 Offset

Register	Offset
SMPL2PKT_CNFG	3_0084h

13.4.10.1.333.2 Diagram



13.4.10.1.333.3 Fields

Field	Function
31-21 reserved_0	

Table continues on the next page...

Clocks And Resets

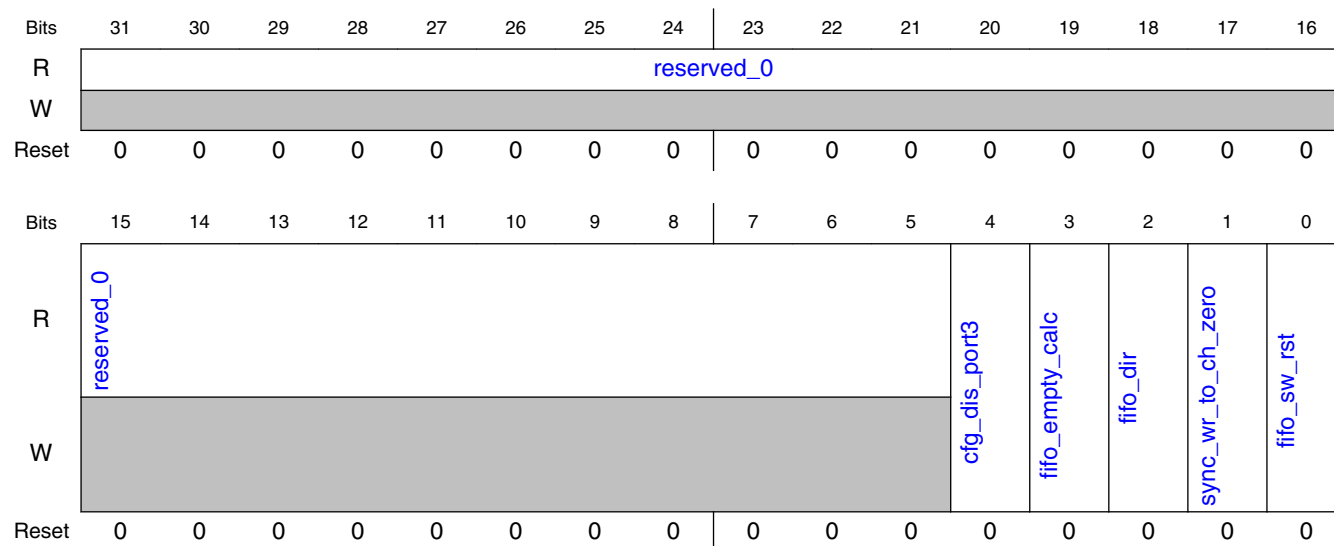
Field	Function
20 cfg_sample_pre sent_force	Force sample present bits
19-16 cfg_sample_pre sent	Sample present bits if force them is active
15 cfg_en_auto_su b_pkt_num	Enable automatics sub packet number. Enable automatics sub packet number. When enabled number of sub-packets will be set according to MEM FIFO number of samples.
14 cfg_block_lpcm_ first_pkt	0 - All packets behave the same. 0 - All packets behave the same. 1- First lpcm audio packet is sent with 1 - SP.
13-11 cfg_sub_pkt_n um	Number of sub-packets in HDMI audio 2-ch packet. Number of sub-packets in HDMI audio 2-ch packet. 00: 1-SP, 01: 2-SP, 10: 3-SP, 11: 4-SP.100-111: NA.
10-7 audio_type	Audio Type setting. Audio Type setting. Packet is structured according to audio type.
6-5 num_of_i2s_port s	Number of active I2S ports. Number of active I2S ports. 00- 1 port, 01-2 ports, 11- 4 ports, 11 -NA.
4-0 max_num_ch	Number of channels to decode. Number of channels to decode. 0: 1 channel, 31: 32 channels

13.4.10.1.334 FIFO control register (FIFO_CNTL)

13.4.10.1.334.1 Offset

Register	Offset
FIFO_CNTL	3_0088h

13.4.10.1.334.2 Diagram



13.4.10.1.334.3 Fields

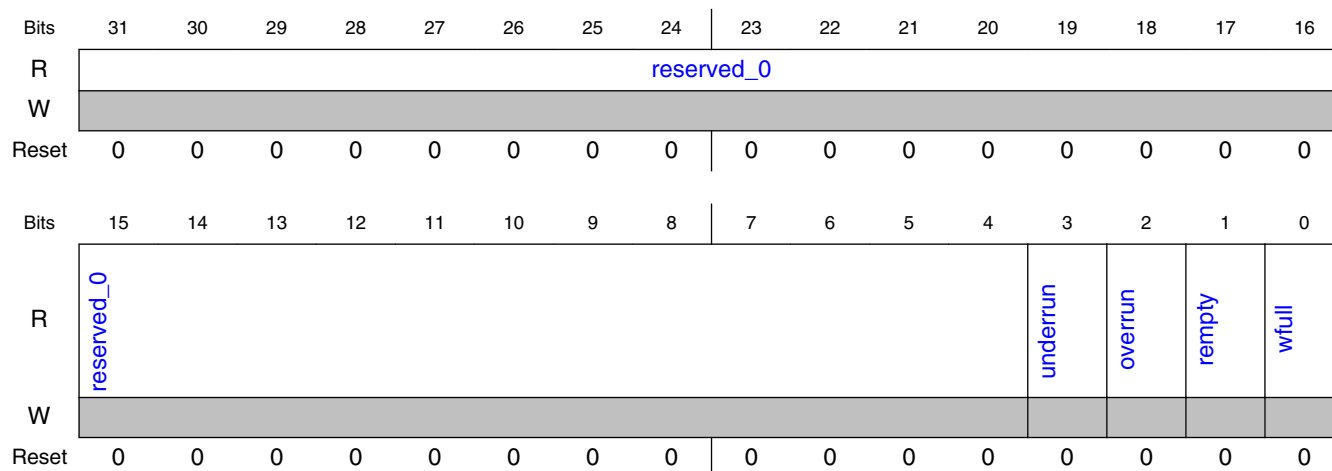
Field	Function
31-5 reserved_0	
4 cfg_dis_port3	0 - Normal Operation. 0 - Normal Operation. 1 - I2S port 3 is disabled (user should ignore its outputs). This allows for 24-ch, 12-ch, 6-ch transfer.
3 fifo_empty_calc	0- Empty is a function of read address. 0- Empty is a function of read address. 1 - Empty is a function of BASE read address.
2 fifo_dir	0 - smpl2pkt (inc_step=number of I2S ports), 1 - pkt2smpl (inc_step=num_ch_per_port)
1 sync_wr_to_ch_zero	When high the last channel index synchronizes the write addresses (to the next channel group)
0 fifo_sw_rst	Resets Fifo's write and read pointers. Resets Fifo's write and read pointers. When FIFO configuration bits change this signal should be high (due to synchronization issues).

13.4.10.1.335 FIFO Status register (FIFO_STTS)

13.4.10.1.335.1 Offset

Register	Offset
FIFO_STTS	3_008Ch

13.4.10.1.335.2 Diagram



13.4.10.1.335.3 Fields

Field	Function
31-4 reserved_0	
3 underrun	Indicates a FIFO underrun error has occurred - FIFO read when it was empty. Indicates a FIFO underrun error has occurred - FIFO read when it was empty. For debug purposes, not synchronized.
2 overrun	Indicates a FIFO overrun error has occurred - FIFO written to when it was full. Indicates a FIFO overrun error has occurred - FIFO written to when it was full. For debug purposes, not synchronized.
1 rempty	Indicates FIFO Empty. Indicates FIFO Empty. For debug purposes, not synchronized.
0 wfull	Indicates FIFO Full. Indicates FIFO Full. For debug purposes, not synchronized.

13.4.10.1.336 SUB Packet Threshold register (SUB_PCKT_THRSH)

13.4.10.1.336.1 Offset

Register	Offset
SUB_PCKT_THRSH	3_0090h

13.4.10.1.336.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_0								cfg_mem_fifo_thrsh3							
W																
Reset	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cfg_mem_fifo_thrsh2								cfg_mem_fifo_thrsh1							
W																
Reset	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0

13.4.10.1.336.3 Fields

Field	Function
31-24 reserved_0	
23-16 cfg_mem_fifo_th rsh3	If number of samples in MEM FIFO is below Threshold 3: Each Packet will contain only 3 subpacket. If number of samples in MEM FIFO is below Threshold 3: Each Packet will contain only 3 subpacket.
15-8 cfg_mem_fifo_th rsh2	If number of samples in MEM FIFO is below Threshold2: Each Packet will contain only 2 subpacket. If number of samples in MEM FIFO is below Threshold2: Each Packet will contain only 2 subpacket.
7-0 cfg_mem_fifo_th rsh1	If number of samples in MEM FIFO is below Threshold 1: Each Packet will contain only 1 subpacket. If number of samples in MEM FIFO is below Threshold 1: Each Packet will contain only 1 subpacket.

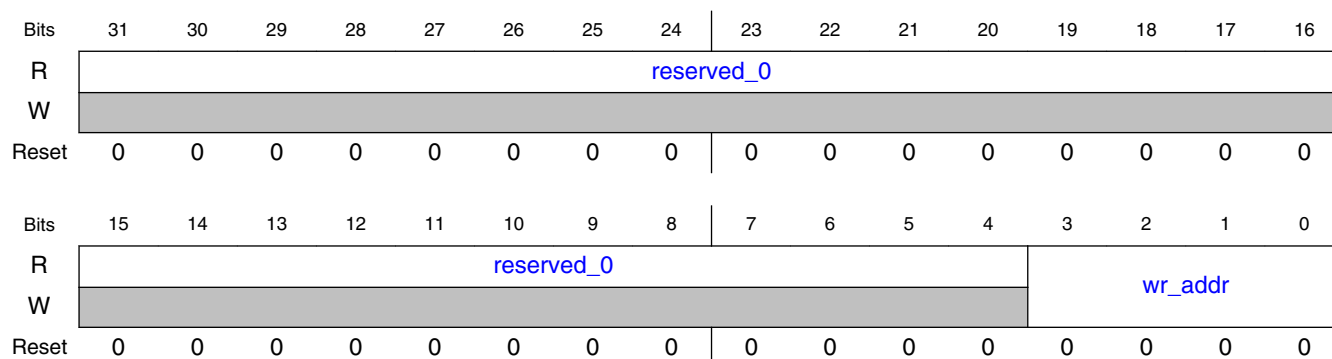
13.4.10.1.337 4 MSB of the packet memory address in which the data is written. (SOURCE_PIF_WR_ADDR)

13.4.10.1.337.1 Offset

Register	Offset
SOURCE_PIF_WR_AD DR	3_0800h

13.4.10.1.337.2 Function

4 MSB of the packet memory address in which the data is written.

13.4.10.1.337.3 Diagram**13.4.10.1.337.4 Fields**

Field	Function
31-4 reserved_0	
3-0 wr_addr	4 MSB of the packet memory address in which the data is written.

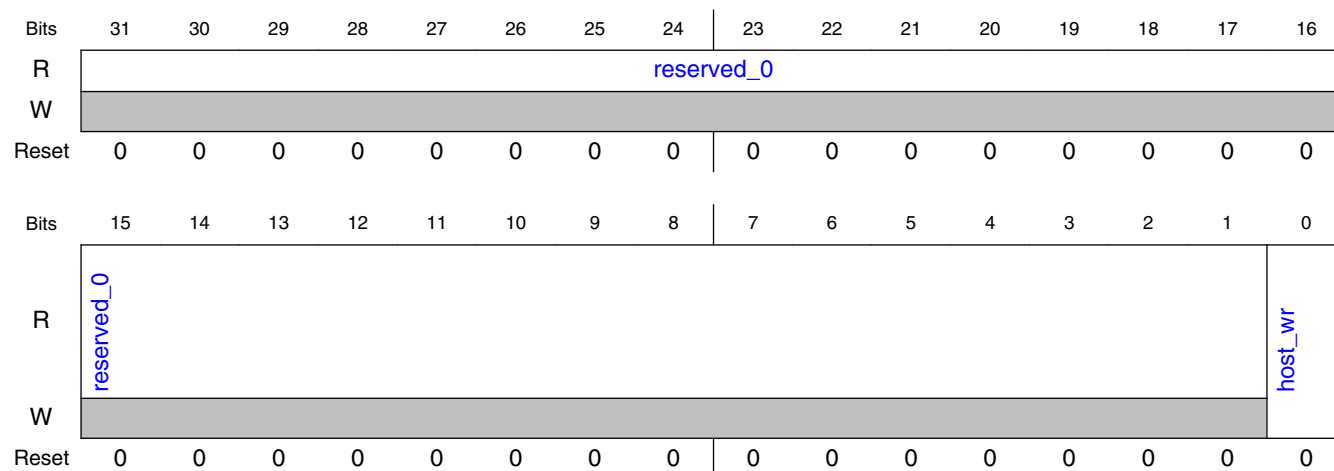
13.4.10.1.338 Write request bit for the host write transaction. (SOURCE_PIF_WR_REQ)**13.4.10.1.338.1 Offset**

Register	Offset
SOURCE_PIF_WR_REQ	3_0804h

13.4.10.1.338.2 Function

Write request bit for the host write transaction.

13.4.10.1.338.3 Diagram



13.4.10.1.338.4 Fields

Field	Function
31-1 reserved_0	
0 host_wr	Write request bit for the host write transaction, active high. Write request bit for the host write transaction, active high. Bit is automatically cleared when operation is completed.

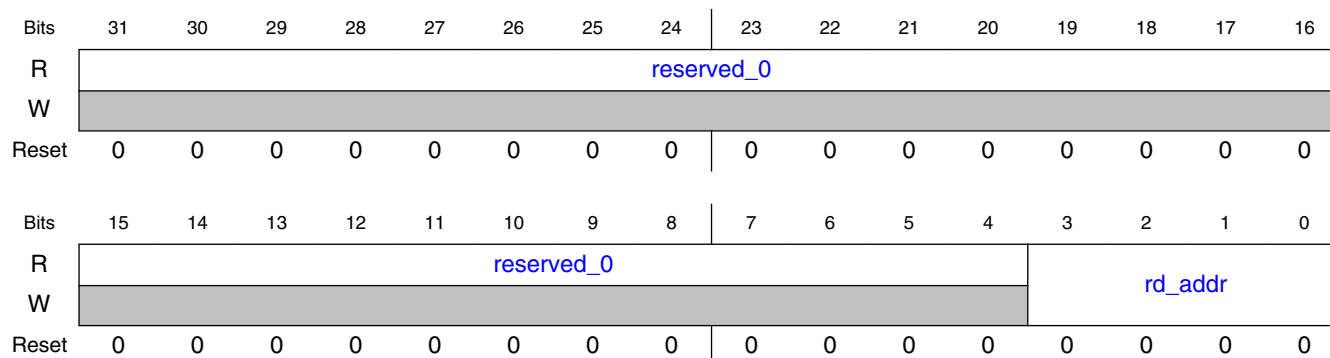
13.4.10.1.339 4 MSB of the packet memory address from which the data is read. (SOURCE_PIF_RD_ADDR)

13.4.10.1.339.1 Offset

Register	Offset
SOURCE_PIF_RD_ADDR	3_0808h

13.4.10.1.339.2 Function

4 MSB of the packet memory address from which the data is read.

13.4.10.1.339.3 Diagram**13.4.10.1.339.4 Fields**

Field	Function
31-4 reserved_0	
3-0 rd_addr	4 MSB of the packet memory address from which the data is read.

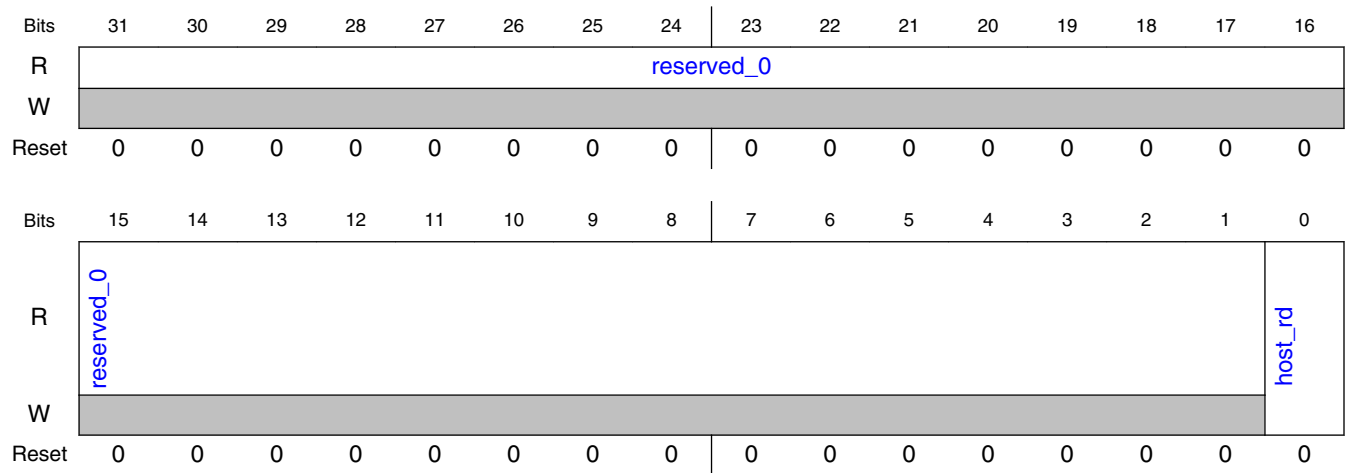
13.4.10.1.340 Read request bit for the host read transaction. (SOURCE_PIF_RD_REQ)**13.4.10.1.340.1 Offset**

Register	Offset
SOURCE_PIF_RD_REQ	3_080Ch

13.4.10.1.340.2 Function

Read request bit for the host read transaction.

13.4.10.1.340.3 Diagram



13.4.10.1.340.4 Fields

Field	Function
31-1 reserved_0	
0 host_rd	Read request bit for the host read transaction, active high. Read request bit for the host read transaction, active high. Bit is automatically cleared when operation is completed.

13.4.10.1.341 The 32 bits of the data to be written to the packet memory.
(SOURCE_PIF_DATA_WR)

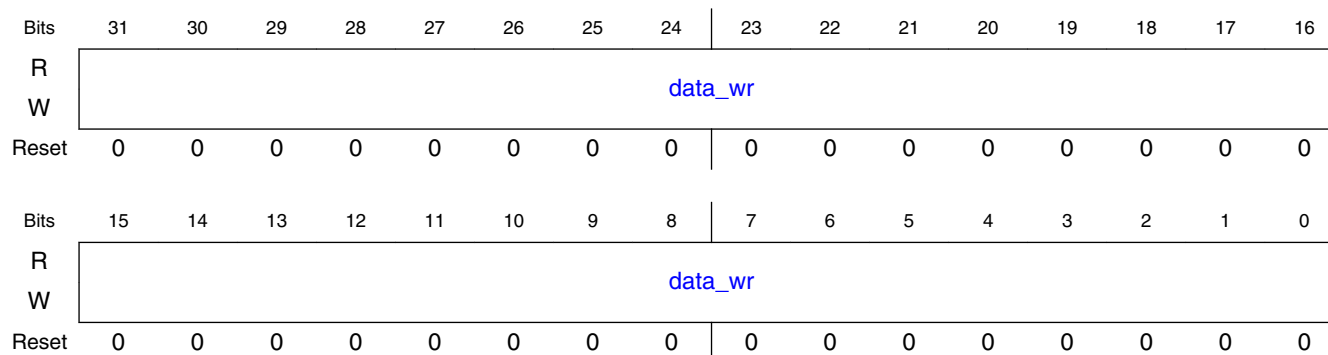
13.4.10.1.341.1 Offset

Register	Offset
SOURCE_PIF_DATA_WR	3_0810h

13.4.10.1.341.2 Function

The 32 bits of the data to be written to the packet memory.

13.4.10.1.341.3 Diagram



13.4.10.1.341.4 Fields

Field	Function
31-0	The 32 bits of the data to be written to the packet memory.
data_wr	The 32 bits of the data to be written to the packet memory. When written to this register fifo1_wr_enable will automatically be asserted and the data is stored in FIFO.

13.4.10.1.342 The 32 bits of the data to be read from the packet memory. (SOURCE_PIF_DATA_RD)

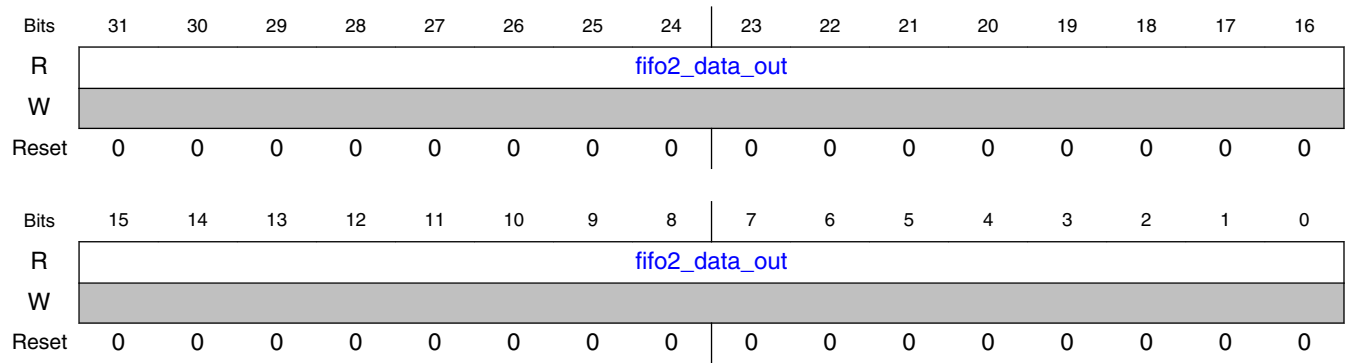
13.4.10.1.342.1 Offset

Register	Offset
SOURCE_PIF_DATA_RD	3_0814h

13.4.10.1.342.2 Function

The 32 bits of the data to be read from the packet memory.

13.4.10.1.342.3 Diagram



13.4.10.1.342.4 Fields

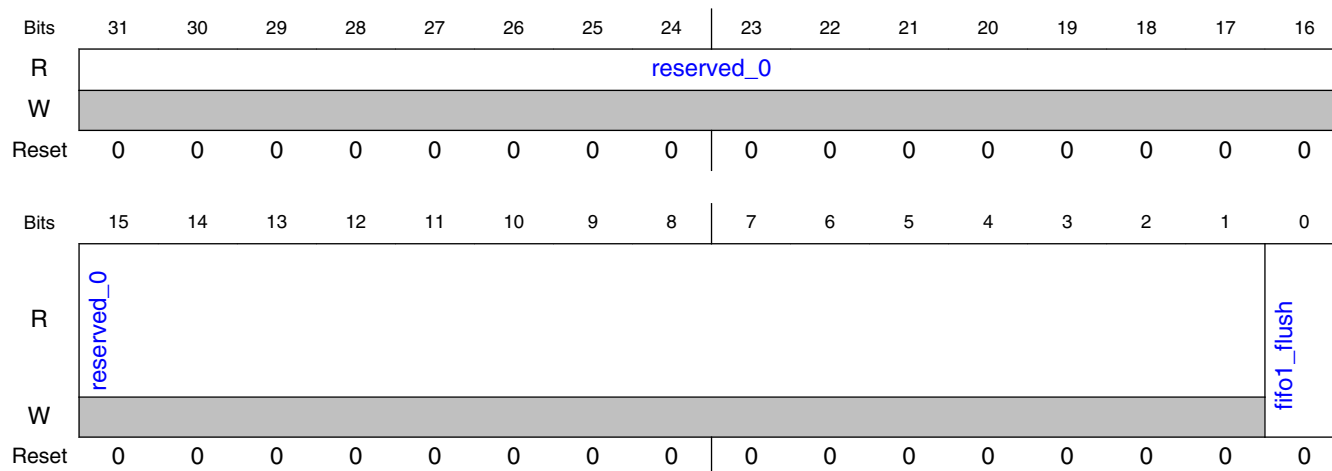
Field	Function
31-0	The 32 bits of the data to be read from the packet memory.
fifo2_data_out	The 32 bits of the data to be read from the packet memory. When read from this register fifo2_rd_enable will automatically be asserted and the data is read from the FIFO.

13.4.10.1.343 Fifo1 flush (SOURCE_PIF_FIFO1_FLUSH)

13.4.10.1.343.1 Offset

Register	Offset
SOURCE_PIF_FIFO1_FLUSH	3_0818h

13.4.10.1.343.2 Diagram



13.4.10.1.343.3 Fields

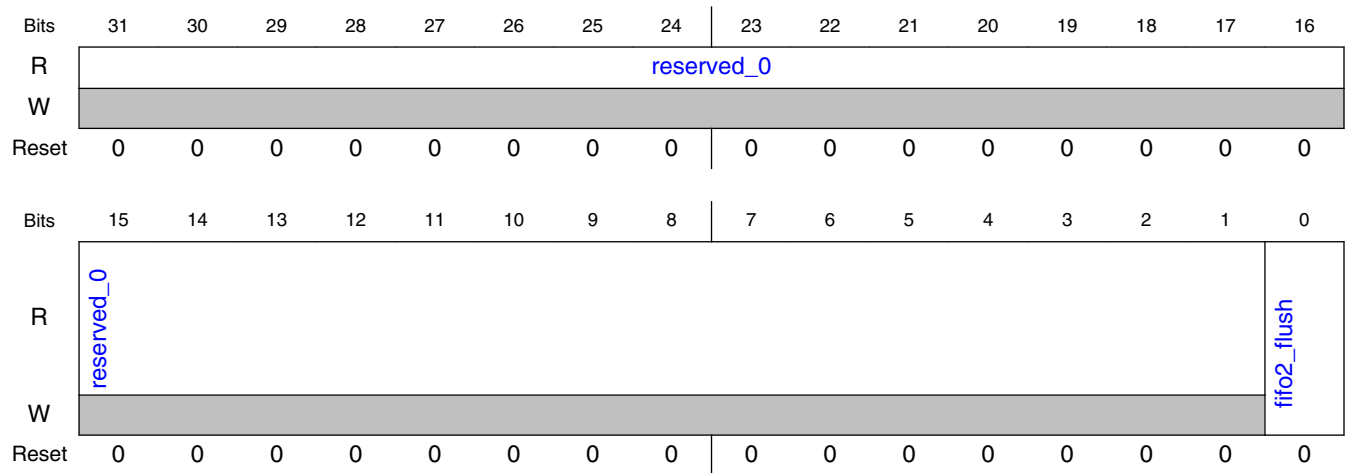
Field	Function
31-1 reserved_0	
0 fifo1_flush	Fifo1 flush bit, active high. Fifo1 flush bit, active high. Bit is automatically cleared when operation is completed.

13.4.10.1.344 Fifo2 flush (SOURCE_PIF_FIFO2_FLUSH)

13.4.10.1.344.1 Offset

Register	Offset
SOURCE_PIF_FIFO2_FLUSH	3_081Ch

13.4.10.1.344.2 Diagram



13.4.10.1.344.3 Fields

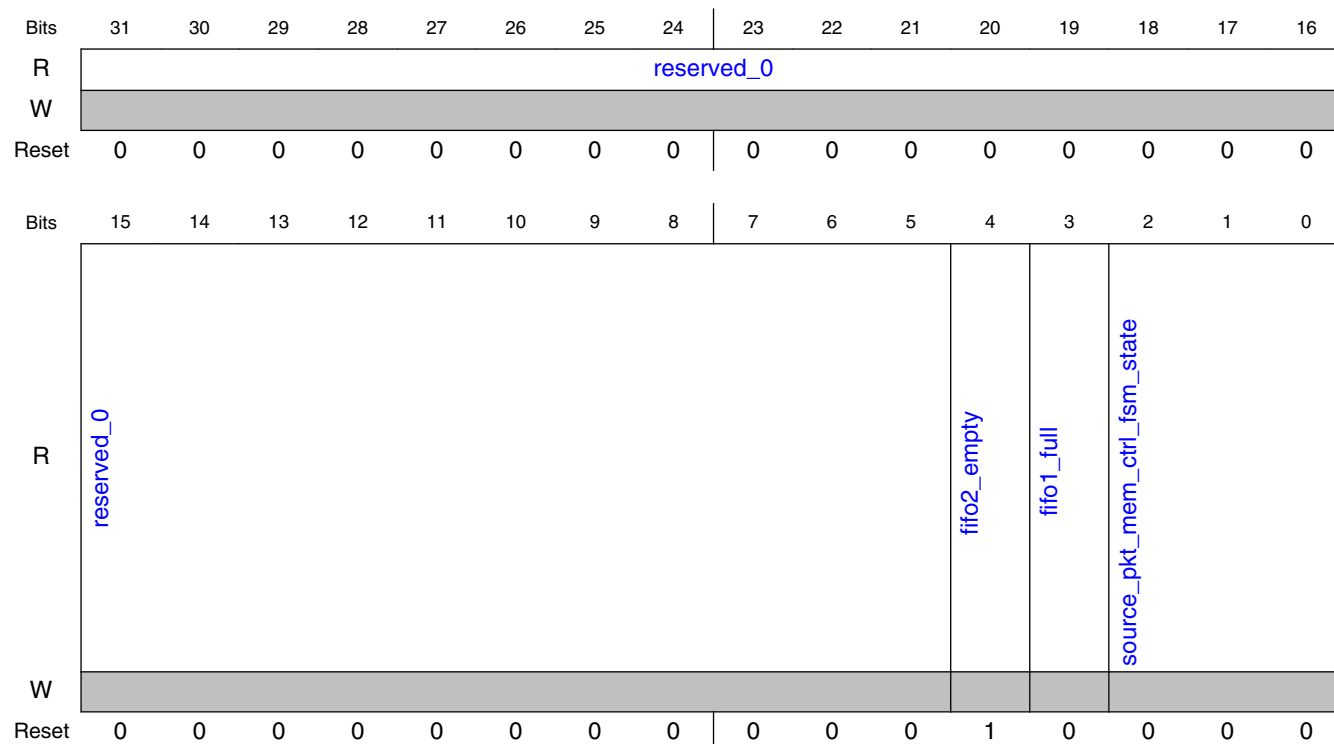
Field	Function
31-1 reserved_0	
0 fifo2_flush	Fifo2 flush bit, active high. Fifo2 flush bit, active high. Bit is automatically cleared when operation is completed.

13.4.10.1.345 Status bits for the PIF module (SOURCE_PIF_STATUS)

13.4.10.1.345.1 Offset

Register	Offset
SOURCE_PIF_STATUS	3_0820h

13.4.10.1.345.2 Diagram



13.4.10.1.345.3 Fields

Field	Function
31-5 reserved_0	
4 fifo2_empty	Fifo2 empty indication, when high indicates that FIFO2 is empty
3 fifo1_full	Fifo1 full indication, when high indicates that FIFO1 is full
2-0 source_pkt_me m_ctrl_fsm_stat e	State of the FSM that controls packet memory transactions. State of the FSM that controls packet memory transactions.

13.4.10.1.346 Interrupt sources of the PIF module, active high. (SOURCE_PIF_INTERRUPT_SOURCE)

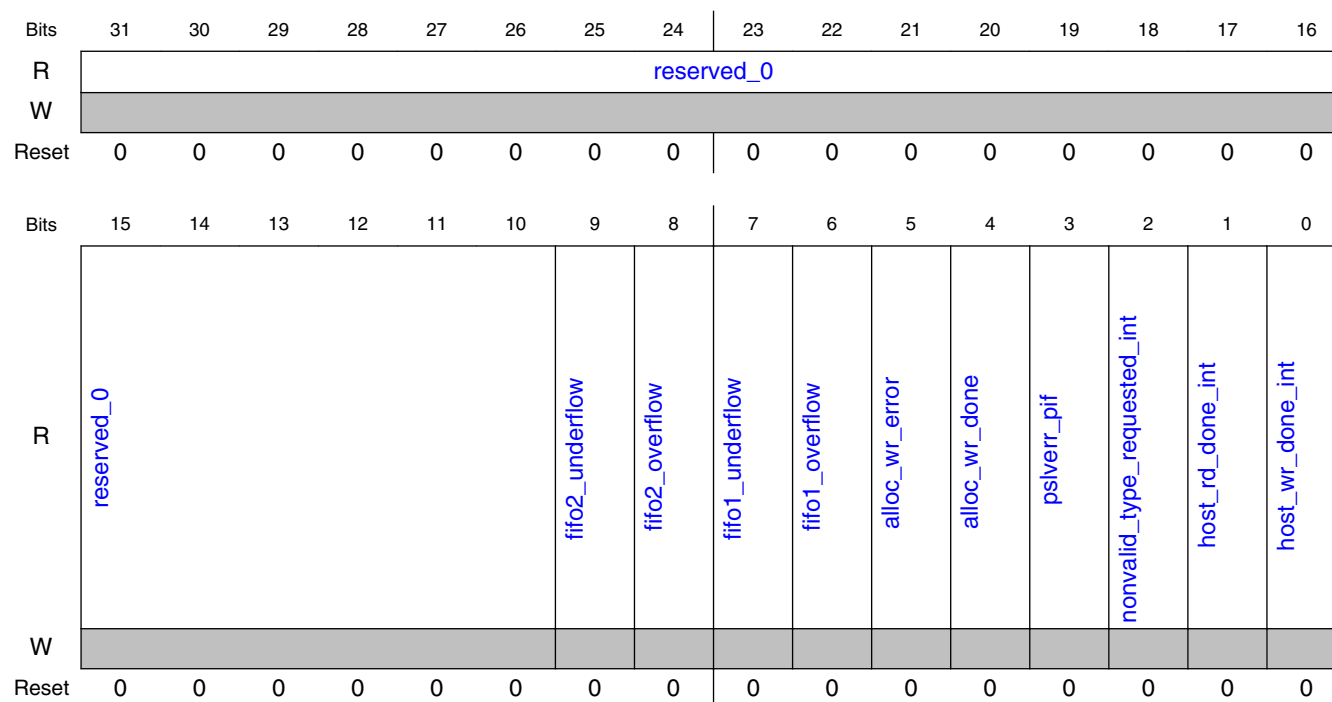
13.4.10.1.346.1 Offset

Register	Offset
SOURCE_PIF_INTER RUPT_SOURCE	3_0824h

13.4.10.1.346.2 Function

Interrupt sources of the PIF module, active high. Automatically cleared on read.

13.4.10.1.346.3 Diagram



13.4.10.1.346.4 Fields

Field	Function
31-10 reserved_0	
9 fifo2_underflow	Fifo2 underflow indication
8 fifo2_overflow	Fifo2 overflow indication

Table continues on the next page...

Clocks And Resets

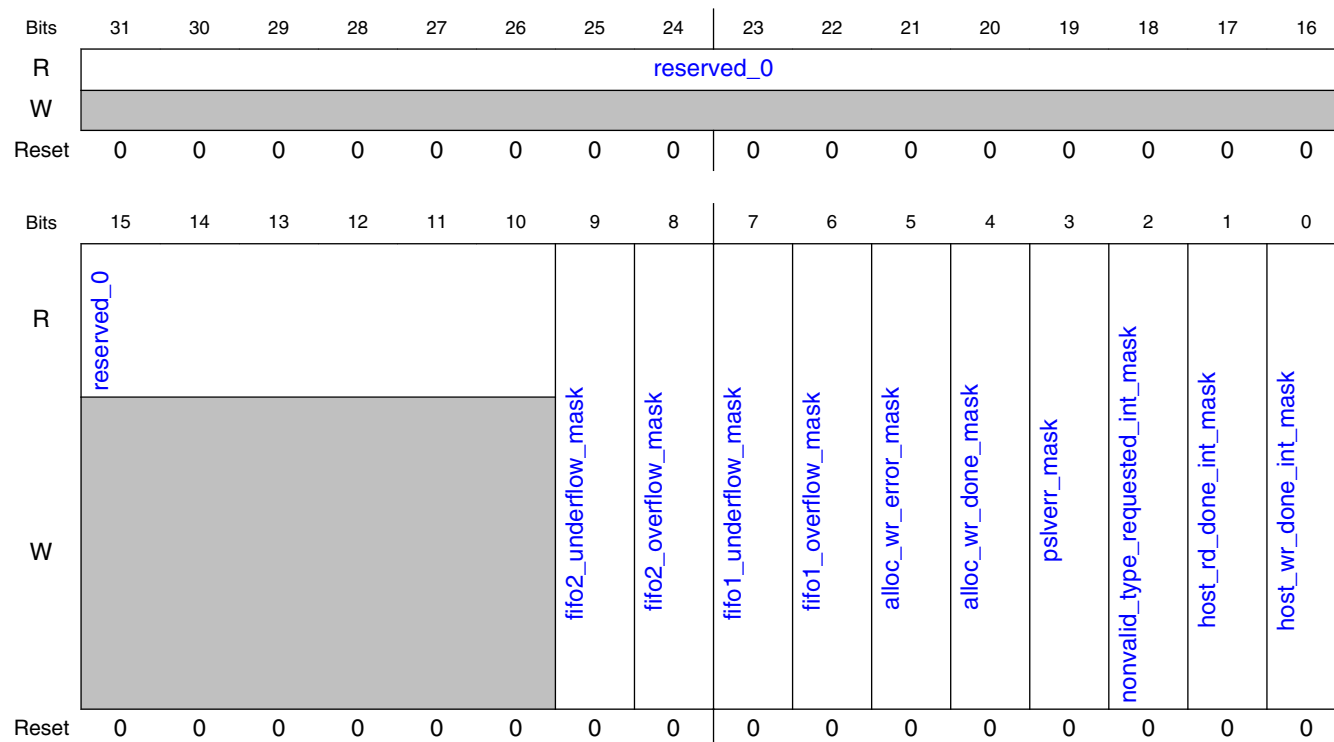
Field	Function
7 fifo1_underflow	Fifo1 underflow indication
6 fifo1_overflow	Fifo1 overflow indication
5 alloc_wr_error	Error happened, invalid write to the allocation table. Error happened, invalid write to the allocation table.
4 alloc_wr_done	Successful write to the allocation table. Successful write to the allocation table.
3 pslverr_pif	APB slave error interrupt from Packet Interface
2 nonvalid_type_requested_int	Indication that nonvalid type of packet is requested by the packet interface. Indication that nonvalid type of packet is requested by the packet interface.
1 host_rd_done_int	Indication that the host read transaction finished. Indication that the host read transaction finished.
0 host_wr_done_int	Indication that the host write transaction finished. Indication that the host write transaction finished.

13.4.10.1.347 Masks for the interrupt sources in the SOURCE_PIF_INTERRUPT_SOURCE register, when set high, these bits disable the corresponding interrupts (SOURCE_PIF_INTERRUPT_MASK)

13.4.10.1.347.1 Offset

Register	Offset
SOURCE_PIF_INTERRUPT_MASK	3_0828h

13.4.10.1.347.2 Diagram



13.4.10.1.347.3 Fields

Field	Function
31-10 reserved_0	
9 fifo2_underflow_mask	Masks the fifo2_underflow interrupt
8 fifo2_overflow_mask	Masks the fifo2_overflow interrupt
7 fifo1_underflow_mask	Masks the fifo1_underflow interrupt
6 fifo1_overflow_mask	Masks the fifo1_overflow interrupt
5 alloc_wr_error_mask	Masks the alloc_wr_error interrupt
4	Masks the alloc_wr_done interrupt

Table continues on the next page...

Clocks And Resets

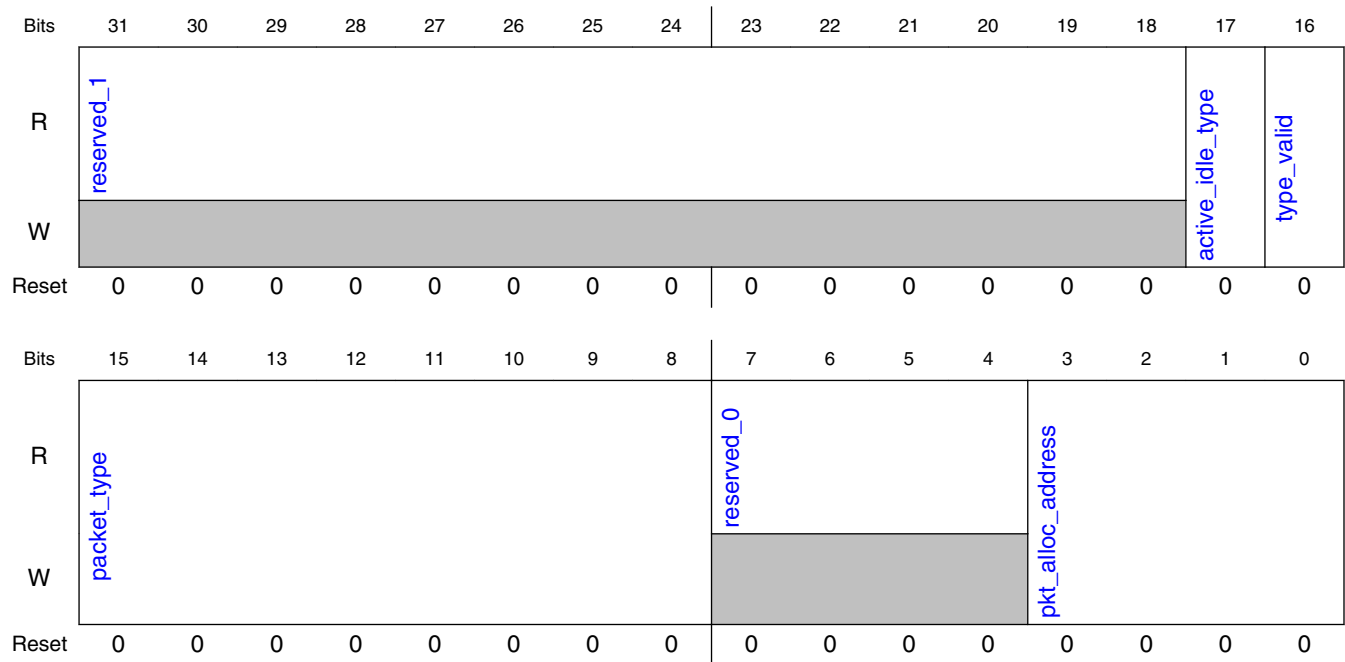
Field	Function
alloc_wr_done_mask	
3 pslverr_mask	Masks the pslverr_pif interrupt
2 nonvalid_type_requested_int_mask	Masks the nonvalid_type_requested_int interrupt
1 host_rd_done_int_mask	Masks the host_rd_done_int interrupt
0 host_wr_done_int_mask	Masks the host_wr_done_int interrupt

13.4.10.1.348 Packet configuration to be stored in the allocation table (SOURCE_PIF_PKT_ALLOC_REG)

13.4.10.1.348.1 Offset

Register	Offset
SOURCE_PIF_PKT_ALLOC_REG	3_082Ch

13.4.10.1.348.2 Diagram



13.4.10.1.348.3 Fields

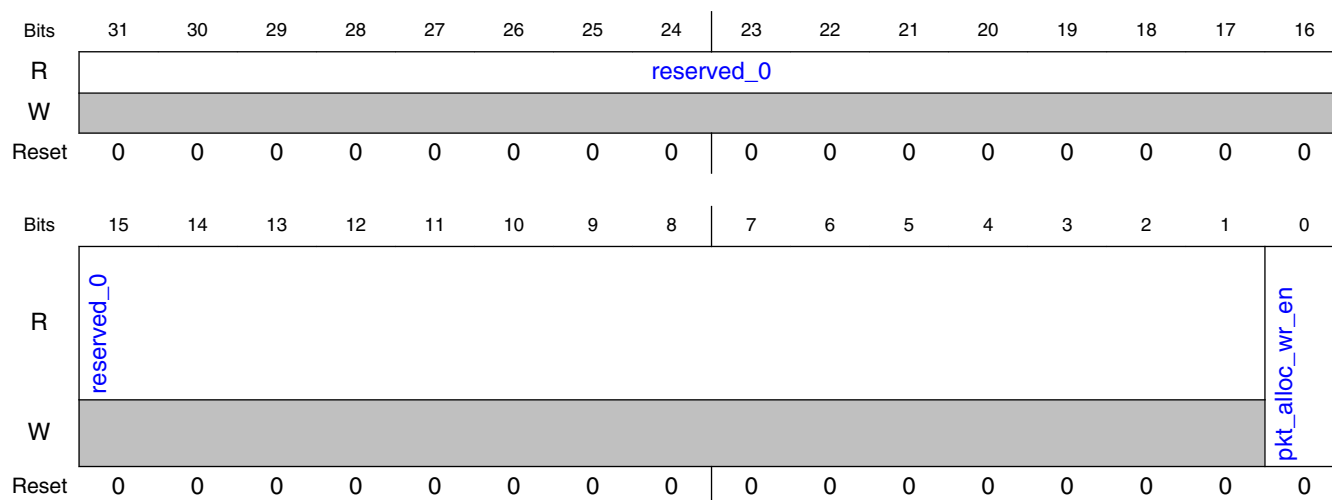
Field	Function
31-18 reserved_1	
17 active_idle_type	Indicates in which mode the SDP will be sent. Indicates in which mode the SDP will be sent. 0- no_video mode, 1- video mode
16 type_valid	1 for valid, 0 for nonvalid
15-8 packet_type	8-bit value of the packet type
7-4 reserved_0	
3-0 pkt_alloc_address	Address of the register in the source allocation table

13.4.10.1.349 Enable bit for writing to the allocation table (SOURCE_PIF_PKT_ALLOC_WR_EN)

13.4.10.1.349.1 Offset

Register	Offset
SOURCE_PIF_PKT_A LLOC_WR_EN	3_0830h

13.4.10.1.349.2 Diagram



13.4.10.1.349.3 Fields

Field	Function
31-1 reserved_0	
0 pkt_alloc_wr_en	Enable bit for writing to the allocation table, active high

13.4.10.1.350 Software reset. (SOURCE_PIF_SW_RESET)

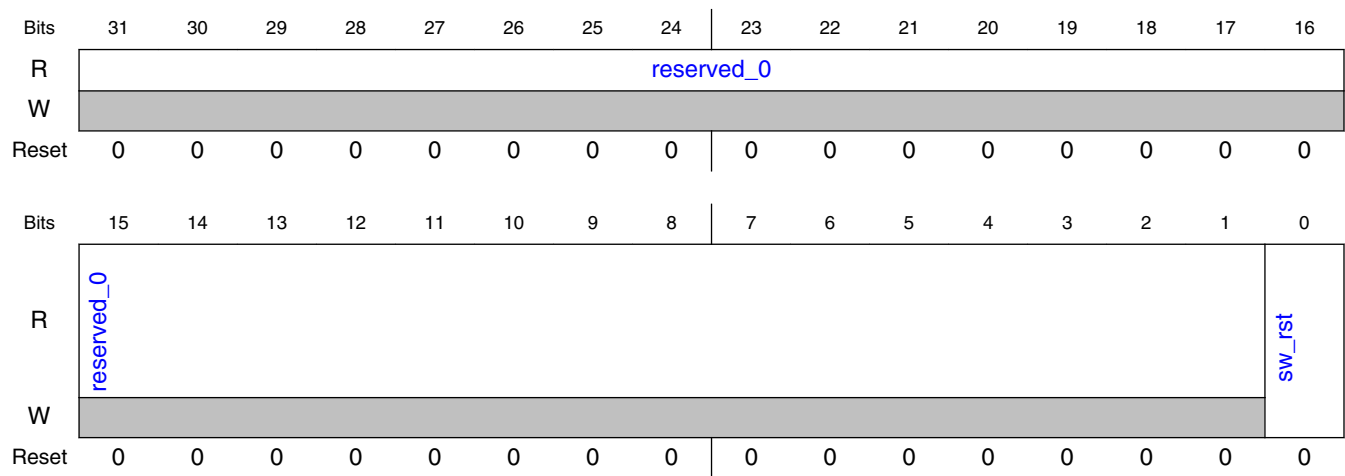
13.4.10.1.350.1 Offset

Register	Offset
SOURCE_PIF_SW_RE SET	3_0834h

13.4.10.1.350.2 **Function**

Software reset.

13.4.10.1.350.3 **Diagram**



13.4.10.1.350.4 **Fields**

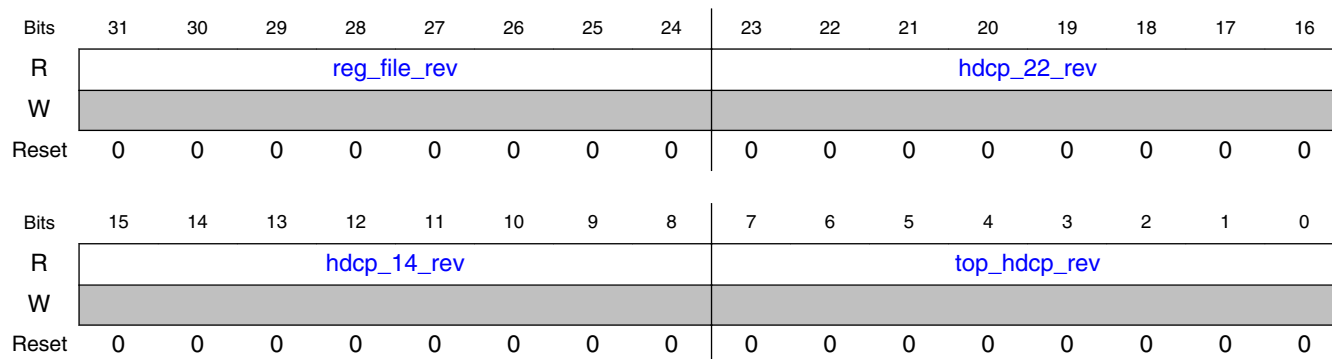
Field	Function
31-1 reserved_0	
0 sw_rst	Software reset, active high. Software reset, active high. Bit is automaticaly cleared when operation is completed.

13.4.10.1.351 **Contains the revision information of the HDCP Cipher module (HDCP_REVISION)**

13.4.10.1.351.1 **Offset**

Register	Offset
HDCP_REVISION	6_0000h

13.4.10.1.351.2 Diagram



13.4.10.1.351.3 Fields

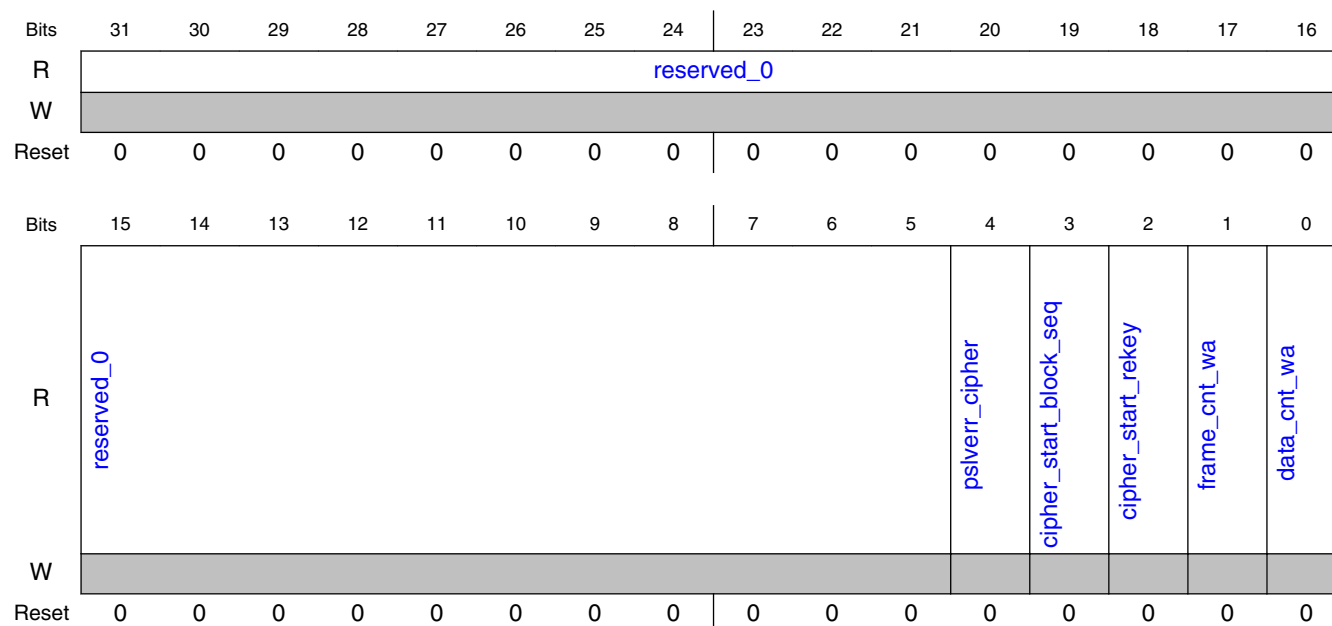
Field	Function
31-24 reg_file_rev	Revision number for the register file
23-16 hdcp_22_rev	Revision number for the HDCP 2. Revision number for the HDCP 2.2 engine
15-8 hdcp_14_rev	Revision number for the HDCP 1. Revision number for the HDCP 1.4 engine
7-0 top_hdcp_rev	Revision number for the top level RTL code

13.4.10.1.352 Contains the status of the HDCP Cipher interrupt sources (INTERRUPT_SOURCE)

13.4.10.1.352.1 Offset

Register	Offset
INTERRUPT_SOURCE	6_0004h

13.4.10.1.352.2 Diagram



13.4.10.1.352.3 Fields

Field	Function
31-5 reserved_0	Reserved Reserved. Writes are ignored. 0x0 when read
4 pslverr_cipher	APB slave error from Cipher module. APB slave error from Cipher module. Generated when APB address is out of the range
3 cipher_start_block_seq	Block start event
2 cipher_start_rekey	Start rekey event
1 frame_cnt_wa	Frame counter wraparound event. Frame counter wraparound event. Active high
0 data_cnt_wa	Data counter wraparound event. Data counter wraparound event. Active high

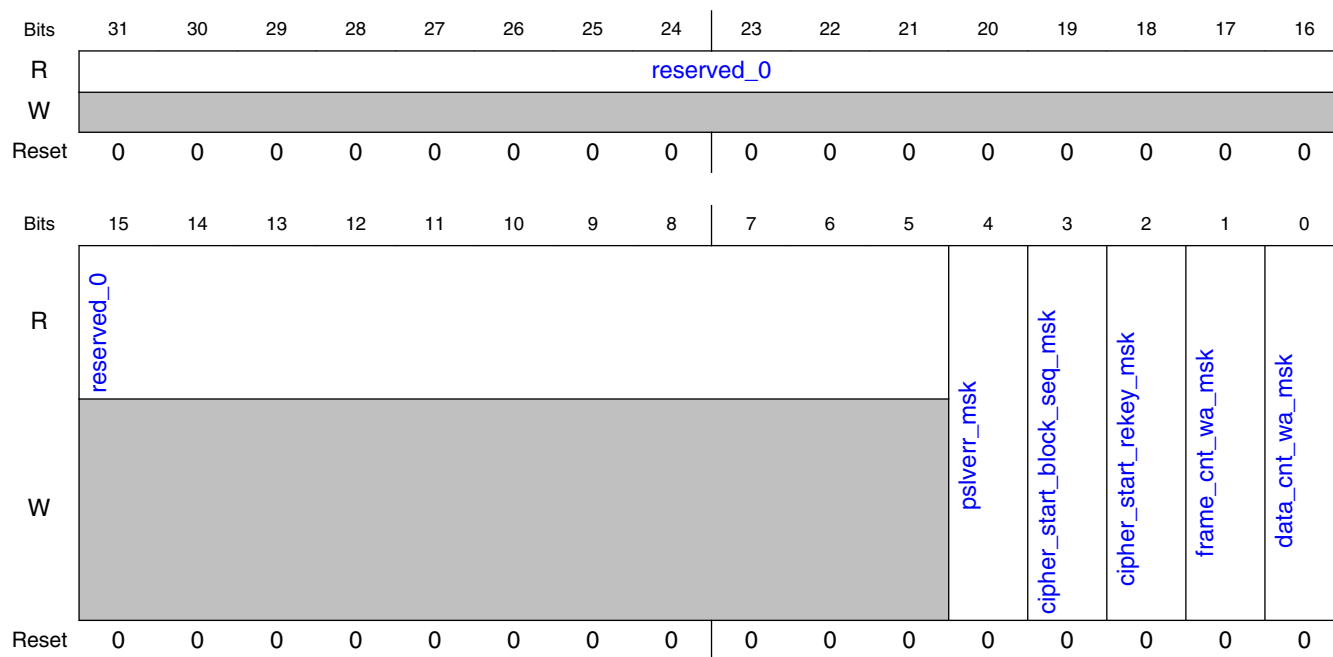
13.4.10.1.353 Contains the mask vector of the HDCP Cipher interrupt sources. (INTERRUPT_MASK)

13.4.10.1.353.1 Offset

Register	Offset
INTERRUPT_MASK	6_0008h

13.4.10.1.353.2 Function

Contains the mask vector of the HDCP Cipher interrupt sources.

13.4.10.1.353.3 Diagram**13.4.10.1.353.4 Fields**

Field	Function
31-5 reserved_0	Reserved Reserved. Writes are ignored. 0x0 when read
4 pslverr_msk	Set to 1 to mask the pslverr_cipher interrupt
3 cipher_start_block_seq_msk	Set to 1 mask the BLOCK_START interrupt
2	Set to 1 mask the START_REKEY interrupt

Table continues on the next page...

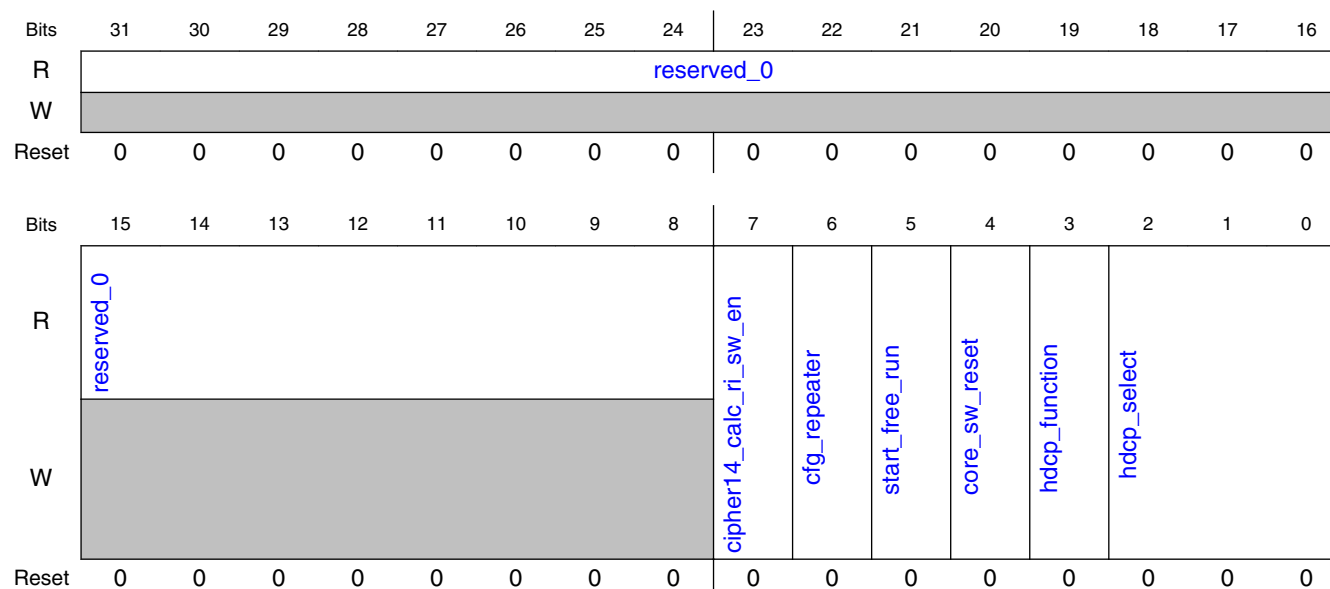
Field	Function
cipher_start_rekey_msk	
1	Set to 1 mask the FRAME_COUNT_WA interrupt
0	Set to 1 mask the DATA_COUNT_WA interrupt
data_cnt_wa_msk	

13.4.10.1.354 Contains the configuration details for the Cipher core module (HDCP_CIPHER_CONFIG)

13.4.10.1.354.1 Offset

Register	Offset
HDCP_CIPHER_CONFIG	6_000Ch

13.4.10.1.354.2 Diagram

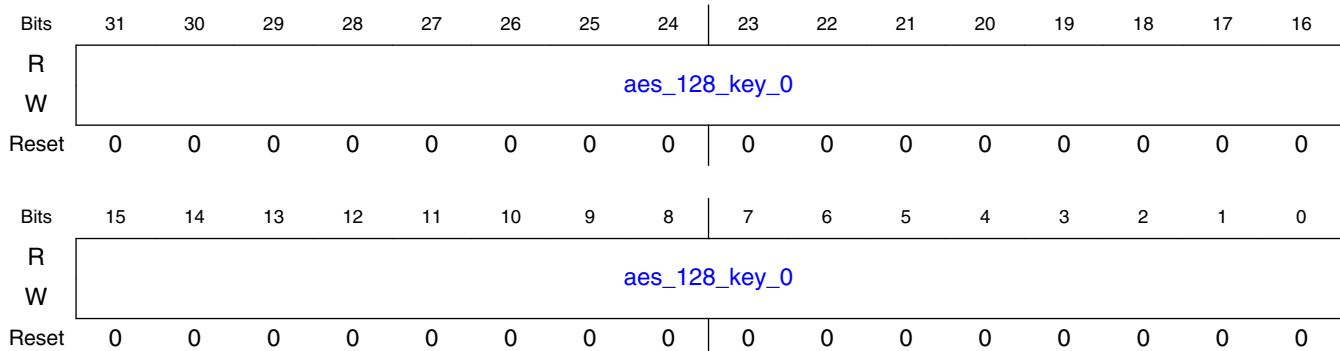


13.4.10.1.354.3 Fields

Field	Function
31-8 reserved_0	Reserved Reserved. Writes are ignored. 0x0 when read
7 cipher14_calc_ri_sw_en	For RX; calculate Ri
6 cfg_repeater	HDCP Repeater capability bit : 0 connection from receiver. HDCP Repeater capability bit : 0 connection from receiver. HDCP receiver is not an HDCP repeater, 1 connection from repeater. HDCP receiver is an HDCP repeater
5 start_free_run	Cipher 1. Cipher 1.4 command to start free running enable for operation hdcpRngCipher
4 core_sw_reset	Software reset signal for the Cipher core. Software reset signal for the Cipher core. Active high
3 hdcp_function	Selects the HDCP function performed. Selects the HDCP function performed.- 0x0 - Transmitter- 0x1 - Receiver
2-0 hdcp_select	Selects the HDCP core version support. Selects the HDCP core version support.- 0x0 - HDCP 1.4- 0x1 - HDCP 2.2- Other: Reserved

13.4.10.1.355 Holds the least significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_KEY_0)**13.4.10.1.355.1 Offset**

Register	Offset
AES_128_KEY_0	6_0010h

13.4.10.1.355.2 Diagram

13.4.10.1.355.3 Fields

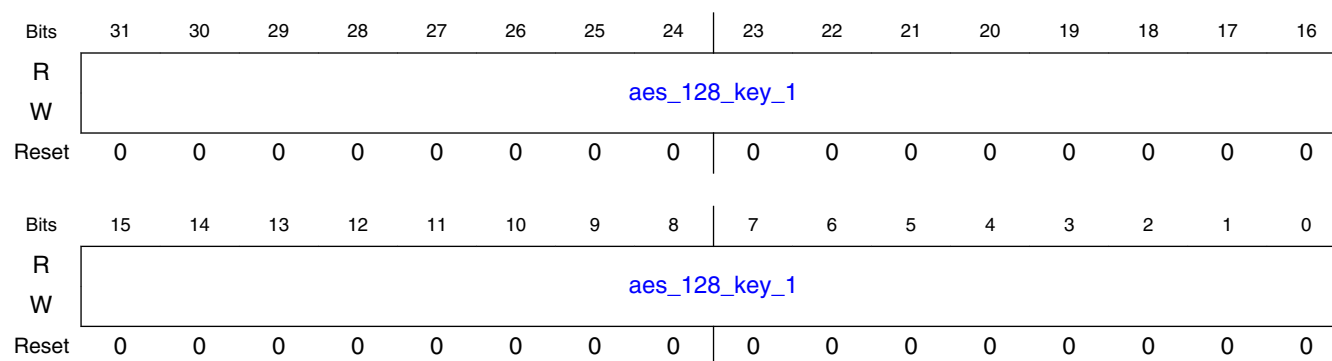
Field	Function
31-0	Holds the [31:0] of the $K_s \wedge L_c$ 128 bits results as defined in HDCP2.
aes_128_key_0	Holds the [31:0] of the $K_s \wedge L_c$ 128 bits results as defined in HDCP2.2 Figure 3-3- input key word of the AES-128 module

13.4.10.1.356 Holds the second significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_KEY_1)

13.4.10.1.356.1 Offset

Register	Offset
AES_128_KEY_1	6_0014h

13.4.10.1.356.2 Diagram



13.4.10.1.356.3 Fields

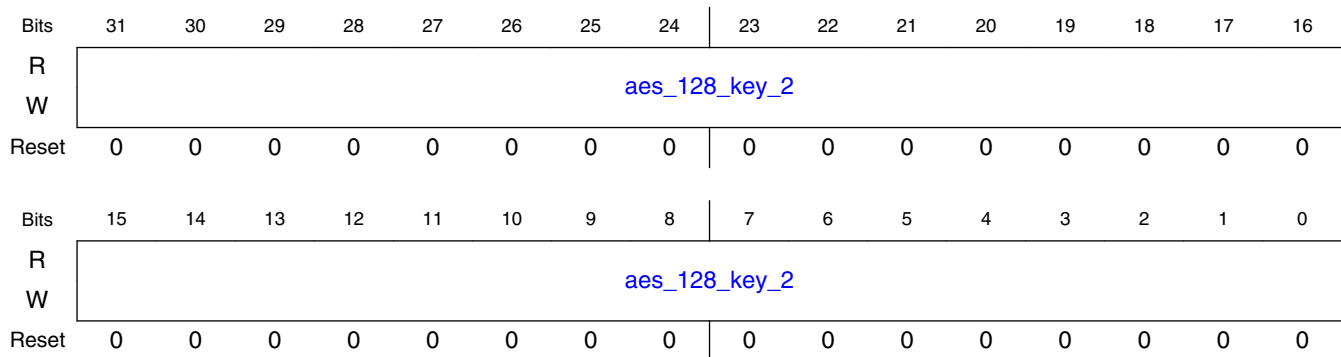
Field	Function
31-0	Holds the [63:32] of the $K_s \wedge L_c$ 128 bits results as defined in HDCP2.
aes_128_key_1	Holds the [63:32] of the $K_s \wedge L_c$ 128 bits results as defined in HDCP2.2 Figure 3-3- input key word of the AES-128 module

13.4.10.1.357 Holds the third significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_KEY_2)

13.4.10.1.357.1 Offset

Register	Offset
AES_128_KEY_2	6_0018h

13.4.10.1.357.2 Diagram



13.4.10.1.357.3 Fields

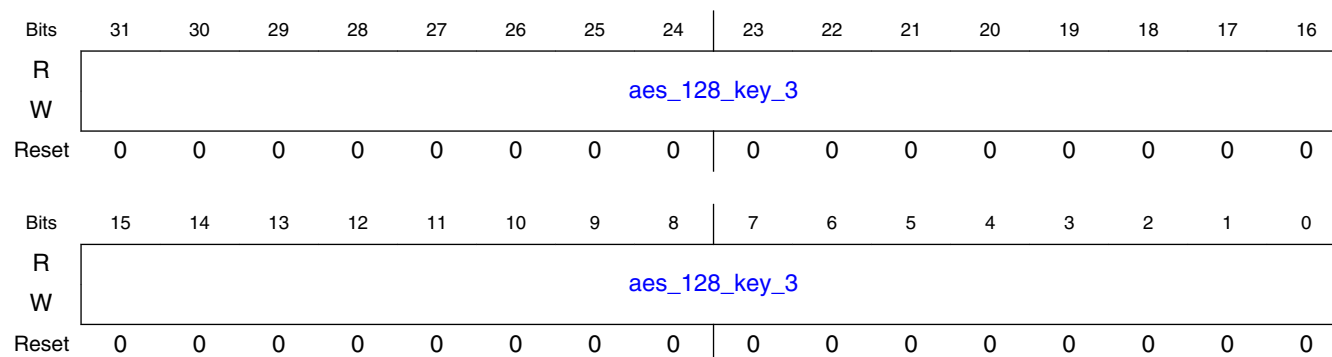
Field	Function
31-0	Holds the [95:64] of the $K_s \wedge L_c$ 128 bits results as defined in HDCP2.
aes_128_key_2	Holds the [95:64] of the $K_s \wedge L_c$ 128 bits results as defined in HDCP2.2 Figure 3-3- input key word of the AES-128 module

13.4.10.1.358 Holds the most significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_KEY_3)

13.4.10.1.358.1 Offset

Register	Offset
AES_128_KEY_3	6_001Ch

13.4.10.1.358.2 Diagram



13.4.10.1.358.3 Fields

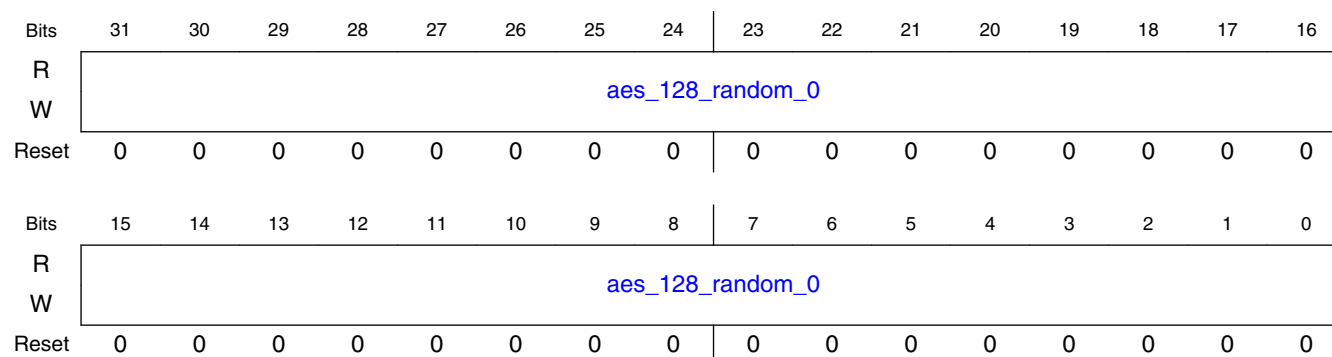
Field	Function
31-0	Holds the [127:96] of the K_s^{Lc128} 128 bits results as defined in HDCP2.
aes_128_key_3	Holds the [127:96] of the K_s^{Lc128} 128 bits results as defined in HDCP2.2 Figure 3-3- input key word of the AES-128 module

13.4.10.1.359 Holds the least significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_RANDOM_0)

13.4.10.1.359.1 Offset

Register	Offset
AES_128_RANDOM_0	6_0020h

13.4.10.1.359.2 Diagram



13.4.10.1.359.3 Fields

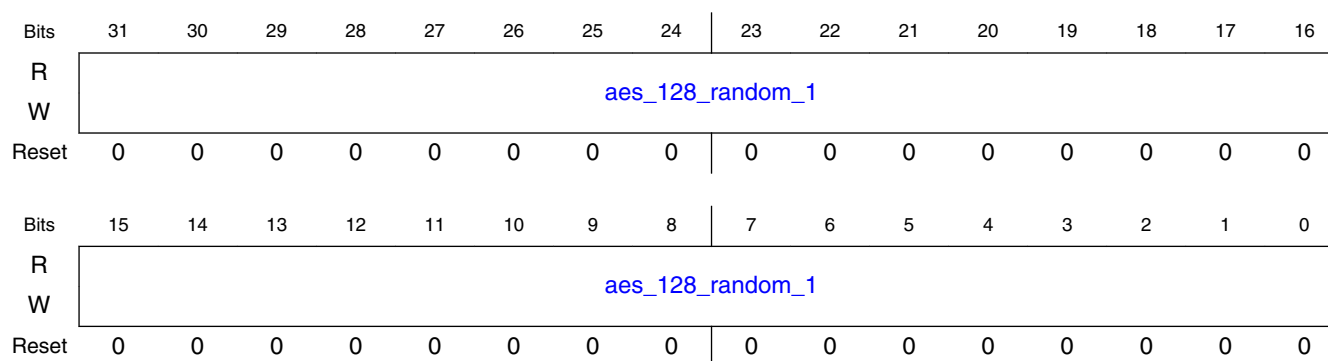
Field	Function
31-0	Holds the RiV[31:0] as defined in HDCP2.
aes_128_random_m_0	Holds the RiV[31:0] as defined in HDCP2.2 Figure 3-3-input random number for the AES-128 module

13.4.10.1.360 Holds the most significant 32-bits word of the 128-bits input key word of the AES-128 module (AES_128_RANDOM_1)

13.4.10.1.360.1 Offset

Register	Offset
AES_128_RANDOM_1	6_0024h

13.4.10.1.360.2 Diagram



13.4.10.1.360.3 Fields

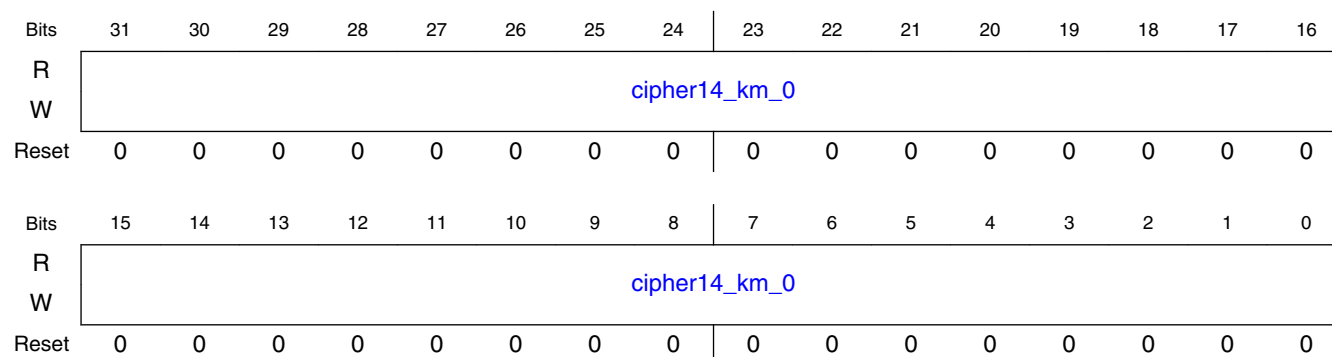
Field	Function
31-0	Holds the RiV[64:32] as defined in HDCP2.
aes_128_random_m_1	Holds the RiV[64:32] as defined in HDCP2.2 Figure 3-3-input random number for the AES-128 module

13.4.10.1.361 Holds the first word of the Km value (CIPHER14_KM_0)

13.4.10.1.361.1 Offset

Register	Offset
CIPHER14_KM_0	6_0028h

13.4.10.1.361.2 Diagram



13.4.10.1.361.3 Fields

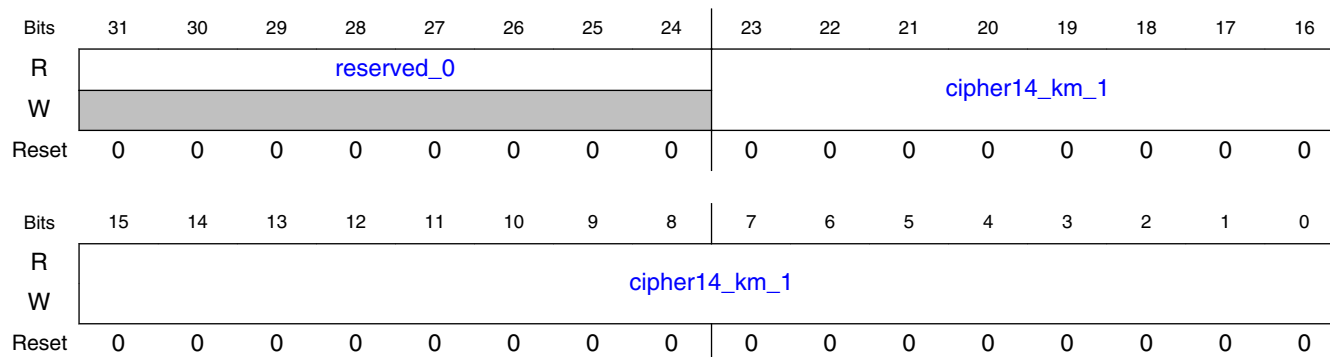
Field	Function
31-0 cipher14_km_0	Holds the first word of the Km value

13.4.10.1.362 Holds the most significant 3 bytes of the Km value (CIPHER14_KM_1)

13.4.10.1.362.1 Offset

Register	Offset
CIPHER14_KM_1	6_002Ch

13.4.10.1.362.2 Diagram



13.4.10.1.362.3 Fields

Field	Function
31-24 reserved_0	Reserved Reserved. Writes are ignored. 0x0 when read
23-0 cipher14_km_1	Holds the most significant 3 bytes of the Km value

13.4.10.1.363 Cipher 1. (CIPHER14_STATUS)

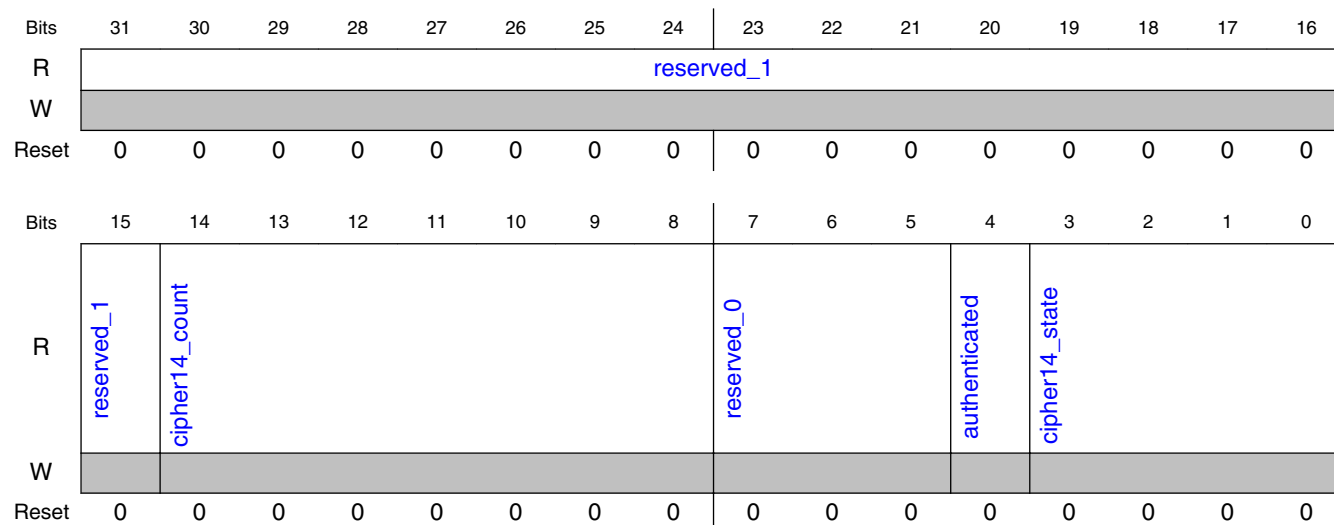
13.4.10.1.363.1 Offset

Register	Offset
CIPHER14_STATUS	6_0030h

13.4.10.1.363.2 Function

Cipher 1.4 status register

13.4.10.1.363.3 Diagram



13.4.10.1.363.4 Fields

Field	Function
31-15 reserved_1	Reserved Reserved. Writes are ignored. 0x0 when read
14-8 cipher14_count	Cipher 1. Cipher 1.4 counter output status.
7-5 reserved_0	Reserved Reserved. Writes are ignored. 0x0 when read
4 authenticated	Authenticated status bit. Authenticated status bit.Set to 1 when first block starts generating R0
3-0 cipher14_state	Current state for Cipher FSM. Current state for Cipher FSM.Possible values:0001 - HDCPRNGCIPHER,0010 - HDCPBLOCKCIPHER, 0100 - HDCPSTREAMCIPHER state,1000 - HDCPREKEYCIPHER

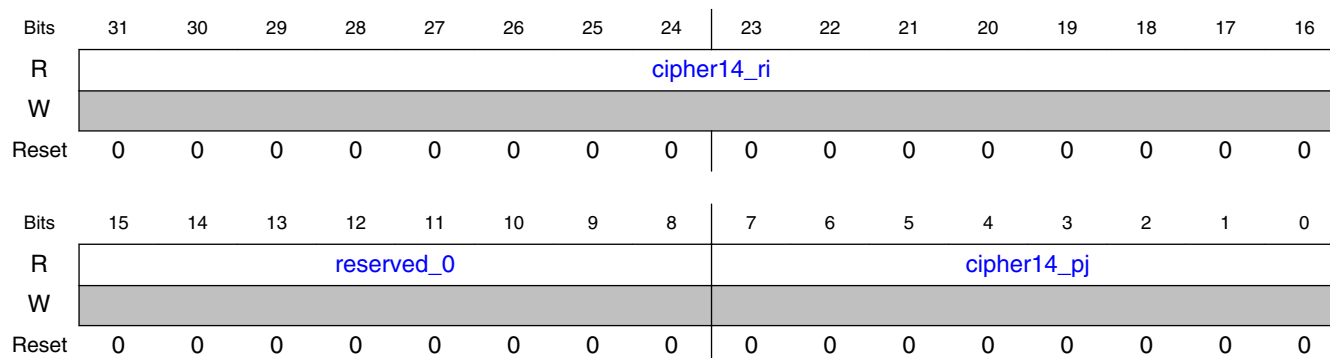
13.4.10.1.364 Cipher 1. (CIPHER14_RI_PJ_STATUS)

13.4.10.1.364.1 Offset

Register	Offset
CIPHER14_RI_PJ_S TATUS	6_0034h

13.4.10.1.364.2 Function

Cipher 1.4 status register for Ri and Pj values

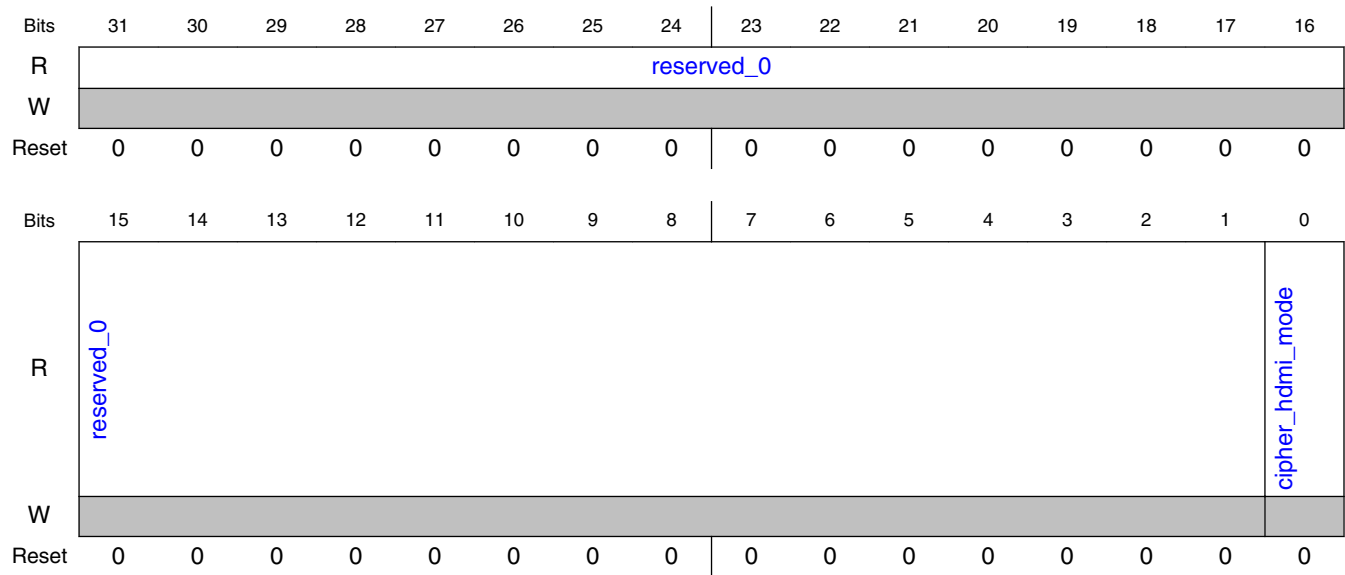
13.4.10.1.364.3 Diagram**13.4.10.1.364.4 Fields**

Field	Function
31-16 cipher14_ri	HDCCP 1. HDCCP 1.4 Ri data.
15-8 reserved_0	Reserved Reserved. Writes are ignored. 0x0 when read
7-0 cipher14_pj	HDCCP 1. HDCCP 1.4 Pj data.

13.4.10.1.365 Cipher mode status register (CIPHER_MODE)**13.4.10.1.365.1 Offset**

Register	Offset
CIPHER_MODE	6_0038h

13.4.10.1.365.2 Diagram



13.4.10.1.365.3 Fields

Field	Function
31-1 reserved_0	Reserved Reserved. Writes are ignored. 0x0 when read
0 cipher_hdmi_mode	HDMI mode status (input cipher_hdmi_mode). HDMI mode status (input cipher_hdmi_mode).When 1 - HDMI -When 0 - DVI

13.4.10.1.366 Holds the First word of An value generated by hdcpRngCipher operation. (CIPHER14_AN_0)

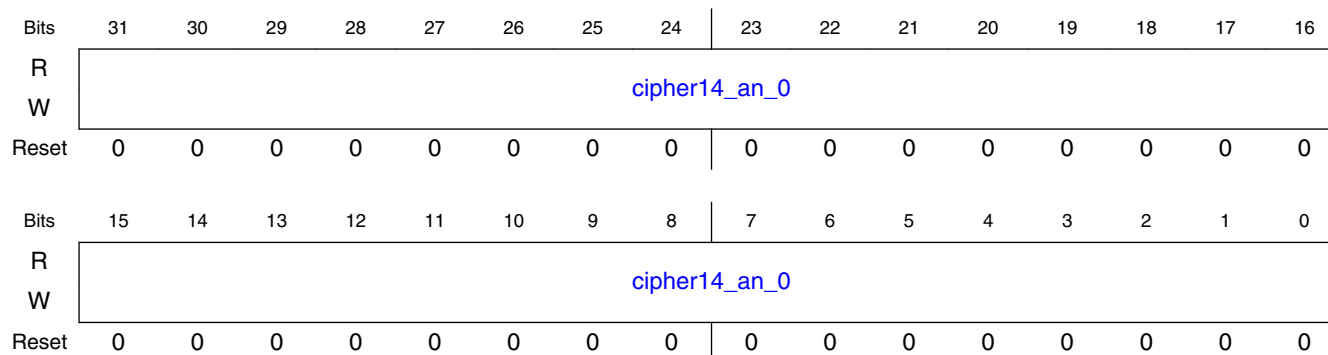
13.4.10.1.366.1 Offset

Register	Offset
CIPHER14_AN_0	6_003Ch

13.4.10.1.366.2 Function

Holds the First word of An value generated by hdcpRngCipher operation.

13.4.10.1.366.3 Diagram



13.4.10.1.366.4 Fields

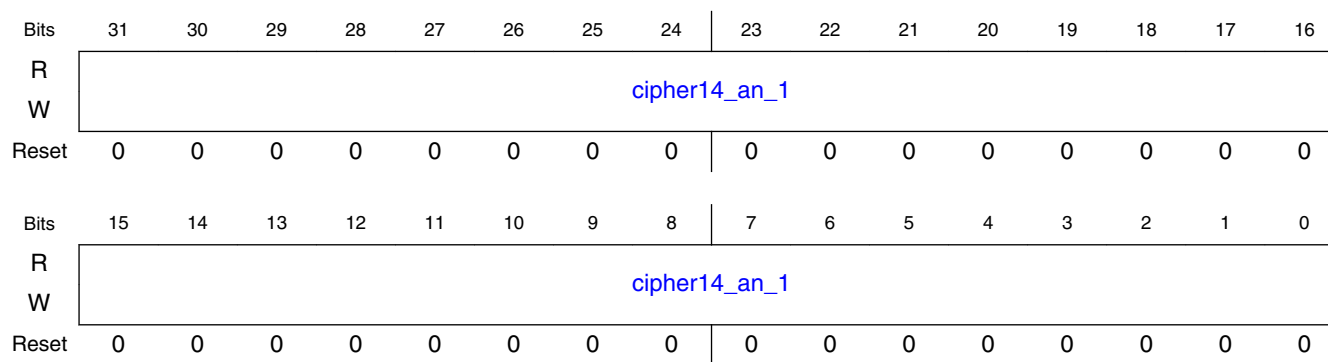
Field	Function
31-0	Holds the First word of An value generated by hdcpRngCipher operation.
cipher14_an_0	Holds the First word of An value generated by hdcpRngCipher operation.

13.4.10.1.367 Holds the second word of An value generated by hdcpRngCipher operation (CIPHER14_AN_1)

13.4.10.1.367.1 Offset

Register	Offset
CIPHER14_AN_1	6_0040h

13.4.10.1.367.2 Diagram



13.4.10.1.367.3 Fields

Field	Function
31-0 cipher14_an_1	Holds the second word of An value generated by hdcpRngCipher operation

13.4.10.1.368 HDCP2. (CIPHER22_AUTH)

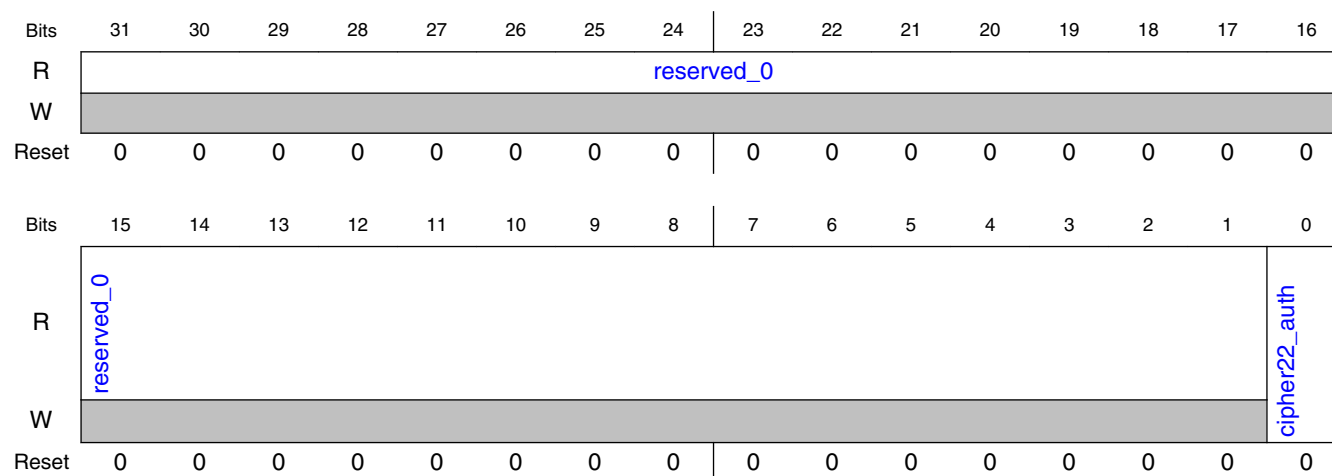
13.4.10.1.368.1 Offset

Register	Offset
CIPHER22_AUTH	6_0044h

13.4.10.1.368.2 Function

HDCP2.2 authentication status register. Set to 1 by firmware when authentication is finished successfully. Cleared by FW when Sink is not authenticated (start of operation, Sink disconnection, authentication restart).

13.4.10.1.368.3 Diagram



13.4.10.1.368.4 Fields

Field	Function
31-1	Reserved

Table continues on the next page...

Clocks And Resets

Field	Function
reserved_0	Reserved. Writes are ignored. 0x0 when read
0	Cipher 2.
cipher22_auth	Cipher 2.2 authentication status. 1 - authenticated, 0 - not authenticated

13.4.10.1.369 This register contains Ri value in A5: Link Integrity Check state [HDCP 1. (CIPHER14_R0_DP_STATUS)]

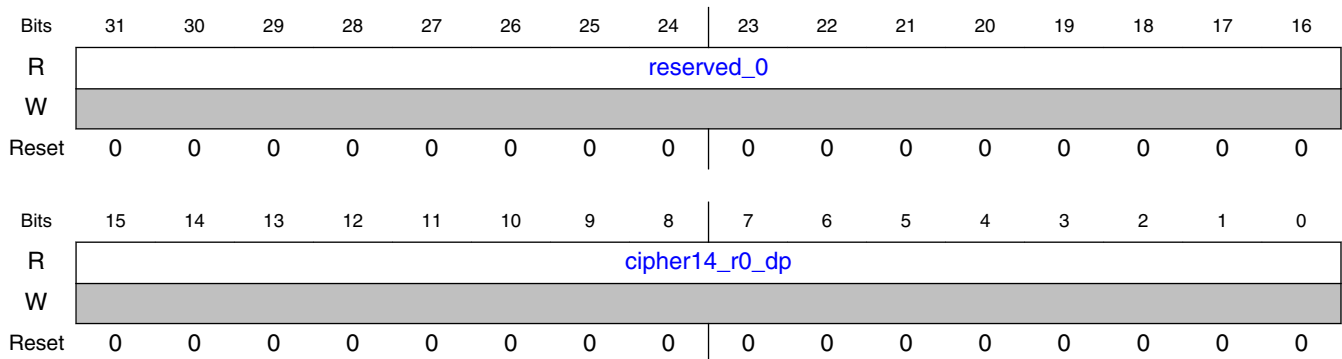
13.4.10.1.369.1 Offset

Register	Offset
CIPHER14_R0_DP_STATUS	6_0048h

13.4.10.1.369.2 Function

This register contains Ri value in A5: Link Integrity Check state [HDCP 1.4]

13.4.10.1.369.3 Diagram



13.4.10.1.369.4 Fields

Field	Function
31-16	Reserved
reserved_0	Reserved. Writes are ignored. 0x0 when read
15-0	Ri value checked in A5 state
cipher14_r0_dp	

13.4.10.1.370 Cipher for HDCP1. (CIPHER14_BOOTSTRAP)

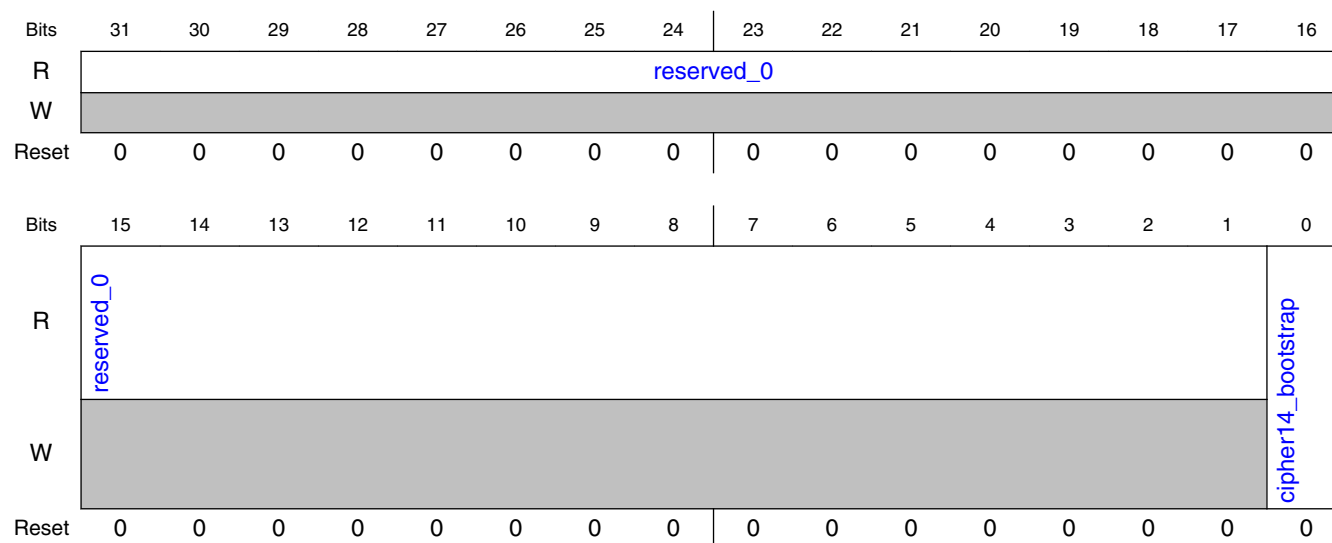
13.4.10.1.370.1 Offset

Register	Offset
CIPHER14_BOOTSTRAP	6_004Ch

13.4.10.1.370.2 Function

Cipher for HDCP1.4 bootstap

13.4.10.1.370.3 Diagram



13.4.10.1.370.4 Fields

Field	Function
31-1 reserved_0	Reserved Reserved. Writes are ignored. 0x0 when read
0 cipher14_bootstrap	Set to 1 when authentication is finished succesfully. Set to 1 when authentication is finished succesfully. Otherwise 0

13.4.10.2 HDMI_TX_PHY register descriptions

13.4.10.2.1 HDMI_TX_PHY memory map

HDMI_TX_PHY base address: 0h

Offset	Register	Width (In bits)	Access	Reset value
4000h	Power state machine control register (lane0_xcvr_psm_ctrl)	16	RW	0201h
4001h	Power state machine reset control register (lane0_xcvr_psm_rctrl)	16	RW	BCFCh
4002h	PSM calibration delay timer register (lane0_xcvr_psm_cal_tmr)	16	RW	0152h
4003h	A0 in delay timer register (lane0_xcvr_psm_a0in_tmr)	16	RW	0152h
4004h	A0 in bypass timer register (lane0_xcvr_psm_a0byp_tmr)	16	RW	0001h
4005h	A1 in delay timer register (lane0_xcvr_psm_a1in_tmr)	16	RW	0008h
4006h	A2 in delay timer register (lane0_xcvr_psm_a2in_tmr)	16	RW	0008h
4007h	A3 in delay timer register (lane0_xcvr_psm_a3in_tmr)	16	RW	0008h
4008h	A4 in delay timer register (lane0_xcvr_psm_a4in_tmr)	16	RW	0008h
4009h	A5 in delay timer register (lane0_xcvr_psm_a5in_tmr)	16	RW	0008h
400Ah	A0 out delay timer register (lane0_xcvr_psm_a0out_tmr)	16	RW	0001h
400Bh	A1 out delay timer register (lane0_xcvr_psm_a1out_tmr)	16	RW	0001h
400Ch	A2 out delay timer register (lane0_xcvr_psm_a2out_tmr)	16	RW	0001h
400Dh	A3 out delay timer register (lane0_xcvr_psm_a3out_tmr)	16	RW	0001h
400Eh	A4 out delay timer register (lane0_xcvr_psm_a4out_tmr)	16	RW	0001h
400Fh	A5 out delay timer register (lane0_xcvr_psm_a5out_tmr)	16	RW	0001h
4010h	Power state machine diagnostic register (lane0_xcvr_psm_diag)	16	RW	0000h
401Fh	Power state machine user defined control register (lane0_xcvr_psm_user_def_ctrl)	16	RW	0002h
4040h	TX coefficient controller control register (lane0_tx_txcc_ctrl)	16	RO	0000h
4041h	TX pre-cursor override register (lane0_tx_txcc_pre_ovrd)	16	RW	0000h
4042h	TX main-cursor override register (lane0_tx_txcc_main_ovrd)	16	RW	0000h
4043h	TX post-cursor override register (lane0_tx_txcc_post_ovrd)	16	RW	0000h
4044h	TX pre-cursor current value register (lane0_tx_txcc_pre_cval)	16	RO	0000h
4045h	TX main-cursor current value register (lane0_tx_txcc_main_cval)	16	RO	0000h
4046h	TX post-cursor current value register (lane0_tx_txcc_post_cval)	16	RO	0000h
4047h	Resistor calibration code scaler multiplier value register (lane0_tx_txcc_cal_cal_sclr_mult)	16	RW	0080h
4048h	Calculated pre emphasis multiplier value 00 register (lane0_tx_txcc_cpre_mult_00)	16	RW	0000h
4049h	Calculated pre emphasis multiplier value 01 register (lane0_tx_txcc_cpre_mult_01)	16	RW	0000h
404Ah	Calculated pre emphasis multiplier value 10 register (lane0_tx_txcc_cpre_mult_10)	16	RW	0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
404Bh	Calculated pre emphasis multiplier value 11 register (lane0_tx_txcc_cpre_mult_11)	16	RW	0000h
404Ch	Calculated post emphasis multiplier value 00 register (lane0_tx_txcc_cpost_mult_00)	16	RW	0020h
404Dh	Calculated post emphasis multiplier value 01 register (lane0_tx_txcc_cpost_mult_01)	16	RW	0015h
404Eh	Calculated post emphasis multiplier value 10 register (lane0_tx_txcc_cpost_mult_10)	16	RW	0000h
404Fh	Calculated post emphasis multiplier value 11 register (lane0_tx_txcc_cpost_mult_11)	16	RW	0000h
4050h	Margin full swing multiplier value 000 register (lane0_tx_txcc_mgnfs_mult_000)	16	RW	0000h
4051h	Margin full swing multiplier value 001 register (lane0_tx_txcc_mgnfs_mult_001)	16	RW	0000h
4052h	Margin full swing multiplier value 010 register (lane0_tx_txcc_mgnfs_mult_010)	16	RW	0001h
4053h	Margin full swing multiplier value 011 register (lane0_tx_txcc_mgnfs_mult_011)	16	RW	0008h
4054h	Margin full swing multiplier value 100 register (lane0_tx_txcc_mgnfs_mult_100)	16	RW	0011h
4055h	Margin full swing multiplier value 101 register (lane0_tx_txcc_mgnfs_mult_101)	16	RW	0019h
4056h	Margin full swing multiplier value 110 register (lane0_tx_txcc_mgnfs_mult_110)	16	RW	0021h
4057h	Margin full swing multiplier value 111 register (lane0_tx_txcc_mgnfs_mult_111)	16	RW	0029h
4058h	Margin half swing multiplier value 000 register (lane0_tx_txcc_mgnls_mult_000)	16	RW	0018h
4059h	Margin half swing multiplier value 001 register (lane0_tx_txcc_mgnls_mult_001)	16	RW	0018h
405Ah	Margin half swing multiplier value 010 register (lane0_tx_txcc_mgnls_mult_010)	16	RW	001Dh
405Bh	Margin half swing multiplier value 011 register (lane0_tx_txcc_mgnls_mult_011)	16	RW	0021h
405Ch	Margin half swing multiplier value 100 register (lane0_tx_txcc_mgnls_mult_100)	16	RW	0026h
405Dh	Margin half swing multiplier value 101 register (lane0_tx_txcc_mgnls_mult_101)	16	RW	002Bh
405Eh	Margin half swing multiplier value 110 register (lane0_tx_txcc_mgnls_mult_110)	16	RW	002Fh
405Fh	Margin half swing multiplier value 111 register (lane0_tx_txcc_mgnls_mult_111)	16	RW	0034h
40E0h	Transceiver PLL data rate clock control register (lane0_xcvr_diag_plldrc_ctrl)	16	RW	5112h
40E1h	Transceiver high speed clock select register (lane0_xcvr_diag_hscclk_sel)	16	RW	1001h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
40E2h	Transceiver high speed clock A divider control register (lane0_xcvt_d iag_hscika_dctrl)	16	RW	0000h
40E3h	Transceiver high speed clock B divider control register (lane0_xcvt_d iag_hscikb_dctrl)	16	RW	0000h
40E7h	Transceiver control reset diagnostic register (lane0_xcvt_diag_rst_ diag)	16	RO	0000h
40E8h	Transceiver bidirectional control register (lane0_xcvt_diag_bidi_ctrl)	16	RW	007Fh
40E9h	Transceiver power island control register (lane0_xcvt_diag_pwr_ctrl)	16	RW	0000h
40EAh	RX Lane calibration reset timer register (lane0_xcvt_diag_rx_lane_ cal_rst_tmr)	16	RW	0095h
40F0h	Lane fast common mode enable timeout register (lane0_xcvt_diag_ lane_fcm_en_to)	16	RW	0000h
40F1h	Lane fast common mode enable sample wait timer register (lane0_xc vr_diag_lane_fcm_en_swait_tmr)	16	RW	0000h
40F2h	Lane fast common mode enable margin timer register (lane0_xcvt_d iag_lane_fcm_en_mgn_tmr)	16	RW	0096h
40F3h	Lane fast common mode enable tuning register (lane0_xcvt_diag_ lane_fcm_en_tune)	16	RW	0031h
40FFh	Transceiver digital cover your alternatives register (lane0_xcvt_diag_ dcya)	16	RW	0000h
4100h	Transmitter A0 power state definition register (lane0_tx_psc_a0)	16	RW	6799h
4101h	Transmitter A1 power state definition register (lane0_tx_psc_a1)	16	RW	6798h
4102h	Transmitter A2 power state definition register (lane0_tx_psc_a2)	16	RW	0098h
4103h	Transmitter A3 power state definition register (lane0_tx_psc_a3)	16	RW	0098h
4104h	Transmitter A4 power state definition register (lane0_tx_psc_a4)	16	RW	0020h
4105h	Transmitter A5 power state definition register (lane0_tx_psc_a5)	16	RW	0000h
4106h	Transmitter calibration power state definition register (lane0_tx_psc_ cal)	16	RW	0000h
4107h	Transmitter ready power state definition register (lane0_tx_psc_rdy)	16	RW	0000h
4120h	Transmit receiver detect control register (lane0_tx_rcvdet_ctrl)	16	RW	0000h
4121h	Transmit receiver detect override register (lane0_tx_rcvdet_ovrd)	16	RW	0000h
4122h	Transmit receiver detect enable timer register (lane0_tx_rcvdet_en_t mr)	16	RW	09C4h
4123h	Transmit receiver detect start timer register (lane0_tx_rcvdet_st_tmr)	16	RW	0032h
4140h	Transmit BIST control register (lane0_tx_bist_ctrl)	16	RW	0000h
4141h	Transmit BIST user defined data write register (lane0_tx_bist_uddwr)	16	WO	0000h
4142h	Transmit BIST PRBS seed 0 register (lane0_tx_bist_seed0)	16	RW	0001h
4143h	Transmit BIST PRBS seed 1 register (lane0_tx_bist_seed1)	16	RW	0000h
41E0h	TX control register (lane0_tx_diag_tx_ctrl)	16	RW	0000h
41E1h	TX driver control register (lane0_tx_diag_tx_drv)	16	RW	0000h
41E2h	TX electrical idle diagnostic register (lane0_tx_diag_elec_idle)	16	RW	0033h
41E3h	TX sync FIFO diagnostic control register (lane0_tx_diag_sfifo_ctrl)	16	RW	0008h
41E4h	TX sync FIFO diagnostic timer register (lane0_tx_diag_sfifo_tmr)	16	RW	060Ch

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
41E5h	TX receiver detect tuning register (lane0_tx_diag_rdvdet_tune)	16	RW	0000h
41E6h	Transmitter control reset diagnostic register (lane0_tx_diag_rst_diag)	16	RO	0000h
41E7h	TX bandgap reference and pre-drive enable delay register (lane0_tx_diag_bgref_predrv_delay)	16	RW	007Dh
41F0h	TX MPHY control register 1 (lane0_tx_diag_mphy_ctrl1)	16	RW	0000h
41F1h	TX MPHY control register 2 (lane0_tx_diag_mphy_ctrl2)	16	RW	0000h
41F4h	TX driver LDO programming register (lane0_tx_diag_drv_ldo_prog)	16	RW	0000h
41F5h	TX extra enable control override register (lane0_tx_diag_ectrl_ovrd)	16	RW	0000h
41FEh	Transmitter digital cover your alternatives register (lane0_tx_diag_dcya)	16	RW	0000h
41FFh	Transmitter analog cover your alternatives register (lane0_tx_diag_acya)	16	RW	0000h
4400h	Power state machine control register (lane1_xcvr_psm_ctrl)	16	RW	0201h
4401h	Power state machine reset control register (lane1_xcvr_psm_rctrl)	16	RW	BCFCCh
4402h	PSM calibration delay timer register (lane1_xcvr_psm_cal_tmr)	16	RW	0152h
4403h	A0 in delay timer register (lane1_xcvr_psm_a0in_tmr)	16	RW	0152h
4404h	A0 in bypass timer register (lane1_xcvr_psm_a0byp_tmr)	16	RW	0001h
4405h	A1 in delay timer register (lane1_xcvr_psm_a1in_tmr)	16	RW	0008h
4406h	A2 in delay timer register (lane1_xcvr_psm_a2in_tmr)	16	RW	0008h
4407h	A3 in delay timer register (lane1_xcvr_psm_a3in_tmr)	16	RW	0008h
4408h	A4 in delay timer register (lane1_xcvr_psm_a4in_tmr)	16	RW	0008h
4409h	A5 in delay timer register (lane1_xcvr_psm_a5in_tmr)	16	RW	0008h
440Ah	A0 out delay timer register (lane1_xcvr_psm_a0out_tmr)	16	RW	0001h
440Bh	A1 out delay timer register (lane1_xcvr_psm_a1out_tmr)	16	RW	0001h
440Ch	A2 out delay timer register (lane1_xcvr_psm_a2out_tmr)	16	RW	0001h
440Dh	A3 out delay timer register (lane1_xcvr_psm_a3out_tmr)	16	RW	0001h
440Eh	A4 out delay timer register (lane1_xcvr_psm_a4out_tmr)	16	RW	0001h
440Fh	A5 out delay timer register (lane1_xcvr_psm_a5out_tmr)	16	RW	0001h
4410h	Power state machine diagnostic register (lane1_xcvr_psm_diag)	16	RW	0000h
441Fh	Power state machine user defined control register (lane1_xcvr_psm_user_def_ctrl)	16	RW	0002h
4440h	TX coefficient controller control register (lane1_tx_txcc_ctrl)	16	RO	0000h
4441h	TX pre-cursor override register (lane1_tx_txcc_pre_ovrd)	16	RW	0000h
4442h	TX main-cursor override register (lane1_tx_txcc_main_ovrd)	16	RW	0000h
4443h	TX post-cursor override register (lane1_tx_txcc_post_ovrd)	16	RW	0000h
4444h	TX pre-cursor current value register (lane1_tx_txcc_pre_cval)	16	RO	0000h
4445h	TX main-cursor current value register (lane1_tx_txcc_main_cval)	16	RO	0000h
4446h	TX post-cursor current value register (lane1_tx_txcc_post_cval)	16	RO	0000h
4447h	Resistor calibration code scaler multiplier value register (lane1_tx_txcc_cal_sclr_mult)	16	RW	0080h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
4448h	Calculated pre emphasis multiplier value 00 register (lane1_tx_txcc_cpre_mult_00)	16	RW	0000h
4449h	Calculated pre emphasis multiplier value 01 register (lane1_tx_txcc_cpre_mult_01)	16	RW	0000h
444Ah	Calculated pre emphasis multiplier value 10 register (lane1_tx_txcc_cpre_mult_10)	16	RW	0000h
444Bh	Calculated pre emphasis multiplier value 11 register (lane1_tx_txcc_cpre_mult_11)	16	RW	0000h
444Ch	Calculated post emphasis multiplier value 00 register (lane1_tx_txcc_cpost_mult_00)	16	RW	0020h
444Dh	Calculated post emphasis multiplier value 01 register (lane1_tx_txcc_cpost_mult_01)	16	RW	0015h
444Eh	Calculated post emphasis multiplier value 10 register (lane1_tx_txcc_cpost_mult_10)	16	RW	0000h
444Fh	Calculated post emphasis multiplier value 11 register (lane1_tx_txcc_cpost_mult_11)	16	RW	0000h
4450h	Margin full swing multiplier value 000 register (lane1_tx_txcc_mgnfs_mult_000)	16	RW	0000h
4451h	Margin full swing multiplier value 001 register (lane1_tx_txcc_mgnfs_mult_001)	16	RW	0000h
4452h	Margin full swing multiplier value 010 register (lane1_tx_txcc_mgnfs_mult_010)	16	RW	0001h
4453h	Margin full swing multiplier value 011 register (lane1_tx_txcc_mgnfs_mult_011)	16	RW	0008h
4454h	Margin full swing multiplier value 100 register (lane1_tx_txcc_mgnfs_mult_100)	16	RW	0011h
4455h	Margin full swing multiplier value 101 register (lane1_tx_txcc_mgnfs_mult_101)	16	RW	0019h
4456h	Margin full swing multiplier value 110 register (lane1_tx_txcc_mgnfs_mult_110)	16	RW	0021h
4457h	Margin full swing multiplier value 111 register (lane1_tx_txcc_mgnfs_mult_111)	16	RW	0029h
4458h	Margin half swing multiplier value 000 register (lane1_tx_txcc_mgnls_mult_000)	16	RW	0018h
4459h	Margin half swing multiplier value 001 register (lane1_tx_txcc_mgnls_mult_001)	16	RW	0018h
445Ah	Margin half swing multiplier value 010 register (lane1_tx_txcc_mgnls_mult_010)	16	RW	001Dh
445Bh	Margin half swing multiplier value 011 register (lane1_tx_txcc_mgnls_mult_011)	16	RW	0021h
445Ch	Margin half swing multiplier value 100 register (lane1_tx_txcc_mgnls_mult_100)	16	RW	0026h
445Dh	Margin half swing multiplier value 101 register (lane1_tx_txcc_mgnls_mult_101)	16	RW	002Bh
445Eh	Margin half swing multiplier value 110 register (lane1_tx_txcc_mgnls_mult_110)	16	RW	002Fh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
445Fh	Margin half swing multiplier value 111 register (lane1_tx_txcc_mgnls_mult_111)	16	RW	0034h
44E0h	Transceiver PLL data rate clock control register (lane1_xcvr_diag_pllsrc_ctrl)	16	RW	5112h
44E1h	Transceiver high speed clock select register (lane1_xcvr_diag_hsclk_sel)	16	RW	1001h
44E2h	Transceiver high speed clock A divider control register (lane1_xcvr_diag_hsclka_dctrl)	16	RW	0000h
44E3h	Transceiver high speed clock B divider control register (lane1_xcvr_diag_hsclkb_dctrl)	16	RW	0000h
44E7h	Transceiver control reset diagnostic register (lane1_xcvr_diag_rst_diag)	16	RO	0000h
44E8h	Transceiver bidirectional control register (lane1_xcvr_diag_bidi_ctrl)	16	RW	007Fh
44E9h	Transceiver power island control register (lane1_xcvr_diag_pwr_ctrl)	16	RW	0000h
44EAh	RX Lane calibration reset timer register (lane1_xcvr_diag_rx_lane_cal_rst_tmr)	16	RW	0095h
44F0h	Lane fast common mode enable timeout register (lane1_xcvr_diag_lane_fcm_en_to)	16	RW	0000h
44F1h	Lane fast common mode enable sample wait timer register (lane1_xcvr_diag_lane_fcm_en_swait_tmr)	16	RW	0000h
44F2h	Lane fast common mode enable margin timer register (lane1_xcvr_diag_lane_fcm_en_mgn_tmr)	16	RW	0096h
44F3h	Lane fast common mode enable tuning register (lane1_xcvr_diag_lane_fcm_en_tune)	16	RW	0031h
44FFh	Transceiver digital cover your alternatives register (lane1_xcvr_diag_dcya)	16	RW	0000h
4500h	Transmitter A0 power state definition register (lane1_tx_psc_a0)	16	RW	6799h
4501h	Transmitter A1 power state definition register (lane1_tx_psc_a1)	16	RW	6798h
4502h	Transmitter A2 power state definition register (lane1_tx_psc_a2)	16	RW	0098h
4503h	Transmitter A3 power state definition register (lane1_tx_psc_a3)	16	RW	0098h
4504h	Transmitter A4 power state definition register (lane1_tx_psc_a4)	16	RW	0020h
4505h	Transmitter A5 power state definition register (lane1_tx_psc_a5)	16	RW	0000h
4506h	Transmitter calibration power state definition register (lane1_tx_psc_cal)	16	RW	0000h
4507h	Transmitter ready power state definition register (lane1_tx_psc_rdy)	16	RW	0000h
4520h	Transmit receiver detect control register (lane1_tx_rcvdet_ctrl)	16	RW	0000h
4521h	Transmit receiver detect override register (lane1_tx_rcvdet_ovrd)	16	RW	0000h
4522h	Transmit receiver detect enable timer register (lane1_tx_rcvdet_en_tmr)	16	RW	09C4h
4523h	Transmit receiver detect start timer register (lane1_tx_rcvdet_st_tmr)	16	RW	0032h
4540h	Transmit BIST control register (lane1_tx_bist_ctrl)	16	RW	0000h
4541h	Transmit BIST user defined data write register (lane1_tx_bist_uddwr)	16	WO	0000h
4542h	Transmit BIST PRBS seed 0 register (lane1_tx_bist_seed0)	16	RW	0001h
4543h	Transmit BIST PRBS seed 1 register (lane1_tx_bist_seed1)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
45E0h	TX control register (lane1_tx_diag_tx_ctrl)	16	RW	0000h
45E1h	TX driver control register (lane1_tx_diag_tx_drv)	16	RW	0000h
45E2h	TX electrical idle diagnostic register (lane1_tx_diag_elec_idle)	16	RW	0033h
45E3h	TX sync FIFO diagnostic control register (lane1_tx_diag_sfifo_ctrl)	16	RW	0008h
45E4h	TX sync FIFO diagnostic timer register (lane1_tx_diag_sfifo_tmr)	16	RW	060Ch
45E5h	TX receiver detect tuning register (lane1_tx_diag_rdvdet_tune)	16	RW	0000h
45E6h	Transmitter control reset diagnostic register (lane1_tx_diag_rst_diag)	16	RO	0000h
45E7h	TX bandgap reference and pre-drive enable delay register (lane1_tx_diag_bgref_predrv_delay)	16	RW	007Dh
45F0h	TX MPHY control register 1 (lane1_tx_diag_mphy_ctrl1)	16	RW	0000h
45F1h	TX MPHY control register 2 (lane1_tx_diag_mphy_ctrl2)	16	RW	0000h
45F4h	TX driver LDO programming register (lane1_tx_diag_drv_ldo_prog)	16	RW	0000h
45F5h	TX extra enable control override register (lane1_tx_diag_ectrl_ovrd)	16	RW	0000h
45FEh	Transmitter digital cover your alternatives register (lane1_tx_diag_dcya)	16	RW	0000h
45FFh	Transmitter analog cover your alternatives register (lane1_tx_diag_acya)	16	RW	0000h
4800h	Power state machine control register (lane2_xcvr_psm_ctrl)	16	RW	0201h
4801h	Power state machine reset control register (lane2_xcvr_psm_rctrl)	16	RW	BCFCh
4802h	PSM calibration delay timer register (lane2_xcvr_psm_cal_tmr)	16	RW	0152h
4803h	A0 in delay timer register (lane2_xcvr_psm_a0in_tmr)	16	RW	0152h
4804h	A0 in bypass timer register (lane2_xcvr_psm_a0byp_tmr)	16	RW	0001h
4805h	A1 in delay timer register (lane2_xcvr_psm_a1in_tmr)	16	RW	0008h
4806h	A2 in delay timer register (lane2_xcvr_psm_a2in_tmr)	16	RW	0008h
4807h	A3 in delay timer register (lane2_xcvr_psm_a3in_tmr)	16	RW	0008h
4808h	A4 in delay timer register (lane2_xcvr_psm_a4in_tmr)	16	RW	0008h
4809h	A5 in delay timer register (lane2_xcvr_psm_a5in_tmr)	16	RW	0008h
480Ah	A0 out delay timer register (lane2_xcvr_psm_a0out_tmr)	16	RW	0001h
480Bh	A1 out delay timer register (lane2_xcvr_psm_a1out_tmr)	16	RW	0001h
480Ch	A2 out delay timer register (lane2_xcvr_psm_a2out_tmr)	16	RW	0001h
480Dh	A3 out delay timer register (lane2_xcvr_psm_a3out_tmr)	16	RW	0001h
480Eh	A4 out delay timer register (lane2_xcvr_psm_a4out_tmr)	16	RW	0001h
480Fh	A5 out delay timer register (lane2_xcvr_psm_a5out_tmr)	16	RW	0001h
4810h	Power state machine diagnostic register (lane2_xcvr_psm_diag)	16	RW	0000h
481Fh	Power state machine user defined control register (lane2_xcvr_psm_user_def_ctrl)	16	RW	0002h
4840h	TX coefficient controller control register (lane2_tx_txcc_ctrl)	16	RO	0000h
4841h	TX pre-cursor override register (lane2_tx_txcc_pre_ovrd)	16	RW	0000h
4842h	TX main-cursor override register (lane2_tx_txcc_main_ovrd)	16	RW	0000h
4843h	TX post-cursor override register (lane2_tx_txcc_post_ovrd)	16	RW	0000h
4844h	TX pre-cursor current value register (lane2_tx_txcc_pre_cval)	16	RO	0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
4845h	TX main-cursor current value register (lane2_tx_txcc_main_cval)	16	RO	0000h
4846h	TX post-cursor current value register (lane2_tx_txcc_post_cval)	16	RO	0000h
4847h	Resistor calibration code scaler multiplier value register (lane2_tx_txcc_cal_sclr_mult)	16	RW	0080h
4848h	Calculated pre emphasis multiplier value 00 register (lane2_tx_txcc_cp_re_mult_00)	16	RW	0000h
4849h	Calculated pre emphasis multiplier value 01 register (lane2_tx_txcc_cp_re_mult_01)	16	RW	0000h
484Ah	Calculated pre emphasis multiplier value 10 register (lane2_tx_txcc_cp_re_mult_10)	16	RW	0000h
484Bh	Calculated pre emphasis multiplier value 11 register (lane2_tx_txcc_cp_re_mult_11)	16	RW	0000h
484Ch	Calculated post emphasis multiplier value 00 register (lane2_tx_txcc_cp_post_mult_00)	16	RW	0020h
484Dh	Calculated post emphasis multiplier value 01 register (lane2_tx_txcc_cp_post_mult_01)	16	RW	0015h
484Eh	Calculated post emphasis multiplier value 10 register (lane2_tx_txcc_cp_post_mult_10)	16	RW	0000h
484Fh	Calculated post emphasis multiplier value 11 register (lane2_tx_txcc_cp_post_mult_11)	16	RW	0000h
4850h	Margin full swing multiplier value 000 register (lane2_tx_txcc_mgnfs_mult_000)	16	RW	0000h
4851h	Margin full swing multiplier value 001 register (lane2_tx_txcc_mgnfs_mult_001)	16	RW	0000h
4852h	Margin full swing multiplier value 010 register (lane2_tx_txcc_mgnfs_mult_010)	16	RW	0001h
4853h	Margin full swing multiplier value 011 register (lane2_tx_txcc_mgnfs_mult_011)	16	RW	0008h
4854h	Margin full swing multiplier value 100 register (lane2_tx_txcc_mgnfs_mult_100)	16	RW	0011h
4855h	Margin full swing multiplier value 101 register (lane2_tx_txcc_mgnfs_mult_101)	16	RW	0019h
4856h	Margin full swing multiplier value 110 register (lane2_tx_txcc_mgnfs_mult_110)	16	RW	0021h
4857h	Margin full swing multiplier value 111 register (lane2_tx_txcc_mgnfs_mult_111)	16	RW	0029h
4858h	Margin half swing multiplier value 000 register (lane2_tx_txcc_mgnls_mult_000)	16	RW	0018h
4859h	Margin half swing multiplier value 001 register (lane2_tx_txcc_mgnls_mult_001)	16	RW	0018h
485Ah	Margin half swing multiplier value 010 register (lane2_tx_txcc_mgnls_mult_010)	16	RW	001Dh
485Bh	Margin half swing multiplier value 011 register (lane2_tx_txcc_mgnls_mult_011)	16	RW	0021h
485Ch	Margin half swing multiplier value 100 register (lane2_tx_txcc_mgnls_mult_100)	16	RW	0026h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
485Dh	Margin half swing multiplier value 101 register (lane2_tx_txcc_mgnls_mult_101)	16	RW	002Bh
485Eh	Margin half swing multiplier value 110 register (lane2_tx_txcc_mgnls_mult_110)	16	RW	002Fh
485Fh	Margin half swing multiplier value 111 register (lane2_tx_txcc_mgnls_mult_111)	16	RW	0034h
48E0h	Transceiver PLL data rate clock control register (lane2_xcvr_diag_pllsrc_ctrl)	16	RW	5112h
48E1h	Transceiver high speed clock select register (lane2_xcvr_diag_hsclk_sel)	16	RW	1001h
48E2h	Transceiver high speed clock A divider control register (lane2_xcvr_diag_hsclka_dctrl)	16	RW	0000h
48E3h	Transceiver high speed clock B divider control register (lane2_xcvr_diag_hsclkb_dctrl)	16	RW	0000h
48E7h	Transceiver control reset diagnostic register (lane2_xcvr_diag_rst_diag)	16	RO	0000h
48E8h	Transceiver bidirectional control register (lane2_xcvr_diag_bidi_ctrl)	16	RW	007Fh
48E9h	Transceiver power island control register (lane2_xcvr_diag_pwr_ctrl)	16	RW	0000h
48EAh	RX Lane calibration reset timer register (lane2_xcvr_diag_rx_lane_cal_rst_tmr)	16	RW	0095h
48F0h	Lane fast common mode enable timeout register (lane2_xcvr_diag_lane_fcm_en_to)	16	RW	0000h
48F1h	Lane fast common mode enable sample wait timer register (lane2_xcvr_diag_lane_fcm_en_swait_tmr)	16	RW	0000h
48F2h	Lane fast common mode enable margin timer register (lane2_xcvr_diag_lane_fcm_en_mgn_tmr)	16	RW	0096h
48F3h	Lane fast common mode enable tuning register (lane2_xcvr_diag_lane_fcm_en_tune)	16	RW	0031h
48FFh	Transceiver digital cover your alternatives register (lane2_xcvr_diag_dcya)	16	RW	0000h
4900h	Transmitter A0 power state definition register (lane2_tx_psc_a0)	16	RW	6799h
4901h	Transmitter A1 power state definition register (lane2_tx_psc_a1)	16	RW	6798h
4902h	Transmitter A2 power state definition register (lane2_tx_psc_a2)	16	RW	0098h
4903h	Transmitter A3 power state definition register (lane2_tx_psc_a3)	16	RW	0098h
4904h	Transmitter A4 power state definition register (lane2_tx_psc_a4)	16	RW	0020h
4905h	Transmitter A5 power state definition register (lane2_tx_psc_a5)	16	RW	0000h
4906h	Transmitter calibration power state definition register (lane2_tx_psc_cal)	16	RW	0000h
4907h	Transmitter ready power state definition register (lane2_tx_psc_rdy)	16	RW	0000h
4920h	Transmit receiver detect control register (lane2_tx_rcvdet_ctrl)	16	RW	0000h
4921h	Transmit receiver detect override register (lane2_tx_rcvdet_ovrd)	16	RW	0000h
4922h	Transmit receiver detect enable timer register (lane2_tx_rcvdet_en_tmr)	16	RW	09C4h
4923h	Transmit receiver detect start timer register (lane2_tx_rcvdet_st_tmr)	16	RW	0032h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
4940h	Transmit BIST control register (lane2_tx_bist_ctrl)	16	RW	0000h
4941h	Transmit BIST user defined data write register (lane2_tx_bist_uddwr)	16	WO	0000h
4942h	Transmit BIST PRBS seed 0 register (lane2_tx_bist_seed0)	16	RW	0001h
4943h	Transmit BIST PRBS seed 1 register (lane2_tx_bist_seed1)	16	RW	0000h
49E0h	TX control register (lane2_tx_diag_tx_ctrl)	16	RW	0000h
49E1h	TX driver control register (lane2_tx_diag_tx_drv)	16	RW	0000h
49E2h	TX electrical idle diagnostic register (lane2_tx_diag_elec_idle)	16	RW	0033h
49E3h	TX sync FIFO diagnostic control register (lane2_tx_diag_sfifo_ctrl)	16	RW	0008h
49E4h	TX sync FIFO diagnostic timer register (lane2_tx_diag_sfifo_tmr)	16	RW	060Ch
49E5h	TX receiver detect tuning register (lane2_tx_diag_rdvdet_tune)	16	RW	0000h
49E6h	Transmitter control reset diagnostic register (lane2_tx_diag_rst_diag)	16	RO	0000h
49E7h	TX bandgap reference and pre-drive enable delay register (lane2_tx_diag_bgref_predrv_delay)	16	RW	007Dh
49F0h	TX MPHY control register 1 (lane2_tx_diag_mphy_ctrl1)	16	RW	0000h
49F1h	TX MPHY control register 2 (lane2_tx_diag_mphy_ctrl2)	16	RW	0000h
49F4h	TX driver LDO programming register (lane2_tx_diag_drv_ldo_prog)	16	RW	0000h
49F5h	TX extra enable control override register (lane2_tx_diag_ectrl_ovrd)	16	RW	0000h
49FEh	Transmitter digital cover your alternatives register (lane2_tx_diag_dc ya)	16	RW	0000h
49FFh	Transmitter analog cover your alternatives register (lane2_tx_diag_ac ya)	16	RW	0000h
4C00h	Power state machine control register (lane3_xcvr_psm_ctrl)	16	RW	0201h
4C01h	Power state machine reset control register (lane3_xcvr_psm_rctrl)	16	RW	BCFCh
4C02h	PSM calibration delay timer register (lane3_xcvr_psm_cal_tmr)	16	RW	0152h
4C03h	A0 in delay timer register (lane3_xcvr_psm_a0in_tmr)	16	RW	0152h
4C04h	A0 in bypass timer register (lane3_xcvr_psm_a0byp_tmr)	16	RW	0001h
4C05h	A1 in delay timer register (lane3_xcvr_psm_a1in_tmr)	16	RW	0008h
4C06h	A2 in delay timer register (lane3_xcvr_psm_a2in_tmr)	16	RW	0008h
4C07h	A3 in delay timer register (lane3_xcvr_psm_a3in_tmr)	16	RW	0008h
4C08h	A4 in delay timer register (lane3_xcvr_psm_a4in_tmr)	16	RW	0008h
4C09h	A5 in delay timer register (lane3_xcvr_psm_a5in_tmr)	16	RW	0008h
4C0Ah	A0 out delay timer register (lane3_xcvr_psm_a0out_tmr)	16	RW	0001h
4C0Bh	A1 out delay timer register (lane3_xcvr_psm_a1out_tmr)	16	RW	0001h
4C0Ch	A2 out delay timer register (lane3_xcvr_psm_a2out_tmr)	16	RW	0001h
4C0Dh	A3 out delay timer register (lane3_xcvr_psm_a3out_tmr)	16	RW	0001h
4C0Eh	A4 out delay timer register (lane3_xcvr_psm_a4out_tmr)	16	RW	0001h
4C0Fh	A5 out delay timer register (lane3_xcvr_psm_a5out_tmr)	16	RW	0001h
4C10h	Power state machine diagnostic register (lane3_xcvr_psm_diag)	16	RW	0000h
4C1Fh	Power state machine user defined control register (lane3_xcvr_psm_user_def_ctrl)	16	RW	0002h
4C40h	TX coefficient controller control register (lane3_tx_txcc_ctrl)	16	RO	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
4C41h	TX pre-cursor override register (lane3_tx_txcc_pre_ovrd)	16	RW	0000h
4C42h	TX main-cursor override register (lane3_tx_txcc_main_ovrd)	16	RW	0000h
4C43h	TX post-cursor override register (lane3_tx_txcc_post_ovrd)	16	RW	0000h
4C44h	TX pre-cursor current value register (lane3_tx_txcc_pre_cval)	16	RO	0000h
4C45h	TX main-cursor current value register (lane3_tx_txcc_main_cval)	16	RO	0000h
4C46h	TX post-cursor current value register (lane3_tx_txcc_post_cval)	16	RO	0000h
4C47h	Resistor calibration code scaler multiplier value register (lane3_tx_txcc_cal_sclr_mult)	16	RW	0080h
4C48h	Calculated pre emphasis multiplier value 00 register (lane3_tx_txcc_cpre_mult_00)	16	RW	0000h
4C49h	Calculated pre emphasis multiplier value 01 register (lane3_tx_txcc_cpre_mult_01)	16	RW	0000h
4C4Ah	Calculated pre emphasis multiplier value 10 register (lane3_tx_txcc_cpre_mult_10)	16	RW	0000h
4C4Bh	Calculated pre emphasis multiplier value 11 register (lane3_tx_txcc_cpre_mult_11)	16	RW	0000h
4C4Ch	Calculated post emphasis multiplier value 00 register (lane3_tx_txcc_cpost_mult_00)	16	RW	0020h
4C4Dh	Calculated post emphasis multiplier value 01 register (lane3_tx_txcc_cpost_mult_01)	16	RW	0015h
4C4Eh	Calculated post emphasis multiplier value 10 register (lane3_tx_txcc_cpost_mult_10)	16	RW	0000h
4C4Fh	Calculated post emphasis multiplier value 11 register (lane3_tx_txcc_cpost_mult_11)	16	RW	0000h
4C50h	Margin full swing multiplier value 000 register (lane3_tx_txcc_mgnfs_mult_000)	16	RW	0000h
4C51h	Margin full swing multiplier value 001 register (lane3_tx_txcc_mgnfs_mult_001)	16	RW	0000h
4C52h	Margin full swing multiplier value 010 register (lane3_tx_txcc_mgnfs_mult_010)	16	RW	0001h
4C53h	Margin full swing multiplier value 011 register (lane3_tx_txcc_mgnfs_mult_011)	16	RW	0008h
4C54h	Margin full swing multiplier value 100 register (lane3_tx_txcc_mgnfs_mult_100)	16	RW	0011h
4C55h	Margin full swing multiplier value 101 register (lane3_tx_txcc_mgnfs_mult_101)	16	RW	0019h
4C56h	Margin full swing multiplier value 110 register (lane3_tx_txcc_mgnfs_mult_110)	16	RW	0021h
4C57h	Margin full swing multiplier value 111 register (lane3_tx_txcc_mgnfs_mult_111)	16	RW	0029h
4C58h	Margin half swing multiplier value 000 register (lane3_tx_txcc_mgnls_mult_000)	16	RW	0018h
4C59h	Margin half swing multiplier value 001 register (lane3_tx_txcc_mgnls_mult_001)	16	RW	0018h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
4C5Ah	Margin half swing multiplier value 010 register (lane3_tx_txcc_mgnls_mult_010)	16	RW	001Dh
4C5Bh	Margin half swing multiplier value 011 register (lane3_tx_txcc_mgnls_mult_011)	16	RW	0021h
4C5Ch	Margin half swing multiplier value 100 register (lane3_tx_txcc_mgnls_mult_100)	16	RW	0026h
4C5Dh	Margin half swing multiplier value 101 register (lane3_tx_txcc_mgnls_mult_101)	16	RW	002Bh
4C5Eh	Margin half swing multiplier value 110 register (lane3_tx_txcc_mgnls_mult_110)	16	RW	002Fh
4C5Fh	Margin half swing multiplier value 111 register (lane3_tx_txcc_mgnls_mult_111)	16	RW	0034h
4CE0h	Transceiver PLL data rate clock control register (lane3_xcvr_diag_pllsrc_ctrl)	16	RW	5112h
4CE1h	Transceiver high speed clock select register (lane3_xcvr_diag_hsclk_sel)	16	RW	1001h
4CE2h	Transceiver high speed clock A divider control register (lane3_xcvr_diag_hsclka_dctrl)	16	RW	0000h
4CE3h	Transceiver high speed clock B divider control register (lane3_xcvr_diag_hsclkb_dctrl)	16	RW	0000h
4CE7h	Transceiver control reset diagnostic register (lane3_xcvr_diag_rst_diag)	16	RO	0000h
4CE8h	Transceiver bidirectional control register (lane3_xcvr_diag_bidi_ctrl)	16	RW	007Fh
4CE9h	Transceiver power island control register (lane3_xcvr_diag_pwr_ctrl)	16	RW	0000h
4CEAh	RX Lane calibration reset timer register (lane3_xcvr_diag_rx_lane_cal_rst_tmr)	16	RW	0095h
4CF0h	Lane fast common mode enable timeout register (lane3_xcvr_diag_lane_fcm_en_to)	16	RW	0000h
4CF1h	Lane fast common mode enable sample wait timer register (lane3_xcvr_diag_lane_fcm_en_swait_tmr)	16	RW	0000h
4CF2h	Lane fast common mode enable margin timer register (lane3_xcvr_diag_lane_fcm_en_mgn_tmr)	16	RW	0096h
4CF3h	Lane fast common mode enable tuning register (lane3_xcvr_diag_lane_fcm_en_tune)	16	RW	0031h
4CFFh	Transceiver digital cover your alternatives register (lane3_xcvr_diag_dcya)	16	RW	0000h
4D00h	Transmitter A0 power state definition register (lane3_tx_psc_a0)	16	RW	6799h
4D01h	Transmitter A1 power state definition register (lane3_tx_psc_a1)	16	RW	6798h
4D02h	Transmitter A2 power state definition register (lane3_tx_psc_a2)	16	RW	0098h
4D03h	Transmitter A3 power state definition register (lane3_tx_psc_a3)	16	RW	0098h
4D04h	Transmitter A4 power state definition register (lane3_tx_psc_a4)	16	RW	0020h
4D05h	Transmitter A5 power state definition register (lane3_tx_psc_a5)	16	RW	0000h
4D06h	Transmitter calibration power state definition register (lane3_tx_psc_cal)	16	RW	0000h
4D07h	Transmitter ready power state definition register (lane3_tx_psc_rdy)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
4D20h	Transmit receiver detect control register (lane3_tx_rcvdet_ctrl)	16	RW	0000h
4D21h	Transmit receiver detect override register (lane3_tx_rcvdet_ovrd)	16	RW	0000h
4D22h	Transmit receiver detect enable timer register (lane3_tx_rcvdet_en_tmr)	16	RW	09C4h
4D23h	Transmit receiver detect start timer register (lane3_tx_rcvdet_st_tmr)	16	RW	0032h
4D40h	Transmit BIST control register (lane3_tx_bist_ctrl)	16	RW	0000h
4D41h	Transmit BIST user defined data write register (lane3_tx_bist_uddwr)	16	WO	0000h
4D42h	Transmit BIST PRBS seed 0 register (lane3_tx_bist_seed0)	16	RW	0001h
4D43h	Transmit BIST PRBS seed 1 register (lane3_tx_bist_seed1)	16	RW	0000h
4DE0h	TX control register (lane3_tx_diag_tx_ctrl)	16	RW	0000h
4DE1h	TX driver control register (lane3_tx_diag_tx_drv)	16	RW	0000h
4DE2h	TX electrical idle diagnostic register (lane3_tx_diag_elec_idle)	16	RW	0033h
4DE3h	TX sync FIFO diagnostic control register (lane3_tx_diag_sfifo_ctrl)	16	RW	0008h
4DE4h	TX sync FIFO diagnostic timer register (lane3_tx_diag_sfifo_tmr)	16	RW	060Ch
4DE5h	TX receiver detect tuning register (lane3_tx_diag_rdvdet_tune)	16	RW	0000h
4DE6h	Transmitter control reset diagnostic register (lane3_tx_diag_rst_diag)	16	RO	0000h
4DE7h	TX bandgap reference and pre-drive enable delay register (lane3_tx_diag_bgref_predrv_delay)	16	RW	007Dh
4DF0h	TX MPHY control register 1 (lane3_tx_diag_mphy_ctrl1)	16	RW	0000h
4DF1h	TX MPHY control register 2 (lane3_tx_diag_mphy_ctrl2)	16	RW	0000h
4DF4h	TX driver LDO programming register (lane3_tx_diag_drv_ldo_prog)	16	RW	0000h
4DF5h	TX extra enable control override register (lane3_tx_diag_ectrl_ovrd)	16	RW	0000h
4DFEh	Transmitter digital cover your alternatives register (lane3_tx_diag_dcya)	16	RW	0000h
4DFFh	Transmitter analog cover your alternatives register (lane3_tx_diag_acya)	16	RW	0000h
8000h	Receiver A0 power state definition register (lane0_rx_psc_a0)	16	RW	8BFDh
8001h	Receiver A1 power state definition register (lane0_rx_psc_a1)	16	RW	8BFDh
8002h	Receiver A2 power state definition register (lane0_rx_psc_a2)	16	RW	8910h
8003h	Receiver A3 power state definition register (lane0_rx_psc_a3)	16	RW	0000h
8004h	Receiver A4 power state definition register (lane0_rx_psc_a4)	16	RW	1000h
8005h	Receiver A5 power state definition register (lane0_rx_psc_a5)	16	RW	0000h
8006h	Receiver calibration power state definition register (lane0_rx_psc_cal)	16	RW	03FFh
8007h	Receiver ready power state definition register (lane0_rx_psc_rdy)	16	RW	0000h
8020h	RX IQ PI ILL calibration control register (lane0_rx_iqpi_ill_cal_ctrl)	16	RW	0018h
8021h	RX IQ PI ILL calibration start point register (lane0_rx_iqpi_ill_cal_start)	16	RW	1018h
8022h	RX IQ PI ILL calibration timer control register (lane0_rx_iqpi_ill_cal_ctrl)	16	RW	0003h
8023h	RX IQ PI ILL calibration override register (lane0_rx_iqpi_ill_cal_ovrd)	16	RW	0000h
8024h	RX IQ PI ILL calibration initialization timer register (lane0_rx_iqpi_ill_cal_init_tmr)	16	RW	02EEh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8025h	RX IQ PI ILL calibration iteration timer register (lane0_rx_iqpi_ill_cal_iter_tmr)	16	RW	0080h
8026h	RX IQ PI ILL lock reference timer start value register (lane0_rx_iqpi_ill_lock_reftmr_start)	16	RW	01FFh
8028h	RX IQ PI ILL lock calibration counter start value standard mode 0 register (lane0_rx_iqpi_ill_lock_calcnt_start_0)	16	RW	007Fh
8029h	RX IQ PI ILL lock calibration counter start value standard mode 1 register (lane0_rx_iqpi_ill_lock_calcnt_start_1)	16	RW	00FFh
802Ah	RX IQ PI ILL lock calibration counter start value standard mode 2 register (lane0_rx_iqpi_ill_lock_calcnt_start_2)	16	RW	00FFh
802Bh	RX IQ PI ILL lock calibration counter start value standard mode 3 register (lane0_rx_iqpi_ill_lock_calcnt_start_3)	16	RW	0000h
8030h	RX E PI ILL calibration control register (lane0_rx_epi_ill_cal_ctrl)	16	RW	0018h
8031h	RX E PI ILL calibration start point register (lane0_rx_epi_ill_cal_start)	16	RW	1018h
8032h	RX E PI ILL calibration timer control register (lane0_rx_epi_ill_cal_tctrl)	16	RW	0003h
8033h	RX E PI ILL calibration override register (lane0_rx_epi_ill_cal_ovrd)	16	RW	0000h
8034h	RX E PI ILL calibration initialization timer register (lane0_rx_epi_ill_cal_init_tmr)	16	RW	02EEh
8035h	RX E PI ILL calibration iteration timer register (lane0_rx_epi_ill_cal_iter_tmr)	16	RW	0080h
8036h	RX E PI ILL lock reference timer start value register (lane0_rx_epi_ill_lock_reftmr_start)	16	RW	01FFh
8038h	RX E PI ILL lock calibration counter start value standard mode 0 register (lane0_rx_epi_ill_lock_calcnt_start_0)	16	RW	007Fh
8039h	RX E PI ILL lock calibration counter start value standard mode 1 register (lane0_rx_epi_ill_lock_calcnt_start_1)	16	RW	00FFh
803Ah	RX E PI ILL lock calibration counter start value standard mode 2 register (lane0_rx_epi_ill_lock_calcnt_start_2)	16	RW	00FFh
803Bh	RX E PI ILL lock calibration counter start value standard mode 3 register (lane0_rx_epi_ill_lock_calcnt_start_3)	16	RW	0000h
8040h	Signal detect calibration 0 control register (lane0_rx_sdcal0_ctrl)	16	RW	0000h
8041h	Signal detect calibration 0 override register (lane0_rx_sdcal0_ovrd)	16	RW	0000h
8042h	Signal detect calibration 0 start register (lane0_rx_sdcal0_start)	16	RW	0000h
8043h	Signal detect calibration 0 tune register (lane0_rx_sdcal0_tune)	16	RW	0000h
8044h	Signal detect calibration 0 initialization timer register (lane0_rx_sdcal0_init_tmr)	16	RW	0019h
8045h	Signal detect calibration 0 iteration timer register (lane0_rx_sdcal0_iter_tmr)	16	RW	007Dh
8048h	Signal detect calibration 1 control register (lane0_rx_sdcal1_ctrl)	16	RW	0000h
8049h	Signal detect calibration 1 override register (lane0_rx_sdcal1_ovrd)	16	RW	0000h
804Ah	Signal detect calibration 1 start register (lane0_rx_sdcal1_start)	16	RW	0000h
804Bh	Signal detect calibration 1 tune register (lane0_rx_sdcal1_tune)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
804Ch	Signal detect calibration 1 initialization timer register (lane0_rx_sdc al1_init_tmr)	16	RW	0019h
804Dh	Signal detect calibration 1 iteration timer register (lane0_rx_sdc al1_iter_tmr)	16	RW	007Dh
8058h	Sampler error DAC control register (lane0_rx_samp_dac_ctrl)	16	RW	0014h
8080h	CDRLF configuration register (lane0_rx_cdrf_cnf)	16	RW	20B3h
8081h	CDRLF configuration register 2 (lane0_rx_cdrf_cnf2)	16	RW	0000h
8082h	CDRLF margin diagnostic register (lane0_rx_cdrf_mgn_diag)	16	RW	0000h
8083h	CDRLF fast phase lock timer value register 0 (lane0_rx_cdrf_fpl_t mr0)	16	RW	0071h
8084h	CDRLF fast phase lock timer value register 1 (lane0_rx_cdrf_fpl_t mr1)	16	RW	0731h
8085h	CDRLF fast frequency lock timer value register (lane0_rx_cdrf_ffl_t mr)	16	RW	0018h
8088h	CDRLF fast frequency lock step 0 control register (lane0_rx_cdrf_f fl0_ctrl)	16	RW	460Ch
8089h	CDRLF fast frequency lock step 1 control register (lane0_rx_cdrf_f fl1_ctrl)	16	RW	4A0Ah
808Ah	CDRLF fast frequency lock step 2 control register (lane0_rx_cdrf_f fl2_ctrl)	16	RW	5414h
808Bh	CDRLF fast frequency lock step 3 control register (lane0_rx_cdrf_f fl3_ctrl)	16	RW	5E1Eh
808Ch	CDRLF fast frequency lock step 4 control register (lane0_rx_cdrf_f fl4_ctrl)	16	RW	9E1Eh
8090h	Receiver signal detect filter high to low filter timer register (lane0_rx_ sigdet_hl_filt_tmr)	16	RW	0019h
8091h	Receiver signal detect filter high to low delay timer register (lane0_rx_ sigdet_hl_dly_tmr)	16	RW	0000h
8092h	Receiver signal detect filter high to low min timer register (lane0_rx_ sigdet_hl_min_tmr)	16	RW	0000h
8093h	Receiver signal detect filter high to low init timer register (lane0_rx_ sigdet_hl_init_tmr)	16	RW	03E8h
8094h	Receiver signal detect filter low to high filter timer register (lane0_rx_ sigdet_lh_filt_tmr)	16	RW	0004h
8095h	Receiver signal detect filter low to high delay timer register (lane0_rx_ sigdet_lh_dly_tmr)	16	RW	0004h
8096h	Receiver signal detect filter low to high min timer register (lane0_rx_ sigdet_lh_min_tmr)	16	RW	0000h
8097h	Receiver signal detect filter low to high init timer register (lane0_rx_ sigdet_lh_init_tmr)	16	RW	03E8h
8098h	Receiver LFPS detect filter filter timer register (lane0_rx_lfpsdet_filt_ tmr)	16	RW	0008h
8099h	Receiver LFPS detect filter delay timer register (lane0_rx_lfpsdet_dly_ tmr)	16	RW	000Bh
809Ah	Receiver LFPS detect min timer register (lane0_rx_lfpsdet_min_tmr)	16	RW	0013h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
809Bh	Receiver LFPS detect init timer register (lane0_rx_lfpsdet_init_tmr)	16	RW	0000h
80A0h	Eye surf control register (lane0_rx_eyesurf_ctrl)	16	RW	0000h
80A4h	Eye surf timer delay low register (lane0_rx_eyesurf_tmr_dellow)	16	RW	0000h
80A5h	Eye surf timer delay high register (lane0_rx_eyesurf_tmr_delhigh)	16	RW	0000h
80A6h	Eye surf timer test low register (lane0_rx_eyesurf_tmr_testlow)	16	RW	0000h
80A7h	Eye surf timer test high register (lane0_rx_eyesurf_tmr_testhigh)	16	RW	0000h
80A8h	Eye surf north south test point coordinate register (lane0_rx_eyesurf_ns_coord)	16	RW	0000h
80A9h	Eye surf east west test point coordinate register (lane0_rx_eyesurf_ew_coord)	16	RW	0000h
80AAh	Eye surf bit error count register (lane0_rx_eyesurf_errcnt)	16	RO	0000h
80B0h	Receiver BIST control register (lane0_rx_bist_ctrl)	16	RW	0000h
80B1h	Receiver BIST sync count register (lane0_rx_bist_syncnt)	16	RW	0000h
80B2h	Receiver BIST user defined data write register (lane0_rx_bist_uddwr)	16	WO	0000h
80B3h	Receiver BIST error count register (lane0_rx_bist_errcnt)	16	RO	0000h
80B4h	Clock frequency measurement control register (lane0_xcvr_cmsmt_clk_freq_msmt_ctrl)	16	RW	0000h
80B5h	Test clock selection register (lane0_xcvr_cmsmt_test_clk_sel)	16	RW	0000h
80B6h	Reference clock timer value register (lane0_xcvr_cmsmt_ref_clk_tmr_value)	16	RW	0000h
80B7h	Test clock counter value register (lane0_xcvr_cmsmt_test_clk_cnt_value)	16	RO	0000h
80E0h	RX sampler latch calibration control register (lane0_rx_slc_ctrl)	16	RW	0EFFh
80E1h	RX sampler latch calibration enable initialization timer value register (lane0_rx_slc_en_init_tmr)	16	RW	0004h
80E2h	RX sampler latch calibration unit initialization timer value register (lane0_rx_slc_cu_init_tmr)	16	RW	0010h
80E3h	RX sampler latch calibration unit iteration timer value register (lane0_rx_slc_cu_iter_tmr)	16	RW	0001h
80E4h	RX sampler latch calibration I even data mask register (lane0_rx_slc_ie_mask)	16	RW	03FFh
80E5h	RX sampler latch calibration I odd data mask register (lane0_rx_slc_io_mask)	16	RW	03FFh
80E6h	RX sampler latch calibration Q even data mask register (lane0_rx_slc_qe_mask)	16	RW	03FFh
80E7h	RX sampler latch calibration Q odd data mask register (lane0_rx_slc_qo_mask)	16	RW	03FFh
80E8h	RX sampler latch calibration E even data mask register (lane0_rx_slc_ee_mask)	16	RW	03FFh
80E9h	RX sampler latch calibration E odd data mask register (lane0_rx_slc_eo_mask)	16	RW	03FFh
80EAh	RX sampler latch calibration data threshold register (lane0_rx_slc_data_thr)	16	RW	0005h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8100h	RX sampler latch I odd positive 0 calibration unit control register (lane0_rx_slc_iop0_ctrl)	16	RW	0021h
8101h	RX sampler latch I odd positive 0 calibration unit override register (lane0_rx_slc_iop0_ovrd)	16	RW	0000h
8102h	RX sampler latch I odd positive 0 calibration unit start register (lane0_rx_slc_iop0_start)	16	RW	0021h
8103h	RX sampler latch I odd positive 0 calibration unit tune register (lane0_rx_slc_iop0_tune)	16	RW	0000h
8104h	RX sampler latch I odd positive 1 calibration unit control register (lane0_rx_slc_iop1_ctrl)	16	RW	001Fh
8105h	RX sampler latch I odd positive 1 calibration unit override register (lane0_rx_slc_iop1_ovrd)	16	RW	0000h
8106h	RX sampler latch I odd positive 1 calibration unit start register (lane0_rx_slc_iop1_start)	16	RW	801Fh
8107h	RX sampler latch I odd positive 1 calibration unit tune register (lane0_rx_slc_iop1_tune)	16	RW	0000h
8108h	RX sampler latch Q odd positive 0 calibration unit control register (lane0_rx_slc_qop0_ctrl)	16	RW	0021h
8109h	RX sampler latch Q odd positive 0 calibration unit override register (lane0_rx_slc_qop0_ovrd)	16	RW	0000h
810Ah	RX sampler latch Q odd positive 0 calibration unit start register (lane0_rx_slc_qop0_start)	16	RW	0021h
810Bh	RX sampler latch Q odd positive 0 calibration unit tune register (lane0_rx_slc_qop0_tune)	16	RW	0000h
810Ch	RX sampler latch Q odd positive 1 calibration unit control register (lane0_rx_slc_qop1_ctrl)	16	RW	001Fh
810Dh	RX sampler latch Q odd positive 1 calibration unit override register (lane0_rx_slc_qop1_ovrd)	16	RW	0000h
810Eh	RX sampler latch Q odd positive 1 calibration unit start register (lane0_rx_slc_qop1_start)	16	RW	801Fh
810Fh	RX sampler latch Q odd positive 1 calibration unit tune register (lane0_rx_slc_qop1_tune)	16	RW	0000h
8110h	RX sampler latch E odd positive 0 calibration unit control register (lane0_rx_slc_eop0_ctrl)	16	RW	0021h
8111h	RX sampler latch E odd positive 0 calibration unit override register (lane0_rx_slc_eop0_ovrd)	16	RW	0000h
8112h	RX sampler latch E odd positive 0 calibration unit start register (lane0_rx_slc_eop0_start)	16	RW	0021h
8113h	RX sampler latch E odd positive 0 calibration unit tune register (lane0_rx_slc_eop0_tune)	16	RW	0000h
8114h	RX sampler latch E odd positive 1 calibration unit control register (lane0_rx_slc_eop1_ctrl)	16	RW	001Fh
8115h	RX sampler latch E odd positive 1 calibration unit override register (lane0_rx_slc_eop1_ovrd)	16	RW	0000h
8116h	RX sampler latch E odd positive 1 calibration unit start register (lane0_rx_slc_eop1_start)	16	RW	801Fh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8117h	RX sampler latch E odd positive 1 calibration unit tune register (lane0_rx_slc_eop1_tune)	16	RW	0000h
8118h	RX sampler latch I odd negative 0 calibration unit control register (lane0_rx_slc_ion0_ctrl)	16	RW	0021h
8119h	RX sampler latch I odd negative 0 calibration unit override register (lane0_rx_slc_ion0_ovrd)	16	RW	0000h
811Ah	RX sampler latch I odd negative 0 calibration unit start register (lane0_rx_slc_ion0_start)	16	RW	0021h
811Bh	RX sampler latch I odd negative 0 calibration unit tune register (lane0_rx_slc_ion0_tune)	16	RW	0000h
811Ch	RX sampler latch I odd negative 1 calibration unit control register (lane0_rx_slc_ion1_ctrl)	16	RW	001Fh
811Dh	RX sampler latch I odd negative 1 calibration unit override register (lane0_rx_slc_ion1_ovrd)	16	RW	0000h
811Eh	RX sampler latch I odd negative 1 calibration unit start register (lane0_rx_slc_ion1_start)	16	RW	801Fh
811Fh	RX sampler latch I odd negative 1 calibration unit tune register (lane0_rx_slc_ion1_tune)	16	RW	0000h
8120h	RX sampler latch Q odd negative 0 calibration unit control register (lane0_rx_slc_qon0_ctrl)	16	RW	0021h
8121h	RX sampler latch Q odd negative 0 calibration unit override register (lane0_rx_slc_qon0_ovrd)	16	RW	0000h
8122h	RX sampler latch Q odd negative 0 calibration unit start register (lane0_rx_slc_qon0_start)	16	RW	0021h
8123h	RX sampler latch Q odd negative 0 calibration unit tune register (lane0_rx_slc_qon0_tune)	16	RW	0000h
8124h	RX sampler latch Q odd negative 1 calibration unit control register (lane0_rx_slc_qon1_ctrl)	16	RW	001Fh
8125h	RX sampler latch Q odd negative 1 calibration unit override register (lane0_rx_slc_qon1_ovrd)	16	RW	0000h
8126h	RX sampler latch Q odd negative 1 calibration unit start register (lane0_rx_slc_qon1_start)	16	RW	801Fh
8127h	RX sampler latch Q odd negative 1 calibration unit tune register (lane0_rx_slc_qon1_tune)	16	RW	0000h
8128h	RX sampler latch E odd negative 0 calibration unit control register (lane0_rx_slc_eon0_ctrl)	16	RW	0021h
8129h	RX sampler latch E odd negative 0 calibration unit override register (lane0_rx_slc_eon0_ovrd)	16	RW	0000h
812Ah	RX sampler latch E odd negative 0 calibration unit start register (lane0_rx_slc_eon0_start)	16	RW	0021h
812Bh	RX sampler latch E odd negative 0 calibration unit tune register (lane0_rx_slc_eon0_tune)	16	RW	0000h
812Ch	RX sampler latch E odd negative 1 calibration unit control register (lane0_rx_slc_eon1_ctrl)	16	RW	001Fh
812Dh	RX sampler latch E odd negative 1 calibration unit override register (lane0_rx_slc_eon1_ovrd)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
812Eh	RX sampler latch E odd negative 1 calibration unit start register (lane0_rx_slc_eon1_start)	16	RW	801Fh
812Fh	RX sampler latch E odd negative 1 calibration unit tune register (lane0_rx_slc_eon1_tune)	16	RW	0000h
8130h	RX sampler latch I even positive 0 calibration unit control register (lane0_rx_slc_iep0_ctrl)	16	RW	0021h
8131h	RX sampler latch I even positive 0 calibration unit override register (lane0_rx_slc_iep0_ovrd)	16	RW	0000h
8132h	RX sampler latch I even positive 0 calibration unit start register (lane0_rx_slc_iep0_start)	16	RW	0021h
8133h	RX sampler latch I even positive 0 calibration unit tune register (lane0_rx_slc_iep0_tune)	16	RW	0000h
8134h	RX sampler latch I even positive 1 calibration unit control register (lane0_rx_slc_iep1_ctrl)	16	RW	001Fh
8135h	RX sampler latch I even positive 1 calibration unit override register (lane0_rx_slc_iep1_ovrd)	16	RW	0000h
8136h	RX sampler latch I even positive 1 calibration unit start register (lane0_rx_slc_iep1_start)	16	RW	801Fh
8137h	RX sampler latch I even positive 1 calibration unit tune register (lane0_rx_slc_iep1_tune)	16	RW	0000h
8138h	RX sampler latch Q even positive 0 calibration unit control register (lane0_rx_slc_qep0_ctrl)	16	RW	0021h
8139h	RX sampler latch Q even positive 0 calibration unit override register (lane0_rx_slc_qep0_ovrd)	16	RW	0000h
813Ah	RX sampler latch Q even positive 0 calibration unit start register (lane0_rx_slc_qep0_start)	16	RW	0021h
813Bh	RX sampler latch Q even positive 0 calibration unit tune register (lane0_rx_slc_qep0_tune)	16	RW	0000h
813Ch	RX sampler latch Q even positive 1 calibration unit control register (lane0_rx_slc_qep1_ctrl)	16	RW	001Fh
813Dh	RX sampler latch Q even positive 1 calibration unit override register (lane0_rx_slc_qep1_ovrd)	16	RW	0000h
813Eh	RX sampler latch Q even positive 1 calibration unit start register (lane0_rx_slc_qep1_start)	16	RW	801Fh
813Fh	RX sampler latch Q even positive 1 calibration unit tune register (lane0_rx_slc_qep1_tune)	16	RW	0000h
8140h	RX sampler latch E even positive 0 calibration unit control register (lane0_rx_slc_eep0_ctrl)	16	RW	0021h
8141h	RX sampler latch E even positive 0 calibration unit override register (lane0_rx_slc_eep0_ovrd)	16	RW	0000h
8142h	RX sampler latch E even positive 0 calibration unit start register (lane0_rx_slc_eep0_start)	16	RW	0021h
8143h	RX sampler latch E even positive 0 calibration unit tune register (lane0_rx_slc_eep0_tune)	16	RW	0000h
8144h	RX sampler latch E even positive 1 calibration unit control register (lane0_rx_slc_eep1_ctrl)	16	RW	001Fh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8145h	RX sampler latch E even positive 1 calibration unit override register (lane0_rx_slc_eep1_ovrd)	16	RW	0000h
8146h	RX sampler latch E even positive 1 calibration unit start register (lane0_rx_slc_eep1_start)	16	RW	801Fh
8147h	RX sampler latch E even positive 1 calibration unit tune register (lane0_rx_slc_eep1_tune)	16	RW	0000h
8148h	RX sampler latch I even negative 0 calibration unit control register (lane0_rx_slc_ien0_ctrl)	16	RW	0021h
8149h	RX sampler latch I even negative 0 calibration unit override register (lane0_rx_slc_ien0_ovrd)	16	RW	0000h
814Ah	RX sampler latch I even negative 0 calibration unit start register (lane0_rx_slc_ien0_start)	16	RW	0021h
814Bh	RX sampler latch I even negative 0 calibration unit tune register (lane0_rx_slc_ien0_tune)	16	RW	0000h
814Ch	RX sampler latch I even negative 1 calibration unit control register (lane0_rx_slc_ien1_ctrl)	16	RW	001Fh
814Dh	RX sampler latch I even negative 1 calibration unit override register (lane0_rx_slc_ien1_ovrd)	16	RW	0000h
814Eh	RX sampler latch I even negative 1 calibration unit start register (lane0_rx_slc_ien1_start)	16	RW	801Fh
814Fh	RX sampler latch I even negative 1 calibration unit tune register (lane0_rx_slc_ien1_tune)	16	RW	0000h
8150h	RX sampler latch Q even negative 0 calibration unit control register (lane0_rx_slc_qen0_ctrl)	16	RW	0021h
8151h	RX sampler latch Q even negative 0 calibration unit override register (lane0_rx_slc_qen0_ovrd)	16	RW	0000h
8152h	RX sampler latch Q even negative 0 calibration unit start register (lane0_rx_slc_qen0_start)	16	RW	0021h
8153h	RX sampler latch Q even negative 0 calibration unit tune register (lane0_rx_slc_qen0_tune)	16	RW	0000h
8154h	RX sampler latch Q even negative 1 calibration unit control register (lane0_rx_slc_qen1_ctrl)	16	RW	001Fh
8155h	RX sampler latch Q even negative 1 calibration unit override register (lane0_rx_slc_qen1_ovrd)	16	RW	0000h
8156h	RX sampler latch Q even negative 1 calibration unit start register (lane0_rx_slc_qen1_start)	16	RW	801Fh
8157h	RX sampler latch Q even negative 1 calibration unit tune register (lane0_rx_slc_qen1_tune)	16	RW	0000h
8158h	RX sampler latch E even negative 0 calibration unit control register (lane0_rx_slc_een0_ctrl)	16	RW	0021h
8159h	RX sampler latch E even negative 0 calibration unit override register (lane0_rx_slc_een0_ovrd)	16	RW	0000h
815Ah	RX sampler latch E even negative 0 calibration unit start register (lane0_rx_slc_een0_start)	16	RW	0021h
815Bh	RX sampler latch E even negative 0 calibration unit tune register (lane0_rx_slc_een0_tune)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
815Ch	RX sampler latch E even negative 1 calibration unit control register (lane0_rx_slc_eeen1_ctrl)	16	RW	001Fh
815Dh	RX sampler latch E even negative 1 calibration unit override register (lane0_rx_slc_eeen1_ovrd)	16	RW	0000h
815Eh	RX sampler latch E even negative 1 calibration unit start register (lane0_rx_slc_eeen1_start)	16	RW	801Fh
815Fh	RX sampler latch E even negative 1 calibration unit tune register (lane0_rx_slc_eeen1_tune)	16	RW	0000h
8160h	REE USB 3 general control state machine control register (lane0_rx_ree_u3gcsn_ctrl)	16	RW	0001h
8161h	REE USB 3 general control state machine phase 1 equalization enable mask register (lane0_rx_ree_u3gcsn_eqenm_ph1)	16	RW	00C7h
8162h	REE USB 3 general control state machine phase 2 equalization enable mask register (lane0_rx_ree_u3gcsn_eqenm_ph2)	16	RW	00C7h
8163h	REE USB 3 general control state machine start timer value register (lane0_rx_ree_u3gcsn_start_tmr)	16	RW	0125h
8164h	REE USB 3 general control state machine run phase 1 timer value register (lane0_rx_ree_u3gcsn_run_ph1_tmr)	16	RW	07A2h
8165h	REE USB 3 general control state machine run phase 2 timer value register (lane0_rx_ree_u3gcsn_run_ph2_tmr)	16	RW	009Dh
8168h	REE PCIe Gen 2 general control state machine control register (lane0_rx_ree_g2gcsn_ctrl)	16	RW	0001h
8169h	REE PCIe Gen 2 general control state machine phase 1 equalization enable mask register (lane0_rx_ree_g2gcsn_eqenm_ph1)	16	RW	00C7h
816Ah	REE PCIe Gen 2 general control state machine phase 2 equalization enable mask register (lane0_rx_ree_g2gcsn_eqenm_ph2)	16	RW	00C7h
816Bh	REE PCIe Gen 2 general control state machine start timer value register (lane0_rx_ree_g2gcsn_start_tmr)	16	RW	0000h
816Ch	REE PCIe Gen 2 general control state machine run phase 1 timer value register (lane0_rx_ree_g2gcsn_run_ph1_tmr)	16	RW	0F43h
816Dh	REE PCIe Gen 2 general control state machine run phase 2 timer value register (lane0_rx_ree_g2gcsn_run_ph2_tmr)	16	RW	009Dh
8178h	REE periodic general control state machine control register (lane0_rx_ree_pergcsn_ctrl)	16	RW	0001h
8179h	REE periodic general control state machine phase 1 equalization enable mask register (lane0_rx_ree_pergcsn_eqenm_ph1)	16	RW	0080h
817Ah	REE periodic general control state machine phase 2 equalization enable mask register (lane0_rx_ree_pergcsn_eqenm_ph2)	16	RW	0080h
817Bh	REE periodic general control state machine start timer value register (lane0_rx_ree_pergcsn_start_tmr)	16	RW	0000h
817Ch	REE periodic general control state machine run phase 1 timer value register (lane0_rx_ree_pergcsn_run_ph1_tmr)	16	RW	FFFFh
817Dh	REE periodic general control state machine run phase 2 timer value register (lane0_rx_ree_pergcsn_run_ph2_tmr)	16	RW	009Dh
8180h	REE tap 1 control register (lane0_rx_ree_tap1_ctrl)	16	RW	0000h
8181h	REE tap 1 override register (lane0_rx_ree_tap1_ovrd)	16	RW	0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8182h	REE tap 1 diagnostics register (lane0_rx_ree_tap1_diag)	16	RW	0000h
8184h	REE tap 2 control register (lane0_rx_ree_tap2_ctrl)	16	RW	0400h
8185h	REE tap 2 override register (lane0_rx_ree_tap2_ovrd)	16	RW	0000h
8186h	REE tap 2 diagnostics register (lane0_rx_ree_tap2_diag)	16	RW	0000h
8188h	REE tap 3 control register (lane0_rx_ree_tap3_ctrl)	16	RW	0400h
8189h	REE tap 3 override register (lane0_rx_ree_tap3_ovrd)	16	RW	0000h
818Ah	REE tap 3 diagnostics register (lane0_rx_ree_tap3_diag)	16	RW	0000h
8194h	REE analog enable control state machine delay timer value register (lane0_rx_ree_anaensm_del_tmr)	16	RW	02EEh
8198h	REE peaking amp control register (lane0_rx_ree_peak_ctrl)	16	RW	0701h
8199h	REE peaking amp code control register (lane0_rx_ree_peak_code_ctrl)	16	RW	2A0Eh
819Ah	REE peaking amp upper threshold register (lane0_rx_ree_peak_uthr)	16	RW	0004h
819Bh	REE peaking amp lower threshold register (lane0_rx_ree_peak_lthr)	16	RW	0000h
819Ch	REE peaking amp input override register (lane0_rx_ree_peak_iovrd)	16	RW	0000h
819Dh	REE peaking amp code override register (lane0_rx_ree_peak_covrd)	16	RW	0000h
819Eh	REE peaking amp diagnostics register (lane0_rx_ree_peak_diag)	16	RW	0020h
81A0h	REE attenuation control register (lane0_rx_ree_atten_ctrl)	16	RW	0005h
81A1h	REE attenuation threshold register (lane0_rx_ree_atten_thr)	16	RW	0C02h
81A2h	REE attenuation counter register (lane0_rx_ree_atten_cnt)	16	RW	2100h
81A3h	REE attenuation override register (lane0_rx_ree_atten_ovrd)	16	RW	0000h
81A4h	REE attenuation diagnostics register (lane0_rx_ree_atten_diag)	16	RO	0000h
81A8h	REE low frequency equalizer control register (lane0_rx_ree_lfeq_ctrl)	16	RW	0000h
81A9h	REE low frequency equalizer override register (lane0_rx_ree_lfeq_ovrd)	16	RW	0000h
81AAh	REE low frequency equalizer diagnostics register (lane0_rx_ree_lfeq_diag)	16	RW	0000h
81ACh	REE VGA gain control register (lane0_rx_ree_vga_gain_ctrl)	16	RW	1001h
81ADh	REE VGA gain override register (lane0_rx_ree_vga_gain_ovrd)	16	RW	0000h
81AEh	REE VGA gain diagnostics register (lane0_rx_ree_vga_gain_diag)	16	RW	000Ah
81AFh	REE VGA gain target adjust diagnostics register (lane0_rx_ree_vga_gain_tgt_diag)	16	RO	0000h
81B0h	REE offset correction control register (lane0_rx_ree_off_cor_ctrl)	16	RW	0001h
81B1h	REE offset correction override register (lane0_rx_ree_off_cor_ovrd)	16	RW	0000h
81B2h	REE offset correction diagnostics register (lane0_rx_ree_off_cor_diag)	16	RW	0000h
81B8h	REE adder configuration register (lane0_rx_ree_addr_cfg)	16	RW	0001h
81B9h	REE tap 1 clip control register (lane0_rx_ree_tap1_clip)	16	RW	0519h
81BAh	REE taps 2 and 3 clip control register (lane0_rx_ree_tap2ton_clip)	16	RW	0519h
81BBh	REE control data mask register (lane0_rx_ree_ctrl_data_mask)	16	RW	4000h
81BCh	REE diagnostic control register (lane0_rx_ree_diag_ctrl)	16	RW	0040h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
81BDh	REE control state machine gen mode control register 1 (lane0_rx_ree_smgm_ctrl1)	16	RW	0702h
81BEh	REE control state machine gen mode control register 2 (lane0_rx_ree_smgm_ctrl2)	16	RW	0000h
81C0h	RX ILL diagnostic control register (lane0_rx_diag_ill_ctrl)	16	RW	0000h
81C1h	RX ILL IQ trim 0 register (lane0_rx_diag_ill_iq_trim0)	16	RW	533Fh
81C2h	RX ILL E trim 0 register (lane0_rx_diag_ill_e_trim0)	16	RW	533Fh
81C3h	RX ILL IQ trim 1 register (lane0_rx_diag_ill_iq_trim1)	16	RW	0002h
81C4h	RX ILL E trim 1 register (lane0_rx_diag_ill_e_trim1)	16	RW	0002h
81C5h	RX ILL IQ E trim 2 register (lane0_rx_diag_ill_iqe_trim2)	16	RW	3E00h
81C6h	RX ILL IQ E trim 3 register (lane0_rx_diag_ill_iqe_trim3)	16	RW	199Fh
81C7h	RX ILL IQ E trim 4 register (lane0_rx_diag_ill_iqe_trim4)	16	RW	0014h
81C8h	RX ILL IQ E trim 5 register (lane0_rx_diag_ill_iqe_trim5)	16	RW	1504h
81C9h	RX ILL IQ E trim 6 register (lane0_rx_diag_ill_iqe_trim6)	16	RW	0015h
81D0h	DFE amp fine tuning register (lane0_rx_diag_dfe_amp_tune)	16	RW	4CCCh
81D1h	DFE amp fine tuning 2 register (lane0_rx_diag_dfe_amp_tune_2)	16	RW	0C21h
81D2h	REE DAC control register (lane0_rx_diag_ree_dac_ctrl)	16	RW	0004h
81D3h	Receiver DFE control register 1 (lane0_rx_diag_dfe_ctrl1)	16	RW	0007h
81D4h	Receiver DFE control register 2 (lane0_rx_diag_dfe_ctrl2)	16	RW	003Eh
81D5h	Receiver DFE control register 3 (lane0_rx_diag_dfe_ctrl3)	16	RW	0ACAh
81D6h	Nyquist control register (lane0_rx_diag_nqst_ctrl)	16	RW	0998h
81D7h	Low frequency equalizer tuning register (lane0_rx_diag_lfeq_tune)	16	RW	0014h
81D8h	RX control register (lane0_rx_diag_rxctrl)	16	RW	0060h
81D9h	Receiver control reset diagnostic register (lane0_rx_diag_rst_diag)	16	RO	0000h
81DCh	RX signal detect tuning and control register (lane0_rx_diag_sigdet_tune)	16	RW	1005h
81DDh	RX LFPS detect tuning and control register (lane0_rx_diag_lfpsdet_tune)	16	RW	2410h
81DEh	Signal detect test register (lane0_rx_diag_sd_test)	16	RW	0000h
81E0h	RX sampler diagnostic control register (lane0_rx_diag_samp_ctrl)	16	RW	0001h
81E1h	RX Sampler CML TO CMOS enable delay register (lane0_rx_diag_sc2c_delay)	16	RW	012Ch
81E4h	MPHY control register 1 (lane0_rx_diag_mphy_ctrl_1)	16	RW	0000h
81E5h	MPHY control register 2 (lane0_rx_diag_mphy_ctrl_2)	16	RW	0000h
81E8h	RX loopback controller register (lane0_rx_diag_lpbk_ctrl)	16	RW	0000h
81E9h	RX extra enable control override register (lane0_rx_diag_ectrl_ovrd)	16	RW	0000h
81F0h	CML to CMOS bias trim register (lane0_rx_diag_cml2cmos_btrim)	16	RW	0000h
81F1h	RX bias gen control register 1 (lane0_rx_diag_bias_gen_ctrl1)	16	RW	0000h
81F2h	RX bias gen control register 2 (lane0_rx_diag_bias_gen_ctrl2)	16	RW	0000h
81F3h	RX bias gen control register 3 (lane0_rx_diag_bias_gen_ctrl3)	16	RW	0000h
81F4h	RX bias gen control register 4 (lane0_rx_diag_bias_gen_ctrl4)	16	RW	0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
81F5h	RX boundary scan test mode register (lane0_rx_diag_bs_tm)	16	RW	0000h
81F6h	RX receiver front end test mode register 1 (lane0_rx_diag_rxfe_tm1)	16	RW	0000h
81F7h	RX receiver front end test mode register 2 (lane0_rx_diag_rxfe_tm2)	16	RW	0000h
81FEh	Receiver digital cover your alternatives register (lane0_rx_diag_dcya)	16	RW	0000h
81FFh	Receiver analog cover your alternatives register (lane0_rx_diag_acya)	16	RW	0000h
8400h	Receiver A0 power state definition register (lane1_rx_psc_a0)	16	RW	8BFDh
8401h	Receiver A1 power state definition register (lane1_rx_psc_a1)	16	RW	8BFDh
8402h	Receiver A2 power state definition register (lane1_rx_psc_a2)	16	RW	8910h
8403h	Receiver A3 power state definition register (lane1_rx_psc_a3)	16	RW	0000h
8404h	Receiver A4 power state definition register (lane1_rx_psc_a4)	16	RW	1000h
8405h	Receiver A5 power state definition register (lane1_rx_psc_a5)	16	RW	0000h
8406h	Receiver calibration power state definition register (lane1_rx_psc_cal)	16	RW	03FFh
8407h	Receiver ready power state definition register (lane1_rx_psc_rdy)	16	RW	0000h
8420h	RX IQ PI ILL calibration control register (lane1_rx_iqpi_ill_cal_ctrl)	16	RW	0018h
8421h	RX IQ PI ILL calibration start point register (lane1_rx_iqpi_ill_cal_start)	16	RW	1018h
8422h	RX IQ PI ILL calibration timer control register (lane1_rx_iqpi_ill_cal_ctrl)	16	RW	0003h
8423h	RX IQ PI ILL calibration override register (lane1_rx_iqpi_ill_cal_ovrd)	16	RW	0000h
8424h	RX IQ PI ILL calibration initialization timer register (lane1_rx_iqpi_ill_cal_init_tmr)	16	RW	02EEh
8425h	RX IQ PI ILL calibration iteration timer register (lane1_rx_iqpi_ill_cal_iter_tmr)	16	RW	0080h
8426h	RX IQ PI ILL lock reference timer start value register (lane1_rx_iqpi_ill_lock_ref_tmr_start)	16	RW	01FFh
8428h	RX IQ PI ILL lock calibration counter start value standard mode 0 register (lane1_rx_iqpi_ill_lock_calcnt_start_0)	16	RW	007Fh
8429h	RX IQ PI ILL lock calibration counter start value standard mode 1 register (lane1_rx_iqpi_ill_lock_calcnt_start_1)	16	RW	00FFh
842Ah	RX IQ PI ILL lock calibration counter start value standard mode 2 register (lane1_rx_iqpi_ill_lock_calcnt_start_2)	16	RW	00FFh
842Bh	RX IQ PI ILL lock calibration counter start value standard mode 3 register (lane1_rx_iqpi_ill_lock_calcnt_start_3)	16	RW	0000h
8430h	RX E PI ILL calibration control register (lane1_rx_epi_ill_cal_ctrl)	16	RW	0018h
8431h	RX E PI ILL calibration start point register (lane1_rx_epi_ill_cal_start)	16	RW	1018h
8432h	RX E PI ILL calibration timer control register (lane1_rx_epi_ill_cal_ctrl)	16	RW	0003h
8433h	RX E PI ILL calibration override register (lane1_rx_epi_ill_cal_ovrd)	16	RW	0000h
8434h	RX E PI ILL calibration initialization timer register (lane1_rx_epi_ill_cal_init_tmr)	16	RW	02EEh
8435h	RX E PI ILL calibration iteration timer register (lane1_rx_epi_ill_cal_iter_tmr)	16	RW	0080h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8436h	RX E PI ILL lock reference timer start value register (lane1_rx_epi_ill_lock_reftrmr_start)	16	RW	01FFh
8438h	RX E PI ILL lock calibration counter start value standard mode 0 register (lane1_rx_epi_ill_lock_calcnt_start_0)	16	RW	007Fh
8439h	RX E PI ILL lock calibration counter start value standard mode 1 register (lane1_rx_epi_ill_lock_calcnt_start_1)	16	RW	00FFh
843Ah	RX E PI ILL lock calibration counter start value standard mode 2 register (lane1_rx_epi_ill_lock_calcnt_start_2)	16	RW	00FFh
843Bh	RX E PI ILL lock calibration counter start value standard mode 3 register (lane1_rx_epi_ill_lock_calcnt_start_3)	16	RW	0000h
8440h	Signal detect calibration 0 control register (lane1_rx_sdcal0_ctrl)	16	RW	0000h
8441h	Signal detect calibration 0 override register (lane1_rx_sdcal0_ovrd)	16	RW	0000h
8442h	Signal detect calibration 0 start register (lane1_rx_sdcal0_start)	16	RW	0000h
8443h	Signal detect calibration 0 tune register (lane1_rx_sdcal0_tune)	16	RW	0000h
8444h	Signal detect calibration 0 initialization timer register (lane1_rx_sdcal0_init_tmr)	16	RW	0019h
8445h	Signal detect calibration 0 iteration timer register (lane1_rx_sdcal0_iter_tmr)	16	RW	007Dh
8448h	Signal detect calibration 1 control register (lane1_rx_sdcal1_ctrl)	16	RW	0000h
8449h	Signal detect calibration 1 override register (lane1_rx_sdcal1_ovrd)	16	RW	0000h
844Ah	Signal detect calibration 1 start register (lane1_rx_sdcal1_start)	16	RW	0000h
844Bh	Signal detect calibration 1 tune register (lane1_rx_sdcal1_tune)	16	RW	0000h
844Ch	Signal detect calibration 1 initialization timer register (lane1_rx_sdcal1_init_tmr)	16	RW	0019h
844Dh	Signal detect calibration 1 iteration timer register (lane1_rx_sdcal1_iter_tmr)	16	RW	007Dh
8458h	Sampler error DAC control register (lane1_rx_samp_dac_ctrl)	16	RW	0014h
8480h	CDRLF configuration register (lane1_rx_cdrflf_cnfg)	16	RW	20B3h
8481h	CDRLF configuration register 2 (lane1_rx_cdrflf_cnfg2)	16	RW	0000h
8482h	CDRLF margin diagnostic register (lane1_rx_cdrflf_mgn_diag)	16	RW	0000h
8483h	CDRLF fast phase lock timer value register 0 (lane1_rx_cdrflf_fpl_tmr0)	16	RW	0071h
8484h	CDRLF fast phase lock timer value register 1 (lane1_rx_cdrflf_fpl_tmr1)	16	RW	0731h
8485h	CDRLF fast frequency lock timer value register (lane1_rx_cdrflf_ffl_tmr)	16	RW	0018h
8488h	CDRLF fast frequency lock step 0 control register (lane1_rx_cdrflf_fl0_ctrl)	16	RW	460Ch
8489h	CDRLF fast frequency lock step 1 control register (lane1_rx_cdrflf_fl1_ctrl)	16	RW	4A0Ah
848Ah	CDRLF fast frequency lock step 2 control register (lane1_rx_cdrflf_fl2_ctrl)	16	RW	5414h
848Bh	CDRLF fast frequency lock step 3 control register (lane1_rx_cdrflf_fl3_ctrl)	16	RW	5E1Eh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
848Ch	CDRLF fast frequency lock step 4 control register (lane1_rx_cdrf_fl4_ctrl)	16	RW	9E1Eh
8490h	Receiver signal detect filter high to low filter timer register (lane1_rx_sigdet_hl_filt_tmr)	16	RW	0019h
8491h	Receiver signal detect filter high to low delay timer register (lane1_rx_sigdet_hl_dly_tmr)	16	RW	0000h
8492h	Receiver signal detect filter high to low min timer register (lane1_rx_sigdet_hl_min_tmr)	16	RW	0000h
8493h	Receiver signal detect filter high to low init timer register (lane1_rx_sigdet_hl_init_tmr)	16	RW	03E8h
8494h	Receiver signal detect filter low to high filter timer register (lane1_rx_sigdet_lh_filt_tmr)	16	RW	0004h
8495h	Receiver signal detect filter low to high delay timer register (lane1_rx_sigdet_lh_dly_tmr)	16	RW	0004h
8496h	Receiver signal detect filter low to high min timer register (lane1_rx_sigdet_lh_min_tmr)	16	RW	0000h
8497h	Receiver signal detect filter low to high init timer register (lane1_rx_sigdet_lh_init_tmr)	16	RW	03E8h
8498h	Receiver LFPS detect filter filter timer register (lane1_rx_lfpsdet_filt_tmr)	16	RW	0008h
8499h	Receiver LFPS detect filter delay timer register (lane1_rx_lfpsdet_dly_tmr)	16	RW	000Bh
849Ah	Receiver LFPS detect min timer register (lane1_rx_lfpsdet_min_tmr)	16	RW	0013h
849Bh	Receiver LFPS detect init timer register (lane1_rx_lfpsdet_init_tmr)	16	RW	0000h
84A0h	Eye surf control register (lane1_rx_eyesurf_ctrl)	16	RW	0000h
84A4h	Eye surf timer delay low register (lane1_rx_eyesurf_tmr_dellow)	16	RW	0000h
84A5h	Eye surf timer delay high register (lane1_rx_eyesurf_tmr_delhigh)	16	RW	0000h
84A6h	Eye surf timer test low register (lane1_rx_eyesurf_tmr_testlow)	16	RW	0000h
84A7h	Eye surf timer test high register (lane1_rx_eyesurf_tmr_testhigh)	16	RW	0000h
84A8h	Eye surf north south test point coordinate register (lane1_rx_eyesurf_ns_coord)	16	RW	0000h
84A9h	Eye surf east west test point coordinate register (lane1_rx_eyesurf_ew_coord)	16	RW	0000h
84AAh	Eye surf bit error count register (lane1_rx_eyesurf_errcnt)	16	RO	0000h
84B0h	Receiver BIST control register (lane1_rx_bist_ctrl)	16	RW	0000h
84B1h	Receiver BIST sync count register (lane1_rx_bist_synccnt)	16	RW	0000h
84B2h	Receiver BIST user defined data write register (lane1_rx_bist_uddwr)	16	WO	0000h
84B3h	Receiver BIST error count register (lane1_rx_bist_errcnt)	16	RO	0000h
84B4h	Clock frequency measurement control register (lane1_xcvr_cmsmt_clk_freq_msmt_ctrl)	16	RW	0000h
84B5h	Test clock selection register (lane1_xcvr_cmsmt_test_clk_sel)	16	RW	0000h
84B6h	Reference clock timer value register (lane1_xcvr_cmsmt_ref_clk_tmr_value)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
84B7h	Test clock counter value register (lane1_xcvt_cmsmt_test_clk_cnt_value)	16	RO	0000h
84E0h	RX sampler latch calibration control register (lane1_rx_slc_ctrl)	16	RW	0EFFh
84E1h	RX sampler latch calibration enable initialization timer value register (lane1_rx_slc_en_init_tmr)	16	RW	0004h
84E2h	RX sampler latch calibration unit initialization timer value register (lane1_rx_slc_cu_init_tmr)	16	RW	0010h
84E3h	RX sampler latch calibration unit iteration timer value register (lane1_rx_slc_cu_iter_tmr)	16	RW	0001h
84E4h	RX sampler latch calibration I even data mask register (lane1_rx_slc_ie_mask)	16	RW	03FFh
84E5h	RX sampler latch calibration I odd data mask register (lane1_rx_slc_io_mask)	16	RW	03FFh
84E6h	RX sampler latch calibration Q even data mask register (lane1_rx_slc_qe_mask)	16	RW	03FFh
84E7h	RX sampler latch calibration Q odd data mask register (lane1_rx_slc_qo_mask)	16	RW	03FFh
84E8h	RX sampler latch calibration E even data mask register (lane1_rx_slc_ee_mask)	16	RW	03FFh
84E9h	RX sampler latch calibration E odd data mask register (lane1_rx_slc_eo_mask)	16	RW	03FFh
84EAh	RX sampler latch calibration data threshold register (lane1_rx_slc_data_thr)	16	RW	0005h
8500h	RX sampler latch I odd positive 0 calibration unit control register (lane1_rx_slc_iop0_ctrl)	16	RW	0021h
8501h	RX sampler latch I odd positive 0 calibration unit override register (lane1_rx_slc_iop0_ovrd)	16	RW	0000h
8502h	RX sampler latch I odd positive 0 calibration unit start register (lane1_rx_slc_iop0_start)	16	RW	0021h
8503h	RX sampler latch I odd positive 0 calibration unit tune register (lane1_rx_slc_iop0_tune)	16	RW	0000h
8504h	RX sampler latch I odd positive 1 calibration unit control register (lane1_rx_slc_iop1_ctrl)	16	RW	001Fh
8505h	RX sampler latch I odd positive 1 calibration unit override register (lane1_rx_slc_iop1_ovrd)	16	RW	0000h
8506h	RX sampler latch I odd positive 1 calibration unit start register (lane1_rx_slc_iop1_start)	16	RW	801Fh
8507h	RX sampler latch I odd positive 1 calibration unit tune register (lane1_rx_slc_iop1_tune)	16	RW	0000h
8508h	RX sampler latch Q odd positive 0 calibration unit control register (lane1_rx_slc_qop0_ctrl)	16	RW	0021h
8509h	RX sampler latch Q odd positive 0 calibration unit override register (lane1_rx_slc_qop0_ovrd)	16	RW	0000h
850Ah	RX sampler latch Q odd positive 0 calibration unit start register (lane1_rx_slc_qop0_start)	16	RW	0021h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
850Bh	RX sampler latch Q odd positive 0 calibration unit tune register (lane 1_rx_slc_qop0_tune)	16	RW	0000h
850Ch	RX sampler latch Q odd positive 1 calibration unit control register (lane1_rx_slc_qop1_ctrl)	16	RW	001Fh
850Dh	RX sampler latch Q odd positive 1 calibration unit override register (lane1_rx_slc_qop1_ovrd)	16	RW	0000h
850Eh	RX sampler latch Q odd positive 1 calibration unit start register (lane 1_rx_slc_qop1_start)	16	RW	801Fh
850Fh	RX sampler latch Q odd positive 1 calibration unit tune register (lane 1_rx_slc_qop1_tune)	16	RW	0000h
8510h	RX sampler latch E odd positive 0 calibration unit control register (lane1_rx_slc_eop0_ctrl)	16	RW	0021h
8511h	RX sampler latch E odd positive 0 calibration unit override register (lane1_rx_slc_eop0_ovrd)	16	RW	0000h
8512h	RX sampler latch E odd positive 0 calibration unit start register (lane 1_rx_slc_eop0_start)	16	RW	0021h
8513h	RX sampler latch E odd positive 0 calibration unit tune register (lane 1_rx_slc_eop0_tune)	16	RW	0000h
8514h	RX sampler latch E odd positive 1 calibration unit control register (lane1_rx_slc_eop1_ctrl)	16	RW	001Fh
8515h	RX sampler latch E odd positive 1 calibration unit override register (lane1_rx_slc_eop1_ovrd)	16	RW	0000h
8516h	RX sampler latch E odd positive 1 calibration unit start register (lane 1_rx_slc_eop1_start)	16	RW	801Fh
8517h	RX sampler latch E odd positive 1 calibration unit tune register (lane 1_rx_slc_eop1_tune)	16	RW	0000h
8518h	RX sampler latch I odd negative 0 calibration unit control register (lane1_rx_slc_ion0_ctrl)	16	RW	0021h
8519h	RX sampler latch I odd negative 0 calibration unit override register (lane1_rx_slc_ion0_ovrd)	16	RW	0000h
851Ah	RX sampler latch I odd negative 0 calibration unit start register (lane 1_rx_slc_ion0_start)	16	RW	0021h
851Bh	RX sampler latch I odd negative 0 calibration unit tune register (lane 1_rx_slc_ion0_tune)	16	RW	0000h
851Ch	RX sampler latch I odd negative 1 calibration unit control register (lane1_rx_slc_ion1_ctrl)	16	RW	001Fh
851Dh	RX sampler latch I odd negative 1 calibration unit override register (lane1_rx_slc_ion1_ovrd)	16	RW	0000h
851Eh	RX sampler latch I odd negative 1 calibration unit start register (lane 1_rx_slc_ion1_start)	16	RW	801Fh
851Fh	RX sampler latch I odd negative 1 calibration unit tune register (lane 1_rx_slc_ion1_tune)	16	RW	0000h
8520h	RX sampler latch Q odd negative 0 calibration unit control register (lane1_rx_slc_qon0_ctrl)	16	RW	0021h
8521h	RX sampler latch Q odd negative 0 calibration unit override register (lane1_rx_slc_qon0_ovrd)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8522h	RX sampler latch Q odd negative 0 calibration unit start register (lane 1_rx_slc_qon0_start)	16	RW	0021h
8523h	RX sampler latch Q odd negative 0 calibration unit tune register (lane 1_rx_slc_qon0_tune)	16	RW	0000h
8524h	RX sampler latch Q odd negative 1 calibration unit control register (lane1_rx_slc_qon1_ctrl)	16	RW	001Fh
8525h	RX sampler latch Q odd negative 1 calibration unit override register (lane1_rx_slc_qon1_ovrd)	16	RW	0000h
8526h	RX sampler latch Q odd negative 1 calibration unit start register (lane 1_rx_slc_qon1_start)	16	RW	801Fh
8527h	RX sampler latch Q odd negative 1 calibration unit tune register (lane 1_rx_slc_qon1_tune)	16	RW	0000h
8528h	RX sampler latch E odd negative 0 calibration unit control register (lane1_rx_slc_eon0_ctrl)	16	RW	0021h
8529h	RX sampler latch E odd negative 0 calibration unit override register (lane1_rx_slc_eon0_ovrd)	16	RW	0000h
852Ah	RX sampler latch E odd negative 0 calibration unit start register (lane 1_rx_slc_eon0_start)	16	RW	0021h
852Bh	RX sampler latch E odd negative 0 calibration unit tune register (lane 1_rx_slc_eon0_tune)	16	RW	0000h
852Ch	RX sampler latch E odd negative 1 calibration unit control register (lane1_rx_slc_eon1_ctrl)	16	RW	001Fh
852Dh	RX sampler latch E odd negative 1 calibration unit override register (lane1_rx_slc_eon1_ovrd)	16	RW	0000h
852Eh	RX sampler latch E odd negative 1 calibration unit start register (lane 1_rx_slc_eon1_start)	16	RW	801Fh
852Fh	RX sampler latch E odd negative 1 calibration unit tune register (lane 1_rx_slc_eon1_tune)	16	RW	0000h
8530h	RX sampler latch I even positive 0 calibration unit control register (lane1_rx_slc_iep0_ctrl)	16	RW	0021h
8531h	RX sampler latch I even positive 0 calibration unit override register (lane1_rx_slc_iep0_ovrd)	16	RW	0000h
8532h	RX sampler latch I even positive 0 calibration unit start register (lane 1_rx_slc_iep0_start)	16	RW	0021h
8533h	RX sampler latch I even positive 0 calibration unit tune register (lane 1_rx_slc_iep0_tune)	16	RW	0000h
8534h	RX sampler latch I even positive 1 calibration unit control register (lane1_rx_slc_iep1_ctrl)	16	RW	001Fh
8535h	RX sampler latch I even positive 1 calibration unit override register (lane1_rx_slc_iep1_ovrd)	16	RW	0000h
8536h	RX sampler latch I even positive 1 calibration unit start register (lane 1_rx_slc_iep1_start)	16	RW	801Fh
8537h	RX sampler latch I even positive 1 calibration unit tune register (lane 1_rx_slc_iep1_tune)	16	RW	0000h
8538h	RX sampler latch Q even positive 0 calibration unit control register (lane1_rx_slc_qep0_ctrl)	16	RW	0021h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8539h	RX sampler latch Q even positive 0 calibration unit override register (lane1_rx_slc_qep0_ovrd)	16	RW	0000h
853Ah	RX sampler latch Q even positive 0 calibration unit start register (lane1_rx_slc_qep0_start)	16	RW	0021h
853Bh	RX sampler latch Q even positive 0 calibration unit tune register (lane1_rx_slc_qep0_tune)	16	RW	0000h
853Ch	RX sampler latch Q even positive 1 calibration unit control register (lane1_rx_slc_qep1_ctrl)	16	RW	001Fh
853Dh	RX sampler latch Q even positive 1 calibration unit override register (lane1_rx_slc_qep1_ovrd)	16	RW	0000h
853Eh	RX sampler latch Q even positive 1 calibration unit start register (lane1_rx_slc_qep1_start)	16	RW	801Fh
853Fh	RX sampler latch Q even positive 1 calibration unit tune register (lane1_rx_slc_qep1_tune)	16	RW	0000h
8540h	RX sampler latch E even positive 0 calibration unit control register (lane1_rx_slc_eep0_ctrl)	16	RW	0021h
8541h	RX sampler latch E even positive 0 calibration unit override register (lane1_rx_slc_eep0_ovrd)	16	RW	0000h
8542h	RX sampler latch E even positive 0 calibration unit start register (lane1_rx_slc_eep0_start)	16	RW	0021h
8543h	RX sampler latch E even positive 0 calibration unit tune register (lane1_rx_slc_eep0_tune)	16	RW	0000h
8544h	RX sampler latch E even positive 1 calibration unit control register (lane1_rx_slc_eep1_ctrl)	16	RW	001Fh
8545h	RX sampler latch E even positive 1 calibration unit override register (lane1_rx_slc_eep1_ovrd)	16	RW	0000h
8546h	RX sampler latch E even positive 1 calibration unit start register (lane1_rx_slc_eep1_start)	16	RW	801Fh
8547h	RX sampler latch E even positive 1 calibration unit tune register (lane1_rx_slc_eep1_tune)	16	RW	0000h
8548h	RX sampler latch I even negative 0 calibration unit control register (lane1_rx_slc_ien0_ctrl)	16	RW	0021h
8549h	RX sampler latch I even negative 0 calibration unit override register (lane1_rx_slc_ien0_ovrd)	16	RW	0000h
854Ah	RX sampler latch I even negative 0 calibration unit start register (lane1_rx_slc_ien0_start)	16	RW	0021h
854Bh	RX sampler latch I even negative 0 calibration unit tune register (lane1_rx_slc_ien0_tune)	16	RW	0000h
854Ch	RX sampler latch I even negative 1 calibration unit control register (lane1_rx_slc_ien1_ctrl)	16	RW	001Fh
854Dh	RX sampler latch I even negative 1 calibration unit override register (lane1_rx_slc_ien1_ovrd)	16	RW	0000h
854Eh	RX sampler latch I even negative 1 calibration unit start register (lane1_rx_slc_ien1_start)	16	RW	801Fh
854Fh	RX sampler latch I even negative 1 calibration unit tune register (lane1_rx_slc_ien1_tune)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8550h	RX sampler latch Q even negative 0 calibration unit control register (lane1_rx_slc_qen0_ctrl)	16	RW	0021h
8551h	RX sampler latch Q even negative 0 calibration unit override register (lane1_rx_slc_qen0_ovrd)	16	RW	0000h
8552h	RX sampler latch Q even negative 0 calibration unit start register (lane1_rx_slc_qen0_start)	16	RW	0021h
8553h	RX sampler latch Q even negative 0 calibration unit tune register (lane1_rx_slc_qen0_tune)	16	RW	0000h
8554h	RX sampler latch Q even negative 1 calibration unit control register (lane1_rx_slc_qen1_ctrl)	16	RW	001Fh
8555h	RX sampler latch Q even negative 1 calibration unit override register (lane1_rx_slc_qen1_ovrd)	16	RW	0000h
8556h	RX sampler latch Q even negative 1 calibration unit start register (lane1_rx_slc_qen1_start)	16	RW	801Fh
8557h	RX sampler latch Q even negative 1 calibration unit tune register (lane1_rx_slc_qen1_tune)	16	RW	0000h
8558h	RX sampler latch E even negative 0 calibration unit control register (lane1_rx_slc_eeen0_ctrl)	16	RW	0021h
8559h	RX sampler latch E even negative 0 calibration unit override register (lane1_rx_slc_eeen0_ovrd)	16	RW	0000h
855Ah	RX sampler latch E even negative 0 calibration unit start register (lane1_rx_slc_eeen0_start)	16	RW	0021h
855Bh	RX sampler latch E even negative 0 calibration unit tune register (lane1_rx_slc_eeen0_tune)	16	RW	0000h
855Ch	RX sampler latch E even negative 1 calibration unit control register (lane1_rx_slc_eeen1_ctrl)	16	RW	001Fh
855Dh	RX sampler latch E even negative 1 calibration unit override register (lane1_rx_slc_eeen1_ovrd)	16	RW	0000h
855Eh	RX sampler latch E even negative 1 calibration unit start register (lane1_rx_slc_eeen1_start)	16	RW	801Fh
855Fh	RX sampler latch E even negative 1 calibration unit tune register (lane1_rx_slc_eeen1_tune)	16	RW	0000h
8560h	REE USB 3 general control state machine control register (lane1_rx_ree_u3gcsn_ctrl)	16	RW	0001h
8561h	REE USB 3 general control state machine phase 1 equalization enable mask register (lane1_rx_ree_u3gcsn_eqenm_ph1)	16	RW	00C7h
8562h	REE USB 3 general control state machine phase 2 equalization enable mask register (lane1_rx_ree_u3gcsn_eqenm_ph2)	16	RW	00C7h
8563h	REE USB 3 general control state machine start timer value register (lane1_rx_ree_u3gcsn_start_tmr)	16	RW	0125h
8564h	REE USB 3 general control state machine run phase 1 timer value register (lane1_rx_ree_u3gcsn_run_ph1_tmr)	16	RW	07A2h
8565h	REE USB 3 general control state machine run phase 2 timer value register (lane1_rx_ree_u3gcsn_run_ph2_tmr)	16	RW	009Dh
8568h	REE PCIe Gen 2 general control state machine control register (lane1_rx_ree_g2gcsn_ctrl)	16	RW	0001h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8569h	REE PCIe Gen 2 general control state machine phase 1 equalization enable mask register (lane1_rx_ree_g2gcsn_eqenm_ph1)	16	RW	00C7h
856Ah	REE PCIe Gen 2 general control state machine phase 2 equalization enable mask register (lane1_rx_ree_g2gcsn_eqenm_ph2)	16	RW	00C7h
856Bh	REE PCIe Gen 2 general control state machine start timer value register (lane1_rx_ree_g2gcsn_start_tmr)	16	RW	0000h
856Ch	REE PCIe Gen 2 general control state machine run phase 1 timer value register (lane1_rx_ree_g2gcsn_run_ph1_tmr)	16	RW	0F43h
856Dh	REE PCIe Gen 2 general control state machine run phase 2 timer value register (lane1_rx_ree_g2gcsn_run_ph2_tmr)	16	RW	009Dh
8578h	REE periodic general control state machine control register (lane1_rx_ree_pergcsn_ctrl)	16	RW	0001h
8579h	REE periodic general control state machine phase 1 equalization enable mask register (lane1_rx_ree_pergcsn_eqenm_ph1)	16	RW	0080h
857Ah	REE periodic general control state machine phase 2 equalization enable mask register (lane1_rx_ree_pergcsn_eqenm_ph2)	16	RW	0080h
857Bh	REE periodic general control state machine start timer value register (lane1_rx_ree_pergcsn_start_tmr)	16	RW	0000h
857Ch	REE periodic general control state machine run phase 1 timer value register (lane1_rx_ree_pergcsn_run_ph1_tmr)	16	RW	FFFFh
857Dh	REE periodic general control state machine run phase 2 timer value register (lane1_rx_ree_pergcsn_run_ph2_tmr)	16	RW	009Dh
8580h	REE tap 1 control register (lane1_rx_ree_tap1_ctrl)	16	RW	0000h
8581h	REE tap 1 override register (lane1_rx_ree_tap1_ovrd)	16	RW	0000h
8582h	REE tap 1 diagnostics register (lane1_rx_ree_tap1_diag)	16	RW	0000h
8584h	REE tap 2 control register (lane1_rx_ree_tap2_ctrl)	16	RW	0400h
8585h	REE tap 2 override register (lane1_rx_ree_tap2_ovrd)	16	RW	0000h
8586h	REE tap 2 diagnostics register (lane1_rx_ree_tap2_diag)	16	RW	0000h
8588h	REE tap 3 control register (lane1_rx_ree_tap3_ctrl)	16	RW	0400h
8589h	REE tap 3 override register (lane1_rx_ree_tap3_ovrd)	16	RW	0000h
858Ah	REE tap 3 diagnostics register (lane1_rx_ree_tap3_diag)	16	RW	0000h
8594h	REE analog enable control state machine delay timer value register (lane1_rx_ree_anaensm_del_tmr)	16	RW	02EEh
8598h	REE peaking amp control register (lane1_rx_ree_peak_ctrl)	16	RW	0701h
8599h	REE peaking amp code control register (lane1_rx_ree_peak_code_ctrl)	16	RW	2A0Eh
859Ah	REE peaking amp upper threshold register (lane1_rx_ree_peak_uthr)	16	RW	0004h
859Bh	REE peaking amp lower threshold register (lane1_rx_ree_peak_lthr)	16	RW	0000h
859Ch	REE peaking amp input override register (lane1_rx_ree_peak_iovrd)	16	RW	0000h
859Dh	REE peaking amp code override register (lane1_rx_ree_peak_covrd)	16	RW	0000h
859Eh	REE peaking amp diagnostics register (lane1_rx_ree_peak_diag)	16	RW	0020h
85A0h	REE attenuation control register (lane1_rx_ree_atten_ctrl)	16	RW	0005h
85A1h	REE attenuation threshold register (lane1_rx_ree_atten_thr)	16	RW	0C02h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
85A2h	REE attenuation counter register (lane1_rx_ree_atten_cnt)	16	RW	2100h
85A3h	REE attenuation override register (lane1_rx_ree_atten_ovrd)	16	RW	0000h
85A4h	REE attenuation diagnostics register (lane1_rx_ree_atten_diag)	16	RO	0000h
85A8h	REE low frequency equalizer control register (lane1_rx_ree_lfeq_ctrl)	16	RW	0000h
85A9h	REE low frequency equalizer override register (lane1_rx_ree_lfeq_ovrd)	16	RW	0000h
85AAh	REE low frequency equalizer diagnostics register (lane1_rx_ree_lfeq_diag)	16	RW	0000h
85ACh	REE VGA gain control register (lane1_rx_ree_vga_gain_ctrl)	16	RW	1001h
85ADh	REE VGA gain override register (lane1_rx_ree_vga_gain_ovrd)	16	RW	0000h
85AEh	REE VGA gain diagnostics register (lane1_rx_ree_vga_gain_diag)	16	RW	000Ah
85AFh	REE VGA gain target adjust diagnostics register (lane1_rx_ree_vga_gain_tgt_diag)	16	RO	0000h
85B0h	REE offset correction control register (lane1_rx_ree_off_cor_ctrl)	16	RW	0001h
85B1h	REE offset correction override register (lane1_rx_ree_off_cor_ovrd)	16	RW	0000h
85B2h	REE offset correction diagnostics register (lane1_rx_ree_off_cor_diag)	16	RW	0000h
85B8h	REE adder configuration register (lane1_rx_ree_addr_cfg)	16	RW	0001h
85B9h	REE tap 1 clip control register (lane1_rx_ree_tap1_clip)	16	RW	0519h
85BAh	REE taps 2 and 3 clip control register (lane1_rx_ree_tap2ton_clip)	16	RW	0519h
85BBh	REE control data mask register (lane1_rx_ree_ctrl_data_mask)	16	RW	4000h
85BCh	REE diagnostic control register (lane1_rx_ree_diag_ctrl)	16	RW	0040h
85BDh	REE control state machine gen mode control register 1 (lane1_rx_ree_smgm_ctrl1)	16	RW	0702h
85BEh	REE control state machine gen mode control register 2 (lane1_rx_ree_smgm_ctrl2)	16	RW	0000h
85C0h	RX ILL diagnostic control register (lane1_rx_diag_ill_ctrl)	16	RW	0000h
85C1h	RX ILL IQ trim 0 register (lane1_rx_diag_ill_iq_trim0)	16	RW	533Fh
85C2h	RX ILL E trim 0 register (lane1_rx_diag_ill_e_trim0)	16	RW	533Fh
85C3h	RX ILL IQ trim 1 register (lane1_rx_diag_ill_iq_trim1)	16	RW	0002h
85C4h	RX ILL E trim 1 register (lane1_rx_diag_ill_e_trim1)	16	RW	0002h
85C5h	RX ILL IQ E trim 2 register (lane1_rx_diag_ill_iqe_trim2)	16	RW	3E00h
85C6h	RX ILL IQ E trim 3 register (lane1_rx_diag_ill_iqe_trim3)	16	RW	199Fh
85C7h	RX ILL IQ E trim 4 register (lane1_rx_diag_ill_iqe_trim4)	16	RW	0014h
85C8h	RX ILL IQ E trim 5 register (lane1_rx_diag_ill_iqe_trim5)	16	RW	1504h
85C9h	RX ILL IQ E trim 6 register (lane1_rx_diag_ill_iqe_trim6)	16	RW	0015h
85D0h	DFE amp fine tuning register (lane1_rx_diag_dfe_amp_tune)	16	RW	4CCCh
85D1h	DFE amp fine tuning 2 register (lane1_rx_diag_dfe_amp_tune_2)	16	RW	0C21h
85D2h	REE DAC control register (lane1_rx_diag_ree_dac_ctrl)	16	RW	0004h
85D3h	Receiver DFE control register 1 (lane1_rx_diag_dfe_ctrl1)	16	RW	0007h
85D4h	Receiver DFE control register 2 (lane1_rx_diag_dfe_ctrl2)	16	RW	003Eh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
85D5h	Receiver DFE control register 3 (lane1_rx_diag_dfe_ctrl3)	16	RW	0ACAh
85D6h	Nyquist control register (lane1_rx_diag_nqst_ctrl)	16	RW	0998h
85D7h	Low frequency equalizer tuning register (lane1_rx_diag_lfeq_tune)	16	RW	0014h
85D8h	RX control register (lane1_rx_diag_rxctrl)	16	RW	0060h
85D9h	Receiver control reset diagnostic register (lane1_rx_diag_rst_diag)	16	RO	0000h
85DCh	RX signal detect tuning and control register (lane1_rx_diag_sigdet_tune)	16	RW	1005h
85DDh	RX LFPS detect tuning and control register (lane1_rx_diag_lfpsdet_tune)	16	RW	2410h
85DEh	Signal detect test register (lane1_rx_diag_sd_test)	16	RW	0000h
85E0h	RX sampler diagnostic control register (lane1_rx_diag_samp_ctrl)	16	RW	0001h
85E1h	RX Sampler CML TO CMOS enable delay register (lane1_rx_diag_sc2c_delay)	16	RW	012Ch
85E4h	MPHY control register 1 (lane1_rx_diag_mphy_ctrl_1)	16	RW	0000h
85E5h	MPHY control register 2 (lane1_rx_diag_mphy_ctrl_2)	16	RW	0000h
85E8h	RX loopback controller register (lane1_rx_diag_lpbk_ctrl)	16	RW	0000h
85E9h	RX extra enable control override register (lane1_rx_diag_ectrl_ovrd)	16	RW	0000h
85F0h	CML to CMOS bias trim register (lane1_rx_diag_cml2cmos_btrim)	16	RW	0000h
85F1h	RX bias gen control register 1 (lane1_rx_diag_bias_gen_ctrl1)	16	RW	0000h
85F2h	RX bias gen control register 2 (lane1_rx_diag_bias_gen_ctrl2)	16	RW	0000h
85F3h	RX bias gen control register 3 (lane1_rx_diag_bias_gen_ctrl3)	16	RW	0000h
85F4h	RX bias gen control register 4 (lane1_rx_diag_bias_gen_ctrl4)	16	RW	0000h
85F5h	RX boundary scan test mode register (lane1_rx_diag_bs_tm)	16	RW	0000h
85F6h	RX receiver front end test mode register 1 (lane1_rx_diag_rxfe_tm1)	16	RW	0000h
85F7h	RX receiver front end test mode register 2 (lane1_rx_diag_rxfe_tm2)	16	RW	0000h
85FEh	Receiver digital cover your alternatives register (lane1_rx_diag_dcya)	16	RW	0000h
85FFh	Receiver analog cover your alternatives register (lane1_rx_diag_acya)	16	RW	0000h
8800h	Receiver A0 power state definition register (lane2_rx_psc_a0)	16	RW	8BFDh
8801h	Receiver A1 power state definition register (lane2_rx_psc_a1)	16	RW	8BFDh
8802h	Receiver A2 power state definition register (lane2_rx_psc_a2)	16	RW	8910h
8803h	Receiver A3 power state definition register (lane2_rx_psc_a3)	16	RW	0000h
8804h	Receiver A4 power state definition register (lane2_rx_psc_a4)	16	RW	1000h
8805h	Receiver A5 power state definition register (lane2_rx_psc_a5)	16	RW	0000h
8806h	Receiver calibration power state definition register (lane2_rx_psc_cal)	16	RW	03FFh
8807h	Receiver ready power state definition register (lane2_rx_psc_rdy)	16	RW	0000h
8820h	RX IQ PI ILL calibration control register (lane2_rx_iqpi_ill_cal_ctrl)	16	RW	0018h
8821h	RX IQ PI ILL calibration start point register (lane2_rx_iqpi_ill_cal_start)	16	RW	1018h
8822h	RX IQ PI ILL calibration timer control register (lane2_rx_iqpi_ill_cal_ctrl)	16	RW	0003h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8823h	RX IQ PI ILL calibration override register (lane2_rx_iqpi_ill_cal_ovrd)	16	RW	0000h
8824h	RX IQ PI ILL calibration initialization timer register (lane2_rx_iqpi_ill_cal_init_tmr)	16	RW	02EEh
8825h	RX IQ PI ILL calibration iteration timer register (lane2_rx_iqpi_ill_cal_iter_tmr)	16	RW	0080h
8826h	RX IQ PI ILL lock reference timer start value register (lane2_rx_iqpi_ill_lock_reftmr_start)	16	RW	01FFh
8828h	RX IQ PI ILL lock calibration counter start value standard mode 0 register (lane2_rx_iqpi_ill_lock_calcnt_start_0)	16	RW	007Fh
8829h	RX IQ PI ILL lock calibration counter start value standard mode 1 register (lane2_rx_iqpi_ill_lock_calcnt_start_1)	16	RW	00FFh
882Ah	RX IQ PI ILL lock calibration counter start value standard mode 2 register (lane2_rx_iqpi_ill_lock_calcnt_start_2)	16	RW	00FFh
882Bh	RX IQ PI ILL lock calibration counter start value standard mode 3 register (lane2_rx_iqpi_ill_lock_calcnt_start_3)	16	RW	0000h
8830h	RX E PI ILL calibration control register (lane2_rx_epi_ill_cal_ctrl)	16	RW	0018h
8831h	RX E PI ILL calibration start point register (lane2_rx_epi_ill_cal_start)	16	RW	1018h
8832h	RX E PI ILL calibration timer control register (lane2_rx_epi_ill_cal_tctrl)	16	RW	0003h
8833h	RX E PI ILL calibration override register (lane2_rx_epi_ill_cal_ovrd)	16	RW	0000h
8834h	RX E PI ILL calibration initialization timer register (lane2_rx_epi_ill_cal_init_tmr)	16	RW	02EEh
8835h	RX E PI ILL calibration iteration timer register (lane2_rx_epi_ill_cal_iter_tmr)	16	RW	0080h
8836h	RX E PI ILL lock reference timer start value register (lane2_rx_epi_ill_lock_reftmr_start)	16	RW	01FFh
8838h	RX E PI ILL lock calibration counter start value standard mode 0 register (lane2_rx_epi_ill_lock_calcnt_start_0)	16	RW	007Fh
8839h	RX E PI ILL lock calibration counter start value standard mode 1 register (lane2_rx_epi_ill_lock_calcnt_start_1)	16	RW	00FFh
883Ah	RX E PI ILL lock calibration counter start value standard mode 2 register (lane2_rx_epi_ill_lock_calcnt_start_2)	16	RW	00FFh
883Bh	RX E PI ILL lock calibration counter start value standard mode 3 register (lane2_rx_epi_ill_lock_calcnt_start_3)	16	RW	0000h
8840h	Signal detect calibration 0 control register (lane2_rx_sdc0_ctrl)	16	RW	0000h
8841h	Signal detect calibration 0 override register (lane2_rx_sdc0_ovrd)	16	RW	0000h
8842h	Signal detect calibration 0 start register (lane2_rx_sdc0_start)	16	RW	0000h
8843h	Signal detect calibration 0 tune register (lane2_rx_sdc0_tune)	16	RW	0000h
8844h	Signal detect calibration 0 initialization timer register (lane2_rx_sdc0_init_tmr)	16	RW	0019h
8845h	Signal detect calibration 0 iteration timer register (lane2_rx_sdc0_iter_tmr)	16	RW	007Dh
8848h	Signal detect calibration 1 control register (lane2_rx_sdc1_ctrl)	16	RW	0000h
8849h	Signal detect calibration 1 override register (lane2_rx_sdc1_ovrd)	16	RW	0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
884Ah	Signal detect calibration 1 start register (lane2_rx_sdcal1_start)	16	RW	0000h
884Bh	Signal detect calibration 1 tune register (lane2_rx_sdcal1_tune)	16	RW	0000h
884Ch	Signal detect calibration 1 initialization timer register (lane2_rx_sdcal1_init_tmr)	16	RW	0019h
884Dh	Signal detect calibration 1 iteration timer register (lane2_rx_sdcal1_iter_tmr)	16	RW	007Dh
8858h	Sampler error DAC control register (lane2_rx_samp_dac_ctrl)	16	RW	0014h
8880h	CDRLF configuration register (lane2_rx_cdrflf_cfg)	16	RW	20B3h
8881h	CDRLF configuration register 2 (lane2_rx_cdrflf_cfg2)	16	RW	0000h
8882h	CDRLF margin diagnostic register (lane2_rx_cdrflf_mgn_diag)	16	RW	0000h
8883h	CDRLF fast phase lock timer value register 0 (lane2_rx_cdrflf_fpl_tmr0)	16	RW	0071h
8884h	CDRLF fast phase lock timer value register 1 (lane2_rx_cdrflf_fpl_tmr1)	16	RW	0731h
8885h	CDRLF fast frequency lock timer value register (lane2_rx_cdrflf_ffl_tmr)	16	RW	0018h
8888h	CDRLF fast frequency lock step 0 control register (lane2_rx_cdrflf_fl0_ctrl)	16	RW	460Ch
8889h	CDRLF fast frequency lock step 1 control register (lane2_rx_cdrflf_fl1_ctrl)	16	RW	4A0Ah
888Ah	CDRLF fast frequency lock step 2 control register (lane2_rx_cdrflf_fl2_ctrl)	16	RW	5414h
888Bh	CDRLF fast frequency lock step 3 control register (lane2_rx_cdrflf_fl3_ctrl)	16	RW	5E1Eh
888Ch	CDRLF fast frequency lock step 4 control register (lane2_rx_cdrflf_fl4_ctrl)	16	RW	9E1Eh
8890h	Receiver signal detect filter high to low filter timer register (lane2_rx_sigdet_hl_filt_tmr)	16	RW	0019h
8891h	Receiver signal detect filter high to low delay timer register (lane2_rx_sigdet_hl_dly_tmr)	16	RW	0000h
8892h	Receiver signal detect filter high to low min timer register (lane2_rx_sigdet_hl_min_tmr)	16	RW	0000h
8893h	Receiver signal detect filter high to low init timer register (lane2_rx_sigdet_hl_init_tmr)	16	RW	03E8h
8894h	Receiver signal detect filter low to high filter timer register (lane2_rx_sigdet_lh_filt_tmr)	16	RW	0004h
8895h	Receiver signal detect filter low to high delay timer register (lane2_rx_sigdet_lh_dly_tmr)	16	RW	0004h
8896h	Receiver signal detect filter low to high min timer register (lane2_rx_sigdet_lh_min_tmr)	16	RW	0000h
8897h	Receiver signal detect filter low to high init timer register (lane2_rx_sigdet_lh_init_tmr)	16	RW	03E8h
8898h	Receiver LFPS detect filter filter timer register (lane2_rx_lfpsdet_filt_tmr)	16	RW	0008h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8899h	Receiver LFPS detect filter delay timer register (lane2_rx_lfpsdet_dly_tmr)	16	RW	000Bh
889Ah	Receiver LFPS detect min timer register (lane2_rx_lfpsdet_min_tmr)	16	RW	0013h
889Bh	Receiver LFPS detect init timer register (lane2_rx_lfpsdet_init_tmr)	16	RW	0000h
88A0h	Eye surf control register (lane2_rx_eyesurf_ctrl)	16	RW	0000h
88A4h	Eye surf timer delay low register (lane2_rx_eyesurf_tmr_dellow)	16	RW	0000h
88A5h	Eye surf timer delay high register (lane2_rx_eyesurf_tmr_delhigh)	16	RW	0000h
88A6h	Eye surf timer test low register (lane2_rx_eyesurf_tmr_testlow)	16	RW	0000h
88A7h	Eye surf timer test high register (lane2_rx_eyesurf_tmr_testhigh)	16	RW	0000h
88A8h	Eye surf north south test point coordinate register (lane2_rx_eyesurf_ns_coord)	16	RW	0000h
88A9h	Eye surf east west test point coordinate register (lane2_rx_eyesurf_ew_coord)	16	RW	0000h
88AAh	Eye surf bit error count register (lane2_rx_eyesurf_errcnt)	16	RO	0000h
88B0h	Receiver BIST control register (lane2_rx_bist_ctrl)	16	RW	0000h
88B1h	Receiver BIST sync count register (lane2_rx_bist_syncnt)	16	RW	0000h
88B2h	Receiver BIST user defined data write register (lane2_rx_bist_uddwr)	16	WO	0000h
88B3h	Receiver BIST error count register (lane2_rx_bist_errcnt)	16	RO	0000h
88B4h	Clock frequency measurement control register (lane2_xcvr_cmsmt_clk_freq_msmt_ctrl)	16	RW	0000h
88B5h	Test clock selection register (lane2_xcvr_cmsmt_test_clk_sel)	16	RW	0000h
88B6h	Reference clock timer value register (lane2_xcvr_cmsmt_ref_clk_tmr_value)	16	RW	0000h
88B7h	Test clock counter value register (lane2_xcvr_cmsmt_test_clk_cnt_value)	16	RO	0000h
88E0h	RX sampler latch calibration control register (lane2_rx_slc_ctrl)	16	RW	0EFFh
88E1h	RX sampler latch calibration enable initialization timer value register (lane2_rx_slc_en_init_tmr)	16	RW	0004h
88E2h	RX sampler latch calibration unit initialization timer value register (lane2_rx_slc_cu_init_tmr)	16	RW	0010h
88E3h	RX sampler latch calibration unit iteration timer value register (lane2_rx_slc_cu_iter_tmr)	16	RW	0001h
88E4h	RX sampler latch calibration I even data mask register (lane2_rx_slc_ie_mask)	16	RW	03FFh
88E5h	RX sampler latch calibration I odd data mask register (lane2_rx_slc_io_mask)	16	RW	03FFh
88E6h	RX sampler latch calibration Q even data mask register (lane2_rx_slc_qe_mask)	16	RW	03FFh
88E7h	RX sampler latch calibration Q odd data mask register (lane2_rx_slc_qo_mask)	16	RW	03FFh
88E8h	RX sampler latch calibration E even data mask register (lane2_rx_slc_ee_mask)	16	RW	03FFh
88E9h	RX sampler latch calibration E odd data mask register (lane2_rx_slc_eo_mask)	16	RW	03FFh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
88EAh	RX sampler latch calibration data threshold register (lane2_rx_slc_data_thr)	16	RW	0005h
8900h	RX sampler latch I odd positive 0 calibration unit control register (lane2_rx_slc_iop0_ctrl)	16	RW	0021h
8901h	RX sampler latch I odd positive 0 calibration unit override register (lane2_rx_slc_iop0_ovrd)	16	RW	0000h
8902h	RX sampler latch I odd positive 0 calibration unit start register (lane2_rx_slc_iop0_start)	16	RW	0021h
8903h	RX sampler latch I odd positive 0 calibration unit tune register (lane2_rx_slc_iop0_tune)	16	RW	0000h
8904h	RX sampler latch I odd positive 1 calibration unit control register (lane2_rx_slc_iop1_ctrl)	16	RW	001Fh
8905h	RX sampler latch I odd positive 1 calibration unit override register (lane2_rx_slc_iop1_ovrd)	16	RW	0000h
8906h	RX sampler latch I odd positive 1 calibration unit start register (lane2_rx_slc_iop1_start)	16	RW	801Fh
8907h	RX sampler latch I odd positive 1 calibration unit tune register (lane2_rx_slc_iop1_tune)	16	RW	0000h
8908h	RX sampler latch Q odd positive 0 calibration unit control register (lane2_rx_slc_qop0_ctrl)	16	RW	0021h
8909h	RX sampler latch Q odd positive 0 calibration unit override register (lane2_rx_slc_qop0_ovrd)	16	RW	0000h
890Ah	RX sampler latch Q odd positive 0 calibration unit start register (lane2_rx_slc_qop0_start)	16	RW	0021h
890Bh	RX sampler latch Q odd positive 0 calibration unit tune register (lane2_rx_slc_qop0_tune)	16	RW	0000h
890Ch	RX sampler latch Q odd positive 1 calibration unit control register (lane2_rx_slc_qop1_ctrl)	16	RW	001Fh
890Dh	RX sampler latch Q odd positive 1 calibration unit override register (lane2_rx_slc_qop1_ovrd)	16	RW	0000h
890Eh	RX sampler latch Q odd positive 1 calibration unit start register (lane2_rx_slc_qop1_start)	16	RW	801Fh
890Fh	RX sampler latch Q odd positive 1 calibration unit tune register (lane2_rx_slc_qop1_tune)	16	RW	0000h
8910h	RX sampler latch E odd positive 0 calibration unit control register (lane2_rx_slc_eop0_ctrl)	16	RW	0021h
8911h	RX sampler latch E odd positive 0 calibration unit override register (lane2_rx_slc_eop0_ovrd)	16	RW	0000h
8912h	RX sampler latch E odd positive 0 calibration unit start register (lane2_rx_slc_eop0_start)	16	RW	0021h
8913h	RX sampler latch E odd positive 0 calibration unit tune register (lane2_rx_slc_eop0_tune)	16	RW	0000h
8914h	RX sampler latch E odd positive 1 calibration unit control register (lane2_rx_slc_eop1_ctrl)	16	RW	001Fh
8915h	RX sampler latch E odd positive 1 calibration unit override register (lane2_rx_slc_eop1_ovrd)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8916h	RX sampler latch E odd positive 1 calibration unit start register (lane 2_rx_slc_eop1_start)	16	RW	801Fh
8917h	RX sampler latch E odd positive 1 calibration unit tune register (lane 2_rx_slc_eop1_tune)	16	RW	0000h
8918h	RX sampler latch I odd negative 0 calibration unit control register (lane2_rx_slc_ion0_ctrl)	16	RW	0021h
8919h	RX sampler latch I odd negative 0 calibration unit override register (lane2_rx_slc_ion0_ovrd)	16	RW	0000h
891Ah	RX sampler latch I odd negative 0 calibration unit start register (lane 2_rx_slc_ion0_start)	16	RW	0021h
891Bh	RX sampler latch I odd negative 0 calibration unit tune register (lane 2_rx_slc_ion0_tune)	16	RW	0000h
891Ch	RX sampler latch I odd negative 1 calibration unit control register (lane2_rx_slc_ion1_ctrl)	16	RW	001Fh
891Dh	RX sampler latch I odd negative 1 calibration unit override register (lane2_rx_slc_ion1_ovrd)	16	RW	0000h
891Eh	RX sampler latch I odd negative 1 calibration unit start register (lane 2_rx_slc_ion1_start)	16	RW	801Fh
891Fh	RX sampler latch I odd negative 1 calibration unit tune register (lane 2_rx_slc_ion1_tune)	16	RW	0000h
8920h	RX sampler latch Q odd negative 0 calibration unit control register (lane2_rx_slc_qon0_ctrl)	16	RW	0021h
8921h	RX sampler latch Q odd negative 0 calibration unit override register (lane2_rx_slc_qon0_ovrd)	16	RW	0000h
8922h	RX sampler latch Q odd negative 0 calibration unit start register (lane 2_rx_slc_qon0_start)	16	RW	0021h
8923h	RX sampler latch Q odd negative 0 calibration unit tune register (lane 2_rx_slc_qon0_tune)	16	RW	0000h
8924h	RX sampler latch Q odd negative 1 calibration unit control register (lane2_rx_slc_qon1_ctrl)	16	RW	001Fh
8925h	RX sampler latch Q odd negative 1 calibration unit override register (lane2_rx_slc_qon1_ovrd)	16	RW	0000h
8926h	RX sampler latch Q odd negative 1 calibration unit start register (lane 2_rx_slc_qon1_start)	16	RW	801Fh
8927h	RX sampler latch Q odd negative 1 calibration unit tune register (lane 2_rx_slc_qon1_tune)	16	RW	0000h
8928h	RX sampler latch E odd negative 0 calibration unit control register (lane2_rx_slc_eon0_ctrl)	16	RW	0021h
8929h	RX sampler latch E odd negative 0 calibration unit override register (lane2_rx_slc_eon0_ovrd)	16	RW	0000h
892Ah	RX sampler latch E odd negative 0 calibration unit start register (lane 2_rx_slc_eon0_start)	16	RW	0021h
892Bh	RX sampler latch E odd negative 0 calibration unit tune register (lane 2_rx_slc_eon0_tune)	16	RW	0000h
892Ch	RX sampler latch E odd negative 1 calibration unit control register (lane2_rx_slc_eon1_ctrl)	16	RW	001Fh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
892Dh	RX sampler latch E odd negative 1 calibration unit override register (lane2_rx_slc_eon1_ovrd)	16	RW	0000h
892Eh	RX sampler latch E odd negative 1 calibration unit start register (lane2_rx_slc_eon1_start)	16	RW	801Fh
892Fh	RX sampler latch E odd negative 1 calibration unit tune register (lane2_rx_slc_eon1_tune)	16	RW	0000h
8930h	RX sampler latch I even positive 0 calibration unit control register (lane2_rx_slc_iep0_ctrl)	16	RW	0021h
8931h	RX sampler latch I even positive 0 calibration unit override register (lane2_rx_slc_iep0_ovrd)	16	RW	0000h
8932h	RX sampler latch I even positive 0 calibration unit start register (lane2_rx_slc_iep0_start)	16	RW	0021h
8933h	RX sampler latch I even positive 0 calibration unit tune register (lane2_rx_slc_iep0_tune)	16	RW	0000h
8934h	RX sampler latch I even positive 1 calibration unit control register (lane2_rx_slc_iep1_ctrl)	16	RW	001Fh
8935h	RX sampler latch I even positive 1 calibration unit override register (lane2_rx_slc_iep1_ovrd)	16	RW	0000h
8936h	RX sampler latch I even positive 1 calibration unit start register (lane2_rx_slc_iep1_start)	16	RW	801Fh
8937h	RX sampler latch I even positive 1 calibration unit tune register (lane2_rx_slc_iep1_tune)	16	RW	0000h
8938h	RX sampler latch Q even positive 0 calibration unit control register (lane2_rx_slc_qep0_ctrl)	16	RW	0021h
8939h	RX sampler latch Q even positive 0 calibration unit override register (lane2_rx_slc_qep0_ovrd)	16	RW	0000h
893Ah	RX sampler latch Q even positive 0 calibration unit start register (lane2_rx_slc_qep0_start)	16	RW	0021h
893Bh	RX sampler latch Q even positive 0 calibration unit tune register (lane2_rx_slc_qep0_tune)	16	RW	0000h
893Ch	RX sampler latch Q even positive 1 calibration unit control register (lane2_rx_slc_qep1_ctrl)	16	RW	001Fh
893Dh	RX sampler latch Q even positive 1 calibration unit override register (lane2_rx_slc_qep1_ovrd)	16	RW	0000h
893Eh	RX sampler latch Q even positive 1 calibration unit start register (lane2_rx_slc_qep1_start)	16	RW	801Fh
893Fh	RX sampler latch Q even positive 1 calibration unit tune register (lane2_rx_slc_qep1_tune)	16	RW	0000h
8940h	RX sampler latch E even positive 0 calibration unit control register (lane2_rx_slc_eep0_ctrl)	16	RW	0021h
8941h	RX sampler latch E even positive 0 calibration unit override register (lane2_rx_slc_eep0_ovrd)	16	RW	0000h
8942h	RX sampler latch E even positive 0 calibration unit start register (lane2_rx_slc_eep0_start)	16	RW	0021h
8943h	RX sampler latch E even positive 0 calibration unit tune register (lane2_rx_slc_eep0_tune)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8944h	RX sampler latch E even positive 1 calibration unit control register (lane2_rx_slc_eep1_ctrl)	16	RW	001Fh
8945h	RX sampler latch E even positive 1 calibration unit override register (lane2_rx_slc_eep1_ovrd)	16	RW	0000h
8946h	RX sampler latch E even positive 1 calibration unit start register (lane2_rx_slc_eep1_start)	16	RW	801Fh
8947h	RX sampler latch E even positive 1 calibration unit tune register (lane2_rx_slc_eep1_tune)	16	RW	0000h
8948h	RX sampler latch I even negative 0 calibration unit control register (lane2_rx_slc_ien0_ctrl)	16	RW	0021h
8949h	RX sampler latch I even negative 0 calibration unit override register (lane2_rx_slc_ien0_ovrd)	16	RW	0000h
894Ah	RX sampler latch I even negative 0 calibration unit start register (lane2_rx_slc_ien0_start)	16	RW	0021h
894Bh	RX sampler latch I even negative 0 calibration unit tune register (lane2_rx_slc_ien0_tune)	16	RW	0000h
894Ch	RX sampler latch I even negative 1 calibration unit control register (lane2_rx_slc_ien1_ctrl)	16	RW	001Fh
894Dh	RX sampler latch I even negative 1 calibration unit override register (lane2_rx_slc_ien1_ovrd)	16	RW	0000h
894Eh	RX sampler latch I even negative 1 calibration unit start register (lane2_rx_slc_ien1_start)	16	RW	801Fh
894Fh	RX sampler latch I even negative 1 calibration unit tune register (lane2_rx_slc_ien1_tune)	16	RW	0000h
8950h	RX sampler latch Q even negative 0 calibration unit control register (lane2_rx_slc_qen0_ctrl)	16	RW	0021h
8951h	RX sampler latch Q even negative 0 calibration unit override register (lane2_rx_slc_qen0_ovrd)	16	RW	0000h
8952h	RX sampler latch Q even negative 0 calibration unit start register (lane2_rx_slc_qen0_start)	16	RW	0021h
8953h	RX sampler latch Q even negative 0 calibration unit tune register (lane2_rx_slc_qen0_tune)	16	RW	0000h
8954h	RX sampler latch Q even negative 1 calibration unit control register (lane2_rx_slc_qen1_ctrl)	16	RW	001Fh
8955h	RX sampler latch Q even negative 1 calibration unit override register (lane2_rx_slc_qen1_ovrd)	16	RW	0000h
8956h	RX sampler latch Q even negative 1 calibration unit start register (lane2_rx_slc_qen1_start)	16	RW	801Fh
8957h	RX sampler latch Q even negative 1 calibration unit tune register (lane2_rx_slc_qen1_tune)	16	RW	0000h
8958h	RX sampler latch E even negative 0 calibration unit control register (lane2_rx_slc_een0_ctrl)	16	RW	0021h
8959h	RX sampler latch E even negative 0 calibration unit override register (lane2_rx_slc_een0_ovrd)	16	RW	0000h
895Ah	RX sampler latch E even negative 0 calibration unit start register (lane2_rx_slc_een0_start)	16	RW	0021h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
895Bh	RX sampler latch E even negative 0 calibration unit tune register (lane2_rx_slc_eeen0_tune)	16	RW	0000h
895Ch	RX sampler latch E even negative 1 calibration unit control register (lane2_rx_slc_eeen1_ctrl)	16	RW	001Fh
895Dh	RX sampler latch E even negative 1 calibration unit override register (lane2_rx_slc_eeen1_ovrd)	16	RW	0000h
895Eh	RX sampler latch E even negative 1 calibration unit start register (lane2_rx_slc_eeen1_start)	16	RW	801Fh
895Fh	RX sampler latch E even negative 1 calibration unit tune register (lane2_rx_slc_eeen1_tune)	16	RW	0000h
8960h	REE USB 3 general control state machine control register (lane2_rx_ree_u3gcsn_ctrl)	16	RW	0001h
8961h	REE USB 3 general control state machine phase 1 equalization enable mask register (lane2_rx_ree_u3gcsn_eqenm_ph1)	16	RW	00C7h
8962h	REE USB 3 general control state machine phase 2 equalization enable mask register (lane2_rx_ree_u3gcsn_eqenm_ph2)	16	RW	00C7h
8963h	REE USB 3 general control state machine start timer value register (lane2_rx_ree_u3gcsn_start_tmr)	16	RW	0125h
8964h	REE USB 3 general control state machine run phase 1 timer value register (lane2_rx_ree_u3gcsn_run_ph1_tmr)	16	RW	07A2h
8965h	REE USB 3 general control state machine run phase 2 timer value register (lane2_rx_ree_u3gcsn_run_ph2_tmr)	16	RW	009Dh
8968h	REE PCIe Gen 2 general control state machine control register (lane2_rx_ree_g2gcsn_ctrl)	16	RW	0001h
8969h	REE PCIe Gen 2 general control state machine phase 1 equalization enable mask register (lane2_rx_ree_g2gcsn_eqenm_ph1)	16	RW	00C7h
896Ah	REE PCIe Gen 2 general control state machine phase 2 equalization enable mask register (lane2_rx_ree_g2gcsn_eqenm_ph2)	16	RW	00C7h
896Bh	REE PCIe Gen 2 general control state machine start timer value register (lane2_rx_ree_g2gcsn_start_tmr)	16	RW	0000h
896Ch	REE PCIe Gen 2 general control state machine run phase 1 timer value register (lane2_rx_ree_g2gcsn_run_ph1_tmr)	16	RW	0F43h
896Dh	REE PCIe Gen 2 general control state machine run phase 2 timer value register (lane2_rx_ree_g2gcsn_run_ph2_tmr)	16	RW	009Dh
8978h	REE periodic general control state machine control register (lane2_rx_ree_pergcsn_ctrl)	16	RW	0001h
8979h	REE periodic general control state machine phase 1 equalization enable mask register (lane2_rx_ree_pergcsn_eqenm_ph1)	16	RW	0080h
897Ah	REE periodic general control state machine phase 2 equalization enable mask register (lane2_rx_ree_pergcsn_eqenm_ph2)	16	RW	0080h
897Bh	REE periodic general control state machine start timer value register (lane2_rx_ree_pergcsn_start_tmr)	16	RW	0000h
897Ch	REE periodic general control state machine run phase 1 timer value register (lane2_rx_ree_pergcsn_run_ph1_tmr)	16	RW	FFFFh
897Dh	REE periodic general control state machine run phase 2 timer value register (lane2_rx_ree_pergcsn_run_ph2_tmr)	16	RW	009Dh

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8980h	REE tap 1 control register (lane2_rx_ree_tap1_ctrl)	16	RW	0000h
8981h	REE tap 1 override register (lane2_rx_ree_tap1_ovrd)	16	RW	0000h
8982h	REE tap 1 diagnostics register (lane2_rx_ree_tap1_diag)	16	RW	0000h
8984h	REE tap 2 control register (lane2_rx_ree_tap2_ctrl)	16	RW	0400h
8985h	REE tap 2 override register (lane2_rx_ree_tap2_ovrd)	16	RW	0000h
8986h	REE tap 2 diagnostics register (lane2_rx_ree_tap2_diag)	16	RW	0000h
8988h	REE tap 3 control register (lane2_rx_ree_tap3_ctrl)	16	RW	0400h
8989h	REE tap 3 override register (lane2_rx_ree_tap3_ovrd)	16	RW	0000h
898Ah	REE tap 3 diagnostics register (lane2_rx_ree_tap3_diag)	16	RW	0000h
8994h	REE analog enable control state machine delay timer value register (lane2_rx_ree_anaensm_del_tmr)	16	RW	02EEh
8998h	REE peaking amp control register (lane2_rx_ree_peak_ctrl)	16	RW	0701h
8999h	REE peaking amp code control register (lane2_rx_ree_peak_code_ctrl)	16	RW	2A0Eh
899Ah	REE peaking amp upper threshold register (lane2_rx_ree_peak_uthr)	16	RW	0004h
899Bh	REE peaking amp lower threshold register (lane2_rx_ree_peak_lthr)	16	RW	0000h
899Ch	REE peaking amp input override register (lane2_rx_ree_peak_iovrd)	16	RW	0000h
899Dh	REE peaking amp code override register (lane2_rx_ree_peak_covrd)	16	RW	0000h
899Eh	REE peaking amp diagnostics register (lane2_rx_ree_peak_diag)	16	RW	0020h
89A0h	REE attenuation control register (lane2_rx_ree_atten_ctrl)	16	RW	0005h
89A1h	REE attenuation threshold register (lane2_rx_ree_atten_thr)	16	RW	0C02h
89A2h	REE attenuation counter register (lane2_rx_ree_atten_cnt)	16	RW	2100h
89A3h	REE attenuation override register (lane2_rx_ree_atten_ovrd)	16	RW	0000h
89A4h	REE attenuation diagnostics register (lane2_rx_ree_atten_diag)	16	RO	0000h
89A8h	REE low frequency equalizer control register (lane2_rx_ree_lfeq_ctrl)	16	RW	0000h
89A9h	REE low frequency equalizer override register (lane2_rx_ree_lfeq_ovrd)	16	RW	0000h
89AAh	REE low frequency equalizer diagnostics register (lane2_rx_ree_lfeq_diag)	16	RW	0000h
89ACh	REE VGA gain control register (lane2_rx_ree_vga_gain_ctrl)	16	RW	1001h
89ADh	REE VGA gain override register (lane2_rx_ree_vga_gain_ovrd)	16	RW	0000h
89AEh	REE VGA gain diagnostics register (lane2_rx_ree_vga_gain_diag)	16	RW	000Ah
89AFh	REE VGA gain target adjust diagnostics register (lane2_rx_ree_vga_gain_tgt_diag)	16	RO	0000h
89B0h	REE offset correction control register (lane2_rx_ree_off_cor_ctrl)	16	RW	0001h
89B1h	REE offset correction override register (lane2_rx_ree_off_cor_ovrd)	16	RW	0000h
89B2h	REE offset correction diagnostics register (lane2_rx_ree_off_cor_diag)	16	RW	0000h
89B8h	REE adder configuration register (lane2_rx_ree_addr_cfg)	16	RW	0001h
89B9h	REE tap 1 clip control register (lane2_rx_ree_tap1_clip)	16	RW	0519h
89BAh	REE taps 2 and 3 clip control register (lane2_rx_ree_tap2ton_clip)	16	RW	0519h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
89BBh	REE control data mask register (lane2_rx_ree_ctrl_data_mask)	16	RW	4000h
89BCh	REE diagnostic control register (lane2_rx_ree_diag_ctrl)	16	RW	0040h
89BDh	REE control state machine gen mode control register 1 (lane2_rx_ree_smgm_ctrl1)	16	RW	0702h
89BEh	REE control state machine gen mode control register 2 (lane2_rx_ree_smgm_ctrl2)	16	RW	0000h
89C0h	RX ILL diagnostic control register (lane2_rx_diag_ill_ctrl)	16	RW	0000h
89C1h	RX ILL IQ trim 0 register (lane2_rx_diag_ill_iq_trim0)	16	RW	533Fh
89C2h	RX ILL E trim 0 register (lane2_rx_diag_ill_e_trim0)	16	RW	533Fh
89C3h	RX ILL IQ trim 1 register (lane2_rx_diag_ill_iq_trim1)	16	RW	0002h
89C4h	RX ILL E trim 1 register (lane2_rx_diag_ill_e_trim1)	16	RW	0002h
89C5h	RX ILL IQ E trim 2 register (lane2_rx_diag_ill_iqe_trim2)	16	RW	3E00h
89C6h	RX ILL IQ E trim 3 register (lane2_rx_diag_ill_iqe_trim3)	16	RW	199Fh
89C7h	RX ILL IQ E trim 4 register (lane2_rx_diag_ill_iqe_trim4)	16	RW	0014h
89C8h	RX ILL IQ E trim 5 register (lane2_rx_diag_ill_iqe_trim5)	16	RW	1504h
89C9h	RX ILL IQ E trim 6 register (lane2_rx_diag_ill_iqe_trim6)	16	RW	0015h
89D0h	DFE amp fine tuning register (lane2_rx_diag_dfe_amp_tune)	16	RW	4CCCh
89D1h	DFE amp fine tuning 2 register (lane2_rx_diag_dfe_amp_tune_2)	16	RW	0C21h
89D2h	REE DAC control register (lane2_rx_diag_ree_dac_ctrl)	16	RW	0004h
89D3h	Receiver DFE control register 1 (lane2_rx_diag_dfe_ctrl1)	16	RW	0007h
89D4h	Receiver DFE control register 2 (lane2_rx_diag_dfe_ctrl2)	16	RW	003Eh
89D5h	Receiver DFE control register 3 (lane2_rx_diag_dfe_ctrl3)	16	RW	0ACAh
89D6h	Nyquist control register (lane2_rx_diag_nqst_ctrl)	16	RW	0998h
89D7h	Low frequency equalizer tuning register (lane2_rx_diag_lfeq_tune)	16	RW	0014h
89D8h	RX control register (lane2_rx_diag_rxctrl)	16	RW	0060h
89D9h	Receiver control reset diagnostic register (lane2_rx_diag_rst_diag)	16	RO	0000h
89DCh	RX signal detect tuning and control register (lane2_rx_diag_sigdet_tune)	16	RW	1005h
89DDh	RX LFPS detect tuning and control register (lane2_rx_diag_lfpsdet_tune)	16	RW	2410h
89DEh	Signal detect test register (lane2_rx_diag_sd_test)	16	RW	0000h
89E0h	RX sampler diagnostic control register (lane2_rx_diag_samp_ctrl)	16	RW	0001h
89E1h	RX Sampler CML TO CMOS enable delay register (lane2_rx_diag_sc2c_delay)	16	RW	012Ch
89E4h	MPHY control register 1 (lane2_rx_diag_mphy_ctrl_1)	16	RW	0000h
89E5h	MPHY control register 2 (lane2_rx_diag_mphy_ctrl_2)	16	RW	0000h
89E8h	RX loopback controller register (lane2_rx_diag_lpbk_ctrl)	16	RW	0000h
89E9h	RX extra enable control override register (lane2_rx_diag_ectrl_ovrd)	16	RW	0000h
89F0h	CML to CMOS bias trim register (lane2_rx_diag_cml2cmos_btrim)	16	RW	0000h
89F1h	RX bias gen control register 1 (lane2_rx_diag_bias_gen_ctrl1)	16	RW	0000h
89F2h	RX bias gen control register 2 (lane2_rx_diag_bias_gen_ctrl2)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
89F3h	RX bias gen control register 3 (lane2_rx_diag_bias_gen_ctrl3)	16	RW	0000h
89F4h	RX bias gen control register 4 (lane2_rx_diag_bias_gen_ctrl4)	16	RW	0000h
89F5h	RX boundary scan test mode register (lane2_rx_diag_bs_tm)	16	RW	0000h
89F6h	RX receiver front end test mode register 1 (lane2_rx_diag_rxfe_tm1)	16	RW	0000h
89F7h	RX receiver front end test mode register 2 (lane2_rx_diag_rxfe_tm2)	16	RW	0000h
89FEh	Receiver digital cover your alternatives register (lane2_rx_diag_dcya)	16	RW	0000h
89FFh	Receiver analog cover your alternatives register (lane2_rx_diag_acya)	16	RW	0000h
8C00h	Receiver A0 power state definition register (lane3_rx_psc_a0)	16	RW	8BFDh
8C01h	Receiver A1 power state definition register (lane3_rx_psc_a1)	16	RW	8BFDh
8C02h	Receiver A2 power state definition register (lane3_rx_psc_a2)	16	RW	8910h
8C03h	Receiver A3 power state definition register (lane3_rx_psc_a3)	16	RW	0000h
8C04h	Receiver A4 power state definition register (lane3_rx_psc_a4)	16	RW	1000h
8C05h	Receiver A5 power state definition register (lane3_rx_psc_a5)	16	RW	0000h
8C06h	Receiver calibration power state definition register (lane3_rx_psc_cal)	16	RW	03FFh
8C07h	Receiver ready power state definition register (lane3_rx_psc_rdy)	16	RW	0000h
8C20h	RX IQ PI ILL calibration control register (lane3_rx_iqpi_ill_cal_ctrl)	16	RW	0018h
8C21h	RX IQ PI ILL calibration start point register (lane3_rx_iqpi_ill_cal_start)	16	RW	1018h
8C22h	RX IQ PI ILL calibration timer control register (lane3_rx_iqpi_ill_cal_ctrl)	16	RW	0003h
8C23h	RX IQ PI ILL calibration override register (lane3_rx_iqpi_ill_cal_ovrd)	16	RW	0000h
8C24h	RX IQ PI ILL calibration initialization timer register (lane3_rx_iqpi_ill_cal_init_tmr)	16	RW	02EEh
8C25h	RX IQ PI ILL calibration iteration timer register (lane3_rx_iqpi_ill_cal_iter_tmr)	16	RW	0080h
8C26h	RX IQ PI ILL lock reference timer start value register (lane3_rx_iqpi_ill_lock_ref_tmr_start)	16	RW	01FFh
8C28h	RX IQ PI ILL lock calibration counter start value standard mode 0 register (lane3_rx_iqpi_ill_lock_calcnt_start_0)	16	RW	007Fh
8C29h	RX IQ PI ILL lock calibration counter start value standard mode 1 register (lane3_rx_iqpi_ill_lock_calcnt_start_1)	16	RW	00FFh
8C2Ah	RX IQ PI ILL lock calibration counter start value standard mode 2 register (lane3_rx_iqpi_ill_lock_calcnt_start_2)	16	RW	00FFh
8C2Bh	RX IQ PI ILL lock calibration counter start value standard mode 3 register (lane3_rx_iqpi_ill_lock_calcnt_start_3)	16	RW	0000h
8C30h	RX E PI ILL calibration control register (lane3_rx_epi_ill_cal_ctrl)	16	RW	0018h
8C31h	RX E PI ILL calibration start point register (lane3_rx_epi_ill_cal_start)	16	RW	1018h
8C32h	RX E PI ILL calibration timer control register (lane3_rx_epi_ill_cal_ctrl)	16	RW	0003h
8C33h	RX E PI ILL calibration override register (lane3_rx_epi_ill_cal_ovrd)	16	RW	0000h
8C34h	RX E PI ILL calibration initialization timer register (lane3_rx_epi_ill_cal_init_tmr)	16	RW	02EEh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8C35h	RX E PI ILL calibration iteration timer register (lane3_rx_epi_ill_cal_iter_tmr)	16	RW	0080h
8C36h	RX E PI ILL lock reference timer start value register (lane3_rx_epi_ill_lock_reftmr_start)	16	RW	01FFh
8C38h	RX E PI ILL lock calibration counter start value standard mode 0 register (lane3_rx_epi_ill_lock_calcnt_start_0)	16	RW	007Fh
8C39h	RX E PI ILL lock calibration counter start value standard mode 1 register (lane3_rx_epi_ill_lock_calcnt_start_1)	16	RW	00FFh
8C3Ah	RX E PI ILL lock calibration counter start value standard mode 2 register (lane3_rx_epi_ill_lock_calcnt_start_2)	16	RW	00FFh
8C3Bh	RX E PI ILL lock calibration counter start value standard mode 3 register (lane3_rx_epi_ill_lock_calcnt_start_3)	16	RW	0000h
8C40h	Signal detect calibration 0 control register (lane3_rx_sdc0_ctrl)	16	RW	0000h
8C41h	Signal detect calibration 0 override register (lane3_rx_sdc0_ovrd)	16	RW	0000h
8C42h	Signal detect calibration 0 start register (lane3_rx_sdc0_start)	16	RW	0000h
8C43h	Signal detect calibration 0 tune register (lane3_rx_sdc0_tune)	16	RW	0000h
8C44h	Signal detect calibration 0 initialization timer register (lane3_rx_sdc0_init_tmr)	16	RW	0019h
8C45h	Signal detect calibration 0 iteration timer register (lane3_rx_sdc0_iter_tmr)	16	RW	007Dh
8C48h	Signal detect calibration 1 control register (lane3_rx_sdc1_ctrl)	16	RW	0000h
8C49h	Signal detect calibration 1 override register (lane3_rx_sdc1_ovrd)	16	RW	0000h
8C4Ah	Signal detect calibration 1 start register (lane3_rx_sdc1_start)	16	RW	0000h
8C4Bh	Signal detect calibration 1 tune register (lane3_rx_sdc1_tune)	16	RW	0000h
8C4Ch	Signal detect calibration 1 initialization timer register (lane3_rx_sdc1_init_tmr)	16	RW	0019h
8C4Dh	Signal detect calibration 1 iteration timer register (lane3_rx_sdc1_iter_tmr)	16	RW	007Dh
8C58h	Sampler error DAC control register (lane3_rx_samp_dac_ctrl)	16	RW	0014h
8C80h	CDRLF configuration register (lane3_rx_cdrif_cfg)	16	RW	20B3h
8C81h	CDRLF configuration register 2 (lane3_rx_cdrif_cfg2)	16	RW	0000h
8C82h	CDRLF margin diagnostic register (lane3_rx_cdrif_mgn_diag)	16	RW	0000h
8C83h	CDRLF fast phase lock timer value register 0 (lane3_rx_cdrif_fpl_tmr0)	16	RW	0071h
8C84h	CDRLF fast phase lock timer value register 1 (lane3_rx_cdrif_fpl_tmr1)	16	RW	0731h
8C85h	CDRLF fast frequency lock timer value register (lane3_rx_cdrif_ffl_tmr)	16	RW	0018h
8C88h	CDRLF fast frequency lock step 0 control register (lane3_rx_cdrif_ffl0_ctrl)	16	RW	460Ch
8C89h	CDRLF fast frequency lock step 1 control register (lane3_rx_cdrif_ffl1_ctrl)	16	RW	4A0Ah
8C8Ah	CDRLF fast frequency lock step 2 control register (lane3_rx_cdrif_ffl2_ctrl)	16	RW	5414h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8C8Bh	CDRLF fast frequency lock step 3 control register (lane3_rx_cdrf_fl3_ctrl)	16	RW	5E1Eh
8C8Ch	CDRLF fast frequency lock step 4 control register (lane3_rx_cdrf_fl4_ctrl)	16	RW	9E1Eh
8C90h	Receiver signal detect filter high to low filter timer register (lane3_rx_sigdet_hl_filt_tmr)	16	RW	0019h
8C91h	Receiver signal detect filter high to low delay timer register (lane3_rx_sigdet_hl_dly_tmr)	16	RW	0000h
8C92h	Receiver signal detect filter high to low min timer register (lane3_rx_sigdet_hl_min_tmr)	16	RW	0000h
8C93h	Receiver signal detect filter high to low init timer register (lane3_rx_sigdet_hl_init_tmr)	16	RW	03E8h
8C94h	Receiver signal detect filter low to high filter timer register (lane3_rx_sigdet_lh_filt_tmr)	16	RW	0004h
8C95h	Receiver signal detect filter low to high delay timer register (lane3_rx_sigdet_lh_dly_tmr)	16	RW	0004h
8C96h	Receiver signal detect filter low to high min timer register (lane3_rx_sigdet_lh_min_tmr)	16	RW	0000h
8C97h	Receiver signal detect filter low to high init timer register (lane3_rx_sigdet_lh_init_tmr)	16	RW	03E8h
8C98h	Receiver LFPS detect filter filter timer register (lane3_rx_lfpsdet_filt_tmr)	16	RW	0008h
8C99h	Receiver LFPS detect filter delay timer register (lane3_rx_lfpsdet_dly_tmr)	16	RW	000Bh
8C9Ah	Receiver LFPS detect min timer register (lane3_rx_lfpsdet_min_tmr)	16	RW	0013h
8C9Bh	Receiver LFPS detect init timer register (lane3_rx_lfpsdet_init_tmr)	16	RW	0000h
8CA0h	Eye surf control register (lane3_rx_eyesurf_ctrl)	16	RW	0000h
8CA4h	Eye surf timer delay low register (lane3_rx_eyesurf_tmr_dellow)	16	RW	0000h
8CA5h	Eye surf timer delay high register (lane3_rx_eyesurf_tmr_delhigh)	16	RW	0000h
8CA6h	Eye surf timer test low register (lane3_rx_eyesurf_tmr_testlow)	16	RW	0000h
8CA7h	Eye surf timer test high register (lane3_rx_eyesurf_tmr_testhigh)	16	RW	0000h
8CA8h	Eye surf north south test point coordinate register (lane3_rx_eyesurf_ns_coord)	16	RW	0000h
8CA9h	Eye surf east west test point coordinate register (lane3_rx_eyesurf_ew_coord)	16	RW	0000h
8CAAh	Eye surf bit error count register (lane3_rx_eyesurf_errcnt)	16	RO	0000h
8CB0h	Receiver BIST control register (lane3_rx_bist_ctrl)	16	RW	0000h
8CB1h	Receiver BIST sync count register (lane3_rx_bist_synccnt)	16	RW	0000h
8CB2h	Receiver BIST user defined data write register (lane3_rx_bist_uddwr)	16	WO	0000h
8CB3h	Receiver BIST error count register (lane3_rx_bist_errcnt)	16	RO	0000h
8CB4h	Clock frequency measurement control register (lane3_xcvr_cmsmt_clk_freq_msmt_ctrl)	16	RW	0000h
8CB5h	Test clock selection register (lane3_xcvr_cmsmt_test_clk_sel)	16	RW	0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8CB6h	Reference clock timer value register (lane3_xcvt_cmsmt_ref_clk_tmr_value)	16	RW	0000h
8CB7h	Test clock counter value register (lane3_xcvt_cmsmt_test_clk_cnt_value)	16	RO	0000h
8CE0h	RX sampler latch calibration control register (lane3_rx_slc_ctrl)	16	RW	0EFFh
8CE1h	RX sampler latch calibration enable initialization timer value register (lane3_rx_slc_en_init_tmr)	16	RW	0004h
8CE2h	RX sampler latch calibration unit initialization timer value register (lane3_rx_slc_cu_init_tmr)	16	RW	0010h
8CE3h	RX sampler latch calibration unit iteration timer value register (lane3_rx_slc_cu_iter_tmr)	16	RW	0001h
8CE4h	RX sampler latch calibration I even data mask register (lane3_rx_slc_ie_mask)	16	RW	03FFh
8CE5h	RX sampler latch calibration I odd data mask register (lane3_rx_slc_io_mask)	16	RW	03FFh
8CE6h	RX sampler latch calibration Q even data mask register (lane3_rx_slc_qe_mask)	16	RW	03FFh
8CE7h	RX sampler latch calibration Q odd data mask register (lane3_rx_slc_qo_mask)	16	RW	03FFh
8CE8h	RX sampler latch calibration E even data mask register (lane3_rx_slc_ee_mask)	16	RW	03FFh
8CE9h	RX sampler latch calibration E odd data mask register (lane3_rx_slc_eo_mask)	16	RW	03FFh
8CEAh	RX sampler latch calibration data threshold register (lane3_rx_slc_data_thr)	16	RW	0005h
8D00h	RX sampler latch I odd positive 0 calibration unit control register (lane3_rx_slc_iop0_ctrl)	16	RW	0021h
8D01h	RX sampler latch I odd positive 0 calibration unit override register (lane3_rx_slc_iop0_ovrd)	16	RW	0000h
8D02h	RX sampler latch I odd positive 0 calibration unit start register (lane3_rx_slc_iop0_start)	16	RW	0021h
8D03h	RX sampler latch I odd positive 0 calibration unit tune register (lane3_rx_slc_iop0_tune)	16	RW	0000h
8D04h	RX sampler latch I odd positive 1 calibration unit control register (lane3_rx_slc_iop1_ctrl)	16	RW	001Fh
8D05h	RX sampler latch I odd positive 1 calibration unit override register (lane3_rx_slc_iop1_ovrd)	16	RW	0000h
8D06h	RX sampler latch I odd positive 1 calibration unit start register (lane3_rx_slc_iop1_start)	16	RW	801Fh
8D07h	RX sampler latch I odd positive 1 calibration unit tune register (lane3_rx_slc_iop1_tune)	16	RW	0000h
8D08h	RX sampler latch Q odd positive 0 calibration unit control register (lane3_rx_slc_qop0_ctrl)	16	RW	0021h
8D09h	RX sampler latch Q odd positive 0 calibration unit override register (lane3_rx_slc_qop0_ovrd)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8D0Ah	RX sampler latch Q odd positive 0 calibration unit start register (lane 3_rx_slc_qop0_start)	16	RW	0021h
8D0Bh	RX sampler latch Q odd positive 0 calibration unit tune register (lane 3_rx_slc_qop0_tune)	16	RW	0000h
8D0Ch	RX sampler latch Q odd positive 1 calibration unit control register (lane3_rx_slc_qop1_ctrl)	16	RW	001Fh
8D0Dh	RX sampler latch Q odd positive 1 calibration unit override register (lane3_rx_slc_qop1_ovrd)	16	RW	0000h
8D0Eh	RX sampler latch Q odd positive 1 calibration unit start register (lane 3_rx_slc_qop1_start)	16	RW	801Fh
8D0Fh	RX sampler latch Q odd positive 1 calibration unit tune register (lane 3_rx_slc_qop1_tune)	16	RW	0000h
8D10h	RX sampler latch E odd positive 0 calibration unit control register (lane3_rx_slc_eop0_ctrl)	16	RW	0021h
8D11h	RX sampler latch E odd positive 0 calibration unit override register (lane3_rx_slc_eop0_ovrd)	16	RW	0000h
8D12h	RX sampler latch E odd positive 0 calibration unit start register (lane 3_rx_slc_eop0_start)	16	RW	0021h
8D13h	RX sampler latch E odd positive 0 calibration unit tune register (lane 3_rx_slc_eop0_tune)	16	RW	0000h
8D14h	RX sampler latch E odd positive 1 calibration unit control register (lane3_rx_slc_eop1_ctrl)	16	RW	001Fh
8D15h	RX sampler latch E odd positive 1 calibration unit override register (lane3_rx_slc_eop1_ovrd)	16	RW	0000h
8D16h	RX sampler latch E odd positive 1 calibration unit start register (lane 3_rx_slc_eop1_start)	16	RW	801Fh
8D17h	RX sampler latch E odd positive 1 calibration unit tune register (lane 3_rx_slc_eop1_tune)	16	RW	0000h
8D18h	RX sampler latch I odd negative 0 calibration unit control register (lane3_rx_slc_ion0_ctrl)	16	RW	0021h
8D19h	RX sampler latch I odd negative 0 calibration unit override register (lane3_rx_slc_ion0_ovrd)	16	RW	0000h
8D1Ah	RX sampler latch I odd negative 0 calibration unit start register (lane 3_rx_slc_ion0_start)	16	RW	0021h
8D1Bh	RX sampler latch I odd negative 0 calibration unit tune register (lane 3_rx_slc_ion0_tune)	16	RW	0000h
8D1Ch	RX sampler latch I odd negative 1 calibration unit control register (lane3_rx_slc_ion1_ctrl)	16	RW	001Fh
8D1Dh	RX sampler latch I odd negative 1 calibration unit override register (lane3_rx_slc_ion1_ovrd)	16	RW	0000h
8D1Eh	RX sampler latch I odd negative 1 calibration unit start register (lane 3_rx_slc_ion1_start)	16	RW	801Fh
8D1Fh	RX sampler latch I odd negative 1 calibration unit tune register (lane 3_rx_slc_ion1_tune)	16	RW	0000h
8D20h	RX sampler latch Q odd negative 0 calibration unit control register (lane3_rx_slc_qon0_ctrl)	16	RW	0021h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8D21h	RX sampler latch Q odd negative 0 calibration unit override register (lane3_rx_slc_qon0_ovrd)	16	RW	0000h
8D22h	RX sampler latch Q odd negative 0 calibration unit start register (lane3_rx_slc_qon0_start)	16	RW	0021h
8D23h	RX sampler latch Q odd negative 0 calibration unit tune register (lane3_rx_slc_qon0_tune)	16	RW	0000h
8D24h	RX sampler latch Q odd negative 1 calibration unit control register (lane3_rx_slc_qon1_ctrl)	16	RW	001Fh
8D25h	RX sampler latch Q odd negative 1 calibration unit override register (lane3_rx_slc_qon1_ovrd)	16	RW	0000h
8D26h	RX sampler latch Q odd negative 1 calibration unit start register (lane3_rx_slc_qon1_start)	16	RW	801Fh
8D27h	RX sampler latch Q odd negative 1 calibration unit tune register (lane3_rx_slc_qon1_tune)	16	RW	0000h
8D28h	RX sampler latch E odd negative 0 calibration unit control register (lane3_rx_slc_eon0_ctrl)	16	RW	0021h
8D29h	RX sampler latch E odd negative 0 calibration unit override register (lane3_rx_slc_eon0_ovrd)	16	RW	0000h
8D2Ah	RX sampler latch E odd negative 0 calibration unit start register (lane3_rx_slc_eon0_start)	16	RW	0021h
8D2Bh	RX sampler latch E odd negative 0 calibration unit tune register (lane3_rx_slc_eon0_tune)	16	RW	0000h
8D2Ch	RX sampler latch E odd negative 1 calibration unit control register (lane3_rx_slc_eon1_ctrl)	16	RW	001Fh
8D2Dh	RX sampler latch E odd negative 1 calibration unit override register (lane3_rx_slc_eon1_ovrd)	16	RW	0000h
8D2Eh	RX sampler latch E odd negative 1 calibration unit start register (lane3_rx_slc_eon1_start)	16	RW	801Fh
8D2Fh	RX sampler latch E odd negative 1 calibration unit tune register (lane3_rx_slc_eon1_tune)	16	RW	0000h
8D30h	RX sampler latch I even positive 0 calibration unit control register (lane3_rx_slc_iep0_ctrl)	16	RW	0021h
8D31h	RX sampler latch I even positive 0 calibration unit override register (lane3_rx_slc_iep0_ovrd)	16	RW	0000h
8D32h	RX sampler latch I even positive 0 calibration unit start register (lane3_rx_slc_iep0_start)	16	RW	0021h
8D33h	RX sampler latch I even positive 0 calibration unit tune register (lane3_rx_slc_iep0_tune)	16	RW	0000h
8D34h	RX sampler latch I even positive 1 calibration unit control register (lane3_rx_slc_iep1_ctrl)	16	RW	001Fh
8D35h	RX sampler latch I even positive 1 calibration unit override register (lane3_rx_slc_iep1_ovrd)	16	RW	0000h
8D36h	RX sampler latch I even positive 1 calibration unit start register (lane3_rx_slc_iep1_start)	16	RW	801Fh
8D37h	RX sampler latch I even positive 1 calibration unit tune register (lane3_rx_slc_iep1_tune)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8D38h	RX sampler latch Q even positive 0 calibration unit control register (lane3_rx_slc_qep0_ctrl)	16	RW	0021h
8D39h	RX sampler latch Q even positive 0 calibration unit override register (lane3_rx_slc_qep0_ovrd)	16	RW	0000h
8D3Ah	RX sampler latch Q even positive 0 calibration unit start register (lane3_rx_slc_qep0_start)	16	RW	0021h
8D3Bh	RX sampler latch Q even positive 0 calibration unit tune register (lane3_rx_slc_qep0_tune)	16	RW	0000h
8D3Ch	RX sampler latch Q even positive 1 calibration unit control register (lane3_rx_slc_qep1_ctrl)	16	RW	001Fh
8D3Dh	RX sampler latch Q even positive 1 calibration unit override register (lane3_rx_slc_qep1_ovrd)	16	RW	0000h
8D3Eh	RX sampler latch Q even positive 1 calibration unit start register (lane3_rx_slc_qep1_start)	16	RW	801Fh
8D3Fh	RX sampler latch Q even positive 1 calibration unit tune register (lane3_rx_slc_qep1_tune)	16	RW	0000h
8D40h	RX sampler latch E even positive 0 calibration unit control register (lane3_rx_slc_eep0_ctrl)	16	RW	0021h
8D41h	RX sampler latch E even positive 0 calibration unit override register (lane3_rx_slc_eep0_ovrd)	16	RW	0000h
8D42h	RX sampler latch E even positive 0 calibration unit start register (lane3_rx_slc_eep0_start)	16	RW	0021h
8D43h	RX sampler latch E even positive 0 calibration unit tune register (lane3_rx_slc_eep0_tune)	16	RW	0000h
8D44h	RX sampler latch E even positive 1 calibration unit control register (lane3_rx_slc_eep1_ctrl)	16	RW	001Fh
8D45h	RX sampler latch E even positive 1 calibration unit override register (lane3_rx_slc_eep1_ovrd)	16	RW	0000h
8D46h	RX sampler latch E even positive 1 calibration unit start register (lane3_rx_slc_eep1_start)	16	RW	801Fh
8D47h	RX sampler latch E even positive 1 calibration unit tune register (lane3_rx_slc_eep1_tune)	16	RW	0000h
8D48h	RX sampler latch I even negative 0 calibration unit control register (lane3_rx_slc_ien0_ctrl)	16	RW	0021h
8D49h	RX sampler latch I even negative 0 calibration unit override register (lane3_rx_slc_ien0_ovrd)	16	RW	0000h
8D4Ah	RX sampler latch I even negative 0 calibration unit start register (lane3_rx_slc_ien0_start)	16	RW	0021h
8D4Bh	RX sampler latch I even negative 0 calibration unit tune register (lane3_rx_slc_ien0_tune)	16	RW	0000h
8D4Ch	RX sampler latch I even negative 1 calibration unit control register (lane3_rx_slc_ien1_ctrl)	16	RW	001Fh
8D4Dh	RX sampler latch I even negative 1 calibration unit override register (lane3_rx_slc_ien1_ovrd)	16	RW	0000h
8D4Eh	RX sampler latch I even negative 1 calibration unit start register (lane3_rx_slc_ien1_start)	16	RW	801Fh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8D4Fh	RX sampler latch I even negative 1 calibration unit tune register (lane3_rx_slc_ien1_tune)	16	RW	0000h
8D50h	RX sampler latch Q even negative 0 calibration unit control register (lane3_rx_slc_qen0_ctrl)	16	RW	0021h
8D51h	RX sampler latch Q even negative 0 calibration unit override register (lane3_rx_slc_qen0_ovrd)	16	RW	0000h
8D52h	RX sampler latch Q even negative 0 calibration unit start register (lane3_rx_slc_qen0_start)	16	RW	0021h
8D53h	RX sampler latch Q even negative 0 calibration unit tune register (lane3_rx_slc_qen0_tune)	16	RW	0000h
8D54h	RX sampler latch Q even negative 1 calibration unit control register (lane3_rx_slc_qen1_ctrl)	16	RW	001Fh
8D55h	RX sampler latch Q even negative 1 calibration unit override register (lane3_rx_slc_qen1_ovrd)	16	RW	0000h
8D56h	RX sampler latch Q even negative 1 calibration unit start register (lane3_rx_slc_qen1_start)	16	RW	801Fh
8D57h	RX sampler latch Q even negative 1 calibration unit tune register (lane3_rx_slc_qen1_tune)	16	RW	0000h
8D58h	RX sampler latch E even negative 0 calibration unit control register (lane3_rx_slc_een0_ctrl)	16	RW	0021h
8D59h	RX sampler latch E even negative 0 calibration unit override register (lane3_rx_slc_een0_ovrd)	16	RW	0000h
8D5Ah	RX sampler latch E even negative 0 calibration unit start register (lane3_rx_slc_een0_start)	16	RW	0021h
8D5Bh	RX sampler latch E even negative 0 calibration unit tune register (lane3_rx_slc_een0_tune)	16	RW	0000h
8D5Ch	RX sampler latch E even negative 1 calibration unit control register (lane3_rx_slc_een1_ctrl)	16	RW	001Fh
8D5Dh	RX sampler latch E even negative 1 calibration unit override register (lane3_rx_slc_een1_ovrd)	16	RW	0000h
8D5Eh	RX sampler latch E even negative 1 calibration unit start register (lane3_rx_slc_een1_start)	16	RW	801Fh
8D5Fh	RX sampler latch E even negative 1 calibration unit tune register (lane3_rx_slc_een1_tune)	16	RW	0000h
8D60h	REE USB 3 general control state machine control register (lane3_rx_ree_u3gcsn_ctrl)	16	RW	0001h
8D61h	REE USB 3 general control state machine phase 1 equalization enable mask register (lane3_rx_ree_u3gcsn_eqenm_ph1)	16	RW	00C7h
8D62h	REE USB 3 general control state machine phase 2 equalization enable mask register (lane3_rx_ree_u3gcsn_eqenm_ph2)	16	RW	00C7h
8D63h	REE USB 3 general control state machine start timer value register (lane3_rx_ree_u3gcsn_start_tmr)	16	RW	0125h
8D64h	REE USB 3 general control state machine run phase 1 timer value register (lane3_rx_ree_u3gcsn_run_ph1_tmr)	16	RW	07A2h
8D65h	REE USB 3 general control state machine run phase 2 timer value register (lane3_rx_ree_u3gcsn_run_ph2_tmr)	16	RW	009Dh

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8D68h	REE PCIe Gen 2 general control state machine control register (lane3_rx_ree_g2gcsn_ctrl)	16	RW	0001h
8D69h	REE PCIe Gen 2 general control state machine phase 1 equalization enable mask register (lane3_rx_ree_g2gcsn_eqnm_ph1)	16	RW	00C7h
8D6Ah	REE PCIe Gen 2 general control state machine phase 2 equalization enable mask register (lane3_rx_ree_g2gcsn_eqnm_ph2)	16	RW	00C7h
8D6Bh	REE PCIe Gen 2 general control state machine start timer value register (lane3_rx_ree_g2gcsn_start_tmr)	16	RW	0000h
8D6Ch	REE PCIe Gen 2 general control state machine run phase 1 timer value register (lane3_rx_ree_g2gcsn_run_ph1_tmr)	16	RW	0F43h
8D6Dh	REE PCIe Gen 2 general control state machine run phase 2 timer value register (lane3_rx_ree_g2gcsn_run_ph2_tmr)	16	RW	009Dh
8D78h	REE periodic general control state machine control register (lane3_rx_ree_pergcsn_ctrl)	16	RW	0001h
8D79h	REE periodic general control state machine phase 1 equalization enable mask register (lane3_rx_ree_pergcsn_eqnm_ph1)	16	RW	0080h
8D7Ah	REE periodic general control state machine phase 2 equalization enable mask register (lane3_rx_ree_pergcsn_eqnm_ph2)	16	RW	0080h
8D7Bh	REE periodic general control state machine start timer value register (lane3_rx_ree_pergcsn_start_tmr)	16	RW	0000h
8D7Ch	REE periodic general control state machine run phase 1 timer value register (lane3_rx_ree_pergcsn_run_ph1_tmr)	16	RW	FFFFh
8D7Dh	REE periodic general control state machine run phase 2 timer value register (lane3_rx_ree_pergcsn_run_ph2_tmr)	16	RW	009Dh
8D80h	REE tap 1 control register (lane3_rx_ree_tap1_ctrl)	16	RW	0000h
8D81h	REE tap 1 override register (lane3_rx_ree_tap1_ovrd)	16	RW	0000h
8D82h	REE tap 1 diagnostics register (lane3_rx_ree_tap1_diag)	16	RW	0000h
8D84h	REE tap 2 control register (lane3_rx_ree_tap2_ctrl)	16	RW	0400h
8D85h	REE tap 2 override register (lane3_rx_ree_tap2_ovrd)	16	RW	0000h
8D86h	REE tap 2 diagnostics register (lane3_rx_ree_tap2_diag)	16	RW	0000h
8D88h	REE tap 3 control register (lane3_rx_ree_tap3_ctrl)	16	RW	0400h
8D89h	REE tap 3 override register (lane3_rx_ree_tap3_ovrd)	16	RW	0000h
8D8Ah	REE tap 3 diagnostics register (lane3_rx_ree_tap3_diag)	16	RW	0000h
8D94h	REE analog enable control state machine delay timer value register (lane3_rx_ree_anaensm_del_tmr)	16	RW	02EEh
8D98h	REE peaking amp control register (lane3_rx_ree_peak_ctrl)	16	RW	0701h
8D99h	REE peaking amp code control register (lane3_rx_ree_peak_code_ctrl)	16	RW	2A0Eh
8D9Ah	REE peaking amp upper threshold register (lane3_rx_ree_peak_uthr)	16	RW	0004h
8D9Bh	REE peaking amp lower threshold register (lane3_rx_ree_peak_lthr)	16	RW	0000h
8D9Ch	REE peaking amp input override register (lane3_rx_ree_peak_iovrd)	16	RW	0000h
8D9Dh	REE peaking amp code override register (lane3_rx_ree_peak_covrd)	16	RW	0000h
8D9Eh	REE peaking amp diagnostics register (lane3_rx_ree_peak_diag)	16	RW	0020h
8DA0h	REE attenuation control register (lane3_rx_ree_atten_ctrl)	16	RW	0005h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8DA1h	REE attenuation threshold register (lane3_rx_ree_atten_thr)	16	RW	0C02h
8DA2h	REE attenuation counter register (lane3_rx_ree_atten_cnt)	16	RW	2100h
8DA3h	REE attenuation override register (lane3_rx_ree_atten_ovrd)	16	RW	0000h
8DA4h	REE attenuation diagnostics register (lane3_rx_ree_atten_diag)	16	RO	0000h
8DA8h	REE low frequency equalizer control register (lane3_rx_ree_lfeq_ctrl)	16	RW	0000h
8DA9h	REE low frequency equalizer override register (lane3_rx_ree_lfeq_ovrd)	16	RW	0000h
8DAAh	REE low frequency equalizer diagnostics register (lane3_rx_ree_lfeq_diag)	16	RW	0000h
8DACH	REE VGA gain control register (lane3_rx_ree_vga_gain_ctrl)	16	RW	1001h
8DADh	REE VGA gain override register (lane3_rx_ree_vga_gain_ovrd)	16	RW	0000h
8DAEh	REE VGA gain diagnostics register (lane3_rx_ree_vga_gain_diag)	16	RW	000Ah
8DAFh	REE VGA gain target adjust diagnostics register (lane3_rx_ree_vga_gain_tgt_diag)	16	RO	0000h
8DB0h	REE offset correction control register (lane3_rx_ree_off_cor_ctrl)	16	RW	0001h
8DB1h	REE offset correction override register (lane3_rx_ree_off_cor_ovrd)	16	RW	0000h
8DB2h	REE offset correction diagnostics register (lane3_rx_ree_off_cor_diag)	16	RW	0000h
8DB8h	REE adder configuration register (lane3_rx_ree_addr_cfg)	16	RW	0001h
8DB9h	REE tap 1 clip control register (lane3_rx_ree_tap1_clip)	16	RW	0519h
8DBAh	REE taps 2 and 3 clip control register (lane3_rx_ree_tap2ton_clip)	16	RW	0519h
8DBBh	REE control data mask register (lane3_rx_ree_ctrl_data_mask)	16	RW	4000h
8DBCCh	REE diagnostic control register (lane3_rx_ree_diag_ctrl)	16	RW	0040h
8DBDh	REE control state machine gen mode control register 1 (lane3_rx_ree_smgm_ctrl1)	16	RW	0702h
8DBEh	REE control state machine gen mode control register 2 (lane3_rx_ree_smgm_ctrl2)	16	RW	0000h
8DC0h	RX ILL diagnostic control register (lane3_rx_diag_ill_ctrl)	16	RW	0000h
8DC1h	RX ILL IQ trim 0 register (lane3_rx_diag_ill_iq_trim0)	16	RW	533Fh
8DC2h	RX ILL E trim 0 register (lane3_rx_diag_ill_e_trim0)	16	RW	533Fh
8DC3h	RX ILL IQ trim 1 register (lane3_rx_diag_ill_iq_trim1)	16	RW	0002h
8DC4h	RX ILL E trim 1 register (lane3_rx_diag_ill_e_trim1)	16	RW	0002h
8DC5h	RX ILL IQ E trim 2 register (lane3_rx_diag_ill_iqe_trim2)	16	RW	3E00h
8DC6h	RX ILL IQ E trim 3 register (lane3_rx_diag_ill_iqe_trim3)	16	RW	199Fh
8DC7h	RX ILL IQ E trim 4 register (lane3_rx_diag_ill_iqe_trim4)	16	RW	0014h
8DC8h	RX ILL IQ E trim 5 register (lane3_rx_diag_ill_iqe_trim5)	16	RW	1504h
8DC9h	RX ILL IQ E trim 6 register (lane3_rx_diag_ill_iqe_trim6)	16	RW	0015h
8DD0h	DFE amp fine tuning register (lane3_rx_diag_dfe_amp_tune)	16	RW	4CCCh
8DD1h	DFE amp fine tuning 2 register (lane3_rx_diag_dfe_amp_tune_2)	16	RW	0C21h
8DD2h	REE DAC control register (lane3_rx_diag_ree_dac_ctrl)	16	RW	0004h
8DD3h	Receiver DFE control register 1 (lane3_rx_diag_dfe_ctrl1)	16	RW	0007h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8DD4h	Receiver DFE control register 2 (lane3_rx_diag_dfe_ctrl2)	16	RW	003Eh
8DD5h	Receiver DFE control register 3 (lane3_rx_diag_dfe_ctrl3)	16	RW	0ACAh
8DD6h	Nyquist control register (lane3_rx_diag_nqst_ctrl)	16	RW	0998h
8DD7h	Low frequency equalizer tuning register (lane3_rx_diag_lfq_tune)	16	RW	0014h
8DD8h	RX control register (lane3_rx_diag_rxctrl)	16	RW	0060h
8DD9h	Receiver control reset diagnostic register (lane3_rx_diag_rst_diag)	16	RO	0000h
8DDCh	RX signal detect tuning and control register (lane3_rx_diag_sigdet_tune)	16	RW	1005h
8DDdh	RX LFPS detect tuning and control register (lane3_rx_diag_lfpsdet_tune)	16	RW	2410h
8DDEh	Signal detect test register (lane3_rx_diag_sd_test)	16	RW	0000h
8DE0h	RX sampler diagnostic control register (lane3_rx_diag_samp_ctrl)	16	RW	0001h
8DE1h	RX Sampler CML TO CMOS enable delay register (lane3_rx_diag_sc2c_delay)	16	RW	012Ch
8DE4h	MPHY control register 1 (lane3_rx_diag_mphy_ctrl_1)	16	RW	0000h
8DE5h	MPHY control register 2 (lane3_rx_diag_mphy_ctrl_2)	16	RW	0000h
8DE8h	RX loopback controller register (lane3_rx_diag_lpbk_ctrl)	16	RW	0000h
8DE9h	RX extra enable control override register (lane3_rx_diag_ectrl_ovrd)	16	RW	0000h
8DF0h	CML to CMOS bias trim register (lane3_rx_diag_cml2cmos_btrim)	16	RW	0000h
8DF1h	RX bias gen control register 1 (lane3_rx_diag_bias_gen_ctrl1)	16	RW	0000h
8DF2h	RX bias gen control register 2 (lane3_rx_diag_bias_gen_ctrl2)	16	RW	0000h
8DF3h	RX bias gen control register 3 (lane3_rx_diag_bias_gen_ctrl3)	16	RW	0000h
8DF4h	RX bias gen control register 4 (lane3_rx_diag_bias_gen_ctrl4)	16	RW	0000h
8DF5h	RX boundary scan test mode register (lane3_rx_diag_bs_tm)	16	RW	0000h
8DF6h	RX receiver front end test mode register 1 (lane3_rx_diag_rxfe_tm1)	16	RW	0000h
8DF7h	RX receiver front end test mode register 2 (lane3_rx_diag_rxfe_tm2)	16	RW	0000h
8DFEh	Receiver digital cover your alternatives register (lane3_rx_diag_dcya)	16	RW	0000h
8DFFh	Receiver analog cover your alternatives register (lane3_rx_diag_acya)	16	RW	0000h
C408h	HDP Lane Configuration register (lane0_phy_hdp_tx_ctl)	16	RW	0000h
C41Ch	DP Tx data low isolation register (lane0_phy_dp_iso_tx_data_lo)	16	RW	0000h
C41Dh	DP Tx data high isolation register (lane0_phy_dp_iso_tx_data_hi)	16	RW	0000h
C448h	HDP Lane Configuration register (lane1_phy_hdp_tx_ctl)	16	RW	0000h
C45Ch	DP Tx data low isolation register (lane1_phy_dp_iso_tx_data_lo)	16	RW	0000h
C45Dh	DP Tx data high isolation register (lane1_phy_dp_iso_tx_data_hi)	16	RW	0000h
C488h	HDP Lane Configuration register (lane2_phy_hdp_tx_ctl)	16	RW	0000h
C49Ch	DP Tx data low isolation register (lane2_phy_dp_iso_tx_data_lo)	16	RW	0000h
C49Dh	DP Tx data high isolation register (lane2_phy_dp_iso_tx_data_hi)	16	RW	0000h
C4C8h	HDP Lane Configuration register (lane3_phy_hdp_tx_ctl)	16	RW	0000h
C4DCh	DP Tx data low isolation register (lane3_phy_dp_iso_tx_data_lo)	16	RW	0000h
C4DDh	DP Tx data high isolation register (lane3_phy_dp_iso_tx_data_hi)	16	RW	0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
CC00h	PMA transceiver control register (lane0_phy_pma_xcvr_ctrl)	16	RW	0000h
CC01h	PMA loopback control register (lane0_phy_pma_xcvr_lpbk)	16	RW	0000h
CC04h	PMA PSM current state lower register (lane0_phy_pma_psm_state_lo)	16	RO	0000h
CC05h	PMA PSM current state higher register (lane0_phy_pma_psm_state_hi)	16	RO	0000h
CC10h	PMA Isolation Transceiver control register (lane0_phy_pma_iso_xcvr_ctrl)	16	RW	0000h
CC11h	PMA TX configuration register (lane0_phy_pma_iso_tx_cfg)	16	RW	0000h
CC12h	PMA Isolation mode control register (lane0_phy_pma_iso_link_mode)	16	RW	C000h
CC13h	PMA Isolation power state control register (lane0_phy_pma_iso_pwrst_ctrl)	16	RW	0000h
CC14h	PMA transmit low data isolation register (lane0_phy_pma_iso_tx_data_lo)	16	RW	0000h
CC15h	PMA transmit high data isolation register (lane0_phy_pma_iso_tx_data_hi)	16	RW	0000h
CC40h	PMA transceiver control register (lane1_phy_pma_xcvr_ctrl)	16	RW	0000h
CC41h	PMA loopback control register (lane1_phy_pma_xcvr_lpbk)	16	RW	0000h
CC44h	PMA PSM current state lower register (lane1_phy_pma_psm_state_lo)	16	RO	0000h
CC45h	PMA PSM current state higher register (lane1_phy_pma_psm_state_hi)	16	RO	0000h
CC50h	PMA Isolation Transceiver control register (lane1_phy_pma_iso_xcvr_ctrl)	16	RW	0000h
CC51h	PMA TX configuration register (lane1_phy_pma_iso_tx_cfg)	16	RW	0000h
CC52h	PMA Isolation mode control register (lane1_phy_pma_iso_link_mode)	16	RW	C000h
CC53h	PMA Isolation power state control register (lane1_phy_pma_iso_pwrst_ctrl)	16	RW	0000h
CC54h	PMA transmit low data isolation register (lane1_phy_pma_iso_tx_data_lo)	16	RW	0000h
CC55h	PMA transmit high data isolation register (lane1_phy_pma_iso_tx_data_hi)	16	RW	0000h
CC80h	PMA transceiver control register (lane2_phy_pma_xcvr_ctrl)	16	RW	0000h
CC81h	PMA loopback control register (lane2_phy_pma_xcvr_lpbk)	16	RW	0000h
CC84h	PMA PSM current state lower register (lane2_phy_pma_psm_state_lo)	16	RO	0000h
CC85h	PMA PSM current state higher register (lane2_phy_pma_psm_state_hi)	16	RO	0000h
CC90h	PMA Isolation Transceiver control register (lane2_phy_pma_iso_xcvr_ctrl)	16	RW	0000h
CC91h	PMA TX configuration register (lane2_phy_pma_iso_tx_cfg)	16	RW	0000h
CC92h	PMA Isolation mode control register (lane2_phy_pma_iso_link_mode)	16	RW	C000h
CC93h	PMA Isolation power state control register (lane2_phy_pma_iso_pwrst_ctrl)	16	RW	0000h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
CC94h	PMA transmit low data isolation register (lane2_phy_pma_iso_tx_data_lo)	16	RW	0000h
CC95h	PMA transmit high data isolation register (lane2_phy_pma_iso_tx_data_hi)	16	RW	0000h
CCC0h	PMA transceiver control register (lane3_phy_pma_xcvr_ctrl)	16	RW	0000h
CCC1h	PMA loopback control register (lane3_phy_pma_xcvr_lpbk)	16	RW	0000h
CCC4h	PMA PSM current state lower register (lane3_phy_pma_psm_state_lo)	16	RO	0000h
CCC5h	PMA PSM current state higher register (lane3_phy_pma_psm_state_hi)	16	RO	0000h
CCD0h	PMA Isolation Transceiver control register (lane3_phy_pma_iso_xcvr_ctrl)	16	RW	0000h
CCD1h	PMA TX configuration register (lane3_phy_pma_iso_tx_cfg)	16	RW	0000h
CCD2h	PMA Isolation mode control register (lane3_phy_pma_iso_link_mode)	16	RW	C000h
CCD3h	PMA Isolation power state control register (lane3_phy_pma_iso_pwrst_ctrl)	16	RW	0000h
CCD4h	PMA transmit low data isolation register (lane3_phy_pma_iso_tx_data_lo)	16	RW	0000h
CCD5h	PMA transmit high data isolation register (lane3_phy_pma_iso_tx_data_hi)	16	RW	0000h
8_0000h	Product type ID register (cmn_pid_type)	16	RO	7364h
8_0001h	Product number 1 2 ID register (cmn_pid_num12)	16	RO	0000h
8_0002h	Product number 3 4 ID register (cmn_pid_num34)	16	RO	0000h
8_0003h	Product number 5 6 ID register (cmn_pid_num56)	16	RO	0000h
8_0004h	Product number 7 8 ID register (cmn_pid_num78)	16	RO	0000h
8_0005h	Product number 9 10 ID register (cmn_pid_num910)	16	RO	0000h
8_0006h	Product number 11 12 ID register (cmn_pid_num1112)	16	RO	0000h
8_0007h	Product revision ID register (cmn_pid_rev)	16	RO	0100h
8_0008h	Product technology manufacturer ID register (cmn_pid_mfg)	16	RO	0074h
8_0009h	Product technology process node ID register (cmn_pid_node)	16	RO	0028h
8_000Ah	Product technology process flavor ID register 0 (cmn_pid_flv0)	16	RO	0000h
8_000Bh	Product technology process flavor ID register 1 (cmn_pid_flv1)	16	RO	7400h
8_000Ch	Product I/O voltage ID register (cmn_pid_iov)	16	RO	0180h
8_000Dh	Product SerDes lanes ID register (cmn_pid_lanes)	16	RO	0202h
8_0010h	Product metal layers ID register 0 (cmn_pid_metal0)	16	RO	0000h
8_0011h	Product metal layers ID register 1 (cmn_pid_metal1)	16	RO	0000h
8_0012h	Product metal layers ID register 2 (cmn_pid_metal2)	16	RO	0000h
8_0013h	Product metal layers ID register 3 (cmn_pid_metal3)	16	RO	0000h
8_0014h	Product metal layer direction ID register (cmn_pid_metald)	16	RO	0000h
8_0020h	Startup state machine control register (cmn_ssm_sm_ctrl)	16	RW	0001h
8_0021h	Bandgap enable timer register (cmn_ssm_bandgap_tmr)	16	RW	0003h
8_0022h	Bias enable timer register (cmn_ssm_bias_tmr)	16	RW	007Dh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8_0027h	Startup state machine user defined control register (cmn_ssm_user_def_ctrl)	16	RW	0000h
8_0028h	PLL 0 control state machine control register (cmn_pllsm0_sm_ctrl)	16	RW	0000h
8_0029h	PLL 0 enable timer register (cmn_pllsm0_pllen_tmr)	16	RW	0019h
8_002Ah	PLL 0 pre-charge timer register (cmn_pllsm0_pllpre_tmr)	16	RW	0032h
8_002Bh	PLL 0 VREF delay timer register (cmn_pllsm0_pllref_tmr)	16	RW	0019h
8_002Ch	PLL 0 lock delay timer register (cmn_pllsm0_plllock_tmr)	16	RW	0064h
8_002Dh	PLL 0 clock disable delay timer register (cmn_pllsm0_pllclkdis_tmr)	16	RW	0001h
8_002Fh	PLL 0 control state machine user defined control register (cmn_pllsm0_user_def_ctrl)	16	RW	1000h
8_0030h	PLL 1 control state machine control register (cmn_pllsm1_sm_ctrl)	16	RW	0000h
8_0031h	PLL 1 enable timer register (cmn_pllsm1_pllen_tmr)	16	RW	0019h
8_0032h	PLL 1 pre-charge timer register (cmn_pllsm1_pllpre_tmr)	16	RW	0032h
8_0033h	PLL 1 VREF delay timer register (cmn_pllsm1_pllref_tmr)	16	RW	0019h
8_0034h	PLL 1 lock delay timer register (cmn_pllsm1_plllock_tmr)	16	RW	0064h
8_0035h	PLL 1 clock disable delay timer register (cmn_pllsm1_pllclkdis_tmr)	16	RW	0001h
8_0037h	PLL 1 control state machine user defined control register (cmn_pllsm1_user_def_ctrl)	16	RW	1000h
8_0060h	Common control power island control register (cmn_cdiag_pwr_ctrl)	16	RW	0012h
8_0061h	Common PSM clock control register (cmn_psm_clk_ctrl)	16	RW	0006h
8_0062h	Reference clock receiver control register (cmn_cdiag_refclk_ctrl)	16	RW	1000h
8_0064h	Common control CDB diagnostic register (cmn_cdiag_cdb_diag)	16	WO	0000h
8_0066h	Common control reset diagnostic register (cmn_cdiag_rst_diag)	16	RO	0000h
8_007Fh	Common control cover your alternatives register (cmn_cdiag_dcya)	16	RW	0000h
8_0080h	PLL 0 VCO calibration control register (cmn_pll0_vcocal_ctrl)	16	RW	00A0h
8_0081h	PLL 0 VCO calibration start point register (cmn_pll0_vcocal_start)	16	RW	30A0h
8_0082h	PLL 0 VCO calibration timer control register (cmn_pll0_vcocal_tctrl)	16	RW	0003h
8_0083h	PLL 0 VCO calibration override register (cmn_pll0_vcocal_ovrd)	16	RW	0000h
8_0084h	PLL 0 VCO calibration initialization timer register (cmn_pll0_vcocal_init_tmr)	16	RW	03E8h
8_0085h	PLL 0 VCO calibration iteration timer register (cmn_pll0_vcocal_iter_tmr)	16	RW	0064h
8_0086h	PLL 0 VCO calibration reference clock timer start value register (cmn_pll0_vcocal_reftim_start)	16	RW	0200h
8_0088h	PLL 0 VCO calibration PLL clock counter start value register (cmn_pll0_vcocal_pllcnt_start)	16	RW	0200h
8_0090h	PLL 0 lock reference counter start value register (cmn_pll0_lock_refcnt_start)	16	RW	00C8h
8_0091h	PLL 0 lock reference counter idle value register (cmn_pll0_lock_refcnt_idle)	16	RW	0004h
8_0092h	PLL 0 lock PLL counter start value register (cmn_pll0_lock_pllcnt_start)	16	RW	00C8h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8_0093h	PLL 0 lock PLL counter threshold value register (cmn_pll0_lock_pllcnt_thr)	16	RW	0003h
8_0094h	PLL 0 feedback divider integer register (cmn_pll0_intdiv)	16	RW	0000h
8_0095h	PLL 0 feedback divider fractional register (cmn_pll0_fracdiv)	16	RW	0000h
8_0096h	PLL 0 feedback divider high threshold register (cmn_pll0_high_thr)	16	RW	0000h
8_0097h	PLL 0 delta sigma modulator diagnostics register (cmn_pll0_dsm_diag)	16	RW	8020h
8_0098h	PLL 0 spread spectrum control register 1 (cmn_pll0_ss_ctrl1)	16	RW	8000h
8_0099h	PLL 0 spread spectrum control register 2 (cmn_pll0_ss_ctrl2)	16	RW	0000h
8_009Ch	PLL 0 VCO calibration V2I control register (cmn_pll0_vcocal_v2i_ctrl)	16	RW	0003h
8_00A0h	PLL 1 VCO calibration control register (cmn_pll1_vcocal_ctrl)	16	RW	00A0h
8_00A1h	PLL 1 VCO calibration start point register (cmn_pll1_vcocal_start)	16	RW	30A0h
8_00A2h	PLL 1 VCO calibration timer control register (cmn_pll1_vcocal_tctrl)	16	RW	0003h
8_00A3h	PLL 1 VCO calibration override register (cmn_pll1_vcocal_ovrd)	16	RW	0000h
8_00A4h	PLL 1 VCO calibration initialization timer register (cmn_pll1_vcocal_init_tmr)	16	RW	03E8h
8_00A5h	PLL 1 VCO calibration iteration timer register (cmn_pll1_vcocal_iter_tmr)	16	RW	0064h
8_00A6h	PLL 1 VCO calibration reference clock timer start value register (cmn_pll1_vcocal_reftim_start)	16	RW	0200h
8_00A8h	PLL 1 VCO calibration PLL clock counter start value register (cmn_pll1_vcocal_pllcnt_start)	16	RW	0200h
8_00B0h	PLL 1 lock reference counter start value register (cmn_pll1_lock_refcnt_start)	16	RW	00C8h
8_00B1h	PLL 1 lock reference counter idle value register (cmn_pll1_lock_refcnt_idle)	16	RW	0004h
8_00B2h	PLL 1 lock PLL counter start value register (cmn_pll1_lock_pllcnt_start)	16	RW	00C8h
8_00B3h	PLL 1 lock PLL counter threshold value register (cmn_pll1_lock_pllcnt_thr)	16	RW	0003h
8_00B4h	PLL 1 feedback divider integer register (cmn_pll1_intdiv)	16	RW	0000h
8_00B5h	PLL 1 feedback divider fractional register (cmn_pll1_fracdiv)	16	RW	0000h
8_00B6h	PLL 1 feedback divider high threshold register (cmn_pll1_high_thr)	16	RW	0000h
8_00B7h	PLL 1 delta sigma modulator diagnostics register (cmn_pll1_dsm_diag)	16	RW	8020h
8_00B8h	PLL 1 spread spectrum control register 1 (cmn_pll1_ss_ctrl1)	16	RW	8000h
8_00B9h	PLL 1 spread spectrum control register 2 (cmn_pll1_ss_ctrl2)	16	RW	0000h
8_00BCh	PLL 1 VCO calibration V2I control register (cmn_pll1_vcocal_v2i_ctrl)	16	RW	0003h
8_00C0h	Current calibration control register (cmn_ical_ctrl)	16	RW	0010h
8_00C1h	Current calibration override register (cmn_ical_ovrd)	16	RW	0000h
8_00C2h	Current calibration start register (cmn_ical_start)	16	RW	0010h
8_00C3h	Current calibration tune register (cmn_ical_tune)	16	RW	0000h
8_00C4h	Current calibration initialization timer register (cmn_ical_init_tmr)	16	RW	003Fh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8_00C5h	Current calibration iteration timer register (cmn_ical_iter_tmr)	16	RW	0006h
8_00D0h	RX resistor calibration control register (cmn_rxcal_ctrl)	16	RW	0008h
8_00D1h	RX resistor calibration override register (cmn_rxcal_ovrd)	16	RW	0000h
8_00D2h	RX resistor calibration start register (cmn_rxcal_start)	16	RW	0008h
8_00D3h	RX resistor calibration tune register (cmn_rxcal_tune)	16	RW	0000h
8_00D4h	RX resistor calibration initialization timer register (cmn_rxcal_init_tmr)	16	RW	0020h
8_00D5h	RX resistor calibration iteration timer register (cmn_rxcal_iter_tmr)	16	RW	0006h
8_00E0h	TX pull-up resistor calibration control register (cmn_txpucal_ctrl)	16	RW	0028h
8_00E1h	TX pull-up resistor calibration override register (cmn_txpucal_ovrd)	16	RW	0000h
8_00E2h	TX pull-up resistor calibration start register (cmn_txpucal_start)	16	RW	0028h
8_00E3h	TX pull-up resistor calibration tune register (cmn_txpucal_tune)	16	RW	0000h
8_00E4h	TX pull-up resistor calibration initialization timer register (cmn_txpucal_init_tmr)	16	RW	0020h
8_00E5h	TX pull-up resistor calibration iteration timer register (cmn_txpucal_iter_tmr)	16	RW	0006h
8_00F0h	TX pull-down resistor calibration control register (cmn_tpxdcal_ctrl)	16	RW	0028h
8_00F1h	TX pull-down resistor calibration override register (cmn_tpxdcal_ovrd)	16	RW	0000h
8_00F2h	TX pull-down resistor calibration start register (cmn_tpxdcal_start)	16	RW	0028h
8_00F3h	TX pull-down resistor calibration tune register (cmn_tpxdcal_tune)	16	RW	0000h
8_00F4h	TX pull-down resistor calibration initialization timer register (cmn_tpxdcal_init_tmr)	16	RW	0020h
8_00F5h	TX pull-down resistor calibration iteration timer register (cmn_tpxdcal_iter_tmr)	16	RW	0006h
8_0100h	Current calibration adjust control register (cmn_ical_adj_ctrl)	16	RW	0000h
8_0101h	Current calibration adjust count register (cmn_ical_adj_cnt)	16	RW	0001h
8_0102h	Current calibration adjust initialization timer register (cmn_ical_adj_init_tmr)	16	RW	0020h
8_0103h	Current calibration adjust iteration timer register (cmn_ical_adj_iter_tmr)	16	RW	0006h
8_0104h	RX resistor calibration adjust control register (cmn_rx_adj_ctrl)	16	RW	0000h
8_0105h	RX resistor calibration adjust count register (cmn_rx_adj_cnt)	16	RW	0001h
8_0106h	RX resistor calibration adjust initialization timer register (cmn_rx_adj_init_tmr)	16	RW	0020h
8_0107h	RX resistor calibration adjust iteration timer register (cmn_rx_adj_iter_tmr)	16	RW	0006h
8_0108h	TX pull up resistor calibration adjust control register (cmn_txpu_adj_ctrl)	16	RW	0000h
8_0109h	TX pull up resistor calibration adjust count register (cmn_txpu_adj_cnt)	16	RW	0001h
8_010Ah	TX pull up resistor calibration adjust initialization timer register (cmn_txpu_adj_init_tmr)	16	RW	0020h
8_010Bh	TX pull up resistor calibration adjust iteration timer register (cmn_txpu_adj_iter_tmr)	16	RW	0006h

Table continues on the next page...

Clocks And Resets

Offset	Register	Width (In bits)	Access	Reset value
8_010Ch	TX pull down resistor calibration adjust control register (cmn_txpd_adj_ctrl)	16	RW	0000h
8_010Dh	TX pull down resistor calibration adjust count register (cmn_txpd_adj_cnt)	16	RW	0001h
8_010Eh	TX pull down resistor calibration adjust initialization timer register (cmn_txpd_adj_init_tmr)	16	RW	0020h
8_010Fh	TX pull down resistor calibration adjust iteration timer register (cmn_txpd_adj_iter_tmr)	16	RW	0006h
8_01A0h	Clock frequency measurement control register (cmn_cmsmt_clk_fr_eq_msmt_ctrl)	16	RW	0000h
8_01A1h	Test clock selection register (cmn_cmsmt_test_clk_sel)	16	RW	0000h
8_01A2h	Reference clock timer value register (cmn_cmsmt_ref_clk_tmr_value)	16	RW	0000h
8_01A3h	Test clock counter value register (cmn_cmsmt_test_clk_cnt_value)	16	RO	0000h
8_01C0h	PLL 0 feedback divider value high override register (cmn_diag_pll0_fbh_ovrd)	16	RW	8016h
8_01C1h	PLL 0 feedback divider value low override register (cmn_diag_pll0_fbl_ovrd)	16	RW	8018h
8_01C2h	PLL 0 override register (cmn_diag_pll0_ovrd)	16	RW	0010h
8_01C3h	PLL 0 LDO control register (cmn_diag_pll0_ldo_ctrl)	16	RW	0000h
8_01C4h	PLL 0 test mode register (cmn_diag_pll0_test_mode)	16	RW	0000h
8_01C5h	PLL 0 voltage to current unit tuning register (cmn_diag_pll0_v2i_tune)	16	RW	0037h
8_01C6h	PLL 0 charge pump tuning register (cmn_diag_pll0_cp_tune)	16	RW	0072h
8_01C7h	PLL 0 loop filter programmability register (cmn_diag_pll0_lf_prog)	16	RW	0004h
8_01C8h	PLL 0 PTAT current slope tuning register 1 (cmn_diag_pll0_ptatis_tune1)	16	RW	0001h
8_01C9h	PLL 0 PTAT current slope tuning register 2 (cmn_diag_pll0_ptatis_tune2)	16	RW	0001h
8_01CAh	PLL 0 input clock control register (cmn_diag_pll0_inclk_ctrl)	16	RW	0001h
8_01D0h	PLL 1 feedback divider value high override register (cmn_diag_pll1_fbh_ovrd)	16	RW	8026h
8_01D1h	PLL 1 feedback divider value low override register (cmn_diag_pll1_fbl_ovrd)	16	RW	8026h
8_01D2h	PLL 1 override register (cmn_diag_pll1_ovrd)	16	RW	0010h
8_01D3h	PLL 1 LDO control register (cmn_diag_pll1_ld1_ctrl)	16	RW	0000h
8_01D4h	PLL 1 test mode register (cmn_diag_pll1_test_mode)	16	RW	0000h
8_01D5h	PLL 1 voltage to current unit tuning register (cmn_diag_pll1_v2i_tune)	16	RW	0037h
8_01D6h	PLL 1 charge pump tuning register (cmn_diag_pll1_cp_tune)	16	RW	0072h
8_01D7h	PLL 1 loop filter programmability register (cmn_diag_pll1_lf_prog)	16	RW	0004h
8_01D8h	PLL 1 PTAT current slope tuning register 1 (cmn_diag_pll1_ptatis_tune1)	16	RW	0001h
8_01D9h	PLL 1 PTAT current slope tuning register 2 (cmn_diag_pll1_ptatis_tune2)	16	RW	0001h
8_01DAh	PLL 1 input clock control register (cmn_diag_pll1_inclk_ctrl)	16	RW	0001h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8_01E0h	Common high speed clock select register (cmn_diag_hscclk_sel)	16	RW	0010h
8_01E1h	Common calibration clock control register (cmn_diag_calclk_ctrl)	16	RW	0004h
8_01E2h	Common high speed reset release state machine control register (cmn_diag_hsrsm_ctrl)	16	RW	0001h
8_01E3h	Common functions reset diagnostic register (cmn_diag_rst_diag)	16	RO	0000h
8_01E8h	Bandgap override register (cmn_diag_bandgap_ovrd)	16	RW	0000h
8_01E9h	Bias override register (cmn_diag_bias_ovrd)	16	RW	0000h
8_01ECh	Common periodic calibration adjust control register (cmn_diag_per_cal_adj)	16	RW	00F4h
8_01EDh	Common calibration control register (cmn_diag_cal_ctrl)	16	RW	0000h
8_01F0h	ATB control register 1 (cmn_diag_atb_ctrl1)	16	RW	0000h
8_01F1h	ATB control register 2 (cmn_diag_atb_ctrl2)	16	RW	0000h
8_01F4h	Common bandgap PTAT current control register (cmn_diag_bgi_pta_t_ctrl)	16	RW	0000h
8_01F5h	Common bandgap current control register 1 (cmn_diag_bgi_ctrl1)	16	RW	0000h
8_01F6h	Common bandgap current control register 2 (cmn_diag_bgi_ctrl2)	16	RW	0000h
8_01F7h	Common control current control register 1 (cmn_diag_ctrli_ctrl1)	16	RW	0000h
8_01F8h	Common control current control register 2 (cmn_diag_ctrli_ctrl2)	16	RW	0000h
8_01FEh	Common digital functions cover your alternatives register (cmn_diag_dcya)	16	RW	0000h
8_01FFh	Common analog cover your alternatives register (cmn_diag_acya)	16	RW	0000h
8_5020h	DP Aux analog control 1 (tx_ana_ctrl_reg_1)	16	RW	0000h
8_5021h	DP Aux analog control 2 (tx_ana_ctrl_reg_2)	16	RW	0000h
8_5022h	Tx_coef_calc module Inputs (txda_coeff_calc_ctrl)	16	RW	0000h
8_5023h	Tx dig control reg 1 (tx_dig_ctrl_reg_1)	16	RW	0000h
8_5024h	Tx dig control reg 2 (tx_dig_ctrl_reg_2)	16	RW	0000h
8_5025h	DP Aux analog control 3 (txda_cya_auxda_cya)	16	RW	0000h
8_5026h	DP Aux analog control 4 (tx_ana_ctrl_reg_3)	16	RW	0000h
8_5027h	DP Aux analog control 5 (tx_ana_ctrl_reg_4)	16	RW	0000h
8_5028h	DP Aux analog status 1 (tx_ana_status_reg_1)	16	RO	0000h
8_5029h	DP Aux analog status 6 (tx_ana_ctrl_reg_5)	16	RW	0000h
8_9060h	rx_aux analog control1 (rx_ana_ctrl_reg_1)	16	RW	0000h
8_C008h	HDP Mode Control register (phy_hdp_mode_ctl)	16	RW	C004h
8_C009h	HDP Clock Control register (phy_hdp_clk_ctl)	16	RW	1205h
8_C00Fh	PHY status register (phy_sts)	16	RO	0000h
8_C010h	PHY common control signal isolation register (phy_iso_cmn_ctrl)	16	RW	0000h
8_C800h	PMA common control1 register (phy_pma_cmn_ctrl1)	16	RW	0000h
8_C801h	PMA common control2 register (phy_pma_cmn_ctrl2)	16	RO	000Ch
8_C802h	PMA SSM current state register (phy_pma_ssm_state)	16	RO	0020h
8_C803h	PMA PLL State Machine current state register (phy_pma_iso_pll_sm_state)	16	RO	0300h

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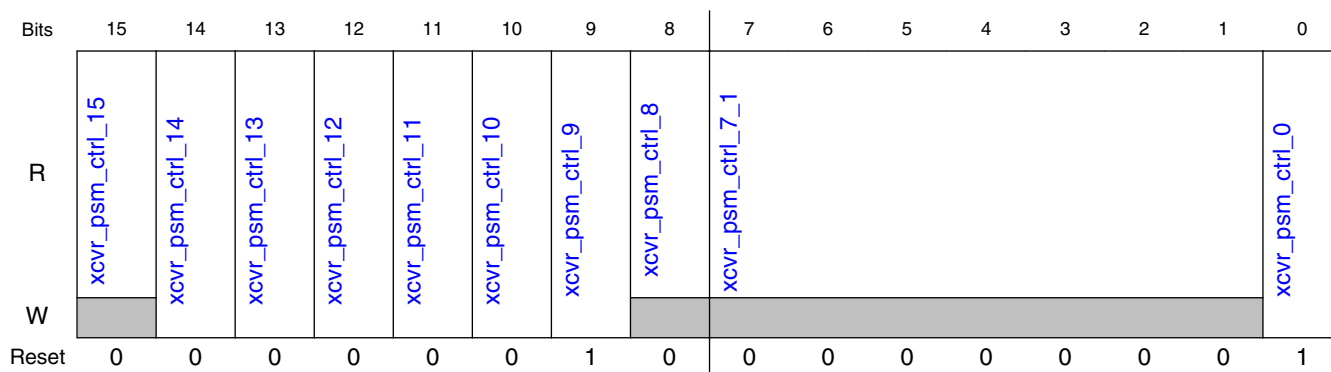
Offset	Register	Width (In bits)	Access	Reset value
8_C810h	PMA common control signal isolation register (phy_pma_iso_cm_n_ctrl)	16	RW	0001h
8_C811h	PMA PLL control0 isolation register (phy_pma_iso_pll_ctrl0)	16	RW	000Fh
8_C812h	PMA PLL control1 isolation register (phy_pma_iso_pll_ctrl1)	16	RW	1122h
8_C81Fh	Isolation control register (phy_pma_isolation_ctrl)	16	RW	0000h

13.4.10.2.2 Power state machine control register (lane0_xcvr_psm_ctrl - lane3_xcvr_psm_ctrl)

13.4.10.2.2.1 Offset

Register	Offset
lane0_xcvr_psm_ctrl	4000h
lane1_xcvr_psm_ctrl	4400h
lane2_xcvr_psm_ctrl	4800h
lane3_xcvr_psm_ctrl	4C00h

13.4.10.2.2.2 Diagram



13.4.10.2.2.3 Fields

Field	Function
15 xcvr_psm_ctrl_15	Reserved

Table continues on the next page...

Field	Function
14 xcvr_psm_ctrl_1_4	Bypass A0 in delay from PSM ready : When this bit is active (1b1), the A0 input delay is bypassed when transitioning from the PSM ready state to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the
13 xcvr_psm_ctrl_1_3	Bypass A0 in delay from A5 : When this bit is active (1b1), the A0 input delay is bypassed when transitioning from the A5 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the
12 xcvr_psm_ctrl_1_2	Bypass A0 in delay from A4 : When this bit is active (1b1), the A0 input delay is bypassed when transitioning from the A4 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the
11 xcvr_psm_ctrl_1_1	Bypass A0 in delay from A3 : When this bit is active (1b1), the A0 input delay is bypassed when transitioning from the A3 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the
10 xcvr_psm_ctrl_1_0	Bypass A0 in delay from A2 : When this bit is active (1b1), the A0 input delay is bypassed when transitioning from the A2 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the
9 xcvr_psm_ctrl_9	Bypass A0 in delay from A1 : When this bit is active (1b1), the A0 input delay is bypassed when transitioning from the A1 to the A0 power state. The result of this is the amount of time spent in the A0 in delay state is the value specified in the
8 xcvr_psm_ctrl_8	Reserved
7-1 xcvr_psm_ctrl_7_1	Reserved
0 xcvr_psm_ctrl_0	Skip lane re-calibration : When this bit is active (1b1), the lane calibration state will be skipped if it was previously run, unless the macro is disabled or reset.

13.4.10.2.3 Power state machine reset control register (lane0_xcvr_psm_rctrl - lane3_xcvr_psm_rctrl)

13.4.10.2.3.1 Offset

Register	Offset
lane0_xcvr_psm_rctrl	4001h
lane1_xcvr_psm_rctrl	4401h
lane2_xcvr_psm_rctrl	4801h
lane3_xcvr_psm_rctrl	4C01h

13.4.10.2.3.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W	xcsr_psm_rctrl_15	xcsr_psm_rctrl_14	xcsr_psm_rctrl_13	xcsr_psm_rctrl_12	xcsr_psm_rctrl_11	xcsr_psm_rctrl_10	xcsr_psm_rctrl_9	xcsr_psm_rctrl_8	xcsr_psm_rctrl_7	xcsr_psm_rctrl_6	xcsr_psm_rctrl_5	xcsr_psm_rctrl_4	xcsr_psm_rctrl_3	xcsr_psm_rctrl_2	xcsr_psm_rctrl_1	xcsr_psm_rctrl_0
Reset	1	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0

13.4.10.2.3.3 Fields

Field	Function
15 xcsr_psm_rctrl_15	RX reset active ready : Controls the state the receiver reset is changed to when in the ready power state.
14 xcsr_psm_rctrl_14	RX reset active calibration : Controls the state the receiver reset is changed to when in the calibration power state.
13 xcsr_psm_rctrl_13	RX reset active A5 : Controls the state the receiver reset is changed to when in the A5 entry power state.
12 xcsr_psm_rctrl_12	RX reset active A4 : Controls the state the receiver reset is changed to when in the A4 entry power state.
11 xcsr_psm_rctrl_11	RX reset active A3 : Controls the state the receiver reset is changed to when in the A3 entry power state.
10 xcsr_psm_rctrl_10	RX reset active A2 : Controls the state the receiver reset is changed to when in the A2 entry power state.
9 xcsr_psm_rctrl_9	RX reset active A1 : Controls the state the receiver reset is changed to when in the A1 entry power state.
8 xcsr_psm_rctrl_8	RX reset active A0 : Controls the state the receiver reset is changed to when in the A0 entry power state.
7 xcsr_psm_rctrl_7	TX reset active ready : Controls the state the transmitter reset is changed to when in the ready power state.
6	TX reset active calibration : Controls the state the transmitter reset is changed to when in the calibration power state.

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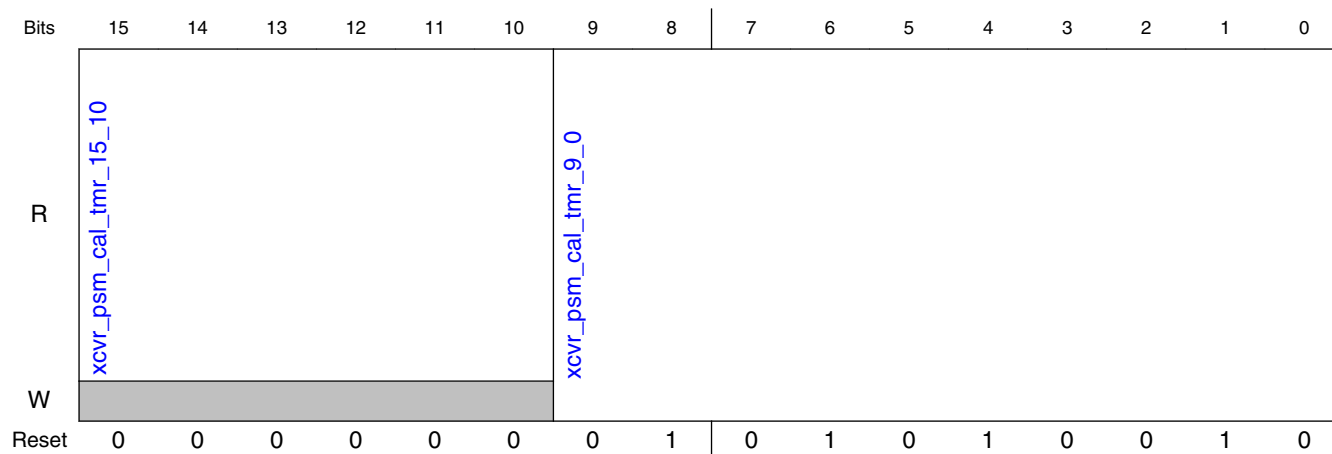
Field	Function
xcvr_psm_rctrl_6	
5 xcvr_psm_rctrl_5	TX reset active A5 : Controls the state the transmitter reset is changed to when in the A5 entry power state.
4 xcvr_psm_rctrl_4	TX reset active A4 : Controls the state the transmitter reset is changed to when in the A4 entry power state.
3 xcvr_psm_rctrl_3	TX reset active A3 : Controls the state the transmitter reset is changed to when in the A3 entry power state.
2 xcvr_psm_rctrl_2	TX reset active A2 : Controls the state the transmitter reset is changed to when in the A2 entry power state.
1 xcvr_psm_rctrl_1	TX reset active A1 : Controls the state the transmitter reset is changed to when in the A1 entry power state.
0 xcvr_psm_rctrl_0	TX reset active A0 : Controls the state the transmitter reset is changed to when in the A0 entry power state.

13.4.10.2.4 PSM calibration delay timer register (lane0_xcvr_psm_cal_tmr - lane3_xcvr_psm_cal_tmr)

13.4.10.2.4.1 Offset

Register	Offset
lane0_xcvr_psm_cal_tmr	4002h
lane1_xcvr_psm_cal_tmr	4402h
lane2_xcvr_psm_cal_tmr	4802h
lane3_xcvr_psm_cal_tmr	4C02h

13.4.10.2.4.2 Diagram



13.4.10.2.4.3 Fields

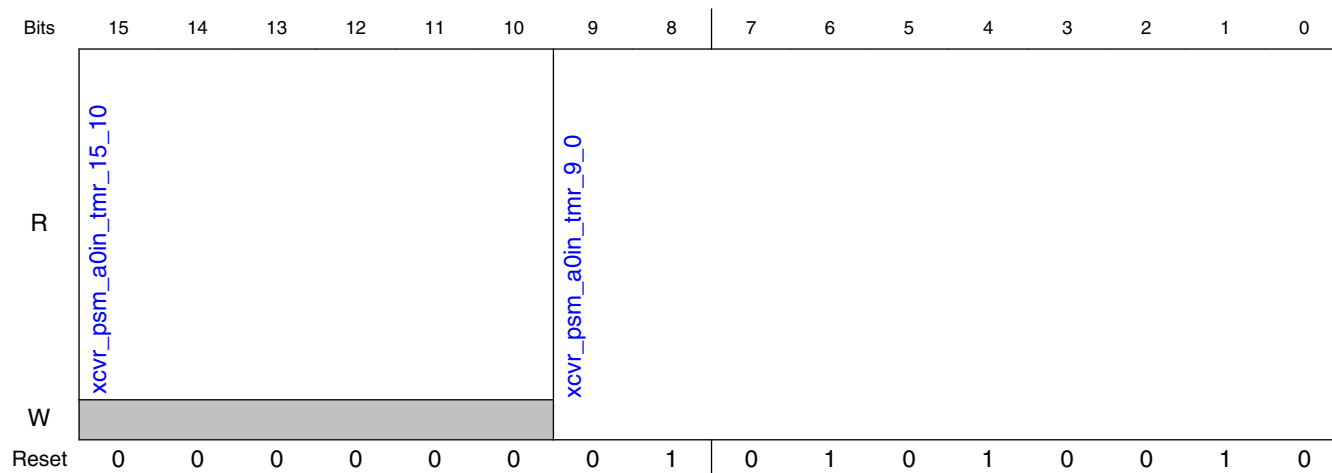
Field	Function
15-10 <code>xcvr_psm_cal_tmr_15_10</code>	Reserved
9-0 <code>xcvr_psm_cal_tmr_9_0</code>	PSM calibration delay state timer value : Value used for the timer when the power state machine is in the PSM calibration delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for <code>tpsm_lane_cal_del</code> .

13.4.10.2.5 A0 in delay timer register (lane0_xcvr_psm_a0in_tmr - lane3_xcvr_psm_a0in_tmr)

13.4.10.2.5.1 Offset

Register	Offset
<code>lane0_xcvr_psm_a0in_tmr</code>	4003h
<code>lane1_xcvr_psm_a0in_tmr</code>	4403h
<code>lane2_xcvr_psm_a0in_tmr</code>	4803h
<code>lane3_xcvr_psm_a0in_tmr</code>	4C03h

13.4.10.2.5.2 Diagram



13.4.10.2.5.3 Fields

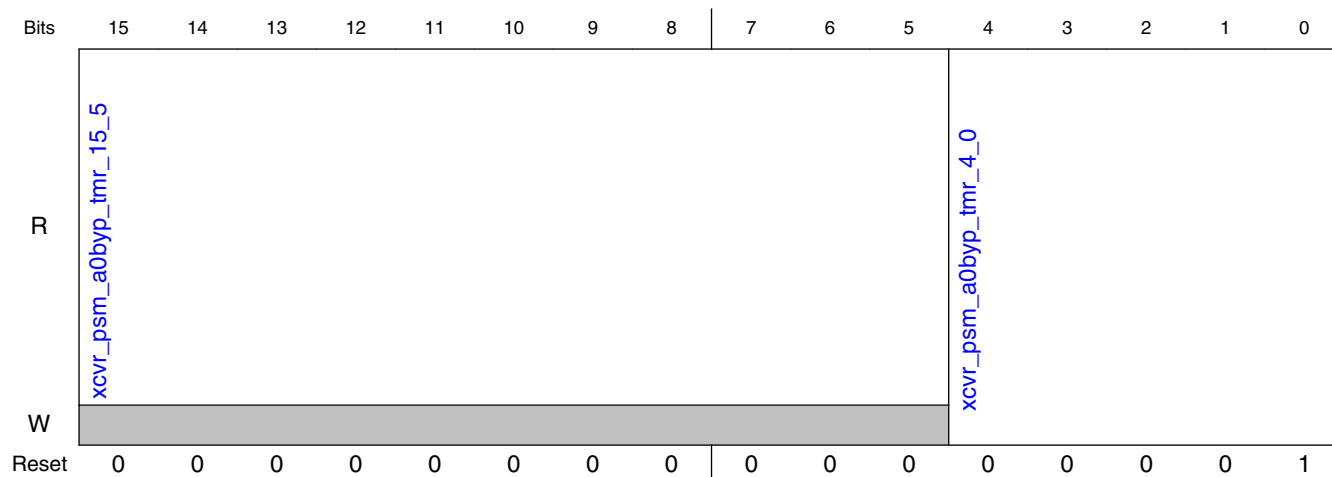
Field	Function
15-10 xcvr_psm_a0in_tmr_15_10	Reserved
9-0 xcvr_psm_a0in_tmr_9_0	A0 in delay state timer value : Value used for the timer when the power state machine is in the A0 in delay state, unless the timer is bypassed under the control of the bypass A0 bits in the

13.4.10.2.6 A0 in bypass timer register (lane0_xcvr_psm_a0byp_tmr - lane3_xcvr_psm_a0byp_tmr)

13.4.10.2.6.1 Offset

Register	Offset
lane0_xcvr_psm_a0byp_tmr	4004h
lane1_xcvr_psm_a0byp_tmr	4404h
lane2_xcvr_psm_a0byp_tmr	4804h
lane3_xcvr_psm_a0byp_tmr	4C04h

13.4.10.2.6.2 Diagram



13.4.10.2.6.3 Fields

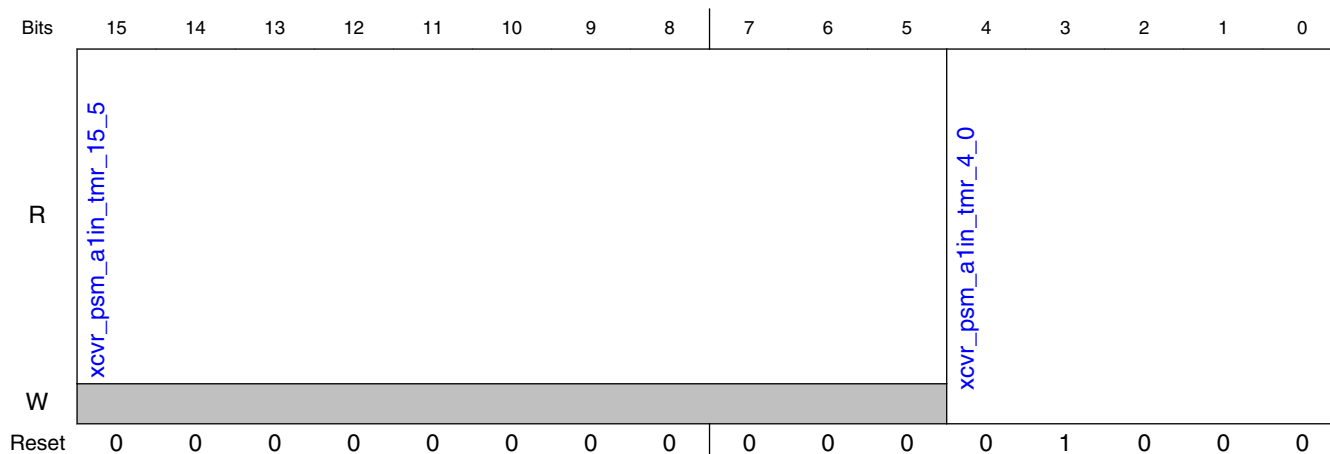
Field	Function
15-5 xcvr_psm_a0byp_tmr_15_5	Reserved
4-0 xcvr_psm_a0byp_tmr_4_0	A0 in delay state bypass timer value : Value used for the timer when the power state machine is in the A0 in delay state and the timer is bypassed under the control of the bypass A0 bits in the

13.4.10.2.7 A1 in delay timer register (lane0_xcvr_psm_a1in_tmr - lane3_xcvr_psm_a1in_tmr)

13.4.10.2.7.1 Offset

Register	Offset
lane0_xcvr_psm_a1in_tmr	4005h
lane1_xcvr_psm_a1in_tmr	4405h
lane2_xcvr_psm_a1in_tmr	4805h
lane3_xcvr_psm_a1in_tmr	4C05h

13.4.10.2.7.2 Diagram



13.4.10.2.7.3 Fields

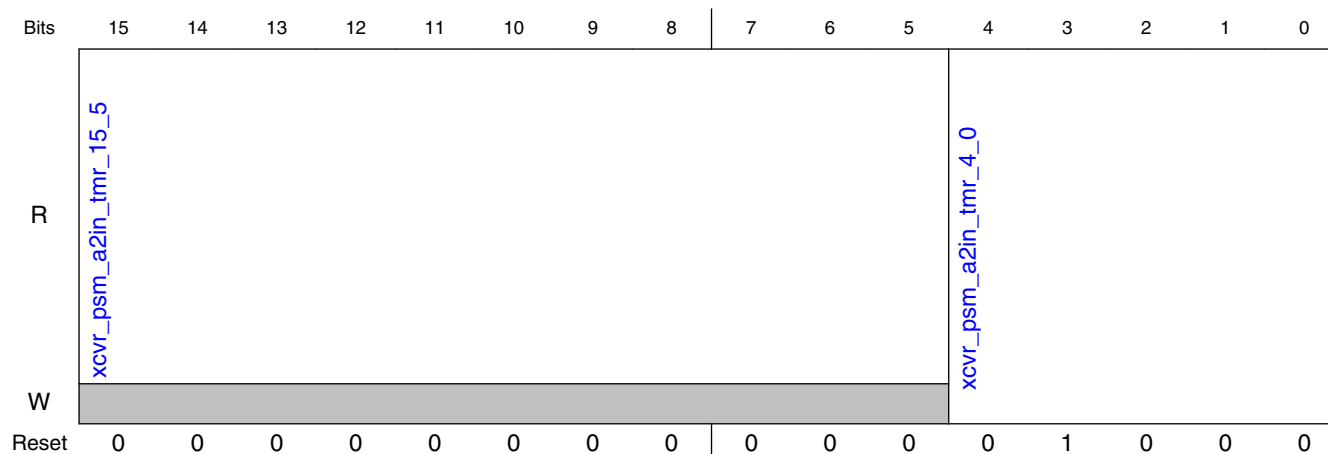
Field	Function
15-5 <code>xcvr_psm_a1in_tmr_15_5</code>	Reserved
4-0 <code>xcvr_psm_a1in_tmr_4_0</code>	A1 in delay state timer value : Value used for the timer when the power state machine is in the A1 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for <code>tpsm_A1_in_del</code> .

13.4.10.2.8 A2 in delay timer register (lane0_xcvr_psm_a2in_tmr - lane3_xcvr_psm_a2in_tmr)

13.4.10.2.8.1 Offset

Register	Offset
<code>lane0_xcvr_psm_a2in_tmr</code>	4006h
<code>lane1_xcvr_psm_a2in_tmr</code>	4406h
<code>lane2_xcvr_psm_a2in_tmr</code>	4806h
<code>lane3_xcvr_psm_a2in_tmr</code>	4C06h

13.4.10.2.8.2 Diagram



13.4.10.2.8.3 Fields

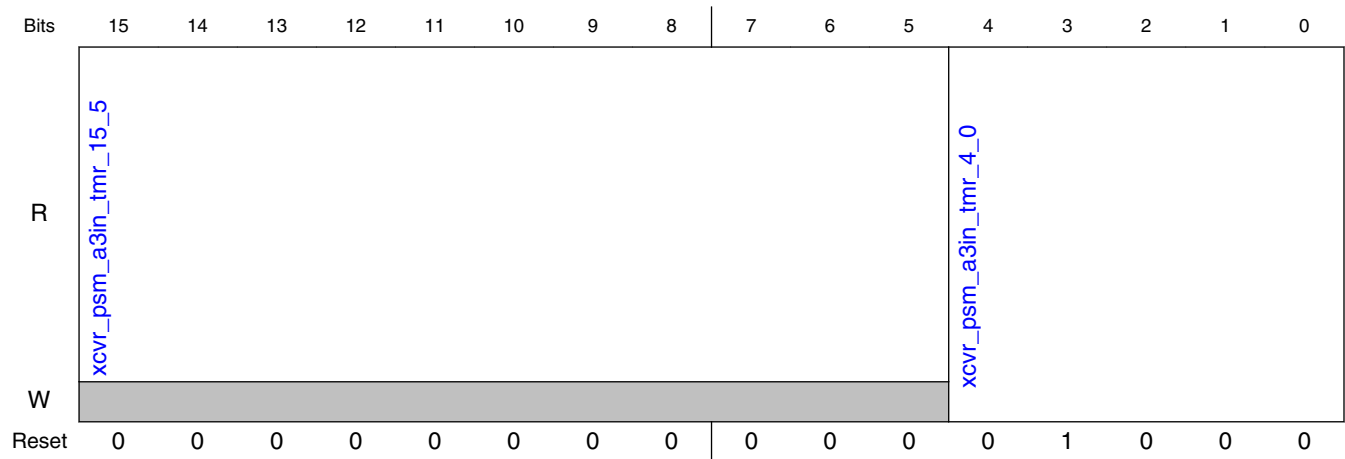
Field	Function
15-5 xcvr_psm_a2in_tmr_15_5	Reserved
4-0 xcvr_psm_a2in_tmr_4_0	A2 in delay state timer value : Value used for the timer when the power state machine is in the A2 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A2_in_del.

13.4.10.2.9 A3 in delay timer register (lane0_xcvr_psm_a3in_tmr - lane3_xcvr_psm_a3in_tmr)

13.4.10.2.9.1 Offset

Register	Offset
lane0_xcvr_psm_a3in_tmr	4007h
lane1_xcvr_psm_a3in_tmr	4407h
lane2_xcvr_psm_a3in_tmr	4807h
lane3_xcvr_psm_a3in_tmr	4C07h

13.4.10.2.9.2 Diagram



13.4.10.2.9.3 Fields

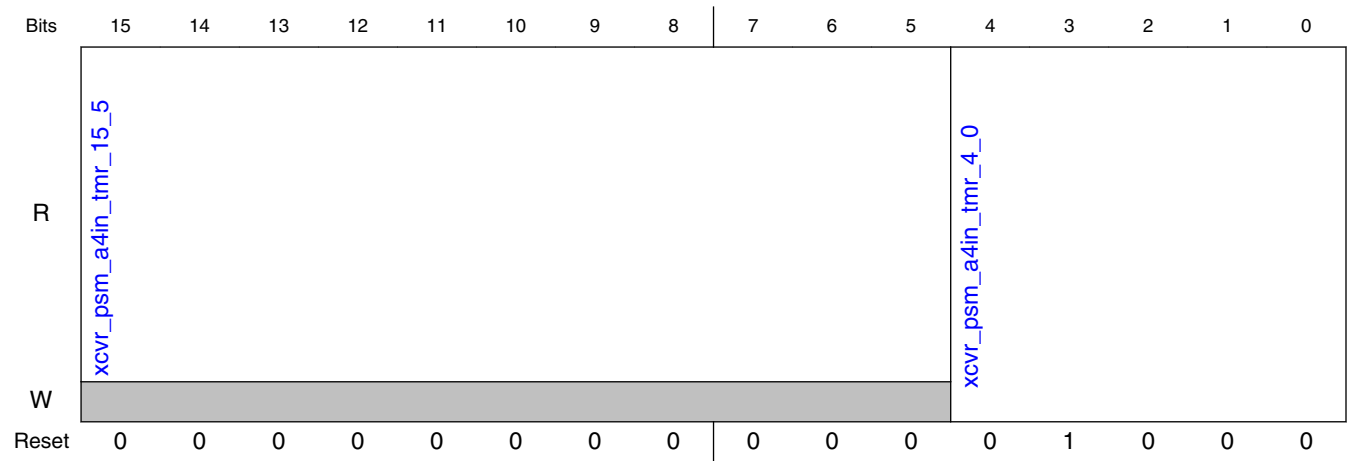
Field	Function
15-5 <code>xcvr_psm_a3in_tmr_15_5</code>	Reserved
4-0 <code>xcvr_psm_a3in_tmr_4_0</code>	A3 in delay state timer value : Value used for the timer when the power state machine is in the A3 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for <code>tpsm_A3_in_del</code> .

13.4.10.2.10 A4 in delay timer register (lane0_xcvr_psm_a4in_tmr - lane3_xcvr_psm_a4in_tmr)

13.4.10.2.10.1 Offset

Register	Offset
<code>lane0_xcvr_psm_a4in_tmr</code>	4008h
<code>lane1_xcvr_psm_a4in_tmr</code>	4408h
<code>lane2_xcvr_psm_a4in_tmr</code>	4808h
<code>lane3_xcvr_psm_a4in_tmr</code>	4C08h

13.4.10.2.10.2 Diagram



13.4.10.2.10.3 Fields

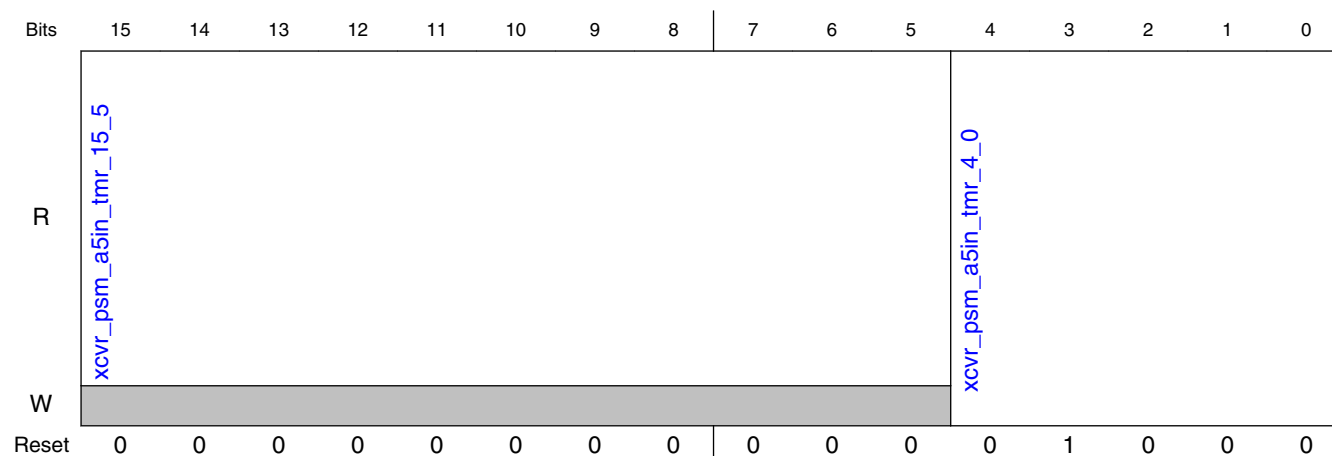
Field	Function
15-5 xcvr_psm_a4in_tmr_15_5	Reserved
4-0 xcvr_psm_a4in_tmr_4_0	A4 in delay state timer value : Value used for the timer when the power state machine is in the A4 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A4_in_del.

13.4.10.2.11 A5 in delay timer register (lane0_xcvr_psm_a5in_tmr - lane3_xcvr_psm_a5in_tmr)

13.4.10.2.11.1 Offset

Register	Offset
lane0_xcvr_psm_a5in_tmr	4009h
lane1_xcvr_psm_a5in_tmr	4409h
lane2_xcvr_psm_a5in_tmr	4809h
lane3_xcvr_psm_a5in_tmr	4C09h

13.4.10.2.11.2 Diagram



13.4.10.2.11.3 Fields

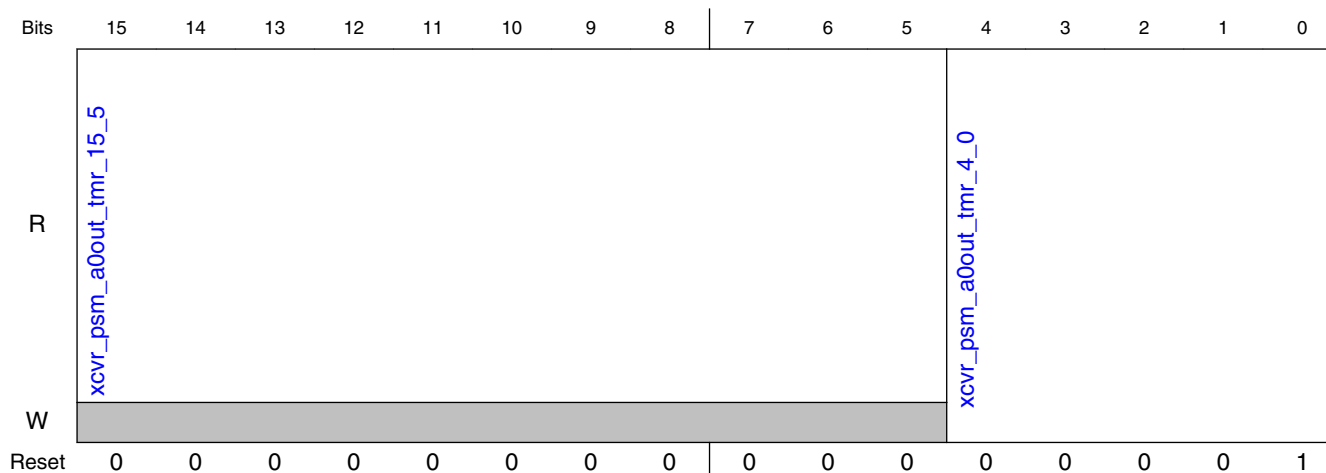
Field	Function
15-5 xcvr_psm_a5in_tmr_15_5	Reserved
4-0 xcvr_psm_a5in_tmr_4_0	A5 in delay state timer value : Value used for the timer when the power state machine is in the A5 in delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A5_in_del.

13.4.10.2.12 A0 out delay timer register (lane0_xcvr_psm_a0out_tmr - lane3_xcvr_psm_a0out_tmr)

13.4.10.2.12.1 Offset

Register	Offset
lane0_xcvr_psm_a0out_tmr	400Ah
lane1_xcvr_psm_a0out_tmr	440Ah
lane2_xcvr_psm_a0out_tmr	480Ah
lane3_xcvr_psm_a0out_tmr	4C0Ah

13.4.10.2.12.2 Diagram



13.4.10.2.12.3 Fields

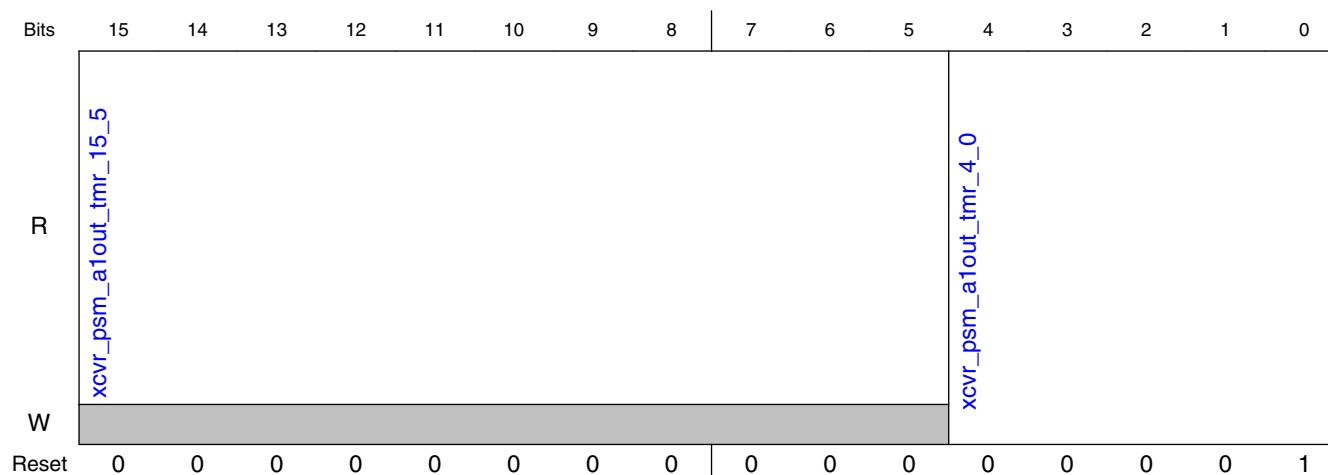
Field	Function
15-5 xcvr_psm_a0out_tmr_15_5	Reserved
4-0 xcvr_psm_a0out_tmr_4_0	A0 out delay state timer value : Value used for the timer when the power state machine is in the A0 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A0_out_del.

13.4.10.2.13 A1 out delay timer register (lane0_xcvr_psm_a1out_tmr - lane3_xcvr_psm_a1out_tmr)

13.4.10.2.13.1 Offset

Register	Offset
lane0_xcvr_psm_a1out_tmr	400Bh
lane1_xcvr_psm_a1out_tmr	440Bh
lane2_xcvr_psm_a1out_tmr	480Bh
lane3_xcvr_psm_a1out_tmr	4C0Bh

13.4.10.2.13.2 Diagram



13.4.10.2.13.3 Fields

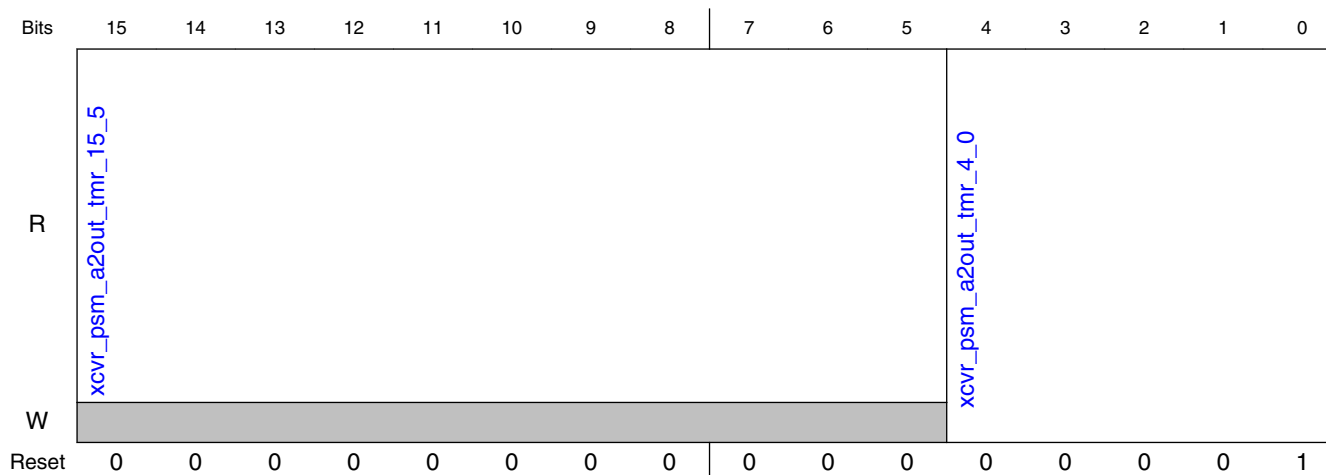
Field	Function
15-5 xcvr_psm_a1out_tmr_15_5	Reserved
4-0 xcvr_psm_a1out_tmr_4_0	A1 out delay state timer value : Value used for the timer when the power state machine is in the A1 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A1_out_del.

13.4.10.2.14 A2 out delay timer register (lane0_xcvr_psm_a2out_tmr - lane3_xcvr_psm_a2out_tmr)

13.4.10.2.14.1 Offset

Register	Offset
lane0_xcvr_psm_a2out_tmr	400Ch
lane1_xcvr_psm_a2out_tmr	440Ch
lane2_xcvr_psm_a2out_tmr	480Ch
lane3_xcvr_psm_a2out_tmr	4C0Ch

13.4.10.2.14.2 Diagram



13.4.10.2.14.3 Fields

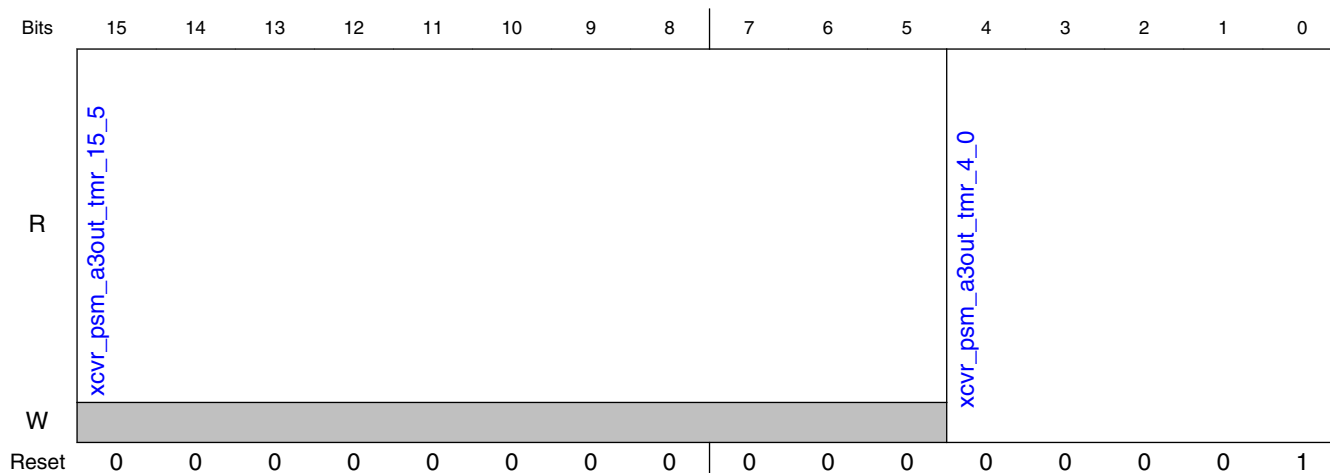
Field	Function
15-5 xcvr_psm_a2out_tmr_15_5	Reserved
4-0 xcvr_psm_a2out_tmr_4_0	A2 out delay state timer value : Value used for the timer when the power state machine is in the A2 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A2_out_del.

13.4.10.2.15 A3 out delay timer register (lane0_xcvr_psm_a3out_tmr - lane3_xcvr_psm_a3out_tmr)

13.4.10.2.15.1 Offset

Register	Offset
lane0_xcvr_psm_a3out_tmr	400Dh
lane1_xcvr_psm_a3out_tmr	440Dh
lane2_xcvr_psm_a3out_tmr	480Dh
lane3_xcvr_psm_a3out_tmr	4C0Dh

13.4.10.2.15.2 Diagram



13.4.10.2.15.3 Fields

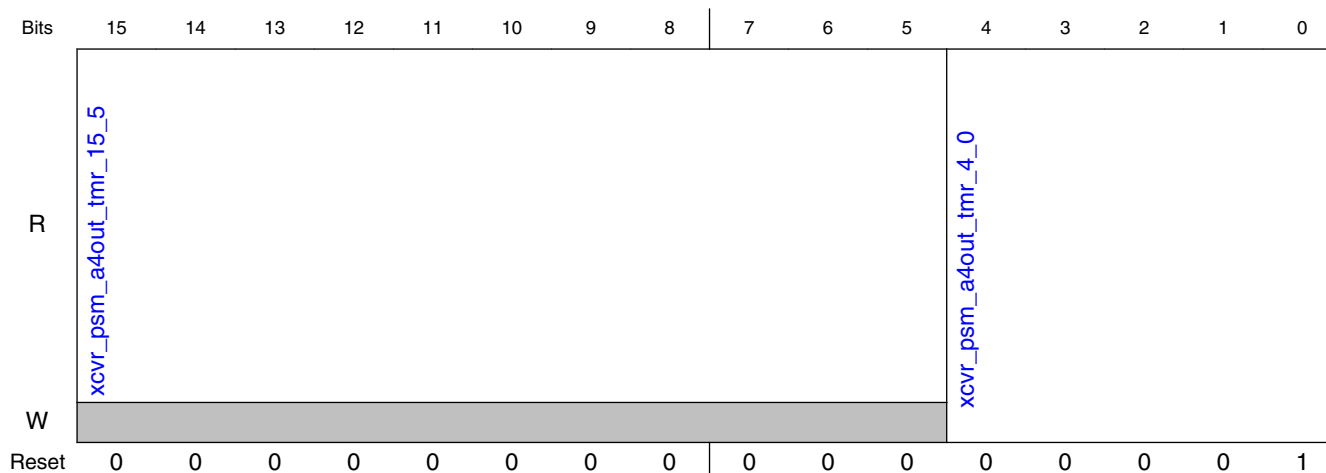
Field	Function
15-5 xcvr_psm_a3out_tmr_15_5	Reserved
4-0 xcvr_psm_a3out_tmr_4_0	A3 out delay state timer value : Value used for the timer when the power state machine is in the A3 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A3_out_del.

13.4.10.2.16 A4 out delay timer register (lane0_xcvr_psm_a4out_tmr - lane3_xcvr_psm_a4out_tmr)

13.4.10.2.16.1 Offset

Register	Offset
lane0_xcvr_psm_a4out_tmr	400Eh
lane1_xcvr_psm_a4out_tmr	440Eh
lane2_xcvr_psm_a4out_tmr	480Eh
lane3_xcvr_psm_a4out_tmr	4C0Eh

13.4.10.2.16.2 Diagram



13.4.10.2.16.3 Fields

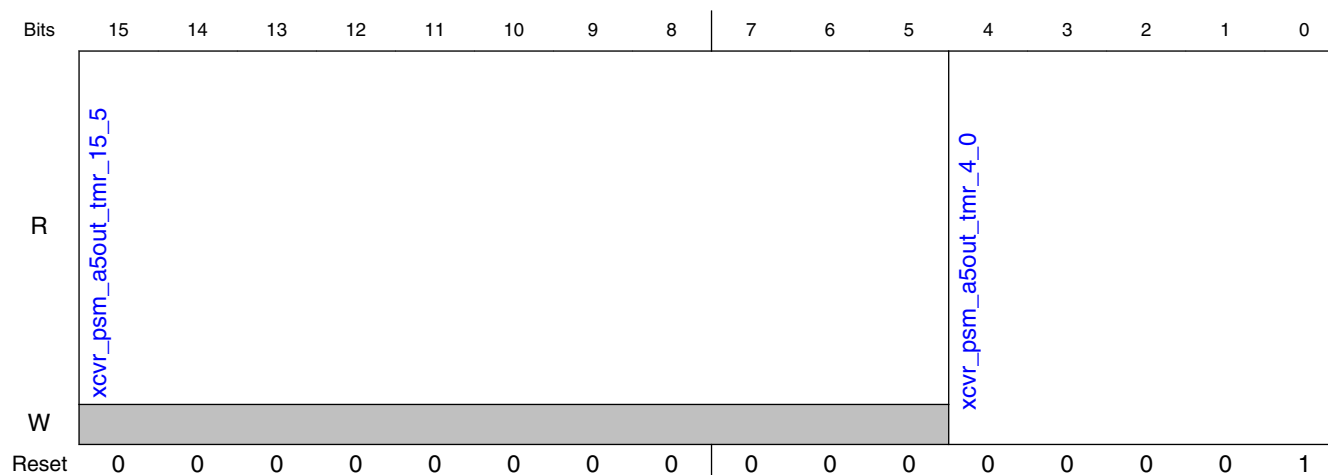
Field	Function
15-5 xcvr_psm_a4out_tmr_15_5	Reserved
4-0 xcvr_psm_a4out_tmr_4_0	A4 out delay state timer value : Value used for the timer when the power state machine is in the A4 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A4_out_del.

13.4.10.2.17 A5 out delay timer register (lane0_xcvr_psm_a5out_tmr - lane3_xcvr_psm_a5out_tmr)

13.4.10.2.17.1 Offset

Register	Offset
lane0_xcvr_psm_a5out_tmr	400Fh
lane1_xcvr_psm_a5out_tmr	440Fh
lane2_xcvr_psm_a5out_tmr	480Fh
lane3_xcvr_psm_a5out_tmr	4C0Fh

13.4.10.2.17.2 Diagram



13.4.10.2.17.3 Fields

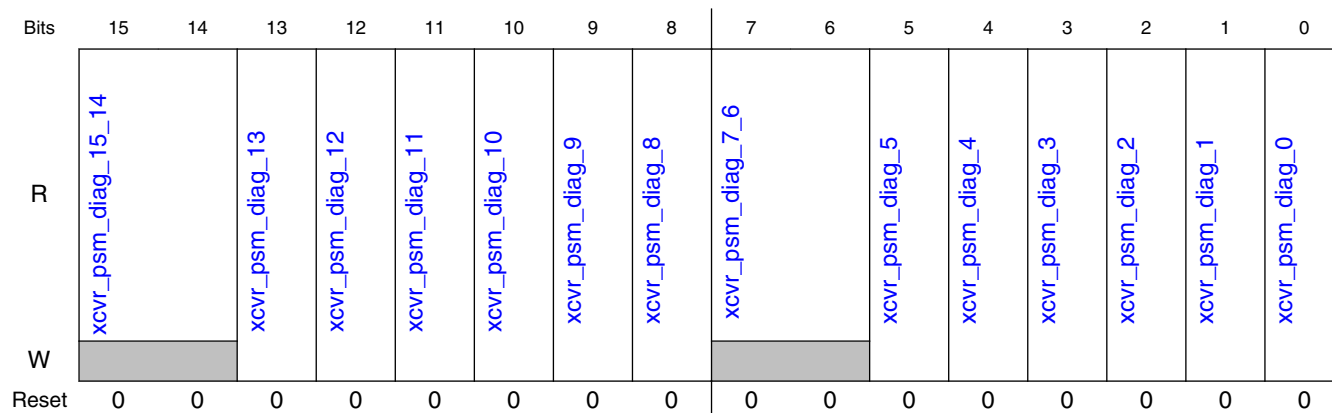
Field	Function
15-5 xcvr_psm_a5out_tmr_15_5	Reserved
4-0 xcvr_psm_a5out_tmr_4_0	A5 out delay state timer value : Value used for the timer when the power state machine is in the A5 out delay state. This timer delay is specified as the number of PSM clocks to count to implement the time required for tpsm_A5_out_del.

13.4.10.2.18 Power state machine diagnostic register (lane0_xcvr_psm_diag - lane3_xcvr_psm_diag)

13.4.10.2.18.1 Offset

Register	Offset
lane0_xcvr_psm_diag	4010h
lane1_xcvr_psm_diag	4410h
lane2_xcvr_psm_diag	4810h
lane3_xcvr_psm_diag	4C10h

13.4.10.2.18.2 Diagram



13.4.10.2.18.3 Fields

Field	Function
15-14 xcvr_psm_diag_15_14	Reserved
13 xcvr_psm_diag_13	Force A5 exit acknowledge : Setting this bit to 1b1 forces the
12 xcvr_psm_diag_12	Force A4 exit acknowledge : Setting this bit to 1b1 forces the
11 xcvr_psm_diag_11	Force A3 exit acknowledge : Setting this bit to 1b1 forces the
10 xcvr_psm_diag_10	Force A2 exit acknowledge : Setting this bit to 1b1 forces the
9 xcvr_psm_diag_9	Force A1 exit acknowledge : Setting this bit to 1b1 forces the
8 xcvr_psm_diag_8	Force A0 exit acknowledge : Setting this bit to 1b1 forces the
7-6 xcvr_psm_diag_7_6	Reserved
5 xcvr_psm_diag_5	Force A5 entry acknowledge : Setting this bit to 1b1 forces the

Table continues on the next page...

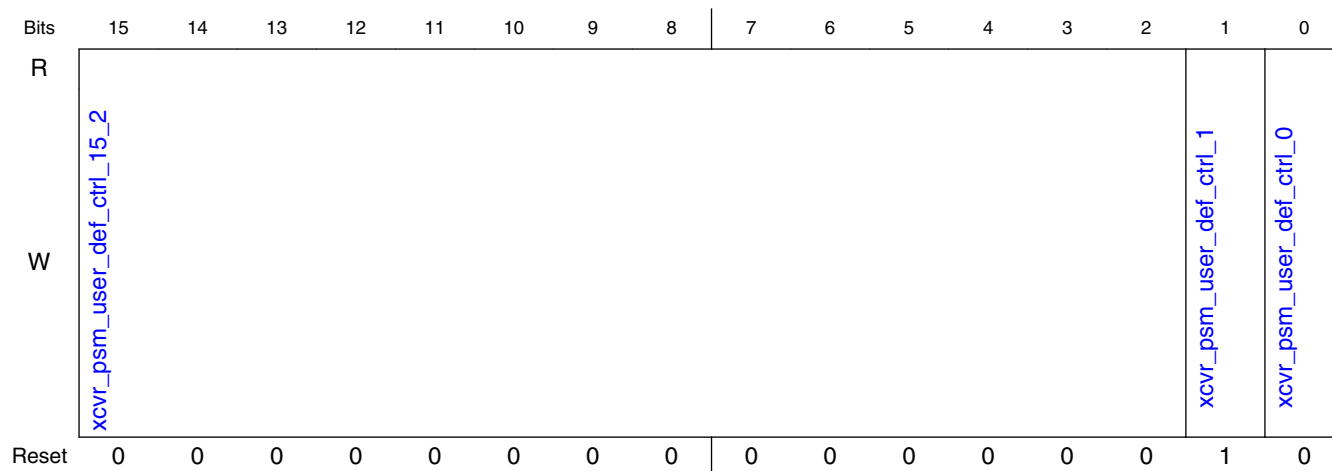
Field	Function
4 xcvr_psm_diag_ 4	Force A4 entry acknowledge : Setting this bit to 1b1 forces the
3 xcvr_psm_diag_ 3	Force A3 entry acknowledge : Setting this bit to 1b1 forces the
2 xcvr_psm_diag_ 2	Force A2 entry acknowledge : Setting this bit to 1b1 forces the
1 xcvr_psm_diag_ 1	Force A1 entry acknowledge : Setting this bit to 1b1 forces the
0 xcvr_psm_diag_ 0	Force A0 entry acknowledge : Setting this bit to 1b1 forces the

13.4.10.2.19 Power state machine user defined control register (lane0_xc vr_psm_user_def_ctrl - lane3_xcvr_psm_user_def_ctrl)

13.4.10.2.19.1 Offset

Register	Offset
lane0_xcvr_psm_user_ def_ctrl	401Fh
lane1_xcvr_psm_user_ def_ctrl	441Fh
lane2_xcvr_psm_user_ def_ctrl	481Fh
lane3_xcvr_psm_user_ def_ctrl	4C1Fh

13.4.10.2.19.2 Diagram



13.4.10.2.19.3 Fields

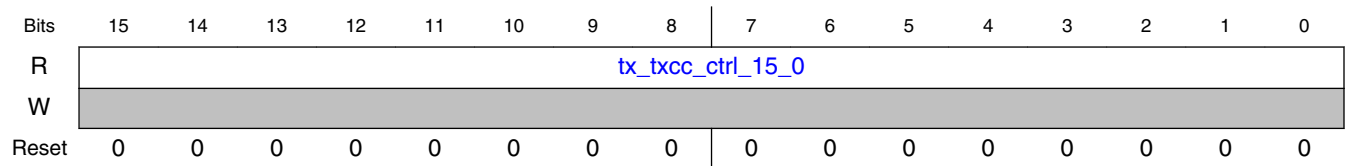
Field	Function
15-2 xcvr_psm_user_def_ctrl_15_2	Reserved
1 xcvr_psm_user_def_ctrl_1	Disable PSM clock gating: This bit controls the PSM clock gating function. By disabling the PSM clock gating, faster state transition times can be achieved, at the expense of this clock always being enabled.
0 xcvr_psm_user_def_ctrl_0	Disable early A0 acknowledge response : Setting this bit to 1b1, will disable the function where an A0 acknowledge is generated early, when transitioning from A1 to A0, relative to the PSM actually being in the A0 power state.

13.4.10.2.20 TX coefficient controller control register (lane0_tx_txcc_ctrl - lane3_tx_txcc_ctrl)

13.4.10.2.20.1 Offset

Register	Offset
lane0_tx_txcc_ctrl	4040h
lane1_tx_txcc_ctrl	4440h
lane2_tx_txcc_ctrl	4840h
lane3_tx_txcc_ctrl	4C40h

13.4.10.2.20.2 Diagram



13.4.10.2.20.3 Fields

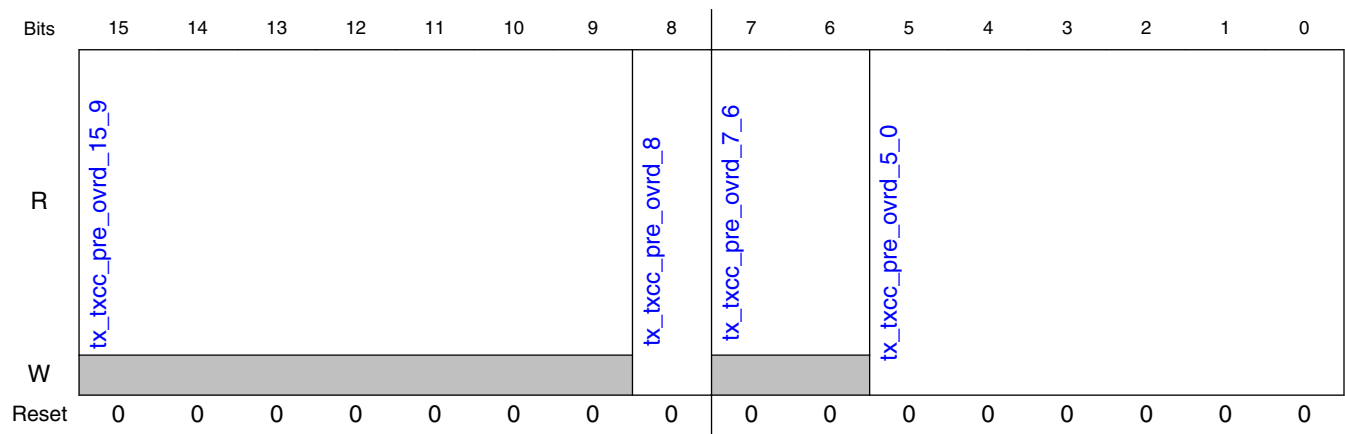
Field	Function
15-0 tx_txcc_ctrl_15_0	Reserved

13.4.10.2.21 TX pre-cursor override register (lane0_tx_txcc_pre_ovrd - lane3_tx_txcc_pre_ovrd)

13.4.10.2.21.1 Offset

Register	Offset
lane0_tx_txcc_pre_ovrd	4041h
lane1_tx_txcc_pre_ovrd	4441h
lane2_tx_txcc_pre_ovrd	4841h
lane3_tx_txcc_pre_ovrd	4C41h

13.4.10.2.21.2 Diagram



13.4.10.2.21.3 Fields

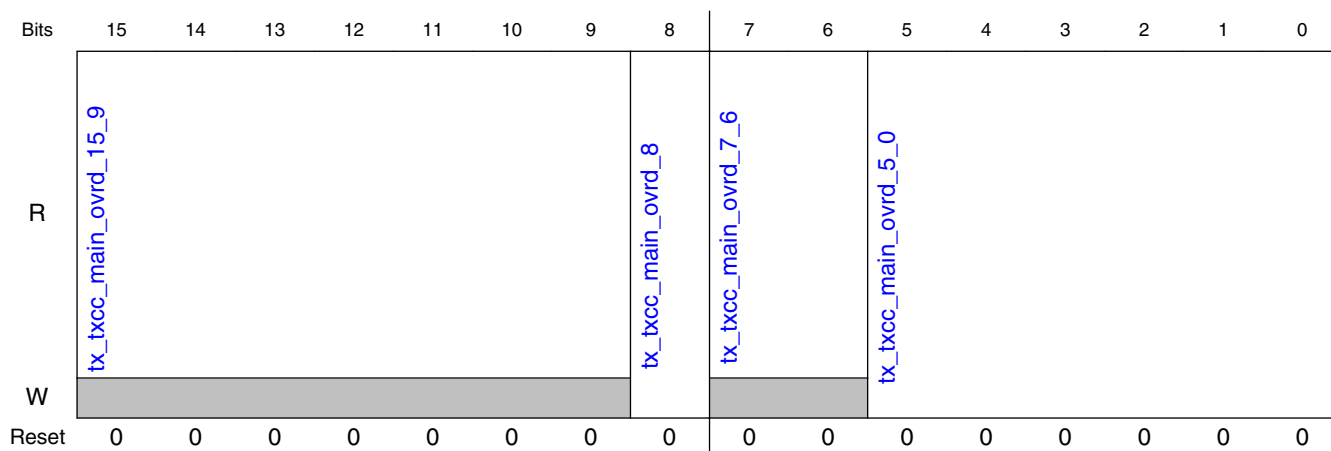
Field	Function
15-9 tx_txcc_pre_ovrd_15_9	Reserved
8 tx_txcc_pre_ovrd_8	Pre-cursor override enable: When enabled, the pre-cursor field in this register is used to override the pre-cursor value.
7-6 tx_txcc_pre_ovrd_7_6	Reserved
5-0 tx_txcc_pre_ovrd_5_0	Pre-cursor override value: When enabled by the pre-cursor override enable bit in this register, the value in this field is used to override the pre-cursor value. Note that this field is 6 bits wide, to match the pre-cursor data input width on the

13.4.10.2.22 TX main-cursor override register (lane0_tx_txcc_main_ovrd - lane3_tx_txcc_main_ovrd)

13.4.10.2.22.1 Offset

Register	Offset
lane0_tx_txcc_main_ovrd	4042h
lane1_tx_txcc_main_ovrd	4442h
lane2_tx_txcc_main_ovrd	4842h
lane3_tx_txcc_main_ovrd	4C42h

13.4.10.2.22.2 Diagram



13.4.10.2.22.3 Fields

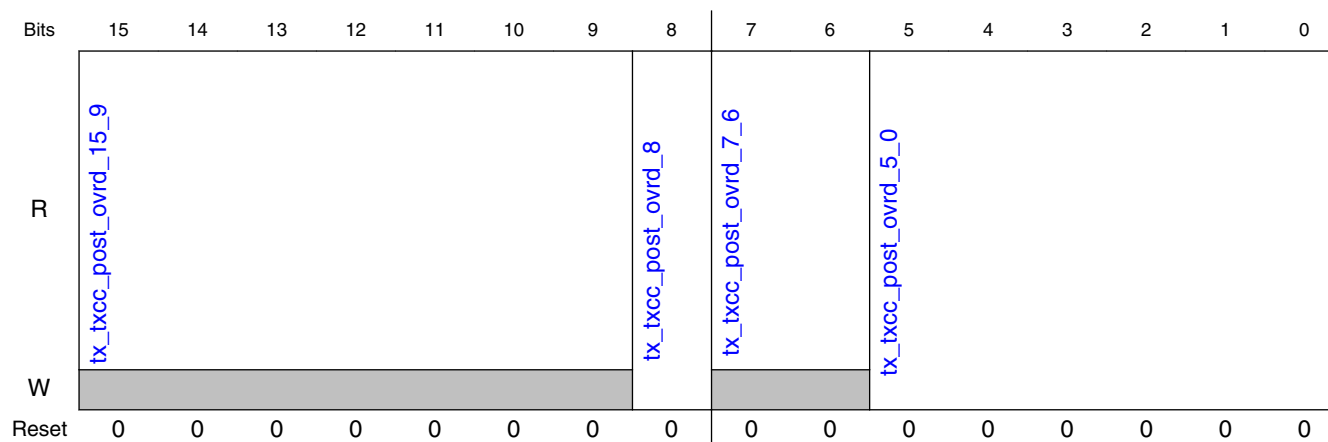
Field	Function
15-9 tx_txcc_main_ov rd_15_9	Reserved
8 tx_txcc_main_ov rd_8	Main-cursor override enable: When enabled, the main-cursor field in this register is used to override the main-cursor value.
7-6 tx_txcc_main_ov rd_7_6	Reserved
5-0 tx_txcc_main_ov rd_5_0	Main-cursor override value: When enabled by the main-cursor override enable bit in this register, the value in this field is used to override the main-cursor value. Note that this field is 6 bits wide, to match the main-cursor data input width on the

13.4.10.2.23 TX post-cursor override register (lane0_tx_txcc_post_ovrd - lane3_tx_txcc_post_ovrd)

13.4.10.2.23.1 Offset

Register	Offset
lane0_tx_txcc_post_ovrd	4043h
lane1_tx_txcc_post_ovrd	4443h
lane2_tx_txcc_post_ovrd	4843h
lane3_tx_txcc_post_ovrd	4C43h

13.4.10.2.23.2 Diagram



13.4.10.2.23.3 Fields

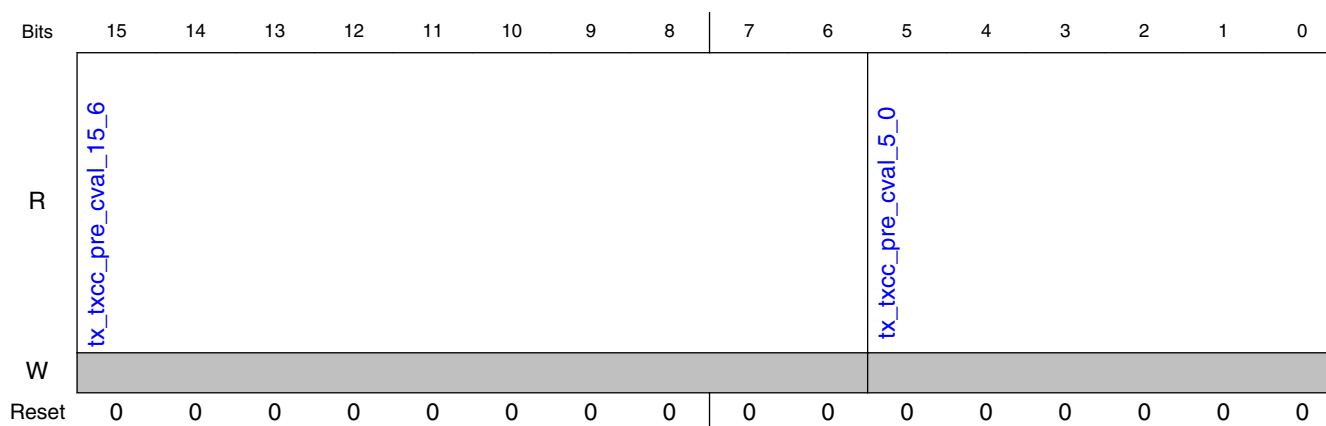
Field	Function
15-9 tx_txcc_post_ovrd_15_9	Reserved
8 tx_txcc_post_ovrd_8	Post-cursor override enable: When enabled, the post-cursor field in this register is used to override the post-cursor value.
7-6 tx_txcc_post_ovrd_7_6	Reserved
5-0 tx_txcc_post_ovrd_5_0	Post-cursor override value: When enabled by the post-cursor override enable bit in this register, the value in this field is used to override the post-cursor value. Note that this field is 6 bits wide, to match the post-cursor data input width on the

13.4.10.2.24 TX pre-cursor current value register (lane0_tx_txcc_pre_cval - lane3_tx_txcc_pre_cval)

13.4.10.2.24.1 Offset

Register	Offset
lane0_tx_txcc_pre_cval	4044h
lane1_tx_txcc_pre_cval	4444h
lane2_tx_txcc_pre_cval	4844h
lane3_tx_txcc_pre_cval	4C44h

13.4.10.2.24.2 Diagram



13.4.10.2.24.3 Fields

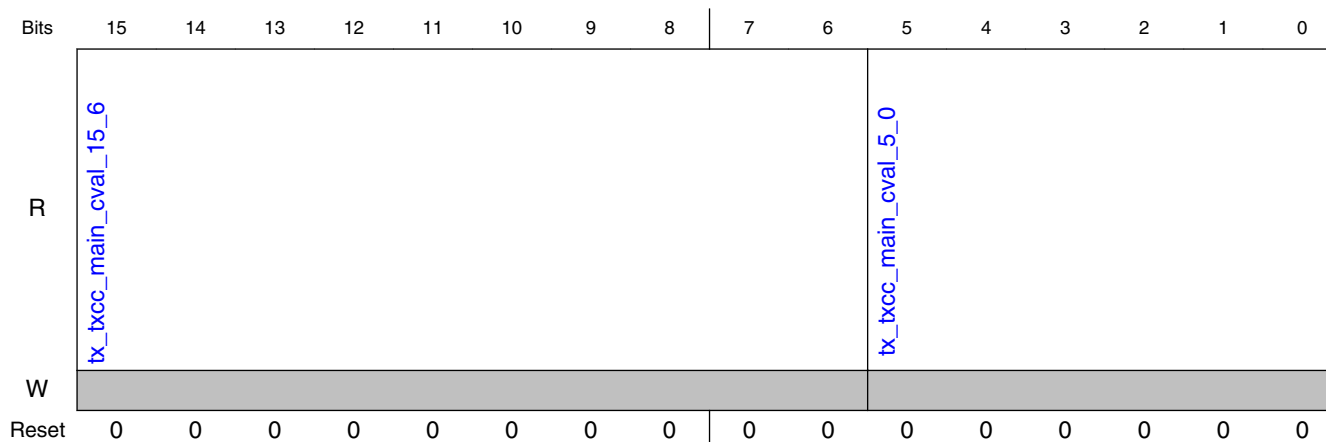
Field	Function
15-6 tx_txcc_pre_cval_15_6	Reserved
5-0 tx_txcc_pre_cval_5_0	Pre-cursor value: The value in this field indicates the current value of the pre-cursor (C-1) coefficient. The value of this field can be any of the following, depending on the current mode of operation.

13.4.10.2.25 TX main-cursor current value register (lane0_tx_txcc_main_cval - lane3_tx_txcc_main_cval)

13.4.10.2.25.1 Offset

Register	Offset
lane0_tx_txcc_main_cval	4045h
lane1_tx_txcc_main_cval	4445h
lane2_tx_txcc_main_cval	4845h
lane3_tx_txcc_main_cval	4C45h

13.4.10.2.25.2 Diagram



13.4.10.2.25.3 Fields

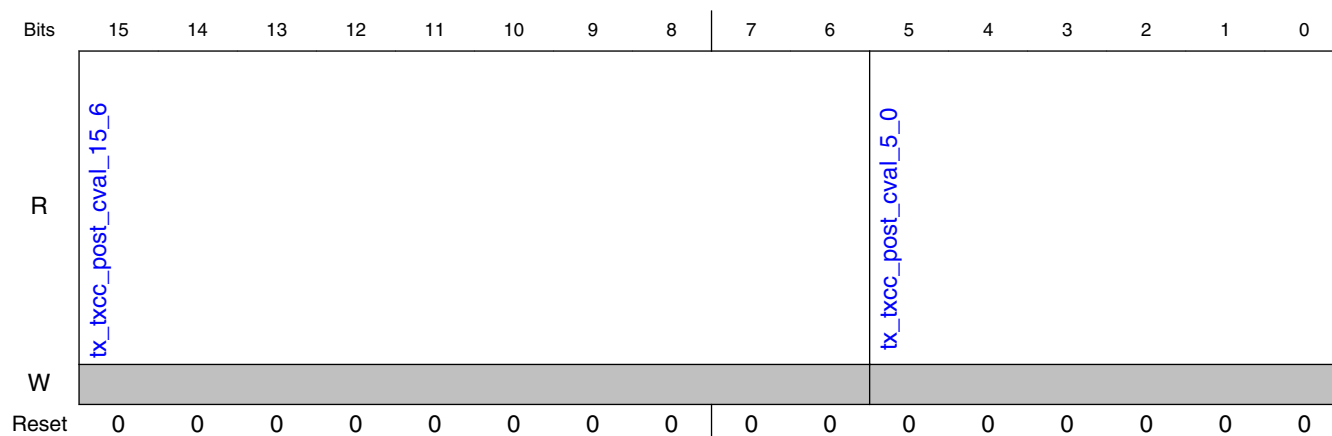
Field	Function
15-6 tx_txcc_main_cval_15_6	Reserved
5-0 tx_txcc_main_cval_5_0	Main-cursor value: The value in this field indicates the current value of the main-cursor (C0) coefficient. The value of this field can be any of the following, depending on the current mode of operation.

13.4.10.2.26 TX post-cursor current value register (lane0_tx_txcc_post_cval - lane3_tx_txcc_post_cval)

13.4.10.2.26.1 Offset

Register	Offset
lane0_tx_txcc_post_cval	4046h
lane1_tx_txcc_post_cval	4446h
lane2_tx_txcc_post_cval	4846h
lane3_tx_txcc_post_cval	4C46h

13.4.10.2.26.2 Diagram



13.4.10.2.26.3 Fields

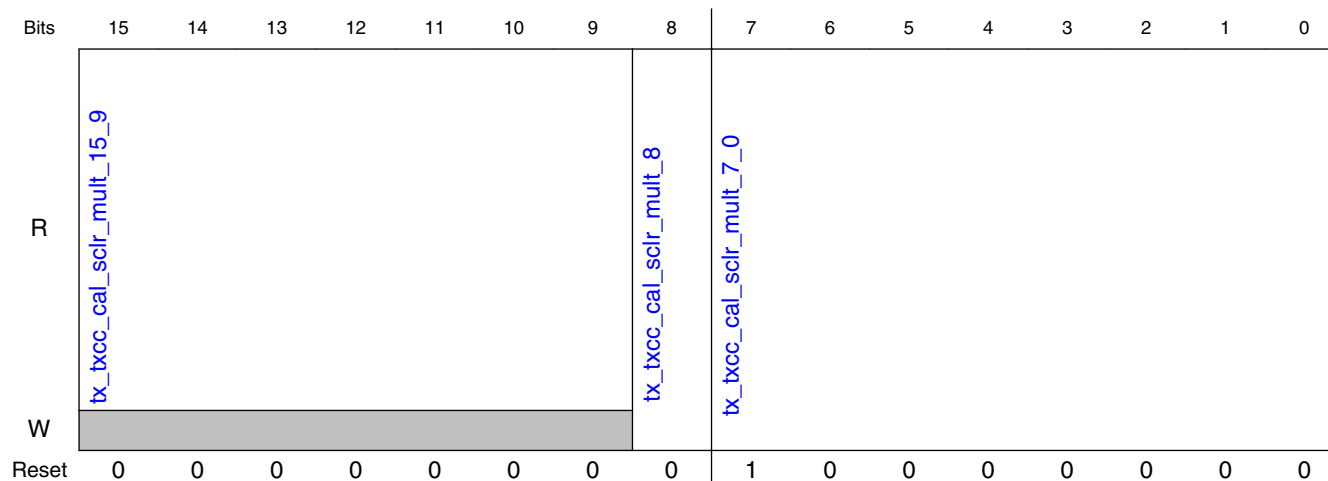
Field	Function
15-6 <code>tx_txcc_post_cval_15_6</code>	Reserved
5-0 <code>tx_txcc_post_cval_5_0</code>	Post-cursor value: The value in this field indicates the current value of the post-cursor (C+1) coefficient. The value of this field can be any of the following, depending on the current mode of operation.

13.4.10.2.27 Resistor calibration code scaler multiplier value register (lane 0_tx_txcc_cal_sclr_mult - lane3_tx_txcc_cal_sclr_mult)

13.4.10.2.27.1 Offset

Register	Offset
<code>lane0_tx_txcc_cal_sclr_mult</code>	4047h
<code>lane1_tx_txcc_cal_sclr_mult</code>	4447h
<code>lane2_tx_txcc_cal_sclr_mult</code>	4847h
<code>lane3_tx_txcc_cal_sclr_mult</code>	4C47h

13.4.10.2.27.2 Diagram



13.4.10.2.27.3 Fields

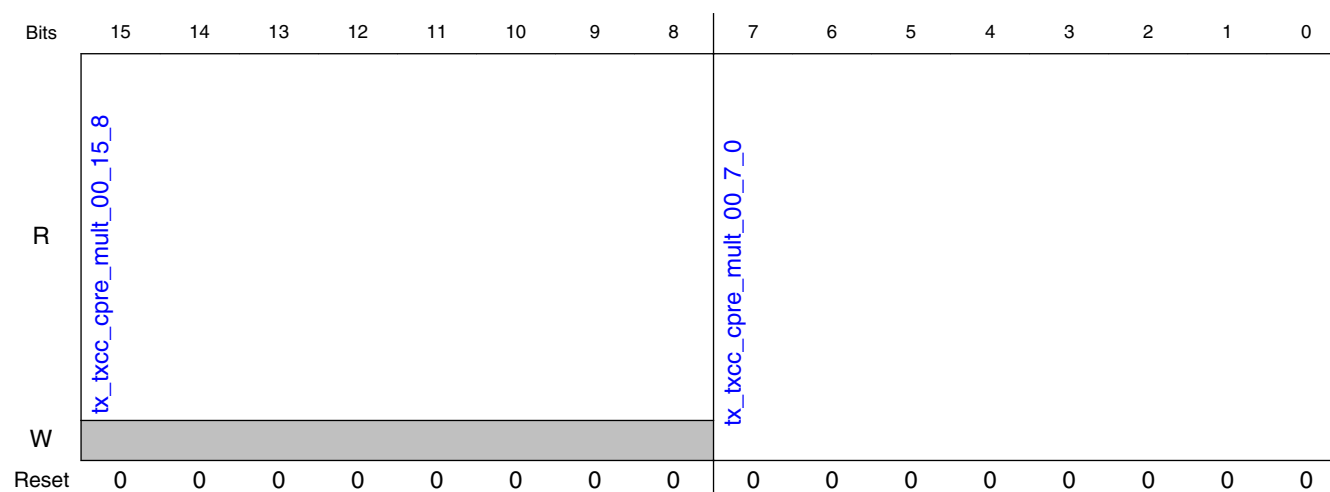
Field	Function
15-9 tx_txcc_cal_sclr_mult_15_9	Reserved
8 tx_txcc_cal_sclr_mult_8	Scaled resistor calibration code add: When this bit is set to
7-0 tx_txcc_cal_sclr_mult_7_0	Resistor calibration multiplier value: The value in this field specifies the multiplier value used to scale the resistor calibration code by. The following describes the multiplier value each bit in this field corresponds to.

13.4.10.2.28 Calculated pre emphasis multiplier value 00 register (lane0_tx_txcc_cpre_mult_00 - lane3_tx_txcc_cpre_mult_00)

13.4.10.2.28.1 Offset

Register	Offset
lane0_tx_txcc_cpre_mult_00	4048h
lane1_tx_txcc_cpre_mult_00	4448h
lane2_tx_txcc_cpre_mult_00	4848h
lane3_tx_txcc_cpre_mult_00	4C48h

13.4.10.2.28.2 Diagram



13.4.10.2.28.3 Fields

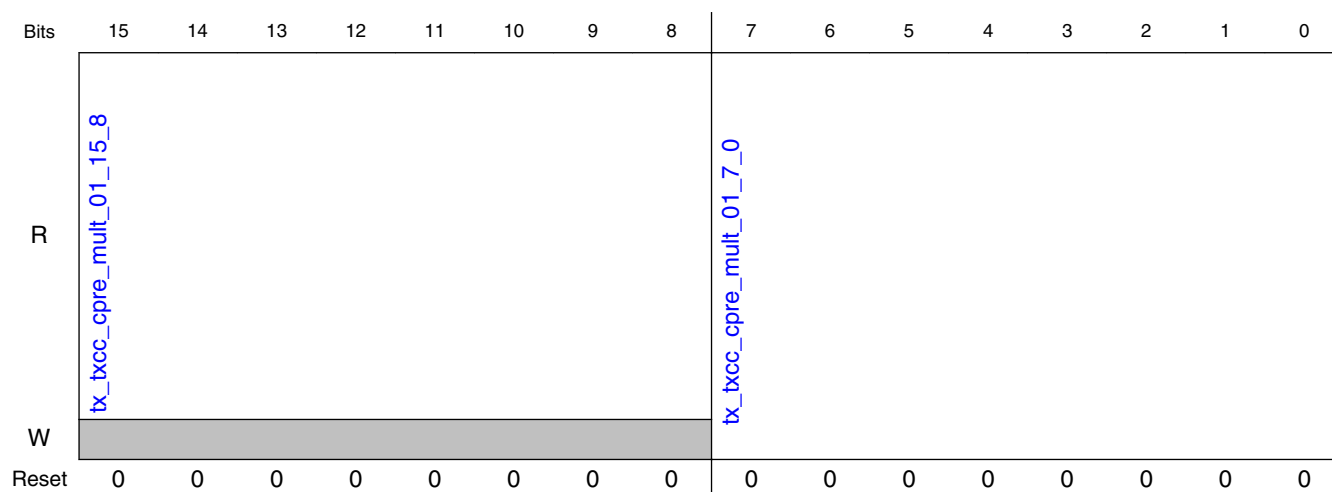
Field	Function
15-8 tx_txcc_cpre_mult_00_15_8	Reserved
7-0 tx_txcc_cpre_mult_00_7_0	Calculated pre emphasis multiplier value 00: The value in this field specifies the multiplier value used to generate the calculated pre emphasis value from the FS value when

13.4.10.2.29 Calculated pre emphasis multiplier value 01 register (lane0_tx_txcc_cpre_mult_01 - lane3_tx_txcc_cpre_mult_01)

13.4.10.2.29.1 Offset

Register	Offset
lane0_tx_txcc_cpre_mult_01	4049h
lane1_tx_txcc_cpre_mult_01	4449h
lane2_tx_txcc_cpre_mult_01	4849h
lane3_tx_txcc_cpre_mult_01	4C49h

13.4.10.2.29.2 Diagram



13.4.10.2.29.3 Fields

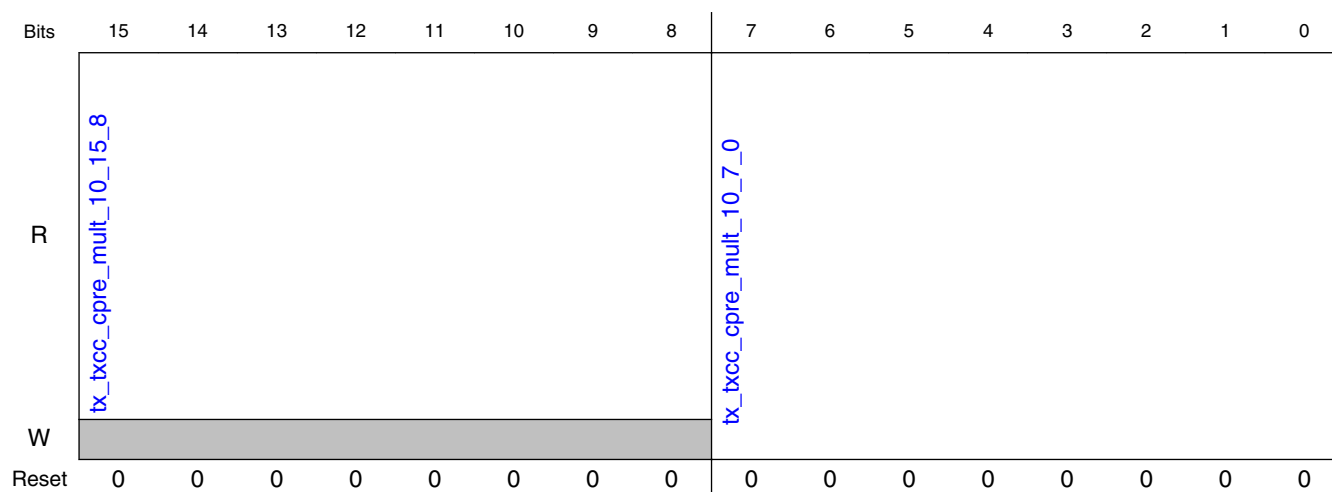
Field	Function
15-8 tx_txcc_cpre_mult_01_15_8	Reserved
7-0 tx_txcc_cpre_mult_01_7_0	Calculated pre emphasis multiplier value 01: The value in this field specifies the multiplier value used to generate the calculated pre emphasis value from the FS value when

13.4.10.2.30 Calculated pre emphasis multiplier value 10 register (lane0_tx_txcc_cpre_mult_10 - lane3_tx_txcc_cpre_mult_10)

13.4.10.2.30.1 Offset

Register	Offset
lane0_tx_txcc_cpre_mult_10	404Ah
lane1_tx_txcc_cpre_mult_10	444Ah
lane2_tx_txcc_cpre_mult_10	484Ah
lane3_tx_txcc_cpre_mult_10	4C4Ah

13.4.10.2.30.2 Diagram



13.4.10.2.30.3 Fields

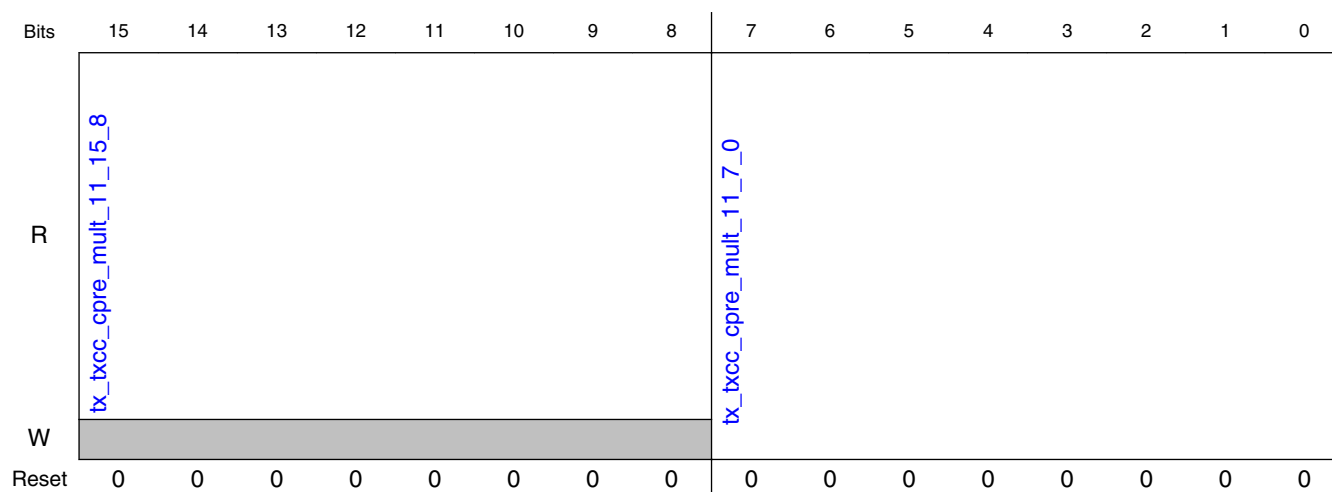
Field	Function
15-8 tx_txcc_cpre_mult_10_15_8	Reserved
7-0 tx_txcc_cpre_mult_10_7_0	Calculated pre emphasis multiplier value 10: The value in this field specifies the multiplier value used to generate the calculated pre emphasis value from the FS value when

13.4.10.2.31 Calculated pre emphasis multiplier value 11 register (lane0_tx_txcc_cpre_mult_11 - lane3_tx_txcc_cpre_mult_11)

13.4.10.2.31.1 Offset

Register	Offset
lane0_tx_txcc_cpre_mult_11	404Bh
lane1_tx_txcc_cpre_mult_11	444Bh
lane2_tx_txcc_cpre_mult_11	484Bh
lane3_tx_txcc_cpre_mult_11	4C4Bh

13.4.10.2.31.2 Diagram



13.4.10.2.31.3 Fields

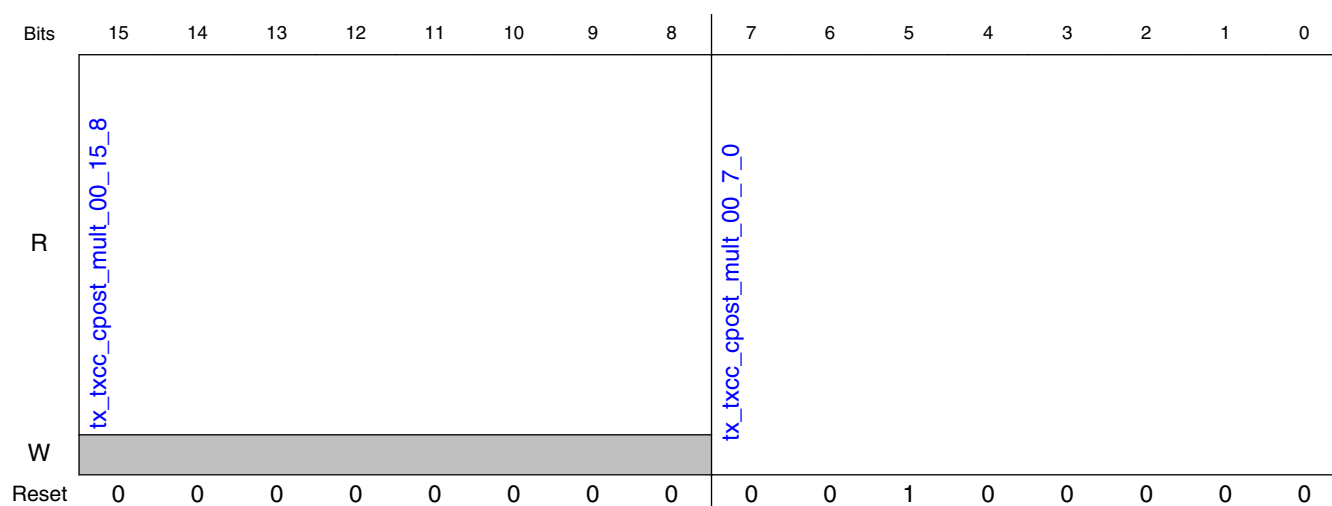
Field	Function
15-8 tx_txcc_cpre_mult_11_15_8	Reserved
7-0 tx_txcc_cpre_mult_11_7_0	Calculated pre emphasis multiplier value 11: The value in this field specifies the multiplier value used to generate the calculated pre emphasis value from the FS value when

13.4.10.2.32 Calculated post emphasis multiplier value 00 register (lane0_tx_txcc_cpost_mult_00 - lane3_tx_txcc_cpost_mult_00)

13.4.10.2.32.1 Offset

Register	Offset
lane0_tx_txcc_cpost_mult_00	404Ch
lane1_tx_txcc_cpost_mult_00	444Ch
lane2_tx_txcc_cpost_mult_00	484Ch
lane3_tx_txcc_cpost_mult_00	4C4Ch

13.4.10.2.32.2 Diagram



13.4.10.2.32.3 Fields

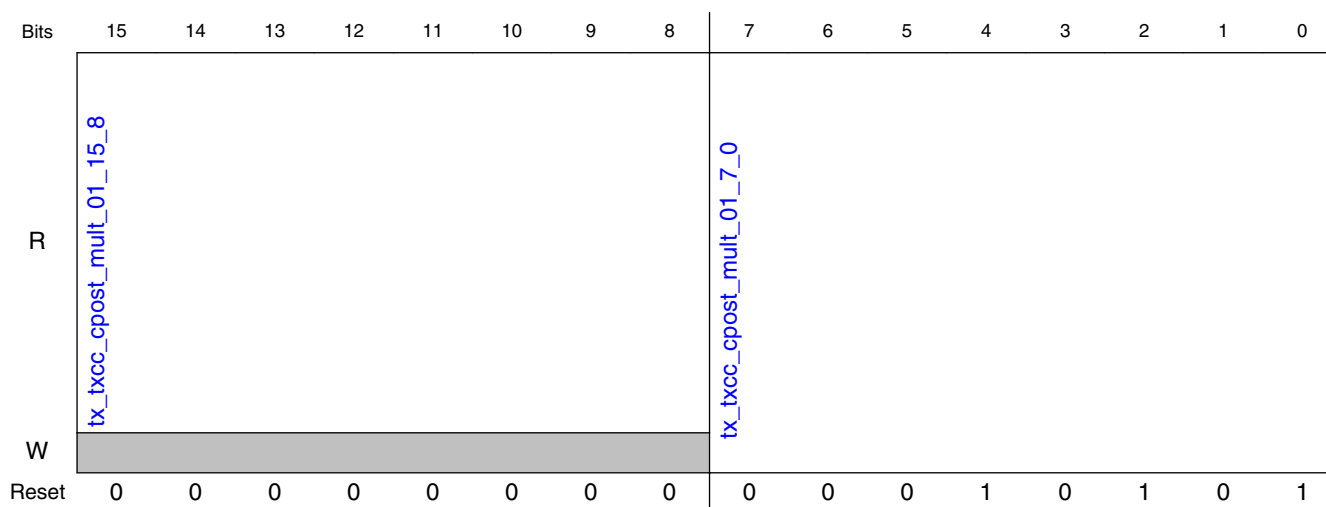
Field	Function
15-8 tx_txcc_cpost_mult_00_15_8	Reserved
7-0 tx_txcc_cpost_mult_00_7_0	Calculated post emphasis multiplier value 00: The value in this field specifies the multiplier value used to generate the calculated post emphasis value from the FS value when

13.4.10.2.33 Calculated post emphasis multiplier value 01 register (lane0_tx_txcc_cpost_mult_01 - lane3_tx_txcc_cpost_mult_01)

13.4.10.2.33.1 Offset

Register	Offset
lane0_tx_txcc_cpost_mult_01	404Dh
lane1_tx_txcc_cpost_mult_01	444Dh
lane2_tx_txcc_cpost_mult_01	484Dh
lane3_tx_txcc_cpost_mult_01	4C4Dh

13.4.10.2.33.2 Diagram



13.4.10.2.33.3 Fields

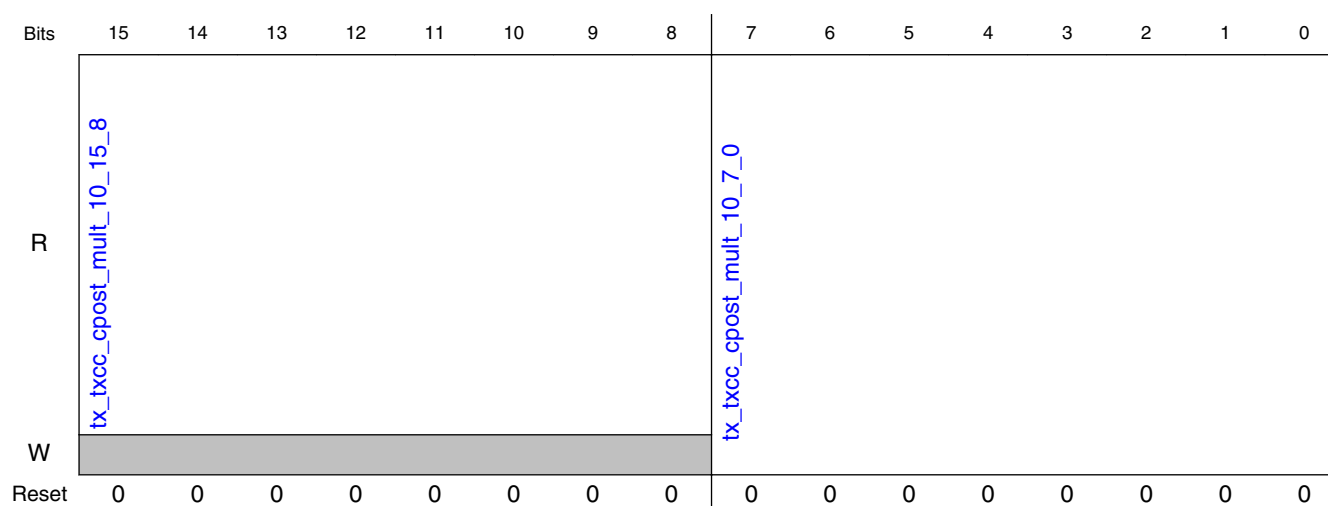
Field	Function
15-8 tx_txcc_cpost_mult_01_15_8	Reserved
7-0 tx_txcc_cpost_mult_01_7_0	Calculated post emphasis multiplier value 01: The value in this field specifies the multiplier value used to generate the calculated post emphasis value from the FS value when

13.4.10.2.34 Calculated post emphasis multiplier value 10 register (lane0_tx_txcc_cpost_mult_10 - lane3_tx_txcc_cpost_mult_10)

13.4.10.2.34.1 Offset

Register	Offset
lane0_tx_txcc_cpost_mult_10	404Eh
lane1_tx_txcc_cpost_mult_10	444Eh
lane2_tx_txcc_cpost_mult_10	484Eh
lane3_tx_txcc_cpost_mult_10	4C4Eh

13.4.10.2.34.2 Diagram



13.4.10.2.34.3 Fields

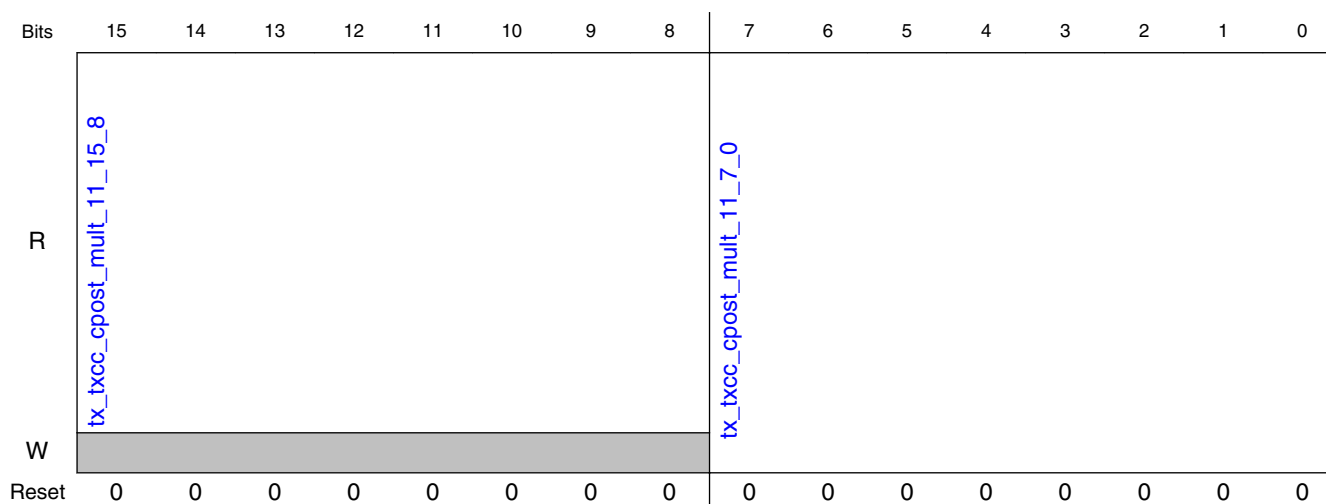
Field	Function
15-8 tx_txcc_cpost_mult_10_15_8	Reserved
7-0 tx_txcc_cpost_mult_10_7_0	Calculated post emphasis multiplier value 10: The value in this field specifies the multiplier value used to generate the calculated post emphasis value from the FS value when

13.4.10.2.35 Calculated post emphasis multiplier value 11 register (lane0_tx_txcc_cpost_mult_11 - lane3_tx_txcc_cpost_mult_11)

13.4.10.2.35.1 Offset

Register	Offset
lane0_tx_txcc_cpost_mult_11	404Fh
lane1_tx_txcc_cpost_mult_11	444Fh
lane2_tx_txcc_cpost_mult_11	484Fh
lane3_tx_txcc_cpost_mult_11	4C4Fh

13.4.10.2.35.2 Diagram



13.4.10.2.35.3 Fields

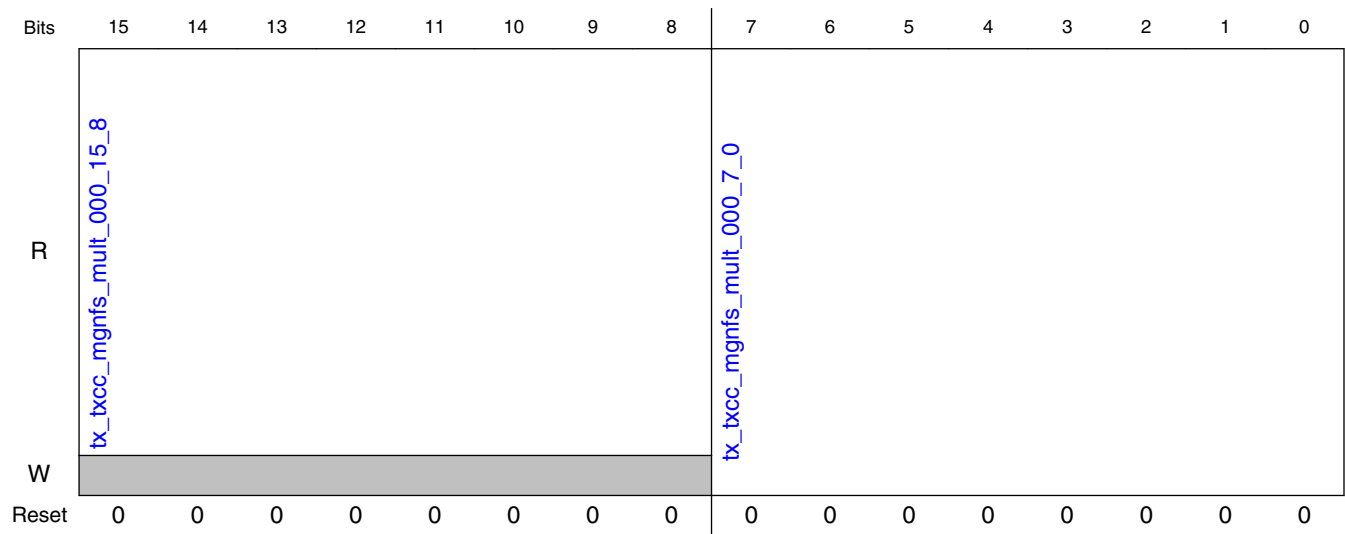
Field	Function
15-8 tx_txcc_cpost_mult_11_15_8	Reserved
7-0 tx_txcc_cpost_mult_11_7_0	Calculated post emphasis multiplier value 11: The value in this field specifies the multiplier value used to generate the calculated post emphasis value from the FS value when

13.4.10.2.36 Margin full swing multiplier value 000 register (lane0_tx_txcc_mgnfs_mult_000 - lane3_tx_txcc_mgnfs_mult_000)

13.4.10.2.36.1 Offset

Register	Offset
lane0_tx_txcc_mgnfs_mult_000	4050h
lane1_tx_txcc_mgnfs_mult_000	4450h
lane2_tx_txcc_mgnfs_mult_000	4850h
lane3_tx_txcc_mgnfs_mult_000	4C50h

13.4.10.2.36.2 Diagram



13.4.10.2.36.3 Fields

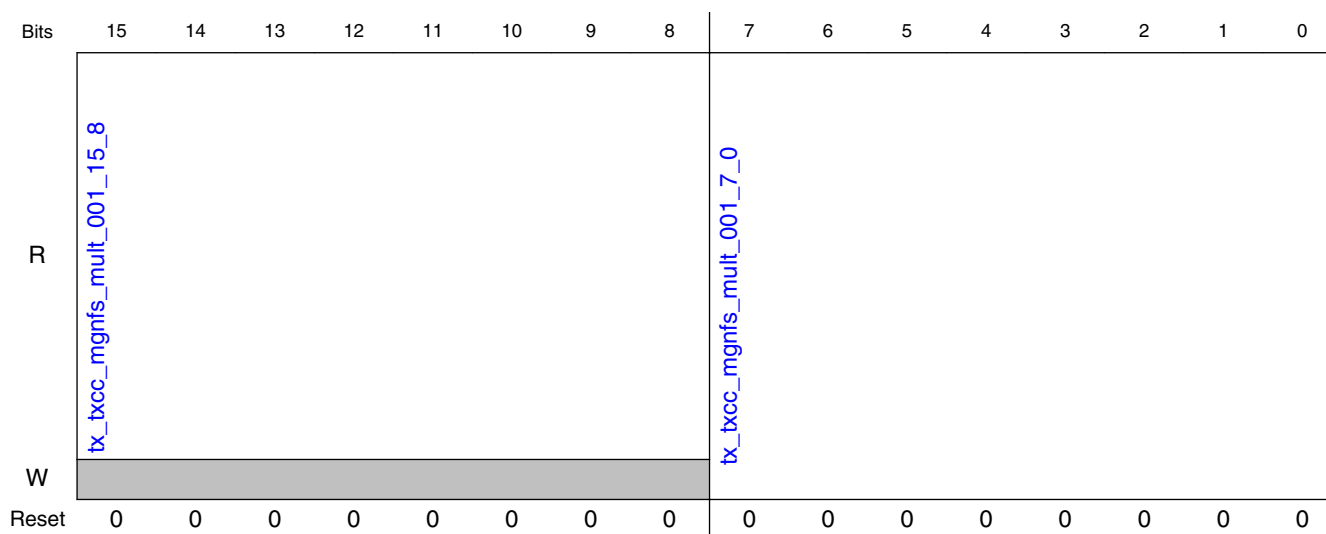
Field	Function
15-8 tx_txcc_mgnfs_mult_000_15_8	Reserved
7-0 tx_txcc_mgnfs_mult_000_7_0	Margin full swing multiplier value 000: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.37 Margin full swing multiplier value 001 register (lane0_tx_txcc_mgnfs_mult_001 - lane3_tx_txcc_mgnfs_mult_001)

13.4.10.2.37.1 Offset

Register	Offset
lane0_tx_txcc_mgnfs_mult_001	4051h
lane1_tx_txcc_mgnfs_mult_001	4451h
lane2_tx_txcc_mgnfs_mult_001	4851h
lane3_tx_txcc_mgnfs_mult_001	4C51h

13.4.10.2.37.2 Diagram



13.4.10.2.37.3 Fields

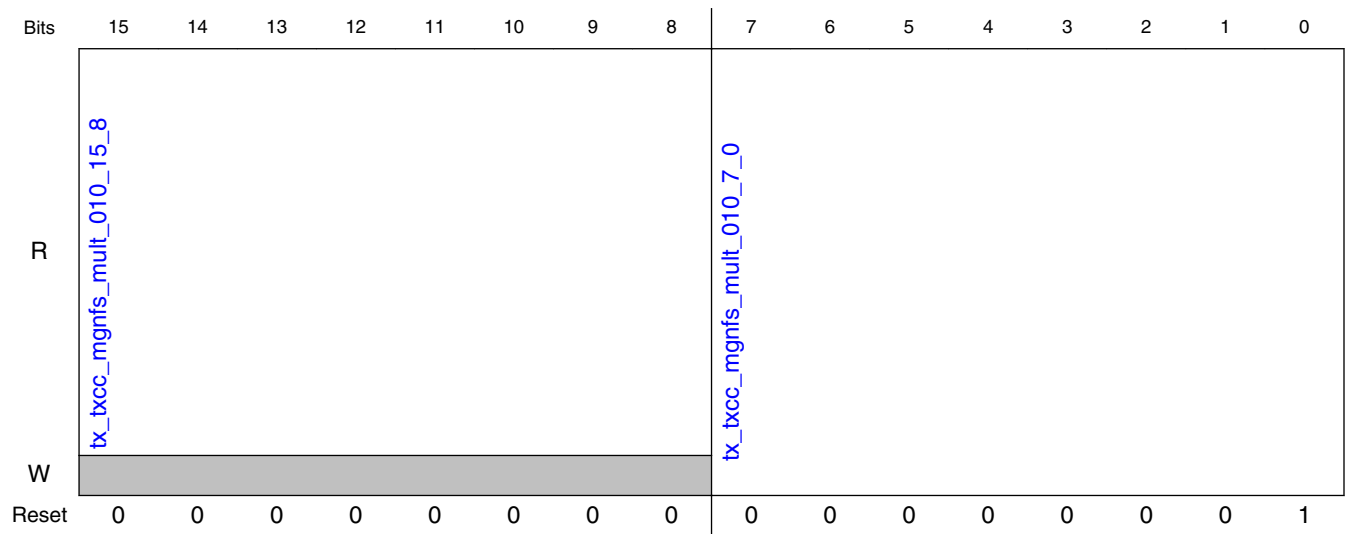
Field	Function
15-8 tx_txcc_mgnfs_mult_001_15_8	Reserved
7-0 tx_txcc_mgnfs_mult_001_7_0	Margin full swing multiplier value 001: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.38 Margin full swing multiplier value 010 register (lane0_tx_txcc_mgnfs_mult_010 - lane3_tx_txcc_mgnfs_mult_010)

13.4.10.2.38.1 Offset

Register	Offset
lane0_tx_txcc_mgnfs_mult_010	4052h
lane1_tx_txcc_mgnfs_mult_010	4452h
lane2_tx_txcc_mgnfs_mult_010	4852h
lane3_tx_txcc_mgnfs_mult_010	4C52h

13.4.10.2.38.2 Diagram



13.4.10.2.38.3 Fields

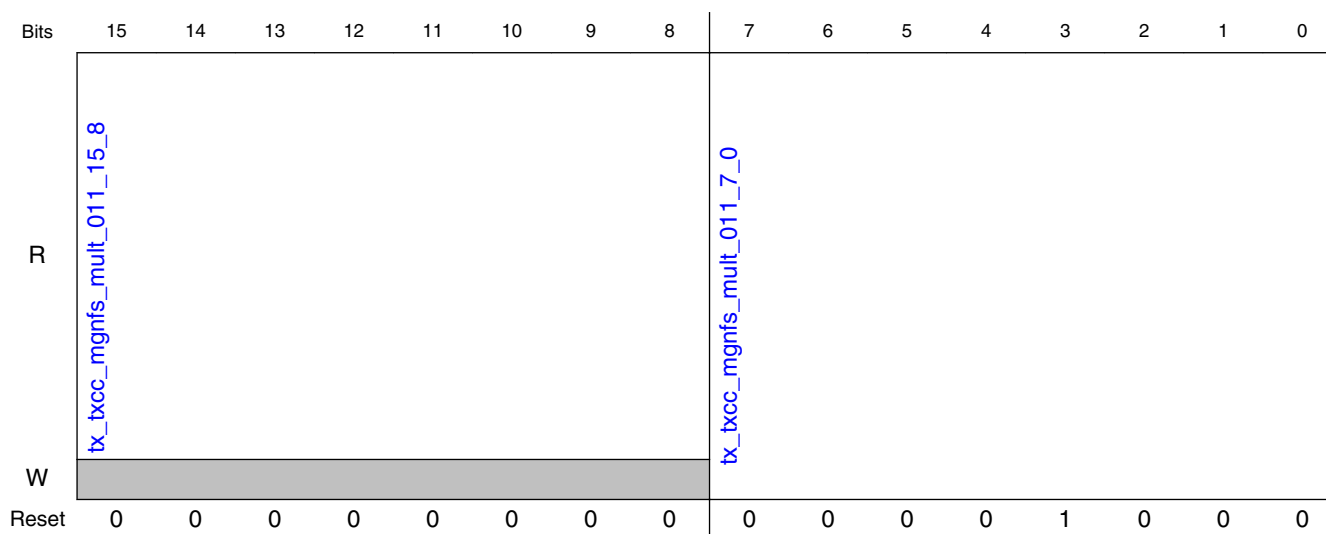
Field	Function
15-8 tx_txcc_mgnfs_mult_010_15_8	Reserved
7-0 tx_txcc_mgnfs_mult_010_7_0	Margin full swing multiplier value 010: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.39 Margin full swing multiplier value 011 register (lane0_tx_txcc_mgnfs_mult_011 - lane3_tx_txcc_mgnfs_mult_011)

13.4.10.2.39.1 Offset

Register	Offset
lane0_tx_txcc_mgnfs_mult_011	4053h
lane1_tx_txcc_mgnfs_mult_011	4453h
lane2_tx_txcc_mgnfs_mult_011	4853h
lane3_tx_txcc_mgnfs_mult_011	4C53h

13.4.10.2.39.2 Diagram



13.4.10.2.39.3 Fields

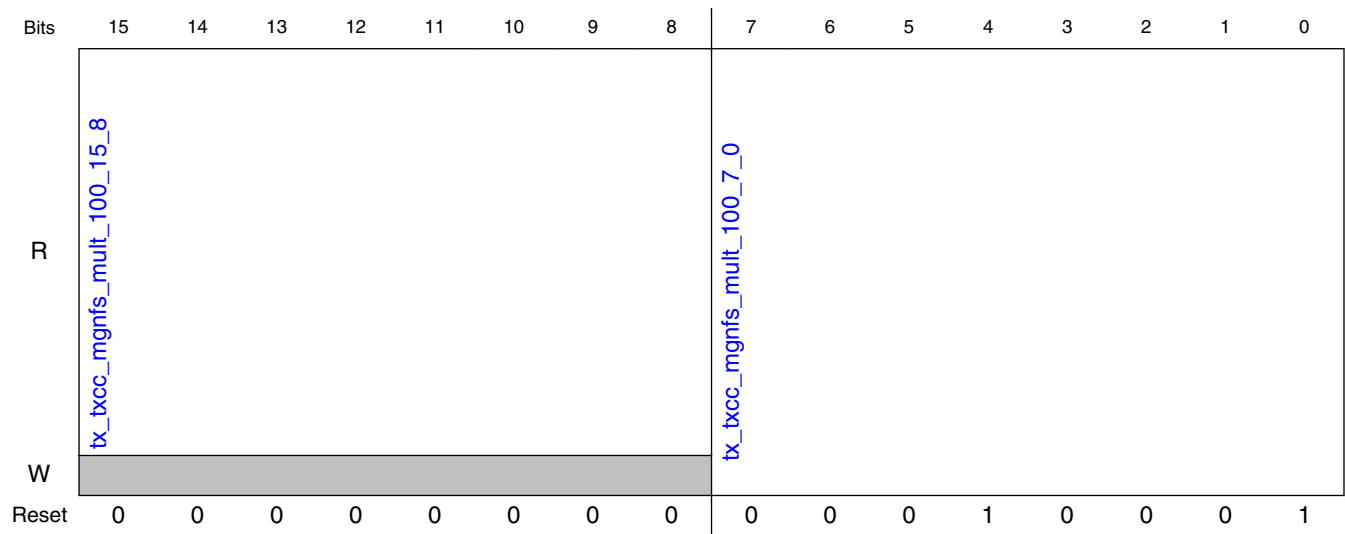
Field	Function
15-8 <code>tx_txcc_mgnfs_mult_011_15_8</code>	Reserved
7-0 <code>tx_txcc_mgnfs_mult_011_7_0</code>	Margin full swing multiplier value 011: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.40 Margin full swing multiplier value 100 register (`lane0_tx_txcc_mgnfs_mult_100` - `lane3_tx_txcc_mgnfs_mult_100`)

13.4.10.2.40.1 Offset

Register	Offset
<code>lane0_tx_txcc_mgnfs_mult_100</code>	4054h
<code>lane1_tx_txcc_mgnfs_mult_100</code>	4454h
<code>lane2_tx_txcc_mgnfs_mult_100</code>	4854h
<code>lane3_tx_txcc_mgnfs_mult_100</code>	4C54h

13.4.10.2.40.2 Diagram



13.4.10.2.40.3 Fields

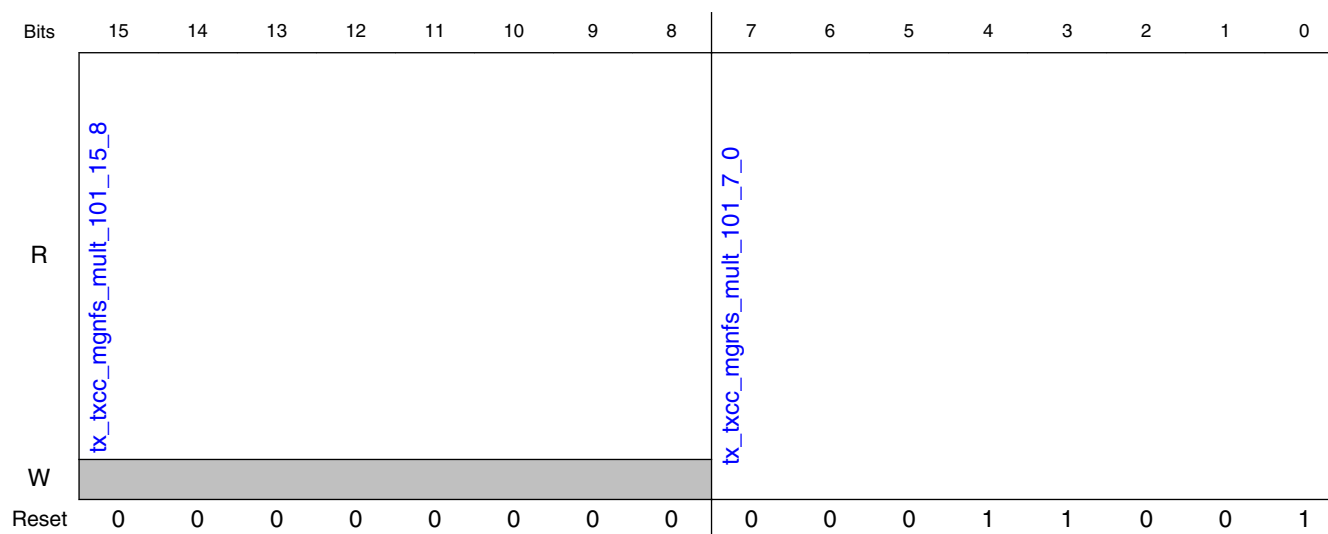
Field	Function
15-8 tx_txcc_mgnfs_mult_100_15_8	Reserved
7-0 tx_txcc_mgnfs_mult_100_7_0	Margin full swing multiplier value 100: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.41 Margin full swing multiplier value 101 register (lane0_tx_txcc_mgnfs_mult_101 - lane3_tx_txcc_mgnfs_mult_101)

13.4.10.2.41.1 Offset

Register	Offset
lane0_tx_txcc_mgnfs_mult_101	4055h
lane1_tx_txcc_mgnfs_mult_101	4455h
lane2_tx_txcc_mgnfs_mult_101	4855h
lane3_tx_txcc_mgnfs_mult_101	4C55h

13.4.10.2.41.2 Diagram



13.4.10.2.41.3 Fields

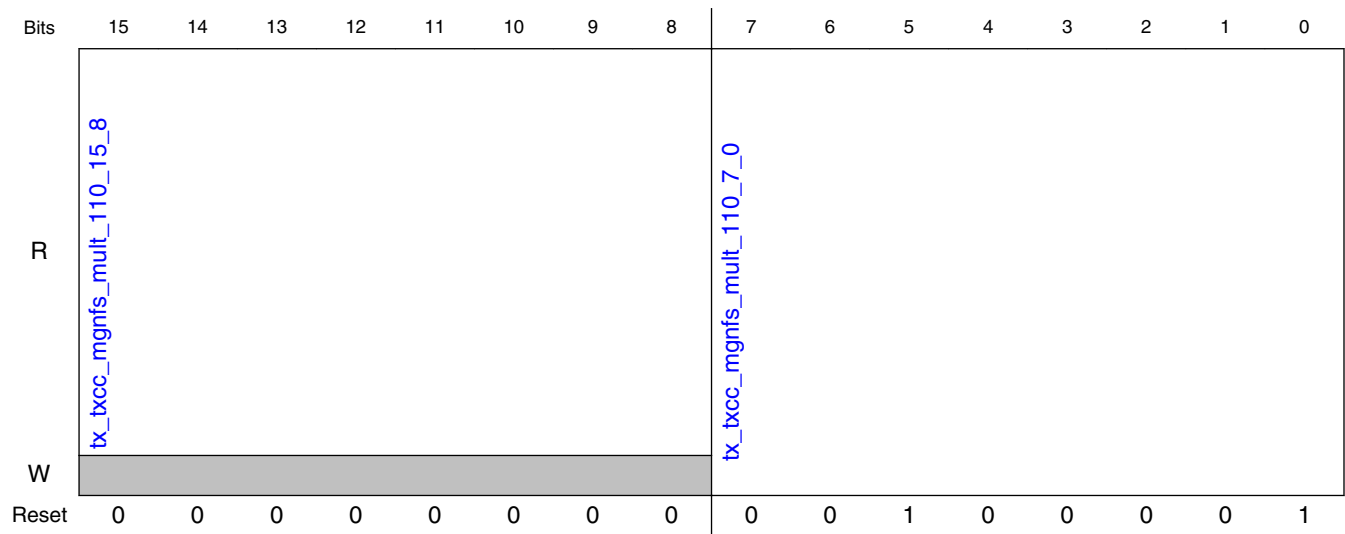
Field	Function
15-8 <code>tx_txcc_mgnfs_mult_101_15_8</code>	Reserved
7-0 <code>tx_txcc_mgnfs_mult_101_7_0</code>	Margin full swing multiplier value 101: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.42 Margin full swing multiplier value 110 register (lane0_tx_txcc_mgnfs_mult_110 - lane3_tx_txcc_mgnfs_mult_110)

13.4.10.2.42.1 Offset

Register	Offset
<code>lane0_tx_txcc_mgnfs_mult_110</code>	4056h
<code>lane1_tx_txcc_mgnfs_mult_110</code>	4456h
<code>lane2_tx_txcc_mgnfs_mult_110</code>	4856h
<code>lane3_tx_txcc_mgnfs_mult_110</code>	4C56h

13.4.10.2.42.2 Diagram



13.4.10.2.42.3 Fields

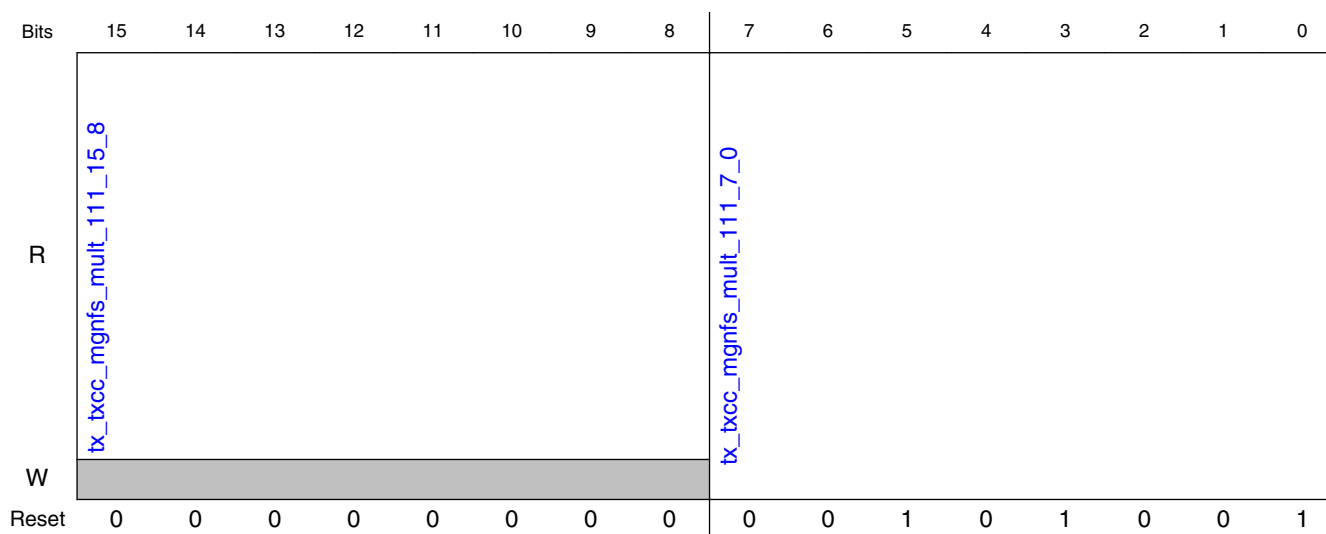
Field	Function
15-8 tx_txcc_mgnfs_mult_110_15_8	Reserved
7-0 tx_txcc_mgnfs_mult_110_7_0	Margin full swing multiplier value 110: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.43 Margin full swing multiplier value 111 register (lane0_tx_txcc_mgnfs_mult_111 - lane3_tx_txcc_mgnfs_mult_111)

13.4.10.2.43.1 Offset

Register	Offset
lane0_tx_txcc_mgnfs_mult_111	4057h
lane1_tx_txcc_mgnfs_mult_111	4457h
lane2_tx_txcc_mgnfs_mult_111	4857h
lane3_tx_txcc_mgnfs_mult_111	4C57h

13.4.10.2.43.2 Diagram



13.4.10.2.43.3 Fields

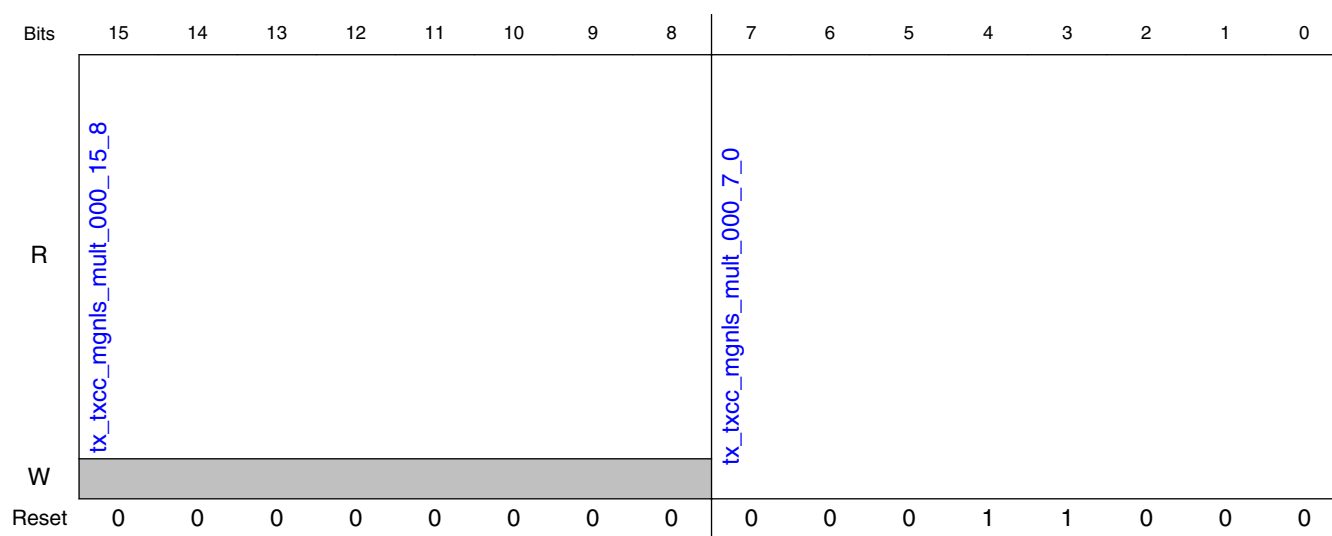
Field	Function
15-8 <code>tx_txcc_mgnfs_mult_111_15_8</code>	Reserved
7-0 <code>tx_txcc_mgnfs_mult_111_7_0</code>	Margin full swing multiplier value 111: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.44 Margin half swing multiplier value 000 register (lane0_tx_txc_c_mgnls_mult_000 - lane3_tx_txcc_mgnls_mult_000)

13.4.10.2.44.1 Offset

Register	Offset
<code>lane0_tx_txcc_mgnls_mult_000</code>	4058h
<code>lane1_tx_txcc_mgnls_mult_000</code>	4458h
<code>lane2_tx_txcc_mgnls_mult_000</code>	4858h
<code>lane3_tx_txcc_mgnls_mult_000</code>	4C58h

13.4.10.2.44.2 Diagram



13.4.10.2.44.3 Fields

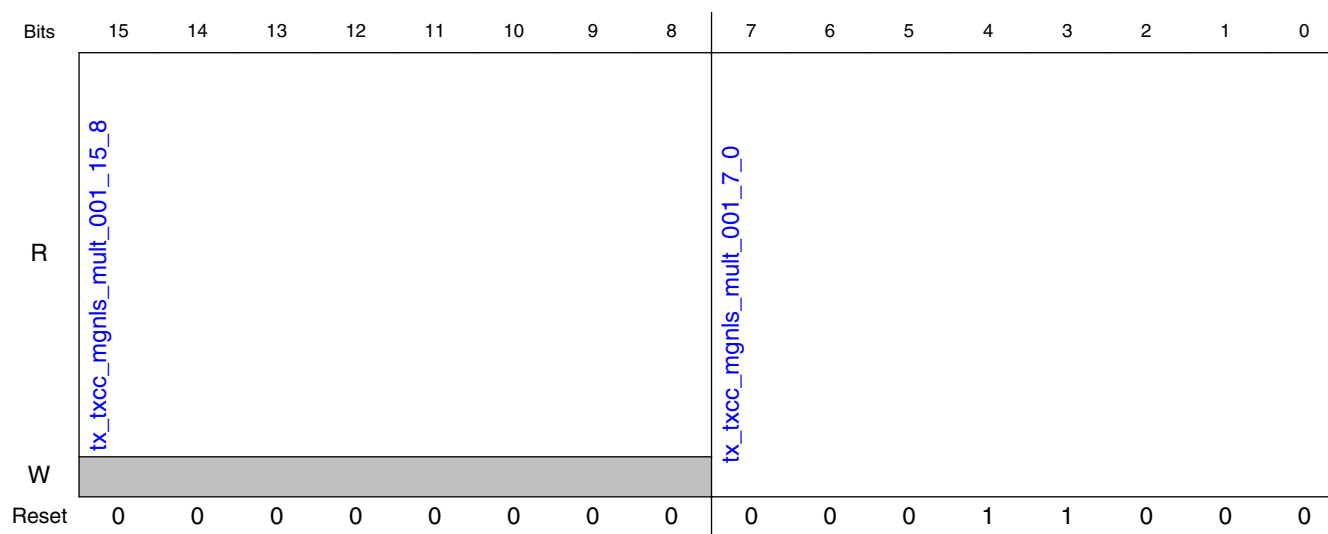
Field	Function
15-8 tx_txcc_mgnls_mult_000_15_8	Reserved
7-0 tx_txcc_mgnls_mult_000_7_0	Margin half swing multiplier value 000: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.45 Margin half swing multiplier value 001 register (lane0_tx_txcc_mgnls_mult_001 - lane3_tx_txcc_mgnls_mult_001)

13.4.10.2.45.1 Offset

Register	Offset
lane0_tx_txcc_mgnls_mult_001	4059h
lane1_tx_txcc_mgnls_mult_001	4459h
lane2_tx_txcc_mgnls_mult_001	4859h
lane3_tx_txcc_mgnls_mult_001	4C59h

13.4.10.2.45.2 Diagram



13.4.10.2.45.3 Fields

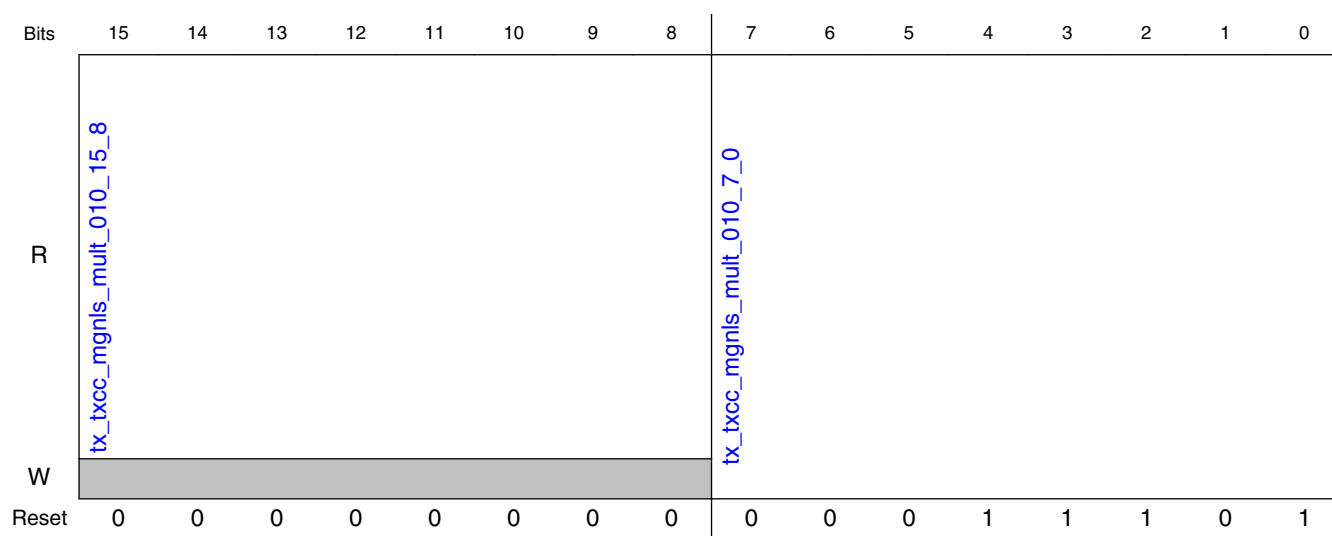
Field	Function
15-8 tx_txcc_mgnls_mult_001_15_8	Reserved
7-0 tx_txcc_mgnls_mult_001_7_0	Margin half swing multiplier value 001: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.46 Margin half swing multiplier value 010 register (lane0_tx_txcc_mgnls_mult_010 - lane3_tx_txcc_mgnls_mult_010)

13.4.10.2.46.1 Offset

Register	Offset
lane0_tx_txcc_mgnls_mult_010	405Ah
lane1_tx_txcc_mgnls_mult_010	445Ah
lane2_tx_txcc_mgnls_mult_010	485Ah
lane3_tx_txcc_mgnls_mult_010	4C5Ah

13.4.10.2.46.2 Diagram



13.4.10.2.46.3 Fields

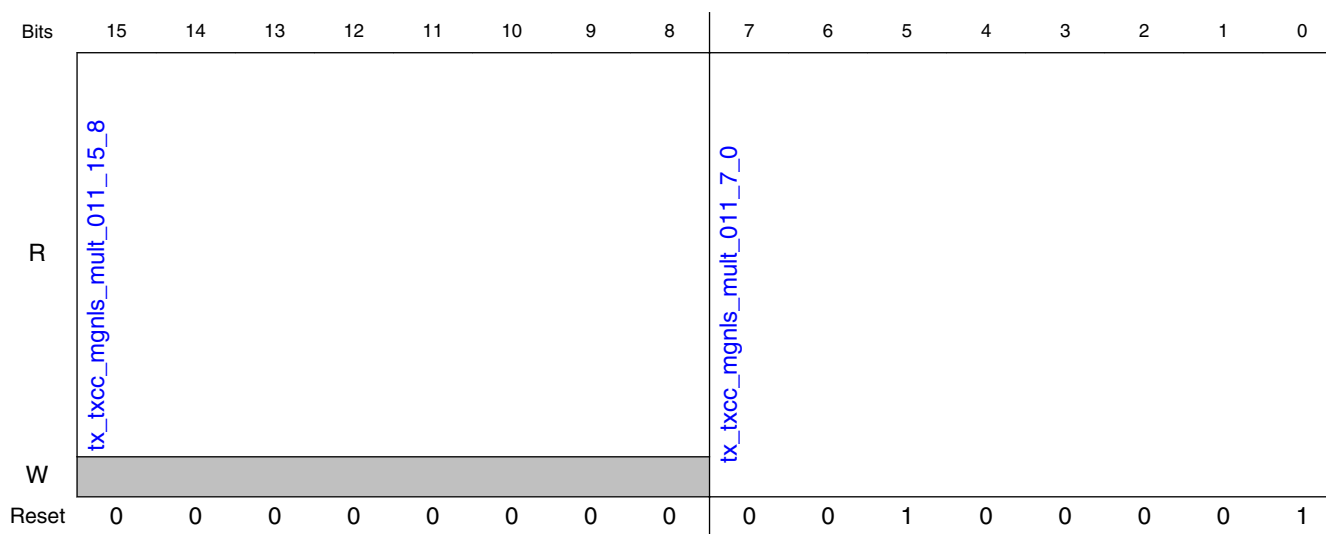
Field	Function
15-8 tx_txcc_mgnls_mult_010_15_8	Reserved
7-0 tx_txcc_mgnls_mult_010_7_0	Margin half swing multiplier value 010: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.47 Margin half swing multiplier value 011 register (lane0_tx_txcc_mgnls_mult_011 - lane3_tx_txcc_mgnls_mult_011)

13.4.10.2.47.1 Offset

Register	Offset
lane0_tx_txcc_mgnls_mult_011	405Bh
lane1_tx_txcc_mgnls_mult_011	445Bh
lane2_tx_txcc_mgnls_mult_011	485Bh
lane3_tx_txcc_mgnls_mult_011	4C5Bh

13.4.10.2.47.2 Diagram



13.4.10.2.47.3 Fields

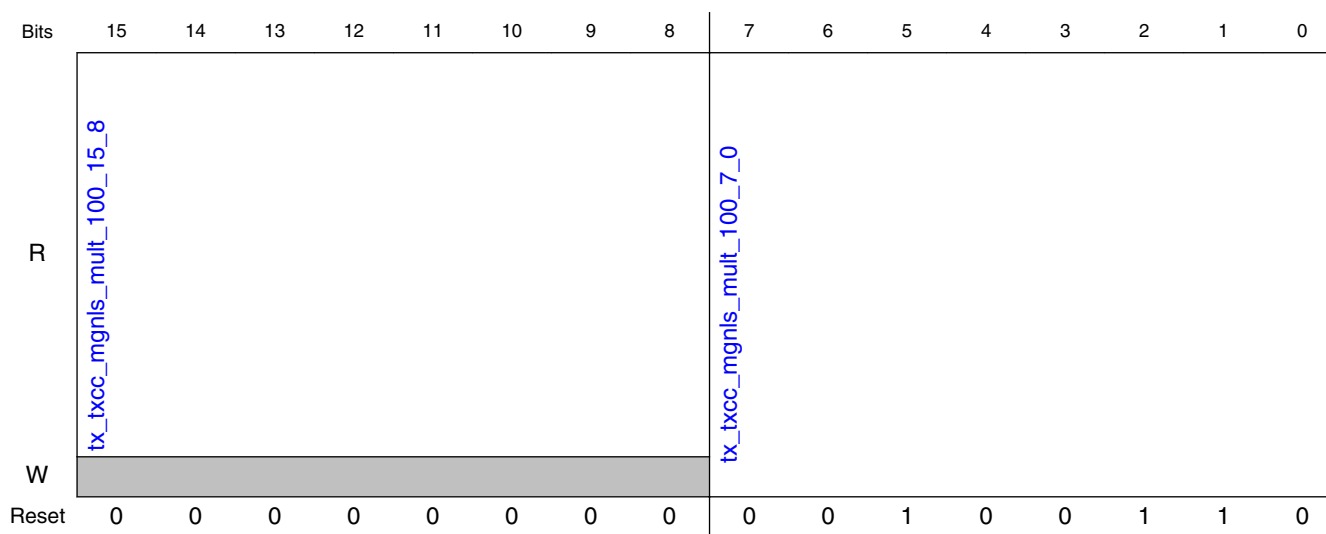
Field	Function
15-8 tx_txcc_mgnls_mult_011_15_8	Reserved
7-0 tx_txcc_mgnls_mult_011_7_0	Margin half swing multiplier value 011: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.48 Margin half swing multiplier value 100 register (lane0_tx_txcc_mgnls_mult_100 - lane3_tx_txcc_mgnls_mult_100)

13.4.10.2.48.1 Offset

Register	Offset
lane0_tx_txcc_mgnls_mult_100	405Ch
lane1_tx_txcc_mgnls_mult_100	445Ch
lane2_tx_txcc_mgnls_mult_100	485Ch
lane3_tx_txcc_mgnls_mult_100	4C5Ch

13.4.10.2.48.2 Diagram



13.4.10.2.48.3 Fields

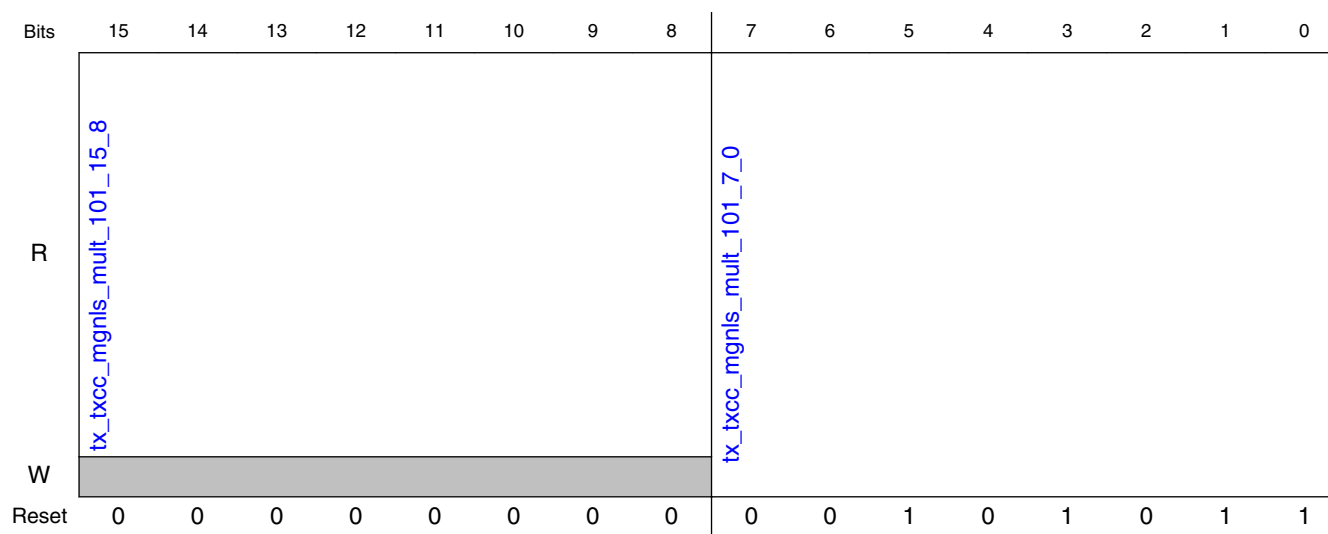
Field	Function
15-8 tx_txcc_mgnls_mult_100_15_8	Reserved
7-0 tx_txcc_mgnls_mult_100_7_0	Margin half swing multiplier value 100: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.49 Margin half swing multiplier value 101 register (lane0_tx_txcc_mgnls_mult_101 - lane3_tx_txcc_mgnls_mult_101)

13.4.10.2.49.1 Offset

Register	Offset
lane0_tx_txcc_mgnls_mult_101	405Dh
lane1_tx_txcc_mgnls_mult_101	445Dh
lane2_tx_txcc_mgnls_mult_101	485Dh
lane3_tx_txcc_mgnls_mult_101	4C5Dh

13.4.10.2.49.2 Diagram



13.4.10.2.49.3 Fields

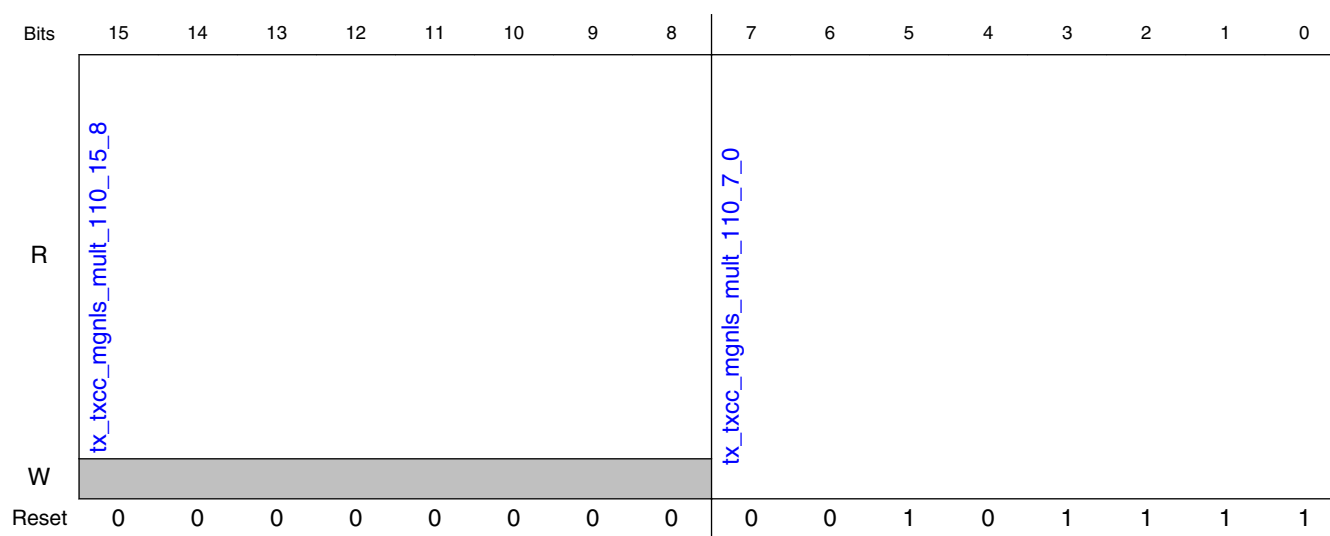
Field	Function
15-8 tx_txcc_mgnls_mult_101_15_8	Reserved
7-0 tx_txcc_mgnls_mult_101_7_0	Margin half swing multiplier value 101: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.50 Margin half swing multiplier value 110 register (lane0_tx_txcc_mgnls_mult_110 - lane3_tx_txcc_mgnls_mult_110)

13.4.10.2.50.1 Offset

Register	Offset
lane0_tx_txcc_mgnls_mult_110	405Eh
lane1_tx_txcc_mgnls_mult_110	445Eh
lane2_tx_txcc_mgnls_mult_110	485Eh
lane3_tx_txcc_mgnls_mult_110	4C5Eh

13.4.10.2.50.2 Diagram



13.4.10.2.50.3 Fields

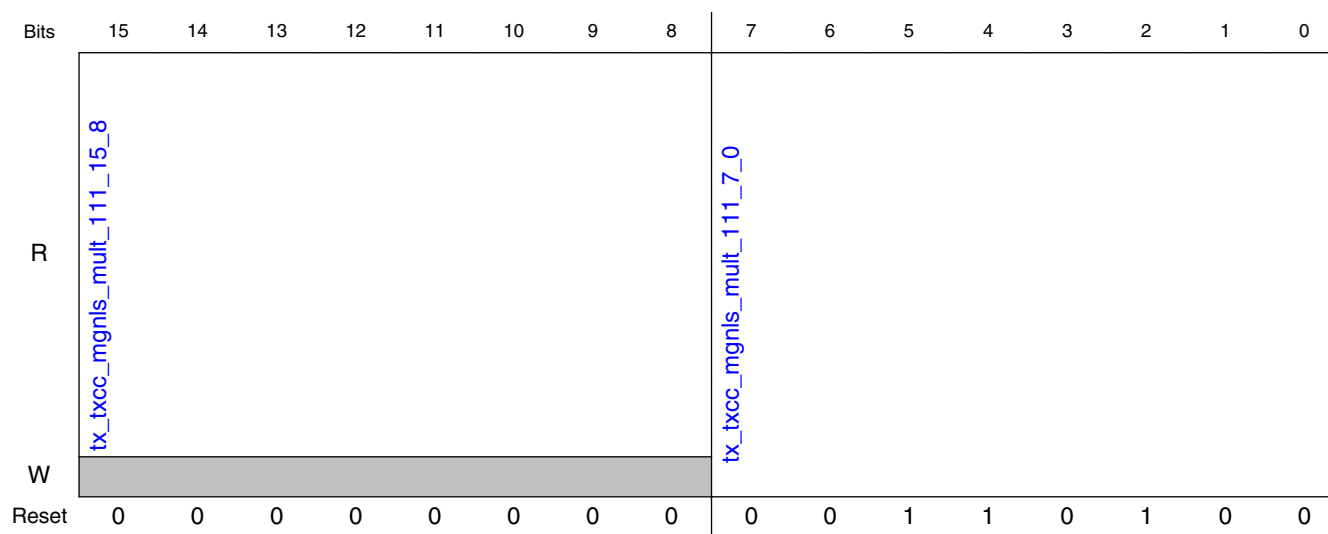
Field	Function
15-8 tx_txcc_mgnls_mult_110_15_8	Reserved
7-0 tx_txcc_mgnls_mult_110_7_0	Margin half swing multiplier value 110: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.51 Margin half swing multiplier value 111 register (lane0_tx_txcc_mgnls_mult_111 - lane3_tx_txcc_mgnls_mult_111)

13.4.10.2.51.1 Offset

Register	Offset
lane0_tx_txcc_mgnls_mult_111	405Fh
lane1_tx_txcc_mgnls_mult_111	445Fh
lane2_tx_txcc_mgnls_mult_111	485Fh
lane3_tx_txcc_mgnls_mult_111	4C5Fh

13.4.10.2.51.2 Diagram



13.4.10.2.51.3 Fields

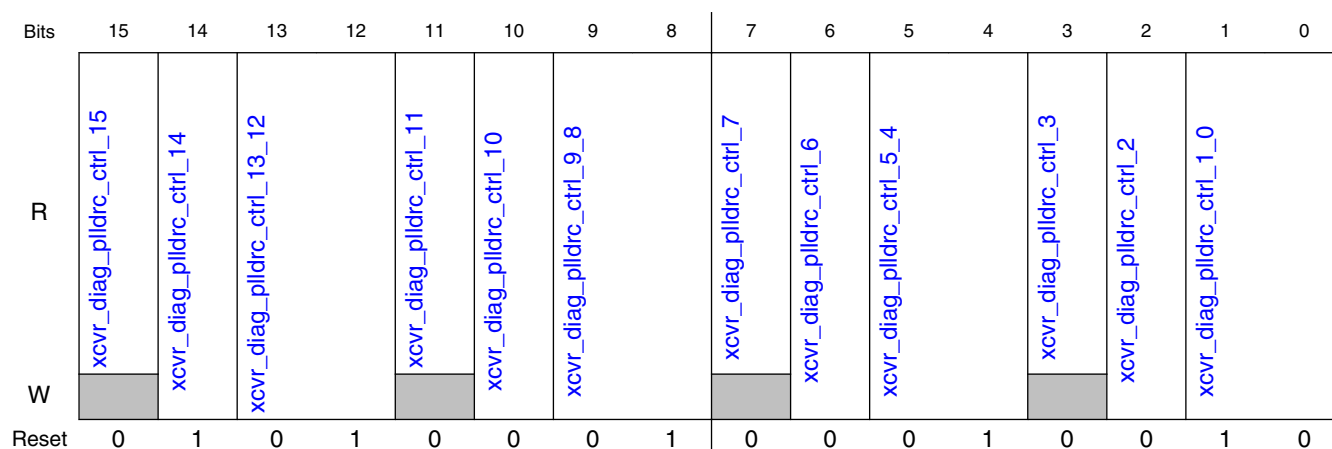
Field	Function
15-8 tx_txcc_mgnls_mult_111_15_8	Reserved
7-0 tx_txcc_mgnls_mult_111_7_0	Margin half swing multiplier value 111: The value in this field specifies the multiplier value used to generate the margin value from the resistor calibration value when

13.4.10.2.52 Transceiver PLL data rate clock control register (lane0_xcvr_diag_plldrc_ctrl - lane3_xcvr_diag_plldrc_ctrl)

13.4.10.2.52.1 Offset

Register	Offset
lane0_xcvr_diag_plldrc_ctrl	40E0h
lane1_xcvr_diag_plldrc_ctrl	44E0h
lane2_xcvr_diag_plldrc_ctrl	48E0h
lane3_xcvr_diag_plldrc_ctrl	4CE0h

13.4.10.2.52.2 Diagram



13.4.10.2.52.3 Fields

Field	Function
15 xcvr_diag_plldrc_ctrl_15	Reserved
14 xcvr_diag_plldrc_ctrl_14	Digital PLL clock select standard mode 3: This bit controls which 500 MHz PLL clock is selected, when
13-12 xcvr_diag_plldrc_ctrl_13_12	Digital PLL data rate divider standard mode 3 value: This field will directly control the xcvr_pll_clk_datart_div signal, which controls which divided clock is selected when generating the
11 xcvr_diag_plldrc_ctrl_11	Reserved
10 xcvr_diag_plldrc_ctrl_10	Digital PLL clock select standard mode 2: This bit controls which 500 MHz PLL clock is selected, when
9-8 xcvr_diag_plldrc_ctrl_9_8	Digital PLL data rate divider standard mode 2 value: This field will directly control the xcvr_pll_clk_datart_div signal, which controls which divided clock is selected when generating the
7 xcvr_diag_plldrc_ctrl_7	Reserved
6 xcvr_diag_plldrc_ctrl_6	Digital PLL clock select standard mode 1: This bit controls which 500 MHz PLL clock is selected, when

Table continues on the next page...

Clocks And Resets

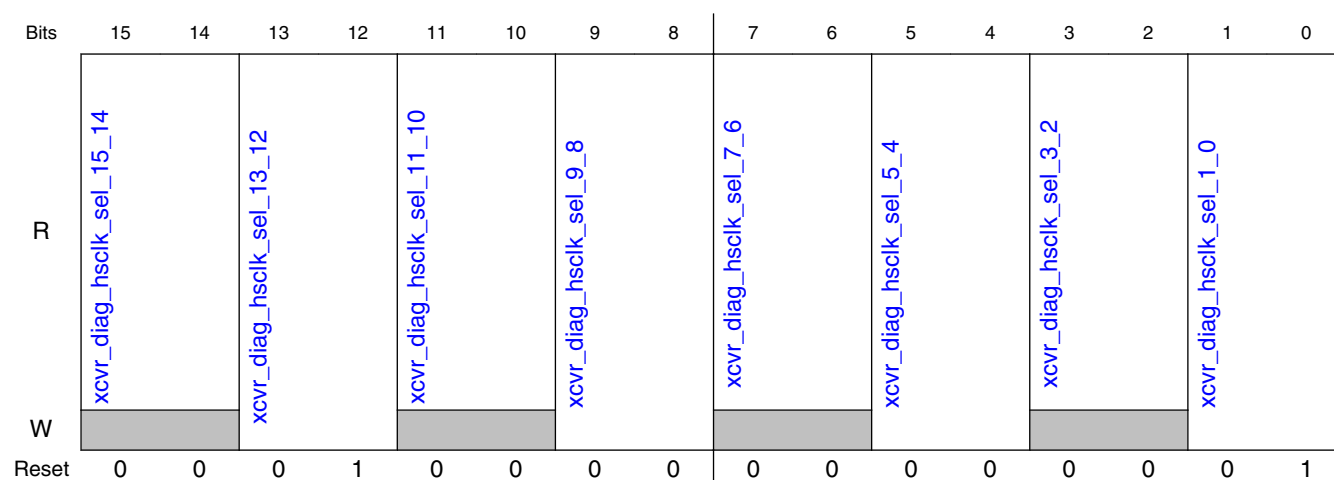
Field	Function
5-4 xcvr_diag_plldrc _ctrl_5_4	Digital PLL data rate divider standard mode 1 value: This field will directly control the xcvr_pll_clk_datart_div signal, which controls which divided clock is selected when generating the
3 xcvr_diag_plldrc _ctrl_3	Reserved
2 xcvr_diag_plldrc _ctrl_2	Digital PLL clock select standard mode 0: This bit controls which 500 MHz PLL clock is selected, when
1-0 xcvr_diag_plldrc _ctrl_1_0	Digital PLL data rate divider standard mode 0 value: This field will directly control the xcvr_pll_clk_datart_div signal, which controls which divided clock is selected when generating the

13.4.10.2.53 Transceiver high speed clock select register (lane0_xcvr_diag_hsclock_sel - lane3_xcvr_diag_hsclock_sel)

13.4.10.2.53.1 Offset

Register	Offset
lane0_xcvr_diag_hsclock_sel	40E1h
lane1_xcvr_diag_hsclock_sel	44E1h
lane2_xcvr_diag_hsclock_sel	48E1h
lane3_xcvr_diag_hsclock_sel	4CE1h

13.4.10.2.53.2 Diagram



13.4.10.2.53.3 Fields

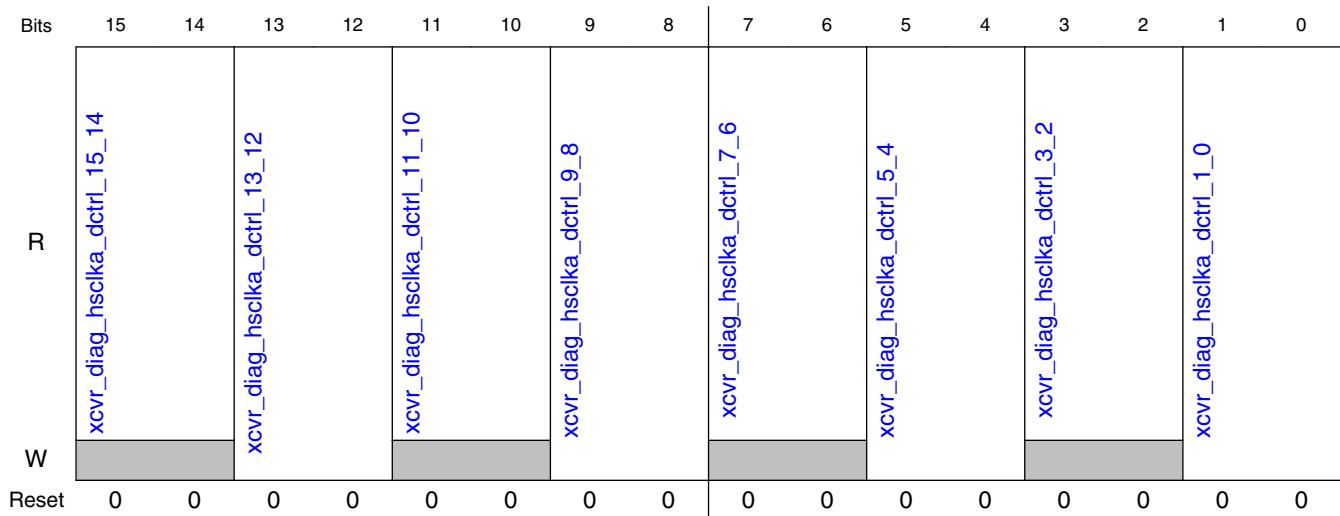
Field	Function
15-14 xcvr_diag_hscclk_sel_15_14	Reserved
13-12 xcvr_diag_hscclk_sel_13_12	High speed clock select standard mode 3: This specifies which analog high speed clock is selected when
11-10 xcvr_diag_hscclk_sel_11_10	Reserved
9-8 xcvr_diag_hscclk_sel_9_8	High speed clock select standard mode 2: This specifies which analog high speed clock is selected when
7-6 xcvr_diag_hscclk_sel_7_6	Reserved
5-4 xcvr_diag_hscclk_sel_5_4	High speed clock select standard mode 1: This specifies which analog high speed clock is selected when
3-2 xcvr_diag_hscclk_sel_3_2	Reserved
1-0 xcvr_diag_hscclk_sel_1_0	High speed clock select standard mode 0: This specifies which analog high speed clock is selected when

13.4.10.2.54 Transceiver high speed clock A divider control register (lane 0_xcvr_diag_hscika_dctrl - lane3_xcvr_diag_hscika_dctrl)

13.4.10.2.54.1 Offset

Register	Offset
lane0_xcvr_diag_hscika_dctrl	40E2h
lane1_xcvr_diag_hscika_dctrl	44E2h
lane2_xcvr_diag_hscika_dctrl	48E2h
lane3_xcvr_diag_hscika_dctrl	4CE2h

13.4.10.2.54.2 Diagram



13.4.10.2.54.3 Fields

Field	Function
15-14 xcvr_diag_hscika_dctrl_15_14	Reserved
13-12 xcvr_diag_hscika_dctrl_13_12	Transceiver clock A (transmitter) divider control standard mode 3: Controls the divider for clock A (which is the transmitter clock) in the analog transceiver clock distribution module, when

Table continues on the next page...

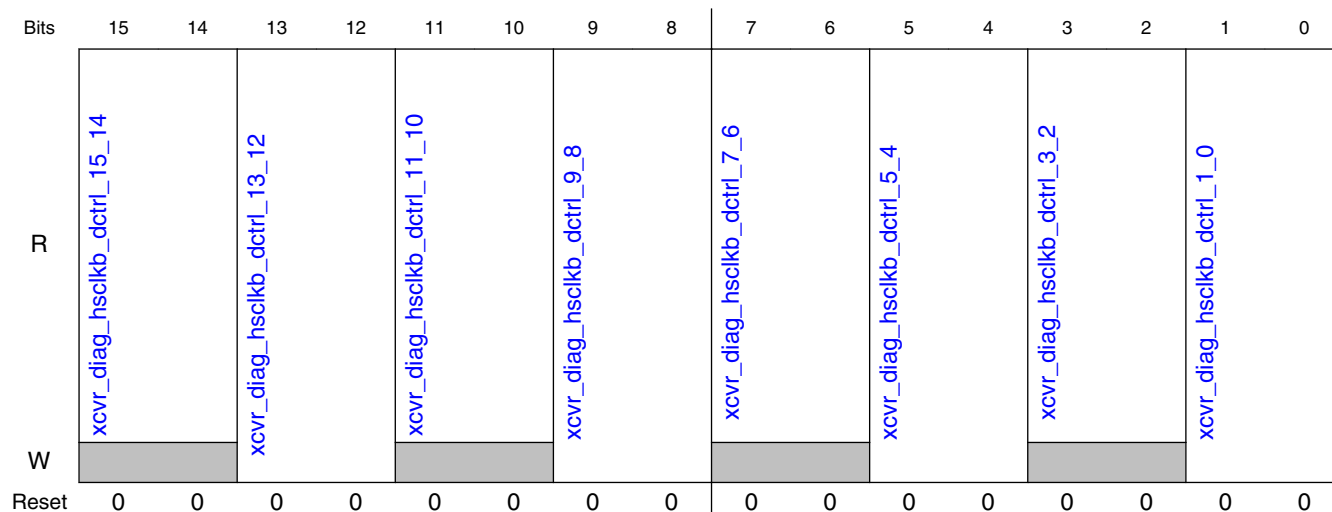
Field	Function
11-10 xcvr_diag_hscclk a_dctrl_11_10	Reserved
9-8 xcvr_diag_hscclk a_dctrl_9_8	Transceiver clock A (transmitter) divider control standard mode 2: Controls the divider for clock A (which is the transmitter clock) in the analog transceiver clock distribution module, when
7-6 xcvr_diag_hscclk a_dctrl_7_6	Reserved
5-4 xcvr_diag_hscclk a_dctrl_5_4	Transceiver clock A (transmitter) divider control standard mode 1: Controls the divider for clock A (which is the transmitter clock) in the analog transceiver clock distribution module, when
3-2 xcvr_diag_hscclk a_dctrl_3_2	Reserved
1-0 xcvr_diag_hscclk a_dctrl_1_0	Transceiver clock A (transmitter) divider control standard mode 0: Controls the divider for clock A (which is the transmitter clock) in the analog transceiver clock distribution module, when

13.4.10.2.55 Transceiver high speed clock B divider control register (lane 0_xcvr_diag_hscclkb_dctrl - lane3_xcvr_diag_hscclkb_dctrl)

13.4.10.2.55.1 Offset

Register	Offset
lane0_xcvr_diag_hscclkb_dctrl	40E3h
lane1_xcvr_diag_hscclkb_dctrl	44E3h
lane2_xcvr_diag_hscclkb_dctrl	48E3h
lane3_xcvr_diag_hscclkb_dctrl	4CE3h

13.4.10.2.55.2 Diagram



13.4.10.2.55.3 Fields

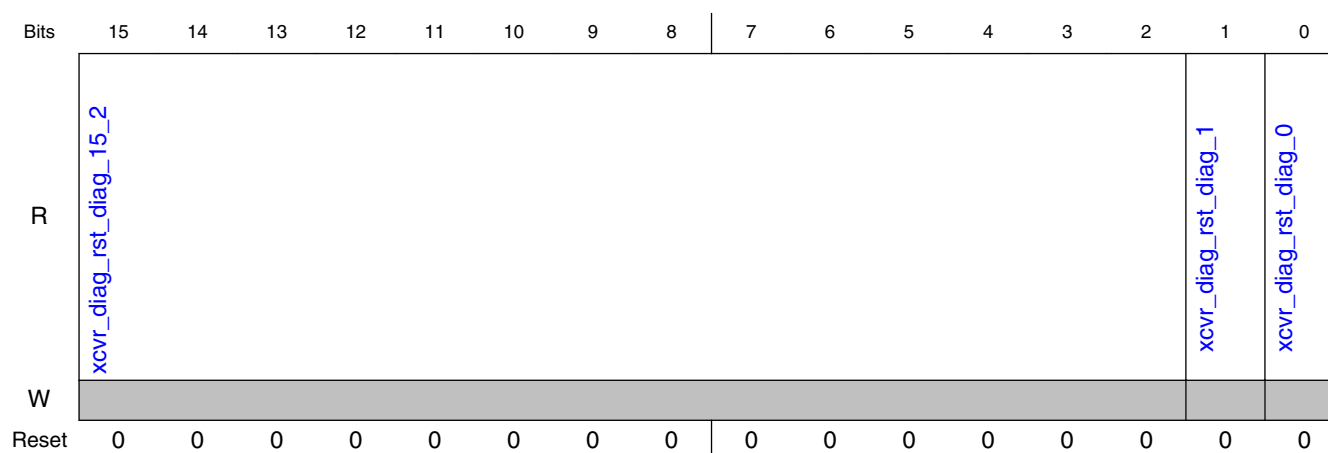
Field	Function
15-14 xcvr_diag_hscldb_dctrl_15_14	Reserved
13-12 xcvr_diag_hscldb_dctrl_13_12	Transceiver clock B (receiver) divider control standard mode 3: Controls the divider for clock B (which is the receiver clock) in the analog transceiver clock distribution module, when
11-10 xcvr_diag_hscldb_dctrl_11_10	Reserved
9-8 xcvr_diag_hscldb_dctrl_9_8	Transceiver clock B (receiver) divider control standard mode 2: Controls the divider for clock B (which is the receiver clock) in the analog transceiver clock distribution module, when
7-6 xcvr_diag_hscldb_dctrl_7_6	Reserved
5-4 xcvr_diag_hscldb_dctrl_5_4	Transceiver clock B (receiver) divider control standard mode 1: Controls the divider for clock B (which is the receiver clock) in the analog transceiver clock distribution module, when
3-2 xcvr_diag_hscldb_dctrl_3_2	Reserved
1-0 xcvr_diag_hscldb_dctrl_1_0	Transceiver clock B (receiver) divider control standard mode 0: Controls the divider for clock B (which is the receiver clock) in the analog transceiver clock distribution module, when

13.4.10.2.56 Transceiver control reset diagnostic register (lane0_xcvr_diag_rst_diag - lane3_xcvr_diag_rst_diag)

13.4.10.2.56.1 Offset

Register	Offset
lane0_xcvr_diag_rst_diag	40E7h
lane1_xcvr_diag_rst_diag	44E7h
lane2_xcvr_diag_rst_diag	48E7h
lane3_xcvr_diag_rst_diag	4CE7h

13.4.10.2.56.2 Diagram



13.4.10.2.56.3 Fields

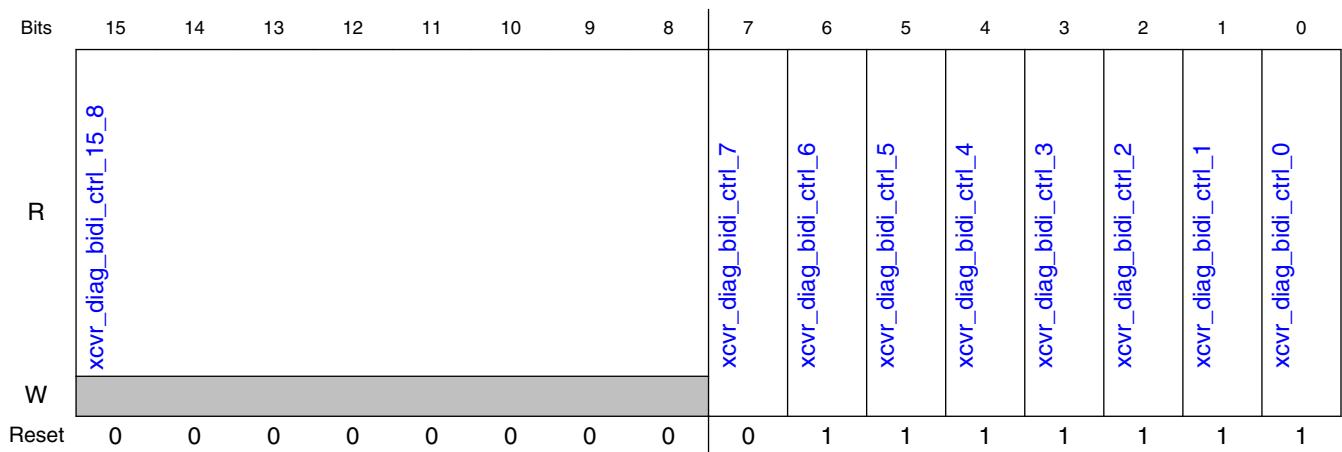
Field	Function
15-2 xcvr_diag_rst_diag_15_2	Reserved
1 xcvr_diag_rst_diag_1	Current state of the xcvr_psm_reset_n reset.
0 xcvr_diag_rst_diag_0	Current state of the xcvr_ref_clk_reset_n reset.

13.4.10.2.57 Transceiver bidirectional control register (lane0_xcvr_diag_bidi_ctrl - lane3_xcvr_diag_bidi_ctrl)

13.4.10.2.57.1 Offset

Register	Offset
lane0_xcvr_diag_bidi_ctrl	40E8h
lane1_xcvr_diag_bidi_ctrl	44E8h
lane2_xcvr_diag_bidi_ctrl	48E8h
lane3_xcvr_diag_bidi_ctrl	4CE8h

13.4.10.2.57.2 Diagram



13.4.10.2.57.3 Fields

Field	Function
15-8 xcvr_diag_bidi_ctrl_15_8	Reserved
7 xcvr_diag_bidi_ctrl_7	Receiver enable standard mode 3: This bit is a global enable for the receiver function, when
6 xcvr_diag_bidi_ctrl_6	Receiver enable standard mode 2: This bit is a global enable for the receiver function, when
5 xcvr_diag_bidi_ctrl_5	Receiver enable standard mode 1: This bit is a global enable for the receiver function, when

Table continues on the next page...

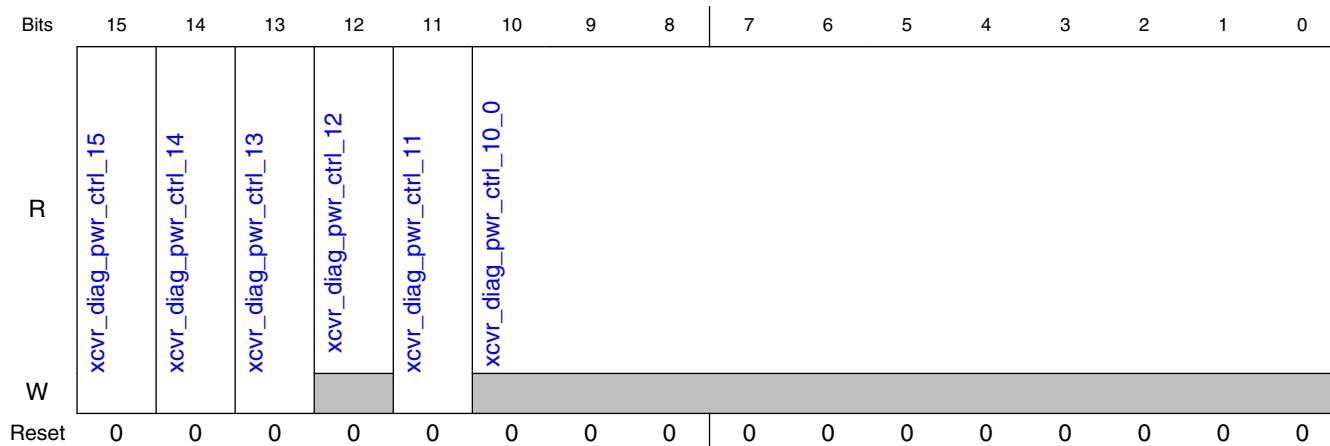
Field	Function
4 xcvr_diag_bidi_c trl_4	Receiver enable standard mode 0: This bit is a global enable for the receiver function, when
3 xcvr_diag_bidi_c trl_3	Transmitter enable standard mode 3: This bit is a global enable for the transmitter function, when
2 xcvr_diag_bidi_c trl_2	Transmitter enable standard mode 2: This bit is a global enable for the transmitter function, when
1 xcvr_diag_bidi_c trl_1	Transmitter enable standard mode 1: This bit is a global enable for the transmitter function, when
0 xcvr_diag_bidi_c trl_0	Transmitter enable standard mode 0: This bit is a global enable for the transmitter function, when

13.4.10.2.58 Transceiver power island control register (lane0_xcvr_diag_pwr_ctrl - lane3_xcvr_diag_pwr_ctrl)

13.4.10.2.58.1 Offset

Register	Offset
lane0_xcvr_diag_pwr_ctrl	40E9h
lane1_xcvr_diag_pwr_ctrl	44E9h
lane2_xcvr_diag_pwr_ctrl	48E9h
lane3_xcvr_diag_pwr_ctrl	4CE9h

13.4.10.2.58.2 Diagram



13.4.10.2.58.3 Fields

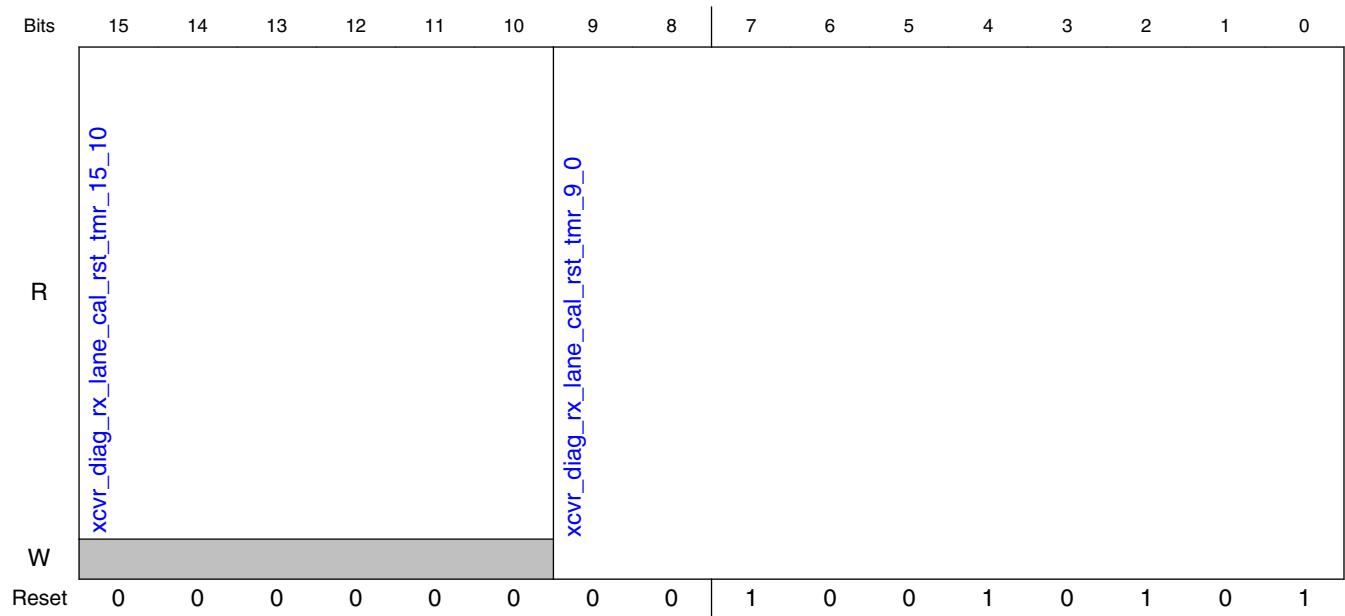
Field	Function
15 xcvr_diag_pwr_ctrl_15	Transceiver Dsync power down disable: Setting this bit to 1b1 will disable turning off the transceiver dsync power island when the macro is in a state that would normally switch the power island off.
14 xcvr_diag_pwr_ctrl_14	Transceiver calibration one time power down disable: Setting this bit to 1b1 will disable turning off the transceiver calibration one time power island when the macro is in a state that would normally switch the power island off.
13 xcvr_diag_pwr_ctrl_13	Transceiver calibration multiples time power down disable: Setting this bit to 1b1 will disable turning off the transceiver calibration multiple times power island when the macro is in a state that would normally switch the power island off.
12 xcvr_diag_pwr_ctrl_12	Reserved
11 xcvr_diag_pwr_ctrl_11	Transceiver test functions power enable: Controls the enabling of the transceiver test functions power island.
10-0 xcvr_diag_pwr_ctrl_10_0	Reserved

13.4.10.2.59 RX Lane calibration reset timer register (lane0_xcvr_diag_rx_lane_cal_rst_tmr - lane3_xcvr_diag_rx_lane_cal_rst_tmr)

13.4.10.2.59.1 Offset

Register	Offset
lane0_xcvr_diag_rx_lane_cal_rst_tmr	40EAh
lane1_xcvr_diag_rx_lane_cal_rst_tmr	44EAh
lane2_xcvr_diag_rx_lane_cal_rst_tmr	48EAh
lane3_xcvr_diag_rx_lane_cal_rst_tmr	4CEAh

13.4.10.2.59.2 Diagram



13.4.10.2.59.3 Fields

Field	Function
15-10 xcvr_diag_rx_lane_cal_rst_tmr_15_10	Reserved
9-0 xcvr_diag_rx_lane_cal_rst_tmr_9_0	Lane calibration receiver reset timer value: Value of the lane calibration receiver reset timer. This specifies the number of

13.4.10.2.60 Lane fast common mode enable timeout register (lane0_xcvr_diag_lane_fcm_en_to - lane3_xcvr_diag_lane_fcm_en_to)

13.4.10.2.60.1 Offset

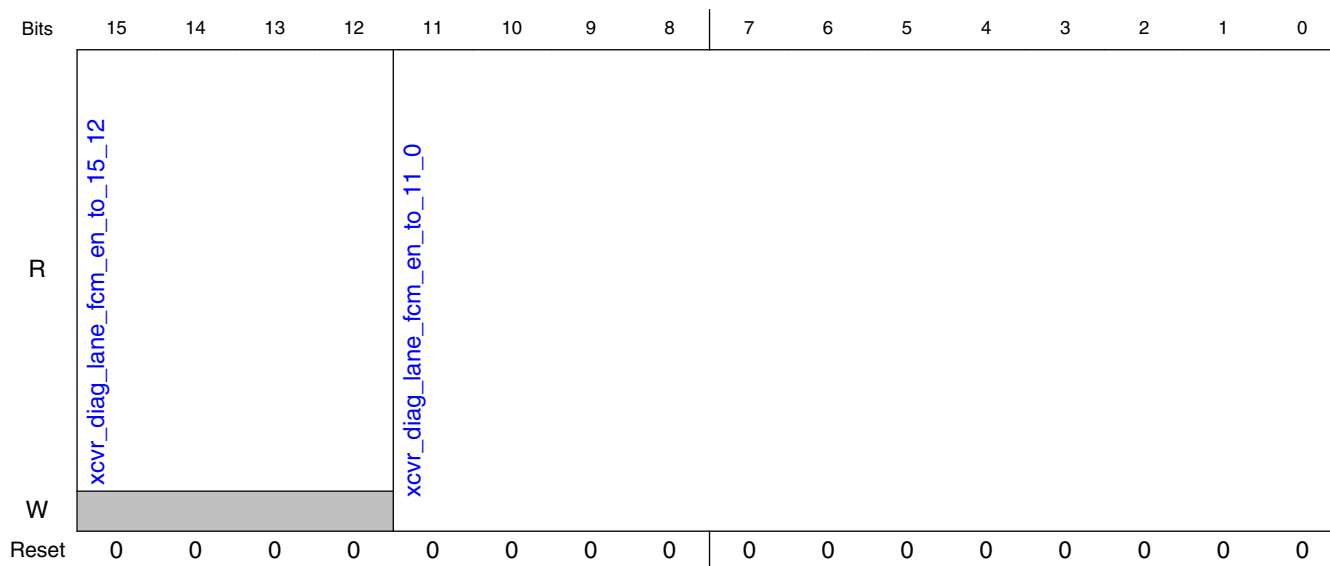
Register	Offset
lane0_xcvr_diag_lane_fcm_en_to	40F0h
lane1_xcvr_diag_lane_fcm_en_to	44F0h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane2_xcvr_diag_lane_fcm_en_to	48F0h
lane3_xcvr_diag_lane_fcm_en_to	4CF0h

13.4.10.2.60.2 Diagram



13.4.10.2.60.3 Fields

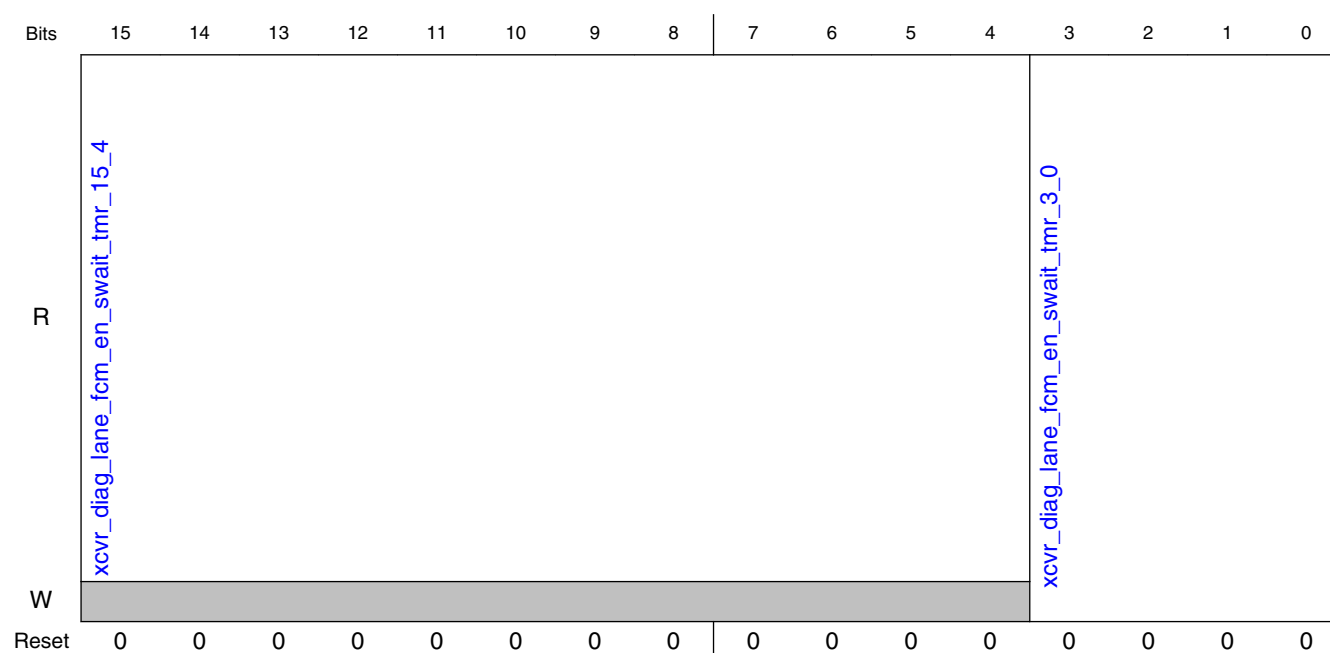
Field	Function
15-12 xcvr_diag_lane_fcm_en_to_15_12	Reserved
11-0 xcvr_diag_lane_fcm_en_to_11_0	Lane fast common mode enable timeout value: This specifies the maximum number of reference clock cycles the fast establishment of common mode process will wait for the

13.4.10.2.61 Lane fast common mode enable sample wait timer register (lane0_xcvr_diag_lane_fcm_en_swait_tmr - lane3_xcvr_diag_lane_fcm_en_swait_tmr)

13.4.10.2.61.1 Offset

Register	Offset
lane0_xcvr_diag_lane_fcm_en_swait_tmr	40F1h
lane1_xcvr_diag_lane_fcm_en_swait_tmr	44F1h
lane2_xcvr_diag_lane_fcm_en_swait_tmr	48F1h
lane3_xcvr_diag_lane_fcm_en_swait_tmr	4CF1h

13.4.10.2.61.2 Diagram



13.4.10.2.61.3 Fields

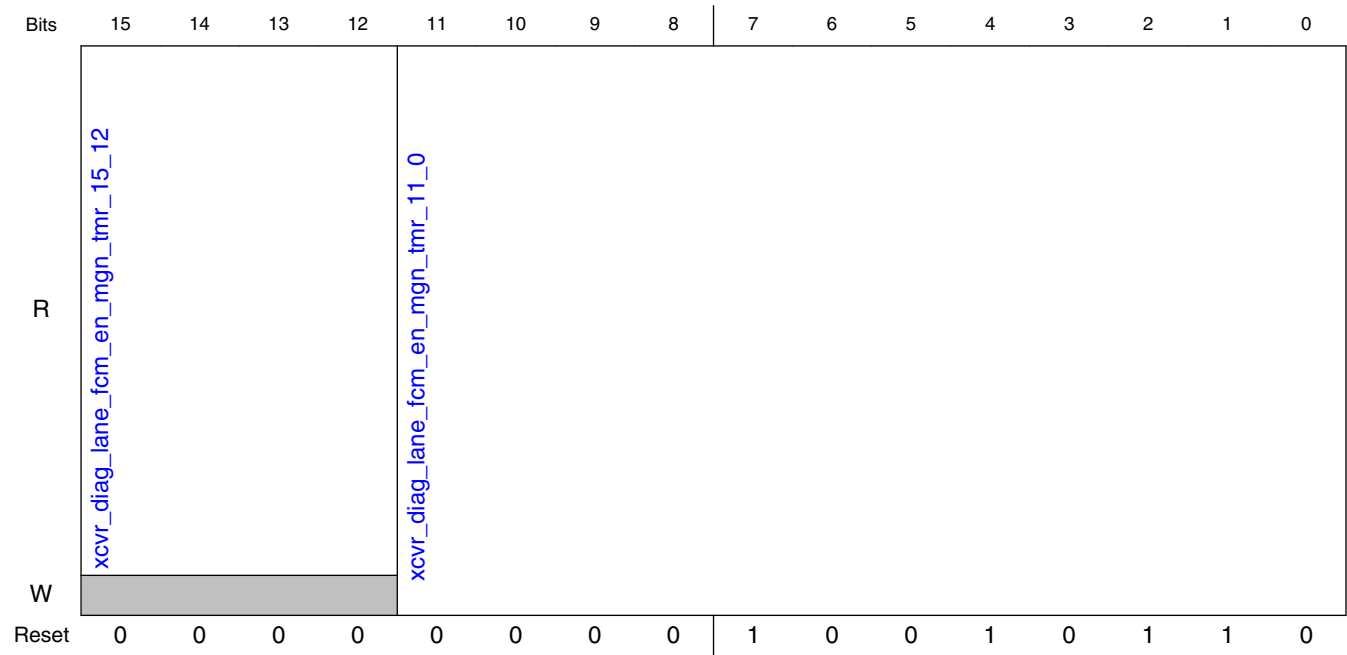
Field	Function
15-4 xcvr_diag_lane_fcm_en_swait_tmr_15_4	Reserved
3-0 xcvr_diag_lane_fcm_en_swait_tmr_3_0	Lane fast common mode enable sample wait timer value: This specifies the number of reference clock cycles the fast establishment of common mode process will wait between changing the state of the signals controlling the analog common mode sense circuits, and testing the results of the new state.

13.4.10.2.62 Lane fast common mode enable margin timer register (lane0_xcvr_diag_lane_fcm_en_mgn_tmr - lane3_xcvr_diag_lane_fcm_en_mgn_tmr)

13.4.10.2.62.1 Offset

Register	Offset
lane0_xcvr_diag_lane_fcm_en_mgn_tmr	40F2h
lane1_xcvr_diag_lane_fcm_en_mgn_tmr	44F2h
lane2_xcvr_diag_lane_fcm_en_mgn_tmr	48F2h
lane3_xcvr_diag_lane_fcm_en_mgn_tmr	4CF2h

13.4.10.2.62.2 Diagram



13.4.10.2.62.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

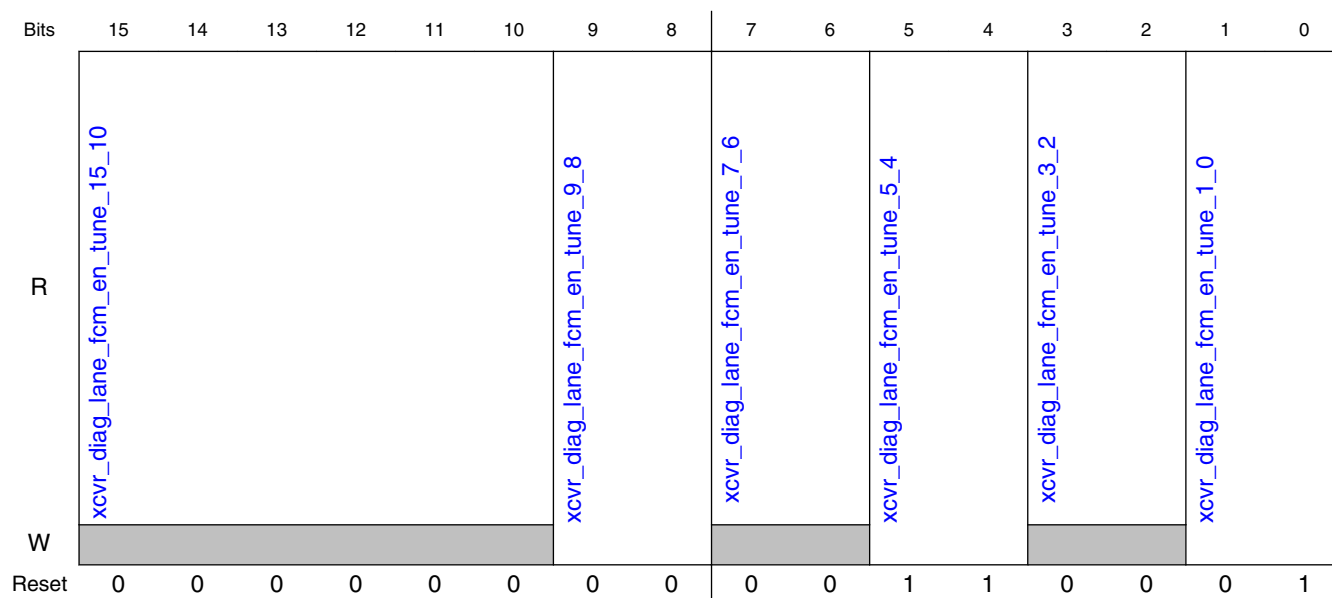
Field	Function
xcvr_diag_lane_fcm_en_mgn_tm_r_15_12	
11-0 xcvr_diag_lane_fcm_en_mgn_tm_r_11_0	Lane fast common mode enable margin timer value: This specifies the number of reference clock cycles the fast establishment of common mode process will enable all the margin segments for.

13.4.10.2.63 Lane fast common mode enable tuning register (lane0_xcvr_diag_lane_fcm_en_tune - lane3_xcvr_diag_lane_fcm_en_tune)

13.4.10.2.63.1 Offset

Register	Offset
lane0_xcvr_diag_lane_fcm_en_tune	40F3h
lane1_xcvr_diag_lane_fcm_en_tune	44F3h
lane2_xcvr_diag_lane_fcm_en_tune	48F3h
lane3_xcvr_diag_lane_fcm_en_tune	4CF3h

13.4.10.2.63.2 Diagram



13.4.10.2.63.3 Fields

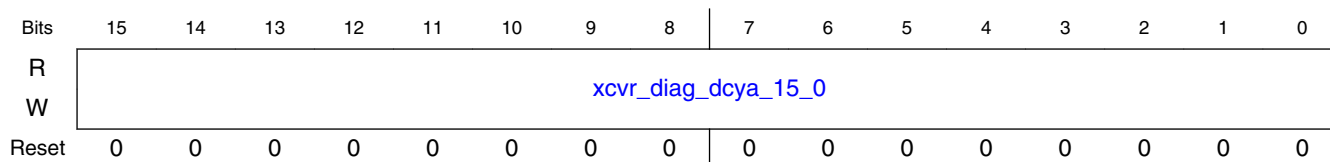
Field	Function
15-10 xcvr_diag_lane_ fcm_en_tune_15 _10	Reserved
9-8 xcvr_diag_lane_ fcm_en_tune_9 _8	Common mode sense reference DAC voltage initial test: This field sets the common mode detect reference voltage for the common mode detect comparator, when the fast establishment of common mode function is checking the initial state of the
7-6 xcvr_diag_lane_ fcm_en_tune_7 _6	Reserved
5-4 xcvr_diag_lane_ fcm_en_tune_5 _4	Common mode sense reference DAC voltage high test: This field sets the common mode detect reference voltage for the common mode detect comparator, when the fast establishment of common mode function is waiting for the
3-2 xcvr_diag_lane_ fcm_en_tune_3 _2	Reserved
1-0 xcvr_diag_lane_ fcm_en_tune_1 _0	Common mode sense reference DAC voltage low test: This field sets the common mode detect reference voltage for the common mode detect comparator, when the fast establishment of common mode function is waiting for the

13.4.10.2.64 Transceiver digital cover your alternatives register (lane0_xc vr_diag_dcya - lane3_xcvr_diag_dcya)

13.4.10.2.64.1 Offset

Register	Offset
lane0_xcvr_diag_dcya	40FFh
lane1_xcvr_diag_dcya	44FFh
lane2_xcvr_diag_dcya	48FFh
lane3_xcvr_diag_dcya	4CFFh

13.4.10.2.64.2 Diagram



13.4.10.2.64.3 Fields

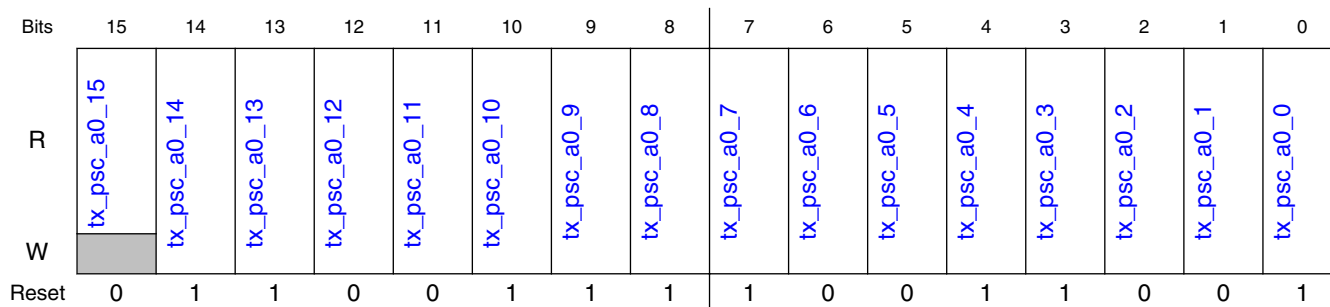
Field	Function
15-0 xcvr_diag_dcya_15_0	Reserved

13.4.10.2.65 Transmitter A0 power state definition register (lane0_tx_psc_a0 - lane3_tx_psc_a0)

13.4.10.2.65.1 Offset

Register	Offset
lane0_tx_psc_a0	4100h
lane1_tx_psc_a0	4500h
lane2_tx_psc_a0	4900h
lane3_tx_psc_a0	4D00h

13.4.10.2.65.2 Diagram



13.4.10.2.65.3 Fields

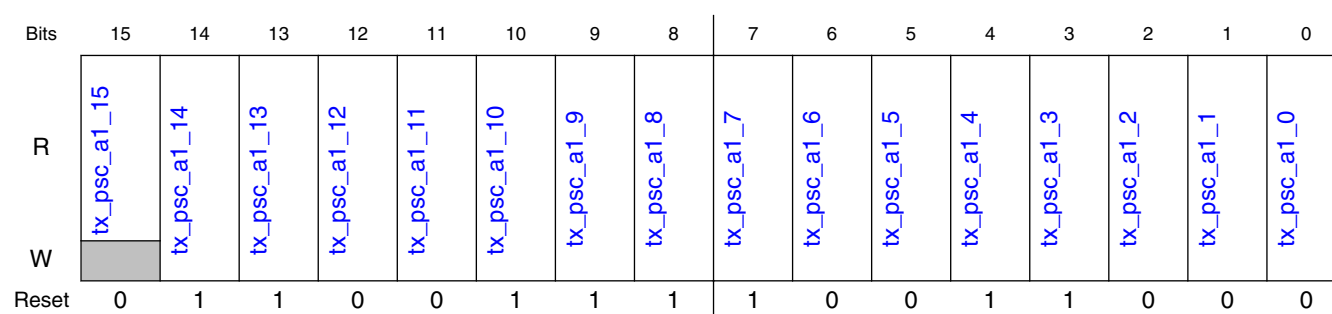
Field	Function
15 tx_psc_a0_15	Reserved
14 tx_psc_a0_14	TX DCAP enable: Enables the transmitter DCAP, via the
13 tx_psc_a0_13	TX UPHY supply enable: Enables the transmitter UPHY components supply, via the
12 tx_psc_a0_12	LFPS controller enable: Enables the LFPS controller function.
11 tx_psc_a0_11	MPHY pre-driver enable: Enables the MPHY pre-driver components.
10 tx_psc_a0_10	TX serializer clock enable: Enables the serializer clock divider circuits.
9 tx_psc_a0_9	TX serializer enable: Enables the serializer circuits.
8 tx_psc_a0_8	TX clock enable: Enables the transmitter clock and MUX circuits in the transceiver clock distribution, via the
7 tx_psc_a0_7	Transmitter low current mode: Enables a low current consumption mode within the common mode voltage circuit in the driver, via the
6 tx_psc_a0_6	Transmitter mission mode enable: Enables the analog circuits in the driver required to run in mission mode, via the
5 tx_psc_a0_5	TX driver common mode enable extend control: Specifies which power states the
4 tx_psc_a0_4	TX driver common mode enable: Enables the common mode voltage circuits in the driver.
3 tx_psc_a0_3	TX post-emphasis enable: Enables the transmitter circuits related to the post-emphasis function.
2 tx_psc_a0_2	TX pre-emphasis enable: Enables the transmitter circuits related to the pre-emphasis function.
1 tx_psc_a0_1	TX driver LDO enable: Enables the transmitter circuits related to the driver LDO function, but driving the
0 tx_psc_a0_0	TX driver enable: Enables the transmitter driver, via the H bridge driver controller. Note that this also controls whether the

13.4.10.2.66 Transmitter A1 power state definition register (lane0_tx_psc_a1 - lane3_tx_psc_a1)

13.4.10.2.66.1 Offset

Register	Offset
lane0_tx_psc_a1	4101h
lane1_tx_psc_a1	4501h
lane2_tx_psc_a1	4901h
lane3_tx_psc_a1	4D01h

13.4.10.2.66.2 Diagram



13.4.10.2.66.3 Fields

Field	Function
15 tx_psc_a1_15	Reserved
14 tx_psc_a1_14	TX DCAP enable
13 tx_psc_a1_13	TX UPHY supply enable
12 tx_psc_a1_12	LFPS controller enable
11 tx_psc_a1_11	MPHY pre-driver enable
10 tx_psc_a1_10	TX serializer clock enable
9 tx_psc_a1_9	TX serializer enable
8 tx_psc_a1_8	TX clock enable
7	Transmitter low current mode

Table continues on the next page...

Clocks And Resets

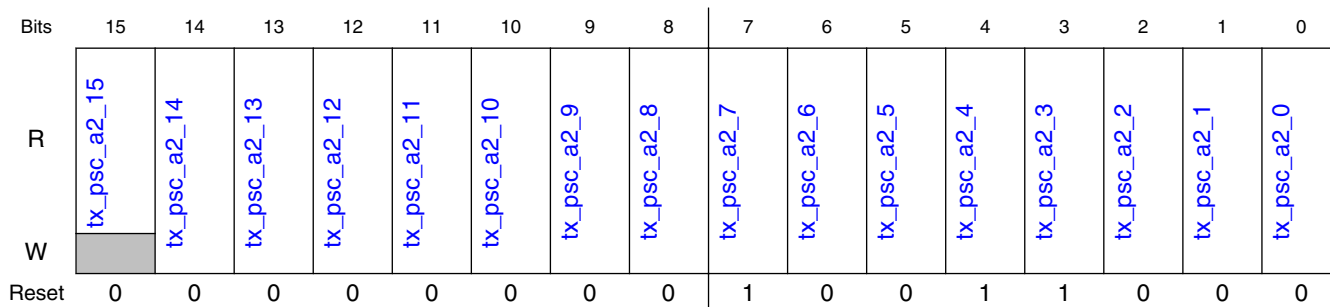
Field	Function
tx_psc_a1_7	
6 tx_psc_a1_6	Transmitter mission mode enable
5 tx_psc_a1_5	TX driver common mode enable extend control
4 tx_psc_a1_4	TX driver common mode enable
3 tx_psc_a1_3	TX post-emphasis enable
2 tx_psc_a1_2	TX pre-emphasis enable
1 tx_psc_a1_1	TX driver LDO enable
0 tx_psc_a1_0	TX driver enable

13.4.10.2.67 Transmitter A2 power state definition register (lane0_tx_psc_a2 - lane3_tx_psc_a2)

13.4.10.2.67.1 Offset

Register	Offset
lane0_tx_psc_a2	4102h
lane1_tx_psc_a2	4502h
lane2_tx_psc_a2	4902h
lane3_tx_psc_a2	4D02h

13.4.10.2.67.2 Diagram



13.4.10.2.67.3 Fields

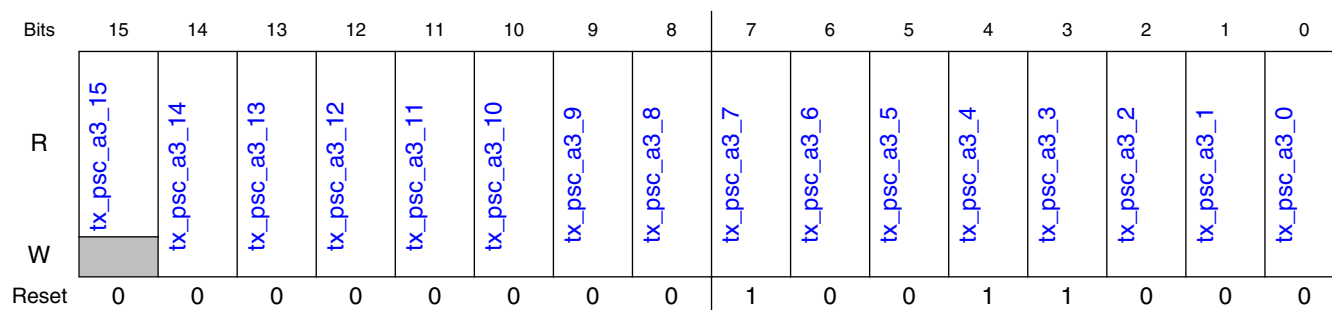
Field	Function
15 tx_psc_a2_15	Reserved
14 tx_psc_a2_14	TX DCAP enable
13 tx_psc_a2_13	TX UPHY supply enable
12 tx_psc_a2_12	LFPS controller enable
11 tx_psc_a2_11	MPHY pre-driver enable
10 tx_psc_a2_10	TX serializer clock enable
9 tx_psc_a2_9	TX serializer enable
8 tx_psc_a2_8	TX clock enable
7 tx_psc_a2_7	Transmitter low current mode
6 tx_psc_a2_6	Transmitter mission mode enable
5 tx_psc_a2_5	TX driver common mode enable extend control
4 tx_psc_a2_4	TX driver common mode enable
3 tx_psc_a2_3	TX post-emphasis enable
2 tx_psc_a2_2	TX pre-emphasis enable
1 tx_psc_a2_1	TX driver LDO enable
0 tx_psc_a2_0	TX driver enable

13.4.10.2.68 Transmitter A3 power state definition register (lane0_tx_psc_a3 - lane3_tx_psc_a3)

13.4.10.2.68.1 Offset

Register	Offset
lane0_tx_psc_a3	4103h
lane1_tx_psc_a3	4503h
lane2_tx_psc_a3	4903h
lane3_tx_psc_a3	4D03h

13.4.10.2.68.2 Diagram



13.4.10.2.68.3 Fields

Field	Function
15 tx_psc_a3_15	Reserved
14 tx_psc_a3_14	TX DCAP enable
13 tx_psc_a3_13	TX UPHY supply enable
12 tx_psc_a3_12	LFPS controller enable
11 tx_psc_a3_11	MPHY pre-driver enable
10 tx_psc_a3_10	TX serializer clock enable
9 tx_psc_a3_9	TX serializer enable
8 tx_psc_a3_8	TX clock enable
7	Transmitter low current mode

Table continues on the next page...

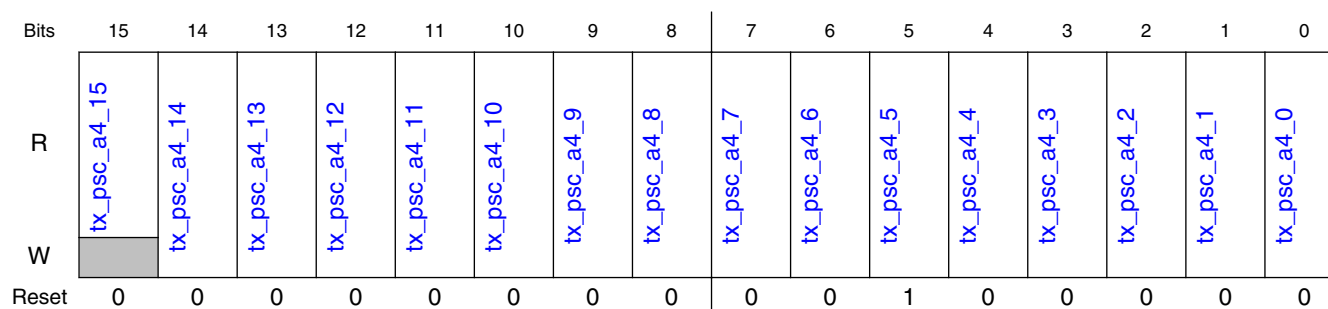
Field	Function
tx_psc_a3_7	
6 tx_psc_a3_6	Transmitter mission mode enable
5 tx_psc_a3_5	TX driver common mode enable extend control
4 tx_psc_a3_4	TX driver common mode enable
3 tx_psc_a3_3	TX post-emphasis enable
2 tx_psc_a3_2	TX pre-emphasis enable
1 tx_psc_a3_1	TX driver LDO enable
0 tx_psc_a3_0	TX driver enable

13.4.10.2.69 Transmitter A4 power state definition register (lane0_tx_psc_a4 - lane3_tx_psc_a4)

13.4.10.2.69.1 Offset

Register	Offset
lane0_tx_psc_a4	4104h
lane1_tx_psc_a4	4504h
lane2_tx_psc_a4	4904h
lane3_tx_psc_a4	4D04h

13.4.10.2.69.2 Diagram



13.4.10.2.69.3 Fields

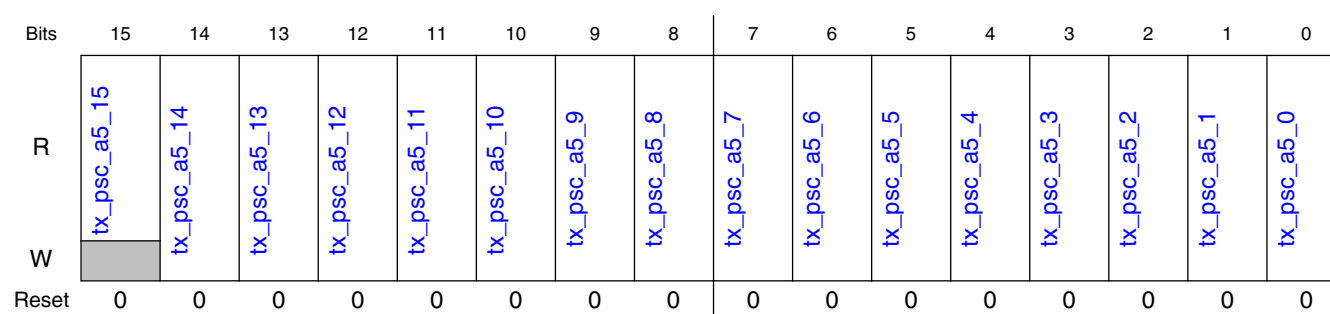
Field	Function
15 tx_psc_a4_15	Reserved
14 tx_psc_a4_14	TX DCAP enable
13 tx_psc_a4_13	TX UPHY supply enable
12 tx_psc_a4_12	LFPS controller enable
11 tx_psc_a4_11	MPHY pre-driver enable
10 tx_psc_a4_10	TX serializer clock enable
9 tx_psc_a4_9	TX serializer enable
8 tx_psc_a4_8	TX clock enable
7 tx_psc_a4_7	Transmitter low current mode
6 tx_psc_a4_6	Transmitter mission mode enable
5 tx_psc_a4_5	TX driver common mode enable extend control
4 tx_psc_a4_4	TX driver common mode enable
3 tx_psc_a4_3	TX post-emphasis enable
2 tx_psc_a4_2	TX pre-emphasis enable
1 tx_psc_a4_1	TX driver LDO enable
0 tx_psc_a4_0	TX driver enable

13.4.10.2.70 Transmitter A5 power state definition register (lane0_tx_psc_a5 - lane3_tx_psc_a5)

13.4.10.2.70.1 Offset

Register	Offset
lane0_tx_psc_a5	4105h
lane1_tx_psc_a5	4505h
lane2_tx_psc_a5	4905h
lane3_tx_psc_a5	4D05h

13.4.10.2.70.2 Diagram



13.4.10.2.70.3 Fields

Field	Function
15 tx_psc_a5_15	Reserved
14 tx_psc_a5_14	TX DCAP enable
13 tx_psc_a5_13	TX UPHY supply enable
12 tx_psc_a5_12	LFPS controller enable
11 tx_psc_a5_11	MPHY pre-driver enable
10 tx_psc_a5_10	TX serializer clock enable
9 tx_psc_a5_9	TX serializer enable
8 tx_psc_a5_8	TX clock enable
7	Transmitter low current mode

Table continues on the next page...

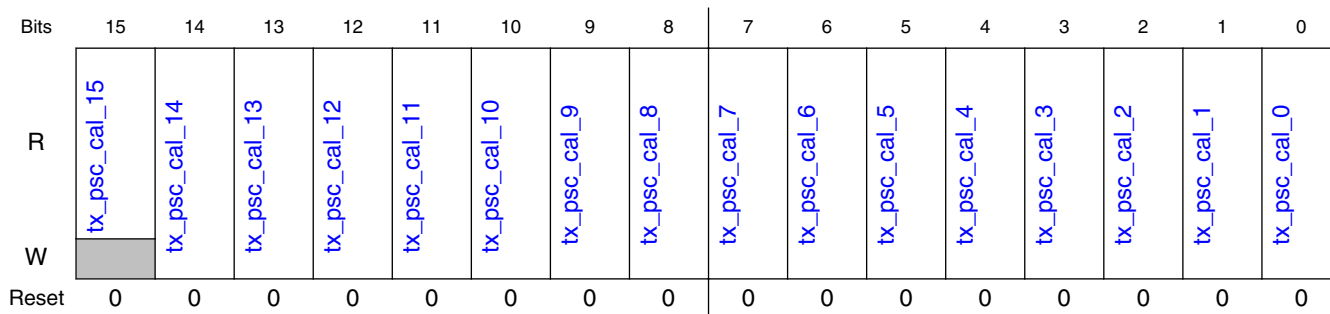
Field	Function
tx_psc_a5_7	
6 tx_psc_a5_6	Transmitter mission mode enable
5 tx_psc_a5_5	TX driver common mode enable extend control
4 tx_psc_a5_4	TX driver common mode enable
3 tx_psc_a5_3	TX post-emphasis enable
2 tx_psc_a5_2	TX pre-emphasis enable
1 tx_psc_a5_1	TX driver LDO enable
0 tx_psc_a5_0	TX driver enable

13.4.10.2.71 Transmitter calibration power state definition register (lane0_tx_psc_cal - lane3_tx_psc_cal)

13.4.10.2.71.1 Offset

Register	Offset
lane0_tx_psc_cal	4106h
lane1_tx_psc_cal	4506h
lane2_tx_psc_cal	4906h
lane3_tx_psc_cal	4D06h

13.4.10.2.71.2 Diagram



13.4.10.2.71.3 Fields

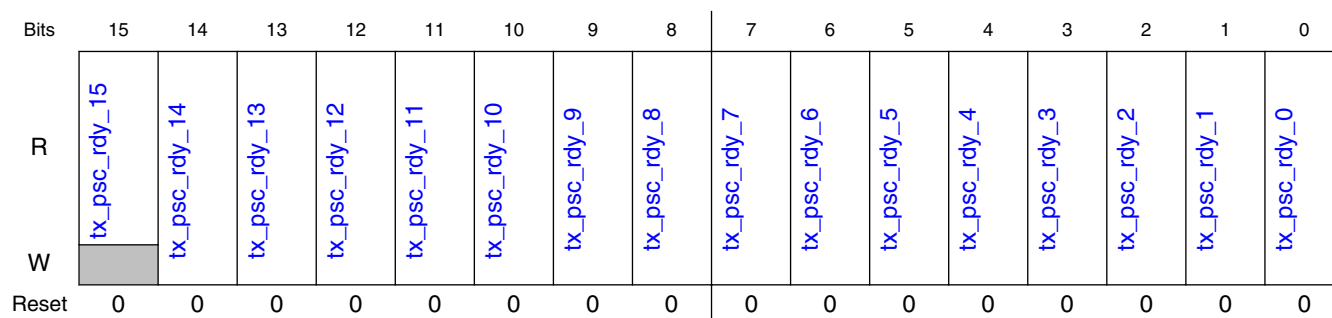
Field	Function
15 tx_psc_cal_15	Reserved
14 tx_psc_cal_14	TX DCAP enable
13 tx_psc_cal_13	TX UPHY supply enable
12 tx_psc_cal_12	LFPS controller enable
11 tx_psc_cal_11	MPHY pre-driver enable
10 tx_psc_cal_10	TX serializer clock enable
9 tx_psc_cal_9	TX serializer enable
8 tx_psc_cal_8	TX clock enable
7 tx_psc_cal_7	Transmitter low current mode
6 tx_psc_cal_6	Transmitter mission mode enable
5 tx_psc_cal_5	TX driver common mode enable extend control
4 tx_psc_cal_4	TX driver common mode enable
3 tx_psc_cal_3	TX post-emphasis enable
2 tx_psc_cal_2	TX pre-emphasis enable
1 tx_psc_cal_1	TX driver LDO enable
0 tx_psc_cal_0	TX driver enable

13.4.10.2.72 Transmitter ready power state definition register (lane0_tx_psc_rdy - lane3_tx_psc_rdy)

13.4.10.2.72.1 Offset

Register	Offset
lane0_tx_psc_rdy	4107h
lane1_tx_psc_rdy	4507h
lane2_tx_psc_rdy	4907h
lane3_tx_psc_rdy	4D07h

13.4.10.2.72.2 Diagram



13.4.10.2.72.3 Fields

Field	Function
15 tx_psc_rdy_15	Reserved
14 tx_psc_rdy_14	TX DCAP enable
13 tx_psc_rdy_13	TX UPHY supply enable
12 tx_psc_rdy_12	LFPS controller enable
11 tx_psc_rdy_11	MPHY pre-driver enable
10 tx_psc_rdy_10	TX serializer clock enable
9 tx_psc_rdy_9	TX serializer enable
8 tx_psc_rdy_8	TX clock enable
7	Transmitter low current mode

Table continues on the next page...

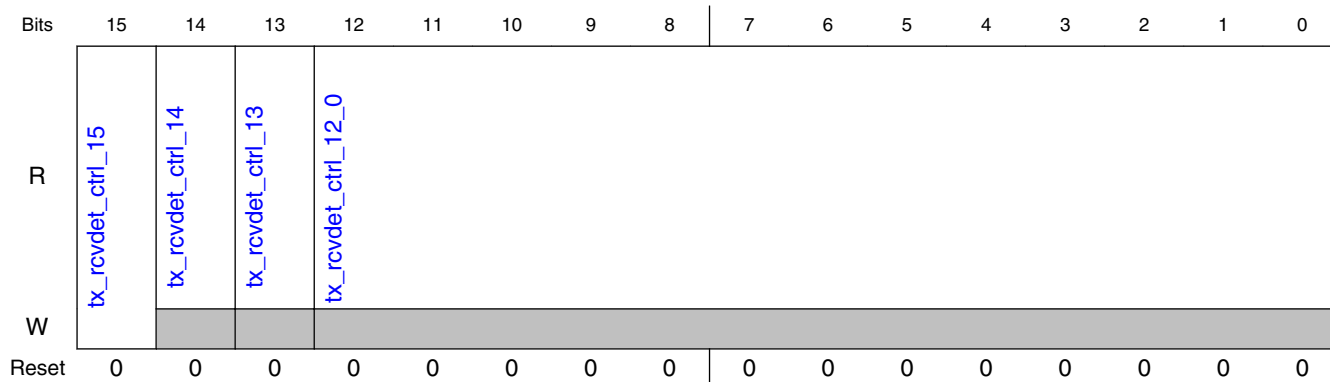
Field	Function
tx_psc_rdy_7	
6 tx_psc_rdy_6	Transmitter mission mode enable
5 tx_psc_rdy_5	TX driver common mode enable extend control
4 tx_psc_rdy_4	TX driver common mode enable
3 tx_psc_rdy_3	TX post-emphasis enable
2 tx_psc_rdy_2	TX pre-emphasis enable
1 tx_psc_rdy_1	TX driver LDO enable
0 tx_psc_rdy_0	TX driver enable

13.4.10.2.73 Transmit receiver detect control register (lane0_tx_rcvdet_ctrl - lane3_tx_rcvdet_ctrl)

13.4.10.2.73.1 Offset

Register	Offset
lane0_tx_rcvdet_ctrl	4120h
lane1_tx_rcvdet_ctrl	4520h
lane2_tx_rcvdet_ctrl	4920h
lane3_tx_rcvdet_ctrl	4D20h

13.4.10.2.73.2 Diagram



13.4.10.2.73.3 Fields

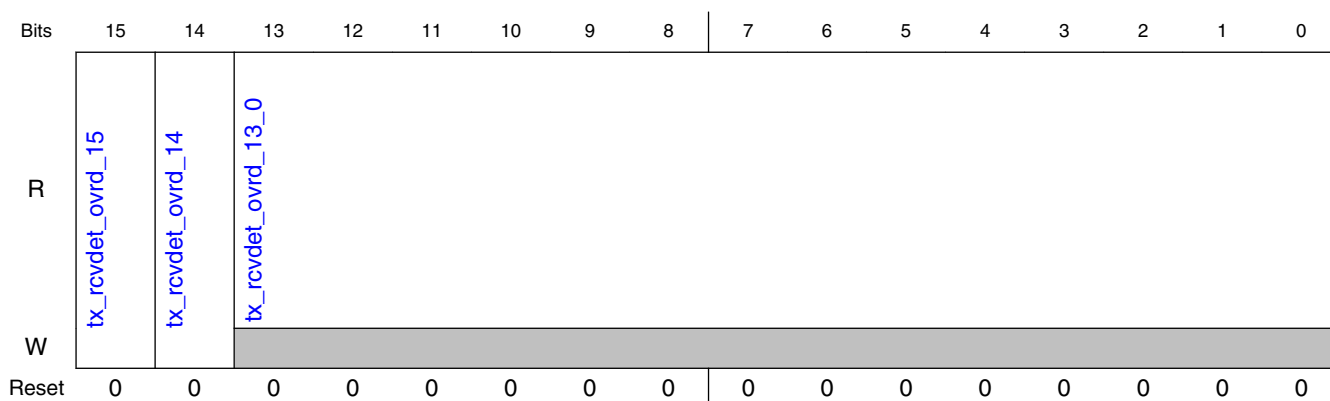
Field	Function
15 tx_rcvdet_ctrl_1 5	Start receiver detect: Activating (1b1) this bit will start the receiver detect process. This bit must remain active until the receiver detect process is complete, as indicated by the receiver detect process done bit in this register. To start another receiver detect process, this register must first be set inactive (1b0) until the receiver detect process done bit in this register is cleared.
14 tx_rcvdet_ctrl_1 4	Receiver detect process done: This bit will be set to 1b1 when the receiver detect process is complete. It will be cleared by
13 tx_rcvdet_ctrl_1 3	Receiver detected: When the receiver detect process is complete, this register bit will indicate the current state of the
12-0 tx_rcvdet_ctrl_1 2_0	Reserved

13.4.10.2.74 Transmit receiver detect override register (lane0_tx_rcvdet_ovrd - lane3_tx_rcvdet_ovrd)

13.4.10.2.74.1 Offset

Register	Offset
lane0_tx_rcvdet_ovrd	4121h
lane1_tx_rcvdet_ovrd	4521h
lane2_tx_rcvdet_ovrd	4921h
lane3_tx_rcvdet_ovrd	4D21h

13.4.10.2.74.2 Diagram



13.4.10.2.74.3 Fields

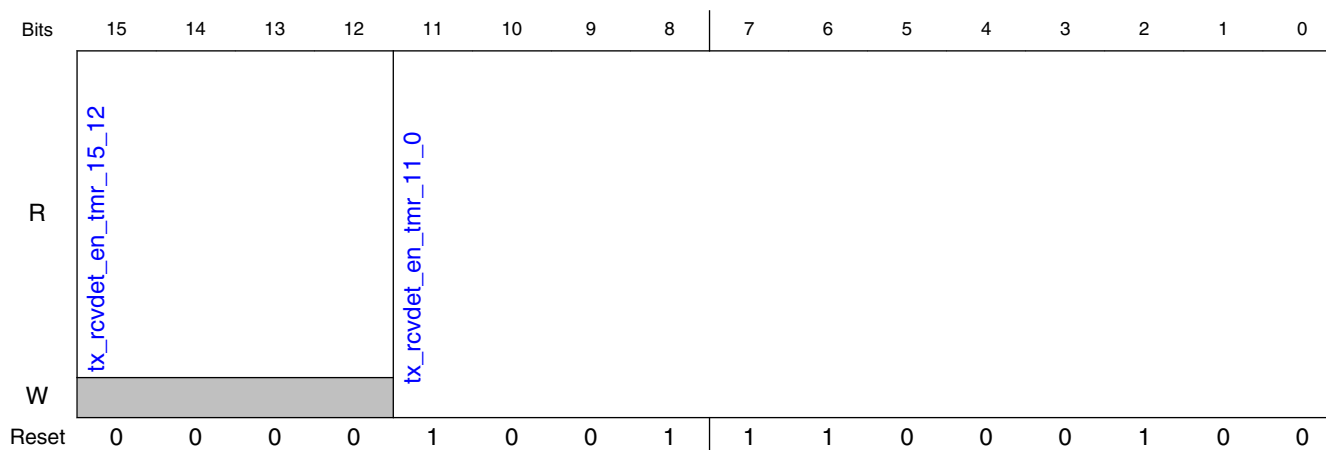
Field	Function
15 tx_rcvdet_ovrd_ 15	Receiver detect override enable: Activation (1b1) of this register bit enables the
14 tx_rcvdet_ovrd_ 14	Receiver detect override: When the receiver detect override enable bit in this register is active (1b1), this bit will directly control the
13-0 tx_rcvdet_ovrd_ 13_0	Reserved

13.4.10.2.75 Transmit receiver detect enable timer register (lane0_tx_rcv det_en_tmr - lane3_tx_rcvdet_en_tmr)

13.4.10.2.75.1 Offset

Register	Offset
lane0_tx_rcvdet_en_tmr	4122h
lane1_tx_rcvdet_en_tmr	4522h
lane2_tx_rcvdet_en_tmr	4922h
lane3_tx_rcvdet_en_tmr	4D22h

13.4.10.2.75.2 Diagram



13.4.10.2.75.3 Fields

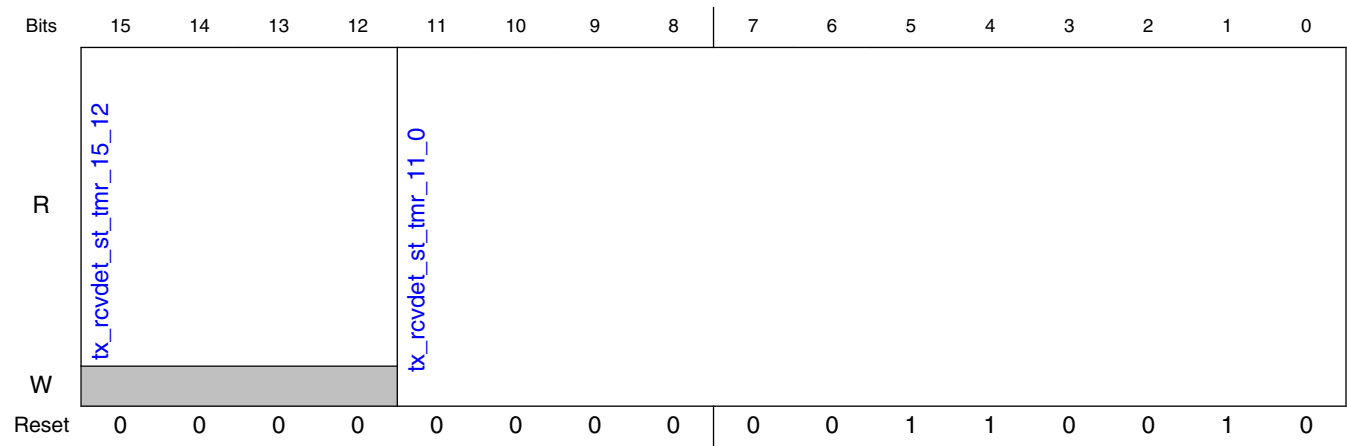
Field	Function
15-12 tx_rcvdet_en_tm r_15_12	Reserved
11-0 tx_rcvdet_en_tm r_11_0	Enable wait time value: This is the number of clocks the receiver detect state machine waits between driving the

13.4.10.2.76 Transmit receiver detect start timer register (lane0_tx_rcvdet_st_tmr - lane3_tx_rcvdet_st_tmr)

13.4.10.2.76.1 Offset

Register	Offset
lane0_tx_rcvdet_st_tmr	4123h
lane1_tx_rcvdet_st_tmr	4523h
lane2_tx_rcvdet_st_tmr	4923h
lane3_tx_rcvdet_st_tmr	4D23h

13.4.10.2.76.2 Diagram



13.4.10.2.76.3 Fields

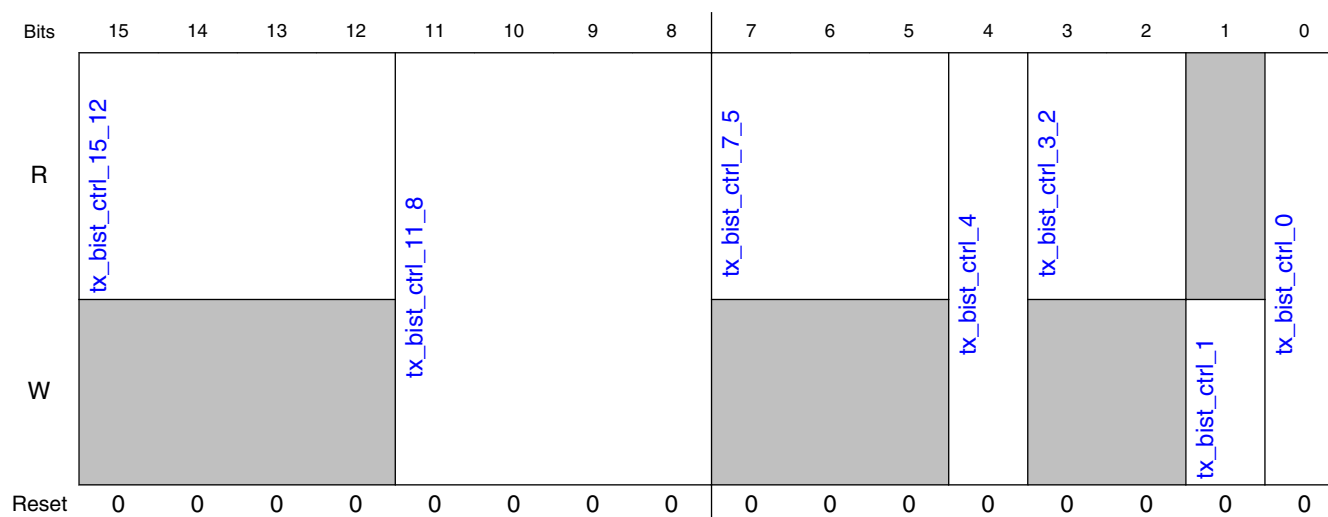
Field	Function
15-12 tx_rcvdet_st_tmr_15_12	Reserved
11-0 tx_rcvdet_st_tmr_11_0	Start wait time value: This is the number of clocks the receiver detect state machine waits between driving the

13.4.10.2.77 Transmit BIST control register (lane0_tx_bist_ctrl - lane3_tx_bist_ctrl)

13.4.10.2.77.1 Offset

Register	Offset
lane0_tx_bist_ctrl	4140h
lane1_tx_bist_ctrl	4540h
lane2_tx_bist_ctrl	4940h
lane3_tx_bist_ctrl	4D40h

13.4.10.2.77.2 Diagram



13.4.10.2.77.3 Fields

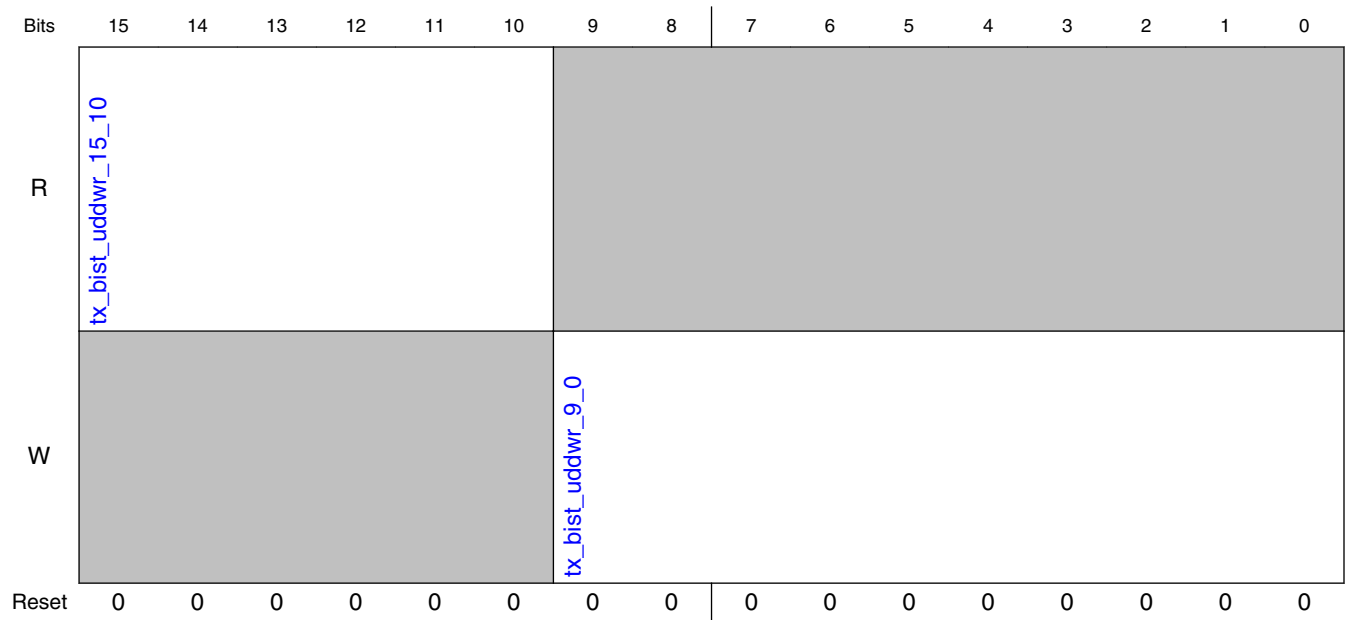
Field	Function
15-12 tx_bist_ctrl_15_12	Reserved
11-8 tx_bist_ctrl_11_8	Transmitter BIST mode: Controls which mode the BIST will operate in. The value of this field must match the corresponding field for the receive BIST controller. The following are the values used for this field, and what BIST mode they correspond to.
7-5 tx_bist_ctrl_7_5	Reserved
4 tx_bist_ctrl_4	Transmitter BIST force error: When this bit transitions from 1b0 to 1b1, the transmit BIST controller will force an error to be transmitted from the BIST logic, by inverting one of the parallel data bits.
3-2 tx_bist_ctrl_3_2	Reserved
1 tx_bist_ctrl_1	Transmitter BIST user defined data FIFO clear: Writing a 1b1 to this bit will clear the transmitter BIST user defined data FIFO.
0 tx_bist_ctrl_0	Transmitter BIST enable: This bit enables the transmitter BIST function.

13.4.10.2.78 Transmit BIST user defined data write register (lane0_tx_bist_uddwr - lane3_tx_bist_uddwr)

13.4.10.2.78.1 Offset

Register	Offset
lane0_tx_bist_uddwr	4141h
lane1_tx_bist_uddwr	4541h
lane2_tx_bist_uddwr	4941h
lane3_tx_bist_uddwr	4D41h

13.4.10.2.78.2 Diagram



13.4.10.2.78.3 Fields

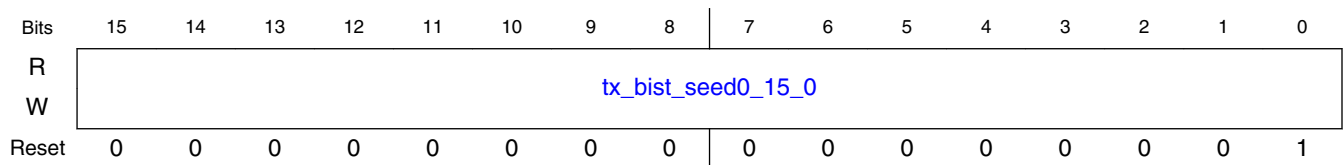
Field	Function
15-10 tx_bist_uddwr_15_10	Reserved
9-0 tx_bist_uddwr_9_0	Transmitter BIST user defined data: Writing a data word to this field will result in that data word being placed in the next available position in the transmitter BIST user defined data FIFO. Note, when in 20 bit mode, all 10 of these bits are used. When in 16 bit mode, only the least significant 8 bits are used. Note that the FIFO in this implementation is 18 words deep. It is up to the user to not write more than 18 words of data into this FIFO. Exceeding this amount of data will generate unpredictable results.

13.4.10.2.79 Transmit BIST PRBS seed 0 register (lane0_tx_bist_seed0 - lane3_tx_bist_seed0)

13.4.10.2.79.1 Offset

Register	Offset
lane0_tx_bist_seed0	4142h
lane1_tx_bist_seed0	4542h
lane2_tx_bist_seed0	4942h
lane3_tx_bist_seed0	4D42h

13.4.10.2.79.2 Diagram



13.4.10.2.79.3 Fields

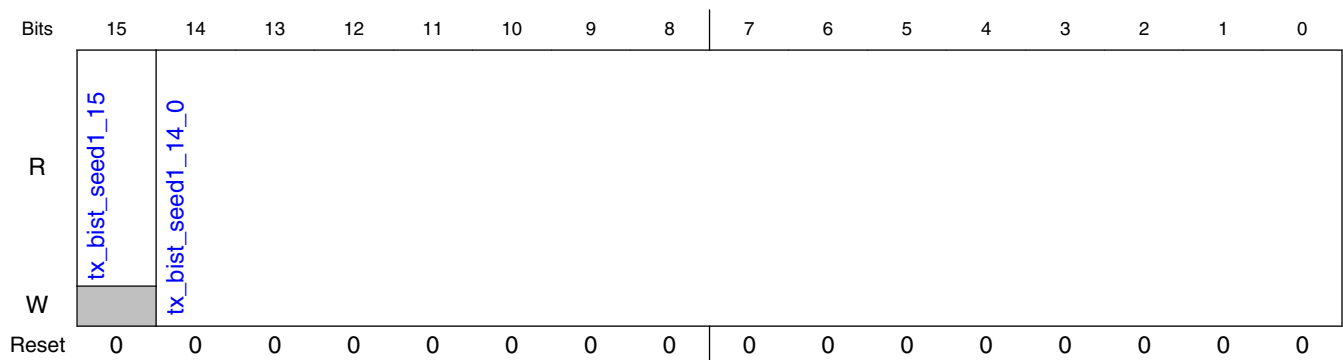
Field	Function
tx_bist_seed0_15_0	Transmitter BIST PRBS seed (15:0): When the BIST is in PRBS mode, this field provides a seed for the PRBS, such that different lanes can have different BIST patterns. Note that this field contains the least significant 16 bits of the full 31 bit seed value.

13.4.10.2.80 Transmit BIST PRBS seed 1 register (lane0_tx_bist_seed1 - lane3_tx_bist_seed1)

13.4.10.2.80.1 Offset

Register	Offset
lane0_tx_bist_seed1	4143h
lane1_tx_bist_seed1	4543h
lane2_tx_bist_seed1	4943h
lane3_tx_bist_seed1	4D43h

13.4.10.2.80.2 Diagram



13.4.10.2.80.3 Fields

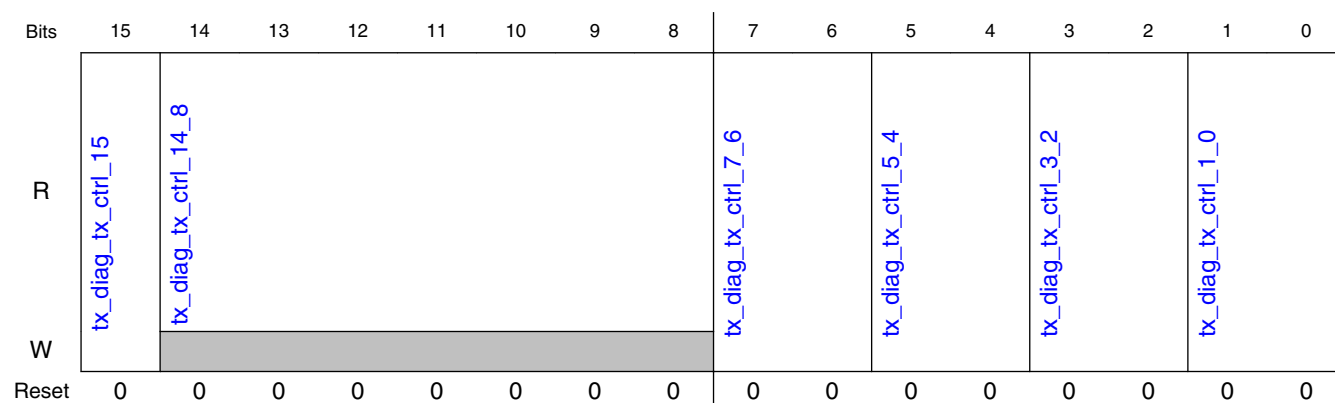
Field	Function
15 tx_bist_seed1_15	Reserved
14-0 tx_bist_seed1_14_0	Transmitter BIST PRBS seed (30:16): When the BIST is in PRBS mode, this field provides a seed for the PRBS, such that different lanes can have different BIST patterns. Note that this field contains the most significant 15 bits of the full 31 bit seed value.

13.4.10.2.81 TX control register (lane0_tx_diag_tx_ctrl - lane3_tx_diag_tx_ctrl)

13.4.10.2.81.1 Offset

Register	Offset
lane0_tx_diag_tx_ctrl	41E0h
lane1_tx_diag_tx_ctrl	45E0h
lane2_tx_diag_tx_ctrl	49E0h
lane3_tx_diag_tx_ctrl	4DE0h

13.4.10.2.81.2 Diagram



13.4.10.2.81.3 Fields

Field	Function
15	TX serializer clock invert: This bit is used to optionally invert the serializer clock (

Table continues on the next page...

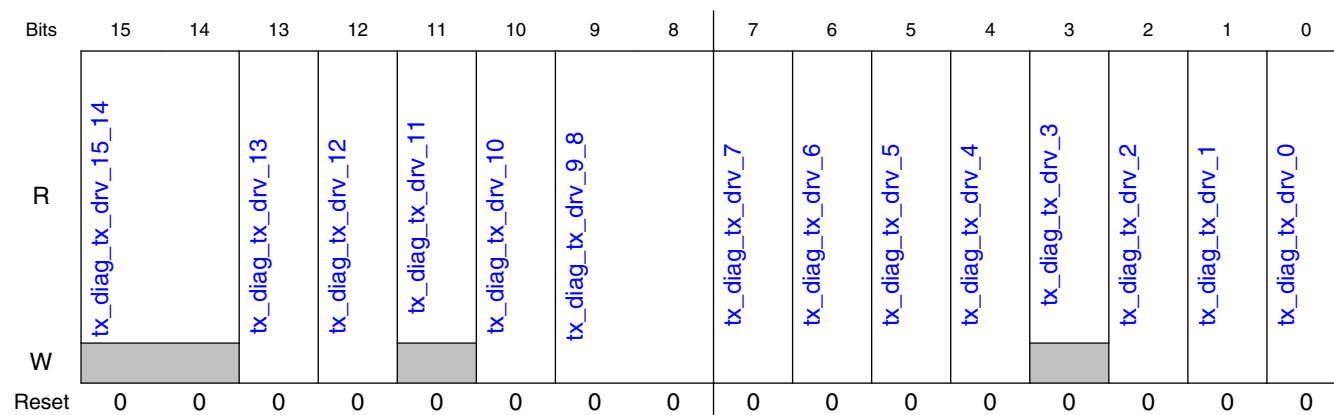
Field	Function
tx_diag_tx_ctrl_15	
14-8 tx_diag_tx_ctrl_14_8	Reserved
7-6 tx_diag_tx_ctrl_7_6	TX interface subrate standard mode 3: This field specifies the transmitter subrate when generating the
5-4 tx_diag_tx_ctrl_5_4	TX interface subrate standard mode 2: This field specifies the transmitter subrate when generating the
3-2 tx_diag_tx_ctrl_3_2	TX interface subrate standard mode 1: This field specifies the transmitter subrate when generating the
1-0 tx_diag_tx_ctrl_1_0	TX interface subrate standard mode 0: This field specifies the transmitter subrate when generating the

13.4.10.2.82 TX driver control register (lane0_tx_diag_tx_drv - lane3_tx_diag_tx_drv)

13.4.10.2.82.1 Offset

Register	Offset
lane0_tx_diag_tx_drv	41E1h
lane1_tx_diag_tx_drv	45E1h
lane2_tx_diag_tx_drv	49E1h
lane3_tx_diag_tx_drv	4DE1h

13.4.10.2.82.2 Diagram



13.4.10.2.82.3 Fields

Field	Function
15-14 tx_diag_tx_drv_15_14	Reserved
13 tx_diag_tx_drv_13	Transmitter reset pull down override enable: When this bit is active (1b1), the transmitter reset pull down override bit in this register will drive the
12 tx_diag_tx_drv_12	Transmitter reset pull down override : When enabled by the transmitter reset pull down override enable bit in this register, this bit will drive the
11 tx_diag_tx_drv_11	Reserved
10 tx_diag_tx_drv_10	TX driver programmable boost enable: Enables the programmable boost function.
9-8 tx_diag_tx_drv_9_8	TX driver programmable boost level: When the programmable boost function is enabled, this signal controls the level of the boost.
7 tx_diag_tx_drv_7	TX driver LDO bandgap dependent feedback reference enable: Enables BG dependent feedback reference to LDO.
6 tx_diag_tx_drv_6	TX driver LDO bandgap dependent reference enable: Enables BG dependent reference to LDO.
5 tx_diag_tx_drv_5	TX driver LDO VDD dependent feedback reference enable: Enables VDD dependent feedback reference to LDO.

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Clocks And Resets

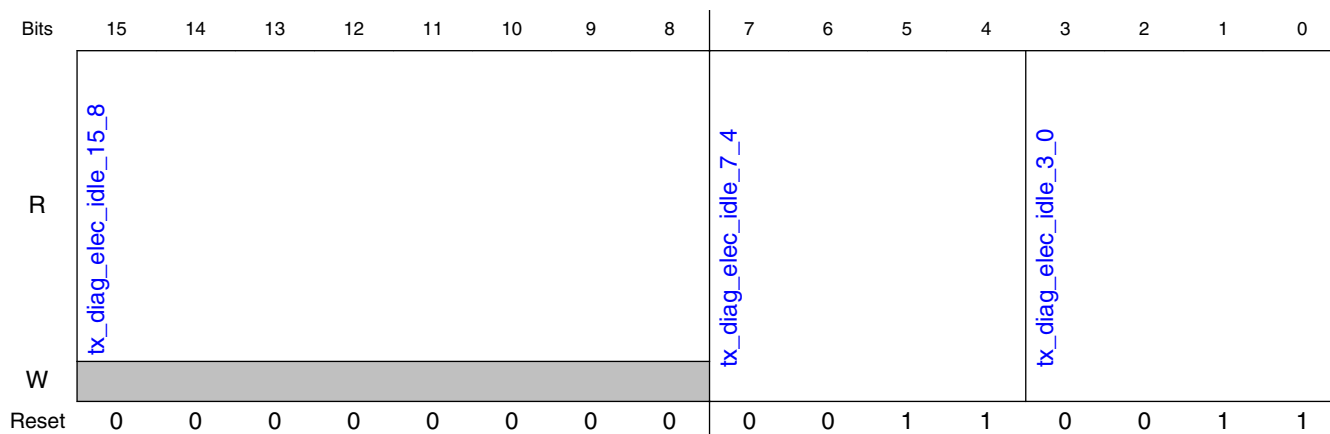
Field	Function
4 tx_diag_tx_drv_ 4	TX driver LDO VDD dependent reference enable: Enables VDD dependent reference to LDO.
3 tx_diag_tx_drv_ 3	Reserved
2 tx_diag_tx_drv_ 2	TD driver polarity control: Controls the polarity of the driver pull-up/down.
1 tx_diag_tx_drv_ 1	TX pre-driver pull up control: When the pre-driver is disabled, this bit controls the state of the pre-driver output, by controlling the
0 tx_diag_tx_drv_ 0	TX driver margin type: Selects the margining type the driver will operate in, by controlling the

13.4.10.2.83 TX electrical idle diagnostic register (lane0_tx_diag_elec_idle - lane3_tx_diag_elec_idle)

13.4.10.2.83.1 Offset

Register	Offset
lane0_tx_diag_elec_idle	41E2h
lane1_tx_diag_elec_idle	45E2h
lane2_tx_diag_elec_idle	49E2h
lane3_tx_diag_elec_idle	4DE2h

13.4.10.2.83.2 Diagram



13.4.10.2.83.3 Fields

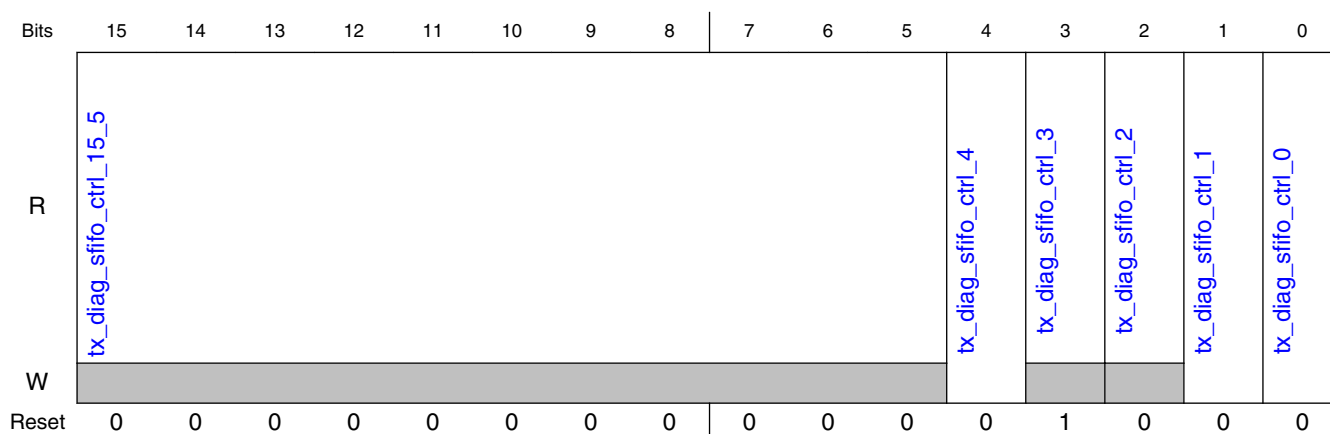
Field	Function
15-8 tx_diag_elec_idl e_15_8	Reserved
7-4 tx_diag_elec_idl e_7_4	TX electrical idle exit delay : This field controls the amount of additional delay added to the electrical idle signal after the sync FIFO, when exiting the electrical idle state. The following are the valid values that can be used for this field, and the corresponding delays. Note that in the cases of 1/2 clock cycle delays, this is implemented on the falling edge of the clock, so the result is a function of the duty cycle of the clock.
3-0 tx_diag_elec_idl e_3_0	TX electrical idle entry delay : This field controls the amount of additional delay added to the electrical idle signal after the sync FIFO, when entering the electrical idle state. The following are the valid values that can be used for this field, and the corresponding delays. Note that in the cases of 1/2 clock cycle delays, this is implemented on the falling edge of the clock, so the result is a function of the duty cycle of the clock.

13.4.10.2.84 TX sync FIFO diagnostic control register (lane0_tx_diag_sfifo_ctrl - lane3_tx_diag_sfifo_ctrl)

13.4.10.2.84.1 Offset

Register	Offset
lane0_tx_diag_sfifo_ctrl	41E3h
lane1_tx_diag_sfifo_ctrl	45E3h
lane2_tx_diag_sfifo_ctrl	49E3h
lane3_tx_diag_sfifo_ctrl	4DE3h

13.4.10.2.84.2 Diagram



13.4.10.2.84.3 Fields

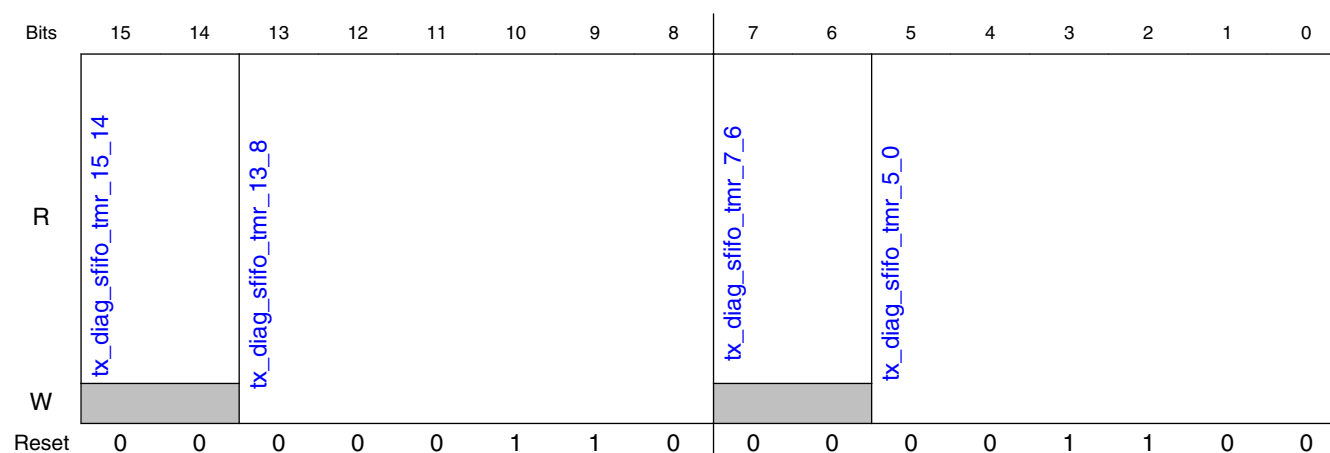
Field	Function
15-5 tx_diag_sfifo_ctl_15_5	Reserved
4 tx_diag_sfifo_ctl_4	FIFO enqueue pointer bump: This bit can be used to decrement the enqueue pointer relative to the dequeue pointer, for diagnostic purposes. Changing this bit from a value of 1b0 to 1b1 will trigger a single decrement of the enqueue pointer.
3 tx_diag_sfifo_ctl_3	FIFO alignment error: This bit indicates if a FIFO overflow or FIFO underflow condition currently exists. Note that during the FIFO alignment process, this bit will toggle. This bit is driven directly by the
2 tx_diag_sfifo_ctl_2	FIFO alignment acknowledge: This bit indicates that the FIFO alignment process is complete, as initiated either automatically by the hardware of the FIFO alignment enable override bits in this register. This bit is driven directly by the
1 tx_diag_sfifo_ctl_1	FIFO alignment enable override enable: This bit enables the FIFO alignment enable override register to drive the
0 tx_diag_sfifo_ctl_0	FIFO alignment enable override: When enabled by the FIFO alignment enable override enable bit in this register, this bit directly controls the

13.4.10.2.85 TX sync FIFO diagnostic timer register (lane0_tx_diag_sfifo_tmr - lane3_tx_diag_sfifo_tmr)

13.4.10.2.85.1 Offset

Register	Offset
lane0_tx_diag_sfifo_tmr	41E4h
lane1_tx_diag_sfifo_tmr	45E4h
lane2_tx_diag_sfifo_tmr	49E4h
lane3_tx_diag_sfifo_tmr	4DE4h

13.4.10.2.85.2 Diagram



13.4.10.2.85.3 Fields

Field	Function
15-14 tx_diag_sfifo_tm r_15_14	Reserved
13-8 tx_diag_sfifo_tm r_13_8	FIFO alignment settle delay: This field specifies the number of clocks to wait for a prior change to the enqueue pointer to complete before initiating the check phase of the alignment procedure in the sync FIFO. It drives the
7-6 tx_diag_sfifo_tm r_7_6	Reserved
5-0 tx_diag_sfifo_tm r_5_0	FIFO alignment detect delay: This field specifies the number of clocks to wait in the delay state for each phase of the alignment procedure in the sync FIFO. It drives the

13.4.10.2.86 TX receiver detect tuning register (lane0_tx_diag_rdvdet_tune - lane3_tx_diag_rdvdet_tune)

13.4.10.2.86.1 Offset

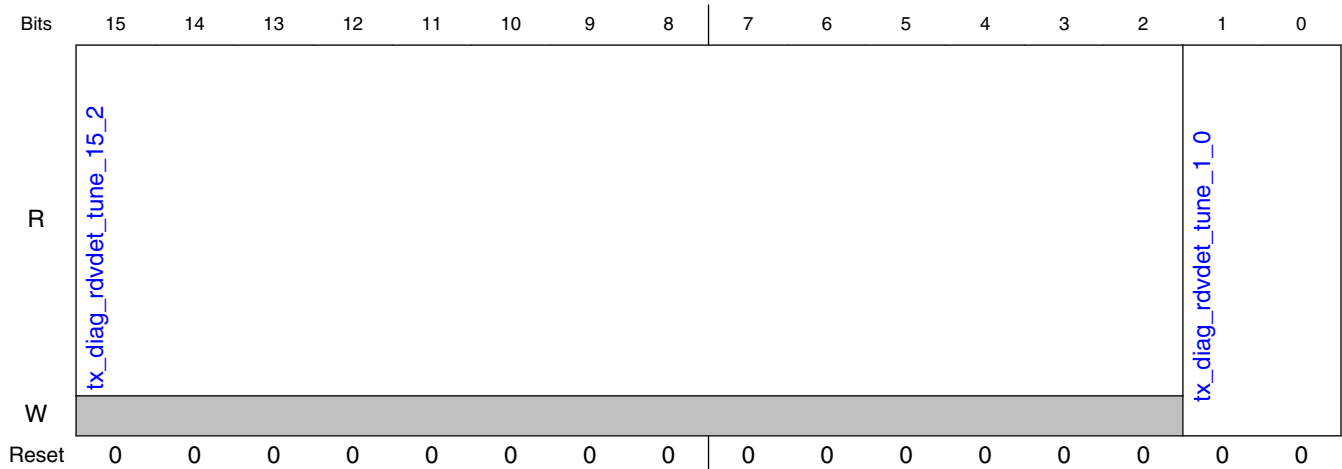
Register	Offset
lane0_tx_diag_rdvdet_tune	41E5h
lane1_tx_diag_rdvdet_tune	45E5h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane2_tx_diag_rdvdet_tune	49E5h
lane3_tx_diag_rdvdet_tune	4DE5h

13.4.10.2.86.2 Diagram



13.4.10.2.86.3 Fields

Field	Function
15-2 tx_diag_rdvdet_tune_15_2	Reserved
1-0 tx_diag_rdvdet_tune_1_0	Receiver detect reference DAC voltage: This field sets the receiver detect reference voltage for the receiver detect comparator. This field drives the

13.4.10.2.87 Transmitter control reset diagnostic register (lane0_tx_diag_rst_diag - lane3_tx_diag_rst_diag)

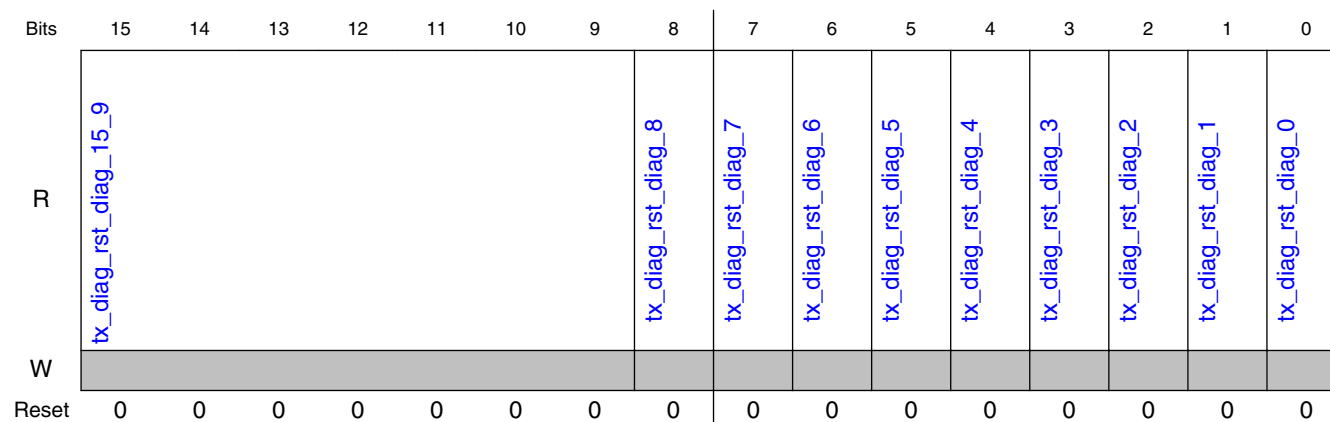
13.4.10.2.87.1 Offset

Register	Offset
lane0_tx_diag_rst_diag	41E6h
lane1_tx_diag_rst_diag	45E6h

Table continues on the next page...

Register	Offset
lane2_tx_diag_rst_diag	49E6h
lane3_tx_diag_rst_diag	4DE6h

13.4.10.2.87.2 Diagram



13.4.10.2.87.3 Fields

Field	Function
15-9 tx_diag_rst_diag_15_9	Reserved
8 tx_diag_rst_diag_8	Current state of the dsync_power_reset_n reset. Note that when the power islands are enabled by driving
7 tx_diag_rst_diag_7	Current state of the tfunc_power_reset_n reset. Note that when the power islands are enabled by driving
6 tx_diag_rst_diag_6	Current state of the xcal1_power_reset_n reset. Note that when the power islands are enabled by driving
5 tx_diag_rst_diag_5	Current state of the xcaln_power_reset_n reset. Note that when the power islands are enabled by driving
4 tx_diag_rst_diag_4	Current state of the txda_tx_clk_reset_n reset.
3	Current state of the tx_dig_reset_n reset.

Table continues on the next page...

Clocks And Resets

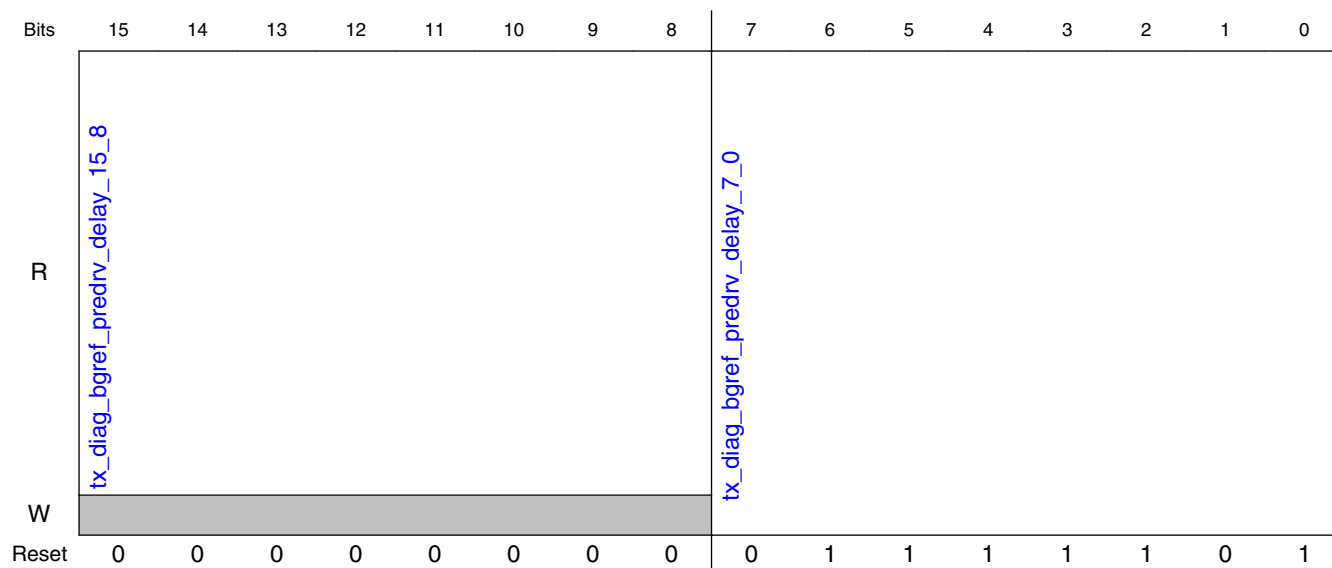
Field	Function
tx_diag_rst_diag_3	
2 tx_diag_rst_diag_2	Current state of the tx_sync_fifo_deq_rst_n reset.
1 tx_diag_rst_diag_1	Current state of the tx_sync_fifo_enq_rst_n reset.
0 tx_diag_rst_diag_0	Current state of the tx_lfps_reset_n reset.

13.4.10.2.88 TX bandgap reference and pre-drive enable delay register (lane0_tx_diag_bgref_predrv_delay - lane3_tx_diag_bgref_predrv_delay)

13.4.10.2.88.1 Offset

Register	Offset
lane0_tx_diag_bgref_predrv_delay	41E7h
lane1_tx_diag_bgref_predrv_delay	45E7h
lane2_tx_diag_bgref_predrv_delay	49E7h
lane3_tx_diag_bgref_predrv_delay	4DE7h

13.4.10.2.88.2 Diagram



13.4.10.2.88.3 Fields

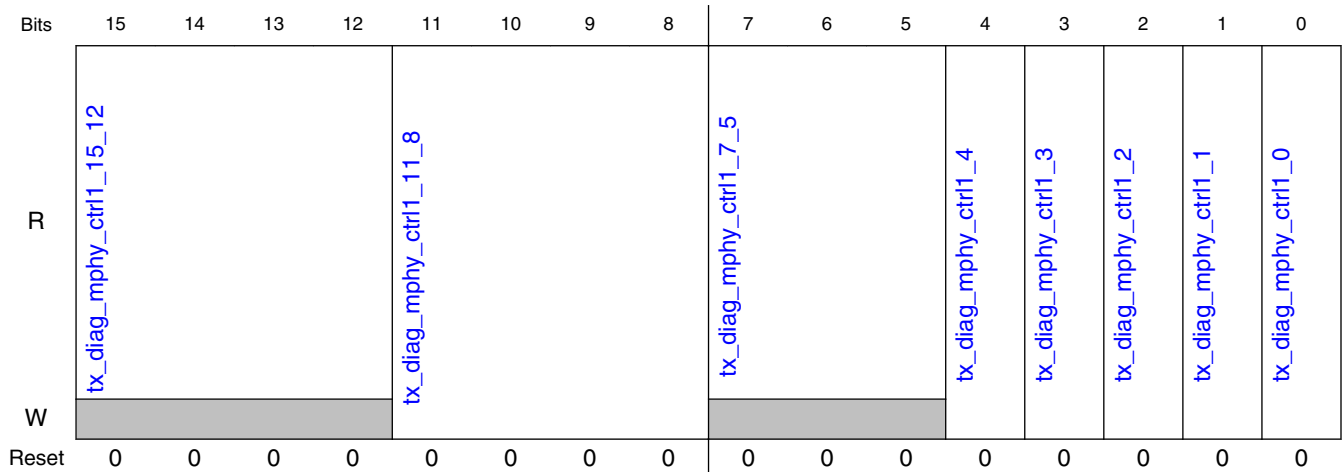
Field	Function
15-8 <code>tx_diag_bgreg_p redrv_delay_15_8</code>	Reserved
7-0 <code>tx_diag_bgreg_p redrv_delay_7_0</code>	TX bandgap reference and pre-drive enable delay: This field specifies the number of PSM clocks between when

13.4.10.2.89 TX MPHY control register 1 (lane0_tx_diag_mphy_ctrl1 - lane3_tx_diag_mphy_ctrl1)

13.4.10.2.89.1 Offset

Register	Offset
<code>lane0_tx_diag_mphy_ctrl1</code>	41F0h
<code>lane1_tx_diag_mphy_ctrl1</code>	45F0h
<code>lane2_tx_diag_mphy_ctrl1</code>	49F0h
<code>lane3_tx_diag_mphy_ctrl1</code>	4DF0h

13.4.10.2.89.2 Diagram



13.4.10.2.89.3 Fields

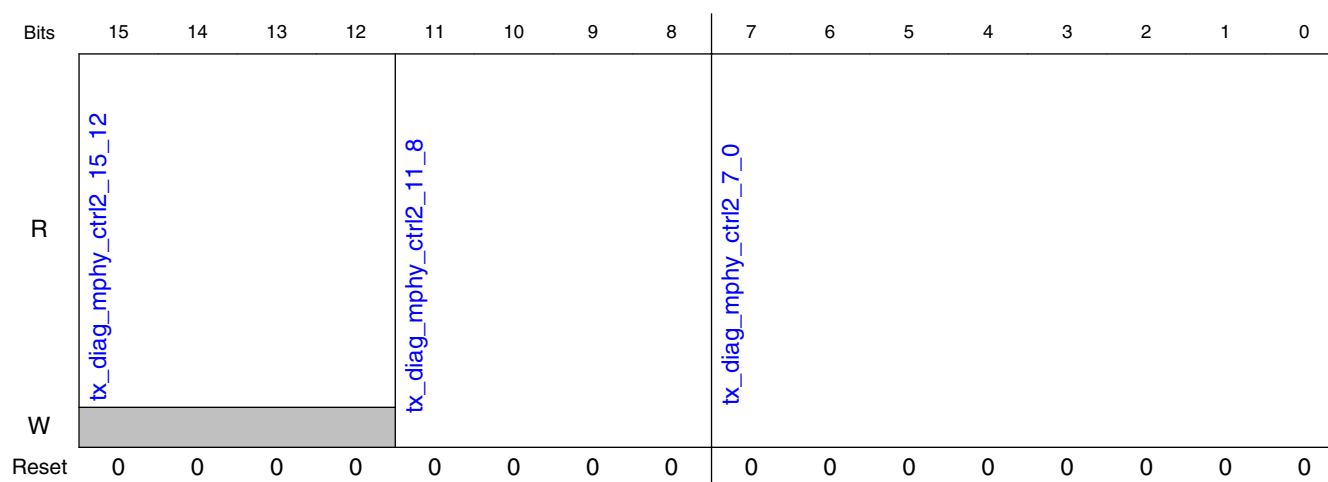
Field	Function
15-12 tx_diag_mphy_ctrl1_15_12	Reserved
11-8 tx_diag_mphy_ctrl1_11_8	Register definition to be provided by the analog team (Jira CCUPHY-2654)
7-5 tx_diag_mphy_ctrl1_7_5	Reserved
4 tx_diag_mphy_ctrl1_4	MPHY small amplitude mode: Enables the driver LDO small amplitude mode, which is used for eDP, AUX, ARC and MPHY, by driving the
3 tx_diag_mphy_ctrl1_3	AUX bias current enable: Enables the bias from the bias block of the analog boost circuit for the AUX receiver, by driving the
2 tx_diag_mphy_ctrl1_2	LDO no-load current reduce: Reduces the no-load current from the driver LDO in order to reduce the settling time, by driving the
1 tx_diag_mphy_ctrl1_1	Register definition to be provided by the analog team (Jira CCUPHY-2654)
0 tx_diag_mphy_ctrl1_0	MPHY high load current mode enable: Enables the high load current mode for the driver LDO. This is intended for non-terminated (NT) HS mode operation, by driving the

13.4.10.2.90 TX MPHY control register 2 (lane0_tx_diag_mphy_ctrl2 - lane3_tx_diag_mphy_ctrl2)

13.4.10.2.90.1 Offset

Register	Offset
lane0_tx_diag_mphy_ctrl2	41F1h
lane1_tx_diag_mphy_ctrl2	45F1h
lane2_tx_diag_mphy_ctrl2	49F1h
lane3_tx_diag_mphy_ctrl2	4DF1h

13.4.10.2.90.2 Diagram



13.4.10.2.90.3 Fields

Field	Function
tx_diag_mphy_ctrl2_15_12	Reserved
tx_diag_mphy_ctrl2_11_8	Register definition to be provided by the analog team (Jira CCUPHY-2654)

Table continues on the next page...

Clocks And Resets

Field	Function
7-0 tx_diag_mphy_c trl2_7_0	MPHY slew rate control: Controls the slew rate of the line in MPHY-G1 mode and MHL-eCBUS mode, by driving

13.4.10.2.91 TX driver LDO programming register (lane0_tx_diag_drv_ldo_prog - lane3_tx_diag_drv_ldo_prog)

13.4.10.2.91.1 Offset

Register	Offset
lane0_tx_diag_drv_ldo_prog	41F4h
lane1_tx_diag_drv_ldo_prog	45F4h
lane2_tx_diag_drv_ldo_prog	49F4h
lane3_tx_diag_drv_ldo_prog	4DF4h

13.4.10.2.91.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	tx_diag_drv_ldo_prog_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.91.3 Fields

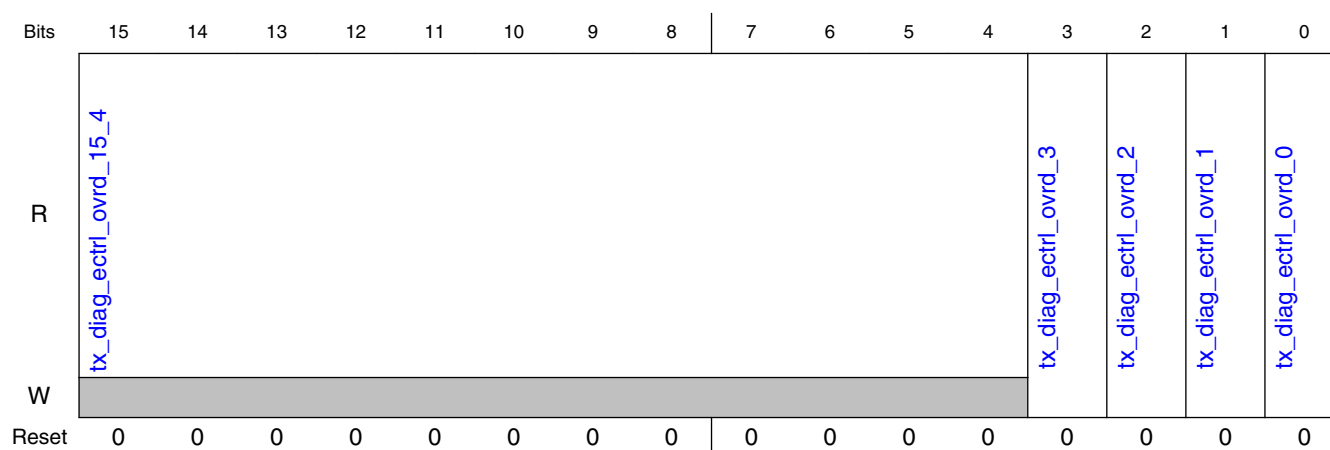
Field	Function
15-0 tx_diag_drv_ldo _prog_15_0	Register definition to be provided by the analog team (Jira CCUPHY-2654)

13.4.10.2.92 TX extra enable control override register (lane0_tx_diag_ectrl_ovrd - lane3_tx_diag_ectrl_ovrd)

13.4.10.2.92.1 Offset

Register	Offset
lane0_tx_diag_ectrl_ovrd	41F5h
lane1_tx_diag_ectrl_ovrd	45F5h
lane2_tx_diag_ectrl_ovrd	49F5h
lane3_tx_diag_ectrl_ovrd	4DF5h

13.4.10.2.92.2 Diagram



13.4.10.2.92.3 Fields

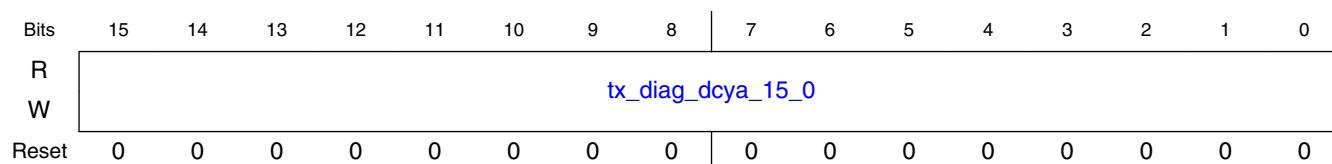
Field	Function
15-4 tx_diag_ectrl_ovrd_15_4	Reserved
3 tx_diag_ectrl_ovrd_3	Driver pre-drive enable override enable : When active (1b1), the driver pre-drive enable override bit in this register will drive the
2 tx_diag_ectrl_ovrd_2	Driver pre-drive enable override : When enabled by the driver pre-drive enable override enable bit in this register, this bit will drive the
1 tx_diag_ectrl_ovrd_1	Bandgap reference enable override enable : When active (1b1), the bandgap reference enable override bit in this register will drive the
0 tx_diag_ectrl_ovrd_0	Bandgap reference enable override : When enabled by the bandgap reference enable override enable bit in this register, this bit will drive the

13.4.10.2.93 Transmitter digital cover your alternatives register (lane0_tx_diag_dcya - lane3_tx_diag_dcya)

13.4.10.2.93.1 Offset

Register	Offset
lane0_tx_diag_dcya	41FEh
lane1_tx_diag_dcya	45FEh
lane2_tx_diag_dcya	49FEh
lane3_tx_diag_dcya	4DFEh

13.4.10.2.93.2 Diagram



13.4.10.2.93.3 Fields

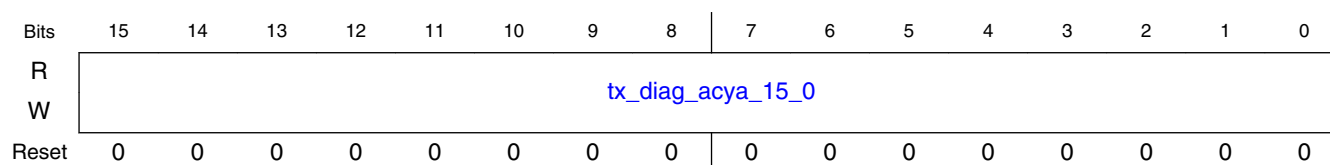
Field	Function
15-0 tx_diag_dcya_15_0	Reserved

13.4.10.2.94 Transmitter analog cover your alternatives register (lane0_tx_diag_acya - lane3_tx_diag_acya)

13.4.10.2.94.1 Offset

Register	Offset
lane0_tx_diag_acya	41FFh
lane1_tx_diag_acya	45FFh
lane2_tx_diag_acya	49FFh
lane3_tx_diag_acya	4DFFh

13.4.10.2.94.2 Diagram



13.4.10.2.94.3 Fields

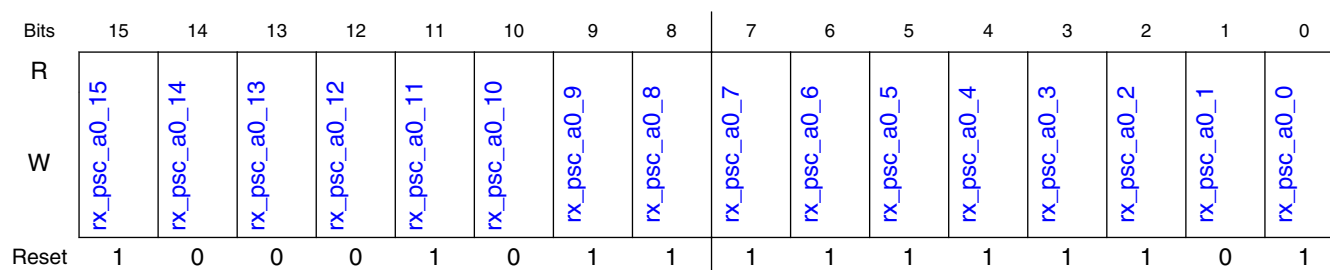
Field	Function
15-0 tx_diag_acya_15_0	Reserved

13.4.10.2.95 Receiver A0 power state definition register (lane0_rx_psc_a0 - lane3_rx_psc_a0)

13.4.10.2.95.1 Offset

Register	Offset
lane0_rx_psc_a0	8000h
lane1_rx_psc_a0	8400h
lane2_rx_psc_a0	8800h
lane3_rx_psc_a0	8C00h

13.4.10.2.95.2 Diagram



13.4.10.2.95.3 Fields

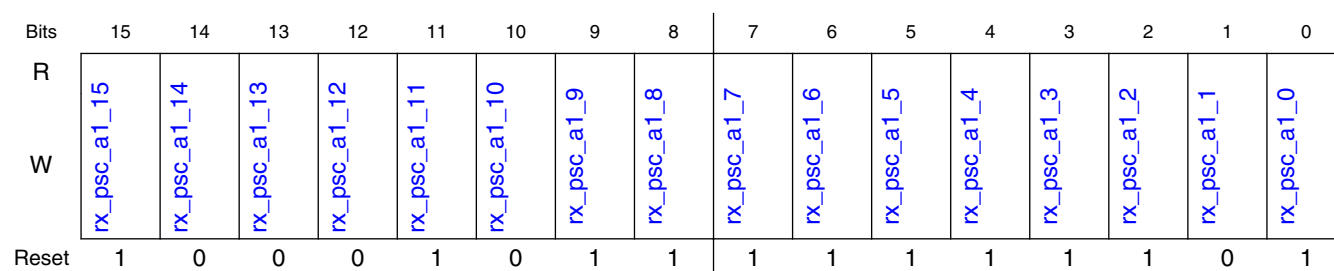
Field	Function
15 rx_psc_a0_15	RX LFPS / signal detect filter mode: Selects the mode of operation of the LFPS and signal detect filters.
14 rx_psc_a0_14	RX squelch enable: Enables the receiver squelch function. This drives the
13 rx_psc_a0_13	RX LFPS detect enable: Enables the receiver LFPS detect function. This drives the
12 rx_psc_a0_12	RX signal detect extend valid: Indicates the power state where the
11 rx_psc_a0_11	RX signal detect filter enable: Enables the receiver signal detect filter function in the digital receiver controller.
10 rx_psc_a0_10	RX LFPS detect filter enable: Enables the receiver LFPS detect filter function in the digital receiver controller.
9 rx_psc_a0_9	RX clock enable: Enables the receiver circuits related to the high speed clocks.
8 rx_psc_a0_8	RX signal detect enable: Enables the receiver signal detect function. This drives the
7 rx_psc_a0_7	RX equalization enable: Enables the receiver equalization circuits, via the
6 rx_psc_a0_6	RX sampler enable: Enables the receiver circuits related to the sampler.
5 rx_psc_a0_5	RX CDRLF enable: Enables the receiver circuits related to the CDRLF.
4 rx_psc_a0_4	RX bias enable: Enables the receiver circuits related to the bias.
3 rx_psc_a0_3	RX DFE equalization enable: Enables the receiver DFE equalization circuits, via the
2 rx_psc_a0_2	RX PI enable: Enables the receiver circuits related to the PI and associated clocking components.
1 rx_psc_a0_1	RX e path enable (calibration and eye surf only) : Enables the receiver circuits related to the eye plot PI and e path deserializer for calibration and eye surf.
0 rx_psc_a0_0	RX enable: Enables the receiver circuits related to the FE, and Deserializer.

13.4.10.2.96 Receiver A1 power state definition register (lane0_rx_psc_a1 - lane3_rx_psc_a1)

13.4.10.2.96.1 Offset

Register	Offset
lane0_rx_psc_a1	8001h
lane1_rx_psc_a1	8401h
lane2_rx_psc_a1	8801h
lane3_rx_psc_a1	8C01h

13.4.10.2.96.2 Diagram



13.4.10.2.96.3 Fields

Field	Function
15 rx_psc_a1_15	RX LFPS / signal detect filter mode
14 rx_psc_a1_14	RX squelch enable
13 rx_psc_a1_13	RX LFPS detect enable
12 rx_psc_a1_12	RX signal detect extend valid
11 rx_psc_a1_11	RX signal detect filter enable
10 rx_psc_a1_10	RX LFPS detect filter enable
9 rx_psc_a1_9	RX clock enable
8 rx_psc_a1_8	RX signal detect enable
7 rx_psc_a1_7	RX equalization enable

Table continues on the next page...

Clocks And Resets

Field	Function
6 rx_psc_a1_6	RX sampler enable
5 rx_psc_a1_5	RX CDRLF enable
4 rx_psc_a1_4	RX bias enable
3 rx_psc_a1_3	RX DFE equalization enable
2 rx_psc_a1_2	RX PI enable
1 rx_psc_a1_1	RX e path enable (calibration and eye surf only)
0 rx_psc_a1_0	RX enable

13.4.10.2.97 Receiver A2 power state definition register (lane0_rx_psc_a2 - lane3_rx_psc_a2)

13.4.10.2.97.1 Offset

Register	Offset
lane0_rx_psc_a2	8002h
lane1_rx_psc_a2	8402h
lane2_rx_psc_a2	8802h
lane3_rx_psc_a2	8C02h

13.4.10.2.97.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W	rx_psc_a2_15	rx_psc_a2_14	rx_psc_a2_13	rx_psc_a2_12	rx_psc_a2_11	rx_psc_a2_10	rx_psc_a2_9	rx_psc_a2_8	rx_psc_a2_7	rx_psc_a2_6	rx_psc_a2_5	rx_psc_a2_4	rx_psc_a2_3	rx_psc_a2_2	rx_psc_a2_1	rx_psc_a2_0
Reset	1	0	0	0	1	0	0	1	0	0	0	1	0	0	0	0

13.4.10.2.97.3 Fields

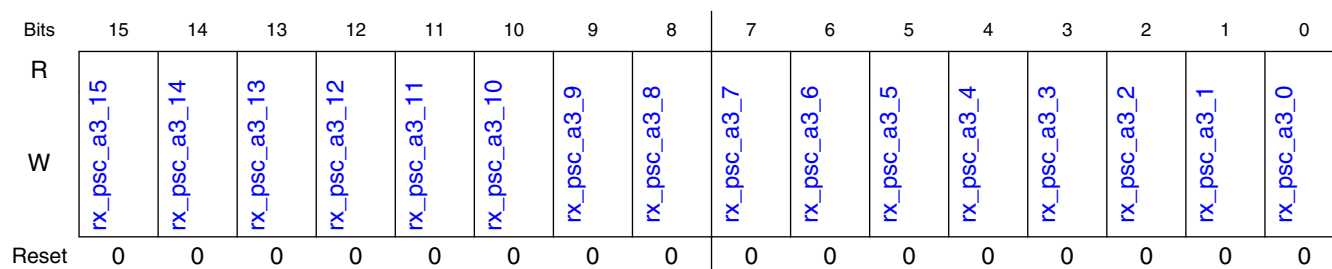
Field	Function
15 rx_psc_a2_15	RX LFPS / signal detect filter mode
14 rx_psc_a2_14	RX squelch enable
13 rx_psc_a2_13	RX LFPS detect enable
12 rx_psc_a2_12	RX signal detect extend valid
11 rx_psc_a2_11	RX signal detect filter enable
10 rx_psc_a2_10	RX LFPS detect filter enable
9 rx_psc_a2_9	RX clock enable
8 rx_psc_a2_8	RX signal detect enable
7 rx_psc_a2_7	RX equalization enable
6 rx_psc_a2_6	RX sampler enable
5 rx_psc_a2_5	RX CDRLF enable
4 rx_psc_a2_4	RX bias enable
3 rx_psc_a2_3	RX DFE equalization enable
2 rx_psc_a2_2	RX PI enable
1 rx_psc_a2_1	RX e path enable (calibration and eye surf only)
0 rx_psc_a2_0	RX enable

13.4.10.2.98 Receiver A3 power state definition register (lane0_rx_psc_a3 - lane3_rx_psc_a3)

13.4.10.2.98.1 Offset

Register	Offset
lane0_rx_psc_a3	8003h
lane1_rx_psc_a3	8403h
lane2_rx_psc_a3	8803h
lane3_rx_psc_a3	8C03h

13.4.10.2.98.2 Diagram



13.4.10.2.98.3 Fields

Field	Function
15 rx_psc_a3_15	RX LFPS / signal detect filter mode
14 rx_psc_a3_14	RX squelch enable
13 rx_psc_a3_13	RX LFPS detect enable
12 rx_psc_a3_12	RX signal detect extend valid
11 rx_psc_a3_11	RX signal detect filter enable
10 rx_psc_a3_10	RX LFPS detect filter enable
9 rx_psc_a3_9	RX clock enable
8 rx_psc_a3_8	RX signal detect enable
7 rx_psc_a3_7	RX equalization enable

Table continues on the next page...

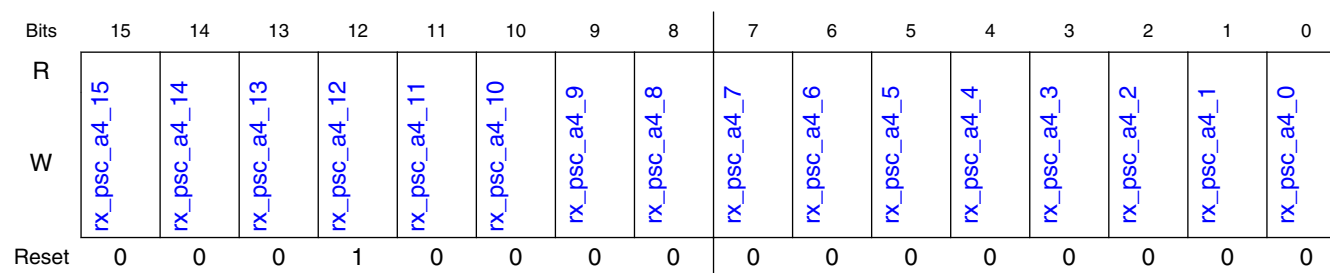
Field	Function
6 rx_psc_a3_6	RX sampler enable
5 rx_psc_a3_5	RX CDRLF enable
4 rx_psc_a3_4	RX bias enable
3 rx_psc_a3_3	RX DFE equalization enable
2 rx_psc_a3_2	RX PI enable
1 rx_psc_a3_1	RX e path enable (calibration and eye surf only)
0 rx_psc_a3_0	RX enable

13.4.10.2.99 Receiver A4 power state definition register (lane0_rx_psc_a4 - lane3_rx_psc_a4)

13.4.10.2.99.1 Offset

Register	Offset
lane0_rx_psc_a4	8004h
lane1_rx_psc_a4	8404h
lane2_rx_psc_a4	8804h
lane3_rx_psc_a4	8C04h

13.4.10.2.99.2 Diagram



13.4.10.2.99.3 Fields

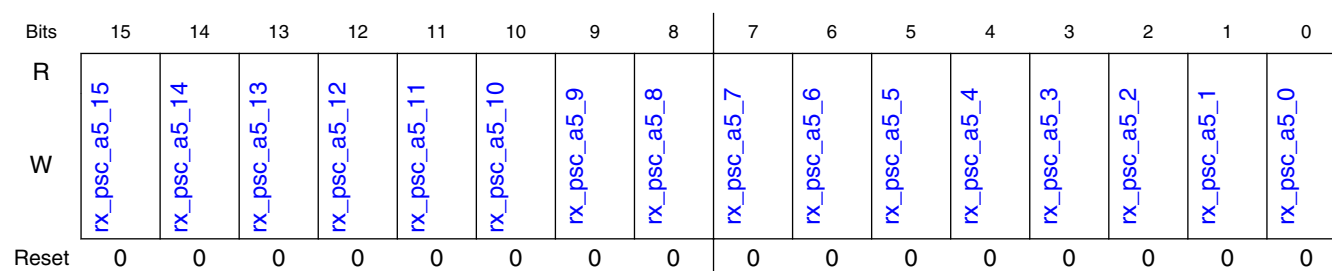
Field	Function
15 rx_psc_a4_15	RX LFPS / signal detect filter mode
14 rx_psc_a4_14	RX squelch enable
13 rx_psc_a4_13	RX LFPS detect enable
12 rx_psc_a4_12	RX signal detect extend valid
11 rx_psc_a4_11	RX signal detect filter enable
10 rx_psc_a4_10	RX LFPS detect filter enable
9 rx_psc_a4_9	RX clock enable
8 rx_psc_a4_8	RX signal detect enable
7 rx_psc_a4_7	RX equalization enable
6 rx_psc_a4_6	RX sampler enable
5 rx_psc_a4_5	RX CDRLF enable
4 rx_psc_a4_4	RX bias enable
3 rx_psc_a4_3	RX DFE equalization enable
2 rx_psc_a4_2	RX PI enable
1 rx_psc_a4_1	RX e path enable (calibration and eye surf only)
0 rx_psc_a4_0	RX enable

13.4.10.2.100 Receiver A5 power state definition register (lane0_rx_psc_a5 - lane3_rx_psc_a5)

13.4.10.2.100.1 Offset

Register	Offset
lane0_rx_psc_a5	8005h
lane1_rx_psc_a5	8405h
lane2_rx_psc_a5	8805h
lane3_rx_psc_a5	8C05h

13.4.10.2.100.2 Diagram



13.4.10.2.100.3 Fields

Field	Function
15 rx_psc_a5_15	RX LFPS / signal detect filter mode
14 rx_psc_a5_14	RX squelch enable
13 rx_psc_a5_13	RX LFPS detect enable
12 rx_psc_a5_12	RX signal detect extend valid
11 rx_psc_a5_11	RX signal detect filter enable
10 rx_psc_a5_10	RX LFPS detect filter enable
9 rx_psc_a5_9	RX clock enable
8 rx_psc_a5_8	RX signal detect enable
7 rx_psc_a5_7	RX equalization enable

Table continues on the next page...

Clocks And Resets

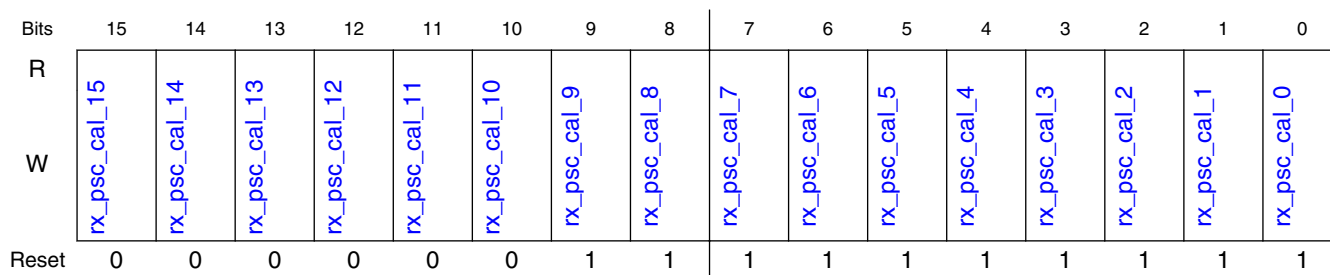
Field	Function
6 rx_psc_a5_6	RX sampler enable
5 rx_psc_a5_5	RX CDRLF enable
4 rx_psc_a5_4	RX bias enable
3 rx_psc_a5_3	RX DFE equalization enable
2 rx_psc_a5_2	RX PI enable
1 rx_psc_a5_1	RX e path enable (calibration and eye surf only)
0 rx_psc_a5_0	RX enable

13.4.10.2.101 Receiver calibration power state definition register (lane0_rx_psc_cal - lane3_rx_psc_cal)

13.4.10.2.101.1 Offset

Register	Offset
lane0_rx_psc_cal	8006h
lane1_rx_psc_cal	8406h
lane2_rx_psc_cal	8806h
lane3_rx_psc_cal	8C06h

13.4.10.2.101.2 Diagram



13.4.10.2.101.3 Fields

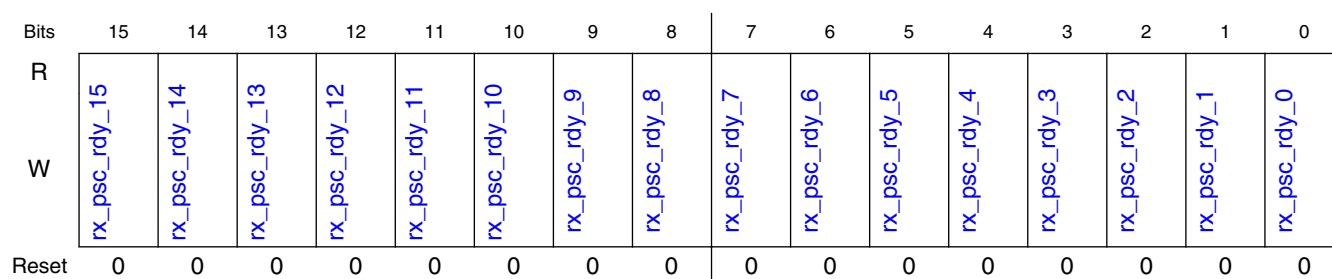
Field	Function
15 rx_psc_cal_15	RX LFPS / signal detect filter mode
14 rx_psc_cal_14	RX squelch enable
13 rx_psc_cal_13	RX LFPS detect enable
12 rx_psc_cal_12	RX signal detect extend valid
11 rx_psc_cal_11	RX signal detect filter enable
10 rx_psc_cal_10	RX LFPS detect filter enable
9 rx_psc_cal_9	RX clock enable
8 rx_psc_cal_8	RX signal detect enable
7 rx_psc_cal_7	RX equalization enable
6 rx_psc_cal_6	RX sampler enable
5 rx_psc_cal_5	RX CDRLF enable
4 rx_psc_cal_4	RX bias enable
3 rx_psc_cal_3	RX DFE equalization enable
2 rx_psc_cal_2	RX PI enable
1 rx_psc_cal_1	RX e path enable (calibration and eye surf only)
0 rx_psc_cal_0	RX enable

13.4.10.2.102 Receiver ready power state definition register (lane0_rx_psc_rdy - lane3_rx_psc_rdy)

13.4.10.2.102.1 Offset

Register	Offset
lane0_rx_psc_rdy	8007h
lane1_rx_psc_rdy	8407h
lane2_rx_psc_rdy	8807h
lane3_rx_psc_rdy	8C07h

13.4.10.2.102.2 Diagram



13.4.10.2.102.3 Fields

Field	Function
15 rx_psc_rdy_15	RX LFPS / signal detect filter mode
14 rx_psc_rdy_14	RX squelch enable
13 rx_psc_rdy_13	RX LFPS detect enable
12 rx_psc_rdy_12	RX signal detect extend valid
11 rx_psc_rdy_11	RX signal detect filter enable
10 rx_psc_rdy_10	RX LFPS detect filter enable
9 rx_psc_rdy_9	RX clock enable
8 rx_psc_rdy_8	RX signal detect enable
7 rx_psc_rdy_7	RX equalization enable

Table continues on the next page...

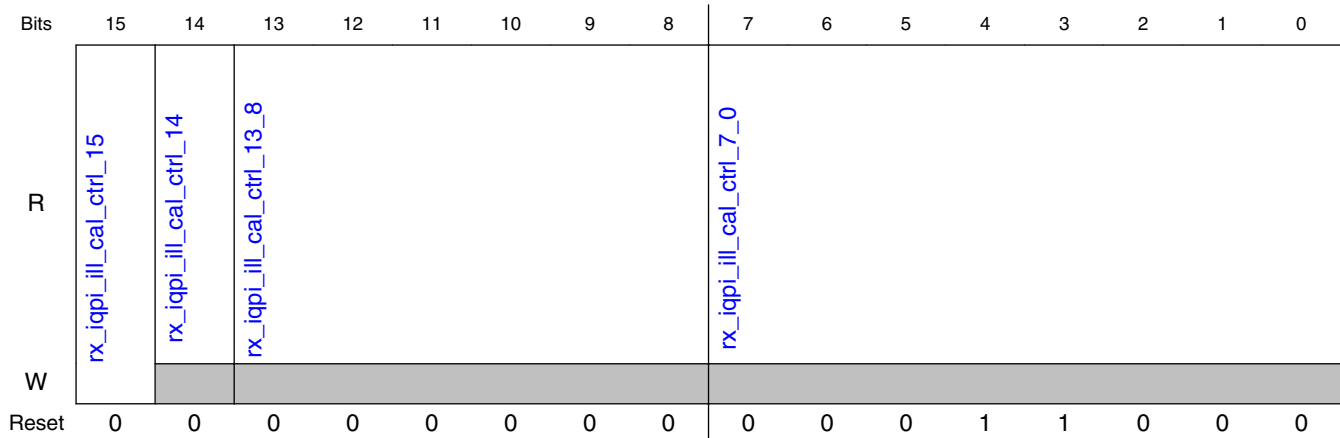
Field	Function
6 rx_psc_rdy_6	RX sampler enable
5 rx_psc_rdy_5	RX CDRLF enable
4 rx_psc_rdy_4	RX bias enable
3 rx_psc_rdy_3	RX DFE equalization enable
2 rx_psc_rdy_2	RX PI enable
1 rx_psc_rdy_1	RX e path enable (calibration and eye surf only)
0 rx_psc_rdy_0	RX enable

13.4.10.2.103 RX IQ PI ILL calibration control register (lane0_rx_iqpi_ill_cal_ctrl - lane3_rx_iqpi_ill_cal_ctrl)

13.4.10.2.103.1 Offset

Register	Offset
lane0_rx_iqpi_ill_cal_ctrl	8020h
lane1_rx_iqpi_ill_cal_ctrl	8420h
lane2_rx_iqpi_ill_cal_ctrl	8820h
lane3_rx_iqpi_ill_cal_ctrl	8C20h

13.4.10.2.103.2 Diagram



13.4.10.2.103.3 Fields

Field	Function
15 rx_iqpi_ill_cal_ctrl_15	Start ILL calibration: Activating (1b1) this bit will start an ILL calibration process. This bit must remain active until the ILL calibration process is complete (as indicated by the ILL calibration process done bit in this register). To start another ILL calibration process, the ILL calibration process done bit must have gone inactive from any prior calibration process.
14 rx_iqpi_ill_cal_ctrl_14	ILL calibration process done: This bit will be set to 1b1 when the ILL calibration process is complete. It will be cleared by the receiver being reset, or by the deactivation of the Start ILL calibration bit in this register.
13-8 rx_iqpi_ill_cal_ctrl_13_8	Reserved
7-0 rx_iqpi_ill_cal_ctrl_7_0	ILL calibration code: This is the calibration code that was determined by the ILL calibration process. This signal is valid when the ILL calibration process is complete. This field specifies the number of resistors that are switched in. The following are the values for this code:

13.4.10.2.104 RX IQ PI ILL calibration start point register (lane0_rx_iqpi_ill_cal_start - lane3_rx_iqpi_ill_cal_start)

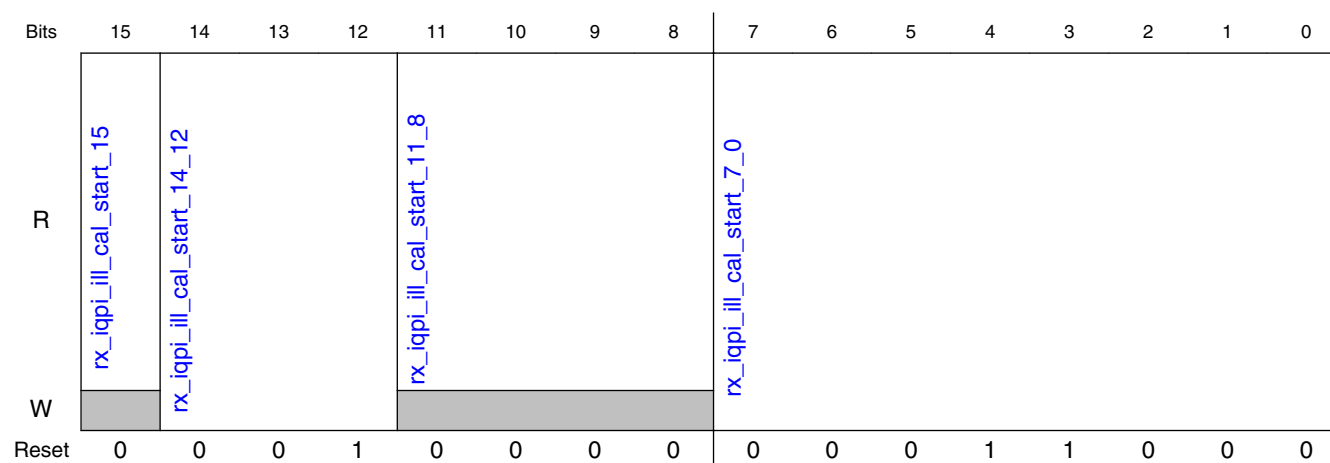
13.4.10.2.104.1 Offset

Register	Offset
lane0_rx_iqpi_ill_cal_start	8021h
lane1_rx_iqpi_ill_cal_start	8421h

Table continues on the next page...

Register	Offset
lane2_rx_iqpi_ill_cal_start	8821h
lane3_rx_iqpi_ill_cal_start	8C21h

13.4.10.2.104.2 Diagram



13.4.10.2.104.3 Fields

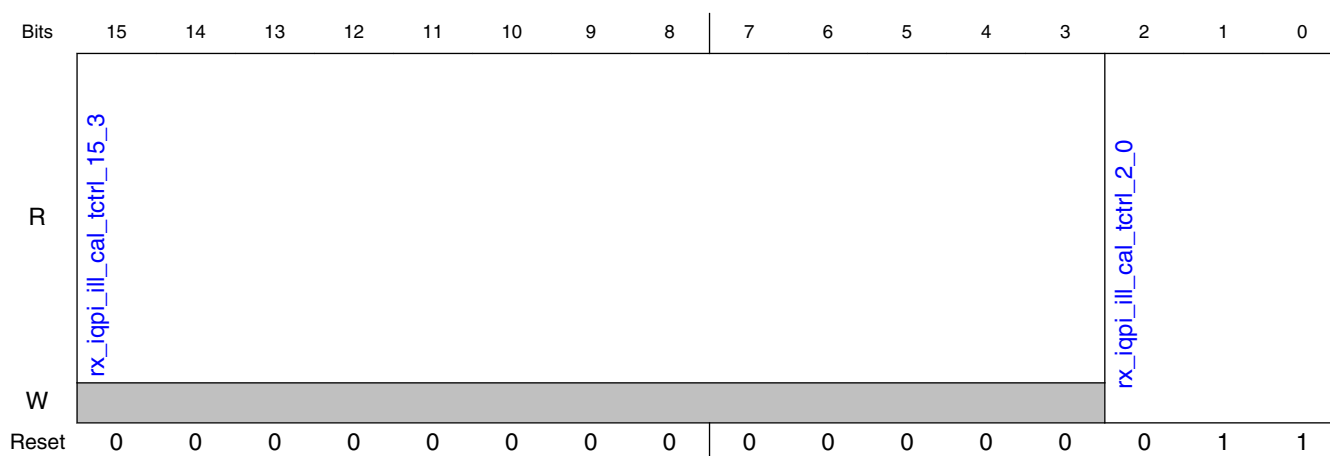
Field	Function
15 rx_iqpi_ill_cal_start_15	Reserved
14-12 rx_iqpi_ill_cal_start_14_12	ILL calibration initial step size control: This field specifies the initial step size for the ILL calibration state machine. The following are the values that can be used in this field, and the corresponding step sizes.
11-8 rx_iqpi_ill_cal_start_11_8	Reserved
7-0 rx_iqpi_ill_cal_start_7_0	ILL calibration code starting point value: This field specifies the starting ILL code that is used by the ILL calibration state machine. The purpose of this value is such that the ILL calibration process starts at a point that is, on average, relatively close to the final calibration point. This allows the calibration time to be reduced, on average. The following are the values for this code:

13.4.10.2.105 RX IQ PI ILL calibration timer control register (lane0_rx_iqpi_ill_cal_tctrl - lane3_rx_iqpi_ill_cal_tctrl)

13.4.10.2.105.1 Offset

Register	Offset
lane0_rx_iqpi_ill_cal_tctrl	8022h
lane1_rx_iqpi_ill_cal_tctrl	8422h
lane2_rx_iqpi_ill_cal_tctrl	8822h
lane3_rx_iqpi_ill_cal_tctrl	8C22h

13.4.10.2.105.2 Diagram



13.4.10.2.105.3 Fields

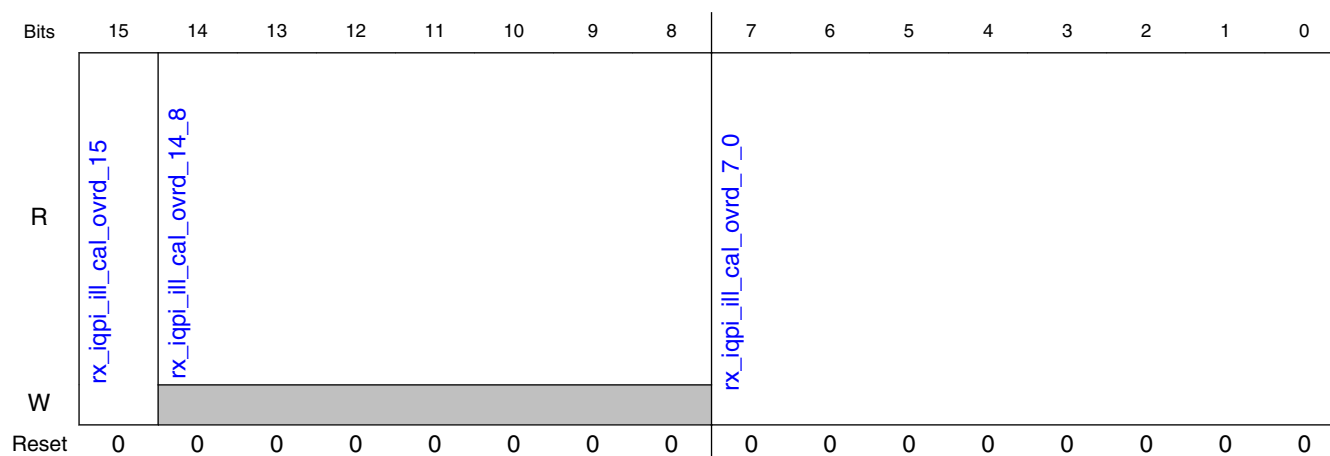
Field	Function
15-3 rx_iqpi_ill_cal_tctrl_15_3	Reserved
2-0 rx_iqpi_ill_cal_tctrl_2_0	ILL calibration initial time scale control: This field specifies the calibration start time scaling factor applied to the ILL calibration when running the initial step size for the calibration code is not set to 1. Setting this value to a value other than 1 will reduce the calibration measurement time by the amount specified below. Then when the final calibration steps are made the full calibration time will be used to get the appropriate amount of resolution.

13.4.10.2.106 RX IQ PI ILL calibration override register (lane0_rx_iqpi_ill_cal_ovrd - lane3_rx_iqpi_ill_cal_ovrd)

13.4.10.2.106.1 Offset

Register	Offset
lane0_rx_iqpi_ill_cal_ovrd	8023h
lane1_rx_iqpi_ill_cal_ovrd	8423h
lane2_rx_iqpi_ill_cal_ovrd	8823h
lane3_rx_iqpi_ill_cal_ovrd	8C23h

13.4.10.2.106.2 Diagram



13.4.10.2.106.3 Fields

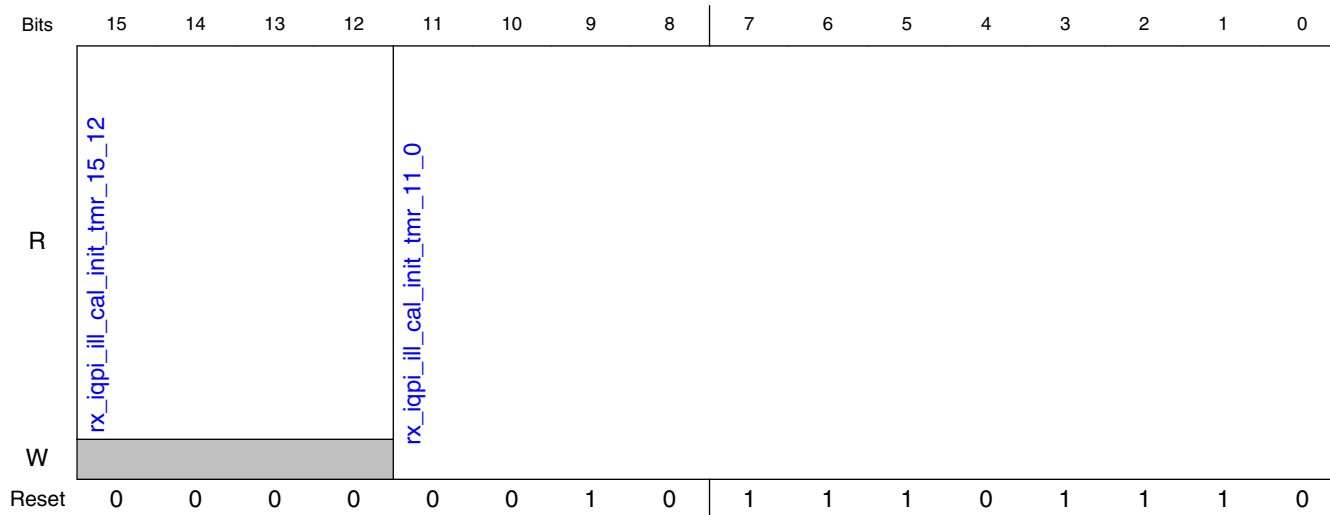
Field	Function
15 rx_iqpi_ill_cal_ovrd_15	ILL calibration code override enable: Activating (1b1) this bit allows the ILL code determined during the automatic ILL calibration process to be overridden by the value driven by the ILL calibration code override value field in this register.
14-8 rx_iqpi_ill_cal_ovrd_14_8	Reserved
7-0 rx_iqpi_ill_cal_ovrd_7_0	ILL calibration code override value: This field is used to override the ILL code determined during the automatic ILL calibration process. The ILL code driven on this field is valid when the ILL calibration code override enable bit in this register is active. The following are the values for this code:

13.4.10.2.107 RX IQ PI ILL calibration initialization timer register (lane0_rx_iqpi_ill_cal_init_tmr - lane3_rx_iqpi_ill_cal_init_tmr)

13.4.10.2.107.1 Offset

Register	Offset
lane0_rx_iqpi_ill_cal_init_tmr	8024h
lane1_rx_iqpi_ill_cal_init_tmr	8424h
lane2_rx_iqpi_ill_cal_init_tmr	8824h
lane3_rx_iqpi_ill_cal_init_tmr	8C24h

13.4.10.2.107.2 Diagram



13.4.10.2.107.3 Fields

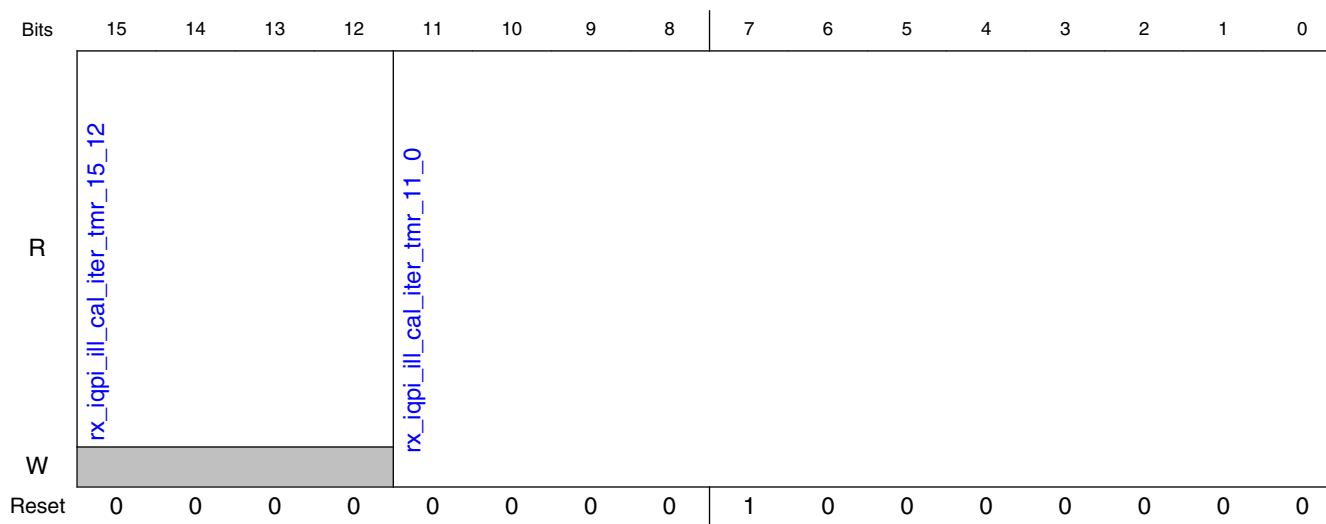
Field	Function
15-12 rx_iqpi_ill_cal_init_tmr_15_12	Reserved
11-0 rx_iqpi_ill_cal_init_tmr_11_0	Initialization wait timer value: This is the number of

13.4.10.2.108 RX IQ PI ILL calibration iteration timer register (lane0_rx_iqpi_ill_cal_iter_tmr - lane3_rx_iqpi_ill_cal_iter_tmr)

13.4.10.2.108.1 Offset

Register	Offset
lane0_rx_iqpi_ill_cal_iter_tmr	8025h
lane1_rx_iqpi_ill_cal_iter_tmr	8425h
lane2_rx_iqpi_ill_cal_iter_tmr	8825h
lane3_rx_iqpi_ill_cal_iter_tmr	8C25h

13.4.10.2.108.2 Diagram



13.4.10.2.108.3 Fields

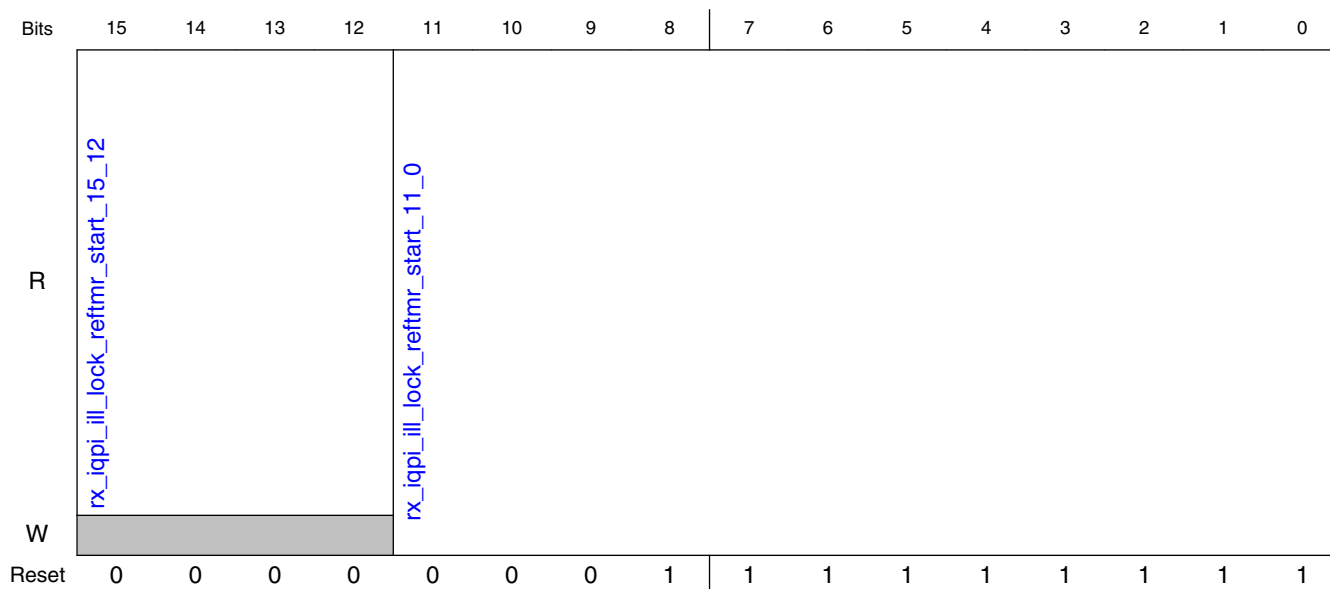
Field	Function
15-12 rx_iqpi_ill_cal_iter_tmr_15_12	Reserved
11-0 rx_iqpi_ill_cal_iter_tmr_11_0	Iteration wait timer value; This is the number of

13.4.10.2.109 RX IQ PI ILL lock reference timer start value register (lane0_rx_iqpi_ill_lock_reftmr_start - lane3_rx_iqpi_ill_lock_reftmr_start)

13.4.10.2.109.1 Offset

Register	Offset
lane0_rx_iqpi_ill_lock_reftmr_start	8026h
lane1_rx_iqpi_ill_lock_reftmr_start	8426h
lane2_rx_iqpi_ill_lock_reftmr_start	8826h
lane3_rx_iqpi_ill_lock_reftmr_start	8C26h

13.4.10.2.109.2 Diagram



13.4.10.2.109.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

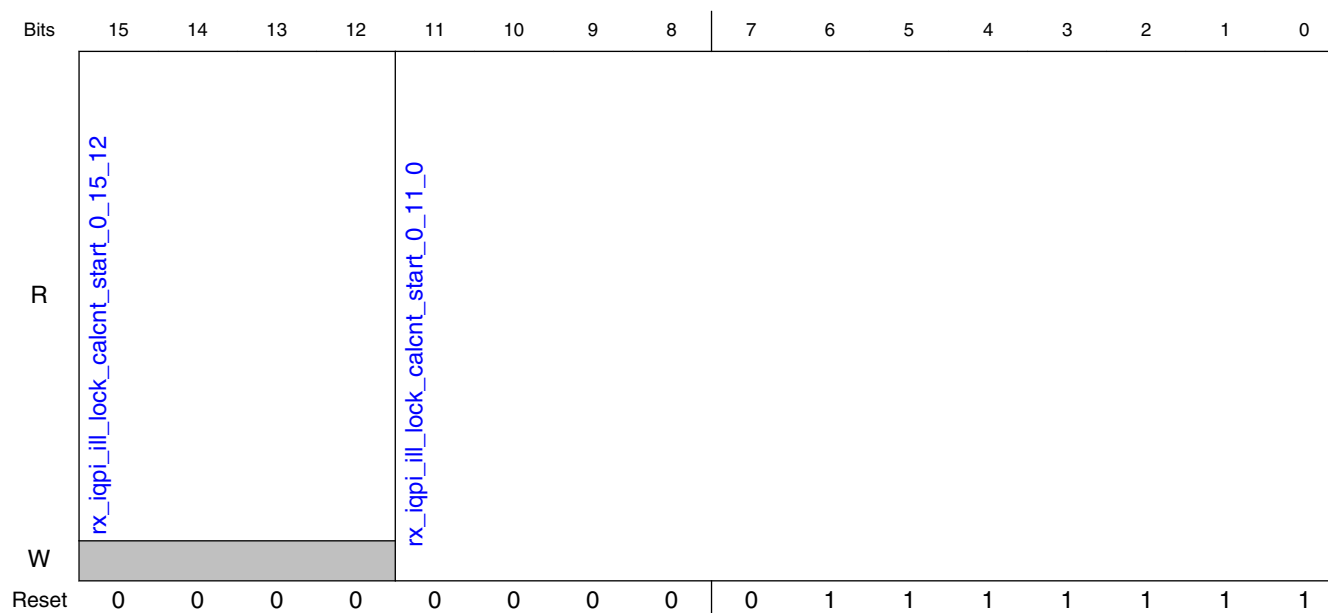
Field	Function
rx_iqpi_ill_lock_ref_tmr_start_15_12	
rx_iqpi_ill_lock_ref_tmr_start_11_0	ILL lock reference timer start value : This is the value that is loaded into the ILL lock reference timer as the starting point for that timer, when checking for ILL lock. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes. Note that the counter counts down to 0, therefore this is a value of 1 less than value required for the timer function.

13.4.10.2.110 RX IQ PI ILL lock calibration counter start value standard mode 0 register (lane0_rx_iqpi_ill_lock_calcnt_start_0 - lane3_rx_iqpi_ill_lock_calcnt_start_0)

13.4.10.2.110.1 Offset

Register	Offset
lane0_rx_iqpi_ill_lock_calcnt_start_0	8028h
lane1_rx_iqpi_ill_lock_calcnt_start_0	8428h
lane2_rx_iqpi_ill_lock_calcnt_start_0	8828h
lane3_rx_iqpi_ill_lock_calcnt_start_0	8C28h

13.4.10.2.110.2 Diagram



13.4.10.2.110.3 Fields

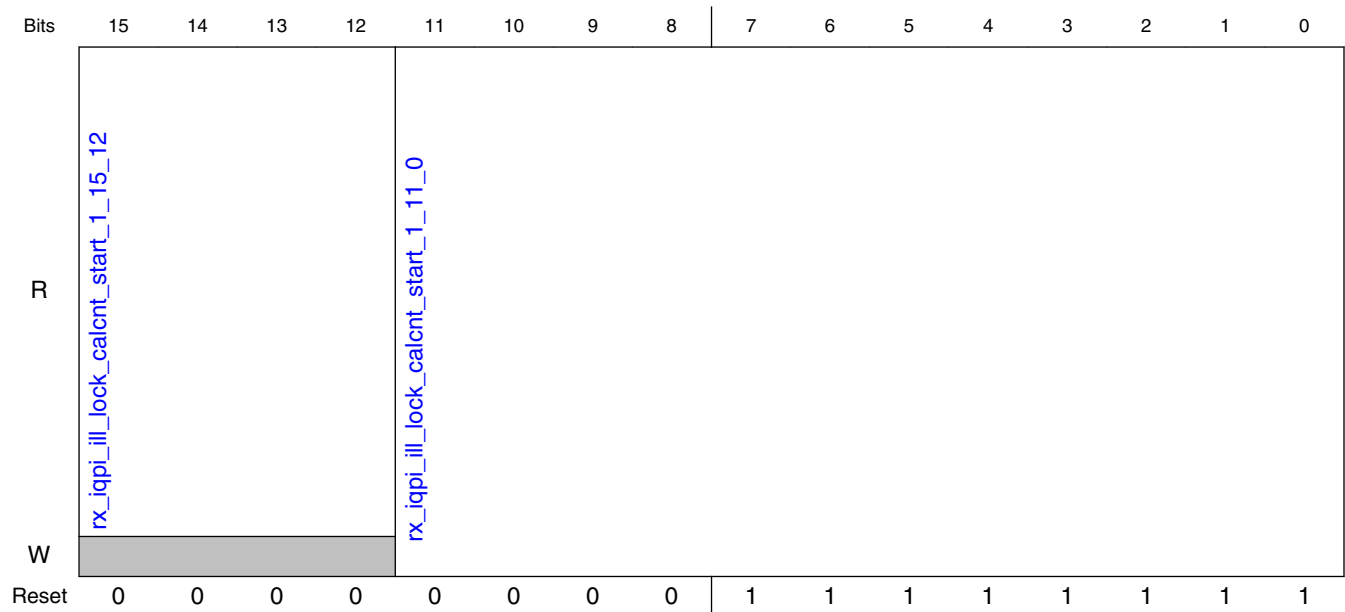
Field	Function
15-12 rx_iqpi_ill_lock_calcnt_start_0_15_12	Reserved
11-0 rx_iqpi_ill_lock_calcnt_start_0_11_0	ILL lock calibration counter start value : This is the value that is loaded into the ILL lock calibration counter as the starting point for that counter, when checking for ILL lock when

13.4.10.2.111 RX IQ PI ILL lock calibration counter start value standard mode 1 register (lane0_rx_iqpi_ill_lock_calcnt_start_1 - lane3_rx_iqpi_ill_lock_calcnt_start_1)

13.4.10.2.111.1 Offset

Register	Offset
lane0_rx_iqpi_ill_lock_calcnt_start_1	8029h
lane1_rx_iqpi_ill_lock_calcnt_start_1	8429h
lane2_rx_iqpi_ill_lock_calcnt_start_1	8829h
lane3_rx_iqpi_ill_lock_calcnt_start_1	8C29h

13.4.10.2.111.2 Diagram



13.4.10.2.111.3 Fields

Field	Function
15-12 rx_iqpi_ill_lock_calcnt_start_1_15_12	Reserved
11-0 rx_iqpi_ill_lock_calcnt_start_1_11_0	ILL lock calibration counter start value : This is the value that is loaded into the ILL lock calibration counter as the starting point for that counter, when checking for ILL lock when

13.4.10.2.112 RX IQ PI ILL lock calibration counter start value standard mode 2 register (lane0_rx_iqpi_ill_lock_calcnt_start_2 - lane3_rx_iqpi_ill_lock_calcnt_start_2)

13.4.10.2.112.1 Offset

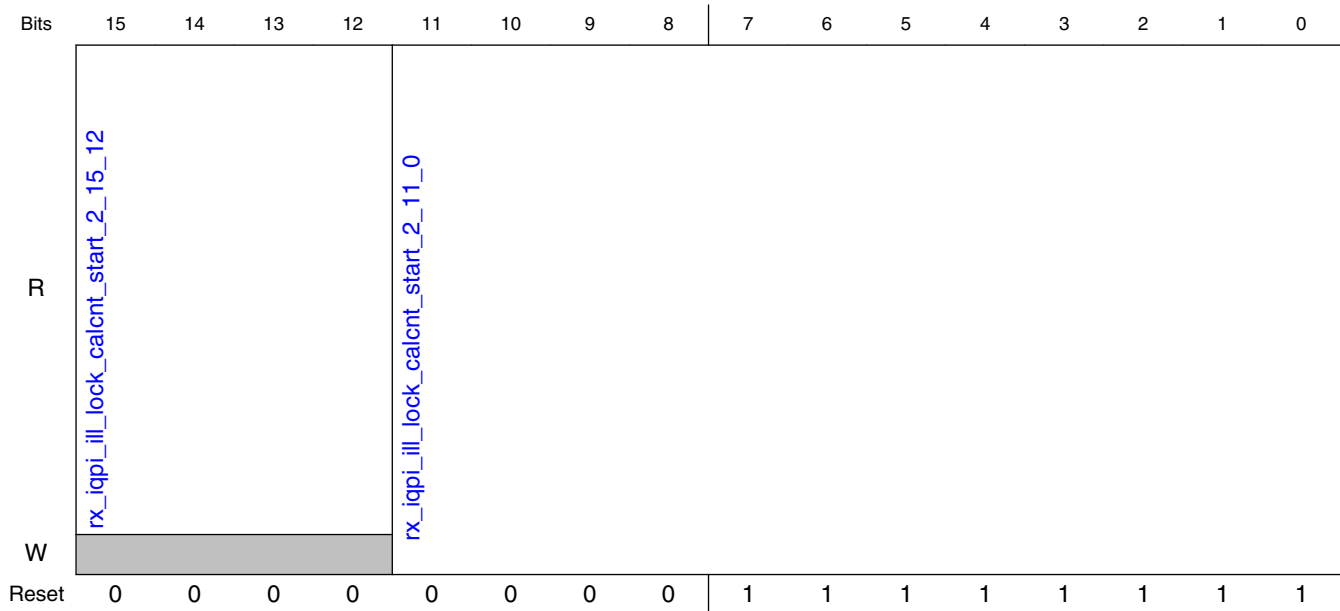
Register	Offset
lane0_rx_iqpi_ill_lock_calcnt_start_2	802Ah
lane1_rx_iqpi_ill_lock_calcnt_start_2	842Ah

Table continues on the next page...

Clocks And Resets

Register	Offset
lane2_rx_iqpi_ill_lock_calcnt_start_2	882Ah
lane3_rx_iqpi_ill_lock_calcnt_start_2	8C2Ah

13.4.10.2.112.2 Diagram



13.4.10.2.112.3 Fields

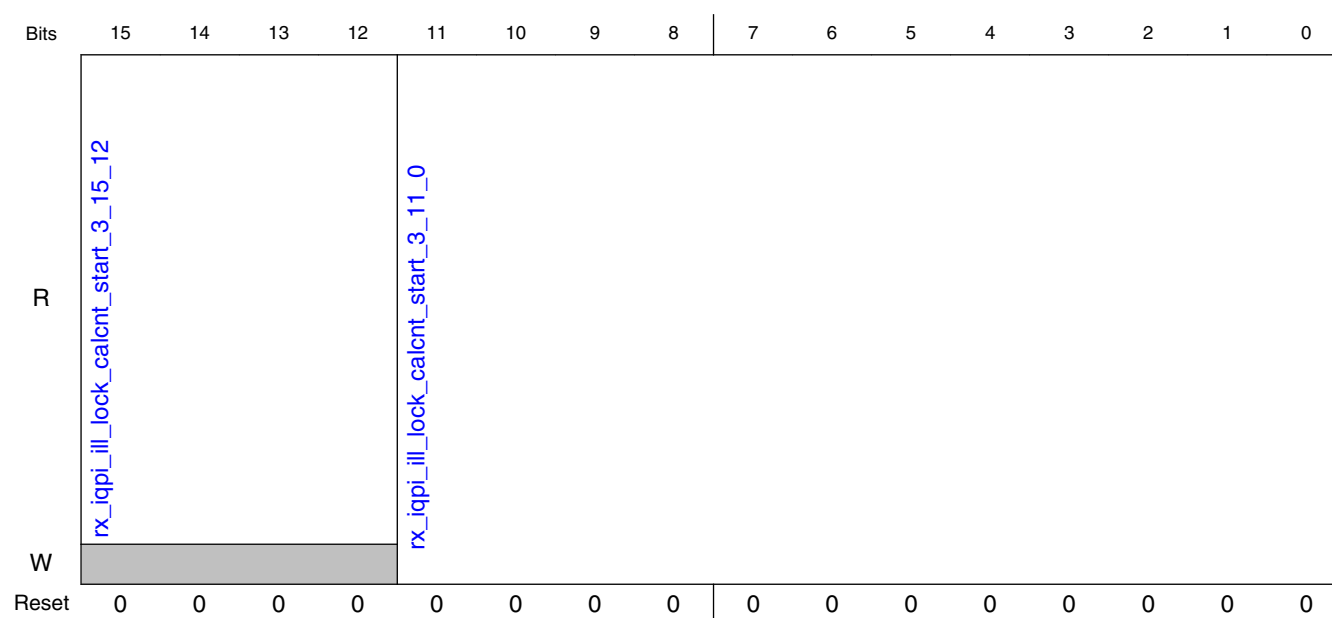
Field	Function
15-12 <code>rx_iqpi_ill_lock_calcnt_start_2_15_12</code>	Reserved
11-0 <code>rx_iqpi_ill_lock_calcnt_start_2_11_0</code>	ILL lock calibration counter start value : This is the value that is loaded into the ILL lock calibration counter as the starting point for that counter, when checking for ILL lock when

13.4.10.2.113 RX IQ PI ILL lock calibration counter start value standard mode 3 register (lane0_rx_iqpi_ill_lock_calcnt_start_3 - lane3_rx_iqpi_ill_lock_calcnt_start_3)

13.4.10.2.113.1 Offset

Register	Offset
lane0_rx_iqpi_ill_lock_calcnt_start_3	802Bh
lane1_rx_iqpi_ill_lock_calcnt_start_3	842Bh
lane2_rx_iqpi_ill_lock_calcnt_start_3	882Bh
lane3_rx_iqpi_ill_lock_calcnt_start_3	8C2Bh

13.4.10.2.113.2 Diagram



13.4.10.2.113.3 Fields

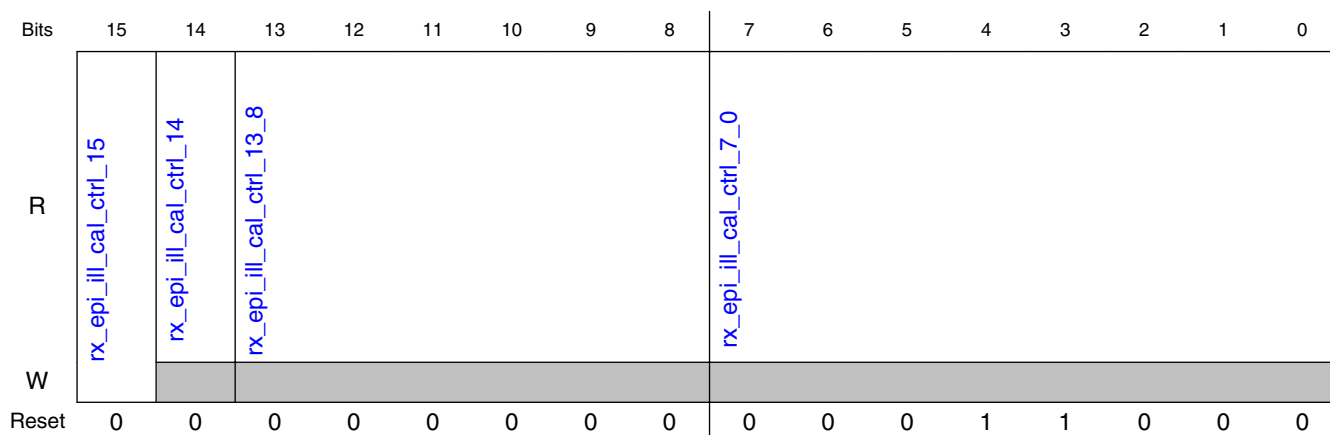
Field	Function
15-12 rx_iqpi_ill_lock_calcnt_start_3_15_12	Reserved
11-0 rx_iqpi_ill_lock_calcnt_start_3_11_0	ILL lock calibration counter start value : This is the value that is loaded into the ILL lock calibration counter as the starting point for that counter, when checking for ILL lock when

13.4.10.2.114 RX E PI ILL calibration control register (lane0_rx_epi_ill_cal_ctrl - lane3_rx_epi_ill_cal_ctrl)

13.4.10.2.114.1 Offset

Register	Offset
lane0_rx_epi_ill_cal_ctrl	8030h
lane1_rx_epi_ill_cal_ctrl	8430h
lane2_rx_epi_ill_cal_ctrl	8830h
lane3_rx_epi_ill_cal_ctrl	8C30h

13.4.10.2.114.2 Diagram



13.4.10.2.114.3 Fields

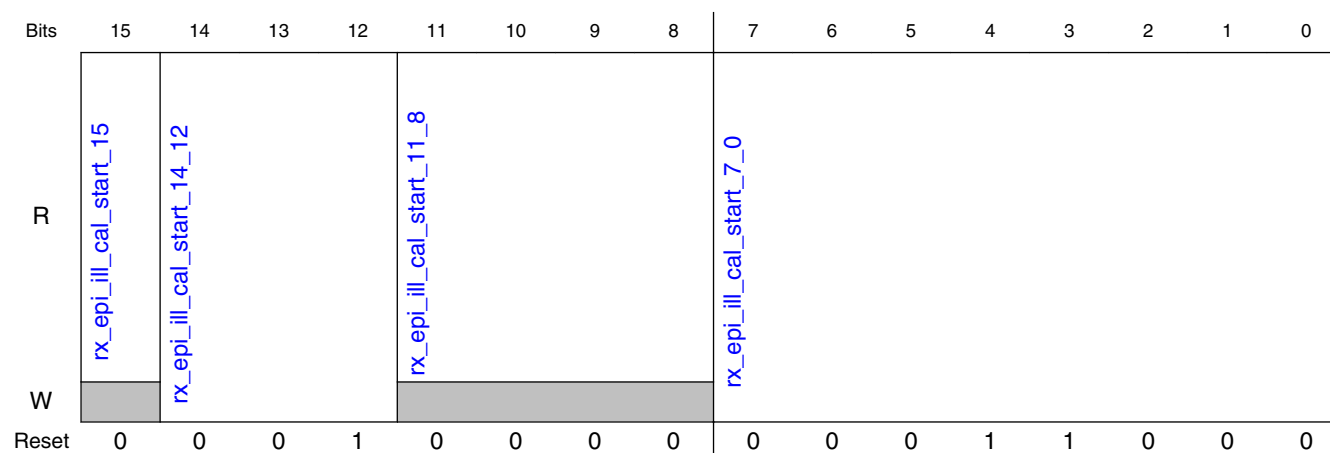
Field	Function
15 rx_epi_ill_cal_ctrl_15	Start ILL calibration: Activating (1b1) this bit will start an ILL calibration process. This bit must remain active until the ILL calibration process is complete (as indicated by the ILL calibration process done bit in this register). To start another ILL calibration process, the ILL calibration process done bit must have gone inactive from any prior calibration process.
14 rx_epi_ill_cal_ctrl_14	ILL calibration process done: This bit will be set to 1b1 when the ILL calibration process is complete. It will be cleared by the receiver being reset, or by the deactivation of the Start ILL calibration bit in this register.
13-8 rx_epi_ill_cal_ctrl_13_8	Reserved
7-0 rx_epi_ill_cal_ctrl_7_0	ILL calibration code: This is the calibration code that was determined by the ILL calibration process. This signal is valid when the ILL calibration process is complete. This field specifies the number of resistors that are switched in. The following are the values for this code:

13.4.10.2.115 RX E PI ILL calibration start point register (lane0_rx_epi_ill_cal_start - lane3_rx_epi_ill_cal_start)

13.4.10.2.115.1 Offset

Register	Offset
lane0_rx_epi_ill_cal_start	8031h
lane1_rx_epi_ill_cal_start	8431h
lane2_rx_epi_ill_cal_start	8831h
lane3_rx_epi_ill_cal_start	8C31h

13.4.10.2.115.2 Diagram



13.4.10.2.115.3 Fields

Field	Function
15 rx_epi_ill_cal_start_15	Reserved
14-12 rx_epi_ill_cal_start_14_12	ILL calibration initial step size control: This field specifies the initial step size for the ILL calibration state machine. The following are the values that can be used in this field, and the corresponding step sizes.
11-8 rx_epi_ill_cal_start_11_8	Reserved

Table continues on the next page...

Clocks And Resets

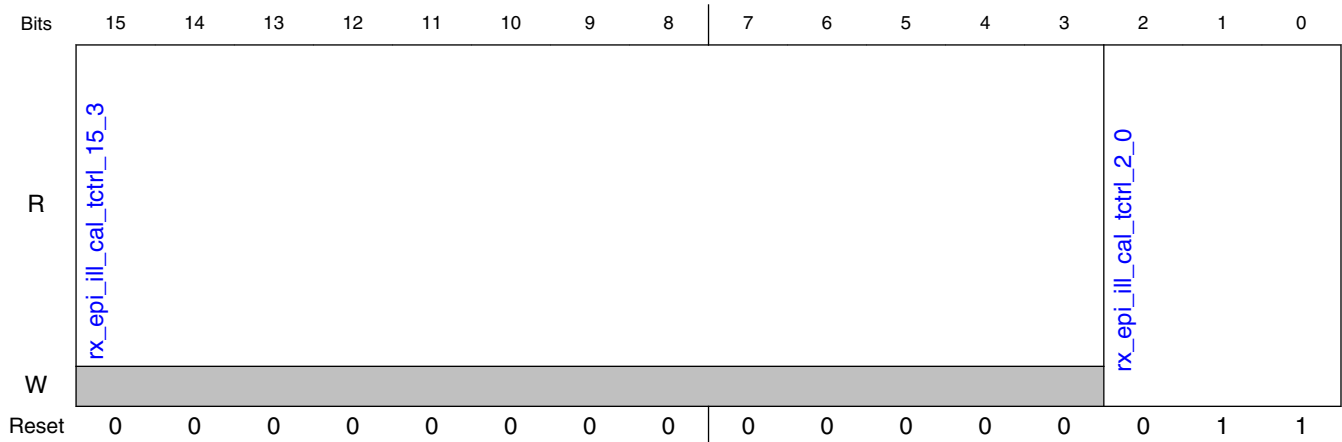
Field	Function
7-0 rx_epi_ill_cal_start_7_0	ILL calibration code starting point value: This field specifies the starting ILL code that is used by the ILL calibration state machine. The purpose of this value is such that the ILL calibration process starts at a point that is, on average, relatively close to the final calibration point. This allows the calibration time to be reduced, on average. The following are the values for this code:

13.4.10.2.116 RX E PI ILL calibration timer control register (lane0_rx_epi_ill_cal_tctrl - lane3_rx_epi_ill_cal_tctrl)

13.4.10.2.116.1 Offset

Register	Offset
lane0_rx_epi_ill_cal_tctrl	8032h
lane1_rx_epi_ill_cal_tctrl	8432h
lane2_rx_epi_ill_cal_tctrl	8832h
lane3_rx_epi_ill_cal_tctrl	8C32h

13.4.10.2.116.2 Diagram



13.4.10.2.116.3 Fields

Field	Function
15-3 rx_epi_ill_cal_tctrl_15_3	Reserved
2-0	ILL calibration initial time scale control: This field specifies the calibration start time scaling factor applied to the ILL calibration when running the initial step size for the calibration code is not set to 1. Setting this

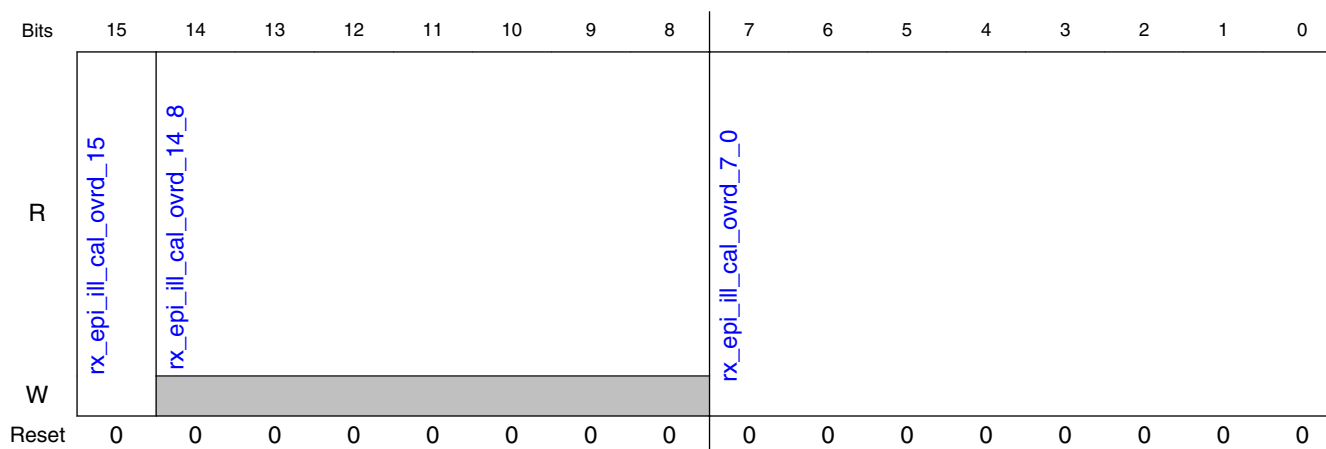
Field	Function
rx_epi_ill_cal_tct rl_2_0	value to a value other than 1 will reduce the calibration measurement time by the amount specified below. Then when the final calibration steps are made the full calibration time will be used to get the appropriate amount of resolution.

13.4.10.2.117 RX E PI ILL calibration override register (lane0_rx_epi_ill_cal_ovrd - lane3_rx_epi_ill_cal_ovrd)

13.4.10.2.117.1 Offset

Register	Offset
lane0_rx_epi_ill_cal_ovrd	8033h
lane1_rx_epi_ill_cal_ovrd	8433h
lane2_rx_epi_ill_cal_ovrd	8833h
lane3_rx_epi_ill_cal_ovrd	8C33h

13.4.10.2.117.2 Diagram



13.4.10.2.117.3 Fields

Field	Function
15 rx_epi_ill_cal_ovrd_15	ILL calibration code override enable: Activating (1b1) this bit allows the ILL code determined during the automatic ILL calibration process to be overridden by the value driven by the ILL calibration code override value field in this register.
14-8	Reserved

Table continues on the next page...

Clocks And Resets

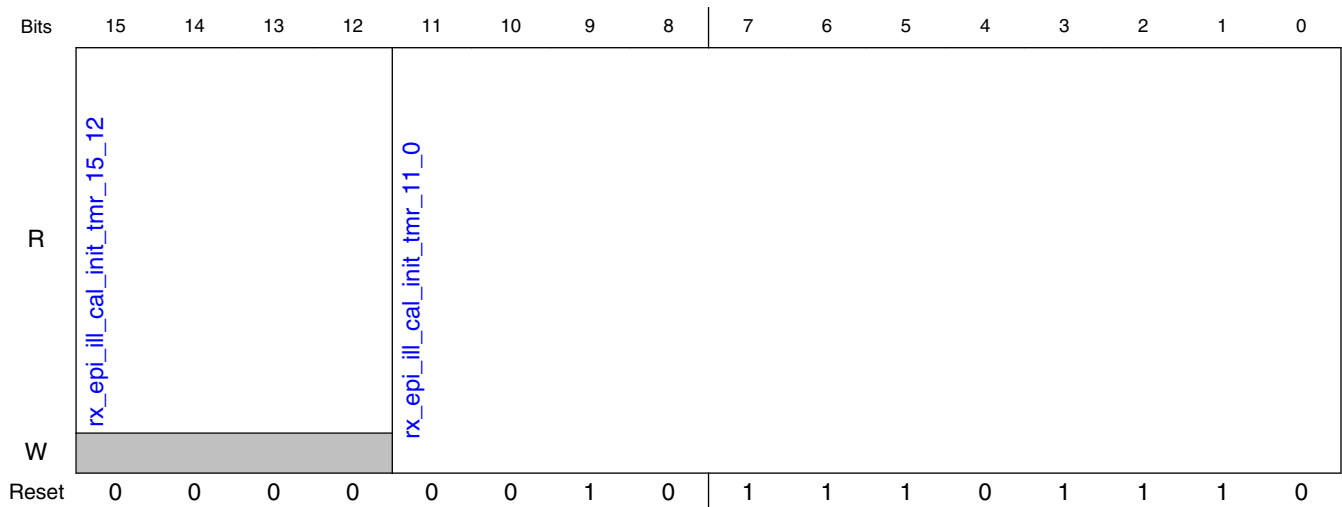
Field	Function
rx_epi_ill_cal_ov rd_14_8	
7-0 rx_epi_ill_cal_ov rd_7_0	ILL calibration code override value: This field is used to override the ILL code determined during the automatic ILL calibration process. The ILL code driven on this field is valid when the ILL calibration code override enable bit in this register is active. The following are the values for this code:

13.4.10.2.118 RX E PI ILL calibration initialization timer register (lane0_rx_epi_ill_cal_init_tmr - lane3_rx_epi_ill_cal_init_tmr)

13.4.10.2.118.1 Offset

Register	Offset
lane0_rx_epi_ill_cal_init_tmr	8034h
lane1_rx_epi_ill_cal_init_tmr	8434h
lane2_rx_epi_ill_cal_init_tmr	8834h
lane3_rx_epi_ill_cal_init_tmr	8C34h

13.4.10.2.118.2 Diagram



13.4.10.2.118.3 Fields

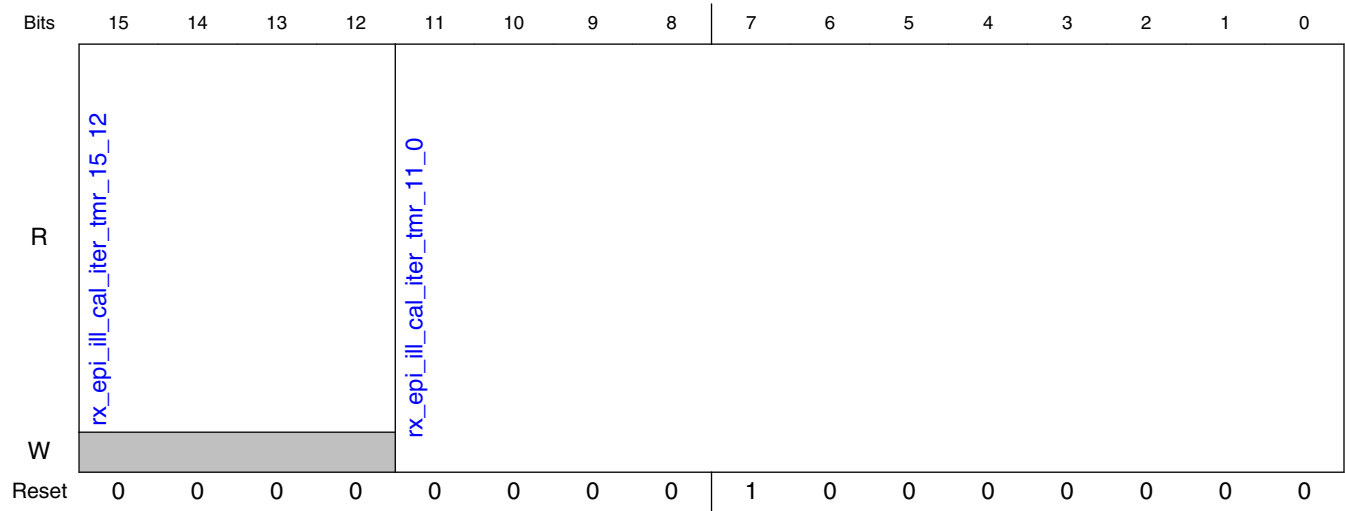
Field	Function
15-12 rx_epi_ill_cal_init_tmr_15_12	Reserved
11-0 rx_epi_ill_cal_init_tmr_11_0	Initialization wait timer value: This is the number of

13.4.10.2.119 RX E PI ILL calibration iteration timer register (lane0_rx_epi_ill_cal_iter_tmr - lane3_rx_epi_ill_cal_iter_tmr)

13.4.10.2.119.1 Offset

Register	Offset
lane0_rx_epi_ill_cal_iter_tmr	8035h
lane1_rx_epi_ill_cal_iter_tmr	8435h
lane2_rx_epi_ill_cal_iter_tmr	8835h
lane3_rx_epi_ill_cal_iter_tmr	8C35h

13.4.10.2.119.2 Diagram



13.4.10.2.119.3 Fields

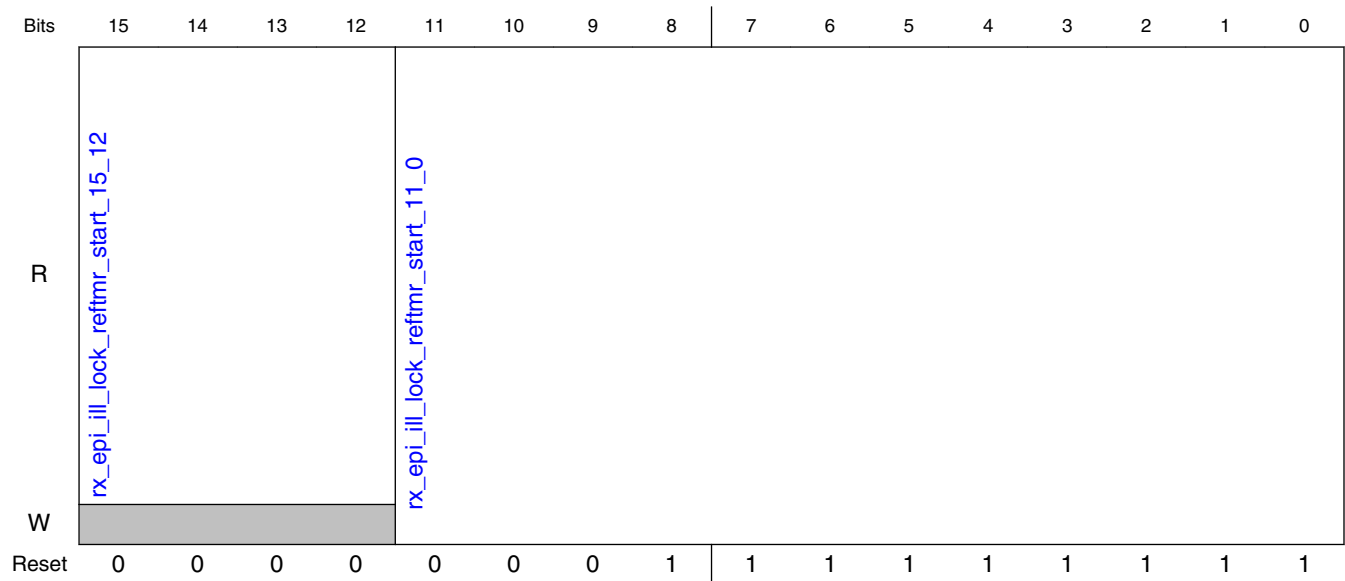
Field	Function
15-12 rx_epi_ill_cal_ite r_tmr_15_12	Reserved
11-0 rx_epi_ill_cal_ite r_tmr_11_0	Iteration wait timer value: This is the number of

13.4.10.2.120 RX E PI ILL lock reference timer start value register (lane0_rx_epi_ill_lock_reftmr_start - lane3_rx_epi_ill_lock_reftmr_start)

13.4.10.2.120.1 Offset

Register	Offset
lane0_rx_epi_ill_lock_reftmr_start	8036h
lane1_rx_epi_ill_lock_reftmr_start	8436h
lane2_rx_epi_ill_lock_reftmr_start	8836h
lane3_rx_epi_ill_lock_reftmr_start	8C36h

13.4.10.2.120.2 Diagram



13.4.10.2.120.3 Fields

Field	Function
15-12 rx_epi_ill_lock_refmr_start_15_12	Reserved
11-0 rx_epi_ill_lock_refmr_start_11_0	ILL lock reference timer start value : This is the value that is loaded into the ILL lock timer as the starting point for that timer, when checking for ILL lock. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes. Note that the counter counts down to 0, therefore this is a value of 1 less than value required for the timer function.

13.4.10.2.121 RX E PI ILL lock calibration counter start value standard mode 0 register (lane0_rx_epi_ill_lock_calcnt_start_0 - lane 3_rx_epi_ill_lock_calcnt_start_0)

13.4.10.2.121.1 Offset

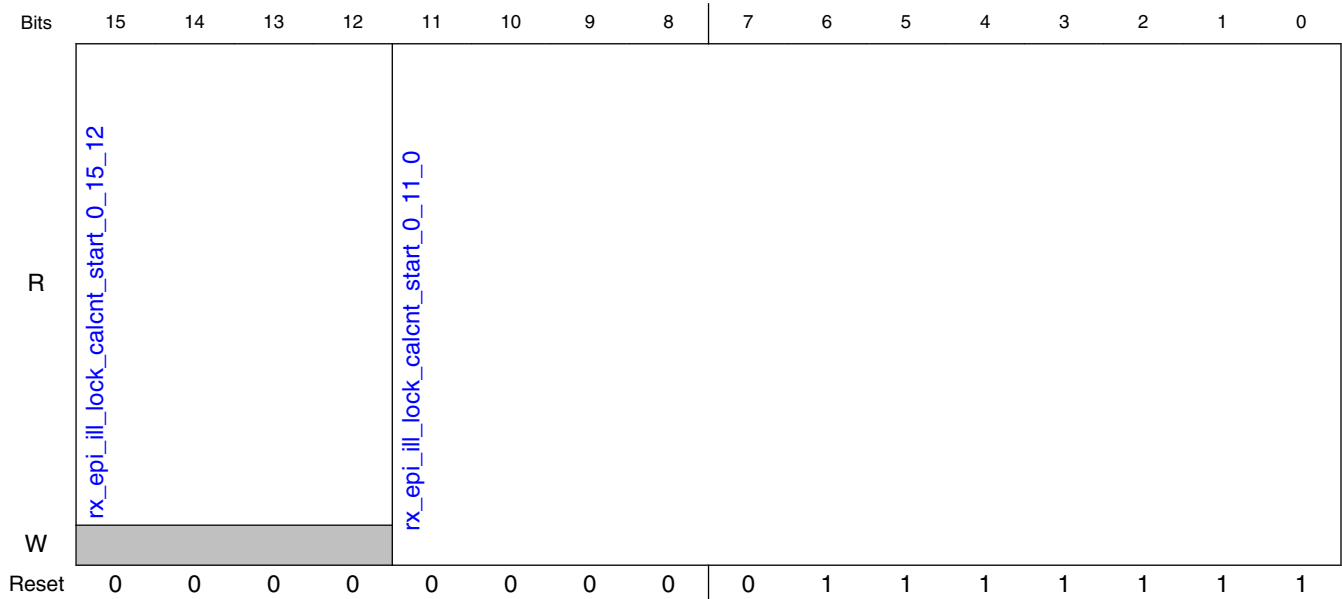
Register	Offset
lane0_rx_epi_ill_lock_calcnt_start_0	8038h
lane1_rx_epi_ill_lock_calcnt_start_0	8438h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane2_rx_epi_ill_lock_calcnt_start_0	8838h
lane3_rx_epi_ill_lock_calcnt_start_0	8C38h

13.4.10.2.121.2 Diagram



13.4.10.2.121.3 Fields

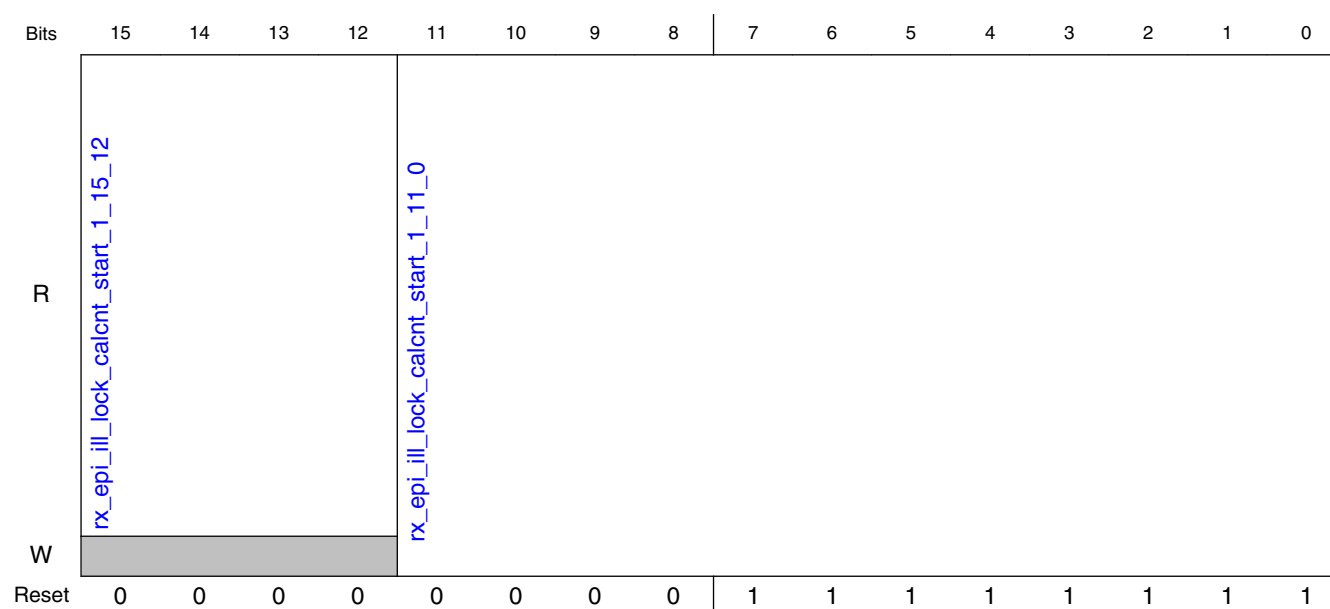
Field	Function
15-12 rx_epi_ill_lock_calcnt_start_0_15_12	Reserved
11-0 rx_epi_ill_lock_calcnt_start_0_11_0	ILL lock calibration counter start value : This is the value that is loaded into the ILL lock calibration counter as the starting point for that counter, when checking for ILL lock when

13.4.10.2.122 RX E PI ILL lock calibration counter start value standard mode 1 register (lane0_rx_epi_ill_lock_calcnt_start_1 - lane3_rx_epi_ill_lock_calcnt_start_1)

13.4.10.2.122.1 Offset

Register	Offset
lane0_rx_epi_ill_lock_calcnt_start_1	8039h
lane1_rx_epi_ill_lock_calcnt_start_1	8439h
lane2_rx_epi_ill_lock_calcnt_start_1	8839h
lane3_rx_epi_ill_lock_calcnt_start_1	8C39h

13.4.10.2.122.2 Diagram



13.4.10.2.122.3 Fields

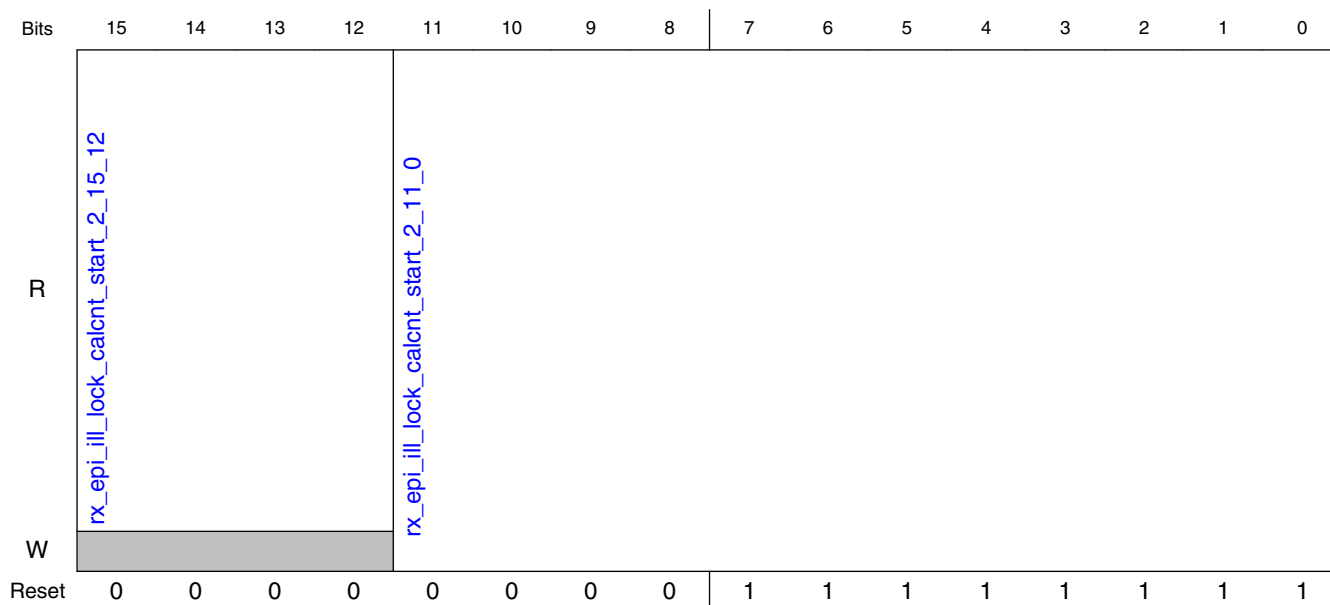
Field	Function
15-12 <code>rx_epi_ill_lock_calcnt_start_1_15_12</code>	Reserved
11-0 <code>rx_epi_ill_lock_calcnt_start_1_11_0</code>	ILL lock calibration counter start value : This is the value that is loaded into the ILL lock calibration counter as the starting point for that counter, when checking for ILL lock when

13.4.10.2.123 RX EPI ILL lock calibration counter start value standard mode 2 register (lane0_rx_epi_ill_lock_calcnt_start_2 - lane 3_rx_epi_ill_lock_calcnt_start_2)

13.4.10.2.123.1 Offset

Register	Offset
lane0_rx_epi_ill_lock_calcnt_start_2	803Ah
lane1_rx_epi_ill_lock_calcnt_start_2	843Ah
lane2_rx_epi_ill_lock_calcnt_start_2	883Ah
lane3_rx_epi_ill_lock_calcnt_start_2	8C3Ah

13.4.10.2.123.2 Diagram



13.4.10.2.123.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

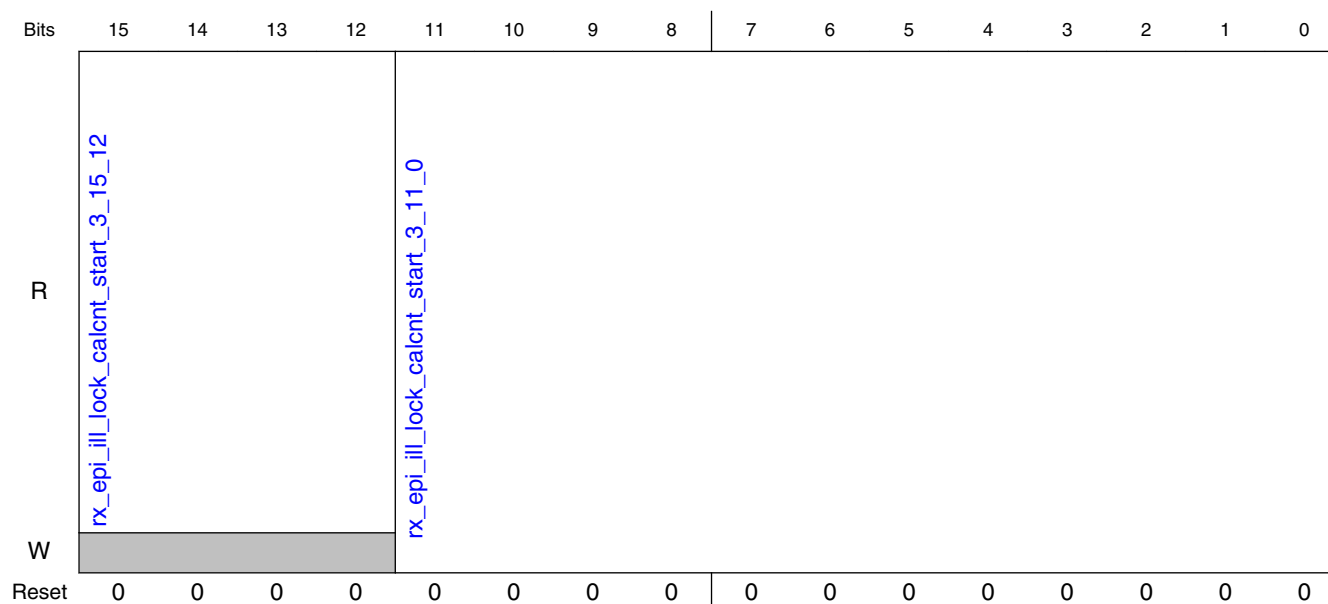
Field	Function
rx_epi_ill_lock_calcnt_start_2_15_12	
11-0 rx_epi_ill_lock_calcnt_start_2_11_0	ILL lock calibration counter start value : This is the value that is loaded into the ILL lock calibration counter as the starting point for that counter, when checking when

13.4.10.2.124 RX E PI ILL lock calibration counter start value standard mode 3 register (lane0_rx_epi_ill_lock_calcnt_start_3 - lane 3_rx_epi_ill_lock_calcnt_start_3)

13.4.10.2.124.1 Offset

Register	Offset
lane0_rx_epi_ill_lock_calcnt_start_3	803Bh
lane1_rx_epi_ill_lock_calcnt_start_3	843Bh
lane2_rx_epi_ill_lock_calcnt_start_3	883Bh
lane3_rx_epi_ill_lock_calcnt_start_3	8C3Bh

13.4.10.2.124.2 Diagram



13.4.10.2.124.3 Fields

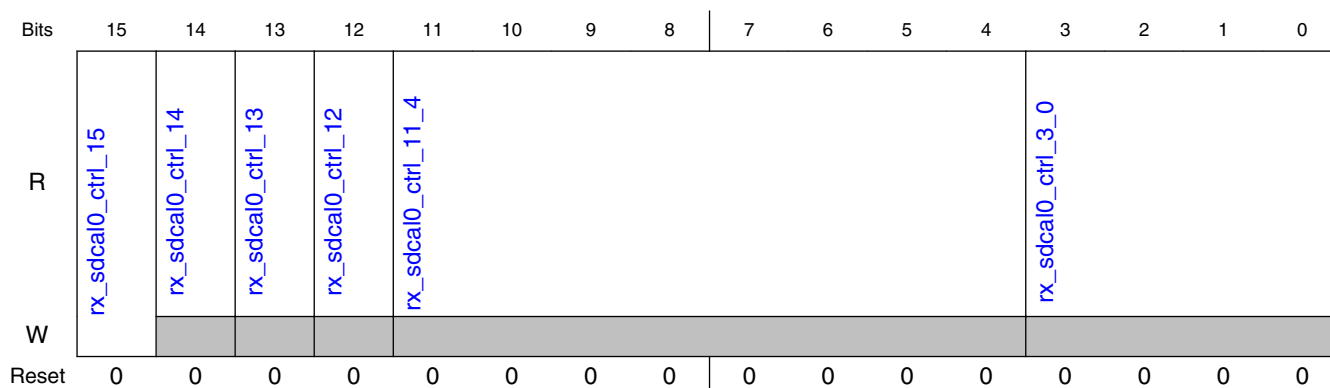
Field	Function
15-12 rx_epi_ill_lock_c alcnt_start_3_15 _12	Reserved
11-0 rx_epi_ill_lock_c alcnt_start_3_11 _0	ILL lock calibration counter start value : This is the value that is loaded into the ILL lock calibration counter as the starting point for that counter, when checking when

13.4.10.2.125 Signal detect calibration 0 control register (lane0_rx_sdcal0_ctrl - lane3_rx_sdcal0_ctrl)

13.4.10.2.125.1 Offset

Register	Offset
lane0_rx_sdcal0_ctrl	8040h
lane1_rx_sdcal0_ctrl	8440h
lane2_rx_sdcal0_ctrl	8840h
lane3_rx_sdcal0_ctrl	8C40h

13.4.10.2.125.2 Diagram



13.4.10.2.125.3 Fields

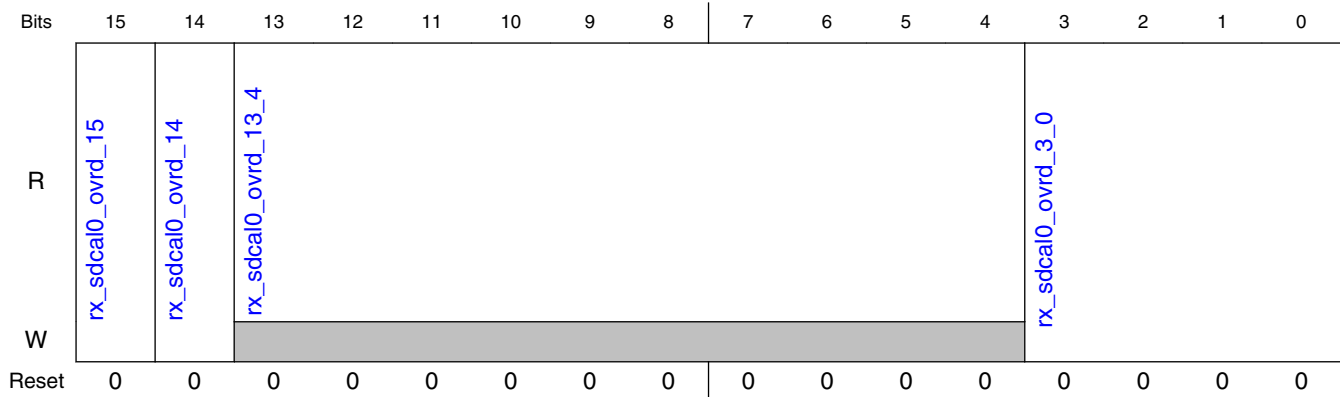
Field	Function
15 rx_sdcal0_ctrl_1_5	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the bandbap calibration process done bit in this register is cleared.
14 rx_sdcal0_ctrl_1_4	Calibration process done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_sdcal0_ctrl_1_3	No analog calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached.
12 rx_sdcal0_ctrl_1_2	Current analog comparator response: This is the current state of the analog comparator response signal (
11-4 rx_sdcal0_ctrl_1_1_4	Reserved
3-0 rx_sdcal0_ctrl_3_0	Calibration code: This is the calibration code that was determined by the calibration process. The following is the encoding of this field, and how it maps to the

13.4.10.2.126 Signal detect calibration 0 override register (lane0_rx_sdcal0_ovrd - lane3_rx_sdcal0_ovrd)

13.4.10.2.126.1 Offset

Register	Offset
lane0_rx_sdcal0_ovrd	8041h
lane1_rx_sdcal0_ovrd	8441h
lane2_rx_sdcal0_ovrd	8841h
lane3_rx_sdcal0_ovrd	8C41h

13.4.10.2.126.2 Diagram



13.4.10.2.126.3 Fields

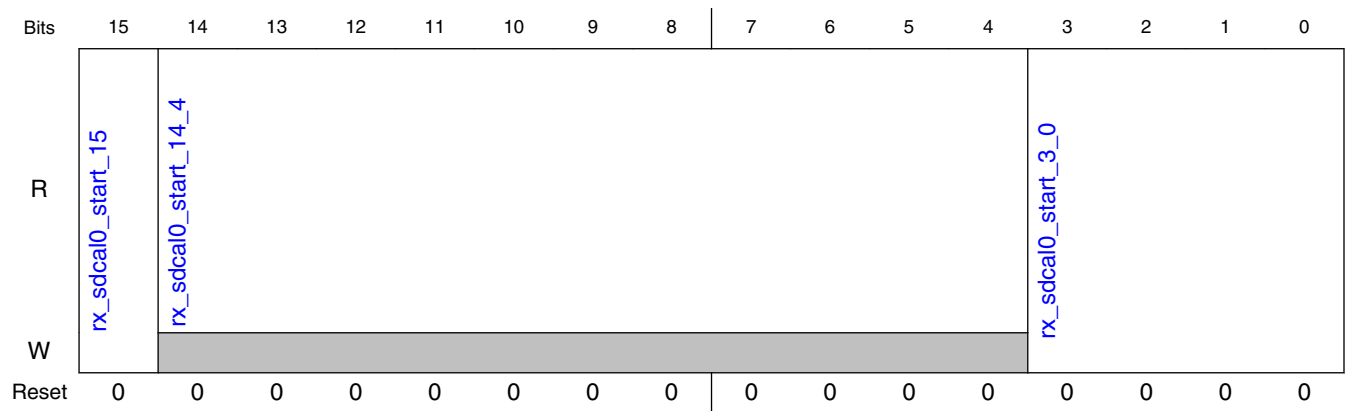
Field	Function
15 rx_sdcal0_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the codes determined during the automatic calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_sdcal0_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-4 rx_sdcal0_ovrd_13_4	Reserved
3-0 rx_sdcal0_ovrd_3_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic calibration process. The code written to these bits is valid when the calibration code override enable bit in this register is active.

13.4.10.2.127 Signal detect calibration 0 start register (lane0_rx_sdcal0_start - lane3_rx_sdcal0_start)

13.4.10.2.127.1 Offset

Register	Offset
lane0_rx_sdcal0_start	8042h
lane1_rx_sdcal0_start	8442h
lane2_rx_sdcal0_start	8842h
lane3_rx_sdcal0_start	8C42h

13.4.10.2.127.2 Diagram



13.4.10.2.127.3 Fields

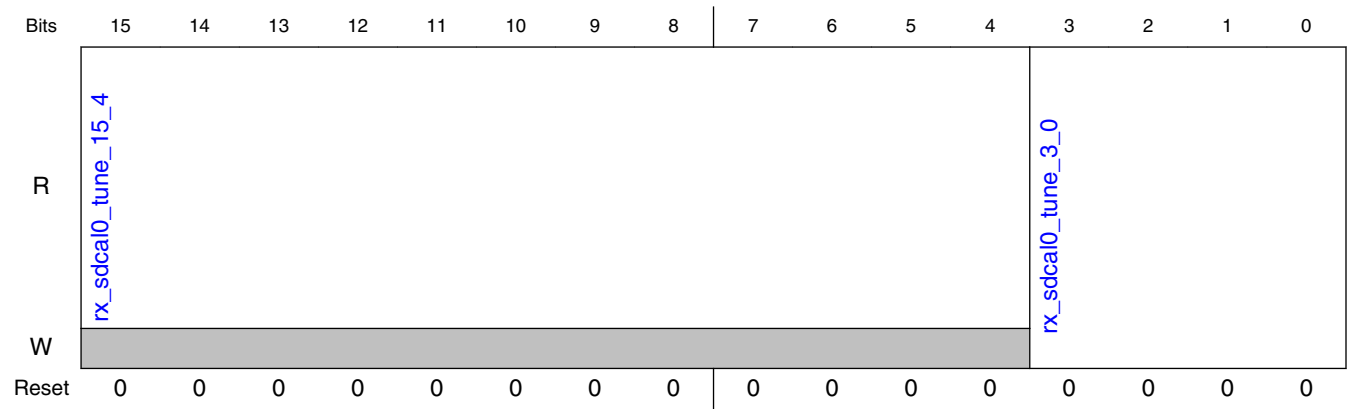
Field	Function
15 rx_sdcal0_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-4 rx_sdcal0_start_14_4	Reserved
3-0 rx_sdcal0_start_3_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run.

13.4.10.2.128 Signal detect calibration 0 tune register (lane0_rx_sdcal0_tune - lane3_rx_sdcal0_tune)

13.4.10.2.128.1 Offset

Register	Offset
lane0_rx_sdcal0_tune	8043h
lane1_rx_sdcal0_tune	8443h
lane2_rx_sdcal0_tune	8843h
lane3_rx_sdcal0_tune	8C43h

13.4.10.2.128.2 Diagram



13.4.10.2.128.3 Fields

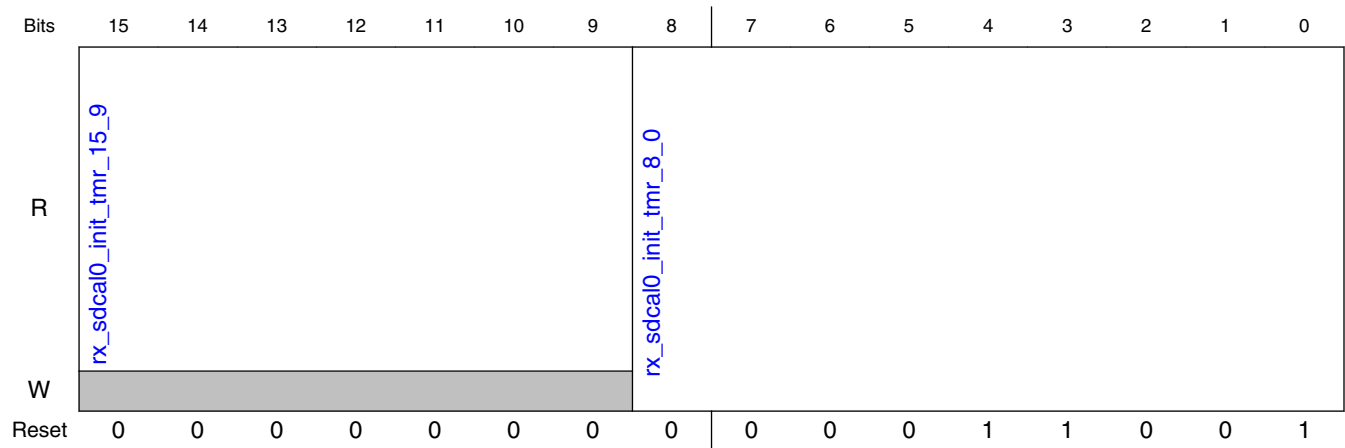
Field	Function
15-4 rx_sdcal0_tune_15_4	Reserved
3-0 rx_sdcal0_tune_3_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.129 Signal detect calibration 0 initialization timer register (lane0_rx_sdcal0_init_tmr - lane3_rx_sdcal0_init_tmr)

13.4.10.2.129.1 Offset

Register	Offset
lane0_rx_sdcal0_init_tmr	8044h
lane1_rx_sdcal0_init_tmr	8444h
lane2_rx_sdcal0_init_tmr	8844h
lane3_rx_sdcal0_init_tmr	8C44h

13.4.10.2.129.2 Diagram



13.4.10.2.129.3 Fields

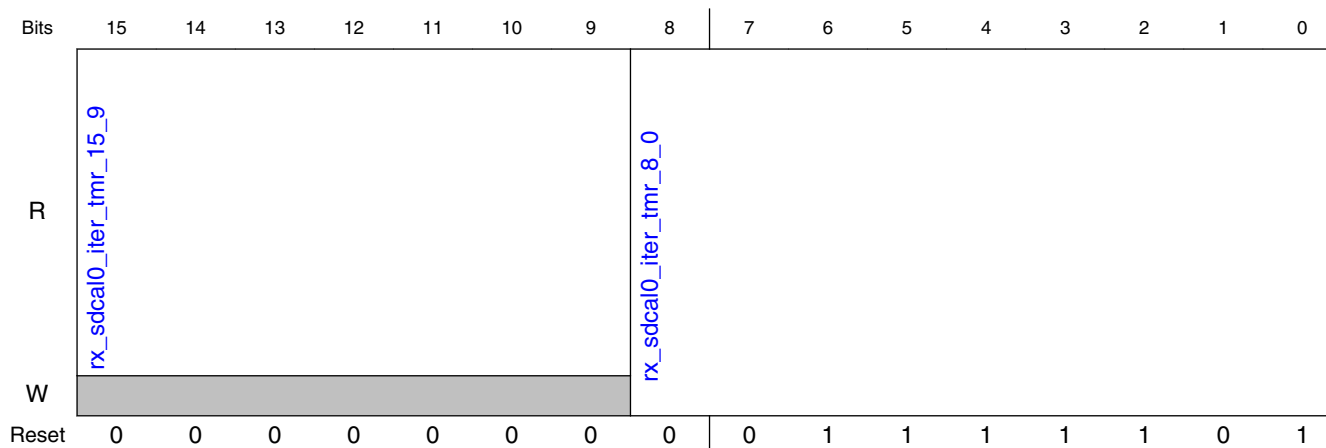
Field	Function
15-9 rx_sdcal0_init_tmr_15_9	Reserved
8-0 rx_sdcal0_init_tmr_8_0	Initialization wait timer value: This is the number of

13.4.10.2.130 Signal detect calibration 0 iteration timer register (lane0_rx_sdcal0_iter_tmr - lane3_rx_sdcal0_iter_tmr)

13.4.10.2.130.1 Offset

Register	Offset
lane0_rx_sdcal0_iter_tmr	8045h
lane1_rx_sdcal0_iter_tmr	8445h
lane2_rx_sdcal0_iter_tmr	8845h
lane3_rx_sdcal0_iter_tmr	8C45h

13.4.10.2.130.2 Diagram



13.4.10.2.130.3 Fields

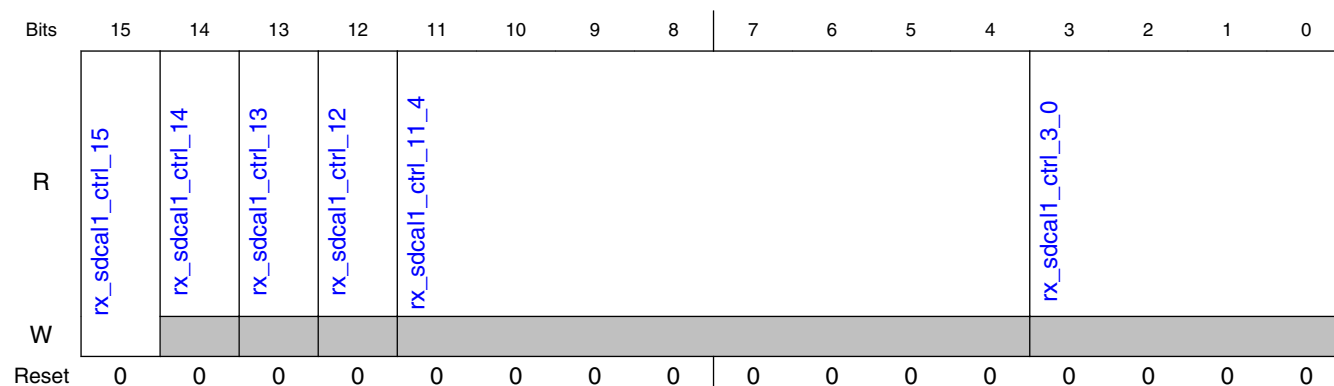
Field	Function
15-9 rx_sdcal0_iter_tmr_15_9	Reserved
8-0 rx_sdcal0_iter_tmr_8_0	Iteration wait timer value: This is the number of

13.4.10.2.131 Signal detect calibration 1 control register (lane0_rx_sdcal1_ctrl - lane3_rx_sdcal1_ctrl)

13.4.10.2.131.1 Offset

Register	Offset
lane0_rx_sdcal1_ctrl	8048h
lane1_rx_sdcal1_ctrl	8448h
lane2_rx_sdcal1_ctrl	8848h
lane3_rx_sdcal1_ctrl	8C48h

13.4.10.2.131.2 Diagram



13.4.10.2.131.3 Fields

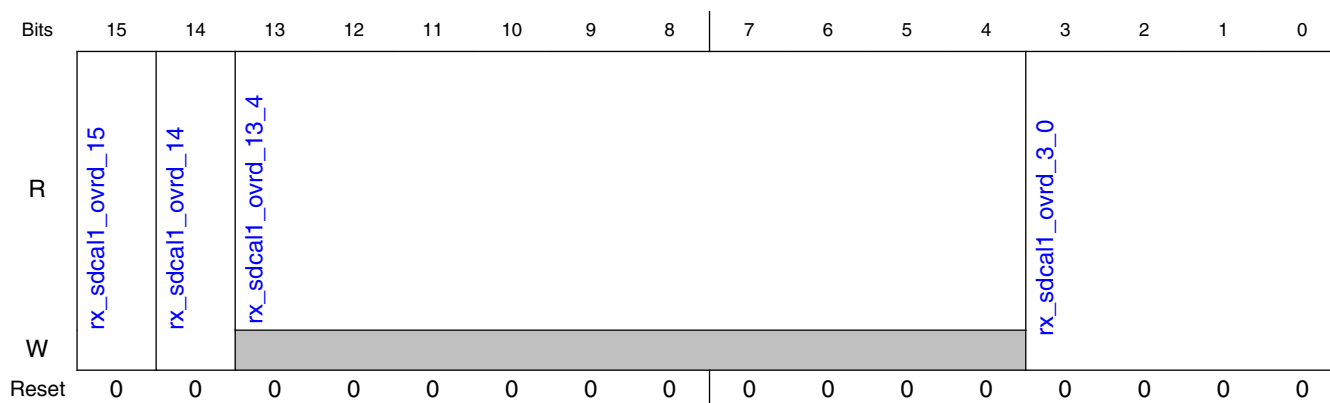
Field	Function
15 rx_sdcal1_ctrl_1_5	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the bandbap calibration process done bit in this register is cleared.
14 rx_sdcal1_ctrl_1_4	Calibration process done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_sdcal1_ctrl_1_3	No analog calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached.
12 rx_sdcal1_ctrl_1_2	Current analog comparator response: This is the current state of the analog comparator response signal (
11-4 rx_sdcal1_ctrl_1_1_4	Reserved
3-0 rx_sdcal1_ctrl_3_0	Calibration code: This is the calibration code that was determined by the calibration process. The following is the encoding of this field, and how it maps to the

13.4.10.2.132 Signal detect calibration 1 override register (lane0_rx_sdcal1_ovrd - lane3_rx_sdcal1_ovrd)

13.4.10.2.132.1 Offset

Register	Offset
lane0_rx_sdcal1_ovrd	8049h
lane1_rx_sdcal1_ovrd	8449h
lane2_rx_sdcal1_ovrd	8849h
lane3_rx_sdcal1_ovrd	8C49h

13.4.10.2.132.2 Diagram



13.4.10.2.132.3 Fields

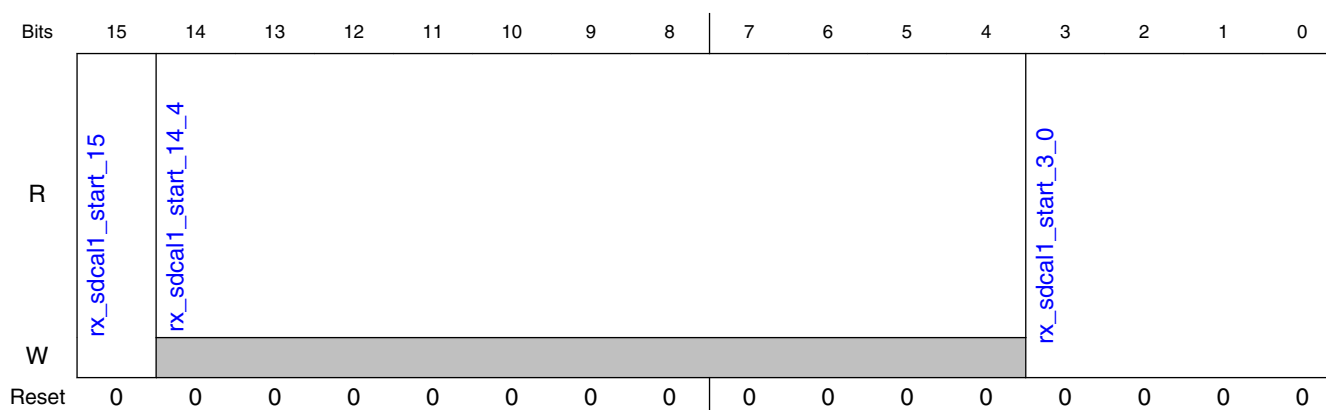
Field	Function
15 rx_sdcal1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_sdcal1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-4 rx_sdcal1_ovrd_13_4	Reserved
3-0 rx_sdcal1_ovrd_3_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic calibration process. The code written to these bits is valid when the calibration code override enable bit in this register is active.

13.4.10.2.133 Signal detect calibration 1 start register (lane0_rx_sdcal1_start - lane3_rx_sdcal1_start)

13.4.10.2.133.1 Offset

Register	Offset
lane0_rx_sdcal1_start	804Ah
lane1_rx_sdcal1_start	844Ah
lane2_rx_sdcal1_start	884Ah
lane3_rx_sdcal1_start	8C4Ah

13.4.10.2.133.2 Diagram



13.4.10.2.133.3 Fields

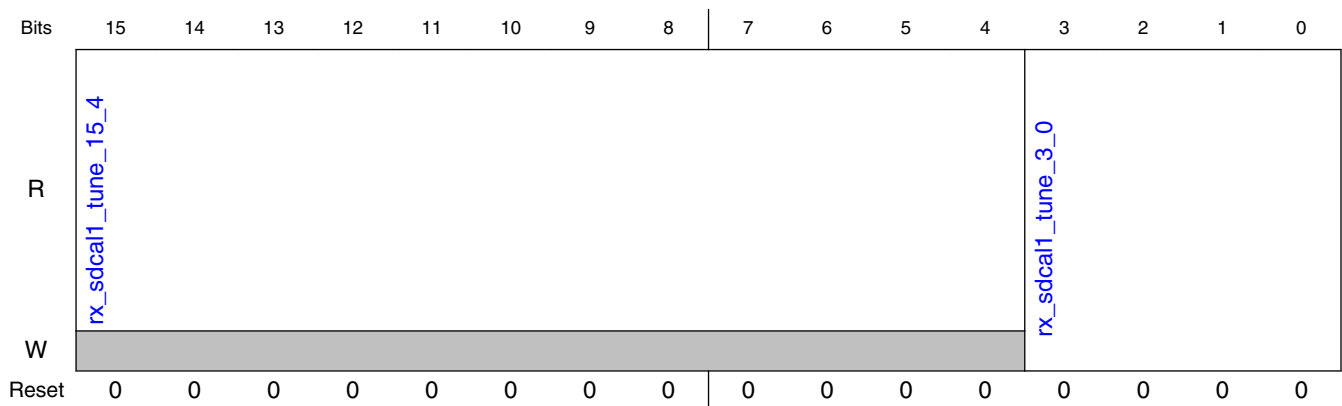
Field	Function
15 rx_sdcal1_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-4 rx_sdcal1_start_14_4	Reserved
3-0 rx_sdcal1_start_3_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run.

13.4.10.2.134 Signal detect calibration 1 tune register (lane0_rx_sdcal1_tune - lane3_rx_sdcal1_tune)

13.4.10.2.134.1 Offset

Register	Offset
lane0_rx_sdcal1_tune	804Bh
lane1_rx_sdcal1_tune	844Bh
lane2_rx_sdcal1_tune	884Bh
lane3_rx_sdcal1_tune	8C4Bh

13.4.10.2.134.2 Diagram



13.4.10.2.134.3 Fields

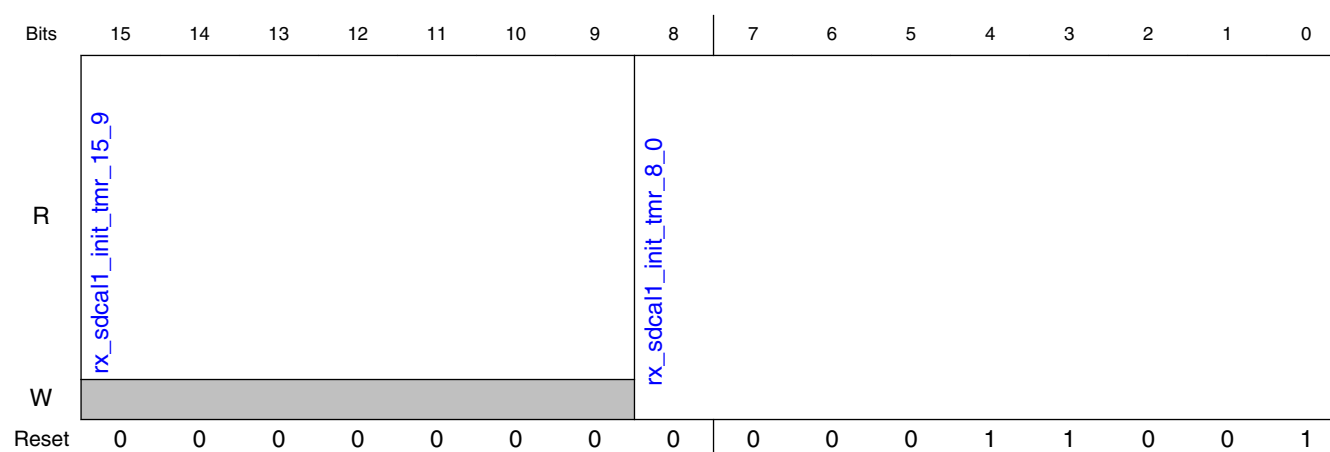
Field	Function
15-4 rx_sdcal1_tune_15_4	Reserved
3-0 rx_sdcal1_tune_3_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.135 Signal detect calibration 1 initialization timer register (lane0_rx_sdcal1_init_tmr - lane3_rx_sdcal1_init_tmr)

13.4.10.2.135.1 Offset

Register	Offset
lane0_rx_sdcal1_init_tmr	804Ch
lane1_rx_sdcal1_init_tmr	844Ch
lane2_rx_sdcal1_init_tmr	884Ch
lane3_rx_sdcal1_init_tmr	8C4Ch

13.4.10.2.135.2 Diagram



13.4.10.2.135.3 Fields

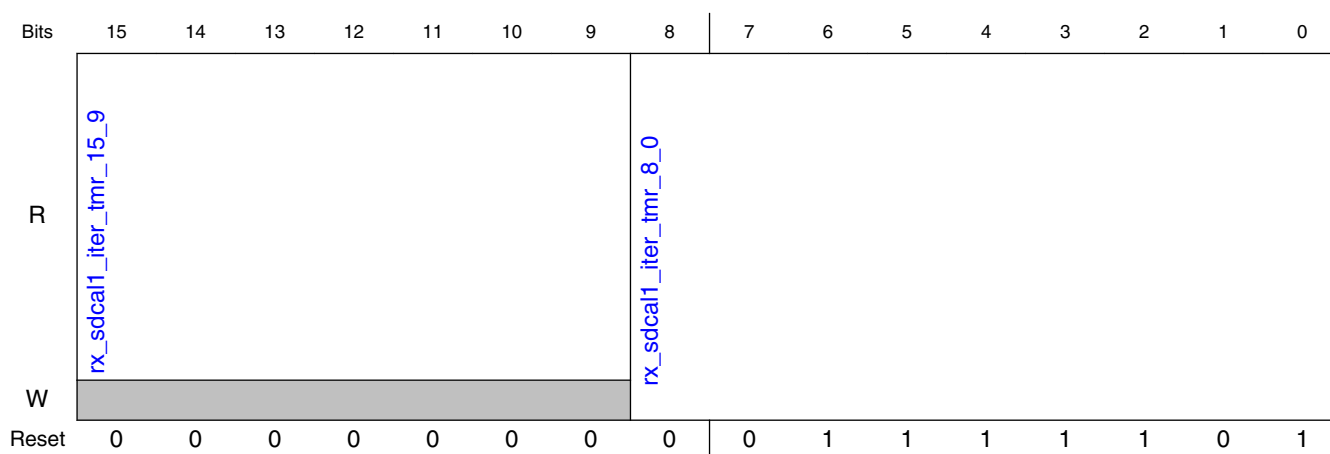
Field	Function
15-9 rx_sdcal1_init_tmr_15_9	Reserved
8-0 rx_sdcal1_init_tmr_8_0	Initialization wait timer value: This is the number of

13.4.10.2.136 Signal detect calibration 1 iteration timer register (lane0_rx_sdcal1_iter_tmr - lane3_rx_sdcal1_iter_tmr)

13.4.10.2.136.1 Offset

Register	Offset
lane0_rx_sdcal1_iter_tmr	804Dh
lane1_rx_sdcal1_iter_tmr	844Dh
lane2_rx_sdcal1_iter_tmr	884Dh
lane3_rx_sdcal1_iter_tmr	8C4Dh

13.4.10.2.136.2 Diagram



13.4.10.2.136.3 Fields

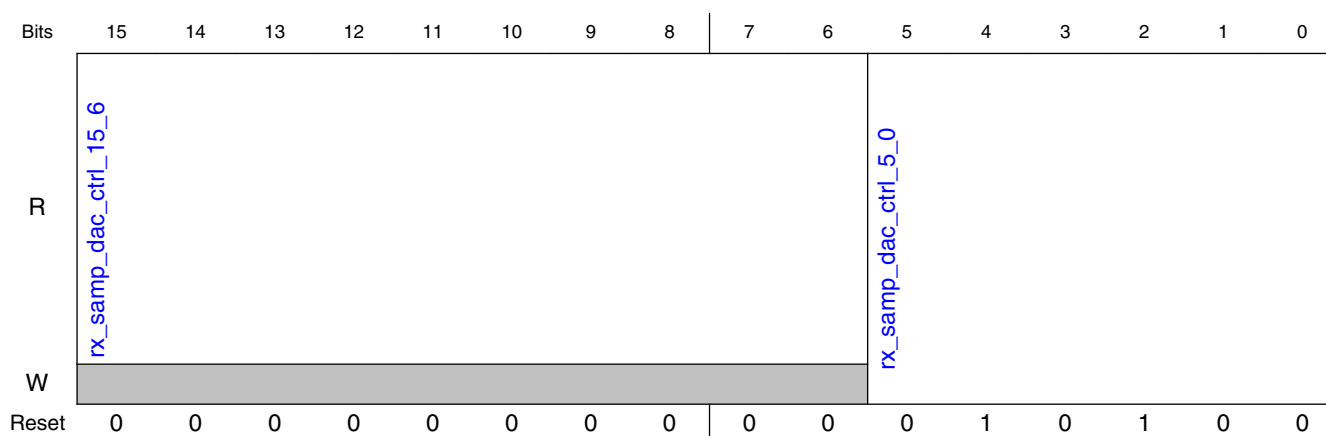
Field	Function
15-9 <code>rx_sdcal1_iter_tmr_15_9</code>	Reserved
8-0 <code>rx_sdcal1_iter_tmr_8_0</code>	Iteration wait timer value: This is the number of

13.4.10.2.137 Sampler error DAC control register (lane0_rx_samp_dac_ctrl - lane3_rx_samp_dac_ctrl)

13.4.10.2.137.1 Offset

Register	Offset
lane0_rx_samp_dac_ctrl	8058h
lane1_rx_samp_dac_ctrl	8458h
lane2_rx_samp_dac_ctrl	8858h
lane3_rx_samp_dac_ctrl	8C58h

13.4.10.2.137.2 Diagram



13.4.10.2.137.3 Fields

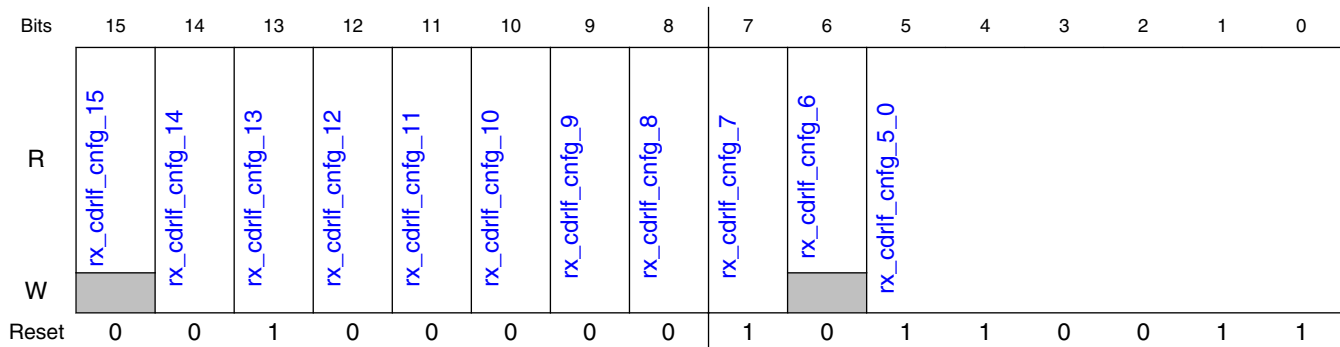
Field	Function
15-6 rx_samp_dac_ctrl_15_6	Reserved
5-0 rx_samp_dac_ctrl_5_0	Sampler error DAC value: Specifies the input value to the sampler error DAC. This value is a twos complement binary number, with the range specified below.

13.4.10.2.138 CDRLF configuration register (lane0_rx_cdrlf_cnfg - lane3_rx_cdrlf_cnfg)

13.4.10.2.138.1 Offset

Register	Offset
lane0_rx_cdrif_cfg	8080h
lane1_rx_cdrif_cfg	8480h
lane2_rx_cdrif_cfg	8880h
lane3_rx_cdrif_cfg	8C80h

13.4.10.2.138.2 Diagram



13.4.10.2.138.3 Fields

Field	Function
15 rx_cdrif_cfg_15	CDRLF fast phase lock locked detected: This register bit is the current status of the
14 rx_cdrif_cfg_14	CDRLF fast phase lock diagnostic enable: Setting this bit will force the
13 rx_cdrif_cfg_13	CDRLF fast phase lock enable: Controls the enable of the fast phase lock function.
12 rx_cdrif_cfg_12	CDRLF fast frequency lock enable: Controls the enable of the fast frequency lock function.
11 rx_cdrif_cfg_11	CDRLF second order loop integrator max clear enable: This signal enables the function in the CDRLF where the second order loop integrator is cleared when it reaches the maximum value.
10 rx_cdrif_cfg_10	CDRLF reset on CDRLF PM Accumulator Max: Activating (1b1) this bit will force the CDRLF to be reset when the PM accumulator in the CDRLF reaches is maximum absolute value (the largest positive or negative value).
9 rx_cdrif_cfg_9	CDRLF freeze on electrical idle detect: Activating (1b1) this bit will force the CDRLF to be freeze in its current state when the receiver signal detect detects an electrical idle. When high speed data is detected, the freeze will be released.

Table continues on the next page...

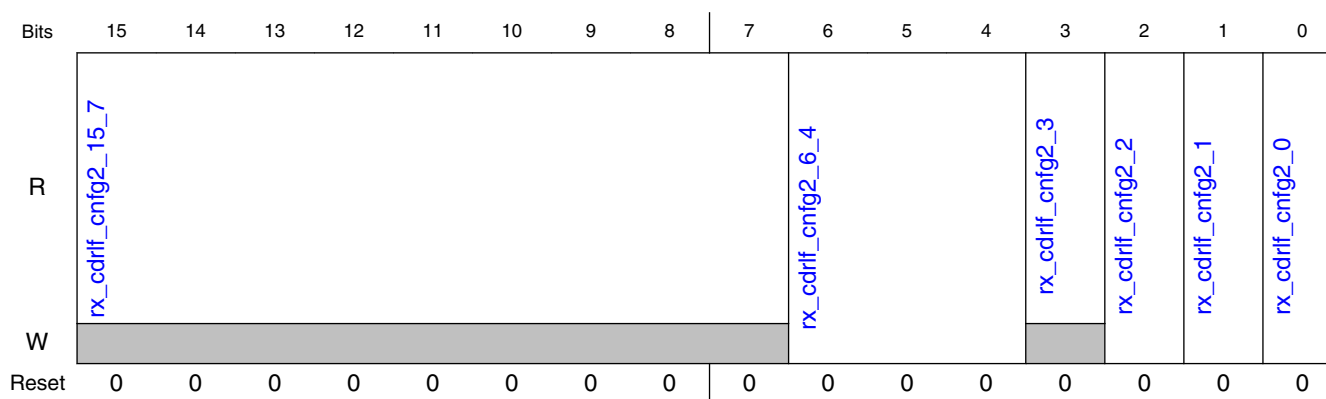
Field	Function
8 rx_cdrif_cfg_8	CDRLF reset on electrical idle detect: Activating (1b1) this bit will force the CDRLF to be reset when the receiver signal detect detects an electrical idle. When high speed data is detected, the reset will be released.
7 rx_cdrif_cfg_7	CDRLF data filter enable : Enables the filter function for the data that feeds into the CDRLF from the deserializer, using the
6 rx_cdrif_cfg_6	Reserved
5-0 rx_cdrif_cfg_5_0	CDRLF second order loop integrator threshold : This value is the maximum magnitude the CDRLF second order loop integrator will be allowed to go to.

13.4.10.2.139 CDRLF configuration register 2 (lane0_rx_cdrif_cfg2 - lane3_rx_cdrif_cfg2)

13.4.10.2.139.1 Offset

Register	Offset
lane0_rx_cdrif_cfg2	8081h
lane1_rx_cdrif_cfg2	8481h
lane2_rx_cdrif_cfg2	8881h
lane3_rx_cdrif_cfg2	8C81h

13.4.10.2.139.2 Diagram



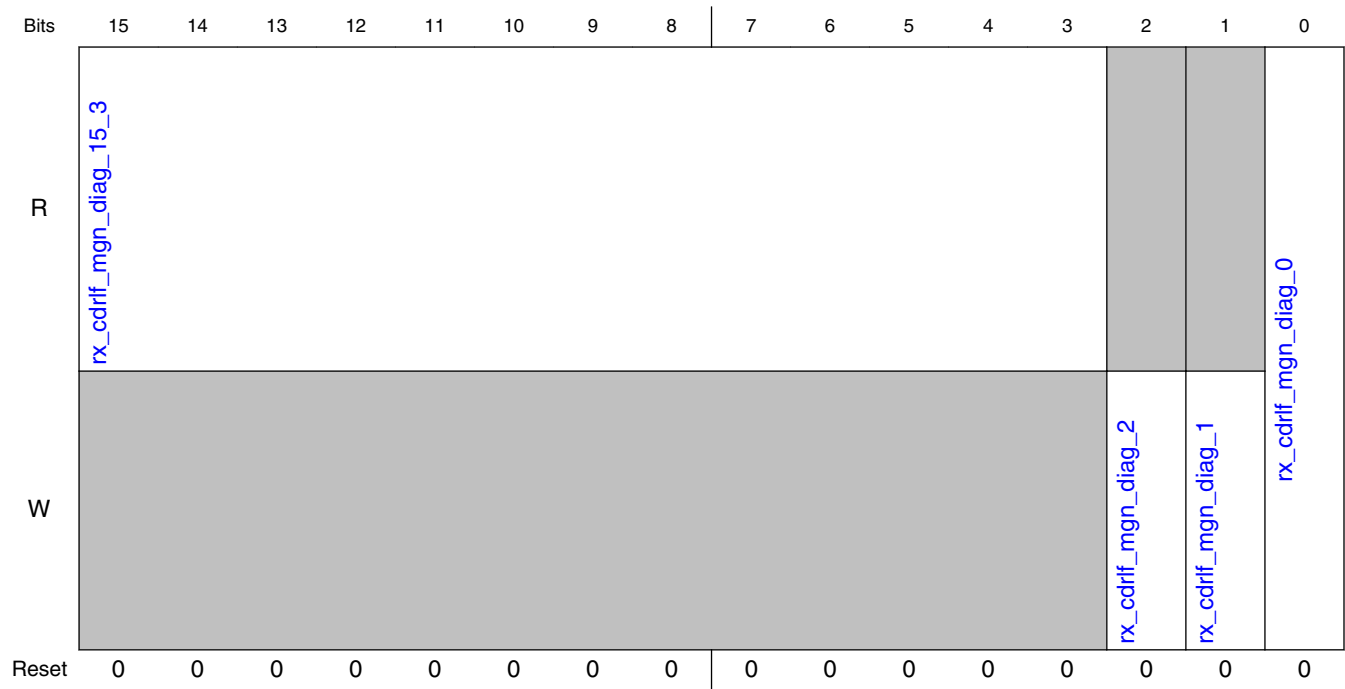
13.4.10.2.139.3 Fields

Field	Function
15-7 rx_cdrif_cfg2_1 5_7	Reserved
6-4 rx_cdrif_cfg2_6 _4	CDRLF diagnostic mode control: This field controls the information driven on the rx_pi_val_in_{15:0}[7:0] signal, when in diagnostics mode.
3 rx_cdrif_cfg2_3	Reserved
2 rx_cdrif_cfg2_2	CDRLF reset hold: When active (1b1), the CDRLF will be held in reset beyond the time that it would normally be released by its asynchronous release signals. The CDRLF will be held in reset until this bit is deactivated (1b0).
1 rx_cdrif_cfg2_1	CDRLF second order loop disable: Activating (1b1) this bit will disable the CDRLF second order loop.
0 rx_cdrif_cfg2_0	CDRLF first order loop disable: Activating (1b1) this bit will disable the CDRLF first order loop.

**13.4.10.2.140 CDRLF margin diagnostic register (lane0_rx_cdrif_mgn_d
iag - lane3_rx_cdrif_mgn_diag)****13.4.10.2.140.1 Offset**

Register	Offset
lane0_rx_cdrif_mgn_diag	8082h
lane1_rx_cdrif_mgn_diag	8482h
lane2_rx_cdrif_mgn_diag	8882h
lane3_rx_cdrif_mgn_diag	8C82h

13.4.10.2.140.2 Diagram



13.4.10.2.140.3 Fields

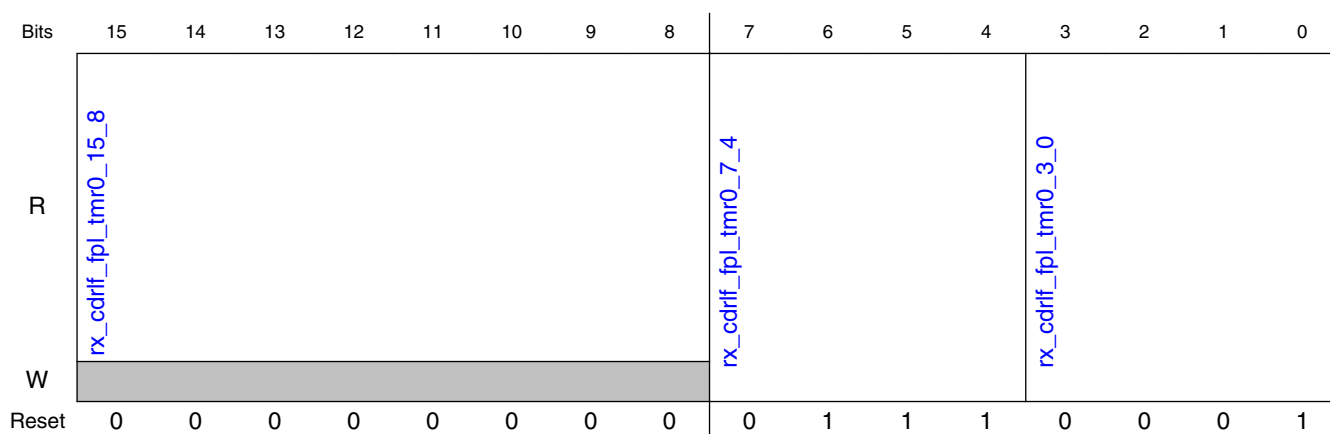
Field	Function
15-3 rx_cdrif_mgn_di ag_15_3	Reserved
2 rx_cdrif_mgn_di ag_2	CDRLF PI override down : When the CDRLF PI override enable function is enabled, writing a 1b1 to this bit will force a down to be generated in the CDRLF PI interface logic.
1 rx_cdrif_mgn_di ag_1	CDRLF PI override up : When the CDRLF PI override enable function is enabled, writing a 1b1 to this bit will force an up to be generated in the CDRLF PI interface logic.
0 rx_cdrif_mgn_di ag_0	CDRLF PI override enable : Setting this bit to 1b1 will enable the CDRLF PI override function, which will allow ups and downs to be forced to the CDRLF PI interface logic from the up and down override bits in this register.

13.4.10.2.141 CDRLF fast phase lock timer value register 0 (lane0_rx_cdr
lf_fpl_tmr0 - lane3_rx_cdrif_fpl_tmr0)

13.4.10.2.141.1 Offset

Register	Offset
lane0_rx_cdrif_fpl_tmr0	8083h
lane1_rx_cdrif_fpl_tmr0	8483h
lane2_rx_cdrif_fpl_tmr0	8883h
lane3_rx_cdrif_fpl_tmr0	8C83h

13.4.10.2.141.2 Diagram



13.4.10.2.141.3 Fields

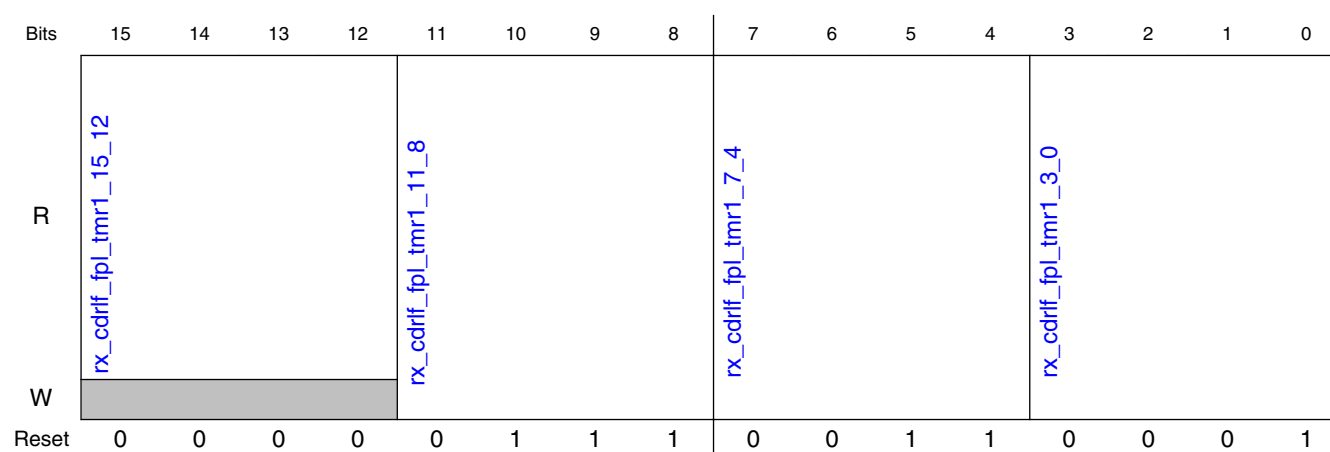
Field	Function
15-8 rx_cdrif_fpl_tmr0_15_8	Reserved
7-4 rx_cdrif_fpl_tmr0_7_4	Fast phase lock timer accumulate state time value : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the accumulate state.
3-0 rx_cdrif_fpl_tmr0_3_0	Fast phase lock timer delay state time value : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the delay state.

13.4.10.2.142 CDRLF fast phase lock timer value register 1 (lane0_rx_cdrif_fpl_tmr1 - lane3_rx_cdrif_fpl_tmr1)

13.4.10.2.142.1 Offset

Register	Offset
lane0_rx_cdrif_fpl_tmr1	8084h
lane1_rx_cdrif_fpl_tmr1	8484h
lane2_rx_cdrif_fpl_tmr1	8884h
lane3_rx_cdrif_fpl_tmr1	8C84h

13.4.10.2.142.2 Diagram



13.4.10.2.142.3 Fields

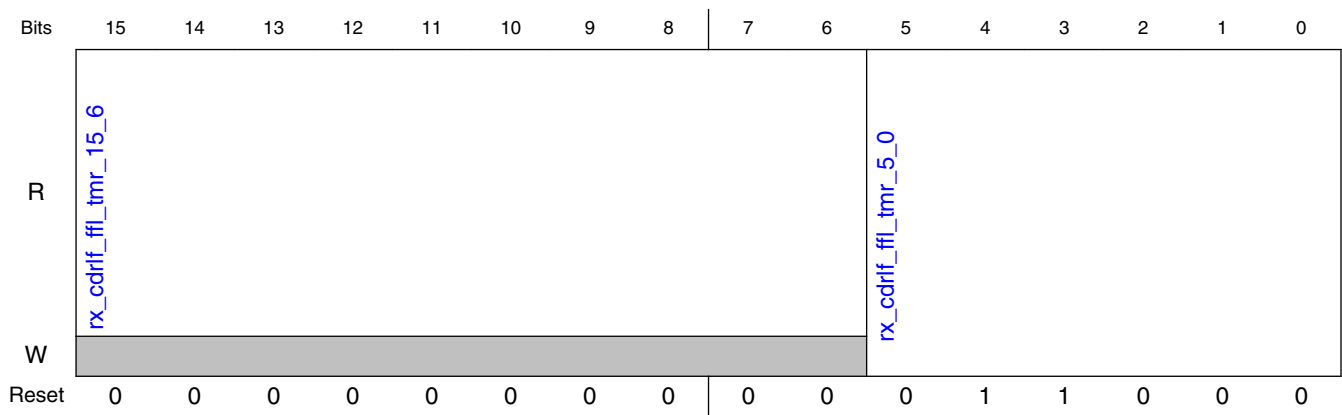
Field	Function
15-12 <code>rx_cdrif_fpl_tmr1_15_12</code>	Reserved
11-8 <code>rx_cdrif_fpl_tmr1_11_8</code>	Fast phase lock timer trigger 1 state time value : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the trigger state the first time it is in that state. The recommended value for this is the number of PI steps required to move the PI 1/4 UI.
7-4 <code>rx_cdrif_fpl_tmr1_7_4</code>	Fast phase lock timer trigger 2 state time value : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the trigger state the second time it is in that state. The recommended value for this is the number of PI steps required to move the PI 1/8 UI.
3-0 <code>rx_cdrif_fpl_tmr1_3_0</code>	Fast phase lock timer trigger 3 state time value : Specifies the number of clock cycles minus 1 that the fast phase lock state machine will remain in the trigger state the third time it is in that state. The recommended value for this is the number of PI steps required to move the PI 1/16 UI.

13.4.10.2.143 CDRLF fast frequency lock timer value register (lane0_rx_cdrif_ffl_tmr - lane3_rx_cdrif_ffl_tmr)

13.4.10.2.143.1 Offset

Register	Offset
lane0_rx_cdrif_ffl_tmr	8085h
lane1_rx_cdrif_ffl_tmr	8485h
lane2_rx_cdrif_ffl_tmr	8885h
lane3_rx_cdrif_ffl_tmr	8C85h

13.4.10.2.143.2 Diagram



13.4.10.2.143.3 Fields

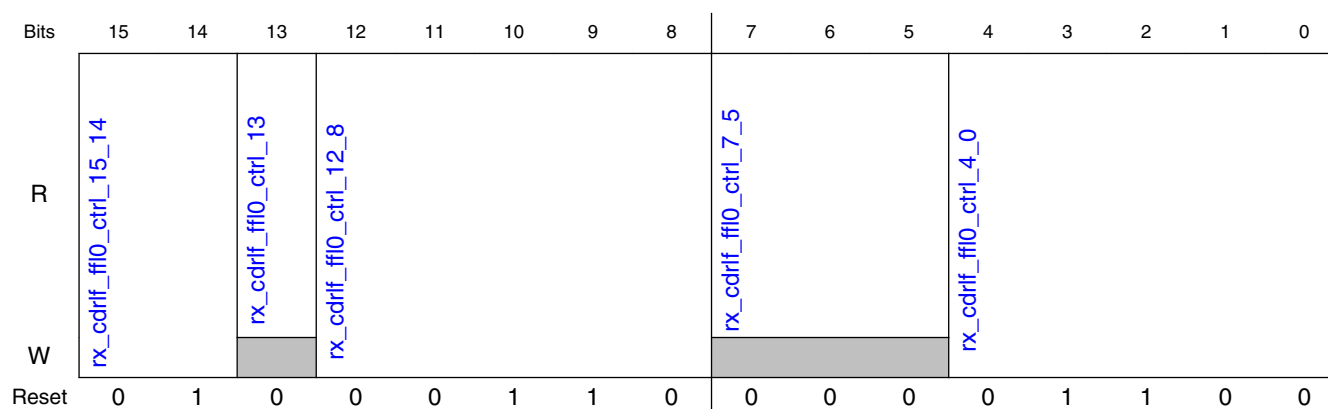
Field	Function
15-6 rx_cdrif_ffl_tmr_15_6	Reserved
5-0 rx_cdrif_ffl_tmr_5_0	Fast frequency lock step timer value : Specifies the number of clock cycles for each step of the fast frequency lock process.

13.4.10.2.144 CDRLF fast frequency lock step 0 control register (lane0_rx_cdrif_ffl0_ctrl - lane3_rx_cdrif_ffl0_ctrl)

13.4.10.2.144.1 Offset

Register	Offset
lane0_rx_cdrif_ffl0_ctrl	8088h
lane1_rx_cdrif_ffl0_ctrl	8488h
lane2_rx_cdrif_ffl0_ctrl	8888h
lane3_rx_cdrif_ffl0_ctrl	8C88h

13.4.10.2.144.2 Diagram



13.4.10.2.144.3 Fields

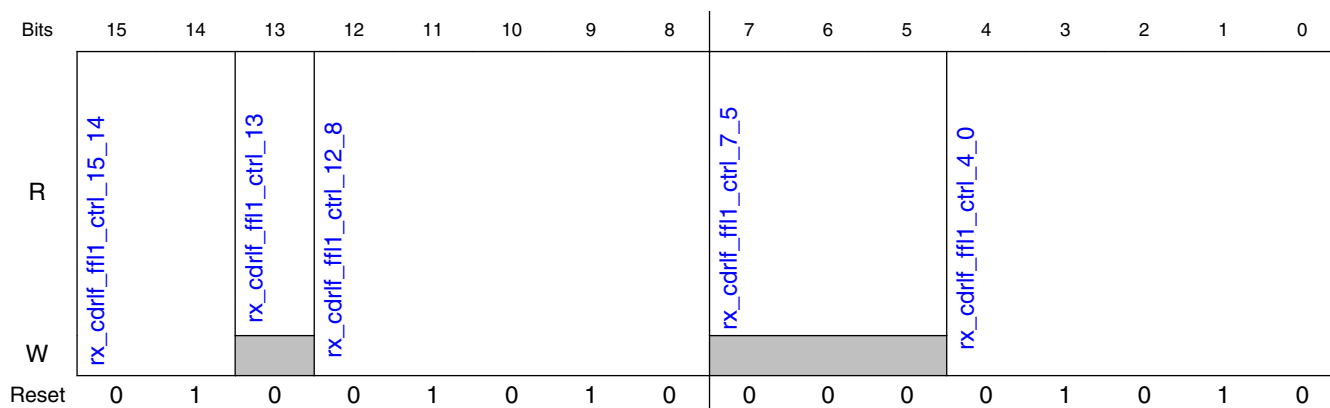
Field	Function
15-14 rx_cdrif_ffl0_ctrl_15_14	FFL step 0 CDRLF second order loop integrator scaler: Scaler value for the second order loop integrator input. The output of the second order loop sigma delta accumulator is multiplied by the value specified by this signal before it is added to the second order loop integrator.
13 rx_cdrif_ffl0_ctrl_13	Reserved
12-8 rx_cdrif_ffl0_ctrl_12_8	FFL step 0 CDRLF second order loop sigma delta update rate: This is the value that is added to or subtracted from the second order loop accumulator register when the serial data sample clock is detected as being out of phase with the serial data on a given parallel data word cycle. This is part of the CDRLF averaging function.
7-5 rx_cdrif_ffl0_ctrl_7_5	Reserved
4-0 rx_cdrif_ffl0_ctrl_4_0	FFL step 0 CDRLF first order loop sigma delta update rate: This is the value that is added to or subtracted from the first order loop accumulator register when the serial data sample clock is detected as being out of phase with the serial data on a given parallel data word cycle. This is part of the CDRLF averaging function.

13.4.10.2.145 CDRLF fast frequency lock step 1 control register (lane0_rx_cdrif_ffl1_ctrl - lane3_rx_cdrif_ffl1_ctrl)

13.4.10.2.145.1 Offset

Register	Offset
lane0_rx_cdrif_ffl1_ctrl	8089h
lane1_rx_cdrif_ffl1_ctrl	8489h
lane2_rx_cdrif_ffl1_ctrl	8889h
lane3_rx_cdrif_ffl1_ctrl	8C89h

13.4.10.2.145.2 Diagram



13.4.10.2.145.3 Fields

Field	Function
15-14 rx_cdrif_ffl1_ctrl_15_14	FFL step 1 CDRLF second order loop integrator scaler
13 rx_cdrif_ffl1_ctrl_13	Reserved
12-8 rx_cdrif_ffl1_ctrl_12_8	FFL step 1 CDRLF second order loop sigma delta update rate
7-5 rx_cdrif_ffl1_ctrl_7_5	Reserved

Table continues on the next page...

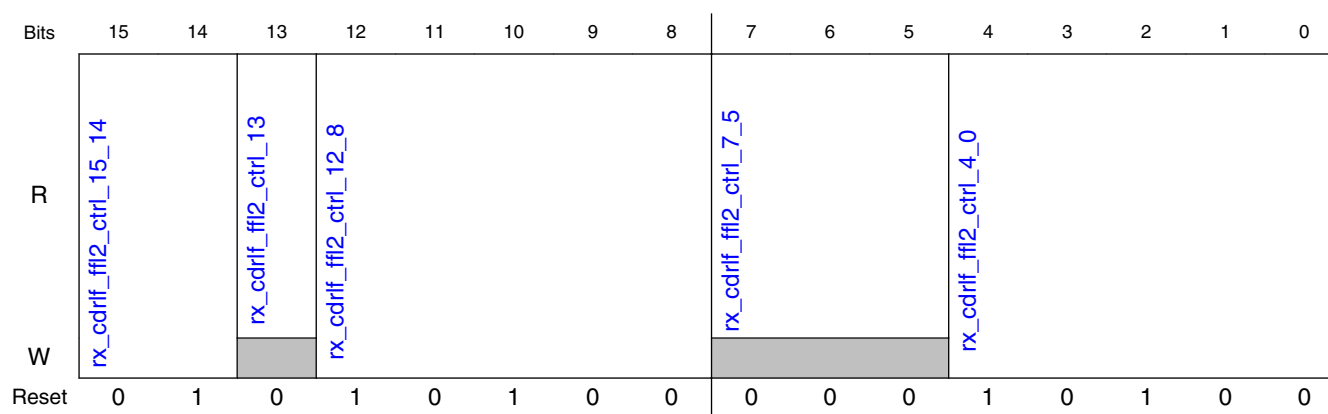
Field	Function
4-0 rx_cdrif_ffl1_ctrl_4_0	FFL step 1 CDRLF first order loop sigma delta update rate

13.4.10.2.146 CDRLF fast frequency lock step 2 control register (lane0_rx_cdrif_ffl2_ctrl - lane3_rx_cdrif_ffl2_ctrl)

13.4.10.2.146.1 Offset

Register	Offset
lane0_rx_cdrif_ffl2_ctrl	808Ah
lane1_rx_cdrif_ffl2_ctrl	848Ah
lane2_rx_cdrif_ffl2_ctrl	888Ah
lane3_rx_cdrif_ffl2_ctrl	8C8Ah

13.4.10.2.146.2 Diagram



13.4.10.2.146.3 Fields

Field	Function
15-14 rx_cdrif_ffl2_ctrl_15_14	FFL step 2 CDRLF second order loop integrator scaler
13 rx_cdrif_ffl2_ctrl_13	Reserved

Table continues on the next page...

Clocks And Resets

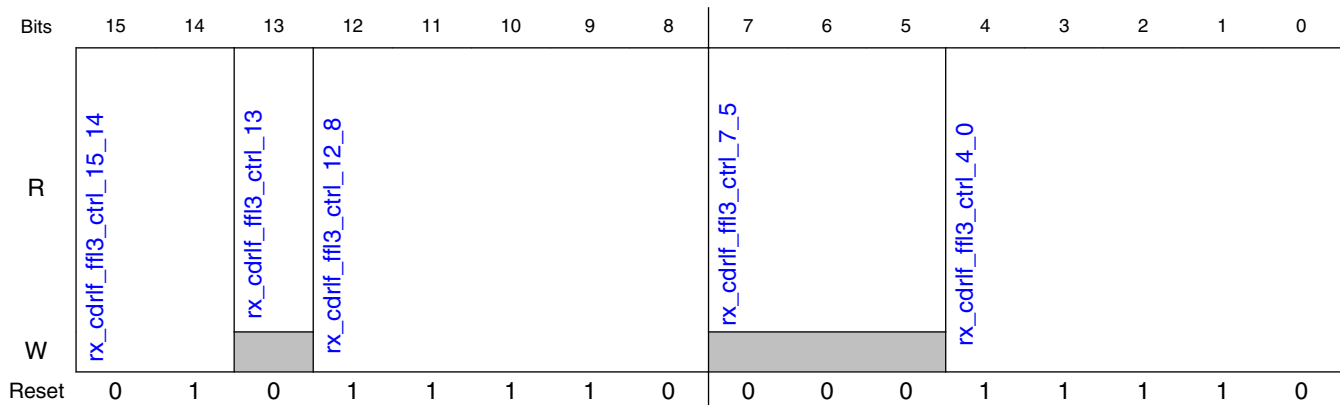
Field	Function
12-8 rx_cdrif_ffl2_ctrl_12_8	FFL step 2 CDRLF second order loop sigma delta update rate
7-5 rx_cdrif_ffl2_ctrl_7_5	Reserved
4-0 rx_cdrif_ffl2_ctrl_4_0	FFL step 2 CDRLF first order loop sigma delta update rate

13.4.10.2.147 CDRLF fast frequency lock step 3 control register (lane0_rx_cdrif_ffl3_ctrl - lane3_rx_cdrif_ffl3_ctrl)

13.4.10.2.147.1 Offset

Register	Offset
lane0_rx_cdrif_ffl3_ctrl	808Bh
lane1_rx_cdrif_ffl3_ctrl	848Bh
lane2_rx_cdrif_ffl3_ctrl	888Bh
lane3_rx_cdrif_ffl3_ctrl	8C8Bh

13.4.10.2.147.2 Diagram



13.4.10.2.147.3 Fields

Field	Function
15-14	FFL step 3 CDRLF second order loop integrator scaler

Table continues on the next page...

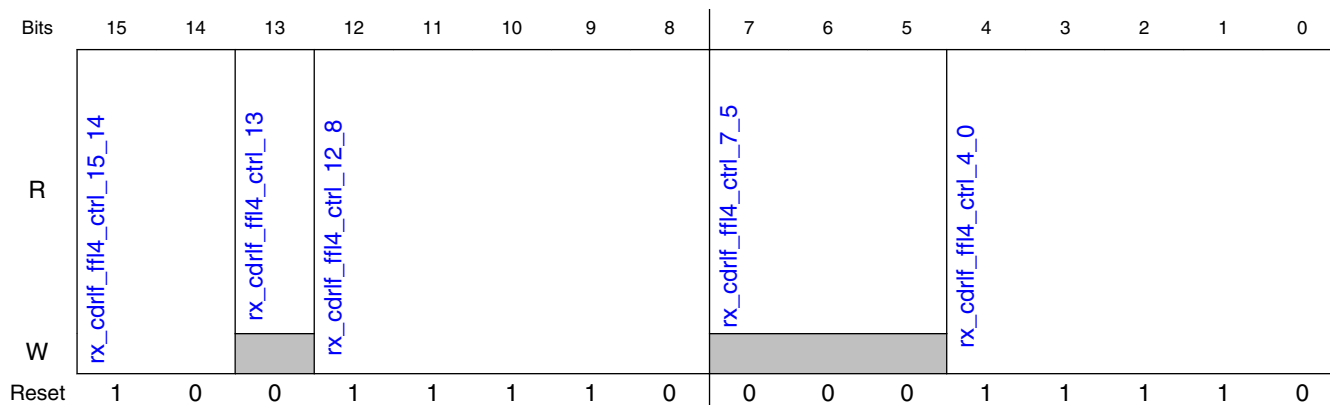
Field	Function
rx_cdrif_ffl3_ctrl_15_14	
13 rx_cdrif_ffl3_ctrl_13	Reserved
12-8 rx_cdrif_ffl3_ctrl_12_8	FFL step 3 CDRLF second order loop sigma delta update rate
7-5 rx_cdrif_ffl3_ctrl_7_5	Reserved
4-0 rx_cdrif_ffl3_ctrl_4_0	FFL step 3 CDRLF first order loop sigma delta update rate

13.4.10.2.148 CDRLF fast frequency lock step 4 control register (lane0_rx_cdrif_ffl4_ctrl - lane3_rx_cdrif_ffl4_ctrl)

13.4.10.2.148.1 Offset

Register	Offset
lane0_rx_cdrif_ffl4_ctrl	808Ch
lane1_rx_cdrif_ffl4_ctrl	848Ch
lane2_rx_cdrif_ffl4_ctrl	888Ch
lane3_rx_cdrif_ffl4_ctrl	8C8Ch

13.4.10.2.148.2 Diagram



13.4.10.2.148.3 Fields

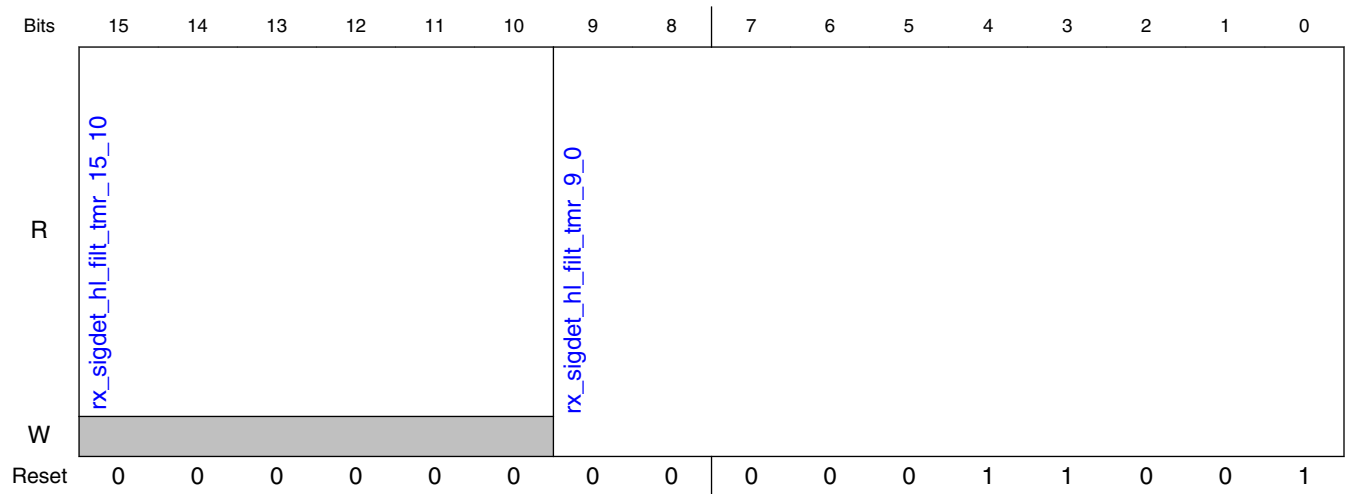
Field	Function
15-14 rx_cdrif_ffl4_ctrl _15_14	FFL step 4 CDRLF second order loop integrator scaler
13 rx_cdrif_ffl4_ctrl _13	Reserved
12-8 rx_cdrif_ffl4_ctrl _12_8	FFL step 4 CDRLF second order loop sigma delta update rate
7-5 rx_cdrif_ffl4_ctrl _7_5	Reserved
4-0 rx_cdrif_ffl4_ctrl _4_0	FFL step 4 CDRLF first order loop sigma delta update rate

13.4.10.2.149 Receiver signal detect filter high to low filter timer register (lane0_rx_sigdet_hl_filt_tmr - lane3_rx_sigdet_hl_filt_tmr)

13.4.10.2.149.1 Offset

Register	Offset
lane0_rx_sigdet_hl_filt_tmr	8090h
lane1_rx_sigdet_hl_filt_tmr	8490h
lane2_rx_sigdet_hl_filt_tmr	8890h
lane3_rx_sigdet_hl_filt_tmr	8C90h

13.4.10.2.149.2 Diagram



13.4.10.2.149.3 Fields

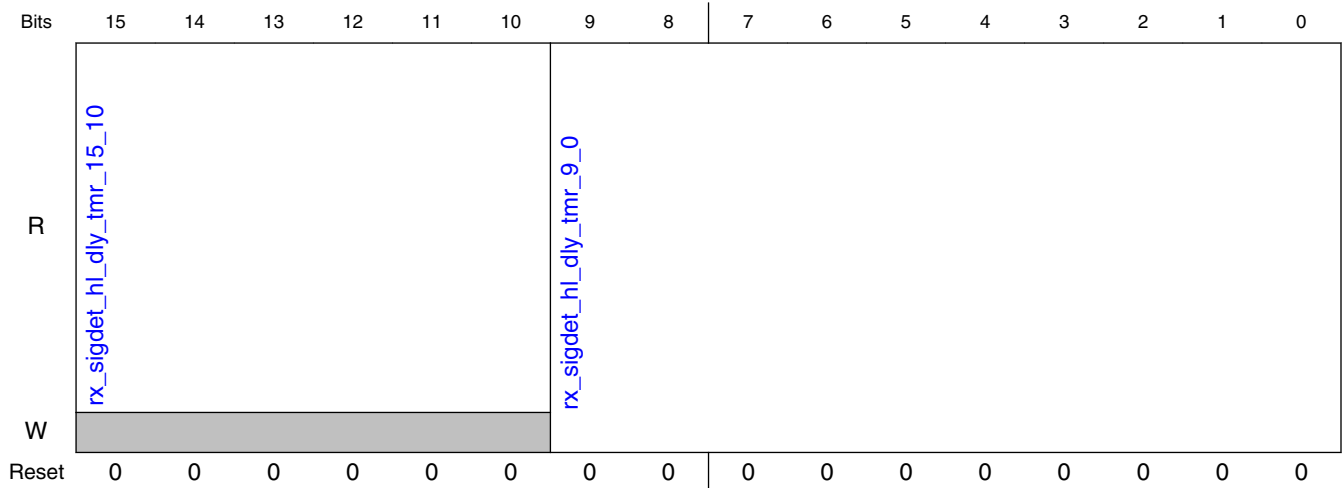
Field	Function
15-10 rx_sigdet_hl_filt_tmr_15_10	Reserved
9-0 rx_sigdet_hl_filt_tmr_9_0	Signal detect filter high to low filter timer value: This is the value loaded into the filter timer in the signal detect high to low filter circuit. This should be set to 1 less than the number of clocks of desired filter time.

13.4.10.2.150 Receiver signal detect filter high to low delay timer register (lane0_rx_sigdet_hl_dly_tmr - lane3_rx_sigdet_hl_dly_tmr)

13.4.10.2.150.1 Offset

Register	Offset
lane0_rx_sigdet_hl_dly_tmr	8091h
lane1_rx_sigdet_hl_dly_tmr	8491h
lane2_rx_sigdet_hl_dly_tmr	8891h
lane3_rx_sigdet_hl_dly_tmr	8C91h

13.4.10.2.150.2 Diagram



13.4.10.2.150.3 Fields

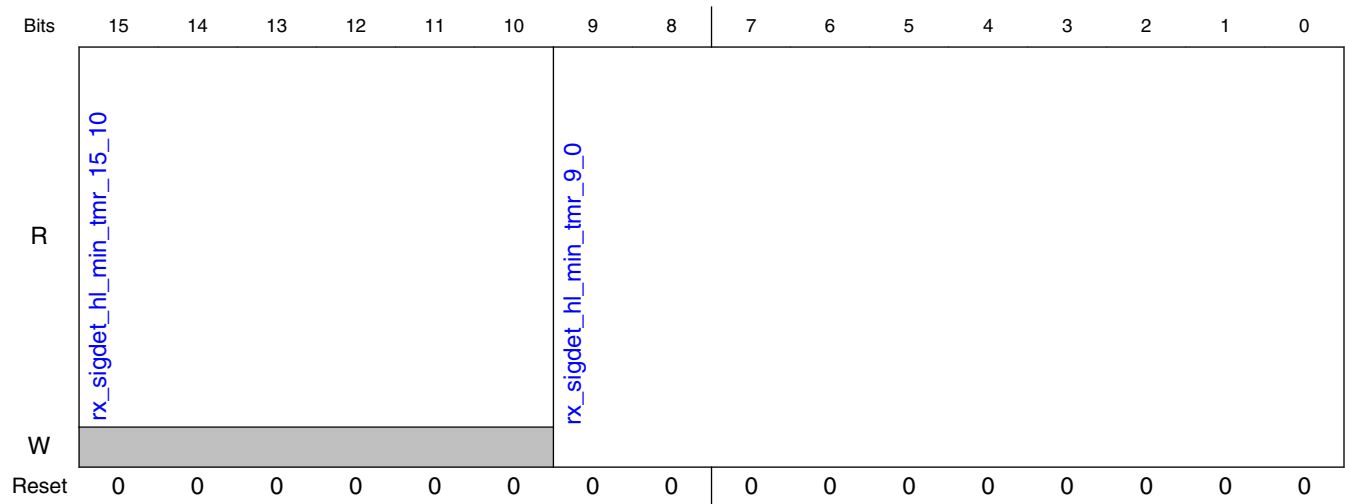
Field	Function
15-10 <code>rx_sigdet_hl_dly_tmr_15_10</code>	Reserved
9-0 <code>rx_sigdet_hl_dly_tmr_9_0</code>	Signal detect filter high to low delay timer value: This is the value loaded into the delay timer in the signal detect high to low filter circuit. This should be set to 1 less than the number of clocks of desired delay time.

13.4.10.2.151 Receiver signal detect filter high to low min timer register (lane0_rx_sigdet_hl_min_tmr - lane3_rx_sigdet_hl_min_tmr)

13.4.10.2.151.1 Offset

Register	Offset
<code>lane0_rx_sigdet_hl_min_tmr</code>	8092h
<code>lane1_rx_sigdet_hl_min_tmr</code>	8492h
<code>lane2_rx_sigdet_hl_min_tmr</code>	8892h
<code>lane3_rx_sigdet_hl_min_tmr</code>	8C92h

13.4.10.2.151.2 Diagram



13.4.10.2.151.3 Fields

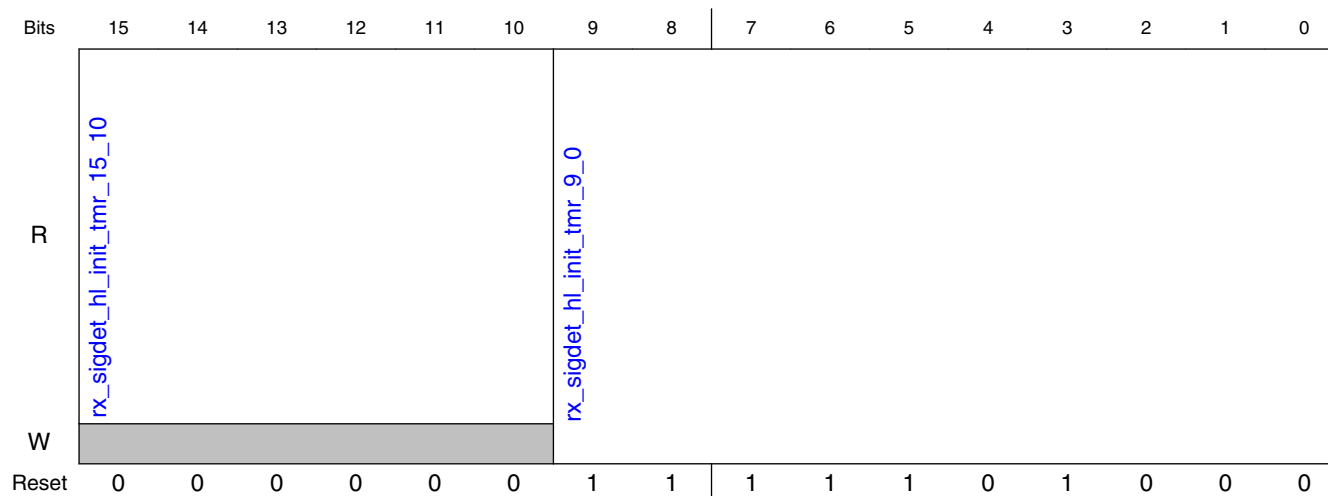
Field	Function
15-10 rx_sigdet_hl_min_tmr_15_10	Reserved
9-0 rx_sigdet_hl_min_tmr_9_0	Signal detect filter high to low min timer value: This is the value loaded into the min timer in the signal detect high to low filter circuit. This should be set to 1 less than the number of clocks of desired min time.

13.4.10.2.152 Receiver signal detect filter high to low init timer register (lane0_rx_sigdet_hl_init_tmr - lane3_rx_sigdet_hl_init_tmr)

13.4.10.2.152.1 Offset

Register	Offset
lane0_rx_sigdet_hl_init_tmr	8093h
lane1_rx_sigdet_hl_init_tmr	8493h
lane2_rx_sigdet_hl_init_tmr	8893h
lane3_rx_sigdet_hl_init_tmr	8C93h

13.4.10.2.152.2 Diagram



13.4.10.2.152.3 Fields

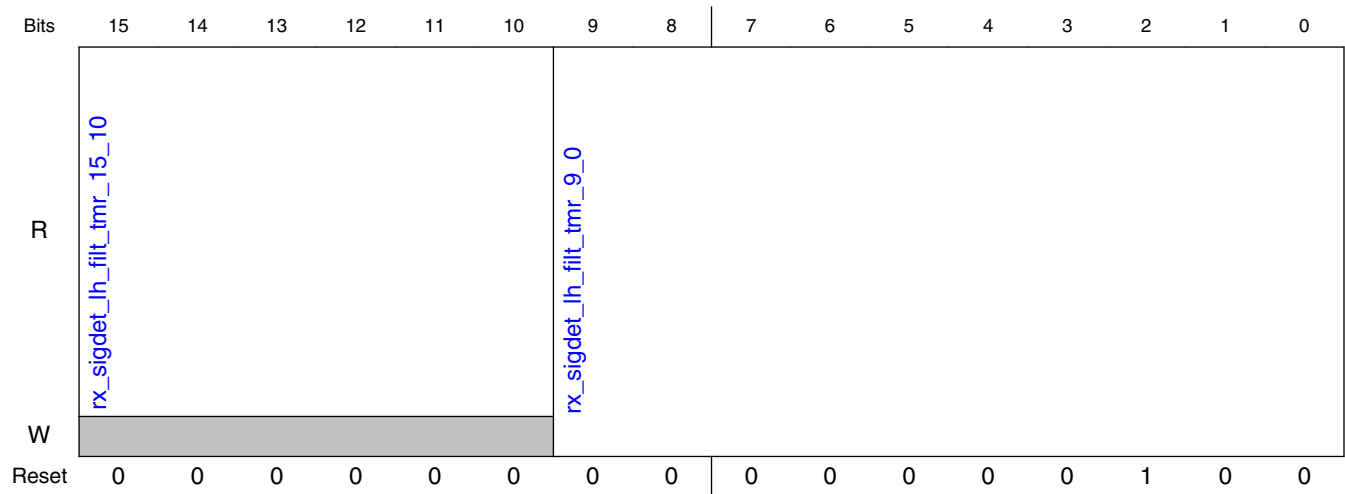
Field	Function
15-10 rx_sigdet_lh_init_tmr_15_10	Reserved
9-0 rx_sigdet_lh_init_tmr_9_0	Signal detect init timer value: This is the value loaded into the initialization timer in the signal detect filter high to low filter circuit.

13.4.10.2.153 Receiver signal detect filter low to high filter timer register (lane0_rx_sigdet_lh_filt_tmr - lane3_rx_sigdet_lh_filt_tmr)

13.4.10.2.153.1 Offset

Register	Offset
lane0_rx_sigdet_lh_filt_tmr	8094h
lane1_rx_sigdet_lh_filt_tmr	8494h
lane2_rx_sigdet_lh_filt_tmr	8894h
lane3_rx_sigdet_lh_filt_tmr	8C94h

13.4.10.2.153.2 Diagram



13.4.10.2.153.3 Fields

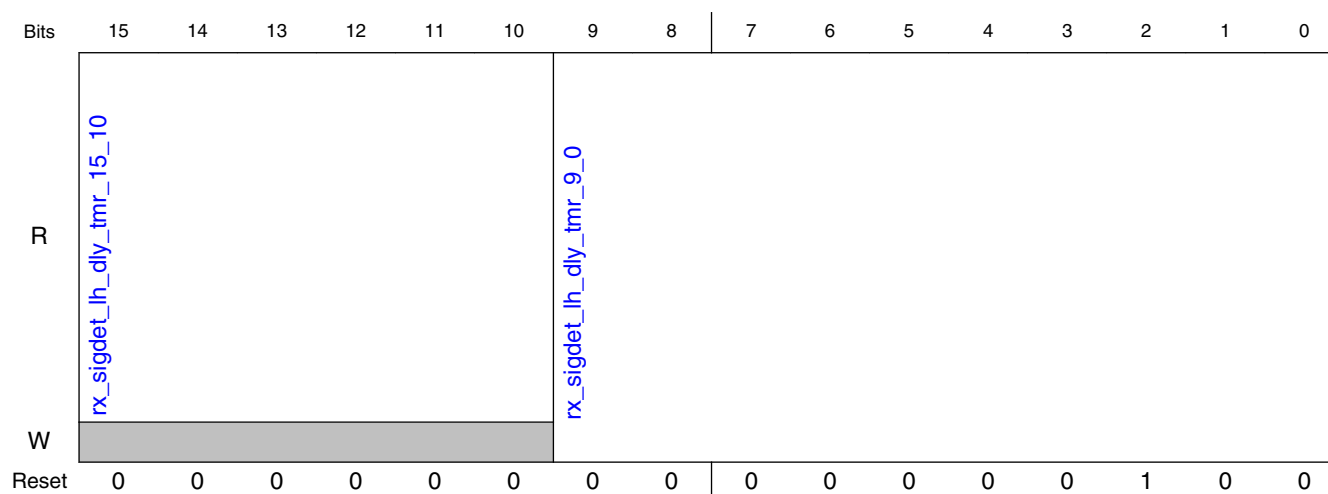
Field	Function
15-10 rx_sigdet_lh_filt_tmr_15_10	Reserved
9-0 rx_sigdet_lh_filt_tmr_9_0	Signal detect filter low to high filter timer value: This is the value loaded into the filter timer in the signal detect low to high filter circuit. This should be set to 6 less than the number of clocks of desired filter time. This takes into account the synchronizing of the signal to be filtered (4 clocks), the time to load the filter timer (1 clock), and that the timer counts to 0. Therefore, the smallest granularity of filtering available here is 6 clocks. This is more of an issue of latency through the filter circuit as the signal initially rises.

13.4.10.2.154 Receiver signal detect filter low to high delay timer register (lane0_rx_sigdet_lh_dly_tmr - lane3_rx_sigdet_lh_dly_tmr)

13.4.10.2.154.1 Offset

Register	Offset
lane0_rx_sigdet_lh_dly_tmr	8095h
lane1_rx_sigdet_lh_dly_tmr	8495h
lane2_rx_sigdet_lh_dly_tmr	8895h
lane3_rx_sigdet_lh_dly_tmr	8C95h

13.4.10.2.154.2 Diagram



13.4.10.2.154.3 Fields

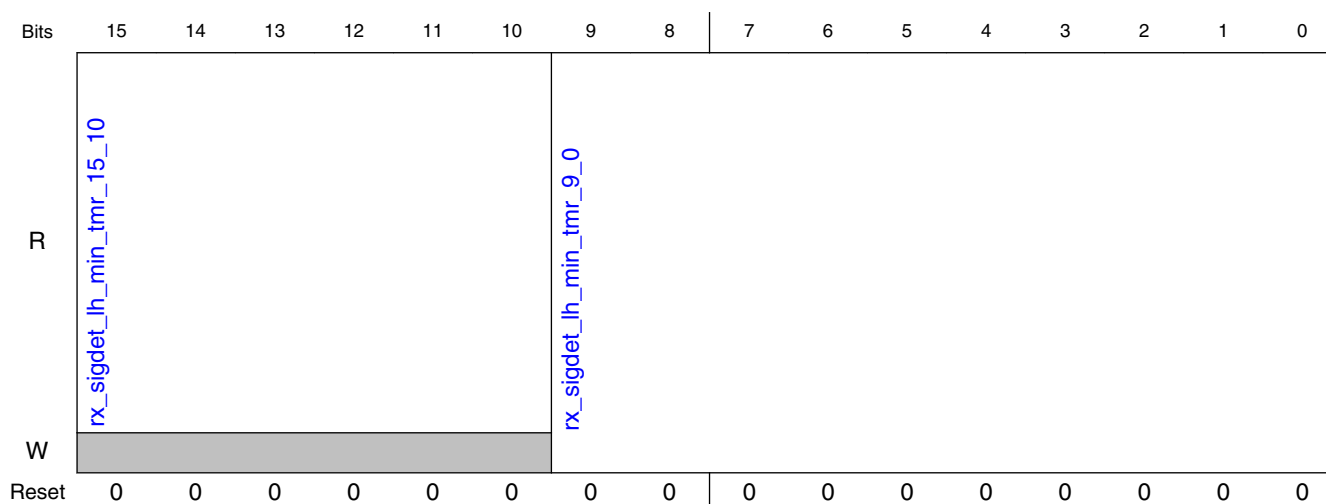
Field	Function
15-10 rx_sigdet_lh_dly_tmr_15_10	Reserved
9-0 rx_sigdet_lh_dly_tmr_9_0	Signal detect filter low to high delay timer value: This is the value loaded into the delay timer in the signal detect low to high filter circuit. This should be set to 1 less than the number of clocks of desired delay time.

13.4.10.2.155 Receiver signal detect filter low to high min timer register (lane0_rx_sigdet_lh_min_tmr - lane3_rx_sigdet_lh_min_tmr)

13.4.10.2.155.1 Offset

Register	Offset
lane0_rx_sigdet_lh_min_tmr	8096h
lane1_rx_sigdet_lh_min_tmr	8496h
lane2_rx_sigdet_lh_min_tmr	8896h
lane3_rx_sigdet_lh_min_tmr	8C96h

13.4.10.2.155.2 Diagram



13.4.10.2.155.3 Fields

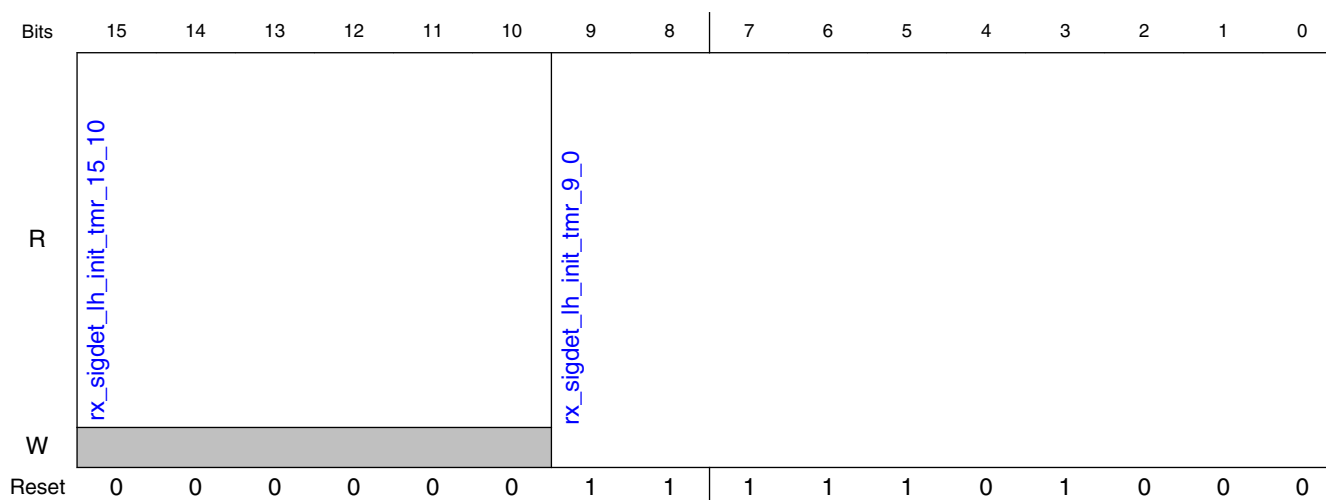
Field	Function
15-10 rx_sigdet_lh_min_tmr_15_10	Reserved
9-0 rx_sigdet_lh_min_tmr_9_0	Signal detect filter low to high min timer value: This is the value loaded into the min timer in the signal detect low to high filter circuit. This should be set to 1 less than the number of clocks of desired min time.

13.4.10.2.156 Receiver signal detect filter low to high init timer register (lane0_rx_sigdet_lh_init_tmr - lane3_rx_sigdet_lh_init_tmr)

13.4.10.2.156.1 Offset

Register	Offset
lane0_rx_sigdet_lh_init_tmr	8097h
lane1_rx_sigdet_lh_init_tmr	8497h
lane2_rx_sigdet_lh_init_tmr	8897h
lane3_rx_sigdet_lh_init_tmr	8C97h

13.4.10.2.156.2 Diagram



13.4.10.2.156.3 Fields

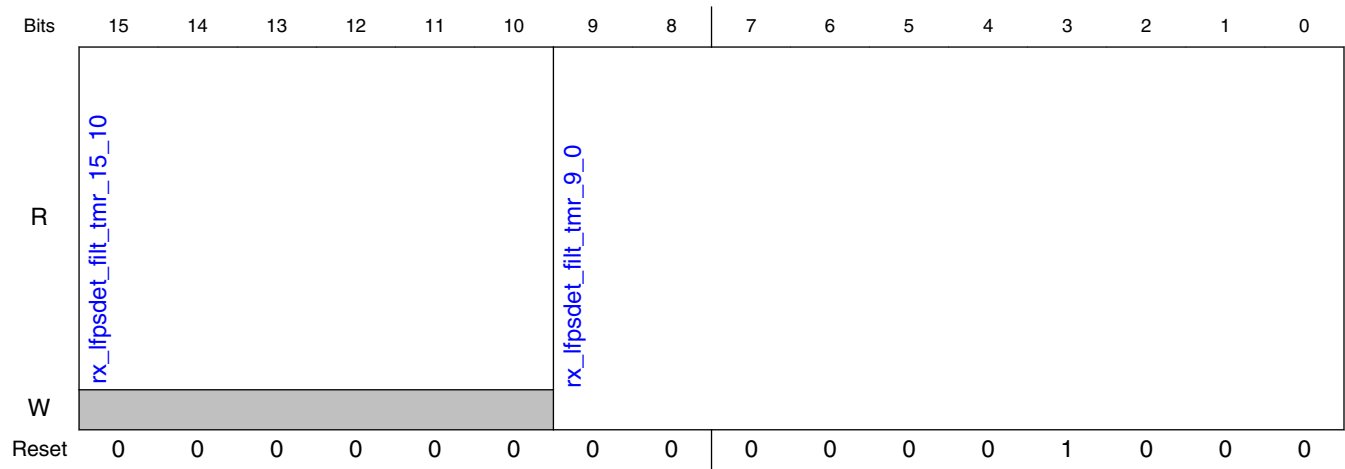
Field	Function
15-10 rx_sigdet_lh_init_tmr_15_10	Reserved
9-0 rx_sigdet_lh_init_tmr_9_0	Signal detect init timer value: This is the value loaded into the initialization timer in the signal detect filter low to high filter circuit.

13.4.10.2.157 Receiver LFPS detect filter filter timer register (lane0_rx_lfpsdet_filt_tmr - lane3_rx_lfpsdet_filt_tmr)

13.4.10.2.157.1 Offset

Register	Offset
lane0_rx_lfpsdet_filt_tmr	8098h
lane1_rx_lfpsdet_filt_tmr	8498h
lane2_rx_lfpsdet_filt_tmr	8898h
lane3_rx_lfpsdet_filt_tmr	8C98h

13.4.10.2.157.2 Diagram



13.4.10.2.157.3 Fields

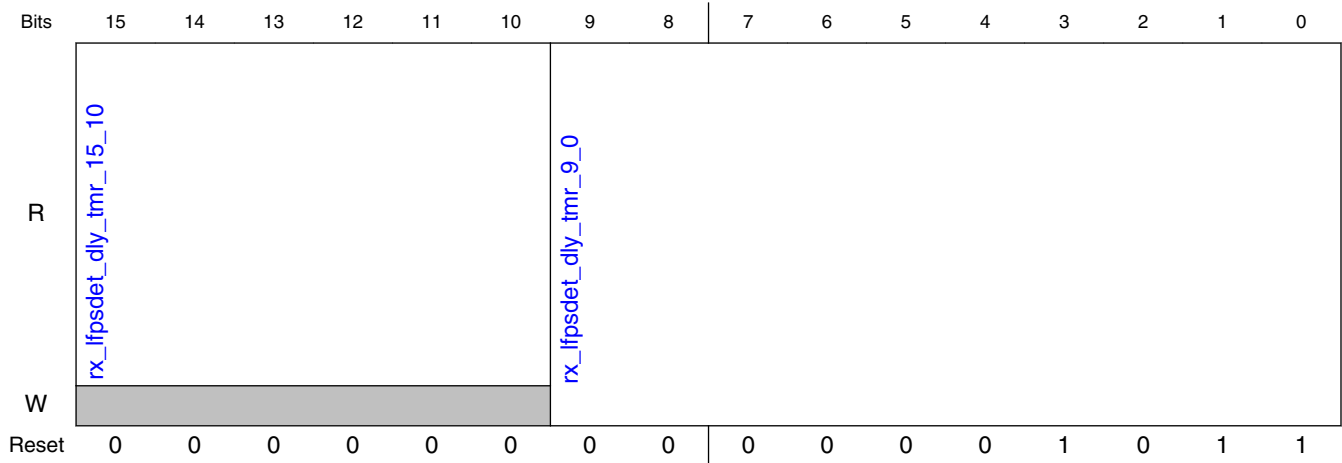
Field	Function
15-10 rx_lfpsdet_filt_tmr_15_10	Reserved
9-0 rx_lfpsdet_filt_tmr_9_0	LFPS detect filter timer value: This is the value loaded into the filter timer in the signal detect filter low circuit. This should be set to 6 less than the number of clocks of desired filter time.

13.4.10.2.158 Receiver LFPS detect filter delay timer register (lane0_rx_lfpsdet_dly_tmr - lane3_rx_lfpsdet_dly_tmr)

13.4.10.2.158.1 Offset

Register	Offset
lane0_rx_lfpsdet_dly_tmr	8099h
lane1_rx_lfpsdet_dly_tmr	8499h
lane2_rx_lfpsdet_dly_tmr	8899h
lane3_rx_lfpsdet_dly_tmr	8C99h

13.4.10.2.158.2 Diagram



13.4.10.2.158.3 Fields

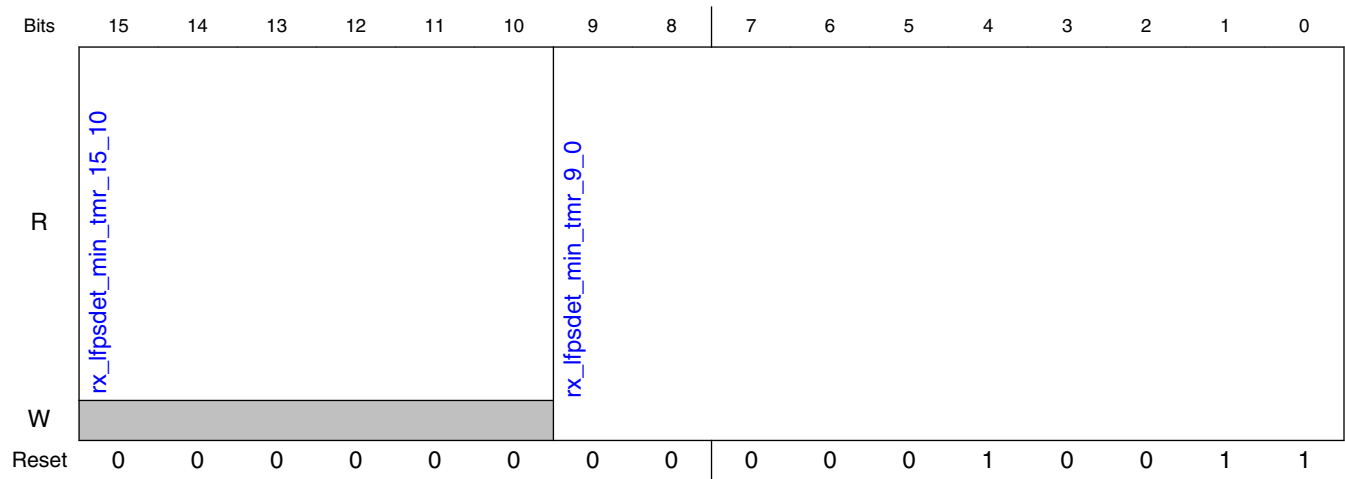
Field	Function
15-10 rx_lfpsdet_dly_tmr_15_10	Reserved
9-0 rx_lfpsdet_dly_tmr_9_0	LFPS detect filter delay timer value: This is the value loaded into the delay timer in the signal detect low to high filter circuit. This should be set to 1 less than the number of clocks of desired delay time.

13.4.10.2.159 Receiver LFPS detect min timer register (lane0_rx_lfpsdet_min_tmr - lane3_rx_lfpsdet_min_tmr)

13.4.10.2.159.1 Offset

Register	Offset
lane0_rx_lfpsdet_min_tmr	809Ah
lane1_rx_lfpsdet_min_tmr	849Ah
lane2_rx_lfpsdet_min_tmr	889Ah
lane3_rx_lfpsdet_min_tmr	8C9Ah

13.4.10.2.159.2 Diagram



13.4.10.2.159.3 Fields

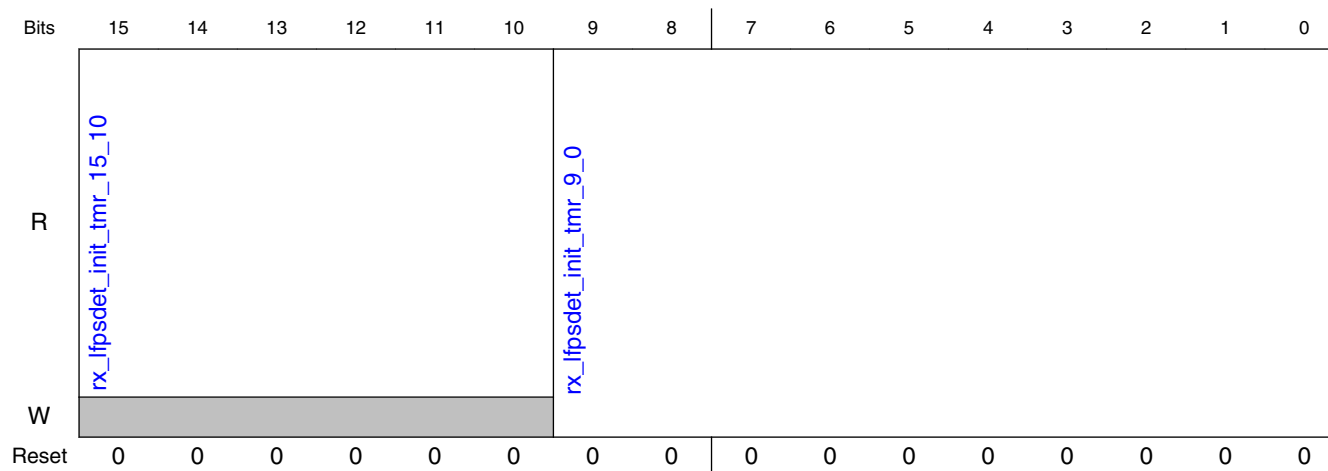
Field	Function
15-10 rx_lfpsdet_min_tmr_15_10	Reserved
9-0 rx_lfpsdet_min_tmr_9_0	LFPS detect min timer value: This is the value loaded into the min timer in the signal detect filter low circuit. This should be set to 1 less than the number of clocks of desired min time.

13.4.10.2.160 Receiver LFPS detect init timer register (lane0_rx_lfpsdet_init_tmr - lane3_rx_lfpsdet_init_tmr)

13.4.10.2.160.1 Offset

Register	Offset
lane0_rx_lfpsdet_init_tmr	809Bh
lane1_rx_lfpsdet_init_tmr	849Bh
lane2_rx_lfpsdet_init_tmr	889Bh
lane3_rx_lfpsdet_init_tmr	8C9Bh

13.4.10.2.160.2 Diagram



13.4.10.2.160.3 Fields

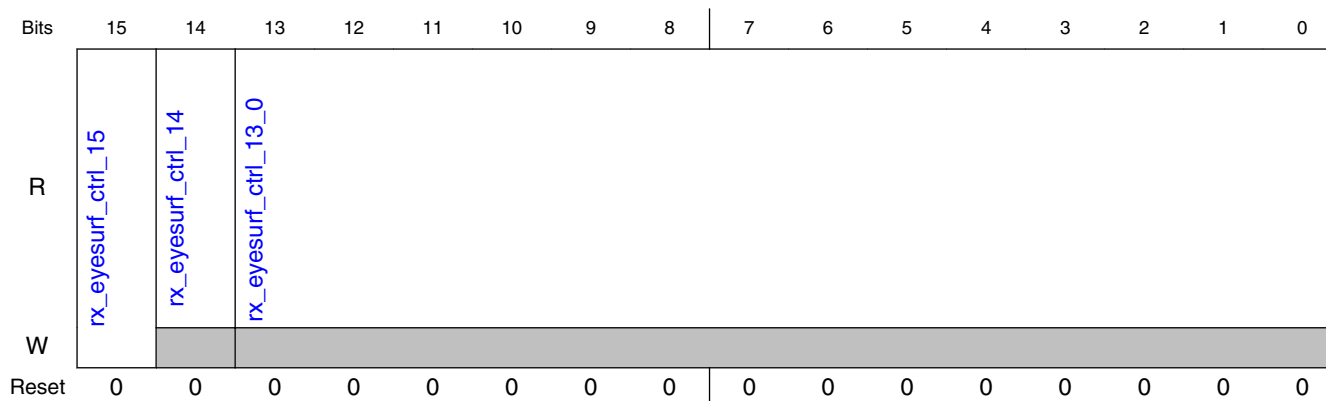
Field	Function
15-10 rx_lfpsdet_init_tmr_15_10	Reserved
9-0 rx_lfpsdet_init_tmr_9_0	LFPS detect init timer value: This is the value loaded into the initialization timer in the signal detect filter low circuit.

13.4.10.2.161 Eye surf control register (lane0_rx_eyesurf_ctrl - lane3_rx_eyesurf_ctrl)

13.4.10.2.161.1 Offset

Register	Offset
lane0_rx_eyesurf_ctrl	80A0h
lane1_rx_eyesurf_ctrl	84A0h
lane2_rx_eyesurf_ctrl	88A0h
lane3_rx_eyesurf_ctrl	8CA0h

13.4.10.2.161.2 Diagram



13.4.10.2.161.3 Fields

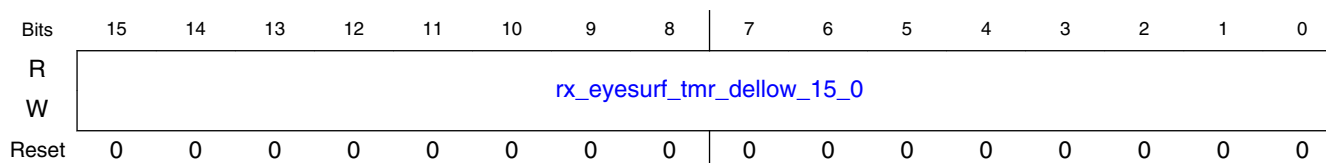
Field	Function
15 rx_eyesurf_ctrl_15	Eye surf run: Setting this bit to 1b1 will initiate the eye surf process. This bit must remain set to 1b1 until the eye surf done bit is set.
14 rx_eyesurf_ctrl_14	Eye surf done: When this bit is set to 1b1, the eye surf process has completed. This bit will be cleared after the eye surf run bit is cleared.
13-0 rx_eyesurf_ctrl_13_0	Reserved

13.4.10.2.162 Eye surf timer delay low register (lane0_rx_eyesurf_tmr_dellow - lane3_rx_eyesurf_tmr_dellow)

13.4.10.2.162.1 Offset

Register	Offset
lane0_rx_eyesurf_tmr_dellow	80A4h
lane1_rx_eyesurf_tmr_dellow	84A4h
lane2_rx_eyesurf_tmr_dellow	88A4h
lane3_rx_eyesurf_tmr_dellow	8CA4h

13.4.10.2.162.2 Diagram



13.4.10.2.162.3 Fields

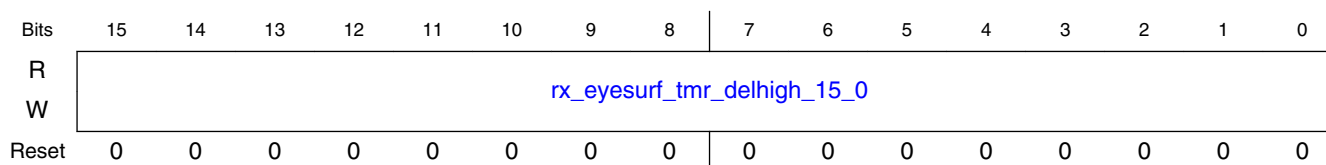
Field	Function
15-0 rx_eyesurf_tmr_dellow_15_0	Least significant 16 bits of the delay time: The delay time specifies the number of clock cycles to wait between when a coordinate test point is set, and when to start testing the i and e data.

13.4.10.2.163 Eye surf timer delay high register (lane0_rx_eyesurf_tmr_delhigh - lane3_rx_eyesurf_tmr_delhigh)

13.4.10.2.163.1 Offset

Register	Offset
lane0_rx_eyesurf_tmr_delhigh	80A5h
lane1_rx_eyesurf_tmr_delhigh	84A5h
lane2_rx_eyesurf_tmr_delhigh	88A5h
lane3_rx_eyesurf_tmr_delhigh	8CA5h

13.4.10.2.163.2 Diagram



13.4.10.2.163.3 Fields

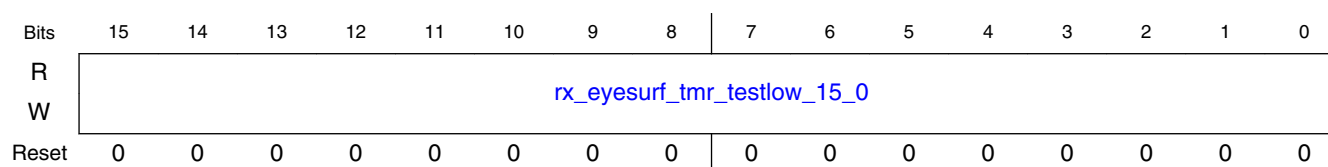
Field	Function
15-0 rx_eyesurf_tmr_ delhigh_15_0	Most significant 16 bits of the delay time: The delay time specifies the number of clock cycles to wait between when a coordinate test point is set, and when to start testing the i and e data.

13.4.10.2.164 Eye surf timer test low register (lane0_rx_eyesurf_tmr_testlow - lane3_rx_eyesurf_tmr_testlow)

13.4.10.2.164.1 Offset

Register	Offset
lane0_rx_eyesurf_tmr_ testlow	80A6h
lane1_rx_eyesurf_tmr_ testlow	84A6h
lane2_rx_eyesurf_tmr_ testlow	88A6h
lane3_rx_eyesurf_tmr_ testlow	8CA6h

13.4.10.2.164.2 Diagram



13.4.10.2.164.3 Fields

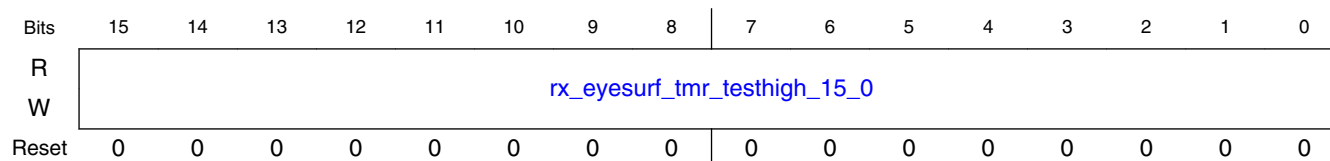
Field	Function
15-0 rx_eyesurf_tmr_ testlow_15_0	Least significant 16 bits of the test time: The test time specifies the number of clock cycles to test the i and e data at a given coordinate test point.

13.4.10.2.165 Eye surf timer test high register (lane0_rx_eyesurf_tmr_testhigh - lane3_rx_eyesurf_tmr_testhigh)

13.4.10.2.165.1 Offset

Register	Offset
lane0_rx_eyesurf_tmr_testhigh	80A7h
lane1_rx_eyesurf_tmr_testhigh	84A7h
lane2_rx_eyesurf_tmr_testhigh	88A7h
lane3_rx_eyesurf_tmr_testhigh	8CA7h

13.4.10.2.165.2 Diagram



13.4.10.2.165.3 Fields

Field	Function
15-0 rx_eyesurf_tmr_testhigh_15_0	Most significant 16 bits of the test time: The test time specifies the number of clock cycles to test the i and e data at a given coordinate test point.

13.4.10.2.166 Eye surf north south test point coordinate register (lane0_rx_eyesurf_ns_coord - lane3_rx_eyesurf_ns_coord)

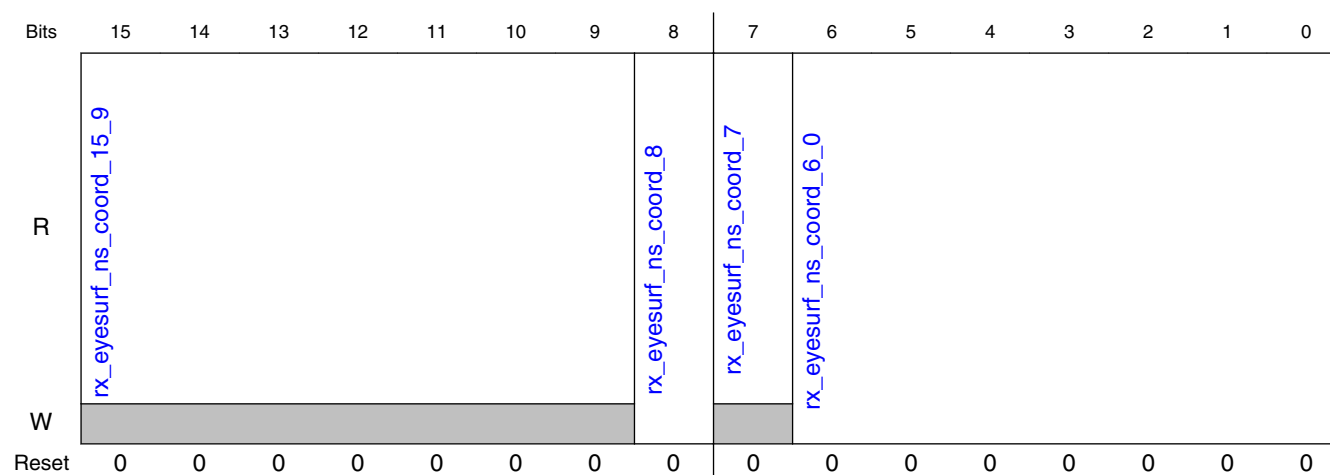
13.4.10.2.166.1 Offset

Register	Offset
lane0_rx_eyesurf_ns_coord	80A8h
lane1_rx_eyesurf_ns_coord	84A8h

Table continues on the next page...

Register	Offset
lane2_rx_eyesurf_ns_coord	88A8h
lane3_rx_eyesurf_ns_coord	8CA8h

13.4.10.2.166.2 Diagram



13.4.10.2.166.3 Fields

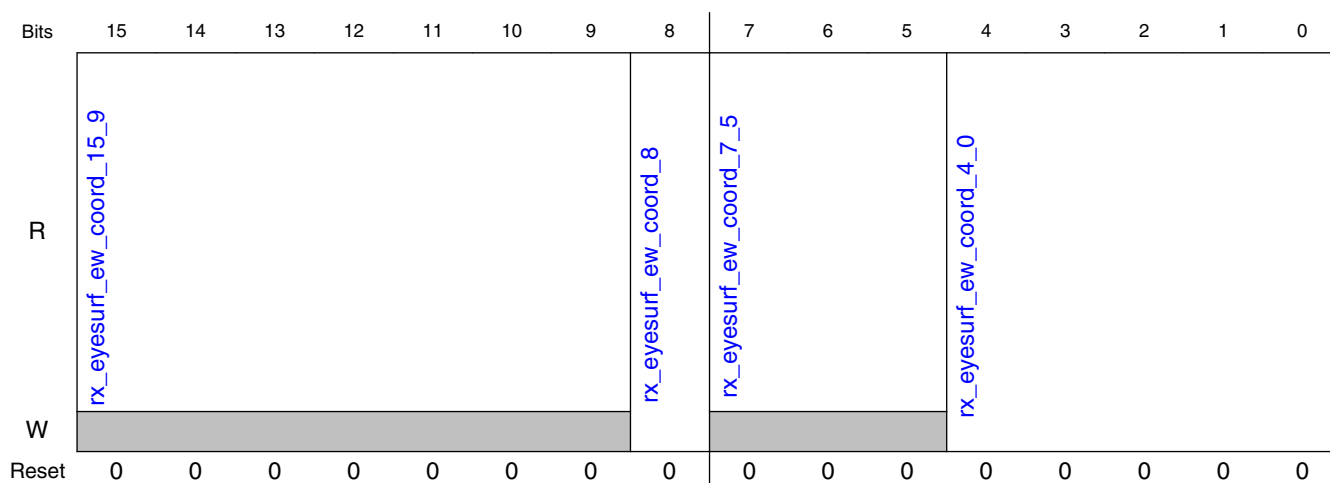
Field	Function
15-9 rx_eyesurf_ns_coord_15_9	Reserved
8 rx_eyesurf_ns_coord_8	Test point coordinate north south direction : Indicates whether the desired test point is in the
7 rx_eyesurf_ns_coord_7	Reserved
6-0 rx_eyesurf_ns_coord_6_0	Test point coordinate north south offset : Indicates how many steps in the north or south

13.4.10.2.167 Eye surf east west test point coordinate register (lane0_rx_eyesurf_ew_coord - lane3_rx_eyesurf_ew_coord)

13.4.10.2.167.1 Offset

Register	Offset
lane0_rx_eyesurf_ew_coord	80A9h
lane1_rx_eyesurf_ew_coord	84A9h
lane2_rx_eyesurf_ew_coord	88A9h
lane3_rx_eyesurf_ew_coord	8CA9h

13.4.10.2.167.2 Diagram



13.4.10.2.167.3 Fields

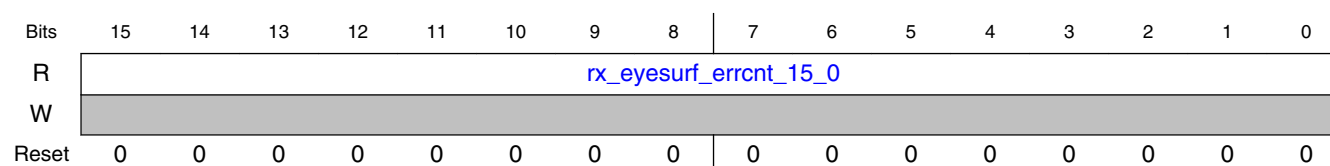
Field	Function
15-9 rx_eyesurf_ew_coord_15_9	Reserved
8 rx_eyesurf_ew_coord_8	Test point coordinate east west direction : Indicates whether the desired test point is in the
7-5 rx_eyesurf_ew_coord_7_5	Reserved
4-0 rx_eyesurf_ew_coord_4_0	Test point coordinate east west offset : Indicates how many steps in the east or west

13.4.10.2.168 Eye surf bit error count register (lane0_rx_eyesurf_errcnt - lane3_rx_eyesurf_errcnt)

13.4.10.2.168.1 Offset

Register	Offset
lane0_rx_eyesurf_errcnt	80AAh
lane1_rx_eyesurf_errcnt	84AAh
lane2_rx_eyesurf_errcnt	88AAh
lane3_rx_eyesurf_errcnt	8CAAh

13.4.10.2.168.2 Diagram



13.4.10.2.168.3 Fields

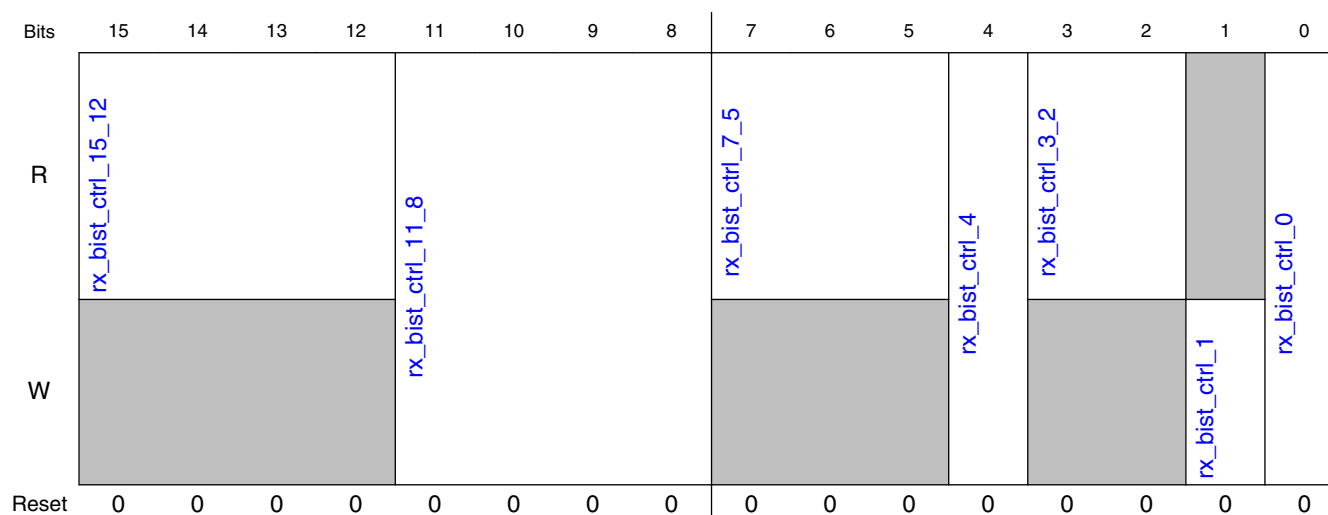
Field	Function
15-0 rx_eyesurf_errcnt_15_0	Test point bit error count : The total number of bit errors that were detected for a given run of the eye surf function. This register is only valid after the eye surf done bit in the

13.4.10.2.169 Receiver BIST control register (lane0_rx_bist_ctrl - lane3_rx_bist_ctrl)

13.4.10.2.169.1 Offset

Register	Offset
lane0_rx_bist_ctrl	80B0h
lane1_rx_bist_ctrl	84B0h
lane2_rx_bist_ctrl	88B0h
lane3_rx_bist_ctrl	8CB0h

13.4.10.2.169.2 Diagram



13.4.10.2.169.3 Fields

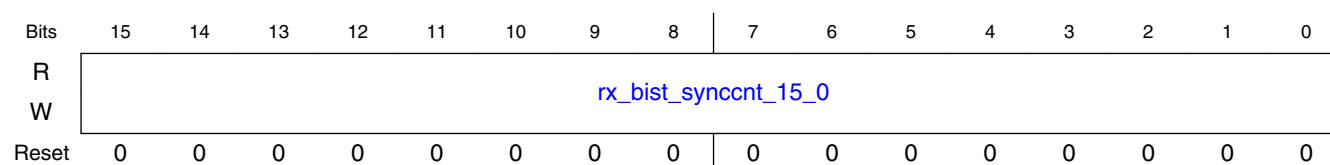
Field	Function
15-12 rx_bist_ctrl_15_12	Reserved
11-8 rx_bist_ctrl_11_8	Receiver BIST mode: Controls which mode the BIST will operate in. The value of this field must match the corresponding field for the receive BIST controller. The following are the values used for this field, and what BIST mode they correspond to.
7-5 rx_bist_ctrl_7_5	Reserved
4 rx_bist_ctrl_4	Receiver BIST error reset: Writing this bit is set to a 1b1 will hold the error indicators in the receive BIST logic in reset. When this signal goes active, the
3-2 rx_bist_ctrl_3_2	Reserved
1 rx_bist_ctrl_1	Receiver BIST user defined data FIFO clear: Writing a 1b1 to this bit will clear the receiver BIST user defined data FIFO.
0 rx_bist_ctrl_0	Receiver BIST enable: This bit enables the receiver BIST function.

13.4.10.2.170 Receiver BIST sync count register (lane0_rx_bist_synccnt - lane3_rx_bist_synccnt)

13.4.10.2.170.1 Offset

Register	Offset
lane0_rx_bist_syncnt	80B1h
lane1_rx_bist_syncnt	84B1h
lane2_rx_bist_syncnt	88B1h
lane3_rx_bist_syncnt	8CB1h

13.4.10.2.170.2 Diagram



13.4.10.2.170.3 Fields

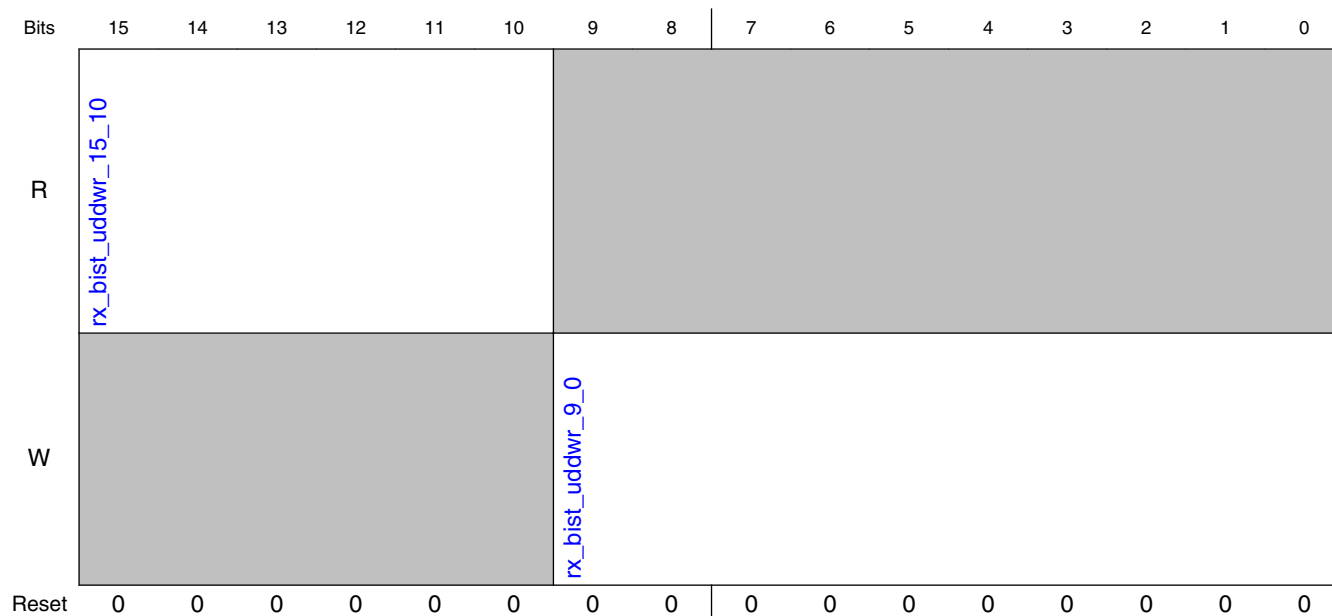
Field	Function
15-0 rx_bist_syncnt_15_0	Receiver BIST sync count: This field controls the value of the RX BIST sync count. The sync count indicates the number of consecutive received data words with no BIST bit errors that must be received in order for the RX BIST module to be considered synced to the input data stream.

13.4.10.2.171 Receiver BIST user defined data write register (lane0_rx_bist_uddwr - lane3_rx_bist_uddwr)

13.4.10.2.171.1 Offset

Register	Offset
lane0_rx_bist_uddwr	80B2h
lane1_rx_bist_uddwr	84B2h
lane2_rx_bist_uddwr	88B2h
lane3_rx_bist_uddwr	8CB2h

13.4.10.2.171.2 Diagram



13.4.10.2.171.3 Fields

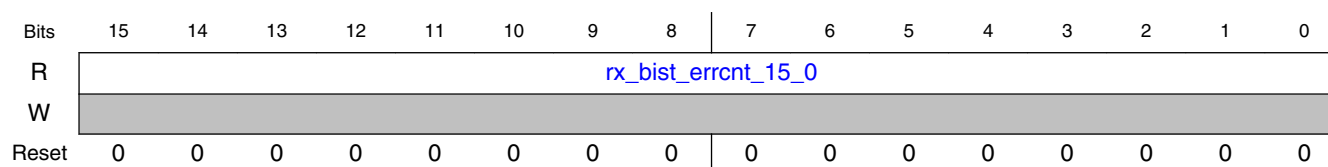
Field	Function
15-10 rx_bist_uddwr_15_10	Reserved
9-0 rx_bist_uddwr_9_0	Receiver BIST user defined data: Writing a data word to this field will result in that data word being placed in the next available position in the receiver BIST user defined data FIFO. Note, when in 20 bit mode, all 10 of these bits are used. When in 16 bit mode, only the least significant 8 bits are used. Note that the FIFO in this implementation is 18 words deep. It is up to the user to not write more than 18 words of data into this FIFO. Exceeding this amount of data will generate unpredictable results.

13.4.10.2.172 Receiver BIST error count register (lane0_rx_bist_errcnt - lane3_rx_bist_errcnt)

13.4.10.2.172.1 Offset

Register	Offset
lane0_rx_bist_errcnt	80B3h
lane1_rx_bist_errcnt	84B3h
lane2_rx_bist_errcnt	88B3h
lane3_rx_bist_errcnt	8CB3h

13.4.10.2.172.2 Diagram



13.4.10.2.172.3 Fields

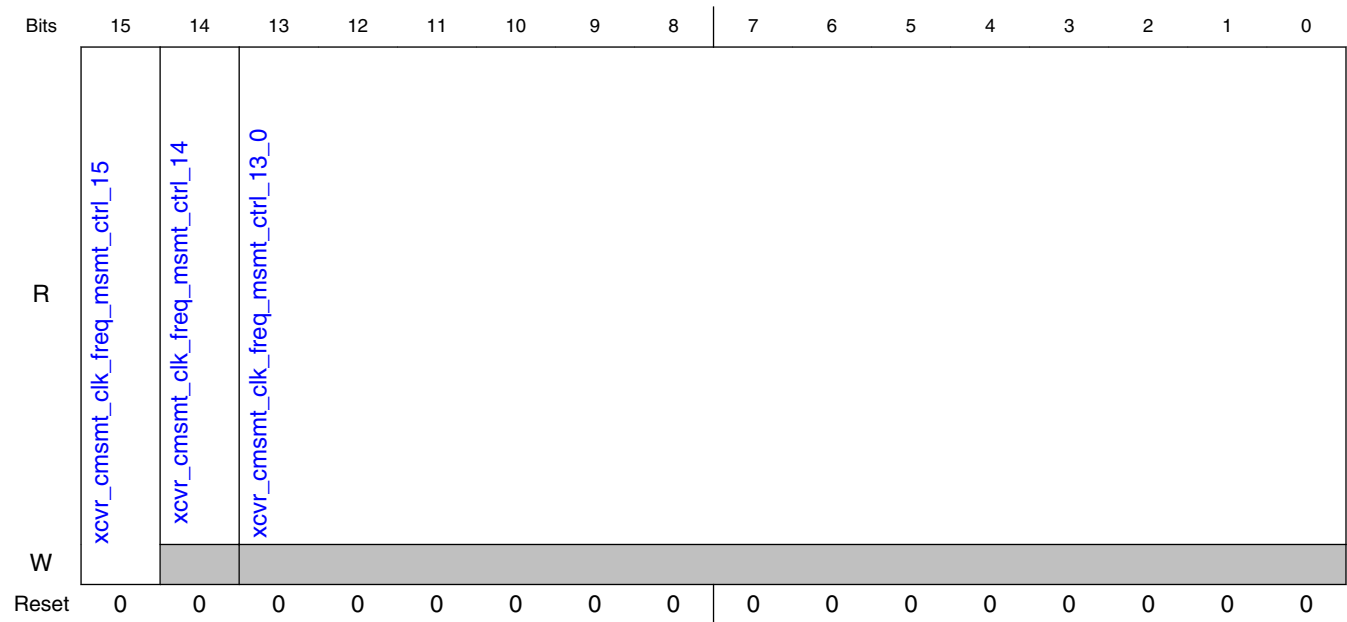
Field	Function
15-0 rx_bist_errcnt_15_0	Receiver BIST error count: Indicates the number of BIST errors that have been observed by the receive BIST logic, since the last time the BIST error indicator logic was reset or restarted. This counter increments up to a maximum value of 16hFFFF.

13.4.10.2.173 Clock frequency measurement control register (lane0_xcvr_cmsmt_clk_freq_msmt_ctrl - lane3_xcvr_cmsmt_clk_freq_msmt_ctrl)

13.4.10.2.173.1 Offset

Register	Offset
lane0_xcvr_cmsmt_clk_freq_msmt_ctrl	80B4h
lane1_xcvr_cmsmt_clk_freq_msmt_ctrl	84B4h
lane2_xcvr_cmsmt_clk_freq_msmt_ctrl	88B4h
lane3_xcvr_cmsmt_clk_freq_msmt_ctrl	8CB4h

13.4.10.2.173.2 Diagram



13.4.10.2.173.3 Fields

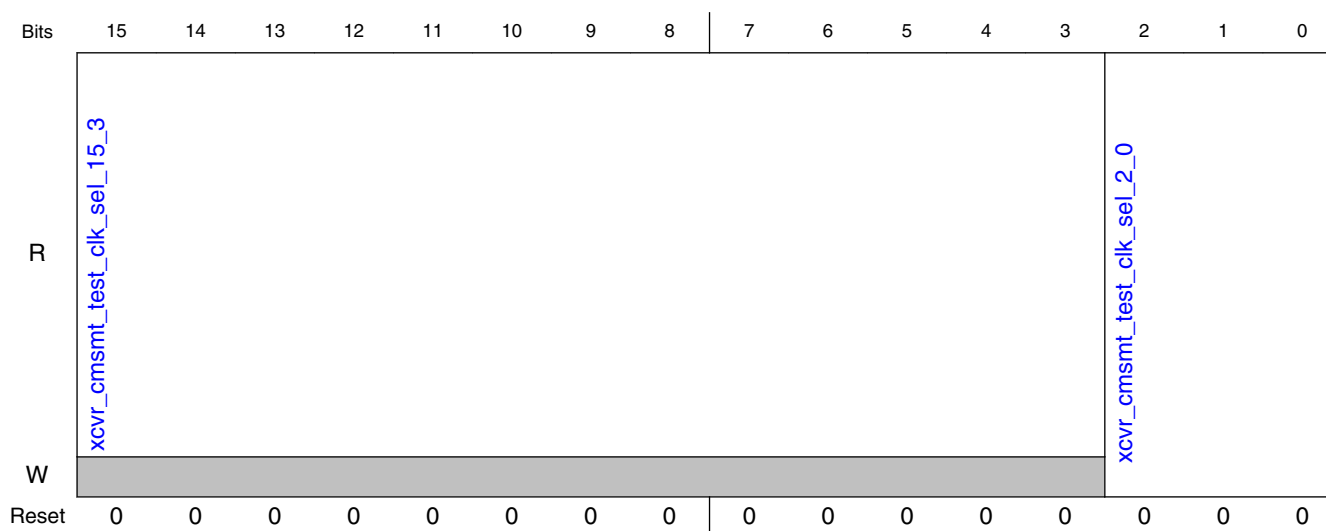
Field	Function
15 <code>xcvr_cmsmt_clk_freq_msmt_ctrl_15</code>	Run test clock measurement: Activating (1b1) this bit will run the test clock measurement process. This bit must remain active until the test clock measurement process is complete, as indicated by the test clock measurement done bit in this register. To start another measurement process, this bit must first be driven inactive then driven active again.
14 <code>xcvr_cmsmt_clk_freq_msmt_ctrl_14</code>	Test clock measurement done: This bit will be set to 1b1 when the test clock measurement process is complete. It will be cleared by the deactivation of the start test clock measurement bit in this register.
13-0 <code>xcvr_cmsmt_clk_freq_msmt_ctrl_13_0</code>	Reserved

13.4.10.2.174 Test clock selection register (lane0_xcvr_cmsmt_test_clk_sel - lane3_xcvr_cmsmt_test_clk_sel)

13.4.10.2.174.1 Offset

Register	Offset
lane0_xcvr_cmsmt_test_clk_sel	80B5h
lane1_xcvr_cmsmt_test_clk_sel	84B5h
lane2_xcvr_cmsmt_test_clk_sel	88B5h
lane3_xcvr_cmsmt_test_clk_sel	8CB5h

13.4.10.2.174.2 Diagram



13.4.10.2.174.3 Fields

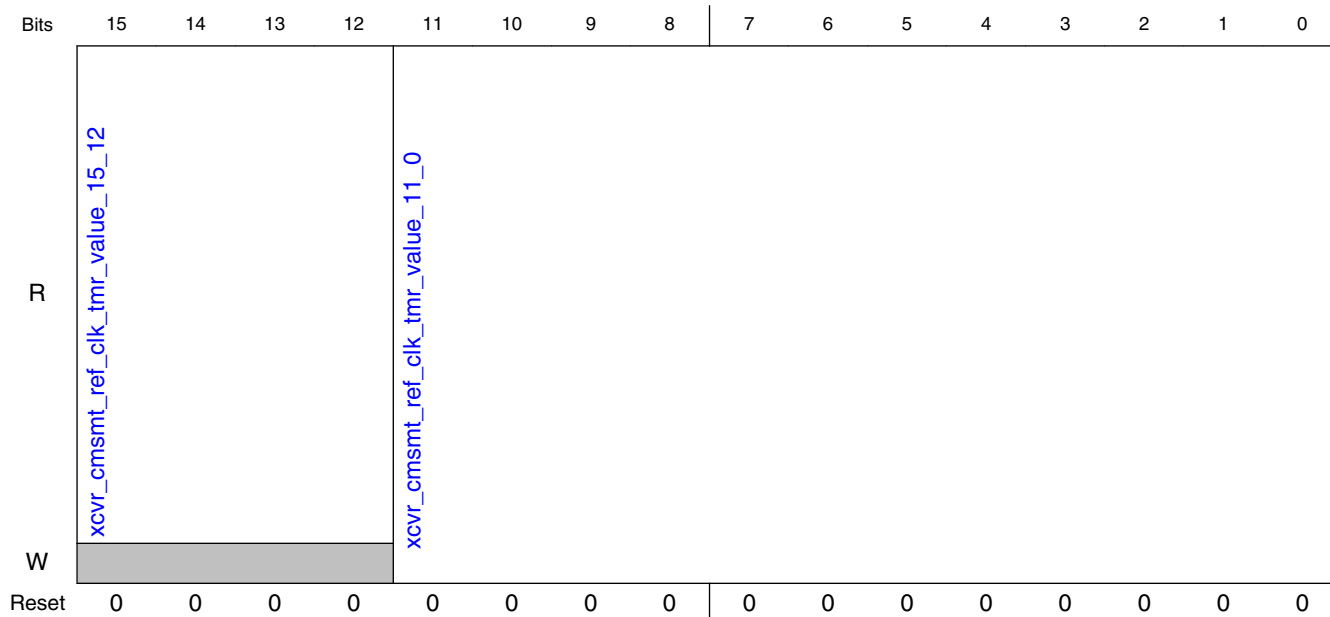
Field	Function
15-3 xcvr_cmsmt_test_clk_sel_15_3	Reserved
2-0 xcvr_cmsmt_test_clk_sel_2_0	Test clock select: This field drives the

13.4.10.2.175 Reference clock timer value register (lane0_xcvr_cmsmt_ref_clk_tmr_value - lane3_xcvr_cmsmt_ref_clk_tmr_value)

13.4.10.2.175.1 Offset

Register	Offset
lane0_xcvr_cmsmt_ref_clk_tmr_value	80B6h
lane1_xcvr_cmsmt_ref_clk_tmr_value	84B6h
lane2_xcvr_cmsmt_ref_clk_tmr_value	88B6h
lane3_xcvr_cmsmt_ref_clk_tmr_value	8CB6h

13.4.10.2.175.2 Diagram



13.4.10.2.175.3 Fields

Field	Function
15-12	Reserved
xcvr_cmsmt_ref_clk_tmr_value_15_12	

Table continues on the next page...

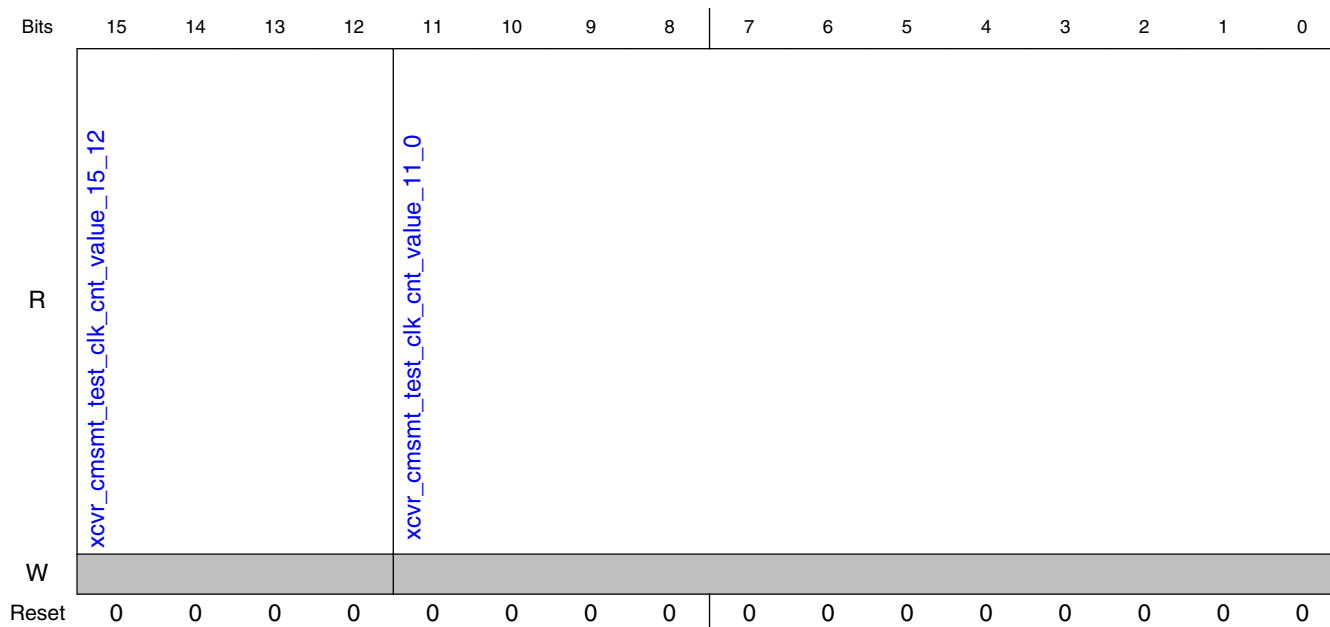
Field	Function
11-0 xcvr_cmsmt_ref_clk_tmr_value_11_0	Reference clock timer value : This specifies the amount of time, in reference clock cycles, to count test clock cycles. This value minus 1 is loaded into the reference clock timer. A value of 0 for this field is not valid when running this function.

13.4.10.2.176 Test clock counter value register (lane0_xcvt_r_cmsmt_test_clk_cnt_value - lane3_xcvt_r_cmsmt_test_clk_cnt_value)

13.4.10.2.176.1 Offset

Register	Offset
lane0_xcvt_r_cmsmt_test_clk_cnt_value	80B7h
lane1_xcvt_r_cmsmt_test_clk_cnt_value	84B7h
lane2_xcvt_r_cmsmt_test_clk_cnt_value	88B7h
lane3_xcvt_r_cmsmt_test_clk_cnt_value	8CB7h

13.4.10.2.176.2 Diagram



13.4.10.2.176.3 Fields

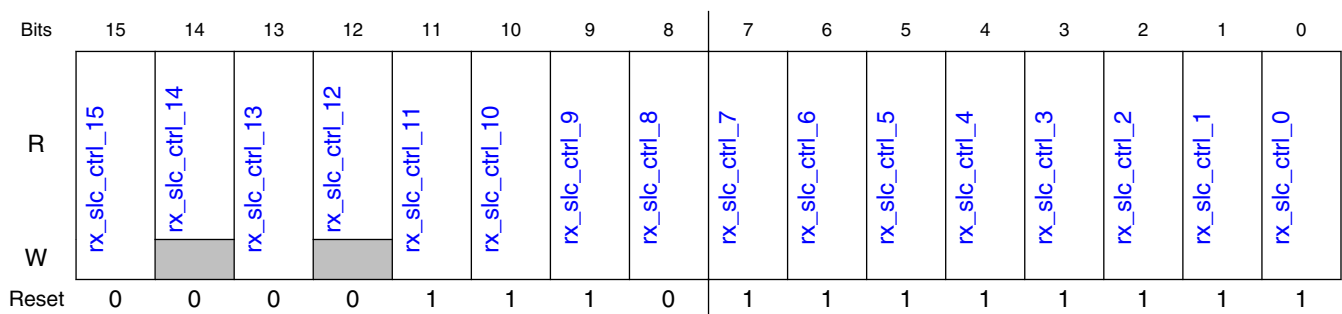
Field	Function
15-12 xcvr_cmsmt_test_clk_cnt_value_15_12	Reserved
11-0 xcvr_cmsmt_test_clk_cnt_value_11_0	Test clock counter value: When the test clock measurement process is complete, the value in this field specifies the number of test clock cycles that were counted in the time specified by the reference clock timer value. This field is only valid while the test clock measurement done bit in the

13.4.10.2.177 RX sampler latch calibration control register (lane0_rx_slc_ctrl - lane3_rx_slc_ctrl)

13.4.10.2.177.1 Offset

Register	Offset
lane0_rx_slc_ctrl	80E0h
lane1_rx_slc_ctrl	84E0h
lane2_rx_slc_ctrl	88E0h
lane3_rx_slc_ctrl	8CE0h

13.4.10.2.177.2 Diagram



13.4.10.2.177.3 Fields

Field	Function
15 rx_slc_ctrl_15	Start RX sampler latch calibration: Activating (1b1) this bit will start the RX sampler latch calibration process. This bit should remain active until the RX sampler latch calibration process is complete. To start

Table continues on the next page...

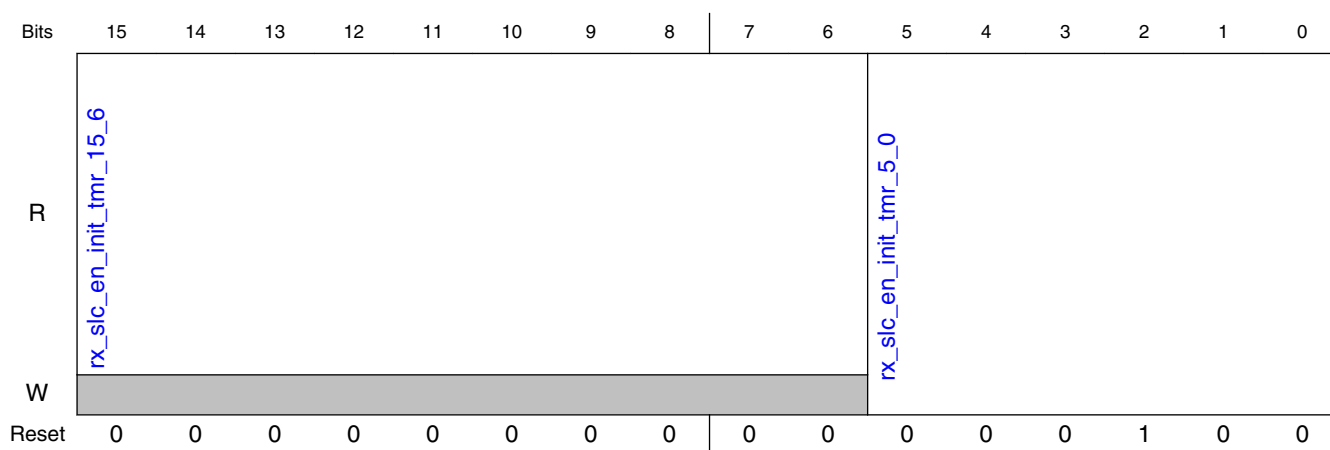
Field	Function
	another RX sampler latch calibration process, this bit must first be set inactive until the RX sampler latch calibration done bit is read as inactive.
14 rx_slc_ctrl_14	RX sampler latch calibration done: This bit will be set to 1b1 when the RX sampler latch calibration process is complete. It will be cleared by
13 rx_slc_ctrl_13	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
12 rx_slc_ctrl_12	Reserved
11 rx_slc_ctrl_11	I odd positive calibration unit enable.
10 rx_slc_ctrl_10	Q odd positive calibration unit enable.
9 rx_slc_ctrl_9	E odd positive calibration unit enable.
8 rx_slc_ctrl_8	I odd negative calibration unit enable.
7 rx_slc_ctrl_7	Q odd negative calibration unit enable.
6 rx_slc_ctrl_6	E odd negative calibration unit enable.
5 rx_slc_ctrl_5	I even positive calibration unit enable.
4 rx_slc_ctrl_4	Q even positive calibration unit enable.
3 rx_slc_ctrl_3	E even positive calibration unit enable.
2 rx_slc_ctrl_2	I even negative calibration unit enable.
1 rx_slc_ctrl_1	Q even negative calibration unit enable.
0 rx_slc_ctrl_0	E even negative calibration unit enable.

13.4.10.2.178 RX sampler latch calibration enable initialization timer value register (lane0_rx_slc_en_init_tmr - lane3_rx_slc_en_init_tmr)

13.4.10.2.178.1 Offset

Register	Offset
lane0_rx_slc_en_init_tmr	80E1h
lane1_rx_slc_en_init_tmr	84E1h
lane2_rx_slc_en_init_tmr	88E1h
lane3_rx_slc_en_init_tmr	8CE1h

13.4.10.2.178.2 Diagram



13.4.10.2.178.3 Fields

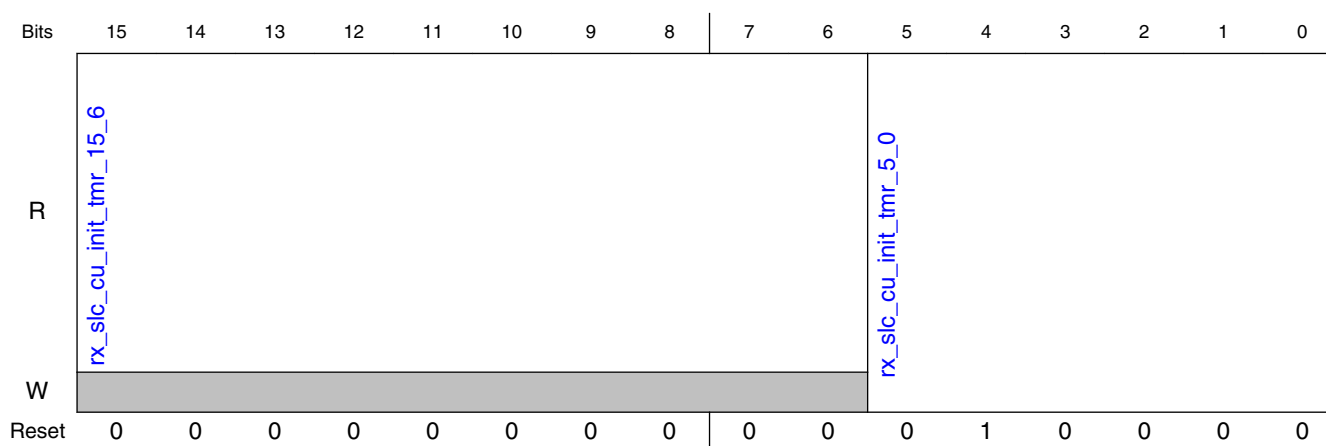
Field	Function
15-6 rx_slc_en_init_tmr_15_6	Reserved
5-0 rx_slc_en_init_tmr_5_0	RX sampler latch calibration enable initialization timer value : This is the value that is used for the RX sampler latch calibration enable initialization timer, which controls the time the

13.4.10.2.179 RX sampler latch calibration unit initialization timer value register (lane0_rx_slc_cu_init_tmr - lane3_rx_slc_cu_init_tmr)

13.4.10.2.179.1 Offset

Register	Offset
lane0_rx_slc_cu_init_tmr	80E2h
lane1_rx_slc_cu_init_tmr	84E2h
lane2_rx_slc_cu_init_tmr	88E2h
lane3_rx_slc_cu_init_tmr	8CE2h

13.4.10.2.179.2 Diagram



13.4.10.2.179.3 Fields

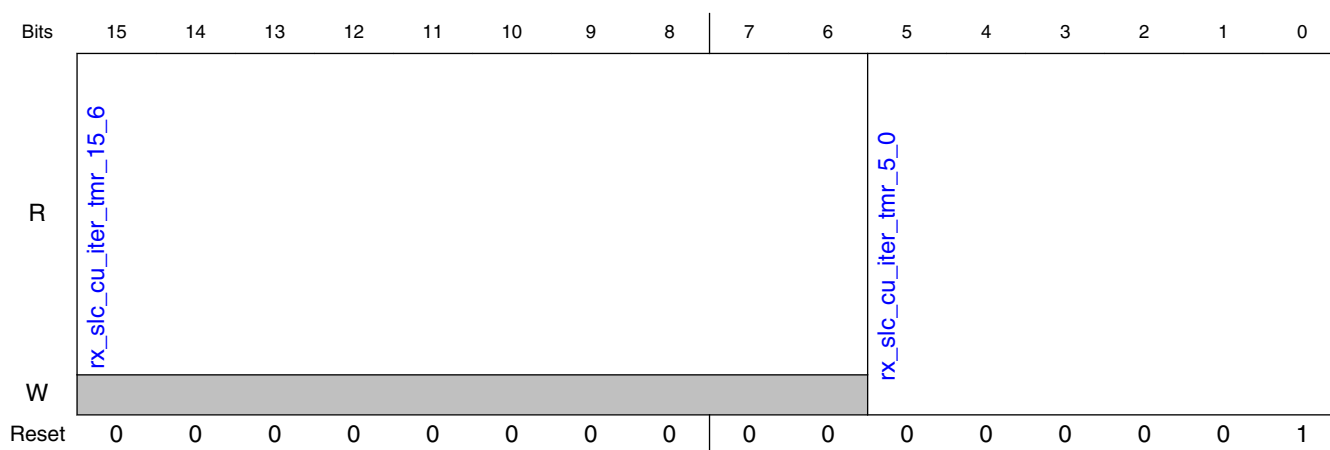
Field	Function
15-6 rx_slc_cu_init_tmr_15_6	Reserved
5-0 rx_slc_cu_init_tmr_5_0	RX sampler latch calibration unit initialization timer value : This is the value that is used for each of the calibration units initialization timers.

13.4.10.2.180 RX sampler latch calibration unit iteration timer value register (lane0_rx_slc_cu_iter_tmr - lane3_rx_slc_cu_iter_tmr)

13.4.10.2.180.1 Offset

Register	Offset
lane0_rx_slc_cu_iter_tmr	80E3h
lane1_rx_slc_cu_iter_tmr	84E3h
lane2_rx_slc_cu_iter_tmr	88E3h
lane3_rx_slc_cu_iter_tmr	8CE3h

13.4.10.2.180.2 Diagram



13.4.10.2.180.3 Fields

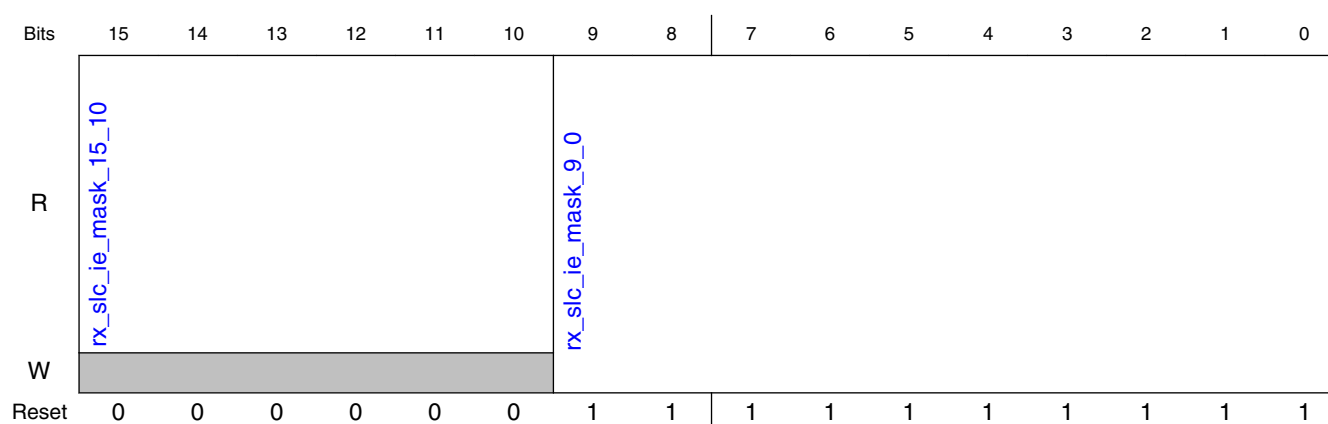
Field	Function
15-6 rx_slc_cu_iter_tmr_15_6	Reserved
5-0 rx_slc_cu_iter_tmr_5_0	RX sampler latch calibration unit iteration timer value : This is the value that is used for each of the calibration units iteration timers.

13.4.10.2.181 RX sampler latch calibration I even data mask register (lane0_rx_slc_ie_mask - lane3_rx_slc_ie_mask)

13.4.10.2.181.1 Offset

Register	Offset
lane0_rx_slc_ie_mask	80E4h
lane1_rx_slc_ie_mask	84E4h
lane2_rx_slc_ie_mask	88E4h
lane3_rx_slc_ie_mask	8CE4h

13.4.10.2.181.2 Diagram



13.4.10.2.181.3 Fields

Field	Function
15-10 rx_slc_ie_mask_15_10	Reserved
9-0 rx_slc_ie_mask_9_0	I even data mask: The bits in this field can be used to mask the I data even bits in the data unit component.

13.4.10.2.182 RX sampler latch calibration I odd data mask register (lane0_rx_slc_io_mask - lane3_rx_slc_io_mask)

13.4.10.2.182.1 Offset

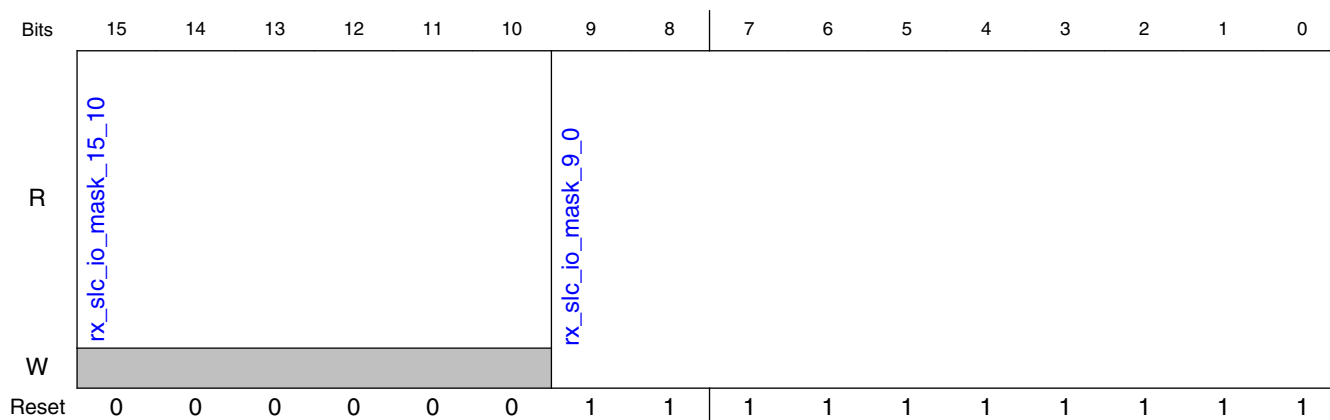
Register	Offset
lane0_rx_slc_io_mask	80E5h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane1_rx_slc_io_mask	84E5h
lane2_rx_slc_io_mask	88E5h
lane3_rx_slc_io_mask	8CE5h

13.4.10.2.182.2 Diagram



13.4.10.2.182.3 Fields

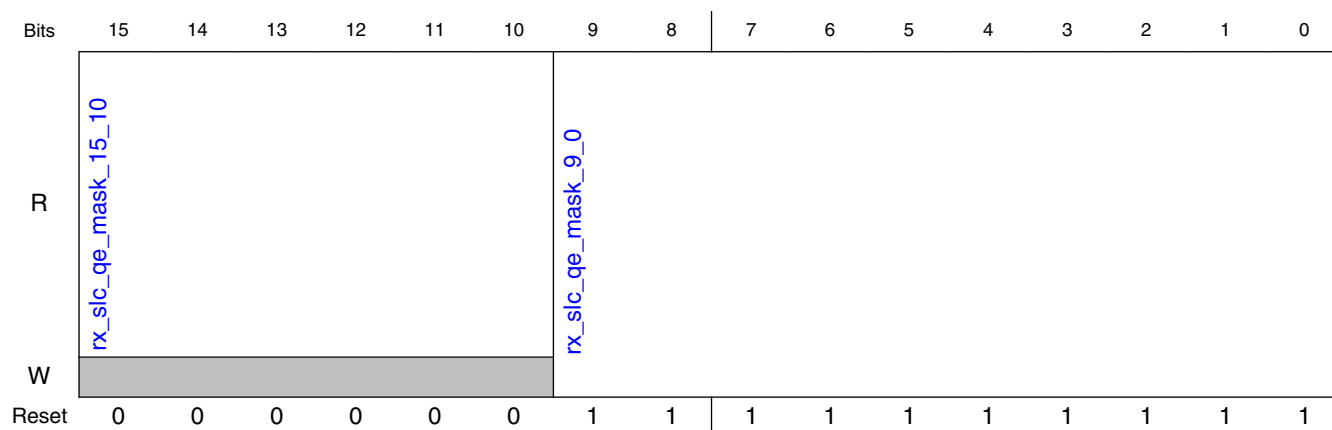
Field	Function
15-10 rx_slc_io_mask_15_10	Reserved
9-0 rx_slc_io_mask_9_0	I odd data mask: The bits in this field can be used to mask the I data odd bits in the data unit component.

13.4.10.2.183 RX sampler latch calibration Q even data mask register (lane 0_rx_slc_qe_mask - lane3_rx_slc_qe_mask)

13.4.10.2.183.1 Offset

Register	Offset
lane0_rx_slc_qe_mask	80E6h
lane1_rx_slc_qe_mask	84E6h
lane2_rx_slc_qe_mask	88E6h
lane3_rx_slc_qe_mask	8CE6h

13.4.10.2.183.2 Diagram



13.4.10.2.183.3 Fields

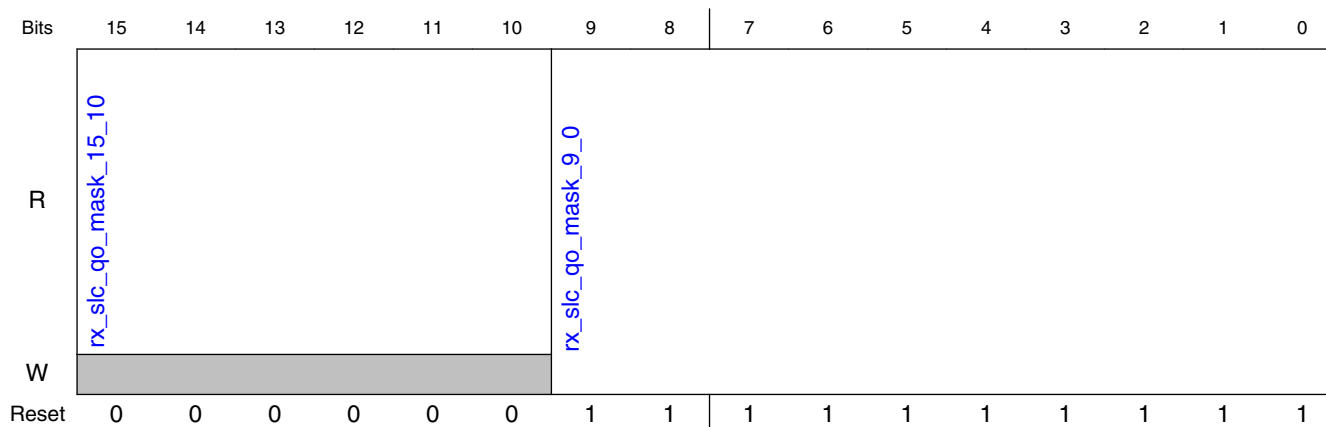
Field	Function
15-10 rx_slc_qe_mask_15_10	Reserved
9-0 rx_slc_qe_mask_9_0	Q even data mask: The bits in this field can be used to mask the Q data even bits in the data unit component.

13.4.10.2.184 RX sampler latch calibration Q odd data mask register (lane 0_rx_slc_qo_mask - lane3_rx_slc_qo_mask)

13.4.10.2.184.1 Offset

Register	Offset
lane0_rx_slc_qo_mask	80E7h
lane1_rx_slc_qo_mask	84E7h
lane2_rx_slc_qo_mask	88E7h
lane3_rx_slc_qo_mask	8CE7h

13.4.10.2.184.2 Diagram



13.4.10.2.184.3 Fields

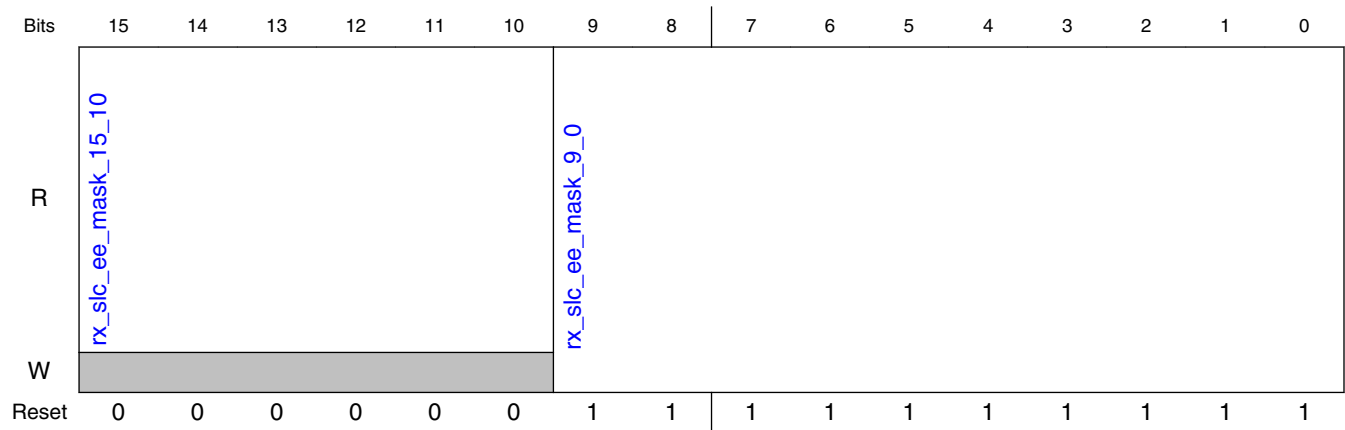
Field	Function
15-10 rx_slc_qo_mask_15_10	Reserved
9-0 rx_slc_qo_mask_9_0	Q odd data mask: The bits in this field can be used to mask the Q data odd bits in the data unit component.

13.4.10.2.185 RX sampler latch calibration E even data mask register (lane 0_rx_slc_ee_mask - lane3_rx_slc_ee_mask)

13.4.10.2.185.1 Offset

Register	Offset
lane0_rx_slc_ee_mask	80E8h
lane1_rx_slc_ee_mask	84E8h
lane2_rx_slc_ee_mask	88E8h
lane3_rx_slc_ee_mask	8CE8h

13.4.10.2.185.2 Diagram



13.4.10.2.185.3 Fields

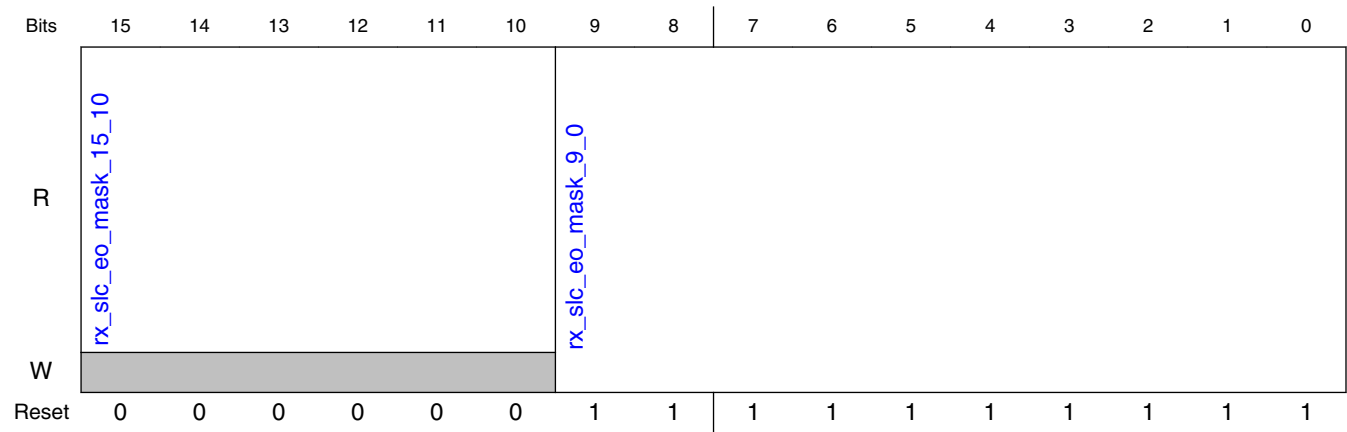
Field	Function
15-10 rx_slc_ee_mask_15_10	Reserved
9-0 rx_slc_ee_mask_9_0	E even data mask: The bits in this field can be used to mask the E data even bits in the data unit component.

13.4.10.2.186 RX sampler latch calibration E odd data mask register (lane 0_rx_slc_eo_mask - lane3_rx_slc_eo_mask)

13.4.10.2.186.1 Offset

Register	Offset
lane0_rx_slc_eo_mask	80E9h
lane1_rx_slc_eo_mask	84E9h
lane2_rx_slc_eo_mask	88E9h
lane3_rx_slc_eo_mask	8CE9h

13.4.10.2.186.2 Diagram



13.4.10.2.186.3 Fields

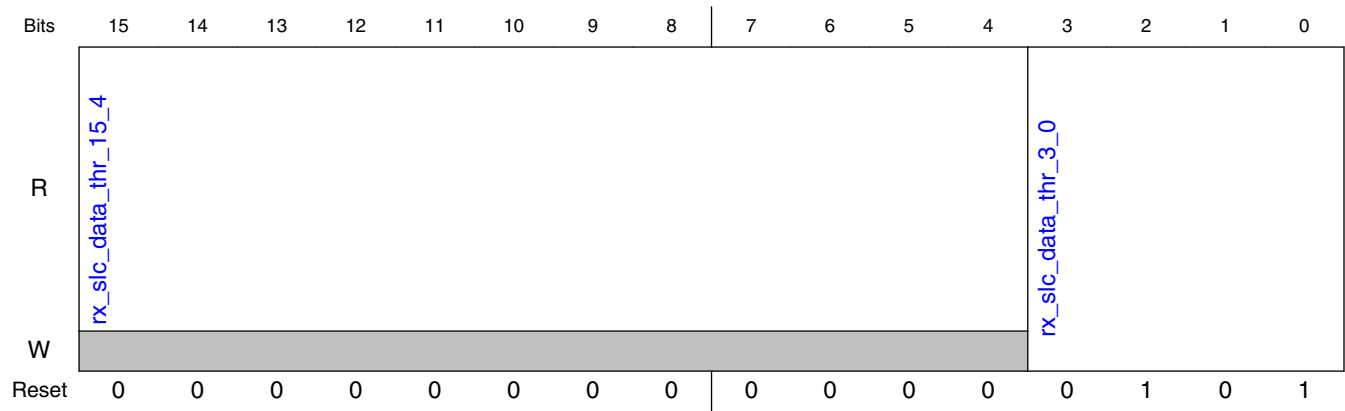
Field	Function
15-10 rx_slc_eo_mask_15_10	Reserved
9-0 rx_slc_eo_mask_9_0	E odd data mask: The bits in this field can be used to mask the E data odd bits in the data unit component.

13.4.10.2.187 RX sampler latch calibration data threshold register (lane0_rx_slc_data_thr - lane3_rx_slc_data_thr)

13.4.10.2.187.1 Offset

Register	Offset
lane0_rx_slc_data_thr	80EAh
lane1_rx_slc_data_thr	84EAh
lane2_rx_slc_data_thr	88EAh
lane3_rx_slc_data_thr	8CEAh

13.4.10.2.187.2 Diagram



13.4.10.2.187.3 Fields

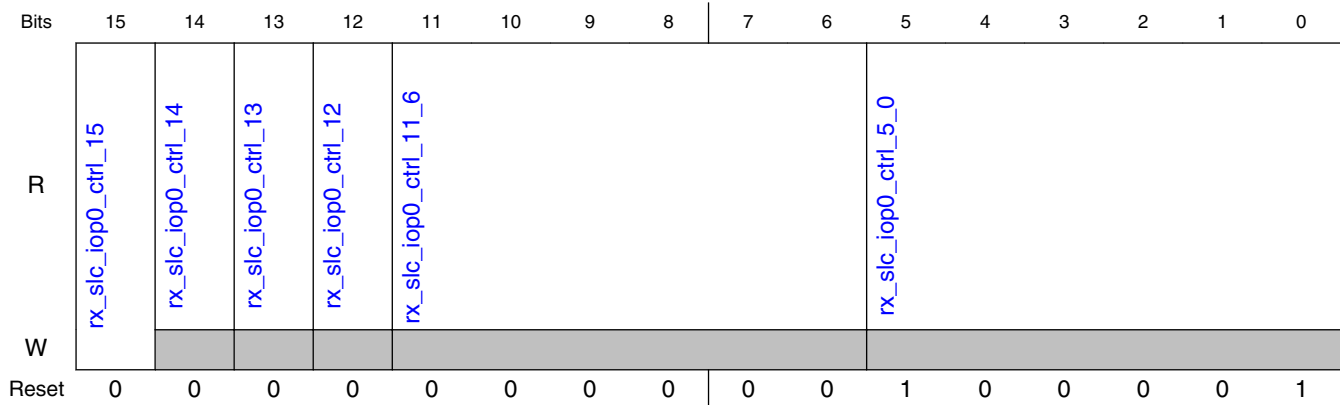
Field	Function
15-4 rx_slc_data_thr_15_4	Reserved
3-0 rx_slc_data_thr_3_0	Data threshold value: Specifies the value that the RX sampler latch calibration data unit compares the sum of 1s for each parallel data input against to create the inputs to each of the calibration units. The range of values for this field is as follows.

13.4.10.2.188 RX sampler latch I odd positive 0 calibration unit control register (lane0_rx_slc_iop0_ctrl - lane3_rx_slc_iop0_ctrl)

13.4.10.2.188.1 Offset

Register	Offset
lane0_rx_slc_iop0_ctrl	8100h
lane1_rx_slc_iop0_ctrl	8500h
lane2_rx_slc_iop0_ctrl	8900h
lane3_rx_slc_iop0_ctrl	8D00h

13.4.10.2.188.2 Diagram



13.4.10.2.188.3 Fields

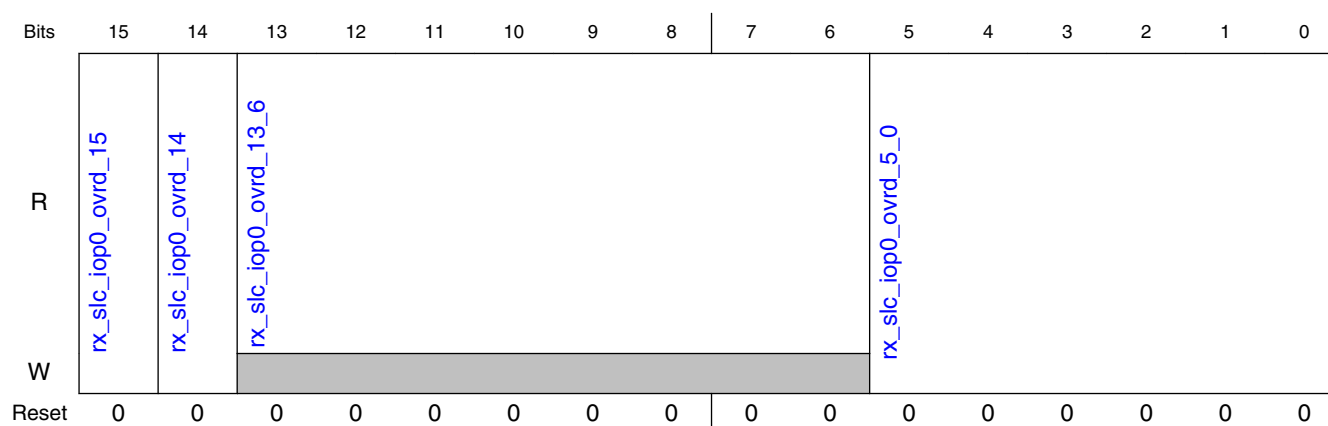
Field	Function
15 rx_slc_iop0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_iop0_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_iop0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_iop0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_iop0_ctrl_11_6	Reserved
5-0 rx_slc_iop0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.189 RX sampler latch I odd positive 0 calibration unit override register (lane0_rx_slc_iop0_ovrd - lane3_rx_slc_iop0_ovrd)

13.4.10.2.189.1 Offset

Register	Offset
lane0_rx_slc_iop0_ovrd	8101h
lane1_rx_slc_iop0_ovrd	8501h
lane2_rx_slc_iop0_ovrd	8901h
lane3_rx_slc_iop0_ovrd	8D01h

13.4.10.2.189.2 Diagram



13.4.10.2.189.3 Fields

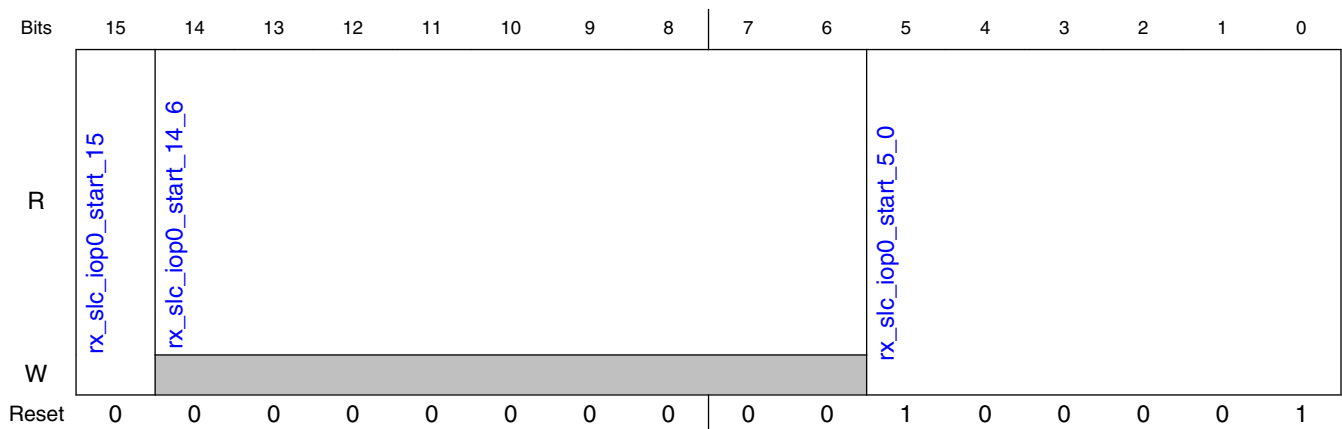
Field	Function
15 <code>rx_slc_iop0_ovrd_15</code>	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 <code>rx_slc_iop0_ovrd_14</code>	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 <code>rx_slc_iop0_ovrd_13_6</code>	Reserved
5-0 <code>rx_slc_iop0_ovrd_5_0</code>	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.190 RX sampler latch I odd positive 0 calibration unit start register (lane0_rx_slc_iop0_start - lane3_rx_slc_iop0_start)

13.4.10.2.190.1 Offset

Register	Offset
lane0_rx_slc_iop0_start	8102h
lane1_rx_slc_iop0_start	8502h
lane2_rx_slc_iop0_start	8902h
lane3_rx_slc_iop0_start	8D02h

13.4.10.2.190.2 Diagram



13.4.10.2.190.3 Fields

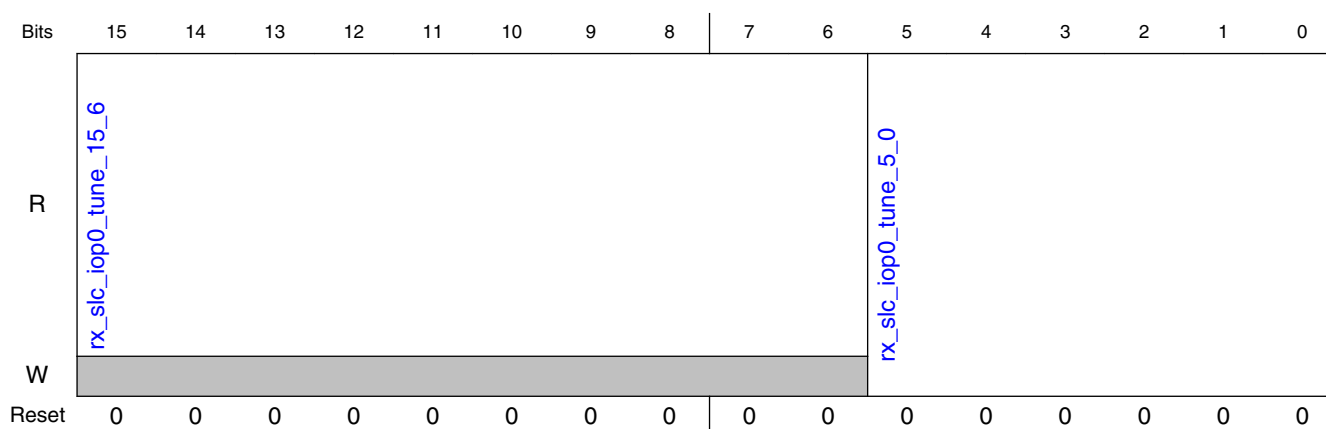
Field	Function
15 rx_slc_iop0_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_iop0_start_14_6	Reserved
5-0 rx_slc_iop0_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.191 RX sampler latch I odd positive 0 calibration unit tune register (lane0_rx_slc_iop0_tune - lane3_rx_slc_iop0_tune)

13.4.10.2.191.1 Offset

Register	Offset
lane0_rx_slc_iop0_tune	8103h
lane1_rx_slc_iop0_tune	8503h
lane2_rx_slc_iop0_tune	8903h
lane3_rx_slc_iop0_tune	8D03h

13.4.10.2.191.2 Diagram



13.4.10.2.191.3 Fields

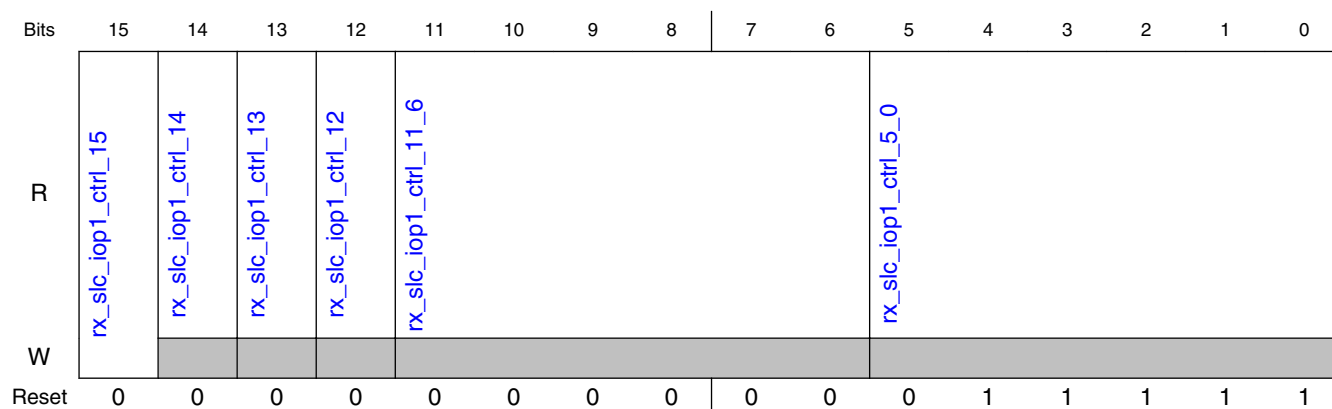
Field	Function
15-6 rx_slc_iop0_tune_15_6	Reserved
5-0 rx_slc_iop0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.192 RX sampler latch I odd positive 1 calibration unit control register (lane0_rx_slc_iop1_ctrl - lane3_rx_slc_iop1_ctrl)

13.4.10.2.192.1 Offset

Register	Offset
lane0_rx_slc_iop1_ctrl	8104h
lane1_rx_slc_iop1_ctrl	8504h
lane2_rx_slc_iop1_ctrl	8904h
lane3_rx_slc_iop1_ctrl	8D04h

13.4.10.2.192.2 Diagram



13.4.10.2.192.3 Fields

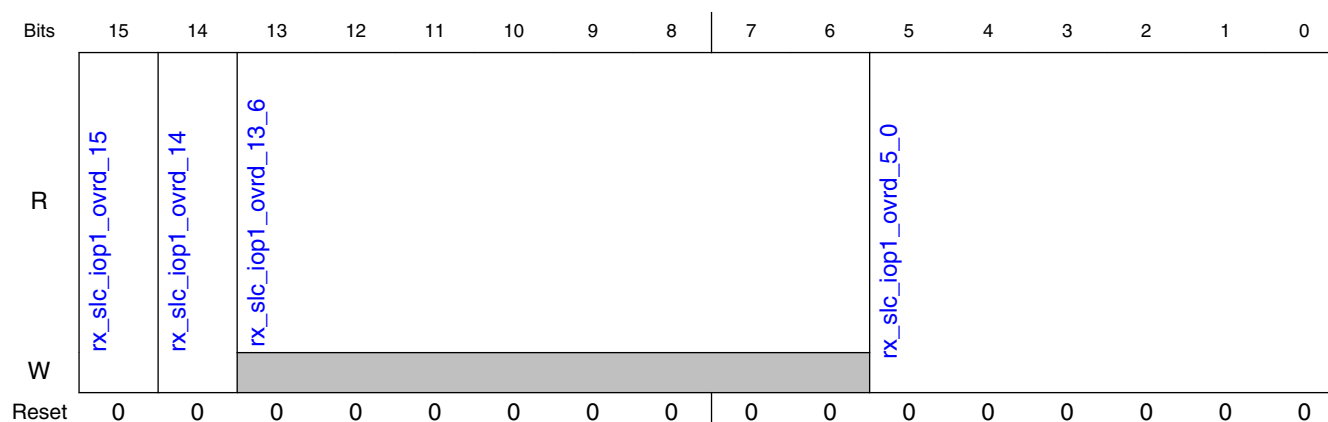
Field	Function
15 <code>rx_slc_iop1_ctrl_15</code>	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 <code>rx_slc_iop1_ctrl_14</code>	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 <code>rx_slc_iop1_ctrl_13</code>	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 <code>rx_slc_iop1_ctrl_12</code>	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 <code>rx_slc_iop1_ctrl_11_6</code>	Reserved
5-0 <code>rx_slc_iop1_ctrl_5_0</code>	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.193 RX sampler latch I odd positive 1 calibration unit override register (lane0_rx_slc_iop1_ovrd - lane3_rx_slc_iop1_ovrd)

13.4.10.2.193.1 Offset

Register	Offset
lane0_rx_slc_iop1_ovrd	8105h
lane1_rx_slc_iop1_ovrd	8505h
lane2_rx_slc_iop1_ovrd	8905h
lane3_rx_slc_iop1_ovrd	8D05h

13.4.10.2.193.2 Diagram



13.4.10.2.193.3 Fields

Field	Function
15 <code>rx_slc_iop1_ovrd_15</code>	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 <code>rx_slc_iop1_ovrd_14</code>	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 <code>rx_slc_iop1_ovrd_13_6</code>	Reserved
5-0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the

Clocks And Resets

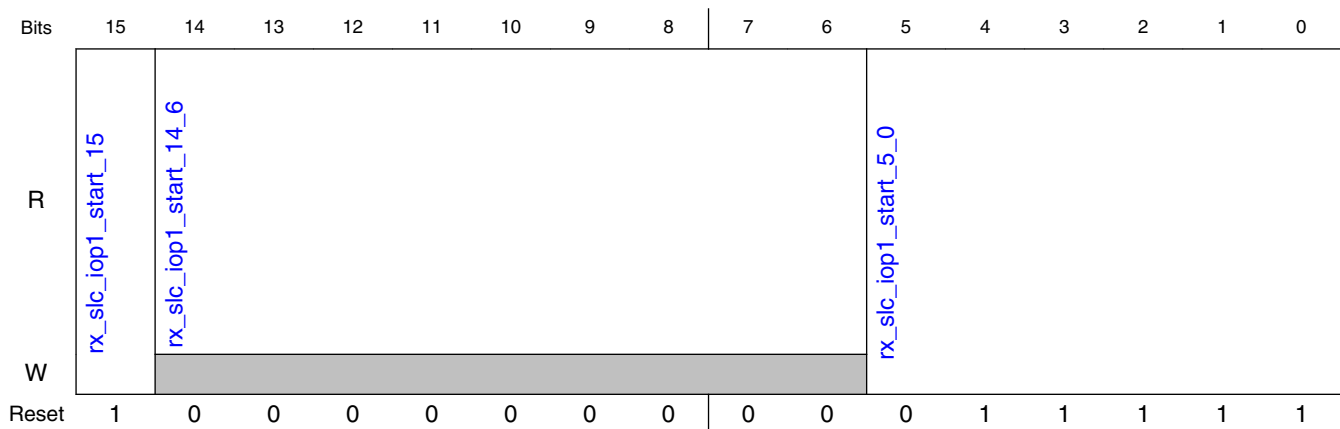
Field	Function
rx_slc_iop1_ovr d_5_0	calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.194 RX sampler latch I odd positive 1 calibration unit start register (lane0_rx_slc_iop1_start - lane3_rx_slc_iop1_start)

13.4.10.2.194.1 Offset

Register	Offset
lane0_rx_slc_iop1_start	8106h
lane1_rx_slc_iop1_start	8506h
lane2_rx_slc_iop1_start	8906h
lane3_rx_slc_iop1_start	8D06h

13.4.10.2.194.2 Diagram



13.4.10.2.194.3 Fields

Field	Function
15 rx_slc_iop1_star t_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_iop1_star t_14_6	Reserved

Table continues on the next page...

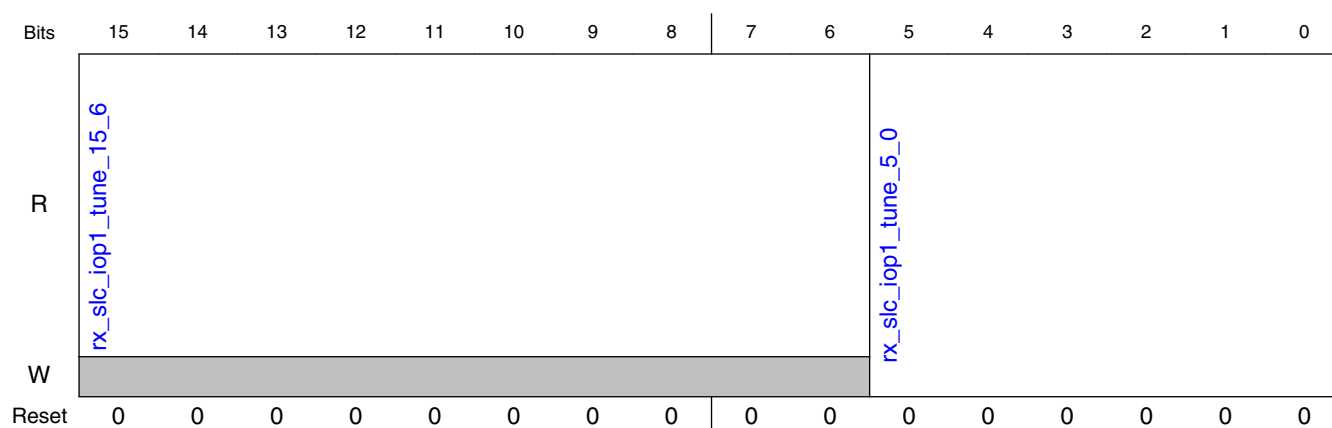
Field	Function
5-0 rx_slc_iop1_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.195 RX sampler latch I odd positive 1 calibration unit tune register (lane0_rx_slc_iop1_tune - lane3_rx_slc_iop1_tune)

13.4.10.2.195.1 Offset

Register	Offset
lane0_rx_slc_iop1_tune	8107h
lane1_rx_slc_iop1_tune	8507h
lane2_rx_slc_iop1_tune	8907h
lane3_rx_slc_iop1_tune	8D07h

13.4.10.2.195.2 Diagram



13.4.10.2.195.3 Fields

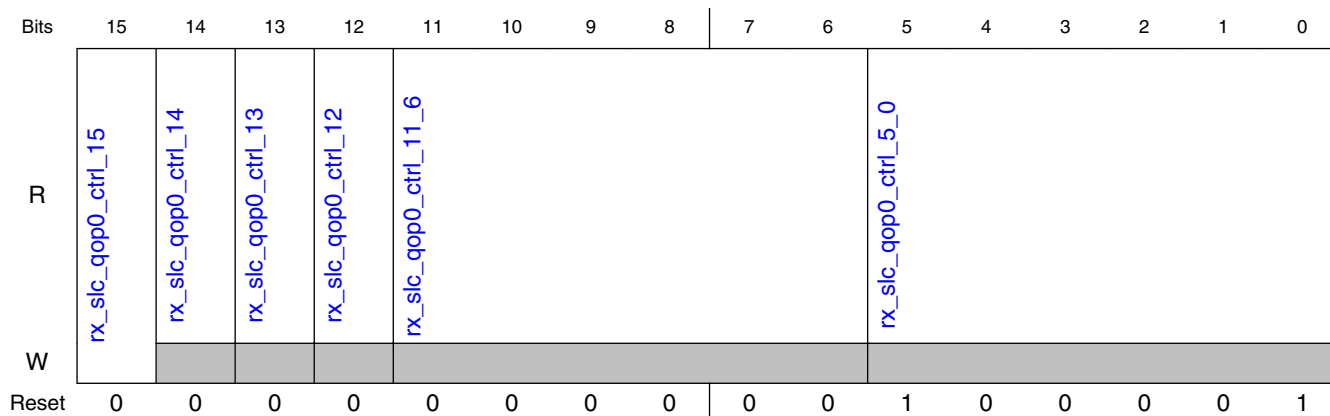
Field	Function
15-6 rx_slc_iop1_tune_15_6	Reserved
5-0 rx_slc_iop1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.196 RX sampler latch Q odd positive 0 calibration unit control register (lane0_rx_slc_qop0_ctrl - lane3_rx_slc_qop0_ctrl)

13.4.10.2.196.1 Offset

Register	Offset
lane0_rx_slc_qop0_ctrl	8108h
lane1_rx_slc_qop0_ctrl	8508h
lane2_rx_slc_qop0_ctrl	8908h
lane3_rx_slc_qop0_ctrl	8D08h

13.4.10.2.196.2 Diagram



13.4.10.2.196.3 Fields

Field	Function
15 rx_slc_qop0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_qop0_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_qop0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.

Table continues on the next page...

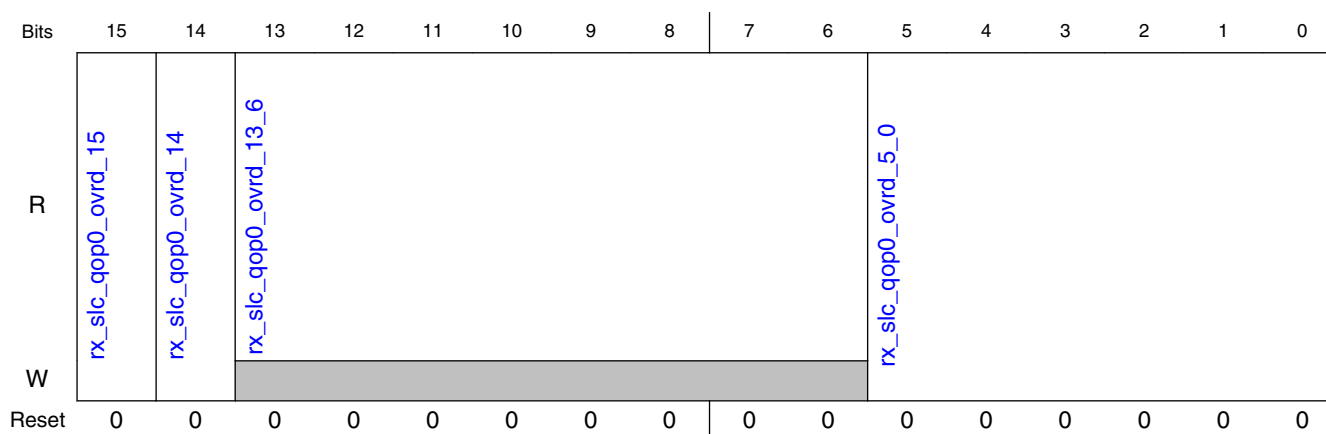
Field	Function
rx_slc_qop0_ctrl_12	
11-6 rx_slc_qop0_ctrl_11_6	Reserved
5-0 rx_slc_qop0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.197 RX sampler latch Q odd positive 0 calibration unit override register (lane0_rx_slc_qop0_ovrd - lane3_rx_slc_qop0_ovrd)

13.4.10.2.197.1 Offset

Register	Offset
lane0_rx_slc_qop0_ovrd	8109h
lane1_rx_slc_qop0_ovrd	8509h
lane2_rx_slc_qop0_ovrd	8909h
lane3_rx_slc_qop0_ovrd	8D09h

13.4.10.2.197.2 Diagram



13.4.10.2.197.3 Fields

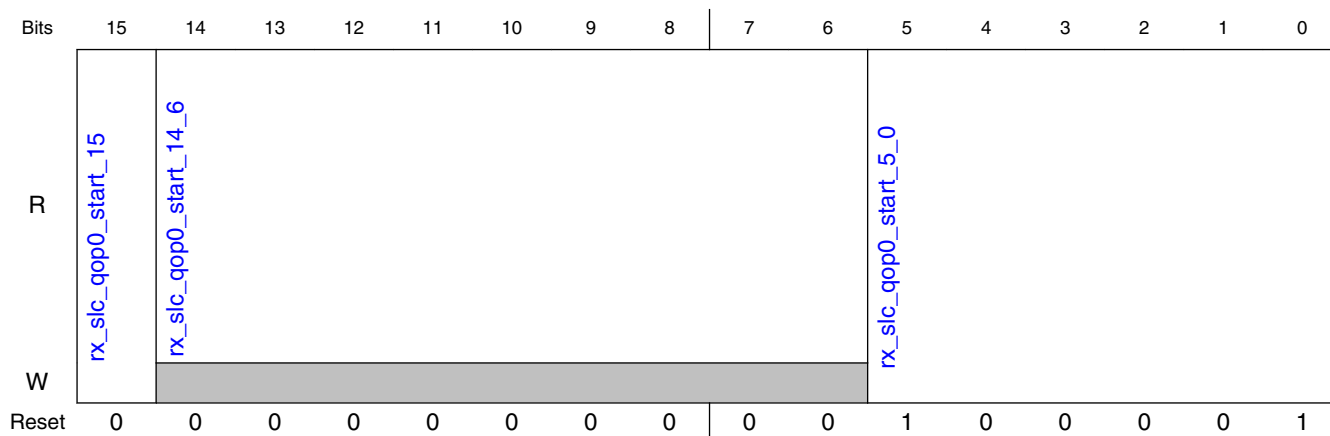
Field	Function
15 rx_slc_qop0_ovr_d_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_qop0_ovr_d_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_qop0_ovr_d_13_6	Reserved
5-0 rx_slc_qop0_ovr_d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.198 RX sampler latch Q odd positive 0 calibration unit start register (lane0_rx_slc_qop0_start - lane3_rx_slc_qop0_start)

13.4.10.2.198.1 Offset

Register	Offset
lane0_rx_slc_qop0_start	810Ah
lane1_rx_slc_qop0_start	850Ah
lane2_rx_slc_qop0_start	890Ah
lane3_rx_slc_qop0_start	8D0Ah

13.4.10.2.198.2 Diagram



13.4.10.2.198.3 Fields

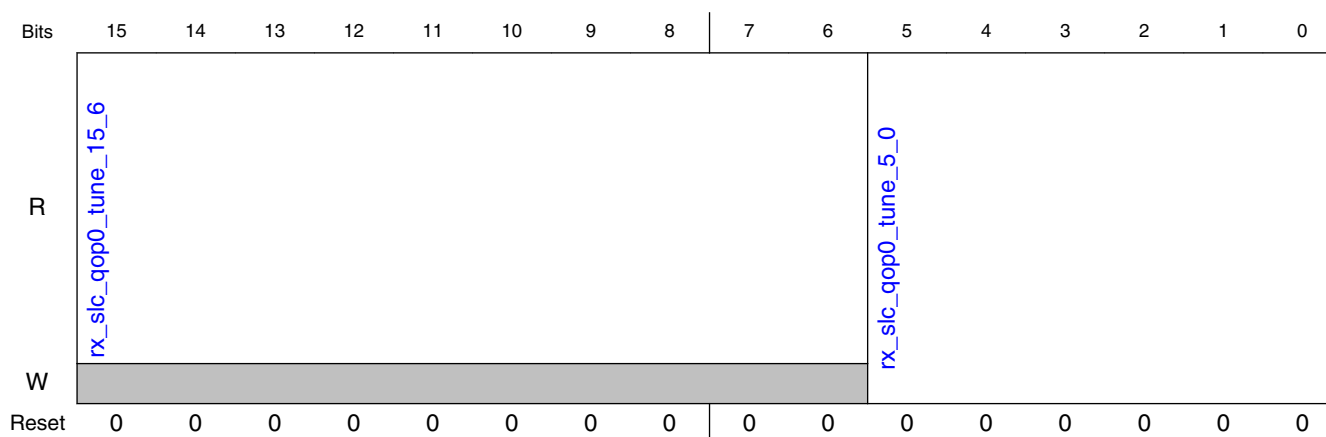
Field	Function
15 rx_slc_qop0_sta rt_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_qop0_sta rt_14_6	Reserved
5-0 rx_slc_qop0_sta rt_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.199 RX sampler latch Q odd positive 0 calibration unit tune register (lane0_rx_slc_qop0_tune - lane3_rx_slc_qop0_tune)

13.4.10.2.199.1 Offset

Register	Offset
lane0_rx_slc_qop0_tune	810Bh
lane1_rx_slc_qop0_tune	850Bh
lane2_rx_slc_qop0_tune	890Bh
lane3_rx_slc_qop0_tune	8D0Bh

13.4.10.2.199.2 Diagram



13.4.10.2.199.3 Fields

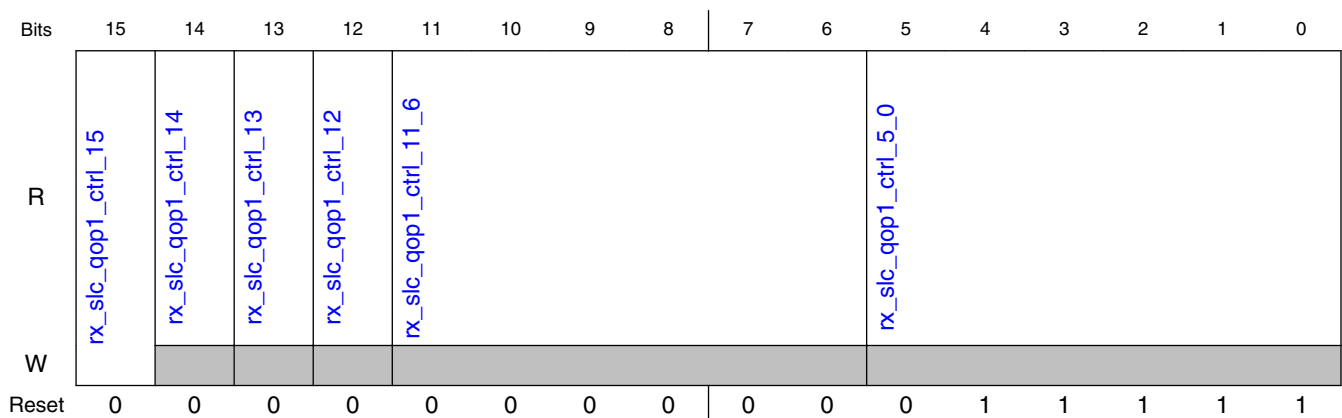
Field	Function
15-6 rx_slc_qop0_tune_15_6	Reserved
5-0 rx_slc_qop0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.200 RX sampler latch Q odd positive 1 calibration unit control register (lane0_rx_slc_qop1_ctrl - lane3_rx_slc_qop1_ctrl)

13.4.10.2.200.1 Offset

Register	Offset
lane0_rx_slc_qop1_ctrl	810Ch
lane1_rx_slc_qop1_ctrl	850Ch
lane2_rx_slc_qop1_ctrl	890Ch
lane3_rx_slc_qop1_ctrl	8D0Ch

13.4.10.2.200.2 Diagram



13.4.10.2.200.3 Fields

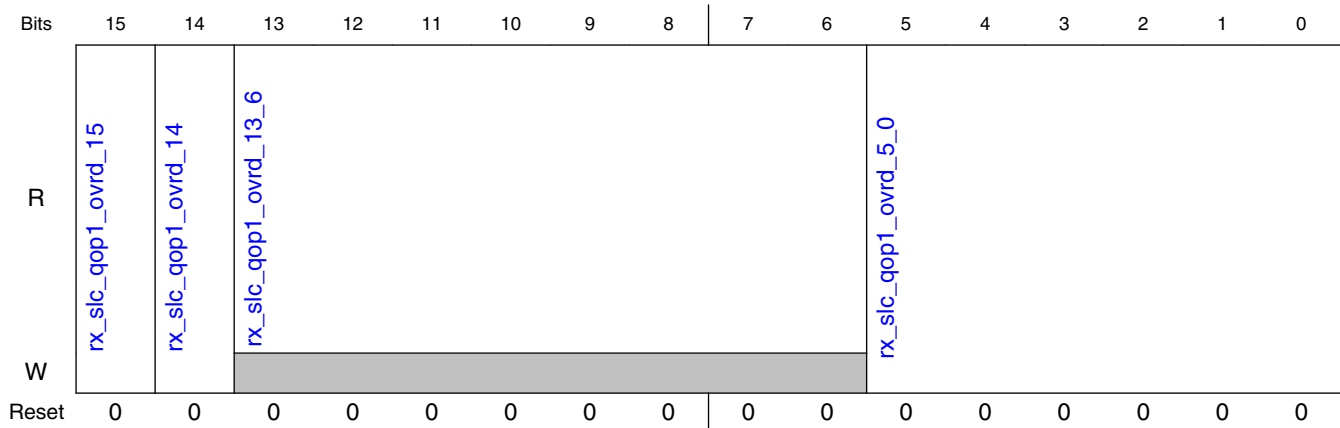
Field	Function
15 rx_slc_qop1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_qop1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_qop1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_qop1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_qop1_ctrl_11_6	Reserved
5-0 rx_slc_qop1_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.201 RX sampler latch Q odd positive 1 calibration unit override register (lane0_rx_slc_qop1_ovrd - lane3_rx_slc_qop1_ovrd)

13.4.10.2.201.1 Offset

Register	Offset
lane0_rx_slc_qop1_ovrd	810Dh
lane1_rx_slc_qop1_ovrd	850Dh
lane2_rx_slc_qop1_ovrd	890Dh
lane3_rx_slc_qop1_ovrd	8D0Dh

13.4.10.2.201.2 Diagram



13.4.10.2.201.3 Fields

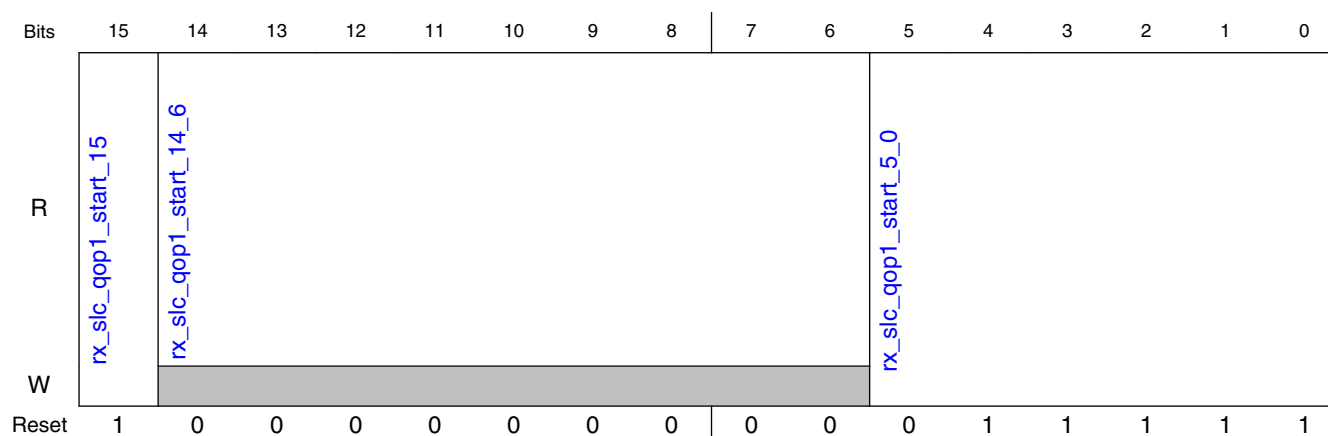
Field	Function
15 rx_slc_qop1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_qop1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_qop1_ovrd_13_6	Reserved
5-0 rx_slc_qop1_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.202 RX sampler latch Q odd positive 1 calibration unit start register (lane0_rx_slc_qop1_start - lane3_rx_slc_qop1_start)

13.4.10.2.202.1 Offset

Register	Offset
lane0_rx_slc_qop1_start	810Eh
lane1_rx_slc_qop1_start	850Eh
lane2_rx_slc_qop1_start	890Eh
lane3_rx_slc_qop1_start	8D0Eh

13.4.10.2.202.2 Diagram



13.4.10.2.202.3 Fields

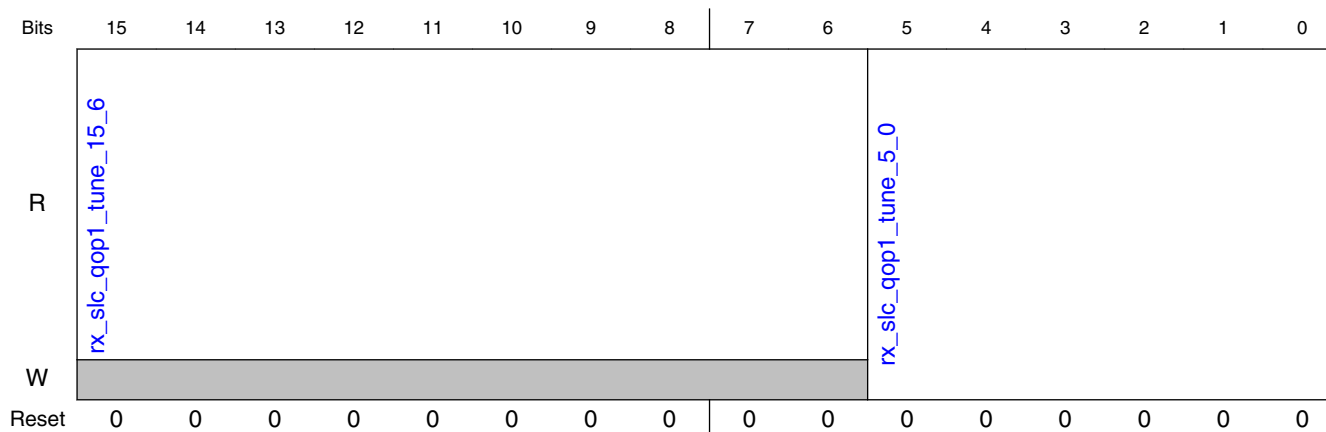
Field	Function
15 rx_slc_qop1_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_qop1_start_14_6	Reserved
5-0 rx_slc_qop1_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.203 RX sampler latch Q odd positive 1 calibration unit tune register (lane0_rx_slc_qop1_tune - lane3_rx_slc_qop1_tune)

13.4.10.2.203.1 Offset

Register	Offset
lane0_rx_slc_qop1_tune	810Fh
lane1_rx_slc_qop1_tune	850Fh
lane2_rx_slc_qop1_tune	890Fh
lane3_rx_slc_qop1_tune	8D0Fh

13.4.10.2.203.2 Diagram



13.4.10.2.203.3 Fields

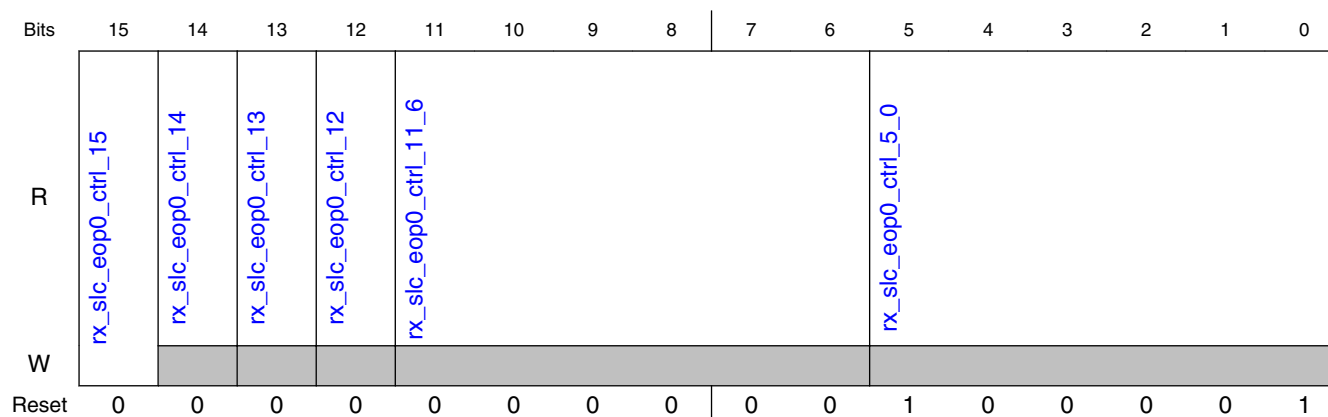
Field	Function
15-6 <code>rx_slc_qop1_tune_15_6</code>	Reserved
5-0 <code>rx_slc_qop1_tune_5_0</code>	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.204 RX sampler latch E odd positive 0 calibration unit control register (lane0_rx_slc_eop0_ctrl - lane3_rx_slc_eop0_ctrl)

13.4.10.2.204.1 Offset

Register	Offset
<code>lane0_rx_slc_eop0_ctrl</code>	8110h
<code>lane1_rx_slc_eop0_ctrl</code>	8510h
<code>lane2_rx_slc_eop0_ctrl</code>	8910h
<code>lane3_rx_slc_eop0_ctrl</code>	8D10h

13.4.10.2.204.2 Diagram



13.4.10.2.204.3 Fields

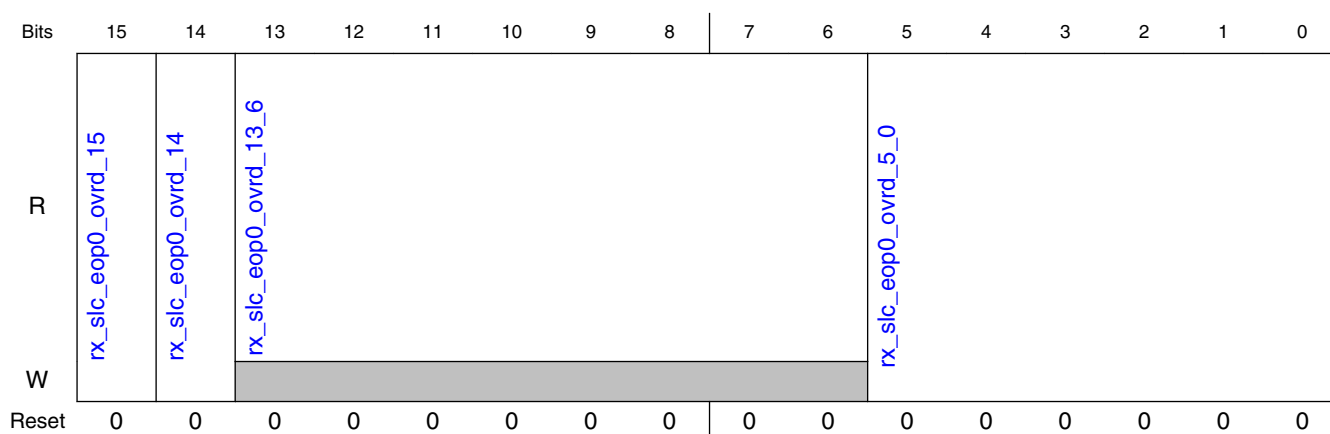
Field	Function
15 rx_slc_eop0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_eop0_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_eop0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_eop0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_eop0_ctrl_11_6	Reserved
5-0 rx_slc_eop0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.205 RX sampler latch E odd positive 0 calibration unit override register (lane0_rx_slc_eop0_ovrd - lane3_rx_slc_eop0_ovrd)

13.4.10.2.205.1 Offset

Register	Offset
lane0_rx_slc_eop0_ovrd	8111h
lane1_rx_slc_eop0_ovrd	8511h
lane2_rx_slc_eop0_ovrd	8911h
lane3_rx_slc_eop0_ovrd	8D11h

13.4.10.2.205.2 Diagram



13.4.10.2.205.3 Fields

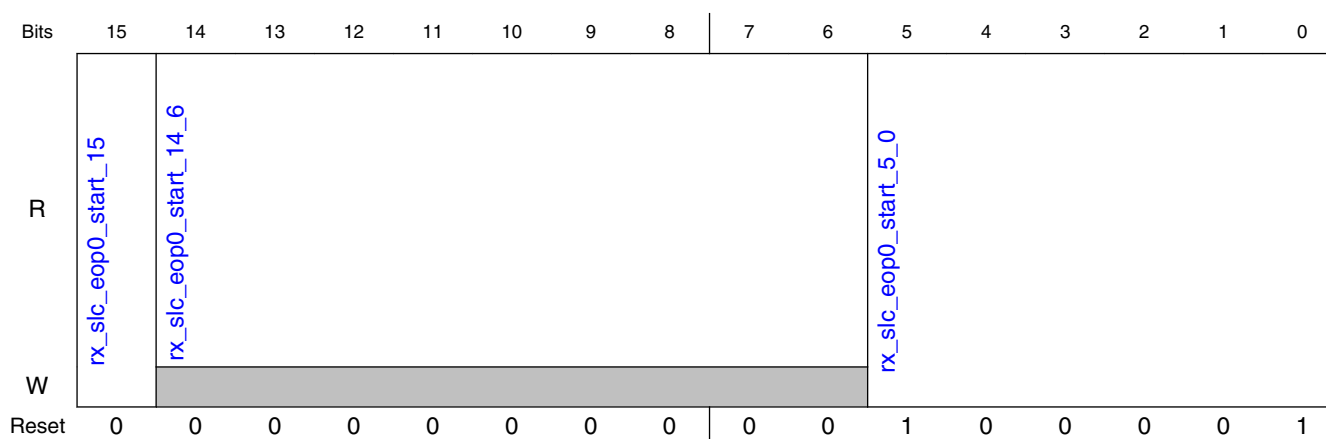
Field	Function
15 rx_slc_eop0_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_eop0_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_eop0_ovrd_13_6	Reserved
5-0 rx_slc_eop0_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.206 RX sampler latch E odd positive 0 calibration unit start register (lane0_rx_slc_eop0_start - lane3_rx_slc_eop0_start)

13.4.10.2.206.1 Offset

Register	Offset
lane0_rx_slc_eop0_start	8112h
lane1_rx_slc_eop0_start	8512h
lane2_rx_slc_eop0_start	8912h
lane3_rx_slc_eop0_start	8D12h

13.4.10.2.206.2 Diagram



13.4.10.2.206.3 Fields

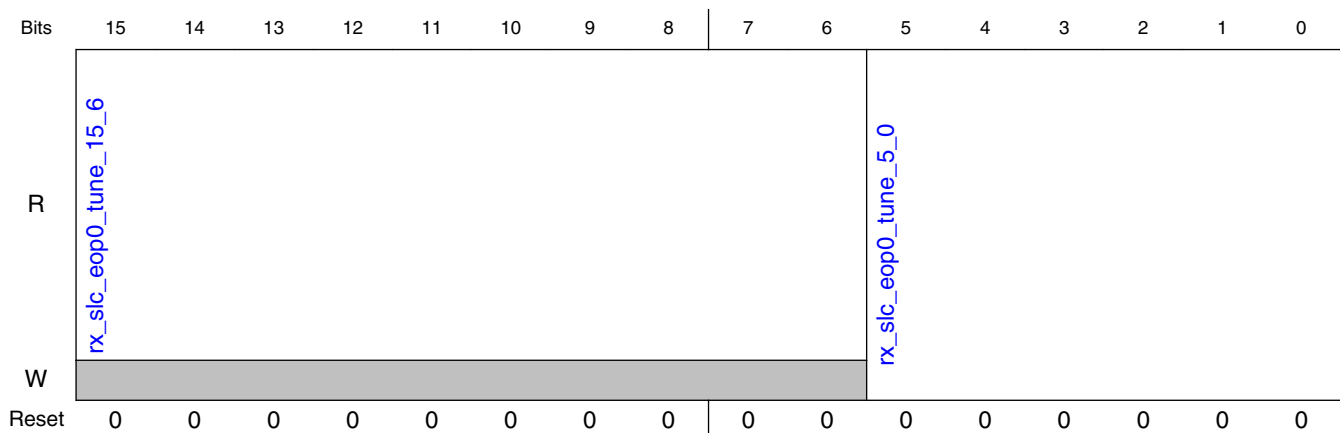
Field	Function
15 rx_slc_eop0_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_eop0_start_14_6	Reserved
5-0 rx_slc_eop0_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.207 RX sampler latch E odd positive 0 calibration unit tune register (lane0_rx_slc_eop0_tune - lane3_rx_slc_eop0_tune)

13.4.10.2.207.1 Offset

Register	Offset
lane0_rx_slc_eop0_tune	8113h
lane1_rx_slc_eop0_tune	8513h
lane2_rx_slc_eop0_tune	8913h
lane3_rx_slc_eop0_tune	8D13h

13.4.10.2.207.2 Diagram



13.4.10.2.207.3 Fields

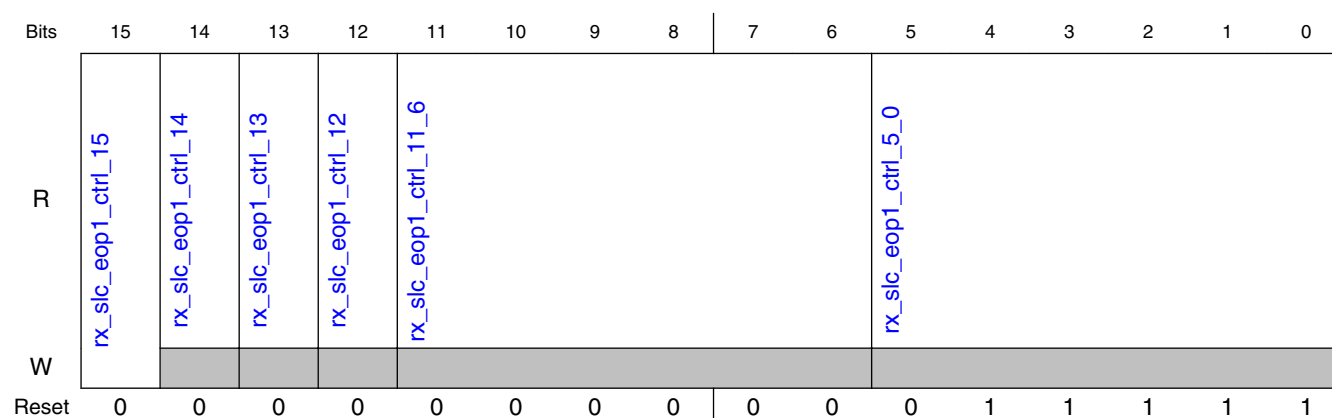
Field	Function
15-6 rx_slc_eop0_tune_15_6	Reserved
5-0 rx_slc_eop0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.208 RX sampler latch E odd positive 1 calibration unit control register (lane0_rx_slc_eop1_ctrl - lane3_rx_slc_eop1_ctrl)

13.4.10.2.208.1 Offset

Register	Offset
lane0_rx_slc_eop1_ctrl	8114h
lane1_rx_slc_eop1_ctrl	8514h
lane2_rx_slc_eop1_ctrl	8914h
lane3_rx_slc_eop1_ctrl	8D14h

13.4.10.2.208.2 Diagram



13.4.10.2.208.3 Fields

Field	Function
15 rx_slc_eop1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_eop1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_eop1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_eop1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_eop1_ctrl_11_6	Reserved
5-0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

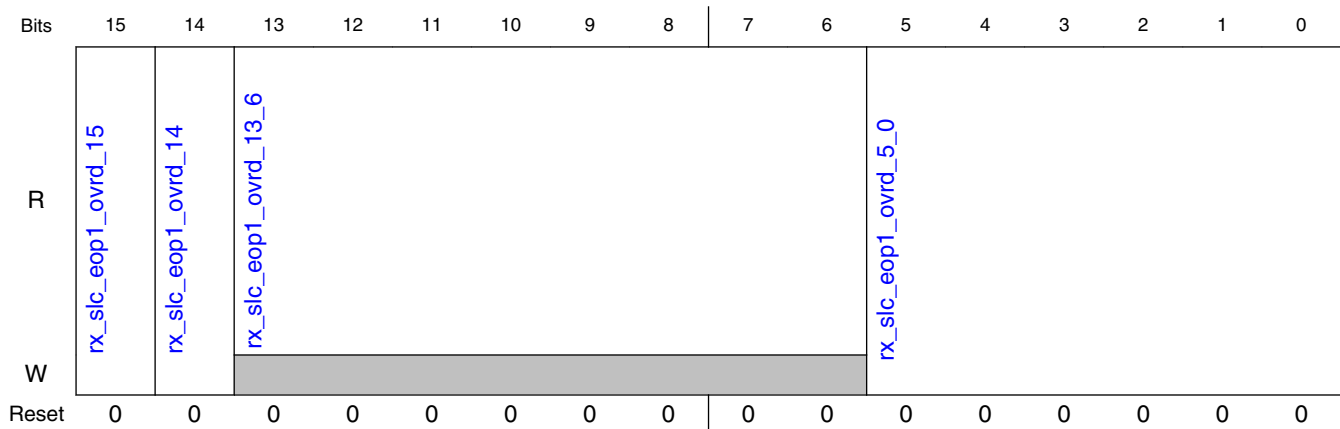
Field	Function
rx_slc_eop1_ctrl_5_0	

13.4.10.2.209 RX sampler latch E odd positive 1 calibration unit override register (lane0_rx_slc_eop1_ovrd - lane3_rx_slc_eop1_ovrd)

13.4.10.2.209.1 Offset

Register	Offset
lane0_rx_slc_eop1_ovrd	8115h
lane1_rx_slc_eop1_ovrd	8515h
lane2_rx_slc_eop1_ovrd	8915h
lane3_rx_slc_eop1_ovrd	8D15h

13.4.10.2.209.2 Diagram



13.4.10.2.209.3 Fields

Field	Function
15 rx_slc_eop1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_eop1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the

Table continues on the next page...

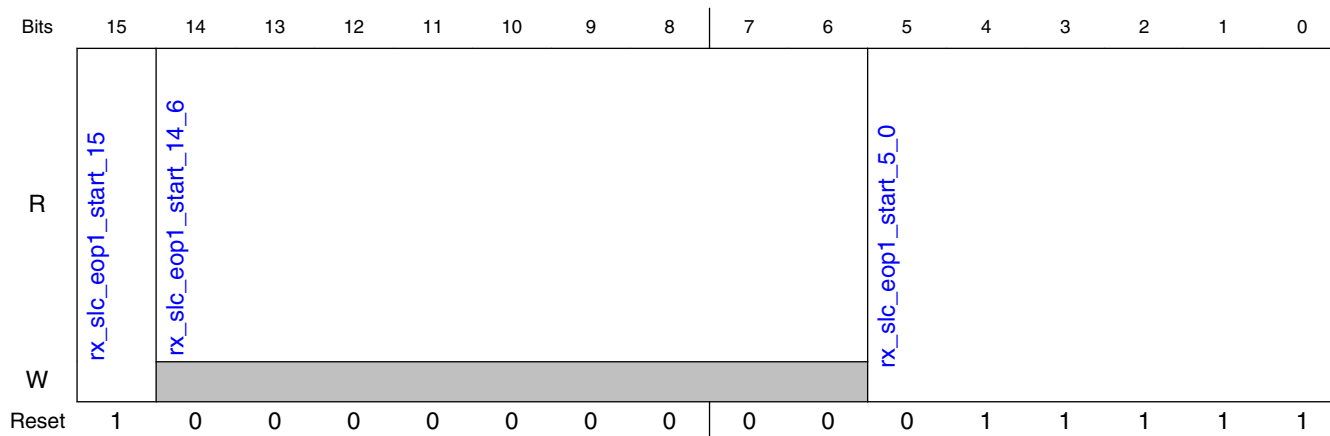
Field	Function
13-6 rx_slc_eop1_ovr d_13_6	Reserved
5-0 rx_slc_eop1_ovr d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.210 RX sampler latch E odd positive 1 calibration unit start register (lane0_rx_slc_eop1_start - lane3_rx_slc_eop1_start)

13.4.10.2.210.1 Offset

Register	Offset
lane0_rx_slc_eop1_start	8116h
lane1_rx_slc_eop1_start	8516h
lane2_rx_slc_eop1_start	8916h
lane3_rx_slc_eop1_start	8D16h

13.4.10.2.210.2 Diagram



13.4.10.2.210.3 Fields

Field	Function
15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.

Table continues on the next page...

Clocks And Resets

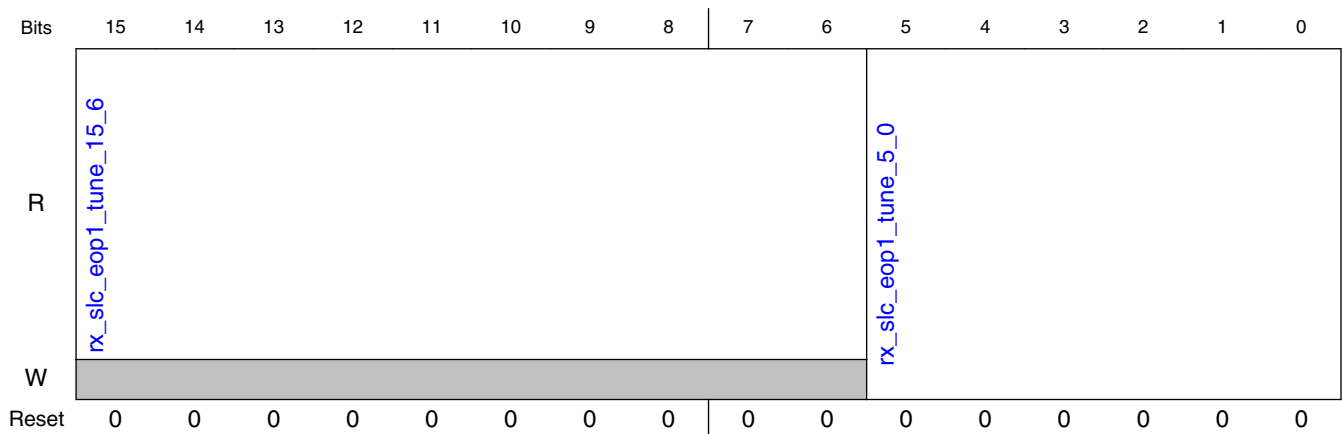
Field	Function
rx_slc_eop1_sta rt_15	
14-6 rx_slc_eop1_sta rt_14_6	Reserved
5-0 rx_slc_eop1_sta rt_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.211 RX sampler latch E odd positive 1 calibration unit tune register (lane0_rx_slc_eop1_tune - lane3_rx_slc_eop1_tune)

13.4.10.2.211.1 Offset

Register	Offset
lane0_rx_slc_eop1_tune	8117h
lane1_rx_slc_eop1_tune	8517h
lane2_rx_slc_eop1_tune	8917h
lane3_rx_slc_eop1_tune	8D17h

13.4.10.2.211.2 Diagram



13.4.10.2.211.3 Fields

Field	Function
15-6	Reserved

Table continues on the next page...

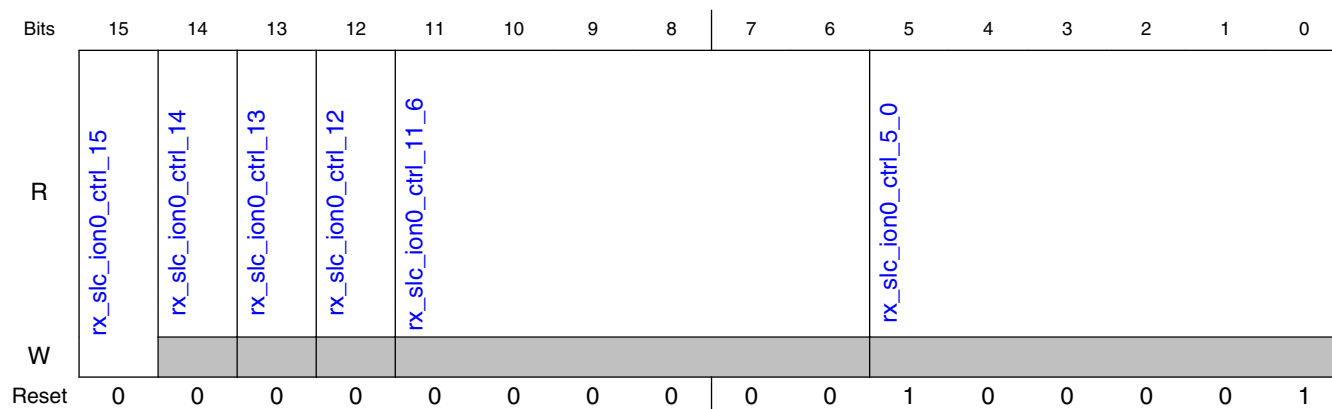
Field	Function
rx_slc_eop1_tune_15_6	
5-0 rx_slc_eop1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a two's complement value, so the calibrated code can be increased or decreased.

13.4.10.2.212 RX sampler latch I odd negative 0 calibration unit control register (lane0_rx_slc_ion0_ctrl - lane3_rx_slc_ion0_ctrl)

13.4.10.2.212.1 Offset

Register	Offset
lane0_rx_slc_ion0_ctrl	8118h
lane1_rx_slc_ion0_ctrl	8518h
lane2_rx_slc_ion0_ctrl	8918h
lane3_rx_slc_ion0_ctrl	8D18h

13.4.10.2.212.2 Diagram



13.4.10.2.212.3 Fields

Field	Function
15 rx_slc_ion0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by

Table continues on the next page...

Clocks And Resets

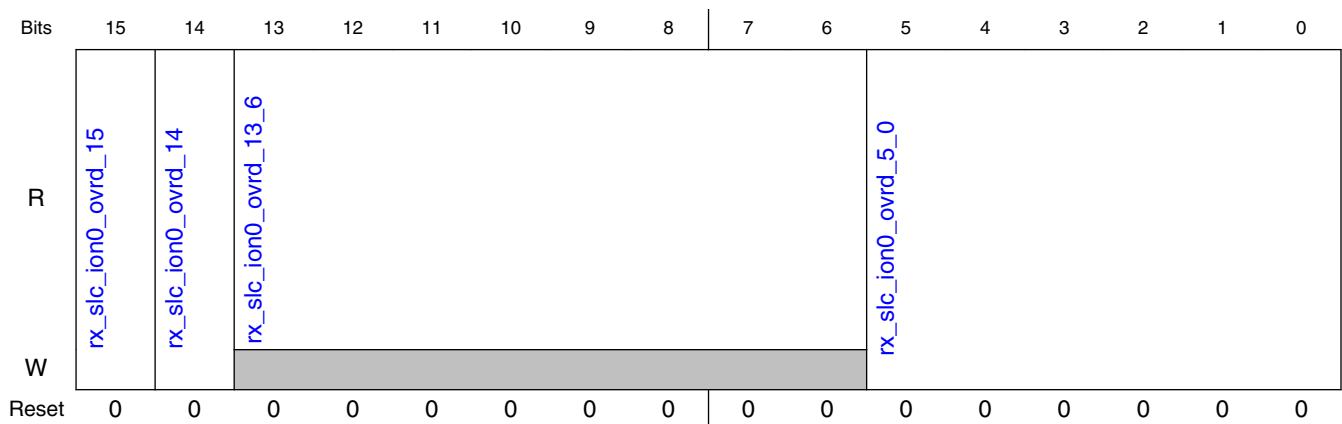
Field	Function
rx_slc_ion0_ctrl_14	
13 rx_slc_ion0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_ion0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_ion0_ctrl_11_6	Reserved
5-0 rx_slc_ion0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.213 RX sampler latch I odd negative 0 calibration unit override register (lane0_rx_slc_ion0_ovrd - lane3_rx_slc_ion0_ovrd)

13.4.10.2.213.1 Offset

Register	Offset
lane0_rx_slc_ion0_ovrd	8119h
lane1_rx_slc_ion0_ovrd	8519h
lane2_rx_slc_ion0_ovrd	8919h
lane3_rx_slc_ion0_ovrd	8D19h

13.4.10.2.213.2 Diagram



13.4.10.2.213.3 Fields

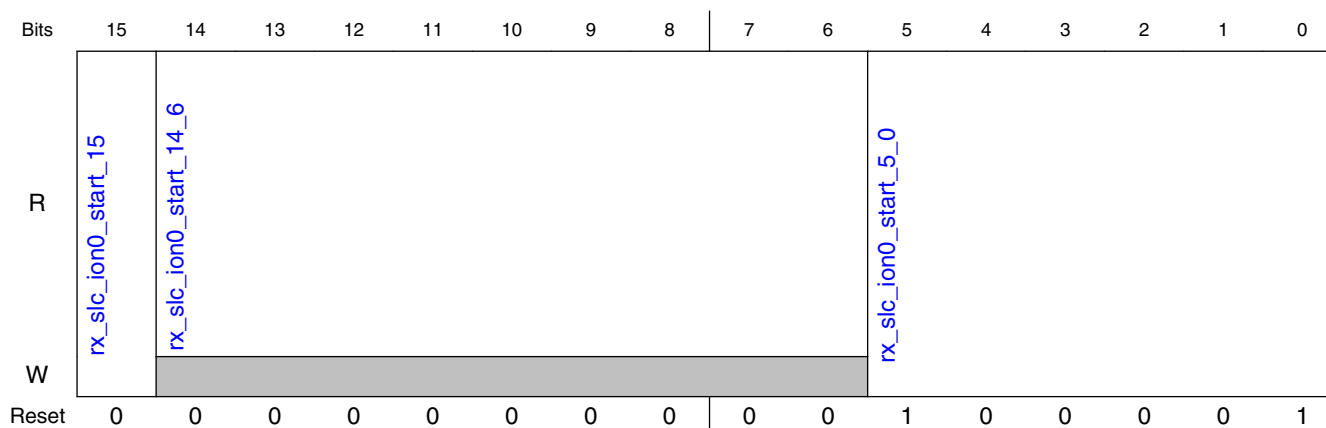
Field	Function
15 rx_slc_ion0_ovr d_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_ion0_ovr d_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_ion0_ovr d_13_6	Reserved
5-0 rx_slc_ion0_ovr d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.214 RX sampler latch I odd negative 0 calibration unit start register (lane0_rx_slc_ion0_start - lane3_rx_slc_ion0_start)

13.4.10.2.214.1 Offset

Register	Offset
lane0_rx_slc_ion0_start	811Ah
lane1_rx_slc_ion0_start	851Ah
lane2_rx_slc_ion0_start	891Ah
lane3_rx_slc_ion0_start	8D1Ah

13.4.10.2.214.2 Diagram



13.4.10.2.214.3 Fields

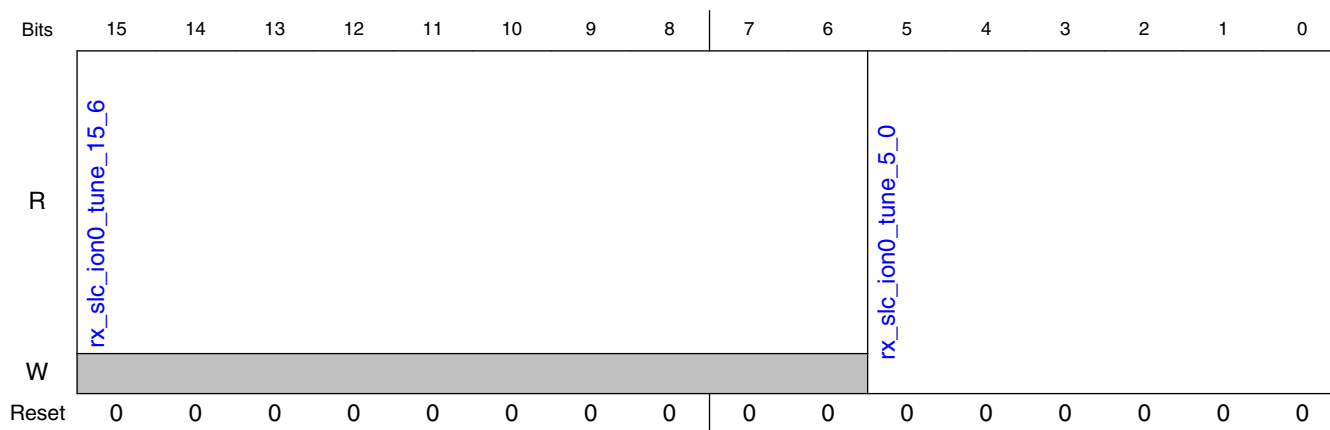
Field	Function
15 rx_slc_ion0_star t_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_ion0_star t_14_6	Reserved
5-0 rx_slc_ion0_star t_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.215 RX sampler latch I odd negative 0 calibration unit tune register (lane0_rx_slc_ion0_tune - lane3_rx_slc_ion0_tune)

13.4.10.2.215.1 Offset

Register	Offset
lane0_rx_slc_ion0_tune	811Bh
lane1_rx_slc_ion0_tune	851Bh
lane2_rx_slc_ion0_tune	891Bh
lane3_rx_slc_ion0_tune	8D1Bh

13.4.10.2.215.2 Diagram



13.4.10.2.215.3 Fields

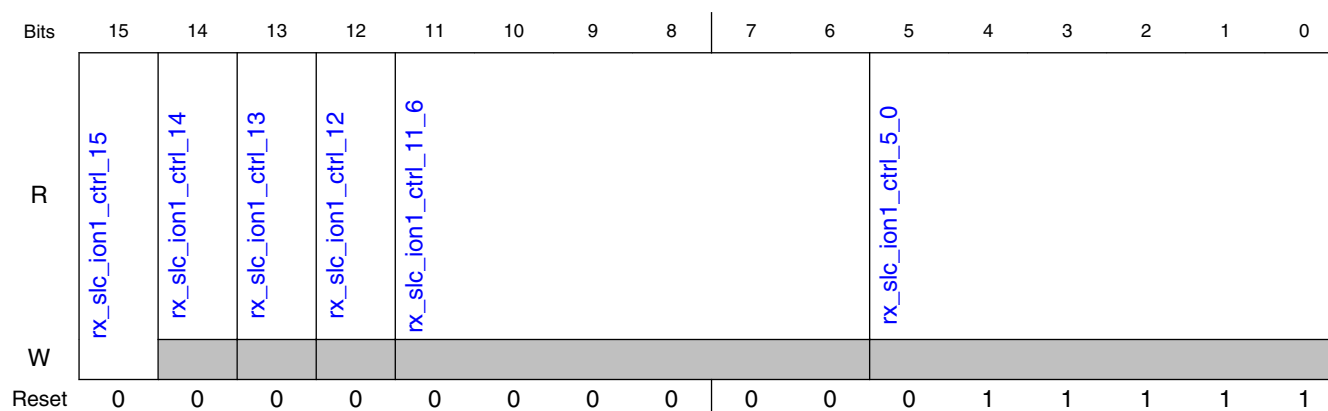
Field	Function
15-6 rx_slc_ion0_tune_15_6	Reserved
5-0 rx_slc_ion0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.216 RX sampler latch I odd negative 1 calibration unit control register (lane0_rx_slc_ion1_ctrl - lane3_rx_slc_ion1_ctrl)

13.4.10.2.216.1 Offset

Register	Offset
lane0_rx_slc_ion1_ctrl	811Ch
lane1_rx_slc_ion1_ctrl	851Ch
lane2_rx_slc_ion1_ctrl	891Ch
lane3_rx_slc_ion1_ctrl	8D1Ch

13.4.10.2.216.2 Diagram



13.4.10.2.216.3 Fields

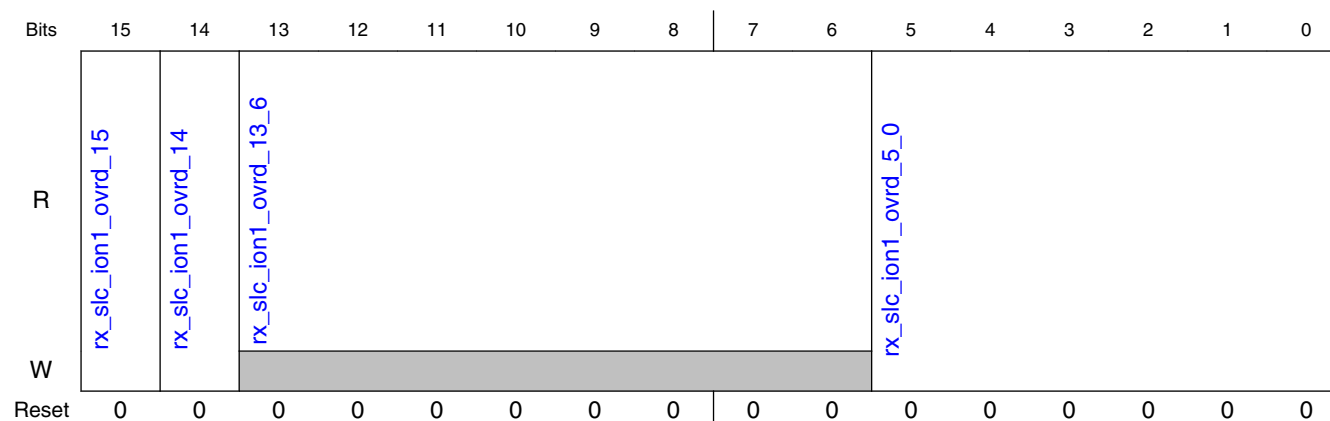
Field	Function
15 rx_slc_ion1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_ion1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_ion1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_ion1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_ion1_ctrl_11_6	Reserved
5-0 rx_slc_ion1_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.217 RX sampler latch I odd negative 1 calibration unit override register (lane0_rx_slc_ion1_ovrd - lane3_rx_slc_ion1_ovrd)

13.4.10.2.217.1 Offset

Register	Offset
lane0_rx_slc_ion1_ovrd	811Dh
lane1_rx_slc_ion1_ovrd	851Dh
lane2_rx_slc_ion1_ovrd	891Dh
lane3_rx_slc_ion1_ovrd	8D1Dh

13.4.10.2.217.2 Diagram



13.4.10.2.217.3 Fields

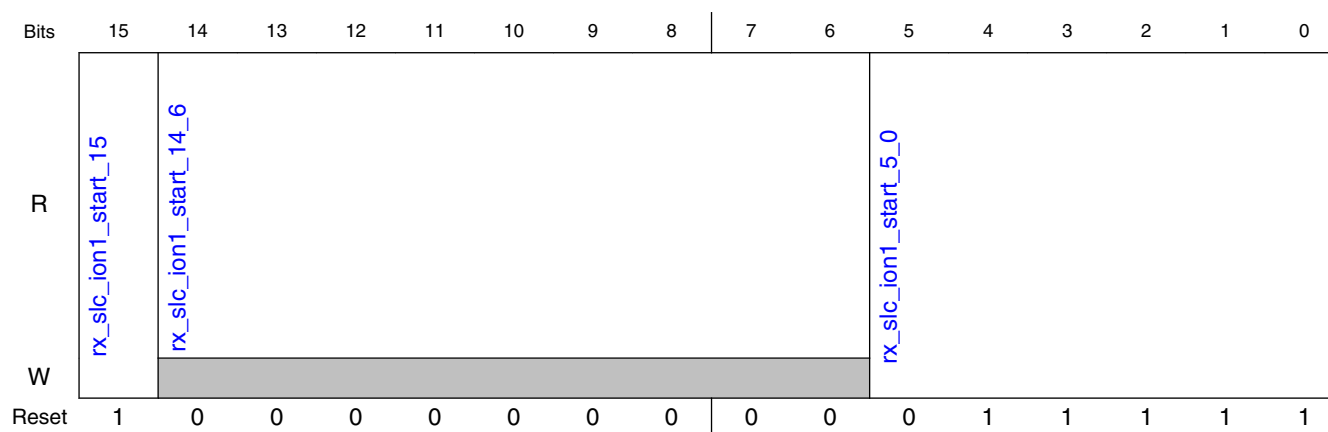
Field	Function
15 rx_slc_ion1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_ion1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_ion1_ovrd_13_6	Reserved
5-0 rx_slc_ion1_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.218 RX sampler latch I odd negative 1 calibration unit start register (lane0_rx_slc_ion1_start - lane3_rx_slc_ion1_start)

13.4.10.2.218.1 Offset

Register	Offset
lane0_rx_slc_ion1_start	811Eh
lane1_rx_slc_ion1_start	851Eh
lane2_rx_slc_ion1_start	891Eh
lane3_rx_slc_ion1_start	8D1Eh

13.4.10.2.218.2 Diagram



13.4.10.2.218.3 Fields

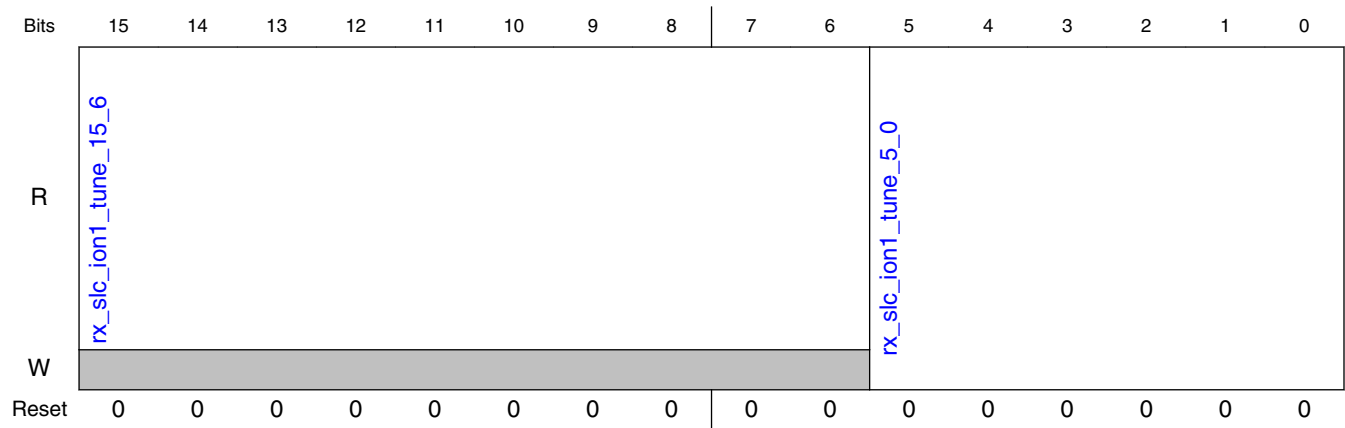
Field	Function
15 rx_slc_ion1_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_ion1_start_14_6	Reserved
5-0 rx_slc_ion1_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.219 RX sampler latch I odd negative 1 calibration unit tune register (lane0_rx_slc_ion1_tune - lane3_rx_slc_ion1_tune)

13.4.10.2.219.1 Offset

Register	Offset
lane0_rx_slc_ion1_tune	811Fh
lane1_rx_slc_ion1_tune	851Fh
lane2_rx_slc_ion1_tune	891Fh
lane3_rx_slc_ion1_tune	8D1Fh

13.4.10.2.219.2 Diagram



13.4.10.2.219.3 Fields

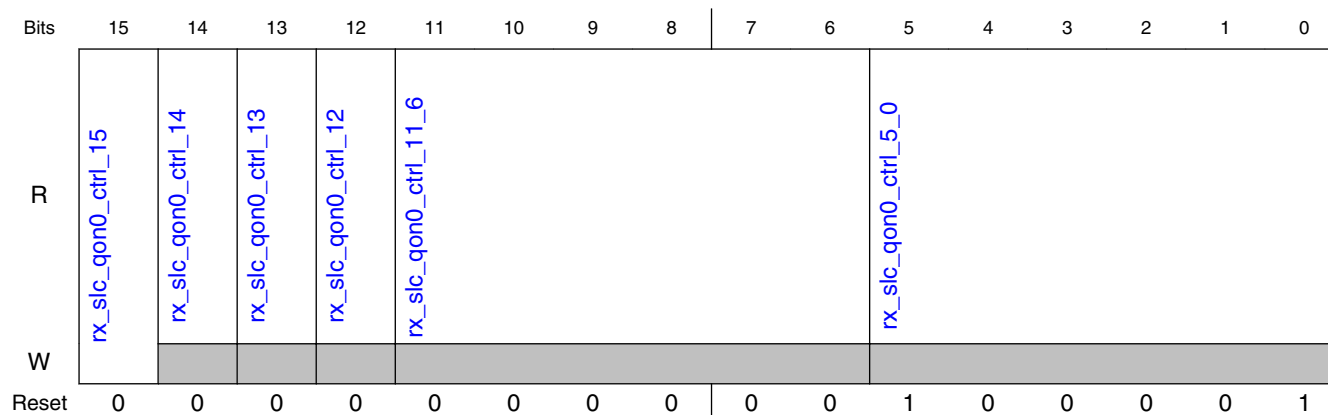
Field	Function
15-6 rx_slc_ion1_tune_15_6	Reserved
5-0 rx_slc_ion1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.220 RX sampler latch Q odd negative 0 calibration unit control register (lane0_rx_slc_qon0_ctrl - lane3_rx_slc_qon0_ctrl)

13.4.10.2.220.1 Offset

Register	Offset
lane0_rx_slc_qon0_ctrl	8120h
lane1_rx_slc_qon0_ctrl	8520h
lane2_rx_slc_qon0_ctrl	8920h
lane3_rx_slc_qon0_ctrl	8D20h

13.4.10.2.220.2 Diagram



13.4.10.2.220.3 Fields

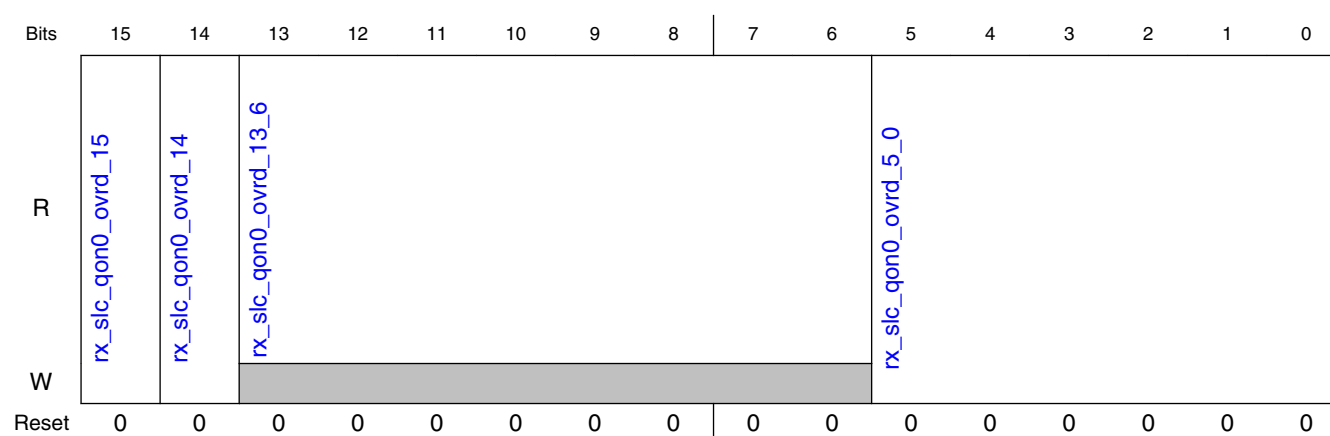
Field	Function
15 rx_slc_qon0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_qon0_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_qon0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_qon0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_qon0_ctrl_11_6	Reserved
5-0 rx_slc_qon0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.221 RX sampler latch Q odd negative 0 calibration unit override register (lane0_rx_slc_qon0_ovrd - lane3_rx_slc_qon0_ovrd)

13.4.10.2.221.1 Offset

Register	Offset
lane0_rx_slc_qon0_ovrd	8121h
lane1_rx_slc_qon0_ovrd	8521h
lane2_rx_slc_qon0_ovrd	8921h
lane3_rx_slc_qon0_ovrd	8D21h

13.4.10.2.221.2 Diagram



13.4.10.2.221.3 Fields

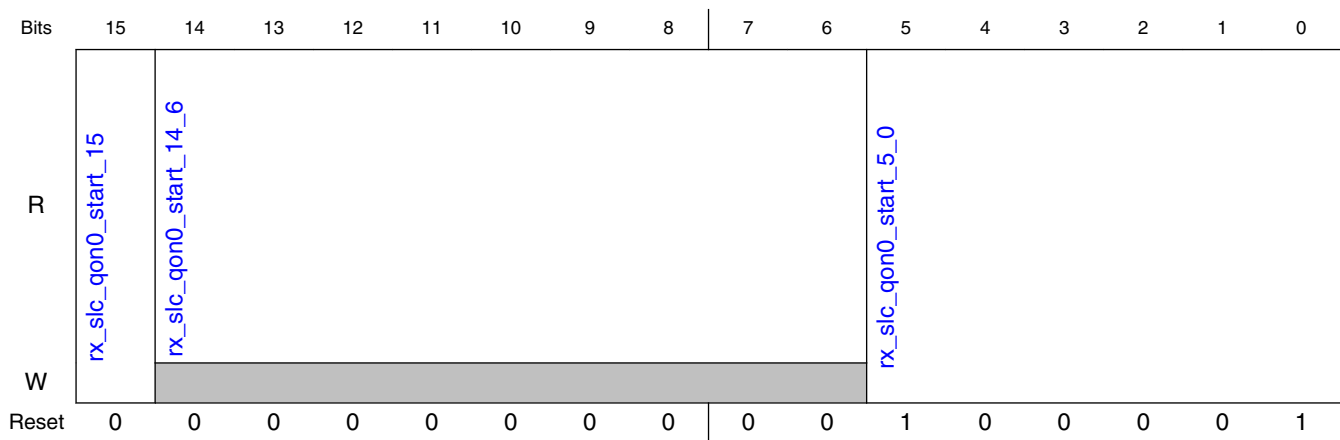
Field	Function
15 rx_slc_qon0_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_qon0_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_qon0_ovrd_13_6	Reserved
5-0 rx_slc_qon0_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.222 RX sampler latch Q odd negative 0 calibration unit start register (lane0_rx_slc_qon0_start - lane3_rx_slc_qon0_start)

13.4.10.2.222.1 Offset

Register	Offset
lane0_rx_slc_qon0_start	8122h
lane1_rx_slc_qon0_start	8522h
lane2_rx_slc_qon0_start	8922h
lane3_rx_slc_qon0_start	8D22h

13.4.10.2.222.2 Diagram



13.4.10.2.222.3 Fields

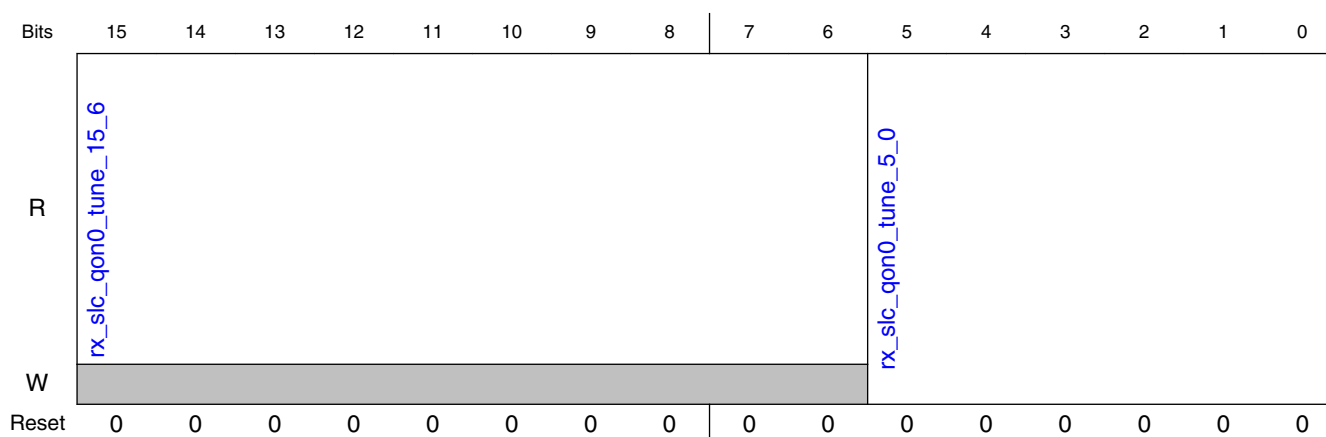
Field	Function
15 rx_slc_qon0_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_qon0_start_14_6	Reserved
5-0 rx_slc_qon0_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.223 RX sampler latch Q odd negative 0 calibration unit tune register (lane0_rx_slc_qon0_tune - lane3_rx_slc_qon0_tune)

13.4.10.2.223.1 Offset

Register	Offset
lane0_rx_slc_qon0_tune	8123h
lane1_rx_slc_qon0_tune	8523h
lane2_rx_slc_qon0_tune	8923h
lane3_rx_slc_qon0_tune	8D23h

13.4.10.2.223.2 Diagram



13.4.10.2.223.3 Fields

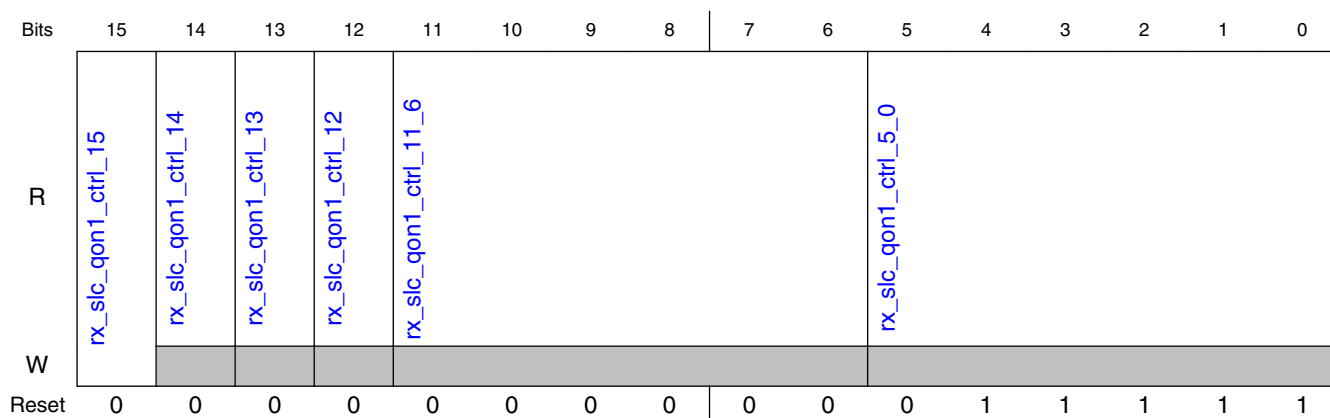
Field	Function
15-6 <code>rx_slc_qon0_tune_15_6</code>	Reserved
5-0 <code>rx_slc_qon0_tune_5_0</code>	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.224 RX sampler latch Q odd negative 1 calibration unit control register (lane0_rx_slc_qon1_ctrl - lane3_rx_slc_qon1_ctrl)

13.4.10.2.224.1 Offset

Register	Offset
lane0_rx_slc_qon1_ctrl	8124h
lane1_rx_slc_qon1_ctrl	8524h
lane2_rx_slc_qon1_ctrl	8924h
lane3_rx_slc_qon1_ctrl	8D24h

13.4.10.2.224.2 Diagram



13.4.10.2.224.3 Fields

Field	Function
15 rx_slc_qon1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_qon1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_qon1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_qon1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_qon1_ctrl_11_6	Reserved
5-0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

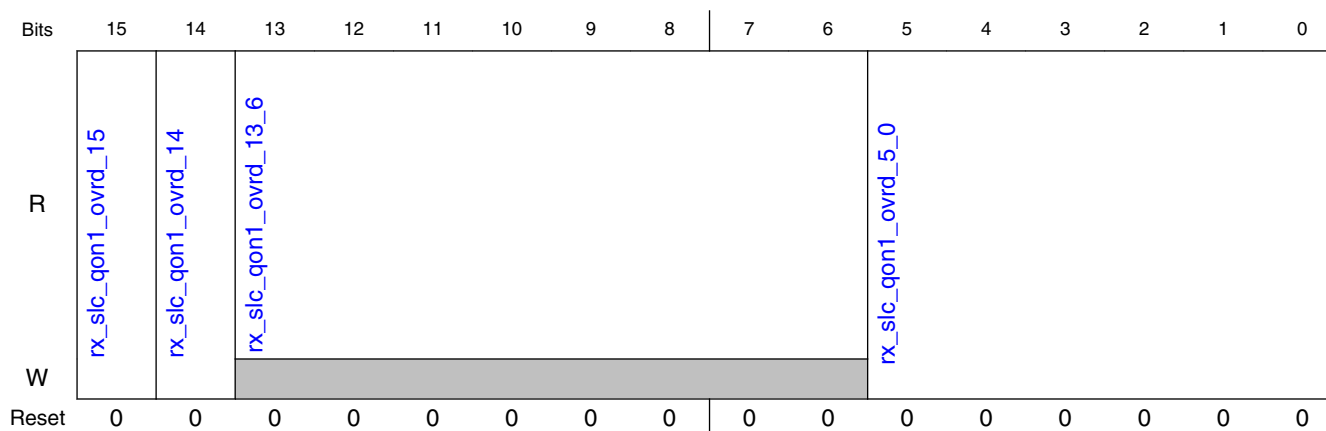
Field	Function
rx_slc_qon1_ctrl_5_0	

13.4.10.2.225 RX sampler latch Q odd negative 1 calibration unit override register (lane0_rx_slc_qon1_ovrd - lane3_rx_slc_qon1_ovrd)

13.4.10.2.225.1 Offset

Register	Offset
lane0_rx_slc_qon1_ovrd	8125h
lane1_rx_slc_qon1_ovrd	8525h
lane2_rx_slc_qon1_ovrd	8925h
lane3_rx_slc_qon1_ovrd	8D25h

13.4.10.2.225.2 Diagram



13.4.10.2.225.3 Fields

Field	Function
15 rx_slc_qon1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_qon1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the

Table continues on the next page...

Clocks And Resets

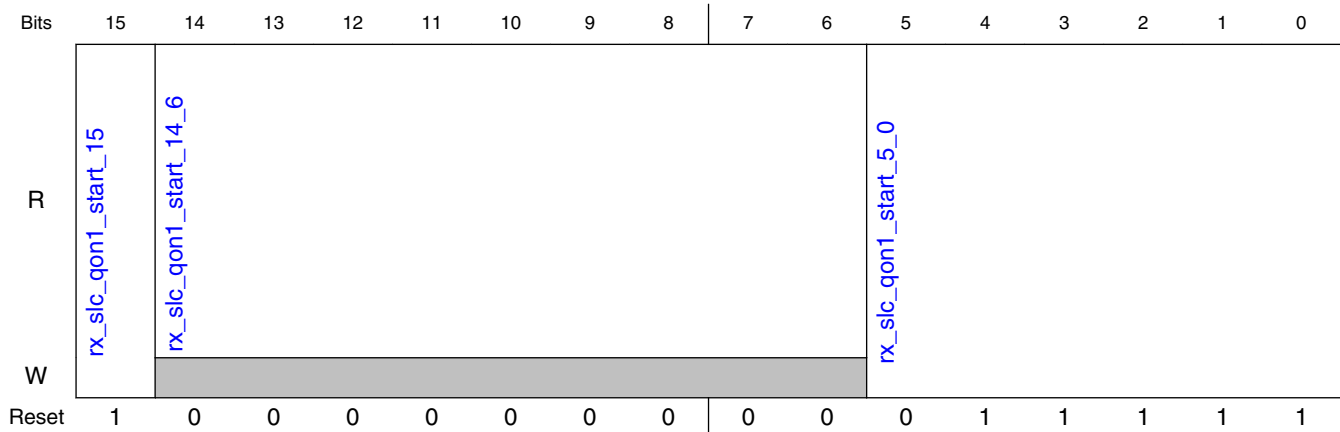
Field	Function
13-6 rx_slc_qon1_ovr d_13_6	Reserved
5-0 rx_slc_qon1_ovr d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.226 RX sampler latch Q odd negative 1 calibration unit start register (lane0_rx_slc_qon1_start - lane3_rx_slc_qon1_start)

13.4.10.2.226.1 Offset

Register	Offset
lane0_rx_slc_qon1_start	8126h
lane1_rx_slc_qon1_start	8526h
lane2_rx_slc_qon1_start	8926h
lane3_rx_slc_qon1_start	8D26h

13.4.10.2.226.2 Diagram



13.4.10.2.226.3 Fields

Field	Function
15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.

Table continues on the next page...

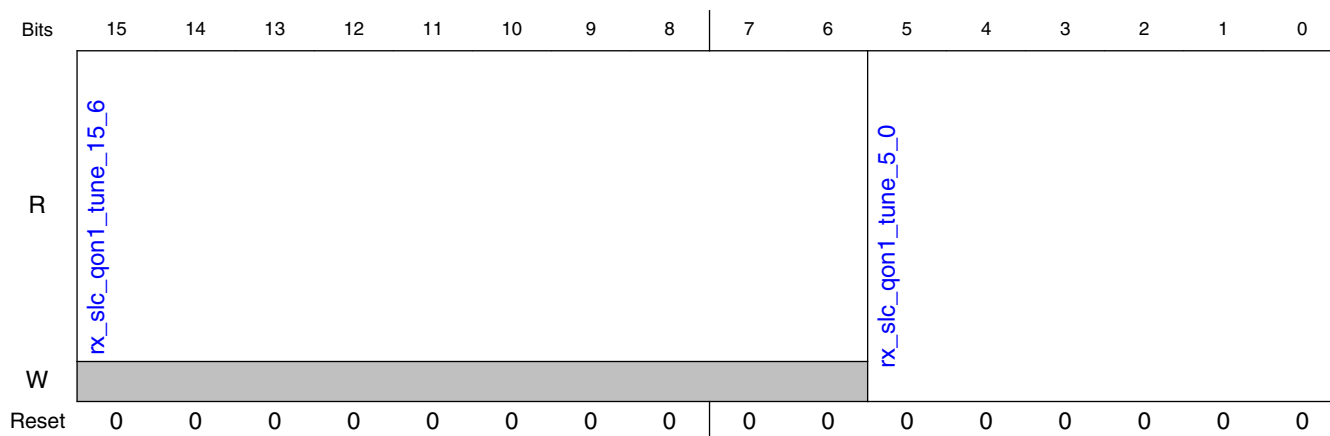
Field	Function
rx_slc_qon1_sta rt_15	
14-6 rx_slc_qon1_sta rt_14_6	Reserved
5-0 rx_slc_qon1_sta rt_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.227 RX sampler latch Q odd negative 1 calibration unit tune register (lane0_rx_slc_qon1_tune - lane3_rx_slc_qon1_tune)

13.4.10.2.227.1 Offset

Register	Offset
lane0_rx_slc_qon1_tune	8127h
lane1_rx_slc_qon1_tune	8527h
lane2_rx_slc_qon1_tune	8927h
lane3_rx_slc_qon1_tune	8D27h

13.4.10.2.227.2 Diagram



13.4.10.2.227.3 Fields

Field	Function
15-6	Reserved

Table continues on the next page...

Clocks And Resets

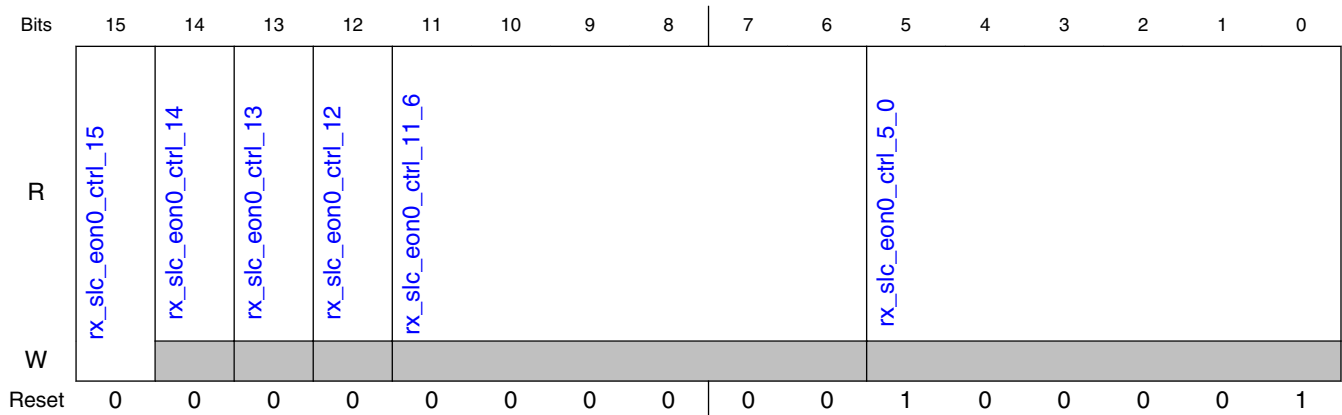
Field	Function
rx_slc_qon1_tune_15_6	
5-0 rx_slc_qon1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a two's complement value, so the calibrated code can be increased or decreased.

13.4.10.2.228 RX sampler latch E odd negative 0 calibration unit control register (lane0_rx_slc_eon0_ctrl - lane3_rx_slc_eon0_ctrl)

13.4.10.2.228.1 Offset

Register	Offset
lane0_rx_slc_eon0_ctrl	8128h
lane1_rx_slc_eon0_ctrl	8528h
lane2_rx_slc_eon0_ctrl	8928h
lane3_rx_slc_eon0_ctrl	8D28h

13.4.10.2.228.2 Diagram



13.4.10.2.228.3 Fields

Field	Function
15 rx_slc_eon0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by

Table continues on the next page...

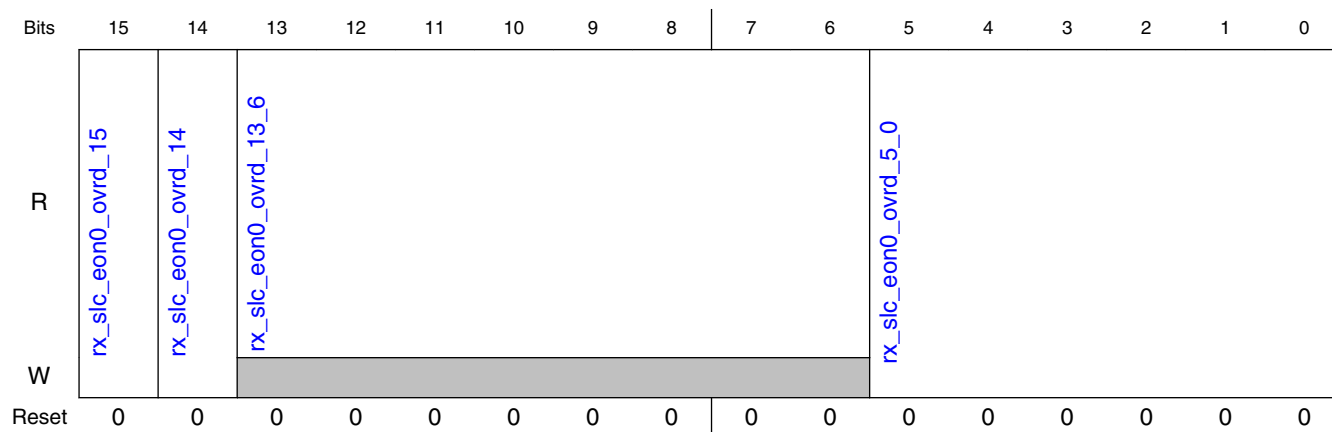
Field	Function
rx_slc_eon0_ctrl_14	
13 rx_slc_eon0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_eon0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_eon0_ctrl_11_6	Reserved
5-0 rx_slc_eon0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.229 RX sampler latch E odd negative 0 calibration unit override register (lane0_rx_slc_eon0_ovrd - lane3_rx_slc_eon0_ovrd)

13.4.10.2.229.1 Offset

Register	Offset
lane0_rx_slc_eon0_ovrd	8129h
lane1_rx_slc_eon0_ovrd	8529h
lane2_rx_slc_eon0_ovrd	8929h
lane3_rx_slc_eon0_ovrd	8D29h

13.4.10.2.229.2 Diagram



13.4.10.2.229.3 Fields

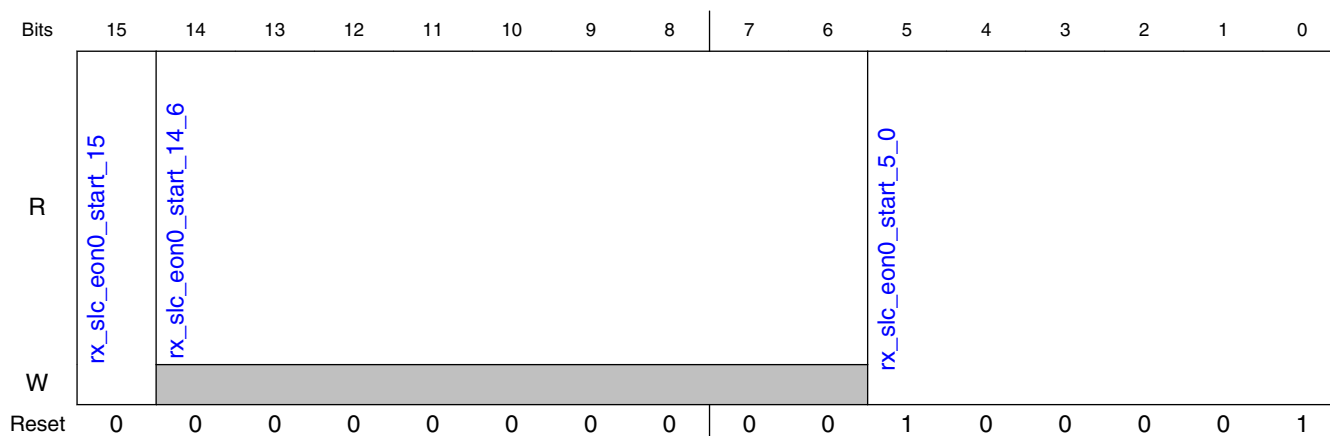
Field	Function
15 rx_slc_eon0_ovr d_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_eon0_ovr d_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_eon0_ovr d_13_6	Reserved
5-0 rx_slc_eon0_ovr d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.230 RX sampler latch E odd negative 0 calibration unit start register (lane0_rx_slc_eon0_start - lane3_rx_slc_eon0_start)

13.4.10.2.230.1 Offset

Register	Offset
lane0_rx_slc_eon0_start	812Ah
lane1_rx_slc_eon0_start	852Ah
lane2_rx_slc_eon0_start	892Ah
lane3_rx_slc_eon0_start	8D2Ah

13.4.10.2.230.2 Diagram



13.4.10.2.230.3 Fields

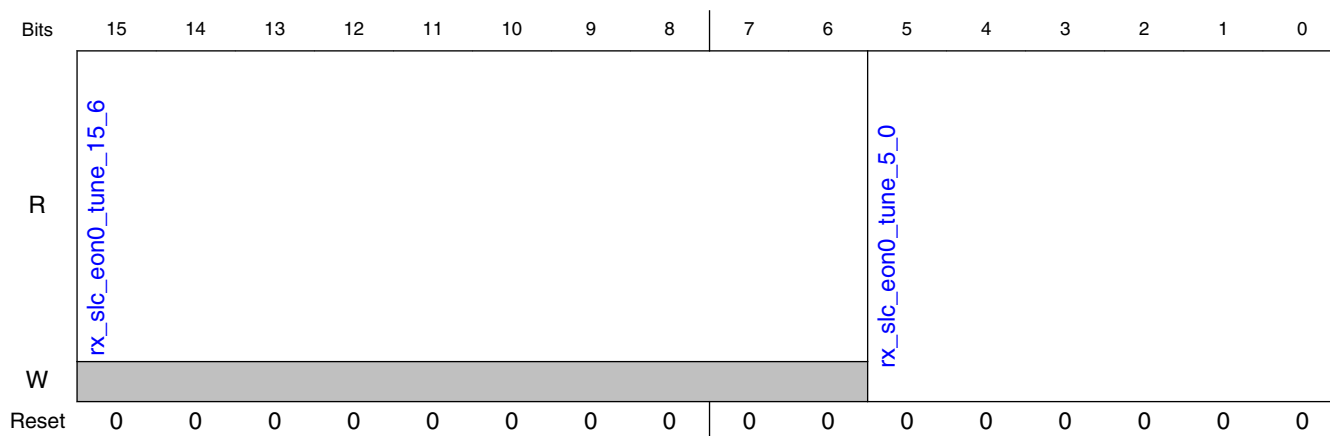
Field	Function
15 rx_slc_eon0_sta rt_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_eon0_sta rt_14_6	Reserved
5-0 rx_slc_eon0_sta rt_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.231 RX sampler latch E odd negative 0 calibration unit tune register (lane0_rx_slc_eon0_tune - lane3_rx_slc_eon0_tune)

13.4.10.2.231.1 Offset

Register	Offset
lane0_rx_slc_eon0_tune	812Bh
lane1_rx_slc_eon0_tune	852Bh
lane2_rx_slc_eon0_tune	892Bh
lane3_rx_slc_eon0_tune	8D2Bh

13.4.10.2.231.2 Diagram



13.4.10.2.231.3 Fields

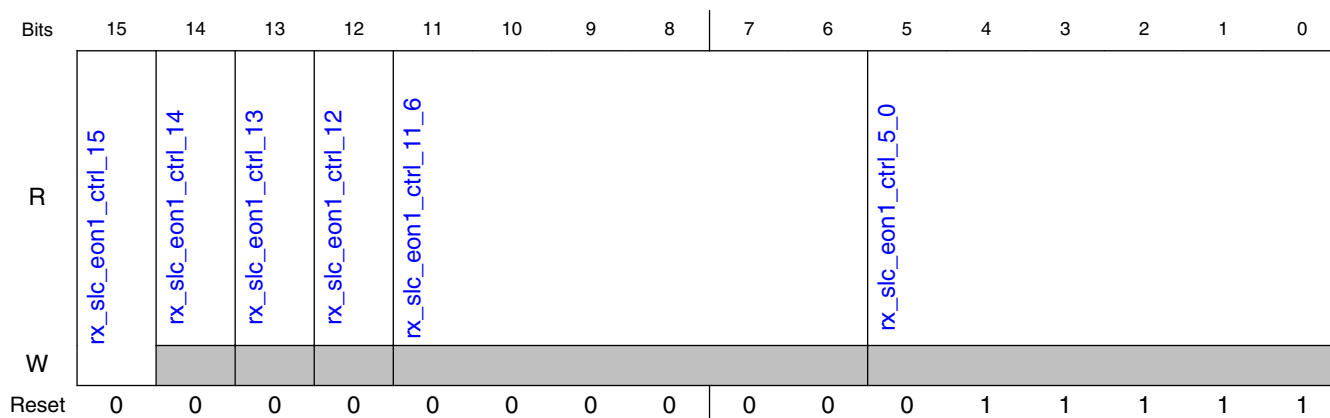
Field	Function
15-6 rx_slc_eon0_tune_15_6	Reserved
5-0 rx_slc_eon0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.232 RX sampler latch E odd negative 1 calibration unit control register (lane0_rx_slc_eon1_ctrl - lane3_rx_slc_eon1_ctrl)

13.4.10.2.232.1 Offset

Register	Offset
lane0_rx_slc_eon1_ctrl	812Ch
lane1_rx_slc_eon1_ctrl	852Ch
lane2_rx_slc_eon1_ctrl	892Ch
lane3_rx_slc_eon1_ctrl	8D2Ch

13.4.10.2.232.2 Diagram



13.4.10.2.232.3 Fields

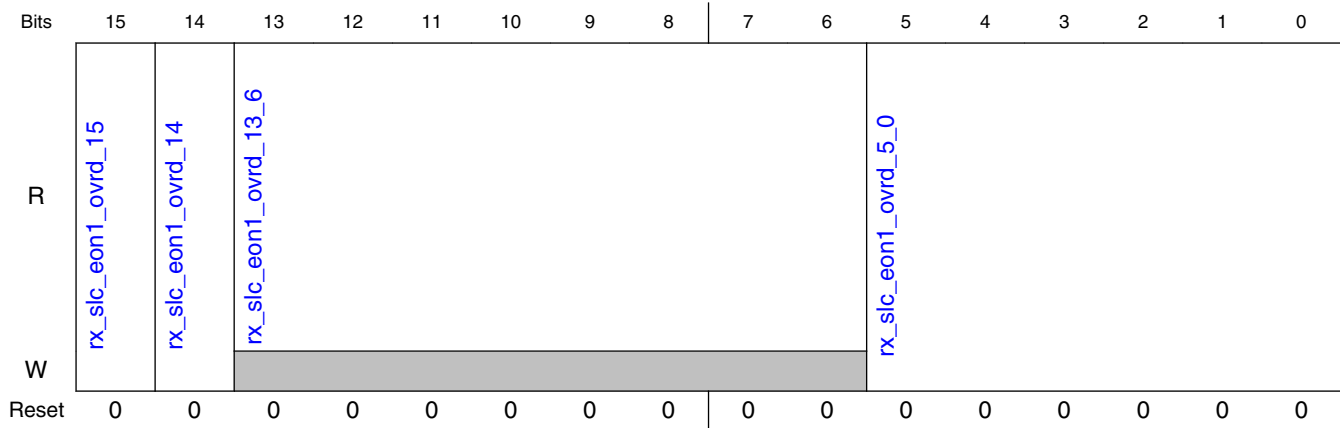
Field	Function
15 rx_slc_eon1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_eon1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_eon1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_eon1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_eon1_ctrl_11_6	Reserved
5-0 rx_slc_eon1_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.233 RX sampler latch E odd negative 1 calibration unit override register (lane0_rx_slc_eon1_ovrd - lane3_rx_slc_eon1_ovrd)

13.4.10.2.233.1 Offset

Register	Offset
lane0_rx_slc_eon1_ovrd	812Dh
lane1_rx_slc_eon1_ovrd	852Dh
lane2_rx_slc_eon1_ovrd	892Dh
lane3_rx_slc_eon1_ovrd	8D2Dh

13.4.10.2.233.2 Diagram



13.4.10.2.233.3 Fields

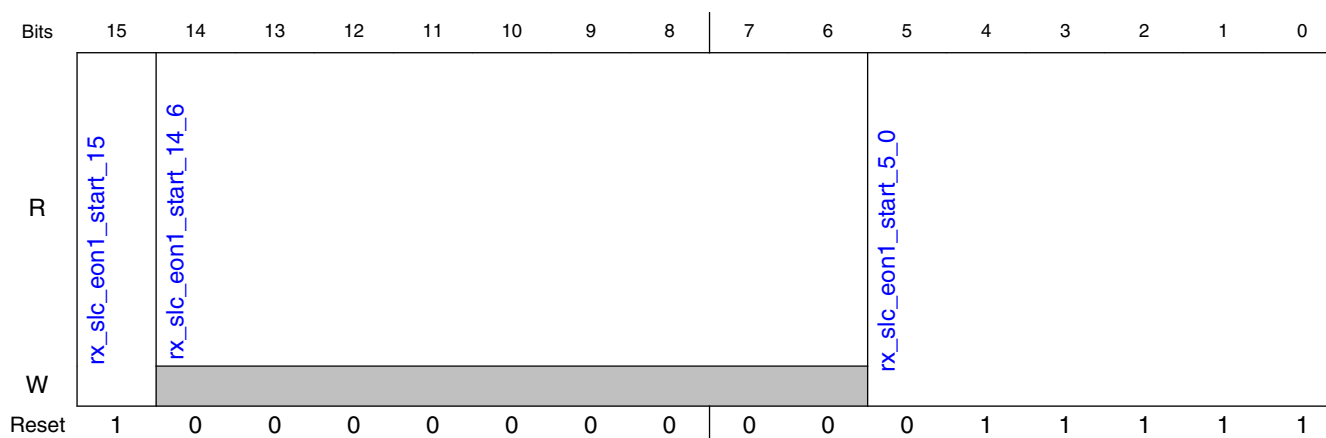
Field	Function
15 rx_slc_eon1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_eon1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_eon1_ovrd_13_6	Reserved
5-0 rx_slc_eon1_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.234 RX sampler latch E odd negative 1 calibration unit start register (lane0_rx_slc_eon1_start - lane3_rx_slc_eon1_start)

13.4.10.2.234.1 Offset

Register	Offset
lane0_rx_slc_eon1_start	812Eh
lane1_rx_slc_eon1_start	852Eh
lane2_rx_slc_eon1_start	892Eh
lane3_rx_slc_eon1_start	8D2Eh

13.4.10.2.234.2 Diagram



13.4.10.2.234.3 Fields

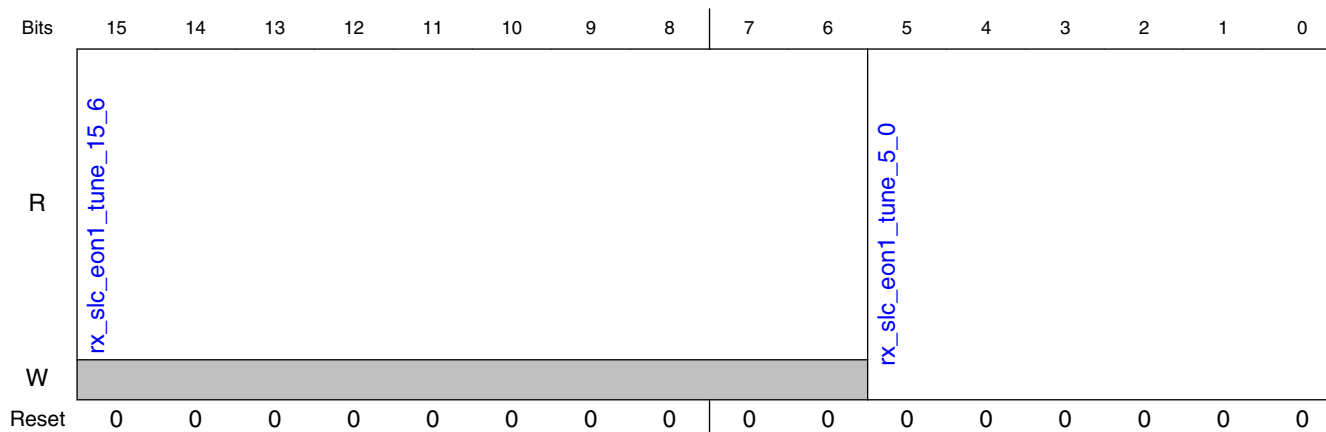
Field	Function
15 rx_slc_eon1_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_eon1_start_14_6	Reserved
5-0 rx_slc_eon1_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.235 RX sampler latch E odd negative 1 calibration unit tune register (lane0_rx_slc_eon1_tune - lane3_rx_slc_eon1_tune)

13.4.10.2.235.1 Offset

Register	Offset
lane0_rx_slc_eon1_tune	812Fh
lane1_rx_slc_eon1_tune	852Fh
lane2_rx_slc_eon1_tune	892Fh
lane3_rx_slc_eon1_tune	8D2Fh

13.4.10.2.235.2 Diagram



13.4.10.2.235.3 Fields

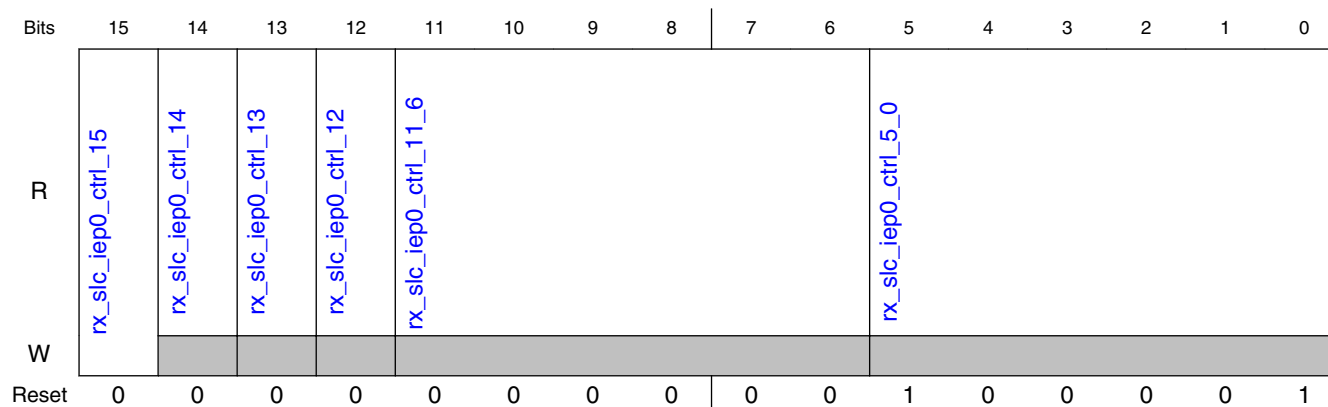
Field	Function
15-6 <code>rx_slc_eon1_tune_15_6</code>	Reserved
5-0 <code>rx_slc_eon1_tune_5_0</code>	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.236 RX sampler latch I even positive 0 calibration unit control register (lane0_rx_slc_iep0_ctrl - lane3_rx_slc_iep0_ctrl)

13.4.10.2.236.1 Offset

Register	Offset
<code>lane0_rx_slc_iep0_ctrl</code>	8130h
<code>lane1_rx_slc_iep0_ctrl</code>	8530h
<code>lane2_rx_slc_iep0_ctrl</code>	8930h
<code>lane3_rx_slc_iep0_ctrl</code>	8D30h

13.4.10.2.236.2 Diagram



13.4.10.2.236.3 Fields

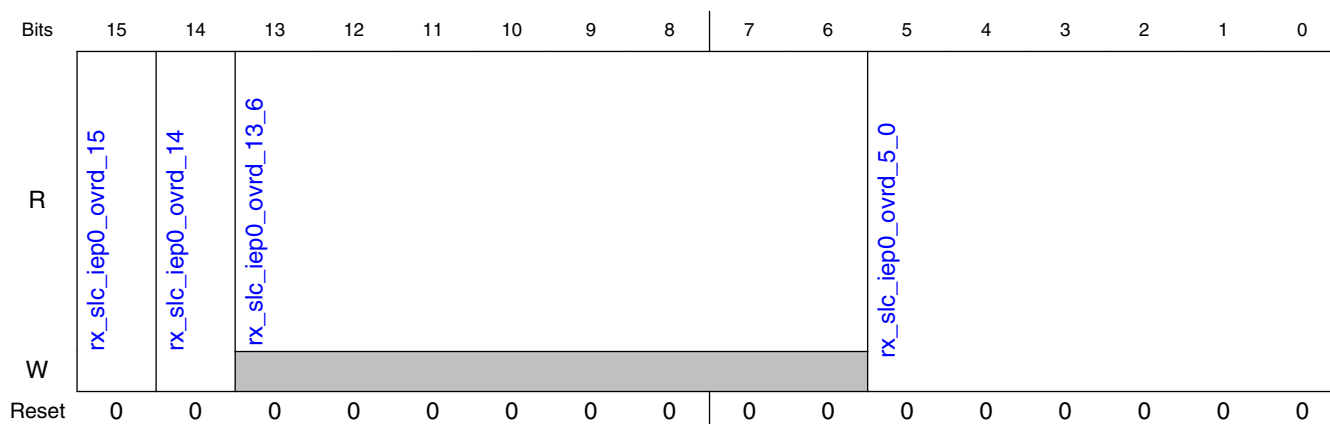
Field	Function
15 rx_slc_iep0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_iep0_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_iep0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_iep0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_iep0_ctrl_11_6	Reserved
5-0 rx_slc_iep0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.237 RX sampler latch I even positive 0 calibration unit override register (lane0_rx_slc_iep0_ovrd - lane3_rx_slc_iep0_ovrd)

13.4.10.2.237.1 Offset

Register	Offset
lane0_rx_slc_iep0_ovrd	8131h
lane1_rx_slc_iep0_ovrd	8531h
lane2_rx_slc_iep0_ovrd	8931h
lane3_rx_slc_iep0_ovrd	8D31h

13.4.10.2.237.2 Diagram



13.4.10.2.237.3 Fields

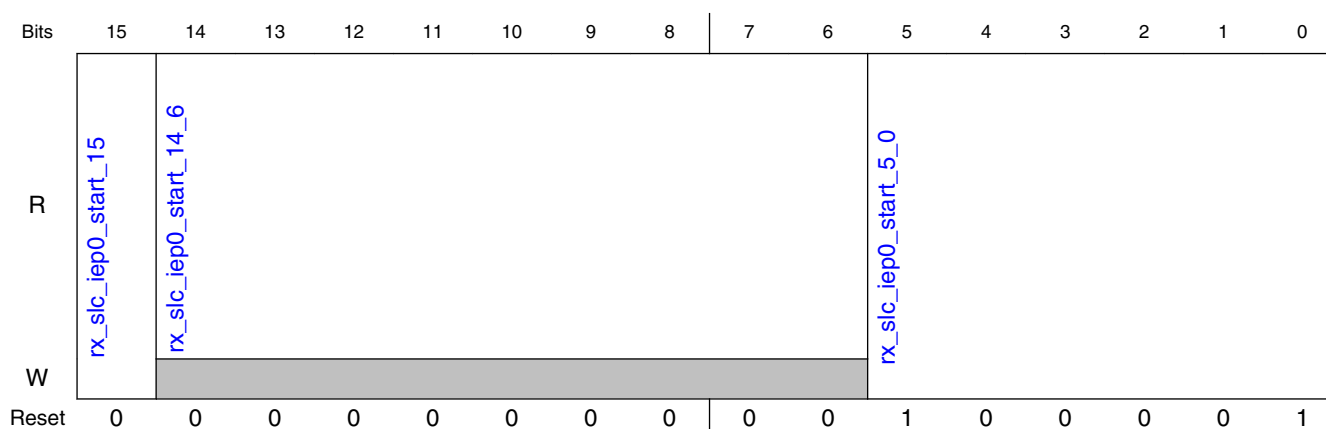
Field	Function
15 rx_slc_iep0_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_iep0_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_iep0_ovrd_13_6	Reserved
5-0 rx_slc_iep0_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.238 RX sampler latch I even positive 0 calibration unit start register (lane0_rx_slc_iep0_start - lane3_rx_slc_iep0_start)

13.4.10.2.238.1 Offset

Register	Offset
lane0_rx_slc_iep0_start	8132h
lane1_rx_slc_iep0_start	8532h
lane2_rx_slc_iep0_start	8932h
lane3_rx_slc_iep0_start	8D32h

13.4.10.2.238.2 Diagram



13.4.10.2.238.3 Fields

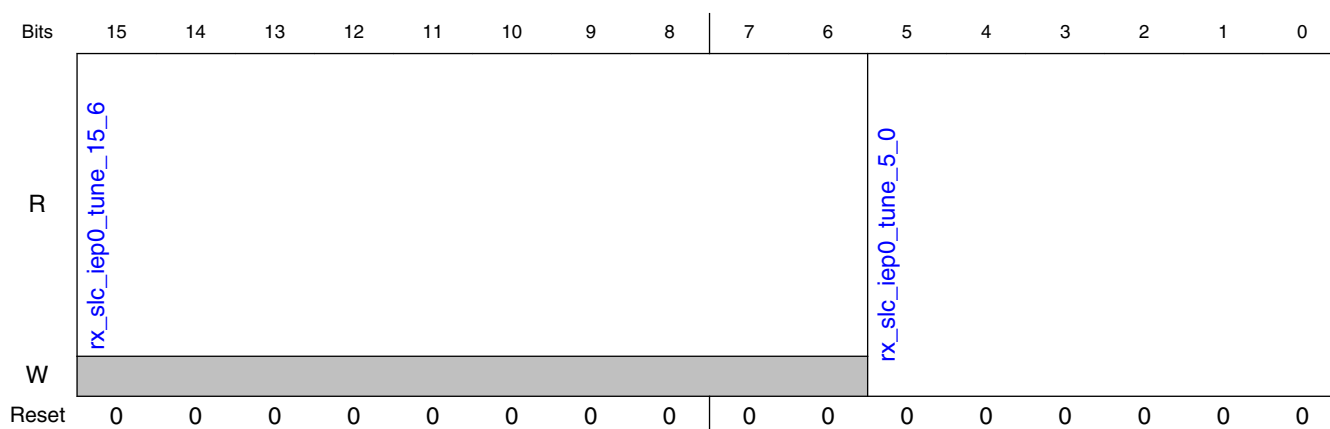
Field	Function
15 rx_slc_iep0_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_iep0_start_14_6	Reserved
5-0 rx_slc_iep0_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.239 RX sampler latch I even positive 0 calibration unit tune register (lane0_rx_slc_iep0_tune - lane3_rx_slc_iep0_tune)

13.4.10.2.239.1 Offset

Register	Offset
lane0_rx_slc_iep0_tune	8133h
lane1_rx_slc_iep0_tune	8533h
lane2_rx_slc_iep0_tune	8933h
lane3_rx_slc_iep0_tune	8D33h

13.4.10.2.239.2 Diagram



13.4.10.2.239.3 Fields

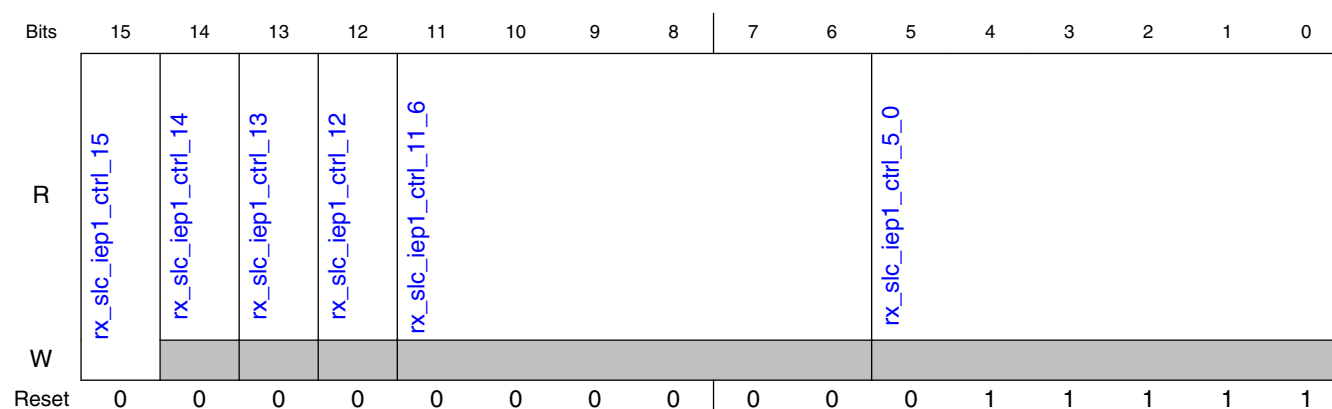
Field	Function
15-6 rx_slc_iep0_tune_15_6	Reserved
5-0 rx_slc_iep0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.240 RX sampler latch I even positive 1 calibration unit control register (lane0_rx_slc_iep1_ctrl - lane3_rx_slc_iep1_ctrl)

13.4.10.2.240.1 Offset

Register	Offset
lane0_rx_slc_iep1_ctrl	8134h
lane1_rx_slc_iep1_ctrl	8534h
lane2_rx_slc_iep1_ctrl	8934h
lane3_rx_slc_iep1_ctrl	8D34h

13.4.10.2.240.2 Diagram



13.4.10.2.240.3 Fields

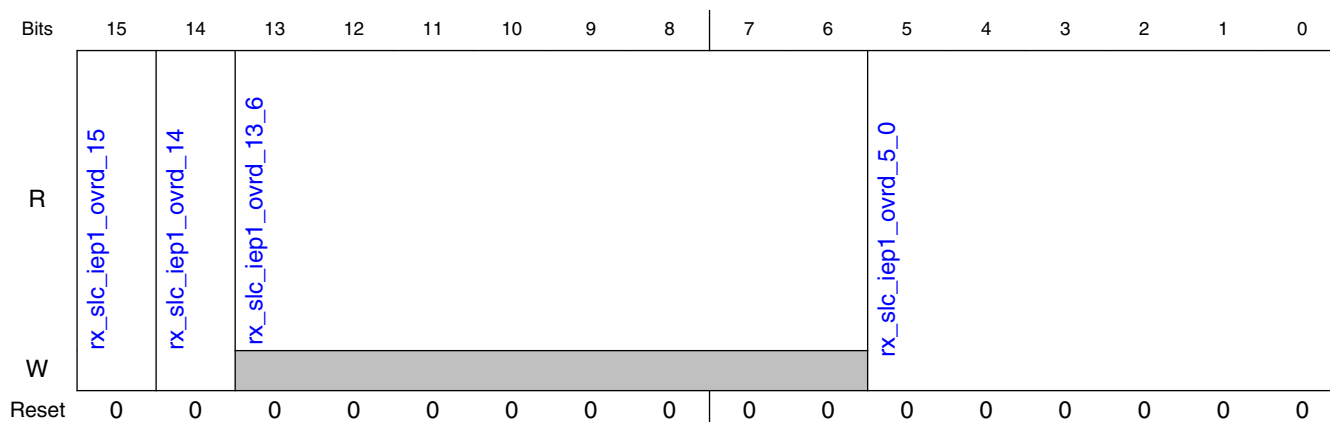
Field	Function
15 rx_slc_iep1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_iep1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_iep1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_iep1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_iep1_ctrl_11_6	Reserved
5-0 rx_slc_iep1_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.241 RX sampler latch I even positive 1 calibration unit override register (lane0_rx_slc_iep1_ovrd - lane3_rx_slc_iep1_ovrd)

13.4.10.2.241.1 Offset

Register	Offset
lane0_rx_slc_iep1_ovrd	8135h
lane1_rx_slc_iep1_ovrd	8535h
lane2_rx_slc_iep1_ovrd	8935h
lane3_rx_slc_iep1_ovrd	8D35h

13.4.10.2.241.2 Diagram



13.4.10.2.241.3 Fields

Field	Function
15 rx_slc_iep1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_iep1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_iep1_ovrd_13_6	Reserved
5-0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the

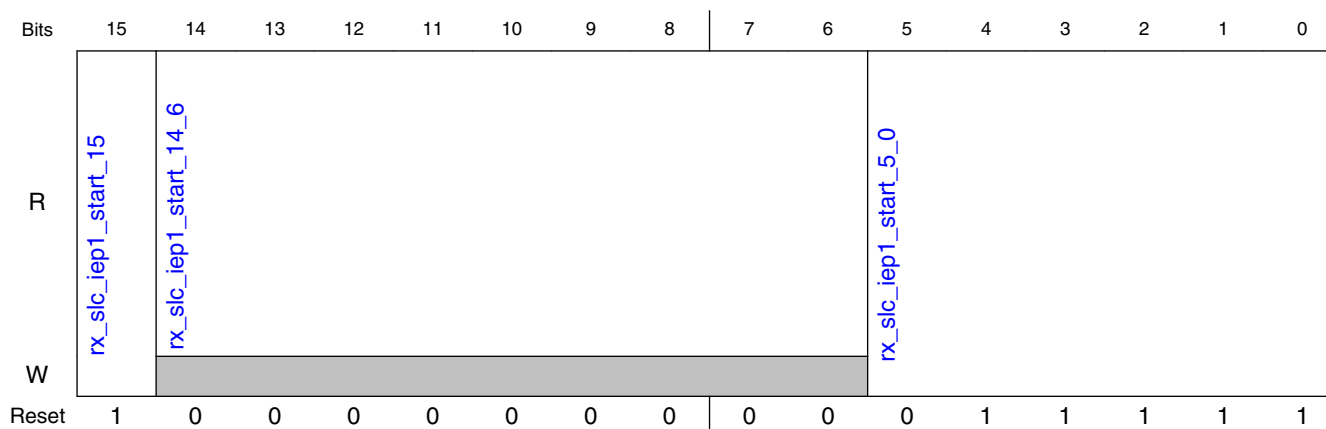
Field	Function
rx_slc_iep1_ovr d_5_0	calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.242 RX sampler latch I even positive 1 calibration unit start register (lane0_rx_slc_iep1_start - lane3_rx_slc_iep1_start)

13.4.10.2.242.1 Offset

Register	Offset
lane0_rx_slc_iep1_start	8136h
lane1_rx_slc_iep1_start	8536h
lane2_rx_slc_iep1_start	8936h
lane3_rx_slc_iep1_start	8D36h

13.4.10.2.242.2 Diagram



13.4.10.2.242.3 Fields

Field	Function
15 rx_slc_iep1_star t_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_iep1_star t_14_6	Reserved

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Clocks And Resets

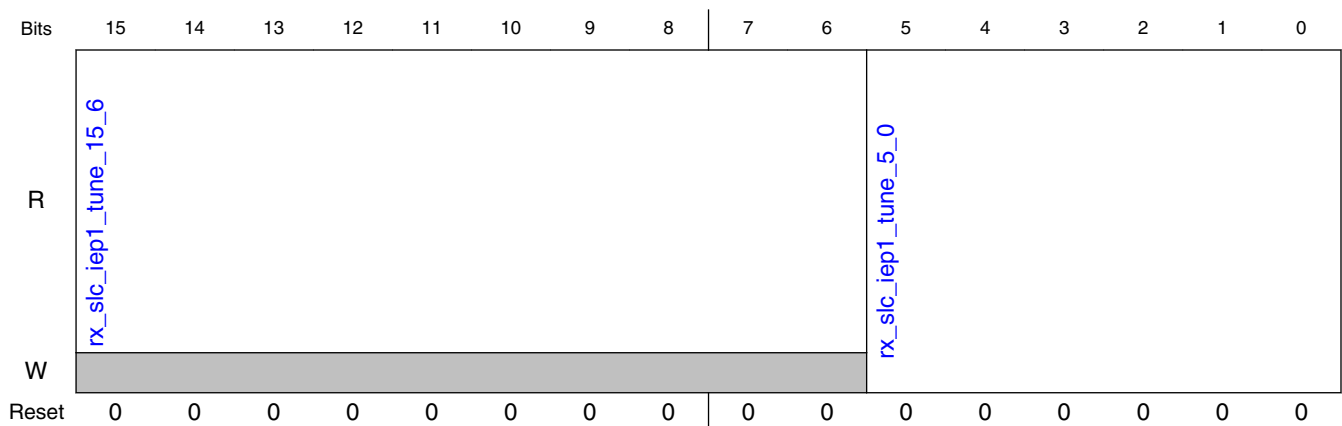
Field	Function
5-0 rx_slc_iep1_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.243 RX sampler latch I even positive 1 calibration unit tune register (lane0_rx_slc_iep1_tune - lane3_rx_slc_iep1_tune)

13.4.10.2.243.1 Offset

Register	Offset
lane0_rx_slc_iep1_tune	8137h
lane1_rx_slc_iep1_tune	8537h
lane2_rx_slc_iep1_tune	8937h
lane3_rx_slc_iep1_tune	8D37h

13.4.10.2.243.2 Diagram



13.4.10.2.243.3 Fields

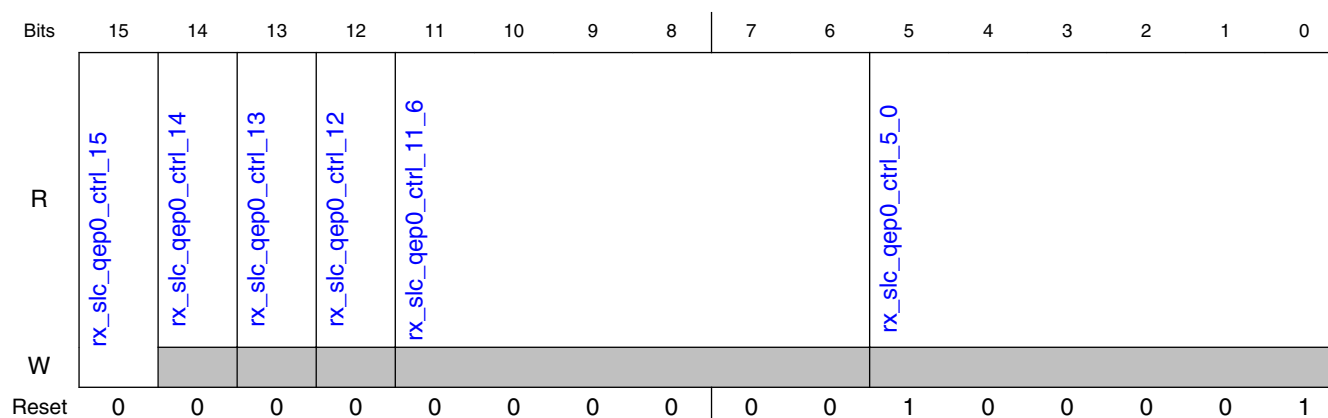
Field	Function
15-6 rx_slc_iep1_tune_15_6	Reserved
5-0 rx_slc_iep1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.244 RX sampler latch Q even positive 0 calibration unit control register (lane0_rx_slc_qep0_ctrl - lane3_rx_slc_qep0_ctrl)

13.4.10.2.244.1 Offset

Register	Offset
lane0_rx_slc_qep0_ctrl	8138h
lane1_rx_slc_qep0_ctrl	8538h
lane2_rx_slc_qep0_ctrl	8938h
lane3_rx_slc_qep0_ctrl	8D38h

13.4.10.2.244.2 Diagram



13.4.10.2.244.3 Fields

Field	Function
15 rx_slc_qep0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_qep0_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_qep0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.

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Clocks And Resets

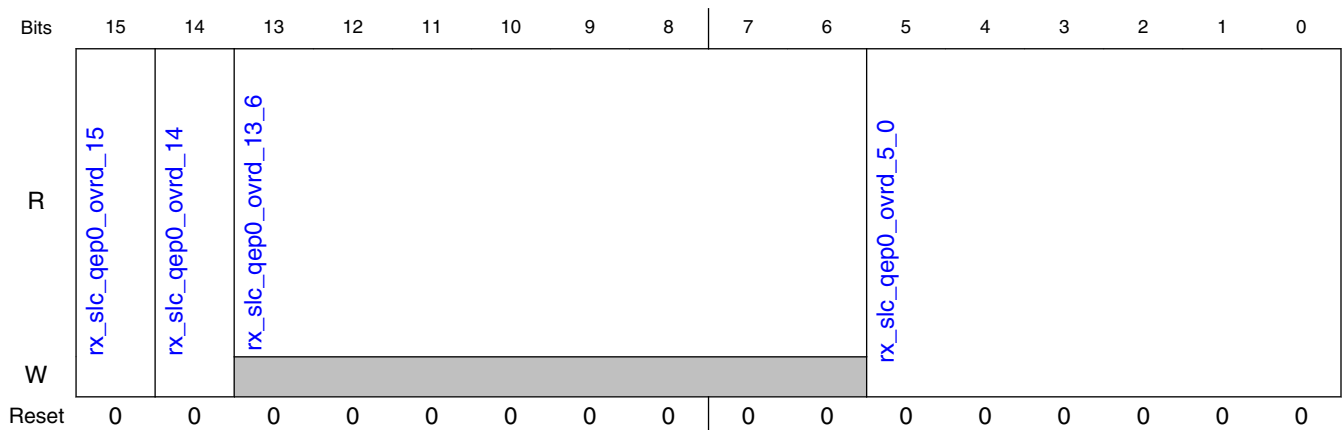
Field	Function
rx_slc_qep0_ctrl_12	
11-6 rx_slc_qep0_ctrl_11_6	Reserved
5-0 rx_slc_qep0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.245 RX sampler latch Q even positive 0 calibration unit override register (lane0_rx_slc_qep0_ovrd - lane3_rx_slc_qep0_ovrd)

13.4.10.2.245.1 Offset

Register	Offset
lane0_rx_slc_qep0_ovrd	8139h
lane1_rx_slc_qep0_ovrd	8539h
lane2_rx_slc_qep0_ovrd	8939h
lane3_rx_slc_qep0_ovrd	8D39h

13.4.10.2.245.2 Diagram



13.4.10.2.245.3 Fields

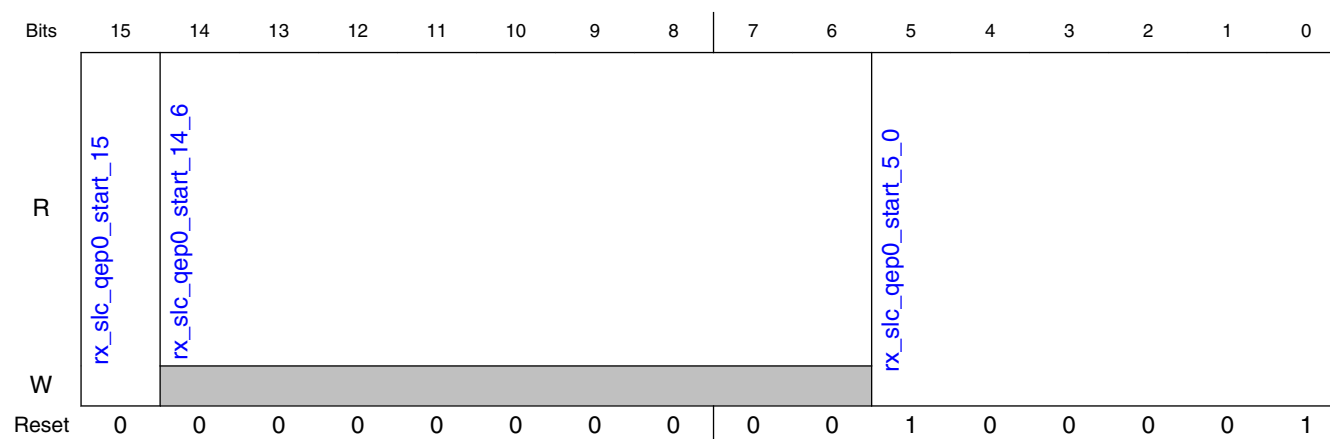
Field	Function
15 rx_slc_qep0_ovr_d_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_qep0_ovr_d_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_qep0_ovr_d_13_6	Reserved
5-0 rx_slc_qep0_ovr_d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.246 RX sampler latch Q even positive 0 calibration unit start register (lane0_rx_slc_qep0_start - lane3_rx_slc_qep0_start)

13.4.10.2.246.1 Offset

Register	Offset
lane0_rx_slc_qep0_start	813Ah
lane1_rx_slc_qep0_start	853Ah
lane2_rx_slc_qep0_start	893Ah
lane3_rx_slc_qep0_start	8D3Ah

13.4.10.2.246.2 Diagram



13.4.10.2.246.3 Fields

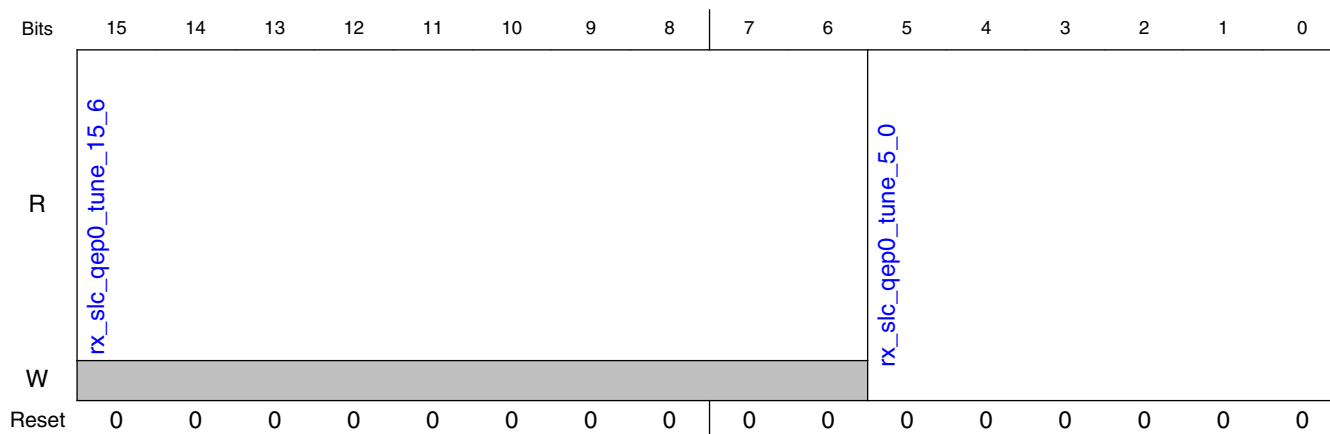
Field	Function
15 rx_slc_qep0_sta rt_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_qep0_sta rt_14_6	Reserved
5-0 rx_slc_qep0_sta rt_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.247 RX sampler latch Q even positive 0 calibration unit tune register (lane0_rx_slc_qep0_tune - lane3_rx_slc_qep0_tune)

13.4.10.2.247.1 Offset

Register	Offset
lane0_rx_slc_qep0_tune	813Bh
lane1_rx_slc_qep0_tune	853Bh
lane2_rx_slc_qep0_tune	893Bh
lane3_rx_slc_qep0_tune	8D3Bh

13.4.10.2.247.2 Diagram



13.4.10.2.247.3 Fields

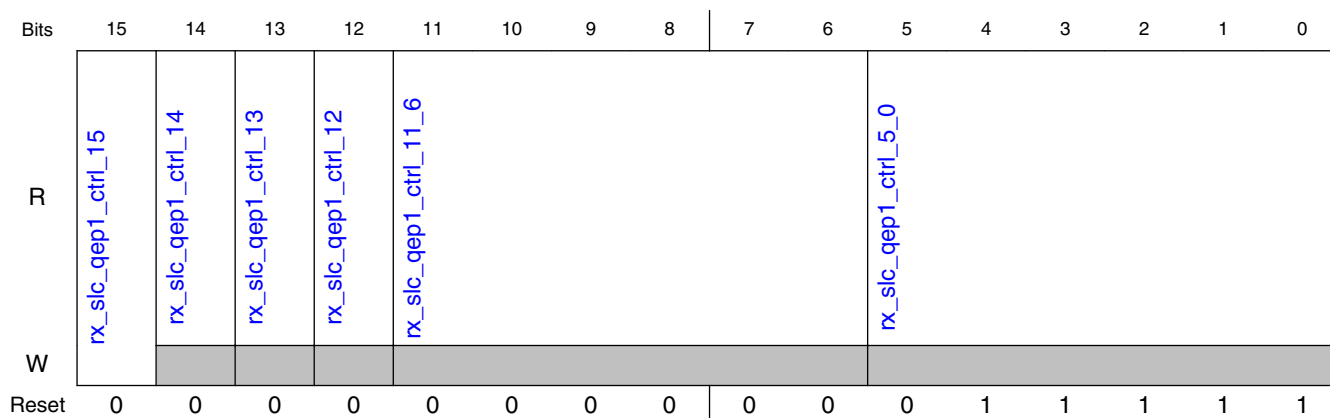
Field	Function
15-6 rx_slc_qep0_tune_15_6	Reserved
5-0 rx_slc_qep0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.248 RX sampler latch Q even positive 1 calibration unit control register (lane0_rx_slc_qep1_ctrl - lane3_rx_slc_qep1_ctrl)

13.4.10.2.248.1 Offset

Register	Offset
lane0_rx_slc_qep1_ctrl	813Ch
lane1_rx_slc_qep1_ctrl	853Ch
lane2_rx_slc_qep1_ctrl	893Ch
lane3_rx_slc_qep1_ctrl	8D3Ch

13.4.10.2.248.2 Diagram



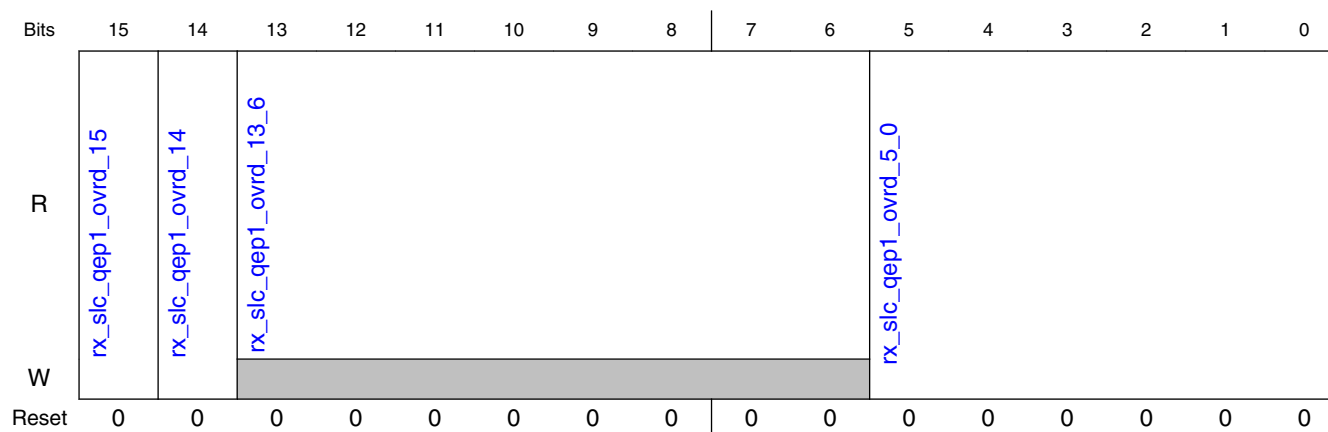
13.4.10.2.248.3 Fields

Field	Function
15 rx_slc_qep1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_qep1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_qep1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_qep1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_qep1_ctrl_11_6	Reserved
5-0 rx_slc_qep1_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.249 RX sampler latch Q even positive 1 calibration unit override register (lane0_rx_slc_qep1_ovrd - lane3_rx_slc_qep1_ovrd)**13.4.10.2.249.1 Offset**

Register	Offset
lane0_rx_slc_qep1_ovrd	813Dh
lane1_rx_slc_qep1_ovrd	853Dh
lane2_rx_slc_qep1_ovrd	893Dh
lane3_rx_slc_qep1_ovrd	8D3Dh

13.4.10.2.249.2 Diagram



13.4.10.2.249.3 Fields

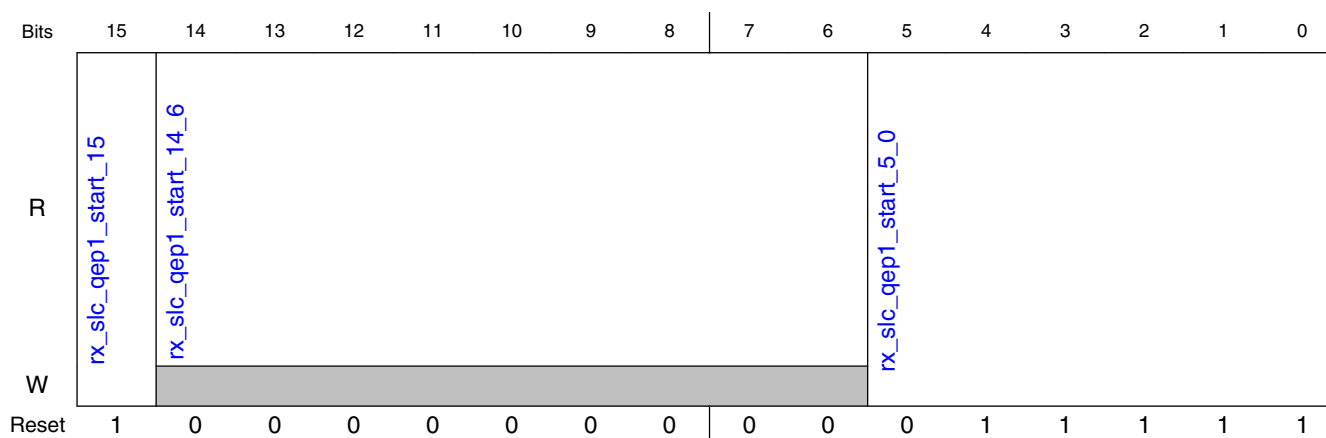
Field	Function
15 rx_slc_qep1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_qep1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_qep1_ovrd_13_6	Reserved
5-0 rx_slc_qep1_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.250 RX sampler latch Q even positive 1 calibration unit start register (lane0_rx_slc_qep1_start - lane3_rx_slc_qep1_start)

13.4.10.2.250.1 Offset

Register	Offset
lane0_rx_slc_qep1_start	813Eh
lane1_rx_slc_qep1_start	853Eh
lane2_rx_slc_qep1_start	893Eh
lane3_rx_slc_qep1_start	8D3Eh

13.4.10.2.250.2 Diagram



13.4.10.2.250.3 Fields

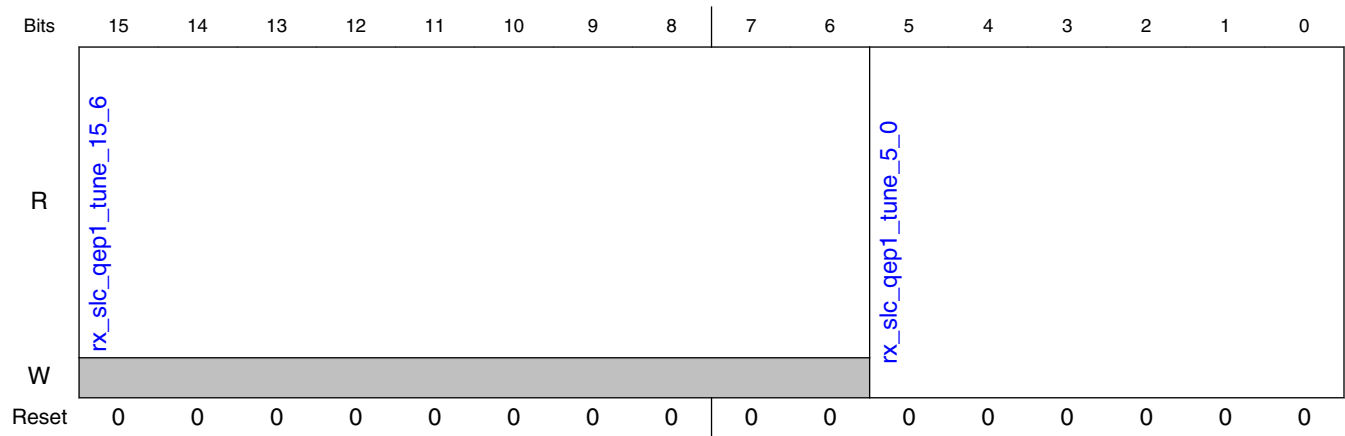
Field	Function
15 rx_slc_qep1_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_qep1_start_14_6	Reserved
5-0 rx_slc_qep1_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.251 RX sampler latch Q even positive 1 calibration unit tune register (lane0_rx_slc_qep1_tune - lane3_rx_slc_qep1_tune)

13.4.10.2.251.1 Offset

Register	Offset
lane0_rx_slc_qep1_tune	813Fh
lane1_rx_slc_qep1_tune	853Fh
lane2_rx_slc_qep1_tune	893Fh
lane3_rx_slc_qep1_tune	8D3Fh

13.4.10.2.251.2 Diagram



13.4.10.2.251.3 Fields

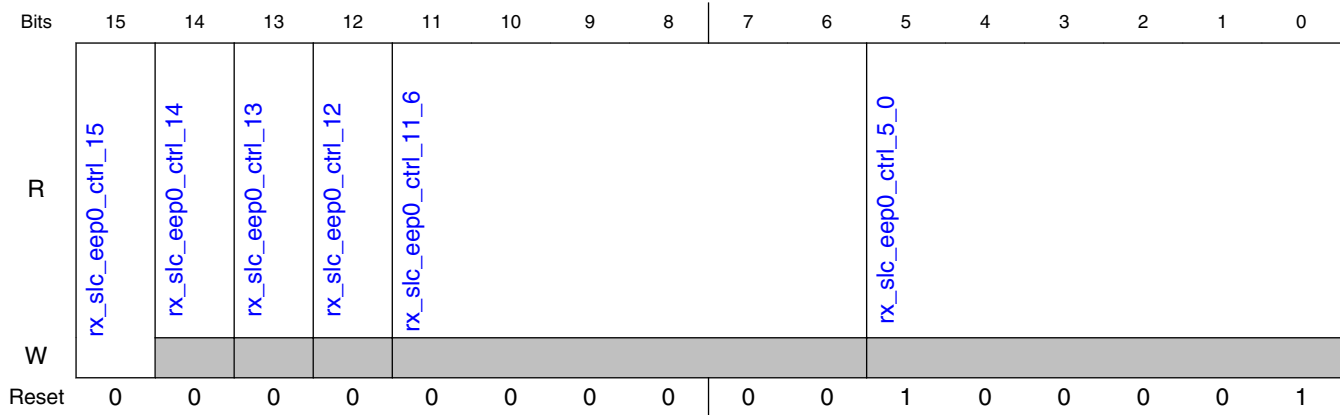
Field	Function
15-6 rx_slc_qep1_tune_15_6	Reserved
5-0 rx_slc_qep1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.252 RX sampler latch E even positive 0 calibration unit control register (lane0_rx_slc_eep0_ctrl - lane3_rx_slc_eep0_ctrl)

13.4.10.2.252.1 Offset

Register	Offset
lane0_rx_slc_eep0_ctrl	8140h
lane1_rx_slc_eep0_ctrl	8540h
lane2_rx_slc_eep0_ctrl	8940h
lane3_rx_slc_eep0_ctrl	8D40h

13.4.10.2.252.2 Diagram



13.4.10.2.252.3 Fields

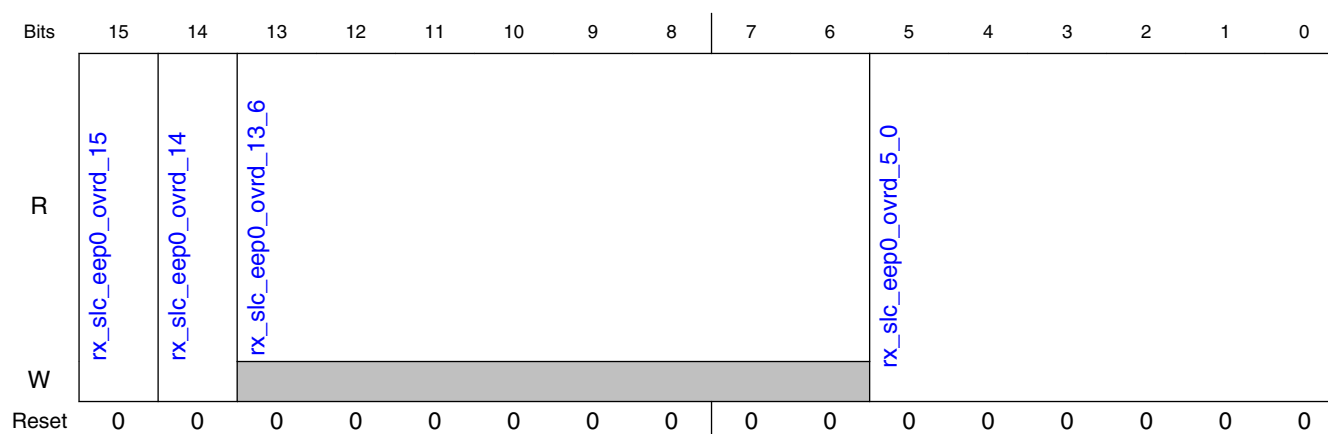
Field	Function
15 rx_slc_eep0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_eep0_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_eep0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_eep0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_eep0_ctrl_11_6	Reserved
5-0 rx_slc_eep0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.253 RX sampler latch E even positive 0 calibration unit override register (lane0_rx_slc_eep0_ovrd - lane3_rx_slc_eep0_ovrd)

13.4.10.2.253.1 Offset

Register	Offset
lane0_rx_slc_eep0_ovrd	8141h
lane1_rx_slc_eep0_ovrd	8541h
lane2_rx_slc_eep0_ovrd	8941h
lane3_rx_slc_eep0_ovrd	8D41h

13.4.10.2.253.2 Diagram



13.4.10.2.253.3 Fields

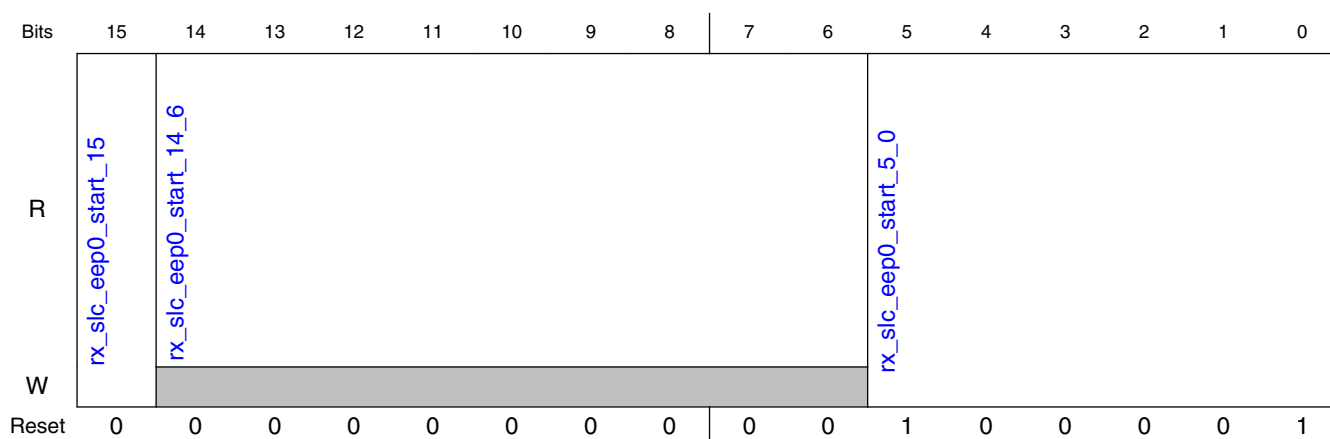
Field	Function
15 <code>rx_slc_eep0_ovrd_15</code>	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 <code>rx_slc_eep0_ovrd_14</code>	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 <code>rx_slc_eep0_ovrd_13_6</code>	Reserved
5-0 <code>rx_slc_eep0_ovrd_5_0</code>	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.254 RX sampler latch E even positive 0 calibration unit start register (lane0_rx_slc_eep0_start - lane3_rx_slc_eep0_start)

13.4.10.2.254.1 Offset

Register	Offset
lane0_rx_slc_eep0_start	8142h
lane1_rx_slc_eep0_start	8542h
lane2_rx_slc_eep0_start	8942h
lane3_rx_slc_eep0_start	8D42h

13.4.10.2.254.2 Diagram



13.4.10.2.254.3 Fields

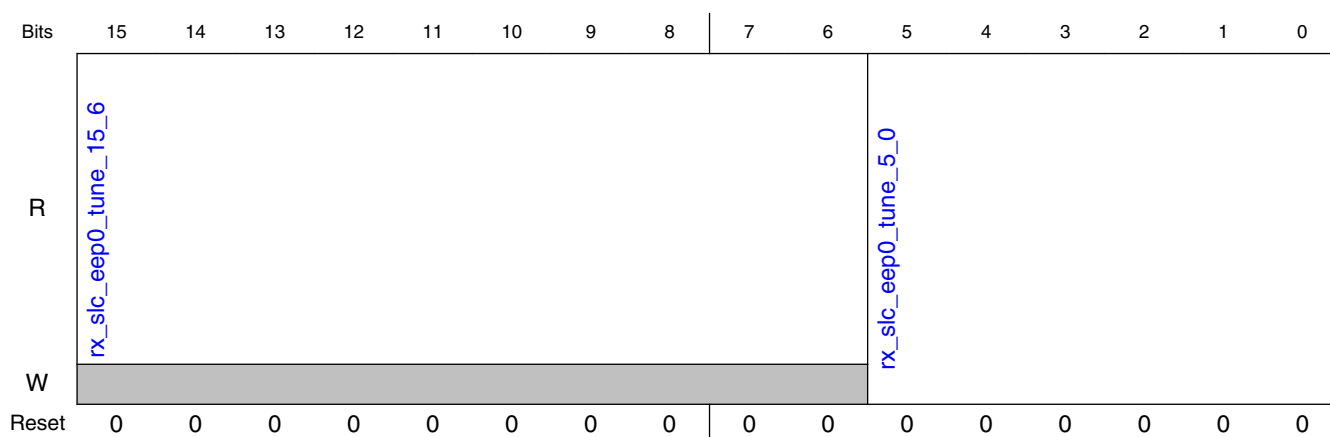
Field	Function
15 rx_slc_eep0_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_eep0_start_14_6	Reserved
5-0 rx_slc_eep0_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.255 RX sampler latch E even positive 0 calibration unit tune register (lane0_rx_slc_eep0_tune - lane3_rx_slc_eep0_tune)

13.4.10.2.255.1 Offset

Register	Offset
lane0_rx_slc_eep0_tune	8143h
lane1_rx_slc_eep0_tune	8543h
lane2_rx_slc_eep0_tune	8943h
lane3_rx_slc_eep0_tune	8D43h

13.4.10.2.255.2 Diagram



13.4.10.2.255.3 Fields

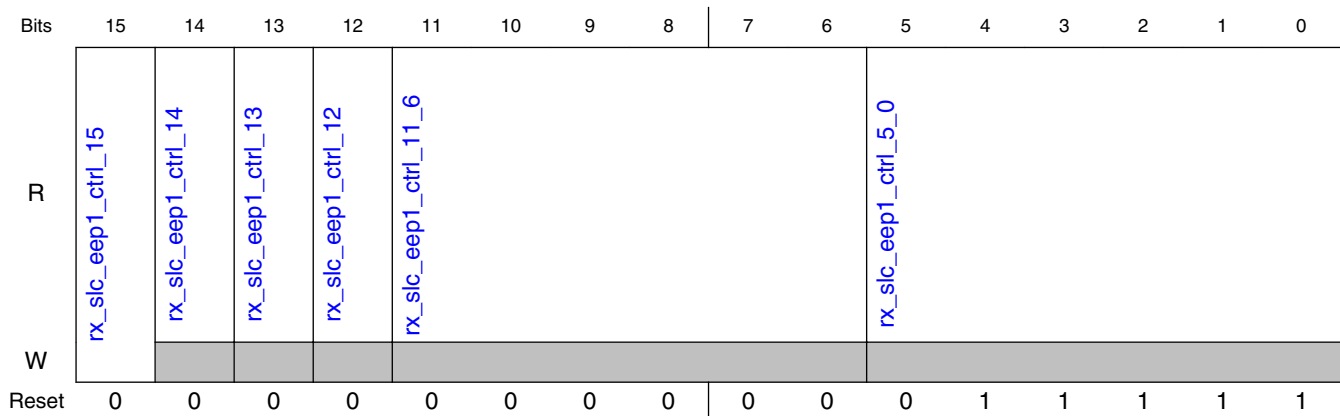
Field	Function
15-6 rx_slc_eep0_tune_15_6	Reserved
5-0 rx_slc_eep0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.256 RX sampler latch E even positive 1 calibration unit control register (lane0_rx_slc_eep1_ctrl - lane3_rx_slc_eep1_ctrl)

13.4.10.2.256.1 Offset

Register	Offset
lane0_rx_slc_eep1_ctrl	8144h
lane1_rx_slc_eep1_ctrl	8544h
lane2_rx_slc_eep1_ctrl	8944h
lane3_rx_slc_eep1_ctrl	8D44h

13.4.10.2.256.2 Diagram



13.4.10.2.256.3 Fields

Field	Function
15 rx_slc_eep1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_eep1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_eep1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_eep1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_eep1_ctrl_11_6	Reserved
5-0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

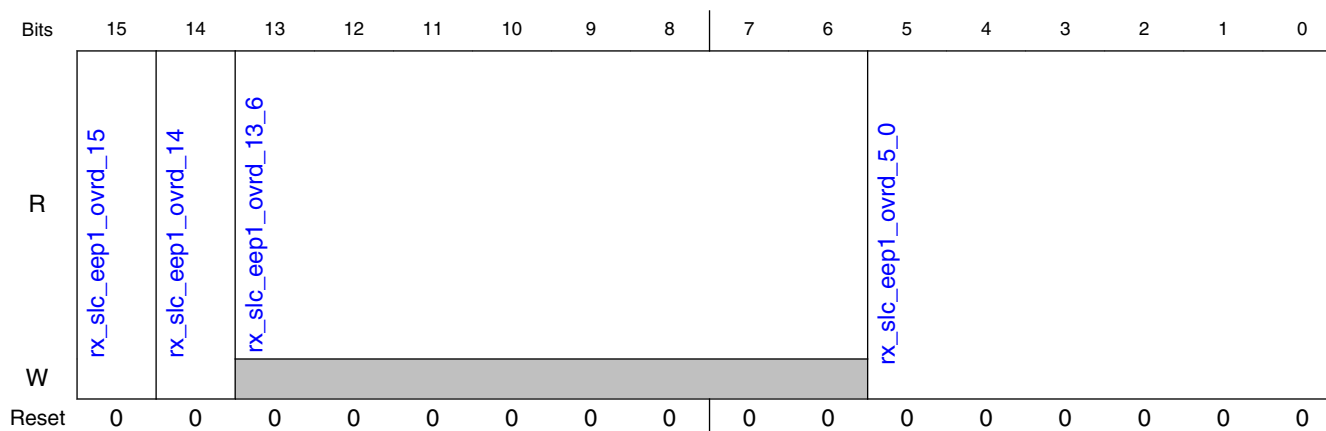
Field	Function
rx_slc_eep1_ctrl_5_0	

13.4.10.2.257 RX sampler latch E even positive 1 calibration unit override register (lane0_rx_slc_eep1_ovrd - lane3_rx_slc_eep1_ovrd)

13.4.10.2.257.1 Offset

Register	Offset
lane0_rx_slc_eep1_ovrd	8145h
lane1_rx_slc_eep1_ovrd	8545h
lane2_rx_slc_eep1_ovrd	8945h
lane3_rx_slc_eep1_ovrd	8D45h

13.4.10.2.257.2 Diagram



13.4.10.2.257.3 Fields

Field	Function
15 rx_slc_eep1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_eep1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the

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Clocks And Resets

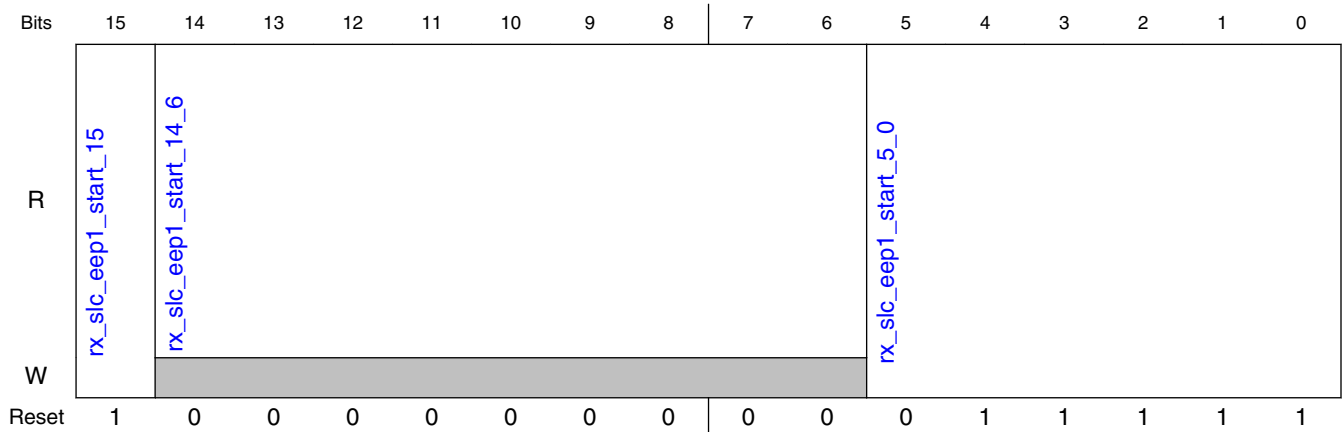
Field	Function
13-6 rx_slc_eep1_ovr d_13_6	Reserved
5-0 rx_slc_eep1_ovr d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.258 RX sampler latch E even positive 1 calibration unit start register (lane0_rx_slc_eep1_start - lane3_rx_slc_eep1_start)

13.4.10.2.258.1 Offset

Register	Offset
lane0_rx_slc_eep1_start	8146h
lane1_rx_slc_eep1_start	8546h
lane2_rx_slc_eep1_start	8946h
lane3_rx_slc_eep1_start	8D46h

13.4.10.2.258.2 Diagram



13.4.10.2.258.3 Fields

Field	Function
15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.

Table continues on the next page...

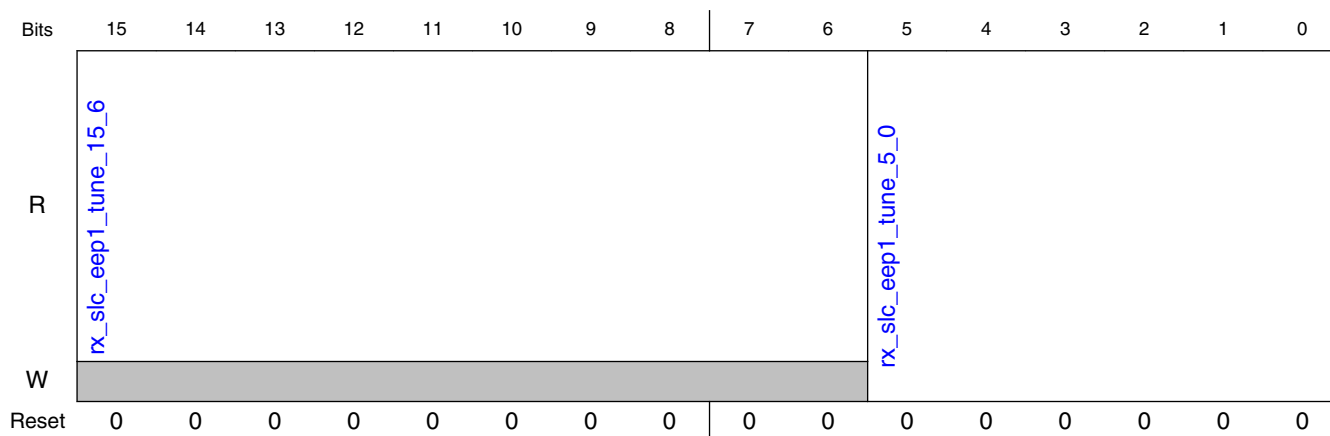
Field	Function
rx_slc_eep1_sta rt_15	
14-6 rx_slc_eep1_sta rt_14_6	Reserved
5-0 rx_slc_eep1_sta rt_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.259 RX sampler latch E even positive 1 calibration unit tune register (lane0_rx_slc_eep1_tune - lane3_rx_slc_eep1_tune)

13.4.10.2.259.1 Offset

Register	Offset
lane0_rx_slc_eep1_tune	8147h
lane1_rx_slc_eep1_tune	8547h
lane2_rx_slc_eep1_tune	8947h
lane3_rx_slc_eep1_tune	8D47h

13.4.10.2.259.2 Diagram



13.4.10.2.259.3 Fields

Field	Function
15-6	Reserved

Table continues on the next page...

Clocks And Resets

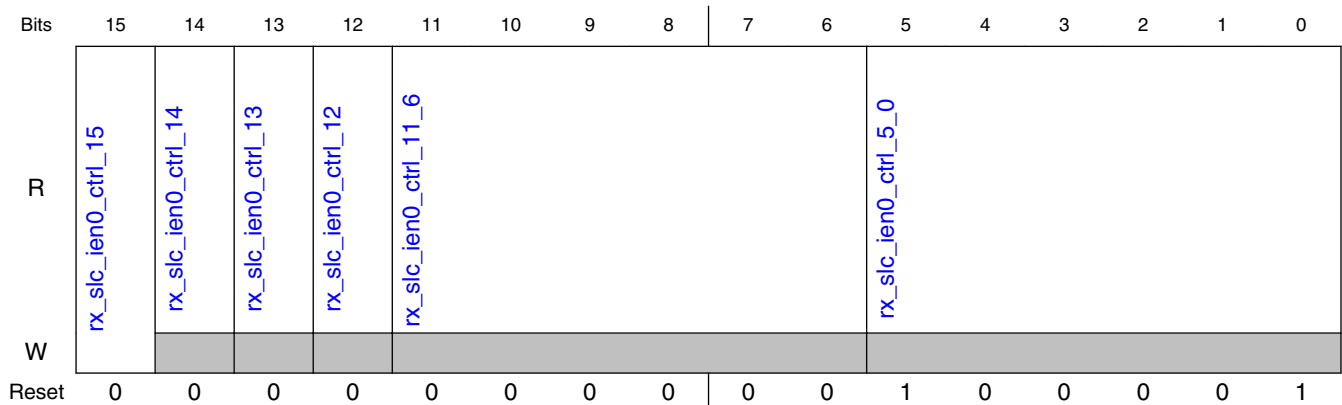
Field	Function
rx_slc_eep1_tune_15_6	
5-0 rx_slc_eep1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a two's complement value, so the calibrated code can be increased or decreased.

13.4.10.2.260 RX sampler latch I even negative 0 calibration unit control register (lane0_rx_slc_ien0_ctrl - lane3_rx_slc_ien0_ctrl)

13.4.10.2.260.1 Offset

Register	Offset
lane0_rx_slc_ien0_ctrl	8148h
lane1_rx_slc_ien0_ctrl	8548h
lane2_rx_slc_ien0_ctrl	8948h
lane3_rx_slc_ien0_ctrl	8D48h

13.4.10.2.260.2 Diagram



13.4.10.2.260.3 Fields

Field	Function
15 rx_slc_ien0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by

Table continues on the next page...

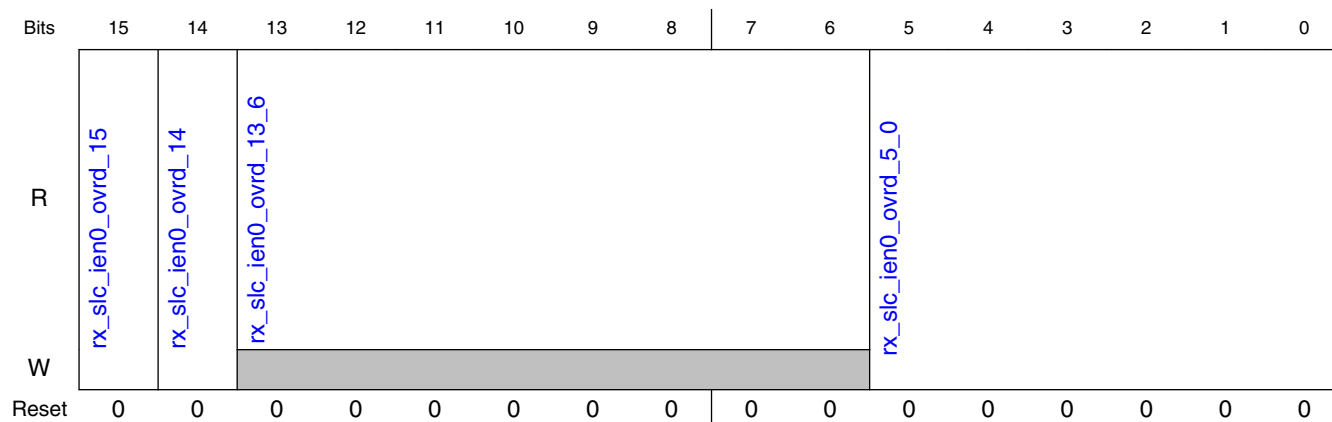
Field	Function
rx_slc_ien0_ctrl_14	
13 rx_slc_ien0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_ien0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_ien0_ctrl_11_6	Reserved
5-0 rx_slc_ien0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.261 RX sampler latch I even negative 0 calibration unit override register (lane0_rx_slc_ien0_ovrd - lane3_rx_slc_ien0_ovrd)

13.4.10.2.261.1 Offset

Register	Offset
lane0_rx_slc_ien0_ovrd	8149h
lane1_rx_slc_ien0_ovrd	8549h
lane2_rx_slc_ien0_ovrd	8949h
lane3_rx_slc_ien0_ovrd	8D49h

13.4.10.2.261.2 Diagram



13.4.10.2.261.3 Fields

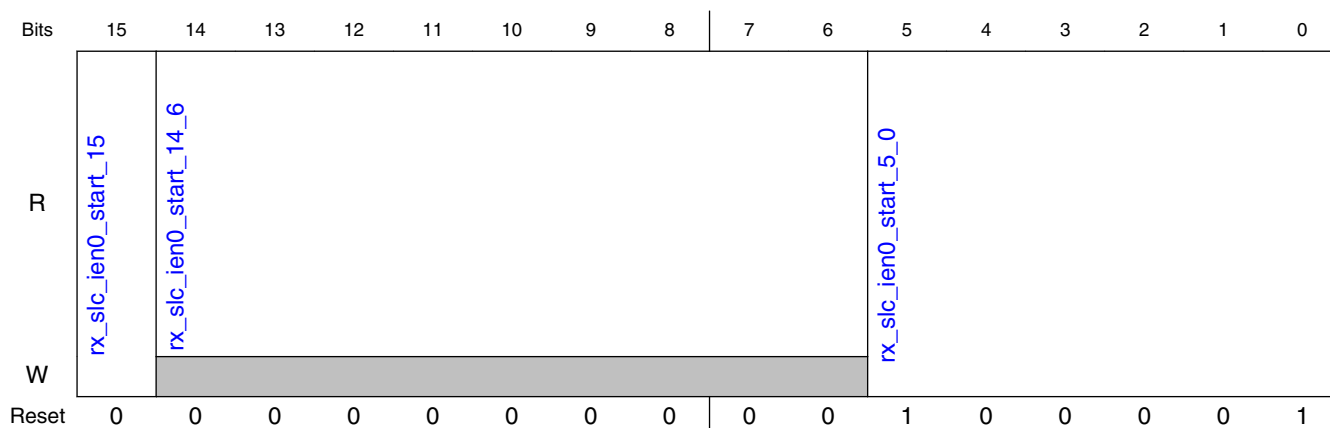
Field	Function
15 rx_slc_ien0_ovr_d_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_ien0_ovr_d_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_ien0_ovr_d_13_6	Reserved
5-0 rx_slc_ien0_ovr_d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.262 RX sampler latch I even negative 0 calibration unit start register (lane0_rx_slc_ien0_start - lane3_rx_slc_ien0_start)

13.4.10.2.262.1 Offset

Register	Offset
lane0_rx_slc_ien0_start	814Ah
lane1_rx_slc_ien0_start	854Ah
lane2_rx_slc_ien0_start	894Ah
lane3_rx_slc_ien0_start	8D4Ah

13.4.10.2.262.2 Diagram



13.4.10.2.262.3 Fields

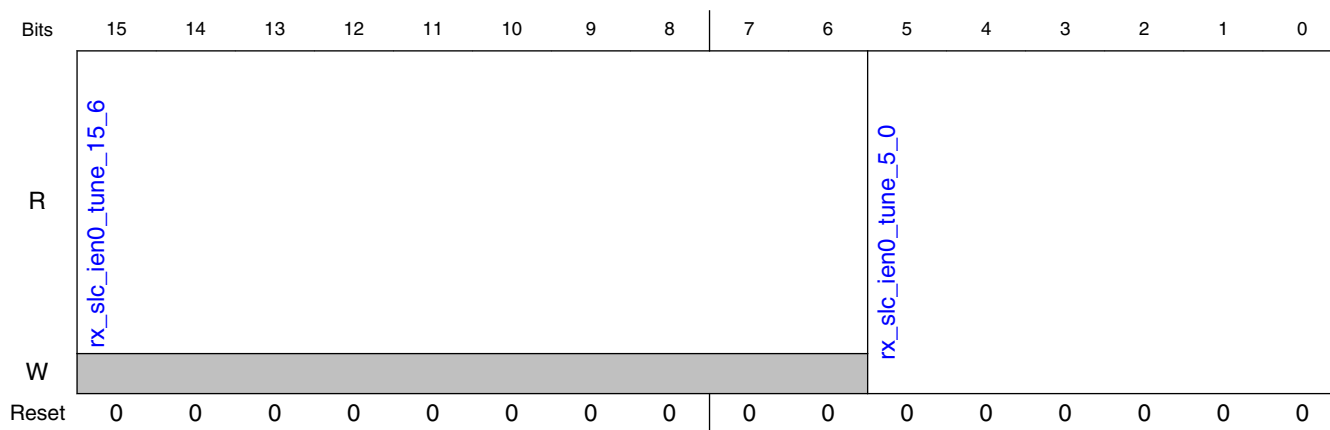
Field	Function
15 rx_slc_ien0_star t_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_ien0_star t_14_6	Reserved
5-0 rx_slc_ien0_star t_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.263 RX sampler latch I even negative 0 calibration unit tune register (lane0_rx_slc_ien0_tune - lane3_rx_slc_ien0_tune)

13.4.10.2.263.1 Offset

Register	Offset
lane0_rx_slc_ien0_tune	814Bh
lane1_rx_slc_ien0_tune	854Bh
lane2_rx_slc_ien0_tune	894Bh
lane3_rx_slc_ien0_tune	8D4Bh

13.4.10.2.263.2 Diagram



13.4.10.2.263.3 Fields

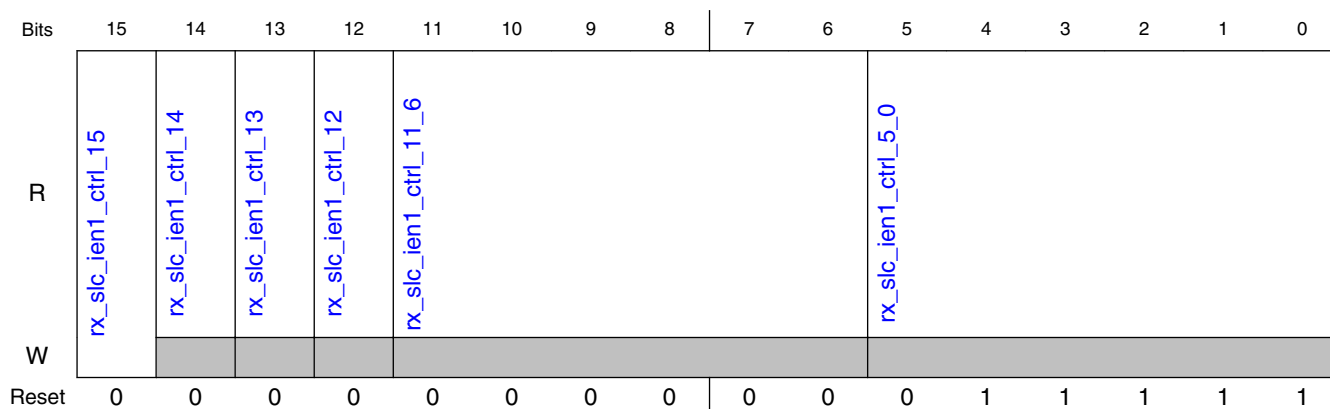
Field	Function
15-6 rx_slc_ien0_tune_15_6	Reserved
5-0 rx_slc_ien0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.264 RX sampler latch I even negative 1 calibration unit control register (lane0_rx_slc_ien1_ctrl - lane3_rx_slc_ien1_ctrl)

13.4.10.2.264.1 Offset

Register	Offset
lane0_rx_slc_ien1_ctrl	814Ch
lane1_rx_slc_ien1_ctrl	854Ch
lane2_rx_slc_ien1_ctrl	894Ch
lane3_rx_slc_ien1_ctrl	8D4Ch

13.4.10.2.264.2 Diagram



13.4.10.2.264.3 Fields

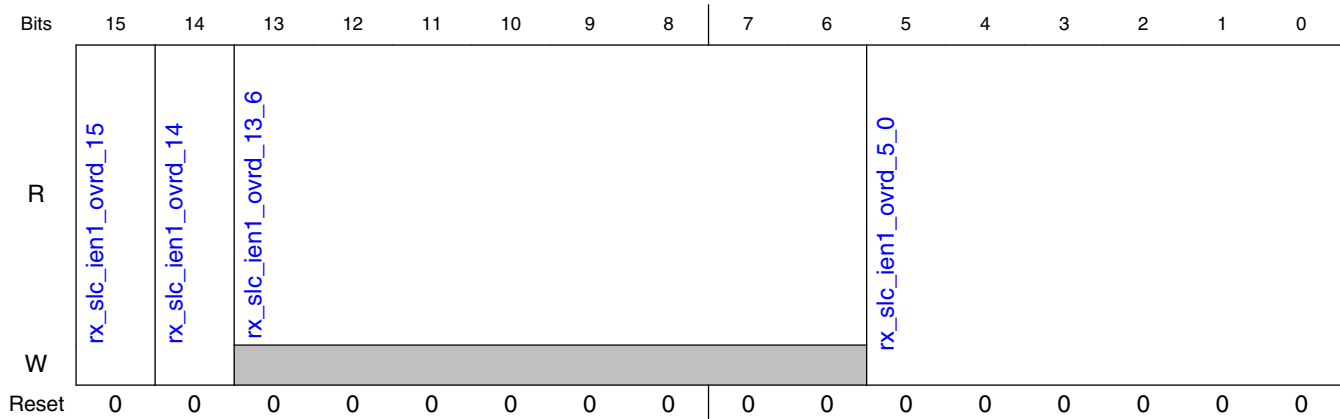
Field	Function
15 rx_slc_ien1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_ien1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_ien1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_ien1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_ien1_ctrl_11_6	Reserved
5-0 rx_slc_ien1_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.265 RX sampler latch I even negative 1 calibration unit override register (lane0_rx_slc_ien1_ovrd - lane3_rx_slc_ien1_ovrd)

13.4.10.2.265.1 Offset

Register	Offset
lane0_rx_slc_ien1_ovrd	814Dh
lane1_rx_slc_ien1_ovrd	854Dh
lane2_rx_slc_ien1_ovrd	894Dh
lane3_rx_slc_ien1_ovrd	8D4Dh

13.4.10.2.265.2 Diagram



13.4.10.2.265.3 Fields

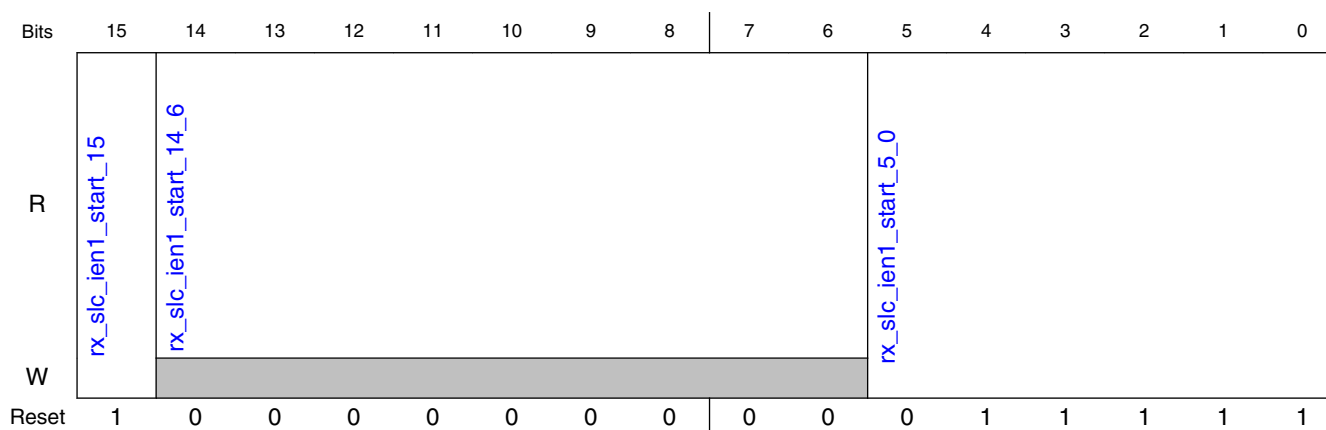
Field	Function
15 rx_slc_ien1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_ien1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_ien1_ovrd_13_6	Reserved
5-0 rx_slc_ien1_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.266 RX sampler latch I even negative 1 calibration unit start register (lane0_rx_slc_ien1_start - lane3_rx_slc_ien1_start)

13.4.10.2.266.1 Offset

Register	Offset
lane0_rx_slc_ien1_start	814Eh
lane1_rx_slc_ien1_start	854Eh
lane2_rx_slc_ien1_start	894Eh
lane3_rx_slc_ien1_start	8D4Eh

13.4.10.2.266.2 Diagram



13.4.10.2.266.3 Fields

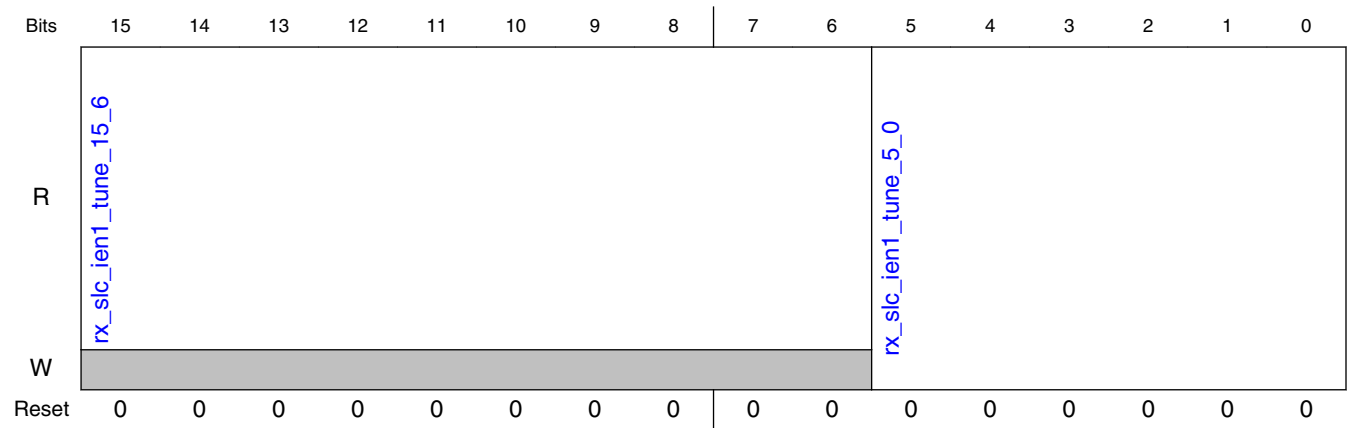
Field	Function
15 rx_slc_ien1_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_ien1_start_14_6	Reserved
5-0 rx_slc_ien1_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.267 RX sampler latch I even negative 1 calibration unit tune register (lane0_rx_slc_ien1_tune - lane3_rx_slc_ien1_tune)

13.4.10.2.267.1 Offset

Register	Offset
lane0_rx_slc_ien1_tune	814Fh
lane1_rx_slc_ien1_tune	854Fh
lane2_rx_slc_ien1_tune	894Fh
lane3_rx_slc_ien1_tune	8D4Fh

13.4.10.2.267.2 Diagram



13.4.10.2.267.3 Fields

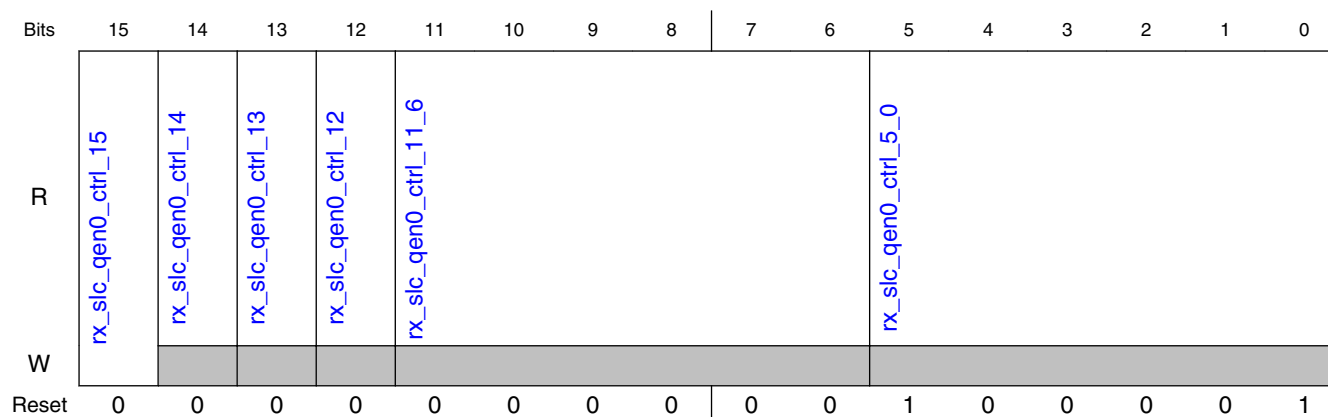
Field	Function
15-6 <code>rx_slc_ien1_tune_15_6</code>	Reserved
5-0 <code>rx_slc_ien1_tune_5_0</code>	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.268 RX sampler latch Q even negative 0 calibration unit control register (lane0_rx_slc_qen0_ctrl - lane3_rx_slc_qen0_ctrl)

13.4.10.2.268.1 Offset

Register	Offset
<code>lane0_rx_slc_qen0_ctrl</code>	8150h
<code>lane1_rx_slc_qen0_ctrl</code>	8550h
<code>lane2_rx_slc_qen0_ctrl</code>	8950h
<code>lane3_rx_slc_qen0_ctrl</code>	8D50h

13.4.10.2.268.2 Diagram



13.4.10.2.268.3 Fields

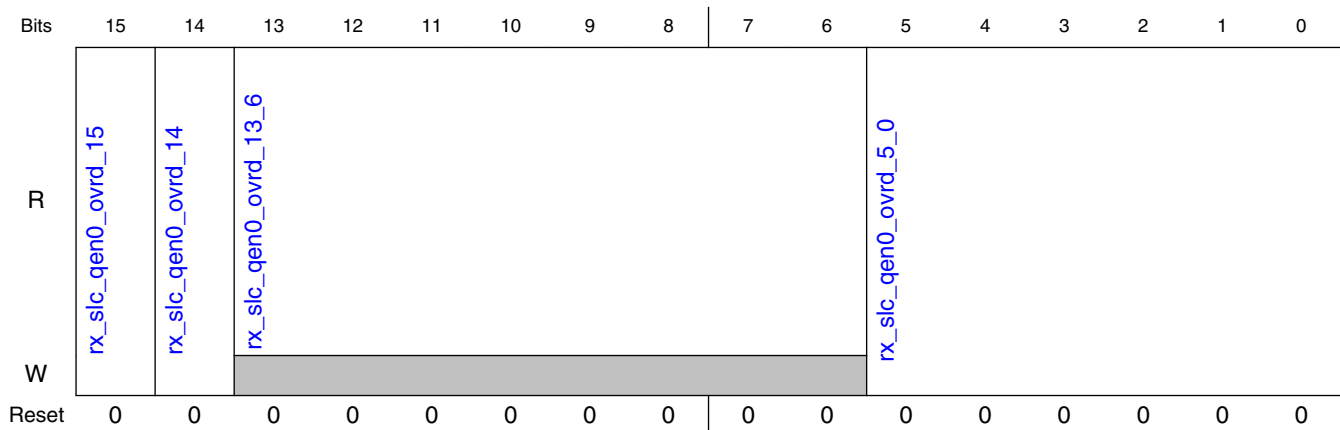
Field	Function
15 rx_slc_qen0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_qen0_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_qen0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_qen0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_qen0_ctrl_11_6	Reserved
5-0 rx_slc_qen0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.269 RX sampler latch Q even negative 0 calibration unit override register (lane0_rx_slc_qen0_ovrd - lane3_rx_slc_qen0_ovrd)

13.4.10.2.269.1 Offset

Register	Offset
lane0_rx_slc_qen0_ovrd	8151h
lane1_rx_slc_qen0_ovrd	8551h
lane2_rx_slc_qen0_ovrd	8951h
lane3_rx_slc_qen0_ovrd	8D51h

13.4.10.2.269.2 Diagram



13.4.10.2.269.3 Fields

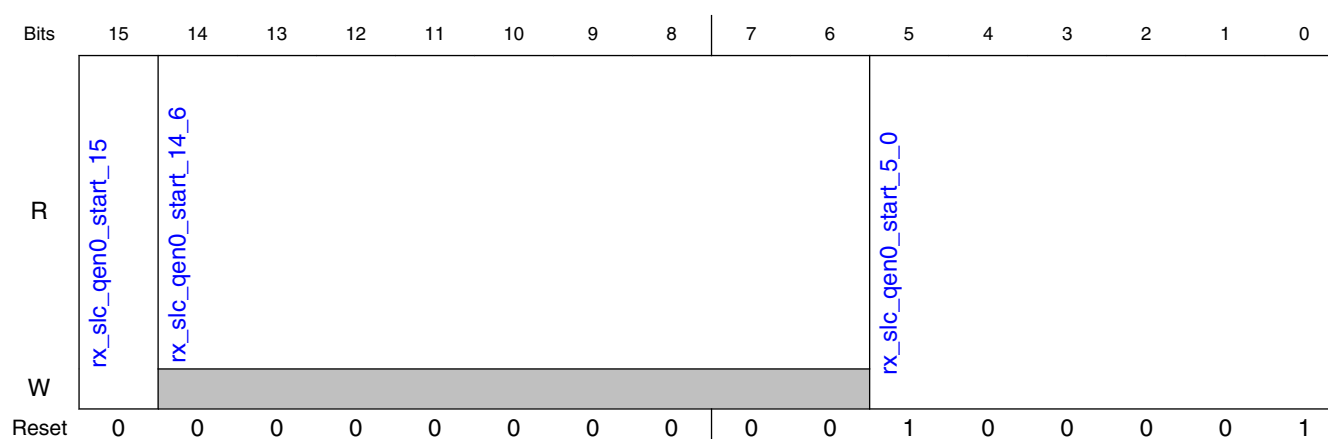
Field	Function
15 <code>rx_slc_qen0_ovrd_15</code>	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 <code>rx_slc_qen0_ovrd_14</code>	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 <code>rx_slc_qen0_ovrd_13_6</code>	Reserved
5-0 <code>rx_slc_qen0_ovrd_5_0</code>	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.270 RX sampler latch Q even negative 0 calibration unit start register (lane0_rx_slc_qen0_start - lane3_rx_slc_qen0_start)

13.4.10.2.270.1 Offset

Register	Offset
lane0_rx_slc_qen0_start	8152h
lane1_rx_slc_qen0_start	8552h
lane2_rx_slc_qen0_start	8952h
lane3_rx_slc_qen0_start	8D52h

13.4.10.2.270.2 Diagram



13.4.10.2.270.3 Fields

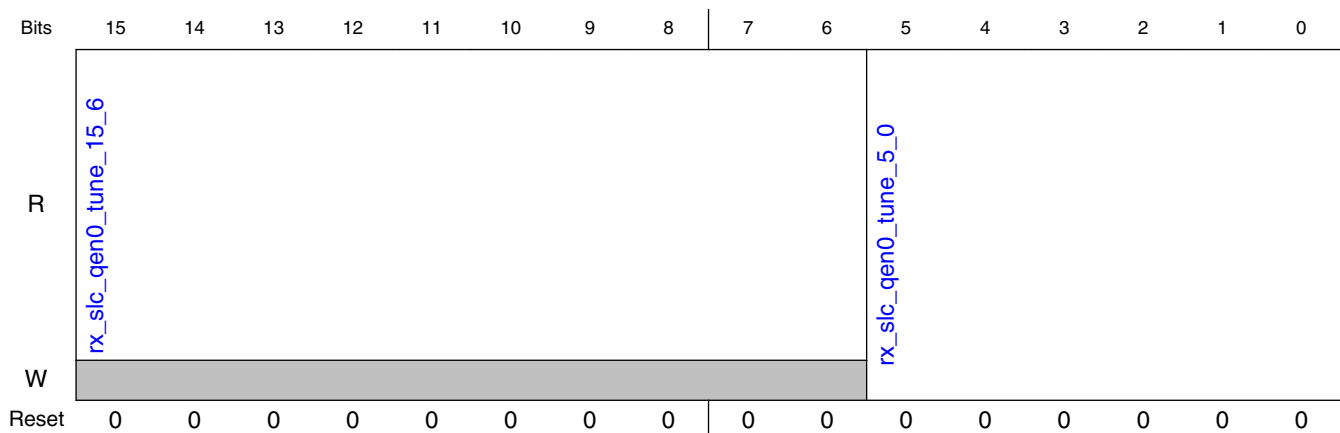
Field	Function
15 rx_slc_qen0_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_qen0_start_14_6	Reserved
5-0 rx_slc_qen0_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.271 RX sampler latch Q even negative 0 calibration unit tune register (lane0_rx_slc_qen0_tune - lane3_rx_slc_qen0_tune)

13.4.10.2.271.1 Offset

Register	Offset
lane0_rx_slc_qen0_tune	8153h
lane1_rx_slc_qen0_tune	8553h
lane2_rx_slc_qen0_tune	8953h
lane3_rx_slc_qen0_tune	8D53h

13.4.10.2.271.2 Diagram



13.4.10.2.271.3 Fields

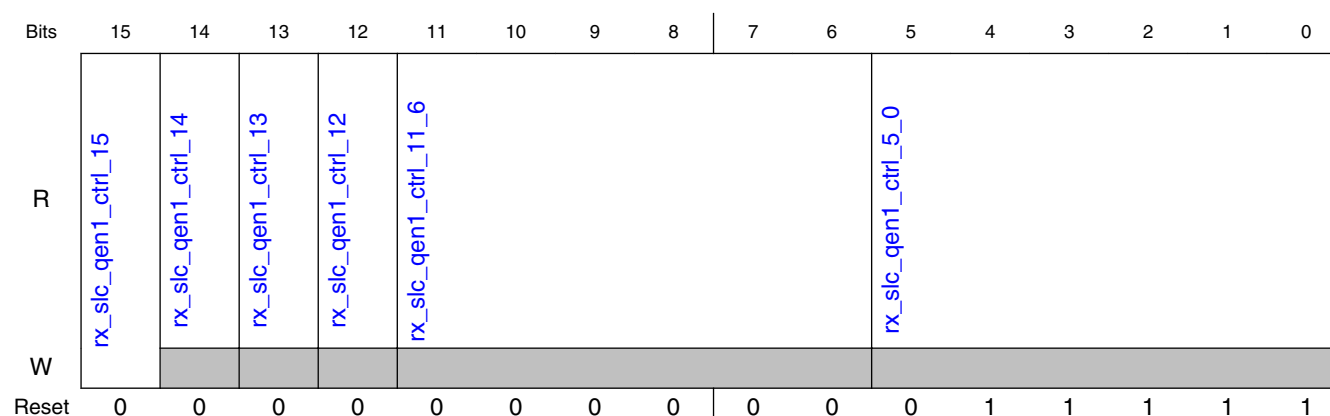
Field	Function
15-6 <code>rx_slc_qen0_tune_15_6</code>	Reserved
5-0 <code>rx_slc_qen0_tune_5_0</code>	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.272 RX sampler latch Q even negative 1 calibration unit control register (lane0_rx_slc_qen1_ctrl - lane3_rx_slc_qen1_ctrl)

13.4.10.2.272.1 Offset

Register	Offset
lane0_rx_slc_qen1_ctrl	8154h
lane1_rx_slc_qen1_ctrl	8554h
lane2_rx_slc_qen1_ctrl	8954h
lane3_rx_slc_qen1_ctrl	8D54h

13.4.10.2.272.2 Diagram



13.4.10.2.272.3 Fields

Field	Function
15 rx_slc_qen1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_qen1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_qen1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_qen1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_qen1_ctrl_11_6	Reserved
5-0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

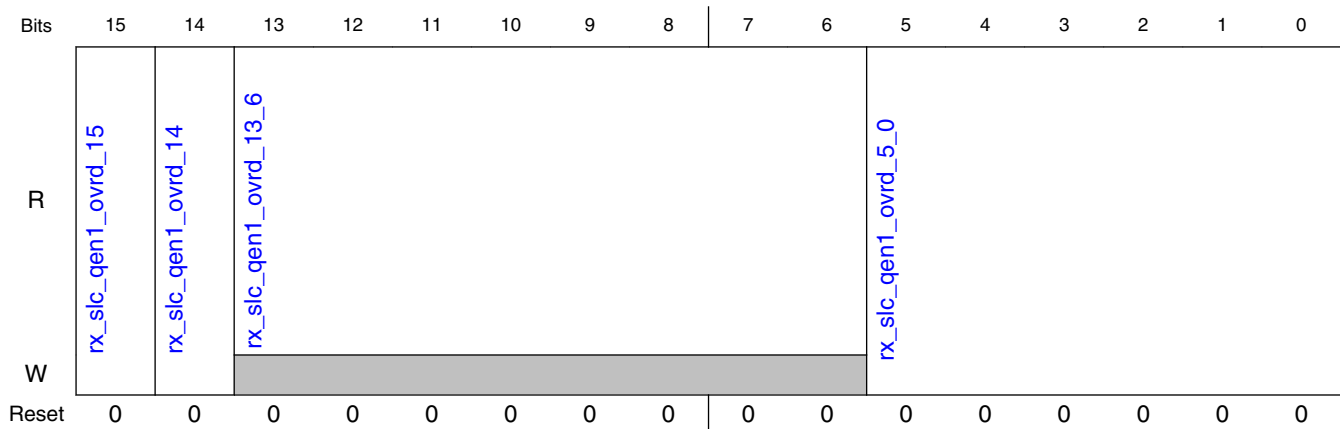
Field	Function
rx_slc_qen1_ctrl_5_0	

13.4.10.2.273 RX sampler latch Q even negative 1 calibration unit override register (lane0_rx_slc_qen1_ovrd - lane3_rx_slc_qen1_ovrd)

13.4.10.2.273.1 Offset

Register	Offset
lane0_rx_slc_qen1_ovrd	8155h
lane1_rx_slc_qen1_ovrd	8555h
lane2_rx_slc_qen1_ovrd	8955h
lane3_rx_slc_qen1_ovrd	8D55h

13.4.10.2.273.2 Diagram



13.4.10.2.273.3 Fields

Field	Function
15 rx_slc_qen1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_qen1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the

Table continues on the next page...

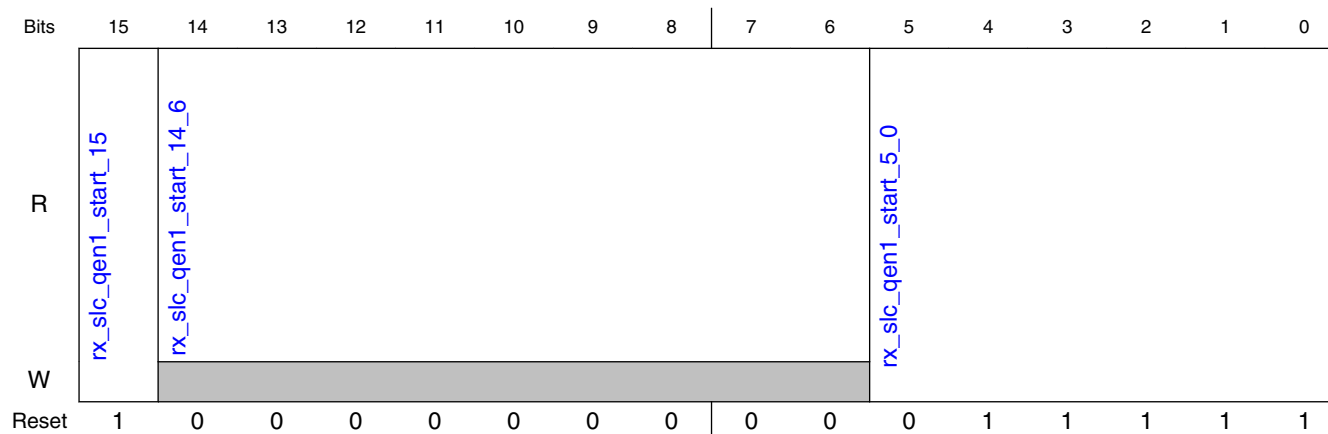
Field	Function
13-6 rx_slc_qen1_ovr d_13_6	Reserved
5-0 rx_slc_qen1_ovr d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.274 RX sampler latch Q even negative 1 calibration unit start register (lane0_rx_slc_qen1_start - lane3_rx_slc_qen1_start)

13.4.10.2.274.1 Offset

Register	Offset
lane0_rx_slc_qen1_start	8156h
lane1_rx_slc_qen1_start	8556h
lane2_rx_slc_qen1_start	8956h
lane3_rx_slc_qen1_start	8D56h

13.4.10.2.274.2 Diagram



13.4.10.2.274.3 Fields

Field	Function
15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.

Table continues on the next page...

Clocks And Resets

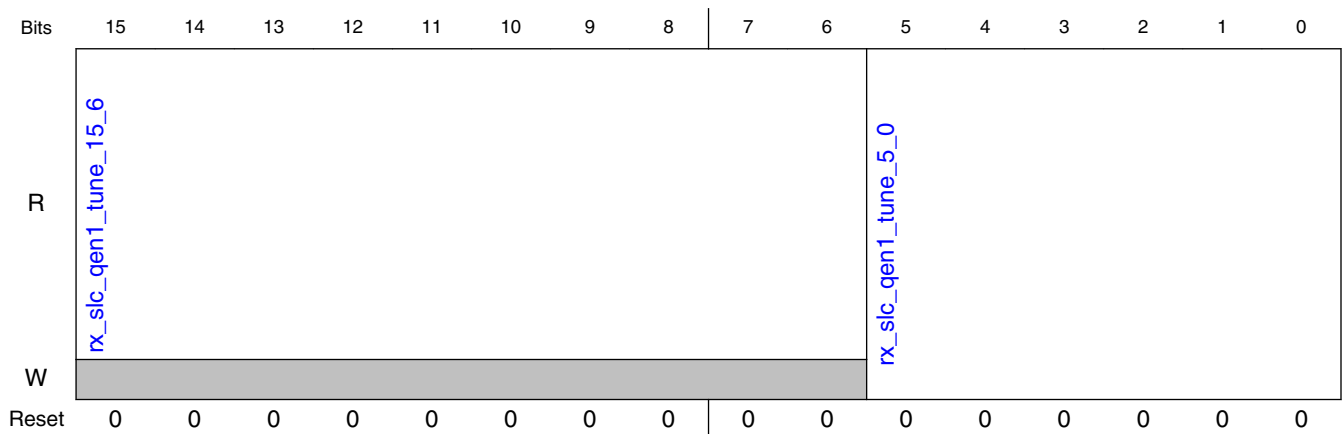
Field	Function
rx_slc_qen1_sta rt_15	Reserved
14-6 rx_slc_qen1_sta rt_14_6	
5-0 rx_slc_qen1_sta rt_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.275 RX sampler latch Q even negative 1 calibration unit tune register (lane0_rx_slc_qen1_tune - lane3_rx_slc_qen1_tune)

13.4.10.2.275.1 Offset

Register	Offset
lane0_rx_slc_qen1_tune	8157h
lane1_rx_slc_qen1_tune	8557h
lane2_rx_slc_qen1_tune	8957h
lane3_rx_slc_qen1_tune	8D57h

13.4.10.2.275.2 Diagram



13.4.10.2.275.3 Fields

Field	Function
15-6	Reserved

Table continues on the next page...

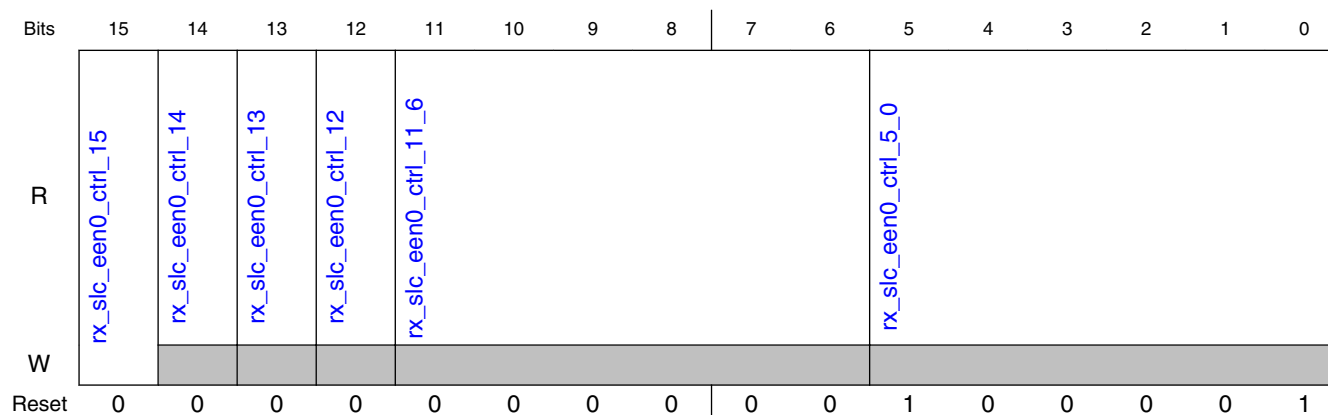
Field	Function
rx_slc_qen1_tune_15_6	
5-0 rx_slc_qen1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a two's complement value, so the calibrated code can be increased or decreased.

13.4.10.2.276 RX sampler latch E even negative 0 calibration unit control register (lane0_rx_slc_een0_ctrl - lane3_rx_slc_een0_ctrl)

13.4.10.2.276.1 Offset

Register	Offset
lane0_rx_slc_een0_ctrl	8158h
lane1_rx_slc_een0_ctrl	8558h
lane2_rx_slc_een0_ctrl	8958h
lane3_rx_slc_een0_ctrl	8D58h

13.4.10.2.276.2 Diagram



13.4.10.2.276.3 Fields

Field	Function
15 rx_slc_een0_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by

Table continues on the next page...

Clocks And Resets

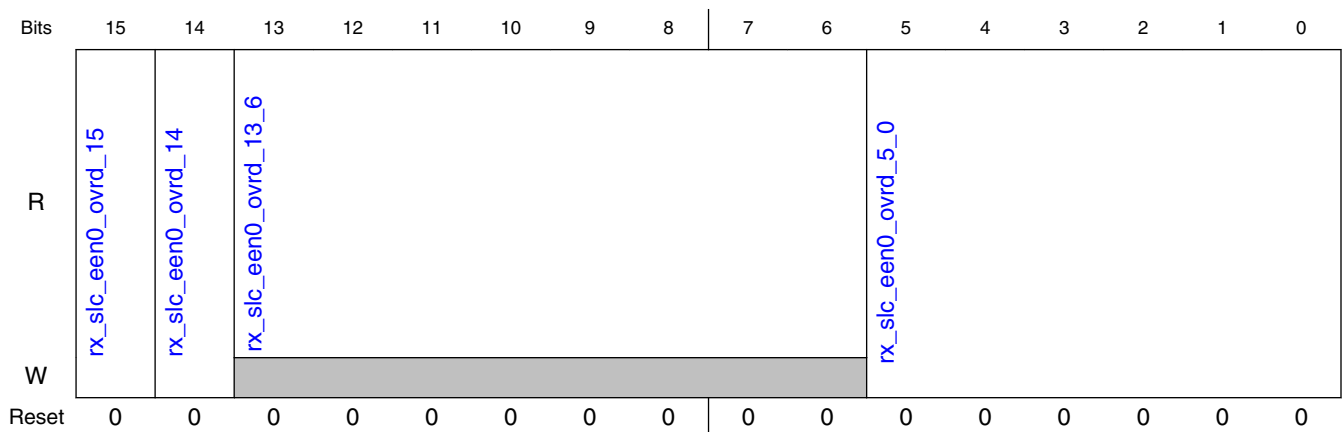
Field	Function
rx_slc_eeen0_ctrl_14	
13 rx_slc_eeen0_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_eeen0_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_eeen0_ctrl_11_6	Reserved
5-0 rx_slc_eeen0_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.277 RX sampler latch E even negative 0 calibration unit override register (lane0_rx_slc_eeen0_ovrd - lane3_rx_slc_eeen0_ovrd)

13.4.10.2.277.1 Offset

Register	Offset
lane0_rx_slc_eeen0_ovrd	8159h
lane1_rx_slc_eeen0_ovrd	8559h
lane2_rx_slc_eeen0_ovrd	8959h
lane3_rx_slc_eeen0_ovrd	8D59h

13.4.10.2.277.2 Diagram



13.4.10.2.277.3 Fields

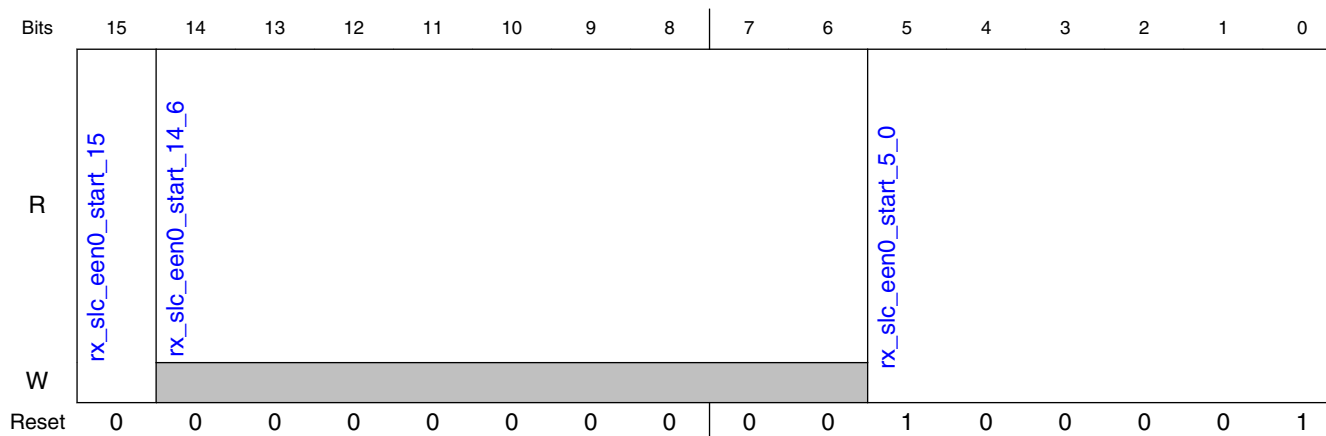
Field	Function
15 rx_slc_eeen0_ovr d_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_eeen0_ovr d_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_eeen0_ovr d_13_6	Reserved
5-0 rx_slc_eeen0_ovr d_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.278 RX sampler latch E even negative 0 calibration unit start register (lane0_rx_slc_eeen0_start - lane3_rx_slc_eeen0_start)

13.4.10.2.278.1 Offset

Register	Offset
lane0_rx_slc_eeen0_start	815Ah
lane1_rx_slc_eeen0_start	855Ah
lane2_rx_slc_eeen0_start	895Ah
lane3_rx_slc_eeen0_start	8D5Ah

13.4.10.2.278.2 Diagram



13.4.10.2.278.3 Fields

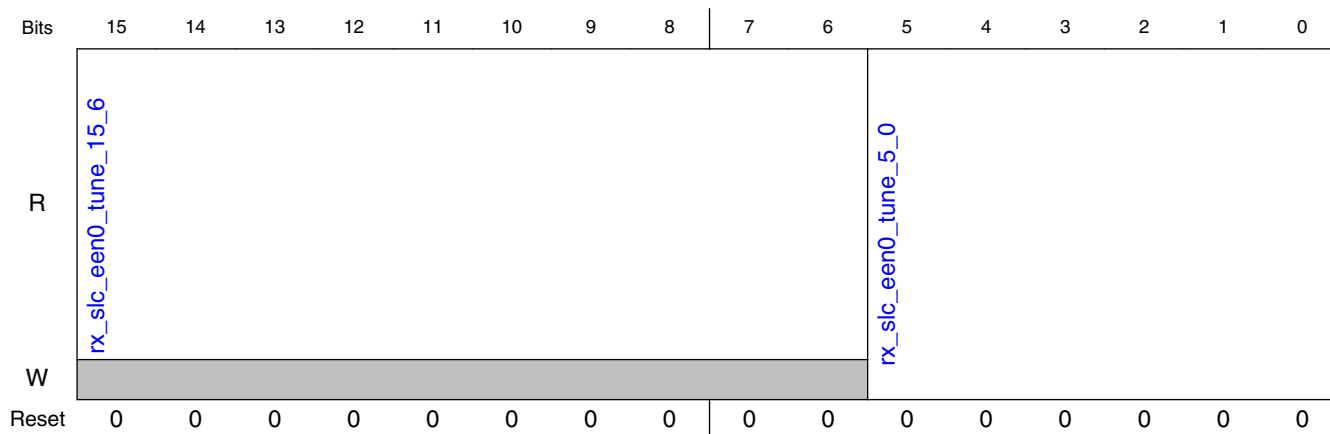
Field	Function
15 rx_slc_een0_sta rt_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_een0_sta rt_14_6	Reserved
5-0 rx_slc_een0_sta rt_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.279 RX sampler latch E even negative 0 calibration unit tune register (lane0_rx_slc_een0_tune - lane3_rx_slc_een0_tune)

13.4.10.2.279.1 Offset

Register	Offset
lane0_rx_slc_een0_tune	815Bh
lane1_rx_slc_een0_tune	855Bh
lane2_rx_slc_een0_tune	895Bh
lane3_rx_slc_een0_tune	8D5Bh

13.4.10.2.279.2 Diagram



13.4.10.2.279.3 Fields

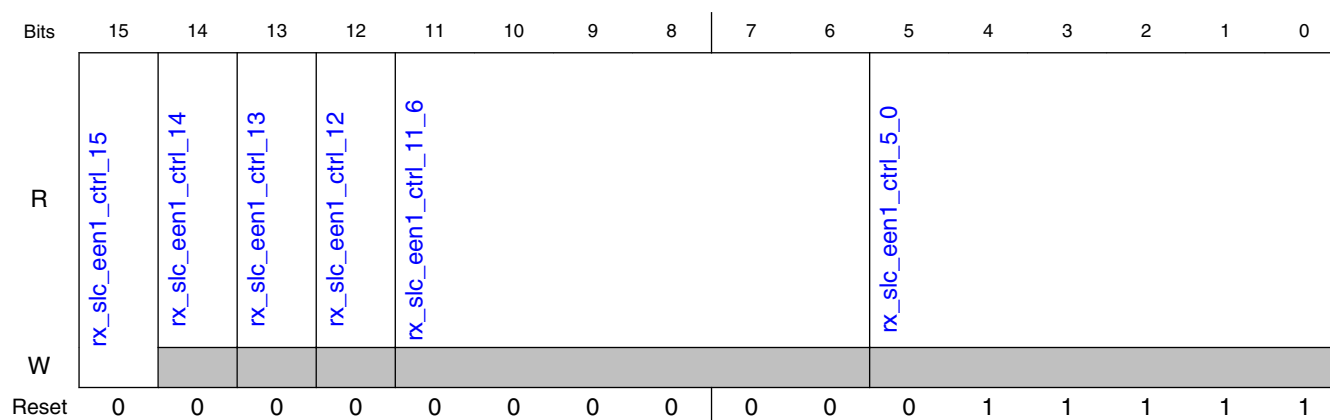
Field	Function
15-6 rx_slc_eeen0_tune_15_6	Reserved
5-0 rx_slc_eeen0_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.280 RX sampler latch E even negative 1 calibration unit control register (lane0_rx_slc_eeen1_ctrl - lane3_rx_slc_eeen1_ctrl)

13.4.10.2.280.1 Offset

Register	Offset
lane0_rx_slc_eeen1_ctrl	815Ch
lane1_rx_slc_eeen1_ctrl	855Ch
lane2_rx_slc_eeen1_ctrl	895Ch
lane3_rx_slc_eeen1_ctrl	8D5Ch

13.4.10.2.280.2 Diagram



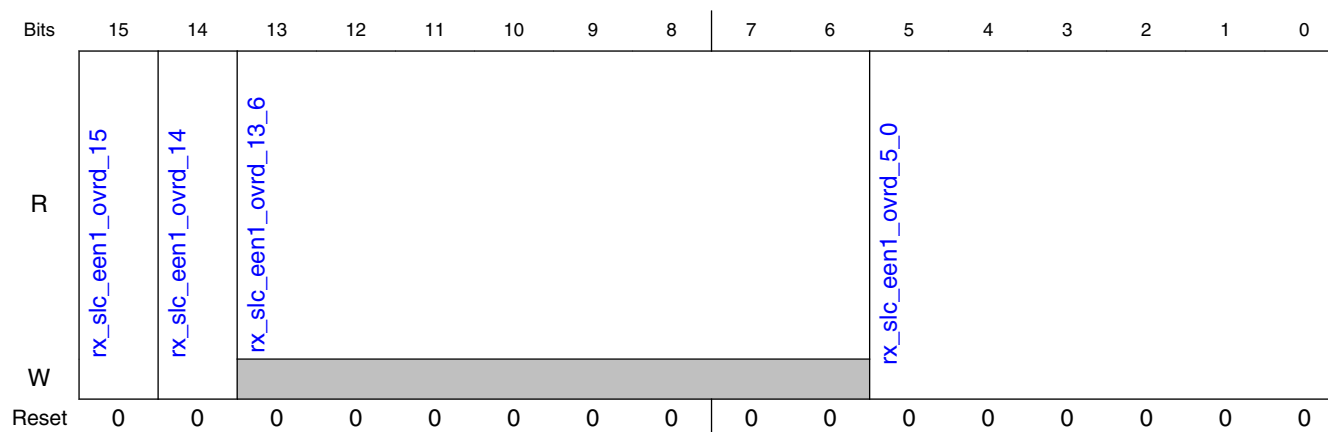
13.4.10.2.280.3 Fields

Field	Function
15 rx_slc_eeen1_ctrl_15	Start calibration: Activating (1b1) this bit will start the calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the calibration process done bit in this register is cleared.
14 rx_slc_eeen1_ctrl_14	Calibration done: This bit will be set to 1b1 when the calibration process is complete. It will be cleared by
13 rx_slc_eeen1_ctrl_13	No calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the comparator has not responded indicating that a valid calibration value has been reached.
12 rx_slc_eeen1_ctrl_12	Current comparator response: This is the current state of the comparator response signal. This signal is not synchronized, and is provided for diagnostic purposes only.
11-6 rx_slc_eeen1_ctrl_11_6	Reserved
5-0 rx_slc_eeen1_ctrl_5_0	Calibration code: This is the calibration code that was determined by the calibration process. The following are the values for the code:

13.4.10.2.281 RX sampler latch E even negative 1 calibration unit override register (lane0_rx_slc_eeen1_ovrd - lane3_rx_slc_eeen1_ovrd)**13.4.10.2.281.1 Offset**

Register	Offset
lane0_rx_slc_eeen1_ovrd	815Dh
lane1_rx_slc_eeen1_ovrd	855Dh
lane2_rx_slc_eeen1_ovrd	895Dh
lane3_rx_slc_eeen1_ovrd	8D5Dh

13.4.10.2.281.2 Diagram



13.4.10.2.281.3 Fields

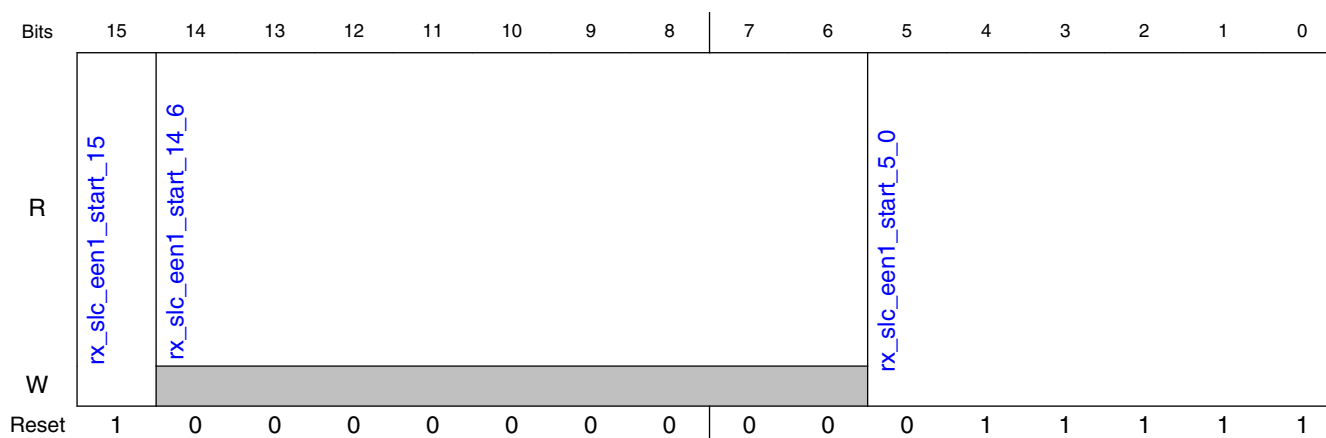
Field	Function
15 rx_slc_eeen1_ovrd_15	Calibration code override enable: Activation (1b1) of this register bit allows the calibration code determined during the automatic resistor calibration process to be overridden. The override value is specified using the calibration code override value field of this register.
14 rx_slc_eeen1_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-6 rx_slc_eeen1_ovrd_13_6	Reserved
5-0 rx_slc_eeen1_ovrd_5_0	Calibration code override value: These bits are used to override the calibration code determined during the automatic resistor calibration process. The calibration code written to these bits is valid when the calibration code override enable bit in this register is active. See the control register description for valid codes.

13.4.10.2.282 RX sampler latch E even negative 1 calibration unit start register (lane0_rx_slc_eeen1_start - lane3_rx_slc_eeen1_start)

13.4.10.2.282.1 Offset

Register	Offset
lane0_rx_slc_eeen1_start	815Eh
lane1_rx_slc_eeen1_start	855Eh
lane2_rx_slc_eeen1_start	895Eh
lane3_rx_slc_eeen1_start	8D5Eh

13.4.10.2.282.2 Diagram



13.4.10.2.282.3 Fields

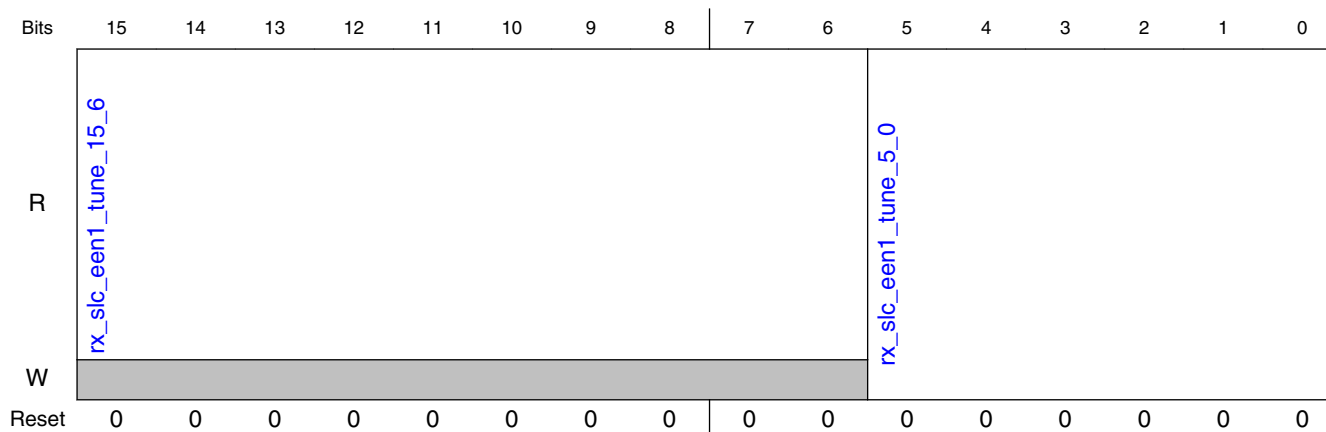
Field	Function
15 rx_slc_een1_start_15	Calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 rx_slc_een1_start_14_6	Reserved
5-0 rx_slc_een1_start_5_0	Start calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run. See the control register description for valid codes.

13.4.10.2.283 RX sampler latch E even negative 1 calibration unit tune register (lane0_rx_slc_een1_tune - lane3_rx_slc_een1_tune)

13.4.10.2.283.1 Offset

Register	Offset
lane0_rx_slc_een1_tune	815Fh
lane1_rx_slc_een1_tune	855Fh
lane2_rx_slc_een1_tune	895Fh
lane3_rx_slc_een1_tune	8D5Fh

13.4.10.2.283.2 Diagram



13.4.10.2.283.3 Fields

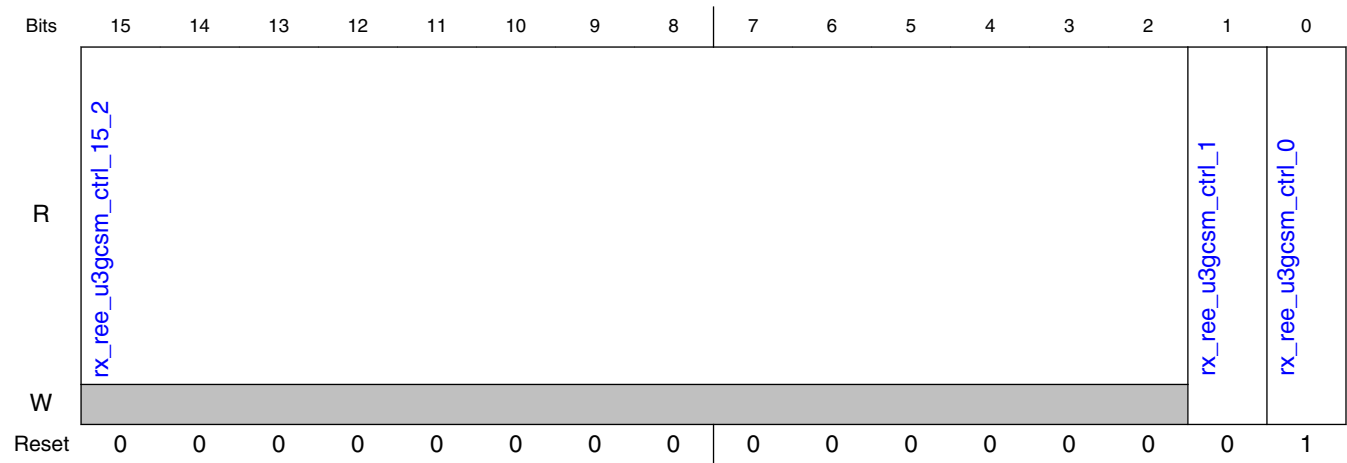
Field	Function
15-6 rx_slc_eeen1_tune_15_6	Reserved
5-0 rx_slc_eeen1_tune_5_0	Calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.284 REE USB 3 general control state machine control register (lane0_rx_ree_u3gcsn_ctrl - lane3_rx_ree_u3gcsn_ctrl)

13.4.10.2.284.1 Offset

Register	Offset
lane0_rx_ree_u3gcsn_ctrl	8160h
lane1_rx_ree_u3gcsn_ctrl	8560h
lane2_rx_ree_u3gcsn_ctrl	8960h
lane3_rx_ree_u3gcsn_ctrl	8D60h

13.4.10.2.284.2 Diagram



13.4.10.2.284.3 Fields

Field	Function
15-2 rx_ree_u3gcsm_ctrl_15_2	Reserved
1 rx_ree_u3gcsm_ctrl_1	Force run equalization: Setting this bit to a 1b1, will force the general control state machine to run, independent of the macro functions that normally run the equalization.
0 rx_ree_u3gcsm_ctrl_0	Enable: This bit enables the general control state machine function.

13.4.10.2.285 REE USB 3 general control state machine phase 1 equalization enable mask register (lane0_rx_ree_u3gcsm_eqenm_ph1 - lane3_rx_ree_u3gcsm_eqenm_ph1)

13.4.10.2.285.1 Offset

Register	Offset
lane0_rx_ree_u3gcsm_eqenm_ph1	8161h
lane1_rx_ree_u3gcsm_eqenm_ph1	8561h
lane2_rx_ree_u3gcsm_eqenm_ph1	8961h

Table continues on the next page...

Register	Offset
lane3_rx_ree_u3gcsn_eqenm_ph1	8D61h

13.4.10.2.285.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																
Reset	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1	1

13.4.10.2.285.3 Fields

Field	Function
15 rx_ree_u3gcsn_eqenm_ph1_15	Reserved
14 rx_ree_u3gcsn_eqenm_ph1_14	Ignore 1010 controller : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
13 rx_ree_u3gcsn_eqenm_ph1_13	Reserved
12 rx_ree_u3gcsn_eqenm_ph1_12	Reserved
11 rx_ree_u3gcsn_eqenm_ph1_11	Reserved
10 rx_ree_u3gcsn_eqenm_ph1_10	Reserved

Table continues on the next page...

Field	Function
9 rx_ree_u3gcsn_ eqenm_ph1_9	RX attenuation : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
8 rx_ree_u3gcsn_ eqenm_ph1_8	RX VGA gain : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
7 rx_ree_u3gcsn_ eqenm_ph1_7	RX offset correction coefficient : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
6 rx_ree_u3gcsn_ eqenm_ph1_6	RX peaking amp gain : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
5 rx_ree_u3gcsn_ eqenm_ph1_5	RX low frequency equalizer adaptive control : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
4 rx_ree_u3gcsn_ eqenm_ph1_4	Reserved
3 rx_ree_u3gcsn_ eqenm_ph1_3	Reserved
2 rx_ree_u3gcsn_ eqenm_ph1_2	RX tap 3 : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
1 rx_ree_u3gcsn_ eqenm_ph1_1	RX tap 2 : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
0 rx_ree_u3gcsn_ eqenm_ph1_0	RX tap 1 : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.

13.4.10.2.286 REE USB 3 general control state machine phase 2 equalization enable mask register (lane0_rx_ree_u3gcsn_eqenm_ph2 - lane3_rx_ree_u3gcsn_eqenm_ph2)

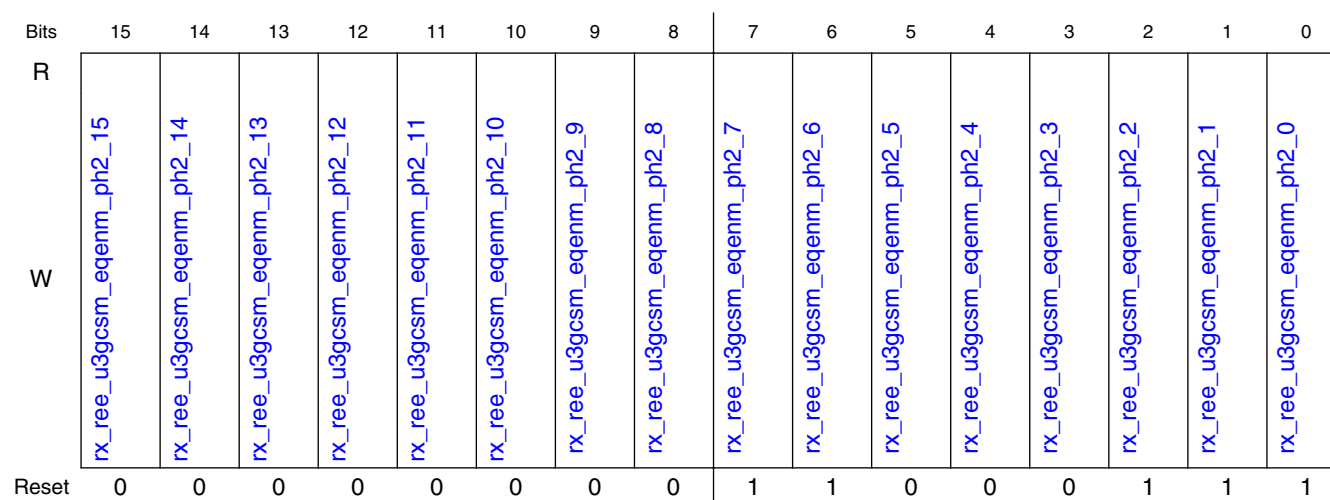
13.4.10.2.286.1 Offset

Register	Offset
lane0_rx_ree_u3gcsn_eqenm_ph2	8162h
lane1_rx_ree_u3gcsn_eqenm_ph2	8562h

Table continues on the next page...

Register	Offset
lane2_rx_ree_u3gcsn_eqenm_ph2	8962h
lane3_rx_ree_u3gcsn_eqenm_ph2	8D62h

13.4.10.2.286.2 Diagram



13.4.10.2.286.3 Fields

Field	Function
15 rx_ree_u3gcsn_eqenm_ph2_15	Reserved
14 rx_ree_u3gcsn_eqenm_ph2_14	Ignore 1010 controller : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
13 rx_ree_u3gcsn_eqenm_ph2_13	Reserved
12 rx_ree_u3gcsn_eqenm_ph2_12	Reserved
11 rx_ree_u3gcsn_eqenm_ph2_11	Reserved
10	Reserved

Table continues on the next page...

Clocks And Resets

Field	Function
rx_ree_u3gcsn_eqnph2_10	
9 rx_ree_u3gcsn_eqnph2_9	RX attenuation : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
8 rx_ree_u3gcsn_eqnph2_8	RX VGA gain : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
7 rx_ree_u3gcsn_eqnph2_7	RX offset correction coefficient : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
6 rx_ree_u3gcsn_eqnph2_6	RX peaking amp gain : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
5 rx_ree_u3gcsn_eqnph2_5	RX low frequency equalizer adaptive control : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
4 rx_ree_u3gcsn_eqnph2_4	Reserved
3 rx_ree_u3gcsn_eqnph2_3	Reserved
2 rx_ree_u3gcsn_eqnph2_2	RX tap 3 : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
1 rx_ree_u3gcsn_eqnph2_1	RX tap 2 : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.
0 rx_ree_u3gcsn_eqnph2_0	RX tap 1 : When set to 1b1, this function is enabled when the USB 3.0 general control state machine is controlling the REE.

13.4.10.2.287 REE USB 3 general control state machine start timer value register (lane0_rx_ree_u3gcsn_start_tmr - lane3_rx_ree_u3gcsn_start_tmr)

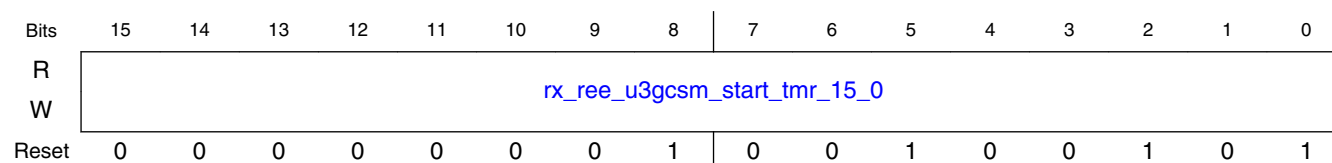
13.4.10.2.287.1 Offset

Register	Offset
lane0_rx_ree_u3gcsn_start_tmr	8163h

Table continues on the next page...

Register	Offset
lane1_rx_ree_u3gcsn_start_tmr	8563h
lane2_rx_ree_u3gcsn_start_tmr	8963h
lane3_rx_ree_u3gcsn_start_tmr	8D63h

13.4.10.2.287.2 Diagram



13.4.10.2.287.3 Fields

Field	Function
15-0 rx_ree_u3gcsn_start_tmr_15_0	Start timer value : The number of clock cycles the state machine will wait in the Start Delay state.

13.4.10.2.288 REE USB 3 general control state machine run phase 1 timer value register (lane0_rx_ree_u3gcsn_run_ph1_tmr - lane3_rx_ree_u3gcsn_run_ph1_tmr)

13.4.10.2.288.1 Offset

Register	Offset
lane0_rx_ree_u3gcsn_run_ph1_tmr	8164h
lane1_rx_ree_u3gcsn_run_ph1_tmr	8564h
lane2_rx_ree_u3gcsn_run_ph1_tmr	8964h
lane3_rx_ree_u3gcsn_run_ph1_tmr	8D64h

13.4.10.2.288.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	rx_ree_u3gcsn_run_ph1_tmr_15_0															
W																
Reset	0	0	0	0	0	1	1	1	1	0	1	0	0	0	1	0

13.4.10.2.288.3 Fields

Field	Function
15-0 rx_ree_u3gcsn_run_ph1_tmr_15_0	Run phase 1 timer value : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 1 state.

13.4.10.2.289 REE USB 3 general control state machine run phase 2 timer value register (lane0_rx_ree_u3gcsn_run_ph2_tmr - lane3_rx_ree_u3gcsn_run_ph2_tmr)

13.4.10.2.289.1 Offset

Register	Offset
lane0_rx_ree_u3gcsn_run_ph2_tmr	8165h
lane1_rx_ree_u3gcsn_run_ph2_tmr	8565h
lane2_rx_ree_u3gcsn_run_ph2_tmr	8965h
lane3_rx_ree_u3gcsn_run_ph2_tmr	8D65h

13.4.10.2.289.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	rx_ree_u3gcsn_run_ph2_tmr_15_0															
W																
Reset	0	0	0	0	0	0	0	0	1	0	0	1	1	1	0	1

13.4.10.2.289.3 Fields

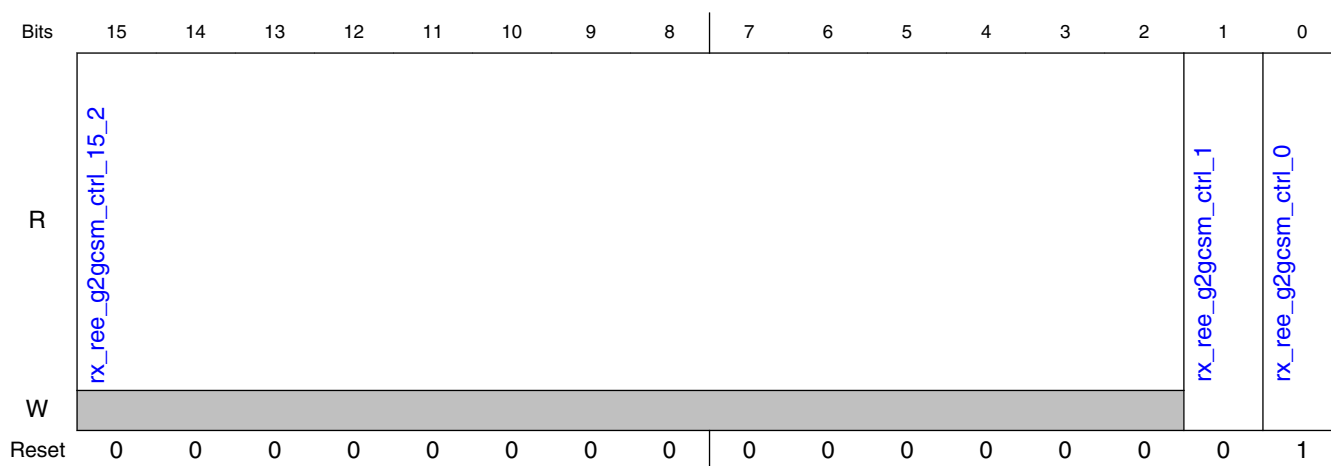
Field	Function
15-0 rx_ree_u3gcsn_ run_ph2_tmr_15 _0	Run phase 2 timer value : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 2 state.

13.4.10.2.290 REE PCIe Gen 2 general control state machine control register (lane0_rx_ree_g2gcsn_ctrl - lane3_rx_ree_g2gcsn_ctrl)

13.4.10.2.290.1 Offset

Register	Offset
lane0_rx_ree_g2gcsn_ctrl	8168h
lane1_rx_ree_g2gcsn_ctrl	8568h
lane2_rx_ree_g2gcsn_ctrl	8968h
lane3_rx_ree_g2gcsn_ctrl	8D68h

13.4.10.2.290.2 Diagram



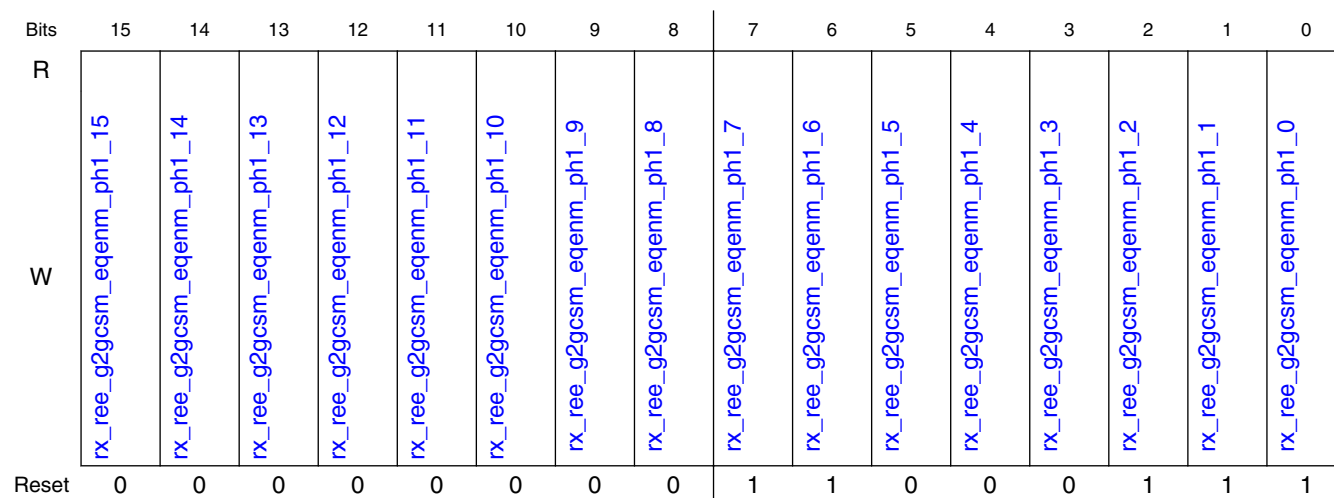
13.4.10.2.290.3 Fields

Field	Function
15-2 rx_ree_g2gcsn_ ctrl_15_2	Reserved
1 rx_ree_g2gcsn_ ctrl_1	Force run equalization: Setting this bit to a 1b1, will force the general control state machine to run, independent of the macro functions that normally run the equalization.
0 rx_ree_g2gcsn_ ctrl_0	Enable: This bit enables the general control state machine function.

13.4.10.2.291 REE PCIe Gen 2 general control state machine phase 1 equalization enable mask register (lane0_rx_ree_g2gcsn_eqenm_ph1 - lane3_rx_ree_g2gcsn_eqenm_ph1)
13.4.10.2.291.1 Offset

Register	Offset
lane0_rx_ree_g2gcsn_eqenm_ph1	8169h
lane1_rx_ree_g2gcsn_eqenm_ph1	8569h
lane2_rx_ree_g2gcsn_eqenm_ph1	8969h
lane3_rx_ree_g2gcsn_eqenm_ph1	8D69h

13.4.10.2.291.2 Diagram



13.4.10.2.291.3 Fields

Field	Function
15 rx_ree_g2gcsn_eqenm_ph1_15	Reserved
14 rx_ree_g2gcsn_eqenm_ph1_14	Ignore 1010 controller : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
13 rx_ree_g2gcsn_eqenm_ph1_13	Reserved
12 rx_ree_g2gcsn_eqenm_ph1_12	Reserved
11 rx_ree_g2gcsn_eqenm_ph1_11	Reserved
10 rx_ree_g2gcsn_eqenm_ph1_10	Reserved
9 rx_ree_g2gcsn_eqenm_ph1_9	RX attenuation : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
8 rx_ree_g2gcsn_eqenm_ph1_8	RX VGA gain : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.

Table continues on the next page...

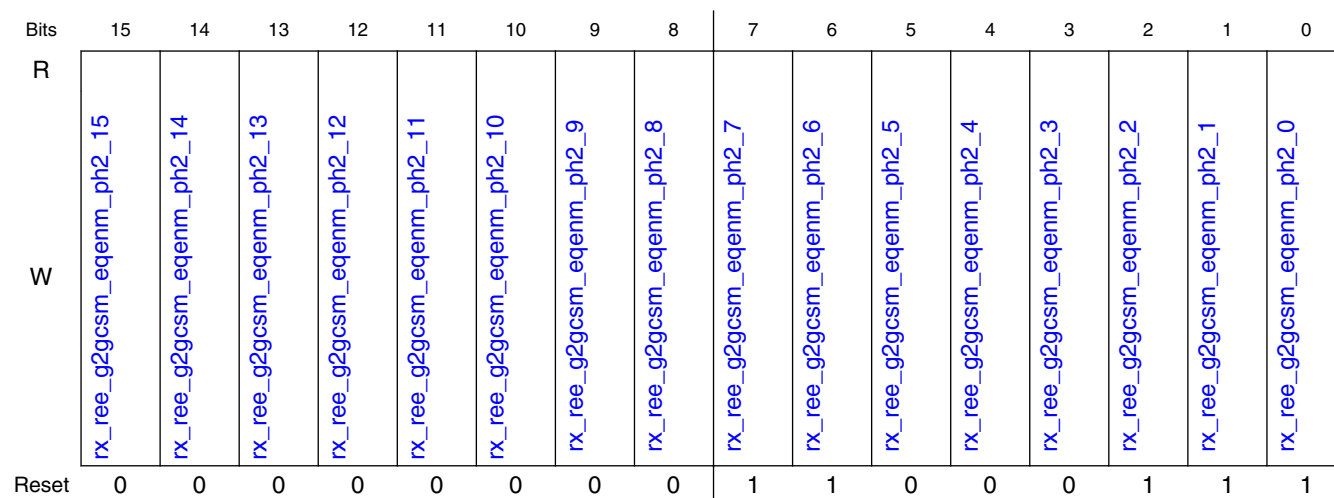
Field	Function
7 rx_ree_g2gcsn_eqnph1_7	RX offset correction coefficient : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
6 rx_ree_g2gcsn_eqnph1_6	RX peaking amp gain : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
5 rx_ree_g2gcsn_eqnph1_5	RX low frequency equalizer adaptive control : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
4 rx_ree_g2gcsn_eqnph1_4	Reserved
3 rx_ree_g2gcsn_eqnph1_3	Reserved
2 rx_ree_g2gcsn_eqnph1_2	RX tap 3 : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
1 rx_ree_g2gcsn_eqnph1_1	RX tap 2 : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
0 rx_ree_g2gcsn_eqnph1_0	RX tap 1 : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.

13.4.10.2.292 REE PCIe Gen 2 general control state machine phase 2 equalization enable mask register (lane0_rx_ree_g2gcsn_eqnph2 - lane3_rx_ree_g2gcsn_eqnph2)

13.4.10.2.292.1 Offset

Register	Offset
lane0_rx_ree_g2gcsn_eqnph2	816Ah
lane1_rx_ree_g2gcsn_eqnph2	856Ah
lane2_rx_ree_g2gcsn_eqnph2	896Ah
lane3_rx_ree_g2gcsn_eqnph2	8D6Ah

13.4.10.2.292.2 Diagram



13.4.10.2.292.3 Fields

Field	Function
15 rx_ree_g2gcsn_eqenm_ph2_15	Reserved
14 rx_ree_g2gcsn_eqenm_ph2_14	Ignore 1010 controller : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
13 rx_ree_g2gcsn_eqenm_ph2_13	Reserved
12 rx_ree_g2gcsn_eqenm_ph2_12	Reserved
11 rx_ree_g2gcsn_eqenm_ph2_11	Reserved
10 rx_ree_g2gcsn_eqenm_ph2_10	Reserved
9 rx_ree_g2gcsn_eqenm_ph2_9	RX attenuation : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
8 rx_ree_g2gcsn_eqenm_ph2_8	RX VGA gain : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.

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Clocks And Resets

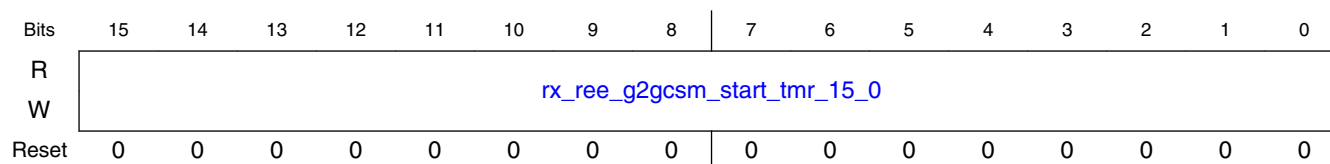
Field	Function
7 rx_ree_g2gcsn_eqenm_ph2_7	RX offset correction coefficient : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
6 rx_ree_g2gcsn_eqenm_ph2_6	RX peaking amp gain : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
5 rx_ree_g2gcsn_eqenm_ph2_5	RX low frequency equalizer adaptive control : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
4 rx_ree_g2gcsn_eqenm_ph2_4	Reserved
3 rx_ree_g2gcsn_eqenm_ph2_3	Reserved
2 rx_ree_g2gcsn_eqenm_ph2_2	RX tap 3 : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
1 rx_ree_g2gcsn_eqenm_ph2_1	RX tap 2 : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.
0 rx_ree_g2gcsn_eqenm_ph2_0	RX tap 1 : When set to 1b1, this function is enabled when the PCIe Gen 2 general control state machine is controlling the REE.

13.4.10.2.293 REE PCIe Gen 2 general control state machine start timer value register (lane0_rx_ree_g2gcsn_start_tmr - lane3_rx_ree_g2gcsn_start_tmr)

13.4.10.2.293.1 Offset

Register	Offset
lane0_rx_ree_g2gcsn_start_tmr	816Bh
lane1_rx_ree_g2gcsn_start_tmr	856Bh
lane2_rx_ree_g2gcsn_start_tmr	896Bh
lane3_rx_ree_g2gcsn_start_tmr	8D6Bh

13.4.10.2.293.2 Diagram



13.4.10.2.293.3 Fields

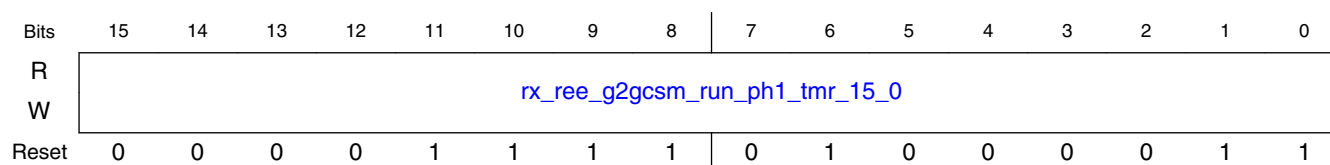
Field	Function
15-0 rx_ree_g2gcsn_start_tmr_15_0	Start timer value : The number of clock cycles the state machine will wait in the Start Delay state.

13.4.10.2.294 REE PCIe Gen 2 general control state machine run phase 1 timer value register (lane0_rx_ree_g2gcsn_run_ph1_tmr - lane3_rx_ree_g2gcsn_run_ph1_tmr)

13.4.10.2.294.1 Offset

Register	Offset
lane0_rx_ree_g2gcsn_run_ph1_tmr	816Ch
lane1_rx_ree_g2gcsn_run_ph1_tmr	856Ch
lane2_rx_ree_g2gcsn_run_ph1_tmr	896Ch
lane3_rx_ree_g2gcsn_run_ph1_tmr	8D6Ch

13.4.10.2.294.2 Diagram



13.4.10.2.294.3 Fields

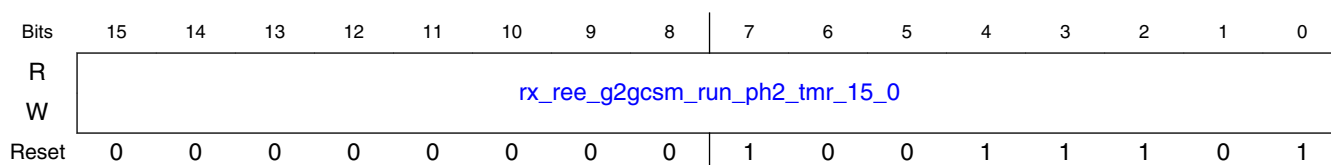
Field	Function
15-0 rx_ree_g2gcsn_run_ph1_tmr_15_0	Run phase 1 timer value : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 1 state.

13.4.10.2.295 REE PCIe Gen 2 general control state machine run phase 2 timer value register (lane0_rx_ree_g2gcsn_run_ph2_tmr - lane3_rx_ree_g2gcsn_run_ph2_tmr)

13.4.10.2.295.1 Offset

Register	Offset
lane0_rx_ree_g2gcsn_run_ph2_tmr	816Dh
lane1_rx_ree_g2gcsn_run_ph2_tmr	856Dh
lane2_rx_ree_g2gcsn_run_ph2_tmr	896Dh
lane3_rx_ree_g2gcsn_run_ph2_tmr	8D6Dh

13.4.10.2.295.2 Diagram



13.4.10.2.295.3 Fields

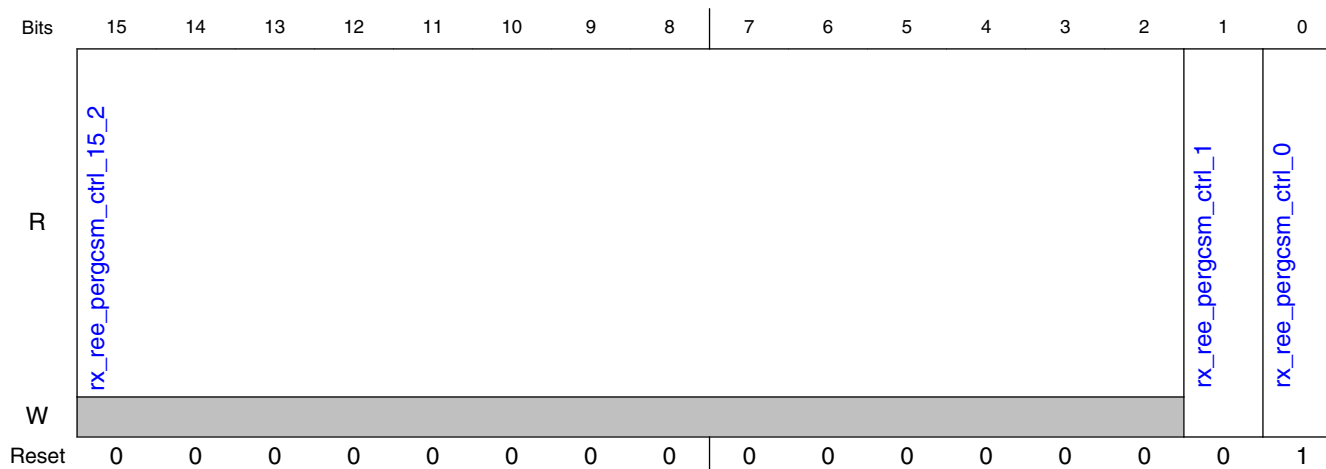
Field	Function
15-0 rx_ree_g2gcsn_run_ph2_tmr_15_0	Run phase 2 timer value : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 2 state.

13.4.10.2.296 REE periodic general control state machine control register (lane0_rx_ree_pergcsm_ctrl - lane3_rx_ree_pergcsm_ctrl)

13.4.10.2.296.1 Offset

Register	Offset
lane0_rx_ree_pergcsm_ctrl	8178h
lane1_rx_ree_pergcsm_ctrl	8578h
lane2_rx_ree_pergcsm_ctrl	8978h
lane3_rx_ree_pergcsm_ctrl	8D78h

13.4.10.2.296.2 Diagram



13.4.10.2.296.3 Fields

Field	Function
15-2 rx_ree_pergcsm_ctrl_15_2	Reserved
1 rx_ree_pergcsm_ctrl_1	Force run equalization: Setting this bit to a 1b1, will force the general control state machine to run, independent of the macro functions that normally run the equalization.
0 rx_ree_pergcsm_ctrl_0	Enable: This bit enables the general control state machine function.

13.4.10.2.297 REE periodic general control state machine phase 1 equalization enable mask register (lane0_rx_ree_pergcsm_eqenm_ph1 - lane3_rx_ree_pergcsm_eqenm_ph1)

13.4.10.2.297.1 Offset

Register	Offset
lane0_rx_ree_pergcsm_eqenm_ph1	8179h
lane1_rx_ree_pergcsm_eqenm_ph1	8579h
lane2_rx_ree_pergcsm_eqenm_ph1	8979h
lane3_rx_ree_pergcsm_eqenm_ph1	8D79h

13.4.10.2.297.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W	rx_ree_pergcsm_eqenm_ph1_15	rx_ree_pergcsm_eqenm_ph1_14	rx_ree_pergcsm_eqenm_ph1_13	rx_ree_pergcsm_eqenm_ph1_12	rx_ree_pergcsm_eqenm_ph1_11	rx_ree_pergcsm_eqenm_ph1_10	rx_ree_pergcsm_eqenm_ph1_9	rx_ree_pergcsm_eqenm_ph1_8	rx_ree_pergcsm_eqenm_ph1_7	rx_ree_pergcsm_eqenm_ph1_6	rx_ree_pergcsm_eqenm_ph1_5	rx_ree_pergcsm_eqenm_ph1_4	rx_ree_pergcsm_eqenm_ph1_3	rx_ree_pergcsm_eqenm_ph1_2	rx_ree_pergcsm_eqenm_ph1_1	rx_ree_pergcsm_eqenm_ph1_0
Reset	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

13.4.10.2.297.3 Fields

Field	Function
15 rx_ree_pergcsm_eqenm_ph1_15	Reserved
14 rx_ree_pergcsm_eqenm_ph1_14	
13 rx_ree_pergcsm_eqenm_ph1_13	
12 rx_ree_pergcsm_eqenm_ph1_12	
11 rx_ree_pergcsm_eqenm_ph1_11	

Table continues on the next page...

Field	Function
14 rx_ree_pergcsm _eqenm_ph1_1 4	Ignore 1010 controller : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
13 rx_ree_pergcsm _eqenm_ph1_1 3	Reserved
12 rx_ree_pergcsm _eqenm_ph1_1 2	Reserved
11 rx_ree_pergcsm _eqenm_ph1_1 1	Reserved
10 rx_ree_pergcsm _eqenm_ph1_1 0	Reserved
9 rx_ree_pergcsm _eqenm_ph1_9	RX attenuation : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
8 rx_ree_pergcsm _eqenm_ph1_8	RX VGA gain : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
7 rx_ree_pergcsm _eqenm_ph1_7	RX offset correction coefficient : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
6 rx_ree_pergcsm _eqenm_ph1_6	RX peaking amp gain : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
5 rx_ree_pergcsm _eqenm_ph1_5	RX low frequency equalizer adaptive control : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
4 rx_ree_pergcsm _eqenm_ph1_4	Reserved
3 rx_ree_pergcsm _eqenm_ph1_3	Reserved
2 rx_ree_pergcsm _eqenm_ph1_2	RX tap 3 : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.

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Clocks And Resets

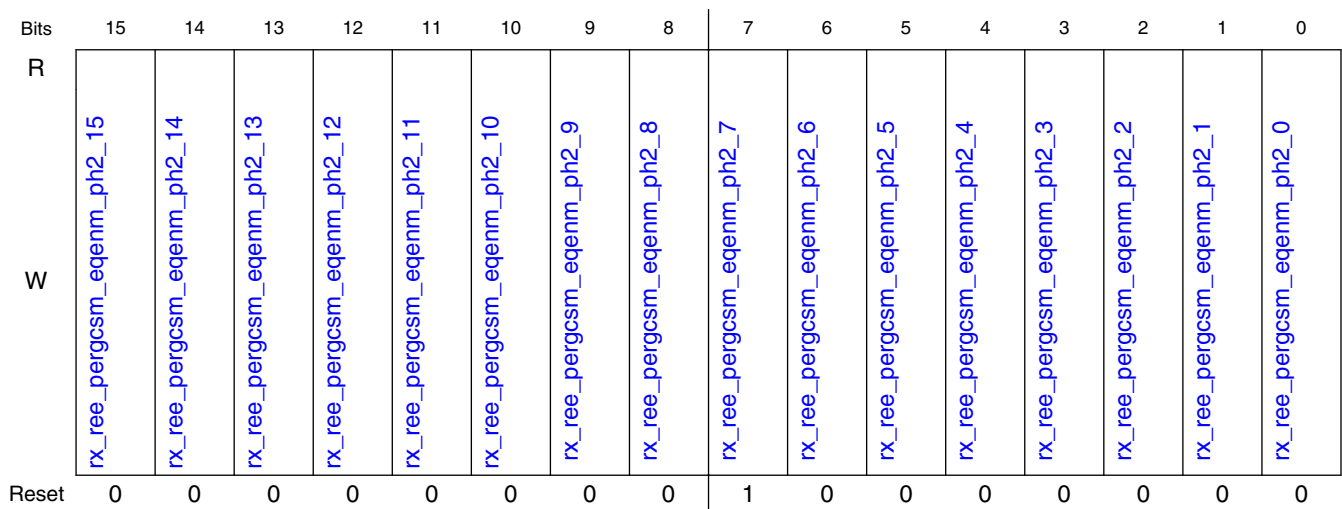
Field	Function
1 rx_ree_pergcsm_eqenm_ph1_1	RX tap 2 : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
0 rx_ree_pergcsm_eqenm_ph1_0	RX tap 1 : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.

13.4.10.2.298 REE periodic general control state machine phase 2 equalization enable mask register (lane0_rx_ree_pergcsm_eqenm_ph2 - lane3_rx_ree_pergcsm_eqenm_ph2)

13.4.10.2.298.1 Offset

Register	Offset
lane0_rx_ree_pergcsm_eqenm_ph2	817Ah
lane1_rx_ree_pergcsm_eqenm_ph2	857Ah
lane2_rx_ree_pergcsm_eqenm_ph2	897Ah
lane3_rx_ree_pergcsm_eqenm_ph2	8D7Ah

13.4.10.2.298.2 Diagram



13.4.10.2.298.3 Fields

Field	Function
15 rx_ree_pergcsm _eqenm_ph2_1 5	Reserved
14 rx_ree_pergcsm _eqenm_ph2_1 4	Ignore 1010 controller : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
13 rx_ree_pergcsm _eqenm_ph2_1 3	Reserved
12 rx_ree_pergcsm _eqenm_ph2_1 2	Reserved
11 rx_ree_pergcsm _eqenm_ph2_1 1	Reserved
10 rx_ree_pergcsm _eqenm_ph2_1 0	Reserved
9 rx_ree_pergcsm _eqenm_ph2_9	RX attenuation : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
8 rx_ree_pergcsm _eqenm_ph2_8	RX VGA gain : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
7 rx_ree_pergcsm _eqenm_ph2_7	RX offset correction coefficient : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
6 rx_ree_pergcsm _eqenm_ph2_6	RX peaking amp gain : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
5 rx_ree_pergcsm _eqenm_ph2_5	RX low frequency equalizer adaptive control : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
4 rx_ree_pergcsm _eqenm_ph2_4	Reserved
3	Reserved

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Clocks And Resets

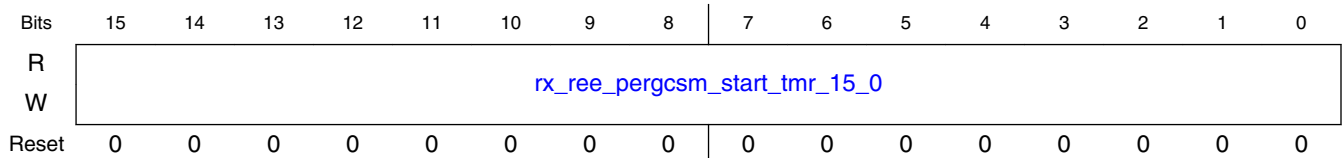
Field	Function
rx_ree_pergcsm_eqenm_ph2_3	
2 rx_ree_pergcsm_eqenm_ph2_2	RX tap 3 : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
1 rx_ree_pergcsm_eqenm_ph2_1	RX tap 2 : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.
0 rx_ree_pergcsm_eqenm_ph2_0	RX tap 1 : When set to 1b1, this function is enabled when the periodic general control state machine is controlling the REE.

13.4.10.2.299 REE periodic general control state machine start timer value register (lane0_rx_ree_pergcsm_start_tmr - lane3_rx_ree_pergcsm_start_tmr)

13.4.10.2.299.1 Offset

Register	Offset
lane0_rx_ree_pergcsm_start_tmr	817Bh
lane1_rx_ree_pergcsm_start_tmr	857Bh
lane2_rx_ree_pergcsm_start_tmr	897Bh
lane3_rx_ree_pergcsm_start_tmr	8D7Bh

13.4.10.2.299.2 Diagram



13.4.10.2.299.3 Fields

Field	Function
15-0	Start timer value : The number of clock cycles the state machine will wait in the Start Delay state.

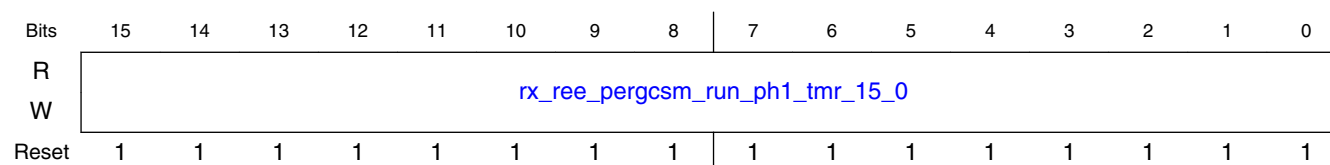
Field	Function
rx_ree_pergcsm_start_tmr_15_0	

13.4.10.2.300 REE periodic general control state machine run phase 1 timer value register (lane0_rx_ree_pergcsm_run_ph1_tmr - lane3_rx_ree_pergcsm_run_ph1_tmr)

13.4.10.2.300.1 Offset

Register	Offset
lane0_rx_ree_pergcsm_run_ph1_tmr	817Ch
lane1_rx_ree_pergcsm_run_ph1_tmr	857Ch
lane2_rx_ree_pergcsm_run_ph1_tmr	897Ch
lane3_rx_ree_pergcsm_run_ph1_tmr	8D7Ch

13.4.10.2.300.2 Diagram



13.4.10.2.300.3 Fields

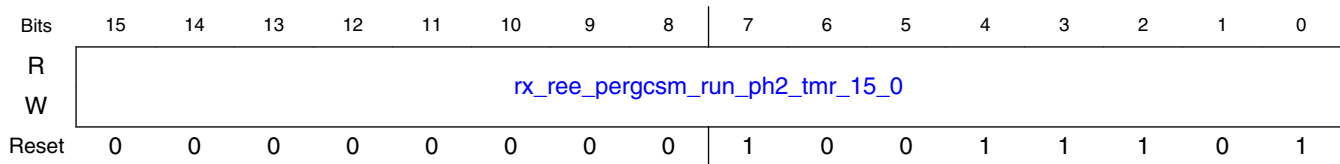
Field	Function
15-0 rx_ree_pergcsm_run_ph1_tmr_15_0	Run phase 1 timer value : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 1 state.

13.4.10.2.301 REE periodic general control state machine run phase 2 timer value register (lane0_rx_ree_pergcsm_run_ph2_tmr - lane3_rx_ree_pergcsm_run_ph2_tmr)

13.4.10.2.301.1 Offset

Register	Offset
lane0_rx_ree_pergcsm_run_ph2_tmr	817Dh
lane1_rx_ree_pergcsm_run_ph2_tmr	857Dh
lane2_rx_ree_pergcsm_run_ph2_tmr	897Dh
lane3_rx_ree_pergcsm_run_ph2_tmr	8D7Dh

13.4.10.2.301.2 Diagram



13.4.10.2.301.3 Fields

Field	Function
15-0 rx_ree_pergcsm_run_ph2_tmr_15_0	Run phase 2 timer value : This specifies the number of clock cycles the state machine will wait in the Run Equalization Phase 2 state.

13.4.10.2.302 REE tap 1 control register (lane0_rx_ree_tap1_ctrl - lane3_rx_ree_tap1_ctrl)

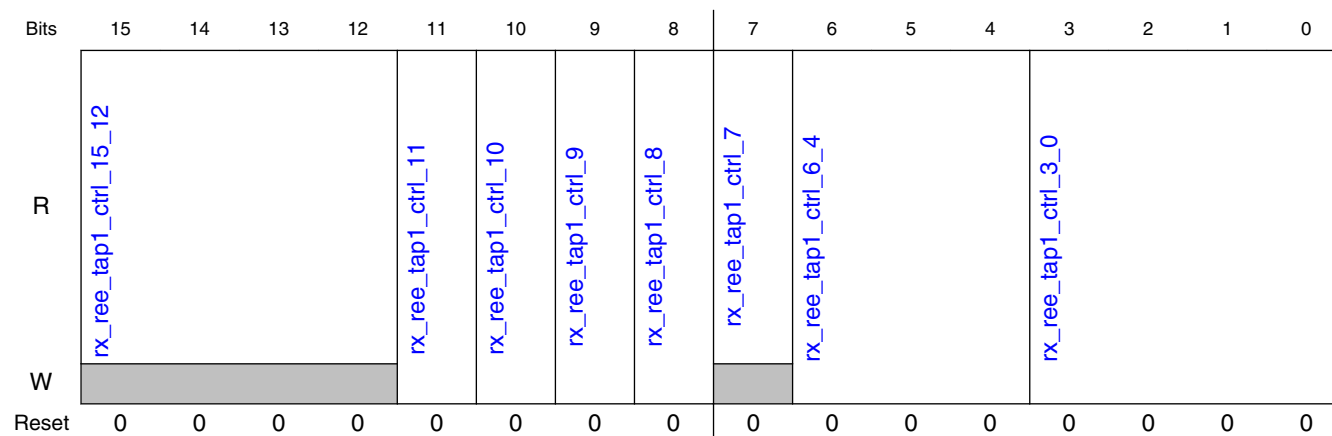
13.4.10.2.302.1 Offset

Register	Offset
lane0_rx_ree_tap1_ctrl	8180h
lane1_rx_ree_tap1_ctrl	8580h

Table continues on the next page...

Register	Offset
lane2_rx_ree_tap1_ctrl	8980h
lane3_rx_ree_tap1_ctrl	8D80h

13.4.10.2.302.2 Diagram



13.4.10.2.302.3 Fields

Field	Function
15-12 rx_ree_tap1_ctrl_15_12	Reserved
11 rx_ree_tap1_ctrl_11	Tap coefficient combinational logic zero crossing enable:
10 rx_ree_tap1_ctrl_10	Tap coefficient combinational logic non zero crossing enable:
9 rx_ree_tap1_ctrl_9	Tap coefficient combinational logic bit 0 only enable:
8 rx_ree_tap1_ctrl_8	Receiver DFE tap coefficient disable: This bit disables the
7 rx_ree_tap1_ctrl_7	Reserved
6-4	Tap integrator accumulator scaler value: Specifies the amount to scale the input to the tap integrator accumulator by. The following are the valid settings for this field:

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Clocks And Resets

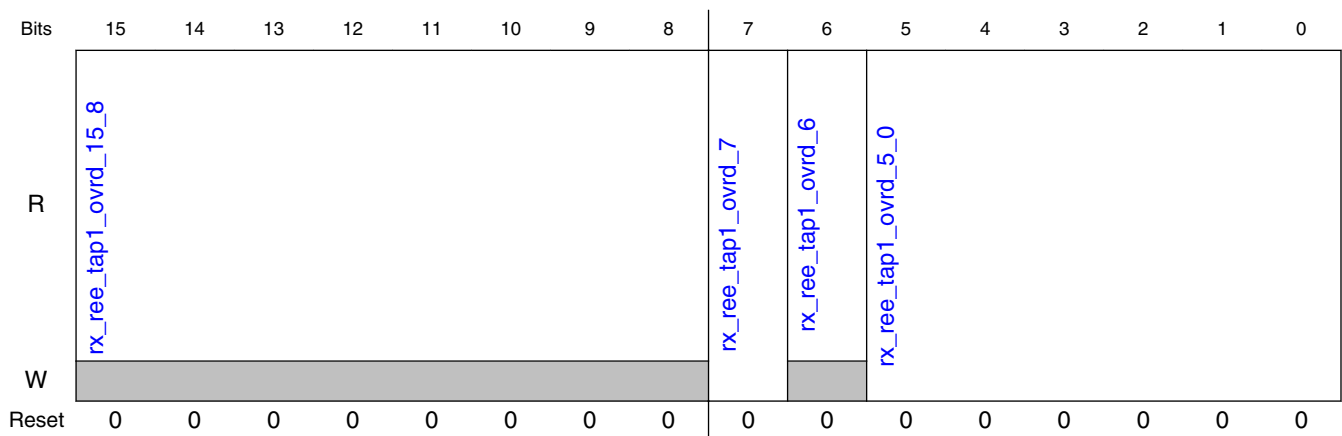
Field	Function
rx_ree_tap1_ctrl_6_4	
3-0 rx_ree_tap1_ctrl_3_0	Tap sigma delta accumulator scaler value: Specifies the amount to scale the input to the tap sigma delta accumulator by. The following are the valid settings for this field:

13.4.10.2.303 REE tap 1 override register (lane0_rx_ree_tap1_ovrd - lane3_rx_ree_tap1_ovrd)

13.4.10.2.303.1 Offset

Register	Offset
lane0_rx_ree_tap1_ovrd	8181h
lane1_rx_ree_tap1_ovrd	8581h
lane2_rx_ree_tap1_ovrd	8981h
lane3_rx_ree_tap1_ovrd	8D81h

13.4.10.2.303.2 Diagram



13.4.10.2.303.3 Fields

Field	Function
15-8 rx_ree_tap1_ovrd_15_8	Reserved

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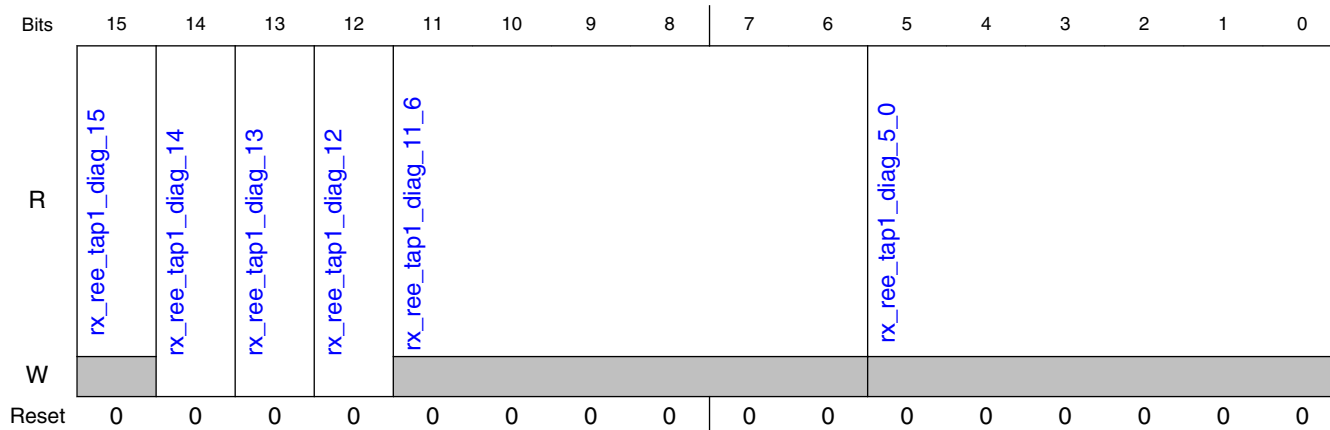
Field	Function
7 rx_ree_tap1_ovr d_7	Tap override enable: Setting this bit to a 1b1 will enable the tap override field in this register to override the tap integrator accumulator functions.
6 rx_ree_tap1_ovr d_6	Reserved
5-0 rx_ree_tap1_ovr d_5_0	Tap override value: When the tap override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

13.4.10.2.304 REE tap 1 diagnostics register (lane0_rx_ree_tap1_diag - lane3_rx_ree_tap1_diag)

13.4.10.2.304.1 Offset

Register	Offset
lane0_rx_ree_tap1_diag	8182h
lane1_rx_ree_tap1_diag	8582h
lane2_rx_ree_tap1_diag	8982h
lane3_rx_ree_tap1_diag	8D82h

13.4.10.2.304.2 Diagram



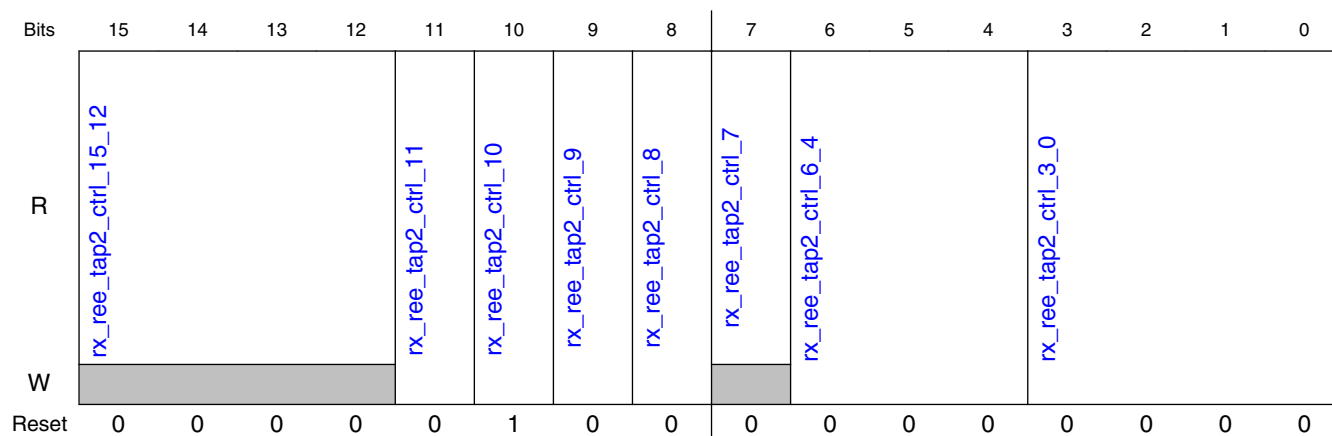
13.4.10.2.304.3 Fields

Field	Function
15 rx_ree_tap1_dia g_15	Reserved
14 rx_ree_tap1_dia g_14	Voter override neg : Writing a 1b1 in this register bit will force the tap voter function to activate the voter neg signal for a single clock cycle.
13 rx_ree_tap1_dia g_13	Voter override pos : Writing a 1b1 in this register bit will force the tap voter function to activate the voter pos signal for a single clock cycle.
12 rx_ree_tap1_dia g_12	Voter override enable : Setting this bit to a 1b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the tap.
11-6 rx_ree_tap1_dia g_11_6	Reserved
5-0 rx_ree_tap1_dia g_5_0	Current tap integrator accumulator: Current value of the tap integrator accumulator.

13.4.10.2.305 REE tap 2 control register (lane0_rx_ree_tap2_ctrl - lane3_rx_ree_tap2_ctrl)**13.4.10.2.305.1 Offset**

Register	Offset
lane0_rx_ree_tap2_ctrl	8184h
lane1_rx_ree_tap2_ctrl	8584h
lane2_rx_ree_tap2_ctrl	8984h
lane3_rx_ree_tap2_ctrl	8D84h

13.4.10.2.305.2 Diagram



13.4.10.2.305.3 Fields

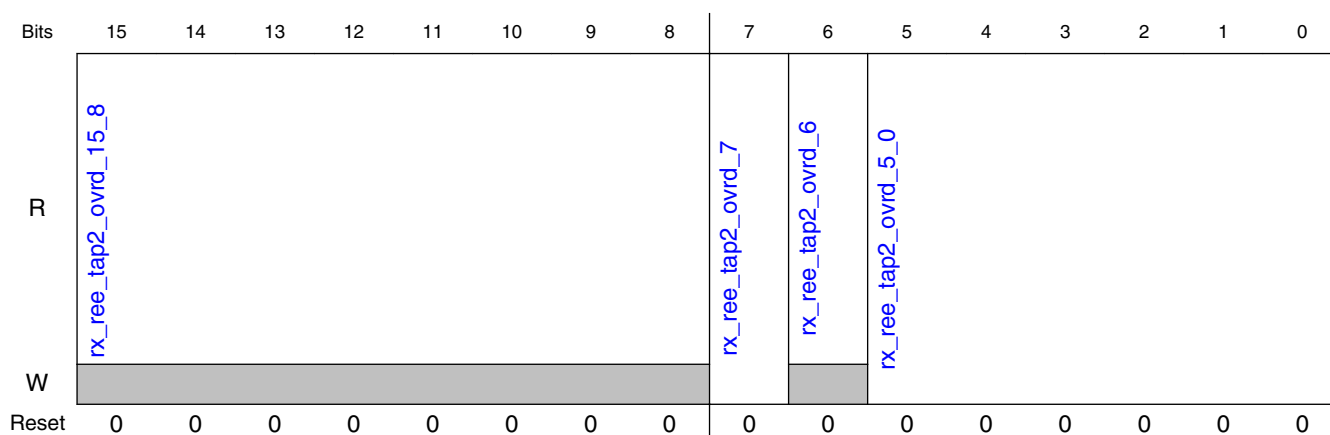
Field	Function
15-12 rx_ree_tap2_ctrl_15_12	Reserved
11 rx_ree_tap2_ctrl_11	Tap coefficient combinational logic zero crossing enable:
10 rx_ree_tap2_ctrl_10	Tap coefficient combinational logic non zero crossing enable:
9 rx_ree_tap2_ctrl_9	Tap coefficient combinational logic bit 0 only enable:
8 rx_ree_tap2_ctrl_8	Receiver DFE tap coefficient disable: This bit disables the
7 rx_ree_tap2_ctrl_7	Reserved
6-4 rx_ree_tap2_ctrl_6_4	Tap integrator accumulator scaler value: Specifies the amount to scale the input to the tap integrator accumulator by. The following are the valid settings for this field:
3-0 rx_ree_tap2_ctrl_3_0	Tap sigma delta accumulator scaler value: Specifies the amount to scale the input to the tap sigma delta accumulator by. The following are the valid settings for this field:

13.4.10.2.306 REE tap 2 override register (lane0_rx_ree_tap2_ovrd - lane3_rx_ree_tap2_ovrd)

13.4.10.2.306.1 Offset

Register	Offset
lane0_rx_ree_tap2_ovrd	8185h
lane1_rx_ree_tap2_ovrd	8585h
lane2_rx_ree_tap2_ovrd	8985h
lane3_rx_ree_tap2_ovrd	8D85h

13.4.10.2.306.2 Diagram



13.4.10.2.306.3 Fields

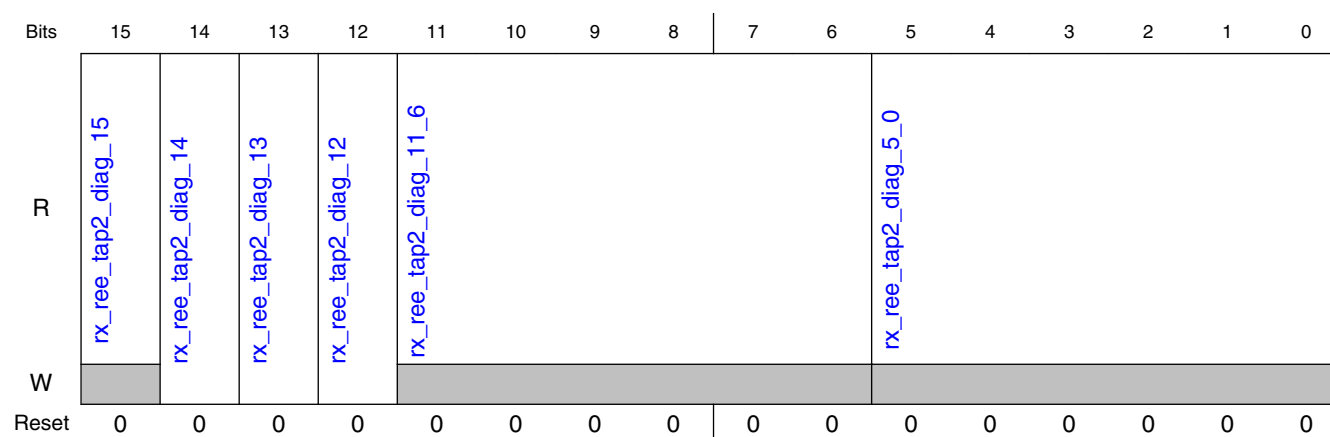
Field	Function
15-8 rx_ree_tap2_ovrd_15_8	Reserved
7 rx_ree_tap2_ovrd_7	Tap override enable: Setting this bit to a 1b1 will enable the tap override field in this register to override the tap integrator accumulator functions.
6 rx_ree_tap2_ovrd_6	Reserved
5-0 rx_ree_tap2_ovrd_5_0	Tap override value: When the tap override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

13.4.10.2.307 REE tap 2 diagnostics register (lane0_rx_ree_tap2_diag - lane3_rx_ree_tap2_diag)

13.4.10.2.307.1 Offset

Register	Offset
lane0_rx_ree_tap2_diag	8186h
lane1_rx_ree_tap2_diag	8586h
lane2_rx_ree_tap2_diag	8986h
lane3_rx_ree_tap2_diag	8D86h

13.4.10.2.307.2 Diagram



13.4.10.2.307.3 Fields

Field	Function
15 rx_ree_tap2_diag_15	Reserved
14 rx_ree_tap2_diag_14	Voter override neg : Writing a 1b1 in this register bit will force the tap voter function to activate the voter neg signal for a single clock cycle.
13 rx_ree_tap2_diag_13	Voter override pos : Writing a 1b1 in this register bit will force the tap voter function to activate the voter pos signal for a single clock cycle.
12	Voter override enable : Setting this bit to a 1b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the tap.

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Clocks And Resets

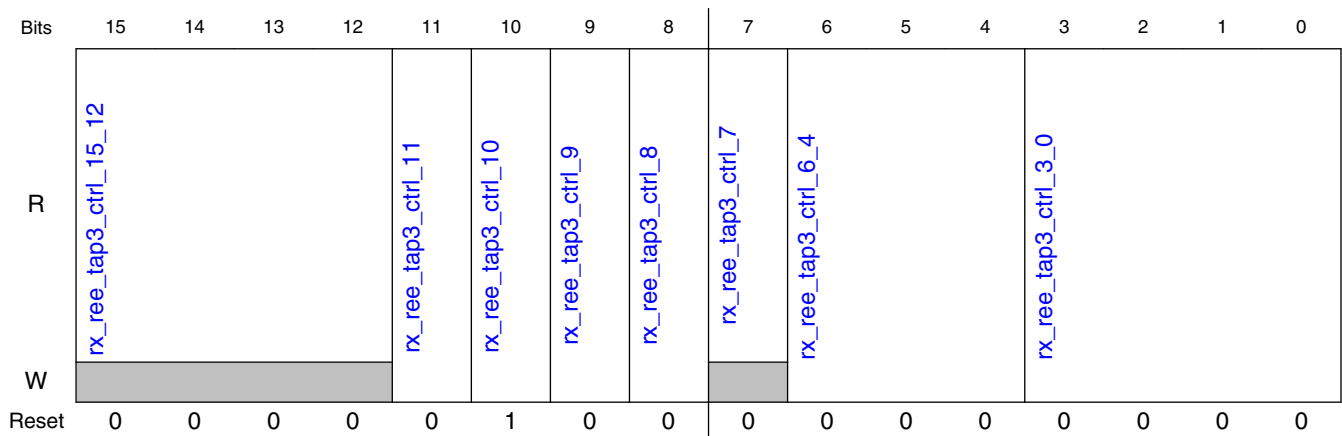
Field	Function
rx_ree_tap2_dia_g_12	
11-6 rx_ree_tap2_dia_g_11_6	Reserved
5-0 rx_ree_tap2_dia_g_5_0	Current tap integrator accumulator: Current value of the tap integrator accumulator.

13.4.10.2.308 REE tap 3 control register (lane0_rx_ree_tap3_ctrl - lane3_rx_ree_tap3_ctrl)

13.4.10.2.308.1 Offset

Register	Offset
lane0_rx_ree_tap3_ctrl	8188h
lane1_rx_ree_tap3_ctrl	8588h
lane2_rx_ree_tap3_ctrl	8988h
lane3_rx_ree_tap3_ctrl	8D88h

13.4.10.2.308.2 Diagram



13.4.10.2.308.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

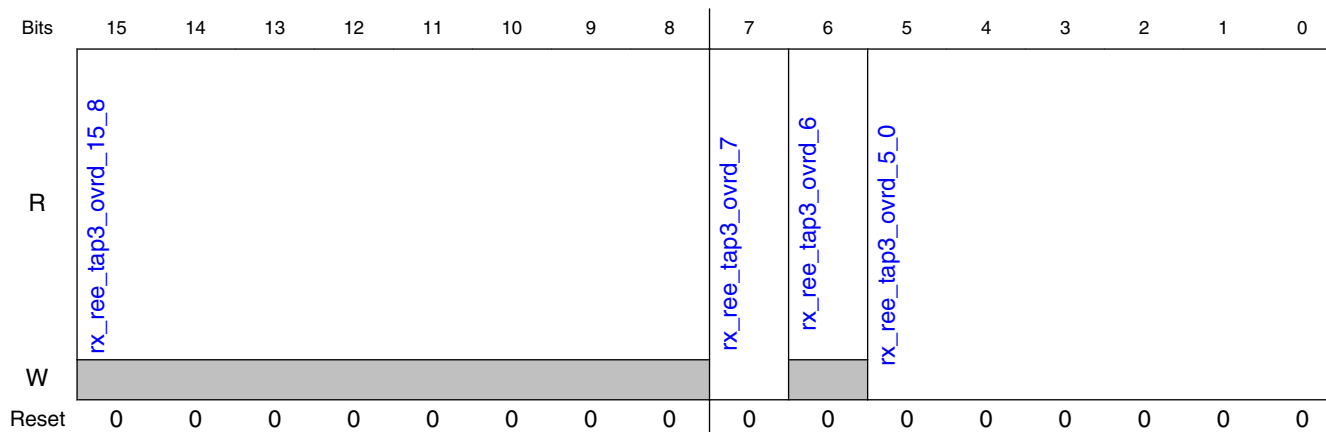
Field	Function
rx_ree_tap3_ctrl_15_12	
11 rx_ree_tap3_ctrl_11	Tap coefficient combinational logic zero crossing enable:
10 rx_ree_tap3_ctrl_10	Tap coefficient combinational logic non zero crossing enable:
9 rx_ree_tap3_ctrl_9	Tap coefficient combinational logic bit 0 only enable:
8 rx_ree_tap3_ctrl_8	Receiver DFE tap coefficient disable: This bit disables the
7 rx_ree_tap3_ctrl_7	Reserved
6-4 rx_ree_tap3_ctrl_6_4	Tap integrator accumulator scaler value: Specifies the amount to scale the input to the tap integrator accumulator by. The following are the valid settings for this field:
3-0 rx_ree_tap3_ctrl_3_0	Tap sigma delta accumulator scaler value: Specifies the amount to scale the input to the tap sigma delta accumulator by. The following are the valid settings for this field:

13.4.10.2.309 REE tap 3 override register (lane0_rx_ree_tap3_ovrd - lane3_rx_ree_tap3_ovrd)

13.4.10.2.309.1 Offset

Register	Offset
lane0_rx_ree_tap3_ovrd	8189h
lane1_rx_ree_tap3_ovrd	8589h
lane2_rx_ree_tap3_ovrd	8989h
lane3_rx_ree_tap3_ovrd	8D89h

13.4.10.2.309.2 Diagram



13.4.10.2.309.3 Fields

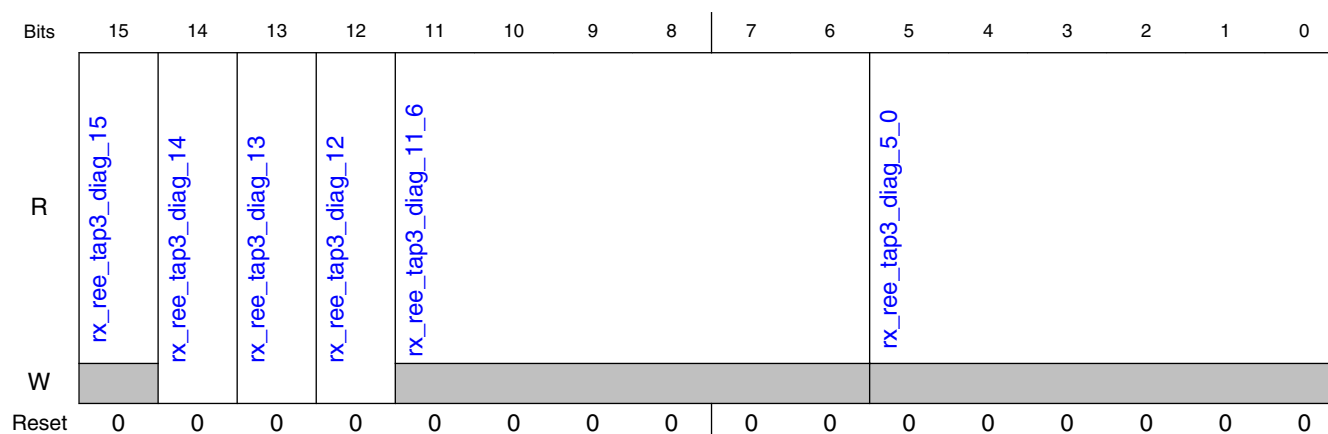
Field	Function
15-8 rx_ree_tap3_ovrd_15_8	Reserved
7 rx_ree_tap3_ovrd_7	Tap override enable: Setting this bit to a 1b1 will enable the tap override field in this register to override the tap integrator accumulator functions.
6 rx_ree_tap3_ovrd_6	Reserved
5-0 rx_ree_tap3_ovrd_5_0	Tap override value: When the tap override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

13.4.10.2.310 REE tap 3 diagnostics register (lane0_rx_ree_tap3_diag - lane3_rx_ree_tap3_diag)

13.4.10.2.310.1 Offset

Register	Offset
lane0_rx_ree_tap3_diag	818Ah
lane1_rx_ree_tap3_diag	858Ah
lane2_rx_ree_tap3_diag	898Ah
lane3_rx_ree_tap3_diag	8D8Ah

13.4.10.2.310.2 Diagram



13.4.10.2.310.3 Fields

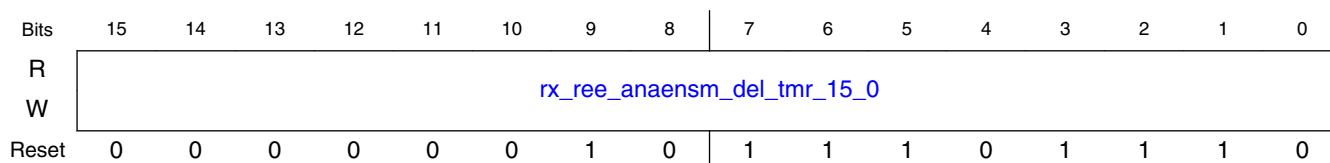
Field	Function
15 rx_ree_tap3_diag_15	Reserved
14 rx_ree_tap3_diag_14	Voter override neg : Writing a 1b1 in this register bit will force the tap voter function to activate the voter neg signal for a single clock cycle.
13 rx_ree_tap3_diag_13	Voter override pos : Writing a 1b1 in this register bit will force the tap voter function to activate the voter pos signal for a single clock cycle.
12 rx_ree_tap3_diag_12	Voter override enable : Setting this bit to a 1b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the tap.
11-6 rx_ree_tap3_diag_11_6	Reserved
5-0 rx_ree_tap3_diag_5_0	Current tap integrator accumulator: Current value of the tap integrator accumulator.

13.4.10.2.311 REE analog enable control state machine delay timer value register (lane0_rx_ree_anaensm_del_tmr - lane3_rx_ree_anaensm_del_tmr)

13.4.10.2.311.1 Offset

Register	Offset
lane0_rx_ree_anaensm_del_tmr	8194h
lane1_rx_ree_anaensm_del_tmr	8594h
lane2_rx_ree_anaensm_del_tmr	8994h
lane3_rx_ree_anaensm_del_tmr	8D94h

13.4.10.2.311.2 Diagram



13.4.10.2.311.3 Fields

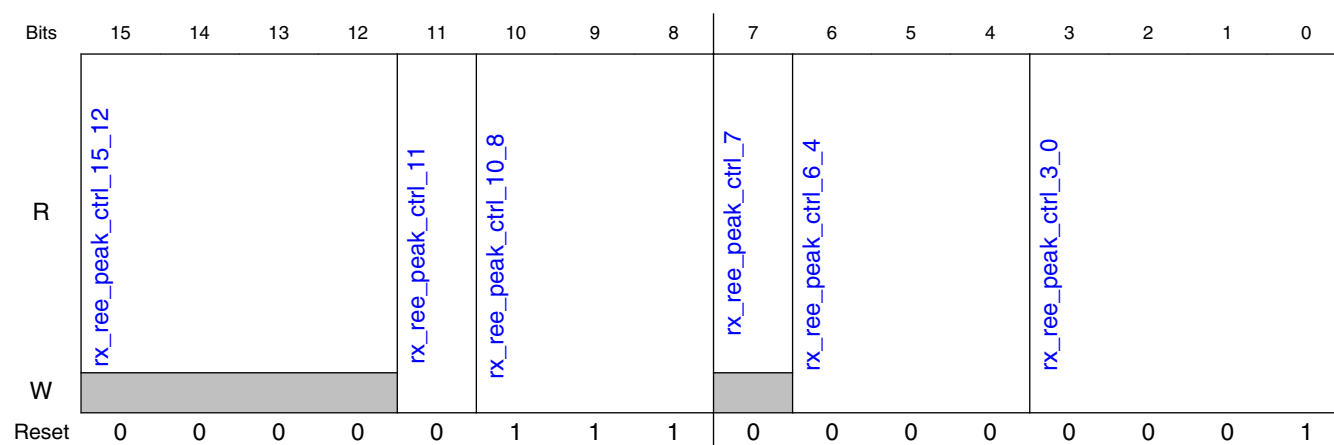
Field	Function
15-0 rx_ree_anaensm_del_tmr_15_0	Analog enable delay timer value : The number of clock cycles the state machine will wait in the Analog Enable Delay state. The time specified here is the number of clock cycles to wait between when the analog enable signal (

13.4.10.2.312 REE peaking amp control register (lane0_rx_ree_peak_ctrl - lane3_rx_ree_peak_ctrl)

13.4.10.2.312.1 Offset

Register	Offset
lane0_rx_ree_peak_ctrl	8198h
lane1_rx_ree_peak_ctrl	8598h
lane2_rx_ree_peak_ctrl	8998h
lane3_rx_ree_peak_ctrl	8D98h

13.4.10.2.312.2 Diagram



13.4.10.2.312.3 Fields

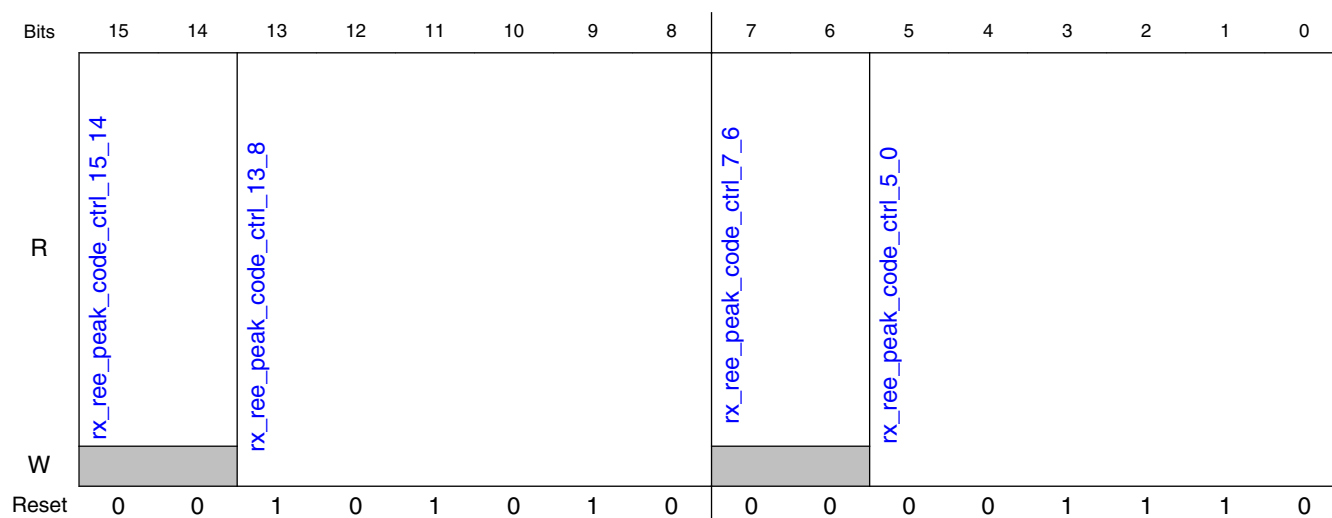
Field	Function
15-12 rx_ree_peak_ctrl_15_12	Reserved
11 rx_ree_peak_ctrl_11	Peaking amp feedback path enable: Enables the peaking amp feedback path.
10-8 rx_ree_peak_ctrl_10_8	Peaking amp feedback scaler value: Specifies the amount to scale the peaking amp feedback by. The following are the valid settings for this signal:
7 rx_ree_peak_ctrl_7	Reserved
6-4 rx_ree_peak_ctrl_6_4	Peaking amp integrator accumulator scaler value: Specifies the amount to scale the input to the peaking amp integrator accumulator by. The following are the valid settings for this field:
3-0 rx_ree_peak_ctrl_3_0	Peaking amp sigma delta accumulator scaler value: Specifies the amount to scale the input to the peaking amp sigma delta accumulator by. The following are the valid settings for this field:

13.4.10.2.313 REE peaking amp code control register (lane0_rx_ree_peak_code_ctrl - lane3_rx_ree_peak_code_ctrl)

13.4.10.2.313.1 Offset

Register	Offset
lane0_rx_ree_peak_code_ctrl	8199h
lane1_rx_ree_peak_code_ctrl	8599h
lane2_rx_ree_peak_code_ctrl	8999h
lane3_rx_ree_peak_code_ctrl	8D99h

13.4.10.2.313.2 Diagram



13.4.10.2.313.3 Fields

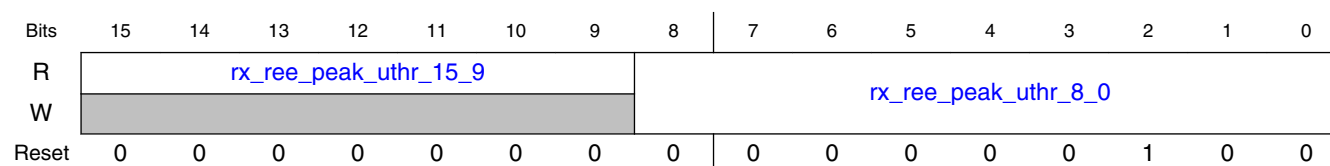
Field	Function
15-14 rx_ree_peak_code_ctrl_15_14	Reserved
13-8 rx_ree_peak_code_ctrl_13_8	Peaking amp code maximum value: This is the maximum value that the peaking amp code will be allowed to increase to.
7-6 rx_ree_peak_code_ctrl_7_6	Reserved
5-0 rx_ree_peak_code_ctrl_5_0	Peaking amp initial code: Initial value the peaking amp code is set to when training starts.

13.4.10.2.314 REE peaking amp upper threshold register (lane0_rx_ree_peak_uthr - lane3_rx_ree_peak_uthr)

13.4.10.2.314.1 Offset

Register	Offset
lane0_rx_ree_peak_uthr	819Ah
lane1_rx_ree_peak_uthr	859Ah
lane2_rx_ree_peak_uthr	899Ah
lane3_rx_ree_peak_uthr	8D9Ah

13.4.10.2.314.2 Diagram



13.4.10.2.314.3 Fields

Field	Function
15-9 rx_ree_peak_uthr_15_9	Reserved
8-0 rx_ree_peak_uthr_8_0	Peaking amp algorithm upper threshold: This is the upper threshold value used in the peaking amp algorithm.

13.4.10.2.315 REE peaking amp lower threshold register (lane0_rx_ree_peak_lthr - lane3_rx_ree_peak_lthr)

13.4.10.2.315.1 Offset

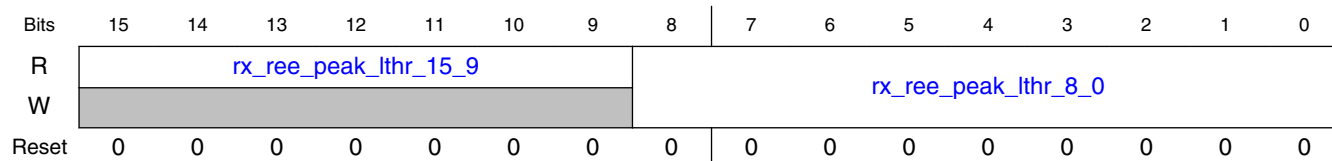
Register	Offset
lane0_rx_ree_peak_lthr	819Bh
lane1_rx_ree_peak_lthr	859Bh

Table continues on the next page...

Clocks And Resets

Register	Offset
lane2_rx_ree_peak_lthr	899Bh
lane3_rx_ree_peak_lthr	8D9Bh

13.4.10.2.315.2 Diagram



13.4.10.2.315.3 Fields

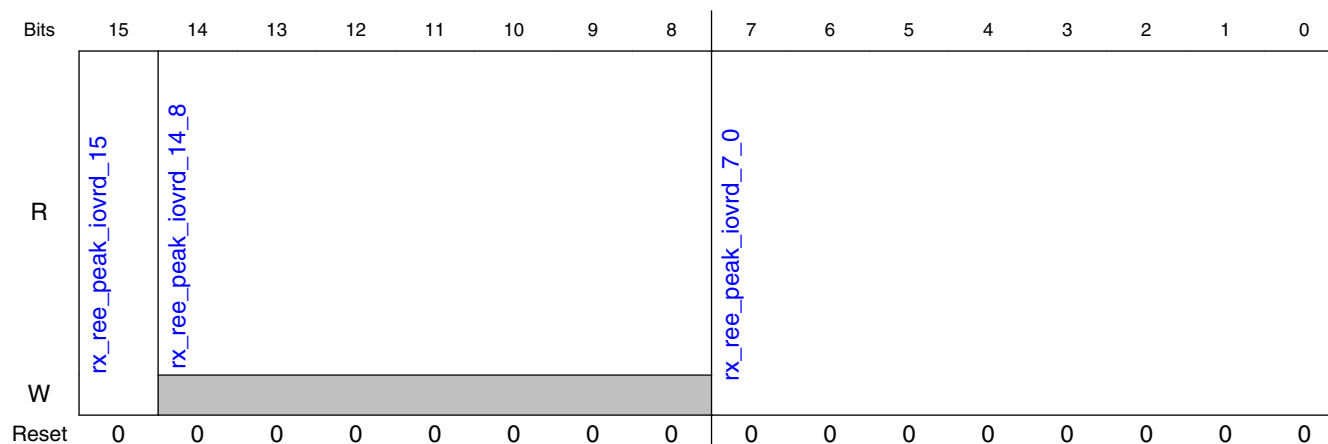
Field	Function
15-9 rx_ree_peak_lthr_15_9	Reserved
8-0 rx_ree_peak_lthr_8_0	Peaking amp algorithm lower threshold: This is the lower threshold value used in the peaking amp algorithm.

13.4.10.2.316 REE peaking amp input override register (lane0_rx_ree_peak_iovrd - lane3_rx_ree_peak_iovrd)

13.4.10.2.316.1 Offset

Register	Offset
lane0_rx_ree_peak_iovrd	819Ch
lane1_rx_ree_peak_iovrd	859Ch
lane2_rx_ree_peak_iovrd	899Ch
lane3_rx_ree_peak_iovrd	8D9Ch

13.4.10.2.316.2 Diagram



13.4.10.2.316.3 Fields

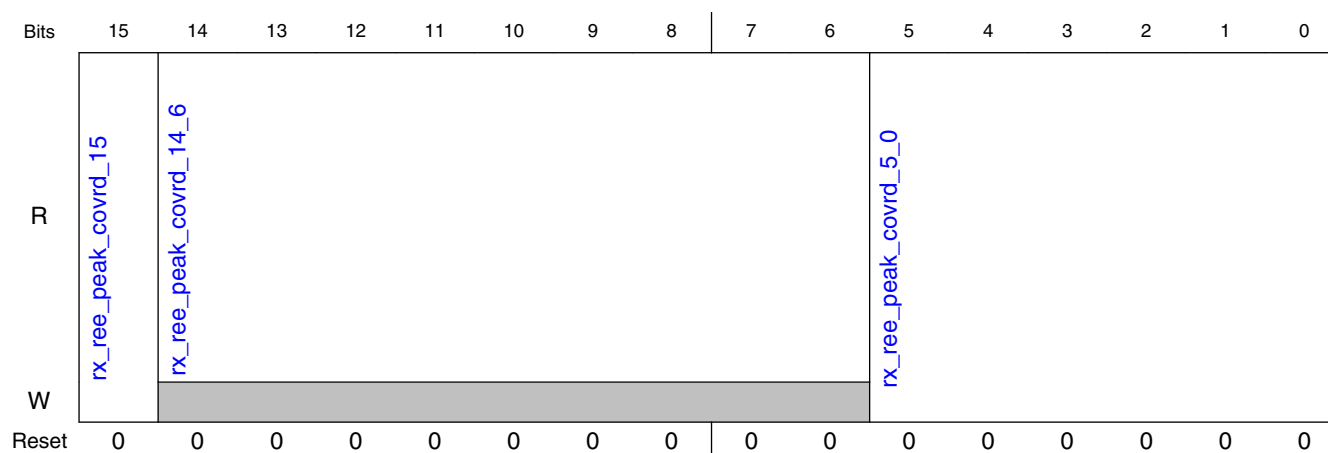
Field	Function
15 <code>rx_ree_peak_covrd_15</code>	Peaking amp tap accumulator input override enable: Setting this bit to a 1b1 will allow the tap accumulator input in the peaking amp gain algorithm to be overridden by the peaking amp tap accumulator input override field in this register.
14-8 <code>rx_ree_peak_covrd_14_8</code>	Reserved
7-0 <code>rx_ree_peak_covrd_7_0</code>	Peaking amp tap accumulator input override : Value that will override the tap accumulator input in the peaking amp gain algorithm, when the Peaking amp tap accumulator input override enable bit is active.

13.4.10.2.317 REE peaking amp code override register (lane0_rx_ree_peak_covrd - lane3_rx_ree_peak_covrd)

13.4.10.2.317.1 Offset

Register	Offset
<code>lane0_rx_ree_peak_covrd</code>	819Dh
<code>lane1_rx_ree_peak_covrd</code>	859Dh
<code>lane2_rx_ree_peak_covrd</code>	899Dh
<code>lane3_rx_ree_peak_covrd</code>	8D9Dh

13.4.10.2.317.2 Diagram



13.4.10.2.317.3 Fields

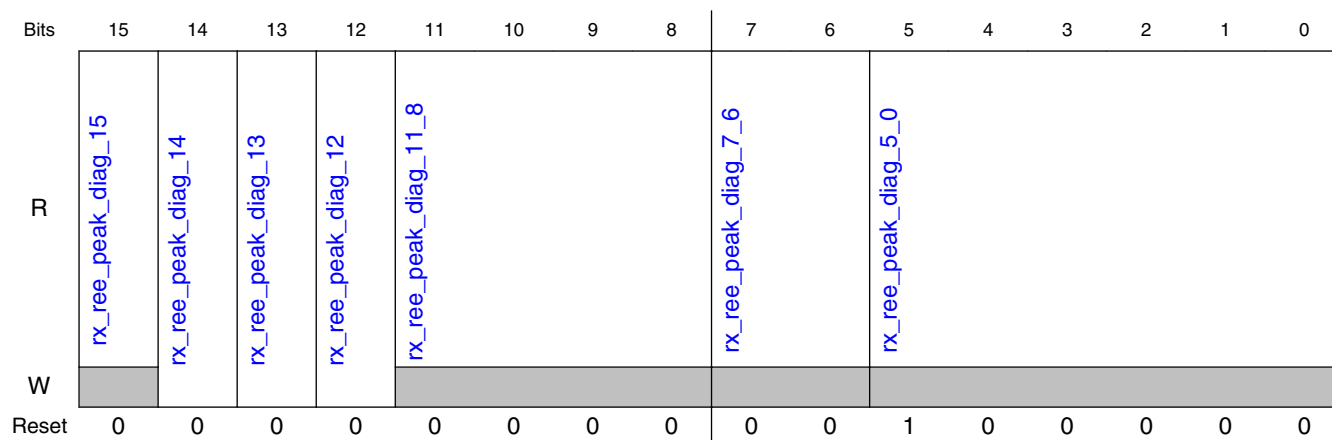
Field	Function
15 rx_ree_peak_covrd_15	Peaking amp code override enable: Setting this bit to a 1b1 will allow the peaking amp code to be overridden by the peaking amp code override value field in this register. This allows the
14-6 rx_ree_peak_covrd_14_6	Reserved
5-0 rx_ree_peak_covrd_5_0	Peaking amp code override value: Value that will override the peaking amp code when the peaking amp code override enable bit in this register is active.

13.4.10.2.318 REE peaking amp diagnostics register (lane0_rx_ree_peak_diag - lane3_rx_ree_peak_diag)

13.4.10.2.318.1 Offset

Register	Offset
lane0_rx_ree_peak_diag	819Eh
lane1_rx_ree_peak_diag	859Eh
lane2_rx_ree_peak_diag	899Eh
lane3_rx_ree_peak_diag	8D9Eh

13.4.10.2.318.2 Diagram



13.4.10.2.318.3 Fields

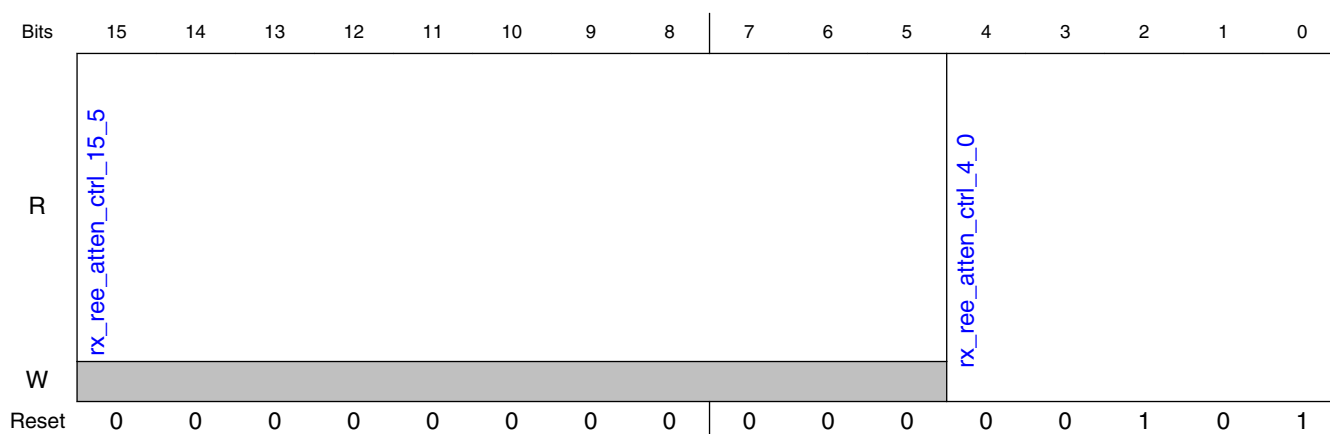
Field	Function
15 rx_ree_peak_diag_15	Reserved
14 rx_ree_peak_diag_14	Voter override neg : Writing a 1b1 in this register bit will force the peaking amp voter function to activate the voter neg signal for a single clock cycle.
13 rx_ree_peak_diag_13	Voter override pos : Writing a 1b1 in this register bit will force the peaking amp voter function to activate the voter pos signal for a single clock cycle.
12 rx_ree_peak_diag_12	Voter override enable : Setting this bit to a 1b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the peaking amp.
11-8 rx_ree_peak_diag_11_8	Reserved
7-6 rx_ree_peak_diag_7_6	Reserved
5-0 rx_ree_peak_diag_5_0	Current peaking amp integrator accumulator: Current value of the tap integrator accumulator, without the unused sign bit.

13.4.10.2.319 REE attenuation control register (lane0_rx_ree_atten_ctrl - lane3_rx_ree_atten_ctrl)

13.4.10.2.319.1 Offset

Register	Offset
lane0_rx_ree_atten_ctrl	81A0h
lane1_rx_ree_atten_ctrl	85A0h
lane2_rx_ree_atten_ctrl	89A0h
lane3_rx_ree_atten_ctrl	8DA0h

13.4.10.2.319.2 Diagram



13.4.10.2.319.3 Fields

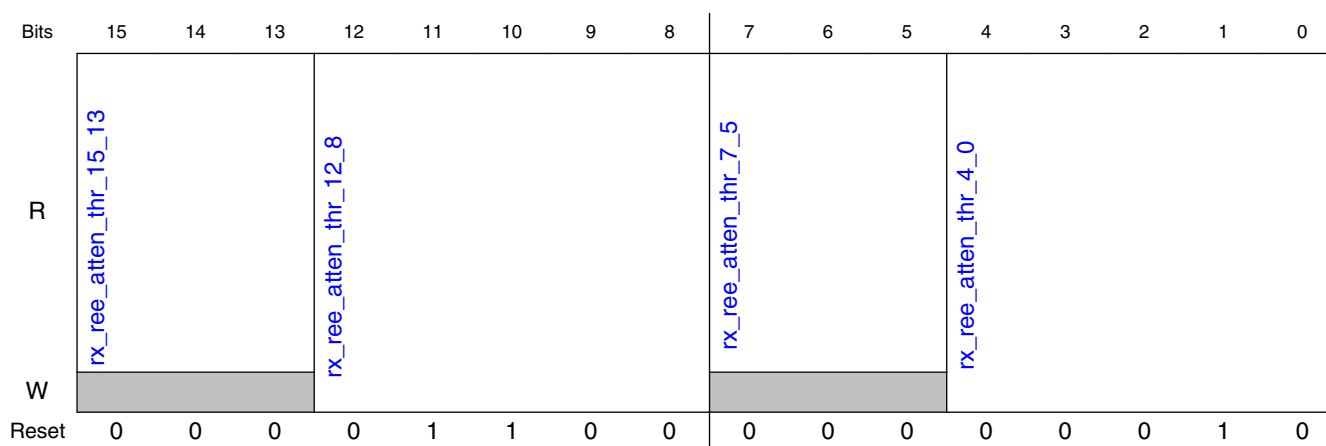
Field	Function
15-5 rx_ree_atten_ctrl_15_5	Reserved
4-0 rx_ree_atten_ctrl_4_0	Receiver DFE attenuation maximum value: The maximum value the

13.4.10.2.320 REE attenuation threshold register (lane0_rx_ree_atten_thr - lane3_rx_ree_atten_thr)

13.4.10.2.320.1 Offset

Register	Offset
lane0_rx_ree_atten_thr	81A1h
lane1_rx_ree_atten_thr	85A1h
lane2_rx_ree_atten_thr	89A1h
lane3_rx_ree_atten_thr	8DA1h

13.4.10.2.320.2 Diagram



13.4.10.2.320.3 Fields

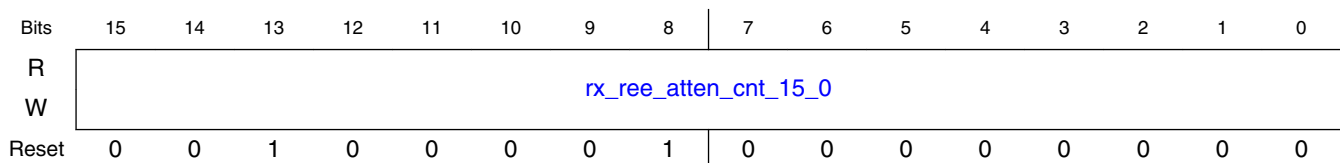
Field	Function
15-13 <code>rx_ree_atten_thr_15_13</code>	Reserved
12-8 <code>rx_ree_atten_thr_12_8</code>	Attenuation high threshold value: High threshold value to compare against the VGA gain accumulator value. Note that the value on this field is not a twos complement value (There is no sign bit and it is always a positive number).
7-5 <code>rx_ree_atten_thr_7_5</code>	Reserved
4-0 <code>rx_ree_atten_thr_4_0</code>	Attenuation low threshold value: Low threshold value to compare against the VGA gain accumulator value. Note that the value on this field is not a twos complement value (There is no sign bit and it is always a positive number).

13.4.10.2.321 REE attenuation counter register (lane0_rx_ree_atten_cnt - lane3_rx_ree_atten_cnt)

13.4.10.2.321.1 Offset

Register	Offset
lane0_rx_ree_atten_cnt	81A2h
lane1_rx_ree_atten_cnt	85A2h
lane2_rx_ree_atten_cnt	89A2h
lane3_rx_ree_atten_cnt	8DA2h

13.4.10.2.321.2 Diagram



13.4.10.2.321.3 Fields

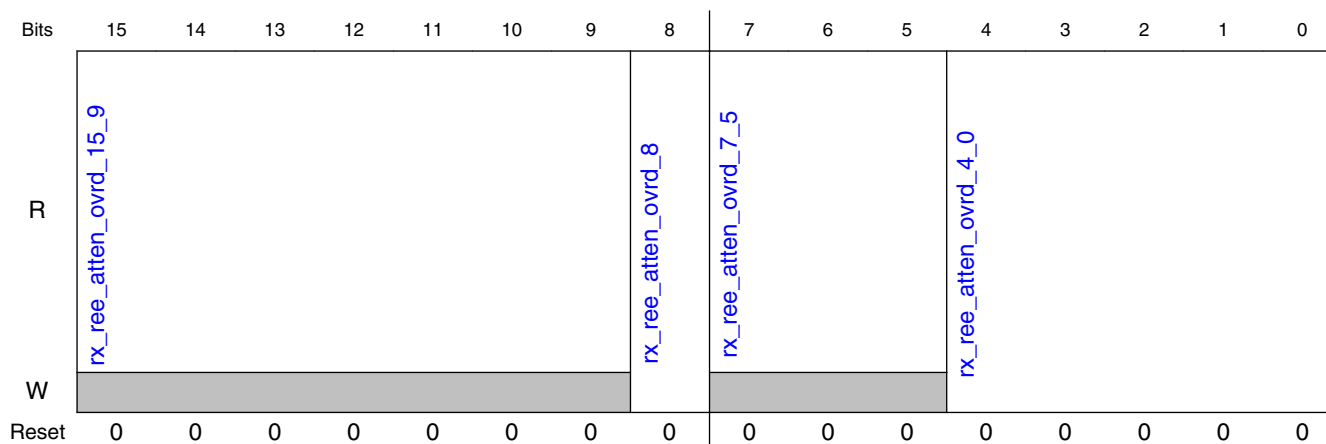
Field	Function
15-0 rx_ree_atten_cnt_15_0	Attenuation counter max: Value used to specify the maximum number of consecutive words above or below the specified thresholds which will result in triggering an increase or decrease in the

13.4.10.2.322 REE attenuation override register (lane0_rx_ree_atten_ovrd - lane3_rx_ree_atten_ovrd)

13.4.10.2.322.1 Offset

Register	Offset
lane0_rx_ree_atten_ovrd	81A3h
lane1_rx_ree_atten_ovrd	85A3h
lane2_rx_ree_atten_ovrd	89A3h
lane3_rx_ree_atten_ovrd	8DA3h

13.4.10.2.322.2 Diagram



13.4.10.2.322.3 Fields

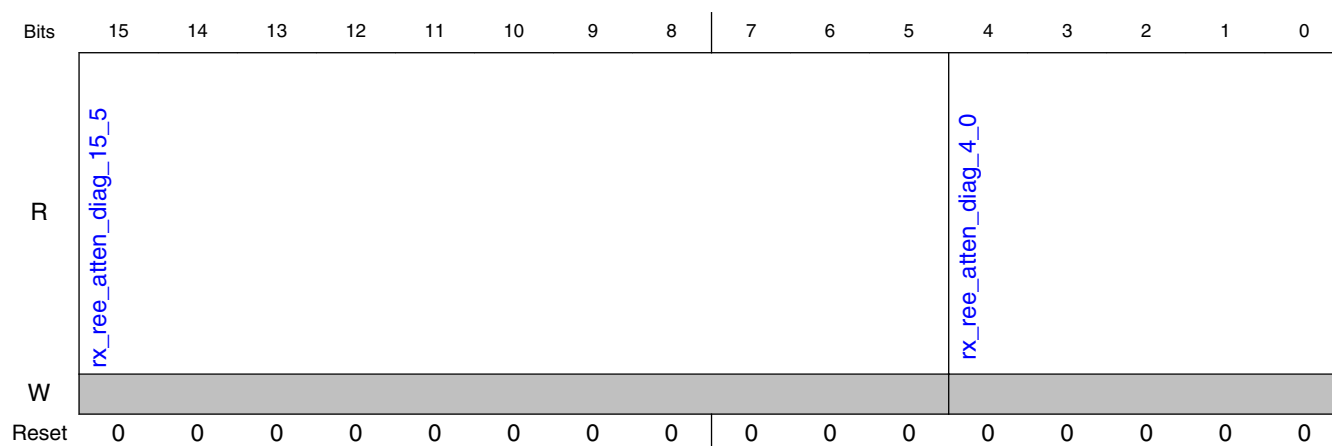
Field	Function
15-9 rx_ree_atten_ovrd_15_9	Reserved
8 rx_ree_atten_ovrd_8	Attenuation override enable: Setting this bit to a 1b1 will allow the
7-5 rx_ree_atten_ovrd_7_5	Reserved
4-0 rx_ree_atten_ovrd_4_0	Attenuation override value: When enabled by the attenuation override enable bit in this register, this value will override the current attenuation value on the

13.4.10.2.323 REE attenuation diagnostics register (lane0_rx_ree_atten_diag - lane3_rx_ree_atten_diag)

13.4.10.2.323.1 Offset

Register	Offset
lane0_rx_ree_atten_diag	81A4h
lane1_rx_ree_atten_diag	85A4h
lane2_rx_ree_atten_diag	89A4h
lane3_rx_ree_atten_diag	8DA4h

13.4.10.2.323.2 Diagram



13.4.10.2.323.3 Fields

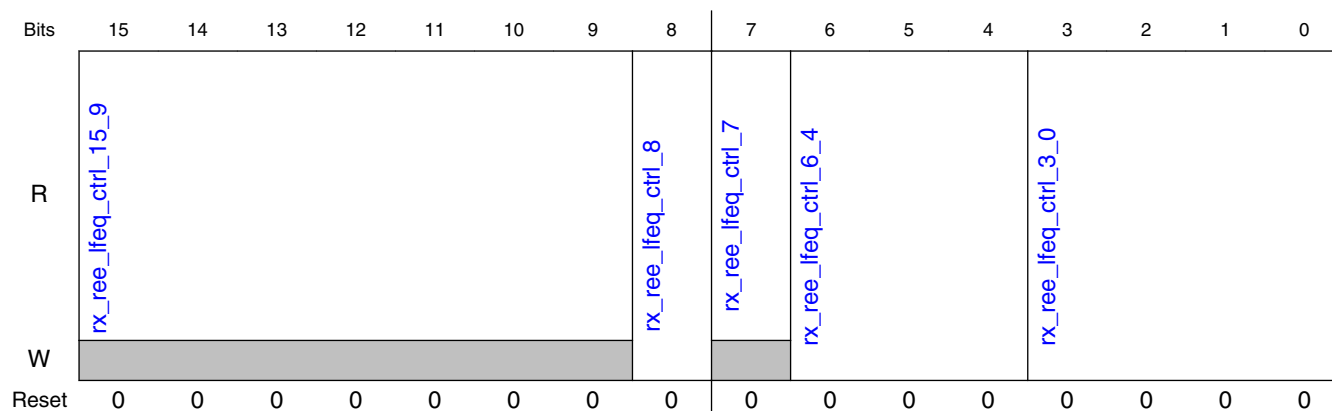
Field	Function
15-5 rx_ree_atten_diag_15_5	Reserved
4-0 rx_ree_atten_diag_4_0	Current attenuation value: Current value of the attenuation.

13.4.10.2.324 REE low frequency equalizer control register (lane0_rx_ree_lfeq_ctrl - lane3_rx_ree_lfeq_ctrl)

13.4.10.2.324.1 Offset

Register	Offset
lane0_rx_ree_lfeq_ctrl	81A8h
lane1_rx_ree_lfeq_ctrl	85A8h
lane2_rx_ree_lfeq_ctrl	89A8h
lane3_rx_ree_lfeq_ctrl	8DA8h

13.4.10.2.324.2 Diagram



13.4.10.2.324.3 Fields

Field	Function
15-9 rx_ree_lfeq_ctrl_15_9	Reserved
8 rx_ree_lfeq_ctrl_8	Receiver DFE coefficient disable: This bit disables the
7 rx_ree_lfeq_ctrl_7	Reserved
6-4 rx_ree_lfeq_ctrl_6_4	Integrator accumulator scaler value: Specifies the amount to scale the input to the integrator accumulator by. The following are the valid settings for this field:
3-0 rx_ree_lfeq_ctrl_3_0	Sigma delta accumulator scaler value: Specifies the amount to scale the input to the sigma delta accumulator by. The following are the valid settings for this field:

13.4.10.2.325 REE low frequency equalizer override register (lane0_rx_ree_lfeq_ovrd - lane3_rx_ree_lfeq_ovrd)

13.4.10.2.325.1 Offset

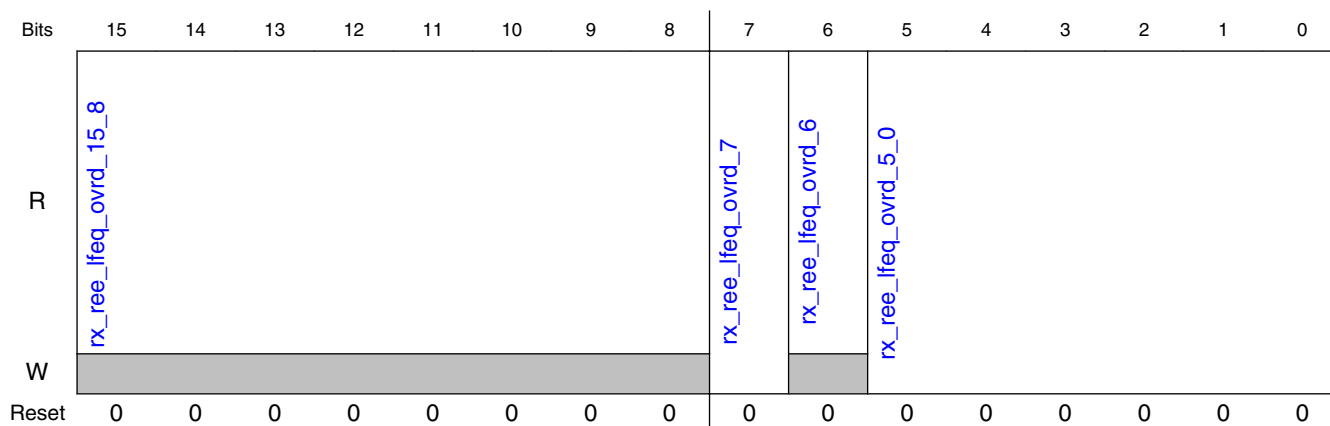
Register	Offset
lane0_rx_ree_lfeq_ovrd	81A9h
lane1_rx_ree_lfeq_ovrd	85A9h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane2_rx_ree_lfeq_ovrd	89A9h
lane3_rx_ree_lfeq_ovrd	8DA9h

13.4.10.2.325.2 Diagram



13.4.10.2.325.3 Fields

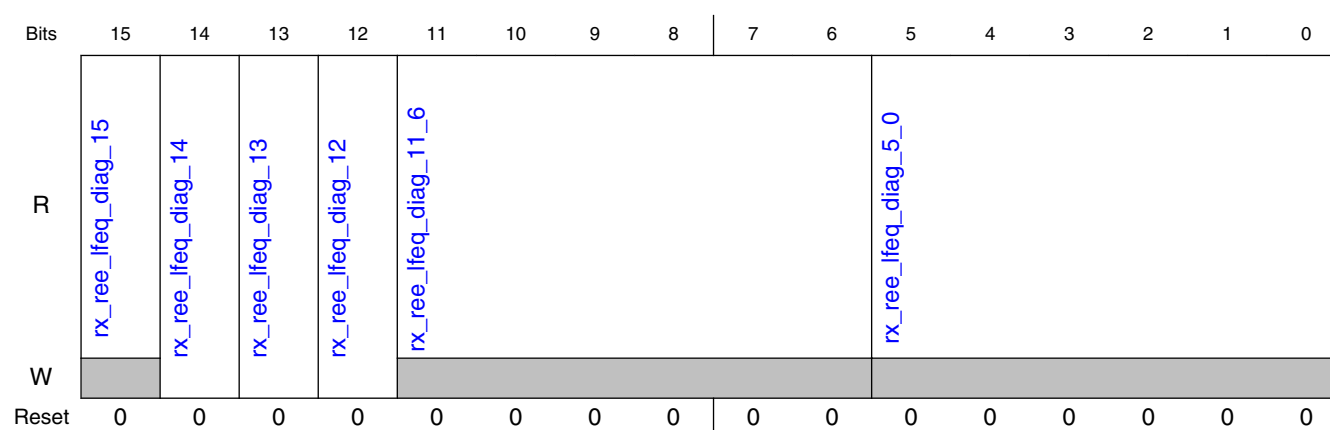
Field	Function
15-8 rx_ree_lfeq_ovrd_15_8	Reserved
7 rx_ree_lfeq_ovrd_7	Override enable: Setting this bit to a 1b1 will enable the override field in this register to override the integrator accumulator functions.
6 rx_ree_lfeq_ovrd_6	Reserved
5-0 rx_ree_lfeq_ovrd_5_0	Override value: When the override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

13.4.10.2.326 REE low frequency equalizer diagnostics register (lane0_rx_ree_lfeq_diag - lane3_rx_ree_lfeq_diag)

13.4.10.2.326.1 Offset

Register	Offset
lane0_rx_ree_lfeq_diag	81AAh
lane1_rx_ree_lfeq_diag	85AAh
lane2_rx_ree_lfeq_diag	89AAh
lane3_rx_ree_lfeq_diag	8DAAh

13.4.10.2.326.2 Diagram



13.4.10.2.326.3 Fields

Field	Function
15 rx_ree_lfeq_diag_15	Reserved
14 rx_ree_lfeq_diag_14	Voter override neg : Writing a 1b1 in this register bit will force the voter function to activate the voter neg signal for a single clock cycle.
13 rx_ree_lfeq_diag_13	Voter override pos : Writing a 1b1 in this register bit will force the voter function to activate the voter pos signal for a single cycle.
12 rx_ree_lfeq_diag_12	Voter override enable : Setting this bit to a 1b1 will allow only the voter override pos and voter override neg bits in this register to control the voter.
11-6 rx_ree_lfeq_diag_11_6	Reserved
5-0	Current integrator accumulator: Current value of the integrator accumulator.

Clocks And Resets

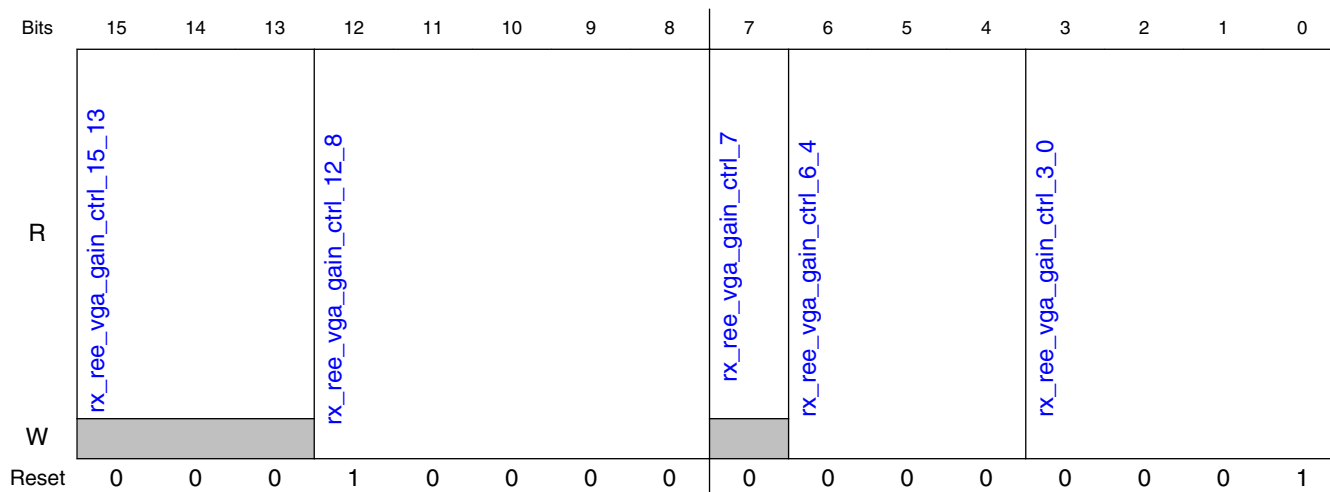
Field	Function
rx_ree_lfeq_diag_5_0	

13.4.10.2.327 REE VGA gain control register (lane0_rx_ree_vga_gain_ctrl - lane3_rx_ree_vga_gain_ctrl)

13.4.10.2.327.1 Offset

Register	Offset
lane0_rx_ree_vga_gain_ctrl	81ACh
lane1_rx_ree_vga_gain_ctrl	85ACh
lane2_rx_ree_vga_gain_ctrl	89ACh
lane3_rx_ree_vga_gain_ctrl	8DACh

13.4.10.2.327.2 Diagram



13.4.10.2.327.3 Fields

Field	Function
15-13	Reserved

Table continues on the next page...

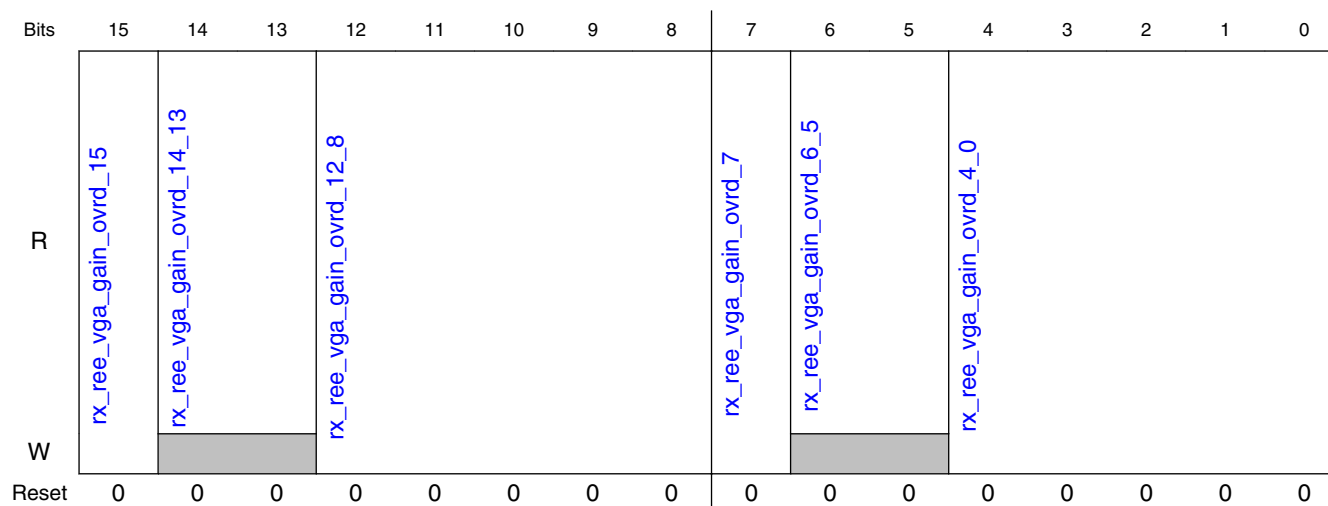
Field	Function
rx_ree_vga_gain_ctrl_15_13	
12-8 rx_ree_vga_gain_ctrl_12_8	VGA gain max: Specifies the maximum value of the VGA gain integrator accumulator, and therefore also the maximum number of bits in the rxda_dfe_vga_gain thermometer code that will be set.
7 rx_ree_vga_gain_ctrl_7	Reserved
6-4 rx_ree_vga_gain_ctrl_6_4	VGA gain integrator accumulator scaler value: Specifies the amount to scale the input to the VGA gain integrator accumulator by. The following are the valid settings for this field:
3-0 rx_ree_vga_gain_ctrl_3_0	VGA gain sigma delta accumulator scaler value: Specifies the amount to scale the input to the VGA gain sigma delta accumulator by. The following are the valid settings for this field:

13.4.10.2.328 REE VGA gain override register (lane0_rx_ree_vga_gain_ovrd - lane3_rx_ree_vga_gain_ovrd)

13.4.10.2.328.1 Offset

Register	Offset
lane0_rx_ree_vga_gain_ovrd	81ADh
lane1_rx_ree_vga_gain_ovrd	85ADh
lane2_rx_ree_vga_gain_ovrd	89ADh
lane3_rx_ree_vga_gain_ovrd	8DADh

13.4.10.2.328.2 Diagram



13.4.10.2.328.3 Fields

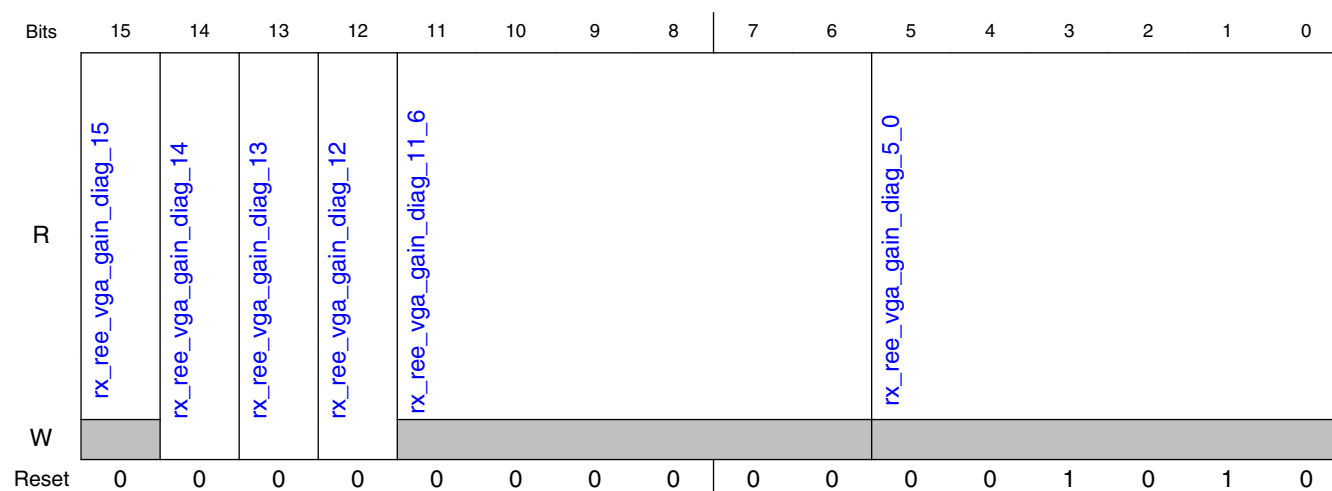
Field	Function
15 rx_ree_vga_gain_ovrd_15	VGA gain target adjust override enable: Setting this bit to a 1b1 will enable the VGA gain target adjust override field in this register to override the VGA gain target adjust accumulator functions.
14-13 rx_ree_vga_gain_ovrd_14_13	Reserved
12-8 rx_ree_vga_gain_ovrd_12_8	VGA gain target adjust override value: When the VGA gain target adjust override enable bit in this register is active, the value in this field will override the accumulator value used to drive the
7 rx_ree_vga_gain_ovrd_7	VGA gain override enable: Setting this bit to a 1b1 will enable the VGA gain override field in this register to override the VGA gain integrator accumulator functions.
6-5 rx_ree_vga_gain_ovrd_6_5	Reserved
4-0 rx_ree_vga_gain_ovrd_4_0	VGA gain override value: When the VGA gain override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

13.4.10.2.329 REE VGA gain diagnostics register (lane0_rx_ree_vga_gain_diag - lane3_rx_ree_vga_gain_diag)

13.4.10.2.329.1 Offset

Register	Offset
lane0_rx_ree_vga_gain_diag	81AEh
lane1_rx_ree_vga_gain_diag	85AEh
lane2_rx_ree_vga_gain_diag	89AEh
lane3_rx_ree_vga_gain_diag	8DAEh

13.4.10.2.329.2 Diagram



13.4.10.2.329.3 Fields

Field	Function
15 rx_ree_vga_gain_diag_15	Reserved
14 rx_ree_vga_gain_diag_14	Voter override neg : Writing a 1b1 in this register bit will force the VGA gain voter function to activate the voter neg signal for a single clock cycle.
13 rx_ree_vga_gain_diag_13	Voter override pos : Writing a 1b1 in this register bit will force the VGA gain voter function to activate the voter pos signal for a single clock cycle.
12 rx_ree_vga_gain_diag_12	Voter override enable : Setting this bit to a 1b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the VGA gain.

Table continues on the next page...

Clocks And Resets

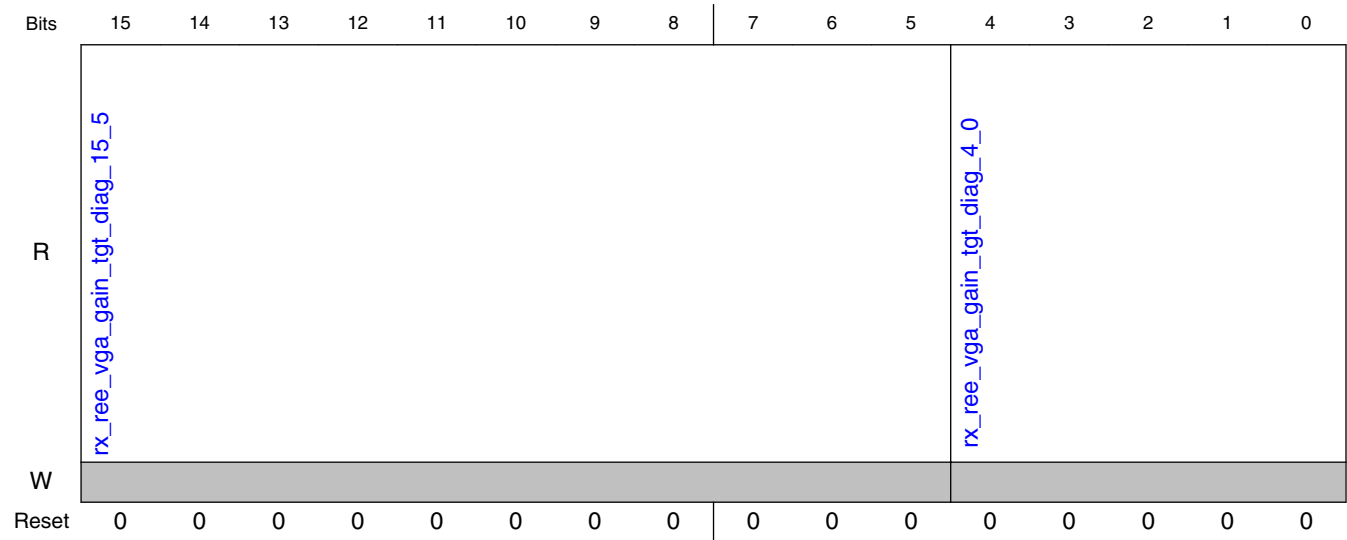
Field	Function
11-6 rx_ree_vga_gain _diag_11_6	Reserved
5-0 rx_ree_vga_gain _diag_5_0	Current VGA gain integrator accumulator: Current value of the VGA gain integrator accumulator.

13.4.10.2.330 REE VGA gain target adjust diagnostics register (lane0_rx_ree_vga_gain_tgt_diag - lane3_rx_ree_vga_gain_tgt_diag)

13.4.10.2.330.1 Offset

Register	Offset
lane0_rx_ree_vga_gain_tgt_diag	81AFh
lane1_rx_ree_vga_gain_tgt_diag	85AFh
lane2_rx_ree_vga_gain_tgt_diag	89AFh
lane3_rx_ree_vga_gain_tgt_diag	8DAFh

13.4.10.2.330.2 Diagram



13.4.10.2.330.3 Fields

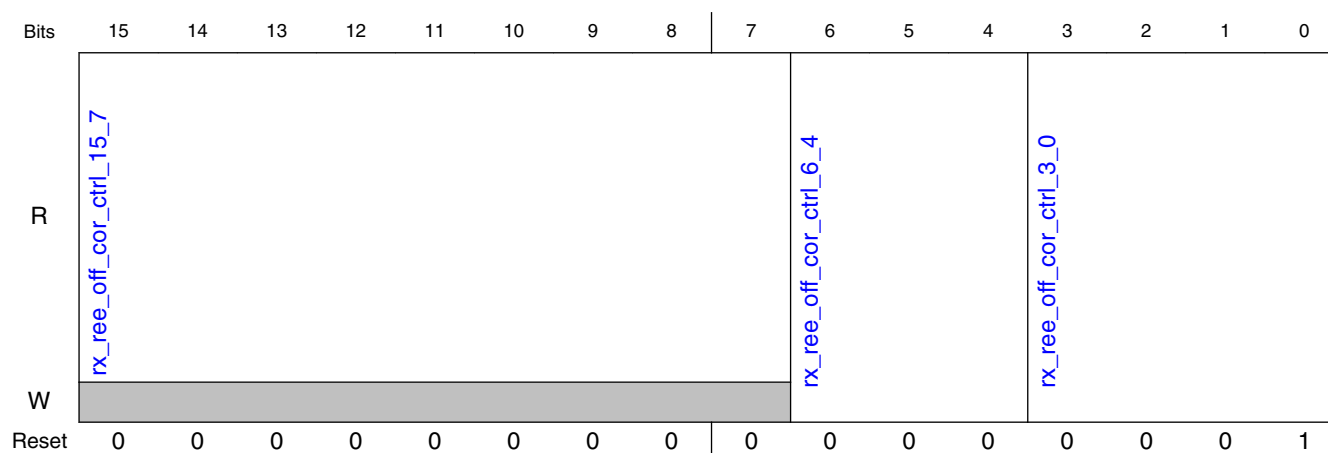
Field	Function
15-5 rx_ree_vga_gain _tgt_diag_15_5	Reserved
4-0 rx_ree_vga_gain _tgt_diag_4_0	Current VGA gain integrator accumulator: Current value of the VGA gain integrator accumulator.

13.4.10.2.331 REE offset correction control register (lane0_rx_ree_off_cor_ctrl - lane3_rx_ree_off_cor_ctrl)

13.4.10.2.331.1 Offset

Register	Offset
lane0_rx_ree_off_cor_ctrl	81B0h
lane1_rx_ree_off_cor_ctrl	85B0h
lane2_rx_ree_off_cor_ctrl	89B0h
lane3_rx_ree_off_cor_ctrl	8DB0h

13.4.10.2.331.2 Diagram



13.4.10.2.331.3 Fields

Field	Function
15-7	Reserved

Table continues on the next page...

Clocks And Resets

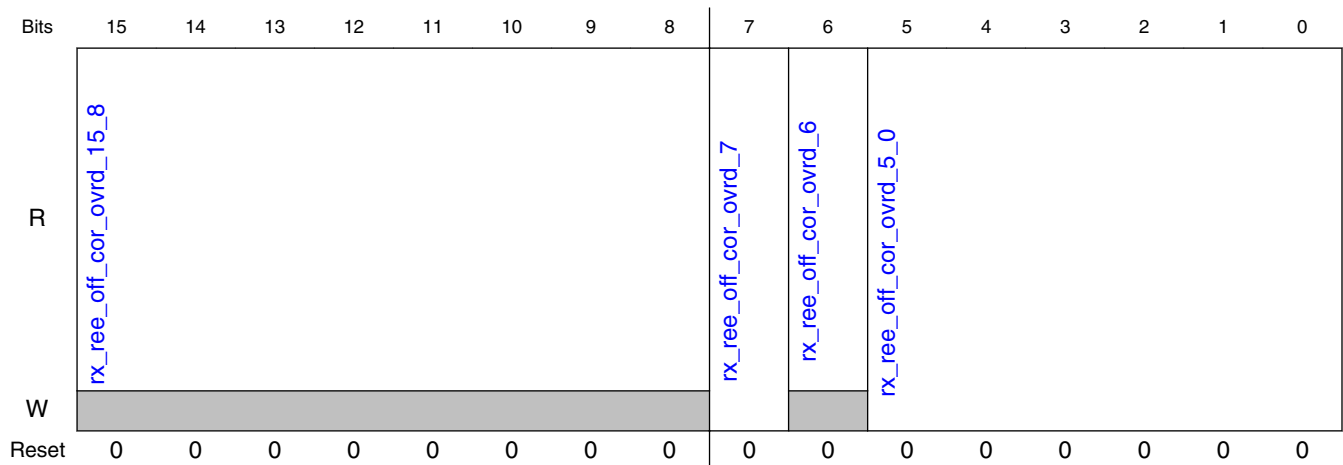
Field	Function
rx_ree_off_cor_ctrl_15_7	
6-4 rx_ree_off_cor_ctrl_6_4	Offset correction integrator accumulator scaler value: Specifies the amount to scale the input to the offset correction integrator accumulator by. The following are the valid settings for this field:
3-0 rx_ree_off_cor_ctrl_3_0	Offset correction sigma delta accumulator scaler value: Specifies the amount to scale the input to the offset correction sigma delta accumulator by. The following are the valid settings for this field:

13.4.10.2.332 REE offset correction override register (lane0_rx_ree_off_cor_ovrd - lane3_rx_ree_off_cor_ovrd)

13.4.10.2.332.1 Offset

Register	Offset
lane0_rx_ree_off_cor_ovrd	81B1h
lane1_rx_ree_off_cor_ovrd	85B1h
lane2_rx_ree_off_cor_ovrd	89B1h
lane3_rx_ree_off_cor_ovrd	8DB1h

13.4.10.2.332.2 Diagram



13.4.10.2.332.3 Fields

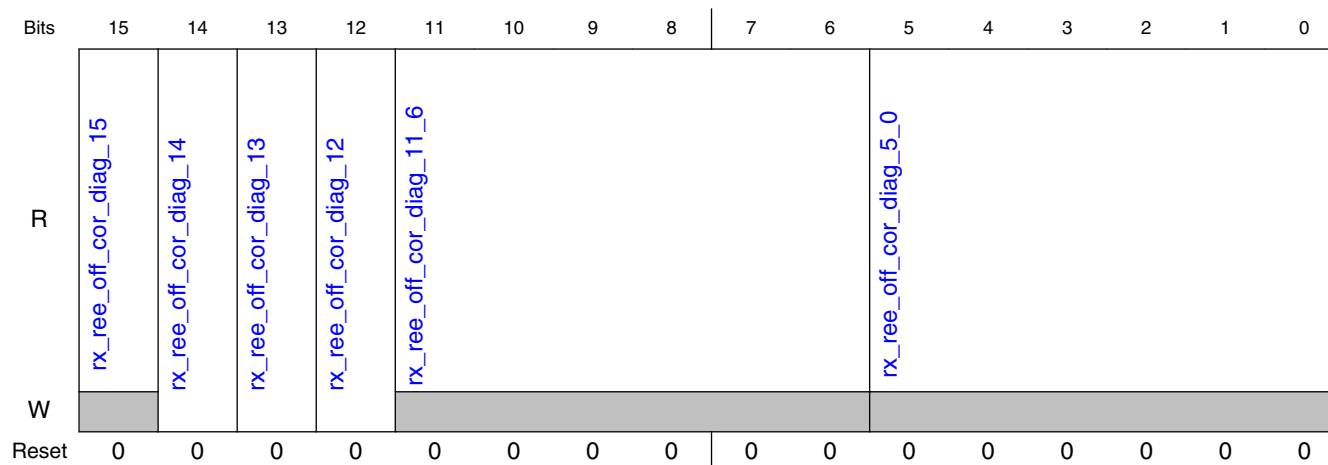
Field	Function
15-8 rx_ree_off_cor_ovrd_15_8	Reserved
7 rx_ree_off_cor_ovrd_7	Offset correction override enable: Setting this bit to a 1b1 will enable the offset correction override field in this register to override the offset correction integrator accumulator functions.
6 rx_ree_off_cor_ovrd_6	Reserved
5-0 rx_ree_off_cor_ovrd_5_0	Offset correction override value: When the offset correction override enable bit in this register is active, the value in this field will override the integrator accumulator value, as well as the input to the binary to thermometer encoder.

13.4.10.2.333 REE offset correction diagnostics register (lane0_rx_ree_off_cor_diag - lane3_rx_ree_off_cor_diag)

13.4.10.2.333.1 Offset

Register	Offset
lane0_rx_ree_off_cor_diag	81B2h
lane1_rx_ree_off_cor_diag	85B2h
lane2_rx_ree_off_cor_diag	89B2h
lane3_rx_ree_off_cor_diag	8DB2h

13.4.10.2.333.2 Diagram



13.4.10.2.333.3 Fields

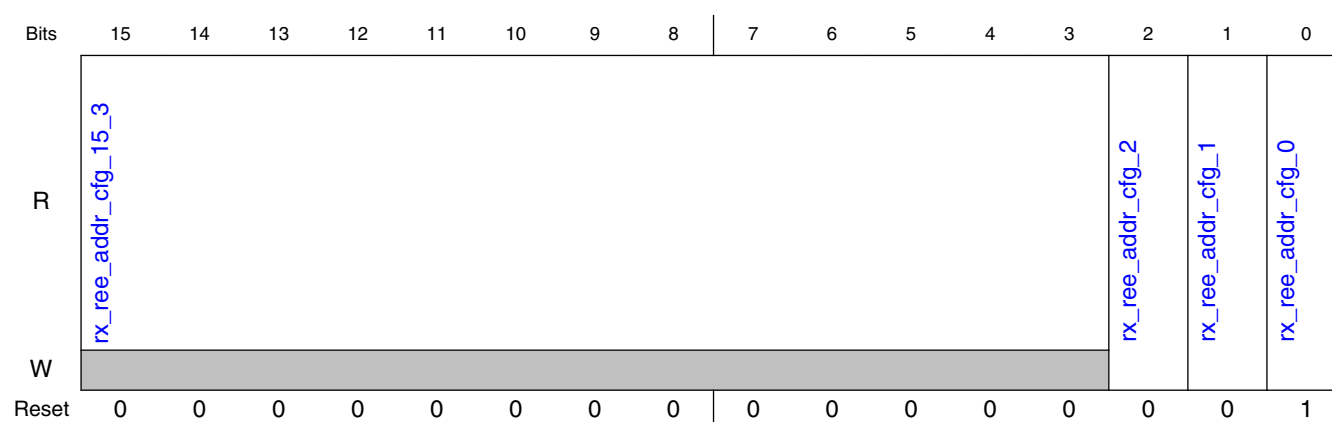
Field	Function
15 rx_ree_off_cor_diag_15	Reserved
14 rx_ree_off_cor_diag_14	Voter override neg : Writing a 1b1 in this register bit will force the offset correction voter function to activate the voter neg signal for a single clock cycle.
13 rx_ree_off_cor_diag_13	Voter override pos : Writing a 1b1 in this register bit will force the offset correction voter function to activate the voter pos signal for a single clock cycle.
12 rx_ree_off_cor_diag_12	Voter override enable : Setting this bit to a 1b1 will allow only the voter override pos and voter override neg bits in this register to control the voter in the offset correction.
11-6 rx_ree_off_cor_diag_11_6	Reserved
5-0 rx_ree_off_cor_diag_5_0	Current offset correction integrator accumulator: Current value of the offset correction integrator accumulator.

13.4.10.2.334 REE adder configuration register (lane0_rx_ree_addr_cfg - lane3_rx_ree_addr_cfg)

13.4.10.2.334.1 Offset

Register	Offset
lane0_rx_ree_addr_cfg	81B8h
lane1_rx_ree_addr_cfg	85B8h
lane2_rx_ree_addr_cfg	89B8h
lane3_rx_ree_addr_cfg	8DB8h

13.4.10.2.334.2 Diagram



13.4.10.2.334.3 Fields

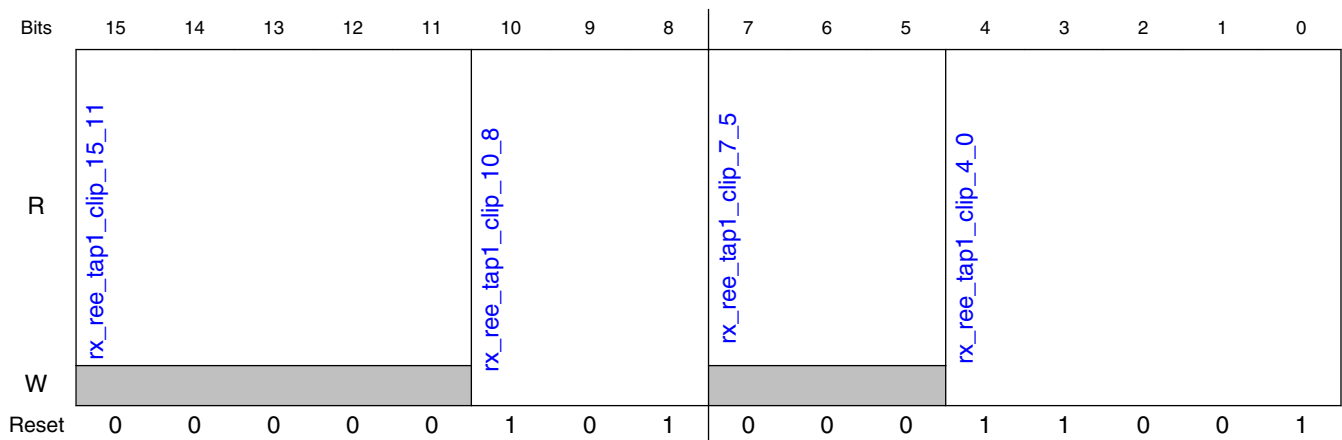
Field	Function
15-3 <code>rx_ree_addr_cfg_15_3</code>	Reserved
2 <code>rx_ree_addr_cfg_2</code>	RX peaking tap 3 adder enable: Setting this bit to 1b1, enables the results of tap 3 to be added to the RX peaking amp gain input.
1 <code>rx_ree_addr_cfg_1</code>	RX peaking tap 2 adder enable: Setting this bit to 1b1, enables the results of tap 2 to be added to the RX peaking amp gain input.
0 <code>rx_ree_addr_cfg_0</code>	RX peaking tap 1 adder enable: Setting this bit to 1b1, enables the results of tap 1 to be added to the RX peaking amp gain input.

13.4.10.2.335 REE tap 1 clip control register (lane0_rx_ree_tap1_clip - lane3_rx_ree_tap1_clip)

13.4.10.2.335.1 Offset

Register	Offset
lane0_rx_ree_tap1_clip	81B9h
lane1_rx_ree_tap1_clip	85B9h
lane2_rx_ree_tap1_clip	89B9h
lane3_rx_ree_tap1_clip	8DB9h

13.4.10.2.335.2 Diagram



13.4.10.2.335.3 Fields

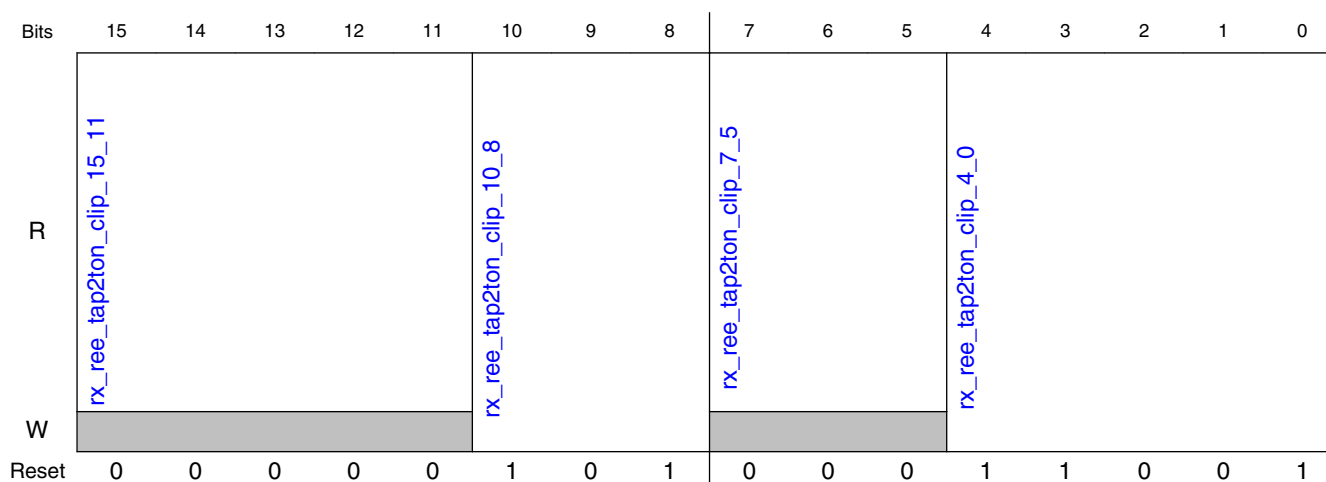
Field	Function
15-11 rx_ree_tap1_clip_15_11	Reserved
10-8 rx_ree_tap1_clip_10_8	VGA target gain adjust multiplier: Controls how much to multiply the VGA target gain adjust by, when calculating the tap threshold. The following are valid values, and the corresponding multiplier values.
7-5 rx_ree_tap1_clip_7_5	Reserved
4-0 rx_ree_tap1_clip_4_0	Threshold adjust: Controls how much the threshold can be adjusted by, after multiplying the VGA target gain adjust multiplier with the VGA target gain adjust. Note that this field is a positive number (not twos complement).

13.4.10.2.336 REE taps 2 and 3 clip control register (lane0_rx_ree_tap2ton_clip - lane3_rx_ree_tap2ton_clip)

13.4.10.2.336.1 Offset

Register	Offset
lane0_rx_ree_tap2ton_clip	81BAh
lane1_rx_ree_tap2ton_clip	85BAh
lane2_rx_ree_tap2ton_clip	89BAh
lane3_rx_ree_tap2ton_clip	8DBAh

13.4.10.2.336.2 Diagram



13.4.10.2.336.3 Fields

Field	Function
15-11 rx_ree_tap2ton_clip_15_11	Reserved
10-8 rx_ree_tap2ton_clip_10_8	VGA target gain adjust multiplier: Controls how much to multiply the VGA target gain adjust by, when calculating the tap threshold. The following are valid values, and the corresponding multiplier values.
7-5	Reserved

Table continues on the next page...

Clocks And Resets

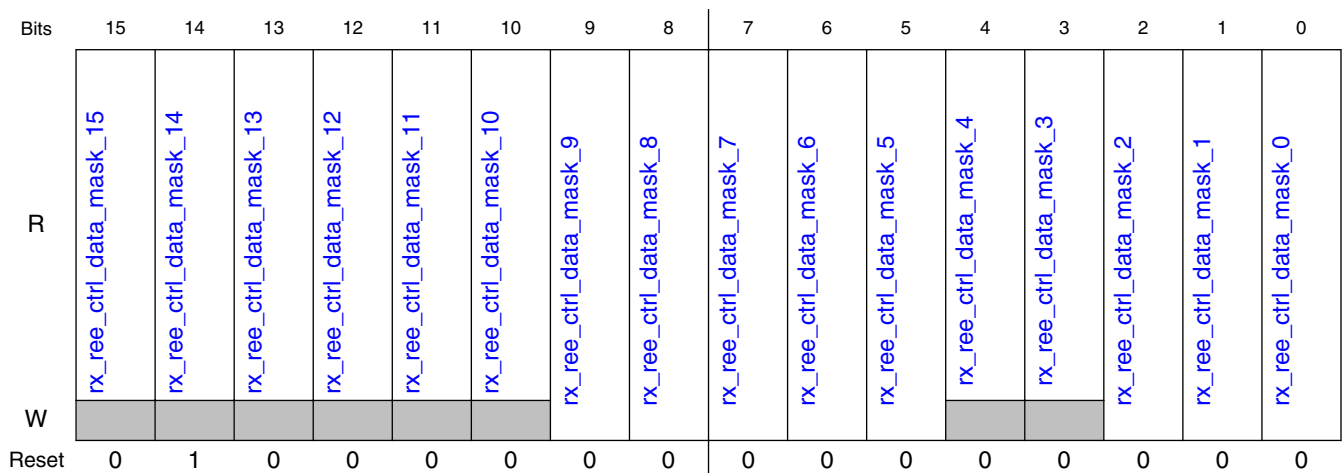
Field	Function
rx_ree_tap2ton_clip_7_5	
4-0 rx_ree_tap2ton_clip_4_0	Threshold adjust: Controls how much the threshold can be adjusted by, after multiplying the VGA target gain adjust multiplier with the VGA target gain adjust. Note that this field is a positive number (not twos complement).

13.4.10.2.337 REE control data mask register (lane0_rx_ree_ctrl_data_mask - lane3_rx_ree_ctrl_data_mask)

13.4.10.2.337.1 Offset

Register	Offset
lane0_rx_ree_ctrl_data_mask	81BBh
lane1_rx_ree_ctrl_data_mask	85BBh
lane2_rx_ree_ctrl_data_mask	89BBh
lane3_rx_ree_ctrl_data_mask	8DBBh

13.4.10.2.337.2 Diagram



13.4.10.2.337.3 Fields

Field	Function
15 rx_ree_ctrl_data_mask_15	Reserved
14 rx_ree_ctrl_data_mask_14	Ignore 1010 controller - Note that this is read only. This should never be disabled.
13 rx_ree_ctrl_data_mask_13	Reserved
12 rx_ree_ctrl_data_mask_12	Reserved
11 rx_ree_ctrl_data_mask_11	Reserved
10 rx_ree_ctrl_data_mask_10	Reserved
9 rx_ree_ctrl_data_mask_9	RX attenuation: When set to 1b0, this REE component will be turned off when control data is being received, as indicated by either the
8 rx_ree_ctrl_data_mask_8	RX VGA gain: When set to 1b0, this REE component will be turned off when control data is being received, as indicated by either the
7 rx_ree_ctrl_data_mask_7	RX offset correction coefficient: When set to 1b0, this REE component will be turned off when control data is being received, as indicated by either the
6 rx_ree_ctrl_data_mask_6	RX peaking amp gain: When set to 1b0, this REE component will be turned off when control data is being received, as indicated by either the
5 rx_ree_ctrl_data_mask_5	RX low frequency equalizer adaptive control: When set to 1b0, this REE component will be turned off when control data is being received, as indicated by either the
4 rx_ree_ctrl_data_mask_4	Reserved
3 rx_ree_ctrl_data_mask_3	Reserved
2 rx_ree_ctrl_data_mask_2	RX tap 3: When set to 1b0, this REE component will be turned off when control data is being received, as indicated by either the

Table continues on the next page...

Clocks And Resets

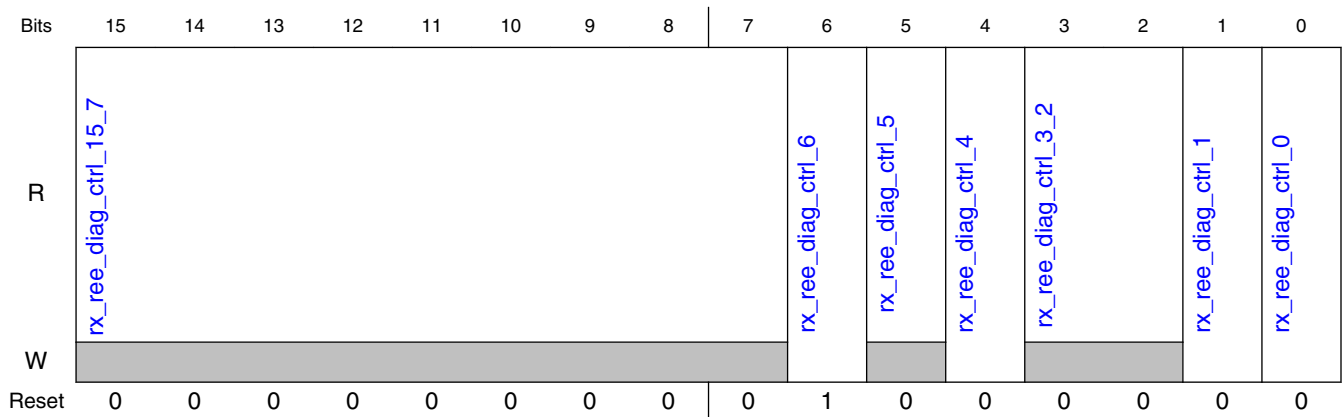
Field	Function
1 rx_ree_ctrl_data_mask_1	RX tap 2: When set to 1b0, this REE component will be turned off when control data is being received, as indicated by either the
0 rx_ree_ctrl_data_mask_0	RX tap 1: When set to 1b0, this REE component will be turned off when control data is being received, as indicated by either the

13.4.10.2.338 REE diagnostic control register (lane0_rx_ree_diag_ctrl - lane3_rx_ree_diag_ctrl)

13.4.10.2.338.1 Offset

Register	Offset
lane0_rx_ree_diag_ctrl	81BCh
lane1_rx_ree_diag_ctrl	85BCh
lane2_rx_ree_diag_ctrl	89BCh
lane3_rx_ree_diag_ctrl	8DBCh

13.4.10.2.338.2 Diagram



13.4.10.2.338.3 Fields

Field	Function
15-7 rx_ree_diag_ctrl_15_7	Reserved

Table continues on the next page...

Field	Function
6 rx_ree_diag_ctrl_6	Hold periodic equalization while RX idle: When this bit is set to 1b1, a detection of electrical idle on the receiver (
5 rx_ree_diag_ctrl_5	Reserved
4 rx_ree_diag_ctrl_4	Hold gen 2 equalization while RX idle: When this bit is set to 1b1, a detection of electrical idle on the receiver (
3-2 rx_ree_diag_ctrl_3_2	Reserved
1 rx_ree_diag_ctrl_1	Force REE controller clock on : When active, the REE controller clock gate will allow the clock to run.
0 rx_ree_diag_ctrl_0	Force REE function clock on : When active, the REE function clock gate will allow the clock to run.

13.4.10.2.339 REE control state machine gen mode control register 1 (lane 0_rx_ree_smgm_ctrl1 - lane3_rx_ree_smgm_ctrl1)

13.4.10.2.339.1 Offset

Register	Offset
lane0_rx_ree_smgm_ctrl1	81BDh
lane1_rx_ree_smgm_ctrl1	85BDh
lane2_rx_ree_smgm_ctrl1	89BDh
lane3_rx_ree_smgm_ctrl1	8DBDh

13.4.10.2.339.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W	rx_ree_smgm_ctrl1_15	rx_ree_smgm_ctrl1_14	rx_ree_smgm_ctrl1_13	rx_ree_smgm_ctrl1_12	rx_ree_smgm_ctrl1_11	rx_ree_smgm_ctrl1_10	rx_ree_smgm_ctrl1_9	rx_ree_smgm_ctrl1_8	rx_ree_smgm_ctrl1_7	rx_ree_smgm_ctrl1_6	rx_ree_smgm_ctrl1_5	rx_ree_smgm_ctrl1_4	rx_ree_smgm_ctrl1_3	rx_ree_smgm_ctrl1_2	rx_ree_smgm_ctrl1_1	rx_ree_smgm_ctrl1_0
Reset	0	0	0	0	0	1	1	1	0	0	0	0	0	0	1	0

13.4.10.2.339.3 Fields

Field	Function
15 rx_ree_smgm_ctrl1_15	REE Periodic general control state machine E path enable standard mode 3: This bit controls when the REE periodic general control state machine will enable the analog E path when running REE functions, when
14 rx_ree_smgm_ctrl1_14	REE Periodic general control state machine E path enable standard mode 2: This bit controls when the REE periodic general control state machine will enable the analog E path when running REE functions, when
13 rx_ree_smgm_ctrl1_13	REE Periodic general control state machine E path enable standard mode 1: This bit controls when the REE periodic general control state machine will enable the analog E path when running REE functions, when
12 rx_ree_smgm_ctrl1_12	REE Periodic general control state machine E path enable standard mode 0: This bit controls when the REE periodic general control state machine will enable the analog E path when running REE functions, when
11 rx_ree_smgm_ctrl1_11	REE Periodic general control state machine enable standard mode 3: This bit will control if the REE periodic general control state machine will run, when
10 rx_ree_smgm_ctrl1_10	REE Periodic general control state machine enable standard mode 2: This bit will control if the REE periodic general control state machine will run, when
9 rx_ree_smgm_ctrl1_9	REE Periodic general control state machine enable standard mode 1: This bit will control if the REE periodic general control state machine will run, when
8 rx_ree_smgm_ctrl1_8	REE Periodic general control state machine enable standard mode 0: This bit will control if the REE periodic general control state machine will run, when
7 rx_ree_smgm_ctrl1_7	REE Gen 2 general control state machine E path enable standard mode 3: This bit controls when the REE Gen 2 general control state machine will enable the analog E path when running REE functions, when

Table continues on the next page...

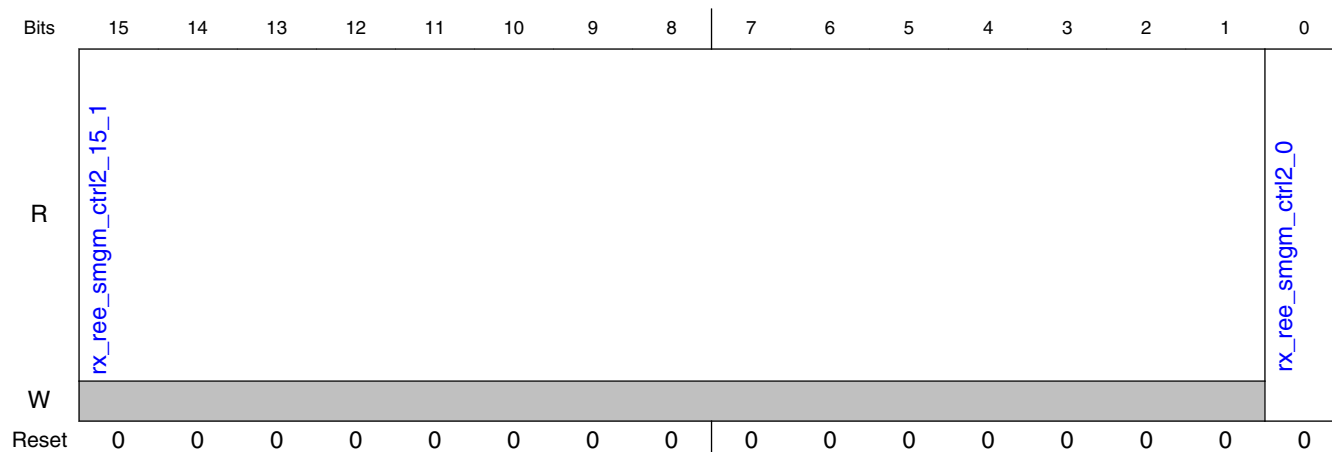
Field	Function
6 rx_ree_smgm_ctl1_6	REE Gen 2 general control state machine E path enable standard mode 2: This bit controls when the REE Gen 2 general control state machine will enable the analog E path when running REE functions, when
5 rx_ree_smgm_ctl1_5	REE Gen 2 general control state machine E path enable standard mode 1: This bit controls when the REE Gen 2 general control state machine will enable the analog E path when running REE functions, when
4 rx_ree_smgm_ctl1_4	REE Gen 2 general control state machine E path enable standard mode 0: This bit controls when the REE Gen 2 general control state machine will enable the analog E path when running REE functions, when
3 rx_ree_smgm_ctl1_3	REE Gen 2 general control state machine enable standard mode 3: This bit will control if the REE Gen 2 general control state machine will run, when
2 rx_ree_smgm_ctl1_2	REE Gen 2 general control state machine enable standard mode 2: This bit will control if the REE Gen 2 general control state machine will run, when
1 rx_ree_smgm_ctl1_1	REE Gen 2 general control state machine enable standard mode 1: This bit will control if the REE Gen 2 general control state machine will run, when
0 rx_ree_smgm_ctl1_0	REE Gen 2 general control state machine enable standard mode 0: This bit will control if the REE Gen 2 general control state machine will run, when

13.4.10.2.340 REE control state machine gen mode control register 2 (lane 0_rx_ree_smgm_ctrl2 - lane3_rx_ree_smgm_ctrl2)

13.4.10.2.340.1 Offset

Register	Offset
lane0_rx_ree_smgm_ctrl2	81BEh
lane1_rx_ree_smgm_ctrl2	85BEh
lane2_rx_ree_smgm_ctrl2	89BEh
lane3_rx_ree_smgm_ctrl2	8DBEh

13.4.10.2.340.2 Diagram



13.4.10.2.340.3 Fields

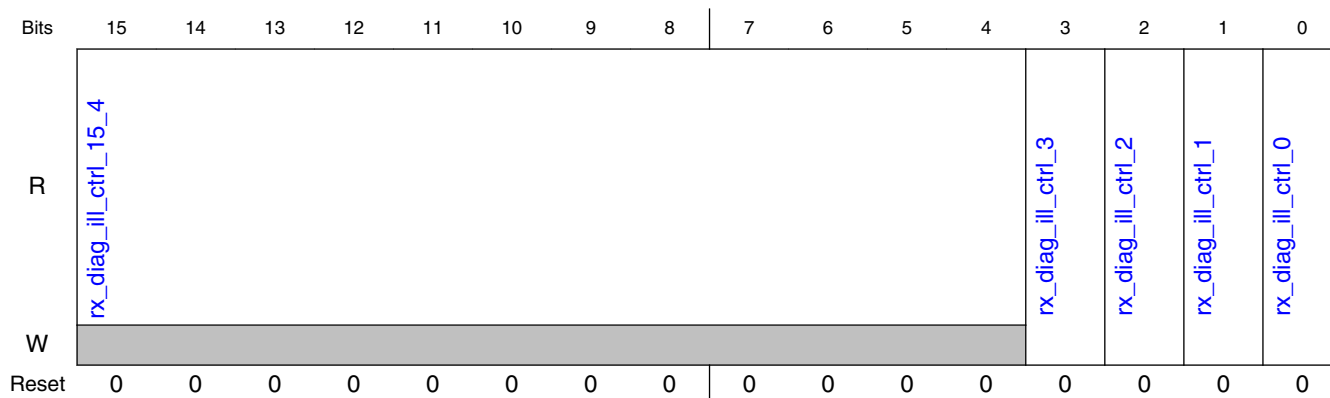
Field	Function
15-1 rx_ree_smgm_ctrl2_15_1	Reserved
0 rx_ree_smgm_ctrl2_0	REE USB 3 general control state machine E path enable: This bit controls when the REE USB 3 general control state machine will enable the analog E path when running REE functions.

13.4.10.2.341 RX ILL diagnostic control register (lane0_rx_diag_ill_ctrl - lane3_rx_diag_ill_ctrl)

13.4.10.2.341.1 Offset

Register	Offset
lane0_rx_diag_ill_ctrl	81C0h
lane1_rx_diag_ill_ctrl	85C0h
lane2_rx_diag_ill_ctrl	89C0h
lane3_rx_diag_ill_ctrl	8DC0h

13.4.10.2.341.2 Diagram



13.4.10.2.341.3 Fields

Field	Function
15-4 rx_diag_ill_ctrl_15_4	Reserved
3 rx_diag_ill_ctrl_3	IQ PI ILL calibration enable override enable: When active (1b1), the IQ PI ILL calibration enable override bit in this register, can be used to directly control the enable of the IQ PI ILL calibration function in the IQ PI ILL (instead of the IQ PI ILL calibration module).
2 rx_diag_ill_ctrl_2	IQ PI ILL calibration enable override: When enabled by the IQ PI ILL calibration enable override enable bit in this register, this bit will directly control the enable of the IQ PI ILL calibration function in the IQ PI ILL.
1 rx_diag_ill_ctrl_1	E PI ILL calibration enable override enable: When active (1b1), the E PI ILL calibration enable override bit in this register, can be used to directly control the enable of the E PI ILL calibration function in the E PI ILL (instead of the E PI ILL calibration module).
0 rx_diag_ill_ctrl_0	E PI ILL calibration enable override: When enabled by the E PI ILL calibration enable override enable bit in this register, this bit will directly control the enable of the E PI ILL calibration function in the E PI ILL.

13.4.10.2.342 RX ILL IQ trim 0 register (lane0_rx_diag_ill_iq_trim0 - lane3_rx_diag_ill_iq_trim0)

13.4.10.2.342.1 Offset

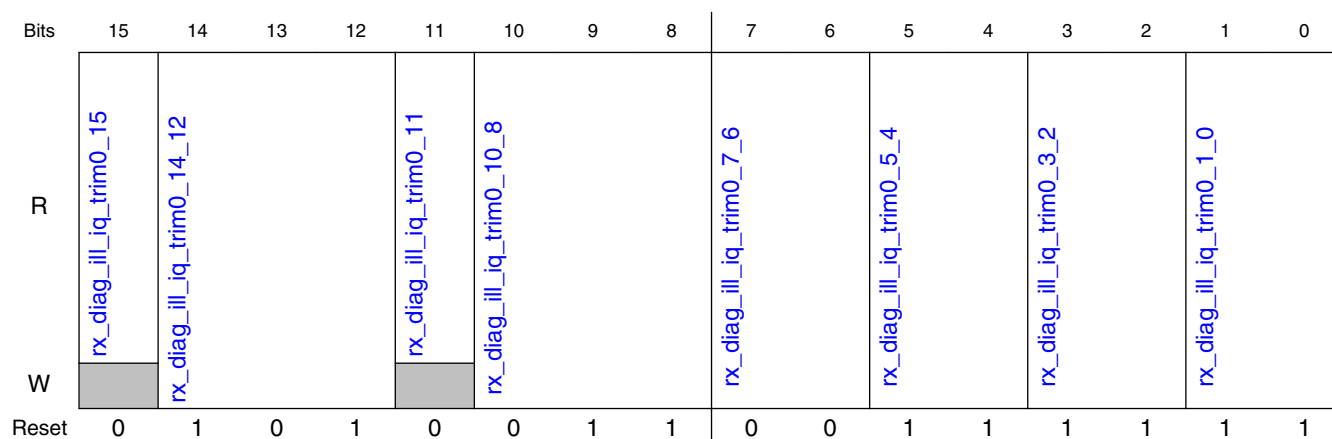
Register	Offset
lane0_rx_diag_ill_iq_trim0	81C1h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane1_rx_diag_ill_iq_trim0_m0	85C1h
lane2_rx_diag_ill_iq_trim0_m0	89C1h
lane3_rx_diag_ill_iq_trim0_m0	8DC1h

13.4.10.2.342.2 Diagram



13.4.10.2.342.3 Fields

Field	Function
15 rx_diag_ill_iq_trim0_m0_15	Reserved
14-12 rx_diag_ill_iq_trim0_m0_14_12	rx_diag_ill_iq_trim0_14_12
11 rx_diag_ill_iq_trim0_m0_11	Reserved
10-8 rx_diag_ill_iq_trim0_m0_10_8	rx_diag_ill_iq_trim0_10_8
7-6 rx_diag_ill_iq_trim0_m0_7_6	rx_diag_ill_iq_trim0_7_6
5-4 rx_diag_ill_iq_trim0_m0_5_4	rx_diag_ill_iq_trim0_5_4

Table continues on the next page...

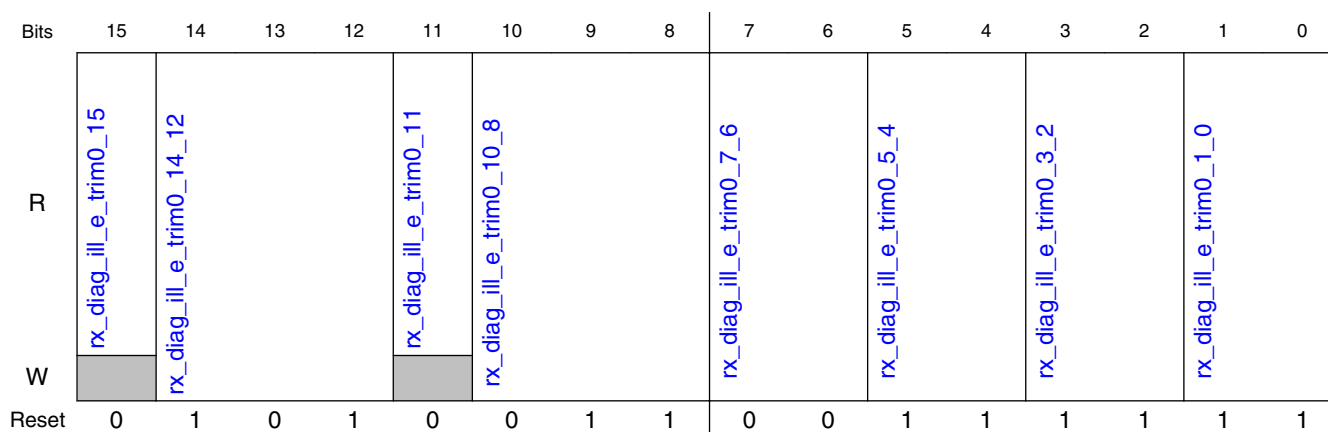
Field	Function
rx_diag_ill_iq_trim0_5_4	
3-2 rx_diag_ill_iq_trim0_3_2	rx_diag_ill_iq_trim0_3_2
1-0 rx_diag_ill_iq_trim0_1_0	rx_diag_ill_iq_trim0_1_0

13.4.10.2.343 RX ILL E trim 0 register (lane0_rx_diag_ill_e_trim0 - lane3_rx_diag_ill_e_trim0)

13.4.10.2.343.1 Offset

Register	Offset
lane0_rx_diag_ill_e_trim0	81C2h
lane1_rx_diag_ill_e_trim0	85C2h
lane2_rx_diag_ill_e_trim0	89C2h
lane3_rx_diag_ill_e_trim0	8DC2h

13.4.10.2.343.2 Diagram



13.4.10.2.343.3 Fields

Field	Function
15	Reserved

Table continues on the next page...

Clocks And Resets

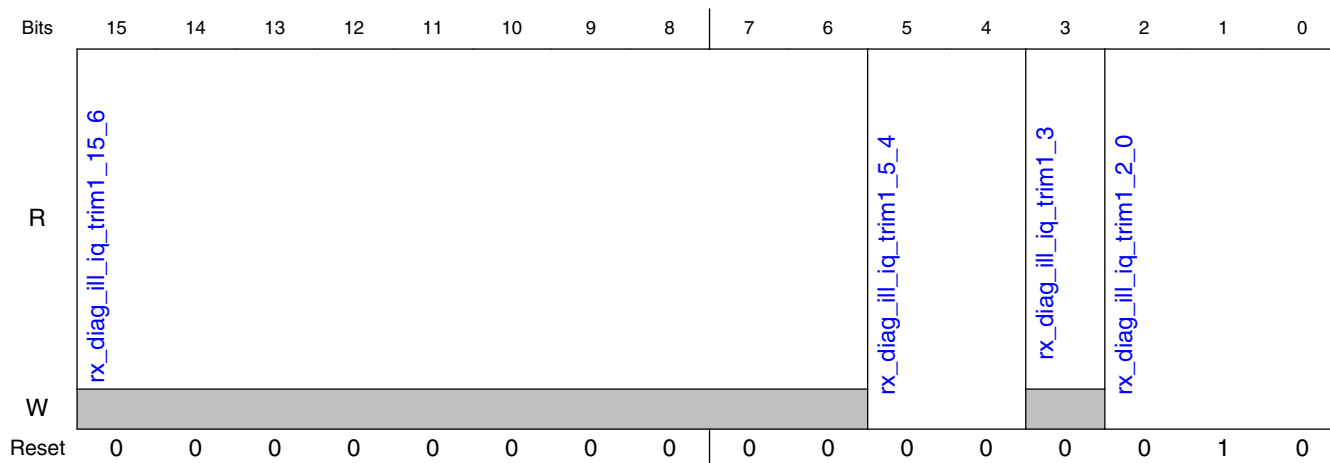
Field	Function
rx_diag_ill_e_trim0_15	
14-12 rx_diag_ill_e_trim0_14_12	rx_diag_ill_e_trim0_14_12
11 rx_diag_ill_e_trim0_11	Reserved
10-8 rx_diag_ill_e_trim0_10_8	rx_diag_ill_e_trim0_10_8
7-6 rx_diag_ill_e_trim0_7_6	rx_diag_ill_e_trim0_7_6
5-4 rx_diag_ill_e_trim0_5_4	rx_diag_ill_e_trim0_5_4
3-2 rx_diag_ill_e_trim0_3_2	rx_diag_ill_e_trim0_3_2
1-0 rx_diag_ill_e_trim0_1_0	rx_diag_ill_e_trim0_1_0

13.4.10.2.344 RX ILL IQ trim 1 register (lane0_rx_diag_ill_iq_trim1 - lane3_rx_diag_ill_iq_trim1)

13.4.10.2.344.1 Offset

Register	Offset
lane0_rx_diag_ill_iq_trim1	81C3h
lane1_rx_diag_ill_iq_trim1	85C3h
lane2_rx_diag_ill_iq_trim1	89C3h
lane3_rx_diag_ill_iq_trim1	8DC3h

13.4.10.2.344.2 Diagram



13.4.10.2.344.3 Fields

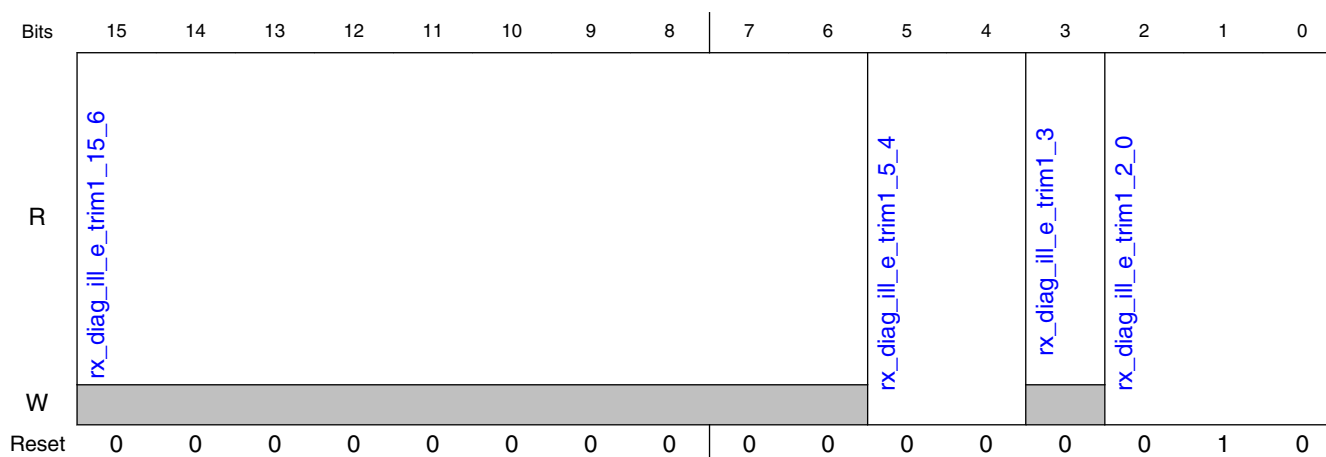
Field	Function
15-6 rx_diag_ill_iq_trim1_15_6	Reserved
5-4 rx_diag_ill_iq_trim1_5_4	Drives the
3 rx_diag_ill_iq_trim1_3	Reserved
2-0 rx_diag_ill_iq_trim1_2_0	Drives the

13.4.10.2.345 RX ILL E trim 1 register (lane0_rx_diag_ill_e_trim1 - lane3_rx_diag_ill_e_trim1)

13.4.10.2.345.1 Offset

Register	Offset
lane0_rx_diag_ill_e_trim1	81C4h
lane1_rx_diag_ill_e_trim1	85C4h
lane2_rx_diag_ill_e_trim1	89C4h
lane3_rx_diag_ill_e_trim1	8DC4h

13.4.10.2.345.2 Diagram



13.4.10.2.345.3 Fields

Field	Function
15-6 rx_diag_ill_e_trim1_15_6	Reserved
5-4 rx_diag_ill_e_trim1_5_4	Drives the
3 rx_diag_ill_e_trim1_3	Reserved
2-0 rx_diag_ill_e_trim1_2_0	Drives the

13.4.10.2.346 RX ILL IQ E trim 2 register (lane0_rx_diag_ill_iqe_trim2 - lane3_rx_diag_ill_iqe_trim2)

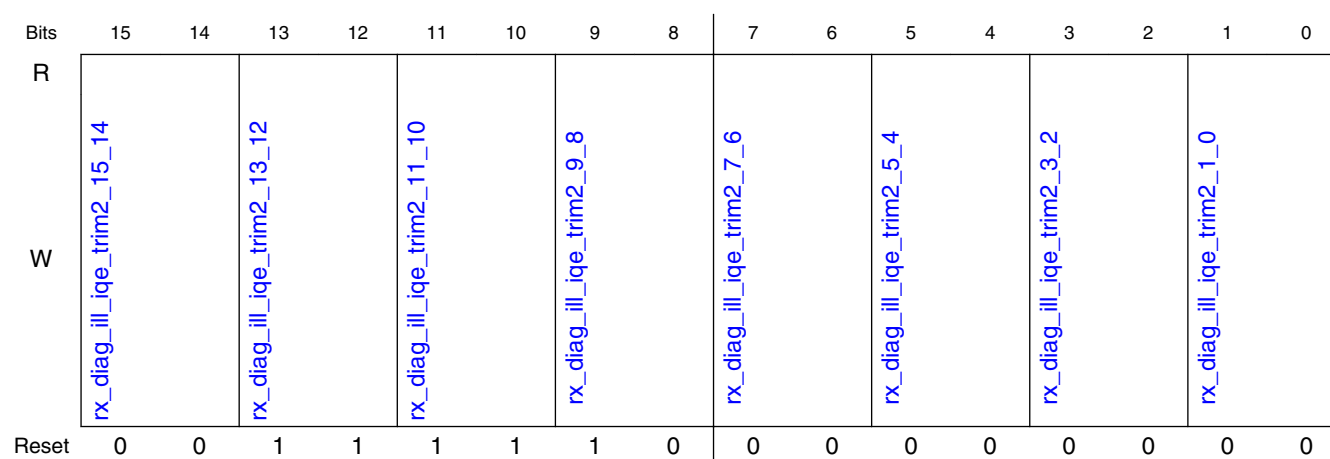
13.4.10.2.346.1 Offset

Register	Offset
lane0_rx_diag_ill_iqe_trim2	81C5h

Table continues on the next page...

Register	Offset
lane1_rx_diag_ill_iqe_trim2	85C5h
lane2_rx_diag_ill_iqe_trim2	89C5h
lane3_rx_diag_ill_iqe_trim2	8DC5h

13.4.10.2.346.2 Diagram



13.4.10.2.346.3 Fields

Field	Function
15-14 rx_diag_ill_iqe_trim2_15_14	rx_diag_ill_iqe_trim2_15_14
13-12 rx_diag_ill_iqe_trim2_13_12	rx_diag_ill_iqe_trim2_13_12
11-10 rx_diag_ill_iqe_trim2_11_10	rx_diag_ill_iqe_trim2_11_10
9-8 rx_diag_ill_iqe_trim2_9_8	rx_diag_ill_iqe_trim2_9_8
7-6 rx_diag_ill_iqe_trim2_7_6	rx_diag_ill_iqe_trim2_7_6
5-4 rx_diag_ill_iqe_trim2_5_4	rx_diag_ill_iqe_trim2_5_4

Table continues on the next page...

Clocks And Resets

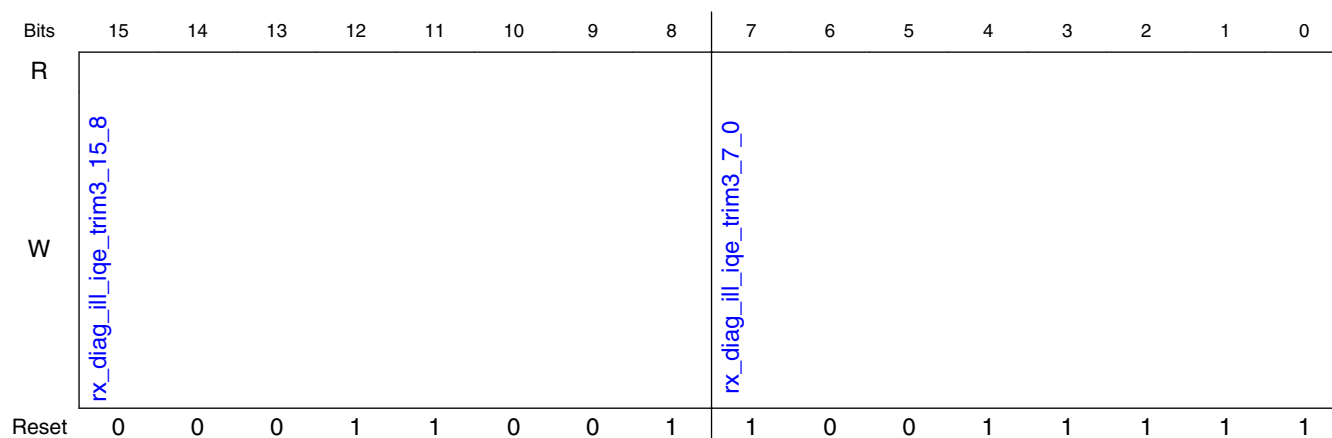
Field	Function
rx_diag_ill_iqe_trim2_5_4	
3-2 rx_diag_ill_iqe_trim2_3_2	rx_diag_ill_iqe_trim2_3_2
1-0 rx_diag_ill_iqe_trim2_1_0	rx_diag_ill_iqe_trim2_1_0

13.4.10.2.347 RX ILL IQ E trim 3 register (lane0_rx_diag_ill_iqe_trim3 - lane3_rx_diag_ill_iqe_trim3)

13.4.10.2.347.1 Offset

Register	Offset
lane0_rx_diag_ill_iqe_trim3	81C6h
lane1_rx_diag_ill_iqe_trim3	85C6h
lane2_rx_diag_ill_iqe_trim3	89C6h
lane3_rx_diag_ill_iqe_trim3	8DC6h

13.4.10.2.347.2 Diagram



13.4.10.2.347.3 Fields

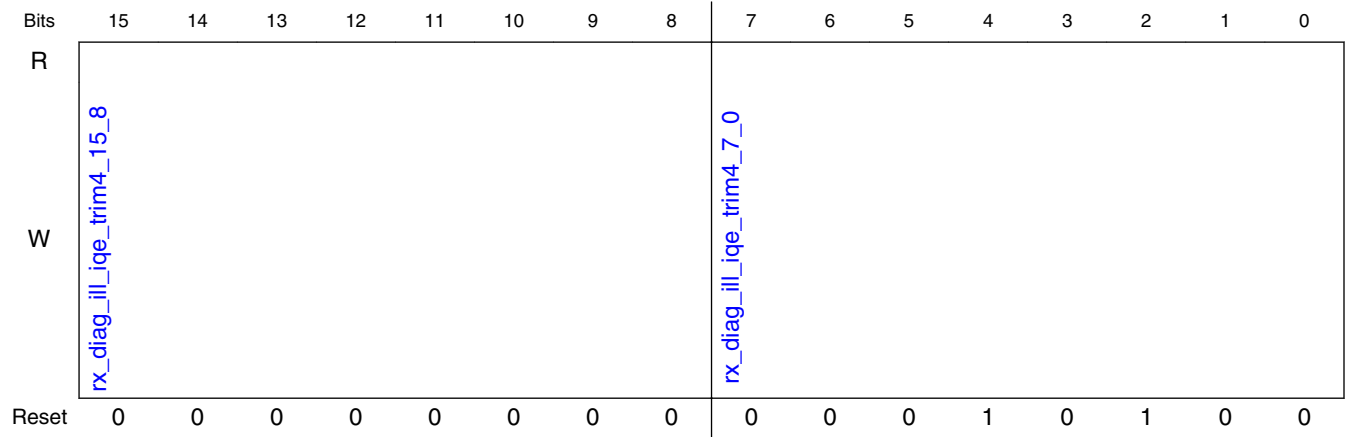
Field	Function
15-8 rx_diag_ill_iqe_trim3_15_8	rx_diag_ill_iqe_trim3_15_8
7-0 rx_diag_ill_iqe_trim3_7_0	rx_diag_ill_iqe_trim3_7_0

13.4.10.2.348 RX ILL IQ E trim 4 register (lane0_rx_diag_ill_iqe_trim4 - lane3_rx_diag_ill_iqe_trim4)

13.4.10.2.348.1 Offset

Register	Offset
lane0_rx_diag_ill_iqe_trim4	81C7h
lane1_rx_diag_ill_iqe_trim4	85C7h
lane2_rx_diag_ill_iqe_trim4	89C7h
lane3_rx_diag_ill_iqe_trim4	8DC7h

13.4.10.2.348.2 Diagram



13.4.10.2.348.3 Fields

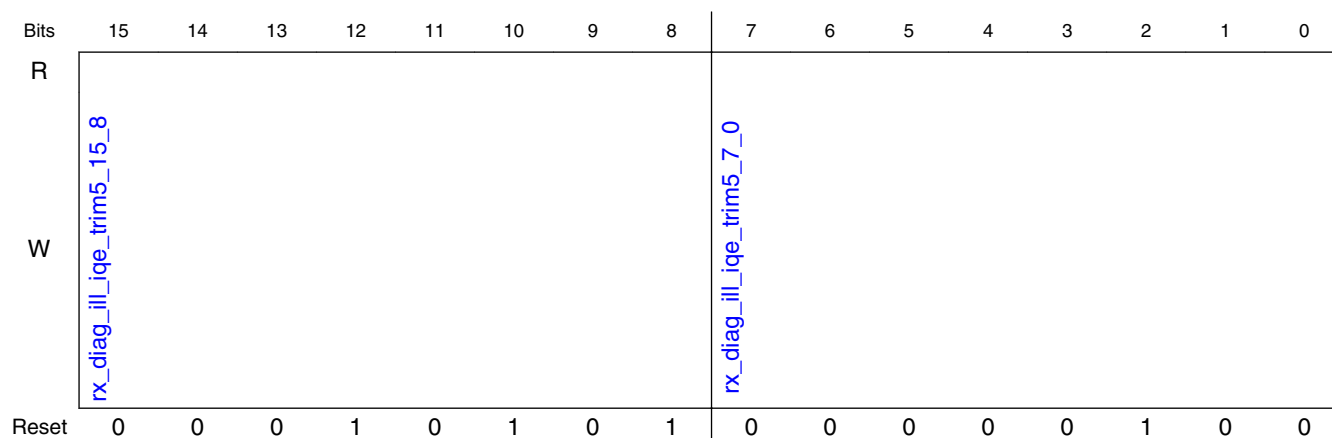
Field	Function
15-8 rx_diag_ill_iqe_trim4_15_8	rx_diag_ill_iqe_trim4_15_8
7-0 rx_diag_ill_iqe_trim4_7_0	rx_diag_ill_iqe_trim4_7_0

13.4.10.2.349 RX ILL IQ E trim 5 register (lane0_rx_diag_ill_iqe_trim5 - lane3_rx_diag_ill_iqe_trim5)

13.4.10.2.349.1 Offset

Register	Offset
lane0_rx_diag_ill_iqe_trim5	81C8h
lane1_rx_diag_ill_iqe_trim5	85C8h
lane2_rx_diag_ill_iqe_trim5	89C8h
lane3_rx_diag_ill_iqe_trim5	8DC8h

13.4.10.2.349.2 Diagram



13.4.10.2.349.3 Fields

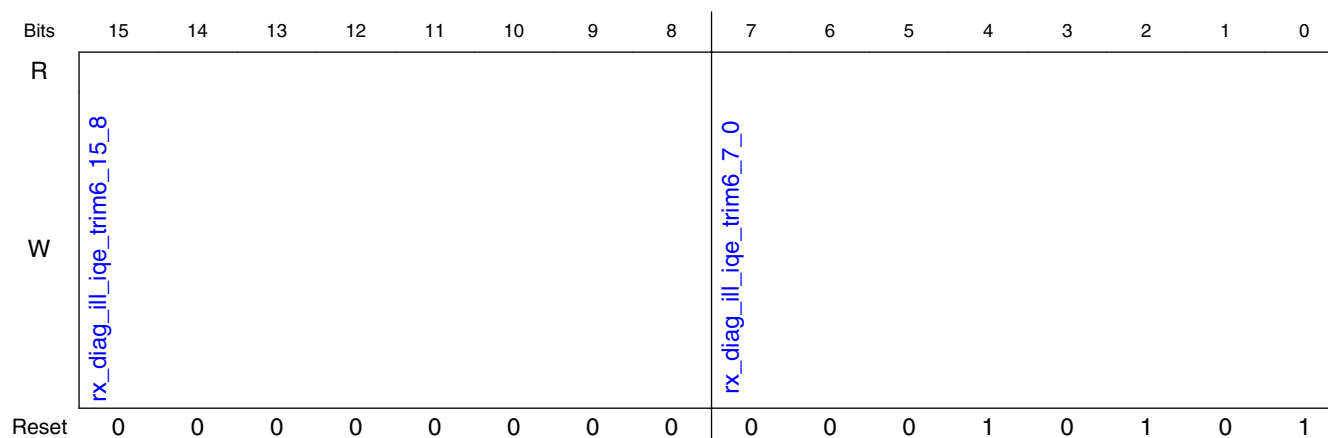
Field	Function
15-8 rx_diag_ill_iqe_trim5_15_8	rx_diag_ill_iqe_trim5_15_8
7-0 rx_diag_ill_iqe_trim5_7_0	rx_diag_ill_iqe_trim5_7_0

13.4.10.2.350 RX ILL IQ E trim 6 register (lane0_rx_diag_ill_iqe_trim6 - lane3_rx_diag_ill_iqe_trim6)

13.4.10.2.350.1 Offset

Register	Offset
lane0_rx_diag_ill_iqe_trim6	81C9h
lane1_rx_diag_ill_iqe_trim6	85C9h
lane2_rx_diag_ill_iqe_trim6	89C9h
lane3_rx_diag_ill_iqe_trim6	8DC9h

13.4.10.2.350.2 Diagram



13.4.10.2.350.3 Fields

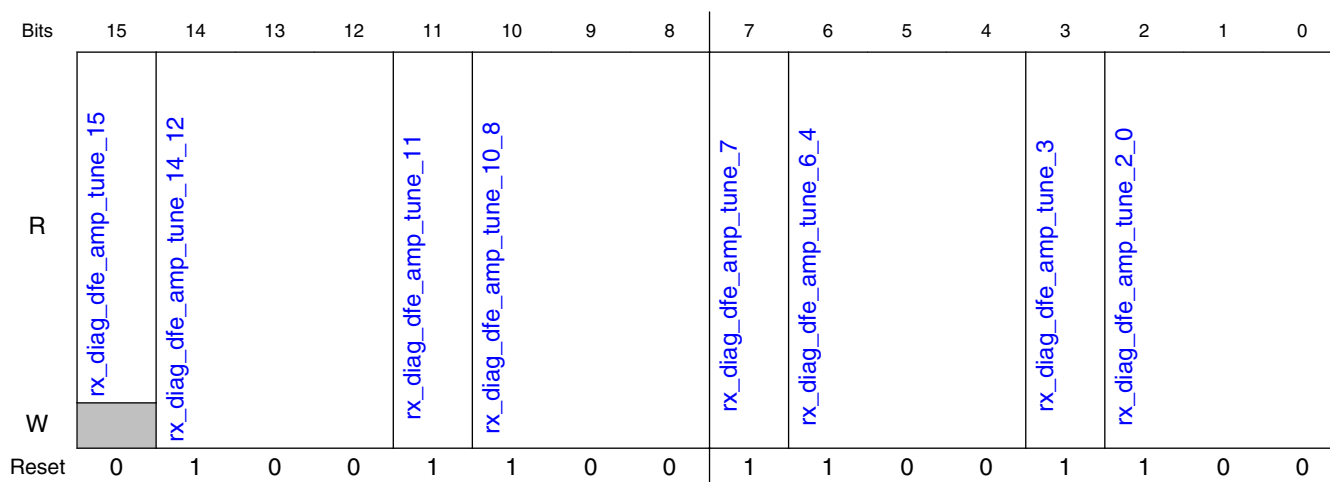
Field	Function
15-8 rx_diag_ill_iqe_trim6_15_8	rx_diag_ill_iqe_trim6_15_8
7-0 rx_diag_ill_iqe_trim6_7_0	rx_diag_ill_iqe_trim6_7_0

13.4.10.2.351 DFE amp fine tuning register (lane0_rx_diag_dfe_amp_tune - lane3_rx_diag_dfe_amp_tune)

13.4.10.2.351.1 Offset

Register	Offset
lane0_rx_diag_dfe_amp_tune	81D0h
lane1_rx_diag_dfe_amp_tune	85D0h
lane2_rx_diag_dfe_amp_tune	89D0h
lane3_rx_diag_dfe_amp_tune	8DD0h

13.4.10.2.351.2 Diagram



13.4.10.2.351.3 Fields

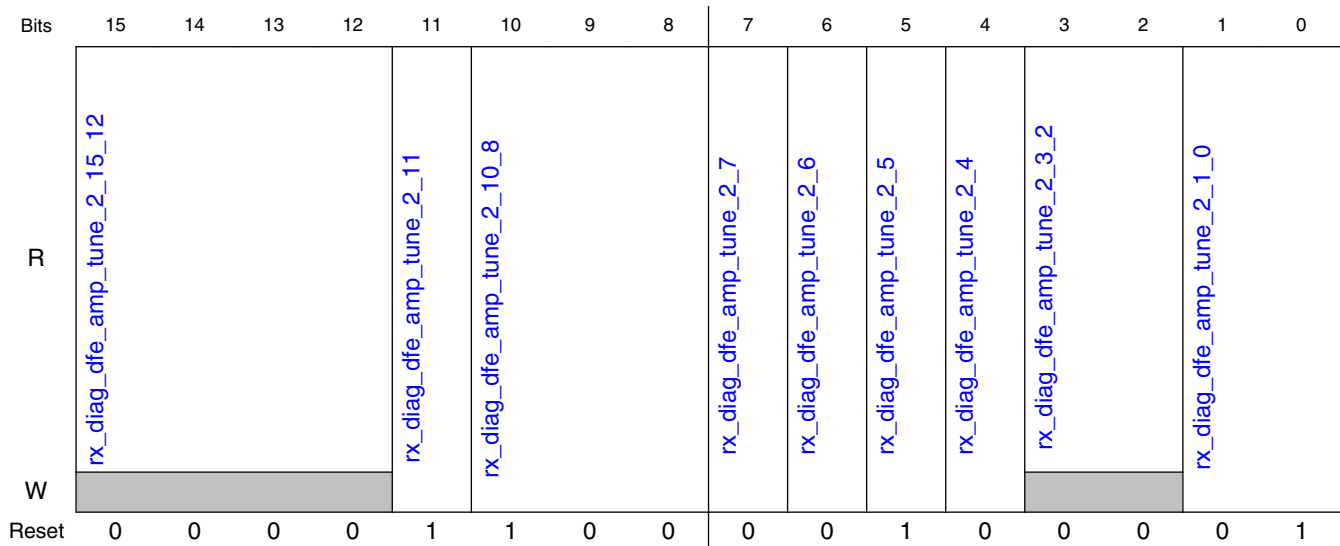
Field	Function
15 rx_diag_dfe_amp_p_tune_15	Reserved
14-12 rx_diag_dfe_amp_p_tune_14_12	DFE constant gm bias tune: Adjusts the constant gm bias.
11 rx_diag_dfe_amp_p_tune_11	DFE VGA constant gm bias enable: Enables the VGA constant gm bias. This bit drives the
10-8 rx_diag_dfe_amp_p_tune_10_8	DFE VGA amp current adjust: Adjusts the current for the DFE VGA amp, using the
7 rx_diag_dfe_amp_p_tune_7	DFE peaking constant gm bias enable: Enables the peaking constant gm bias. This bit drives the
6-4 rx_diag_dfe_amp_p_tune_6_4	DFE peaking amp current adjust: Adjusts the current for the DFE peaking amp, using the
3 rx_diag_dfe_amp_p_tune_3	DFE summing constant gm bias enable: Enables the summing constant gm bias. This bit drives the
2-0 rx_diag_dfe_amp_p_tune_2_0	DFE summing amp current adjust: Adjusts the current for the DFE summing amp, using the

13.4.10.2.352 DFE amp fine tuning 2 register (lane0_rx_diag_dfe_amp_tune_2 - lane3_rx_diag_dfe_amp_tune_2)

13.4.10.2.352.1 Offset

Register	Offset
lane0_rx_diag_dfe_amp_tune_2	81D1h
lane1_rx_diag_dfe_amp_tune_2	85D1h
lane2_rx_diag_dfe_amp_tune_2	89D1h
lane3_rx_diag_dfe_amp_tune_2	8DD1h

13.4.10.2.352.2 Diagram



13.4.10.2.352.3 Fields

Field	Function
15-12 rx_diag_dfe_amp_tune_2_15_12	Reserved
11 rx_diag_dfe_amp_tune_2_11	DFE low frequency equalizer constant gm bias enable: Enables the low frequency equalizer constant gm bias. This bit drives the
10-8 rx_diag_dfe_amp_tune_2_10_8	DFE low frequency equalizer current adjust: Adjusts the current for the DFE low frequency equalizer, using the
7 rx_diag_dfe_amp_tune_2_7	DFE peaking amp boost: Enables the active inductors boost function in the peaking amp, for high data rates. This bit controls the
6 rx_diag_dfe_amp_tune_2_6	DFE VGA stage 1 boost: Enables the active inductors boost function in stage 1 of the VGA, for high data rates. This bit controls the
5 rx_diag_dfe_amp_tune_2_5	DFE VGA stage 2 boost: Enables the active inductors boost function in stage 2 of the VGA, for high data rates. This bit controls the
4 rx_diag_dfe_amp_tune_2_4	DFE RX Tap 1 DAC Range Select: Controls the tap 1 DAC range in the analog DFE. This bit controls the

Table continues on the next page...

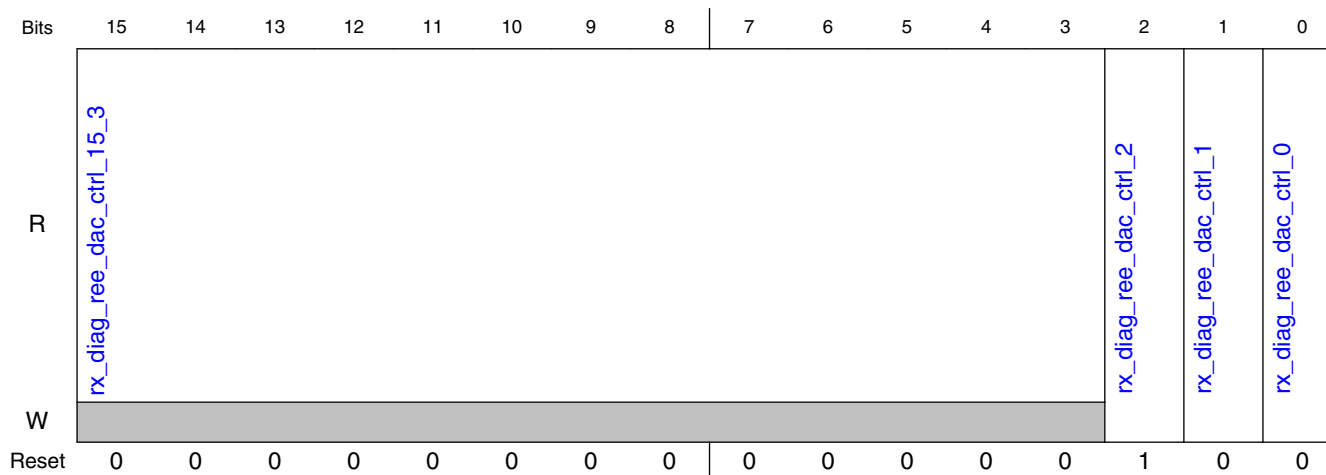
Field	Function
3-2 rx_diag_dfe_amp_tune_2_3_2	Reserved
1-0 rx_diag_dfe_amp_tune_2_1_0	DFE RX amp current adjust: Adjusts the mix of constant-gm and External Current for RX front end amplifiers. This controls the

13.4.10.2.353 REE DAC control register (lane0_rx_diag_ree_dac_ctrl - lane3_rx_diag_ree_dac_ctrl)

13.4.10.2.353.1 Offset

Register	Offset
lane0_rx_diag_ree_dac_ctrl	81D2h
lane1_rx_diag_ree_dac_ctrl	85D2h
lane2_rx_diag_ree_dac_ctrl	89D2h
lane3_rx_diag_ree_dac_ctrl	8DD2h

13.4.10.2.353.2 Diagram



13.4.10.2.353.3 Fields

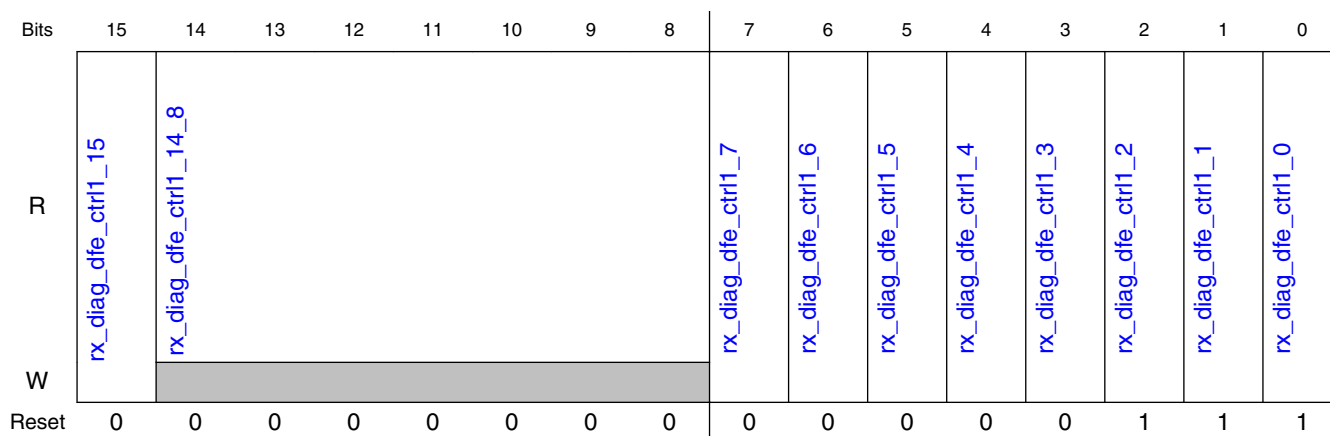
Field	Function
15-3 rx_diag_ree_dac_ctrl_15_3	Reserved
2 rx_diag_ree_dac_ctrl_2	DFE Offset DAC enable: Enables the DFE offset DAC associated with the VGA amp.
1 rx_diag_ree_dac_ctrl_1	DFE Offset DAC attenuation: Adds attenuation to the DFE offset DAC associated with the VGA amp.
0 rx_diag_ree_dac_ctrl_0	DFE DAC attenuation: Adds attenuation to the DFE DACs associated with the summing amp.

13.4.10.2.354 Receiver DFE control register 1 (lane0_rx_diag_dfe_ctrl1 - lane3_rx_diag_dfe_ctrl1)

13.4.10.2.354.1 Offset

Register	Offset
lane0_rx_diag_dfe_ctrl1	81D3h
lane1_rx_diag_dfe_ctrl1	85D3h
lane2_rx_diag_dfe_ctrl1	89D3h
lane3_rx_diag_dfe_ctrl1	8DD3h

13.4.10.2.354.2 Diagram



13.4.10.2.354.3 Fields

Field	Function
15 rx_diag_dfe_ctrl 1_15	DFE Tap 1 deserializer MUX select: When the DFE is enabled, there are two samplers that ultimately provide the input data to the deserializer. In cases where the DFE is not enabled, only one of the samplers can be used to provide data to the deserializer. This bit specifies which of these paths provides this data, by driving the
14-8 rx_diag_dfe_ctrl 1_14_8	Reserved
7 rx_diag_dfe_ctrl 1_7	Receiver DFE low frequency equalization enable value standard mode 3: This bit controls the
6 rx_diag_dfe_ctrl 1_6	Receiver DFE low frequency equalization enable value standard mode 2: This bit controls the
5 rx_diag_dfe_ctrl 1_5	Receiver DFE low frequency equalization enable value standard mode 1: This bit controls the
4 rx_diag_dfe_ctrl 1_4	Receiver DFE low frequency equalization enable value standard mode 0: This bit controls the
3 rx_diag_dfe_ctrl 1_3	Receiver DFE equalization enable mask value standard mode 3: This bit will controls the rx_dfe_eq_enable_std_mask digital signal, which can be used to mask off the
2 rx_diag_dfe_ctrl 1_2	Receiver DFE equalization enable mask value standard mode 2: This bit will controls the rx_dfe_eq_enable_std_mask digital signal, which can be used to mask off the
1 rx_diag_dfe_ctrl 1_1	Receiver DFE equalization enable mask value standard mode 1: This bit will controls the rx_dfe_eq_enable_std_mask digital signal, which can be used to mask off the
0 rx_diag_dfe_ctrl 1_0	Receiver DFE equalization enable mask value standard mode 0: This bit will controls the rx_dfe_eq_enable_std_mask digital signal, which can be used to mask off the

13.4.10.2.355 Receiver DFE control register 2 (lane0_rx_diag_dfe_ctrl2 - lane3_rx_diag_dfe_ctrl2)

13.4.10.2.355.1 Offset

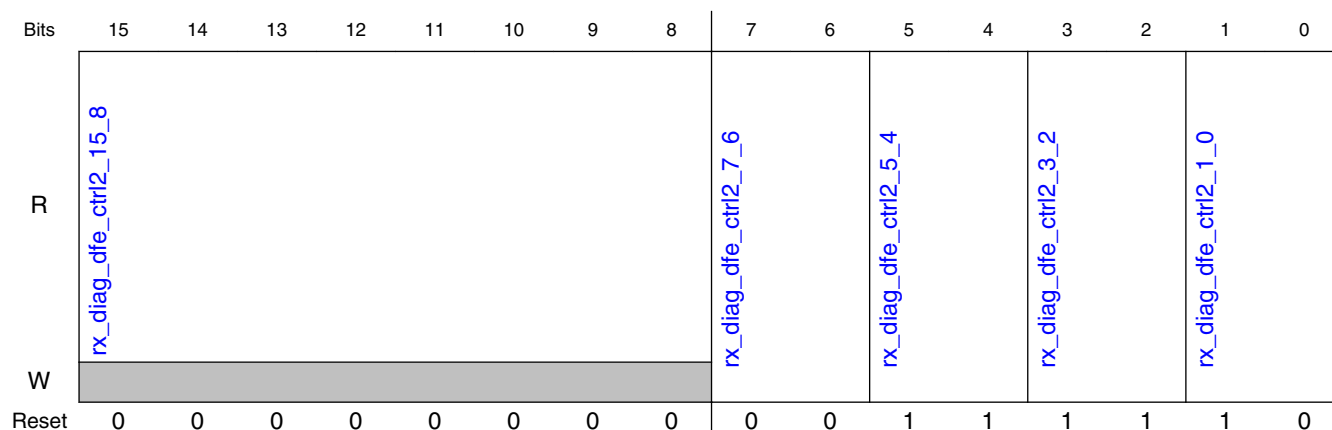
Register	Offset
lane0_rx_diag_dfe_ctrl2	81D4h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane1_rx_diag_dfe_ctrl2	85D4h
lane2_rx_diag_dfe_ctrl2	89D4h
lane3_rx_diag_dfe_ctrl2	8DD4h

13.4.10.2.355.2 Diagram



13.4.10.2.355.3 Fields

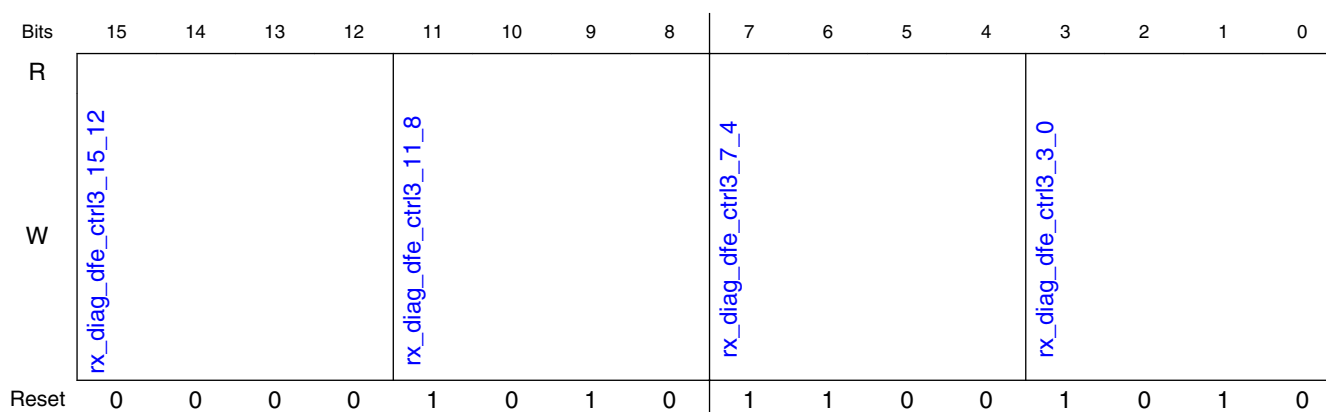
Field	Function
15-8 rx_diag_dfe_ctrl2_15_8	Reserved
7-6 rx_diag_dfe_ctrl2_7_6	RX equalizer range select standard mode 3: Thermometer encoded field that sets the equalizer to the required operating frequency range, by driving the
5-4 rx_diag_dfe_ctrl2_5_4	RX equalizer range select standard mode 2: Thermometer encoded field that sets the equalizer to the required operating frequency range, by driving the
3-2 rx_diag_dfe_ctrl2_3_2	RX equalizer range select standard mode 1: Thermometer encoded field that sets the equalizer to the required operating frequency range, by driving the
1-0 rx_diag_dfe_ctrl2_1_0	RX equalizer range select standard mode 0: Thermometer encoded field that sets the equalizer to the required operating frequency range, by driving the

13.4.10.2.356 Receiver DFE control register 3 (lane0_rx_diag_dfe_ctrl3 - lane3_rx_diag_dfe_ctrl3)

13.4.10.2.356.1 Offset

Register	Offset
lane0_rx_diag_dfe_ctrl3	81D5h
lane1_rx_diag_dfe_ctrl3	85D5h
lane2_rx_diag_dfe_ctrl3	89D5h
lane3_rx_diag_dfe_ctrl3	8DD5h

13.4.10.2.356.2 Diagram



13.4.10.2.356.3 Fields

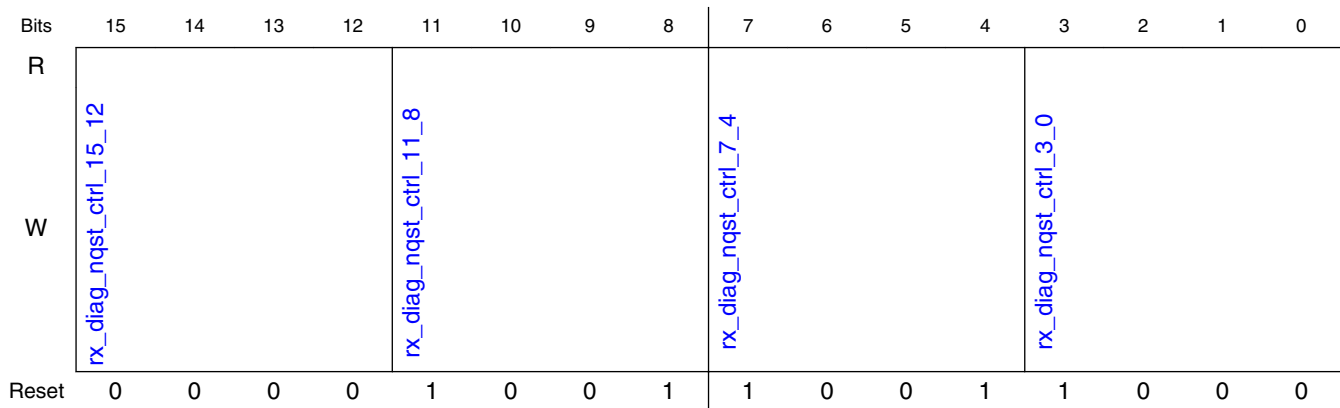
Field	Function
15-12 rx_diag_dfe_ctrl3_15_12	RX DFE peaking resistor code select standard mode 3: Thermometer encoded field used to program the DC gain/relative peaking of the equalizer, by driving the
11-8 rx_diag_dfe_ctrl3_11_8	RX DFE peaking resistor code select standard mode 2: Thermometer encoded field used to program the DC gain/relative peaking of the equalizer, by driving the
7-4 rx_diag_dfe_ctrl3_7_4	RX DFE peaking resistor code select standard mode 1: Thermometer encoded field used to program the DC gain/relative peaking of the equalizer, by driving the
3-0 rx_diag_dfe_ctrl3_3_0	RX DFE peaking resistor code select standard mode 0: Thermometer encoded field used to program the DC gain/relative peaking of the equalizer, by driving the

13.4.10.2.357 Nyquist control register (lane0_rx_diag_nqst_ctrl - lane3_rx_diag_nqst_ctrl)

13.4.10.2.357.1 Offset

Register	Offset
lane0_rx_diag_nqst_ctrl	81D6h
lane1_rx_diag_nqst_ctrl	85D6h
lane2_rx_diag_nqst_ctrl	89D6h
lane3_rx_diag_nqst_ctrl	8DD6h

13.4.10.2.357.2 Diagram



13.4.10.2.357.3 Fields

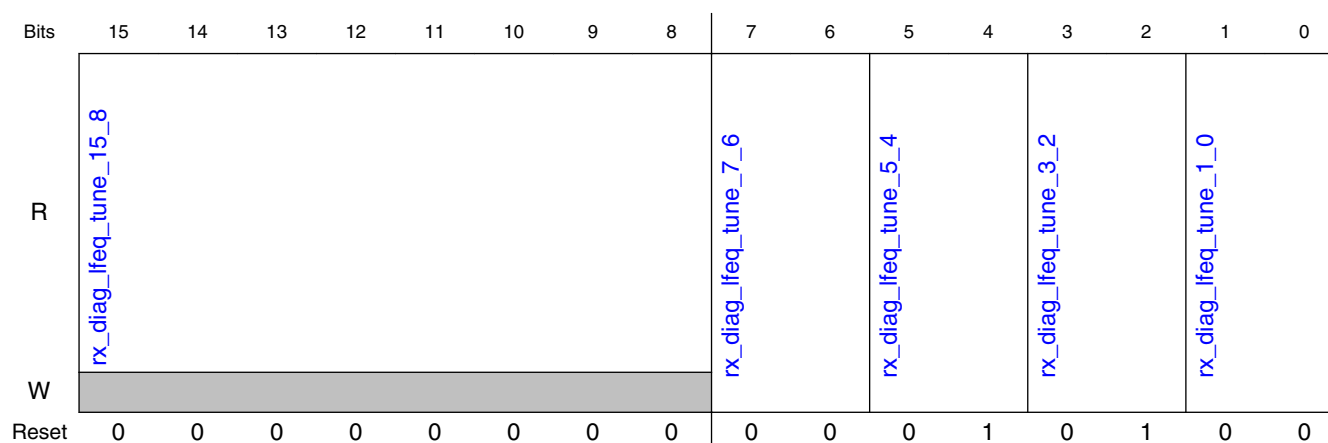
Field	Function
15-12 rx_diag_nqst_ctrl_15_12	RX nyquist select value standard mode 3: This field specifies the receiver DFE nyquist select value on the
11-8 rx_diag_nqst_ctrl_11_8	RX nyquist select value standard mode 2: This field specifies the receiver DFE nyquist select value on the
7-4 rx_diag_nqst_ctrl_7_4	RX nyquist select value standard mode 1: This field specifies the receiver DFE nyquist select value on the
3-0 rx_diag_nqst_ctrl_3_0	RX nyquist select value standard mode 0: This field specifies the receiver DFE nyquist select value on the

13.4.10.2.358 Low frequency equalizer tuning register (lane0_rx_diag_lfeq_tune - lane3_rx_diag_lfeq_tune)

13.4.10.2.358.1 Offset

Register	Offset
lane0_rx_diag_lfeq_tune	81D7h
lane1_rx_diag_lfeq_tune	85D7h
lane2_rx_diag_lfeq_tune	89D7h
lane3_rx_diag_lfeq_tune	8DD7h

13.4.10.2.358.2 Diagram



13.4.10.2.358.3 Fields

Field	Function
15-8 rx_diag_lfeq_tune_15_8	Reserved
7-6 rx_diag_lfeq_tune_7_6	RX low frequency equalization zero frequency value standard mode 3: This field specifies the receiver zero frequency setting for the low frequency equalization function, by driving the
5-4 rx_diag_lfeq_tune_5_4	RX low frequency equalization zero frequency value standard mode 2: This field specifies the receiver zero frequency setting for the low frequency equalization function, by driving the
3-2	RX low frequency equalization zero frequency value standard mode 1: This field specifies the receiver zero frequency setting for the low frequency equalization function, by driving the

Table continues on the next page...

Clocks And Resets

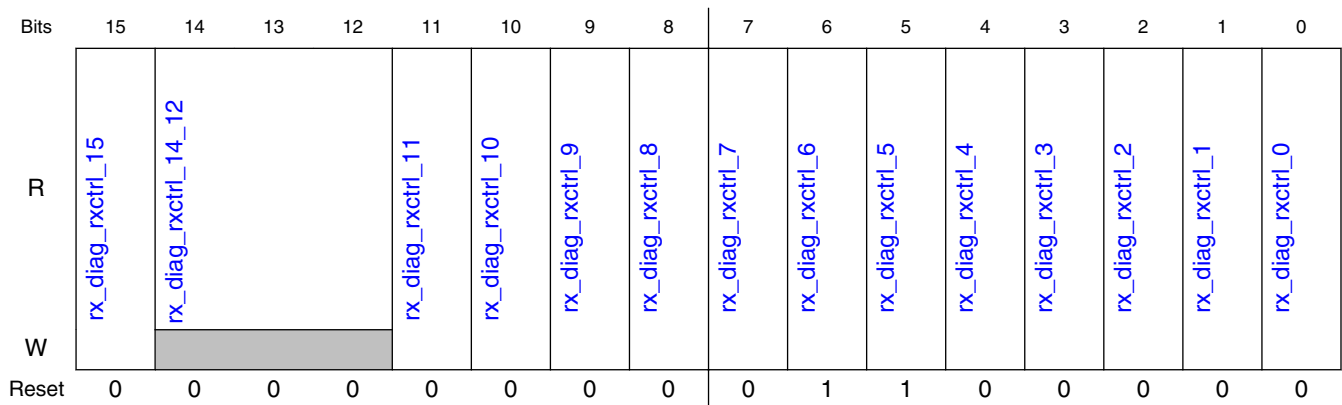
Field	Function
rx_diag_lfeq_tune_3_2	
1-0 rx_diag_lfeq_tune_1_0	RX low frequency equalization zero frequency value standard mode 0: This field specifies the receiver zero frequency setting for the low frequency equalization function, by driving the

13.4.10.2.359 RX control register (lane0_rx_diag_rxctrl - lane3_rx_diag_rxctrl)

13.4.10.2.359.1 Offset

Register	Offset
lane0_rx_diag_rxctrl	81D8h
lane1_rx_diag_rxctrl	85D8h
lane2_rx_diag_rxctrl	89D8h
lane3_rx_diag_rxctrl	8DD8h

13.4.10.2.359.2 Diagram



13.4.10.2.359.3 Fields

Field	Function
15 rx_diag_rxctrl_15	RX deserializer clock invert: This bit is used to optionally invert the deserializer clock (
14-12	Reserved

Table continues on the next page...

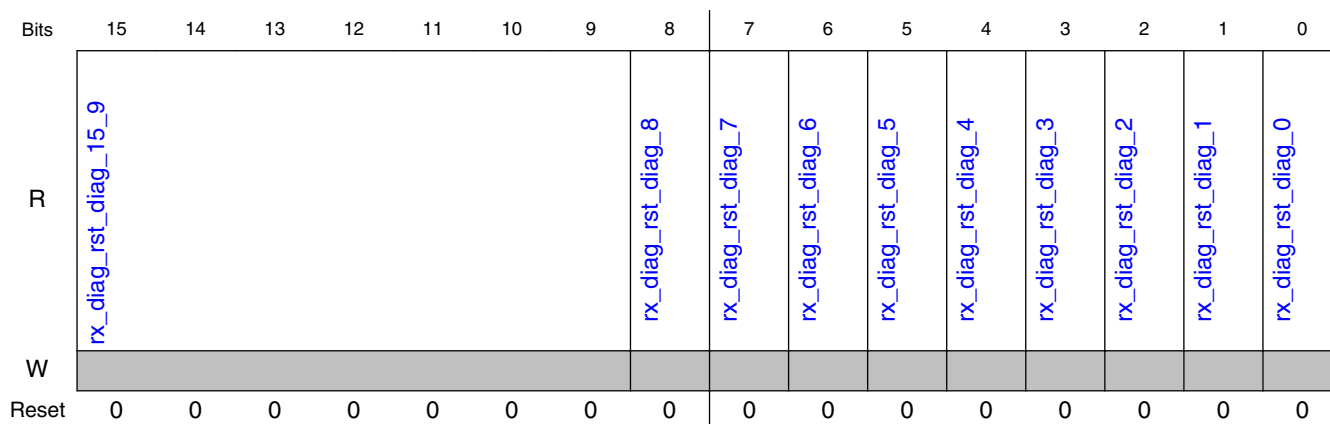
Field	Function
rx_diag_rxctrl_14_12	
11 rx_diag_rxctrl_11	PI output clock divider enable standard mode 3: This bit controls the
10 rx_diag_rxctrl_10	PI output clock divider enable standard mode 2: This bit controls the
9 rx_diag_rxctrl_9	PI output clock divider enable standard mode 1: This bit controls the
8 rx_diag_rxctrl_8	PI output clock divider enable standard mode 0: This bit controls the
7 rx_diag_rxctrl_7	Receiver CML to CMOS rate select value standard mode 3: This bit will drive the
6 rx_diag_rxctrl_6	Receiver CML to CMOS rate select value standard mode 2: This bit will drive the
5 rx_diag_rxctrl_5	Receiver CML to CMOS rate select value standard mode 1: This bit will drive the
4 rx_diag_rxctrl_4	Receiver CML to CMOS rate select value standard mode 0: This bit will drive the
3 rx_diag_rxctrl_3	RX interface subrate standard mode 3: This field specifies the receiver subrate, when
2 rx_diag_rxctrl_2	RX interface subrate standard mode 2: This field specifies the receiver subrate, when
1 rx_diag_rxctrl_1	RX interface subrate standard mode 1: This field specifies the receiver subrate, when
0 rx_diag_rxctrl_0	RX interface subrate standard mode 0: This field specifies the receiver subrate, when

13.4.10.2.360 Receiver control reset diagnostic register (lane0_rx_diag_rst_diag - lane3_rx_diag_rst_diag)

13.4.10.2.360.1 Offset

Register	Offset
lane0_rx_diag_rst_diag	81D9h
lane1_rx_diag_rst_diag	85D9h
lane2_rx_diag_rst_diag	89D9h
lane3_rx_diag_rst_diag	8DD9h

13.4.10.2.360.2 Diagram



13.4.10.2.360.3 Fields

Field	Function
15-9 rx_diag_rst_diag_15_9	Reserved
8 rx_diag_rst_diag_8	Current state of the rxda_clk_reset_n reset.
7 rx_diag_rst_diag_7	Current state of the rx_dig_reset_n reset.
6 rx_diag_rst_diag_6	Current state of the rxda_cdr1f_reset_n reset.
5 rx_diag_rst_diag_5	Current state of the rx_ree_reset_n reset.
4 rx_diag_rst_diag_4	Current state of the rx_lfps_det_filter_reset_n reset.
3 rx_diag_rst_diag_3	Current state of the rx_epi_ill_cal_lock_det_clk_reset_n reset.
2 rx_diag_rst_diag_2	Current state of the rx_epi_ill_cal_ref_clk_reset_n reset.
1	Current state of the rx_iqpi_ill_cal_lock_det_clk_reset_n reset.

Table continues on the next page...

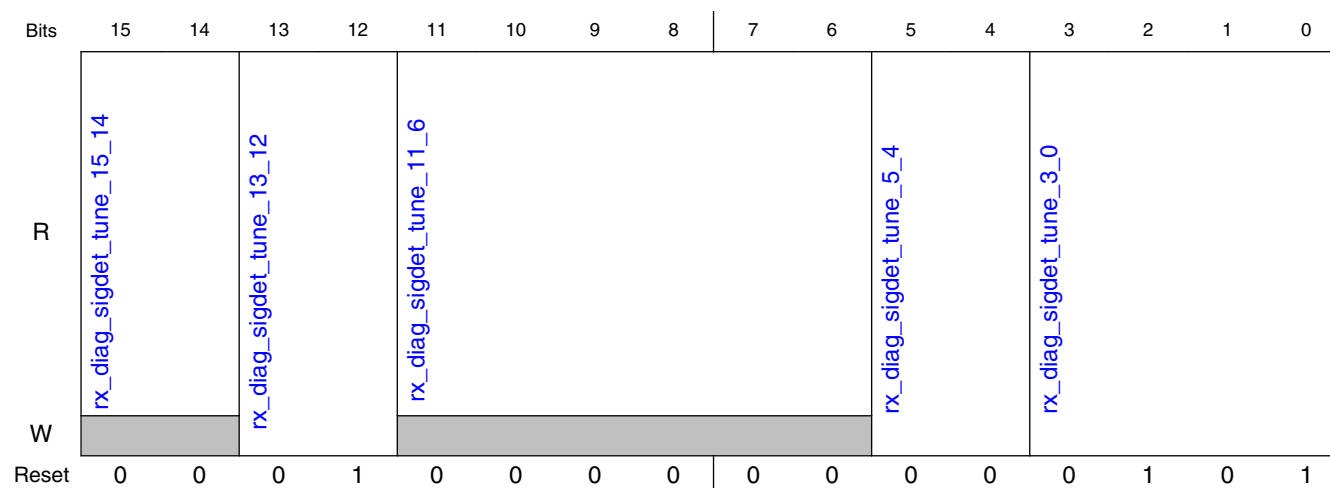
Field	Function
rx_diag_rst_diag_1	Current state of the rx_iqpi_ill_cal_ref_clk_reset_n reset.
rx_diag_rst_diag_0	

13.4.10.2.361 RX signal detect tuning and control register (lane0_rx_diag_sigdet_tune - lane3_rx_diag_sigdet_tune)

13.4.10.2.361.1 Offset

Register	Offset
lane0_rx_diag_sigdet_tune	81DCh
lane1_rx_diag_sigdet_tune	85DCh
lane2_rx_diag_sigdet_tune	89DCh
lane3_rx_diag_sigdet_tune	8DDCh

13.4.10.2.361.2 Diagram



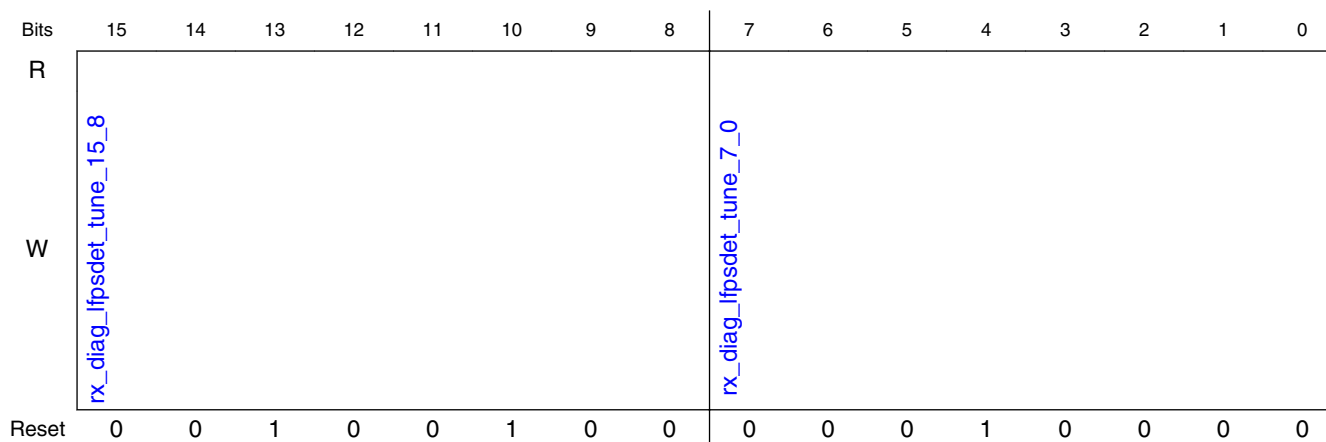
13.4.10.2.361.3 Fields

Field	Function
15-14 rx_diag_sigdet_tune_15_14	Reserved
13-12 rx_diag_sigdet_tune_13_12	Signal detect filter function select: Selects which of the two RX signal detect filter functions are enabled. As specified below.
11-6 rx_diag_sigdet_tune_11_6	Reserved
5-4 rx_diag_sigdet_tune_5_4	Signal definition to be provided by the analog team (Jira CCUPHY-2954)
3-0 rx_diag_sigdet_tune_3_0	Signal detect level: Sets the reference voltage level at which the comparators will detect a signal by driving the

13.4.10.2.362 RX LFPS detect tuning and control register (lane0_rx_diag_lfpsdet_tune - lane3_rx_diag_lfpsdet_tune)**13.4.10.2.362.1 Offset**

Register	Offset
lane0_rx_diag_lfpsdet_tune	81DDh
lane1_rx_diag_lfpsdet_tune	85DDh
lane2_rx_diag_lfpsdet_tune	89DDh
lane3_rx_diag_lfpsdet_tune	8DDDh

13.4.10.2.362.2 Diagram



13.4.10.2.362.3 Fields

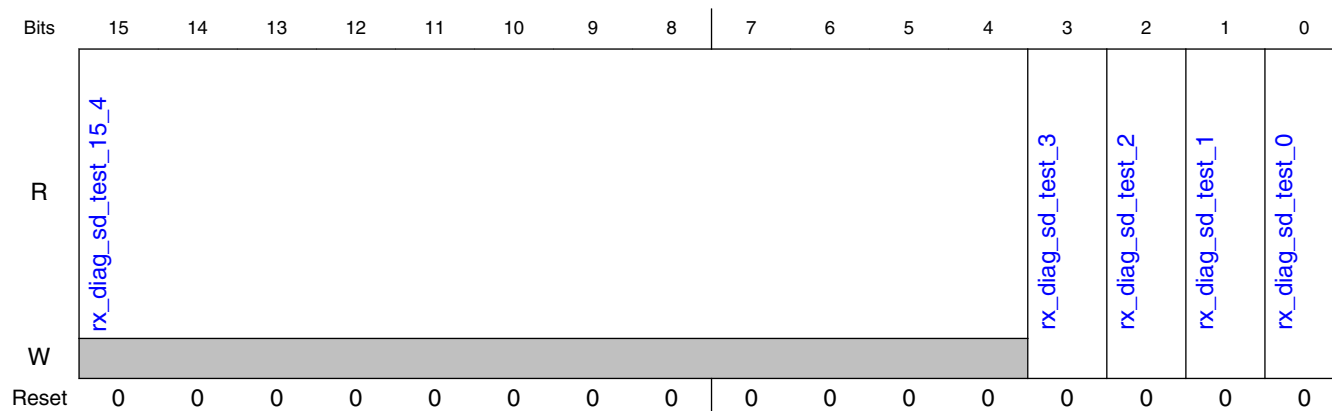
Field	Function
15-8 rx_diag_lfpsdet_tune_15_8	Signal definition to be provided by the analog team (Jira CCUPHY-2954)
7-0 rx_diag_lfpsdet_tune_7_0	LFPS detect level: Sets the reference voltage level at which the LFPS detect comparators will detect a signal. This register field drives the

13.4.10.2.363 Signal detect test register (lane0_rx_diag_sd_test - lane3_rx_diag_sd_test)

13.4.10.2.363.1 Offset

Register	Offset
lane0_rx_diag_sd_test	81DEh
lane1_rx_diag_sd_test	85DEh
lane2_rx_diag_sd_test	89DEh
lane3_rx_diag_sd_test	8DDEh

13.4.10.2.363.2 Diagram



13.4.10.2.363.3 Fields

Field	Function
15-4 rx_diag_sd_test_15_4	Reserved
3 rx_diag_sd_test_3	LFPS detected low test bit: This bit can be used to detect if the
2 rx_diag_sd_test_2	LFPS detected high test bit: This bit can be used to detect if the
1 rx_diag_sd_test_1	Signal detected low test bit: This bit can be used to detect if the
0 rx_diag_sd_test_0	Signal detected high test bit: This bit can be used to detect if the

13.4.10.2.364 RX sampler diagnostic control register (lane0_rx_diag_samp_ctrl - lane3_rx_diag_samp_ctrl)

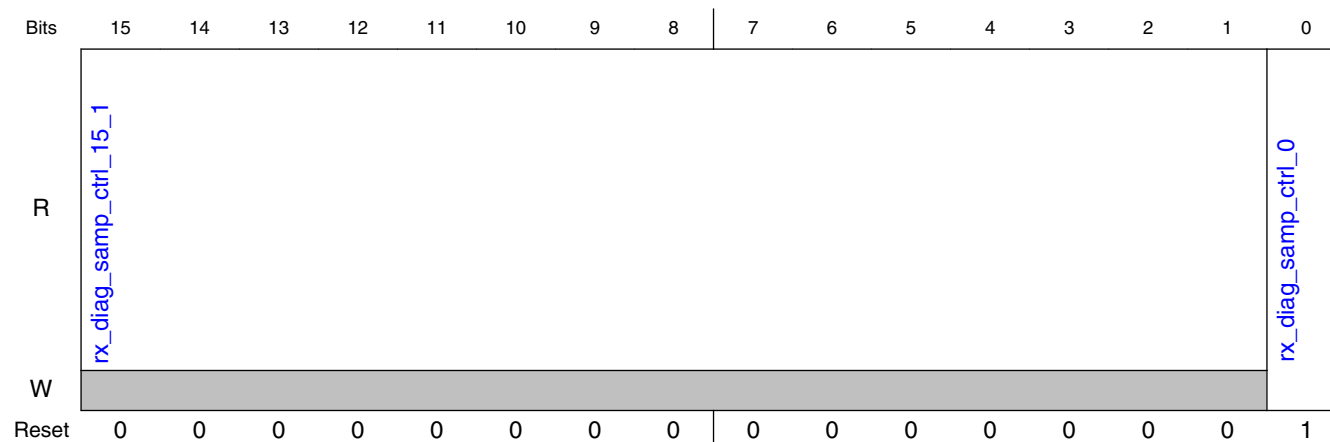
13.4.10.2.364.1 Offset

Register	Offset
lane0_rx_diag_samp_ctrl	81E0h
lane1_rx_diag_samp_ctrl	85E0h

Table continues on the next page...

Register	Offset
lane2_rx_diag_samp_ctrl	89E0h
lane3_rx_diag_samp_ctrl	8DE0h

13.4.10.2.364.2 Diagram



13.4.10.2.364.3 Fields

Field	Function
15-1 rx_diag_samp_ctrl_15_1	Reserved
0 rx_diag_samp_ctrl_0	Analog sampler

13.4.10.2.365 RX Sampler CML TO CMOS enable delay register (lane0_rx_diag_sc2c_delay - lane3_rx_diag_sc2c_delay)

13.4.10.2.365.1 Offset

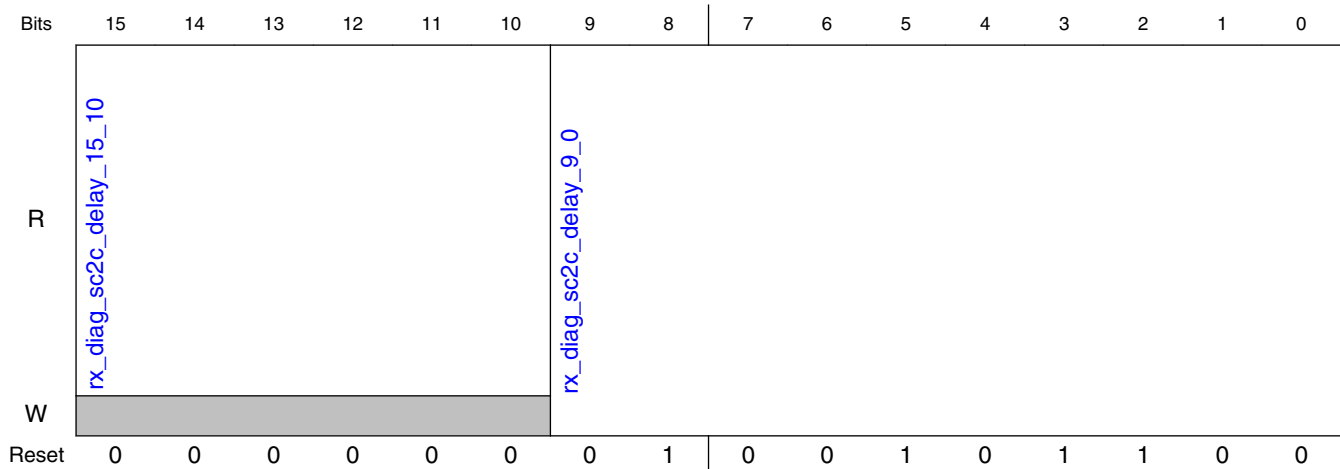
Register	Offset
lane0_rx_diag_sc2c_delay	81E1h
lane1_rx_diag_sc2c_delay	85E1h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane2_rx_diag_sc2c_delay	89E1h
lane3_rx_diag_sc2c_delay	8DE1h

13.4.10.2.365.2 Diagram



13.4.10.2.365.3 Fields

Field	Function
15-10 rx_diag_sc2c_delay_15_10	Reserved
9-0 rx_diag_sc2c_delay_9_0	Sampler CML to CMOS enable delay: This field specifies the number of PSM clocks between when

13.4.10.2.366 MPHY control register 1 (lane0_rx_diag_mphy_ctrl_1 - lane3_rx_diag_mphy_ctrl_1)

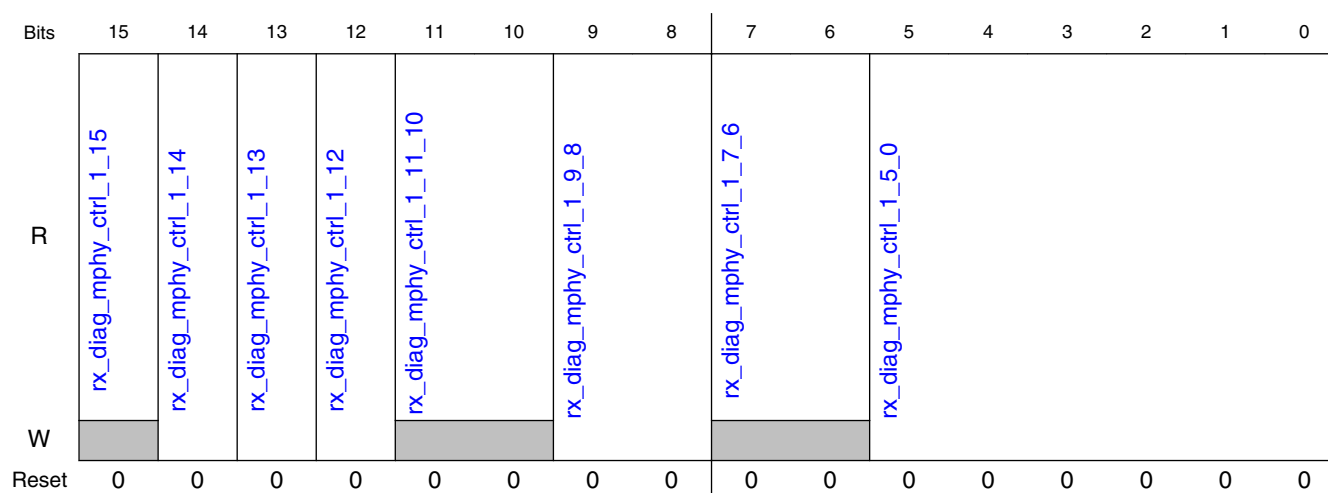
13.4.10.2.366.1 Offset

Register	Offset
lane0_rx_diag_mphy_ctrl_1	81E4h

Table continues on the next page...

Register	Offset
lane1_rx_diag_mphy_ctl_1	85E4h
lane2_rx_diag_mphy_ctl_1	89E4h
lane3_rx_diag_mphy_ctl_1	8DE4h

13.4.10.2.366.2 Diagram



13.4.10.2.366.3 Fields

Field	Function
15 rx_diag_mphy_ctl_1_15	Reserved
14 rx_diag_mphy_ctl_1_14	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
13 rx_diag_mphy_ctl_1_13	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
12 rx_diag_mphy_ctl_1_12	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
11-10 rx_diag_mphy_ctl_1_11_10	Reserved

Table continues on the next page...

Clocks And Resets

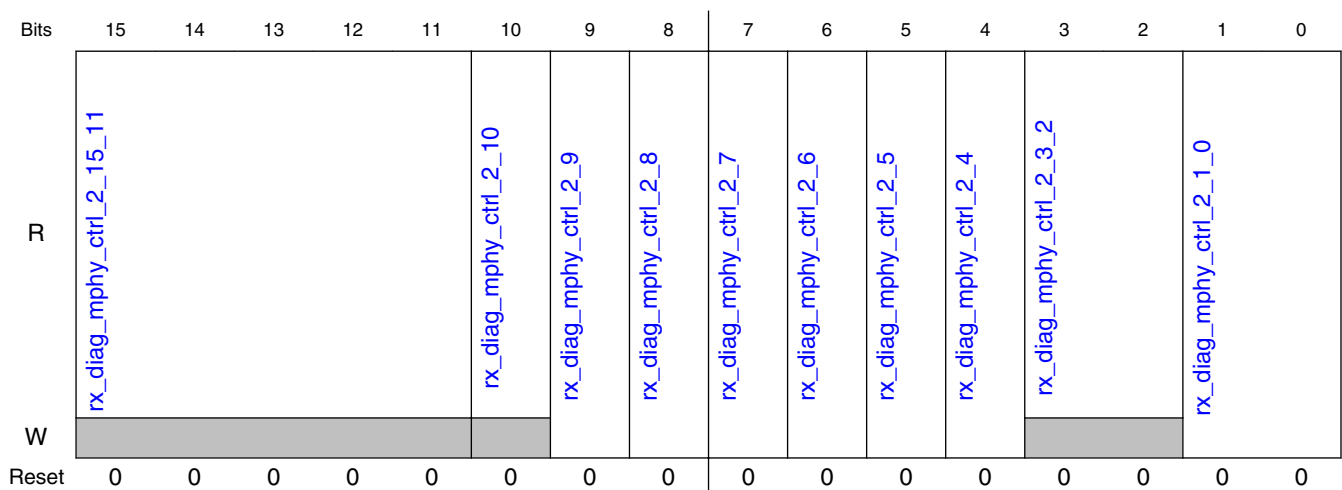
Field	Function
9-8 rx_diag_mphy_ctrl_1_9_8	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
7-6 rx_diag_mphy_ctrl_1_7_6	Reserved
5-0 rx_diag_mphy_ctrl_1_5_0	Signal definition to be provided by the analog team (Jira CCUPHY-2673)

13.4.10.2.367 MPHY control register 2 (lane0_rx_diag_mphy_ctrl_2 - lane3_rx_diag_mphy_ctrl_2)

13.4.10.2.367.1 Offset

Register	Offset
lane0_rx_diag_mphy_ctrl_2	81E5h
lane1_rx_diag_mphy_ctrl_2	85E5h
lane2_rx_diag_mphy_ctrl_2	89E5h
lane3_rx_diag_mphy_ctrl_2	8DE5h

13.4.10.2.367.2 Diagram



13.4.10.2.367.3 Fields

Field	Function
15-11 rx_diag_mphy_c trl_2_15_11	Reserved
10 rx_diag_mphy_c trl_2_10	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
9 rx_diag_mphy_c trl_2_9	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
8 rx_diag_mphy_c trl_2_8	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
7 rx_diag_mphy_c trl_2_7	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
6 rx_diag_mphy_c trl_2_6	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
5 rx_diag_mphy_c trl_2_5	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
4 rx_diag_mphy_c trl_2_4	Signal definition to be provided by the analog team (Jira CCUPHY-2673)
3-2 rx_diag_mphy_c trl_2_3_2	Reserved
1-0 rx_diag_mphy_c trl_2_1_0	Signal definition to be provided by the analog team (Jira CCUPHY-2673)

13.4.10.2.368 RX loopback controller register (lane0_rx_diag_lpbk_ctrl - lane3_rx_diag_lpbk_ctrl)

13.4.10.2.368.1 Offset

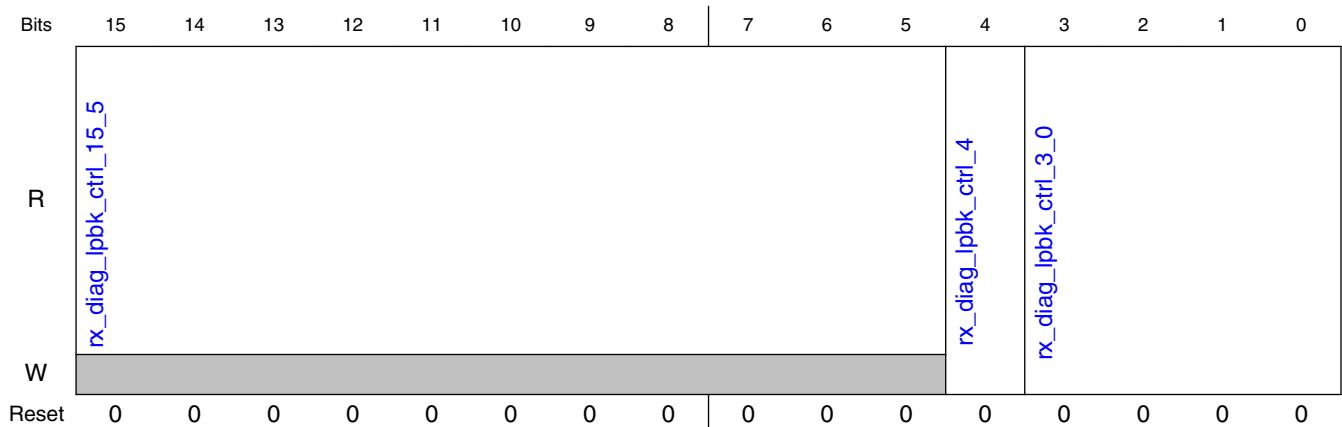
Register	Offset
lane0_rx_diag_lpbk_ctrl	81E8h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane1_rx_diag_lpbk_ctrl	85E8h
lane2_rx_diag_lpbk_ctrl	89E8h
lane3_rx_diag_lpbk_ctrl	8DE8h

13.4.10.2.368.2 Diagram



13.4.10.2.368.3 Fields

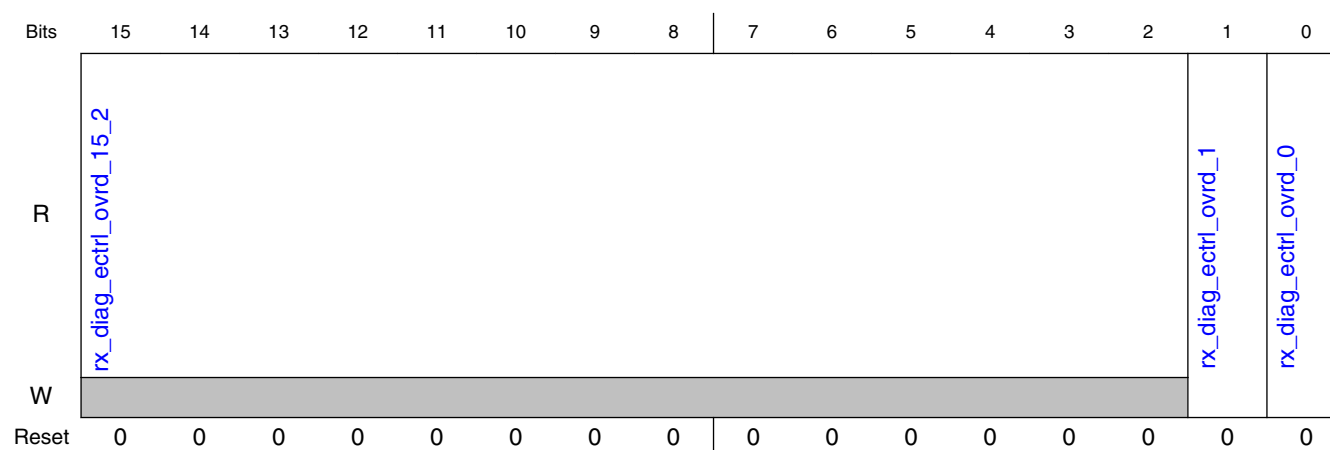
Field	Function
15-5 rx_diag_lpbk_ctrl_15_5	Reserved
4 rx_diag_lpbk_ctrl_4	Recovered clock loopback select: Selects which recovered clock to use when recovered clock loopback is enabled.
3-0 rx_diag_lpbk_ctrl_3_0	Attenuation settings: Sets the attenuation for the ISI generation loopback filter, as specified below.

13.4.10.2.369 RX extra enable control override register (lane0_rx_diag_ectrl_ovrd - lane3_rx_diag_ectrl_ovrd)

13.4.10.2.369.1 Offset

Register	Offset
lane0_rx_diag_ectrl_ovrd	81E9h
lane1_rx_diag_ectrl_ovrd	85E9h
lane2_rx_diag_ectrl_ovrd	89E9h
lane3_rx_diag_ectrl_ovrd	8DE9h

13.4.10.2.369.2 Diagram



13.4.10.2.369.3 Fields

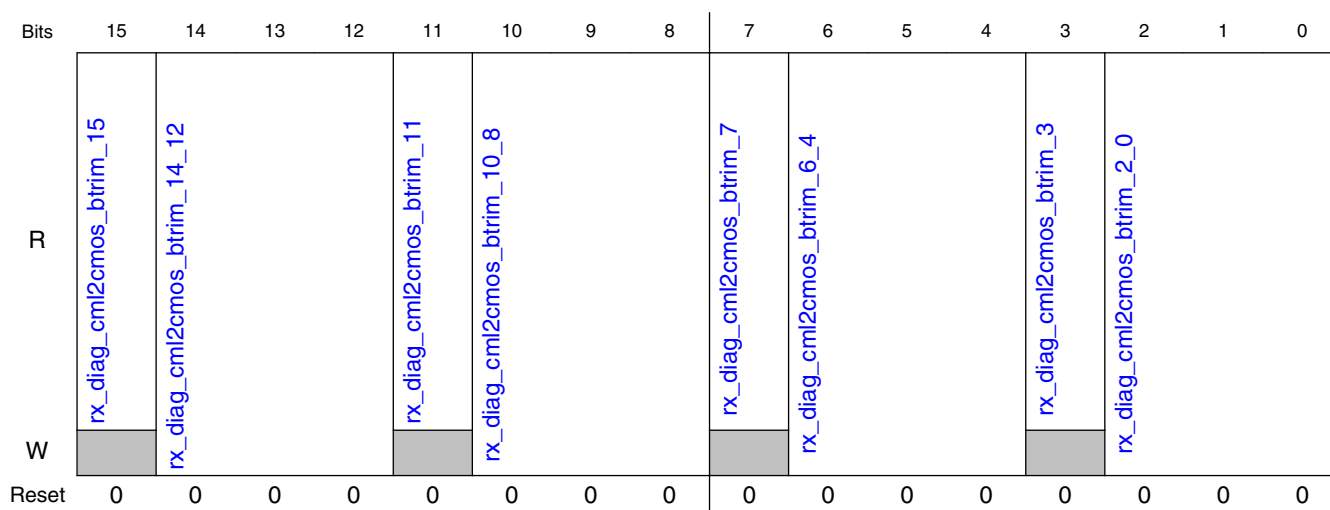
Field	Function
15-2 rx_diag_ectrl_ovrd_15_2	Reserved
1 rx_diag_ectrl_ovrd_1	Sampler CML to CMOS enable override enable : When active (1b1), the sampler CML to CMOS enable override bit in this register will drive the
0 rx_diag_ectrl_ovrd_0	Sampler CML to CMOS enable override : When enabled by the sampler CML to CMOS enable override enable bit in this register, this bit will drive the

13.4.10.2.370 CML to CMOS bias trim register (lane0_rx_diag_cml2cmos_btrim - lane3_rx_diag_cml2cmos_btrim)

13.4.10.2.370.1 Offset

Register	Offset
lane0_rx_diag_cml2cm os_btrim	81F0h
lane1_rx_diag_cml2cm os_btrim	85F0h
lane2_rx_diag_cml2cm os_btrim	89F0h
lane3_rx_diag_cml2cm os_btrim	8DF0h

13.4.10.2.370.2 Diagram



13.4.10.2.370.3 Fields

Field	Function
15 rx_diag_cml2cm os_btrim_15	Reserved
14-12 rx_diag_cml2cm os_btrim_14_12	CML to CMOS IQ bias sink current trim: Controls the
11 rx_diag_cml2cm os_btrim_11	Reserved
10-8	CML to CMOS IQ bias source current trim: Controls the

Table continues on the next page...

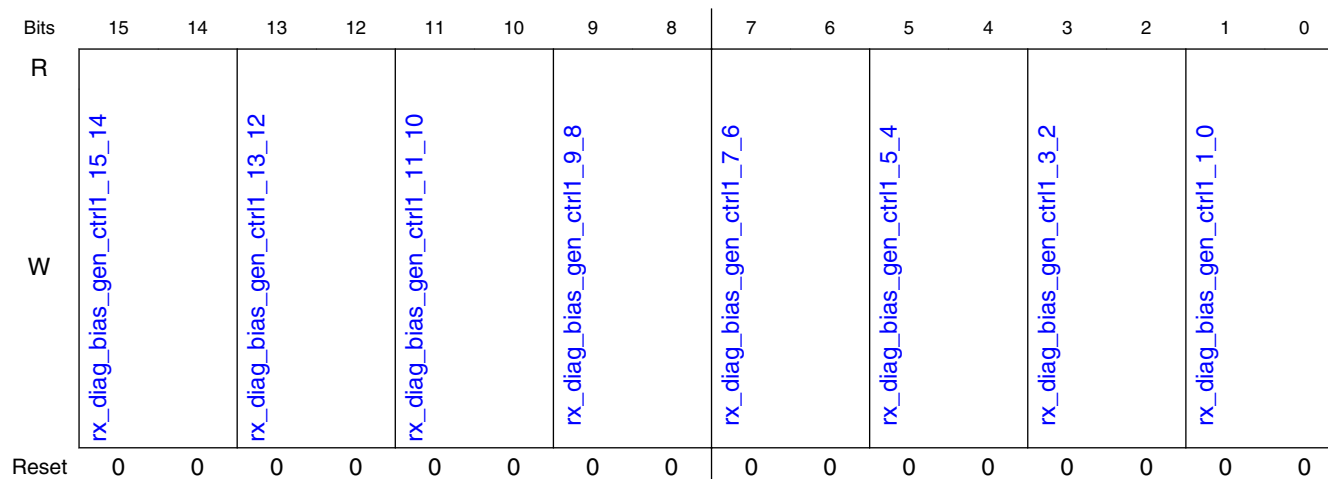
Field	Function
rx_diag_cml2cm os_btrim_10_8	
7 rx_diag_cml2cm os_btrim_7	Reserved
6-4 rx_diag_cml2cm os_btrim_6_4	CML to CMOS E bias sink current trim: Controls the
3 rx_diag_cml2cm os_btrim_3	Reserved
2-0 rx_diag_cml2cm os_btrim_2_0	CML to CMOS E bias source current trim: Controls the

13.4.10.2.371 RX bias gen control register 1 (lane0_rx_diag_bias_gen_ctrl1 - lane3_rx_diag_bias_gen_ctrl1)

13.4.10.2.371.1 Offset

Register	Offset
lane0_rx_diag_bias_gen_ctrl1	81F1h
lane1_rx_diag_bias_gen_ctrl1	85F1h
lane2_rx_diag_bias_gen_ctrl1	89F1h
lane3_rx_diag_bias_gen_ctrl1	8DF1h

13.4.10.2.371.2 Diagram



13.4.10.2.371.3 Fields

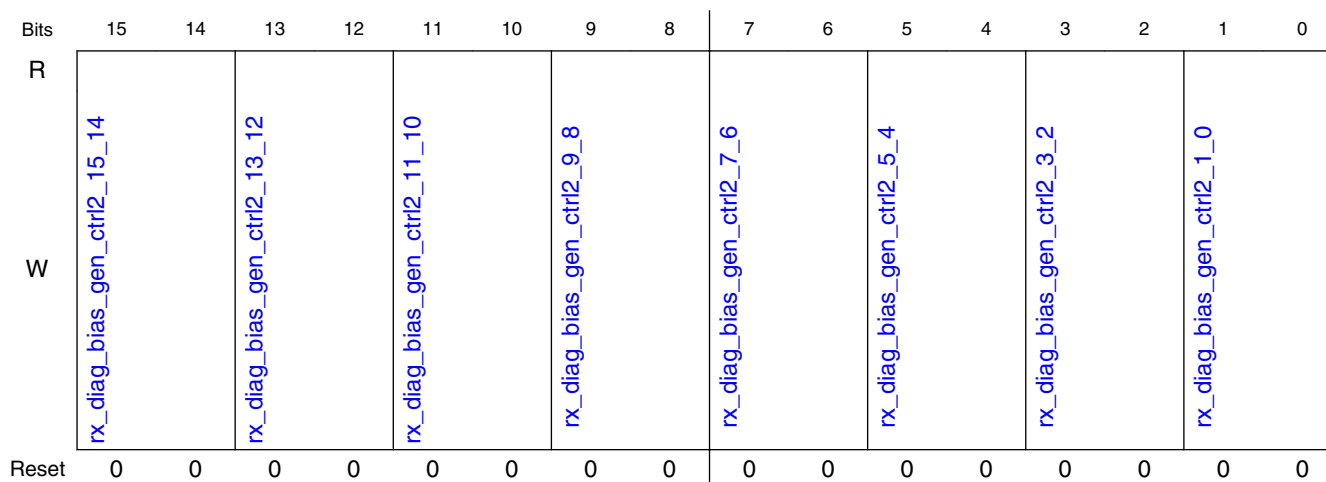
Field	Function
15-14 rx_diag_bias_gen_ctrl1_15_14	Current programmability on the biasgen current supplying block xxx, by driving
13-12 rx_diag_bias_gen_ctrl1_13_12	Current programmability on the biasgen current supplying block xxx, by driving
11-10 rx_diag_bias_gen_ctrl1_11_10	Current programmability on the biasgen current supplying block xxx, by driving
9-8 rx_diag_bias_gen_ctrl1_9_8	Current programmability on the biasgen current supplying block xxx, by driving
7-6 rx_diag_bias_gen_ctrl1_7_6	Current programmability on the biasgen current supplying block xxx, by driving
5-4 rx_diag_bias_gen_ctrl1_5_4	Current programmability on the biasgen current supplying block xxx, by driving
3-2 rx_diag_bias_gen_ctrl1_3_2	Current programmability on the biasgen current supplying block xxx, by driving
1-0 rx_diag_bias_gen_ctrl1_1_0	Current programmability on the biasgen current supplying block xxx, by driving

13.4.10.2.372 RX bias gen control register 2 (lane0_rx_diag_bias_gen_ctrl2 - lane3_rx_diag_bias_gen_ctrl2)

13.4.10.2.372.1 Offset

Register	Offset
lane0_rx_diag_bias_gen_ctrl2	81F2h
lane1_rx_diag_bias_gen_ctrl2	85F2h
lane2_rx_diag_bias_gen_ctrl2	89F2h
lane3_rx_diag_bias_gen_ctrl2	8DF2h

13.4.10.2.372.2 Diagram



13.4.10.2.372.3 Fields

Field	Function
15-14 rx_diag_bias_gen_ctrl2_15_14	Current programmability on the biasgen current supplying block xxx, by driving
13-12 rx_diag_bias_gen_ctrl2_13_12	Current programmability on the biasgen current supplying block xxx, by driving
11-10	Current programmability on the biasgen current supplying block xxx, by driving

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Clocks And Resets

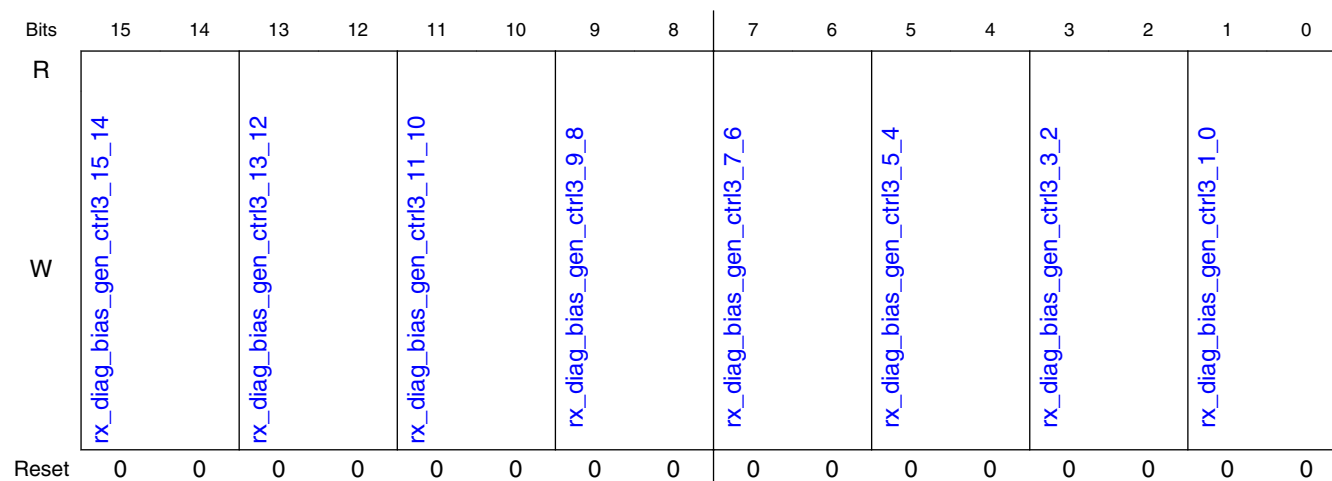
Field	Function
rx_diag_bias_gen_ctrl2_11_10	
9-8 rx_diag_bias_gen_ctrl2_9_8	Current programmability on the biasgen current supplying block xxx, by driving
7-6 rx_diag_bias_gen_ctrl2_7_6	Current programmability on the biasgen current supplying block xxx, by driving
5-4 rx_diag_bias_gen_ctrl2_5_4	Current programmability on the biasgen current supplying block xxx, by driving
3-2 rx_diag_bias_gen_ctrl2_3_2	Current programmability on the biasgen current supplying block xxx, by driving
1-0 rx_diag_bias_gen_ctrl2_1_0	Current programmability on the biasgen current supplying block xxx, by driving

13.4.10.2.373 RX bias gen control register 3 (lane0_rx_diag_bias_gen_ctrl3 - lane3_rx_diag_bias_gen_ctrl3)

13.4.10.2.373.1 Offset

Register	Offset
lane0_rx_diag_bias_gen_ctrl3	81F3h
lane1_rx_diag_bias_gen_ctrl3	85F3h
lane2_rx_diag_bias_gen_ctrl3	89F3h
lane3_rx_diag_bias_gen_ctrl3	8DF3h

13.4.10.2.373.2 Diagram



13.4.10.2.373.3 Fields

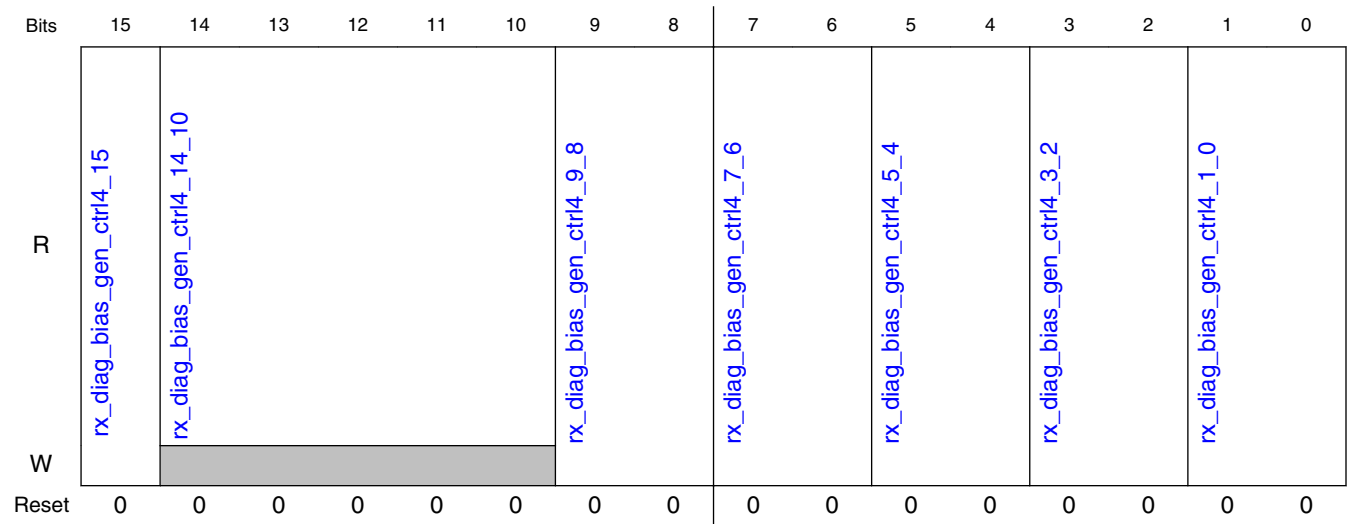
Field	Function
15-14 rx_diag_bias_gen_ctrl3_15_14	Current programmability on the biasgen current supplying block xxx, by driving
13-12 rx_diag_bias_gen_ctrl3_13_12	Current programmability on the biasgen current supplying block xxx, by driving
11-10 rx_diag_bias_gen_ctrl3_11_10	Current programmability on the biasgen current supplying block xxx, by driving
9-8 rx_diag_bias_gen_ctrl3_9_8	Current programmability on the biasgen current supplying block xxx, by driving
7-6 rx_diag_bias_gen_ctrl3_7_6	Current programmability on the biasgen current supplying block xxx, by driving
5-4 rx_diag_bias_gen_ctrl3_5_4	Current programmability on the biasgen current supplying block xxx, by driving
3-2 rx_diag_bias_gen_ctrl3_3_2	Current programmability on the biasgen current supplying block xxx, by driving
1-0 rx_diag_bias_gen_ctrl3_1_0	Current programmability on the biasgen current supplying block xxx, by driving

13.4.10.2.374 RX bias gen control register 4 (lane0_rx_diag_bias_gen_ctrl4 - lane3_rx_diag_bias_gen_ctrl4)

13.4.10.2.374.1 Offset

Register	Offset
lane0_rx_diag_bias_gen_ctrl4	81F4h
lane1_rx_diag_bias_gen_ctrl4	85F4h
lane2_rx_diag_bias_gen_ctrl4	89F4h
lane3_rx_diag_bias_gen_ctrl4	8DF4h

13.4.10.2.374.2 Diagram



13.4.10.2.374.3 Fields

Field	Function
15 rx_diag_bias_gen_ctrl4_15	Enable the base unit on all the current outputs from the RX bias generation block, by driving
14-10 rx_diag_bias_gen_ctrl4_14_10	Reserved

Table continues on the next page...

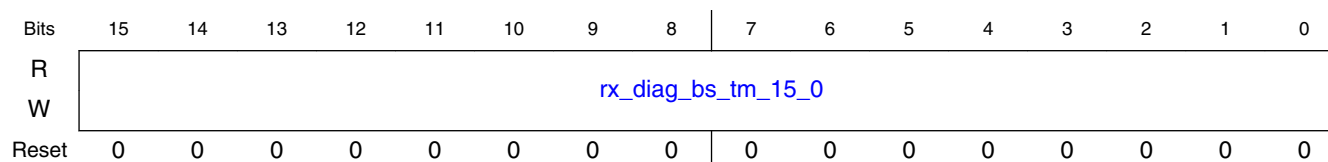
Field	Function
9-8 rx_diag_bias_ge n_ctrl4_9_8	Current programmability on the biasgen current supplying block xxx, by driving
7-6 rx_diag_bias_ge n_ctrl4_7_6	Current programmability on the biasgen current supplying block xxx, by driving
5-4 rx_diag_bias_ge n_ctrl4_5_4	Current programmability on the biasgen current supplying block xxx, by driving
3-2 rx_diag_bias_ge n_ctrl4_3_2	Current programmability on the biasgen current supplying block xxx, by driving
1-0 rx_diag_bias_ge n_ctrl4_1_0	Current programmability on the biasgen current supplying block xxx, by driving

13.4.10.2.375 RX boundary scan test mode register (lane0_rx_diag_bs_tm - lane3_rx_diag_bs_tm)

13.4.10.2.375.1 Offset

Register	Offset
lane0_rx_diag_bs_tm	81F5h
lane1_rx_diag_bs_tm	85F5h
lane2_rx_diag_bs_tm	89F5h
lane3_rx_diag_bs_tm	8DF5h

13.4.10.2.375.2 Diagram



13.4.10.2.375.3 Fields

Field	Function
15-0	Register definition to be provided by the analog team (Jira CCUPHY-2954)

Clocks And Resets

Field	Function
rx_diag_bs_tm_15_0	

13.4.10.2.376 RX receiver front end test mode register 1 (lane0_rx_diag_rxfe_tm1 - lane3_rx_diag_rxfe_tm1)

13.4.10.2.376.1 Offset

Register	Offset
lane0_rx_diag_rxfe_tm1	81F6h
lane1_rx_diag_rxfe_tm1	85F6h
lane2_rx_diag_rxfe_tm1	89F6h
lane3_rx_diag_rxfe_tm1	8DF6h

13.4.10.2.376.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	rx_diag_rxfe_tm1_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.376.3 Fields

Field	Function
15-0 rx_diag_rxfe_tm1_15_0	Register definition to be provided by the analog team (Jira CCUPHY-2954)

13.4.10.2.377 RX receiver front end test mode register 2 (lane0_rx_diag_rxfe_tm2 - lane3_rx_diag_rxfe_tm2)

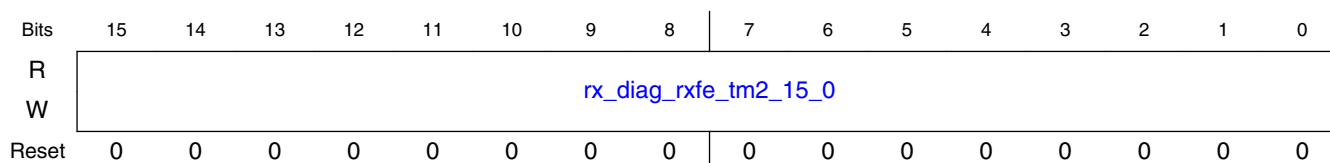
13.4.10.2.377.1 Offset

Register	Offset
lane0_rx_diag_rxfe_tm2	81F7h

Table continues on the next page...

Register	Offset
lane1_rx_diag_rxfe_tm2	85F7h
lane2_rx_diag_rxfe_tm2	89F7h
lane3_rx_diag_rxfe_tm2	8DF7h

13.4.10.2.377.2 Diagram



13.4.10.2.377.3 Fields

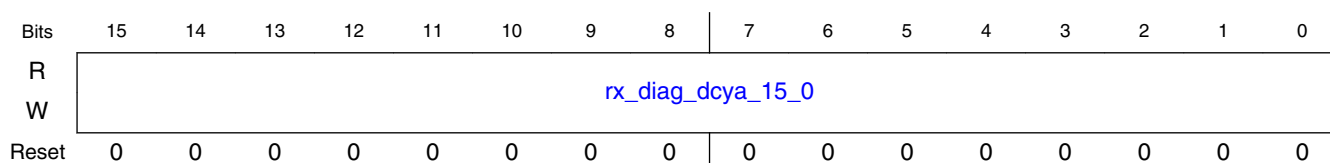
Field	Function
15-0 rx_diag_rxfe_tm2_15_0	Register definition to be provided by the analog team (Jira CCUPHY-2954)

13.4.10.2.378 Receiver digital cover your alternatives register (lane0_rx_diag_dcya - lane3_rx_diag_dcya)

13.4.10.2.378.1 Offset

Register	Offset
lane0_rx_diag_dcya	81FEh
lane1_rx_diag_dcya	85FEh
lane2_rx_diag_dcya	89FEh
lane3_rx_diag_dcya	8DFEh

13.4.10.2.378.2 Diagram



13.4.10.2.378.3 Fields

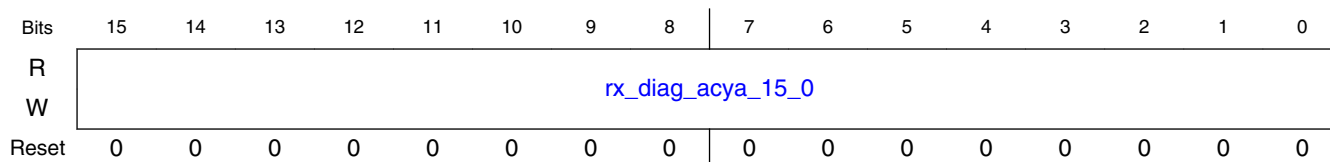
Field	Function
15-0 rx_diag_dcya_15_0	Reserved

13.4.10.2.379 Receiver analog cover your alternatives register (lane0_rx_diag_acya - lane3_rx_diag_acya)

13.4.10.2.379.1 Offset

Register	Offset
lane0_rx_diag_acya	81FFh
lane1_rx_diag_acya	85FFh
lane2_rx_diag_acya	89FFh
lane3_rx_diag_acya	8DFFh

13.4.10.2.379.2 Diagram



13.4.10.2.379.3 Fields

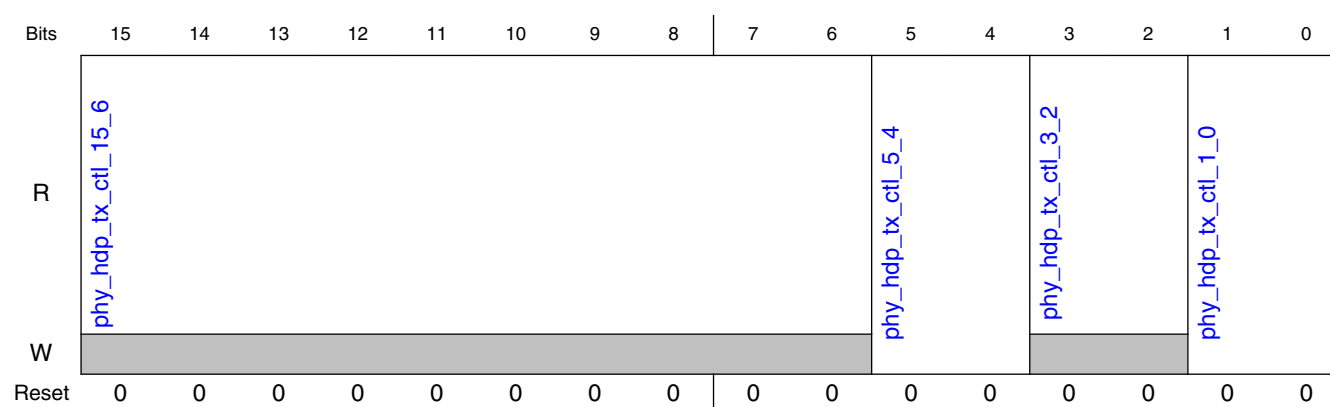
Field	Function
15-0 rx_diag_acya_15_0	Reserved

13.4.10.2.380 HDP Lane Configuration register (lane0_phy_hdp_tx_ctl - lane3_phy_hdp_tx_ctl)

13.4.10.2.380.1 Offset

Register	Offset
lane0_phy_hdp_tx_ctl	C408h
lane1_phy_hdp_tx_ctl	C448h
lane2_phy_hdp_tx_ctl	C488h
lane3_phy_hdp_tx_ctl	C4C8h

13.4.10.2.380.2 Diagram



13.4.10.2.380.3 Fields

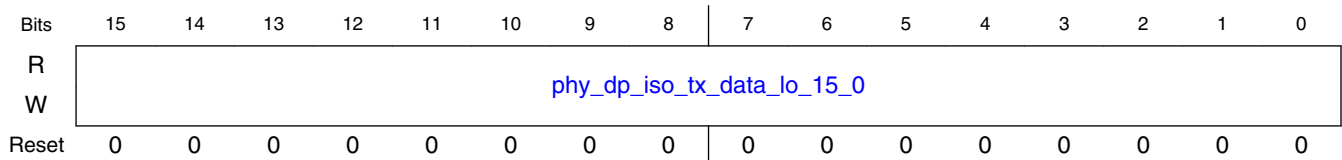
Field	Function
15-6 phy_hdp_tx_ctl_15_6	Reserved
5-4 phy_hdp_tx_ctl_5_4	Tx Voltage Level - Drives tx_vmargin PMA input for the mapped PMA lane (for functional and isolation modes). This field is used to set the DP Voltage Swing Level (0b00 = Level 0, 0b01 = Level 1, 0b10 = Level 2 and 0b11 = Level 3).
3-2 phy_hdp_tx_ctl_3_2	Reserved
1-0 phy_hdp_tx_ctl_1_0	Tx De-emphasis setting - Drives tx_deemphasis PMA input for the mapped PMA lane (for functional and isolation modes). This field is used to set the DP Pre-emphasis Level (0b00 = Level 0, 0b01 = Level 1, 0b10 = Level 2 and 0b11 = Level 3).

13.4.10.2.381 DP Tx data low isolation register (lane0_phy_dp_iso_tx_data_lo - lane3_phy_dp_iso_tx_data_lo)

13.4.10.2.381.1 Offset

Register	Offset
lane0_phy_dp_iso_tx_data_lo	C41Ch
lane1_phy_dp_iso_tx_data_lo	C45Ch
lane2_phy_dp_iso_tx_data_lo	C49Ch
lane3_phy_dp_iso_tx_data_lo	C4DCh

13.4.10.2.381.2 Diagram



13.4.10.2.381.3 Fields

Field	Function
15-0 phy_dp_iso_tx_data_lo_15_0	Drives phy_pma_tx_data_In_XX[15:0] PHY input for the associated lane when in PHY and PMA isolation modes. (Not re-synchronized)

13.4.10.2.382 DP Tx data high isolation register (lane0_phy_dp_iso_tx_data_hi - lane3_phy_dp_iso_tx_data_hi)

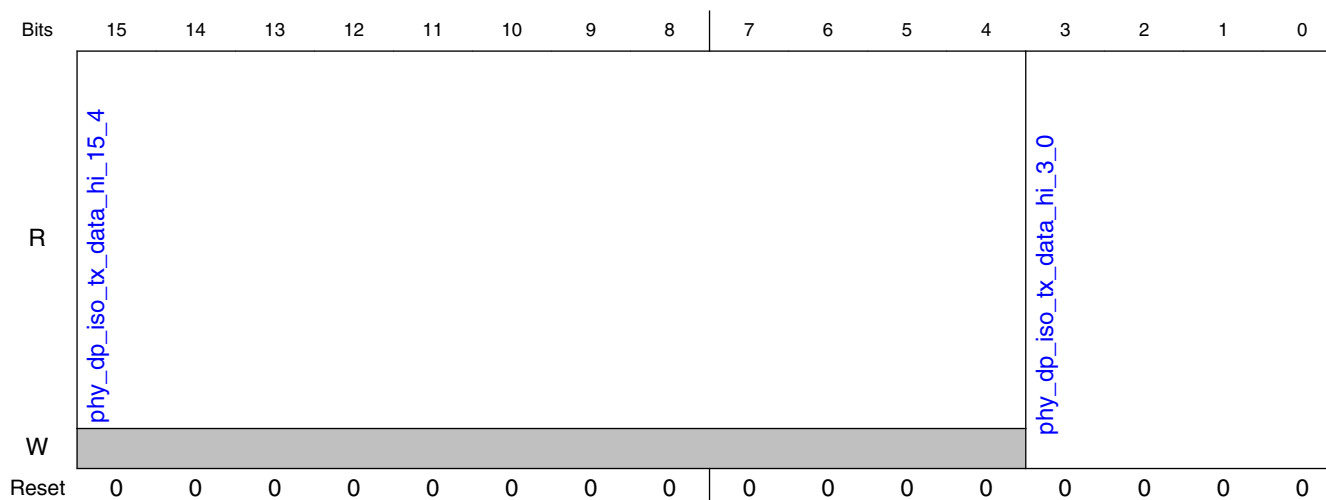
13.4.10.2.382.1 Offset

Register	Offset
lane0_phy_dp_iso_tx_data_hi	C41Dh
lane1_phy_dp_iso_tx_data_hi	C45Dh

Table continues on the next page...

Register	Offset
lane2_phy_dp_iso_tx_data_hi	C49Dh
lane3_phy_dp_iso_tx_data_hi	C4DDh

13.4.10.2.382.2 Diagram



13.4.10.2.382.3 Fields

Field	Function
15-4 phy_dp_iso_tx_data_hi_15_4	Reserved
3-0 phy_dp_iso_tx_data_hi_3_0	Drives phy_pma_tx_data_In_XX[19:16] PHY input for the associated lane when in PHY isolation modes. (Not re-synchronized)

13.4.10.2.383 PMA transceiver control register (lane0_phy_pma_xcvr_ctrl - lane3_phy_pma_xcvr_ctrl)

13.4.10.2.383.1 Offset

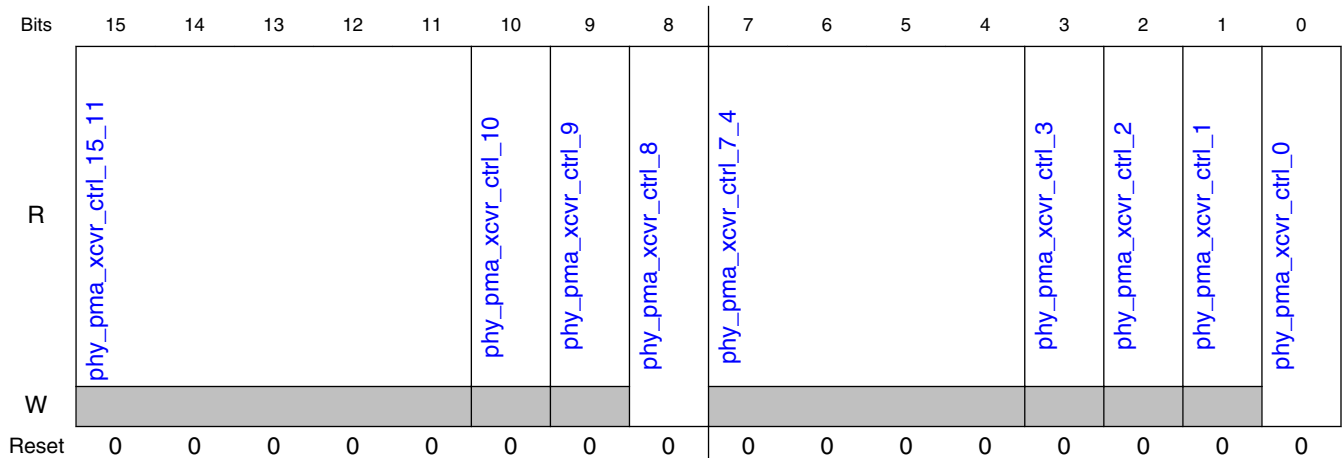
Register	Offset
lane0_phy_pma_xcvr_ctrl	CC00h

Table continues on the next page...

Clocks And Resets

Register	Offset
lane1_phy_pma_xcvr_ctrl	CC40h
lane2_phy_pma_xcvr_ctrl	CC80h
lane3_phy_pma_xcvr_ctrl	CCC0h

13.4.10.2.383.2 Diagram



13.4.10.2.383.3 Fields

Field	Function
15-11 phy_pma_xcvr_ctrl_15_11	Reserved
10 phy_pma_xcvr_ctrl_10	Current value of xcvr_lane_en_ack PMA output for the associated lane
9 phy_pma_xcvr_ctrl_9	Reserved
8 phy_pma_xcvr_ctrl_8	Drives the tx_differential_invert PMA input for the associated lane.
7-4 phy_pma_xcvr_ctrl_7_4	Reserved
3 phy_pma_xcvr_ctrl_3	Current value of rx_bist_status_ln_{nnn} PMA output for the associated lane.

Table continues on the next page...

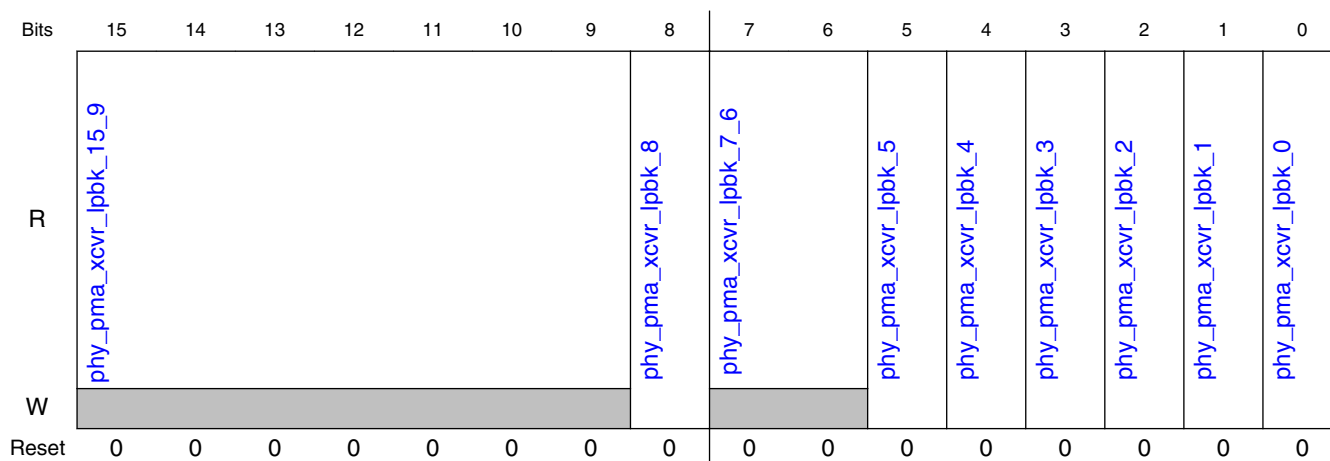
Field	Function
2 phy_pma_xcvr_ctrl_2	Current value of rx_bist_err_toggle_ln_{nnn} PMA output for the associated lane.
1 phy_pma_xcvr_ctrl_1	Current value of rx_bist_sync_ln_{nnn} PMA output for the associated lane.
0 phy_pma_xcvr_ctrl_0	Reserved

13.4.10.2.384 PMA loopback control register (lane0_phy_pma_xcvr_lpbk - lane3_phy_pma_xcvr_lpbk)

13.4.10.2.384.1 Offset

Register	Offset
lane0_phy_pma_xcvr_lpbk	CC01h
lane1_phy_pma_xcvr_lpbk	CC41h
lane2_phy_pma_xcvr_lpbk	CC81h
lane3_phy_pma_xcvr_lpbk	CCC1h

13.4.10.2.384.2 Diagram



13.4.10.2.384.3 Fields

Field	Function
15-9 phy_pma_xcvr_lpbk_15_9	Reserved
8 phy_pma_xcvr_lpbk_8	Drives the tx_bist_hold PMA input for all lanes in the associated link. Synchronized to transmit data rate clock. This signal can be used during test to start each lanes BIST engine on the same clock cycle to test lane to lane transmit skew.
7-6 phy_pma_xcvr_lpbk_7_6	Reserved
5 phy_pma_xcvr_lpbk_5	Drives the txrx_fe_parallel_lpbk_en_in_{nnn} PMA input for the associated lane.
4 phy_pma_xcvr_lpbk_4	Drives the txrx_ne_parallel_lpbk_en_in_{nnn} PMA input for the associated lane.
3 phy_pma_xcvr_lpbk_3	Drives the txrx_recovered_clk_lpbk_en_in_{nnn} PMA input for the associated lane.
2 phy_pma_xcvr_lpbk_2	Drives the txrx_line_lpbk_en_in_{nnn} PMA input for the associated lane.
1 phy_pma_xcvr_lpbk_1	Drives the txrx_isi_gen_lpbk_en_in_{nnn} PMA input for the associated lane.
0 phy_pma_xcvr_lpbk_0	Drives the txrx_serial_lpbk_en_in_{nnn} PMA input for the associated lane.

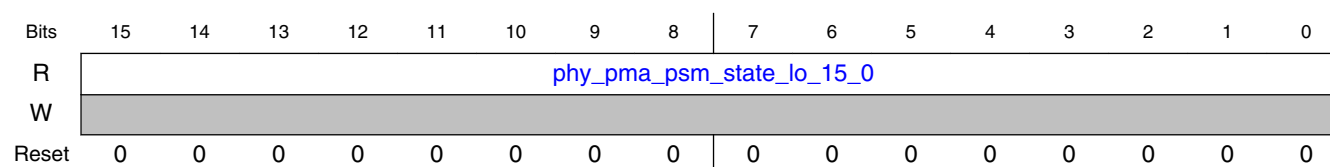
13.4.10.2.385 PMA PSM current state lower register (lane0_phy_pma_psm_state_lo - lane3_phy_pma_psm_state_lo)**13.4.10.2.385.1 Offset**

Register	Offset
lane0_phy_pma_psm_state_lo	CC04h
lane1_phy_pma_psm_state_lo	CC44h
lane2_phy_pma_psm_state_lo	CC84h

Table continues on the next page...

Register	Offset
lane3_phy_pma_psm_state_lo	CCC4h

13.4.10.2.385.2 Diagram



13.4.10.2.385.3 Fields

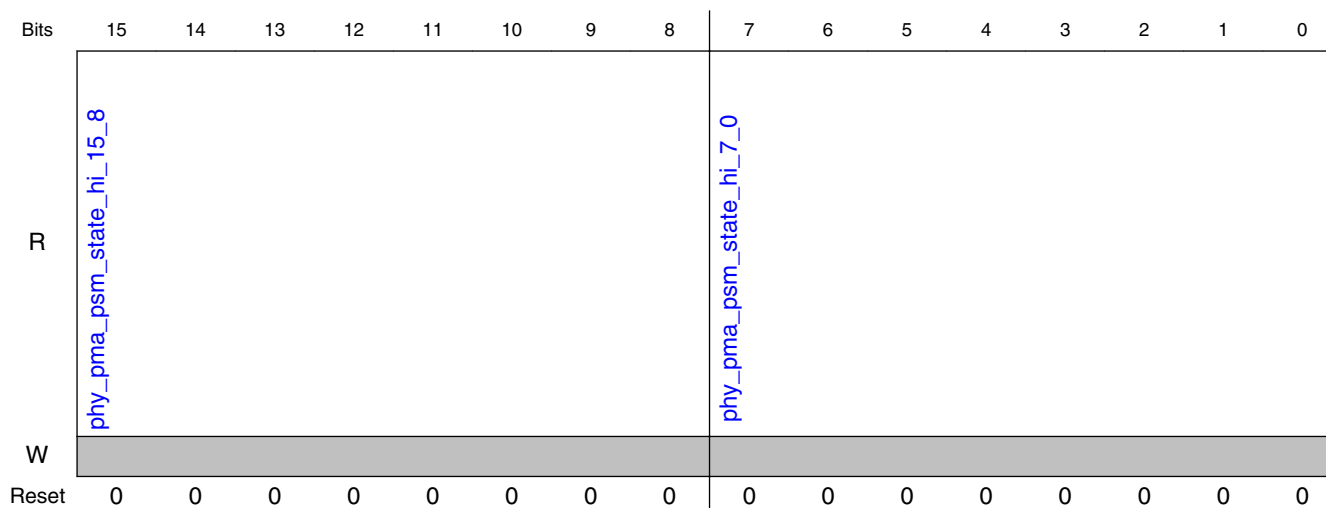
Field	Function
15-0 phy_pma_psm_state_lo_15_0	PMA PSM[15:0] : Current state of the PMA power state machine. (Not re-synchronized to apb_pclk)

13.4.10.2.386 PMA PSM current state higher register (lane0_phy_pma_psm_state_hi - lane3_phy_pma_psm_state_hi)

13.4.10.2.386.1 Offset

Register	Offset
lane0_phy_pma_psm_state_hi	CC05h
lane1_phy_pma_psm_state_hi	CC45h
lane2_phy_pma_psm_state_hi	CC85h
lane3_phy_pma_psm_state_hi	CCC5h

13.4.10.2.386.2 Diagram



13.4.10.2.386.3 Fields

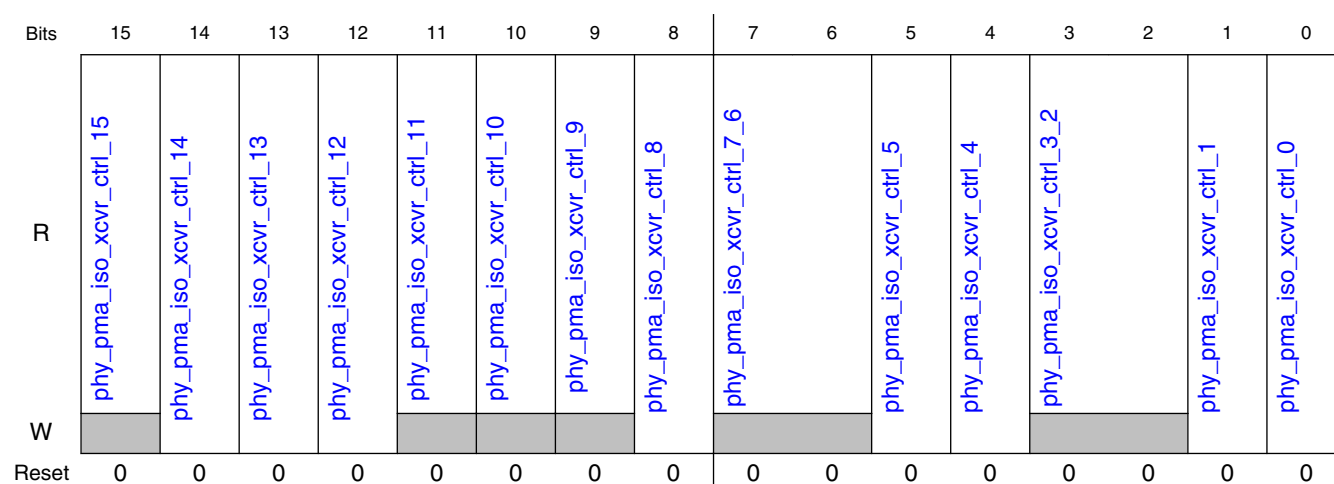
Field	Function
15-8 <code>phy_pma_psm_state_hi_15_8</code>	Reserved
7-0 <code>phy_pma_psm_state_hi_7_0</code>	PMA PSM[23:16] : Current state of the PMA power state machine. (Not re-synchronized to <code>apb_pclk</code>)

13.4.10.2.387 PMA Isolation Transceiver control register (lane0_phy_pma_iso_xcvr_ctrl - lane3_phy_pma_iso_xcvr_ctrl)

13.4.10.2.387.1 Offset

Register	Offset
<code>lane0_phy_pma_iso_xcvr_ctrl</code>	CC10h
<code>lane1_phy_pma_iso_xcvr_ctrl</code>	CC50h
<code>lane2_phy_pma_iso_xcvr_ctrl</code>	CC90h
<code>lane3_phy_pma_iso_xcvr_ctrl</code>	CCD0h

13.4.10.2.387.2 Diagram



13.4.10.2.387.3 Fields

Field	Function
15 phy_pma_iso_xcvr_ctrl_15	Current value of xcvr_pll_clk_en_ack PMA output for the associated lane.
14 phy_pma_iso_xcvr_ctrl_14	Drives xcvr_pll_clk_en PMA input for the associated lane when in PMA isolation mode.
13 phy_pma_iso_xcvr_ctrl_13	Drives tx_lfps_en PMA input for the associated lane when in PMA isolation mode.
12 phy_pma_iso_xcvr_ctrl_12	Drives tx_elec_idle PMA input for the associated lane when in PMA isolation mode.
11 phy_pma_iso_xcvr_ctrl_11	Current value of xcvr_psm_ready PMA output for the associated lane.
10 phy_pma_iso_xcvr_ctrl_10	Current value of tx_rcv_detected PMA output for the associated lane when in PMA isolation mode. (Not re-synchronized to apb_pclk)
9 phy_pma_iso_xcvr_ctrl_9	Current value of tx_rcv_detect_done PMA output for the associated lane.
8 phy_pma_iso_xcvr_ctrl_8	Drives tx_rcv_detect_en PMA input for the associated lane when in PMA isolation mode.

Table continues on the next page...

Clocks And Resets

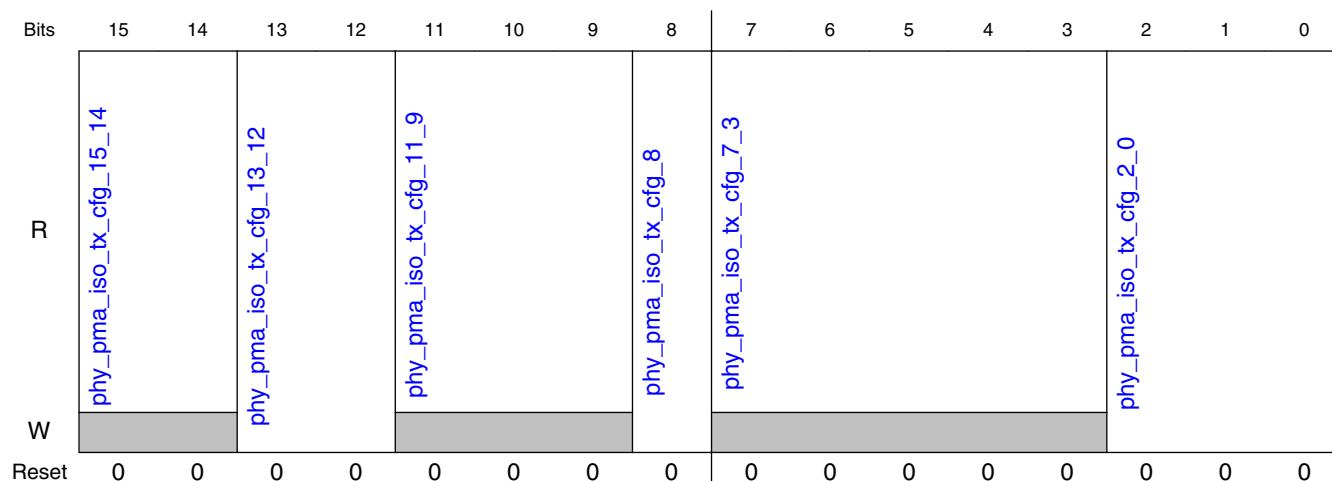
Field	Function
7-6 phy_pma_iso_x cvt_ctrl_7_6	Reserved
5 phy_pma_iso_x cvt_ctrl_5	Drives xcvt_link_reset_n PMA input for the associated lane when in PMA isolation mode.
4 phy_pma_iso_x cvt_ctrl_4	Drives xcvt_lane_suspend PMA input for the associated lane when in PMA isolation mode.
3-2 phy_pma_iso_x cvt_ctrl_3_2	Reserved
1 phy_pma_iso_x cvt_ctrl_1	Reserved
0 phy_pma_iso_x cvt_ctrl_0	Drives xcvt_lane_en PMA input for the associated lane when in PMA isolation mode.

13.4.10.2.388 PMA TX configuration register (lane0_phy_pma_iso_tx_cfg - lane3_phy_pma_iso_tx_cfg)

13.4.10.2.388.1 Offset

Register	Offset
lane0_phy_pma_iso_tx_cfg	CC11h
lane1_phy_pma_iso_tx_cfg	CC51h
lane2_phy_pma_iso_tx_cfg	CC91h
lane3_phy_pma_iso_tx_cfg	CCD1h

13.4.10.2.388.2 Diagram



13.4.10.2.388.3 Fields

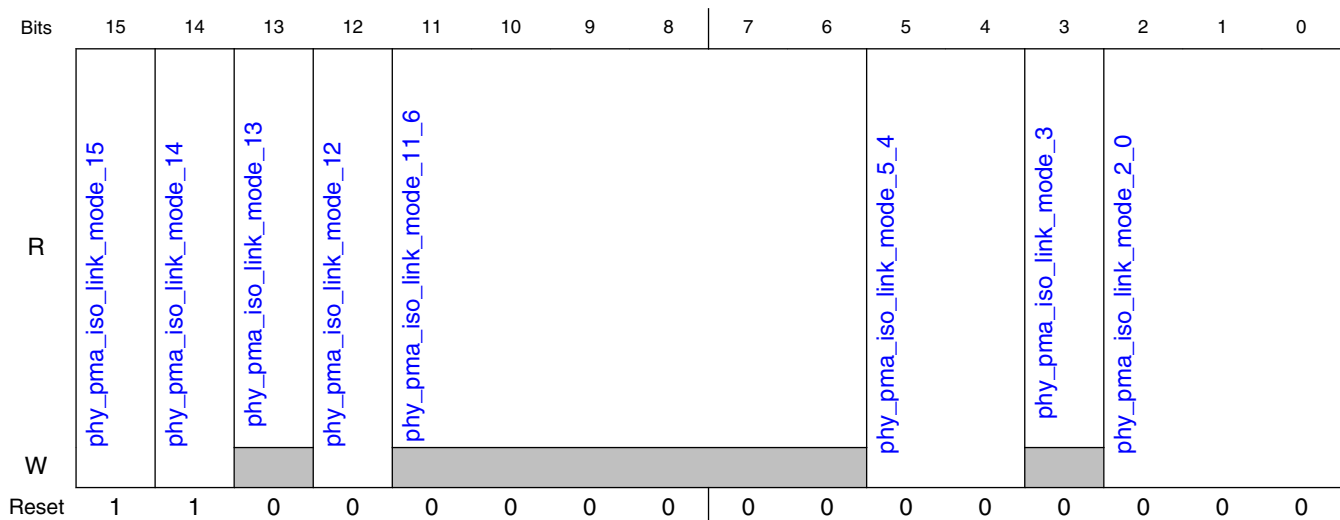
Field	Function
15-14 phy_pma_iso_tx_cfg_15_14	Reserved
13-12 phy_pma_iso_tx_cfg_13_12	Drives tx_deemphasis PMA input for the associated lane when in PMA isolation mode.
11-9 phy_pma_iso_tx_cfg_11_9	Reserved
8 phy_pma_iso_tx_cfg_8	Drives tx_low_power_swing_en PMA input for the associated lane when in PMA isolation mode.
7-3 phy_pma_iso_tx_cfg_7_3	Reserved
2-0 phy_pma_iso_tx_cfg_2_0	Drives tx_vmargin PMA input for the associated lane when in PMA isolation mode.

13.4.10.2.389 PMA Isolation mode control register (lane0_phy_pma_iso_link_mode - lane3_phy_pma_iso_link_mode)

13.4.10.2.389.1 Offset

Register	Offset
lane0_phy_pma_iso_link_mode	CC12h
lane1_phy_pma_iso_link_mode	CC52h
lane2_phy_pma_iso_link_mode	CC92h
lane3_phy_pma_iso_link_mode	CCD2h

13.4.10.2.389.2 Diagram



13.4.10.2.389.3 Fields

Field	Function
15 phy_pma_iso_link_mode_15	Drives tx_reset_n PMA input for the associated lane when in PMA isolation mode.
14 phy_pma_iso_link_mode_14	Reserved
13 phy_pma_iso_link_mode_13	Reserved
12	Drives the tx_high_z PMA input for the associated lane when in PMA isolation mode

Table continues on the next page...

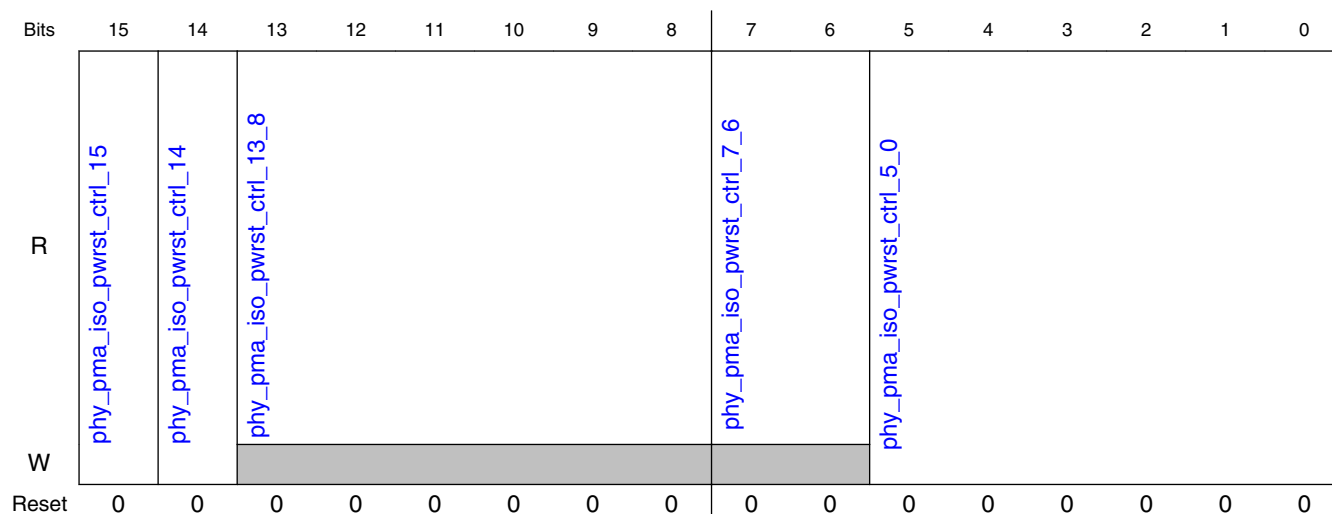
Field	Function
phy_pma_iso_link_mode_12	
11-6 phy_pma_iso_link_mode_11_6	Reserved
5-4 phy_pma_iso_link_mode_5_4	Drives xcvr_standard_mode PMA input for the associated lane when in PMA isolation mode.
3 phy_pma_iso_link_mode_3	Reserved
2-0 phy_pma_iso_link_mode_2_0	Drives xcvr_data_width PMA input for the associated lane when in PMA isolation mode.

13.4.10.2.390 PMA Isolation power state control register (lane0_phy_pma_iso_pwrst_ctrl - lane3_phy_pma_iso_pwrst_ctrl)

13.4.10.2.390.1 Offset

Register	Offset
lane0_phy_pma_iso_pwrst_ctrl	CC13h
lane1_phy_pma_iso_pwrst_ctrl	CC53h
lane2_phy_pma_iso_pwrst_ctrl	CC93h
lane3_phy_pma_iso_pwrst_ctrl	CCD3h

13.4.10.2.390.2 Diagram



13.4.10.2.390.3 Fields

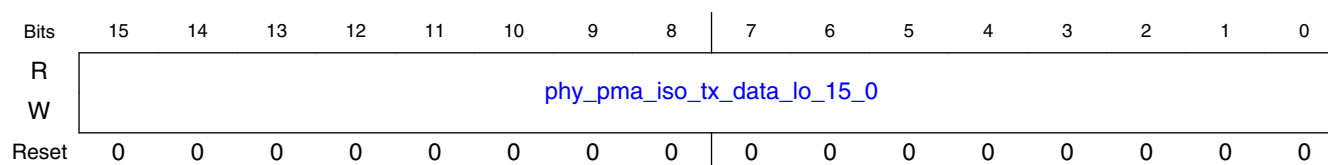
Field	Function
15 phy_pma_iso_pwrst_ctrl_15	Reserved
14 phy_pma_iso_pwrst_ctrl_14	tx_cmn_mode_en_ext PMA input for the associated lane when in PMA isolation mode. (Used for PCIe)
13-8 phy_pma_iso_pwrst_ctrl_13_8	Current value of xcvr_power_state_ack PMA output for the associated lane.
7-6 phy_pma_iso_pwrst_ctrl_7_6	Reserved
5-0 phy_pma_iso_pwrst_ctrl_5_0	Drives xcvr_power_state_req PMA input for the associated lane when in PMA isolation mode.

13.4.10.2.391 PMA transmit low data isolation register (lane0_phy_pma_iso_tx_data_lo - lane3_phy_pma_iso_tx_data_lo)

13.4.10.2.391.1 Offset

Register	Offset
lane0_phy_pma_iso_tx_data_lo	CC14h
lane1_phy_pma_iso_tx_data_lo	CC54h
lane2_phy_pma_iso_tx_data_lo	CC94h
lane3_phy_pma_iso_tx_data_lo	CCD4h

13.4.10.2.391.2 Diagram



13.4.10.2.391.3 Fields

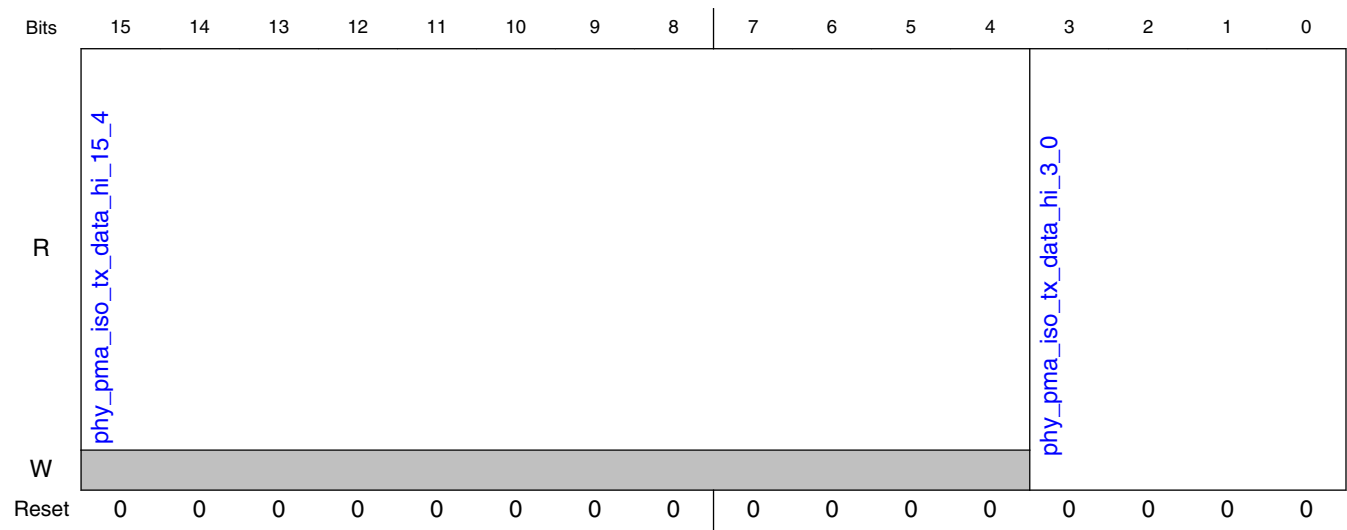
Field	Function
15-0 phy_pma_iso_tx_data_lo_15_0	Drives tx_td[15:0] PMA input for the associated lane when in PMA isolation mode. (Not re-synchronized to apb_pclk).

13.4.10.2.392 PMA transmit high data isolation register (lane0_phy_pma_iso_tx_data_hi - lane3_phy_pma_iso_tx_data_hi)

13.4.10.2.392.1 Offset

Register	Offset
lane0_phy_pma_iso_tx_data_hi	CC15h
lane1_phy_pma_iso_tx_data_hi	CC55h
lane2_phy_pma_iso_tx_data_hi	CC95h
lane3_phy_pma_iso_tx_data_hi	CCD5h

13.4.10.2.392.2 Diagram



13.4.10.2.392.3 Fields

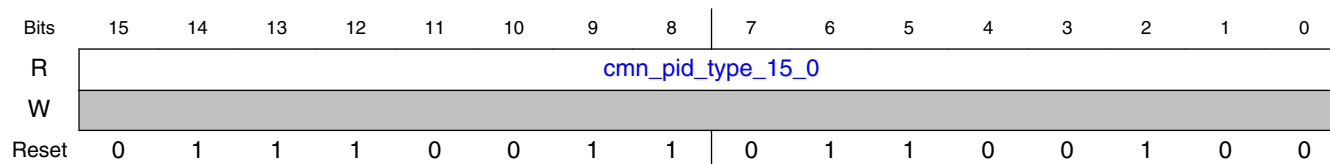
Field	Function
15-4 phy_pma_iso_tx_data_hi_15_4	Reserved
3-0 phy_pma_iso_tx_data_hi_3_0	Drives tx_td[19:16] PMA input for the associated lane when in PMA isolation mode. (Not re-synchronized to apb_pclk).

13.4.10.2.393 Product type ID register (cmn_pid_type)

13.4.10.2.393.1 Offset

Register	Offset
cmn_pid_type	8_0000h

13.4.10.2.393.2 Diagram



13.4.10.2.393.3 Fields

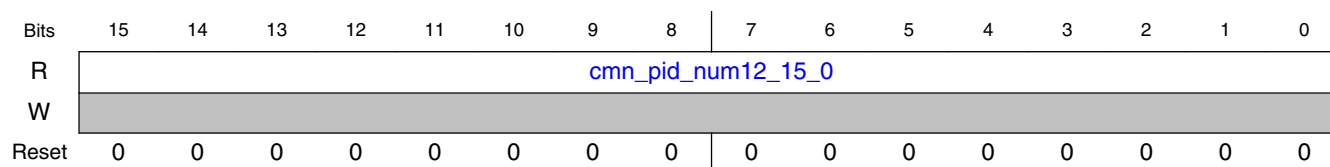
Field	Function
15-0 cmn_pid_type_15_0	Product type : This field contains the ASCII codes that represent the product type sd for SerDes.

13.4.10.2.394 Product number 1 2 ID register (cmn_pid_num12)

13.4.10.2.394.1 Offset

Register	Offset
cmn_pid_num12	8_0001h

13.4.10.2.394.2 Diagram



13.4.10.2.394.3 Fields

Field	Function
15-0 cmn_pid_num12_15_0	Product number : This field contains the ASCII codes that represent the characters 1 and 2 of the product part number.

13.4.10.2.395 Product number 3 4 ID register (cmn_pid_num34)

13.4.10.2.395.1 Offset

Register	Offset
cmn_pid_num34	8_0002h

13.4.10.2.395.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_num34_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.395.3 Fields

Field	Function
15-0 cmn_pid_num34_15_0	Product number : If necessary, this field contains the ASCII codes that represent the characters 3 and 4 of the product part number.

13.4.10.2.396 Product number 5 6 ID register (cmn_pid_num56)

13.4.10.2.396.1 Offset

Register	Offset
cmn_pid_num56	8_0003h

13.4.10.2.396.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_num56_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.396.3 Fields

Field	Function
15-0 cmn_pid_num56_15_0	Product number : If necessary, this field contains the ASCII codes that represent the characters 5 and 6 of the product part number.

13.4.10.2.397 Product number 7 8 ID register (cmn_pid_num78)

13.4.10.2.397.1 Offset

Register	Offset
cmn_pid_num78	8_0004h

13.4.10.2.397.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_num78_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.397.3 Fields

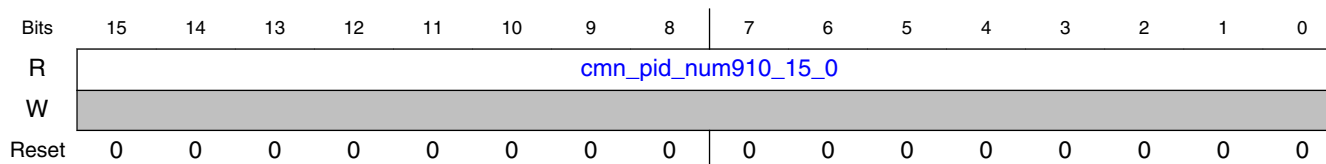
Field	Function
15-0 cmn_pid_num78_15_0	Product number : If necessary, this field contains the ASCII codes that represent the characters 7 and 8 of the product part number.

13.4.10.2.398 Product number 9 10 ID register (cmn_pid_num910)

13.4.10.2.398.1 Offset

Register	Offset
cmn_pid_num910	8_0005h

13.4.10.2.398.2 Diagram



13.4.10.2.398.3 Fields

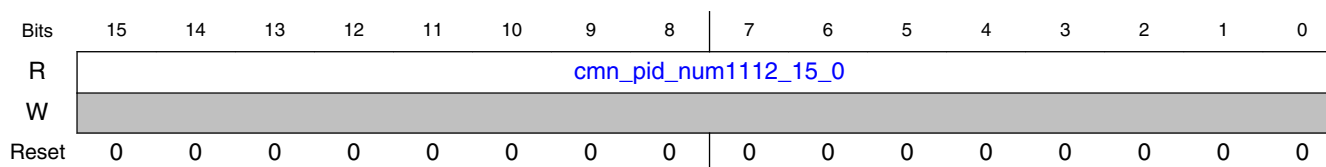
Field	Function
15-0 cmn_pid_num910_15_0	Product number : If necessary, this field contains the ASCII codes that represent the characters 9 and 10 of the product part number.

13.4.10.2.399 Product number 11 12 ID register (cmn_pid_num1112)

13.4.10.2.399.1 Offset

Register	Offset
cmn_pid_num1112	8_0006h

13.4.10.2.399.2 Diagram



13.4.10.2.399.3 Fields

Field	Function
15-0 cmn_pid_num1112_15_0	Product number : If necessary, this field contains the ASCII codes that represent the characters 11 and 12 of the product part number.

13.4.10.2.400 Product revision ID register (cmn_pid_rev)

13.4.10.2.400.1 Offset

Register	Offset
cmn_pid_rev	8_0007h

13.4.10.2.400.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_rev_15_0															
W																
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

13.4.10.2.400.3 Fields

Field	Function
15-0 cmn_pid_rev_15_0	Product revision : This field contains the binary coded decimal numbers that represent the product revision 0100.

13.4.10.2.401 Product technology manufacturer ID register (cmn_pid_mfg)

13.4.10.2.401.1 Offset

Register	Offset
cmn_pid_mfg	8_0008h

13.4.10.2.401.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_mfg_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	1	1	1	0	1	0	0

13.4.10.2.401.3 Fields

Field	Function
15-0 cmn_pid_mfg_15_0	Product technology manufacturer : This field contains the ASCII codes that represent the product technology manufacturer.

13.4.10.2.402 Product technology process node ID register (cmn_pid_node)

13.4.10.2.402.1 Offset

Register	Offset
cmn_pid_node	8_0009h

13.4.10.2.402.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_node_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0

13.4.10.2.402.3 Fields

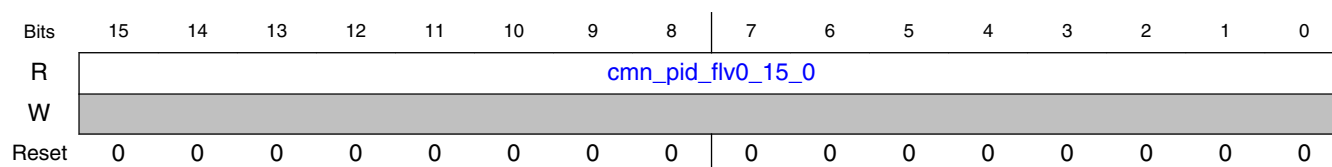
Field	Function
15-0 cmn_pid_node_15_0	Product technology process node : This field contains the binary coded decimal numbers that represent the product technology node 28nm

13.4.10.2.403 Product technology process flavor ID register 0 (cmn_pid_flv0)

13.4.10.2.403.1 Offset

Register	Offset
cmn_pid_flv0	8_000Ah

13.4.10.2.403.2 Diagram



13.4.10.2.403.3 Fields

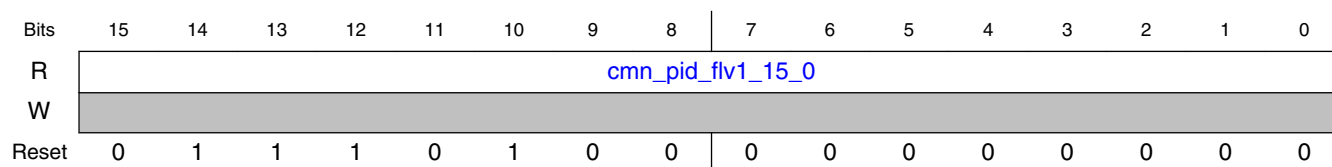
Field	Function
15-0 cmn_pid_flv0_15_0	Product technology flavor : This field contains the ASCII codes that represent the first two characters of the product technology flavor.

13.4.10.2.404 Product technology process flavor ID register 1 (cmn_pid_flv1)

13.4.10.2.404.1 Offset

Register	Offset
cmn_pid_flv1	8_000Bh

13.4.10.2.404.2 Diagram



13.4.10.2.404.3 Fields

Field	Function
15-0 cmn_pid_flv1_15_0	Product technology flavor : This field contains the ASCII codes that represent the second two characters of the product technology flavor (when applicable).

13.4.10.2.405 Product I/O voltage ID register (cmn_pid_iov)

13.4.10.2.405.1 Offset

Register	Offset
cmn_pid_iov	8_000Ch

13.4.10.2.405.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_iov_15_0															
W																
Reset	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0

13.4.10.2.405.3 Fields

Field	Function
15-0 cmn_pid_iov_15_0	Product I/O voltage : This field contains the binary coded decimal numbers that represent the product I/O voltage. The most significant byte represents the value to the left of the decimal point, and the least significant byte represents the value to the right of the decimal point. For example, 1.5V is represented as 0x0150.

13.4.10.2.406 Product SerDes lanes ID register (cmn_pid_lanes)

13.4.10.2.406.1 Offset

Register	Offset
cmn_pid_lanes	8_000Dh

13.4.10.2.406.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_lanes_15_8								cmn_pid_lanes_7_0							
W																
Reset	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0

13.4.10.2.406.3 Fields

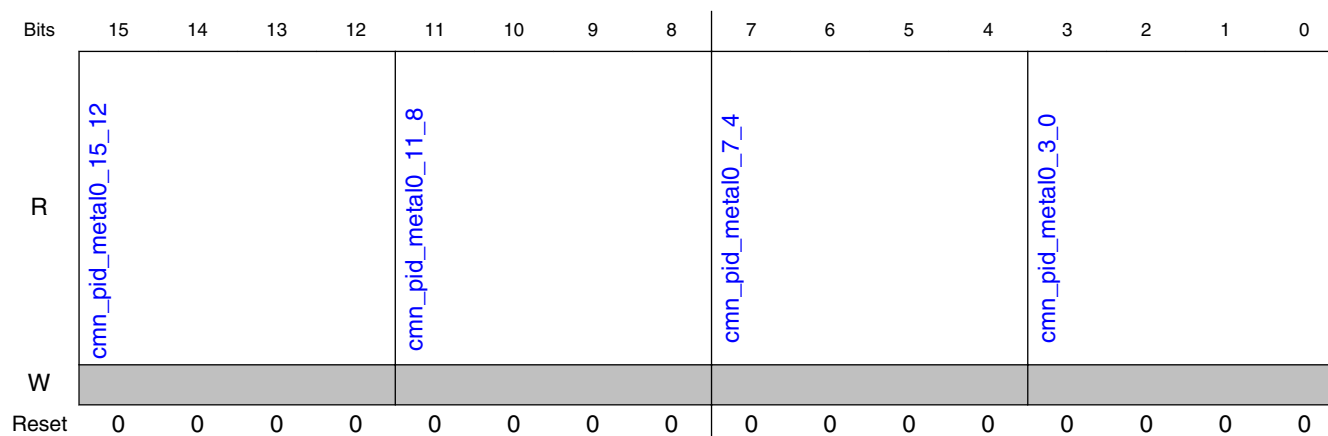
Field	Function
15-8 cmn_pid_lanes_15_8	Product SerDes lanes left of common : This field contains the binary coded decimal numbers that represent the number of lanes implemented in this SerDes product on the left side of the common module.
7-0 cmn_pid_lanes_7_0	Product SerDes lanes right of common : This field contains the binary coded decimal numbers that represent the number of lanes implemented in this SerDes product on the right side of the common module.

13.4.10.2.407 Product metal layers ID register 0 (cmn_pid_metal0)

13.4.10.2.407.1 Offset

Register	Offset
cmn_pid_metal0	8_0010h

13.4.10.2.407.2 Diagram



13.4.10.2.407.3 Fields

Field	Function
15-12 cmn_pid_metal0_15_12	Product X metal layers : This field contains the binary coded decimal number that represent the number of xd metal layers used for this product.

Table continues on the next page...

Clocks And Resets

Field	Function
11-8 cmn_pid_metal0_11_8	Product Y metal layers : This field contains the binary coded decimal number that represent the number of xc metal layers used for this product.
7-4 cmn_pid_metal0_7_4	Product Z metal layers : This field contains the binary coded decimal number that represent the number of xa metal layers used for this product.
3-0 cmn_pid_metal0_3_0	Product R/U metal layers : This field contains the binary coded decimal number that represent the number of x metal layers used for this product.

13.4.10.2.408 Product metal layers ID register 1 (cmn_pid_metal1)

13.4.10.2.408.1 Offset

Register	Offset
cmn_pid_metal1	8_0011h

13.4.10.2.408.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_metal1_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.408.3 Fields

Field	Function
15-0 cmn_pid_metal1_15_0	Reserved

13.4.10.2.409 Product metal layers ID register 2 (cmn_pid_metal2)

13.4.10.2.409.1 Offset

Register	Offset
cmn_pid_metal2	8_0012h

13.4.10.2.409.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_metal2_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.409.3 Fields

Field	Function
15-0 cmn_pid_metal2_15_0	Reserved

13.4.10.2.410 Product metal layers ID register 3 (cmn_pid_metal3)

13.4.10.2.410.1 Offset

Register	Offset
cmn_pid_metal3	8_0013h

13.4.10.2.410.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_metal3_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.410.3 Fields

Field	Function
15-0 cmn_pid_metal3_15_0	Reserved

13.4.10.2.411 Product metal layer direction ID register (cmn_pid_metald)

13.4.10.2.411.1 Offset

Register	Offset
cmn_pid_metald	8_0014h

13.4.10.2.411.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pid_metald_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.411.3 Fields

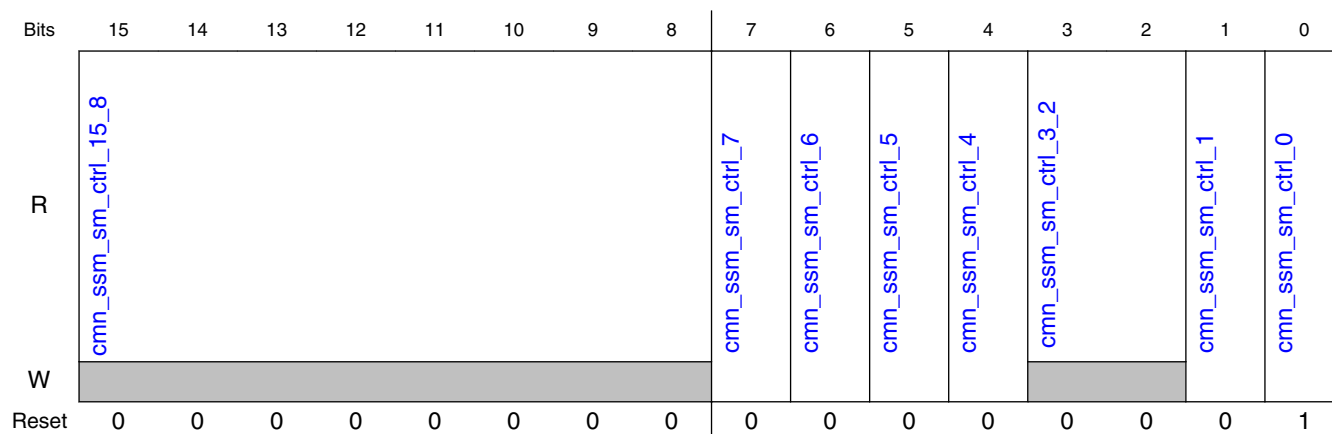
Field	Function
15-0 cmn_pid_metald_15_0	Reserved

13.4.10.2.412 Startup state machine control register (cmn_ssm_sm_ctrl)

13.4.10.2.412.1 Offset

Register	Offset
cmn_ssm_sm_ctrl	8_0020h

13.4.10.2.412.2 Diagram



13.4.10.2.412.3 Fields

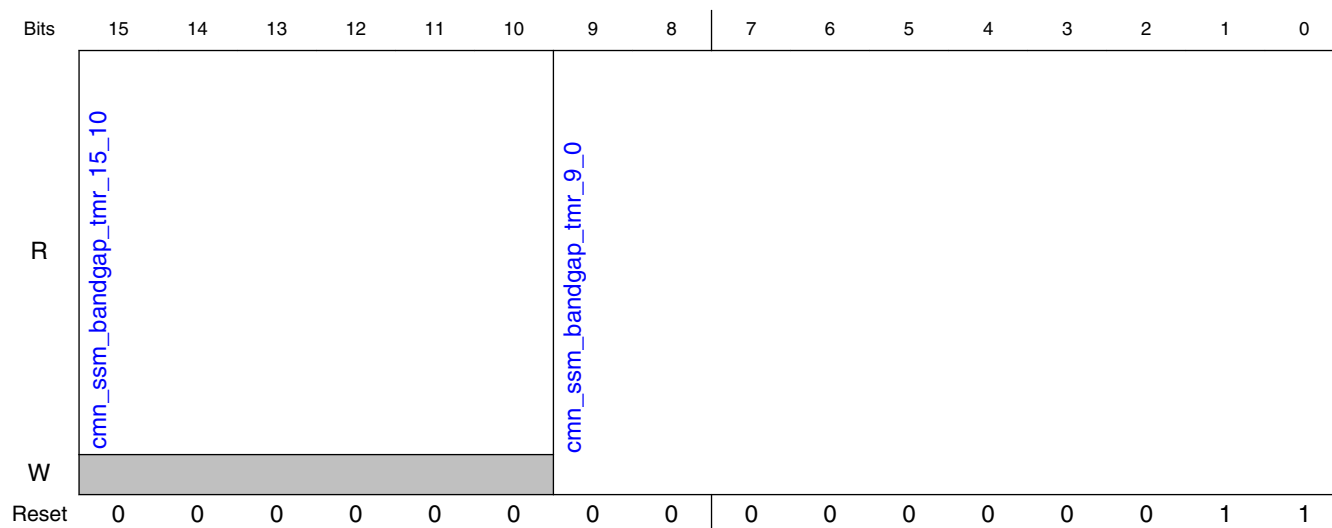
Field	Function
15-8 <code>cmn_ssm_sm_ctrl_15_8</code>	Reserved
7 <code>cmn_ssm_sm_ctrl_7</code>	Bandgap enable override enable : When active (1b1), the bandgap enable override bit in this register will drive the
6 <code>cmn_ssm_sm_ctrl_6</code>	Bandgap enable override : When enabled by the bandgap enable override enable bit in this register, this bit will drive the
5 <code>cmn_ssm_sm_ctrl_5</code>	Bias enable override enable : When active (1b1), the bias enable override bit in this register will drive the
4 <code>cmn_ssm_sm_ctrl_4</code>	Bias enable override : When enabled by the bias enable override enable bit in this register, this bit will drive the
3-2 <code>cmn_ssm_sm_ctrl_3_2</code>	Reserved
1 <code>cmn_ssm_sm_ctrl_1</code>	Skip post bandgap enable re-calibration : When this bit is active (1b1), the post bandgap enable calibration state will be skipped if it was previously run, unless the macro is disabled or reset.
0 <code>cmn_ssm_sm_ctrl_0</code>	Skip auto re-calibration : When this bit is active (1b1), the auto calibration state will be skipped if it was previously run, unless the macro is disabled or reset.

13.4.10.2.413 Bandgap enable timer register (cmn_ssm_bandgap_tmr)

13.4.10.2.413.1 Offset

Register	Offset
cmn_ssm_bandgap_tmr	8_0021h

13.4.10.2.413.2 Diagram



13.4.10.2.413.3 Fields

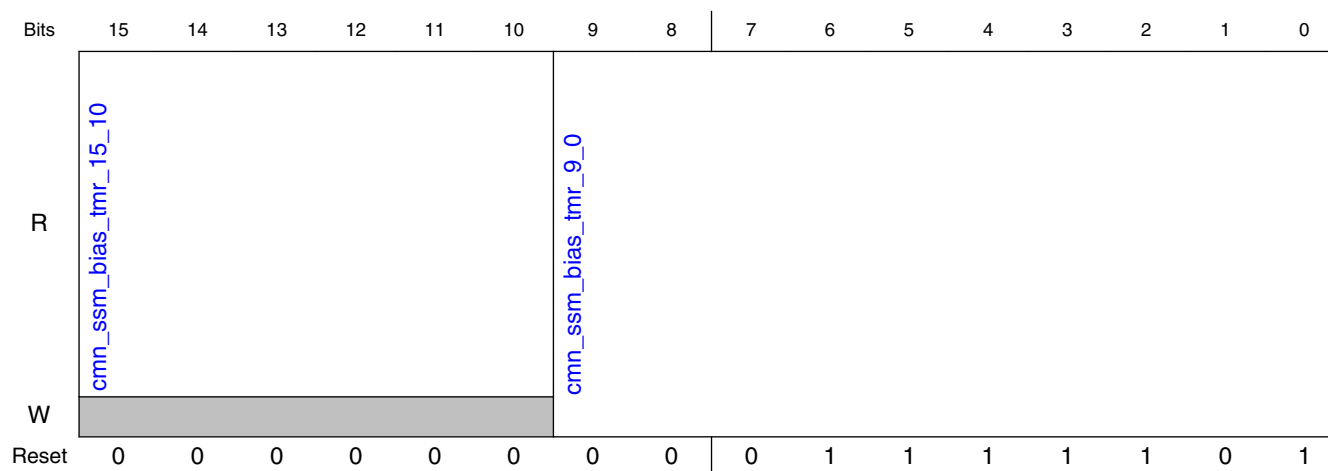
Field	Function
15-10 cmn_ssm_bandgap_tmr_15_10	Reserved
9-0 cmn_ssm_bandgap_tmr_9_0	Bandgap enable state timer value : Value used for the timer when the startup state machine is in the bandgap enable state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.414 Bias enable timer register (cmn_ssm_bias_tmr)

13.4.10.2.414.1 Offset

Register	Offset
cmn_ssm_bias_tmr	8_0022h

13.4.10.2.414.2 Diagram



13.4.10.2.414.3 Fields

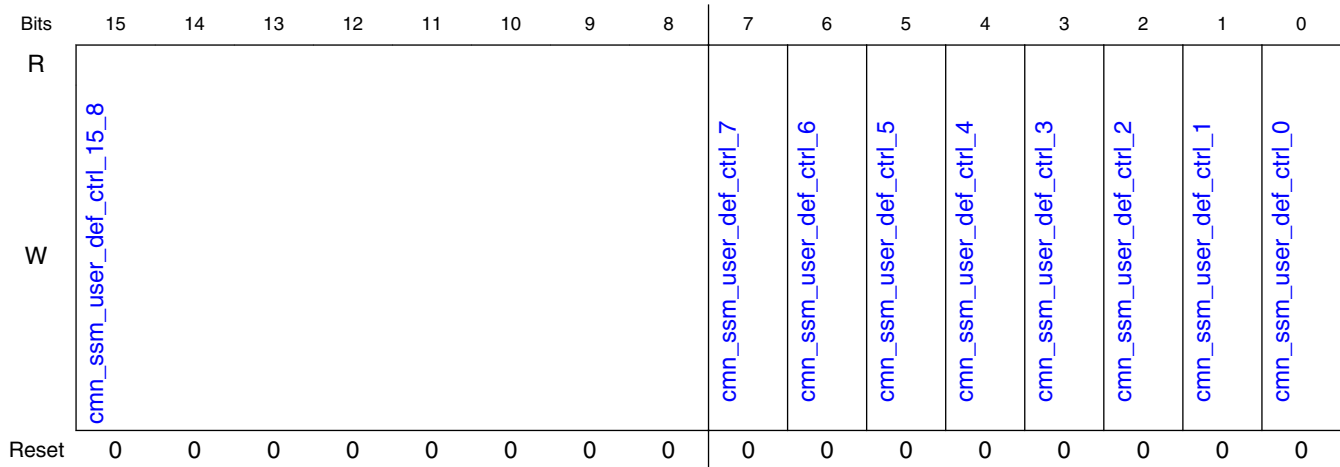
Field	Function
15-10 cmn_ssm_bias_tmr_15_10	Reserved
9-0 cmn_ssm_bias_tmr_9_0	Bias enable state timer value : Value used for the timer when the startup state machine is in the bias enable state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.415 Startup state machine user defined control register (cmn_ssm_user_def_ctrl)

13.4.10.2.415.1 Offset

Register	Offset
cmn_ssm_user_def_ctrl	8_0027h

13.4.10.2.415.2 Diagram



13.4.10.2.415.3 Fields

Field	Function
15-8 cmn_ssm_user_def_ctrl_15_8	Reserved
7 cmn_ssm_user_def_ctrl_7	Analog reference clock enable override enable : When active (1b1), the analog reference clock enable override bit in this register will drive the
6 cmn_ssm_user_def_ctrl_6	Analog reference clock enable override : When enabled by the analog reference clock enable override enable bit in this register, this bit will drive the
5 cmn_ssm_user_def_ctrl_5	Calibration iconst enable override enable : When active (1b1), the calibration iconst enable override bit in this register will drive the
4 cmn_ssm_user_def_ctrl_4	Calibration iconst enable override : When enabled by the calibration power enable override enable bit in this register, this bit will drive the
3 cmn_ssm_user_def_ctrl_3	Calibration power enable override enable : When active (1b1), the calibration iconst enable override bit in this register will drive the
2 cmn_ssm_user_def_ctrl_2	Calibration power enable override : When enabled by the calibration power enable override enable bit in this register, this bit will drive the
1 cmn_ssm_user_def_ctrl_1	Bandgap enable override enable : When active (1b1), the bandgap enable override bit in this register will drive the
0	Bandgap enable override : When enabled by the bandgap enable override enable bit in this register, this bit will drive the

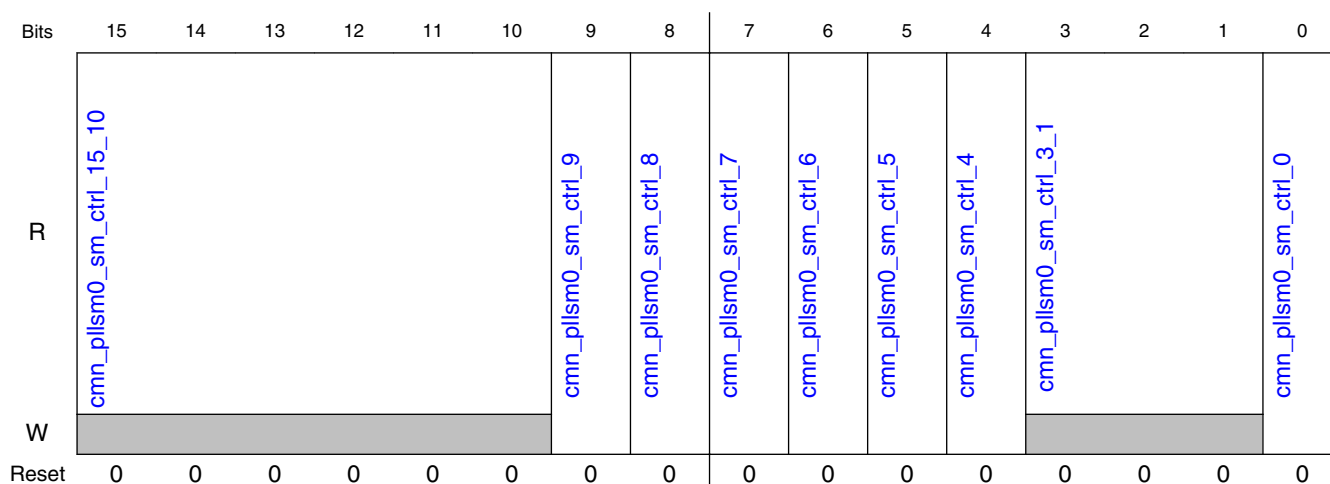
Field	Function
cmn_ssm_user_def_ctrl_0	

13.4.10.2.416 PLL 0 control state machine control register (cmn_pllsm0_sm_ctrl)

13.4.10.2.416.1 Offset

Register	Offset
cmn_pllsm0_sm_ctrl	8_0028h

13.4.10.2.416.2 Diagram



13.4.10.2.416.3 Fields

Field	Function
15-10 cmn_pllsm0_sm_ctrl_15_10	Reserved
9 cmn_pllsm0_sm_ctrl_9	PLL enable override enable : When active (1b1), the PLL enable override bit in this register will drive the
8	PLL enable override : When enabled by the PLL enable override enable bit in this register, this bit will drive the

Table continues on the next page...

Clocks And Resets

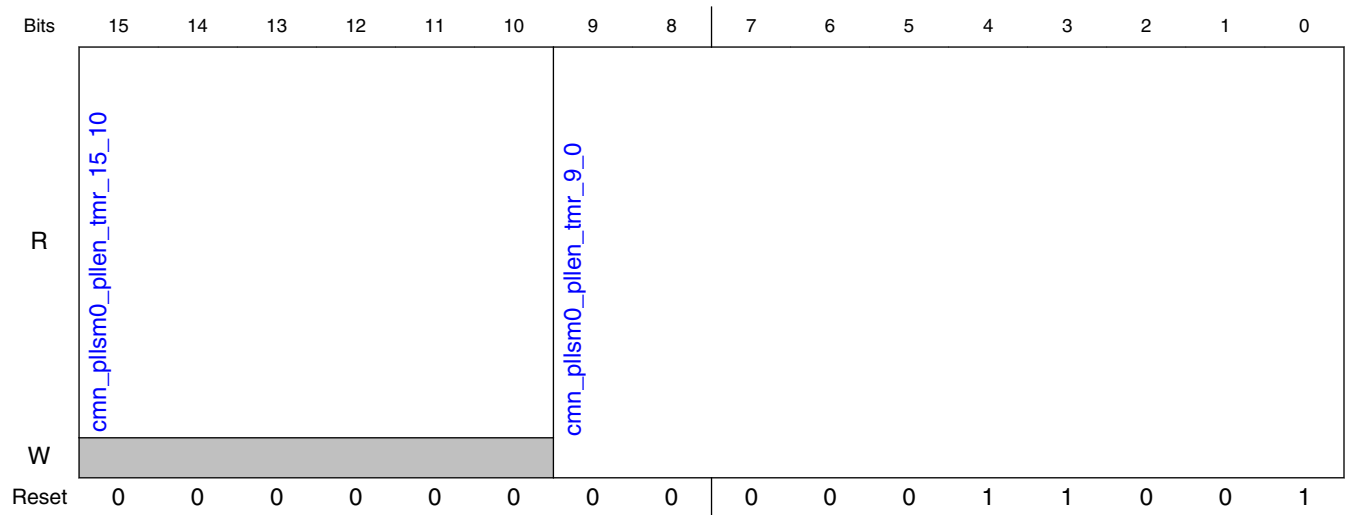
Field	Function
cmn_pllsm0_sm_ctrl_8	
7 cmn_pllsm0_sm_ctrl_7	PLL VCO LDO reference override enable : When active (1b1), the PLL VCO LDO reference override bit in this register will drive the
6 cmn_pllsm0_sm_ctrl_6	PLL VCO LDO reference override : When enabled by the PLL PLL VCO LDO reference override enable bit in this register, this bit will drive the
5 cmn_pllsm0_sm_ctrl_5	PLL VCO LDO reference charge pulse override enable : When active (1b1), the PLL VCO LDO reference charge pulse override bit in this register will drive the
4 cmn_pllsm0_sm_ctrl_4	PLL VCO LDO reference charge pulse override : When enabled by the PLL VCO LDO reference charge pulse override enable bit in this register, this bit will drive the
3-1 cmn_pllsm0_sm_ctrl_3_1	Reserved
0 cmn_pllsm0_sm_ctrl_0	Skip PLL re-calibration : When this bit is active (1b1), the PLL calibration state will be skipped if it was previously run, unless the PLL is disabled or resetting the state machine.

13.4.10.2.417 PLL 0 enable timer register (cmn_pllsm0_pllen_tmr)

13.4.10.2.417.1 Offset

Register	Offset
cmn_pllsm0_pllen_tmr	8_0029h

13.4.10.2.417.2 Diagram



13.4.10.2.417.3 Fields

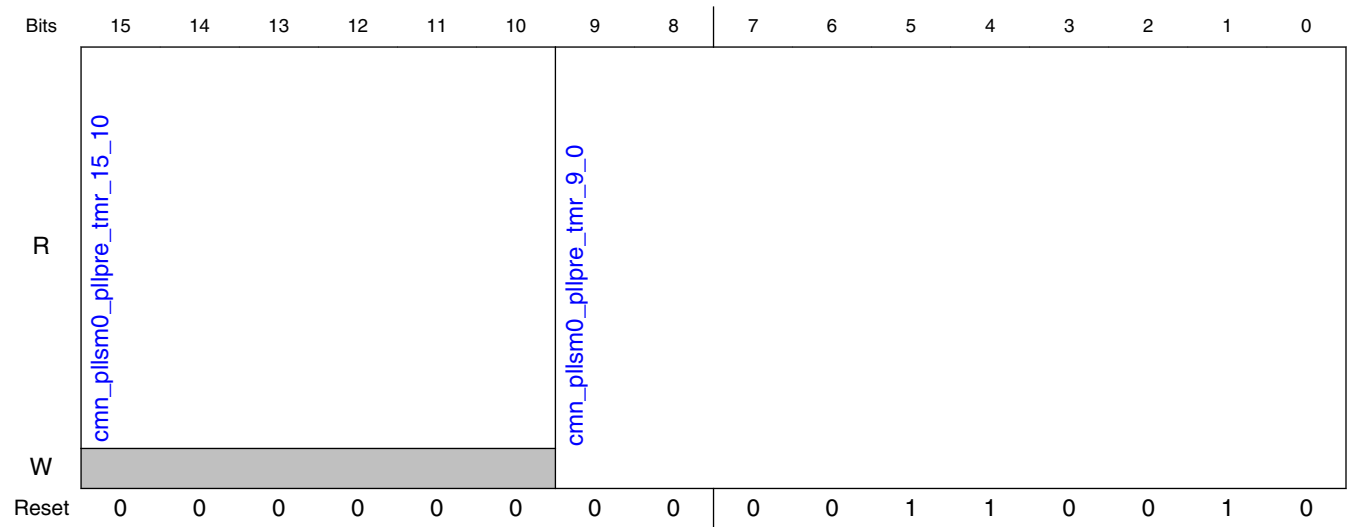
Field	Function
15-10 cmn_pllsm0_pllen_tmr_15_10	Reserved
9-0 cmn_pllsm0_pllen_tmr_9_0	PLL enable state timer value : Value used for the timer when the startup state machine is in the PLL enable state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.418 PLL 0 pre-charge timer register (cmn_pllsm0_pllpre_tmr)

13.4.10.2.418.1 Offset

Register	Offset
cmn_pllsm0_pllpre_tmr	8_002Ah

13.4.10.2.418.2 Diagram



13.4.10.2.418.3 Fields

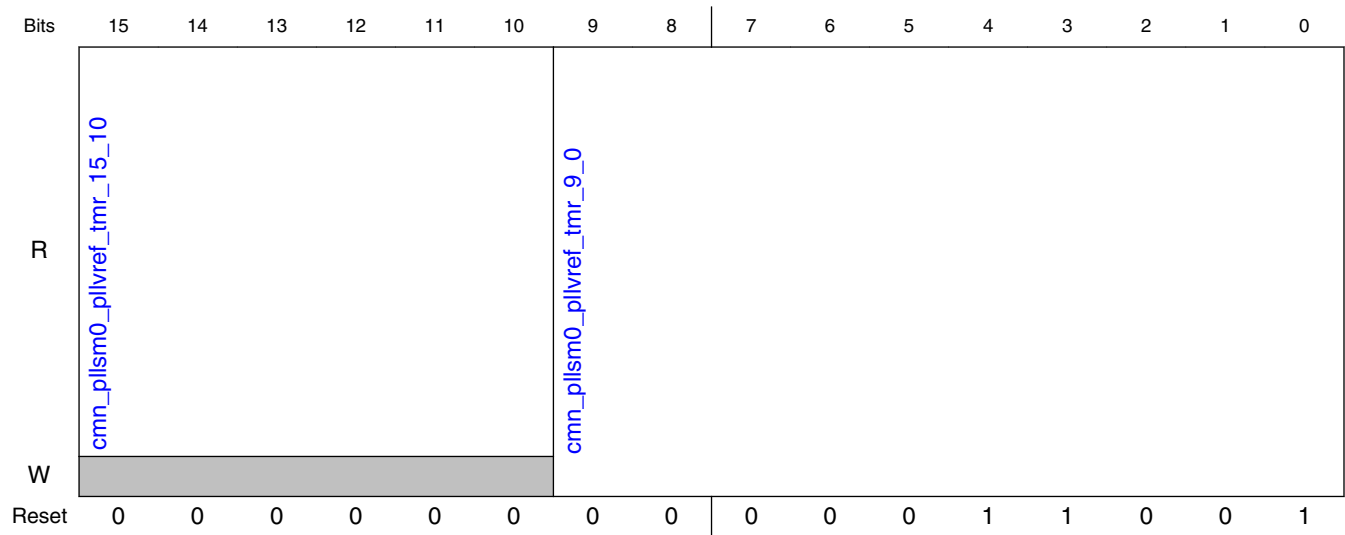
Field	Function
15-10 cmn_pllsm0_pllpre_tmr_15_10	Reserved
9-0 cmn_pllsm0_pllpre_tmr_9_0	PLL pre-charge state timer value : Value used for the timer when the startup state machine is in the PLL pre-charge state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.419 PLL 0 VREF delay timer register (cmn_pllsm0_pllvref_tmr)

13.4.10.2.419.1 Offset

Register	Offset
cmn_pllsm0_pllvref_tmr	8_002Bh

13.4.10.2.419.2 Diagram



13.4.10.2.419.3 Fields

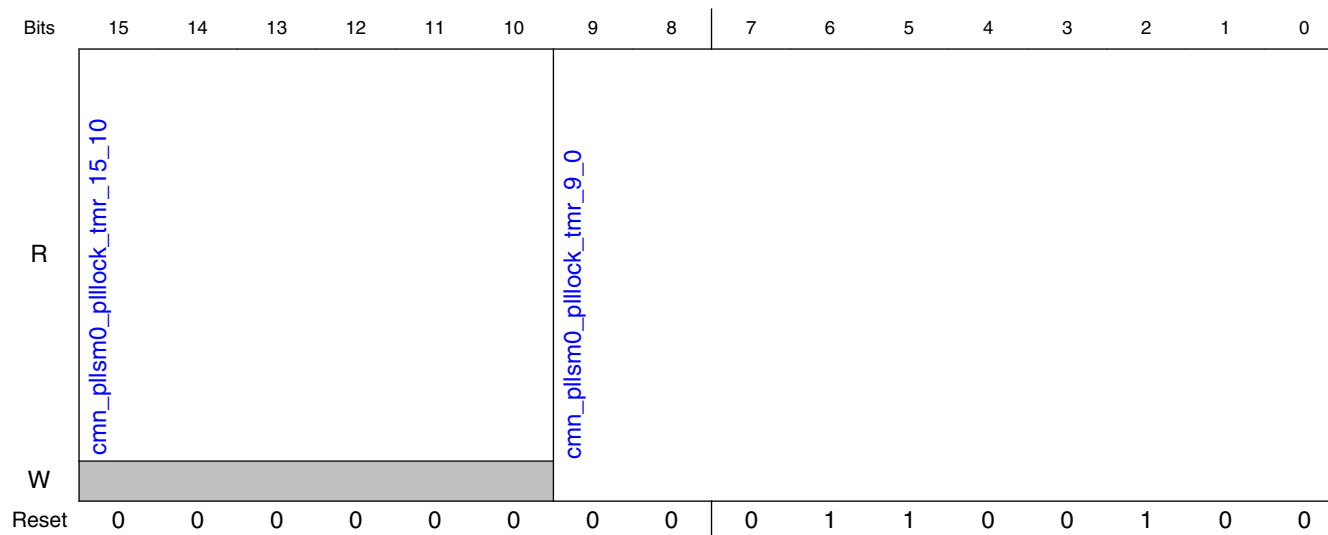
Field	Function
15-10 cmn_pllsm0_pllvref_tmr_15_10	Reserved
9-0 cmn_pllsm0_pllvref_tmr_9_0	PLL VREF delay state timer value : Value used for the timer when the startup state machine is in the PLL VREF delay state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.420 PLL 0 lock delay timer register (cmn_pllsm0_plllock_tmr)

13.4.10.2.420.1 Offset

Register	Offset
cmn_pllsm0_plllock_tmr	8_002Ch

13.4.10.2.420.2 Diagram



13.4.10.2.420.3 Fields

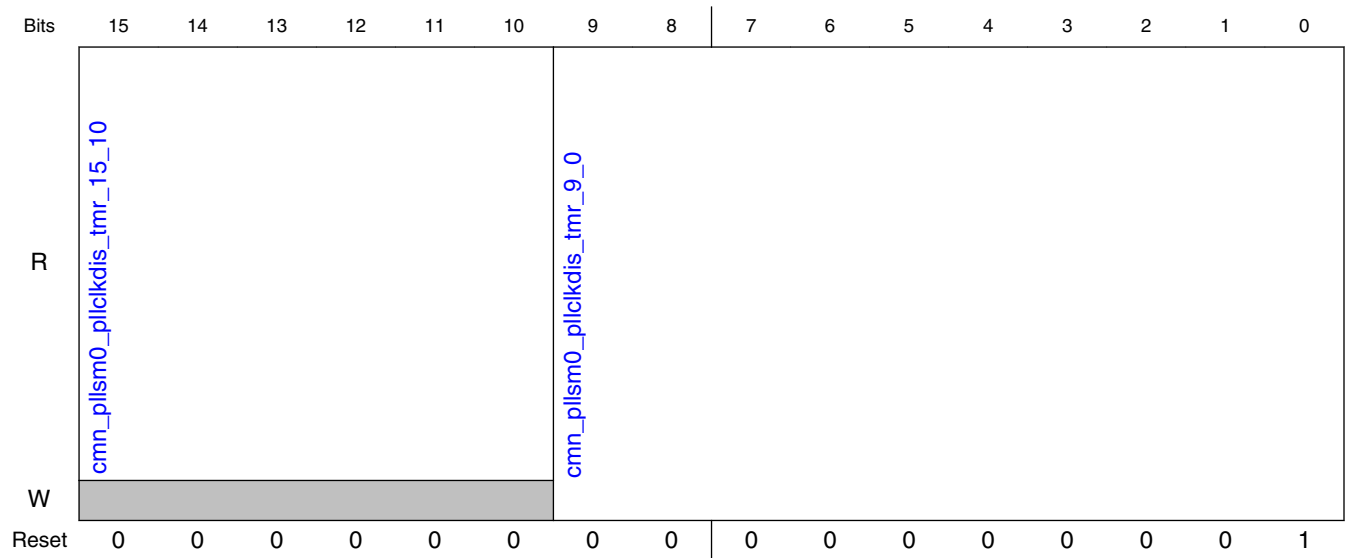
Field	Function
15-10 cmn_pllsm0_plllock_tmr_15_10	Reserved
9-0 cmn_pllsm0_plllock_tmr_9_0	PLL lock delay state timer value : Value used for the timer when the startup state machine is in the PLL lock delay state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.421 PLL 0 clock disable delay timer register (cmn_pllsm0_pllckdis_tmr)

13.4.10.2.421.1 Offset

Register	Offset
cmn_pllsm0_pllckdis_tmr	8_002Dh

13.4.10.2.421.2 Diagram



13.4.10.2.421.3 Fields

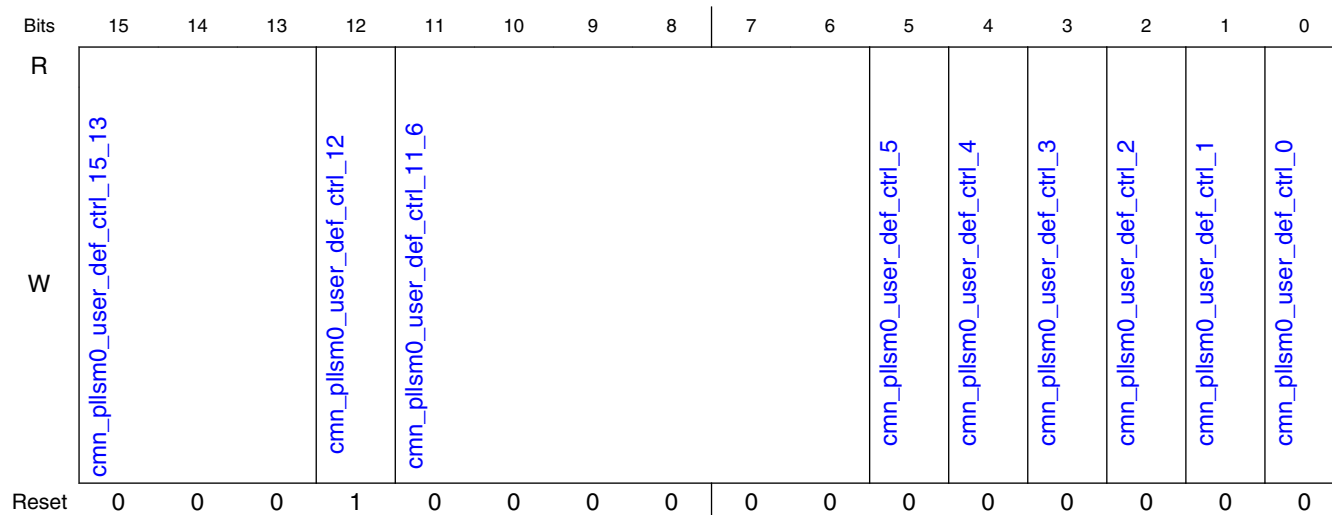
Field	Function
15-10 <code>cmn_pllsm0_pllckdis_tmr_15_10</code>	Reserved
9-0 <code>cmn_pllsm0_pllckdis_tmr_9_0</code>	PLL clock disable delay state timer value : Value used for the timer when the startup state machine is in the PLL clock disable delay state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.422 PLL 0 control state machine user defined control register (`cmn_pllsm0_user_def_ctrl`)

13.4.10.2.422.1 Offset

Register	Offset
<code>cmn_pllsm0_user_def_ctrl</code>	8_002Fh

13.4.10.2.422.2 Diagram



13.4.10.2.422.3 Fields

Field	Function
15-13 cmn_pllsm0_user_def_ctrl_15_13	Reserved
12 cmn_pllsm0_user_def_ctrl_12	PLL oscillator ring select: Selects which oscillator ring is enabled.
11-6 cmn_pllsm0_user_def_ctrl_11_6	Reserved
5 cmn_pllsm0_user_def_ctrl_5	PLL bias enable override enable : When active (1b1), the PLL bias enable override bit in this register will drive the
4 cmn_pllsm0_user_def_ctrl_4	PLL bias enable override : When enabled by the PLL bias enable override enable bit in this register, this bit will drive the
3 cmn_pllsm0_user_def_ctrl_3	PLL 0 start loop override enable : When active (1b1), the PLL 0 start loop override bit in this register will drive the
2 cmn_pllsm0_user_def_ctrl_2	PLL 0 start loop enable override : When enabled by the PLL 0 start loop override enable bit in this register, this bit will drive the
1 cmn_pllsm0_user_def_ctrl_1	PLL 0 VCO LDO enable override enable : When active (1b1), the PLL 0 VCO LDO enable override bit in this register will drive the

Table continues on the next page...

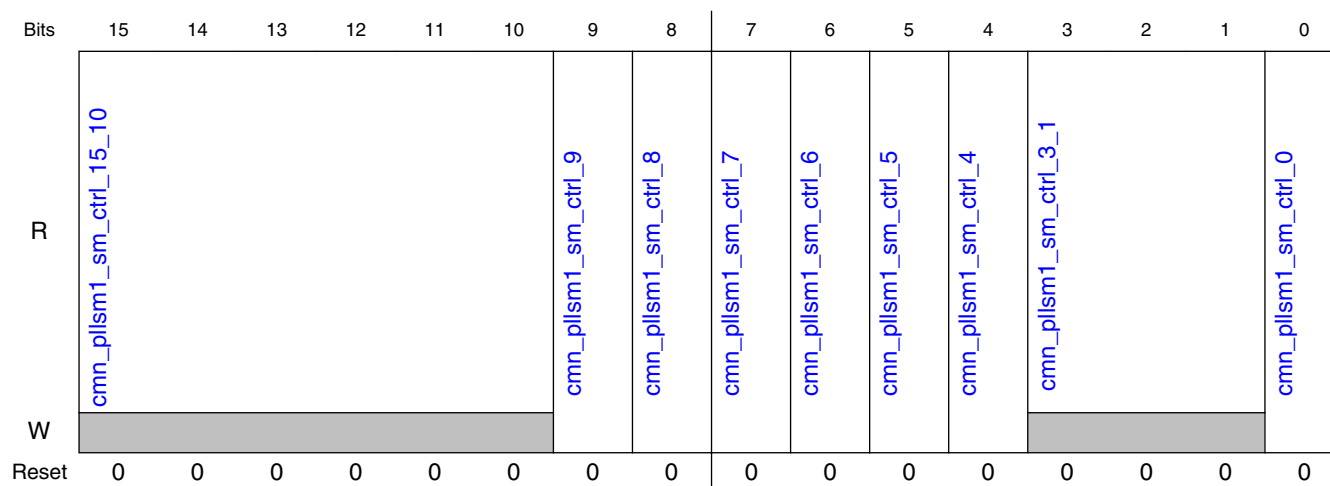
Field	Function
cmn_pllsm0_us er_def_ctrl_1	
0 cmn_pllsm0_us er_def_ctrl_0	PLL 0 VCO LDO enable override : When enabled by the PLL 0 VCO LDO enable override enable bit in this register, this bit will drive the

13.4.10.2.423 PLL 1 control state machine control register (cmn_pllsm1_sm_ctrl)

13.4.10.2.423.1 Offset

Register	Offset
cmn_pllsm1_sm_ctrl	8_0030h

13.4.10.2.423.2 Diagram



13.4.10.2.423.3 Fields

Field	Function
15-10 cmn_pllsm1_sm _ctrl_15_10	Reserved
9	PLL enable override enable : When active (1b1), the PLL enable override bit in this register will drive the

Table continues on the next page...

Clocks And Resets

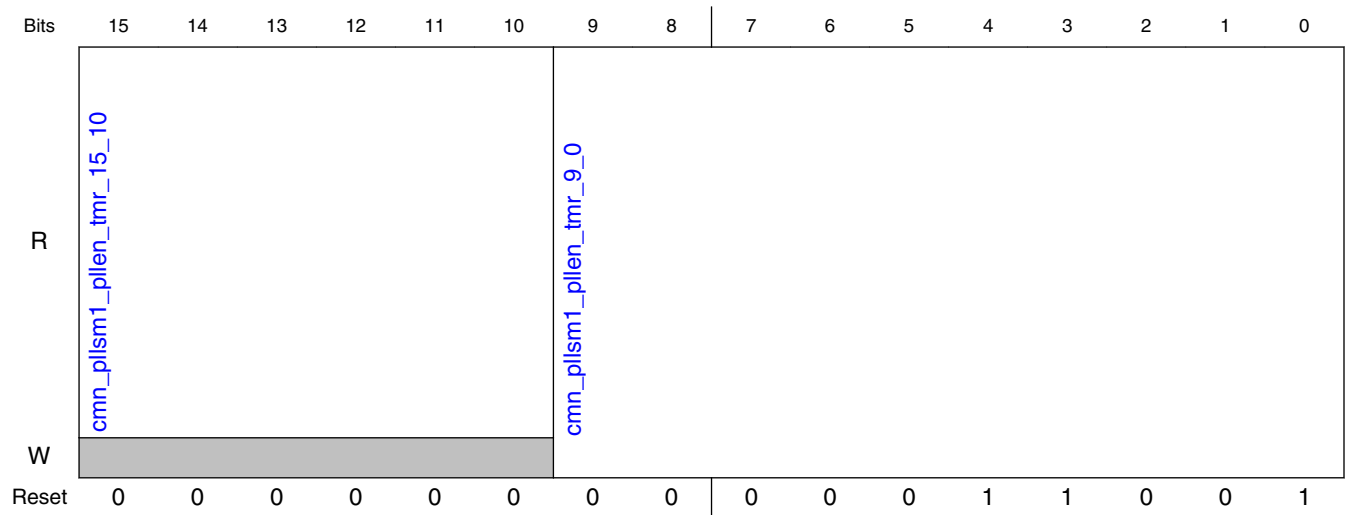
Field	Function
cmn_pllsm1_sm_ctrl_9	
8 cmn_pllsm1_sm_ctrl_8	PLL enable override : When enabled by the PLL enable override enable bit in this register, this bit will drive the
7 cmn_pllsm1_sm_ctrl_7	PLL VCO LDO reference override enable : When active (1b1), the PLL VCO LDO reference override bit in this register will drive the
6 cmn_pllsm1_sm_ctrl_6	PLL VCO LDO reference override : When enabled by the PLL PLL VCO LDO reference override enable bit in this register, this bit will drive the
5 cmn_pllsm1_sm_ctrl_5	PLL VCO LDO reference charge pulse override enable : When active (1b1), the PLL VCO LDO reference charge pulse override bit in this register will drive the
4 cmn_pllsm1_sm_ctrl_4	PLL VCO LDO reference charge pulse override : When enabled by the PLL VCO LDO reference charge pulse override enable bit in this register, this bit will drive the
3-1 cmn_pllsm1_sm_ctrl_3_1	Reserved
0 cmn_pllsm1_sm_ctrl_0	Skip PLL re-calibration : When this bit is active (1b1), the PLL calibration state will be skipped if it was previously run, unless the PLL is disabled or resetting the state machine.

13.4.10.2.424 PLL 1 enable timer register (cmn_pllsm1_pllen_tmr)

13.4.10.2.424.1 Offset

Register	Offset
cmn_pllsm1_pllen_tmr	8_0031h

13.4.10.2.424.2 Diagram



13.4.10.2.424.3 Fields

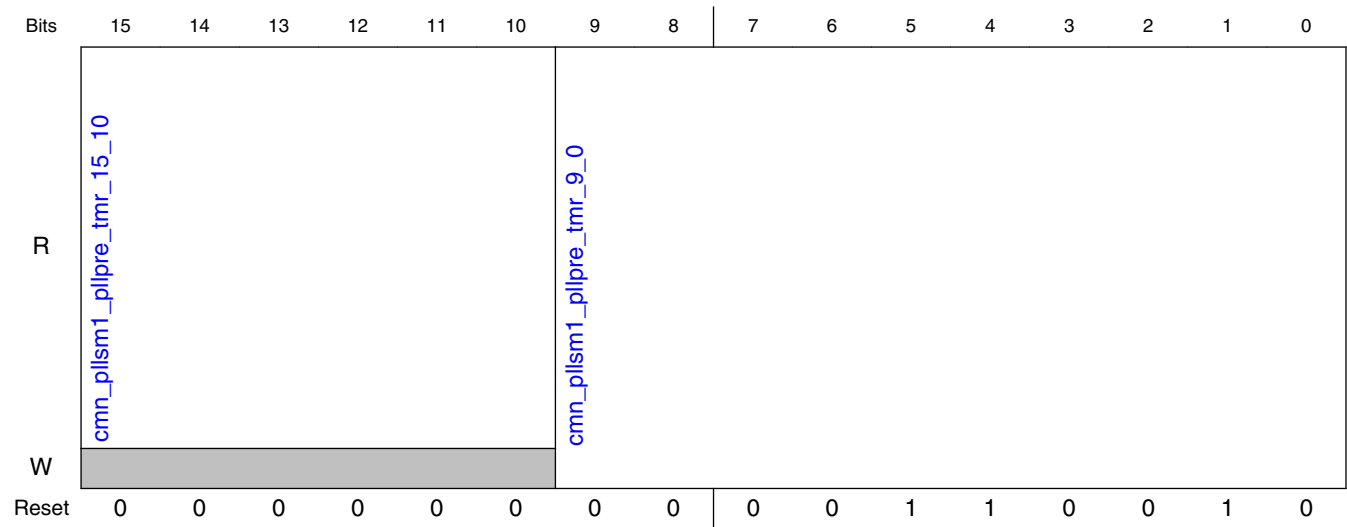
Field	Function
15-10 cmn_pllsm1_pllpre_tmr_15_10	Reserved
9-0 cmn_pllsm1_pllpre_tmr_9_0	PLL enable state timer value : Value used for the timer when the startup state machine is in the PLL enable state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.425 PLL 1 pre-charge timer register (cmn_pllsm1_pllpre_tmr)

13.4.10.2.425.1 Offset

Register	Offset
cmn_pllsm1_pllpre_tmr	8_0032h

13.4.10.2.425.2 Diagram



13.4.10.2.425.3 Fields

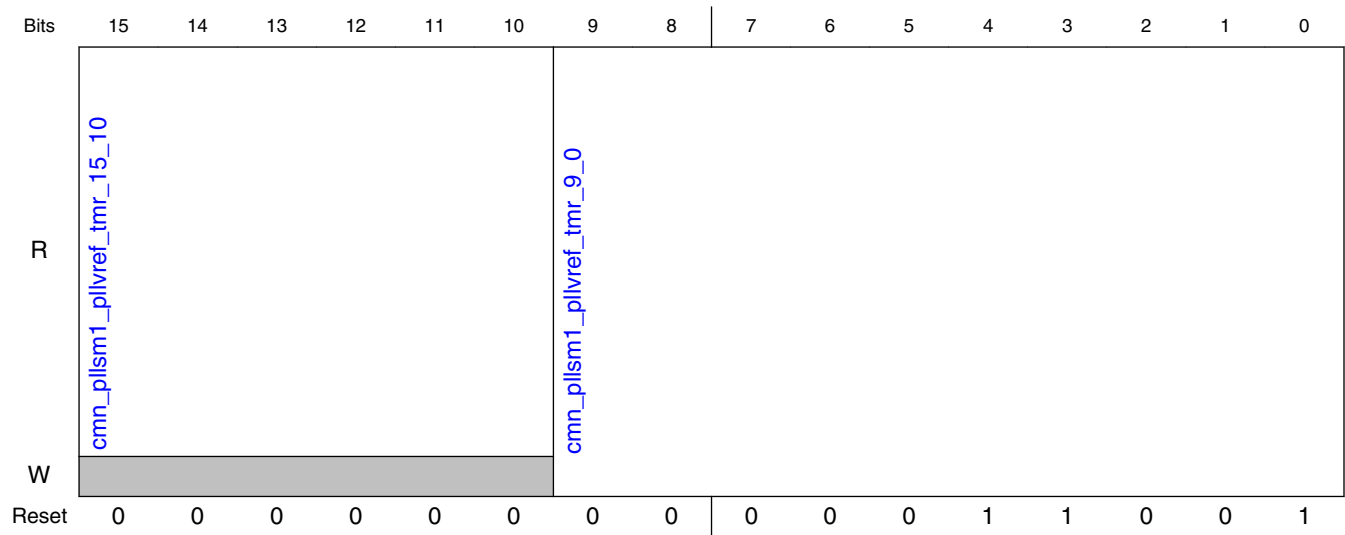
Field	Function
15-10 cmn_pllsm1_pllpre_tmr_15_10	Reserved
9-0 cmn_pllsm1_pllpre_tmr_9_0	PLL pre-charge state timer value : Value used for the timer when the startup state machine is in the PLL pre-charge state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.426 PLL 1 VREF delay timer register (cmn_pllsm1_pllvref_tmr)

13.4.10.2.426.1 Offset

Register	Offset
cmn_pllsm1_pllvref_tmr	8_0033h

13.4.10.2.426.2 Diagram



13.4.10.2.426.3 Fields

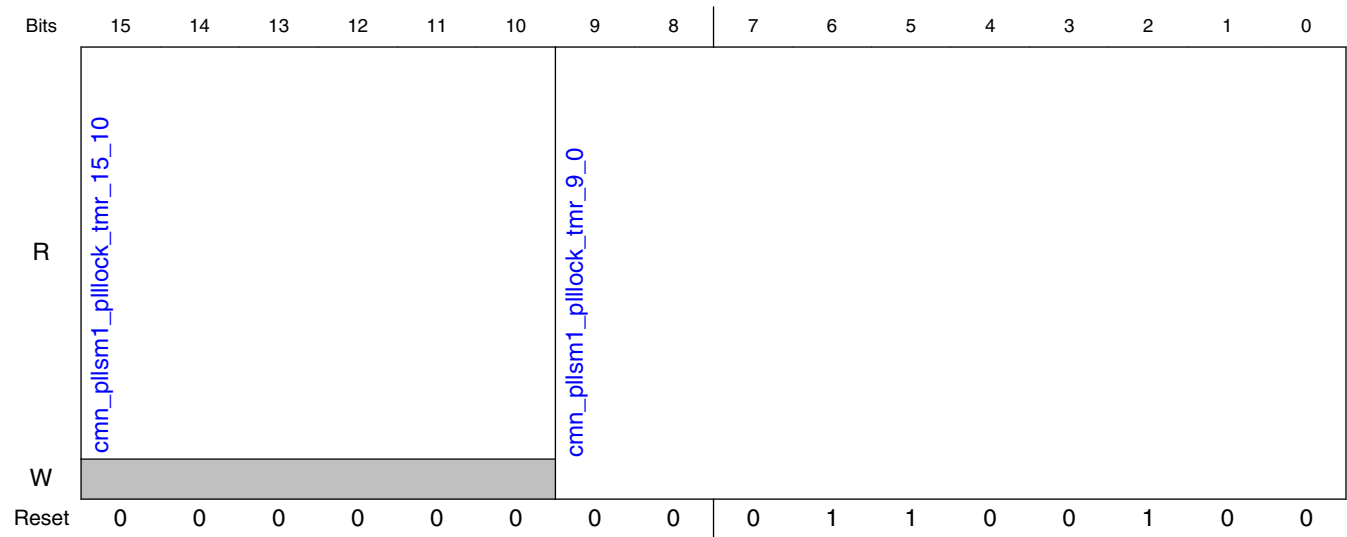
Field	Function
15-10 cmn_pllsm1_pll ref_tmr_15_10	Reserved
9-0 cmn_pllsm1_pll ref_tmr_9_0	PLL VREF delay state timer value : Value used for the timer when the startup state machine is in the PLL VREF delay state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.427 PLL 1 lock delay timer register (cmn_pllsm1_plllock_tmr)

13.4.10.2.427.1 Offset

Register	Offset
cmn_pllsm1_plllock_tmr	8_0034h

13.4.10.2.427.2 Diagram



13.4.10.2.427.3 Fields

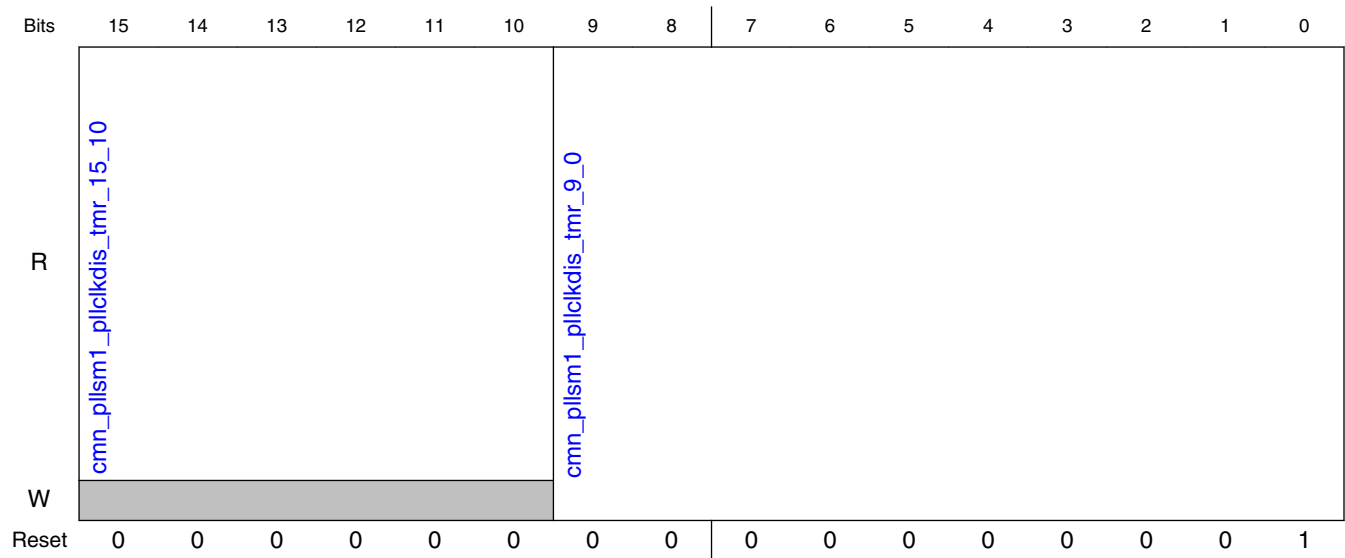
Field	Function
15-10 cmn_pllsm1_plllock_tmr_15_10	Reserved
9-0 cmn_pllsm1_plllock_tmr_9_0	PLL lock delay state timer value : Value used for the timer when the startup state machine is in the PLL lock delay state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.428 PLL 1 clock disable delay timer register (cmn_pllsm1_pllckdis_tmr)

13.4.10.2.428.1 Offset

Register	Offset
cmn_pllsm1_pllckdis_tmr	8_0035h

13.4.10.2.428.2 Diagram



13.4.10.2.428.3 Fields

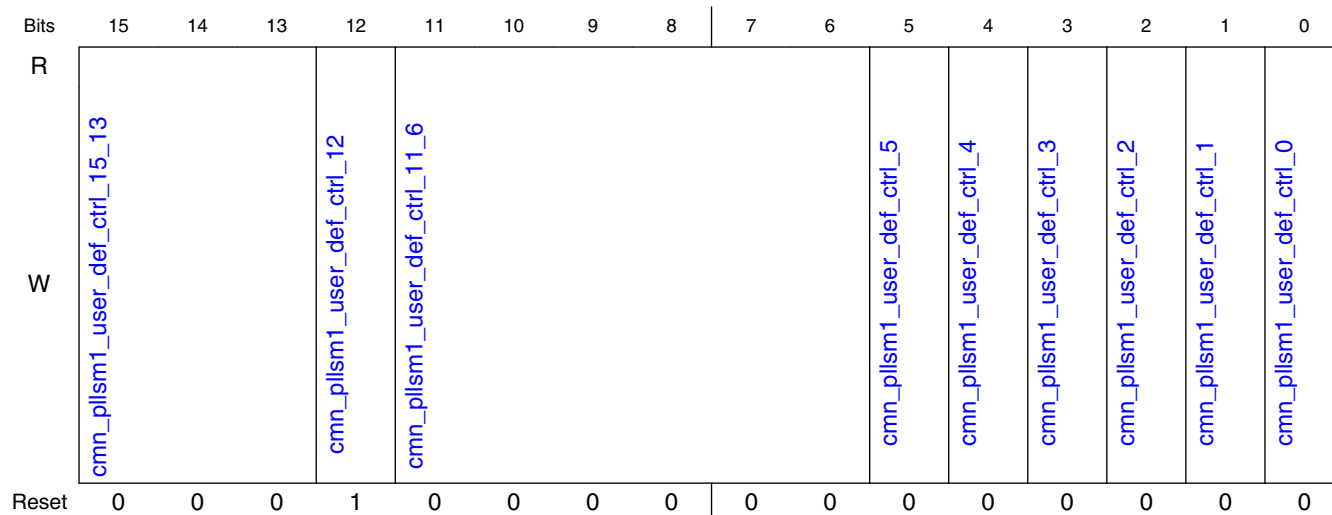
Field	Function
15-10 cmn_pllsm1_pllclkdis_tmr_15_10	Reserved
9-0 cmn_pllsm1_pllclkdis_tmr_9_0	PLL clock disable delay state timer value : Value used for the timer when the startup state machine is in the PLL clock disable delay state. This timer delay is specified as the number of reference clocks to count.

13.4.10.2.429 PLL 1 control state machine user defined control register (cmn_pllsm1_user_def_ctrl)

13.4.10.2.429.1 Offset

Register	Offset
cmn_pllsm1_user_def_ctrl	8_0037h

13.4.10.2.429.2 Diagram



13.4.10.2.429.3 Fields

Field	Function
15-13 cmn_pllsm1_user_def_ctrl_15_13	Reserved
12 cmn_pllsm1_user_def_ctrl_12	PLL oscillator ring select: Selects which oscillator ring is enabled.
11-6 cmn_pllsm1_user_def_ctrl_11_6	Reserved
5 cmn_pllsm1_user_def_ctrl_5	PLL bias enable override enable : When active (1b1), the PLL bias enable override bit in this register will drive the
4 cmn_pllsm1_user_def_ctrl_4	PLL bias enable override : When enabled by the PLL bias enable override enable bit in this register, this bit will drive the
3 cmn_pllsm1_user_def_ctrl_3	PLL 1 start loop override enable : When active (1b1), the PLL 1 start loop override bit in this register will drive the
2 cmn_pllsm1_user_def_ctrl_2	PLL 1 start loop enable override : When enabled by the PLL 1 start loop override enable bit in this register, this bit will drive the
1 cmn_pllsm1_user_def_ctrl_1	PLL 1 VCO LDO enable override enable : When active (1b1), the PLL 1 VCO LDO enable override bit in this register will drive the

Table continues on the next page...

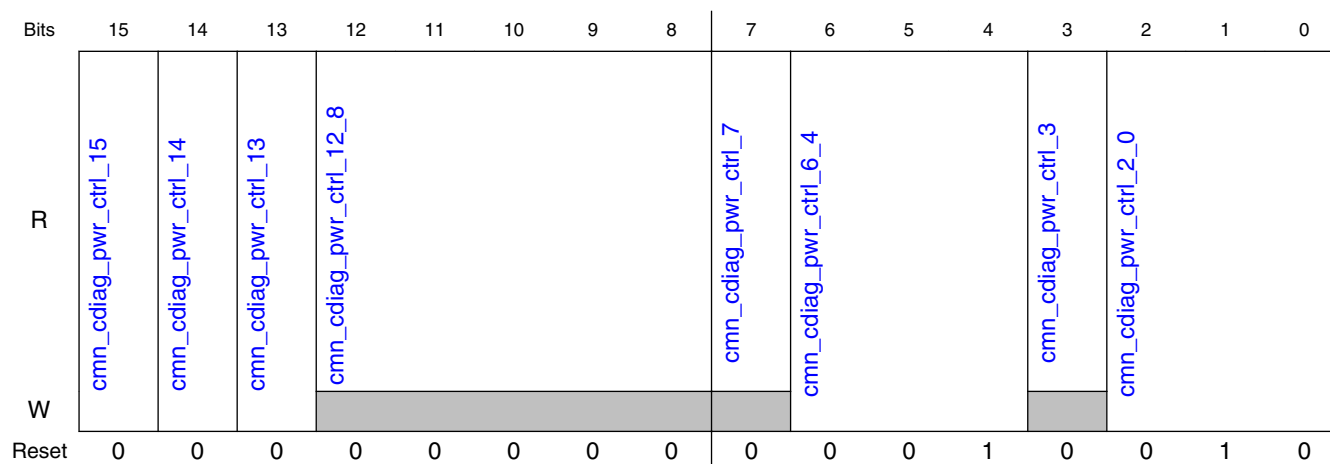
Field	Function
cmn_pllsm1_us er_def_ctrl_1	
0 cmn_pllsm1_us er_def_ctrl_0	PLL 1 VCO LDO enable override : When enabled by the PLL 1 VCO LDO enable override enable bit in this register, this bit will drive the

13.4.10.2.430 Common control power island control register (cmn_cdiag_pwr_ctrl)

13.4.10.2.430.1 Offset

Register	Offset
cmn_cdiag_pwr_ctrl	8_0060h

13.4.10.2.430.2 Diagram



13.4.10.2.430.3 Fields

Field	Function
15 cmn_cdiag_pwr _ctrl_15	Startup State Machine Auto Calibration Power down disable: Setting this bit to 1b1 will disable turning off the startup state machine auto calibration power island when the macro is in a state that would normally switch the power islands off.
14 cmn_cdiag_pwr _ctrl_14	PLL Control 1 State Machine Auto Calibration Power down disable: Setting this bit to 1b1 will disable turning off the PLL control 1 state machine calibration power island when the macro is in a state that would normally switch the power islands off.

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Clocks And Resets

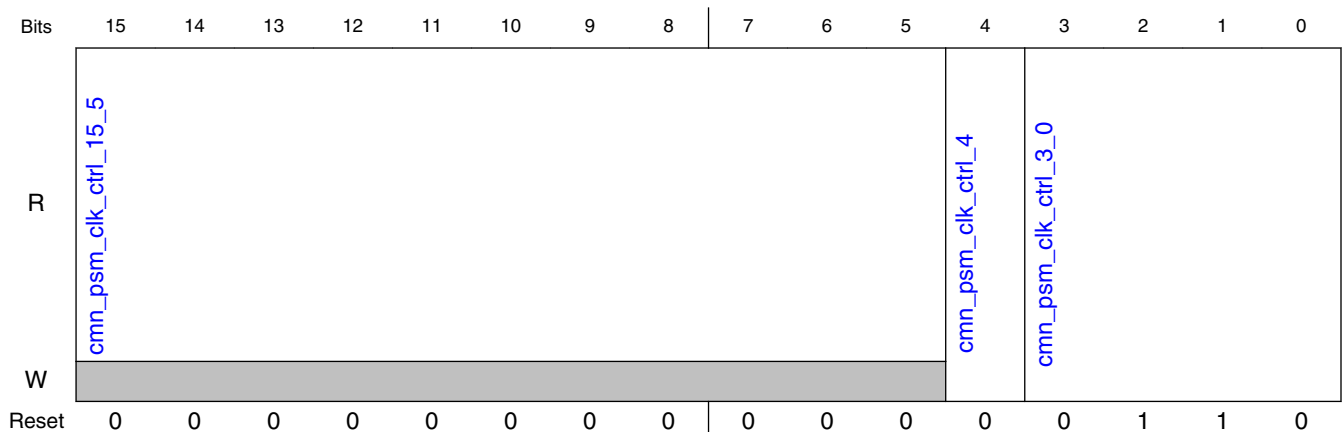
Field	Function
13 cmn_cdiag_pwr_ctrl_13	PLL Control 0 State Machine Auto Calibration Power down disable: Setting this bit to 1b1 will disable turning off the PLL control 0 state machine calibration power island when the macro is in a state that would normally switch the power islands off.
12-8 cmn_cdiag_pwr_ctrl_12_8	Reserved
7 cmn_cdiag_pwr_ctrl_7	Reserved
6-4 cmn_cdiag_pwr_ctrl_6_4	Power enable phase 2 timer value: This specifies the number of reference clock cycles all the power island control state machines will wait in the power phase 2 enable states, in order to allow enough time for the second phase of the switched domain to power up, before deactivating the isolation functions.
3 cmn_cdiag_pwr_ctrl_3	Reserved
2-0 cmn_cdiag_pwr_ctrl_2_0	Power enable phase 1 timer value: This specifies the number of reference clock cycles all the power island control state machine will wait in the power phase 1 enable states, in order to allow enough time for the first phase of the switched domain to power up, before enabling the second phase of the power up.

13.4.10.2.431 Common PSM clock control register (cmn_psm_clk_ctrl)

13.4.10.2.431.1 Offset

Register	Offset
cmn_psm_clk_ctrl	8_0061h

13.4.10.2.431.2 Diagram



13.4.10.2.431.3 Fields

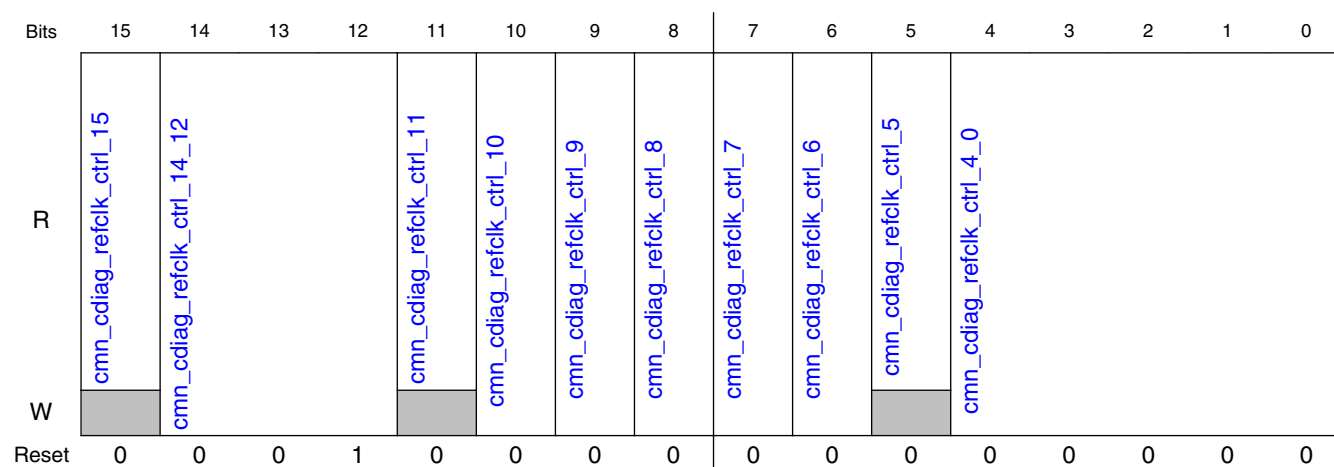
Field	Function
15-5 cmn_psm_clk_ctl rl_15_5	Reserved
4 cmn_psm_clk_ctl rl_4	PSM clock select: This bit selects the source of the PSM clock.
3-0 cmn_psm_clk_ctl rl_3_0	PLL clock divider value for PSM clock: This specifies the value of the clock divider associated with PLL 0 that is used to generate the PSM clock.

13.4.10.2.432 Reference clock receiver control register (cmn_cdiag_refclk_ctrl)

13.4.10.2.432.1 Offset

Register	Offset
cmn_cdiag_refclk_ctrl	8_0062h

13.4.10.2.432.2 Diagram



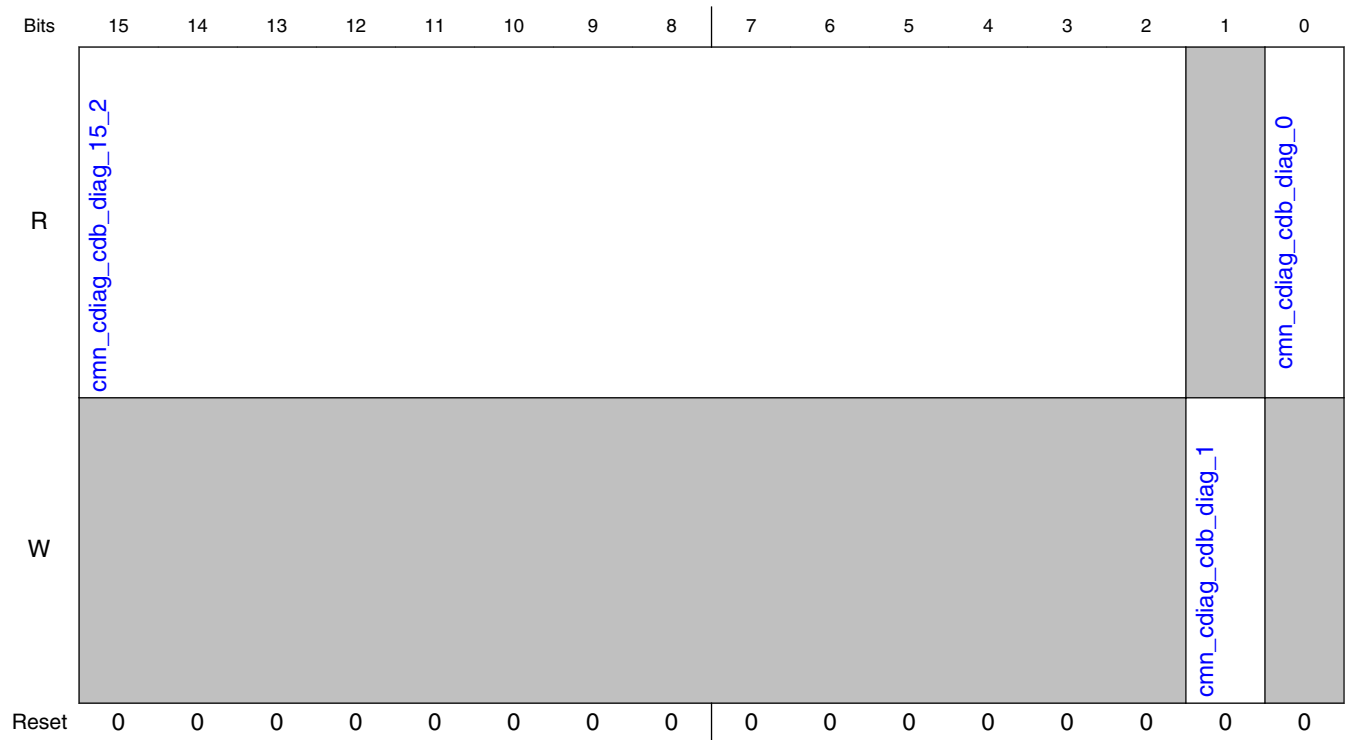
13.4.10.2.432.3 Fields

Field	Function
15 cmn_cdiag_refcl k_ctrl_15	Reserved
14-12 cmn_cdiag_refcl k_ctrl_14_12	Digital reference clock divider scaler: Scales the reference clock digital divider control signal (
11 cmn_cdiag_refcl k_ctrl_11	Reserved
10 cmn_cdiag_refcl k_ctrl_10	Reference clock analog clock test mode: Test mode signal that allows the analog path reference clock to drive the digital, by driving the
9 cmn_cdiag_refcl k_ctrl_9	Reference clock digital clock test mode: Test mode signal that allows the digital path reference clock to drive the PLL, by driving the
8 cmn_cdiag_refcl k_ctrl_8	Reference clock hysteresis enable: Enables the hysteresis in the digital slicer output, by driving the
7 cmn_cdiag_refcl k_ctrl_7	Reference clock termination enable override enable: When this bit is set, the reference clock termination enable override bit in this register will override the reference clock termination enable, by driving the
6 cmn_cdiag_refcl k_ctrl_6	Reference clock termination enable override: When the reference clock termination enable override enable bit in this register is set to 1b1, this bit will override the reference clock termination enable, by driving the
5 cmn_cdiag_refcl k_ctrl_5	Reserved
4-0 cmn_cdiag_refcl k_ctrl_4_0	Reference clock termination code: Controls the integrated reference clock termination impedance, by driving the

13.4.10.2.433 Common control CDB diagnostic register (cmn_cdiag_cdb_diag)**13.4.10.2.433.1 Offset**

Register	Offset
cmn_cdiag_cdb_diag	8_0064h

13.4.10.2.433.2 Diagram



13.4.10.2.433.3 Fields

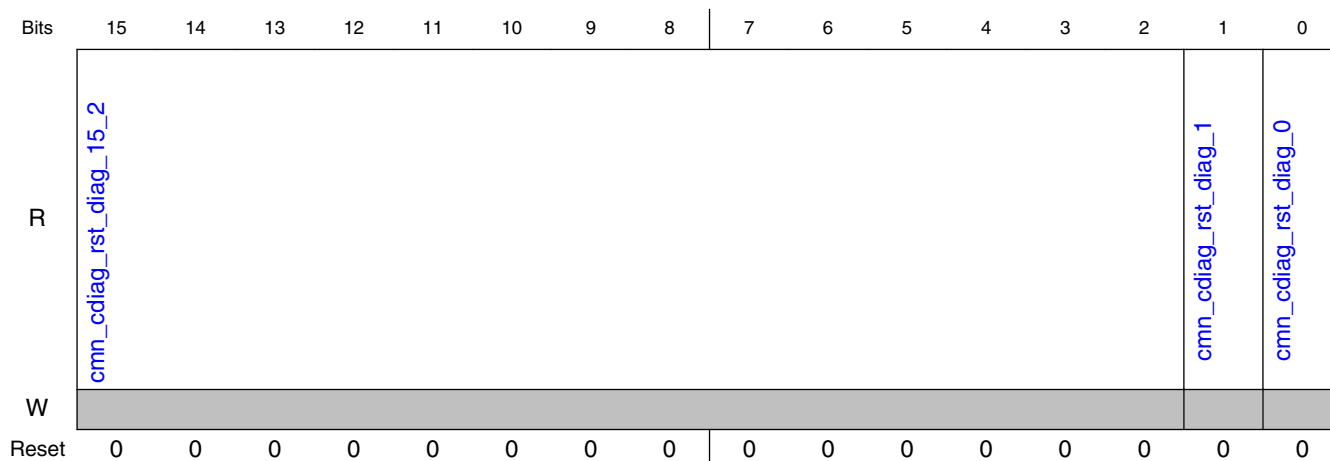
Field	Function
15-2 <code>cmn_cdiag_cdb_diag_15_2</code>	Reserved
1 <code>cmn_cdiag_cdb_diag_1</code>	Force clear of the CDB first access detected register bit : Writing a 1b1 to this register will force the clearing of the CDB first access detected register bit, which is the register bit that overrides the
0 <code>cmn_cdiag_cdb_diag_0</code>	CDB bus error: This bit will be set when the internal CDB watchdog timer expires. It is automatically cleared on a read of this register.

13.4.10.2.434 Common control reset diagnostic register (`cmn_cdiag_rst_diag`)

13.4.10.2.434.1 Offset

Register	Offset
cmn_cdiag_rst_diag	8_0066h

13.4.10.2.434.2 Diagram



13.4.10.2.434.3 Fields

Field	Function
15-2 cmn_cdiag_rst_diag_15_2	Reserved
1 cmn_cdiag_rst_diag_1	Current state of the macro_pwr_reset_sync_n reset.
0 cmn_cdiag_rst_diag_0	Current state of the cmn_reset_sync_n reset.

13.4.10.2.435 Common control cover your alternatives register (cmn_cdiag_dcya)

13.4.10.2.435.1 Offset

Register	Offset
cmn_cdiag_dcya	8_007Fh

13.4.10.2.435.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_cdiag_dcy_a_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.435.3 Fields

Field	Function
15-0 cmn_cdiag_dcy_a_15_0	Reserved

13.4.10.2.436 PLL 0 VCO calibration control register (cmn_pll0_vcocal_ctrl)

13.4.10.2.436.1 Offset

Register	Offset
cmn_pll0_vcocal_ctrl	8_0080h

13.4.10.2.436.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pll0_vcocal_ctrl_15	cmn_pll0_vcocal_ctrl_14	cmn_pll0_vcocal_ctrl_13_9					cmn_pll0_vcocal_ctrl_8_0								
W																
Reset	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

13.4.10.2.436.3 Fields

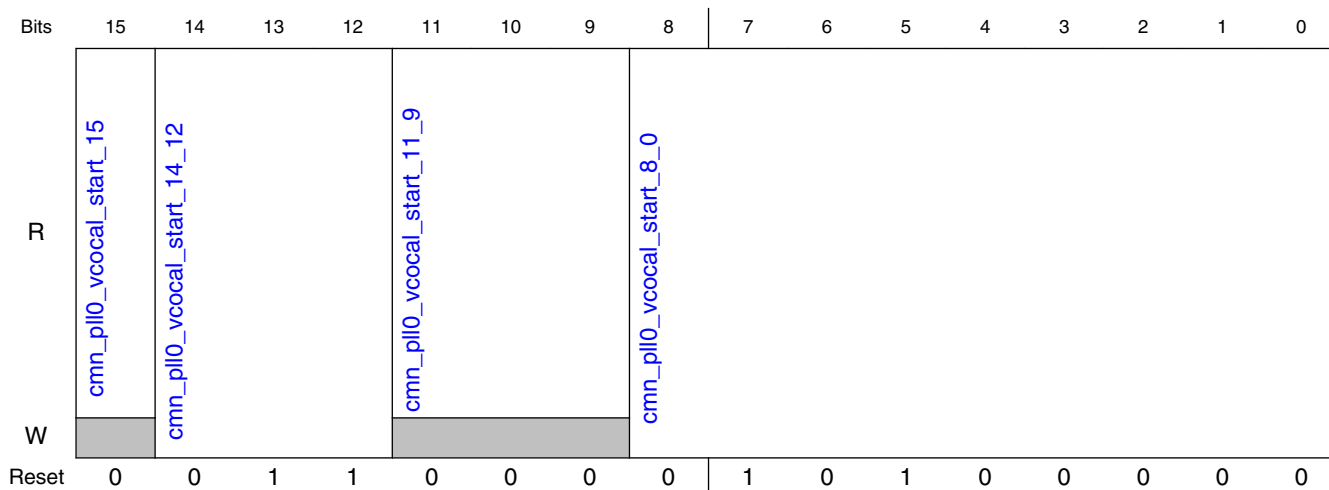
Field	Function
15 cmn_pll0_vcocal_ctrl_15	Start VCO calibration: Activating (1b1) this bit will start a VCO calibration process. This bit must remain active until the VCO calibration process is complete (as indicated by the VCO calibration process done bit in this register). To start another VCO calibration process, the VCO calibration process done bit must have gone inactive from any prior calibration process.
14 cmn_pll0_vcocal_ctrl_14	VCO calibration process done: This bit will be set to 1b1 when the VCO calibration process is complete. It will be cleared by the deactivation of the Start VCO calibration bit in this register.
13-9 cmn_pll0_vcocal_ctrl_13_9	Reserved
8-0 cmn_pll0_vcocal_ctrl_8_0	VCO calibration code: This is the calibration code that was determined by the VCO calibration process. This signal is valid when the VCO calibration process is complete. The values of this field correspond to different frequency bands the VCO will operate in. The frequency bands are controlled by the number of capacitors that are switched in the VCO analog circuit. This field specifies the number of capacitors that are switched in. The following are the values for this code:

13.4.10.2.437 PLL 0 VCO calibration start point register (cmn_pll0_vcocal_start)

13.4.10.2.437.1 Offset

Register	Offset
cmn_pll0_vcocal_start	8_0081h

13.4.10.2.437.2 Diagram



13.4.10.2.437.3 Fields

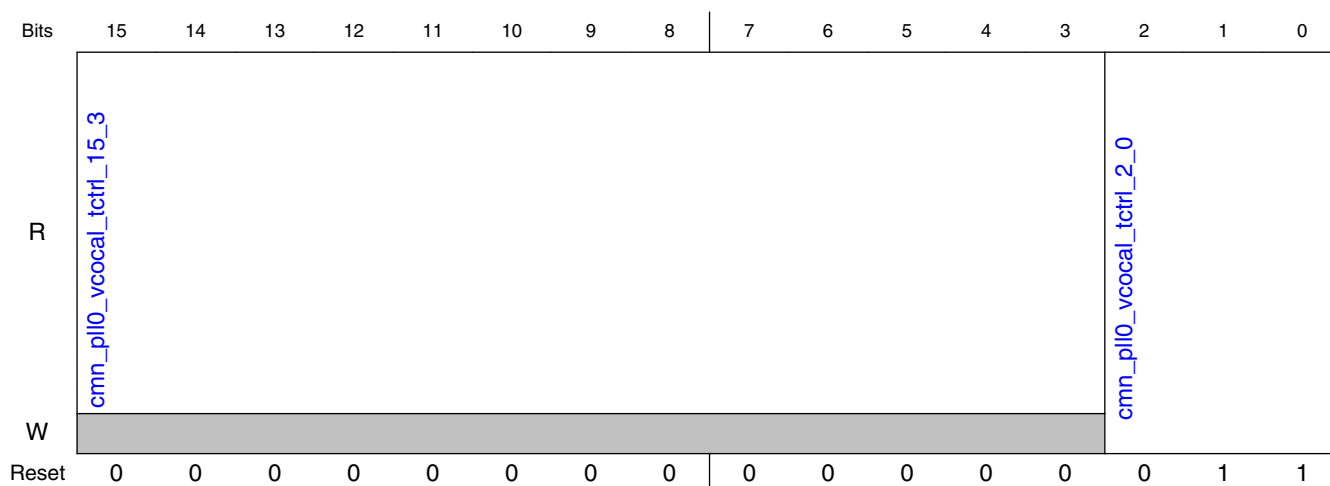
Field	Function
15 cmn_pll0_vcocal_start_15	Reserved
14-12 cmn_pll0_vcocal_start_14_12	VCO calibration initial step size control: This field specifies the initial step size for the VCO calibration state machine. The following are the values that can be used in this field, and the corresponding step sizes.
11-9 cmn_pll0_vcocal_start_11_9	Reserved
8-0 cmn_pll0_vcocal_start_8_0	VCO calibration code starting point value: This field specifies the starting VCO code that is used by the VCO calibration state machine. The purpose of this value is such that the VCO calibration process starts at a point that is, on average, relatively close to the final calibration point. This allows the calibration time to be reduced, on average.

13.4.10.2.438 PLL 0 VCO calibration timer control register (cmn_pll0_vcocal_tctrl)

13.4.10.2.438.1 Offset

Register	Offset
cmn_pll0_vcocal_tctrl	8_0082h

13.4.10.2.438.2 Diagram



13.4.10.2.438.3 Fields

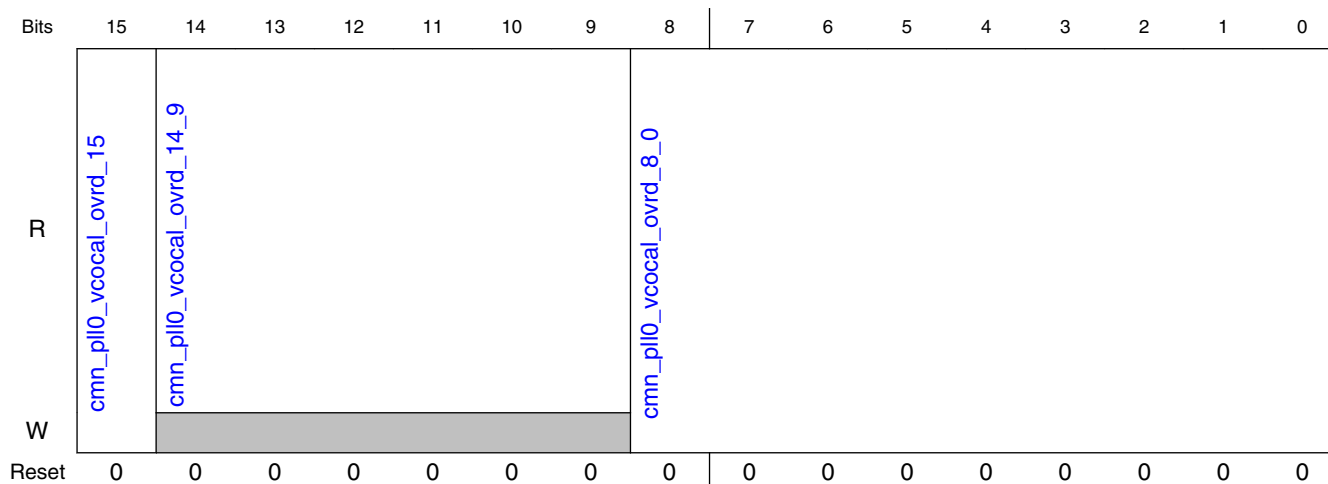
Field	Function
15-3 cmn_pll0_vcocal_tctrl_15_3	Reserved
2-0 cmn_pll0_vcocal_tctrl_2_0	VCO calibration initial time scale control: This field specifies the calibration start time scaling factor applied to the VCO calibration when running the initial step size for the calibration code is not set to 1. Setting this value to a value other than 1 will reduce the calibration measurement time by the amount specified below. Then when the final calibration steps are made the full calibration time will be used to get the appropriate amount of resolution.

13.4.10.2.439 PLL 0 VCO calibration override register (cmn_pll0_vcocal_ovrd)

13.4.10.2.439.1 Offset

Register	Offset
cmn_pll0_vcocal_ovrd	8_0083h

13.4.10.2.439.2 Diagram



13.4.10.2.439.3 Fields

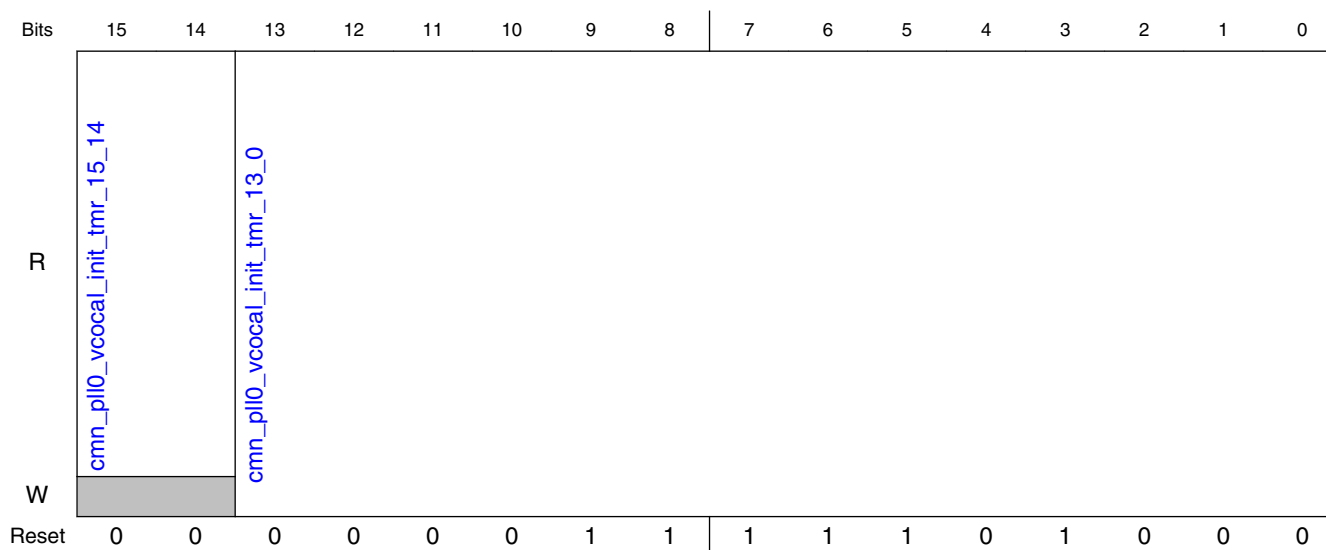
Field	Function
15 cmn_pll0_vcocal_ovrd_15	VCO calibration code override enable: Activating (1b1) this bit allows the VCO code determined during the automatic VCO calibration process to be overridden by the value driven by the VCO calibration code override value field in this register.
14-9 cmn_pll0_vcocal_ovrd_14_9	Reserved
8-0 cmn_pll0_vcocal_ovrd_8_0	VCO calibration code override value: This field is used to override the VCO code determined during the automatic VCO calibration process. The VCO code driven on this field is valid when the VCO calibration code override enable bit in this register is active.

13.4.10.2.440 PLL 0 VCO calibration initialization timer register (cmn_pll0_vcocal_init_tmr)

13.4.10.2.440.1 Offset

Register	Offset
cmn_pll0_vcocal_init_tmr	8_0084h

13.4.10.2.440.2 Diagram



13.4.10.2.440.3 Fields

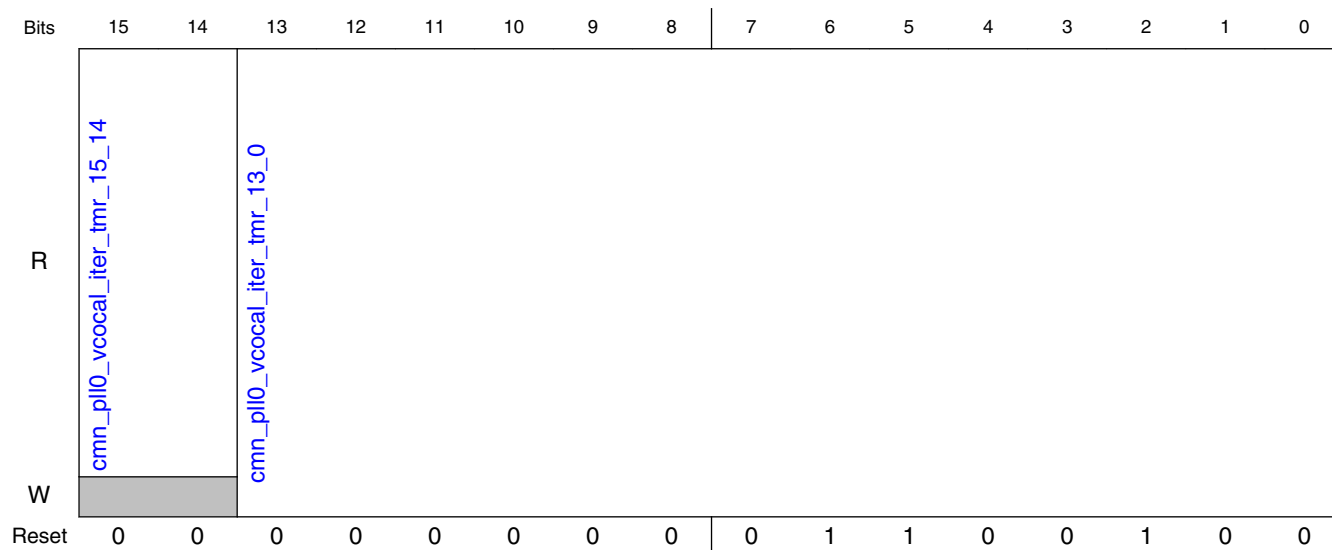
Field	Function
15-14 cmn_pll0_vcocal_init_tmr_15_14	Reserved
13-0 cmn_pll0_vcocal_init_tmr_13_0	Initialization wait timer value: This is the number of clocks to wait between when the analog VCO calibration circuits are enabled, and when the first calibration code is driven to the analog.

13.4.10.2.441 PLL 0 VCO calibration iteration timer register (cmn_pll0_vcocal_iter_tmr)

13.4.10.2.441.1 Offset

Register	Offset
cmn_pll0_vcocal_iter_tmr	8_0085h

13.4.10.2.441.2 Diagram



13.4.10.2.441.3 Fields

Field	Function
15-14	Reserved

Table continues on the next page...

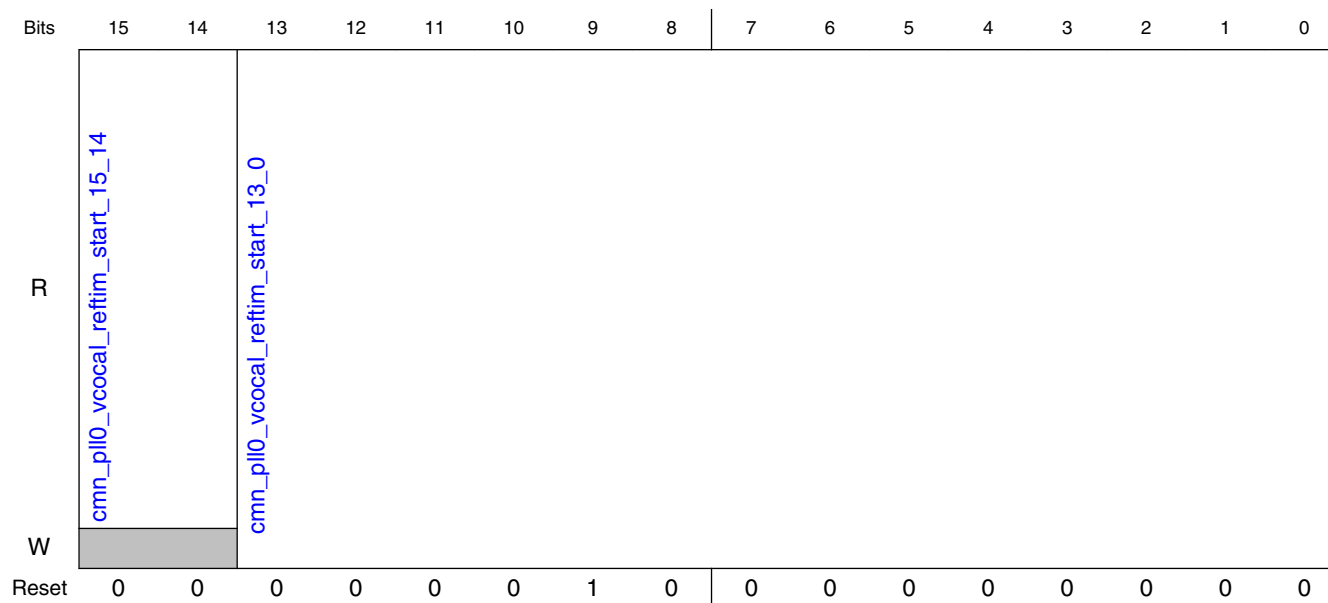
Field	Function
cmn_pll0_vcocal_iter_tmr_15_14	
13-0 cmn_pll0_vcocal_iter_tmr_13_0	Iteration wait timer value: This is the number of clocks to wait between when a calibration code is driven to the analog, and when the clock rates are measured.

13.4.10.2.442 PLL 0 VCO calibration reference clock timer start value register (cmn_pll0_vcocal_reftim_start)

13.4.10.2.442.1 Offset

Register	Offset
cmn_pll0_vcocal_reftim_start	8_0086h

13.4.10.2.442.2 Diagram



13.4.10.2.442.3 Fields

Field	Function
15-14	Reserved

Table continues on the next page...

Clocks And Resets

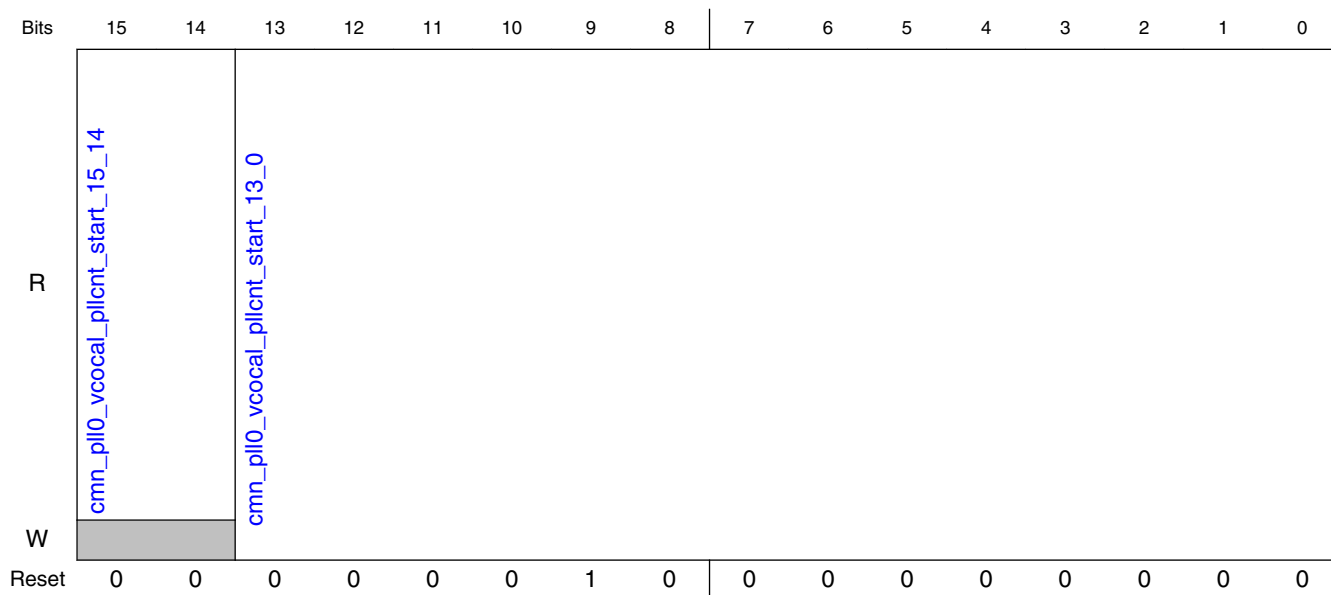
Field	Function
cmn_pll0_vcocal_refim_start_15_14	
13-0 cmn_pll0_vcocal_refim_start_13_0	PLL VCO calibration reference clock timer start value : This is the value that is loaded into the reference clock timer as the starting point for that timer, when running VCO calibration. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes.

13.4.10.2.443 PLL 0 VCO calibration PLL clock counter start value register (cmn_pll0_vcocal_pllcnt_start)

13.4.10.2.443.1 Offset

Register	Offset
cmn_pll0_vcocal_pllcnt_start	8_0088h

13.4.10.2.443.2 Diagram



13.4.10.2.443.3 Fields

Field	Function
15-14	Reserved

Table continues on the next page...

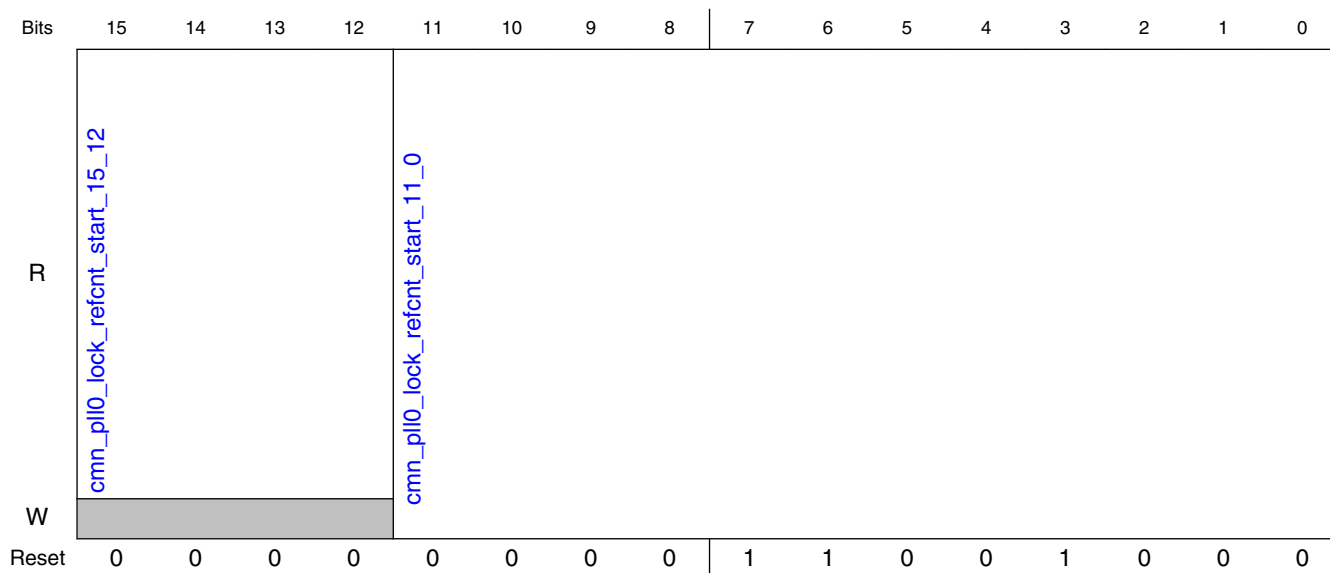
Field	Function
cmn_pll0_vcocal _pllcnt_start_15 _14	
13-0 cmn_pll0_vcocal _pllcnt_start_13 _0	PLL VCO calibration PLL clock counter start value : This is the value that is loaded into the PLL clock counter as the starting point for that counter, when running VCO calibration. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes.

13.4.10.2.444 PLL 0 lock reference counter start value register (cmn_pll0_lock_refcnt_start)

13.4.10.2.444.1 Offset

Register	Offset
cmn_pll0_lock_refcnt_start	8_0090h

13.4.10.2.444.2 Diagram



13.4.10.2.444.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

Clocks And Resets

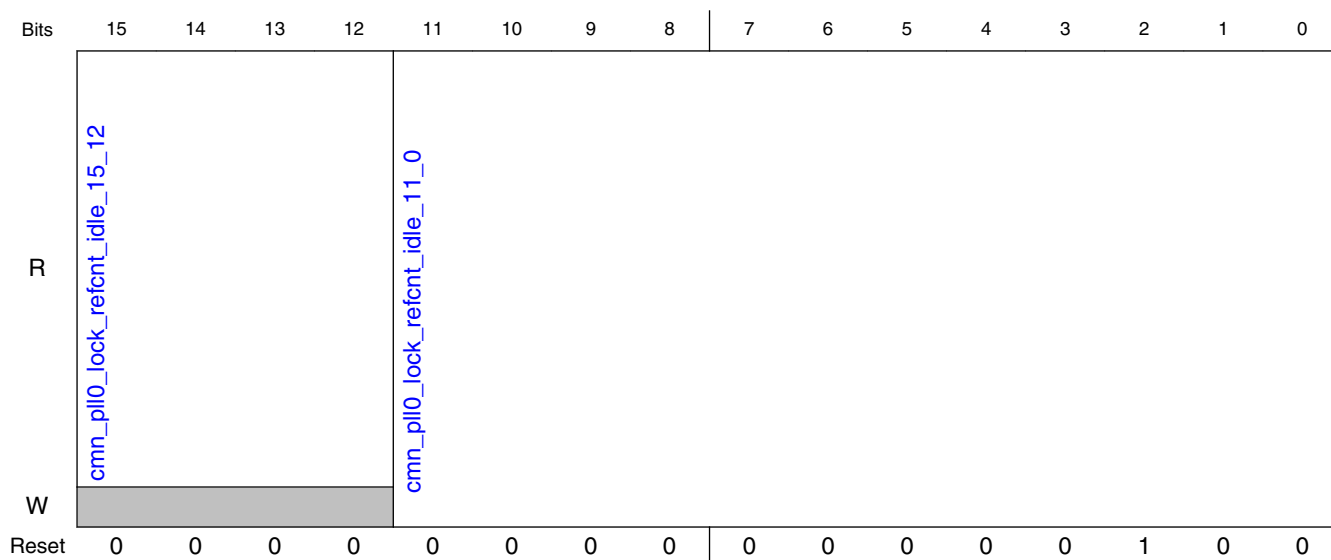
Field	Function
cmn_pll0_lock_refcnt_start_15_12	
11-0 cmn_pll0_lock_refcnt_start_11_0	PLL lock reference counter start value : This is the value that is loaded into the PLL lock detect reference counter as the starting point for that counter, when checking for PLL lock. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes. This is set to 200 clocks by default.

13.4.10.2.445 PLL 0 lock reference counter idle value register (cmn_pll0_lock_refcnt_idle)

13.4.10.2.445.1 Offset

Register	Offset
cmn_pll0_lock_refcnt_idle	8_0091h

13.4.10.2.445.2 Diagram



13.4.10.2.445.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

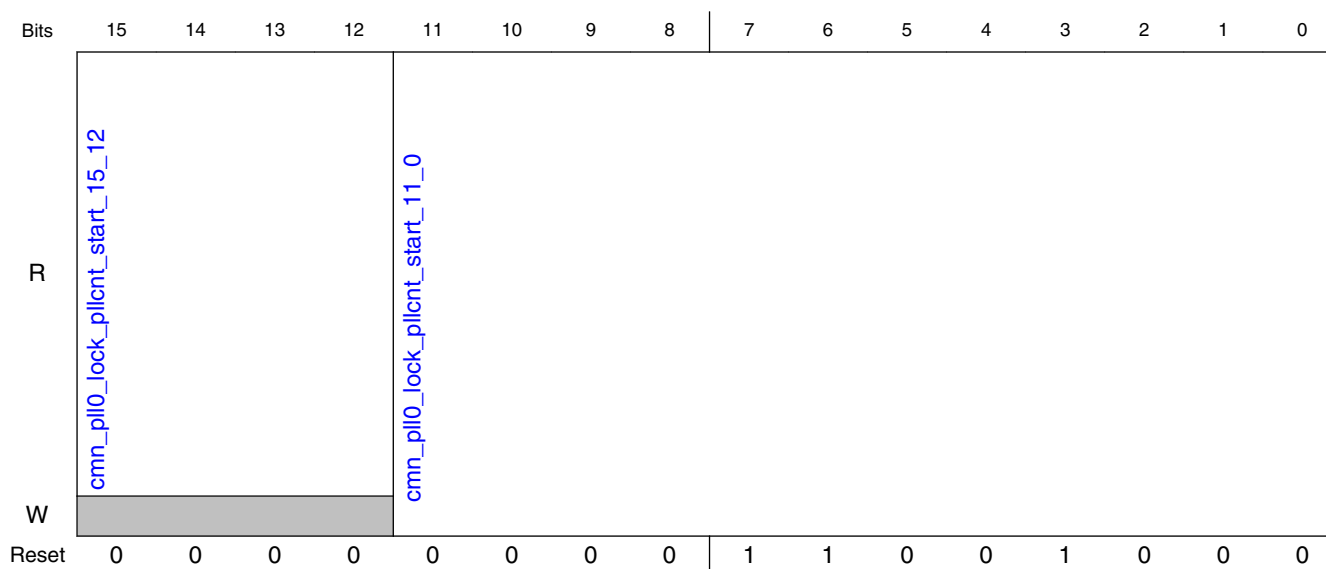
Field	Function
cmn_pll0_lock_refcnt_idle_15_12	
11-0 cmn_pll0_lock_refcnt_idle_11_0	PLL lock reference counter idle value : This is the value used by the PLL lock detection logic to specify the number of reference clocks between each phase of counting PLL clocks.

13.4.10.2.446 PLL 0 lock PLL counter start value register (cmn_pll0_lock_pllcnt_start)

13.4.10.2.446.1 Offset

Register	Offset
cmn_pll0_lock_pllcnt_start	8_0092h

13.4.10.2.446.2 Diagram



13.4.10.2.446.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

Clocks And Resets

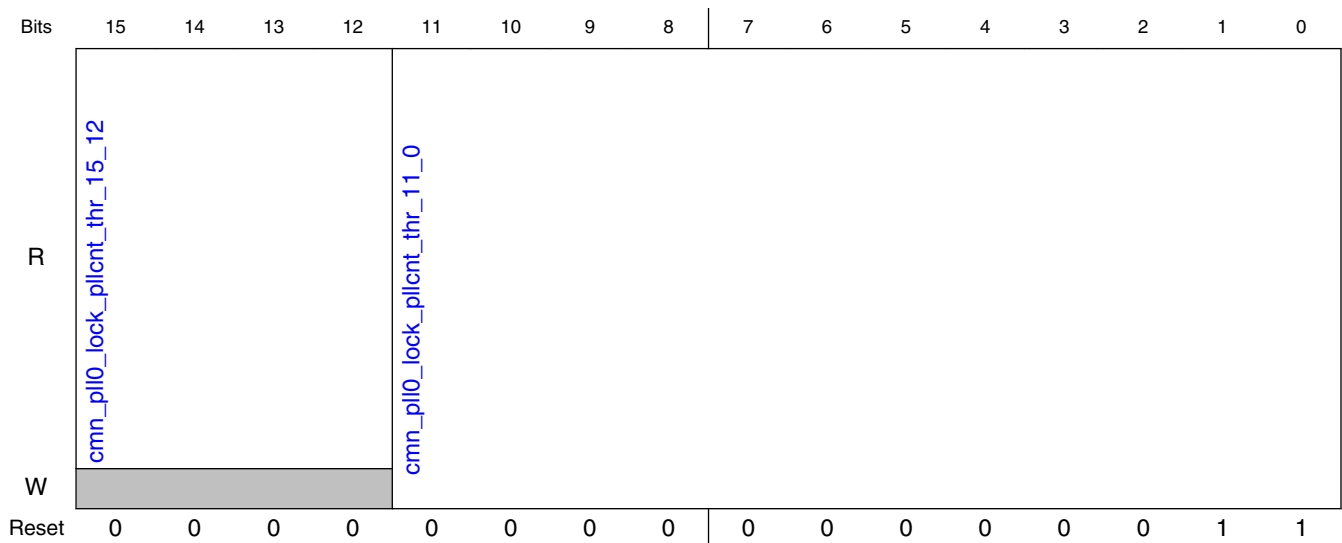
Field	Function
cmn_pll0_lock_pllcnt_start_15_12	
11-0 cmn_pll0_lock_pllcnt_start_11_0	PLL lock PLL counter start value : This is the value that is loaded into the PLL lock detect PLL counter as the starting point for that counter, when checking for PLL lock. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes. This is set to 200 clocks by default.

13.4.10.2.447 PLL 0 lock PLL counter threshold value register (cmn_pll0_lock_pllcnt_thr)

13.4.10.2.447.1 Offset

Register	Offset
cmn_pll0_lock_pllcnt_thr	8_0093h

13.4.10.2.447.2 Diagram



13.4.10.2.447.3 Fields

Field	Function
15-12 cmn_pll0_lock_pllcnt_thr_15_12	Reserved

Table continues on the next page...

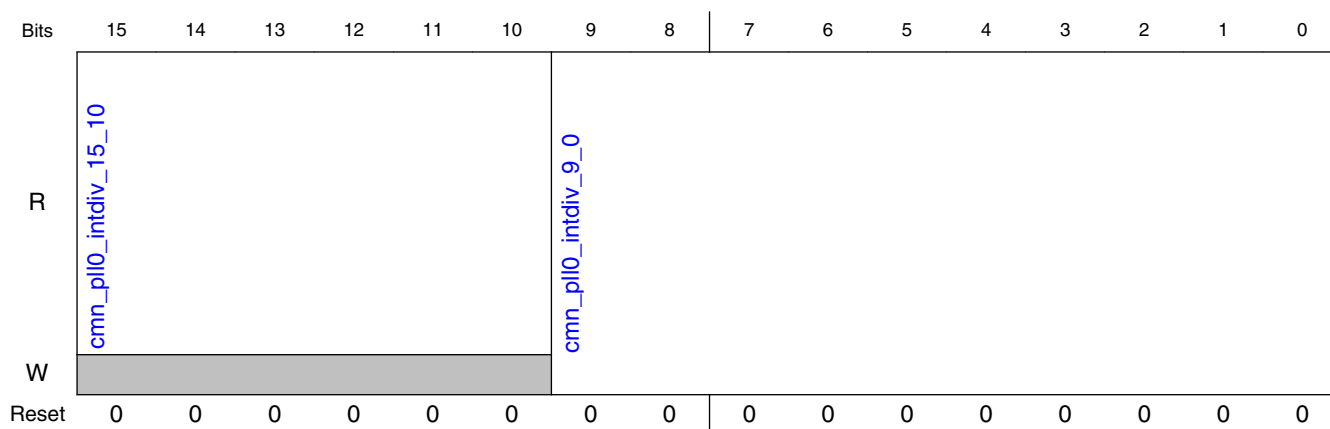
Field	Function
11-0 cmn_pll0_lock_pllcnt_thr_11_0	PLL lock counter threshold value : This is the value used by the PLL lock detection logic to determine if the PLL has locked. If the two counters in the PLL lock detection logic differ by less than this value, the PLL is considered locked. This is set to 3 clocks by default.

13.4.10.2.448 PLL 0 feedback divider integer register (cmn_pll0_intdiv)

13.4.10.2.448.1 Offset

Register	Offset
cmn_pll0_intdiv	8_0094h

13.4.10.2.448.2 Diagram



13.4.10.2.448.3 Fields

Field	Function
15-10 cmn_pll0_intdiv_15_10	Reserved
9-0 cmn_pll0_intdiv_9_0	pll_fb_div_integer value: Value of the pll_fb_div_integer signal.

13.4.10.2.449 PLL 0 feedback divider fractional register (cmn_pll0_fracdiv)**13.4.10.2.449.1 Offset**

Register	Offset
cmn_pll0_fracdiv	8_0095h

13.4.10.2.449.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_pll0_fracdiv_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

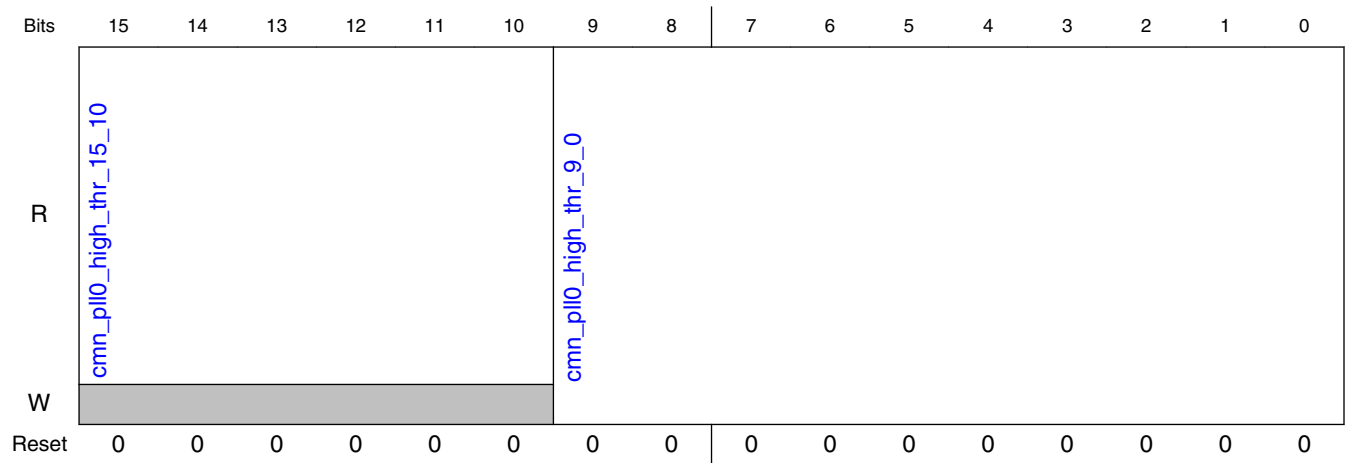
13.4.10.2.449.3 Fields

Field	Function
15-0 cmn_pll0_fracdiv_15_0	pll_fb_div_fractional: Value of the pll_fb_div_fractional signal.

13.4.10.2.450 PLL 0 feedback divider high threshold register (cmn_pll0_high_thr)**13.4.10.2.450.1 Offset**

Register	Offset
cmn_pll0_high_thr	8_0096h

13.4.10.2.450.2 Diagram



13.4.10.2.450.3 Fields

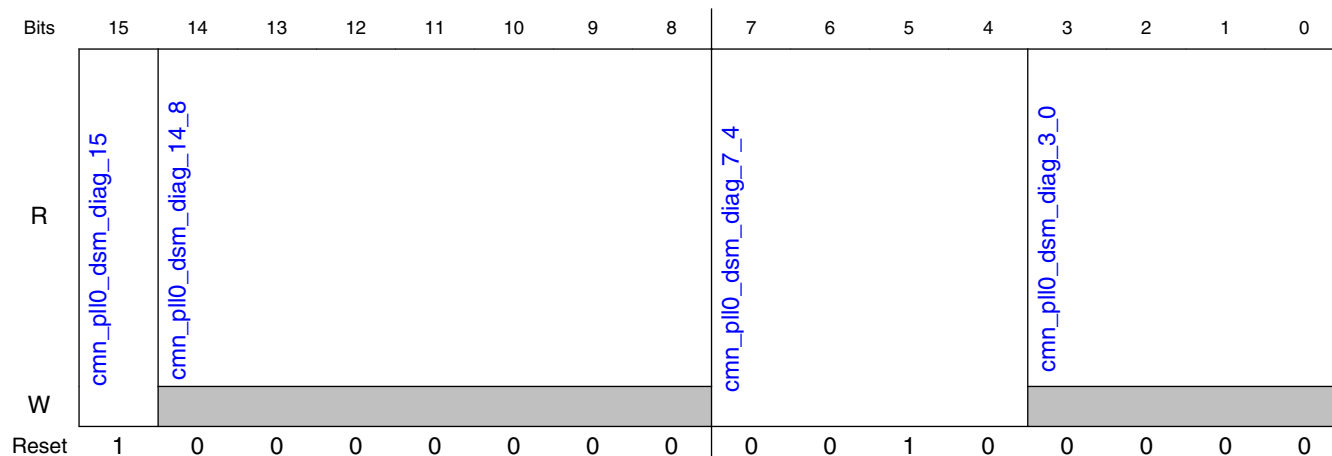
Field	Function
15-10 cmn_pll0_high_thr_15_10	Reserved
9-0 cmn_pll0_high_thr_9_0	pll_fb_div_high_theshold: Value of the pll_fb_div_high_threshold signal.

13.4.10.2.451 PLL 0 delta sigma modulator diagnostics register (cmn_pll0_dsm_diag)

13.4.10.2.451.1 Offset

Register	Offset
cmn_pll0_dsm_diag	8_0097h

13.4.10.2.451.2 Diagram



13.4.10.2.451.3 Fields

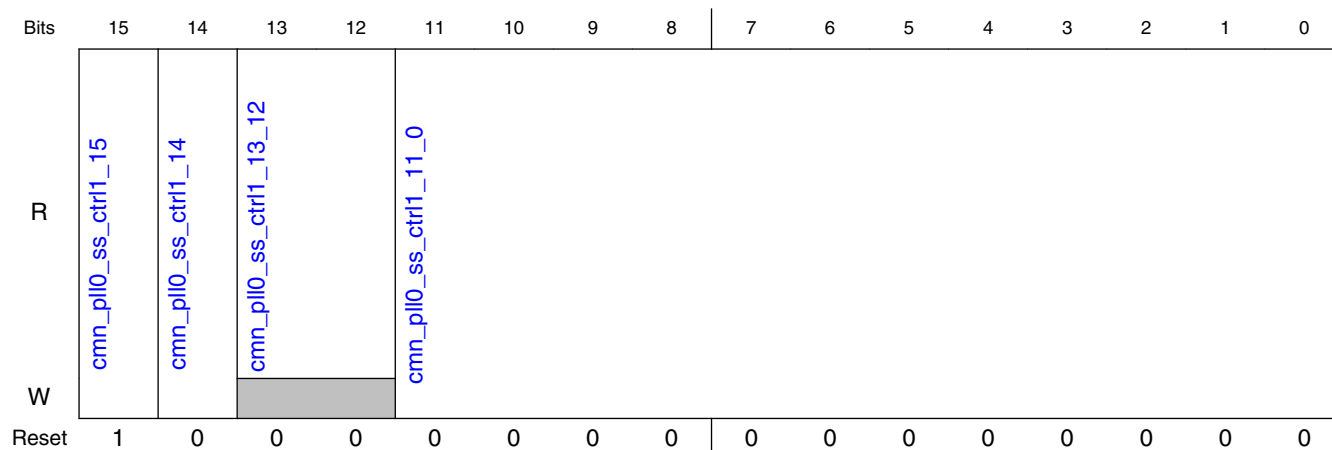
Field	Function
15 cmn_pll0_dsm_diag_15	Delta sigma bypass enable: When set to 1b1, the delta sigma modulator will be bypassed, and the output will be the value specified for the internal pll_fb_div_integer signal.
14-8 cmn_pll0_dsm_diag_14_8	Reserved
7-4 cmn_pll0_dsm_diag_7_4	PLL feedback divider latency adjustment: This signal specifies a value to be subtracted from the feedback divider settings before they are output on the
3-0 cmn_pll0_dsm_diag_3_0	Reserved

13.4.10.2.452 PLL 0 spread spectrum control register 1 (cmn_pll0_ss_ctrl1)

13.4.10.2.452.1 Offset

Register	Offset
cmn_pll0_ss_ctrl1	8_0098h

13.4.10.2.452.2 Diagram



13.4.10.2.452.3 Fields

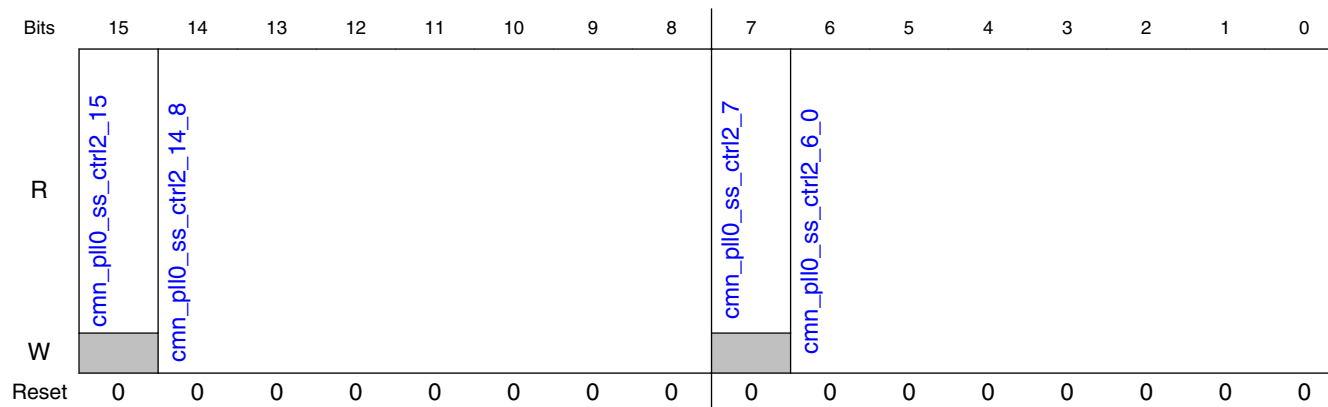
Field	Function
15 cmn_pll0_ss_ctrl1_15	Spread spectrum waveform generator disable: Setting this bit to a 1b1 will disable the spread spectrum waveform generator.
14 cmn_pll0_ss_ctrl1_14	Spread spectrum enable during VCO calibration : Setting this bit to a 1b1 will enable the spread spectrum function while VCO calibration is taking place.
13-12 cmn_pll0_ss_ctrl1_13_12	Reserved
11-0 cmn_pll0_ss_ctrl1_11_0	Amplitude step size: Value of the amplitude_step_size pin on the spread spectrum waveform generator.

13.4.10.2.453 PLL 0 spread spectrum control register 2 (cmn_pll0_ss_ctrl2)

13.4.10.2.453.1 Offset

Register	Offset
cmn_pll0_ss_ctrl2	8_0099h

13.4.10.2.453.2 Diagram



13.4.10.2.453.3 Fields

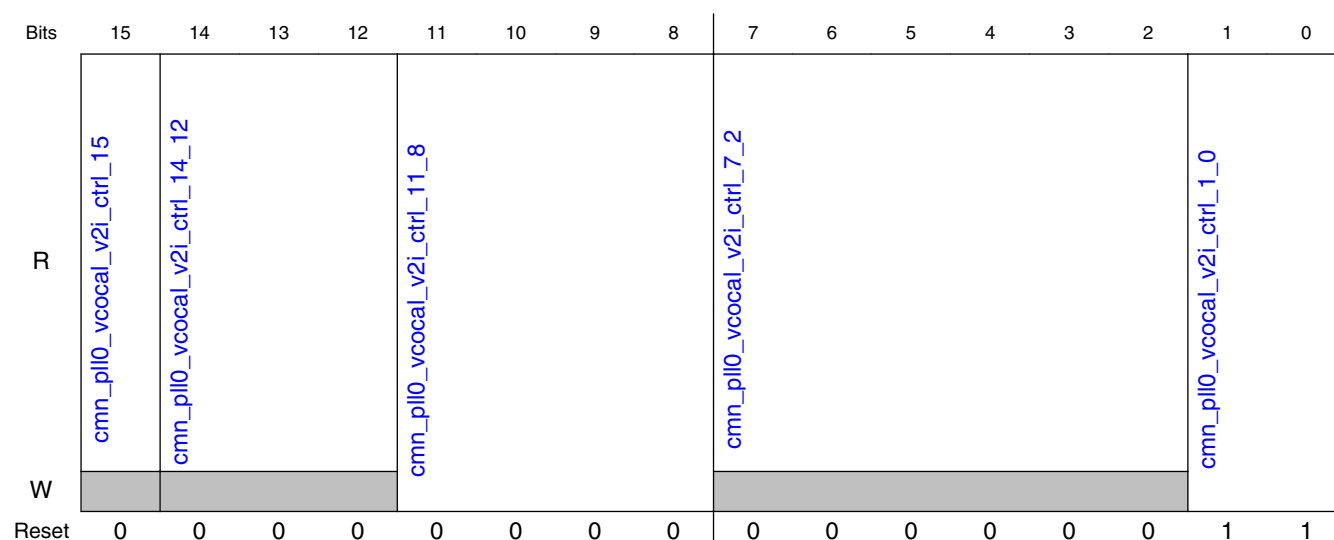
Field	Function
15 cmn_pll0_ss_ctrl2_15	Reserved
14-8 cmn_pll0_ss_ctrl2_14_8	Number of steps: Value of the num_steps pin on the spread spectrum waveform generator.
7 cmn_pll0_ss_ctrl2_7	Reserved
6-0 cmn_pll0_ss_ctrl2_6_0	Time step size: Value for the time_step_size pin on the spread spectrum waveform generator.

13.4.10.2.454 PLL 0 VCO calibration V2I control register (cmn_pll0_vco_cal_v2i_ctrl)

13.4.10.2.454.1 Offset

Register	Offset
cmn_pll0_vco_cal_v2i_ctrl	8_009Ch

13.4.10.2.454.2 Diagram



13.4.10.2.454.3 Fields

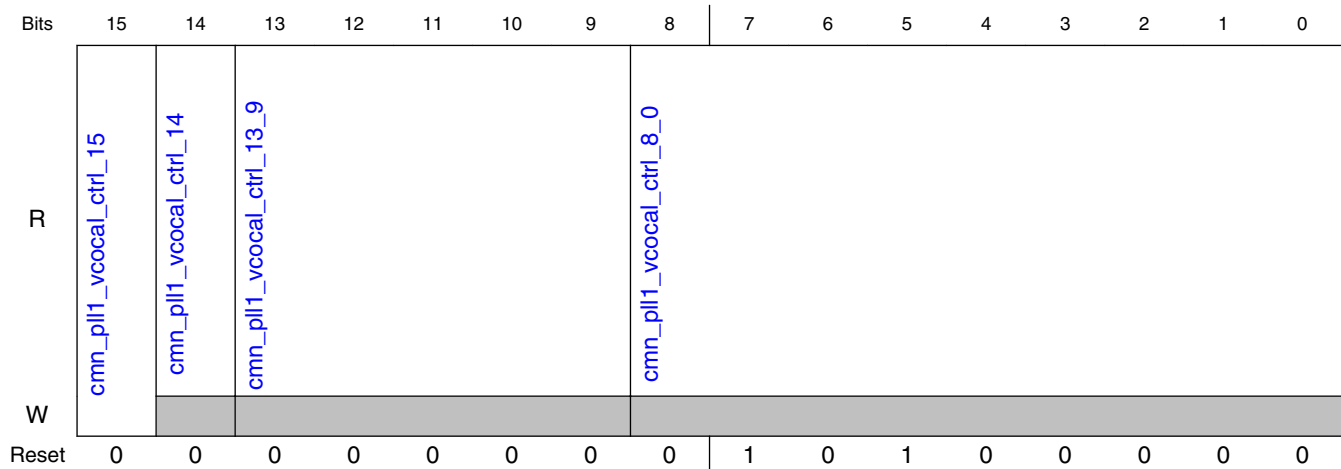
Field	Function
15 cmn_pll0_vcocal_v2i_ctrl_15	VCO calibration code for voltage to current DAC override enable: Activating (1b1) this bit allows the
14-12 cmn_pll0_vcocal_v2i_ctrl_14_12	Reserved
11-8 cmn_pll0_vcocal_v2i_ctrl_11_8	VCO calibration code for voltage to current DAC override value: This field is used to override the
7-2 cmn_pll0_vcocal_v2i_ctrl_7_2	Reserved
1-0 cmn_pll0_vcocal_v2i_ctrl_1_0	VCO calibration code for voltage to current DAC scaling factor: This field is used to control the scaling function in the logic used in the internal calculations related to the VCO calibration code for voltage to current DAC. The following is the encoding used for this field.

13.4.10.2.455 PLL 1 VCO calibration control register (cmn_pll1_vcocal_ctrl)

13.4.10.2.455.1 Offset

Register	Offset
cmn_pll1_vcocal_ctrl	8_00A0h

13.4.10.2.455.2 Diagram



13.4.10.2.455.3 Fields

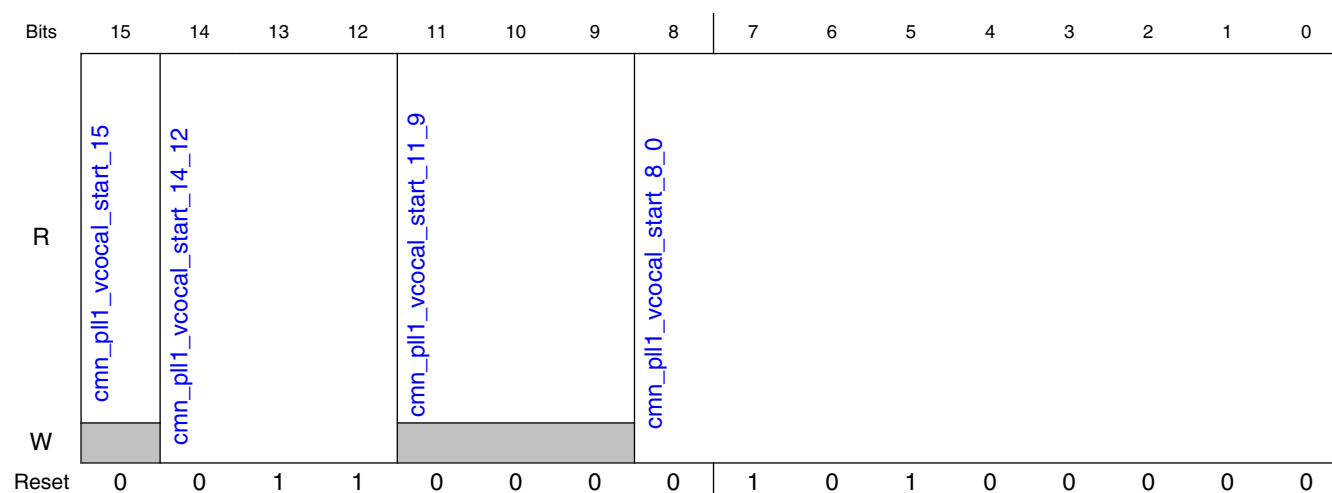
Field	Function
15 cmn_pll1_vcocal_ctrl_15	Start VCO calibration: Activating (1b1) this bit will start a VCO calibration process. This bit must remain active until the VCO calibration process is complete (as indicated by the VCO calibration process done bit in this register). To start another VCO calibration process, the VCO calibration process done bit must have gone inactive from any prior calibration process.
14 cmn_pll1_vcocal_ctrl_14	VCO calibration process done: This bit will be set to 1b1 when the VCO calibration process is complete. It will be cleared by the deactivation of the Start VCO calibration bit in this register.
13-9 cmn_pll1_vcocal_ctrl_13_9	Reserved
8-0 cmn_pll1_vcocal_ctrl_8_0	VCO calibration code: This is the calibration code that was determined by the VCO calibration process. This signal is valid when the VCO calibration process is complete. The values of this field correspond to different frequency bands the VCO will operate in. The frequency bands are controlled by the number of capacitors that are switched in the VCO analog circuit. This field specifies the number of capacitors that are switched in. The following are the values for this code:

13.4.10.2.456 PLL 1 VCO calibration start point register (cmn_pll1_vcocal_start)

13.4.10.2.456.1 Offset

Register	Offset
cmn_pll1_vcocal_start	8_00A1h

13.4.10.2.456.2 Diagram



13.4.10.2.456.3 Fields

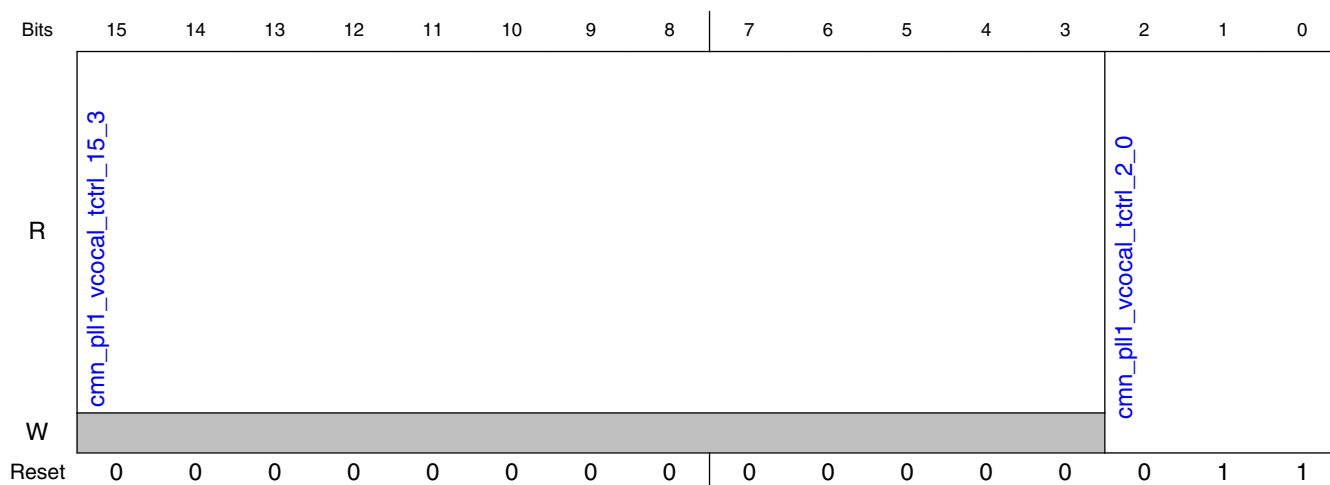
Field	Function
15 cmn_pll1_vcocal_start_15	Reserved
14-12 cmn_pll1_vcocal_start_14_12	VCO calibration initial step size control: This field specifies the initial step size for the VCO calibration state machine. The following are the values that can be used in this field, and the corresponding step sizes.
11-9 cmn_pll1_vcocal_start_11_9	Reserved
8-0 cmn_pll1_vcocal_start_8_0	VCO calibration code starting point value: This field specifies the starting VCO code that is used by the VCO calibration state machine. The purpose of this value is such that the VCO calibration process starts at a point that is, on average, relatively close to the final calibration point. This allows the calibration time to be reduced, on average.

13.4.10.2.457 PLL 1 VCO calibration timer control register (cmn_pll1_vcocal_tctrl)

13.4.10.2.457.1 Offset

Register	Offset
cmn_pll1_vcocal_tctrl	8_00A2h

13.4.10.2.457.2 Diagram



13.4.10.2.457.3 Fields

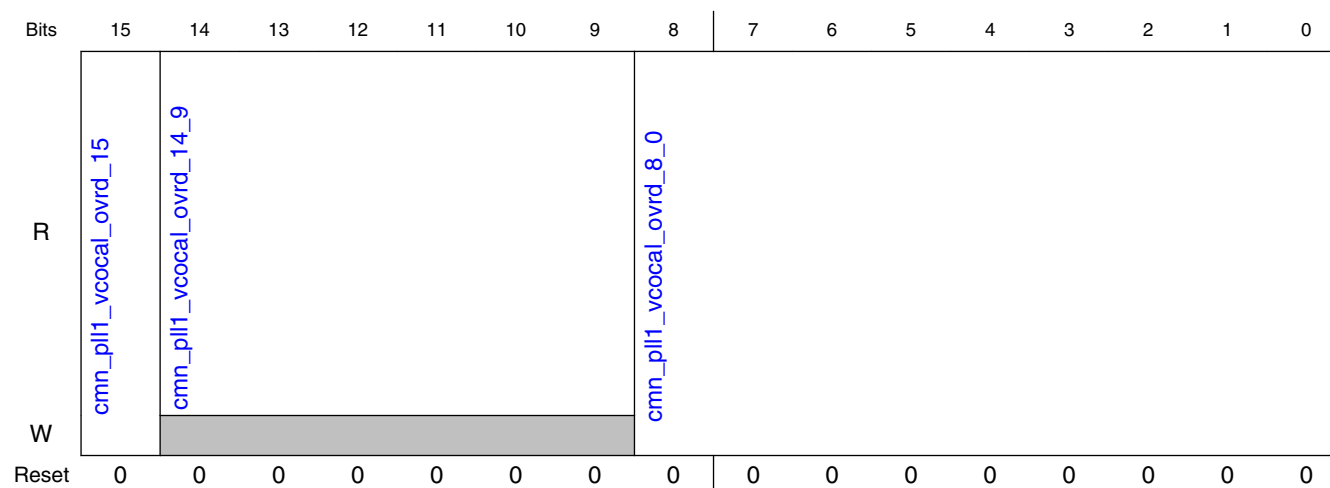
Field	Function
15-3 cmn_pll1_vcocal_tctrl_15_3	Reserved
2-0 cmn_pll1_vcocal_tctrl_2_0	VCO calibration initial time scale control: This field specifies the calibration start time scaling factor applied to the VCO calibration when running the initial step size for the calibration code is not set to 1. Setting this value to a value other than 1 will reduce the calibration measurement time by the amount specified below. Then when the final calibration steps are made the full calibration time will be used to get the appropriate amount of resolution.

13.4.10.2.458 PLL 1 VCO calibration override register (cmn_pll1_vcocal_ovrd)

13.4.10.2.458.1 Offset

Register	Offset
cmn_pll1_vcocal_ovrd	8_00A3h

13.4.10.2.458.2 Diagram



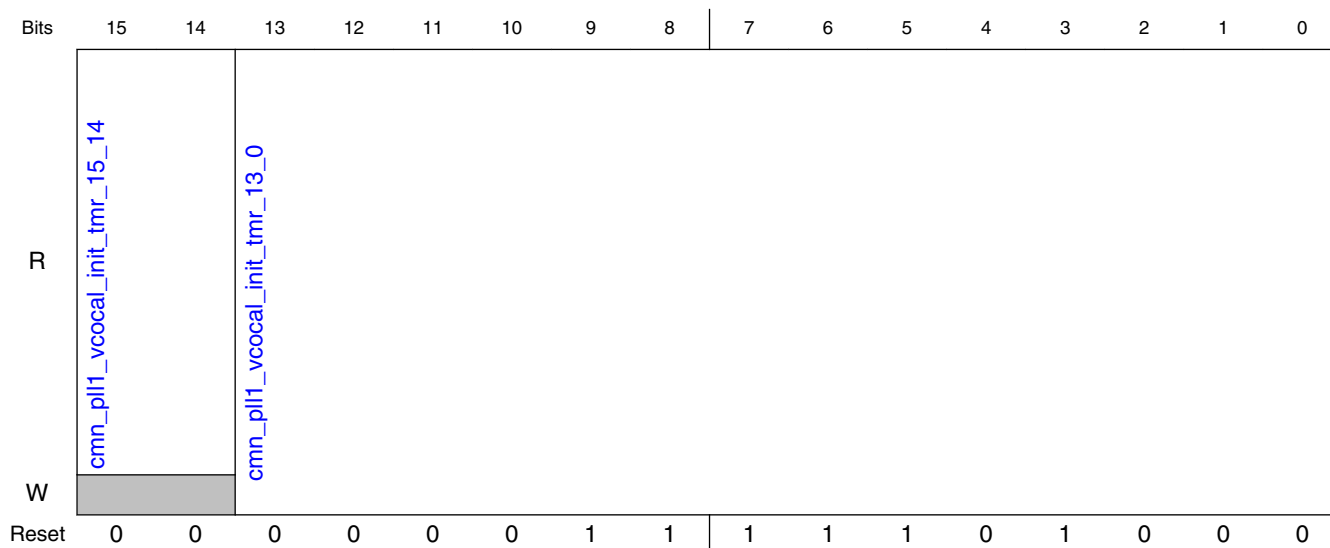
13.4.10.2.458.3 Fields

Field	Function
15 cmn_pll1_vcocal_ovrd_15	VCO calibration code override enable: Activating (1b1) this bit allows the VCO code determined during the automatic VCO calibration process to be overridden by the value driven by the VCO calibration code override value field in this register.
14-9 cmn_pll1_vcocal_ovrd_14_9	Reserved
8-0 cmn_pll1_vcocal_ovrd_8_0	VCO calibration code override value: This field is used to override the VCO code determined during the automatic VCO calibration process. The VCO code driven on this field is valid when the VCO calibration code override enable bit in this register is active.

13.4.10.2.459 PLL 1 VCO calibration initialization timer register (cmn_pll1_vcocal_init_tmr)

13.4.10.2.459.1 Offset

Register	Offset
cmn_pll1_vcocal_init_tmr	8_00A4h

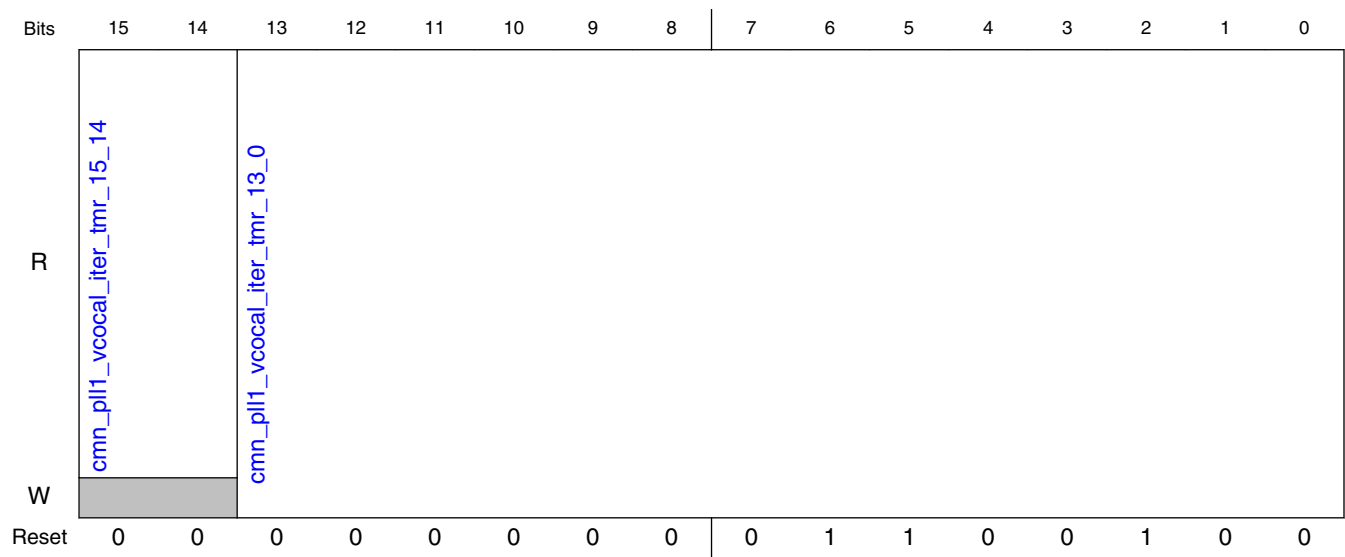
13.4.10.2.459.2 Diagram**13.4.10.2.459.3 Fields**

Field	Function
15-14 cmn_pll1_vcocal_init_tmr_15_14	Reserved
13-0 cmn_pll1_vcocal_init_tmr_13_0	Initialization wait timer value: This is the number of clocks to wait between when the analog VCO calibration circuits are enabled, and when the first calibration code is driven to the analog.

13.4.10.2.460 PLL 1 VCO calibration iteration timer register (cmn_pll1_vcocal_iter_tmr)**13.4.10.2.460.1 Offset**

Register	Offset
cmn_pll1_vcocal_iter_tmr	8_00A5h

13.4.10.2.460.2 Diagram



13.4.10.2.460.3 Fields

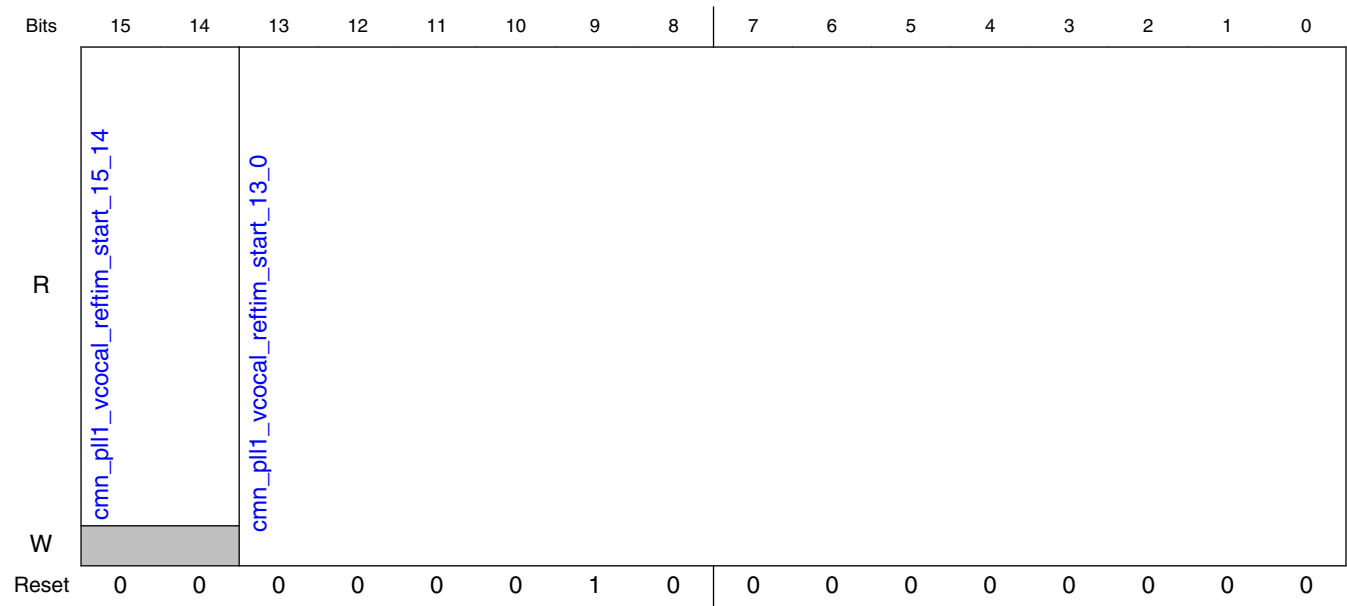
Field	Function
15-14 <code>cmn_pll1_vcocal_iter_tmr_15_14</code>	Reserved
13-0 <code>cmn_pll1_vcocal_iter_tmr_13_0</code>	Iteration wait timer value: This is the number of clocks to wait between when a calibration code is driven to the analog, and when the clock rates are measured.

13.4.10.2.461 PLL 1 VCO calibration reference clock timer start value register (`cmn_pll1_vcocal_reftim_start`)

13.4.10.2.461.1 Offset

Register	Offset
<code>cmn_pll1_vcocal_reftim_start</code>	8_00A6h

13.4.10.2.461.2 Diagram



13.4.10.2.461.3 Fields

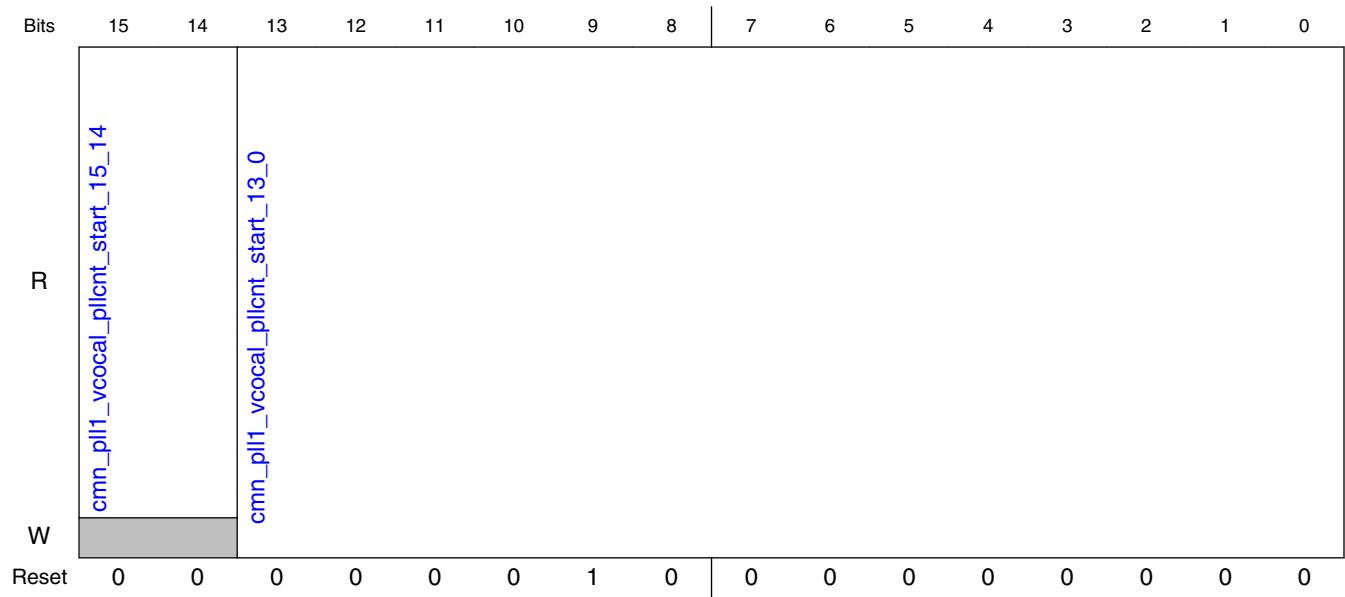
Field	Function
15-14 cmn_pll1_vcocal_reftim_start_15_14	Reserved
13-0 cmn_pll1_vcocal_reftim_start_13_0	PLL VCO calibration reference clock timer start value : This is the value that is loaded into the reference clock timer as the starting point for that timer, when running VCO calibration. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes.

13.4.10.2.462 PLL 1 VCO calibration PLL clock counter start value register (cmn_pll1_vcocal_pllcnt_start)

13.4.10.2.462.1 Offset

Register	Offset
cmn_pll1_vcocal_pllcnt_start	8_00A8h

13.4.10.2.462.2 Diagram



13.4.10.2.462.3 Fields

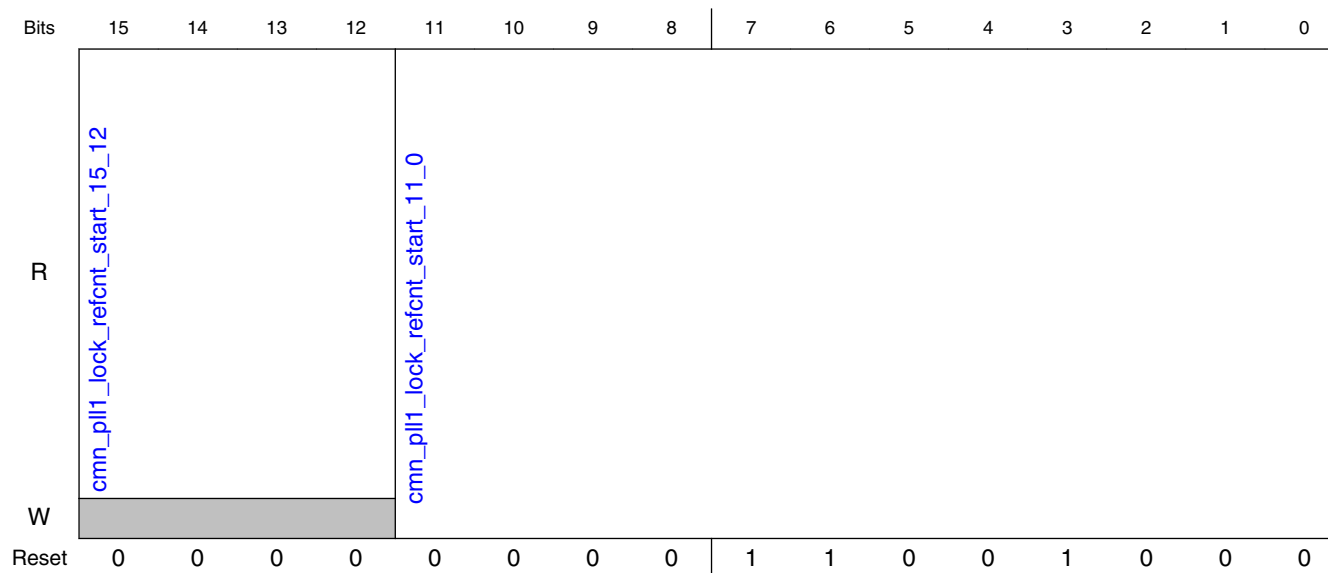
Field	Function
15-14 cmn_pll1_vcocal_pllcnt_start_15_14	Reserved
13-0 cmn_pll1_vcocal_pllcnt_start_13_0	PLL VCO calibration PLL clock counter start value : This is the value that is loaded into the PLL clock counter as the starting point for that counter, when running VCO calibration. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes.

13.4.10.2.463 PLL 1 lock reference counter start value register (cmn_pll1_lock_refcnt_start)

13.4.10.2.463.1 Offset

Register	Offset
cmn_pll1_lock_refcnt_start	8_00B0h

13.4.10.2.463.2 Diagram



13.4.10.2.463.3 Fields

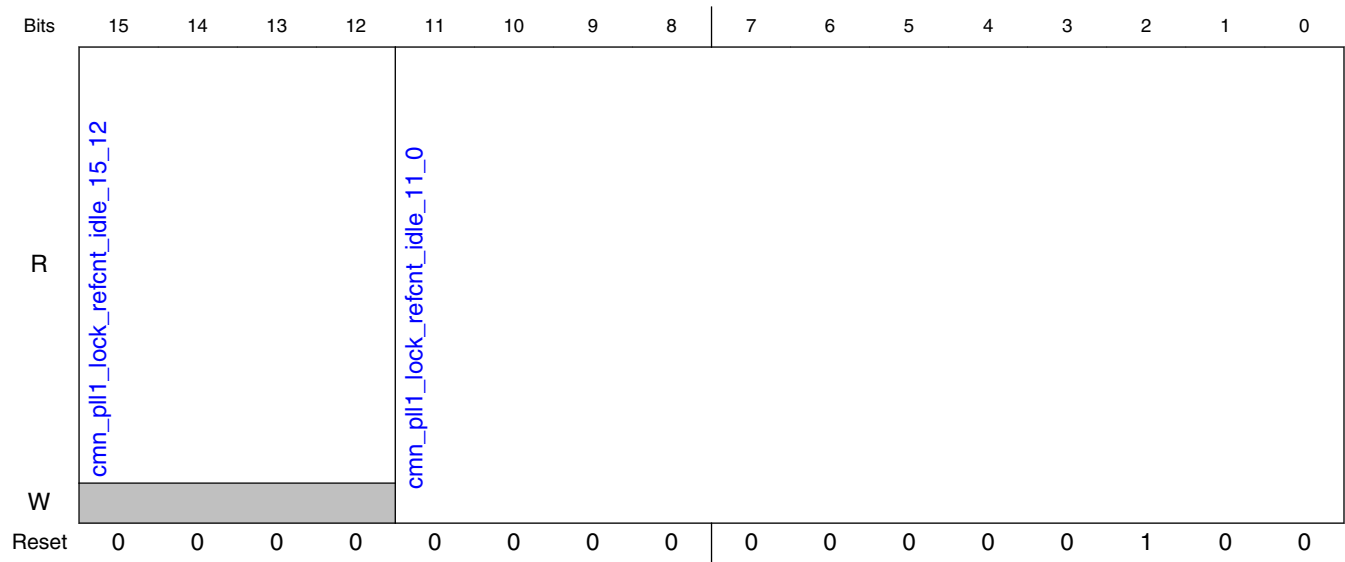
Field	Function
15-12 <code>cmn_pll1_lock_refcnt_start_15_12</code>	Reserved
11-0 <code>cmn_pll1_lock_refcnt_start_11_0</code>	PLL lock reference counter start value : This is the value that is loaded into the PLL lock detect reference counter as the starting point for that counter, when checking for PLL lock. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes. This is set to 200 clocks by default.

13.4.10.2.464 PLL 1 lock reference counter idle value register (`cmn_pll1_lock_refcnt_idle`)

13.4.10.2.464.1 Offset

Register	Offset
<code>cmn_pll1_lock_refcnt_idle</code>	8_00B1h

13.4.10.2.464.2 Diagram



13.4.10.2.464.3 Fields

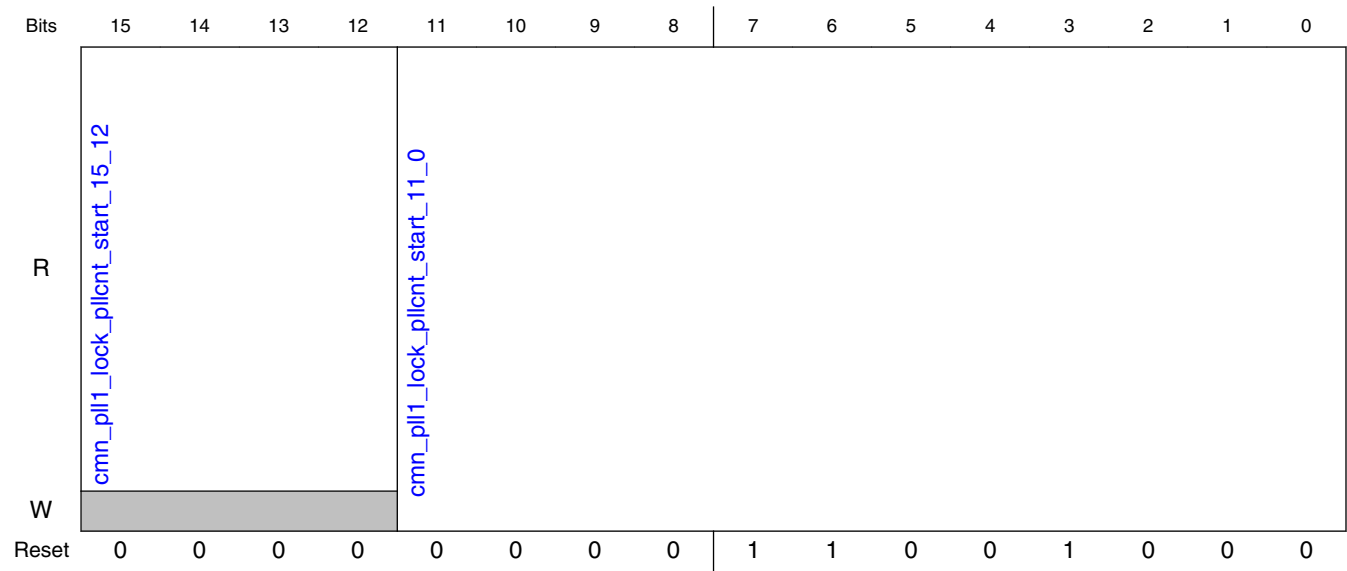
Field	Function
15-12 cmn_pll1_lock_refcnt_idle_15_12	Reserved
11-0 cmn_pll1_lock_refcnt_idle_11_0	PLL lock reference counter idle value : This is the value used by the PLL lock detection logic to specify the number of reference clocks between each phase of counting PLL clocks.

13.4.10.2.465 PLL 1 lock PLL counter start value register (cmn_pll1_lock_pllcnt_start)

13.4.10.2.465.1 Offset

Register	Offset
cmn_pll1_lock_pllcnt_start	8_00B2h

13.4.10.2.465.2 Diagram



13.4.10.2.465.3 Fields

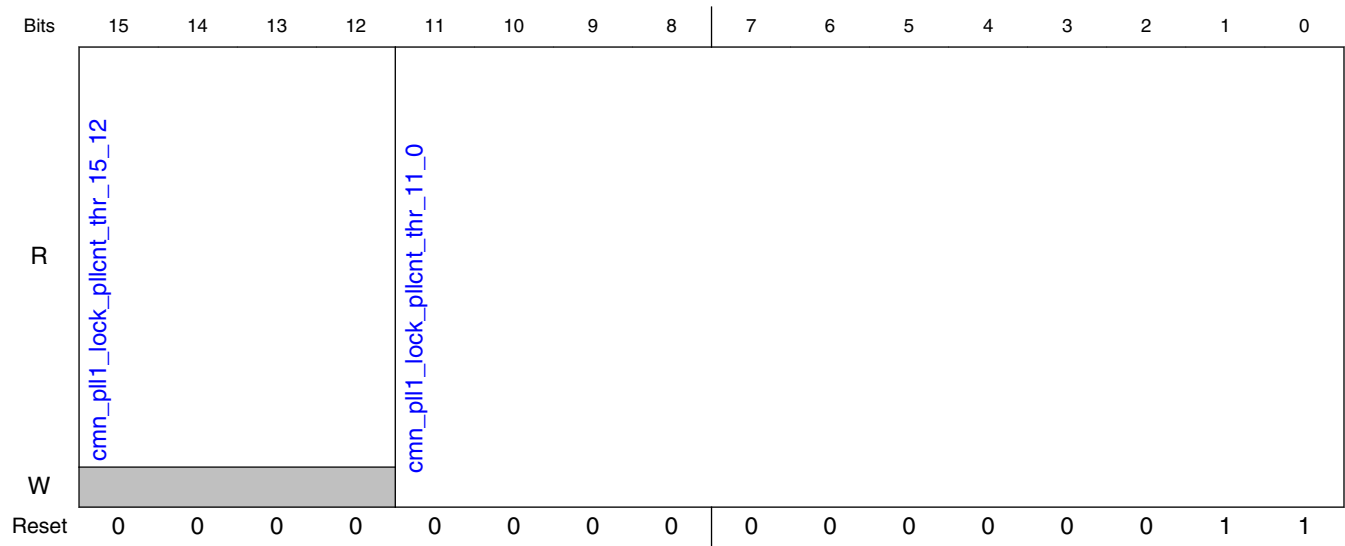
Field	Function
15-12 cmn_pll1_lock_pllcnt_start_15_12	Reserved
11-0 cmn_pll1_lock_pllcnt_start_11_0	PLL lock PLL counter start value : This is the value that is loaded into the PLL lock detect PLL counter as the starting point for that counter, when checking for PLL lock. This register is provided for simulation speedup and diagnostic purposes only. It is not intended to be for functional purposes. This is set to 200 clocks by default.

13.4.10.2.466 PLL 1 lock PLL counter threshold value register (cmn_pll1_lock_pllcnt_thr)

13.4.10.2.466.1 Offset

Register	Offset
cmn_pll1_lock_pllcnt_thr	8_00B3h

13.4.10.2.466.2 Diagram



13.4.10.2.466.3 Fields

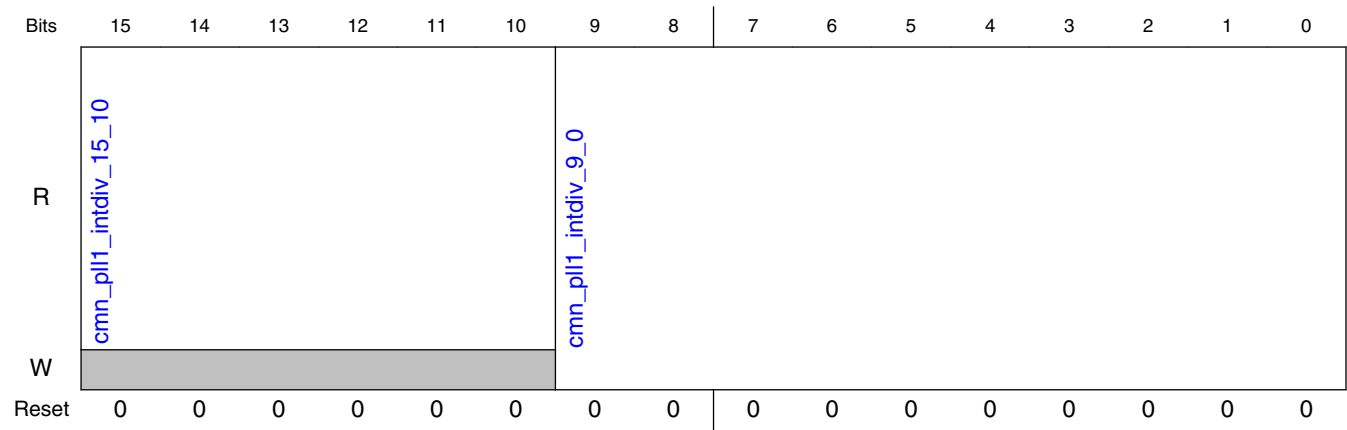
Field	Function
15-12	Reserved
cmn_pll1_lock_pllcnt_thr_15_12	
11-0	PLL lock counter threshold value : This is the value used by the PLL lock detection logic to determine if the PLL has locked. If the two counters in the PLL lock detection logic differ by less than this value, the PLL is considered locked. This is set to 3 clocks by default.
cmn_pll1_lock_pllcnt_thr_11_0	

13.4.10.2.467 PLL 1 feedback divider integer register (cmn_pll1_intdiv)

13.4.10.2.467.1 Offset

Register	Offset
cmn_pll1_intdiv	8_00B4h

13.4.10.2.467.2 Diagram



13.4.10.2.467.3 Fields

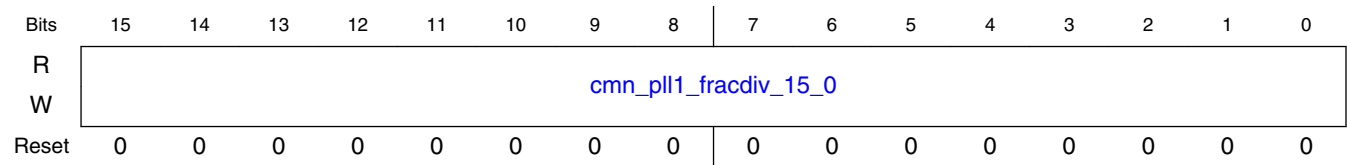
Field	Function
15-10 cmn_pll1_intdiv_15_10	Reserved
9-0 cmn_pll1_intdiv_9_0	pll_fb_div_integer value: Value of the pll_fb_div_integer signal.

13.4.10.2.468 PLL 1 feedback divider fractional register (cmn_pll1_fracdiv)

13.4.10.2.468.1 Offset

Register	Offset
cmn_pll1_fracdiv	8_00B5h

13.4.10.2.468.2 Diagram



13.4.10.2.468.3 Fields

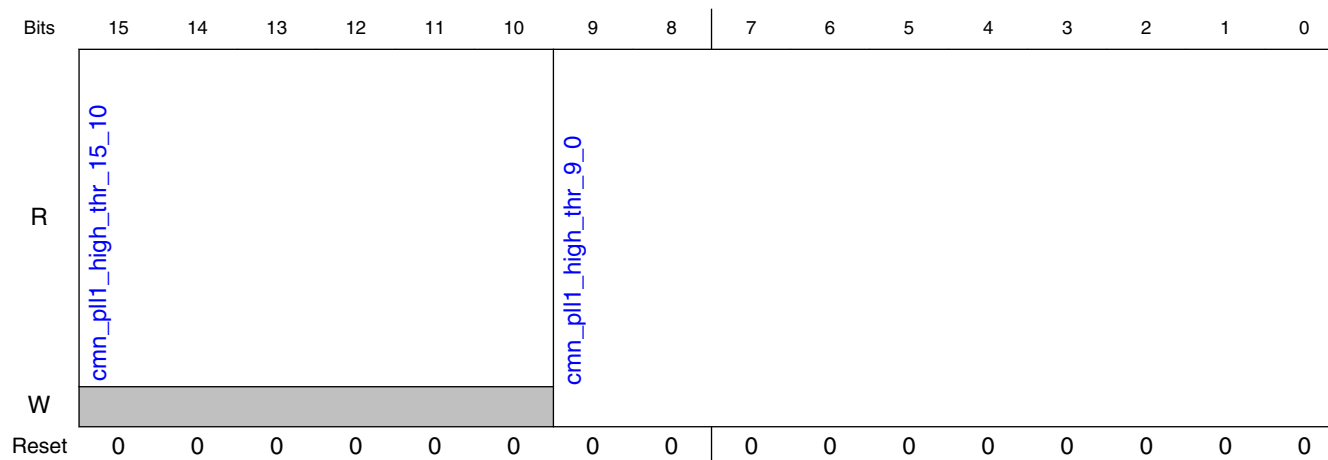
Field	Function
15-0 cmn_pll1_fracdi v_15_0	pll_fb_div_fractional: Value of the pll_fb_div_fractional signal.

13.4.10.2.469 PLL 1 feedback divider high threshold register (cmn_pll1_high_thr)

13.4.10.2.469.1 Offset

Register	Offset
cmn_pll1_high_thr	8_00B6h

13.4.10.2.469.2 Diagram



13.4.10.2.469.3 Fields

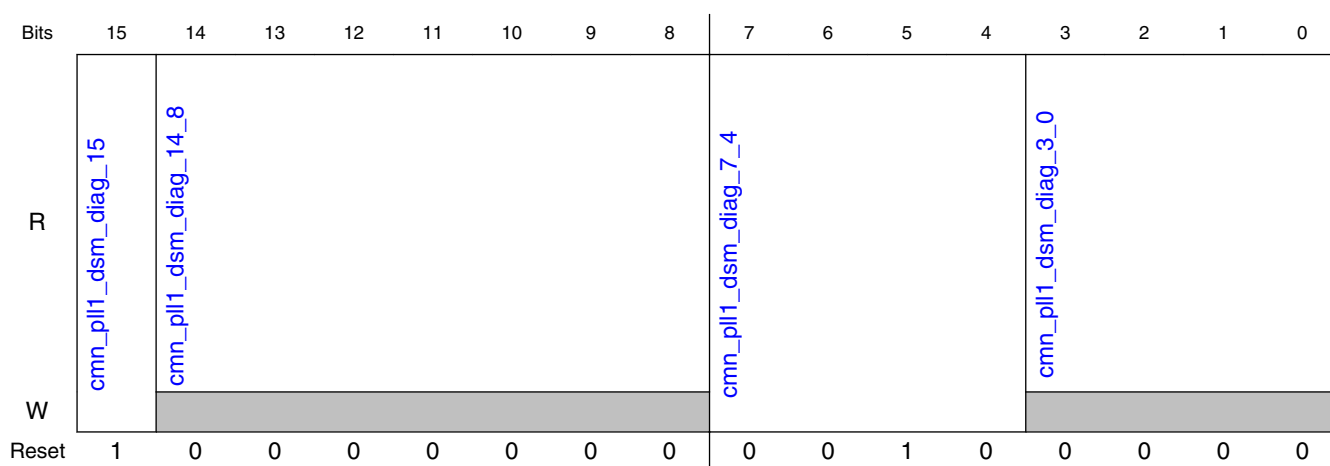
Field	Function
15-10 cmn_pll1_high_t hr_15_10	Reserved
9-0 cmn_pll1_high_t hr_9_0	pll_fb_div_high_theshold: Value of the pll_fb_div_high_threshold signal.

13.4.10.2.470 PLL 1 delta sigma modulator diagnostics register (cmn_pll1_dsm_diag)

13.4.10.2.470.1 Offset

Register	Offset
cmn_pll1_dsm_diag	8_00B7h

13.4.10.2.470.2 Diagram



13.4.10.2.470.3 Fields

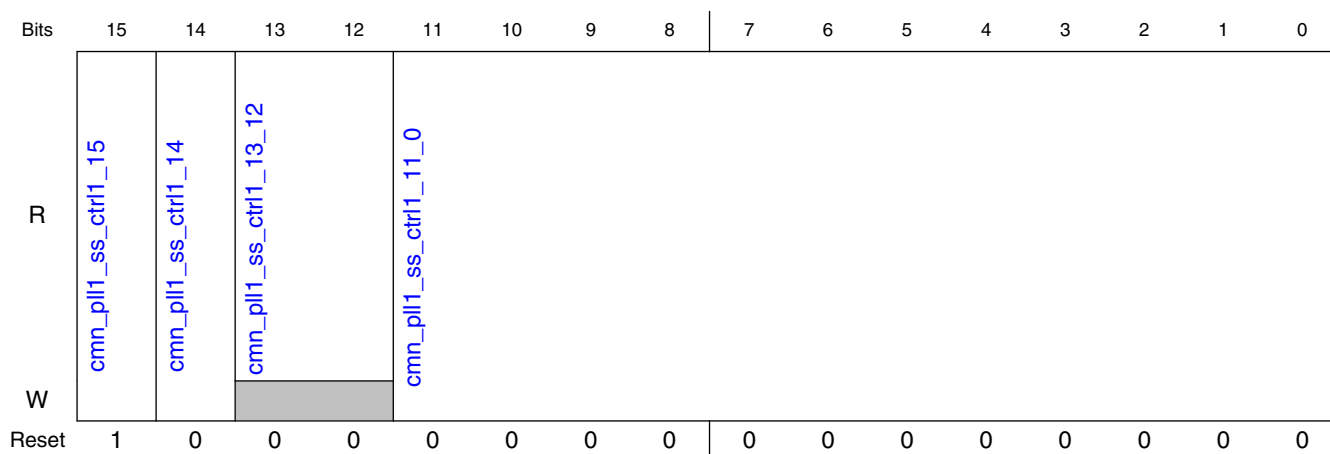
Field	Function
15 cmn_pll1_dsm_diag_15	Delta sigma bypass enable: When set to 1b1, the delta sigma modulator will be bypassed, and the output will be the value specified for the internal pll_fb_div_integer signal.
14-8 cmn_pll1_dsm_diag_14_8	Reserved
7-4 cmn_pll1_dsm_diag_7_4	PLL feedback divider latency adjustment: This signal specifies a value to be subtracted from the feedback divider settings before they are output on the
3-0 cmn_pll1_dsm_diag_3_0	Reserved

13.4.10.2.471 PLL 1 spread spectrum control register 1 (cmn_pll1_ss_ctrl 1)

13.4.10.2.471.1 Offset

Register	Offset
cmn_pll1_ss_ctrl1	8_00B8h

13.4.10.2.471.2 Diagram



13.4.10.2.471.3 Fields

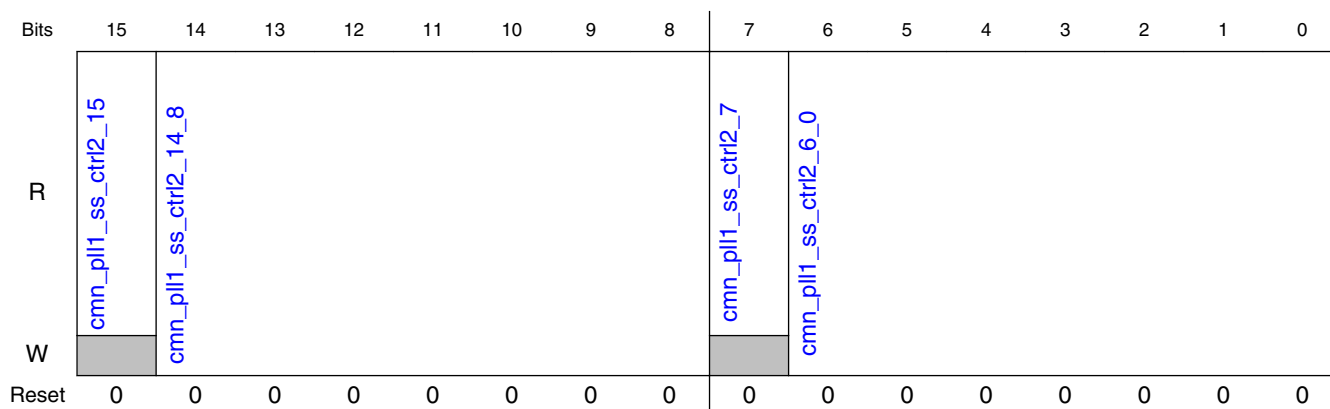
Field	Function
15 cmn_pll1_ss_ctrl1_15	Spread spectrum waveform generator disable: Setting this bit to a 1b1 will disable the spread spectrum waveform generator.
14 cmn_pll1_ss_ctrl1_14	Spread spectrum enable during VCO calibration : Setting this bit to a 1b1 will enable the spread spectrum function while VCO calibration is taking place.
13-12 cmn_pll1_ss_ctrl1_13_12	Reserved
11-0 cmn_pll1_ss_ctrl1_11_0	Amplitude step size: Value of the amplitude_step_size pin on the spread spectrum waveform generator.

13.4.10.2.472 PLL 1 spread spectrum control register 2 (cmn_pll1_ss_ctrl2)

13.4.10.2.472.1 Offset

Register	Offset
cmn_pll1_ss_ctrl2	8_00B9h

13.4.10.2.472.2 Diagram



13.4.10.2.472.3 Fields

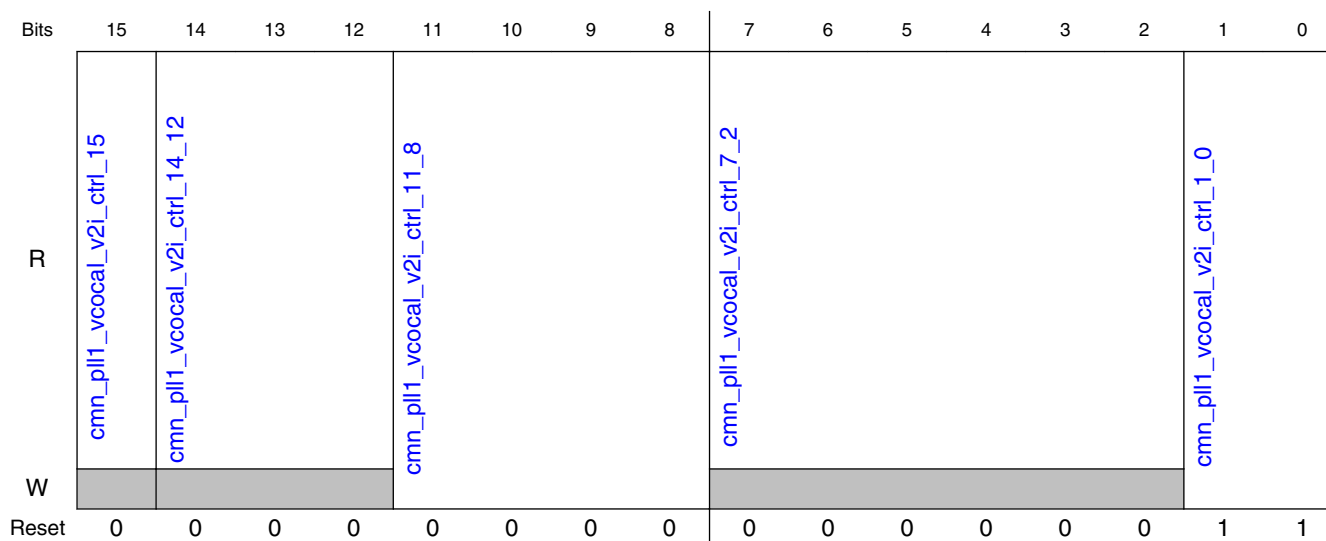
Field	Function
15 cmn_pll1_ss_ctrl2_15	Reserved
14-8 cmn_pll1_ss_ctrl2_14_8	Number of steps: Value of the num_steps pin on the spread spectrum waveform generator.
7 cmn_pll1_ss_ctrl2_7	Reserved
6-0 cmn_pll1_ss_ctrl2_6_0	Time step size: Value for the time_step_size pin on the spread spectrum waveform generator.

13.4.10.2.473 PLL 1 VCO calibration V2I control register (cmn_pll1_vcocal_v2i_ctrl)

13.4.10.2.473.1 Offset

Register	Offset
cmn_pll1_vcocal_v2i_ctrl	8_00BCh

13.4.10.2.473.2 Diagram



13.4.10.2.473.3 Fields

Field	Function
15 cmn_pll1_vcocal_v2i_ctrl_15	VCO calibration code for voltage to current DAC override enable: Activating (1b1) this bit allows the
14-12 cmn_pll1_vcocal_v2i_ctrl_14_12	Reserved
11-8 cmn_pll1_vcocal_v2i_ctrl_11_8	VCO calibration code for voltage to current DAC override value: This field is used to override the
7-2 cmn_pll1_vcocal_v2i_ctrl_7_2	Reserved

Table continues on the next page...

Clocks And Resets

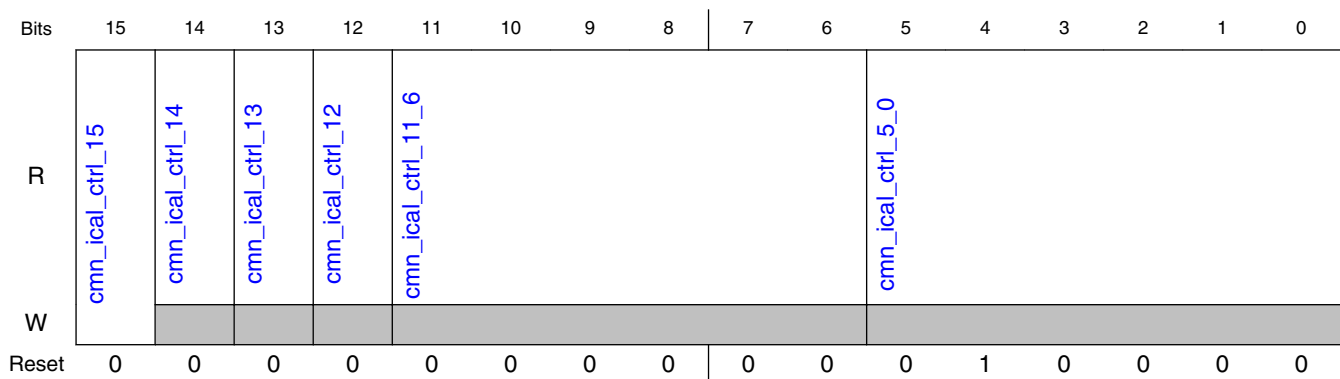
Field	Function
1-0 cmn_pll1_vcocal_v2i_ctrl_1_0	VCO calibration code for voltage to current DAC scaling factor: This field is used to control the scaling function in the logic used in the internal calculations related to the VCO calibration code for voltage to current DAC. The following is the encoding used for this field.

13.4.10.2.474 Current calibration control register (cmn_ical_ctrl)

13.4.10.2.474.1 Offset

Register	Offset
cmn_ical_ctrl	8_00C0h

13.4.10.2.474.2 Diagram



13.4.10.2.474.3 Fields

Field	Function
15 cmn_ical_ctrl_15	Start current calibration: Activating (1b1) this bit will start the current calibration process. This signal must remain active until the calibration process is complete. To start another calibration process, this register must first be set inactive (1b0) until the bandbap calibration process done bit in this register is cleared.
14 cmn_ical_ctrl_14	Current calibration process done: This bit will be set to 1b1 when the current calibration process is complete. It will be cleared by
13 cmn_ical_ctrl_13	No analog calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached.
12	Current analog comparator response: This is the current state of the analog comparator response signal (

Table continues on the next page...

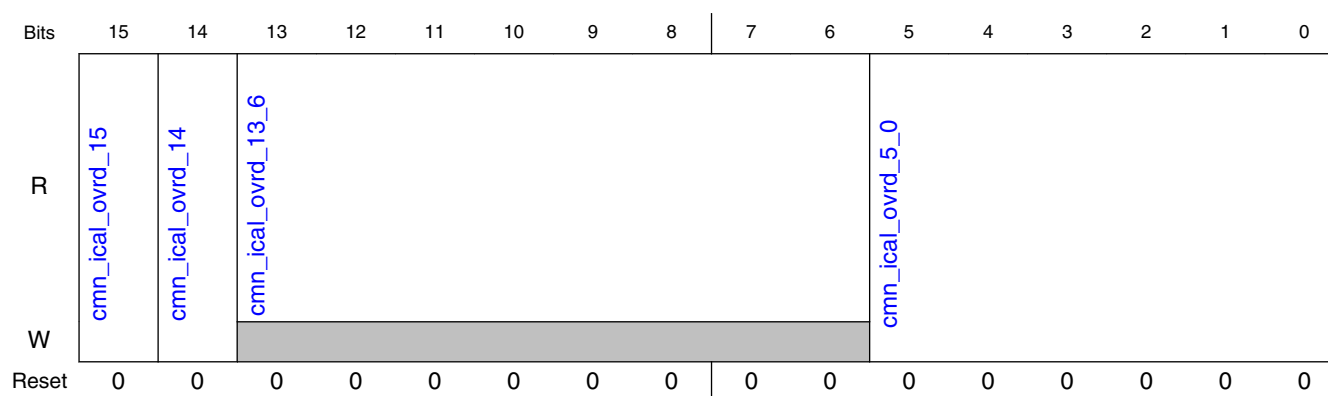
Field	Function
cmn_ical_ctrl_1_2	
11-6 cmn_ical_ctrl_1_6	Reserved
5-0 cmn_ical_ctrl_5_0	Current calibration code: This is the calibration code that was determined by the current calibration process. The following are the values for the code.

13.4.10.2.475 Current calibration override register (cmn_ical_ovrd)

13.4.10.2.475.1 Offset

Register	Offset
cmn_ical_ovrd	8_00C1h

13.4.10.2.475.2 Diagram



13.4.10.2.475.3 Fields

Field	Function
15 cmn_ical_ovrd_15	Current code override enable: Activation (1b1) of this register bit allows the current codes determined during the automatic calibration process to be overridden. The override value is specified using the current code override value field of this register.
14 cmn_ical_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the

Table continues on the next page...

Clocks And Resets

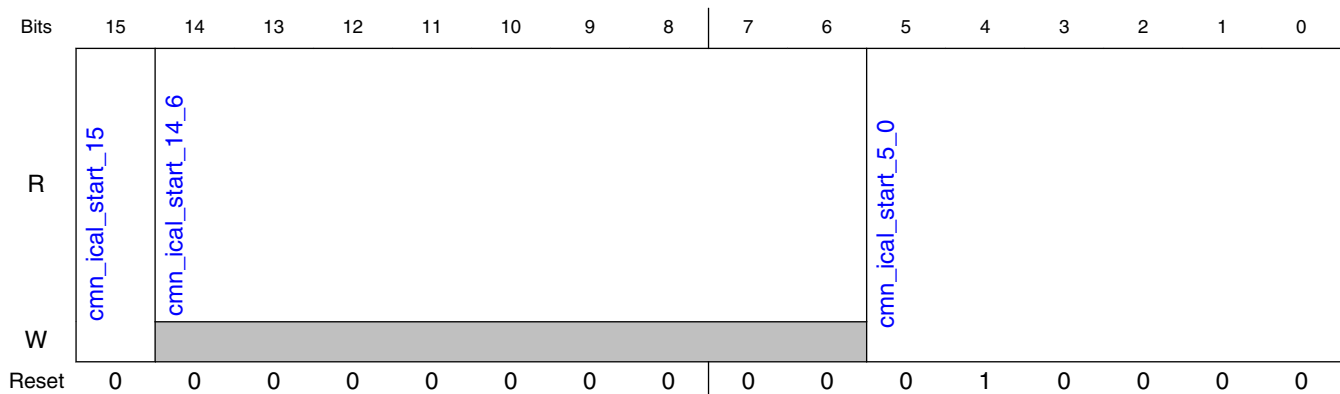
Field	Function
13-6 cmn_ical_ovrd_13_6	Reserved
5-0 cmn_ical_ovrd_5_0	Current code override value: These bits are used to override the current code determined during the automatic calibration process. The code written to these bits is valid when the current code override enable bit in this register is active.

13.4.10.2.476 Current calibration start register (cmn_ical_start)

13.4.10.2.476.1 Offset

Register	Offset
cmn_ical_start	8_00C2h

13.4.10.2.476.2 Diagram



13.4.10.2.476.3 Fields

Field	Function
15 cmn_ical_start_15	Current calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-6 cmn_ical_start_14_6	Reserved
5-0	Start current calibration code: This is the calibration code that the calibration process starts with when automatic calibration is run.

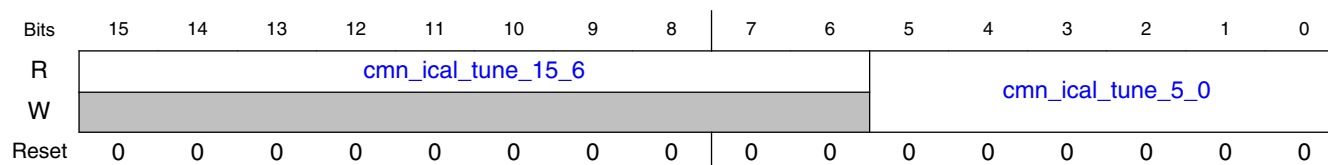
Field	Function
cmn_ical_start_5_0	

13.4.10.2.477 Current calibration tune register (cmn_ical_tune)

13.4.10.2.477.1 Offset

Register	Offset
cmn_ical_tune	8_00C3h

13.4.10.2.477.2 Diagram



13.4.10.2.477.3 Fields

Field	Function
15-6 cmn_ical_tune_15_6	Reserved
5-0 cmn_ical_tune_5_0	Current calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.478 Current calibration initialization timer register (cmn_ical_init_tmr)

13.4.10.2.478.1 Offset

Register	Offset
cmn_ical_init_tmr	8_00C4h

13.4.10.2.478.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_ical_init_tmr_15_8								cmn_ical_init_tmr_7_0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1

13.4.10.2.478.3 Fields

Field	Function
15-8 cmn_ical_init_tmr_15_8	Reserved
7-0 cmn_ical_init_tmr_7_0	Initialization wait timer value: This is the number of

13.4.10.2.479 Current calibration iteration timer register (cmn_ical_iter_tmr)

13.4.10.2.479.1 Offset

Register	Offset
cmn_ical_iter_tmr	8_00C5h

13.4.10.2.479.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_ical_iter_tmr_15_8								cmn_ical_iter_tmr_7_0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0

13.4.10.2.479.3 Fields

Field	Function
15-8	Reserved

Table continues on the next page...

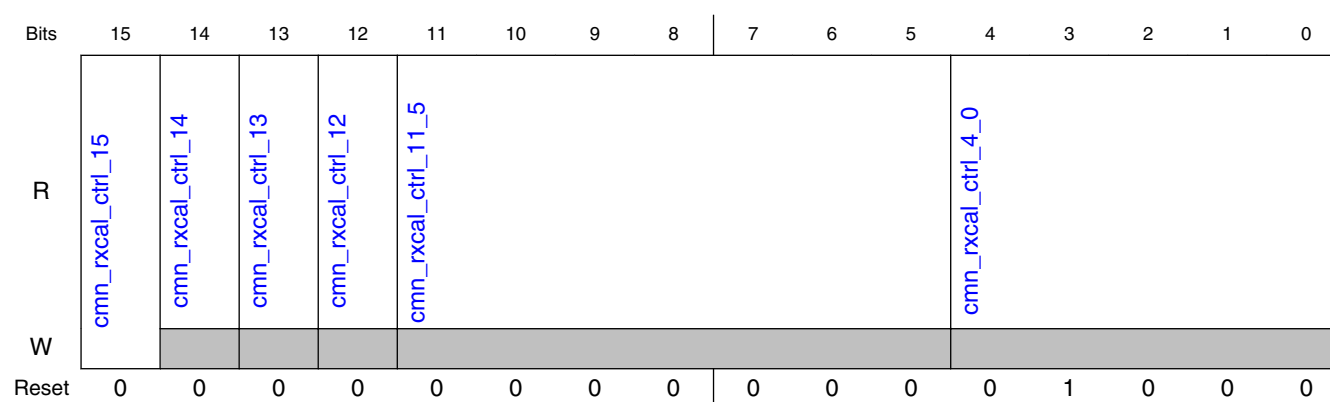
Field	Function
cmn_ical_iter_t mr_15_8	
7-0 cmn_ical_iter_t mr_7_0	Iteration wait timer value: This is the number of

13.4.10.2.480 RX resistor calibration control register (cmn_rxcal_ctrl)

13.4.10.2.480.1 Offset

Register	Offset
cmn_rxcal_ctrl	8_00D0h

13.4.10.2.480.2 Diagram



13.4.10.2.480.3 Fields

Field	Function
15 cmn_rxcal_ctrl_15	Start resistor calibration: Activating (1b1) this bit will start the resistor calibration process. This signal must remain active until the resistor calibration process is complete. To start another resistor calibration process, this register must first be set inactive (1b0) until the resistor calibration process done bit in this register is cleared.
14 cmn_rxcal_ctrl_14	Resistor calibration process done: This bit will be set to 1b1 when the resistor calibration process is complete. It will be cleared by
13	No analog calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached.

Table continues on the next page...

Clocks And Resets

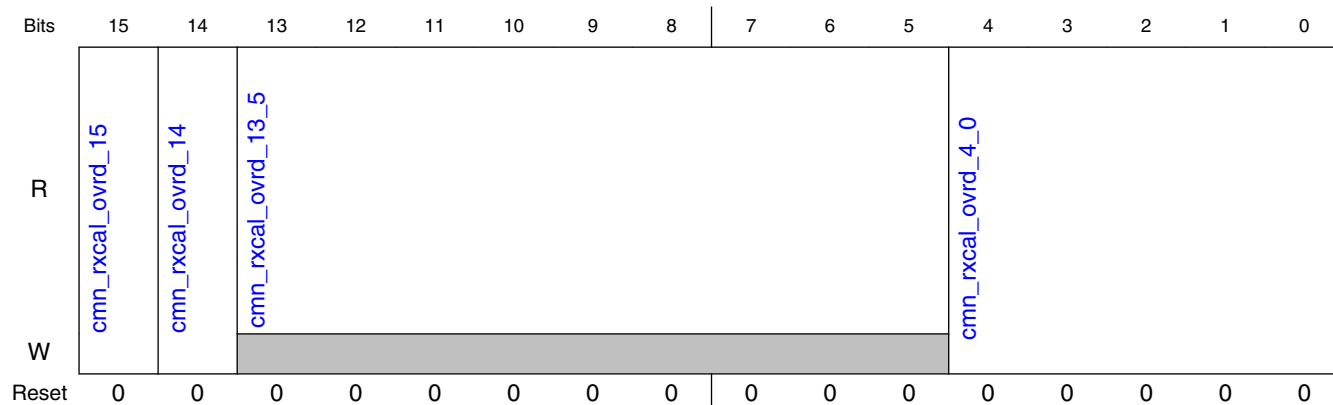
Field	Function
cmn_rxcal_ctrl_13	
12 cmn_rxcal_ctrl_12	Current analog comparator response: This is the current state of the analog comparator response signal (
11-5 cmn_rxcal_ctrl_11_5	Reserved
4-0 cmn_rxcal_ctrl_4_0	Resistor calibration code: This is the calibration code that was determined by the resistor calibration process. The following are the values for the code:

13.4.10.2.481 RX resistor calibration override register (cmn_rxcal_ovrd)

13.4.10.2.481.1 Offset

Register	Offset
cmn_rxcal_ovrd	8_00D1h

13.4.10.2.481.2 Diagram



13.4.10.2.481.3 Fields

Field	Function
15	Resistor code override enable: Activation (1b1) of this register bit allows the resistor codes determined during the automatic resistor calibration process to be overridden. The override value is specified using the resistor code override value field of this register.

Table continues on the next page...

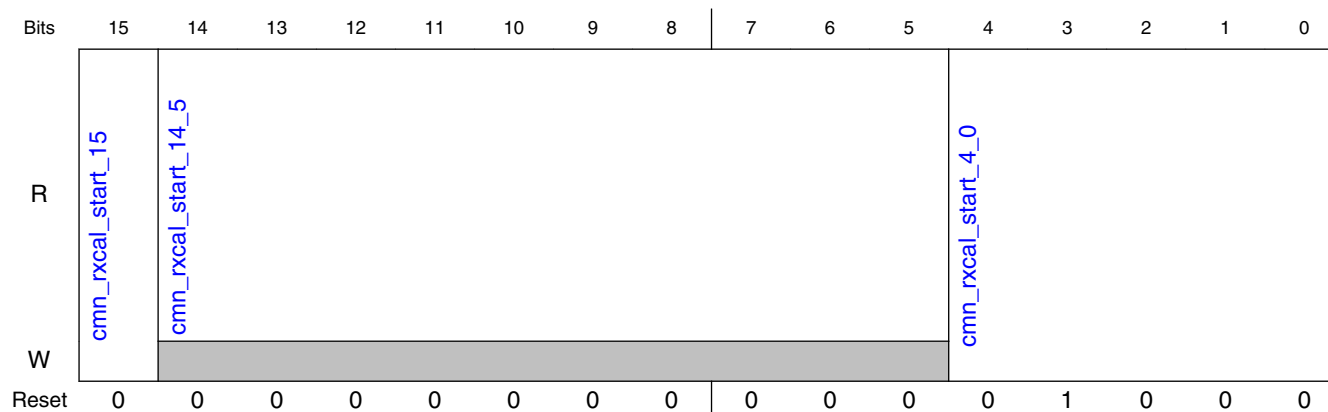
Field	Function
cmn_rxcal_ovrd_15	
14 cmn_rxcal_ovrd_14	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-5 cmn_rxcal_ovrd_13_5	Reserved
4-0 cmn_rxcal_ovrd_4_0	Resistor code override value: These bits are used to override the resistor code determined during the automatic resistor calibration process. The resistor code written to these bits is valid when the resistor code override enable bit in this register is active. The following are the values for the code:

13.4.10.2.482 RX resistor calibration start register (cmn_rxcal_start)

13.4.10.2.482.1 Offset

Register	Offset
cmn_rxcal_start	8_00D2h

13.4.10.2.482.2 Diagram



13.4.10.2.482.3 Fields

Field	Function
15	Resistor calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.

Table continues on the next page...

Clocks And Resets

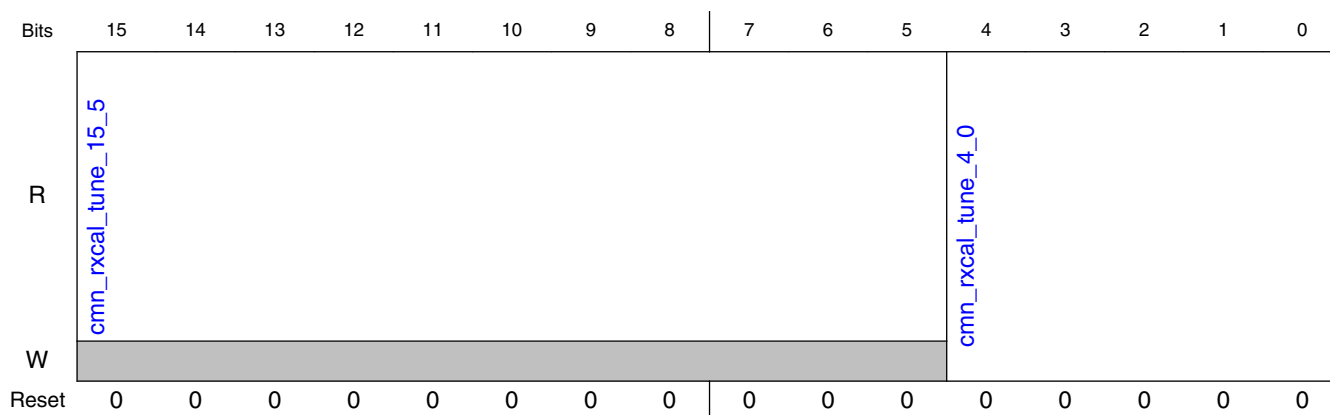
Field	Function
cmn_rxcal_start_15	
14-5 cmn_rxcal_start_14_5	Reserved
4-0 cmn_rxcal_start_4_0	Start resistor calibration code: This is the calibration code that the resistor calibration process starts with when automatic calibration is run. The following are the values for the code.

13.4.10.2.483 RX resistor calibration tune register (cmn_rxcal_tune)

13.4.10.2.483.1 Offset

Register	Offset
cmn_rxcal_tune	8_00D3h

13.4.10.2.483.2 Diagram



13.4.10.2.483.3 Fields

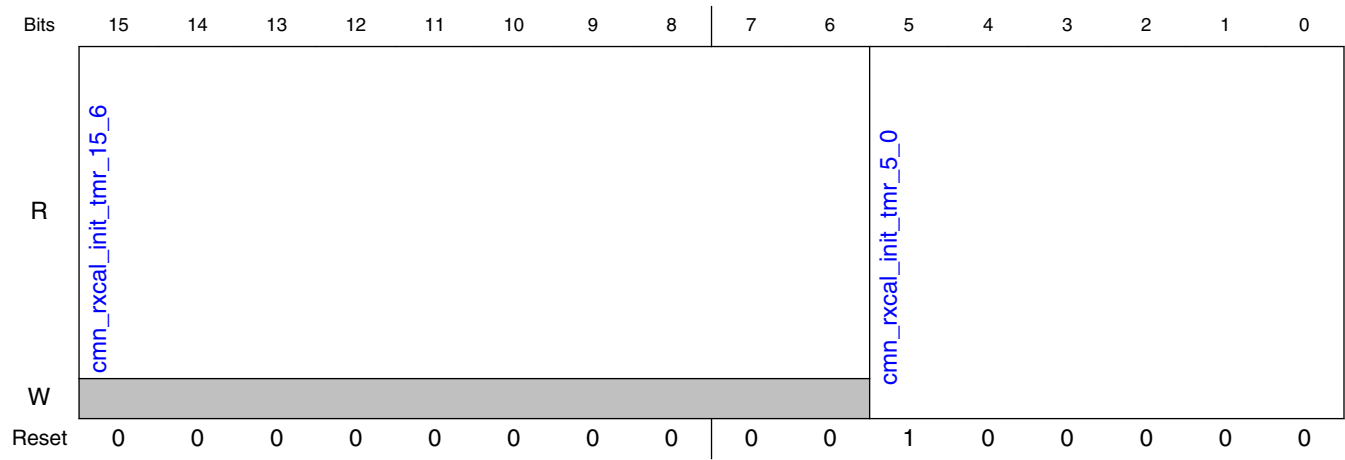
Field	Function
15-5 cmn_rxcal_tune_15_5	Reserved
4-0 cmn_rxcal_tune_4_0	Resistor calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.484 RX resistor calibration initialization timer register (cmn_rxcal_init_tmr)

13.4.10.2.484.1 Offset

Register	Offset
cmn_rxcal_init_tmr	8_00D4h

13.4.10.2.484.2 Diagram



13.4.10.2.484.3 Fields

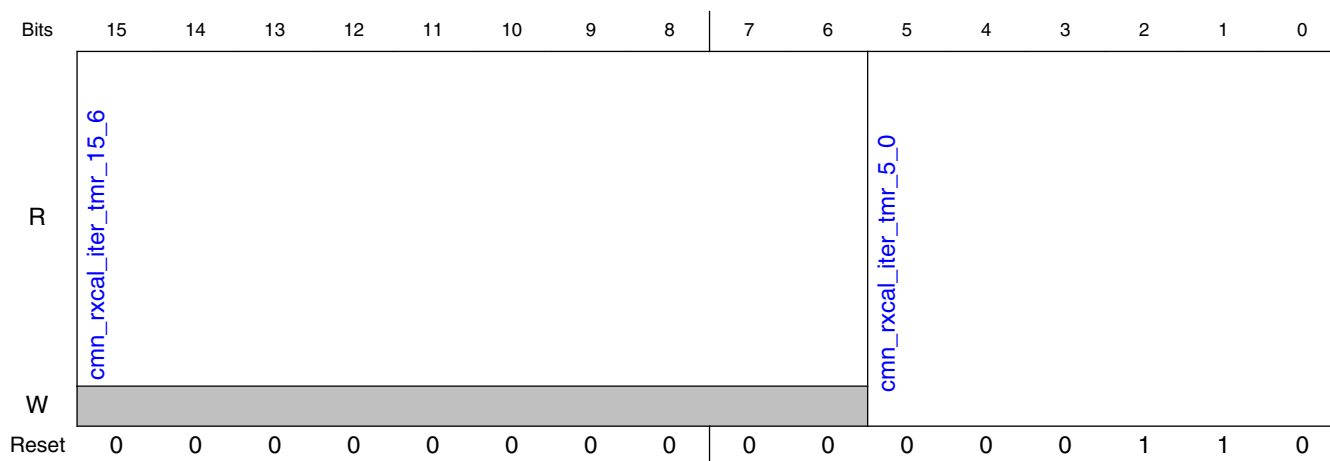
Field	Function
15-6 cmn_rxcal_init_tmr_15_6	Reserved
5-0 cmn_rxcal_init_tmr_5_0	Initialization wait timer value: This is the number of

13.4.10.2.485 RX resistor calibration iteration timer register (cmn_rxcal_iter_tmr)

13.4.10.2.485.1 Offset

Register	Offset
cmn_rxcal_iter_tmr	8_00D5h

13.4.10.2.485.2 Diagram



13.4.10.2.485.3 Fields

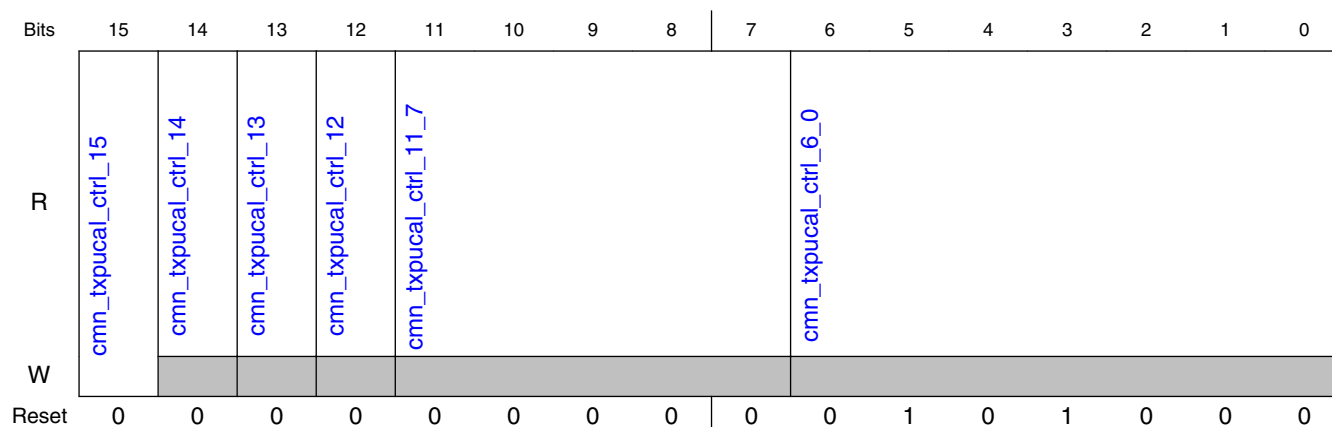
Field	Function
15-6 cmn_rxcal_iter_tmr_15_6	Reserved
5-0 cmn_rxcal_iter_tmr_5_0	Iteration wait timer value: This is the number of

13.4.10.2.486 TX pull-up resistor calibration control register (cmn_txpucal_ctrl)

13.4.10.2.486.1 Offset

Register	Offset
cmn_txpucal_ctrl	8_00E0h

13.4.10.2.486.2 Diagram



13.4.10.2.486.3 Fields

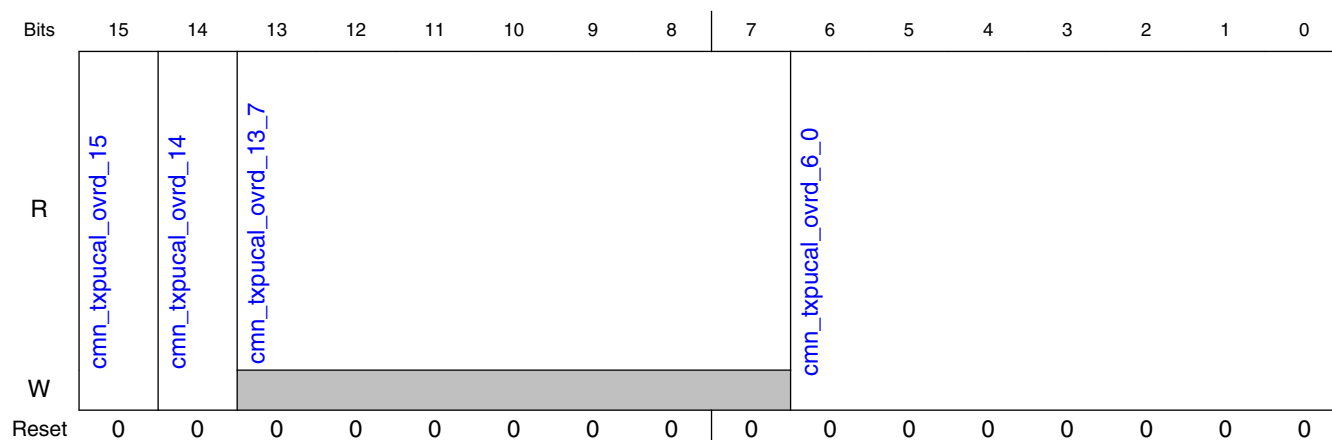
Field	Function
15 cmn_txpucal_ctrl_15	Start resistor calibration: Activating (1b1) this bit will start the resistor calibration process. This signal must remain active until the resistor calibration process is complete. To start another resistor calibration process, this register must first be set inactive (1b0) until the resistor calibration process done bit in this register is cleared.
14 cmn_txpucal_ctrl_14	Resistor calibration process done: This bit will be set to 1b1 when the resistor calibration process is complete. It will be cleared by
13 cmn_txpucal_ctrl_13	No analog calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached.
12 cmn_txpucal_ctrl_12	Current analog comparator response: This is the current state of the analog comparator response signal (
11-7 cmn_txpucal_ctrl_11_7	Reserved
6-0 cmn_txpucal_ctrl_6_0	Resistor calibration code: This is the calibration code that was determined by the resistor calibration process. The following are the values for the code.

13.4.10.2.487 TX pull-up resistor calibration override register (cmn_txpucal_ovrd)

13.4.10.2.487.1 Offset

Register	Offset
cmn_txpucal_ovrd	8_00E1h

13.4.10.2.487.2 Diagram



13.4.10.2.487.3 Fields

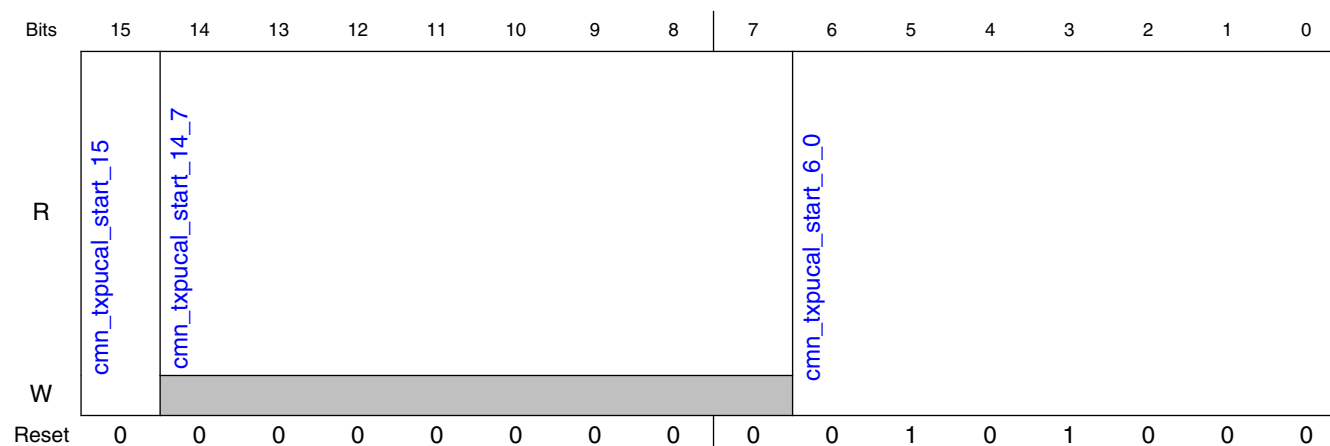
Field	Function
15 <code>cmn_txpucal_ovrd_15</code>	Resistor code override enable: Activation (1b1) of this register bit allows the resistor codes determined during the automatic resistor calibration process to be overridden. The override value is specified using the resistor code override value field of this register.
14 <code>cmn_txpucal_ovrd_14</code>	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-7 <code>cmn_txpucal_ovrd_13_7</code>	Reserved
6-0 <code>cmn_txpucal_ovrd_6_0</code>	Resistor code override value: These bits are used to override the resistor code determined during the automatic resistor calibration process. The resistor code written to these bits is valid when the resistor code override enable bit in this register is active.

13.4.10.2.488 TX pull-up resistor calibration start register (`cmn_txpucal_start`)

13.4.10.2.488.1 Offset

Register	Offset
cmn_txpucal_start	8_00E2h

13.4.10.2.488.2 Diagram



13.4.10.2.488.3 Fields

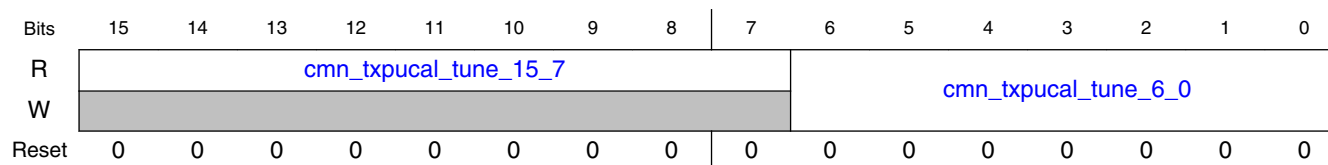
Field	Function
15 cmn_txpucal_start_15	Resistor calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-7 cmn_txpucal_start_14_7	Reserved
6-0 cmn_txpucal_start_6_0	Start resistor calibration code: This is the calibration code that the resistor calibration process starts with when automatic calibration is run. The following are the values for the code.

13.4.10.2.489 TX pull-up resistor calibration tune register (cmn_txpucal_tune)

13.4.10.2.489.1 Offset

Register	Offset
cmn_txpucal_tune	8_00E3h

13.4.10.2.489.2 Diagram



13.4.10.2.489.3 Fields

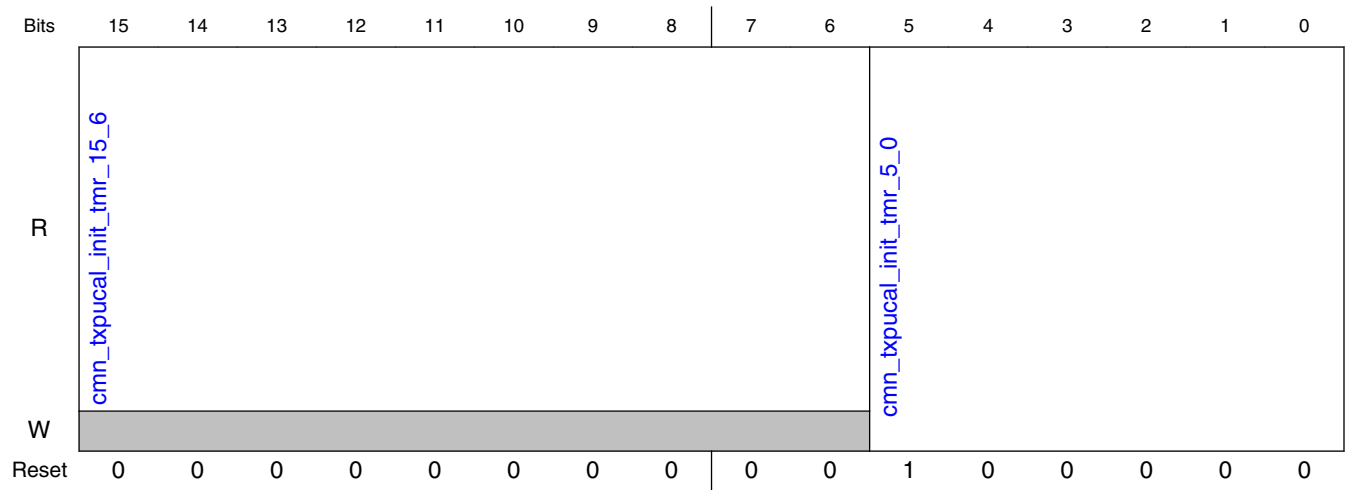
Field	Function
15-7 cmn_txpucal_tune_15_7	Reserved
6-0 cmn_txpucal_tune_6_0	Resistor calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.490 TX pull-up resistor calibration initialization timer register (cmn_txpucal_init_tmr)

13.4.10.2.490.1 Offset

Register	Offset
cmn_txpucal_init_tmr	8_00E4h

13.4.10.2.490.2 Diagram



13.4.10.2.490.3 Fields

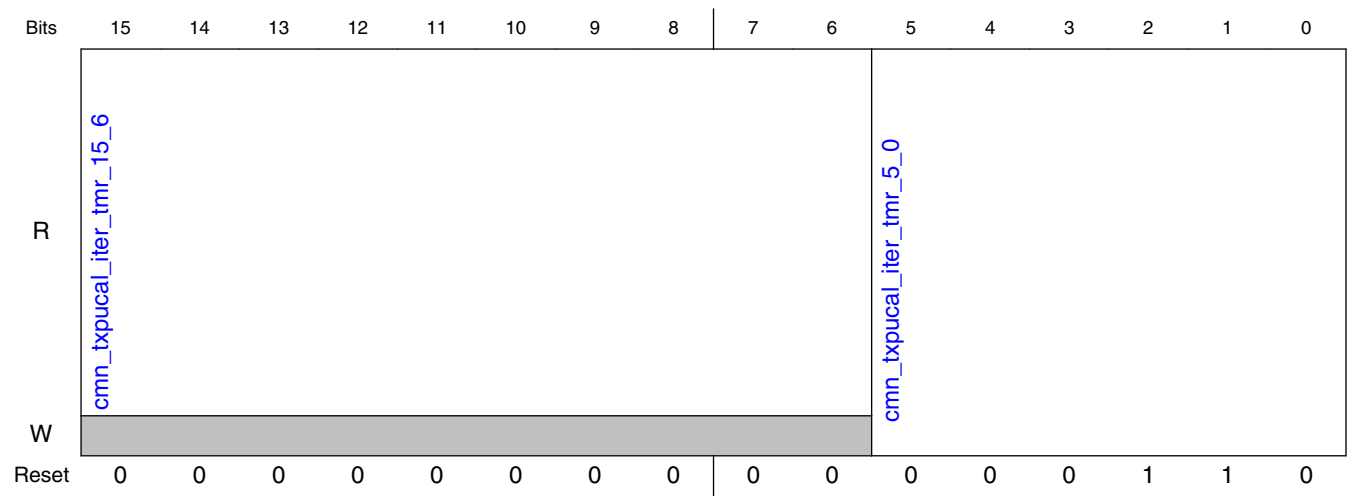
Field	Function
15-6 cmn_txpucal_init_tmr_15_6	Reserved
5-0 cmn_txpucal_init_tmr_5_0	Initialization wait timer value: This is the number of

13.4.10.2.491 TX pull-up resistor calibration iteration timer register (cmn_txpucal_iter_tmr)

13.4.10.2.491.1 Offset

Register	Offset
cmn_txpucal_iter_tmr	8_00E5h

13.4.10.2.491.2 Diagram



13.4.10.2.491.3 Fields

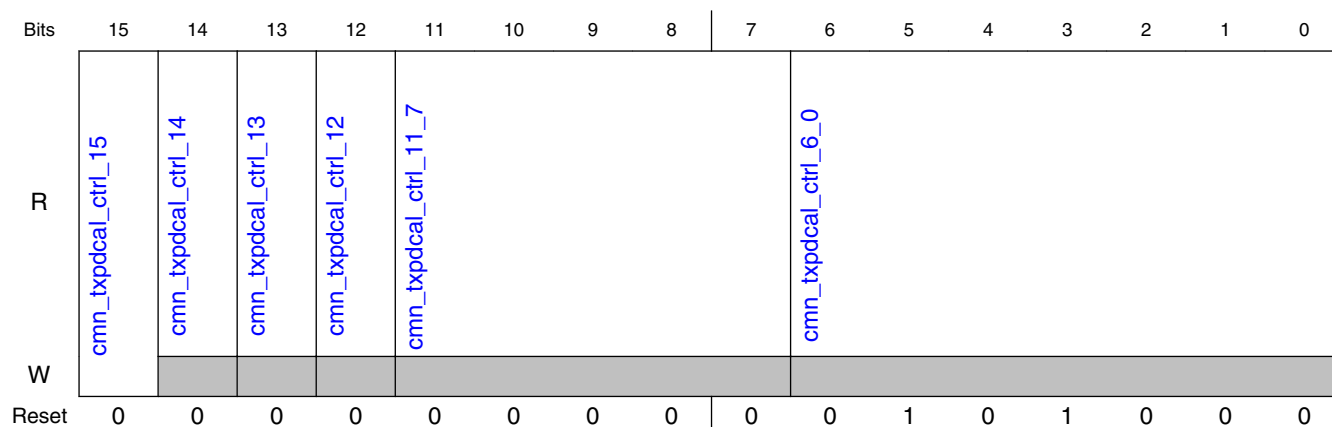
Field	Function
15-6 <code>cmn_txpucal_iter_tmr_15_6</code>	Reserved
5-0 <code>cmn_txpucal_iter_tmr_5_0</code>	Iteration wait timer value: This is the number of

13.4.10.2.492 TX pull-down resistor calibration control register (`cmn_txpdcal_ctrl`)

13.4.10.2.492.1 Offset

Register	Offset
<code>cmn_txpdcal_ctrl</code>	8_00F0h

13.4.10.2.492.2 Diagram



13.4.10.2.492.3 Fields

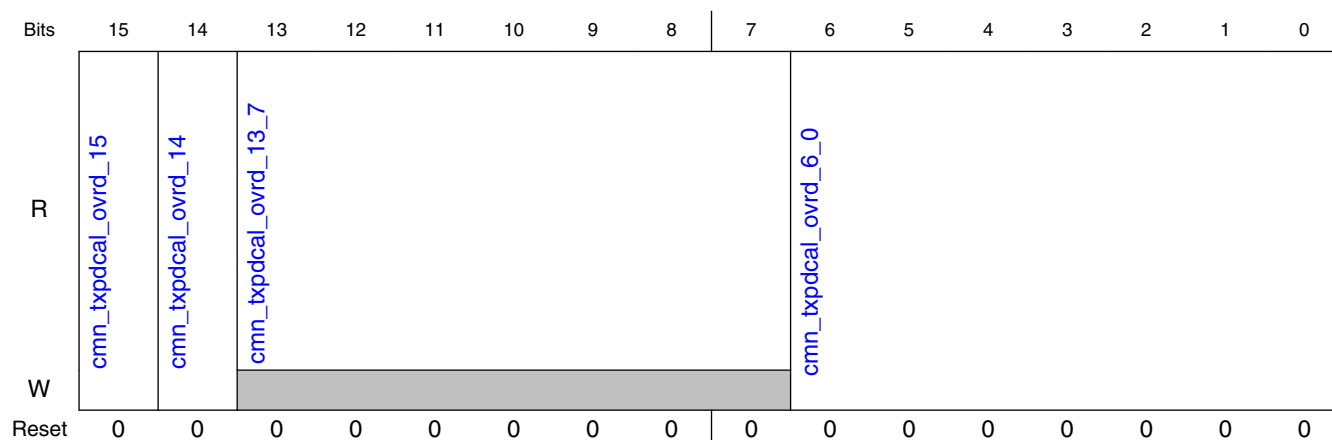
Field	Function
15 cmn_txpdcal_ctrl_15	Start resistor calibration: Activating (1b1) this bit will start the resistor calibration process. This signal must remain active until the resistor calibration process is complete. To start another resistor calibration process, this register must first be set inactive (1b0) until the resistor calibration process done bit in this register is cleared.
14 cmn_txpdcal_ctrl_14	Resistor calibration process done: This bit will be set to 1b1 when the resistor calibration process is complete. It will be cleared by
13 cmn_txpdcal_ctrl_13	No analog calibration response : This signal indicates that the calibration function has gone through the entire calibration process, reached the final calibration value, and the analog has not responded indicating that a valid calibration value has been reached.
12 cmn_txpdcal_ctrl_12	Current analog comparator response: This is the current state of the analog comparator response signal (
11-7 cmn_txpdcal_ctrl_11_7	Reserved
6-0 cmn_txpdcal_ctrl_6_0	Resistor calibration code: This is the calibration code that was determined by the resistor calibration process. The following are the values for the code.

13.4.10.2.493 TX pull-down resistor calibration override register (cmn_txpdcal_ovrd)

13.4.10.2.493.1 Offset

Register	Offset
cmn_txpdcal_ovrd	8_00F1h

13.4.10.2.493.2 Diagram



13.4.10.2.493.3 Fields

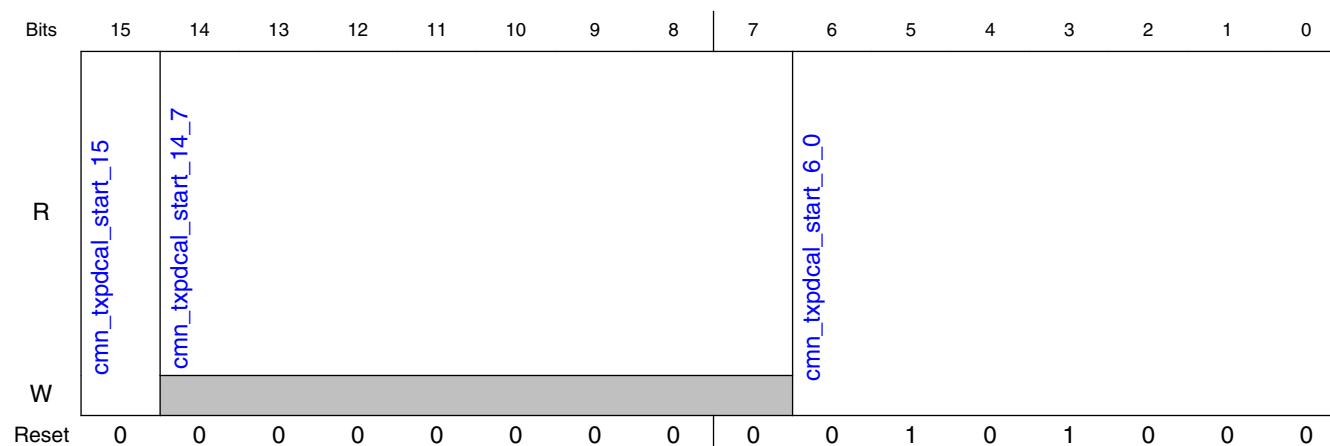
Field	Function
15 <code>cmn_txpdcal_ovrd_15</code>	Resistor code override enable: Activation (1b1) of this register bit allows the resistor codes determined during the automatic resistor calibration process to be overridden. The override value is specified using the resistor code override value field of this register.
14 <code>cmn_txpdcal_ovrd_14</code>	Analog calibration enable override: Activation (1b1) of this register bit will force the analog calibration circuits to be enabled by activating the
13-7 <code>cmn_txpdcal_ovrd_13_7</code>	Reserved
6-0 <code>cmn_txpdcal_ovrd_6_0</code>	Resistor code override value: These bits are used to override the resistor code determined during the automatic resistor calibration process. The resistor code written to these bits is valid when the resistor code override enable bit in this register is active. The following are the values for the code.

13.4.10.2.494 TX pull-down resistor calibration start register (`cmn_txpdcal_start`)

13.4.10.2.494.1 Offset

Register	Offset
cmn_txpdcal_start	8_00F2h

13.4.10.2.494.2 Diagram



13.4.10.2.494.3 Fields

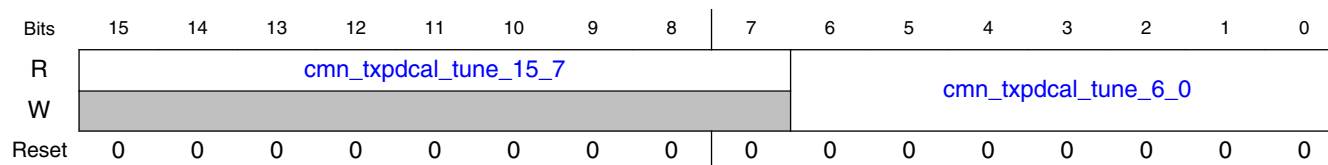
Field	Function
15 cmn_txpdcal_start_15	Resistor calibration direction: This controls the direction that the automatic calibration process steps the calibration codes in.
14-7 cmn_txpdcal_start_14_7	Reserved
6-0 cmn_txpdcal_start_6_0	Start resistor calibration code: This is the calibration code that the resistor calibration process starts with when automatic calibration is run. The following are the values for the code.

13.4.10.2.495 TX pull-down resistor calibration tune register (cmn_txpdcal_tune)

13.4.10.2.495.1 Offset

Register	Offset
cmn_txpdcal_tune	8_00F3h

13.4.10.2.495.2 Diagram



13.4.10.2.495.3 Fields

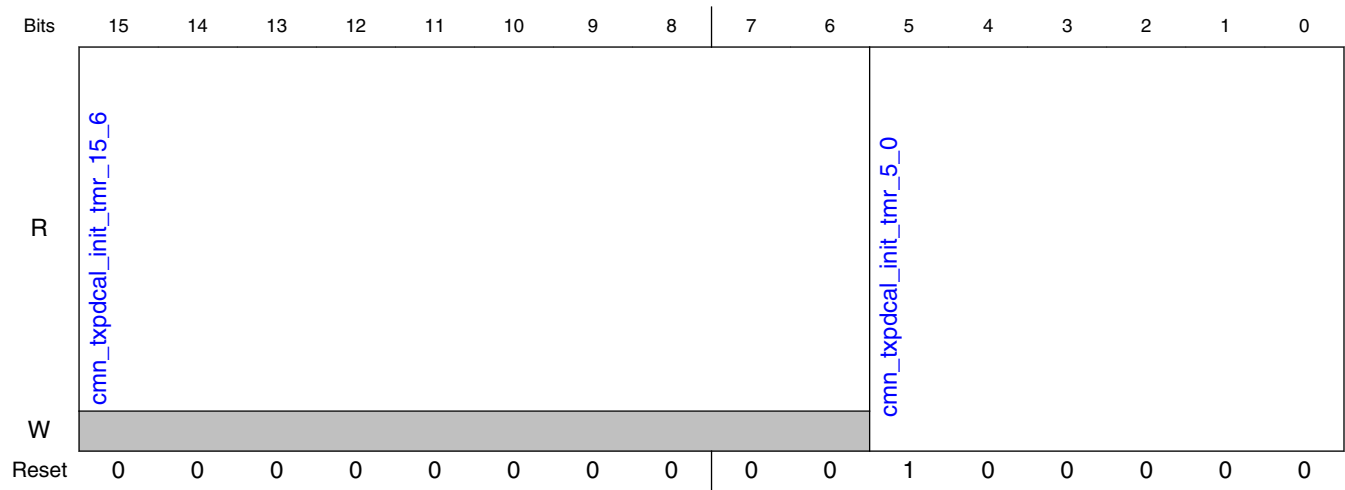
Field	Function
15-7 cmn_txpdcal_tune_15_7	Reserved
6-0 cmn_txpdcal_tune_6_0	Resistor calibration tune value: The value of this field is added to the automatically calibrated code, or the override code if override is enabled. Note that this value is a twos complement value, so the calibrated code can be increased or decreased.

13.4.10.2.496 TX pull-down resistor calibration initialization timer register (cmn_txpdcal_init_tmr)

13.4.10.2.496.1 Offset

Register	Offset
cmn_txpdcal_init_tmr	8_00F4h

13.4.10.2.496.2 Diagram



13.4.10.2.496.3 Fields

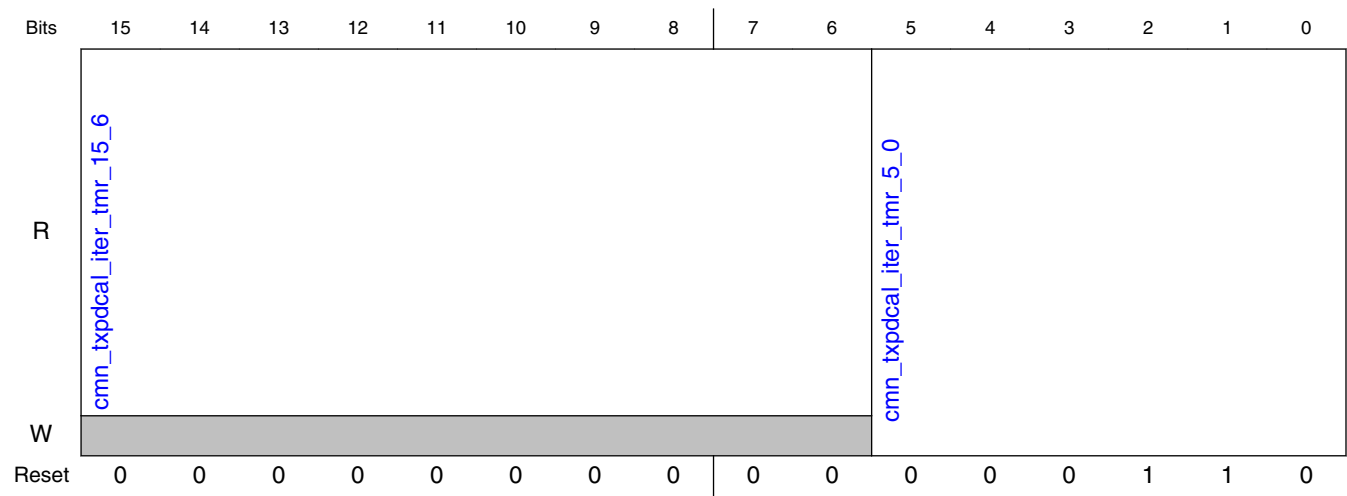
Field	Function
15-6 <code>cmn_txpdcal_init_tmr_15_6</code>	Reserved
5-0 <code>cmn_txpdcal_init_tmr_5_0</code>	Initialization wait timer value: This is the number of

13.4.10.2.497 TX pull-down resistor calibration iteration timer register (`cmn_txpdcal_iter_tmr`)

13.4.10.2.497.1 Offset

Register	Offset
<code>cmn_txpdcal_iter_tmr</code>	8_00F5h

13.4.10.2.497.2 Diagram



13.4.10.2.497.3 Fields

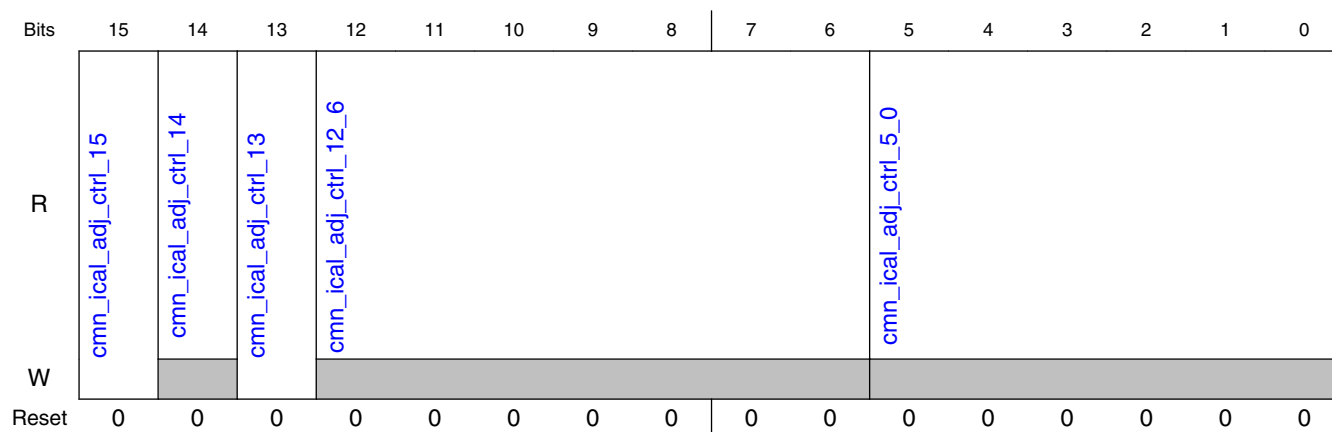
Field	Function
15-6 cmn_txdcal_iter_tmr_15_6	Reserved
5-0 cmn_txdcal_iter_tmr_5_0	Iteration wait timer value: This is the number of

13.4.10.2.498 Current calibration adjust control register (cmn_ical_adj_ctrl)

13.4.10.2.498.1 Offset

Register	Offset
cmn_ical_adj_ctrl	8_0100h

13.4.10.2.498.2 Diagram



13.4.10.2.498.3 Fields

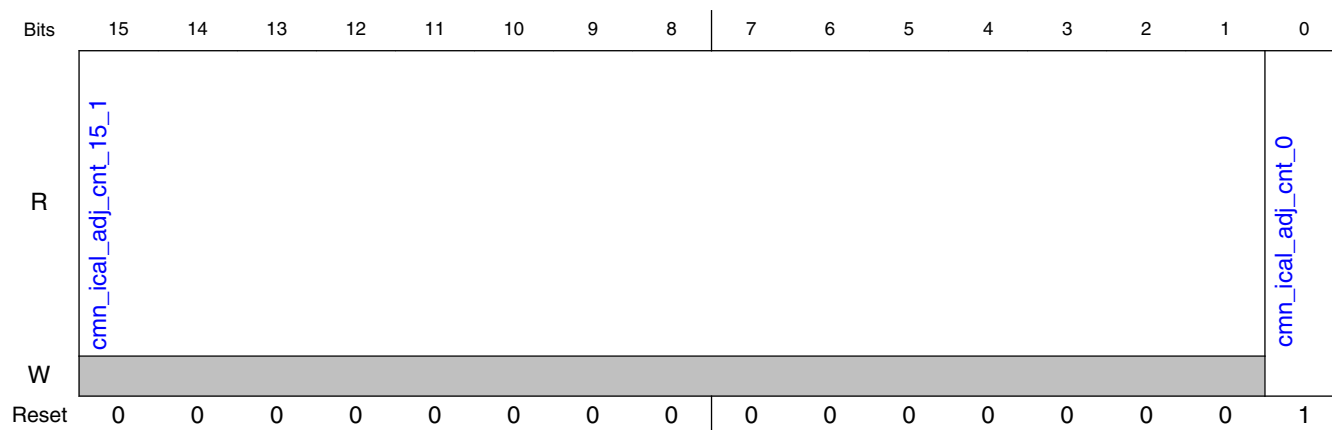
Field	Function
15 <code>cmn_ical_adj_ctrl_15</code>	Start calibration adjust: Activating (1b1) this bit will start the calibration adjust process. This signal must remain active until the calibration adjust process is complete. To start another calibration adjust process, this register must first be set inactive (1b0) until the calibration adjust process done bit in this register is cleared.
14 <code>cmn_ical_adj_ctrl_14</code>	Calibration adjust process done: This bit will be set to 1b1 when the calibration adjust process is complete. It will be cleared by
13 <code>cmn_ical_adj_ctrl_13</code>	Calibration adjust direction: This controls the direction that the automatic calibration process steps the calibration codes in.
12-6 <code>cmn_ical_adj_ctrl_12_6</code>	Reserved
5-0 <code>cmn_ical_adj_ctrl_5_0</code>	Calibration adjust code: This is the value that the calibration adjust function is currently attempting to adjust the calibration code by. Note that this is a signed two's complement value. The following are the values for the code:

13.4.10.2.499 Current calibration adjust count register (`cmn_ical_adj_cnt`)

13.4.10.2.499.1 Offset

Register	Offset
<code>cmn_ical_adj_cnt</code>	8_0101h

13.4.10.2.499.2 Diagram



13.4.10.2.499.3 Fields

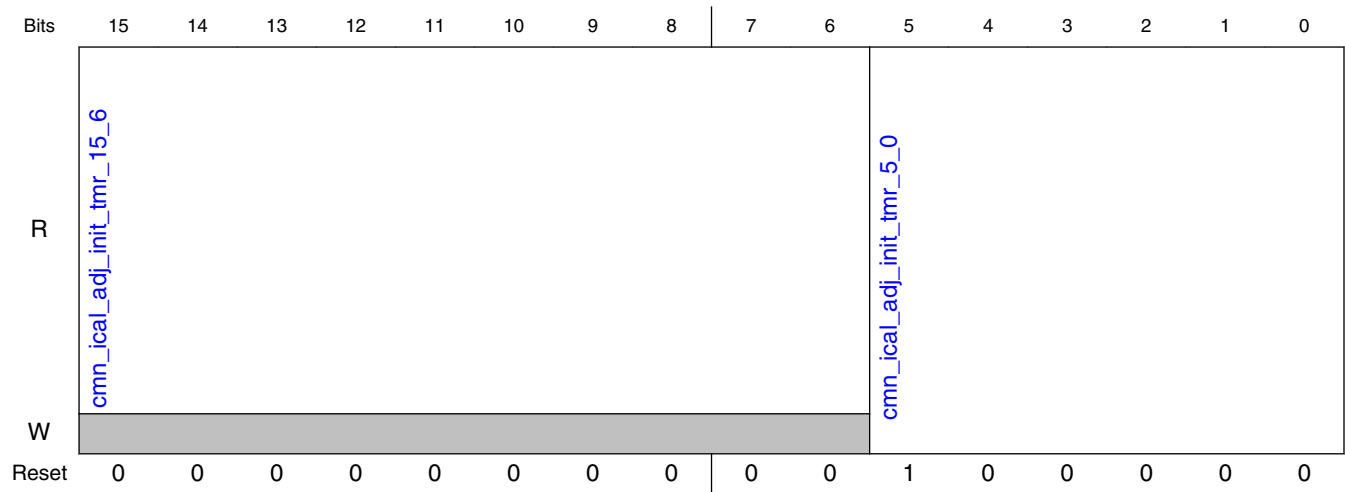
Field	Function
15-1 cmn_ical_adj_cn t_15_1	Reserved
0 cmn_ical_adj_cn t_0	Calibration adjust count value: Specifies the number of calibration adjust iterations that will be performed every time a calibration adjust is requested.

13.4.10.2.500 Current calibration adjust initialization timer register (cmn_ical_adj_init_tmr)

13.4.10.2.500.1 Offset

Register	Offset
cmn_ical_adj_init_tmr	8_0102h

13.4.10.2.500.2 Diagram



13.4.10.2.500.3 Fields

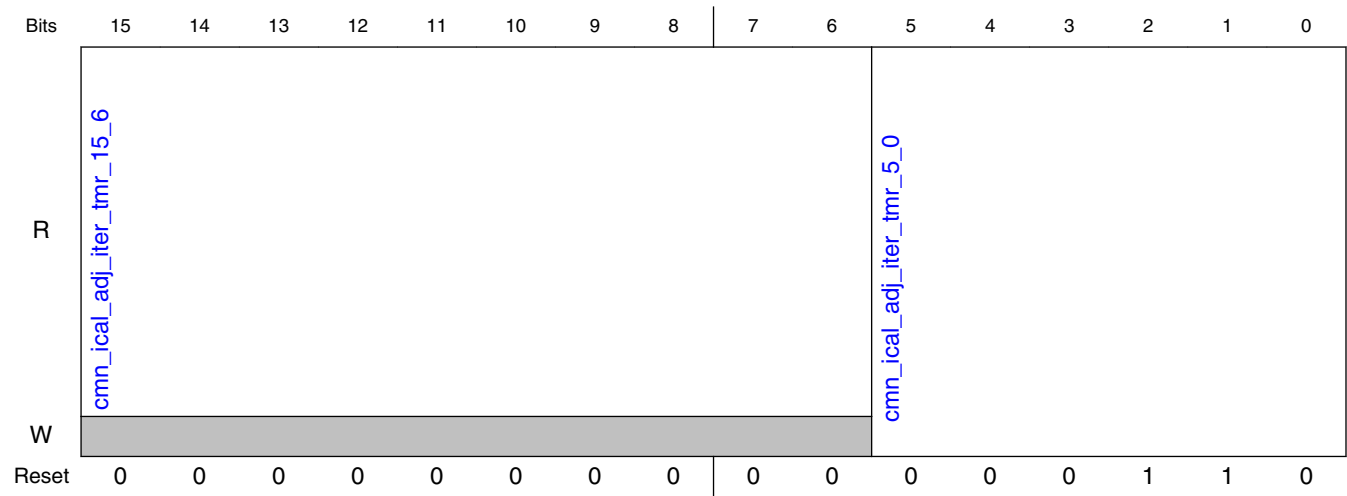
Field	Function
15-6 cmn_ical_adj_init_tmr_15_6	Reserved
5-0 cmn_ical_adj_init_tmr_5_0	Initialization wait timer value: This is the number of clock cycles to wait between when the analog calibration circuits are enabled, and when the first calibration adjust test value is placed on the calibration code signal, going to the analog.

13.4.10.2.501 Current calibration adjust iteration timer register (cmn_ical_adj_iter_tmr)

13.4.10.2.501.1 Offset

Register	Offset
cmn_ical_adj_iter_tmr	8_0103h

13.4.10.2.501.2 Diagram



13.4.10.2.501.3 Fields

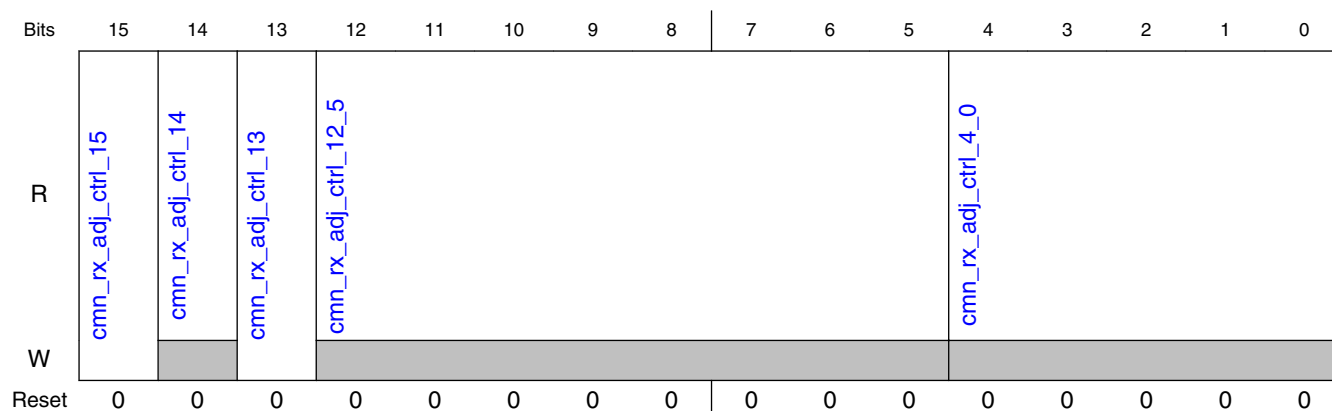
Field	Function
15-6 cmn_ical_adj_iter_tmr_15_6	Reserved
5-0 cmn_ical_adj_iter_tmr_5_0	Iteration wait timer value: This is the number of clock cycles to wait between when an adjusted calibration value is placed on the calibration signal going to the analog, and when the comparator value coming from the analog circuits can be checked.

13.4.10.2.502 RX resistor calibration adjust control register (cmn_rx_adj_ctrl)

13.4.10.2.502.1 Offset

Register	Offset
cmn_rx_adj_ctrl	8_0104h

13.4.10.2.502.2 Diagram



13.4.10.2.502.3 Fields

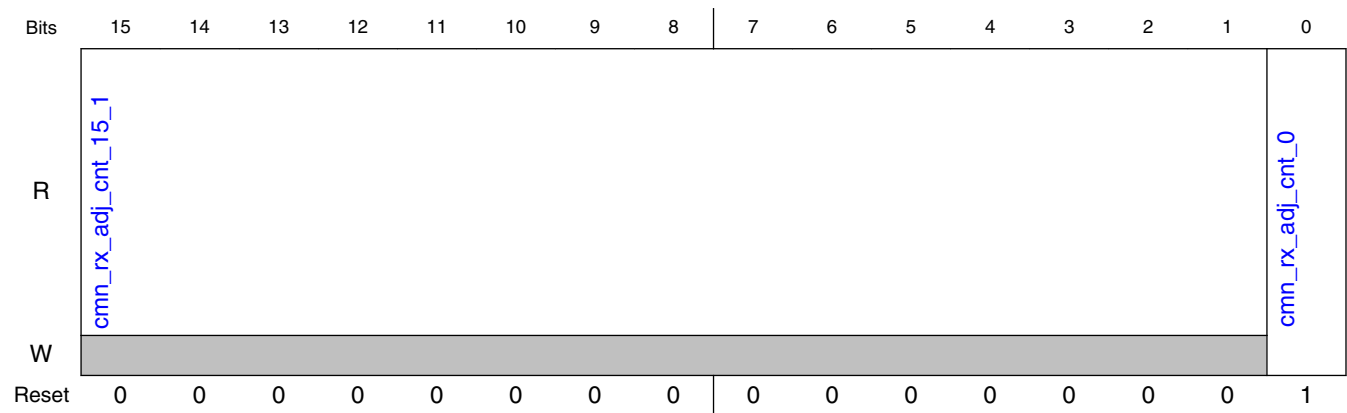
Field	Function
15 cmn_rx_adj_ctrl_15	Start calibration adjust: Activating (1b1) this bit will start the calibration adjust process. This signal must remain active until the calibration adjust process is complete. To start another calibration adjust process, this register must first be set inactive (1b0) until the calibration adjust process done bit in this register is cleared.
14 cmn_rx_adj_ctrl_14	Calibration adjust process done: This bit will be set to 1b1 when the calibration adjust process is complete. It will be cleared by
13 cmn_rx_adj_ctrl_13	Calibration adjust direction: This controls the direction that the automatic calibration process steps the calibration codes in.
12-5 cmn_rx_adj_ctrl_12_5	Reserved
4-0 cmn_rx_adj_ctrl_4_0	Calibration adjust code: This is the value that the calibration adjust function is currently attempting to adjust the calibration code by. Note that this is a signed two's complement value. The following are the values for the code:

13.4.10.2.503 RX resistor calibration adjust count register (cmn_rx_adj_cnt)

13.4.10.2.503.1 Offset

Register	Offset
cmn_rx_adj_cnt	8_0105h

13.4.10.2.503.2 Diagram



13.4.10.2.503.3 Fields

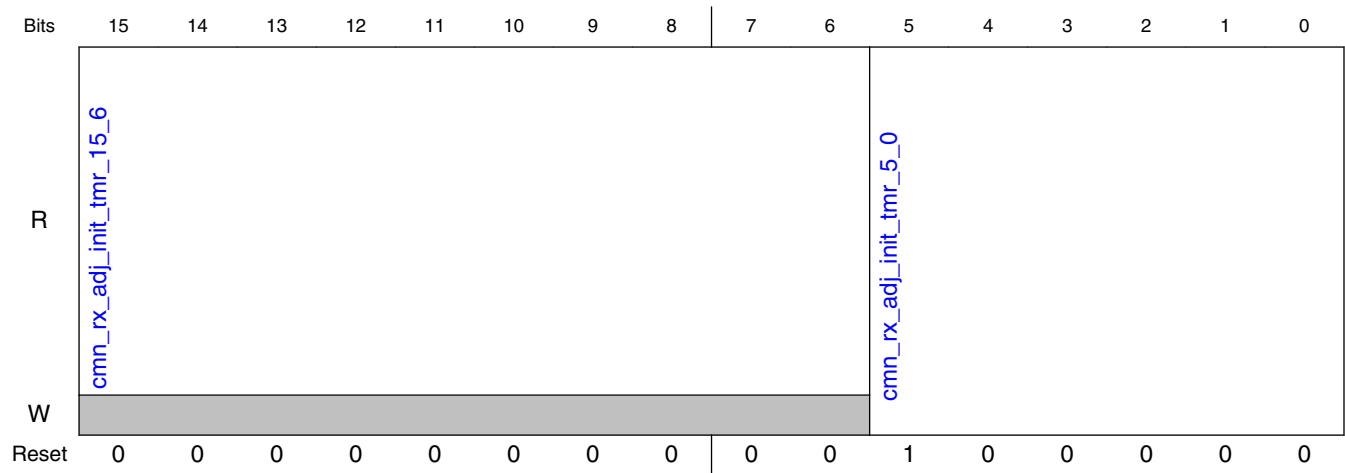
Field	Function
15-1 <code>cmn_rx_adj_cnt_15_1</code>	Reserved
0 <code>cmn_rx_adj_cnt_0</code>	Calibration adjust count value: Specifies the number of calibration adjust iterations that will be performed every time a calibration adjust is requested.

13.4.10.2.504 RX resistor calibration adjust initialization timer register (`cmn_rx_adj_init_tmr`)

13.4.10.2.504.1 Offset

Register	Offset
<code>cmn_rx_adj_init_tmr</code>	8_0106h

13.4.10.2.504.2 Diagram



13.4.10.2.504.3 Fields

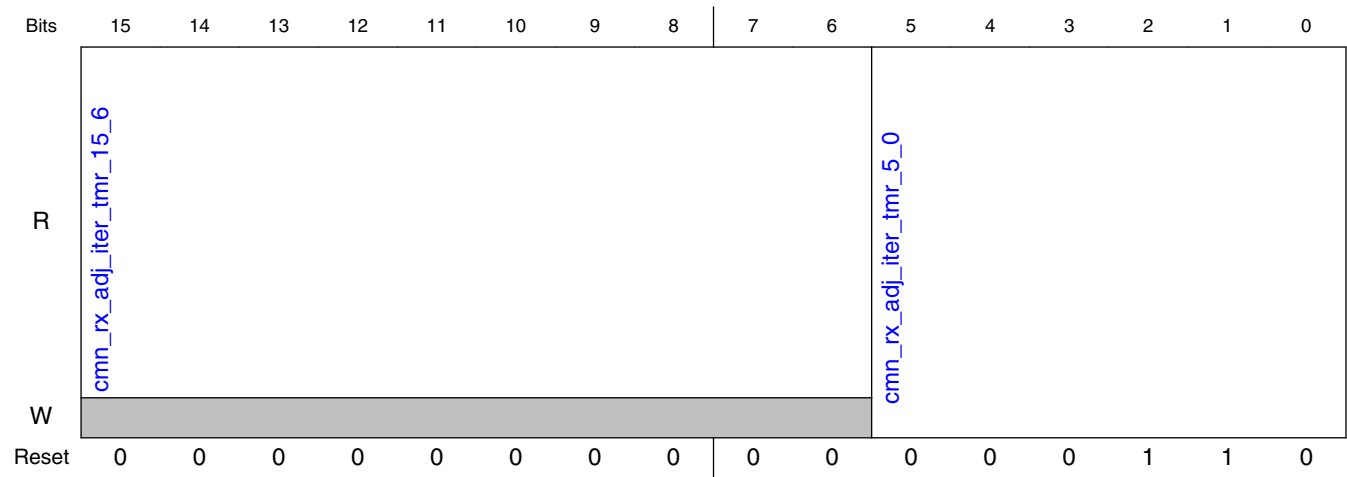
Field	Function
15-6 cmn_rx_adj_init_tmr_15_6	Reserved
5-0 cmn_rx_adj_init_tmr_5_0	Initialization wait timer value: This is the number of clock cycles to wait between when the analog calibration circuits are enabled, and when the first calibration adjust test value is placed on the calibration code signal, going to the analog.

13.4.10.2.505 RX resistor calibration adjust iteration timer register (cmn_rx_adj_iter_tmr)

13.4.10.2.505.1 Offset

Register	Offset
cmn_rx_adj_iter_tmr	8_0107h

13.4.10.2.505.2 Diagram



13.4.10.2.505.3 Fields

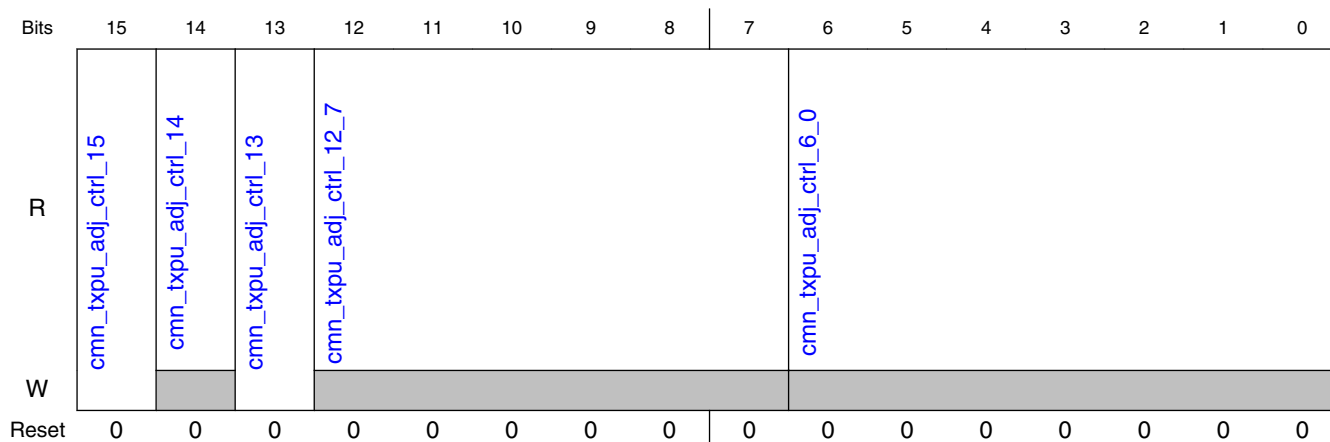
Field	Function
15-6 cmn_rx_adj_iter_tmr_15_6	Reserved
5-0 cmn_rx_adj_iter_tmr_5_0	Iteration wait timer value: This is the number of clock cycles to wait between when an adjusted calibration value is placed on the calibration signal going to the analog, and when the comparator value coming from the analog circuits can be checked.

13.4.10.2.506 TX pull up resistor calibration adjust control register (cmn_txpu_adj_ctrl)

13.4.10.2.506.1 Offset

Register	Offset
cmn_txpu_adj_ctrl	8_0108h

13.4.10.2.506.2 Diagram



13.4.10.2.506.3 Fields

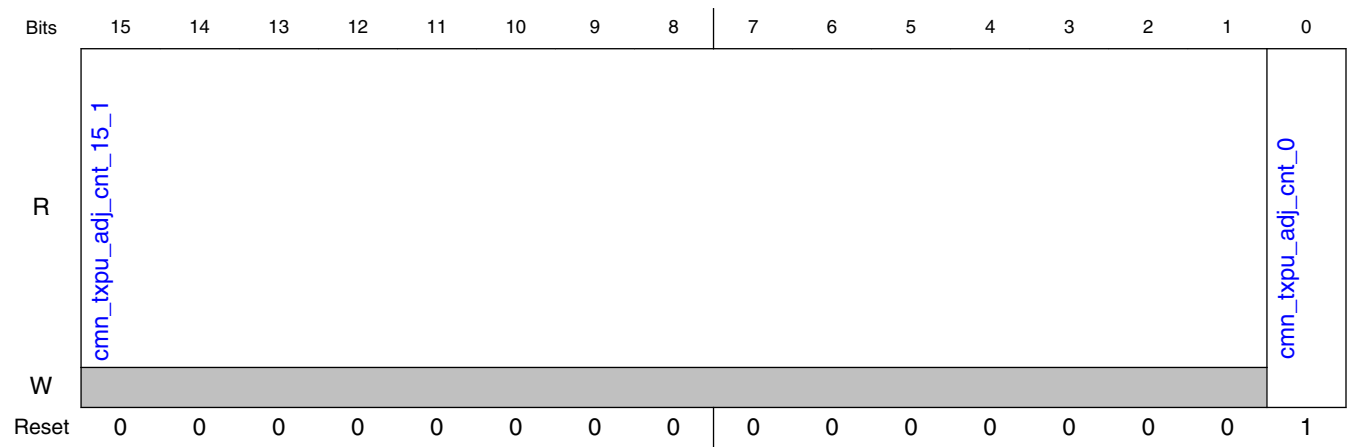
Field	Function
15 cmn_txpu_adj_ctrl_15	Start calibration adjust: Activating (1b1) this bit will start the calibration adjust process. This signal must remain active until the calibration adjust process is complete. To start another calibration adjust process, this register must first be set inactive (1b0) until the calibration adjust process done bit in this register is cleared.
14 cmn_txpu_adj_ctrl_14	Calibration adjust process done: This bit will be set to 1b1 when the calibration adjust process is complete. It will be cleared by
13 cmn_txpu_adj_ctrl_13	Calibration adjust direction: This controls the direction that the automatic calibration process steps the calibration codes in.
12-7 cmn_txpu_adj_ctrl_12_7	Reserved
6-0 cmn_txpu_adj_ctrl_6_0	Calibration adjust code: This is the value that the calibration adjust function is currently attempting to adjust the calibration code by. Note that this is a signed two's complement value. The following are the values for the code:

13.4.10.2.507 TX pull up resistor calibration adjust count register (cmn_txpu_adj_cnt)

13.4.10.2.507.1 Offset

Register	Offset
cmn_txpu_adj_cnt	8_0109h

13.4.10.2.507.2 **Diagram**



13.4.10.2.507.3 **Fields**

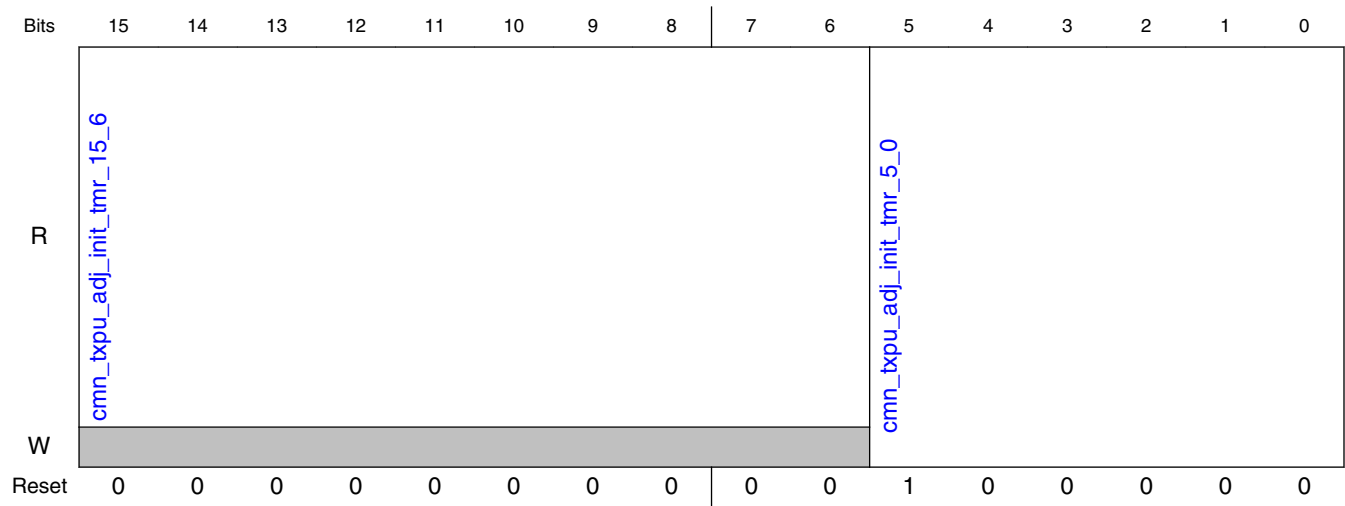
Field	Function
15-1 cmn_txpu_adj_cnt_15_1	Reserved
0 cmn_txpu_adj_cnt_0	Calibration adjust count value: Specifies the number of calibration adjust iterations that will be performed every time a calibration adjust is requested.

13.4.10.2.508 **TX pull up resistor calibration adjust initialization timer register (cmn_txpu_adj_init_tmr)**

13.4.10.2.508.1 **Offset**

Register	Offset
cmn_txpu_adj_init_tmr	8_010Ah

13.4.10.2.508.2 Diagram



13.4.10.2.508.3 Fields

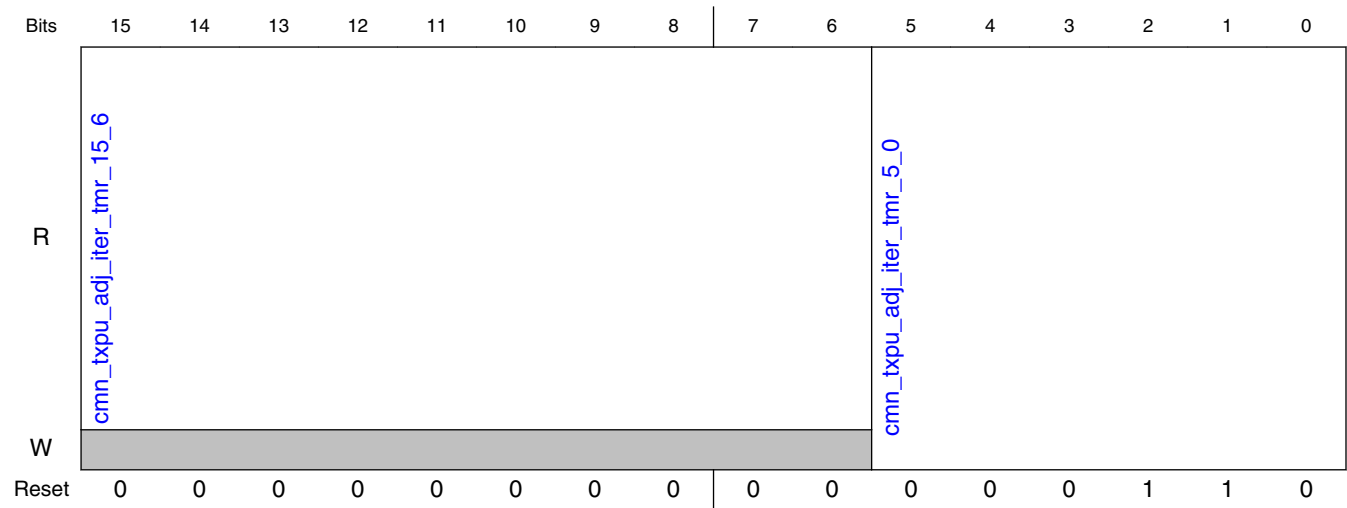
Field	Function
15-6 <code>cmn_txpu_adj_i nit_tmr_15_6</code>	Reserved
5-0 <code>cmn_txpu_adj_i nit_tmr_5_0</code>	Initialization wait timer value: This is the number of clock cycles to wait between when the analog calibration circuits are enabled, and when the first calibration adjust test value is placed on the calibration code signal, going to the analog.

13.4.10.2.509 TX pull up resistor calibration adjust iteration timer register (`cmn_txpu_adj_iter_tmr`)

13.4.10.2.509.1 Offset

Register	Offset
<code>cmn_txpu_adj_iter_tmr</code>	8_010Bh

13.4.10.2.509.2 Diagram



13.4.10.2.509.3 Fields

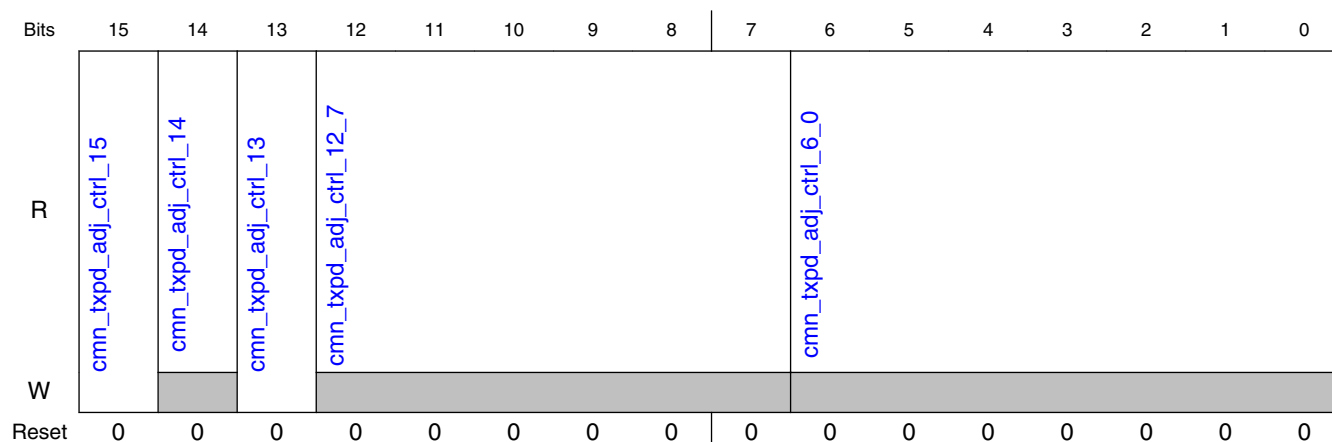
Field	Function
15-6 cmn_txpu_adj_iter_tmr_15_6	Reserved
5-0 cmn_txpu_adj_iter_tmr_5_0	Iteration wait timer value: This is the number of clock cycles to wait between when an adjusted calibration value is placed on the calibration signal going to the analog, and when the comparator value coming from the analog circuits can be checked.

13.4.10.2.510 TX pull down resistor calibration adjust control register (cmn_txpd_adj_ctrl)

13.4.10.2.510.1 Offset

Register	Offset
cmn_txpd_adj_ctrl	8_010Ch

13.4.10.2.510.2 Diagram



13.4.10.2.510.3 Fields

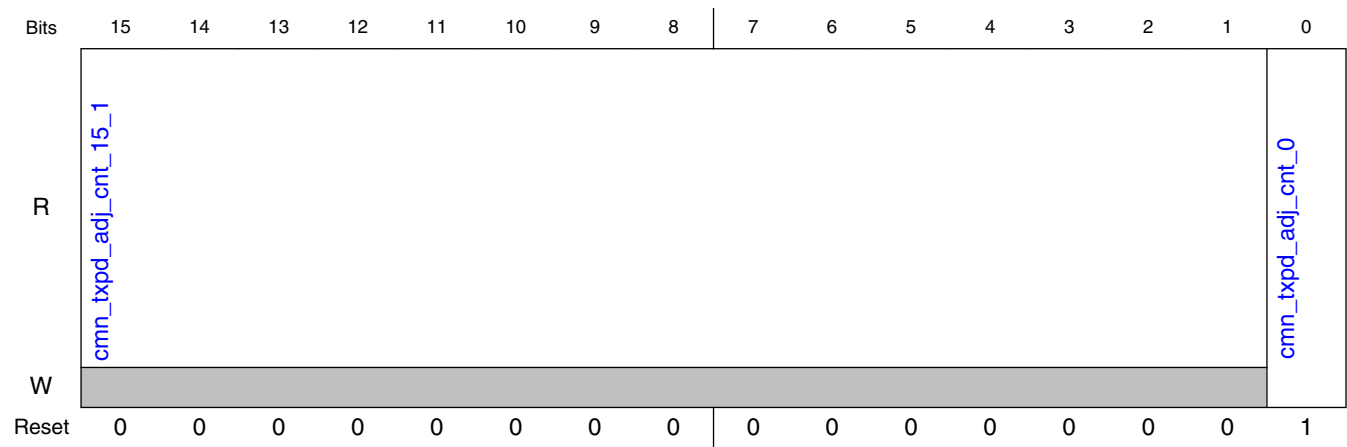
Field	Function
15 cmn_txpd_adj_ctrl_15	Start calibration adjust: Activating (1b1) this bit will start the calibration adjust process. This signal must remain active until the calibration adjust process is complete. To start another calibration adjust process, this register must first be set inactive (1b0) until the calibration adjust process done bit in this register is cleared.
14 cmn_txpd_adj_ctrl_14	Calibration adjust process done: This bit will be set to 1b1 when the calibration adjust process is complete. It will be cleared by
13 cmn_txpd_adj_ctrl_13	Calibration adjust direction: This controls the direction that the automatic calibration process steps the calibration codes in.
12-7 cmn_txpd_adj_ctrl_12_7	Reserved
6-0 cmn_txpd_adj_ctrl_6_0	Calibration adjust code: This is the value that the calibration adjust function is currently attempting to adjust the calibration code by. Note that this is a signed two's complement value. The following are the values for the code:

13.4.10.2.511 TX pull down resistor calibration adjust count register (cmn_txpd_adj_cnt)

13.4.10.2.511.1 Offset

Register	Offset
cmn_txpd_adj_cnt	8_010Dh

13.4.10.2.511.2 Diagram



13.4.10.2.511.3 Fields

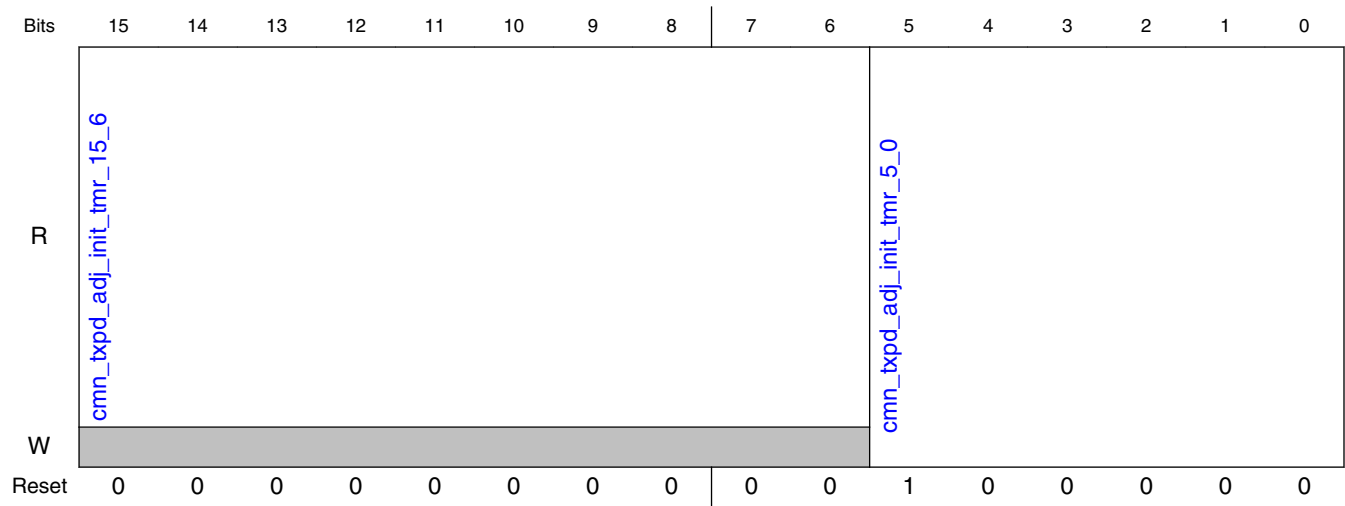
Field	Function
15-1 cmn_txpd_adj_cnt_15_1	Reserved
0 cmn_txpd_adj_cnt_0	Calibration adjust count value: Specifies the number of calibration adjust iterations that will be performed every time a calibration adjust is requested.

13.4.10.2.512 TX pull down resistor calibration adjust initialization timer register (cmn_txpd_adj_init_tmr)

13.4.10.2.512.1 Offset

Register	Offset
cmn_txpd_adj_init_tmr	8_010Eh

13.4.10.2.512.2 Diagram



13.4.10.2.512.3 Fields

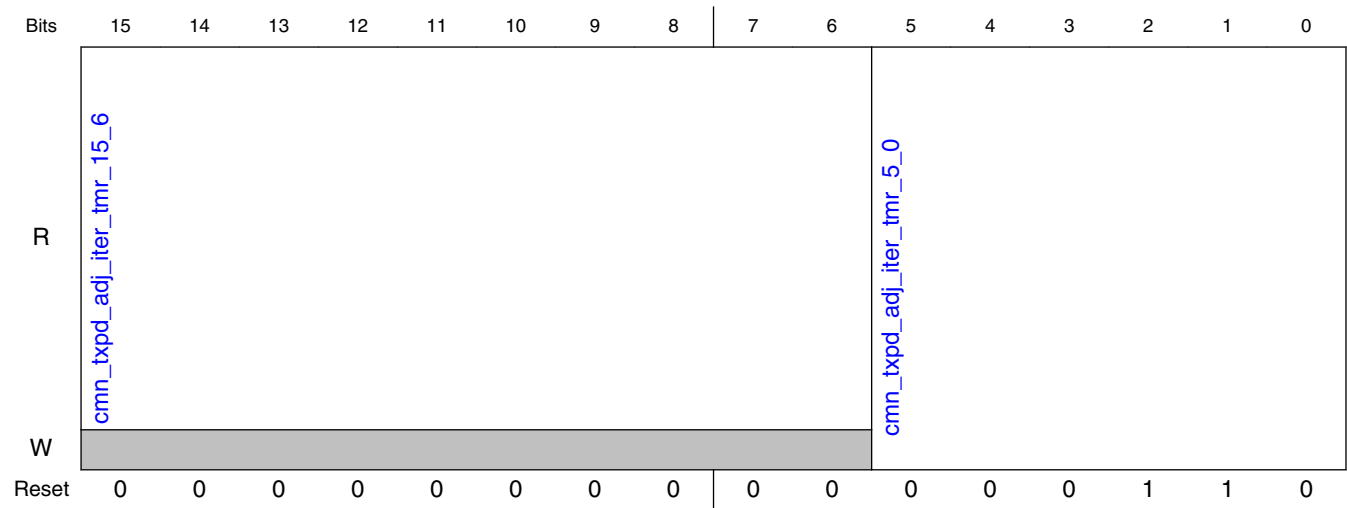
Field	Function
15-6 <code>cmn_txpd_adj_i nit_tmr_15_6</code>	Reserved
5-0 <code>cmn_txpd_adj_i nit_tmr_5_0</code>	Initialization wait timer value: This is the number of clock cycles to wait between when the analog calibration circuits are enabled, and when the first calibration adjust test value is placed on the calibration code signal, going to the analog.

13.4.10.2.513 TX pull down resistor calibration adjust iteration timer register (`cmn_txpd_adj_iter_tmr`)

13.4.10.2.513.1 Offset

Register	Offset
<code>cmn_txpd_adj_iter_tmr</code>	8_010Fh

13.4.10.2.513.2 Diagram



13.4.10.2.513.3 Fields

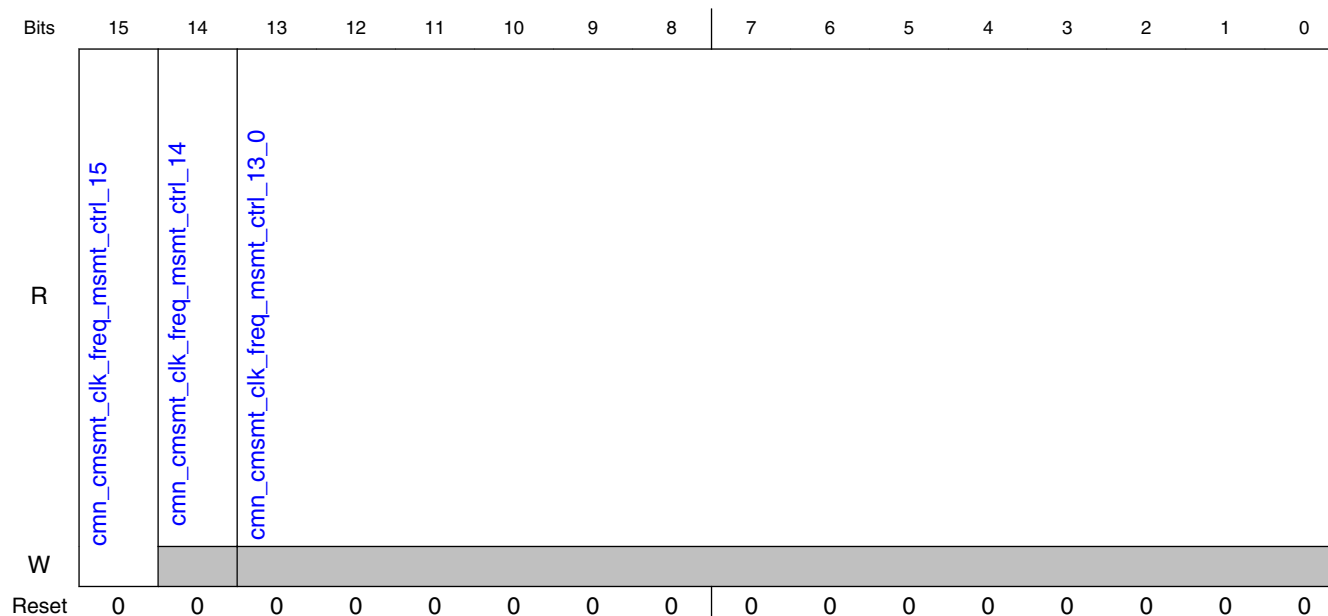
Field	Function
15-6 <code>cmn_txpd_adj_iter_tmr_15_6</code>	Reserved
5-0 <code>cmn_txpd_adj_iter_tmr_5_0</code>	Iteration wait timer value: This is the number of clock cycles to wait between when an adjusted calibration value is placed on the calibration signal going to the analog, and when the comparator value coming from the analog circuits can be checked.

13.4.10.2.514 Clock frequency measurement control register (`cmn_cmsmt_clk_freq_msmt_ctrl`)

13.4.10.2.514.1 Offset

Register	Offset
<code>cmn_cmsmt_clk_freq_msmt_ctrl</code>	8_01A0h

13.4.10.2.514.2 Diagram



13.4.10.2.514.3 Fields

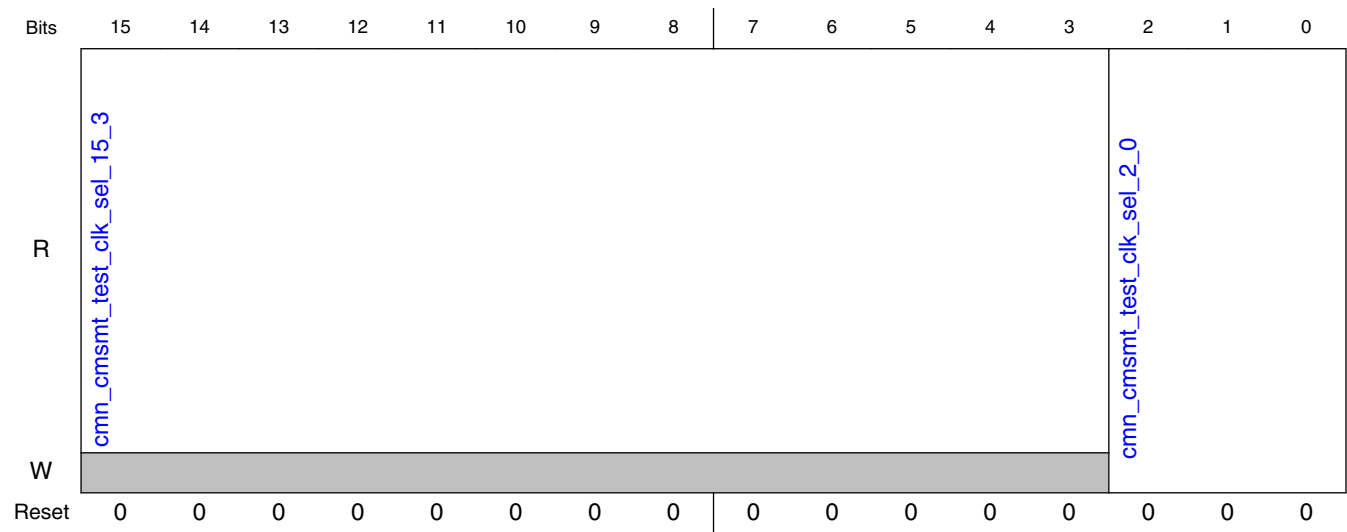
Field	Function
15 <code>cmn_cmsmt_clk_freq_msmt_ctrl_15</code>	Run test clock measurement: Activating (1b1) this bit will run the test clock measurement process. This bit must remain active until the test clock measurement process is complete, as indicated by the test clock measurement done bit in this register. To start another measurement process, this bit must first be driven inactive then driven active again.
14 <code>cmn_cmsmt_clk_freq_msmt_ctrl_14</code>	Test clock measurement done: This bit will be set to 1b1 when the test clock measurement process is complete. It will be cleared by the deactivation of the start test clock measurement bit in this register.
13-0 <code>cmn_cmsmt_clk_freq_msmt_ctrl_13_0</code>	Reserved

13.4.10.2.515 Test clock selection register (`cmn_cmsmt_test_clk_sel`)

13.4.10.2.515.1 Offset

Register	Offset
<code>cmn_cmsmt_test_clk_sel</code>	8_01A1h

13.4.10.2.515.2 Diagram



13.4.10.2.515.3 Fields

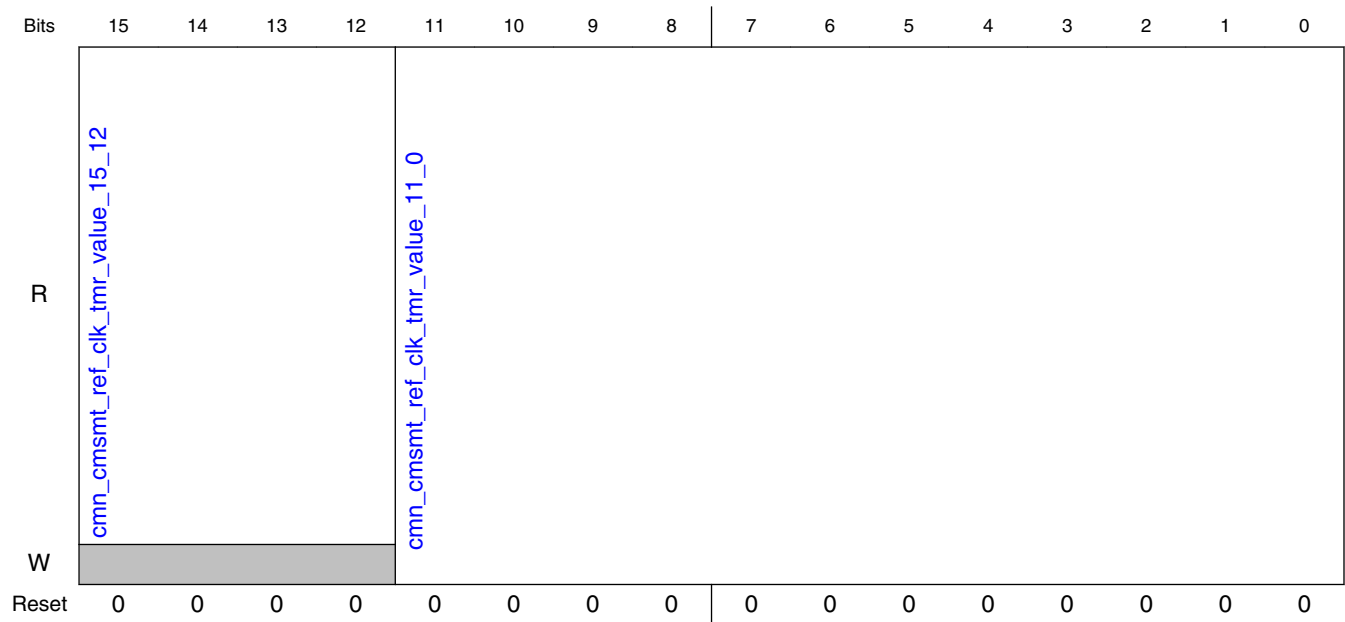
Field	Function
15-3 cmn_cmsmt_test_clk_sel_15_3	Reserved
2-0 cmn_cmsmt_test_clk_sel_2_0	Test clock select: This field drives the

13.4.10.2.516 Reference clock timer value register (cmn_cmsmt_ref_clk_tmr_value)

13.4.10.2.516.1 Offset

Register	Offset
cmn_cmsmt_ref_clk_tmr_value	8_01A2h

13.4.10.2.516.2 Diagram



13.4.10.2.516.3 Fields

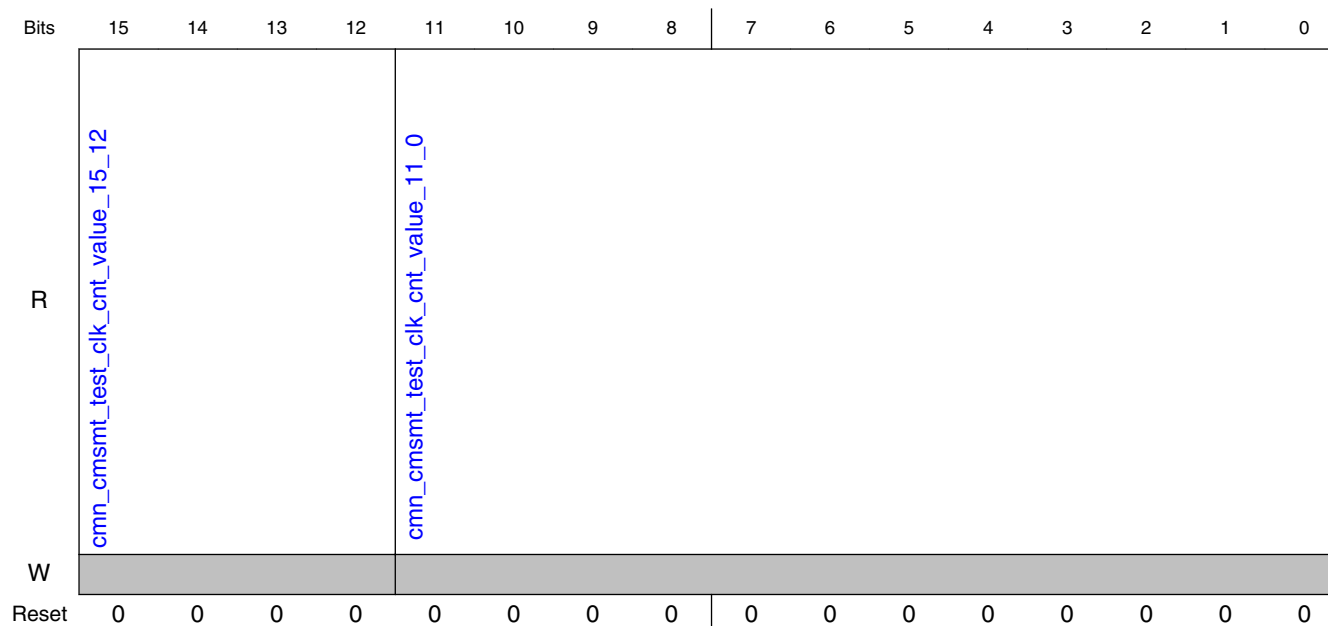
Field	Function
15-12 cmn_cmsmt_ref_clk_tmr_value_15_12	Reserved
11-0 cmn_cmsmt_ref_clk_tmr_value_11_0	Reference clock timer value : This specifies the amount of time, in reference clock cycles, to count test clock cycles. This value minus 1 is loaded into the reference clock timer. A value of 0 for this field is not valid when running this function.

13.4.10.2.517 Test clock counter value register (cmn_cmsmt_test_clk_cnt_value)

13.4.10.2.517.1 Offset

Register	Offset
cmn_cmsmt_test_clk_cnt_value	8_01A3h

13.4.10.2.517.2 Diagram



13.4.10.2.517.3 Fields

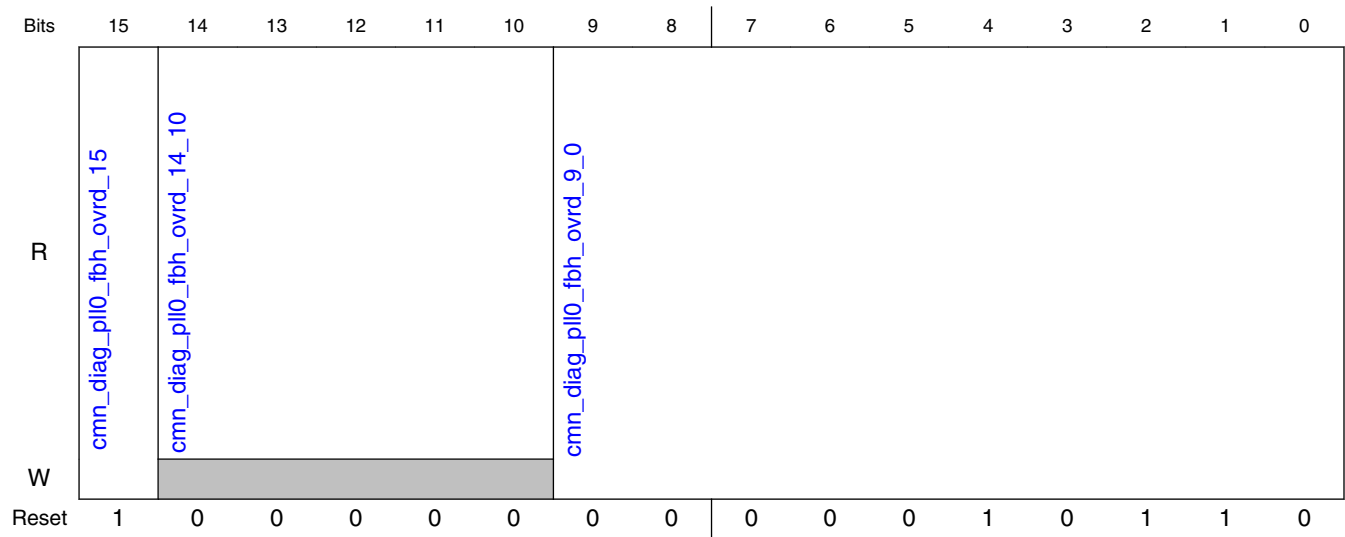
Field	Function
15-12 <code>cmn_cmsmt_test_clk_cnt_value_15_12</code>	Reserved
11-0 <code>cmn_cmsmt_test_clk_cnt_value_11_0</code>	Test clock counter value: When the test clock measurement process is complete, the value in this field specifies the number of test clock cycles that were counted in the time specified by the reference clock timer value. This field is only valid while the test clock measurement done bit in the

13.4.10.2.518 PLL 0 feedback divider value high override register (`cmn_diag_pll0_fbh_ovrd`)

13.4.10.2.518.1 Offset

Register	Offset
<code>cmn_diag_pll0_fbh_ovrd</code>	8_01C0h

13.4.10.2.518.2 Diagram



13.4.10.2.518.3 Fields

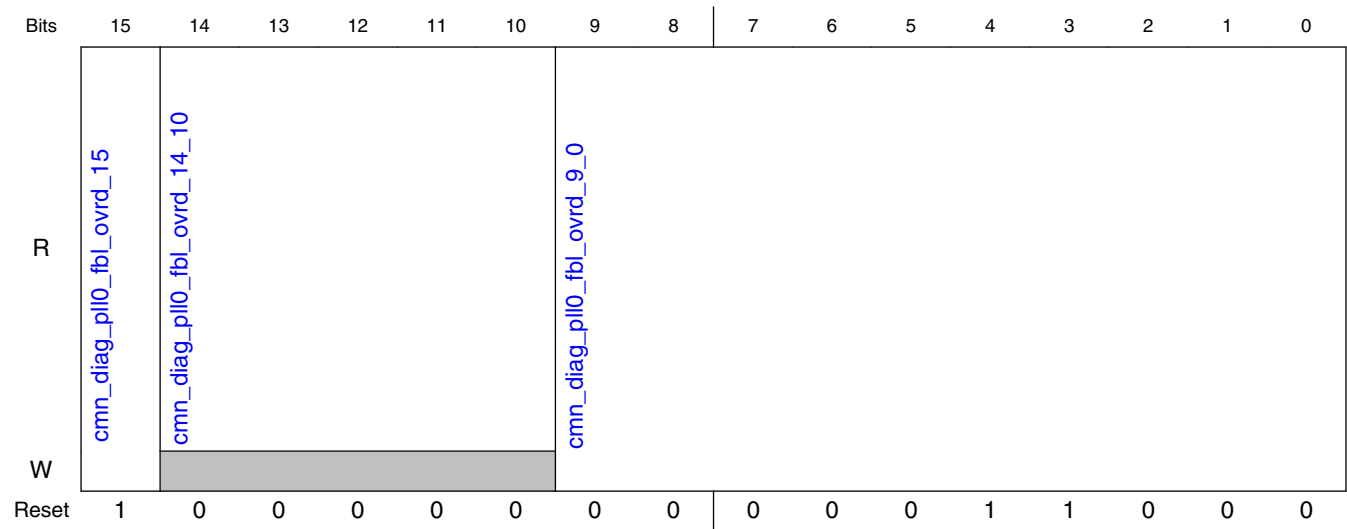
Field	Function
15 cmn_diag_pll0_fbh_ovrd_15	PLL feedback divider high override enable : When active (1b1), the PLL feedback divider high override value field of this register will be used to override the value on the
14-10 cmn_diag_pll0_fbh_ovrd_14_10	Reserved
9-0 cmn_diag_pll0_fbh_ovrd_9_0	PLL feedback divider high override value : When enabled, the value in this field will be used to override the value on the

13.4.10.2.519 PLL 0 feedback divider value low override register (cmn_diag_pll0_fbl_ovrd)

13.4.10.2.519.1 Offset

Register	Offset
cmn_diag_pll0_fbl_ovrd	8_01C1h

13.4.10.2.519.2 Diagram



13.4.10.2.519.3 Fields

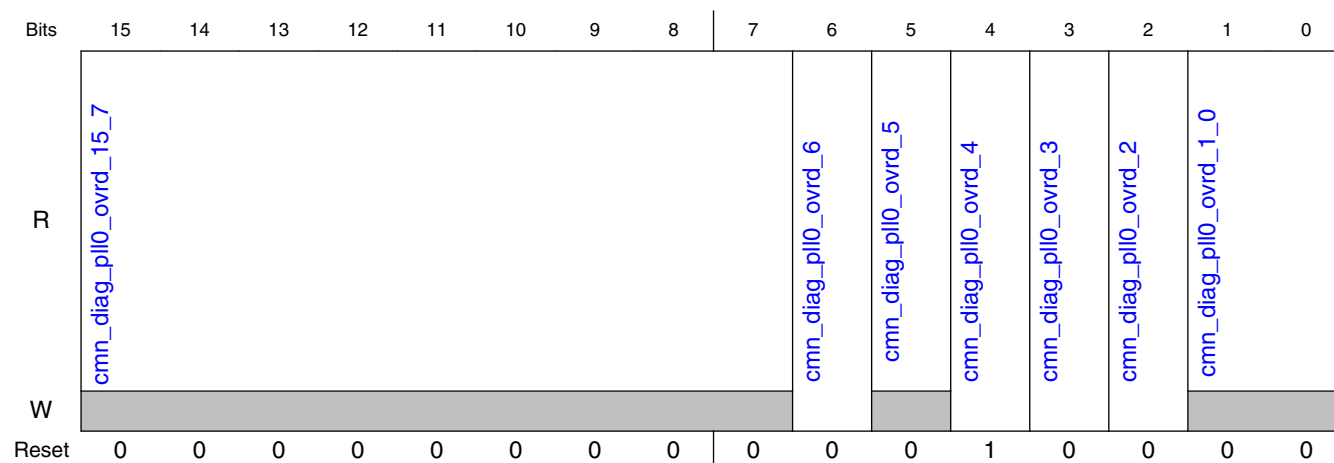
Field	Function
15 cmn_diag_pll0_fbl_ovrd_15	PLL feedback divider low override enable : When active (1b1), the PLL feedback divider low override value field of this register will be used to override the value on the
14-10 cmn_diag_pll0_fbl_ovrd_14_10	Reserved
9-0 cmn_diag_pll0_fbl_ovrd_9_0	PLL feedback divider low override value : When enabled, the value in this field will be used to override the value on the

13.4.10.2.520 PLL 0 override register (cmn_diag_pll0_ovrd)

13.4.10.2.520.1 Offset

Register	Offset
cmn_diag_pll0_ovrd	8_01C2h

13.4.10.2.520.2 Diagram



13.4.10.2.520.3 Fields

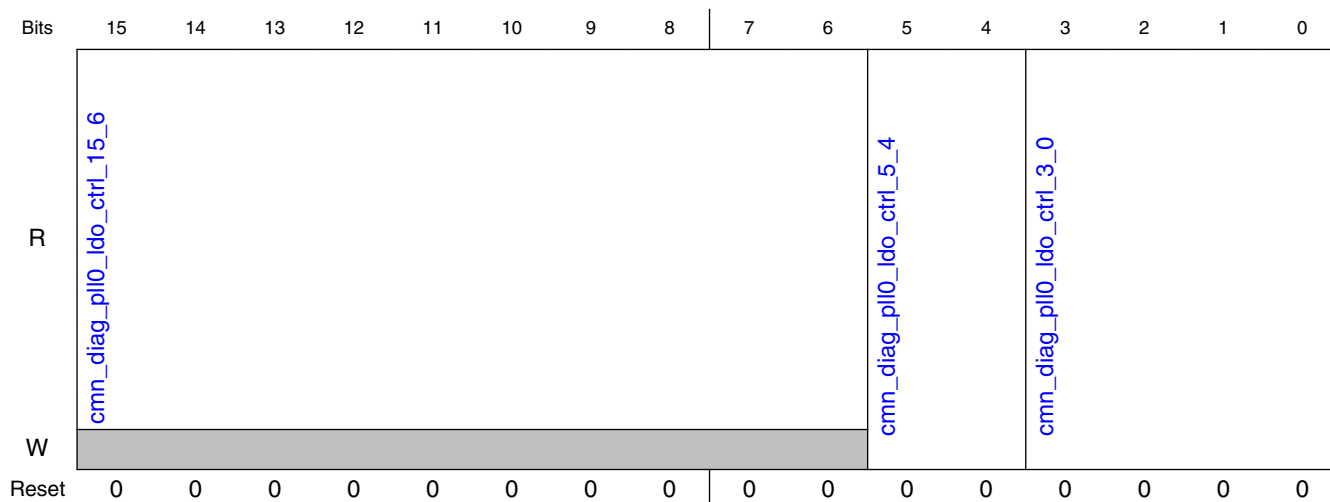
Field	Function
15-7 cmn_diag_pll0_ovrd_15_7	Reserved
6 cmn_diag_pll0_ovrd_6	PLL lock override: When active (1b1), this bit will force the PLL lock indication active.
5 cmn_diag_pll0_ovrd_5	Reserved
4 cmn_diag_pll0_ovrd_4	PLL PCIe mode: Selects corresponding PCIe CP and Loop filter block sections, by driving the
3 cmn_diag_pll0_ovrd_3	PLL VCO calibration enable override enable: When active (1b1), the PLL VCO calibration enable override bit in this register, can be used to directly control the enable of the VCO calibration function in the PLL (instead of the VCO calibration module).
2 cmn_diag_pll0_ovrd_2	PLL VCO calibration enable override: When enabled by the PLL VCO calibration enable override enable bit in this register, this bit will directly control the enable of the VCO calibration function in the PLL.
1-0 cmn_diag_pll0_ovrd_1_0	Reserved

13.4.10.2.521 PLL 0 LDO control register (cmn_diag_pll0_ldo_ctrl)

13.4.10.2.521.1 Offset

Register	Offset
cmn_diag_pll0_ldo_ctrl	8_01C3h

13.4.10.2.521.2 Diagram



13.4.10.2.521.3 Fields

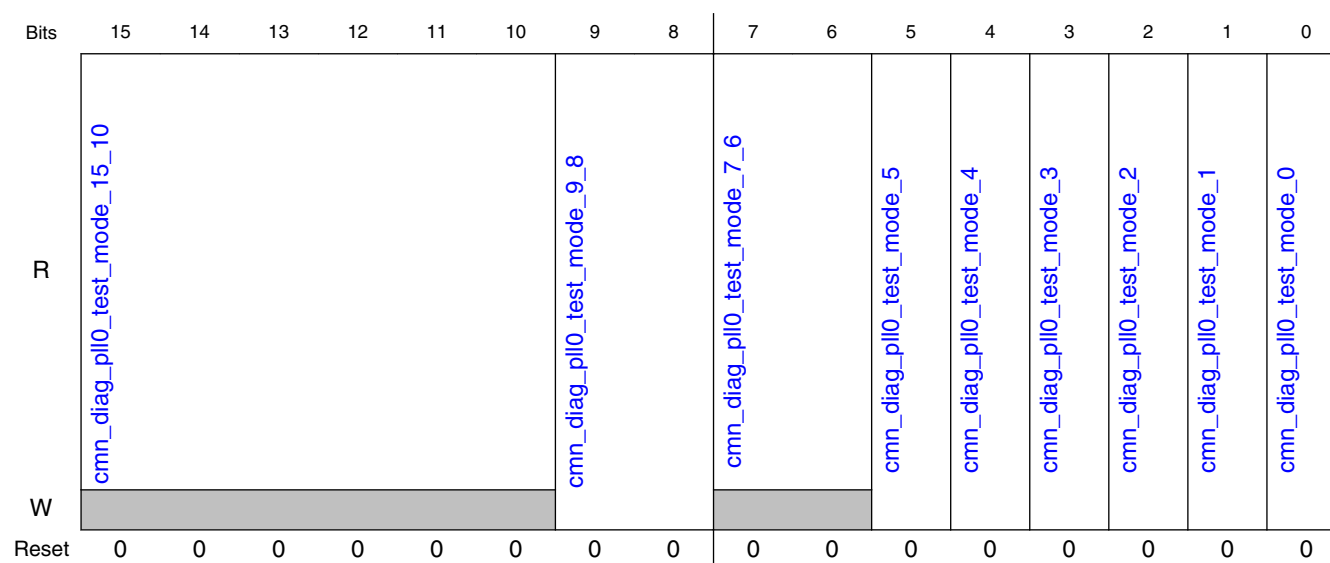
Field	Function
15-6 cmn_diag_pll0_ldo_ctrl_15_6	Reserved
5-4 cmn_diag_pll0_ldo_ctrl_5_4	PLL 0 LDO output bypass selection: When the optional PLL voltage regulator is present, this binary code controls the output bypass selection.
3-0 cmn_diag_pll0_ldo_ctrl_3_0	PLL 0 LDO nominal output voltage: When the optional PLL voltage regulator is present in the design, this code controls the regulator's nominal output voltage.

13.4.10.2.522 PLL 0 test mode register (cmn_diag_pll0_test_mode)

13.4.10.2.522.1 Offset

Register	Offset
cmn_diag_pll0_test_mode	8_01C4h

13.4.10.2.522.2 Diagram



13.4.10.2.522.3 Fields

Field	Function
15-10 cmn_diag_pll0_test_mode_15_10	Reserved
9-8 cmn_diag_pll0_test_mode_9_8	PLL 0 secondary loop bandwidth control: Controls the bandwidth of the second order loop, by driving the
7-6 cmn_diag_pll0_test_mode_7_6	Reserved
5 cmn_diag_pll0_test_mode_5	PLL 0 VCO clock divider test mode select: During VCO calibration, the VCO clock is divided by 2 before being sent to the feedback divider, by driving the
4 cmn_diag_pll0_test_mode_4	PLL 0 secondary loop bypass: Controls the bypass secondary loop output filter, by driving the

Table continues on the next page...

Clocks And Resets

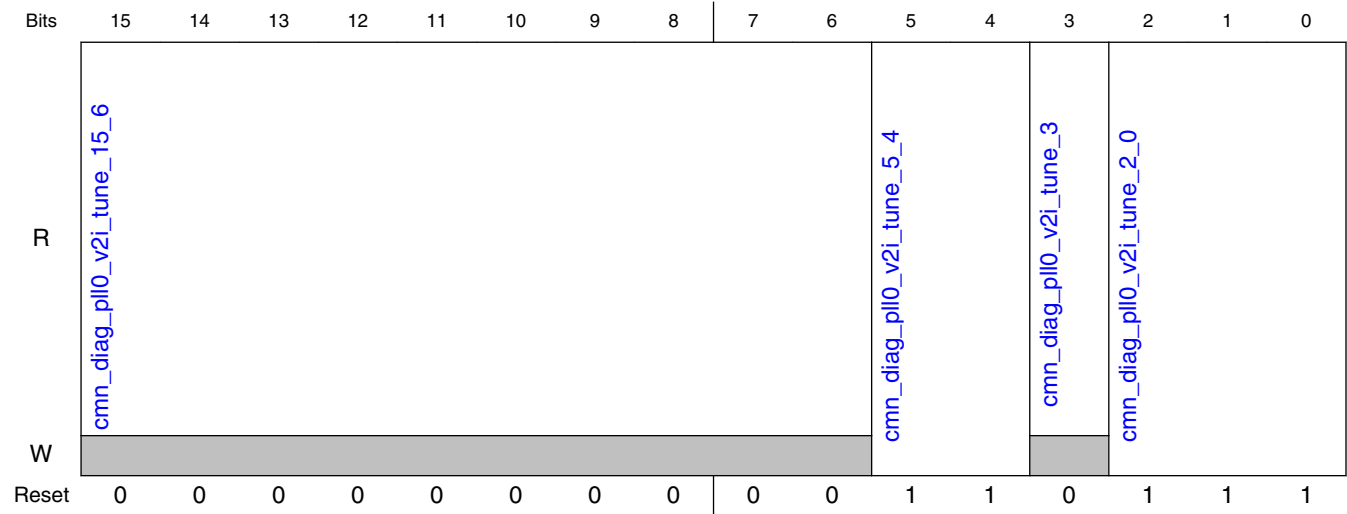
Field	Function
3 cmn_diag_pll0_t est_mode_3	PLL 0 high speed ripple divider select: Test mode control of the VCO/2 clock to HS ripple divider, by driving the
2 cmn_diag_pll0_t est_mode_2	PLL 0 secondary loop amp bypass enable: Test mode control of the secondary loop amp, by driving the
1 cmn_diag_pll0_t est_mode_1	PLL 0 secondary loop gain control: Controls the secondary loop gain, by driving the
0 cmn_diag_pll0_t est_mode_0	PLL 0 test mode charge pump current select: Test mode control of the charge pump current for the ATB cell, by driving the

13.4.10.2.523 PLL 0 voltage to current unit tuning register (cmn_diag_pll0_v2i_tune)

13.4.10.2.523.1 Offset

Register	Offset
cmn_diag_pll0_v2i_tune	8_01C5h

13.4.10.2.523.2 Diagram



13.4.10.2.523.3 Fields

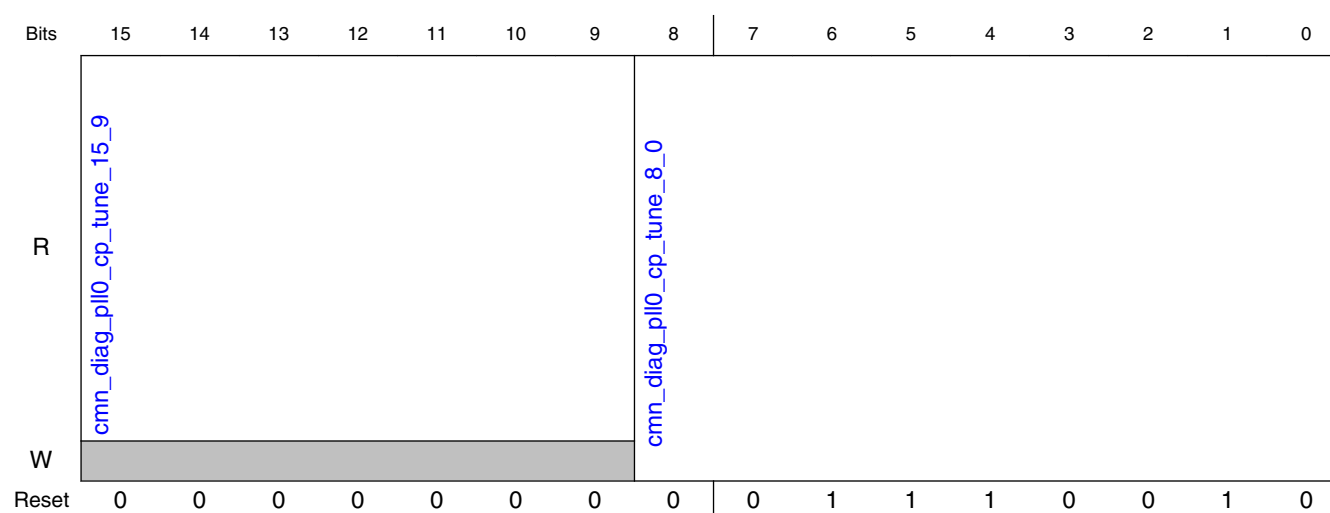
Field	Function
15-6 cmn_diag_pll0_v2i_tune_15_6	Reserved
5-4 cmn_diag_pll0_v2i_tune_5_4	PLL 0 voltage to current program code: Voltage to current program code used to multiply the V2I unit current, by driving the
3 cmn_diag_pll0_v2i_tune_3	Reserved
2-0 cmn_diag_pll0_v2i_tune_2_0	PLL 0 voltage to current coarse program code: Voltage to current coarse program code used to program CDAC/V2I unit current, by driving the

13.4.10.2.524 PLL 0 charge pump tuning register (cmn_diag_pll0_cp_tune)

13.4.10.2.524.1 Offset

Register	Offset
cmn_diag_pll0_cp_tune	8_01C6h

13.4.10.2.524.2 Diagram



13.4.10.2.524.3 Fields

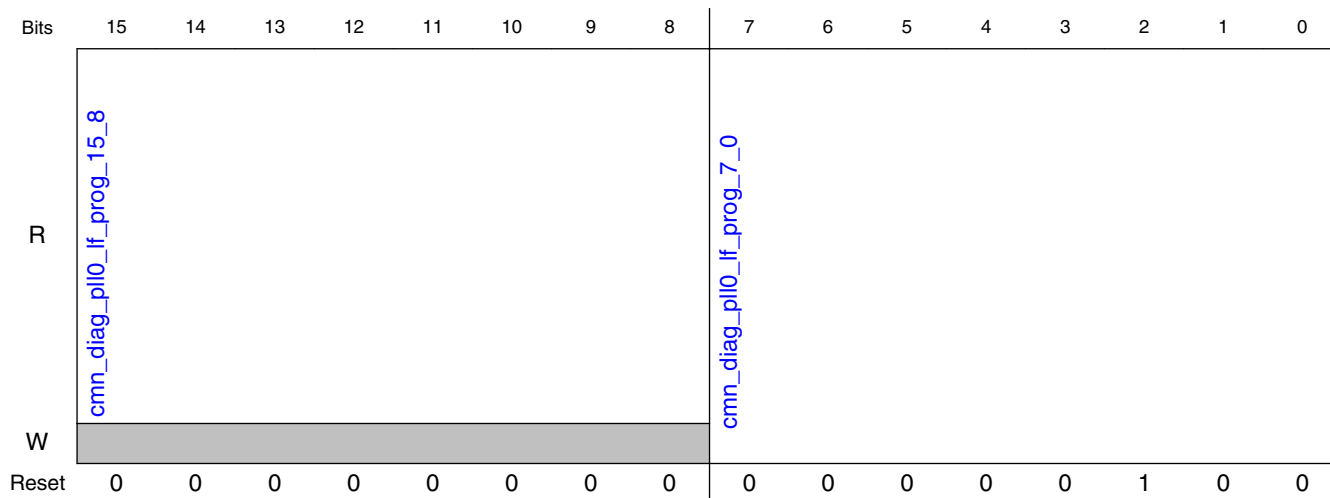
Field	Function
15-9 cmn_diag_pll0_ cp_tune_15_9	Reserved
8-0 cmn_diag_pll0_ cp_tune_8_0	PLL 0 charge pump gain: Controls the charge pump gain in PLL loop, by driving the

13.4.10.2.525 PLL 0 loop filter programmability register (cmn_diag_pll0_lf_prog)

13.4.10.2.525.1 Offset

Register	Offset
cmn_diag_pll0_lf_prog	8_01C7h

13.4.10.2.525.2 Diagram



13.4.10.2.525.3 Fields

Field	Function
15-8	Reserved

Table continues on the next page...

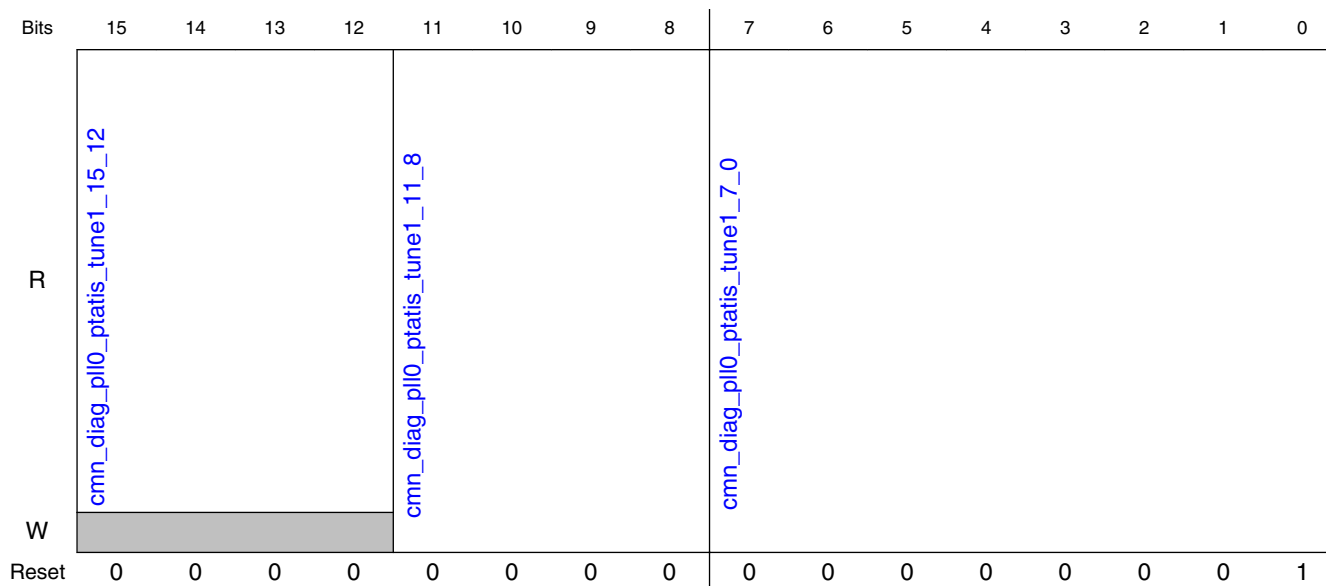
Field	Function
cmn_diag_pll0_lf_prog_15_8	
7-0 cmn_diag_pll0_lf_prog_7_0	PLL 0 loop filter programmability: Loop filter binary programmability is used for following purposes.

13.4.10.2.526 PLL 0 PTAT current slope tuning register 1 (cmn_diag_pll0_ptatis_tune1)

13.4.10.2.526.1 Offset

Register	Offset
cmn_diag_pll0_ptatis_tune1	8_01C8h

13.4.10.2.526.2 Diagram



13.4.10.2.526.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

Clocks And Resets

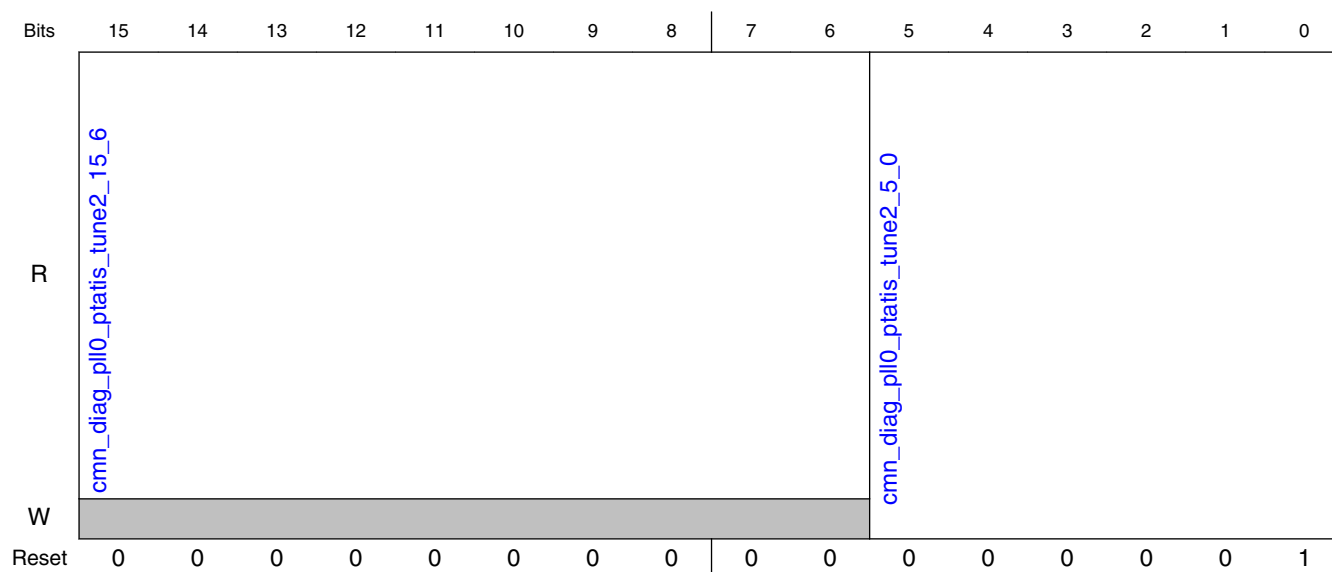
Field	Function
cmn_diag_pll0_ptatis_tune1_15_12	
11-8 cmn_diag_pll0_ptatis_tune1_11_8	PLL 0 NDAC control: PTAT current slope is varied by adding/subtracting bandgap VBGbyR current to/from bandgap PTAT current. Amount of subtracted bandgap VBGbyR current is controlled by this field, by driving the
7-0 cmn_diag_pll0_ptatis_tune1_7_0	PLL 0 PMOS control: PTAT current slope is varied by adding/subtracting bandgap VBGbyR current to/from bandgap PTAT current. Amount of subtracted bandgap VBGbyR current is controlled by this field, by driving the

13.4.10.2.527 PLL 0 PTAT current slope tuning register 2 (cmn_diag_pll0_ptatis_tune2)

13.4.10.2.527.1 Offset

Register	Offset
cmn_diag_pll0_ptatis_tune2	8_01C9h

13.4.10.2.527.2 Diagram



13.4.10.2.527.3 Fields

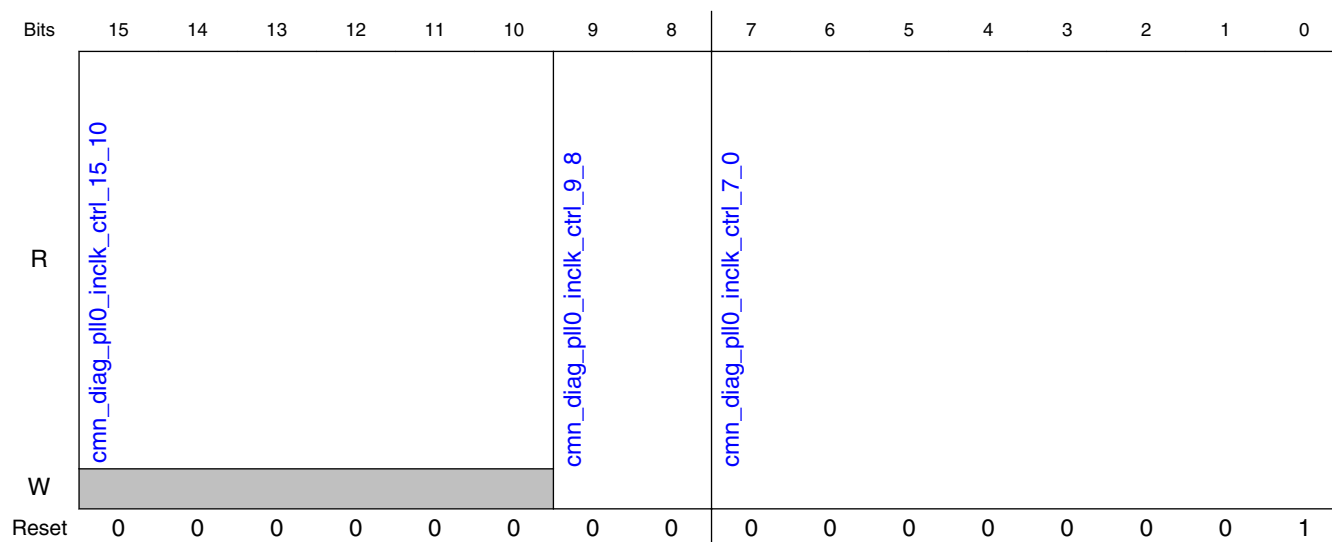
Field	Function
15-6 cmn_diag_pll0_ptatis_tune2_15_6	Reserved
5-0 cmn_diag_pll0_ptatis_tune2_5_0	PLL 0 PTAT NDAC control: PTAT current slope is varied by adding/subtracting bandgap VBGbyR current to/from bandgap PTAT current. After addition/subtraction the final current sent PTAT NDAC DAC to make sure output typical current is 10uA. This field controls this function, by driving the

13.4.10.2.528 PLL 0 input clock control register (cmn_diag_pll0_inclk_ctrl)

13.4.10.2.528.1 Offset

Register	Offset
cmn_diag_pll0_inclk_ctrl	8_01CAh

13.4.10.2.528.2 Diagram



13.4.10.2.528.3 Fields

Field	Function
15-10	Reserved

Table continues on the next page...

Clocks And Resets

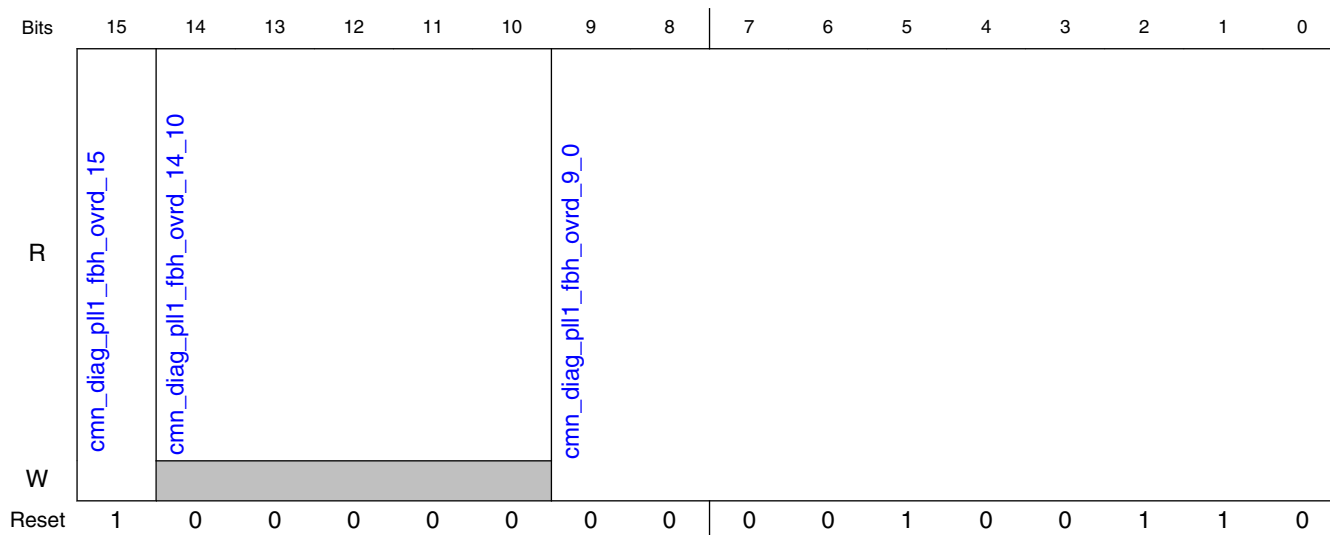
Field	Function
cmn_diag_pll0_i nclk_ctrl_15_10	
9-8 cmn_diag_pll0_i nclk_ctrl_9_8	PLL 0 low speed output clock divider control: Controls the division value for the input clock for the divide by 5 function used to generate the
7-0 cmn_diag_pll0_i nclk_ctrl_7_0	PLL 0 input clock divider control: Controls the division value for the input reference clock divider which provides a divided reference clock to the PLL, by driving the

13.4.10.2.529 PLL 1 feedback divider value high override register (cmn_diag_pll1_fbh_ovrd)

13.4.10.2.529.1 Offset

Register	Offset
cmn_diag_pll1_fbh_ovrd	8_01D0h

13.4.10.2.529.2 Diagram



13.4.10.2.529.3 Fields

Field	Function
15	PLL feedback divider high override enable : When active (1b1), the PLL feedback divider high override value field of this register will be used to override the value on the

Table continues on the next page...

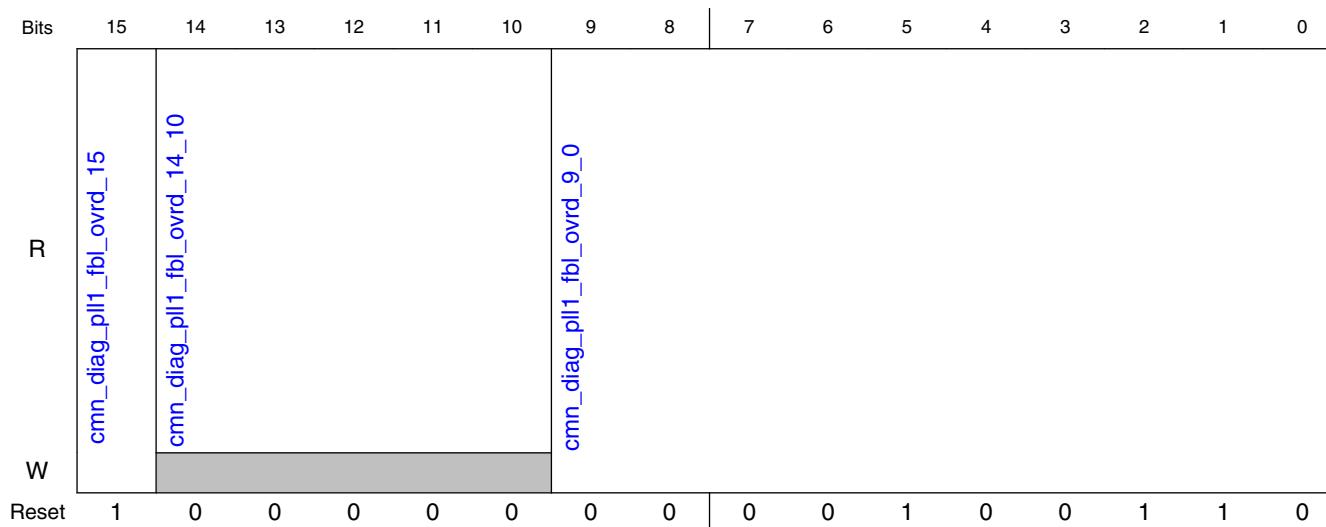
Field	Function
cmn_diag_pll1_fbh_ovrd_15	
14-10 cmn_diag_pll1_fbh_ovrd_14_10	Reserved
9-0 cmn_diag_pll1_fbh_ovrd_9_0	PLL feedback divider high override value : When enabled, the value in this field will be used to override the value on the

13.4.10.2.530 PLL 1 feedback divider value low override register (cmn_diag_pll1_fbl_ovrd)

13.4.10.2.530.1 Offset

Register	Offset
cmn_diag_pll1_fbl_ovrd	8_01D1h

13.4.10.2.530.2 Diagram



13.4.10.2.530.3 Fields

Field	Function
15	PLL feedback divider low override enable : When active (1b1), the PLL feedback divider low override value field of this register will be used to override the value on the

Table continues on the next page...

Clocks And Resets

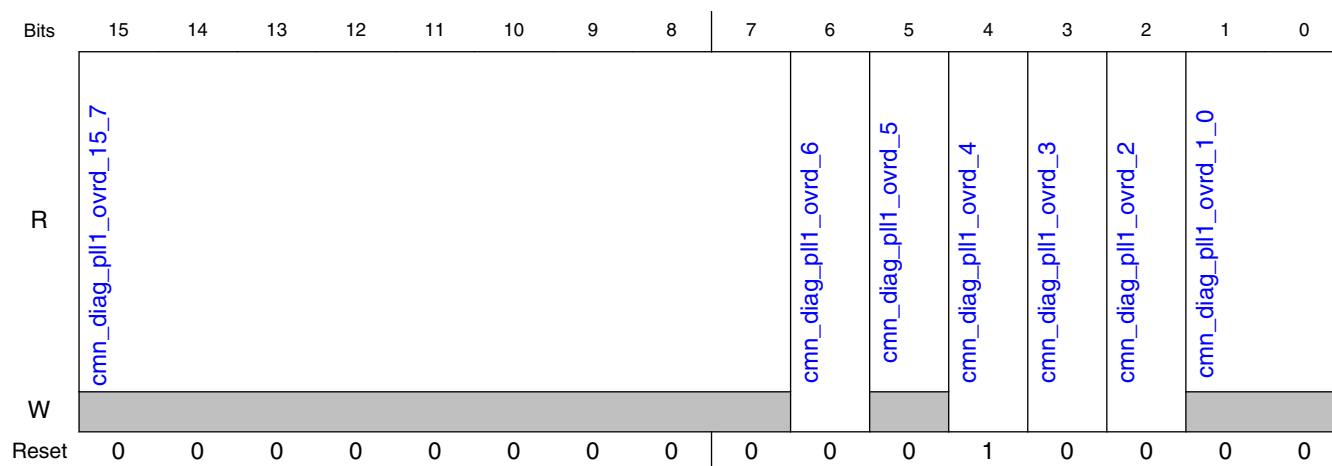
Field	Function
cmn_diag_pll1_f bl_ovrd_15	
14-10 cmn_diag_pll1_f bl_ovrd_14_10	Reserved
9-0 cmn_diag_pll1_f bl_ovrd_9_0	PLL feedback divider low override value : When enabled, the value in this field will be used to override the value on the

13.4.10.2.531 PLL 1 override register (cmn_diag_pll1_ovrd)

13.4.10.2.531.1 Offset

Register	Offset
cmn_diag_pll1_ovrd	8_01D2h

13.4.10.2.531.2 Diagram



13.4.10.2.531.3 Fields

Field	Function
15-7 cmn_diag_pll1_ ovrd_15_7	Reserved
6	PLL lock override: When active (1b1), this bit will force the PLL lock indication active.

Table continues on the next page...

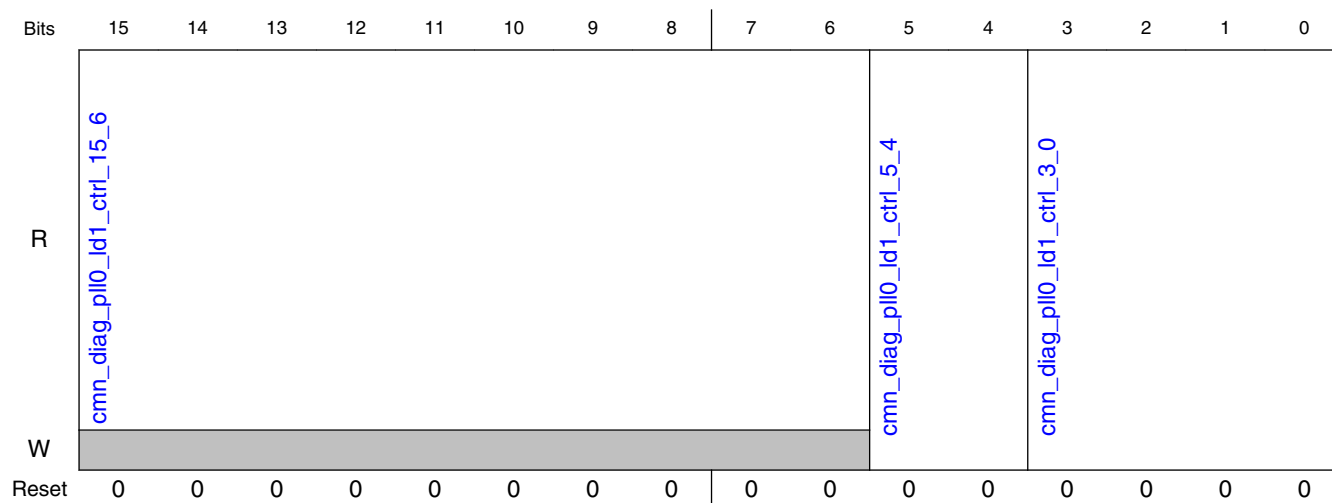
Field	Function
cmn_diag_pll1_ovrd_6	
5 cmn_diag_pll1_ovrd_5	Reserved
4 cmn_diag_pll1_ovrd_4	PLL PCIe mode: Selects corresponding PCIe CP and Loop filter block sections, by driving the
3 cmn_diag_pll1_ovrd_3	PLL VCO calibration enable override enable: When active (1b1), the PLL VCO calibration enable override bit in this register, can be used to directly control the enable of the VCO calibration function in the PLL (instead of the VCO calibration module).
2 cmn_diag_pll1_ovrd_2	PLL VCO calibration enable override: When enabled by the PLL VCO calibration enable override enable bit in this register, this bit will directly control the enable of the VCO calibration function in the PLL.
1-0 cmn_diag_pll1_ovrd_1_0	Reserved

13.4.10.2.532 PLL 1 LDO control register (cmn_diag_pll0_ld1_ctrl)

13.4.10.2.532.1 Offset

Register	Offset
cmn_diag_pll0_ld1_ctrl	8_01D3h

13.4.10.2.532.2 Diagram



13.4.10.2.532.3 Fields

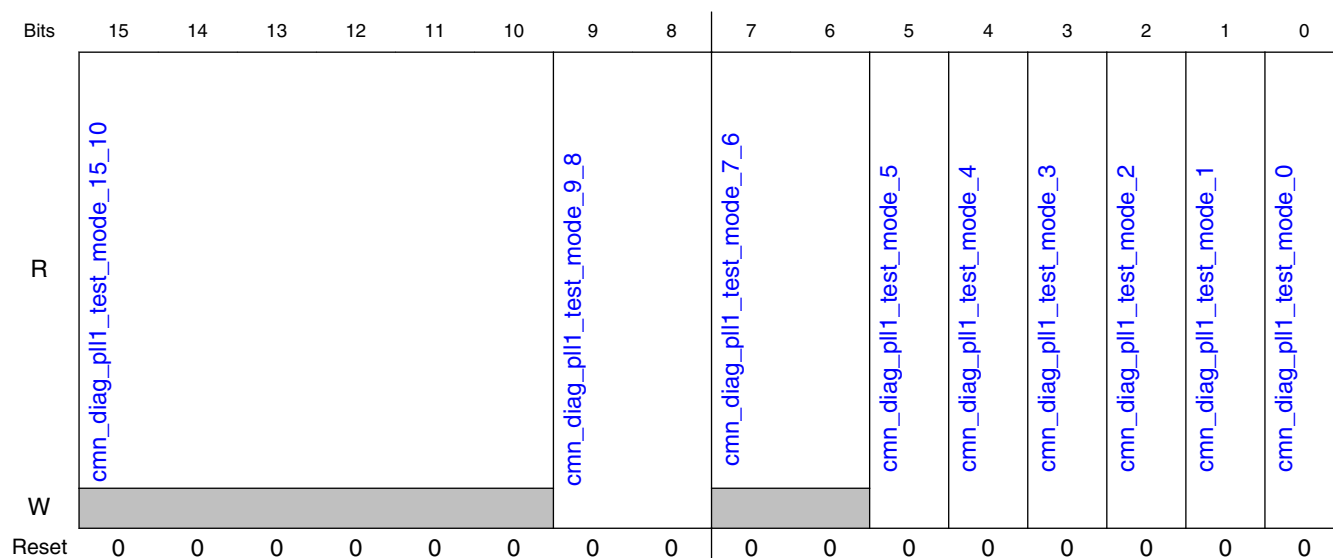
Field	Function
15-6 cmn_diag_pll0_id1_ctrl_15_6	Reserved
5-4 cmn_diag_pll0_id1_ctrl_5_4	PLL 1 LDO output bypass selection: When the optional PLL voltage regulator is present, this binary code controls the output bypass selection.
3-0 cmn_diag_pll0_id1_ctrl_3_0	PLL 1 LDO nominal output voltage: When the optional PLL voltage regulator is present in the design, this code controls the regulator's nominal output voltage.

13.4.10.2.533 PLL 1 test mode register (cmn_diag_pll1_test_mode)

13.4.10.2.533.1 Offset

Register	Offset
cmn_diag_pll1_test_mode	8_01D4h

13.4.10.2.533.2 Diagram



13.4.10.2.533.3 Fields

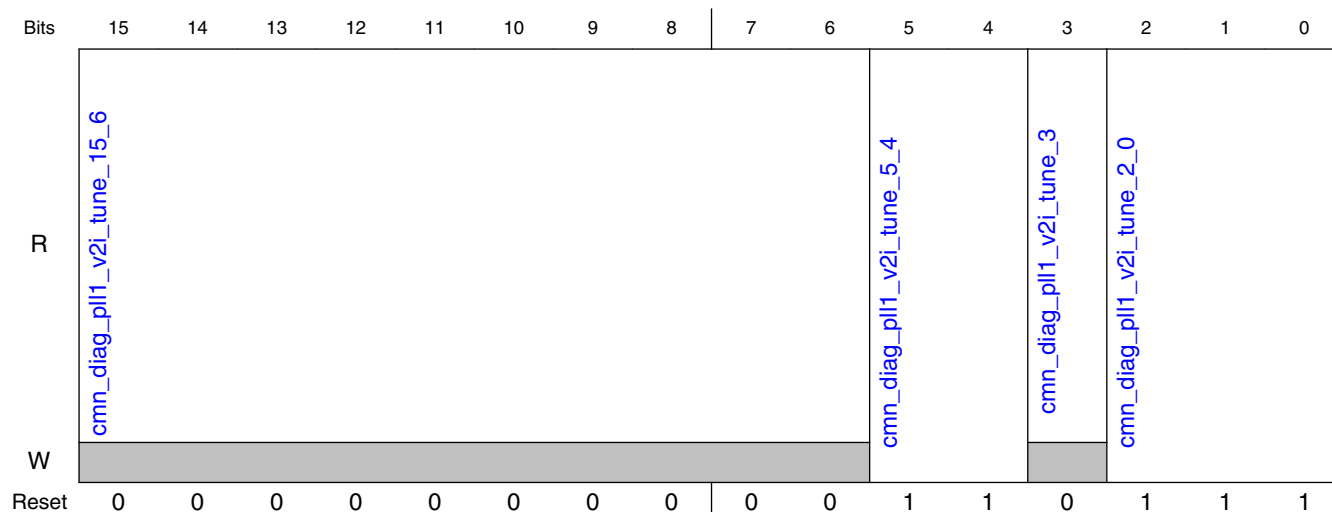
Field	Function
15-10 cmn_diag_pll1_test_mode_15_10	Reserved
9-8 cmn_diag_pll1_test_mode_9_8	PLL 1 secondary loop bandwidth control: Controls the bandwidth of the second order loop, by driving the
7-6 cmn_diag_pll1_test_mode_7_6	Reserved
5 cmn_diag_pll1_test_mode_5	PLL 1 VCO clock divider test mode select: During VCO calibration, the VCO clock is divided by 2 before being sent to the feedback divider, by driving the
4 cmn_diag_pll1_test_mode_4	PLL 1 secondary loop bypass: Controls the bypass secondary loop output filter, by driving the
3 cmn_diag_pll1_test_mode_3	PLL 1 high speed ripple divider select: Test mode control of the VCO/2 clock to HS ripple divider, by driving the
2 cmn_diag_pll1_test_mode_2	PLL 1 secondary loop amp bypass enable: Test mode control of the secondary loop amp, by driving the
1 cmn_diag_pll1_test_mode_1	PLL 1 secondary loop gain control: Controls the secondary loop gain, by driving the
0 cmn_diag_pll1_test_mode_0	PLL 1 test mode charge pump current select: Test mode control of the charge pump current for the ATB cell, by driving the

13.4.10.2.534 PLL 1 voltage to current unit tuning register (cmn_diag_pll1_v2i_tune)

13.4.10.2.534.1 Offset

Register	Offset
cmn_diag_pll1_v2i_tune	8_01D5h

13.4.10.2.534.2 Diagram



13.4.10.2.534.3 Fields

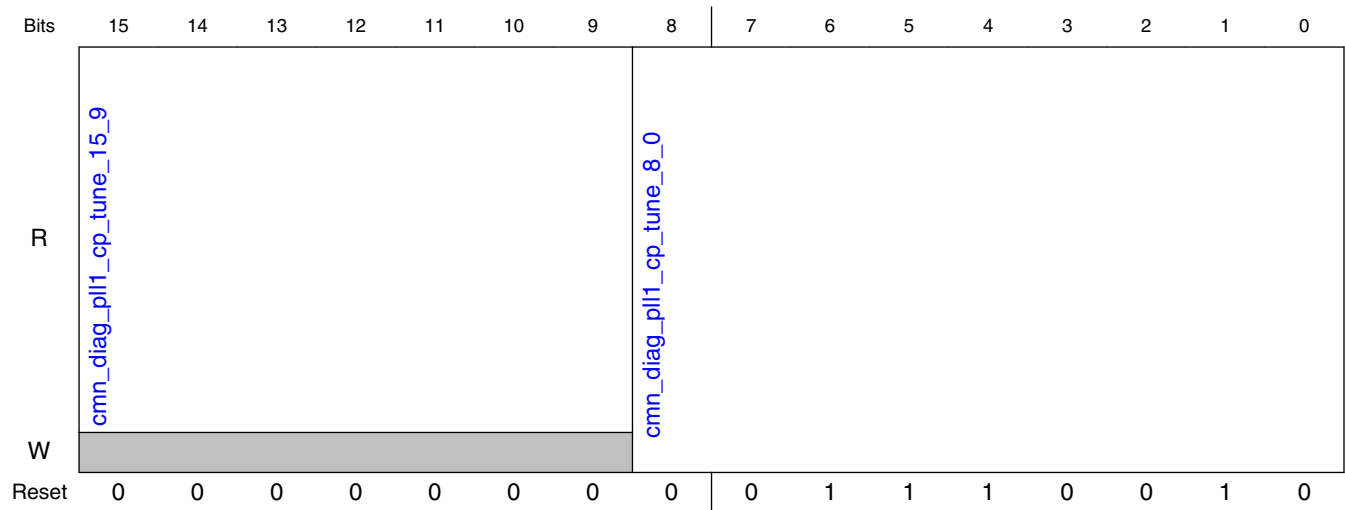
Field	Function
15-6 cmn_diag_pll1_v2i_tune_15_6	Reserved
5-4 cmn_diag_pll1_v2i_tune_5_4	PLL 0 voltage to current program code: Voltage to current program code used to multiply the V2I unit current, by driving the
3 cmn_diag_pll1_v2i_tune_3	Reserved
2-0 cmn_diag_pll1_v2i_tune_2_0	PLL 0 voltage to current coarse program code: Voltage to current coarse program code used to program CDAC/V2I unit current, by driving the

13.4.10.2.535 PLL 1 charge pump tuning register (cmn_diag_pll1_cp_tune)

13.4.10.2.535.1 Offset

Register	Offset
cmn_diag_pll1_cp_tune	8_01D6h

13.4.10.2.535.2 Diagram



13.4.10.2.535.3 Fields

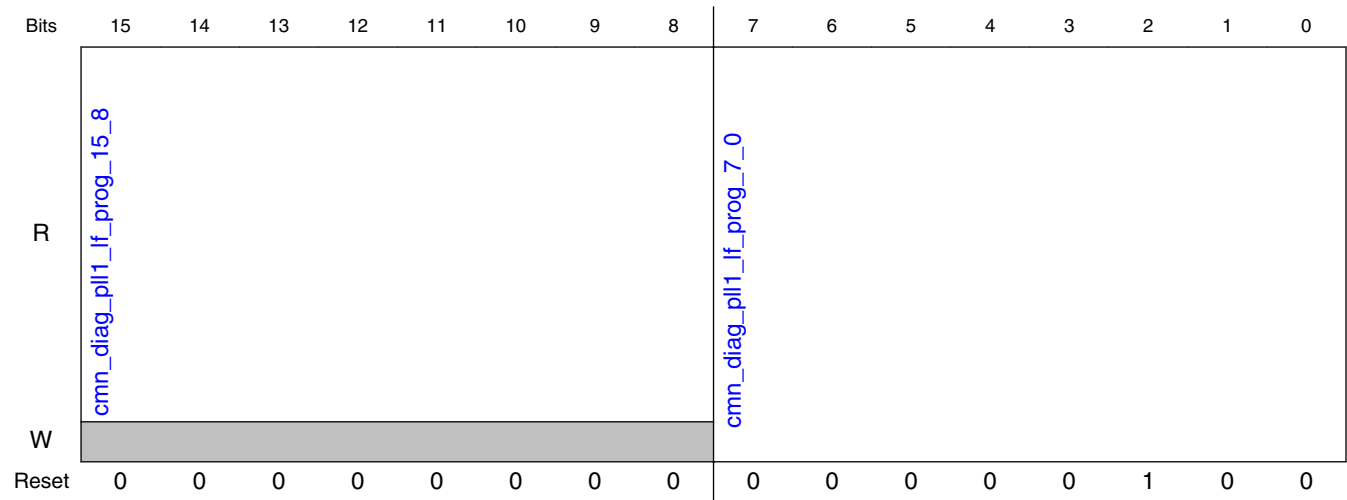
Field	Function
15-9 cmn_diag_pll1_cp_tune_15_9	Reserved
8-0 cmn_diag_pll1_cp_tune_8_0	PLL 1 charge pump gain: Controls the charge pump gain in PLL loop, by driving the

13.4.10.2.536 PLL 1 loop filter programmability register (cmn_diag_pll1_if_prog)

13.4.10.2.536.1 Offset

Register	Offset
cmn_diag_pll1_if_prog	8_01D7h

13.4.10.2.536.2 Diagram



13.4.10.2.536.3 Fields

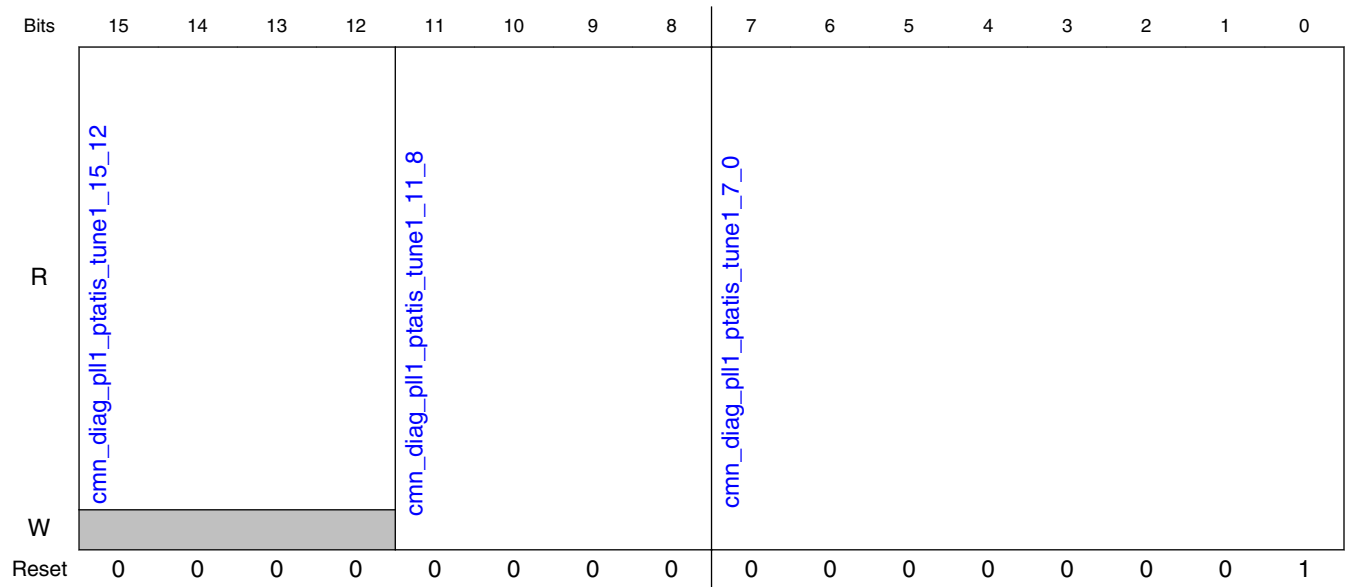
Field	Function
15-8 cmn_diag_pll1_if_prog_15_8	Reserved
7-0 cmn_diag_pll1_if_prog_7_0	PLL 1 loop filter programmability: Loop filter binary programmability is used for following purposes.

13.4.10.2.537 PLL 1 PTAT current slope tuning register 1 (cmn_diag_pll1_ptatis_tune1)

13.4.10.2.537.1 Offset

Register	Offset
cmn_diag_pll1_ptatis_tune1	8_01D8h

13.4.10.2.537.2 Diagram



13.4.10.2.537.3 Fields

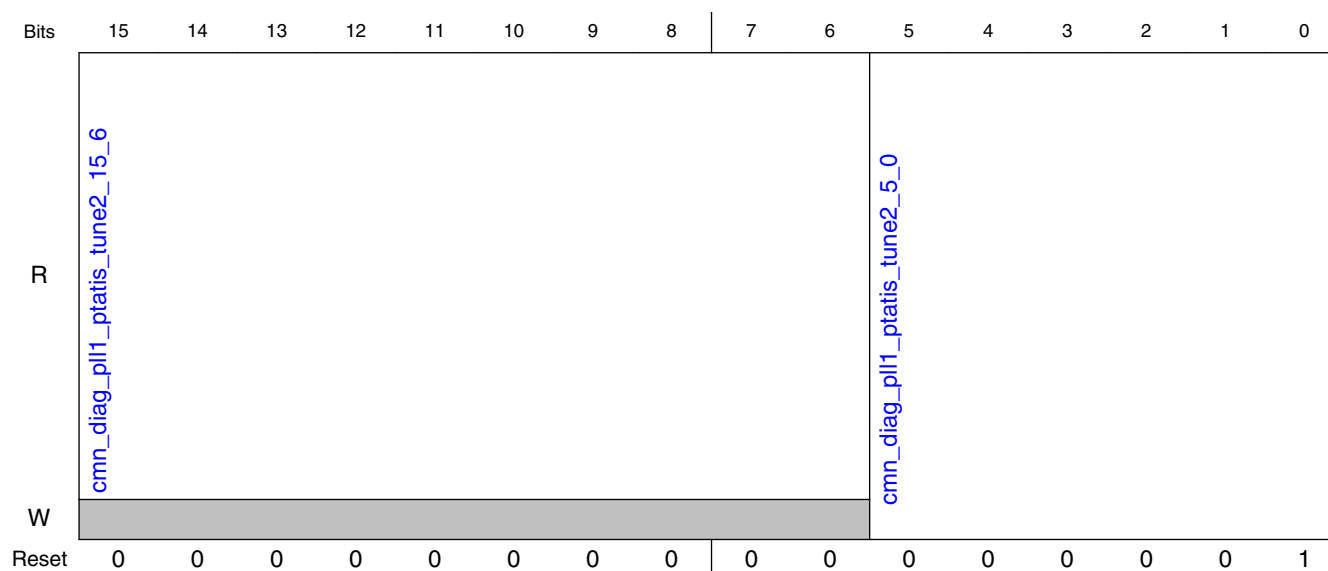
Field	Function
15-12 cmn_diag_pll1_ptatis_tune1_15_12	Reserved
11-8 cmn_diag_pll1_ptatis_tune1_11_8	PLL 1 NDAC control: PTAT current slope is varied by adding/subtracting bandgap VBGbyR current to/from bandgap PTAT current. Amount of subtracted bandgap VBGbyR current is controlled by this field, by driving the
7-0 cmn_diag_pll1_ptatis_tune1_7_0	PLL 1 PMOS control: PTAT current slope is varied by adding/subtracting bandgap VBGbyR current to/from bandgap PTAT current. Amount of subtracted bandgap VBGbyR current is controlled by this field, by driving the

13.4.10.2.538 PLL 1 PTAT current slope tuning register 2 (cmn_diag_pll1_ptatis_tune2)

13.4.10.2.538.1 Offset

Register	Offset
cmn_diag_pll1_ptatis_tune2	8_01D9h

13.4.10.2.538.2 Diagram



13.4.10.2.538.3 Fields

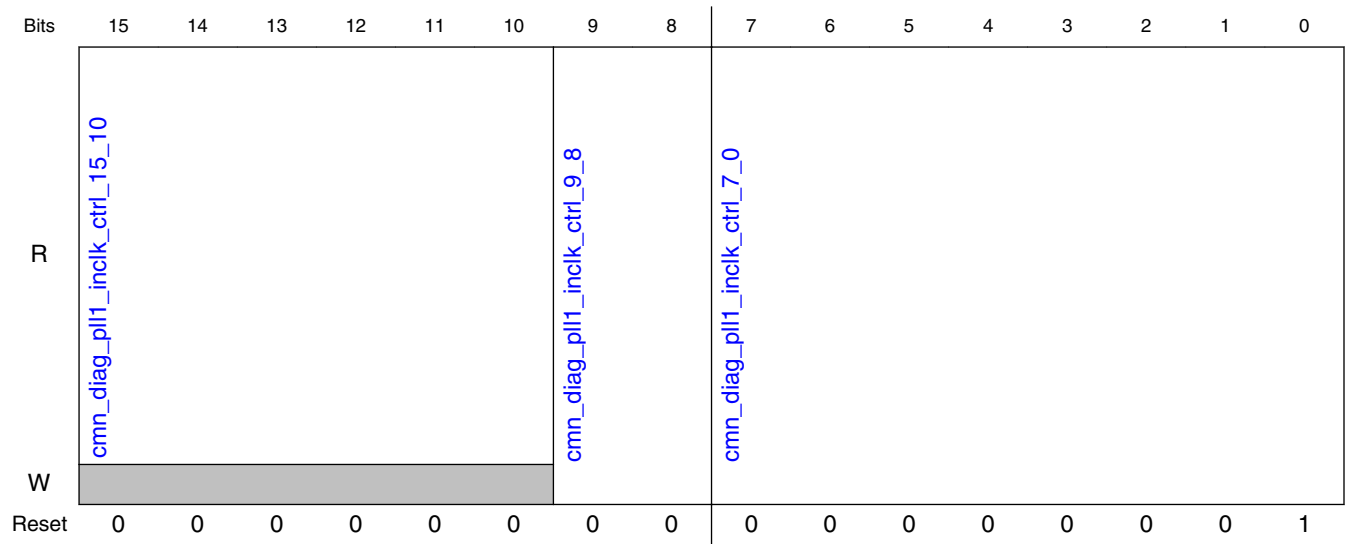
Field	Function
15-6 cmn_diag_pll1_ptatis_tune2_15_6	Reserved
5-0 cmn_diag_pll1_ptatis_tune2_5_0	PLL 1 PTAT NDAC control: PTAT current slope is varied by adding/subtracting bandgap VBGbyR current to/from bandgap PTAT current. After addition/subtraction the final current sent PTAT NDAC DAC to make sure output typical current is 10uA. This field controls this function, by driving the

13.4.10.2.539 PLL 1 input clock control register (cmn_diag_pll1_inclk_ctrl)

13.4.10.2.539.1 Offset

Register	Offset
cmn_diag_pll1_inclk_ctrl	8_01DAh

13.4.10.2.539.2 Diagram



13.4.10.2.539.3 Fields

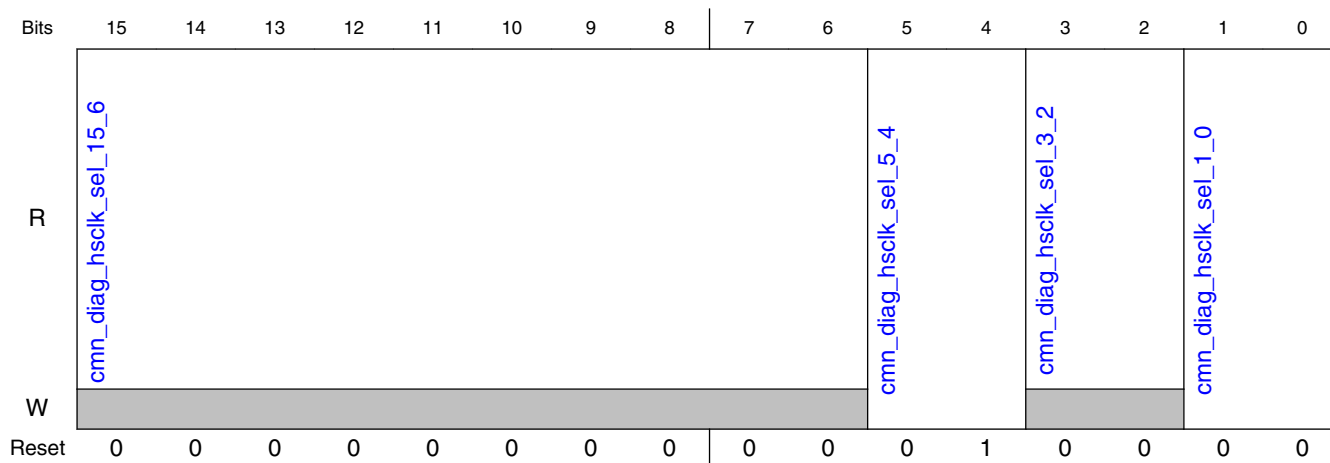
Field	Function
15-10 <code>cmn_diag_pll1_inclk_ctrl_15_10</code>	Reserved
9-8 <code>cmn_diag_pll1_inclk_ctrl_9_8</code>	PLL 1 low speed output clock divider control: Controls the division value for the input clock for the divide by 5 function used to generate the
7-0 <code>cmn_diag_pll1_inclk_ctrl_7_0</code>	PLL 1 input clock divider control: Controls the division value for the input reference clock divider which provides a divided reference clock to the PLL, by driving the

13.4.10.2.540 Common high speed clock select register (`cmn_diag_hsclk_sel`)

13.4.10.2.540.1 Offset

Register	Offset
<code>cmn_diag_hsclk_sel</code>	8_01E0h

13.4.10.2.540.2 Diagram



13.4.10.2.540.3 Fields

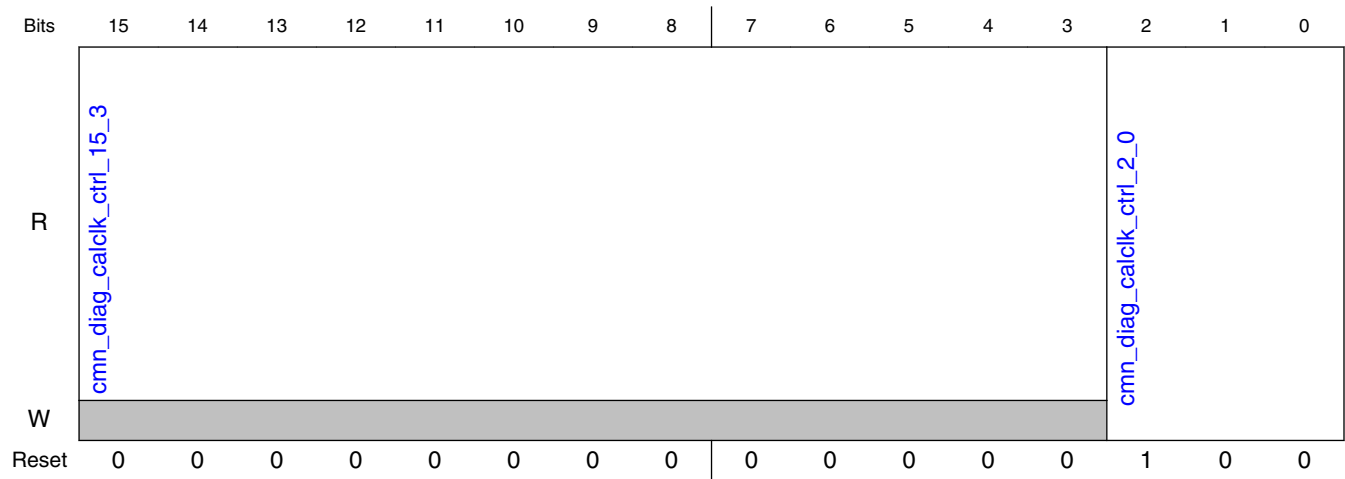
Field	Function
15-6 <code>cmn_diag_hscclk_sel_15_6</code>	Reserved
5-4 <code>cmn_diag_hscclk_sel_5_4</code>	High speed clock 1 select: Selects which PLL clock will be driven on the analog high speed clock 1.
3-2 <code>cmn_diag_hscclk_sel_3_2</code>	Reserved
1-0 <code>cmn_diag_hscclk_sel_1_0</code>	High speed clock 0 select: Selects which PLL clock will be driven on the analog high speed clock 0.

13.4.10.2.541 Common calibration clock control register (`cmn_diag_calclk_ctrl`)

13.4.10.2.541.1 Offset

Register	Offset
<code>cmn_diag_calclk_ctrl</code>	8_01E1h

13.4.10.2.541.2 Diagram



13.4.10.2.541.3 Fields

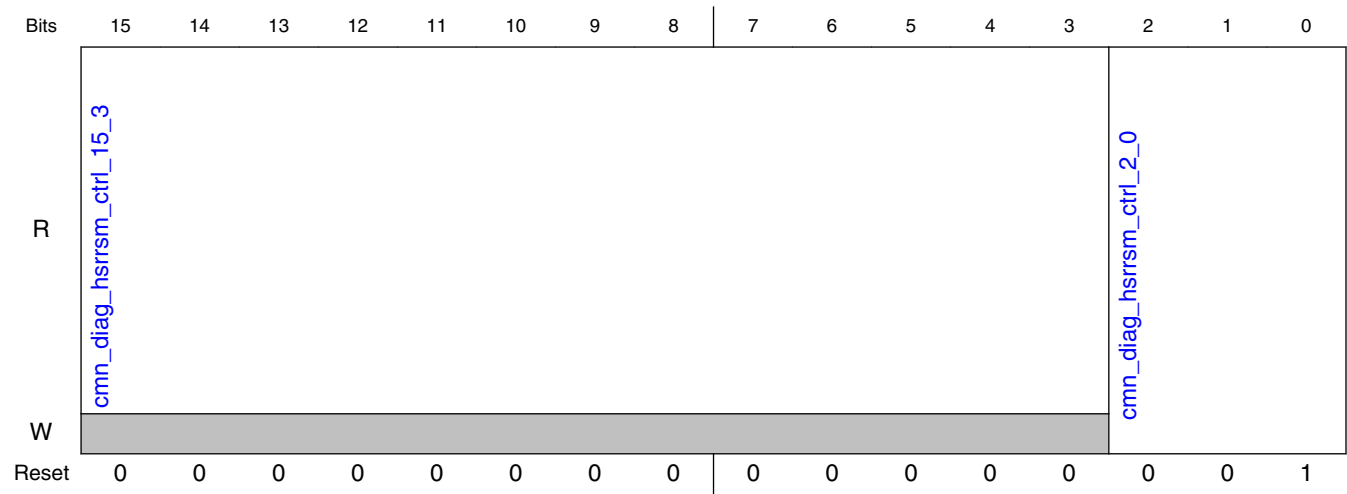
Field	Function
15-3 <code>cmn_diag_calclk_ctrl_15_3</code>	Reserved
2-0 <code>cmn_diag_calclk_ctrl_2_0</code>	Calibration clock divider select: Selects the divider setting for the calibration clock.

13.4.10.2.542 Common high speed reset release state machine control register (`cmn_diag_hsrrsm_ctrl`)

13.4.10.2.542.1 Offset

Register	Offset
<code>cmn_diag_hsrrsm_ctrl</code>	8_01E2h

13.4.10.2.542.2 Diagram



13.4.10.2.542.3 Fields

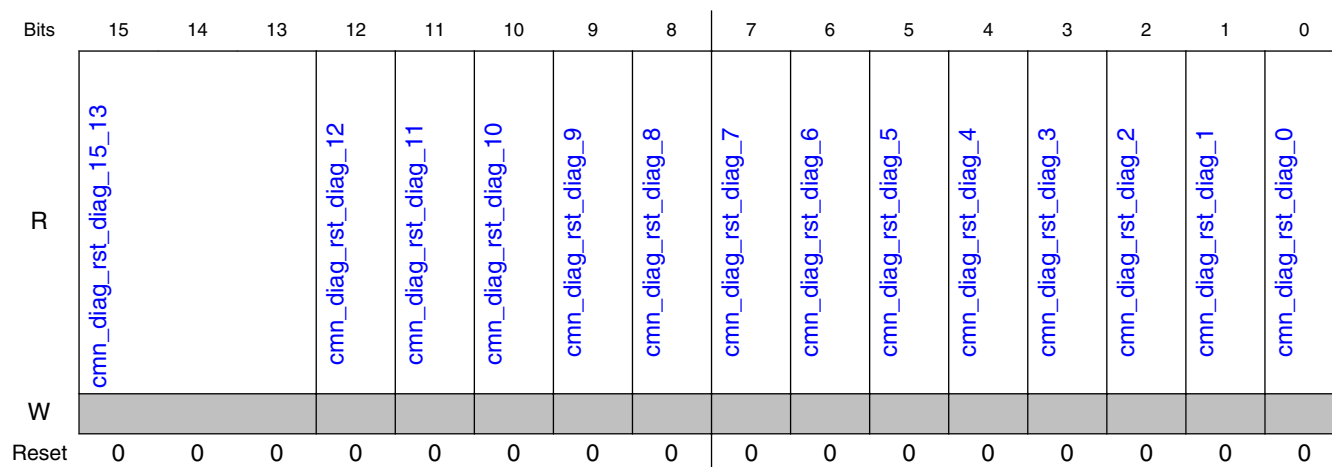
Field	Function
15-3 cmn_diag_hsrrsm_ctrl_15_3	Reserved
2-0 cmn_diag_hsrrsm_ctrl_2_0	Transmitter reset delay : Species the number of PSM clock cycles the transmitter common high speed reset state machine stays in the delay state.

13.4.10.2.543 Common functions reset diagnostic register (cmn_diag_rst_diag)

13.4.10.2.543.1 Offset

Register	Offset
cmn_diag_rst_diag	8_01E3h

13.4.10.2.543.2 Diagram



13.4.10.2.543.3 Fields

Field	Function
15-13 cmn_diag_rst_diag_15_13	Reserved
12 cmn_diag_rst_diag_12	Current state of the ssmac_power_reset_n reset.
11 cmn_diag_rst_diag_11	Reserved
10 cmn_diag_rst_diag_10	Reserved
9 cmn_diag_rst_diag_9	Current state of the cmn_pll1_dsm_reset_n reset.
8 cmn_diag_rst_diag_8	Current state of the cmn_pll0_dsm_reset_n reset.
7 cmn_diag_rst_diag_7	Current state of the cmn_pll1_vco_cal_fbdiv_clk_reset_n reset.
6 cmn_diag_rst_diag_6	Current state of the cmn_pll1_lock_det_fbdiv_clk_reset_n reset.
5	Current state of the cmn_pll1_vco_cal_ref_clk_reset_n reset.

Table continues on the next page...

Clocks And Resets

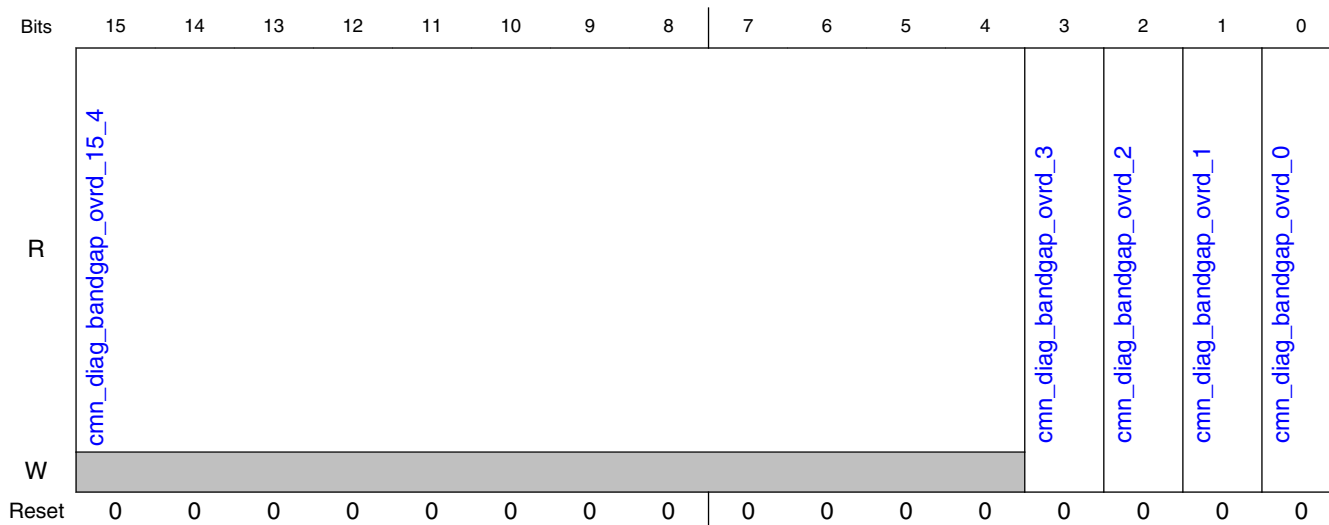
Field	Function
cmn_diag_rst_diag_5	
4	Current state of the cmn_pll1_lock_det_ref_clk_reset_n reset.
cmn_diag_rst_diag_4	
3	Current state of the cmn_pll0_vco_cal_fbdiv_clk_reset_n reset.
cmn_diag_rst_diag_3	
2	Current state of the cmn_pll0_lock_det_fbdiv_clk_reset_n reset.
cmn_diag_rst_diag_2	
1	Current state of the cmn_pll0_vco_cal_ref_clk_reset_n reset.
cmn_diag_rst_diag_1	
0	Current state of the cmn_pll0_lock_det_ref_clk_reset_n reset.
cmn_diag_rst_diag_0	

13.4.10.2.544 Bandgap override register (cmn_diag_bandgap_ovrd)

13.4.10.2.544.1 Offset

Register	Offset
cmn_diag_bandgap_ovrd	8_01E8h

13.4.10.2.544.2 Diagram



13.4.10.2.544.3 Fields

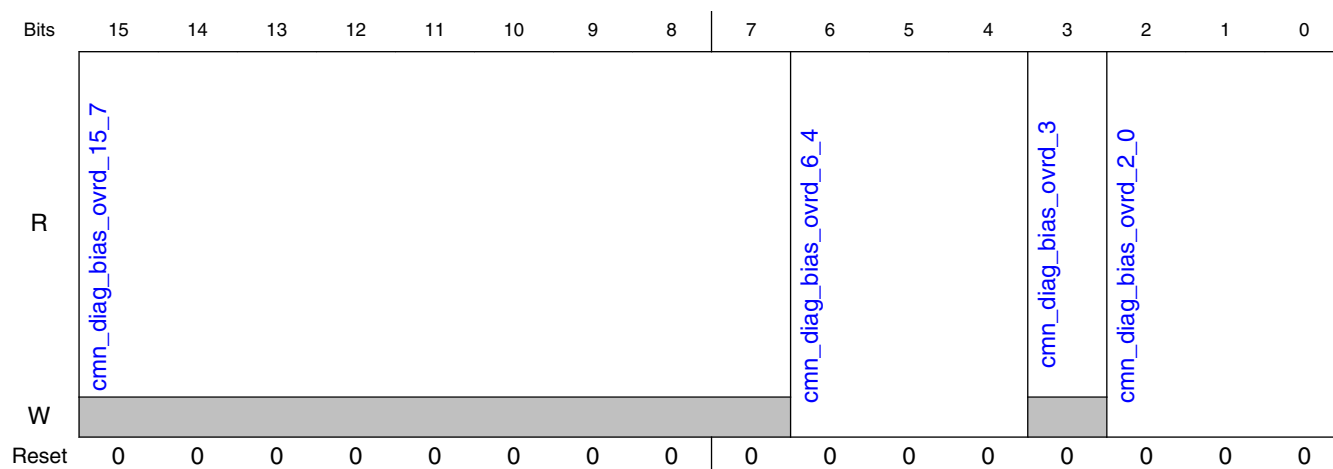
Field	Function
15-4 cmn_diag_band gap_ovrd_15_4	Reserved
3 cmn_diag_band gap_ovrd_3	Bandgap reference current increase: Increases the value of the crude reference current used in the bandgap startup block to determine when the bandgap startup circuitry can be disabled.
2 cmn_diag_band gap_ovrd_2	Bandgap reference current decrease: Decreases the value of the crude reference current used in the bandgap startup block to determine when the bandgap startup circuitry can be disabled.
1 cmn_diag_band gap_ovrd_1	Bandgap startup force on: Forces the bandgap startup circuitry to turn on. When the startup circuitry is turned on in this manner, the bandgap startup force off bit in this register shall be used to turn it off after a delay of at least 2 uSec.
0 cmn_diag_band gap_ovrd_0	Bandgap startup force off: Forces the bandgap startup circuitry to turn off. This signal has higher priority than bandgap startup force on bit in this register.

13.4.10.2.545 Bias override register (cmn_diag_bias_ovrd)

13.4.10.2.545.1 Offset

Register	Offset
cmn_diag_bias_ovrd	8_01E9h

13.4.10.2.545.2 Diagram



13.4.10.2.545.3 Fields

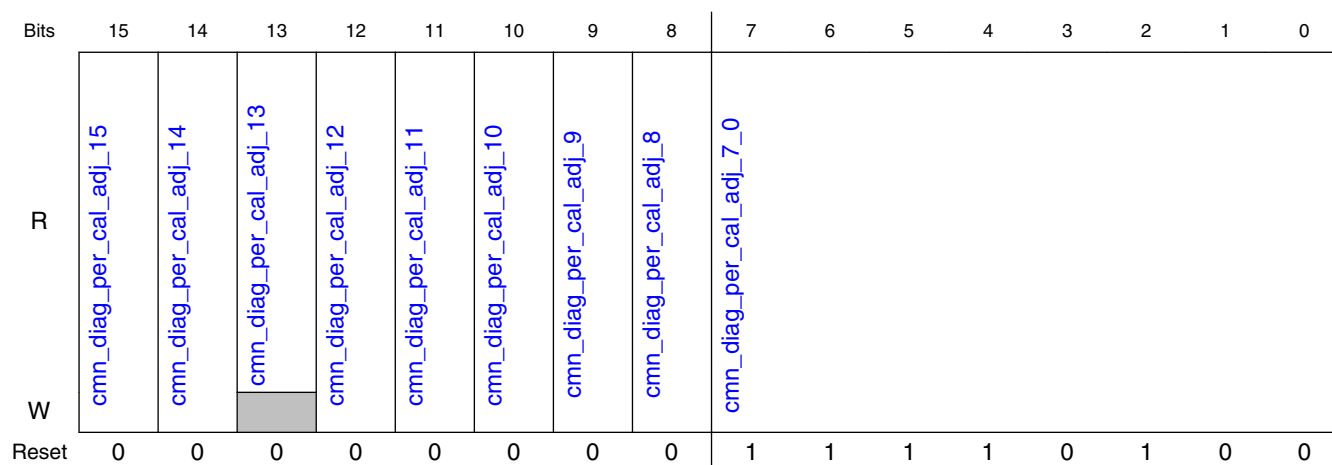
Field	Function
15-7 cmn_diag_bias_ovrd_15_7	Reserved
6-4 cmn_diag_bias_ovrd_6_4	Receiver resistor calibration current adjust: This field is used to adjust the receiver resistor calibration bias current. It drives the
3 cmn_diag_bias_ovrd_3	Reserved
2-0 cmn_diag_bias_ovrd_2_0	Transmitter resistor calibration current adjust: This field is used to adjust the transmitter resistor calibration bias current. It drives the

13.4.10.2.546 Common periodic calibration adjust control register (cmn_diag_per_cal_adj)

13.4.10.2.546.1 Offset

Register	Offset
cmn_diag_per_cal_adj	8_01ECh

13.4.10.2.546.2 Diagram



13.4.10.2.546.3 Fields

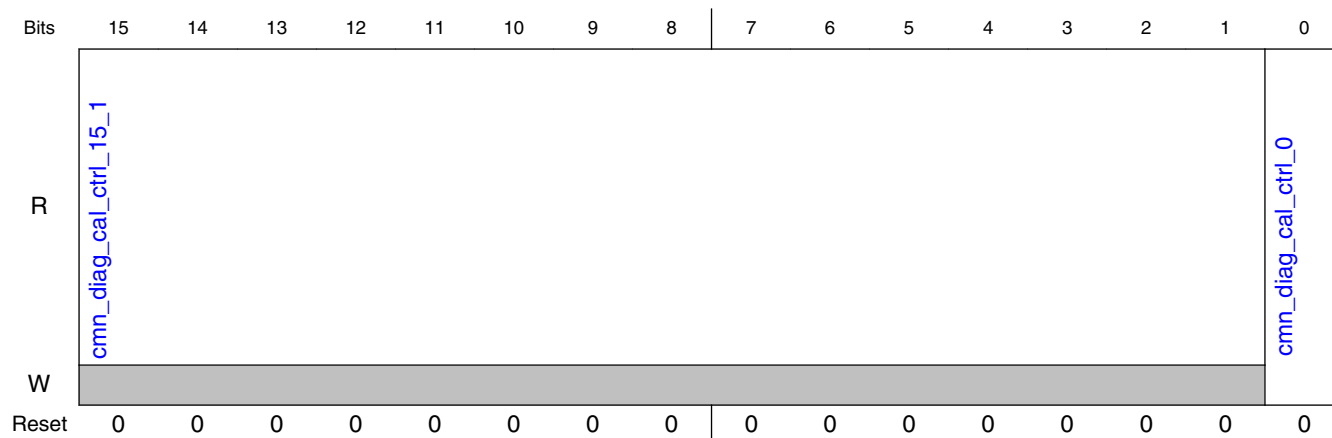
Field	Function
15 cmn_diag_per_cal_adj_15	Calibration latch enable override enable : When active (1b1), the calibration latch enable override bit in this register will drive the
14 cmn_diag_per_cal_adj_14	Calibration latch enable override : When enabled by the calibration latch enable override enable bit in this register, this bit will drive the
13 cmn_diag_per_cal_adj_13	Reserved
12 cmn_diag_per_cal_adj_12	Transmitter termination pull up calibration adjust enable: Enables the transmitter pull up termination calibration adjust function, when the periodic calibration adjust function is enabled.
11 cmn_diag_per_cal_adj_11	Transmitter termination pull down calibration adjust enable: Enables the transmitter pull down termination calibration adjust function, when the periodic calibration adjust function is enabled.
10 cmn_diag_per_cal_adj_10	Receiver termination calibration adjust enable: Enables the receiver termination calibration adjust function, when the periodic calibration adjust function is enabled.
9 cmn_diag_per_cal_adj_9	Current calibration adjust enable: Enables the current calibration adjust function, when the periodic calibration adjust function is enabled.
8 cmn_diag_per_cal_adj_8	Periodic calibration adjust enable: Enables the periodic calibration adjust function.
7-0 cmn_diag_per_cal_adj_7_0	Periodic calibration timer value: This is the value of the most significant 8 bits of the 16 bit periodic calibration timer.

13.4.10.2.547 Common calibration control register (cmn_diag_cal_ctrl)

13.4.10.2.547.1 Offset

Register	Offset
cmn_diag_cal_ctrl	8_01EDh

13.4.10.2.547.2 Diagram



13.4.10.2.547.3 Fields

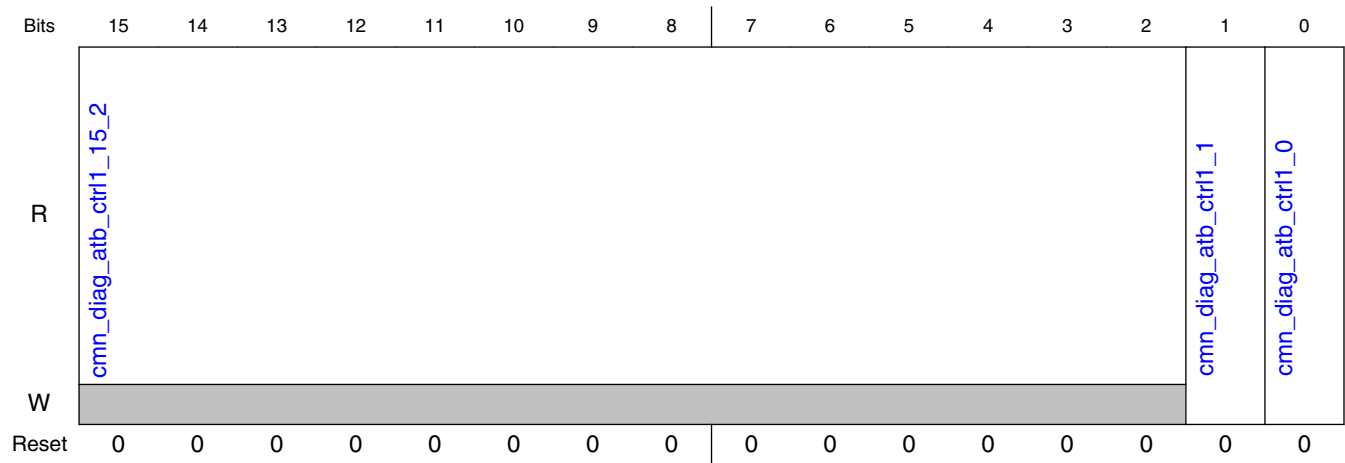
Field	Function
15-1 <code>cmn_diag_cal_ctrl_15_1</code>	Reserved
0 <code>cmn_diag_cal_ctrl_0</code>	Resistor calibration low swing select: In some protocols the TX calibration needs to be performed for a low swing output. This bit indicates that low swing calibration should be performed, by driving the

13.4.10.2.548 ATB control register 1 (`cmn_diag_atb_ctrl1`)

13.4.10.2.548.1 Offset

Register	Offset
<code>cmn_diag_atb_ctrl1</code>	8_01F0h

13.4.10.2.548.2 Diagram



13.4.10.2.548.3 Fields

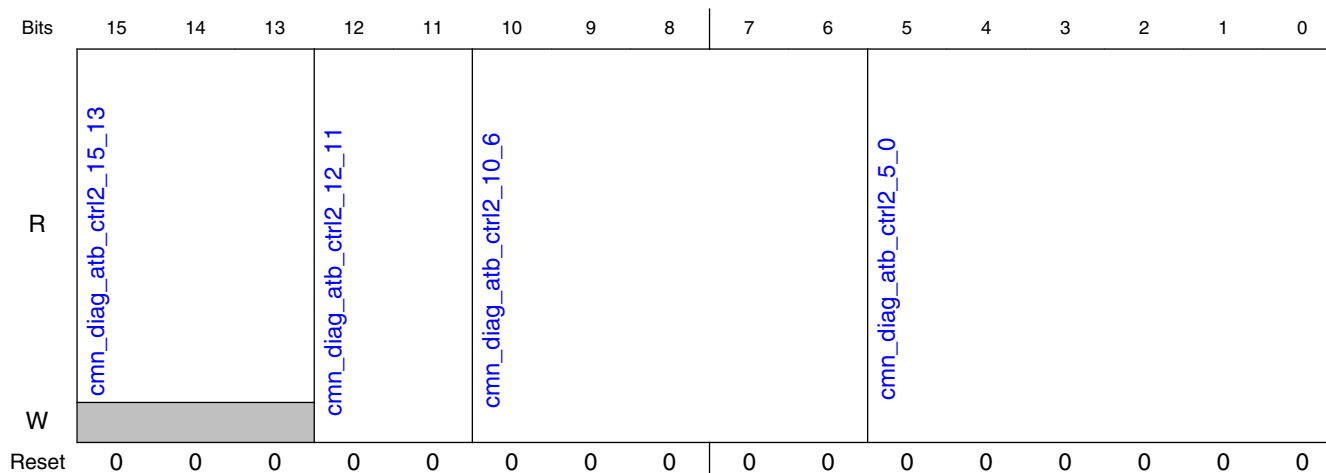
Field	Function
15-2 <code>cmn_diag_atb_ctrl1_15_2</code>	Reserved
1 <code>cmn_diag_atb_ctrl1_1</code>	ATB source select: Selects between the two source domains that can drive the ATB signals, by driving the
0 <code>cmn_diag_atb_ctrl1_0</code>	ATB enable: When active (1b1), the ATB test function is enabled.

13.4.10.2.549 ATB control register 2 (`cmn_diag_atb_ctrl2`)

13.4.10.2.549.1 Offset

Register	Offset
<code>cmn_diag_atb_ctrl2</code>	8_01F1h

13.4.10.2.549.2 Diagram



13.4.10.2.549.3 Fields

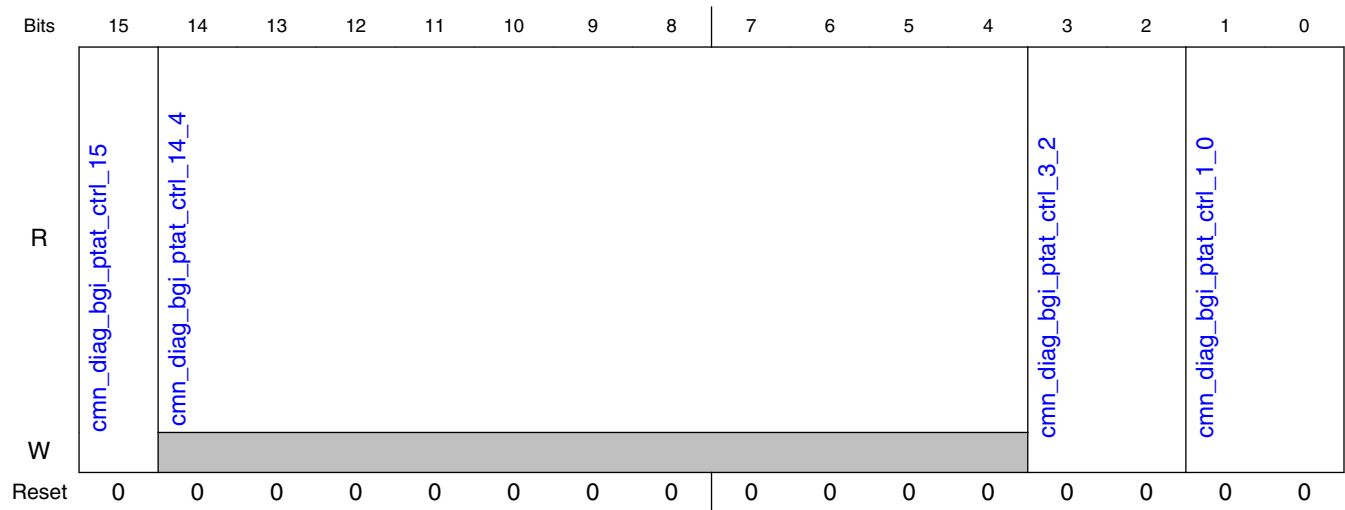
Field	Function
15-13 <code>cmn_diag_atb_ctrl2_15_13</code>	Reserved
12-11 <code>cmn_diag_atb_ctrl2_12_11</code>	ATB component type select: These bits specify which component type is currently selected by the ATB, as specified below.
10-6 <code>cmn_diag_atb_ctrl2_10_6</code>	ATB component sub address: Specifies the sub address of the component being selected. In this design, the sub address must be 0 when accessing the common, and the lane number when accessing a transmitter or receiver.
5-0 <code>cmn_diag_atb_ctrl2_5_0</code>	ATB test point address: Specifies the exact point in the selected analog component to be observed.

13.4.10.2.550 Common bandgap PTAT current control register (`cmn_diag_bgi_ptat_ctrl`)

13.4.10.2.550.1 Offset

Register	Offset
<code>cmn_diag_bgi_ptat_ctrl</code>	8_01F4h

13.4.10.2.550.2 Diagram



13.4.10.2.550.3 Fields

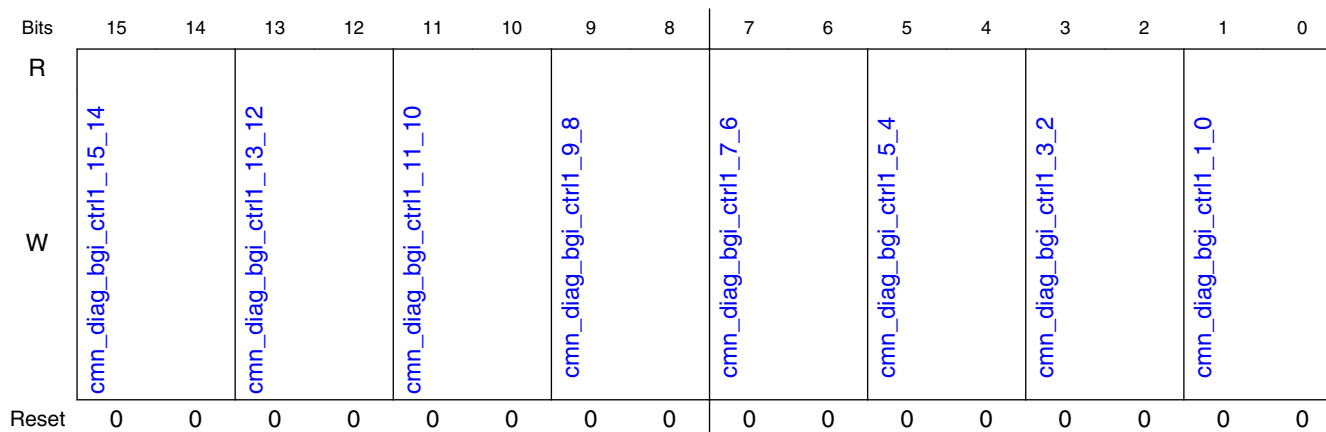
Field	Function
15 <code>cmn_diag_bgi_ptat_ctrl_15</code>	Bandgap PTAT current base unit enable: Enables the base unit on all the PTAT current outputs from the bandgap.
14-4 <code>cmn_diag_bgi_ptat_ctrl_14_4</code>	Reserved
3-2 <code>cmn_diag_bgi_ptat_ctrl_3_2</code>	Current programmability on the bandgap PTAT current supplying block xxx, by driving
1-0 <code>cmn_diag_bgi_ptat_ctrl_1_0</code>	Current programmability on the bandgap PTAT current supplying block xxx, by driving

13.4.10.2.551 Common bandgap current control register 1 (`cmn_diag_bgi_ctrl1`)

13.4.10.2.551.1 Offset

Register	Offset
<code>cmn_diag_bgi_ctrl1</code>	8_01F5h

13.4.10.2.551.2 Diagram



13.4.10.2.551.3 Fields

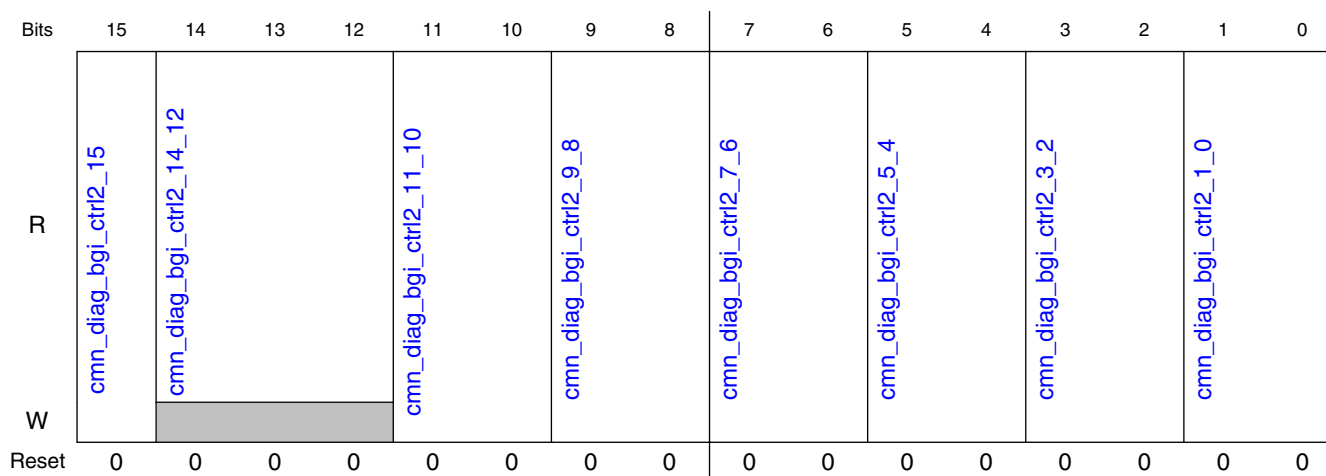
Field	Function
15-14 cmn_diag_bgi_ctrl1_15_14	Current programmability on the bandgap normal current supplying block xxx, by driving
13-12 cmn_diag_bgi_ctrl1_13_12	Current programmability on the bandgap normal current supplying block xxx, by driving
11-10 cmn_diag_bgi_ctrl1_11_10	Current programmability on the bandgap normal current supplying block xxx, by driving
9-8 cmn_diag_bgi_ctrl1_9_8	Current programmability on the bandgap normal current supplying block xxx, by driving
7-6 cmn_diag_bgi_ctrl1_7_6	Current programmability on the bandgap normal current supplying block xxx, by driving
5-4 cmn_diag_bgi_ctrl1_5_4	Current programmability on the bandgap normal current supplying block xxx, by driving
3-2 cmn_diag_bgi_ctrl1_3_2	Current programmability on the bandgap normal current supplying block xxx, by driving
1-0 cmn_diag_bgi_ctrl1_1_0	Current programmability on the bandgap normal current supplying block xxx, by driving

13.4.10.2.552 Common bandgap current control register 2 (cmn_diag_bgi_ctrl2)

13.4.10.2.552.1 Offset

Register	Offset
cmn_diag_bgi_ctrl2	8_01F6h

13.4.10.2.552.2 Diagram



13.4.10.2.552.3 Fields

Field	Function
15 cmn_diag_bgi_ctrl2_15	Bandgap normal current base unit enable: Enables the base unit on all the normal current outputs from the bandgap.
14-12 cmn_diag_bgi_ctrl2_14_12	Reserved
11-10 cmn_diag_bgi_ctrl2_11_10	Current programmability on the bandgap normal current supplying block xxx, by driving
9-8 cmn_diag_bgi_ctrl2_9_8	Current programmability on the bandgap normal current supplying block xxx, by driving
7-6 cmn_diag_bgi_ctrl2_7_6	Current programmability on the bandgap normal current supplying block xxx, by driving

Table continues on the next page...

Clocks And Resets

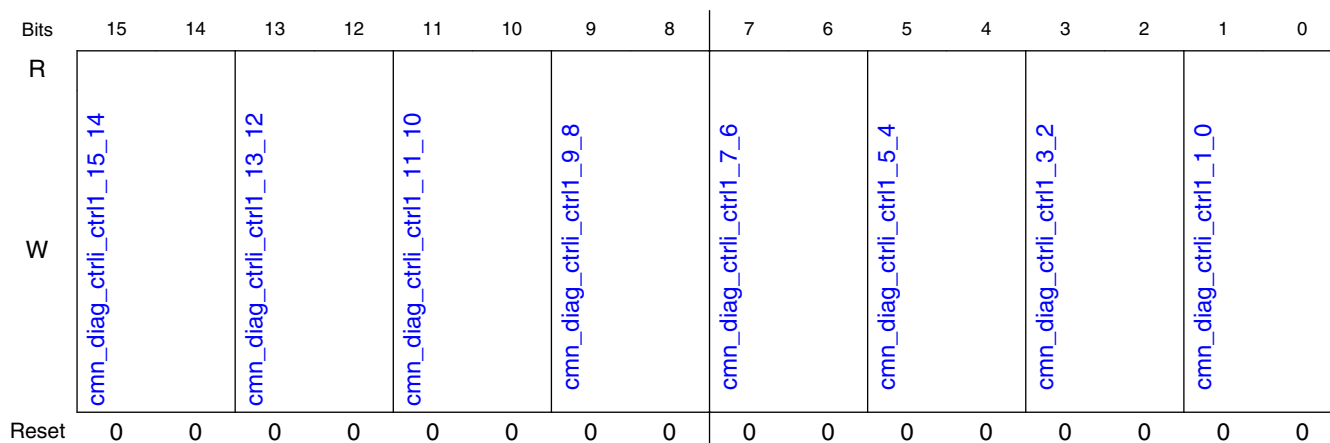
Field	Function
5-4 cmn_diag_bgi_c trl2_5_4	Current programmability on the bandgap normal current supplying block xxx, by driving
3-2 cmn_diag_bgi_c trl2_3_2	Current programmability on the bandgap normal current supplying block xxx, by driving
1-0 cmn_diag_bgi_c trl2_1_0	Current programmability on the bandgap normal current supplying block xxx, by driving

13.4.10.2.553 Common control current control register 1 (cmn_diag_ctrl1_ctrl1)

13.4.10.2.553.1 Offset

Register	Offset
cmn_diag_ctrl1_ctrl1	8_01F7h

13.4.10.2.553.2 Diagram



13.4.10.2.553.3 Fields

Field	Function
15-14	Current programmability on the constant current from the calibration block supplying block xxx, by driving

Table continues on the next page...

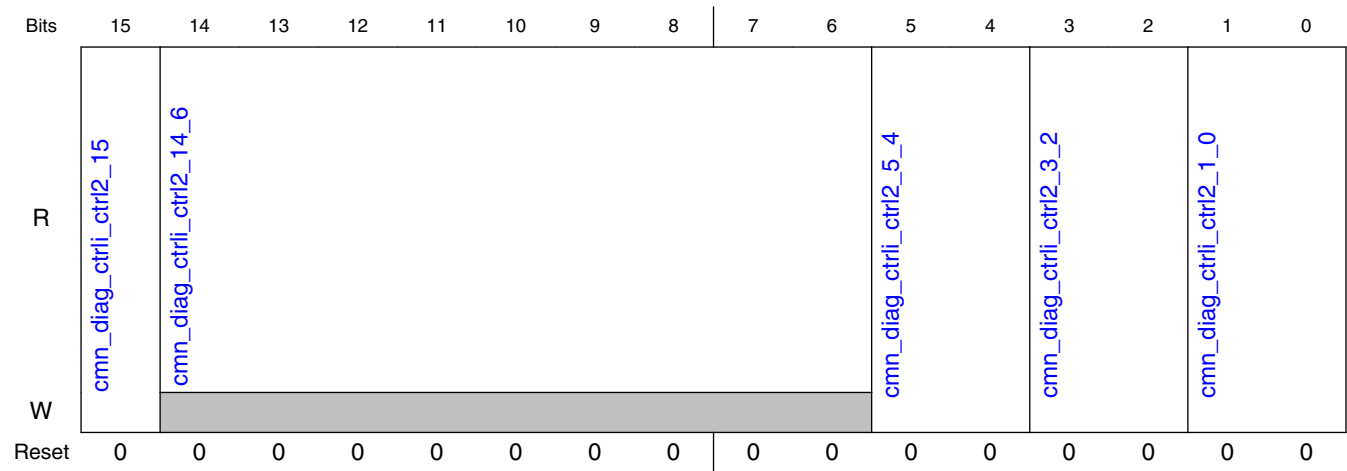
Field	Function
cmn_diag_ctrl1_15_14	
13-12 cmn_diag_ctrl1_13_12	Current programmability on the constant current from the calibration block supplying block xxx, by driving
11-10 cmn_diag_ctrl1_11_10	Current programmability on the constant current from the calibration block supplying block xxx, by driving
9-8 cmn_diag_ctrl1_9_8	Current programmability on the constant current from the calibration block supplying block xxx, by driving
7-6 cmn_diag_ctrl1_7_6	Current programmability on the constant current from the calibration block supplying block xxx, by driving
5-4 cmn_diag_ctrl1_5_4	Current programmability on the constant current from the calibration block supplying block xxx, by driving
3-2 cmn_diag_ctrl1_3_2	Current programmability on the constant current from the calibration block supplying block xxx, by driving
1-0 cmn_diag_ctrl1_1_0	Current programmability on the constant current from the calibration block supplying block xxx, by driving

13.4.10.2.554 Common control current control register 2 (cmn_diag_ctrl1_ctrl2)

13.4.10.2.554.1 Offset

Register	Offset
cmn_diag_ctrl1_ctrl2	8_01F8h

13.4.10.2.554.2 Diagram



13.4.10.2.554.3 Fields

Field	Function
15 <code>cmn_diag_ctrl2_15</code>	Constant current base unit enable: Enables the base unit on all the normal current outputs from the bandgap.
14-6 <code>cmn_diag_ctrl2_14_6</code>	Reserved
5-4 <code>cmn_diag_ctrl2_5_4</code>	Current programmability on the constant current from the calibration block supplying block xxx, by driving
3-2 <code>cmn_diag_ctrl2_3_2</code>	Current programmability on the constant current from the calibration block supplying block xxx, by driving
1-0 <code>cmn_diag_ctrl2_1_0</code>	Current programmability on the constant current from the calibration block supplying block xxx, by driving

13.4.10.2.555 Common digital functions cover your alternatives register (`cmn_diag_dcya`)

13.4.10.2.555.1 Offset

Register	Offset
<code>cmn_diag_dcya</code>	8_01FEh

13.4.10.2.555.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_diag_dcya_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.555.3 Fields

Field	Function
15-0 cmn_diag_dcya_15_0	Reserved

13.4.10.2.556 Common analog cover your alternatives register (cmn_diag_acya)

13.4.10.2.556.1 Offset

Register	Offset
cmn_diag_acya	8_01FFh

13.4.10.2.556.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	cmn_diag_acya_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.556.3 Fields

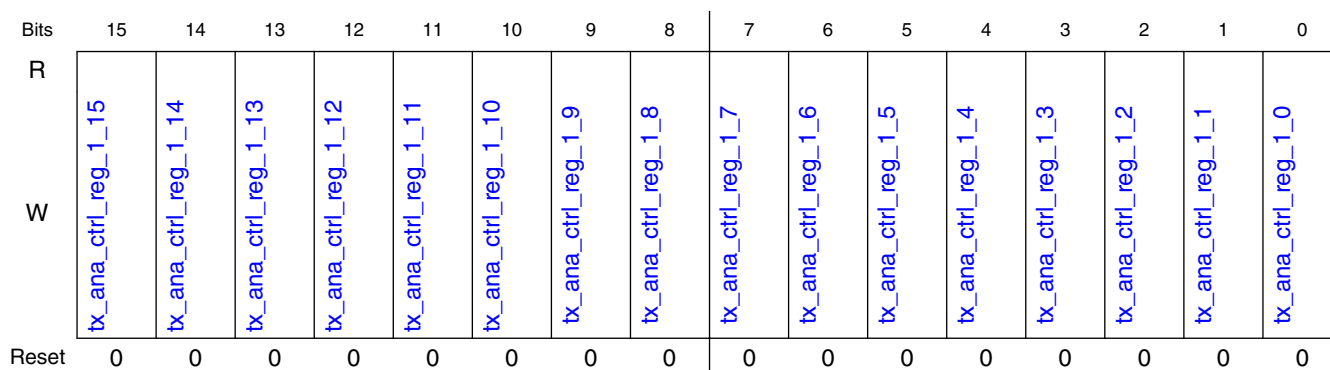
Field	Function
15-0 cmn_diag_acya_15_0	Reserved

13.4.10.2.557 DP Aux analog control 1 (tx_ana_ctrl_reg_1)

13.4.10.2.557.1 Offset

Register	Offset
tx_ana_ctrl_reg_1	8_5020h

13.4.10.2.557.2 Diagram



13.4.10.2.557.3 Fields

Field	Function
15 tx_ana_ctrl_reg_1_15	Controls txda_dp_aux_en,
14 tx_ana_ctrl_reg_1_14	Controls auxda_se_en
13 tx_ana_ctrl_reg_1_13	Controls txda_cal_latch_en, Enable signal for latch that sample and holds calibration values. Activate this signal for 1 clock cycle to sample new calibration values.
12 tx_ana_ctrl_reg_1_12	Controls auxda_polarity, which selects the polarity of the xcvr
11 tx_ana_ctrl_reg_1_11	Controls txda_drv_power_isolation_en, Power island isolation signal.This signal is currently unused.
10	Controls txda_drv_power_en_ph_2_n, Power island enable signals.This signal is currently unused.

Table continues on the next page...

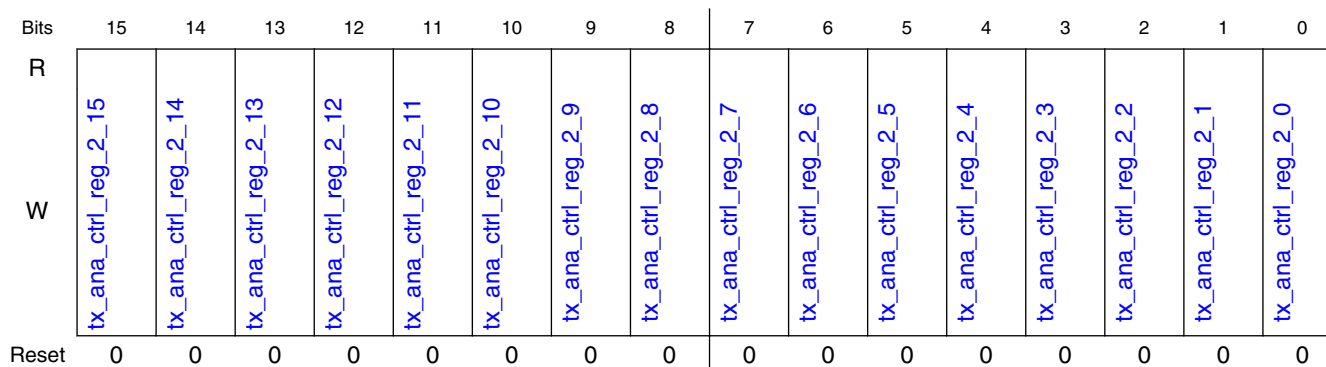
Field	Function
tx_ana_ctrl_reg_1_10	
9 tx_ana_ctrl_reg_1_9	Controls txda_drv_power_en_ph_1_n, Power island enable signals. This signal is currently unused.
8 tx_ana_ctrl_reg_1_8	Controls txda_bgref_en, which enables Bandgap Reference Circuit. Must be asserted after (X=5us) txda_drv_ldo_en, else held to default value
7 tx_ana_ctrl_reg_1_7	Controls txda_drv_ldo_en, which enables the transmitter circuits related to the driver LDO function
6 tx_ana_ctrl_reg_1_6	Controls txda_decap_en_del, delayed version (tX=100ns) of tx_decap_en
5 tx_ana_ctrl_reg_1_5	Controls txda_decap_en, which enables the analog decap circuits.
4 tx_ana_ctrl_reg_1_4	Controls txda_uphy_supply_en_del, delayed version (tX=100ns) of txda_uphy_supply_en
3 tx_ana_ctrl_reg_1_3	Controls txda_uphy_supply_en, which enables the transmitter UPHY components supply
2 tx_ana_ctrl_reg_1_2	Controls txda_low_leakage_en,
1 tx_ana_ctrl_reg_1_1	Controls txda_drv_idle_lowi_en,
0 tx_ana_ctrl_reg_1_0	Controls txda_drv_cmn_mode_en, which enables the analog circuits required, to hold the driver output at a common mode voltage when required.

13.4.10.2.558 DP Aux analog control 2 (tx_ana_ctrl_reg_2)

13.4.10.2.558.1 Offset

Register	Offset
tx_ana_ctrl_reg_2	8_5021h

13.4.10.2.558.2 Diagram



13.4.10.2.558.3 Fields

Field	Function
15 tx_ana_ctrl_reg_2_15	Controls auxda_debouncing_clk, required in MHL mode, for data reception
14 tx_ana_ctrl_reg_2_14	Controls txda_lpbk_recovered_clk_en,
13 tx_ana_ctrl_reg_2_13	Controls txda_lpbk_isi_gen_en
12 tx_ana_ctrl_reg_2_12	Controls txda_lpbk_serial_en, when enabled (1'b1), the serial loopback path from a lanes serializer to its deserializer is activated.
11 tx_ana_ctrl_reg_2_11	Controls txda_lpbk_line_en, when enabled (1'b1), the line side loopback path from a lanes serial receiver to its serial transmitter is activated.
10 tx_ana_ctrl_reg_2_10	Controls txda_drv_ldo_redc_sinkiq, when enabled (1'b1) reduces the no-load current from the driver LDO in order to reduce the settling time
9 tx_ana_ctrl_reg_2_9	Controls xcvr_decap_en_del. Delayed version (tX=100ns) of xcvr_decap_en
8 tx_ana_ctrl_reg_2_8	Controls xcvr_decap_en, which must be enabled when the common module initially powers up, or exits the suspend mode.
7 tx_ana_ctrl_reg_2_7	Controls txda_mphy_enable_hs_nt
6	Controls txda_mphy_sa_mode, which enables the driver LDO small amplitude mode.

Table continues on the next page...

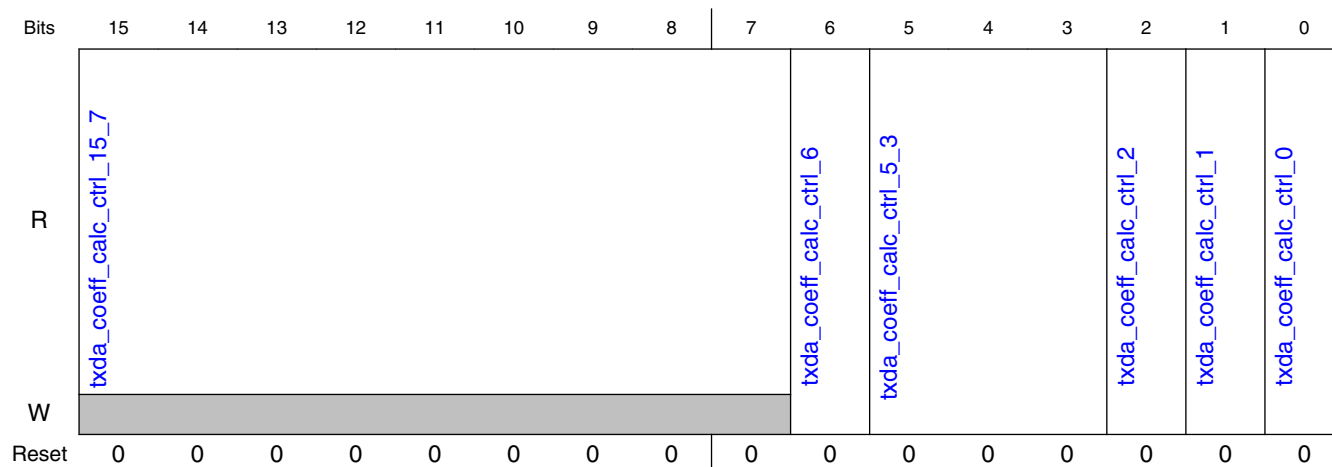
Field	Function
tx_ana_ctrl_reg_2_6	
5 tx_ana_ctrl_reg_2_5	Controls txda_drv_ldo_rbyr_fb_en, TX driver LDO VDD dependent feedback reference enable: Enables VDD dependent feedback reference to LDO.
4 tx_ana_ctrl_reg_2_4	Controls txda_drv_rst_pull_down
3 tx_ana_ctrl_reg_2_3	Controls txda_drv_ldo_bg_fb_en,
2 tx_ana_ctrl_reg_2_2	Controls txda_drv_ldo_bg_ref_en,
1 tx_ana_ctrl_reg_2_1	Controls txda_drv_predrv_en_del, delayed version (tX=100ns) of txda_predrv_en
0 tx_ana_ctrl_reg_2_0	Controls txda_drv_predrv_en,

13.4.10.2.559 Tx_coef_calc module Inputs (txda_coeff_calc_ctrl)

13.4.10.2.559.1 Offset

Register	Offset
txda_coeff_calc_ctrl	8_5022h

13.4.10.2.559.2 Diagram



13.4.10.2.559.3 Fields

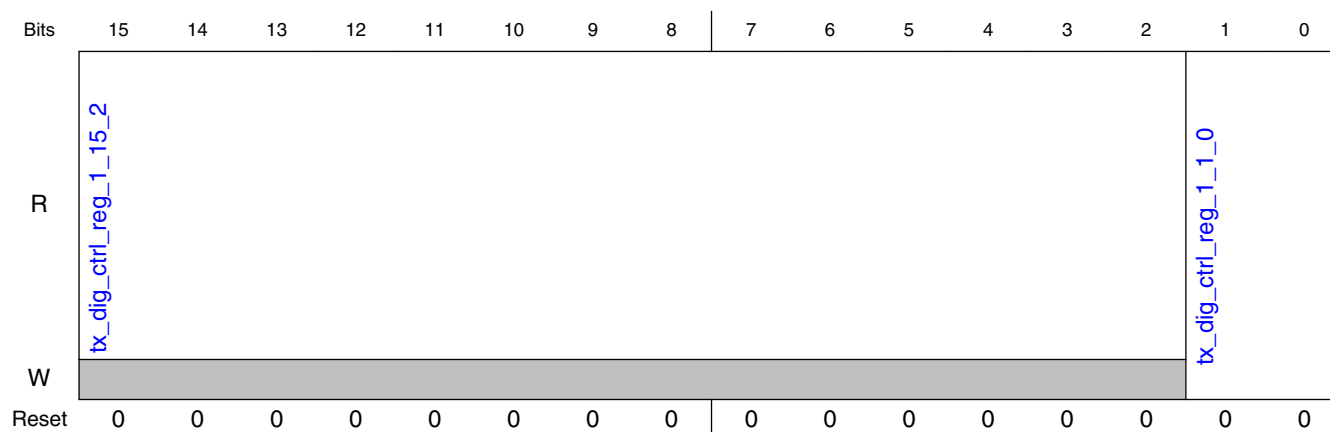
Field	Function
15-7 txda_coeff_calc_ctrl_15_7	Reserved
6 txda_coeff_calc_ctrl_6	Controls tx_high_z, when enabled (1'b1), puts the analog driver in a high impedance state.
5-3 txda_coeff_calc_ctrl_5_3	Controls, tx_vmargin, the voltage margining of the transmitter for the lane, as specified below.
2 txda_coeff_calc_ctrl_2	Controls low_power_swing_en, the voltage swing of the driver. The values below are peak to peak (differential) values.
1 txda_coeff_calc_ctrl_1	Controls tx_fcm_drv_main_en, which enable the driver with no emphasis
0 txda_coeff_calc_ctrl_0	Controls tx_fcm_full_margin, which enables the maximum margining possible

13.4.10.2.560 Tx dig control reg 1 (tx_dig_ctrl_reg_1)

13.4.10.2.560.1 Offset

Register	Offset
tx_dig_ctrl_reg_1	8_5023h

13.4.10.2.560.2 Diagram



13.4.10.2.560.3 Fields

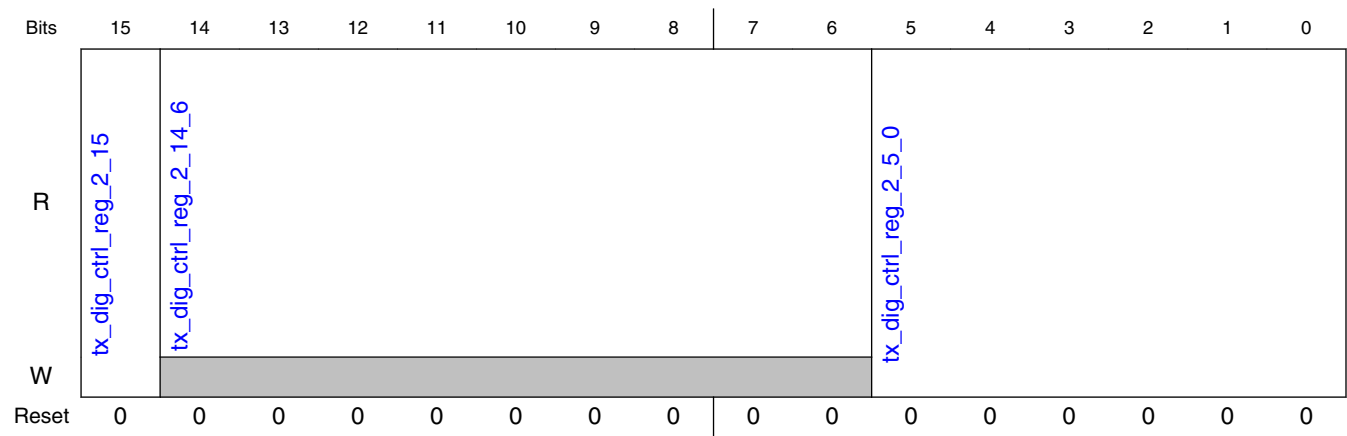
Field	Function
15-2 tx_dig_ctrl_reg_1_15_2	Reserved
1-0 tx_dig_ctrl_reg_1_1_0	Controls tx_deemphasis[1:0], the de-emphasis level of the transmitter as specified below.

13.4.10.2.561 Tx dig control reg 2 (tx_dig_ctrl_reg_2)

13.4.10.2.561.1 Offset

Register	Offset
tx_dig_ctrl_reg_2	8_5024h

13.4.10.2.561.2 Diagram



13.4.10.2.561.3 Fields

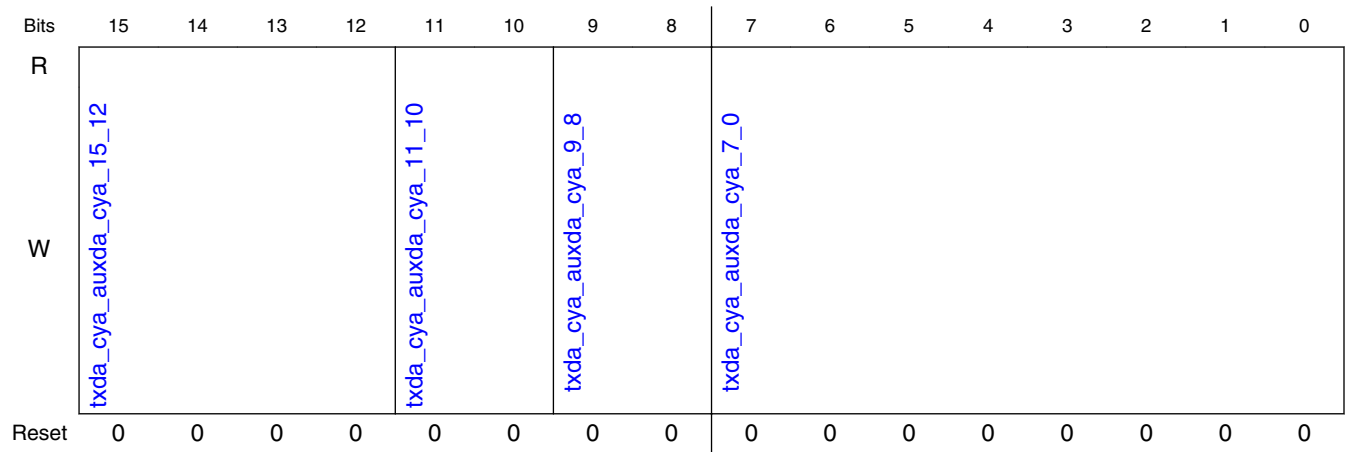
Field	Function
15 tx_dig_ctrl_reg_2_15	Controls tx_high_z_tm_en,
14-6 tx_dig_ctrl_reg_2_14_6	Reserved
5-0 tx_dig_ctrl_reg_2_5_0	tx_rescal_code required for hbdc and tx_coef_calc, average of the pull up and pull down resistor calibration values.

13.4.10.2.562 DP Aux analog control 3 (txda_cya_auxda_cya)

13.4.10.2.562.1 Offset

Register	Offset
txda_cya_auxda_cya	8_5025h

13.4.10.2.562.2 Diagram



13.4.10.2.562.3 Fields

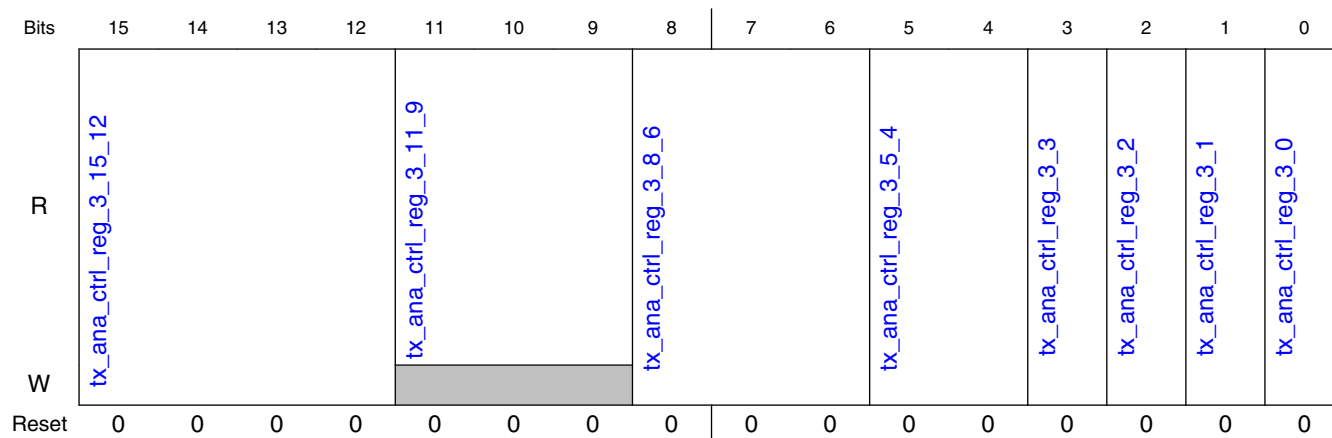
Field	Function
15-12 txda_cya_auxda_cya_15_12	txda_cya_auxda_cya_15_12
11-10 txda_cya_auxda_cya_11_10	txda_cya_auxda_cya_11_10
9-8 txda_cya_auxda_cya_9_8	txda_cya_auxda_cya_9_8
7-0 txda_cya_auxda_cya_7_0	Controls txda_cya, the TX testmode signal (cover your alternatives signals).These signals are reserved for reserved for last minute analog modifications that require CDB register control

13.4.10.2.563 DP Aux analog control 4 (tx_ana_ctrl_reg_3)

13.4.10.2.563.1 Offset

Register	Offset
tx_ana_ctrl_reg_3	8_5026h

13.4.10.2.563.2 Diagram



13.4.10.2.563.3 Fields

Field	Function
15-12 tx_ana_ctrl_reg_3_15_12	Programs txda_vcmhold_prog[3:0]
11-9 tx_ana_ctrl_reg_3_11_9	Reserved
8-6 tx_ana_ctrl_reg_3_8_6	txda_drv_boost_prog[2:0], When the programmable boost function is enabled, this signal controls the level of the boost.
5-4 tx_ana_ctrl_reg_3_5_4	Programs txda_cm_sense_vref_dac[1:0]
3 tx_ana_ctrl_reg_3_3	Controls txda_drv_mission_en,
2 tx_ana_ctrl_reg_3_2	Controls txda_drv_predrv_pullup, when the pre-driver is disabled via the txda_drv_predrv_en signal, this signal controls the state of the predriver output.
1 tx_ana_ctrl_reg_3_1	Controls txda_drv_margin_type, TX driver margin type: Selects the margining type the driver will operate in
0 tx_ana_ctrl_reg_3_0	Controls txda_cm_sense_en, Common mode sense enable, which enables the common mode sense analog circuits. This signal must be active during the entire fast establishment of common mode process.

13.4.10.2.564 DP Aux analog control 5 (tx_ana_ctrl_reg_4)

13.4.10.2.564.1 Offset

Register	Offset
tx_ana_ctrl_reg_4	8_5027h

13.4.10.2.564.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	tx_ana_ctrl_reg_4_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.4.10.2.564.3 Fields

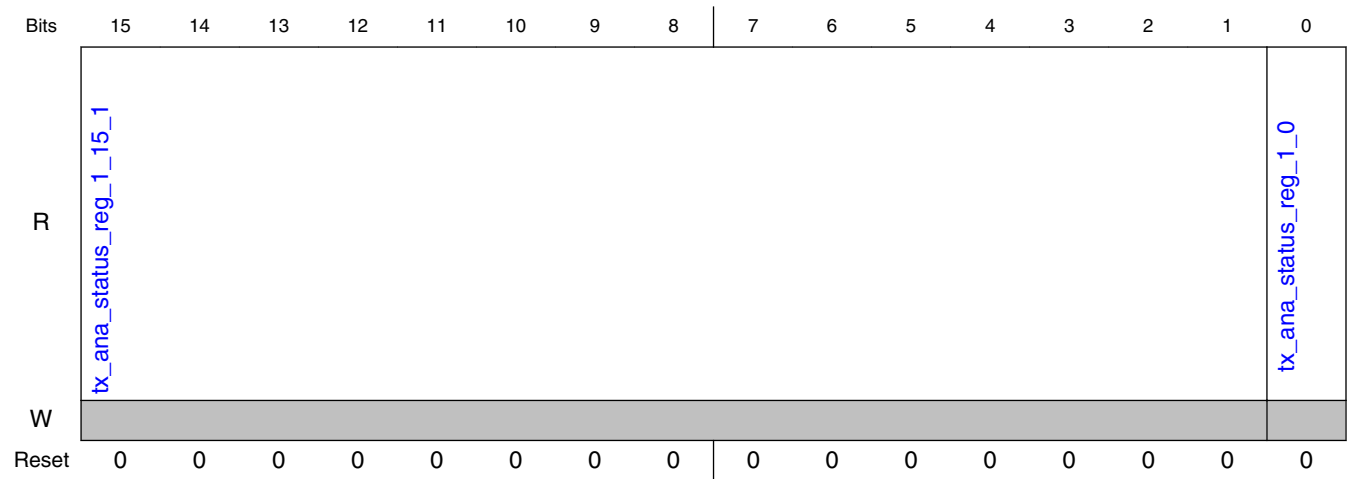
Field	Function
15-0 tx_ana_ctrl_reg_4_15_0	Programs txda_drv_ldo_prog[15:0], Sets driver LDO voltage

13.4.10.2.565 DP Aux analog status 1 (tx_ana_status_reg_1)

13.4.10.2.565.1 Offset

Register	Offset
tx_ana_status_reg_1	8_5028h

13.4.10.2.565.2 Diagram



13.4.10.2.565.3 Fields

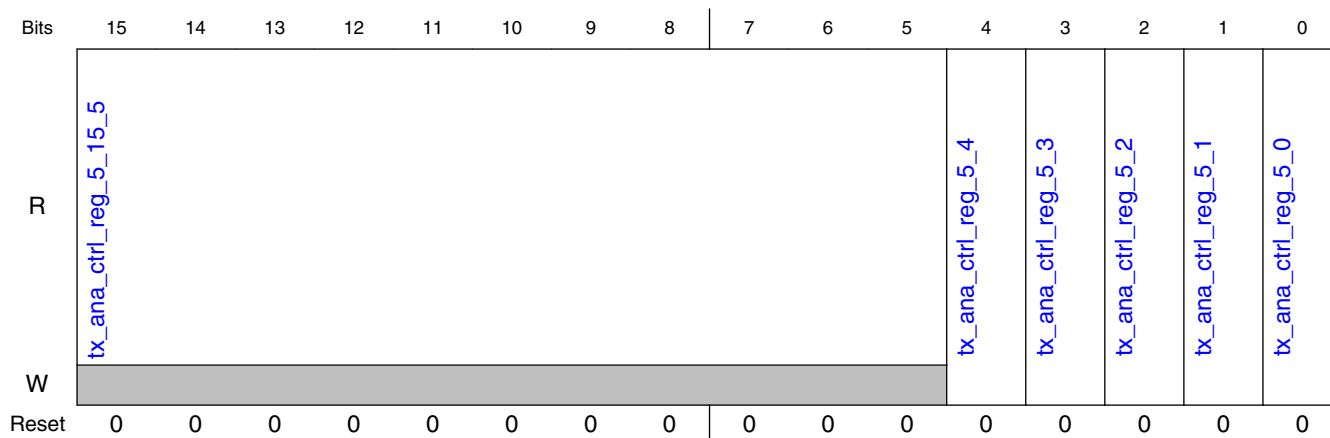
Field	Function
15-1 tx_ana_status_reg_1_15_1	Reserved
0 tx_ana_status_reg_1_0	Gives out txda_cm_sense_out, Common mode sense comparator output: Indicates the level of the tx_{p,m} signals relative to the reference voltage, controlled by the txda_cm_sense_vref_dac signal.

13.4.10.2.566 DP Aux analog status 6 (tx_ana_ctrl_reg_5)

13.4.10.2.566.1 Offset

Register	Offset
tx_ana_ctrl_reg_5	8_5029h

13.4.10.2.566.2 Diagram



13.4.10.2.566.3 Fields

Field	Function
15-5 tx_ana_ctrl_reg_5_15_5	Reserved
4 tx_ana_ctrl_reg_5_4	Programs auxda_arc_en, HDMI_ARC mode transmission/reception
3 tx_ana_ctrl_reg_5_3	Programs auxda_mhl_en, MHL/eCBUS mode transmission/reception
2 tx_ana_ctrl_reg_5_2	Programs txda_bidi_term_en
1 tx_ana_ctrl_reg_5_1	Controls txda_drv_boost_prog_en, which enables the programmable boost function
0 tx_ana_ctrl_reg_5_0	Controls txda_drv_ldo_vdd_ref_en,

13.4.10.2.567 rx_aux analog control1 (rx_ana_ctrl_reg_1)

13.4.10.2.567.1 Offset

Register	Offset
rx_ana_ctrl_reg_1	8_9060h

13.4.10.2.567.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	rx_ana_ctrl_reg_1_15_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

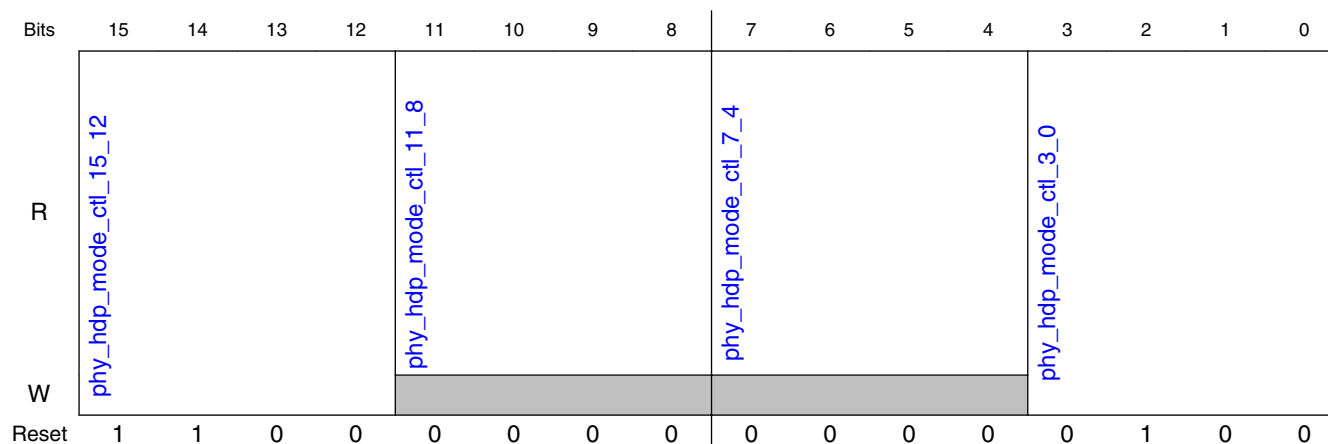
13.4.10.2.567.3 Fields

Field	Function
15-0 rx_ana_ctrl_reg_1_15_0	Unused

13.4.10.2.568 HDP Mode Control register (phy_hdp_mode_ctl)**13.4.10.2.568.1 Offset**

Register	Offset
phy_hdp_mode_ctl	8_C008h

13.4.10.2.568.2 Diagram



13.4.10.2.568.3 Fields

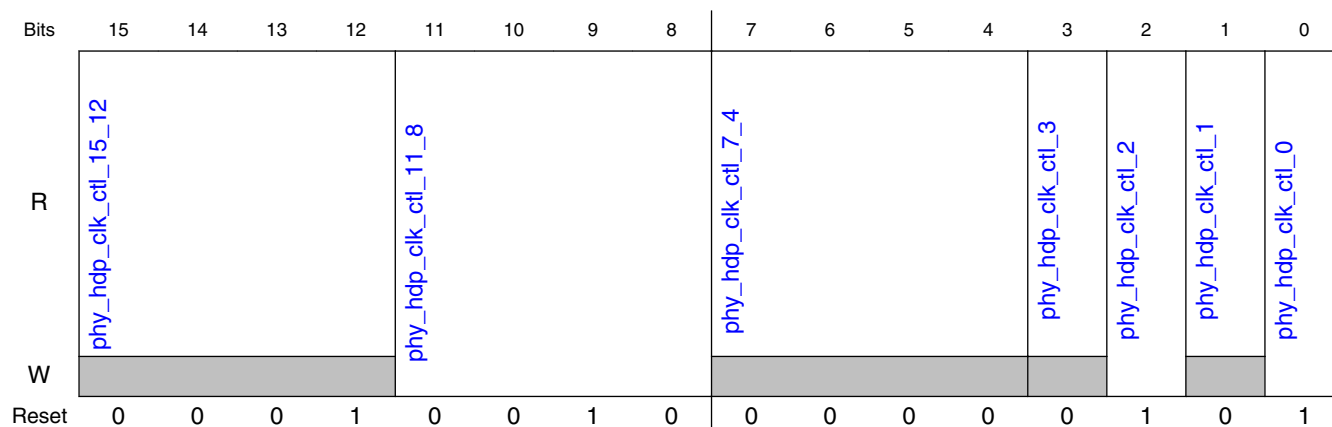
Field	Function
15-12 phy_hdp_mode_ctl_15_12	PHY HDP lane disable - 0 = enable associated PHY HDP lane; 1 = disable/powerdown the associated PHY DP lane. This field is used to disable PHY lanes when not used. For example, DP can operate on a sub-set of the lanes (1 or 2) instead of all 4. HDMI requires all 4 lanes at all times. The PMA lane(s) mapped to the disabled HDP lane(s) will be powered down as well. Value can only be changed when phy_reset_n is asserted low or as described by the DP lane configuration change procedure.
11-8 phy_hdp_mode_ctl_11_8	Reserved
7-4 phy_hdp_mode_ctl_7_4	HDP Power State Acknowledgement - power state acknowledgement for HDP lanes. (Re-synchronized to APB clock.)
3-0 phy_hdp_mode_ctl_3_0	HDP Power State - power state for HDP lanes. Direct mapping to the PMAs A0 to A3 power states (A0 = 0b0001, A1 = 0b0010, A2 = 0b0100 and A3 = 0b1000). (Re-synchronized to PSM clock.)

13.4.10.2.569 HDP Clock Control register (phy_hdp_clk_ctl)

13.4.10.2.569.1 Offset

Register	Offset
phy_hdp_clk_ctl	8_C009h

13.4.10.2.569.2 Diagram



13.4.10.2.569.3 Fields

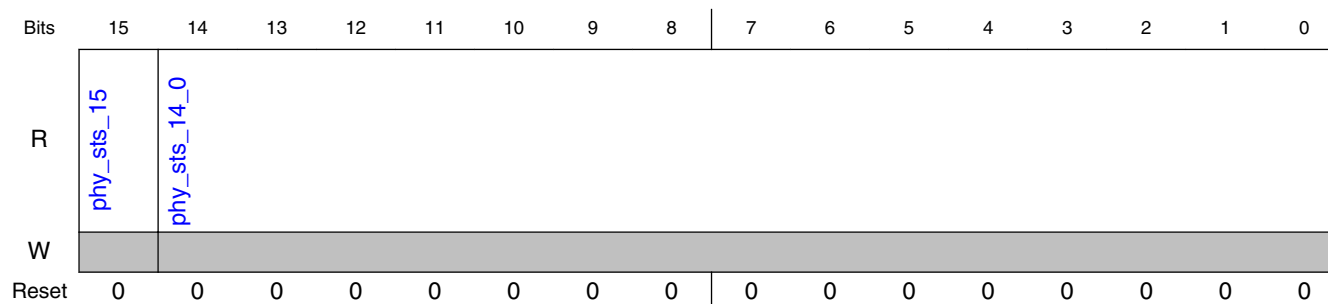
Field	Function
15-12 phy_hdp_clk_ctl_15_12	DP PLL data rate 1 clock divider value. Divider value for the PLL clock to generate phy_pma_char_clk_out. (HBR2 = 1, RBR/HBR = 2.)
11-8 phy_hdp_clk_ctl_11_8	DP PLL data rate 0 clock divider value. Divider value for the PLL clock to generate phy_pma_tx_data_clk_out. (HBR2 = 2, RBR/HBR = 4.)
7-4 phy_hdp_clk_ctl_7_4	Reserved
3 phy_hdp_clk_ctl_3	HDP PLL clock enable acknowledge - Indicates whether HDP PLLs data rate and full rate clocks are active/enabled. 1 = clocks enabled/active, 0 = clocks disabled/gated.
2 phy_hdp_clk_ctl_2	HDP PLL clock enable - Clock enable for HDP PLLs data rate and full rate clocks out of PMA. 1 = enable PLL data rate and full rate clocks, 0 = gate PLL data rate and full rate clocks
1 phy_hdp_clk_ctl_1	HDP PLL ready - HDP PLLs ready indication for high speed clocks. 1 = PLL ready, 0 = PLL not ready.
0 phy_hdp_clk_ctl_0	HDP PLL enable - HDP PLLs enable for high speed clocks. 1 = enabled, 0 = disabled.

13.4.10.2.570 PHY status register (phy_sts)

13.4.10.2.570.1 Offset

Register	Offset
phy_sts	8_C00Fh

13.4.10.2.570.2 Diagram



13.4.10.2.570.3 Fields

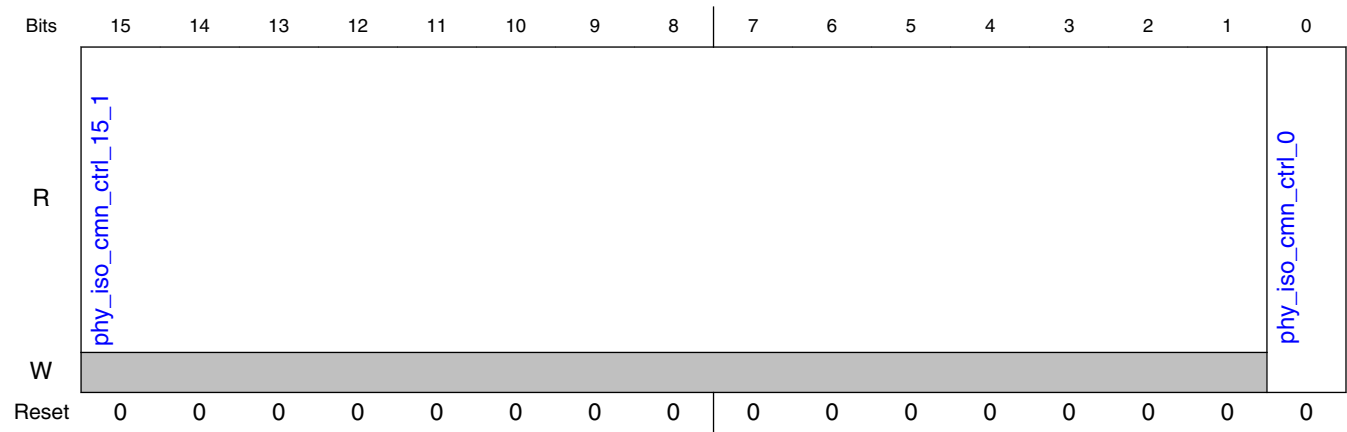
Field	Function
15 phy_sts_15	PHY APB access timeout: When set, an APB read/write request to PHY level registers failed (i.e. timed out). When set, this bit is cleared upon read.
14-0 phy_sts_14_0	Reserved

13.4.10.2.571 PHY common control signal isolation register (phy_iso_cm_n_ctrl)

13.4.10.2.571.1 Offset

Register	Offset
phy_iso_cm_n_ctrl	8_C010h

13.4.10.2.571.2 Diagram



13.4.10.2.571.3 Fields

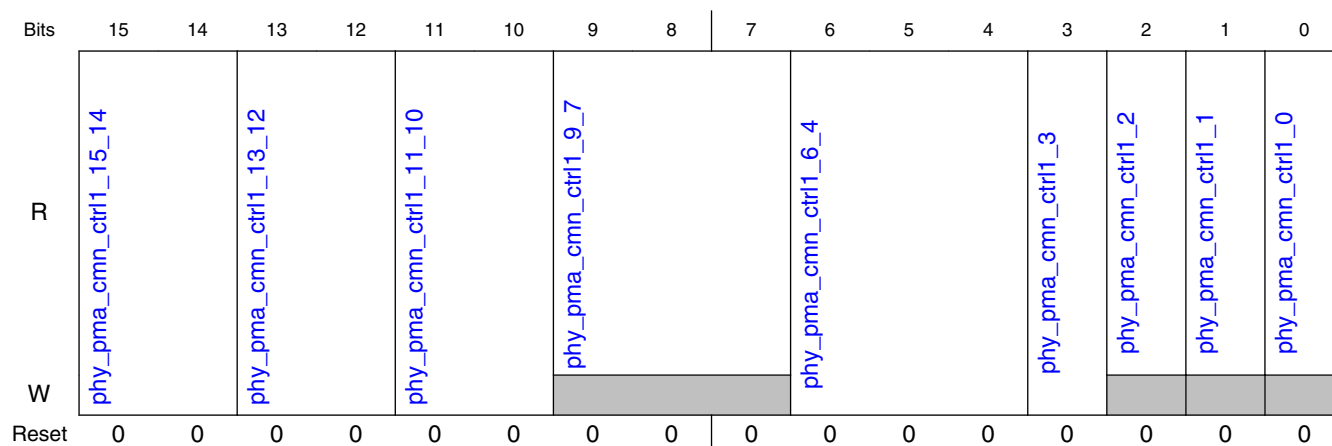
Field	Function
15-1 phy_iso_cmn_ctrl_15_1	Reserved
0 phy_iso_cmn_ctrl_0	Drives phy_reset_n PHY input when in PHY and PMA isolation modes.

13.4.10.2.572 PMA common control1 register (phy_pma_cmn_ctrl1)

13.4.10.2.572.1 Offset

Register	Offset
phy_pma_cmn_ctrl1	8_C800h

13.4.10.2.572.2 Diagram



13.4.10.2.572.3 Fields

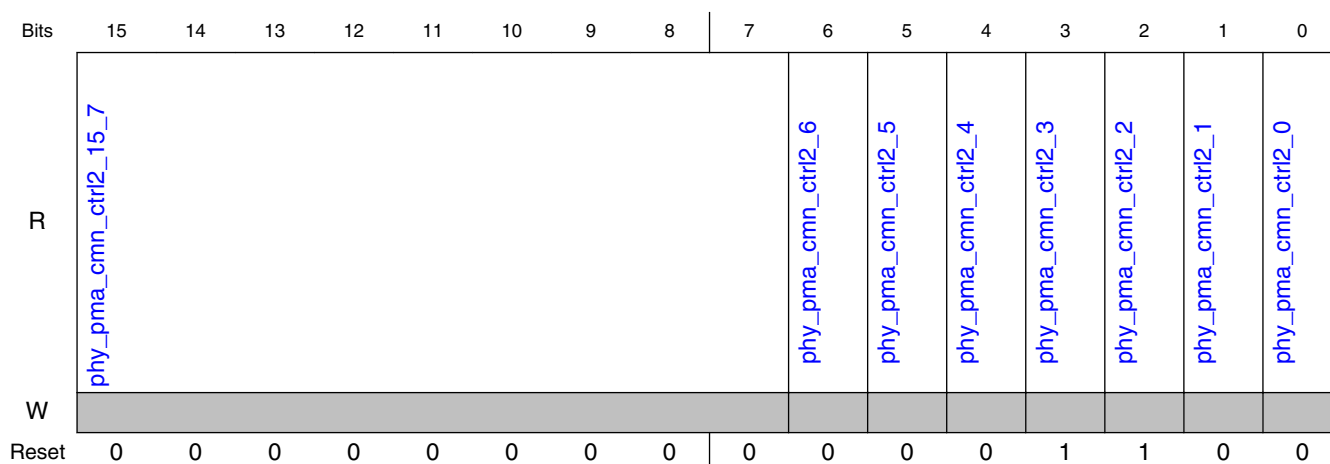
Field	Function
15-14 phy_pma_cmn_ctrl1_15_14	Drives cmn_ref_clk_ana_div PMA input
13-12 phy_pma_cmn_ctrl1_13_12	Drives cmn_ref_clk_dig_div PMA input
11-10 phy_pma_cmn_ctrl1_11_10	Drives cmn_psm_clk_dig_div PMA input
9-7 phy_pma_cmn_ctrl1_9_7	Reserved
6-4 phy_pma_cmn_ctrl1_6_4	Drives cmn_ref_clk_sel PMA input
3 phy_pma_cmn_ctrl1_3	Drives cmn_ref_clk_rcv_en PMA input
2 phy_pma_cmn_ctrl1_2	Current value of cmn_macro_suspend_ack PMA output
1 phy_pma_cmn_ctrl1_1	Current value of cmn_refclk_active PMA output
0 phy_pma_cmn_ctrl1_0	Current value of cmn_ready PMA output

13.4.10.2.573 PMA common control2 register (phy_pma_cm_n_ctrl2)

13.4.10.2.573.1 Offset

Register	Offset
phy_pma_cm_n_ctrl2	8_C801h

13.4.10.2.573.2 Diagram



13.4.10.2.573.3 Fields

Field	Function
15-7 phy_pma_cm_n_ctrl2_15_7	Reserved
6 phy_pma_cm_n_ctrl2_6	Current value of cmn_pll0_locked PMA output
5 phy_pma_cm_n_ctrl2_5	Reserved
4 phy_pma_cm_n_ctrl2_4	Current value of cmn_pll0_clk_datart_en_ack PMA output
3	Reserved

Table continues on the next page...

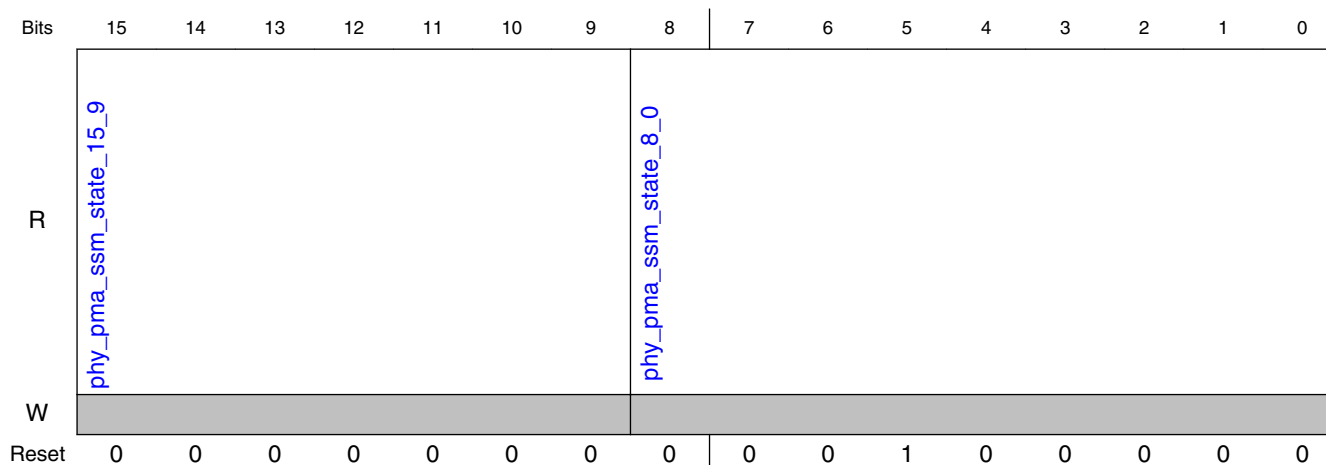
Field	Function
phy_pma_cm_n_ ctrl2_3	
2 phy_pma_cm_n_ ctrl2_2	Current value of cmn_pll0_disabled PMA output
1 phy_pma_cm_n_ ctrl2_1	Reserved
0 phy_pma_cm_n_ ctrl2_0	Current value of cmn_pll0_ready PMA output

13.4.10.2.574 PMA SSM current state register (phy_pma_ssm_state)

13.4.10.2.574.1 Offset

Register	Offset
phy_pma_ssm_state	8_C802h

13.4.10.2.574.2 Diagram



13.4.10.2.574.3 Fields

Field	Function
15-9	Reserved

Table continues on the next page...

Clocks And Resets

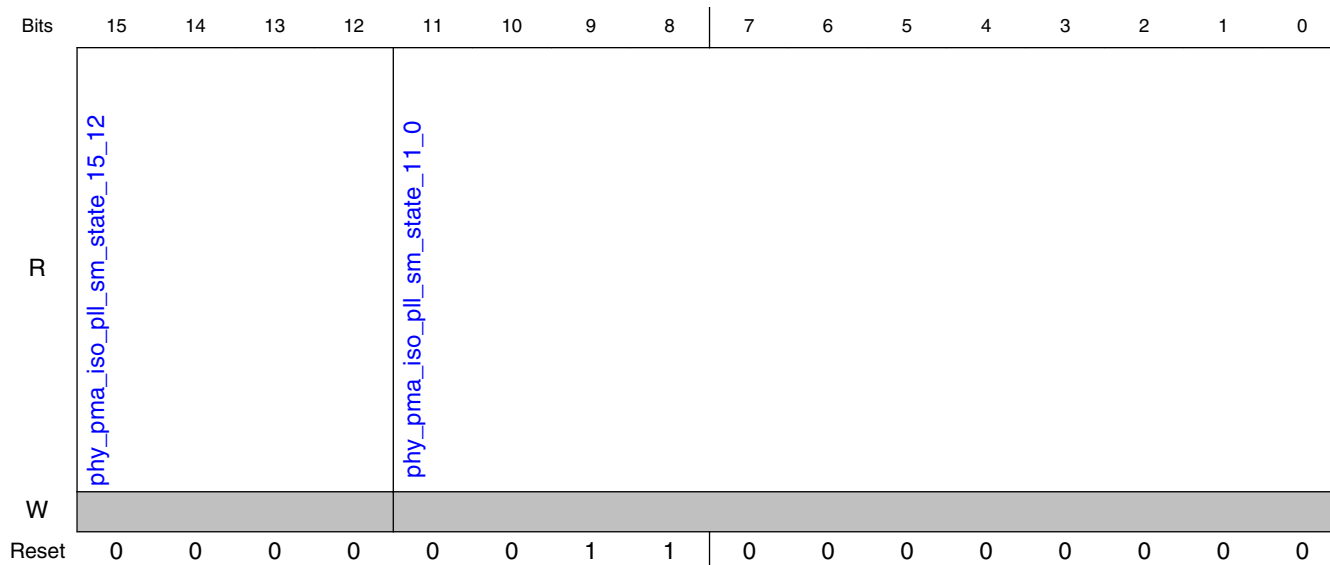
Field	Function
phy_pma_ssm_state_15_9	
8-0 phy_pma_ssm_state_8_0	PMA SSM : Current state of the PMA startup state machine. PMA output (Not re-synchronized to apb_pclk)

13.4.10.2.575 PMA PLL State Machine current state register (phy_pma_iso_pll_sm_state)

13.4.10.2.575.1 Offset

Register	Offset
phy_pma_iso_pll_sm_state	8_C803h

13.4.10.2.575.2 Diagram



13.4.10.2.575.3 Fields

Field	Function
15-12	Reserved

Table continues on the next page...

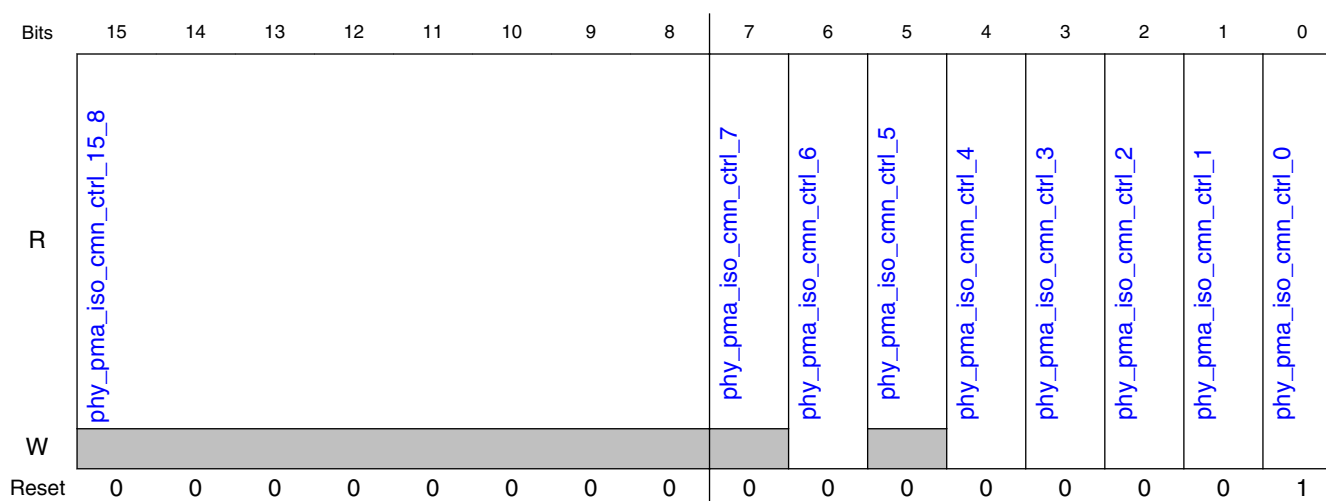
Field	Function
phy_pma_iso_pll_sm_state_15_12	Current value of cmn_pllsm0_state[11:0]. PMA output (Debug only: Not re-synchronized)
phy_pma_iso_pll_sm_state_11_0	

13.4.10.2.576 PMA common control signal isolation register (phy_pma_iso_cm_n_ctrl)

13.4.10.2.576.1 Offset

Register	Offset
phy_pma_iso_cm_n_ctrl	8_C810h

13.4.10.2.576.2 Diagram



13.4.10.2.576.3 Fields

Field	Function
15-8 phy_pma_iso_cm_n_ctrl_15_8	Reserved

Table continues on the next page...

Clocks And Resets

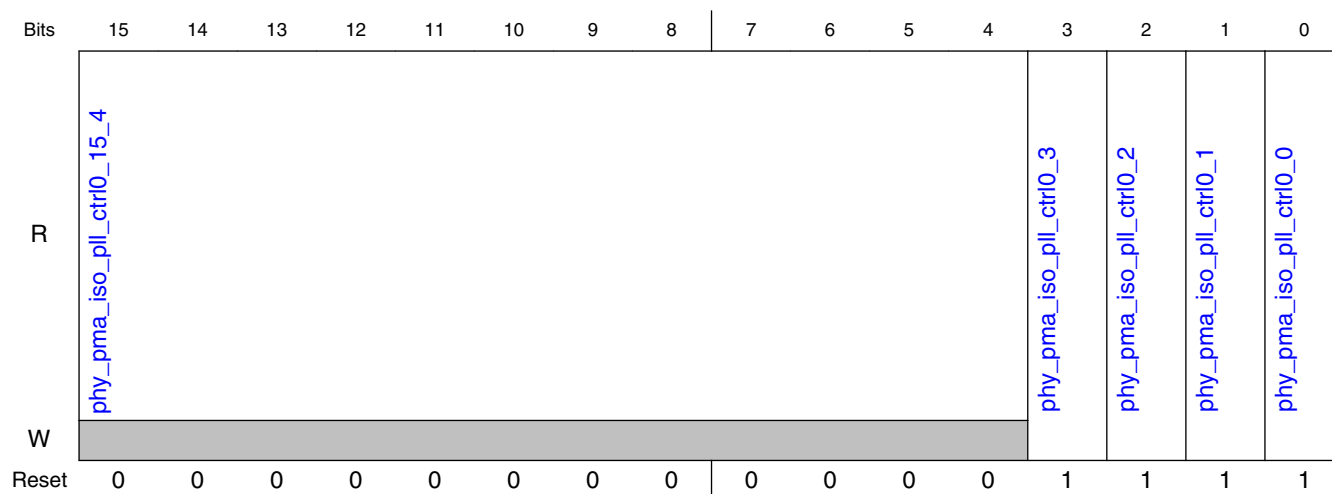
Field	Function
7 phy_pma_iso_cmn_ctrl_7	Current value of cmn_clock_stop_ack PMA output.
6 phy_pma_iso_cmn_ctrl_6	Drives cmn_clock_stop_req PMA input when in PMA isolation mode
5 phy_pma_iso_cmn_ctrl_5	Current value of cmn_macro_pwr_en_ack PMA output.
4 phy_pma_iso_cmn_ctrl_4	Drives cmn_macro_pwr_en PMA input when in PHY macro and PMA isolation modes
3 phy_pma_iso_cmn_ctrl_3	Drives cmn_refclk_disable PMA input when in PMA isolation mode.
2 phy_pma_iso_cmn_ctrl_2	Drives macro_suspend_req PMA input when in PMA isolation mode.
1 phy_pma_iso_cmn_ctrl_1	Drives cmn_macro_en PMA input when in PMA isolation mode.
0 phy_pma_iso_cmn_ctrl_0	Drives cmn_reset_n PMA input when in PMA isolation mode.

13.4.10.2.577 PMA PLL control0 isolation register (phy_pma_iso_pll_ctrl0)

13.4.10.2.577.1 Offset

Register	Offset
phy_pma_iso_pll_ctrl0	8_C811h

13.4.10.2.577.2 Diagram



13.4.10.2.577.3 Fields

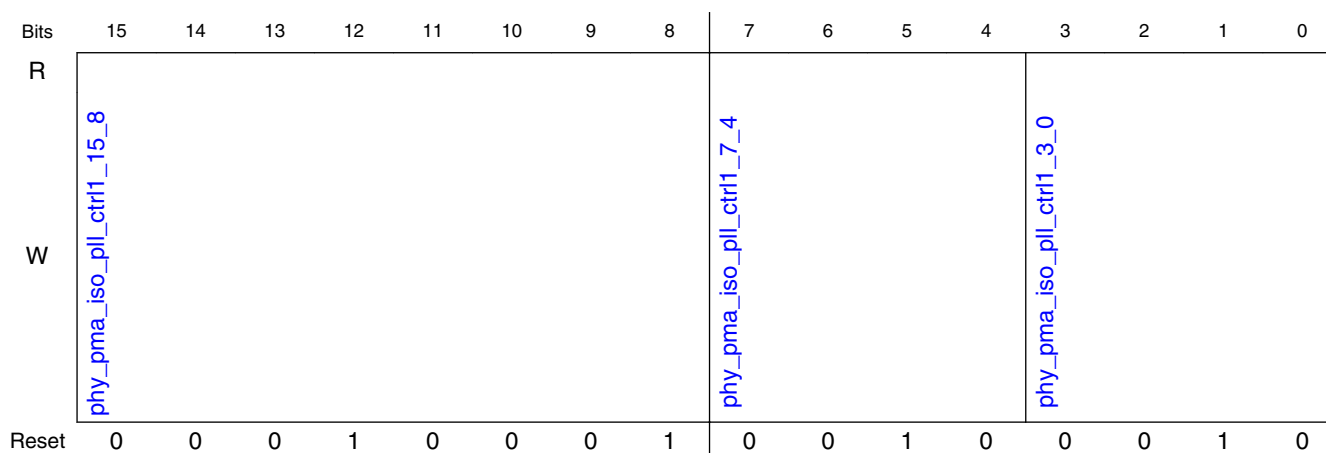
Field	Function
15-4 phy_pma_iso_pll_ctrl0_15_4	Reserved
3 phy_pma_iso_pll_ctrl0_3	Reserved
2 phy_pma_iso_pll_ctrl0_2	Drives cmn_pll0_clk_datart_en PMA input when in PMA isolation mode
1 phy_pma_iso_pll_ctrl0_1	Reserved
0 phy_pma_iso_pll_ctrl0_0	Drives cmn_pll0_en PMA input when in PMA isolation mode

13.4.10.2.578 PMA PLL control1 isolation register (phy_pma_iso_pll_ctrl1)

13.4.10.2.578.1 Offset

Register	Offset
phy_pma_iso_pll_ctrl1	8_C812h

13.4.10.2.578.2 Diagram



13.4.10.2.578.3 Fields

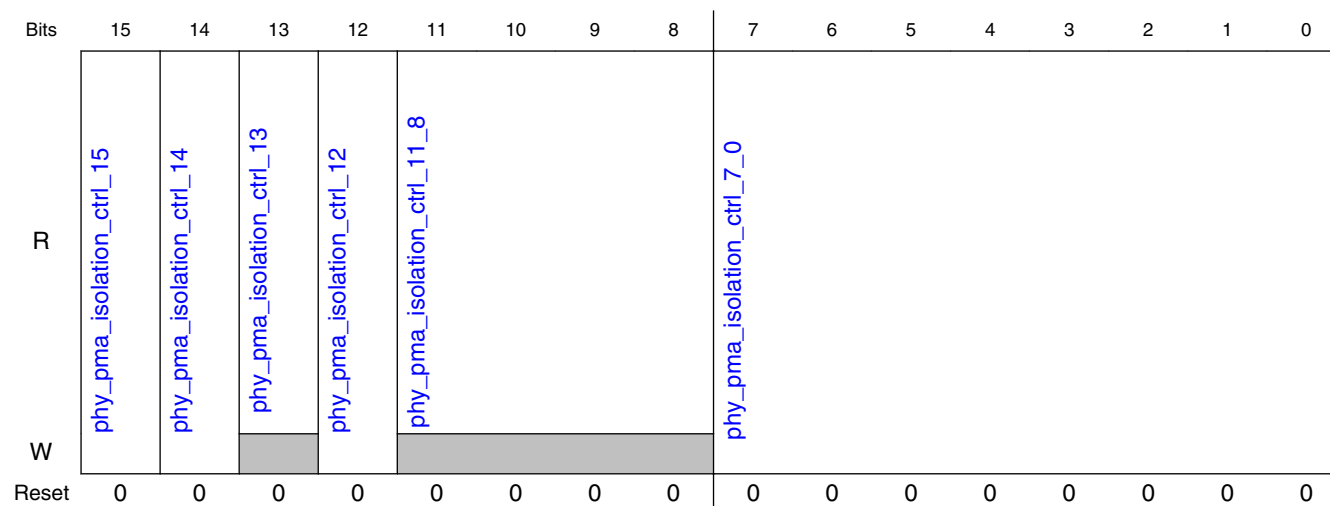
Field	Function
15-8 phy_pma_iso_pll_ctrl1_15_8	Reserved
7-4 phy_pma_iso_pll_ctrl1_7_4	Drives cmn_pll0_clk_datart1_div PMA input when in PMA isolation mode
3-0 phy_pma_iso_pll_ctrl1_3_0	Drives cmn_pll0_clk_datart0_div PMA input when in PMA isolation mode

13.4.10.2.579 Isolation control register (phy_pma_isolation_ctrl)

13.4.10.2.579.1 Offset

Register	Offset
phy_pma_isolation_ctrl	8_C81Fh

13.4.10.2.579.2 Diagram



13.4.10.2.579.3 Fields

Field	Function
15 phy_pma_isolation_ctrl_15	PHY/PMA isolation enable (isolation_en) - When set, enables isolation (PHY or PMA).
14 phy_pma_isolation_ctrl_14	PHY/PMA common isolation enable (cmn_isolation_en) - When in PHY Macro Isolation Mode, the PHY common isolation register(s) are selected. When in PMA Isolation Mode, the PMA common isolation register(s) are selected.
13 phy_pma_isolation_ctrl_13	Reserved
12 phy_pma_isolation_ctrl_12	PHY/PMA isolation mode select (isolation_mode_sel) - When isolation_en is set, this bit selects between PHY isolation and PMA isolation mode. 0 = PHY isolation mode; 1 = PMA isolation mode.
11-8 phy_pma_isolation_ctrl_11_8	Reserved
7-0 phy_pma_isolation_ctrl_7_0	PHY/PMA lane isolation enable (ln_isolation_en) - When in PHY Macro Isolation Mode, the selected PHY lane(s) isolation registers are selected. When in PMA Isolation Mode, the selected PMA lane(s) isolation registers are selected.

13.5 HDMI / DisplayPort Transmit PHY (HDMI PHY)

13.5.1 Introduction

13.5.1.1 Overview

The HDMI and DisplayPort Transmit PHY consisting of a multi-protocol PMA, AUX/ARC channel, and supporting components (APB interface, TAP controller, PHY and PMA isolation and control registers and clock and reset logic). In regards to the top level PHY specification, it is assumed that the reader of this document is familiar with these specifications.

[Figure 13-33](#) illustrates a top-level block diagram for the PHY macro, and how it interfaces with an external MAC. The following are high level points worth noting in this figure.

- The PMA isolation and control logic
 - It contains registers for controlling the PHY/PMA's power state and mode of operation.
- The Async APB Interface Slave module is part of a two module system that allows the internal CDB registers to be written and read by either the external APB interface or the integrated TAP interface. Both interfaces can access the internal CDB, with the TAP having higher priority.

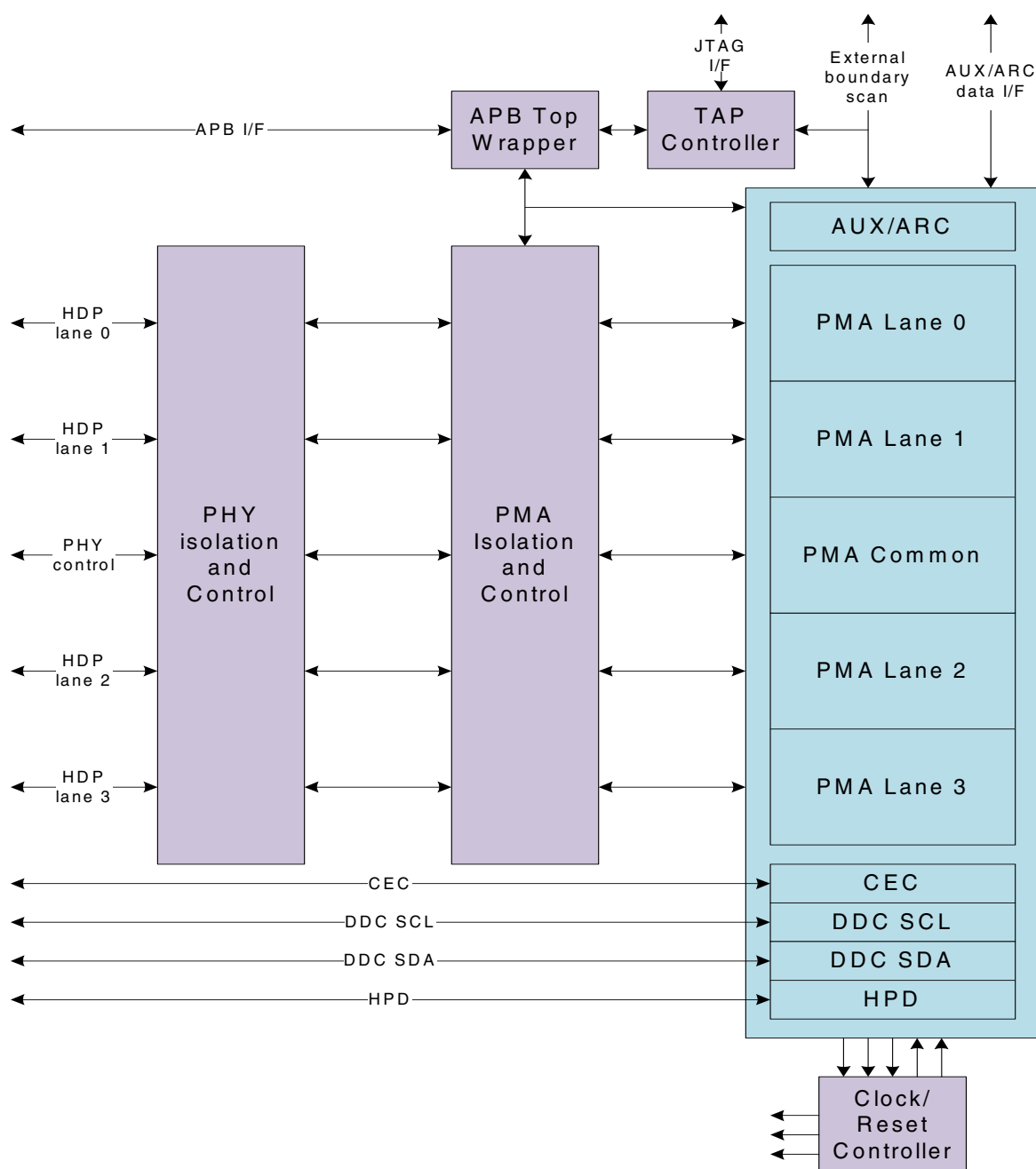


Figure 13-33. HDP PHY block diagram

13.5.1.2 Features

The features of the HDMI and DisplayPort Transmit PHY are listed below:

- Supported standards:
 - HDMI v2.0/1.4
 - DisplayPort 1.3 (RDR, HDR and HDR2 rates)
 - Embedded DisplayPort v1.4

NOTE

DP 1.3 HBR3, 8.1 Gbps, data rate not supported.

- 4 PMA TX lanes
- Supports DisplayPort AUX channel and HDMI ARC channel
- PMA and AUX/ARC
- 20-bit data interface width for both DP and HDMI
- Includes HPD, DDC SDA and SCL and CEC pads

13.5.2 External Signals

The external signals are listed in the table below:

Table 13-15. External signals

Signal Name	Direction	Type	Description
refclk_p refclk_m	input	Analog	PMA external reference clock. Only valid for DP. The HDMI pixel clock cannot be driven on these pins. For DP, supports nominal frequencies of 19.2, 20, 24, 27, 54 and 108 MHz. The following external reference clock sources are supported: • AC coupled differential low swing clock (HCSL levels) • DC single ended clock on refclk_p pin. In this mode refclk_m should be tied to ground. This mode is for test purposes only. For DP, a reference clock must be provided on either these external pins or on the refclock internal SoC-side pin.
tx_p_ln_{3:0} tx_m_ln_{3:0}	output	Diff	PMA transmitter serial data
rext	input	Analog	PMA external calibration resistor
aux_p aux_m	inout	Analog	AUX differential Tx/Rx serial data. For ARC, aux_p pin is used as input only.
hpd	inout	Analog	HDMI Hot Plug Detect
ddc_sda	inout	Analog	HDMI DDC serial data
ddc_scl	inout	Analog	HDMI DDC serial clock
cec	inout	Analog	HDMI CEC

13.5.3 On-Chip Functional Pin List

On-chip functional signals are listed in the table below:

Table 13-16. On-chip functional pin list

Signal Name	Direction	Type	Timing	Description
External APB Interface				
apb_pclk	input	CMOS	n/a	APB bus clock.
apb_preset_n	input	CMOS	async	APB bus reset.
apb_penable	input	CMOS	apb_pclk	APB bus enable.
apb_psel	input	CMOS	apb_pclk	APB bus device select.
apb_pwrite	input	CMOS	apb_pclk	APB bus write.
apb_paddr[15:0]	input	CMOS	apb_pclk	APB bus address.
apb_pwdata[15:0]	input	CMOS	apb_pclk	APB bus write data.
apb_prdata[15:0]	output	CMOS	apb_pclk	APB bus read data.
apb_pready	output	CMOS	apb_pclk	APB bus ready.
HDP Interface				
phy_reset_n	input	CMOS	async	PHY reset : Asserting this signal low will reset the PHY, with the exception of the APB registers.
phy_pma_tx_data_clk_in	input	CMOS	n/a	Tx Data clock. Transmit data clock. The Tx data is synchronous to this clock.
phy_pma_tx_data_clk_out	output	CMOS	n/a	PHY data rate clock out - 20-bit data rate clock out from PHY which provides the clock source for phy_pma_tx_data_clk_in, so that the PHY transmit interface is source synchronous.
phy_pma_tx_data_in_{3:0}[19:0]	input	CMOS	phy_pma_tx_data_clk_in	Transmit data. HDMI/Display Tx display data. Synchronous to phy_pma_tx_data_clk_in.
phy_pma_char_clk_out	output	CMOS	n/a	PHY character clock out - 10-bit free running clock from PHY. Runs at 2x frequency of phy_pma_tx_data_clk_out. Frequency aligned, but not edge aligned

Table continues on the next page...

Table 13-16. On-chip functional pin list (continued)

Signal Name	Direction	Type	Timing	Description
				(not synchronous) to phy_pma_tx_data_clk_out.
AUX / ARC Interface				
source_aux_arc_oen	input	CMOS	async	AUX/ARC output enable : 1 : Receive mode 0 : Transmit mode For ARC, the AUX/ARC block will always be in receive mode. Therefore, this pin shall be tied high for HDMI operation.
source_aux_data_in	input	CMOS	async	AUX transmit data - For HDMI Tx, ARC is only used as a receiver and thus there is not ARC transmit data.
source_aux_arc_data_out	output	CMOS	async	AUX/ARC receive data
PMA Interface				
refclock	input	CMOS	n/a	PMA internal reference clock input - Single-ended alternate reference clock input. For DP, supports nominal frequencies of 19.2, 20, 24, 27, 54 and 108 MHz. For HDMI, the pixel clock must be driven on this pin with a frequency range from 25 to 600 MHz. PHY_PMA_CMN_CTR L1[6:4] is used to select between refclock and refclk_<p/m> as the reference clock source. refclock must be selected as the reference clock source for HDMI.
pma_refclk_rcv_out	output	CMOS	n/a	PMA common reference clock receiver output - Clock output from the analog reference clock receiver circuits. Frequency is always the same as the selected reference

Table continues on the next page...

Table 13-16. On-chip functional pin list (continued)

Signal Name	Direction	Type	Timing	Description
				clock input. Once enabled, this clock is free-running and is active in all power states (provided the input reference clock is running). This clock is enabled by setting PHY_PMA_CMN_CTRL[3]. This clock is provided for use by the SoC, if needed.
HPD Interface				
hpd_in_from_pad	output	CMOS	n/a	HPD pad data in
hpd_out_high_to_pad	input	CMOS	n/a	HPD pad output drive high
hpd_out_low_to_pad	input	CMOS	n/a	HPD pad output drive low
hpd_out_drv_strength[1:0]	input	CMOS	static	HPD pad output drive strength - Selects output drive strength for HPD pad.
hpd_out_en_to_pad	input	CMOS	n/a	HPD pad output enable - 1 = output mode 0 = input mode
hpd_power_down_to_pad	input	CMOS	n/a	HPD pad power down - 1 = power down HPD pad 0 = normal mode
phy_pma_pixel_clk_out	output			PHY Pixel clock out - Pixel clock output from PHY.
DDC SDA Interface				
ddc_sda_in_from_pad	output	CMOS	n/a	DDC SDA pad in
ddc_sda_out_high_to_pad	input	CMOS	n/a	DDC SDA pad output drive high
ddc_sda_out_low_to_pad	input	CMOS	n/a	DDC SDA pad output drive low
ddc_sda_out_drv_strength[1:0]	input	CMOS	n/a	DDC SDA pad output drive strength - Selects output drive strength for DDC SDA pad.
ddc_sda_out_en_to_pad	input	CMOS	n/a	DDC SDA pad output enable -

Table continues on the next page...

Table 13-16. On-chip functional pin list (continued)

Signal Name	Direction	Type	Timing	Description
				1 = output mode 0 = input mode
ddc_sda_power_down_to_pad	input	CMOS	n/a	DDC SDA pad power down - 1 = power down DDC SDA pad 0 = normal mode
DDC SCL Interface				
ddc_scl_in_from_pad	output	CMOS	n/a	DDC SCL pad in
ddc_scl_out_high_to_pad	input	CMOS	n/a	DDC SCL pad output drive high
ddc_scl_out_low_to_pad	input	CMOS	n/a	DDC SCL pad output drive low
ddc_scl_out_drv_strength[1:0]	input	CMOS	n/a	DDC SCL pad output drive strength - Selects output drive strength for DDC SDA pad.
ddc_scl_out_en_to_pad	input	CMOS	n/a	DDC SCL pad output enable - 1 = output mode 0 = input mode
ddc_scl_power_down_to_pad	input	CMOS	n/a	DDC SCL pad power down - 1 = power down DDC SCL pad 0 = normal mode
CEC Interface				
cec_in_from_pad	output	CMOS	n/a	CEC pad in
cec_out_high_to_pad	input	CMOS	n/a	CEC pad output drive high
cec_out_low_to_pad	input	CMOS	n/a	CEC pad output drive low
cec_out_drv_strength[1:0]	input	CMOS	n/a	CEC pad output drive strength - Selects output drive strength for DDC SDA pad.
cec_out_en_to_pad	input	CMOS	n/a	CEC pad output enable - 1 = output mode 0 = input mode
cec_power_down_to_pad	input	CMOS	n/a	CEC pad power down -

Table 13-16. On-chip functional pin list

Signal Name	Direction	Type	Timing	Description
				1 = power down CEC pad 0 = normal mode

13.5.4 PHY Clocking and Reset

This section describes the details of the PHY clocking and reset functions. Note that what is described here is the top level function.

13.5.4.1 Reference and Data Path Clocking

[Figure 13-34](#) illustrates the clocking associated with the reference clock and data path clocks. The following points should be noted about this function.

- The selection of the internal (refclock) or external (refclk_p / refclk_m) clock as the source for the PMA clock, and ultimately generates cmn_pllclk_datart0 PMA clock output is controlled by a PMA APB register.
- phy_pma_tx_data_clk_in must be driven from phy_pma_tx_data_clk_out. The DP/HDMI transmit data for all lanes (phy_pma_tx_data_ln_<>[19:0]) is synchronous to phy_pma_tx_data_clk_in.

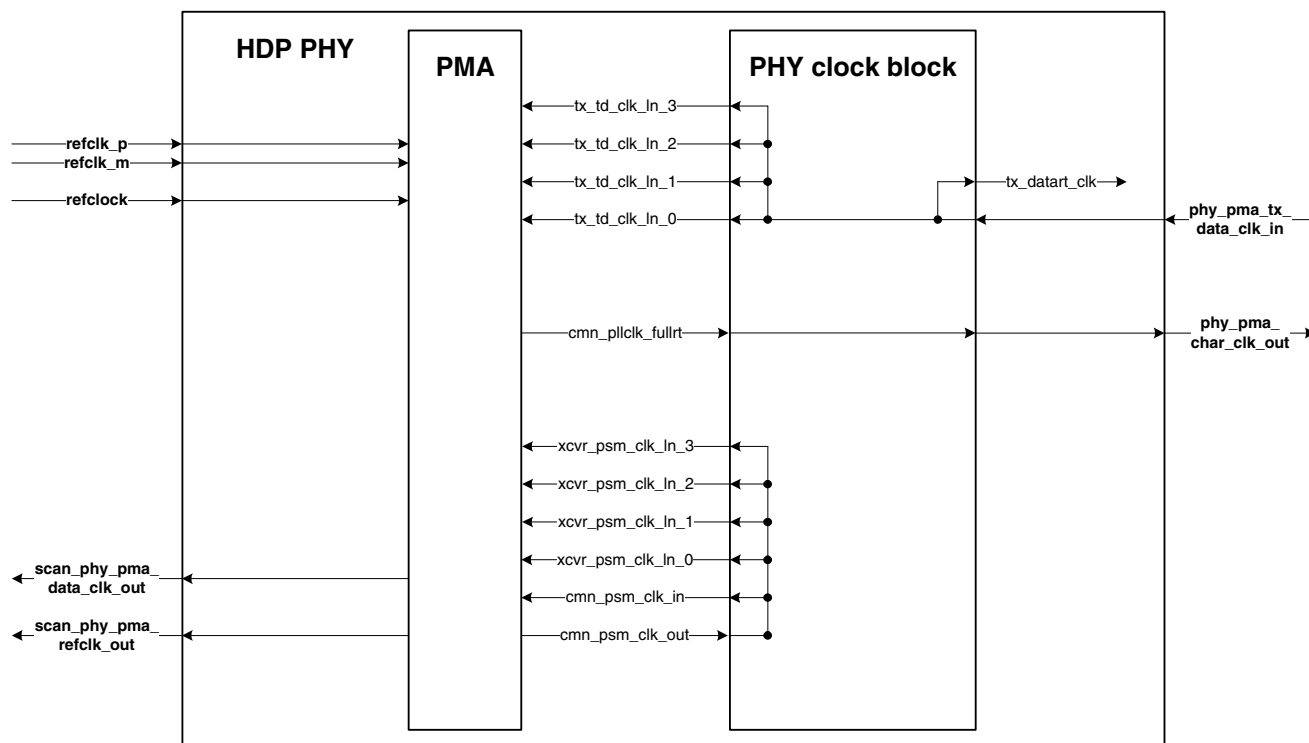


Figure 13-34. PHY clocking block diagram

13.5.4.2 Reset Distribution

All resets are active low, asynchronously activated, and synchronously deactivated (synchronized internally).

phy_reset_n is master reset for entire PHY. Asserting phy_reset_n will drive reset to entire PHY, except for APB and TAP controller registers.

When phy_reset_n is released, a reset sequencer controls the release of the various resets for the entire macro, as follows:

- After phy_reset_n is released, cmn_reset_n going to the PMA is released.
- The PHY reset sequencer waits for cmn_ready from the PMA to go active, indicating common modules are powered up, calibrated, and the PLL has locked.
- At this point, all other resets going to logic in the PHY are synchronously released (1 for each internal PHY clock domain).

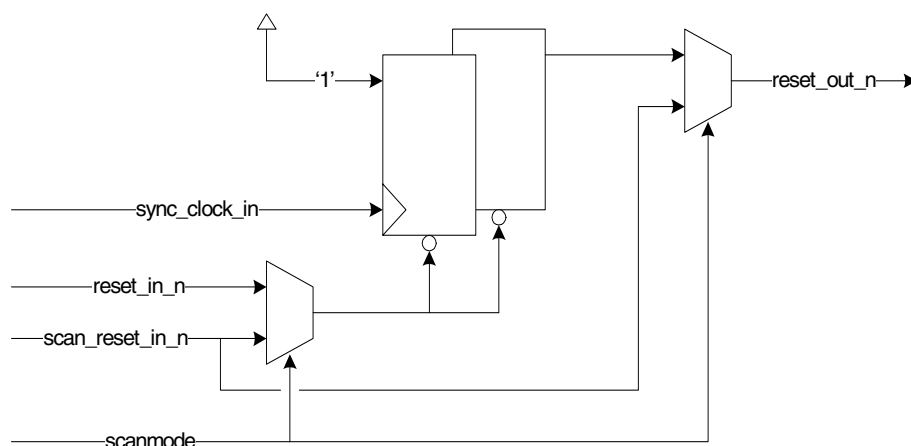


Figure 13-35. Reset synchronizer

13.5.4.3 PHY Clocking

The following table shows the clock names and their associated frequencies in PHY.

Table 13-17. Clock and frequency details

Clock Name	Maximum Frequency (MHz)
refclk_p	600
refclk_m	600
apb_pclk	200
phy_pma_tx_data_clk_in	300
phy_pma_tx_data_clk_out	300
phy_pma_char_clk_out	600
refclock	600
pma_refclk_rcv_out	600
tap_tck	100
scan_phy_pma_refclk_in	200
scan_phy_pma_data_clk_in	300
scan_phy_pma_refclk_out	600
scan_phy_pma_data_clk_out	300
tx_bscan_ext_clockdr	100
tx_bscan_ext_updatedr	100
phy_pma_pixel_clk_out	600

13.5.4.4 Reset Pins

The following table shows the reset names and their polarities.

Table 13-18. Reset pins and their polarities

Reset Name	Polarity
apb_preset_n	Active Low
phy_reset_n	Active Low
tap_trst_n	Active Low

13.5.5 PHY Register Read through TAP

- As required, write PHY macro diagnostic control register with 0x3 to source APB clock from TAP clock.
- Write address of APB register to PHY macro diagnostic APB access address register.
- Write 0x2 to PHY macro diagnostic APB access control register.
- Read PHY macro diagnostic APB access control register. If [0] is set high, continue. Otherwise, read PHY macro diagnostic APB access control register again.
- Read PHY macro diagnostic APB access read data register, which contains the read data.

13.5.6 PHY Register Write through TAP

- As required, write PHY macro diagnostic control register with 0x3 to source APB clock from TAP clock.
- Write address of APB register to PHY macro diagnostic APB access address register.
- Write data to be written to PHY macro diagnostic APB access write data register.
- Write 0x6 to PHY macro diagnostic APB access control register.
- Read PHY macro diagnostic APB access control register. If [0] is set high, the write access is complete. Otherwise, read PHY macro diagnostic APB access control register again.

13.5.7 Functional Description

This section describes PHY functional operation, such as start-up, disable, transmitting data, changing power state and changing the data rate.

13.5.7.1 Start-up Sequence Timing Diagram

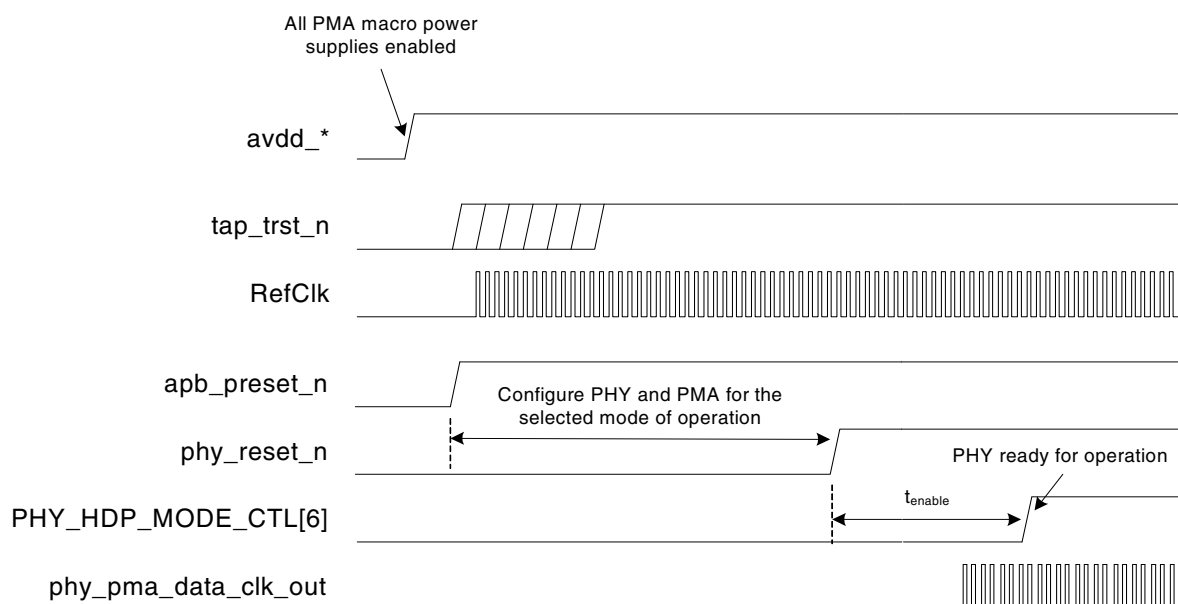


Figure 13-36. Start-up Sequence

Prior to de-assertion of phy_reset_n, write PHY_HDP_MODE_CTL[3:0] with 0x4 (0b0100). Upon start-up, the initial power state for each PMA lane must be A2. The PHY indicates ready for operation upon assertion (high) of the corresponding PHY_HDP_MODE_CTL[6] acknowledgement bit.

13.5.7.2 PHY Disable Description

To disable the PHY, assert apb_preset_n and phy_reset_n low and disable PMA reference clock (refclk_<p/m>/refclock) and apb_pclk. Note: apb_preset_n must be asserted at the same time as or after phy_reset_n is asserted. To re-enable the PHY follow the start-up sequence described above.

13.5.7.3 DP Transmit Data Path Timing Diagram

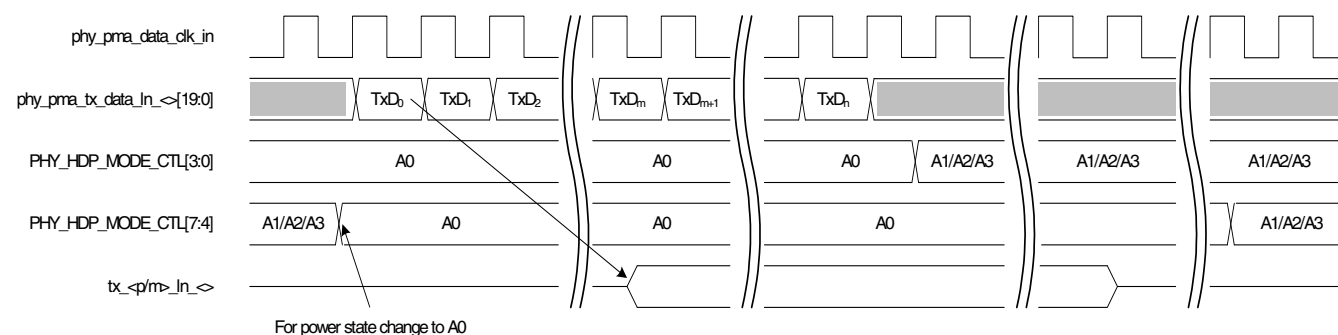


Figure 13-37. DP transmit data path

The HDP PHY must be in the A0 power state in order to transmit data. When in the A0 power state, the data provided on `phy_pma_tx_data_ln_*` is transmitted on `tx_<p/m>_ln_*`. When exiting the A0 power state, there is a race condition between the last data provided on `phy_pma_tx_data_ln_*` traversing the PMA and the PMA changing its power state. This may result in this data not being transmitted. Therefore, it is recommend to send a minimum of 4 dummy words into the PHY after the last valid data word prior to changing the PHY's power state.

13.5.7.3.1 HDMI TMDS Clock Lane

For the HDMI TMDS clock lane, the 20-bit data pattern depends on the HDMI data rate.

- For HDMI 1.4 data rates, a fixed 20-bit data pattern shall be driven on `phy_pma_tx_data_ln_<>` : 20'b1111_1000_0011_1110_0000 (5 1s, 5 0s, 5 1s, 5 0s).
- For HDMI 2.0 data rates, a repeating pattern that toggles every `phy_pma_tx_data_clk_in` clock cycle shall be driven on `phy_pma_tx_data_ln_<>` : toggle between 20'hF_FFFF and 20'h0_0000 (repeating all 1s, then all 0s).

13.5.7.4 Power State Change

The procedure for changing the PHY's power state is:

- Write `PHY_HDP_MODE_CTL[3:0]` with one-hot encoded value for the new power state (0b0001 = A0, 0b0010 = A1, 0b0100 = A2 and 0b1000 = A3).
- Wait for `PHY_HDP_MODE_CTL[7:4] = value` for the selected power state.

A0 is the active power state. A1, A2 and A3 are various power down states, with A1 having the fastest recovery to A0 time, but the lowest power savings and A3 having the slowest recovery time, but largest power savings.

13.5.7.5 Data Rate/DP Lane Configuration Change

The procedure for changing the PHY's power state and/or the DisplayPort lane configuration is:

- Write PHY_HDP_MODE_CTL[3:0] with 0b1000. (Place the PHY lanes in the A3 power state.)
- Wait for PHY_HDP_MODE_CTL[7:4] == 0b1000.
- Clear PHY_HDP_CLK_CTL[2]. Gate the PLL clocks from PMA.
- Wait for PHY_HDP_CLK_CTL[3] == 0.
- Clear PHY_HDP_CLK_CTL[0]. Disable the PLL.
- Wait for PHY_HDP_CLK_CTL[1] == 0.
- Re-configure PMA registers for the new data rate (as defined in the PMA specification / programmer's guide) and/or DP lane configuration.
- Set PHY_HDP_CLK_CTL[0]. Enable the PLL.
- Wait for PHY_HDP_CLK_CTL[1] == 1.
- Set PHY_HDP_CLK_CTL[2]. Enable PMA PLL clocks.
- Wait for PHY_HDP_CLK_CTL[3] == 1.
- Write PHY_HDP_MODE_CTL[3:0] with 0b0100 (A2 power state). The PMA must go through the A2 power state upon a data rate change.
- Wait for PHY_HDP_MODE_CTL[7:4] == 0b0100.
- As required, change the PHY power state to A0.

13.5.7.6 Lane Mapping

For the HDP PHY, the PHY lanes have a fixed mapping to the PMA lanes as follows:

- PHY HDP lane 0 → PMA lane 0 (master lane)
- PHY HDP lane 1 → PMA lane 1
- PHY HDP lane 2 → PMA lane 2
- PHY HDP lane 3 → PMA lane 3

13.5.7.7 PMA TX Sync FIFO control

Each PMA lane outputs the following Tx FIFO control signals:

tx_sfifo_enq_reset_n_out_ln_<> and tx_sfifo_align_bump_out_ln_<>. Similarly, each PMA lane has a tx_sfifo_enq_reset_n_in_ln_<> and tx_sfifo_align_bump_in_ln_<>.

These signals are used to properly align the Tx Sync FIFOs for lanes combined into multi-lane links in order to meet the Tx inter-lane skew requirements. There are 2

aspects to this: (1) the Tx FIFOs for each DP lane must be released from reset on the same clock cycle (through use of a common, re-synchronized `tx_sfifo_enq_reset_n_out_ln_<>` signal) and (2) the write pointers must be adjusted by the same amount offset from the read pointer by a common `tx_sfifo_align_bump_in_<>` signal.

To accomplish this, one of the PMA lanes must be selected as the master lanes. The lowest numbered enabled PMA lane (as configured in `PHY_DP_MODE_CTL[15:12]`) is used as the master lane.

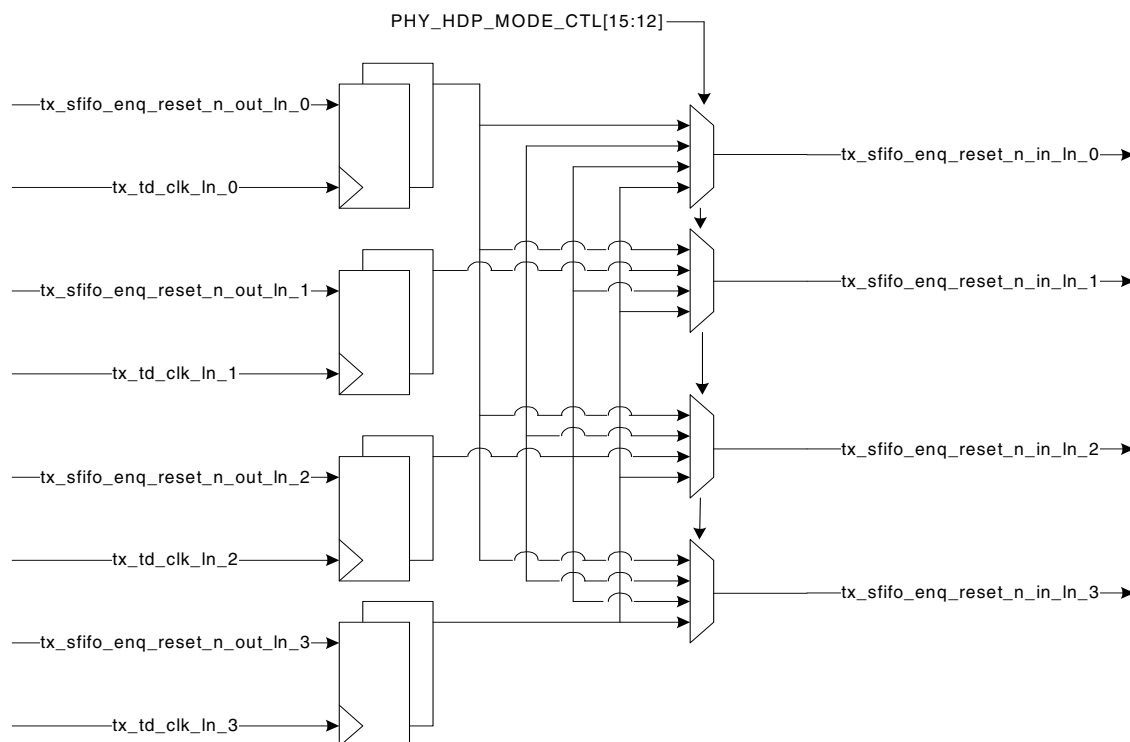


Figure 13-38. tx_sfifo_enq_reset_n_in_ln_<> generation

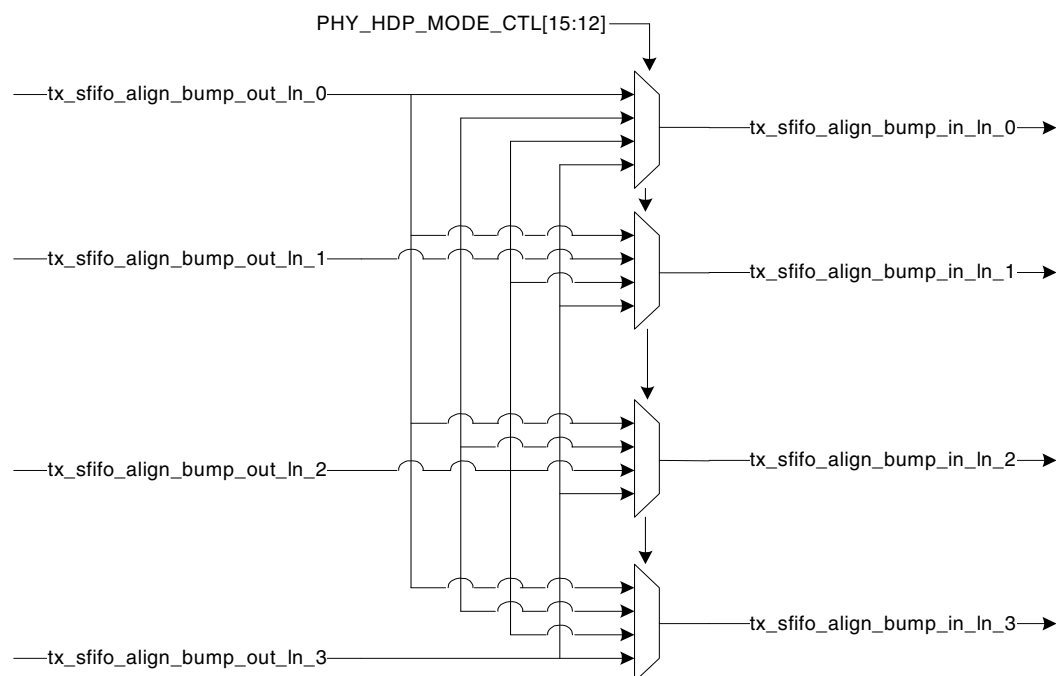


Figure 13-39. tx_sfifo_align_bump_in_ln_<> generation

13.6 MIPI DSI Host Controller (MIPI_DSI)

13.6.1 Overview

The Mobile Industry Processor Interface (MIPI) Display Serial Interface (DSI) controller is a flexible, high-performance, and easy-to-use digital core that implements all protocol functions defined in the MIPI DSI Specification. The MIPI DSI controller provides an interface that allows communication with MIPI DSI-compliant peripherals.

The MIPI DSI D-PHY is a high frequency, low power, low-cost, source-synchronous, physical layer supporting the MIPI Alliance standard for D-PHY.

13.6.2 Features

Key features of the MIPI DSI Controller Core include:

- Implements all three DSI Layers (Pixel to Byte packing, Low Level Protocol, Lane Management)
- Support for Command and Video Modes
- Host Version
- Scalable data lane support, 1 to 4 Data Lanes

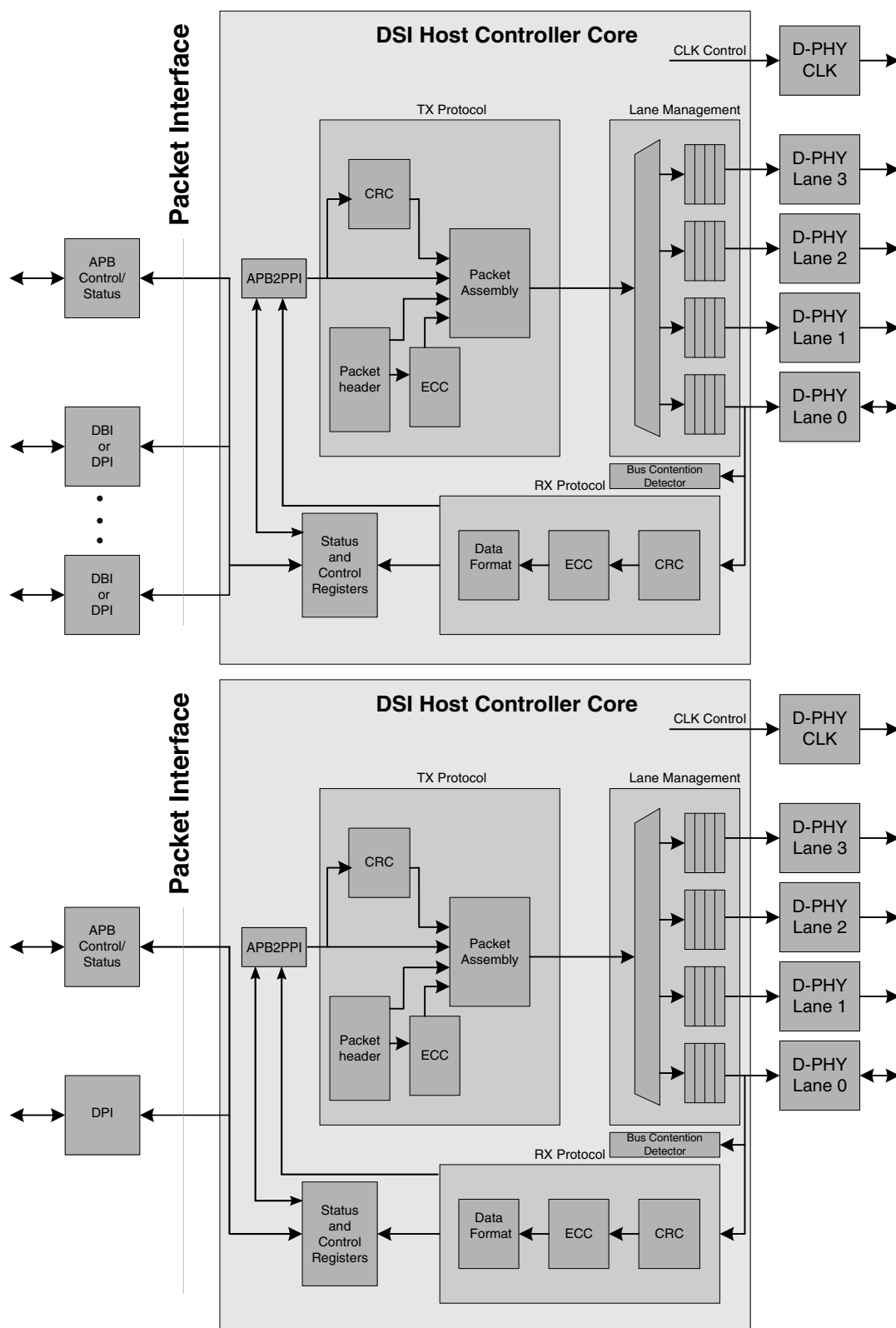
- Optional bidirectional support on lane 0
- Supports High Speed and Low Power operation
- Support for all DSI data types and formats
- Virtual Channel support
- Supports ULPS mode
- Full Low-Level Protocol Error and Contention detection and reporting
- Supports continuous and non-continuous Clock Lane operation
- Supports multiple packets per transmission
- Support for all three Video Mode packet sequences
 - Non-Burst Mode with Sync Pulses
 - Non-Burst Mode with Sync Events
 - Burst mode
- Support for bus turnaround signaling
- Flexible packet based user interface
 - APB interface option (status and control)
 - Display Pixel Interface Core (DPI-2) option
 - Display Bus Interface Core (DBI-2) option
- Supports PHY Protocol Interface (PPI) compatible MIPI D-PHYs
- MIPI Alliance Specification for Display Serial Interface Version 1.1 compliant

The following are the key features of the MIPI DSI D-PHY:

- 1 Clock lane
- 4 data lanes
- Supports MIPI Standard for D-PHY
- Supports both high speed and low power modes
- High Speed Serializers and Deserializers
- Resistance termination calibrator

13.6.3 DSI Host Controller Core

The figure below illustrates the DSI Host Controller Core structure. The DSI Host Controller Core operates on the host (transmit) side of a DSI link.



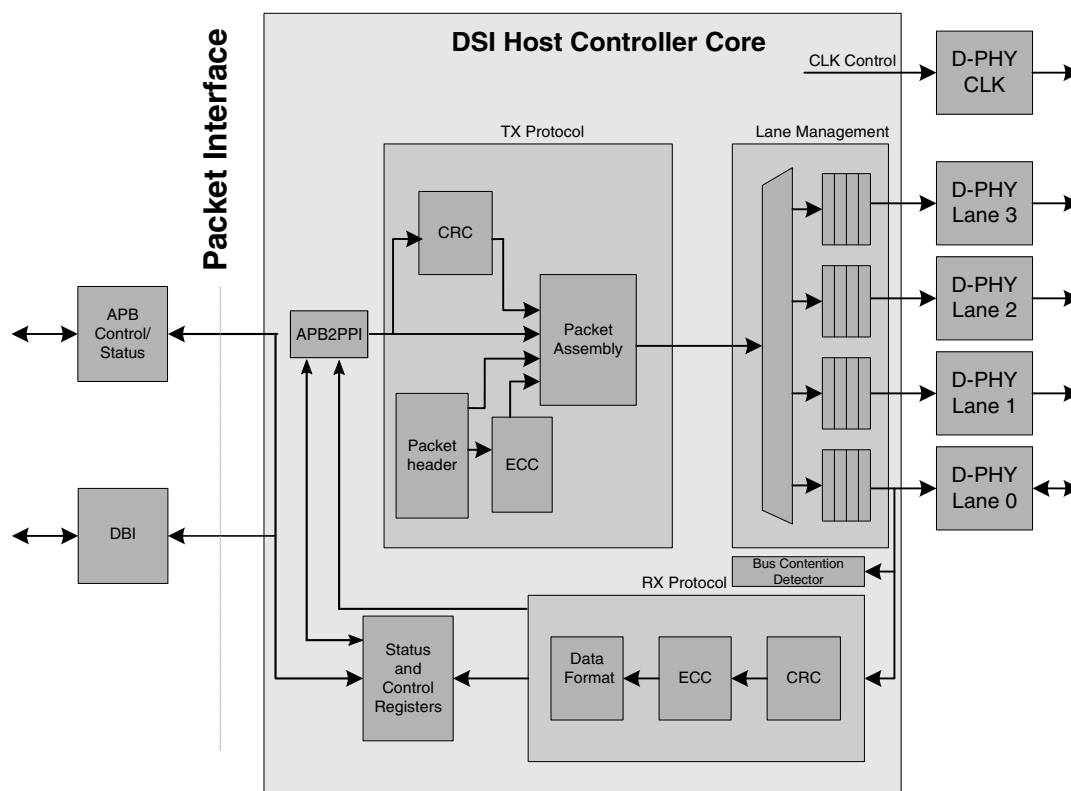


Figure 13-40. DSI Host Controller Core Block Diagram

The DSI Host Controller Core implements all three layers defined by the DSI Specification: Pixel to Byte Packing in the Application layer, Low Level Protocol, and Lane Management. The DSI Host Controller Core sends and receives DSI commands via the Packet Interface. The Packet Interface can be connected to a DBI/DPI translator or to an APB to PPI gasket.

The D-PHY interface of the DSI Host Controller Core supports up to four PHY Protocol Interface (PPI) compatible MIPI D-PHYs.

The Packet Interface is an easy-to-use data interface that accepts commands and data, and sends it over the DSI link. It supports 1 to 4 virtual channels, and the use of 1-4 D-PHY lanes. The DSI Host Controller Core takes care of all packet formatting details and transmission over the MIPI bus.

The DBI/DPI Translator connects to the DSI Host Controller Core via the Packet Interface. DBI/DPI masters may connect directly to the DBI/DPI Translator to send commands across the DSI link.

Sending commands and receiving information through the DSI link is also possible through the APB to PPI gasket. See MIPI DSI Host APB PKT IF Memory Map/Register section for details about the DSI registers which allow the APB to PPI setup and communication.

13.6.3.1 D-PHY Interface

The DSI Host Controller Core D-PHY Interface connects directly to MIPI PPI compliant D-PHYs. The DSI Host Controller Core uses this interface to transmit data, send Escape sequences, receive and transmit triggers, and detect and report D-PHY error conditions.

13.6.3.1.1 High-Speed Transmit Interface

The High-Speed Transmit interface is used to transmit High-Speed data across the MIPI interface. All signals are synchronous to the PPI TX byte clock, TxByteClkHS, and are PPI compliant.

13.6.3.1.2 Escape Mode Transmit Interface

The DSI Host Controller Core uses the Escape Mode Transmit Interface to generate Escape sequences on the MIPI Interface via the D-PHY. Escape sequences are used to change the DPHY transmit mode, go into low-power mode, or initiate a bus turnaround. The Controller generates Escape sequence requests when necessary to implement the DSI protocol. These signals are synchronous to the clk_esc.

13.6.3.1.3 Escape Mode Receive Interface

The DSI Host Controller Core uses the Escape Mode Receive Interface to receive Escape sequences on the MIPI Interface via the D-PHY. The Controller receives Escape sequence requests and Low Power data when necessary to implement the DSI protocol.

The DSI protocol uses Lane 0 in LP mode for returning data from the peripheral to the host. Receive Interface signals are only applicable to the MIPI D-PHY Lane 0. These signals are synchronous to the RxClkEsc_ln0.

13.6.3.2 Packet Interface

The DSI Host Controller Core Packet Interface consists of Transmit, Receive, and Control and Status sections. Through these interfaces, the user application can take complete control of the DSI interface, sending all video timing, sending DSI commands, receiving DSI reads, monitoring the status of the interface and responding to error reporting.

User application may have access to DSI Host Controller Core Packet Interface through DSI_HOST_APB_PKT_IF registers. For details refer to the section [MIPI_DSI_HOST_APB_PKT_IF](#)

13.6.3.2.1 Transmit Ports

The Transmit Packet Interface is the mechanism with which the user creates packets to send over the MIPI Interface.

For Long Packets, the user provides the Virtual Channel (VC) number, Data Type (DT), and Word Count (WC) to the controller. The Controller then creates a packet header and pulls the packet data from the Packet Interface and out to the D-PHY to transmit.

For Short Packets, the user provides the Virtual Channel (VC) number, Data Type (DT), and required parameters (if any) to the controller. The Controller then creates the short packet and sends it to the D-PHY to transmit. This interface enables the user application to transmit and receive any type of DSI packet.

The Packet Interface Transmit Interface ports are listed in the table below. Signals are synchronous to the Tx Byte clock.

NOTE

Only a subset of the Packet Interface Signals are accessible by user application through DSI_HOST_APB_PKT_IFx registers or chip-specific registers. For reset_byte_n and reset_esc_n signals, refer to chip-specific MIPI DSI information. For tx_payload, tx_cmd_data_type, tx_cmd_vc, and tx_cmd_byte_count signals, refer to DSI_HOST_APB_PKT_IFx registers.

Table 13-19. DSI Host Controller Core Transmit Packet Interface

PORT	TYPE	DESCRIPTION
clk_byte	Input	Byte clock input. The D-PHY PPI interface, the tx_cmd and tx_payload interfaces are synchronous to clk_byte. This clock can be independent and unrelated to clk_esc.
clk_esc	Input	The low-speed D-PHY Escape Mode clock, which becomes TxClkEsc on the D-PHY interface. This clock can be independent and unrelated to clk_byte.
reset_byte_n	Input	Asynchronous reset, active low. This reset applies to all logic in the clk_byte clock domain.
reset_esc_n	Input	Asynchronous reset, active low. This reset applies to all logic in the clk_esc clock domain.
tx_payload[31:0]	Input	Packet data input.

Table continues on the next page...

Table 13-19. DSI Host Controller Core Transmit Packet Interface (continued)

PORT	TYPE	DESCRIPTION
tx_payload_en	Output	Packet data read enable. This active high signal indicates that the controller requires a valid packet during the next clk_byte period.
tx_payload_en_last	Output	Last packet read enable, active high signals last cycle of tx_payload_en.
tx_cmd_data_type[5:0]	Input	Transmit packet DSI data type. It is written into the command buffer when tx_cmd_ack is asserted high.
tx_cmd_vc[1:0]	Input	Transmit packet command virtual channel. It is written into the command buffer when tx_cmd_ack is asserted high.
tx_cmd_byte_count[15:0]	Input	Transmit packet payload byte count. It is written into the command buffer when tx_cmd_ack is asserted high. For DSI Long packet types, tx_cmd_byte_count defines the number of bytes of packet data to pull from the tx_payload port. For DSI Short packets, the format of tx_cmd_byte_count contains any optional parameters. If the SDI Short packet type does not have any parameters, it is recommended to set tx_cmd_byte_count to all 0s.
tx_cmd_req	Input	Transmit packet command request. This active high signal informs the controller that the packet command is valid. The packet command consists of the ports tx_cmd_data_type, tx_cmd_vc, and tx_cmd_byte_count. The controller will assert tx_cmd_ack when it accepts the command, after which, the user should either update port values for the next transmit packet command or deassert tx_cmd_req.
tx_cmd_ack	Output	Transmit packet command request acknowledge. This active high signal indicates that the controller has accepted the TX packet request and the user logic should either submit a new request or deassert tx_cmd_req on the next rising edge of clk_byte.
trigger_req	Input	Transmit trigger request. This active high signal informs the controller that the trigger number on trigger_send is valid. The controller will assert trigger_ack when it accepts the command, after which, the user should either put update trigger_ack with the values for the next transmit packet or deassert trigger_req.
trigger_ack	Output	Transmit trigger request acknowledge. This active high signal indicates that the controller has accepted the trigger request and the user logic should either submit a new request or deassert trigger_req on the next rising edge of user_clk.
trigger_send[1:0]	Input	Transmit trigger. The trigger number on trigger_send is sampled when trigger_ack is asserted high. The format of trigger_send is as follows: 1'b00 = Trigger 0 (Reset-Trigger) 1'b01 = Trigger 1 ([Reserved]) 1'b10 = Trigger 2 ([Reserved]) 1'b11 = Trigger 3 ([Reserved])

13.6.3.2.2 Receive Ports

The Receive Packet Interface returns data from the Peripheral to the user. The user is provided the Virtual Channel (VC) number, Data Type (DT), Word Count (WC), and Rx Payload Data.

This interface enables the user application to receive any type of DSI packet. These signals are synchronous to the `clk_byte` clock. The Packet Interface Receive Interface ports are listed in the table below.

NOTE

Only a subset of the Packet Interface signals are accessible by user application through `DSI_HOST_APB_PKT_IFx` registers and `DSI_HOST` device registers:

- `rx_payload`
- `rx_cmd_vc`
- `rx_cmd_data_type`
- `rx_cmd_byte_count`
- `ecc_one_bit_error`
- `ecc_two_bit_error`
- `ecc_one_bit_error_pos[4:0]`
- `ecc_err`
- `ecc_err_pos[2:0]`
- `crc_err`

See `DSI_HOST_APB_PKT_IFx` and `DSI_HOST` device sections for details.

Table 13-20. DSI Host Controller Core Receive Packet interface

PORT	TYPE	DESCRIPTION
<code>rx_payload[31:0]</code>	Output	Received packet data output. The Host Receive Packet Interface presents 4 bytes at a time. Bytes are valid in this interface according to the <code>rx_cmd_byte_count</code> signal, beginning with the lowest byte.
<code>rx_payload_valid</code>	Output	Packet data valid. This active high signal indicates that the controller is presenting valid packet during the next <code>clk_byte</code> period.
<code>rx_payload_valid_last</code>	Output	This data is the last of the packet, active high signals last cycle of <code>rx_payload_valid</code> .
<code>rx_cmd_valid</code>	Output	Packet header data is valid on the packet header ports below when this signal is asserted.
<code>rx_cmd_vc[1:0]</code>	Output	Packet virtual channel number, valid when <code>rx_cmd_valid</code> is asserted.
<code>rx_cmd_data_type[5:0]</code>	Output	Packet data type, valid when <code>rx_cmd_valid</code> is asserted. See the MIPI DSI-2 specification for a definition of possible values.

Table continues on the next page...

Table 13-20. DSI Host Controller Core Receive Packet interface (continued)

PORT	TYPE	DESCRIPTION
rx_cmd_byte_count[15:0]	Output	Packet Word Count (byte count). Contains the number of bytes of data in the received packet. Valid when rx_cmd_valid is asserted.
ecc_one_bit_error	Output	Single bit error in the packet header was detected and corrected. Active high. Valid when rx_cmd_valid is high.
ecc_two_bit_error	Output	Two packet header bit errors were detected and not corrected, active high. Valid when rx_cmd_valid is high.
ecc_one_bit_error_pos[4: 0]	Output	Position of the corrected single bit error in the packet header. Valid when ecc_one_bit_error is high.
ecc_err	Output	Error detected in the ECC bits. Active high. Valid when rx_cmd_valid is high.
ecc_err_pos[2:0]	Output	Position of the erroneous bit in the ECC bits, valid when ecc_err is asserted.
crc_err	Output	Asserts high when the CRC calculated on the received data does not reach the end of the packet.

13.6.3.3 Status Interface Ports

The Status Interface Core provides for reading the Status of the DSI Host Controller Core via a simple interface. All signals in the Status Interface are synchronous to the clk_byte clock. The Status Interface ports are listed in the table below.

NOTE

Only a subset of the Packet Interface signals are accessible by user application through DSI_HOST_APB_PKT_IFx registers and DSI_HOST device registers:

- status_out
- hs_tx_timeout
- lp_rx_timeout

Refer to DSI_HOST_APB_PKT_IFx and DSI_HOST device sections for details.

Table 13-21. DSI Host Controller Status Interface Port

PORT	TYPE	DESCRIPTION
status_rd	Input	Initiates a read of the status register. Asserting this input high will result in the contents of the status register being presented on status_out[15:0] along with status_out_valid asserting high to indicate that status_out[15:0] contains a valid value.

Table continues on the next page...

Table 13-21. DSI Host Controller Status Interface Port (continued)

PORT	TYPE	DESCRIPTION
status_addr[3:0]	Input	When status_rd is asserted, indicates which status register is to be returned. Address 0x0 is a valid address while 0x1-0xF are reserved.
status_out[31:0]	Output	Contains the contents of the addressed status register.
status_out_valid	Output	When asserted high, indicates that the data on status_out is valid.
tx_active	Output	tx_active asserts high when the Host Controller is actively transmitting data or when it has accepted a request from the user but has not yet started transmitting.
hs_tx_timeout	Output	Asserts high for one clk_byte period to indicate that a High Speed transmit has timed out.
lp_rx_timeout	Output	Asserts high for one clk_byte period to indicate that a Low Power RX curred.

13.6.3.4 Control and Status Ports

The Control and Status Packet Interface allows the user to control and receive the status of the underlying link between the DSI Host and DSI Peripheral. It works on the same clk_byte clock as the Host Transmit Packet Interface. The Packet Interface Control and Status Interface ports are listed in the table below.

NOTE

Only a subset of the Packet Interface signals may be accessible by user application through DSI_HOST_APB_PKT_IFx registers and DSI_HOST device registers:

- dphy_direction
- tx_hs_mode
- dphy_turnaround

Please refer to DSI_HOST_APB_PKT_IFx and DSI_HOST device sections for details.

Table 13-22. DSI Host Controller Core Control And Status Packet Interface

PORT	TYPE	DESCRIPTION
tx_hs_mode	Input	Switches the DPHY into High Speed Data Transfer mode or Low Power Data Transfer mode. 1'b1 = request HS mode 1'b0 = request LP mode. The Packet interface will not acknowledge packet commands or data while switching modes.

Table continues on the next page...

Table 13-22. DSI Host Controller Core Control And Status Packet Interface (continued)

PORT	TYPE	DESCRIPTION
dphy_turnaround	Input	Requests bus turnaround. 1'b1 = Request reverse direction LP mode, from Host TX to Host RX. 1'b0 = No effect. This signal is ignored if the bus is already in Reverse (Host RX) direction.
dphy_direction	Output	Reports the current bus direction. 1'b1 = Bus is in Reverse direction (Host RX). 1'b0 = Bus is in Forward direction (Host TX).

13.6.3.5 Interface Cores

There are several interface cores for the DSI Host Controller Core. These interface cores provide industry standard interfaces to the DSI Host Controller Core to simplify interfacing and integration the DSI Host Controller Core to your application. When selected, these interface cores provide additional ports to the DSI Host Controller core.

13.6.3.5.1 DPI-2 Interface Host Bridge Core

The DPI-2 Interface Host Bridge Core provides a DPI-2 compliant interface to the DSI Host controller. The DPI-2 Interface Host Bridge Core handles converting the DPI-2 interface into a packet format that is compatible with the DSI Host Controller Core's Packet Interface, and that adheres to the DSI specification. The DPI-2 Interface Core handles all pixel-to-byte packing for all DPI data types.

The DSI Host Controller DPI-2 Interface Core provides the following features:

- Support for Type 2,3, and 4 displays
- Support for 16-, 18- and 24- bit Pixel data and all alignment configurations
- Support for 16, 18, 24, and 18 bit loosely packed DSI data types
- Supports DSI video modes
 - Non-Burst Mode with Sync Pulses
 - Non-Burst Mode with Sync Events
 - Burst Mode
- Supports normal or inverted HSYNC and VSYNC signals
- Handles clock domain crossing from DPI Pixel clock to the Host controller TX Byte clock
- Interfaces directly to the Host Controller's DSI Packet Interface
- Comes already integrated with the DSI Host Controller

A block diagram of the DPI-2 Interface Host Bridge Core is shown below. The DPI-2 Host Bridge Core accepts a MIPI compliant DPI-2 set of signals (vsync, hsync, pixel data), creates DSI packets for the video timing events and video data and transmits those packets via the DSI Host Controller’s Packet Interface.

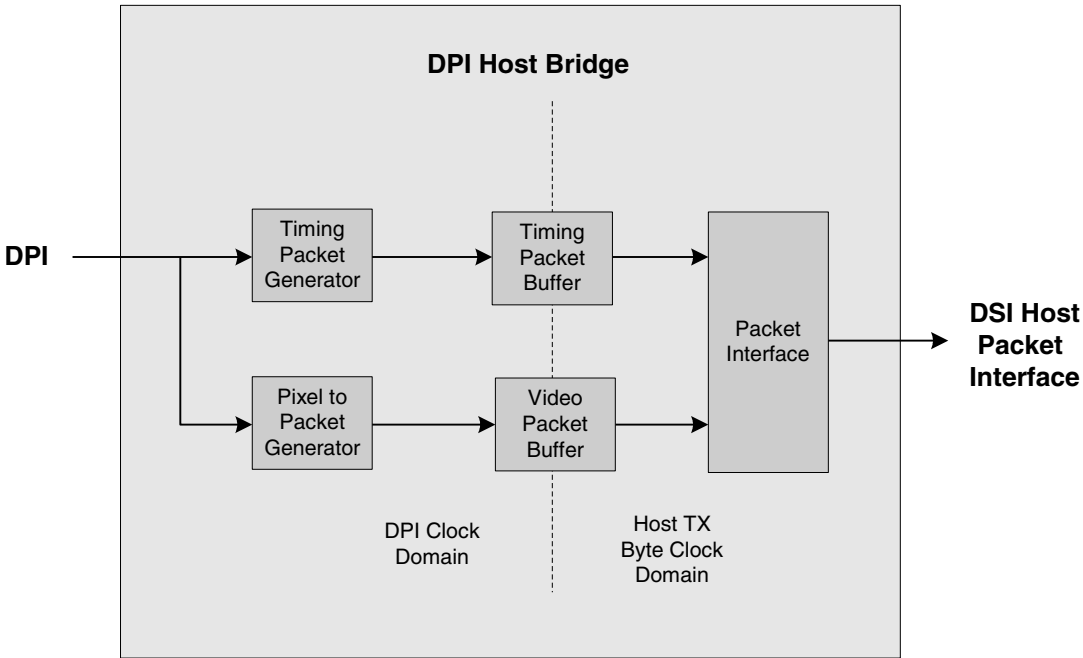


Figure 13-41. DPI-2 Host Bridge Core Block Diagram

13.6.3.5.1.1 DPI-2 Interface Ports

The DPI-2 Interface module provides a MIPI compliant DPI-2 interface to the DSI Host Controller Core. In addition to the standard DPI-2 interface signals (prefixed with a dpi_) there are several configuration and status ports that allow the user to adjust the behavior of the DPI-2 Interface module to better suit a particular application.

NOTE

The DPI configuration signals herein described are accessible to User Application through chip-specific registers (please refer to chip-specific MIPI-DSI information section for details) and DSI_HOST_DPI_INTFC device registers (please refer to MIPI DSI Host Memory Map/Register section for details).

Table 13-23. DSI Host Controller Core Receive Packet interface

PORT	TYPE	DESCRIPTION
dpi_sd	Input	Shut Down – Control to shutdown display (type 4 only) 1'b1= Send shutdown command. 1'b0= No effect

Table continues on the next page...

Table 13-23. DSI Host Controller Core Receive Packet interface (continued)

PORT	TYPE	DESCRIPTION
dpi_cm	Input	Color Mode control. 1'b0== Normal Mode 1'b1== Low-color Mode
cfg_dpi_pixel_payload_size[15:0]	Input	Maximum number of pixels that should send as one DSI packet. Recommended to be evenly divisible by the line size (in pixels).
cfg_dpi_pixel_fifo_send_level[15:0]	Input	In order to optimize DSI utility, the DPI bridge buffers a certain number of DPI pixels before initiating a DSI packet. This configuration port controls the level at which the DPI Host bridge begins sending pixels.
cfg_dpi_interface_color_coding[2:0]	Input	Sets the distribution of RGB bits within the 24-bit d bus, as specified by the DPI specification. 0= 16-bit Configuration 1 1= 16-bit Configuration 2 2= 16-bit Configuration 3 3= 18-bit Configuration 1 4= 18-bit Configuration 2 5= 24-bit
cfg_dpi_pixel_format[1:0]	Input	Sets the DSI packet type of the pixels. 0= 16-bit 1= 18-bit 2= 18-bit loosely packed, 3= 24-bit
dpi_host_underrun_err	Output	During DSI Host transmission of DPI data insufficient DPI data was received. This may indicate that DPI_CLK is too slow, or that the cfg_dpi_* parameters are incorrectly set.
cfg_dpi_vsync_polarity	Input	Sets polarity of dpi_vsync input, 0 – active low, 1 active high
cfg_dpi_hsync_polarity	Input	Sets Polarity of dpi_hsync input, 0 – active low, 1 – active high
cfg_host_dpi_video_mode[1:0]	Input	Select DSI video mode that the host DPI module should generate packets for. 2'b00 – Non-Burst mode with Sync Pulses 2'b01 – Non-Burst mode with Sync Events 2'b10 – Burst mode 2'b11 – Reserved, not valid
cfg_host_dpi_hfp[15:0]	Input	Sets the DSI packet payload size, in bytes, of the horizontal front porch blanking packet.
cfg_host_dpi_hbp[15:0]	Input	Sets the DSI packet payload size, in bytes, of the horizontal back porch blanking packet.
cfg_host_dpi_hsa[15:0]	Input	Sets the DSI packet payload size, in bytes, of the horizontal sync g packet.
cfg_enable_mult_packets	Input	Enable Multiple packets per video line. When enabled, cfg_dpi_pixel_payload_size must be set to exactly half the size of the video line. 0 – Video Line is sent in a single packet 1 – Video Line is sent in two packets
cfg_blip_mode	Input	Optimize blip periods to Low Power mode when possible 0 – blanking packets are sent during BLLP periods 1 – LP mode is used for BLLP periods

13.6.3.5.1.2 DPI-2 Video Timing Recreation over DSI

The DSI Host DPI-2 module is designed to preserve the receive video timing by properly spacing the DSI packets that carry the DPI video information and by inserting blanking packets into the packet stream. In order to achieve this goal, the DSI Host controller needs to have the cfg_host_dpi_hfp, cfg_host_dpi_hbp, and cfg_host_dpi_hsa ports

properly set. The values of these ports reflect the size of the hfp, hbp, and has in terms of DSI packet bytes, values such that the number of DSI bytes when transmitted takes approximately the same amount of absolute time as the video event took on the DPI interface.

To relate pixel sizes on the DPI interface to DSI packet bytes, use the following relationship:

$$\text{Time of DPI event} - \text{time to transmit} \times \text{number of bytes on the DSI interface}$$

$$\text{dpi_event_size} * \text{dpi_pclk_period} = \text{number_of_dsi_bytes} * 8 * \text{dsi_hs_bit_period} / \text{cfg_num_lanes}$$

Solving for number_of_dsi_bytes:

$$\text{number_of_dsi_bytes} = \text{dpi_event_size} * \text{dpi_pclk_period} * \text{cfg_num_lanes} / (8 * \text{dsi_hs_bit_period})$$

number_of_dsi_bytes - value to set the cfg_host_dpi_hfp (hbp, or hsa) port to
dpi_event_size - size of the video event in pixels (hfp, hbp, hsa)
dpi_pclk_period - period of dpi_pclk
dsi_hs_bit_period - bit period of the mipi data lanes in High Speed mode
cfg_num_lanes - host controller setting for number of active lanes, 0 = 1 lane, 1 = 2 lanes, 2 = 3 lanes, 3 = 4 lanes)

13.6.3.5.2 DBI-2 Interface Core

The DBI-2 Interface Core provides a DBI-2 compliant interface that attaches to a Packet Interface Port on the DSI Host controller. The DBI-2 Interface Core handles converting the DBI-2 interface into a packet format that is compatible with the DSI Host Controller Core's Packet Interface and adheres to the DSI specification. The DBI-2 Interface Core handles all pixel-to-byte packing for all DBI data types.

The DSI Host Controller DBI-2 Interface Core provides the following features:

- Support for Type 1, 2, and 3 displays
- Support for DBI-2 Type B and C Interfaces
- Support for 8-, 9- and 16- bit data (Type B)
- Interfaces directly to the Host Controller's Packet Interface
- Comes already integrated with the DSI Host Controller
- Source Code License available (Verilog)
- Provided with expert technical support
- Customization and integration services available

13.6.3.5.2.1 DBI-2 Host Bridge Module Interface Ports

The DBI-2 Interface module provides a MIPI compliant DBI-2 interface to the DSI Host Controller Core. In addition to the standard DBI-2 interface signals (prefixed with a dbi_) there are several configuration and status ports that allow the user to tune adjust the behavior of the DBI-2 Interface module to better suit a particular application.

Table 13-24. DSI Host Controller Core Receive Packet interface

PORT	TYPE	DESCRIPTION
dbi_csx	Input	DBI Chip Select, active low.
dbi_wrx	Input	DBI Write enable. Write data/command is registered at the rising edge of dbi_wrx.
dbi_rdx	Input	DBI Read enable. Read data is read at the rising edge of dbi_rdx.
dbi_d_write[15:0]	Input	8-,9- or 16-bit Input Data for Writes
dbi_d_read[15:0]	Output	8-,9- or 16-bit Output Data for Reads
dbi_dcx	Input	Data/Command indicator. 0 – Command cycle 1 – Data cycle
dbi_resx	Input	Active low DBI Display Module reset
dbi_te	Output	Tearing Effect (optional support)
dbi_host_underrun_err	Output	During DSI Host transmission of DBI data insufficient DBI data was received. This may indicate that the rate of data received on dbi_d_in was insufficient to keep up with the outgoing data rate on the MIPI interface, or that the cfg_dbi_* parameters are incorrectly set.
cfg_dbi_pixel_payload_size[15:0]	Input	Maximum number of pixels that should send as one DSI packet. Recommended to be evenly divisible by the line size (in pixels).
cfg_dbi_pixel_fifo_send_level[15:0]	Input	In order to optimize DSI utility, the DBI bridge buffers a certain number of DBI pixels before initiating a DSI packet. This configuration port controls the level at which the DBI Host bridge begins sending pixels.

13.6.3.5.2.2 DBI-2 Host Bridge Module Write and Read Waveforms

The figure below shows a DBI-2 Write transaction. The command portion of the write is registered with the rising edge of dbi_wrx when dbi_dcx and dbi_csx are low. The write data is registered with the rising edge of dbi_wrx when dbi_dcx is high and dbi_csx is low.

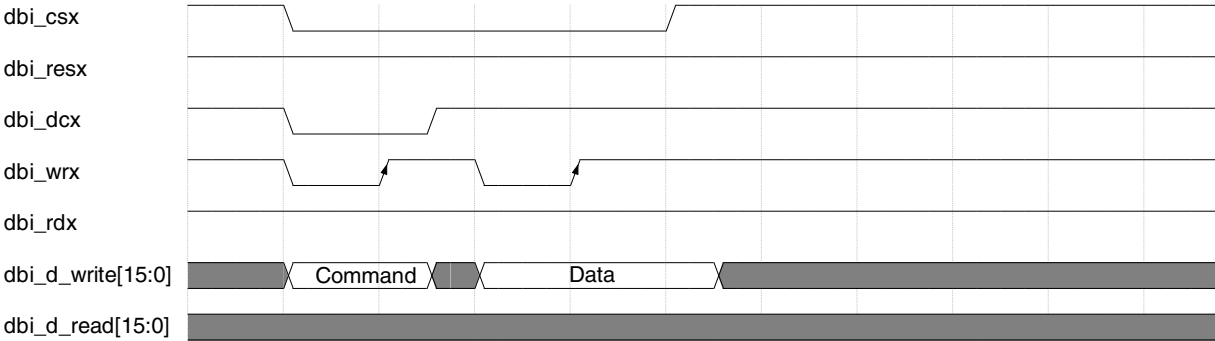


Figure 13-42. DBI-2 Write Waveform

The figure below shows a DBI-2 Read Transaction. The command portion of the read is registered with the rising edge of dbi_wrx when dbi_dcx and dbi_csx are low. The first cycle of read data is invalid with the first rising edge of dbi_rdx while the second cycle of read data is valid with the rising edge of dbi_rdx.

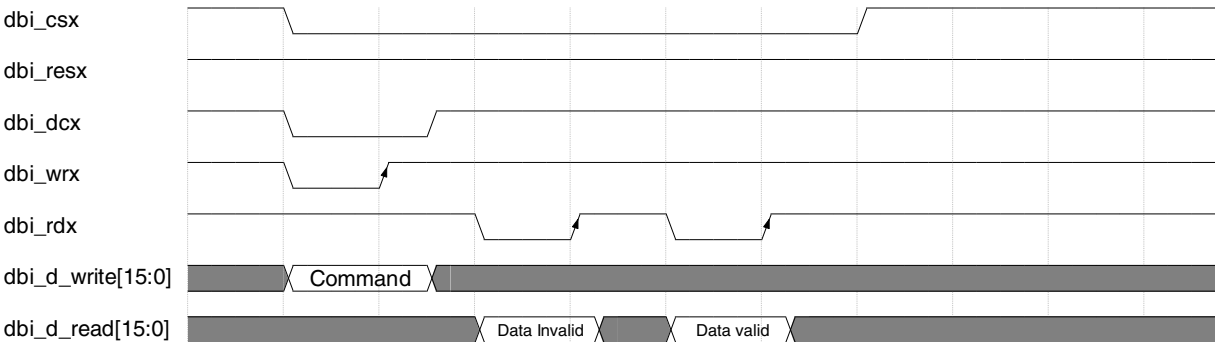


Figure 13-43. DBI-2 Read Waveform

13.6.3.6 RAM Usage

The DSI Host Controller Core requires on-chip RAM arrays. These RAM arrays and their corresponding sizes are listed in the table below.

All RAM arrays used in this core are configured as two-port, having the following attributes:

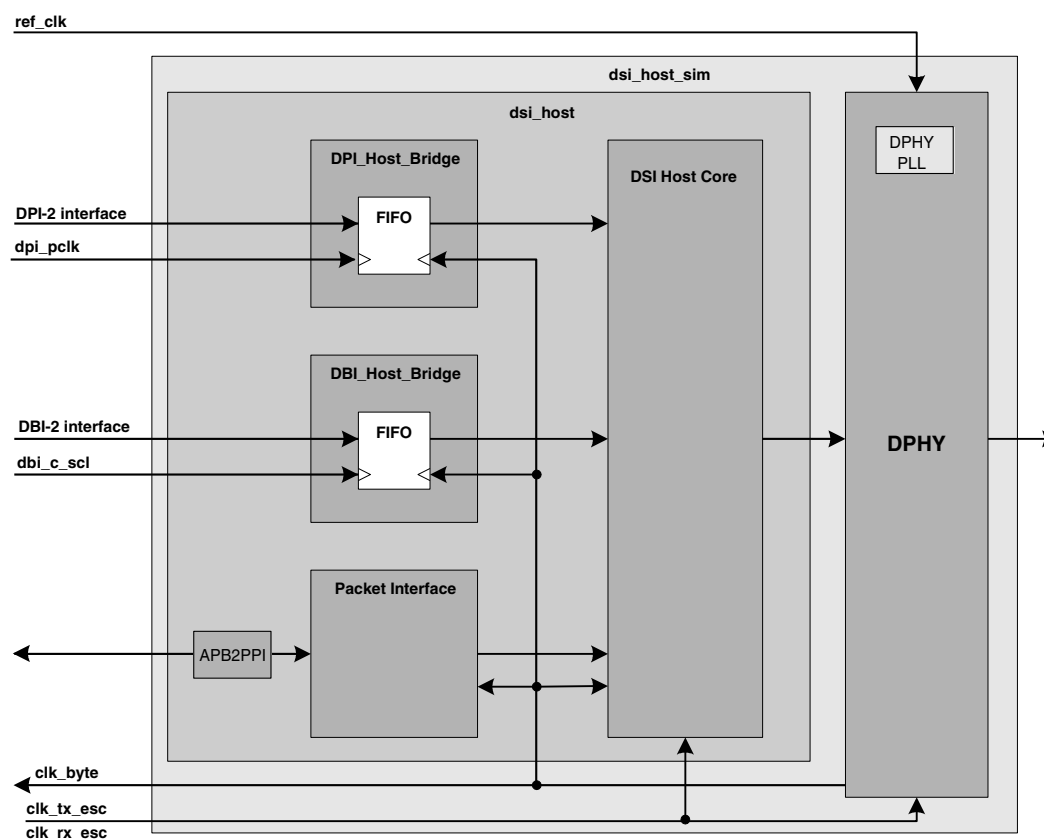
- Two address busses, one for the write port and one for read port
- Two data busses, one for write data on the write port and one for read data on the read port
- Two clock inputs, one for the write port and one for the read port. (Note: In some cases, the core logic will connect the same clock to both the read and write clock inputs of the RAM.)

Ram Purpose	Size (Depth x Width)
DPI Core Pixel Data FIFO	2048 x 32
APB to Packet Interface TX FIFO	64 x 32
APB to Packet Interface RX FIFO	64 x 32

13.6.3.7 DSI Host Clocking

The DSI Host Controller requires the following clocks:

- Reference clock (ref_clk) - used by the DPHY (DPHY PLL generates byte clock)
- Byte clock (clk_byte) - generated by the DPHY, used for packet generation and transfer to the DPHY
- TX escape mode clock (clk_tx_esc) - for internal LPDT state machines and low-power transmissions;
- RX escape mode clock (clk_rx_esc) - used by DPHY for reverse low-power reception.
- Pixel clock (dpi_pclk)
- Serial clock (dbi_c_scl)



MIPI DSI Host Controller (MIPI_DSI)

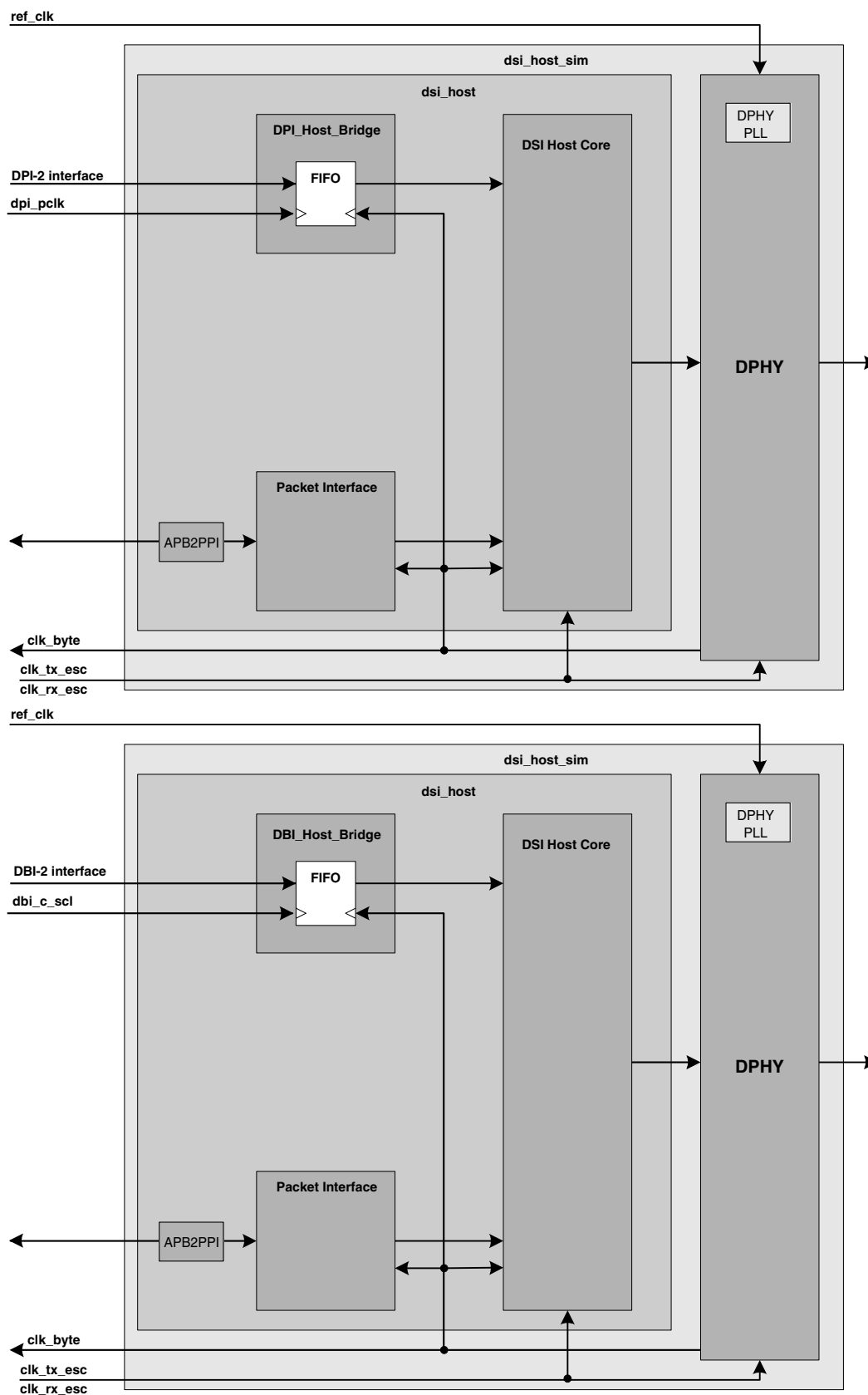


Figure 13-44. DSI Host Controller Clocking Block Diagram

13.6.3.7.1 ref_clk

The reference clock is used by the DPHY (DPHY PLL) to generate the High Speed MIPI clock and High Speed Data Lane signaling. The frequency of the resulting clock generated from DPHY PLL is typically equal to the high speed bit rate of the data lanes. The DPHY will also use this clock to generate the clk_byte clock output used by the Host DSI Controller and user application logic.

The MIPI DPHY PLL must be setup through the MIPI_DSI_HOST_FSL_IP1_DPHY INTFCx registers, prior to starting DSI Host Controller and DPHY operation. For details, please refer to MIPI DSI Host IP1 DPHY INTFC Memory section.

13.6.3.7.2 clk_byte

The clk_byte clock is generated by the DPHY (DPHY PLL) and the frequency is 1/8th the data rate of the DPHY Data Lane High Speed data rate. A DPHY configured to generate High Speed data at 1Gbps/lane would generate a 125MHz clk_byte clock.

The clk_byte clock is used by the DSI Host Controller to interface to the MIPI Tx DPHY High Speed PPI interface. The DSI Host Controller's Packet Interface is synchronous to the clk_byte clock.

13.6.3.7.3 dpi_pclk

The dpi_pclk clock is used on the Host DPI-2 interface. All of the Host DPI-2 signals are synchronous to this clock. The DSI Host Controller's DPI Bridge module handles transferring video data received in the dpi_pclk clock domain over to the clk_byte clock domain.

The dpi_pclk and clk_byte frequencies are related by the following formula:

$$\text{clk_byte_freq} \geq \text{dpi_pclk_freq} * \text{DPI_pixel_size} / (8 * (\text{cfg_num_lanes} + 1))$$

cfg_num_lanes = the configuration port setting that selects the number of active MIPI DPHY data lanes
 clk_byte_freq = frequency of clk_byte which is 1/8th the High Speed data lane rate.
 dpi_pclk_freq = frequency of the dpi_pclk clock on the DPI-2 interface.
 DPI_pixel_size = size of pixels, in bits, on the DPI-2 interface

If the clk_byte frequency does not meet this requirement, the MIPI interface will not be able to keep up with the video stream on the DPI-2 interface resulting in dropped video lines.

13.6.3.7.4 clk_tx_esc

The clk_tx_esc clock is used by the MIPI Tx DPHY for state control and low power data transmission. The DSI Host Controller also uses clk_tx_esc for the portion of controller logic that interfaces to the MIPI Tx DPHY that are synchronous to clk_tx_esc. The frequency of clk_tx_esc is defined by the requirements of the MIPI interface and the MIPI Tx DPHY. For details about the clk_tx_esc source, please refer to the chip-specific MIPI-DSI clocking section.

13.6.3.7.5 clk_rx_esc

The clk_rx_esc clock is used by the MIPI Tx DPHY for reverse low power data reception. The frequency of clk_rx_esc is defined by the Tx DPHY and MIPI DPHY interface timing requirements.

13.6.3.7.6 dbi_c_scl

The dbi_c_scl clock is used by the optional DBI-C interface. The MIPI DBI-2 specification defines the minimum period for dbi_c_scl to be 80ns. The DSI Host Controller DBI-2 bridge module handles the clock domain crossing from dbi_c_scl clock domain to the clk_byte clock domain.

13.6.3.8 Reset and Initialization

Reset and Initialization procedure for the DSI Host Controller is as follows:

1. Assert all resets through the chip-specific registers provided for this purpose:
 - reset_dpi_n
 - reset_byte_n
 - reset_esc_n
2. Setup DSI related clocks through chip-specific registers provided for this purpose (refer to CCM for details).
3. Setup MIPI DSI application by programming all DSI Host Controller and DSI DPHY required parameters (refer to MIPI DSI Host Memory Map/Register for details).
4. Deassert all remaining resets through the Chip-specific registers provided for this purpose:
 - reset_dpi_n
 - reset_byte_n
 - reset_esc_n
5. DSI Host Controller should be ready for use

13.6.4 DSI D-PHY

The figure below shows the block diagram of the MIPI DSI D-PHY. It consists of a clock lane module and four data lane modules. In addition to the data lanes, MIPI DSI D-PHY includes a calibrator module for calibration of termination resistance. Each of these PHY Lane Modules communicates through a differential line to a complementary PHY at the other side of the lane interconnect.

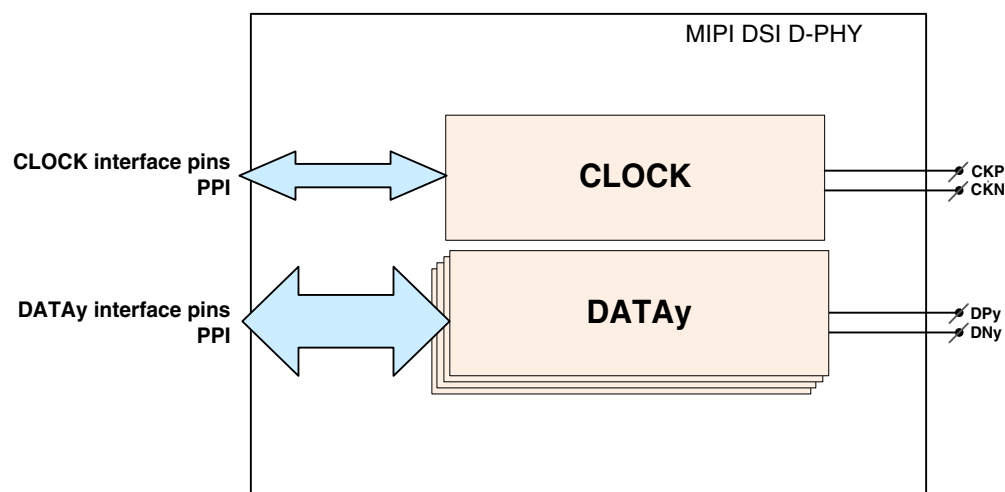


Figure 13-45. MIPI D-PHY Block Diagram

The figure below shows the D-PHY Lane components:

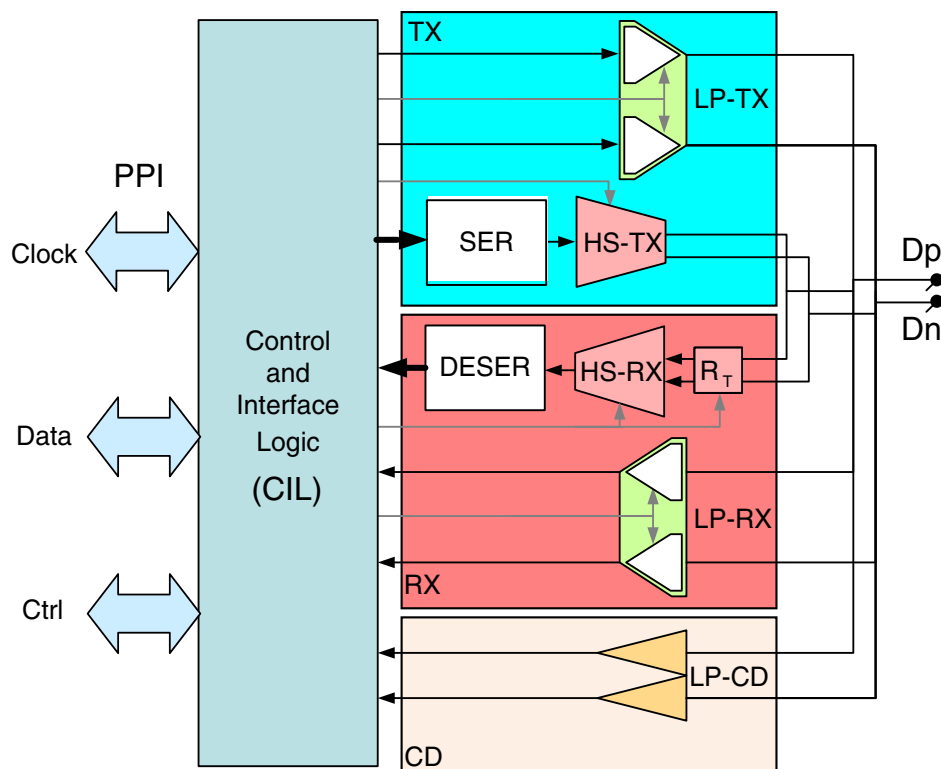


Figure 13-46. D-PHY Lane Diagram

The DATAy are bidirectional lanes and consist of High-Speed Receiver (HS-RX), Low-Power Receiver (LP-RX), High-Speed Transmitter (HS-TX), Low-Power Transmitter (LP-TX), Serializer (SER), De-Serializer (DESER), and Low-Power Contention Detector (LP-CD).

The Clock lane is a bidirectional lane and consists of High-Speed Receiver (HS-RX), Low-Power Receiver (LP-RX), High-Speed Transmitter (HS-TX), and Low-Power Transmitter (LP-TX). The B is optional and is used to calibrate the HS-TX and HS-RX termination, which requires connecting the pin REXT with a precise external resistor.

The Control and Interface Logic (CIL) module interfaces with the Protocol and determines the global operation of the Lane Module. The interface between the D-PHY and the protocol is called the PHY-Protocol Interface (PPI).

During normal operation, a Lane switches between Low-Power and High-Speed mode. Bidirectional Lanes can also switch communication direction. The change of operating mode or direction requires enabling and disabling of certain electrical functions. These enable and disable events do not cause glitches on the Lines that would result in a detection of an incorrect signal level. Therefore, all mode and direction changes are smooth to always ensure a proper detection of the Line signals.

13.6.4.1 High-Speed Transmitter (HS-TX)

The differential output voltage V_{OD} is defined as the difference of the voltages V_{DP} and V_{DN} at the Dp and Dn pins, respectively.

$$V_{OD} = V_{DP} - V_{DN}$$

The output voltages V_{DP} and V_{DN} at the Dp and Dn pins will not exceed the high-speed output high voltage V_{OHHS} . The common-mode voltage V_{CMTX} is defined as the arithmetic mean value of the voltages at the Dp and Dn pins:

$$V_{CMTX} = (V_{DP} + V_{DN})/2$$

V_{OD} and V_{CMTX} are graphically shown in [Figure 13-47](#) for ideal HS signals. [Figure 13-48](#) shows single ended HS signals with the possible kinds of distortion of the differential output and common mode voltages. V_{OD} and V_{CMTX} may be slightly different for driving a Differential-1 or a Differential-0 on the pins.

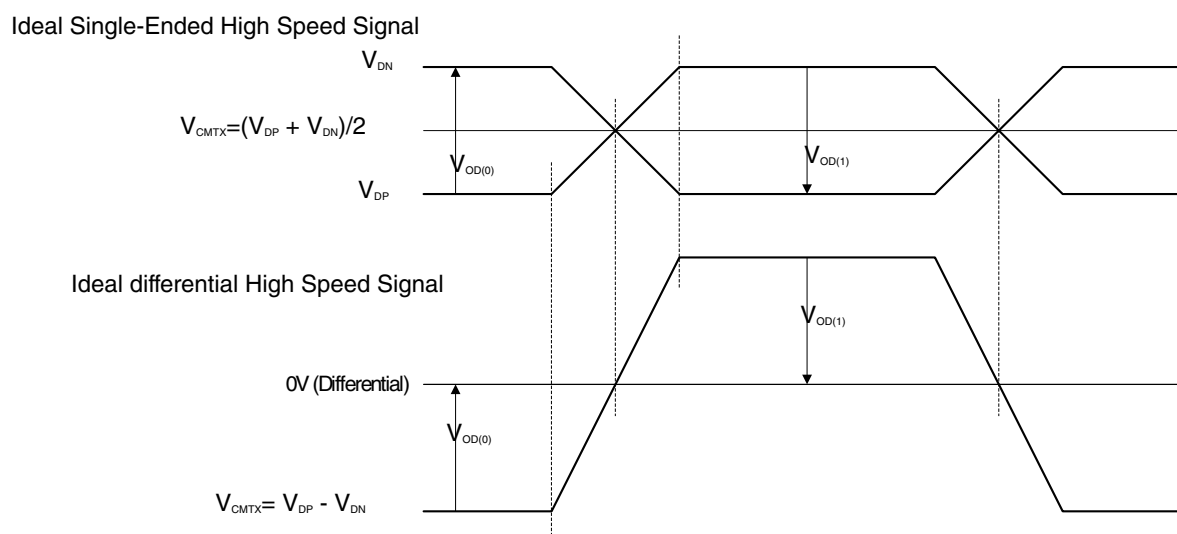


Figure 13-47. Ideal Single-ended and Resulting Differential HS Signals

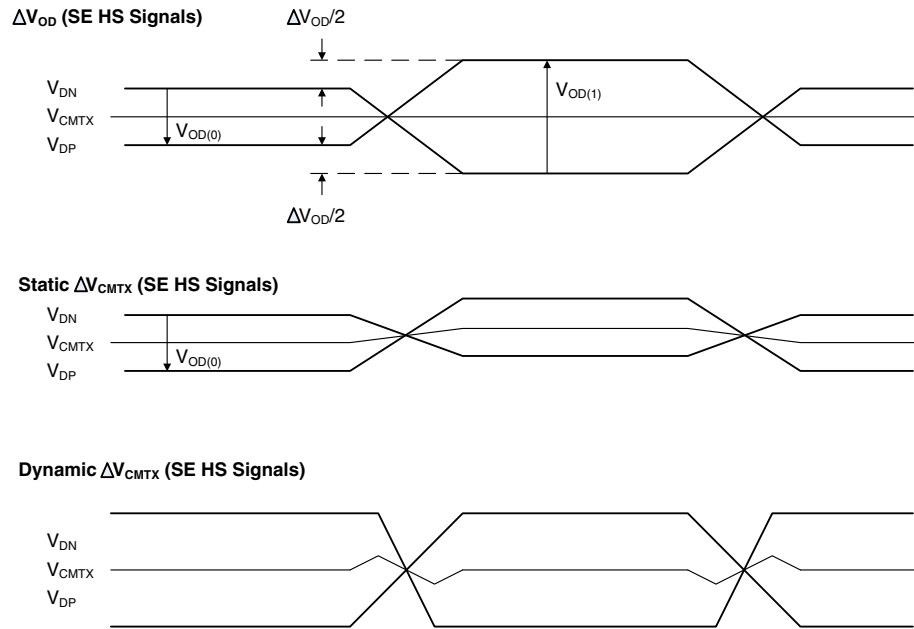


Figure 13-48. Possible ΔV_{cmx} and ΔV_{od} Distortions of the Single-Ended HS Signals

The output differential voltage mismatch ΔV_{OD} is defined as the difference of the absolute values of the differential output voltage in the Differential-1 state $V_{OD(1)}$ and the differential output voltage in the Differential-0 state $V_{OD(0)}$. This is expressed by:

$$\Delta V = |V_{OD(1)}| - |V_{OD(0)}|$$

If $V_{CMTX(1)}$ and $V_{CMTX(0)}$ are the common-mode voltages for static Differential-1 and Differential-0 states respectively, then the common-mode reference voltage is defined by:

$$V_{CMTX, REF} = (V_{CMTX(1)} + V_{CMTX(0)})/2$$

The transient common-mode voltage variation is defined by:

$$\Delta V_{CMTX(t)} = V_{CMTX(t)} - V_{CMTX, REF(t)}$$

The static common-mode voltage mismatch between the Differential-1 and Differential-0 state is given by:

$$\Delta V_{CMTX(1,0)} = (V_{CMTX(1)} - V_{CMTX(0)})/2$$

A test circuit for the measurement of V_{OD} and V_{CMTX} is shown:

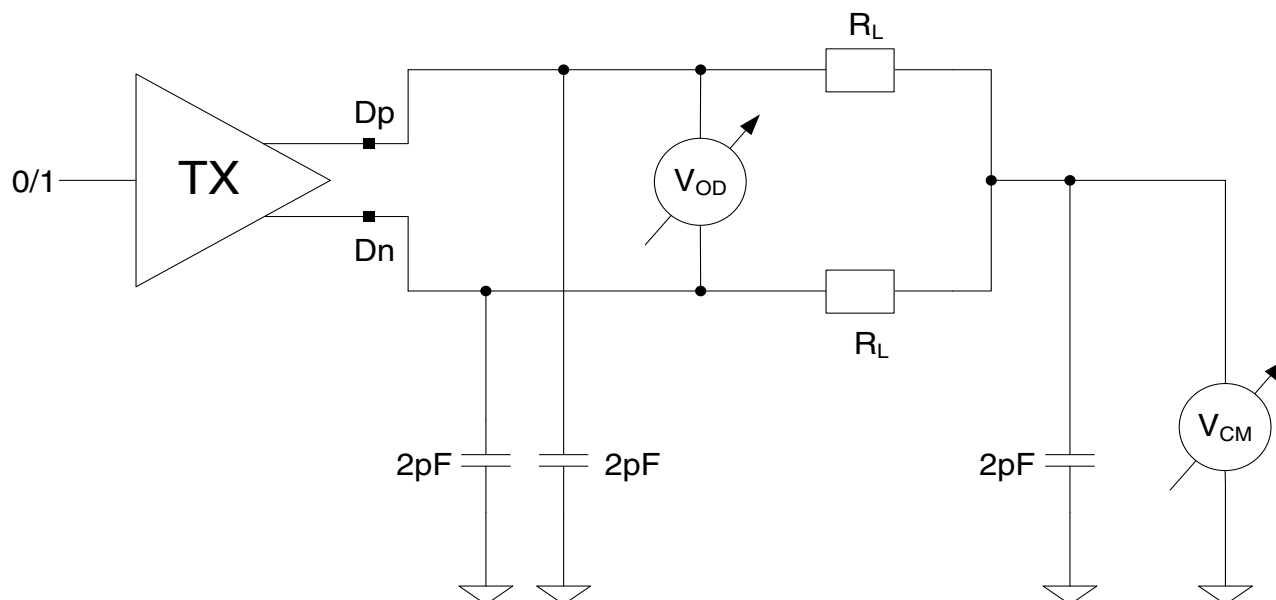


Figure 13-49. Test Circuit for Vcmx and Vod measurement

The single-ended output impedance of the transmitter at both the Dp and Dn pins is denoted by Z_{OS} . ΔZ_{OS} is the mismatch of the single ended output impedances at the Dp and Dn pins, denoted by Z_{OSDP} and Z_{OSDN} respectively. This mismatch is defined as the ratio of the absolute value of the difference of Z_{OSDP} and Z_{OSDN} and the average of those impedances:

$$\Delta Z_{OS} = (2 * |Z_{OSDP} - Z_{OSDN}|) / (Z_{OSDP} + Z_{OSDN})$$

The output impedance Z_{OS} and the output impedance mismatch ΔZ_{OS} is compliant with High Speed Transmitter DC Specifications for both the Differential-0 and Differential-1 states for all allowed loading conditions. It is recommended the design keep the output impedance during state transitions as close as possible to the steady state value. The output impedance Z_{OS} can be determined by injecting an AC current into the Dp and Dn pins and measuring the peak-to-peak voltage amplitude. The test circuit for such a measurement is shown in the following figure, where R_L are load resistors and $V_{TEST,PP}$ is a test signal applied at the common-mode node via an AC coupling capacitor CAC. The frequency of the test signal is f_{TEST} .

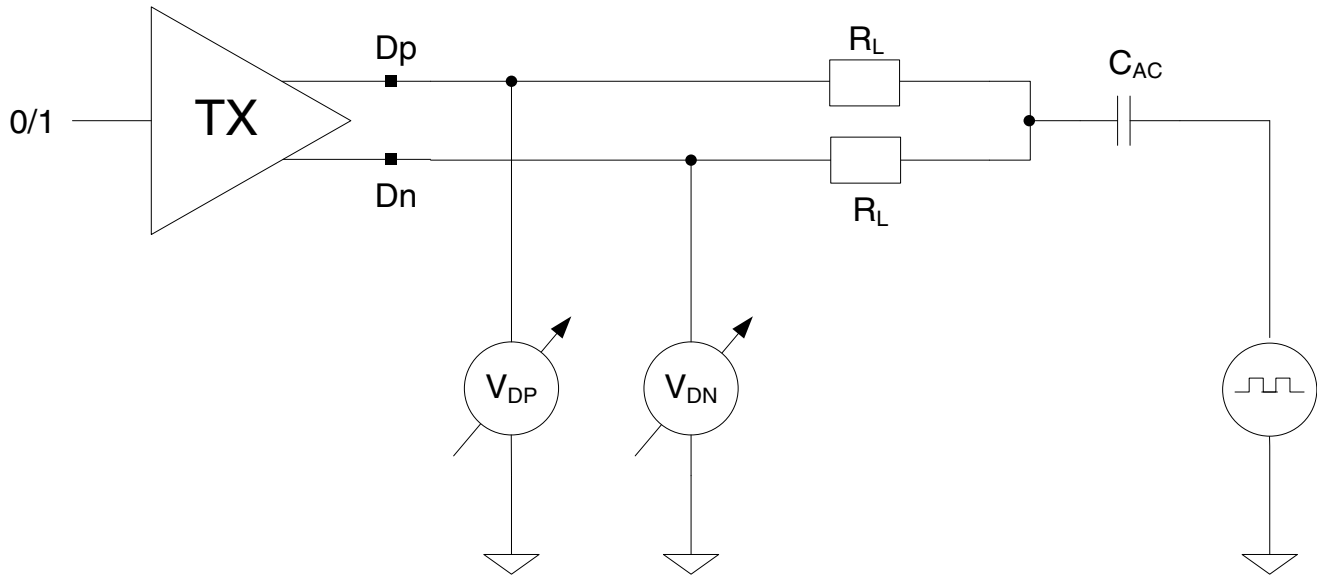


Figure 13-50. Test Circuit for Output Impedance measurements

Assuming negligible effect of the coupling capacitor, the example circuit gives the following relation between peak-to-peak pin voltage $V_{PIN,PP}$ and Z_{OS} . This equation can be used for both Dn and Dp pins individually.

$$Z_{OS} = (V_{PIN,PP} * R_L) / (V_{TEST,PP} - Z_{PIN,PP})$$

Where $V_{PIN,PP}$ is the peak-to-peak voltage at the pin Dp or Dn.

13.6.4.2 High-Speed Receiver (HS-RX)

The HS receiver is a differential line receiver. It contains an on-die switchable parallel input termination, Z_{ID} , between the positive input pin Dp and the negative input pin Dn.

The differential input high and low threshold voltages of the HS receiver are denoted by V_{IDTH} and V_{IDTL} , respectively. V_{CMRXDC} is the differential input common mode voltage. The HS receiver is able to detect differential signals at its Dp and Dn input signal pins when both signal voltages, V_{DP} and V_{DN} , are within the common mode voltage range and if the voltage difference of V_{DP} and V_{DN} exceeds either V_{IDTH} or V_{IDTL} .

During operation of the HS receiver, termination impedance is required between the Dx_P and Dx_N pins of the HS receiver. Z_{ID} is disabled when the module is not in the HS receive mode. CCM is the common mode AC termination, which ensures a proper termination of the receiver at higher frequencies.

13.6.4.3 Low-Power Transmitter (LP-TX)

The Low-Power transmitter is a slew-rate controlled push-pull driver. It is used for driving the Lines in all Low-Power operating modes. The LP transmitter has been optimized to minimize static power consumption. The slew-rate of signal transitions is bounded in order to keep EMI low. The LP transmitter will not drive the pad pin potential statically beyond the maximum value of V_{OH} . V_{OH} is the thevenin output, high-level voltage in the high-level state, when the pad pin is not loaded. V_{OL} is the thevenin output, low-level voltage in the LP transmit mode. This is the voltage at an unloaded pad pin in the low-level state.

13.6.4.4 Low-Power Receiver (LP-RX)

The low power receiver is an un-terminated, single-ended receiver circuit. The LP receiver is used to detect the Low-Power state on each pin. For high robustness, the LP receiver filters out noise pulses and RF interference. The LP receiver is optimized for low power operation.

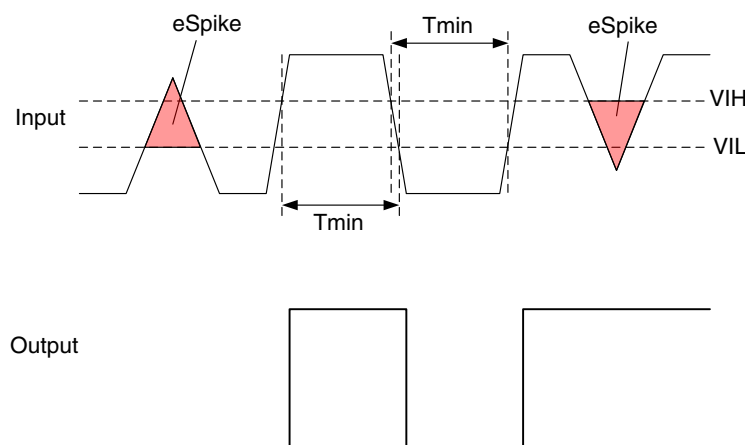


Figure 13-51. Input Glitch Rejection of Low-Power Receiver

The input low-level voltage, V_{IL} , is the voltage at which the receiver is required to detect a low state in the input signal. V_{IL} is larger than the maximum single-ended Line voltage during HS transmission. Therefore, both LP receivers will detect low during HS signaling. The input high-level voltage, V_{IH} , is the voltage at which the receiver is required to detect a high state in the input signal. In order to reduce noise sensitivity on the received signal, the LP receiver incorporates a hysteresis, V_{HYST} .

The LP receiver also rejects any input glitch when the glitch is smaller than eSPIKE. The filter allows pulses wider than T_{MIN} to propagate through the LP receiver. Additionally, the LP receivers are tolerant of super-positioned RF interference on top of the wanted

Line signals. The LP receiver meets all specifications for interference with amplitude VINT and frequency fINT. The interference does not cause glitches or incorrect operation during signal transitions.

13.6.4.5 Low-Power Contention Detector (LP-CD)

Contention can be inferred from any of the following conditions:

- LP high fault is detected when the LP transmitter is driving high and the pin voltage is less than VIL.
- LP low fault is detected when the LP transmitter is driving low and the pad pin voltage is greater than VIHCD. An LP low fault will not be detected when the pin voltage is less than VILCD.

The general operation of a contention detector is similar to that of an LP receiver with lower threshold voltages. Although the DC specifications differ, the AC specifications of the LP-CD are defined to match those of the LP receiver. The LP-CD sufficiently filters the input signal to avoid false triggering on short events

The LP-CD threshold voltages (VILCD, VIHCD) are shown along with the normal signaling voltages:

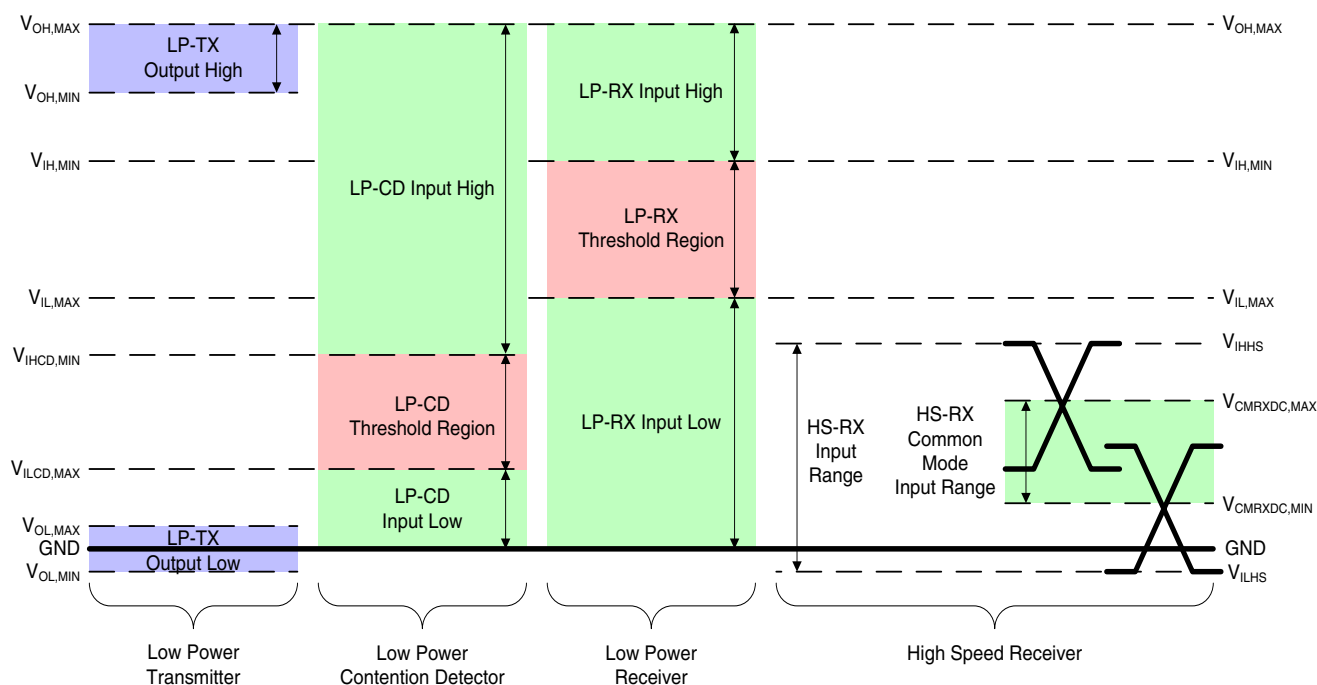


Figure 13-52. Signaling and MIPI Contention Voltage Levels

13.6.4.6 Input Characteristics

No structure within the PHY may be damaged when a DC signal that is within the signal voltage range V_{PIN} is applied to a pad pin for an indefinite period of time. $V_{PIN(absmax)}$ is the maximum transient output voltage at the transmitter pin. The voltage on the transmitter's output pin will not exceed $V_{PIN,MAX}$ for a period greater than $T_{VPIN(absmax)}$. When the PHY is in the low-power receive mode the pad pin leakage current will be I_{LEAK} when the pad signal voltage is within the signal voltage range of V_{PIN} . The specification of I_{LEAK} assures interoperability of any PHY in the LP mode by restricting the maximum load current of an LP transmitter. The test circuit for leakage current measurement is shown:

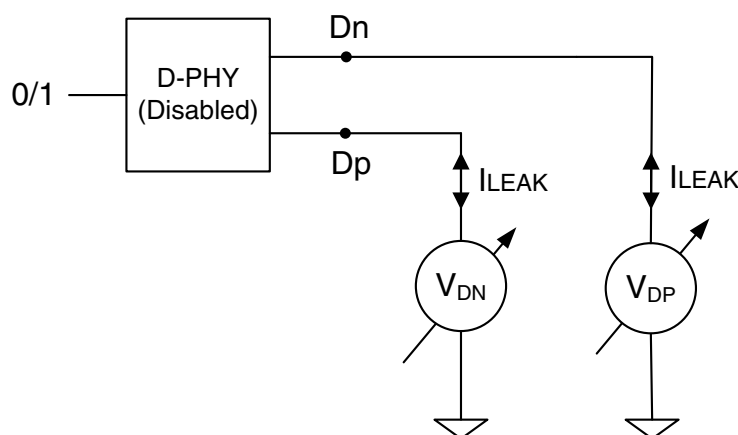


Figure 13-53. Pin-Leakage Measurement DC specification

13.6.4.7 High-Speed Serializer

The HS-Serializer converts 8-bits parallel data into 1 bit data stream to be transmitted by HSTX module. The CIL engine should clock out 8-bits parallel data, `HSTX_DATA[7:0]`, on `D0_HS_BYTE_CLKS` rising edges. Then the Serializer will capture `HSTX_DATA[7:0]` on `D0_HS_BYTE_CLKS` rising edges and the signal connected to `HSTX_DATA[0]` is transmitted first. There is setup timing constraint and hold timing constraint designated to the rising edge of `D0_HS_BYTE_CLKS` as constrained in the liberty view provided.

The figure shows the relationship between `HS_TXCLKP`, `HS_TXCLKN`, `HS_SER_LD`, `HS_BYTE_CLKS` and `HSTX_DATA[7:0]`. It also shows how the 8-bits of parallel bits are mapped into the serial link:

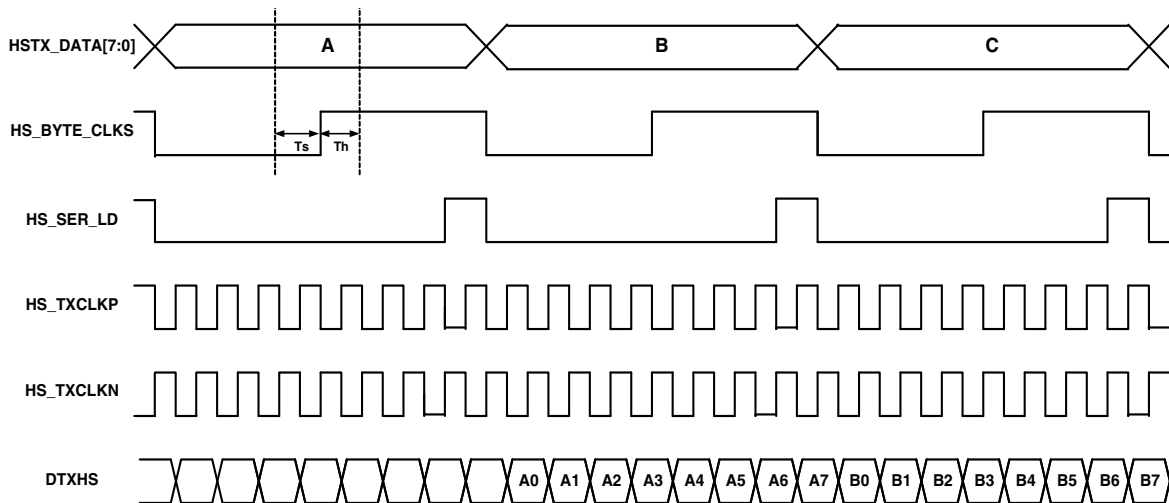


Figure 13-54. Serializer Timing Diagram

13.6.4.8 High-Speed Deserializer

The HS-Deserializer includes a Token Detector to detect SoT leader sequence ‘011101’. After proper match found (any single bit error allowed) for Sync sequence in HS input data stream, Deserializer then converts 1 bit data stream received by HS-RX module into 8-bits parallel data and HS_BYTE_CLKD becomes active.

The High speed input data stream from HS-RX module, DRXHSP/DRXHSN, is sampled by a differential DDR (half-rate) clock, HS_RXCLKP/HS_RXCLKN. The following bits after the Sync leader sequence are payload bits. The signal connected to HSRX_DATA[0] was received first and HSRX_DATA[7:0] can be captured on rising edges of HS_BYTE_CLKD. The transmitter will ensure that a rising edge of the DDR clock is sent during the first payload bit of a transmission burst such that the first payload bit can be sampled by the receiver on the rising clock edge, the second bit can be sampled on the falling edge, and all following bits can be sampled on alternating rising and falling edges.

HS-Deserializer also provides 3 synchronization condition signals. If sync sequence is properly matched, “SYNC” signal will be asserted. If sync sequence is matched with single bit error, “ERRSYNC” signal will be asserted. If the leader sequence is corrupted in a way that proper synchronization cannot be achieved (more than single-bit error), “NOSYNC” signal will be asserted.

The figures show the relationship between HS_RXCLKP/HS_RXCLKN, DRXHSP/DRXHSN, DRXLPP/DRXLPN, SYNC/ERRSYNC/NOSYNC, HS_BYTE_CLKD and HSRX_DATA[7:0]:

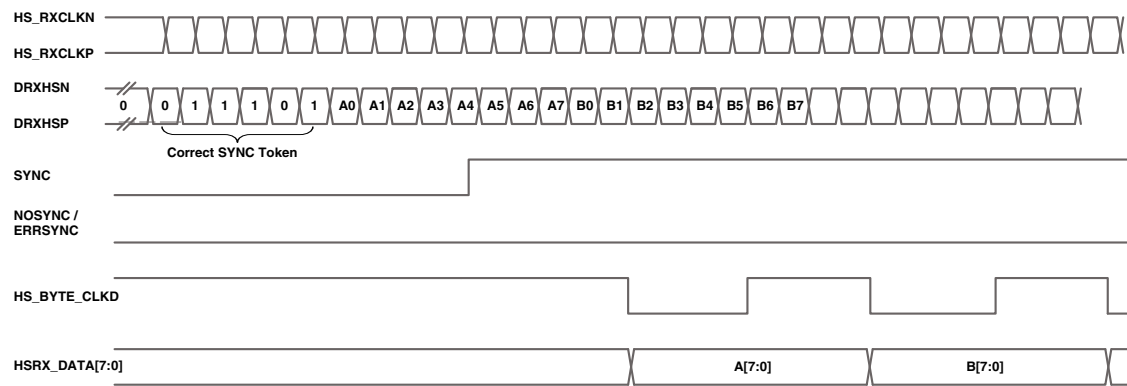


Figure 13-55. Deserializer Timing Diagram with Correct Sync Token

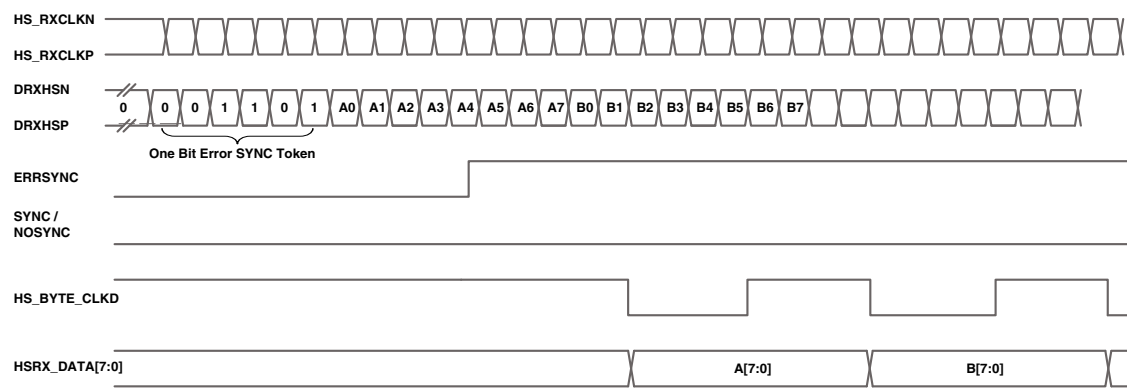


Figure 13-56. Deserializer Timing Diagram with One-Bit-Error Sync Token

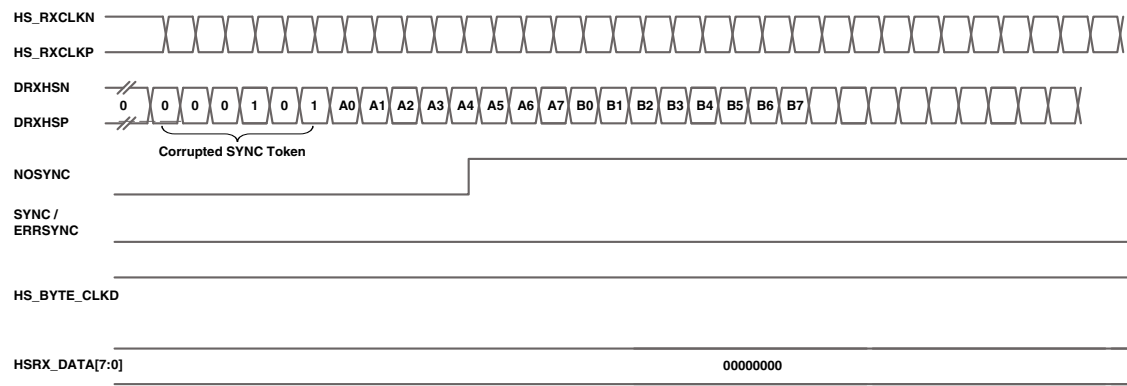


Figure 13-57. Deserializer Timing Diagram with Corrupted Sync Token

13.6.4.9 Termination Calibrator

An optional calibration circuit is implemented inside D-PHY to ensure that the HS-TX and HS-RX meet the required specifications with process corner and temperature variations. A precise external resistor is used to calibrate a scaled replica of the termination resistor, inside the calibrator. The auto calibration procedure is performed every time the D-PHY is powered-up.

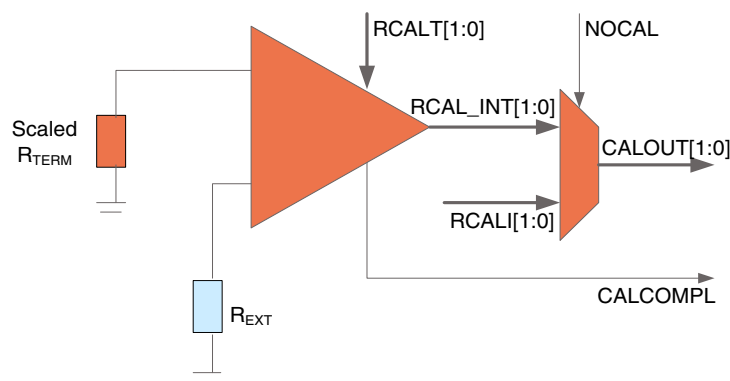


Figure 13-58. Calibrator Block Diagram

The D-PHY supports one of the following receiver termination schemes:

- **Auto Calibration mode:** Termination resistors are automatically configured by calibration circuit with reference resistor connected between REXT and VSSA.
- **User programmable mode:** RCALI[1:0] is programmed by the user through chip registers. In this case, the user has to control NOCAL signal for manual calibration.

13.6.4.10 DSI D-PHY Signals

This section details the interface signals for MIPI DSI D-PHY. For detailed signal spec, please refer to the chip datasheet.

Table 13-25. DSI DPHY PPI Signals

Signal	Type	Description
TxCkEsc	Clock	Escape mode Transmit Clock. This clock is directly used to generate escape sequences. The period of this clock determines the symbol time for low power signals. It is therefore constrained by the normative part of the DPHY specification. NOTE: The max frequency of TxCkEsc is 20 MHz NOTE: The Min frequency of TxCkEsc is 12 MHz. TxCkEsc is synchronous with RxClkInEsc.

Table continues on the next page...

**Table 13-25. DSI DPHY PPI Signals
(continued)**

Signal	Type	Description
RxCkInEsc_In0	Clock	Escape mode Clk for RX. Minimum Clk frequency should be 60 MHz in order to correctly detect the LP states, which are minimum 50 ns long but the minimum LP pulse duration could be as low as 20 ns as per D-PHY CTS. RxCkInEsc is synchronous with TxClkEsc
TxByteClkHS	Clock	High-Speed Transmit Byte Clock. This is used to synchronize PPI signals in the High-Speed transmit clock domain. It is recommended that all transmitting Data Lane Modules share on TxByteClkHS signal. The frequency of TxByteClkHS is exactly 1/8 the High-Speed bit rate.

13.6.4.11 Operating Modes

13.6.4.11.1 High-Speed Transmit from the Master Side

The figure shows an example of a High-Speed transmission on the Master side. While TxRequestHS is low, the Lane Module ignores the value of TxDataHS. To begin transmission, the protocol drives TxDataHS with the first byte of data and asserts TxRequestHS. This data byte is accepted by the PHY on the first rising edge of TxByteClkHS with TxReadyHS also asserted. At this point, the protocol logic drives the next data byte onto TxDataHS. After every rising clock cycle with TxReadyHS active, the protocol supplies a new valid data byte or ends the transmission. After the last data byte has been transferred to the Lane Module, TxRequestHS is driven low to cause the Lane Module to stop the transmission and enter Stop state. The minimum number of bytes transmitted could be as small as one.

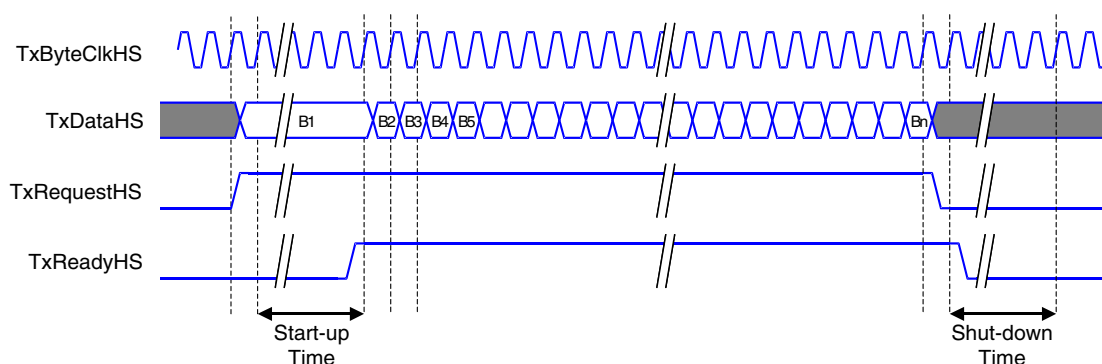


Figure 13-59. High-Speed Transmission from the Master Side

13.6.4.11.2 Low-Power Data Transmission

For Low-Power data transmission the TxClkEsc is used. The Protocol directs the Data Lane to enter Low-Power data transmission Escape mode by asserting TxRequestEsc with TxLpdtEsc high. The Low-Power transmit data is transferred on the TxDataEsc lines when TxValidEsc and TxReadyEsc are both active at a rising edge of TxClkEsc. The byte is transmitted in the time after the TxDataEsc is accepted by the Lane Module (TxValidEsc = TxReadyEsc = high). The Protocol knows the byte transmission is finished when TxReadyEsc is asserted. After the last byte has been transmitted, the protocol de-asserts TxRequestEsc to end the Low-Power data transmission. This causes TxReadyEsc to return low. Whenever TxRequestEsc transitions from high-to-low, it always remains in the low state for a minimum of two TxClkEsc clock cycles. The figure shows an example Low-Power data transmission operation.

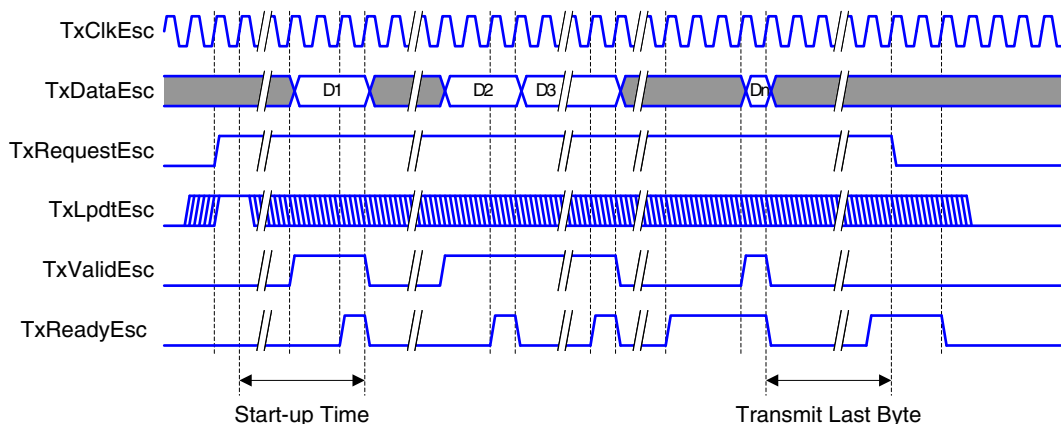


Figure 13-60. Low-Power Data Transmission

13.6.4.11.3 Low-Power Data Reception

The figure shows an example Low-Power data reception. In this example, a Low-Power escape “clock” is generated from the Lane Interconnect by the logical exclusive-OR of the Dp and Dn lines. This “clock” is used within the Lane Module to capture the transmitted data. In this example, the “clock” is also used to generate RxClkEsc. The signal RxLpdtEsc is asserted when the escape entry command is detected and stays high until the Lane returns to Stop state, indicating that the transmission has finished. It is important to note that because of the asynchronous nature of Escape mode transmission, the RxClkEsc signal can stop at anytime in either the high or low state. This is most likely to happen just after a byte has been received, but it could happen at other times as well.

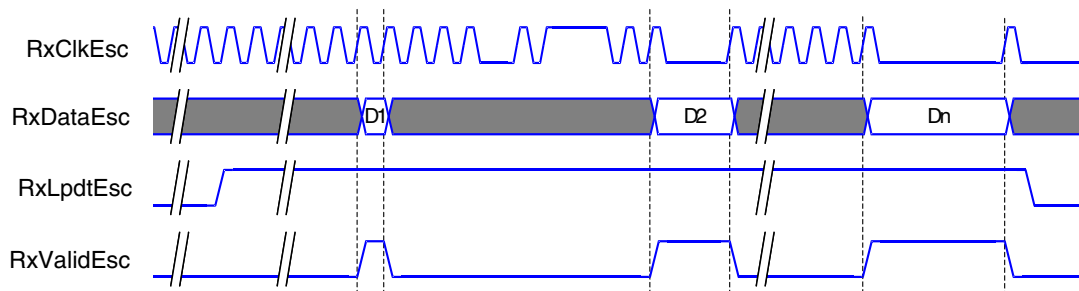


Figure 13-61. Low-Power Data Reception

13.6.4.11.4 Turn-around

If the Master side and Slave side Lane Modules are both bi-directional, it is possible to turn around the Link for Escape mode signaling. Which side is allowed to transmit is determined by passing a “token” back and forth. That is, the side currently transmitting passes the token to the receiving side. If the receiving side acknowledges the turn-around request, as indicated by driving the appropriate line state, the direction is switched. The figure shows an example of two turn-around events. At the beginning, the local side is the transmitter, as shown by $\text{Direction}=0$. When the protocol on this side wishes to turn the Lane around (i.e. give the token to the other side), it asserts TurnRequest for at least one cycle of TxClkEsc . This initiates the turn around procedure. The remote side acknowledges the turn-around request by driving the appropriate states on the Lines. When this happens, the local Direction signal changes from transmit (0) to receive (1). The remote side initiates a turn-around request, passing the token back to the local side. When this happens, the local Direction signal changes back to transmit (0). The transmitter is in control of the Link direction and decides when to turn the Link around, passing control to the receiver. If the remote side does not acknowledge the turnaround request, the Direction signal does not change.

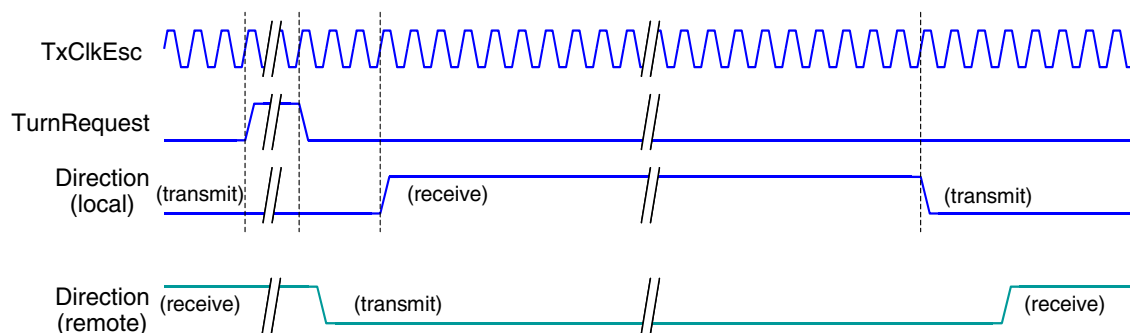


Figure 13-62. Turn-around Actions Transmit-to-Receive and Back to Transmit

13.6.5 MIPI DSI DPHY PLL

The MIPI DSI DPHY PLL is a high performance PLL based frequency synthesizer that incorporates a lock detector, independent output divider, and supports power down modes. The PLL block diagram is shown below.

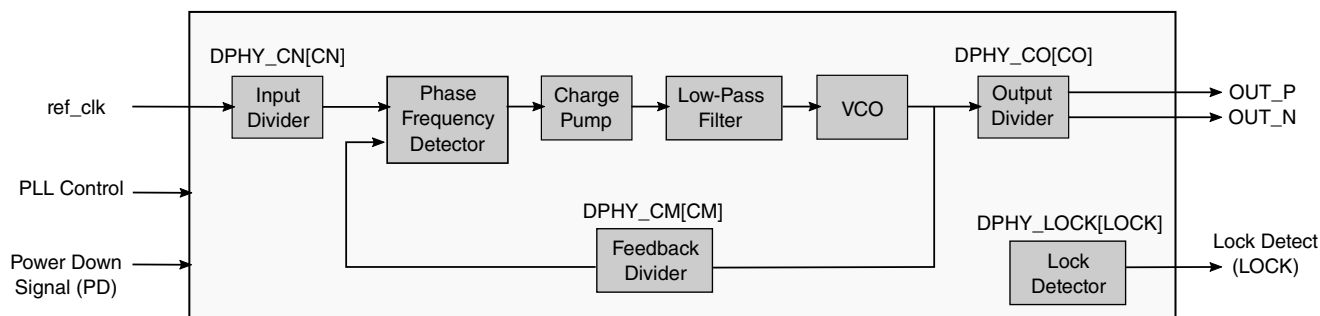


Figure 13-63. MIPI DSI DPHY PLL Block Diagram

The DPHY PLL input clock is `ref_clk`. The PLL output multiplies the input frequency by $(CM / (CN * CO))$.

NOTE

The input frequency ranges from 24 MHz till 200 MHz. The input divider has to be programmed such that the frequency after the input divider ranges from 24 MHz till 30 MHz.

NOTE

The VCO maximum output frequency is 1.5 GHz.

The power down signal (PD) is active-high and used to power down the PLL. Asserting PD will reset the PLL to its initial state. To ensure proper PLL functionality, `ref_clk` needs to be stable before PLL power up. The expected power-on sequence for the DSI DPHY PLL is illustrated below.

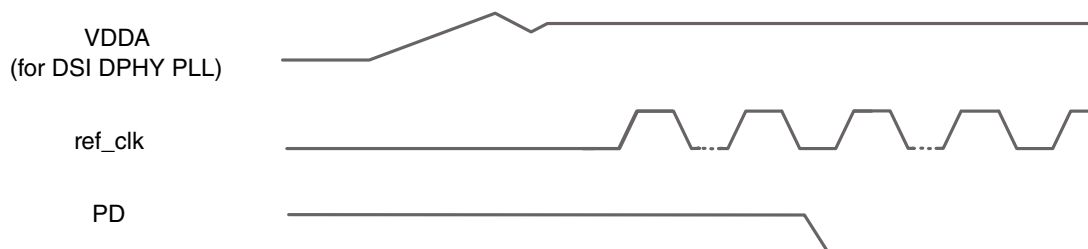


Figure 13-64. MIPI DSI DPHY PLL Power-on Sequence

13.6.6 DPHY Programming

The following sections detail the programming of the MIPI DSI DPHY high-speed timers.

13.6.6.1 High-Speed Prepare Timer ($T_{\text{HS-PREPARE}}$)

Calculate the value for clock lane high-speed prepare timer (mc_PRG_HS_PREPARE) using the formula below:

Assuming TxClkEsc = 20 MHz ($T_{\text{per}} = 50 \text{ ns}$)

When mc_PRG_HS_PREPARE = 0, $T_{\text{CLK-PREPARE}} = 1 * \text{TxClkEsc Period} = 50 \text{ ns}$

When mc_PRG_HS_PREPARE = 1, $T_{\text{CLK-PREPARE}} = 1.5 * \text{TxClkEsc Period} = 75 \text{ ns}$

Calculate the value for data lane high-speed prepare timer (m_PRG_HS_PREPARE) using the formula below:

Assuming TxClkEsc (TLPX) = 20 MHz ($T_{\text{per}} = 50 \text{ ns}$)

When m_PRG_HS_PREPARE = 00, $T_{\text{HS-PREPARE}} = 1 * \text{TxClkEsc Period} = 50 \text{ ns}$

When m_PRG_HS_PREPARE = 01, $T_{\text{HS-PREPARE}} = 1.5 * \text{TxClkEsc Period} = 75 \text{ ns}$

When m_PRG_HS_PREPARE = 10, $T_{\text{HS-PREPARE}} = 2 * \text{TxClkEsc Period} = 100 \text{ ns}$

When m_PRG_HS_PREPARE = 11, $T_{\text{HS-PREPARE}} = 2.5 * \text{TxClkEsc Period} = 125 \text{ ns}$

13.6.6.2 High-Speed Zero Timer($T_{\text{HS-ZERO}}$)

Calculate the value for clock lane high-speed zero timer (mc_PRG_HS_ZERO) using the formula below:

$T_{\text{CLK-ZERO}} = (\text{mc_PRG_HS_ZERO} + 3) * (\text{TxByteClkHS Period})$

$\text{TxByteClkHS} = 1/8 \text{ Data Rate}$

When mc_PRG_HS_ZERO = 000110 @ 250 Mb/s, $T_{\text{CLK-ZERO}} = (6+3) * 32 \text{ ns} = 288 \text{ ns}$

When mc_PRG_HS_ZERO = 100000 @ 1 Gb/s, $T_{\text{CLK-ZERO}} = (32+3) * 8 \text{ ns} = 280 \text{ ns}$

Calculate the value for data lane high-speed zero timer (m_PRG_HS_ZERO) using the formula below:

$T_{\text{HS-ZERO}} = (\text{m_PRG_HS_ZERO} + 6) * (\text{TxByteClkHS Period})$

$\text{TxByteClkHS} = 1/8 \text{ Data Rate}$

When m_PRG_HS_ZERO = 00001 @ 250 Mb/s, $T_{HS-ZERO} = (1+6) * 32 \text{ ns} = 224 \text{ ns}$

When m_PRG_HS_ZERO = 01001 @ 1 Gb/s, $T_{HS-ZERO} = (9+6) * 8 \text{ ns} = 120 \text{ ns}$

13.6.6.3 High-Speed Trail Timer ($T_{HS-TRAIL}$)

Calculate the value for the high-speed trail timer (PRG_HS_TRAIL) using the formula below:

$T_{HS-TRAIL} = (PRG_HS_TRAIL) * (TxByteClkHS \text{ Period})$

$TxByteClkHS = 1/8 \text{ Data Rate}$

When PRG_HS_TRAIL = 0100 @ 250 Mb/s, $>T_{HS-TRAIL} = 4 * 32 \text{ ns} = 128 \text{ ns}$

When PRG_HS_TRAIL = 1100 @ 1 Gb/s, $>T_{HS-TRAIL} = 12 * 8 \text{ ns} = 96 \text{ ns}$

13.6.7 Packet Data and Pixel Formats

The DSI Host Controller Core and the external DSI Peripheral explicitly handle and exchanges 32-bit words encapsulated as packet data. The mapping between pixel formats and pixel data must be done on both sides by user application on both sides of the DSI link. The DSI spec gives many examples of common pixel formats. The DSI Host Controller Core and external DSI Peripheral have no preference or limitation on which pixel formats it will handle. The IP which provides video source packet data through DPI-2 interface (e.g. LCDIF) onto DSI should also be configured by the user application to match the pixel formats expected by the DSI Host Controller.

The DPI-2/DBI-2 interface core handles pixel to byte packing and byte to pixel unpacking according to the MIPI DSI specification.

13.6.8 MIPI DSI Host Memory Map/Register Definition

MIPI_DSI_HOST memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
30A0_0000	MIPI_DSI_HOST_DSI_HOST_CFG_NUM_LANES	32	R	0000_0000h	13.6.8.1/ 5089
30A0_0004	MIPI_DSI_HOST_DSI_HOST_CFG_NONCONTINUOUS_CLK	32	R	0000_0000h	13.6.8.2/ 5090
30A0_0008	MIPI_DSI_HOST_DSI_HOST_CFG_T_PRE	32	R	0000_0000h	13.6.8.3/ 5091

Table continues on the next page...

MIPI_DSI_HOST memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30A0_000C	MIPI_DSI_HOST_DSI_HOST_CFG_T_POST	32	R	0000_0000h	13.6.8.4/5091
30A0_0010	MIPI_DSI_HOST_DSI_HOST_CFG_TX_GAP	32	R	0000_0000h	13.6.8.5/5092
30A0_0014	MIPI_DSI_HOST_DSI_HOST_CFG_AUTOINSERT_EOTP	32	R	0000_0000h	13.6.8.6/5093
30A0_0018	MIPI_DSI_HOST_DSI_HOST_CFG_EXTRA_CMDS_AFTE R_EOTP	32	R	0000_0000h	13.6.8.7/5093
30A0_001C	MIPI_DSI_HOST_DSI_HOST_CFG_HTX_TO_COUNT	32	R	0000_0000h	13.6.8.8/5094
30A0_0020	MIPI_DSI_HOST_DSI_HOST_CFG_LRX_H_TO_COUNT	32	R	0000_0000h	13.6.8.9/5094
30A0_0024	MIPI_DSI_HOST_DSI_HOST_CFG_BTA_H_TO_COUNT	32	R	0000_0000h	13.6.8.10/5095
30A0_0028	MIPI_DSI_HOST_DSI_HOST_CFG_TWAKEUP	32	R	0000_0000h	13.6.8.11/5095
30A0_002C	MIPI_DSI_HOST_DSI_HOST_CFG_STATUS_OUT	32	R	0000_0000h	13.6.8.12/5096
30A0_0030	MIPI_DSI_HOST_DSI_HOST_RX_ERROR_STATUS	32	R	0000_0000h	13.6.8.13/5098

13.6.8.1 MIPI_DSI_HOST_DSI_HOST_CFG_NUM_LANES

Address: 30A0_0000h base + 0h offset = 30A0_0000h

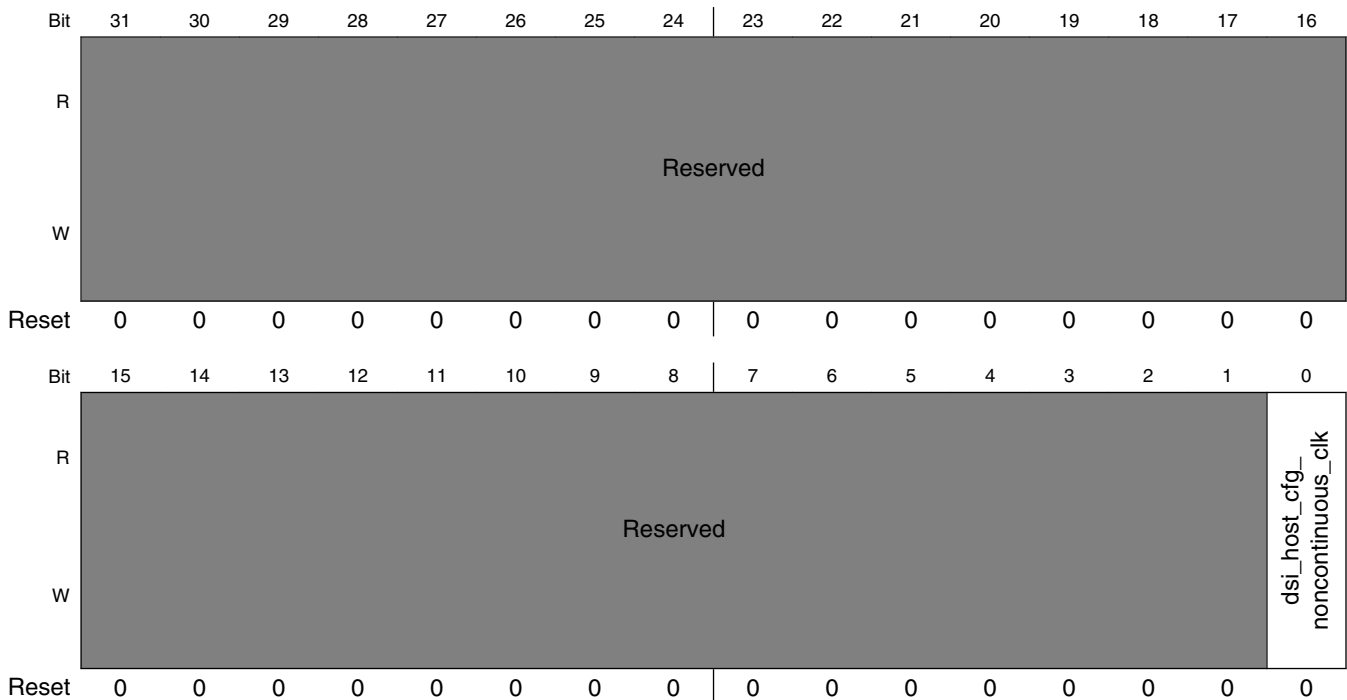
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																dsi_host_cfg_num_lanes
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DSI_HOST_CFG_NUM_LANES field descriptions

Field	Description
31–2 -	This field is reserved.
dsi_host_cfg_num_lanes	Sets the number of active lanes that are to be used for transmitting data. • 2'b00 - 1 Lane • 2'b01 - 2 Lanes • 2'b10 - 3 Lanes • 2'b11 - 4 Lanes

13.6.8.2 MIPI_DSI_HOST_DSI_HOST_CFG_NONCONTINUOUS_CLK

Address: 30A0_0000h base + 4h offset = 30A0_0004h



MIPI_DSI_HOST_DSI_HOST_CFG_NONCONTINUOUS_CLK field descriptions

Field	Description
31–1 -	This field is reserved.
0 dsi_host_cfg_ noncontinuous_clk	Sets the Host Controller into non-continuous MIPI clock mode. When in non-continuous clock mode, the high speed clock will transistion into low power mode between transmissions. 1'b0 - Continuous high speed clock 1'b1 - Non-Continuous high speed clock

13.6.8.3 MIPI_DSI_HOST_DSI_HOST_CFG_T_PRE

NOTE

Please refer to the MIPI Alliance Specification for D-PHY (Global Operation Timing Parameters) for details on calculating the parameter described below

Address: 30A0_0000h base + 8h offset = 30A0_0008h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dsi_host_cfg_t_pre															
W	Reserved																dsi_host_cfg_t_pre															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DSI_HOST_CFG_T_PRE field descriptions

Field	Description
31–7 -	This field is reserved.
dsi_host_cfg_t_pre	Sets the number of byte clock periods ('clk_byte' input) that the controller will wait after enabling the clock lane for HS operation before enabling the data lanes for HS operation. This setting represents the TCLK-PRE parameter. The minimum value for this port is 1.

13.6.8.4 MIPI_DSI_HOST_DSI_HOST_CFG_T_POST

NOTE

Please refer to the MIPI Alliance Specification for D-PHY (Global Operation Timing Parameters) for details on calculating the parameter described below

Address: 30A0_0000h base + Ch offset = 30A0_000Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																ds_i_host_cfg_t_post															
W	Reserved																ds_i_host_cfg_t_post															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DSI_HOST_CFG_T_POST field descriptions

Field	Description
31–7 -	This field is reserved.

Table continues on the next page...

MIPI_DSI_HOST_DSI_HOST_CFG_T_POST field descriptions (continued)

Field	Description
dsi_host_cfg_t_post	Sets the number of byte clock periods ('clk_byte' input) to wait before putting the clock lane into LP mode after the data lanes have been detected to be in Stop State. This setting represents the DPHY timing parameters TLPX (TxClkEsc) + TCLK-PREPARE + TCLK-ZERO + TCLK-PRE requirement for the clock lane before the data lane is allowed to change from LP11 to start a high speed transmission. The minimum value for this port is 1.

13.6.8.5 MIPI_DSI_HOST_DSI_HOST_CFG_TX_GAP

NOTE

Please refer to the MIPI Alliance Specification for D-PHY (Global Operation Timing Parameters) for details on calculating the parameter described below

Address: 30A0_0000h base + 10h offset = 30A0_0010h

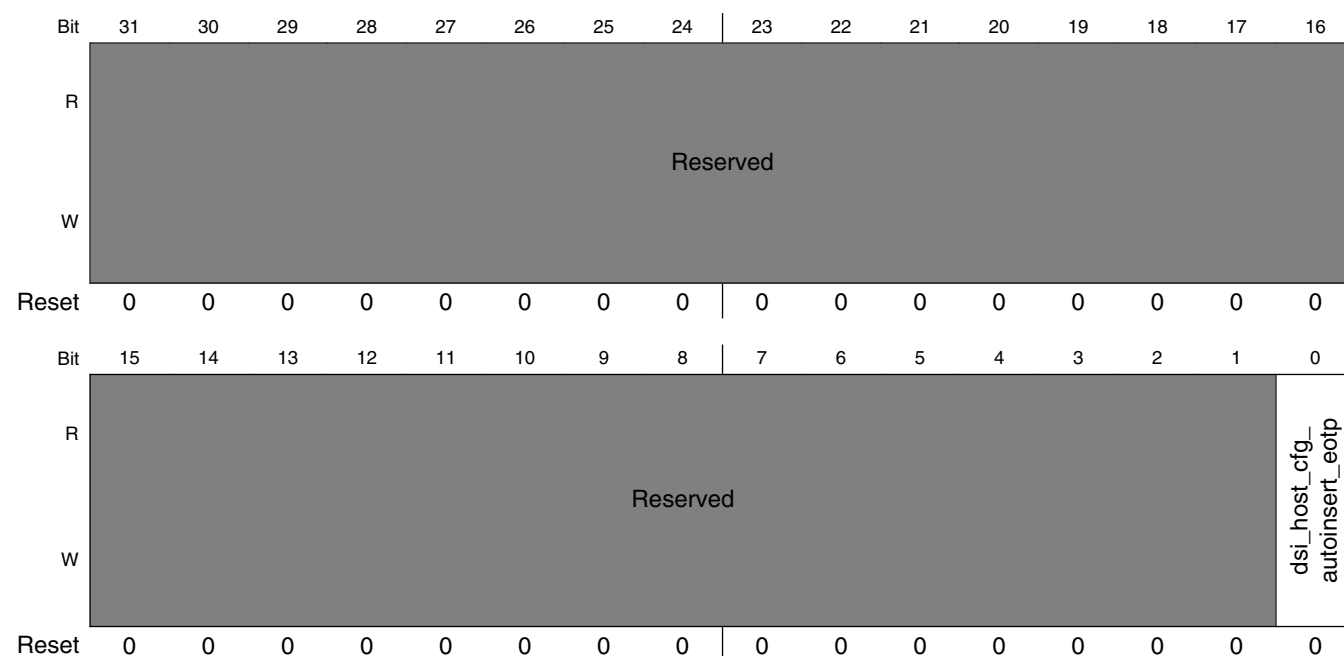
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dsi_host_cfg_tx_gap															
W	Reserved																dsi_host_cfg_tx_gap															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DSI_HOST_CFG_TX_GAP field descriptions

Field	Description
31–7 -	This field is reserved.
dsi_host_cfg_tx_gap	Sets the number of byte clock periods ('clk_byte' input) that the controller will wait after the clock lane has been put into LP mode before enabling the clock lane for HS mode again. This setting represents the THS-EXIT parameter. The minimum value for this port is 1.

13.6.8.6 MIPI_DSI_HOST_DSI_HOST_CFG_AUTOINSERT_EOTP

Address: 30A0_0000h base + 14h offset = 30A0_0014h

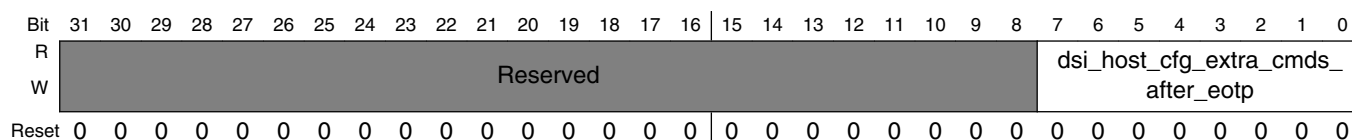


MIPI_DSI_HOST_DSI_HOST_CFG_AUTOINSERT_EOTP field descriptions

Field	Description
31–1 -	This field is reserved.
0 dsi_host_cfg_ autoinsert_eotp	Enables the Host Controller to automatically insert an EoTp short packet when switching from HS to LP mode. 1'b0 - EoTp is not automatically inserted 1'b1 - EoTp is automatically inserted

13.6.8.7 MIPI_DSI_HOST_DSI_HOST_CFG_EXTRA_CMDS_AFTER_EOTP

Address: 30A0_0000h base + 18h offset = 30A0_0018h



MIPI_DSI_HOST_DSI_HOST_CFG_EXTRA_CMDS_AFTER_EOTP field descriptions

Field	Description
31–8 -	This field is reserved.
dsi_host_cfg_extra_cmds_after_eotp	Configures the DSI Host Controller to send extra End Of Transmission Packets after the end of a packet. The value is the number of extra EOTP packets sent.

13.6.8.8 MIPI_DSI_HOST_DSI_HOST_CFG_HTX_TO_COUNT

Address: 30A0_0000h base + 1Ch offset = 30A0_001Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W	Reserved								dsi_host_cfg_htx_to_count																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DSI_HOST_CFG_HTX_TO_COUNT field descriptions

Field	Description
31–24 -	This field is reserved.
dsi_host_cfg_htx_to_count	Sets the value of the DSI Host High Speed TX timeout count in clk_byte clock periods that once reached will initiate a timeout error and follow the recovery procedure documented in the DSI specification. This timeout parameter should be configured to represent the time taken to transmit the biggest HS data payload. If this timeout is reached the DSI byte count is cleared and the HS transmission is aborted. This timer can be also disabled, when set to 0.

13.6.8.9 MIPI_DSI_HOST_DSI_HOST_CFG_LRX_H_TO_COUNT

Address: 30A0_0000h base + 20h offset = 30A0_0020h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W	Reserved								dsi_host_cfg_lrx_h_to_count																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DSI_HOST_CFG_LRX_H_TO_COUNT field descriptions

Field	Description
31–24 -	This field is reserved.
dsi_host_cfg_lrx_h_to_count	Sets the value of the DSI Host low power RX timeout count in clk_byte clock periods that once reached will initiate a timeout error and follow the recovery procedure documented in the DSI specification. This timeout parameter should be configured to represent the time taken to receive the biggest LP (Escape mode) data payload. If this timeout is reached, the DSI byte count is cleared and the LP reception is aborted. This timer can be also disabled, when set to 0

13.6.8.10 MIPI_DSI_HOST_DSI_HOST_CFG_BTA_H_TO_COUNT

Address: 30A0_0000h base + 24h offset = 30A0_0024h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0								
R																																								
W																																								
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0								

MIPI_DSI_HOST_DSI_HOST_CFG_BTA_H_TO_COUNT field descriptions

Field	Description
31–24 -	This field is reserved.
dsi_host_cfg_bta_h_to_count	Sets the value of the DSI Host Bus Turn Around (BTA) timeout in clk_byte clock periods that once reached will initiate a timeout error.

13.6.8.11 MIPI_DSI_HOST_DSI_HOST_CFG_TWAKEUP

Address: 30A0_0000h base + 28h offset = 30A0_0028h

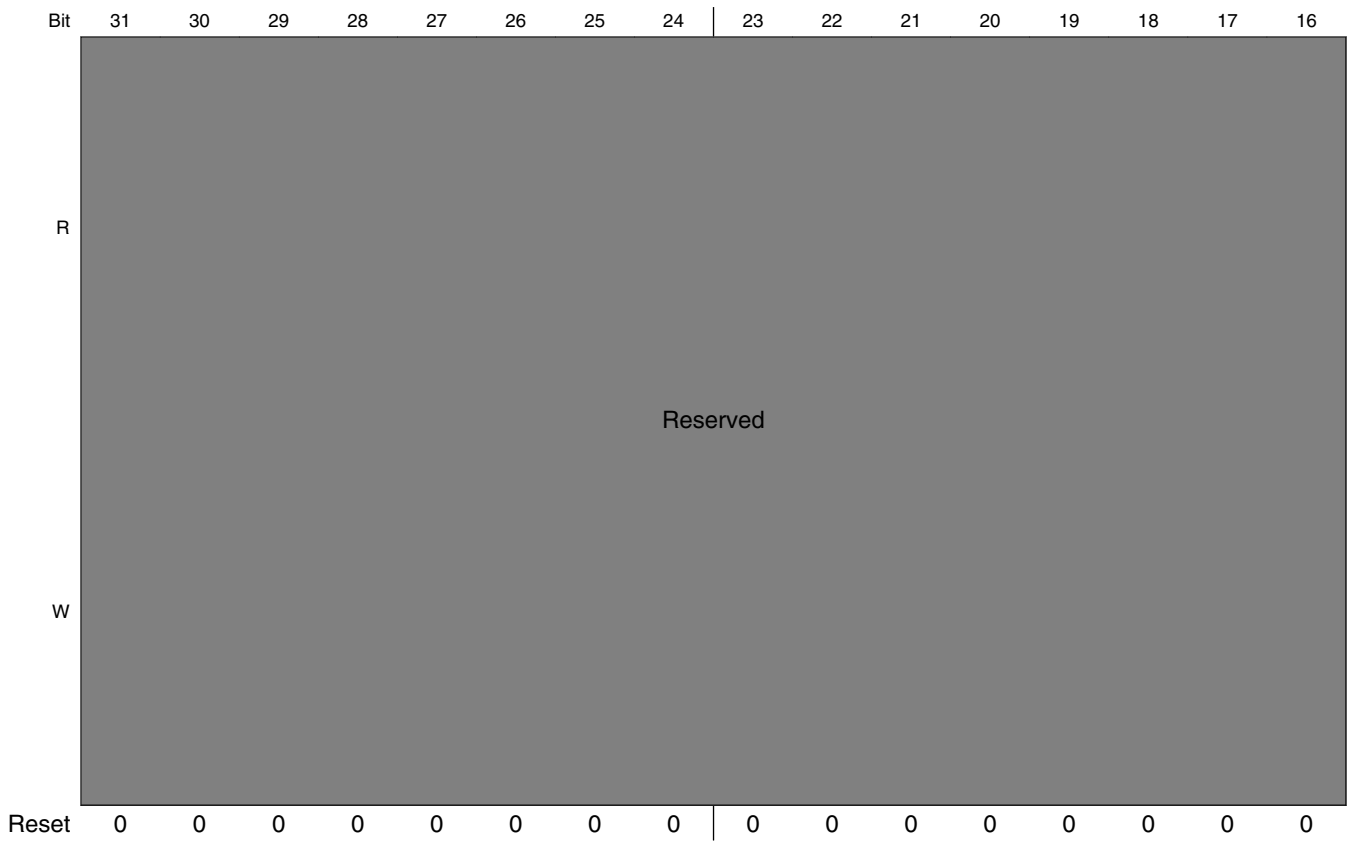
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved													ds_i_host_cfg_twakeup																		
W	Reserved													ds_i_host_cfg_twakeup																		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

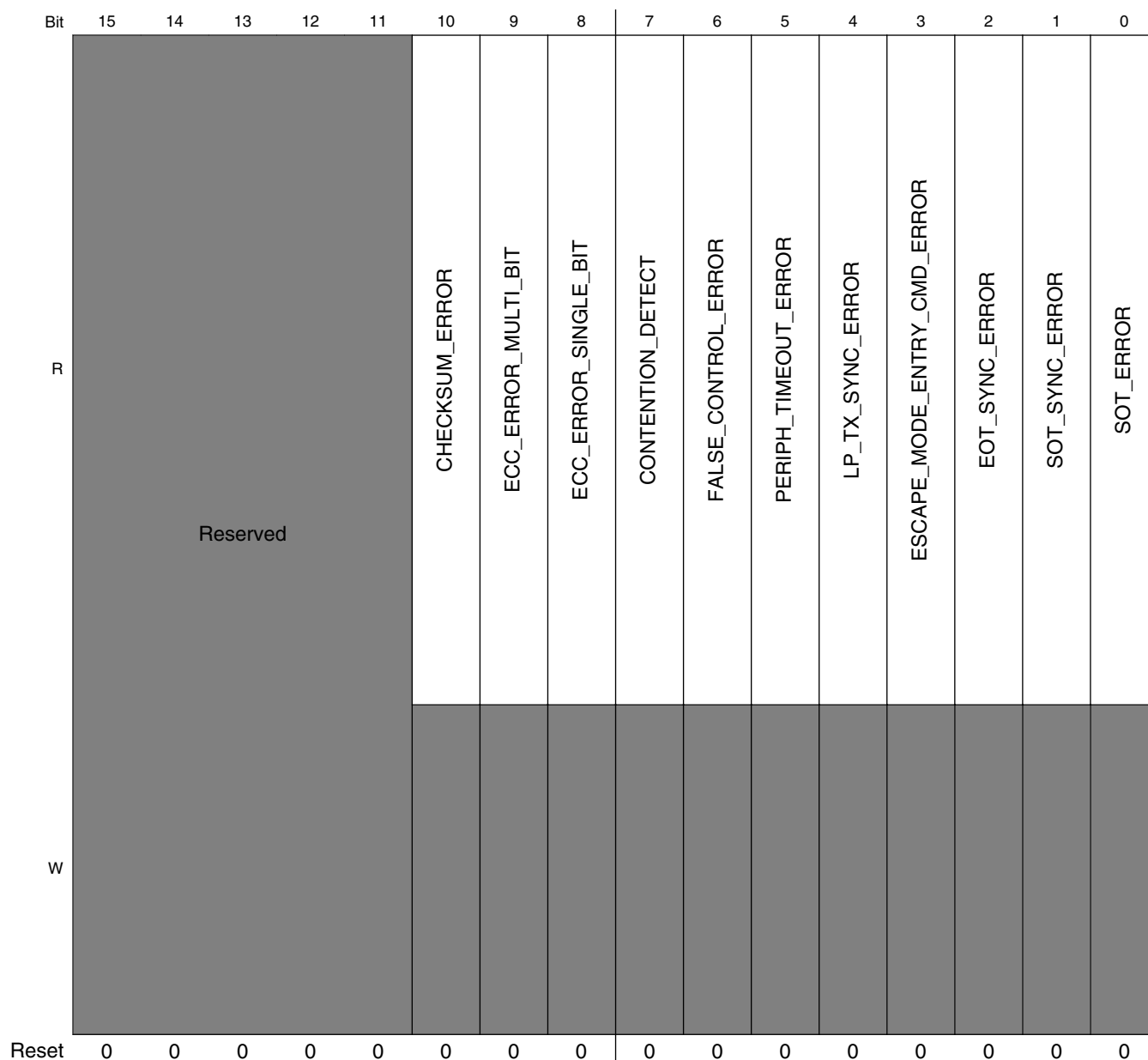
MIPI_DSI_HOST_DSI_HOST_CFG_TWAKEUP field descriptions

Field	Description
31–19 -	This field is reserved.
dsi_host_cfg_twakeup	DPHY Twakeup timing parameter. Sets the number of clk_esc clock periods to keep a clock or data lane in Mark-1 state after exiting ULPS. The MIPI DPHY spec requires a minimum of 1ms in Mark-1 state after leaving ULPS.

13.6.8.12 MIPI_DSI_HOST_DSI_HOST_CFG_STATUS_OUT

Address: 30A0_0000h base + 2Ch offset = 30A0_002Ch





MIPI_DSI_HOST_DSI_HOST_CFG_STATUS_OUT field descriptions

Field	Description
31–11 -	This field is reserved. Reserved
10 CHECKSUM_ ERROR	Checksum Error (long packet only) – Checksum error from peripheral error report, cleared upon read
9 ECC_ERROR_ MULTI_BIT	ECC Error, multi-bit (detected, not corrected) – ECC multi-bit error from peripheral error report, cleared upon read

Table continues on the next page...

MIPI_DSI_HOST_DSI_HOST_CFG_STATUS_OUT field descriptions (continued)

Field	Description
8 ECC_ERROR_SINGLE_BIT	ECC single bit error from peripheral error report, cleared upon read.
7 CONTENTION_DETECT	Contention Detected – Contention Detection from peripheral error report, cleared upon read
6 FALSE_CONTROL_ERROR	False Control Error – False Control Error from peripheral error report, cleared upon read
5 PERIPH_TIMEOUT_ERROR	Peripheral Timeout Error – Peripheral Timeout error from peripheral error report, cleared upon read
4 LP_TX_SYNC_ERROR	Low-Power Transmit Sync Error – Low Power Transmit Sync error from peripheral error report, cleared upon read.
3 ESCAPE_MODE_ENTRY_CMD_ERROR	Escape Mode Entry Command Error – Escape Mode Entry Command Error from peripheral error report, cleared upon read
2 EOT_SYNC_ERROR	EoT Sync Error – End of Transmission (EoT) Sync Error from peripheral error report, cleared upon read
1 SOT_SYNC_ERROR	SoT Sync Error – Start of Transmission (SoT) Sync Error from peripheral error report, cleared upon read
0 SOT_ERROR	SoT Error – Start of Transmission (SoT) Error from peripheral error report, cleared upon read.

13.6.8.13 MIPI_DSI_HOST_DSI_HOST_RX_ERROR_STATUS

Address: 30A0_0000h base + 30h offset = 30A0_0030h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dsi_host_rx_error_status															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DSI_HOST_RX_ERROR_STATUS field descriptions

Field	Description
31–11 -	This field is reserved.
dsi_host_rx_error_status	Status Register for Host receive error detection, ECC errors, CRC errors and for timeout indicators

Table continues on the next page...

MIPI_DSI_HOST_DSI_HOST_RX_ERROR_STATUS field descriptions (continued)

Field	Description
	• [0] ECC single bit error detected • [1] ECC multi bit error detected • [6:2] Errored bit position for single bit ECC error • [7] CRC error detected • [8] High Speed forward TX timeout detected • [9] Reverse Low power data receive timeout detected • [10] BTA timeout detected

13.6.9 MIPI DSI HOST DPI INTFC Memory Map/Register Definition**MIPI_DSI_HOST_DPI_INTFC memory map**

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
30A0_0200	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_PAYLOAD_SIZE	32	R	0000_0000h	13.6.9.1/5100
30A0_0204	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_FIFO_SEND_LEVEL	32	R	0000_0000h	13.6.9.2/5100
30A0_0208	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_INTERFACE_COLOR_CODING	32	R	0000_0000h	13.6.9.3/5101
30A0_020C	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_FORMAT	32	R	0000_0000h	13.6.9.4/5101
30A0_0210	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VSYNC_POLARITY	32	R	0000_0000h	13.6.9.5/5102
30A0_0214	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HSYNC_POLARITY	32	R	0000_0000h	13.6.9.6/5103
30A0_0218	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VIDEO_MODE	32	R	0000_0000h	13.6.9.7/5104
30A0_021C	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HFP	32	R	0000_0000h	13.6.9.8/5104
30A0_0220	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HBP	32	R	0000_0000h	13.6.9.9/5104
30A0_0224	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HSA	32	R	0000_0000h	13.6.9.10/5105
30A0_0228	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_ENABLE_MULT_PKTS	32	R	0000_0000h	13.6.9.11/5106
30A0_022C	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VBP	32	R	0000_0000h	13.6.9.12/5106
30A0_0230	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VFP	32	R	0000_0000h	13.6.9.13/5107
30A0_0234	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_BLLP_MODE	32	R	0000_0000h	13.6.9.14/5107
30A0_0238	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_USE_NULL_PKT_BLLP	32	R	0000_0000h	13.6.9.15/5108

Table continues on the next page...

MIPI_DSI_HOST_DPI_INTFC memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
30A0_023C	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VACTIVE	32	R	0000_0000h	13.6.9.16/5109
30A0_0240	MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VC	32	R	0000_0000h	13.6.9.17/5109

13.6.9.1 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_PAYLOAD

Address: 30A0_0000h base + 200h offset = 30A0_0200h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																ds1_host_cfg_dpi_pixel_payload_size															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_PAYLOAD_SIZE field descriptions

Field	Description
31–16 -	This field is reserved.
dsi_host_cfg_dpi_pixel_payload_size	Maximum number of pixels that should be sent as one DSI packet. Recommended that the line size (in pixels) is evenly divisible by this parameter (packet payload size in pixels).

13.6.9.2 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_FIFO_SEND

Address: 30A0_0000h base + 204h offset = 30A0_0204h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																ds1_host_cfg_dpi_pixel_fifo_send_level															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_FIFO_SEND_LEVEL field descriptions

Field	Description
31–16 -	This field is reserved.
dsi_host_cfg_dpi_pixel_fifo_send_level	In order to optimize DSI utility, the DPI bridge buffers a cerntain number of DPI pixels before initiating a DSI packet. This configuration port controls the level at which the DPI Host bridge begins sending pixels.

13.6.9.3 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_INTERFACE_COLOR_CODING

Address: 30A0_0000h base + 208h offset = 30A0_0208h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved													dsi_host_cfg_dpi_interface_color_coding		
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_INTERFACE_COLOR_CODING field descriptions

Field	Description
31–3 -	This field is reserved.
dsi_host_cfg_dpi_interface_color_coding	Sets the distribution of RGB bits within the 24-bit d bus, as specified by the DPI specification. 0= 16-bit Configuration 1 1= 16-bit Configuration 2 2= 16-bit Configuration 3 3= 18-bit Configuration 1 4= 18-bit Configuration 2 5= 24-bit

13.6.9.4 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_FORMAT

Address: 30A0_0000h base + 20Ch offset = 30A0_020Ch

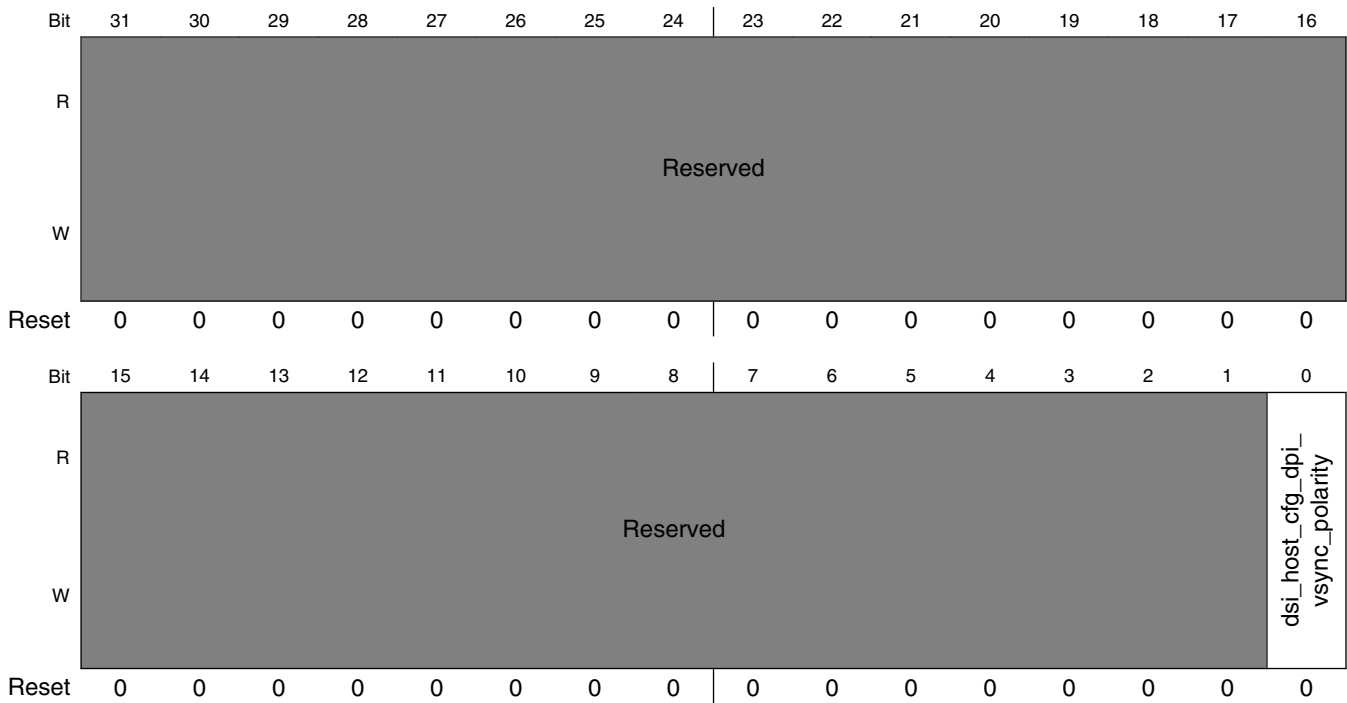
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved													dsi_host_cfg_dpi_pixel_format		
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_PIXEL_FORMAT field descriptions

Field	Description
31–2 -	This field is reserved.
dsi_host_cfg_dpi_pixel_format	Sets the DSI packet type of the pixels. 0 - 16 bit 1 - 18 bit 2 - 18 bit loosely packed 3 - 24 bit

13.6.9.5 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VSYNC_POLARITY

Address: 30A0_0000h base + 210h offset = 30A0_0210h

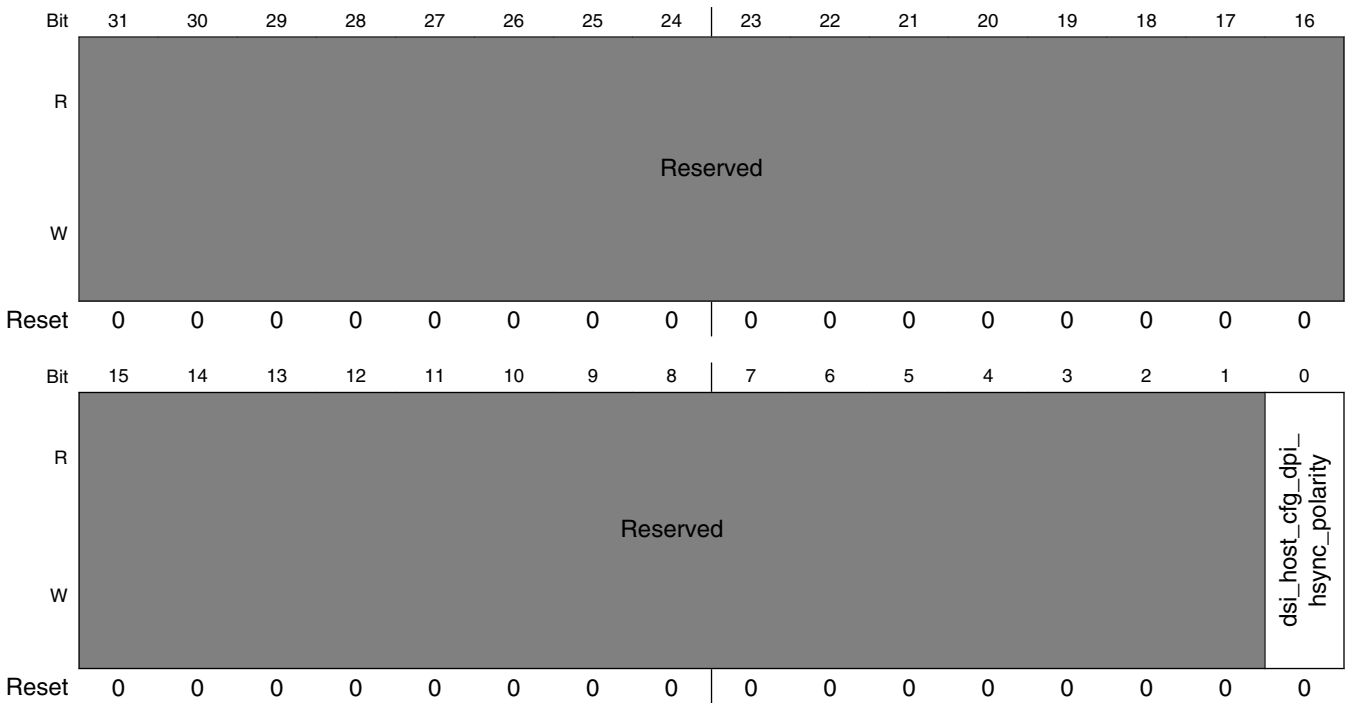


MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VSYNC_POLARITY field descriptions

Field	Description
31–1 -	This field is reserved.
0 dsi_host_cfg_dpi_vsync_polarity	Sets polarity of dpi_vsync_input 0 - active low 1 - active high

13.6.9.6 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HSYNC_POLARITY

Address: 30A0_0000h base + 214h offset = 30A0_0214h



MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HSYNC_POLARITY field descriptions

Field	Description
31–1 -	This field is reserved.
0 dsi_host_cfg_dpi_hsync_polarity	Sets polarity of dpi_hsync_input 0 - active low 1 - active high

13.6.9.7 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VIDEO_MODE

Address: 30A0_0000h base + 218h offset = 30A0_0218h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved														dsi_host_ cfg_dpi_ video_mode	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VIDEO_MODE field descriptions

Field	Description
31–2 -	This field is reserved.
dsi_host_cfg_ dpi_video_mode	Select DSI video mode that the host DPI module should generate packets for. 2'b00 - Non-Burst mode with Sync Pulses 2'b01 - Non-Burst mode with Sync Events 2'b10 - Burst mode 2'b11 - Reserved, not valid

13.6.9.8 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HFP

Address: 30A0_0000h base + 21Ch offset = 30A0_021Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dsi_host_cfg_dpi_hfp															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HFP field descriptions

Field	Description
31–16 -	This field is reserved.
dsi_host_cfg_ dpi_hfp	Sets the DSI packet payload size, in bytes, of the horizontal front porch blanking packet.

13.6.9.9 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HBP

Address: 30A0_0000h base + 220h offset = 30A0_0220h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dsi_host_cfg_dpi_hbp															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HBP field descriptions

Field	Description
31–16 -	This field is reserved.
dsi_host_cfg_ dpi_hbp	Sets the DSI packet payload size, in bytes, of the horizontal back porch blanking packet.

13.6.9.10 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HSA

Address: 30A0_0000h base + 224h offset = 30A0_0224h

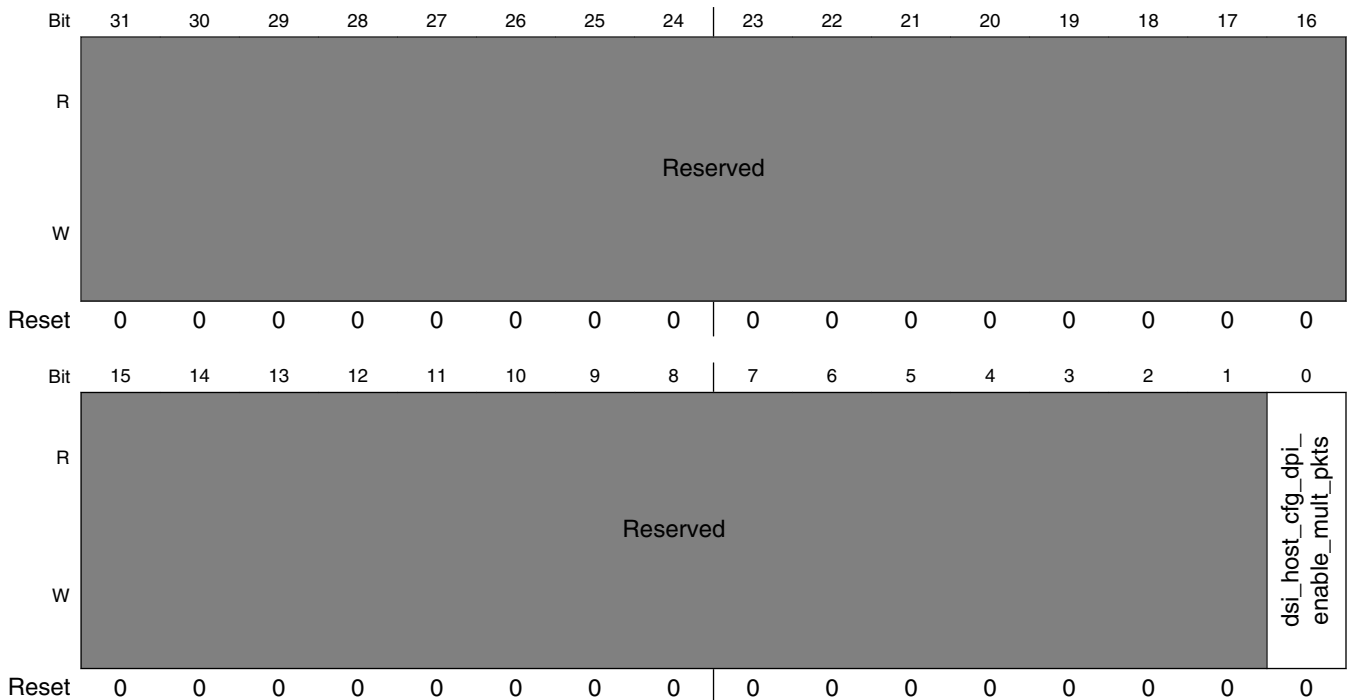
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dsi_host_cfg_dpi_hsa															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_HSA field descriptions

Field	Description
31–16 -	This field is reserved.
dsi_host_cfg_ dpi_hsa	Sets the DSI packet payload size, in bytes, of the horizontal sync width filler blanking packet.

13.6.9.11 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_ENABLE_MULT_PKT

Address: 30A0_0000h base + 228h offset = 30A0_0228h

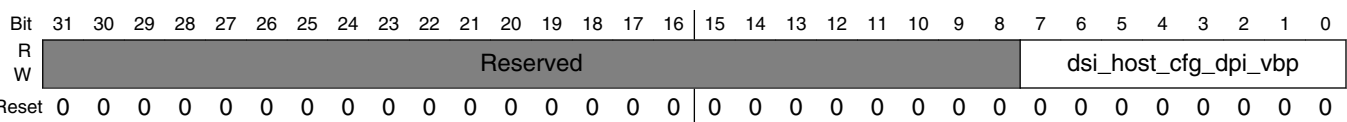


MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_ENABLE_MULT_PKTS field descriptions

Field	Description
31–1 -	This field is reserved.
0 dsi_host_cfg_dpi_enable_mult_pkts	Enable Multiple packets per video line. When enabled, cfg_dpi_pixel_payload_size must be set to exactly half the size of the video line. 0 - Video Line is sent in a single packet 1 - Video Line is sent in two packets

13.6.9.12 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VBP

Address: 30A0_0000h base + 22Ch offset = 30A0_022Ch



MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VBP field descriptions

Field	Description
31–8 -	This field is reserved.
dsi_host_cfg_ dpi_vbp	Sets the number of lines in the vertical back porch.

13.6.9.13 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VFP

Address: 30A0_0000h base + 230h offset = 30A0_0230h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
R																												dsi_host_cfg_dpi_vfp									
W	Reserved																											dsi_host_cfg_dpi_vfp									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VFP field descriptions

Field	Description
31–8 -	This field is reserved.
dsi_host_cfg_ dpi_vfp	Sets the number of lines in the vertical front porch.

13.6.9.14 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_BLLP_MODE

Address: 30A0_0000h base + 234h offset = 30A0_0234h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_BLLP_MODE field descriptions

Field	Description
31–1 -	This field is reserved.
0 dsi_host_cfg_ dpi_bllp_mode	Optimize bllp periods to Low Power mode when possible 0 - blanking packets are sent during BLLP periods 1 - LP mode is used for BLLP periods

13.6.9.15 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_USE_NULL_PKT

Address: 30A0_0000h base + 238h offset = 30A0_0238h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															dsi_host_cfg_dpi_use_ null_pkt_bllp
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_USE_NULL_PKT_BLLP field descriptions

Field	Description
31–1 -	This field is reserved.
0 dsi_host_cfg_ dpi_use_null_ pkt_bllp	Selects type of blanking packet to be sent during bllp region 0 - Blanking packet used in bllp region 1 - Null packet used in bllp region

13.6.9.16 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VACTIVE

Address: 30A0_0000h base + 23Ch offset = 30A0_023Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W	Reserved																ds_i_host_cfg_dpi_vactive															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VACTIVE field descriptions

Field	Description
31–14 -	This field is reserved.
dsi_host_cfg_dpi_vactive	Sets the number of lines in the vertical active area. This field is equivalent to (real vertical size) - 1. For example, for an image of size 640x480, the bit field should be set as 479.

13.6.9.17 MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VC

Address: 30A0_0000h base + 240h offset = 30A0_0240h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_DPI_INTFC_DSI_HOST_CFG_DPI_VC field descriptions

Field	Description
31–2 -	This field is reserved.
dsi_host_cfg_dpi_vc	Sets the Virtual Channel (VC) of packets that will be sent to the receive packet interface. Packets with VC not equal to this value are discarded and the "DSI VC ID Invalid" bit (bit 12) in the DSI error report is set.

13.6.10 MIPI DSI Host APB PKT IF Memory Map/Register Definition

MIPI_DSI_HOST_APB_PKT_IF memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
30A0_0280	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_TX_PAYLOAD	32	R/W	0000_0000h	13.6.10.1/ 5110
30A0_0284	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_CONTROL	32	R/W	0000_0000h	13.6.10.2/ 5111
30A0_0288	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_SEND_PACKET	32	R/W	0000_0000h	13.6.10.3/ 5111
30A0_028C	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_STATUS	32	R	0000_0000h	13.6.10.4/ 5112
30A0_0290	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_FIFO_WRITE_LEVEL	32	R	0000_0000h	13.6.10.5/ 5112
30A0_0294	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_FIFO_READ_LEVEL	32	R	0000_0000h	13.6.10.6/ 5113
30A0_0298	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_RX_PAYLOAD	32	R	0000_0000h	13.6.10.7/ 5113
30A0_029C	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_RX_PKT_HEADER	32	R	0000_0000h	13.6.10.8/ 5114
30A0_02A0	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_STATUS	32	R	0000_0000h	13.6.10.9/ 5114
30A0_02A4	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_STATUS2	32	R	0000_0000h	13.6.10.10/ 5115
30A0_02A8	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_MASK	32	R/W	0000_0000h	13.6.10.11/ 5115
30A0_02AC	MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_MASK2	32	R/W	0000_0000h	13.6.10.12/ 5116

13.6.10.1 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_TX_PAYLOAD

Address: 30A0_0000h base + 280h offset = 30A0_0280h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_TX_PAYLOAD field descriptions

Field	Description
dsi_host_tx_payload	Tx Payload data write register. Writes to this registers load the payload fifo with 32 bit values.

13.6.10.2 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_CONTROL

Address: 30A0_0000h base + 284h offset = 30A0_0284h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved					dsi_host_pkt_control																										
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_CONTROL field descriptions

Field	Description
31–27 -	This field is reserved.
dsi_host_pkt_control	Tx packet control register. • [15:0] - Packet word count • [17:16] - Packet Virtual Channel • [23:18] - Packet Header DSI Data Type • [24] - Lp or HS select. 0 - LP mode, 1 - HS mode • [25] - perform BTA after packet is sent • [26] - perform BTA only, no packet tx

13.6.10.3 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_SEND_PACKET

Address: 30A0_0000h base + 288h offset = 30A0_0288h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															dsi_host_send_packet
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_SEND_PACKET field descriptions

Field	Description
31–1 -	This field is reserved.
0 dsi_host_send_packet	Tx send packet. Writing to this register causes the packet described in dsi_host_pkt_control to be sent.

13.6.10.4 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_STATUS

Address: 30A0_0000h base + 28Ch offset = 30A0_028Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
R	Reserved																							dsi_host_pkt_status													
W	Reserved																																				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_STATUS field descriptions

Field	Description
31–9 -	This field is reserved.
dsi_host_pkt_status	Status of APB to packet interface • [0] - state machine not idle • [1] - Tx packet done • [2] - dphy direction 0 - tx had control, 1 - rx has control • [3] - tx fifo overflow • [4] - tx fifo underflow • [5] - rx fifo overflow • [6] - rx fifo underflow • [7] - rx packet header has been received • [8] - all rx packet payload data has been receive d

13.6.10.5 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_FIFO_WR_LEVEL

Address: 30A0_0000h base + 290h offset = 30A0_0290h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dsi_host_pkt_fifo_wr_level															
W	Reserved																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_FIFO_WR_LEVEL field descriptions

Field	Description
31–16 -	This field is reserved.
dsi_host_pkt_fifo_wr_level	Write level of APB to pkt interface fifo

13.6.10.6 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_FIFO_RD_LEVEL

Address: 30A0_0000h base + 294h offset = 30A0_0294h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dsi_host_pkt_fifo_rd_level															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_FIFO_RD_LEVEL field descriptions

Field	Description
31–16 -	This field is reserved.
dsi_host_pkt_fifo_rd_level	Read level of APB to pkt interface fifo

13.6.10.7 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_RX_PAYLOAD

Address: 30A0_0000h base + 298h offset = 30A0_0298h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	dsi_host_pkt_rx_payload																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_RX_PAYLOAD field descriptions

Field	Description
dsi_host_pkt_rx_payload	APB to pkt interface rx payload read

13.6.10.8 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_RX_PKT_HEADER

Address: 30A0_0000h base + 29Ch offset = 30A0_029Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								dsi_host_pkt_rx_pkt_header																							
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_PKT_RX_PKT_HEADER field descriptions

Field	Description
31–24 -	This field is reserved.
dsi_host_pkt_rx_pkt_header	APB to pkt interface rx packet header [15:0] word count [21:16] data type [23:22] Virtual Channel

13.6.10.9 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_STATUS

Address: 30A0_0000h base + 2A0h offset = 30A0_02A0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	dsi_host_irq_status																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_STATUS field descriptions

Field	Description
dsi_host_irq_status	Status of APB to packet interface [0] - state machine not idle [1] - Tx packet done [2] - dphy direction 0 - tx had control, 1 - rx has control [3] - tx fifo overflow [4] - tx fifo underflow [5] - rx fifo overflow [6] - rx fifo underflow [7] - rx packet header has been received [8] - all rx packet payload data has been received [28:9] - map directory to dsi host controller status_out port bit descriptions [29] - host bta timeout, host controller host_bta_timeout port [30] - low power rx timeout, host controller lp_rx_timeout port [31] - high speed tx timeout, host controller hs_tx_timeout port

13.6.10.10 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_STATUS2

Address: 30A0_0000h base + 2A4h offset = 30A0_02A4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved													dsi_host_irq_status2		
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_STATUS2 field descriptions

Field	Description
31–3 -	This field is reserved.
dsi_host_irq_status2	Status of APB to packet interface part 2. Read part 2 first then dsi_host_irq_status. Reading dsi_host_irq_status will clear both status and status 2. • [0] - single bit ecc error • [1] - multi bit ecc error • [2] - crc error

13.6.10.11 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_MASK

Address: 30A0_0000h base + 2A8h offset = 30A0_02A8h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	dsi_host_irq_mask																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_MASK field descriptions

Field	Description
dsi_host_irq_mask	irq mask • [0] - state machine not idle • [1] - Tx packet done • [2] - dphy direction 0 - tx had control, 1 - rx has control • [3] - tx fifo overflow • [4] - tx fifo underflow • [5] - rx fifo overflow • [6] - rx fifo underflow • [7] - rx packet header has been received • [8] - all rx packet payload data has been received • [28:9] - map directory to dsi host controller status_out port bit descriptions

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_MASK field descriptions (continued)

Field	Description
	[29] - host bta timeout, host controller host_bta_timeout port [30] - low power rx timeout, host controller lp_rx_timeout port [31] - high speed tx timeout, host controller hs_tx_timeout port

13.6.10.12 MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_MASK2

Address: 30A0_0000h base + 2ACh offset = 30A0_02ACh

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_APB_PKT_IF_DSI_HOST_IRQ_MASK2 field descriptions

Field	Description
31–3 -	This field is reserved.
dsi_host_irq_mask2	irq mask 2 [0] - single bit ecc error [1] - multi bit ecc error [2] - crc error

13.6.11 MIPI DSI Host IP1 DPHY INTFC Memory Map/Register Definition

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
30A0_0300	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_PD_DPHY	32	R	0000_0001h	13.6.11.1/ 5118
30A0_0304	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_PREPARE	32	R	0000_0000h	13.6.11.2/ 5118
30A0_0308	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_PREPARE	32	R	0000_0000h	13.6.11.3/ 5119
30A0_030C	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_ZERO	32	R	0000_0000h	13.6.11.4/ 5120
30A0_0310	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_ZERO	32	R	0000_0000h	13.6.11.5/ 5121

Table continues on the next page...

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
30A0_0314	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_TRAIL	32	R	0000_0000h	13.6.11.6/5122
30A0_0318	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_TRAIL	32	R	0000_0000h	13.6.11.7/5122
30A0_031C	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_PD_PLL	32	R	0000_0001h	13.6.11.8/5123
30A0_0320	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_TST	32	R	0000_0025h	13.6.11.9/5124
30A0_0324	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CN	32	R	0000_0000h	13.6.11.10/5124
30A0_0328	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CM	32	R	0000_0000h	13.6.11.11/5125
30A0_032C	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CO	32	R	0000_0000h	13.6.11.12/5126
30A0_0330	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_LOCK	32	R	0000_0000h	13.6.11.13/5127
30A0_0334	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_LOCK_BYP	32	R	0000_0000h	13.6.11.14/5128
30A0_0338	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RTERM_SEL	32	R	0000_0000h	13.6.11.15/5128
30A0_033C	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_AUTO_PD_EN	32	R	0000_0000h	13.6.11.16/5129
30A0_0340	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RXLPRP	32	R	0000_0000h	13.6.11.17/5130
30A0_0344	MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RXCDR_P	32	R	0000_0000h	13.6.11.18/5130

13.6.11.1 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_PD_DPHY

Address: 30A0_0000h base + 300h offset = 30A0_0300h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															dphy_pd_dphy
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_PD_DPHY field descriptions

Field	Description
31–1 -	This field is reserved.
0 dphy_pd_dphy	DPHY PD_DPHY input control. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.6.11.2 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_PREPARE

Address: 30A0_0000h base + 304h offset = 30A0_0304h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																															dphy_m_prg_hs_prepare
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_PREPARE field descriptions

Field	Description
31–2 -	This field is reserved.

Table continues on the next page...

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_PREPARE field descriptions (continued)

Field	Description
dphy_m_prg_hs_prepare	DPHY m_PRG_HS_PREPARE input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section. 00 1 01 1.5 10 2 11 2.5

13.6.11.3 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_PREF

Address: 30A0_0000h base + 308h offset = 30A0_0308h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															dphy_mc_prg_hs_prepare
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_PREPARE field descriptions

Field	Description
31–1 -	This field is reserved.
0 dphy_mc_prg_hs_prepare	DPHY mc_PRG_HS_PREPARE input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section. 0 1 1 1.5

13.6.11.4 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_ZERO

Address: 30A0_0000h base + 30Ch offset = 30A0_030Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_ZERO field descriptions

Field	Description
31–5 -	This field is reserved.
dphy_m_prg_hs_zero	DPHY m_PRG_HS_ZERO input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.
	00000 0
	00001 1
	00010 2
	00011 3
	00100 4
	00101 5
	00110 6
	00111 7
	01000 8
	01001 9
	01010 10
	01011 11
	01100 12
	01101 13
	01110 14
	01111 15
	10000 16
	10001 17
	10010 18
	10011 19
	10100 20
	10101 21
	10110 22
	10111 23
	11000 24
	11001 25
	11010 26
	11011 27
	11100 28
	11101 29
	11110 30
	11111 31

13.6.11.5 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_ZERO

Address: 30A0_0000h base + 310h offset = 30A0_0310h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																dphy_mc_prg_hs_zero															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_ZERO field descriptions

Field	Description
31–6 -	This field is reserved.
dphy_mc_prg_hs_zero	DPHY mc_PRG_HS_ZERO input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section. 100000 32 100001 33 100010 34 100011 35 100100 36 100101 37 100110 38 100111 39 101000 40 101001 41 101010 42 101011 43 101100 44 101101 45 101110 46 101111 47 110000 48 110001 49 110010 50 110011 51 110100 52 110101 53 110110 54 110111 55 111000 56 111001 57 111010 58 111011 59 111100 60 111101 61 111110 62 111111 63

13.6.11.6 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_TRAIL

Address: 30A0_0000h base + 314h offset = 30A0_0314h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_M_PRG_HS_TRAIL field descriptions

Field	Description
31–4 -	This field is reserved.
dphy_m_prg_hs_trail	DPHY m_PRG_HS_TRAIL input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.
	0000 0
	0001 1
	0010 2
	0011 3
	0100 4
	0101 5
	0110 6
	0111 7
	1000 8
	1001 9
	1010 10
	1011 11
	1100 12
	1101 13
	1110 14
	1111 15

13.6.11.7 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_TRAIL

Address: 30A0_0000h base + 318h offset = 30A0_0318h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_MC_PRG_HS_TRAIL field descriptions

Field	Description
31–4 -	This field is reserved.
dphy_mc_prg_ hs_trail	DPHY mc_PRG_HS_TRAIL input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.
	0000 0
	0001 1
	0010 2
	0011 3
	0100 4
	0101 5
	0110 6
	0111 7
	1000 8
	1001 9
	1010 10
	1011 11
	1100 12
	1101 13
	1110 14
	1111 15

13.6.11.8 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_PD_PLL

Address: 30A0_0000h base + 31Ch offset = 30A0_031Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															PD
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_PD_PLL field descriptions

Field	Description
31–1 -	This field is reserved.
0 PD	DPHY PD_PLL input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.6.11.9 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_TST

Address: 30A0_0000h base + 320h offset = 30A0_0320h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																TST															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_TST field descriptions

Field	Description
31–6 -	This field is reserved.
TST	DPHY TST input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.6.11.10 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CN

Address: 30A0_0000h base + 324h offset = 30A0_0324h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																CN															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CN field descriptions

Field	Description																												
31–5 -	This field is reserved.																												
CN	DPHY PLL Input Divider. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section. <table> <tr><td>11111</td><td>Divide by 1</td></tr> <tr><td>00000</td><td>Divide by 2</td></tr> <tr><td>10000</td><td>Divide by 3</td></tr> <tr><td>11000</td><td>Divide by 4</td></tr> <tr><td>11100</td><td>Divide by 5</td></tr> <tr><td>01110</td><td>Divide by 6</td></tr> <tr><td>00111</td><td>Divide by 7</td></tr> <tr><td>10011</td><td>Divide by 8</td></tr> <tr><td>01001</td><td>Divide by 9</td></tr> <tr><td>00100</td><td>Divide by 10</td></tr> <tr><td>00010</td><td>Divide by 11</td></tr> <tr><td>10001</td><td>Divide by 12</td></tr> <tr><td>01000</td><td>Divide by 13</td></tr> <tr><td>10100</td><td>Divide by 14</td></tr> </table>	11111	Divide by 1	00000	Divide by 2	10000	Divide by 3	11000	Divide by 4	11100	Divide by 5	01110	Divide by 6	00111	Divide by 7	10011	Divide by 8	01001	Divide by 9	00100	Divide by 10	00010	Divide by 11	10001	Divide by 12	01000	Divide by 13	10100	Divide by 14
11111	Divide by 1																												
00000	Divide by 2																												
10000	Divide by 3																												
11000	Divide by 4																												
11100	Divide by 5																												
01110	Divide by 6																												
00111	Divide by 7																												
10011	Divide by 8																												
01001	Divide by 9																												
00100	Divide by 10																												
00010	Divide by 11																												
10001	Divide by 12																												
01000	Divide by 13																												
10100	Divide by 14																												

Table continues on the next page...

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CN field descriptions (continued)

Field	Description
01010	Divide by 15
10101	Divide by 16
11010	Divide by 17
11101	Divide by 18
11110	Divide by 19
01111	Divide by 20
10111	Divide by 21
11011	Divide by 22
01101	Divide by 23
10110	Divide by 24
01011	Divide by 25
00101	Divide by 26
10010	Divide by 27
11001	Divide by 28
01100	Divide by 29
00110	Divide by 30
00011	Divide by 31
00001	Divide by 32

13.6.11.11 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CM

Address: 30A0_0000h base + 328h offset = 30A0_0328h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																CM															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CM field descriptions

Field	Description
31–8 -	This field is reserved.
CM	DPHY PLL Feedback Divider. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section. 111x0000 : 111x1111 : 11000000 : 11011111 : 10000000 : 10111111 : 00000000 Divide by 16 Divide by 31 Divide by 32 Divide by 63 Divide by 64 Divide by 127 Divide by 128

Table continues on the next page...

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CM field descriptions (continued)

Field	Description
	: : 01111111 Divide by 255

13.6.11.12 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CO

Address: 30A0_0000h base + 32Ch offset = 30A0_032Ch

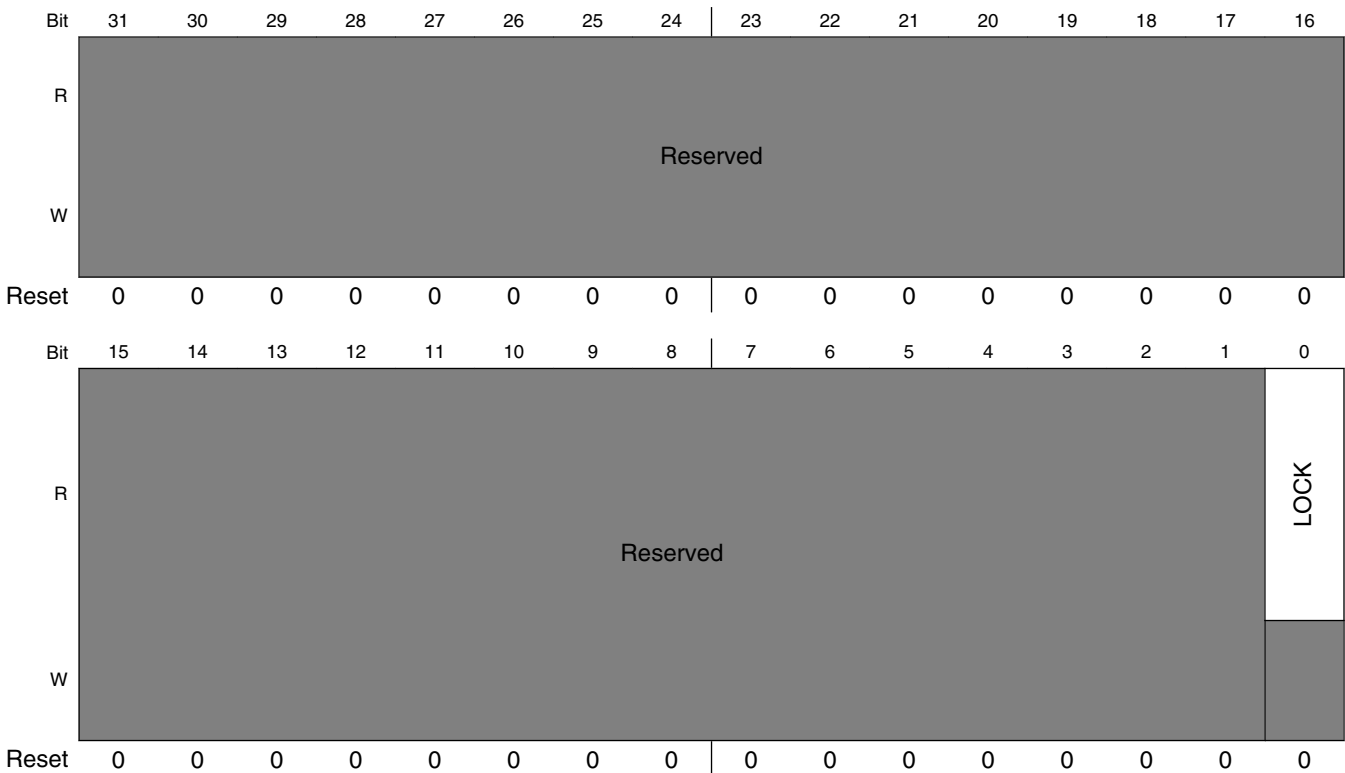
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_CO field descriptions

Field	Description
31–2 -	This field is reserved.
CO	DPHY PLL Output Divider. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section. 00 Divide by 1 01 Divide by 2 10 Divide by 4 11 Divide by 8

13.6.11.13 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_LOCK

Address: 30A0_0000h base + 330h offset = 30A0_0330h



MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_LOCK field descriptions

Field	Description
31–1 -	This field is reserved.
0 LOCK	DPHY PLL LOCK output. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.6.11.14 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_LOCK_BYP

Address: 30A0_0000h base + 334h offset = 30A0_0334h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															dphy_lock_byp
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_LOCK_BYP field descriptions

Field	Description
31–1 -	This field is reserved.
0 dphy_lock_byp	DPHY LOCK_BYP input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.6.11.15 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RTERM_SEL

Address: 30A0_0000h base + 338h offset = 30A0_0338h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															dphy_rterm_sel
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RTERM_SEL field descriptions

Field	Description
31–1 -	This field is reserved.
0 dphy_rterm_sel	DPHY RTERM_SEL input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.6.11.16 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_AUTO_PD_EN

Address: 30A0_0000h base + 33Ch offset = 30A0_033Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															dphy_auto_pd_en
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_AUTO_PD_EN field descriptions

Field	Description
31–1 -	This field is reserved.
0 dphy_auto_pd_en	DPHY AUTO_PD_EN input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.6.11.17 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RXLPRP

Address: 30A0_0000h base + 340h offset = 30A0_0340h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																dphy_rxlprp
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RXLPRP field descriptions

Field	Description
31–2 -	This field is reserved.
dphy_rxlprp	DPHY RXLPRP input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.6.11.18 MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RXCDRP

Address: 30A0_0000h base + 344h offset = 30A0_0344h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																dphy_rxcdrp
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MIPI_DSI_HOST_FSL_IP1_DPHY_INTFC_DPHY_RXCDRP field descriptions

Field	Description
31–2 -	This field is reserved.
dphy_rxcdrp	DPHY RXCDRP input. Detailed information about this parameter programming is available in the MIPI-DSI DPHY section.

13.7 CSI Bridge (CSI)

13.7.1 Overview

This chapter presents the CMOS Sensor Interface (CSI) architecture, operation principles, and programming model.

The CSI enables the chip to connect directly to external CMOS image sensors. CMOS image sensors are separated into two classes, dumb and smart. Dumb sensors are those that support only traditional sensor timing (Vertical SYNC and Horizontal SYNC) and output only Bayer and statistics data, while smart sensors support CCIR656 video decoder formats and perform additional processing of the image (for example, image compression, image pre-filtering, and various data output formats).

The capabilities of the CSI include:

- Configurable interface logic to support most commonly available CMOS sensors.
- Support for CCIR656 video interface as well as traditional sensor interface.
- 8-bit / 16-bit / 24-bit data port for YCbCr, YUV, or RGB data input.
- 8-bit / 10-bit / 16-bit data port for Bayer data input.
- Full control of 8-bit/pixel, 10-bit/pixel or 16-bit / pixel data format to 64-bit receive FIFO packing.
- 256 x 64 FIFO to store received image pixel data.
- Receive FIFO overrun protection mechanism.
- Embedded DMA controllers to transfer data from receive FIFO or statistic FIFO through AHB bus.
- Support 2D DMA transfer from the receive FIFO to the frame buffers in the external memory.
- Support double buffering two frames in the external memory.
- Single interrupt source to interrupt controller from maskable interrupt sources: Start of Frame, End of Frame, Change of Field, FIFO full, FIFO overrun, DMA transfer done, CCIR error and AHB bus response error.
- Configurable master clock frequency output to sensor.
- Statistic data generation for Auto Exposure (AE) and Auto White Balance (AWB) control of the camera (only for Bayer data and 8-bit/pixel format).
- Supports simple deinterlacing of interlaced input.

13.7.2 Clocks

The following table describes the clock sources for CSI.

Table 13-26. CSI Clocks

Clock name	Clock Root	Description
csi_hclk		Module clock
ipg_clk		Peripheral clock
ipg_clk_s		Peripheral access clock
ipg_clk_s_raw		Peripheral raw data clock

13.7.3 Principles of Operation

The information found here describes the modes of operation of the sensor interface.

The CSI is designed to support generic sensor interface timing as well as CCIR656 video interface timing. Traditional CMOS sensors typically use VSYNC (SOF), HSYNC (BLANK), and PIXCLK signals to output Bayer or YUV data. Smart CMOS sensors, that come with on-chip imaging processing, usually support video mode transfer. They use an embedded timing codec to replace the VSYNC and HSYNC signal. The timing codec is defined by the CCIR656 standard.

The CSI can support connection with the sensor as follows.

- To connect with one 8-bit sensor, the sensor data interface should connect to CSI_DATA[9:2].
- To connect with one 10-bit sensor, the sensor data interface should connect to CSI_DATA[9:0].
- To connect with one 16-bit sensor, the sensor data interface should connect to CSI_DATA[15:0].

Table 13-27. CSI input data format

Signal Name	TVdecoder YCbCr 1 Cycle	RGB888 1 Cycle	RGB888/ YUV4444 3 Cycle	RGB666 1 Cycle	RGB565 1 Cycle	YCbCr422 1 Cycle	YCbCr422 2 Cycle	Generic 10 bit	CCIR656
ipp_csi_d[23]	Y[7]	R[7]		R[5]					
ipp_csi_d[22]	Y[6]	R[6]		R[4]					
ipp_csi_d[21]	Y[5]	R[5]		R[3]					

Table continues on the next page...

Table 13-27. CSI input data format (continued)

Signal Name	TVdecoder YCbCr 1 Cycle	RGB888 1 Cycle	RGB888/ YUV4444 3 Cycle	RGB666 1 Cycle	RGB565 1 Cycle	YCbCr422 1 Cycle	YCbCr422 2 Cycle	Generic 10 bit	CCIR656
ipp_csi_d[20]	Y[4]	R[4]		R[2]					
ipp_csi_d[19]	Y[3]	R[3]		R[1]					
ipp_csi_d[18]	Y[2]	R[2]		R[0]					
ipp_csi_d[17]	Y[1]	R[1]		Y[5]					
ipp_csi_d[16]	Y[0]	R[0]		R[4]					
ipp_csi_d[15]	Cb[7]	G[7]		G[5]	R[4]	Y[7]			
ipp_csi_d[14]	Cb[6]	G[6]		G[4]	R[3]	Y[6]			
ipp_csi_d[13]	Cb[5]	G[5]		G[3]	R[2]	Y[5]			
ipp_csi_d[12]	Cb[4]	G[4]		G[2]	R[1]	Y[4]			
ipp_csi_d[11]	Cb[3]	G[3]		G[1]	R[0]	Y[3]			
ipp_csi_d[10]	Cb[2]	G[2]		G[0]	G[5]	Y[2]			
ipp_csi_d[9]	Cb[1]	G[1]	R/G/B[7]	G[5]	G[4]	Y[1]	Y/C[7]	Ge[9]	C/Y[7]
ipp_csi_d[8]	Cb[0]	G[0]	R/G/B[6]	G[4]	G[3]	Y[0]	Y/C[6]	Ge[8]	C/Y[6]
ipp_csi_d[7]	Cr[7]	B[7]	R/G/B[5]	B[5]	G[2]	C[7]	Y/C[5]	Ge[7]	C/Y[5]
ipp_csi_d[6]	Cr[6]	B[6]	R/G/B[4]	B[4]	G[1]	C[6]	Y/C[4]	Ge[6]	C/Y[4]
ipp_csi_d[5]	Cr[5]	B[5]	R/G/B[3]	B[3]	G[0]	C[5]	Y/C[3]	Ge[5]	C/Y[3]
ipp_csi_d[4]	Cr[4]	B[4]	R/G/B[2]	B[2]	B[4]	C[4]	Y/C[2]	Ge[4]	C/Y[2]
ipp_csi_d[3]	Cr[3]	B[3]	R/G/B[1]	B[1]	B[3]	C[3]	Y/C[1]	Ge[3]	C/Y[1]
ipp_csi_d[2]	Cr[2]	B[2]	R/G/B[0]	B[0]	B[2]	C[2]	Y/C[0]	Ge[2]	C/Y[0]
ipp_csi_d[1]	Cr[1]	B[1]		B[5]	B[1]	C[1]		Ge[1]	
ipp_csi_d[0]	Cr[0]	B[0]		B[4]	B[0]	C[0]		Ge[0]	

13.7.3.1 Data Transfer with the Embedded DMA Controllers

The CSI has two embedded DMA controllers, one for the receive FIFO and the other for the statistic FIFO. It supports 2D DMA transfer from the receive FIFO to the frame buffers in the external memory and linear DMA transfer from the statistic FIFO.

To transfer data from the RxFIFO to the external memory, the user should set the start address in the frame buffer where the transferred data is stored, the parameters of the frame buffers, and the parameters of the image coming from the sensor. The user can have two frame buffers in the external memory. Each one will store a frame of image coming from the sensor. The embedded DMA controller will first write the frame buffer1 and then frame buffer2. These two frame buffers will be written by turns. The start address should be aligned in double words and set in the CSIDMASA-FB1 and CSIDMASA-FB2 registers. In the CSIFBUF_PARA register, the user should set the stride of the frame buffer to show how many double words to skip before starting to write the next row of the image. In the CSIIMAG_PARA register, the user should set the width and height of the image coming from the sensor. The RxFF_LEVEL and DMA_REQ_EN_RFF bits in CSICR3 registers also need to be set before the data transfer starts. When the number of the data in the RxFIFO reaches the trigger level, a DMA request will be sent to the embedded DMA controller and the data will be read out from the RxFIFO and written through AHB bus into the external frame buffers. The burst type of transfer can be INCR4, INCR8 and INCR16 by setting DMA_BURST_TYPE_RFF bits in CSICR2 register. After all data in an image frame are transferred, the DMA_TSF_DONE_FB1 or DMA_TSF_DONE_FB2 bit will be set in CSISR register and the interrupt can be triggered if the corresponding enable bit is set in CSICR1 register. The DMA_REFLASH_RFF bit in CSICR3 can be used to activate or restart the embedded DMA controller.

The RxFIFO has the overrun protection mechanism in case the RxFIFO is overrun during data transfer. If the RxFIFO is full and more data needs to be received during the data transfer, the RxFIFO will be overwritten continuously and all 128 words of data in the RxFIFO before overrun occurred will be discarded; the corresponding 128 words memory space in the frame buffer will keep the previous values.

To transfer data from the statistic FIFO to the external memory, the user should set the start address of the external memory where the transferred data is stored and the total transfer sizes. The start address and the transfer sizes are all aligned in double words and should be set in the CSIDMASA-STATFIFO and CSIDMATS-STATFIFO registers. The STATFF_LEVEL and DMA_REQ_EN_SFF bits in CSICR3 registers should also be set before the data transfer starts. When the number of the data in the STATFIFO reaches the trigger level, a dma request will be sent to the embedded DMA controller and the data will be read out from the STATFIFO and written through AHB bus into the external

memory. The burst type of transfer can be INCR4, INCR8 and INCR16 by setting DMA_BURST_TYPE_SFF bits in CSICR2 register. After all expected data (defined by the total transfer sizes) are transferred, the DMA_TSF_DONE_SFF bit will be set in CSISR register and an interrupt can be triggered if the SFF_DMA_DONE_INTEN is enabled in CSICR1 register. The DMA_REFLASH_SFF bit in CSICR3 can be used to activate or re-start the embedded DMA controller.

13.7.3.2 Gated Clock Mode

VSYNC, HSYNC, and PIXCLK signals are used in gated clock mode.

A frame starts with an active edge on VSYNC, then HSYNC asserts and holds for the entire line. The Pixel clock is valid as long as HSYNC is asserted. Data is latched at the active edge of the valid pixel clocks. HSYNC deasserts at the end of line. Pixel clocks then become invalid and CSI stops receiving data from the stream. For the next line the HSYNC timing repeats. For the next frame the VSYNC timing repeats.

13.7.3.3 Non-Gated Clock Mode

In non-gated clock mode, only the VSYNC and PIXCLK signals are used; the HSYNC signal is ignored.

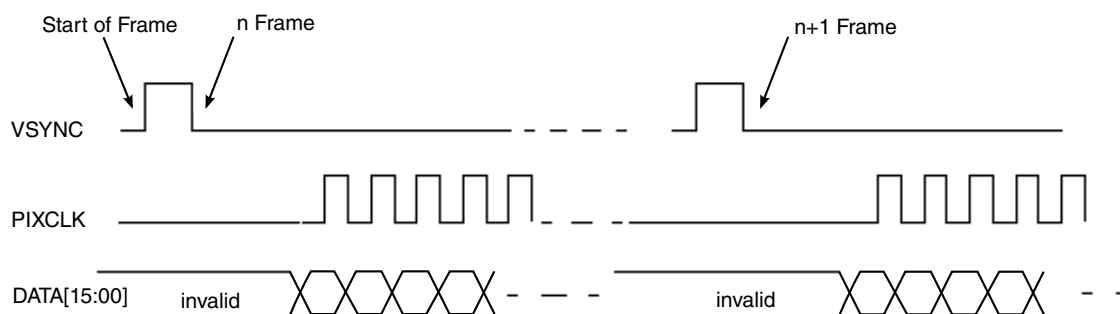


Figure 13-65. Non-Gated Clock Mode Timing Diagram

The overall timing of non-gated mode is the same as the gated-clock mode, except for the HSYNC signal. HSYNC signal is ignored by the CSI. All incoming pixel clocks are valid and cause data to be latched into Rx FIFO. The PIXCLK signal is inactive (states low) until valid data is ready to be transmitted over the bus.

Figure 13-65 shows the timing of a typical sensor. Other sensors may have the slightly different timing from that shown. The CSI can be programmed to support rising/falling-edge triggered VSYNC, active-high/low HSYNC, and rising/falling-edge triggered PIXCLK.

13.7.3.4 CCIR656 Interlace Mode

In CCIR656 interlace mode, only the PIXCLK and CSI_DATA[9:2] signals are used. The start of frame and blank signals are replaced by a timing codec which is embedded in the data stream. Each active line starts with an Start of Active Video (SAV) code and ends with an End of Active Video (EAV) code. In some cases, digital blanking is inserted in between EAV and SAV code. The CSI decodes and filters out the timing-coding from the data stream, recovering VSYNC and HSYNC signals for internal use, such as statistical block control. Data is forwarded to the data receive and packing block in a sequential manner without reordering—that is, field 1 followed by field 2. The fields must be reordered in software to get back the original image.

Change of Field (COF) interrupt is triggered upon every field change. The interrupt service routine reads the status register to check for the current field.

According to the CCIR656 specification, the image must be in 625/50 PAL or 525/60 NTSC format. In addition, the image is interlaced into odd and even fields with vertical and horizontal blank data being filled into certain lines. Data must be in YCbCr422 format, each pixel contains 2 bytes, either Y + Cr or Y + Cb. These requirements are set for TV systems. The CSI module supports PAL and NTSC format only.

The following figure describes the frame structure in PAL system, showing vertical and horizontal blanking.

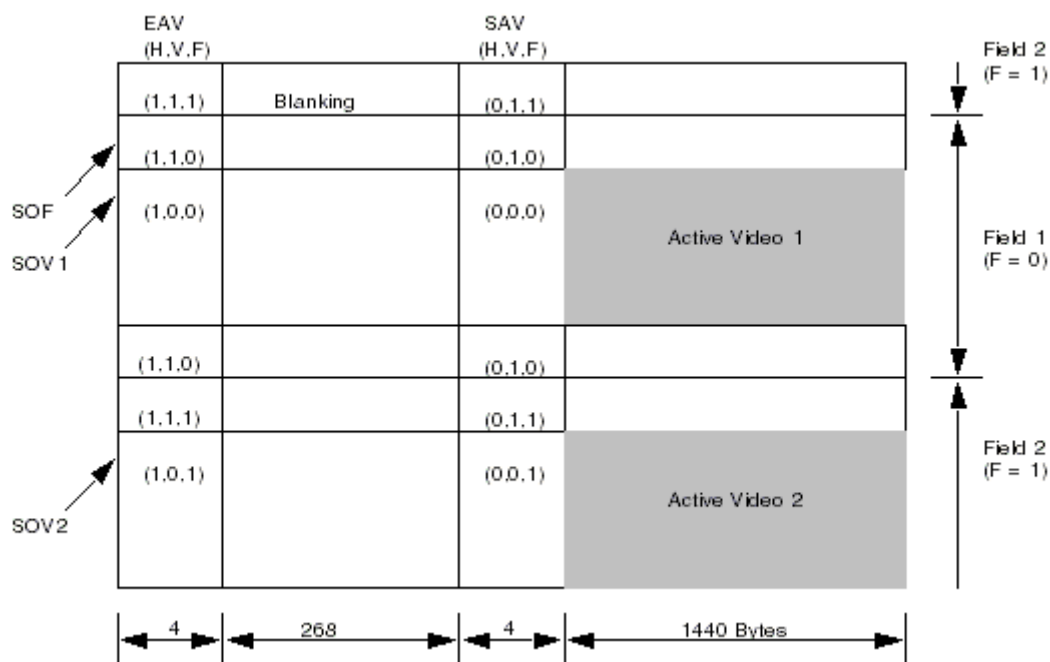


Figure 13-66. CCIR656 Interlace Mode (PAL)

The following figure describes the general timing for a single line, showing SAV and EAV.

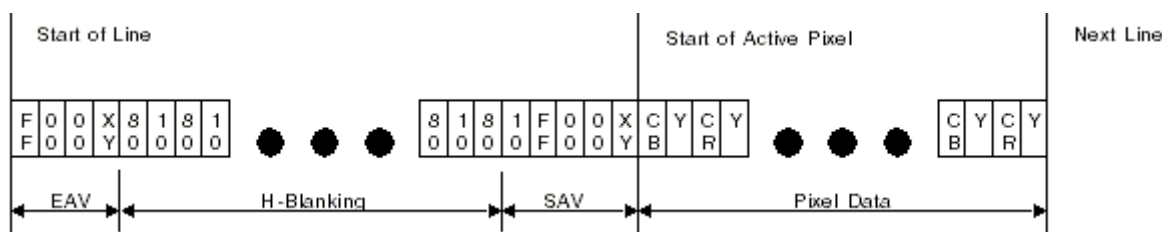


Figure 13-67. CCIR656 General Line Timing

The coding tables recommended by the CCIR656 specification are shown below. It is used in the CCIR656 mode to decode the video stream. An interrupt is generated for SOF, which is decoded from the embedded timing codec.

Table 13-28. Coding for SAV and EAV

Data Bit Number	1st Byte 0xFF	2nd Byte 0x00	3rd Byte 0x00	4th Byte 0xXY
7 (MSB)	1	0	0	1
6	1	0	0	F
5	1	0	0	V
4	1	0	0	H
3	1	0	0	P3
2	1	0	0	P2

Table continues on the next page...

Table 13-28. Coding for SAV and EAV (continued)

Data Bit Number	1st Byte 0xFF	2nd Byte 0x00	3rd Byte 0x00	4th Byte 0xXY
1	1	0	0	P1
0	1	0	0	P0

Table 13-29. Codes with Protection bits for Error Detection/Correction

F	V	H	P3	P2	P1	P0
0	0	0	0	0	0	0
0	0	1	1	1	0	1
0	1	0	1	0	1	1
0	1	1	0	1	1	0
1	0	0	0	1	1	1
1	0	1	1	0	1	0
1	1	0	1	1	0	0
1	1	1	0	0	0	1

Table 13-30. Representations by F-Bit

F-Bit	Representations
0	ODD FIELD (FIELD 1)
1	EVEN FIELD (FIELD 2)

13.7.3.5 CCIR656 Progressive Mode

For a CMOS camera system of VGA or CIF resolution, strict adherence to the interlace requirements stated in the CIR standard is not required.

The image is considered to have only 1 active field which is scanned in a progressive manner. This active field is regarded as field 1 and the F-bit in the timing codec is ignored by the decoder. Most sensors support CCIR timing in this mode (progressive) by default.

The following figure shows the typical flow of progressive mode.

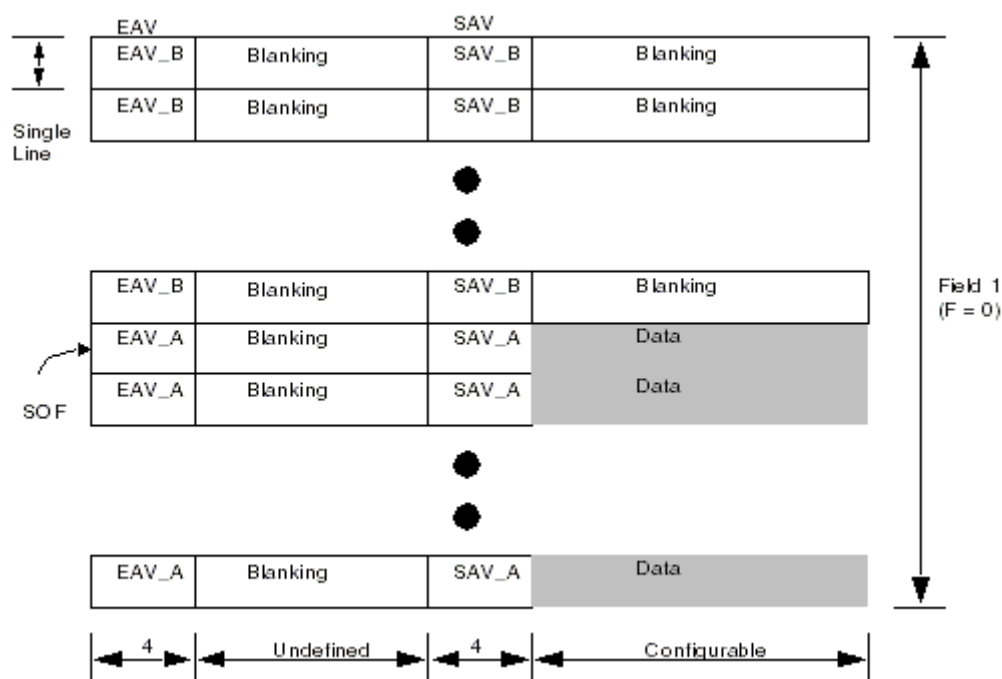


Figure 13-68. CCIR656 Progressive Mode (General Case)

An interrupt is generated for SOF but not for COF. In the general case, when SOF information is retrieved from the embedded coding, it is known as internal VSYNC mode. In other cases, when the VSYNC signal is provided by the sensor, it is known as external VSYNC mode. The CSI can be operated in internal or external VSYNC mode.

13.7.3.6 Error Correction for CCIR656 Coding

According to the algorithm for CCIR coding, protection bits in the SAV and EAV are encoded in the way that allows a 1-bit error to be corrected, or a 2-bit error to be detected by the decoder. This feature is supported by the interlace mode CCIR decoder in CSI.

For the 1-bit error case, users can select the error to be corrected automatically, or simply shown as a status flag instead. For the 2-bit error case, because the decoder is unable to make a correction, the error would be shown as a status flag only.

An interrupt can be generated upon the detection of an error. This signal can be enabled or disabled without affecting the operation of the status bit.

13.7.3.7 Deinterlacer

Deinterlacing is the process of converting interlaced video, such as CCIR656 input, into a non-interlaced form.

The CSI uses the weaving method to do deinterlacing. Weaving is done by adding consecutive fields together. CSI uses top-field detection function. No matter input is NTSC or PAL mode, the combined frame will always put the top-field first and then bottom-field.

13.7.4 Interrupt Generation

The information found here describes CSI events that generate interrupts.

13.7.4.1 Start Of Frame Interrupt (SOF_INT)

The source of an SOF interrupt is dependent on the mode of operation.

In traditional mode, VSYNC signal is taken from sensor and SOF_INT is generated at the rising or falling edge (programmable) of VSYNC.

In CCIR interlace mode, the SOF interrupt information is retrieved from the embedded coding and SOF_INT is generated.

In CCIR progressive mode, there are two sources of an SOF interrupt:

- In *internal* VSYNC mode, SOF is retrieved from the embedded coding.
- In *external* VSYNC mode, VSYNC is taken from the sensor and SOF is generated at the rising edge of VSYNC.

13.7.4.2 End Of Frame Interrupt (EOF_INT)

An EOF interrupt is generated when the frame ends and the complete frame data in RXFIFO is read.

The EOF event triggering works with the RX count register (CSIRXCNT). Software sets the RX count register to the frame size (in words). The CSI RX logic then counts the number of pixel data being received and compares it with the RX count. If the preset value is reached, an EOF interrupt is generated and the data in the RXFIFO are read. If a SOF event is detected before this happens, the EOF interrupt is not generated.

13.7.4.3 Change Of Field Interrupt (COF_INT)

The Change of Field interrupt is only valid in CCIR Interlace mode. The COF interrupt is generated when the field toggles, either from field 1 to field 2, or field 2 to field 1.

Software should first check COF_INT bit in the CSI Status Register (CSISTAT) before checking that F1_INT or F2_INT is turned on.

In PAL systems, the field changes at the beginning of the frame and coincides with SOF. For the first field, a COF interrupt is not generated, only an SOF. The COF interrupt is generated for the second field.

13.7.4.4 CCIR Error Interrupt (ECC_INT)

The CCIR Error Interrupt is only valid for CCIR Interlace mode. An ECC interrupt is generated when an error is found on the SAV or EAV codes in the incoming stream. When this happens, the ECC_INT status bit is set.

13.7.4.5 RxFIFO Full Interrupt (RxFF_INT)

A RxFIFO full interrupt is generated when the number of data in RXFIFO reaches the water mark defined by RxFF_LEVEL in CSICR3.

13.7.4.6 Statistic FIFO Full Interrupt (STATFF_INT)

A StatFIFO full interrupt is generated when the number of data in STATFIFO reaches the water mark defined by STATFF_LEVEL in CSICR3.

13.7.4.7 RxFIFO Overrun Interrupt (RFF_OR_INT)

A RxFIFO Overrun interrupt is generated when the RxFIFO has 128 words data and more data is being written in.

13.7.4.8 Statistic FIFO Overrun Interrupt (SFF_OR_INT)

A StatFIFO Overrun interrupt is generated when the STATFIFO has 64 words data and more data is being written in.

13.7.4.9 Frame Buffer1 DMA Transfer Done Interrupt (DMA_TSF_DONE_FB1)

A DMA transfer done interrupt of frame buffer1 is generated when one frame of data are transferred from RxFIFO to the frame buffer1 in the external memory.

13.7.4.10 Frame Buffer2 DMA Transfer Done Interrupt (DMA_TSF_DONE_FB2)

A DMA transfer done interrupt of frame buffer2 is generated when one frame of data are transferred from RxFIFO to the frame buffer2 in the external memory.

13.7.4.11 Statistic FIFO DMA Transfer Done Interrupt (DMA_TSF_DONE_SFF)

A StatFIFO DMA transfer done interrupt is generated when all the data are transferred from StatFIFO to the external memory. The transfer size is defined in the STATFIFO DMA Transfer Size Register.

13.7.4.12 AHB Bus Response Error Interrupt (HRESP_ERR_INT)

An AHB Bus response error interrupt is generated when a bus error is detected.

13.7.5 Data Packing Style

Careful attention to endianness is needed given the different port sizes at different stages of the image capture path.

To enable flexible packing of image data before storage in the FIFOs, the CSI module can swap data fields by use of the PACK_DIR and the SWAP16_EN bit in CSI Control Register 1 (CSICR1).

The CSI module accepts 8-bit, 10-bit or 16-bit data from the sensor by configuring PIXEL_BIT bit in CSI Control Register 1 (CSICR1) and TWO_8BIT_SENSOR bit in CSI Control Register3 (CSICR3). The input data is packed according to the setting of PACK_DIR bit. The packed data is stored in the RX FIFO according to the setting of the SWAP16_EN bit.

For 10-bit per pixel data format, each pixel is expanded to 16 bits by appending 6 zeros bits to the most significant bit.

13.7.5.1 RX FIFO Path

13.7.5.1.1 Bayer Data

Bayer data is a type of raw data from the image sensor. This byte-wide data must be converted to the RGB space or YUV space by software. The data path for Bayer data is from the CSI to memory. If the system is in little endian, then the PACK_DIR bit should be set to 0. 8-bit data format from a sensor is packed to 64 bits as

P7.P6.P5.P4.P3.P2.P1.P0, where P0 is the pixel coming in time slot 0 (first data) and P3 is the pixel coming in time slot 3 (the last data in the 64-bit word). When the data is addressed as bytes by software, P0 is transferred first, P1 is transferred next, and so on. 10-bit data format is packed to 64 bits as 000000.P3.000000.P2.000000.P1.000000.P0, where P0 is the 10-bit data coming in time slot 0 (first pixel) and P3 is the 10-bit data coming in time slot 3 (fourth pixel). 16-bit data is packed to 64 bits as P7.P6.P5.P4.P3.P2.P1.P0.

13.7.5.1.2 RGB565 Data

RGB565 data is processed data from the image sensor, which can be put directly into the display buffer. The data is 16 bits wide. The data path is from CSI to memory to the display controller. On the sensor side, data must be transmitted as P0 first, followed by P1, and so on. For each pixel, whether the MSB or LSB is sent first depends on the endianness of the sensor. Data is 16 bits wide with the MSB labeled RG, and the LSB labeled GB. P0 is represented as RG0 and GB0.

CSI receives data in one of the following sequence:

- RG0, GB0, RG1, GB1, while RG0 comes out at time slot 0 (first data), and GB1 comes out at time slot 3 (last data)
- GB0, RG0, GB1, RG1

Using the first sequence as an example, and assuming the system is running in little endian, the data is presented as:

- 8-bit data from sensor: RG0, GB0, RG1, GB1, ...
- 64-bit data before storage in the CSI RX FIFO (PACK_DIR bit = 1):
RG0GB0RG1GB1RG2GB2RG3GB3
- 64-bit data in CSI RX FIFO (SWAP16_EN bit enabled):
RG3GB3RG2GB2RG1GB1RG0GB0
- 64-bit transfer to system memory: RG3GB3RG2GB2RG1GB1RG0GB0
- 16-bit read by display controller: RG0GB0, RG1GB1

13.7.5.1.3 RGB888 Data

This is another kind of processed data from image sensor, which can be used for further image processing directly. Each of the data consist of 8-bit Red, 8-bit Green, and 8-bit Blue data. An example of timing scheme is shown in the following figure.

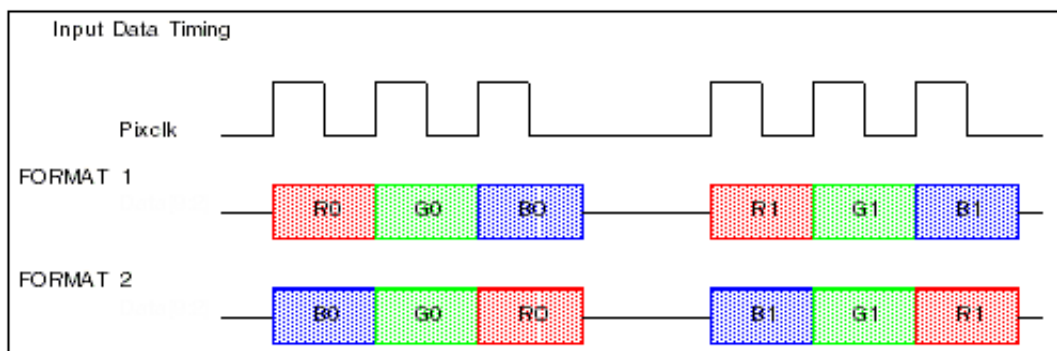
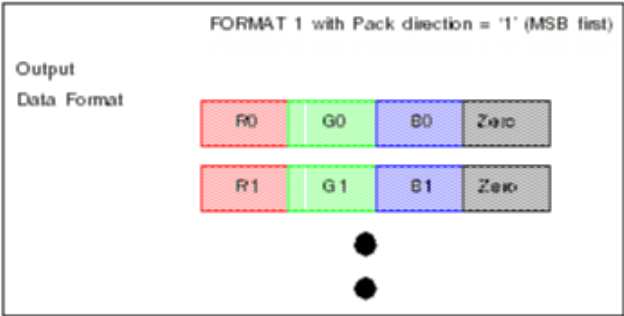


Figure 13-69. Sample Timing Diagram for RGB888 8 bits/cycle Data

An optional scheme to pack a dummy byte is provided. For every group of 3 bytes data, a dummy zero is packed to form a 32-bit word as shown in the following figure. The dummy zero can be packed at the LSB position or MSB position. Using RGB888A_FORMAT_SEL in CSI_CSICR18[18] to determine to put the dummy bytes packed at LSB or MSB position.



13.7.5.2 STAT FIFO Path

Statistics only works for Bayer data in 8-bit per pixel format. It generates 16-bit statistical output from the 8-bit Bayer input (CSI_DATA[13:6]). The outputs are Sum of Green (G), Sum of Red (R), Sum of Blue (B), and Auto Focus (F). Each output is 16-bits wide.

The settings of PACK_DIR and SWAP16_EN bits in the CSICR1 register have no effect on the input path. The PACK_DIR only controls how the 16-bit stat output is packed into the 32-bit STAT FIFO.

When the PACK_DIR bit = 1, the stat data is packed as:

First 32-bit: RG

Second 32-bit: BF

...

When the PACK_DIR bit = 0, the stat data is packed as:

First 32-bit GR

Second 32-bit: FB

...

13.7.6 CSI Memory Map/Register Definition

All the 32-bit registers of the CSI module are summarized in the Memory Map below:

CSI memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
0	CSI Control Register 1 (CSI_CSICR1)	32	R/W	4000_0800h	13.7.6.1/ 5148
4	CSI Control Register 2 (CSI_CSICR2)	32	R/W	0000_0000h	13.7.6.2/ 5152
8	CSI Control Register 3 (CSI_CSICR3)	32	R/W	0000_0000h	13.7.6.3/ 5154
C	CSI Statistic FIFO Register (CSI_CSISTATFIFO)	32	R	0000_0000h	13.7.6.4/ 5156

Table continues on the next page...

CSI memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
10	CSI RX FIFO Register (CSI_CSIRFIFO)	32	R	0000_0000h	13.7.6.5/5156
14	CSI RX Count Register (CSI_CSIRXCNT)	32	R/W	0000_9600h	13.7.6.6/5157
18	CSI Status Register (CSI_CSISR)	32	R/W	0000_4000h	13.7.6.7/5158
20	CSI DMA Start Address Register - for STATFIFO (CSI_CSIDMASA_STATFIFO)	32	R/W	0000_0000h	13.7.6.8/5161
24	CSI DMA Transfer Size Register - for STATFIFO (CSI_CSIDMATS_STATFIFO)	32	R/W	0000_0000h	13.7.6.9/5161
28	CSI DMA Start Address Register - for Frame Buffer1 (CSI_CSIDMASA_FB1)	32	R/W	0000_0000h	13.7.6.10/5162
2C	CSI DMA Transfer Size Register - for Frame Buffer2 (CSI_CSIDMASA_FB2)	32	R/W	0000_0000h	13.7.6.11/5163
30	CSI Frame Buffer Parameter Register (CSI_CSIFBUF_PARA)	32	R/W	0000_0000h	13.7.6.12/5163
34	CSI Image Parameter Register (CSI_CSIMAG_PARA)	32	R/W	0000_0000h	13.7.6.13/5164
48	CSI Control Register 18 (CSI_CSICR18)	32	R/W	0002_D000h	13.7.6.14/5165

13.7.6.1 CSI Control Register 1 (CSI_CSICR1)

This register controls the sensor interface timing and interrupt generation. The interrupt enable bits in this register control the interrupt signals and the status bits. That means status bits will only function when the corresponding interrupt bits are enabled.

Address: 0h base + 0h offset = 0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R									Reserved							
W	SWAP16_EN	EXT_VSYNC	EOF_INT_EN	PRP_IF_EN	VIDEO_MODE	COF_INT_EN	SF_OR_INT_EN	RF_OR_INT_EN		SFF_DMA_DONE_INTEN	STATFF_INTEN	FB2_DMA_DONE_INTEN	FB1_DMA_DONE_INTEN	RXFF_INTEN	SOF_POL	SOF_INTEN
Reset	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W					HSYNC_POL	CCIR_EN	Reserved	FCC	PACK_DIR	CLR_STATFIFO	CLR_RXFIFO	GCLK_MODE	INV_DATA	INV_PCLK	REDGE	PIXEL_BIT
Reset	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0

CSI_CSICR1 field descriptions

Field	Description
31 SWAP16_EN	SWAP 16-Bit Enable. This bit enables the swapping of 16-bit data. Data is packed from 8-bit or 10-bit to 32-bit first (according to the setting of PACK_DIR) and then swapped as 16-bit words before being put into the RX FIFO. The action of the bit only affects the RX FIFO and has no affect on the STAT FIFO. NOTE: Example of swapping enabled: Data input to FIFO = 0x11223344 Data in RX FIFO = 0x 33441122 NOTE: Example of swapping disabled: Data input to FIFO = 0x11223344 Data in RX FIFO = 0x11223344

Table continues on the next page...

CSI_CSICR1 field descriptions (continued)

Field	Description
	0 Disable swapping 1 Enable swapping
30 EXT_VSYNC	External VSYNC Enable. This bit controls the operational VSYNC mode. NOTE: This only works when the CSI is in CCIR progressive mode. 0 Internal VSYNC mode 1 External VSYNC mode
29 EOF_INT_EN	End-of-Frame Interrupt Enable. This bit enables and disables the EOF interrupt. 0 EOF interrupt is disabled. 1 EOF interrupt is generated when RX count value is reached.
28 PrP_IF_EN	CSI-PrP Interface Enable. This bit controls the CSI to PrP bus. When enabled the RxFIFO is detached from the AHB bus and connected to PrP. All CPU reads or DMA accesses to the RxFIFO register are ignored. All CSI interrupts are also masked. 0 CSI to PrP bus is disabled 1 CSI to PrP bus is enabled
27 VIDEO_MODE	Video mode select. This bit controls the video mode in CCIR mode. 0 Progressive mode is selected 1 Interlace mode is selected
26 COF_INT_EN	Change Of Image Field (COF) Interrupt Enable. This bit enables the COF interrupt. This bit works only in CCIR interlace mode which is when CCIR_EN = 1 and CCIR_MODE = 1. 0 COF interrupt is disabled 1 COF interrupt is enabled
25 SF_OR_INTEN	STAT FIFO Overrun Interrupt Enable. This bit enables the STATFIFO overrun interrupt. 0 STATFIFO overrun interrupt is disabled 1 STATFIFO overrun interrupt is enabled
24 RF_OR_INTEN	RxFIFO Overrun Interrupt Enable. This bit enables the RX FIFO overrun interrupt. 0 RxFIFO overrun interrupt is disabled 1 RxFIFO overrun interrupt is enabled
23 -	This field is reserved. Reserved. This bit is reserved and should read 0.
22 SFF_DMA_DONE_INTEN	STATFIFO DMA Transfer Done Interrupt Enable. This bit enables the interrupt of STATFIFO DMA transfer done. 0 STATFIFO DMA Transfer Done interrupt disable 1 STATFIFO DMA Transfer Done interrupt enable
21 STATFF_INTEN	STATFIFO Full Interrupt Enable. This bit enables the STAT FIFO interrupt. 0 STATFIFO full interrupt disable 1 STATFIFO full interrupt enable
20 FB2_DMA_DONE_INTEN	Frame Buffer2 DMA Transfer Done Interrupt Enable. This bit enables the interrupt of Frame Buffer2 DMA transfer done.

Table continues on the next page...

CSI_CSICR1 field descriptions (continued)

Field	Description
	0 Frame Buffer2 DMA Transfer Done interrupt disable 1 Frame Buffer2 DMA Transfer Done interrupt enable
19 FB1_DMA_DONE_INTEN	Frame Buffer1 DMA Transfer Done Interrupt Enable. This bit enables the interrupt of Frame Buffer1 DMA transfer done. 0 Frame Buffer1 DMA Transfer Done interrupt disable 1 Frame Buffer1 DMA Transfer Done interrupt enable
18 RXFF_INTEN	RxFIFO Full Interrupt Enable. This bit enables the RxFIFO full interrupt. 0 RxFIFO full interrupt disable 1 RxFIFO full interrupt enable
17 SOF_POL	SOF Interrupt Polarity. This bit controls the condition that generates an SOF interrupt. 0 SOF interrupt is generated on SOF falling edge 1 SOF interrupt is generated on SOF rising edge
16 SOF_INTEN	Start Of Frame (SOF) Interrupt Enable. This bit enables the SOF interrupt. 0 SOF interrupt disable 1 SOF interrupt enable
15–12 Reserved	This field is reserved. Reserved. This field is reserved.
11 HSYNC_POL	HSYNC Polarity Select. This bit controls the polarity of HSYNC. This bit only works in gated-clock-that is, GCLK_MODE = 1 and CCIR_EN = 0. 0 HSYNC is active low 1 HSYNC is active high
10 CCIR_EN	CCIR656 Interface Enable. This bit selects the type of interface used. When the CCIR656 timing decoder is enabled, it replaces the function of timing interface logic. 0 Traditional interface is selected. Timing interface logic is used to latch data. 1 CCIR656 interface is selected.
9 Reserved	This field is reserved. This field is reserved.
8 FCC	FIFO Clear Control. This bit determines how the RXFIFO and STATFIFO are cleared. When Synchronous FIFO clear is selected the RXFIFO and STATFIFO are cleared, and STAT block is reset, on every SOF. FIFOs and STAT block restarts immediately after reset. For information on the operation when Asynchronous FIFO clear is selected, refer to the descriptions for the CLR_RXFIFO and CLR_STATFIFO bits. 0 Asynchronous FIFO clear is selected. 1 Synchronous FIFO clear is selected.
7 PACK_DIR	Data Packing Direction. This bit Controls how 8-bit/10-bit image data is packed into 32-bit RX FIFO, and how 16-bit statistical data is packed into 32-bit STAT FIFO. 0 Pack from LSB first. For image data, 0x11, 0x22, 0x33, 0x44, it will appear as 0x44332211 in RX FIFO. For stat data, 0xAAAA, 0xBBBB, it will appear as 0xB BBBBAAAA in STAT FIFO. 1 Pack from MSB first. For image data, 0x11, 0x22, 0x33, 0x44, it will appear as 0x11223344 in RX FIFO. For stat data, 0xAAAA, 0xBBBB, it will appear as 0xAAAABBBB in STAT FIFO.

Table continues on the next page...

CSI_CSICR1 field descriptions (continued)

Field	Description
6 CLR_STATFIFO	Asynchronous STATFIFO Clear. This bit clears the STATFIFO and Reset STAT block. This bit works only in async FIFO clear mode-that is, FCC = 0. Otherwise this bit is ignored. Writing 1 will clear STATFIFO and reset STAT block immediately, STATFIFO and STAT block then wait and restart after the arrival of next SOF. The bit is restored to 0 automatically after finish. Normally reads 0.
5 CLR_RXFIFO	Asynchronous RXFIFO Clear. This bit clears the RXFIFO. This bit works only in async FIFO clear mode-that is, FCC = 0. Otherwise this bit is ignored. Writing 1 clears the RXFIFO immediately, RXFIFO restarts immediately after that. The bit is restored to 0 automatically after finish. Normally reads 0.
4 GCLK_MODE	Gated Clock Mode Enable. Controls if CSI is working in gated or non-gated mode. This bit works only in traditional mode-that is, CCIR_EN = 0. Otherwise this bit is ignored. 0 Non-gated clock mode. All incoming pixel clocks are valid. HSYNC is ignored. 1 Gated clock mode. Pixel clock signal is valid only when HSYNC is active.
3 INV_DATA	Invert Data Input. This bit enables or disables internal inverters on the data lines. 0 CSI_D[7:0] data lines are directly applied to internal circuitry 1 CSI_D[7:0] data lines are inverted before applied to internal circuitry
2 INV_PCLK	Invert Pixel Clock Input. This bit determines if the Pixel Clock (CSI_PIXCLK) is inverted before it is applied to the CSI module. 0 CSI_PIXCLK is directly applied to internal circuitry 1 CSI_PIXCLK is inverted before applied to internal circuitry
1 REDGE	Valid Pixel Clock Edge Select. Selects which edge of the CSI_PIXCLK is used to latch the pixel data. 0 Pixel data is latched at the falling edge of CSI_PIXCLK 1 Pixel data is latched at the rising edge of CSI_PIXCLK
0 PIXEL_BIT	Pixel Bit. This bit indicates the bayer data width for each pixel. This bit should be configured before activating or re-starting the embedded DMA controller. 0 8-bit data for each pixel 1 10-bit data for each pixel

13.7.6.2 CSI Control Register 2 (CSI_CSICR2)

This register provides the statistic block with data about which live view resolution is being used, and the starting sensor pixel of the Bayer pattern. It also contains the horizontal and vertical count used to determine the number of pixels to skip between the 64 x 64 blocks of statistics when generating statistics on live view image that are greater than 512 x 384.

Address: 0h base + 4h offset = 4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DMA_BURST_TYPE_RFF		DMA_BURST_TYPE_SFF		Reserved	DRM	AFS		SCE	Reserved		BTS		LVRM		
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	VSC								HSC							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSICR2 field descriptions

Field	Description
31–30 DMA_BURST_TYPE_RFF	Burst Type of DMA Transfer from RxFIFO. Selects the burst type of DMA transfer from RxFIFO. X0 INCR8 01 INCR4 11 INCR16
29–28 DMA_BURST_TYPE_SFF	Burst Type of DMA Transfer from STATFIFO. Selects the burst type of DMA transfer from STATFIFO. X0 INCR8 01 INCR4 11 INCR16

Table continues on the next page...

CSI_CSICR2 field descriptions (continued)

Field	Description
27 -	This field is reserved. Reserved. These bit is reserved and should read 0.
26 DRM	Double Resolution Mode. Controls size of statistics grid. 0 Stats grid of 8 x 6 1 Stats grid of 8 x 12
25–24 AFS	Auto Focus Spread. Selects which green pixels are used for auto-focus. 00 Abs Diff on consecutive green pixels 01 Abs Diff on every third green pixels 1x Abs Diff on every four green pixels
23 SCE	Skip Count Enable. Enables or disables the skip count feature. 0 Skip count disable 1 Skip count enable
22–21 -	This field is reserved. Reserved. These bits are reserved and should read 0.
20–19 BTS	Bayer Tile Start. Controls the Bayer pattern starting point. 00 GR 01 RG 10 BG 11 GB
18–16 LVRM	Live View Resolution Mode. Selects the grid size used for live view resolution. 0 512 x 384 1 448 x 336 2 384 x 288 3 384 x 256 4 320 x 240 5 288 x 216 6 400 x 300
15–8 VSC	Vertical Skip Count. Contains the number of rows to skip. SCE must be 1, otherwise VSC is ignored. 0-255 Number of rows to skip minus 1
HSC	Horizontal Skip Count. Contains the number of pixels to skip. SCE must be 1, otherwise HSC is ignored. 0-255 Number of pixels to skip minus 1

13.7.6.3 CSI Control Register 3 (CSI_CSICR3)

This read/write register acts as an extension of the functionality of the CSI Control register 1, adding additional control and features.

Address: 0h base + 8h offset = 8h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	FRMCNT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FRMCNT_RST	DMA_REFLASH_RFF	DMA_REFLASH_SFF	DMA_REQ_EN_RFF	DMA_REQ_EN_SFF	STATFF_LEVEL			HRESP_ERR_EN	RxFF_LEVEL			TWO_8BIT_SENSOR	ZERO_PACK_EN	ECC_INT_EN	ECC_AUTO_EN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSICR3 field descriptions

Field	Description
31–16 FRMCNT	Frame Counter. This is a 16-bit Frame Counter (Wraps around automatically after reaching the maximum)
15 FRMCNT_RST	Frame Count Reset. Resets the Frame Counter. (Cleared automatically after reset is done) 0 Do not reset 1 Reset frame counter immediately
14 DMA_REFLASH_RFF	Reflash DMA Controller for RxFIFO. This bit reflash the embedded DMA controller for RxFIFO. It should be reflash before the embedded DMA controller starts to work. (Cleared automatically after reflash is done) 0 No reflash 1 Reflash the embedded DMA controller
13 DMA_REFLASH_SFF	Reflash DMA Controller for STATFIFO. This bit reflash the embedded DMA controller for STATFIFO. It should be reflash before the embedded DMA controller starts to work. (Cleared automatically after reflash is done) 0 No reflash 1 Reflash the embedded DMA controller

Table continues on the next page...

CSI_CSICR3 field descriptions (continued)

Field	Description
12 DMA_REQ_EN_RFF	DMA Request Enable for RxFIFO. This bit enables the dma request from RxFIFO to the embedded DMA controller. 0 Disable the dma request 1 Enable the dma request
11 DMA_REQ_EN_SFF	DMA Request Enable for STATFIFO. This bit enables the dma request from STATFIFO to the embedded DMA controller. 0 Disable the dma request 1 Enable the dma request
10–8 STATFF_LEVEL	STATFIFO Full Level. When the number of data in STATFIFO reach this level, STATFIFO full interrupt is generated, or STATFIFO DMA request is sent. 000 4 Double words 001 8 Double words 010 12 Double words 011 16 Double words 100 24 Double words 101 32 Double words 110 48 Double words 111 64 Double words
7 HRESP_ERR_EN	Hresponse Error Enable. This bit enables the hresponse error interrupt. 0 Disable hresponse error interrupt 1 Enable hresponse error interrupt
6–4 RxFF_LEVEL	RxFIFO Full Level. When the number of data in RxFIFO reaches this level, a RxFIFO full interrupt is generated, or an RxFIFO DMA request is sent. 000 4 Double words 001 8 Double words 010 16 Double words 011 24 Double words 100 32 Double words 101 48 Double words 110 64 Double words 111 96 Double words
3 TWO_8BIT_SENSOR	16-bit Sensor Mode. This bit indicates one 16-bit sensor connected to the 16-bit data ports. This bit should be set if there is one 16-bit sensor connected. This bit should be configured before activating or restarting the embedded DMA controller. 0 Only one 8-bit sensor is connected. 1 One 16-bit sensor is connected.
2 ZERO_PACK_EN	Dummy Zero Packing Enable. This bit causes a dummy zero to be packed with every 3 incoming bytes, forming a 32-bit word. The dummy zero is always packed to the LSB position. This packing function is only available in 8-bit/pixel mode. 0 Zero packing disabled 1 Zero packing enabled

Table continues on the next page...

CSI_CSICR3 field descriptions (continued)

Field	Description
1 ECC_INT_EN	Error Detection Interrupt Enable. This bit enables and disables the error detection interrupt. This feature only works in CCIR interlace mode. 0 No interrupt is generated when error is detected. Only the status bit ECC_INT is set. 1 Interrupt is generated when error is detected.
0 ECC_AUTO_EN	Automatic Error Correction Enable. This bit enables and disables the automatic error correction. If an error occurs and error correction is disabled only the ECC_INT status bit is set. This feature only works in CCIR interlace mode. 0 Auto Error correction is disabled. 1 Auto Error correction is enabled.

13.7.6.4 CSI Statistic FIFO Register (CSI_CSISTATFIFO)

The StatFIFO is a read-only register containing statistic data from the sensor. Writing to this register has no effect.

Address: 0h base + Ch offset = Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	STAT																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSISTATFIFO field descriptions

Field	Description
STAT	Static data from sensor

13.7.6.5 CSI RX FIFO Register (CSI_CSIRFIFO)

This read-only register contains received image data. Writing to this register has no effect.

Address: 0h base + 10h offset = 10h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	IMAGE																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSIRFIFO field descriptions

Field	Description
IMAGE	Received image data

13.7.6.6 CSI RX Count Register (CSI_CSIRXCNT)

This register works for EOF interrupt generation. It should be set to the number of words to receive that would generate an EOF interrupt.

There is an internal counter that counts the number of words read from the RX FIFO. Whenever the RX FIFO is being read, by either the CPU or the embedded DMA controller, the counter value is updated and compared with this register. If the values match, then an EOF interrupt is triggered.

Address: 0h base + 14h offset = 14h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0

CSI_CSIRXCNT field descriptions

Field	Description
31–22 -	This field is reserved. Reserved. These bits are reserved and should read 0.
RXCNT	RxFIFO Count. This 22-bit counter for RxFIFO is updated each time the RxFIFO is read by CPU or DMA. This counter should be set to the expected number of words to receive that would generate an EOF interrupt.

13.7.6.7 CSI Status Register (CSI_CSISR)

This read/write register shows sensor interface status, and which kind of interrupt is being generated. The corresponding interrupt bits must be set for the status bit to function. Status bits should function normally even if the corresponding interrupt enable bits are not enabled.

Address: 0h base + 18h offset = 18h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	-			BASEADDR_CHHANGE_ERROR	DMA_FIELD0_DONE	DMA_FIELD1_DONE	SF_OR_INT	RF_OR_INT	Reserved	DMA_TSF_DONE_SFF	STATFF_INT	DMA_TSF_DONE_FB2	DMA_TSF_DONE_FB1	RxFF_INT	EOF_INT	SOF_INT
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	F2_INT	F1_INT	COF_INT	Reserved					HRESP_ERR_INT	Reserved					ECC_INT	DRDY
W																
Reset	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSISR field descriptions

Field	Description
31–29 -	Reserved. These bits are reserved and should read 0.
28 BASEADDR_CHHANGE_ERROR	When using base address switching enable, this bit will be 1 when switching occur before DMA complete. This bit will be clear by writing 1. When this interrupt happens, follow the steps listed below. 1. Unassert the CSI enable, CSIx_CSICR18 bit31, 2. Reflash the DMA, assert the CSIX_CSICR3 bit 14, 3. Assert the CSI enable, CSIx_CSICR18 bit31.
27 DMA_FIELD0_DONE	When DMA field 0 is complete, this bit will be set to 1 (clear by writing 1).

Table continues on the next page...

CSI_CSISR field descriptions (continued)

Field	Description
26 DMA_FIELD1_DONE	When DMA field 0 is complete, this bit will be set to 1 (clear by writing 1).
25 SF_OR_INT	STATFIFO Overrun Interrupt Status. Indicates the overflow status of the STATFIFO register. (Cleared by writing 1) 0 STATFIFO has not overflowed. 1 STATFIFO has overflowed.
24 RF_OR_INT	RxFIFO Overrun Interrupt Status. Indicates the overflow status of the RxFIFO register. (Cleared by writing 1) 0 RxFIFO has not overflowed. 1 RxFIFO has overflowed.
23 -	This field is reserved. Reserved. This bit is reserved and should read 0.
22 DMA_TSF_DONE_SFF	DMA Transfer Done from StatFIFO. Indicates that the dma transfer from StatFIFO is completed. It can trigger an interrupt if the corresponding enable bit is set in CSICR1. This bit can be cleared by writing 1 or reflashing the StatFIFO dma controller in CSICR3. (Cleared by writing 1) 0 DMA transfer is not completed. 1 DMA transfer is completed.
21 STATFF_INT	STATFIFO Full Interrupt Status. Indicates the number of data in the STATFIFO reaches the trigger level. (this bit is cleared automatically by reading the STATFIFO) 0 STATFIFO is not full. 1 STATFIFO is full.
20 DMA_TSF_DONE_FB2	DMA Transfer Done in Frame Buffer2. Indicates that the DMA transfer from RxFIFO to Frame Buffer2 is completed. It can trigger an interrupt if the corresponding enable bit is set in CSICR1. This bit can be cleared by by writing 1 or reflashing the RxFIFO dma controller in CSICR3. (Cleared by writing 1) 0 DMA transfer is not completed. 1 DMA transfer is completed.
19 DMA_TSF_DONE_FB1	DMA Transfer Done in Frame Buffer1. Indicates that the DMA transfer from RxFIFO to Frame Buffer1 is completed. It can trigger an interrupt if the corresponding enable bit is set in CSICR1. This bit can be cleared by by writing 1 or reflashing the RxFIFO dma controller in CSICR3. (Cleared by writing 1) 0 DMA transfer is not completed. 1 DMA transfer is completed.
18 RxFF_INT	RxFIFO Full Interrupt Status. Indicates the number of data in the RxFIFO reaches the trigger level. (this bit is cleared automatically by reading the RxFIFO) 0 RxFIFO is not full. 1 RxFIFO is full.
17 EOF_INT	End of Frame (EOF) Interrupt Status. Indicates when EOF is detected. (Cleared by writing 1) 0 EOF is not detected. 1 EOF is detected.
16 SOF_INT	Start of Frame Interrupt Status. Indicates when SOF is detected. (Cleared by writing 1)

Table continues on the next page...

CSI_CSISR field descriptions (continued)

Field	Description
	0 SOF is not detected. 1 SOF is detected.
15 F2_INT	CCIR Field 2 Interrupt Status. Indicates the presence of field 2 of video in CCIR mode. (Cleared automatically when current field does not match) NOTE: Only works in CCIR Interlace mode. 0 Field 2 of video is not detected 1 Field 2 of video is about to start
14 F1_INT	CCIR Field 1 Interrupt Status. Indicates the presence of field 1 of video in CCIR mode. (Cleared automatically when current field does not match) NOTE: Only works in CCIR Interlace mode. 0 Field 1 of video is not detected. 1 Field 1 of video is about to start.
13 COF_INT	Change Of Field Interrupt Status. Indicates that a change of the video field has been detected. Only works in CCIR Interlace mode. Software should read this bit first and then dispatch the new field from F1_INT and F2_INT. (Cleared by writing 1) 0 Video field has no change. 1 Change of video field is detected.
12–8 -	This field is reserved. Reserved. These bits are reserved and should read 0.
7 HRESP_ERR_INT	Hresponse Error Interrupt Status. Indicates that a hresponse error has been detected. (Cleared by writing 1) 0 No hresponse error. 1 Hresponse error is detected.
6–2 -	This field is reserved. Reserved. These bits are reserved and should read 0.
1 ECC_INT	CCIR Error Interrupt. This bit indicates an error has occurred. This only works in CCIR Interlace mode. (Cleared by writing 1) 0 No error detected 1 Error is detected in CCIR coding
0 DRDY	RXFIFO Data Ready. Indicates the presence of data that is ready for transfer in the RxFIFO. (Cleared automatically by reading FIFO) 0 No data (word) is ready 1 At least 1 datum (word) is ready in RXFIFO.

13.7.6.8 CSI DMA Start Address Register - for STATFIFO (CSI_CSIDMASA_STATFIFO)

This register provides the start address for the embedded DMA controller of STATFIFO. The embedded DMA controller will read data from STATFIFO and write it to the external memory from the start address. This register should be configured before activating or restarting the embedded DMA controller.

Address: 0h base + 20h offset = 20h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DMA_START_ADDR_SFF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DMA_START_ADDR_SFF														Reserved	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSIDMASA_STATFIFO field descriptions

Field	Description
31–2 DMA_START_ADDR_SFF	DMA Start Address for STATFIFO. Indicates the start address to write data. The embedded DMA controller will read data from STATFIFO and write it from this address through AHB bus. The address should be double words aligned.
-	This field is reserved. Reserved. These bits are reserved and should read 0.

13.7.6.9 CSI DMA Transfer Size Register - for STATFIFO (CSI_CSIDMATS_STATFIFO)

This register provides the total transfer size for the embedded DMA controller of STATFIFO. This register should be configured before activating or restarting the embedded DMA controller.

Address: 0h base + 24h offset = 24h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DMA_TSF_SIZE_SFF																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSIDMATS_STATFIFO field descriptions

Field	Description
DMA_TSF_SIZE_SFF	DMA Transfer Size for STATFIFO. Indicates how many words to be transfered by the embedded DMA controller. The size should be double words aligned.

13.7.6.10 CSI DMA Start Address Register - for Frame Buffer1 (CSI_CSIDMASA_FB1)

This register provides the start address in the frame buffer1 for the embedded DMA controller of RxFIFO. The embedded DMA controller will read data from RxFIFO and write it to the frame buffer1 from the start address. This register should be configured before activating or restarting the embedded DMA controller.

Address: 0h base + 28h offset = 28h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DMA_START_ADDR_FB1															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DMA_START_ADDR_FB1															Reserved
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSIDMASA_FB1 field descriptions

Field	Description
31–2 DMA_START_ADDR_FB1	DMA Start Address in Frame Buffer1. Indicates the start address to write data. The embedded DMA controller will read data from RxFIFO and write it from this address through AHB bus. The address should be double words aligned.
-	This field is reserved. Reserved. These bits are reserved and should read 0.

13.7.6.11 CSI DMA Transfer Size Register - for Frame Buffer2 (CSI_CSIDMASA_FB2)

This register provides the start address in the frame buffer2 for the embedded DMA controller of RxFIFO. The embedded DMA controller will read data from RxFIFO and write it to the frame buffer2 from the start address. This register should be configured before activating or restarting the embedded DMA controller.

Address: 0h base + 2Ch offset = 2Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DMA_START_ADDR_FB2															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DMA_START_ADDR_FB2														Reserved	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSIDMASA_FB2 field descriptions

Field	Description
31–2 DMA_START_ADDR_FB2	DMA Start Address in Frame Buffer2. Indicates the start address to write data. The embedded DMA controller will read data from RxFIFO and write it from this address through AHB bus. The address should be double words aligned.
-	This field is reserved. Reserved. These bits are reserved and should read 0.

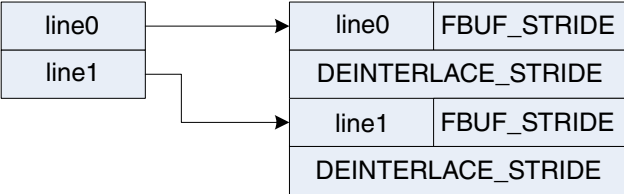
13.7.6.12 CSI Frame Buffer Parameter Register (CSI_CSIFBUF_PARA)

This register provides the stride of the frame buffer to show how many words to skip before starting to write the next row of the image. The width of the frame buffer minus the width of the image is the stride. This register should be configured before activating or restarting the embedded DMA controller.

Address: 0h base + 30h offset = 30h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEINTERLACE_STRIDE																FBUF_STRIDE															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSIFBUF_PARA field descriptions

Field	Description
31–16 DEINTERLACE_STRIDE	DEINTERLACE_STRIDE is only used in the deinterlace mode. If line stride feature is supported in deinterlace mode, FBUF_STRIDE and DEINTERLACE_STRIDE need to be configured at the same time. DEINTERLACE_STRIDE is configured the same as line width. In normal line stride feature, only FBUF_STRIDE needs to be configured.  The diagram illustrates the relationship between line0 and line1 and their respective stride parameters. Line0 is mapped to FBUF_STRIDE and DEINTERLACE_STRIDE. Line1 is mapped to FBUF_STRIDE and DEINTERLACE_STRIDE.
FBUF_STRIDE	Frame Buffer Parameter. Indicates the stride of the frame buffer. The width of the frame buffer(in double words) minus the width of the image(in double words) is the stride. The stride should be double words aligned. The embedded DMA controller will skip the stride before starting to write the next row of the image.

13.7.6.13 CSI Image Parameter Register (CSI_CSIIIMAG_PARA)

This register provides the width and the height of the image from the sensor. The width and height should be aligned in pixel. The width of the image multiplied by the height is the total pixel size that will be transferred in a frame by the embedded DMA controller. This register should be configured before activating or restarting the embedded DMA controller.

Address: 0h base + 34h offset = 34h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	IMAGE_WIDTH																IMAGE_HEIGHT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

CSI_CSIIIMAG_PARA field descriptions

Field	Description
31–16 IMAGE_WIDTH	Image Width. Indicates how many pixels in a line of the image from the sensor. If the input data from the sensor is 8-bit/pixel format, the IMAGE_WIDTH should be a multiple of 8 pixels. If the input data from the sensor is 10-bit/pixel or 16-bit/pixel format, the IMAGE_WIDTH should be a multiple of 4 pixels.
IMAGE_HEIGHT	Image Height. Indicates how many pixels in a column of the image from the sensor.

13.7.6.14 CSI Control Register 18 (CSI_CSICR18)

This read/write register acts as an extension of the functionality of the CSI Control register 1

Address: 0h base + 48h offset = 48h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
R	MIPI_DATA_FORMAT							LINE_STRIDE_EN	-		DATA_FROM_MIPI	MIPI_YU_SWAP	MIPI_DOUBLE_CMPNT	MASK_OPTION		CSI_LCDIF_BUFFER_LINES	
W	CSI_ENABLE																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	AHB_HPROT				Reserved	RGB888A_FORMAT_SEL	BASEADDR_CHANGE_ERROR_IE	LAST_DMA_REQ_SEL	DMA_FIELD1_DONE_IE	FIELD0_DONE_IE	BASEADDR_SWITCH_SEL	BASEADDR_SWITCH_EN	PARALLEL24_EN	DEINTERLACE_EN	Reserved	
W																
Reset	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0

CSI_CSICR18 field descriptions

Field	Description
31 CSI_ENABLE	CSI global enable signal. Only when this bit is 1, CSI can start to receive the data and store to memory.
30–25 MIPI_DATA_FORMAT	Image Data Format Generic Short Packet: 0x08 ~ 0x0F Embedded 8-bit non Image: 0x12 YUV420 (8-bit) 0x18 YUV420 (10-bit) 0x19 YUV420 (8-bit legacy) 0x1A YUV420 (8-bit CSPS) 0x1C YUV420 (10-bit CSPS) 0x1D YUV422 (8-bit) 0x1E YUV422 (10-bit) 0x1F RGB444 0x20 (Not support) RGB555 0x21 (Not support) RGB565 0x22 RGB666 0x23 RGB888 0x24 RAW6 0x28 RAW7 0x29 RAW8 0x2A RAW10 0x2B RAW12 0x2C RAW14 0x2D User defined 1 0x30 User defined 2 0x31 User defined 3 0x32 User defined 4 0x33 User defined 5 0x34 User defined 6 0x35 User defined 7 0x36 User defined 8 0x37
24 LINE_STRIDE_EN	When the line width are not the multiple of the burst length, assert this bit.
23 -	Reserved
22 DATA_FROM_MIPI	0 Data from parallel sensor 1 Data from MIPI
21 MIPI_YU_SWAP	It only works in MIPI CSI YUV422 double component mode.
20 MIPI_DOUBLE_CMPNT	Double component per clock cycle in YUV422 formats. 0 Single component per clock cycle

Table continues on the next page...

CSI_CSICR18 field descriptions (continued)

Field	Description
	(half pixel per clock cycle) 1 Double component per clock cycle (a pixel per clock cycle)
19–18 MASK_OPTION	These bits used to choose the method to mask the CSI input. 00 Writing to memory from first completely frame, when using this option, the CSI_ENABLE should be 1. 01 Writing to memory when CSI_ENABLE is 1. 02 Writing to memory from second completely frame, when using this option, the CSI_ENABLE should be 1. 03 Writing to memory when data comes in, not matter the CSI_ENABLE is 1 or 0.
17–16 CSI_LCDIF_ BUFFER_LINES	The number of lines are used in handshake mode with LCDIF. 00 4 lines 01 8 lines 02 16 lines 03 16 lines
15–12 AHB_HPROT	Hprot value in AHB bus protocol.
11 -	This field is reserved.
10 RGB888A_ FORMAT_SEL	Output is 32-bit format. 0 {8'h0, data[23:0]} 1 {data[23:0], 8'h0}
9 BASEADDR_ CHANGE_ ERROR_IE	Base address change error interrupt enable signal.
8 LAST_DMA_ REQ_SEL	Choosing the last DMA request condition. 0 fifo_full_level 1 hburst_length
7 DMA_FIELD1_ DONE_IE	When in interlace mode, field 1 done interrupt enable. 0 Interrupt disabled 1 Interrupt enabled
6 FIELD0_DONE_ IE	In interlace mode, field 0 means interrupt enabled. 0 Interrupt disabled 1 Interrupt enabled
5 BASEADDR_ SWITCH_SEL	CSI 2 base addresses switching method. When using this bit, BASEADDR_SWITCH_EN is 1. 0 Switching base address at the edge of the vsync 1 Switching base address at the edge of the first data of each frame
4 BASEADDR_ SWITCH_EN	When this bit is enabled, CSI DMA will switch the base address according to BASEADDR_SWITCH_SEL rather than atomically by DMA completed.

Table continues on the next page...

CSI_CSICR18 field descriptions (continued)

Field	Description
3 PARALLEL24_ EN	When input is parallel rgb888/yuv444 24bit, this bit can be enabled.
2 DEINTERLACE_ EN	This bit is used to select the output method When input is standard CCIR656 video. 0 Deinterlace disabled 1 Deinterlace enabled
-	This field is reserved. Reserved.

13.8 MIPI CSI Host Controller (MIPI_CSI)

13.8.1 Overview

This section introduces the MIPI CSI-2 RX subsystem with the CSI-2 RX PHY and host controller. This subsystem handles the sensor/image input and process for all the input imaging devices.

13.8.1.1 Features

The MIPI-CSI2 Controller has the following key features:

- Implements all three CSI-2 MIPI layers (Pixel to byte packing, low level protocol, Lane management)
- Supports unidirectional Master operation
- Transmitter and receiver versions
- Scalable data lane support, 1 to 4 Data Lanes
- Supports high speed mode for the 4 lanes
- Supports 4Kp30, 1080p60/30, 5MP at 15 fps, 720p60, VGA at 60 fps
- Supports 10Mbps data rate in low power mode
- Includes high speed deserializers
- Loopback testability support
- Support for all CSI-2 data types
- Support for DPHY Ultra Low Power State (ULPS)
- Error collection support (Rx Only)
- Flexible pixel-based user interface
 - Supports user generated packets
 - Supports single interface

- Supports PHY Protocol Interface (PPI) compatible MIPI D-PHYs
 - Delivered fully integrate and verified with target MIPI D-PHY
- RX Video Interface
- APB Control and Status Register (CSR) interface with IRQ support
- Easy configuration and control via core ports

13.8.2 Reset

Reset and Initialization procedure for the CSI-2 Rx Controller is as follows:

1. Assert all resets.
2. Wait until clocks to the Rx Controller are stable and ensure DPHY interface is idle (Stop State on all lanes)
3. De-assert reset to CSR, `pclk_reset_n`, if CSR is included with the controller. If no CSR go to step 4.
4. Program CSR registers if CSR is present. If CSR is not present, set values to all configuration ports (`cfg_*` inputs) to appropriate values.
5. De-assert all remaining resets.
6. CSI-2 Rx Controller is ready for use by the next rising edge of `clk_ui`.

13.8.3 Functional description

The CSI-2 Rx Controller Core implements all three layers defined by the CSI-2 Specification: Pixel to Byte Packing, Low Level Protocol, and Lane Management. The D-PHY interface of the CSI-2 Rx Controller Core supports PHY Protocol Interface (PPI) compatible MIPI D-PHYs. The Local Interface is an easy to use pixel based interface that supports all data types. The Local interface runs at the User Interface clock rate for all implementations. The CSI-2 Rx Controller Core takes care of all packet formatting details and transmission over the MIPI bus. The CSI-2 Rx Controller Core also includes an Rx Video Interface. This interface watches the local interface of the controller and generates a simplified video output interface. Image packet types which are supported by the Pixel Format block are assembled into a Video output bus which carries the pixel data which is received by the controller. Valid video output data is indicated using the data enable output(s). Horizontal and vertical sync outputs are also provided.

13.8.3.1 CSI-2 RX Block diagram

The block diagram for the MIPI CSI-2 RX block is given below:

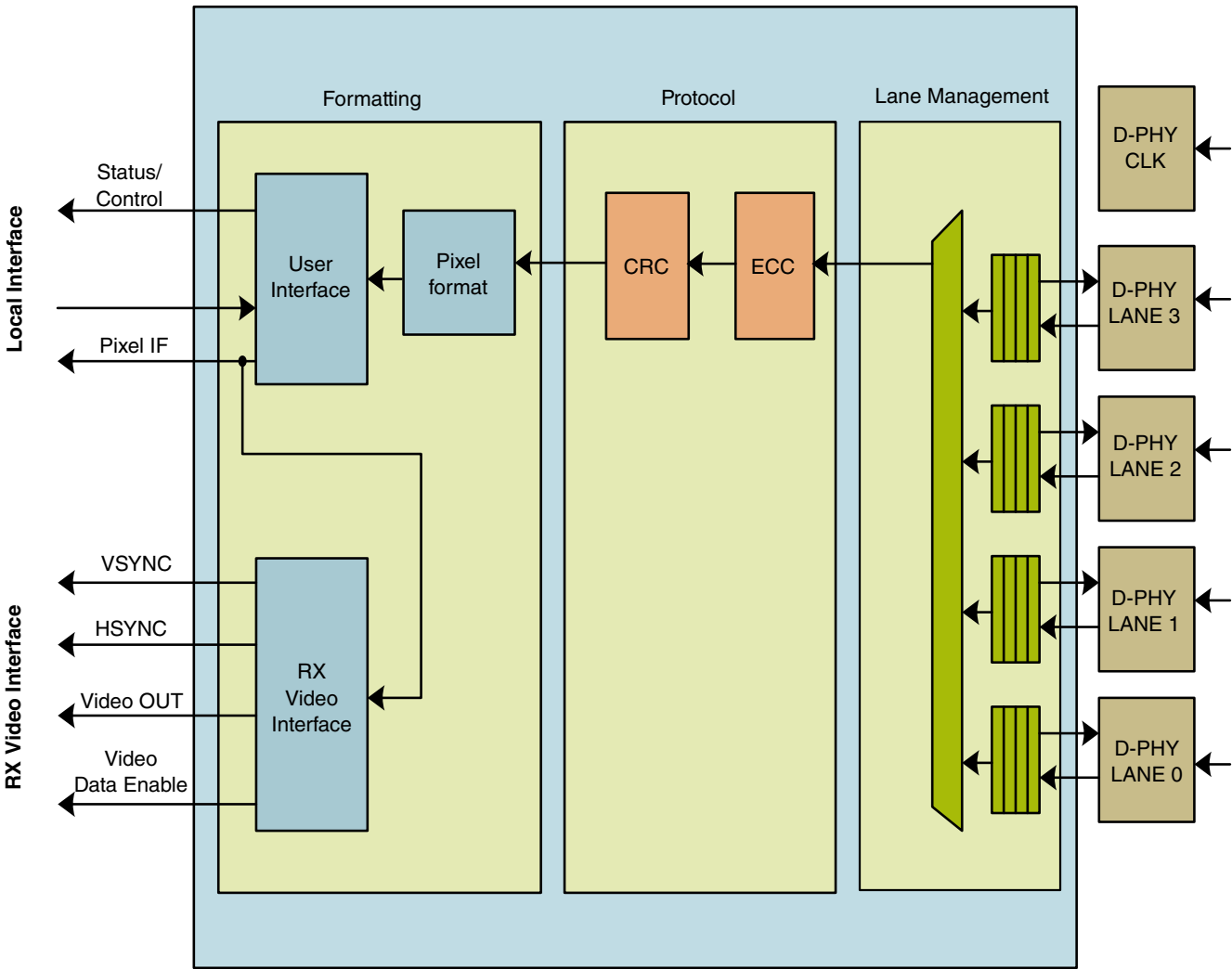


Figure 13-71. MIPI CSI-2 RX Block Diagram

13.8.3.2 Rx Controller Core Configuration Port Descriptions

The CSI-2 Rx Controller Core has configuration ports that can control the configuration of the core in real time. The ports are listed and described in the table below. When the CSR module is included with the CSI-2 RX Controller, the `cfg_*` ports are removed from the top level and connected to ports on the CSR module, making them accessible through the APB interface.

Table 13-31. Rx Controller Core Configuration Port Descriptions

Port	Type	Description
<code>cfg_num_lanes[1:0]</code>	Input	Sets the number of active lanes that are to be used for receiving MIPI data.

Table continues on the next page...

Table 13-31. Rx Controller Core Configuration Port Descriptions (continued)

		2'b00 – 1 Lane 2'b01 – 2 Lanes 2'b10 – 3 Lanes 2'b11 – 4 Lanes
cfg_disable_payload[48:0]	Input	Disables payload data for the selected data type. When a bit is set that represents a supported data type, only the packet header will be presented at the user interface, along with the sop and eop indication where the payload data would have appeared if enabled. [0] – NULL [1] –BLANK [2] –EMBEDDED [9:3] –Reserved [10] – Legacy YUV420 8 bit [13:11] – Reserved [14] – YUV422 8 bit [15] – YUV422 10 bit [16] – RGB444 [17] – RGB555 [18] – RGB565 [19] – RGB666 [20] – RGB888 [23:21] – Reserved [24] –RAW6 [25] –RAW7 [26] –RAW8 [27] –RAW10 [28] –RAW12 [29] –RAW14 [31:30] – Reserved [32] – User defined type 0x30 [33] – User defined type 0x31 [34] – User defined type 0x32 [35] – User defined type 0x33 [36] – User defined type 0x34 [37] – User defined type 0x35 [38] – User defined type 0x36 [39] – User defined type 0x37 [47:40] – Reserved [48] – Unsupported data types
cfg_disable_data_lanes	[3:0]	Setting bits to a '1' value causes the DPHY Enable signal to deassert. [0] – data lane 0 [1] – data lane 1 [2] – data lane 2 [3] – data lane 3

13.8.3.3 CSI-2 RX Controller Core Local Interface

The CSI-2 Rx Controller Core Local Interface consists of a Receive Interface.

Packets are received from the MIPI Interface via the Receive Local Interface. The Receive Interface is designed to easily interface to a FIFO on the user side that holds the pixels that are received until the user application can process them.

13.8.3.3.1 RX Local Interface Description

The Receive Local Interface is the mechanism with which the user receives packets from the MIPI Interface.

The Receive Local Interface first announces that a packet is being or has been received by asserting the `pkt_hdr_valid` signal with the other packet information output ports containing details about the packet. Next, the `sop_out` signal will assert indicating that data is about to arrive at the Receive Local Interface. `dav_out` will assert with every cycle of pixel data on the `data_out` port. The last cycle of receive data is signaled by the assertion of `eop_out`.

With a single pixel interface, `data_out` contains a single pixel every clock that `dav_out` (data valid) is asserted. The single pixel interface must run at a frequency that is equal to number of MIPI lanes times the byte lane frequency.

Table 13-32. Receive Local Interface

Port	Type	Description
<code>clk</code>	Input	RX Controller Core Clock input. This clock must be exactly equal to or faster than the receive byteclock, <code>RxByteClkHS_In0</code> , from the RX DPHY.
<code>clk_ui</code>	Input	User interface clock. The frequency of <code>clk_ui</code> must be such that the data received on the <code>data_out</code> output is greater than or equal to the total bandwidth of the physical MIPI interface. <code>Clk_ui</code> has no relationship requirement with regards to 'clk' other than the bandwidth requirement mentioned previously.
<code>clk_esc</code>	Input	Rx Escape Clock. This must be the same escape clock that the RX DPHY receives.
<code>reset_n</code>	Input	Aysnc reset, active low. This reset applies to all logic in the Controller Core that is in the 'clk' clock domain.

Table continues on the next page...

Table 13-32. Receive Local Interface (continued)

clk_ui_reset_n	Input	Async reset, active low. This reset applies to all logic in the Controller Core that is in the 'clk_ui' clock domain.
clk_esc_reset_n	Input	Async reset, active low. This reset applies to all logic in the Controller Core that is in the clk_esc domain.
byte_clk_reset_n	Input	Asnyc reset, active low. This reset applies to all logic in the Controller Core that is in the byte clock (RxByteClkHS_Inx clock inputs) that is received from the RX DPHY.
data_out[n-1:0]	Output	Pixel Data output. The Rx Controller can present pixel data in single pixels or in pairs. The width of the data_out port is dependent on the largest pixel size possible and whether the controller is configured for single, double, or quad pixel mode. Pixel data types narrower than 'n' bits are justified down into the least significant bits (ie, bits [7:0] in a 12 bit field [11:0]). When in double pixel mode, the first pixel resides at [n-1:0] while the second pixel received is at [n*2-1:n]. When in quad pixel mode, the first pixel resides at [n-1:0], second pixel received is at [n*2-1:n], the third pixel is at [n*3-1:n*2] and the fourth is at [n*4-1:n*3] where 'n' is the maximum pixel width supported.
dav_out	Output	Data Valid Out. When asserted high, data_out, sop_out, and eop_out outputs are valid.
sop_out	Output	Start of packet signal. This active high signal indicates the first cycle of data on dataout. Valid when dav_out asserts high
eop_out[3:0]	Output	Last pixel indicator. Asserts to indicate the last pixel in a transmission. When in single pixel mode, eop_out[0] is used. When in double pixel mode two bits are used, [1:0]. 2'b01 indicates the last pixel is on the lower pixel path. 2'b10 indicates the last pixel is on the upper pixel path. When in quad pixel mode, all four bits are utilized. A value of 4'b0001 indicates the last pixel is on the lowest pixel path portion of data_out. 4'b0010 indicates the last pixel is on the second pixel path of data_out. 4'b0100 indicates the last pixel is on the third pixel path of data_out. 4'b1000 indicates the last pixel is on the fourth and highest pixel path of data_out. Valid when dav_out asserts high.

Table continues on the next page...

Table 13-32. Receive Local Interface (continued)

pkt_hdr_valid	Output	Packet header data is valid on the packet header ports below when this signal is asserted.
pkt_hdr_data_type[4:0]	Output	Packet data type, valid when pkt_hdr_valid is asserted. See the CSI-2 specification for a definition of possible values.
pkt_hdr_wc[15:0]	Output	Packet Word Count. Number of bytes of pixel data in the received payload for long packet data types. See the MIPI CSI-2 specification for meaning for short packet. The value on pkt_hdr_wc is the Word count field from the received packet header. The RX controller does not modify the value in any way. Valid when pkt_hdr_valid is asserted.
pixel_cnt_out[15:0]	Output	Number of pixels in receive packet. The pixel_cnt_out value is updated one clock after eop_out asserts and represents the packet of pixels there were in the just received packet. The pixel_cnt_out value is only valid for long packets.
ecc_one_bit_error	Output	Single bit error in the packet header was detected and corrected. Active high and is valid when pkt_hdr_valid is high.
ecc_two_bit_error	Output	Two packet header bit errors were detected and not corrected, active high and is valid when pkt_hdr_is high..
ecc_one_bit_error_pos[4:0]	Output	Position of the corrected single bit error in the packet header. Valid when pkt_hdr_valid is asserted high.
ecc_err	Output	Error detected in the ECC bits. This signal is no longer supported.
ecc_err_pos[2:0]	Output	Position of the bit in the ECC bits, valid when ecc_err is asserted. This signal is no longer supported.
crc_err	Output	Asserts high when the CRC calculated on the received data does not match the CRC the transmitter sent at the end of the packet. Valid when the eop_out output asserts.
ulps_active[4:0]	Output	Receive UltraLow Power State active. Bits assert high when corresponding clock or data lane is in ULPS mode. Bit [0] – Clock Lane Bit [1] – Data Lane 0 Bit [2] – Data Lane 1 Bit [3] – Data Lane 2 Bit [4] – Data Lane 3

Table continues on the next page...

Table 13-32. Receive Local Interface (continued)

ulps_mark_active[4:0]	Output	Receive UltraLow Power State Mark status. Bits assert high when corresponding clock or data lane has exited ULPS and entered Mark-1 state. Lanes will remain in Mark-1 state for a minimum of 1ms as per the MIPI DPHY specification. While in Mark-1 state, no other active, like High Speed data transmission, is allowed. Bit [0] – Clock Lane Bit [1] – Data Lane 0 Bit [2] – Data Lane 1 Bit [3] – Data Lane 2 Bit [4] – Data Lane 3
rx_enable	Input	Receive Enable, active high. When deasserted, the RX controller will pause data reception at the next packet boundary. Since the CSI-2 protocol does not allow the Receiver to pause data, when rx_enable is deasserted the RX Controller still receives data from the RX DPHY but it does not forward the receive packets to the user interface but instead discards the received data. When rx_enable is asserted, the RX controller will wait for the start of the next packet before allowing data to be sent out over the user interface.

13.8.3.3.2 Receive Interface Example Transaction

High Speed receive packets appear immediately on the Receive local interface as they are received.

13.8.3.3.2.1 Long packet receive with RAW 8 data and one DPHY

Initially, pkt_hdr_valid asserts along with the packet header information. The packet header ports, pkt_hdr_data_type, pkt_hdr_wc, and pkt_hdr_vc indicate that the packet contains RAW8 data, has 4 btes total and a Virtual Channel of zero. Several clocks later sop_out asserts with dav_out indicating the first cycle of receive data. Dav_out asserts for 4 clocks, framing the 4 bytes (pixels) of RAW8 pixel data on data_out[7:0]. During the last cycle of dav_out eop_out[0] asserts indicating the last cycle of received RAW8 data. When the pixel data has a width less than the 24 bit width of the data_out port the data is

justified down into the lower bits. In the example previously described, the RAW8 data will appear on data_out[7:0]. If the data had been RAW10 pixels, the RAW10 data would be contained on data_out[9:0].

The figure below shows a waveform of the received packet on the local interface.

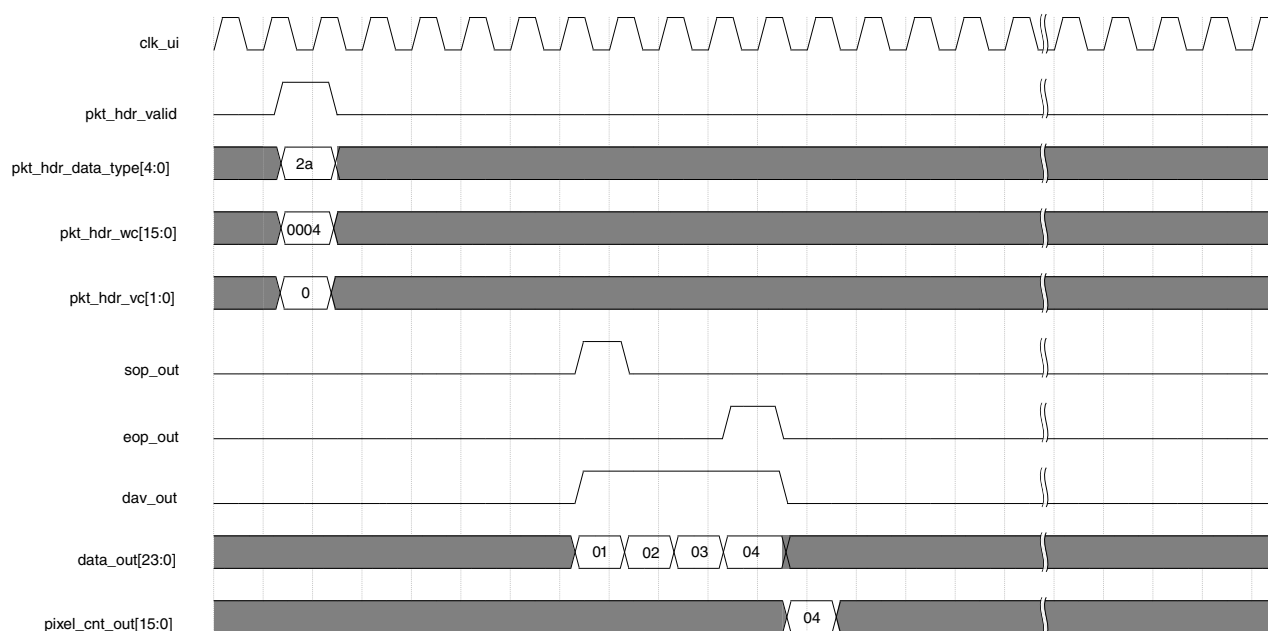


Figure 13-72. Receive Packet with a payload of 8 RAW8 pixels, single DPHY lane, and Virtual Channel = 0 single pixel configuration

13.8.3.3.2.2 Two back to back long packet request with RAW8 data and two DPHY lanes

The figure shows two packet requests that occur back to back. L_tx_pkt_cmd_req is asserted along with the packet parameters on l_tx_pkt_cmd. The request is acknowledged by the controller during the next clock period when l_tx_pkt_cmd_ack is asserted. The second request is immediately submitted by keeping l_tx_pkt_cmd_req asserted and changing the value on l_tx_pkt_cmd, indicating that the next packet will contain 4 bytes rather than the 3 that the previous packet contained. Note that it is not necessary to change l_tx_pkt_cmd in order to request another packet immediately following the first packet.

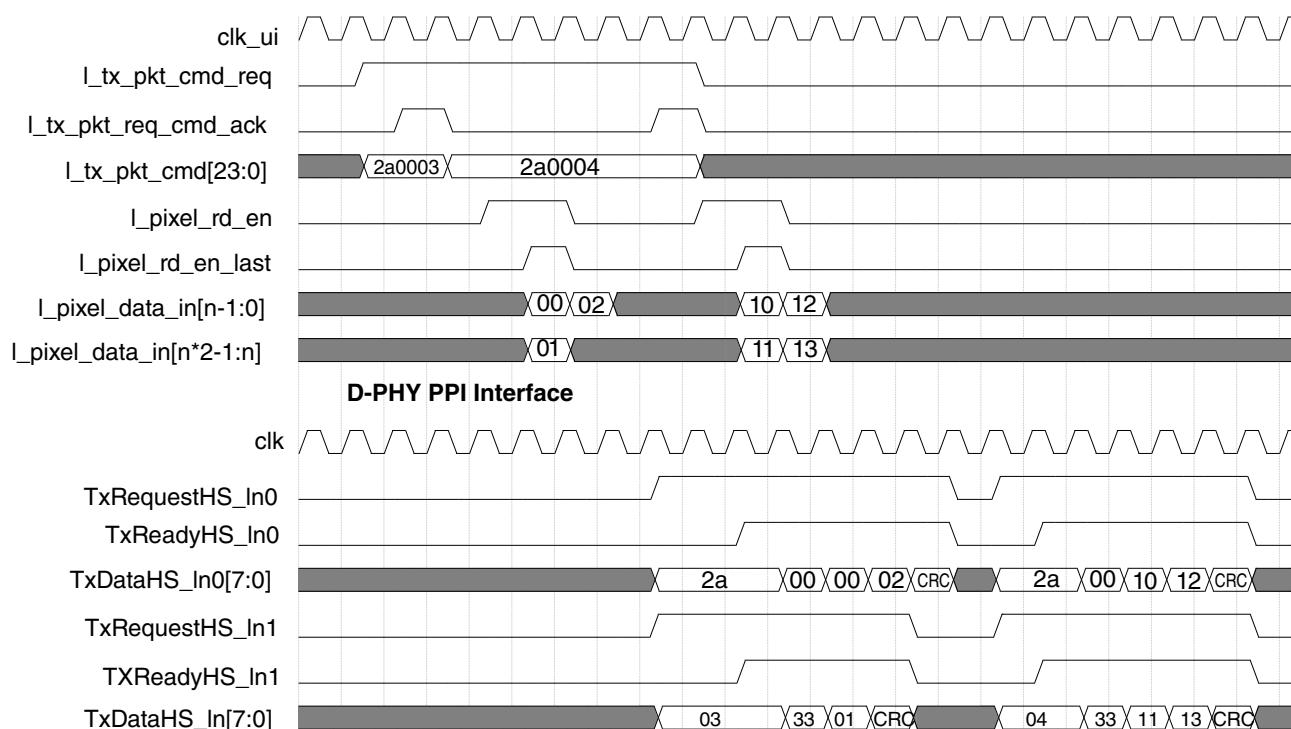


Figure 13-73. Two Transmit Packets, payloads of 3 and 4 pixels, two DPHY lanes, Virtual Channel=0, Double Pixel Mode

NOTE

The relationship of `l_tx_pkt_cmd_req`, `l_tx_pkt_cmd_ack` to `TxRequestHS`, `TxReadyHS` and `l_pixel_data_in` to `TxDataHS` is not defined. Furthermore, the relationship shown in the above figure is compressed for formatting considerations.

13.8.3.3.2.3 Short Packet Receive

The short packets do not contain any data. All the information in a short packet is contained in the 32 bit header. `Pkt_hdr_valid` asserts indicating that the packet header information contained on `pkt_hdr_data_type`, `pkt_hdr_wc`, and `pkt_hdr_vc` is valid. Short packets are defined as having a `pkt_hdr_data_type` value that is less than 0x10. With a short packet the CSI-2 Rx Controller Core does not assert `sop_out`, `eop_out`, nor `dav_out`.

The figure below shows the reception of a CSI-2 Short Packet type.

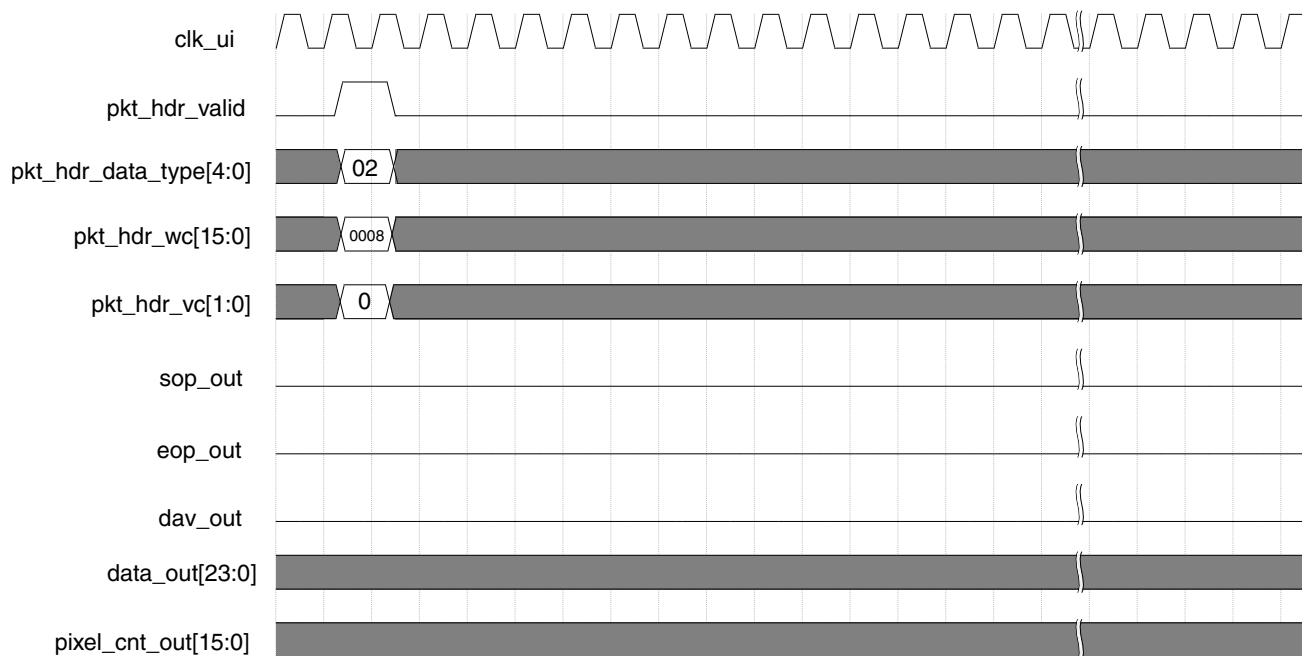


Figure 13-74. Short Packet receive with a data type of Line Start, Line Number of 8 , and Virtual Channel = 0 single pixel configuration

In this example, the data type is 0x30, a user defined byte data type. Initially, `pkt_hdr_valid` asserts along with the packet header information. The packet header ports, `pkt_hdr_data_type`, `pkt_hdr_wc`, and `pkt_hdr_vc` indicate that the packet is a user defined data type with a payload of 4 bytes. For unsupported types, the Rx Controller outputs the entire packet minus the two byte CRC value at the end of the packet. The first four bytes are the packet header followed immediately by the 4 bytes of payload data.

The figure below shows a waveform diagram of a received packet on the local interface for an unsupported data type.

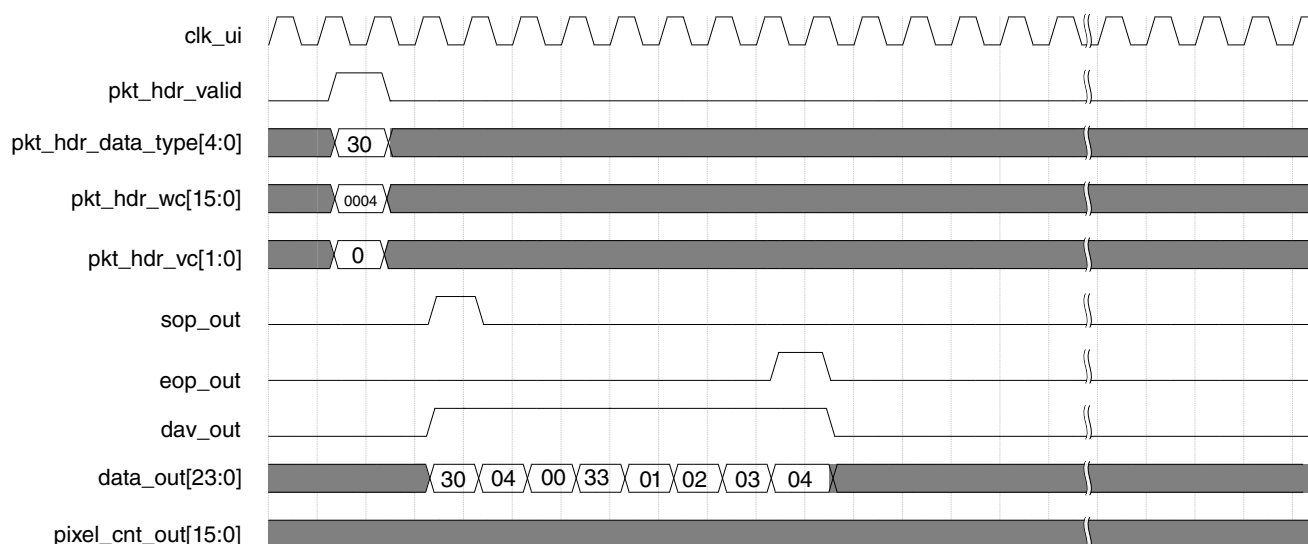


Figure 13-75. Received packet of an unsupported data type, 0x30, user defined data, single pixel configuration

13.8.3.4 CSI-2 Rx Controller Core Clocking

The RX controller core requires two clocks for proper operation `clk_ui` and `clk`. The `clk_ui` clock is the clock that the user interface is synchronous to, with all inputs and outputs referenced to the rising edge of `clk_ui`. The `clk` input is the clock that the CSI-2 Rx Controller Core uses to process data received from the D-PHY. `Clk_ui` must be of a sufficient frequency to keep up with the incoming data from the CSI-2 MIPI interface.

The data from the D-PHY is clocked into shallow FIFOs (one per DPHY lane) using a receive clock that the D-PHY generates. There is no direct timing requirements between the D-PHY generated receive clock and the CSI-2 Rx Controller Core's `clk` or `clk_ui` input. The frequency of the clock used for the RX controller's `clk` input must be of a sufficient frequency that the RX controller can keep up with the data from DPHY. The Rx Controller processes the Rx D-PHY data in a 32 bit wide data sufficient frequency that the Rx Controller can keep up with the receive data from the D-PHY. The Rx Controller processes the Rx D-PHY data in a 32 bit wide data path regardless of whether the CSI-2 MIPI receive data is coming across one, two, three, or four lanes. The frequency of `clk` must be exactly equal to or greater than the RX byte clock coming from the RX DPHY.

The CSI-2 Rx Controller Core can supply pixel data to the User application either one, two or four pixels per clock.

When in single Pixel Mode, the minimum frequency for `clk_ui` must be high enough to keep up with the incoming data rate of the CSI-2 MIPI interface. It is important to realize that the user interface is pixel based and therefore may be wider than the incoming byte data from the MIPI RX DPHY. As an example, if RGB888 data is being received, the user interface data width will be 24 bits wide (1 pixel per `clk_ui` rising edge and each pixel is 24 bits). As an example, let the MIPI interface run at 1000Mbps with a single lane. The data rate into the CSI-2 RX controller will be 125MBps so `clk` must run at 125MHz or higher. Since the user interface is 3 bytes wide, `clk_ui` must run at 125MHz/3 or higher to keep up with the incoming data.

the figure below shows one possible clocking methodology. In this example, the user `Clk_gen` module generates the edge aligned `clk_ui` and `clk` clocks. It also generates a reference clock that the D-PHY might require. The Reference clock has no relationship to any of the CSI-2 Rx Controller Core's clocks.

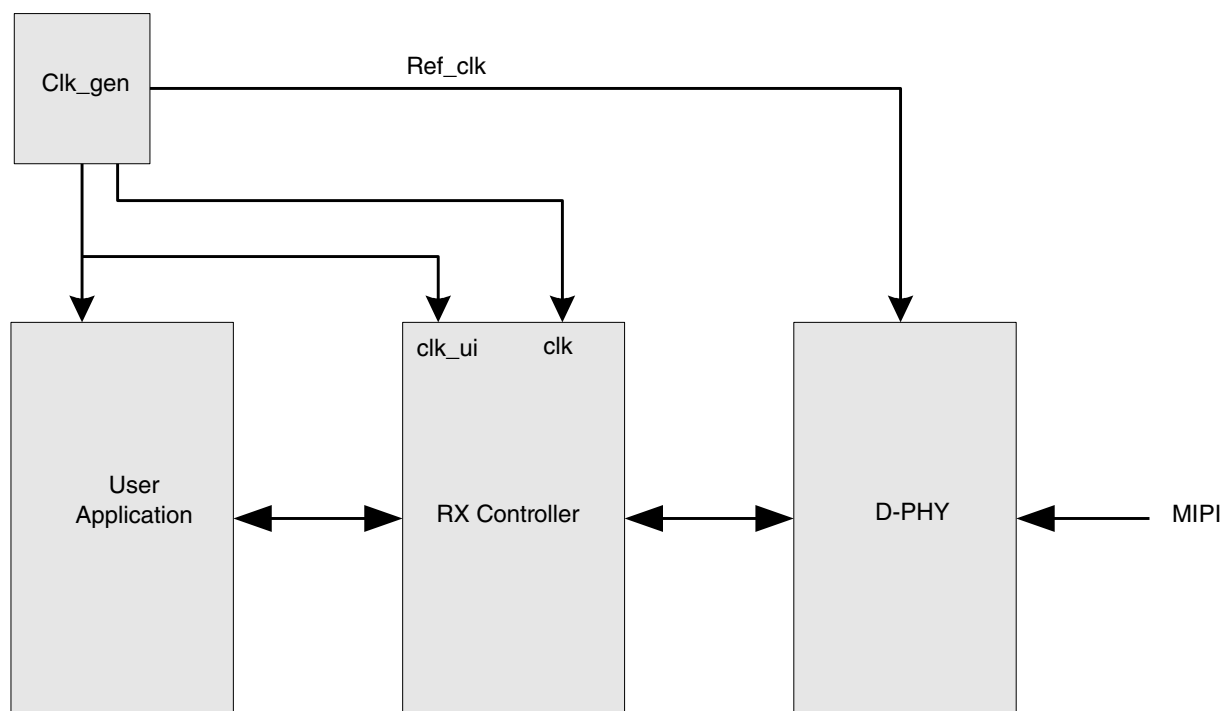


Figure 13-76. CSI-2 Rx Controller Core clocking example

13.8.3.5 CSI-2 Controller Core Configurations

The CSI-2 RX Controller Core User Interface supports either a single, double, or quad pixel wide data path. The double and quad wide pixel modes can help reduce the frequency the user interface must run at while the single pixel wide mode can be easier to interface to for some applications.

This chip supports the following:

- Single Pixel Configuration

13.8.3.5.1 Single Pixel Configuration

The CSI-2 Rx Controller Core User Interface supports a configuration where a single pixel of received data is processed every clock period.

The CSI-2 Rx Controller Core contains minimal pixel buffering (16 pixels or less) so the frequency of that the user interface runs at must be at least high enough to receive the data from the controller at a rate that is equal to or greater than the highest transmission data rate on the MIPI interface for all lanes combined. For example, if the total bandwidth of the physical MIPI interface will be 800Mbps with a data type of 8RAW then the user interface must be clocked at a minimum of 100MHz.

13.8.3.6 CSI-2 Rx Controller Core Data Types Formatting

The CSI-2 Rx Controller Core supports the various CSI-2 Data Types via the data_out port. The following subsections describe how the controller presents the various received data types on the data_out output port.

The CSI2 Controller supports the Single Pixel Configuration.

13.8.3.6.1 RAW6 (Data Type = 0x28)

With RAW6 data type (0x28) and the Rx Controller configured for single pixel mode, RAW6 data is contained on data_out[5:0].

13.8.3.6.2 RAW7 (Data Type = 0x29)

With RAW7 data type (0x29) and the Rx Controller configured for single pixel mode, RAW7 data is contained on data_out[6:0].

13.8.3.6.3 RAW8 (Data Type = 0x2A)

With RAW8 data type (0x2A) and the Rx Controller configured for single pixel mode, RAW8 data is contained on data_out[7:0].

13.8.3.6.4 RAW10 (Data Type = 0x2B)

With RAW10 data type (0x2B) and the Rx Controller configured for single pixel mode, the received RAW10 data is contained on data_out[9:0].

13.8.3.6.5 RAW12 (Data Type = 0x2C)

With RAW12 data type (0x2C) and the Rx Controller configured for single pixel mode, the received RAW12 data is contained on data_out[11:0].

13.8.3.6.6 RAW14 (Data Type = 0x2D)

With RAW14 data type (0x2D) and the Rx Controller configured for single pixel mode, the received RAW14 data is contained on data_out[13:0].

13.8.3.6.7 RGB444 (Data Type = 0x20)

With RGB444 data type (0x20) and the Rx Controller configured for single pixel mode, the blue data is on data_out[3:0], green on data_out[7:4], and red on data_out[11:8].

13.8.3.6.8 RGB555 (Data Type = 0x21)

With RGB555 data type (0x21) and the Rx Controller configured for single pixel mode the blue data is on data_out[4:0], green on data_out[9:5], and red on data_out[14:10].

13.8.3.6.9 RGB565 (Data Type = 0x22)

With RGB565 data type (0x22) and the Rx Controller configured for single pixel mode the blue data is on data_out[4:0], green on data_out[10:5], and red on data_out[15:11].

13.8.3.6.10 RGB666 (Data Type = 0x23)

With RGB666 data type (0x23) and the Rx Controller configured for single pixel mode, the blue data is on data_out[5:0], green on data_out[11:6], and red on data_out[17:12].

13.8.3.6.11 RGB888 (Data Type = 0x24)

With RGB888 data type (0x24) and the Rx Controller configured for single pixel mode, the blue data is on data_out[7:0], green on data_out[15:8], and red on data_out[23:16].

13.8.3.6.12 Legacy YUV420 8 bit (Data Type = 0x1A)

With Legacy YUV420 8 bit and the Rx Controller configured for single pixel mode the video data is received in UYY.../VYY sequences in odd/even lines. The U component is received with YY in a UYY group on odd lines while the V component is received with the same YY pairs in VYY groups on the even lines. See the MIPI CSI-2 Specification for a more complete description. When receiving an odd line data_out[7:0] contains the U component, data_out[15:8] Y1 and data_out[23:16] Y2. On even lines data_out[7:0] contains the V component, data_out[15:8] Y1 and data_out[23:16] Y2.

13.8.3.6.13 YUV420 8 bit (Data Type = 0x18,1C)

With YUV420 8 bit and YUV420 8 bit chroma shifted is received as YYYY (odd lines) UYVY (even lines) sequence. With the Rx controller configured for single pixel mode and receiving even numbered lines, data_out[7:0] contains the U1 component, bits [15:8] Y1. On the next data valid cycle bits [7:0] V1, bits [15:8] Y2. When receiving odd lines, data_out[7:0] contains the Y1 component, bits [15:8] Y2. On the next data valid cycle bits [7:0] Y3, bits [15:8] Y4.

13.8.3.6.14 YUV420 10 bit (Data Type = 0x19, 0x1D)

With YUV420 10 bit and YUV420 10 bit chroma shifted is received as YYYY (odd numbered lines) and UYVY (even numbered lines) sequence with each component represented by 10 bits. With the Rx controller configured for single pixel mode and receiving even numbered lines, data_out[9:0] contains the U1 component, bits [19:10] Y1. On the next data valid cycle bits [9:0] V1, bits [19:10] Y2. When receiving odd lines, data_out[9:0] contains the Y1 component, bits [19:10] Y2. On the next data valid cycle bits [9:0] Y3, bits [19:10] Y4.

13.8.3.6.15 YUV422 8 bit (Data Type = 0x1E)

With YUV422 8 bit is received as UYVY sequence. With the Rx controller configured for single pixel mode, data_out[7:0] contains the U1 component, bits [15:8] Y1. On the next data valid cycle bits [7:0] V1, bits [15:8] Y2.

13.8.3.6.16 YUV422 10 bit (Data Type = 0x1F)

With YUV422 10 bit is received as a UYVY sequence with each component represented by 10 bits. With the Rx controller configured for single pixel mode, data_out[9:0] contains the U1 component, bits [19:10] Y1. On the next RX data valid cycle, bits [9:0] V1, bits [19:10] Y2.

13.8.3.6.17 User Defined Data Types, (Data Type = 0x30,0x31,0x32,0x33,0x34,0x35,0x36,0x37)

With User Defined Data Types and the Rx Controller configured for single pixel mode, User Defined data is contained on data_out[7:0].

13.8.3.6.18 Null Data Type, (Data Type = 0x10)

With User Null Data Type and the Rx Controller configured for single pixel mode, Null data is contained on data_out[7:0].

13.8.3.6.19 Blanking Data Type, (Data Type = 0x11)

With Blanking Data Type and the Rx Controller configured for single pixel mode, Blanking data is contained on data_out[7:0].

13.8.3.6.20 Embedded Data Type, (Data Type = 0x12)

With User Null Data Type and the Rx Controller configured for single pixel mode, Embedded data is contained on data_out[7:0].

When the Rx Controller is configured for double pixel mode, the Embedded data that is received is contained on data_out[7:0] and data_out[MAX_PIXEL_WIDTH+7:MAX_PIXEL_WIDTH] with data_out[7:0] having been received first.

When the Rx Controller is configured for quad pixel mode, the first Embedded data that is received is contained on data_out[7:0], the second on data_out[MAX_PIXEL_WIDTH+7:MAX_PIXEL_WIDTH], the third on data_out[MAX_PIXEL_WIDTH*2+7:MAX_PIXEL_WIDTH*2],

13.8.3.6.21 Unsupported Data Types

With unsupported data types and the Rx Controller configured for single pixel mode, the raw packet data (including the header but minus the CRC) is contained on data_out[7:0].

13.8.3.7 RX Video Interface

The CSI-2 Rx Video Interface is connected internally to the local interface outputs of the CSI-2 Rx Controller Core. This interface provides both the packet header information and the pixel data (after processing by the Pixel Format block). CSI-2 Rx Controller Cores may be built with any number of Pixel Formats supported. The CSI-2 Rx Video Interface only recognizes video formats for which the CSI-2 Rx Controller Core has been configured to support.

The general timing of the video interface is determined by the timing of the packets received by the CSI-2 RX controller which are used to determine when to generate a Vertical Sync and when to output a line of video data.

The CSI-2 Rx Video Interface may receive several different types of packet data types as shown in the table below.

Packet Type	Data ID	Function
Frame Start Code (FS)	0x00	Indicates start of frame
Frame End Code (FE)	0x01	Indicates end of frame
Line Start Code (LS)	0x02	Indicates start of line
Line End Code (LS)	0x02	Indicates end of line
Image Data Codes (IMG)	various	Image data packets carry the sample data. The CSI-2 Rx Video Interface will respond to those Data types which are enabled in the CSI-2 Rx Controller Core. See CSI-2 Rx Controller Core Data Types Formatting

13.8.3.7.1 Video Interface Output Raster Timing

The output timing of the raster is formed by the packets sent to the CSI-2 Rx Controller Core and is determined by the relative timing of these packets. This timing is controlled by the transmitter and is not determined by the CSI-2 Rx Controller Core. As shown in Figure 4.8 below, the basic sequence of packets sent by the transmitter (often a sensor or camera) consists of a Frame Start packet (short packet), followed by some number of Image Data packets (long packets) with one Image data packet per video line and the followed by a Frame End packet (short packet). The Frame End packet is followed by pause which determines the Blanking Lines interval. The length of the Image Data packets is determined by the horizontal resolution of the sensor and each line's samples are carried by a single Image Data packet.

NOTE

Note that each Image Data packet may optionally be preceded by a Line Start packet and followed by an optional Line End packet. The Line Start and Line End packets are ignored by the CSI-2 Rx Video Interface.

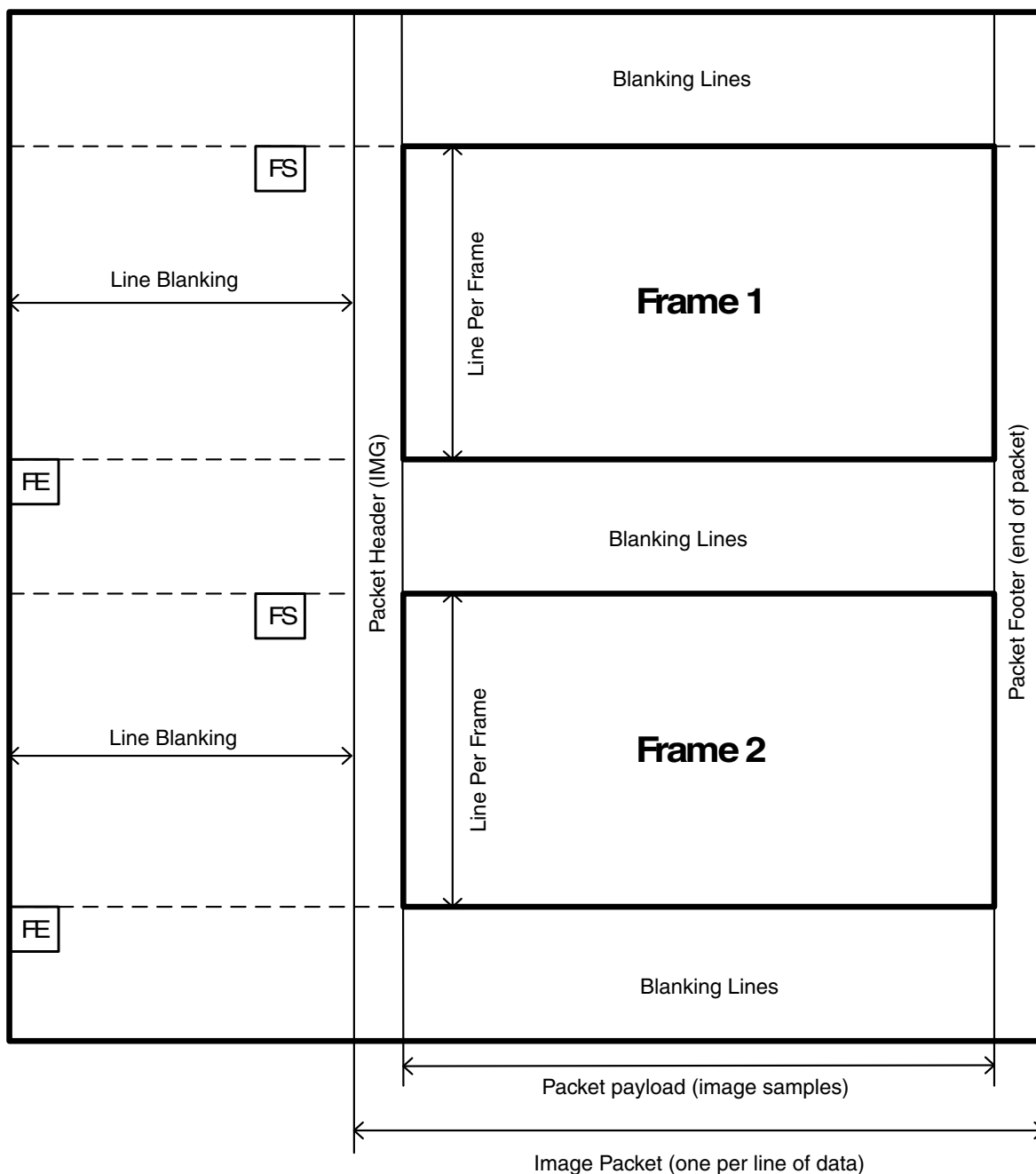


Figure 13-77. Rx Video Interface Raster Timing

NOTE

- FS- Frame Start Packet

- FE- Frame End Packet
- IMG- Image Data Packet

Note that the interface does not generate Horizontal sync pulses during the Blanking Lines.

The CSI-2 Rx Video controller generates a Vertical sync pulse upon the occurrence of every Frame Start packet. Each line of video sample data is preceded by a Horizontal sync pulse.

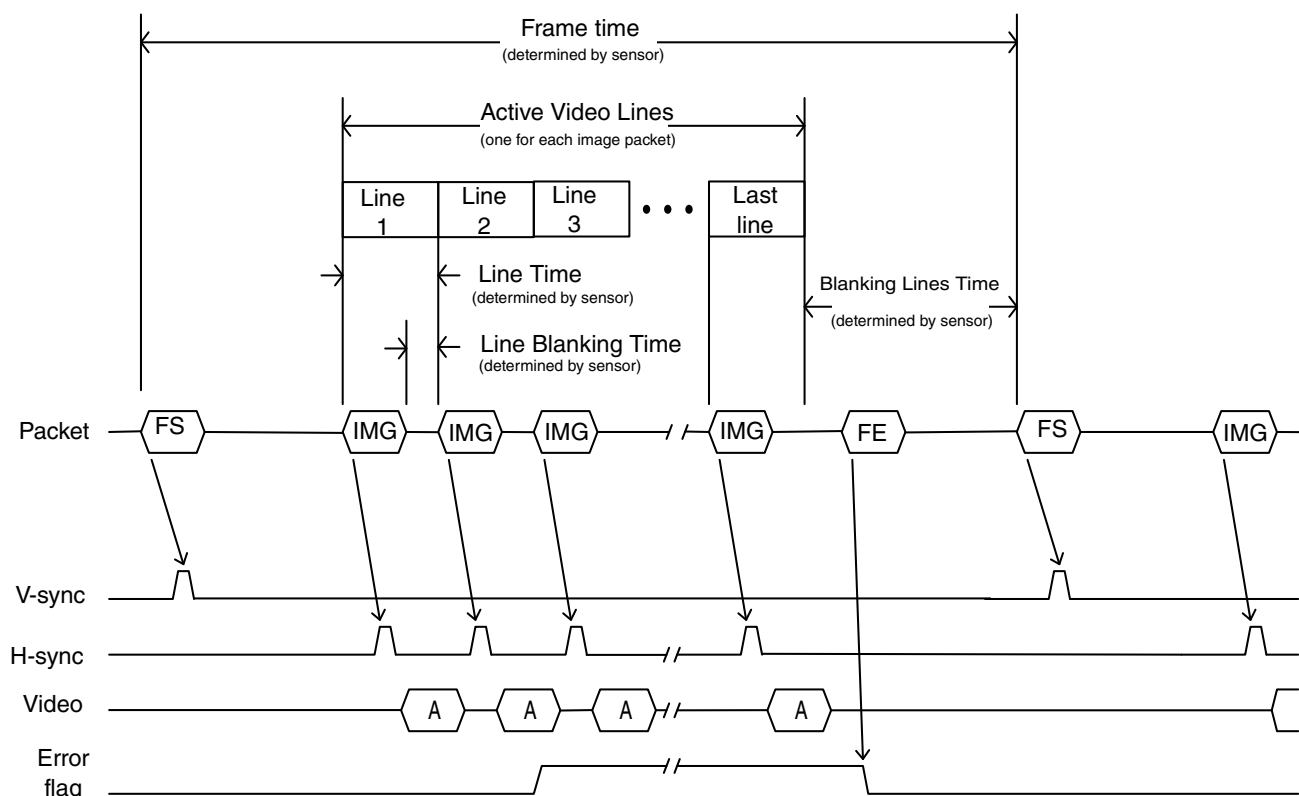


Figure 13-78. Relationship of Packet Stream to Video Interface

NOTE

- FS - Frame start Packet (causes V-sync)
- IMG - Image Data Packet (causes H-sync and Video after FIFO reaches send level)
- FE - Frame End Packet (causes any pending Error Flags which were set in the previous frame)
- A - Time period when video samples are valid (follows H-sync)

The video sample output bus (vid_out) is logically divided up into pixel lanes. The number of pixel lanes is determined by the pixel width of the CSI-2 Controller Core. This width may be one, two or four pixels wide (single, double or quad pixel interface). The bit width of each pixel lane is determined the widest pixel width supported by the core controller's Pixel Format block. When transferring image data whose pixels are narrower than the maximum required width unused more significant bits in each pixel lane will be output as zero.

To determine the validity of the data presented on each pixel lane the CSI-2 Rx Video Interface supplies a data enable output (vid_de) whose width is equal to the number of pixel lanes in the interface. When a data enable bit is high its pixel lane is valid.

Thus as a design example:

1. Controller supports a double pixel interface and data types RAW8, RAW10 and RAW12. In this case the maximum pixel width is 12 bits.
2. For this interface the total width of the video output bus is 24 bits ($2 * 12$) This bus is logically divided into two pixel lanes each supplying 12 bits. That sample which arrived earlier is delivered on the low half (bits[11:0]) of this bus while the following sample is delivered on the high half (bits[23:12]).
3. The vid_de bus would be 2 bits with the least significant bit enabling the low pixel lane and the most significant bit enabling the high pixel lane

The situation is similar for a single pixel controller interface. In this case there is only a single pixel lane in the video output bus and a single video enable bit. For a four pixel controller interface there would be four pixel lanes and a four bit vid_de bus.

The only time at which all vid_de bits will not be asserted is on the final transfer of video pixels in a scan line. At this time only some of the lower order bits in the vid_de bus will be asserted only if the number of samples per line is not evenly divisible by the number of pixel lanes.

Thus as a design example:

1. Assume a double pixel interface which results in a two video data lanes. Image data packets are 1023 samples long.
2. This would result in 511 transfers of video out with both bits of vid_de set (1022 samples). The last transfer would only have the least significant vid_de bits set to transfer the final sample to achieve the 1023 samples in the scan line.

13.8.3.7.2 Video Interface Video Port Descriptions

The Video output interface outputs all the signals associated with video port. All output signals in the interface are synchronous to clk_vid.

PORT	TYPE	DESCRIPTION
clk_vid	Input	Clock for all signals associated with the video port
clk_vid_reset_n	Input	Asynchronous active low reset for all logic clocked by clk_vid
vid_intfc_enb	Input	Active high enable for the video interface.
vid_h_sync	Output	Video H Sync pulse
vid_v_sync	Output	Video V Sync pulse
vid_out [VID_PIXEL_WIDTH*RX_PIXELS_PER_CLK-1:0]	Output	Video data bus. Contains 1, 2 or 4 pixel lanes as set by the CSI-2 Rx Controller port configuration (single, double or quad pixel interface). The width of each pixel lane is set by the widest pixel with of all the data types enabled in the controller. See CSI-2 Rx Controller Core Data Types Formatting <u>SinglePixelInterface:</u> Pixel lane 0= vid_out[VID_PIXEL_WIDTH-1:0] <u>Dual Pixel Interface:</u> Pixel lane 0= vid_out[VID_PIXEL_WIDTH-1:0] Pixel lane 1= vid_out[(2*VID_PIXEL_WIDTH)-1:VID_PIXEL_WIDTH] <u>Quad Pixel Interface:</u> Pixel lane 0= vid_out[VID_PIXEL_WIDTH-1:0] Pixel lane 1= vid_out[(2*VID_PIXEL_WIDTH)-1:VID_PIXEL_WIDTH] Pixel lane 2 = vid_out[(3*VID_PIXEL_WIDTH)-1:2*VID_PIXEL_WIDTH] Pixel lane 3 = vid_out[(4*VID_PIXEL_WIDTH)-1:3*VID_PIXEL_WIDTH]
vid_de [0]	Output	Video data pixel lane valid. One bit per Pixel lane. Logic 1 indicates a valid pixel value. <u>SinglePixelInterface:</u> Bit 0: Pixel lane 0 <u>Dual Pixel Interface:</u> Bit 0 : Pixel lane 0 Bit 1: Pixel lane 1 <u>Dual Pixel Interface:</u> Bit 0 : Pixel lane 0 Bit 1: Pixel lane 1 Bit 2: Pixel lane 2 Bit 3: Pixel lane 3

13.8.3.7.3 Video Output Port Waveforms

The vid_de port is used to enable capture of valid video pixels from the vid_out bus. The vid_de will have a single bit for each pixel lane in the vid_out bus. Thus for a single pixel interface (one pixel lane) this bus will be a single signal.

The vid_de port will assert all its signals high for all transfers except possibly the last transfer. This last transfer may only assert some number of lower order bits to one and the remaining higher order bits to zero. The number of valid pixel lanes in the last transfer is determined by the number of pixels transferred by the image packet which generated the video line.

The ending value may be predicted by taking number of pixels transferred in an image packet modulo number of pixel lanes. Any non-zero remainder indicates some number of vid_de bits will be zero on the last transfer.

The following table shows the last vid_de value as a function of number of pixel lanes and packet size:

MODULO	
	Single Pixel Interface
0	1
1	NA
2	NA
3	NA

NOTE

MODULO = Number_of_pixels_in_image_packet MOD
number_of_pixel_lanes

13.8.3.7.4 Video Interface Error Port Descriptions

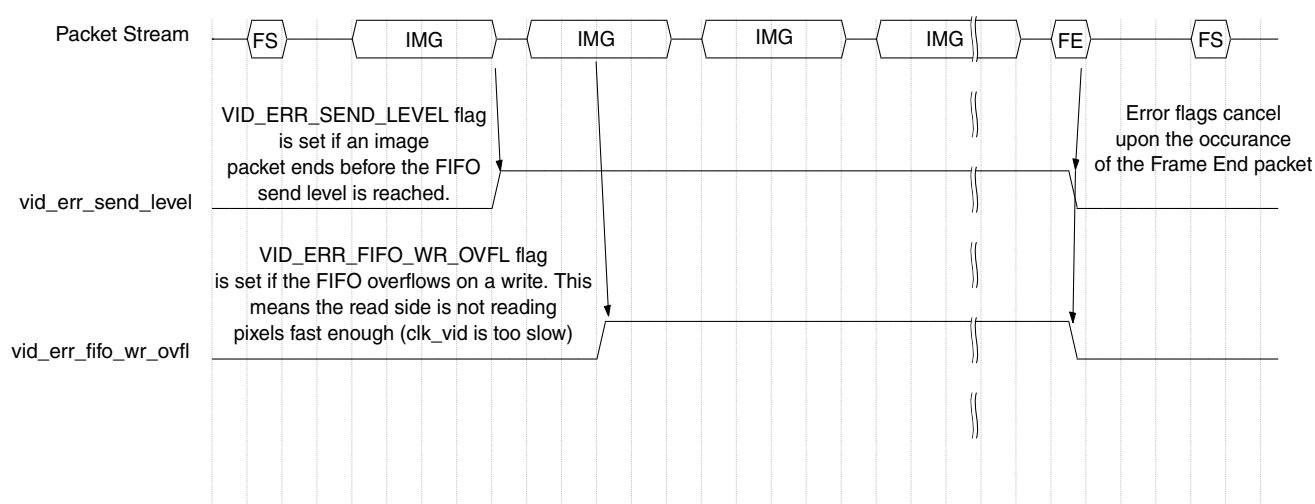
The video interface outputs three error flags which can assist in hardware debugging or be used to notify software of problems. Whenever one of these error conditions occurs the video interface stops responding to input packets until such time a Frame End packet is detected. At this time the error flags are reset and the interface tries again on the next frame. Assertion of these error conditions can mean a misconfigured interface or bad input packets.

All these flags are synchronous to the controller clk_ui. The interface must be enabled (vid_intf_enb = 1) for these errors to be detected.

Table 13-33. CSI-2 Rx Video Interface Error Port Descriptions

Port	Type	Description
vid_err_send_level	Output	This flag indicates the video interface pixel fifo did not accumulate enough sample to trigger its send level setting before the end of an image data packet was detected. This flag would generally mean that the cfg_vid_p_fifo_send_level is set to high.
vid_err_fifo_wr_ovfl	Output	This flag indicates the video interface pixel fifo overflowed on a pixel write. This means the output video port is not removing pixels quickly enough to keep up with the input data rate. This flag would generally mean the clk_vid rate is too low.

The figure below shows the error flag interface waveforms:

**Figure 13-79. Error Flag Interface Waveforms**

13.8.3.8 Rx Control and Status Register (CSR) APB Interface

The CSR provides an APB compatible interface that enables control of the controller's configuration inputs via registers accessible via the APB interface.

13.8.4 Memory Map and register definition

This section includes the memory map and detailed descriptions of all registers.

13.8.4.1 register descriptions

13.8.4.1.1 CSI2RX memory map

csi2_rx_csr_1 base address: 30A7_0000h

csi2_rx_csr_2 base address: 30B6_0000h

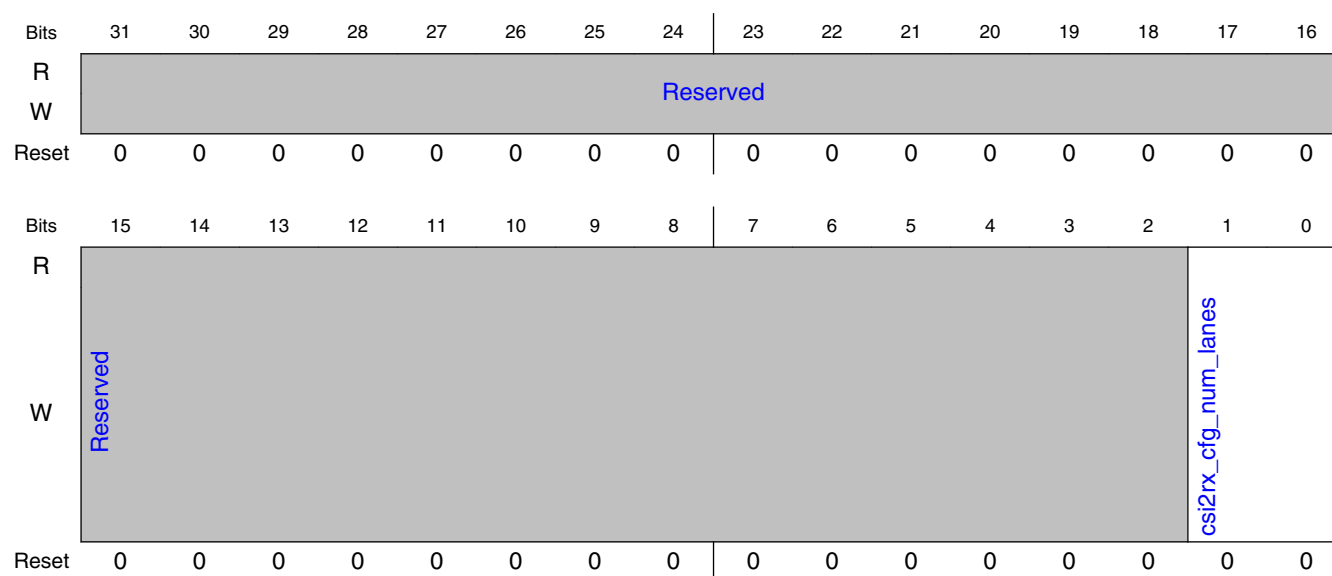
Offset	Register	Width (In bits)	Access	Reset value
100h	Lane Configuration Register (CSI2RX_CFG_NUM_LANES)	32	RW	0000_0000h
104h	Disable Data Lane Register (CSI2RX_CFG_DISABLE_DATA_LANE_S)	32	RW	0000_000Fh
108h	ECC and CRC Error Status Register (CSI2RX_BIT_ERR)	32	RO	0000_0000h
10Ch	IRQ Status Register (CSI2RX_IRQ_STATUS)	32	RO	0000_0000h
110h	IRQ Mask Setting Register (CSI2RX_IRQ_MASK)	32	RW	0000_0000h
114h	ULPS Status Register (CSI2RX_ULPS_STATUS)	32	RO	0000_0000h
118h	ERRSot HS Status Register (CSI2RX_PPI_ERRSOT_HS)	32	RO	0000_0000h
11Ch	ErrSotSync HS Status Register (CSI2RX_PPI_ERRSOTSYNC_HS)	32	RO	0000_0000h
120h	ErrEsc Status Register (CSI2RX_PPI_ERRESC)	32	RO	0000_0000h
124h	ErrSyncEsc Status Register (CSI2RX_PPI_ERRSYNCESC)	32	RO	0000_0000h
128h	ErrControl Status Register (CSI2RX_PPI_ERRCONTROL)	32	RO	0000_0000h
12Ch	Disable Payload 0 Register (CSI2RX_CFG_DISABLE_PAYLOAD_0)	32	RW	0000_0000h
130h	Disable Payload 1 Register (CSI2RX_CFG_DISABLE_PAYLOAD_1)	32	RW	0000_0000h
188h	FIFO Send Level Configuration Register (CSI2RX_CFG_VID_P_FIFO_SEND_LEVEL)	32	RW	0000_0000h
18Ch	VSYNC Configuration Register (CSI2RX_CFG_VID_VSYNC)	32	RW	0000_0000h
190h	Start of HSYNC Delay control Register (CSI2RX_CFG_VID_HSYNC_FP)	32	RW	0000_0000h
194h	HSYNC Configuration Register (CSI2RX_CFG_VID_HSYNC)	32	RW	0000_0000h
198h	End of HSYNC Delay Control Register (CSI2RX_CFG_VID_HSYNC_BP)	32	RW	0000_0000h

13.8.4.1.2 Lane Configuration Register (CSI2RX_CFG_NUM_LANES)

13.8.4.1.2.1 Offset

Register	Offset
CSI2RX_CFG_NUM_LANES	100h

13.8.4.1.2.2 Diagram



13.8.4.1.2.3 Fields

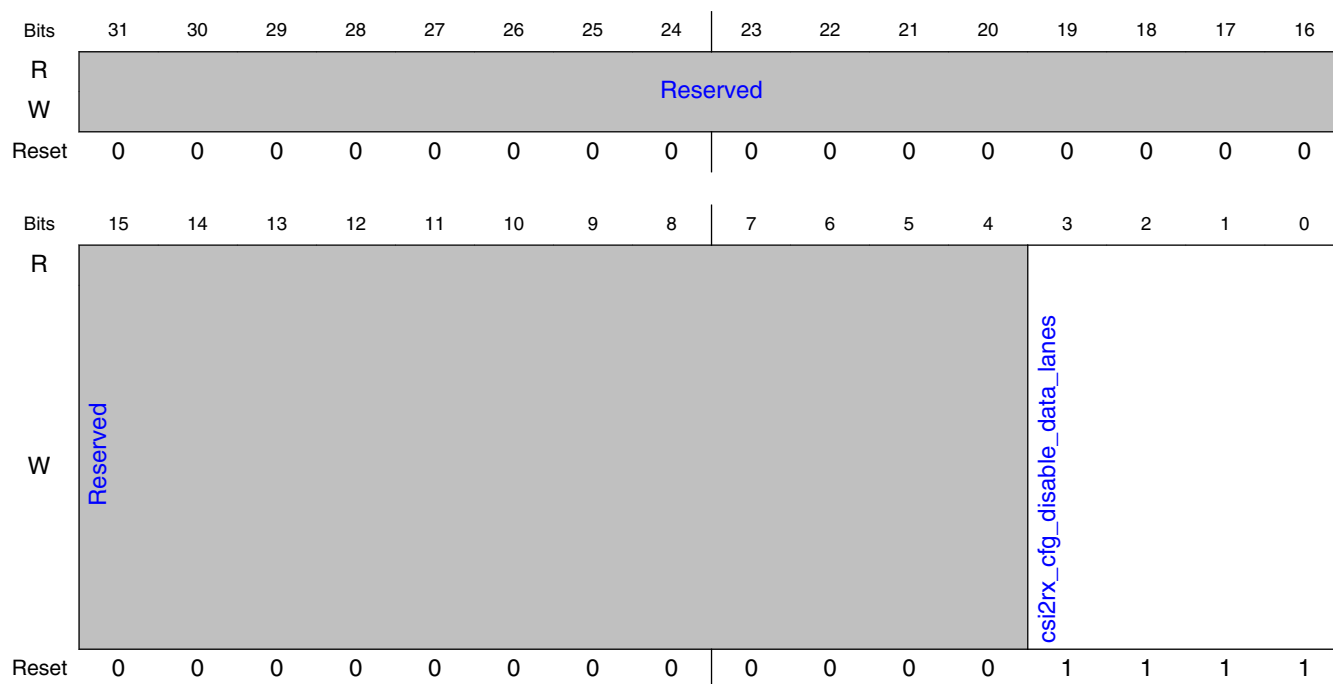
Field	Function
31-2 —	Reserved
1-0 csi2rx_cfg_num_lanes	Sets the number of active lanes that are to be used for receiving data. 00b - 1 Lane 01b - 2 Lane 10b - 3 Lane 11b - 4 Lane

13.8.4.1.3 Disable Data Lane Register (CSI2RX_CFG_DISABLE_DATA_LANES)

13.8.4.1.3.1 Offset

Register	Offset
CSI2RX_CFG_DISAB LE_DATA_LANES	104h

13.8.4.1.3.2 Diagram



13.8.4.1.3.3 Fields

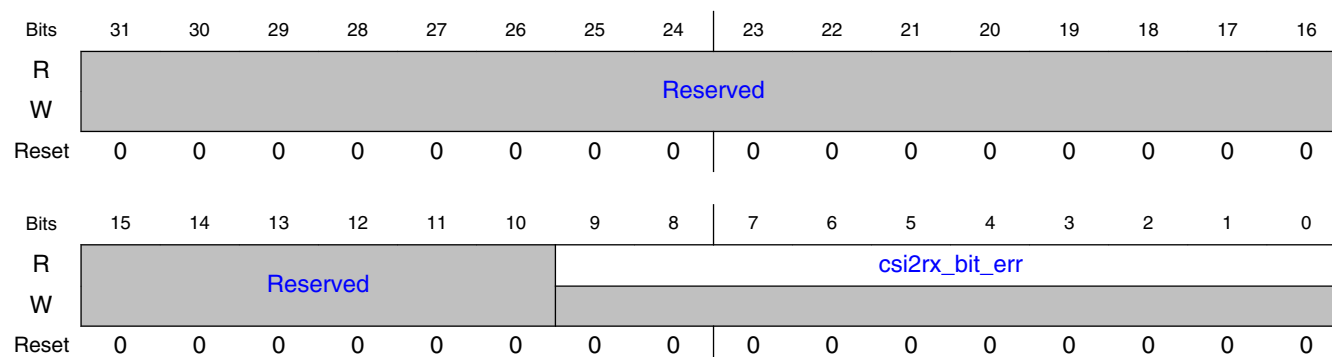
Field	Function
31-4 —	Reserved
3-0 csi2rx_cfg_disab le_data_lanes	Setting bits to a '1' value causes the DPHY Enable signal to deassert. 0001b - Data Lane 0 0010b - Data Lane 1 0100b - Data Lane 2 1000b - Data Lane 3

13.8.4.1.4 ECC and CRC Error Status Register (CSI2RX_BIT_ERR)

13.8.4.1.4.1 Offset

Register	Offset
CSI2RX_BIT_ERR	108h

13.8.4.1.4.2 Diagram



13.8.4.1.4.3 Fields

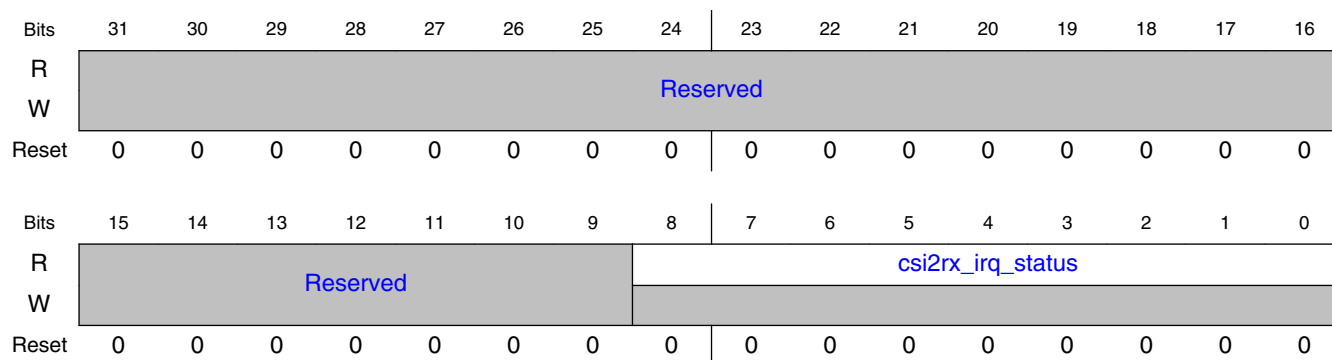
Field	Function
31-10 —	Reserved
9-0 csi2rx_bit_err	BIT_ERR: CSI-2 RX Controller ECC and CRC error status. Bit [0] – value of '1' indicates ECC two bit error has occurred since the last read of this register Bit [1] – value of '1' indicates ECC one bit error has occurred since the last read of this register Bit [6:2] – ECC one bit error bit position that occurred since the last read of this register Bit [7] – value of '1' indicates a CRC error has occurred since the last read of this register Bit [8] – value of '1' indicates that rx video interface signal vid_err_send_level has asserted since the last read of this register. Register reads 0 when rx video interface included. Bit [9] – value of '1' indicates that rx video interface signal vid_err_fifo_wr_ovfl has asserted since the last read of this register. Register reads 0 when rx video interface included.

13.8.4.1.5 IRQ Status Register (CSI2RX_IRQ_STATUS)

13.8.4.1.5.1 Offset

Register	Offset
CSI2RX_IRQ_STATUS	10Ch

13.8.4.1.5.2 Diagram



13.8.4.1.5.3 Fields

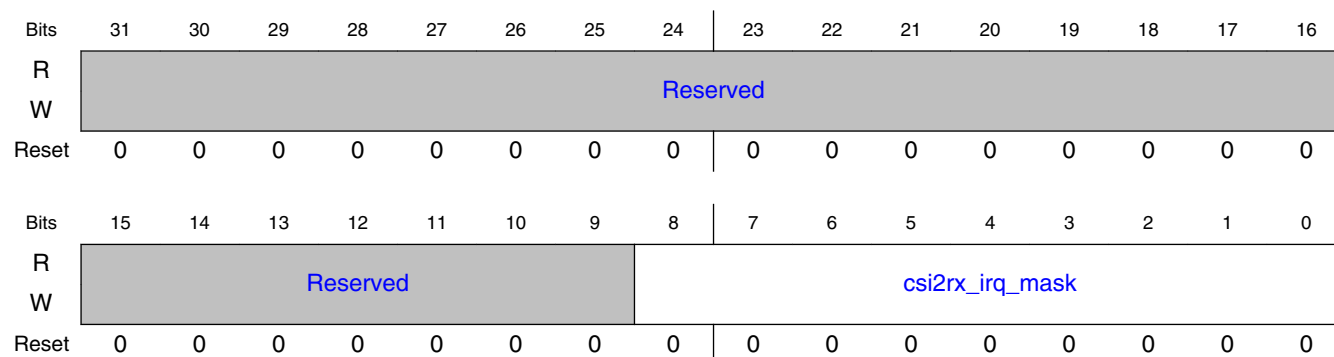
Field	Function
31-9 —	Reserved
8-0 csi2rx_irq_status	CSI2 RX IRQ status [0] – crc error [1] – one bit ecc error [2] – two bit ecc error [3] – ULPS status change [4] – DPHY ErrSotHS has occurred [5] – DPHY ErrSotSync_HS has occurred [6] – DPHY ErrEsc has occurred [7] – DPHY ErrSyncEsc has occurred [8] – DPHY ErrControl has occurred

13.8.4.1.6 IRQ Mask Setting Register (CSI2RX_IRQ_MASK)

13.8.4.1.6.1 Offset

Register	Offset
CSI2RX_IRQ_MASK	110h

13.8.4.1.6.2 Diagram



13.8.4.1.6.3 Fields

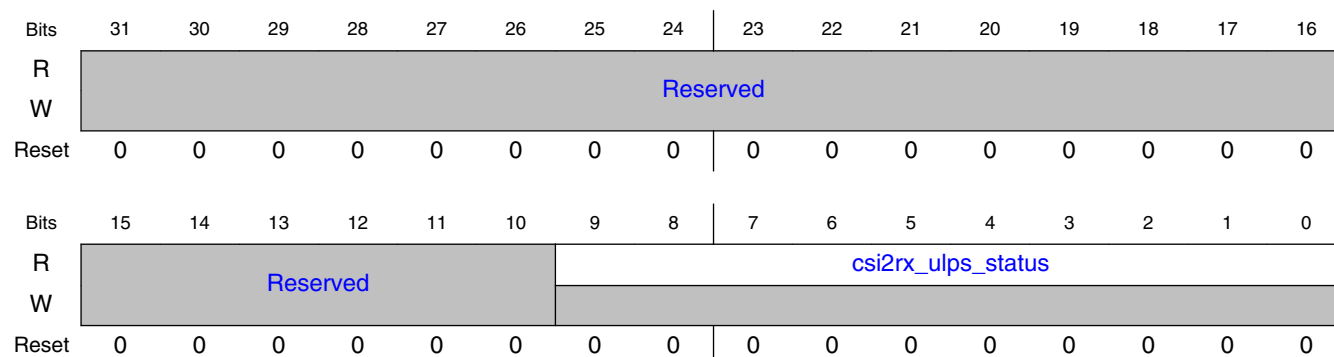
Field	Function
31-9 —	Reserved
8-0 csi2rx_irq_mask	CSI2 RX IRQ Mask setting Each bit in IRQ_MASK corresponds to each bit in IRQ_STATUS. Setting a bit in IRQ_MASK to 1 will mask the corresponding bit in IRQ_STATUS from causing irq_out to assert.

13.8.4.1.7 ULPS Status Register (CSI2RX_ULPS_STATUS)

13.8.4.1.7.1 Offset

Register	Offset
CSI2RX_ULPS_STATUS	114h

13.8.4.1.7.2 Diagram



13.8.4.1.7.3 Fields

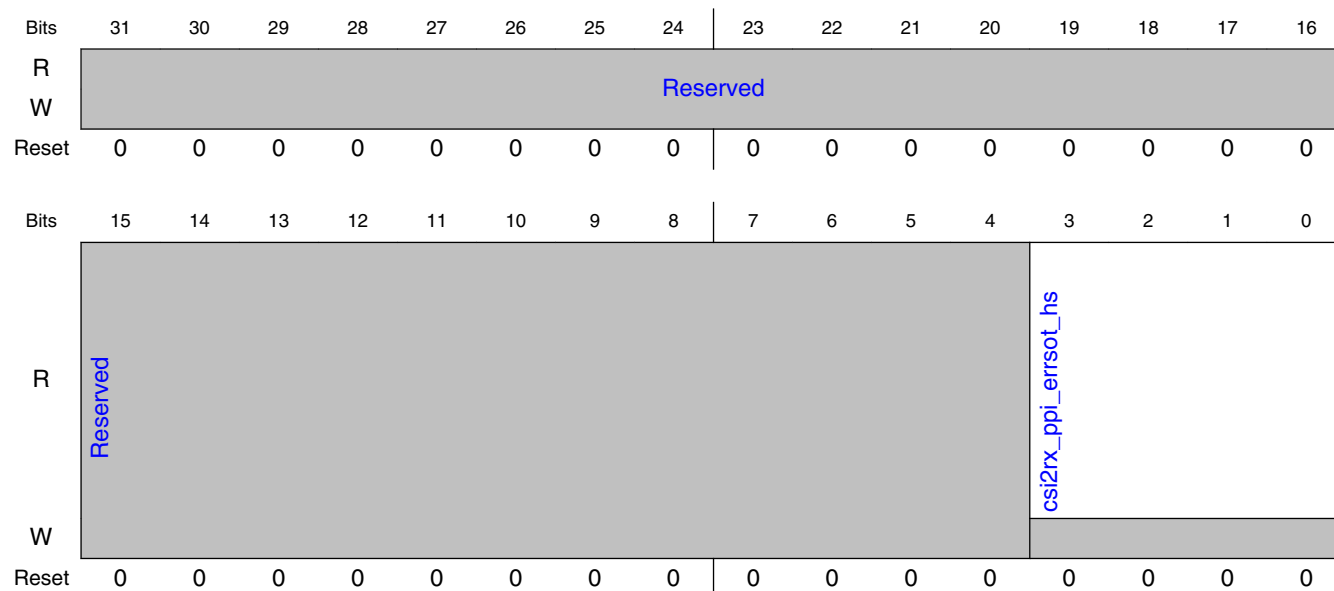
Field	Function
31-10 —	Reserved
9-0 csi2rx_ulps_status	Status of RX DPHY ULPS state [0] – Value '1' indicates Clock Lane in ULPS state [1] – Value '1' indicates Data Lane 0 in ULPS state [2] – Value '1' indicates Data Lane 1 in ULPS state [3] – Value '1' indicates Data Lane 2 in ULPS state [4] – Value '1' indicates Data Lane 3 in ULPS state [5] – Value '1' indicates Clock Lane in Mark state [6] – Value '1' indicates Data Lane 0 in Mark state [7] – Value '1' indicates Data Lane 1 in Mark state [8] – Value '1' indicates Data Lane 2 in Mark state [9] – Value '1' indicates Data Lane 3 in Mark state

13.8.4.1.8 ERRSot HS Status Register (CSI2RX_PPI_ERRSOT_HS)

13.8.4.1.8.1 Offset

Register	Offset
CSI2RX_PPI_ERRSOT_HS	118h

13.8.4.1.8.2 Diagram



13.8.4.1.8.3 Fields

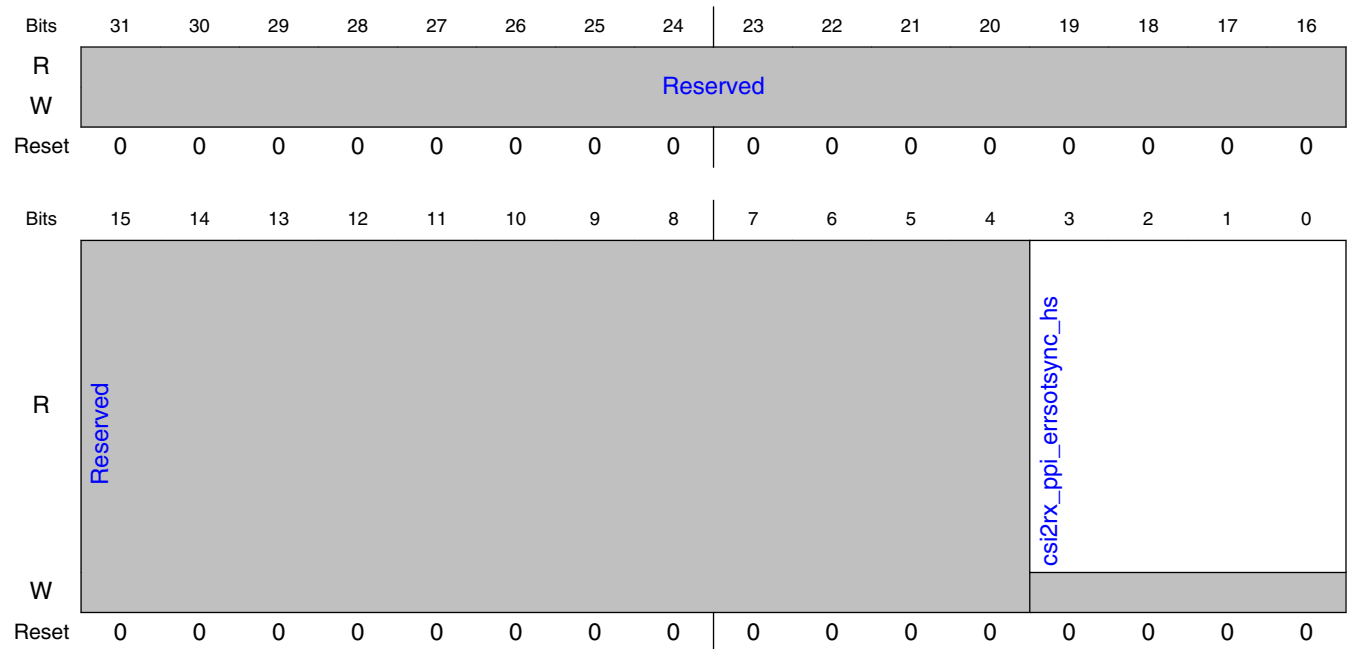
Field	Function
31-4 —	Reserved
3-0 csi2rx_ppi_errsot_hs	CSI2 RX DPHY PPI ErrSotHS captured status from the DPHY. [0] – data lane 0 [1] – data lane 1 [2] – data lane 2 [3] – data lane 3

13.8.4.1.9 ErrSotSync HS Status Register (CSI2RX_PPI_ERRSOTSYNC_HS)

13.8.4.1.9.1 Offset

Register	Offset
CSI2RX_PPI_ERRSO TSYNC_HS	11Ch

13.8.4.1.9.2 Diagram



13.8.4.1.9.3 Fields

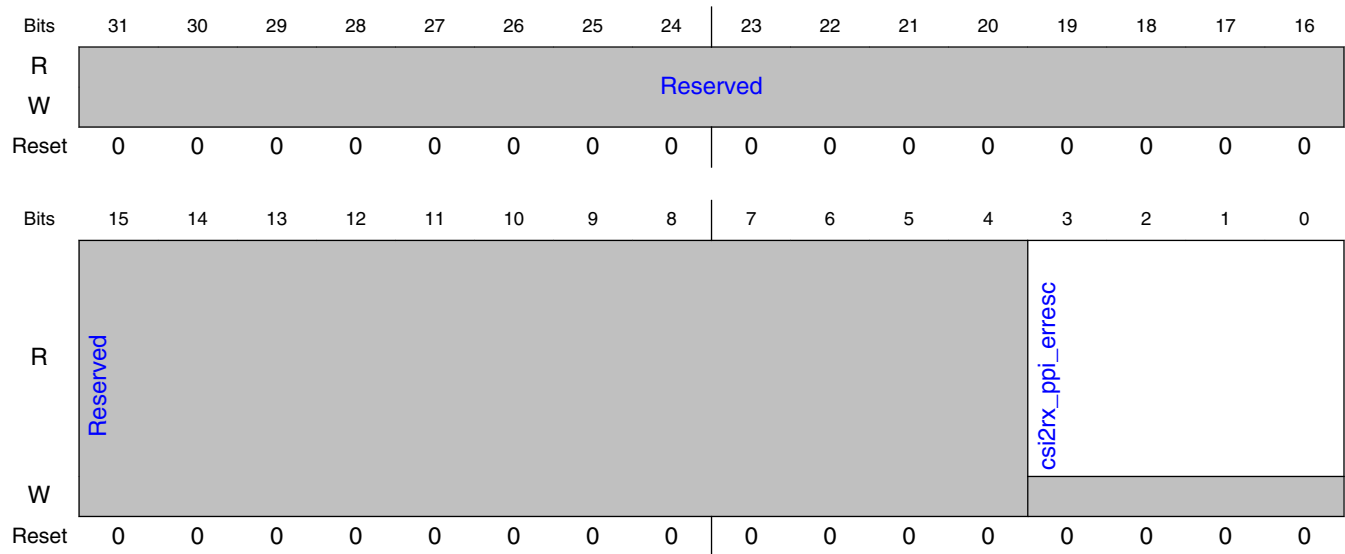
Field	Function
31-4 —	Reserved
3-0 csi2rx_ppi_errsotsync_hs	CSI2 RX DPHY PPI ErrSotSync_HS captured status from the DPHY. [0] – data lane 0 [1] – data lane 1 [2] – data lane 2 [3] – data lane 3

13.8.4.1.10 ErrEsc Status Register (CSI2RX_PPI_ERRESC)

13.8.4.1.10.1 Offset

Register	Offset
CSI2RX_PPI_ERRESC	120h

13.8.4.1.10.2 Diagram



13.8.4.1.10.3 Fields

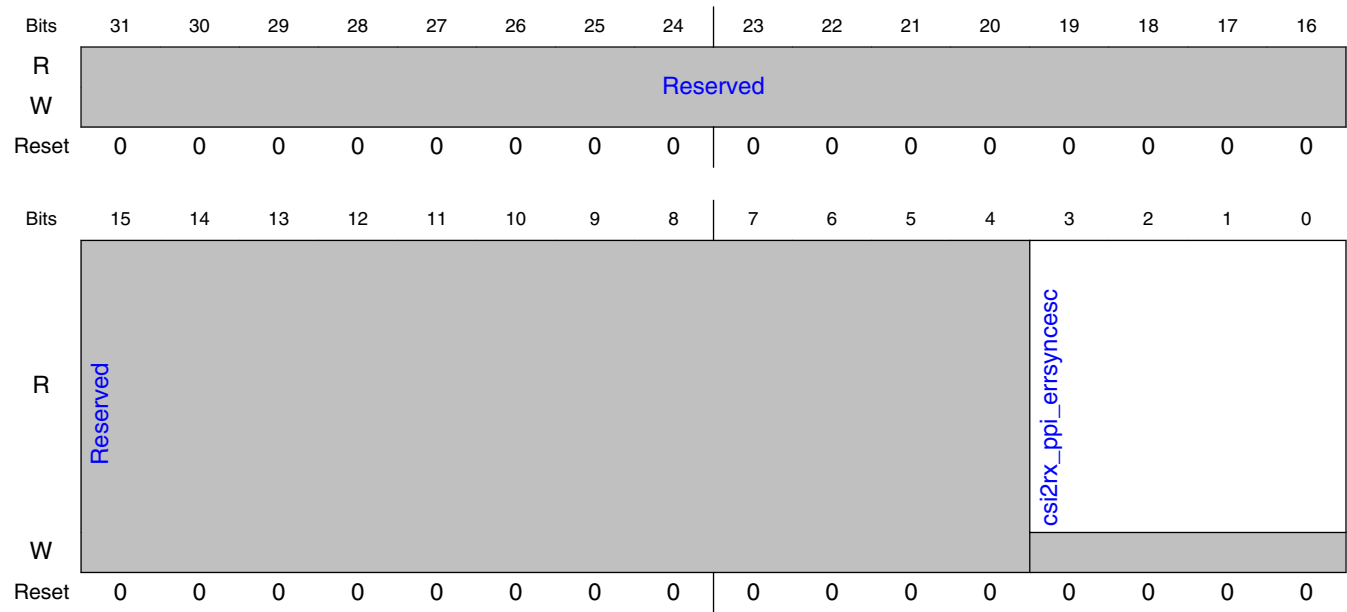
Field	Function
31-4 —	Reserved
3-0 csi2rx_ppi_erresc	CSI2 RX DPHY PPI ErrEsc captured status from the DPHY. [0] – data lane 0 [1] – data lane 1 [2] – data lane 2 [3] – data lane 3

13.8.4.1.11 ErrSyncEsc Status Register (CSI2RX_PPI_ERRSYNCESC)

13.8.4.1.11.1 Offset

Register	Offset
CSI2RX_PPI_ERRSYNCESC	124h

13.8.4.1.11.2 Diagram



13.8.4.1.11.3 Fields

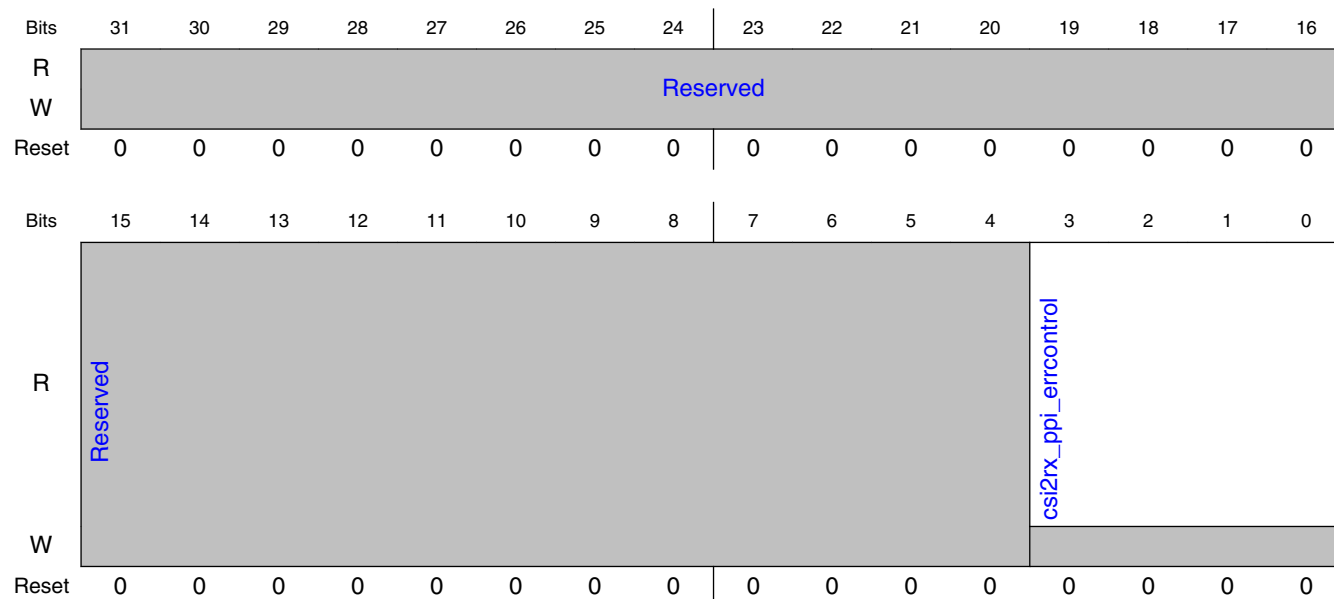
Field	Function
31-4	Reserved
—	
3-0	CSI2 RX DPHY PPI ErrSyncEsc captured status from the DPHY.
csi2rx_ppi_errsyncesc	[0] – data lane 0 [1] – data lane 1 [2] – data lane 2 [3] – data lane 3

13.8.4.1.12 ErrControl Status Register (CSI2RX_PPI_ERRCONTROL)

13.8.4.1.12.1 Offset

Register	Offset
CSI2RX_PPI_ERRCONTROL	128h

13.8.4.1.12.2 Diagram



13.8.4.1.12.3 Fields

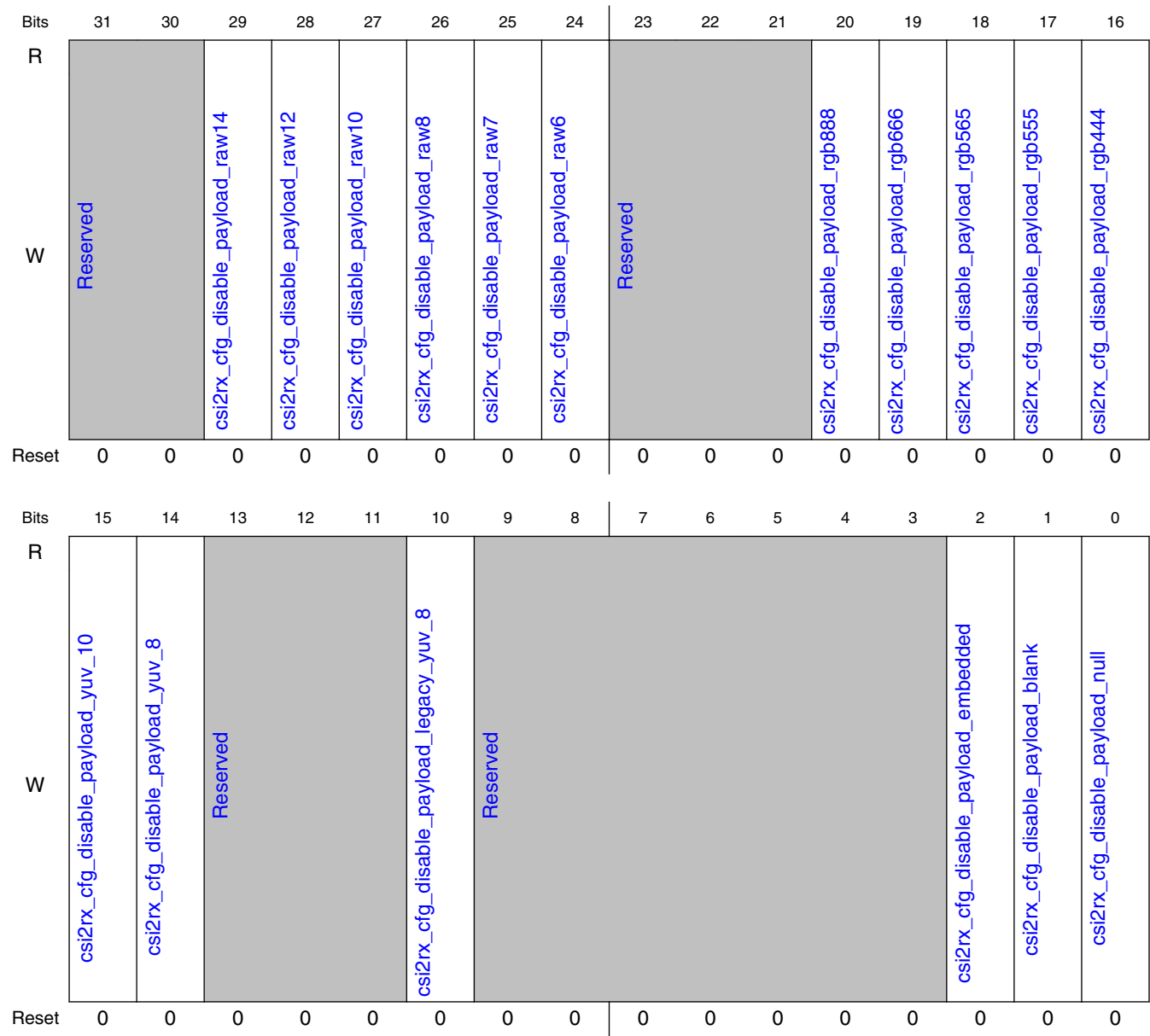
Field	Function
31-4 —	Reserved
3-0 csi2rx_ppi_errcontrol	CSI2 RX DPHY PPI ErrControl captured status from the DPHY. [0] – data lane 0 [1] – data lane 1 [2] – data lane 2 [3] – data lane 3

13.8.4.1.13 Disable Payload 0 Register (CSI2RX_CFG_DISABLE_PAYLOAD_0)

13.8.4.1.13.1 Offset

Register	Offset
CSI2RX_CFG_DISABLE_PAYLOAD_0	12Ch

13.8.4.1.13.2 Diagram



13.8.4.1.13.3 Fields

Field	Function
31-30 —	Reserved
29 csi2rx_cfg_disable_payload_raw14	RAW14

Table continues on the next page...

Field	Function
28 csi2rx_cfg_disable_payload_raw12	RAW12
27 csi2rx_cfg_disable_payload_raw10	RAW10
26 csi2rx_cfg_disable_payload_raw8	RAW8
25 csi2rx_cfg_disable_payload_raw7	RAW7
24 csi2rx_cfg_disable_payload_raw6	RAW6
23-21 —	Reserved
20 csi2rx_cfg_disable_payload_rgb888	RGB888
19 csi2rx_cfg_disable_payload_rgb666	RGB666
18 csi2rx_cfg_disable_payload_rgb565	RGB565
17 csi2rx_cfg_disable_payload_rgb555	RGB555
16 csi2rx_cfg_disable_payload_rgb444	RGB444
15 csi2rx_cfg_disable_payload_yuv10	YUV422 10 bit

Table continues on the next page...

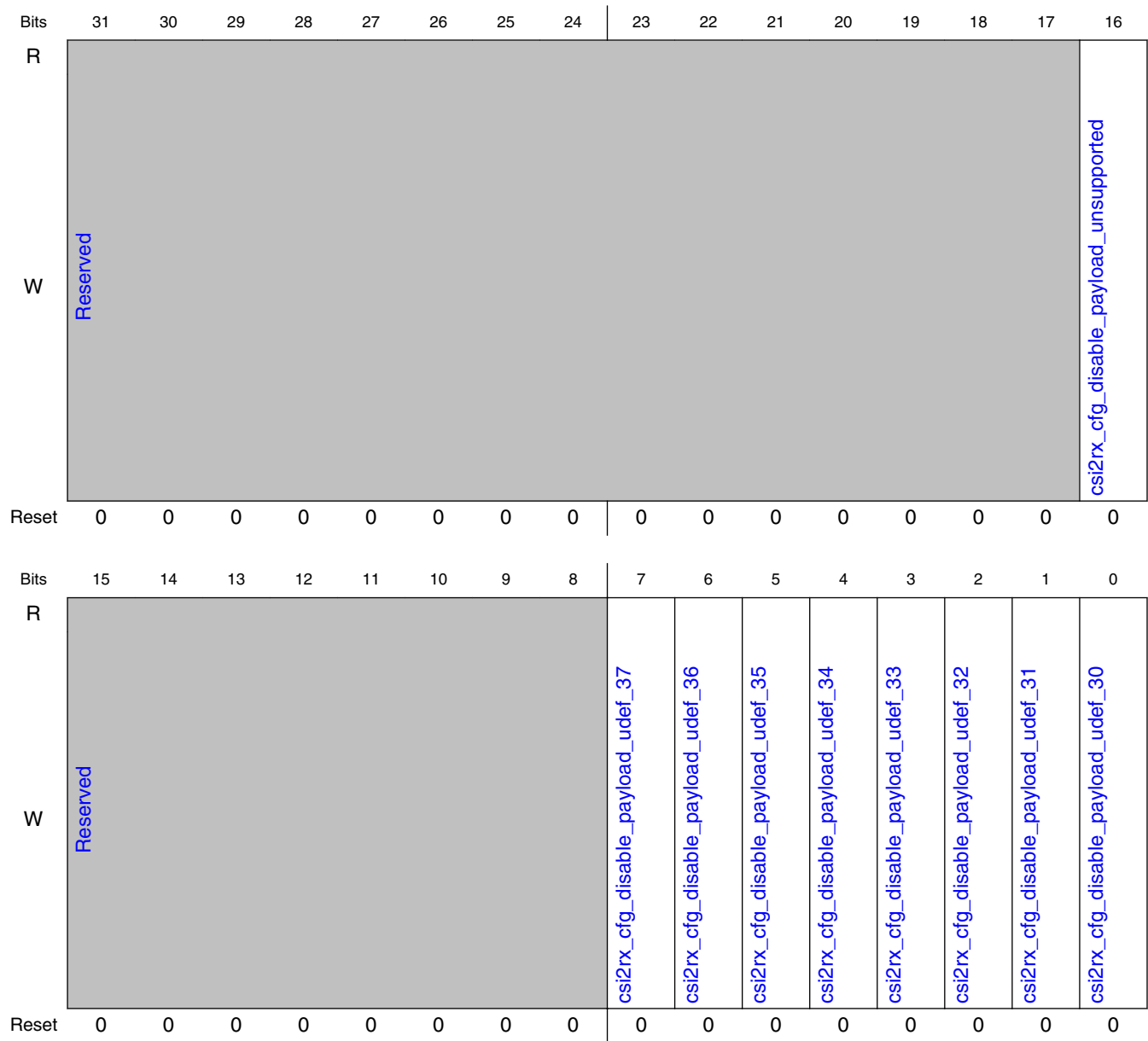
Field	Function
14 csi2rx_cfg_disable_payload_yuv_8	YUV422 8 bit
13-11 —	Reserved
10 csi2rx_cfg_disable_payload_legacy_yuv_8	Legacy YUV 420 8 bit
9-3 —	Reserved
2 csi2rx_cfg_disable_payload_embedded	Embedded
1 csi2rx_cfg_disable_payload_blank	Blank
0 csi2rx_cfg_disable_payload_null	Null

13.8.4.1.14 Disable Payload 1 Register (CSI2RX_CFG_DISABLE_PAYLOAD_1)

13.8.4.1.14.1 Offset

Register	Offset
CSI2RX_CFG_DISABLE_PAYLOAD_1	130h

13.8.4.1.14.2 Diagram



13.8.4.1.14.3 Fields

Field	Function
31-17	Reserved
—	
16	Unsupported Data Types
csi2rx_cfg_disab le_payload_uns upported	

Table continues on the next page...

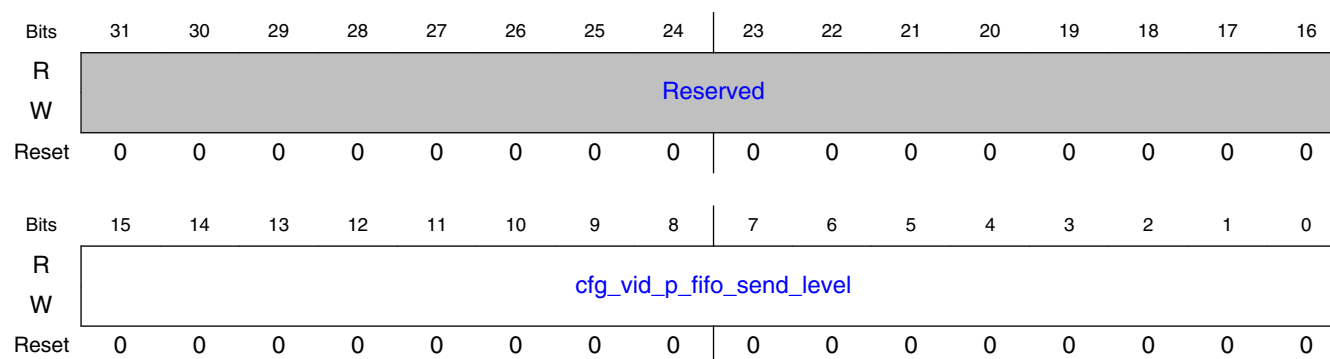
Field	Function
15-8 —	Reserved
7 csi2rx_cfg_disable_payload_undef_37	User defined type 0x37
6 csi2rx_cfg_disable_payload_undef_36	User defined type 0x36
5 csi2rx_cfg_disable_payload_undef_35	User defined type 0x35
4 csi2rx_cfg_disable_payload_undef_34	User defined type 0x35
3 csi2rx_cfg_disable_payload_undef_33	User defined type 0x34
2 csi2rx_cfg_disable_payload_undef_32	User defined type 0x33
1 csi2rx_cfg_disable_payload_undef_31	User defined type 0x32
0 csi2rx_cfg_disable_payload_undef_30	User defined type 0x31

13.8.4.1.15 FIFO Send Level Configuration Register (CSI2RX_CFG_VID_P_FIFO_SEND_LEVEL)

13.8.4.1.15.1 Offset

Register	Offset
CSI2RX_CFG_VID_P_FIFO_SEND_LEVEL	188h

13.8.4.1.15.2 Diagram



13.8.4.1.15.3 Fields

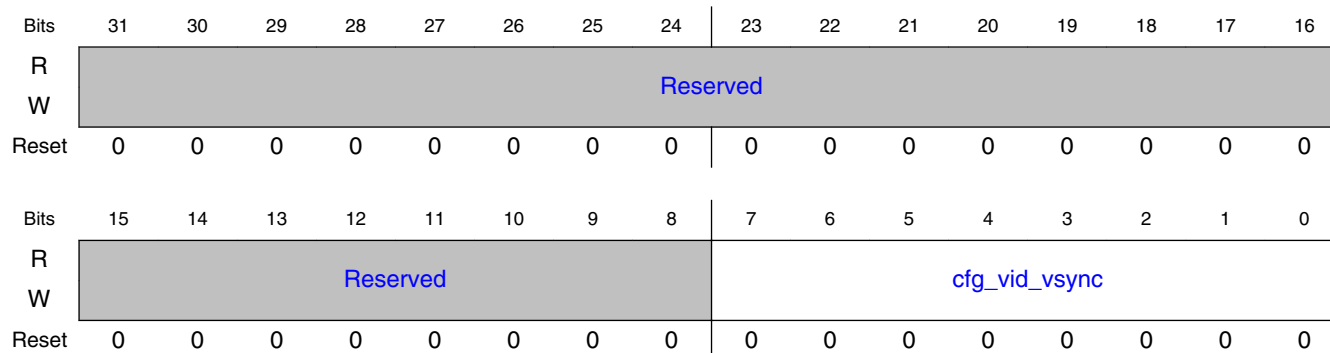
Field	Function
31-16 —	Reserved
15-0 cfg_vid_p_fifo_send_level	This input configures the number of entries that must accumulate in the Pixel FIFO before the data will be transferred to the video output. The exact value needed for this configuration is dependent on the rate at which the sensor transfers the data to the RX controller and the user video clock.

13.8.4.1.16 VSYNC Configuration Register (CSI2RX_CFG_VID_VSYNC)

13.8.4.1.16.1 Offset

Register	Offset
CSI2RX_CFG_VID_VSYNC	18Ch

13.8.4.1.16.2 Diagram



13.8.4.1.16.3 Fields

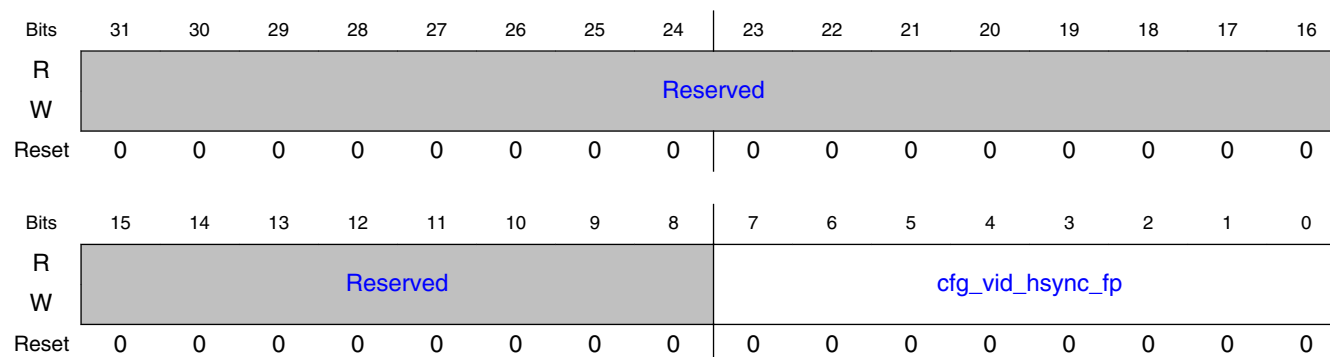
Field	Function
31-8 —	Reserved
7-0 cfg_vid_vsync	Width of VSYNC This input controls the width of the interface VSYNC pulse in video clocks. Pulse width = N+1 video clocks where N is the value of cfg_vid_vsync.

13.8.4.1.17 Start of HSYNC Delay control Register (CSI2RX_CFG_VID_H SYNC_FP)

13.8.4.1.17.1 Offset

Register	Offset
CSI2RX_CFG_VID_H SYNC_FP	190h

13.8.4.1.17.2 Diagram



13.8.4.1.17.3 Fields

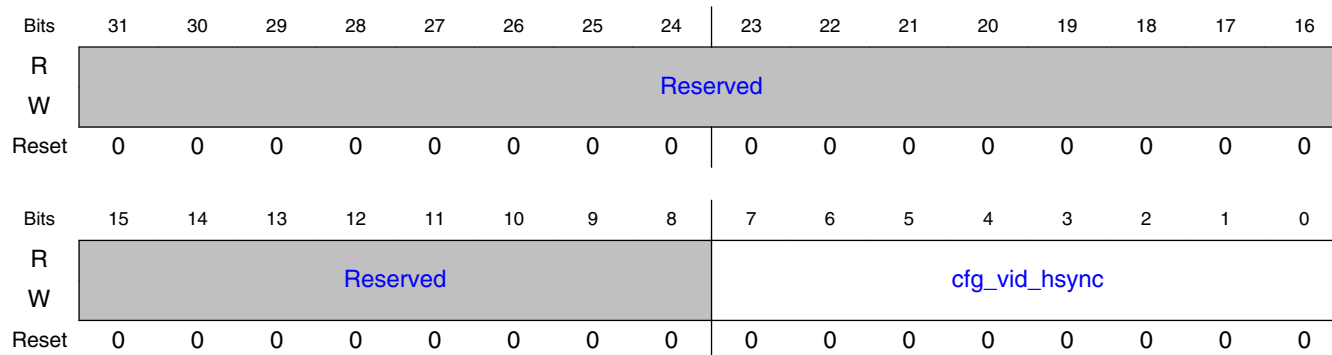
Field	Function
31-8 —	Reserved
7-0 cfg_vid_hsync_fp	Delay control for beginning of HSYNC pulse This input controls an arbitrary delay from when the fifo send level calls for a line of samples to be output and beginning of the HSYNC associated with the line. Pulse width = N+7 video clocks where N is the value of cfg_vid_hsync_fp.

13.8.4.1.18 HSYNC Configuration Register (CSI2RX_CFG_VID_HSYNC)

13.8.4.1.18.1 Offset

Register	Offset
CSI2RX_CFG_VID_HSYNC	194h

13.8.4.1.18.2 Diagram



13.8.4.1.18.3 Fields

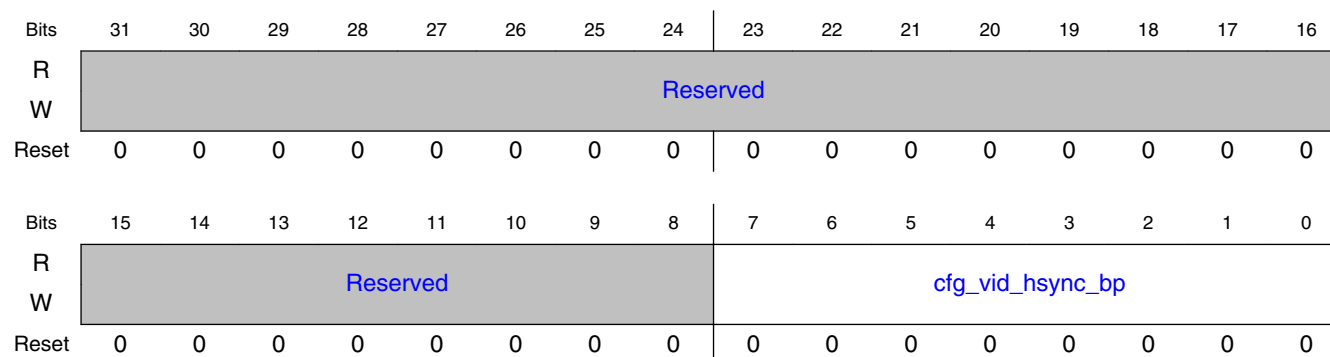
Field	Function
31-8 —	Reserved
7-0 cfg_vid_hsync	Width of HSYNC This input controls the width of the interface HSYNC pulse in video clocks. Pulse width = N+1 video clocks where N is the value of cfg_vid_hsync.

13.8.4.1.19 End of HSYNC Delay Control Register (CSI2RX_CFG_VID_H SYNC_BP)

13.8.4.1.19.1 Offset

Register	Offset
CSI2RX_CFG_VID_H SYNC_BP	198h

13.8.4.1.19.2 Diagram



13.8.4.1.19.3 Fields

Field	Function
31-8 —	Reserved
7-0 cfg_vid_hsync_bp	Delay Control for end of HSYNC pulse This input controls an arbitrary delay from the end of the H sync pulse to the assertion of the first data enable(EN). Pulse width = N+5 video clocks where N is the value of cfg_vid_hsync_bp.

13.9 Sony/Philips Digital Interface (SPDIF)

13.9.1 Overview

The Sony/Philips Digital Interface (SPDIF) audio block is a stereo transceiver that allows the processor to receive and transmit digital audio.

The SPDIF transceiver allows the handling of both SPDIF channel status (CS) and User (U) data and includes a frequency measurement block that allows the precise measurement of an incoming sampling frequency.

A recovered clock is provided to drive both internal components in the system, such as SAI ports, and external components, such as A/Ds or D/As, with clocking control provided via related registers.

As the SPDIF internal data width is 24-bit, the eight most-significant bits of all registers return zeros.

The figure below shows a block diagram of the SPDIF transceiver data paths (receiver and transmitter) and its interface.

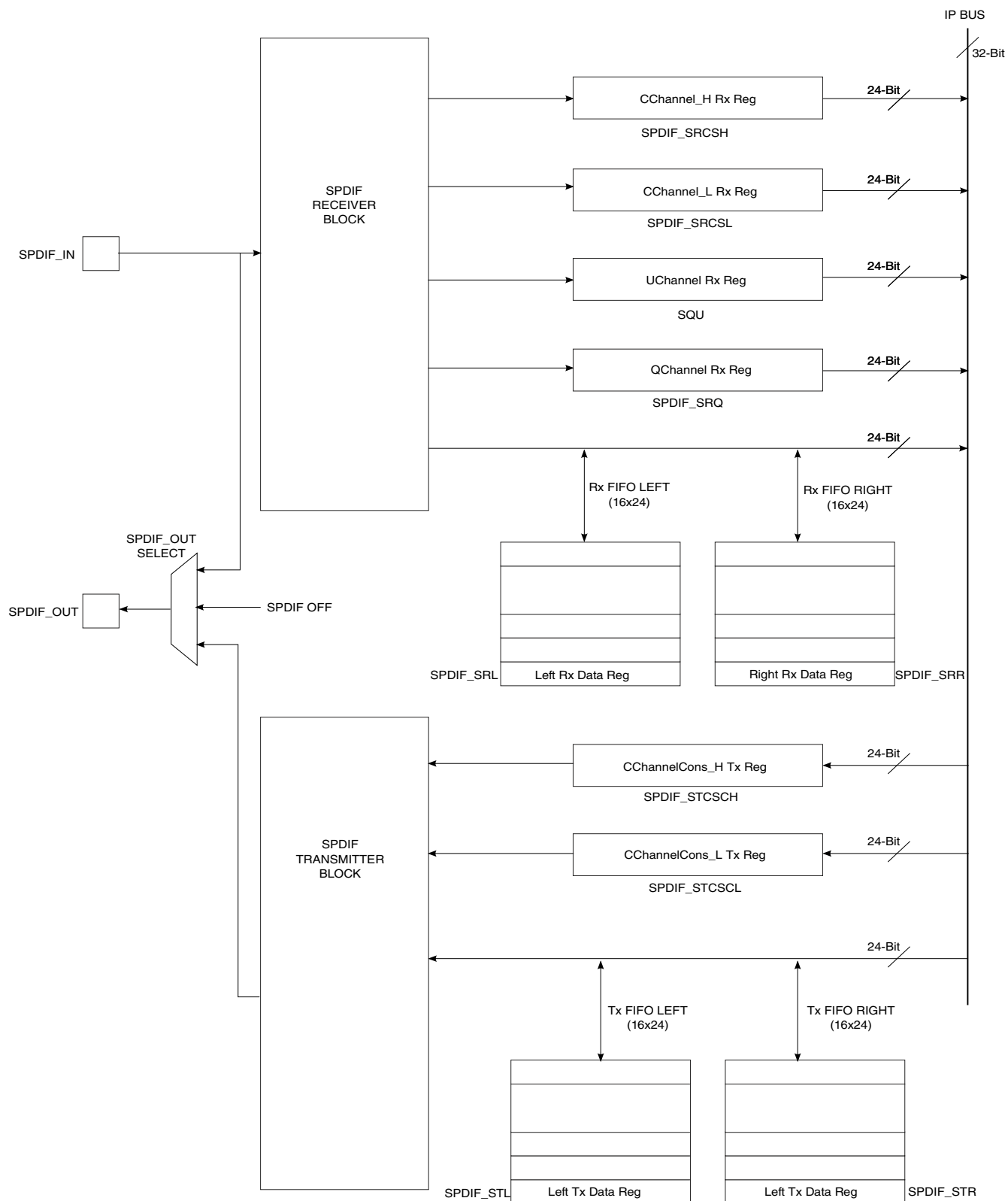


Figure 13-80. SPDIF Transceiver Data Interface Block Diagram

13.9.2 Clocks

The table found here describes the clock sources for SPDIF.

Please see clock control block for clock setting, configuration and gating information.

Table 13-34. SPDIF Clocks

Clock name	Clock Root	Description
gclkw_t0	ipg_clk_root	Global clock
ipg_clk_s	ipg_clk_root	Peripheral access clock
tx_clk	spdif0_clk_root	Module Tx clock

13.9.3 Functional Description

The SPDIF is composed of two parts: SPDIF Receiver and SPDIF Transmitter.

The SPDIF receiver extracts the audio data from each SPDIF frame and places the data in the SPDIF Rx left and right FIFOs. The Channel Status and User Bits are also extracted from each frame and placed in the corresponding registers. The SPDIF receiver also provides a bypass option for direct transfer of the SPDIF input signal to the SPDIF transmitter.

For the SPDIF transmitter, the audio data is provided by the processor via the SPDIFTxLeft and SPDIFTxRight registers. The Channel Status bits are also provided via the corresponding registers. The SPDIF transmitter generates a SPDIF output bitstream in the biphase mark format (IEC60958), which consists of audio data, channel status and user bits.

In the SPDIF transmitter, the IEC60958 biphase bit stream is generated on both edges of the SPDIF Transmit clock. The SPDIF Transmit clock is generated by the SPDIF internal clock generate block and the sources are from outside of the SPDIF block. For the SPDIF receiver, it can recover the SPDIF Rx clock. [Figure 1](#) shows the clock structure of the SPDIF transceiver.

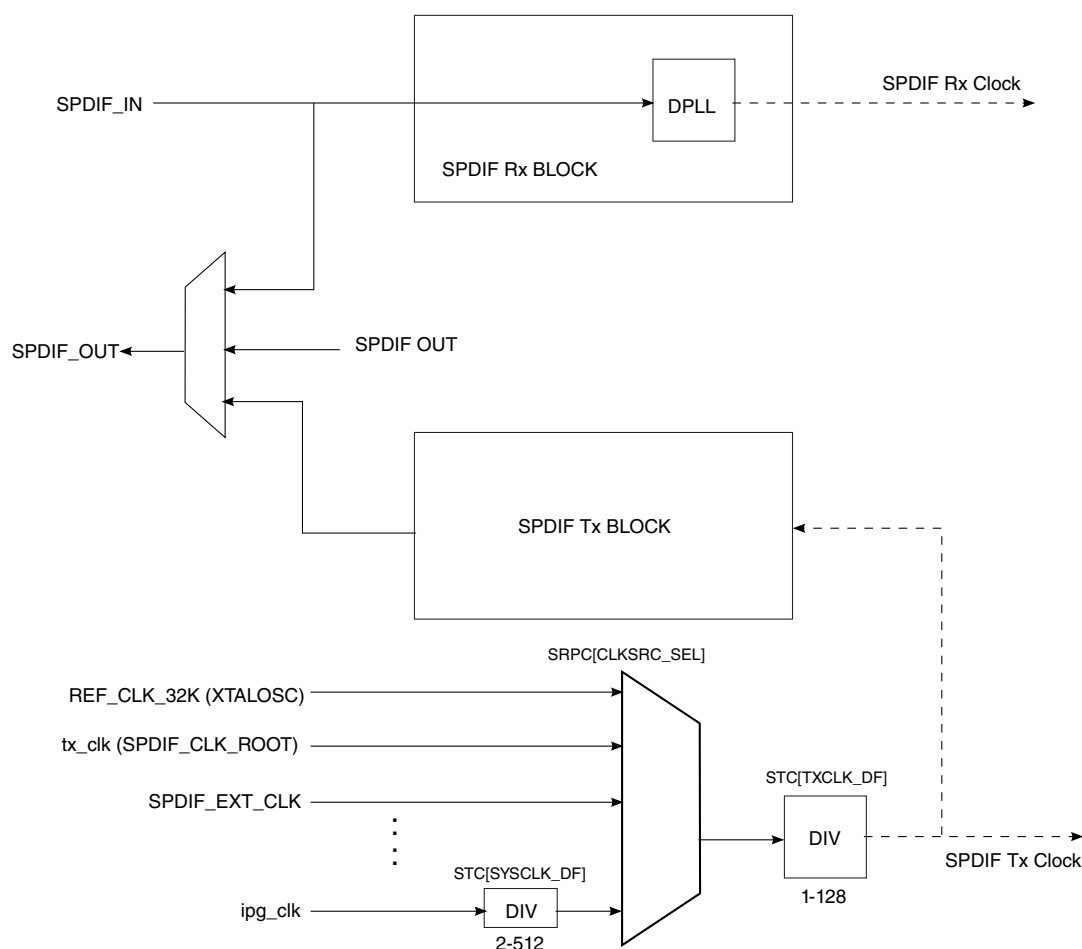


Figure 13-81. SPDIF Transceiver Clock Diagram

13.9.3.1 SPDIF Receiver

The SPDIF receiver extracts the audio data from each SPDIF frame and places the data in Rx left and right FIFOs.

The Tx left and right FIFOs are 16-deep and 24-bit-wide (equal to the audio data width). The Channel Status and User Bits are also extracted from each frame and placed in corresponding registers. The SPDIF receiver also provides a bypass option for direct transfer of the SPDIF input signal to the SPDIF transmitter.

The SPDIF receiver handles the main data audio stream and recovers the bit clock from the SPDIF input signal. The sample rate can be determined from the frequency measuring block. Additionally, the receiver supports the SPDIF C and U channels. The SPDIF C and U channel data is interfaced directly to memory-mapped registers.

All the data registers are controlled by the Interrupt Control Block and transferred to the memory-mapped IP bus.

The following functions are performed by the SPDIF receiver:

- Audio Data Reception see [Audio Data Reception](#)
- Channel Status bits Reception see [Channel Status Reception](#)
- U Channel bits Reception see [User Bit Reception](#)
- Validity Flag Reception see [Validity Flag Reception](#)
- SPDIF Receiver Exception support see [SPDIF Receiver](#)
- SPDIF Lock Detection

13.9.3.1.1 Audio Data Reception

The SPDIF Receiver block extracts the audio data from the IEC60958 stream, and outputs this via Rx left and right FIFOs to the memory-mapped registers SPDIFRxLeft and SPDIFRxRight.

Data from the SPDIF receiver is buffered in receive FIFO, and can be read by the processor from the memory-mapped registers.

- **SPDIF receiver data registers - Behavior on overrun, underrun**

The SPDIF Data Receive registers (SPDIFRxLeft and SPDIFRxRight) have individual FIFOs for left and right channel. As a result, there is always the possibility that left and right FIFOs may go out of sync due to FIFO underruns and FIFO overruns that affect only one part (left or right) of any FIFOs. To prevent this from happening, hardware has been added to the device. Two mechanisms to prevent mismatch between the FIFOs are available.

If a SPDIF Data Rx FIFO overrun occurs on e.g. the right half of the FIFO, the sample that caused the overrun is not written to the right half (due to overrun). Special hardware will make sure the next sample is not written to the left half of the FIFO. If the overrun occurs on the left half of the FIFO, the next sample is not written to the right half of the FIFO.

- **SPDIF receiver data registers - Automatic resynchronization of FIFOs**

An automatic FIFO resynchronization feature is available. It can be enabled and disabled separately for every FIFO. If it is enabled, the hardware will check to see if the left and right FIFOs are in sync. If that is not the case, it will set the filling pointer of the right FIFO to be equal to the filling pointer of the left FIFO.

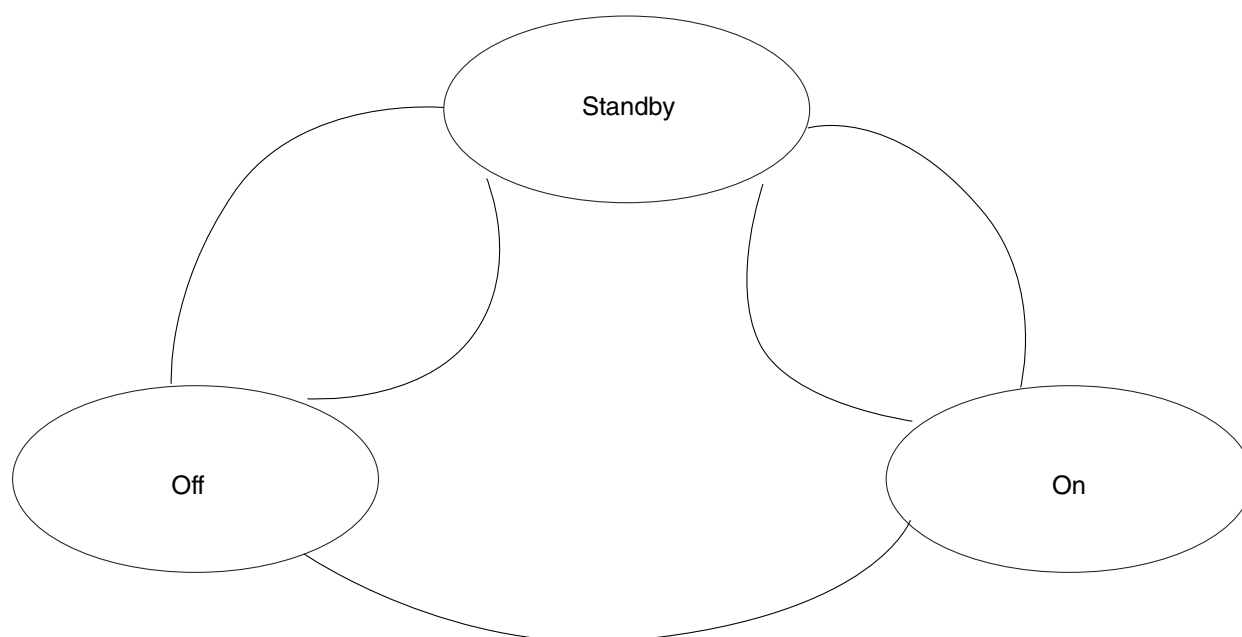


Figure 13-82. FIFO Auto-resync Controller State Machine

The operation is explained from the state diagram shown above. Every FIFO auto-resync controller has a state machine with 3 states: Off, StandBy and On. In the On state, the filling of the left FIFO is compared with the filling of right, and if they are not equal, right is made equal to left, and an interrupt is generated.

The controller will stay in Off state when the feature is disabled. When not disabled, the state machine will go to Off state on any processor read or write to the FIFO. It will go from On or Off to Standby on any left sample read from SPDIF Tx FIFOs, or on any left sample write to SPDIF Rx FIFOs. The controller will go from Standby to On on any right sample read from SPDIF Tx FIFO, or on any right sample write to SPDIF Rx FIFO. There is a control bit in the SPDIFConfig register to enable/disable the feature for the SPDIF Rx FIFO and SPDIF Tx FIFO.

13.9.3.1.1 Application Note

The automatic FIFO resynchronization can be switched on, and will avoid all mismatches between left and right FIFOs, if the software obeys the following rules: 1. When the left data is read or written to the left FIFO, in the same place of the program, data must be read or written to the right FIFO. Maximum time difference between left and right is 1/2 sample clock. (E.g. if sample frequency is 44 KHz, approximately 10 micro-seconds. For 88 KHz, approximately 5 micro-seconds.) 2. Write/read data to FIFO s at least 2 samples

at the time. If there is a mismatch Left-Right, the resync logic may go on only 1 sample clock after last data is read/written to the FIFO. Also acceptable is polling the FIFO, if at least part of the time 2 samples will be read/written to it.

- **SPDIF receiver - Additional features**

There are three exceptions associated with the SPDIF Receivers FIFOs

- full
- under/overflow
- resync

When the "full" condition is set for processor data input registers, the processor should read data from the FIFO, before overflow occurs. When "full" is set, and the FIFO contains e.g. 6 samples, it is acceptable for the software to read first 6 samples from the LEFT address, followed by 6 samples from the RIGHT address, or 6 samples from the RIGHT address, followed by 6 samples from the LEFT address, or 1 sample LEFT, followed by 1 sample RIGHT repeated 6 times. There is no order specified.

The implementation for SPDIF Rx is a double FIFO, one for left and one for right. "full" is set when both FIFOs are full. "underrun, overflow" are set when one of the FIFOs do underrun or do overflow. The resync interrupt means hardware took special action to resynchronize left and right FIFOs.

The FIFO level at which the "full" interrupt is generated, is programmable via the Full Select field in the SPDIFConfigReg register.

Rx FIFO on and Rx FIFO reset.

Two additional control fields of the SPDIF Rx FIFO are the on/off select and FIFO reset fields.

If on/off select is set to off, all-zero will be read from the FIFO, irrespective of the data received over the SPDIF interface.

If FIFO reset is set, the FIFO is blocked at "1 sample in FIFO". In this, the full interrupt will be on if FullSelect is set to "00". If FullSelect is set to any other value, interrupt will be off. The other interrupts are always off.

13.9.3.1.2 Channel Status Reception

A total of 48 channel status bits are received in two registers. No interpretation is performed by the SPDIF receiver block.

Channel Status Bits are ordered first bit left. CS-channel MSB bit "0" is located in bit position 23 in the memory-mapped register SPDIFRxCCChannel_h. CS-channel bit "23" is considered the LSB bit 0 in the register. C-channel bit 24 to 47 is seen as [23:0] bits of register SPDIFRxCCChannel_l.

13.9.3.1.2.1 Channel Status Interrupt

When the value of a new SPDIF "CS" channel status frame is loaded in the register, an interrupt is generated. The interrupt is cleared when the processor writes the corresponding bit in the InterruptStat register.

13.9.3.1.3 User Bit Reception

There are two modes for U Channel reception, CD and non-CD. As is decided by USyncMode (bit 1 of CDText_Control register).

- **Behavior of U Channel receive interface on incoming CD U Channel Sub-code in SPDIF receiver.**

This mode is selected if UsyncMode, bit 1 in register CD Text control is set "1".

The CD sub-code stream embedded into the SPDIF U channel consists of a sequence of packets. Every packet is made up 98 "symbols". The first two symbols of every packet are "sync symbols", the other 96 symbols are "data symbols".

Any sequence found in the SPDIF U channel stream starting with a leading one, followed by 7 information bits, is recognized as a "data symbol". Subsequent data symbols are separated by "pauses". During the "pause", "zero bits" are seen on the SPDIF U channel.

Data symbols are coming in MSB first. The MSB is the leading one.

When a "long pause" is seen between 2 subsequent "data symbols", the SPDIF receiver will assume the reception of one or more "sync symbols". Table below gives details.

Table 13-35. Sync Control Bits

Number of U Channel zero bits	Corresponding number of sync symbols
0-1	Unpredictable, not allowed
2-10	0
11-22	1
23-34	2
35-46	3
>45	Unpredictable, not allowed

The recognition of the number of sync symbols derives from the fact that the U channel transmitter in the CD channel decoder will transmit one symbol on average every 12 SPDIF channel bits. On this average rate, there is a maximum tolerance of 5%.

The SPDIF receiver is tolerant of symbol errors. Due to the physical nature of the transmission of the data over the CD disc, not more than 1 out of any 5 consecutive user channel symbols may be in error. The error may cause a change in data value, which is not detected by this interface, or it may cause a data symbol to be seen as a sync symbol, or a sync symbol to be seen as a data symbol. However, not more than 1 out of any 5 consecutive user channel symbols should be affected in this way.

The SPDIF U channel circuitry recognizes the 98-symbol packet structure, and sends the 96 symbol payload to the processor application. The 96 symbol payload is transmitted to the processor via 2 registers:

- The SPDIFRxUChannel register. In this register, data is presented 3 symbols at the time to the processor. Every time 3 new valid symbols, received on the SPDIF U Channel are present, the UChannelRxFull interrupt is asserted. For one 98-symbol packet, 96 symbols are carried across SPDIFRxUChannel. To transfer all this data, 32 UChannelRxFull interrupts are generated.
- The QChannelReceive register. In this register, only the Q bit of the packet is accumulated. Operation is similar to UChannelReceive. Because only Q-bit is transferred, only 96 Q-bits are transferred for any 98-symbol packet. To transfer this data, 4 QChannelRxFull interrupts are generated. When QChannelRxFull occurs, it is coincident with UChannelRxFull. There is only one QChannelRxFull for every 8 UChannelRxFull. The convention is that most significant data is transmitted first, and is left-aligned in the registers.
- Timing regarding packet boundary is extracted by hardware. The last UChannelRxFull corresponding to a given packet should be coincident with the last QChannelRxFull. In this last U, Q channel interrupt, symbols 95-98 are received, Q channel bits 67-98. The interrupts are coincident with UQSyncFound, flagging last symbols of the current frame.
- When the start of the new packet is found before the current packet is complete (less than 98 symbols in the packet), the UQFrameError interrupt is set. The application software should read out UChannelReceive and QchannelReceive registers, discard the value, and assume the start of a new packet.
- As already said, packet sync extraction is tolerant for single-symbol errors. Packet sync detection is based on the recognition of the sequence data-sync-sync-data in the symbol stream, because this is the only syncing sequence that is not affected by single errors. If the sync symbols are not found 98 symbols after the previous

occurrence, it is assumed to be destroyed by channel error, and a new sync symbols is interpolated.

- Normally, only data bytes are passed to the application software. Every databyte will have its most significant bit set. If sync symbols are passed to the application software, they are seen as all-zero symbols. Sync symbols can only end up in the data stream due to channel error.

- **Behavior of U Channel receive interface on incoming non-CD data.**

This mode is selected if UsyncMode, bit 1 in register CD Text control is set '0'.

In non-CD mode, the SPDIF U channel stream is recognized as a sequence of "data symbols". No packet recognition is done.

Any sequence found in the SPDIF U channel stream starting with a leading one, followed by 7 information bits, is recognized as a "data symbol". Subsequent data symbols are separated by "pauses". During the "pause", "zero bits" are seen on the SPDIF U channel.

3 consecutive data symbols seen in the SPDIF U Channel stream are grouped together into the SPDIFRxUChannel register. First symbol is left, last symbol is right aligned. When SPDIFRxUChannel contains 3 new data symbols, UChannelRxFull is asserted.

In this mode, the operation of QchannelRx and associated interrupt QchannelRxFull is reserved, undefined. And the operation of UQFrameError and UQSyncFound is also reserved, undefined.

The U channel is extracted, and output by the SPDIF Rx on SPDIFRxUChannel-Stream.

When incoming SPDIF data parity error or bit error is detected, and if the next SPDIF word for that channel is error-free, the SPDIF word in error is replaced with the average of the previous word and next word. When incoming SPDIF data parity error or bit error is detected, and the next SPDIF word is in error, the previous SPDIF word is repeated.

13.9.3.1.4 Validity Flag Reception

An interrupt is associated with the Validity flag. (interrupt 16 - SPDIFValNoGood). This interrupt is set every time a frame is seen on the SPDIF interface with the validity bit set to "invalid".

13.9.3.1.5 SPDIF Receiver Interrupt Exception Definition

Several SPDIF exceptions can trigger an interrupt.

They are:

- Control Status channel change. Set when SPDIFRxChannel_1 register is updated. The register is updated for every new C-Channel received. The exception is reset on write to InterruptClear register.
- SPDIF Illegal Symbol. Set on reception of illegal symbol during SPDIF receive. Reset by writing register InterruptClear.¹
- SPDIF bit error. Set on reception of bit error. (Parity bit does not match). Reset on write to InterruptClear register.
- Receive data FIFO full. Set when SPDIF receive data FIFO is full.
- Receive data FIFO underrun/overflow. Set when there is a underrun/overflow on the SPDIF receive data FIFO.
- Receive data FIFO resynchronization. Set when a resynchronization event occurs on the SPDIF receive data FIFO.
- Receive U Channel buffer full. Set when next 24 bits of U channel code are available.
- Receive Q Channel buffer overflow. Set when Q channel buffer overflow.
- Receive U Channel buffer overflow. Set on U channel buffer overflow.
- Receive Q Channel buffer full. Set when next 24 bits of Q channel code are available.
- Receive UQ sync found. Set when UQ channel sync found.
- Receive UQ frame error. Set when UQ frame error found.

13.9.3.1.6 Standards Compliance

The SPDIF interface is compatible with the Tech 3250-E standard of the European Broadcasting Union, except clause 6.3.3 and the IEC60958-3 Ed2 for relevant topics.

Supported input frequency range is 12 Khz up to 96 Khz. (fully compliant) and 96 Khz up to 176 Khz (Can interface with compliant SPDIF transmitter within same cabinet, making reasonable assumptions on jitter added due to interconnecting wire.)

Tolerated jitter on SPDIF input signals are 0.25 bit peak-peak for high frequencies. There is no jitter limit for low frequencies. The user channel extraction in CD mode is capable of coping with single-symbol errors, and still retrieve U channel frames on correct boundaries. This capability is required for reliable reception of CD-Text from some Philips CD channel decoders. This capability was deemed more important than compliance with the IEC60958 annex A.3 standard, and for this reason user channel

1. The SPDIF input is a biphasemark modulated signal. The time between any two successive transitions of the SPDIF signal is always 1, 2 or 3 SPDIF symbol periods long. The SPDIF receiver will parse the stream, and split it in so-called symbols. It recognizes s1, s2 and s3 symbols, depending on the length of the symbols. Not all sequences of these symbols are allowed. To give an example, a sequence s2-s1-s1-s1-s2 cannot occur in a no-error SPDIF signal. If the receiver finds such an illegal sequence, the illegal symbol interrupt is set. No corrective action is undertaken. When the interrupt occurs, this means that(a) The SPDIF signal is destroyed by noise (b) The SPDIF frequency changed.

reception is not compliant with IEC60958 annex A.3. However, the interface is capable to receive U channel inserted by a typical CD channel decoder. Also, in this case, it is more robust and tolerant for channel error than what is required by IEC60958 annex A.3.

13.9.3.1.7 SPDIF PLOCK Detection and Rxclk Output

Using the high speed system clock, the internal DPLL can extract the bit clock (advanced pulse) from the input bitstream. When this internal DPLL is locked, the LOCK bit of PhaseConfig Register will be set, and the SPDIF Lock output pin SPDIF_LOCK will be asserted.

After DPLL has locked, the pulses are generated, and the average pulse rate is 128 x the sampling frequency. (For a 44.1 KHz input sampling frequency, the average pulse rate = 128 x 44.1 KHz.) The pulse signal is used in the FreqMeas circuit to generate the frequency measurement result.

13.9.3.1.8 Measuring Frequency of SPDIF_RxClk

The internal DPLL can extract the bit clock (advanced plus) from the input bitstream. To do that, it is necessary to measure the frequency of the incoming signal in relationship with the system clock (BUS_CLK).

Associated with it are two registers, PhaseConfig and FreqMeas. The circuit will measure the frequency of the incoming clock as a function of the BUS_CLK. The circuit is a second-order filter. The output is a value represented by an unsigned number stored in the 24-bit FreqMeas register, giving the frequency of the source as a function of the BUS_CLK.

$\text{FreqMeas}[23:0] = \text{FreqMeas_CLK} / \text{BUS_CLK} * 2^{10} * \text{GAIN}.$

For example, if the GAIN is selected as 8*(2**10) (PhaseConfig[5:3] = 3'b011), the actual result

$\text{FreqMeas_CLK} / \text{BUS_CLK}$ is equal to $\text{FreqMeas}[23:0] / 2^{23}.$

13.9.3.2 SPDIF Transmitter

Audio data for the SPDIF transmitter is provided by processor via the SPDIFTxLeft and SPDIFTxRight registers.

Clocking for SPDIF transmitter is selected through a multiplexer from several clock sources (see [TxClk_Source](#) for clock source inputs). The SPDIF transmitter clock source can be divided down as needed using Txclk_DF. The SPDIF transmitter output can be chosen from either the SPDIF transmitter block, directly from the SPDIF receiver (via the output multiplexer), or disabled.

The SPDIF transmitter generates a SPDIF output bitstream in IEC60958 biphas mark format, consisting of audio data, channel status.

13.9.3.2.1 Audio Data Transmission

Audio data for the SPDIF transmitter is provided by the processor via SPDIFTxLeft and SPDIFTxRight registers. They send audio data to Tx left and right FIFOs. The Tx left and right FIFOs are also 16-deep and 24-width (equal to the audio data width).

- **SPDIF transmitter data registers - Behavior on overrun, underrun**

The SPDIF Data Transmit registers (SPDIFTxLeft and SPDIFTxRight) have individual FIFOs for left and right channel. As a result, there is always the possibility that left and right FIFOs may go out of sync due to FIFO underruns and FIFO overruns that affect only one part (left or right) of any FIFO. To prevent this from happening, hardware has been added on the device. Two mechanisms to prevent mismatch between the FIFOs are available.

If SPDIF Tx FIFO underruns on the right half of the FIFO, no sample leaves that FIFO (because it was already empty). Special hardware will make sure that the next sample read from the left FIFO will not leave the FIFO (no read strobe is generated). If the underrun occurs on the left half of the FIFO, next read strobe to the right FIFO is blocked.

- **SPDIF transmitter data registers - Automatic resynchronization of FIFOs**

See [Audio Data Reception](#).

- **SPDIFTxLeft, SPDIFTxRight details**

With SPDIF Tx FIFOs three exceptions are associated.

- empty
- under/overrun
- resync

When the empty condition is set for processor data output registers, the processor should write data to the FIFO, before underrun occurs. When empty is set and, for instance, 6 samples need to be written, it is acceptable for the software to write first 6 samples from the LEFT address, followed by 6 samples from the RIGHT address, or 1 sample LEFT,

followed by 1 sample RIGHT repeated 6 times. Left should be written before right. The implementation of all data out FIFOs is a double FIFO, one for left and one for right. Empty is set when both FIFOs are empty. Underrun, overrun are set when one of the FIFOs do underrun or do overrun. Resync is set when the hardware resynchronizes left and right FIFOs.

On receiving underrun, overrun interrupt, synchronization between Left and Right words in the FIFOs may be lost. Synchronization will not be lost when the underrun or overrun comes from the IEC60958 side of the FIFO. If the processor reads or writes more data from, for example, left than from right, synchronization will be lost. If automatic resynchronization is enabled, and if the software obeys the rules to let this work, resynchronization will be automatic.

13.9.3.2.2 Channel Status Transmission

A total of 48 Consumer channel status bits are transmitted from two registers. Channel Status Bits are ordered first bit left.

CS-channel MSB bit "0" is located in bit position 23 in the memory-mapped register SPDIFTxChannelCons_h. CS-channel bit "23" is considered bit 0 in the register. C-channel bits 24-47 are seen as MSB-LSB bits of register SPDIFTxChannelCons_l.

13.9.3.2.3 Validity Flag Transmission

The validity bit setting is performed via bit 5 of the SPDIF_SCR register.

13.9.4 SPDIF Memory Map/Register Definition

SPDIF memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
3081_0000	SPDIF Configuration Register (SPDIF1_SCR)	32	R/W	0000_0400h	13.9.4.1/ 5229
3081_0004	CDText Control Register (SPDIF1_SRCDD)	32	R/W	0000_0000h	13.9.4.2/ 5231
3081_0008	PhaseConfig Register (SPDIF1_SRPC)	32	R/W	0000_0000h	13.9.4.3/ 5232
3081_000C	InterruptEn Register (SPDIF1_SIE)	32	R/W	0000_0000h	13.9.4.4/ 5233
3081_0010	InterruptStat Register (SPDIF1_SIS)	32	R	0000_0002h	13.9.4.5/ 5235

Table continues on the next page...

SPDIF memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
3081_0010	InterruptClear Register (SPDIF1_SIC)	32	W	0000_0000h	13.9.4.6/5237
3081_0014	SPDIFRxLeft Register (SPDIF1_SRL)	32	R	0000_0000h	13.9.4.7/5238
3081_0018	SPDIFRxRight Register (SPDIF1_SRR)	32	R	0000_0000h	13.9.4.8/5239
3081_001C	SPDIFRxCCChannel_h Register (SPDIF1_SRC SH)	32	R	0000_0000h	13.9.4.9/5239
3081_0020	SPDIFRxCCChannel_l Register (SPDIF1_SRC SL)	32	R	0000_0000h	13.9.4.10/5240
3081_0024	UchannelRx Register (SPDIF1_SRU)	32	R	0000_0000h	13.9.4.11/5240
3081_0028	QchannelRx Register (SPDIF1_SRQ)	32	R	0000_0000h	13.9.4.12/5241
3081_002C	SPDIFTxLeft Register (SPDIF1_STL)	32	W	0000_0000h	13.9.4.13/5241
3081_0030	SPDIFTxRight Register (SPDIF1_STR)	32	W	0000_0000h	13.9.4.14/5242
3081_0034	SPDIFTxCCChannelCons_h Register (SPDIF1_STC SCH)	32	R/W	0000_0000h	13.9.4.15/5242
3081_0038	SPDIFTxCCChannelCons_l Register (SPDIF1_STC SCL)	32	R/W	0000_0000h	13.9.4.16/5243
3081_0044	FreqMeas Register (SPDIF1_SRFM)	32	R	0000_0000h	13.9.4.17/5243
3081_0050	SPDIFTxCk Register (SPDIF1_STC)	32	R/W	0002_0F00h	13.9.4.18/5244
308A_0000	SPDIF Configuration Register (SPDIF2_SCR)	32	R/W	0000_0400h	13.9.4.1/5229
308A_0004	CDText Control Register (SPDIF2_SRC D)	32	R/W	0000_0000h	13.9.4.2/5231
308A_0008	PhaseConfig Register (SPDIF2_SRPC)	32	R/W	0000_0000h	13.9.4.3/5232
308A_000C	InterruptEn Register (SPDIF2_SIE)	32	R/W	0000_0000h	13.9.4.4/5233
308A_0010	InterruptStat Register (SPDIF2_SIS)	32	R	0000_0002h	13.9.4.5/5235
308A_0010	InterruptClear Register (SPDIF2_SIC)	32	W	0000_0000h	13.9.4.6/5237
308A_0014	SPDIFRxLeft Register (SPDIF2_SRL)	32	R	0000_0000h	13.9.4.7/5238
308A_0018	SPDIFRxRight Register (SPDIF2_SRR)	32	R	0000_0000h	13.9.4.8/5239
308A_001C	SPDIFRxCCChannel_h Register (SPDIF2_SRC SH)	32	R	0000_0000h	13.9.4.9/5239

Table continues on the next page...

SPDIF memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
308A_0020	SPDIFRxCChannel_I Register (SPDIF2_SRC SL)	32	R	0000_0000h	13.9.4.10/5240
308A_0024	UchannelRx Register (SPDIF2_SR U)	32	R	0000_0000h	13.9.4.11/5240
308A_0028	QchannelRx Register (SPDIF2_SR Q)	32	R	0000_0000h	13.9.4.12/5241
308A_002C	SPDIFTxLeft Register (SPDIF2_ST L)	32	W	0000_0000h	13.9.4.13/5241
308A_0030	SPDIFTxRight Register (SPDIF2_ST R)	32	W	0000_0000h	13.9.4.14/5242
308A_0034	SPDIFTxCChannelCons_h Register (SPDIF2_ST CSCH)	32	R/W	0000_0000h	13.9.4.15/5242
308A_0038	SPDIFTxCChannelCons_l Register (SPDIF2_ST CSCL)	32	R/W	0000_0000h	13.9.4.16/5243
308A_0044	FreqMeas Register (SPDIF2_SR FM)	32	R	0000_0000h	13.9.4.17/5243
308A_0050	SPDIFTxCk Register (SPDIF2_ST C)	32	R/W	0002_0F00h	13.9.4.18/5244

13.9.4.1 SPDIF Configuration Register (SPDIFx_SCR)

Address: Base address + 0h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								RxFIFO_Ctrl	RxFIFO_Off_On	RxFIFO_Rst	RxFIFOFull_Sel	RxAutoSync	TxAutoSync	TxFIFOEmpty_Sel	
W																
Reset									0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TxFIFOEmpty_Sel	Reserved	LOW_POWER	soft_reset	TxFIFO_Ctrl	DMA_Rx_En	DMA_TX_En	Reserved	ValCtrl	TxSel	USrc_Sel					
W																
Reset	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0

SPDIFx_SCR field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
23 RxFIFO_Ctrl	0 Normal operation 1 Always read zero from Rx data register
22 RxFIFO_Off_On	0 SPDIF Rx FIFO is on 1 SPDIF Rx FIFO is off. Does not accept data from interface
21 RxFIFO_Rst	0 Normal operation 1 Reset register to 1 sample remaining
20–19 RxFIFOFull_Sel	00 Full interrupt if at least 1 sample in Rx left and right FIFOs 01 Full interrupt if at least 4 sample in Rx left and right FIFOs 10 Full interrupt if at least 8 sample in Rx left and right FIFOs 11 Full interrupt if at least 16 sample in Rx left and right FIFO
18 RxAutoSync	0 Rx FIFO auto sync off 1 Rx FIFO auto sync on
17 TxAutoSync	0 Tx FIFO auto sync off 1 Tx FIFO auto sync on
16–15 TxFIFOEmpty_Sel	00 Empty interrupt if 0 sample in Tx left and right FIFOs 01 Empty interrupt if at most 4 sample in Tx left and right FIFOs 10 Empty interrupt if at most 8 sample in Tx left and right FIFOs 11 Empty interrupt if at most 12 sample in Tx left and right FIFOs
14 -	This field is reserved. Reserved
13 LOW_POWER	When write 1 to this bit, it will cause SPDIF enter low-power mode. return 1 when SPDIF in Low-Power mode.
12 soft_reset	When write 1 to this bit, it will cause SPDIF software reset. The software reset will last 8 cycles. When in the reset process, return 1 when read. else return 0 when read.
11–10 TxFIFO_Ctrl	00 Send out digital zero on SPDIF Tx 01 Tx Normal operation 10 Reset to 1 sample remaining 11 Reserved
9 DMA_Rx_En	DMA Receive Request Enable (RX FIFO full)
8 DMA_TX_En	DMA Transmit Request Enable (Tx FIFO empty)
7–6 -	This field is reserved. Reserved
5 ValCtrl	0 Outgoing Validity always set 1 Outgoing Validity always clear
4–2 TxSel	000 Off and output 0 001 Feed-through SPDIFIN

Table continues on the next page...

SPDIFx_SCR field descriptions (continued)

Field	Description
	101 Tx Normal operation Others Reserved
USrc_Sel	00 No embedded U channel 01 U channel from SPDIF receive block (CD mode) 10 Reserved 11 U channel from on chip transmitter

13.9.4.2 CDText Control Register (SPDIFx_SRCD)

Address: Base address + 4h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								0							
W																
Reset									0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0	Reserved							0					Reserved	USyncMode	Reserved
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SPDIFx_SRCD field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
23–15 Reserved	This read-only field is reserved and always has the value 0.
14–8 -	This field is reserved. Reserved. set to zero.
7–3 Reserved	This read-only field is reserved and always has the value 0.

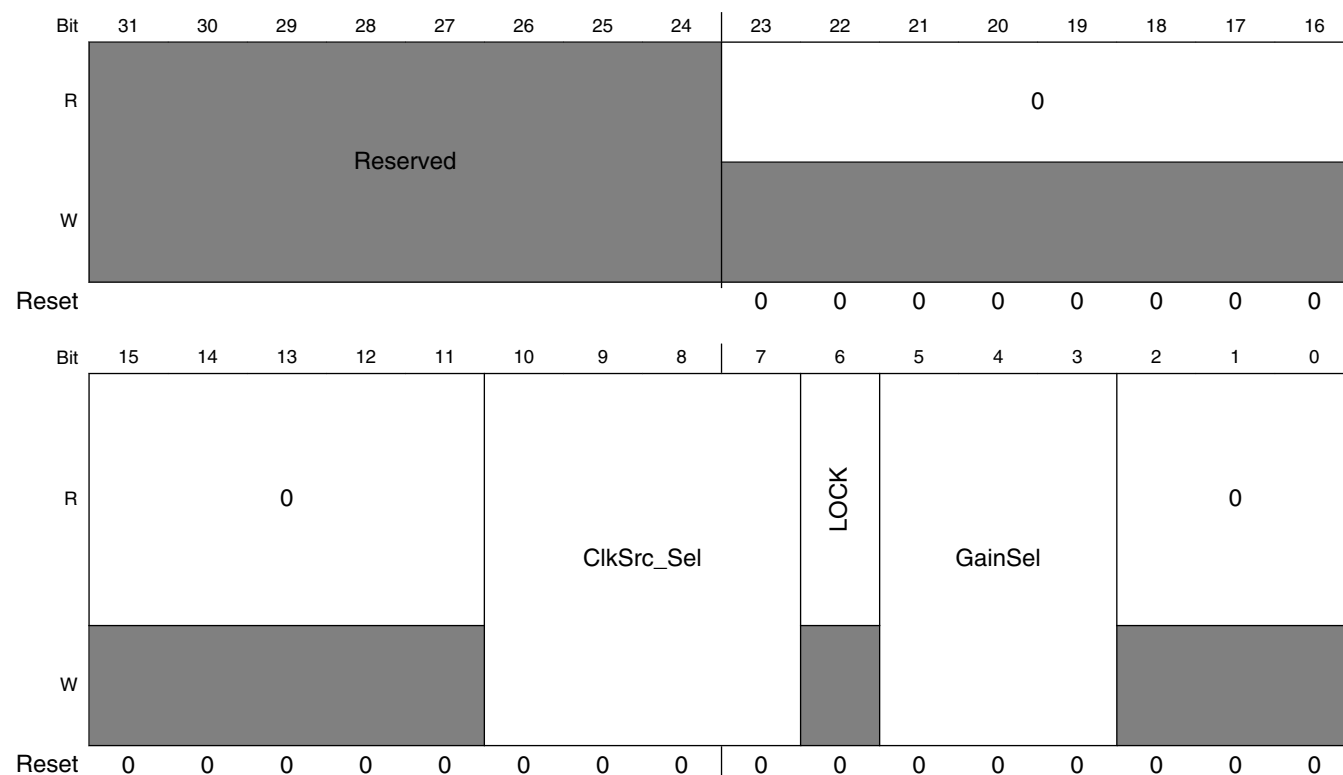
Table continues on the next page...

SPDIFx_SRCD field descriptions (continued)

Field	Description
2 -	This field is reserved. Reserved.
1 USyncMode	0 Non-CD data 1 CD user channel subcode
0 -	This field is reserved. Reserved.

13.9.4.3 PhaseConfig Register (SPDIFx_SRPC)

Address: Base address + 8h offset

**SPDIFx_SRPC field descriptions**

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
23–11 Reserved	This read-only field is reserved and always has the value 0.
10–7 ClkSrc_Sel	Clock source selection, all other settings not shown are reserved: 0000 if (DPLL Locked) SPDIF_RxCIk else REF_CLK_32K (XTALOSC)

Table continues on the next page...

SPDIFx_SRPC field descriptions (continued)

Field	Description
	0001 if (DPLL Locked) SPDIF_RxClk else tx_clk (SPDIF0_CLK_ROOT) 0011 if (DPLL Locked) SPDIF_RxClk else SPDIF_EXT_CLK 0101 REF_CLK_32K (XTALOSC) 0110 tx_clk (SPDIF0_CLK_ROOT) 1000 SPDIF_EXT_CLK
6 LOCK	LOCK bit to show that the internal DPLL is locked, read only
5–3 GainSel	Gain selection: 000 $24 \times (2^{**10})$ 001 $16 \times (2^{**10})$ 010 $12 \times (2^{**10})$ 011 $8 \times (2^{**10})$ 100 $6 \times (2^{**10})$ 101 $4 \times (2^{**10})$ 110 $3 \times (2^{**10})$
Reserved	This read-only field is reserved and always has the value 0.

13.9.4.4 InterruptEn Register (SPDIFx_SIE)

The InterruptEn register (SPDIF_SIE) provides control over the enabling of interrupts.

Address: Base address + Ch offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
R	Reserved								0	Reserved			Lock	TxUnOv	TxResyn	CNew	ValNoGood
W																	
Reset									0	0	0	0	0	0	0	0	

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
R	SymErr	BitErr	Reserved				URxFul	URxOv	QRxFul	QRxOv	UQSync	UQErr	RxFIFOUnOv	RxFIFOResyn	LockLoss	TxEm	RxFIFOFull
W																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

SPDIFx_SIE field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
23 Reserved	This read-only field is reserved and always has the value 0.
22–21 -	This field is reserved. Reserved. set to zero.
20 Lock	SPDIF receiver's DPLL is locked
19 TxUnOv	SPDIF Tx FIFO under/overflow
18 TxResyn	SPDIF Tx FIFO resync
17 CNew	SPDIF receive change in value of control channel
16 ValNoGood	SPDIF validity flag no good
15 SymErr	SPDIF receiver found illegal symbol
14 BitErr	SPDIF receiver found parity bit error
13–11 -	This field is reserved. Reserved. set to zero.
10 URxFul	U Channel receive register full, can't be cleared with reg. IntClear. To clear it, read from U Rx reg.
9 URxOv	U Channel receive register overrun
8 QRxFul	Q Channel receive register full, can't be cleared with reg. IntClear. To clear it, read from Q Rx reg.
7 QRxOv	Q Channel receive register overrun
6 UQSync	U/Q Channel sync found
5 UQErr	U/Q Channel framing error
4 RxFIFOUnOv	Rx FIFO underrun/overflow
3 RxFIFOResyn	Rx FIFO resync
2 LockLoss	SPDIF receiver loss of lock
1 TxEm	SPDIF Tx FIFO empty, can't be cleared with reg. IntClear. To clear it, write to Tx FIFO.
0 RxFIFOFull	SPDIF Rx FIFO full, can't be cleared with reg. IntClear. To clear it, read from Rx FIFO.

13.9.4.5 InterruptStat Register (SPDIFx_SIS)

The InterruptStat (SPDIF_SIS) register is a read only register that provides the status on interrupt operations.

Address: Base address + 10h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								0			Lock	TxUnOv	TxResyn	CNew	ValNoGood
W																
Reset									0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SymErr	BitErr	0			URxFul	URxOv	QRxFul	QRxOv	UQSync	UQErr	RxFIFOUnOv	RxFIFOResyn	LockLoss	TxErm	RxFIFOFull
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

SPDIFx_SIS field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
23–21 Reserved	This read-only field is reserved and always has the value 0.
20 Lock	SPDIF receiver's DPLL is locked
19 TxUnOv	SPDIF Tx FIFO under/overflow
18 TxResyn	SPDIF Tx FIFO resync
17 CNew	SPDIF receive change in value of control channel
16 ValNoGood	SPDIF validity flag no good
15 SymErr	SPDIF receiver found illegal symbol
14 BitErr	SPDIF receiver found parity bit error
13–11 Reserved	This read-only field is reserved and always has the value 0.
10 URxFul	U Channel receive register full, can't be cleared with reg. IntClear. To clear it, read from U Rx reg.
9 URxOv	U Channel receive register overrun
8 QRxFul	Q Channel receive register full, can't be cleared with reg. IntClear. To clear it, read from Q Rx reg.
7 QRxOv	Q Channel receive register overrun
6 UQSync	U/Q Channel sync found
5 UQErr	U/Q Channel framing error
4 RxFIFOUnOv	Rx FIFO underrun/overflow
3 RxFIFOResyn	Rx FIFO resync
2 LockLoss	SPDIF receiver loss of lock
1 TxEm	SPDIF Tx FIFO empty, can't be cleared with reg. IntClear. To clear it, write to Tx FIFO.
0 RxFIFOFull	SPDIF Rx FIFO full, can't be cleared with reg. IntClear. To clear it, read from Rx FIFO.

13.9.4.6 InterruptClear Register (SPDIFx_SIC)

The InterruptClear (SPDIF_SIC) register is a write only register and is used to clear interrupts.

Address: Base address + 10h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								0							
W												Lock	TxUnOv	TxResyn	CNew	ValNoGood
Reset									0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R			Reserved											Reserved		
W	SymErr	BitErr						URxOv	.	QRxOv	UQSync	UQErr	RxFIFOUnOv			RxFIFOResyn
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SPDIFx_SIC field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
23–21 Reserved	This read-only field is reserved and always has the value 0.
20 Lock	SPDIF receiver's DPLL is locked
19 TxUnOv	SPDIF Tx FIFO under/overflow

Table continues on the next page...

SPDIFx_SIC field descriptions (continued)

Field	Description
18 TxResyn	SPDIF Tx FIFO resync
17 CNew	SPDIF receive change in value of control channel
16 ValNoGood	SPDIF validity flag no good
15 SymErr	SPDIF receiver found illegal symbol
14 BitErr	SPDIF receiver found parity bit error
13–10 -	This field is reserved. Reserved.
9 URxOv	U Channel receive register overrun
8 -	Reserved
7 QRxOv	Q Channel receive register overrun
6 UQSync	U/Q Channel sync found
5 UQErr	U/Q Channel framing error
4 RxFIFOUnOv	Rx FIFO underrun/overrun
3 RxFIFOResyn	Rx FIFO resync
2 LockLoss	SPDIF receiver loss of lock
-	This field is reserved. Reserved.

13.9.4.7 SPDIFRxLeft Register (SPDIFx_SRL)

SPDIFRxLeft register is an audio data reception register.

Address: Base address + 14h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								RxDataLeft																							
W																																
Reset	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SPDIFx_SRL field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
RxDataLeft	Processor receive SPDIF data left

13.9.4.8 SPDIFRxRight Register (SPDIFx_SRR)

SPDIFRxRight register is an audio data reception register.

Address: Base address + 18h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								RxDataRight																							
W																																
Reset	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

SPDIFx_SRR field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
RxDataRight	Processor receive SPDIF data right

13.9.4.9 SPDIFRxCChannel_h Register (SPDIFx_SRC SH)

SPDIFRxCChannel_h register is a channel status reception register.

Address: Base address + 1Ch offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								RxCChannel_h																							
W																																
Reset	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

SPDIFx_SRC SH field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented.

Table continues on the next page...

SPDIFx_SRC SH field descriptions (continued)

Field	Description
	This field is reserved.
RxCChannel_h	SPDIF receive C channel register, contains first 24 bits of C channel without interpretation

13.9.4.10 SPDIFRxChannel_I Register (SPDIFx_SRC SL)

SPDIFRxChannel_I register is a channel status reception register.

Address: Base address + 20h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								RxCCChannel_I																							
W	Reserved																															
Reset	0								0																							

SPDIFx_SRC SL field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
RxCChannel_I	SPDIF receive C channel register, contains next 24 bits of C channel without interpretation

13.9.4.11 UchannelRx Register (SPDIFx_SR U)

UchannelRx register is a user bits reception register.

Address: Base address + 24h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								RxUChannel																							
W	Reserved																															
Reset	0								0																							

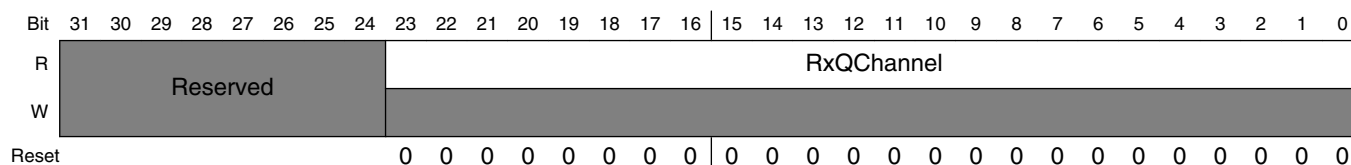
SPDIFx_SR U field descriptions

Field	Description
31–24 [unimplemented]	This field is reserved. This is a 24-bit register the upper byte is unimplemented.
RxUChannel	SPDIF receive U channel register, contains next 3 U channel bytes

13.9.4.12 QchannelRx Register (SPDIFx_SRQ)

QChannelRx register is a user bits reception register.

Address: Base address + 28h offset



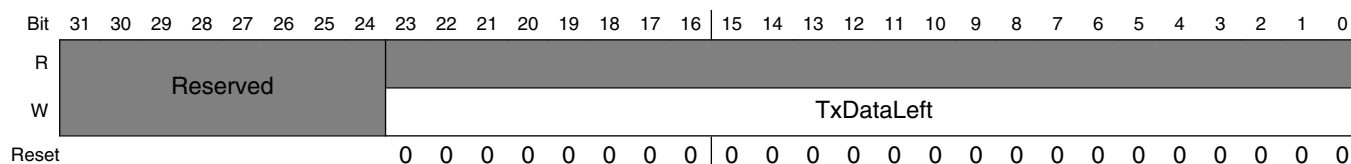
SPDIFx_SRQ field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
RxQChannel	SPDIF receive Q channel register, contains next 3 Q channel bytes

13.9.4.13 SPDIFTxLeft Register (SPDIFx_STL)

SPDIFTxLeft register is an audio data transmission register.

Address: Base address + 2Ch offset



SPDIFx_STL field descriptions

Field	Description
31–24 [unimplemented]	This field is reserved. This is a 24-bit register the upper byte is unimplemented.
TxDataLeft	SPDIF transmit left channel data. It is write-only, and always returns zeros when read

13.9.4.14 SPDIFTxRight Register (SPDIFx_STR)

SPDIFTxRight register is an audio data transmission register.

Address: Base address + 30h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																															
W									TxDataRight																							
Reset									0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

SPDIFx_STR field descriptions

Field	Description
31–24 [unimplemented]	This field is reserved. This is a 24-bit register the upper byte is unimplemented.
TxDataRight	SPDIF transmit right channel data. It is write-only, and always returns zeros when read

13.9.4.15 SPDIFTxCChannelCons_h Register (SPDIFx_STCSCH)

SPDIFTxCChannelCons_h register is a channel status transmission register.

Address: Base address + 34h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								TxCChannelCons_h																							
W																																
Reset	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SPDIFx_STCSCH field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
TxCChannelCons_h	SPDIF transmit Cons. C channel data, contains first 24 bits without interpretation. When read, it returns the latest data written by the processor

13.9.4.16 SPDIFTxChannelCons_I Register (SPDIFx_STCSCL)

SPDIFTxChannelCons_I register is a channel status transmission register.

Address: Base address + 38h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								TxCCChannelCons_I																							
W																																
Reset	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SPDIFx_STCSCL field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
TxChannelCons_I	SPDIF transmit Cons. C channel data, contains next 24 bits without interpretation. When read, it returns the latest data written by the processor

13.9.4.17 FreqMeas Register (SPDIFx_SRFM)

Address: Base address + 44h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								FreqMeas																							
W																																
Reset									0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SPDIFx_SRFM field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
FreqMeas	Frequency measurement data

13.9.4.18 SPDIFTxClk Register (SPDIFx_STC)

The SPDIFTxClk Control register includes the means to select the transmit clock and frequency division.

Address: Base address + 50h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								0				SYSCLK_DF			
W																
Reset									0	0	0	0	0	0	1	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SYSCLK_DF						TxClk_Source		tx_all_clk_en	TxClk_DF						
W																
Reset	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0

SPDIFx_STC field descriptions

Field	Description
31–24 [unimplemented]	This is a 24-bit register the upper byte is unimplemented. This field is reserved.
23–20 Reserved	This read-only field is reserved and always has the value 0.
19–11 SYSCLK_DF	system clock divider factor, 2~512. 0 no clock signal 1 divider factor is 2 ... 511 divider factor is 512
10–8 TxClk_Source	000 REF_CLK_32K input (XTALOSC 32 kHz clock) 001 tx_clk input (from SPDIF0_CLK_ROOT. See CCM.) 011 SPDIF_EXT_CLK, from pads 101 ipg_clk input (frequency divided)
7 tx_all_clk_en	Spdif transfer clock enable. When data is going to be transfered, this bit should be set to 1. 0 disable transfer clock. 1 enable transfer clock.
TxClk_DF	Divider factor (1-128) 0 divider factor is 1

Table continues on the next page...

SPDIFx_STC field descriptions (continued)

Field	Description
1	divider factor is 2
...	...
127	divider factor is 128

13.10 Synchronous Audio Interface (SAI)

13.10.1 Introduction

The I²S (or I2S) module provides a synchronous audio interface (SAI) that supports full-duplex serial interfaces with frame synchronization such as I²S, AC97, TDM, and codec/DSP interfaces.

13.10.1.1 Features

Note that some of the features are not supported across all SAI instances; see the chip-specific information in the first section of this chapter.

- Transmitter with independent bit clock and frame sync supporting 8 data lines
- Receiver with independent bit clock and frame sync supporting 8 data lines
- Each data line can support a maximum Frame size of 32 words
- Word size of between 8-bits and 32-bits
- Word size configured separately for first word and remaining words in frame
- Asynchronous 128 x 32-bit FIFO for each transmit and receive data line
- Supports graceful restart after FIFO error
- Supports automatic restart after FIFO error without software intervention
- Supports packing of 8-bit and 16-bit data into each 32-bit FIFO word
- Supports combining multiple data line FIFOs into single data line FIFO

13.10.1.2 Block diagram

The following block diagram also shows the module clocks.

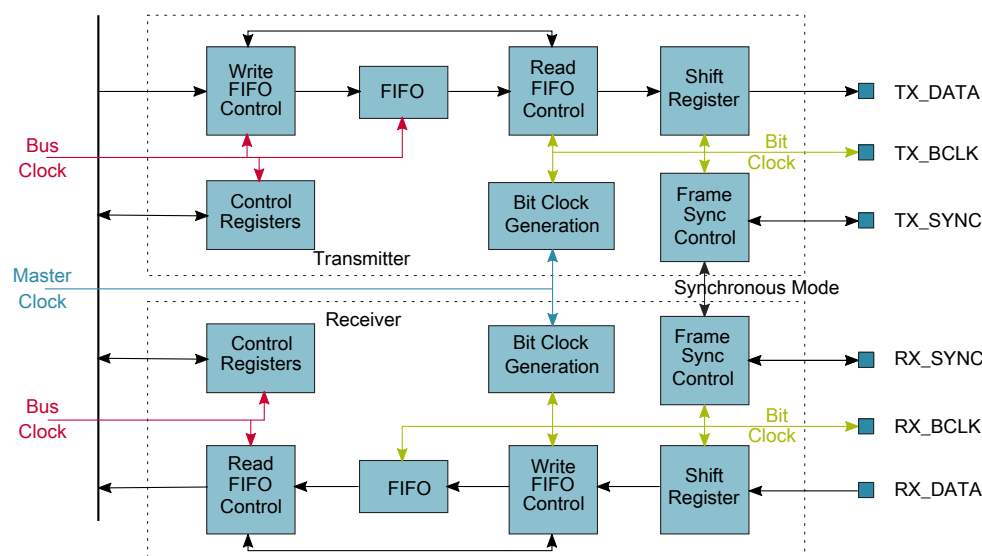


Figure 13-83. I²S/SAI block diagram

13.10.1.3 Modes of operation

Module power modes include Run mode, and Debug mode.

13.10.1.3.1 Run mode

In Run mode, the SAI transmitter and receiver operate normally.

13.10.1.3.2 Debug mode

In Debug mode, the SAI transmitter and/or receiver can continue operating provided the Debug Enable bit is set. When TCSR[DBGE] or RCSR[DBGE] bit is clear and Debug mode is entered, the SAI is disabled after completing the current transmit or receive frame. The transmitter and receiver bit clocks are not affected by Debug mode.

13.10.2 External signals

Name	Function	I/O
TX_BCLK	Transmit Bit Clock. The bit clock is an input when externally generated and an output when internally generated.	I/O

Table continues on the next page...

Name	Function	I/O
TX_SYNC	Transmit Frame Sync. The frame sync is an input sampled synchronously by the bit clock when externally generated and an output generated synchronously by the bit clock when internally generated.	I/O
TX_DATA[7:0]	Transmit Data. The transmit data is generated synchronously by the bit clock and is tristated whenever not transmitting a word.	O
RX_BCLK	Receive Bit Clock. The bit clock is an input when externally generated and an output when internally generated.	I/O
RX_SYNC	Receive Frame Sync. The frame sync is an input sampled synchronously by the bit clock when externally generated and an output generated synchronously by the bit clock when internally generated.	I/O
RX_DATA[7:0]	Receive Data. The receive data is sampled synchronously by the bit clock.	I

13.10.3 Functional description

This section provides a complete functional description of the block.

13.10.3.1 SAI clocking

The SAI clocks include:

- The audio master clock
- The bit clock
- The bus clock

13.10.3.1.1 Audio master clock

The audio master clock is used to generate the bit clock when the receiver or transmitter is configured for an internally generated bit clock. The transmitter and receiver can independently select between the bus clock and up to three audio master clocks to generate the bit clock.

The audio master clock generation and selection is chip-specific. Refer to chip-specific clocking information about how the audio master clocks are generated.

13.10.3.1.2 Bit clock

The SAI transmitter and receiver support asynchronous free-running bit clocks that can be generated internally from an audio master clock or supplied externally. There is also the option for synchronous bit clock and frame sync operation between the receiver and transmitter or between multiple SAI peripherals.

- If both transmitter and receiver are configured for asynchronous operation, then the transmitter and receiver will each use *their own* bit clock and frame sync.
- If the *transmitter* is configured for asynchronous mode and the receiver is configured for synchronous mode, then both transmitter and receiver will use the *transmitter* bit clock and frame sync.
- If the *receiver* is configured for asynchronous mode and the transmitter is configured for synchronous mode, then both transmitter and receiver will use the *receiver* bit clock and frame sync.

Note that the software configures synchronous or asynchronous mode, and that choice selects the bit clock/frame sync used.

Externally generated bit clocks must be:

- Enabled before the SAI transmitter or receiver is enabled
- Disabled after the SAI transmitter or receiver is disabled and completes its current frames

If the SAI transmitter or receiver is using an externally generated bit clock in asynchronous mode and that bit clock is generated by an SAI that is disabled in stop mode, then the transmitter or receiver should be disabled by software before entering stop mode. This issue does not apply when the transmitter or receiver is in a synchronous mode because all synchronous SAIs are enabled and disabled simultaneously.

13.10.3.1.3 Bus clock

The bus clock is used by the control and configuration registers and to generate synchronous interrupts and DMA requests.

NOTE

Although there is no specific minimum bus clock frequency specified, the bus clock frequency must be fast enough (relative to the bit clock frequency) to ensure that the FIFOs can be serviced, without generating either a transmitter FIFO underrun or receiver FIFO overflow condition.

13.10.3.2 SAI resets

The SAI is asynchronously reset on system reset. The SAI has a software reset and a FIFO reset.

13.10.3.2.1 Software reset

The SAI transmitter includes a software reset that resets all transmitter internal logic, including the bit clock generation, status flags, and FIFO pointers. It does not reset the configuration registers. The software reset remains asserted until cleared by software.

The SAI receiver includes a software reset that resets all receiver internal logic, including the bit clock generation, status flags and FIFO pointers. It does not reset the configuration registers. The software reset remains asserted until cleared by software.

13.10.3.2.2 FIFO reset

The SAI transmitter includes a FIFO reset that synchronizes the FIFO write pointer to the same value as the FIFO read pointer. This empties the FIFO contents and is to be used after TCSR[FEF] is set, and before the FIFO is re-initialized and TCSR[FEF] is cleared. The FIFO reset is asserted for one cycle only.

The SAI transmitter can also reset the FIFO of individual data channels by setting the appropriate TCR3[CFR] bit. This should only be done when the corresponding TCR3[TCE] bit is clear.

The SAI receiver includes a FIFO reset that synchronizes the FIFO read pointer to the same value as the FIFO write pointer. This empties the FIFO contents and is to be used after the RCSR[FEF] is set and any remaining data has been read from the FIFO, and before the RCSR[FEF] is cleared. The FIFO reset is asserted for one cycle only.

The SAI receiver can also reset the FIFO of individual data channels by setting the appropriate RCR3[CFR] bit. This should only be done when the corresponding RCR3[RCE] bit is clear.

13.10.3.3 Synchronous modes

The SAI transmitter and receiver can operate synchronously to each other.

13.10.3.3.1 Synchronous mode

The SAI transmitter and receiver can be configured to operate with synchronous bit clock and frame sync.

If the transmitter bit clock and frame sync are to be used by both the transmitter and receiver:

- The transmitter must be configured for asynchronous operation and the receiver for synchronous operation.
- In synchronous mode, the receiver is enabled only when both the transmitter and receiver are enabled.
- It is recommended that the transmitter is the last enabled and the first disabled.

If the receiver bit clock and frame sync are to be used by both the transmitter and receiver:

- The receiver must be configured for asynchronous operation and the transmitter for synchronous operation.
- In synchronous mode, the transmitter is enabled only when both the receiver and transmitter are both enabled.
- It is recommended that the receiver is the last enabled and the first disabled.

When operating in synchronous mode, only the bit clock, frame sync, and transmitter/receiver enable are shared. The transmitter and receiver otherwise operate independently, although configuration registers must be configured consistently across both the transmitter and receiver.

13.10.3.4 Frame sync configuration

When enabled, the SAI continuously transmits and/or receives frames of data. Each frame consists of a fixed number of words and each word consists of a fixed number of bits. Within each frame, any given word can be masked causing the receiver to ignore that word and the transmitter to tri-state for the duration of that word.

The frame sync signal is used to indicate the start of each frame. A valid frame sync requires a rising edge (if active high) or falling edge (if active low) to be detected and the transmitter or receiver cannot be busy with a previous frame. A valid frame sync is also ignored (slave mode) or not generated (master mode) for the first four bit clock cycles after enabling the transmitter or receiver.

The transmitter and receiver frame sync can be configured independently with any of the following options:

- Externally generated or internally generated
- Active high or active low
- Assert with the first bit in frame or asserts one bit early
- Assert for a duration between 1 bit clock and the first word length
- Frame length from 1 to 32 words per frame

- Word length to support 8 to 32 bits per word
 - First word length and remaining word lengths can be configured separately
- Words can be configured to transmit/receive MSB first or LSB first

These configuration options cannot be changed after the SAI transmitter or receiver is enabled.

13.10.3.5 Data FIFO

Each transmit and receive channel includes a FIFO of size 128 x 32-bit. The FIFO data is accessed using the SAI Transmit/Receive Data Registers.

13.10.3.5.1 Data alignment

Data in the FIFO can be aligned anywhere within the 32-bit wide register through the use of the First Bit Shifted configuration field, which selects the bit index (between 31 and 0) of the first bit shifted.

Examples of supported data alignment and the required First Bit Shifted configuration are illustrated in [Figure 13-84](#) for LSB First configurations and [Figure 13-85](#) for MSB First configurations.

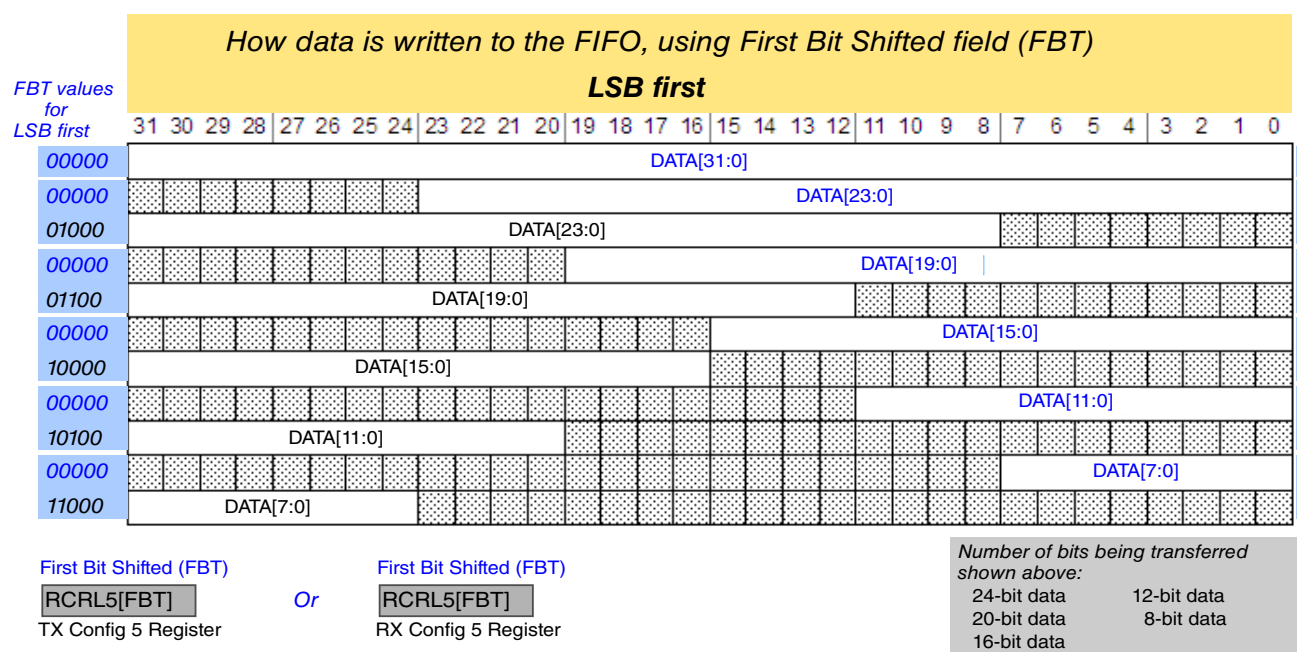


Figure 13-84. SAI first bit shifted, LSB first

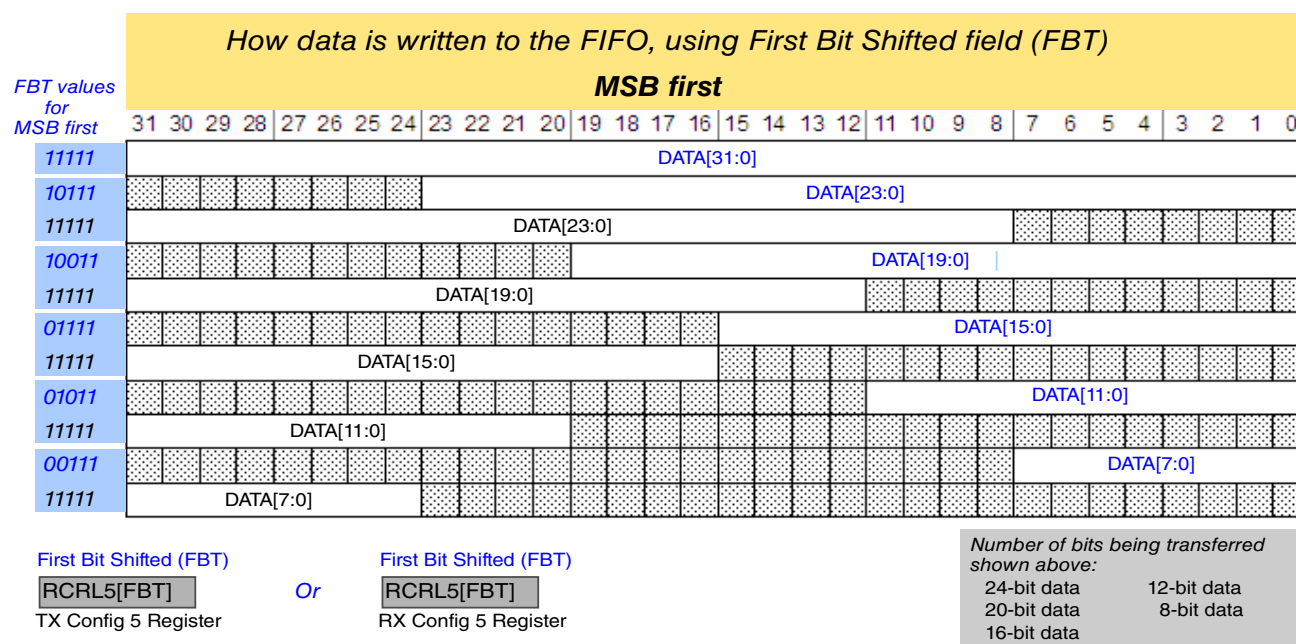


Figure 13-85. SAI first bit shifted, MSB first

13.10.3.5.2 FIFO pointers

When writing to a Transmit Data Register (TDR_n), the Write FIFO Pointer (WFP) of the corresponding Transmit FIFO Register (TFR_n) increments after each valid write. The SAI supports 8-bit, 16-bit and 32-bit writes to the Transmit Data Register and the FIFO pointer will increment after each individual write. Note that 8-bit writes should only be used when transmitting up to 8-bit data; 16-bit writes should only be used when transmitting up to 16-bit data.

- If the Transmit FIFO is full, then writes to a Transmit Data Register are ignored.
- If the Transmit FIFO is empty, then to avoid a FIFO underrun, the Transmit Data Register must be written at least 3 bit clocks before the start of the next unmasked word. Before enabling the transmitter, the Transmit FIFO should be initialized with data (since after the transmitter is enabled, the transmitter will start a new frame, and if no data is in the FIFO, then the transmitter will immediately give an error).

When reading a Receive Data Register (RDR_n), the Read FIFO Pointer (RFP) of the corresponding Receive FIFO Register (RFR_n) increments after each valid read. The SAI supports 8-bit, 16-bit and 32-bit reads from the RDR and the FIFO pointer will increment after each individual read. Note that 8-bit reads should only be used when receiving up to 8-bit data; 16-bit reads should only be used when receiving up to 16-bit data.

- If the Receive FIFO is empty, then reads from a Receive Data Register are ignored.
- If the Receive FIFO is full, then to avoid a FIFO overrun, the Receive Data Register must be read at least 3 bit clocks before the end of an unmasked word.

13.10.3.5.3 FIFO packing

FIFO packing supports storing multiple 8-bit or 16-bit data words in one 32-bit FIFO word for the transmitter and/or receiver. While this can be emulated by adjusting the number of bits per word and number of words per frame (for example, one 32-bit word per frame versus two 16-bit words per frame), FIFO packing does not require even multiples of words per frame and fully supports word masking. When FIFO packing is enabled, the FIFO pointers only increment when the full 32-bit FIFO word has been written (transmit) or read (receive) by software, supporting scenarios where different words within each frame are loaded/stored in different areas of memory.

When 16-bit FIFO packing is enabled for transmit, the transmit shift register is loaded at the start of each frame and after every second unmasked transmit word. The first word transmitted is taken from 16-bit word at byte offset \$0 (first bit is selected by TCFG5[FBT] must be configured within this 16-bit word) and the second word transmitted is taken from the 16-bit word at byte offset \$2 (first bit is selected by TCSR5[FBT][3:0]). The transmitter will transmit logic zero until the start of the next word once the 16-bit word has been transmitted.

When 16-bit FIFO packing is enabled for receive, the receive shift register is stored after every second unmasked received word, and at the end of each frame if there is an odd number of unmasked received words in each frame. The first word received is stored in the 16-bit word at byte offset \$0 (first bit is selected by RCFG5[FBT] and must be configured within this 16-bit word) and the second word received is stored in the 16-bit word at byte offset \$2 (first bit is selected by RCSR5[FBT][3:0]). The receiver will ignore received data until the start of the next word once the 16-bit word has been received.

The 8-bit FIFO packing is similar to 16-bit packing except four words are loaded or stored into each 32-bit FIFO word. The first word is loaded/stored in byte offset \$0, second word in byte offset \$1, third word in byte offset \$2 and fourth word in byte offset \$3. The TCFG5[FBT] and/or RCFG5[FBT] must be configured within byte offset \$0.

13.10.3.5.4 FIFO Combine

FIFO combining mode allows the separate FIFOs for multiple data channels to be used as a single FIFO for either software accesses or a single data channel or both. Note that the enabled data channels must be contiguous and data channel 0 must be enabled when FIFO Combine mode is enabled.

Combining FIFOs for software access (writing transmit FIFO registers, reading receive FIFO registers) allows a DMA controller or software to read or write multiple FIFOs without incrementing the address that is accessed. Once enabled, the first software access

to a FIFO register will access the first enabled channel FIFO, while the second access to a FIFO register will access the second enabled channel FIFO. This continues until software accesses the last enabled channel FIFO and the pointer resets back to the first enabled channel FIFO. To reset the pointer manually, software can reset the FIFOs or disable the FIFO combining on software accesses.

Combining FIFOs for transmit data channels allows one data channel to use the FIFOs of all enabled channel FIFOs, with identical data output on each enabled data channel. The transmit shift registers for all enabled data channels are loaded at the start of each frame and every N unmasked words (where N is the number of enabled data channels). The first word transmitted is loaded from the first enabled channel FIFO, while the second word transmitted is loaded from the second enabled channel FIFO, and so on until the end of the frame. Since the first word in each frame is always loaded from the first enabled data channel, it is recommended that the number of unmasked words in each frame is evenly divisible by the number of enabled data channels.

Combining FIFOs for receive data channels allows one data channel to use the FIFOs of all enabled channel FIFOs, with received data from channel 0 stored into each enabled data channel. The receive shift register for all enabled data channels are stored after every N unmasked words (where N is the number of enabled data channels). The first word received is stored to the first enabled channel FIFO, while the second word received is stored to the second enabled channel FIFO, and so on until the end of the frame. Since the first word in each frame is always stored the first enabled data channel, it is recommended that the number of unmasked words in each frame is evenly divisible by the number of enabled data channels.

Note that combining FIFOs for data channels will load or store each channel FIFO at the same time. This means that FIFO error conditions are only checked every N words (where N is the number of enabled data channels) and that the FIFO warning and request flags will assert if any of the enabled data channel meets the warning flag or request flag conditions.

13.10.3.6 Word mask register

The SAI transmitter and receiver each contain a word mask register, namely TMR and RMR, that can be used to mask any word in the frame. Because the word mask register is double buffered, software can update it before the end of each frame to mask a particular word in the next frame.

The TMR causes the Transmit Data pin to be tri-stated for the length of each selected word and the transmit FIFO is not read for masked words.

The RMR causes the received data for each selected word to be discarded and not written to the receive FIFO.

13.10.3.7 Interrupts and DMA requests

The SAI transmitter and receiver generate separate interrupts and separate DMA requests, but support the same status flags.

13.10.3.7.1 FIFO request flag

The FIFO request flag is set based on the number of entries in the FIFO and the FIFO watermark configuration.

The transmit FIFO request flag is set when the number of entries in any of the enabled transmit FIFOs is less than or equal to the transmit FIFO watermark configuration and is cleared when the number of entries in each enabled transmit FIFO is greater than the transmit FIFO watermark configuration.

The receive FIFO request flag is set when the number of entries in any of the enabled receive FIFOs is greater than the receive FIFO watermark configuration and is cleared when the number of entries in each enabled receive FIFO is less than or equal to the receive FIFO watermark configuration.

The FIFO request flag can generate an interrupt or a DMA request.

13.10.3.7.2 FIFO warning flag

The FIFO warning flag is set based on the number of entries in the FIFO.

The transmit warning flag is set when the number of entries in any of the enabled transmit FIFOs is empty and is cleared when the number of entries in each enabled transmit FIFO is not empty.

The receive warning flag is set when the number of entries in any of the enabled receive FIFOs is full and is cleared when the number of entries in each enabled receive FIFO is not full.

The FIFO warning flag can generate an Interrupt or a DMA request.

13.10.3.7.3 FIFO error flag

The transmit FIFO error flag is set when the any of the enabled transmit FIFOs underflow. After it is set, all enabled transmit channels will transmit zero data before TCSR[FEF] is cleared.

When TCR4[FCONT] is set, the FIFO will continue transmitting data following an underflow without software intervention. To ensure that data is transmitted in the correct order, the transmitter will continue from the same word number in the frame that caused the FIFO to underflow, but only after new data has been written to the transmit FIFO. Software should still clear the TCSR[FEF] flag, but without reinitializing the transmit FIFOs.

RCSR[FEF] is set when any of the enabled receive FIFOs overflow. After it is set, all enabled receive channels discard received data until RCSR[FEF] is cleared and the next receive frame starts. All enabled receive FIFOs should be emptied before RCSR[FEF] is cleared.

When RCR4[FCONT] is set, the FIFO will continue receiving data following an overflow without software intervention. To ensure that data is received in the correct order, the receiver will continue from the same word number in the frame that caused the FIFO to overflow, but only after data has been read from the receive FIFO. Software should still clear the RCSR[FEF] flag, but without emptying the receive FIFOs.

The FIFO error flag can generate only an interrupt.

13.10.3.7.4 Sync error flag

The sync error flag, TCSR[SEF] or RCSR[SEF], is set when configured for an externally generated frame sync and the external frame sync asserts when the transmitter or receiver is busy with the previous frame. The external frame sync assertion is ignored and the sync error flag is set. When the sync error flag is set, the transmitter or receiver continues checking for frame sync assertion when idle or at the end of each frame.

The sync error flag can generate an interrupt only.

13.10.3.7.5 Word start flag

The word start flag is set at the start of the second bit clock for the selected word, as configured by the Word Flag register field.

The word start flag can generate an interrupt only.

13.10.4 Memory map and register definition

A read or write access to an address from offset 0x100 and above will result in a bus error.

13.10.4.1 I2S register descriptions

13.10.4.1.1 I2S Memory map

SAI1 base address: 3001_0000h

SAI2 base address: 308B_0000h

SAI3 base address: 308C_0000h

SAI4 base address: 3005_0000h

SAI5 base address: 3004_0000h

SAI6 base address: 3003_0000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Version ID Register (VERID)	32	RO	0300_0000h
4h	Parameter Register (PARAM)	32	RO	Table 13-223
8h	SAI Transmit Control Register (TCSR)	32	RW	0000_0000h
Ch	SAI Transmit Configuration 1 Register (TCR1)	32	RW	0000_0000h
10h	SAI Transmit Configuration 2 Register (TCR2)	32	RW	0000_0000h
14h	SAI Transmit Configuration 3 Register (TCR3)	32	RW	0000_0000h
18h	SAI Transmit Configuration 4 Register (TCR4)	32	RW	0000_0000h
1Ch	SAI Transmit Configuration 5 Register (TCR5)	32	RW	0000_0000h
20h - 3Ch	SAI Transmit Data Register (TDR0 - TDR7)	32	WORZ	See description
40h - 5Ch	SAI Transmit FIFO Register (TFR0 - TFR7)	32	RO	See description
60h	SAI Transmit Mask Register (TMR)	32	RW	0000_0000h
88h	SAI Receive Control Register (RCSR)	32	RW	0000_0000h
8Ch	SAI Receive Configuration 1 Register (RCR1)	32	RW	0000_0000h
90h	SAI Receive Configuration 2 Register (RCR2)	32	RW	0000_0000h
94h	SAI Receive Configuration 3 Register (RCR3)	32	RW	0000_0000h
98h	SAI Receive Configuration 4 Register (RCR4)	32	RW	0000_0000h
9Ch	SAI Receive Configuration 5 Register (RCR5)	32	RW	0000_0000h
A0h - BCh	SAI Receive Data Register (RDR0 - RDR7)	32	RO	See description
C0h - DCh	SAI Receive FIFO Register (RFR0 - RFR7)	32	RO	See description
E0h	SAI Receive Mask Register (RMR)	32	RW	0000_0000h

13.10.4.1.2 Version ID Register (VERID)

13.10.4.1.2.1 Offset

Register	Offset
VERID	0h

13.10.4.1.2.2 Function

Contains version numbers for the module design and feature set.

13.10.4.1.2.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MAJOR								MINOR							
W																
Reset	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FEATURE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.10.4.1.2.4 Fields

Field	Function
31-24 MAJOR	Major Version Number This read only field returns the major version number for the specification.
23-16 MINOR	Minor Version Number This read only field returns the minor version number for the specification.
15-0 FEATURE	Feature Specification Number This read only field returns the feature set number. 0000000000000000b - Standard feature set.

13.10.4.1.3 Parameter Register (PARAM)

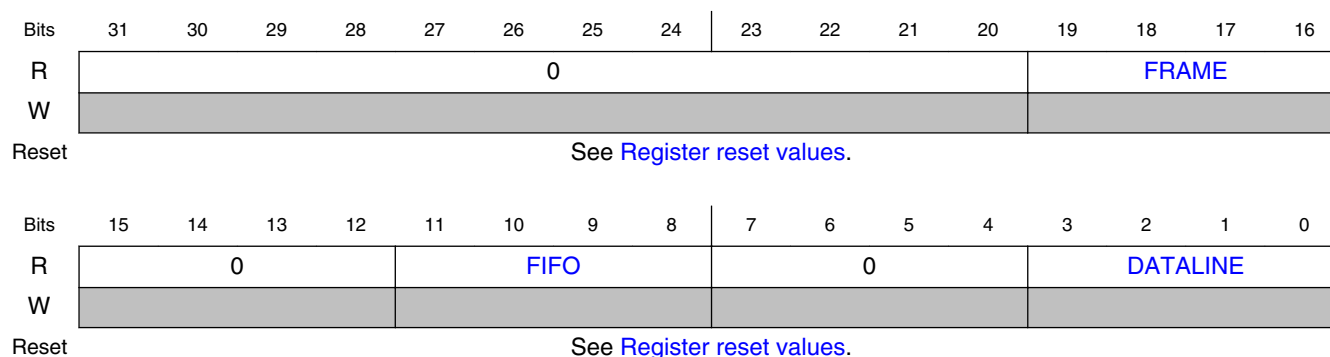
13.10.4.1.3.1 Offset

Register	Offset
PARAM	4h

13.10.4.1.3.2 Function

Contains parameter values that were implemented in the module.

13.10.4.1.3.3 Diagram



13.10.4.1.3.4 Register reset values

Register	Reset value
PARAM	: 0005_0708h —: 0005_0701h

13.10.4.1.3.5 Fields

Field	Function
31-20 —	Reserved
19-16 FRAME	Frame Size The maximum number of slots per frame is 2^{FRAME} .
15-12 —	Reserved
11-8 FIFO	FIFO Size The number of words in each FIFO is 2^{FIFO} .

Table continues on the next page...

Synchronous Audio Interface (SAI)

Field	Function
7-4 —	Reserved
3-0 DATALINE	Number of Datalines The number of datalines implemented.

13.10.4.1.4 SAI Transmit Control Register (TCSR)

13.10.4.1.4.1 Offset

Register	Offset
TCSR	8h

13.10.4.1.4.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R					0		0		0			WS	SE	FE	FW	FR
W	TE	Reserved	DBG _E	BC _E			FR _E	SR				W1C _F	W1C _F	W1C _F		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0								0			0				
W				WSI _E	SEI _E	FEI _E	FWI _E	FRI _E							FWD _E	FRD _E
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.10.4.1.4.3 Fields

Field	Function
31 TE	Transmitter Enable Enables/disables the transmitter. When software clears this field, the transmitter remains enabled, and this bit remains set, until the end of the current frame. 0b - Transmitter is disabled. 1b - Transmitter is enabled, or transmitter has been disabled and has not yet reached end of frame.
30 —	Reserved. Software should only write zero to this reserved bit.
29 DBG _E	Debug Enable

Table continues on the next page...

Field	Function
	Enables/disables transmitter operation in Debug mode. The transmit bit clock is not affected by debug mode. 0b - Transmitter is disabled in Debug mode, after completing the current frame. 1b - Transmitter is enabled in Debug mode.
28 BCE	Bit Clock Enable Enables the transmit bit clock, separately from the TE. This field is automatically set whenever TE is set. When software clears this field, the transmit bit clock remains enabled, and this bit remains set, until the end of the current frame. 0b - Transmit bit clock is disabled. 1b - Transmit bit clock is enabled.
27-26 —	Reserved
25 FR	FIFO Reset Empties the FIFO, and sets the FIFO read and write pointers to the same value, which may or may not be zero. Reading this field will always return zero. FIFO pointers should only be reset when the transmitter is disabled or the FIFO error flag is set. 0b - No effect. 1b - FIFO reset.
24 SR	Software Reset When set, resets the internal transmitter logic including the FIFO read and write pointers. Software-visible registers are not affected, except for the status registers. 0b - No effect. 1b - Software reset.
23-21 —	Reserved
20 WSF	Word Start Flag Indicates that the start of the configured word has been detected. Write a logic 1 to this field to clear this flag. 0b - Start of word not detected. 1b - Start of word detected.
19 SEF	Sync Error Flag Indicates that an error in the externally-generated frame sync has been detected. Write a logic 1 to this field to clear this flag. 0b - Sync error not detected. 1b - Frame sync error detected.
18 FEF	FIFO Error Flag Indicates that an enabled transmit FIFO has underrun. Write a logic 1 to this field to clear this flag. 0b - Transmit underrun not detected. 1b - Transmit underrun detected.
17 FWF	FIFO Warning Flag Indicates that an enabled transmit FIFO is empty. 0b - No enabled transmit FIFO is empty. 1b - Enabled transmit FIFO is empty.
16 FRF	FIFO Request Flag Indicates that the number of words in an enabled transmit channel FIFO is less than or equal to the transmit FIFO watermark. 0b - Transmit FIFO watermark has not been reached. 1b - Transmit FIFO watermark has been reached.

Table continues on the next page...

Synchronous Audio Interface (SAI)

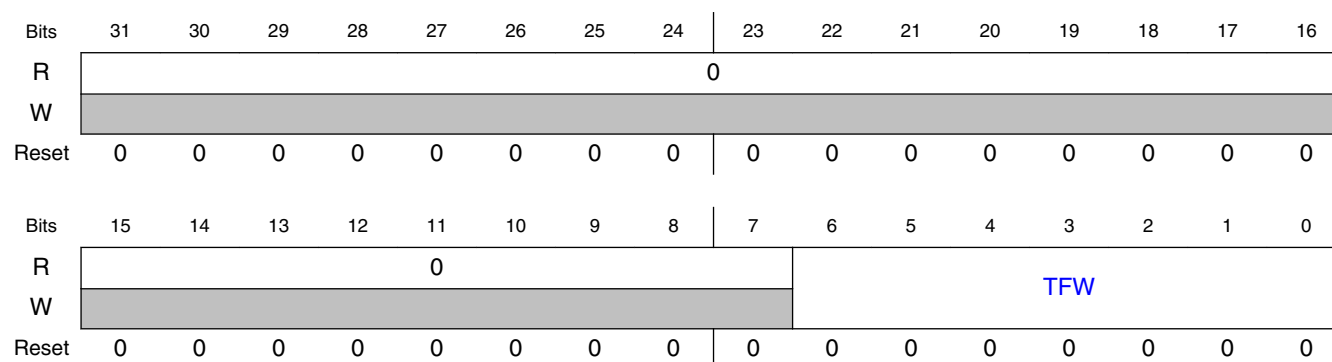
Field	Function
15-13 —	Reserved
12 WSIE	Word Start Interrupt Enable Enables/disables word start interrupts. 0b - Disables interrupt. 1b - Enables interrupt.
11 SEIE	Sync Error Interrupt Enable Enables/disables sync error interrupts. 0b - Disables interrupt. 1b - Enables interrupt.
10 FEIE	FIFO Error Interrupt Enable Enables/disables FIFO error interrupts. 0b - Disables the interrupt. 1b - Enables the interrupt.
9 FWIE	FIFO Warning Interrupt Enable Enables/disables FIFO warning interrupts. 0b - Disables the interrupt. 1b - Enables the interrupt.
8 FRIE	FIFO Request Interrupt Enable Enables/disables FIFO request interrupts. 0b - Disables the interrupt. 1b - Enables the interrupt.
7-5 —	Reserved
4-2 —	Reserved
1 FWDE	FIFO Warning DMA Enable Enables/disables DMA requests. 0b - Disables the DMA request. 1b - Enables the DMA request.
0 FRDE	FIFO Request DMA Enable Enables/disables DMA requests. 0b - Disables the DMA request. 1b - Enables the DMA request.

13.10.4.1.5 SAI Transmit Configuration 1 Register (TCR1)

13.10.4.1.5.1 Offset

Register	Offset
TCR1	Ch

13.10.4.1.5.2 Diagram



13.10.4.1.5.3 Fields

Field	Function
31-7 —	Reserved
6-0 TFW	Transmit FIFO Watermark Configures the watermark level for all enabled transmit channels.

13.10.4.1.6 SAI Transmit Configuration 2 Register (TCR2)

13.10.4.1.6.1 Offset

Register	Offset
TCR2	10h

13.10.4.1.6.2 Function

This register must not be altered when TCSR[TE] is set.

13.10.4.1.6.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SYNC		BCS	BCI	MSEL		BGP	BCD	0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0								DIV							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.10.4.1.6.4 Fields

Field	Function
31-30 SYNC	Synchronous Mode Configures between asynchronous and synchronous modes of operation. When configured for a synchronous mode of operation, the receiver must be configured for asynchronous operation. 00b - Asynchronous mode. 01b - Synchronous with receiver. 10b - Reserved. 11b - Reserved.
29 BCS	Bit Clock Swap This field swaps the bit clock used by the transmitter. When the transmitter is configured in asynchronous mode and this bit is set, the transmitter is clocked by the receiver bit clock (RX_BCLK). This allows the transmitter and receiver to share the same bit clock, but the transmitter continues to use the transmit frame sync (TX_SYNC). When the transmitter is configured in synchronous mode, the transmitter BCS field and receiver BCS field must be set to the same value. When both are set, the transmitter and receiver are both clocked by the transmitter bit clock (TX_BCLK) but use the receiver frame sync (RX_SYNC). 0b - Use the normal bit clock source. 1b - Swap the bit clock source.
28 BCI	Bit Clock Input When this field is set and using an internally generated bit clock in either synchronous or asynchronous mode, the bit clock actually used by the transmitter is delayed by the pad output delay (the transmitter is clocked by the pad input as if the clock was externally generated). This has the effect of decreasing the data input setup time, but increasing the data output valid time. The slave mode timing from the datasheet should be used for the transmitter when this bit is set. In synchronous mode, this bit allows the transmitter to use the slave mode timing from the datasheet, while the receiver uses the master mode timing. This field has no effect when configured for an externally generated bit clock. 0b - No effect. 1b - Internal logic is clocked as if bit clock was externally generated.
27-26 MSEL	MCLK Select Selects the audio Master Clock option used to generate an internally generated bit clock. This field has no effect when configured for an externally generated bit clock.

Table continues on the next page...

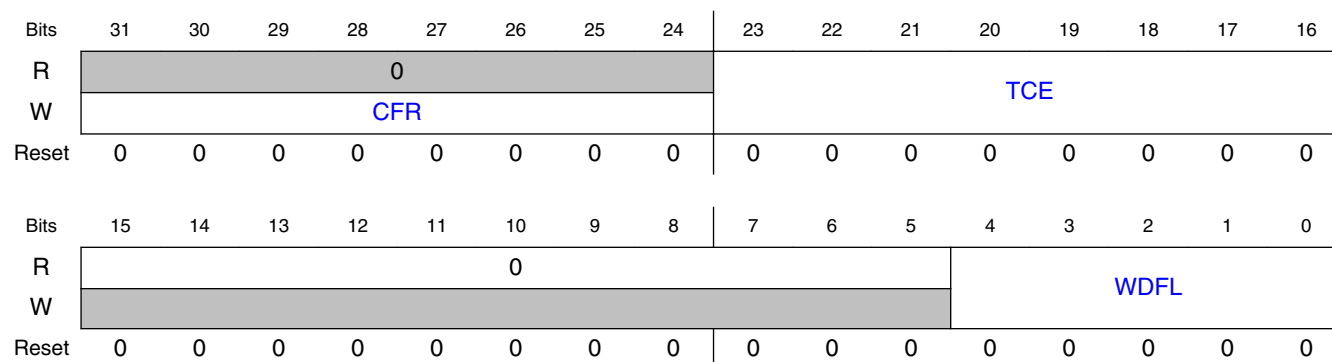
Field	Function
	NOTE: Depending on the device, some Master Clock options might not be available. See the chip-specific information for the meaning of each option. 00b - Bus Clock selected. 01b - Master Clock (MCLK) 1 option selected. 10b - Master Clock (MCLK) 2 option selected. 11b - Master Clock (MCLK) 3 option selected.
25 BCP	Bit Clock Polarity Configures the polarity of the bit clock. 0b - Bit clock is active high with drive outputs on rising edge and sample inputs on falling edge. 1b - Bit clock is active low with drive outputs on falling edge and sample inputs on rising edge.
24 BCD	Bit Clock Direction Configures the direction of the bit clock. 0b - Bit clock is generated externally in Slave mode. 1b - Bit clock is generated internally in Master mode.
23-8 —	Reserved
7-0 DIV	Bit Clock Divide Divides down the audio master clock to generate the bit clock when configured for an internal bit clock. The division value is $(DIV + 1) * 2$.

13.10.4.1.7 SAI Transmit Configuration 3 Register (TCR3)

13.10.4.1.7.1 Offset

Register	Offset
TCR3	14h

13.10.4.1.7.2 Diagram



13.10.4.1.7.3 Fields

Field	Function														
31-24 CFR	Channel FIFO Reset Resets the FIFO pointers for a specific channel. Reading this field will always return zero. FIFO pointers should only be reset when a channel is disabled or the FIFO error flag is set. The width of CFR field = the number of transmit channels (call it N). For example, if CFR is 2 bits wide, then bit position 24 refers to transmit channel 1 FIFO pointer and bit position 25 refers to transmit channel 2 FIFO pointer. Setting bit 24 resets transmit channel 1 FIFO pointer, and setting bit 25 enables transmit channel 2 FIFO pointer. Setting bit N will reset transmit channel N FIFO pointer. 0b - No effect. 1b - Transmit data channel N FIFO is reset. NOTE: This field is not supported in every instance. The following table includes only supported registers. <table> <tr> <th>Field supported in</th><th>Field not supported in</th></tr> <tr> <td>SAI1_TCR3</td><td>—</td></tr> <tr> <td>—</td><td>SAI2_TCR3</td></tr> <tr> <td>—</td><td>SAI3_TCR3</td></tr> <tr> <td>—</td><td>SAI4_TCR3</td></tr> <tr> <td>—</td><td>SAI5_TCR3</td></tr> <tr> <td>—</td><td>SAI6_TCR3</td></tr> </table>	Field supported in	Field not supported in	SAI1_TCR3	—	—	SAI2_TCR3	—	SAI3_TCR3	—	SAI4_TCR3	—	SAI5_TCR3	—	SAI6_TCR3
Field supported in	Field not supported in														
SAI1_TCR3	—														
—	SAI2_TCR3														
—	SAI3_TCR3														
—	SAI4_TCR3														
—	SAI5_TCR3														
—	SAI6_TCR3														
23-16 TCE	Transmit Channel Enable Enables the corresponding data channel for transmit operation. Changing TCE field will take effect immediately for generating the FIFO request and warning flags, but at the end of each frame for transmit operation. The width of TCE field = the number of transmit channels (call it N). For example, if TCE field is 2 bits wide, then bit position 16 refers to transmit channel 1 and bit position 17 refers to transmit channel 2. Setting bit 16 enables transmit channel 1, and setting bit 17 enables transmit channel 2. Setting bit N will enable transmit channel N. 0b - Transmit data channel N is disabled. 1b - Transmit data channel N is enabled. NOTE: This field is not supported in every instance. The following table includes only supported registers. <table> <tr> <th>Field supported in</th><th>Field not supported in</th></tr> <tr> <td>SAI1_TCR3</td><td>—</td></tr> <tr> <td>SAI2_TCR3[16]</td><td>SAI2_TCR3[23–17]</td></tr> <tr> <td>SAI3_TCR3[16]</td><td>SAI3_TCR3[23–17]</td></tr> <tr> <td>SAI4_TCR3[16]</td><td>SAI4_TCR3[23–17]</td></tr> <tr> <td>SAI5_TCR3[16]</td><td>SAI5_TCR3[23–17]</td></tr> <tr> <td>SAI6_TCR3[16]</td><td>SAI6_TCR3[23–17]</td></tr> </table>	Field supported in	Field not supported in	SAI1_TCR3	—	SAI2_TCR3[16]	SAI2_TCR3[23–17]	SAI3_TCR3[16]	SAI3_TCR3[23–17]	SAI4_TCR3[16]	SAI4_TCR3[23–17]	SAI5_TCR3[16]	SAI5_TCR3[23–17]	SAI6_TCR3[16]	SAI6_TCR3[23–17]
Field supported in	Field not supported in														
SAI1_TCR3	—														
SAI2_TCR3[16]	SAI2_TCR3[23–17]														
SAI3_TCR3[16]	SAI3_TCR3[23–17]														
SAI4_TCR3[16]	SAI4_TCR3[23–17]														
SAI5_TCR3[16]	SAI5_TCR3[23–17]														
SAI6_TCR3[16]	SAI6_TCR3[23–17]														
15-5	Reserved														

Table continues on the next page...

Field	Function
—	
4-0 WDFL	Word Flag Configuration Configures which word sets the start of word flag. The value written must be one less than the word number. For example, writing 0 configures the first word in the frame. When configured to a value greater than TCR4[FRSZ], then the start of word flag is never set.

13.10.4.1.8 SAI Transmit Configuration 4 Register (TCR4)

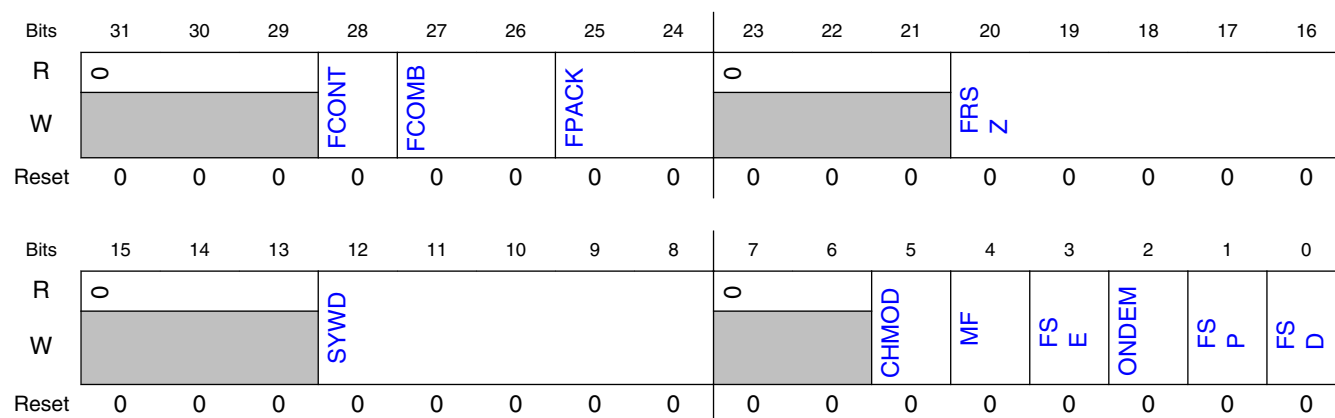
13.10.4.1.8.1 Offset

Register	Offset
TCR4	18h

13.10.4.1.8.2 Function

This register must not be altered when TCSR[TE] is set.

13.10.4.1.8.3 Diagram



13.10.4.1.8.4 Fields

Field	Function
31-29 —	Reserved
28 FCONT	FIFO Continue on Error Configures when the SAI will continue transmitting after a FIFO error has been detected.

Table continues on the next page...

Synchronous Audio Interface (SAI)

Field	Function														
	0b - On FIFO error, the SAI will continue from the start of the next frame after the FIFO error flag has been cleared. 1b - On FIFO error, the SAI will continue from the same word that caused the FIFO error to set after the FIFO warning flag has been cleared.														
27-26 FCOMB	FIFO Combine Mode When FIFO combine mode is enabled for FIFO writes, software writing to any FIFO data register will alternate the write among the enabled data channel FIFOs. For example, if two data channels are enabled then the first write will be performed to the first enabled data channel FIFO and the second write will be performed to the second enabled data channel FIFO. Resetting the FIFO or disabling FIFO combine mode for FIFO writes will reset the pointer back to the first enabled data channel. When FIFO combine mode is enabled for FIFO reads from the transmit shift registers, the transmit data channel output will alternate between the enabled data channel FIFOs. For example, if two data channels are enabled then the first unmasked word will be transmitted from the first enabled data channel FIFO and the second unmasked word will be transmitted from the second enabled data channel FIFO. Since the first word of the frame is always transmitted from the first enabled data channel FIFO, it is recommended that the number of unmasked words per frame is evenly divisible by the number of enabled data channels. NOTE: This field is not supported in every instance. The following table includes only supported registers. <table> <tr> <th>Field supported in</th><th>Field not supported in</th></tr> <tr> <td>SAI1_TCR4</td><td>—</td></tr> <tr> <td>—</td><td>SAI2_TCR4</td></tr> <tr> <td>—</td><td>SAI3_TCR4</td></tr> <tr> <td>—</td><td>SAI4_TCR4</td></tr> <tr> <td>—</td><td>SAI5_TCR4</td></tr> <tr> <td>—</td><td>SAI6_TCR4</td></tr> </table> 00b - FIFO combine mode disabled. 01b - FIFO combine mode enabled on FIFO reads (from transmit shift registers). 10b - FIFO combine mode enabled on FIFO writes (by software). 11b - FIFO combine mode enabled on FIFO reads (from transmit shift registers) and writes (by software).	Field supported in	Field not supported in	SAI1_TCR4	—	—	SAI2_TCR4	—	SAI3_TCR4	—	SAI4_TCR4	—	SAI5_TCR4	—	SAI6_TCR4
Field supported in	Field not supported in														
SAI1_TCR4	—														
—	SAI2_TCR4														
—	SAI3_TCR4														
—	SAI4_TCR4														
—	SAI5_TCR4														
—	SAI6_TCR4														
25-24 FPACK	FIFO Packing Mode Enables packing of 8-bit data or 16-bit data into each 32-bit FIFO word. If the word size is greater than 8-bit or 16-bit then only the first 8-bit or 16-bits are loaded from the FIFO. The first word in each frame always starts with a new 32-bit FIFO word and the first bit shifted must be configured within the first packed word. When FIFO packing is enabled, the FIFO write pointer will only increment when the full 32-bit FIFO word has been written by software. 00b - FIFO packing is disabled 01b - Reserved 10b - 8-bit FIFO packing is enabled 11b - 16-bit FIFO packing is enabled														
23-21 —	Reserved														
20-16 FRSZ	Frame size Configures the number of words in each frame. The value written must be one less than the number of words in the frame. For example, write 0 for one word per frame. The maximum supported frame size is 32 words.														

Table continues on the next page...

Field	Function
15-13 —	Reserved
12-8 SYWD	Sync Width Configures the length of the frame sync in number of bit clocks. The value written must be one less than the number of bit clocks. For example, write 0 for the frame sync to assert for one bit clock only. The sync width cannot be configured longer than the first word of the frame.
7-6 —	Reserved
5 CHMOD	Channel Mode Configures if transmit data pins are configured for TDM mode or Output mode. 0b - TDM mode, transmit data pins are tri-stated when slots are masked or channels are disabled. 1b - Output mode, transmit data pins are never tri-stated and will output zero when slots are masked or channels are disabled.
4 MF	MSB First Configures whether the LSB or the MSB is transmitted first. 0b - LSB is transmitted first. 1b - MSB is transmitted first.
3 FSE	Frame Sync Early 0b - Frame sync asserts with the first bit of the frame. 1b - Frame sync asserts one bit before the first bit of the frame.
2 ONDEM	On Demand Mode When set, and the frame sync is generated internally, a frame sync is only generated when the FIFO warning flag is clear. 0b - Internal frame sync is generated continuously. 1b - Internal frame sync is generated when the FIFO warning flag is clear.
1 FSP	Frame Sync Polarity Configures the polarity of the frame sync. 0b - Frame sync is active high. 1b - Frame sync is active low.
0 FSD	Frame Sync Direction Configures the direction of the frame sync. 0b - Frame sync is generated externally in Slave mode. 1b - Frame sync is generated internally in Master mode.

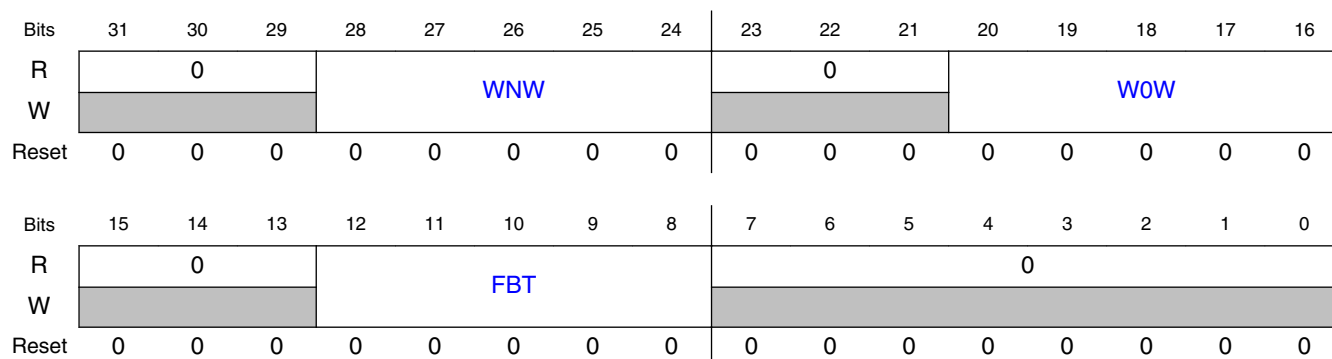
13.10.4.1.9 SAI Transmit Configuration 5 Register (TCR5)

13.10.4.1.9.1 Offset

Register	Offset
TCR5	1Ch

13.10.4.1.9.2 Function

This register must not be altered when TCSR[TE] is set.

13.10.4.1.9.3 Diagram**13.10.4.1.9.4 Fields**

Field	Function
31-29 —	Reserved
28-24 WNW	Word N Width Configures the number of bits in each word, for each word except the first in the frame. The value written must be one less than the number of bits per word. Word width of less than 8 bits is not supported.
23-21 —	Reserved
20-16 WOW	Word 0 Width Configures the number of bits in the first word in each frame. The value written must be one less than the number of bits in the first word. Word width of less than 8 bits is not supported if there is only one word per frame.
15-13 —	Reserved
12-8 FBT	First Bit Shifted Configures the bit index for the first bit transmitted for each word in the frame. If configured for MSB First, the index of the next bit transmitted is one less than the current bit transmitted. If configured for LSB First, the index of the next bit transmitted is one more than the current bit transmitted. The value written must be greater than or equal to the word width when configured for MSB First. The value written must be less than or equal to 31-word width when configured for LSB First.
7-0 —	Reserved

13.10.4.1.10 SAI Transmit Data Register (TDR0 - TDR7)

13.10.4.1.10.1 Offset

For a = 0 to 7:

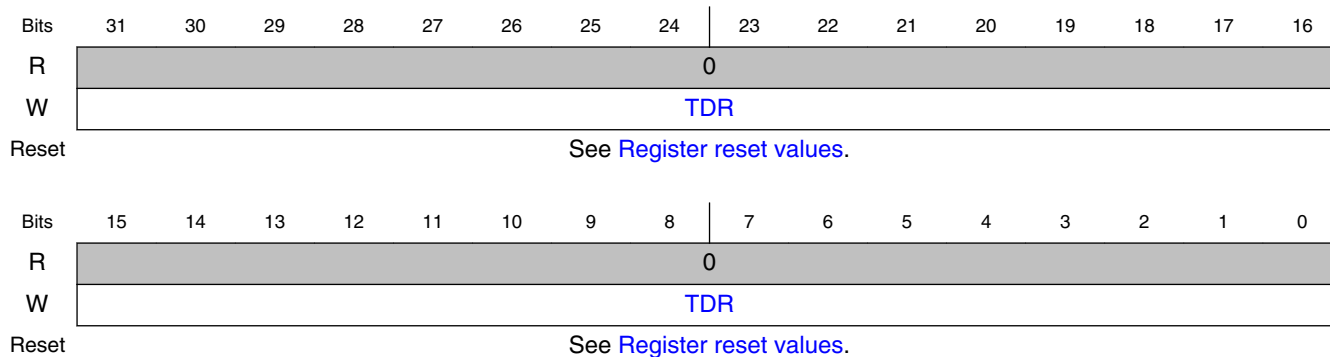
Register	Offset
TDRa	20h + (a × 4h)

NOTE

Each module instance supports a different number of registers.

Register supported	Register not supported
SAI1_T DR0– TDR7	—
SAI2_T DR0	SAI2_T DR1– TDR7
SAI3_T DR0	SAI3_T DR1– TDR7
SAI4_T DR0	SAI4_T DR1– TDR7
SAI5_T DR0	SAI5_T DR1– TDR7
SAI6_T DR0	SAI6_T DR1– TDR7

13.10.4.1.10.2 Diagram



13.10.4.1.10.3 Register reset values

Register	Reset value
TDR0	–: 0000_0000h
TDR1–TDR7	0000_0000h

13.10.4.1.10.4 Fields

Field	Function
31-0	Transmit Data Register
TDR	Writes to this register when the transmit FIFO is not full will push the data written into the transmit data FIFO. Writes to this register when the transmit FIFO is full are ignored.

13.10.4.1.11 SAI Transmit FIFO Register (TFR0 - TFR7)

13.10.4.1.11.1 Offset

For a = 0 to 7:

Register	Offset
TFRa	40h + (a × 4h)

13.10.4.1.11.2 Function

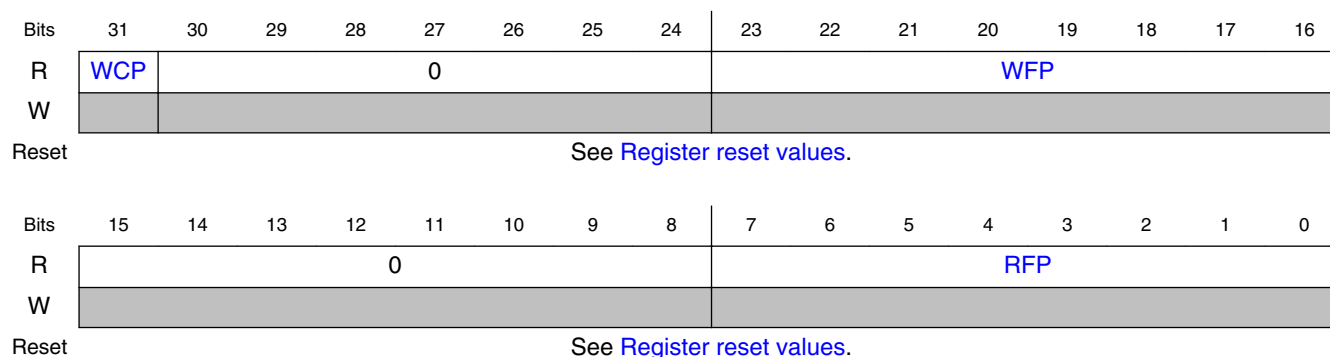
The MSB of the read and write pointers is used to distinguish between FIFO full and empty conditions. If the read and write pointers are identical, then the FIFO is empty. If the read and write pointers are identical except for the MSB, then the FIFO is full.

NOTE

Each module instance supports a different number of registers.

Register supported	Register not supported
SAI1_T FR0– TFR7	—
SAI2_T FR0	SAI2_T FR1– TFR7
SAI3_T FR0	SAI3_T FR1– TFR7
SAI4_T FR0	SAI4_T FR1– TFR7
SAI5_T FR0	SAI5_T FR1– TFR7
SAI6_T FR0	SAI6_T FR1– TFR7

13.10.4.1.11.3 Diagram



13.10.4.1.11.4 Register reset values

Register	Reset value
TFR0	—: 0000_0000h
TFR1–TFR7	0000_0000h

13.10.4.1.11.5 Fields

Field	Function														
31 WCP	Write Channel Pointer When FIFO Combine mode is enabled for writes, indicates that this data channel is the next FIFO to be written. NOTE: This field is not supported in every instance. The following table includes only supported registers. <table> <tr> <th>Field supported in</th><th>Field not supported in</th></tr> <tr> <td>SAI1_TFR0–TFR7</td><td>—</td></tr> <tr> <td>—</td><td>SAI2_TFR0</td></tr> <tr> <td>—</td><td>SAI3_TFR0</td></tr> <tr> <td>—</td><td>SAI4_TFR0</td></tr> <tr> <td>—</td><td>SAI5_TFR0</td></tr> <tr> <td>—</td><td>SAI6_TFR0</td></tr> </table> 0b - No effect. 1b - FIFO combine is enabled for FIFO writes and this FIFO will be written on the next FIFO write.	Field supported in	Field not supported in	SAI1_TFR0–TFR7	—	—	SAI2_TFR0	—	SAI3_TFR0	—	SAI4_TFR0	—	SAI5_TFR0	—	SAI6_TFR0
Field supported in	Field not supported in														
SAI1_TFR0–TFR7	—														
—	SAI2_TFR0														
—	SAI3_TFR0														
—	SAI4_TFR0														
—	SAI5_TFR0														
—	SAI6_TFR0														
30-24 —	Reserved														
23-16 WFP	Write FIFO Pointer FIFO write pointer for transmit data channel.														
15-8 —	Reserved														
7-0 RFP	Read FIFO Pointer FIFO read pointer for transmit data channel.														

13.10.4.1.12 SAI Transmit Mask Register (TMR)

13.10.4.1.12.1 Offset

Register	Offset
TMR	60h

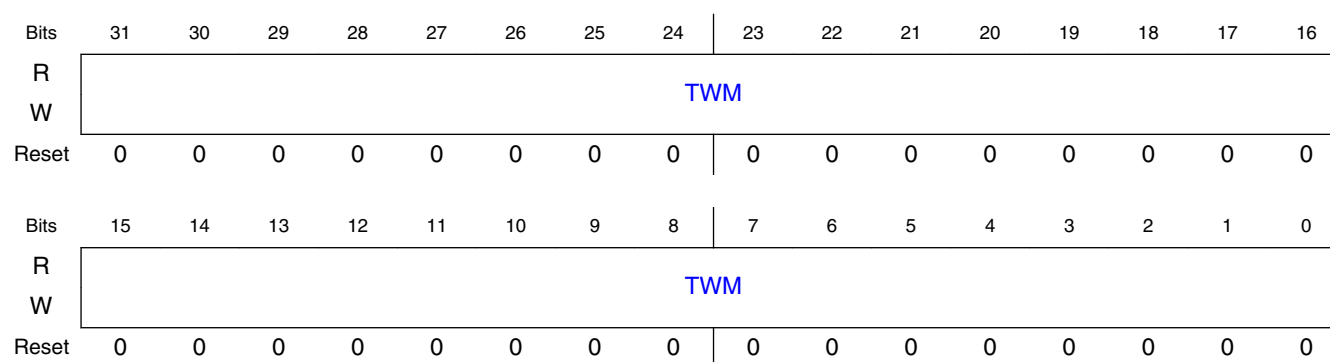
13.10.4.1.12.2 Function

This register is double-buffered and updates:

1. When TCSR[TE] is first set
2. At the end of each frame.

This allows the masked words in each frame to change from frame to frame.

13.10.4.1.12.3 Diagram



13.10.4.1.12.4 Fields

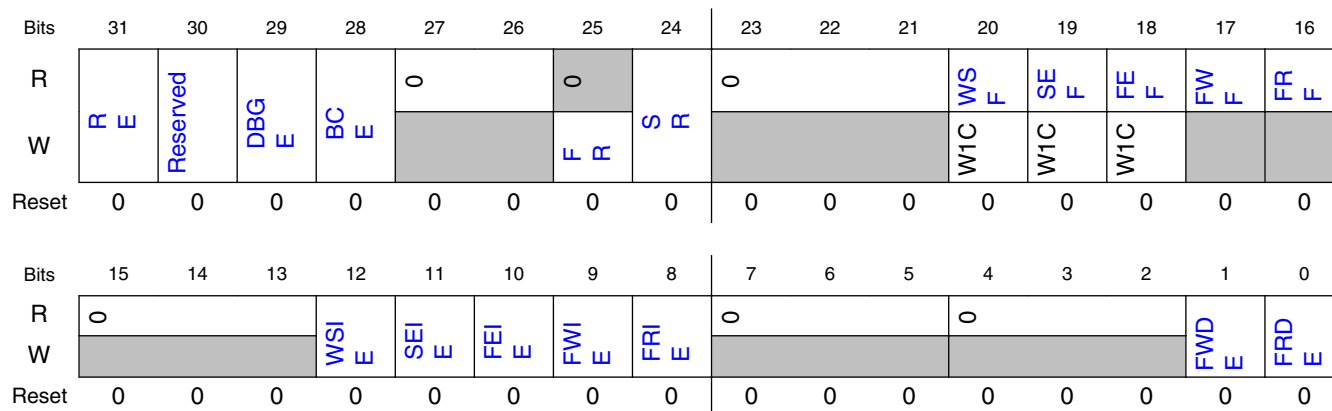
Field	Function
31-0 TWM	Transmit Word Mask Configures whether the transmit word is masked (transmit data pins are tri-stated or drive zero and transmit data not read from FIFO) for the corresponding word in the frame. 00000000000000000000000000000000b - Word N is enabled. 00000000000000000000000000000001b - Word N is masked. The transmit data pins are tri-stated or drive zero when masked.

13.10.4.1.13 SAI Receive Control Register (RCSR)

13.10.4.1.13.1 Offset

Register	Offset
RCSR	88h

13.10.4.1.13.2 Diagram



13.10.4.1.13.3 Fields

Field	Function
31 RE	Receiver Enable Enables/disables the receiver. When software clears this field, the receiver remains enabled, and this bit remains set, until the end of the current frame. 0b - Receiver is disabled. 1b - Receiver is enabled, or receiver has been disabled and has not yet reached end of frame.
30 —	Reserved. Software should only write zero to this reserved bit.
29 DBG	Debug Enable Enables/disables receiver operation in Debug mode. The receive bit clock is not affected by Debug mode. 0b - Receiver is disabled in Debug mode, after completing the current frame. 1b - Receiver is enabled in Debug mode.
28 BCE	Bit Clock Enable Enables the receive bit clock, separately from RE. This field is automatically set whenever RE is set. When software clears this field, the receive bit clock remains enabled, and this field remains set, until the end of the current frame. 0b - Receive bit clock is disabled. 1b - Receive bit clock is enabled.
27-26 —	Reserved
25 FR	FIFO Reset Empties the FIFO, and sets the FIFO read and write pointers to the same value, which may or may not be zero. Reading this field will always return zero. FIFO pointers should only be reset when the receiver is disabled or the FIFO error flag is set. 0b - No effect. 1b - FIFO reset.
24 SR	Software Reset Resets the internal receiver logic including the FIFO pointers. Software-visible registers are not affected, except for the status registers. 0b - No effect.

Table continues on the next page...

Field	Function
	1b - Software reset.
23-21 —	Reserved
20 WSF	Word Start Flag Indicates that the start of the configured word has been detected. Write a logic 1 to this field to clear this flag. 0b - Start of word not detected. 1b - Start of word detected.
19 SEF	Sync Error Flag Indicates that an error in the externally-generated frame sync has been detected. Write a logic 1 to this field to clear this flag. 0b - Sync error not detected. 1b - Frame sync error detected.
18 FEF	FIFO Error Flag Indicates that an enabled receive FIFO has overflowed. Write a logic 1 to this field to clear this flag. 0b - Receive overflow not detected. 1b - Receive overflow detected.
17 FWF	FIFO Warning Flag Indicates that an enabled receive FIFO is full. 0b - No enabled receive FIFO is full. 1b - Enabled receive FIFO is full.
16 FRF	FIFO Request Flag Indicates that the number of words in an enabled receive channel FIFO is greater than the receive FIFO watermark. 0b - Receive FIFO watermark not reached. 1b - Receive FIFO watermark has been reached.
15-13 —	Reserved
12 WSIE	Word Start Interrupt Enable Enables/disables word start interrupts. 0b - Disables interrupt. 1b - Enables interrupt.
11 SEIE	Sync Error Interrupt Enable Enables/disables sync error interrupts. 0b - Disables interrupt. 1b - Enables interrupt.
10 FEIE	FIFO Error Interrupt Enable Enables/disables FIFO error interrupts. 0b - Disables the interrupt. 1b - Enables the interrupt.
9 FWIE	FIFO Warning Interrupt Enable Enables/disables FIFO warning interrupts. 0b - Disables the interrupt. 1b - Enables the interrupt.
8 FRIE	FIFO Request Interrupt Enable Enables/disables FIFO request interrupts.

Table continues on the next page...

Synchronous Audio Interface (SAI)

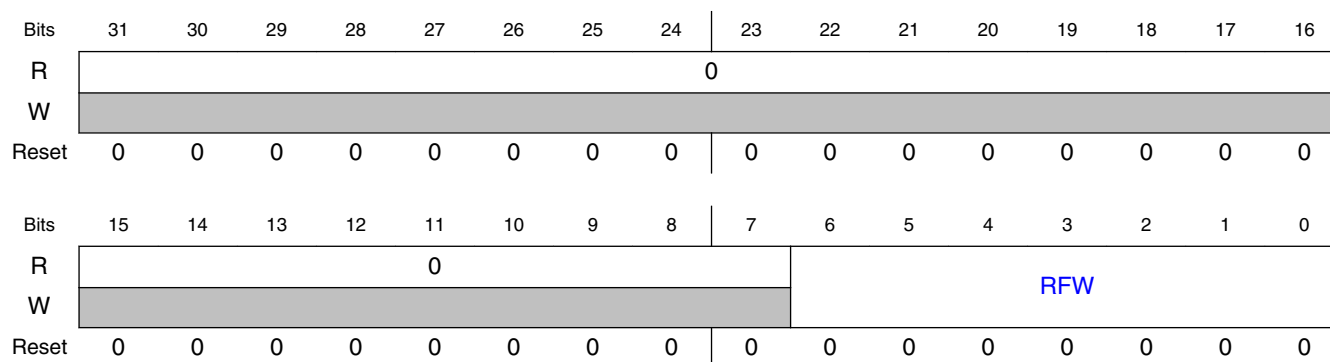
Field	Function
	0b - Disables the interrupt. 1b - Enables the interrupt.
7-5 —	Reserved
4-2 —	Reserved
1 FWDE	FIFO Warning DMA Enable Enables/disables DMA requests. 0b - Disables the DMA request. 1b - Enables the DMA request.
0 FRDE	FIFO Request DMA Enable Enables/disables DMA requests. 0b - Disables the DMA request. 1b - Enables the DMA request.

13.10.4.1.14 SAI Receive Configuration 1 Register (RCR1)

13.10.4.1.14.1 Offset

Register	Offset
RCR1	8Ch

13.10.4.1.14.2 Diagram



13.10.4.1.14.3 Fields

Field	Function
31-7	Reserved

Table continues on the next page...

Field	Function
—	
6-0	Receive FIFO Watermark
RFW	Configures the watermark level for all enabled receiver channels.

13.10.4.1.15 SAI Receive Configuration 2 Register (RCR2)

13.10.4.1.15.1 Offset

Register	Offset
RCR2	90h

13.10.4.1.15.2 Function

This register must not be altered when RCSR[RE] is set.

13.10.4.1.15.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R													0			
W	SYNC		BCS	BCI		MSEL	BCP	BCD								
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W													DIV			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.10.4.1.15.4 Fields

Field	Function
31-30	Synchronous Mode
SYNC	Configures between asynchronous and synchronous modes of operation. When configured for a synchronous mode of operation, the transmitter must be configured for asynchronous operation. 00b - Asynchronous mode. 01b - Synchronous with transmitter. 10b - Reserved. 11b - Reserved.
29	Bit Clock Swap

Table continues on the next page...

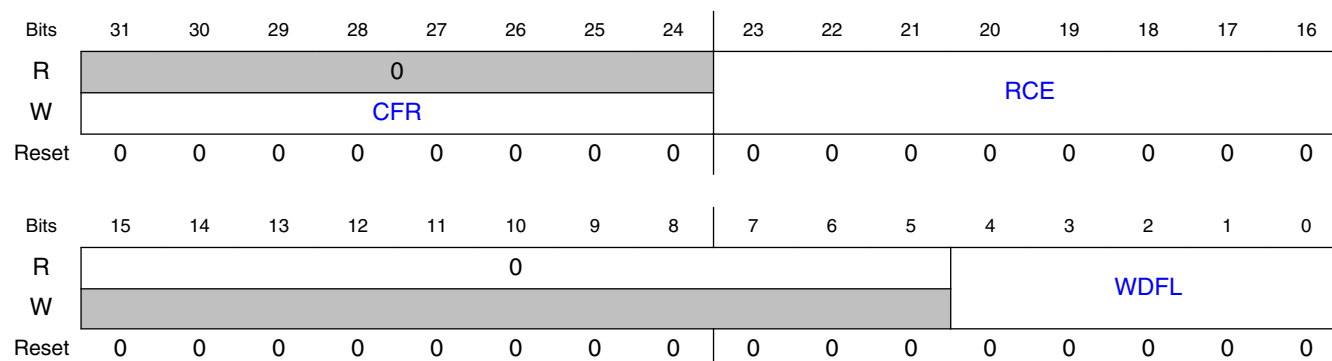
Field	Function
BCS	This field swaps the bit clock used by the receiver. When the receiver is configured in asynchronous mode and this bit is set, the receiver is clocked by the transmitter bit clock (TX_BCLK). This allows the transmitter and receiver to share the same bit clock, but the receiver continues to use the receiver frame sync (RX_SYNC). When the receiver is configured in synchronous mode, the transmitter BCS field and receiver BCS field must be set to the same value. When both are set, the transmitter and receiver are both clocked by the receiver bit clock (RX_BCLK) but use the transmitter frame sync (TX_SYNC). 0b - Use the normal bit clock source. 1b - Swap the bit clock source.
28 BCI	Bit Clock Input When this field is set and using an internally generated bit clock in either synchronous or asynchronous mode, the bit clock actually used by the receiver is delayed by the pad output delay (the receiver is clocked by the pad input as if the clock was externally generated). This has the effect of decreasing the data input setup time, but increasing the data output valid time. The slave mode timing from the datasheet should be used for the receiver when this bit is set. In synchronous mode, this bit allows the receiver to use the slave mode timing from the datasheet, while the transmitter uses the master mode timing. This field has no effect when configured for an externally generated bit clock . 0b - No effect. 1b - Internal logic is clocked as if bit clock was externally generated.
27-26 MSEL	MCLK Select Selects the audio Master Clock option used to generate an internally generated bit clock. This field has no effect when configured for an externally generated bit clock. NOTE: Depending on the device, some Master Clock options might not be available. See the chip-specific information for the availability and chip-specific meaning of each option. 00b - Bus Clock selected. 01b - Master Clock (MCLK) 1 option selected. 10b - Master Clock (MCLK) 2 option selected. 11b - Master Clock (MCLK) 3 option selected.
25 BCP	Bit Clock Polarity Configures the polarity of the bit clock. 0b - Bit Clock is active high with drive outputs on rising edge and sample inputs on falling edge. 1b - Bit Clock is active low with drive outputs on falling edge and sample inputs on rising edge.
24 BCD	Bit Clock Direction Configures the direction of the bit clock. 0b - Bit clock is generated externally in Slave mode. 1b - Bit clock is generated internally in Master mode.
23-8 —	Reserved
7-0 DIV	Bit Clock Divide Divides down the audio master clock to generate the bit clock when configured for an internal bit clock. The division value is $(DIV + 1) * 2$.

13.10.4.1.16 SAI Receive Configuration 3 Register (RCR3)

13.10.4.1.16.1 Offset

Register	Offset
RCR3	94h

13.10.4.1.16.2 Diagram



13.10.4.1.16.3 Fields

Field	Function														
31-24	Channel FIFO Reset														
CFR	Resets the FIFO pointers for a specific channel. Reading this field will always return zero. FIFO pointers should only be reset when a channel is disabled or the FIFO error flag is set. The width of CFR field = the number of receive channels (call it N). For example, if CFR is 2 bits wide, then bit position 24 refers to receive channel 1 FIFO pointer and bit position 25 refers to receive channel 2 FIFO pointer. Setting bit 24 resets receive channel 1 FIFO pointer, and setting bit 25 enables receive channel 2 FIFO pointer. Setting bit N will reset receive channel N FIFO pointer. 0b - No effect. 1b - Receive data channel N FIFO is reset. NOTE: This field is not supported in every instance. The following table includes only supported registers. <table border="1"> <tr> <th>Field supported in</th><th>Field not supported in</th></tr> <tr> <td>SAI1_RCR3</td><td>—</td></tr> <tr> <td>—</td><td>SAI2_RCR3</td></tr> <tr> <td>—</td><td>SAI3_RCR3</td></tr> <tr> <td>—</td><td>SAI4_RCR3</td></tr> <tr> <td>—</td><td>SAI5_RCR3</td></tr> <tr> <td>—</td><td>SAI6_RCR3</td></tr> </table>	Field supported in	Field not supported in	SAI1_RCR3	—	—	SAI2_RCR3	—	SAI3_RCR3	—	SAI4_RCR3	—	SAI5_RCR3	—	SAI6_RCR3
Field supported in	Field not supported in														
SAI1_RCR3	—														
—	SAI2_RCR3														
—	SAI3_RCR3														
—	SAI4_RCR3														
—	SAI5_RCR3														
—	SAI6_RCR3														
23-16	Receive Channel Enable														

Table continues on the next page...

Field	Function														
RCE	Enables the corresponding data channel for receive operation. Changing this field will take effect immediately for generating the FIFO request and warning flags, but at the end of each frame for receive operation. The width of RCE field = the number of receive channels (call it N). For example, if RCE field is 2 bits wide, then bit position 16 refers to receive channel 1 and bit position 17 refers to receive channel 2. Setting bit 16 enables receive channel 1, and setting bit 17 enables receive channel 2. Setting bit N will enable receive channel N. 0b - Receive data channel N is disabled. 1b - Receive data channel N is enabled. NOTE: This field is not supported in every instance. The following table includes only supported registers. <table> <tr> <th>Field supported in</th><th>Field not supported in</th></tr> <tr> <td>SAI1_RCR3</td><td>—</td></tr> <tr> <td>SAI2_RCR3[16]</td><td>SAI2_RCR3[23–17]</td></tr> <tr> <td>SAI3_RCR3[16]</td><td>SAI3_RCR3[23–17]</td></tr> <tr> <td>SAI4_RCR3[16]</td><td>SAI4_RCR3[23–17]</td></tr> <tr> <td>SAI5_RCR3[16]</td><td>SAI5_RCR3[23–17]</td></tr> <tr> <td>SAI6_RCR3[16]</td><td>SAI6_RCR3[23–17]</td></tr> </table>	Field supported in	Field not supported in	SAI1_RCR3	—	SAI2_RCR3[16]	SAI2_RCR3[23–17]	SAI3_RCR3[16]	SAI3_RCR3[23–17]	SAI4_RCR3[16]	SAI4_RCR3[23–17]	SAI5_RCR3[16]	SAI5_RCR3[23–17]	SAI6_RCR3[16]	SAI6_RCR3[23–17]
Field supported in	Field not supported in														
SAI1_RCR3	—														
SAI2_RCR3[16]	SAI2_RCR3[23–17]														
SAI3_RCR3[16]	SAI3_RCR3[23–17]														
SAI4_RCR3[16]	SAI4_RCR3[23–17]														
SAI5_RCR3[16]	SAI5_RCR3[23–17]														
SAI6_RCR3[16]	SAI6_RCR3[23–17]														
15-5 —	Reserved														
4-0 WDFL	Word Flag Configuration Configures which word the start of word flag is set. The value written should be one less than the word number (for example, write zero to configure for the first word in the frame). When configured to a value greater than the Frame Size field, then the start of word flag is never set.														

13.10.4.1.17 SAI Receive Configuration 4 Register (RCR4)

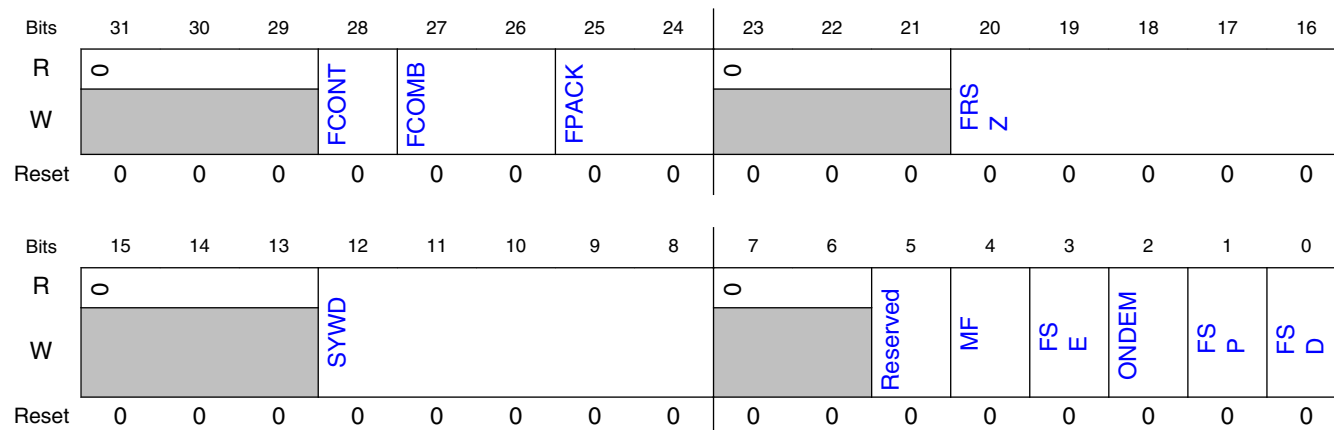
13.10.4.1.17.1 Offset

Register	Offset
RCR4	98h

13.10.4.1.17.2 Function

This register must not be altered when RCSR[RE] is set.

13.10.4.1.17.3 Diagram



13.10.4.1.17.4 Fields

Field	Function								
31-29 —	Reserved								
28 FCONT	FIFO Continue on Error Configures when the SAI will continue receiving after a FIFO error has been detected. 0b - On FIFO error, the SAI will continue from the start of the next frame after the FIFO error flag has been cleared. 1b - On FIFO error, the SAI will continue from the same word that caused the FIFO error to set after the FIFO warning flag has been cleared.								
27-26 FCOMB	FIFO Combine Mode When FIFO combine mode is enabled for FIFO reads, software reading any FIFO data register will alternate the read among the enabled data channel FIFOs. For example, if two data channels are enabled then the first read will be performed to the first enabled data channel FIFO and the second read will be performed to the second enabled data channel FIFO. Resetting the FIFO or disabling FIFO combine mode for FIFO reads will reset the pointer back to the first enabled data channel. When FIFO combine mode is enabled for FIFO writes from the receive shift registers, the first enabled data channel input will alternate between the enabled data channel FIFOs. For example, if two data channels are enabled then the first unmasked received word will be stored in the first enabled data channel FIFO and the second unmasked received word will be stored in the second enabled data channel FIFO. Since the first word of the frame is always stored in the first enabled data channel FIFO, it is recommended that the number of unmasked words per frame is evenly divisible by the number of enabled data channels. NOTE: This field is not supported in every instance. The following table includes only supported registers. <table border="1"> <tr> <th>Field supported in</th><th>Field not supported in</th></tr> <tr> <td>SAI1_RCR4</td><td>—</td></tr> <tr> <td>—</td><td>SAI2_RCR4</td></tr> <tr> <td>—</td><td>SAI3_RCR4</td></tr> </table>	Field supported in	Field not supported in	SAI1_RCR4	—	—	SAI2_RCR4	—	SAI3_RCR4
Field supported in	Field not supported in								
SAI1_RCR4	—								
—	SAI2_RCR4								
—	SAI3_RCR4								

Table continues on the next page...

Synchronous Audio Interface (SAI)

Field	Function								
	<table> <tr> <th>Field supported in</th><th>Field not supported in</th></tr> <tr> <td>—</td><td>SAI4_RCR4</td></tr> <tr> <td>—</td><td>SAI5_RCR4</td></tr> <tr> <td>—</td><td>SAI6_RCR4</td></tr> </table> 00b - FIFO combine mode disabled. 01b - FIFO combine mode enabled on FIFO writes (from receive shift registers). 10b - FIFO combine mode enabled on FIFO reads (by software). 11b - FIFO combine mode enabled on FIFO writes (from receive shift registers) and reads (by software).	Field supported in	Field not supported in	—	SAI4_RCR4	—	SAI5_RCR4	—	SAI6_RCR4
Field supported in	Field not supported in								
—	SAI4_RCR4								
—	SAI5_RCR4								
—	SAI6_RCR4								
25-24 FPACK	FIFO Packing Mode Enables packing of 8-bit data or 16-bit data into each 32-bit FIFO word. If the word size is greater than 8-bit or 16-bit then only the first 8-bit or 16-bits are stored to the FIFO. The first word in each frame always starts with a new 32-bit FIFO word and the first bit shifted must be configured within the first packed word. When FIFO packing is enabled, the FIFO read pointer will only increment when the full 32-bit FIFO word has been read by software. 00b - FIFO packing is disabled 01b - Reserved. 10b - 8-bit FIFO packing is enabled 11b - 16-bit FIFO packing is enabled								
23-21 —	Reserved								
20-16 FRSZ	Frame Size Configures the number of words in each frame. The value written must be one less than the number of words in the frame. For example, write 0 for one word per frame. The maximum supported frame size is 32 words.								
15-13 —	Reserved								
12-8 SYWD	Sync Width Configures the length of the frame sync in number of bit clocks. The value written must be one less than the number of bit clocks. For example, write 0 for the frame sync to assert for one bit clock only. The sync width cannot be configured longer than the first word of the frame.								
7-6 —	Reserved								
5 —	Reserved. Software should only write zero to this bit.								
4 MF	MSB First Configures whether the LSB or the MSB is received first. 0b - LSB is received first. 1b - MSB is received first.								
3 FSE	Frame Sync Early 0b - Frame sync asserts with the first bit of the frame. 1b - Frame sync asserts one bit before the first bit of the frame.								
2 ONDEM	On Demand Mode When set, and the frame sync is generated internally, a frame sync is only generated when the FIFO warning flag is clear.								

Table continues on the next page...

Field	Function
	0b - Internal frame sync is generated continuously. 1b - Internal frame sync is generated when the FIFO warning flag is clear.
1 FSP	Frame Sync Polarity Configures the polarity of the frame sync. 0b - Frame sync is active high. 1b - Frame sync is active low.
0 FSD	Frame Sync Direction Configures the direction of the frame sync. 0b - Frame Sync is generated externally in Slave mode. 1b - Frame Sync is generated internally in Master mode.

13.10.4.1.18 SAI Receive Configuration 5 Register (RCR5)

13.10.4.1.18.1 Offset

Register	Offset
RCR5	9Ch

13.10.4.1.18.2 Function

This register must not be altered when RCSR[RE] is set.

13.10.4.1.18.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0			WNW					0			WOW				
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0			FBT					0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

13.10.4.1.18.4 Fields

Field	Function
31-29	Reserved

Table continues on the next page...

Field	Function
—	
28-24 WNW	Word N Width Configures the number of bits in each word, for each word except the first in the frame. The value written must be one less than the number of bits per word. Word width of less than 8 bits is not supported.
23-21 —	Reserved
20-16 W0W	Word 0 Width Configures the number of bits in the first word in each frame. The value written must be one less than the number of bits in the first word. Word width of less than 8 bits is not supported if there is only one word per frame.
15-13 —	Reserved
12-8 FBT	First Bit Shifted Configures the bit index for the first bit received for each word in the frame. If configured for MSB First, the index of the next bit received is one less than the current bit received. If configured for LSB First, the index of the next bit received is one more than the current bit received. The value written must be greater than or equal to the word width when configured for MSB First. The value written must be less than or equal to 31-word width when configured for LSB First.
7-0 —	Reserved

13.10.4.1.19 SAI Receive Data Register (RDR0 - RDR7)

13.10.4.1.19.1 Offset

For a = 0 to 7:

Register	Offset
RDRa	A0h + (a × 4h)

13.10.4.1.19.2 Function

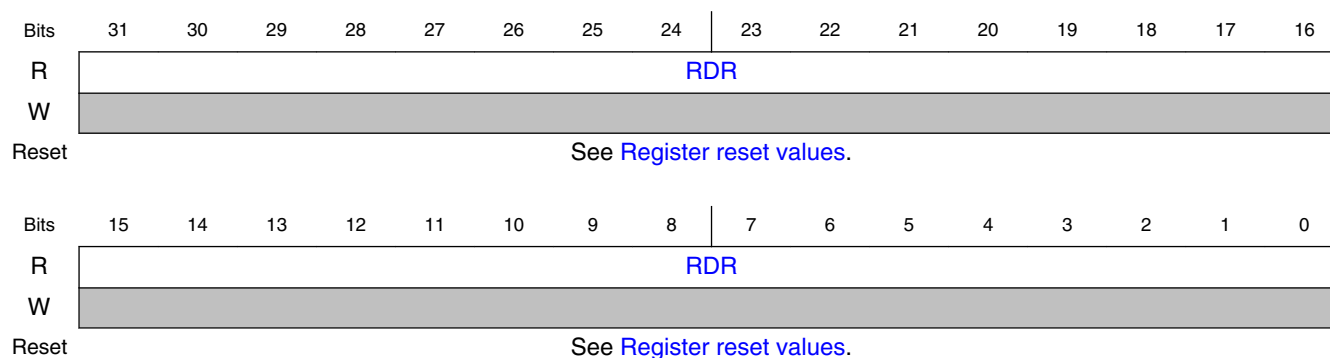
Reading this register introduces one additional peripheral clock wait state on each read.

NOTE

Each module instance supports a different number of registers.

Register supported	Register not supported
SAI1_R DR0– RDR7	—
SAI2_R DR0	SAI2_R DR1– RDR7
SAI3_R DR0	SAI3_R DR1– RDR7
SAI4_R DR0	SAI4_R DR1– RDR7
SAI5_R DR0	SAI5_R DR1– RDR7
SAI6_R DR0	SAI6_R DR1– RDR7

13.10.4.1.19.3 Diagram



13.10.4.1.19.4 Register reset values

Register	Reset value
RDR0	—: 0000_0000h
RDR1–RDR7	0000_0000h

13.10.4.1.19.5 Fields

Field	Function
31-0	Receive Data Register
RDR	Reads from this register when the receive FIFO is not empty will return the data from the top of the receive FIFO. Reads from this register when the receive FIFO is empty are ignored.

13.10.4.1.20 SAI Receive FIFO Register (RFR0 - RFR7)**13.10.4.1.20.1 Offset**

For a = 0 to 7:

Register	Offset
RFRa	C0h + (a × 4h)

13.10.4.1.20.2 Function

The MSB of the read and write pointers is used to distinguish between FIFO full and empty conditions. If the read and write pointers are identical, then the FIFO is empty. If the read and write pointers are identical except for the MSB, then the FIFO is full.

NOTE

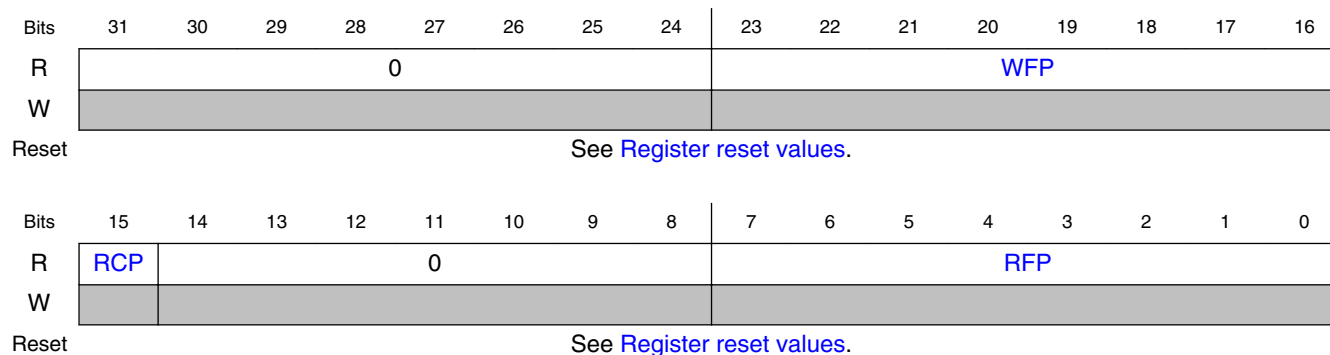
Each module instance supports a different number of registers.

Register supported	Register not supported
SAI1_R FR0– RFR7	—
SAI2_R FR0	SAI2_R FR1– RFR7
SAI3_R FR0	SAI3_R FR1– RFR7
SAI4_R FR0	SAI4_R FR1– RFR7
SAI5_R FR0	SAI5_R FR1– RFR7

Table continues on the next page...

Register supported	Register not supported
SAI6_R FR0	SAI6_R FR1– RFR7

13.10.4.1.20.3 Diagram



13.10.4.1.20.4 Register reset values

Register	Reset value
RFR0	–: 0000_0000h
RFR1–RFR7	0000_0000h

13.10.4.1.20.5 Fields

Field	Function
31-24 —	Reserved
23-16 WFP	Write FIFO Pointer FIFO write pointer for receive data channel.
15 RCP	Receive Channel Pointer When FIFO Combine mode is enabled for reads, indicates that this data channel is the next FIFO to be read. NOTE: This field is not supported in every instance. The following table includes only supported registers.

Table continues on the next page...

Field	Function	
	Field supported in	Field not supported in
	SAI1_RFR0–RFR7	—
	—	SAI2_RFR0
	—	SAI3_RFR0
	—	SAI4_RFR0
	—	SAI5_RFR0
	—	SAI6_RFR0
	0b - No effect. 1b - FIFO combine is enabled for FIFO reads and this FIFO will be read on the next FIFO read.	
14-8 —	Reserved	
7-0 RFP	Read FIFO Pointer FIFO read pointer for receive data channel.	

13.10.4.1.21 SAI Receive Mask Register (RMR)

13.10.4.1.21.1 Offset

Register	Offset
RMR	E0h

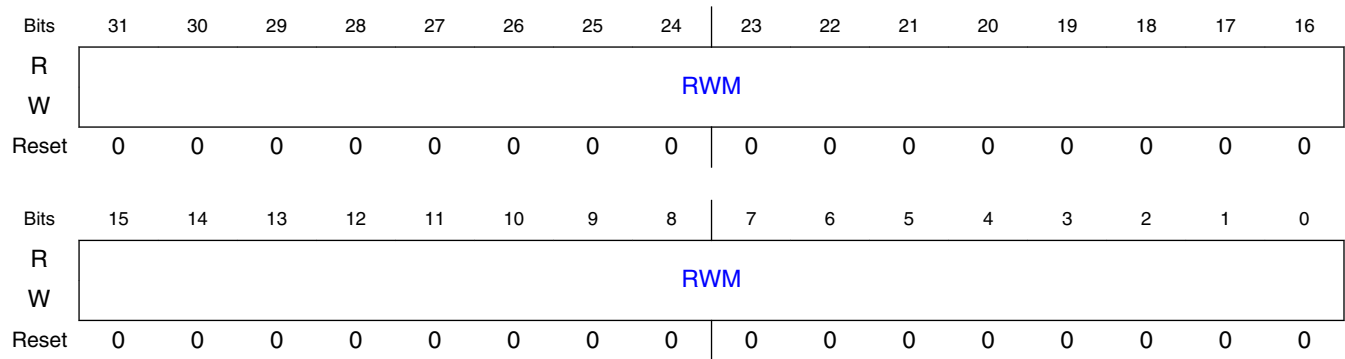
13.10.4.1.21.2 Function

This register is double-buffered and updates:

1. When RCSR[RE] is first set
2. At the end of each frame

This allows the masked words in each frame to change from frame to frame.

13.10.4.1.21.3 Diagram



13.10.4.1.21.4 Fields

Field	Function
31-0	Receive Word Mask
RWM	Configures whether the receive word is masked (received data ignored and not written to receive FIFO) for the corresponding word in the frame. 00000000000000000000000000000000b - Word N is enabled. 00000000000000000000000000000001b - Word N is masked.

Chapter 14

Video Processing Unit (VPU)

14.1 VPU G1 (VPU_G1)

14.1.1 Overview

This block details the VPU G1 hardware based decoders and API. The decoders are able to decode H.264, VP8. The decoders conform to the H.264 Baseline, Main and High profile, and VP8 Main Profile

NOTE

The overall performance and capabilities of the decoders is system dependent. Formats and features that are presented depend on the chip configuration.

14.1.2 Features (G1)

The API is implemented on top of the VPU G1 decoder H.264 and VP8. The features of the decoders are noted in the following sections.

14.1.2.1 Decoder Features

The VPU G1 decoder has the following features:

- H.264 baseline profile, levels 1-4.1 decoding
 - Byte stream input
 - NAL unit input
- VP8 decoding
- Video post-processing features
 - Frame rotation 90 degrees left/right
 - Frame mirroring horizontally/vertically
 - Frame cropping

- Frame conversion from YCbCr formats to 16-bit or 32-bit RGB formats
- Frame scaling with maximum up-scaling factor of 3
- Two rectangular or alpha blending masks for output frame

The post-processing features can be used in pipeline with the decoder. The post-processing features can also be used as stand-alone, without performing any decoding.

- H.264 main and high profile, levels 1-4.1 decoding
- VP8 decoding

14.1.3 Video Frame Storage Format

This chapter describes the different input and output picture storage formats supported by the decoder and post-processor. Notice that some of the formats are only used for input or output.

14.1.3.1 YCbCr 4:2:0 Planar Format

In the planar format each video sample component forms one memory plane. The luminance and both chrominance planes must be stored in a linear and contiguous memory block as shown in Figure 3. The luminance samples are stored in raster-scan order (Y0Y1Y2Y3Y4...). The chrominance samples are stored in two planes also in raster scan order (Cb0Cb1Cb2Cb3... and Cr0Cr1Cr2Cr3...).

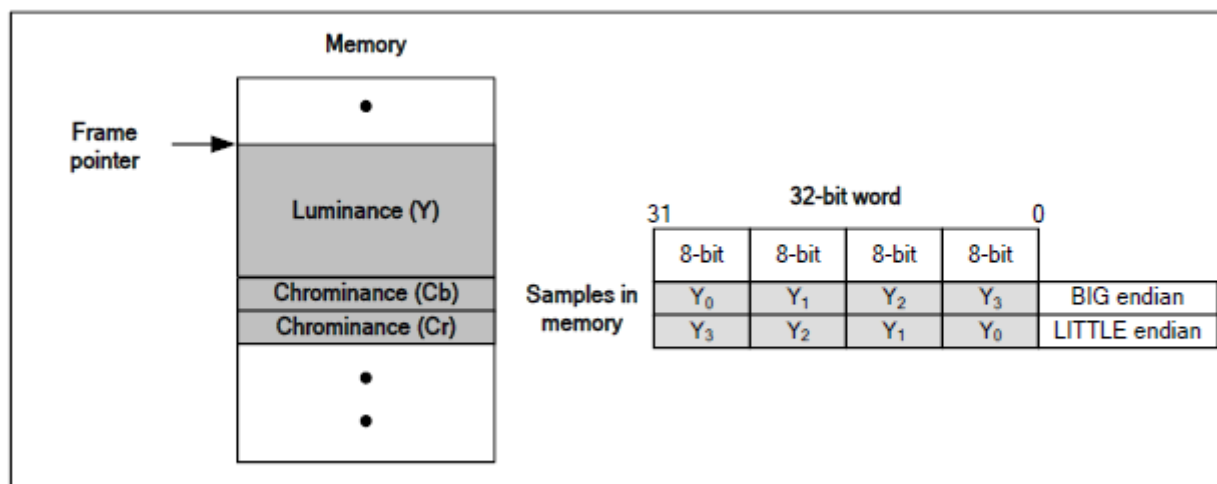


Figure 14-1. YCbCr 4:2:0 Planar Format External Memory Usage

The YCbCr 4:2:0 planar format is supported only as an input format for the postprocessor. In this format each pixel takes 12 bits of memory.

14.1.3.2 YCbCr 4:2:0 Semi-Planar Format

In semi-planar YCbCr 4:2:0 format the luminance samples form one plane in memory, and chrominance samples form another. The luminance and chrominance planes must be stored in a linear and contiguous memory block as presented in the following figure. The luminance pixels are stored in raster-scan order $Y_0Y_1Y_2Y_3Y_4\dots$. The interleaved chrominance CbCr samples are stored in raster-scan order in memory as $Cb_0Cr_0Cb_1Cr_1Cb_2Cr_2Cb_3\dots$.

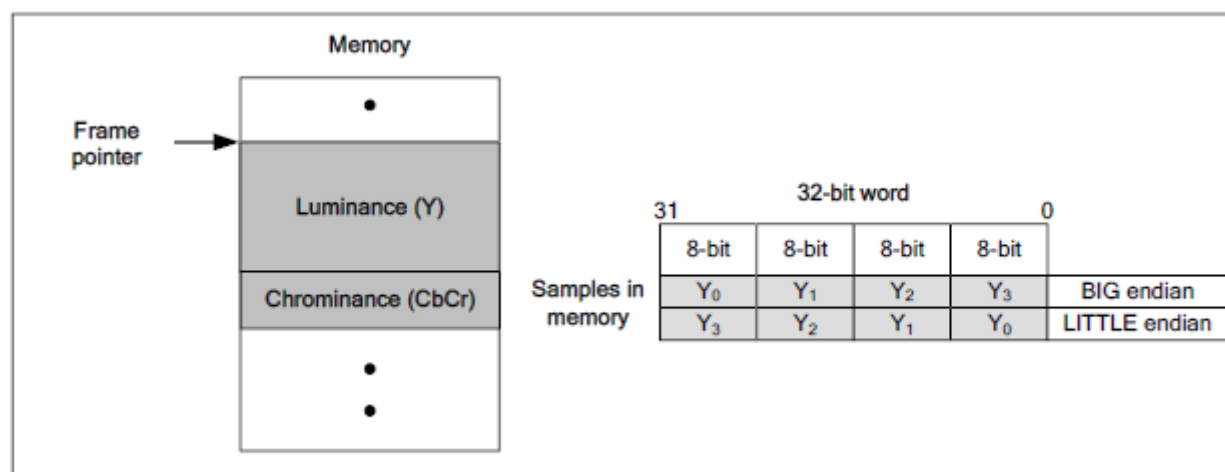


Figure 14-2. YCBCR 4:2:0 Planar Format External Memory Usage

The YCbCr 4:2:0 semi-planar format is supported both as an input and as an output format for the post-processor. In this format each pixel takes 12 bits of memory. As the chrominance components are interleaved, the bus load caused by this format can be slightly lower than with the planar format due to the reduced amount of non-sequential memory addressing.

14.1.3.3 YCbCr 4:2:2 Interleaved Format

In the interleaved YCbCr 4:2:2 format the pixel samples form a single plane in which the data has to be stored linearly and contiguously as shown in Figure 5. The pixel data is in raster scan order and the chrominance samples are interleaved between the luminance samples as $Y_0Cb_0Y_1Cr_0Y_2Cb_1Y_3Cr_1\dots$.

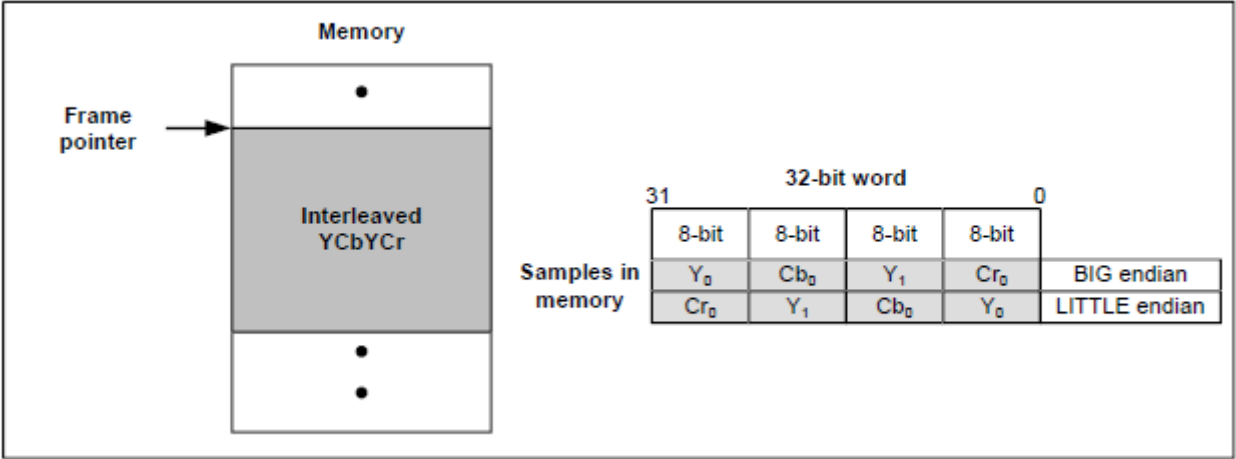


Figure 14-3. YCbCr 4:2:2 Interleaved Format External Memory Usage

The YCbCr 4:2:2 interleaved format is supported both as an input and as an output format for the post-processor. In this format each pixel takes 16 bits of memory. Although there is more data to be transferred than in the 4:2:0 formats, the interleaved format is bus effective as there are no non-sequential memory accesses caused by the plane changes.

14.1.3.4 G2 Decoder Tiled 4x4 Format

The output picture of the decoder is in semi-planar YCbCr 4:2:0 4x4 tiled format, i.e. luminance data forms one plane in memory, and chrominance data forms another. The distinction from raster scan format is that the output picture is grouped into tiles, which are then stored linearly and contiguously in the memory. The chrominance tiles have to be stored right after the luminance tiles in external memory as shown in Figure 6.

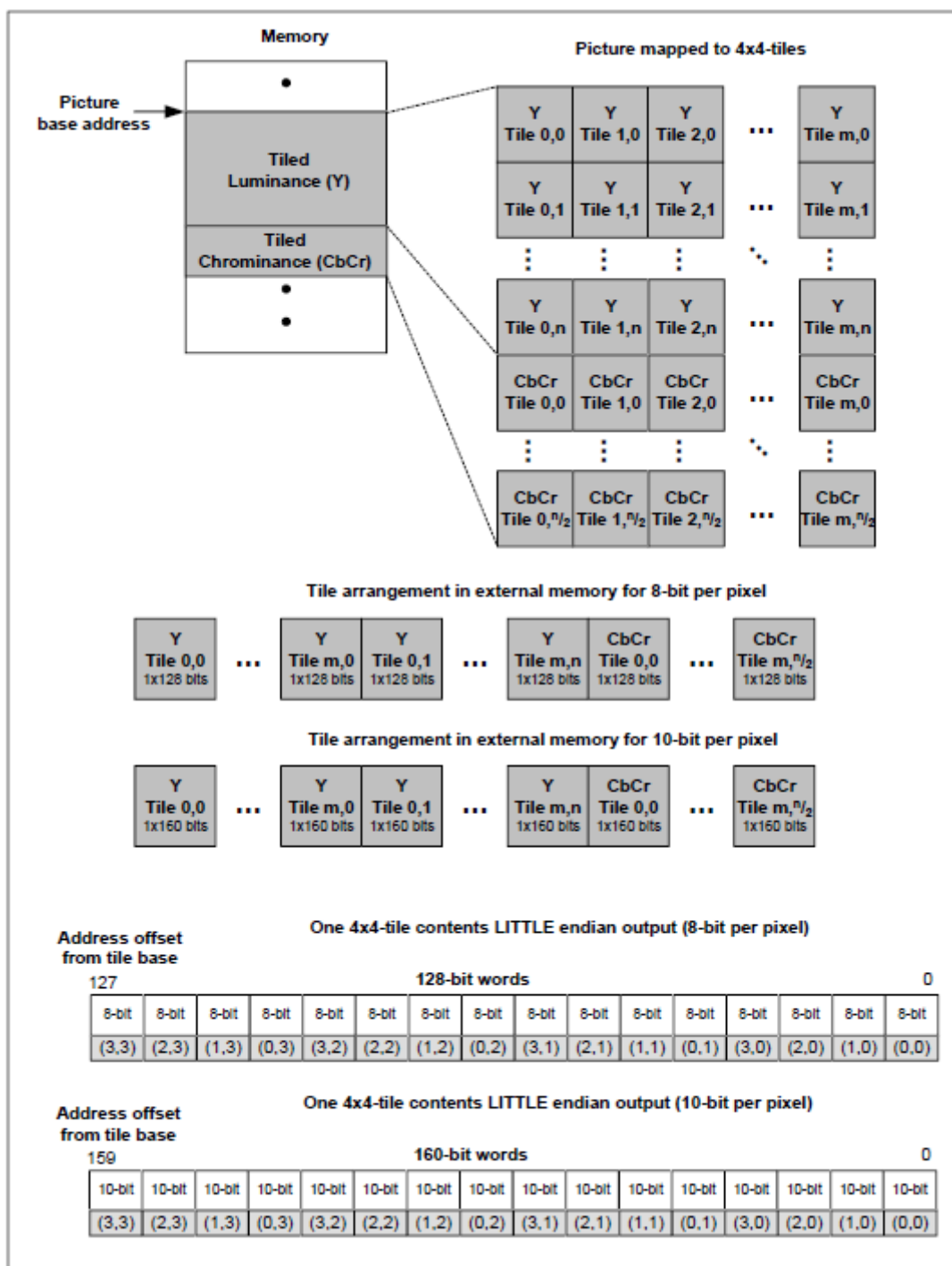


Figure 14-4. Tiled 4x4 Format External Memory Usage

14.1.3.5 RGB 16bpp Format

In this format each pixel is represented by 16 or less bits containing the red, blue and green samples. There are several 16bpp formats which use different number of bits for each sample. For example the RGB 5-5-5 format uses 5 bits for each sample and 1 bit is left unused or can represent a transparency flag, where RGB 5-6-5 uses 6 bits for the G sample and 5 bits for R and B samples. Common for all 16bpp types is that two pixels fit into one 32-bit space.

The data has to be stored linearly and contiguously in the memory as shown in Figure 7. The order of the sample bytes is always defined with a 16 bit word which will define the byte order in the memory. The pixel samples are stored in raster-scan order.

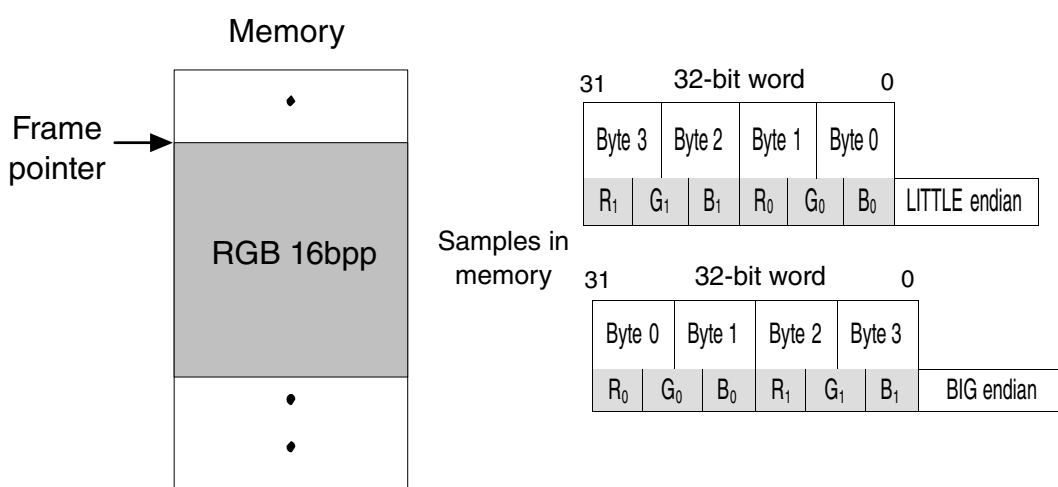


Figure 14-5. RGB 16bpp Format External Memory Usage

The RGB 16bpp formats are supported only as post-processor output formats.

14.1.3.6 RGB 32bpp Format

Any RGB format that has its pixels represented by more than 16 bits each is considered to be of 32bpp type. Typically in this format each pixel is represented by three bytes containing a red, blue and green sample and a 4th byte which can be empty or hold an alpha blending value. Common for all 32bpp types is that only one pixel fit into one 32-bit space.

The data has to be stored linearly and contiguously in the memory as shown in Figure 8. The order of the sample bytes is always defined on a 32 bit word and that will define the byte order in the memory (for example, ARGB order means that on a little endian system

the B sample will be stored in the lowest offset byte followed by the G, R and A samples. In a big endian system the A sample will be on the lowest offset byte). The pixel samples are stored in raster-scan order.

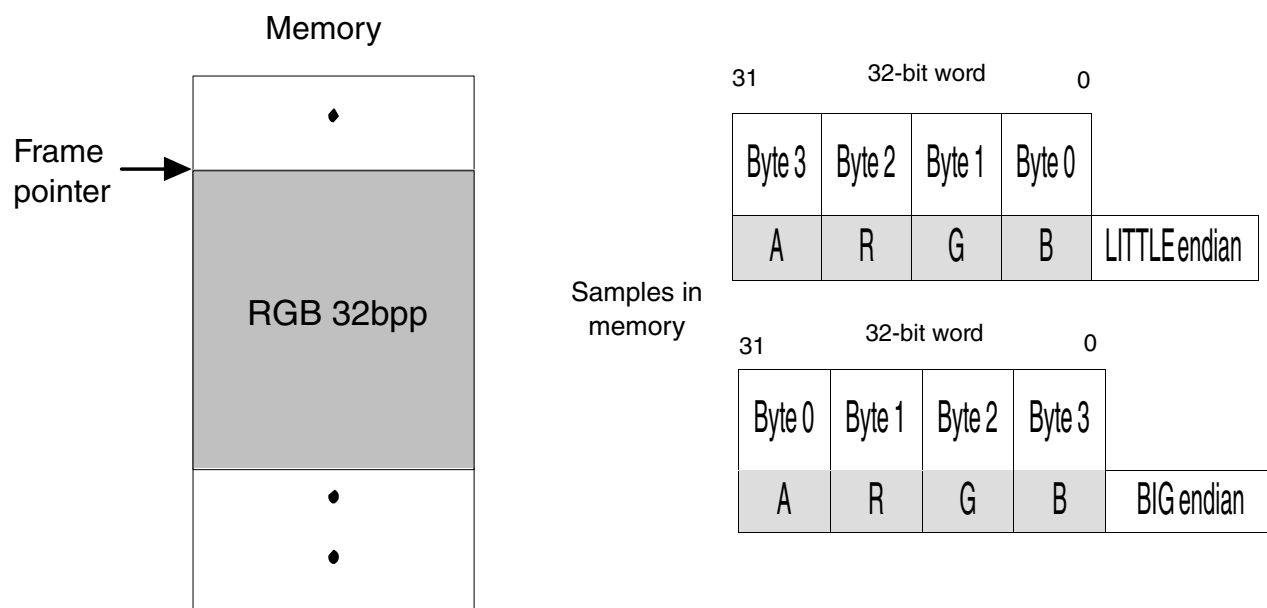


Figure 14-6. RGB 32bpp Format External Memory Usage

14.1.4 G1 Register Decoder

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
swreg0										
31:16	-	x	x	x	x	x	x	x	x	x
15:12	-	x	x	x	x	x	x	x	x	x
11:4	-	x	x	x	x	x	x	x	x	x
3	-	x	x	x	x	x	x	x	x	x
2:0	-	x	x	x	x	x	x	x	x	x
swreg1										
31:25	-									
24	sw_dec_pic_inf	x	x							
23:19	-									
18	sw_dec_timeout	x	x	x	x	x	x	x	x	x
17	sw_dec_slice_int			x						
16	sw_dec_error_int	x	x	x	x	x	x	x	x	x
15	sw_dec_aso_int	x							x	

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
14	sw_dec_buffer_int	x	x	x	x	x	x	x	x	x
13	sw_dec_bus_int	x	x	x	x	x	x	x	x	x
12	sw_dec_rdy_int	x	x	x	x	x	x	x	x	x
11	sw_dec_abort_int	x	x	x	x	x	x	x	x	x
10:9	-									
8	sw_dec_irq	x	x	x	x	x	x	x	x	x
7:6	-									
5	sw_dec_abort_e	x	x	x	x	x	x	x	x	x
4	sw_dec_irq_dis	x	x	x	x	x	x	x	x	x
3:1	-									
0	sw_dec_e	x	x	x	x	x	x	x	x	x
swreg2										
31:24	sw_dec_axi_rd_id	x	x	x	x	x	x	x	x	x
23	sw_dec_timeout_e	x	x	x	x	x	x	x	x	x
22	sw_dec_strswap32_e	x	x	x	x	x	x	x	x	x
21	sw_dec_strendian_e	x	x	x	x	x	x	x	x	x
20	sw_dec_inswap32_e	x	x	x	x	x	x	x	x	x
19	sw_dec_outswap32_e	x	x	x	x	x	x	x	x	x
18	sw_dec_data_disc_e	x	x	x	x	x	x	x	x	x
18	sw_dec_2chan_dis	x	x	x	x	x	x	x	x	x
17	sw_tiled_mode_msb	x	x		x	x	x	x	x	x
17	sw_dec_out_tiled_e	x	x	x	x	x				
16:11	sw_dec_latency	x	x	x	x	x	x	x	x	x
10	sw_dec_clk_gate_e	x	x	x	x	x	x	x	x	x
9	sw_dec_in_endian	x	x	x	x	x	x	x	x	x
8	sw_dec_out_endian	x	x	x	x	x	x	x	x	x
7:5	sw_priority_mode	x	x	x	x	x				
7	sw_tiled_mode_lsb	x	x	x	x	x	x	x	x	x

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
6	sw_dec_adv_pre_dis	x	x		x	x	x	x	x	x
5	sw_dec_scmd_dis	x	x	x	x	x	x	x	x	x
4:0	sw_dec_max_burst	x	x	x	x	x	x	x	x	x
swreg3										
31:28	sw_dec_mode	x	x	x	x	x	x	x	x	x
27	sw_rlc_mode_e	x	x							
26	sw_skip_mode		x						x	x
25	sw_divx3_e		x							
24	sw_pjpeg_e			x						
23	sw_pic_interlace_e	x	x		x	x		x		x
22	sw_pic_fieldmode_e	x	x		x	x				x
21	sw_pic_b_e		x		x	x		x		x
20	sw_pic_inter_e		x		x	x	x		x	x
19	sw_pic_topfield_e	x	x		x	x				x
18	sw_fwd_interlace_e		x		x	x				
17	sw_sorenson_e		x							
16	sw_ref_topfield_e				x					
15	sw_dec_out_dis	x	x	x	x	x	x	x	x	x
14	sw_filtering_dis	x	x		x	x	x	x		x
13	sw_webp_e								x	
13	sw_mvc_e	x								
13	sw_pic_fixed_quant				x					x
12	sw_write_mvs_e	x	x		x	x	x	x	x	x
11	sw_reftopfirst_e				x					
10	sw_seq_mbaff_e	x								
9	sw_picord_count_e	x								
8	sw_dec_ahb_hlock_e	x	x	x	x	x	x	x	x	x
7:0	sw_dec_axi_wrid	x	x	x	x	x	x	x	x	x
swreg4										
31:23	sw_pic_mb_width	x	x	x	x	x	x	x	x	x
22:19	sw_mb_width_off				x					

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
18:11	sw_pic_mb_height_p	x	x	x	x	x	x	x	x	x
10:7	sw_mb_height_of				x					
6	sw_alt_scan_e		x			x				
5	sw_topfieldfirst_e		x		x					
4:0	sw_ref_frames	x			x					
5:3	sw_pic_mb_w_ext			x					x	
2:0	sw_pic_mb_h_ext		x	x		x	x	x	x	
0	sw_pic_refer_flag									x
swreg5										
31:26	sw_strm_start_bit	x	x	x	x	x	x	x	x	x
25	sw_sync_marker_e		x	x	x					
24	sw_type1_quant_e	x	x							
23:19	sw_ch_qp_offset	x	x							
18:14	sw_ch_qp_offset2	x								
13:1	-	x								
0	sw_fieldpic_flag_e	x								
18:16	sw_intradc_vlc_thr		x							
15:0	sw_vop_time_incr		x							
24	sw_dq_profile				x					
23	sw_dqbi_level				x					
22	sw_range_red_frm_e				x					
21	-				x					
20	sw_fast_uvmc_e				x					
19:18	-				x					
17	sw_transdctab		x		x					
16:15	sw_transacfrm		x		x					
14:13	sw_transacfrm2		x		x					
12:10	sw_mb_mode_tab				x					
9:7	sw_mvtab				x					
6:4	sw_cbptab				x					

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
3:2	sw_2mv_blk_pat _tab				x					
1:0	sw_4mv_blk_pat _tab				x					
24	sw_qscales_type					x				
23:5	-					x				
4	sw_con_mv_e					x				
3:2	sw_intra_dc_prec					x				
1	sw_intra_vlc_tab					x				
0	sw_frame_pred_ dct					x				
12:11	sw_jpeg_qtables			x						
10:8	sw_jpeg_mode			x						
7	sw_jpeg_filright_ e			x						
6	sw_jpeg_stream_ all			x						
5	sw_cr_ac_vlctabl e			x						
4	sw_cb_ac_vlctabl e			x						
3	sw_cr_dc_vlctabl e			x						
2	sw_cb_dc_vlctabl e			x						
1	sw_cr_dc_vlctabl e3			x						
0	sw_cb_dc_vlctabl e3			x						
23:18	sw_strm1_start_b it						x		x	
17	sw_huffman_e						x			
16	sw_multistream_ e						x			
15:8	sw_boolean_valu e						x		x	
7:0	sw_boolean_rang e						x		x	
9:5	sw_alpha_offset									x
4:0	sw_beta_offset									x
swreg6										
31	sw_start_code_e	x			x					
30:25	sw_init_qp	x	x		x	x	x			x

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
24	sw_ch_8pix_ileav_e	x								
31:24	sw_stream_len_ext								x	
23:0	sw_stream_len	x	x	x	x	x	x	x	x	x
swreg7										
31	sw_cabac_e	x								
30	sw_blackwhite_e	x								
29	sw_dir_8x8_infer_e	x								
28	sw_weight_pred_e	x								
27:26	sw_weight_bipr_idc	x								
25	sw_avs_h264_h_ext	x								x
24:21	-	x								
20:16	sw_framenum_len	x						x		
15:0	sw_framenum	x								
31	sw_bitplane0_e				x					
30	sw_bitplane1_e				x					
29	sw_bitplane2_e				x					
28:24	sw_alt_pquant				x					
23:20	sw_dq_edges				x					
19	sw_ttmbf				x					
18:14	sw_pqindex				x					
13	sw_vc1_height_ext				x					
12	sw_bilin_mc_e				x		x		x	
11	sw_uniqp_e				x					
10	sw_halfqp_e				x					
9:8	sw_ttfm				x					
7	sw_2nd_byte_emul_e				x					
6	sw_dquant_e				x					
5	sw_vc1_adv_e				x					
31:26	sw_dct1_start_bit								x	
25:20	sw_dct2_start_bit								x	
13	sw_ch_mv_res								x	
11:9	sw_init_dc_matc_h0								x	

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
8:6	sw_init_dc_matc h1								x	
5	sw_vp7_version								x	
swreg8										
31	sw_const_intra_e	x								
30	sw_filt_ctrl_pres	x								
29	sw_rdpic_cnt_pre s	x								
28	sw_8x8trans_flag _e	x								
27:17	sw_refpic_mk_le n	x								
16	sw_idr_pic_e	x								
15:0	sw_idr_pic_id	x								
31:24	sw_mv_scalefact or				x					
23:19	sw_ref_dist_fwd				x					
18:14	sw_ref_dist_bwd				x					
17:14	sw_loop_filt_limit						x			
13	sw_variance_test _e						x			
12:10	sw_mv_threshold						x			
9:0	sw_var_threshold						x			
8	sw_divx_idct_e		x							
7:0	sw_divx3_slice_s ize		x							
31:30	sw_rv_profile							x		
29:28	sw_rv_osv_quant							x		
27:14	sw_rv_fwd_scale							x		
13:0	sw_rv_bwd_scale							x		
31:16	sw_init_dc_comp 0								x	
15:0	sw_init_dc_comp 1								x	
swreg9										
31:24	sw_pps_id	x								
23:19	sw_refidx1_activ e	x								
18:14	sw_refidx0_activ e	x								
7:0	sw_poc_length	x								
24	sw_icomp0_e				x					

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
23:16	sw_iscale0				x					
15:0	sw_ishift0				x					
23:0	sw_stream1_len						x		x	
31:2	sw_mb_ctrl_base	x	x							
12:0	sw_pic_slice_am							x		
27:24	sw_coeffs_part_a m								x	
swreg10										
31:2	sw_diff_mv_base	x	x							
29:25	sw_pinit_rlist_f9	x								
24:20	sw_pinit_rlist_f8	x								
19:15	sw_pinit_rlist_f7	x								
14:10	sw_pinit_rlist_f6	x								
9:5	sw_pinit_rlist_f5	x								
4:0	sw_pinit_rlist_f4	x								
24	sw_icomp1_e				x					
23:16	sw_iscale1				x					
15:0	sw_ishift1				x					
31:2	sw_segment_base								x	
1	sw_segment_upd _e								x	
0	sw_segment_e								x	
swreg11										
31:2	sw_i4x4_or_dc_b ase	x	x							
29:25	sw_pinit_rlist_f15	x								
24:20	sw_pinit_rlist_f14	x								
19:15	sw_pinit_rlist_f13	x								
14:10	sw_pinit_rlist_f12	x								
9:5	sw_pinit_rlist_f11	x								
4:0	sw_pinit_rlist_f10	x								
24	sw_icomp2_e				x					
23:16	sw_iscale2				x					
15:0	sw_ishift2				x					
29:24	sw_dct3_start_bit								x	
23:18	sw_dct4_start_bit								x	
17:12	sw_dct5_start_bit								x	
11:6	sw_dct6_start_bit								x	
5:0	sw_dct7_start_bit								x	

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
swreg12										
31:0	sw_rlc_vlc_base	x	x	x	x	x	x	x	x	x
swreg13										
31:2	sw_dec_out_base	x	x	x	x	x	x	x	x	x
1	sw_dpb_ilace_mode	x	x		x					x
swreg14										
31:2	sw_refer0_base	x	x		x	x	x	x	x	x
1	sw_refer0_field_enable	x								x
0	sw_refer0_topc_enable	x								x
31:2	sw_jpg_ch_out_base			x					x	
swreg15										
31:2	sw_refer1_base	x	x		x	x		x	x	x
1	sw_refer1_field_enable	x								x
0	sw_refer1_topc_enable	x								x
7:0	sw_jpeg_slice_h			x					x	
swreg16										
31:2	sw_refer2_base	x	x		x	x		x		x
1	sw_refer2_field_enable	x								x
0	sw_refer2_topc_enable	x								x
31	-									
30:24	sw_ac1_code6_count			x						
23:22	-									
21:16	sw_ac1_code5_count			x						
15:11	sw_ac1_code4_count			x						
10:7	sw_ac1_code3_count			x						
6	-									
5:3	sw_ac1_code2_count			x						
2	-									
1:0	sw_ac1_code1_count			x						
swreg17										

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
31:2	sw_refer3_base	x	x		x	x		x		x
1	sw_refer3_field_e	x								x
0	sw_refer3_topc_e	x								x
31:24	sw_ac1_code10_cnt			x						
23:16	sw_ac1_code9_cnt			x						
15:8	sw_ac1_code8_cnt			x						
7:0	sw_ac1_code7_cnt			x						
swreg18										
31:2	sw_refer4_base	x					x		x	
1	sw_refer4_field_e	x								
0	sw_refer4_topc_e	x								
31:16	sw_pic_header_len				x					
15:14	-				x					
13	sw_pic_4mv_e				x					
12	-				x					
11	sw_range_red_ref_e				x					
10:9	sw_vc1_difmv_range				x					
8	-				x					
7:6	sw_mv_range				x					
5	sw_overlap_e				x					
4:3	sw_overlap_method				x					
19	sw_alt_scan_flag_e		x			x				
18:15	sw_fcode_fwd_hor		x			x				
14:11	sw_fcode_fwd_ver					x				
10:7	sw_fcode_bwd_hor		x			x				
6:3	sw_fcode_bwd_ver					x				
2	sw_mv_accuracy_fwd		x		x	x				

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
1	sw_mv_accuracy_bwd					x				
1	sw_mpeg4_vc1_rc		x		x					
0	sw_prev_anc_type		x		x			x		x
31:24	sw_ac1_code14_cnt			x						
23:16	sw_ac1_code13_cnt			x						
15:8	sw_ac1_code12_cnt			x						
7:0	sw_ac1_code11_cnt			x						
0	sw_gref_sign_bias								x	
swreg19										
31:2	sw_refer5_base	x							x	
1	sw_refer5_field_enable	x								
0	sw_refer5_topc_enable	x								
26:0	sw_trb_per_trd_d0		x							
24	sw_icomp3_enable				x					
23:16	sw_iscale3				x					
15:0	sw_ishift3				x					
31:27	sw_ac2_code4_cnt			x						
26:23	sw_ac2_code3_cnt			x						
22	-									
21:19	sw_ac2_code2_cnt			x						
18	-									
17:16	sw_ac2_code1_cnt			x						
15:8	sw_ac1_code16_cnt			x						
7:0	sw_ac1_code15_cnt			x						
29:24	sw_scan_map_1						x		x	
23:18	sw_scan_map_2						x		x	
17:12	sw_scan_map_3						x		x	
11:6	sw_scan_map_4						x		x	

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
5:0	sw_scan_map_5						x		x	
0	sw_aref_sign_bia s								x	
swreg20										
31:2	sw_refer6_base	x								
31:2	sw_vp8_dec_ch_ base	x								
1	sw_vp8_stride_e								x	
0	sw_vp8_ch_base_ _e								x	
1	sw_refer6_field_e	x								
0	sw_refer6_topc_ e	x								
26:0	sw_trb_per_trd_d m1		x							
24	sw_icomp4_e				x					
23:16	sw_iscale4				x					
15:0	sw_ishift4				x					
31:24	sw_ac2_code8_c nt			x						
23:16	sw_ac2_code7_c nt			x						
15										
14:8	sw_ac2_code6_c nt			x						
7:6	-									
5:0	sw_ac2_code5_c nt			x						
29:24	sw_scan_map_6						x		x	
23:18	sw_scan_map_7						x		x	
17:12	sw_scan_map_8						x		x	
11:6	sw_scan_map_9						x		x	
5:0	sw_scan_map_1 0						x		x	
swreg21										
31:2	sw_refer7_base	x								
31:27	sw_y_stride_pow 2								x	
26:22	sw_c_stride_pow 2								x	
1	sw_refer7_field_e	x								
0	sw_refer7_topc_ e	x								

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
26:0	sw_trb_per_trd_d1		x							
31:24	sw_ac2_code12_cnt			x						
23:16	sw_ac2_code11_cnt			x						
15:8	sw_ac2_code10_cnt			x						
7:0	sw_ac2_code9_cnt			x						
29:24	sw_scan_map_11						x		x	
23:18	sw_scan_map_12						x		x	
17:12	sw_scan_map_13						x		x	
11:6	sw_scan_map_14						x		x	
5:0	sw_scan_map_15						x		x	
swreg22										
31:2	sw_refer8_base	x								
31:2	sw_dct_strm1_base								x	
1	sw_refer8_field_e	x								
0	sw_refer8_topc_e	x								
31:24	sw_ac2_code16_cnt			x						
23:16	sw_ac2_code15_cnt			x						
15:8	sw_ac2_code14_cnt			x						
7:0	sw_ac2_code13_cnt			x						
29:24	sw_scan_map_16						x			
23:18	sw_scan_map_17						x			
17:12	sw_scan_map_18						x			
11:6	sw_scan_map_19						x			
5:0	sw_scan_map_20						x			

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
swreg23										
31:2	sw_refer9_base	x								
31:2	sw_dct_strm2_base								x	
1	sw_refer9_field_e	x								
0	sw_refer9_topc_e	x								
31:28	sw_dc1_code8_cnt			x						
27:24	sw_dc1_code7_cnt			x						
23:20	sw_dc1_code6_cnt			x						
19:16	sw_dc1_code5_cnt			x						
15:12	sw_dc1_code4_cnt			x						
11:8	sw_dc1_code3_cnt			x						
7	-									
6:4	sw_dc1_code2_cnt			x						
3:2	-									
1:0	sw_dc1_code1_cnt			x						
29:24	sw_scan_map_21						x			
23:18	sw_scan_map_22						x			
17:12	sw_scan_map_23						x			
11:6	sw_scan_map_24						x			
5:0	sw_scan_map_25						x			
swreg24										
31:2	sw_refer10_base	x								
31:2	sw_dct_strm3_base								x	
1	sw_refer10_field_e	x								
0	sw_refer10_topc_e	x								

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
31:28	sw_dc1_code16_cnt			x						
27:24	sw_dc1_code15_cnt			x						
23:20	sw_dc1_code14_cnt			x						
19:16	sw_dc1_code13_cnt			x						
15:12	sw_dc1_code12_cnt			x						
11:8	sw_dc1_code11_cnt			x						
7:4	sw_dc1_code10_cnt			x						
3:0	sw_dc1_code9_cnt			x						
29:24	sw_scan_map_26						x			
23:18	sw_scan_map_27						x			
17:12	sw_scan_map_28						x			
11:6	sw_scan_map_29						x			
5:0	sw_scan_map_30						x			
swreg25										
31:2	sw_refer11_base	x								
31:2	sw_dct_strm4_base								x	
1	sw_refer11_field	x								
0	sw_refer11_topc_e	x								
31:28	sw_dc2_code8_cnt			x						
27:24	sw_dc2_code7_cnt			x						
23:20	sw_dc2_code6_cnt			x						
19:16	sw_dc2_code5_cnt			x						
15:12	sw_dc2_code4_cnt			x						

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
11:8	sw_dc2_code3_cnt			x						
7	-									
6:4	sw_dc2_code2_cnt			x						
3:2	-									
1:0	sw_dc2_code1_cnt			x						
29:24	sw_scan_map_31						x			
23:18	sw_scan_map_32						x			
17:12	sw_scan_map_33						x			
11:6	sw_scan_map_34						x			
5:0	sw_scan_map_35						x			
swreg26										
31:2	sw_refer12_base	x								
31:2	sw_dct_strm5_base								x	
1	sw_refer12_field_e	x								
0	sw_refer12_topc_e	x								
31:28	sw_dc2_code16_cnt			x						
27:24	sw_dc2_code15_cnt			x						
23:20	sw_dc2_code14_cnt			x						
19:16	sw_dc2_code13_cnt			x						
15:12	sw_dc2_code12_cnt			x						
11:8	sw_dc2_code11_cnt			x						
7:4	sw_dc2_code10_cnt			x						
3:0	sw_dc2_code9_cnt			x						
29:24	sw_scan_map_36						x			

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
23:18	sw_scan_map_37						x			
17:12	sw_scan_map_38						x			
11:6	sw_scan_map_39						x			
5:0	sw_scan_map_40						x			
swreg27										
31:2	sw_refer13_base	x								
1	sw_refer13_field_e	x								
0	sw_refer13_topc_e	x								
31:28	sw_dc3_code8_cnt			x						
27:24	sw_dc3_code7_cnt			x						
23:20	sw_dc3_code6_cnt			x						
19:16	sw_dc3_code5_cnt			x						
15:12	sw_dc3_code4_cnt			x						
11:8	sw_dc3_code3_cnt			x						
7	-									
6:4	sw_dc3_code2_cnt			x						
3:2	-									
1:0	sw_dc3_code1_cnt			x						
31:2	sw_bitpl_ctrl_base				x		x		x	
swreg28										
31:2	sw_refer14_base	x								
31:2	sw_dct_strm6_base								x	
1	sw_refer14_field_e	x								
0	sw_refer14_topc_e	x								
31:16	sw_ref_invd_cur_1									x

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
15:0	sw_ref_invd_cur_0									x
31:28	sw_dc3_code16_cnt			x						
27:24	sw_dc3_code15_cnt			x						
23:20	sw_dc3_code14_cnt			x						
19:16	sw_dc3_code13_cnt			x						
15:12	sw_dc3_code12_cnt			x						
11:8	sw_dc3_code11_cnt			x						
7:4	sw_dc3_code10_cnt			x						
3:0	sw_dc3_code9_cnt			x						
29:24	sw_scan_map_41						x			
23:18	sw_scan_map_42						x			
17:12	sw_scan_map_43						x			
11:6	sw_scan_map_44						x			
5:0	sw_scan_map_45						x			
swreg29										
31:2	sw_refer15_base	x								
31:2	sw_dct_strm7_base								x	
1	sw_refer15_field_e	x								
0	sw_refer15_topc_e	x								
31:16	sw_ref_invd_cur_3									x
15:0	sw_ref_invd_cur_2									x
29:24	sw_scan_map_46						x			
23:18	sw_scan_map_47						x			

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
17:12	sw_scan_map_4 8						x			
11:6	sw_scan_map_4 9						x			
5:0	sw_scan_map_5 0						x			
swreg30										
31:16	sw_refer1_nbr	x								
15:0	sw_refer0_nbr	x								
31:16	sw_ref_dist_cur_ 1									x
15:0	sw_ref_dist_cur_ 0									x
31	sw_filt_type								x	
30:28	sw_filt_sharpnes s								x	
27:21	sw_filt_mb_adj_0								x	
20:14	sw_filt_mb_adj_1								x	
13:7	sw_filt_mb_adj_2								x	
6:0	sw_filt_mb_adj_3								x	
swreg31										
31:16	sw_refer3_nbr	x								
15:0	sw_refer2_nbr	x								
29:24	sw_scan_map_5 1						x			
23:18	sw_scan_map_5 2						x			
17:12	sw_scan_map_5 3						x			
11:6	sw_scan_map_5 4						x			
5:0	sw_scan_map_5 5						x			
31:16	sw_ref_dist_cur_ 3									x
15:0	sw_ref_dist_cur_ 2									x
27:21	sw_filt_ref_adj_0								x	
20:14	sw_filt_ref_adj_1								x	
13:7	sw_filt_ref_adj_2								x	
6:0	sw_filt_ref_adj_3								x	
swreg32										
31:16	sw_refer5_nbr	x								

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
15:0	sw_refer4_nbr	x								
29:24	sw_scan_map_5 6						x			
23:18	sw_scan_map_5 7						x			
17:12	sw_scan_map_5 8						x			
11:6	sw_scan_map_5 9						x			
5:0	sw_scan_map_6 0						x			
31:16	sw_ref_invd_col_ 1									x
15:0	sw_ref_invd_col_ 0									x
23:18	sw_filt_level_0								x	
17:12	sw_filt_level_1								x	
11:6	sw_filt_level_2								x	
5:0	sw_filt_level_3								x	
swreg33										
31:16	sw_refer7_nbr	x								
15:0	sw_refer6_nbr	x								
29:24	sw_scan_map_6 1						x			
23:18	sw_scan_map_6 2						x			
17:12	sw_scan_map_6 3						x			
31:16	sw_ref_invd_col_ 3									x
15:0	sw_ref_invd_col_ 2									x
31:27	sw_quant_delta_ 0								x	
26:22	sw_quant_delta_ 1								x	
21:11	sw_quant_0								x	
10:0	sw_quant_1								x	
swreg34										
31:16	sw_refer9_nbr	x								
15:0	sw_refer8_nbr	x								
31:22	sw_pred_bc_tap_ 0_3		x		x		x	x	x	x

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
21:12	sw_pred_bc_tap_1_0				x		x	x	x	x
11:2	sw_pred_bc_tap_1_1				x		x	x	x	x
swreg35										
31:16	sw_refer11_nbr	x								
15:0	sw_refer10_nbr	x								
31:22	sw_pred_bc_tap_1_2				x		x	x	x	x
21:12	sw_pred_bc_tap_1_3				x		x	x	x	x
11:2	sw_pred_bc_tap_2_0				x		x		x	x
swreg36										
31:16	sw_refer13_nbr	x								
15:0	sw_refer12_nbr	x								
31:22	sw_pred_bc_tap_2_1				x		x		x	x
21:12	sw_pred_bc_tap_2_2				x		x		x	x
11:2	sw_pred_bc_tap_2_3				x		x		x	x
swreg37										
31:16	sw_refer15_nbr	x								
15:0	sw_refer14_nbr	x								
31:22	sw_pred_bc_tap_3_0						x		x	
21:12	sw_pred_bc_tap_3_1						x		x	
11:2	sw_pred_bc_tap_3_2						x		x	
swreg38										
31:0	sw_refer_lterm_e	x								
31:22	sw_pred_bc_tap_3_3						x		x	
21:12	sw_pred_bc_tap_4_0						x		x	
11:2	sw_pred_bc_tap_4_1						x		x	
swreg39										
31:0	sw_refer_valid_e	x								
31:22	sw_pred_bc_tap_4_2						x		x	

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
21:12	sw_pred_bc_tap_4_3						x		x	
11:2	sw_pred_bc_tap_5_0						x		x	
swreg40										
31:2	sw_qtable_base	x	x	x		x	x	x	x	
swreg41										
31:2	sw_dir_mv_base	x	x	x	x	x		x	x	x
swreg42										
29:25	sw_binit_rlist_b2	x								
24:20	sw_binit_rlist_f2	x								
19:15	sw_binit_rlist_b1	x								
14:10	sw_binit_rlist_f1	x								
9:5	sw_binit_rlist_b0	x								
4:0	sw_binit_rlist_f0	x								
31:22	sw_pred_bc_tap_5_1						x		x	
21:12	sw_pred_bc_tap_5_2						x		x	
11:2	sw_pred_bc_tap_5_3						x		x	
31:24	sw_weight_qp_1									x
23:21	sw_ref_delta_col_0									x
20:18	sw_ref_delta_col_1									x
17:15	sw_ref_delta_col_2									x
14:12	sw_ref_delta_col_3									x
11:9	sw_ref_delta_cur_0									x
8:6	sw_ref_delta_cur_1									x
5:3	sw_ref_delta_cur_2									x
2:0	sw_ref_delta_cur_3									x
swreg43										
29:25	sw_binit_rlist_b5	x								
24:20	sw_binit_rlist_f5	x								
19:15	sw_binit_rlist_b4	x								
14:10	sw_binit_rlist_f4	x								

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
9:5	sw_binit_rlist_b3	x								
4:0	sw_binit_rlist_f3	x								
31:22	sw_pred_bc_tap_6_0						x		x	
21:12	sw_pred_bc_tap_6_1						x		x	
11:2	sw_pred_bc_tap_6_2						x		x	
31:24	sw_weight_qp_2									x
23:16	sw_weight_qp_3									x
15:8	sw_weight_qp_4									x
7:0	sw_weight_qp_5									x
swreg44										
29:25	sw_binit_rlist_b8	x								
24:20	sw_binit_rlist_f8	x								
19:15	sw_binit_rlist_b7	x								
14:10	sw_binit_rlist_f7	x								
9:5	sw_binit_rlist_b6	x								
4:0	sw_binit_rlist_f6	x								
31:22	sw_pred_bc_tap_6_3						x		x	
21:12	sw_pred_bc_tap_7_0						x		x	
11:2	sw_pred_bc_tap_7_1						x		x	
26	sw_dec_avsp_ena									x
25	sw_weight_qp_e									x
24:23	sw_weight_qp_model									x
22	sw_avs_aec_e									x
21	sw_no_fwd_ref_e									x
20	sw_pb_field_enhanced_e									x
19:14	sw_qp_delta_cb									x
13:8	sw_qp_delta_cr									x
7:0	sw_weight_qp_0									x
swreg45										
29:25	sw_binit_rlist_b11	x								
24:20	sw_binit_rlist_f11	x								

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
19:15	sw_binit_rlist_b1_0	x								
14:10	sw_binit_rlist_f10	x								
9:5	sw_binit_rlist_b9	x								
4:0	sw_binit_rlist_f9	x								
31:22	sw_pred_bc_tap_7_2						x		x	
21:12	sw_pred_bc_tap_7_3						x		x	
11:10	sw_pred_tap_2_m1								x	
9:8	sw_pred_tap_2_4								x	
7:6	sw_pred_tap_4_m1								x	
5:4	sw_pred_tap_4_4								x	
3:2	sw_pred_tap_6_m1								x	
1:0	sw_pred_tap_6_4								x	
31:0	sw_dir_mv_base_2									x
swreg46										
31:30										
29:25	sw_binit_rlist_b1_4	x								
24:20	sw_binit_rlist_f14	x								
19:15	sw_binit_rlist_b1_3	x								
14:10	sw_binit_rlist_f13	x								
9:5	sw_binit_rlist_b1_2	x								
4:0	sw_binit_rlist_f12	x								
31:27	sw_quant_delta_2								x	
26:22	sw_quant_delta_3								x	
21:11	sw_quant_2								x	
10:0	sw_quant_3								x	
swreg47										
31:30	-									
29:25	sw_pinit_rlist_f3	x								

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
24:20	sw_pinit_rlist_f2	x								
19:15	sw_pinit_rlist_f1	x								
14:10	sw_pinit_rlist_f0	x								
9:5	sw_binit_rlist_b1 5	x								
4:0	sw_binit_rlist_f15	x								
31:27	sw_quant_delta_4								x	
21:11	sw_quant_4								x	
10:0	sw_quant_5								x	
swreg48										
31:23	sw_startmb_x								x	
22:14	sw_startmb_y								x	
13:12	sw_error_conc_mode								x	
11:0	-									
swreg49										
31:22	sw_pred_bc_tap_0_0	x	x		x		x	x		x
21:12	sw_pred_bc_tap_0_1	x	x		x		x	x		x
11:2	sw_pred_bc_tap_0_2	x	x		x		x	x		x
1:0	-									
swreg50										
31	SW_DEC_MPEG2_PROF					x				
30:29	SW_DEC_VC1_PROF				x					
28	SW_DEC_JPEG_PROF			x						
27:26	SW_DEC_MPEG4_PROF		x							
25:24	SW_DEC_H264_PROF	x								
23	SW_DEC_VP6_PROF						x			
22	SW_DEC_PJPE_G_EXIST			x						
21	SW_DEC_OBUF_F_LEVEL	x	x		x	x	x	x	x	x
20	SW_REF_BUFF_EXIST	x	x		x	x	x	x	x	x

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
19:16	SW_DEC_BUS_STRD	x	x	x	x	x	x	x	x	x
15:14	SW_DEC_SYNT H_LAN	x	x	x	x	x	x	x	x	x
13:12	SW_DEC_BUS_WIDTH	x	x	x	x	x	x	x	x	x
11	SW_DEC_SOREN_PROF		x							
10:0	SW_DEC_MAX_OWIDTH	x	x	x	x	x	x	x	x	x
swreg51										
31	sw_refbu_e	x	x		x	x	x	x	x	x
30:19	sw_refbu_thr	x	x		x	x	x	x	x	x
18:14	sw_refbu_picid	x	x		x	x	x	x	x	x
13	sw_refbu_eval_e	x	x		x	x	x	x	x	x
12	sw_refbu_fparmod_e	x	x		x	x	x	x	x	x
11:9	-									
8:0	sw_refbu_y_offset	x	x		x	x	x	x	x	x
swreg52										
31:16	sw_refbu_hit_sum	x	x		x	x	x	x	x	x
15:0	sw_refbu_intra_sum	x	x		x	x	x	x	x	x
swreg53										
31:22	-									
21:0	sw_refbu_y_mv_sum	x	x		x	x	x	x	x	x
swreg54										
31	SW_DEC_JPEG_EXTENS			x						
30	SW_DEC_REFBU_ILACE	x	x		x	x				
29	SW_DEC_DIVX_PROF		x							
28:0	SW_REF_BUFF2_EXIST	x	x		x	x	x	x	x	x
27:26	SW_DEC_RV_PROF							x		
25	SW_DEC_RTL_ROM		x		x			x		
24	SW_DEC_VP7_PROF								x	

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
23	SW_DEC_VP8_PROF								x	
22	SW_DEC_AVS_PROF									x
21:20	SW_DEC_MVC_PROF	x								
19	SW_DEC_WEBP_E								x	
18:17	SW_DEC_TILED_L	x	x		x	x	x	x	x	x
16	SW_DEC_VP8S_ARCH								x	
15:14	SW_DEC_MAX_OW_EXT	x	x		x	x	x	x	x	x
13:12	SW_DEC_ERRCO_LEVEL								x	
11	SW_VP8_STRIDE_E								x	
10	SW_DPB_FIELD_E	x	x	x	x	x	x	x	x	x
9:7	SW_DEC_CORE_AM	x	x	x	x	x	x	x	x	x
6:0	-									
swreg55										
31	sw_refbu2_buf_e	x	x		x	x	x	x	x	x
30:19	sw_refbu2_thr	x	x		x	x	x	x	x	x
18:14	sw_refbu2_picid	x	x		x	x	x	x	x	x
13:0	sw_apf_threshold	x	x		x	x	x	x	x	x
swreg56										
31:16	sw_refbu_top_sum	x	x		x	x	x	x	x	x
15:0	sw_refbu_bot_sum	x	x		x	x	x	x	x	x
swreg57										
31	fuse_dec_h264	x								
30	fuse_dec_mpeg4		x							
29	fuse_dec_mpeg2					x				
28	fuse_dec_sorenson		x							
27	fuse_dec_jpeg			x						
26	fuse_dec_vp6						x			
25	fuse_dec_vc1				x					
24	fuse_dec_pjpeg			x						

Table continues on the next page...

VPU G1 (VPU_G1)

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
23	fuse_dec_divx		x							
22	fuse_dec_rv							x		
21	fuse_dec_vp7								x	
20	fuse_dec_vp8								x	
19	fuse_dec_avs									x
18	fuse_dec_mvc	x								x
17	-									
16	fuse_dec_maxw_4k	x	x	x	x	x	x	x	x	x
15	fuse_dec_maxw_1920	x	x		x	x	x	x	x	x
14	fuse_dec_maxw_1280	x	x		x	x	x	x	x	x
13	fuse_dec_maxw_720	x	x		x	x	x	x	x	x
12	fuse_dec_maxw_352	x	x		x	x	x	x	x	x
11:8	-									
7	fuse_dec_refbuffer	x	x		x	x	x	x	x	x
6:0	-									
swreg58										
31	sw_serv_merge_dis	x	x	x	x	x	x	x	x	x
30	sw_dec_multicore_e	x							x	
29	sw_dec_writestat_e	x							x	
28:27	sw_dec_mc_pollmode	x							x	
26:17	sw_dec_mc_polltime	x							x	
16:0	-									
swreg59										
31:2	sw_dec_ch8pix_base	x								
1:0	-									
Post-processor (swreg60-100)										
swreg102										
31:2	sw_dec_ch_base	x								x
1	-									
0	sw_ch_base_e	x								x

Table continues on the next page...

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
swreg103										
31:2	sw_refer0_ch_base	x								x
1:0	-									
swreg104										
31:2	sw_refer1_ch_base	x								x
1:0	-									
swreg105										
31:2	sw_refer2_ch_base	x								x
1:0	-									
swreg106										
31:2	sw_refer3_ch_base	x								x
1:0	-									
swreg107										
31:2	sw_refer4_ch_base	x								x
1:0	-									
swreg108										
31:2	sw_refer5_ch_base	x								x
1:0	-									
swreg109										
31:2	sw_refer6_ch_base	x								x
1:0	-									
swreg110										
31:2	sw_refer7_ch_base	x								x
1:0	-									
swreg111										
31:2	sw_refer8_ch_base	x								x
1:0	-									
swreg112										
31:2	sw_refer9_ch_base	x								x
1:0	-									
swreg113										

Table continues on the next page...

VPU G1 Memory Map/Register Definition

Bit	Name	Dec Modes								
		H.264	H.263	JPEG	VC-1	MPEG2/ MPEG1	VP6	RV	VP7 / VP8	AVS
31:2	sw_refer10_ch_base	x								x
1:0	-									
swreg114										
31:2	sw_refer11_ch_base	x								x
1:0	-									
swreg115										
31:2	sw_refer12_ch_base	x								x
1:0	-									
swreg116										
31:2	sw_refer13_ch_base	x								x
1:0	-									
swreg117										
31:2	sw_refer14_ch_base	x								x
1:0	-									
swreg118										
31:2	sw_refer15_ch_base	x								x
1:0	-									

14.1.5 VPU G1 Memory Map/Register Definition

14.1.5.1 VPU_G1 common registers descriptions

14.1.5.1.1 VPU_G1 common registers memory map

VPU_G1 base address: 3830_0000h

Offset	Register	Width (In bits)	Access	Reset value
4h	Interrupt register decoder (SWREG1)	32	RW	0000_0000h
8h	Device configuration register decoder (SWREG2)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
Ch	Decoder control register 0 (decmode,picture type etc) (SWREG3)	32	RW	0000_0000h
30h	Base address for RLC data (RLC) / stream start address/decoded end addr register (VLC) (SWREG12)	32	RW	0000_0000h
34h	Base address for decoded picture (SWREG13)	32	RW	0000_0000h
A0h	Base address for standard dependent tables (SWREG40)	32	RW	0000_0000h
A4h	Base address for direct mode motion vectors (SWREG41)	32	RW	0000_0000h
C0h	Error concealment register (SWREG48)	32	RW	0000_0000h
C4h	Prediction filter tap register for H264 (SWREG49)	32	RW	0000_0000h
C8h	Synthesis configuration register decoder 0 (SWREG50)	32	RO	Table 14-
CCh	Reference picture buffer control register (SWREG51)	32	RW	0000_0000h
D0h	Reference picture buffer information register 1 (SWREG52)	32	RO	0000_0000h
D4h	Reference picture buffer information register 2 (SWREG53)	32	RO	0000_0000h
D8h	Synthesis configuration register decoder 1 (SWREG54)	32	RO	0000_0000h
DCh	Reference picture buffer 2 / Advanced prefetch control register (SWREG55)	32	RW	0000_0000h
E0h	Reference buffer information register 3 (SWREG56)	32	RO	0000_0000h
E4h	Decoder fuse register (SWREG57)	32	RO	Table 14-
E8h	Device configuration register decoder 2 + Multi core control register (SWREG58)	32	RW	0000_0000h
ECh	H264 Chrominance 8 pixel interleaved data base (SWREG59)	32	RW	0000_0000h
F0h	Interrupt register post-processor (SWREG60)	32	RW	0000_0000h
F4h	Device configuration register post-processor (SWREG61)	32	RW	0000_0100h
F8h	Deinterlace control register (SWREG62)	32	RW	0000_0000h
FCh	Base address for reading post-processing input picture luminance (top field/frame) (SWREG63)	32	RW	0000_0000h
100h	Base address for reading post-processing input picture Cb/Ch (top field/frame) (SWREG64)	32	RW	0000_0000h
104h	Base address for reading post-processing input picture Cr (SWREG65)	32	RW	0000_0000h
108h	Base address for writing post-processed picture luminance/RGB (SWREG66)	32	RW	0000_0000h
10Ch	Base address for writing post-processed picture Ch (SWREG67)	32	RW	0000_0000h
110h	Register for contrast adjusting (SWREG68)	32	RW	0000_0000h
114h	Register for colour conversion and contrast adjusting/YUYV 422 channel orders (SWREG69)	32	RW	0000_0000h
118h	Register for colour conversion 0 (SWREG70)	32	RW	0000_0000h
11Ch	Register for colour conversion 1 + rotation mode (SWREG71)	32	RW	0000_0000h
120h	PP input size and -cropping register (SWREG72)	32	RW	0000_0000h
124h	PP input picture base address for Y bottom field (SWREG73)	32	RW	0000_0000h
128h	PP input picture base for Ch bottom field (SWREG74)	32	RW	0000_0000h
13Ch	Scaling register 0 ratio and padding for R and G (SWREG79)	32	RW	0000_0000h
140h	Scaling ratio register 1 and padding for B (SWREG80)	32	RW	0000_0000h

Table continues on the next page...

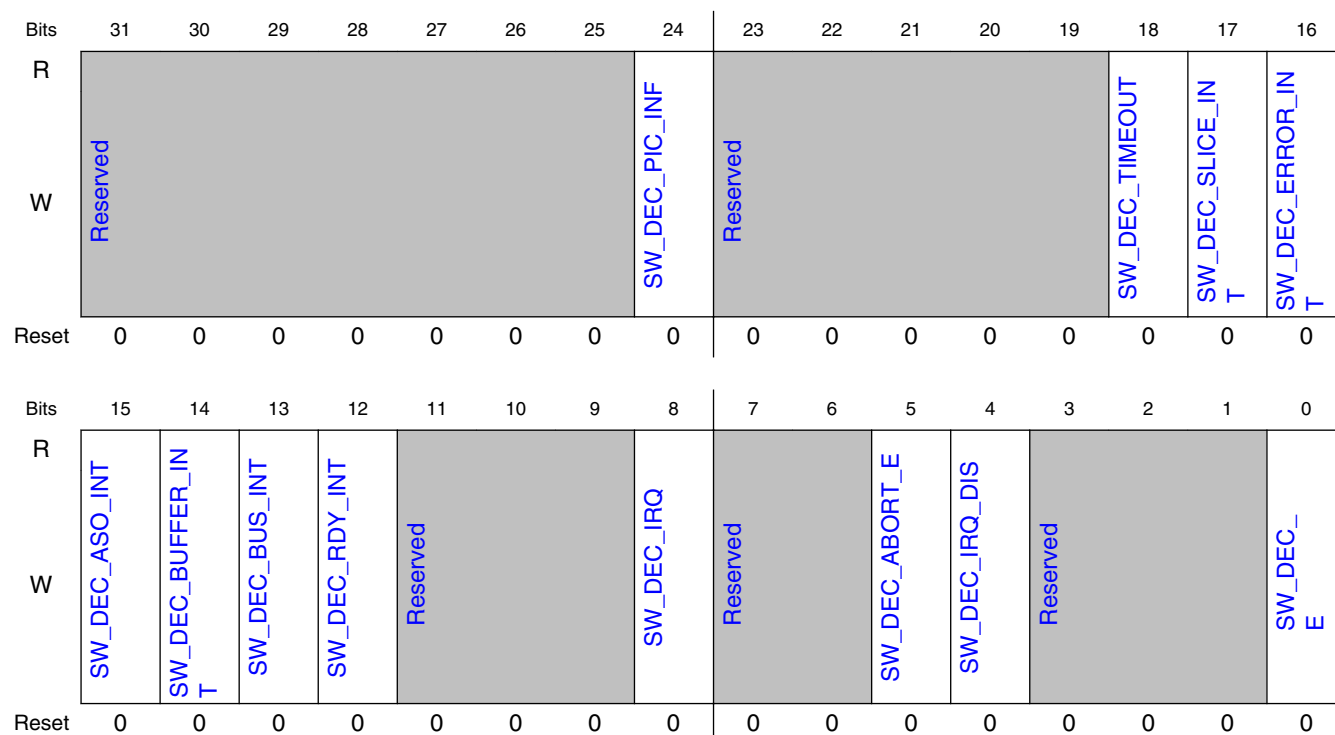
Offset	Register	Width (In bits)	Access	Reset value
144h	Scaling ratio register 2 (SWREG81)	32	RW	0000_0000h
148h	Rmask register (SWREG82)	32	RW	0000_0000h
14Ch	Gmask register (SWREG83)	32	RW	0000_0000h
150h	Bmask register (SWREG84)	32	RW	0000_0000h
154h	Post-processor control register (SWREG85)	32	RW	0000_0000h
158h	Mask 1 start coordinate register (SWREG86)	32	RW	0000_0000h
15Ch	Mask 2 start coordinate register + Mask extensions (SWREG87)	32	RW	0000_0000h
160h	Mask 1 size and PP original width register (SWREG88)	32	RW	0000_0000h
164h	Mask 2 size register + mask extensions (SWREG89)	32	RW	0000_0000h
168h	PiP register 0 (SWREG90)	32	RW	0000_0000h
16Ch	PiP register 1 and dithering control (SWREG91)	32	RW	0000_0000h
170h	Display width and PP input size extension register (SWREG92)	32	RW	0000_0000h
174h	Base address for alpha blend 1 gui component (SWREG93)	32	RW	0000_0000h
178h	Base address for alpha blend 2 gui component (SWREG94)	32	RW	0000_0000h
17Ch	Alpha blend input cropping register (scanline for cropping) (SWREG95)	32	RW	0000_0000h
18Ch	PP fuse register (SWREG99)	32	RO	Table 14-
190h	Synthesis configuration register post-processor (SWREG100)	32	RO	Table 14-
198h	Base address for H264 decoded chroma picture (SWREG102)	32	RW	0000_0000h
19Ch	Base address for reference chroma picture index 0 (SWREG103)	32	RW	0000_0000h
1A0h	Base address for reference chroma picture index 1 (SWREG104)	32	RW	0000_0000h
1A4h	Base address for reference chroma picture index 2 (SWREG105)	32	RW	0000_0000h
1A8h	Base address for reference chroma picture index 3 (SWREG106)	32	RW	0000_0000h
1ACh	Base address for reference chroma picture index 4 (SWREG107)	32	RW	0000_0000h
1B0h	Base address for reference chroma picture index 5 (SWREG108)	32	RW	0000_0000h
1B4h	Base address for reference chroma picture index 6 (SWREG109)	32	RW	0000_0000h
1B8h	Base address for reference chroma picture index 7 (SWREG110)	32	RW	0000_0000h
1BCh	Base address for reference chroma picture index 8 (SWREG111)	32	RW	0000_0000h
1C0h	Base address for reference chroma picture index 9 (SWREG112)	32	RW	0000_0000h
1C4h	Base address for reference chroma picture index 10 (SWREG113)	32	RW	0000_0000h
1C8h	Base address for reference chroma picture index 11 (SWREG114)	32	RW	0000_0000h
1CCh	Base address for reference chroma picture index 12 (SWREG115)	32	RW	0000_0000h
1D0h	Base address for reference chroma picture index 13 (SWREG116)	32	RW	0000_0000h
1D4h	Base address for reference chroma picture index 14 (SWREG117)	32	RW	0000_0000h
1D8h	Base address for reference chroma picture index 15 (SWREG118)	32	RW	0000_0000h

14.1.5.1.2 Interrupt register decoder (SWREG1)

14.1.5.1.2.1 Offset

Register	Offset
SWREG1	4h

14.1.5.1.2.2 Diagram



14.1.5.1.2.3 Fields

Field	Function
31-25 —	Reserved.
24 SW_DEC_PIC_INF	B slice detected. This signal is driven high during picture ready interrupt if B-type slice is found. This bit does not launch interrupt but is used to inform SW about h264 tools.
23-19 —	Reserved.
18 SW_DEC_TIMEOUT	Interrupt status bit decoder timeout. When high the decoder has been idling for too long. HW will self reset. Possible only if timeout interrupt is enabled

Table continues on the next page...

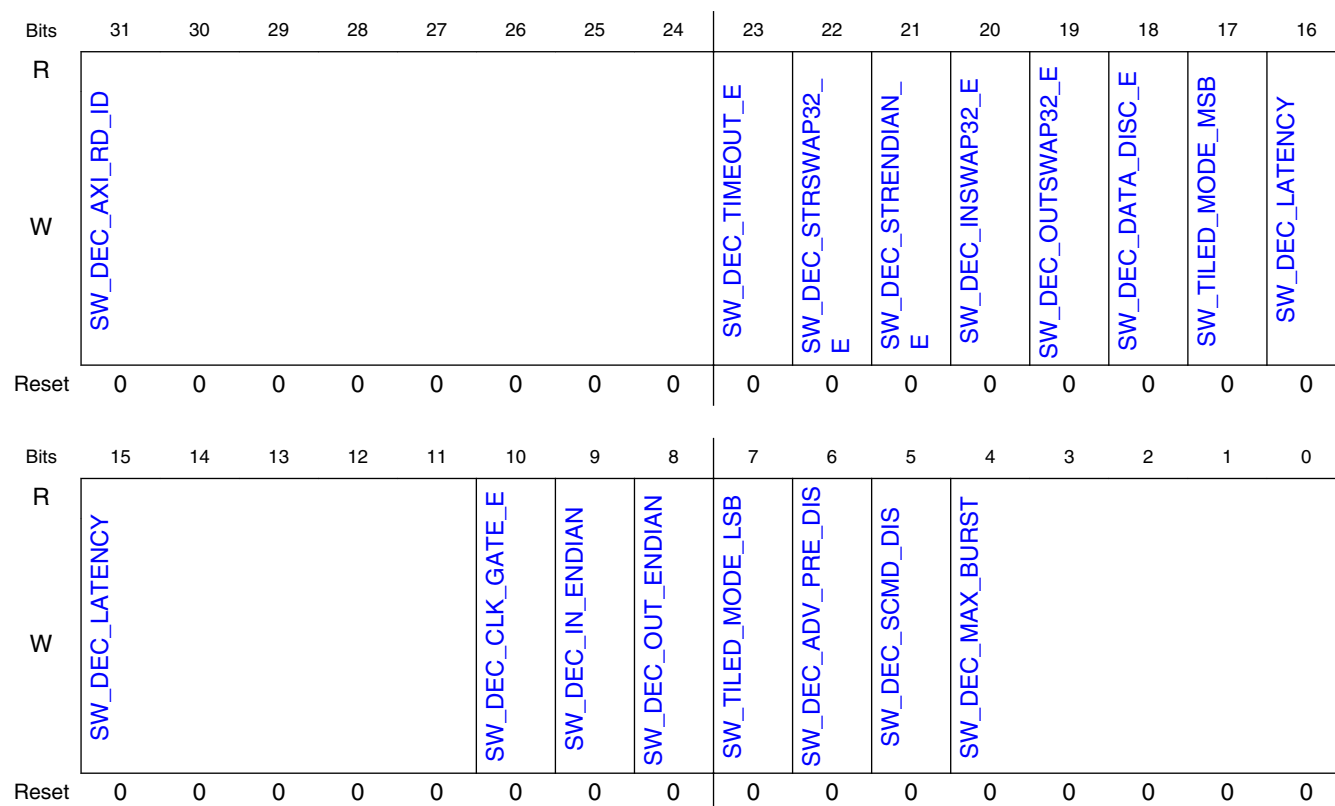
Field	Function
17 SW_DEC_SLICE_INT	Interrupt status bit dec_slice_decoded. When high SW must set new base addresses for sw_dec_out_base and sw_jpg_ch_out_base before resetting this status bit. Used for VP8 web-p modes
16 SW_DEC_ERROR_INT	Interrupt status bit input stream error. When high, an error is found in input data stream decoding. HW will self reset.
15 SW_DEC_ASO_INT	H264: Interrupt status bit ASO (Arbitrary Slice Ordering) detected. When high, ASO detected in input data stream decoding. HW will self reset. VP8: Error detected in Residual data. HW returns MB number in error concealment register for MB it detected it
14 SW_DEC_BUFFER_INT	Interrupt status bit input buffer empty. When high, input stream buffer is empty but picture is not ready. HW will not self reset.
13 SW_DEC_BUS_INT	Interrupt status bit bus. Error response from bus.
12 SW_DEC_RDY_INT	Interrupt status bit decoder. When this bit is high decoder has decoded a picture. HW will self reset.
11-9 —	Reserved.
8 SW_DEC_IRQ	Decoder IRQ. When high, decoder requests an interrupt. SW will reset this after interrupt is handled.
7-6 —	Reserved.
5 SW_DEC_ABORT_E	Abort decoding enable. Setting this bit high will cause HW to abort decoding and safely to reset itself down. After abort is complete the corresponding interrupt status is set and this bit is set low as well as the decoder enable.
4 SW_DEC_IRQ_DIS	Decoder IRQ disable. When high, there are no interrupts concerning decoder from HW. Polling must be used to see the interrupt statuses.
3-1 —	Reserved.
0 SW_DEC_E	Decoder enable. Setting this bit high will start the decoding operation. HW will reset this when picture is processed or ASO or stream error is detected or bus error or timeout interrupt is given.

14.1.5.1.3 Device configuration register decoder (SWREG2)

14.1.5.1.3.1 Offset

Register	Offset
SWREG2	8h

14.1.5.1.3.2 Diagram



14.1.5.1.3.3 Fields

Field	Function
31-24 SW_DEC_AXI_RD_ID	Read ID used for decoder reading services in AXI bus (if connected to AXI).
23 SW_DEC_TIMEOUT_E	Timeout interrupt enable. If enabled HW may return timeout interrupt in case HW gets stuck while decoding picture.
22 SW_DEC_STRSWAP32_E	Decoder input 32bit data swap for stream data (may be used for 64 bit environment): 0b - no swapping of 32 bit words 1b - 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
21 SW_DEC_STRENDIAN_E	Decoder input endian mode for stream data: 0b - Big endian (0-1-2-3 order) 1b - Little endian (3-2-1-0 order)
20	Decoder input 32bit data swap for other than stream data (may be used for 64 bit environment): 0b - no swapping of 32 bit words

Table continues on the next page...

VPU G1 Memory Map/Register Definition

Field	Function
SW_DEC_INSW AP32_E	1b - 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
19 SW_DEC_OUT SWAP32_E	Decoder output 32bit data swap (may be used for 64 bit environment): 0b - no swapping of 32 bit words 1b - 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
18 SW_DEC_DAT A_DISC_E	Data discard enable. Precise burst lengths are used with reading services. Extra data is discarded internally. Note. If AHB maxburst 17 is used data discard cannot be enabled (causes conflict)
17 SW_TILED_MO DE_MSB	Tiled mode msb. Concatenated to Tiled mode lsb which form 2 bit tiled mode. 0b - Tiled mode not enabled 1b - Tiled mode enabled for 8x4 tile size
16-11 SW_DEC_LATE NCY	Decoder master interface additional latency. Can be used to slow down decoder HW between services in steps of 8 clock cycles: 0 = no latency 1 = minimum 8 cycles of IDLE between services 2 = minimum 16 cycles of IDLE between services . . . 63 = minimum latency of 504 cycles of IDLE between services
10 SW_DEC_CLK_ GATE_E	Decoder dynamic clock gating enable: 0b - Clock is running for all structures 1b - Clock is gated for decoder structures that are not used. Note: Clock gating value can be changed only when decoder is disabled.
9 SW_DEC_IN_E NDIAN	Decoder input endian mode for other than stream data: 0b - Big endian (0-1-2-3 order) 1b - Little endian (3-2-1-0 order)
8 SW_DEC_OUT_ ENDIAN	Decoder output endian mode: 0b - Big endian (0-1-2-3 order) 1b - Little endian (3-2-1-0 order)
7 SW_TILED_MO DE_LSB	Tiled mode lsb. Concatenated to Tiled mode msb which form 2 bit tiled mode. Defined in tiled_mode_msb
6 SW_DEC_ADV_ PRE_DIS	Advanced PREFETCH mode disable (advanced reference picture reading mode for video)
5 SW_DEC_SCM D_DIS	9170 decoder and later->: AXI Single Command Multiple Data disable. 9170 axi wrapper supports this mode by default (where only the first addresses of the burst are given from address generator). This bit is used to disable the feature (possible SW workaround if something is not working correctly)
4-0 SW_DEC_MAX_ BURST	Maximum burst length for decoder bus transactions. Valid values: AHB: 1, 4, 8, 16 and 17. Other values will result in INCR type (undefined length) for all transfers. INCR used for lengths 2 and 3.

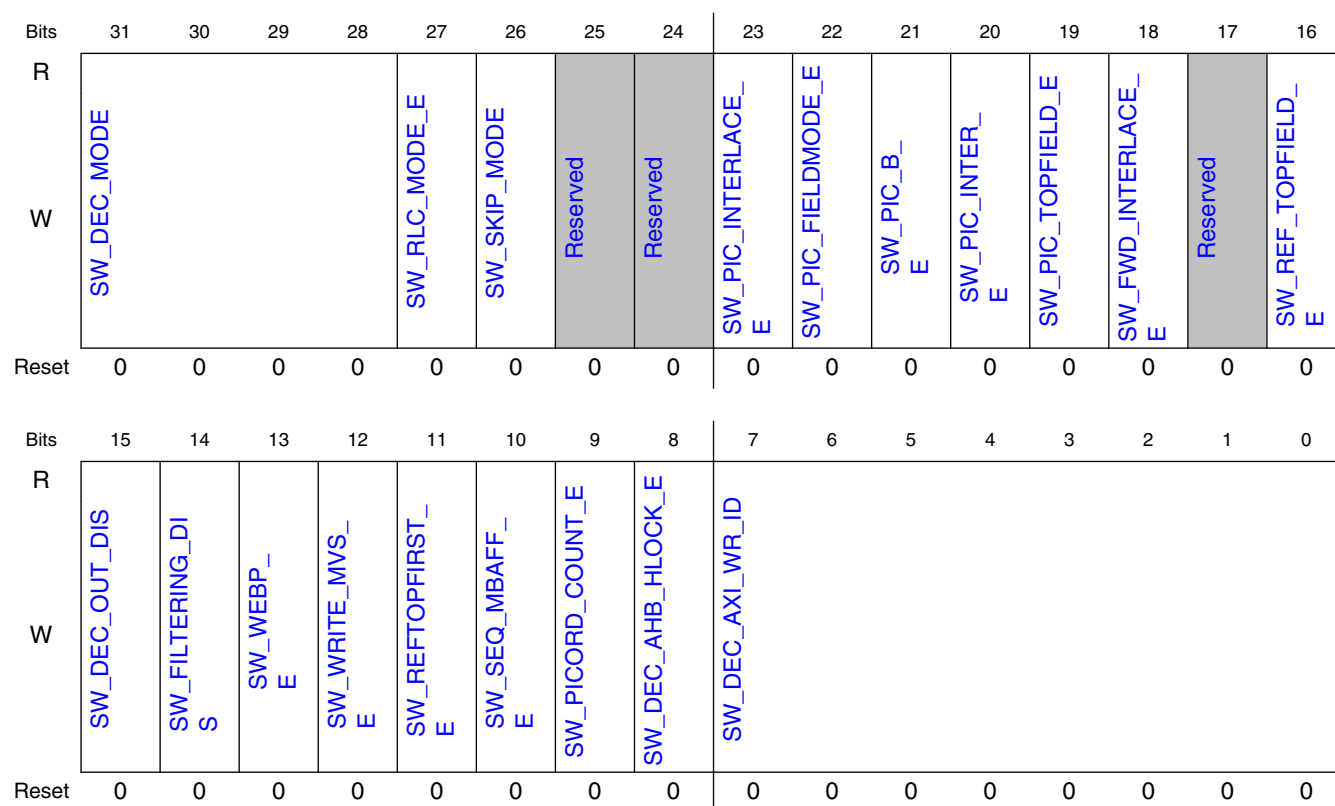
Field	Function
	17 = fixed bursts are used when possible. INCR bursts are used for lengths 2, 3, 5, 6, 7, 9, 10, 11, 12, 13, 14 and 15. OCP: 16-31 AXI: 1-16

14.1.5.1.4 Decoder control register 0 (decmode,picture type etc) (SWREG3)

14.1.5.1.4.1 Offset

Register	Offset
SWREG3	Ch

14.1.5.1.4.2 Diagram



14.1.5.1.4.3 Fields

Field	Function
31-28 SW_DEC_MODE	Decoding mode: 0000b - H.264 0001b - Reserved 0010b - Reserved 0011b - Reserved 0100b - Reserved 0101b - Reserved 0110b - Reserved 0111b - Reserved 1000b - Reserved 1001b - Reserved 1010b - VP8 1011b - Reserved 1100b - Reserved 1101b - Reserved 1110b - Reserved 1111b - Reserved
27 SW_RLC_MODE	RLC mode enable: 0b - HW decodes video from bit stream (VLC mode) + side information 1b - HW decodes video from RLC input data + side information (Differential MV's, separate DC coeffs, Intra 4x4 modes, MB control). Valid only for H.264 Baseline.
26 SW_SKIP_MODE	VP8: 0 : HW decodes mb_coeff_skip -flag 1 : HW does not decode mb_coeff_skip -flag
25 —	Reserved.
24 —	Reserved.
23 SW_PIC_INTERLACE	Coding mode of the current picture: 0b - progressive 1b - interlaced
22 SW_PIC_FIELD_MODE	Structure of the current picture (residual structure) 0b - Frame structure. For H264, this means MBAFF structured picture for interlaced sequence 1b - Field structure
21 SW_PIC_B	B picture enable for current picture: 0b - picture type is I or P depending on sw_pic_inter_e 1b - picture type is B depending on sw_pic_inter_e (not valid for H264 since it is slice based information)
20 SW_PIC_INTER_E	Picture type. Please also see SW_PIC_B. 0b - Intra type (I) 1b - Inter type (P)
19 SW_PIC_TOPFIELD	If field structure is enabled, this bit informs which one of the fields is being decoded: 0b - bottom field 1b - top field
18 SW_FWD_INTERLACE	Coding mode of forward reference picture NOTE: For backward reference picture, the coding mode is always the same as for current picture. 0b - progressive

Table continues on the next page...

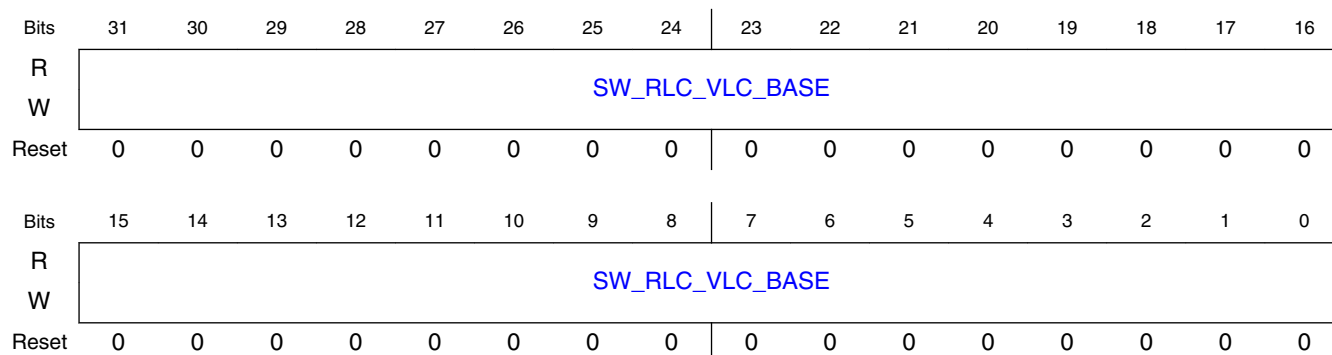
Field	Function
	1b - interlaced
17 —	Reserved.
16 SW_REF_TOPFIELD_E	Indicates which field should be used as reference if sw_ref_frames = '0': 0b - bottom field 1b - top field
15 SW_DEC_OUT_DIS	Disable decoder output picture writing: 0b - Decoder output picture is written to external memory 1b - Decoder output picture is not written to external memory
14 SW_FILTERING_DIS	De-block filtering disable: 0b - filtering is enabled for current picture 1b - filtering is disabled for current picture
13 SW_WEBP_E	SW_WEBP_E: Webp enable for VP8: '0' = Normal VP8 '1' = WEBP picture SW_MVC_E: Multi View Coding enable. Possible for H264 only.
12 SW_WRITE_MVS_E	Direct mode motion vector write enable for current picture / VPX motion vector write enable for error concealment purposes: 0b - Writing disabled for current picture 1b - The direct mode motion vectors are written to external memory. H264 direct mode motion vectors are written to DPB aside with the corresponding reference picture. Other decoding mode dir mode mvs are written to external memory starting from sw_dir_mv_base.
11 SW_REFTOPFIRST_E	Indicates which FWD reference field has been decoded first. 0b - FWD reference bottom field 1b - FWD reference top field
10 SW_SEQ_MBAFF_E	Sequence includes MBAFF coded pictures
9 SW_PIC_ORDER_COUNT_E	h264_high config: Picture order count table read enable. If enabled HW will read picture order counts from memory in the beginning of picture
8 SW_DEC_AHB_HLOCK_E	AHB master HLOCK enable. When high the service is locked to decoder as long as it needs the bus (whenever decoder requests the bus it will be granted)
7-0 SW_DEC_AXI_WR_ID	Write ID used for decoder writing services in AXI bus (if connected to AXI)

14.1.5.1.5 Base address for RLC data (RLC) / stream start address/ decoded end addr register (VLC) (SWREG12)

14.1.5.1.5.1 Offset

Register	Offset
SWREG12	30h

14.1.5.1.5.2 Diagram



14.1.5.1.5.3 Fields

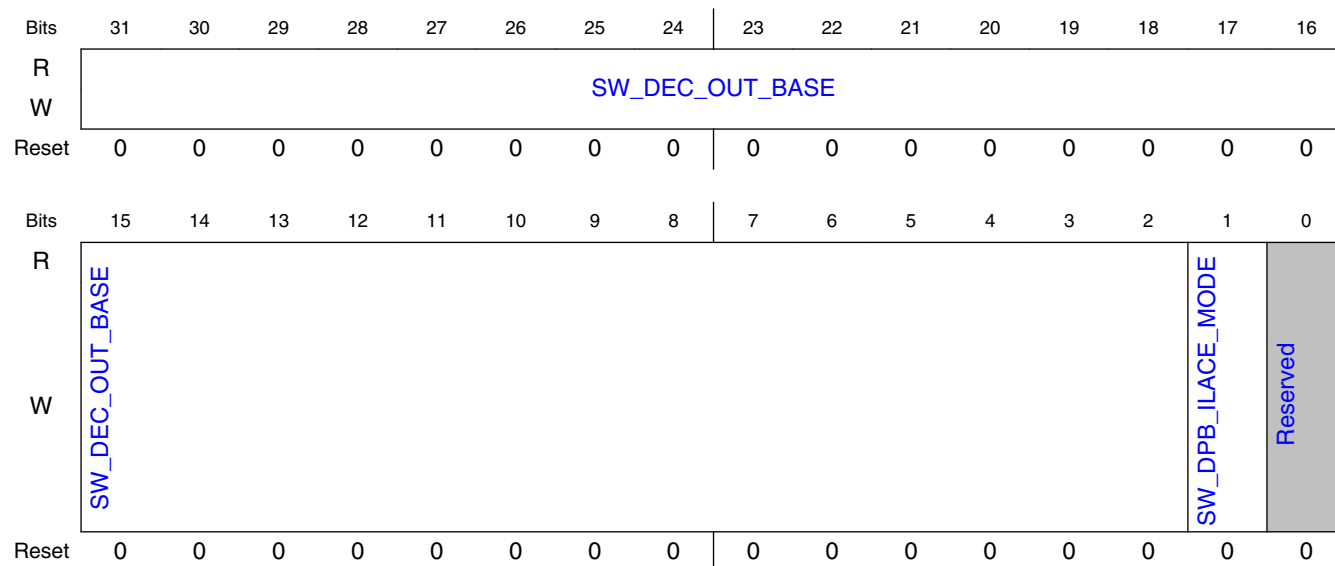
Field	Function
31-0	RLC mode: Base address for RLC data (swreg3.sw_rlc_mode_e = 1).
SW_RLC_VLC_BASE	VLC mode: Stream start address / end address with byte precision (swreg4.rlc_mode_en = 0), start bit number in swreg5.stream_start_bit. When sw_dec_buffer_int is high or sw_dec_e is low this register contains HW return value of last_byte_address where stream has been read (and used) in accuracy of byte. For debug purposes the last_byte_address is also written when stream error/ASO is detected even though it may not be accurate. VP8: This base address is used as sw_dct_strm0_base including DCT stream for MB rows 0,2n.

14.1.5.1.6 Base address for decoded picture (SWREG13)

14.1.5.1.6.1 Offset

Register	Offset
SWREG13	34h

14.1.5.1.6.2 Diagram



14.1.5.1.6.3 Fields

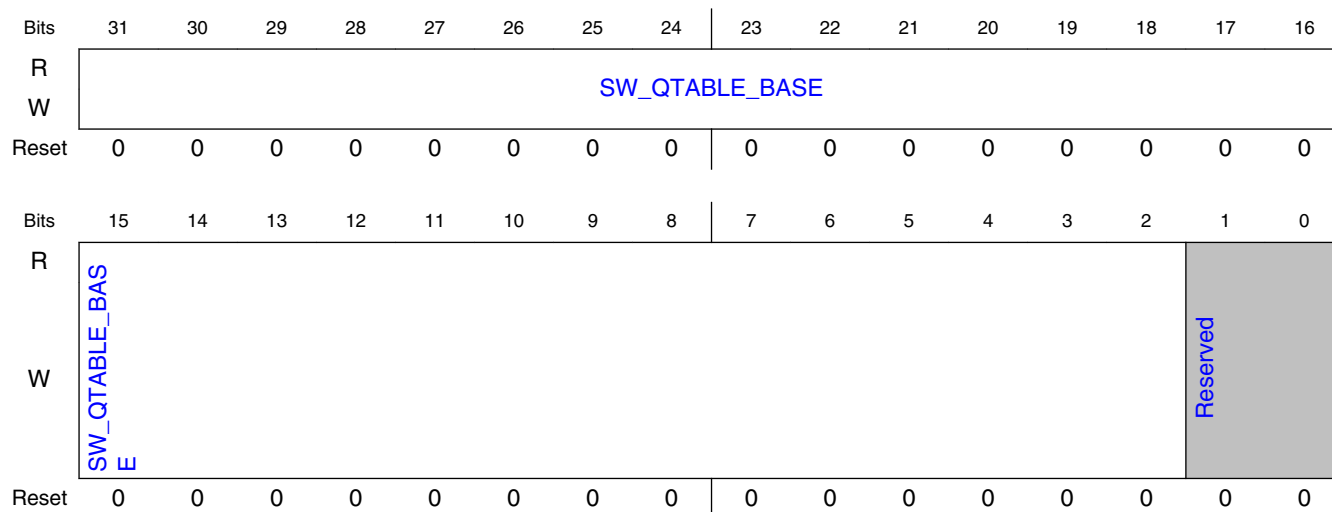
Field	Function
31-2 SW_DEC_OUT_BASE	Video: Base address for decoder output picture. Points directly to start of decoder output picture or field.
1 SW_DPB_ILACE_MODE	DPB ilaced mode: '0' : DPB consist of ilaced/progressive frames '1' : DPB consist of progressive frames / separate fields. This mode requires config support from HW
0 —	Reserved.

14.1.5.1.7 Base address for standard dependent tables (SWREG40)

14.1.5.1.7.1 Offset

Register	Offset
SWREG40	A0h

14.1.5.1.7.2 Diagram



14.1.5.1.7.3 Fields

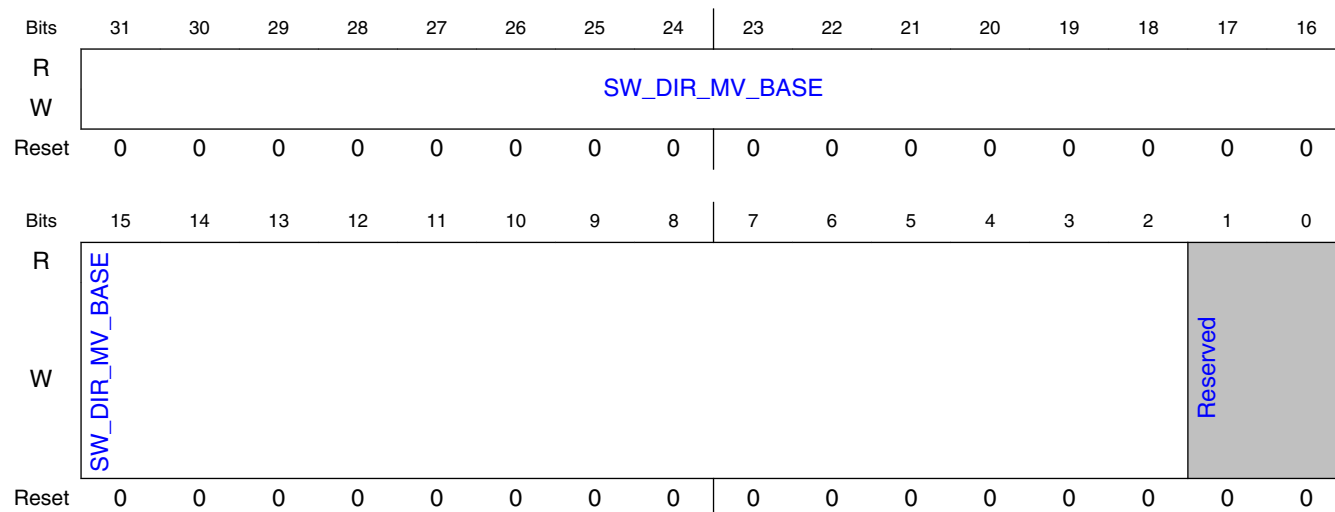
Field	Function
31-2	Base address for standard dependent tables:
SW_QTABLE_BASE	H.264: base address for various tables VP8: base address for stream decoding tables
1-0	Reserved.
—	

14.1.5.1.8 Base address for direct mode motion vectors (SWREG41)

14.1.5.1.8.1 Offset

Register	Offset
SWREG41	A4h

14.1.5.1.8.2 Diagram



14.1.5.1.8.3 Fields

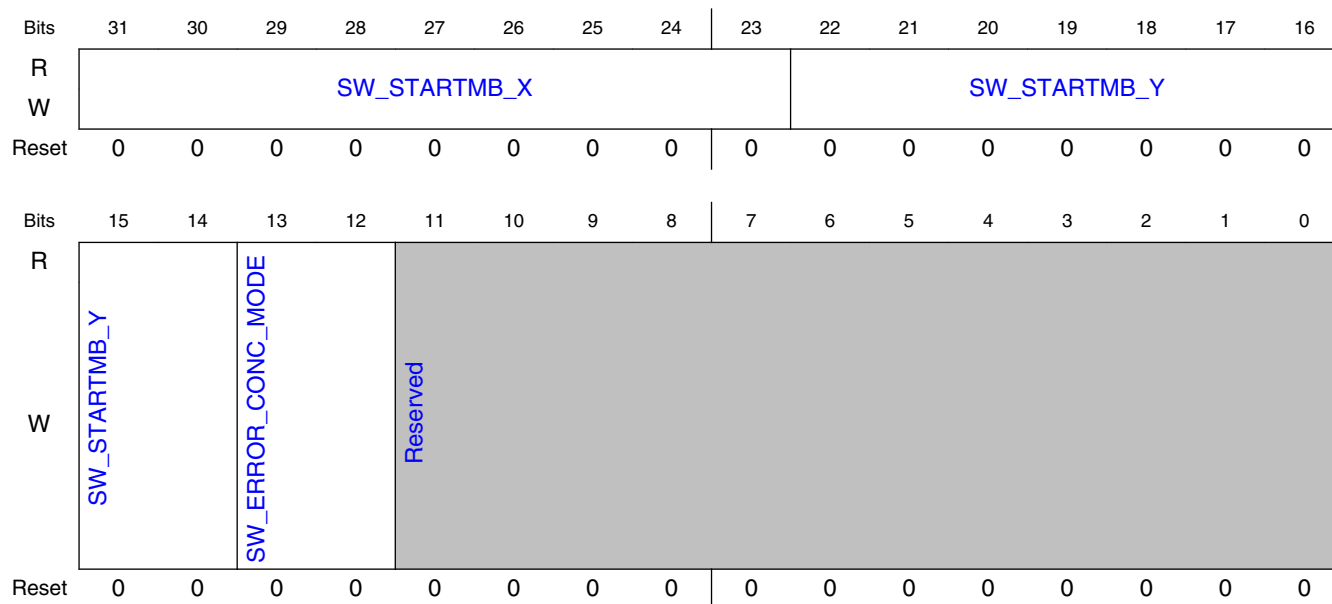
Field	Function
31-2 SW_DIR_MV_BASE	Direct mode motion vector write/read base address. For H264 this is used only for direct mode motion vector write base. VP8: Motion vectors are written for error concealment purposes if sw_write_mvs is high. In error concealment mode motion vectors are read from this base address
1-0 —	Reserved.

14.1.5.1.9 Error concealment register (SWREG48)

14.1.5.1.9.1 Offset

Register	Offset
SWREG48	C0h

14.1.5.1.9.2 Diagram



14.1.5.1.9.3 Fields

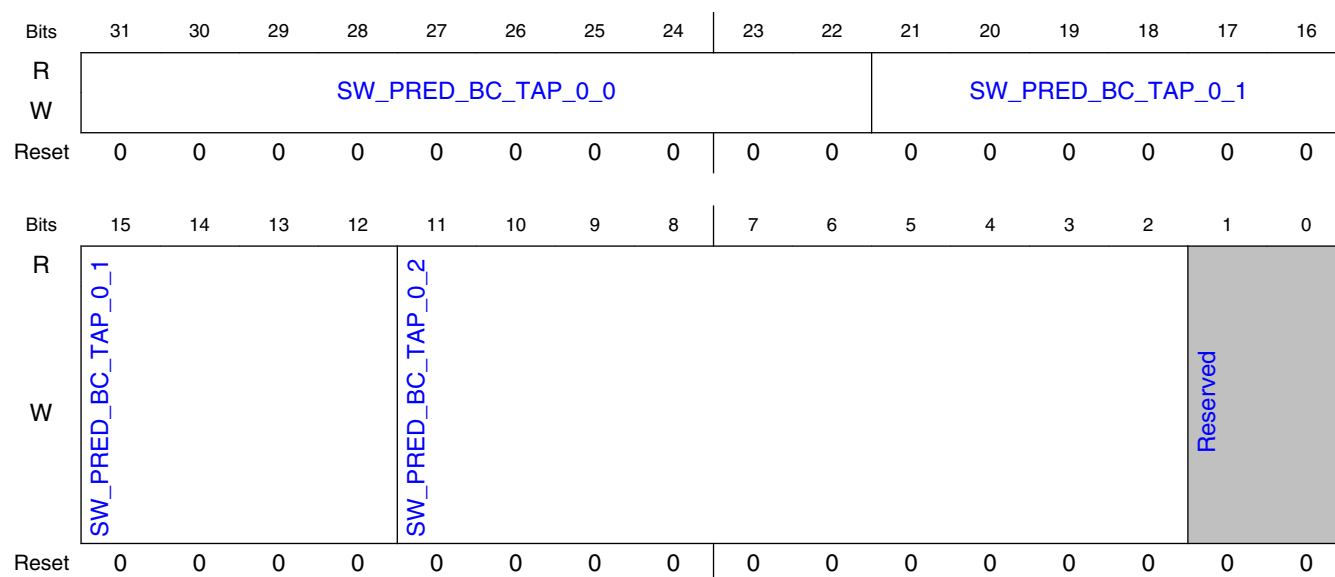
Field	Function
31-23 SW_STARTMB_X	Start MB from SW for X dimension. Used in error concealment case as HW return value if HW finds an error or in HW init mb for error concealment if SW enables error concealment
22-14 SW_STARTMB_Y	Start MB from SW for Y dimension. Used in error concealment case as HW return value if HW finds an error or in HW init mb for error concealment if SW enables error concealment
13-12 SW_ERROR_CONC_MODE	Error concealment mode: 00b - disabled (normal decoding mode) 01b - enabled for direct mode MV usage starting from MB defined by sw_startmb_x, sw_startmb_y
11-0 —	Reserved.

14.1.5.1.10 Prediction filter tap register for H264 (SWREG49)

14.1.5.1.10.1 Offset

Register	Offset
SWREG49	C4h

14.1.5.1.10.2 Diagram



14.1.5.1.10.3 Fields

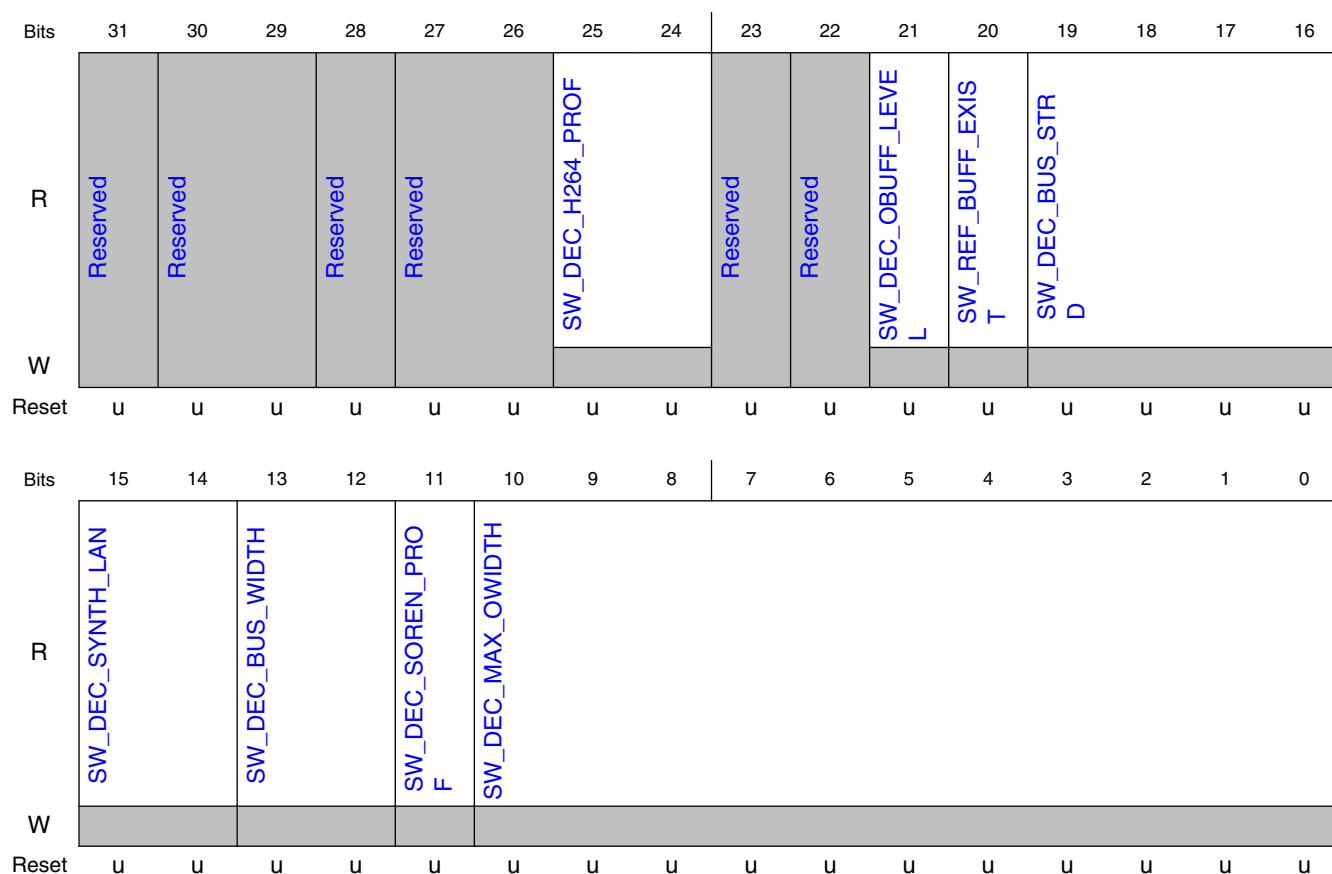
Field	Function
31-22 SW_PRED_BC_TAP_0_0	Prediction filter set 0, tap 0
21-12 SW_PRED_BC_TAP_0_1	Prediction filter set 0, tap 1
11-2 SW_PRED_BC_TAP_0_2	Prediction filter set 0, tap 2
1-0 —	Reserved.

14.1.5.1.11 Synthesis configuration register decoder 0 (SWREG50)

14.1.5.1.11.1 Offset

Register	Offset
SWREG50	C8h

14.1.5.1.11.2 Diagram



14.1.5.1.11.3 Fields

Field	Function
31 —	Reserved.
30-29 —	Reserved.
28 —	Reserved.
27-26 —	Reserved.
25-24 SW_DEC_H264_PROF	Decoding format support, H.264 00b - not supported 01b - supported up to baseline profile 10b - supported up to high profile labeled stream with restricted high profile tools (Tools that are used in Hantro 7280, 8270 encoder) 11b - supported up to high profile

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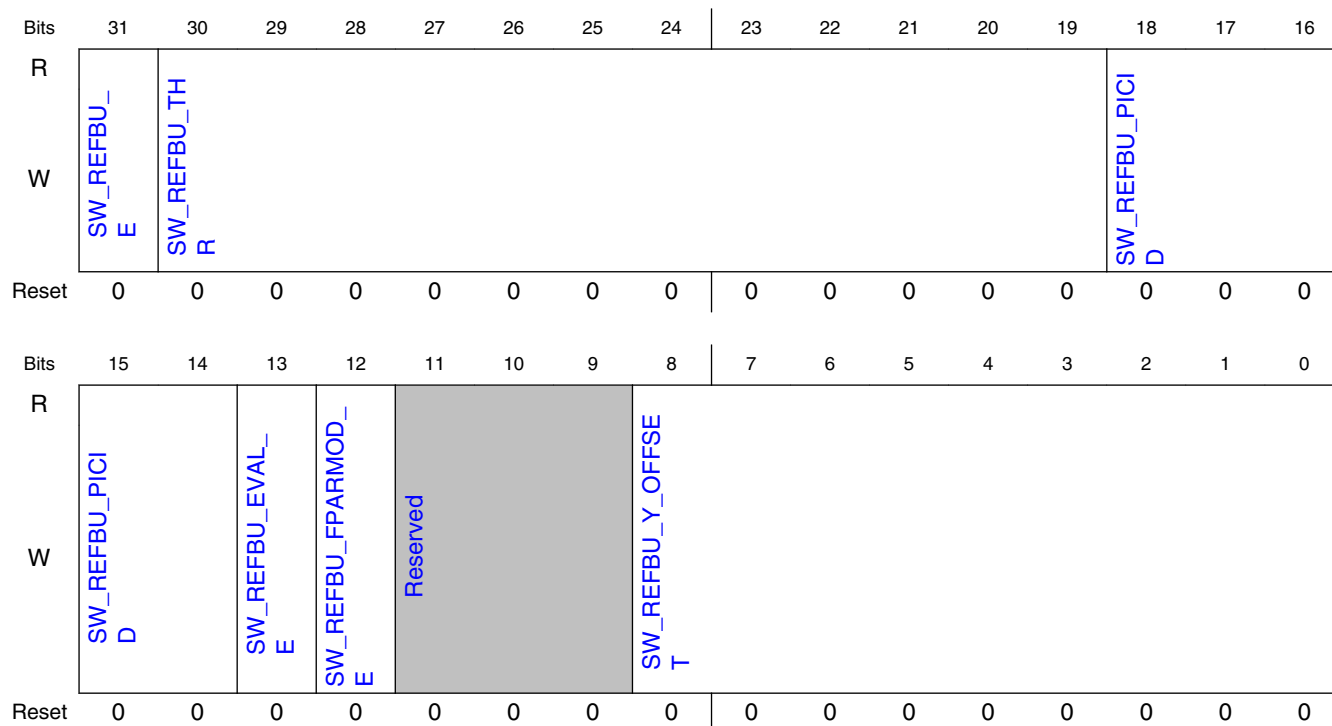
Field	Function
23 —	Reserved.
22 —	Reserved.
21 SW_DEC_OBU FF_LEVEL	Decoder output buffer level: 0b - 1 MB buffering is used 1b - 4 MB buffering is used
20 SW_REF_BUFF _EXIST	Reference picture buffer usage: 0b - not supported 1b - reference buffer is used
19-16 SW_DEC_BUS_ STRD	Connected to standard bus: 0000b - error 0001b - AHB master, AHB slave 0010b - OCP master, OCP slave 0011b - AXI master, AXI slave 0100b - AXI master, APB slave 0101b - AXI master, AHB slave
15-14 SW_DEC_SYN TH_LAN	00b - error 01b - vhdI 10b - verilog
13-12 SW_DEC_BUS_ WIDTH	00b - error 01b - 32 bit bus 10b - 64 bit bus 11b - 128 bit bus
11 SW_DEC_SOR EN_PROF	Decoding format support, Sorenson 0b - not supported 1b - supported
10-0 SW_DEC_MAX _OWIDTH	Max configured decoder video resolution that can be decoded. Informed as width of the picture in pixels.

14.1.5.1.12 Reference picture buffer control register (SWREG51)

14.1.5.1.12.1 Offset

Register	Offset
SWREG51	CCh

14.1.5.1.12.2 Diagram



14.1.5.1.12.3 Fields

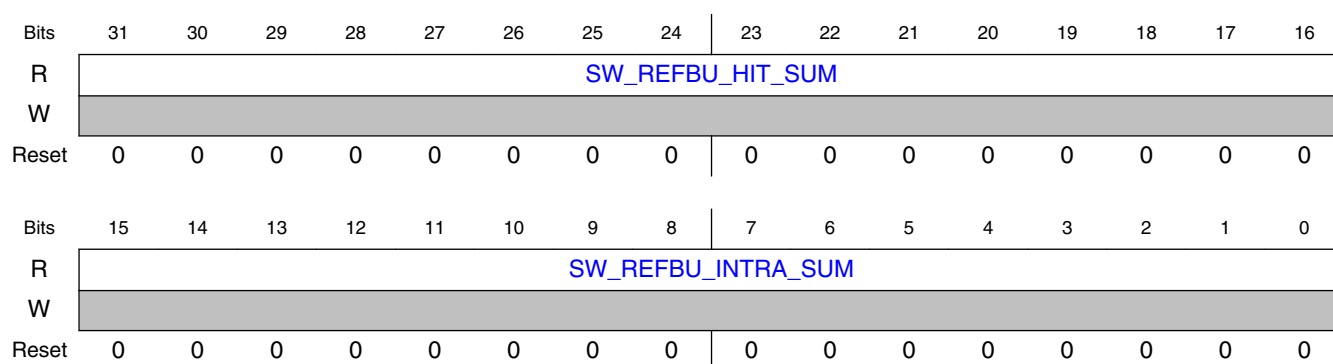
Field	Function
31 SW_REFBU_E	Refer picture buffer enable: 0b - refer picture buffer disabled 1b - refer picture buffer enabled. Valid if picture size is QVGA or more.
30-19 SW_REFBU_TH R	Reference buffer disable threshold value (cache miss amount). Used to buffer shut down (if more misses than allowed)
18-14 SW_REFBU_PICID	The used reference picture ID for reference buffer usage
13 SW_REFBU_EVAL_E	Enable for HW internal reference ID calculation. If given threshold level is reached by any picture_id after first MB row, that picture_id is used for reference buffer fill for rest of the picture
12 SW_REFBU_FPARMOD_E	Field parity mode enable. Used in reffbufferd evaluation mode. 0b - use the result field of the evaluation 1b - use the parity mode field
11-9 —	Reserved.
8-0 SW_REFBU_Y_OFFSET	Y offset for reffbufferd. This coordinate is used to compensate the global motion of the video for better buffer hit rate

14.1.5.1.13 Reference picture buffer information register 1 (SWREG52)

14.1.5.1.13.1 Offset

Register	Offset
SWREG52	D0h

14.1.5.1.13.2 Diagram



14.1.5.1.13.3 Fields

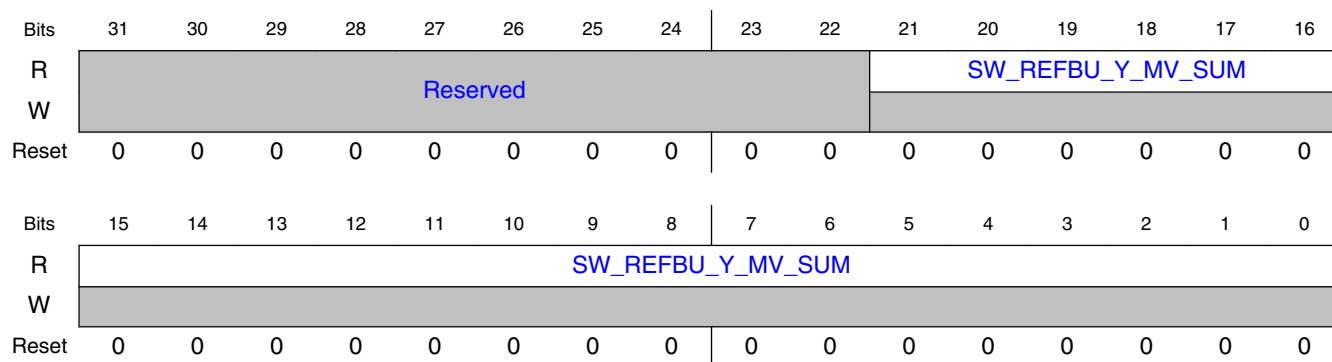
Field	Function
31-16 SW_REFBU_HIT_SUM	The sum of the rebuffered hits of the picture. Determined for each 8x8 luminance partition of the picture. The proceeding of the HW calculation can be read during HW decoding.
15-0 SW_REFBU_INTRA_SUM	The sum of the luminance 8x8 intra partitions of the picture. The proceeding of the HW calculation can be read during HW decoding.

14.1.5.1.14 Reference picture buffer information register 2 (SWREG53)

14.1.5.1.14.1 Offset

Register	Offset
SWREG53	D4h

14.1.5.1.14.2 Diagram



14.1.5.1.14.3 Fields

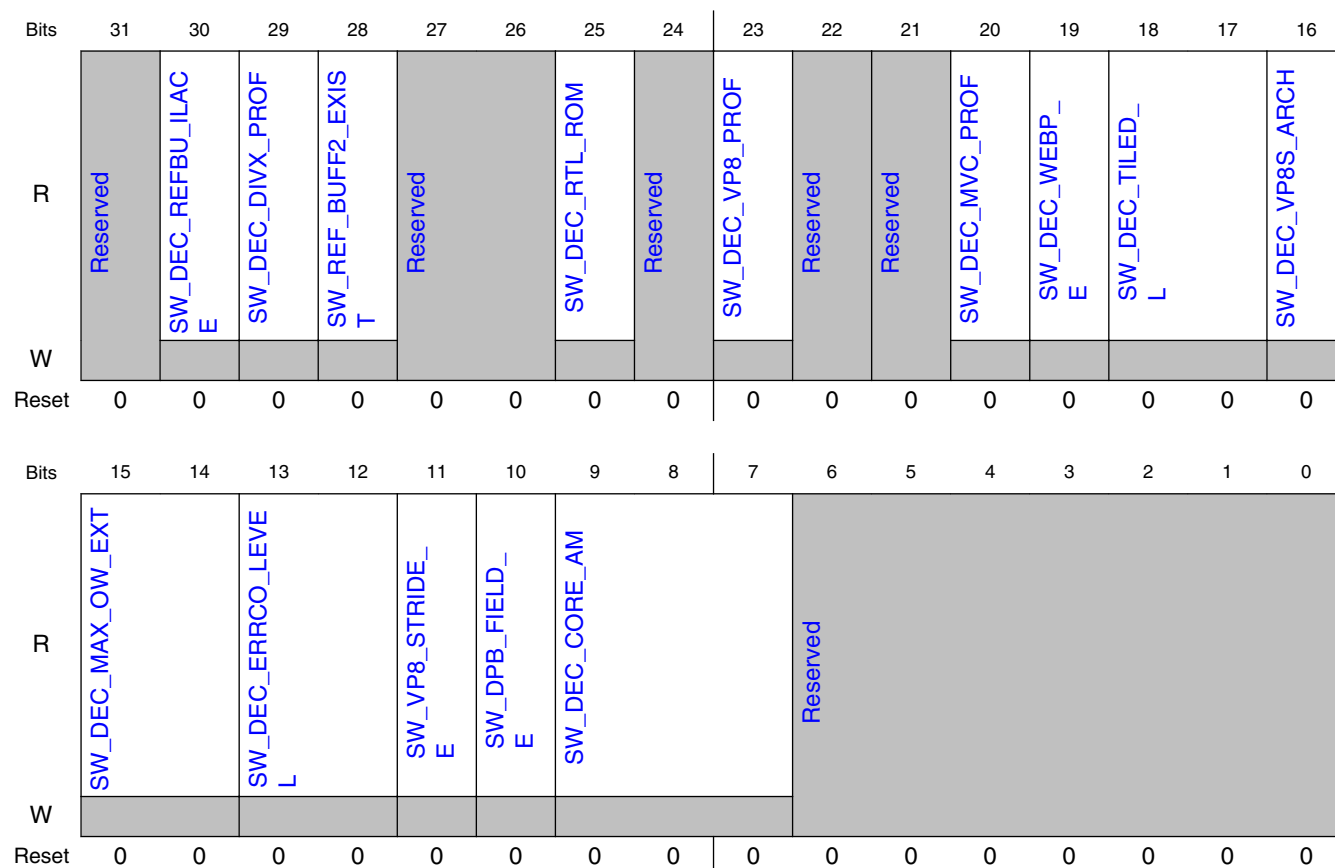
Field	Function
31-22 —	Reserved.
21-0 SW_REFBU_Y_MV_SUM	The sum of the decoded motion vector y-components of the picture. The first luminance motion vector of each MB is used in calculation. Other motion vectors of the MB are discarded. Each motion vector is saturated between -256 - 255 before calculation. The proceeding of the HW calculation can be read during HW decoding.

14.1.5.1.15 Synthesis configuration register decoder 1 (SWREG54)

14.1.5.1.15.1 Offset

Register	Offset
SWREG54	D8h

14.1.5.1.15.2 Diagram



14.1.5.1.15.3 Fields

Field	Function
31 —	Reserved.
30 SW_DEC_REF BU_ILACE	Refbufferd support for interlaced content: 0b - not supported 1b - supported
29 SW_DEC_DIVX _PROF	DIVX Support: 0b - not supported 1b - supported
28 SW_REF_BUFF 2_EXIST	Reference picture buffer 2 usage: 0b - not supported 1b - reference buffer 2 is used
27-26 —	Reserved.
25	ROM implementation type (If design includes ROMs) 0b - ROMs are implemented from actual ROM units

Table continues on the next page...

VPU G1 Memory Map/Register Definition

Field	Function
SW_DEC_RTL_ROM	1b - ROMs are implemented from RTL
24 —	Reserved.
23 SW_DEC_VP8_PROF	Decoding format support, VP8 0b - not supported 1b - supported
22 —	Reserved.
21 —	Reserved.
20 SW_DEC_MVC_PROF	Decoding format support, MVC 0b - not supported 1b - supported
19 SW_DEC_WEBP_E	Decoding format support, Web-p 0b - not supported bigger than 1080p resolution 1b - supported upto 16kx16k pixel resolution (defined max)
18-17 SW_DEC_TILE_D_L	Tiled mode support level 00b - not supported 01b - supported with 8x4 tile size for progressive content 10b - supported with 8x4 tile size for progressive/ilaced content
16 SW_DEC_VP8S_ARCH	VP8 Architecture type (for prediction) 0b - Same prediction architecture as for other decoding formats 1b - Dedicated small architecture for VP8 (refbuffer cannot be used either)
15-14 SW_DEC_MAX_OW_EXT	Max configured decoder video resolution that can be decoded. This is the MSB part of the configuration signal
13-12 SW_DEC_ERR_CO_LEVEL	Decoder error concealment support level: 00b - Error concealment not supported (only error detection) 01b - VP8 direct mode motion vector error concealment supported
11 SW_VP8_STRIDE_E	Decoder output stride support for VP8. Separate base addresses for Y/C data and possibility to set scanline bigger than picture width: 0b - not supported, Y and C tables attached. 1b - supported, Y and C tables can be set freely.
10 SW_DPB_FIELD_E	DPB field separate mode support for ilaced content: 0b - Not supported. For ilaced content, DPB is ilaced frame order. 1b - Supported. For ilaced content, DPB can consist of ilaced frames or separate fields (TOP/BOT).
9-7 SW_DEC_CORE_AM	Decoder core amount. If other than 0, the multicore can be used. Each individual cores can be identified from corresponding core ID register: 000b - single core decoder 001b - dual core decoder 010b - 3 core decoder 011b - 4 core decoder 100b - 5 core decoder 101b - 6 core decoder 110b - 7 core decoder 111b - 8 core decoder

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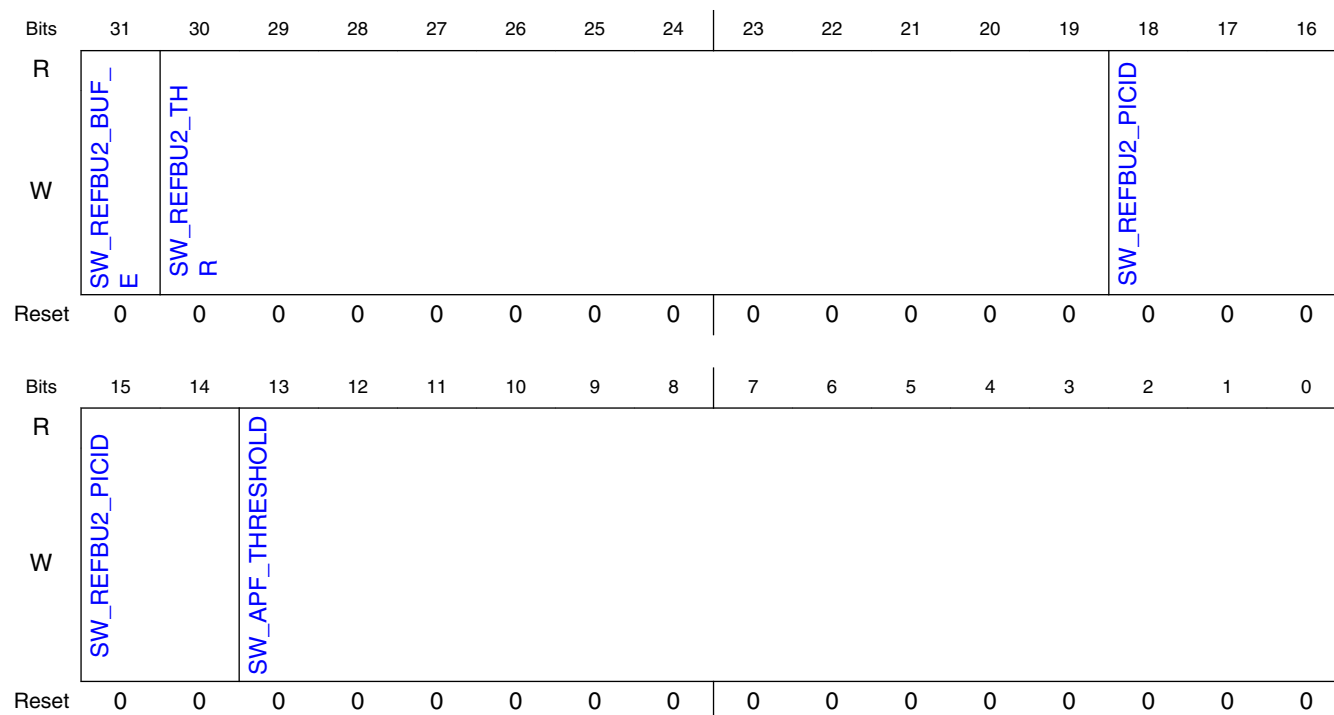
Field	Function
6-0 —	Reserved.

14.1.5.1.16 Reference picture buffer 2 / Advanced prefetch control register (SWREG55)

14.1.5.1.16.1 Offset

Register	Offset
SWREG55	DCh

14.1.5.1.16.2 Diagram



14.1.5.1.16.3 Fields

Field	Function
31 SW_REFBU2_BUF_E	Refer picture buffer 2 enable: 0b - refer picture buffer disabled

Table continues on the next page...

VPU G1 Memory Map/Register Definition

Field	Function
	1b - refer picture buffer enabled. Valid if picture size is QVGA or more (can be turned of by HW if threshold value reached).
30-19 SW_REFBU2_THR	Reference buffer disable threshold value (buffer miss amount). Used to buffer shut down (if more misses than allowed)
18-14 SW_REFBU2_PICID	The used reference picture ID for reference buffer usage
13-0 SW_APF_THRESHOLD	G1 decoder and later :Advanced prefetch threshold value. If current MB exceeds the threshold the advanced mode is not used. Value 0 disables threshold usage and advanced prefetch usage is restricted by internal memory limitation only

14.1.5.1.17 Reference buffer information register 3 (SWREG56)

14.1.5.1.17.1 Offset

Register	Offset
SWREG56	E0h

14.1.5.1.17.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFBU_TOP_SUM															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFBU_BOT_SUM															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.1.5.1.17.3 Fields

Field	Function
31-16 SW_REFBU_TOP_SUM	The sum of the top partitions of the picture
15-0	The sum of the bottom partitions of the picture

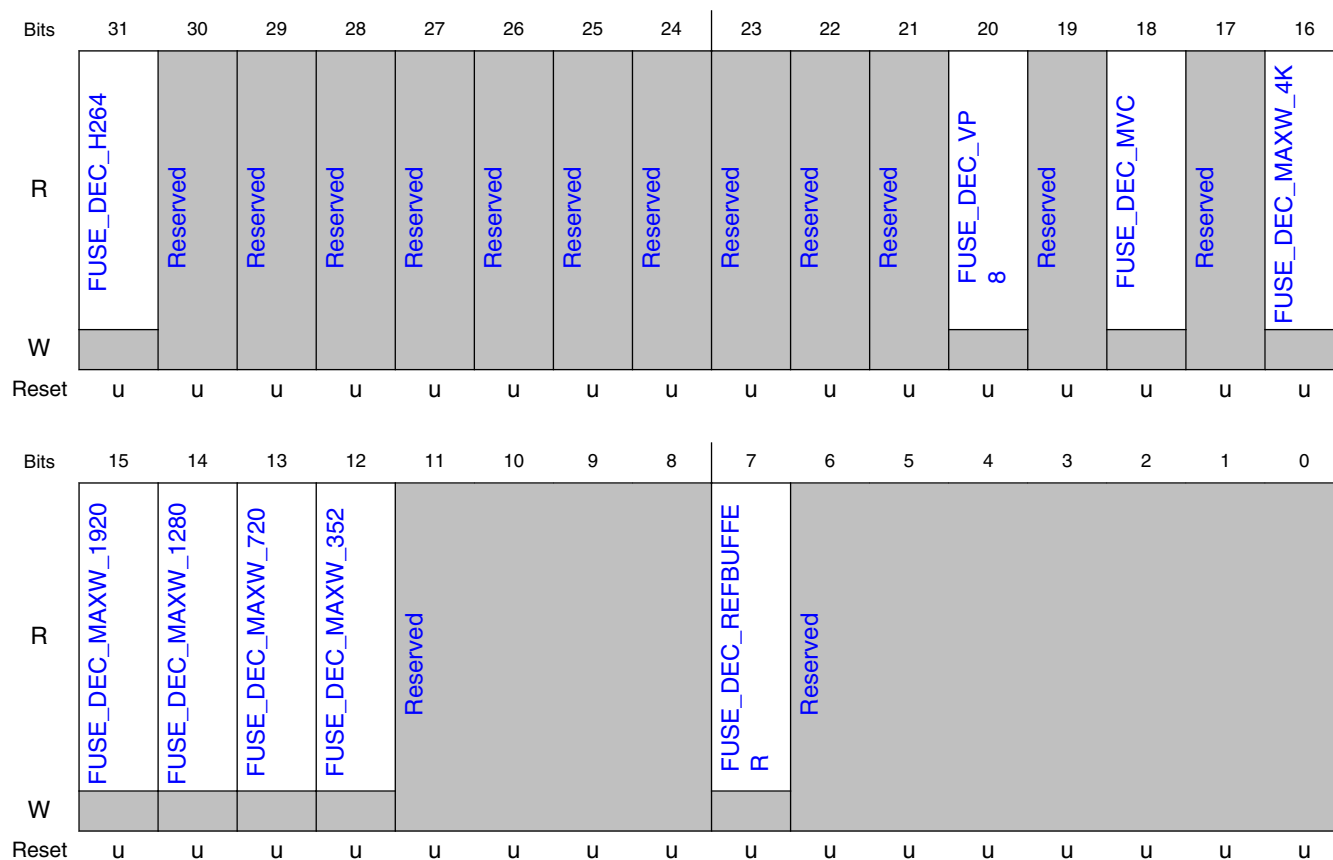
Field	Function
SW_REFBU_B OT_SUM	

14.1.5.1.18 Decoder fuse register (SWREG57)

14.1.5.1.18.1 Offset

Register	Offset
SWREG57	E4h

14.1.5.1.18.2 Diagram



14.1.5.1.18.3 Fields

Field	Function
31 FUSE_DEC_H2 64	1 = H.264 enabled
30 —	Reserved.
29 —	Reserved.
28 —	Reserved.
27 —	Reserved.
26 —	Reserved.
25 —	Reserved.
24 —	Reserved.
23 —	Reserved.
22 —	Reserved.
21 —	Reserved.
20 FUSE_DEC_VP 8	1 = VP8 enabled
19 —	Reserved.
18 FUSE_DEC_MV C	1 = MVC enabled (requires also H264 to be enabled)
17 —	Reserved.
16 FUSE_DEC_MA XW_4K	1 = Max video width up to 4096 pixels enabled. Priority coded with priority 1.
15 FUSE_DEC_MA XW_1920	1 = Max video width up to 1920 pixels enabled. Priority coded with priority 2.

Table continues on the next page...

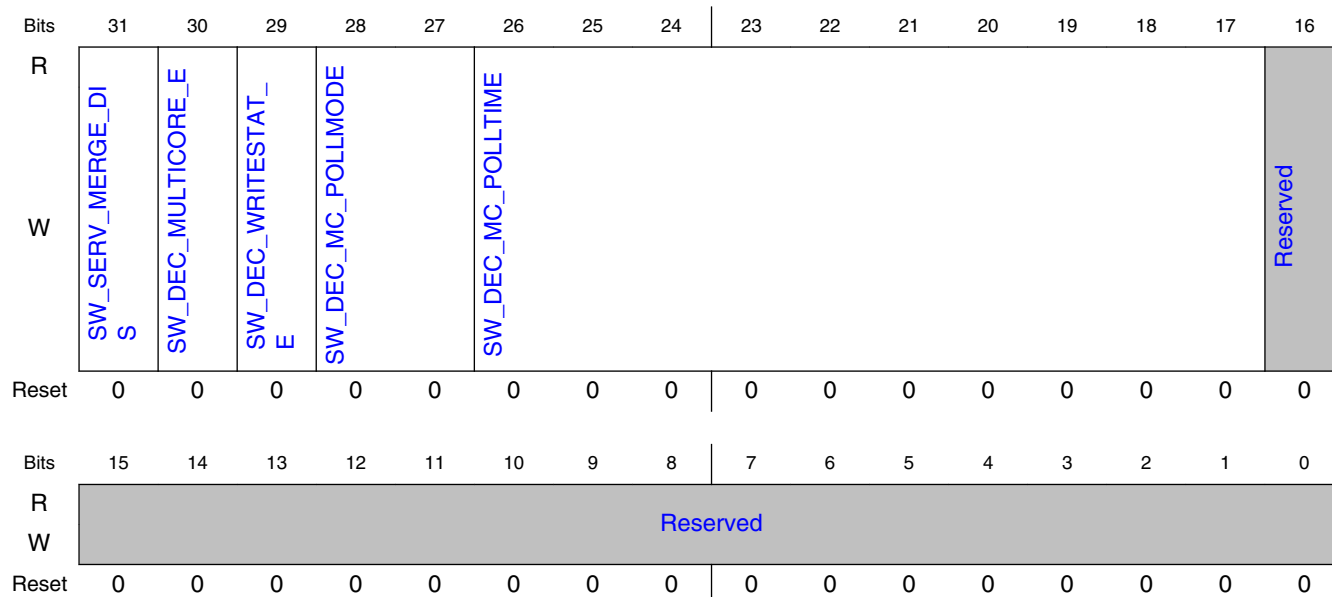
Field	Function
14 FUSE_DEC_MAXW_1280	1 = Max video width up to 1280 pixels enabled. Priority coded with priority 3.
13 FUSE_DEC_MAXW_720	1 = Max video width up to 720 pixels enabled. Priority coded with priority 4.
12 FUSE_DEC_MAXW_352	1 = Max video width up to 352 pixels enabled. Priority coded with priority 5.
11-8 —	Reserved.
7 FUSE_DEC_REFERENCE_BUFFER	1 = reference buffer used
6-0 —	Reserved.

14.1.5.1.19 Device configuration register decoder 2 + Multi core control register (SWREG58)

14.1.5.1.19.1 Offset

Register	Offset
SWREG58	E8h

14.1.5.1.19.2 Diagram



14.1.5.1.19.3 Fields

Field	Function
31 SW_SERV_MERGE_DIS	Decoder service merge disable: 0b - HW merges simultaneous sub-block requests internally if they are same type (read or write). 1b - decoder serves one sub-block per service and merging is disabled.
30 SW_DEC_MULTICORE_E	Decoder multi core enable: 0b - Multi core disabled or only one core exists in design. 1b - Multi core enable. Each reference picture status must be verified from external memory status field before usage. 128 bits status word exists after each reference picture and include picture proceeding coordinates Y and X.
29 SW_DEC_WRITE_STAT_E	Decoder write statusword enable. Must be high if multi core decoding enabled. HW writes output picture data proceeding to external memory after picture data (and after H264 direct mode MVS if they exist)
28-27 SW_DEC_MC_POLLMODE	Decoder multicore status reading mode: 00b - HW internal status polling mechanism is used. Status of reference picture is read only when required coordinate for the reference picture is not big enough. If the status is still not big enough after reading it the HW waits N clock cycles per pixel from the coordinate difference. The N is defined by the sw_dec_mc_polltime (range 0...4). 01b - Dummy status polling mechanism is used for all reference pictures. HW reads status of all reference pictures at frequency defined by sw_dec_mc_polltime.
26-17 SW_DEC_MC_POLLTIME	sw_dec_mc_polltime definition depends on sw_dec_mc_mode. if mode is 0 (HW internal mechanism) the sw_dec_mc_polltime: 0 = 1/4 cycles per pixel 1 = wait 1/2 cycles per pixel 2 = wait one cycle per pixel 3 = wait 2 cycles per pixel 4=wait 4 cycles per pixel

Table continues on the next page...

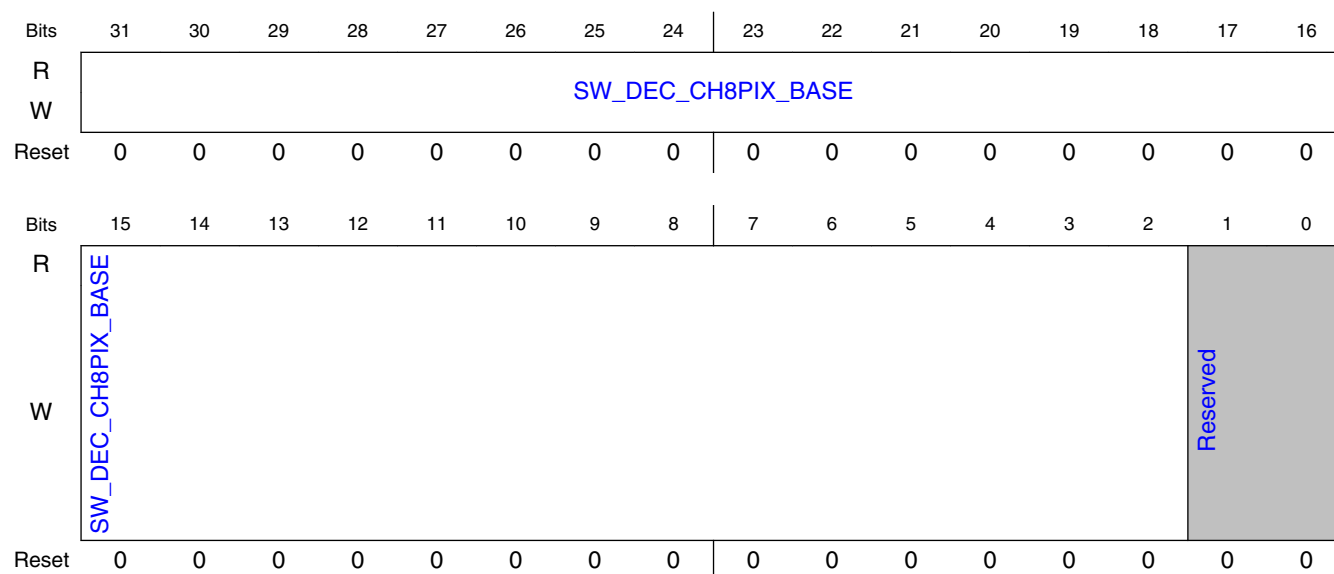
Field	Function
	For dummy polling mechanism the sw_dec_mc_polltime defines the amount cycles between status reads x1024: • 0 = wait 1024 cycles between status reading • 1 = wait 2048 cycles between status reading • 2 = wait 3072 cycles between status reading • N = wait (N+1)x1024 cycles
16-0 —	Reserved.

14.1.5.1.20 H264 Chrominance 8 pixel interleaved data base (SWREG59)

14.1.5.1.20.1 Offset

Register	Offset
SWREG59	ECh

14.1.5.1.20.2 Diagram



14.1.5.1.20.3 Fields

Field	Function
31-2	Base address for additional chrominance data format where chrominance is interleaved in group of 8 pixels. The usage is enabled by sw_ch_8pix_ileav_e

Table continues on the next page...

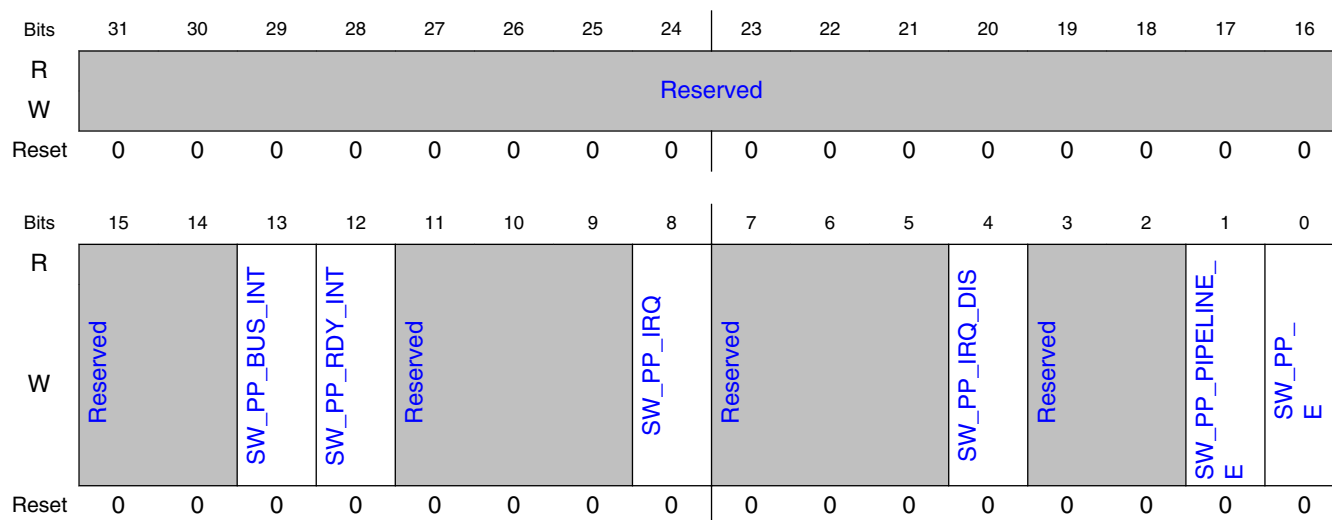
Field	Function
SW_DEC_CH8 PIX_BASE	
1-0 —	Reserved.

14.1.5.1.21 Interrupt register post-processor (SWREG60)

14.1.5.1.21.1 Offset

Register	Offset
SWREG60	F0h

14.1.5.1.21.2 Diagram



14.1.5.1.21.3 Fields

Field	Function
31-14 —	Reserved.
13 SW_PP_BUS_INT	Interrupt status bit bus. Error response from bus. In pipeline mode this bit is not used

Table continues on the next page...

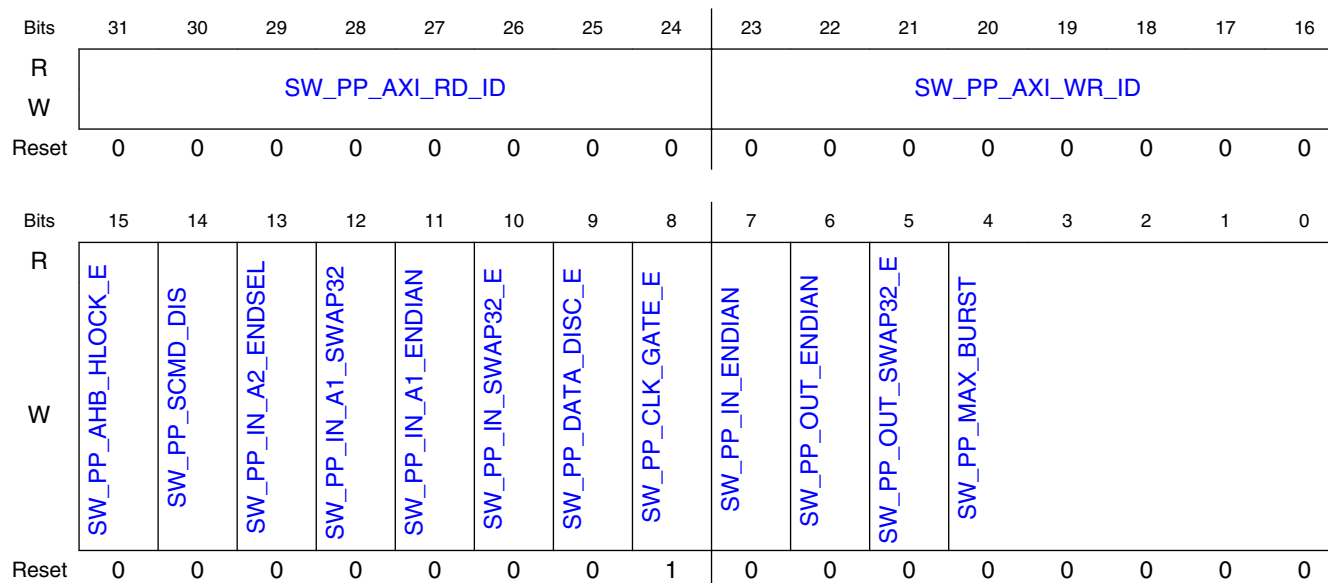
Field	Function
12 SW_PP_RDY_I NT	Interrupt status bit pp. When this bit is high post processor has processed a picture in external mode. In pipeline mode this bit is not used.
11-9 —	Reserved.
8 SW_PP_IRQ	Post-processor IRQ. SW will reset this after interrupt is handled. HINTpp is not used for pp if IRQ disable pp is high (sw_pp_irq_n_e = 1). In pipeline mode this bit is not used
7-5 —	Reserved.
4 SW_PP_IRQ_DI S	Post-processor IRQ disable. When high, there are no interrupts from HW concerning post processing. Polling must be used to see the interrupt
3-2 —	Reserved.
1 SW_PP_PIPELI NE_E	Decoder – post-processing pipeline enable: 0b - Post-processing is processing different picture than decoder or is disabled 1b - Post-processing is performed in pipeline with decoder
0 SW_PP_E	External mode post-processing enable. This bit will start the post-processing operation. Not to be used if PP is in pipeline with decoder (sw_pp_pipeline_e = 1). HW will reset this when picture is post-processed.

14.1.5.1.22 Device configuration register post-processor (SWREG61)

14.1.5.1.22.1 Offset

Register	Offset
SWREG61	F4h

14.1.5.1.22.2 Diagram



14.1.5.1.22.3 Fields

Field	Function
31-24 SW_PP_AXI_RD_ID	Read ID used for AXI PP read services (if connected to AXI)
23-16 SW_PP_AXI_WR_ID	Write ID used for AXI PP write services (if connected to AXI)
15 SW_PP_AHB_HLOCK_E	AHB master HLOCK enable. When high the service is locked to pp as long as it needs the bus (whenever pp requests the bus it will be granted)
14 SW_PP_SCMD_DIS	9170 decoder: AXI Single Command Multiple Data disable. 9170 axi wrapper supports this mode by default (where only the first addresses of the burst are given from address generator). This bit is used to disable the feature (possible SW workaround if something is not working correctly)
13 SW_PP_IN_A2_ENDSEL	Endian/swap select for Alpha blend input source 2: 0b - Use PP in endian/swap definitions (sw_pp_in_endian, sw_pp_in_swap) 1b - Use Ablend source 1 endian/swap definitions (sw_pp_in_a1_endian, sw_pp_in_a1_swap)
12 SW_PP_IN_A1_SWAP32	Alpha blend source 1 input 32bit data swap (may be used for 64 bit environment): 0b - no swapping of 32 bit words 1b - 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
11 SW_PP_IN_A1_ENDIAN	Alpha blend source 1 input data byte endian mode. 0b - Big endian (0-1-2-3 order) 1b - Little endian (3-2-1-0 order)
10	PP input 32bit data swap (may be used for 64 bit environment):

Table continues on the next page...

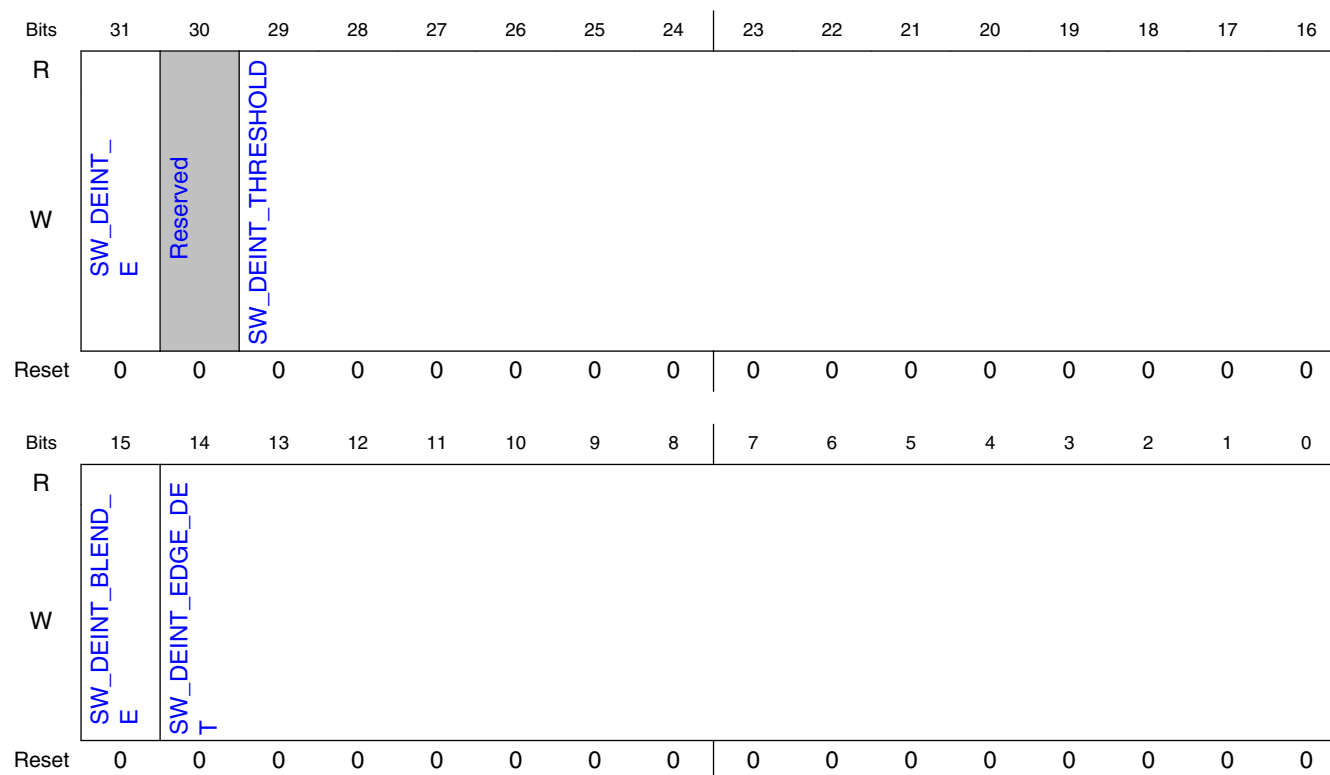
Field	Function
SW_PP_IN_SW AP32_E	0b - no swapping of 32 bit words 1b - 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
9 SW_PP_DATA_ DISC_E	PP data discard enable. Precise burst lengths are used with reading services. Extra data is discarded internally. Note. If AHB maxburst 17 is used data discard cannot be enabled (causes conflict)
8 SW_PP_CLK_G ATE_E	PP dynamic clock gating enable. NOTE: Clock gating value can be changed only when PP is not enabled. 0b - Clock is running for all PP structures 1b - Clock is gated from PP structures that are not used
7 SW_PP_IN_EN DIAN	PP input picture byte endian mode. Used only if PP is in standalone mode. If PP is running pipelined with the decoder, this bit has no effect. 0b - Big endian (0-1-2-3 order) 1b - Little endian (3-2-1-0 order)
6 SW_PP_OUT_E NDIAN	PP output picture endian mode for YCbCr data or for any data if config value SW_PP_OEN_VERSION = 1. NOTE: For SW_PP_OEN_VERSION=0, 16 bit RGB data, this bit works as pixel swapping bit. For 32 bit RGB, this bit has no meaning. 0b - Big endian (0-1-2-3 order) 1b - Little endian (3-2-1-0 order)
5 SW_PP_OUT_S WAP32_E	PP output data word swap (may be used for 64 bit environment): 0b - no swapping of 32 bit words 1b - 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order (also little endian should be enabled))
4-0 SW_PP_MAX_B URST	Maximum burst length for PP bus transactions. Valid values: AHB: 1, 4, 8, 16 and 17. Other values will result in INCR type (undefined length) transfers. 17 = fixed bursts are used when possible. INCR bursts for others. OCP: 16-31 AXI: 1-16

14.1.5.1.23 Deinterlace control register (SWREG62)

14.1.5.1.23.1 Offset

Register	Offset
SWREG62	F8h

14.1.5.1.23.2 Diagram



14.1.5.1.23.3 Fields

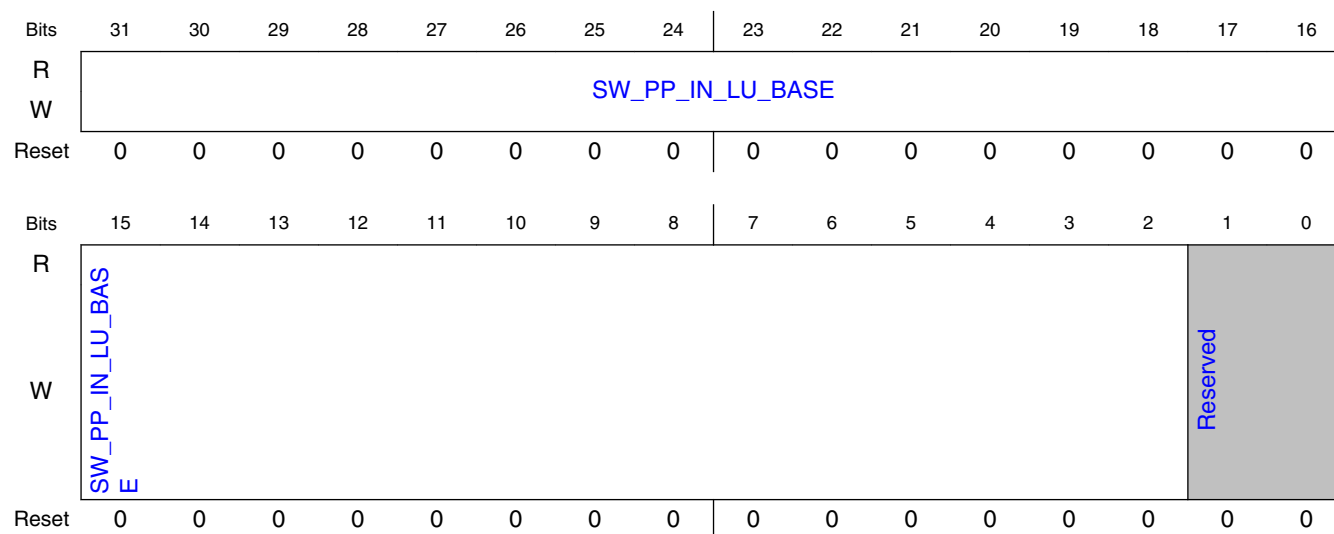
Field	Function
31 SW_DEINT_E	De-interlace enable. Input data is in interlaced format and deinterlacing needs to be performed
30 —	Reserved.
29-16 SW_DEINT_TH RESHOLD	Threshold value used in deinterlacing
15 SW_DEINT_BL END_E	Blend enable for de-interlacing
14-0 SW_DEINT_ED GE_DET	Edge detect value used for deinterlacing

14.1.5.1.24 Base address for reading post-processing input picture luminance (top field/frame) (SWREG63)

14.1.5.1.24.1 Offset

Register	Offset
SWREG63	FCh

14.1.5.1.24.2 Diagram



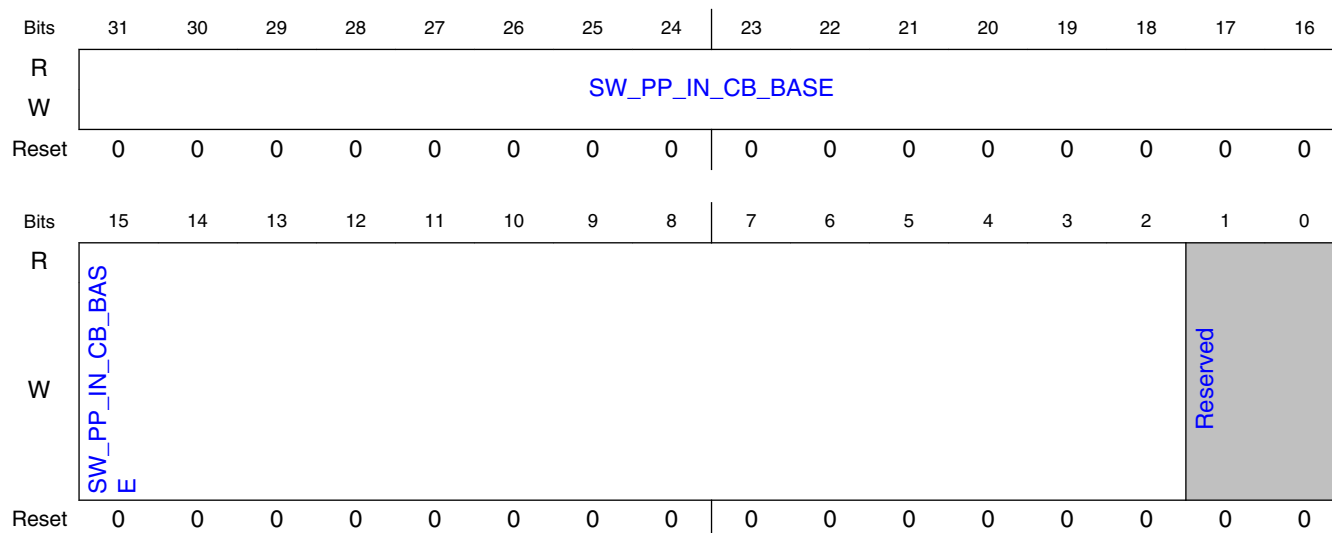
14.1.5.1.24.3 Fields

Field	Function
31-2 SW_PP_IN_LU_BASE	Base address for post-processing input luminance picture. If PP input picture is fetched from fields this base address is used to point to top field of the picture. Used in external mode only.
1-0 —	Reserved.

14.1.5.1.25 Base address for reading post-processing input picture Cb/Ch (top field/frame) (SWREG64)

14.1.5.1.25.1 Offset

Register	Offset
SWREG64	100h

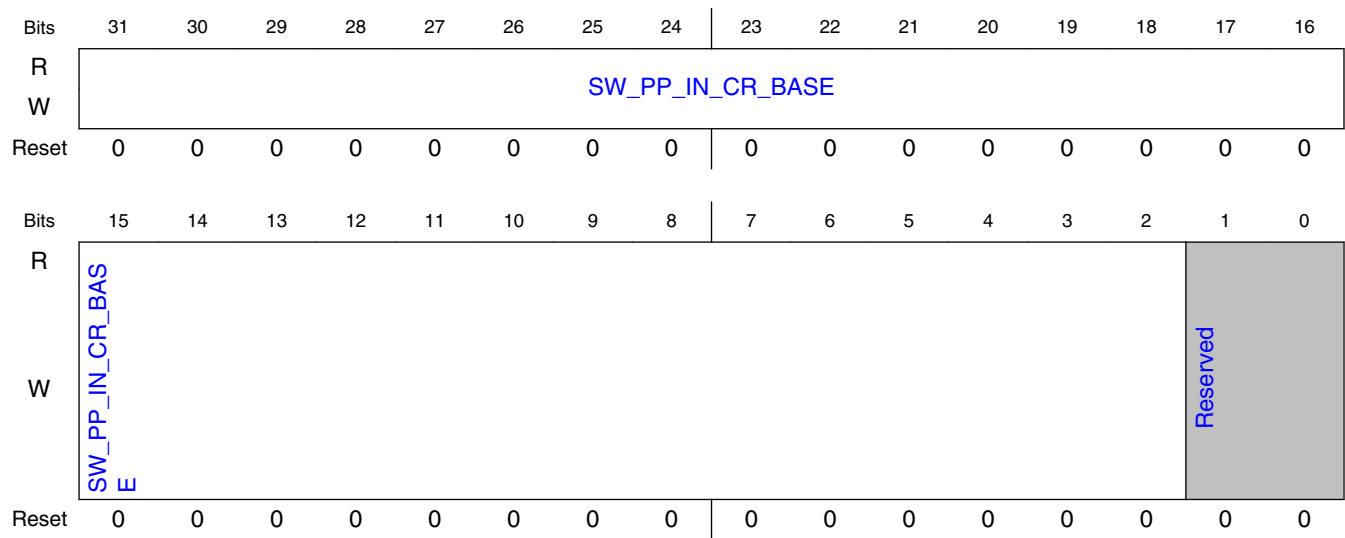
14.1.5.1.25.2 Diagram**14.1.5.1.25.3 Fields**

Field	Function
31-2 SW_PP_IN_CB_BASE	Base address for post-processing input Cb picture or for both chrominance pictures (if chrominances interleaved). If PP input picture is fetched from fields this base address is used to point to top field of the picture. Used in external mode only
1-0 —	Reserved.

14.1.5.1.26 Base address for reading post-processing input picture Cr (SWREG65)**14.1.5.1.26.1 Offset**

Register	Offset
SWREG65	104h

14.1.5.1.26.2 Diagram



14.1.5.1.26.3 Fields

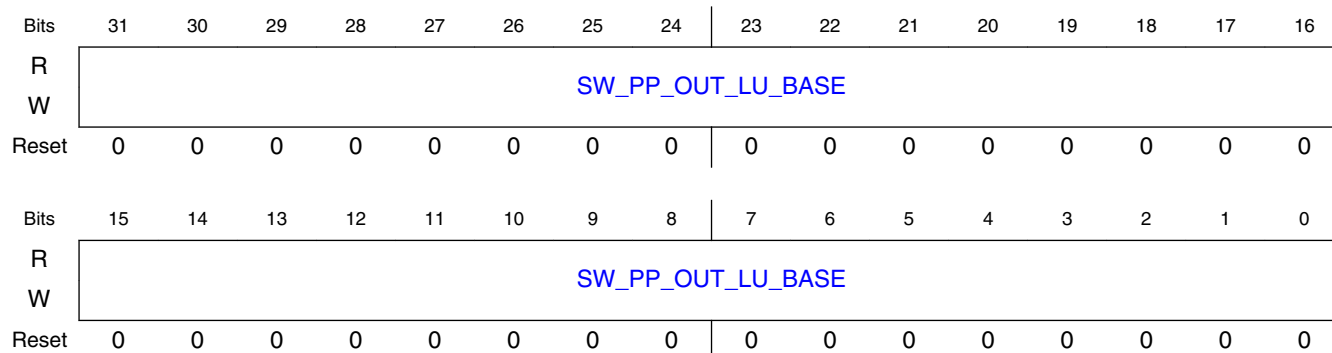
Field	Function
31-2 SW_PP_IN_CR_BASE	Base address for post-processing input cr picture. Used in external mode only
1-0 —	Reserved.

14.1.5.1.27 Base address for writing post-processed picture luminance/RGB (SWREG66)

14.1.5.1.27.1 Offset

Register	Offset
SWREG66	108h

14.1.5.1.27.2 Diagram



14.1.5.1.27.3 Fields

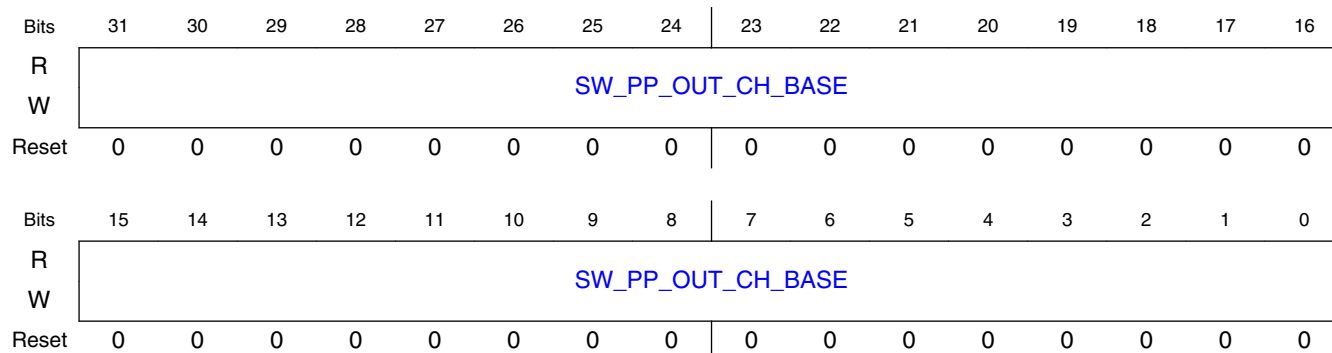
Field	Function
31-0 SW_PP_OUT_L U_BASE	Base address for post-processing output picture (luminance/YUYV/RGB). NOTE: Bits 2:0 are used to adjust the post-processor output to start from zertain byte (1:0 for 32 bit bus). These bits can be other than zero only if Pixel Accurate PP output configuration is enabled

14.1.5.1.28 Base address for writing post-processed picture Ch (SWRE G67)

14.1.5.1.28.1 Offset

Register	Offset
SWREG67	10Ch

14.1.5.1.28.2 Diagram



14.1.5.1.28.3 Fields

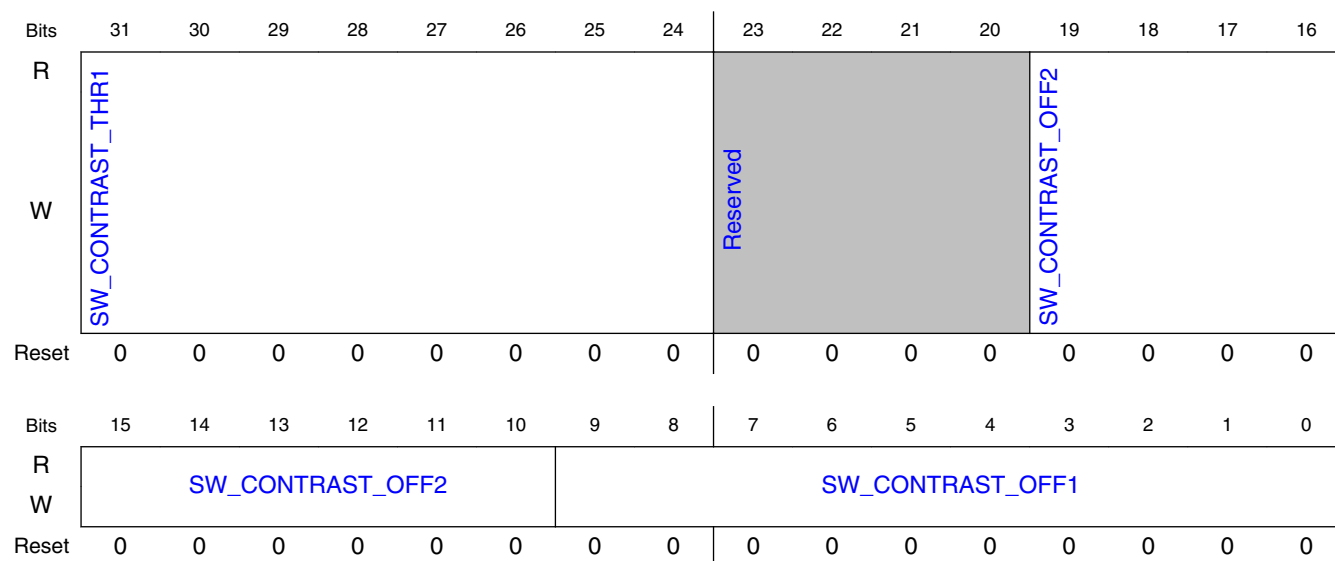
Field	Function
31-0 SW_PP_OUT_C H_BASE	Base address for post-processing output chrominance picture (interleaved chrominance). NOTE: Bits 2:0 are used to adjust the post-processor output to start from certain byte (1:0 for 32 bit bus). These bits can be other than zero only if Pixel Accurate PP output configuration is enabled

14.1.5.1.29 Register for contrast adjusting (SWREG68)

14.1.5.1.29.1 Offset

Register	Offset
SWREG68	110h

14.1.5.1.29.2 Diagram



14.1.5.1.29.3 Fields

Field	Function
31-24 SW_CONTRAS T_THR1	Threshold value 1, used with contrast adjusting

Table continues on the next page...

VPU G1 Memory Map/Register Definition

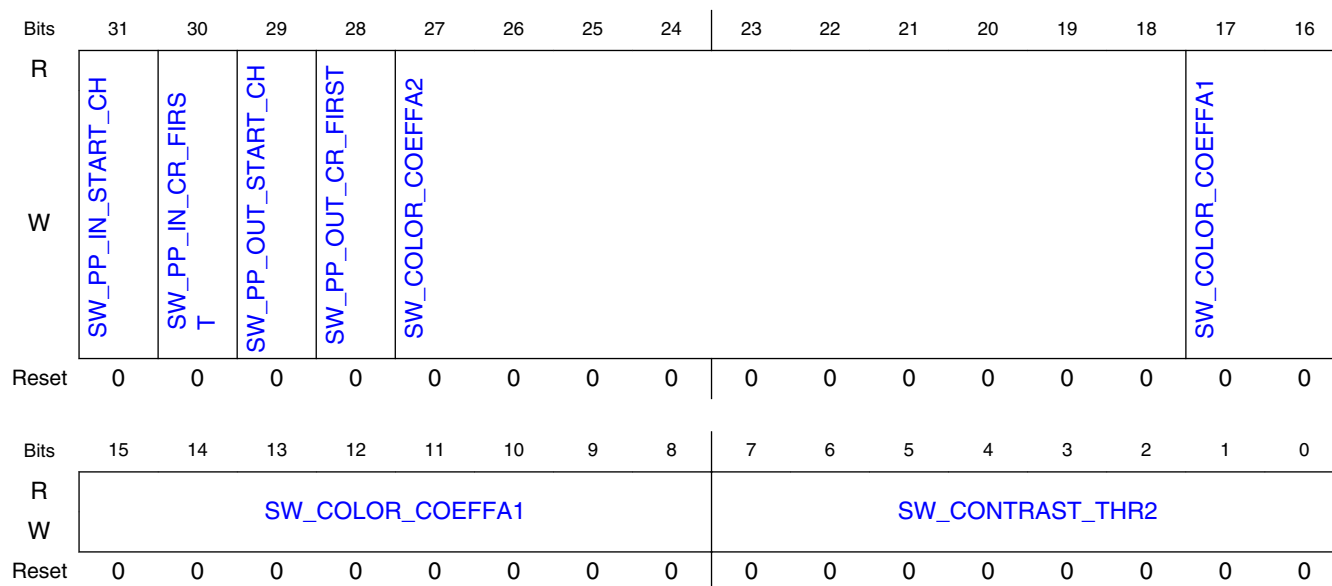
Field	Function
23-20 —	Reserved.
19-10 SW_CONTRAS T_OFF2	Offset value 2, used with contrast adjusting
9-0 SW_CONTRAS T_OFF1	Offset value 1, used with contrast adjusting

14.1.5.1.30 Register for colour conversion and contrast adjusting/YUYV 422 channel orders (SWREG69)

14.1.5.1.30.1 Offset

Register	Offset
SWREG69	114h

14.1.5.1.30.2 Diagram



14.1.5.1.30.3 Fields

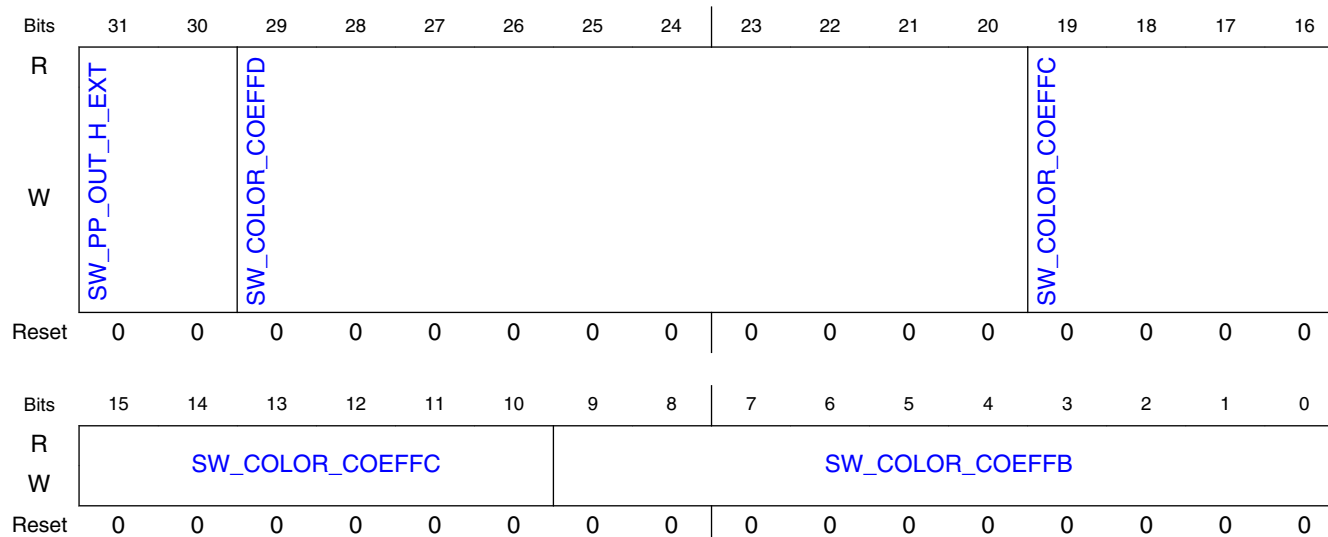
Field	Function
31 SW_PP_IN_ST ART_CH	For YUYV 422 input format. Enable for start_with_chrominance. 0b - the order is Y0CbY0Cr or Y0CrY0Cb 1b - the order is CbY0CrY0 or CrY0CbY0
30 SW_PP_IN_CR _FIRST	For YUYV 422 input format. Enable for Cr first (before Cb). 0b - the order is Y0CbY0Cr or CbY0CrY0 (if 420 semiplanar chrominance: CbCrCbCr) 1b - the order is Y0CrY0Cb or CrY0CbY0 (if 420 semiplanar chrominance: CrCbCrCb)
29 SW_PP_OUT_S TART_CH	For YUYV 422 output format. Enable for start_with_chrominance. 0b - the order is Y0CbY0Cr or Y0CrY0Cb 1b - the order is CbY0CrY0 or CrY0CbY0
28 SW_PP_OUT_C R_FIRST	For YUYV 422 output format. Enable for Cr first (before Cb). 0b - the order is Y0CbY0Cr or CbY0CrY0 1b - the order is Y0CrY0Cb or CrY0CbY0
27-18 SW_COLOR_C OEFFA2	Coefficient a2, used with Y pixel to calculate all color components
17-8 SW_COLOR_C OEFFA1	Coefficient a1, used with Y pixel to calculate all color components
7-0 SW_CONTRAS T_THR2	Threshold value 2, used with contrast adjusting

14.1.5.1.31 Register for colour conversion 0 (SWREG70)

14.1.5.1.31.1 Offset

Register	Offset
SWREG70	118h

14.1.5.1.31.2 Diagram



14.1.5.1.31.3 Fields

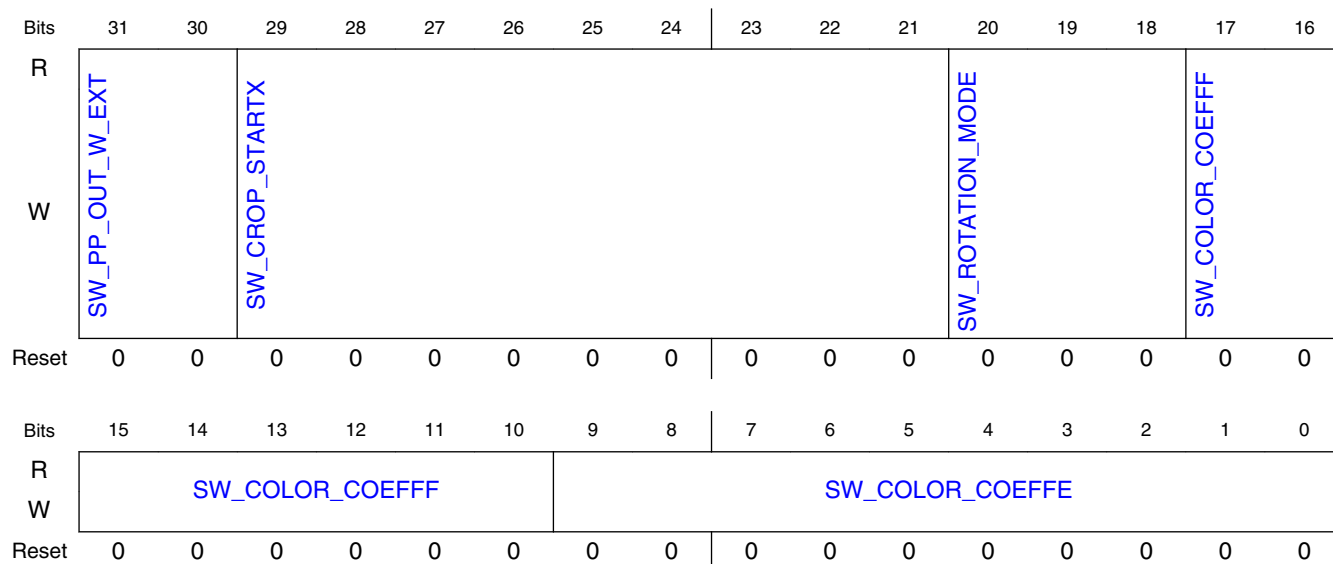
Field	Function
31-30 SW_PP_OUT_H_EXT	Extended output height for 4k resolution
29-20 SW_COLOR_COEFFD	Coefficient d, used with Cb to calculate green component value
19-10 SW_COLOR_COEFFC	Coefficient c, used with Cr to calculate green component value
9-0 SW_COLOR_COEFFB	Coefficient b, used with Cr to calculate red component value

14.1.5.1.32 Register for colour conversion 1 + rotation mode (SWREG71)

14.1.5.1.32.1 Offset

Register	Offset
SWREG71	11Ch

14.1.5.1.32.2 Diagram



14.1.5.1.32.3 Fields

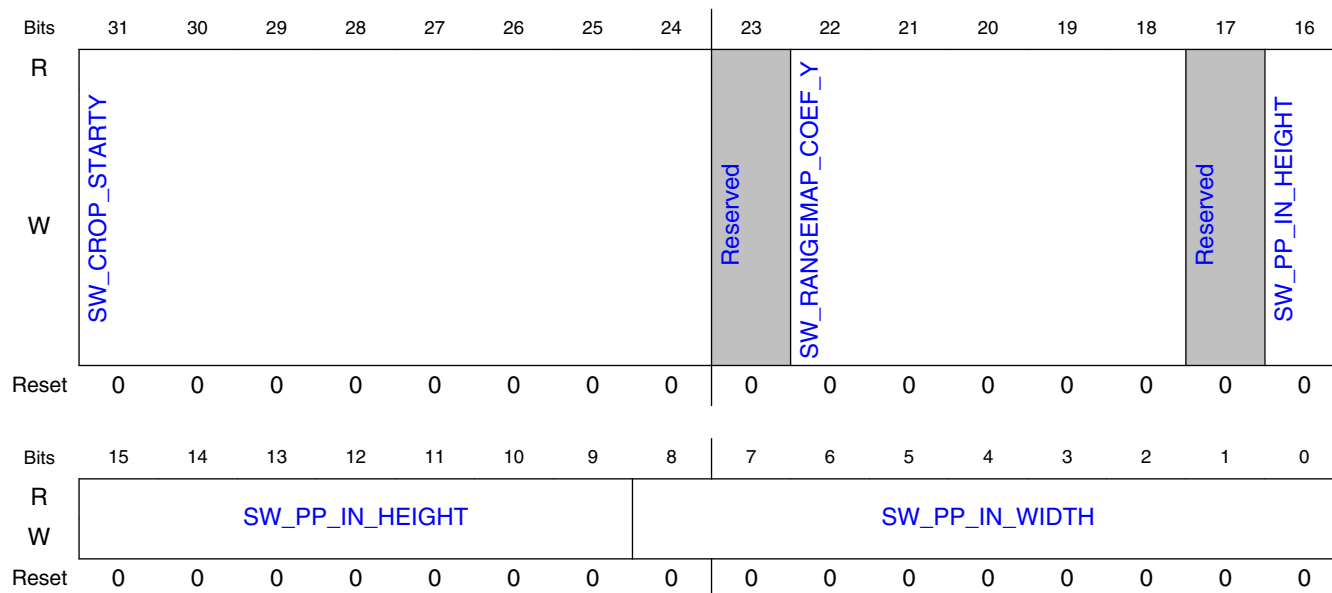
Field	Function
31-30 SW_PP_OUT_W_EXT	Extended output width for 4k resolution
29-21 SW_CROP_STARTX	Start coordinate x for the cropped area in macroblocks.
20-18 SW_ROTATION_MODE	Rotation mode: 000b - rotation disabled 001b - rotate + 90 010b - rotate – 90 011b - horizontal flip (mirror) 100b - vertical flip 101b - rotate 180
17-10 SW_COLOR_COEFFFF	Coefficient f, used with Y to adjust brightness
9-0 SW_COLOR_COEFFE	Coefficient e, used with Cb to calculate blue component value

14.1.5.1.33 PP input size and -cropping register (SWREG72)

14.1.5.1.33.1 Offset

Register	Offset
SWREG72	120h

14.1.5.1.33.2 Diagram



14.1.5.1.33.3 Fields

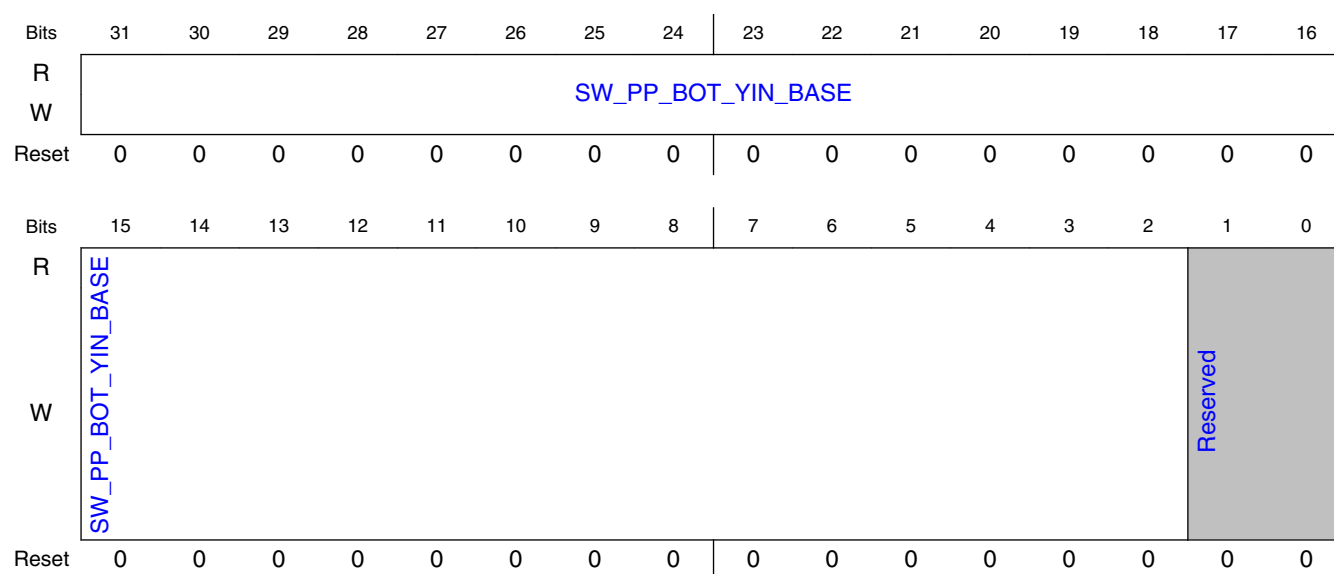
Field	Function
31-24 SW_CROP_ST ARTY	Start coordinate y for the cropped area in macroblocks.
23 —	Reserved.
22-18 SW_RANGEMA P_COEF_Y	Range map value for Y component
17 —	Reserved.
16-9 SW_PP_IN_HEI GHT	PP input picture height in MBs. Can be cropped from a bigger input picture in external mode
8-0 SW_PP_IN_WI DTH	PP input picture width in MBs. Can be cropped from a bigger input picture in external mode

14.1.5.1.34 PP input picture base address for Y bottom field (SWREG73)

14.1.5.1.34.1 Offset

Register	Offset
SWREG73	124h

14.1.5.1.34.2 Diagram



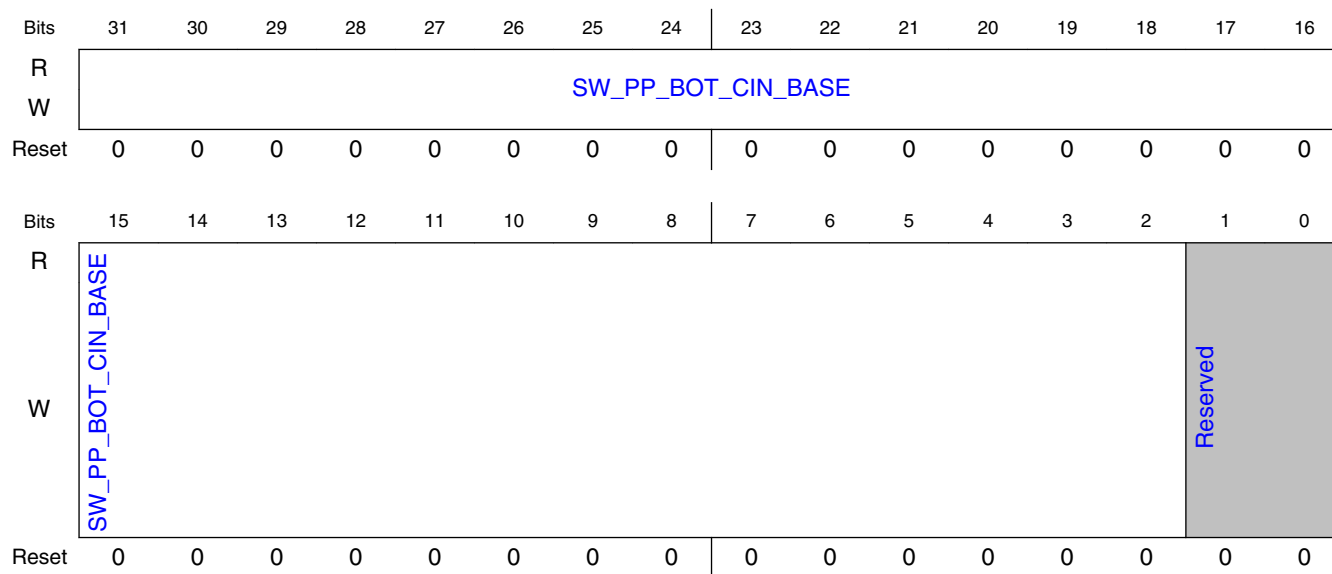
14.1.5.1.34.3 Fields

Field	Function
31-2 SW_PP_BOT_Y IN_BASE	PP input Y base for bottom field
1-0 —	Reserved.

14.1.5.1.35 PP input picture base for Ch bottom field (SWREG74)

14.1.5.1.35.1 Offset

Register	Offset
SWREG74	128h

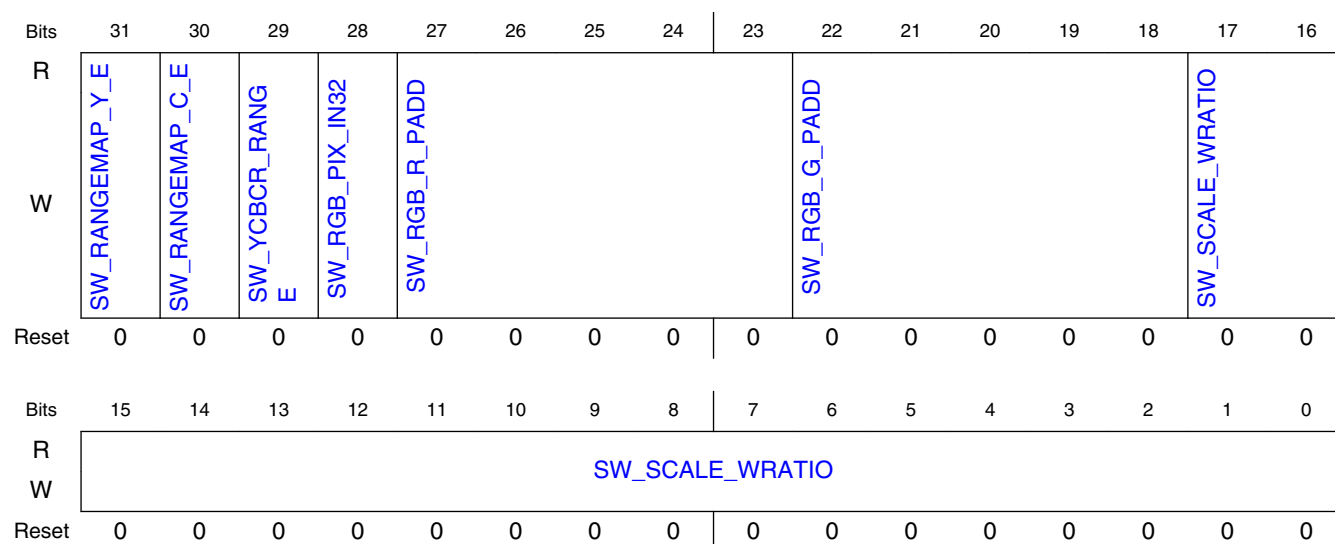
14.1.5.1.35.2 Diagram**14.1.5.1.35.3 Fields**

Field	Function
31-2 SW_PP_BOT_C IN_BASE	PP input C base for bottom field (mixed chrominance)
1-0 —	Reserved.

14.1.5.1.36 Scaling register 0 ratio and padding for R and G (SWREG79)**14.1.5.1.36.1 Offset**

Register	Offset
SWREG79	13Ch

14.1.5.1.36.2 Diagram



14.1.5.1.36.3 Fields

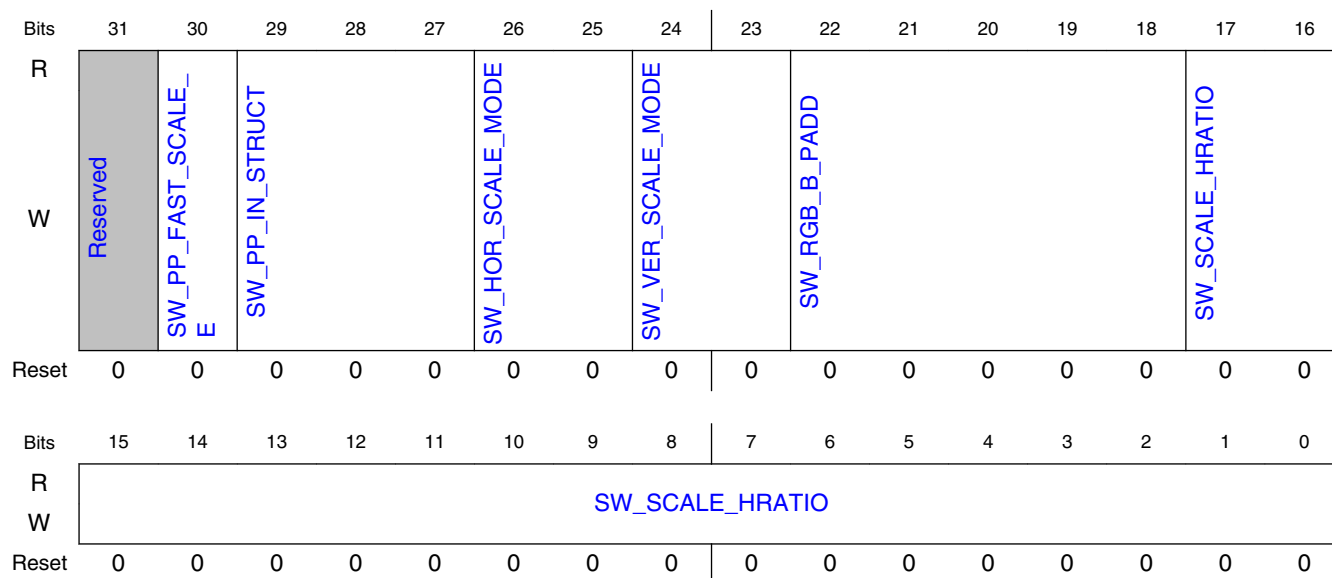
Field	Function
31 SW_RANGEMAP_Y_E	Range map enable for Y component
30 SW_RANGEMAP_C_E	Range map enable for chrominance component
29 SW_YCBCR_RANGE	Defines the YCbCr range in RGB conversion: 0b - 16...235 for Y, 16...240 for Chrominance. 1b - 0...255 for all components
28 SW_RGB_PIX_IN32	RGB pixel amount/ 32 bit word 0b - 1 RGB pixel/32 bit 1b - 2 RGB pixels/32 bit
27-23 SW_RGB_R_PADD	Amount of ones that will be padded in front of the R-component
22-18 SW_RGB_G_PADD	Amount of ones that will be padded in front of the G-component
17-0 SW_SCALE_WRATIO	Scaling ratio for width (outputw-1/inputw-1)

14.1.5.1.37 Scaling ratio register 1 and padding for B (SWREG80)

14.1.5.1.37.1 Offset

Register	Offset
SWREG80	140h

14.1.5.1.37.2 Diagram



14.1.5.1.37.3 Fields

Field	Function
31 —	Reserved.
30 SW_PP_FAST_SCALE_E	0b - fast downscaling is not enabled 1b - fast downscaling is enabled. The quality of the picture is decreased but performance is improved.
29-27 SW_PP_IN_STRUCT	PP input data picture structure: 000b - Top field / progressive frame structure: Read input data from top field base address /frame base address and read every line. 001b - Bottom field structure: Read input data from bottom field base address and read every line. 010b - Interlaced field structure: Read input data from both top and bottom field base address and take every line from each field. 011b - Interlaced frame structure: Read input data from both top and bottom field base address and take every second line from each field. 100b - Ripped top field structure: Read input data from top field base address and read every second line.

Table continues on the next page...

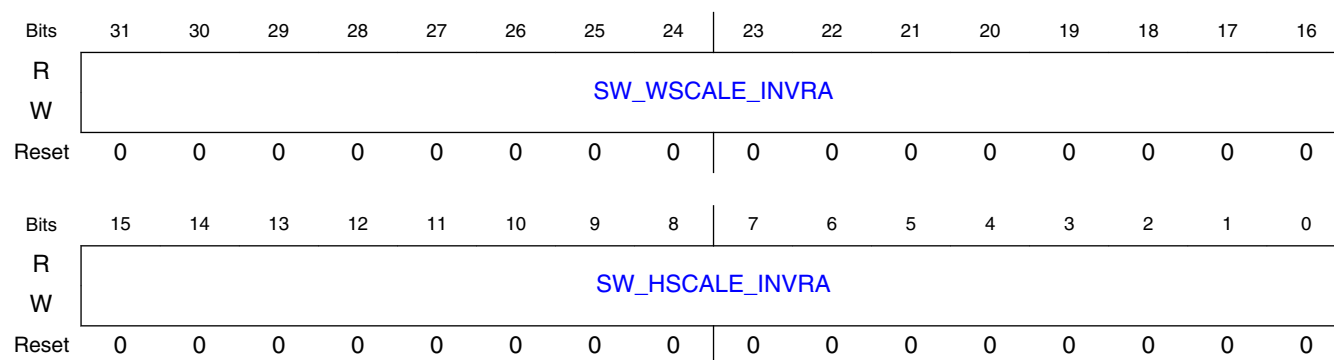
Field	Function
	101b - Ripped bottom field structure: Read input data from bottom field base address and read every second line.
26-25 SW_HOR_SCALE_MODE	Horizontal scaling mode: 00b - Off 01b - Upscale 10b - Downscale
24-23 SW_VER_SCALE_MODE	Vertical scaling mode: 00b - Off 01b - Upscale 10b - Downscale
22-18 SW_RGB_B_PADDING	Amount of ones that will be padded in front of the B-component
17-0 SW_SCALE_HEIGHT_RATIO	Scaling ratio for height (outputw-1/inputw-1)

14.1.5.1.38 Scaling ratio register 2 (SWREG81)

14.1.5.1.38.1 Offset

Register	Offset
SWREG81	144h

14.1.5.1.38.2 Diagram



14.1.5.1.38.3 Fields

Field	Function
31-16	Inverse scaling ratio for width, or ch (inputw-1 / outputw-1)

Table continues on the next page...

Field	Function
SW_WSCALE_I NVRA	Inverse scaling ratio for height or cv (inputh-1 / outputh-1)
15-0 SW_HSCALE_I NVRA	

14.1.5.1.39 Rmask register (SWREG82)

14.1.5.1.39.1 Offset

Register	Offset
SWREG82	148h

14.1.5.1.39.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_R_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_R_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.1.5.1.39.3 Fields

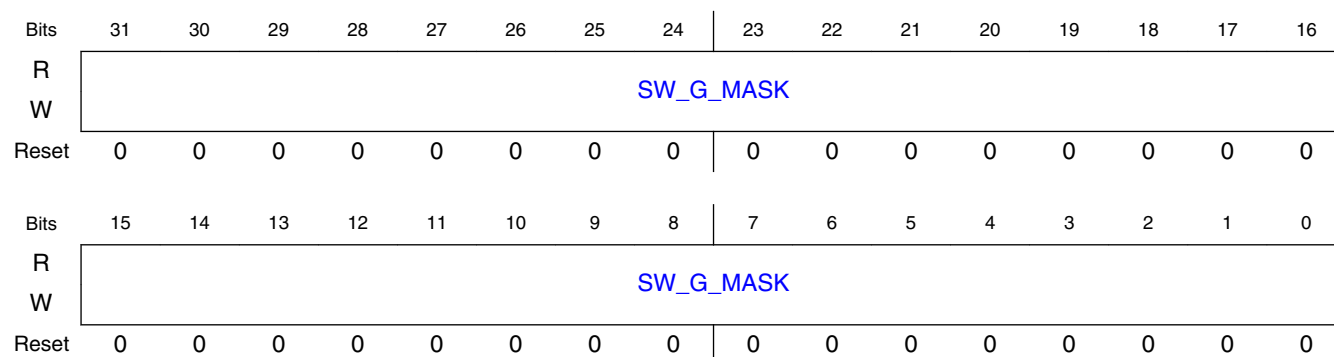
Field	Function
31-0 SW_R_MASK	Bit mask for R component (and alpha channel)

14.1.5.1.40 Gmask register (SWREG83)

14.1.5.1.40.1 Offset

Register	Offset
SWREG83	14Ch

14.1.5.1.40.2 Diagram



14.1.5.1.40.3 Fields

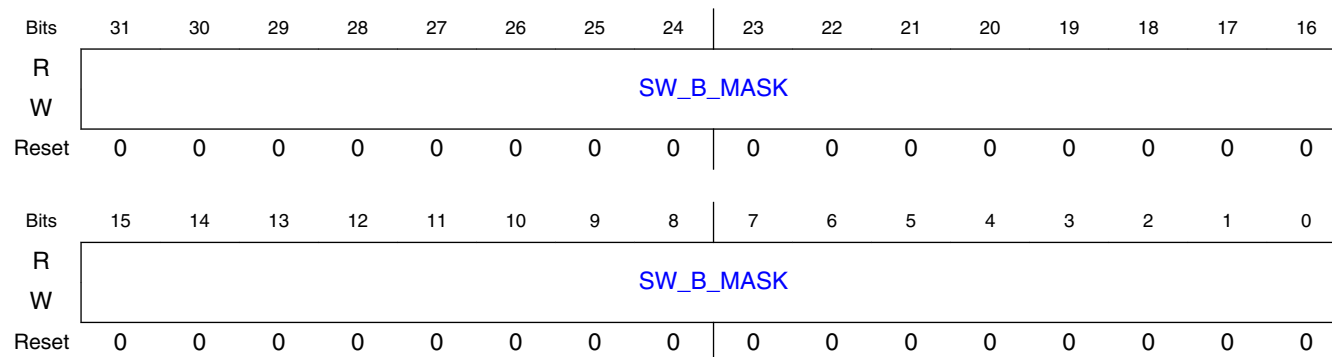
Field	Function
31-0 SW_G_MASK	Bit mask for G component (and alpha channel)

14.1.5.1.41 Bmask register (SWREG84)

14.1.5.1.41.1 Offset

Register	Offset
SWREG84	150h

14.1.5.1.41.2 Diagram



14.1.5.1.41.3 Fields

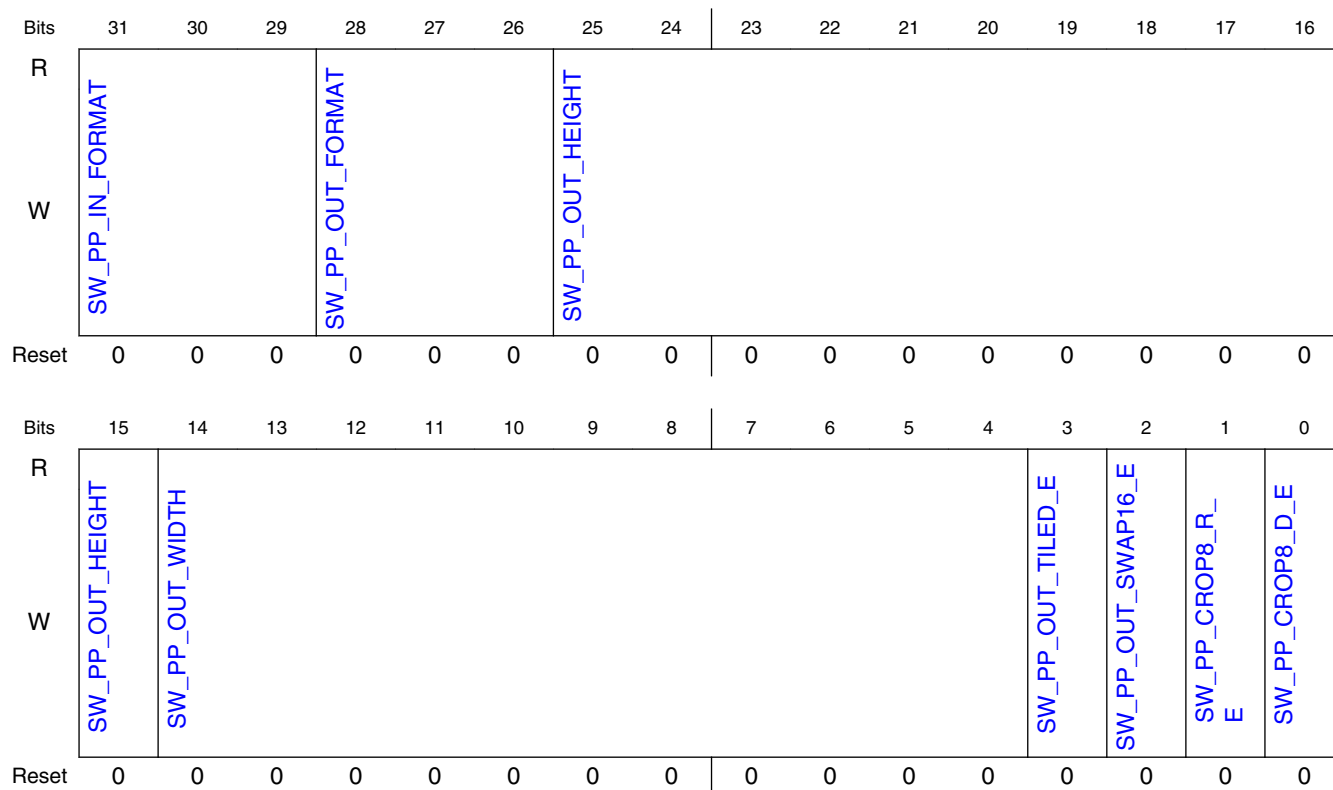
Field	Function
31-0 SW_B_MASK	Bit mask for B component (and alpha channel)

14.1.5.1.42 Post-processor control register (SWREG85)

14.1.5.1.42.1 Offset

Register	Offset
SWREG85	154h

14.1.5.1.42.2 Diagram



14.1.5.1.42.3 Fields

Field	Function
31-29 SW_PP_IN_FORMAT	PP input picture data format 000b - YUYV 4:2:2 interleaved (supported only in external mode) 001b - YCbCr 4:2:0 Semi-planar in linear raster-scan format 010b - YCbCr 4:2:0 planar (supported only in external mode) 011b - YCbCr 4:0:0 (supported only in pipelined mode) 100b - YCbCr 4:2:2 Semi-planar (supported only in pipelined mode) 101b - YCbCr 4:2:0 Semi-planar in tiled format (supported only in external mode (8170 decoder only)) 110b - Reserved 111b - Escape pp input data format. Defined in swreg86.
28-26 SW_PP_OUT_FORMAT	PP output picture data format: 000b - RGB 001b - YCbCr 4:2:0 planar (Not supported) 010b - YCbCr 4:2:2 planar (Not supported) 011b - YUYV 4:2:2 interleaved 100b - YCbCr 4:4:4 planar (Not supported) 101b - YCh 4:2:0 chrominance interleaved 110b - YCh 4:2:2 (Not supported) 111b - YCh 4:4:4 (Not supported)

Table continues on the next page...

VPU G1 Memory Map/Register Definition

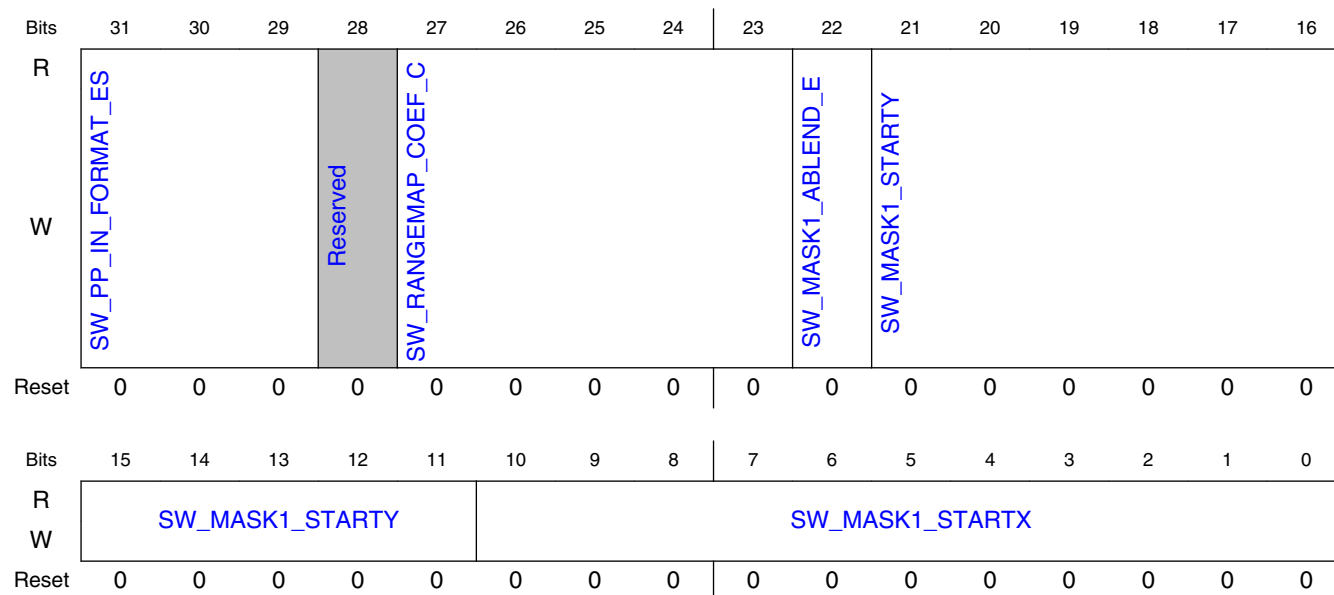
Field	Function
25-15 SW_PP_OUT_HEIGHT	Scaled picture height in pixels (Must be dividable by 2 or by any if Pixel Accurate PP output configuration is enabled) Max scaled picture height is 1920 pixels or maximum three times the input source height minus 8 pixels
14-4 SW_PP_OUT_WIDTH	Scaled picture width in pixels. Must be dividable by 8 or by any if Pixel Accurate PP output configuration is enabled. Max scaled picture width is 1920 pixels or maximum three times the input source width minus 8 pixels
3 SW_PP_OUT_TILED_E	Tiled mode enable for PP output. Can be used only for YCbYCr 422 output format. Can be used only if corresponding configuration supports this feature. Tile size is 4x4 pixels.
2 SW_PP_OUT_SWAP16_E	PP output swap 16, swaps 16 bit half inside of 32 bit word. Can be used for 16 bit RGB to change pixel orders but is valid also for any output format. NOTE: requires that configuration of SW_PPD_OEN_VERSION=1
1 SW_PP_CROP8_R_E	PP input picture width is not 16 pixels multiple. Only 8 pixels of the most right MB of the unrotated input picture is used for PP input.
0 SW_PP_CROP8_D_E	PP input picture height is not 16 pixels multiple. Only 8 pixel rows of the most down MB of the unrotated input picture is used for PP input.

14.1.5.1.43 Mask 1 start coordinate register (SWREG86)

14.1.5.1.43.1 Offset

Register	Offset
SWREG86	158h

14.1.5.1.43.2 Diagram



14.1.5.1.43.3 Fields

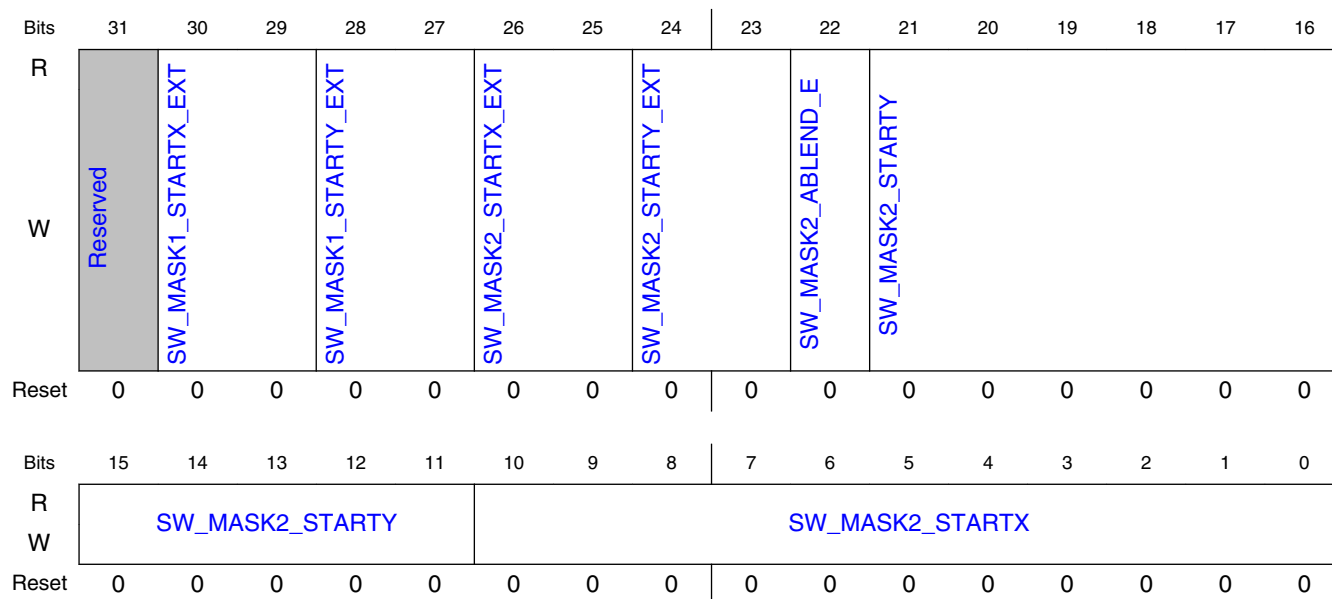
Field	Function
31-29 SW_PP_IN_FORMAT_ES	Escape PP in format. Used if sw_pp_in_format is defined to 7. 000b - YCbCr 4:4:4 001b - YCbCr 4:1:1
28 —	Reserved.
27-23 SW_RANGEMAP_COEF_C	Range map value for chrominance component
22 SW_MASK1_ABLEND_E	Mask 1 alpha blending enable. Instead of masking the output picture the alpha blending is performed. Alpha blending source can be found from alpha blend 1 base address. Alpha blending can be enabled only for RGB/ YUYV 422 data.
21-11 SW_MASK1_STARTY	Vertical start pixel for mask area 1. Defines the y coordinate. Coordinate 0,0 means the up-left corner in PP output luminance picture. See Table 47 for restrictions
10-0 SW_MASK1_STARTX	Horizontal start pixel for mask area 1. Defines the x coordinate. Coordinate 0,0 means the up-left corner in PP output luminance picture. See Table 47 for restrictions

14.1.5.1.44 Mask 2 start coordinate register + Mask extensions (SWREG87)

14.1.5.1.44.1 Offset

Register	Offset
SWREG87	15Ch

14.1.5.1.44.2 Diagram



14.1.5.1.44.3 Fields

Field	Function
31 —	Reserved.
30-29 SW_MASK1_ST ARTX_EXT	Extended coordinate upto 4k resolution
28-27 SW_MASK1_ST ARTY_EXT	Extended coordinate upto 4k resolution
26-25 SW_MASK2_ST ARTX_EXT	Extended coordinate upto 4k resolution

Table continues on the next page...

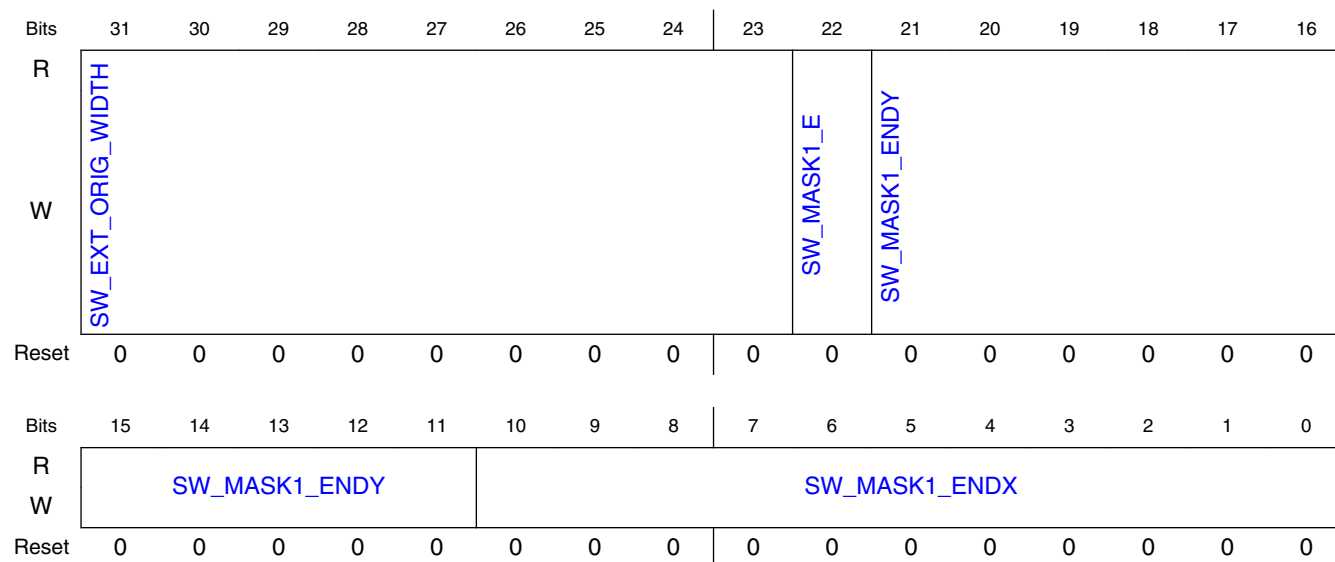
Field	Function
24-23 SW_MASK2_ST ARTY_EXT	Extended coordinate upto 4k resolution
22 SW_MASK2_AB LEND_E	Mask 2 alpha blending enable. Instead of masking the output picture the alpha blending is performed. Alpha blending source can be found from alpha blend 2 base address. Alpha blending can be enabled only for RGB/YUYV 422 data.
21-11 SW_MASK2_ST ARTY	Vertical start pixel for mask area 2. Defines the y coordinate. Coordinate 0,0 means the up-left corner in PP output Y picture. See Table 47 for restrictions
10-0 SW_MASK2_ST ARTX	Horizontal start pixel for mask area 2. Defines the x coordinate. Coordinate 0,0 means the up-left corner in PP output Y picture. See Table 47 for restrictions

14.1.5.1.45 Mask 1 size and PP original width register (SWREG88)

14.1.5.1.45.1 Offset

Register	Offset
SWREG88	160h

14.1.5.1.45.2 Diagram



14.1.5.1.45.3 Fields

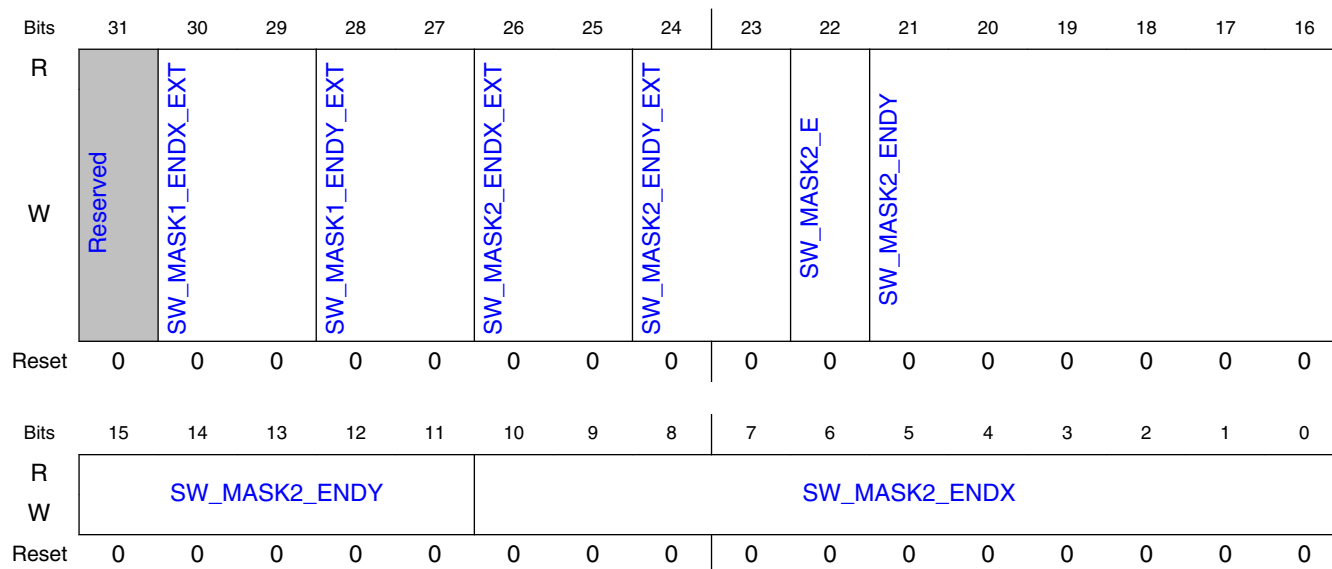
Field	Function
31-23 SW_EXT_ORIG_WIDTH	PP input picture original width in macro blocks.
22 SW_MASK1_E	Mask 1 enable. If mask 1 is used this bit is high
21-11 SW_MASK1_ENDY	Mask 1 end coordinate y in pixels (inside of PPD output picture). Range must be between [Mask1StartCoordinateY, ScaledHeight].
10-0 SW_MASK1_ENDX	Mask 1 end coordinate x in pixels (inside of PPD output picture). Range must be between [Mask1StartCoordinateX, ScaledWidth]

14.1.5.1.46 Mask 2 size register + mask extensions (SWREG89)

14.1.5.1.46.1 Offset

Register	Offset
SWREG89	164h

14.1.5.1.46.2 Diagram



14.1.5.1.46.3 Fields

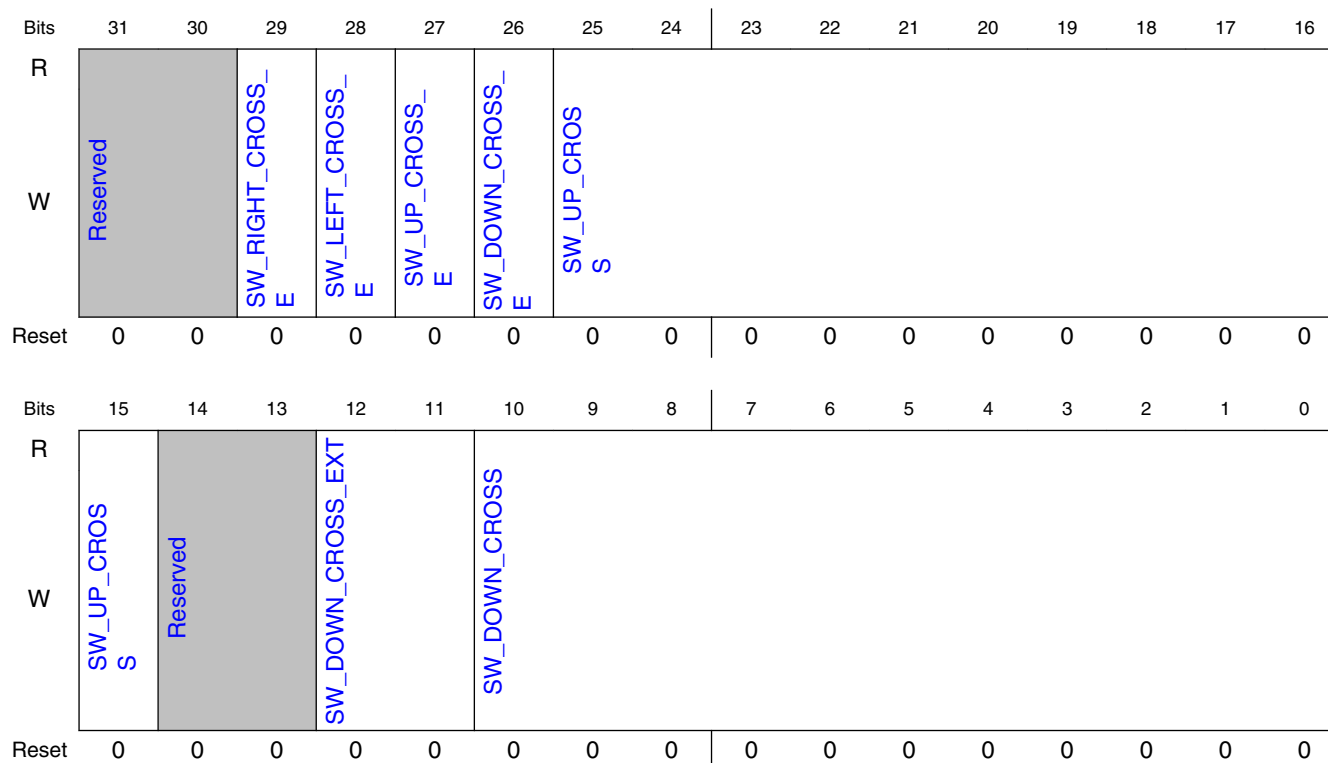
Field	Function
31 —	Reserved.
30-29 SW_MASK1_E NDX_EXT	Extended coordinate upto 4k resolution
28-27 SW_MASK1_E NDY_EXT	Extended coordinate upto 4k resolution
26-25 SW_MASK2_E NDX_EXT	Extended coordinate upto 4k resolution
24-23 SW_MASK2_E NDY_EXT	Extended coordinate upto 4k resolution
22 SW_MASK2_E	Mask 2 enable. If mask 1 is used this bit is high
21-11 SW_MASK2_E NDY	Mask 2 end coordinate y in pixels (inside of PP output picture). Range must be between [Mask2StartCoordinateY, ScaledHeight].
10-0 SW_MASK2_E NDX	Mask 2 end coordinate x in pixels (inside of PP output picture). Range must be between [Mask2StartCoordinateX, ScaledWidth].

14.1.5.1.47 PiP register 0 (SWREG90)

14.1.5.1.47.1 Offset

Register	Offset
SWREG90	168h

14.1.5.1.47.2 Diagram



14.1.5.1.47.3 Fields

Field	Function
31-30 —	Reserved.
29 SW_RIGHT_CR OSS_E	Right side overcross enable. 0b - No right side overcross 1b - Right side overcross
28 SW_LEFT_CRO SS_E	Left side overcross enable. 0b - No left side overcross 1b - Left side overcross
27 SW_UP_CROS S_E	Upward overcross enable. 0b - No upward overcross 1b - Upward overcross
26 SW_DOWN_CR OSS_E	Downward overcross enable. 0b - No downward overcross 1b - Downward overcross
25-15 SW_UP_CROS S	Amount of upward overcross (vertical pixels outside of display from the upper side). Range must be between [0, ScaledHeight].

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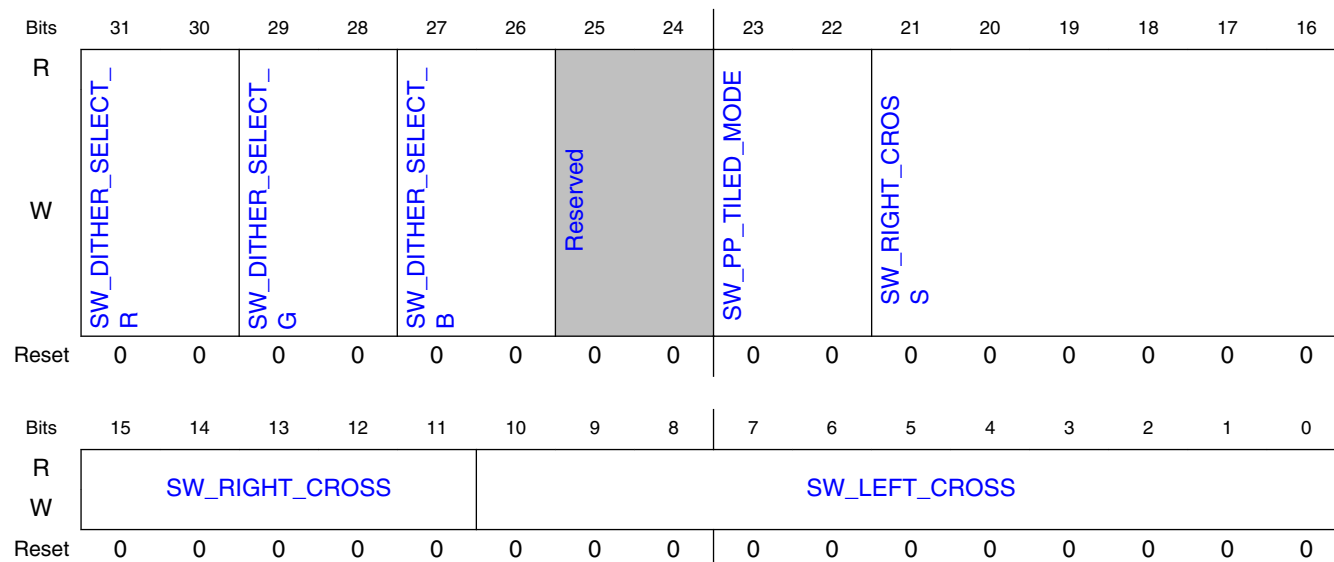
Field	Function
14-13 —	Reserved.
12-11 SW_DOWN_CR OSS_EXT	Extended coordinate for 4k resolution
10-0 SW_DOWN_CR OSS	Amount of downward overcross (vertical pixels outside of display from the down side). Range must be between [0, ScaledHeight].

14.1.5.1.48 PiP register 1 and dithering control (SWREG91)

14.1.5.1.48.1 Offset

Register	Offset
SWREG91	16Ch

14.1.5.1.48.2 Diagram



14.1.5.1.48.3 Fields

Field	Function
31-30	Dithering control for R channel: 00b - dithering disabled

Table continues on the next page...

VPU G1 Memory Map/Register Definition

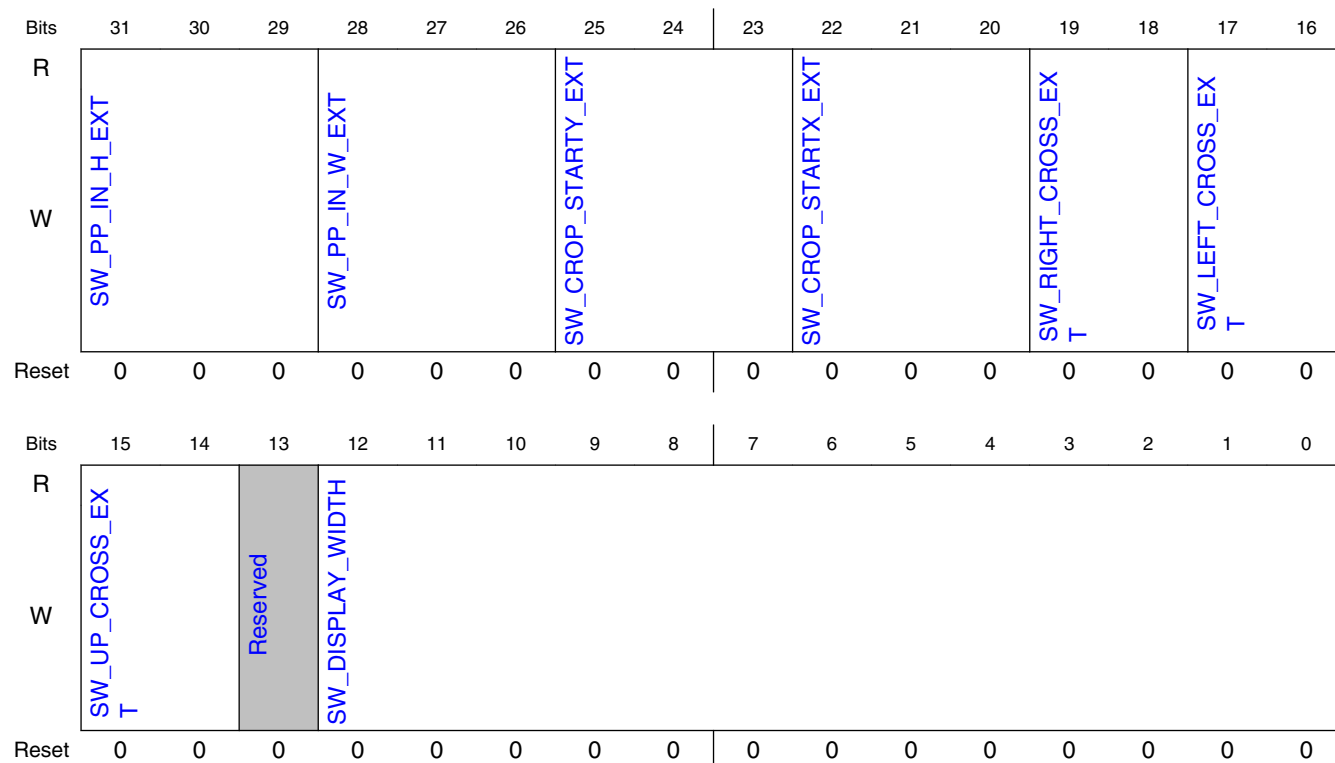
Field	Function
SW_DITHER_SELECT_R	01b - use four-bit dither matrix 10b - use five-bit dither matrix 11b - use six-bit dither matrix
29-28 SW_DITHER_SELECT_G	Dithering control for G channel: 00b - dithering disabled 01b - use four-bit dither matrix 10b - use five-bit dither matrix 11b - use six-bit dither matrix
27-26 SW_DITHER_SELECT_B	Dithering control for B channel: 00b - dithering disabled 01b - use four-bit dither matrix 10b - use five-bit dither matrix 11b - use six-bit dither matrix
25-24 —	Reserved.
23-22 SW_PP_TILED_MODE	Input data is in tiled mode (at the moment valid only for YCbCr 420 data, pipeline or external mode): 00b - Tiled mode not used 01b - Tiled mode enabled for 8x4 sized tiles
21-11 SW_RIGHT_CROSS	Amount of right side overcross (Horizontal pixels outside of display from the right side). Range must be between [0, ScaledWidth].
10-0 SW_LEFT_CROSS	Amount of left side overcross (Horizontal pixels outside of display from the left side). Range must be between [0, ScaledWidth].

14.1.5.1.49 Display width and PP input size extension register (SWREG92)

14.1.5.1.49.1 Offset

Register	Offset
SWREG92	170h

14.1.5.1.49.2 Diagram



14.1.5.1.49.3 Fields

Field	Function
31-29 SW_PP_IN_H_EXT	Extended PP input height. Used with WEBP
28-26 SW_PP_IN_W_EXT	Extended PP input width. Used with WEBP
25-23 SW_CROP_ST ARTY_EXT	Extended PP input crop start coordinate x. Used with WEBP
22-20 SW_CROP_ST ARTX_EXT	Extended PP input crop start coordinate y. Used with WEBP
19-18 SW_RIGHT_CR OSS_EXT	Extended coordinate for 4k resolution
17-16 SW_LEFT_CRO SS_EXT	Extended coordinate for 4k resolution

Table continues on the next page...

Field	Function
15-14 SW_UP_CROSS_EXT	Extended coordinate for 4k resolution
13 —	Reserved.
12-0 SW_DISPLAY_WIDTH	Width of the display in pixels. Max 4k (depends on HW config support)

14.1.5.1.50 Base address for alpha blend 1 gui component (SWREG93)

14.1.5.1.50.1 Offset

Register	Offset
SWREG93	174h

14.1.5.1.50.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_ABLEND1_BASE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_ABLEND1_BASE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.1.5.1.50.3 Fields

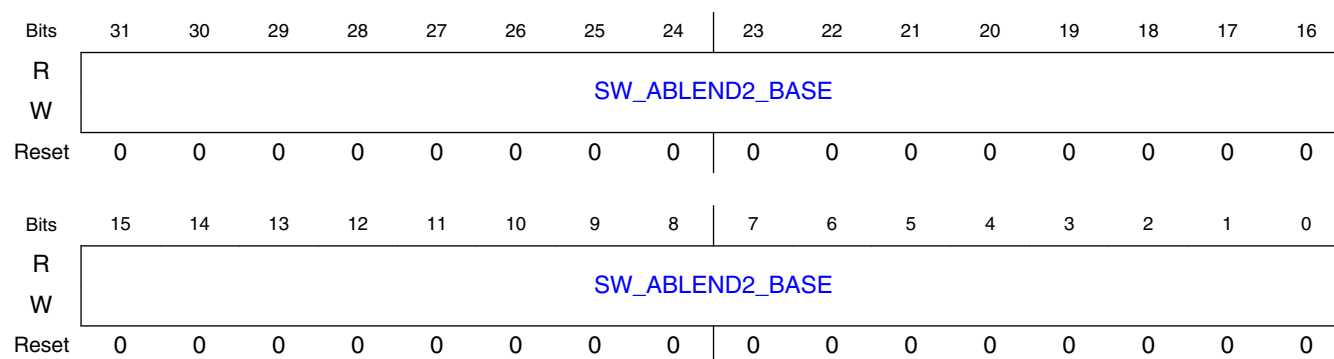
Field	Function
31-0 SW_ABLEND1_BASE	Base address for alpha blending input 1 (if mask1 is used in alpha blending mode). Format of data is 24 bit RGB/ YCbCr and endian/swap -mode is as in PP input. Amount of data is informed with mask 1 size or with ablend1_scanline if ablend cropping is supported in configuration.

14.1.5.1.51 Base address for alpha blend 2 gui component (SWREG94)

14.1.5.1.51.1 Offset

Register	Offset
SWREG94	178h

14.1.5.1.51.2 Diagram



14.1.5.1.51.3 Fields

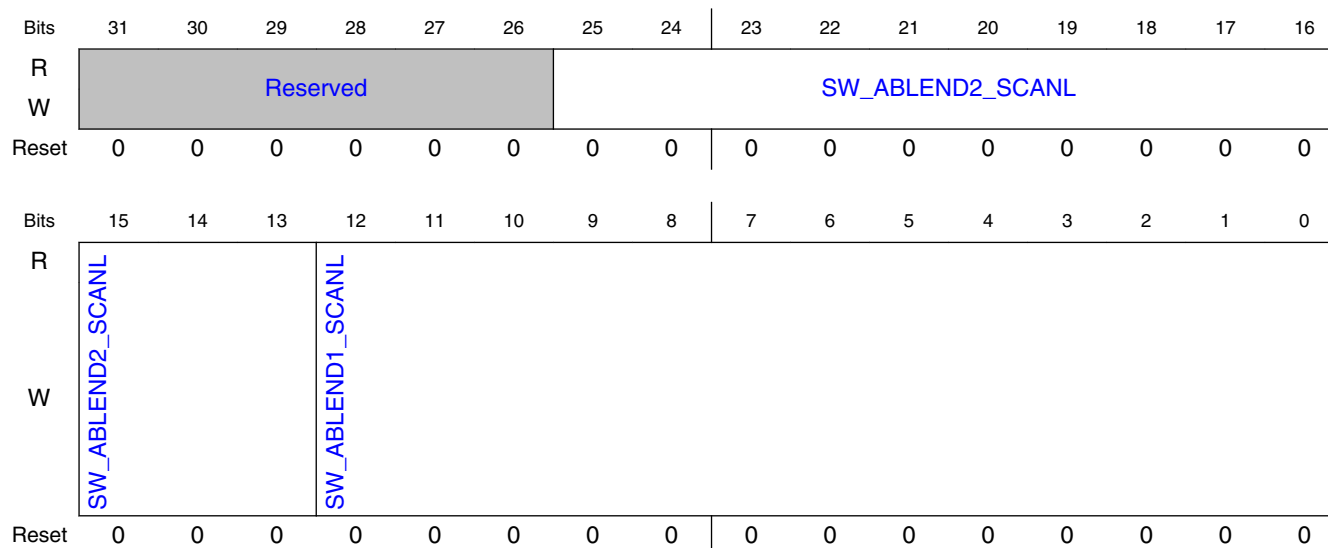
Field	Function
31-0 SW_ABLEND2_BASE	Base address for alpha blending input 2 (if mask2 is used in alpha blending mode). Format of data is 24 bit RGB/ YCbCr and endian/swap -mode is as in PP input. Amount of data is informed with mask 2 size or with ablend2_scanline if ablend cropping is supported in configuration.

14.1.5.1.52 Alpha blend input cropping register (scanline for cropping) (SWREG95)

14.1.5.1.52.1 Offset

Register	Offset
SWREG95	17Ch

14.1.5.1.52.2 Diagram



14.1.5.1.52.3 Fields

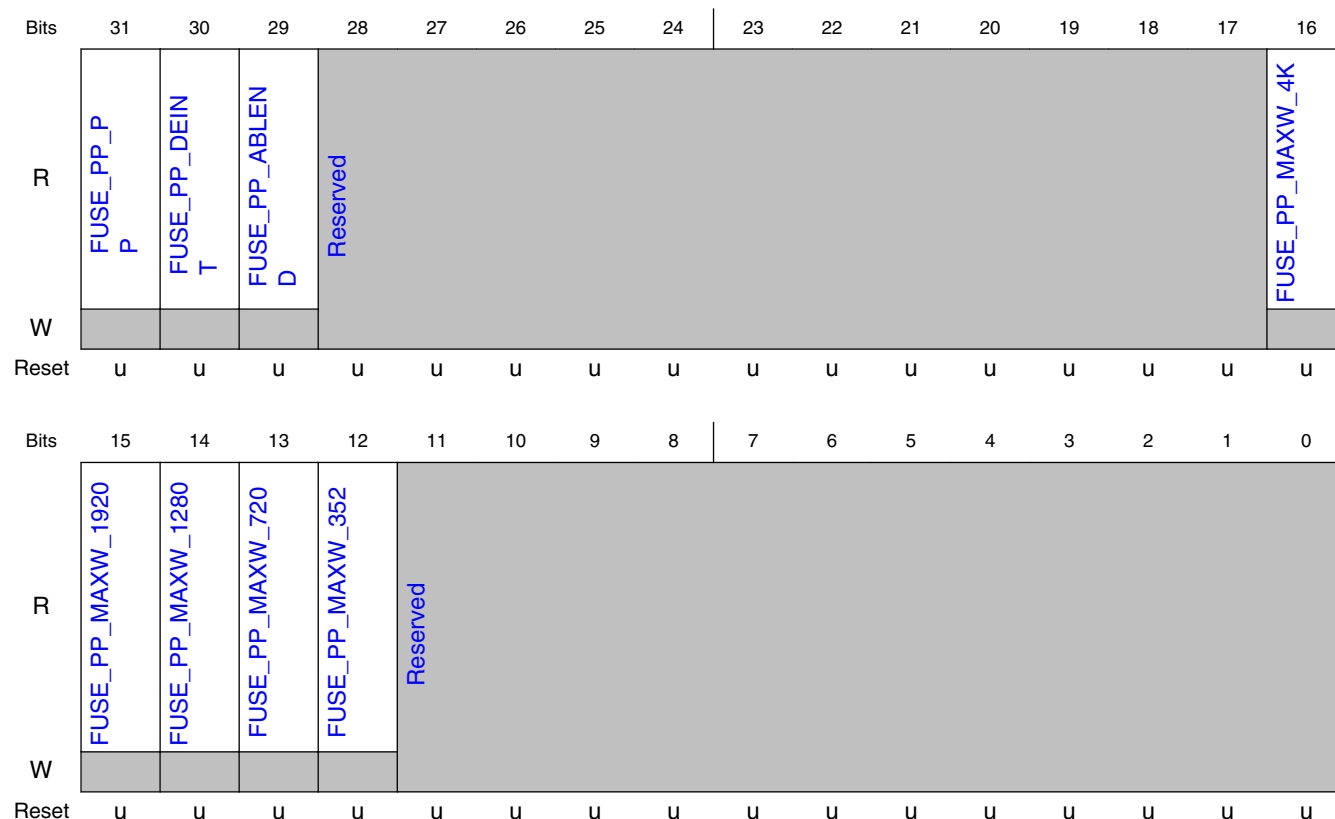
Field	Function
31-26 —	Reserved.
25-13 SW_ABLEND2_SCANL	Scanline width in pixels for Ablend 2. Usage enabled if corresponding configuration bit is enabled
12-0 SW_ABLEND1_SCANL	Scanline width in pixels for Ablend 1. Usage enabled if corresponding configuration bit is enabled

14.1.5.1.53 PP fuse register (SWREG99)

14.1.5.1.53.1 Offset

Register	Offset
SWREG99	18Ch

14.1.5.1.53.2 Diagram



14.1.5.1.53.3 Fields

Field	Function
31 FUSE_PP_PP	1 = PP enabled
30 FUSE_PP_DEINT	1 = Deinterlacing enabled
29 FUSE_PP_ABLEND	1 = Alpha Blending enabled
28-17 —	Reserved.
16 FUSE_PP_MAXW_4K	1 = Max PP output width up to 4096 pixels enabled. Priority coded with priority 1
15 FUSE_PP_MAXW_1920	1 = Max PP output width up to 1920 pixels enabled. Priority coded with priority 2

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VPU G1 Memory Map/Register Definition

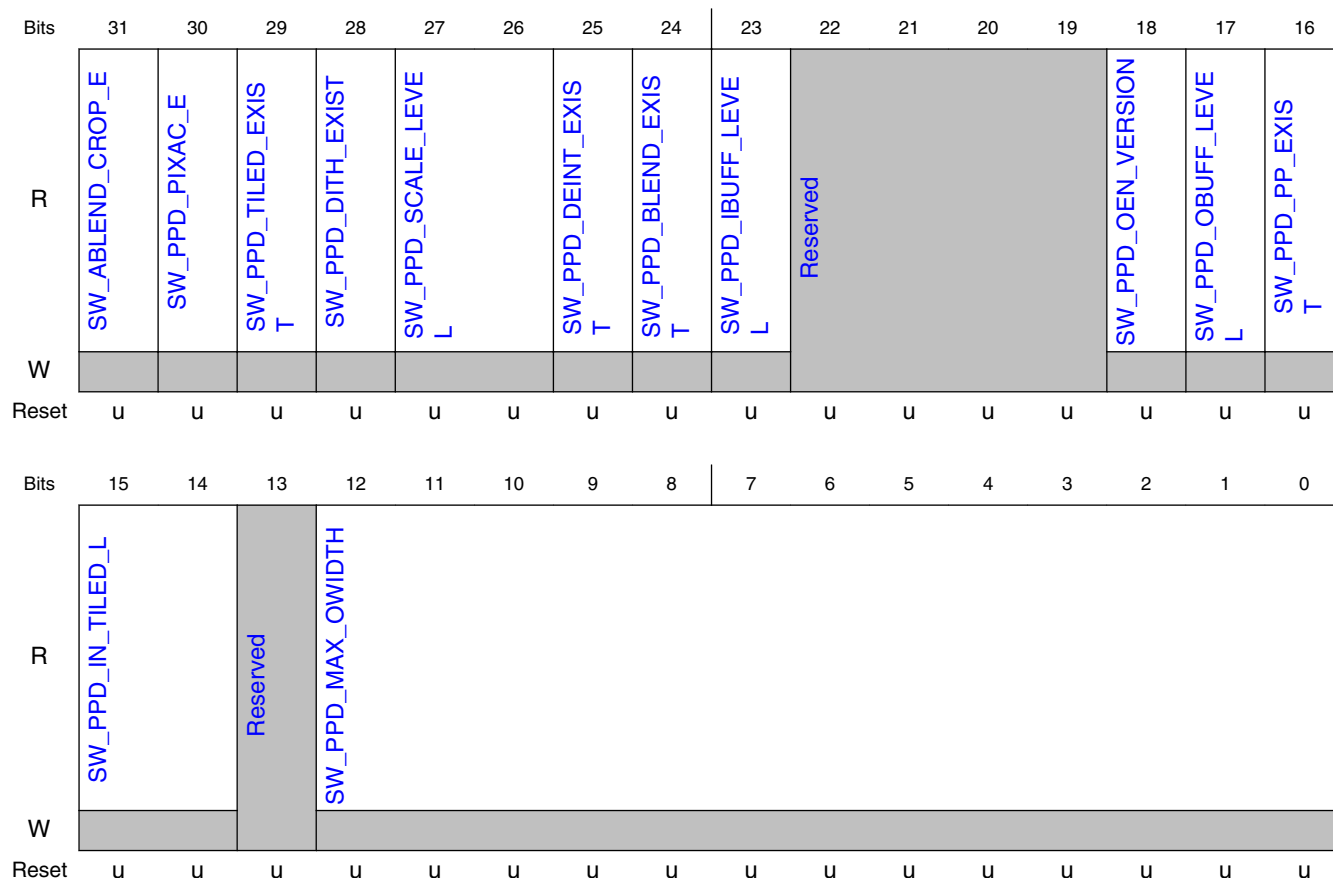
Field	Function
14 FUSE_PP_MAX W_1280	1 = Max PP output width up to 1280 pixels enabled. Priority coded with priority 3
13 FUSE_PP_MAX W_720	1 = Max PP output width up to 720 pixels enabled. Priority coded with priority 4
12 FUSE_PP_MAX W_352	1 = Max PP output width up to 352 pixels enabled. Priority coded with priority 5
11-0 —	Reserved.

14.1.5.1.54 Synthesis configuration register post-processor (SWREG100)

14.1.5.1.54.1 Offset

Register	Offset
SWREG100	190h

14.1.5.1.54.2 Diagram



14.1.5.1.54.3 Fields

Field	Function
31 SW_ABLEND_CROP_E	Alpha blending support for input cropping: 0b - Not supported. External memory must include the exact image of the area being alpha blended. 1b - Supported. External memory can include a picture from blended area can be cropped. Requires usage of swreg95.
30 SW_PPD_PIXAC_E	Pixel Accurate PP output mode exists: 0b - PIP, Scaling and masks can be adjusted by steps of 8 pixels (width) or 2 pixels (height) 1b - PIP, Scaling and masks can be adjusted by steps of 1 pixel for RGB and 2 pixels for subsampled chroma formats (by using bus specific write strobe functionality)
29 SW_PPD_TILE_D_EXIS	PP output YCbYCr 422 tiled support (4x4 pixel tiles) 0b - Not supported 1b - Supported
28 SW_PPD_DITH_EXIS	Dithering exists: 0b - No 1b - Yes
27-26	Scaling support:

Table continues on the next page...

VPU G1 Memory Map/Register Definition

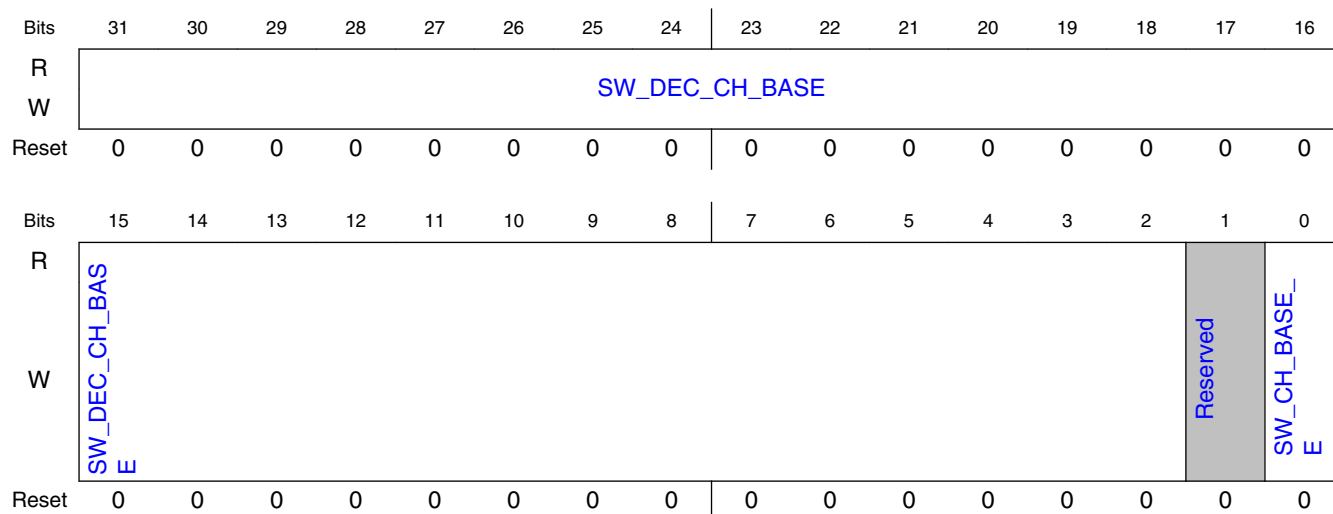
Field	Function
SW_PPD_SCAL E_LEVEL	00b - No scaling 01b - Scaling with lo performance architecture 10b - Scaling with high performance architecture 11b - Scaling with high performance architecture + fast downscaling enabled
25 SW_PPD_DEIN T_EXIST	De-interlacing exists: 0b - No 1b - Yes
24 SW_PPD_BLEN D_EXIST	Alpha blending exists: 0b - No 1b - Yes
23 SW_PPD_IBUF F_LEVEL	PP input buffering level: 0b - 1 MB input buffering is used 1b - 4 MB input buffering is used
22-19 —	Reserved.
18 SW_PPD_OEN _VERSION	PP output endian version: 0b - Endian mode supported for other than RGB 1b - Endian mode supported for any output format
17 SW_PPD_OBU FF_LEVEL	PP output buffering level: 0b - 1 unit output buffering is used 1b - 4 unit output buffering is used
16 SW_PPD_PP_E XIST	PPD exists: 0b - No 1b - Yes
15-14 SW_PPD_IN_TI LED_L	PPD input tiled mode support level 00b - not supported 01b - 8x4 tile size supported
13 —	Reserved.
12-0 SW_PPD_MAX _OWIDTH	Max supported PP output width in pixels

14.1.5.1.55 Base address for H264 decoded chroma picture (SWREG102)

14.1.5.1.55.1 Offset

Register	Offset
SWREG102	198h

14.1.5.1.55.2 Diagram



14.1.5.1.55.3 Fields

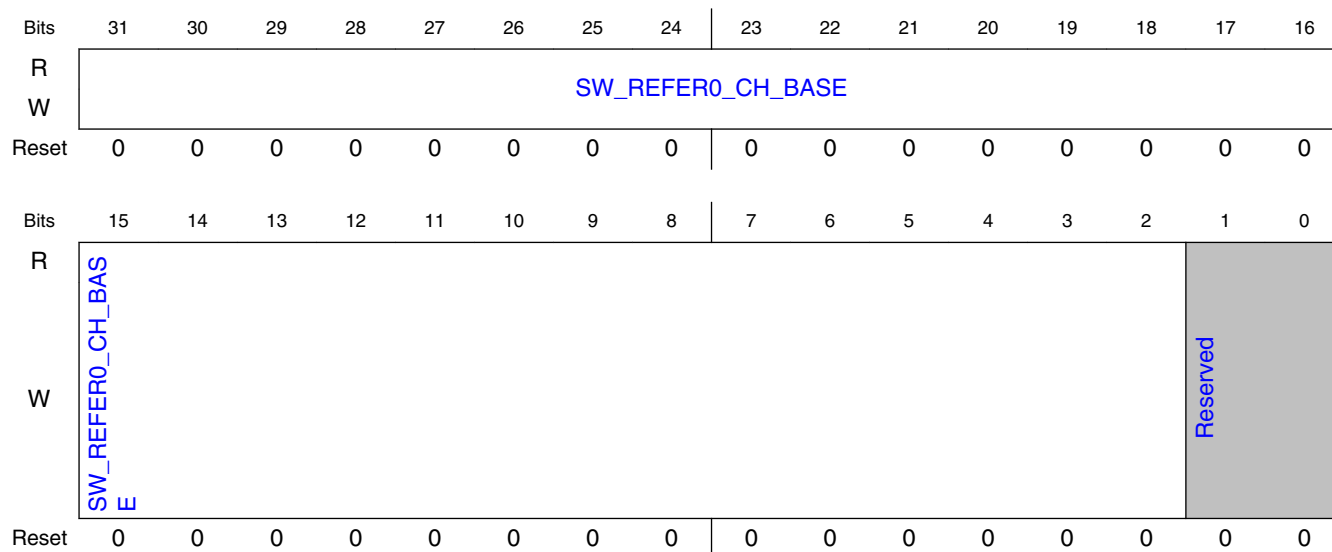
Field	Function
31-2 SW_DEC_CH_BASE	Valid only if chroma address separate mode is enabled. H264: Base address for decoder output chroma picture. Points directly to start of decoder output chroma picture or field.
1 —	Reserved.
0 SW_CH_BASE_E	chroma address separate mode enable: 0b - HW outputs decoded chroma picture to the end of decoded luma picture. HW calculates the chroma picture address according to sw_dec_base and luma data length. 1b - HW outputs decoded chroma picture to independent memory address

14.1.5.1.56 Base address for reference chroma picture index 0 (SWREG103)

14.1.5.1.56.1 Offset

Register	Offset
SWREG103	19Ch

14.1.5.1.56.2 Diagram



14.1.5.1.56.3 Fields

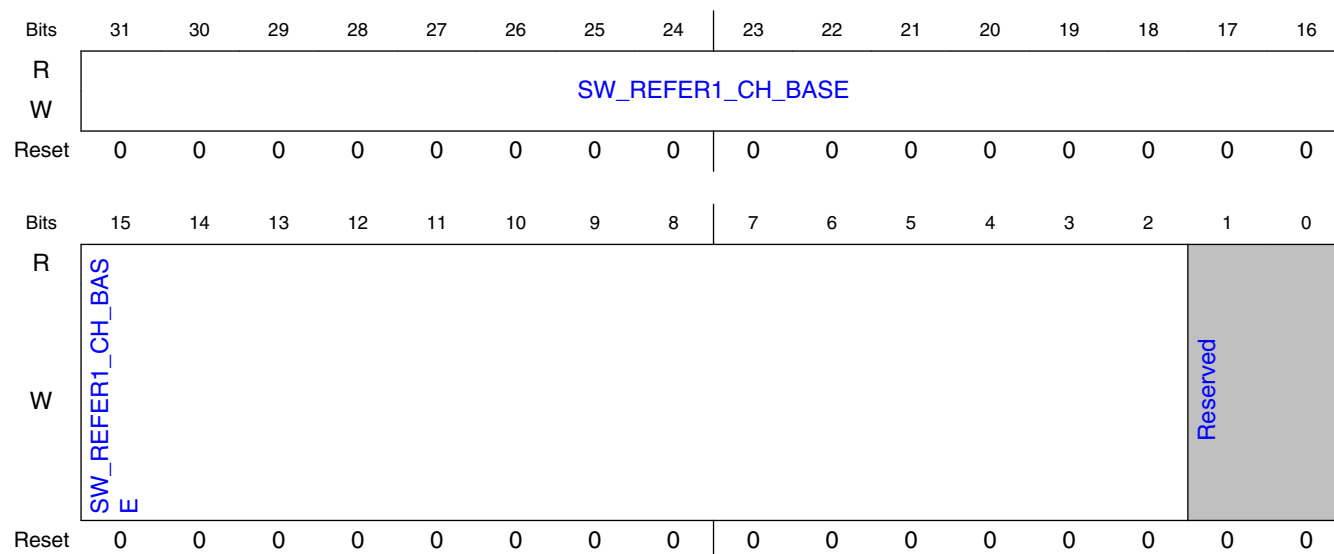
Field	Function
31-2 SW_REFER0_CH_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 0.
1-0 —	Reserved.

14.1.5.1.57 Base address for reference chroma picture index 1 (SWREG104)

14.1.5.1.57.1 Offset

Register	Offset
SWREG104	1A0h

14.1.5.1.57.2 Diagram



14.1.5.1.57.3 Fields

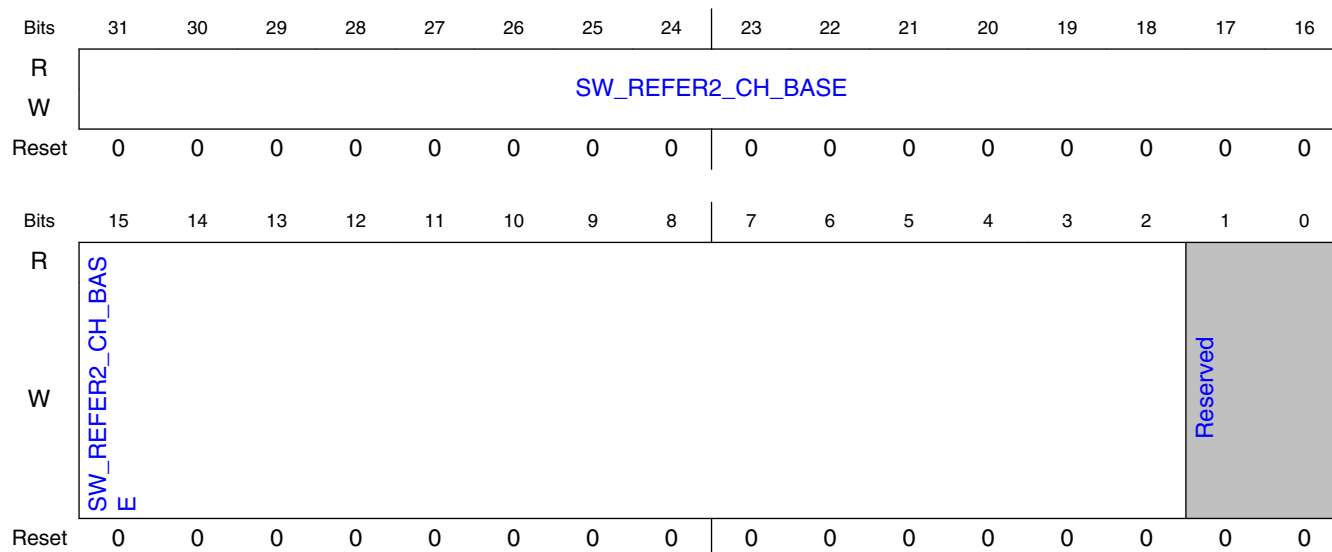
Field	Function
31-2 SW_REFER1_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 1.
1-0 —	Reserved.

14.1.5.1.58 Base address for reference chroma picture index 2 (SWREG105)

14.1.5.1.58.1 Offset

Register	Offset
SWREG105	1A4h

14.1.5.1.58.2 Diagram



14.1.5.1.58.3 Fields

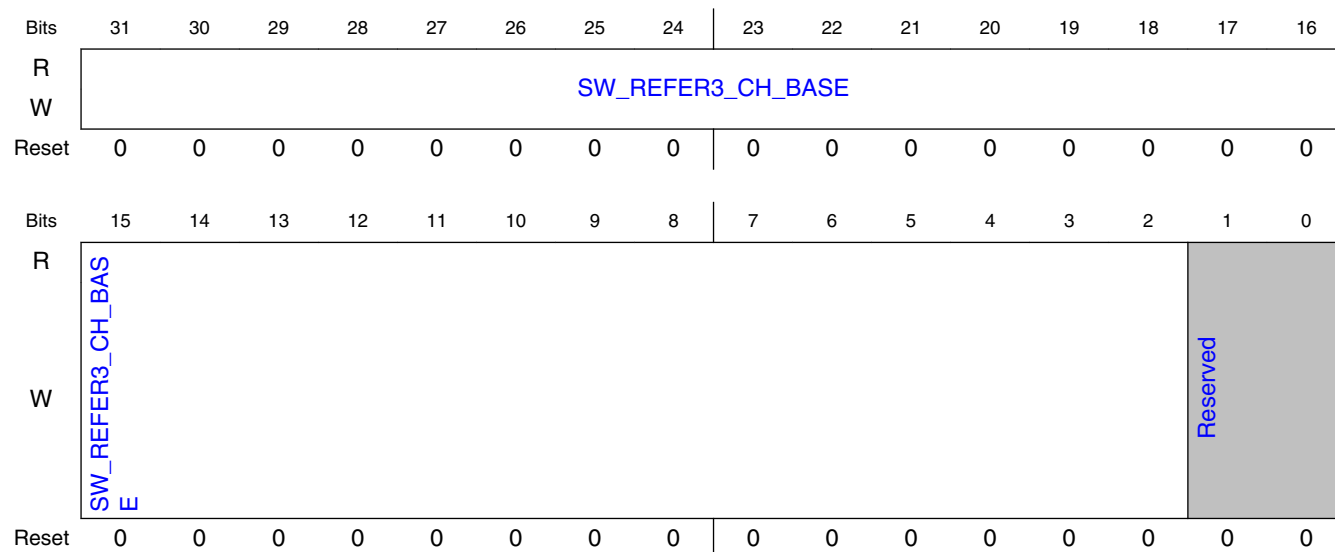
Field	Function
31-2 SW_REFER2_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 2.
1-0 —	Reserved.

14.1.5.1.59 Base address for reference chroma picture index 3 (SWRE G106)

14.1.5.1.59.1 Offset

Register	Offset
SWREG106	1A8h

14.1.5.1.59.2 Diagram



14.1.5.1.59.3 Fields

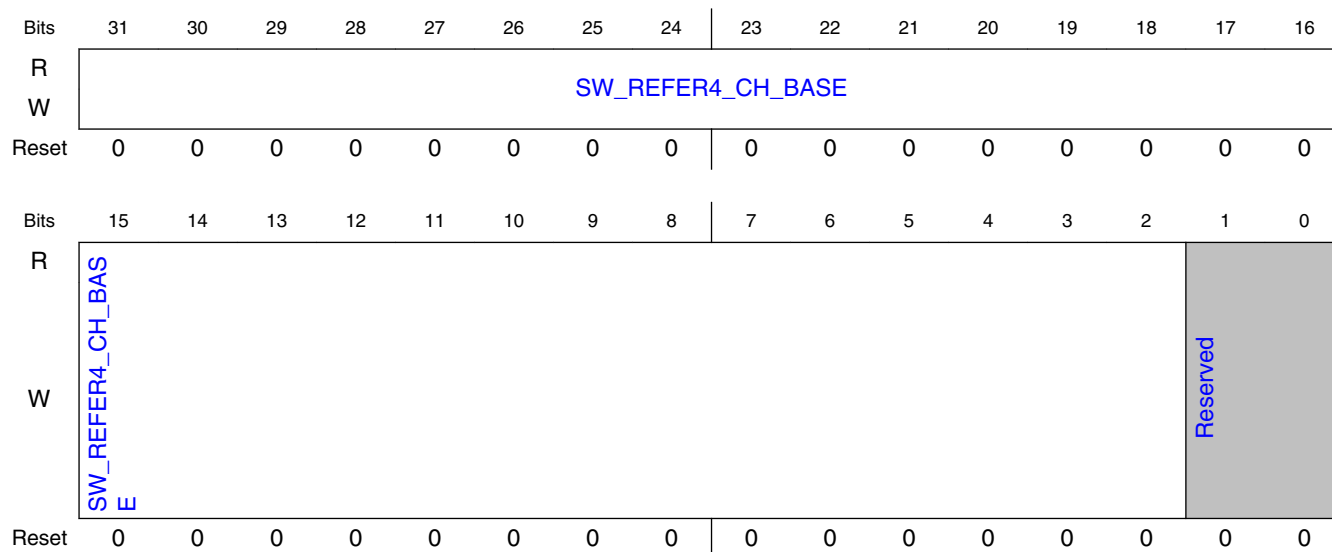
Field	Function
31-2 SW_REFER3_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 3.
1-0 —	Reserved.

14.1.5.1.60 Base address for reference chroma picture index 4 (SWREG107)

14.1.5.1.60.1 Offset

Register	Offset
SWREG107	1ACh

14.1.5.1.60.2 Diagram



14.1.5.1.60.3 Fields

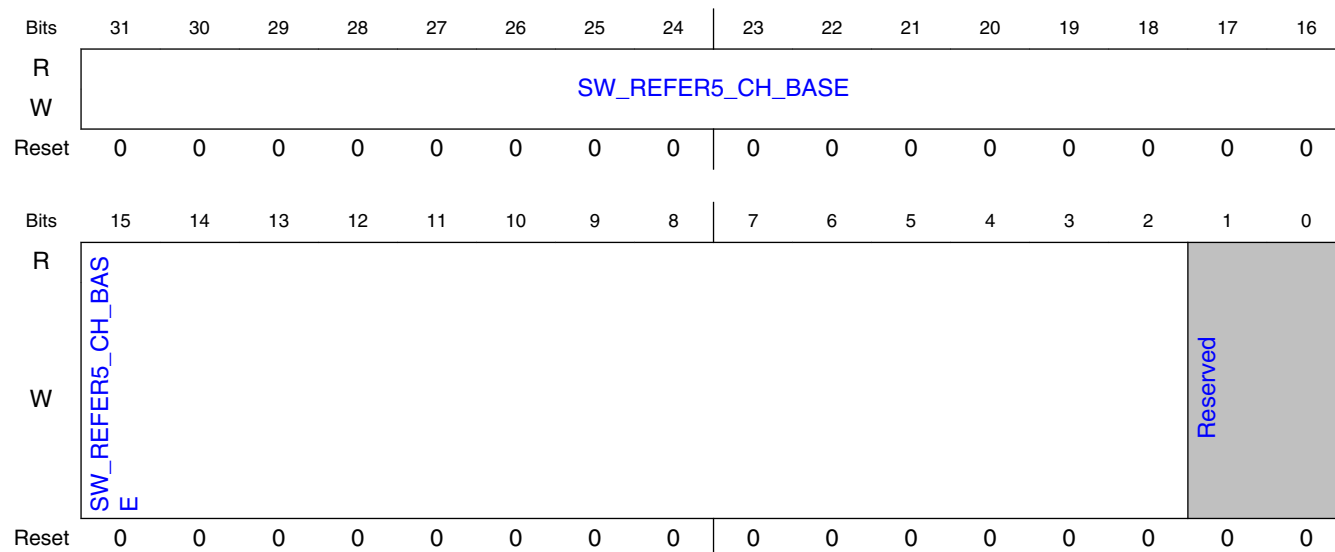
Field	Function
31-2 SW_REFER4_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 4.
1-0 —	Reserved.

14.1.5.1.61 Base address for reference chroma picture index 5 (SWRE G108)

14.1.5.1.61.1 Offset

Register	Offset
SWREG108	1B0h

14.1.5.1.61.2 Diagram



14.1.5.1.61.3 Fields

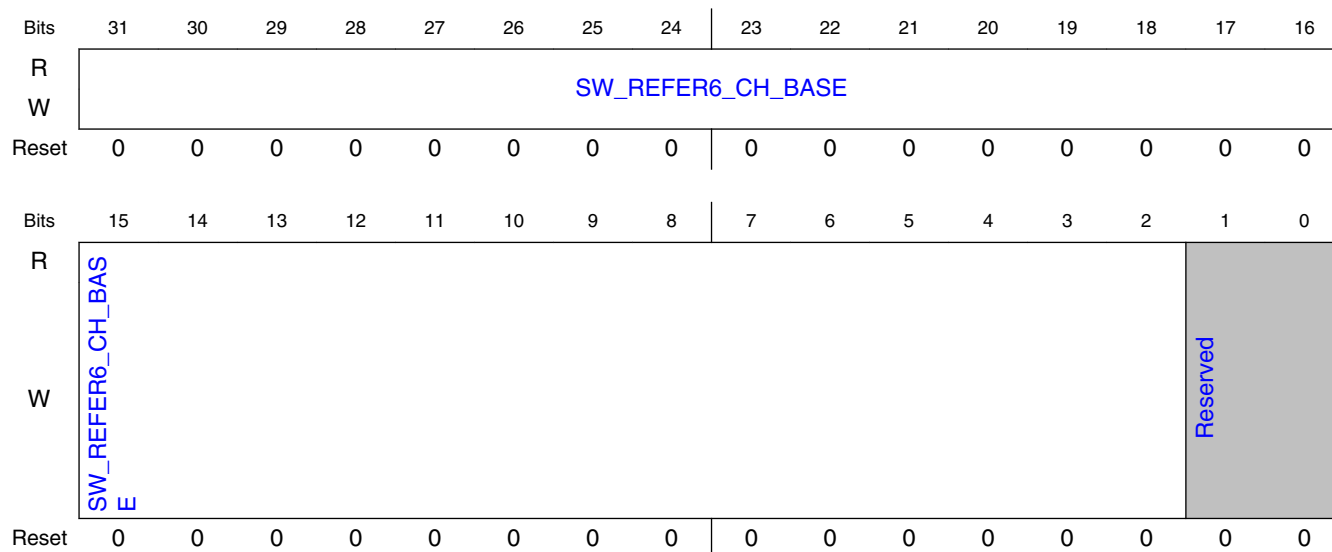
Field	Function
31-2 SW_REFER5_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 5.
1-0 —	Reserved.

14.1.5.1.62 Base address for reference chroma picture index 6 (SWREG109)

14.1.5.1.62.1 Offset

Register	Offset
SWREG109	1B4h

14.1.5.1.62.2 Diagram



14.1.5.1.62.3 Fields

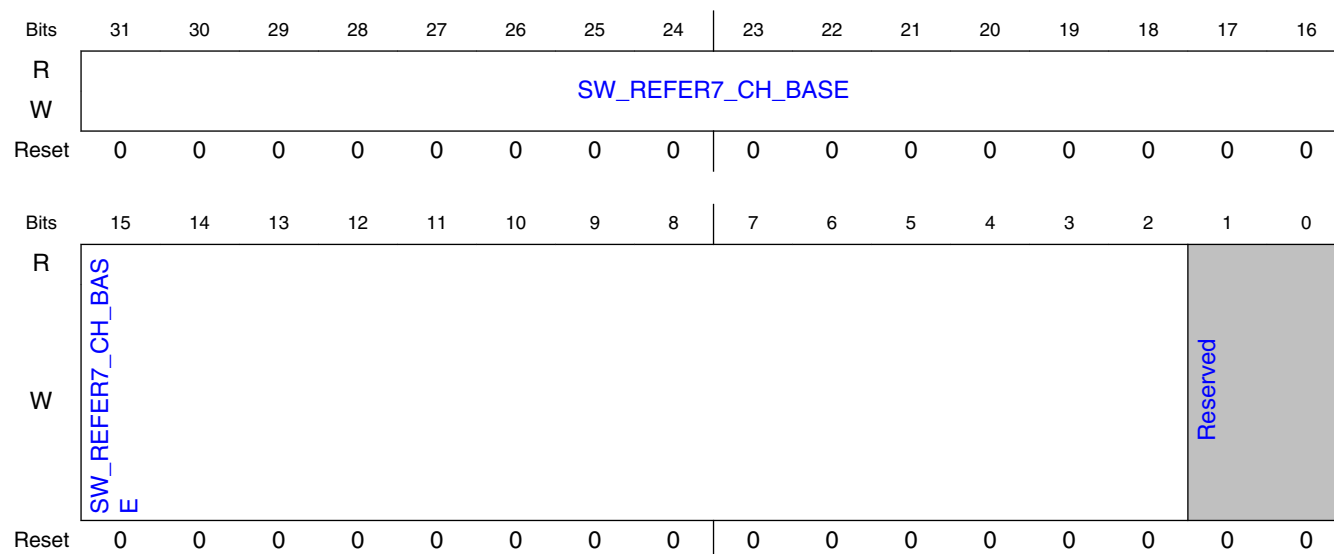
Field	Function
31-2 SW_REFER6_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 6.
1-0 —	Reserved.

14.1.5.1.63 Base address for reference chroma picture index 7 (SWRE G110)

14.1.5.1.63.1 Offset

Register	Offset
SWREG110	1B8h

14.1.5.1.63.2 Diagram



14.1.5.1.63.3 Fields

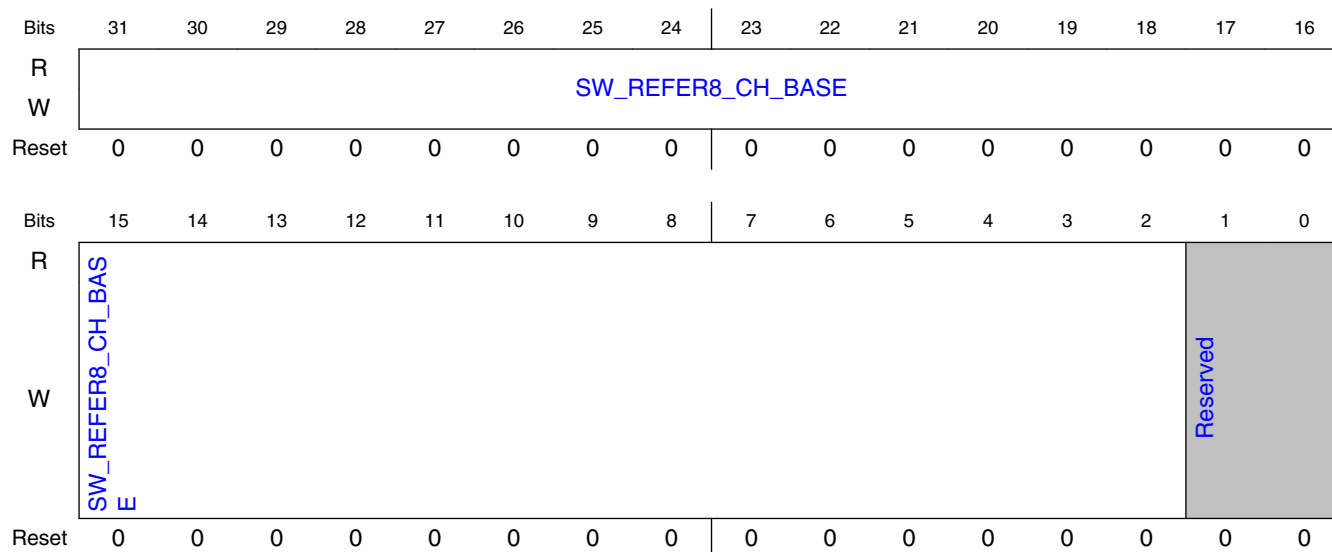
Field	Function
31-2 SW_REFER7_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 7.
1-0 —	Reserved.

14.1.5.1.64 Base address for reference chroma picture index 8 (SWREG111)

14.1.5.1.64.1 Offset

Register	Offset
SWREG111	1BCh

14.1.5.1.64.2 Diagram



14.1.5.1.64.3 Fields

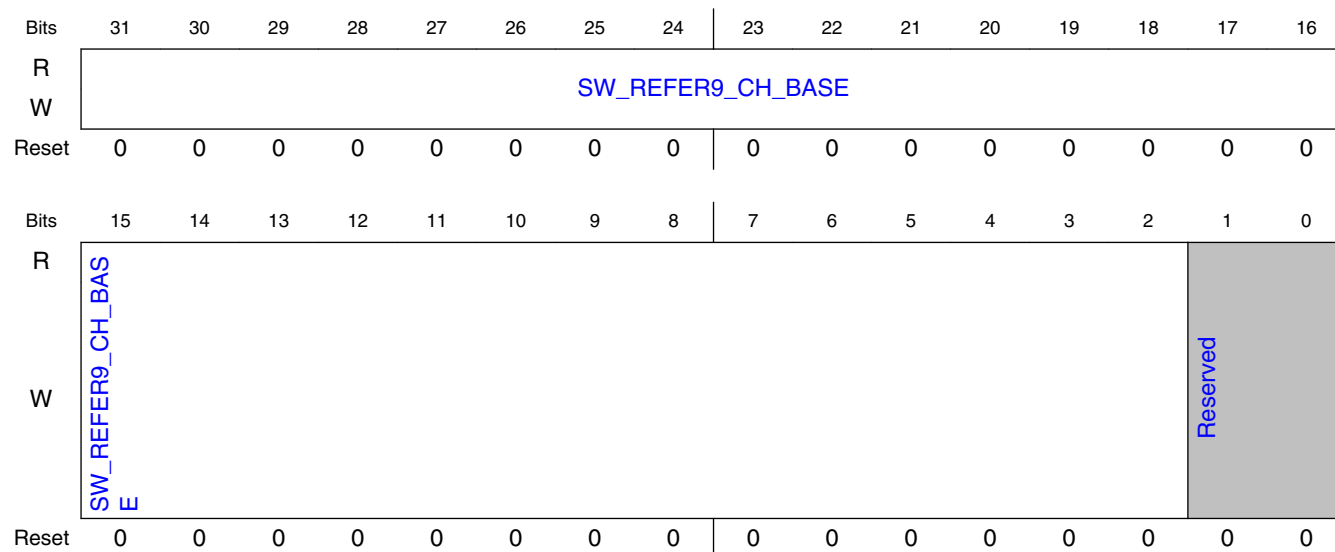
Field	Function
31-2 SW_REFER8_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 8.
1-0 —	Reserved.

14.1.5.1.65 Base address for reference chroma picture index 9 (SWRE G112)

14.1.5.1.65.1 Offset

Register	Offset
SWREG112	1C0h

14.1.5.1.65.2 Diagram



14.1.5.1.65.3 Fields

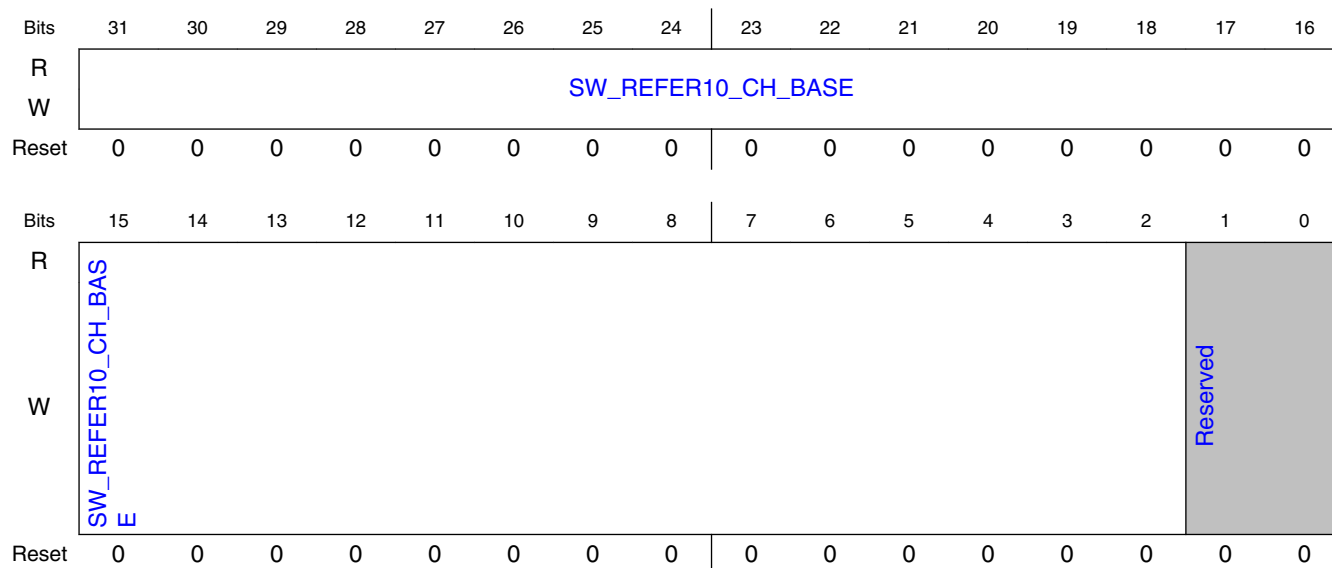
Field	Function
31-2 SW_REFER9_C H_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 9.
1-0 —	Reserved.

14.1.5.1.66 Base address for reference chroma picture index 10 (SWREG113)

14.1.5.1.66.1 Offset

Register	Offset
SWREG113	1C4h

14.1.5.1.66.2 Diagram



14.1.5.1.66.3 Fields

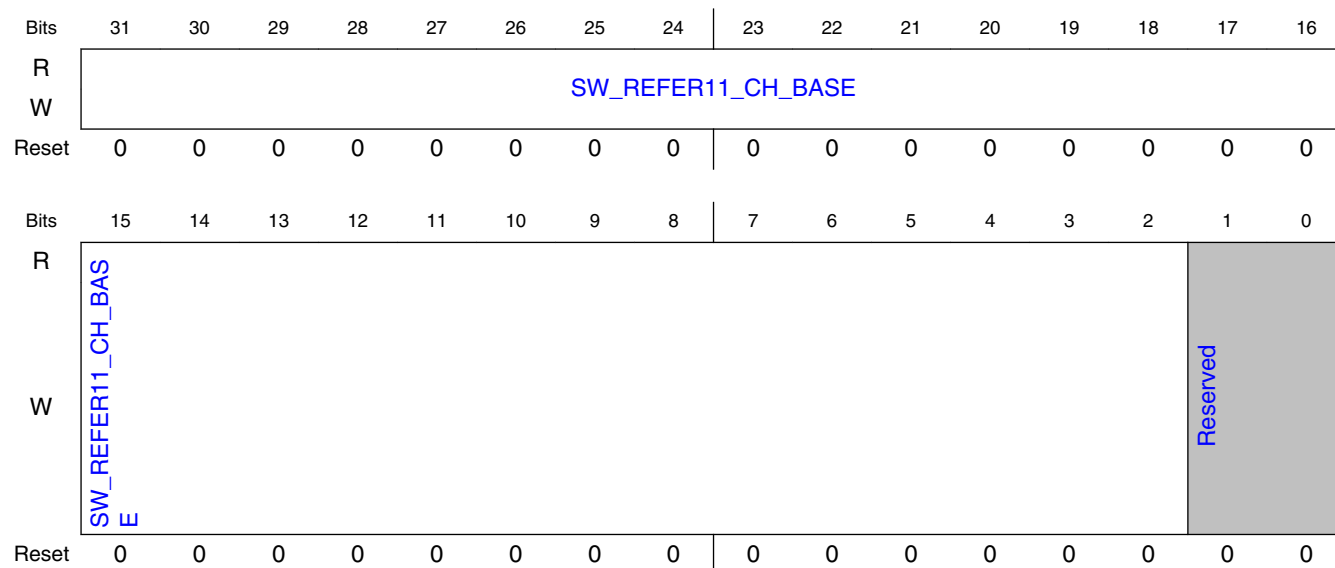
Field	Function
31-2 SW_REFER10_CH_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 10.
1-0 —	Reserved.

14.1.5.1.67 Base address for reference chroma picture index 11 (SWREG114)

14.1.5.1.67.1 Offset

Register	Offset
SWREG114	1C8h

14.1.5.1.67.2 Diagram



14.1.5.1.67.3 Fields

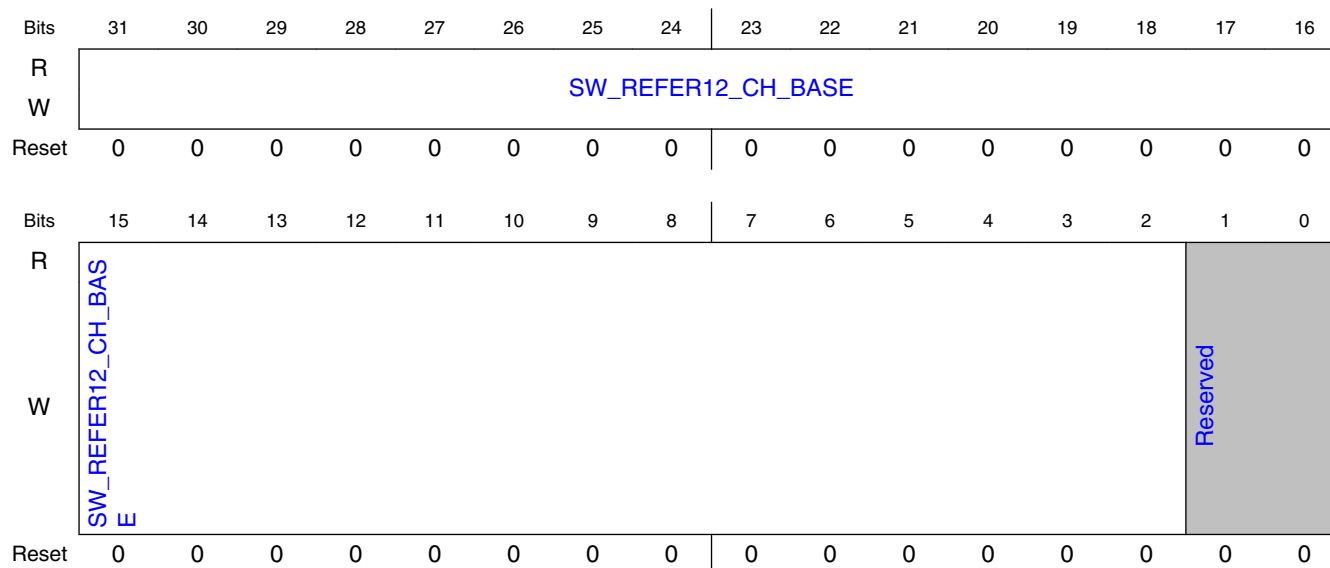
Field	Function
31-2 SW_REFER11_CH_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 11.
1-0 —	Reserved.

14.1.5.1.68 Base address for reference chroma picture index 12 (SWREG115)

14.1.5.1.68.1 Offset

Register	Offset
SWREG115	1CCh

14.1.5.1.68.2 Diagram



14.1.5.1.68.3 Fields

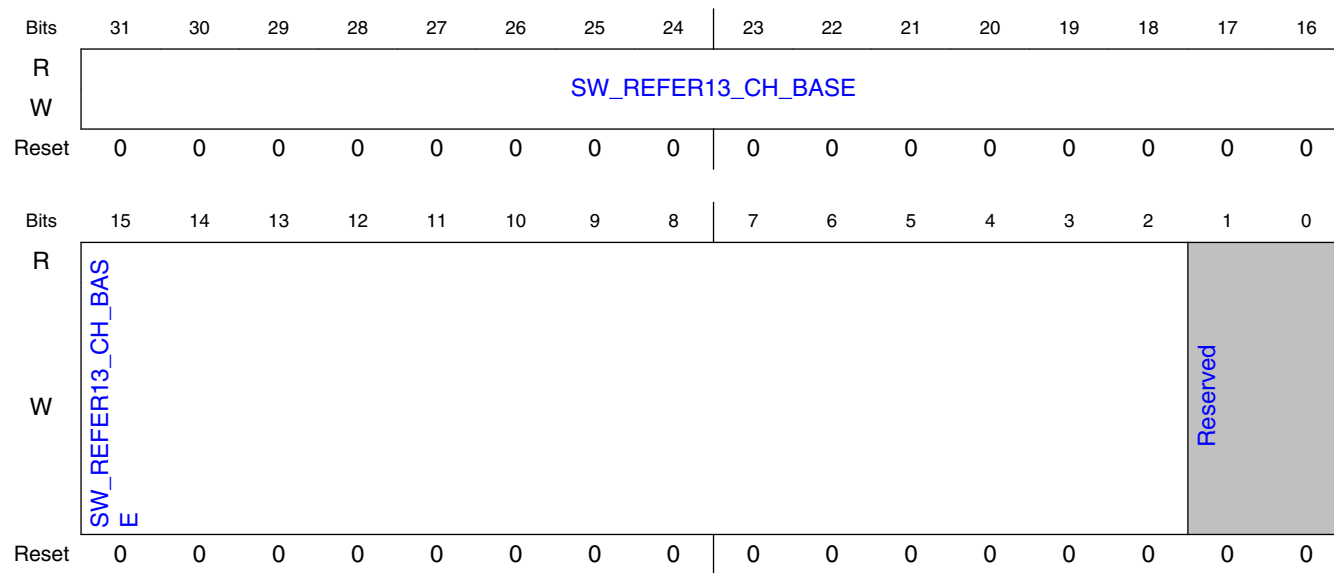
Field	Function
31-2 SW_REFER12_CH_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 12.
1-0 —	Reserved.

14.1.5.1.69 Base address for reference chroma picture index 13 (SWREG116)

14.1.5.1.69.1 Offset

Register	Offset
SWREG116	1D0h

14.1.5.1.69.2 Diagram



14.1.5.1.69.3 Fields

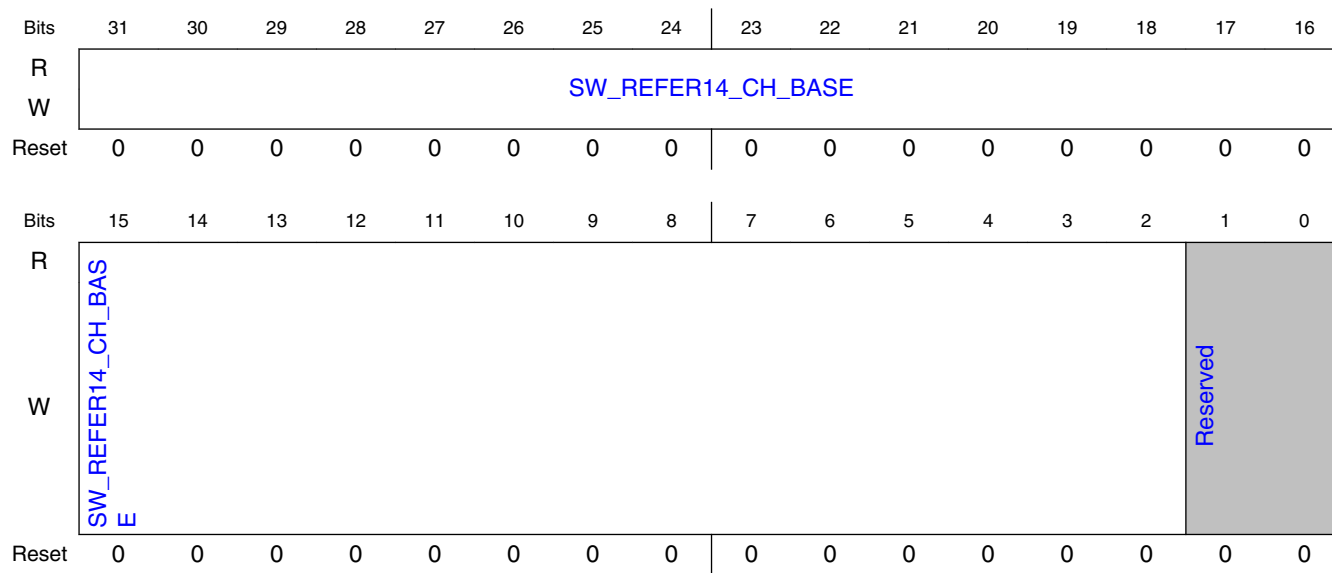
Field	Function
31-2 SW_REFER13_CH_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 13.
1-0 —	Reserved.

14.1.5.1.70 Base address for reference chroma picture index 14 (SWREG117)

14.1.5.1.70.1 Offset

Register	Offset
SWREG117	1D4h

14.1.5.1.70.2 Diagram



14.1.5.1.70.3 Fields

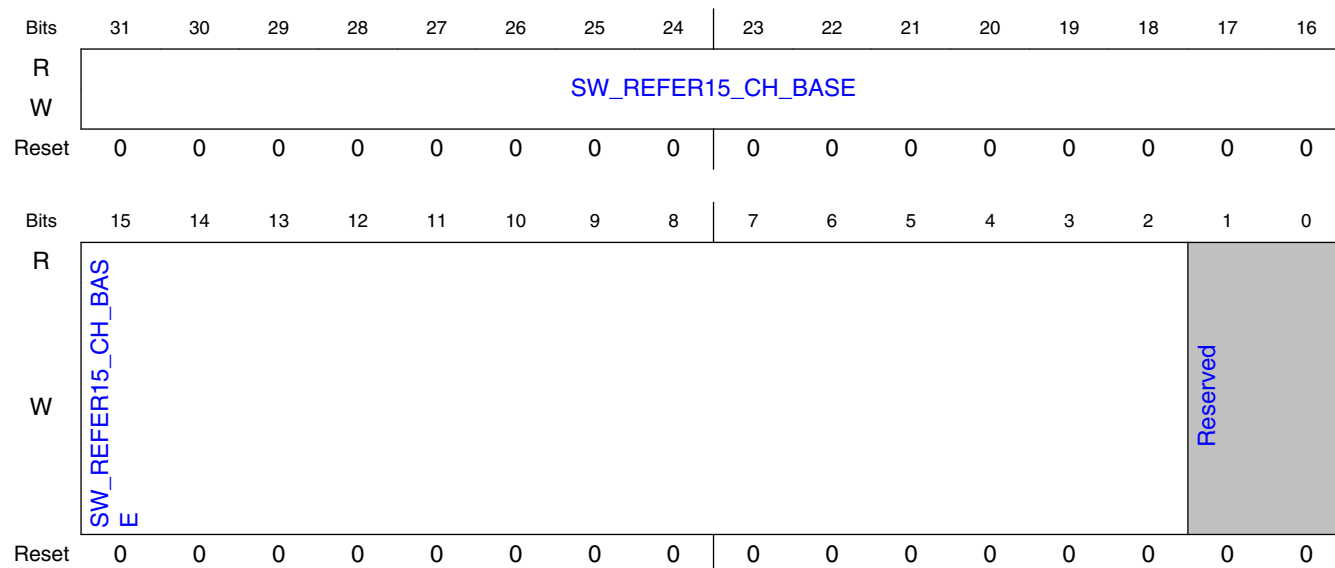
Field	Function
31-2 SW_REFER14_CH_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 14.
1-0 —	Reserved.

14.1.5.1.71 Base address for reference chroma picture index 15 (SWREG118)

14.1.5.1.71.1 Offset

Register	Offset
SWREG118	1D8h

14.1.5.1.71.2 Diagram



14.1.5.1.71.3 Fields

Field	Function
31-2 SW_REFER15_CH_BASE	Valid only if chroma address separate mode is enabled. Base address for reference chroma picture index 15.
1-0 —	Reserved.

14.1.5.2 VPU_G1 H264 decoder register descriptions

14.1.5.2.1 VPU_G1 H264 decoder memory map

VPU_G1_H264 base address: 3830_0000h

Offset	Register	Width (In bits)	Access	Reset value
10h	Decoder control register 1 (picture parameters) (SWREG4)	32	RW	0000_0000h
14h	Decoder control register 2 (stream decoding table selects) (SWREG5)	32	RW	0000_0000h
18h	Decoder control register 3 (stream buffer information) (SWREG6)	32	RW	0000_0000h

Table continues on the next page...

VPU G1 Memory Map/Register Definition

Offset	Register	Width (In bits)	Access	Reset value
1Ch	Decoder control register 4 (H264, VC-1, VP6 and progressive JPEG control) (SWREG7)	32	RW	0000_0000h
20h	Decoder control register 5 (H264, VC-1, VP6, Progressive JPEG and RV control) (SWREG8)	32	RW	0000_0000h
24h	Decoder control register 6 / base address for MB-control (RLC) / VC-1 intensity control 0/ VP6,VP7,VP8 ctrl-stream length/ RV pic slice amount (SWREG9)	32	RW	0000_0000h
28h	Base address for differential motion vector base address (RLC-mode) /H264 P initial fwd ref pic list register (4-9)/ VC-1 intensity control 1/ VP7 and VP8 segmentation base register (SWREG10)	32	RW	0000_0000h
2Ch	Decoder control register 7 (VLC) / base address for H.264 intra prediction 4x4 / base address for MPEG-4 DC component (RLC) / H264 P initial fwd ref pic list register (10-15) / VC-1 intensity control 2 (SWREG11)	32	RW	0000_0000h
38h	Base address for reference picture index 0 / base address for JPEG decoder output chrominance picture (SWREG14)	32	RW	0000_0000h
3Ch	Base address for reference picture index 1 / JPEG control (SWREG15)	32	RW	0000_0000h
40h	Base address for reference picture index 2 / List of VLC code lengths in first JPEG AC table (SWREG16)	32	RW	0000_0000h
44h	Base address for reference picture index 3 / List of VLC code lengths in first JPEG AC table (SWREG17)	32	RW	0000_0000h
48h	Base address for reference picture index 4 / VC1 control / MPEG4 MVD control/ List of VLC code lengths in first JPEG AC table / VC-1 intensity control 4 / VP6/VP7, VP8 Golden refer picture base (SWREG18)	32	RW	0000_0000h
4Ch	Base address for reference picture index 5 / MPEG4 TRB/TRD delta 0 / VC-1 intensity control 3 List of VLC code lengths in first/second JPEG AC table / VP6/VP7 scan maps (SWREG19)	32	RW	0000_0000h
50h	Base address for reference picture index 6 / / MPEG4 TRB/TRD delta -1 / List of VLC code lengths in second JPEG AC table / VP6/VP7 scan maps (SWREG20)	32	RW	0000_0000h
54h	Base address for reference picture index 7 / MPEG4 TRB/TRD delta 1 / List of VLC code lengths in second JPEG AC table / VP6/VP7 scan maps (SWREG21)	32	RW	0000_0000h
58h	Base address for reference picture index 8 / List of VLC code lengths in second JPEG AC table / VP6 scan maps / VP7,VP8 DCT stream 1 base (SWREG22)	32	RW	0000_0000h
5Ch	Base address for reference picture index 9 / List of VLC code lengths in first JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 2 base (SWREG23)	32	RW	0000_0000h
60h	Base address for reference picture index 10 / List of VLC code lengths in first JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 3 base (SWREG24)	32	RW	0000_0000h
64h	Base address for reference picture index 11 / List of VLC code lengths in second JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 4 base (SWREG25)	32	RW	0000_0000h

Table continues on the next page...

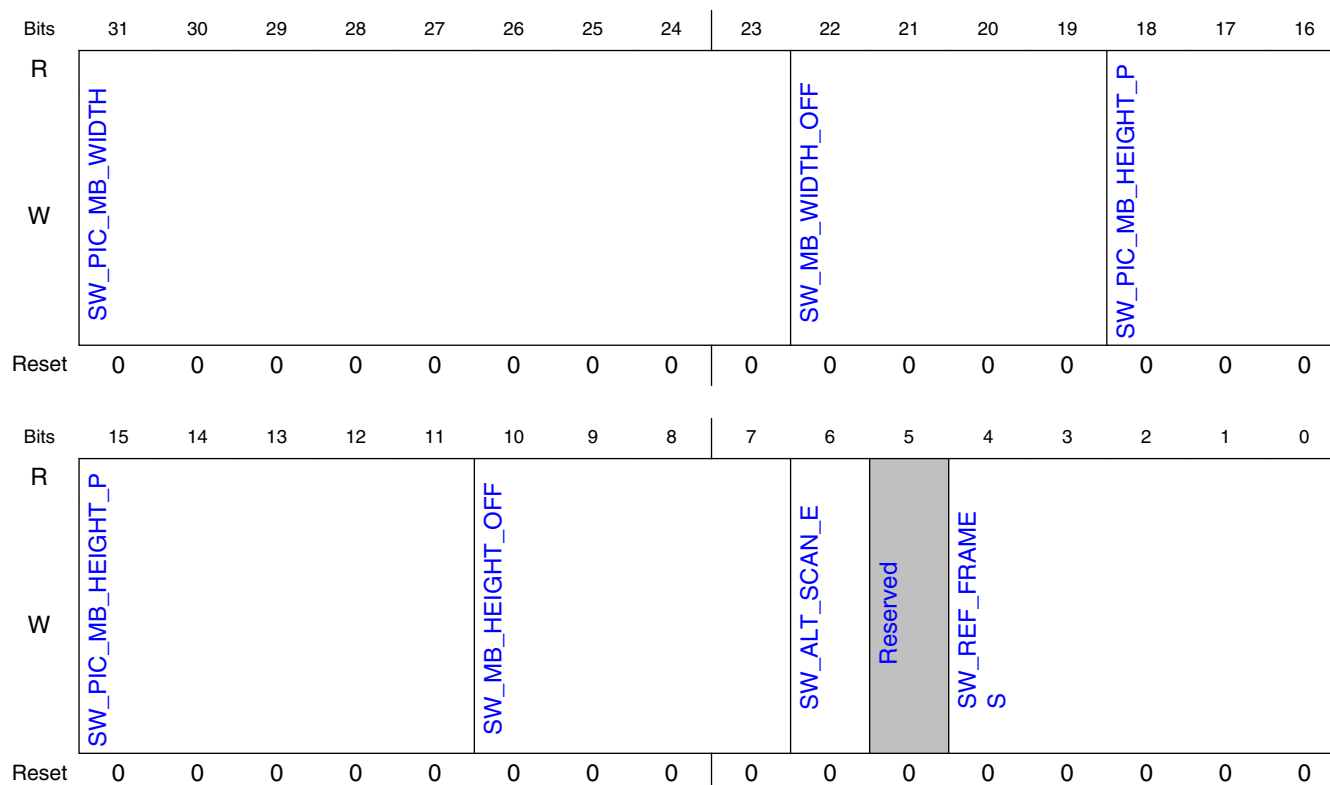
Offset	Register	Width (In bits)	Access	Reset value
68h	Base address for reference picture index 12 / List of VLC code lengths in second JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 5 base (SWREG26)	32	RW	0000_0000h
6Ch	Base address for reference picture index 13 / VC-1 bitpl mbctrl or VP6,VP7,VP8 ctrl stream base /Progressive JPEG DC table (SWREG27)	32	RW	0000_0000h
70h	Base address for reference picture index 14 / VP6 scan maps / Progressive JPEG DC table / VP7,VP8 DCT stream 6 base (SWREG28)	32	RW	0000_0000h
74h	Base address for reference picture index 15 / VP6 scan maps / VP7,VP8 DCT stream 7 base (SWREG29)	32	RW	0000_0000h
78h	Reference picture numbers for index 0 and 1 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter mb level adjusts (SWREG30)	32	RW	0000_0000h
7Ch	Reference picture numbers for index 2 and 3 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter ref pic level adjusts (SWREG31)	32	RW	0000_0000h
80h	Reference picture numbers for index 4 and 5 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter levels (SWREG32)	32	RW	0000_0000h
84h	Reference picture numbers for index 6 and 7 (H264 VLC) / VP6 scan maps / VP7,VP8 quantization values (SWREG33)	32	RW	0000_0000h
88h	Reference picture numbers for index 8 and 9 (H264 VLC) / MPEG4, VC1, VPx prediction filter taps (SWREG34)	32	RW	0000_0000h
8Ch	Reference picture numbers for index 10 and 11 (H264 VLC) / VC1, VPx prediction filter taps (SWREG35)	32	RW	0000_0000h
90h	Reference picture numbers for index 12 and 13 (H264 VLC) / VC1, VPx prediction filter taps (SWREG36)	32	RW	0000_0000h
94h	Reference picture numbers for index 14 and 15 (H264 VLC) / VPx prediction filter taps (SWREG37)	32	RW	0000_0000h
98h	Reference picture long term flags (H264 VLC) / VPx prediction filter taps (SWREG38)	32	RW	0000_0000h
9Ch	Reference picture valid flags (H264 VLC) / VPx prediction filter taps (SWREG39)	32	RW	0000_0000h
A8h	bi_dir initial ref pic list register (0-2) / VP6 prediction filter taps / Progressive JPEG Cb ACDC coefficient base (SWREG42_H264)	32	RW	0000_0000h
ACh	bi-dir initial ref pic list register (3-5) / VP6 prediction filter taps / Progressive JPEG Cr ACDC coefficient base (SWREG43_H264)	32	RW	0000_0000h
B0h	bi-dir initial ref pic list register (6-8) / VP6 prediction filter taps (SWREG44_H264)	32	RW	0000_0000h
B4h	bi-dir initial ref pic list register (9-11) / VP6 prediction filter taps (SWREG45)	32	RW	0000_0000h
B8h	bi-dir initial ref pic list register (12-14) / VP7,VP8 quantization values (SWREG46)	32	RW	0000_0000h
BCh	bi-dir and P fwd initial ref pic list register (15 and P 0-3) / VP7,VP8 quantization values (SWREG47)	32	RW	0000_0000h

14.1.5.2.2 Decoder control register 1 (picture parameters) (SWREG4)

14.1.5.2.2.1 Offset

Register	Offset
SWREG4	10h

14.1.5.2.2.2 Diagram



14.1.5.2.2.3 Fields

Field	Function
31-23 SW_PIC_MB_WIDTH	Picture width in macroblocks = ((width in pixels + 15) / 16)
22-19 SW_MB_WIDTH_OFF	The amount of meaningful horizontal pixels in last MB (width offset) 0 if exactly 16 pixels multiple picture and all the horizontal pixels in last MB are meaningful
18-11	Picture height in macroblocks = ((height in pixels + 15) / 16). Picture height is informed as size of the (progressive) frame also for single field (of interlaced content) is being decoded

Table continues on the next page...

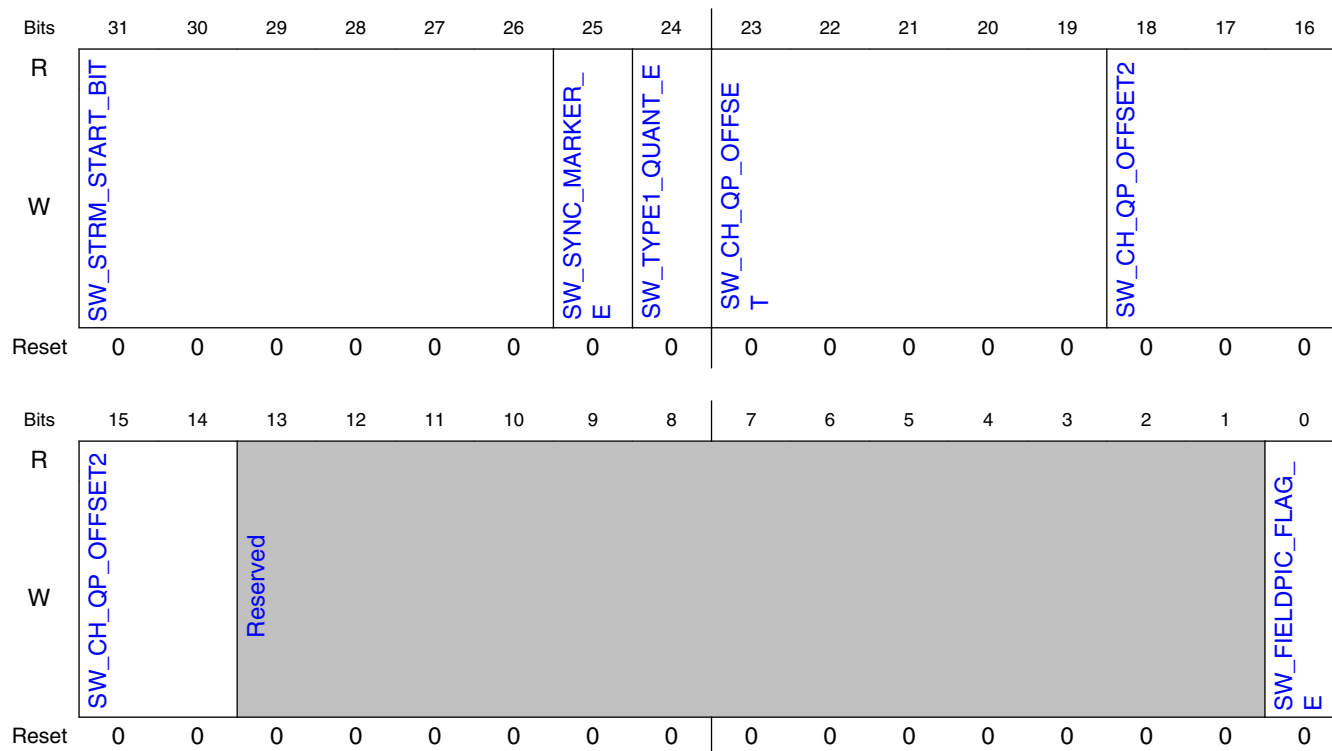
Field	Function
SW_PIC_MB_HEIGHT_P	
10-7 SW_MB_HEIGHT_OFF	The amount of meaningful vertical pixels in last MB (height offset 0 if exactly 16 pixels multiple picture and all the vertical pixels in last MB are meaningful)
6 SW_ALT_SCAN_E	indicates alternative vertical scan method used for interlaced frames
5 —	Reserved.
4-0 SW_REF_FRAMES	H.264: num_ref_frames, maximum number of short and long term reference frames in decoded picture buffer VC-1: num_ref semantics

14.1.5.2.3 Decoder control register 2 (stream decoding table selects) (SWREG5)

14.1.5.2.3.1 Offset

Register	Offset
SWREG5	14h

14.1.5.2.3.2 Diagram



14.1.5.2.3.3 Fields

Field	Function
31-26 SW_STRM_START_BIT	Exact bit of stream start word where decoding can be started (associates with sw_rlc_vlc_base)
25 SW_SYNC_MARKER_E	Sync markers enable For progressive JPEG, this indicates that there are restart markers in the stream after restart interval steps. 0b - synch markers are not used 1b - synch markers are used.
24 SW_TYPE1_QUANT_E	MPEG4: Type 1 quantization enable '0' = type 2 inverse Q method '1' = type 1 inverse Q method (Q-tables used) H264 (h264_high config): scaling matrix enable '0' = normal transform '1' = use scaling matrix for transform (read from external memory)
23-19 SW_CH_QP_OFFSET	Chroma Qp filter offset. (For H.264 this offset concerns Cb only)
18-14	Chroma Qp filter offset for cr type

Table continues on the next page...

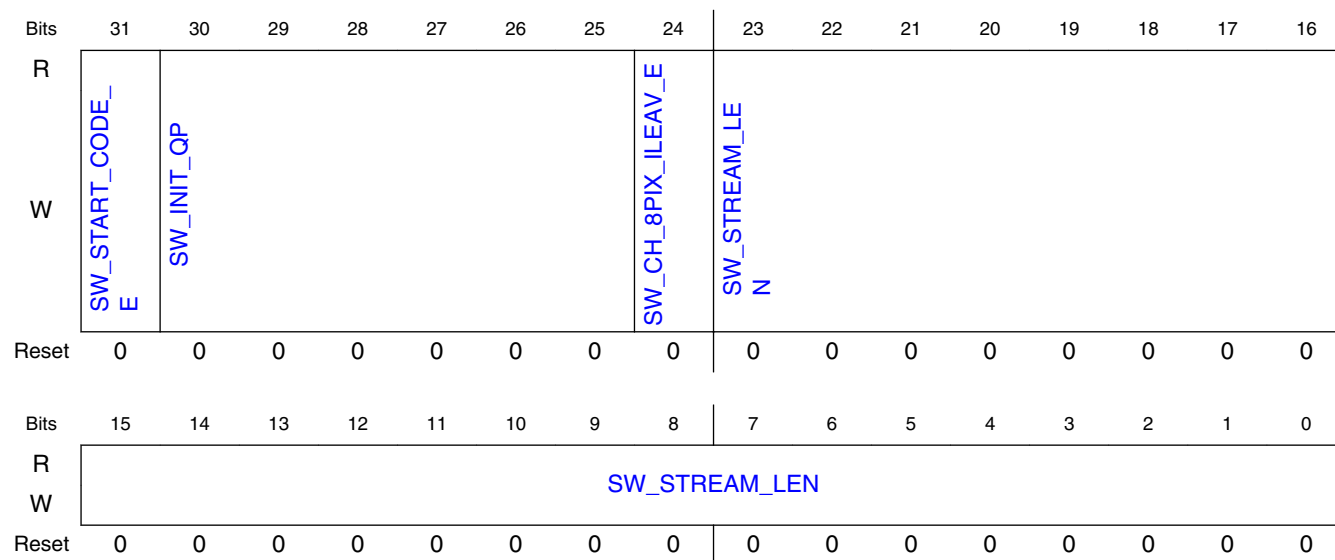
Field	Function
SW_CH_QP_OFFSET2	
13-1 —	Reserved.
0 SW_FIELDPIC_FLAG_E	Flag for streamd that field_pic_flag exists in stream

14.1.5.2.4 Decoder control register 3 (stream buffer information) (SWREG6)

14.1.5.2.4.1 Offset

Register	Offset
SWREG6	18h

14.1.5.2.4.2 Diagram



14.1.5.2.4.3 Fields

Field	Function
31	Bit for indicating stream start code existence: 0b - stream doesn't contain start codes

Table continues on the next page...

VPU G1 Memory Map/Register Definition

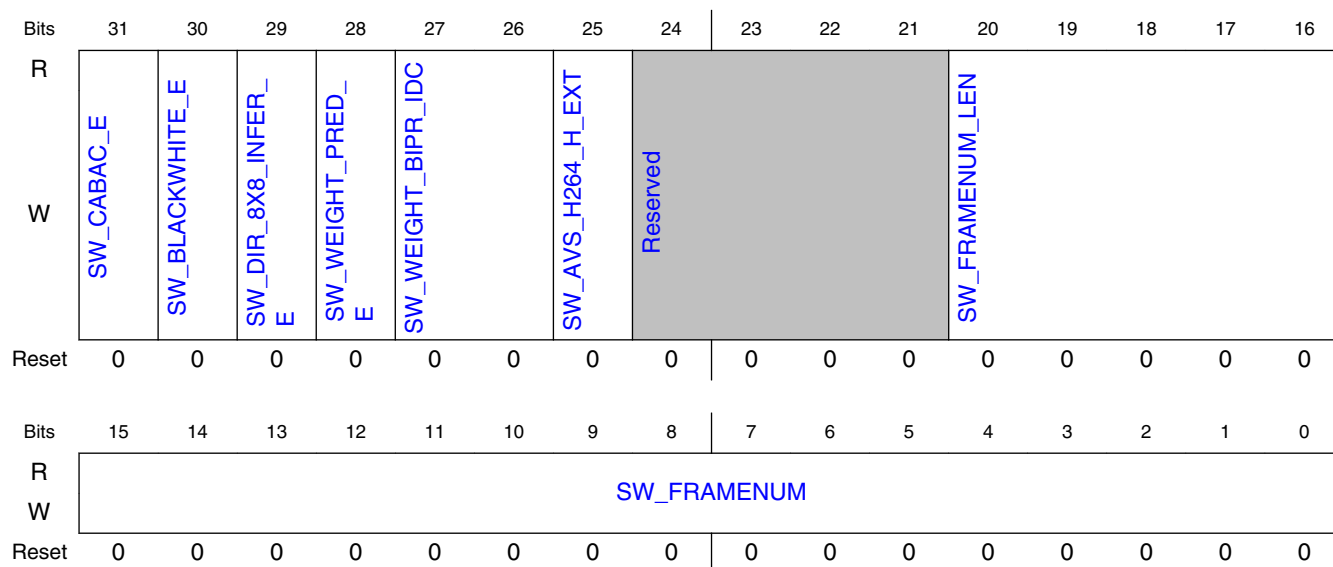
Field	Function
SW_START_C ODE_E	1b - stream contains start codes
30-25 SW_INIT_QP	Initial value for quantization parameter (picture quantizer).
24 SW_CH_8PIX_I LEAV_E	Enable for additional chrominance data format writing where decoder writes chrominance in group of 8 pixels of Cb and then corresponding 8 pixels of Cr. Data is written to sw_dec_ch8pix_base. Cannot be used if tiled mode is enabled
23-0 SW_STREAM_L EN	Amount of stream data bytes in input buffer. If the given buffer size is not enough for finishing the picture the corresponding interrupt is given and new stream buffer base address and stream buffer size information should be given (associates with sw_rlc_vlc_base). For VC-1/VP6 the buffer must include data for one picture/slice of the picture For H264/MPEG4/H263/MPEG2/MPEG1 the buffer must include at least data for one slice/VP of the picture For JPEG the buffer size must be a multiple of 256 bytes or the amount of data for one picture.

14.1.5.2.5 Decoder control register 4 (H264, VC-1, VP6 and progressive JPEG control) (SWREG7)

14.1.5.2.5.1 Offset

Register	Offset
SWREG7	1Ch

14.1.5.2.5.2 Diagram



14.1.5.2.5.3 Fields

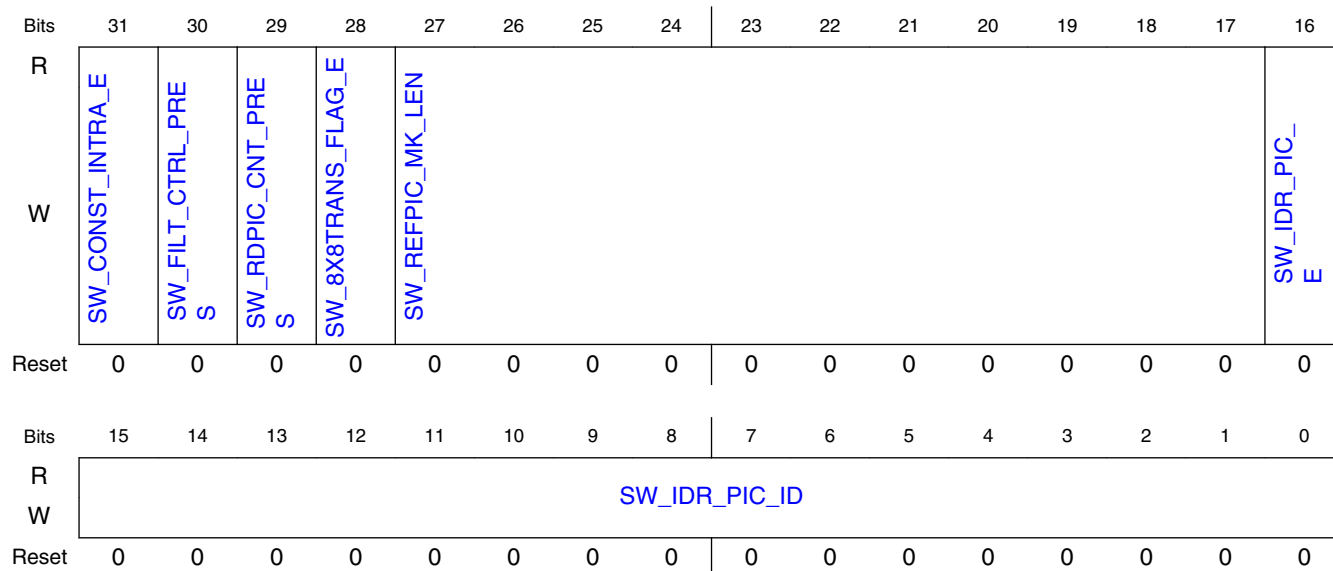
Field	Function
31 SW_CABAC_E	CABAC enable
30 SW_BLACKWHI TE_E	0b - 4:2:0 sampling format 1b - 4:0:0 sampling format (H264 monochrome)
29 SW_DIR_8X8_I NFER_E	Specifies the method to use to derive luma motion vectors in B_skip, B_Direct_16x16 and B_direct_8x8_inference_flag (see direct_8x8_inference flag)
28 SW_WEIGHT_P RED_E	Weighted prediction enable for P slices
27-26 SW_WEIGHT_B IPR_IDC	weighted prediction specification for B slices: 00b - default weighted prediction is applied to B slices 01b - explicit weighted prediction shall be applied to B slices 10b - implicit weighted prediction shall be applied to B slices
25 SW_AVS_H264 _H_EXT	Resolution extension to support 4k resolution for AVS/H264. Used as MSB of sw_pic_mb_height
24-21 —	Reserved.
20-16 SW_FRAMEU M_LEN	H.264: Bit length of frame_num in data stream RV: frame size length. Informs how many bits in stream are used for frame size (HW discards these bits)
15-0 SW_FRAMEU M	current frame_num, used to identify short-term reference frames. Used in reference picture reordering

14.1.5.2.6 Decoder control register 5 (H264, VC-1, VP6, Progressive JPEG and RV control) (SWREG8)

14.1.5.2.6.1 Offset

Register	Offset
SWREG8	20h

14.1.5.2.6.2 Diagram



14.1.5.2.6.3 Fields

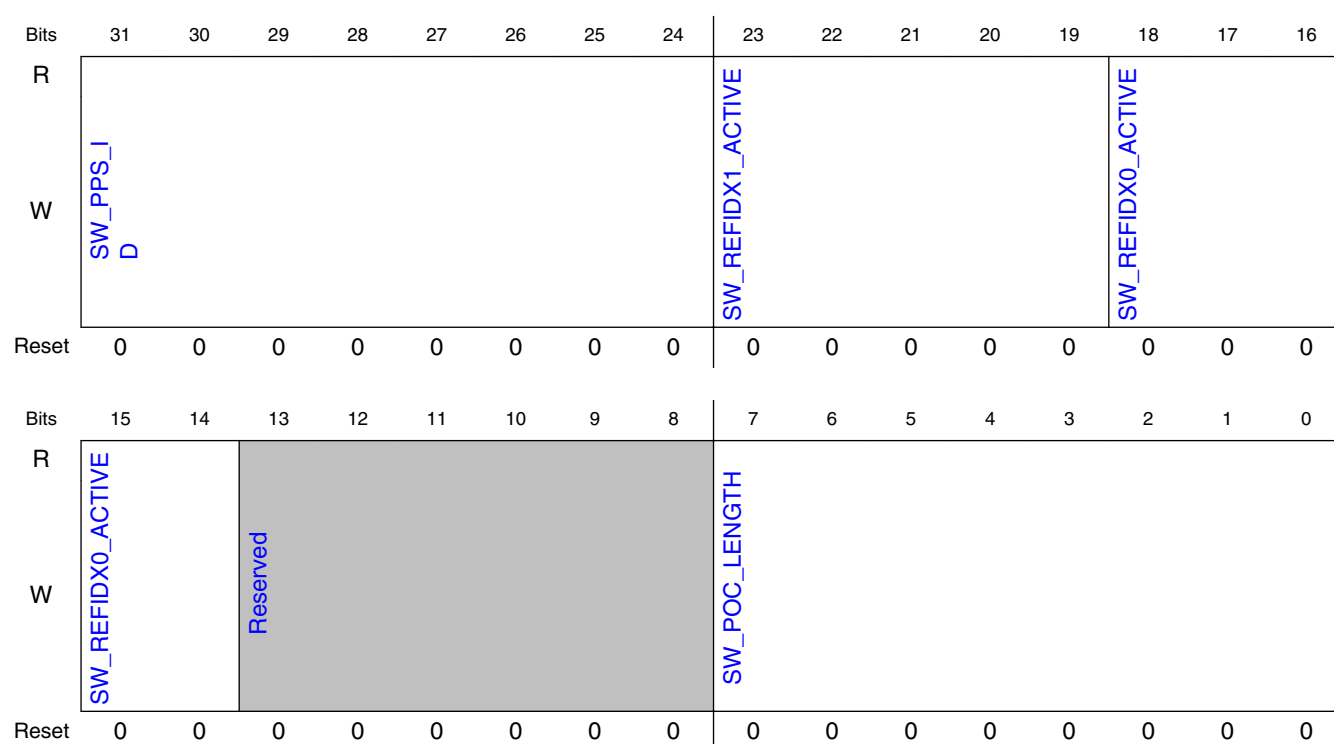
Field	Function
31 SW_CONST_INTRA_E	constrained_intra_pred_flag equal to 1 specifies that intra prediction uses only neighbouring intra macroblocks in prediction. When equal to 0 also neighbouring inter macroblocks are used in intra prediction process.
30 SW_FILT_CTRL_PRE	deblocking_filter_control_present_flag indicates whether extra variables controlling characteristics of the deblocking filter are present in the slice header.
29 SW_RDPIC_CNT_PRE	redundant_pic_cnt_present_flag specifies whether redundant_pic_cnt syntax elements are present in the slice header.
28 SW_8X8TRANS_FLAG_E	8x8 transform flag enable for stream decoding
27-17 SW_REFPIC_MK_LEN	Length of decoded reference picture marking bits
16 SW_IDR_PIC_E	IDR (instantaneous decoding refresh) picture flag.
15-0 SW_IDR_PIC_ID	idr_pic_id, identifies IDR (instantaneous decoding refresh) picture

14.1.5.2.7 Decoder control register 6 / base address for MB-control (RLC) / VC-1 intensity control 0/ VP6,VP7,VP8 ctrl-stream length/ RV pic slice amount (SWREG9)

14.1.5.2.7.1 Offset

Register	Offset
SWREG9	24h

14.1.5.2.7.2 Diagram



14.1.5.2.7.3 Fields

Field	Function
31-24 SW_PPS_ID	pic_parameter_set_id, identifies the picture parameter set that is referred to in the slice header.
23-19 SW_REFIDX1_ACTIVE	Specifies the maximum reference index that can be used while decoding inter predicted macro blocks.

Table continues on the next page...

VPU G1 Memory Map/Register Definition

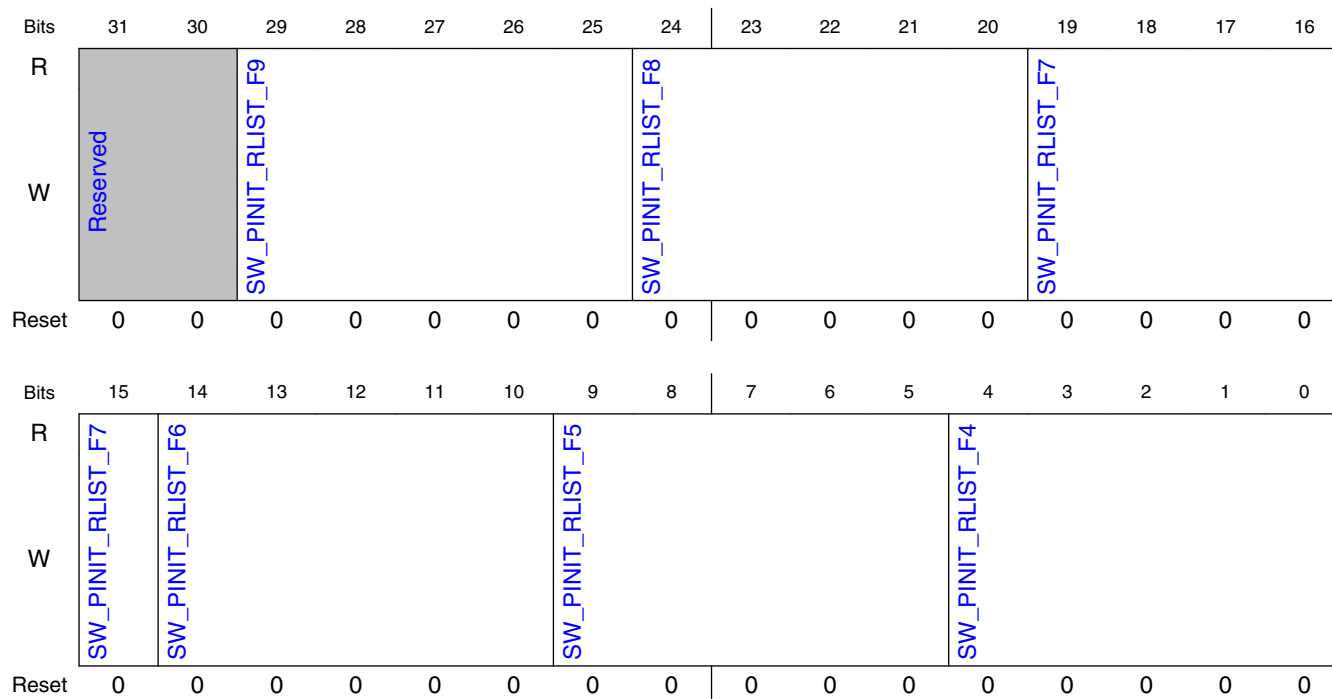
Field	Function
18-14 SW_REFIDX0_ACTIVE	Specifies the maximum reference index that can be used while decoding inter predicted macro blocks. This is same as in previous decoders (width increased with q bit)
13-8 —	Reserved.
7-0 SW_POC_LENGTH	Length of picture order count field in stream

14.1.5.2.8 Base address for differential motion vector base address (RLC-mode) /H264 P initial fwd ref pic list register (4-9)/ VC-1 intensity control 1/ VP7 and VP8 segmentation base register (SWREG10)

14.1.5.2.8.1 Offset

Register	Offset
SWREG10	28h

14.1.5.2.8.2 Diagram



14.1.5.2.8.3 Fields

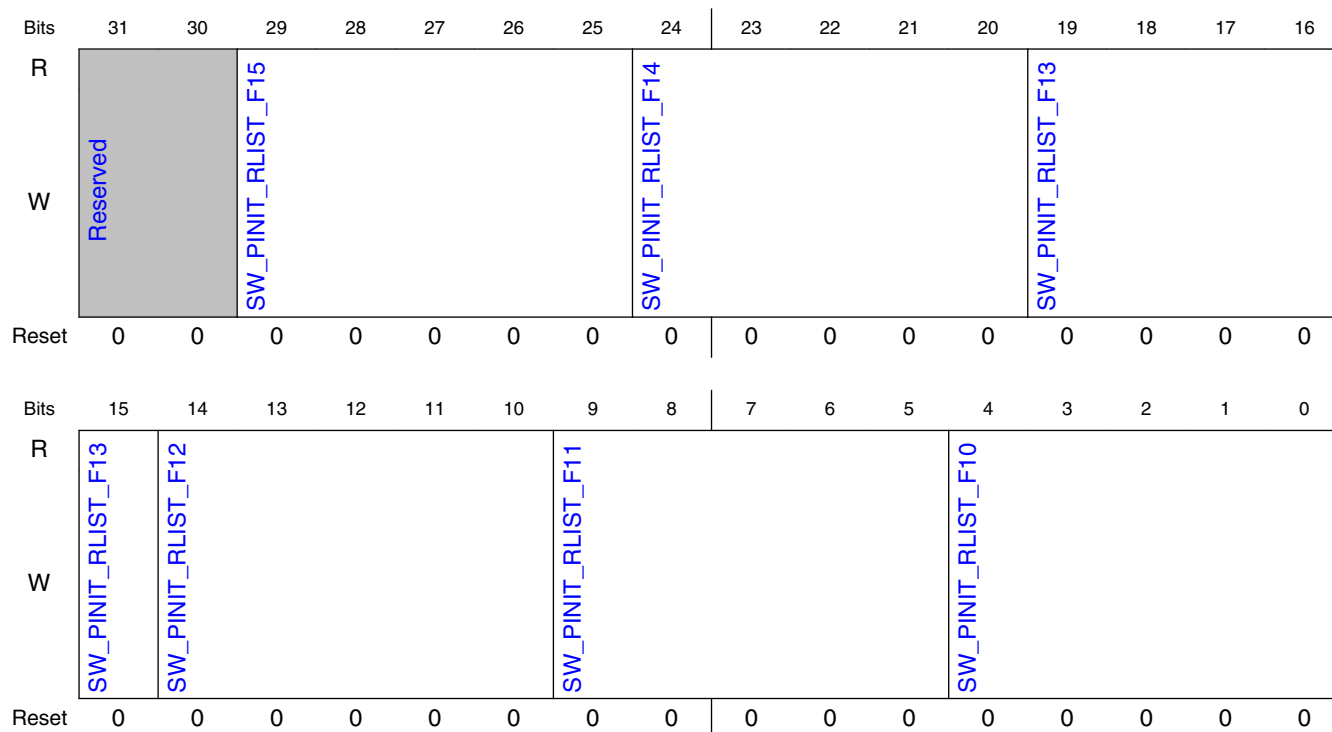
Field	Function
31-30 —	Reserved.
29-25 SW_PINIT_RLI ST_F9	Initial reference picture list for P forward picid 9
24-20 SW_PINIT_RLI ST_F8	Initial reference picture list for P forward picid 8
19-15 SW_PINIT_RLI ST_F7	Initial reference picture list for P forward picid 7
14-10 SW_PINIT_RLI ST_F6	Initial reference picture list for P forward picid 6
9-5 SW_PINIT_RLI ST_F5	Initial reference picture list for P forward picid 5
4-0 SW_PINIT_RLI ST_F4	Initial reference picture list for P forward picid 4

14.1.5.2.9 Decoder control register 7 (VLC) / base address for H.264 intra prediction 4x4 / base address for MPEG-4 DC component (RLC) / H264 P initial fwd ref pic list register (10-15) / VC-1 intensity control 2 (SWREG11)

14.1.5.2.9.1 Offset

Register	Offset
SWREG11	2Ch

14.1.5.2.9.2 Diagram



14.1.5.2.9.3 Fields

Field	Function
31-30 —	Reserved.
29-25 SW_PINIT_RLIST_F15	Initial reference picture list for P forward picid 15
24-20 SW_PINIT_RLIST_F14	Initial reference picture list for P forward picid 14
19-15 SW_PINIT_RLIST_F13	Initial reference picture list for P forward picid 13
14-10 SW_PINIT_RLIST_F12	Initial reference picture list for P forward picid 12
9-5 SW_PINIT_RLIST_F11	Initial reference picture list for P forward picid 11
4-0	Initial reference picture list for P forward picid 10

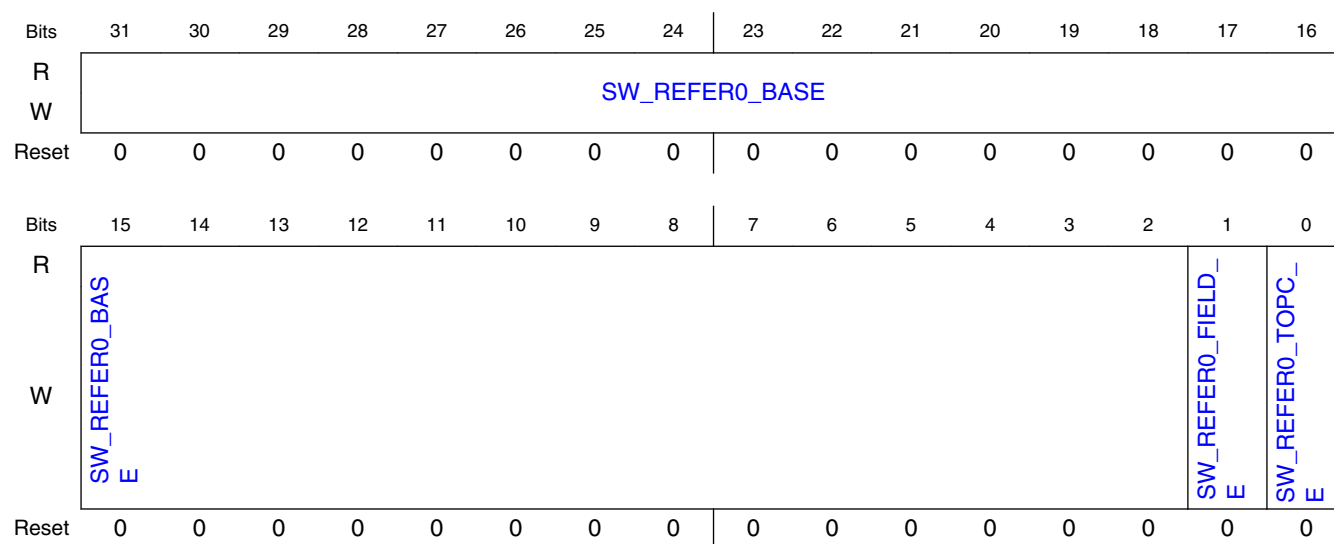
Field	Function
SW_PINIT_RLI ST_F10	

14.1.5.2.10 Base address for reference picture index 0 / base address for JPEG decoder output chrominance picture (SWREG14)

14.1.5.2.10.1 Offset

Register	Offset
SWREG14	38h

14.1.5.2.10.2 Diagram



14.1.5.2.10.3 Fields

Field	Function
31-2 SW_REFERO_B ASE	Base address for reference picture index 0. See picture index definition from toplevel_sp
1 SW_REFERO_F IELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0	Which field of reference picture is closer to current picture:

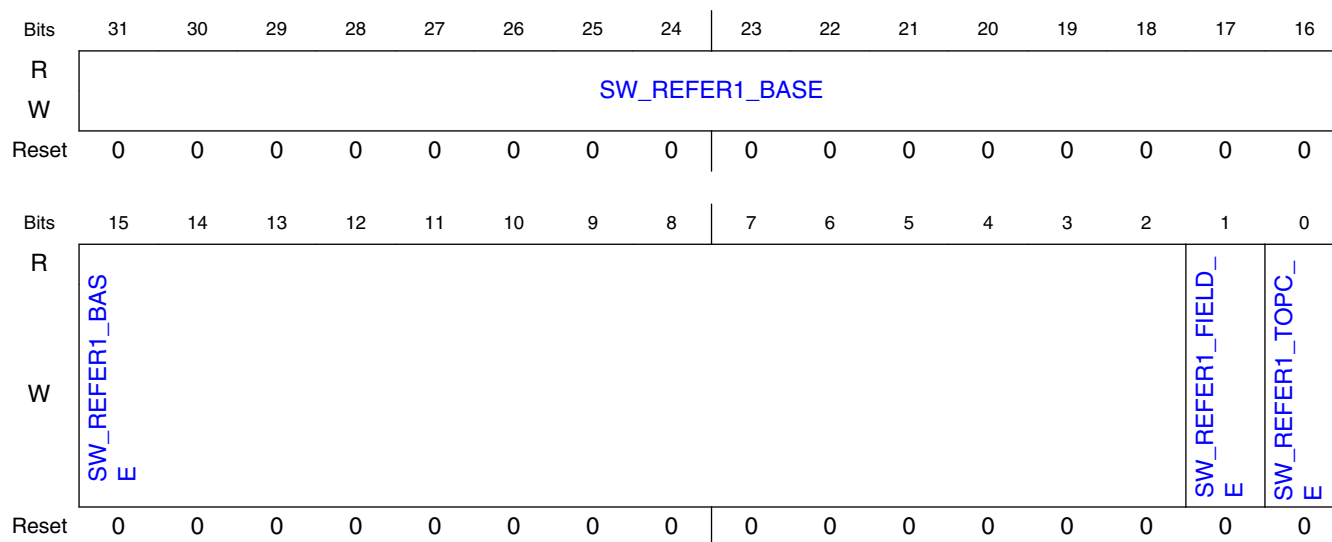
Field	Function
SW_REFER0_T OPC_E	0b - Bottom field is closer to current picture 1b - Top field is closer to current picture

14.1.5.2.11 Base address for reference picture index 1 / JPEG control (SWREG15)

14.1.5.2.11.1 Offset

Register	Offset
SWREG15	3Ch

14.1.5.2.11.2 Diagram



14.1.5.2.11.3 Fields

Field	Function
31-2 SW_REFER1_B ASE	Base address for reference picture index 1. See picture index definition from toplevel_sp. For VP8 this base address is used as Chrominance base address for reference picture 0 (if vp8 stride configuration is enabled)
1 SW_REFER1_F IELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0	Which field of reference picture is closer to current picture:

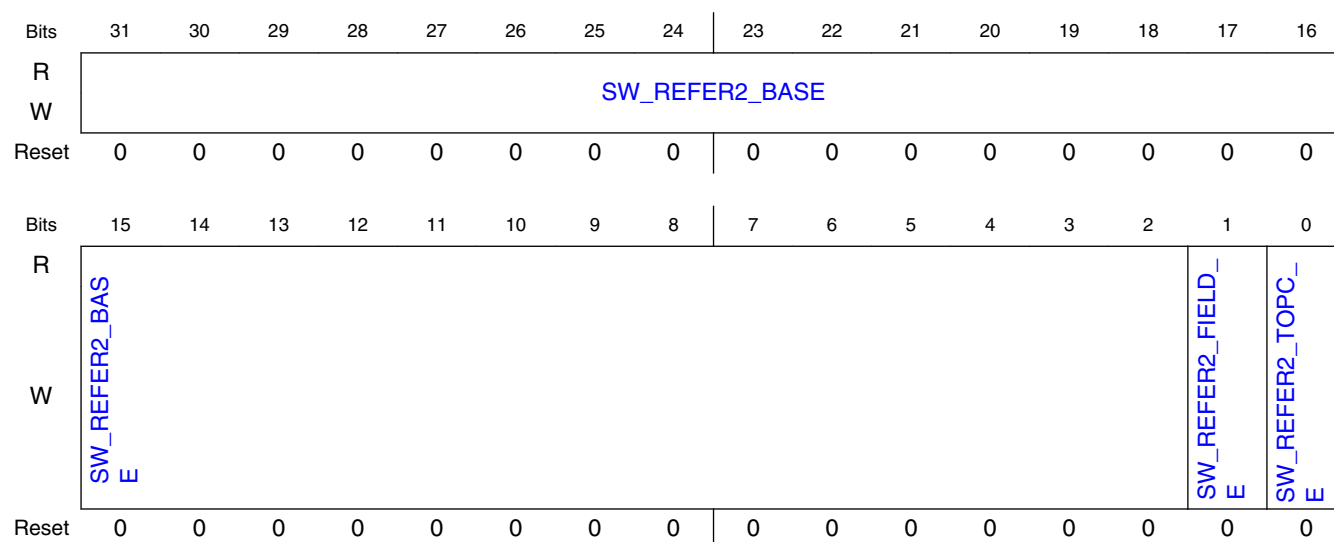
Field	Function
SW_REFER1_T OPC_E	0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.12 Base address for reference picture index 2 / List of VLC code lengths in first JPEG AC table (SWREG16)

14.1.5.2.12.1 Offset

Register	Offset
SWREG16	40h

14.1.5.2.12.2 Diagram



14.1.5.2.12.3 Fields

Field	Function
31-2 SW_REFER2_B ASE	Base address for reference picture index 2. See picture index definition from toplevel_sp. For VP8 video this base address is used as Golden reference chrominance base address (if vp8 stride configuration is enabled)
1 SW_REFER2_F IELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0	Which field of reference picture is closer to current picture:

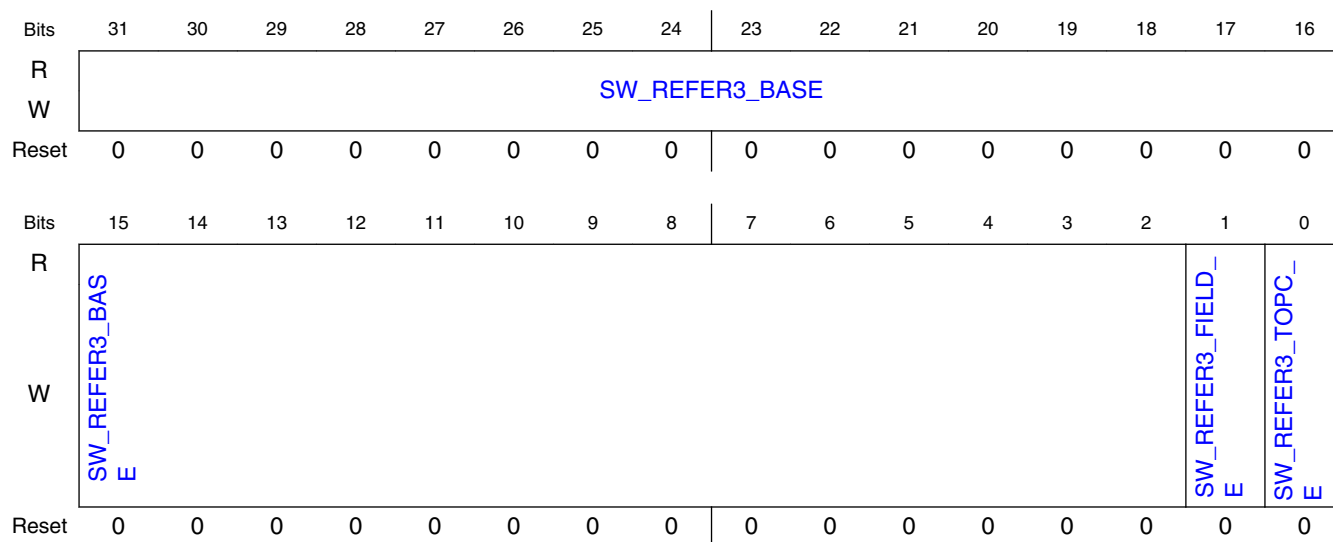
Field	Function
SW_REFER2_T OPC_E	0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.13 Base address for reference picture index 3 / List of VLC code lengths in first JPEG AC table (SWREG17)

14.1.5.2.13.1 Offset

Register	Offset
SWREG17	44h

14.1.5.2.13.2 Diagram



14.1.5.2.13.3 Fields

Field	Function
31-2 SW_REFER3_B ASE	Base address for reference picture index 3. See picture index definition from toplevel_sp. For VP8 video this base address is used as Alternate reference chrominance base address (if vp8 stride configuration is enabled)
1 SW_REFER3_F IELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0	Which field of reference picture is closer to current picture:

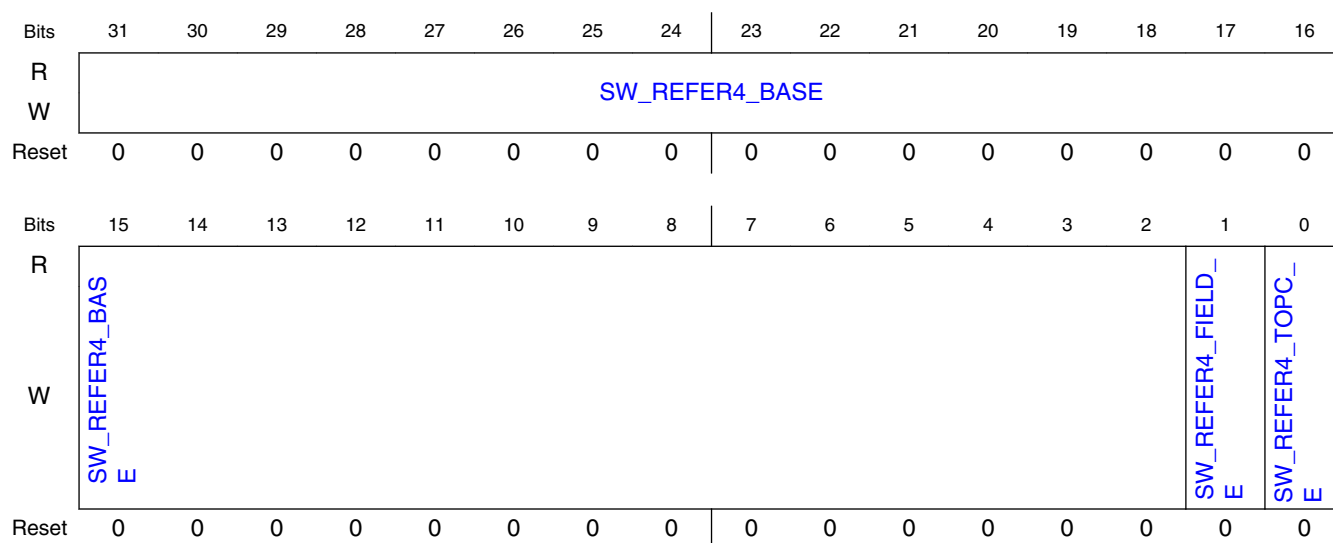
Field	Function
SW_REFER3_T OPC_E	0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.14 Base address for reference picture index 4 / VC1 control / MPEG4 MVD control/ List of VLC code lengths in first JPEG AC table / VC-1 intensity control 4 / VP6/VP7, VP8 Golden refer picture base (SWREG18)

14.1.5.2.14.1 Offset

Register	Offset
SWREG18	48h

14.1.5.2.14.2 Diagram



14.1.5.2.14.3 Fields

Field	Function
31-2 SW_REFER4_B ASE	H264: Base address for reference picture index 4 VP6/VP7/VP8: Base address for Golden reference picture (corresponds picid 4)
1	Refer picture consist of single fields or frame: 0b - reference picture consists of frame

Table continues on the next page...

VPU G1 Memory Map/Register Definition

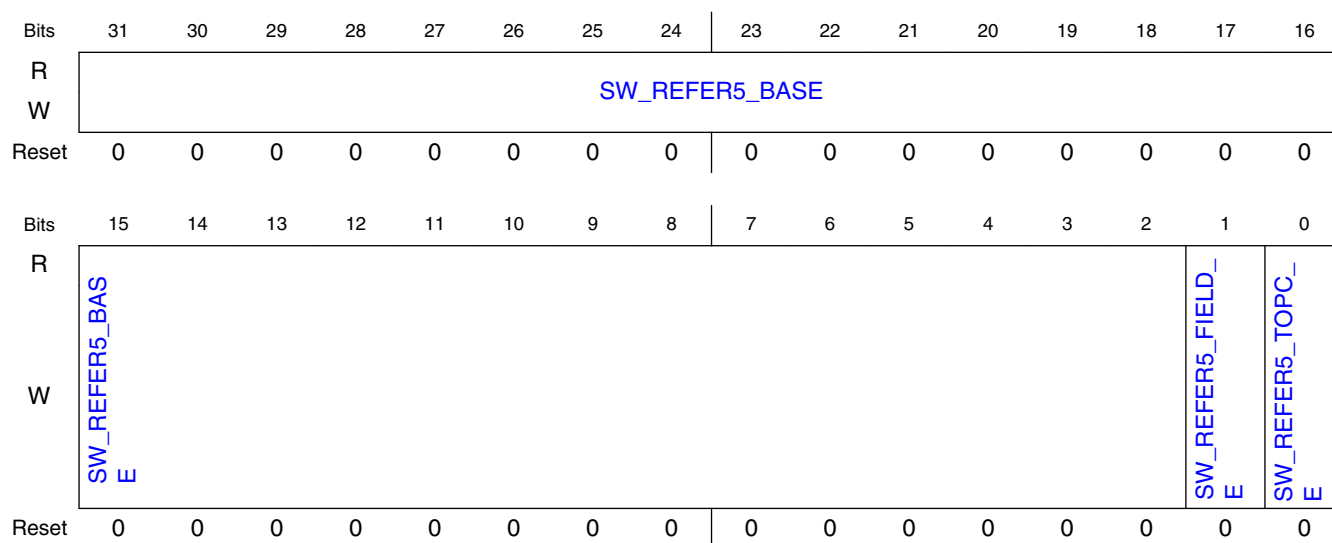
Field	Function
SW_REFER4_F IELD_E	1b - reference picture consists of fields
0 SW_REFER4_T OPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.15 Base address for reference picture index 5 / MPEG4 TRB/TRD delta 0 / VC-1 intensity control 3 List of VLC code lengths in first/second JPEG AC table / VP6/VP7 scan maps (SWREG19)

14.1.5.2.15.1 Offset

Register	Offset
SWREG19	4Ch

14.1.5.2.15.2 Diagram



14.1.5.2.15.3 Fields

Field	Function
31-2 SW_REFER5_B ASE	H.264: Base address for reference picture index 5 VP8: Base address for alternate reference picture (corresponds picid 5)

Table continues on the next page...

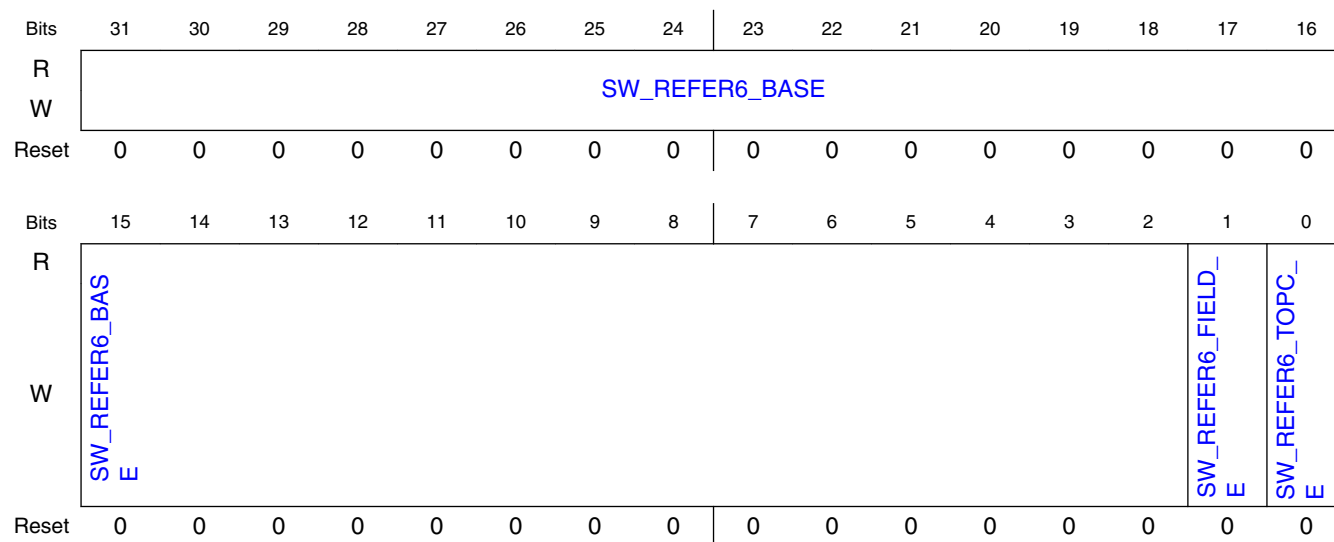
Field	Function
1 SW_REFER5_F IELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER5_T OPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.16 Base address for reference picture index 6 // MPEG4 TRB/TRD delta -1 / List of VLC code lengths in second JPEG AC table / VP6/VP7 scan maps (SWREG20)

14.1.5.2.16.1 Offset

Register	Offset
SWREG20	50h

14.1.5.2.16.2 Diagram



14.1.5.2.16.3 Fields

Field	Function
31-2	Base address for reference picture index 6. For VP8 video this base address is used as decoder output chrominance base address (if vp8 stride configuration is enabled)

Table continues on the next page...

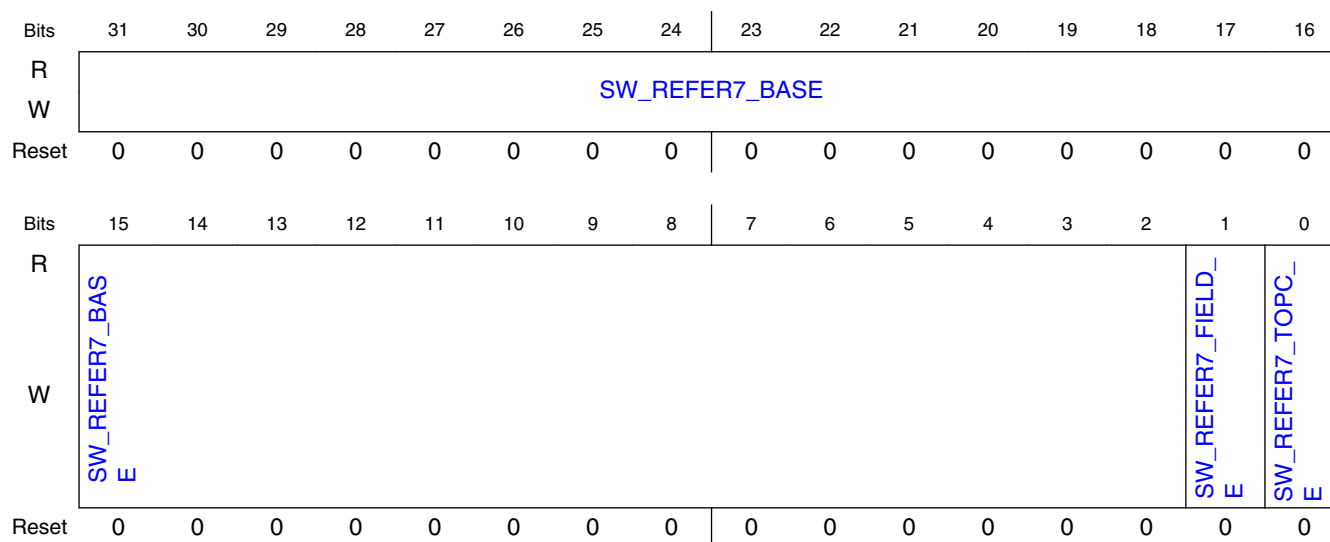
Field	Function
SW_REFER6_BASE	
1 SW_REFER6_FIELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER6_TOPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.17 Base address for reference picture index 7 / MPEG4 TRB/TRD delta 1 / List of VLC code lengths in second JPEG AC table / VP6/VP7 scan maps (SWREG21)

14.1.5.2.17.1 Offset

Register	Offset
SWREG21	54h

14.1.5.2.17.2 Diagram



14.1.5.2.17.3 Fields

Field	Function
31-2	Base address for reference picture index 7

Table continues on the next page...

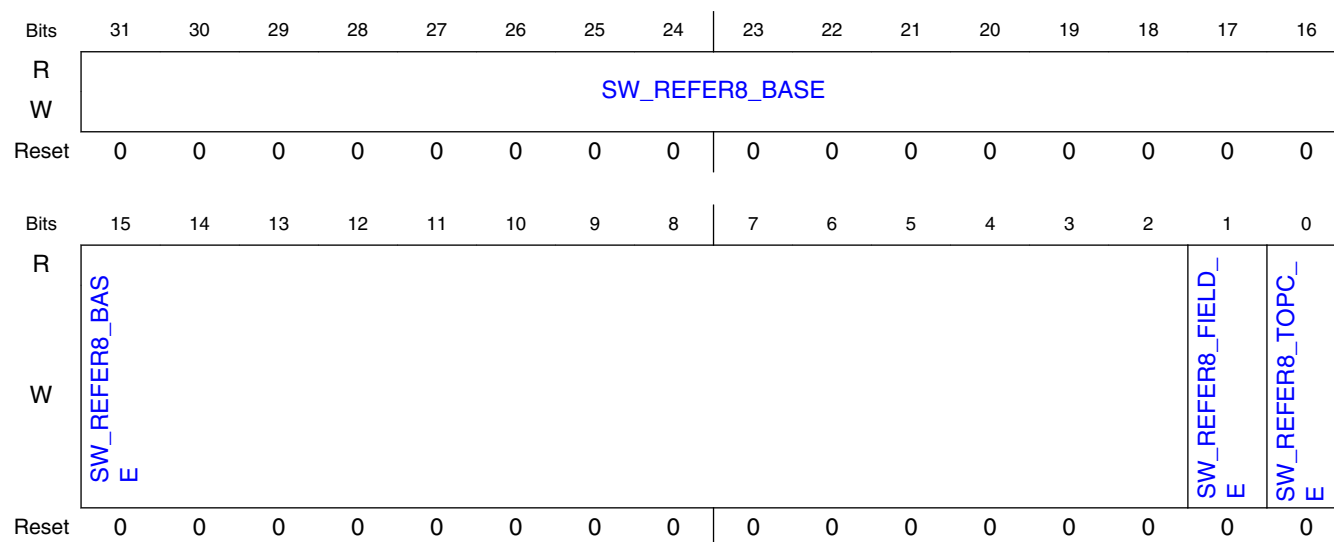
Field	Function
SW_REFER7_B ASE	
1 SW_REFER7_F IELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER7_T OPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.18 Base address for reference picture index 8 / List of VLC code lengths in second JPEG AC table / VP6 scan maps / VP7,VP8 DCT stream 1 base (SWREG22)

14.1.5.2.18.1 Offset

Register	Offset
SWREG22	58h

14.1.5.2.18.2 Diagram



14.1.5.2.18.3 Fields

Field	Function
31-2	Base address for reference picture index 8

Table continues on the next page...

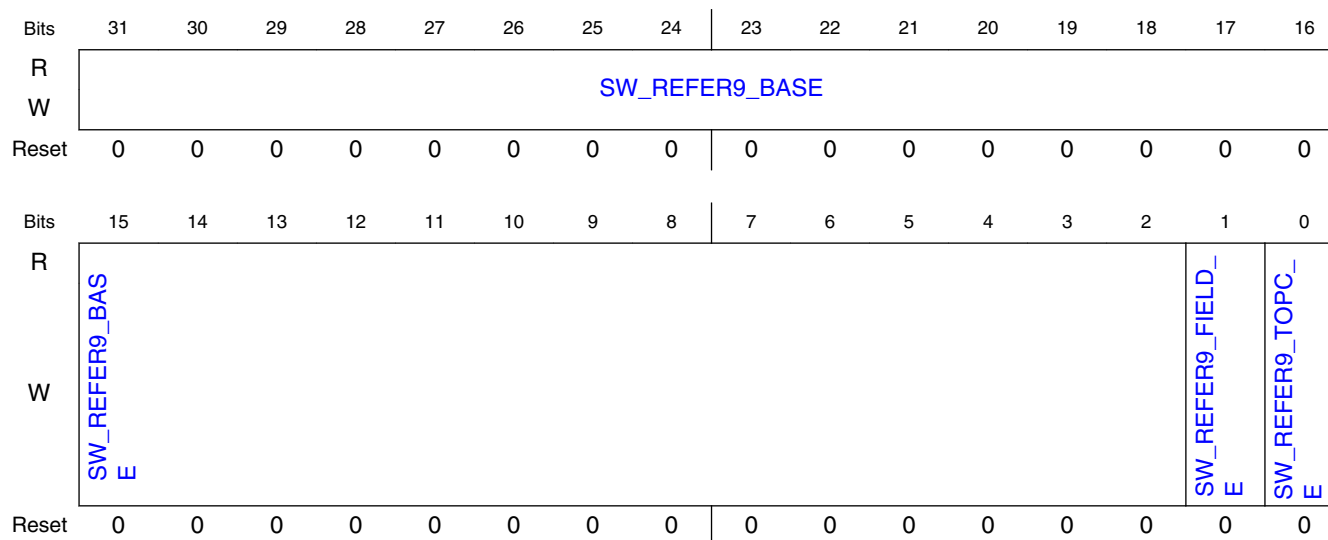
Field	Function
SW_REFER8_B ASE	
1 SW_REFER8_F IELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER8_T OPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.19 Base address for reference picture index 9 / List of VLC code lengths in first JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 2 base (SWREG23)

14.1.5.2.19.1 Offset

Register	Offset
SWREG23	5Ch

14.1.5.2.19.2 Diagram



14.1.5.2.19.3 Fields

Field	Function
31-2	Base address for reference picture index 9

Table continues on the next page...

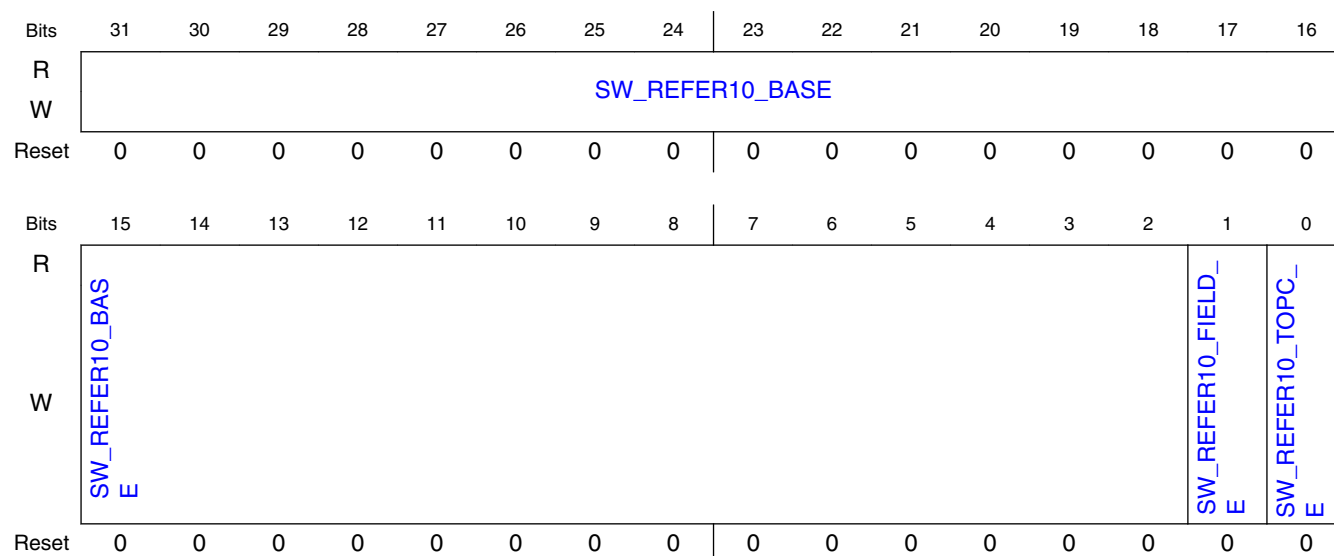
Field	Function
SW_REFER9_B ASE	
1 SW_REFER9_F IELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER9_T OPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.20 Base address for reference picture index 10 / List of VLC code lengths in first JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 3 base (SWREG24)

14.1.5.2.20.1 Offset

Register	Offset
SWREG24	60h

14.1.5.2.20.2 Diagram



14.1.5.2.20.3 Fields

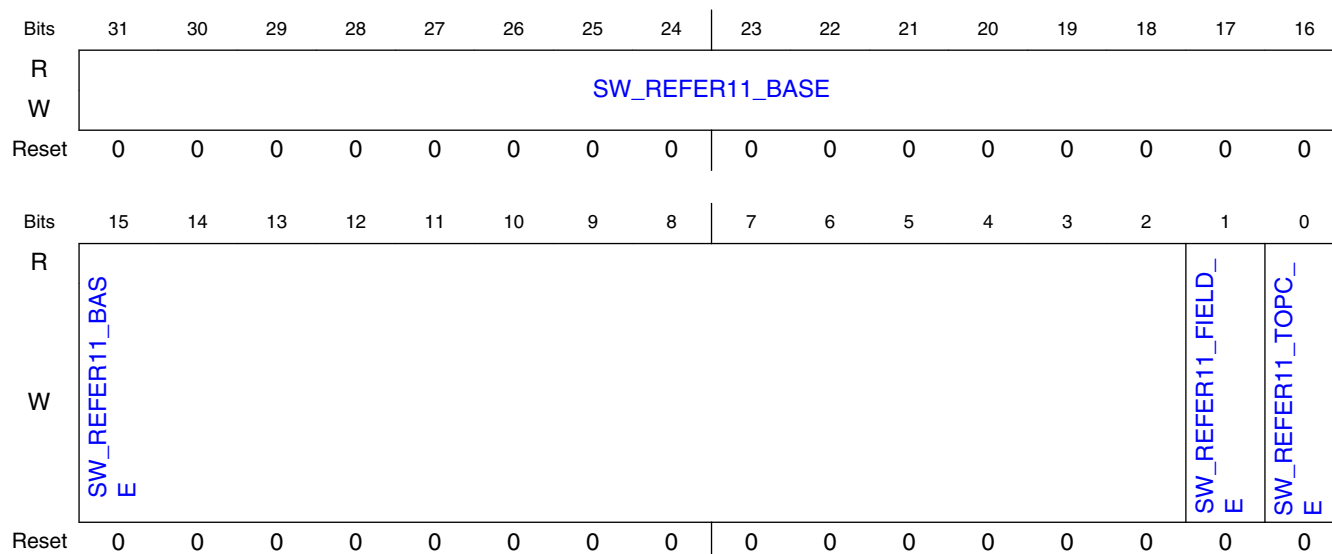
Field	Function
31-2 SW_REFER10_ BASE	Base address for reference picture index 10
1 SW_REFER10_ FIELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER10_ TOPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.21 Base address for reference picture index 11 / List of VLC code lengths in second JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 4 base (SWREG25)

14.1.5.2.21.1 Offset

Register	Offset
SWREG25	64h

14.1.5.2.21.2 Diagram



14.1.5.2.21.3 Fields

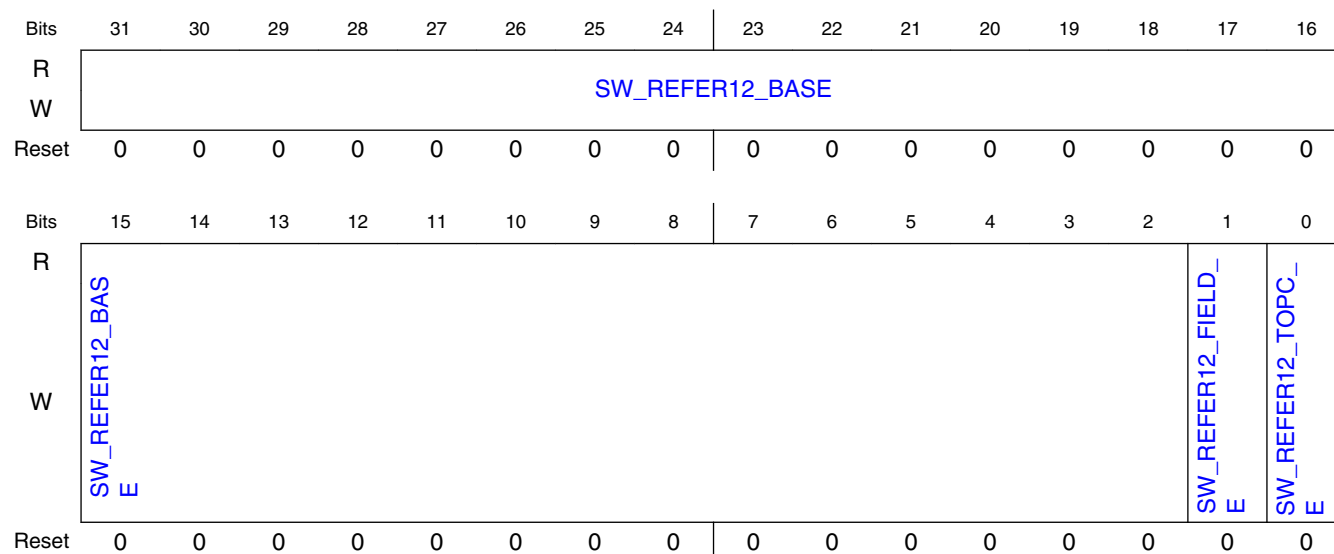
Field	Function
31-2 SW_REFER11_ BASE	Base address for reference picture index 11
1 SW_REFER11_ FIELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER11_ TOPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.22 Base address for reference picture index 12 / List of VLC code lengths in second JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 5 base (SWREG26)

14.1.5.2.22.1 Offset

Register	Offset
SWREG26	68h

14.1.5.2.22.2 Diagram



14.1.5.2.22.3 Fields

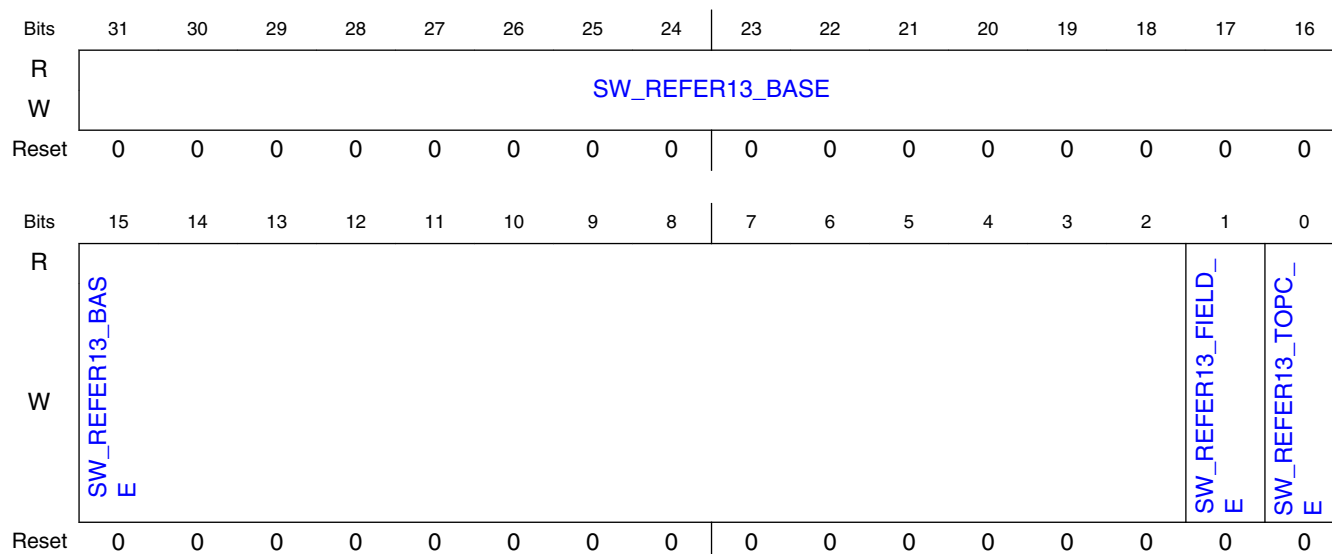
Field	Function
31-2 SW_REFER12_ BASE	Base address for reference picture index 12
1 SW_REFER12_ FIELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER12_ TOPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.23 Base address for reference picture index 13 / VC-1 bitpl mbctrl or VP6,VP7,VP8 ctrl stream base /Progressive JPEG DC table (SWREG27)

14.1.5.2.23.1 Offset

Register	Offset
SWREG27	6Ch

14.1.5.2.23.2 Diagram



14.1.5.2.23.3 Fields

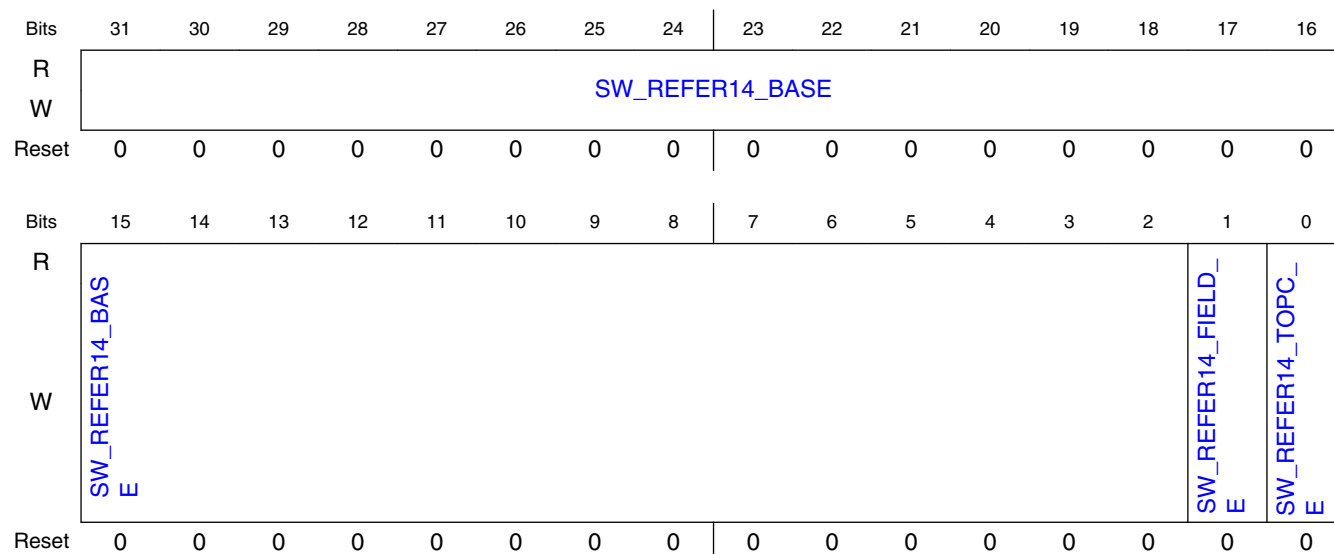
Field	Function
31-2 SW_REFER13_BASE	Base address for reference picture index 13
1 SW_REFER13_FIELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER13_TOPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.24 Base address for reference picture index 14 / VP6 scan maps / Progressive JPEG DC table / VP7,VP8 DCT stream 6 base (SWREG28)

14.1.5.2.24.1 Offset

Register	Offset
SWREG28	70h

14.1.5.2.24.2 Diagram



14.1.5.2.24.3 Fields

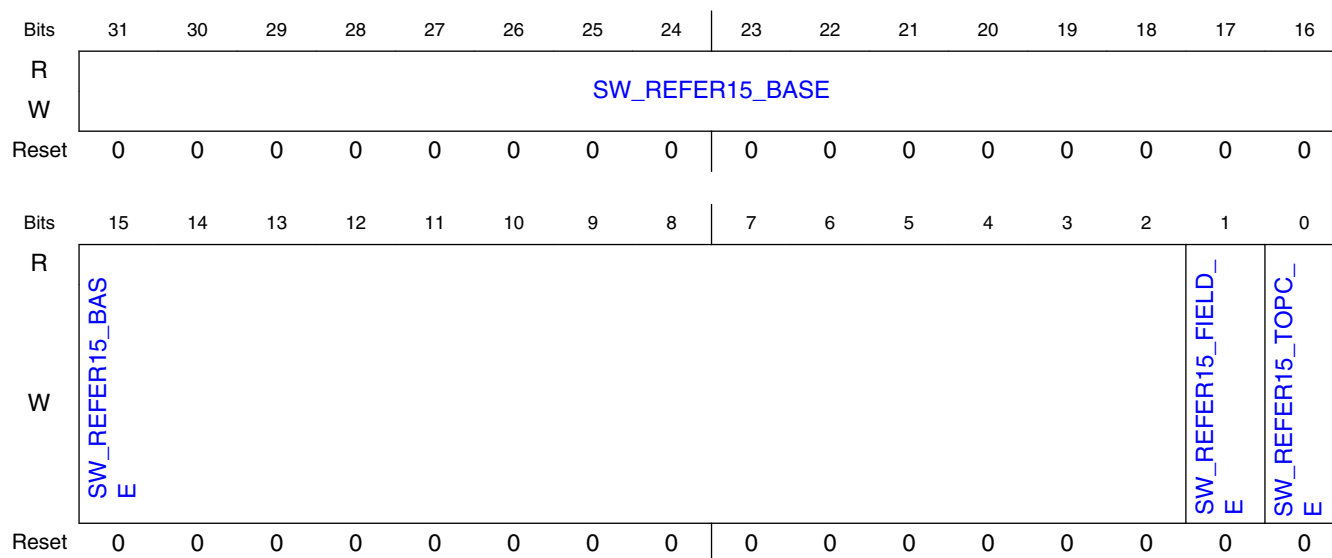
Field	Function
31-2 SW_REFER14_ BASE	Base address for reference picture index 14
1 SW_REFER14_ FIELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER14_ TOPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.25 Base address for reference picture index 15 / VP6 scan maps / VP7,VP8 DCT stream 7 base (SWREG29)

14.1.5.2.25.1 Offset

Register	Offset
SWREG29	74h

14.1.5.2.25.2 Diagram



14.1.5.2.25.3 Fields

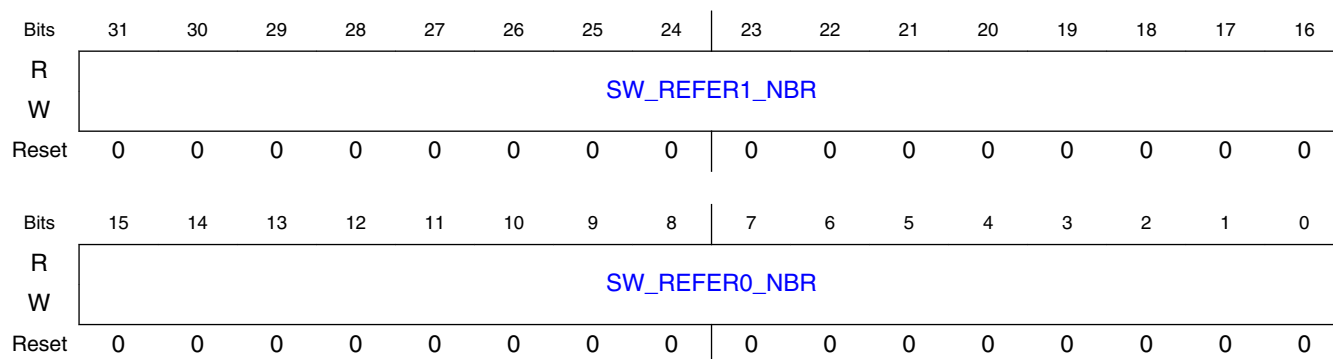
Field	Function
31-2 SW_REFER15_ BASE	Base address for reference picture index 15. For Multi View Coding this base address refers to inter view base address
1 SW_REFER15_ FIELD_E	Refer picture consist of single fields or frame: 0b - reference picture consists of frame 1b - reference picture consists of fields
0 SW_REFER15_ TOPC_E	Which field of reference picture is closer to current picture: 0b - bottom field is closer to current picture 1b - top field is closer to current picture

14.1.5.2.26 Reference picture numbers for index 0 and 1 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter mb level adjusts (SWREG30)

14.1.5.2.26.1 Offset

Register	Offset
SWREG30	78h

14.1.5.2.26.2 Diagram



14.1.5.2.26.3 Fields

Field	Function
31-16 SW_REFER1_N BR	Number for reference picture index 1

Table continues on the next page...

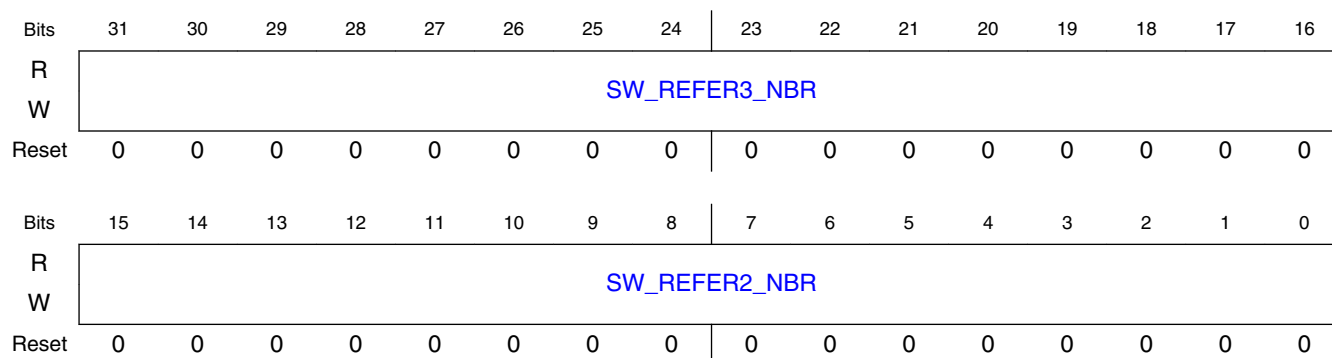
Field	Function
15-0 SW_REFER0_NBR	Number for reference picture index 0

14.1.5.2.27 Reference picture numbers for index 2 and 3 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter ref pic level adjusts (SWREG31)

14.1.5.2.27.1 Offset

Register	Offset
SWREG31	7Ch

14.1.5.2.27.2 Diagram



14.1.5.2.27.3 Fields

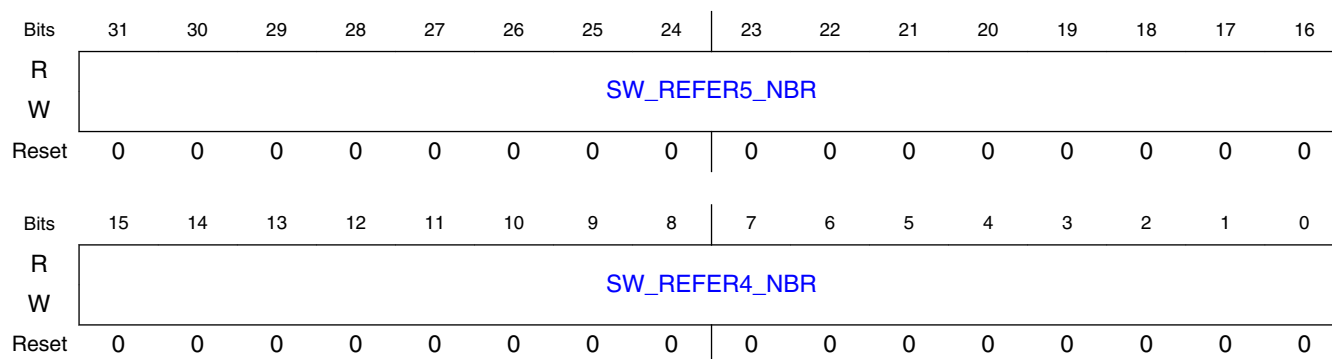
Field	Function
31-16 SW_REFER3_NBR	Number for reference picture index 3
15-0 SW_REFER2_NBR	Number for reference picture index 2

14.1.5.2.28 Reference picture numbers for index 4 and 5 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter levels (SWREG32)

14.1.5.2.28.1 Offset

Register	Offset
SWREG32	80h

14.1.5.2.28.2 Diagram



14.1.5.2.28.3 Fields

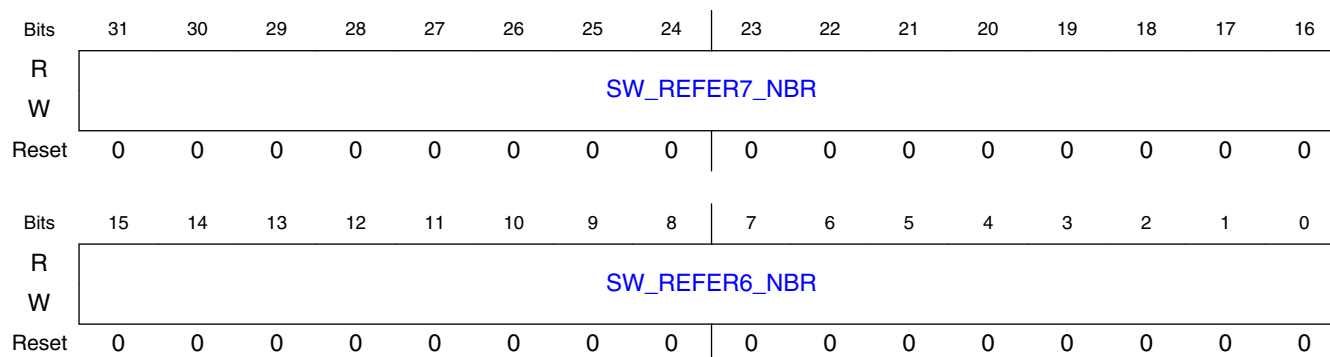
Field	Function
31-16 SW_REFER5_NBR	Number for reference picture index 5
15-0 SW_REFER4_NBR	Number for reference picture index 4

14.1.5.2.29 Reference picture numbers for index 6 and 7 (H264 VLC) / VP6 scan maps / VP7,VP8 quantization values (SWREG33)

14.1.5.2.29.1 Offset

Register	Offset
SWREG33	84h

14.1.5.2.29.2 Diagram



14.1.5.2.29.3 Fields

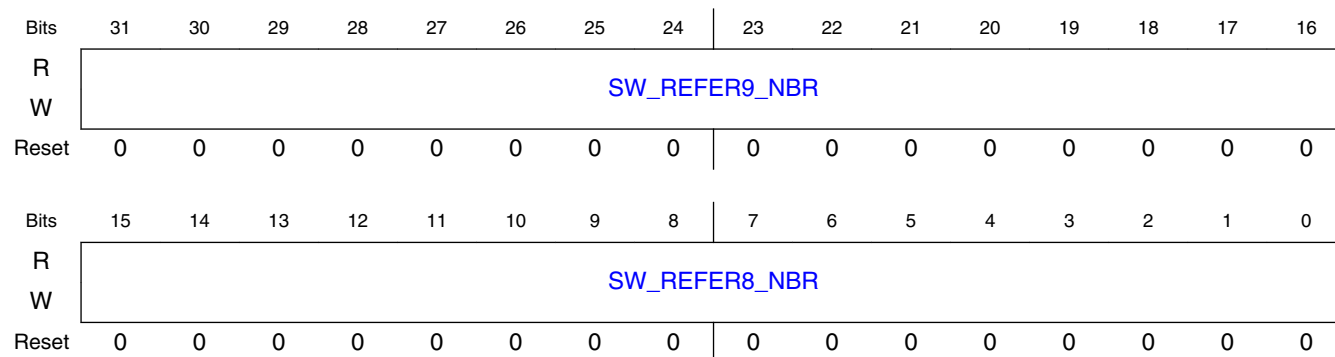
Field	Function
31-16 SW_REFER7_NBR	Number for reference picture index 7
15-0 SW_REFER6_NBR	Number for reference picture index 6

14.1.5.2.30 Reference picture numbers for index 8 and 9 (H264 VLC) / MPEG4, VC1, VPx prediction filter taps (SWREG34)

14.1.5.2.30.1 Offset

Register	Offset
SWREG34	88h

14.1.5.2.30.2 Diagram



14.1.5.2.30.3 Fields

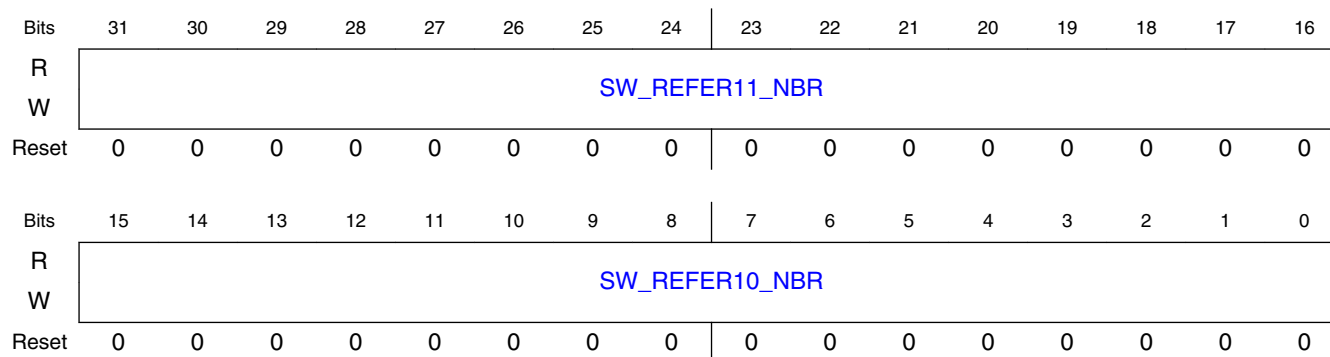
Field	Function
31-16 SW_REFER9_NBR	Number for reference picture index 9
15-0 SW_REFER8_NBR	Number for reference picture index 8

14.1.5.2.31 Reference picture numbers for index 10 and 11 (H264 VLC) / VC1, VPx prediction filter taps (SWREG35)

14.1.5.2.31.1 Offset

Register	Offset
SWREG35	8Ch

14.1.5.2.31.2 Diagram



14.1.5.2.31.3 Fields

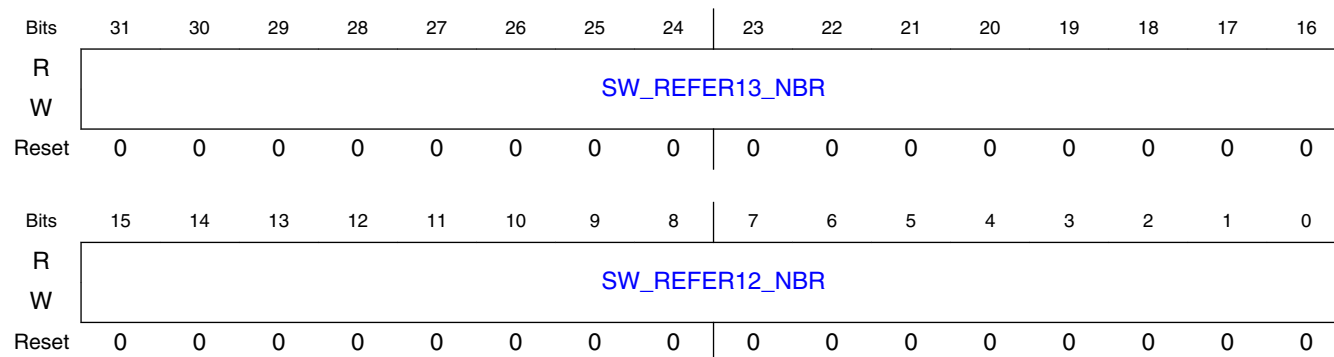
Field	Function
31-16 SW_REFER11_NBR	Number for reference picture index 11
15-0 SW_REFER10_NBR	Number for reference picture index 10

14.1.5.2.32 Reference picture numbers for index 12 and 13 (H264 VLC) / VC1, VPx prediction filter taps (SWREG36)

14.1.5.2.32.1 Offset

Register	Offset
SWREG36	90h

14.1.5.2.32.2 Diagram



14.1.5.2.32.3 Fields

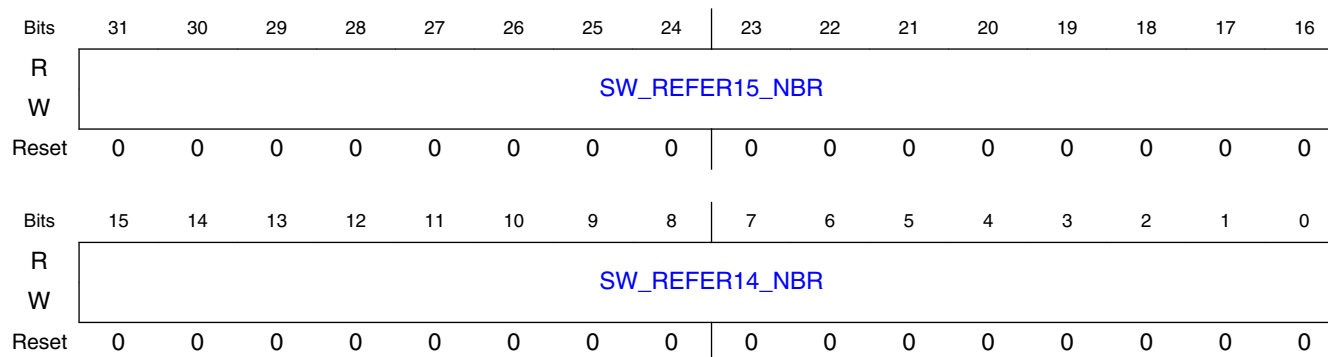
Field	Function
31-16 SW_REFER13_NBR	Number for reference picture index 13
15-0 SW_REFER12_NBR	Number for reference picture index 12

14.1.5.2.33 Reference picture numbers for index 14 and 15 (H264 VLC) / VPx prediction filter taps (SWREG37)

14.1.5.2.33.1 Offset

Register	Offset
SWREG37	94h

14.1.5.2.33.2 Diagram



14.1.5.2.33.3 Fields

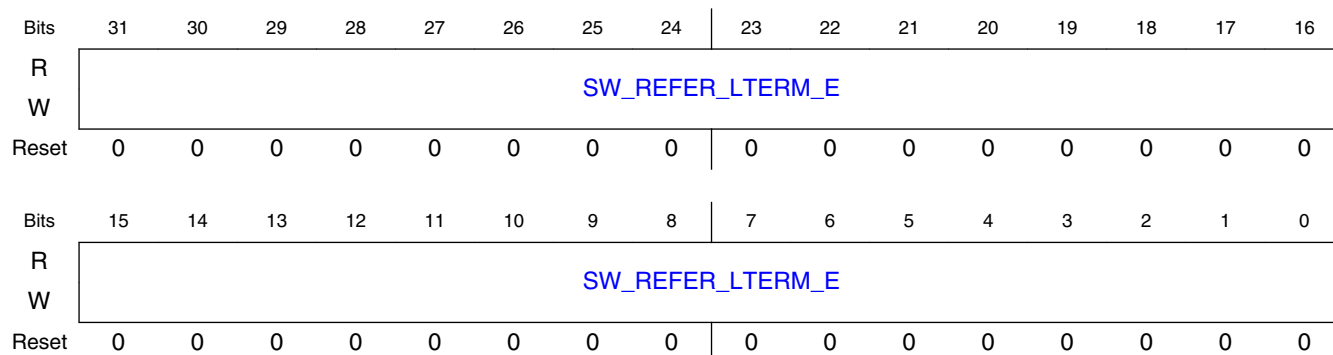
Field	Function
31-16 SW_REFER15_NBR	Number for reference picture index 15
15-0 SW_REFER14_NBR	Number for reference picture index 14

14.1.5.2.34 Reference picture long term flags (H264 VLC) / VPx prediction filter taps (SWREG38)

14.1.5.2.34.1 Offset

Register	Offset
SWREG38	98h

14.1.5.2.34.2 Diagram



14.1.5.2.34.3 Fields

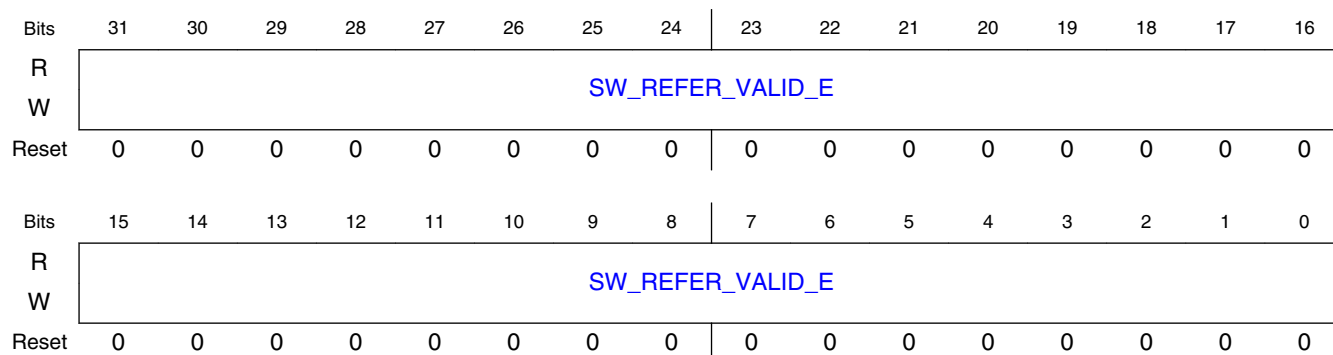
Field	Function
31-0 SW_REFER_LT ERM_E	Long term flag for reference picture index [31:0]. Definition: If frame is being decoded the bits 31:15 are used, Bit 31 for picture index 0, Bit 30 for picture index 1 etc... IF field is being decoded the bits 31:0 are used, Bit 31 for reference picture 0 top field, bit 30 for reference picture 0 bottom field etc...

14.1.5.2.35 Reference picture valid flags (H264 VLC) / VPx prediction filter taps (SWREG39)

14.1.5.2.35.1 Offset

Register	Offset
SWREG39	9Ch

14.1.5.2.35.2 Diagram



14.1.5.2.35.3 Fields

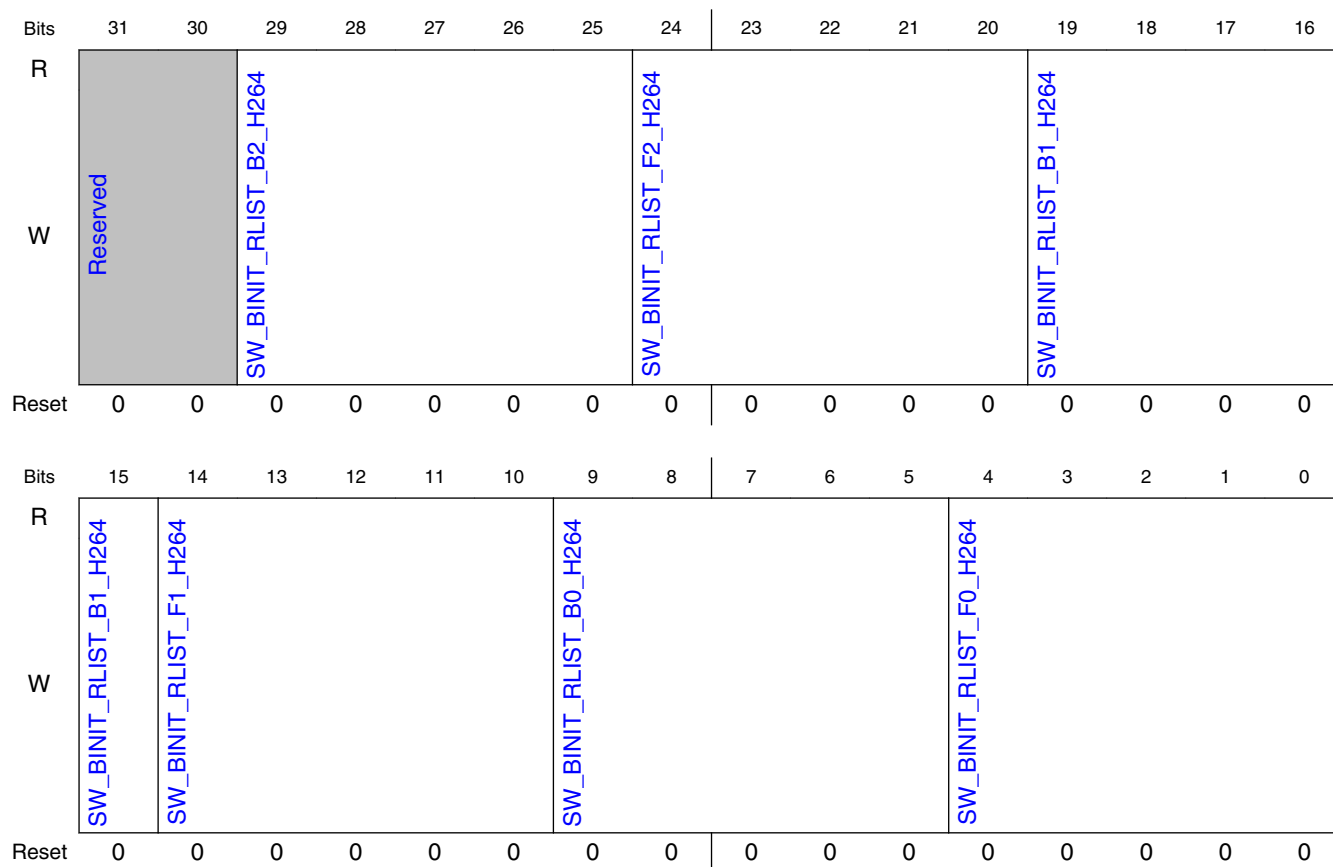
Field	Function
31-0 SW_REFER_VA LID_E	Valid flag for reference picture index [31:0]. Definition: If frame is being decoded the bits 31:15 are used, Bit 31 for picture index 0, Bit 30 for picture index 1 etc... IF field is being decoded the bits 31:0 are used, Bit 31 for reference picture 0 top field, bit 30 for reference picture 0 bottom field etc...

14.1.5.2.36 bi_dir initial ref pic list register (0-2) / VP6 prediction filter taps / Progressive JPEG Cb ACDC coefficient base (SWRE G42_H264)

14.1.5.2.36.1 Offset

Register	Offset
SWREG42_H264	A8h

14.1.5.2.36.2 Diagram



14.1.5.2.36.3 Fields

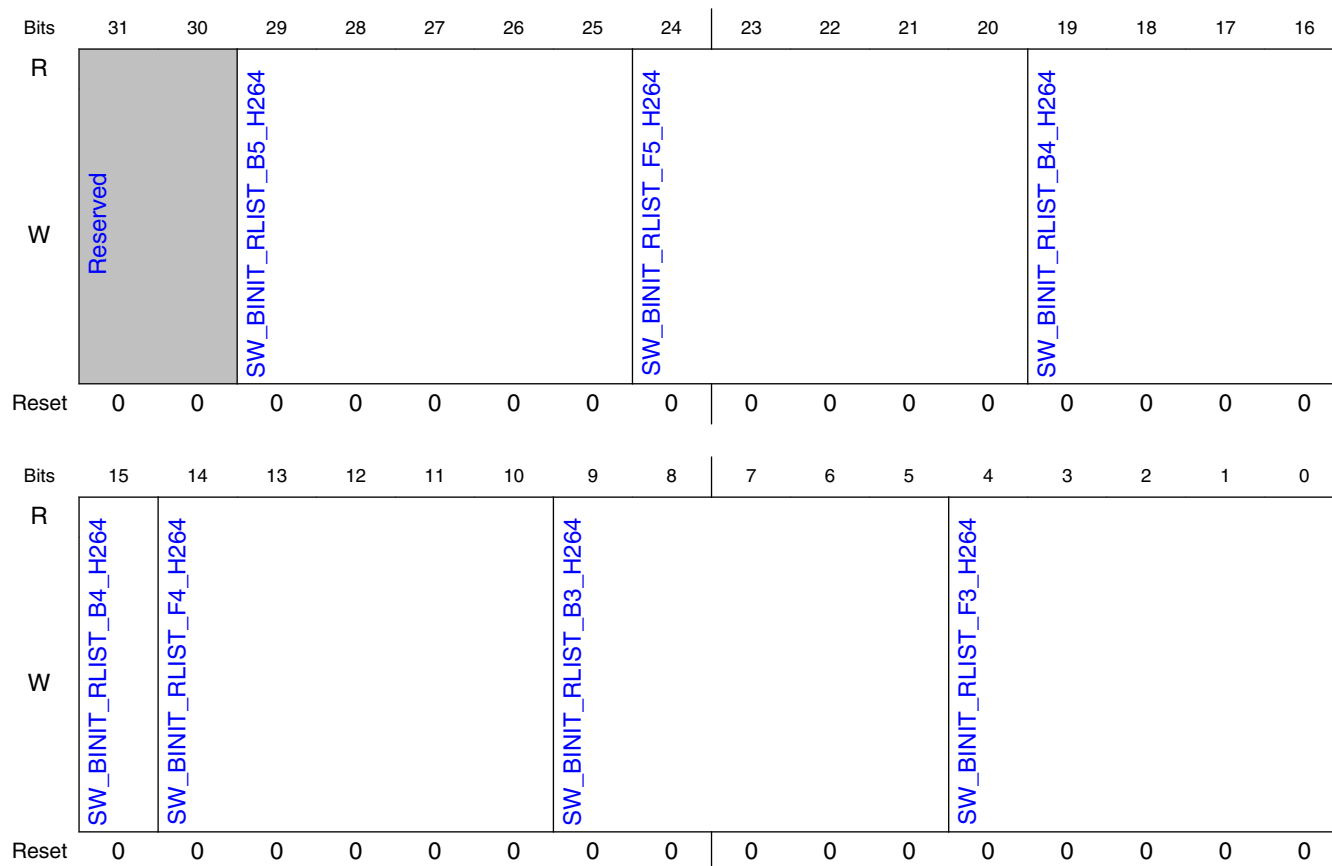
Field	Function
31-30 —	Reserved.
29-25 SW_BINIT_RLI ST_B2_H264	Initial reference picture list for bi-direct backward picid 2
24-20 SW_BINIT_RLI ST_F2_H264	Initial reference picture list for bi-direct forward picid 2
19-15 SW_BINIT_RLI ST_B1_H264	Initial reference picture list for bi-direct backward picid 1
14-10 SW_BINIT_RLI ST_F1_H264	Initial reference picture list for bi-direct forward picid 1
9-5 SW_BINIT_RLI ST_B0_H264	Initial reference picture list for bi-direct backward picid 0
4-0 SW_BINIT_RLI ST_F0_H264	Initial reference picture list for bi-direct forward picid 0

14.1.5.2.37 bi-dir initial ref pic list register (3-5) / VP6 prediction filter taps / Progressive JPEG Cr ACDC coefficient base (SWREG43_H264)

14.1.5.2.37.1 Offset

Register	Offset
SWREG43_H264	ACh

14.1.5.2.37.2 Diagram



14.1.5.2.37.3 Fields

Field	Function
31-30 —	Reserved.
29-25 SW_BINIT_RLIST_B5_H264	Initial reference picture list for bi-direct backward picid 5
24-20 SW_BINIT_RLIST_F5_H264	Initial reference picture list for bi-direct forward picid 5
19-15 SW_BINIT_RLIST_B4_H264	Initial reference picture list for bi-direct backward picid 4
14-10 SW_BINIT_RLIST_F4_H264	Initial reference picture list for bi-direct forward picid 4
9-5	Initial reference picture list for bi-direct backward picid 3

Table continues on the next page...

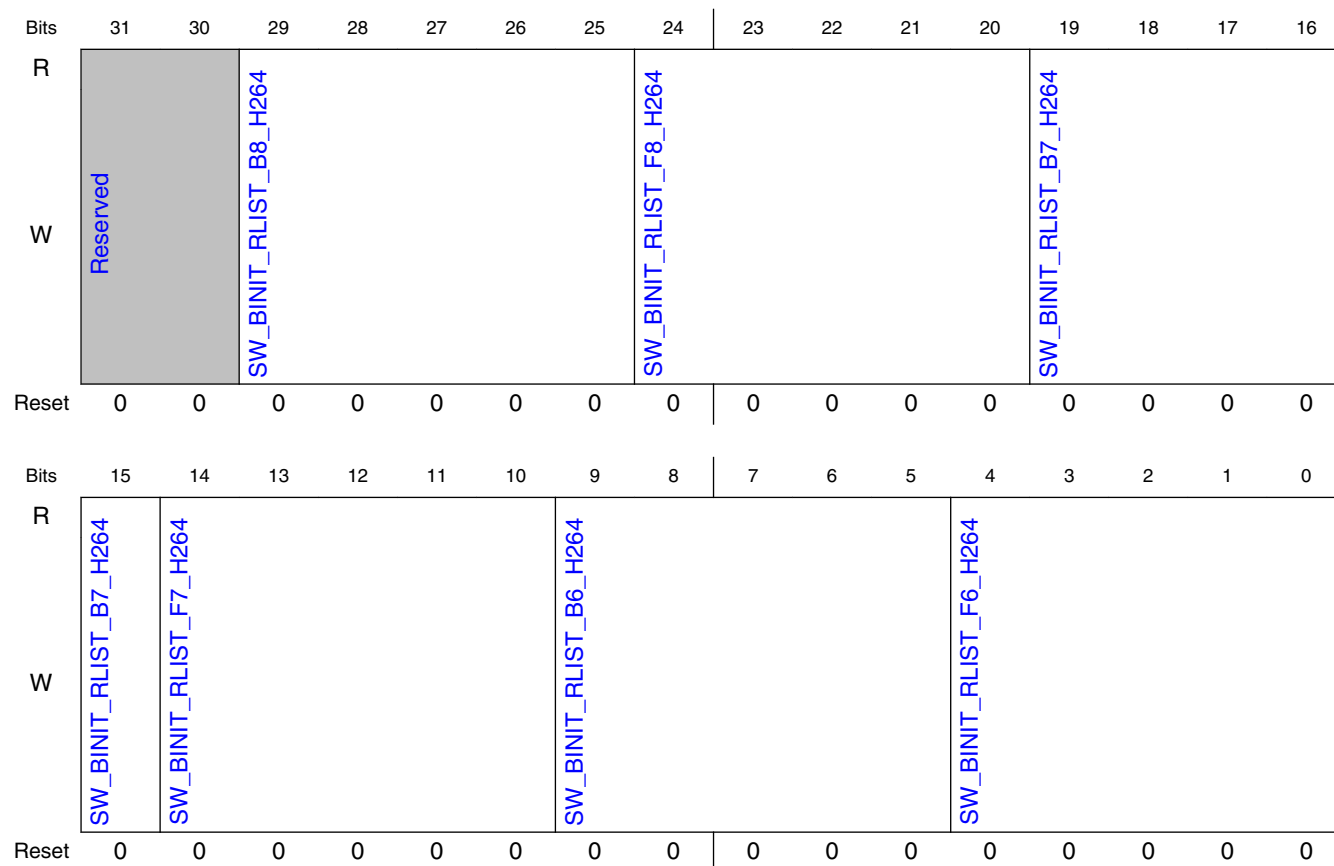
Field	Function
SW_BINIT_RLI ST_B3_H264	
4-0 SW_BINIT_RLI ST_F3_H264	Initial reference picture list for bi-direct forward picid 3

14.1.5.2.38 bi-dir initial ref pic list register (6-8) / VP6 prediction filter taps (SWREG44_H264)

14.1.5.2.38.1 Offset

Register	Offset
SWREG44_H264	B0h

14.1.5.2.38.2 Diagram



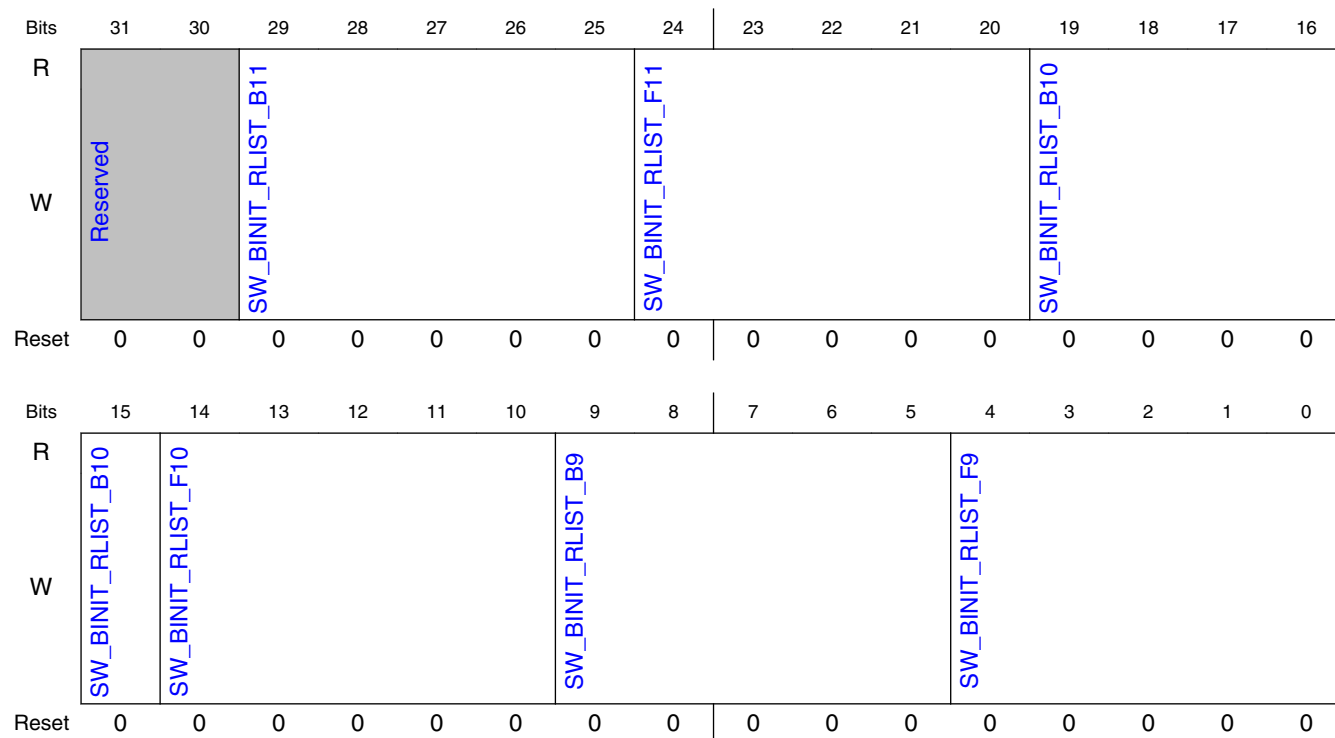
14.1.5.2.38.3 Fields

Field	Function
31-30 —	Reserved.
29-25 SW_BINIT_RLI ST_B8_H264	Initial reference picture list for bi-direct backward picid 8
24-20 SW_BINIT_RLI ST_F8_H264	Initial reference picture list for bi-direct forward picid 8
19-15 SW_BINIT_RLI ST_B7_H264	Initial reference picture list for bi-direct backward picid 7
14-10 SW_BINIT_RLI ST_F7_H264	Initial reference picture list for bi-direct forward picid 7
9-5 SW_BINIT_RLI ST_B6_H264	Initial reference picture list for bi-direct backward picid 6
4-0 SW_BINIT_RLI ST_F6_H264	Initial reference picture list for bi-direct forward picid 6

14.1.5.2.39 bi-dir initial ref pic list register (9-11) / VP6 prediction filter taps (SWREG45)**14.1.5.2.39.1 Offset**

Register	Offset
SWREG45	B4h

14.1.5.2.39.2 Diagram



14.1.5.2.39.3 Fields

Field	Function
31-30 —	Reserved.
29-25 SW_BINIT_RLIST_B11	Initial reference picture list for bi-direct backward picid 11
24-20 SW_BINIT_RLIST_F11	Initial reference picture list for bi-direct forward picid 11
19-15 SW_BINIT_RLIST_B10	Initial reference picture list for bi-direct backward picid 10
14-10 SW_BINIT_RLIST_F10	Initial reference picture list for bi-direct forward picid 10
9-5 SW_BINIT_RLIST_B9	Initial reference picture list for bi-direct backward picid 9
4-0	Initial reference picture list for bi-direct forward picid 9

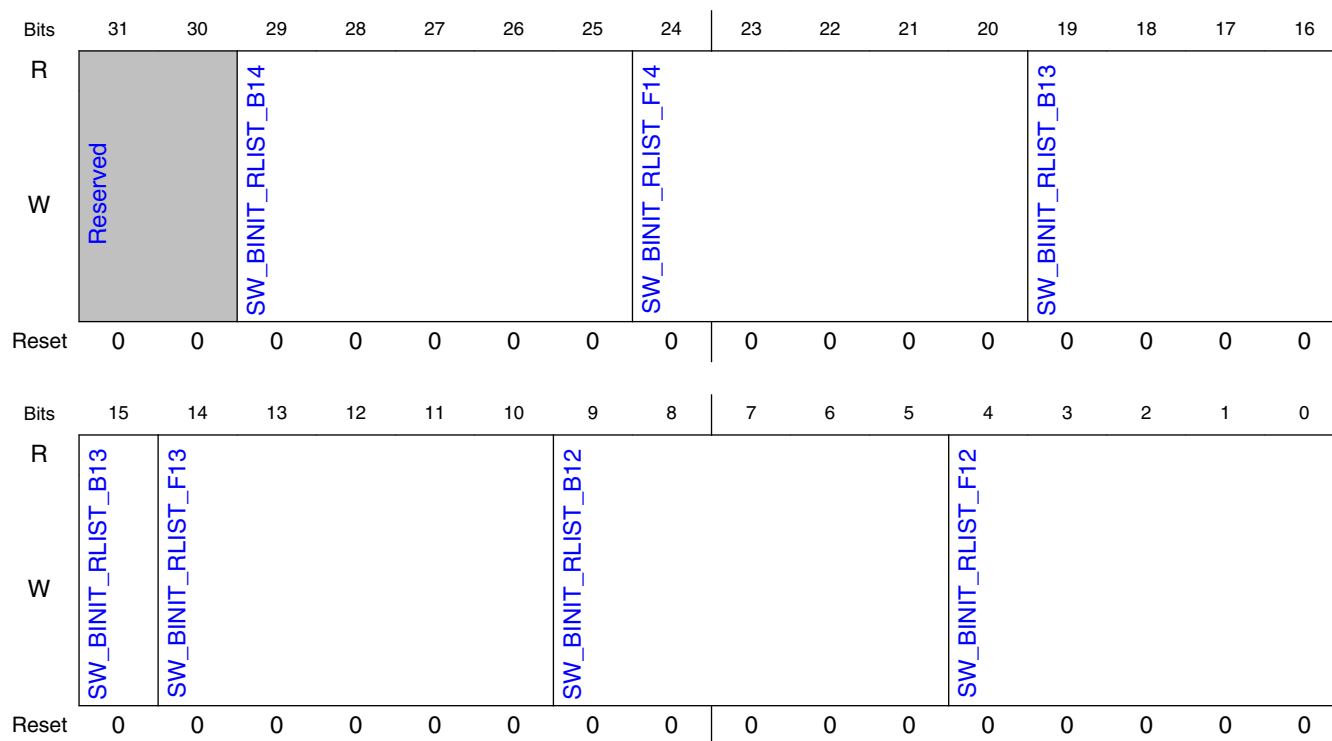
Field	Function
SW_BINIT_RLI ST_F9	

14.1.5.2.40 bi-dir initial ref pic list register (12-14) / VP7,VP8 quantization values (SWREG46)

14.1.5.2.40.1 Offset

Register	Offset
SWREG46	B8h

14.1.5.2.40.2 Diagram



14.1.5.2.40.3 Fields

Field	Function
31-30	Reserved.
—	

Table continues on the next page...

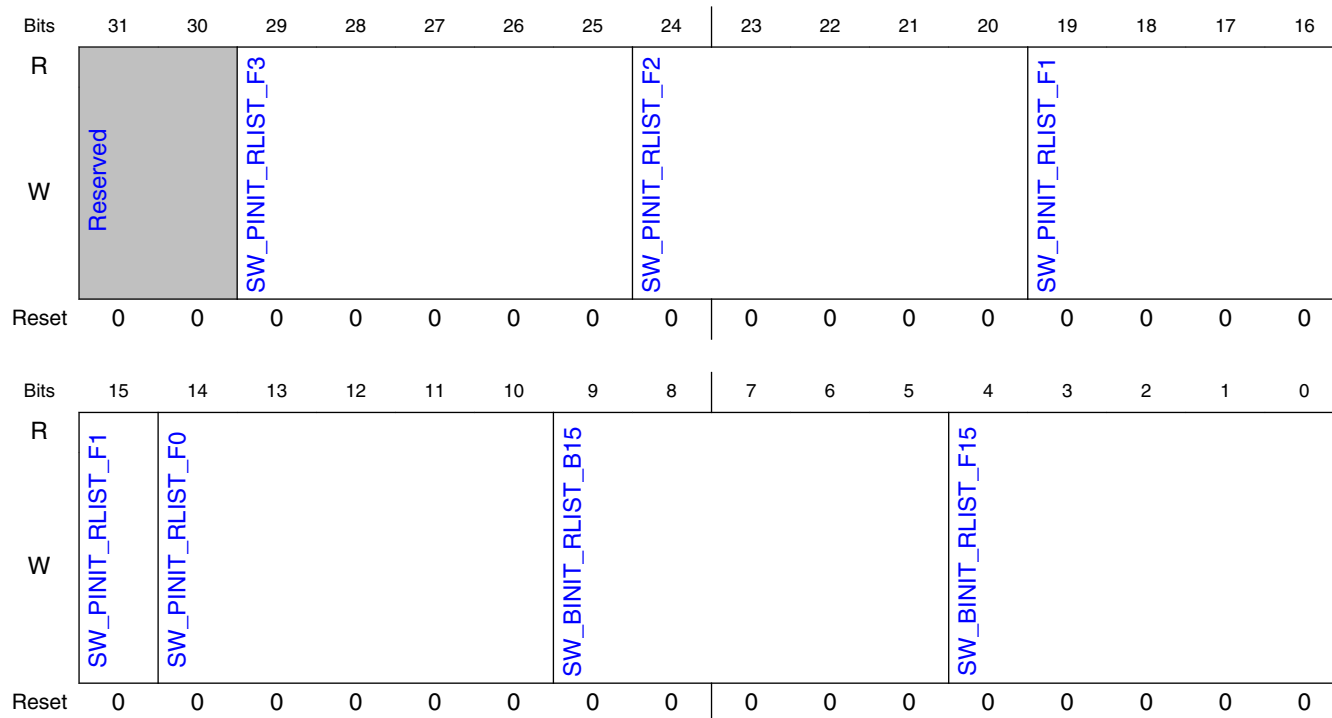
Field	Function
29-25 SW_BINIT_RLI ST_B14	Initial reference picture list for bi-direct backward picid 14
24-20 SW_BINIT_RLI ST_F14	Initial reference picture list for bi-direct forward picid 14
19-15 SW_BINIT_RLI ST_B13	Initial reference picture list for bi-direct backward picid 13
14-10 SW_BINIT_RLI ST_F13	Initial reference picture list for bi-direct forward picid 13
9-5 SW_BINIT_RLI ST_B12	Initial reference picture list for bi-direct backward picid 12
4-0 SW_BINIT_RLI ST_F12	Initial reference picture list for bi-direct forward picid 12

14.1.5.2.41 bi-dir and P fwd initial ref pic list register (15 and P 0-3) / VP7,VP8 quantization values (SWREG47)

14.1.5.2.41.1 Offset

Register	Offset
SWREG47	BCh

14.1.5.2.41.2 Diagram



14.1.5.2.41.3 Fields

Field	Function
31-30 —	Reserved.
29-25 SW_PINIT_RLIST_F3	Initial reference picture list for P forward picid 3
24-20 SW_PINIT_RLIST_F2	Initial reference picture list for P forward picid 2
19-15 SW_PINIT_RLIST_F1	Initial reference picture list for P forward picid 1
14-10 SW_PINIT_RLIST_F0	Initial reference picture list for P forward picid 0
9-5 SW_BINIT_RLIST_B15	Initial reference picture list for bi-direct backward picid 15
4-0	Initial reference picture list for bi-direct forward picid 15

Field	Function
SW_BINIT_RLI ST_F15	

14.1.5.3 VPU_G1 VP8 decoder register descriptions

14.1.5.3.1 VPU_G1 VP8 decoder memory map

VPU_G1_VP8 base address: 3830_0000h

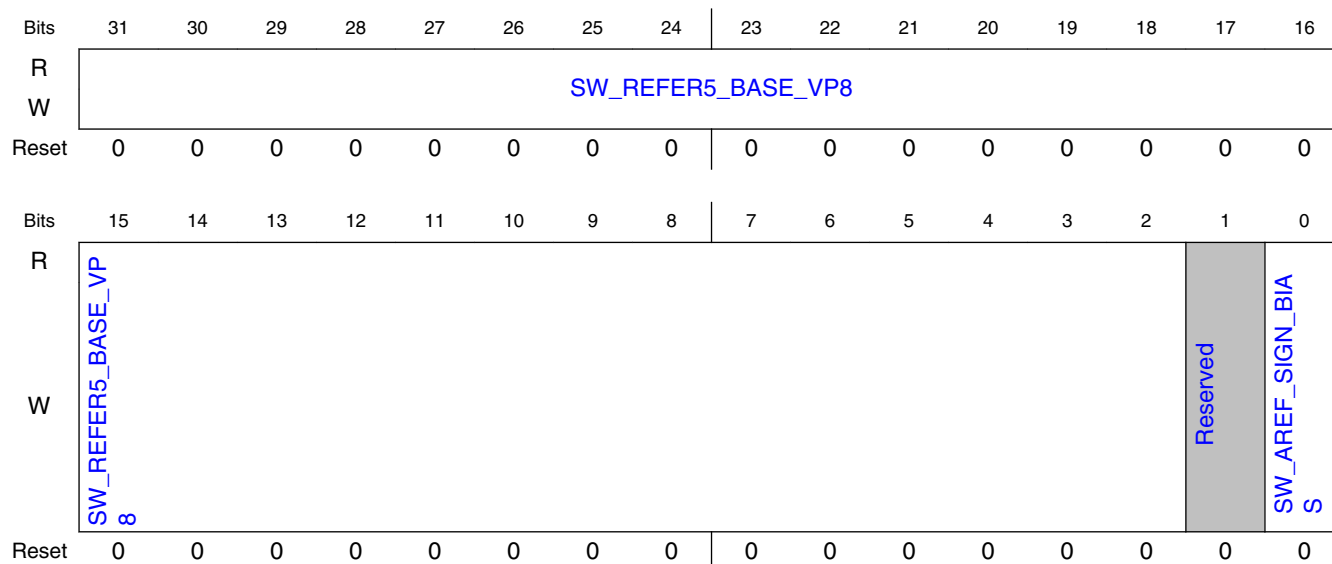
Offset	Register	Width (In bits)	Access	Reset value
4Ch	Base address for reference picture index 5 / MPEG4 TRB/TRD delta 0 / VC-1 intensity control 3 List of VLC code lengths in first/second JPEG AC table / VP6/VP7 scan maps (SWREG19_VP8)	32	RW	0000_0000h
50h	Base address for reference picture index 6 / / MPEG4 TRB/TRD delta -1 / List of VLC code lengths in second JPEG AC table / VP6/VP7 scan maps (SWREG20_VP8)	32	RW	0000_0000h
54h	Base address for reference picture index 7 / MPEG4 TRB/TRD delta 1 / List of VLC code lengths in second JPEG AC table / VP6/VP7 scan maps (SWREG21_VP8)	32	RW	0000_0000h

14.1.5.3.2 Base address for reference picture index 5 / MPEG4 TRB/TRD delta 0 / VC-1 intensity control 3 List of VLC code lengths in first/second JPEG AC table / VP6/VP7 scan maps (SWREG19_VP8)

14.1.5.3.2.1 Offset

Register	Offset
SWREG19_VP8	4Ch

14.1.5.3.2.2 Diagram



14.1.5.3.2.3 Fields

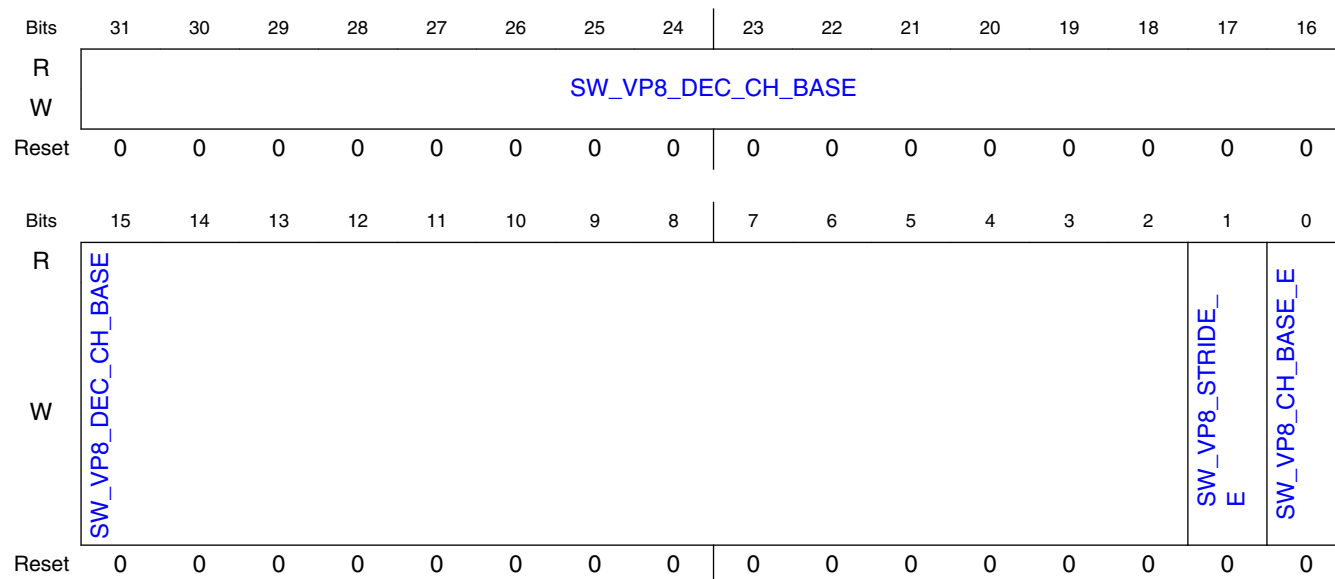
Field	Function
31-2 SW_REFER5_BASE_VP8	H.264: Base address for reference picture index 5 VP8: Base address for alternate reference picture (corresponds picid 5)
1 —	Reserved.
0 SW_AREF_SIGN_BIAS	VP8 only: Reference picture sign bias for Alternate reference frame

14.1.5.3.3 Base address for reference picture index 6 // MPEG4 TRB/TRD delta -1 / List of VLC code lengths in second JPEG AC table / VP6/VP7 scan maps (SWREG20_VP8)

14.1.5.3.3.1 Offset

Register	Offset
SWREG20_VP8	50h

14.1.5.3.3.2 Diagram



14.1.5.3.3.3 Fields

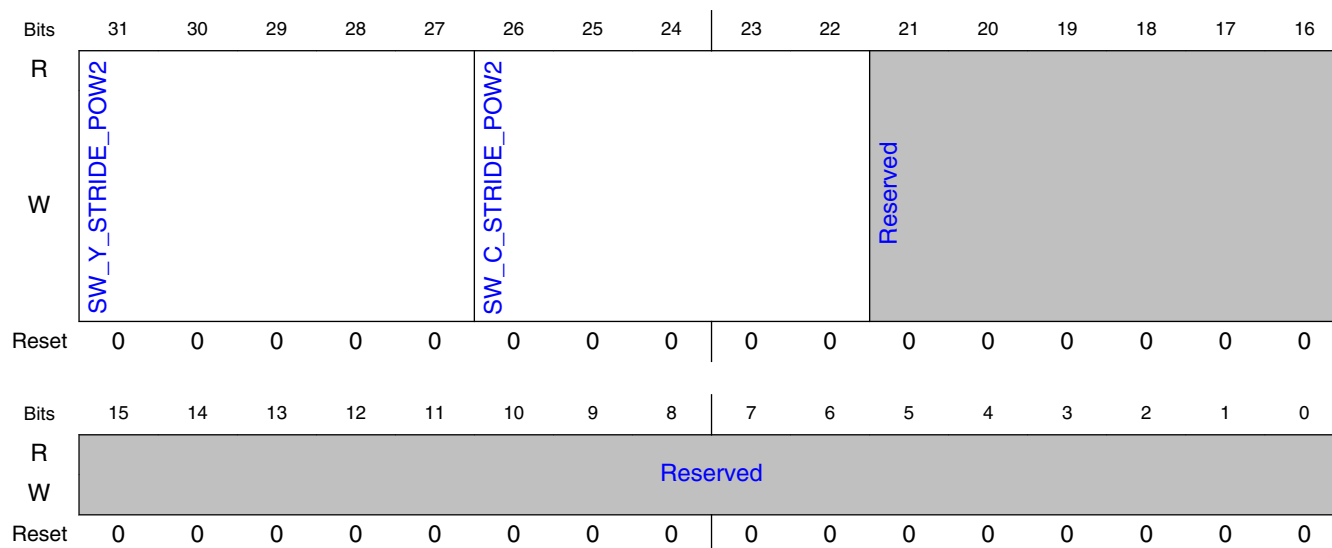
Field	Function
31-2 SW_VP8_DEC_CH_BASE	VP8 video base address for decoder output chrominance data (if vp8 stride configuration is enabled)
1 SW_VP8_STRIDE_E	VP8 stride enable. Can be set high only if HW configuration supports strides. Y and C strides are used instead of picture width. Separate chrominance base addresses are used instead of internal chrominance offsets. 0b - Not enabled 1b - Enabled
0 SW_VP8_CH_BASE_E	VP8 separate chrominance enable: 0b - Write/Read chrominance data from internal offset after the luminance data 1b - Write/Read chrominance data from separate base addresses given by SW

14.1.5.3.4 Base address for reference picture index 7 / MPEG4 TRB/TRD delta 1 / List of VLC code lengths in second JPEG AC table / VP6/VP7 scan maps (SWREG21_VP8)

14.1.5.3.4.1 Offset

Register	Offset
SWREG21_VP8	54h

14.1.5.3.4.2 Diagram



14.1.5.3.4.3 Fields

Field	Function
31-27 SW_Y_STRIDE_POW2	VP8 Y stride length informed by 2^n ($n=sw_y_stride_pow2$). Valid range 10-17 for 32 bit bus and 10-18 for 64 bit bus
26-22 SW_C_STRIDE_POW2	VP8 C stride length informed by 2^n ($n=sw_c_stride_pow2$). Valid range 10-17 for 32 bit bus and 10-18 for 64 bit bus
21-0 —	Reserved.

14.1.5.4 VPU_G1 VP7/VP8 decoder register descriptions

14.1.5.4.1 VPU_G1 VP7/VP8 decoder memory map

VPU_G1_VP7_VP8 base address: 3830_0000h

Offset	Register	Width (In bits)	Access	Reset value
10h	Decoder control register 1 (picture parameters) (SWREG4_JPEG_VP7_VP8)	32	RW	0000_0000h
14h	Decoder control register 2 (stream decoding table selects) (SWREG5_VP7_VP8)	32	RW	0000_0000h
18h	Decoder control register 3 (stream buffer information) (SWREG6_VP7_VP8)	32	RW	0000_0000h
1Ch	Decoder control register 4 (H264, VC-1, VP6 and progressive JPEG control) (SWREG7_VP7_VP8)	32	RW	0000_0000h
28h	Base address for differential motion vector base address (RLC-mode) / H264 P initial fwd ref pic list register (4-9) / VC-1 intensity control 1 / VP7 and VP8 segmentation base register (SWREG10_VP7_VP8)	32	RW	0000_0000h
2Ch	Decoder control register 7 (VLC) / base address for H.264 intra prediction 4x4 / base address for MPEG-4 DC component (RLC) / H264 P initial fwd ref pic list register (10-15) / VC-1 intensity control 2 (SWREG11_VP7_VP8)	32	RW	0000_0000h
38h	Base address for reference picture index 0 / base address for JPEG decoder output chrominance picture (SWREG14_VP7_VP8)	32	RW	0000_0000h
3Ch	Base address for reference picture index 1 / JPEG control (SWREG15_VP7_VP8)	32	RW	0000_0000h
48h	Base address for reference picture index 4 / VC1 control / MPEG4 MVD control / List of VLC code lengths in first JPEG AC table / VC-1 intensity control 4 / VP6/VP7, VP8 Golden refer picture base (SWREG18_VP7_VP8)	32	RW	0000_0000h
58h	Base address for reference picture index 8 / List of VLC code lengths in second JPEG AC table / VP6 scan maps / VP7,VP8 DCT stream 1 base (SWREG22_VP7_VP8)	32	RW	0000_0000h
5Ch	Base address for reference picture index 9 / List of VLC code lengths in first JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 2 base (SWREG23_VP7_VP8)	32	RW	0000_0000h
60h	Base address for reference picture index 10 / List of VLC code lengths in first JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 3 base (SWREG24_VP7_VP8)	32	RW	0000_0000h
64h	Base address for reference picture index 11 / List of VLC code lengths in second JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 4 base (SWREG25_VP7_VP8)	32	RW	0000_0000h
68h	Base address for reference picture index 12 / List of VLC code lengths in second JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 5 base (SWREG26_VP7_VP8)	32	RW	0000_0000h
6Ch	Base address for reference picture index 13 / VC-1 bitpl mbctrl or VP6,VP7,VP8 ctrl stream base / Progressive JPEG DC table (SWREG27_VC1)	32	RW	0000_0000h
70h	Base address for reference picture index 14 / VP6 scan maps / Progressive JPEG DC table / VP7,VP8 DCT stream 6 base (SWREG28_VP7_VP8)	32	RW	0000_0000h
74h	Base address for reference picture index 15 / VP6 scan maps / VP7,VP8 DCT stream 7 base (SWREG29_VP7_VP8)	32	RW	0000_0000h
78h	Reference picture numbers for index 0 and 1 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter mb level adjusts (SWREG30_VP7_VP8)	32	RW	0000_0000h

Table continues on the next page...

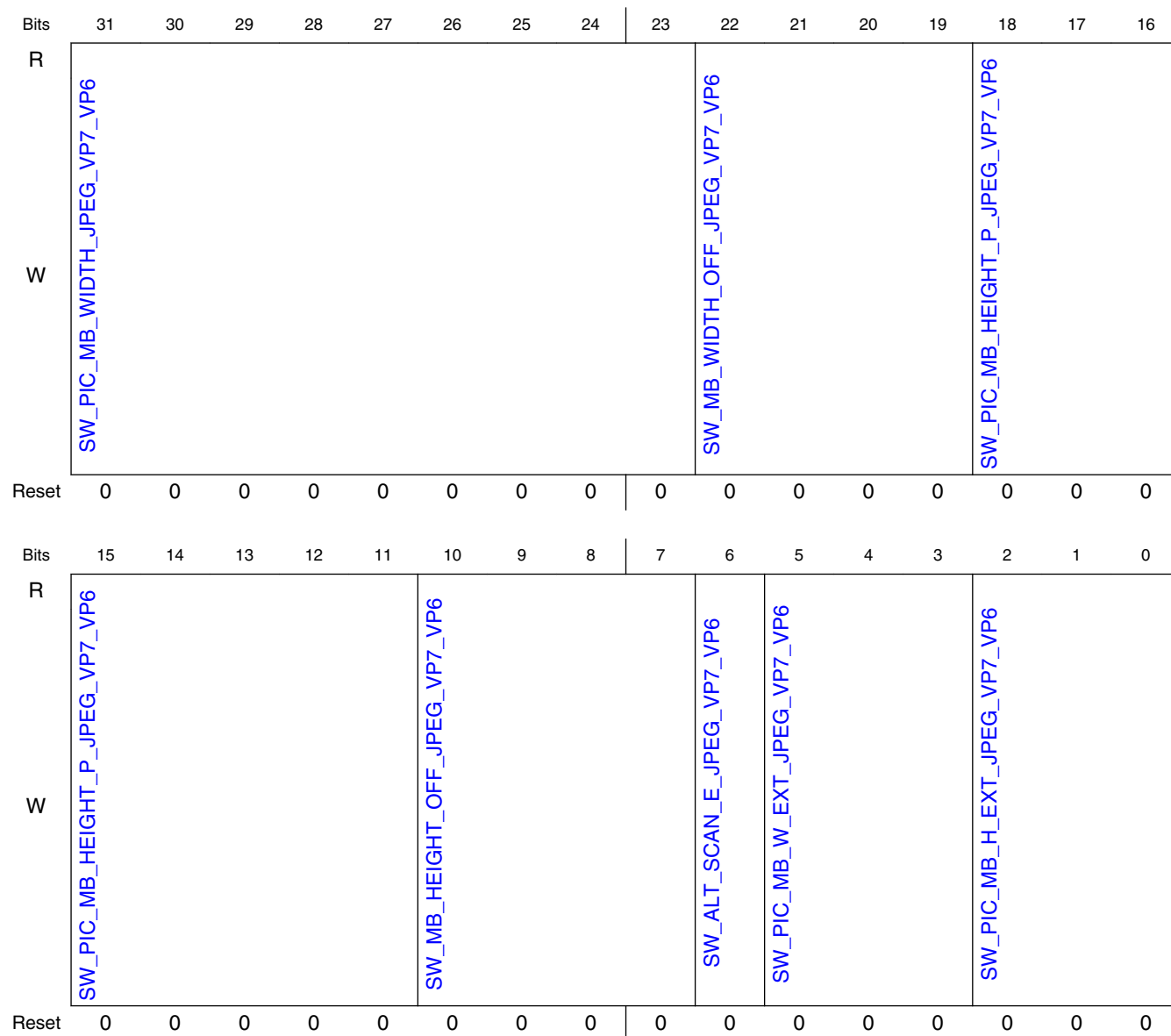
Offset	Register	Width (In bits)	Access	Reset value
7Ch	Reference picture numbers for index 2 and 3 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter ref pic level adjusts (SWREG31_VP7_VP8)	32	RW	0000_0000h
80h	Reference picture numbers for index 4 and 5 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter levels (SWREG32_VP7_VP8)	32	RW	0000_0000h
84h	Reference picture numbers for index 6 and 7 (H264 VLC) / VP6 scan maps / VP7,VP8 quantization values (SWREG33_VP7_VP8)	32	RW	0000_0000h
88h	Reference picture numbers for index 8 and 9 (H264 VLC) / MPEG4, VC1, VPx prediction filter taps (SWREG34_H263)	32	RW	0000_0000h
8Ch	Reference picture numbers for index 10 and 11 (H264 VLC) / VC1, VPx prediction filter taps (SWREG35_VC1)	32	RW	0000_0000h
90h	Reference picture numbers for index 12 and 13 (H264 VLC) / VC1, VPx prediction filter taps (SWREG36_VC1)	32	RW	0000_0000h
94h	Reference picture numbers for index 14 and 15 (H264 VLC) / VPx prediction filter taps (SWREG37_VP6_VP7_VP8)	32	RW	0000_0000h
98h	Reference picture long term flags (H264 VLC) / VPx prediction filter taps (SWREG38_VP6_VP7_VP8)	32	RW	0000_0000h
9Ch	Reference picture valid flags (H264 VLC) / VPx prediction filter taps (SWREG39_VP6_VP7_VP8)	32	RW	0000_0000h
A8h	bi_dir initial ref pic list register (0-2) / VP6 prediction filter taps / Progressive JPEG Cb ACDC coefficient base (SWREG42_VP6)	32	RW	0000_0000h
ACh	bi_dir initial ref pic list register (3-5) / VP6 prediction filter taps / Progressive JPEG Cr ACDC coefficient base (SWREG43_VP7_VP8)	32	RW	0000_0000h
B0h	bi_dir initial ref pic list register (6-8) / VP6 prediction filter taps (SWREG44_VP7_VP8)	32	RW	0000_0000h
B4h	bi_dir initial ref pic list register (9-11) / VP6 prediction filter taps (SWREG45_VP7_VP8)	32	RW	0000_0000h
B8h	bi_dir initial ref pic list register (12-14) / VP7,VP8 quantization values (SWREG46_VP7_VP8)	32	RW	0000_0000h
BCh	bi_dir and P fwd initial ref pic list register (15 and P 0-3) / VP7,VP8 quantization values (SWREG47_VP7_VP8)	32	RW	0000_0000h

14.1.5.4.2 Decoder control register 1 (picture parameters) (SWREG4_JPEG_VP7_VP8)

14.1.5.4.2.1 Offset

Register	Offset
SWREG4_JPEG_VP7_VP8	10h

14.1.5.4.2.2 Diagram



14.1.5.4.2.3 Fields

Field	Function
31-23 SW_PIC_MB_W IDTH_JPEG_VP 7_VP6	Picture width in macroblocks = ((width in pixels + 15) / 16)
22-19	The amount of meaningful horizontal pixels in last MB (width offset) 0 if exactly 16 pixels multiple picture and all the horizontal pixels in last MB are meaningful

Table continues on the next page...

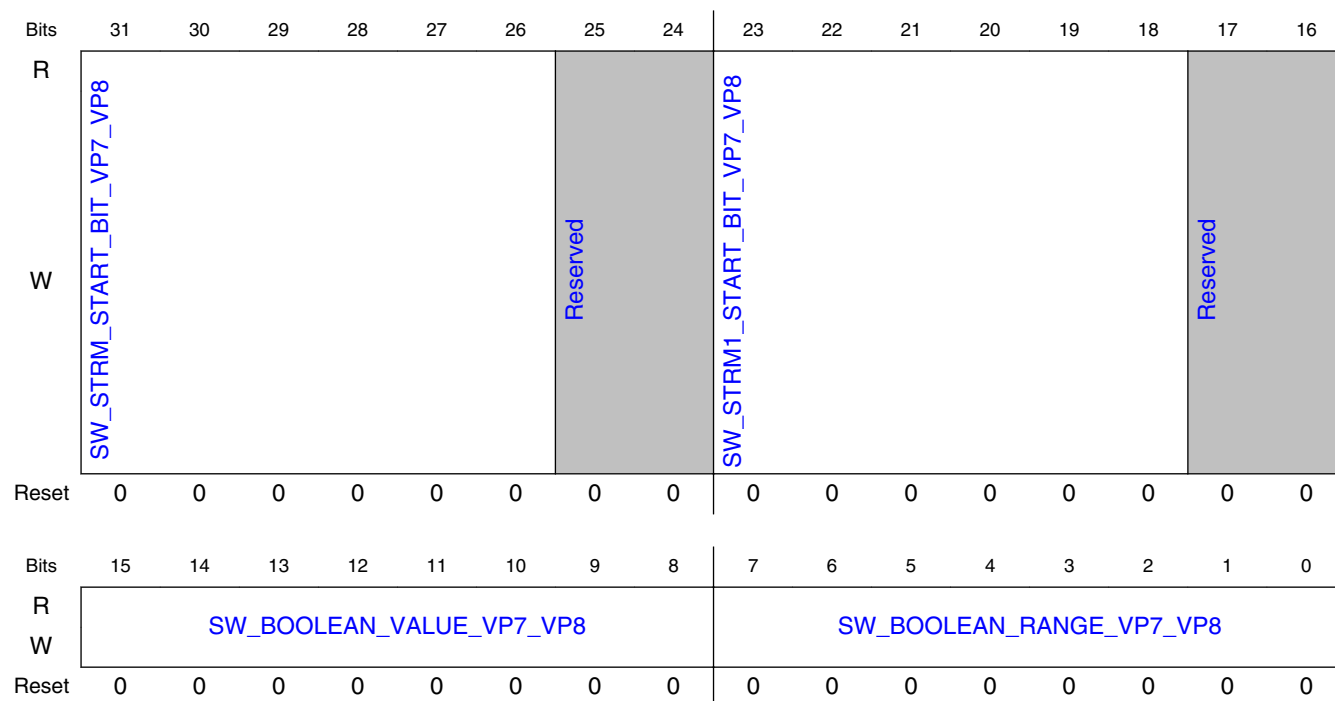
Field	Function
SW_MB_WIDTH _OFF_JPEG_V P7_VP6	
18-11 SW_PIC_MB_H EIGHT_P_JPEG _VP7_VP6	Picture height in macroblocks = ((height in pixels+15)/16). Picture height is informed as size of the (progressive) frame also for single field (of interlaced content) is being decoded
10-7 SW_MB_HEIGHT _OFF_JPEG_V P7_VP6	The amount of meaningful vertical pixels in last MB (height offset 0 if exactly 16 pixels multiple picture and all the vertical pixels in last MB are meaningfull
6 SW_ALT_SCAN _E_JPEG_VP7_VP6	indicates alternative vertical scan method used for interlaced frames
5-3 SW_PIC_MB_W _EXT_JPEG_V P7_VP6	Picture mb width extension. If sw_pic_mb_width does not fit to 9 bits then these bits are used to increase the range upto 11 bits (used as 3 msb)
2-0 SW_PIC_MB_H _EXT_JPEG_V P7_VP6	Picture mb height extension. If sw_pic_mb_height_p does not fit to 9 bits then these bits are used to increase the range upto 11 bits (used as 3 msb). For 4k video one bit is used for extension (bit 0)

14.1.5.4.3 Decoder control register 2 (stream decoding table selects) (SWREG5_VP7_VP8)

14.1.5.4.3.1 Offset

Register	Offset
SWREG5_VP7_VP8	14h

14.1.5.4.3.2 Diagram



14.1.5.4.3.3 Fields

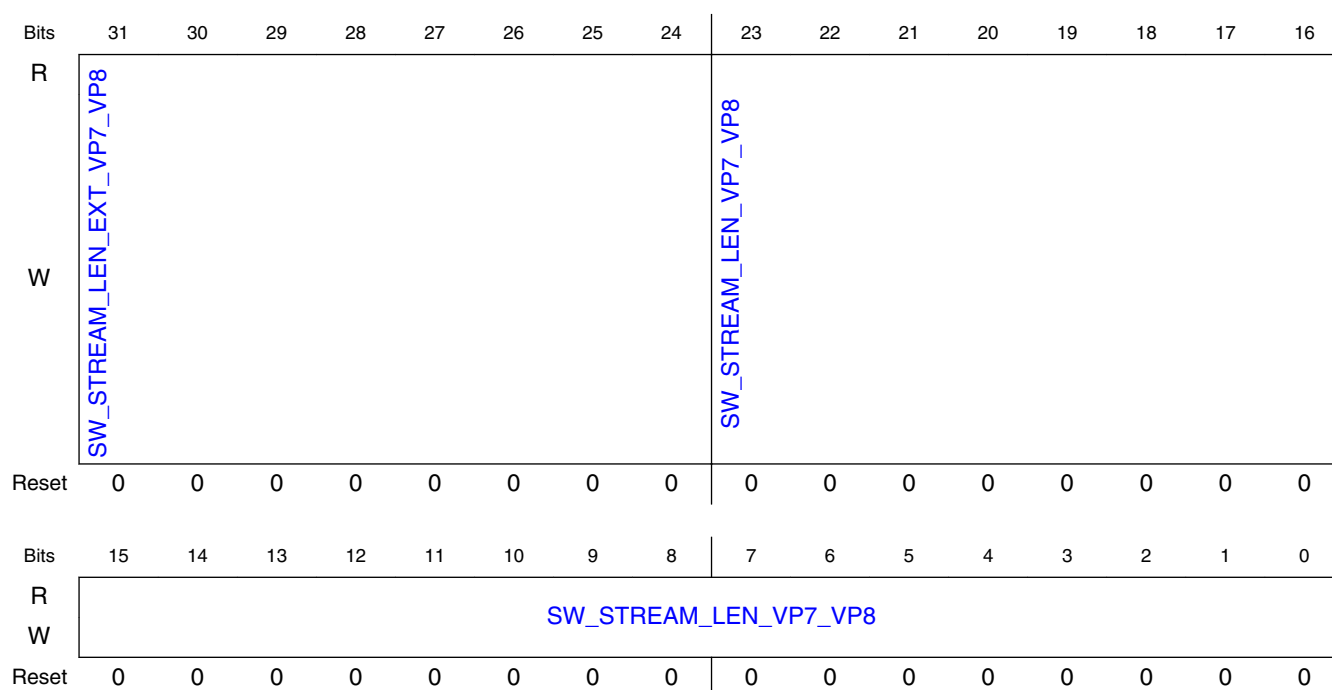
Field	Function
31-26 SW_STRM_START_BIT_VP7_VP8	Exact bit of stream start word where decoding can be started (associates with sw_rlc_vlc_base)
25-24 —	Reserved.
23-18 SW_STRM1_START_BIT_VP7_VP8	Start bit for ctrl-stream (needed if multistream is enabled, associates with sw_bitpl_ctrl_base)
17-16 —	Reserved.
15-8 SW_BOOLEAN_VALUE_VP7_VP8	Initial value for boolean dec
7-0 SW_BOOLEAN_RANGE_VP7_VP8	Initial range for boolean dec

14.1.5.4.4 Decoder control register 3 (stream buffer information) (SWREG6_VP7_VP8)

14.1.5.4.4.1 Offset

Register	Offset
SWREG6_VP7_VP8	18h

14.1.5.4.4.2 Diagram



14.1.5.4.4.3 Fields

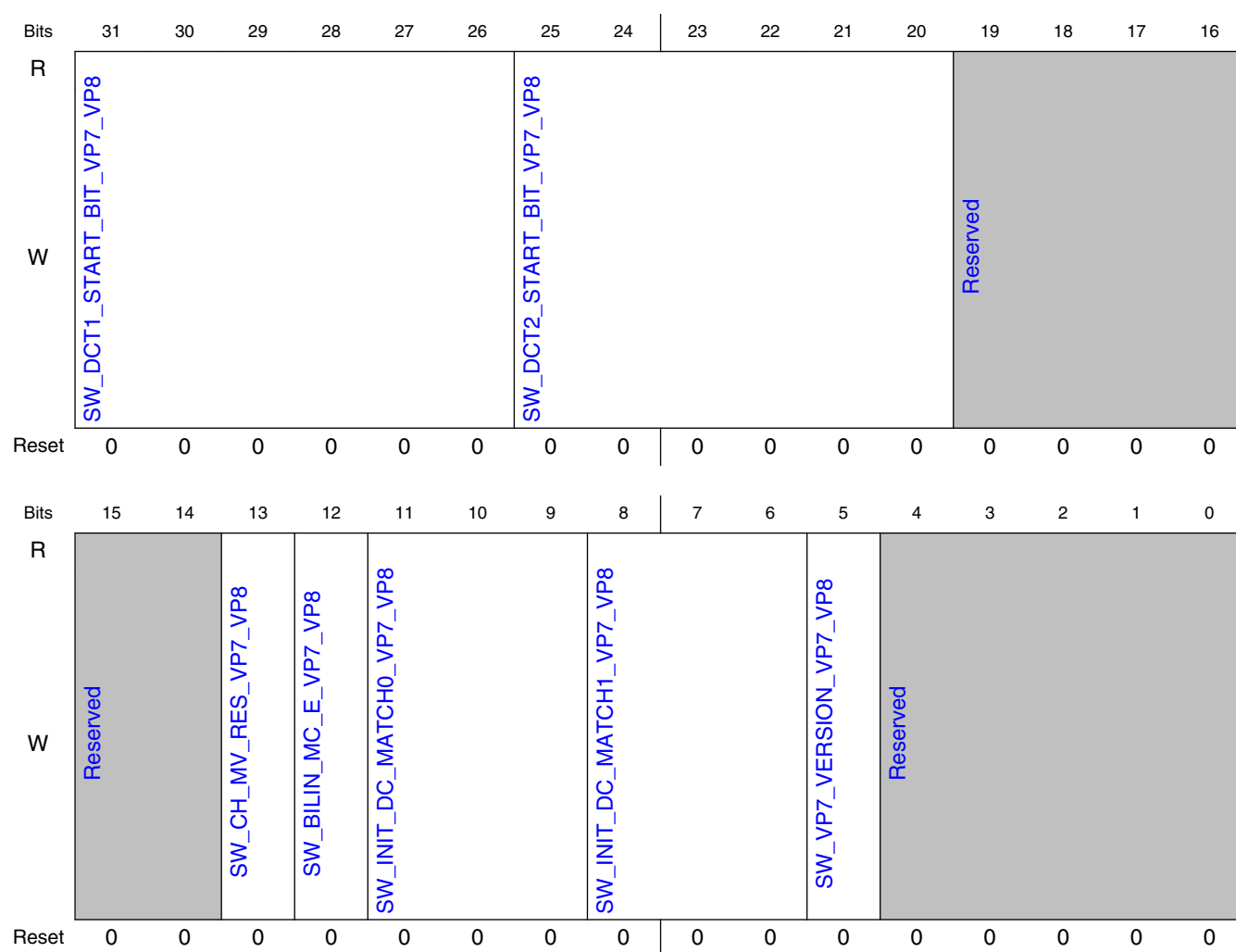
Field	Function
31-24 SW_STREAM_LEN_EXT_VP7_VP8	Extended stream length for WEBP/VP8
23-0 SW_STREAM_LEN_VP7_VP8	Amount of stream data bytes in input buffer. If the given buffer size is not enough for finishing the picture the corresponding interrupt is given and new stream buffer base address and stream buffer size information should be given (associates with sw_rlc_vlc_base). For VC-1/VP6 the buffer must include data for one picture/slice of the picture For H264/MPEG4/H263/MPEG2/MPEG1 the buffer must include at least data for one slice/VP of the picture For JPEG the buffer size must be a multiple of 256 bytes or the amount of data for one picture.

14.1.5.4.5 Decoder control register 4 (H264, VC-1, VP6 and progressive JPEG control) (SWREG7_VP7_VP8)

14.1.5.4.5.1 Offset

Register	Offset
SWREG7_VP7_VP8	1Ch

14.1.5.4.5.2 Diagram



14.1.5.4.5.3 Fields

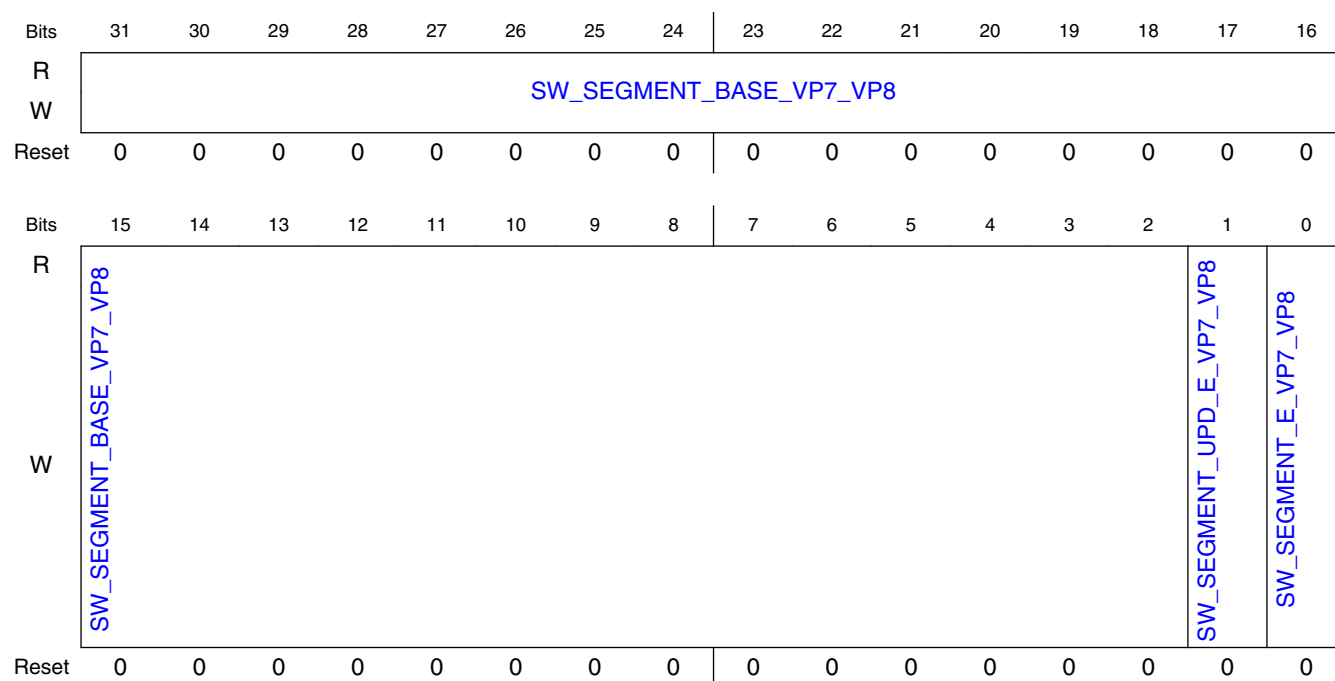
Field	Function
31-26 SW_DCT1_STA RT_BIT_VP7_V P8	Start bit for VP7/VP8 DCT stream partition index 1
25-20 SW_DCT2_STA RT_BIT_VP7_V P8	Start bit for VP7/VP8 DCT stream partition index 2
19-14 —	Reserved.
13 SW_CH_MV_R ES_VP7_VP8	VP7/VP8 Chrominance motion vector resolution: 0b - Full pixel 1b - 1/8 pixel
12 SW_BILIN_MC_ E_VP7_VP8	Bilinear motion compensation enable: 0b - Bicubic interpolation used 1b - Bilinear interpolation used
11-9 SW_INIT_DC_M ATCH0_VP7_V P8	Initial DC prediction mach count 0. After HW has decoded a picture HW returns the final match count0 information which is read by SW
8-6 SW_INIT_DC_M ATCH1_VP7_V P8	Initial DC prediction mach count 1. After HW has decoded a picture HW returns the final match count1 information which is read by SW
5 SW_VP7_VERS ION_VP7_VP8	VP7 version information to streamd: 0b - VP7 version 7.0 1b - VP7 version 7.1 or better
4-0 —	Reserved.

14.1.5.4.6 Base address for differential motion vector base address (RLC-mode) /H264 P initial fwd ref pic list register (4-9)/ VC-1 intensity control 1/ VP7 and VP8 segmentation base register (SWREG10_VP7_VP8)

14.1.5.4.6.1 Offset

Register	Offset
SWREG10_VP7_VP8	28h

14.1.5.4.6.2 Diagram



14.1.5.4.6.3 Fields

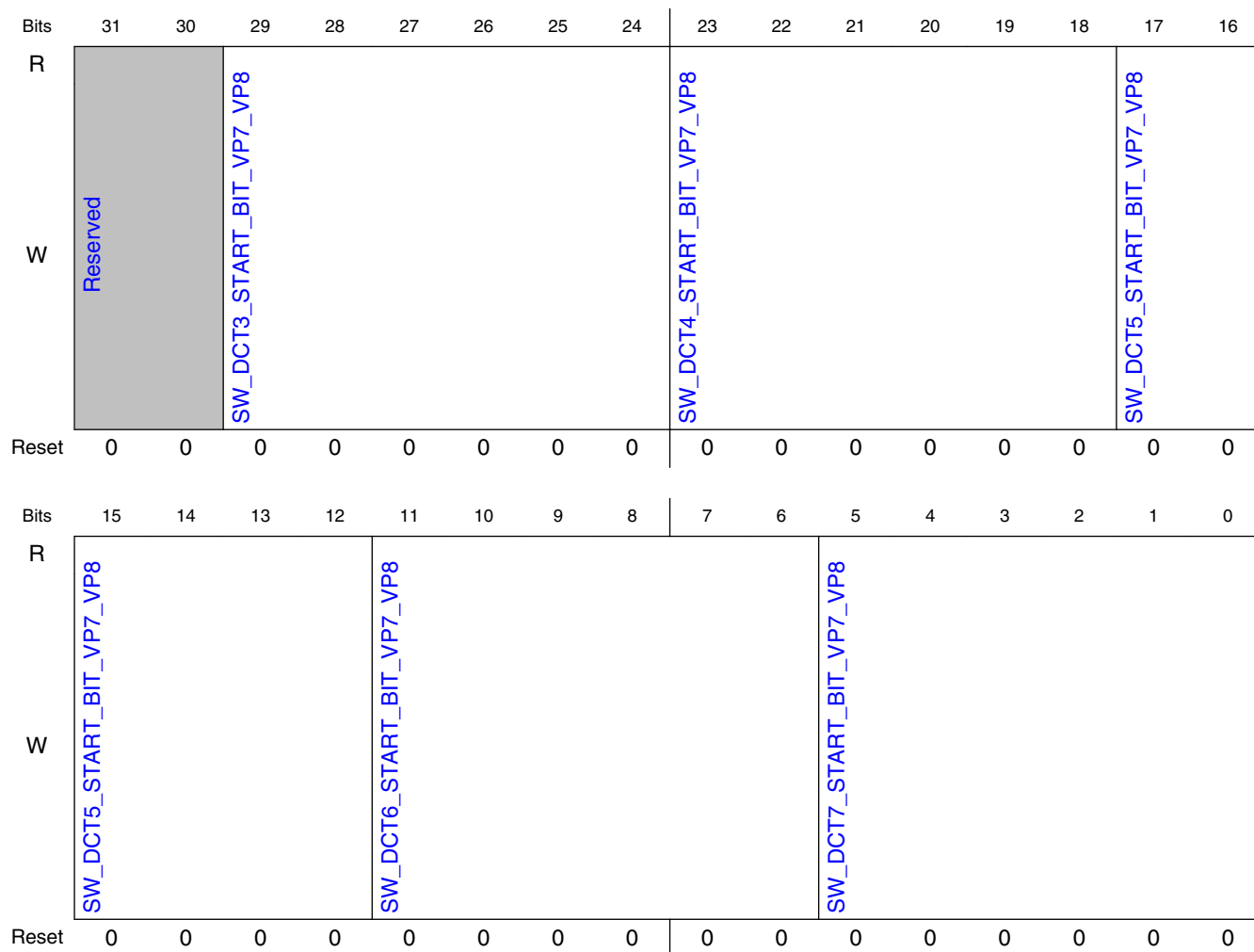
Field	Function
31-2 SW_SEGMENT_BASE_VP7_VP8	VP7/VP8: base address for segmentation map values
1 SW_SEGMENT_UPD_E_VP7_VP8	VP7/VP8 Segmentation map update enable: '0': segmentation values are read from external memory (from segment_base) '1': segmentation update is included in stream
0 SW_SEGMENT_E_VP7_VP8	Segmentation enable: '0': segmentation is not enabled '1': segmentation is enabled (sw_segment_upd_e value is used)

14.1.5.4.7 Decoder control register 7 (VLC) / base address for H.264 intra prediction 4x4 / base address for MPEG-4 DC component (RLC) / H264 P initial fwd ref pic list register (10-15) / VC-1 intensity control 2 (SWREG11_VP7_VP8)

14.1.5.4.7.1 Offset

Register	Offset
SWREG11_VP7_VP8	2Ch

14.1.5.4.7.2 Diagram



14.1.5.4.7.3 Fields

Field	Function
31-30	Reserved.
—	
29-24	Start bit for VP7/VP8 DCT stream partition index 3

Table continues on the next page...

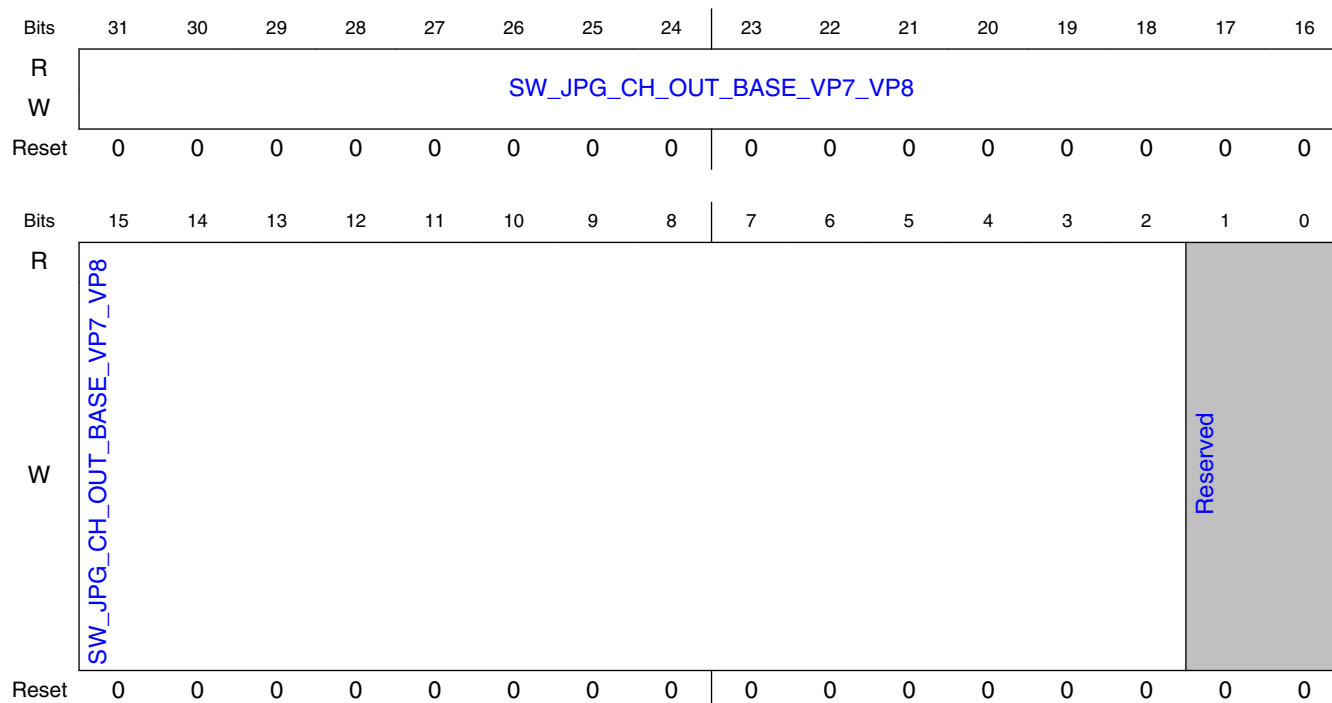
Field	Function
SW_DCT3_STA RT_BIT_VP7_V P8	
23-18 SW_DCT4_STA RT_BIT_VP7_V P8	Start bit for VP7/VP8 DCT stream partition index 4
17-12 SW_DCT5_STA RT_BIT_VP7_V P8	Start bit for VP7/VP8 DCT stream partition index 5
11-6 SW_DCT6_STA RT_BIT_VP7_V P8	Start bit for VP7/VP8 DCT stream partition index 6
5-0 SW_DCT7_STA RT_BIT_VP7_V P8	Start bit for VP7/VP8 DCT stream partition index 7

14.1.5.4.8 Base address for reference picture index 0 / base address for JPEG decoder output chrominance picture (SWREG14_VP7_VP8)

14.1.5.4.8.1 Offset

Register	Offset
SWREG14_VP7_VP8	38h

14.1.5.4.8.2 Diagram



14.1.5.4.8.3 Fields

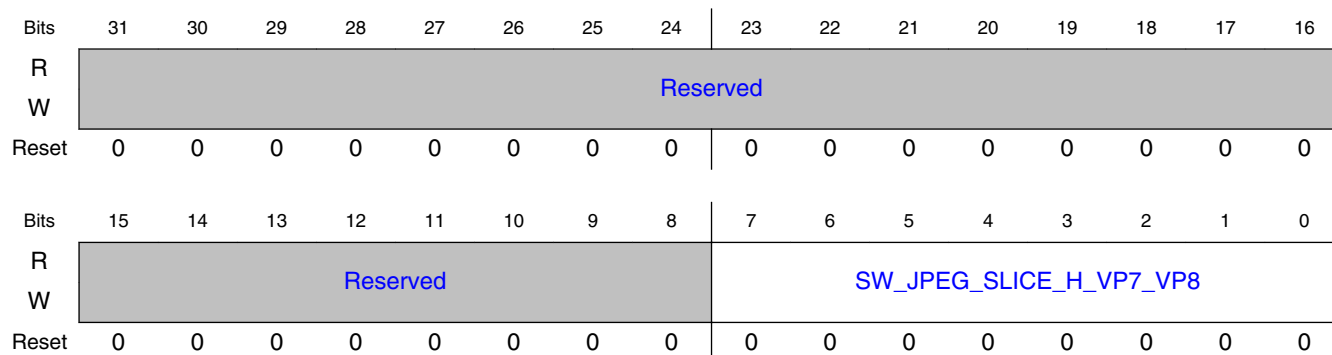
Field	Function
31-2 SW_JPG_CH_OUT_BASE_VP7_VP8	Base address for decoder output chrominance picture. Used in JPEG and web-p picture mode (not needed if decoder output is not written)
1-0 —	Reserved.

14.1.5.4.9 Base address for reference picture index 1 / JPEG control (SWREG15_VP7_VP8)

14.1.5.4.9.1 Offset

Register	Offset
SWREG15_VP7_VP8	3Ch

14.1.5.4.9.2 Diagram



14.1.5.4.9.3 Fields

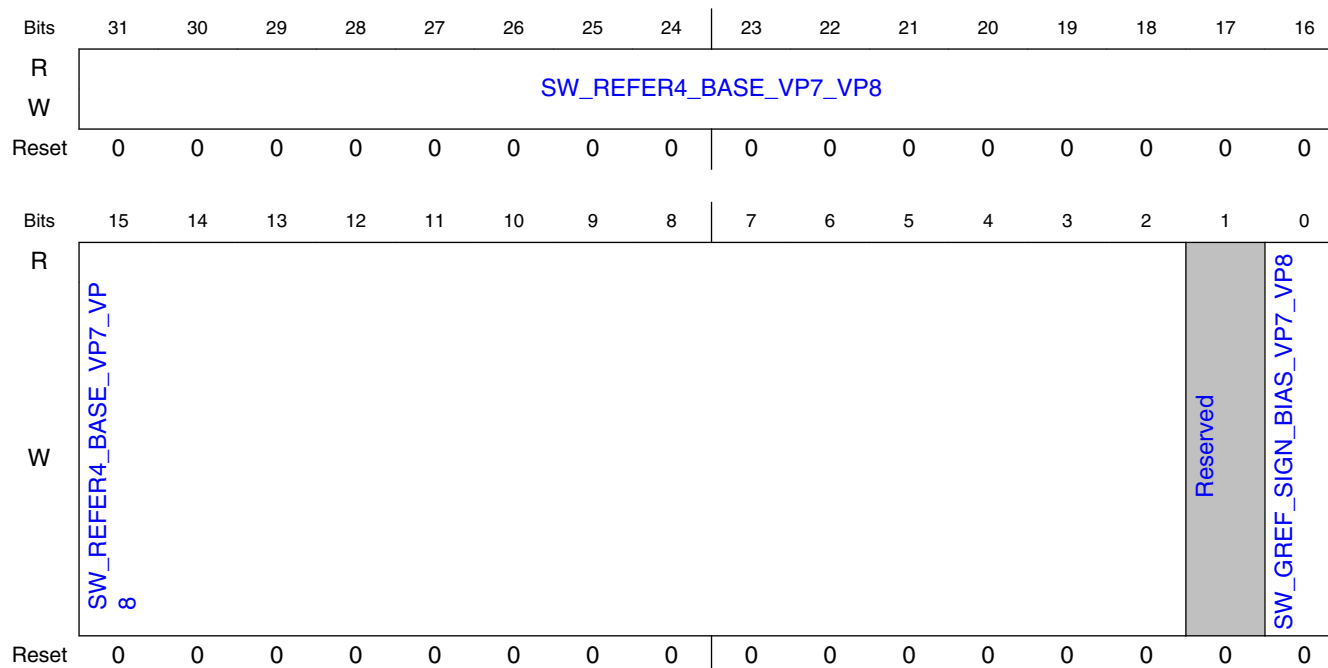
Field	Function
31-8 —	Reserved.
7-0 SW_JPEG_SLICE_H_VP7_VP8	JPEG/Web-p. Height of the slice (multiple of 16 pixels) that HW decodes before interrupt. When slice is decoded HW will rise an interrupt and reset external addresses back to base address. Note, value 0 disables slice mode. Slice mode must be used if picture size is more than 16 Mpixels. However for bigger than 4096 MBs the slice mode usage is recommended.

14.1.5.4.10 Base address for reference picture index 4 / VC1 control / MPEG4 MVD control/ List of VLC code lengths in first JPEG AC table / VC-1 intensity control 4 / VP6/VP7, VP8 Golden refer picture base (SWREG18_VP7_VP8)

14.1.5.4.10.1 Offset

Register	Offset
SWREG18_VP7_VP8	48h

14.1.5.4.10.2 Diagram



14.1.5.4.10.3 Fields

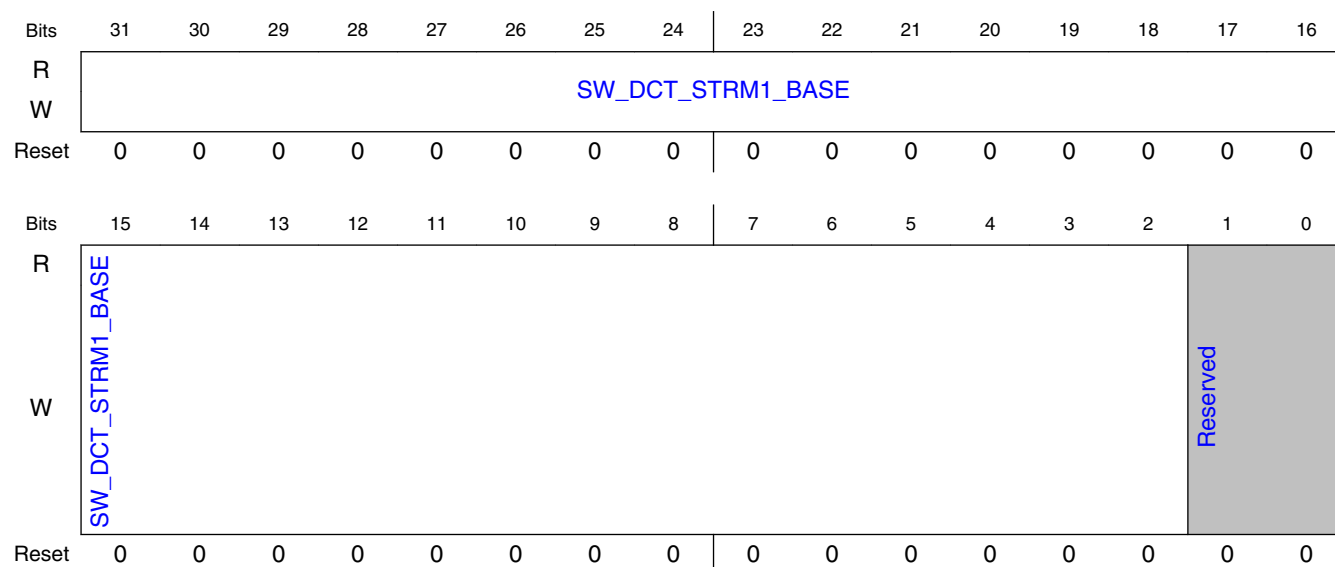
Field	Function
31-2 SW_REFER4_B ASE_VP7_VP8	H264: Base address for reference picture index 4 VP6/VP7/VP8: Base address for Golden reference picture (corresponds picid 4)
1 —	Reserved.
0 SW_GREF_SIG N_BIAS_VP7_V P8	Reference picture sign bias for Golden reference frame

14.1.5.4.11 Base address for reference picture index 8 / List of VLC code lengths in second JPEG AC table / VP6 scan maps / VP7,VP8 DCT stream 1 base (SWREG22_VP7_VP8)

14.1.5.4.11.1 Offset

Register	Offset
SWREG22_VP7_VP8	58h

14.1.5.4.11.2 Diagram



14.1.5.4.11.3 Fields

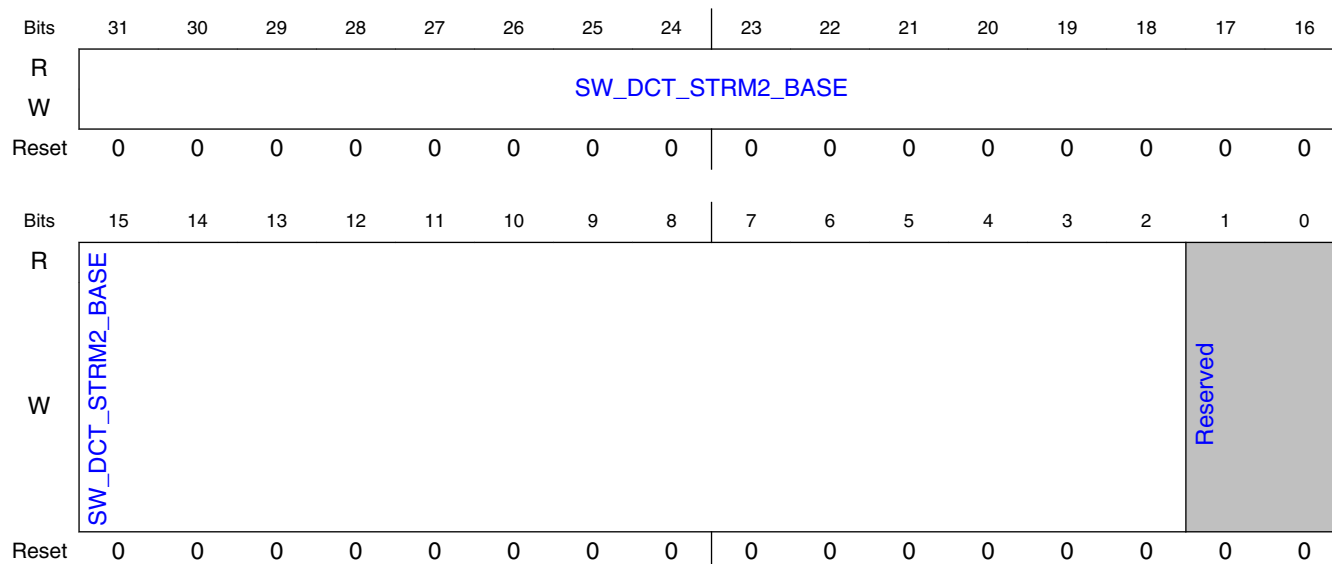
Field	Function
31-2 SW_DCT_STRM1_BASE	Base address for VP7/VP8 DCT stream MB row 1,2n+1
1-0 —	Reserved.

14.1.5.4.12 Base address for reference picture index 9 / List of VLC code lengths in first JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 2 base (SWREG23_VP7_VP8)

14.1.5.4.12.1 Offset

Register	Offset
SWREG23_VP7_VP8	5Ch

14.1.5.4.12.2 Diagram



14.1.5.4.12.3 Fields

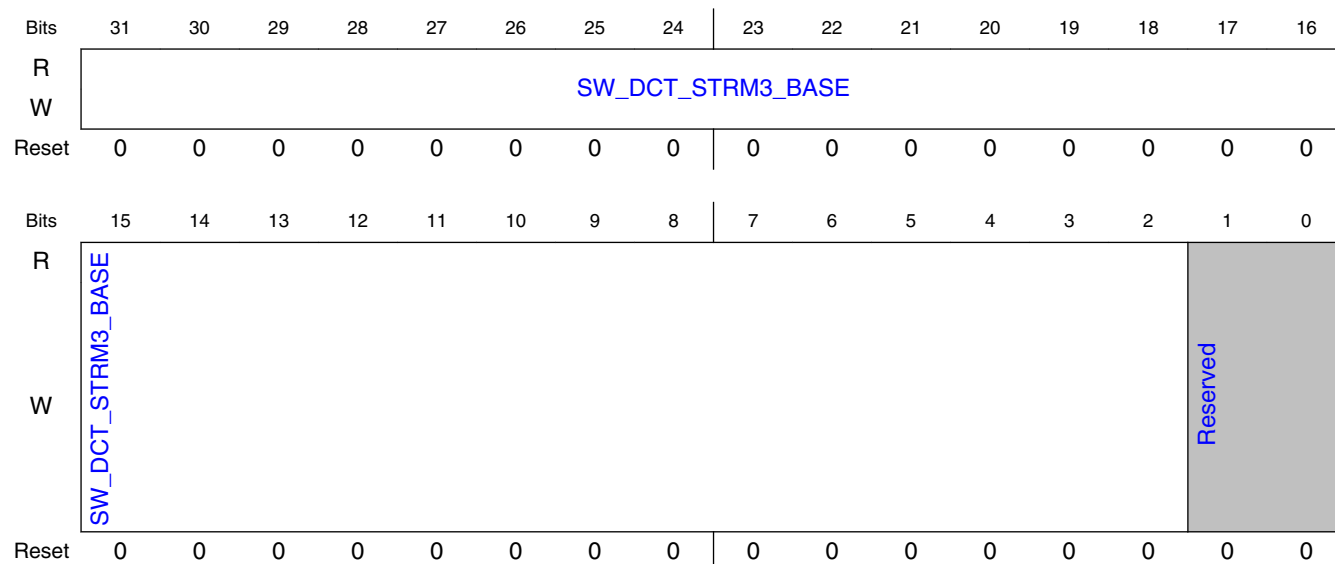
Field	Function
31-2 SW_DCT_STR M2_BASE	Base address for VP7/VP8 DCT stream MB row 2,2n+2
1-0 —	Reserved.

14.1.5.4.13 Base address for reference picture index 10 / List of VLC code lengths in first JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 3 base (SWREG24_VP7_VP8)

14.1.5.4.13.1 Offset

Register	Offset
SWREG24_VP7_VP8	60h

14.1.5.4.13.2 Diagram



14.1.5.4.13.3 Fields

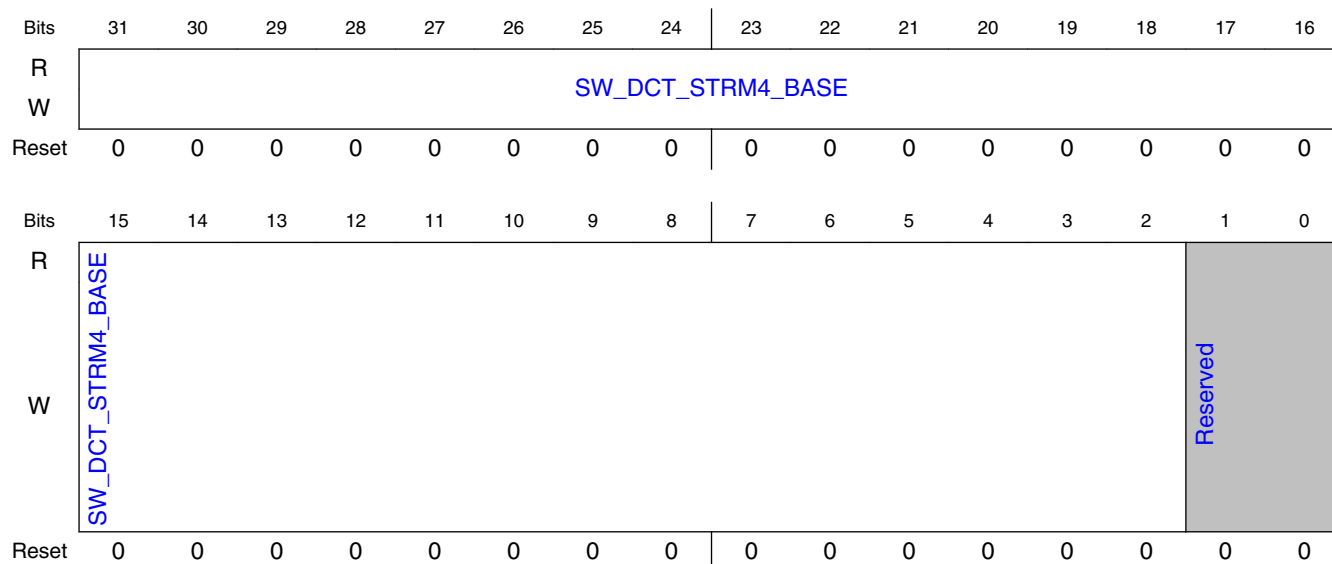
Field	Function
31-2 SW_DCT_STRM3_BASE	Base address for VP7/VP8 DCT stream MB row 3,2n+3
1-0 —	Reserved.

14.1.5.4.14 Base address for reference picture index 11 / List of VLC code lengths in second JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 4 base (SWREG25_VP7_VP8)

14.1.5.4.14.1 Offset

Register	Offset
SWREG25_VP7_VP8	64h

14.1.5.4.14.2 Diagram



14.1.5.4.14.3 Fields

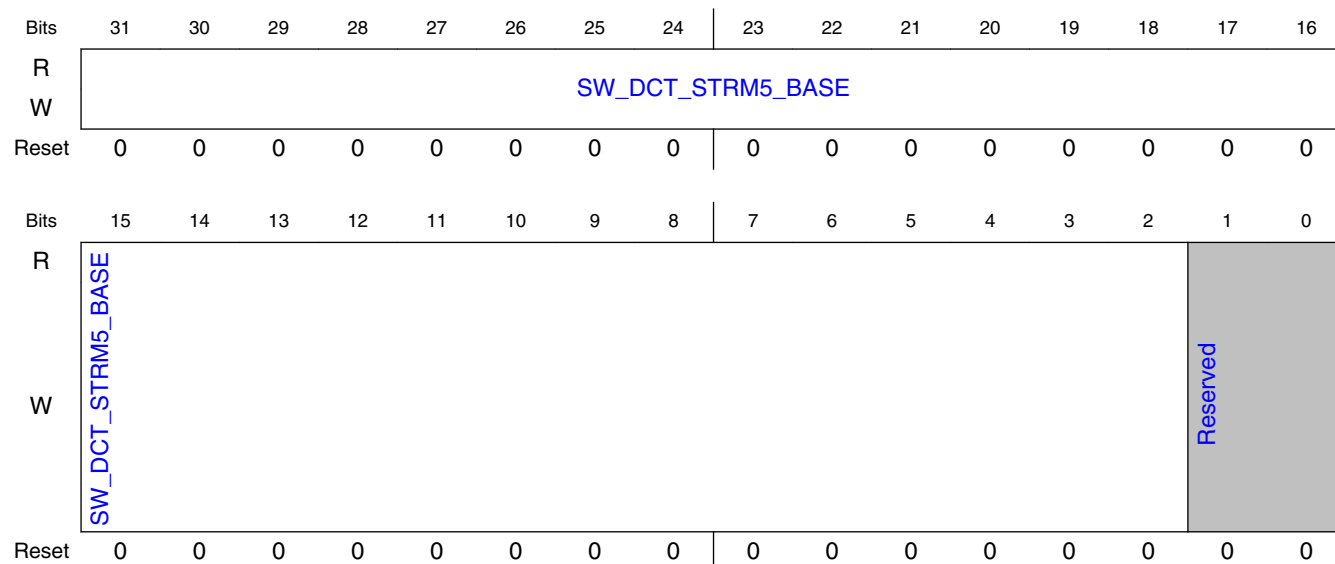
Field	Function
31-2 SW_DCT_STR M4_BASE	Base address for VP7/VP8 DCT stream MB row 4,2n+4
1-0 —	Reserved.

14.1.5.4.15 Base address for reference picture index 12 / List of VLC code lengths in second JPEG DC table / VP6 scan maps / VP7,VP8 DCT stream 5 base (SWREG26_VP7_VP8)

14.1.5.4.15.1 Offset

Register	Offset
SWREG26_VP7_VP8	68h

14.1.5.4.15.2 Diagram



14.1.5.4.15.3 Fields

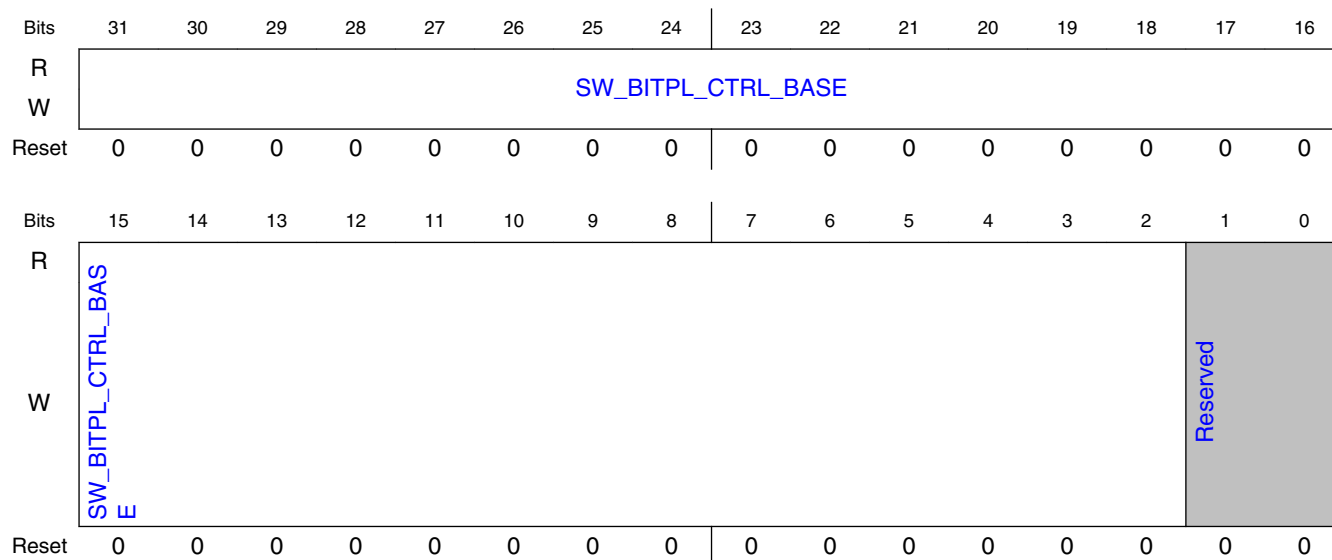
Field	Function
31-2 SW_DCT_STRM5_BASE	Base address for VP7/VP8 DCT stream MB row 5, $2n+5$
1-0 —	Reserved.

14.1.5.4.16 Base address for reference picture index 13 / VC-1 bitpl mbctrl or VP6,VP7,VP8 ctrl stream base /Progressive JPEG DC table (SWREG27_VC1)

14.1.5.4.16.1 Offset

Register	Offset
SWREG27_VC1	6Ch

14.1.5.4.16.2 Diagram



14.1.5.4.16.3 Fields

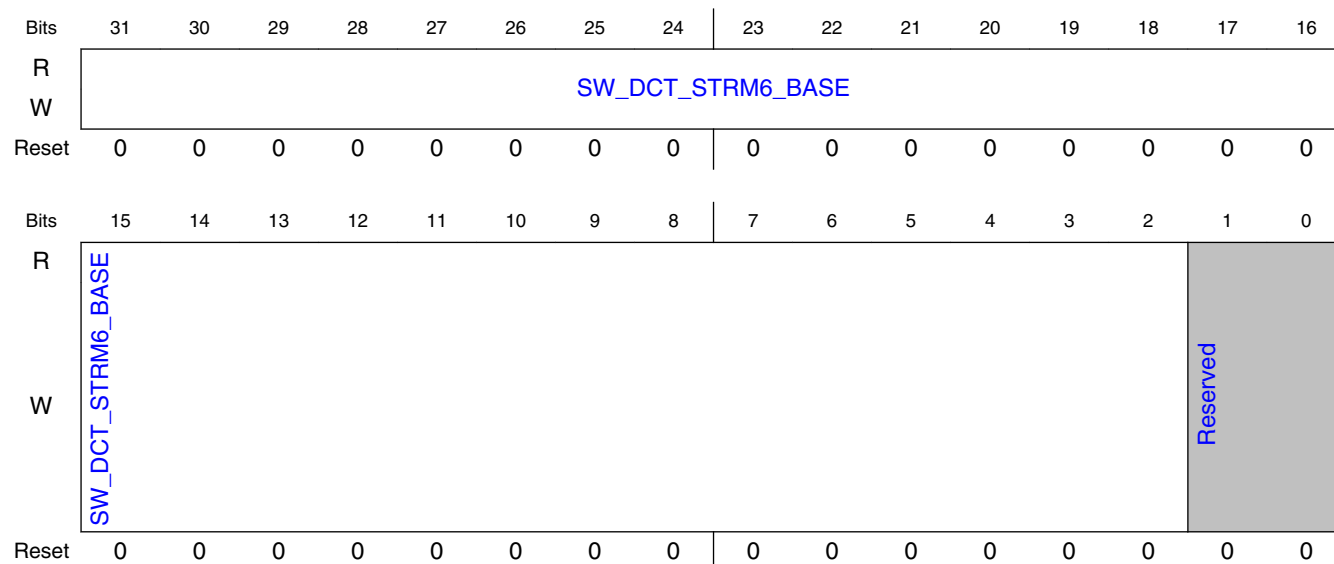
Field	Function
31-2 SW_BITPL_CTRL_BASE	VC-1: Base address for bitplane mb control VP6/VP7/VP8 : Base address for ctrl data stream. Used if multistream is enabled
1-0 —	Reserved.

14.1.5.4.17 Base address for reference picture index 14 / VP6 scan maps / Progressive JPEG DC table / VP7,VP8 DCT stream 6 base (SWREG28_VP7_VP8)

14.1.5.4.17.1 Offset

Register	Offset
SWREG28_VP7_VP8	70h

14.1.5.4.17.2 Diagram



14.1.5.4.17.3 Fields

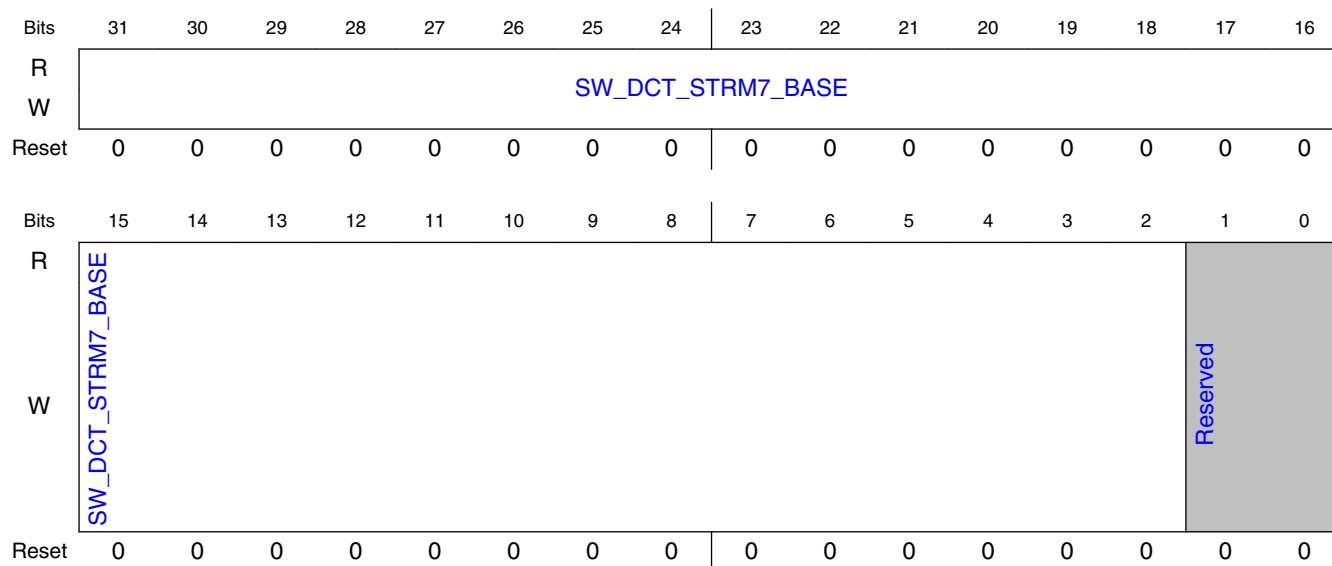
Field	Function
31-2 SW_DCT_STRM6_BASE	Base address for VP7/VP8 DCT stream MB row 6, 2n+6
1-0 —	Reserved.

14.1.5.4.18 Base address for reference picture index 15 / VP6 scan maps / VP7,VP8 DCT stream 7 base (SWREG29_VP7_VP8)

14.1.5.4.18.1 Offset

Register	Offset
SWREG29_VP7_VP8	74h

14.1.5.4.18.2 Diagram



14.1.5.4.18.3 Fields

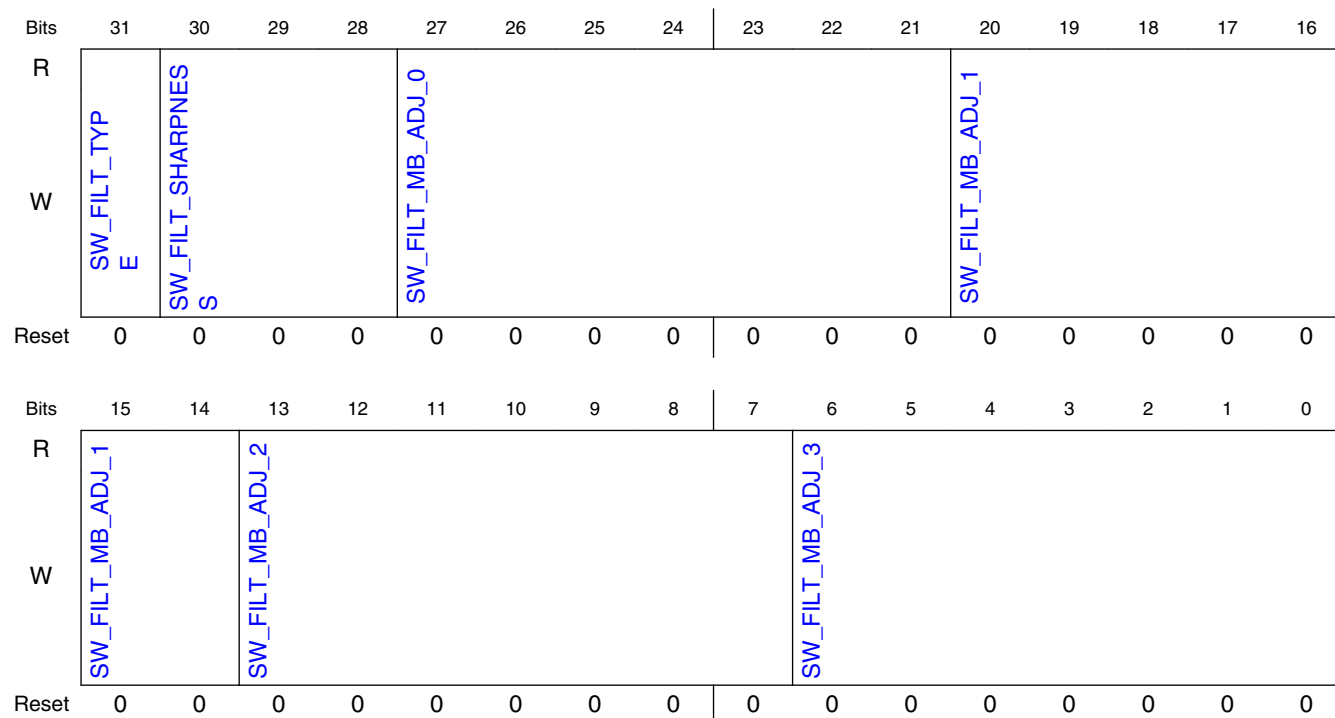
Field	Function
31-2 SW_DCT_STR M7_BASE	Base address for VP7/VP8 DCT stream MB row 7,2n+7
1-0 —	Reserved.

14.1.5.4.19 Reference picture numbers for index 0 and 1 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter mb level adjusts (SWREG30_VP7_VP8)

14.1.5.4.19.1 Offset

Register	Offset
SWREG30_VP7_VP8	78h

14.1.5.4.19.2 Diagram



14.1.5.4.19.3 Fields

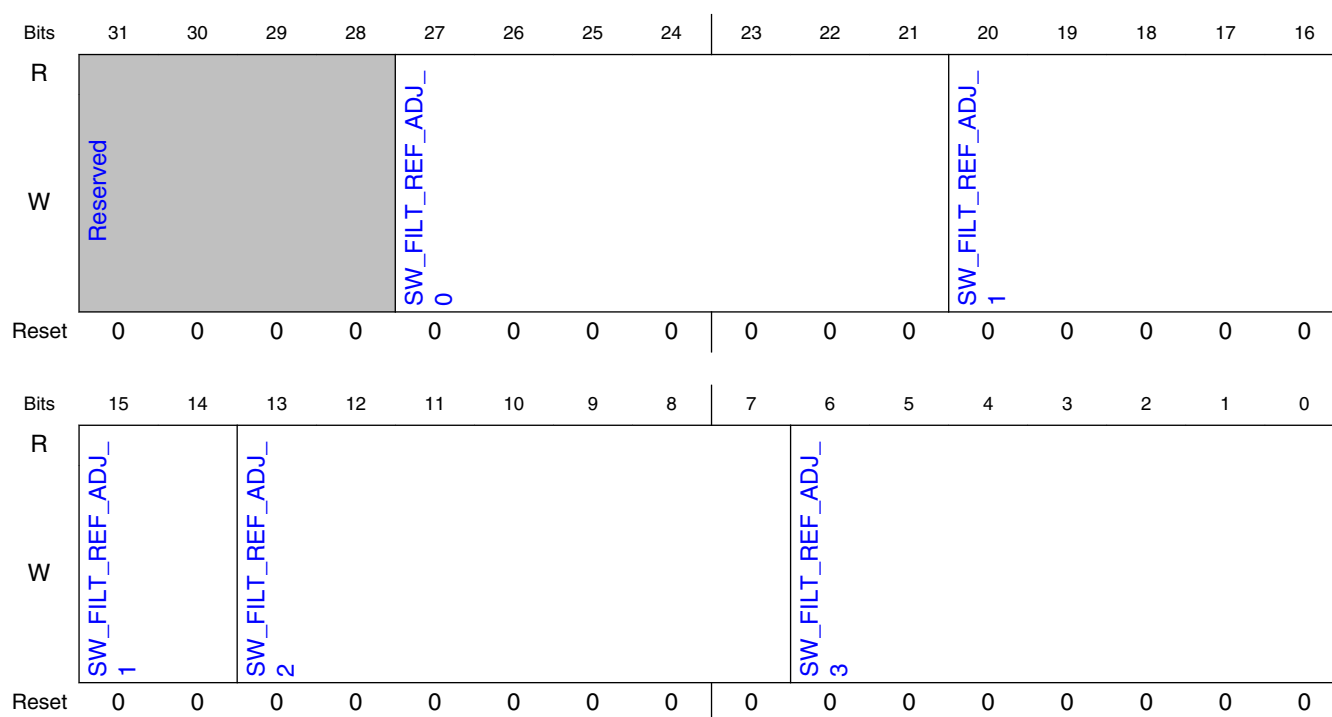
Field	Function
31 SW_FILT_TYPE	VP7/VP8 loop filter type
30-28 SW_FILT_SHARPNESS	VP7/VP8 loop filter sharpness
27-21 SW_FILT_MB_ADJ_0	VP7/VP8 filter level adjustment for MB type 0
20-14 SW_FILT_MB_ADJ_1	VP7/VP8 filter level adjustment for MB type 1
13-7 SW_FILT_MB_ADJ_2	VP7/VP8 filter level adjustment for MB type 2
6-0 SW_FILT_MB_ADJ_3	VP7/VP8 filter level adjustment for MB type 3

14.1.5.4.20 Reference picture numbers for index 2 and 3 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter ref pic level adjusts (SWREG31_VP7_VP8)

14.1.5.4.20.1 Offset

Register	Offset
SWREG31_VP7_VP8	7Ch

14.1.5.4.20.2 Diagram



14.1.5.4.20.3 Fields

Field	Function
31-28 —	Reserved.
27-21 SW_FILT_REF_ADJ_0	VP7/VP8 filter level adjustment for reference frame type 0
20-14	VP7/VP8 filter level adjustment for reference frame type 1

Table continues on the next page...

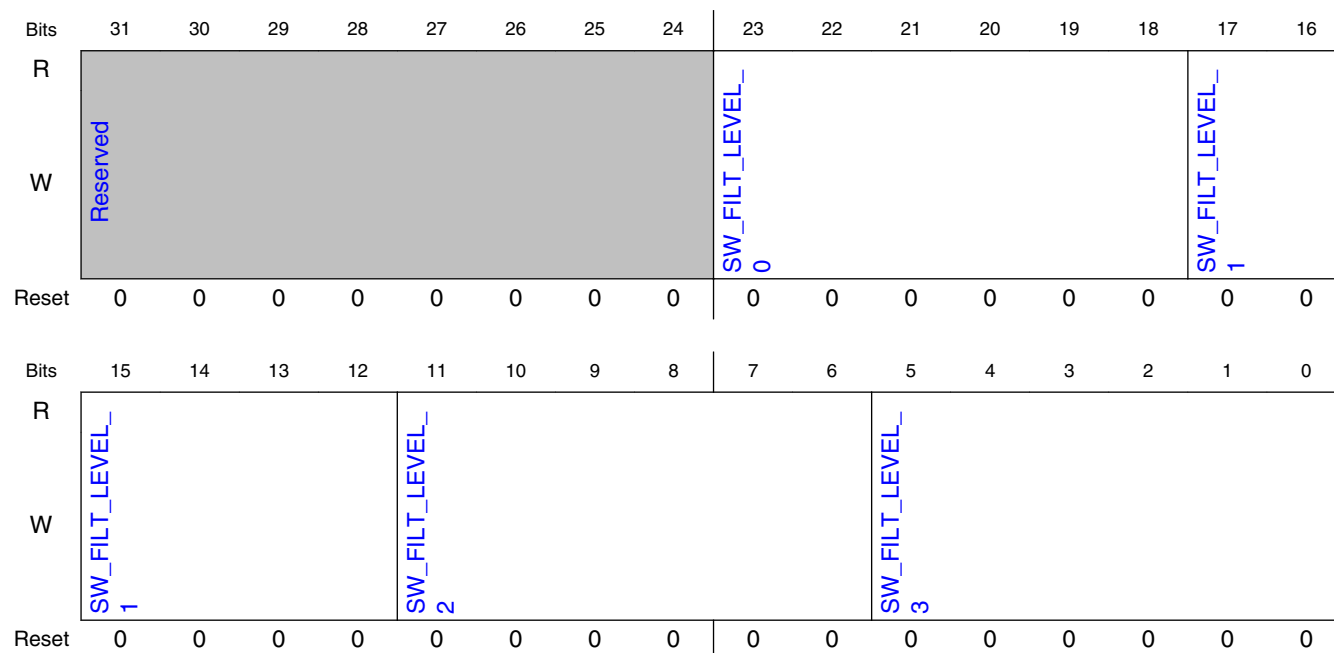
Field	Function
SW_FILT_REF_ADJ_1	
13-7 SW_FILT_REF_ADJ_2	VP7/VP8 filter level adjustment for reference frame type 2
6-0 SW_FILT_REF_ADJ_3	VP7/VP8 filter level adjustment for reference frame type 3

14.1.5.4.21 Reference picture numbers for index 4 and 5 (H264 VLC) / VP6 scan maps / VP7,VP8 loop filter levels (SWREG32_VP7_VP8)

14.1.5.4.21.1 Offset

Register	Offset
SWREG32_VP7_VP8	80h

14.1.5.4.21.2 Diagram



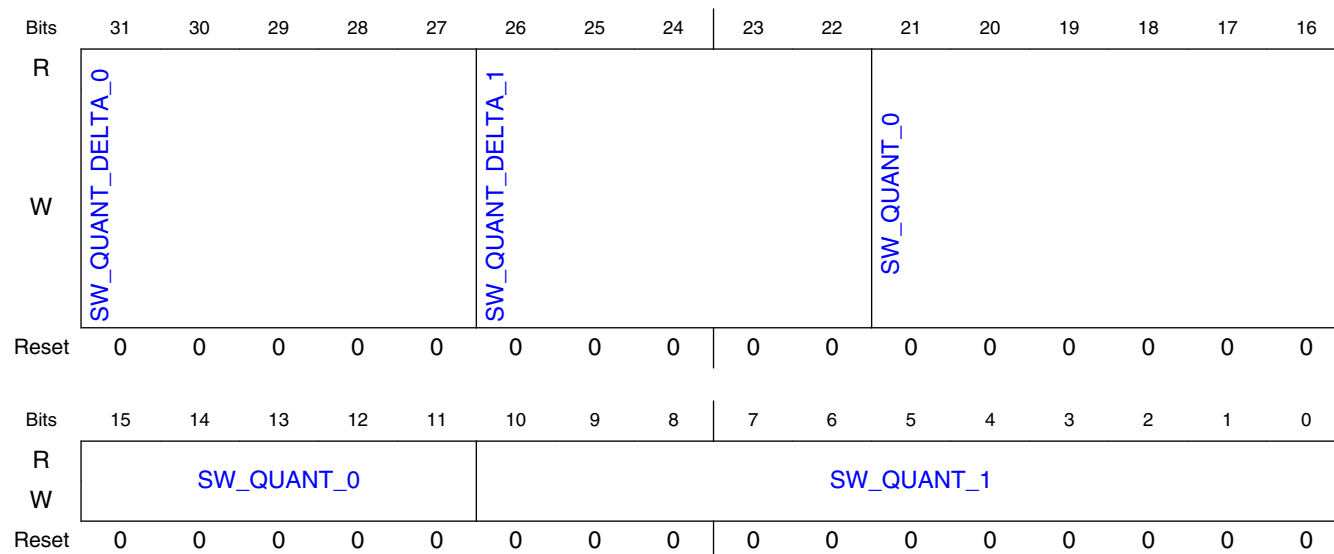
14.1.5.4.21.3 Fields

Field	Function
31-24 —	Reserved.
23-18 SW_FILT_LEVE L_0	VP7/VP8 filter level value for reference frame type 0
17-12 SW_FILT_LEVE L_1	VP7/VP8 filter level value for reference frame type 1
11-6 SW_FILT_LEVE L_2	VP7/VP8 filter level value for reference frame type 2
5-0 SW_FILT_LEVE L_3	VP7/VP8 filter level value for reference frame type 3

14.1.5.4.22 Reference picture numbers for index 6 and 7 (H264 VLC) / VP6 scan maps / VP7,VP8 quantization values (SWREG33_VP7_VP8)**14.1.5.4.22.1 Offset**

Register	Offset
SWREG33_VP7_VP8	84h

14.1.5.4.22.2 Diagram



14.1.5.4.22.3 Fields

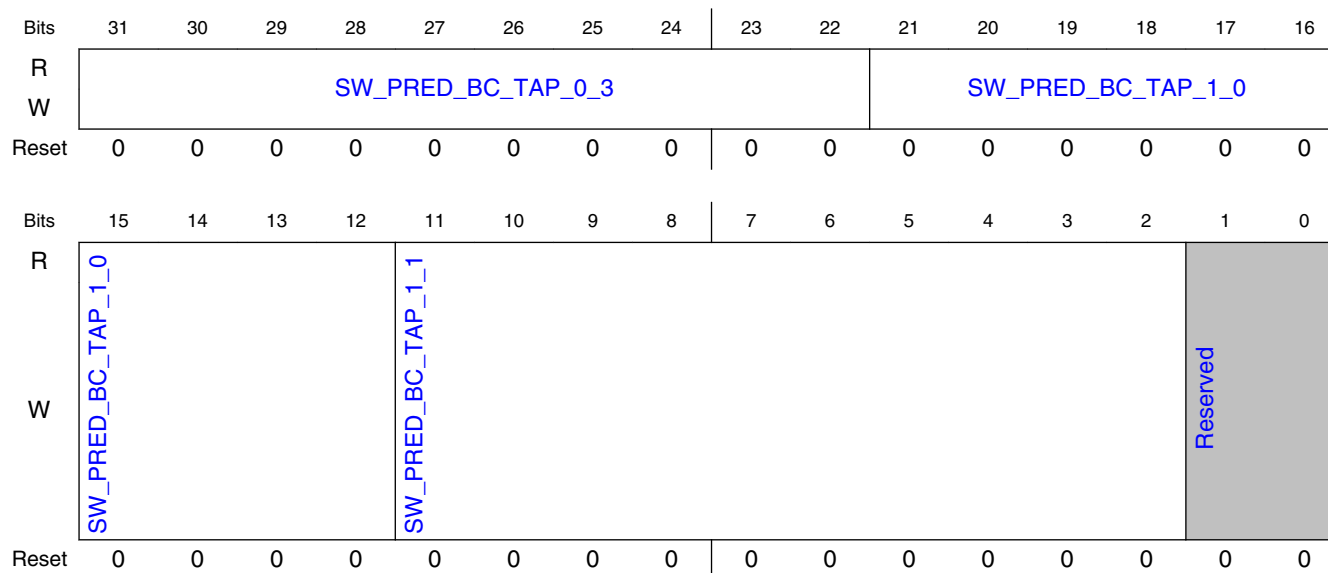
Field	Function
31-27 SW_QUANT_DELTA_0	VP8 quantisizer delta 0
26-22 SW_QUANT_DELTA_1	VP8 quantisizer delta 1
21-11 SW_QUANT_0	VP7: QP (11 bit) VP8: quantisizer value for LUT (7 bit)
10-0 SW_QUANT_1	VP7: QP (11 bit) VP8: quantisizer value for LUT (7 bit)

14.1.5.4.23 Reference picture numbers for index 8 and 9 (H264 VLC) / MPEG4, VC1, VPx prediction filter taps (SWREG34_H263)

14.1.5.4.23.1 Offset

Register	Offset
SWREG34_H263	88h

14.1.5.4.23.2 Diagram



14.1.5.4.23.3 Fields

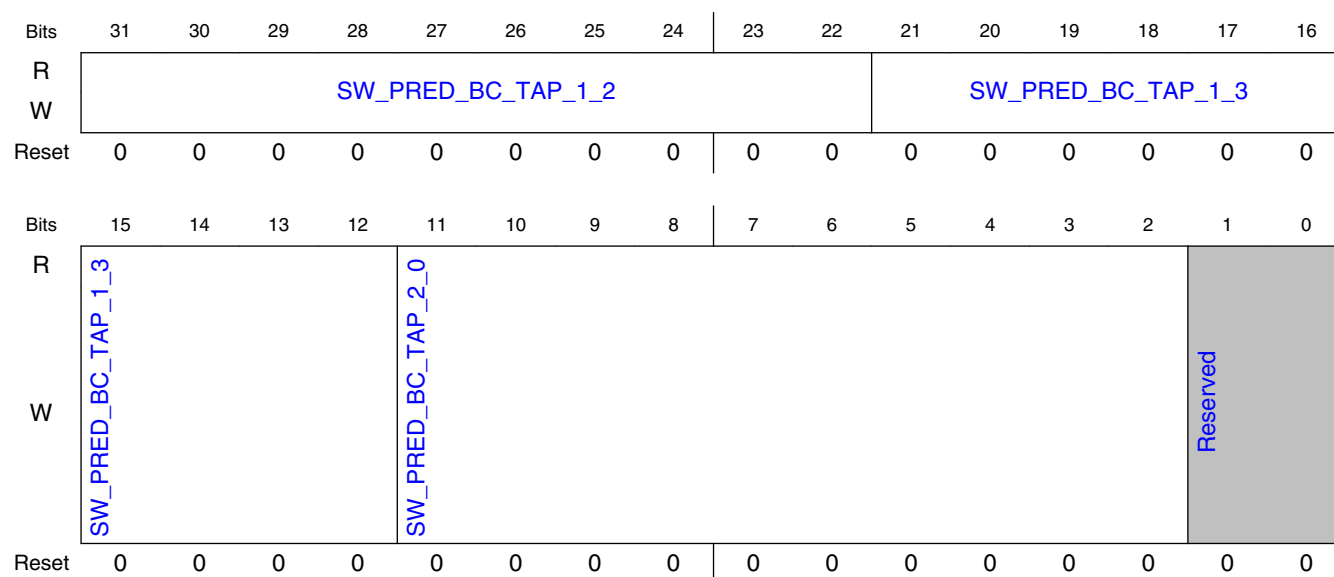
Field	Function
31-22 SW_PRED_BC_TAP_0_3	Prediction filter set 0, tap 3
21-12 SW_PRED_BC_TAP_1_0	Prediction filter set 1, tap 0
11-2 SW_PRED_BC_TAP_1_1	Prediction filter set 1, tap 1
1-0 —	Reserved.

14.1.5.4.24 Reference picture numbers for index 10 and 11 (H264 VLC) / VC1, VPx prediction filter taps (SWREG35_VC1)

14.1.5.4.24.1 Offset

Register	Offset
SWREG35_VC1	8Ch

14.1.5.4.24.2 Diagram



14.1.5.4.24.3 Fields

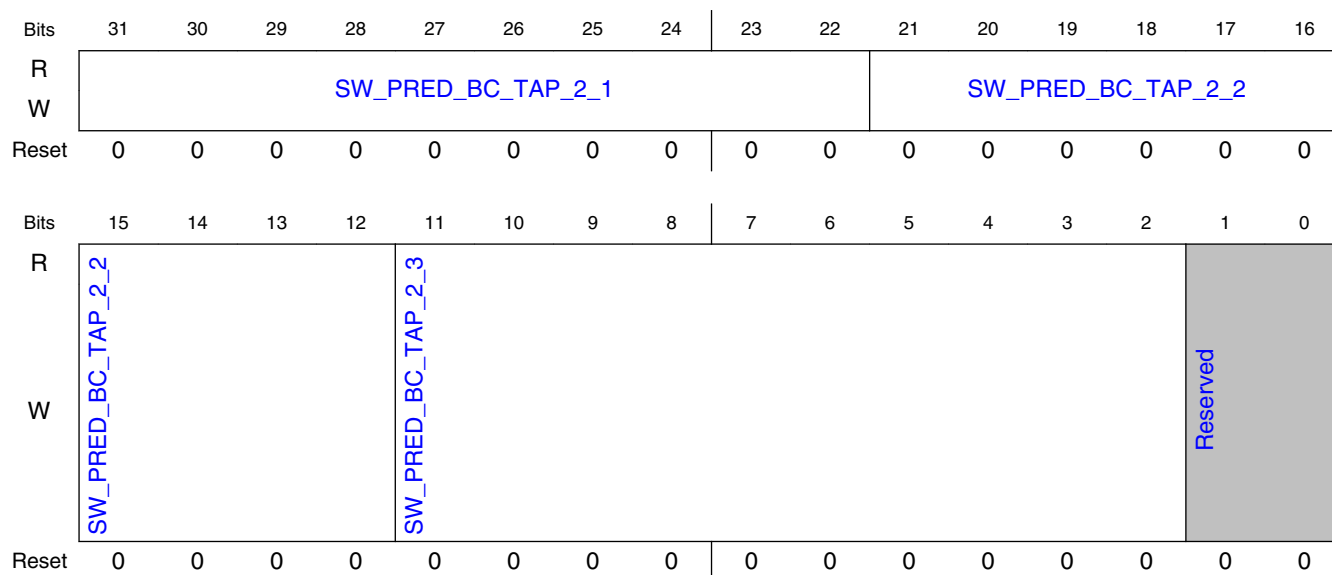
Field	Function
31-22 SW_PRED_BC_TAP_1_2	Prediction filter set 1, tap 2
21-12 SW_PRED_BC_TAP_1_3	Prediction filter set 1, tap 3
11-2 SW_PRED_BC_TAP_2_0	Prediction filter set 2, tap 0
1-0 —	Reserved.

14.1.5.4.25 Reference picture numbers for index 12 and 13 (H264 VLC) / VC1, VPx prediction filter taps (SWREG36_VC1)

14.1.5.4.25.1 Offset

Register	Offset
SWREG36_VC1	90h

14.1.5.4.25.2 Diagram



14.1.5.4.25.3 Fields

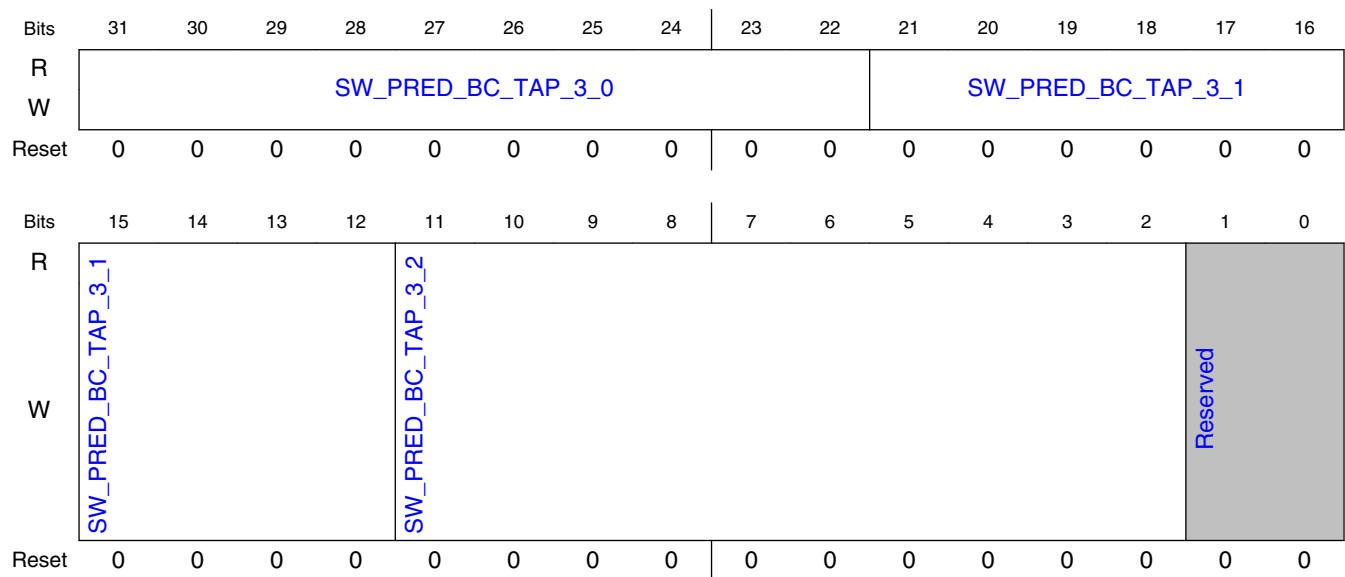
Field	Function
31-22 SW_PRED_BC_TAP_2_1	Prediction filter set 2, tap 1
21-12 SW_PRED_BC_TAP_2_2	Prediction filter set 2, tap 2
11-2 SW_PRED_BC_TAP_2_3	Prediction filter set 2, tap 3
1-0 —	Reserved.

14.1.5.4.26 Reference picture numbers for index 14 and 15 (H264 VLC) / VPx prediction filter taps (SWREG37_VP6_VP7_VP8)

14.1.5.4.26.1 Offset

Register	Offset
SWREG37_VP6_VP7_VP8	94h

14.1.5.4.26.2 Diagram



14.1.5.4.26.3 Fields

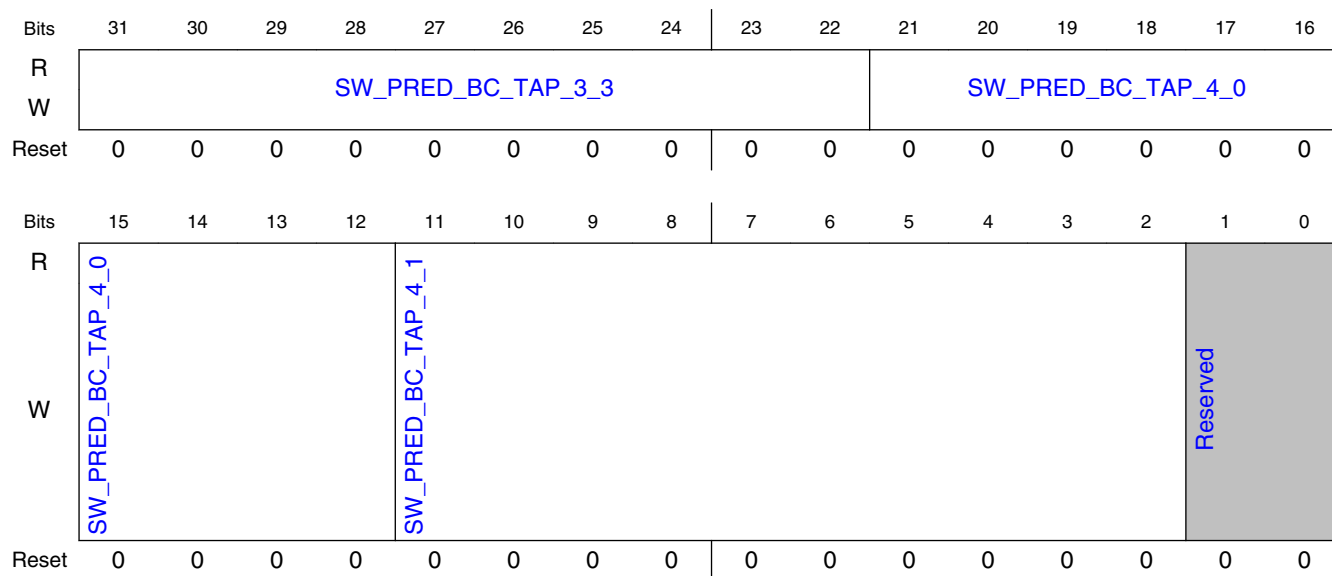
Field	Function
31-22 SW_PRED_BC_TAP_3_0	Prediction filter set 3, tap 0
21-12 SW_PRED_BC_TAP_3_1	Prediction filter set 3, tap 1
11-2 SW_PRED_BC_TAP_3_2	Prediction filter set 3, tap 2
1-0 —	Reserved.

14.1.5.4.27 Reference picture long term flags (H264 VLC) / VPx prediction filter taps (SWREG38_VP6_VP7_VP8)

14.1.5.4.27.1 Offset

Register	Offset
SWREG38_VP6_VP7_VP8	98h

14.1.5.4.27.2 Diagram



14.1.5.4.27.3 Fields

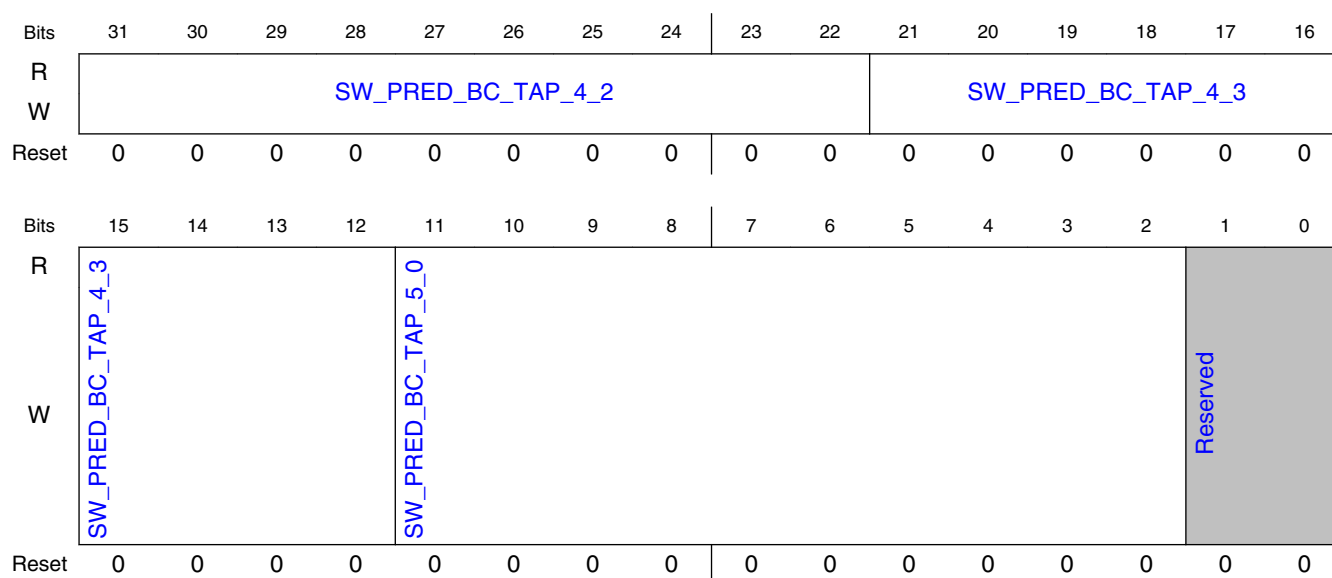
Field	Function
31-22 SW_PRED_BC_TAP_3_3	Prediction filter set 3, tap 3
21-12 SW_PRED_BC_TAP_4_0	Prediction filter set 4, tap 0
11-2 SW_PRED_BC_TAP_4_1	Prediction filter set 4, tap 1
1-0 —	Reserved.

14.1.5.4.28 Reference picture valid flags (H264 VLC) / VPx prediction filter taps (SWREG39_VP6_VP7_VP8)

14.1.5.4.28.1 Offset

Register	Offset
SWREG39_VP6_VP7_VP8	9Ch

14.1.5.4.28.2 Diagram



14.1.5.4.28.3 Fields

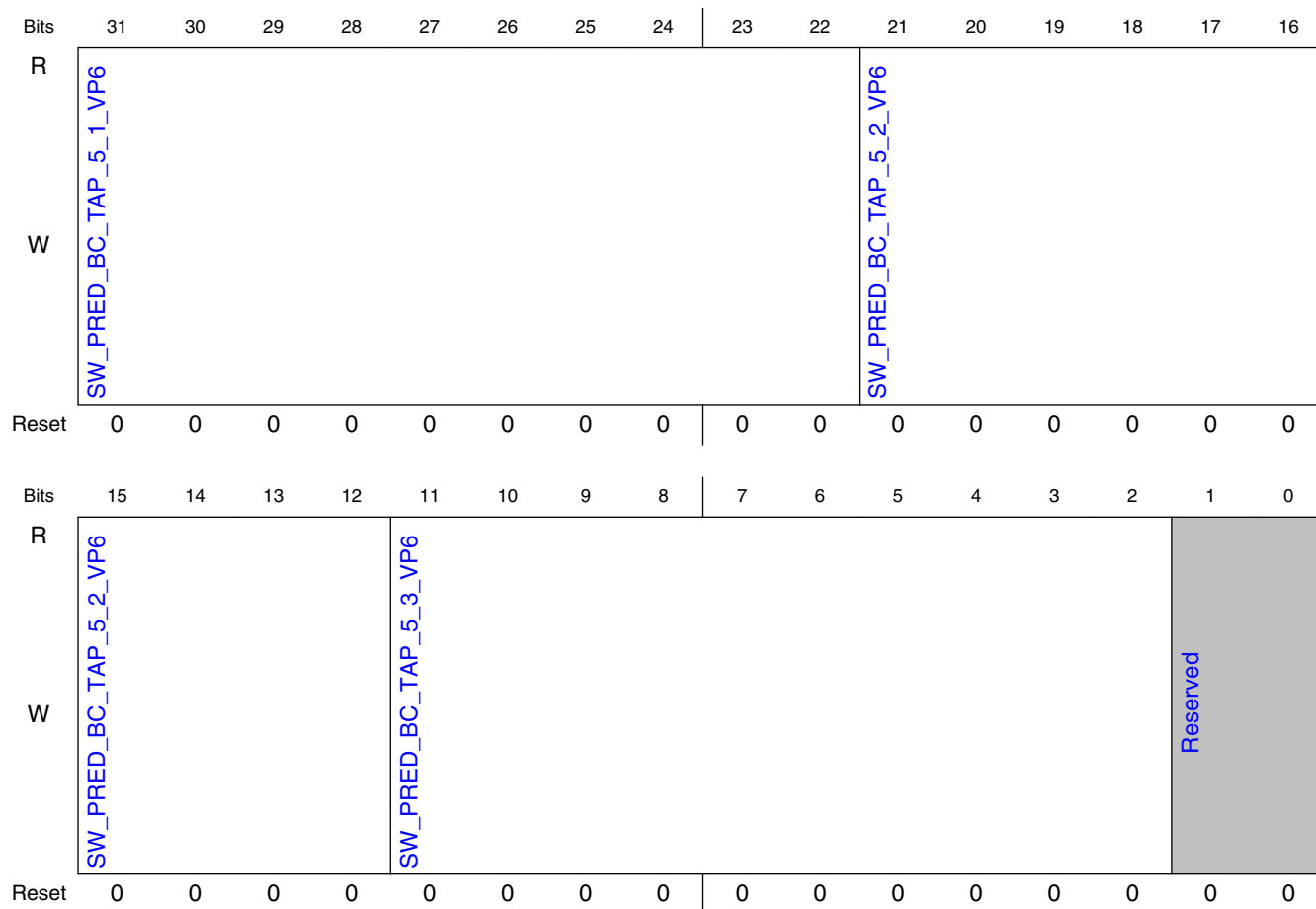
Field	Function
31-22 SW_PRED_BC_TAP_4_2	Prediction filter set 4, tap 2
21-12 SW_PRED_BC_TAP_4_3	Prediction filter set 4, tap 3
11-2 SW_PRED_BC_TAP_5_0	Prediction filter set 5, tap 0
1-0 —	Reserved.

14.1.5.4.29 bi_dir initial ref pic list register (0-2) / VP6 prediction filter taps / Progressive JPEG Cb ACDC coefficient base (SWRE G42_VP6)

14.1.5.4.29.1 Offset

Register	Offset
SWREG42_VP6	A8h

14.1.5.4.29.2 Diagram



14.1.5.4.29.3 Fields

Field	Function
31-22	Prediction filter set 5, tap 1

Table continues on the next page...

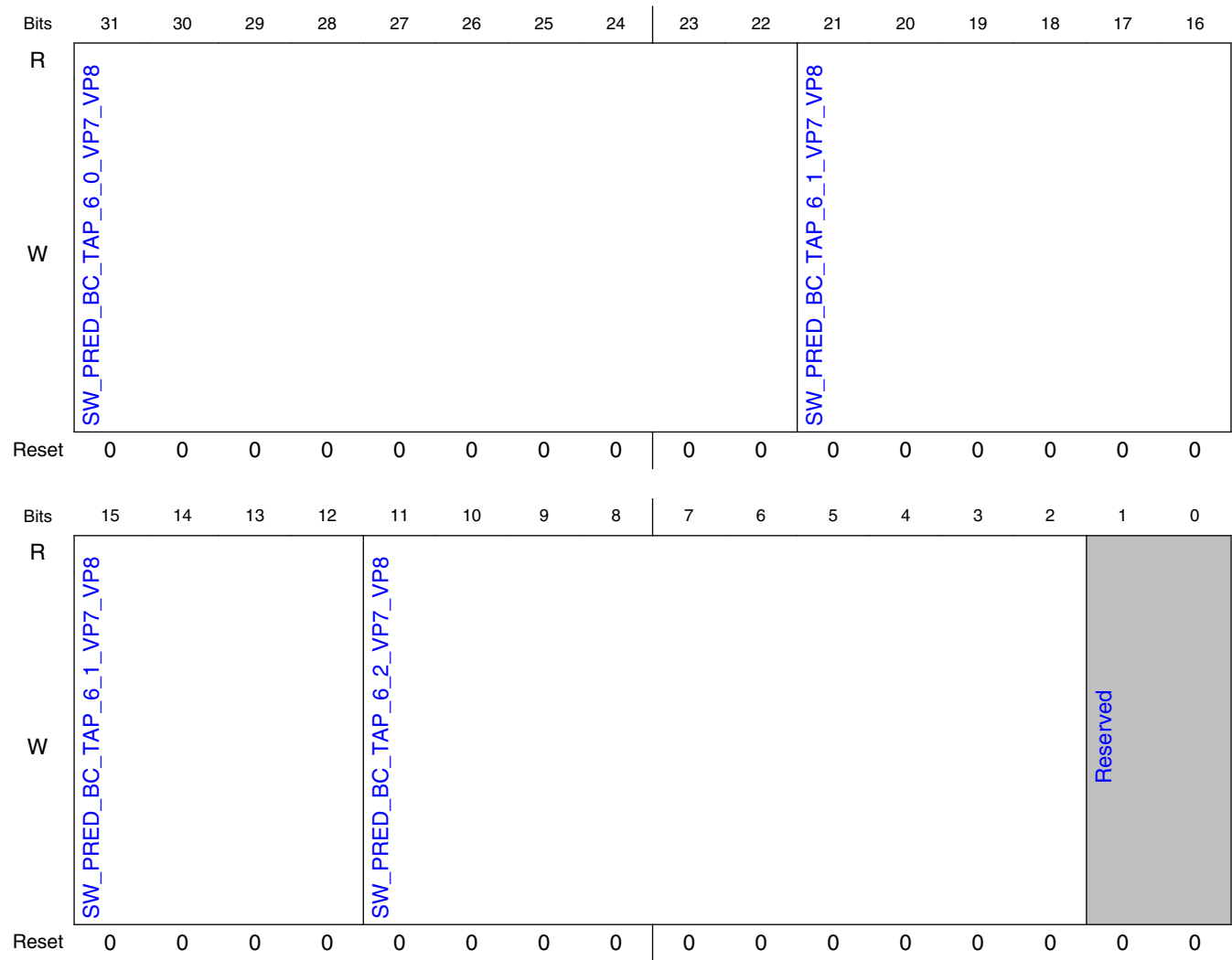
Field	Function
SW_PRED_BC_TAP_5_1_VP6	
21-12 SW_PRED_BC_TAP_5_2_VP6	Prediction filter set 5, tap 2
11-2 SW_PRED_BC_TAP_5_3_VP6	Prediction filter set 5, tap 3
1-0 —	Reserved.

14.1.5.4.30 bi-dir initial ref pic list register (3-5) / VP6 prediction filter taps / Progressive JPEG Cr ACDC coefficient base (SWREG43_VP7_VP8)

14.1.5.4.30.1 Offset

Register	Offset
SWREG43_VP7_VP8	ACh

14.1.5.4.30.2 Diagram



14.1.5.4.30.3 Fields

Field	Function
31-22 SW_PRED_BC_TAP_6_0_VP7_VP8	Prediction filter set 6, tap 0
21-12 SW_PRED_BC_TAP_6_1_VP7_VP8	Prediction filter set 6, tap 1
11-2	Prediction filter set 6, tap 2

Table continues on the next page...

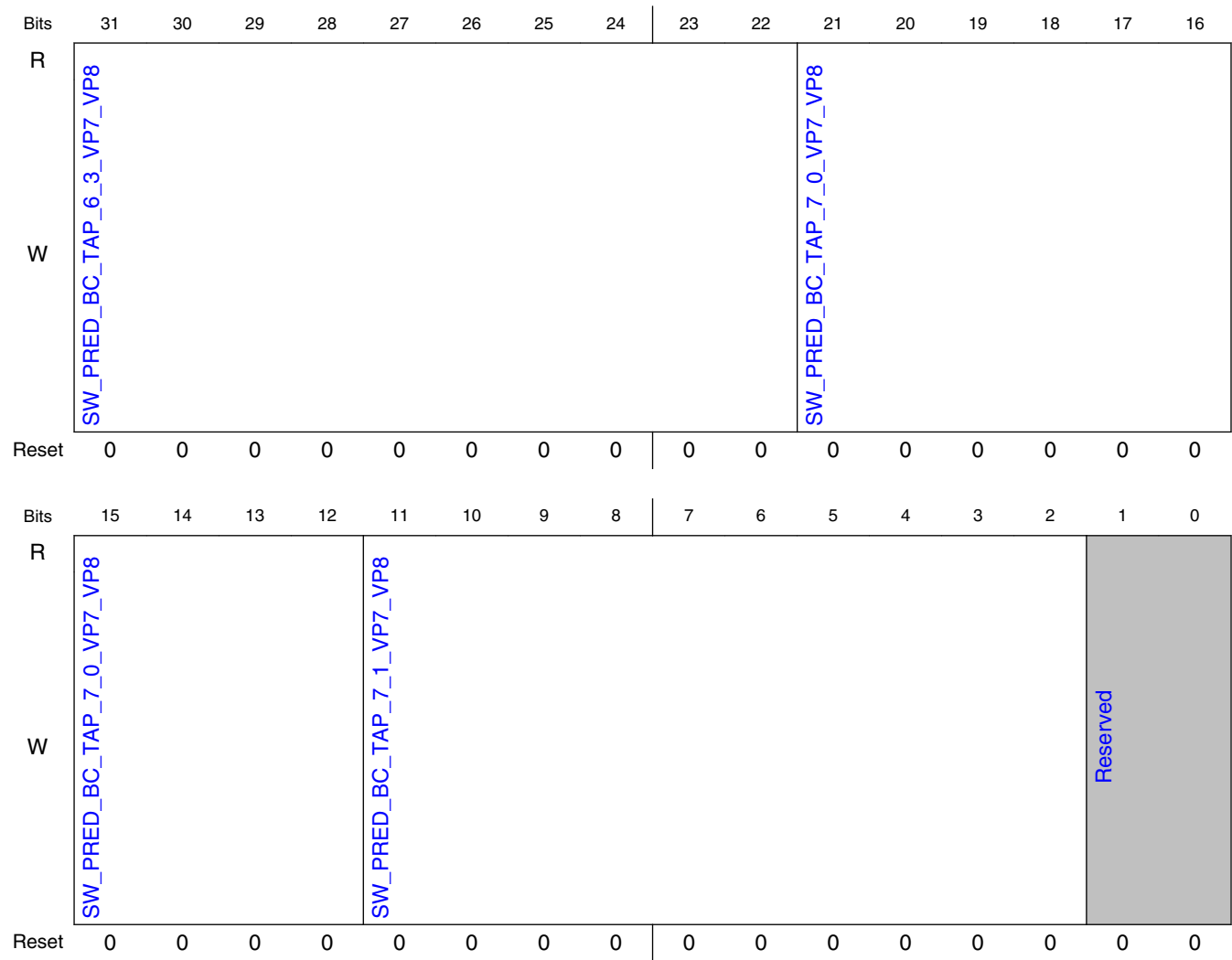
Field	Function
SW_PRED_BC_ TAP_6_2_VP7_ VP8	
1-0 —	Reserved.

14.1.5.4.31 bi-dir initial ref pic list register (6-8) / VP6 prediction filter taps (SWREG44_VP7_VP8)

14.1.5.4.31.1 Offset

Register	Offset
SWREG44_VP7_VP8	B0h

14.1.5.4.31.2 Diagram



14.1.5.4.31.3 Fields

Field	Function
31-22 SW_PRED_BC_TAP_6_3_VP7_VP8	Prediction filter set 6, tap 3
21-12 SW_PRED_BC_TAP_7_0_VP7_VP8	Prediction filter set 7, tap 0
11-2	Prediction filter set 7, tap 1

Table continues on the next page...

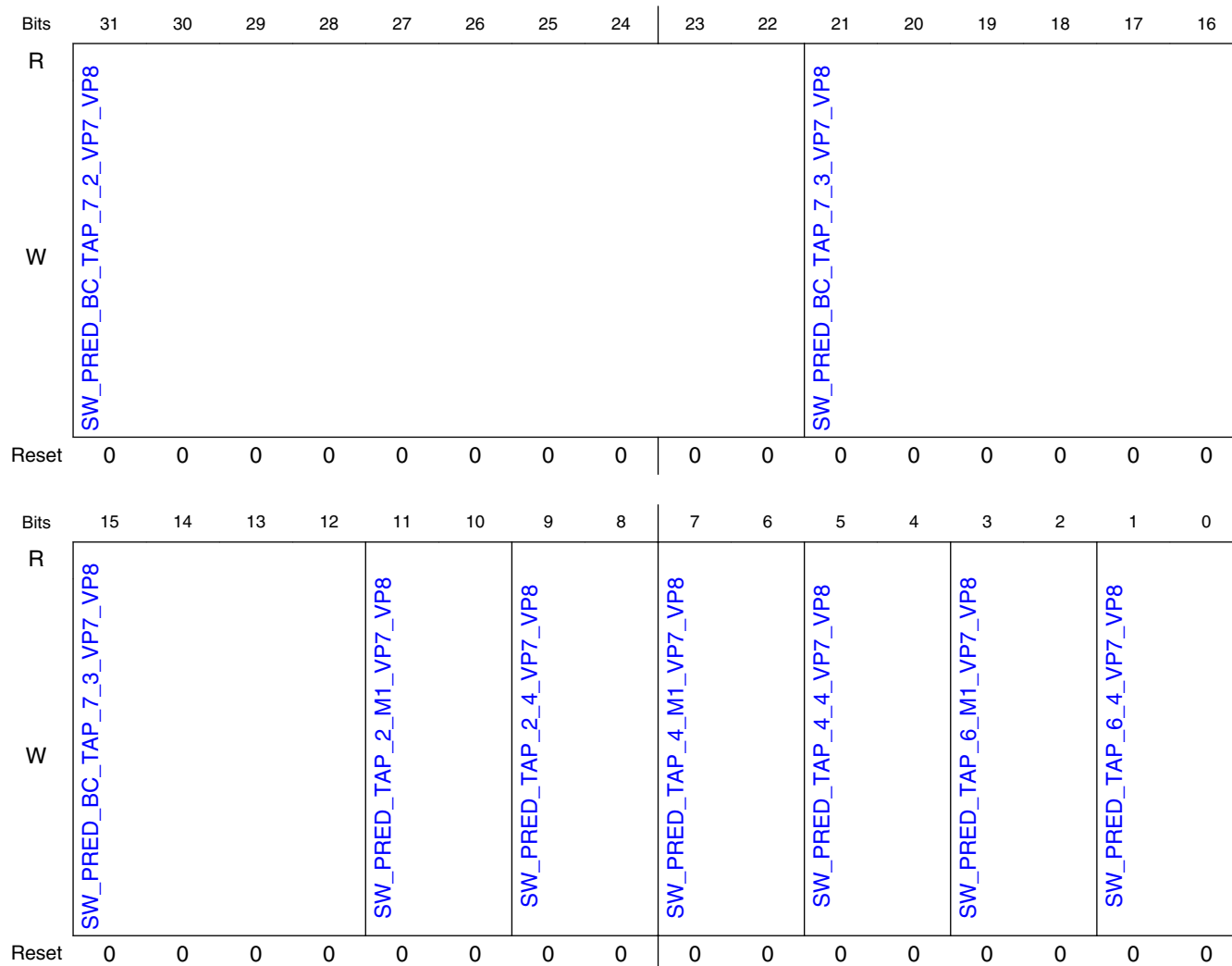
Field	Function
SW_PRED_BC_ TAP_7_1_VP7_ VP8	
1-0 —	Reserved.

14.1.5.4.32 bi-dir initial ref pic list register (9-11) / VP6 prediction filter taps (SWREG45_VP7_VP8)

14.1.5.4.32.1 Offset

Register	Offset
SWREG45_VP7_VP8	B4h

14.1.5.4.32.2 Diagram



14.1.5.4.32.3 Fields

Field	Function
31-22 SW_PRED_BC_TAP_7_2_VP7_VP8	Prediction filter set 7, tap 2
21-12 SW_PRED_BC_TAP_7_3_VP7_VP8	Prediction filter set 7, tap 3
11-10	Additional Prediction filter tap -1 for set 2

Table continues on the next page...

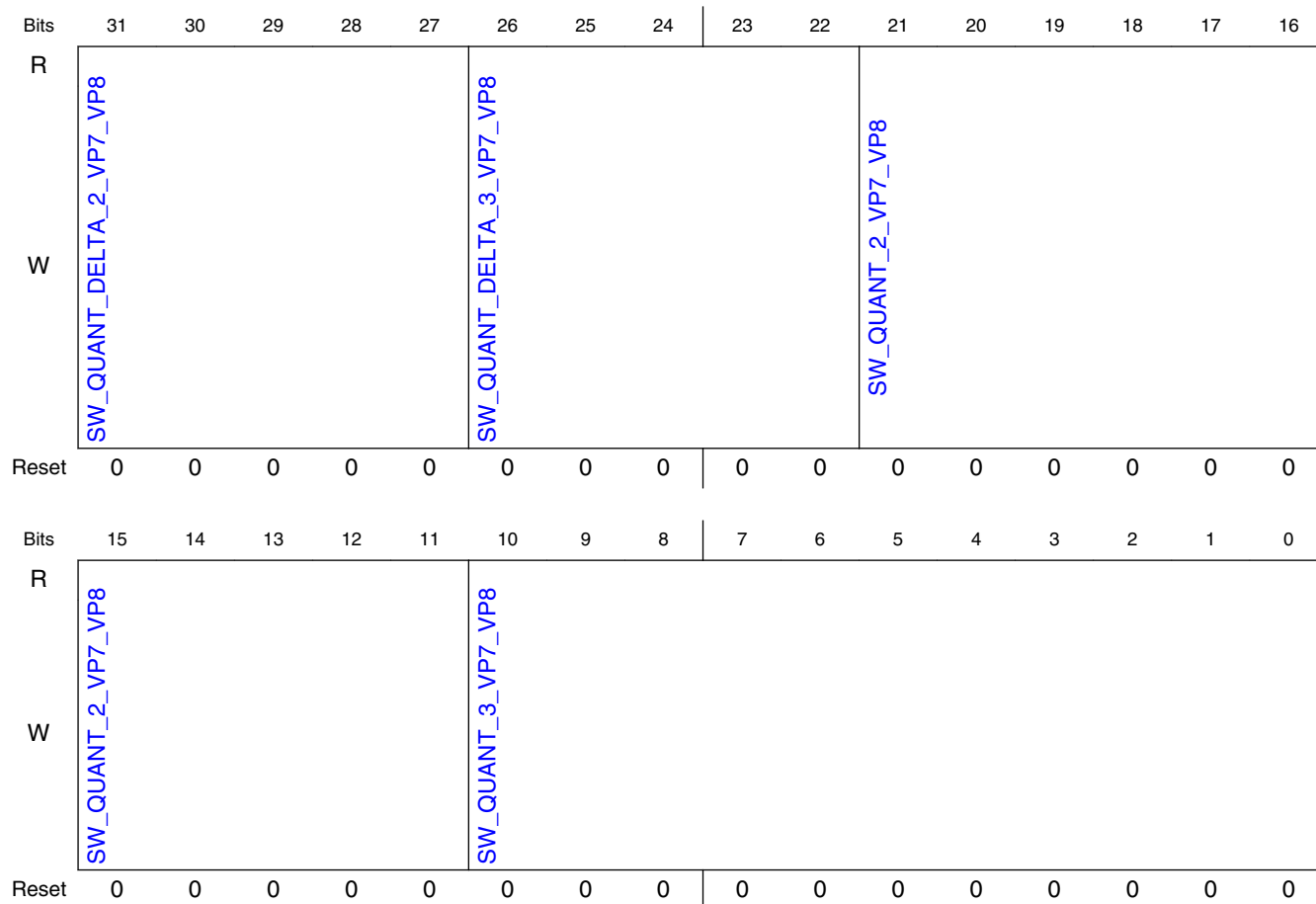
Field	Function
SW_PRED_TAP _2_M1_VP7_VP 8	
9-8 SW_PRED_TAP _2_4_VP7_VP8	Additional Prediction filter tap 4 for set 2
7-6 SW_PRED_TAP _4_M1_VP7_VP 8	Additional Prediction filter tap -1 for set 4
5-4 SW_PRED_TAP _4_4_VP7_VP8	Additional Prediction filter tap 4 for set 4
3-2 SW_PRED_TAP _6_M1_VP7_VP 8	Additional Prediction filter tap -1 for set 6
1-0 SW_PRED_TAP _6_4_VP7_VP8	Additional Prediction filter tap 4 for set 6

14.1.5.4.33 bi-dir initial ref pic list register (12-14) / VP7,VP8 quantization values (SWREG46_VP7_VP8)

14.1.5.4.33.1 Offset

Register	Offset
SWREG46_VP7_VP8	B8h

14.1.5.4.33.2 Diagram



14.1.5.4.33.3 Fields

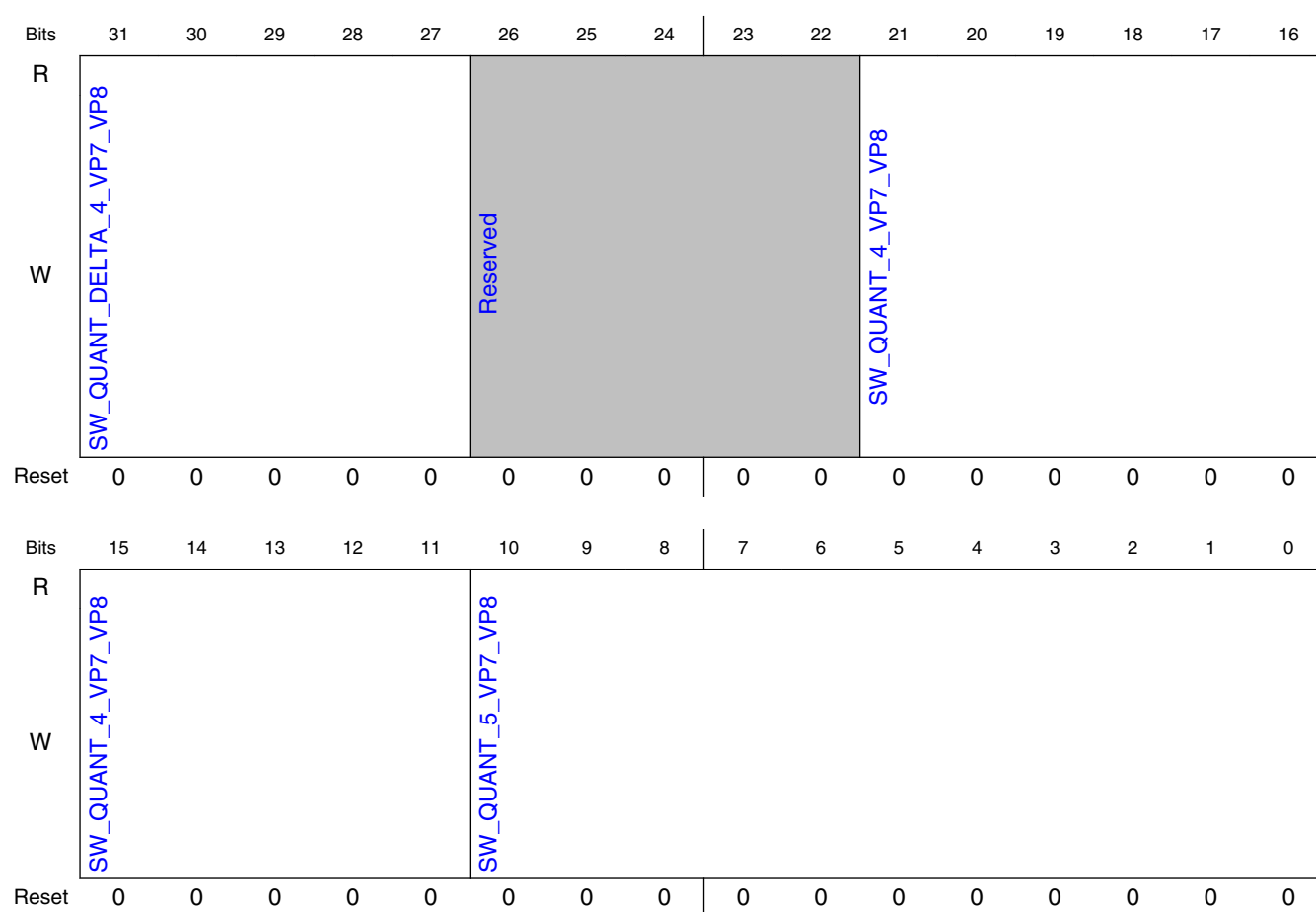
Field	Function
31-27 SW_QUANT_DELTA_2_VP7_VP8	VP8 quantisizer delta 2
26-22 SW_QUANT_DELTA_3_VP7_VP8	VP8 quantisizer delta 3
21-11 SW_QUANT_2_VP7_VP8	VP7: QP (11 bit) VP8: quantisizer value for LUT (7 bit)
10-0 SW_QUANT_3_VP7_VP8	VP7: QP (11 bit) VP8: quantisizer value for LUT (7 bit)

14.1.5.4.34 bi-dir and P fwd initial ref pic list register (15 and P 0-3) / VP7,VP8 quantization values (SWREG47_VP7_VP8)

14.1.5.4.34.1 Offset

Register	Offset
SWREG47_VP7_VP8	BCh

14.1.5.4.34.2 Diagram



14.1.5.4.34.3 Fields

Field	Function
31-27	VP8 quantisizer delta 4

Table continues on the next page...

Field	Function
SW_QUANT_D ELTA_4_VP7_V P8	
26-22 —	Reserved.
21-11 SW_QUANT_4_ VP7_VP8	VP7 QP (11 bit)
10-0 SW_QUANT_5_ VP7_VP8	VP7 QP (11 bit)

14.2 VPU G2 (VPU_G2)

14.2.1 Overview

This block details the G2 hardware-based decoder and API. The decoder is able to decode HEVC standard Main/Main 10 profile compatible video streams, and Google's VP9 Profile 0 compatible video streams. The G2 decoder conforms to the HEVC Main/Main 10 profiles and can decode streams up to level 5.1.

14.2.2 Video Frame Storage Format

The HW decoder may support two storage formats for frames, raster-scan, tiled without being compressed and compressed output. The application using the decoder may specify upon initialization which format is preferred, depending on HW support. The tiled format generally speaking offers better bus utilization and performance, however the decoder output must be converted back to raster-scan format prior to displaying it, except that GPU supports tiled format displaying. This postprocessing is beyond the scope of G2 decoder.

14.2.2.1 Raster-scan format

The output picture of the decoder is in semi-planar YCbCr 4:2:0 format, i.e. luminance data forms one plane in memory, and chrominance data forms the other. The output picture has to be stored linearly and contiguously in the memory. The interleaved

chrominance block has to be stored right after the luminance block in external memory as shown in Figure . The number of bytes in one luminance row must be divisible by 16, which means there will be padded 0 in the right edge when necessary.

The interleaved chrominance is stored raster-scanned in memory in Cb0Cr0Cb1Cr1Cb2Cr2Cb3... format.

The decoder output picture for interlaced sequence is always stored as an interlaced frame where even lines belongs to top field and odd lines belongs to bottom field of the frame.

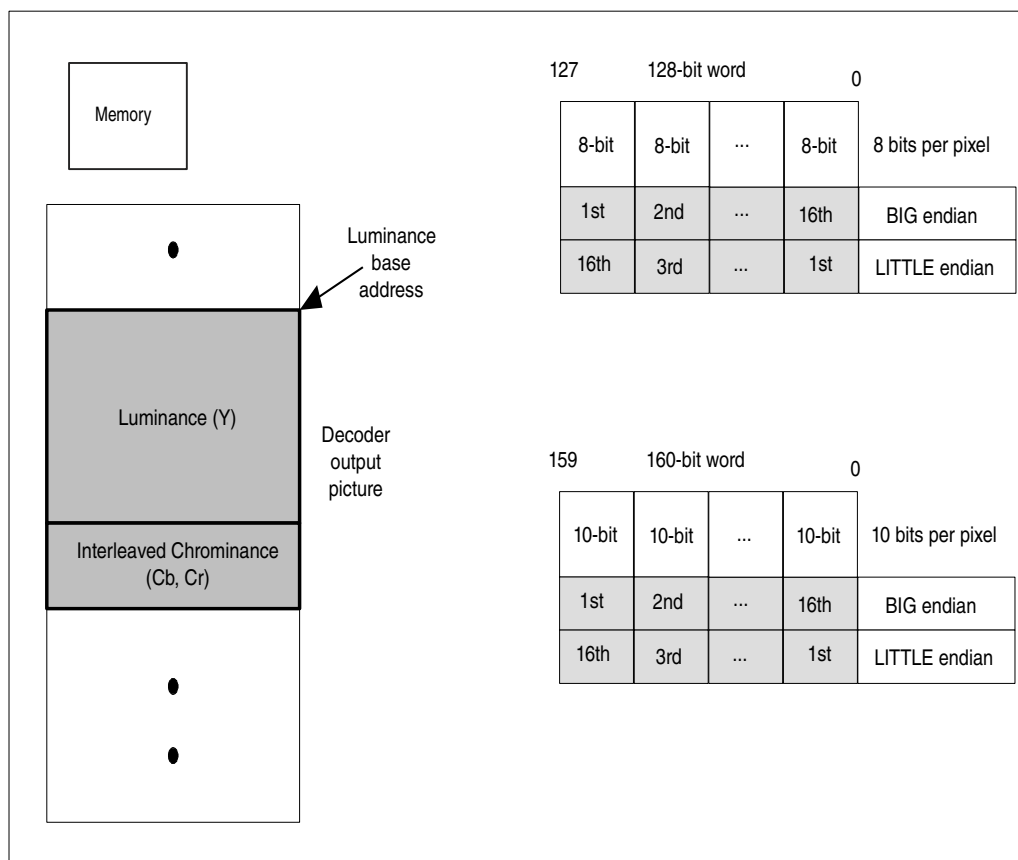


Figure 14-7. External Memory Usage

As an example the luminance pixels of a QCIF sized (176 x 144) frame with each pixel in 8 bits are numbered as presented in Table 1. Table 2 shows the storage of the luminance pixel data of the QCIF sized video frame in raster-scan order (in little endian mode).

Table 14-1. Pixel numbers of the luminance data of the QCIF video frame

0	1	2	3	4	5	6	7	...	172	173	174	175
176	177	178	179	180	181	182	183	...	348	349	350	351
::												::
25200	25201	25202	25203	25204	25205	25206	25207	...	25340	25341	25342	25343

The storage of the luminance pixel data of a QCIF video frame in raster-scan order. All pixels of a pixel row are stored in consecutive memory locations.

Table 14-2. Pixel Memory Locations

Address offset (decimal)	128-bit words(Pixel data of the luminance component. Each byte contains data for one pixel. Number in table are pixel numbers)			
0	15	...	1	0
16	31	...	17	16
32	47	...	33	32
48	63	...	49	48
64	79	...	65	64
:	:			
25296	25311	...	25297	25296
25312	25327	...	25313	25312
25328	25343	...	25329	25328

14.2.2.2 Tiled format

The output picture of the decoder is in semi-planar YCbCr 4:2:0 4x4 tiled format, i.e. luminance data forms one plane in memory, and chrominance data forms another. The distinction from raster scan format is that the output picture is grouped into tiles, which are then stored linearly and contiguously in the memory. The chrominance tiles have to be stored right after the luminance tiles in external memory as shown in Figure . The number of luminance pixels in one row must be divisible by 16.

The chrominance is stored into tiles in interleaved Cb0Cr0Cb1Cr1Cb2Cr2Cb3... format.

Figure contains example external memory usage when using 4x4 tiles, which is used in G2 decoder.

Below is an example code of converting tiled YCbCr 4:2:0 output picture back to raster scan format. The code is in generalized form, i.e. can function on arbitrary tile sizes as long as the picture dimensions are multiple of tile sizes. In the example, chrominance data is interleaved into 4x4 tile so that one 4x4 tile contains one 2x4 block of Cb and Cr each. Output is YCbCr 4:2:0 semiplanar.

```
void ConvertTiledToRaster( DecPicture decPicture, u32 tilewidth,
    u32 tileHeight, u8 *pOutput )
{
    u32 i, j;
    u32 k, l;
    u32 s, t;
    u8 *pIn;
    pIn = decPicture.pOutputPicture;
    /* loop all tile rows for luminance */
    for( i = 0 ; i < decPicture.frameHeight ; i += tileHeight )
```

```

{
t = 0;
s = 0;
/* loop all tiles in one row */
for( j = 0 ; j < decPicture.frameWidth ; j += tileWidth )
{
/* copy one tile */
for( k = 0 ; k < tileHeight ; ++k )
{
for( l = 0 ; l < tileWidth ; ++l )
{
pOutput[ k*decPicture.frameWidth + l + s ] = pIn[t++];
}
}
/* move to next horizontal tile */
s += tileWidth;
}
/* move to next tile row */
pOutput += decPicture.frameWidth * tileHeight;
pIn += decPicture.frameWidth * tileHeight;
}
/* setup pointers to chrominance data */
pIn = decPicture.pOutputPicture +
decPicture.frameWidth * decPicture.frameHeight;
/* loop all tile rows for chrominance */
for( i = 0 ; i < decPicture.frameHeight/2 ; i += tileHeight )
{
t = 0;
s = 0;
/* loop all tiles in one row */
for( j = 0 ; j < decPicture.frameWidth ; j += tileWidth )
{
/* copy one tile */
for( k = 0 ; k < tileHeight ; ++k )
{
for( l = 0 ; l < tileWidth ; ++l )
{
pOutput[ k*decPicture.frameWidth + l + s ] = pIn[t++];
}
}
/* move to next horizontal tile */
s += tileWidth;
}
/* move to next tile row */
pOutput += decPicture.frameWidth * tileHeight;
pIn += decPicture.frameWidth * tileHeight;
}
}

```

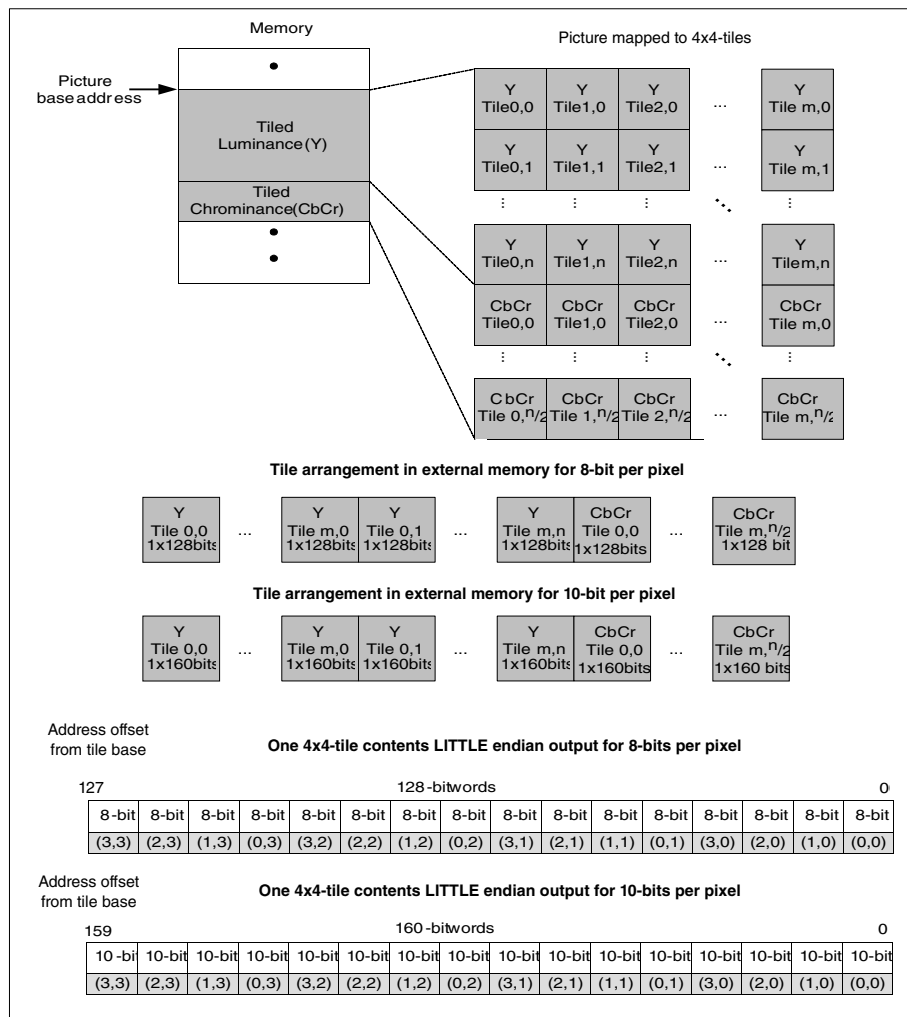


Figure 14-8. External Memory Usage in 4x4 Tiled Reference Picture Mode

14.2.2.3 Compressed format

G2 decoder supports compressed tiled output, when reference buffer compression (RFC) is enabled when synthesis, and user allows compressed reference frame output directly.

Reference frame compression is a feature added in G2 decoder to compress frame buffer so that the bandwidth of storing/loading reference frame can be reduced, especially when the resolution of decoded stream is of high definition.

When compression is enabled, the picture is divided into CBS rows, which is further divided into continuous CBS groups. Each CBS group is composed of 16 CBS. The luminance CBS is composed of 1 8x8 coded block (CB), which one chrominance CBS is composed of two 8x4 coded blocks, with cb CB first then cr CB following. The compression is performed to each CB in the CBS. The first CB's compressed data is

saved from the same offset of that CBS in raster scan buffer. And the compressed output of following CBs in the CBS row is saved continuously. That means there may be gaps between the compressed data of each CBS row.

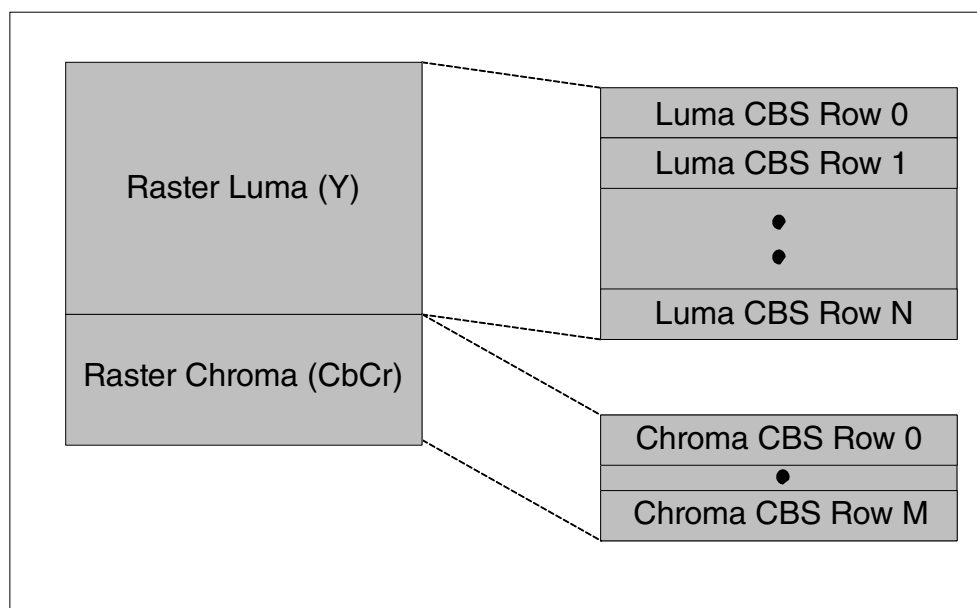


Figure 14-9. Raster scan picture mapped to CBS rows

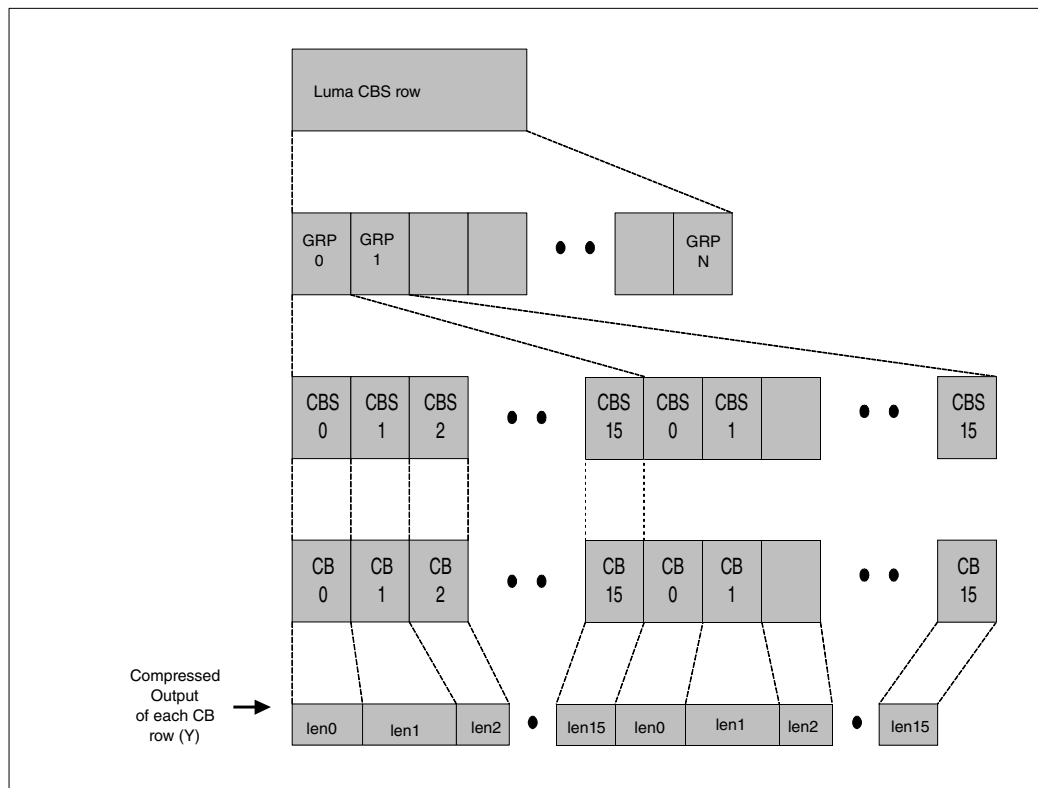


Figure 14-10. Compression performed to each Luma CBS row

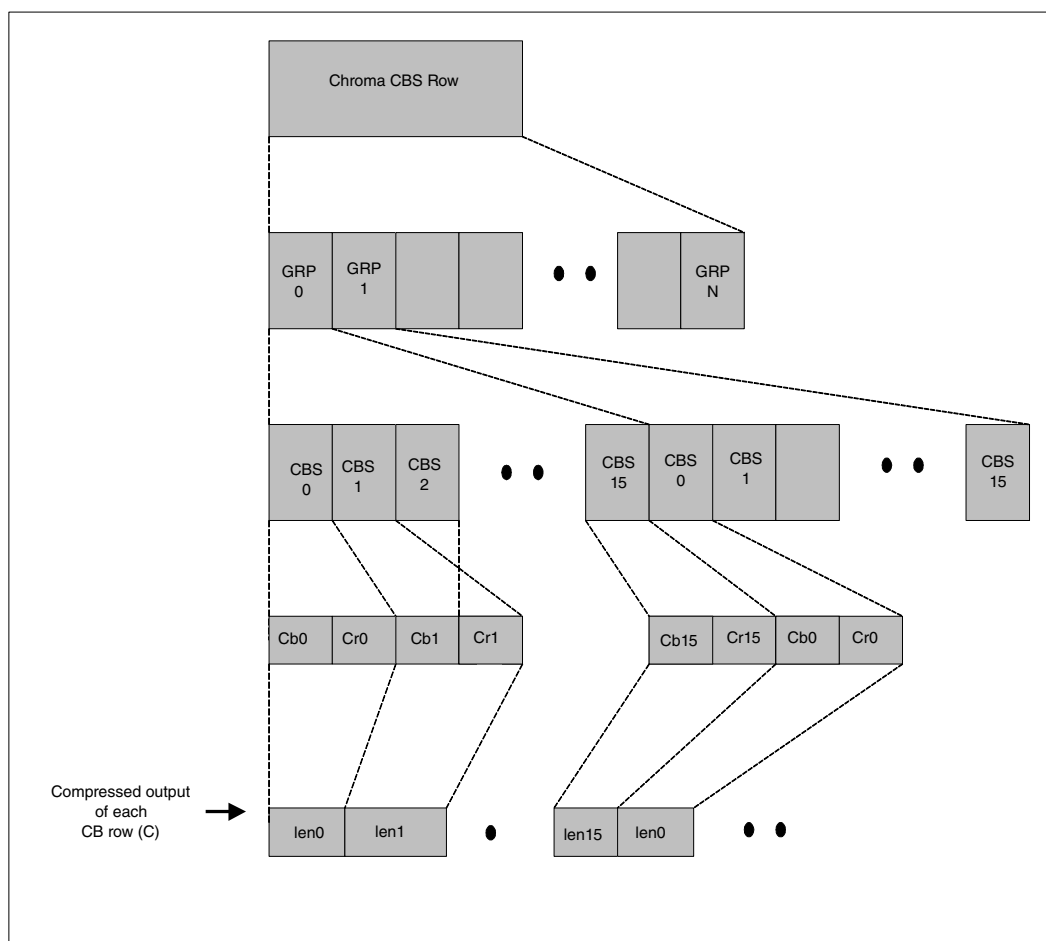


Figure 14-11. Compression performed to each Chroma CBS row

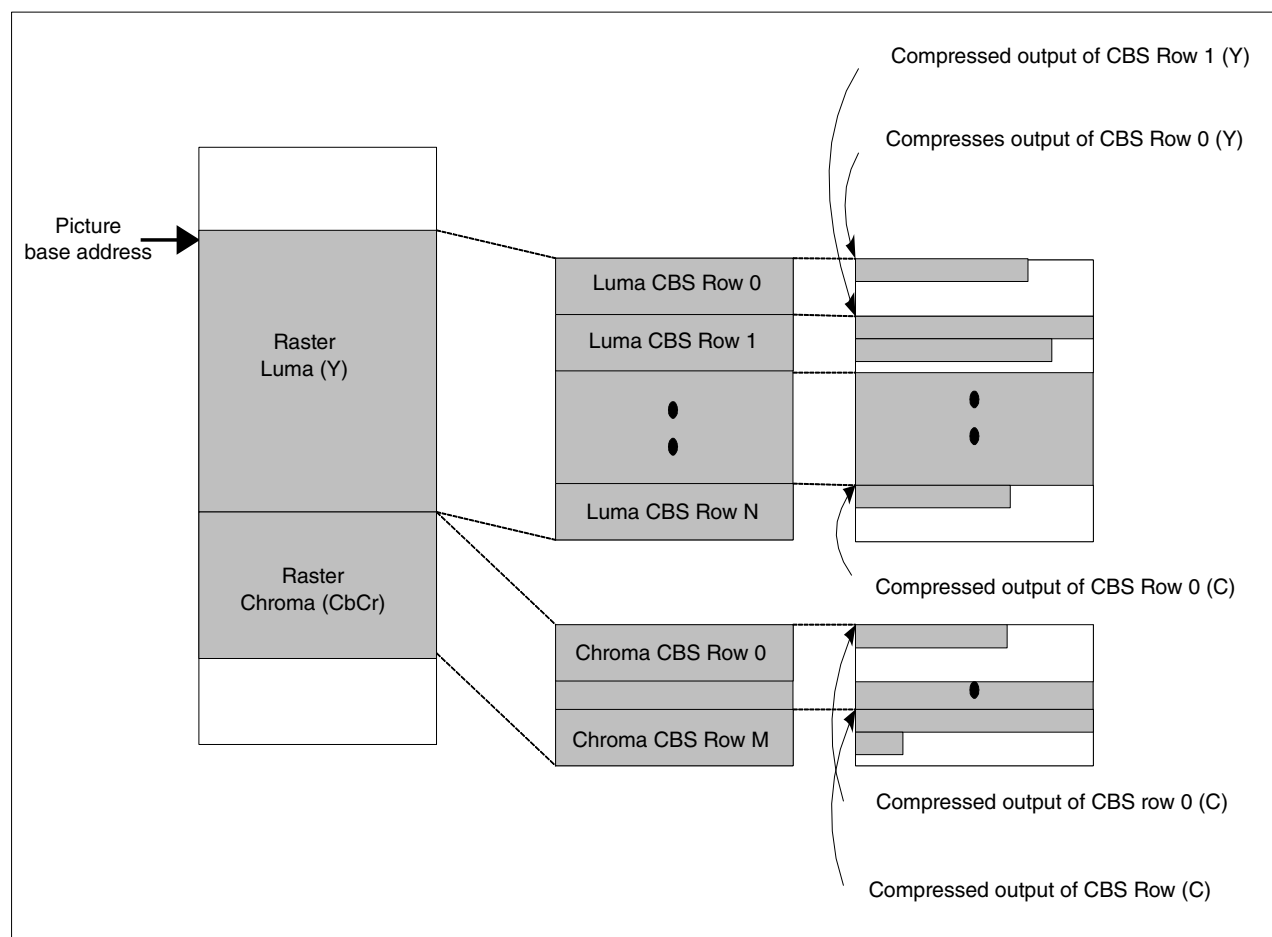


Figure 14-12. Map of external memory in RFC output buffer

Corresponding to each CBS group the starting offset of first compressed CBS output is saved in two bytes in CBS table, followed by the length of each CBS compress output, which is saved in 7 bits. So the table length of each CBS group information is $2 + 7 * 16 / 8 = 16$ bytes.

This 16 bytes for each CBS group in compression table is mapped as shown in the following table. Here, X[n] means the bit-n of the length of CBS X in CBS group.

Table 14-3. Storage of the compression cache table for each CBS group

Byte	Bit7	Bit6	Bit5	Bit4	Bit 3	Bit 2	Bit 1	Bit 0
0	Offset[7:0]							
1	Offset[15:8]							
2	1[0]	0[6]	0[5]	0[4]	0[3]	0[2]	0[1]	0[0]
3	2[1]	2[0]	1[6]	1[5]	1[4]	1[3]	1[2]	1[1]
4	3[2]	3[1]	3[0]	2[6]	2[5]	2[4]	2[3]	2[2]
5	4[3]	4[2]	4[1]	4[0]	3[6]	3[5]	3[4]	3[3]
6	5[4]	5[3]	5[2]	5[1]	5[0]	4[6]	4[5]	4[4]

Table continues on the next page...

Table 14-3. Storage of the compression cache table for each CBS group (continued)

Byte	Bit7	Bit6	Bit5	Bit4	Bit 3	Bit 2	Bit 1	Bit 0
7	6[5]	6[4]	6[3]	6[2]	6[1]	6[0]	5[6]	5[5]
8	7[6]	7[5]	7[4]	7[3]	7[2]	7[1]	7[0]	6[6]
9	9[0]	8[6]	8[5]	8[4]	8[3]	8[2]	8[1]	8[0]
10	10[1]	10[0]	9[6]	9[5]	9[4]	9[3]	9[2]	9[1]
11	11[2]	11[1]	11[0]	10[6]	10[5]	10[4]	10[3]	10[2]
12	12[3]	12[2]	12[1]	12[0]	11[6]	11[5]	11[4]	11[3]
13	13[4]	13[3]	13[2]	13[1]	13[0]	12[6]	12[5]	12[4]
14	14[5]	14[4]	14[3]	14[2]	14[1]	14[0]	13[6]	13[5]
15	15[6]	15[5]	15[4]	15[3]	15[2]	15[1]	15[0]	14[6]

14.2.3 Interface Functions

This chapter describes the API declared in decapi.h, the definition of API in “software\source\common\decapi.c” of the release package.

Common numeric data types are declared in basetype.h as follows:

- **u8** – unsigned 8 bits integer value
- **i8** – signed 8 bits integer value
- **u16** – unsigned 16 bits integer value
- **i16** – signed 16 bits integer value
- **u32** – unsigned 32 bits value
- **i32** – signed 32 bits value

14.2.3.1 DecGetBuild

Syntax

```
struct DecSwHwBuild DecGetBuild(void)
```

Purpose

Returns the hardware and software build information of the decoder. Does not require the creation of a decoder instance.

Parameters

None.

Return value

DecSwHwBuild -- A structure containing the build information of the decoder.

```

struct DecSwHwBuild {
    u32 sw_build;                /* Software build ID */
    u32 hw_build;                /* Hardware build ID */
    struct DecHwConfig hw_config; /* Hardware configuration */
};

```

Function	Description
sw_build	The internal release version of the Hantro HW HEVC software.
hw_build	The internal release version of the Hantro HW hardware.

DecHwConfig -- A structure containing the hardware-supported configuration.

```

struct DecHwConfig {
    u32 mpeg4_support;
    u32 custom_mpeg4_support;
    u32 h264_support;
    u32 vc1_support;
    u32 mpeg2_support;
    u32 jpeg_support;
    u32 jpeg_prog_support;
    u32 max_dec_pic_width;
    u32 max_dec_pic_height;
    u32 pp_support;
    u32 pp_config;
    u32 max_pp_out_pic_width;
    u32 sorenson_spark_support;
    u32 ref_buf_support;
    u32 tiled_mode_support;
    u32 vp6_support;
    u32 vp7_support;
    u32 vp8_support;
    u32 vp9_support;
    u32 avs_support;
    u32 jpeg_esupport;
    u32 rv_support;
    u32 mvc_support;
    u32 webp_support;
    u32 ec_support;
    u32 stride_support;
    u32 field_dpb_support;
    u32 hevc_support;
    u32 double_buffer_support;
    u32 hevc_main10_support;
    u32 vp9_10bit_support;
    u32 ds_support;
    u32 rfc_support;
    u32 ring_buffer_support;
};

```

Function	Description
mpeg4_support	Hardware support for MPEG-4 decoding. Possible values: • MPEG4_NOT_SUPPORTED – decoding not supported • MPEG4_SIMPLE_PROFILE – simple profile decoder • MPEG4_ADVANCED_SIMPLE_PROFILE – advanced simple profile decoder (includes simple profile)
custom_mpeg4_support	Hardware support for DivX® decoding. Possible values:

Table continues on the next page...

Function	Description
	• MPEG4_CUSTOM_NOT_SUPPORTED – decoding not supported • MPEG4_CUSTOM - DivX Home Theater Profile Qualified™ decoder available
h264_support	Hardware support for H.264 decoding. Possible values: • H264_NOT_SUPPORTED – decoding not supported • H264_BASELINE_PROFILE – baseline profile decoder • H264_MAIN_PROFILE – main profile decoder (includes baseline profile) • H264_HIGH_PROFILE – high profile decoder (includes baseline and main profile)
vc1_support	Hardware support for VC-1 decoding. Possible values: • VC1_NOT_SUPPORTED – decoding not supported. • VC1_SIMPLE_PROFILE – simple profile decoder • VC1_MAIN_PROFILE – main profile decoder (includes simple profile) • VC1_ADVANCED_PROFILE – advanced profile decoder (includes main profile)
mpeg2_support	Hardware support for MPEG-2 (includes MPEG-1) decoding. Possible values: • MPEG2_NOT_SUPPORTED – decoding not supported • MPEG2_MAIN_PROFILE – main profile decoder
jpeg_support	Hardware support for JPEG decoding. Possible values: • JPEG_NOT_SUPPORTED – decoding not supported • JPEG_BASELINE – baseline decoder
jpeg_prog_support	Hardware support for progressive JPEG decoding. Possible values: • JPEG_NOT_SUPPORTED – decoding not supported • JPEG_PROGRESSIVE – progressive decoder available
max_dec_pic_width	Maximum decoded picture's width for video decoders. The value is given in pixel units.
max_dec_pic_height	Maximum decoded picture's height for video decoders. The value is given in pixel units.
pp_support	Hardware support for video post-processing. Possible values: • PP_NOT_SUPPORTED – post-processing not supported. • PP_SUPPORTED – post-processing supported in hardware.
pp_config	Post-processing functions available. This is a bitwise list of the following values: • PP_DITHERING, PP_SCALING, PP_DEINTERLACING, and • PP_ALPHA_BLENDING. The availability of all these post-processor functions is dependent on the hardware build.
max_pp_out_pic_width	Maximum output picture's width for the post-processor. The value is given in pixel units.

Table continues on the next page...

Function	Description
sorenson_spark_support	Hardware support for Sorenson Spark decoding. Possible values: • SORENSON_SPARK_NOT_SUPPORTED – decoding not supported • SORENSON_SPARK_SUPPORTED – decoder available
ref_buf_support	Hardware includes reference buffer handling speedup logic. Possible values: • REF_BUF_NOT_SUPPORTED – logic not available • REF_BUF_SUPPORTED – logic available for progressive pictures (flag) • REF_BUF_INTERLACED – logic available for interlaced pictures (flag) • REF_BUF_DOUBLE – logic available for buffering two reference pictures (flag) • Multiple values are possible using bitwise OR operation of valid flags.
tiled_mode_support	Hardware includes support for handling reference pictures in tiled mode to optimize bus utilization. Possible values: • TILED_NOT_SUPPORTED – tiled reference picture support not available • TILED_8x4_SUPPORTED – tiled reference picture support for 8x4 tiles available • TILED_8x4_ILACED_SUPPORTED – tiled reference picture support for 8x4 tiles • available also for interlaced pictures.
vp6_support	Hardware support for VP6 decoding. Possible values: • VP6_NOT_SUPPORTED – decoding not supported • VP6_SUPPORTED – decoder available
vp7_support	Hardware support for VP7 decoding. Possible values: • VP7_NOT_SUPPORTED – decoding not supported • VP7_SUPPORTED – decoder available
vp8_support	Hardware support for VP8 decoding. Possible values: • VP8_NOT_SUPPORTED – decoding not supported • VP8_SUPPORTED – decoder available
vp9_support	Hardware support for VP9 decoding. Possible values: • VP9_NOT_SUPPORTED – decoding not supported • VP9_SUPPORTED – decoder available
avs_support	Hardware support for AVS decoding. Possible values: • AVS_NOT_SUPPORTED – decoding not supported • AVS_SUPPORTED – decoder available
jpeg_esupport	Hardware support for baseline JPEG decoding up to 67 Mpixel images, and 4:1:1 and 4:4:4 sampling. Possible values: • JPEG_EXT_NOT_SUPPORTED – decoding not supported • JPEG_EXT_SUPPORTED – decoder available
rv_support	Hardware support for RV decoding. Possible values:

Table continues on the next page...

Function	Description
	- RV_NOT_SUPPORTED – decoding not supported - RV_SUPPORTED – decoder available
mvc_support	Hardware support for MVC decoding. Possible values: - MVC_NOT_SUPPORTED – decoding not supported - MVC_SUPPORTED – decoder available
webp_support	Hardware support for WebP decoding. Possible values: - WEBP_NOT_SUPPORTED – decoding not supported - WEBP_SUPPORTED – decoder available
ec_support	Hardware support for error concealment. Possible values: - EC_NOT_SUPPORTED – error concealment not supported - EC_SUPPORTED – error concealment available
stride_support	Hardware support for separate Y and C strides. Possible values: - STRIDE_NOT_SUPPORTED – strides not supported - STRIDE_SUPPORTED – strides available
field_dpb_support	Hardware support for field-mode DPB. Possible values: - FIELD_DPB_NOT_SUPPORTED – field-mode DPB not supported - FIELD_DPB_SUPPORTED – field-mode DPB available
hevc_support	Hardware support for HEVC decoding. Possible values: - HEVC_NOT_SUPPORTED – decoding not supported - HEVC_SUPPORTED – decoder available
double_buffer_support	Hardware support for decoder internal reference double buffering. Possible values: - DOUBLE_BUFFER_NOT_SUPPORTED – internal reference double buffering not supported - DOUBLE_BUFFER_SUPPORTED – internal reference double buffering available
hevc_main10_support	Hardware support for HEVC main10 profile decoding. Possible values: 0 – decoding not supported 1 – decoder available
vp9_10bit_support	Hardware support for VP9 10 bits decoding. Possible values: 0 – decoding not supported 1 – decoder available
ds_support	Hardware support for decoder down scale outputting. Possible values: 0 – decoding not supported 1 – decoder available
rfc_support	Hardware support for reference frame compression. Possible values: 0 – decoding not supported 1 – decoder available
ring_buffer_support	Hardware support for ring buffer. Possible values:

Function	Description
	• 0 – decoding not supported • 1 – decoder available

14.2.3.2 Declnit

Syntax

```
enum DecRet Declnit(enum DecCodec codec,
                   DecInst* decoder,
                   struct DecConfig config,
                   struct DecClientHandle callbacks)
```

Purpose

Declnit initializes a new decoder instance based on the given codec type. If parameter noOutputReordering is set to a non-zero value the decoder will not provide output pictures in display order but in decoding order (unless decoding order and display order is the same). Outputting pictures in decode order will lower the memory consumption of the decoder in case the number of reference frames used by the stream is less than the maximum number allowed by the profile and level of the stream.

Parameters

```
enum DecCodec codec
    Codec type to be initialized.
    enum DecCodec {
        DEC_VP9,
        DEC_HEVC
    };
DecInst * decoder
    Pointer to the location where Declnit returns a decoder instance. This
instance is later
    passed to other decoder API functions.
struct DecConfig config
    Decoder initialization parameters.
    struct DecConfig {
        u32 disable_picture_reordering;
        enum DecPictureFormat output_format;
        struct DWL dwl;
        const void* dwl_inst;
        u32 max_num_pics_to_decode;
        enum DecErrorConcealment concealment_mode;
        struct DecDownscaleCfg dscale_cfg;
        u32 use_video_compressor;
        u32 use_ringbuffer;
        u32 use_8bits_output;
    }
u32 disable_picture_reordering
    Flag to disable the reordering of output frames. When set to a non-zero
value the
    decoder will not provide output pictures in display order but in decoding
order (unless
    decoding order and display order is the same).
enum DecPictureFormat output_format
    Format of the output picture.
```

```

enum DecPictureFormat {
    DEC_OUT_FRM_TILED_4X4 = 0,
    DEC_OUT_FRM_RASTER_SCAN = 1,
    DEC_OUT_FRM_PLANAR_420 = 2
};
DEC_OUT_FRM_TILED_4X4 - Tiled 4x4 format
DEC_OUT_FRM_RASTER_SCAN - raster scan format (a.k.a semi-planar)
DEC_OUT_FRM_PLANAR_420 - planar format
struct DWL dwl
    Decoder wrapper layer functionality.
const void* dwl_inst
    Pointer to a DWL instance, which should be instantiated by calling DWLInit
in
    advance.
u32 max_num_pics_to_decode
    Limits the maximum pictures number to be decoded, 0 for unlimited.
enum DecErrorConcealment concealment_mode
    Flag to determine which error concealment method to use. When set to
    DEC_INTRA_FREEZE the decoder will conceal every frame after an error has
been
    detected in the bitstream, until the next IDR (key frame) frame is
decoded. When set to
    DEC_PICTURE_FREEZE, the decoder will conceal only the frames having errors
in the
    bitstream.

enum DecErrorConcealment {
    DEC_PICTURE_FREEZE = 0,
    DEC_INTRA_FREEZE = 1
};
struct DecDownscaleCfg dscale_cfg
    Down scale parameters.
    struct DecDownscaleCfg {
        u32 down_scale_x;
        u32 down_scale_y;
    };
    u32 down_scale_x
        Down scale ratio in horizontal direction. Possible value: 2, 4, 8.
Available
        value:2, 4, 8, which means down scaled to 1/2, 1/4 or 1/8
respectively. 1
        means no scaling.
    u32 down_scale_y
        Down scale ratio in vertical direction. Possible value: 2, 4, 8.
Available value:2,
        4, 8, which means down scaled to 1/2, 1/4 or 1/8 respectively. 1 means
no
        scaling.

```

NOTE

If the down scale ratio in one direction is set to 1, no matter what the value is in the other direction, no down scale will be performed in both directions.

```

u32 use_video_compressor
    Flag to enable reference buffer compression. Set 0 to bypass reference
buffer
    compression, or when reference buffer compression is not supported. When
set to 1,
    the decoded reference buffer will be saved in compressed format.
u32 use_ringbuffer
    Flag to indicate decoder use ring buffer or not. Set 1 to enable ring
buffer. When set to
    0, ring buffer will be disabled.
frame
    Stream ring buffer mode is an alternative storage for stream. In this mode,

```

stream data usually doesn't start from buffer head but follow a previous frame stream data. When stream data goes to stream buffer end, it must turnaround to buffer head to save left stream data. It requires a whole frame stream data saved in stream buffer and stream data start address is byte-aligned. Below figure shows how stream data is stored in stream buffer if ring buffer mode.

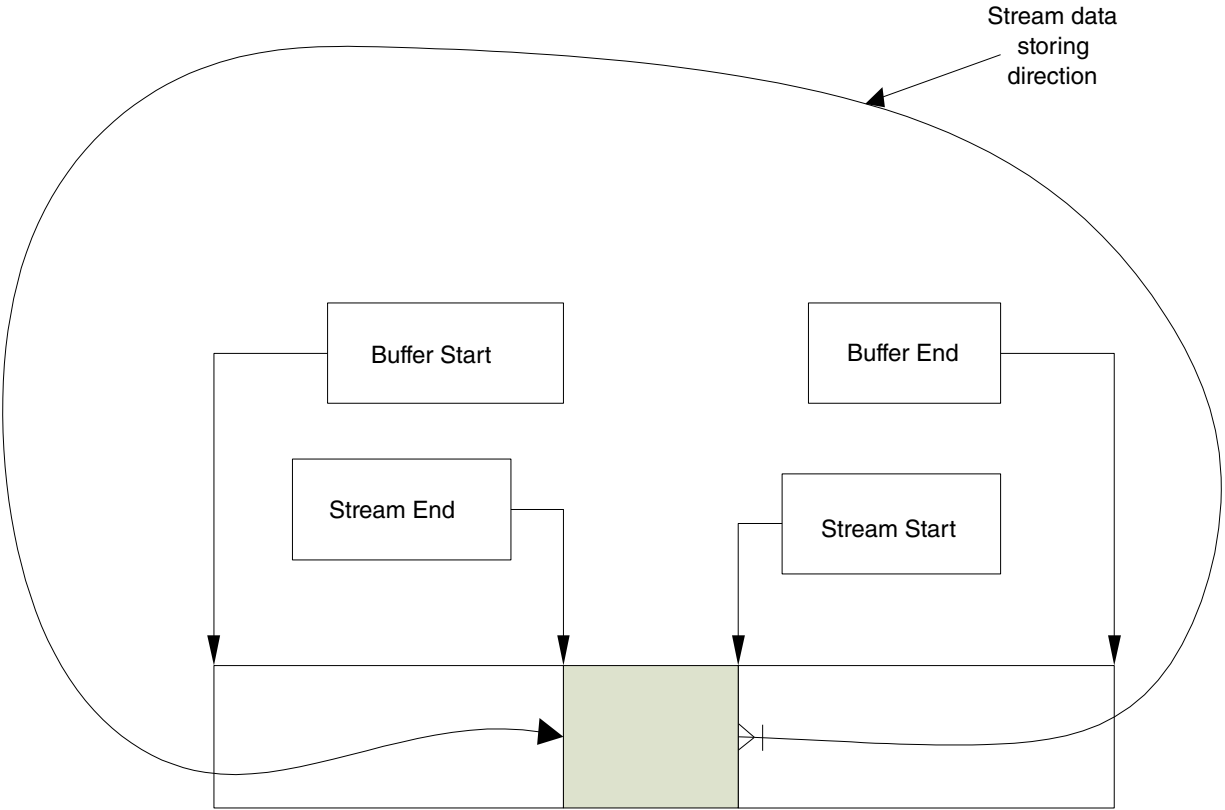


Figure 14-13. Data storage in Stream buffer in Ring Buffer Mode

u32 use_8bits_output
Flag to force 8 bit per pixel output when it's a stream with higher bit depth, e.g., HEVC
Main10 profile stream. The conversion is based on following rule:

Table 14-4. Rule for conversion

Bit depth of original pixel	Original pixel	8bit output
9	XXXXXXXX0	XXXXXXXX
	XXXXXXXX1	XXXXXXXX+1
10	XXXXXXXX00	XXXXXXXX
	XXXXXXXX01	XXXXXXXX
	XXXXXXXX10	XXXXXXXX+1
	XXXXXXXX11	XXXXXXXX+1

struct DecClientHandle callbacks
Callback interfaces that client should provide to facilitate the decoder

to inform the client that there is any events or state changing in decoder.

```

struct DecClientHandle {
    ClientInst client;
    ClientInitialized* Initialized;
    ClientHeadersDecoded* HeadersDecoded;
    ClientBufferDecoded* BufferDecoded;
    ClientPictureReady* PictureReady;
    ClientEndOfStream* EndOfStream;
    ClientReleased* Released;
    ClientNotifyError* NotifyError;
};
typedef const void* ClientInst;
    Opaque pointer to the client.
typedef void ClientInitialized(ClientInst inst);
    Function to notify the client that decoder has successfully
initialized.
    typedef void ClientHeadersDecoded(ClientInst inst,
        struct DecSequenceInfo sequence_info);
        Function to notify about successful decoding of the stream
parameters. Decoder
    expects client to provide needed buffers through
DecSetPictureBuffers function to
    continue decoding the actual stream.
    typedef void ClientBufferDecoded(ClientInst inst,
        struct DecInput* input);
        Function to notify client that a buffer has been consumed by the
decoder and it can be
    handled freely by the client.
    typedef void ClientPictureReady(ClientInst inst,
        struct DecPicture picture);
        Function to notify about picture that is ready to be outputted.
Client is expected to notify
    the decoder when it has finished processing the picture, so
decoder can reuse the
    picture buffer for another picture.
    typedef void ClientEndOfStream(ClientInst inst);
        Function to notify the client that all the pending pictures
have been outputted and
    decoder can be safely shut down.
    typedef void ClientReleased(ClientInst inst);
        Function to notify the client that decoder has shut down.
    typedef void ClientNotifyError(ClientInst inst,
        u32 pic_id,
        enum DecRet rv);
        Function to notify client about error in the decoding process.

```

Return

DEC_OK
API function returned successfully.

DEC_PARAM_ERROR
Bad input parameters.

DEC_MEMFAIL
The decoder was not able to allocate memory.

DEC_FORMAT_NOT_SUPPORTED
Format is not supported in hardware. See DecGetBuild.

14.2.3.3 DecDecode

Syntax

```
enum DecRet DecDecode(DecInst decInst,
                     DecInput *pInput)
```

Purpose

This function will push a COMMAND_DECODE command to the decoding thread, along with the input stream parameters, which will inform the decoding thread to decode one or more NAL units from the current stream for HEVC or element stream for VP9.

The input buffer should contain one of the following:

1. Exactly one NAL unit and nothing else.
2. One or more NAL units in byte stream format, as defined in Annex B of the standard [1].
3. Or for VP9 at least one whole frame bit stream data.

The decoder automatically detects the format of the stream data and decodes NAL units until the whole buffer is processed or decoding of a picture is finished for HEVC. For VP9 if there is not enough bit stream data in input buffer, HW decoder will return TIMEOUT error.

Parameters

DecInst*decInst* - A decoder instance created earlier with an DecInit call.

DecInput **pInput* - Pointer to the decoder input structure.

```
struct DecInput {
    struct DWLLinearMem buffer; /* Pointer to the input buffer. */
    u32 data_len;
};
```

struct DWLLinearMem buffer -- Linear buffer descriptor.

```
struct DWLLinearMem {
    u32 *virtual_address;
    u32 bus_address;
    u32 size;
    u32 logical_size;
};
```

Function	Description
*virtual_address	Pointer to the virtual address of stream linear buffer.
bus_address	Bus address of the linear buffer. NOTE: The start of the stream buffer must be in a 128-bit aligned position or accessible for reading from the previous 128-bit aligned position.
size	Physical size of the linear buffer, which is rounded to page multiple.
logical_size	Physical size of the linear buffer, which is rounded to page multiple.

Table continues on the next page...

Function	Description
<code>dataLen</code>	Number of bytes contained in the buffer stream buffer. The length can be represented on maximum 32 bits. Valid range: $[0, (2^{32}-1)]$

Return value

DEC_OK - The `COMMAND_DECODE` is successfully pushed to decoder thread's input queue.

DEC_PARAM_ERROR - Error in calling parameters.

DEC_NOT_INITIALIZED - Decoder instance is not initialized. Stream decoding is not started. `DecInit` must be called before decoding can be started.

14.2.3.4 DecSetPictureBuffers

Syntax

```
enum DecRet DecSetPictureBuffers(DecInst dec_inst,
                                const struct DWLLinearMem* buffers,
                                u32 num_of_buffers)
```

Purpose

`DecSetPictureBuffers` assigns picture buffers for the decoder. When decoder has finished decoding the stream headers and knows which types of buffers and how many of them it will need, it will inform that through the `HeadersDecoded` callback. Buffers must not be written into until client has successfully called `DecRelease` or decoder has requested new set of buffers through `HeadersDecoded` callback.

Parameters

DecInst*decInst* - A decoder instance created earlier with an `DecInit` call.

const struct DWLLinearMem* *buffers* - Linear buffer to be assigned to the decoder.

u32 *num_of_buffers* - Number of buffers to be assigned to the decoder.

Return value

DEC_OK - Successfully set picture buffers.

DEC_PARAM_ERROR - Error in calling parameters.

14.2.3.5 DecPictureConsumed

Syntax

```
enum DecRet DecPictureConsumed(DecInst decInst,
                               struct DecPicture picture)
```

Purpose

DecPictureConsumed informs the decoder that the client has finished processing a specific picture, which was previously sent to client through the PictureReady callback.

Parameters

DecInst*decInst* - A decoder instance.

DecPicture *pOutput - Pointer to the location used to return the picture parameters. The picture parameters are valid only when the return value indicates that an output picture is available.

```
struct DecPicture {
    struct DecSequenceInfo sequence_info;
    struct DWLLinearMem luma;
    struct DWLLinearMem chroma;
    struct DecPictureInfo picture_info;
    struct DWLLinearMem dscale_luma;
    struct DWLLinearMem dscale_chroma;
    u32 dscale_width;
    u32 dscale_height;
    u32 dscale_stride;
};
struct DecSequenceInfo sequence_info {
    struct DecSequenceInfo {
        u32 pic_width;
        u32 pic_height;
        u32 sar_width;
        u32 sar_height;
        struct DecCropParams crop_params;
        enum DecVideoRange video_range;
        u32 matrix_coefficients;
        u32 is_mono_chrome;
        u32 is_interlaced;
        u32 num_of_ref_frames;
        u32 bit_depth_luma;
        u32 bit_depth_chroma;
        u32 pic_stride;
    };
};
```

Function	Description
picture_width	Decoded picture width in pixels.
picture_height	Decoded picture height in pixels.
sar_width	Horizontal size of the sample aspect ratio.
sar_height	Vertical size of the sample aspect ratio.
struct DecCropParams crop_param	Cropping parameters for the picture.

Table continues on the next page...

Function	Description
	<pre>struct DecCropParams { u32 crop_left_offset; u32 crop_out_width; u32 crop_top_offset; u32 crop_out_height; };</pre>
crop_left_offset	Left offset for cropping process, i.e. number of luminance samples removed from the left edge of the picture (always a multiple of 2).
crop_out_width	Width of the picture in luminance samples after cropping is performed.
crop_top_offset	Top offset for cropping process, i.e. number of luminance samples removed from the top edge of the picture (always a multiple of 2).
crop_out_height	Height of the picture in luminance samples after cropping is performed.
Enum DecVideoRange video_range	YUV sample video range. Possible value: - DEC_VIDEO_RANGE_NORMAL - sample range [16, 235] - DEC_VIDEO_RANGE_FULL - sample range [0, 255]
matrix_coefficients	Matrix coefficients RGB->YUV conversion.
is_mono_chrome	Flag to indicate whether sequence is of monochrome.
is_interlaced	Flag to indicate whether sequence is interlaced.
num_of_ref_frames	Maximum number of reference frames.
bit_depth_luma	Bit depth of stored pixels for luma plane.
bit_depth_chroma	Bit depth of stored pixels for chroma plane.
pic_stride	Stride of the picture as stored in memory in bytes.
struct DWLLinearMem luma	Linear buffer for luma pixels.
struct DWLLinearMem chroma	Linear buffer for chroma pixels.
struct DecPictureInfo picture_info	Picture specific information. <pre>struct DecPictureInfo { enum DecPicCodingType pic_coding_type; u32 is_corrupted; enum DecPictureFormat format; u32 cycles_per_mb; u32 pic_id; };</pre>
enum DecPicCodingType pic_coding_type	Picture coding type. Possible value: - DEC_PIC_TYPE_I - key frame - DEC_PIC_TYPE_P - P frame - DEC_PIC_TYPE_B - B frame
is_corrupted	Flag indicates whether picture is corrupted.
enum DecPictureFormat format	Color format of the picture. Refer to DecPictureFormat in 4.2.
cycles_per_mb	Average decoding time in cycles per mb.

Table continues on the next page...

Function	Description
pic_id	Identifier for the picture to be decoded.
struct DWLLinearMem dscale_luma	Linear buffer for luma pixels in down scaled output.
struct DWLLinearMem dscale_chroma	Linear buffer for chroma pixels in down scaled output.
dscale_width	Width of down scaled output in pixels.
dscale_height	Height of down scaled output in pixels.
dscale_stride	Stride of a pixel line in down scaled output in bytes.

Return value

DEC_OK - No pictures available for display.

DEC_PARAM_ERROR - Error in calling parameters.

DEC_NOT_INITIALIZED - Decoder instance is not initialized. DecInit must be called before decoding can be started.

14.2.3.6 DecEndOfStream

Syntax

```
enum DecRet DecEndOfStream (DecInst decInst)
```

Purpose

DecEndOfStream sends a COMMAND_END_OF_STREAM command to the decoder thread to inform the decoder that it should not be expecting any more input stream and finish decoding and outputting all the buffers that are currently pending in the component. Once decoder has finished outputting the pending pictures it will notify the client about it by calling the EndOfStream callback.

Parameters

DecInst*decInst* - A decoder instance.

Return value

DEC_OK - API function returned successfully

DEC_PARAM_ERROR - Error in calling parameters.

DEC_INITFAIL - Decoder instance is not initialized.

14.2.3.7 DecRelease

Syntax

```
void DecRelease(DecInst decInst)
```

Purpose

DecRelease sends a COMMAND_RELEASE command to decoder thread, which will finally close the decoder instance decInst and releases all internally allocated resources. This function must not be called twice for the same instance.

Parameters

DecInst*decInst* - A decoder instance to be released.

Return value

None.

14.2.4 Decoder Application Examples

Purpose

The purpose of the following examples is to show how to implement client's callback functions, as long as using G2 decoder API to implement a simple decoder with demuxer to decode both HEVC and VP9 streams. InitializedCb, HeadersDecodedCb, BufferDecodedCb, PictureReadyCb, EndOfStreamCb, ReleasedCb, and NotifyErrorCb are the system/application dependent callback functions, which are provided by client to the decoder wrapper.

The examples are simplified, e.g. they do not take errors into consideration.

NOTE

Program code in the examples should be considered as pseudo code and may not compile as such.

```
void SetupDefaultParams(struct TestParams* params);
int ParseParams(int argc, char* argv[], struct TestParams* params);
void* CreateDemuxer(struct Client* client);
void ReleaseDemuxer(struct Client* client);
void PostProcessPicture(struct Client* client, struct DecPicture* picture);

static void DispatchBufferForDecoding(struct Client* client, struct DecInput*
buffer)
{
    enum DecRet rv;
    i32 size = buffer->buffer.size;
    memset(buffer->buffer.virtual_address, 0, size);
    i32 len = client->demuxer.ReadPacket(client->demuxer.inst, (u8*)buffer->
buffer.virtual_address, &size);
    if (len <= 0) { /* EOS or error. */
```

```

/* If reading was due to insufficient buffer size, try to realloc with
sufficient size. */
if (size > buffer->buffer.size) {
    i32 i;
    for (i = 0; i < GetStreamBufferCount(client); i++) {
        if (client->buffers[i].buffer.virtual_address == buffer-
>buffer.virtual_address) {
            DWLFreeLinear(client->dwL, &client->buffers[i].buffer);
            if (DWLMallocLinear(client->dwL, size, &client->buffers[i].buffer))
            {
                DispatchEndOfStream(client);
                return;
            }
            DispatchBufferForDecoding(client, &client->buffers[i]);
            return;
        }
    }
}
else {
    DispatchEndOfStream(client);
    return;
}
}
buffer->data_len = len;
if (client->eos) return; /* Don't dispatch new buffers if EOS already done. */
/* Decode the contents of the input stream buffer. */
switch (rv = DecDecode(client->decoder, buffer)) {
    case DEC_OK:
        /* Everything is good, keep on going. */
        break;
    default:
        DispatchEndOfStream(client);
        break;
}
}

static void InitializedCb(ClientInst inst)
{
    struct Client* client = (struct Client*)inst;
    /* Internal testing feature: Override HW configuration parameters */
    HwconfigOverride(client->decoder, &tb_cfg);
    /* Start the output handling thread, if needed. */
    if (client->test_params.extra_output_thread)
        pthread_create(&client->parallel_output_thread, NULL, ParallelOutput, client);
    for (int i = 0; i < GetStreamBufferCount(client); i++) {
        if (DWLMallocLinear(client->dwL, DEFAULT_STREAM_BUFFER_SIZE, &client-
>buffers[i].buffer)) {
            DecRelease(client->decoder);
            return;
        }
        /* Dispatch the first buffers for decoding. When decoder finished
        * decoding each buffer it will be refilled within the callback. */
        DispatchBufferForDecoding(client, &client->buffers[i]);
    }
}

static void HeadersDecodedCb(ClientInst inst, struct DecSequenceInfo info)
{
    struct Client* client = (struct Client*)inst;
    if (client->yuvsink.inst == NULL) {
        if (client->test_params.out_file_name == NULL) {
            client->yuvsink.inst = CreateSink(client);
        }
        DecSetPictureBuffers(client->decoder, NULL, 0);
    }
    static void BufferDecodedCb(ClientInst inst, struct DecInput* buffer) {
        struct Client* client = (struct Client*)inst;
        if (!client->eos) {
            DispatchBufferForDecoding(client, buffer);
        }
    }
}

```

```

    }
}

static void PictureReadyCb(ClientInst inst, struct DecPicture picture)
{
    static char* pic_types[] = {"IDR", "Non-IDR (P)", "Non-IDR (B)"};
    struct Client* client = (struct Client*)inst;
    client->num_of_output_pics++;
    DEBUG_PRINT(("PIC %2d/%2d, type %s,", client->num_of_output_pics,
picture.picture_info.pic_id, pic_types[picture.picture_info.pic_coding_type]));
    if (picture.picture_info.cycles_per_mb) {
        client->cycle_count += picture.picture_info.cycles_per_mb;
        DEBUG_PRINT((" %4d cycles / mb,", picture.picture_info.cycles_per_mb));
    }
    DEBUG_PRINT((" %d x %d, Crop: (%d, %d), %d x %d %s\n",
picture.sequence_info.pic_width,
picture.sequence_info.pic_height,
picture.sequence_info.crop_params.crop_left_offset,
picture.sequence_info.crop_params.crop_top_offset,
picture.sequence_info.crop_params.crop_out_width,
picture.sequence_info.crop_params.crop_out_height,
picture.picture_info.is_corrupted ? "CORRUPT" : ""));
    if (client->test_params.extra_output_thread) {
        struct DecPicture* copy = malloc(sizeof(struct DecPicture));
        *copy = picture;
        FifoPush(client->pic_fifo, copy, FIFO_EXCEPTION_DISABLE);
    } else {
        PostProcessPicture(client, &picture);
    }
}

static void EndOfStreamCb(ClientInst inst)
{
    struct Client* client = (struct Client*)inst;
    client->eos = 1;
    if (client->test_params.extra_output_thread) {
        /* We're done, wait for the output to Finish it's job. */
        FifoPush(client->pic_fifo, NULL, FIFO_EXCEPTION_DISABLE);
        pthread_join(client->parallel_output_thread, NULL);
    }
    DecRelease(client->decoder);
}

static void ReleasedCb(ClientInst inst)
{
    struct Client* client = (struct Client*)inst;
    for (int i = 0; i < NUM_OF_STREAM_BUFFERS; i++) {
        if (client->buffers[i].buffer.virtual_address) {
            DWLFreeLinear(client->dwL, &client->buffers[i].buffer);
        }
    }
    ReleaseSink(client);
    sem_post(&client->dec_done);
}

static void DispatchEndOfStream(struct Client* client) {
    if (!client->eos) {
        client->eos = 1;
        DecEndOfStream(client->decoder);
    }
}

static void NotifyErrorCb(ClientInst inst, u32 pic_id, enum DecRet rv)
{
    struct Client* client = (struct Client*)inst;
    /* There's serious decoding error, so we'll consider it as end of stream to
    get the pending pictures out of the decoder. */
    DispatchEndOfStream(client);
}

```

```

int main(int argc, char* argv[])
{
    struct Client client;
    memset(&client, 0, sizeof(struct Client));
    struct DecClientHandle client_if = { &client, InitializedCb, HeadersDecodedCb,
BufferDecodedCb,
    PictureReadyCb, EndOfStreamCb, ReleasedCb, NotifyErrorCb, };
    SetupDefaultParams(&client.test_params);
    ParseParams(argc, argv, &client.test_params);
    struct DecSwHwBuild build = DecGetBuild();
    /* Check whether the decoder feature meets our requirements here. */
    ...
    client.demuxer.inst = CreateDemuxer(&client);
    /* Create struct DWL. */
    struct DWLInitParam dwl_params = {DWL_CLIENT_TYPE_HEVC_DEC};
    client.dwl = DWLInit(&dwl_params);
    if (client.test_params.extra_output_thread)
    /* Create the fifo to enable parallel output processing */
    FifoInit(2, &client.pic_fifo);
    sem_init(&client.dec_done, 0, 0);
    enum DecCodec codec;
    switch (client.demuxer.GetVideoFormat(client.demuxer.inst)) {
        case BITSTREAM_HEVC:
            codec = DEC_HEVC;
            break;
        case BITSTREAM_VP9:
            codec = DEC_VP9;
            break;
        default:
            return -1;
    }
    struct DecConfig config;
    config.disable_picture_reordering= client.test_params.disable_display_order;
    config.concealment_mode = client.test_params.concealment_mode;
    switch (client.test_params.hw_format) {
        case DEC_OUT_FRM_TILED_4X4:
            config.output_format = DEC_OUT_FRM_TILED_4X4;
            break;
        case DEC_OUT_FRM_RASTER_SCAN: /* fallthrough */
        case DEC_OUT_FRM_PLANAR_420:
            if (!build.hw_config.pp_support) {
                fprintf(stderr, "Cannot do raster output; No PP support.\n");
                return -1;
            }
            config.output_format = DEC_OUT_FRM_RASTER_SCAN;
            break;
        default:
            return -1;
    }
    config.dwl = dwl;
    config.dwl_inst = client.dwl;
    config.max_num_pics_to_decode = client.test_params.num_of_decoded_pics;
    config.dscale_cfg = client.test_params.dscale_cfg;
    config.use_video_compressor = client.test_params.compress_bypass ? 0 : 1;
    /* Initialize the decoder. */
    DecInit(codec, &client.decoder, config, client_if) != DEC_OK;
    /* The rest is driven by the callbacks and this thread just has to wait
    * until decoder has finished its job. */
    sem_wait(&client.dec_done);
    ReleaseDemuxer(&client);
    if (client.pic_fifo != NULL)
        FifoRelease(client.pic_fifo);
    if (client.dwl != NULL) DWLRelease(client.dwl);
    return 0;
}

```

14.2.5 VPU G2 Memory Map/Register Definition

14.2.5.1 VPU_G2 register descriptions

14.2.5.1.1 VPU Memory map

VPU_G2 base address: 3831_0000h

Offset	Register	Width (In bits)	Access	Reset value
0h	ID register (read only) (SWREG0)	32	RO	Table 14-4
4h	Interrupt register decoder (SWREG1)	32	RW	0000_0000h
8h	Data configuration register decoder (SWREG2)	32	RW	0000_0000h
Ch	Decoder control register 0 (SWREG3)	32	RW	0000_0000h
10h	Decoder control register 1 (SWREG4)	32	RW	0000_0000h
14h	Decoder control register 2 (SWREG5)	32	RW	0000_0000h
18h	Decoder control register 3 (SWREG6)	32	RW	0000_0000h
1Ch	Decoder control register 4 (SWREG7)	32	RW	0000_0000h
20h	Decoder control register 5 (SWREG8)	32	RW	0000_0000h
24h	Decoder control register 6 (SWREG9)	32	RW	0000_0000h
28h	Decoder control register 7 (SWREG10)	32	RW	0000_0000h
2Ch	Decoder control register 8 (SWREG11)	32	RW	0000_0000h
30h	Decoder control register 9 (SWREG12)	32	RW	0000_0000h
34h	Decoder control register 10 (SWREG13)	32	RW	0000_0000h
38h	Initial ref pic list register (0-2) (SWREG14)	32	RW	0000_0000h
3Ch	Initial ref pic list register (3-5) (SWREG15)	32	RW	0000_0000h
40h	Initial ref pic list register (6-8) (SWREG16)	32	RW	0000_0000h
44h	Initial ref pic list register (9-11) (SWREG17)	32	RW	0000_0000h
48h	Initial ref pic list register (12-14) (SWREG18)	32	RW	0000_0000h
4Ch	Initial ref pic list register (15 and P 0-3) (SWREG19)	32	RW	0000_0000h
50h	Decoder control register 11 (SWREG20)	32	RW	0000_0000h
54h	Not used (SWREG21)	32	RU	0000_0000h
58h	Not used (SWREG22)	32	RU	0000_0000h
5Ch	Decoder configure status register (SWREG23)	32	RO	0000_0000h
60h	Not used (SWREG24)	32	RU	0000_0000h
64h	Not used (SWREG25)	32	RU	0000_0000h
68h	Not used (SWREG26)	32	RU	0000_0000h
6Ch	Not used (SWREG27)	32	RU	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
70h	Not used (SWREG28)	32	RU	0000_0000h
74h	Not used (SWREG29)	32	RU	0000_0000h
78h	Not used (SWREG30)	32	RU	0000_0000h
7Ch	VP9 segmentation values (SWREG31)	32	RW	0000_0000h
80h	VP9 segmentation values (SWREG32)	32	RW	0000_0000h
84h	VP9 reference picture scaling register 0 (SWREG33)	32	RW	0000_0000h
88h	VP9 reference picture scaling register 1 (SWREG34)	32	RW	0000_0000h
8Ch	VP9 reference picture scaling register 2 (SWREG35)	32	RW	0000_0000h
90h	VP9 reference picture scaling register 3 (SWREG36)	32	RW	0000_0000h
94h	VP9 reference picture scaling register 4 (SWREG37)	32	RW	0000_0000h
98h	VP9 reference picture scaling register 5 (SWREG38)	32	RW	0000_0000h
9Ch	Not used (SWREG39)	32	RU	0000_0000h
A0h	Not used (SWREG40)	32	RU	0000_0000h
A4h	Not used (SWREG41)	32	RU	0000_0000h
A8h	Not used (SWREG42)	32	RU	0000_0000h
ACh	Not used (SWREG43)	32	RU	0000_0000h
B0h	Not used (SWREG44)	32	RU	0000_0000h
B4h	Timeout control register (SWREG45)	32	RW	0000_0000h
B8h	Picture order count from current pictures for index 0-3 (SWREG46)	32	RW	0000_0000h
BCh	Picture order count from current pictures for index 4-7 (SWREG47)	32	RW	0000_0000h
C0h	Picture order count from current pictures for index 8-11 (SWREG48)	32	RW	0000_0000h
C4h	Picture order count from current pictures for index 12-15 (SWREG49)	32	RW	0000_0000h
C8h	Synthesis configuration register decoder 0 (read only) (SWREG50)	32	RO	Table 14-4
CCh	Reference picture buffer control register (SWREG51)	32	RU	0000_0000h
D0h	Reference picture buffer information register 1 (read only) (SWREG52)	32	RU	0000_0000h
D4h	Reference picture buffer information register 2 (read only) (SWREG53)	32	RU	0000_0000h
D8h	Synthesis configuration register decoder 1 (read only) (SWREG54)	32	RO	0000_0000h
DCh	Advanced prefetch control register (SWREG55)	32	RW	0000_0000h
E0h	Synthesis configuration register decoder 2 (read only) (SWREG56)	32	RO	0000_0000h
E4h	Decoder fuse register (read only) (SWREG57)	32	RU	0000_0000h
E8h	Device configuration register decoder 2 + Multi core control register (SWREG58)	32	RW	0000_0000h
ECh	Device configuration register AXI ID (SWREG59)	32	RW	0000_0000h
F0h	Synthesis configuration register decoder 3 for PP (read only) (SWREG60)	32	RO	0000_0000h
F4h	Not used (SWREG61)	32	RU	0000_0000h
F8h	HW proceed register (CU location) (SWREG62)	32	RW	0000_0000h
FCh	HW performance register (cycles running) (SWREG63)	32	RO	0000_0000h

Table continues on the next page...

VPU G2 Memory Map/Register Definition

Offset	Register	Width (In bits)	Access	Reset value
100h	Base address MSB (bits 63:32) for decoded luminance picture (SWREG64)	32	RW	0000_0000h
104h	Base address LSB (bits 31:0) for decoded luminance picture (SWREG65)	32	RW	0000_0000h
108h	Base address MSB (bits 63:32) for reference luminance picture index 0 (SWREG66)	32	RW	0000_0000h
10Ch	Base address LSB (bits 31:0) for reference luminance picture index 0 (SWREG67)	32	RW	0000_0000h
110h	Base address MSB (bits 63:32) for reference luminance picture index 1 (SWREG68)	32	RW	0000_0000h
114h	Base address LSB (bits 31:0) for reference luminance picture index 1 (SWREG69)	32	RW	0000_0000h
118h	Base address MSB (bits 63:32) for reference luminance picture index 2 (SWREG70)	32	RW	0000_0000h
11Ch	Base address LSB (bits 31:0) for reference luminance picture index 2 (SWREG71)	32	RW	0000_0000h
120h	Base address MSB (bits 63:32) for reference luminance picture index 3 (SWREG72)	32	RW	0000_0000h
124h	Base address LSB (bits 31:0) for reference luminance picture index 3 (SWREG73)	32	RW	0000_0000h
128h	Base address MSB (bits 63:32) for reference luminance picture index 4 (SWREG74)	32	RW	0000_0000h
12Ch	Base address LSB (bits 31:0) for reference luminance picture index 4 (SWREG75)	32	RW	0000_0000h
130h	Base address MSB (bits 63:32) for reference luminance picture index 5 (SWREG76)	32	RW	0000_0000h
134h	Base address LSB (bits 31:0) for reference luminance picture index 5 (SWREG77)	32	RW	0000_0000h
138h	Base address MSB (bits 63:32) for reference luminance picture index 6 /VP9 segment write base MSB (SWREG78)	32	RW	0000_0000h
13Ch	Base address LSB (bits 31:0) for reference luminance picture index 6 /VP9 segment write base LSB (SWREG79)	32	RW	0000_0000h
140h	Base address MSB (bits 63:32) for reference luminance picture index 7 /VP9 segment read base MSB (SWREG80)	32	RW	0000_0000h
144h	Base address LSB (bits 31:0) for reference luminance picture index 7 /VP9 segment read base LSB (SWREG81)	32	RW	0000_0000h
148h	Base address MSB (bits 63:32) for reference luminance picture index 8 (SWREG82)	32	RW	0000_0000h
14Ch	Base address LSB (bits 31:0) for reference luminance picture index 8 (SWREG83)	32	RW	0000_0000h
150h	Base address MSB (bits 63:32) for reference luminance picture index 9 (SWREG84)	32	RW	0000_0000h
154h	Base address LSB (bits 31:0) for reference luminance picture index 9 (SWREG85)	32	RW	0000_0000h
158h	Base address MSB (bits 63:32) for reference luminance picture index 10 (SWREG86)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
15Ch	Base address LSB (bits 31:0) for reference luminance picture index 10 (SWREG87)	32	RW	0000_0000h
160h	Base address MSB (bits 63:32) for reference luminance picture index 11 (SWREG88)	32	RW	0000_0000h
164h	Base address LSB (bits 31:0) for reference luminance picture index 11 (SWREG89)	32	RW	0000_0000h
168h	Base address MSB (bits 63:32) for reference luminance picture index 12 (SWREG90)	32	RW	0000_0000h
16Ch	Base address LSB (bits 31:0) for reference luminance picture index 12 (SWREG91)	32	RW	0000_0000h
170h	Base address MSB (bits 63:32) for reference luminance picture index 13 (SWREG92)	32	RW	0000_0000h
174h	Base address LSB (bits 31:0) for reference luminance picture index 13 (SWREG93)	32	RW	0000_0000h
178h	Base address MSB (bits 63:32) for reference luminance picture index 14 (SWREG94)	32	RW	0000_0000h
17Ch	Base address LSB (bits 31:0) for reference luminance picture index 14 (SWREG95)	32	RW	0000_0000h
180h	Base address MSB (bits 63:32) for reference luminance picture index 15 (SWREG96)	32	RW	0000_0000h
184h	Base address LSB (bits 31:0) for reference luminance picture index 15 (SWREG97)	32	RW	0000_0000h
188h	Base address MSB (bits 63:32) for decoded chrominance picture (SWREG98)	32	RW	0000_0000h
18Ch	Base address LSB (bits 31:0) for decoded chrominance picture (SWREG99)	32	RW	0000_0000h
190h	Base address MSB (bits 63:32) for reference chrominance picture index 0 (SWREG100)	32	RW	0000_0000h
194h	Base address LSB (bits 31:0) for reference chrominance picture index 0 (SWREG101)	32	RW	0000_0000h
198h	Base address MSB (bits 63:32) for reference chrominance picture index 1 (SWREG102)	32	RW	0000_0000h
19Ch	Base address LSB (bits 31:0) for reference chrominance picture index 1 (SWREG103)	32	RW	0000_0000h
1A0h	Base address MSB (bits 63:32) for reference chrominance picture index 2 (SWREG104)	32	RW	0000_0000h
1A4h	Base address LSB (bits 31:0) for reference chrominance picture index 2 (SWREG105)	32	RW	0000_0000h
1A8h	Base address MSB (bits 63:32) for reference chrominance picture index 3 (SWREG106)	32	RW	0000_0000h
1ACh	Base address LSB (bits 31:0) for reference chrominance picture index 3 (SWREG107)	32	RW	0000_0000h
1B0h	Base address MSB (bits 63:32) for reference chrominance picture index 4 (SWREG108)	32	RW	0000_0000h
1B4h	Base address LSB (bits 31:0) for reference chrominance picture index 4 (SWREG109)	32	RW	0000_0000h

Table continues on the next page...

VPU G2 Memory Map/Register Definition

Offset	Register	Width (In bits)	Access	Reset value
1B8h	Base address MSB (bits 63:32) for reference chrominance picture index 5 (SWREG110)	32	RW	0000_0000h
1BCh	Base address LSB (bits 31:0) for reference chrominance picture index 5 (SWREG111)	32	RW	0000_0000h
1C0h	Base address MSB (bits 63:32) for reference chrominance picture index 6 (SWREG112)	32	RW	0000_0000h
1C4h	Base address LSB (bits 31:0) for reference chrominance picture index 6 (SWREG113)	32	RW	0000_0000h
1C8h	Base address MSB (bits 63:32) for reference chrominance picture index 7 (SWREG114)	32	RW	0000_0000h
1CCh	Base address LSB (bits 31:0) for reference chrominance picture index 7 (SWREG115)	32	RW	0000_0000h
1D0h	Base address MSB (bits 63:32) for reference chrominance picture index 8 (SWREG116)	32	RW	0000_0000h
1D4h	Base address LSB (bits 31:0) for reference chrominance picture index 8 (SWREG117)	32	RW	0000_0000h
1D8h	Base address MSB (bits 63:32) for reference chrominance picture index 9 (SWREG118)	32	RW	0000_0000h
1DCh	Base address LSB (bits 31:0) for reference chrominance picture index 9 (SWREG119)	32	RW	0000_0000h
1E0h	Base address MSB (bits 63:32) for reference chrominance picture index 10 (SWREG120)	32	RW	0000_0000h
1E4h	Base address LSB (bits 31:0) for reference chrominance picture index 10 (SWREG121)	32	RW	0000_0000h
1E8h	Base address MSB (bits 63:32) for reference chrominance picture index 11 (SWREG122)	32	RW	0000_0000h
1ECh	Base address LSB (bits 31:0) for reference chrominance picture index 11 (SWREG123)	32	RW	0000_0000h
1F0h	Base address MSB (bits 63:32) for reference chrominance picture index 12 (SWREG124)	32	RW	0000_0000h
1F4h	Base address LSB (bits 31:0) for reference chrominance picture index 12 (SWREG125)	32	RW	0000_0000h
1F8h	Base address MSB (bits 63:32) for reference chrominance picture index 13 (SWREG126)	32	RW	0000_0000h
1FCh	Base address LSB (bits 31:0) for reference chrominance picture index 13 (SWREG127)	32	RW	0000_0000h
200h	Base address MSB (bits 63:32) for reference chrominance picture index 14 (SWREG128)	32	RW	0000_0000h
204h	Base address LSB (bits 31:0) for reference chrominance picture index 14 (SWREG129)	32	RW	0000_0000h
208h	Base address MSB (bits 63:32) for reference chrominance picture index 15 (SWREG130)	32	RW	0000_0000h
20Ch	Base address LSB (bits 31:0) for reference chrominance picture index 15 (SWREG131)	32	RW	0000_0000h
210h	Base address MSB (bits 63:32) for decoded direct mode MVS (SWREG132)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
214h	Base address LSB (bits 31:0) for decoded direct mode MVS (SWREG133)	32	RW	0000_0000h
218h	Base address MSB (bits 63:32) for reference direct mode MVS index 0 (SWREG134)	32	RW	0000_0000h
21Ch	Base address LSB (bits 31:0) for reference direct mode MVS index 0 (SWREG135)	32	RW	0000_0000h
220h	Base address MSB (bits 63:32) for reference direct mode MVS index 1 (SWREG136)	32	RW	0000_0000h
224h	Base address LSB (bits 31:0) for reference direct mode MVS index 1 (SWREG137)	32	RW	0000_0000h
228h	Base address MSB (bits 63:32) for reference direct mode MVS index 2 (SWREG138)	32	RW	0000_0000h
22Ch	Base address LSB (bits 31:0) for reference direct mode MVS index 2 (SWREG139)	32	RW	0000_0000h
230h	Base address MSB (bits 63:32) for reference direct mode MVS index 3 (SWREG140)	32	RW	0000_0000h
234h	Base address LSB (bits 31:0) for reference direct mode MVS index 3 (SWREG141)	32	RW	0000_0000h
238h	Base address MSB (bits 63:32) for reference direct mode MVS index 4 (SWREG142)	32	RW	0000_0000h
23Ch	Base address LSB (bits 31:0) for reference direct mode MVS index 4 (SWREG143)	32	RW	0000_0000h
240h	Base address MSB (bits 63:32) for reference direct mode MVS index 5 (SWREG144)	32	RW	0000_0000h
244h	Base address LSB (bits 31:0) for reference direct mode MVS index 5 (SWREG145)	32	RW	0000_0000h
248h	Base address MSB (bits 63:32) for reference direct mode MVS index 6 (SWREG146)	32	RW	0000_0000h
24Ch	Base address LSB (bits 31:0) for reference direct mode MVS index 6 (SWREG147)	32	RW	0000_0000h
250h	Base address MSB (bits 63:32) for reference direct mode MVS index 7 (SWREG148)	32	RW	0000_0000h
254h	Base address LSB (bits 31:0) for reference direct mode MVS index 7 (SWREG149)	32	RW	0000_0000h
258h	Base address MSB (bits 63:32) for reference direct mode MVS index 8 (SWREG150)	32	RW	0000_0000h
25Ch	Base address LSB (bits 31:0) for reference direct mode MVS index 8 (SWREG151)	32	RW	0000_0000h
260h	Base address MSB (bits 63:32) for reference direct mode mode MVS index 9 (SWREG152)	32	RW	0000_0000h
264h	Base address LSB (bits 31:0) for reference direct mode mode MVS index 9 (SWREG153)	32	RW	0000_0000h
268h	Base address MSB (bits 63:32) for reference direct mode MVS index 10 (SWREG154)	32	RW	0000_0000h
26Ch	Base address LSB (bits 31:0) for reference direct mode MVS index 10 (SWREG155)	32	RW	0000_0000h

Table continues on the next page...

VPU G2 Memory Map/Register Definition

Offset	Register	Width (In bits)	Access	Reset value
270h	Base address MSB (bits 63:32) for reference direct mode MVS index 11 (SWREG156)	32	RW	0000_0000h
274h	Base address LSB (bits 31:0) for reference direct mode MVS index 11 (SWREG157)	32	RW	0000_0000h
278h	Base address MSB (bits 63:32) for reference direct mode MVS index 12 (SWREG158)	32	RW	0000_0000h
27Ch	Base address LSB (bits 31:0) for reference direct mode MVS index 12 (SWREG159)	32	RW	0000_0000h
280h	Base address MSB (bits 63:32) for reference direct mode MVS index 13 (SWREG160)	32	RW	0000_0000h
284h	Base address LSB (bits 31:0) for reference direct mode MVS index 13 (SWREG161)	32	RW	0000_0000h
288h	Base address MSB (bits 63:32) for reference direct mode MVS index 14 (SWREG162)	32	RW	0000_0000h
28Ch	Base address LSB (bits 31:0) for reference direct mode MVS index 14 (SWREG163)	32	RW	0000_0000h
290h	Base address MSB (bits 63:32) for reference direct mode MVS index 15 (SWREG164)	32	RW	0000_0000h
294h	Base address LSB (bits 31:0) for reference direct mode MVS index 15 (SWREG165)	32	RW	0000_0000h
298h	Base address MSB (bits 63:32) for tile sizes (SWREG166)	32	RW	0000_0000h
29Ch	Base address LSB (bits 31:0) for tile sizes (SWREG167)	32	RW	0000_0000h
2A0h	Base address MSB (bits 63:32) for / stream start address/decoded end addr register (SWREG168)	32	RW	0000_0000h
2A4h	Base address LSB (bits 31:0) for / stream start address/decoded end addr register (SWREG169)	32	RW	0000_0000h
2A8h	Base address MSB (bits 63:32) for scaling lists / VP9 CTX counter values (SWREG170)	32	RW	0000_0000h
2ACh	Base address LSB (bits 31:0) for scaling lists / VP9 CTX counter values (SWREG171)	32	RW	0000_0000h
2B0h	Base address MSB (bits 63:32) for stream propability tables (SWREG172)	32	RW	0000_0000h
2B4h	Base address LSB (bits 31:0) for stream propability tables (SWREG173)	32	RW	0000_0000h
2B8h	Base address MSB (bits 63:32) for decoder output raster scan Y picture (SWREG174)	32	RW	0000_0000h
2BCh	Base address LSB (bits 31:0) for decoder output raster scan Y picture (SWREG175)	32	RW	0000_0000h
2C0h	Base address MSB (bits 63:32) for decoder output raster scan C picture (SWREG176)	32	RW	0000_0000h
2C4h	Base address LSB (bits 31:0) for decoder output raster scan C picture (SWREG177)	32	RW	0000_0000h
2C8h	Base address MSB (bits 63:32) for tile border coefficients of filter (SWREG178)	32	RW	0000_0000h
2CCh	Base address LSB (bits 31:0) for tile border coefficients of filter (SWREG179)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
2D0h	Base address MSB (bits 63:32) for tile border coefficients of sao (SWREG180)	32	RW	0000_0000h
2D4h	Base address LSB (bits 31:0) for tile border coefficients of sao (SWREG181)	32	RW	0000_0000h
2D8h	Base address MSB (bits 63:32) for tile border bsd control data (SWREG182)	32	RW	0000_0000h
2DCh	Base address LSB (bits 31:0) for tile border bsd control data (SWREG183)	32	RW	0000_0000h
2E0h	Raster scan down scale control register MSM (SWREG184)	32	RW	0000_0000h
2E4h	Base address MSB (bits 63:32) for decoder output raster scan down scale Y picture (SWREG185)	32	RW	0000_0000h
2E8h	Base address LSB (bits 31:0) for decoder output raster scan down scale Y picture (SWREG186)	32	RW	0000_0000h
2ECh	Base address MSB (bits 63:32) for decoder output raster scan down scale C picture (SWREG187)	32	RW	0000_0000h
2F0h	Base address LSB (bits 31:0) for decoder output raster scan down scale C picture (SWREG188)	32	RW	0000_0000h
2F4h	Base address MSB (bits 63:32) for decoder output compress luminance table (SWREG189)	32	RW	0000_0000h
2F8h	Base address LSB (bits 31:0) for decoder output compress luminance table (SWREG190)	32	RW	0000_0000h
2FCh	Base address MSB (bits 63:32) for reference compress luminance table index 0 (SWREG191)	32	RW	0000_0000h
300h	Base address LSB (bits 31:0) for reference compress luminance table index 0 (SWREG192)	32	RW	0000_0000h
304h	Base address MSB (bits 63:32) for reference compress luminance table index 1 (SWREG193)	32	RW	0000_0000h
308h	Base address LSB (bits 31:0) for reference compress luminance table index 1 (SWREG194)	32	RW	0000_0000h
30Ch	Base address MSB (bits 63:32) for reference compress luminance table index 2 (SWREG195)	32	RW	0000_0000h
310h	Base address LSB (bits 31:0) for reference compress luminance table index 2 (SWREG196)	32	RW	0000_0000h
314h	Base address MSB (bits 63:32) for reference compress luminance table index 3 (SWREG197)	32	RW	0000_0000h
318h	Base address LSB (bits 31:0) for reference compress luminance table index 3 (SWREG198)	32	RW	0000_0000h
31Ch	Base address MSB (bits 63:32) for reference compress luminance table index 4 (SWREG199)	32	RW	0000_0000h
320h	Base address LSB (bits 31:0) for reference compress luminance table index 4 (SWREG200)	32	RW	0000_0000h
324h	Base address MSB (bits 63:32) for reference compress luminance table index 5 (SWREG201)	32	RW	0000_0000h
328h	Base address LSB (bits 31:0) for reference compress luminance table index 5 (SWREG202)	32	RW	0000_0000h

Table continues on the next page...

VPU G2 Memory Map/Register Definition

Offset	Register	Width (In bits)	Access	Reset value
32Ch	Base address MSB (bits 63:32) for reference compress luminance table index 6 (SWREG203)	32	RW	0000_0000h
330h	Base address LSB (bits 31:0) for reference compress luminance table index 6 (SWREG204)	32	RW	0000_0000h
334h	Base address MSB (bits 63:32) for reference compress luminance table index 7 (SWREG205)	32	RW	0000_0000h
338h	Base address LSB (bits 31:0) for reference compress luminance table index 7 (SWREG206)	32	RW	0000_0000h
33Ch	Base address MSB (bits 63:32) for reference compress luminance table index 8 (SWREG207)	32	RW	0000_0000h
340h	Base address LSB (bits 31:0) for reference compress luminance table index 8 (SWREG208)	32	RW	0000_0000h
344h	Base address MSB (bits 63:32) for reference compress luminance table index 9 (SWREG209)	32	RW	0000_0000h
348h	Base address LSB (bits 31:0) for reference compress luminance table index 9 (SWREG210)	32	RW	0000_0000h
34Ch	Base address MSB (bits 63:32) for reference compress luminance table index 10 (SWREG211)	32	RW	0000_0000h
350h	Base address LSB (bits 31:0) for reference compress luminance table index 10 (SWREG212)	32	RW	0000_0000h
354h	Base address MSB (bits 63:32) for reference compress luminance table index 11 (SWREG213)	32	RW	0000_0000h
358h	Base address LSB (bits 31:0) for reference compress luminance table index 11 (SWREG214)	32	RW	0000_0000h
35Ch	Base address MSB (bits 63:32) for reference compress luminance table index 12 (SWREG215)	32	RW	0000_0000h
360h	Base address LSB (bits 31:0) for reference compress luminance table index 12 (SWREG216)	32	RW	0000_0000h
364h	Base address MSB (bits 63:32) for reference compress luminance table index 13 (SWREG217)	32	RW	0000_0000h
368h	Base address LSB (bits 31:0) for reference compress luminance table index 13 (SWREG218)	32	RW	0000_0000h
36Ch	Base address MSB (bits 63:32) for reference compress luminance table index 14 (SWREG219)	32	RW	0000_0000h
370h	Base address LSB (bits 31:0) for reference compress luminance table index 14 (SWREG220)	32	RW	0000_0000h
374h	Base address MSB (bits 63:32) for reference compress luminance table index 15 (SWREG221)	32	RW	0000_0000h
378h	Base address LSB (bits 31:0) for reference compress luminance table index 15 (SWREG222)	32	RW	0000_0000h
37Ch	Base address MSB (bits 63:32) for decoder output compress chrominance table (SWREG223)	32	RW	0000_0000h
380h	Base address LSB (bits 31:0) for decoder output compress chrominance table (SWREG224)	32	RW	0000_0000h
384h	Base address MSB (bits 63:32) for reference compress chrominance table index 0 (SWREG225)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
388h	Base address LSB (bits 31:0) for reference compress chrominance table index 0 (SWREG226)	32	RW	0000_0000h
38Ch	Base address MSB (bits 63:32) for reference compress chrominance table index 1 (SWREG227)	32	RW	0000_0000h
390h	Base address LSB (bits 31:0) for reference compress chrominance table index 1 (SWREG228)	32	RW	0000_0000h
394h	Base address MSB (bits 63:32) for reference compress chrominance table index 2 (SWREG229)	32	RW	0000_0000h
398h	Base address LSB (bits 31:0) for reference compress chrominance table index 2 (SWREG230)	32	RW	0000_0000h
39Ch	Base address MSB (bits 63:32) for reference compress chrominance table index 3 (SWREG231)	32	RW	0000_0000h
3A0h	Base address LSB (bits 31:0) for reference compress chrominance table index 3 (SWREG232)	32	RW	0000_0000h
3A4h	Base address MSB (bits 63:32) for reference compress chrominance table index 4 (SWREG233)	32	RW	0000_0000h
3A8h	Base address LSB (bits 31:0) for reference compress chrominance table index 4 (SWREG234)	32	RW	0000_0000h
3ACh	Base address MSB (bits 63:32) for reference compress chrominance table index 5 (SWREG235)	32	RW	0000_0000h
3B0h	Base address LSB (bits 31:0) for reference compress chrominance table index 5 (SWREG236)	32	RW	0000_0000h
3B4h	Base address MSB (bits 63:32) for reference compress chrominance table index 6 (SWREG237)	32	RW	0000_0000h
3B8h	Base address LSB (bits 31:0) for reference compress chrominance table index 6 (SWREG238)	32	RW	0000_0000h
3BCh	Base address MSB (bits 63:32) for reference compress chrominance table index 7 (SWREG239)	32	RW	0000_0000h
3C0h	Base address LSB (bits 31:0) for reference compress chrominance table index 7 (SWREG240)	32	RW	0000_0000h
3C4h	Base address MSB (bits 63:32) for reference compress chrominance table index 8 (SWREG241)	32	RW	0000_0000h
3C8h	Base address LSB (bits 31:0) for reference compress chrominance table index 8 (SWREG242)	32	RW	0000_0000h
3CCh	Base address MSB (bits 63:32) for reference compress chrominance table index 9 (SWREG243)	32	RW	0000_0000h
3D0h	Base address LSB (bits 31:0) for reference compress chrominance table index 9 (SWREG244)	32	RW	0000_0000h
3D4h	Base address MSB (bits 63:32) for reference compress chrominance table index 10 (SWREG245)	32	RW	0000_0000h
3D8h	Base address LSB (bits 31:0) for reference compress chrominance table index 10 (SWREG246)	32	RW	0000_0000h
3DCh	Base address MSB (bits 63:32) for reference compress chrominance table index 11 (SWREG247)	32	RW	0000_0000h
3E0h	Base address LSB (bits 31:0) for reference compress chrominance table index 11 (SWREG248)	32	RW	0000_0000h

Table continues on the next page...

VPU G2 Memory Map/Register Definition

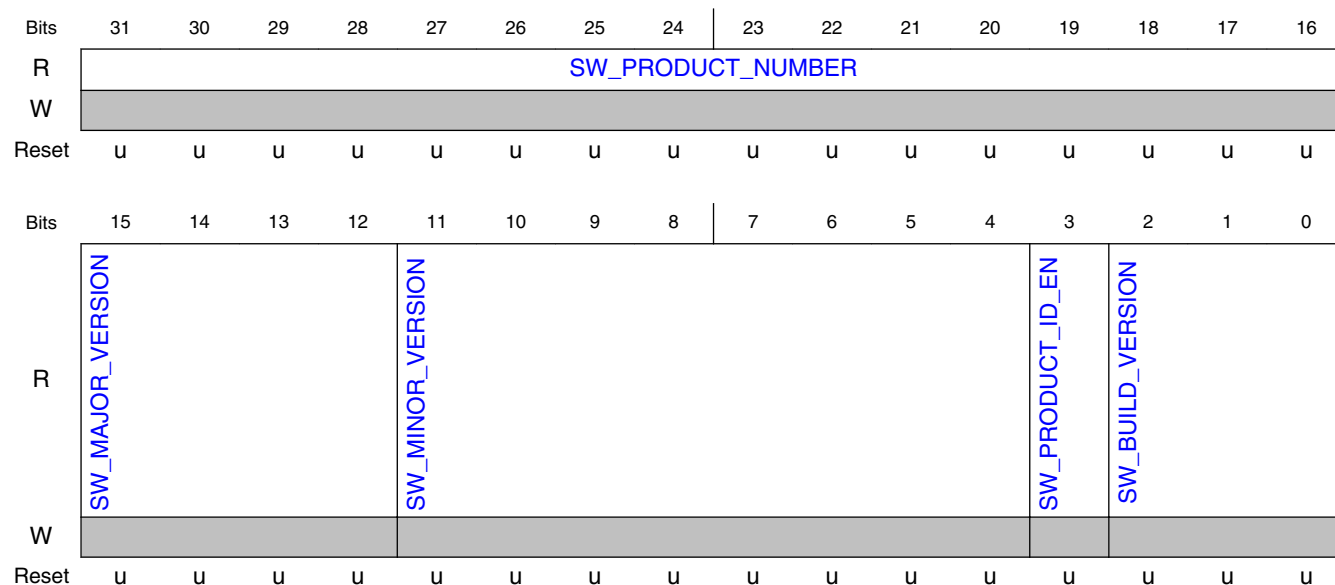
Offset	Register	Width (In bits)	Access	Reset value
3E4h	Base address MSB (bits 63:32) for reference compress chrominance table index 12 (SWREG249)	32	RW	0000_0000h
3E8h	Base address LSB (bits 31:0) for reference compress chrominance table index 12 (SWREG250)	32	RW	0000_0000h
3ECh	Base address MSB (bits 63:32) for reference compress chrominance table index 13 (SWREG251)	32	RW	0000_0000h
3F0h	Base address LSB (bits 31:0) for reference compress chrominance table index 13 (SWREG252)	32	RW	0000_0000h
3F4h	Base address MSB (bits 63:32) for reference compress chrominance table index 14 (SWREG253)	32	RW	0000_0000h
3F8h	Base address LSB (bits 31:0) for reference compress chrominance table index 14 (SWREG254)	32	RW	0000_0000h
3FCh	Base address MSB (bits 63:32) for reference compress chrominance table index 15 (SWREG255)	32	RW	0000_0000h
400h	Base address LSB (bits 31:0) for reference compress chrominance table index 15 (SWREG256)	32	RW	0000_0000h
404h	Not used (SWREG257)	32	RU	0000_0000h
408h	input stream buffer length (SWREG258)	32	RW	0000_0000h
40Ch	input stream buffer start offset (SWREG259)	32	RW	0000_0000h

14.2.5.1.2 ID register (read only) (SWREG0)

14.2.5.1.2.1 Offset

Register	Offset
SWREG0	0h

14.2.5.1.2.2 Diagram



14.2.5.1.2.3 Fields

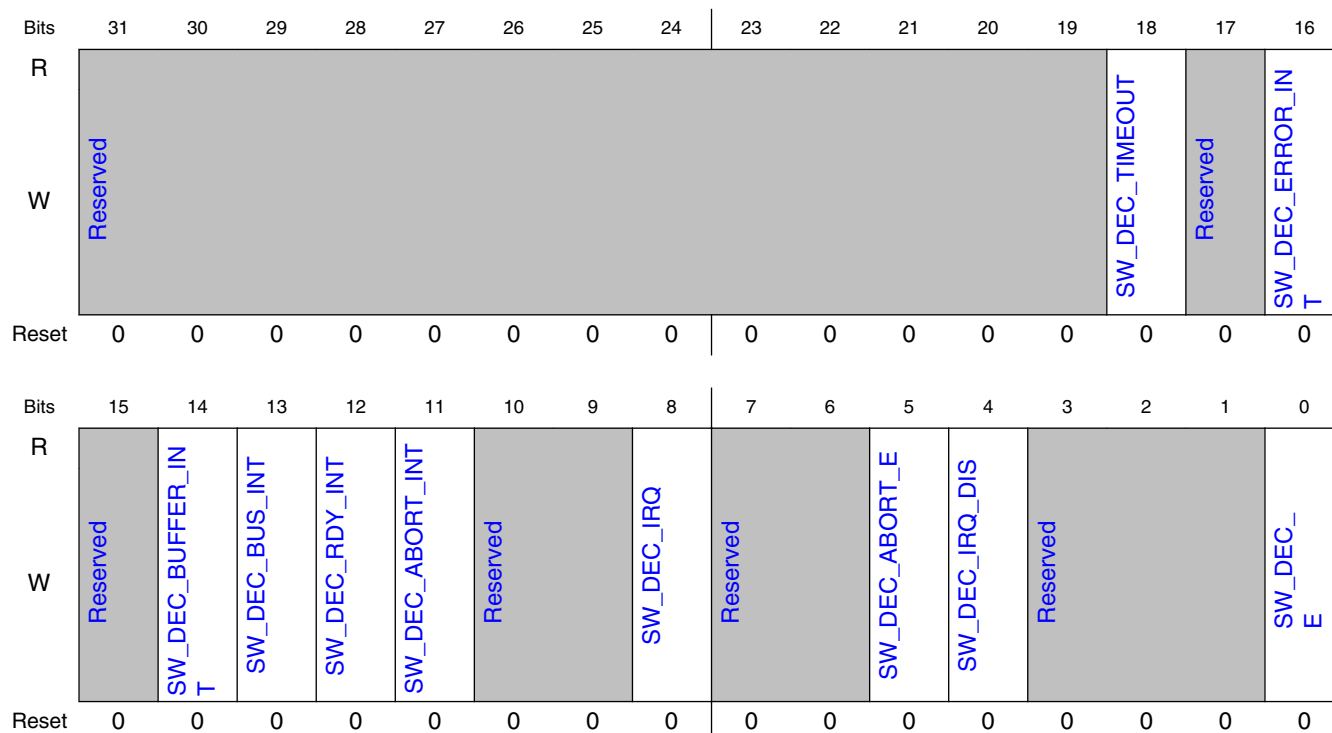
Field	Function
31-16 SW_PRODUCT_NUMBER	Product number (g2)
15-12 SW_MAJOR_VERSION	Major version
11-4 SW_MINOR_VERSION	Minor version
3 SW_PRODUCT_ID_EN	ASCII type product ID enable
2-0 SW_BUILD_VERSION	Build version (core number)

14.2.5.1.3 Interrupt register decoder (SWREG1)

14.2.5.1.3.1 Offset

Register	Offset
SWREG1	4h

14.2.5.1.3.2 Diagram



14.2.5.1.3.3 Fields

Field	Function
31-19 —	Reserved.
18 SW_DEC_TIME OUT	Interrupt status bit decoder timeout. When high the decoder has been idling for too long. HW will self reset. Possible only if timeout interrupt is enabled
17 —	Reserved.
16 SW_DEC_ERR OR_INT	Interrupt status bit input stream error. When high an error is found in input data stream decoding. HW will self reset.
15	Reserved.

Table continues on the next page...

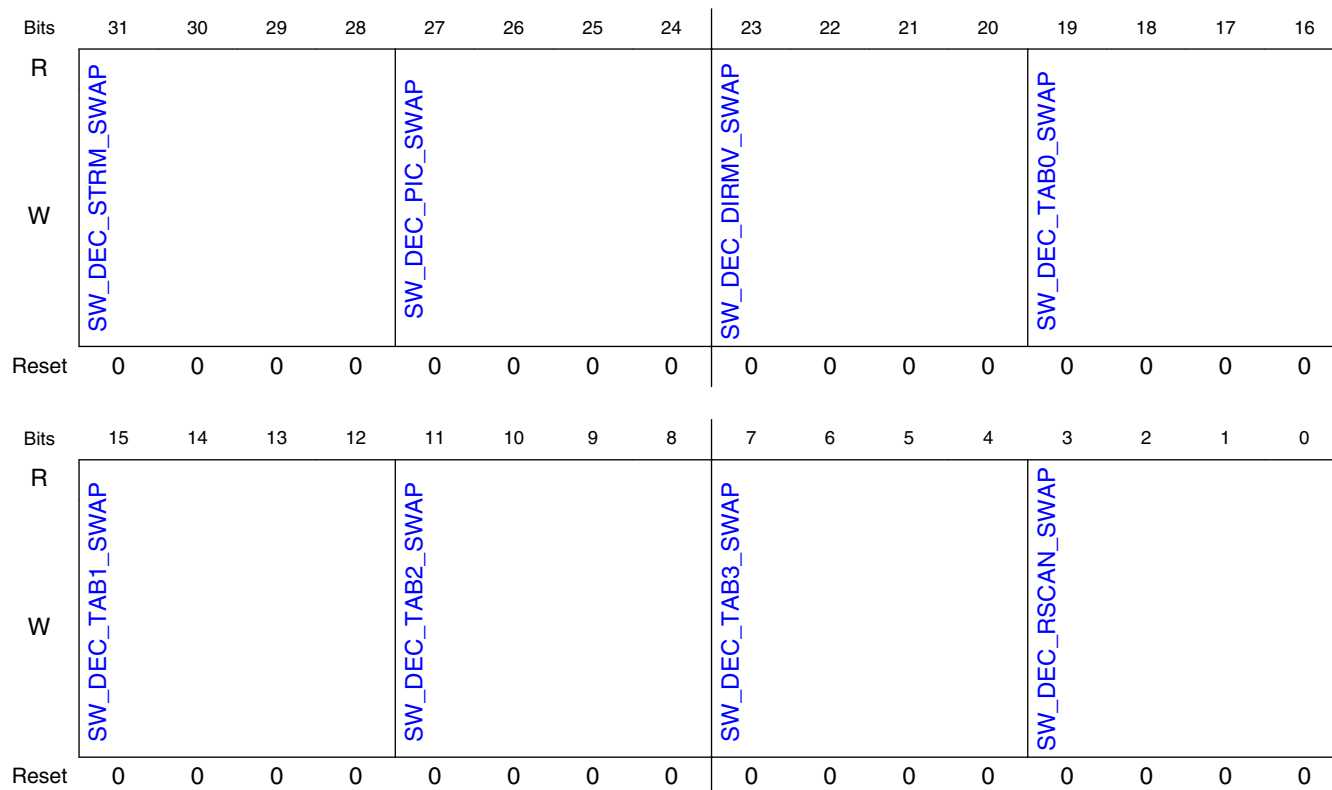
Field	Function
—	
14 SW_DEC_BUFF ER_INT	Interrupt status bit input buffer empty. When high input stream buffer is empty but picture is not ready. HW will not self reset.
13 SW_DEC_BUS_ INT	Interrupt status bit bus. Error response from bus. HW will self reset.
12 SW_DEC_RDY_ _INT	Interrupt status bit decoder. When this bit is high decoder has decoded a picture. HW will self reset.
11 SW_DEC_ABO RT_INT	Interrupt status bit decoding aborted. When this bit is high decoder has aborted the current picture decoding as SW requested (sw_dec_abort_e). Decoder self reset and sw_dec_abort_e written low
10-9 —	Reserved.
8 SW_DEC_IRQ	Decoder IRQ. When high decoder requests an interrupt. SW will reset this after interrupt is handled.
7-6 —	Reserved.
5 SW_DEC_ABO RT_E	Abort decoding enable. Setting this bit high will cause HW to abort decoding and safely to reset itself down. After abort is complete the corresponding interrupt status is set and this bit is set low as well as the decoder enable.
4 SW_DEC_IRQ_ DIS	Decoder IRQ disable. When high there are no interrupts concerning decoder from HW. Polling must be used to see the interrupt statuses.
3-1 —	Reserved.
0 SW_DEC_E	Decoder enable. Setting this bit high will start the decoding operation. HW will reset this when picture is processed or ASO or stream error is detected or bus error or timeout interrupt is given.

14.2.5.1.4 Data configuration register decoder (SWREG2)

14.2.5.1.4.1 Offset

Register	Offset
SWREG2	8h

14.2.5.1.4.2 Diagram



14.2.5.1.4.3 Fields

Field	Function
31-28 SW_DEC_STRM_SWAP	Byte swap configuration for stream data 4 Bit byte order vector to control byte locations inside HW internal 128 bit data vector. For 64 and 32 bit external bus widths, the data is first gathered to 128 bit width and then bytes swapped accordingly: Bit 0: 8 bit swap Bit 1: 16 bit swap Bit 2: 32 bit swap Bit 3: 64 bit swap '0000' = 0-1-2-3-4-5-6-7-8-9-10-11-12-13-14-15 '0001' = 1-0-3-2-5-4-7-6... '0010' = 2-3-0-1-6-7-4-5... '0011' = 3-2-1-0-7-6-5-4... '0100' = 4-5-6-7-0-1-2-3... ... '0111' = 7-6-5-4-3-2-1-0 '1000' = 8-9-10-11-12-13-14-15-0-1-2-3-4-5-6-7... ...

Table continues on the next page...

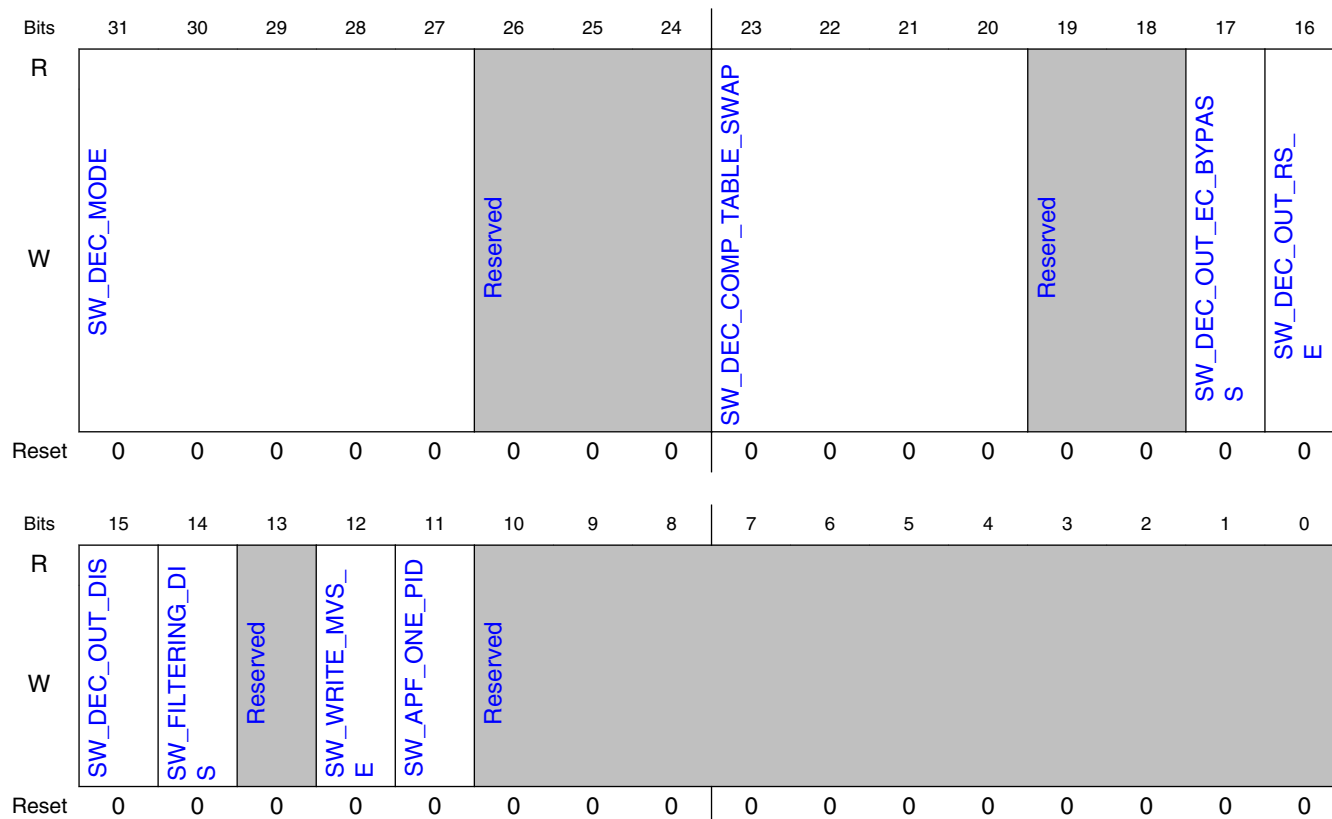
Field	Function
	'1111'=15-14-13-12-11-10-9-8-7-6-5-4-3-2-1-0
27-24 SW_DEC_PIC_SWAP	Byte swap configuration for decoder reference output picture data
23-20 SW_DEC_DIRM_V_SWAP	Byte swap configuration for direct mode MV data (read/write)
19-16 SW_DEC_TAB0_SWAP	Byte swap configuration for VP9 stream propability tables
15-12 SW_DEC_TAB1_SWAP	Byte swap configuration for HEVC scaling lists / VP9 segmentation map read/write
11-8 SW_DEC_TAB2_SWAP	Byte swap configuration for VP9 CTX counter values
7-4 SW_DEC_TAB3_SWAP	Byte swap configuration for tile sizes
3-0 SW_DEC_RSC_AN_SWAP	Byte swap for raster scan output picture data

14.2.5.1.5 Decoder control register 0 (SWREG3)

14.2.5.1.5.1 Offset

Register	Offset
SWREG3	Ch

14.2.5.1.5.2 Diagram



14.2.5.1.5.3 Fields

Field	Function
31-27 SW_DEC_MODE	Decoding mode: 00000-01011b - Reserved 01100b - HEVC 01101b - VP9 01110-11111b - Reserved
26-24 —	Reserved.
23-20 SW_DEC_COMP_TABLE_SWAP	Byte swap configuration for compress table data
19-18 —	Reserved.
17 SW_DEC_OUT_EC_BYPASS	Compress bypass

Table continues on the next page...

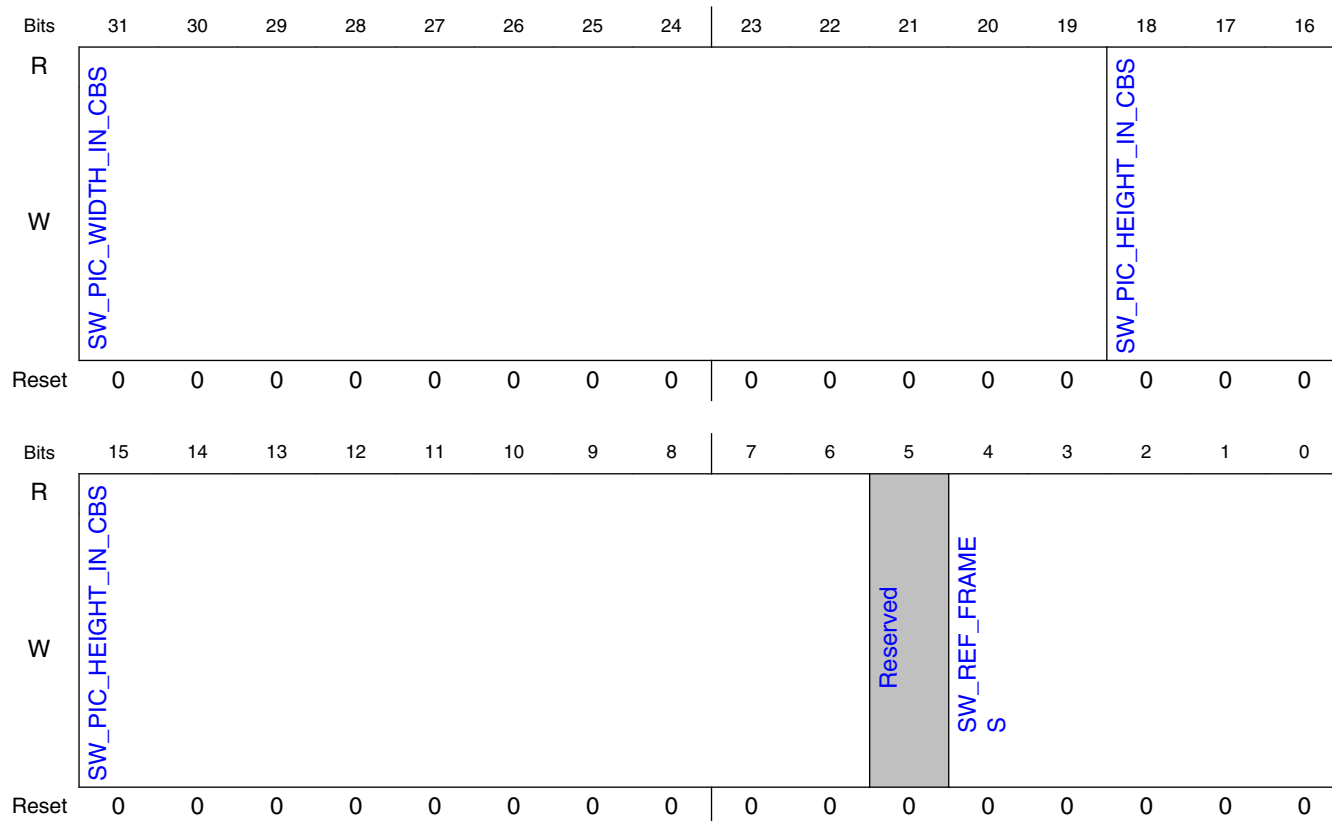
Field	Function
16 SW_DEC_OUT_RS_E	Raster scan output enable. If high decoder writes the raster scan output if the configuration of Decoder includes PP raster scan output
15 SW_DEC_OUT_DIS	Disable decoder output picture writing 0b - Decoder output picture is written to external memory 1b - Decoder output picture is not written to external memory
14 SW_FILTERING_DIS	De-block filtering disable 0b - Filtering is enabled for current picture 1b - Filtering is disabled for current picture
13 —	Reserved.
12 SW_WRITE_MVS_E	Direct mode motion vector write enable for current picture 0b - Writing disabled for current picture. 1b - The direct mode motion vectors are written to external memory. HEVC/VP9 direct mode motion vectors are written to DPB aside with the corresponding reference picture. Other decoding mode dir mode mvs are written to external memory starting from sw_dir_mv_base.
11 SW_APF_ONE_PID	Prefetch partitions that have the same pic_id together
10-0 —	Reserved.

14.2.5.1.6 Decoder control register 1 (SWREG4)

14.2.5.1.6.1 Offset

Register	Offset
SWREG4	10h

14.2.5.1.6.2 Diagram



14.2.5.1.6.3 Fields

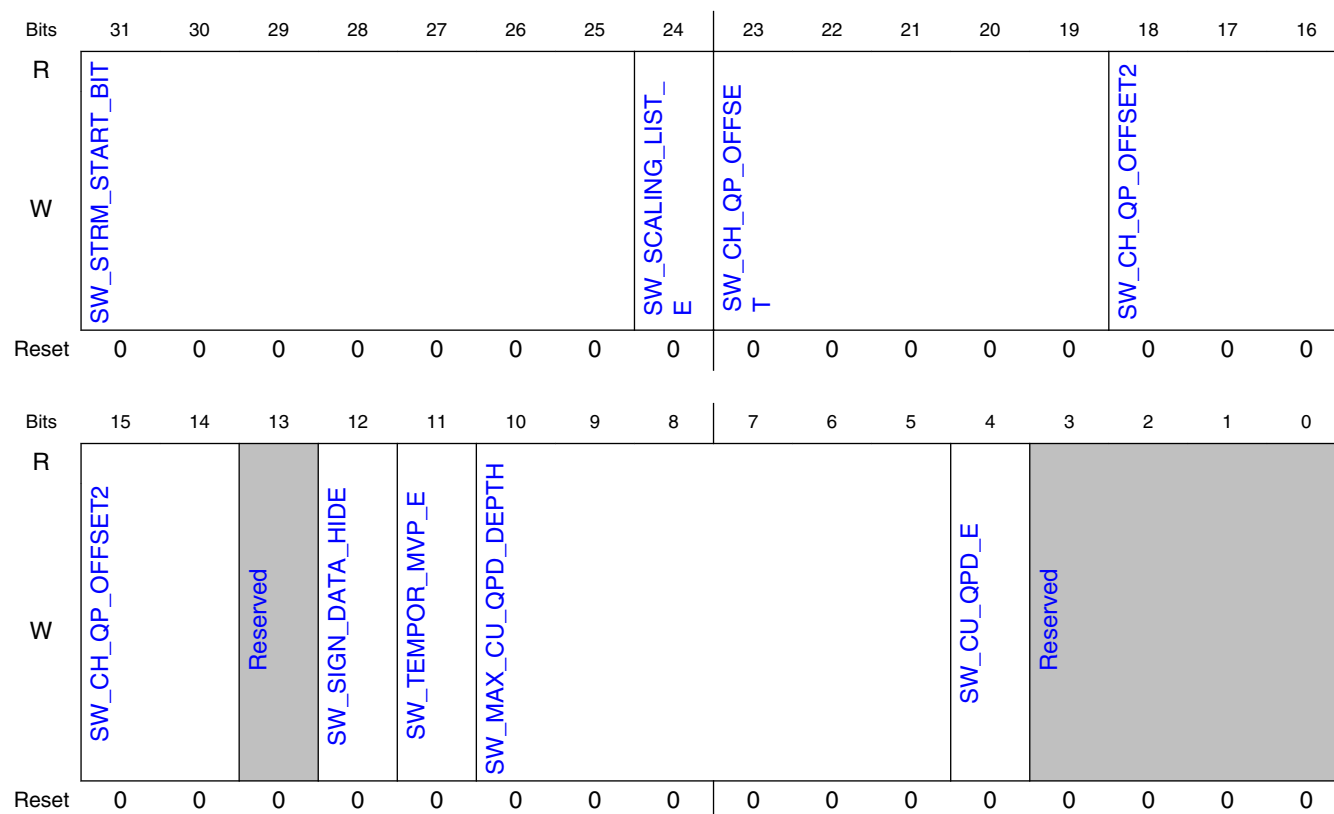
Field	Function
31-19 SW_PIC_WIDT H_IN_CBS	Picture width in min coded blocks (min = 8pix)
18-6 SW_PIC_HEIG HT_IN_CBS	Picture height in min coded blocks (min = 8pix)
5 —	Reserved.
4-0 SW_REF_FRA MES	HEVC: num_ref_frames maximum number of short and long term reference frames in decoded picture buffer

14.2.5.1.7 Decoder control register 2 (SWREG5)

14.2.5.1.7.1 Offset

Register	Offset
SWREG5	14h

14.2.5.1.7.2 Diagram



14.2.5.1.7.3 Fields

Field	Function
31-25 SW_STRM_START_BIT	Exact bit of stream start word where decoding can be started (associates with sw_rlc_vlc_base)
24 SW_SCALING_LIST_E	Scaling matrix enable 0b - Normal transform 1b - Use scaling matrix for transform (read from external memory)
23-19 SW_CH_QP_OFFSET	Chroma Qp filter offset. (For HEVC this offset concerns Cb only)

Table continues on the next page...

VPU G2 Memory Map/Register Definition

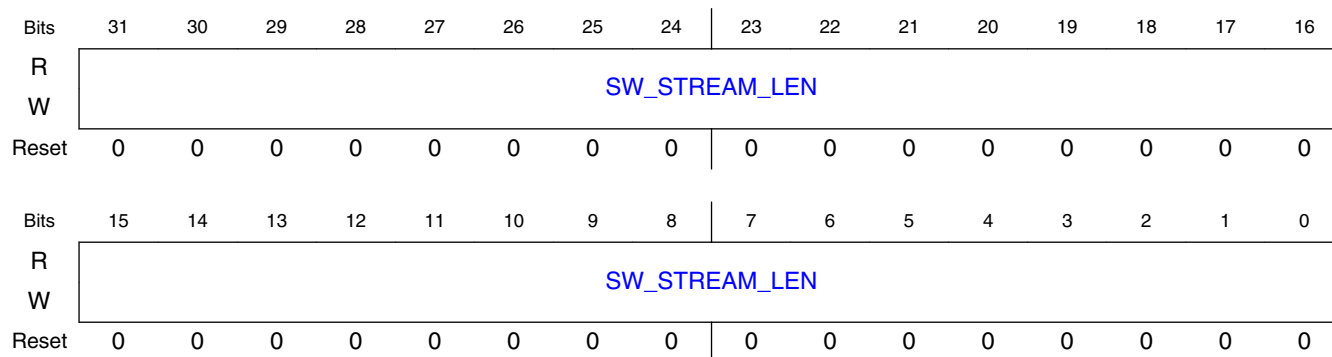
Field	Function
18-14 SW_CH_QP_O FFSET2	Chroma Qp filter offset for cr type
13 —	Reserved.
12 SW_SIGN_DAT A_HIDE	Flag for stream decoding
11 SW_TEMPOR_ MVP_E	Temporal mvp enable
10-5 SW_MAX_CU_ QPD_DEPTH	Max CU qp delta depth
4 SW_CU_QPD_ E	CU qp delta enable
3-0 —	Reserved.

14.2.5.1.8 Decoder control register 3 (SWREG6)

14.2.5.1.8.1 Offset

Register	Offset
SWREG6	18h

14.2.5.1.8.2 Diagram



14.2.5.1.8.3 Fields

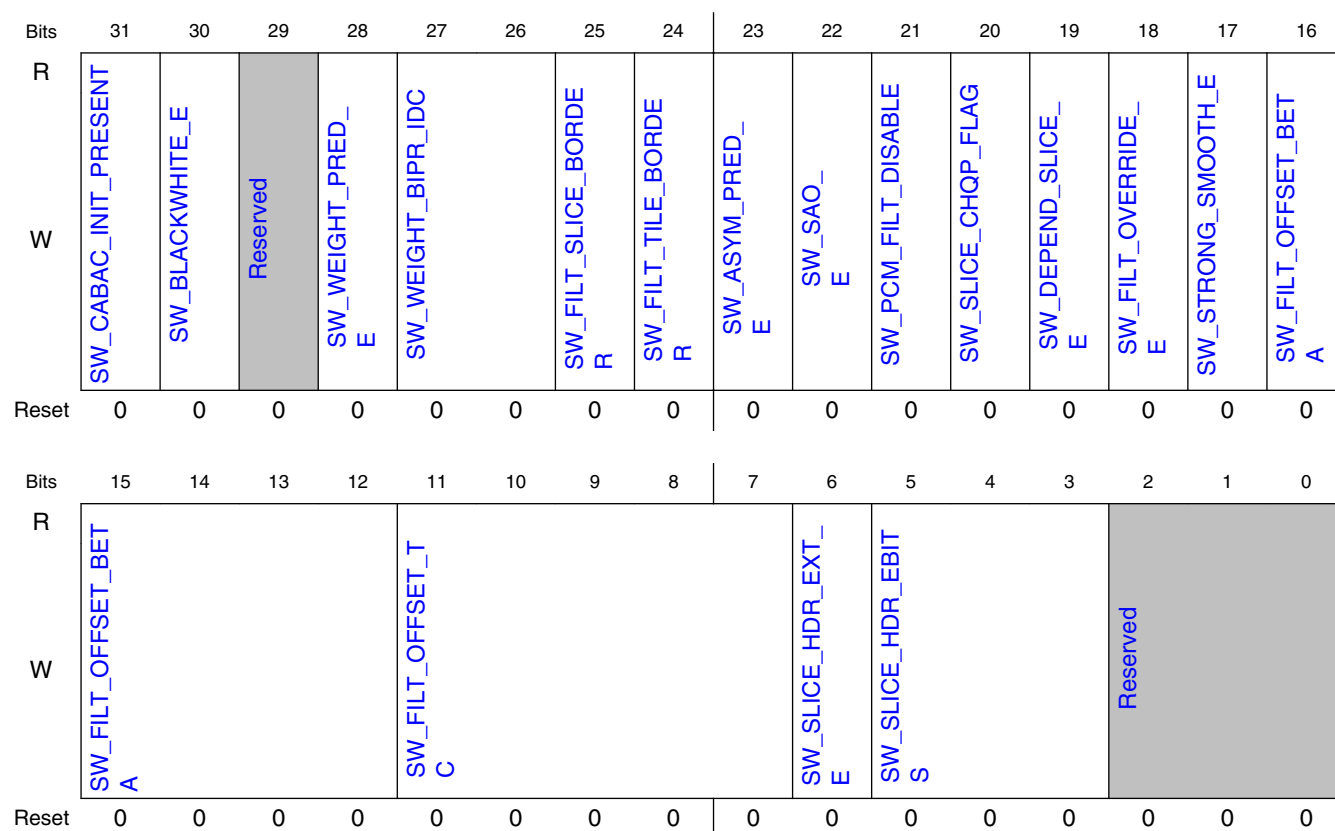
Field	Function
31-0 SW_STREAM_LEN	Amount of stream data bytes in input buffer. If the given buffer size is not enough for finishing the picture the corresponding interrupt is given and new stream buffer base address and stream buffer size information should be given (associates with sw_rlc_vlc_base). For HEVC the buffer must include at least data for one slice/VP of the picture

14.2.5.1.9 Decoder control register 4 (SWREG7)

14.2.5.1.9.1 Offset

Register	Offset
SWREG7	1Ch

14.2.5.1.9.2 Diagram



14.2.5.1.9.3 Fields

Field	Function
31 SW_CABAC_INIT_PRESENT	CABAC init present enable for stream decoding
30 SW_BLACKWHITTE_E	Sampling 0b - 4:2:0 sampling format 1b - 4:0:0 sampling format (H264 monochrome)
29 —	Reserved.
28 SW_WEIGHT_PRED_E	Weighted prediction enable for P slices
27-26 SW_WEIGHT_BIPR_IDC	Weighted prediction specification 00b - Default weighted prediction is applied to B slices 01b - Explicit weighted prediction shall be applied to B slices 10b - NA 11b - NA
25 SW_FILTER_SLICE_BORDER	Filter enable over slice border
24 SW_FILTER_TILE_BORDER	Filter enable over tile border
23 SW_ASYM_PRED_E	Asymmetric prediction flag for stream decoding
22 SW_SAO_E	Sample Adaptive Offset enable for stream decoding
21 SW_PCM_FILTER_DISABLE	Disable for PCM loop filtering
20 SW_SLICE_CHQP_FLAG	Slice header flag for chroma QP present (if it is included in slice header)
19 SW_DEPEND_SLICE_E	Dependent slice enable
18 SW_FILTER_OVERRIDE_E	Filter override enable
17 SW_STRONG_SMOOTH_E	Strong smoothing enable
16-12	Filter beta offset (declared as div2)

Table continues on the next page...

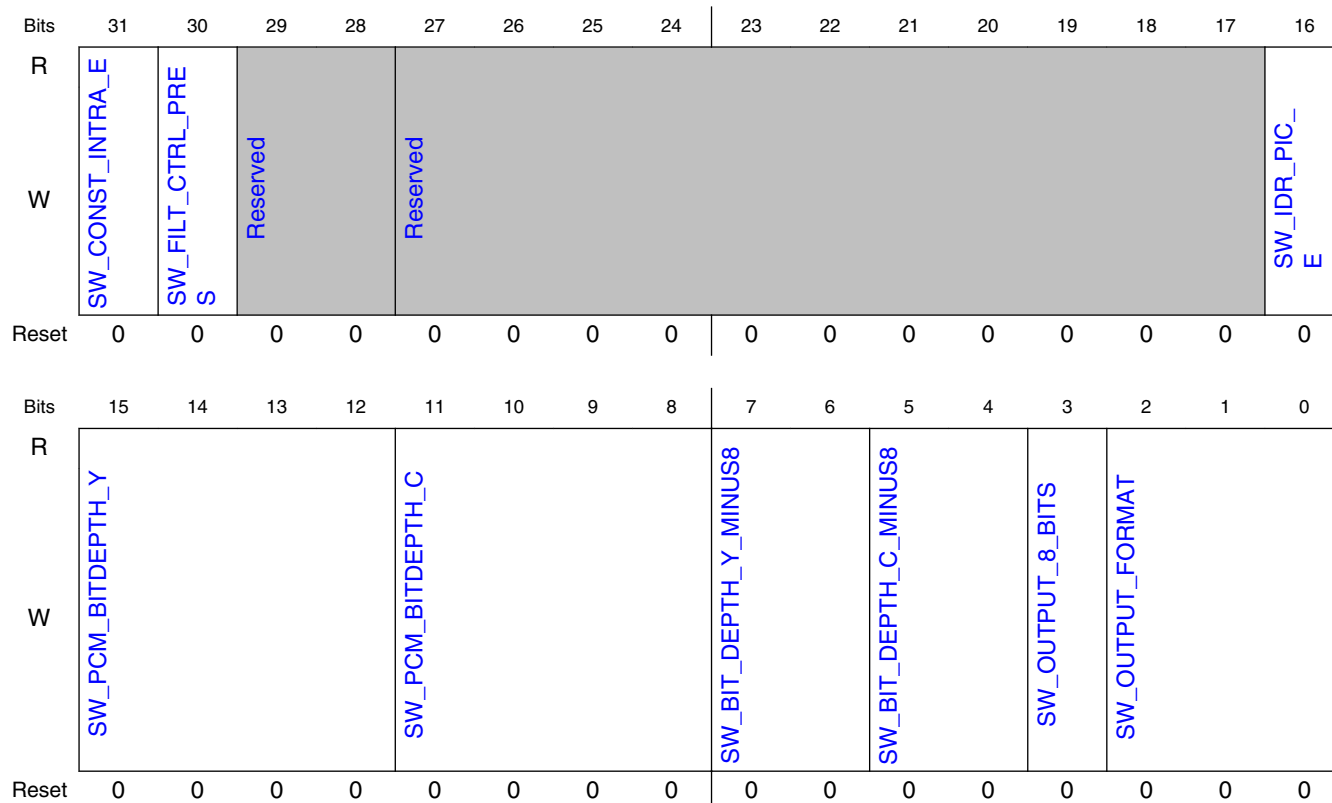
Field	Function
SW_FILT_OFFS ET_BETA	
11-7 SW_FILT_OFFS ET_TC	Filter tc offset (declared as div2)
6 SW_SLICE_HD R_EXT_E	Slice header extension enable. Reserved for future use
5-3 SW_SLICE_HD R_EBITS	Number of extra slice header bits (if enabled slice header extension)
2-0 —	Reserved.

14.2.5.1.10 Decoder control register 5 (SWREG8)

14.2.5.1.10.1 Offset

Register	Offset
SWREG8	20h

14.2.5.1.10.2 Diagram



14.2.5.1.10.3 Fields

Field	Function
31 SW_CONST_INTRA_E	constrained_intra_pred_flag equal to 1 specifies that intra prediction uses only neighbouring intra macroblocks in prediction. When equal to 0 also neighbouring inter macroblocks are used in intra prediction process.
30 SW_FILT_CTRL_PRE	deblocking_filter_control_present_flag indicates whether extra variables controlling characteristics of the deblocking filter are present in the slice header.
29-28 —	Reserved.
27-17 —	Reserved.
16 SW_IDR_PIC_E	IDR (instantaneous decoding refresh) picture flag.
15-12 SW_PCM_BITDEPTH_Y	Bit depth for PCM Y data
11-8	Bit depth for PCM C data

Table continues on the next page...

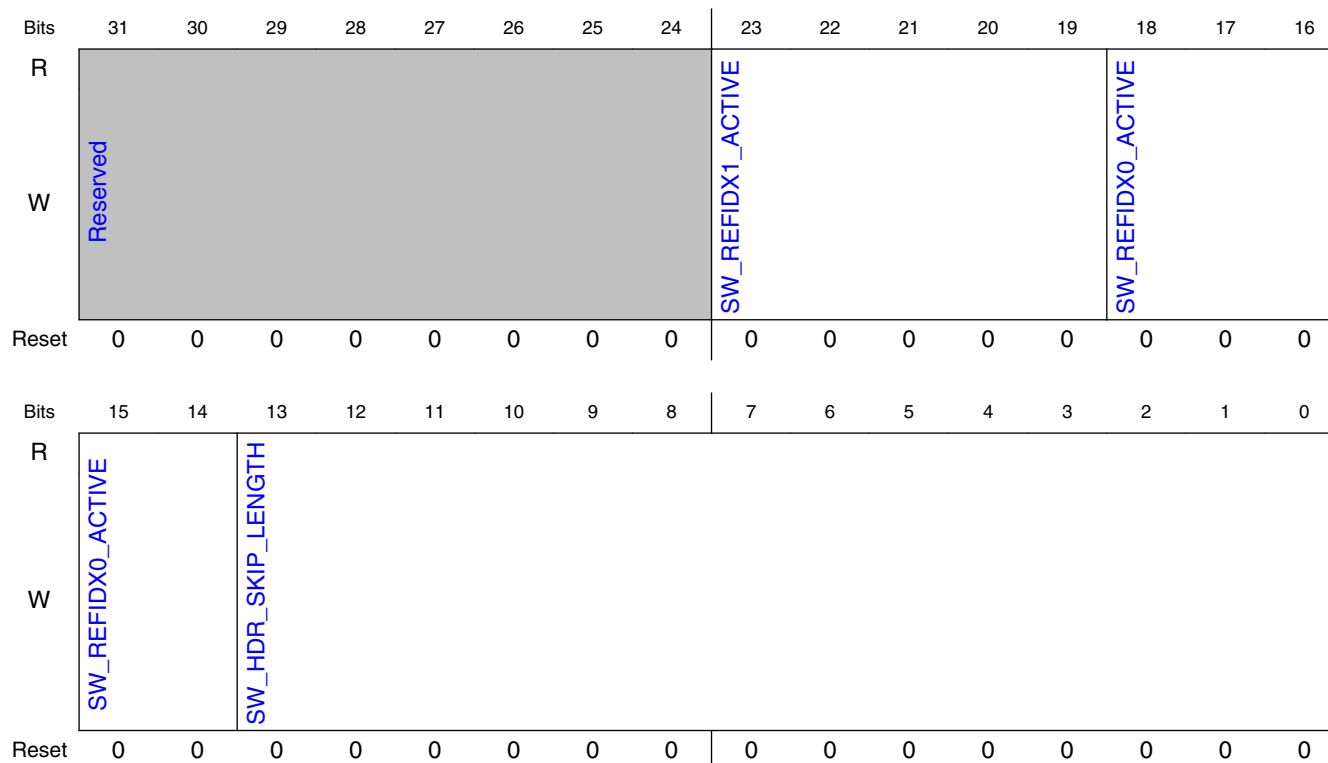
Field	Function
SW_PCM_BITD EPTH_C	
7-6 SW_BIT_DEPT H_Y_MINUS8	Bit depth of luma samples minus 8
5-4 SW_BIT_DEPT H_C_MINUS8	Bit depth of chroma samples minus 8
3 SW_OUTPUT_8 _BITS	enable rasterscan output force to 8 bit(only for hevc main10 and vp9 10bit)
2-0 SW_OUTPUT_F ORMAT	Raster scan and down scale output data format 000b - Each pixel in 10 bits when luma or chroma pixel bit depth is larger than 8; or 8 bits when both luma and chroma pixel bit depth are 8 bits. (default) 001b - Store in P010 format when luma or chroma pixel bit depth is larger than 8. 010b - A customized format: please refer to register SWREG23[6].

14.2.5.1.11 Decoder control register 6 (SWREG9)

14.2.5.1.11.1 Offset

Register	Offset
SWREG9	24h

14.2.5.1.11.2 Diagram



14.2.5.1.11.3 Fields

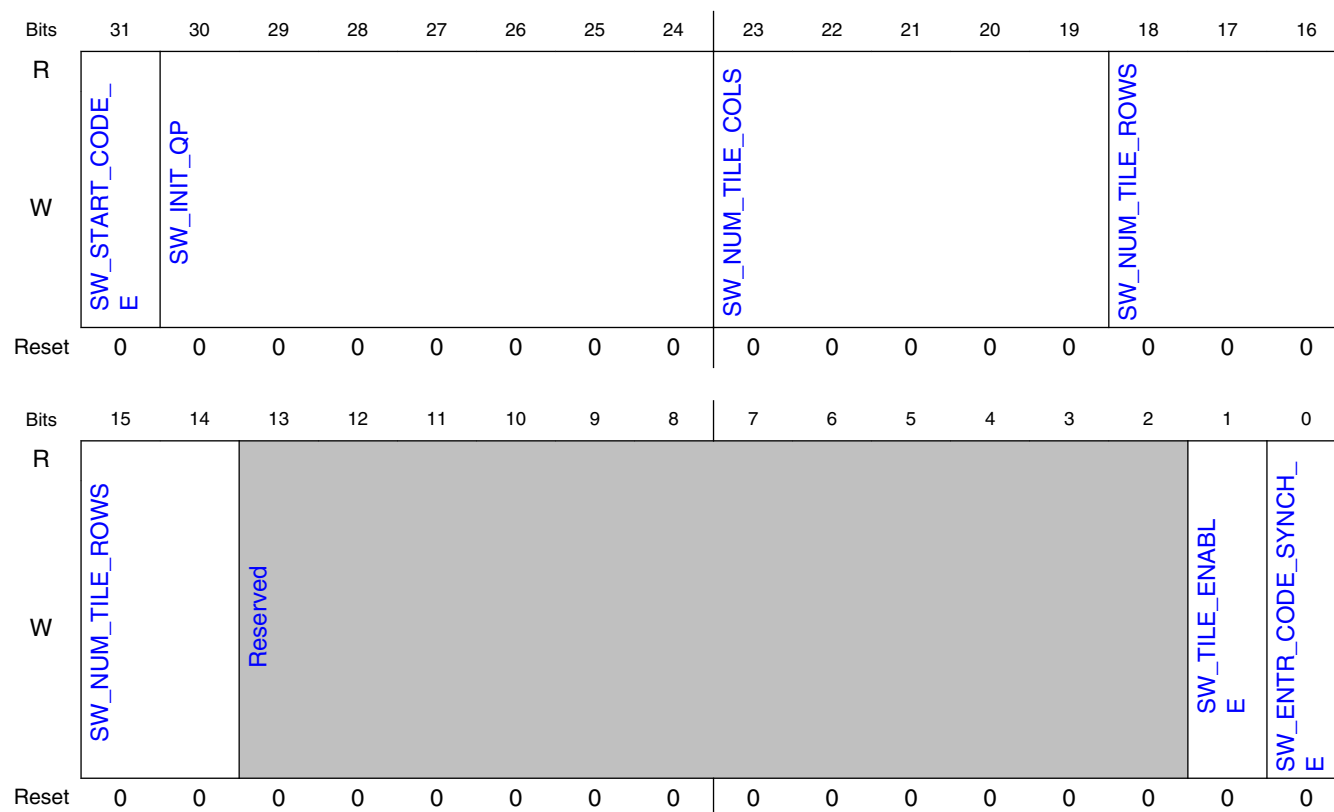
Field	Function
31-24 —	Reserved.
23-19 SW_REFIDX1_ACTIVE	Specifies the maximum reference index that can be used while decoding inter predicted macro blocks.
18-14 SW_REFIDX0_ACTIVE	Specifies the maximum reference index that can be used while decoding inter predicted macro blocks. This is same as in previous decoders (width increased with q bit)
13-0 SW_HDR_SKIP_LENGTH	Length of slice header skip length (bytes used by sw)

14.2.5.1.12 Decoder control register 7 (SWREG10)

14.2.5.1.12.1 Offset

Register	Offset
SWREG10	28h

14.2.5.1.12.2 Diagram



14.2.5.1.12.3 Fields

Field	Function
31 SW_START_CODE_E	Bit for indicating stream start code existence 0b - Stream does not contain start codes 1b - Stream contains start codes
30-24 SW_INIT_QP	Initial value for quantization parameter (picture quantizer).
23-19 SW_NUM_TILE_COLS	Number of tile columns in picture
18-14	Number of tile rows in picture

Table continues on the next page...

VPU G2 Memory Map/Register Definition

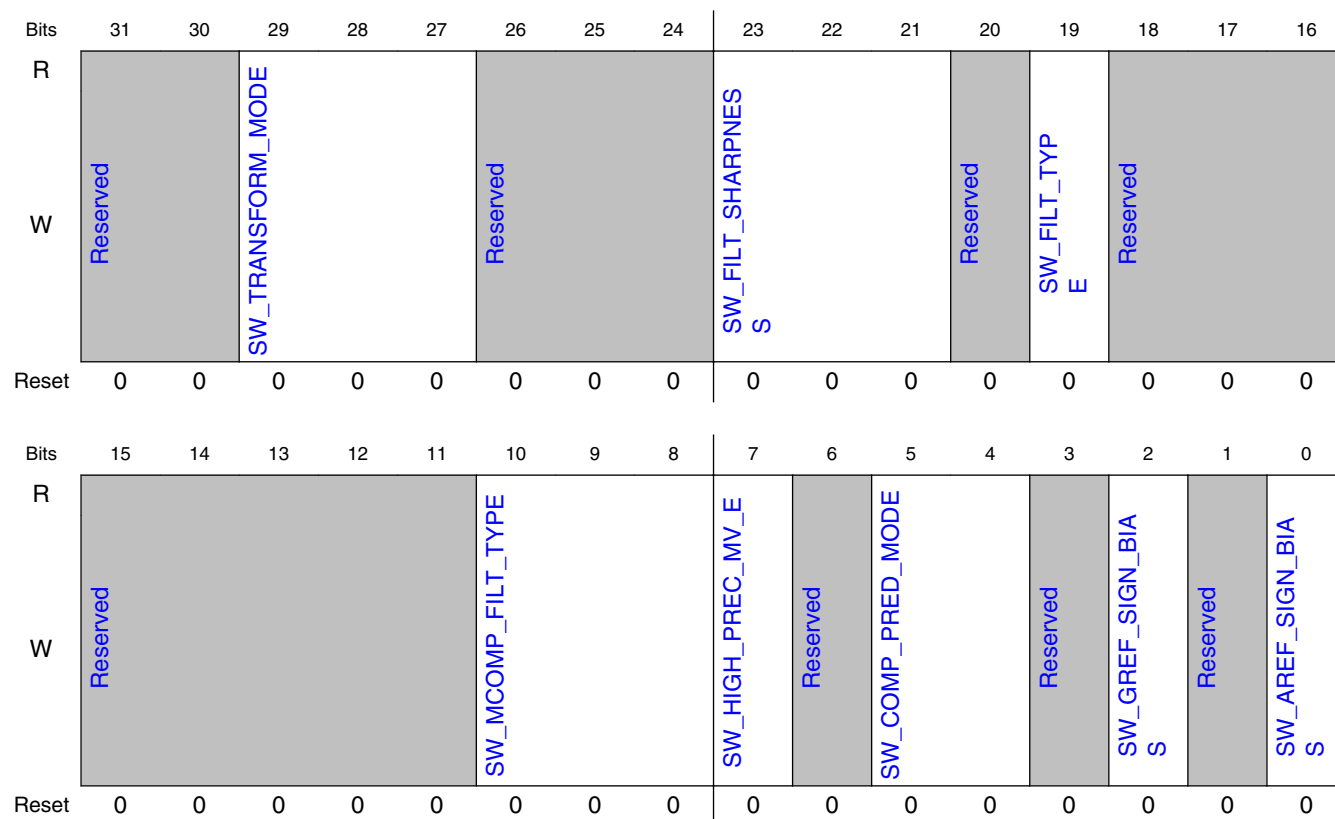
Field	Function
SW_NUM_TILE_ROWS	
13-2 —	Reserved.
1 SW_TILE_ENABLE	Tile enable
0 SW_ENTR_CO DE_SYNCH_E	Entropy coding synchronization enable (Possible parallel cabac decoding)

14.2.5.1.13 Decoder control register 8 (SWREG11)

14.2.5.1.13.1 Offset

Register	Offset
SWREG11	2Ch

14.2.5.1.13.2 Diagram



14.2.5.1.13.3 Fields

Field	Function
31-30 —	Reserved.
29-27 SW_TRANSFORM_MODE	Transform modes 000b - 4x4 only 001b - Allow 8x8 010b - Allow 16x16 011b - Allow 32x32 100b - TX mode select
26-24 —	Reserved.
23-21 SW_FILT_SHARPNESS	Filter sharpness value
20 —	Reserved.
19 SW_FILT_TYPE	Filter Type

Table continues on the next page...

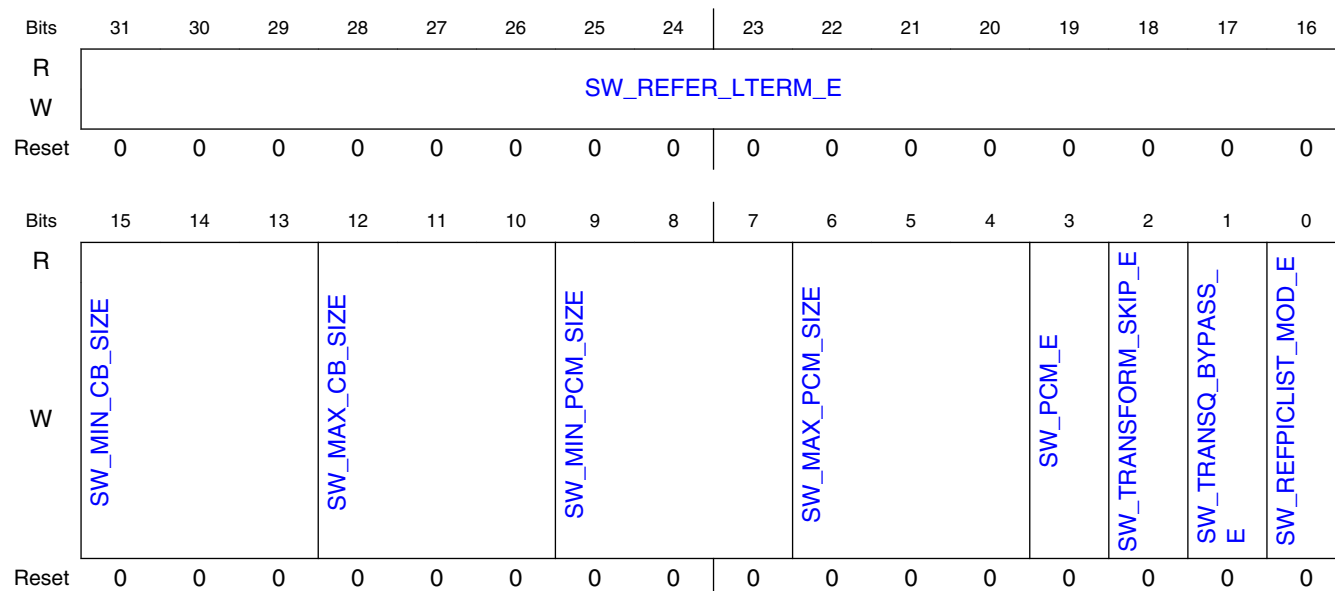
Field	Function
18-11 —	Reserved.
10-8 SW_MCOMP_FILTER_TYPE	Inter prediction filter type to stream decoder 000b - Eight tap smooth 001b - Eight tap 010b - Eight tap sharp 011b - Bilinear 100b - Switchable
7 SW_HIGH_PRECISION_MV_ENABLE	High precision MV prediction enable
6 —	Reserved.
5-4 SW_COMP_PREDICTION_MODE	Prediction Comp Type 00b - Single prediction only 01b - COMP prediction only 10b - Hybrid prediction
3 —	Reserved.
2 SW_GREF_SIGN_BIAS	Golden reference picture sign bias used for motion vector decoding
1 —	Reserved.
0 SW_AREF_SIGN_BIAS	Alternate reference picture sign bias used for motion vector decoding

14.2.5.1.14 Decoder control register 9 (SWREG12)

14.2.5.1.14.1 Offset

Register	Offset
SWREG12	30h

14.2.5.1.14.2 Diagram



14.2.5.1.14.3 Fields

Field	Function
31-16 SW_REFER_LTERM_E	Long term flag for reference picture index Definition: Bit 31 for picture index 0 Bit 30 for picture index 1 etc.
15-13 SW_MIN_CB_SIZE	CodedBlock min size (2^N): 011b - 8 pix 100b - 16 pix 101b - 32 pix 110b - 64 pix
12-10 SW_MAX_CB_SIZE	CodedBlock max size (2^N): 011b - 8 pix 100b - 16 pix 101b - 32 pix 110b - 64 pix
9-7 SW_MIN_PCM_SIZE	PCM min size (2^N): 011b - 8 pix 100b - 16 pix 101b - 32 pix 110b - 64 pix
6-4 SW_MAX_PCM_SIZE	PCM max size (2^N): 011b - 8 pix 100b - 16 pix 101b - 32 pix 110b - 64 pix
3 SW_PCM_E	IPCM MBs flag

Table continues on the next page...

Field	Function
2 SW_TRANSFORM_SKIP_E	Transform skipping flag
1 SW_TRANSQUANT_BYPASS_E	Transform bypass flag (lossless mode)
0 SW_REFPICLIST_MOD_E	Refpic list reordering flag

14.2.5.1.15 Decoder control register 10 (SWREG13)

14.2.5.1.15.1 Offset

Register	Offset
SWREG13	34h

14.2.5.1.15.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved			DEC_CTRL_REG10_BF												
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEC_CTRL_REG10_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.15.3 Fields

Field	Function
31-29 —	Reserved.
28-0 DEC_CTRL_REG10_BF	For HEVC: [31:16] - Reserved - Not used [15:13] - sw_min_trb_size - Transform Block min size (2^N): 3 = 8 pix, 4 = 16 pix, 5 = 32 pix, 6 = 64 pix.

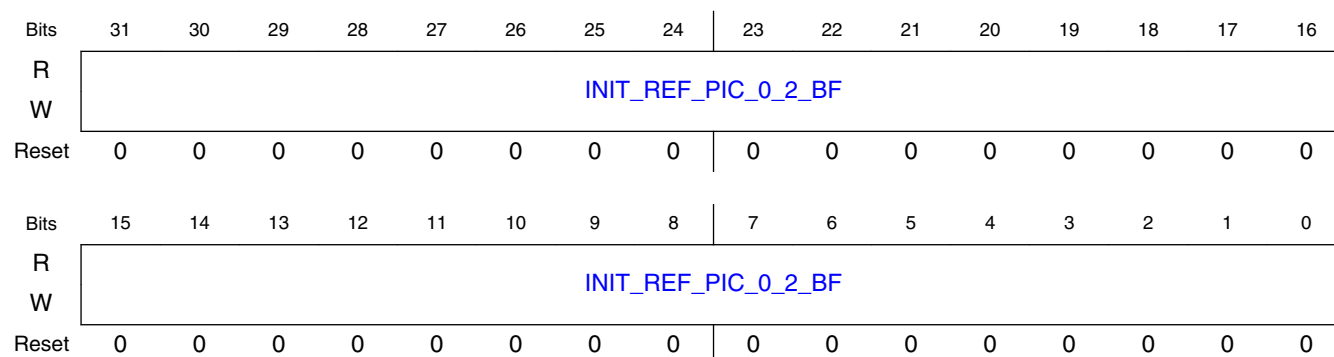
Field	Function
	[12:10] - sw_max_trb_size - Transform Block max size (2^N): 3 = 8 pix, 4 = 16 pix, 5 = 32 pix, 6 = 64 pix. [9:7] - sw_max_intra_hierdepth - Intra max hierarchy dept [6:4] - sw_max_inter_hierdepth - Inter max hierarchy depth [3:0] - sw_parallel_merge - Information about differential MV exist inside of coded block. For VP9: [31:29] - Reserved - Not used [28:23] - sw_qp_delta_y_dc - QP delta value for Y DC [22:17] - sw_qp_delta_ch_dc - QP delta value for C DC [16:11] - sw_qp_delta_ch_ac - QP delta value for C AC [10] - sw_last_sign_bias - Previous reference picture sign bias for motion vector decoding [9] - sw_lossless_e - Lossless transform enable [8:7] - sw_comp_pred_var_ref1 - Compaund prediction parameter to select correct sign bias from bin decoding [6:5] - sw_comp_pred_var_ref0 - Compaund prediction parameter to select correct sign bias from bin decoding [4:3] - sw_comp_pred_fixed_ref - Compaund prediction parameter to select last picture sign bias [2] - sw_segment_temp_upd_e - temporal segmentation update enable [1] - sw_segment_upd_e - segmentation update enable. If high new segmentation values are delivered in stream (and will be written out). If 0 the segmentation values are read from external memory [0] - sw_segment_e - segmentation enable

14.2.5.1.16 Initial ref pic list register (0-2) (SWREG14)

14.2.5.1.16.1 Offset

Register	Offset
SWREG14	38h

14.2.5.1.16.2 Diagram



14.2.5.1.16.3 Fields

Field	Function
31-0 INIT_REF_PIC_0_2_BF	For HEVC: [31:30] - Reserved - Not used [29:25] - sw_init_rlist_b2 - Initial reference picture list for backward picid 2 [24:20] - sw_init_rlist_f2 - Initial reference picture list for forward picid 2 [19:15] - sw_init_rlist_b1 - Initial reference picture list for backward picid 1 [14:10] - sw_init_rlist_f1 - Initial reference picture list for forward picid 1 [9:5] - sw_init_rlist_b0 - Initial reference picture list for backward picid 0 [4:0] - sw_init_rlist_f0 - Initial reference picture list for forward picid 0 For VP9: [31:24] - Reserved - Not used [23:18] - sw_filt_level - Frame filtering level [17:15] - sw_refpic_seg0 - Segment refer picture [14] - sw_skip_seg0 - Segment skip enable [13:8] - sw_filt_level_seg0 - Segment filter level [7:0] - sw_quant_seg0 - Segment quantization parameter

14.2.5.1.17 Initial ref pic list register (3-5) (SWREG15)

14.2.5.1.17.1 Offset

Register	Offset
SWREG15	3Ch

14.2.5.1.17.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	INIT_REF_PIC_3_5_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	INIT_REF_PIC_3_5_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.17.3 Fields

Field	Function
31-0	For HEVC:

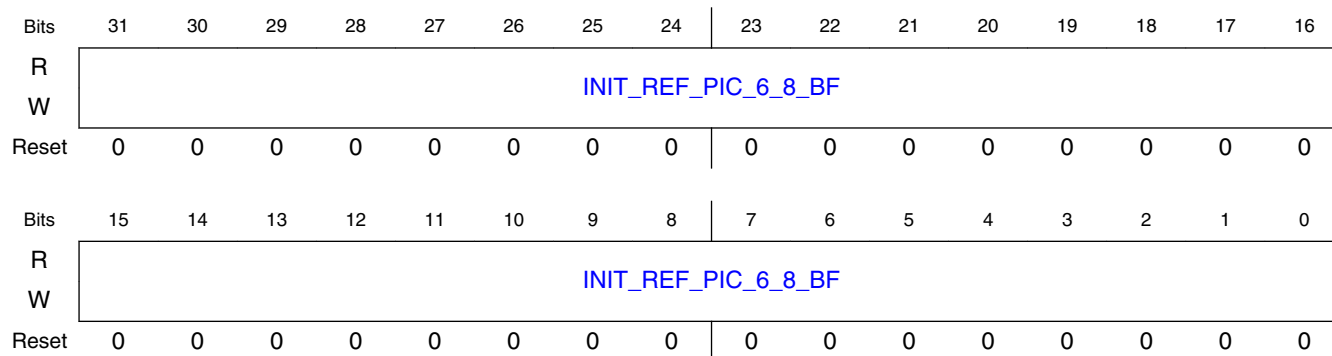
Field	Function
INIT_REF_PIC_3_5_BF	[31:30] - Reserved - Not used [29:25] - sw_init_rlist_b5 - Initial reference picture list for backward picid 5 [24:20] - sw_init_rlist_f5 - Initial reference picture list for forward picid 5 [19:15] - sw_init_rlist_b4 - Initial reference picture list for backward picid 4 [14:10] - sw_init_rlist_f4 - Initial reference picture list for forward picid 4 [9:5] - sw_init_rlist_b3 - Initial reference picture list for backward picid 3 [4:0] - sw_init_rlist_f3 - Initial reference picture list for forward picid 3 For VP9: [31:18] - Reserved - Not used [17:15] - sw_refpic_seg1 - Segment refer picture [14] - sw_skip_seg1 - Segment skip enable [13:8] - sw_filt_level_seg1 - Segment filter level [7:0] - sw_quant_seg1 - Segment quantization parameter

14.2.5.1.18 Initial ref pic list register (6-8) (SWREG16)

14.2.5.1.18.1 Offset

Register	Offset
SWREG16	40h

14.2.5.1.18.2 Diagram



14.2.5.1.18.3 Fields

Field	Function
31-0 INIT_REF_PIC_6_8_BF	For HEVC: [31:30] - Reserved - Not used [29:25] - sw_init_rlist_b8 - Initial reference picture list for backward picid 8 [24:20] - sw_init_rlist_f8 - Initial reference picture list for forward picid 8 [19:15] - sw_init_rlist_b7 - Initial reference picture list for backward picid 7 [14:10] - sw_init_rlist_f7 - Initial reference picture list for forward picid 7

Field	Function
	[9:5] - sw_init_rlist_b6 - Initial reference picture list for backward picid 6 [4:0] - sw_init_rlist_f6 - Initial reference picture list for forward picid 6 For VP9: [31:18] - Reserved - Not used [17:15] - sw_refpic_seg2 - Segment refer picture [14] - sw_skip_seg2 - Segment skip enable [13:8] - sw_filt_level_seg2 - Segment filter level [7:0] - sw_quant_seg2 - Segment quantization parameter

14.2.5.1.19 Initial ref pic list register (9-11) (SWREG17)

14.2.5.1.19.1 Offset

Register	Offset
SWREG17	44h

14.2.5.1.19.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	INIT_REF_PIC_9_11_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	INIT_REF_PIC_9_11_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.19.3 Fields

Field	Function
31-0 INIT_REF_PIC_9_11_BF	For HEVC: [31:30] - Reserved - Not used [29:25] - sw_init_rlist_b11 - Initial reference picture list for backward picid 11 [24:20] - sw_init_rlist_f11 - Initial reference picture list for forward picid 11 [19:15] - sw_init_rlist_b10 - Initial reference picture list for backward picid 10 [14:10] - sw_init_rlist_f10 - Initial reference picture list for forward picid 10 [9:5] - sw_init_rlist_b9 - Initial reference picture list for backward picid 9 [4:0] - sw_init_rlist_f9 - Initial reference picture list for forward picid 9 For VP9: [31:18] - Reserved - Not used

Field	Function
	[17:15] - sw_refpic_seg3 - Segment refer picture [14] - sw_skip_seg3 - Segment skip enable [13:8] - sw_filt_level_seg3 - Segment filter level [7:0] - sw_quant_seg3 - Segment quantization parameter

14.2.5.1.20 Initial ref pic list register (12-14) (SWREG18)

14.2.5.1.20.1 Offset

Register	Offset
SWREG18	48h

14.2.5.1.20.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	INIT_REF_PIC_12_14_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	INIT_REF_PIC_12_14_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.20.3 Fields

Field	Function
31-0 INIT_REF_PIC_12_14_BF	For HEVC: [31:30] - Reserved - Not used [29:25] - sw_init_rlist_b14 - Initial reference picture list for backward picid 14 [24:20] - sw_init_rlist_f14 - Initial reference picture list for forward picid 14 [19:15] - sw_init_rlist_b13 - Initial reference picture list for backward picid 13 [14:10] - sw_init_rlist_f13 - Initial reference picture list for forward picid 13 [9:5] - sw_init_rlist_b12 - Initial reference picture list for backward picid 12 [4:0] - sw_init_rlist_f12 - Initial reference picture list for forward picid 12 For VP9: [31:18] - Reserved - Not used [17:15] - sw_refpic_seg4 - Segment refer picture [14] - sw_skip_seg4 - Segment skip enable [13:8] - sw_filt_level_seg4 - Segment filter level [7:0] - sw_quant_seg4 - Segment quantization parameter

14.2.5.1.21 Initial ref pic list register (15 and P 0-3) (SWREG19)

14.2.5.1.21.1 Offset

Register	Offset
SWREG19	4Ch

14.2.5.1.21.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	INIT_REF_PIC_15_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	INIT_REF_PIC_15_BF															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.21.3 Fields

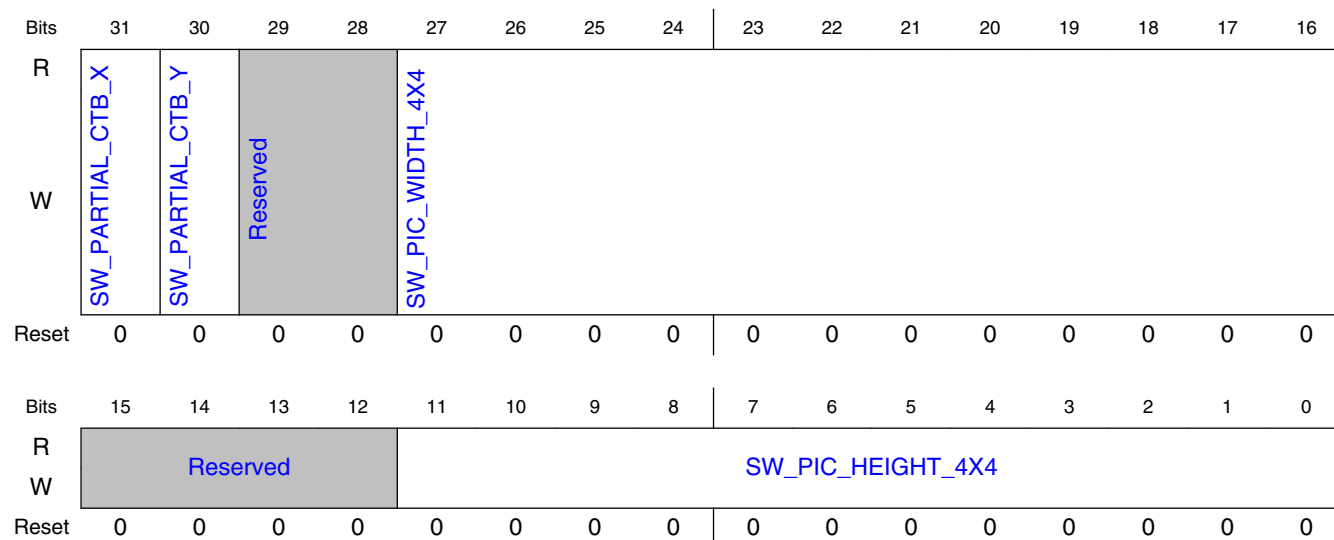
Field	Function
31-0 INIT_REF_PIC_15_BF	For HEVC: [31:10] - Reserved - Not used [9:5] - sw_init_rlist_b15 - Initial reference picture list for backward picid 15 [4:0] - sw_init_rlist_f15 - Initial reference picture list for forward picid 15 For VP9: [31:18] - Reserved - Not used [17:15] - sw_refpic_seg5 - Segment refer picture [14] - sw_skip_seg5 - Segment skip enable [13:8] - sw_filt_level_seg5 - Segment filter level [7:0] - sw_quant_seg5 - Segment quantization parameter

14.2.5.1.22 Decoder control register 11 (SWREG20)

14.2.5.1.22.1 Offset

Register	Offset
SWREG20	50h

14.2.5.1.22.2 Diagram



14.2.5.1.22.3 Fields

Field	Function
31 SW_PARTIAL_CTB_X	Picture width not multiple of CTB size
30 SW_PARTIAL_CTB_Y	Picture height not multiple of CTB size
29-28 —	Reserved.
27-16 SW_PIC_WIDT H_4X4	Current picture width in 4x4 blocks (Needed to reduce overlapping HW conditions in various blocks)
15-12 —	Reserved.
11-0 SW_PIC_HEIG HT_4X4	Current picture height in 4x4 blocks (Needed to reduce overlapping HW conditions in various blocks)

14.2.5.1.23 Not used (SWREG21)

14.2.5.1.23.1 Offset

Register	Offset
SWREG21	54h

14.2.5.1.23.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.23.3 Fields

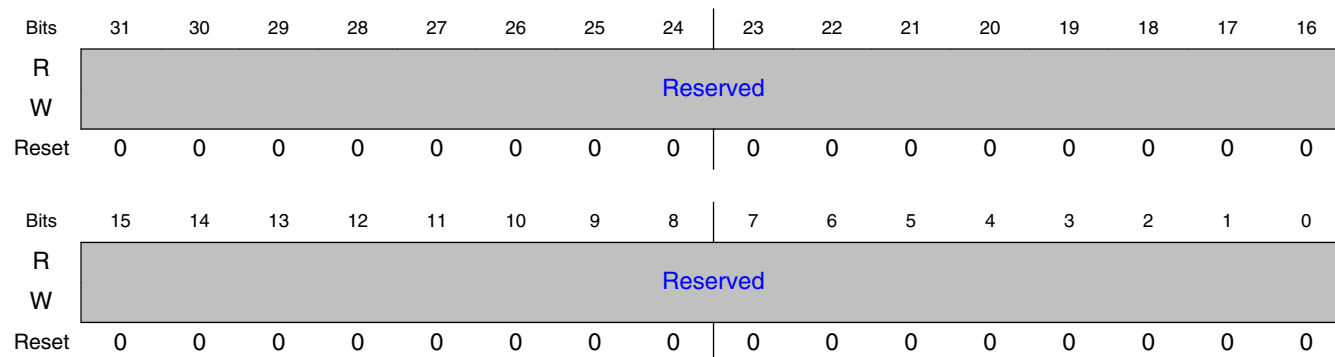
Field	Function
31-0	Reserved.
—	

14.2.5.1.24 Not used (SWREG22)

14.2.5.1.24.1 Offset

Register	Offset
SWREG22	58h

14.2.5.1.24.2 Diagram



14.2.5.1.24.3 Fields

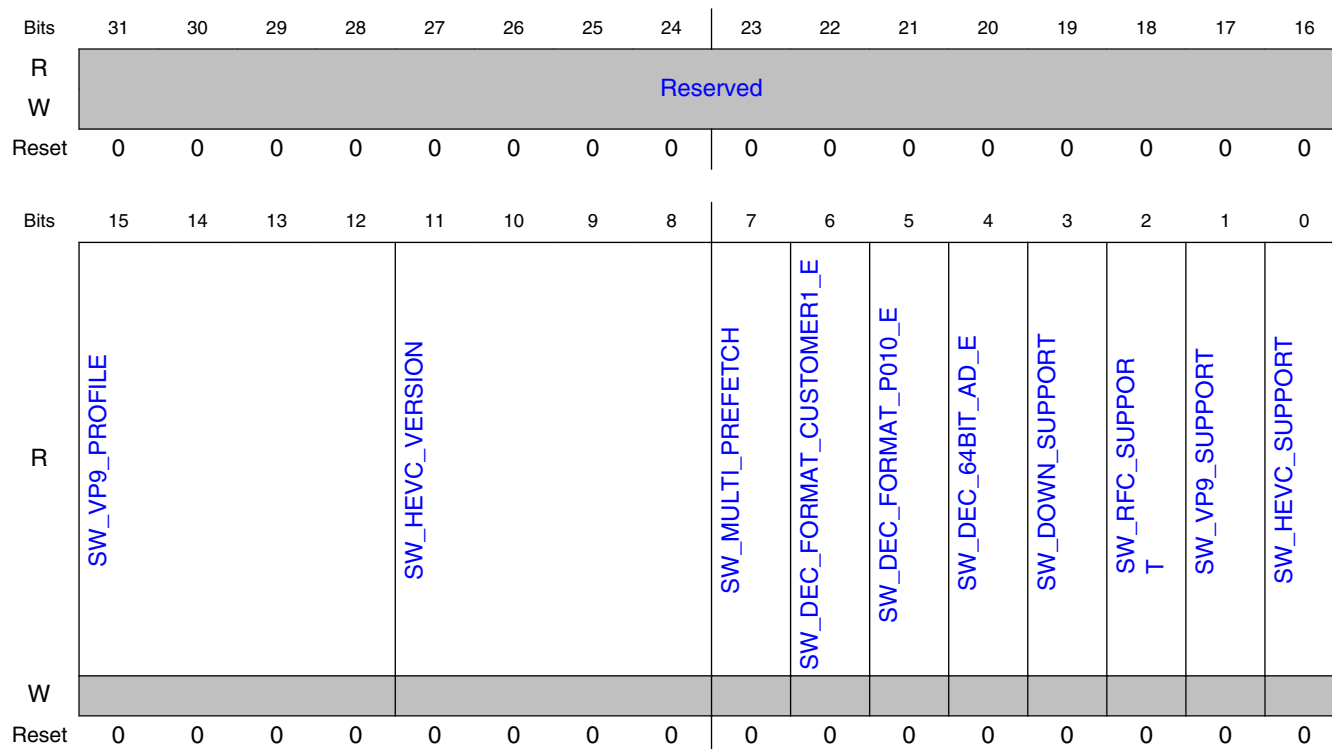
Field	Function
31-0	Reserved.
—	

14.2.5.1.25 Decoder configure status register (SWREG23)

14.2.5.1.25.1 Offset

Register	Offset
SWREG23	5Ch

14.2.5.1.25.2 Diagram



14.2.5.1.25.3 Fields

Field	Function
31-16 —	Reserved.
15-12 SW_VP9_PROFILE	VP9 version 0000b - vp9 profile 0 0001b - vp9 profile 2 - 10bits
11-8 SW_HEVC_VERSION	HEVC version 0000b - main8 0001b - main10
7 SW_MULTI_PREFETCH	Multi-Reference Blocks Prefetch 0b - Not supported 1b - Supported
6 SW_DEC_FORMAT_CUSTOMER1_E	Customized output format support Demonstrated as following - from MSB->LSB in a 128-bit burst. 10-bit pixel 128-bit burst0: Y0 Y1 Y2 Y3 ... Y11 Y12[9:2] 128-bit burst1: Y12[1:0] Y13 Y14 Y15 ... Y24 Y25[9:4] 128-bit burst2: Y25[3:0] Y26 Y27 Y28 ... Y37 Y38[9:6]

Table continues on the next page...

Field	Function
	128-bit burst3: Y38[5:0] Y39 Y40 Y41 ... Y50 Y51[9:8] 128-bit burst4: Y51[7:0] Y52 Y53 Y54 ... Y62 Y63 8-bit pixel 128-bit burst: Y0 Y1 Y2 ... Y15 128-bit burst: Y16 Y17 Y18 ... Y31 0b - Not supported 1b - Supported
5 SW_DEC_FOR MAT_P010_E	P010 output format support 0b - Not supported 1b - Supported
4 SW_DEC_64BI T_AD_E	64 bit addressing of master interface support 0b - Not supported (32 bit addressing) 1b - Supported
3 SW_DOWN_SU PPORT	Downscale support 0b - Do not support downscale 1b - Support downscale
2 SW_RFC_SUP PORT	RFC support 0b - Do not support RFC 1b - Support RFC
1 SW_VP9_SUPP ORT	VP9 support 0b - Do not support VP9 1b - Support VP9
0 SW_HEVC_SU PPORT	HEVC support 0b - Do not support HEVC 1b - Support HEVC

14.2.5.1.26 Not used (SWREG24)

14.2.5.1.26.1 Offset

Register	Offset
SWREG24	60h

14.2.5.1.26.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.26.3 Fields

Field	Function
31-0	Reserved.
—	

14.2.5.1.27 Not used (SWREG25)

14.2.5.1.27.1 Offset

Register	Offset
SWREG25	64h

14.2.5.1.27.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.27.3 Fields

Field	Function
31-0 —	Reserved.

14.2.5.1.28 Not used (SWREG26)

14.2.5.1.28.1 Offset

Register	Offset
SWREG26	68h

14.2.5.1.28.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.28.3 Fields

Field	Function
31-0 —	Reserved.

14.2.5.1.29 Not used (SWREG27)

14.2.5.1.29.1 Offset

Register	Offset
SWREG27	6Ch

14.2.5.1.29.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

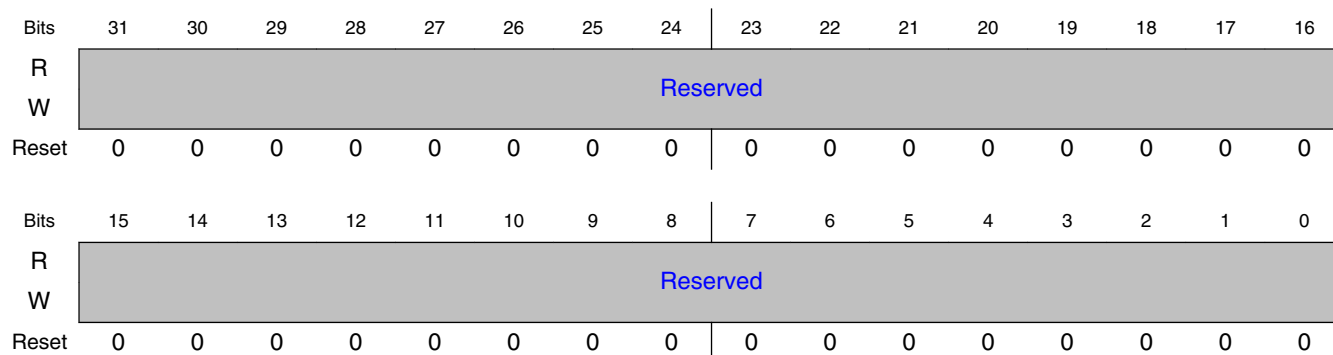
14.2.5.1.29.3 Fields

Field	Function
31-0	Reserved.
—	

14.2.5.1.30 Not used (SWREG28)**14.2.5.1.30.1 Offset**

Register	Offset
SWREG28	70h

14.2.5.1.30.2 Diagram



14.2.5.1.30.3 Fields

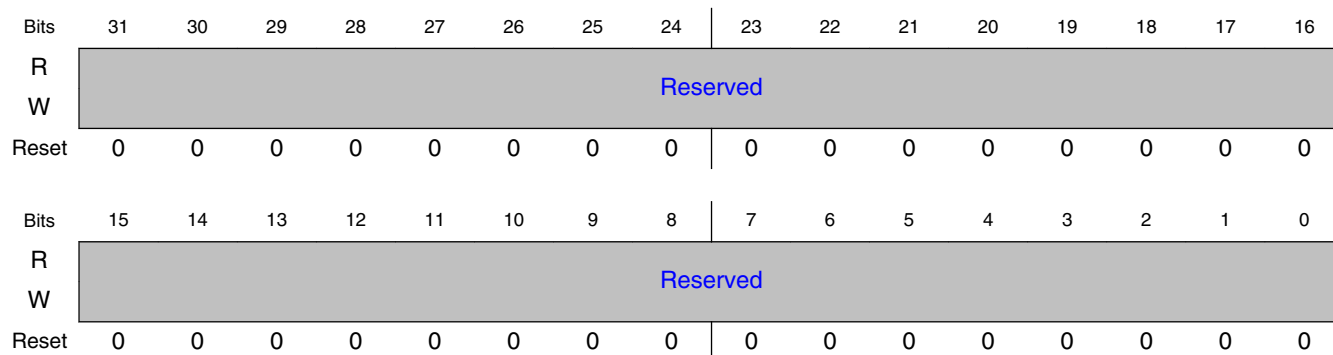
Field	Function
31-0	Reserved.
—	

14.2.5.1.31 Not used (SWREG29)

14.2.5.1.31.1 Offset

Register	Offset
SWREG29	74h

14.2.5.1.31.2 Diagram



14.2.5.1.31.3 Fields

Field	Function
31-0 —	Reserved.

14.2.5.1.32 Not used (SWREG30)**14.2.5.1.32.1 Offset**

Register	Offset
SWREG30	78h

14.2.5.1.32.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.32.3 Fields

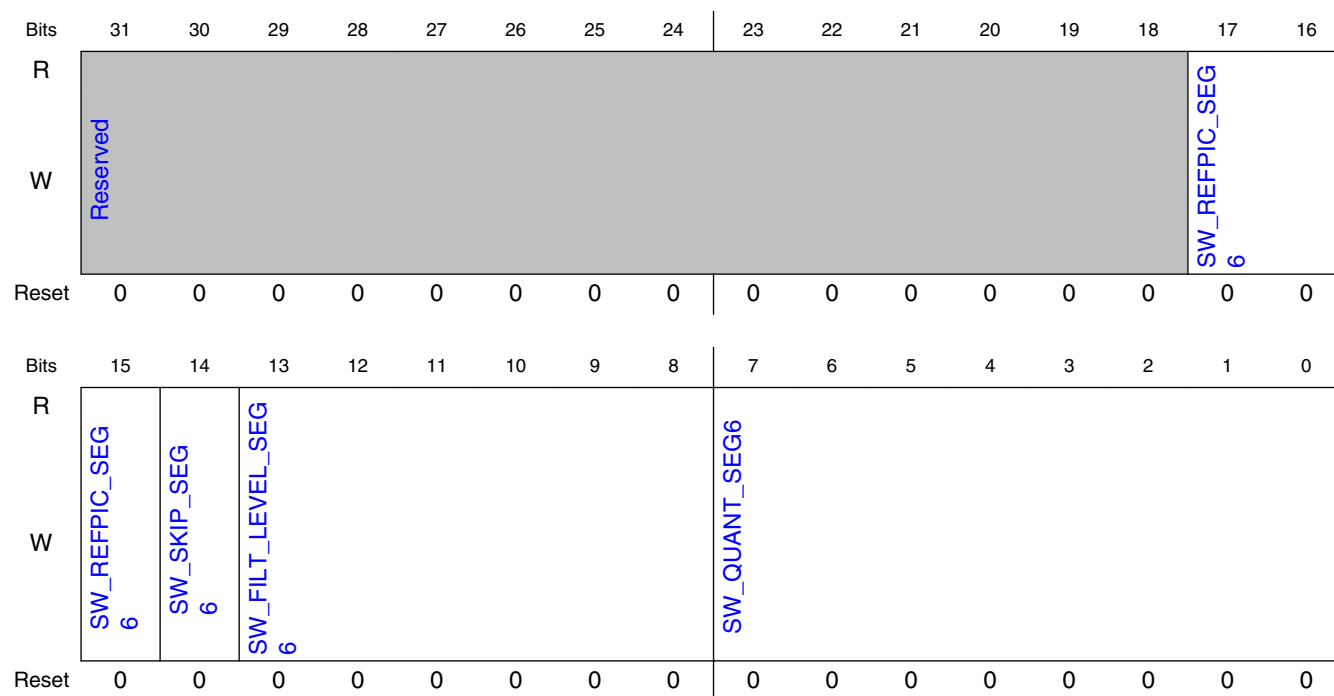
Field	Function
31-0 —	Reserved.

14.2.5.1.33 VP9 segmentation values (SWREG31)

14.2.5.1.33.1 Offset

Register	Offset
SWREG31	7Ch

14.2.5.1.33.2 Diagram



14.2.5.1.33.3 Fields

Field	Function
31-18 —	Reserved.
17-15 SW_REFPIC_SEG6	Segment refer picture
14 SW_SKIP_SEG6	Segment skip enable
13-8 SW_FILT_LEVEL_SEG6	Segment filter level
7-0	Segment quantization parameter

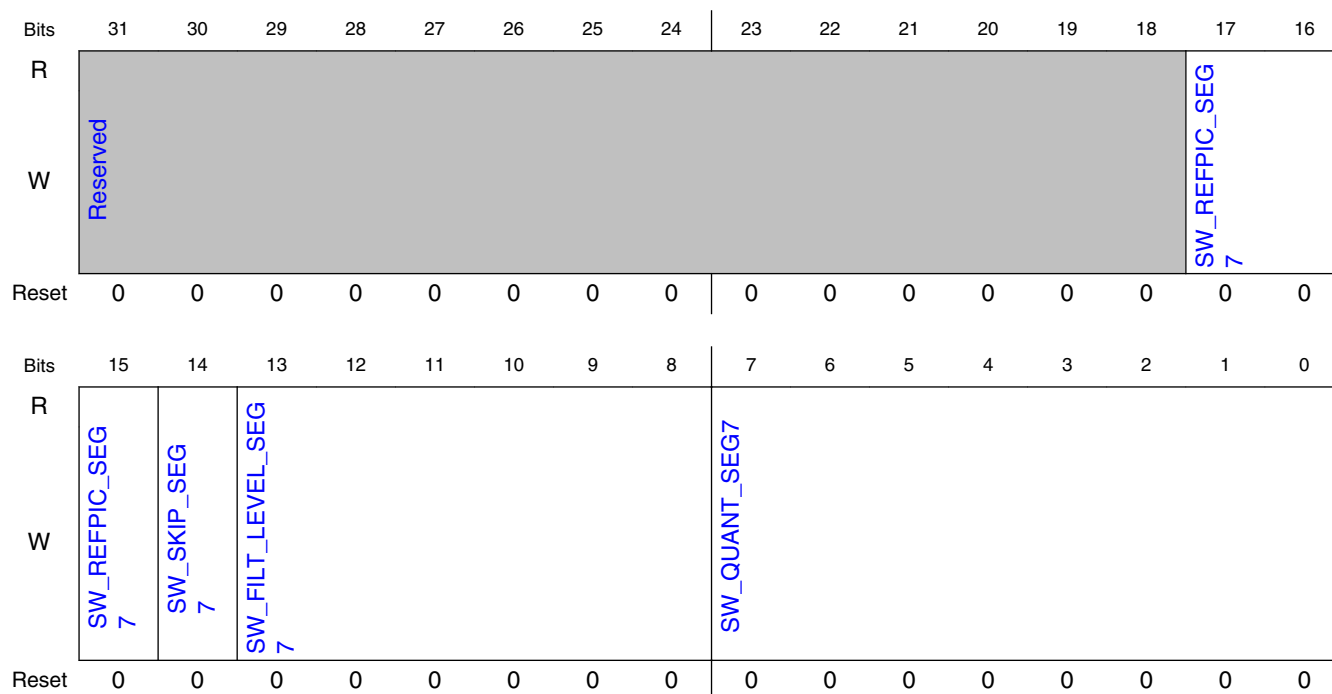
Field	Function
SW_QUANT_SEG6	

14.2.5.1.34 VP9 segmentation values (SWREG32)

14.2.5.1.34.1 Offset

Register	Offset
SWREG32	80h

14.2.5.1.34.2 Diagram



14.2.5.1.34.3 Fields

Field	Function
31-18	Reserved.
—	
17-15	Segment refer picture

Table continues on the next page...

Field	Function
SW_REFPIC_SEG7	
14 SW_SKIP_SEG7	Segment skip enable
13-8 SW_FILT_LEVEL_SEG7	Segment filter level
7-0 SW_QUANT_SEG7	Segment quantization parameter

14.2.5.1.35 VP9 reference picture scaling register 0 (SWREG33)

14.2.5.1.35.1 Offset

Register	Offset
SWREG33	84h

14.2.5.1.35.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_LREF_WIDTH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_LREF_HEIGHT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.35.3 Fields

Field	Function
31-16 SW_LREF_WIDTH	Accurate width of last (previous) reference picture in pixels
15-0	Accurate height of last (previous) reference picture in pixels

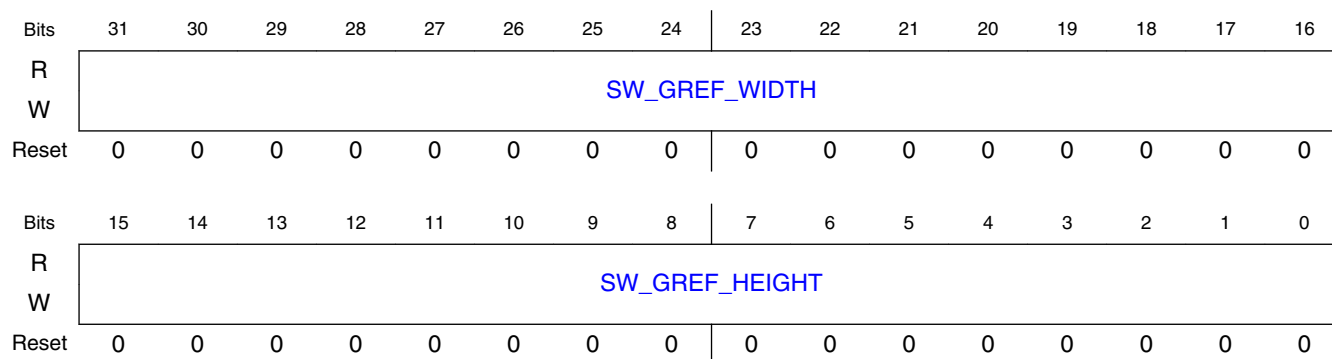
Field	Function
SW_LREF_HEIGHT	

14.2.5.1.36 VP9 reference picture scaling register 1 (SWREG34)

14.2.5.1.36.1 Offset

Register	Offset
SWREG34	88h

14.2.5.1.36.2 Diagram



14.2.5.1.36.3 Fields

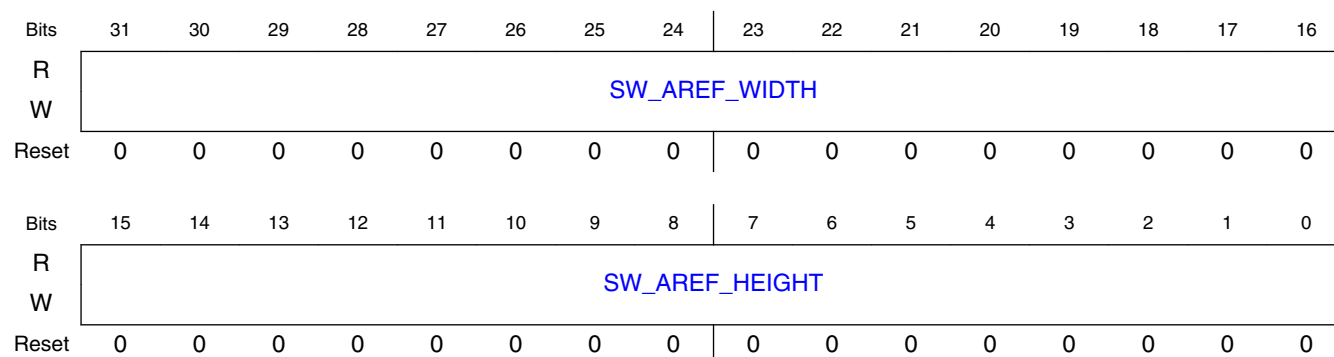
Field	Function
31-16 SW_GREF_WIDTH	Accurate width of golden reference picture in pixels
15-0 SW_GREF_HEIGHT	Accurate height of golden reference picture in pixels

14.2.5.1.37 VP9 reference picture scaling register 2 (SWREG35)

14.2.5.1.37.1 Offset

Register	Offset
SWREG35	8Ch

14.2.5.1.37.2 Diagram



14.2.5.1.37.3 Fields

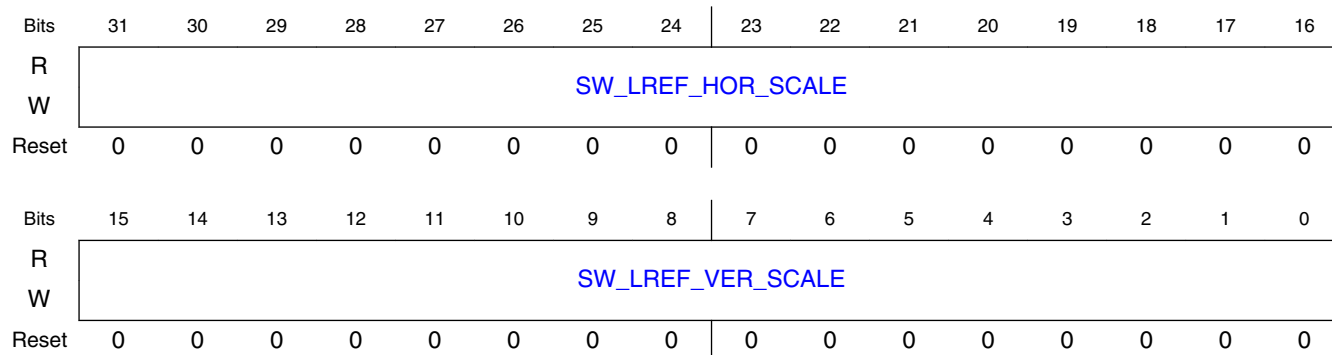
Field	Function
31-16 SW_AREF_WIDTH	Accurate width of alternate reference picture in pixels
15-0 SW_AREF_HEIGHT	Accurate height of alternate reference picture in pixels

14.2.5.1.38 VP9 reference picture scaling register 3 (SWREG36)

14.2.5.1.38.1 Offset

Register	Offset
SWREG36	90h

14.2.5.1.38.2 Diagram



14.2.5.1.38.3 Fields

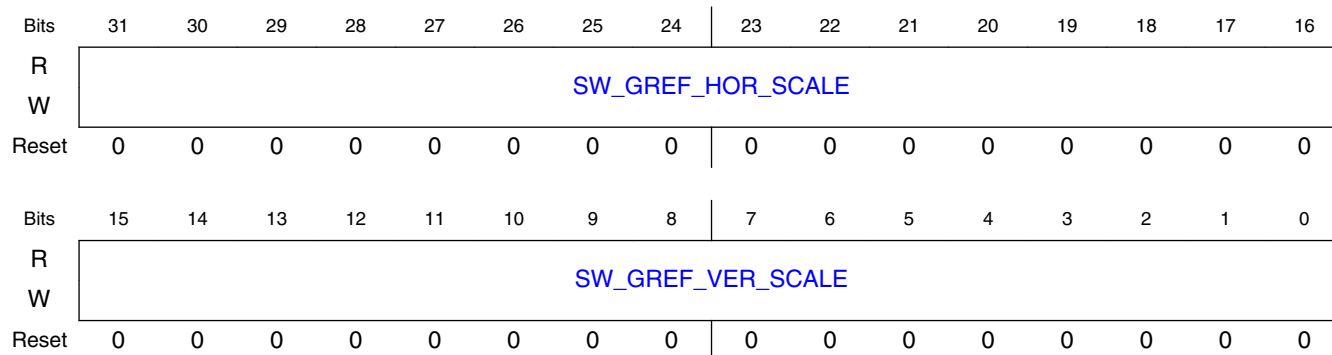
Field	Function
31-16 SW_LREF_HOR_SCALE	Horizontal scaling factor for last (previous) reference picture
15-0 SW_LREF_VER_SCALE	Vertical scaling factor for last (previous) reference picture

14.2.5.1.39 VP9 reference picture scaling register 4 (SWREG37)

14.2.5.1.39.1 Offset

Register	Offset
SWREG37	94h

14.2.5.1.39.2 Diagram



14.2.5.1.39.3 Fields

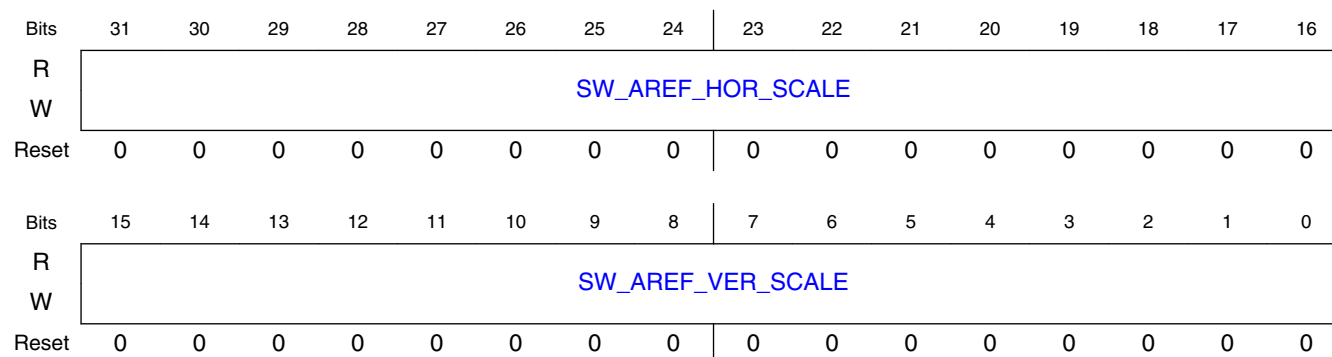
Field	Function
31-16 SW_GREF_HO R_SCALE	Horizontal scaling factor for golden reference picture
15-0 SW_GREF_VE R_SCALE	Vertical scaling factor for golden reference picture

14.2.5.1.40 VP9 reference picture scaling register 5 (SWREG38)

14.2.5.1.40.1 Offset

Register	Offset
SWREG38	98h

14.2.5.1.40.2 Diagram



14.2.5.1.40.3 Fields

Field	Function
31-16 SW_AREF_HO R_SCALE	Horizontal scaling factor for alternate reference picture
15-0 SW_AREF_VER _SCALE	Vertical scaling factor for alternate reference picture

14.2.5.1.41 Not used (SWREG39)

14.2.5.1.41.1 Offset

Register	Offset
SWREG39	9Ch

14.2.5.1.41.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.41.3 Fields

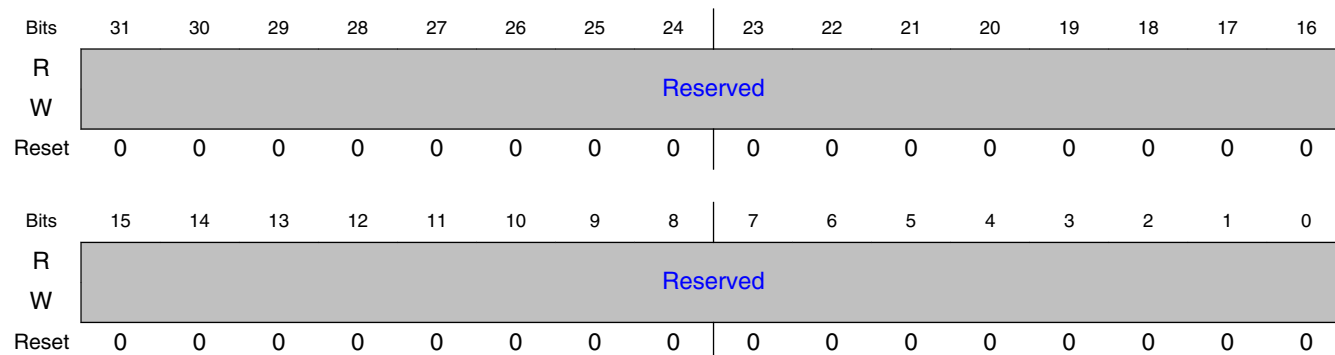
Field	Function
31-0	Reserved.
—	

14.2.5.1.42 Not used (SWREG40)

14.2.5.1.42.1 Offset

Register	Offset
SWREG40	A0h

14.2.5.1.42.2 Diagram



14.2.5.1.42.3 Fields

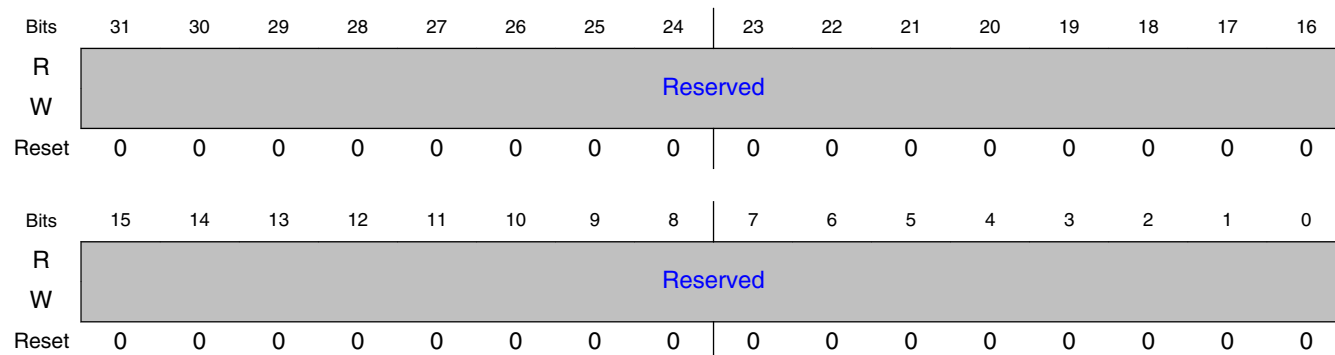
Field	Function
31-0	Reserved.
—	

14.2.5.1.43 Not used (SWREG41)

14.2.5.1.43.1 Offset

Register	Offset
SWREG41	A4h

14.2.5.1.43.2 Diagram



14.2.5.1.43.3 Fields

Field	Function
31-0 —	Reserved.

14.2.5.1.44 Not used (SWREG42)**14.2.5.1.44.1 Offset**

Register	Offset
SWREG42	A8h

14.2.5.1.44.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.44.3 Fields

Field	Function
31-0 —	Reserved.

14.2.5.1.45 Not used (SWREG43)

14.2.5.1.45.1 Offset

Register	Offset
SWREG43	ACh

14.2.5.1.45.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.45.3 Fields

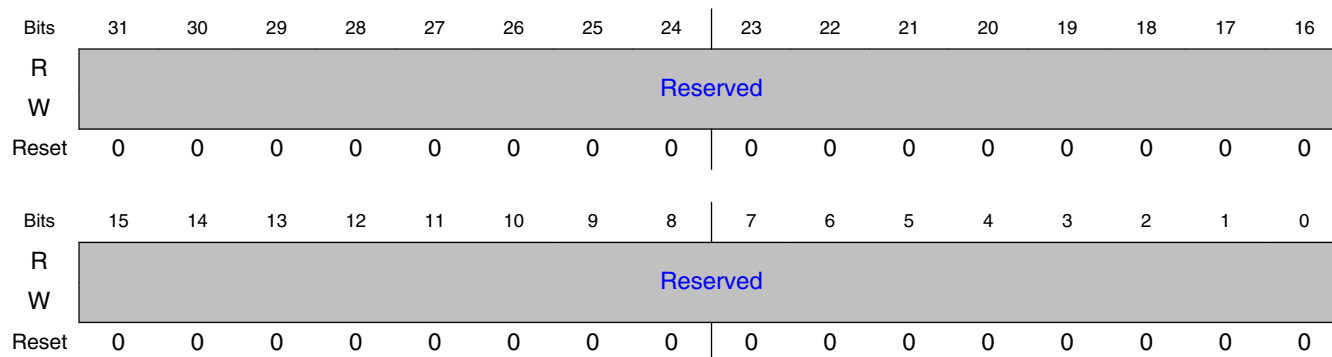
Field	Function
31-0	Reserved.
—	

14.2.5.1.46 Not used (SWREG44)

14.2.5.1.46.1 Offset

Register	Offset
SWREG44	B0h

14.2.5.1.46.2 Diagram



14.2.5.1.46.3 Fields

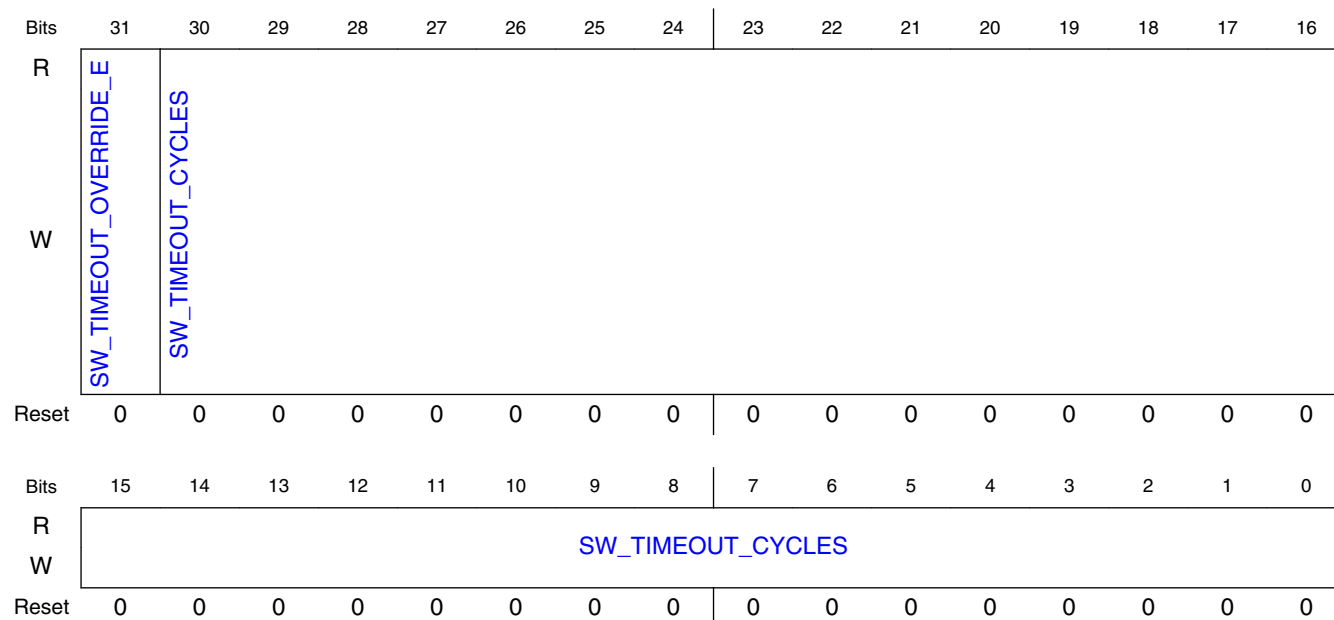
Field	Function
31-0	Reserved.
—	

14.2.5.1.47 Timeout control register (SWREG45)

14.2.5.1.47.1 Offset

Register	Offset
SWREG45	B4h

14.2.5.1.47.2 Diagram



14.2.5.1.47.3 Fields

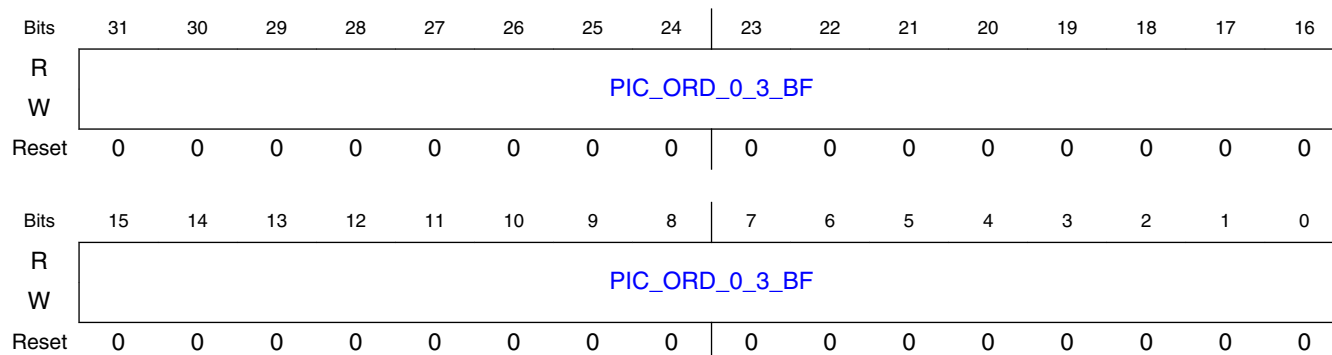
Field	Function
31 SW_TIMEOUT_OVERRIDE_E	Enable for SW controlled timeout. If enabled the sw_timeout_cycles is used to detect HW timeout instead of hard coded HW value
30-0 SW_TIMEOUT_CYCLES	Amount of clock cycles to trigger timeout interrupt if no external master activity acknowledged. Used if sw_timeout_override_e is set

14.2.5.1.48 Picture order count from current pictures for index 0-3 (SWREG46)

14.2.5.1.48.1 Offset

Register	Offset
SWREG46	B8h

14.2.5.1.48.2 Diagram



14.2.5.1.48.3 Fields

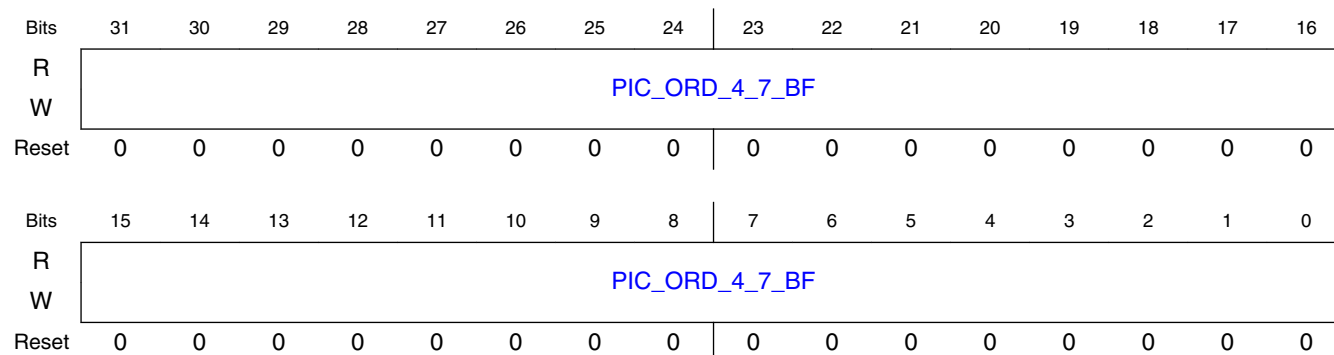
Field	Function
31-0 PIC_ORD_0_3_BF	For HEVC: [31:24] - sw_cur_poc_00 - Picture order count from current picture 0 [23:16] - sw_cur_poc_01 - Picture order count from current picture 1 [15:8] - sw_cur_poc_02 - Picture order count from current picture 2 [7:0] - sw_cur_poc_03 - Picture order count from current picture 3 For VP9: [31] - Reserved - Not used [30:24] - sw_filt_ref_adj_0 - Filter level intra adjustment [23] - Reserved - Not used [22:16] - sw_filt_ref_adj_1 - Filter level last ref pic adjustment [15] - Reserved - Not used [14:8] - sw_filt_ref_adj_2 - Filter level golden pic adjustment [7] - Reserved - Not used [6:0] - sw_filt_ref_adj_3 - Filter level alt ref pic adjustment

14.2.5.1.49 Picture order count from current pictures for index 4-7 (SWREG47)

14.2.5.1.49.1 Offset

Register	Offset
SWREG47	BCh

14.2.5.1.49.2 Diagram



14.2.5.1.49.3 Fields

Field	Function
31-0 PIC_ORD_4_7_BF	For HEVC: [31:24] - sw_cur_poc_04 - Picture order count from current picture 4 [23:16] - sw_cur_poc_05 - Picture order count from current picture 5 [15:8] - sw_cur_poc_06 - Picture order count from current picture 6 [7:0] - sw_cur_poc_07 - Picture order count from current picture 7 For VP9: [31] - Reserved - Not used [30:24] - sw_filt_mb_adj_0 - Filter level ZERO mv adjustment [23] - Reserved - Not used [22:16] - sw_filt_mb_adj_1 - Filter level adjustment [15] - Reserved - Not used [14:8] - sw_filt_mb_adj_2 - Filter level adjustment (Not used) [7] - Reserved - Not used [6:0] - sw_filt_mb_adj_3 - Filter level adjustment (Not used)

14.2.5.1.50 Picture order count from current pictures for index 8-11 (SWREG48)

14.2.5.1.50.1 Offset

Register	Offset
SWREG48	C0h

14.2.5.1.50.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_CUR_POC_08								SW_CUR_POC_09							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_CUR_POC_10								SW_CUR_POC_11							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.50.3 Fields

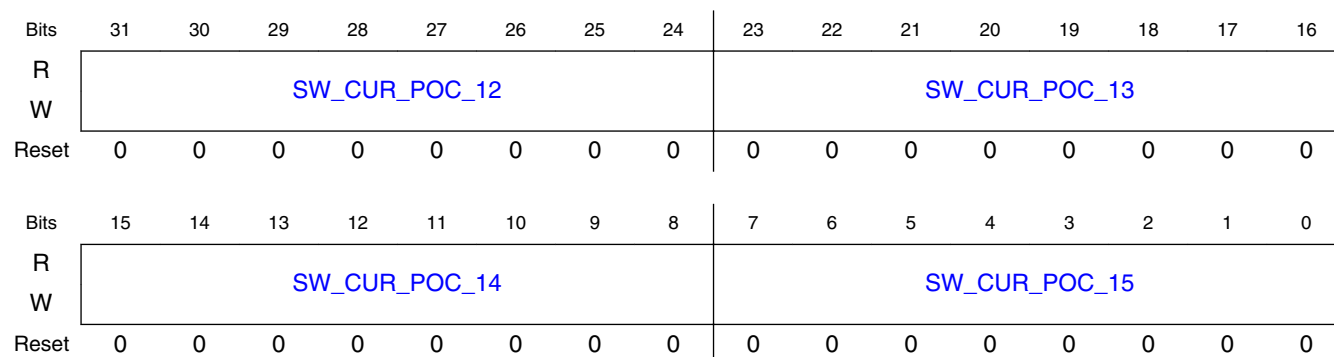
Field	Function
31-24 SW_CUR_POC_08	Picture order count from current picture 8
23-16 SW_CUR_POC_09	Picture order count from current picture 9
15-8 SW_CUR_POC_10	Picture order count from current picture 10
7-0 SW_CUR_POC_11	Picture order count from current picture 11

14.2.5.1.51 Picture order count from current pictures for index 12-15 (SWREG49)

14.2.5.1.51.1 Offset

Register	Offset
SWREG49	C4h

14.2.5.1.51.2 Diagram



14.2.5.1.51.3 Fields

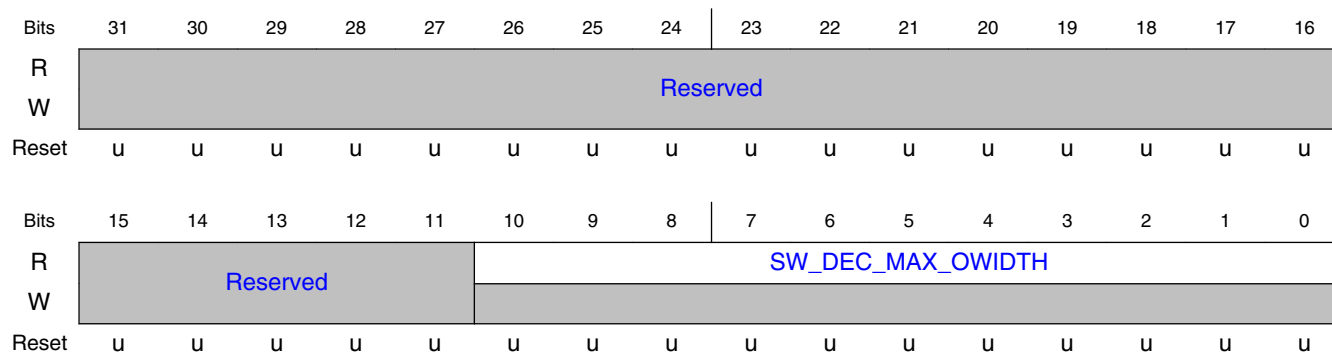
Field	Function
31-24 SW_CUR_POC_12	Picture order count from current picture 12
23-16 SW_CUR_POC_13	Picture order count from current picture 13
15-8 SW_CUR_POC_14	Picture order count from current picture 14
7-0 SW_CUR_POC_15	Picture order count from current picture 15

14.2.5.1.52 Synthesis configuration register decoder 0 (read only) (SWREG50)

14.2.5.1.52.1 Offset

Register	Offset
SWREG50	C8h

14.2.5.1.52.2 Diagram



14.2.5.1.52.3 Fields

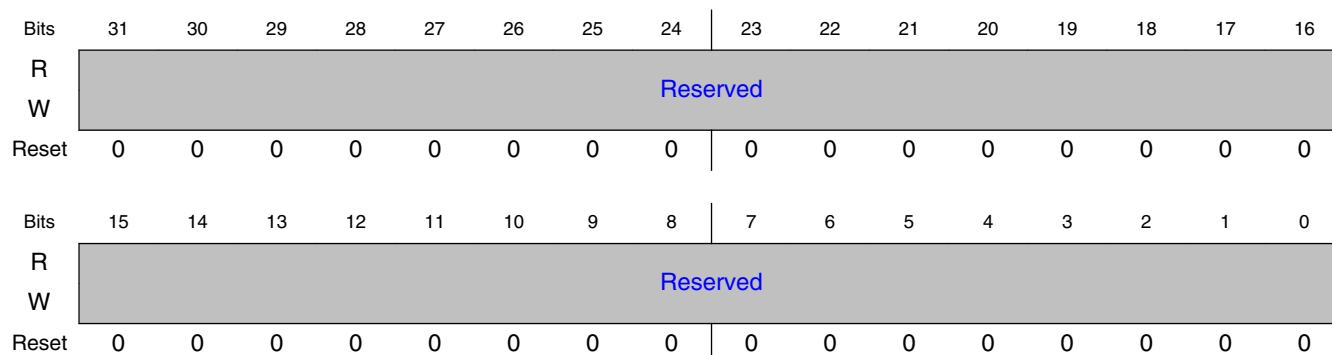
Field	Function
31-11 —	Reserved.
10-0 SW_DEC_MAX_OWIDTH	Max configured decoder video resolution that can be decoded. Informed as width of the picture in pixels

14.2.5.1.53 Reference picture buffer control register (SWREG51)

14.2.5.1.53.1 Offset

Register	Offset
SWREG51	CCh

14.2.5.1.53.2 Diagram



14.2.5.1.53.3 Fields

Field	Function
31-0 —	Reserved.

14.2.5.1.54 Reference picture buffer information register 1 (read only) (SWREG52)

14.2.5.1.54.1 Offset

Register	Offset
SWREG52	D0h

14.2.5.1.54.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.54.3 Fields

Field	Function
31-0 —	Reserved.

14.2.5.1.55 Reference picture buffer information register 2 (read only) (SWREG53)

14.2.5.1.55.1 Offset

Register	Offset
SWREG53	D4h

14.2.5.1.55.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

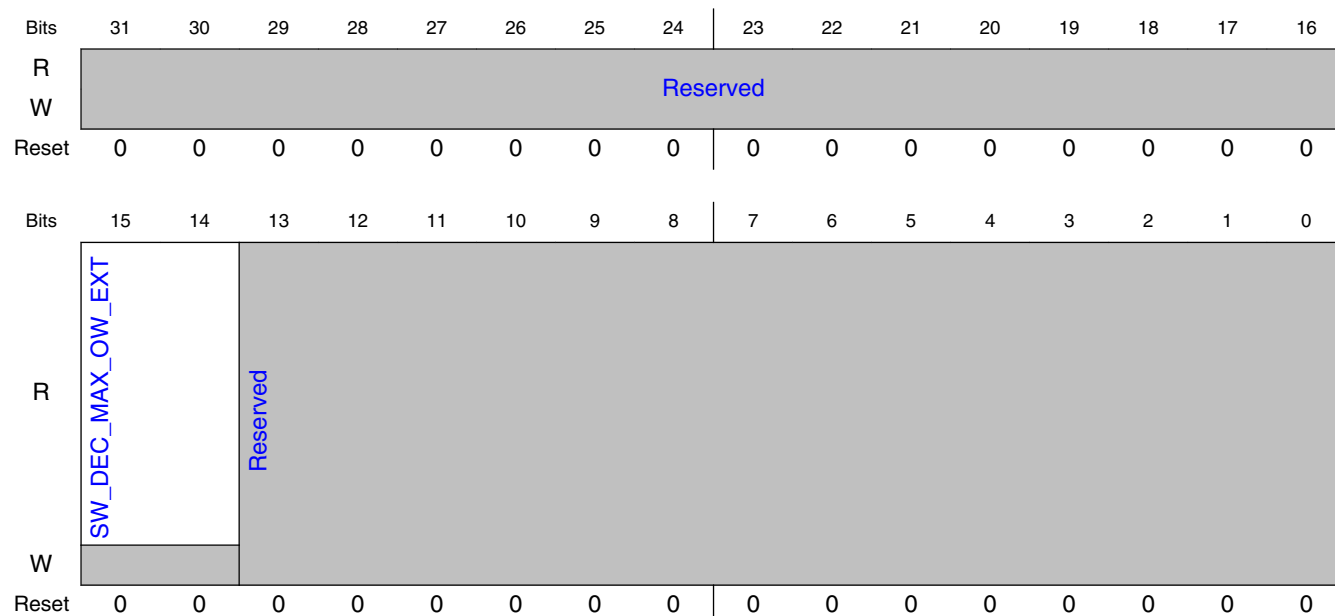
14.2.5.1.55.3 Fields

Field	Function
31-0	Reserved.
—	

14.2.5.1.56 Synthesis configuration register decoder 1 (read only) (SWREG54)**14.2.5.1.56.1 Offset**

Register	Offset
SWREG54	D8h

14.2.5.1.56.2 Diagram



14.2.5.1.56.3 Fields

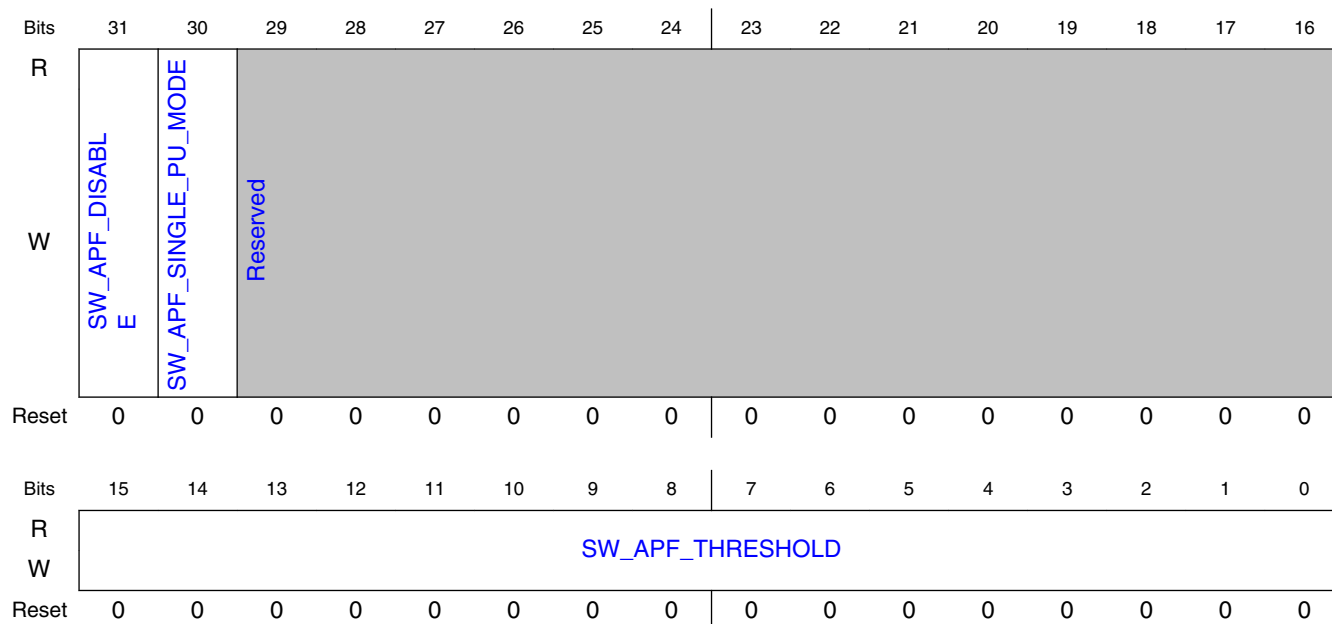
Field	Function
31-16 —	Reserved.
15-14 SW_DEC_MAX_OW_EXT	Max configured decoder video resolution that can be decoded. This is the MSB part of the configuration signal
13-0 —	Reserved.

14.2.5.1.57 Advanced prefetch control register (SWREG55)

14.2.5.1.57.1 Offset

Register	Offset
SWREG55	DCh

14.2.5.1.57.2 Diagram



14.2.5.1.57.3 Fields

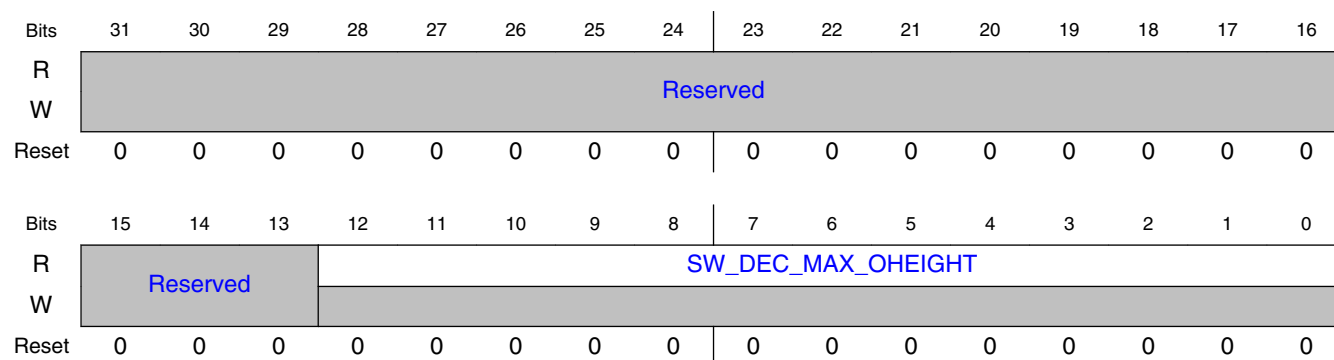
Field	Function
31 SW_APF_DISA BLE	Advanced prefetch disable. If high each partition is read separately
30 SW_APF_SING LE_PU_MODE	APF amount of buffered Pus: can be restricted to buffer one PU at a time
29-16 —	Reserved.
15-0 SW_APF_THRE SHOLD	Advanced prefetch threshold. If current buffered unit exceeds the threshold the advanced mode is not used. Value 0 disables threshold usage and advanced prefetch usage is restricted by internal memory limitation only

14.2.5.1.58 Synthesis configuration register decoder 2 (read only) (SWREG56)

14.2.5.1.58.1 Offset

Register	Offset
SWREG56	E0h

14.2.5.1.58.2 Diagram



14.2.5.1.58.3 Fields

Field	Function
31-13 —	Reserved.
12-0 SW_DEC_MAX_OHEIGHT	Max supported picture height in pixels

14.2.5.1.59 Decoder fuse register (read only) (SWREG57)

14.2.5.1.59.1 Offset

Register	Offset
SWREG57	E4h

14.2.5.1.59.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

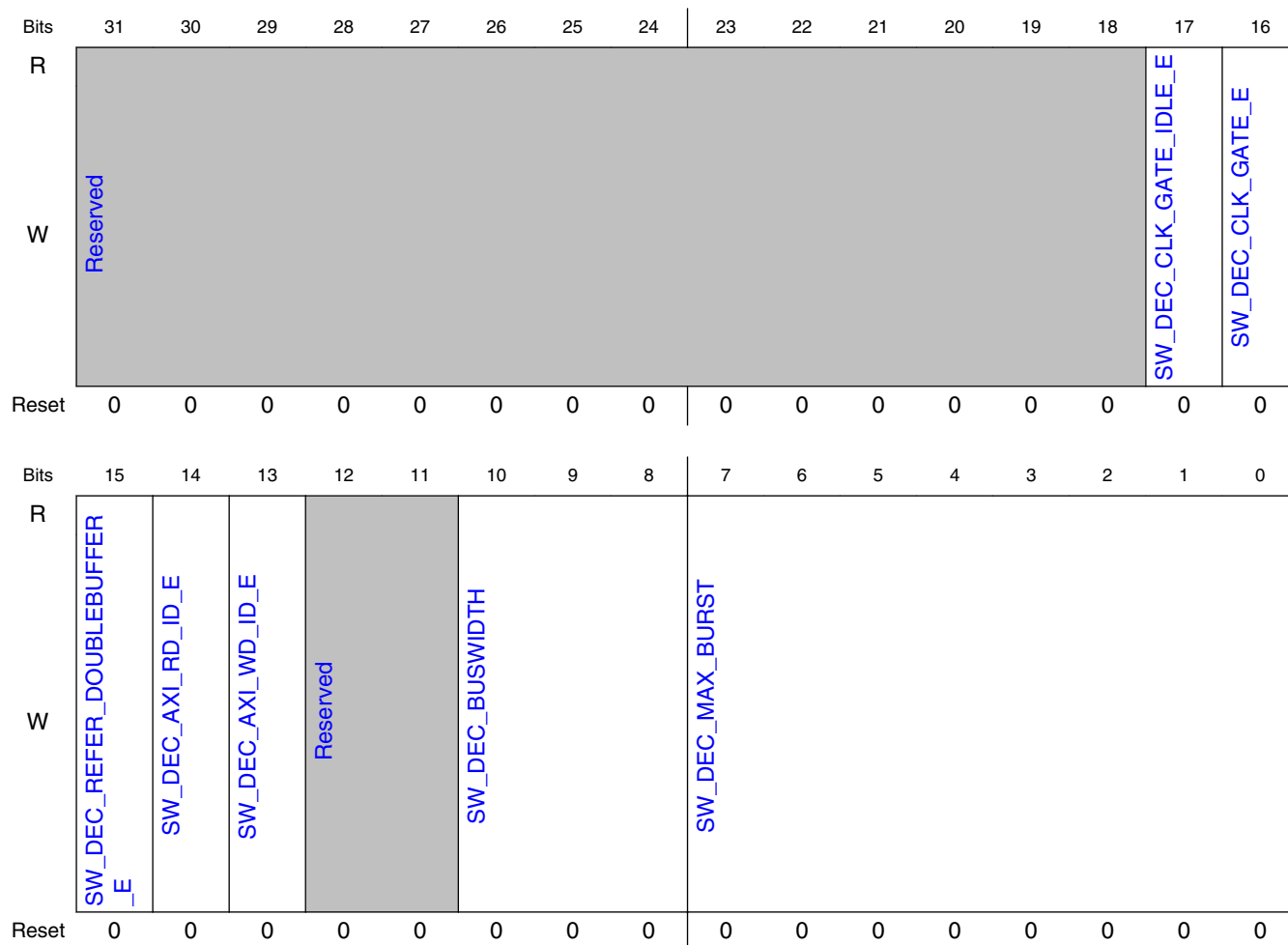
14.2.5.1.59.3 Fields

Field	Function
31-0	Reserved.
—	

14.2.5.1.60 Device configuration register decoder 2 + Multi core control register (SWREG58)**14.2.5.1.60.1 Offset**

Register	Offset
SWREG58	E8h

14.2.5.1.60.2 Diagram



14.2.5.1.60.3 Fields

Field	Function
31-18 —	Reserved.
17 SW_DEC_CLK_GATE_IDLE_E	Clock gating enable for decoder run-time. Generated separate clocks for each block by its own IDLE signal.
16 SW_DEC_CLK_GATE_E	Clock gating enable for picture-wise/decoding format clock gating. Between each picture the clock is gated from HW if this bit is high
15 SW_DEC_REF_DOUBLEBUFFER_E	HW internal double buffering enable for reference data. This enable requires that there are two buffers available at the configured decoder (see configuration register values)

Table continues on the next page...

VPU G2 Memory Map/Register Definition

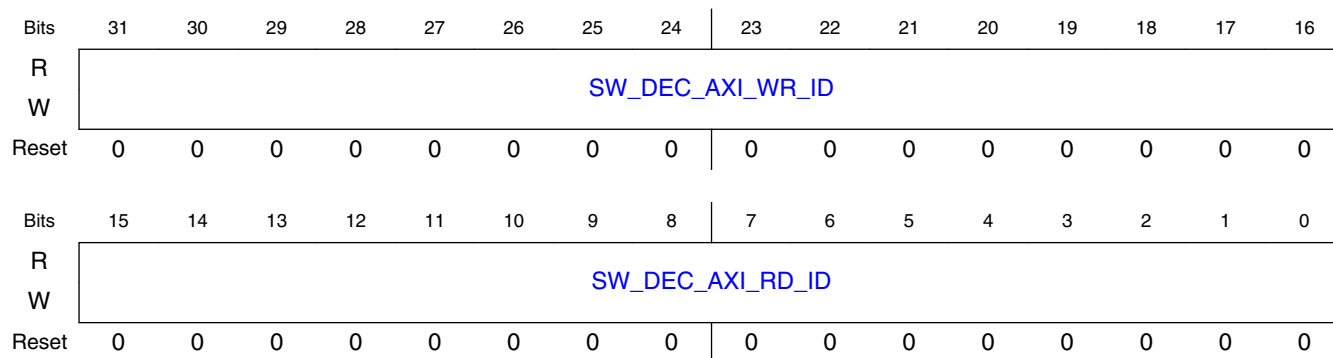
Field	Function
14 SW_DEC_AXI_RD_ID_E	SW axi ID enable. When enabled the given sw_dec_axi_rd_id is used as ID base and each sub-block will use offsets 0...max
13 SW_DEC_AXI_WD_ID_E	SW axi ID enable. When enabled the given sw_dec_axi_wd_id is used as ID base and each sub-block will use offsets 0...max
12-11 —	Reserved.
10-8 SW_DEC_BUS_WIDTH	Decoder master interface buswidth 000b - 32 bit bus 001b - 64 bit bus 010b - 128 bit bus
7-0 SW_DEC_MAX_BURST	Maximum burst length for decoder bus transactions. Valid values: AXI: 1-256

14.2.5.1.61 Device configuration register AXI ID (SWREG59)

14.2.5.1.61.1 Offset

Register	Offset
SWREG59	ECh

14.2.5.1.61.2 Diagram



14.2.5.1.61.3 Fields

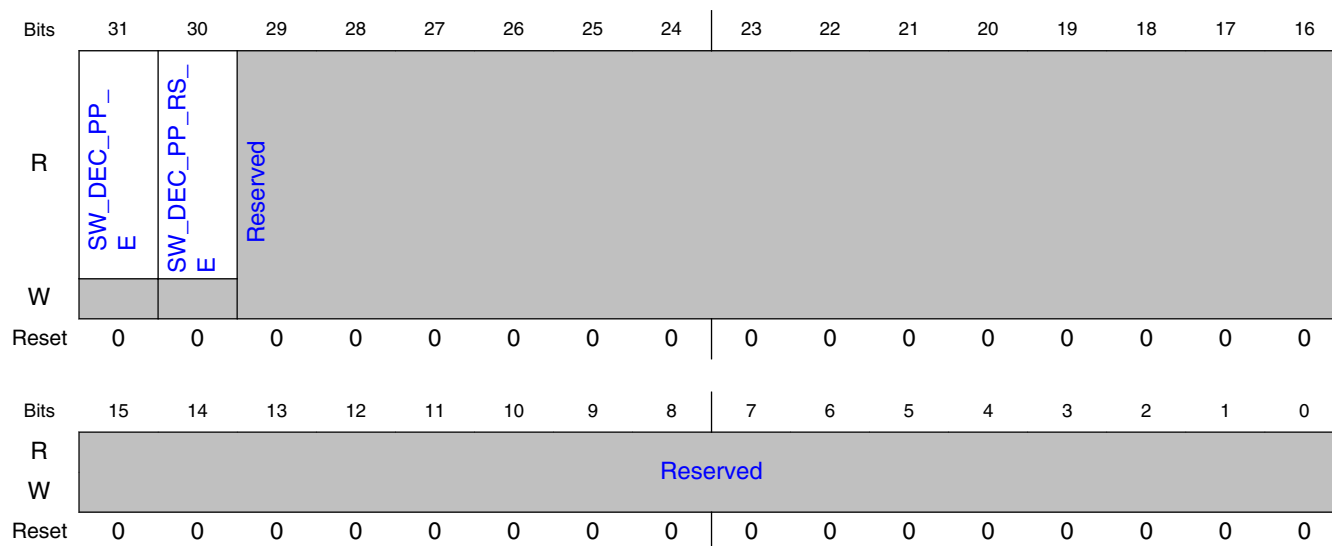
Field	Function
31-16 SW_DEC_AXI_WR_ID	Read ID base for HW write accesses. Each writing device use AXI ID of base+deviceoffset (where device offset is 0 1 2 3...Number of reading sub-blocks)
15-0 SW_DEC_AXI_RD_ID	Write ID base for HW write accesses. Each writing device use AXI ID of base+deviceoffset (where device offset is 0 1 2 3...Number of writing sub-blocks)

14.2.5.1.62 Synthesis configuration register decoder 3 for PP (read only) (SWREG60)

14.2.5.1.62.1 Offset

Register	Offset
SWREG60	F0h

14.2.5.1.62.2 Diagram



14.2.5.1.62.3 Fields

Field	Function
31	Decoder include PP 0b - PP does not exist. None of the PP features can be enabled.

Table continues on the next page...

VPU G2 Memory Map/Register Definition

Field	Function
SW_DEC_PP_E	1b - PP exists
30 SW_DEC_PP_R S_E	Decoder PP raster scan output support 0b - Raster scan output not supported 1b - Raster scan output supported
29-0 —	Reserved.

14.2.5.1.63 Not used (SWREG61)

14.2.5.1.63.1 Offset

Register	Offset
SWREG61	F4h

14.2.5.1.63.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.63.3 Fields

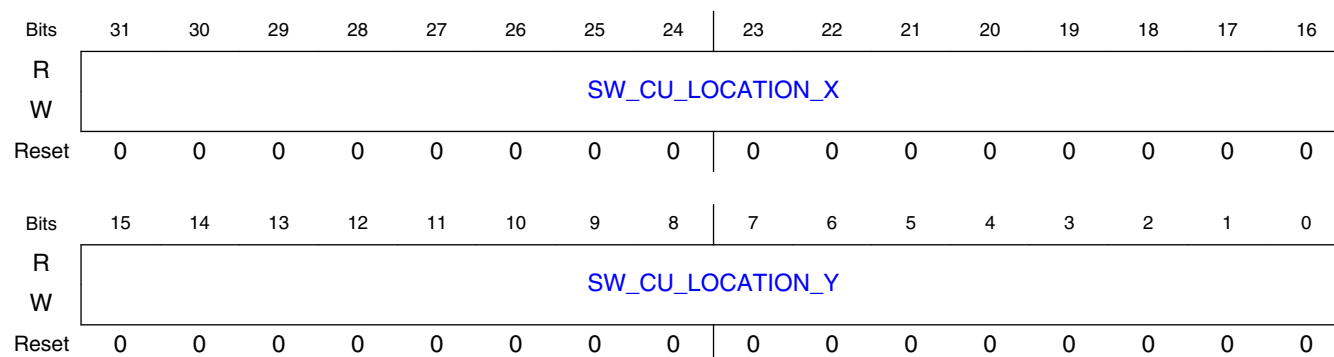
Field	Function
31-0 —	Reserved.

14.2.5.1.64 HW proceed register (CU location) (SWREG62)

14.2.5.1.64.1 Offset

Register	Offset
SWREG62	F8h

14.2.5.1.64.2 Diagram



14.2.5.1.64.3 Fields

Field	Function
31-16 SW_CU_LOCATION_X	Cu horizontal start location X in pixels (returned HW internal position during interrupt)
15-0 SW_CU_LOCATION_Y	Cu vertical start location Y in pixels (returned HW internal position during interrupt)

14.2.5.1.65 HW performance register (cycles running) (SWREG63)

14.2.5.1.65.1 Offset

Register	Offset
SWREG63	FCh

14.2.5.1.65.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_PERF_CYCLE_COUNT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_PERF_CYCLE_COUNT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.65.3 Fields

Field	Function
31-0 SW_PERF_CYCLE_COUNT	HW clock cycle counter return value. Amount of consumed clock cycles returned to this register when interrupt is being made (any kind of interrupt)

14.2.5.1.66 Base address MSB (bits 63:32) for decoded luminance picture (SWREG64)

14.2.5.1.66.1 Offset

Register	Offset
SWREG64	100h

14.2.5.1.66.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_DEC_OUT_YBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_DEC_OUT_YBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.66.3 Fields

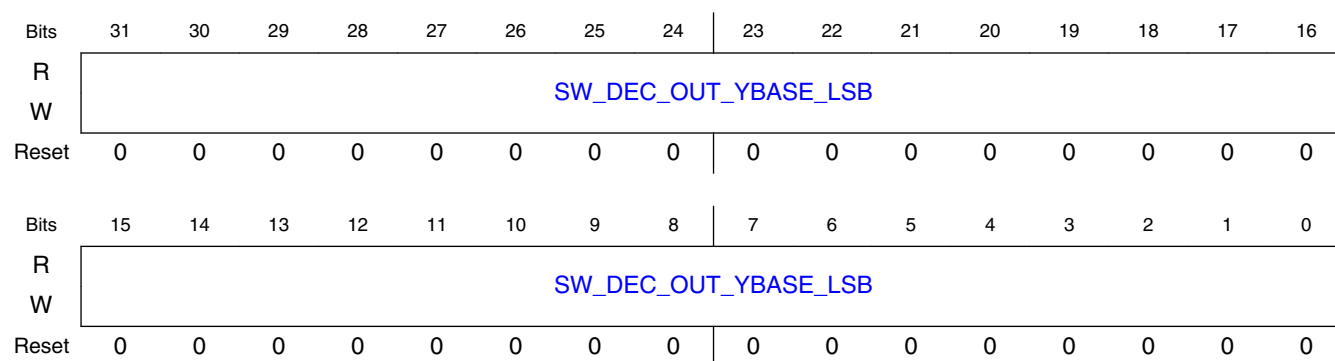
Field	Function
31-0 SW_DEC_OUT_YBASE_MSB	Base address MSB (bits 63:32) for decoded luminance picture

14.2.5.1.67 Base address LSB (bits 31:0) for decoded luminance picture (SWREG65)

14.2.5.1.67.1 Offset

Register	Offset
SWREG65	104h

14.2.5.1.67.2 Diagram



14.2.5.1.67.3 Fields

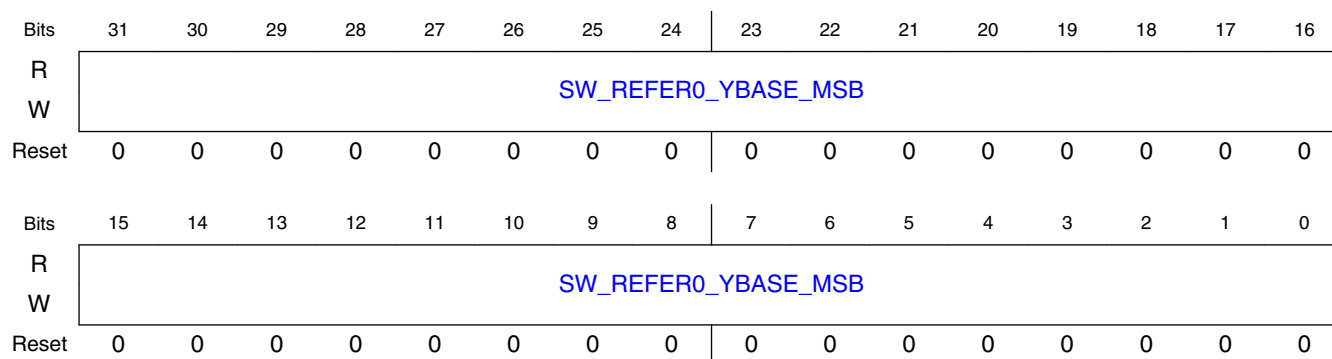
Field	Function
31-0 SW_DEC_OUT_YBASE_LSB	Base address LSB (bits 31:0) for decoded luminance picture

14.2.5.1.68 Base address MSB (bits 63:32) for reference luminance picture index 0 (SWREG66)

14.2.5.1.68.1 Offset

Register	Offset
SWREG66	108h

14.2.5.1.68.2 Diagram



14.2.5.1.68.3 Fields

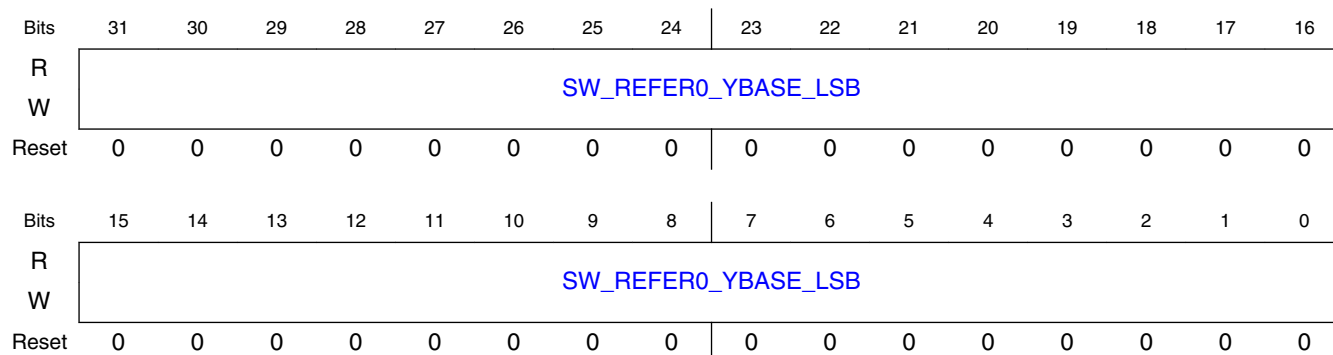
Field	Function
31-0 SW_REFER0_Y BASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 0

14.2.5.1.69 Base address LSB (bits 31:0) for reference luminance picture index 0 (SWREG67)

14.2.5.1.69.1 Offset

Register	Offset
SWREG67	10Ch

14.2.5.1.69.2 Diagram



14.2.5.1.69.3 Fields

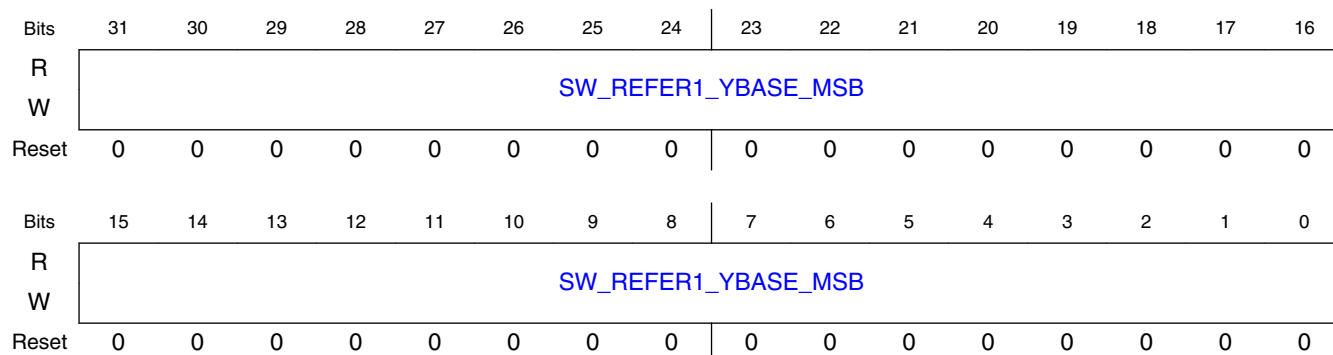
Field	Function
31-0 SW_REFER0_Y BASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 0

14.2.5.1.70 Base address MSB (bits 63:32) for reference luminance picture index 1 (SWREG68)

14.2.5.1.70.1 Offset

Register	Offset
SWREG68	110h

14.2.5.1.70.2 Diagram



14.2.5.1.70.3 Fields

Field	Function
31-0 SW_REFER1_Y BASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 1

14.2.5.1.71 Base address LSB (bits 31:0) for reference luminance picture index 1 (SWREG69)

14.2.5.1.71.1 Offset

Register	Offset
SWREG69	114h

14.2.5.1.71.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER1_YBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER1_YBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.71.3 Fields

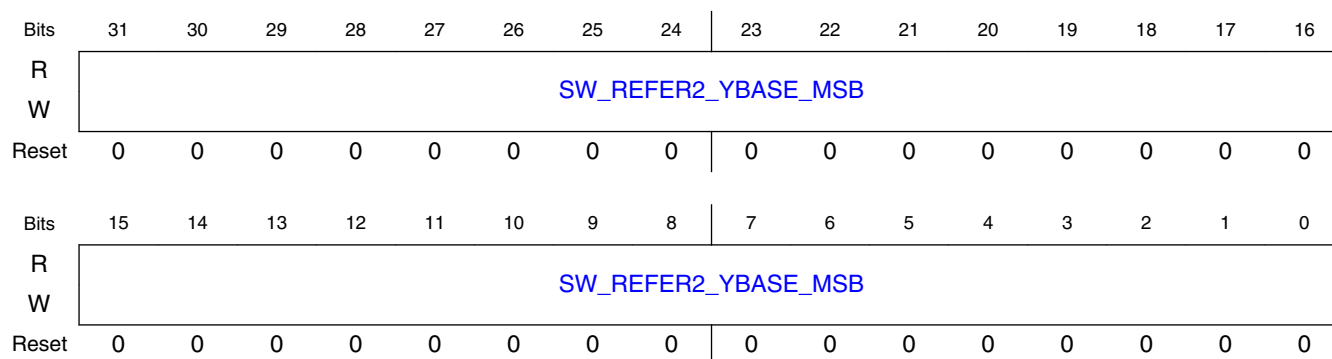
Field	Function
31-0 SW_REFER1_Y BASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 1

14.2.5.1.72 Base address MSB (bits 63:32) for reference luminance picture index 2 (SWREG70)

14.2.5.1.72.1 Offset

Register	Offset
SWREG70	118h

14.2.5.1.72.2 Diagram



14.2.5.1.72.3 Fields

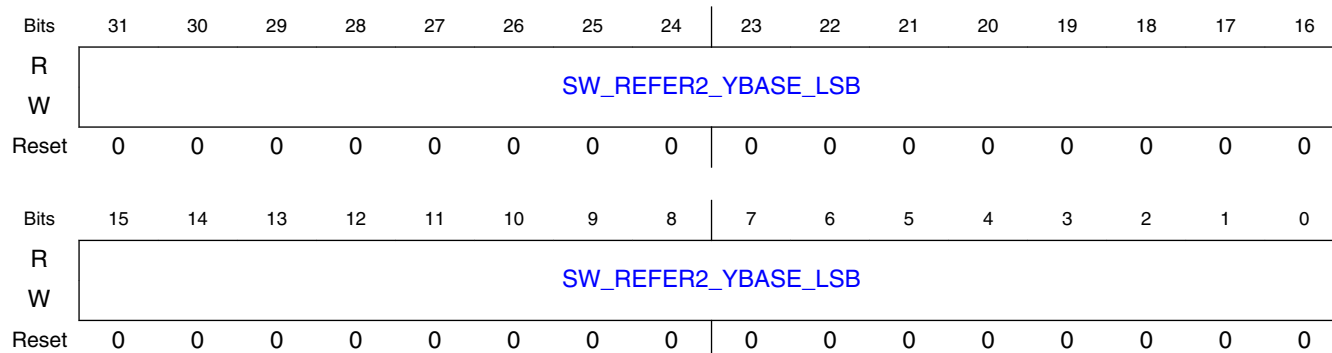
Field	Function
31-0 SW_REFER2_Y BASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 2

14.2.5.1.73 Base address LSB (bits 31:0) for reference luminance picture index 2 (SWREG71)

14.2.5.1.73.1 Offset

Register	Offset
SWREG71	11Ch

14.2.5.1.73.2 Diagram



14.2.5.1.73.3 Fields

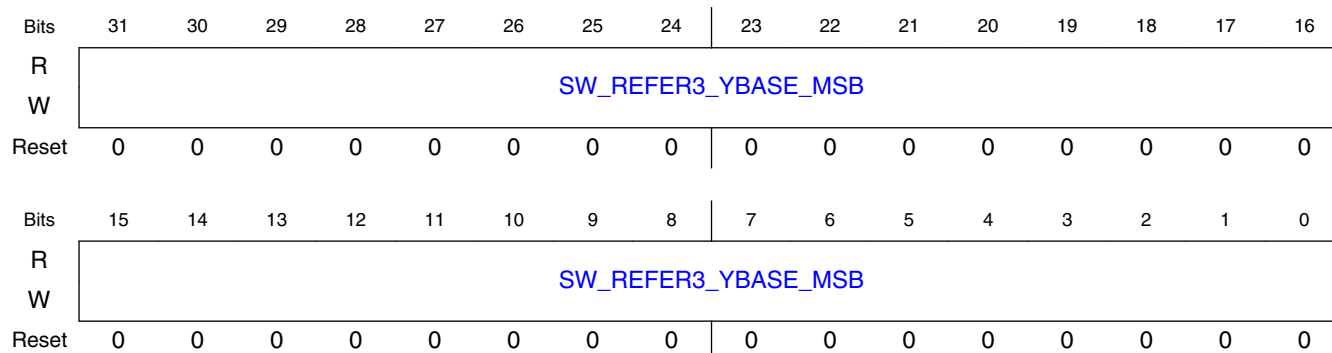
Field	Function
31-0 SW_REFER2_Y BASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 2

14.2.5.1.74 Base address MSB (bits 63:32) for reference luminance picture index 3 (SWREG72)

14.2.5.1.74.1 Offset

Register	Offset
SWREG72	120h

14.2.5.1.74.2 Diagram



14.2.5.1.74.3 Fields

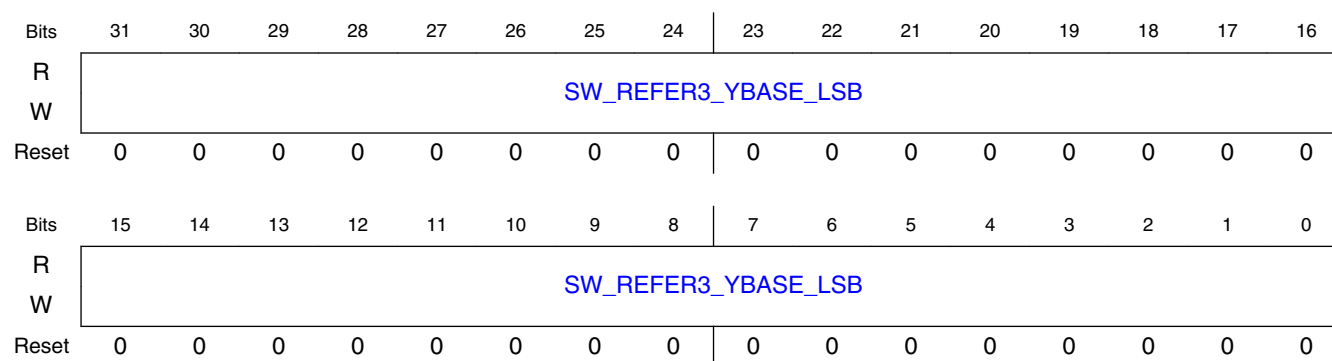
Field	Function
31-0 SW_REFER3_Y BASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 3

14.2.5.1.75 Base address LSB (bits 31:0) for reference luminance picture index 3 (SWREG73)

14.2.5.1.75.1 Offset

Register	Offset
SWREG73	124h

14.2.5.1.75.2 Diagram



14.2.5.1.75.3 Fields

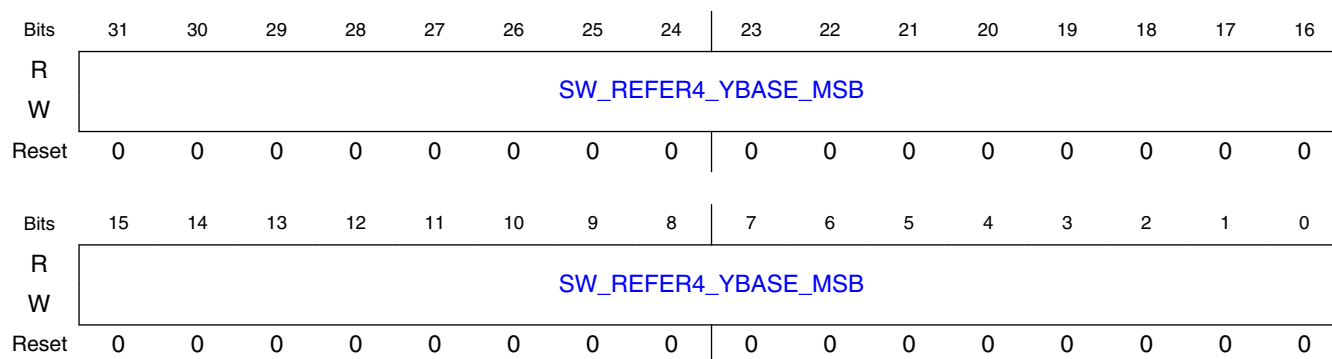
Field	Function
31-0 SW_REFER3_Y BASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 3

14.2.5.1.76 Base address MSB (bits 63:32) for reference luminance picture index 4 (SWREG74)

14.2.5.1.76.1 Offset

Register	Offset
SWREG74	128h

14.2.5.1.76.2 Diagram



14.2.5.1.76.3 Fields

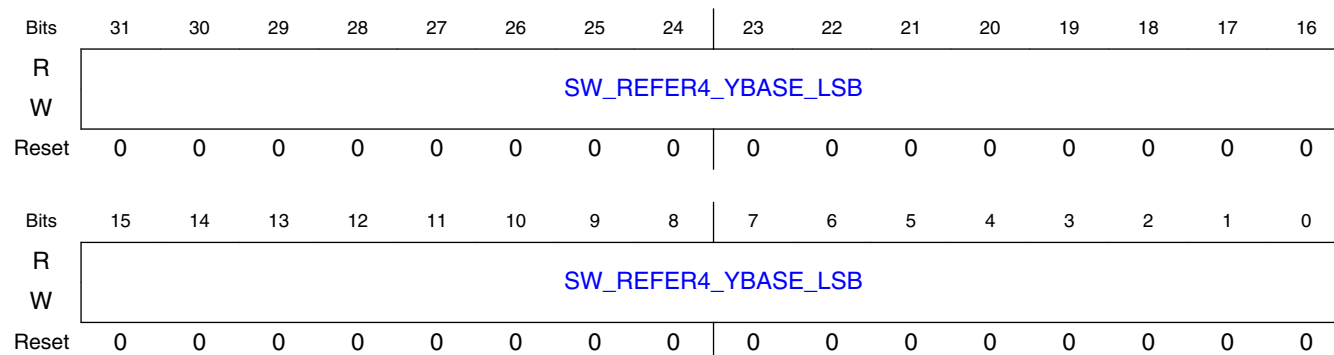
Field	Function
31-0 SW_REFER4_Y BASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 4

14.2.5.1.77 Base address LSB (bits 31:0) for reference luminance picture index 4 (SWREG75)

14.2.5.1.77.1 Offset

Register	Offset
SWREG75	12Ch

14.2.5.1.77.2 Diagram



14.2.5.1.77.3 Fields

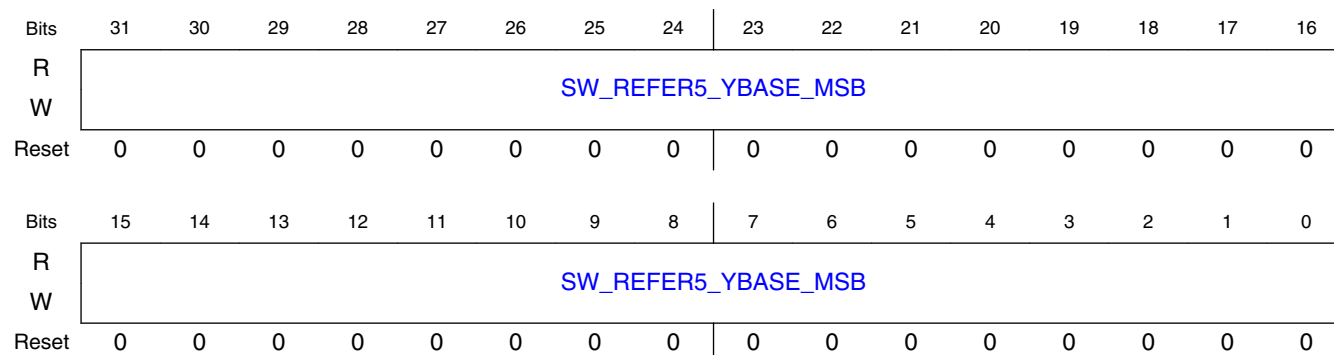
Field	Function
31-0 SW_REFER4_Y BASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 4

14.2.5.1.78 Base address MSB (bits 63:32) for reference luminance picture index 5 (SWREG76)

14.2.5.1.78.1 Offset

Register	Offset
SWREG76	130h

14.2.5.1.78.2 Diagram



14.2.5.1.78.3 Fields

Field	Function
31-0 SW_REFER5_Y BASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 5

14.2.5.1.79 Base address LSB (bits 31:0) for reference luminance picture index 5 (SWREG77)

14.2.5.1.79.1 Offset

Register	Offset
SWREG77	134h

14.2.5.1.79.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER5_YBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER5_YBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.79.3 Fields

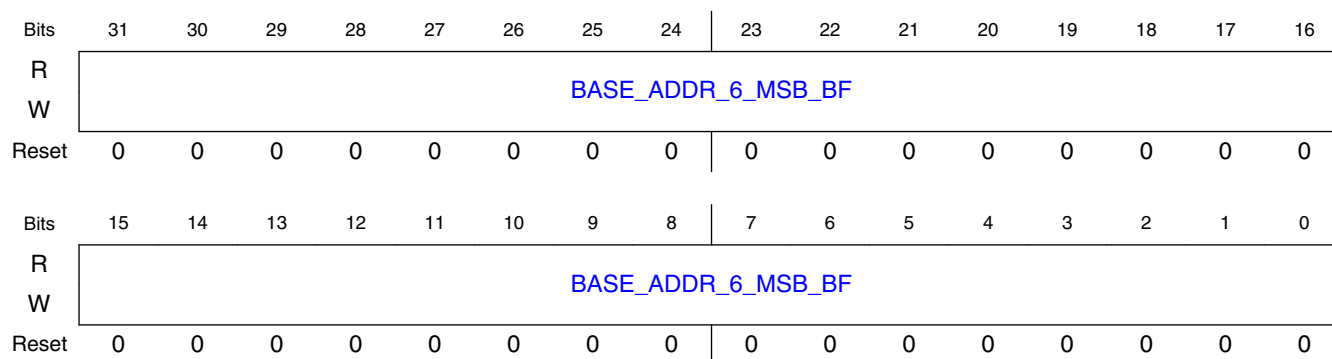
Field	Function
31-0 SW_REFER5_Y BASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 5

14.2.5.1.80 Base address MSB (bits 63:32) for reference luminance picture index 6 /VP9 segment write base MSB (SWREG78)

14.2.5.1.80.1 Offset

Register	Offset
SWREG78	138h

14.2.5.1.80.2 Diagram



14.2.5.1.80.3 Fields

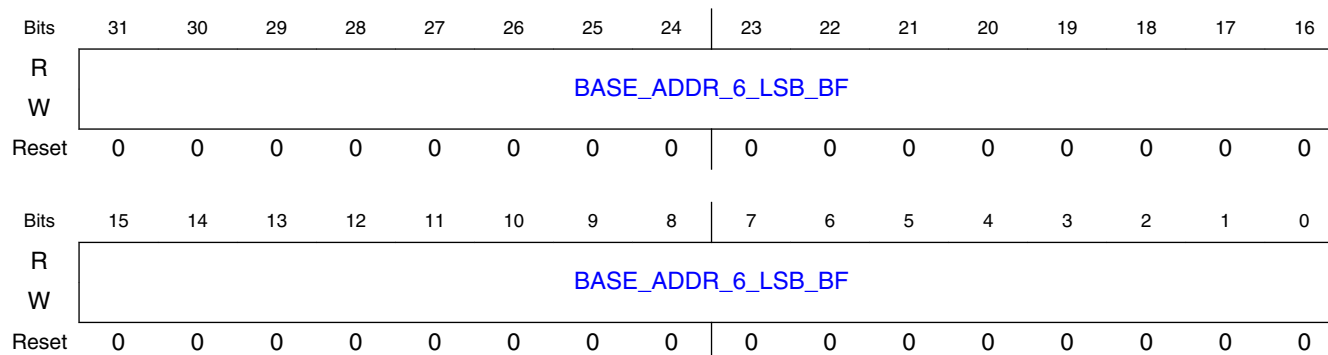
Field	Function
31-0 BASE_ADDR_6_MSB_BF	For HEVC: [31:0] - sw_refer6_ybase_msb - Base address MSB (bits 63:32) for reference luminance picture index 6 For VP9: [31:0] - sw_segment_write_base_msb - VP9 segment write base MSB

14.2.5.1.81 Base address LSB (bits 31:0) for reference luminance picture index 6 /VP9 segment write base LSB (SWREG79)

14.2.5.1.81.1 Offset

Register	Offset
SWREG79	13Ch

14.2.5.1.81.2 Diagram



14.2.5.1.81.3 Fields

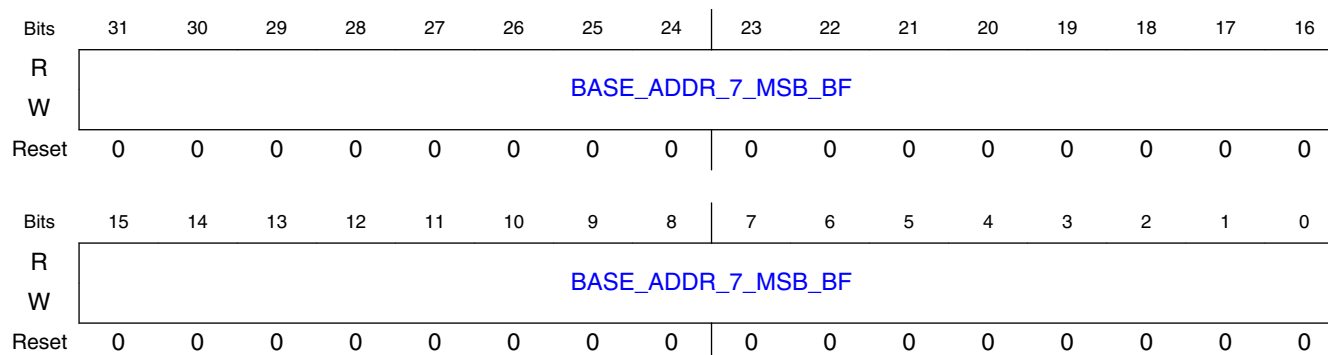
Field	Function
31-0 BASE_ADDR_6_LSB_BF	For HEVC: [31:0] - sw_refer6_ybase_lsb - Base address LSB (bits 31:0) for reference luminance picture index 6 For VP9: [31:0] - sw_segment_write_base_lsb - VP9 segment write base LSB

14.2.5.1.82 Base address MSB (bits 63:32) for reference luminance picture index 7 /VP9 segment read base MSB (SWREG80)

14.2.5.1.82.1 Offset

Register	Offset
SWREG80	140h

14.2.5.1.82.2 Diagram



14.2.5.1.82.3 Fields

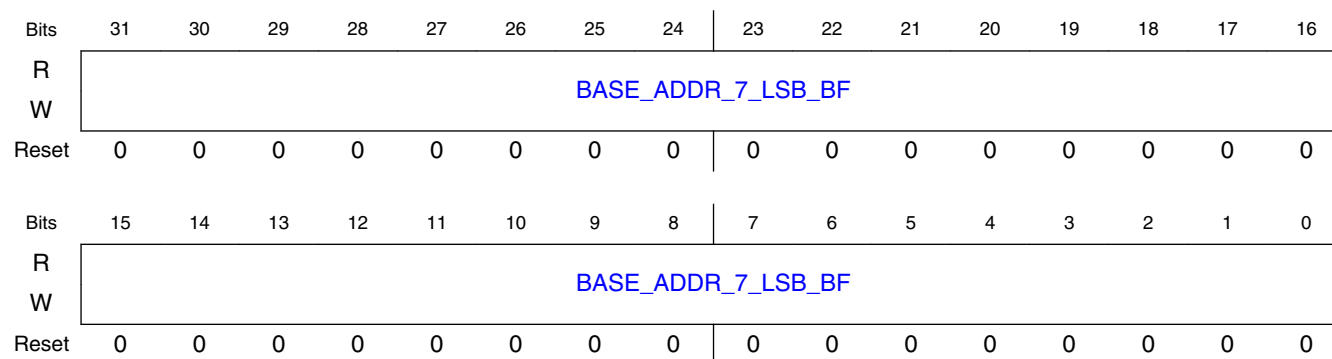
Field	Function
31-0 BASE_ADDR_7_MSB_BF	For HEVC: [31:0] - sw_refer7_ybase_msb - Base address MSB (bits 63:32) for reference luminance picture index 7 For VP9: [31:0] - sw_segment_read_base_msb - VP9 segment read base MSB

14.2.5.1.83 Base address LSB (bits 31:0) for reference luminance picture index 7 /VP9 segment read base LSB (SWREG81)

14.2.5.1.83.1 Offset

Register	Offset
SWREG81	144h

14.2.5.1.83.2 Diagram



14.2.5.1.83.3 Fields

Field	Function
31-0 BASE_ADDR_7_LSB_BF	For HEVC: [31:0] - sw_refer7_ybase_lsb - Base address LSB (bits 31:0) for reference luminance picture index 7 For VP9: [31:0] - sw_segment_read_base_lsb - VP9 segment read base LSB

14.2.5.1.84 Base address MSB (bits 63:32) for reference luminance picture index 8 (SWREG82)

14.2.5.1.84.1 Offset

Register	Offset
SWREG82	148h

14.2.5.1.84.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER8_YBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER8_YBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.84.3 Fields

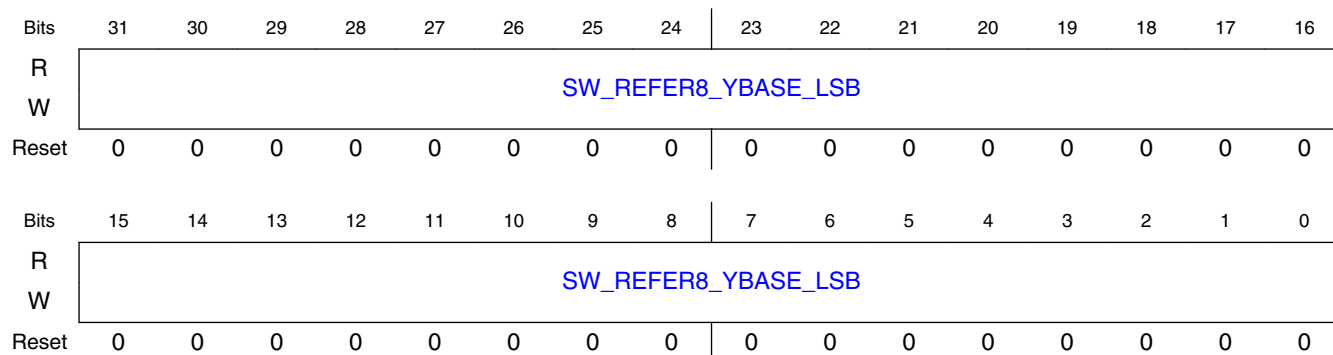
Field	Function
31-0 SW_REFER8_Y BASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 8

14.2.5.1.85 Base address LSB (bits 31:0) for reference luminance picture index 8 (SWREG83)

14.2.5.1.85.1 Offset

Register	Offset
SWREG83	14Ch

14.2.5.1.85.2 Diagram



14.2.5.1.85.3 Fields

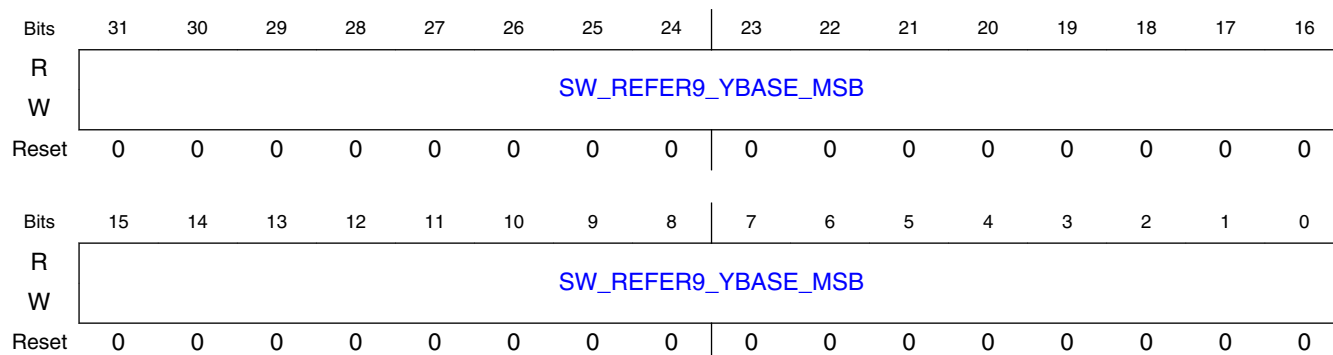
Field	Function
31-0 SW_REFER8_Y BASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 8

14.2.5.1.86 Base address MSB (bits 63:32) for reference luminance picture index 9 (SWREG84)

14.2.5.1.86.1 Offset

Register	Offset
SWREG84	150h

14.2.5.1.86.2 Diagram



14.2.5.1.86.3 Fields

Field	Function
31-0 SW_REFER9_Y BASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 9

14.2.5.1.87 Base address LSB (bits 31:0) for reference luminance picture index 9 (SWREG85)

14.2.5.1.87.1 Offset

Register	Offset
SWREG85	154h

14.2.5.1.87.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER9_YBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER9_YBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.87.3 Fields

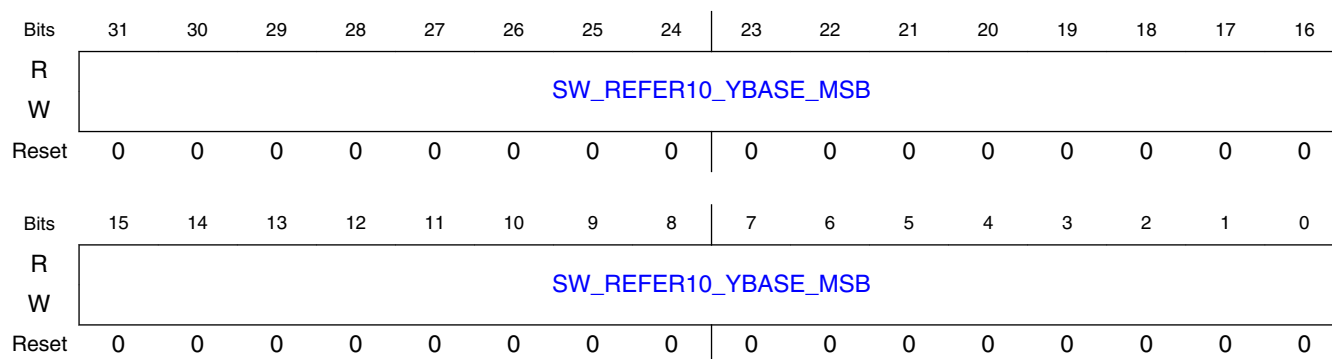
Field	Function
31-0 SW_REFER9_Y BASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 9

14.2.5.1.88 Base address MSB (bits 63:32) for reference luminance picture index 10 (SWREG86)

14.2.5.1.88.1 Offset

Register	Offset
SWREG86	158h

14.2.5.1.88.2 Diagram



14.2.5.1.88.3 Fields

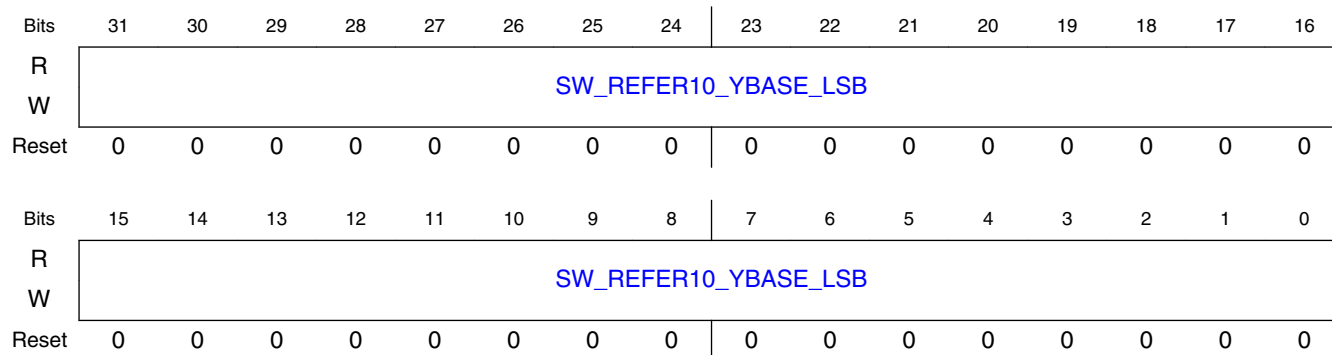
Field	Function
31-0 SW_REFER10_YBASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 10

14.2.5.1.89 Base address LSB (bits 31:0) for reference luminance picture index 10 (SWREG87)

14.2.5.1.89.1 Offset

Register	Offset
SWREG87	15Ch

14.2.5.1.89.2 Diagram



14.2.5.1.89.3 Fields

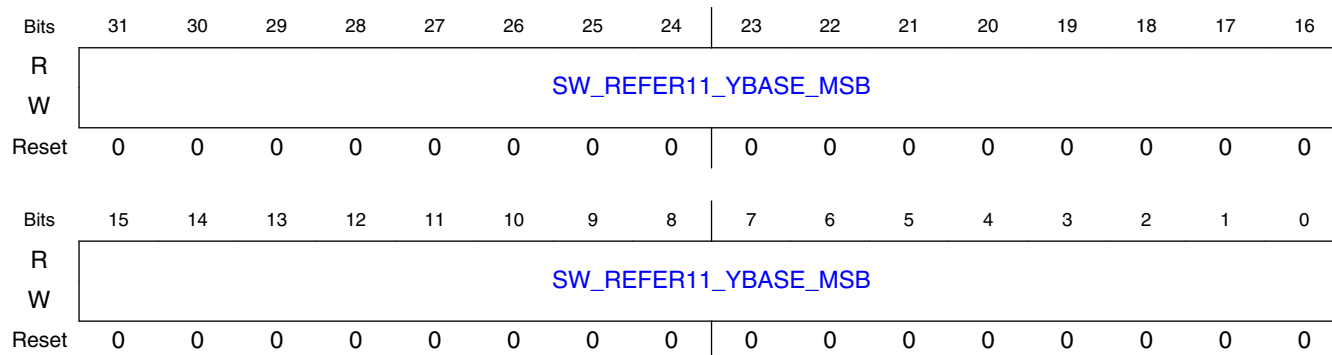
Field	Function
31-0 SW_REFER10_YBASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 10

14.2.5.1.90 Base address MSB (bits 63:32) for reference luminance picture index 11 (SWREG88)

14.2.5.1.90.1 Offset

Register	Offset
SWREG88	160h

14.2.5.1.90.2 Diagram



14.2.5.1.90.3 Fields

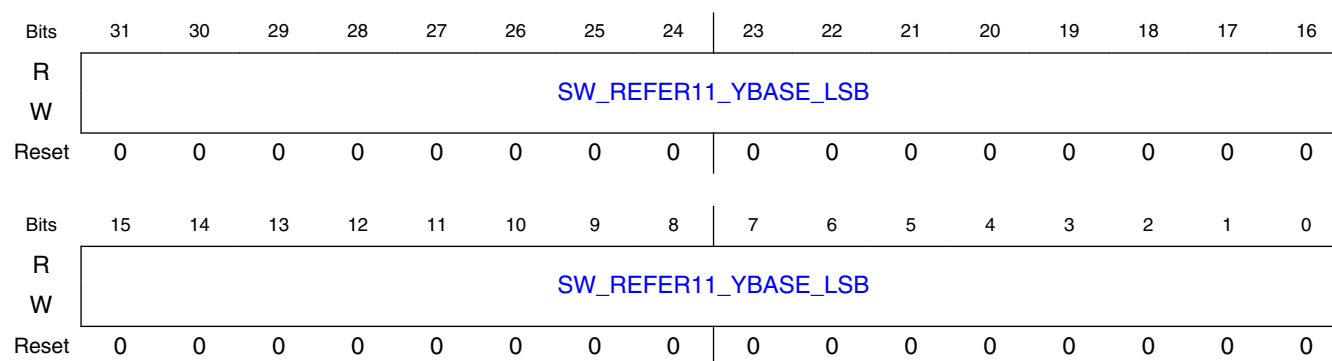
Field	Function
31-0 SW_REFER11_ YBASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 11

14.2.5.1.91 Base address LSB (bits 31:0) for reference luminance picture index 11 (SWREG89)

14.2.5.1.91.1 Offset

Register	Offset
SWREG89	164h

14.2.5.1.91.2 Diagram



14.2.5.1.91.3 Fields

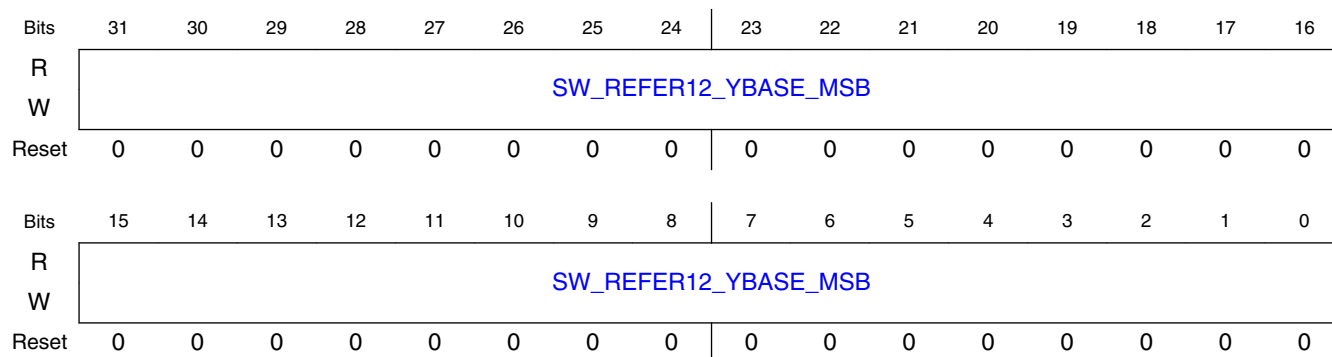
Field	Function
31-0 SW_REFER11_ YBASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 11

14.2.5.1.92 Base address MSB (bits 63:32) for reference luminance picture index 12 (SWREG90)

14.2.5.1.92.1 Offset

Register	Offset
SWREG90	168h

14.2.5.1.92.2 Diagram



14.2.5.1.92.3 Fields

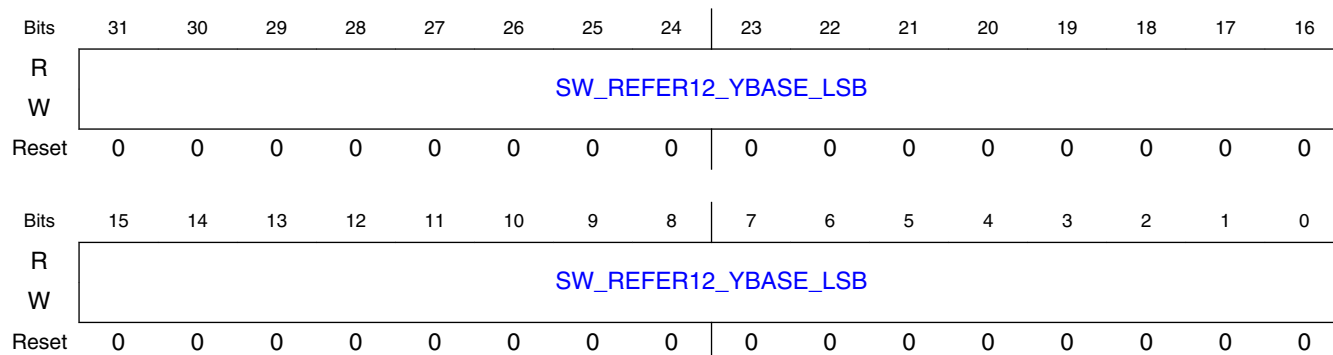
Field	Function
31-0 SW_REFER12_YBASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 12

14.2.5.1.93 Base address LSB (bits 31:0) for reference luminance picture index 12 (SWREG91)

14.2.5.1.93.1 Offset

Register	Offset
SWREG91	16Ch

14.2.5.1.93.2 Diagram



14.2.5.1.93.3 Fields

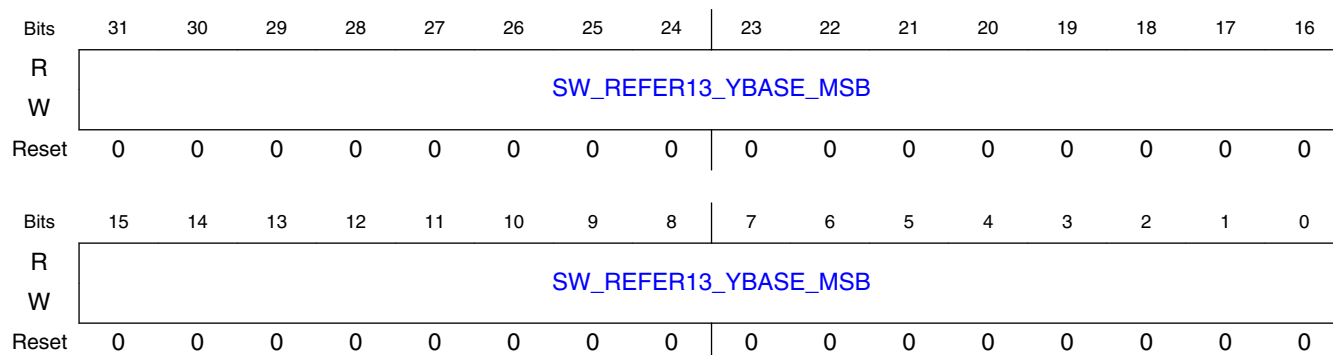
Field	Function
31-0 SW_REFER12_YBASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 12

14.2.5.1.94 Base address MSB (bits 63:32) for reference luminance picture index 13 (SWREG92)

14.2.5.1.94.1 Offset

Register	Offset
SWREG92	170h

14.2.5.1.94.2 Diagram



14.2.5.1.94.3 Fields

Field	Function
31-0 SW_REFER13_ YBASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 13

14.2.5.1.95 Base address LSB (bits 31:0) for reference luminance picture index 13 (SWREG93)

14.2.5.1.95.1 Offset

Register	Offset
SWREG93	174h

14.2.5.1.95.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER13_YBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER13_YBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.95.3 Fields

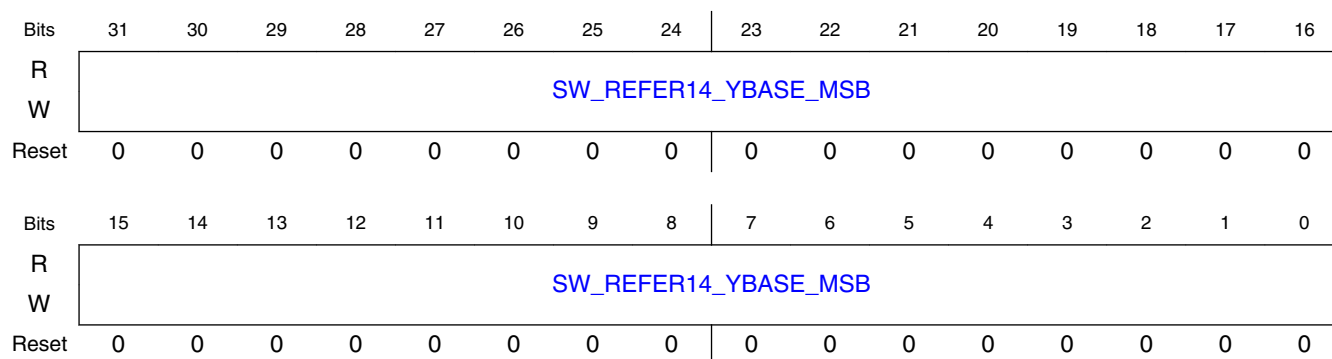
Field	Function
31-0 SW_REFER13_ YBASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 13

14.2.5.1.96 Base address MSB (bits 63:32) for reference luminance picture index 14 (SWREG94)

14.2.5.1.96.1 Offset

Register	Offset
SWREG94	178h

14.2.5.1.96.2 Diagram



14.2.5.1.96.3 Fields

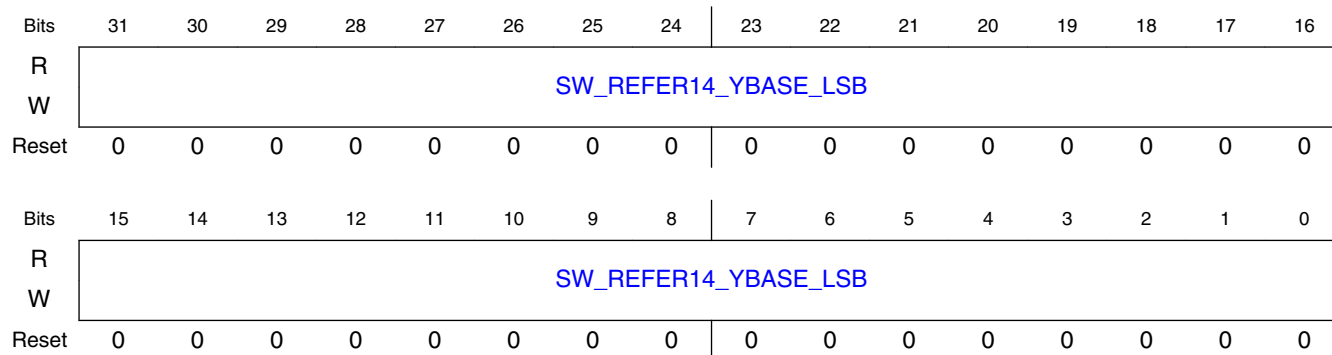
Field	Function
31-0 SW_REFER14_YBASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 14

14.2.5.1.97 Base address LSB (bits 31:0) for reference luminance picture index 14 (SWREG95)

14.2.5.1.97.1 Offset

Register	Offset
SWREG95	17Ch

14.2.5.1.97.2 Diagram



14.2.5.1.97.3 Fields

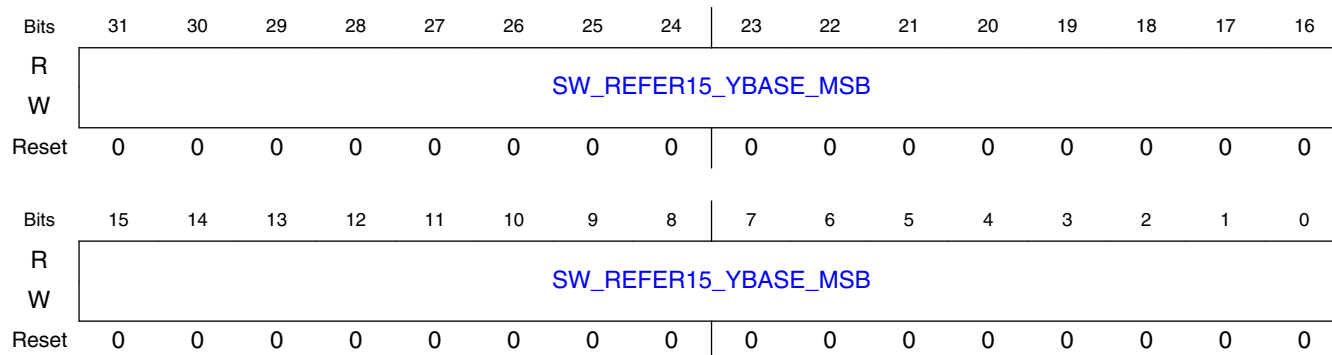
Field	Function
31-0 SW_REFER14_YBASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 14

14.2.5.1.98 Base address MSB (bits 63:32) for reference luminance picture index 15 (SWREG96)

14.2.5.1.98.1 Offset

Register	Offset
SWREG96	180h

14.2.5.1.98.2 Diagram



14.2.5.1.98.3 Fields

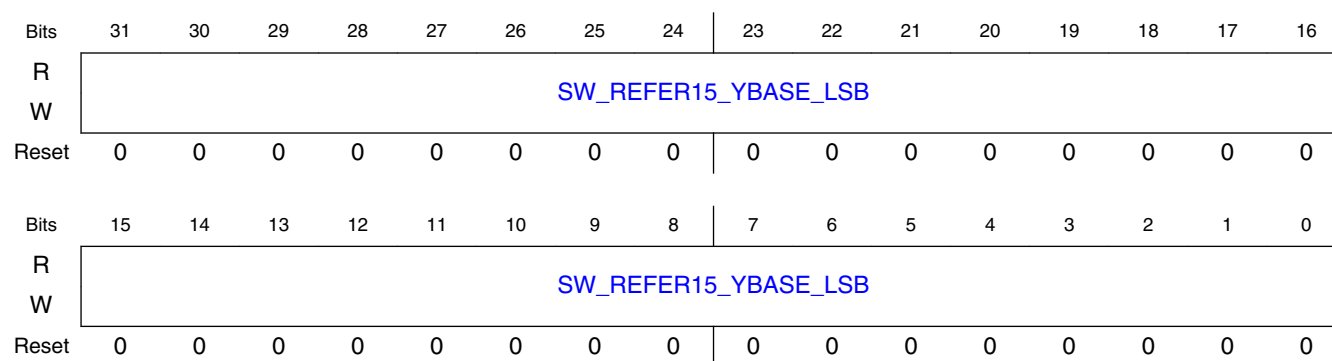
Field	Function
31-0 SW_REFER15_ YBASE_MSB	Base address MSB (bits 63:32) for reference luminance picture index 15

14.2.5.1.99 Base address LSB (bits 31:0) for reference luminance picture index 15 (SWREG97)

14.2.5.1.99.1 Offset

Register	Offset
SWREG97	184h

14.2.5.1.99.2 Diagram



14.2.5.1.99.3 Fields

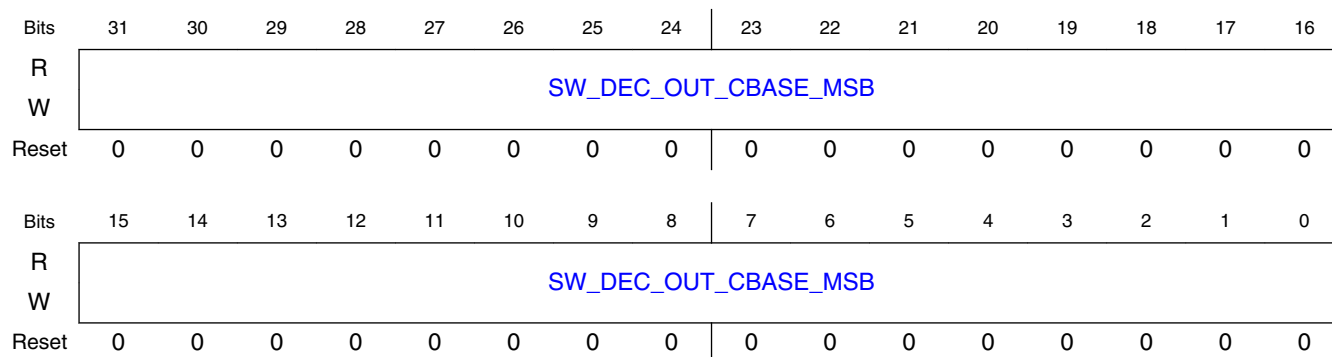
Field	Function
31-0 SW_REFER15_ YBASE_LSB	Base address LSB (bits 31:0) for reference luminance picture index 15

14.2.5.1.100 Base address MSB (bits 63:32) for decoded chrominance picture (SWREG98)

14.2.5.1.100.1 Offset

Register	Offset
SWREG98	188h

14.2.5.1.100.2 Diagram



14.2.5.1.100.3 Fields

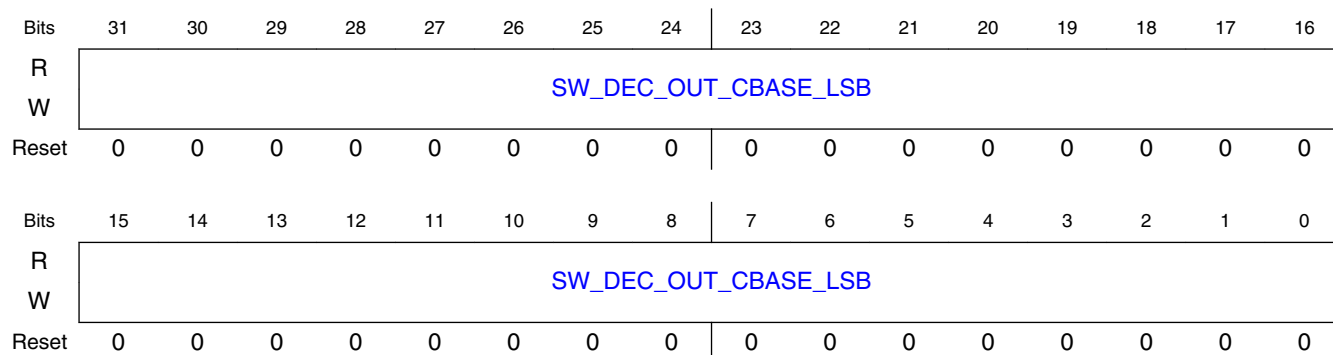
Field	Function
31-0 SW_DEC_OUT_CBASE_MSB	Base address MSB (bits 64:32) for decoded chrominance picture

14.2.5.1.101 Base address LSB (bits 31:0) for decoded chrominance picture (SWREG99)

14.2.5.1.101.1 Offset

Register	Offset
SWREG99	18Ch

14.2.5.1.101.2 Diagram



14.2.5.1.101.3 Fields

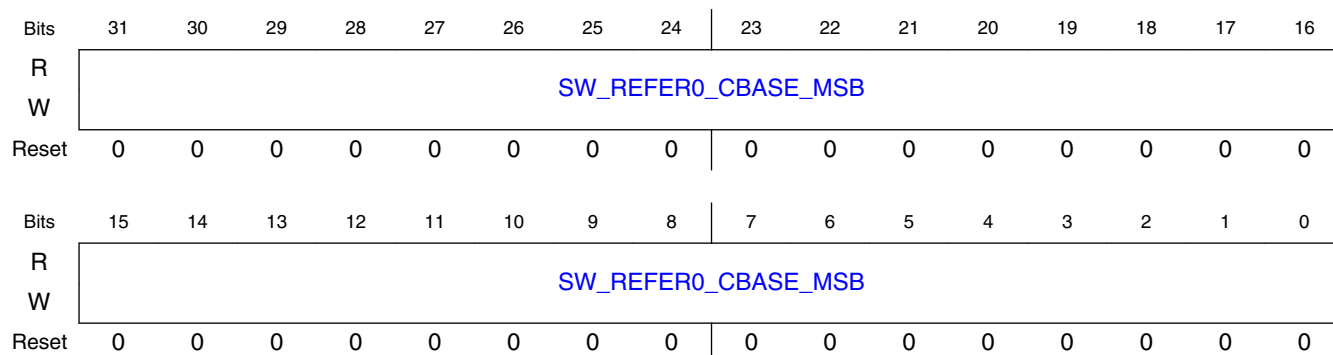
Field	Function
31-0 SW_DEC_OUT_CBASE_LSB	Base address LSB (bits 31:0) for decoded chrominance picture

14.2.5.1.102 Base address MSB (bits 63:32) for reference chrominance picture index 0 (SWREG100)

14.2.5.1.102.1 Offset

Register	Offset
SWREG100	190h

14.2.5.1.102.2 Diagram



14.2.5.1.102.3 Fields

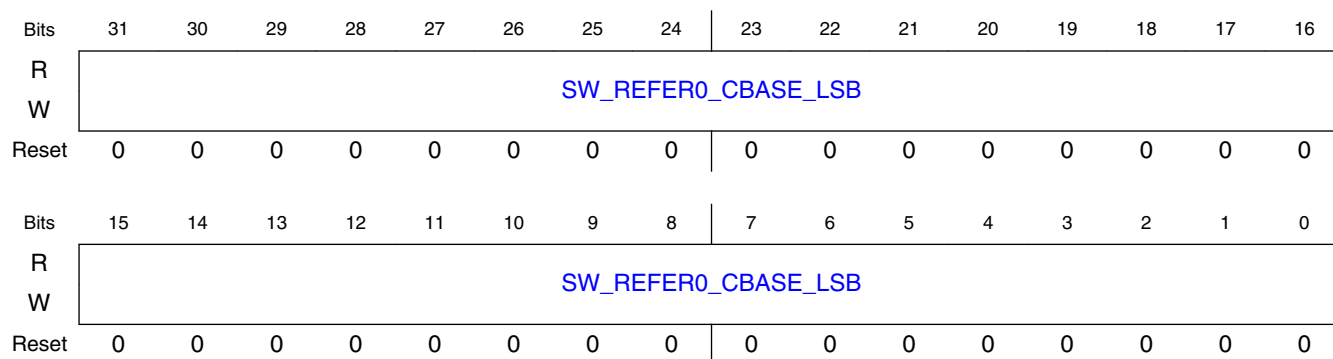
Field	Function
31-0 SW_REFER0_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 0

14.2.5.1.103 Base address LSB (bits 31:0) for reference chrominance picture index 0 (SWREG101)

14.2.5.1.103.1 Offset

Register	Offset
SWREG101	194h

14.2.5.1.103.2 Diagram



14.2.5.1.103.3 Fields

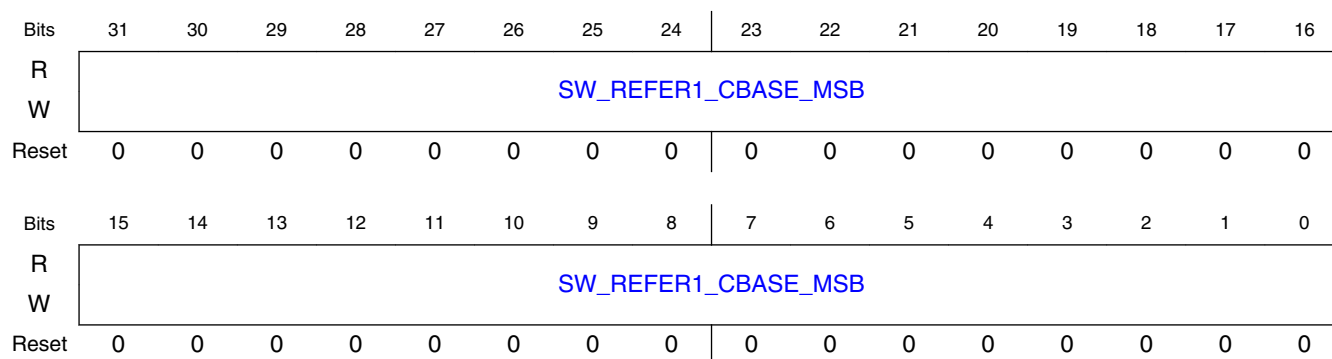
Field	Function
31-0 SW_REFER0_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 0

14.2.5.1.104 Base address MSB (bits 63:32) for reference chrominance picture index 1 (SWREG102)

14.2.5.1.104.1 Offset

Register	Offset
SWREG102	198h

14.2.5.1.104.2 Diagram



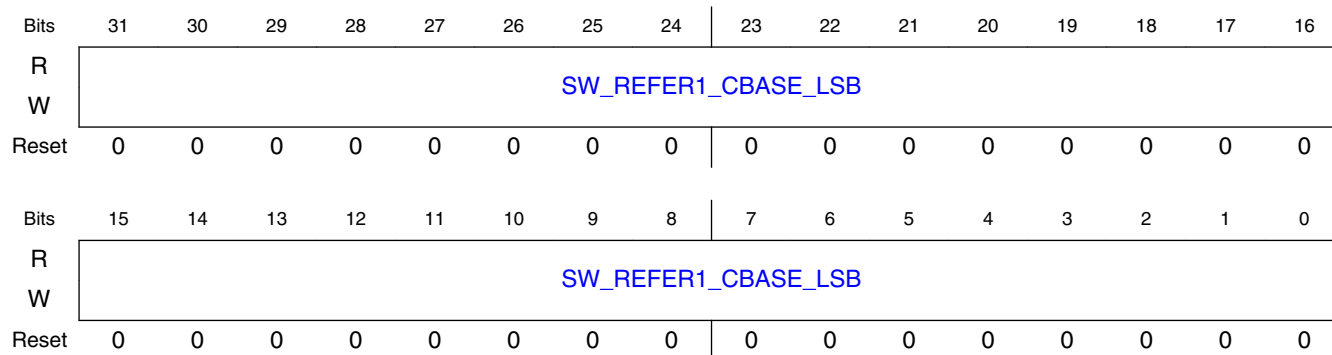
14.2.5.1.104.3 Fields

Field	Function
31-0 SW_REFER1_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 1

14.2.5.1.105 Base address LSB (bits 31:0) for reference chrominance picture index 1 (SWREG103)

14.2.5.1.105.1 Offset

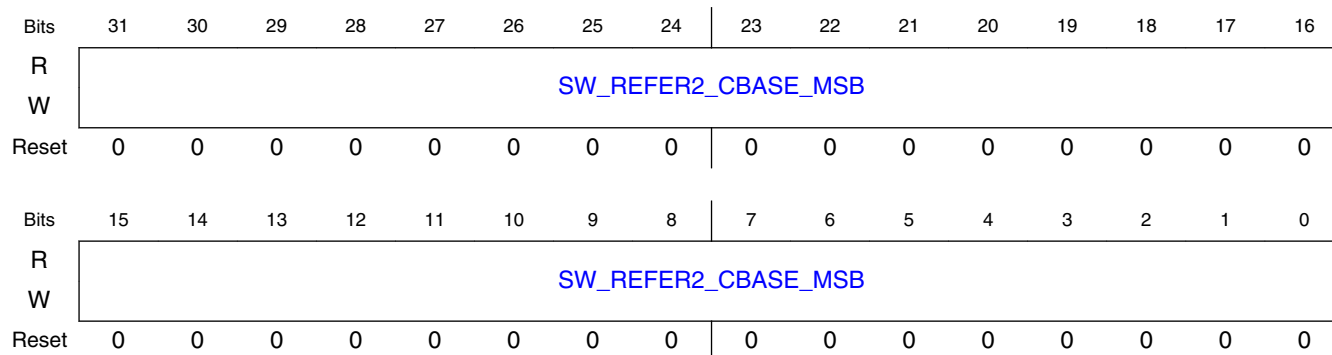
Register	Offset
SWREG103	19Ch

14.2.5.1.105.2 Diagram**14.2.5.1.105.3 Fields**

Field	Function
31-0 SW_REFER1_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 1

14.2.5.1.106 Base address MSB (bits 63:32) for reference chrominance picture index 2 (SWREG104)**14.2.5.1.106.1 Offset**

Register	Offset
SWREG104	1A0h

14.2.5.1.106.2 Diagram

14.2.5.1.106.3 Fields

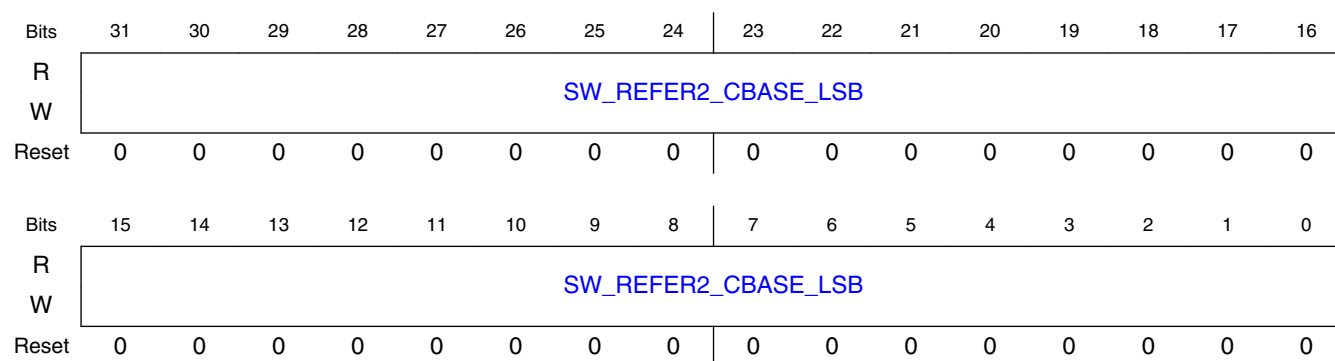
Field	Function
31-0 SW_REFER2_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 2

14.2.5.1.107 Base address LSB (bits 31:0) for reference chrominance picture index 2 (SWREG105)

14.2.5.1.107.1 Offset

Register	Offset
SWREG105	1A4h

14.2.5.1.107.2 Diagram



14.2.5.1.107.3 Fields

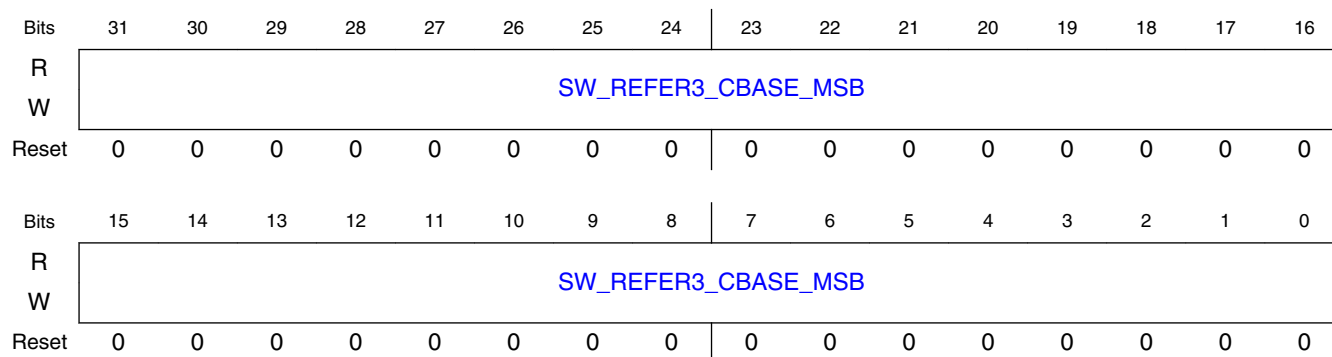
Field	Function
31-0 SW_REFER2_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 2

14.2.5.1.108 Base address MSB (bits 63:32) for reference chrominance picture index 3 (SWREG106)

14.2.5.1.108.1 Offset

Register	Offset
SWREG106	1A8h

14.2.5.1.108.2 Diagram



14.2.5.1.108.3 Fields

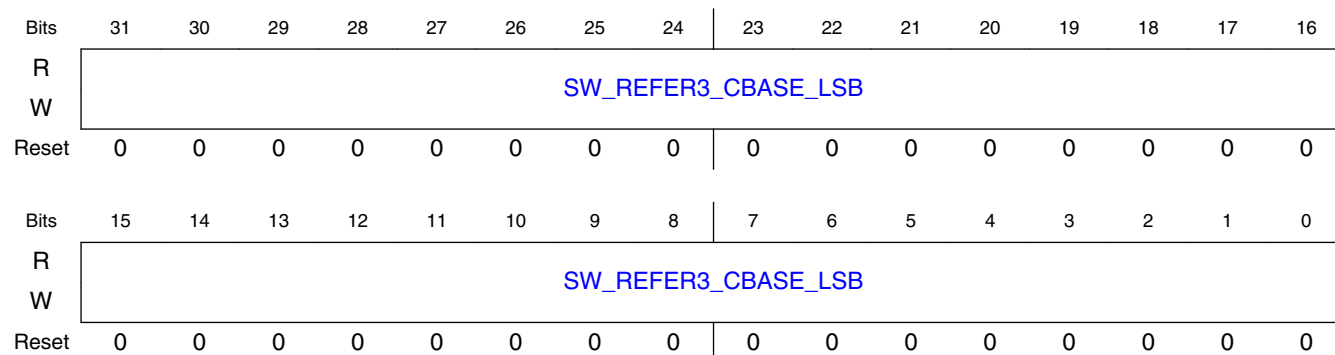
Field	Function
31-0 SW_REFER3_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 3

14.2.5.1.109 Base address LSB (bits 31:0) for reference chrominance picture index 3 (SWREG107)

14.2.5.1.109.1 Offset

Register	Offset
SWREG107	1ACh

14.2.5.1.109.2 Diagram



14.2.5.1.109.3 Fields

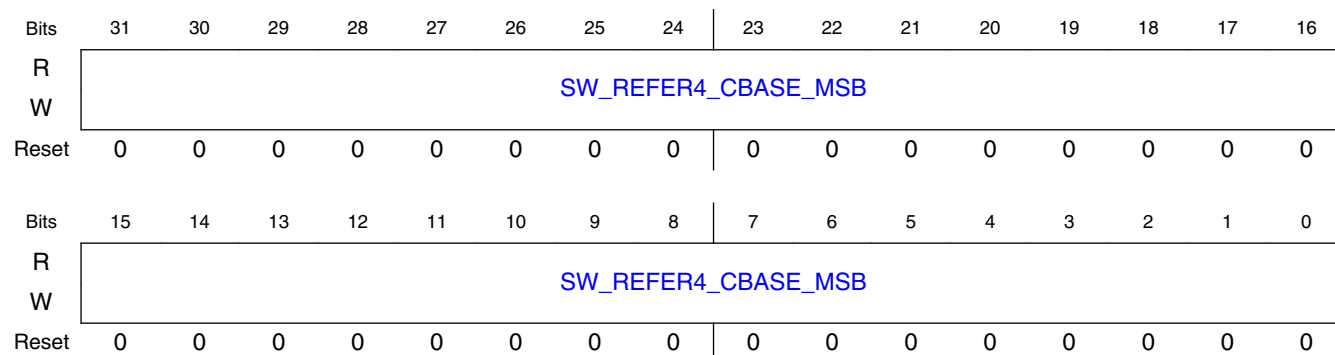
Field	Function
31-0 SW_REFER3_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 3

14.2.5.1.110 Base address MSB (bits 63:32) for reference chrominance picture index 4 (SWREG108)

14.2.5.1.110.1 Offset

Register	Offset
SWREG108	1B0h

14.2.5.1.110.2 Diagram



14.2.5.1.110.3 Fields

Field	Function
31-0 SW_REFER4_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 4

14.2.5.1.111 Base address LSB (bits 31:0) for reference chrominance picture index 4 (SWREG109)

14.2.5.1.111.1 Offset

Register	Offset
SWREG109	1B4h

14.2.5.1.111.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER4_CBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER4_CBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.111.3 Fields

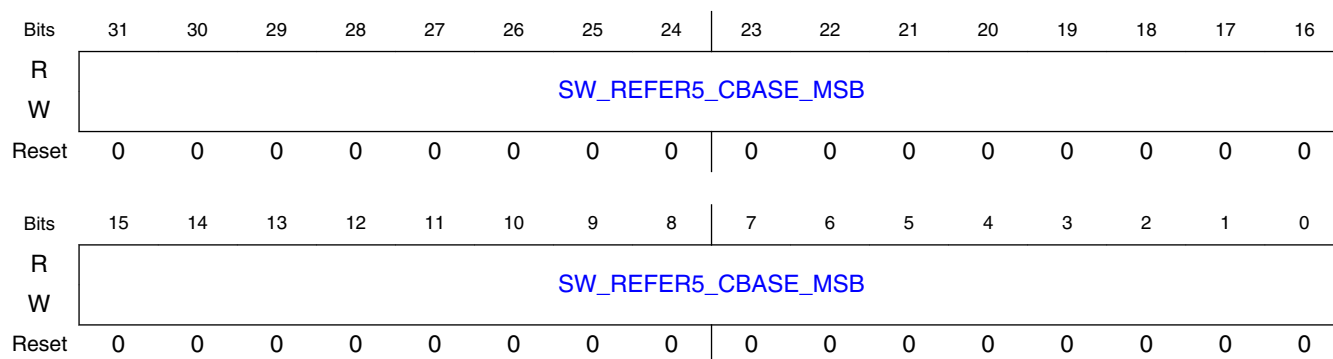
Field	Function
31-0 SW_REFER4_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 4

14.2.5.1.112 Base address MSB (bits 63:32) for reference chrominance picture index 5 (SWREG110)

14.2.5.1.112.1 Offset

Register	Offset
SWREG110	1B8h

14.2.5.1.112.2 Diagram



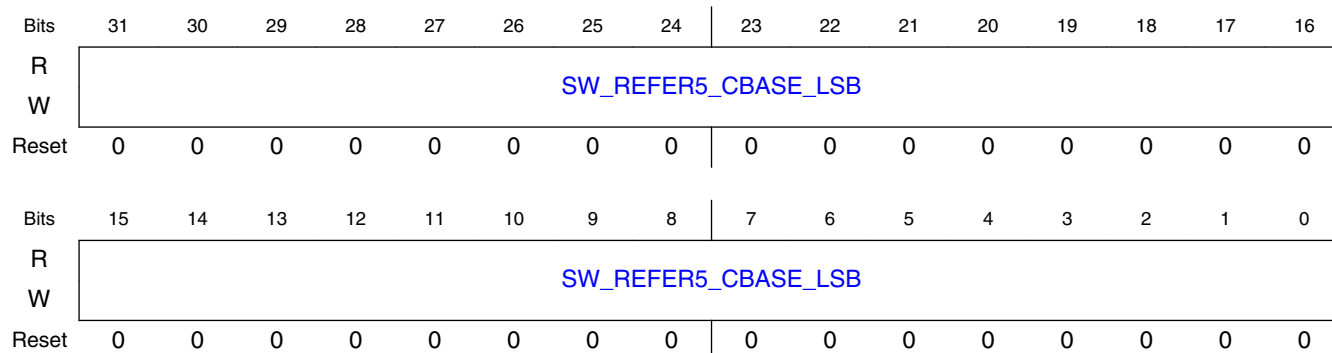
14.2.5.1.112.3 Fields

Field	Function
31-0 SW_REFER5_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 5

14.2.5.1.113 Base address LSB (bits 31:0) for reference chrominance picture index 5 (SWREG111)

14.2.5.1.113.1 Offset

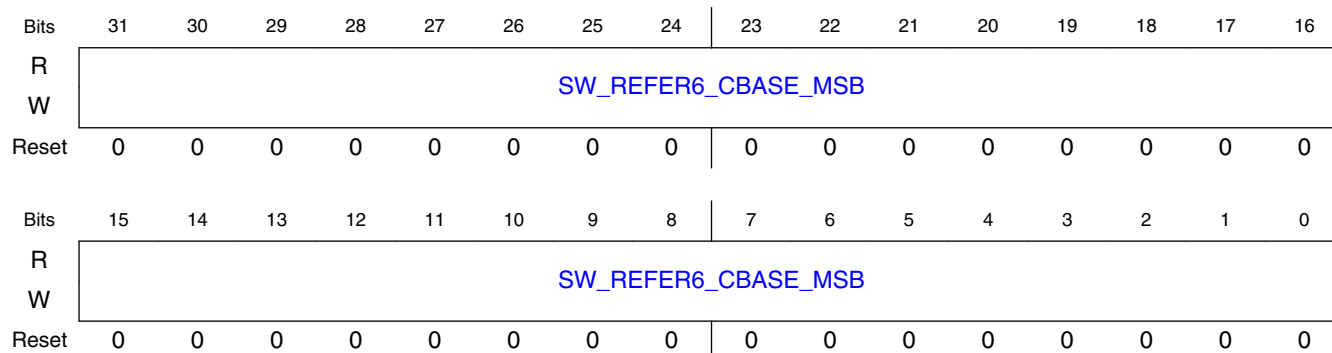
Register	Offset
SWREG111	1BCh

14.2.5.1.113.2 Diagram**14.2.5.1.113.3 Fields**

Field	Function
31-0 SW_REFER5_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 5

14.2.5.1.114 Base address MSB (bits 63:32) for reference chrominance picture index 6 (SWREG112)**14.2.5.1.114.1 Offset**

Register	Offset
SWREG112	1C0h

14.2.5.1.114.2 Diagram

14.2.5.1.114.3 Fields

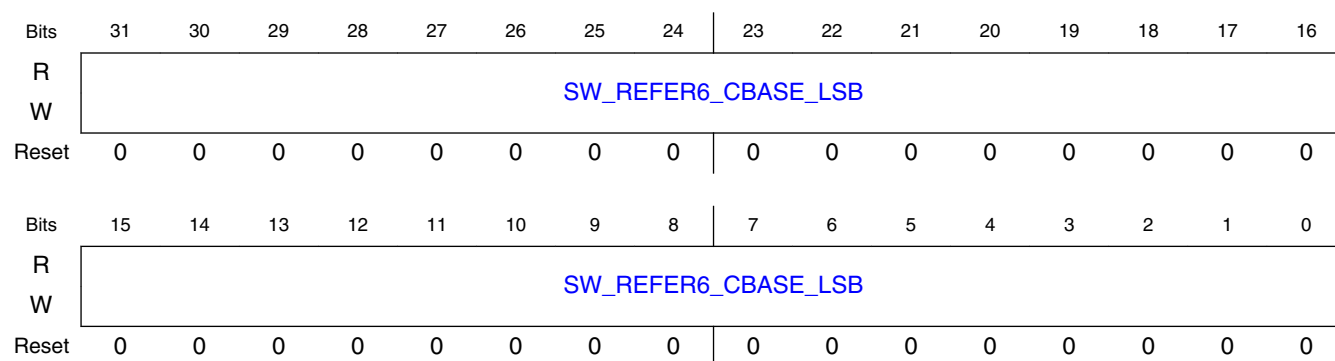
Field	Function
31-0 SW_REFER6_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 6

14.2.5.1.115 Base address LSB (bits 31:0) for reference chrominance picture index 6 (SWREG113)

14.2.5.1.115.1 Offset

Register	Offset
SWREG113	1C4h

14.2.5.1.115.2 Diagram



14.2.5.1.115.3 Fields

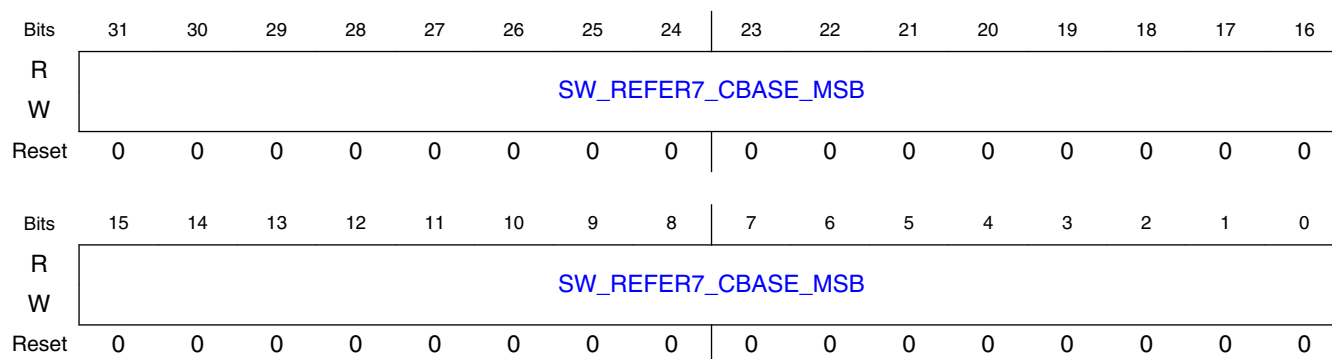
Field	Function
31-0 SW_REFER6_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 6

14.2.5.1.116 Base address MSB (bits 63:32) for reference chrominance picture index 7 (SWREG114)

14.2.5.1.116.1 Offset

Register	Offset
SWREG114	1C8h

14.2.5.1.116.2 Diagram



14.2.5.1.116.3 Fields

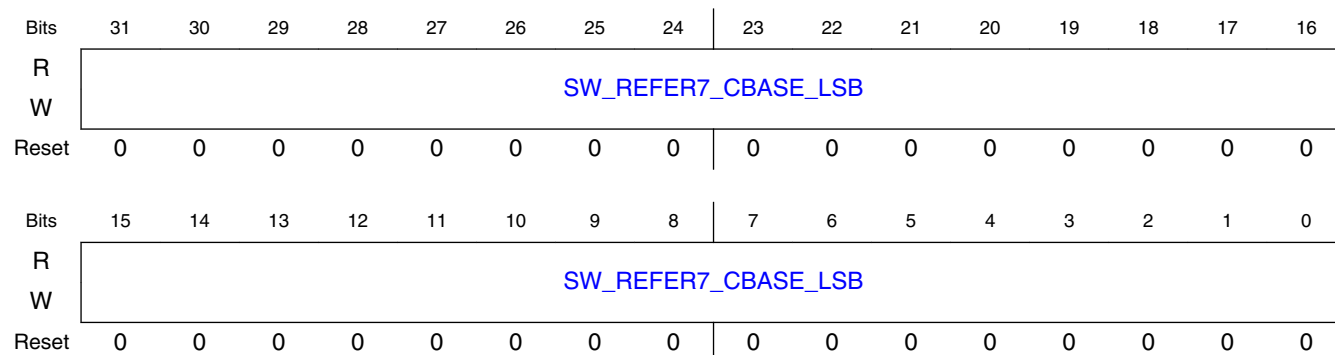
Field	Function
31-0 SW_REFER7_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 7

14.2.5.1.117 Base address LSB (bits 31:0) for reference chrominance picture index 7 (SWREG115)

14.2.5.1.117.1 Offset

Register	Offset
SWREG115	1CCh

14.2.5.1.117.2 Diagram



14.2.5.1.117.3 Fields

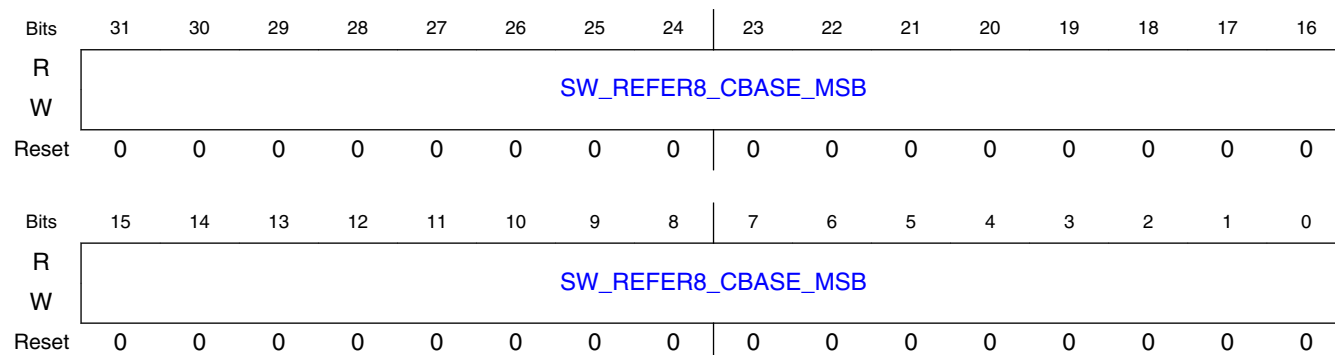
Field	Function
31-0 SW_REFER7_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 7

14.2.5.1.118 Base address MSB (bits 63:32) for reference chrominance picture index 8 (SWREG116)

14.2.5.1.118.1 Offset

Register	Offset
SWREG116	1D0h

14.2.5.1.118.2 Diagram

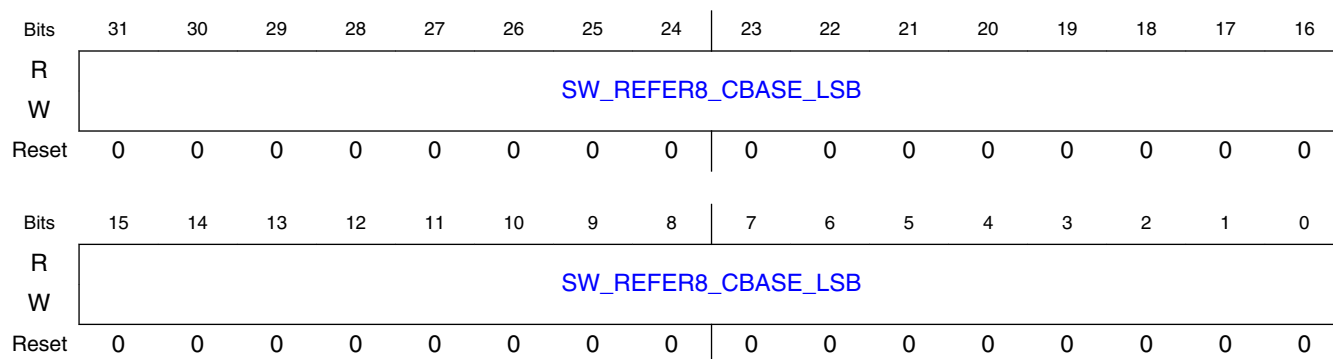


14.2.5.1.118.3 Fields

Field	Function
31-0 SW_REFER8_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 8

14.2.5.1.119 Base address LSB (bits 31:0) for reference chrominance picture index 8 (SWREG117)**14.2.5.1.119.1 Offset**

Register	Offset
SWREG117	1D4h

14.2.5.1.119.2 Diagram**14.2.5.1.119.3 Fields**

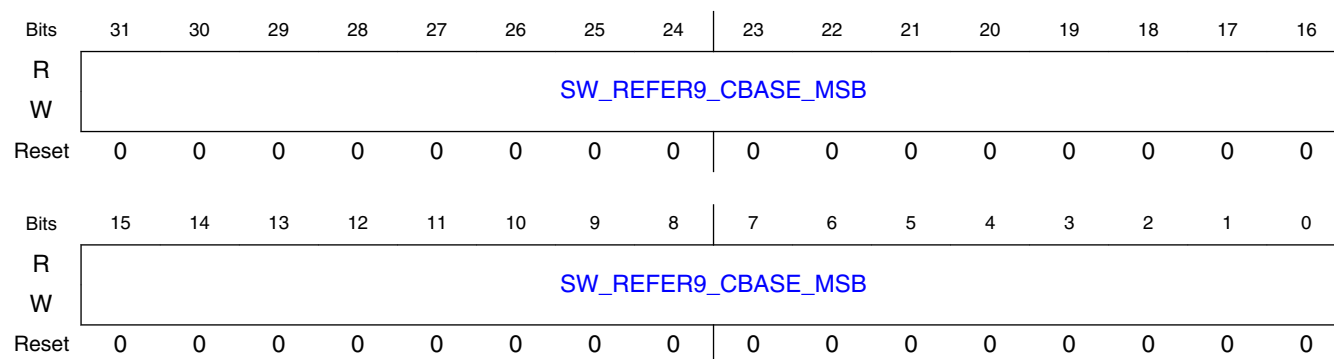
Field	Function
31-0 SW_REFER8_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 8

14.2.5.1.120 Base address MSB (bits 63:32) for reference chrominance picture index 9 (SWREG118)

14.2.5.1.120.1 Offset

Register	Offset
SWREG118	1D8h

14.2.5.1.120.2 Diagram



14.2.5.1.120.3 Fields

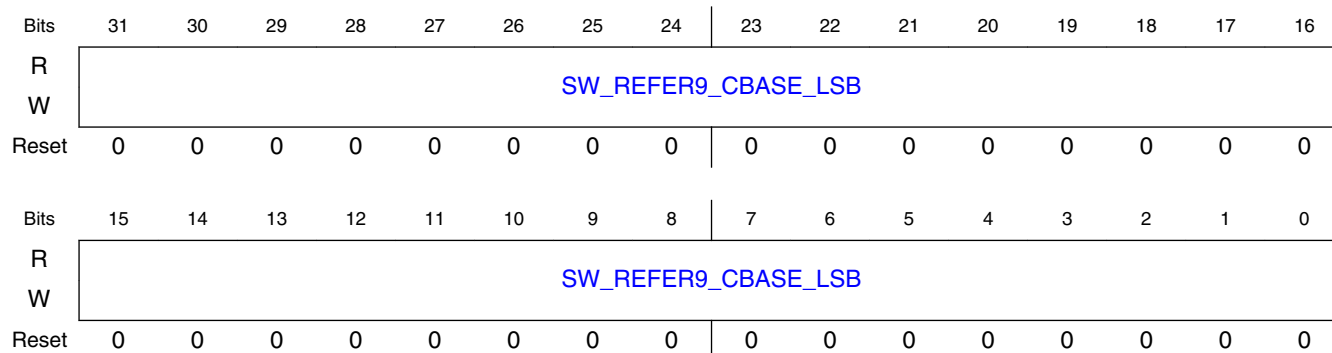
Field	Function
31-0 SW_REFER9_C BASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 9

14.2.5.1.121 Base address LSB (bits 31:0) for reference chrominance picture index 9 (SWREG119)

14.2.5.1.121.1 Offset

Register	Offset
SWREG119	1DCh

14.2.5.1.121.2 Diagram



14.2.5.1.121.3 Fields

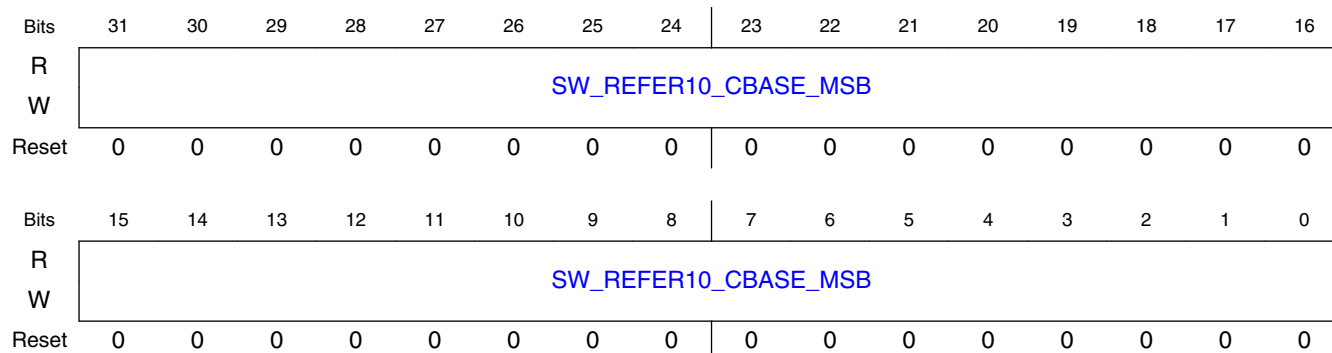
Field	Function
31-0 SW_REFER9_C BASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 9

14.2.5.1.122 Base address MSB (bits 63:32) for reference chrominance picture index 10 (SWREG120)

14.2.5.1.122.1 Offset

Register	Offset
SWREG120	1E0h

14.2.5.1.122.2 Diagram



14.2.5.1.122.3 Fields

Field	Function
31-0 SW_REFER10_ CBASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 10

14.2.5.1.123 Base address LSB (bits 31:0) for reference chrominance picture index 10 (SWREG121)

14.2.5.1.123.1 Offset

Register	Offset
SWREG121	1E4h

14.2.5.1.123.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER10_CBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER10_CBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.123.3 Fields

Field	Function
31-0 SW_REFER10_ CBASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 10

14.2.5.1.124 Base address MSB (bits 63:32) for reference chrominance picture index 11 (SWREG122)

14.2.5.1.124.1 Offset

Register	Offset
SWREG122	1E8h

14.2.5.1.124.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER11_CBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER11_CBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.124.3 Fields

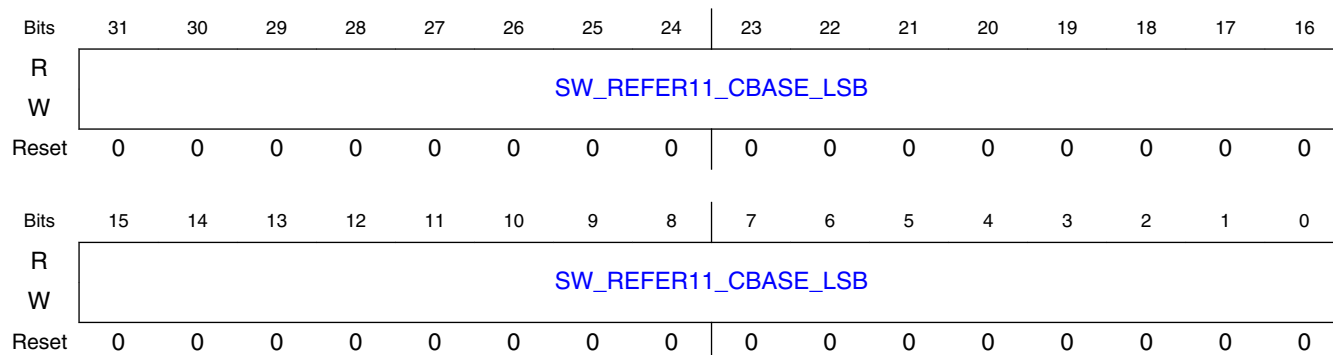
Field	Function
31-0 SW_REFER11_CBASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 11

14.2.5.1.125 Base address LSB (bits 31:0) for reference chrominance picture index 11 (SWREG123)

14.2.5.1.125.1 Offset

Register	Offset
SWREG123	1ECh

14.2.5.1.125.2 Diagram



14.2.5.1.125.3 Fields

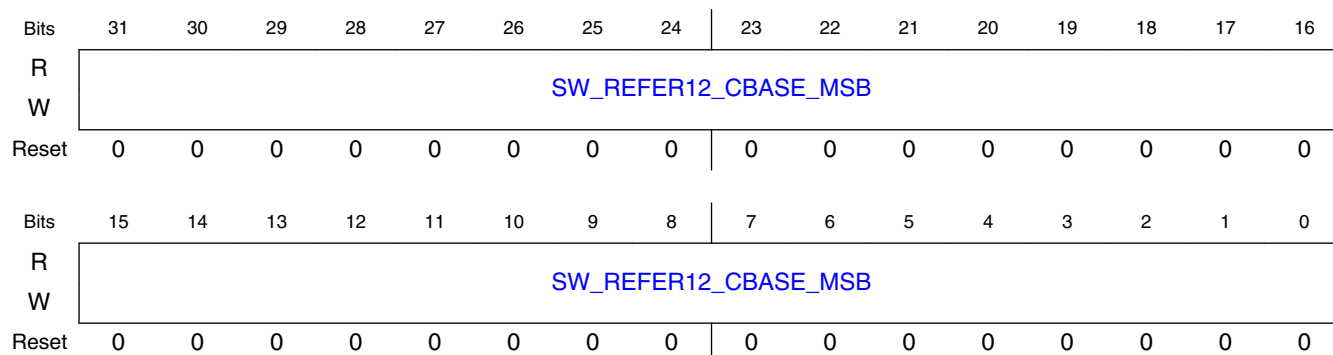
Field	Function
31-0 SW_REFER11_CBASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 11

14.2.5.1.126 Base address MSB (bits 63:32) for reference chrominance picture index 12 (SWREG124)

14.2.5.1.126.1 Offset

Register	Offset
SWREG124	1F0h

14.2.5.1.126.2 Diagram



14.2.5.1.126.3 Fields

Field	Function
31-0 SW_REFER12_ CBASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 12

14.2.5.1.127 Base address LSB (bits 31:0) for reference chrominance picture index 12 (SWREG125)**14.2.5.1.127.1 Offset**

Register	Offset
SWREG125	1F4h

14.2.5.1.127.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER12_CBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER12_CBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.127.3 Fields

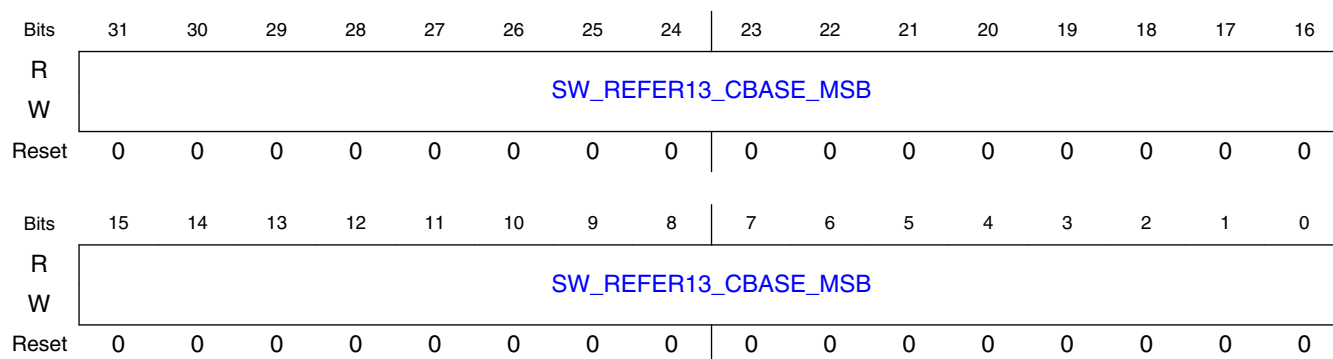
Field	Function
31-0 SW_REFER12_ CBASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 12

14.2.5.1.128 Base address MSB (bits 63:32) for reference chrominance picture index 13 (SWREG126)

14.2.5.1.128.1 Offset

Register	Offset
SWREG126	1F8h

14.2.5.1.128.2 Diagram



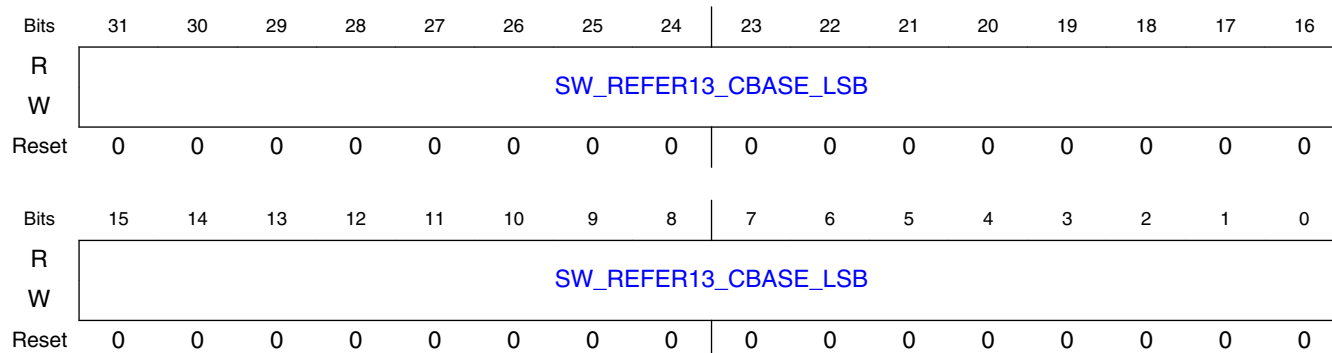
14.2.5.1.128.3 Fields

Field	Function
31-0 SW_REFER13_CBASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 13

14.2.5.1.129 Base address LSB (bits 31:0) for reference chrominance picture index 13 (SWREG127)

14.2.5.1.129.1 Offset

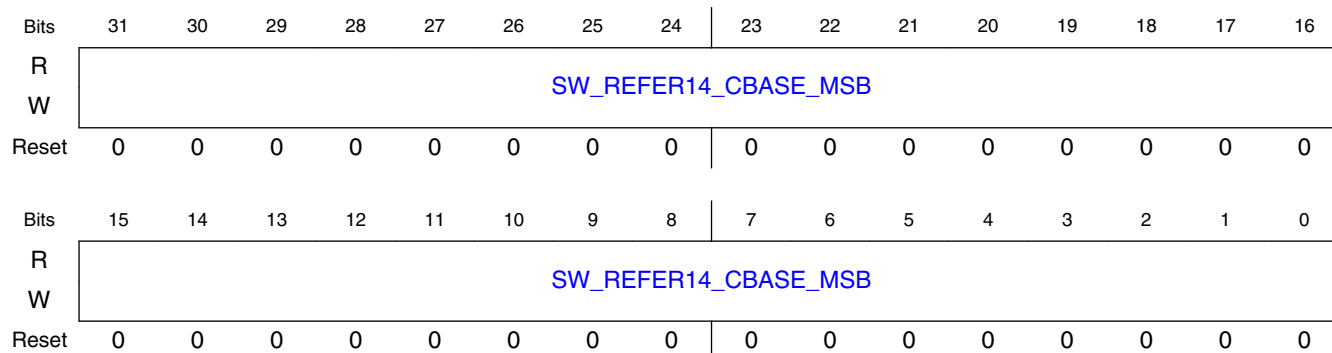
Register	Offset
SWREG127	1FCh

14.2.5.1.129.2 Diagram**14.2.5.1.129.3 Fields**

Field	Function
31-0 SW_REFER13_CBASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 13

14.2.5.1.130 Base address MSB (bits 63:32) for reference chrominance picture index 14 (SWREG128)**14.2.5.1.130.1 Offset**

Register	Offset
SWREG128	200h

14.2.5.1.130.2 Diagram

14.2.5.1.130.3 Fields

Field	Function
31-0 SW_REFER14_ CBASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 14

14.2.5.1.131 Base address LSB (bits 31:0) for reference chrominance picture index 14 (SWREG129)

14.2.5.1.131.1 Offset

Register	Offset
SWREG129	204h

14.2.5.1.131.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER14_CBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER14_CBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.131.3 Fields

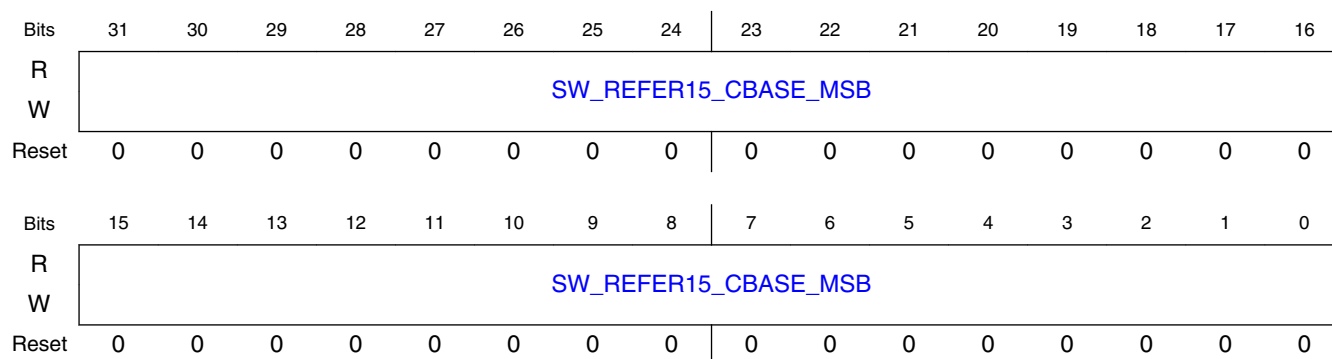
Field	Function
31-0 SW_REFER14_ CBASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 14

14.2.5.1.132 Base address MSB (bits 63:32) for reference chrominance picture index 15 (SWREG130)

14.2.5.1.132.1 Offset

Register	Offset
SWREG130	208h

14.2.5.1.132.2 Diagram



14.2.5.1.132.3 Fields

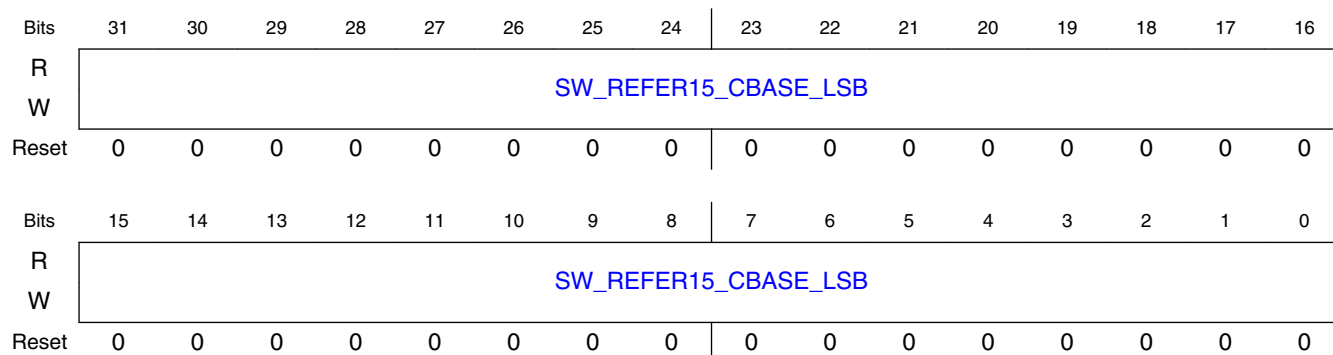
Field	Function
31-0 SW_REFER15_CBASE_MSB	Base address MSB (bits 63:32) for reference chrominance picture index 15

14.2.5.1.133 Base address LSB (bits 31:0) for reference chrominance picture index 15 (SWREG131)

14.2.5.1.133.1 Offset

Register	Offset
SWREG131	20Ch

14.2.5.1.133.2 Diagram



14.2.5.1.133.3 Fields

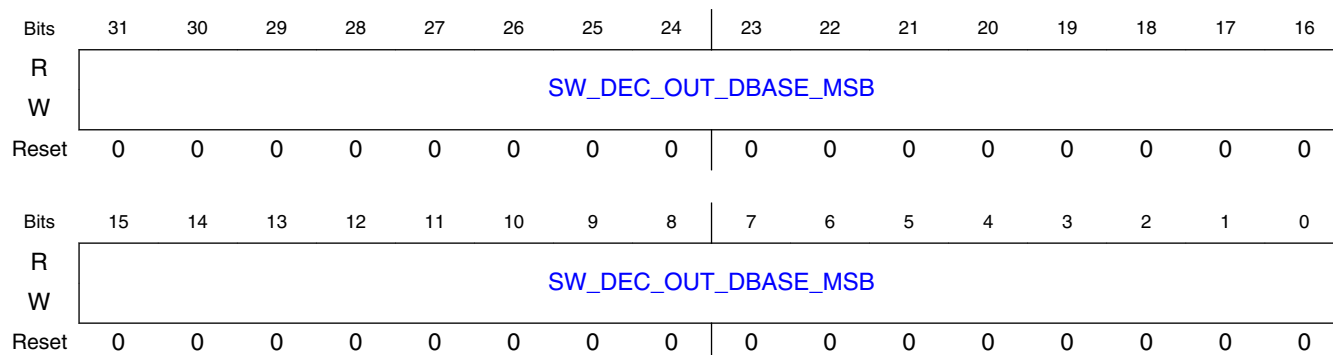
Field	Function
31-0 SW_REFER15_CBASE_LSB	Base address LSB (bits 31:0) for reference chrominance picture index 15

14.2.5.1.134 Base address MSB (bits 63:32) for decoded direct mode MVS (SWREG132)

14.2.5.1.134.1 Offset

Register	Offset
SWREG132	210h

14.2.5.1.134.2 Diagram

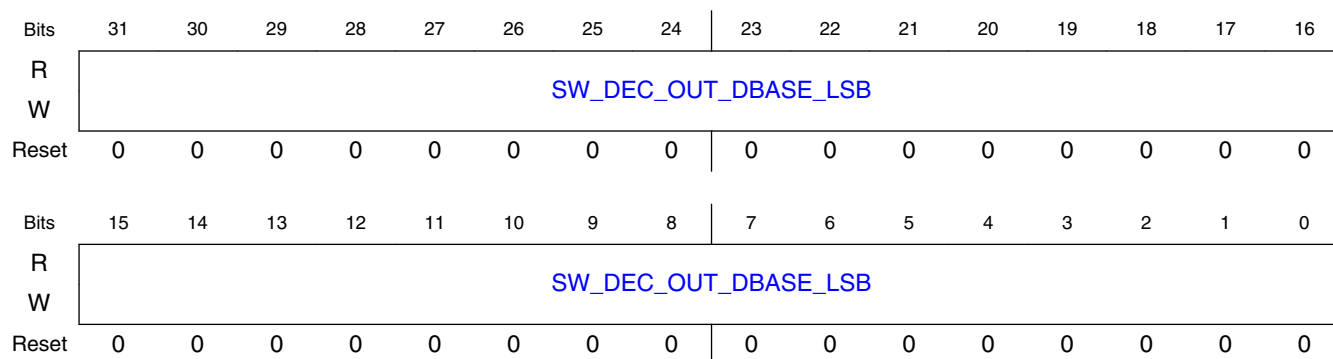


14.2.5.1.134.3 Fields

Field	Function
31-0 SW_DEC_OUT_DBASE_MSB	Base address MSB (bits 63:32) for decoded direct mode MVS

14.2.5.1.135 Base address LSB (bits 31:0) for decoded direct mode MVS (SWREG133)**14.2.5.1.135.1 Offset**

Register	Offset
SWREG133	214h

14.2.5.1.135.2 Diagram**14.2.5.1.135.3 Fields**

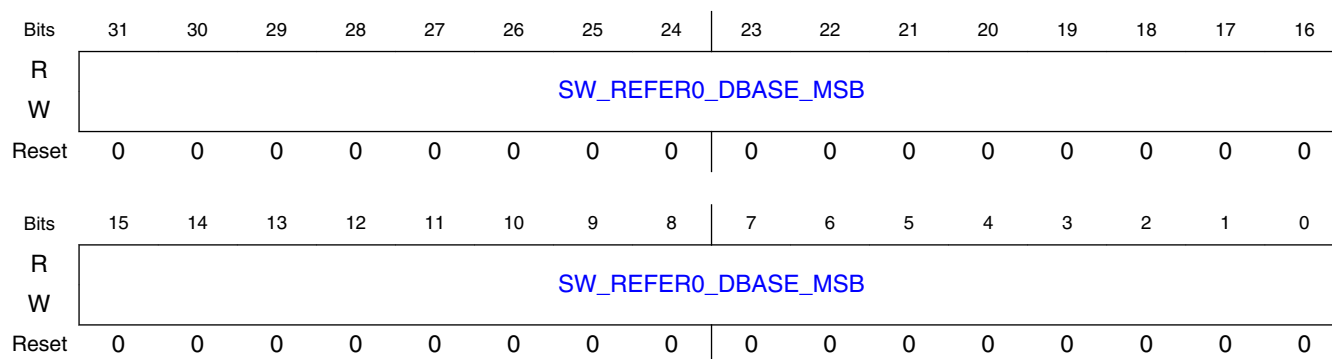
Field	Function
31-0 SW_DEC_OUT_DBASE_LSB	Base address LSB (bits 31:0) for decoded direct mode MVS

14.2.5.1.136 Base address MSB (bits 63:32) for reference direct mode MVS index 0 (SWREG134)

14.2.5.1.136.1 Offset

Register	Offset
SWREG134	218h

14.2.5.1.136.2 Diagram



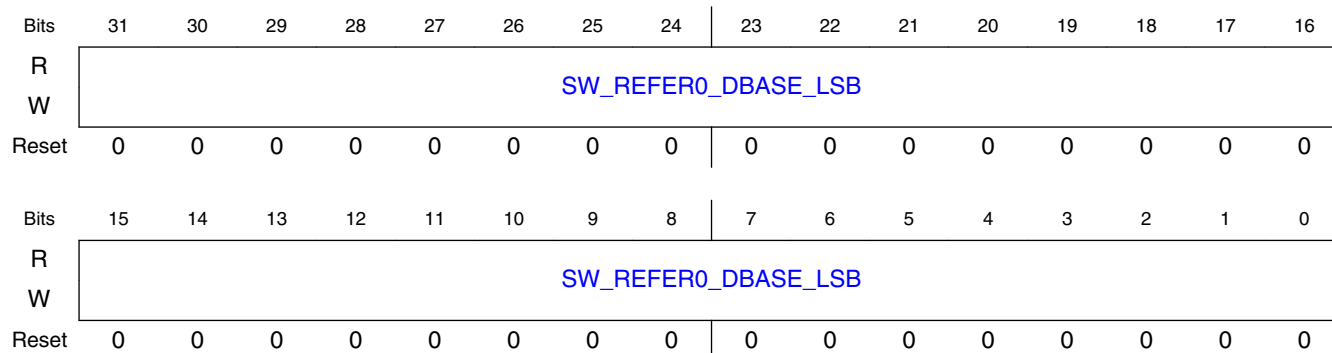
14.2.5.1.136.3 Fields

Field	Function
31-0 SW_REFER0_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 0

14.2.5.1.137 Base address LSB (bits 31:0) for reference direct mode MVS index 0 (SWREG135)

14.2.5.1.137.1 Offset

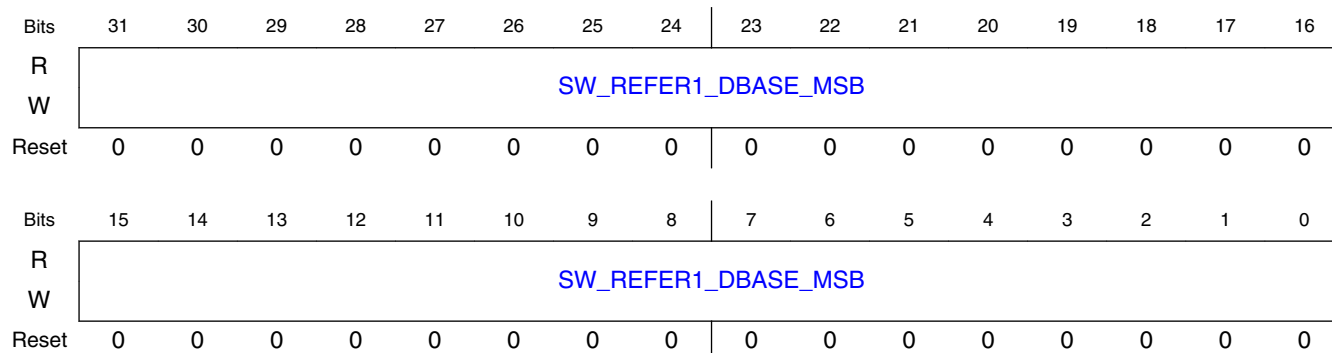
Register	Offset
SWREG135	21Ch

14.2.5.1.137.2 Diagram**14.2.5.1.137.3 Fields**

Field	Function
31-0 SW_REFER0_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 0

14.2.5.1.138 Base address MSB (bits 63:32) for reference direct mode MVS index 1 (SWREG136)**14.2.5.1.138.1 Offset**

Register	Offset
SWREG136	220h

14.2.5.1.138.2 Diagram

14.2.5.1.138.3 Fields

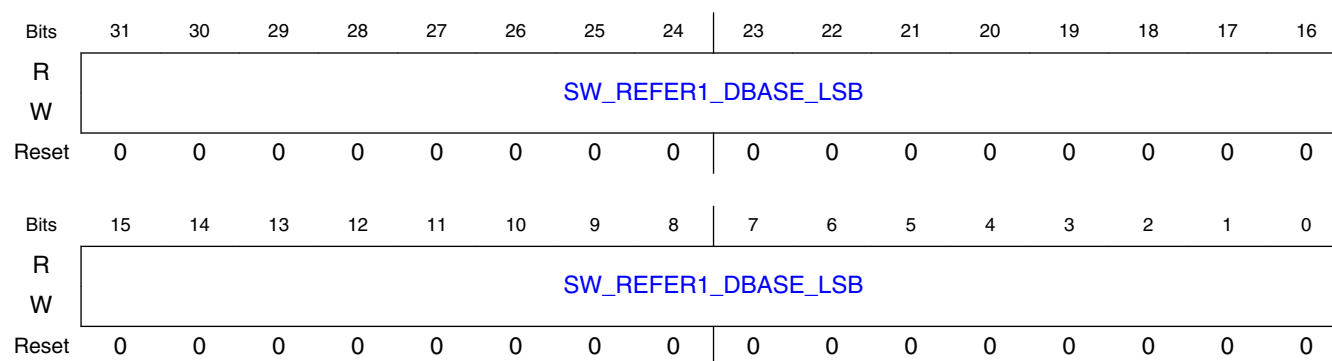
Field	Function
31-0 SW_REFER1_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 1

14.2.5.1.139 Base address LSB (bits 31:0) for reference direct mode MVS index 1 (SWREG137)

14.2.5.1.139.1 Offset

Register	Offset
SWREG137	224h

14.2.5.1.139.2 Diagram



14.2.5.1.139.3 Fields

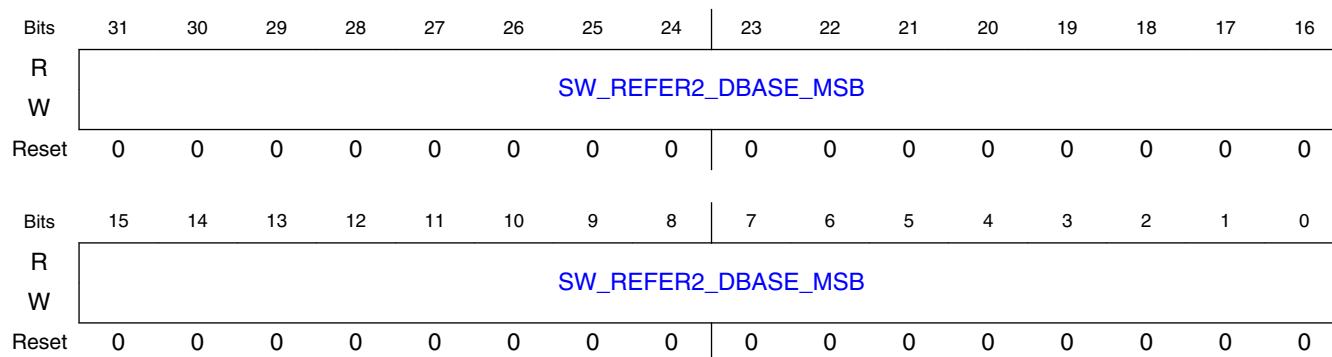
Field	Function
31-0 SW_REFER1_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 1

14.2.5.1.140 Base address MSB (bits 63:32) for reference direct mode MVS index 2 (SWREG138)

14.2.5.1.140.1 Offset

Register	Offset
SWREG138	228h

14.2.5.1.140.2 Diagram



14.2.5.1.140.3 Fields

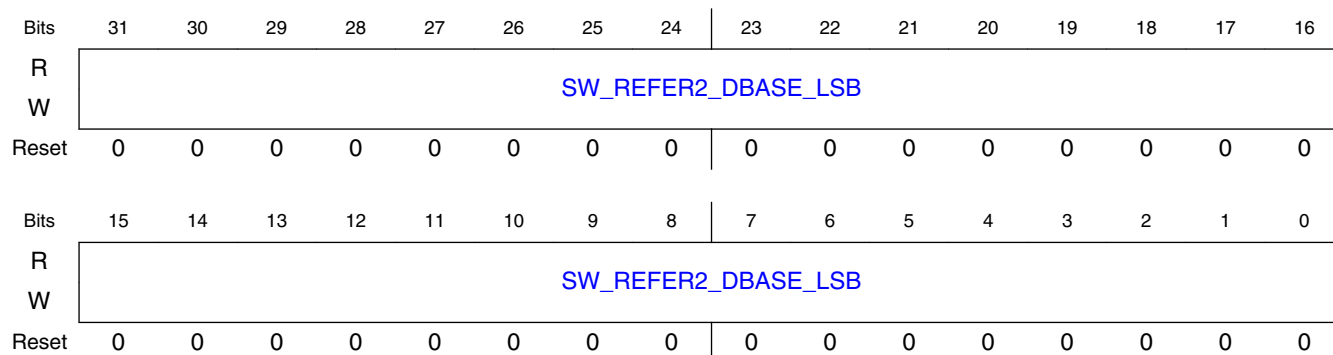
Field	Function
31-0 SW_REFER2_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 2

14.2.5.1.141 Base address LSB (bits 31:0) for reference direct mode MVS index 2 (SWREG139)

14.2.5.1.141.1 Offset

Register	Offset
SWREG139	22Ch

14.2.5.1.141.2 Diagram



14.2.5.1.141.3 Fields

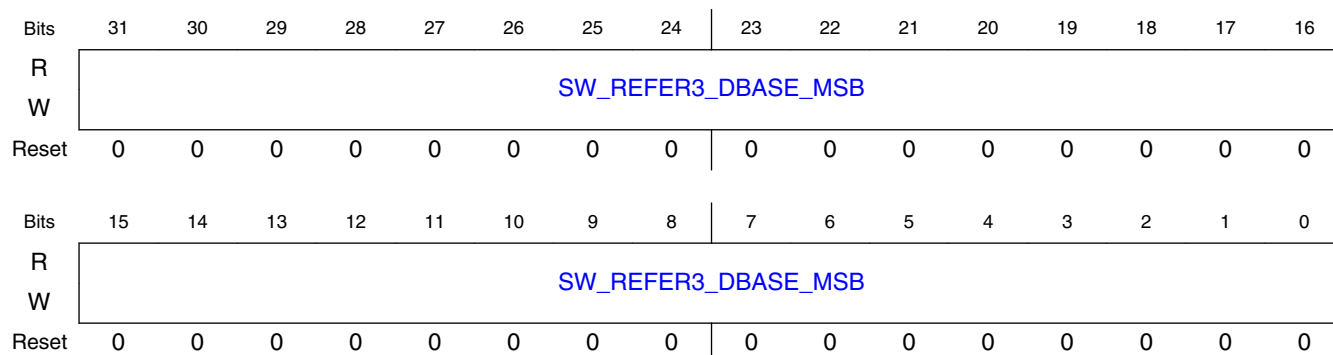
Field	Function
31-0 SW_REFER2_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 2

14.2.5.1.142 Base address MSB (bits 63:32) for reference direct mode MVS index 3 (SWREG140)

14.2.5.1.142.1 Offset

Register	Offset
SWREG140	230h

14.2.5.1.142.2 Diagram

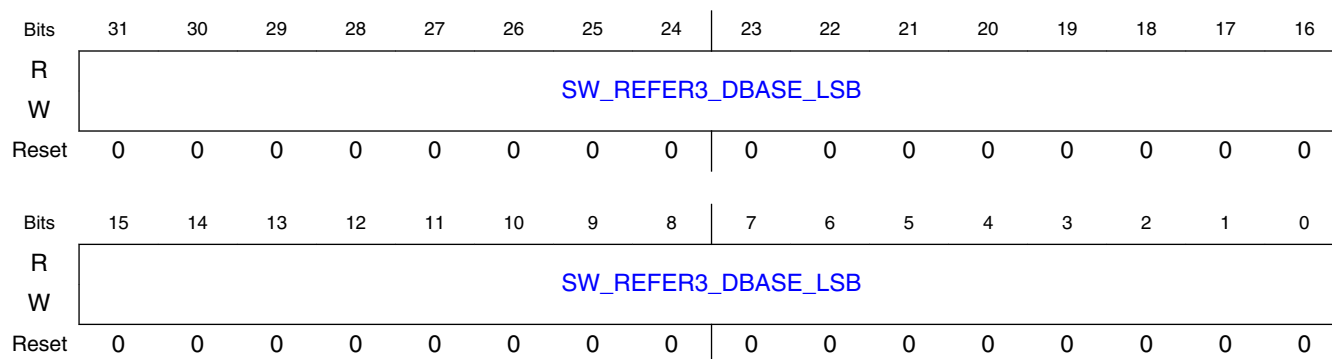


14.2.5.1.142.3 Fields

Field	Function
31-0 SW_REFER3_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 3

14.2.5.1.143 Base address LSB (bits 31:0) for reference direct mode MVS index 3 (SWREG141)**14.2.5.1.143.1 Offset**

Register	Offset
SWREG141	234h

14.2.5.1.143.2 Diagram**14.2.5.1.143.3 Fields**

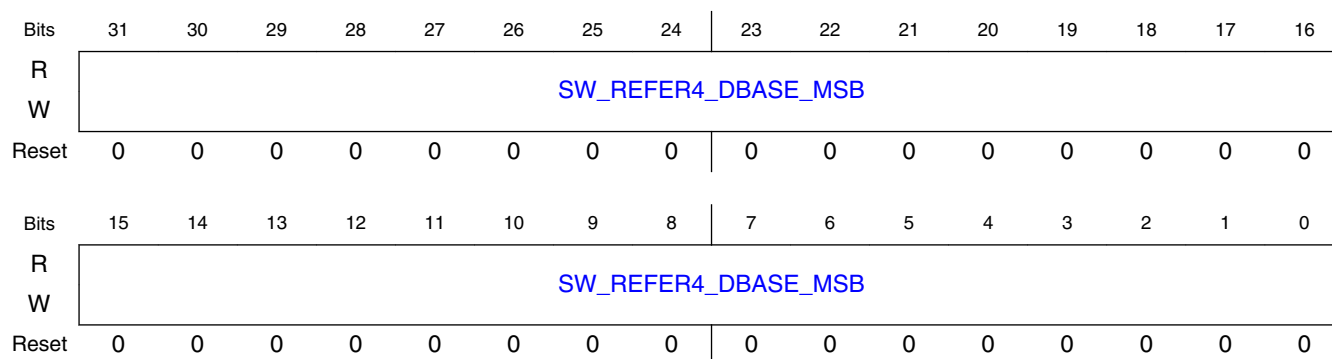
Field	Function
31-0 SW_REFER3_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 3

14.2.5.1.144 Base address MSB (bits 63:32) for reference direct mode MVS index 4 (SWREG142)

14.2.5.1.144.1 Offset

Register	Offset
SWREG142	238h

14.2.5.1.144.2 Diagram



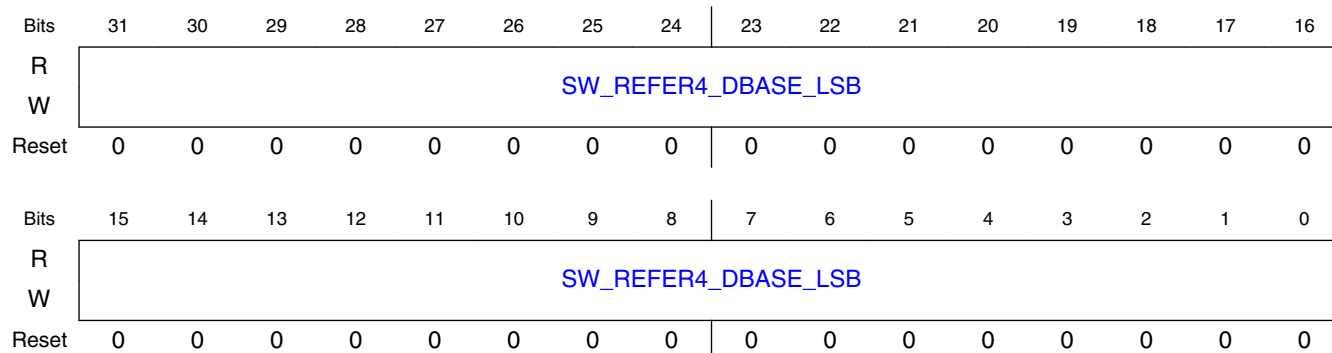
14.2.5.1.144.3 Fields

Field	Function
31-0 SW_REFER4_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 4

14.2.5.1.145 Base address LSB (bits 31:0) for reference direct mode MVS index 4 (SWREG143)

14.2.5.1.145.1 Offset

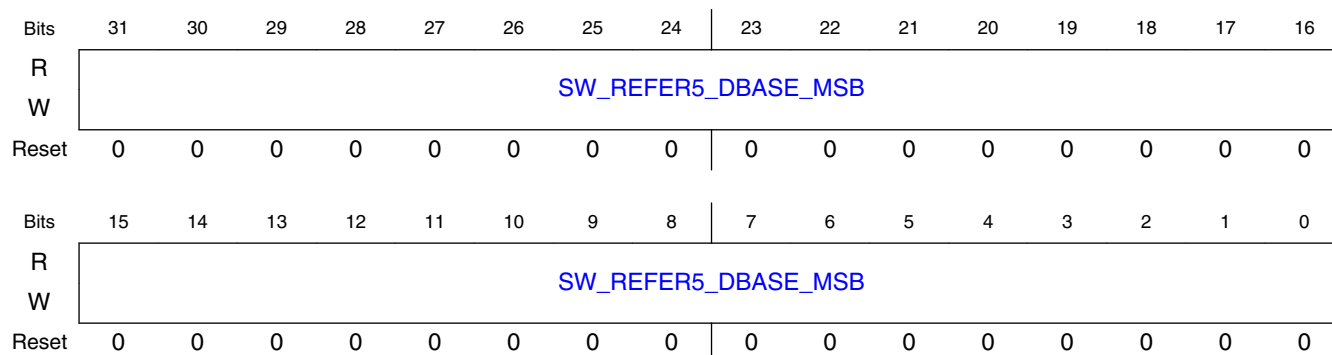
Register	Offset
SWREG143	23Ch

14.2.5.1.145.2 Diagram**14.2.5.1.145.3 Fields**

Field	Function
31-0 SW_REFER4_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 4

14.2.5.1.146 Base address MSB (bits 63:32) for reference direct mode MVS index 5 (SWREG144)**14.2.5.1.146.1 Offset**

Register	Offset
SWREG144	240h

14.2.5.1.146.2 Diagram

14.2.5.1.146.3 Fields

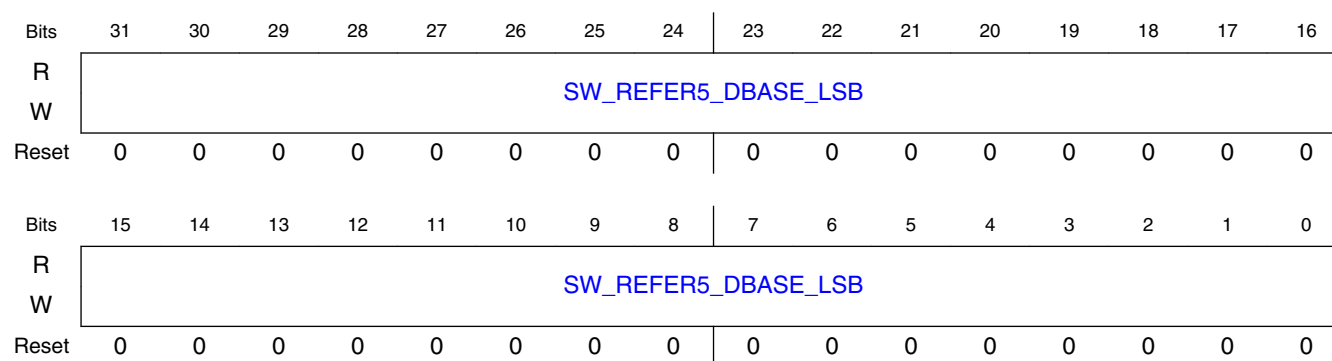
Field	Function
31-0 SW_REFER5_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 5

14.2.5.1.147 Base address LSB (bits 31:0) for reference direct mode MVS index 5 (SWREG145)

14.2.5.1.147.1 Offset

Register	Offset
SWREG145	244h

14.2.5.1.147.2 Diagram



14.2.5.1.147.3 Fields

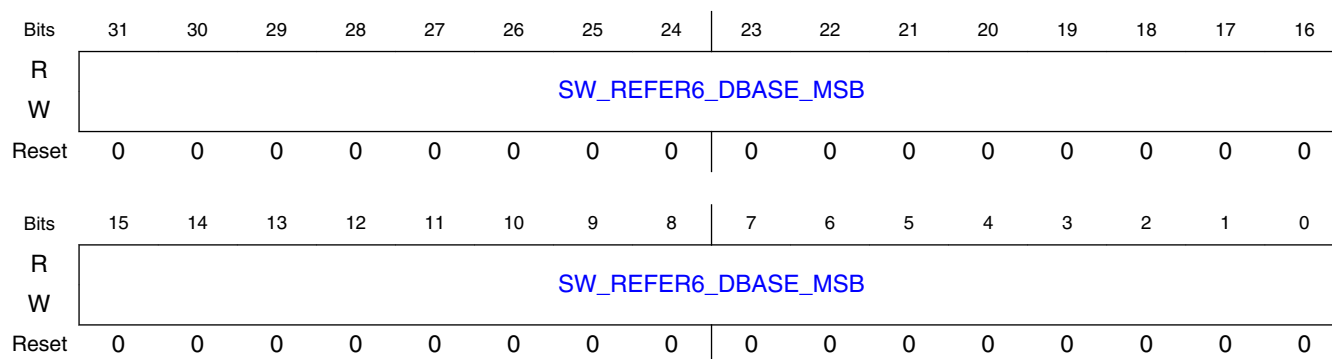
Field	Function
31-0 SW_REFER5_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 5

14.2.5.1.148 Base address MSB (bits 63:32) for reference direct mode MVS index 6 (SWREG146)

14.2.5.1.148.1 Offset

Register	Offset
SWREG146	248h

14.2.5.1.148.2 Diagram



14.2.5.1.148.3 Fields

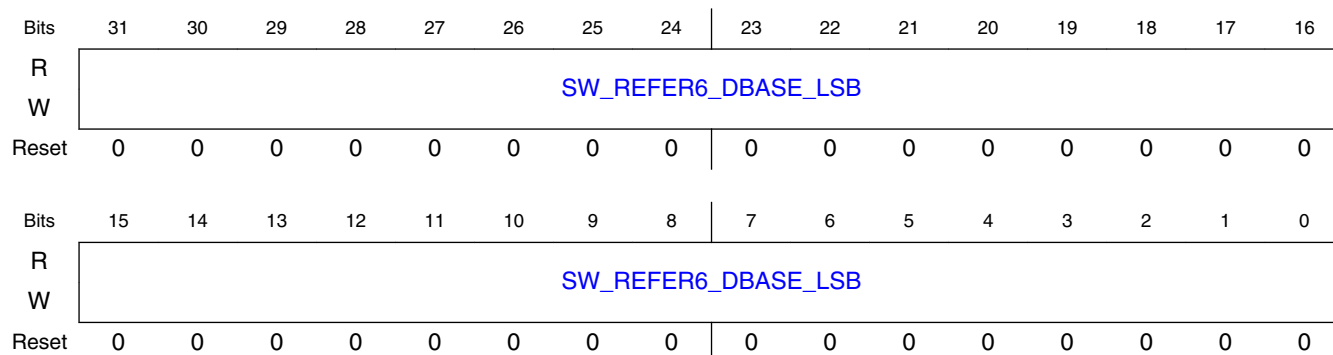
Field	Function
31-0 SW_REFER6_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 6

14.2.5.1.149 Base address LSB (bits 31:0) for reference direct mode MVS index 6 (SWREG147)

14.2.5.1.149.1 Offset

Register	Offset
SWREG147	24Ch

14.2.5.1.149.2 Diagram



14.2.5.1.149.3 Fields

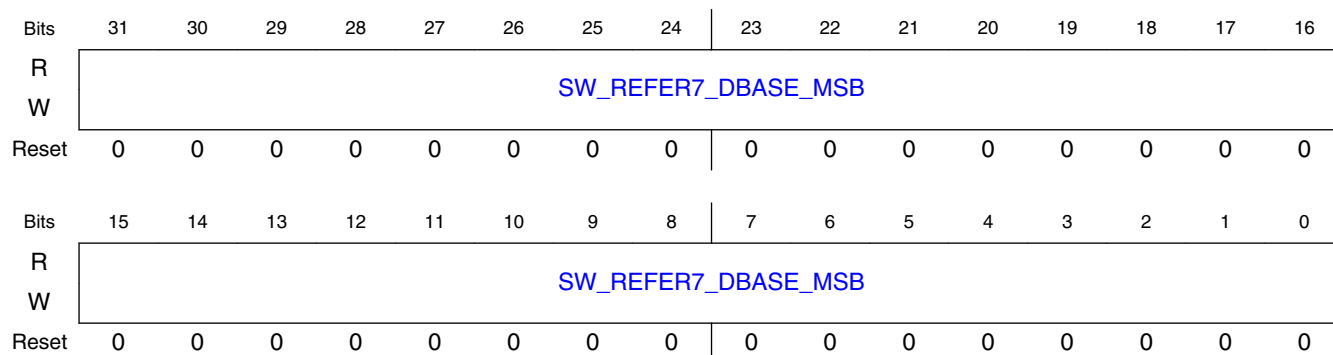
Field	Function
31-0 SW_REFER6_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 6

14.2.5.1.150 Base address MSB (bits 63:32) for reference direct mode MVS index 7 (SWREG148)

14.2.5.1.150.1 Offset

Register	Offset
SWREG148	250h

14.2.5.1.150.2 Diagram



14.2.5.1.150.3 Fields

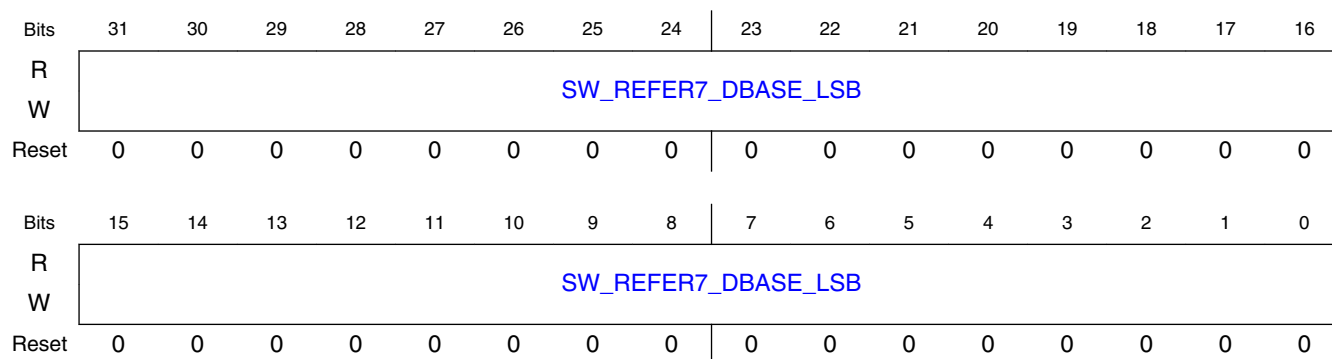
Field	Function
31-0 SW_REFER7_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 7

14.2.5.1.151 Base address LSB (bits 31:0) for reference direct mode MVS index 7 (SWREG149)

14.2.5.1.151.1 Offset

Register	Offset
SWREG149	254h

14.2.5.1.151.2 Diagram



14.2.5.1.151.3 Fields

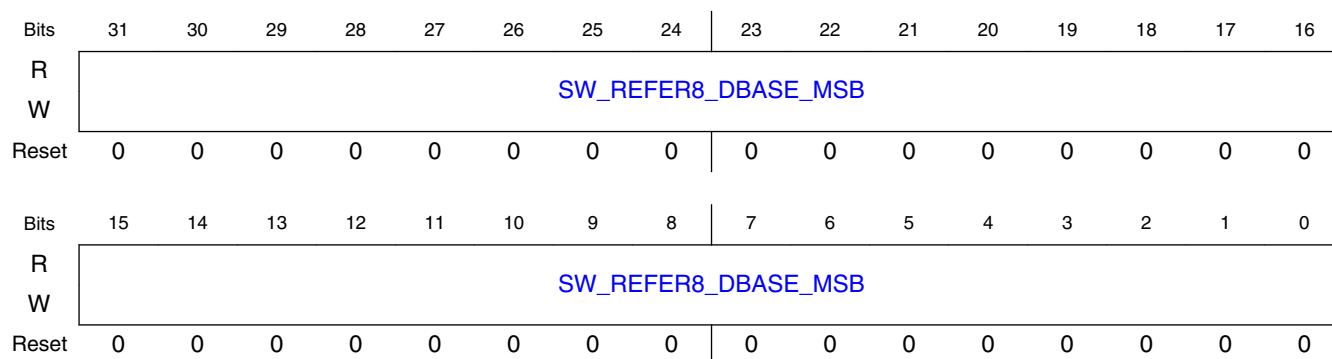
Field	Function
31-0 SW_REFER7_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 7

14.2.5.1.152 Base address MSB (bits 63:32) for reference direct mode MVS index 8 (SWREG150)

14.2.5.1.152.1 Offset

Register	Offset
SWREG150	258h

14.2.5.1.152.2 Diagram



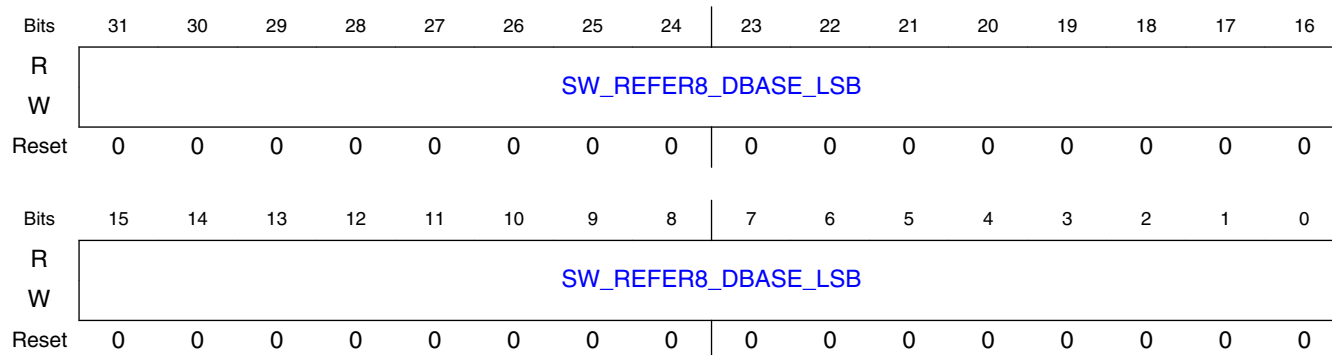
14.2.5.1.152.3 Fields

Field	Function
31-0 SW_REFER8_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 8

14.2.5.1.153 Base address LSB (bits 31:0) for reference direct mode MVS index 8 (SWREG151)

14.2.5.1.153.1 Offset

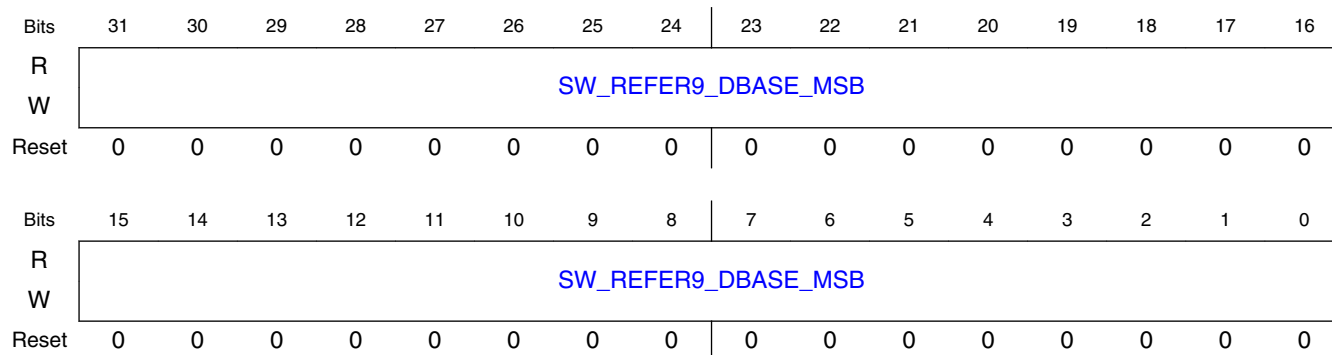
Register	Offset
SWREG151	25Ch

14.2.5.1.153.2 Diagram**14.2.5.1.153.3 Fields**

Field	Function
31-0 SW_REFER8_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode mode MVS index 8

14.2.5.1.154 Base address MSB (bits 63:32) for reference direct mode mode MVS index 9 (SWREG152)**14.2.5.1.154.1 Offset**

Register	Offset
SWREG152	260h

14.2.5.1.154.2 Diagram

14.2.5.1.154.3 Fields

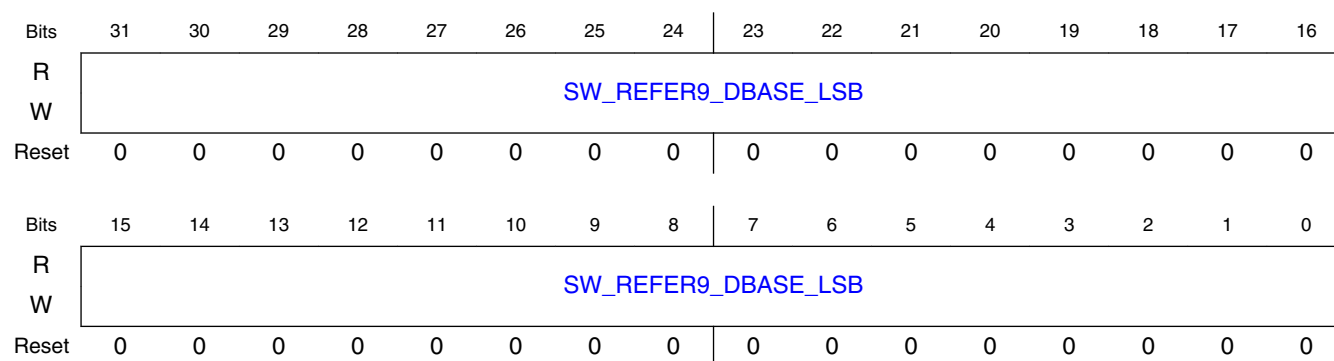
Field	Function
31-0 SW_REFER9_D BASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 9

14.2.5.1.155 Base address LSB (bits 31:0) for reference direct mode mode MVS index 9 (SWREG153)

14.2.5.1.155.1 Offset

Register	Offset
SWREG153	264h

14.2.5.1.155.2 Diagram



14.2.5.1.155.3 Fields

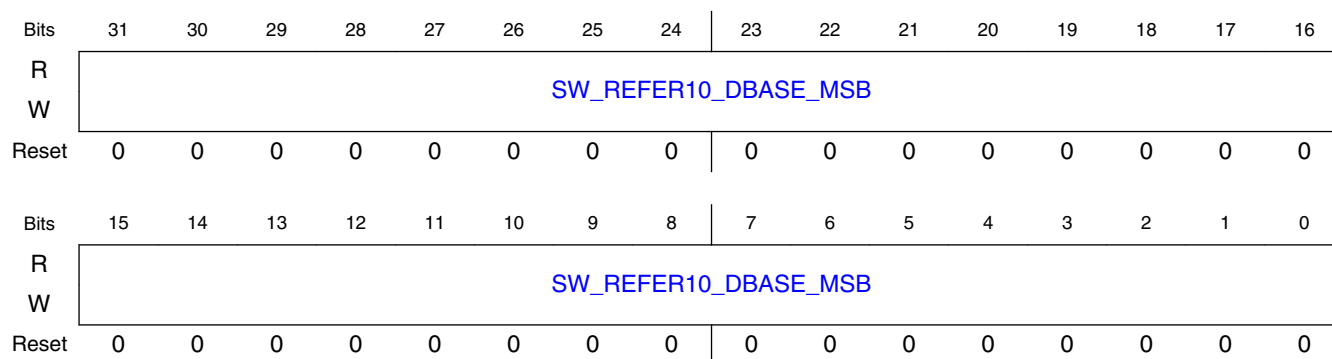
Field	Function
31-0 SW_REFER9_D BASE_LSB	Base address LSB (bits 31:0) for reference direct mode mode MVS index 9

14.2.5.1.156 Base address MSB (bits 63:32) for reference direct mode MVS index 10 (SWREG154)

14.2.5.1.156.1 Offset

Register	Offset
SWREG154	268h

14.2.5.1.156.2 Diagram



14.2.5.1.156.3 Fields

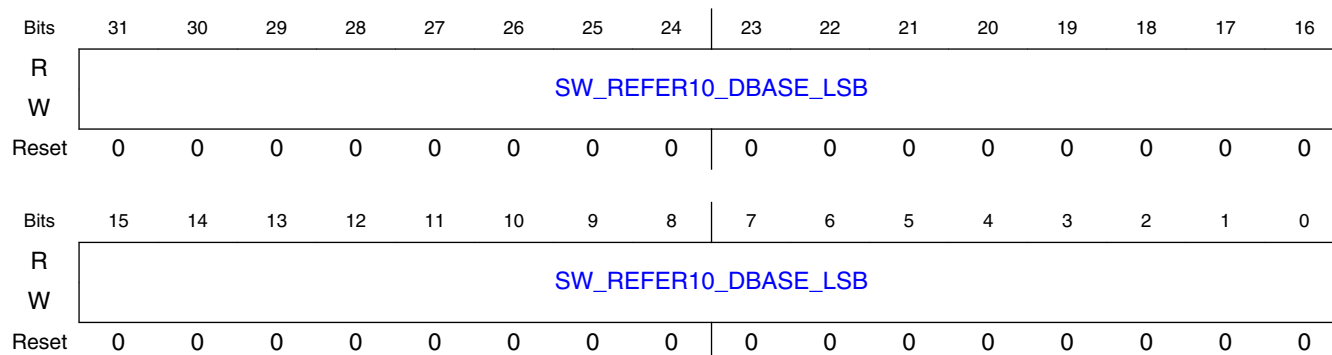
Field	Function
31-0 SW_REFER10_DBASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 10

14.2.5.1.157 Base address LSB (bits 31:0) for reference direct mode MVS index 10 (SWREG155)

14.2.5.1.157.1 Offset

Register	Offset
SWREG155	26Ch

14.2.5.1.157.2 Diagram



14.2.5.1.157.3 Fields

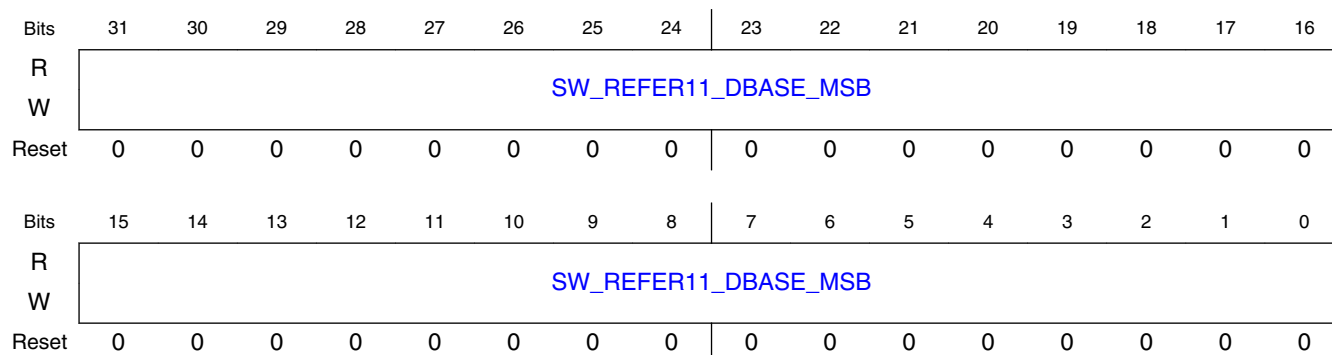
Field	Function
31-0 SW_REFER10_DBASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 10

14.2.5.1.158 Base address MSB (bits 63:32) for reference direct mode MVS index 11 (SWREG156)

14.2.5.1.158.1 Offset

Register	Offset
SWREG156	270h

14.2.5.1.158.2 Diagram

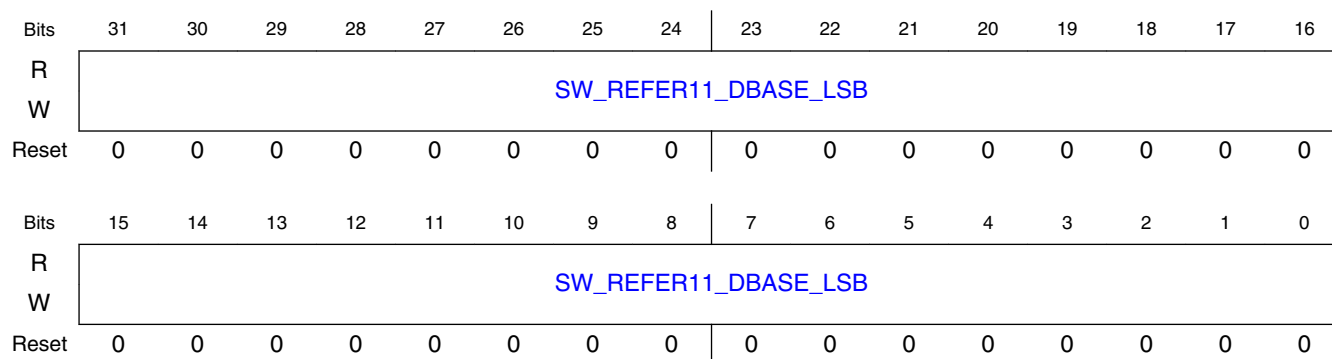


14.2.5.1.158.3 Fields

Field	Function
31-0 SW_REFER11_ DBASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 11

14.2.5.1.159 Base address LSB (bits 31:0) for reference direct mode MVS index 11 (SWREG157)**14.2.5.1.159.1 Offset**

Register	Offset
SWREG157	274h

14.2.5.1.159.2 Diagram**14.2.5.1.159.3 Fields**

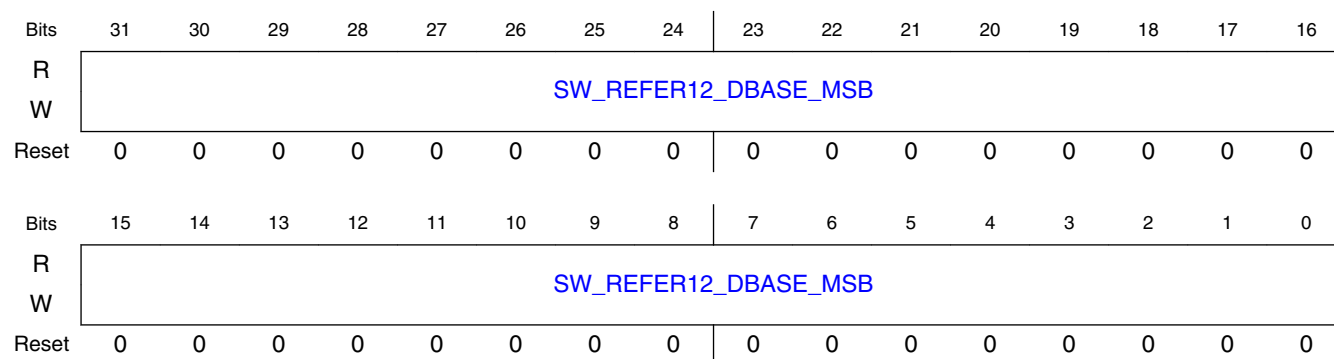
Field	Function
31-0 SW_REFER11_ DBASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 11

14.2.5.1.160 Base address MSB (bits 63:32) for reference direct mode MVS index 12 (SWREG158)

14.2.5.1.160.1 Offset

Register	Offset
SWREG158	278h

14.2.5.1.160.2 Diagram



14.2.5.1.160.3 Fields

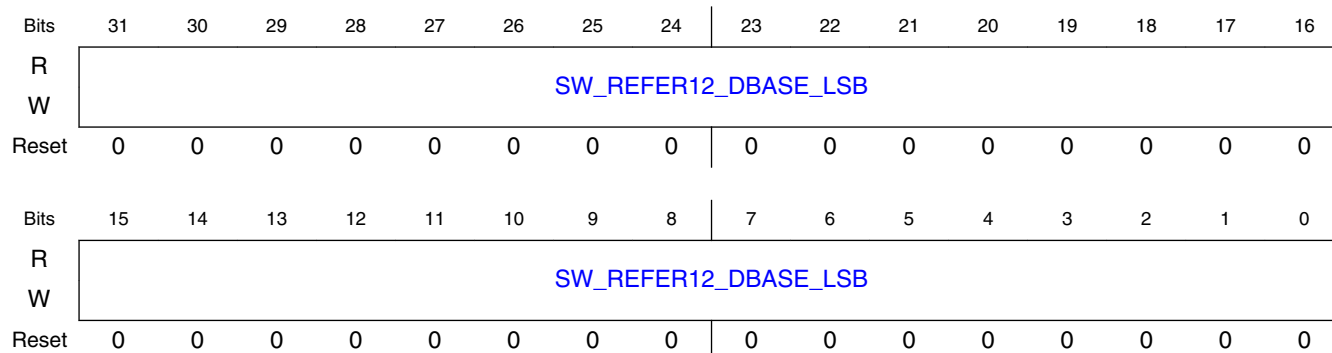
Field	Function
31-0 SW_REFER12_DBASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 12

14.2.5.1.161 Base address LSB (bits 31:0) for reference direct mode MVS index 12 (SWREG159)

14.2.5.1.161.1 Offset

Register	Offset
SWREG159	27Ch

14.2.5.1.161.2 Diagram



14.2.5.1.161.3 Fields

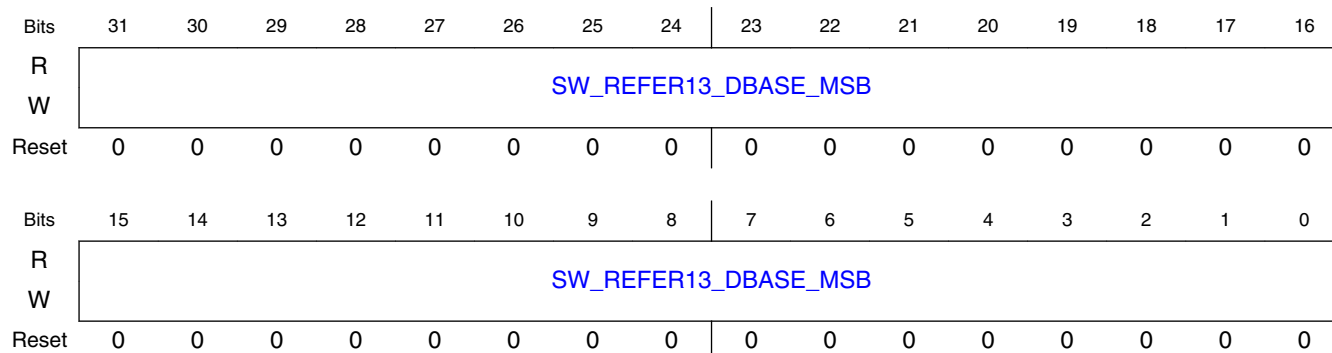
Field	Function
31-0 SW_REFER12_DBASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 12

14.2.5.1.162 Base address MSB (bits 63:32) for reference direct mode MVS index 13 (SWREG160)

14.2.5.1.162.1 Offset

Register	Offset
SWREG160	280h

14.2.5.1.162.2 Diagram



14.2.5.1.162.3 Fields

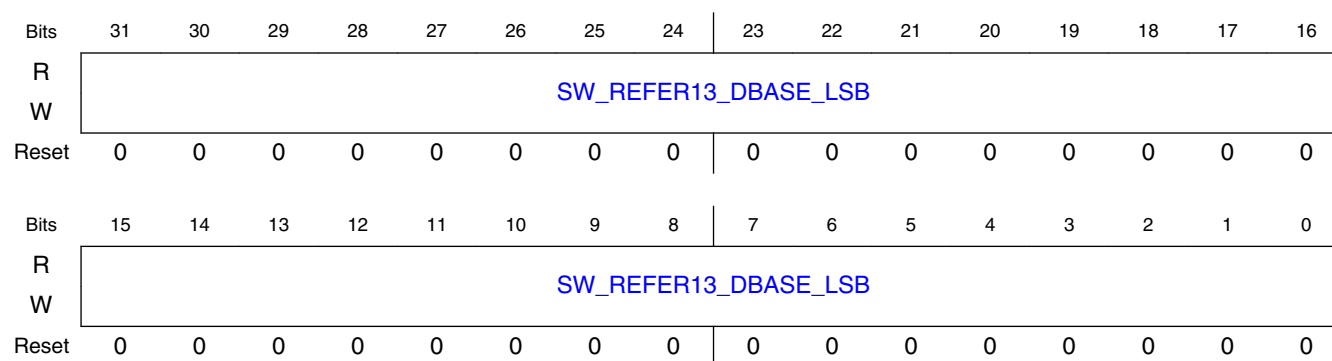
Field	Function
31-0 SW_REFER13_ DBASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 13

14.2.5.1.163 Base address LSB (bits 31:0) for reference direct mode MVS index 13 (SWREG161)

14.2.5.1.163.1 Offset

Register	Offset
SWREG161	284h

14.2.5.1.163.2 Diagram



14.2.5.1.163.3 Fields

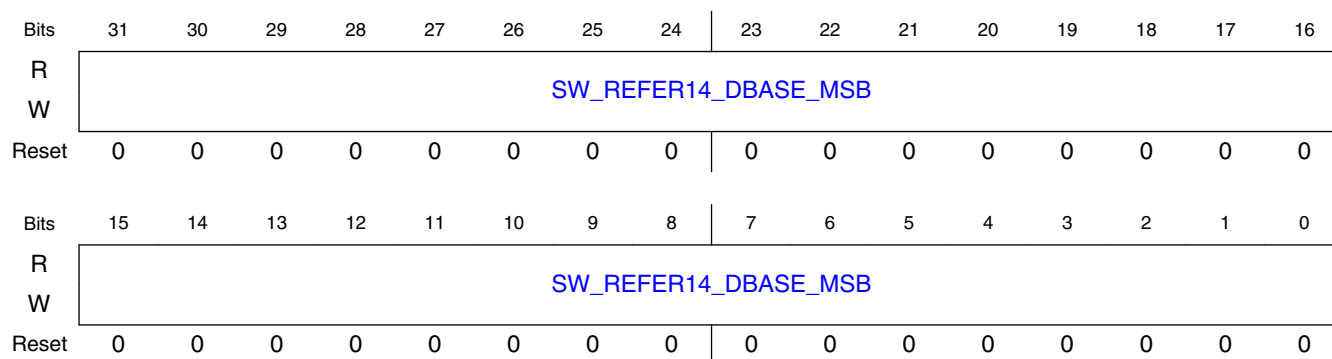
Field	Function
31-0 SW_REFER13_ DBASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 13

14.2.5.1.164 Base address MSB (bits 63:32) for reference direct mode MVS index 14 (SWREG162)

14.2.5.1.164.1 Offset

Register	Offset
SWREG162	288h

14.2.5.1.164.2 Diagram



14.2.5.1.164.3 Fields

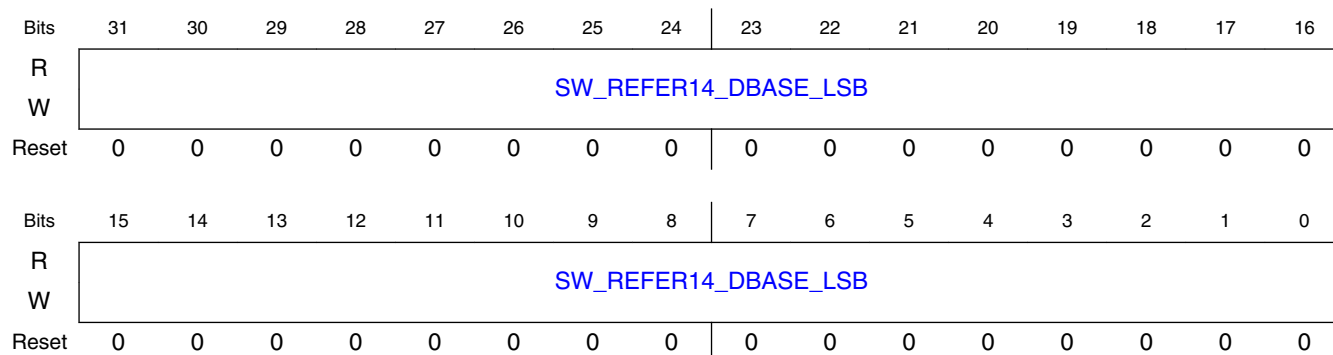
Field	Function
31-0 SW_REFER14_DBASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 14

14.2.5.1.165 Base address LSB (bits 31:0) for reference direct mode MVS index 14 (SWREG163)

14.2.5.1.165.1 Offset

Register	Offset
SWREG163	28Ch

14.2.5.1.165.2 Diagram



14.2.5.1.165.3 Fields

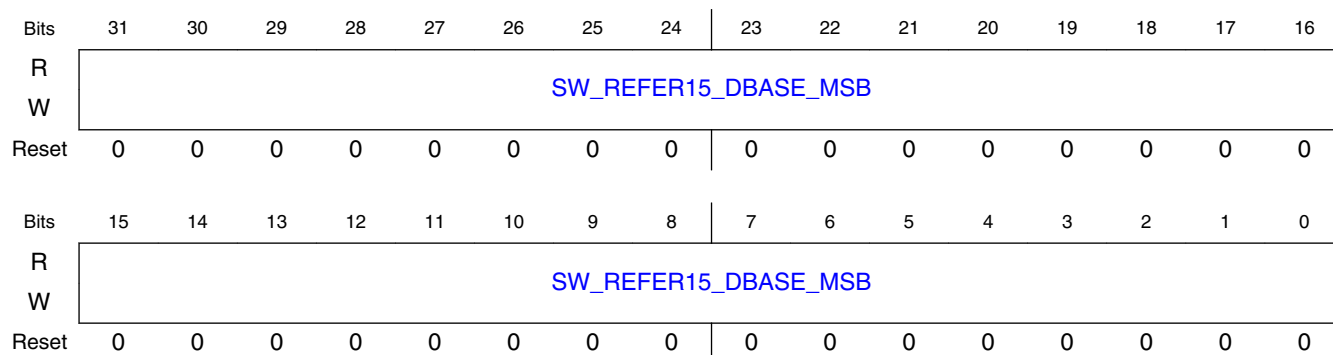
Field	Function
31-0 SW_REFER14_DBASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 14

14.2.5.1.166 Base address MSB (bits 63:32) for reference direct mode MVS index 15 (SWREG164)

14.2.5.1.166.1 Offset

Register	Offset
SWREG164	290h

14.2.5.1.166.2 Diagram



14.2.5.1.166.3 Fields

Field	Function
31-0 SW_REFER15_ DBASE_MSB	Base address MSB (bits 63:32) for reference direct mode MVS index 15

14.2.5.1.167 Base address LSB (bits 31:0) for reference direct mode MVS index 15 (SWREG165)**14.2.5.1.167.1 Offset**

Register	Offset
SWREG165	294h

14.2.5.1.167.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER15_DBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER15_DBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.167.3 Fields

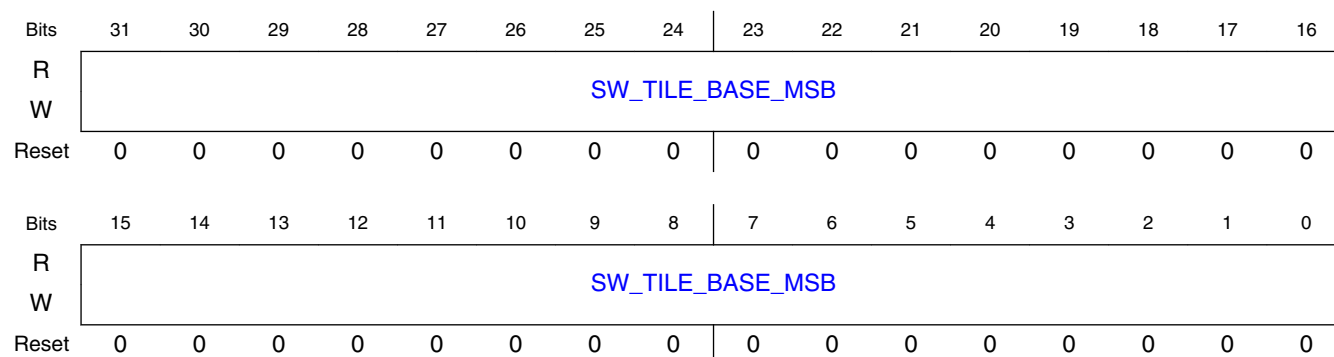
Field	Function
31-0 SW_REFER15_ DBASE_LSB	Base address LSB (bits 31:0) for reference direct mode MVS index 15

14.2.5.1.168 Base address MSB (bits 63:32) for tile sizes (SWREG166)

14.2.5.1.168.1 Offset

Register	Offset
SWREG166	298h

14.2.5.1.168.2 Diagram



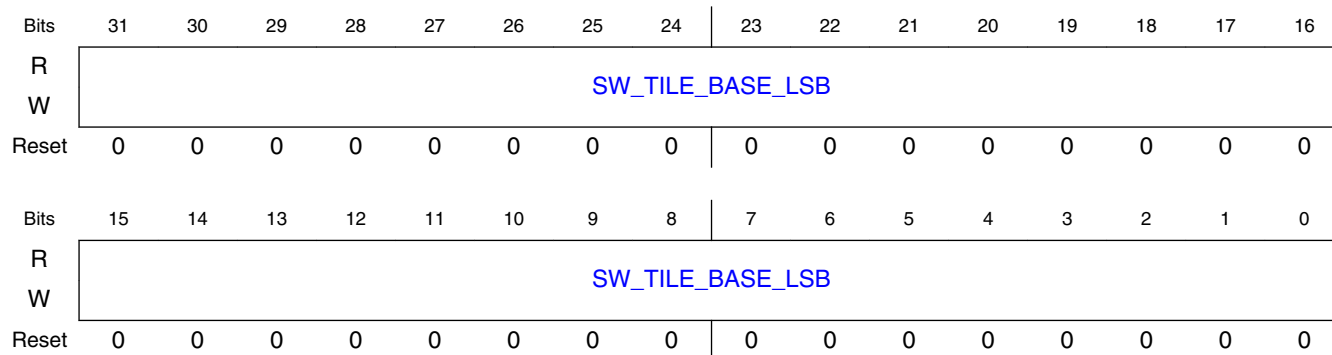
14.2.5.1.168.3 Fields

Field	Function
31-0 SW_TILE_BASE_MSB	Base address MSB (bits 63:32) for tile sizes

14.2.5.1.169 Base address LSB (bits 31:0) for tile sizes (SWREG167)

14.2.5.1.169.1 Offset

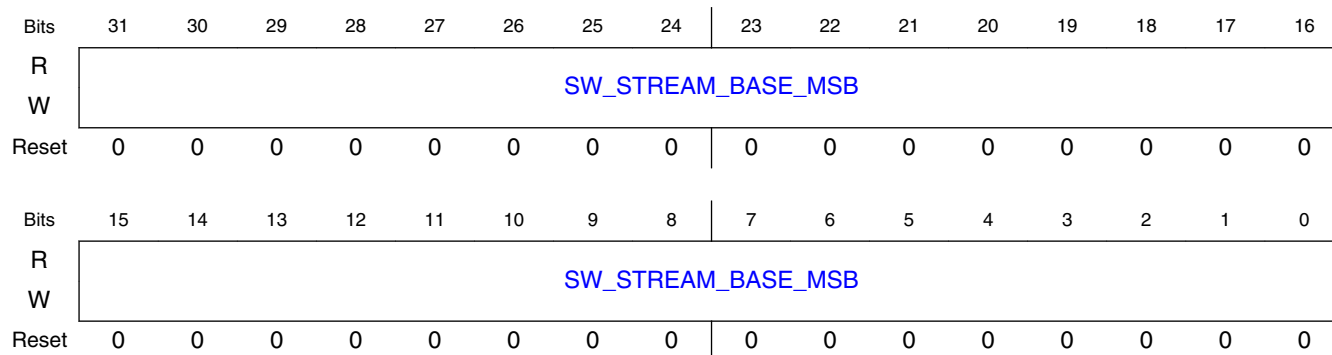
Register	Offset
SWREG167	29Ch

14.2.5.1.169.2 Diagram**14.2.5.1.169.3 Fields**

Field	Function
31-0 SW_TILE_BASE_LSB	Base address LSB (bits 31:0) for tile sizes

14.2.5.1.170 Base address MSB (bits 63:32) for / stream start address/ decoded end addr register (SWREG168)**14.2.5.1.170.1 Offset**

Register	Offset
SWREG168	2A0h

14.2.5.1.170.2 Diagram

14.2.5.1.170.3 Fields

Field	Function
31-0 SW_STREAM_BASE_MSB	Base address MSB (bits 63:32) for / stream start address/decoded end addr register

14.2.5.1.171 Base address LSB (bits 31:0) for / stream start address/decoded end addr register (SWREG169)

14.2.5.1.171.1 Offset

Register	Offset
SWREG169	2A4h

14.2.5.1.171.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_STREAM_BASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_STREAM_BASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.171.3 Fields

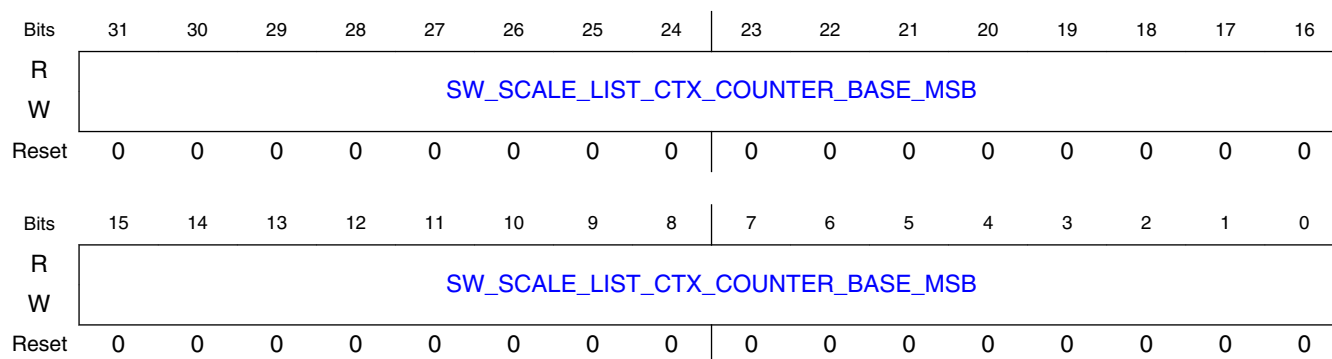
Field	Function
31-0 SW_STREAM_BASE_LSB	Base address LSB (bits 31:0) for / stream start address/decoded end addr register

14.2.5.1.172 Base address MSB (bits 63:32) for scaling lists / VP9 CTX counter values (SWREG170)

14.2.5.1.172.1 Offset

Register	Offset
SWREG170	2A8h

14.2.5.1.172.2 Diagram



14.2.5.1.172.3 Fields

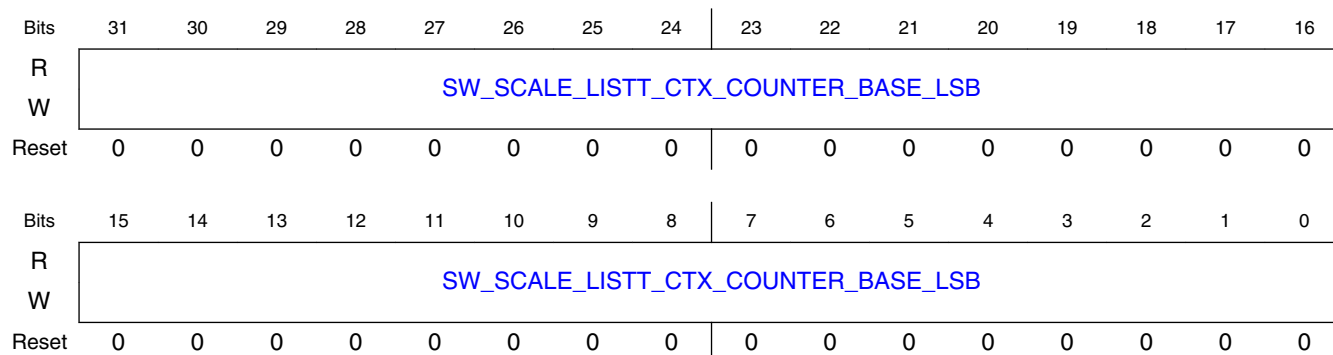
Field	Function
31-0 SW_SCALE_LIST_CTX_COUNTER_BASE_MSB	HEVC: Base address MSB (bits 63:32) for scaling lists VP9: CTX counter values

14.2.5.1.173 Base address LSB (bits 31:0) for scaling lists / VP9 CTX counter values (SWREG171)

14.2.5.1.173.1 Offset

Register	Offset
SWREG171	2ACh

14.2.5.1.173.2 Diagram



14.2.5.1.173.3 Fields

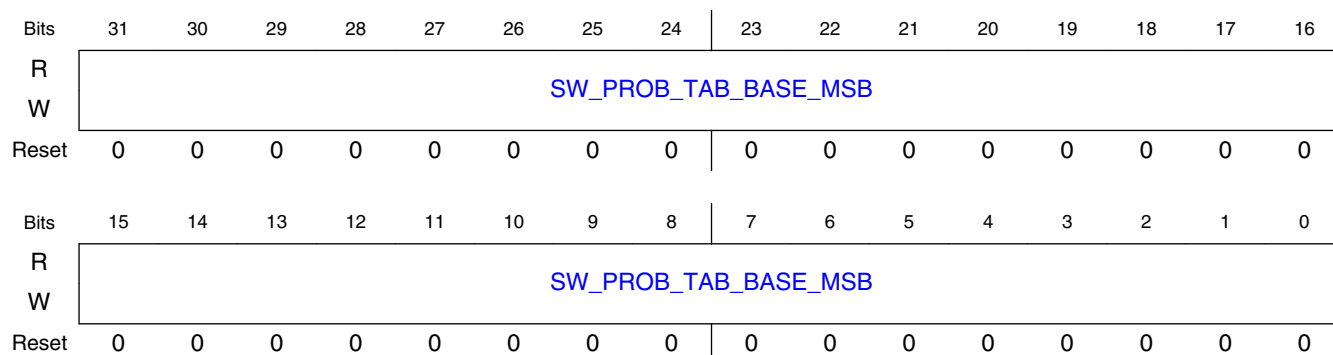
Field	Function
31-0 SW_SCALE_LISTT_CTX_COUNTER_BASE_LSB	HEVC: Base address LSB (bits 31:0) for scaling lists VP9: CTX counter values

14.2.5.1.174 Base address MSB (bits 63:32) for stream propability tables (SWREG172)

14.2.5.1.174.1 Offset

Register	Offset
SWREG172	2B0h

14.2.5.1.174.2 Diagram



14.2.5.1.174.3 Fields

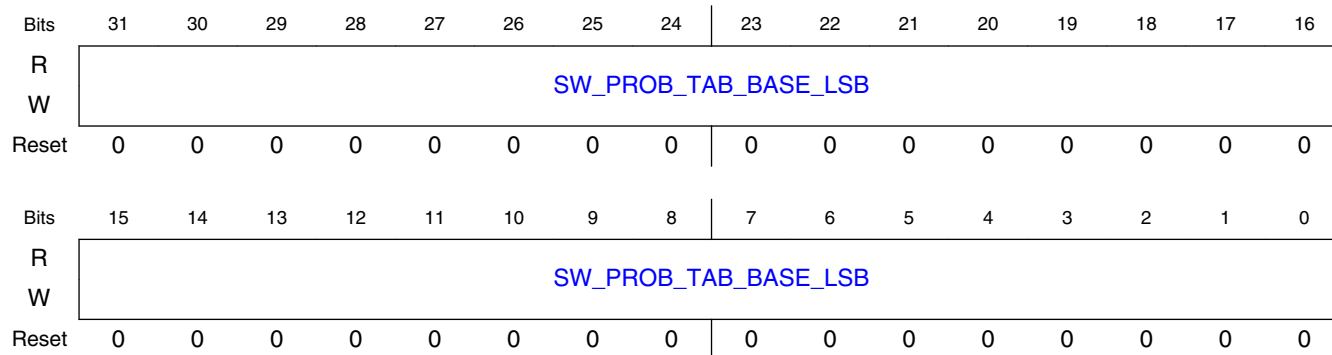
Field	Function
31-0 SW_PROB_TA B_BASE_MSB	Base address MSB (bits 63:32) for stream propability tables

14.2.5.1.175 Base address LSB (bits 31:0) for stream propability tables (SWREG173)

14.2.5.1.175.1 Offset

Register	Offset
SWREG173	2B4h

14.2.5.1.175.2 Diagram



14.2.5.1.175.3 Fields

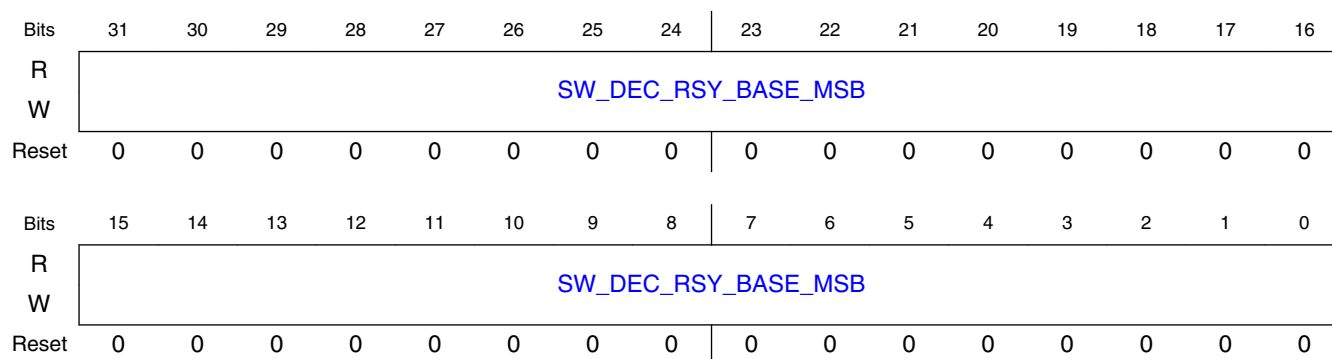
Field	Function
31-0 SW_PROB_TA B_BASE_LSB	Base address LSB (bits 31:0) for stream propability tables

14.2.5.1.176 Base address MSB (bits 63:32) for decoder output raster scan Y picture (SWREG174)

14.2.5.1.176.1 Offset

Register	Offset
SWREG174	2B8h

14.2.5.1.176.2 Diagram



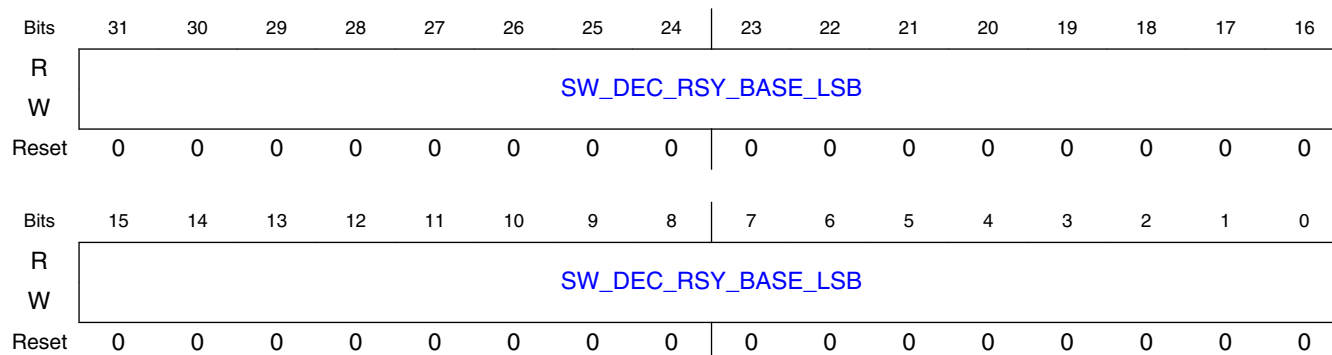
14.2.5.1.176.3 Fields

Field	Function
31-0 SW_DEC_RSY_BASE_MSB	Base address MSB (bits 63:32) for decoder output raster scan Y picture

14.2.5.1.177 Base address LSB (bits 31:0) for decoder output raster scan Y picture (SWREG175)

14.2.5.1.177.1 Offset

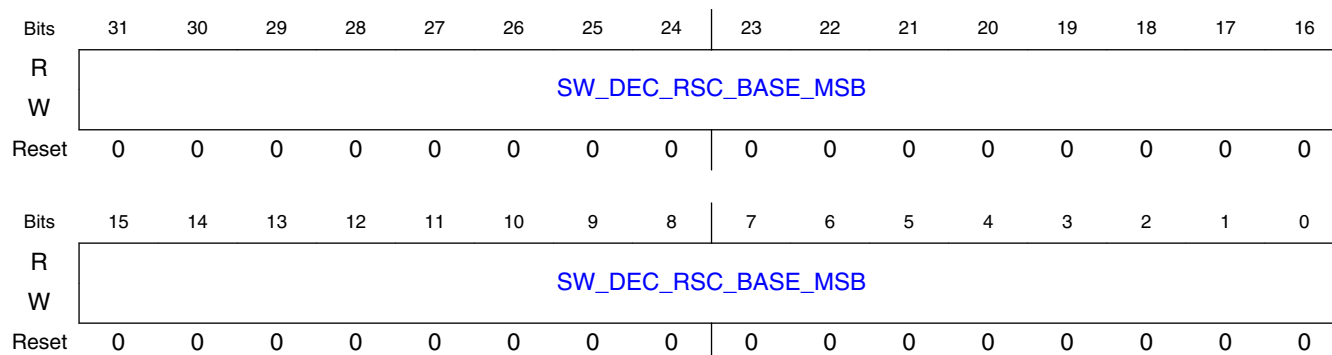
Register	Offset
SWREG175	2BCh

14.2.5.1.177.2 Diagram**14.2.5.1.177.3 Fields**

Field	Function
31-0 SW_DEC_RSY_BASE_LSB	Base address LSB (bits 31:0) for decoder output raster scan Y picture

14.2.5.1.178 Base address MSB (bits 63:32) for decoder output raster scan C picture (SWREG176)**14.2.5.1.178.1 Offset**

Register	Offset
SWREG176	2C0h

14.2.5.1.178.2 Diagram

14.2.5.1.178.3 Fields

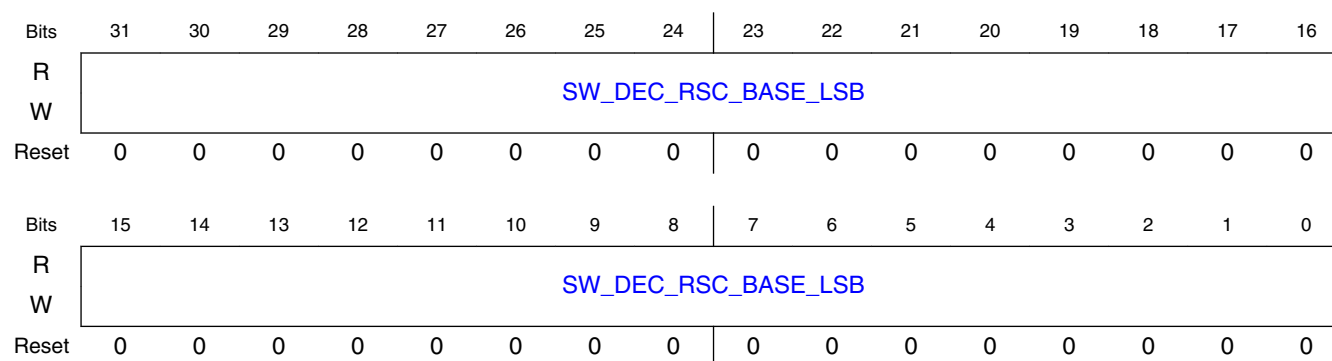
Field	Function
31-0 SW_DEC_RSC_BASE_MSB	Base address MSB (bits 63:32) for decoder output raster scan C picture

14.2.5.1.179 Base address LSB (bits 31:0) for decoder output raster scan C picture (SWREG177)

14.2.5.1.179.1 Offset

Register	Offset
SWREG177	2C4h

14.2.5.1.179.2 Diagram



14.2.5.1.179.3 Fields

Field	Function
31-0 SW_DEC_RSC_BASE_LSB	Base address LSB (bits 31:0) for decoder output raster scan C picture

14.2.5.1.180 Base address MSB (bits 63:32) for tile border coefficients of filter (SWREG178)

14.2.5.1.180.1 Offset

Register	Offset
SWREG178	2C8h

14.2.5.1.180.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_DEC_VERT_FILT_BASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_DEC_VERT_FILT_BASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.180.3 Fields

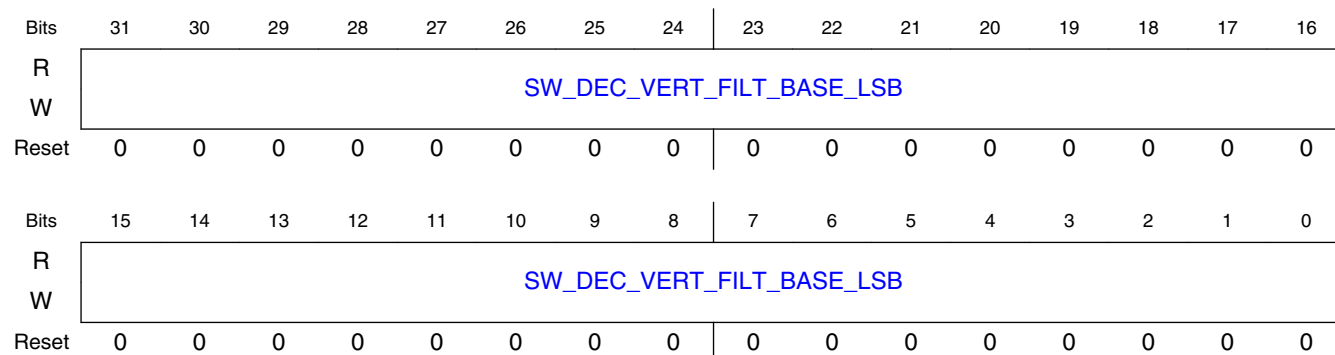
Field	Function
31-0 SW_DEC_VERT_FILT_BASE_MSB	Base address MSB to store/read filtering coefficients of current picture at tile border.

14.2.5.1.181 Base address LSB (bits 31:0) for tile border coefficients of filter (SWREG179)

14.2.5.1.181.1 Offset

Register	Offset
SWREG179	2CCh

14.2.5.1.181.2 Diagram



14.2.5.1.181.3 Fields

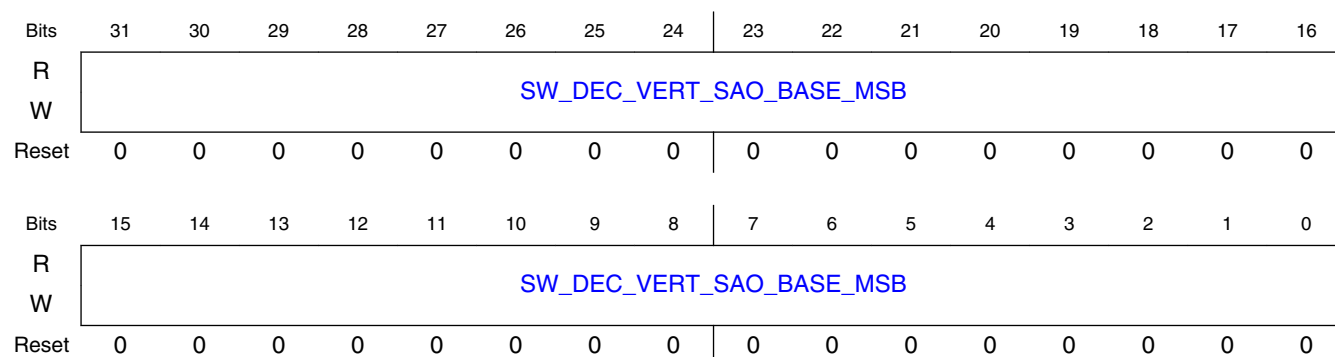
Field	Function
31-0 SW_DEC_VERT_FILT_BASE_LSB	Base address LSB to store/read filtering coefficients of current picture at tile border.

14.2.5.1.182 Base address MSB (bits 63:32) for tile border coefficients of sao (SWREG180)

14.2.5.1.182.1 Offset

Register	Offset
SWREG180	2D0h

14.2.5.1.182.2 Diagram



14.2.5.1.182.3 Fields

Field	Function
31-0 SW_DEC_VER T_SAO_BASE_ MSB	Base address MSB to store/read sao coefficients of current picture at tile border.

14.2.5.1.183 Base address LSB (bits 31:0) for tile border coefficients of sao (SWREG181)

14.2.5.1.183.1 Offset

Register	Offset
SWREG181	2D4h

14.2.5.1.183.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_DEC_VERT_SAO_BASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_DEC_VERT_SAO_BASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.183.3 Fields

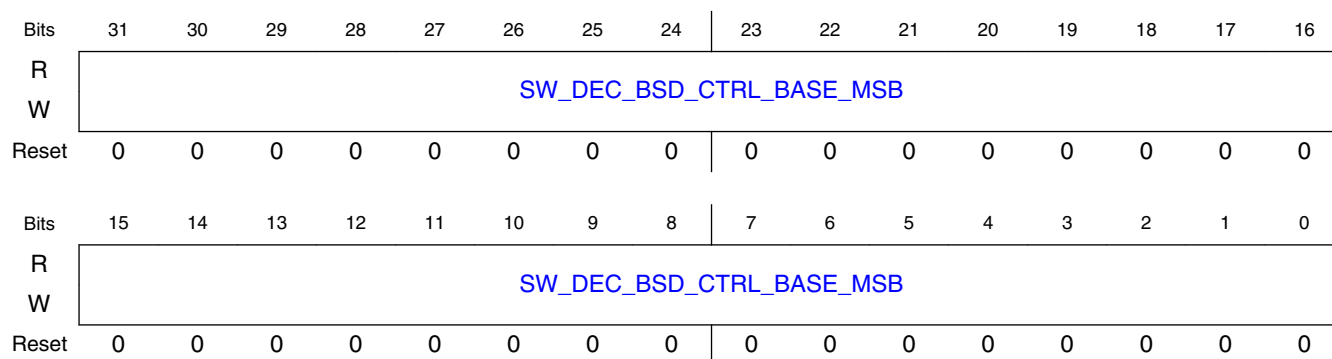
Field	Function
31-0 SW_DEC_VER T_SAO_BASE_ LSB	Base address LSB to store/read sao coefficients of current picture at tile border.

14.2.5.1.184 Base address MSB (bits 63:32) for tile border bsd control data (SWREG182)

14.2.5.1.184.1 Offset

Register	Offset
SWREG182	2D8h

14.2.5.1.184.2 Diagram



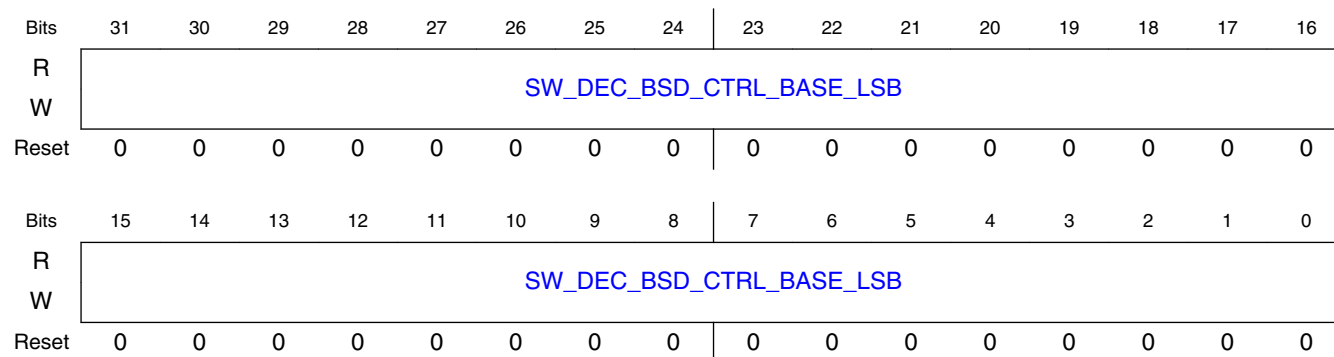
14.2.5.1.184.3 Fields

Field	Function
31-0 SW_DEC_BSD_CTRL_BASE_MSB	Base address MSB to store/read BSD control data of current picture at tile border.

14.2.5.1.185 Base address LSB (bits 31:0) for tile border bsd control data (SWREG183)

14.2.5.1.185.1 Offset

Register	Offset
SWREG183	2DCh

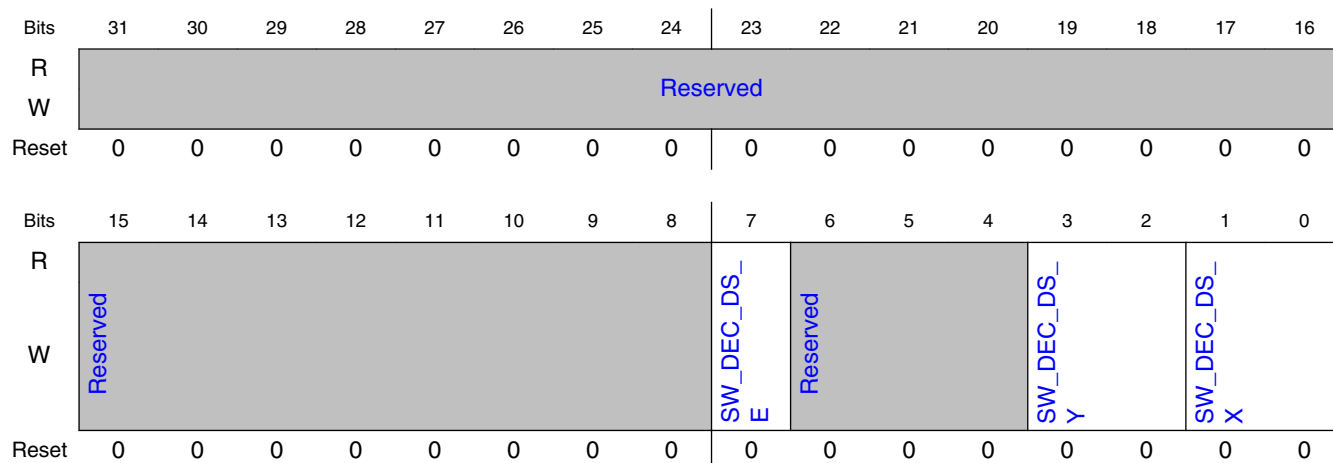
14.2.5.1.185.2 Diagram**14.2.5.1.185.3 Fields**

Field	Function
31-0 SW_DEC_BSD_CTRL_BASE_LSB	Base address LSB to store/read BSD control data of current picture at tile border.

14.2.5.1.186 Raster scan down scale control register MSM (SWREG184)**14.2.5.1.186.1 Offset**

Register	Offset
SWREG184	2E0h

14.2.5.1.186.2 Diagram



14.2.5.1.186.3 Fields

Field	Function
31-8 —	Reserved.
7 SW_DEC_DS_E	Raster scan down scale enable 0b - Disable 1b - Enable
6-4 —	Reserved.
3-2 SW_DEC_DS_Y	Y coordinate down scale times for raster scan output picture data 00b - 1/2 01b - 1/4 10b - 1/8
1-0 SW_DEC_DS_X	X coordinate down scale times for raster scan output picture data 00b - 1/2 01b - 1/4 10b - 1/8

14.2.5.1.187 Base address MSB (bits 63:32) for decoder output raster scan down scale Y picture (SWREG185)

14.2.5.1.187.1 Offset

Register	Offset
SWREG185	2E4h

14.2.5.1.187.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_DEC_DSY_BASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_DEC_DSY_BASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.187.3 Fields

Field	Function
31-0 SW_DEC_DSY_BASE_MSB	Base address MSB (bits 63:32) for decoder output raster scan down scale Y picture

14.2.5.1.188 Base address LSB (bits 31:0) for decoder output raster scan down scale Y picture (SWREG186)

14.2.5.1.188.1 Offset

Register	Offset
SWREG186	2E8h

14.2.5.1.188.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_DEC_DSY_BASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_DEC_DSY_BASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.188.3 Fields

Field	Function
31-0 SW_DEC_DSY_ BASE_LSB	Base address LSB (bits 31:0) for decoder output raster scan down scale Y picture

14.2.5.1.189 Base address MSB (bits 63:32) for decoder output raster scan down scale C picture (SWREG187)

14.2.5.1.189.1 Offset

Register	Offset
SWREG187	2ECh

14.2.5.1.189.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_DEC_DSC_BASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_DEC_DSC_BASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.189.3 Fields

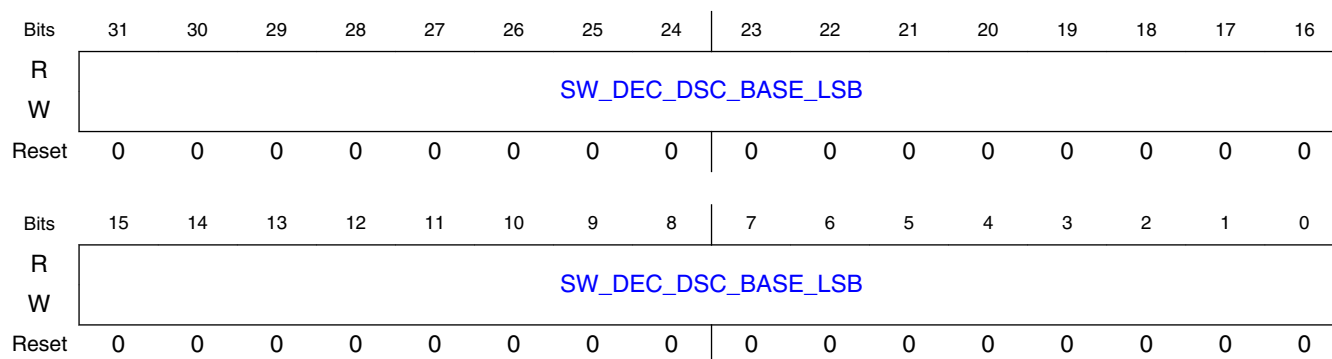
Field	Function
31-0 SW_DEC_DSC_ _BASE_MSB	Base address MSB (bits 63:32) for decoder output raster scan down scale C picture

14.2.5.1.190 Base address LSB (bits 31:0) for decoder output raster scan down scale C picture (SWREG188)

14.2.5.1.190.1 Offset

Register	Offset
SWREG188	2F0h

14.2.5.1.190.2 Diagram



14.2.5.1.190.3 Fields

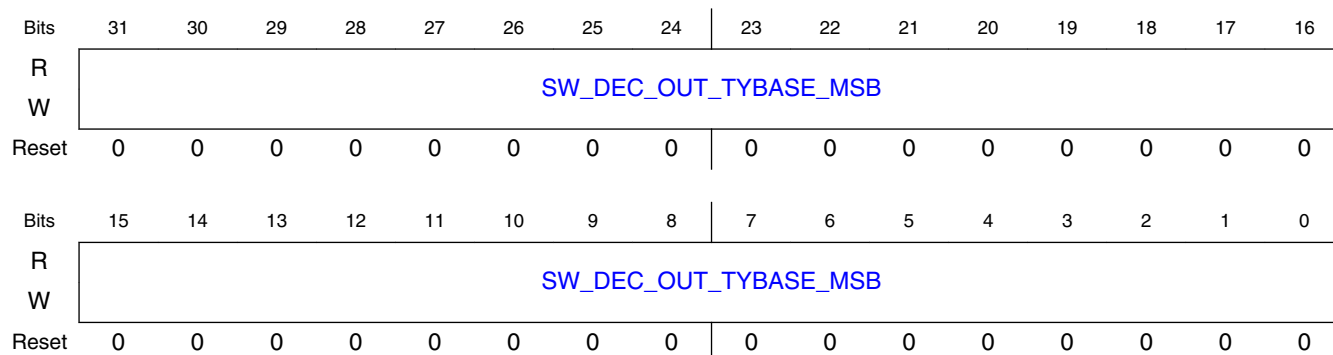
Field	Function
31-0 SW_DEC_DSC_BASE_LSB	Base address LSB (bits 31:0) for decoder output raster scan down scale C picture

14.2.5.1.191 Base address MSB (bits 63:32) for decoder output compress luminance table (SWREG189)

14.2.5.1.191.1 Offset

Register	Offset
SWREG189	2F4h

14.2.5.1.191.2 Diagram



14.2.5.1.191.3 Fields

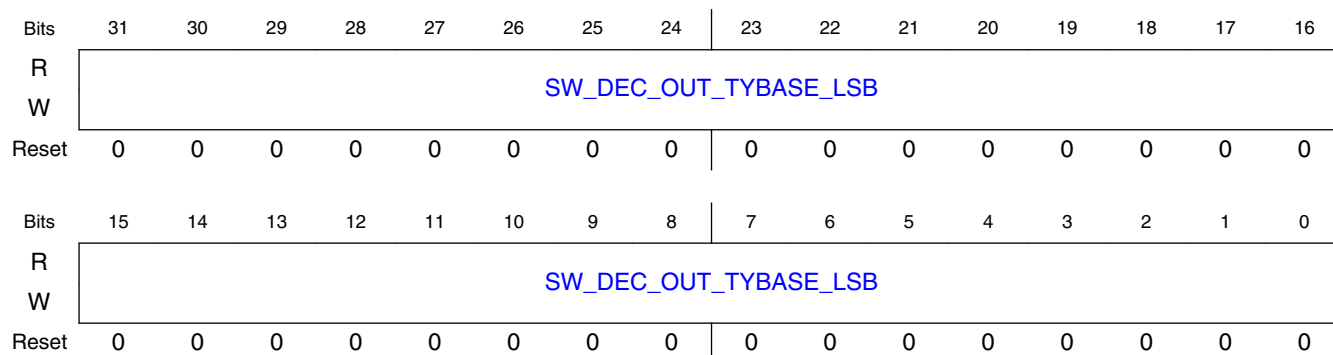
Field	Function
31-0 SW_DEC_OUT_TYBASE_MSB	Base address MSB (bits 63:32) for decoder output compress luminance table

14.2.5.1.192 Base address LSB (bits 31:0) for decoder output compress luminance table (SWREG190)

14.2.5.1.192.1 Offset

Register	Offset
SWREG190	2F8h

14.2.5.1.192.2 Diagram



14.2.5.1.192.3 Fields

Field	Function
31-0 SW_DEC_OUT_TYBASE_LSB	Base address LSB (bits 31:0) for decoder output compress luminance table

14.2.5.1.193 Base address MSB (bits 63:32) for reference compress luminance table index 0 (SWREG191)

14.2.5.1.193.1 Offset

Register	Offset
SWREG191	2FCh

14.2.5.1.193.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER0_TYBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER0_TYBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.193.3 Fields

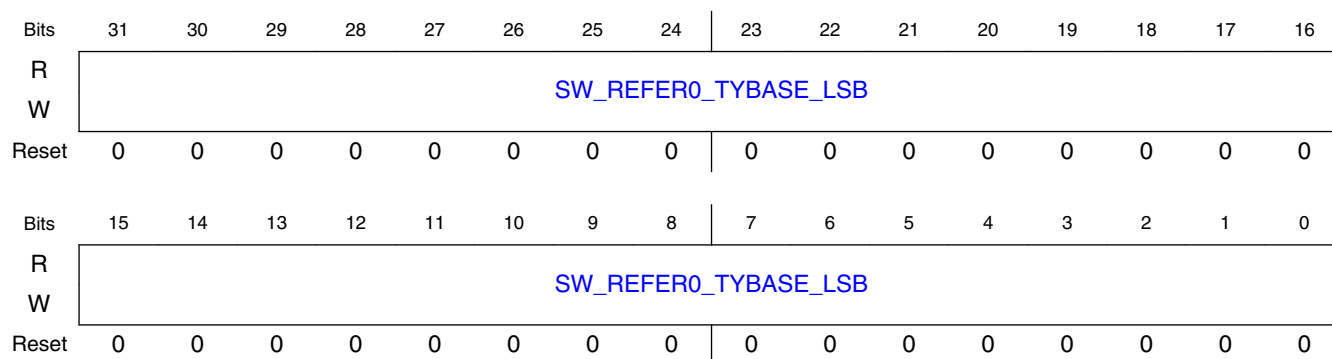
Field	Function
31-0 SW_REFER0_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 0

14.2.5.1.194 Base address LSB (bits 31:0) for reference compress luminance table index 0 (SWREG192)

14.2.5.1.194.1 Offset

Register	Offset
SWREG192	300h

14.2.5.1.194.2 Diagram



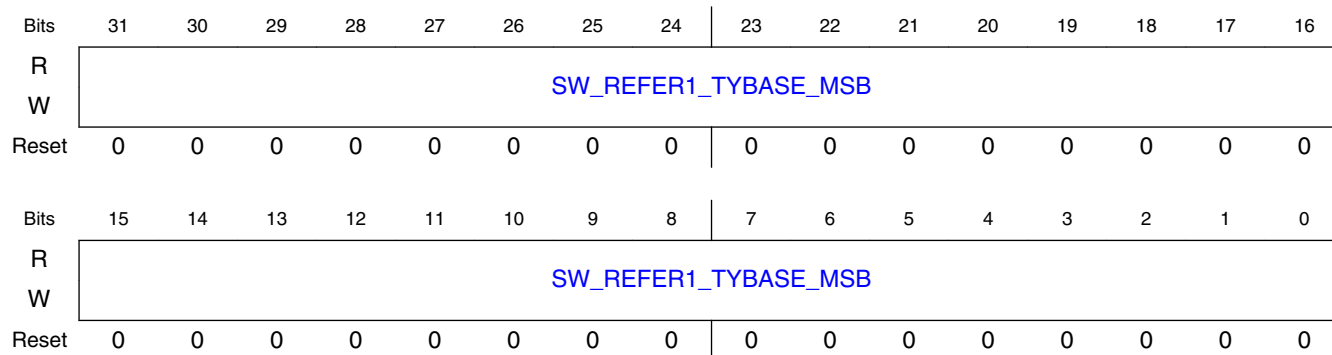
14.2.5.1.194.3 Fields

Field	Function
31-0 SW_REFER0_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 0

14.2.5.1.195 Base address MSB (bits 63:32) for reference compress luminance table index 1 (SWREG193)

14.2.5.1.195.1 Offset

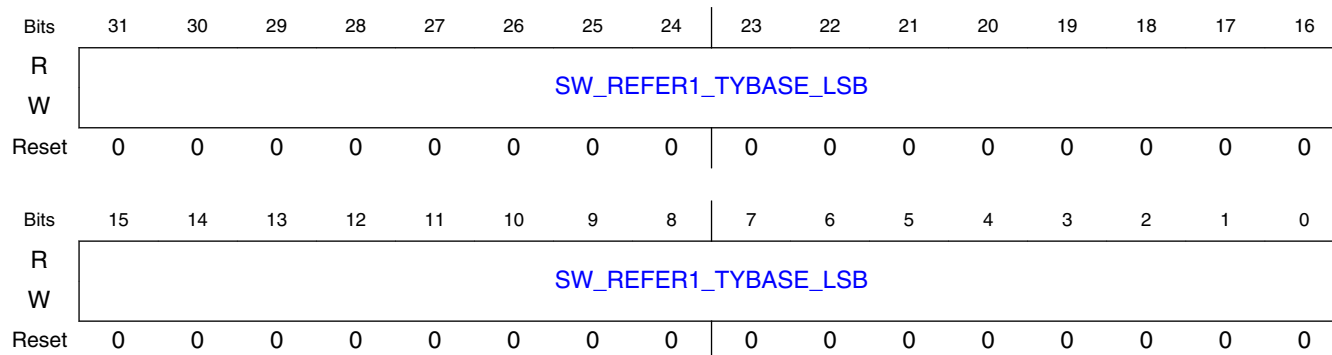
Register	Offset
SWREG193	304h

14.2.5.1.195.2 Diagram**14.2.5.1.195.3 Fields**

Field	Function
31-0 SW_REFER1_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 1

14.2.5.1.196 Base address LSB (bits 31:0) for reference compress luminance table index 1 (SWREG194)**14.2.5.1.196.1 Offset**

Register	Offset
SWREG194	308h

14.2.5.1.196.2 Diagram

14.2.5.1.196.3 Fields

Field	Function
31-0 SW_REFER1_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 1

14.2.5.1.197 Base address MSB (bits 63:32) for reference compress luminance table index 2 (SWREG195)

14.2.5.1.197.1 Offset

Register	Offset
SWREG195	30Ch

14.2.5.1.197.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER2_TYBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER2_TYBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.197.3 Fields

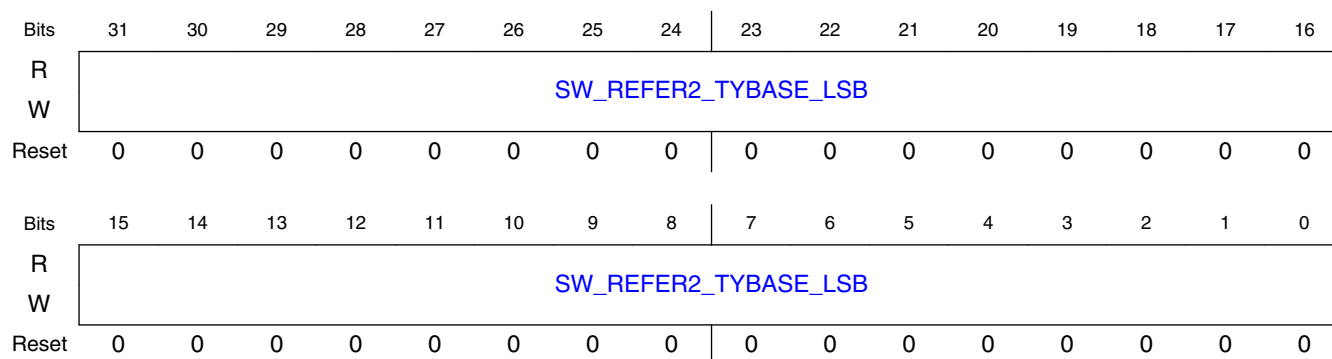
Field	Function
31-0 SW_REFER2_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 2

14.2.5.1.198 Base address LSB (bits 31:0) for reference compress luminance table index 2 (SWREG196)

14.2.5.1.198.1 Offset

Register	Offset
SWREG196	310h

14.2.5.1.198.2 Diagram



14.2.5.1.198.3 Fields

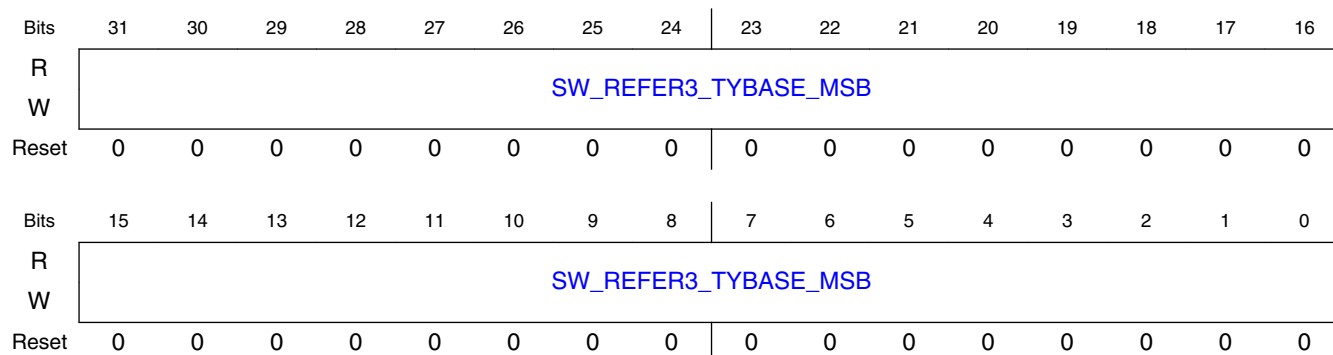
Field	Function
31-0 SW_REFER2_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 2

14.2.5.1.199 Base address MSB (bits 63:32) for reference compress luminance table index 3 (SWREG197)

14.2.5.1.199.1 Offset

Register	Offset
SWREG197	314h

14.2.5.1.199.2 Diagram



14.2.5.1.199.3 Fields

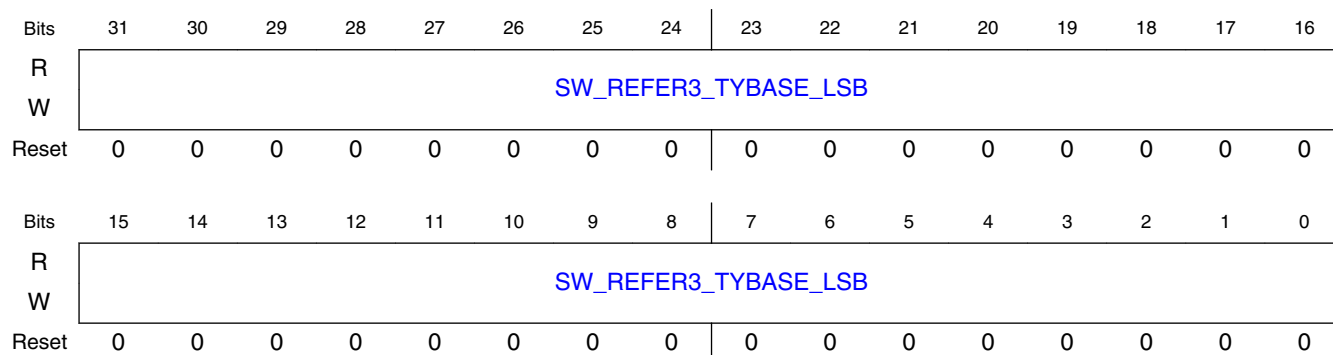
Field	Function
31-0 SW_REFER3_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 3

14.2.5.1.200 Base address LSB (bits 31:0) for reference compress luminance table index 3 (SWREG198)

14.2.5.1.200.1 Offset

Register	Offset
SWREG198	318h

14.2.5.1.200.2 Diagram



14.2.5.1.200.3 Fields

Field	Function
31-0 SW_REFER3_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 3

14.2.5.1.201 Base address MSB (bits 63:32) for reference compress luminance table index 4 (SWREG199)

14.2.5.1.201.1 Offset

Register	Offset
SWREG199	31Ch

14.2.5.1.201.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER4_TYBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER4_TYBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.201.3 Fields

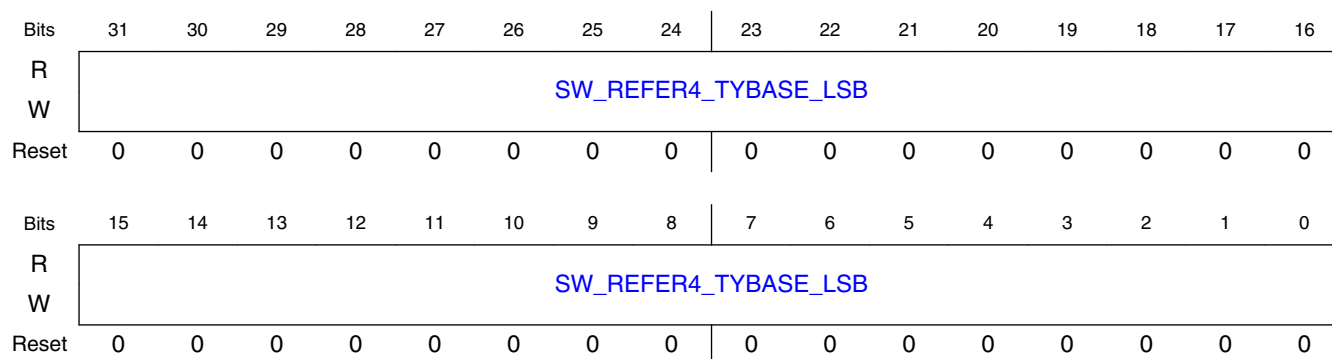
Field	Function
31-0 SW_REFER4_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 4

14.2.5.1.202 Base address LSB (bits 31:0) for reference compress luminance table index 4 (SWREG200)

14.2.5.1.202.1 Offset

Register	Offset
SWREG200	320h

14.2.5.1.202.2 Diagram



14.2.5.1.202.3 Fields

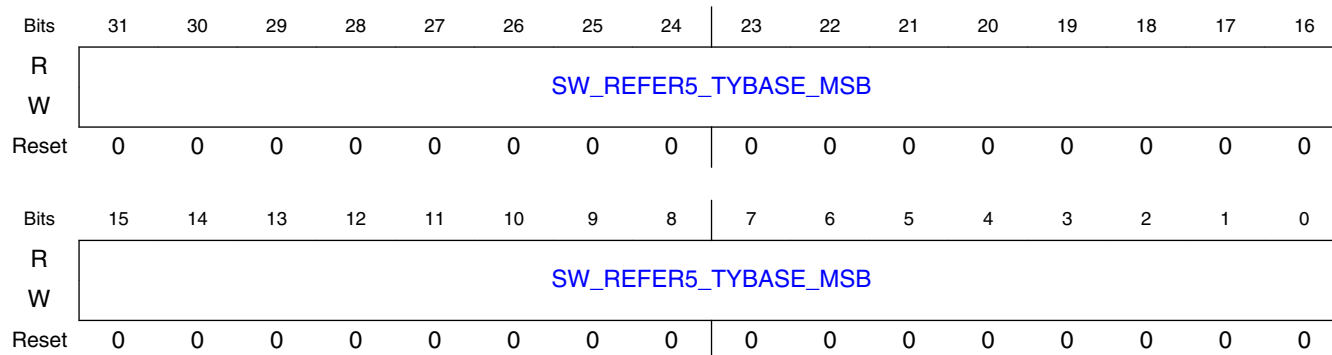
Field	Function
31-0 SW_REFER4_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 4

14.2.5.1.203 Base address MSB (bits 63:32) for reference compress luminance table index 5 (SWREG201)

14.2.5.1.203.1 Offset

Register	Offset
SWREG201	324h

14.2.5.1.203.2 Diagram



14.2.5.1.203.3 Fields

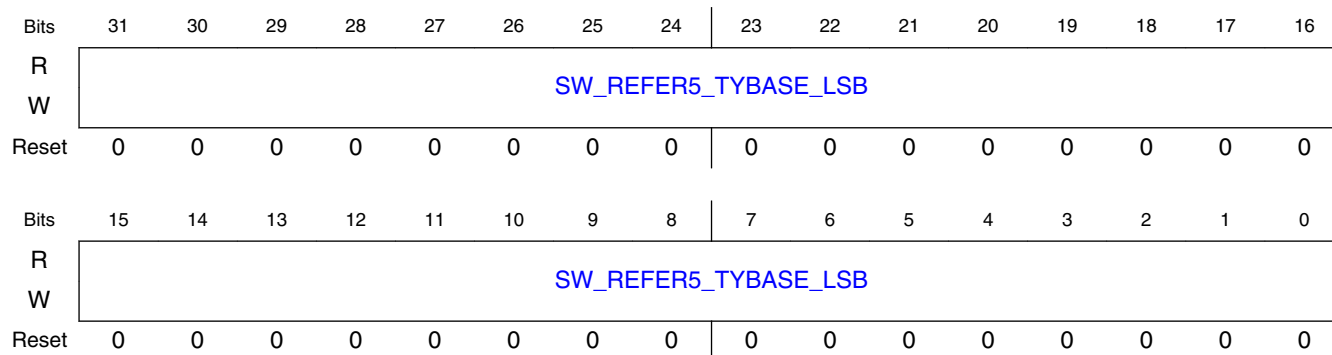
Field	Function
31-0 SW_REFER5_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 5

14.2.5.1.204 Base address LSB (bits 31:0) for reference compress luminance table index 5 (SWREG202)

14.2.5.1.204.1 Offset

Register	Offset
SWREG202	328h

14.2.5.1.204.2 Diagram



14.2.5.1.204.3 Fields

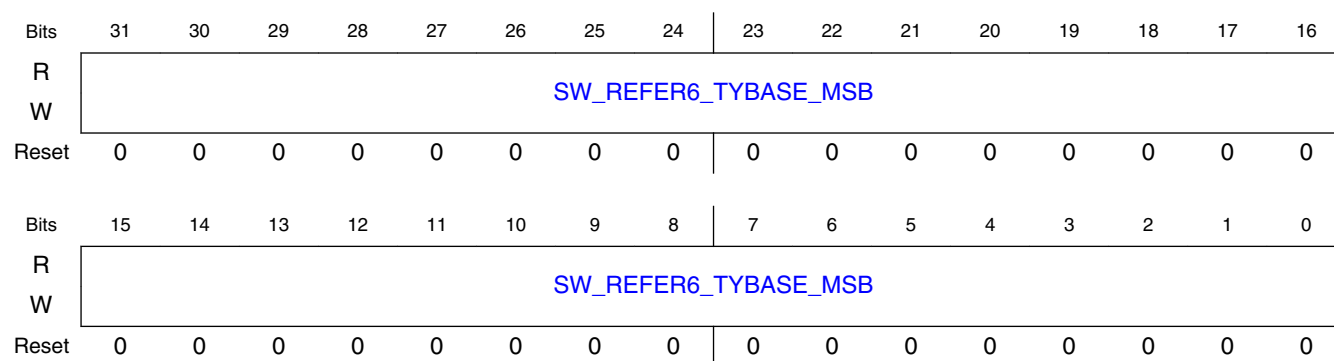
Field	Function
31-0 SW_REFER5_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 5

14.2.5.1.205 Base address MSB (bits 63:32) for reference compress luminance table index 6 (SWREG203)

14.2.5.1.205.1 Offset

Register	Offset
SWREG203	32Ch

14.2.5.1.205.2 Diagram



14.2.5.1.205.3 Fields

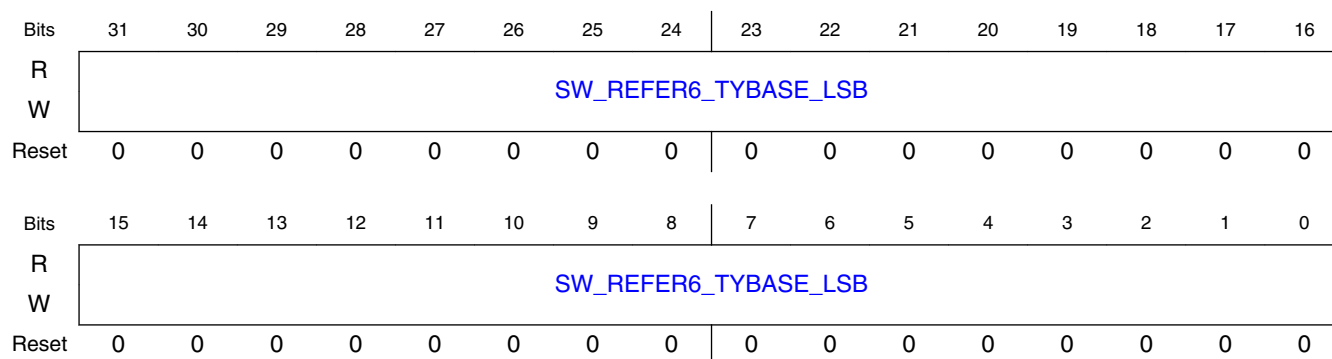
Field	Function
31-0 SW_REFER6_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 6

14.2.5.1.206 Base address LSB (bits 31:0) for reference compress luminance table index 6 (SWREG204)

14.2.5.1.206.1 Offset

Register	Offset
SWREG204	330h

14.2.5.1.206.2 Diagram



14.2.5.1.206.3 Fields

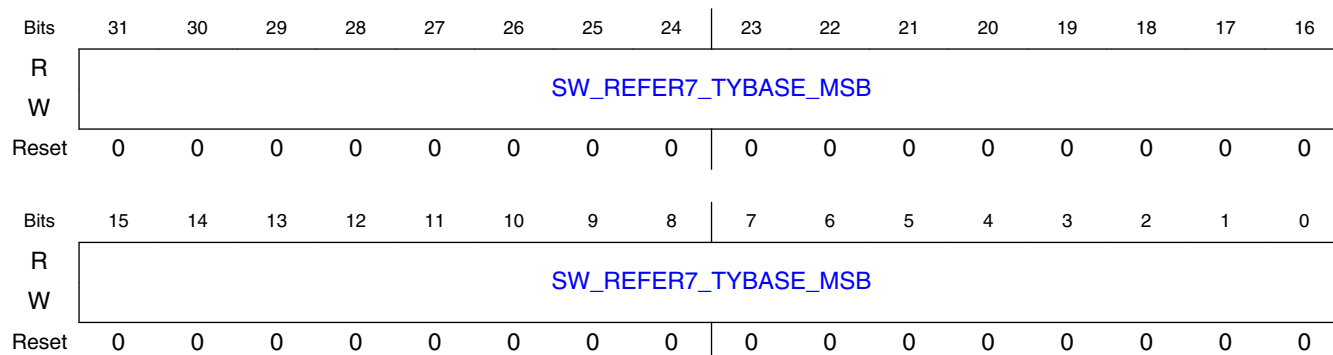
Field	Function
31-0 SW_REFER6_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 6

14.2.5.1.207 Base address MSB (bits 63:32) for reference compress luminance table index 7 (SWREG205)

14.2.5.1.207.1 Offset

Register	Offset
SWREG205	334h

14.2.5.1.207.2 Diagram



14.2.5.1.207.3 Fields

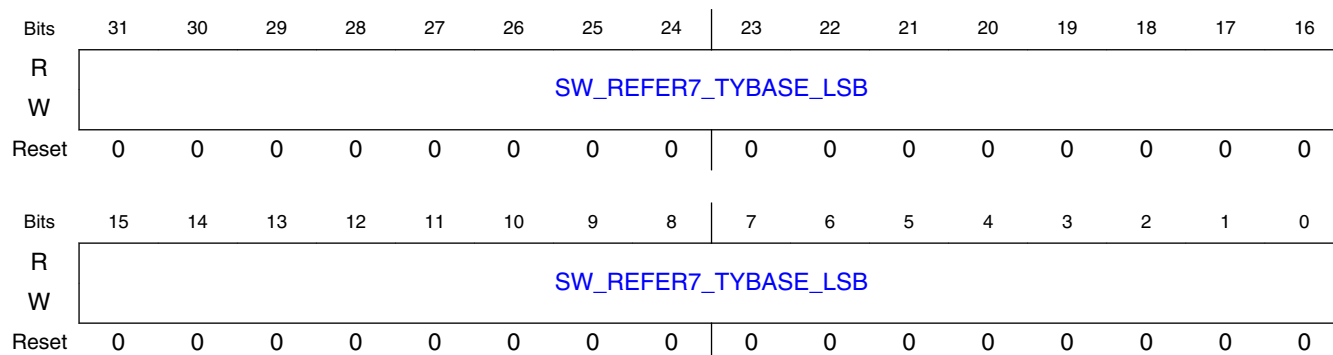
Field	Function
31-0 SW_REFER7_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 7

14.2.5.1.208 Base address LSB (bits 31:0) for reference compress luminance table index 7 (SWREG206)

14.2.5.1.208.1 Offset

Register	Offset
SWREG206	338h

14.2.5.1.208.2 Diagram



14.2.5.1.208.3 Fields

Field	Function
31-0 SW_REFER7_T YBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 7

14.2.5.1.209 Base address MSB (bits 63:32) for reference compress luminance table index 8 (SWREG207)

14.2.5.1.209.1 Offset

Register	Offset
SWREG207	33Ch

14.2.5.1.209.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER8_TYBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER8_TYBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.209.3 Fields

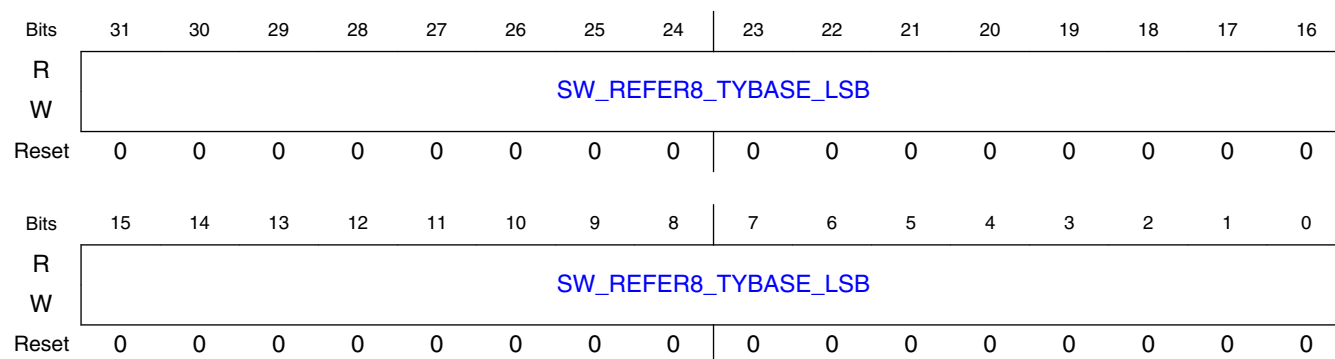
Field	Function
31-0 SW_REFER8_T YBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 8

14.2.5.1.210 Base address LSB (bits 31:0) for reference compress luminance table index 8 (SWREG208)

14.2.5.1.210.1 Offset

Register	Offset
SWREG208	340h

14.2.5.1.210.2 Diagram



14.2.5.1.210.3 Fields

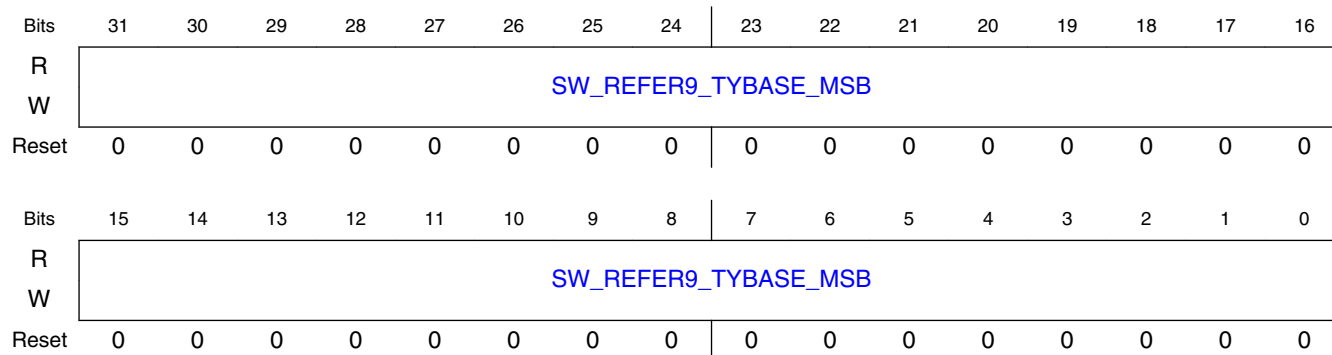
Field	Function
31-0 SW_REFER8_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 8

14.2.5.1.211 Base address MSB (bits 63:32) for reference compress luminance table index 9 (SWREG209)

14.2.5.1.211.1 Offset

Register	Offset
SWREG209	344h

14.2.5.1.211.2 Diagram



14.2.5.1.211.3 Fields

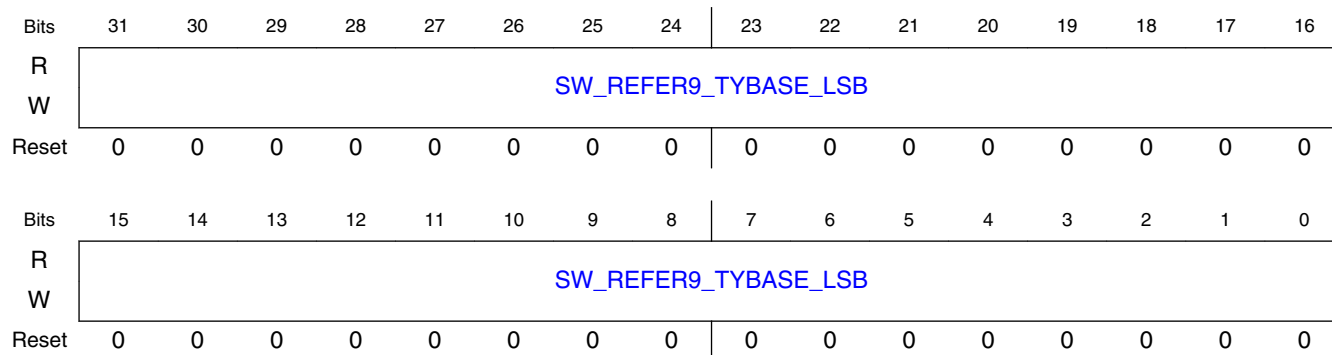
Field	Function
31-0 SW_REFER9_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 9

14.2.5.1.212 Base address LSB (bits 31:0) for reference compress luminance table index 9 (SWREG210)

14.2.5.1.212.1 Offset

Register	Offset
SWREG210	348h

14.2.5.1.212.2 Diagram



14.2.5.1.212.3 Fields

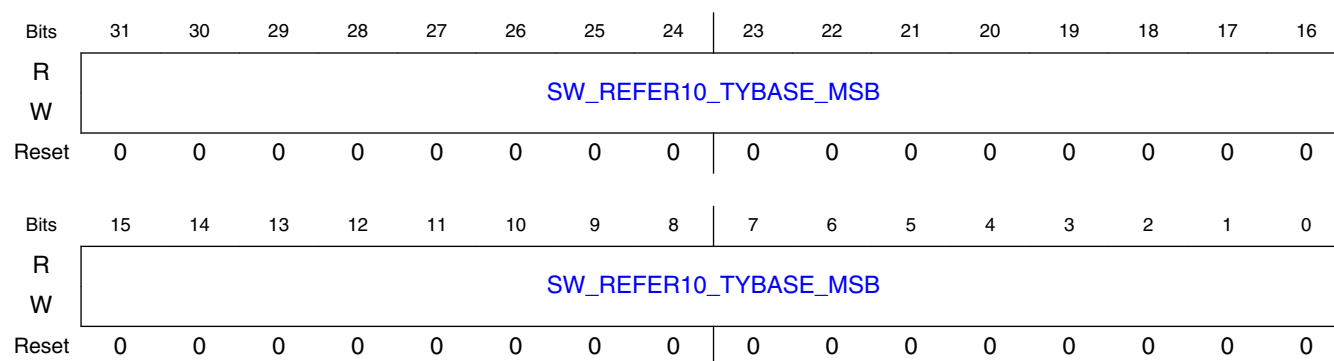
Field	Function
31-0 SW_REFER9_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 9

14.2.5.1.213 Base address MSB (bits 63:32) for reference compress luminance table index 10 (SWREG211)

14.2.5.1.213.1 Offset

Register	Offset
SWREG211	34Ch

14.2.5.1.213.2 Diagram



14.2.5.1.213.3 Fields

Field	Function
31-0 SW_REFER10_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 10

14.2.5.1.214 Base address LSB (bits 31:0) for reference compress luminance table index 10 (SWREG212)

14.2.5.1.214.1 Offset

Register	Offset
SWREG212	350h

14.2.5.1.214.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER10_TYBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER10_TYBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.214.3 Fields

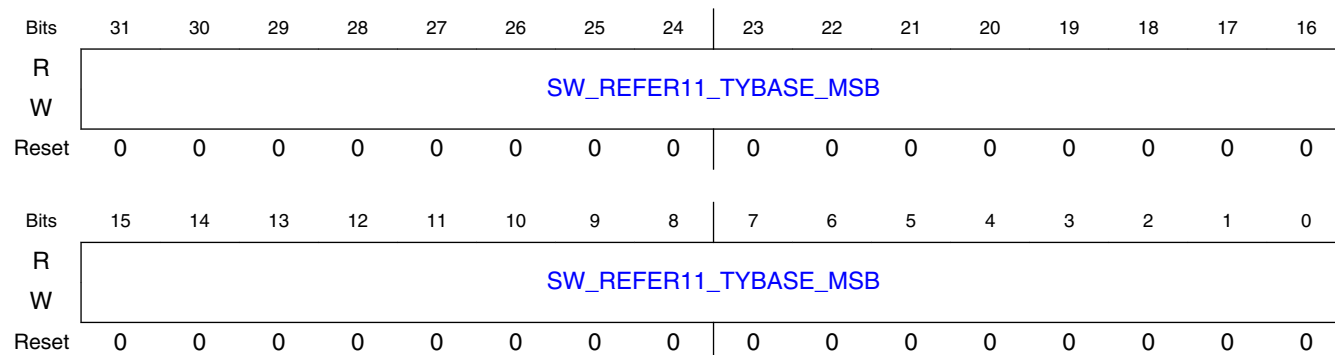
Field	Function
31-0 SW_REFER10_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 10

14.2.5.1.215 Base address MSB (bits 63:32) for reference compress luminance table index 11 (SWREG213)

14.2.5.1.215.1 Offset

Register	Offset
SWREG213	354h

14.2.5.1.215.2 Diagram



14.2.5.1.215.3 Fields

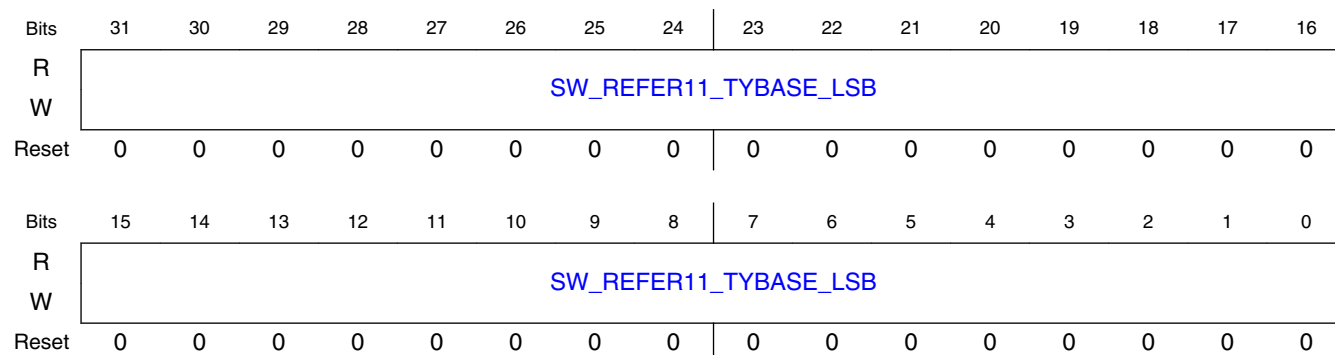
Field	Function
31-0 SW_REFER11_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 11

14.2.5.1.216 Base address LSB (bits 31:0) for reference compress luminance table index 11 (SWREG214)

14.2.5.1.216.1 Offset

Register	Offset
SWREG214	358h

14.2.5.1.216.2 Diagram



14.2.5.1.216.3 Fields

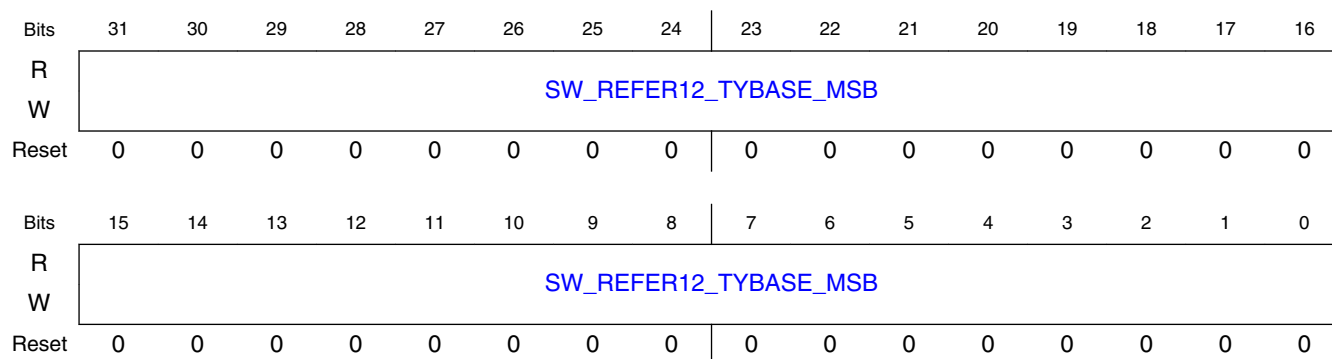
Field	Function
31-0 SW_REFER11_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 11

14.2.5.1.217 Base address MSB (bits 63:32) for reference compress luminance table index 12 (SWREG215)

14.2.5.1.217.1 Offset

Register	Offset
SWREG215	35Ch

14.2.5.1.217.2 Diagram



14.2.5.1.217.3 Fields

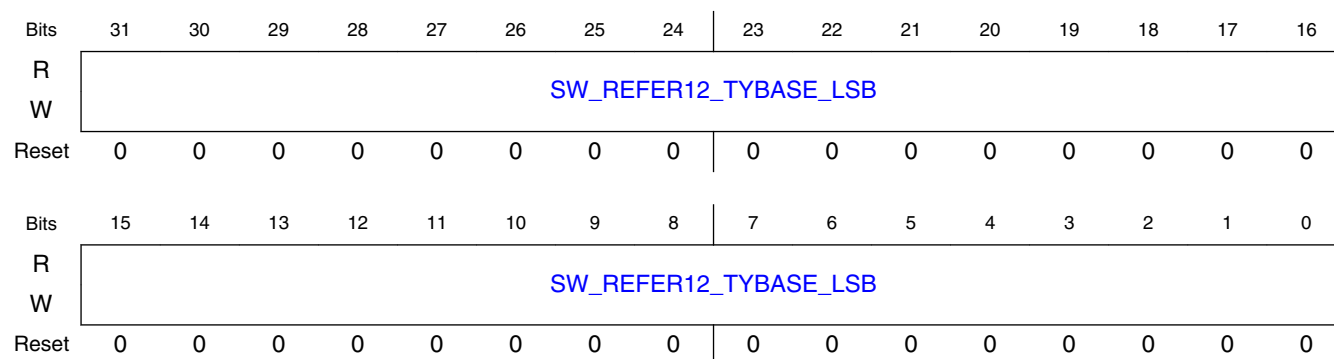
Field	Function
31-0 SW_REFER12_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 12

14.2.5.1.218 Base address LSB (bits 31:0) for reference compress luminance table index 12 (SWREG216)

14.2.5.1.218.1 Offset

Register	Offset
SWREG216	360h

14.2.5.1.218.2 Diagram



14.2.5.1.218.3 Fields

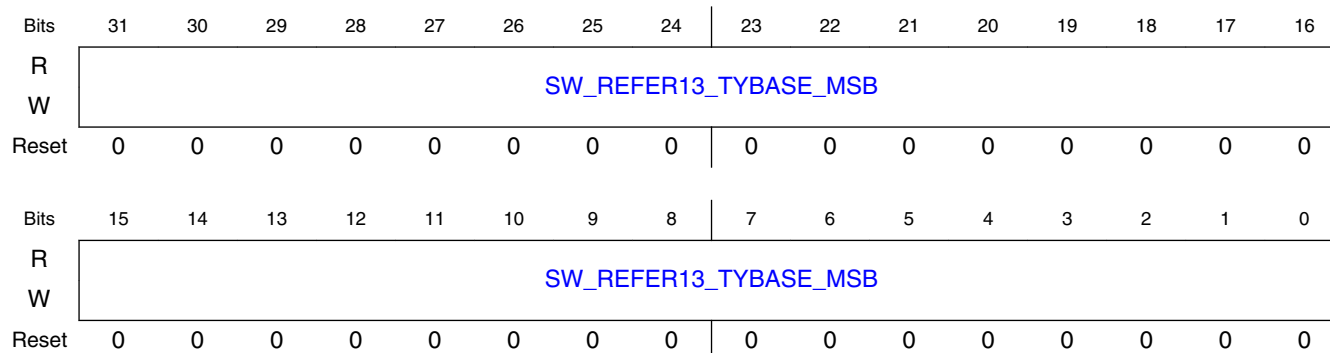
Field	Function
31-0 SW_REFER12_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 12

14.2.5.1.219 Base address MSB (bits 63:32) for reference compress luminance table index 13 (SWREG217)

14.2.5.1.219.1 Offset

Register	Offset
SWREG217	364h

14.2.5.1.219.2 Diagram



14.2.5.1.219.3 Fields

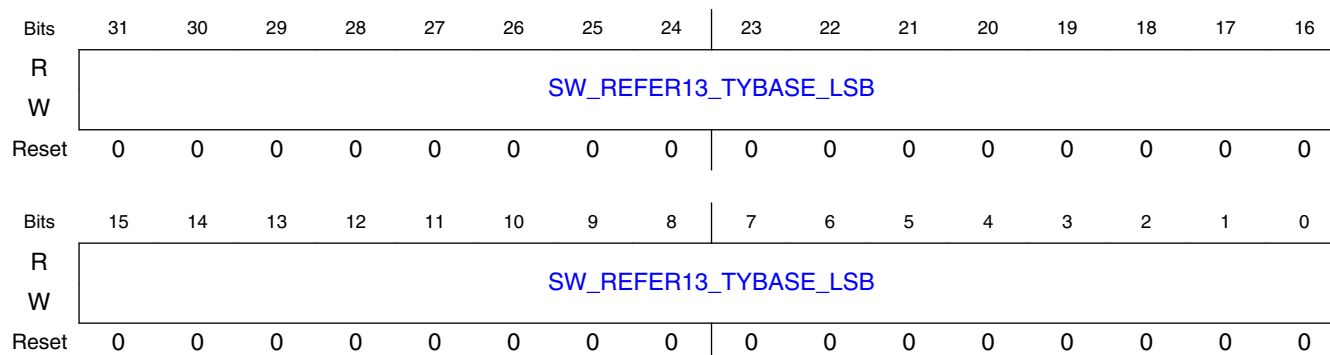
Field	Function
31-0 SW_REFER13_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 13

14.2.5.1.220 Base address LSB (bits 31:0) for reference compress luminance table index 13 (SWREG218)

14.2.5.1.220.1 Offset

Register	Offset
SWREG218	368h

14.2.5.1.220.2 Diagram



14.2.5.1.220.3 Fields

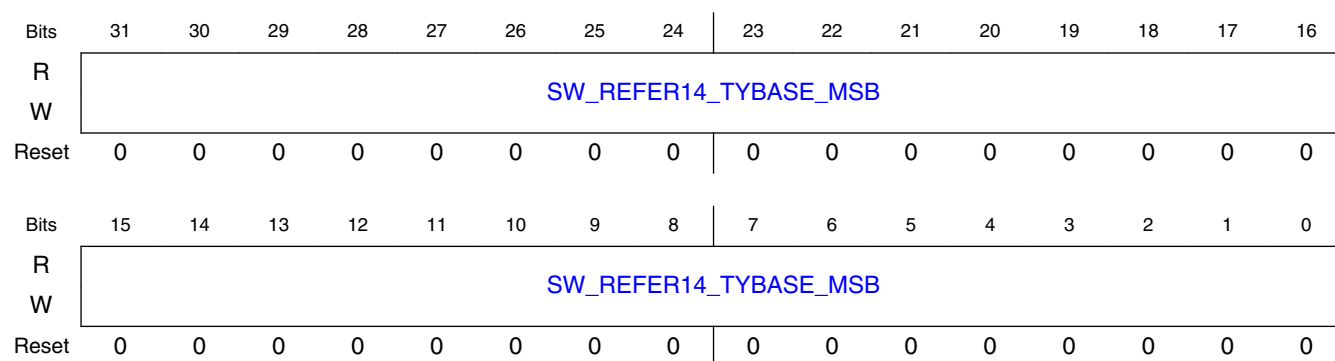
Field	Function
31-0 SW_REFER13_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 13

14.2.5.1.221 Base address MSB (bits 63:32) for reference compress luminance table index 14 (SWREG219)

14.2.5.1.221.1 Offset

Register	Offset
SWREG219	36Ch

14.2.5.1.221.2 Diagram



14.2.5.1.221.3 Fields

Field	Function
31-0 SW_REFER14_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 14

14.2.5.1.222 Base address LSB (bits 31:0) for reference compress luminance table index 14 (SWREG220)

14.2.5.1.222.1 Offset

Register	Offset
SWREG220	370h

14.2.5.1.222.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER14_TYBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER14_TYBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.222.3 Fields

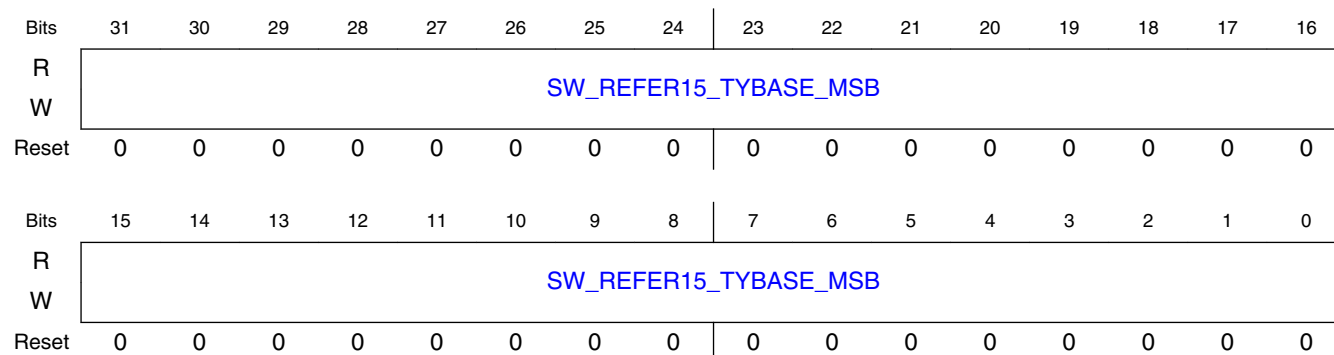
Field	Function
31-0 SW_REFER14_TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 14

14.2.5.1.223 Base address MSB (bits 63:32) for reference compress luminance table index 15 (SWREG221)

14.2.5.1.223.1 Offset

Register	Offset
SWREG221	374h

14.2.5.1.223.2 Diagram



14.2.5.1.223.3 Fields

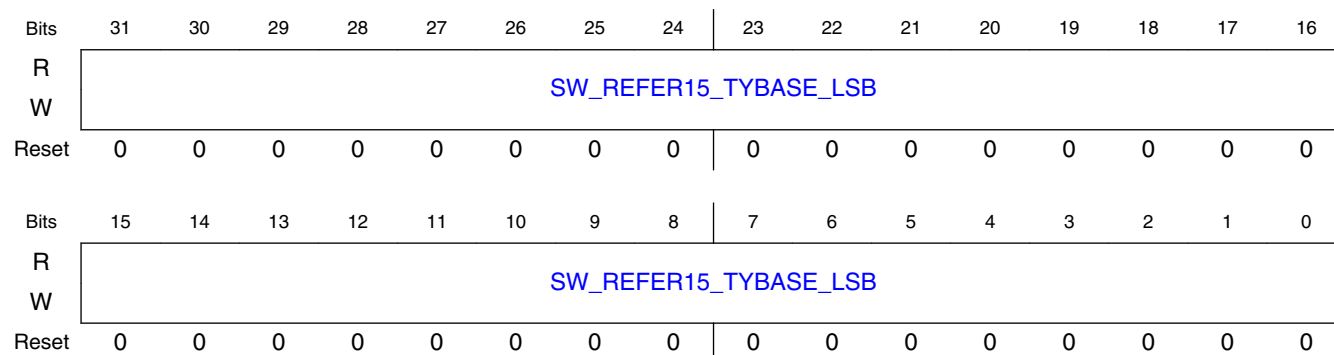
Field	Function
31-0 SW_REFER15_TYBASE_MSB	Base address MSB (bits 63:32) for reference compress luminance table index 15

14.2.5.1.224 Base address LSB (bits 31:0) for reference compress luminance table index 15 (SWREG222)

14.2.5.1.224.1 Offset

Register	Offset
SWREG222	378h

14.2.5.1.224.2 Diagram



14.2.5.1.224.3 Fields

Field	Function
31-0 SW_REFER15_ TYBASE_LSB	Base address LSB (bits 31:0) for reference compress luminance table index 15

14.2.5.1.225 Base address MSB (bits 63:32) for decoder output compress chrominance table (SWREG223)

14.2.5.1.225.1 Offset

Register	Offset
SWREG223	37Ch

14.2.5.1.225.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_DEC_OUT_TCBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_DEC_OUT_TCBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.225.3 Fields

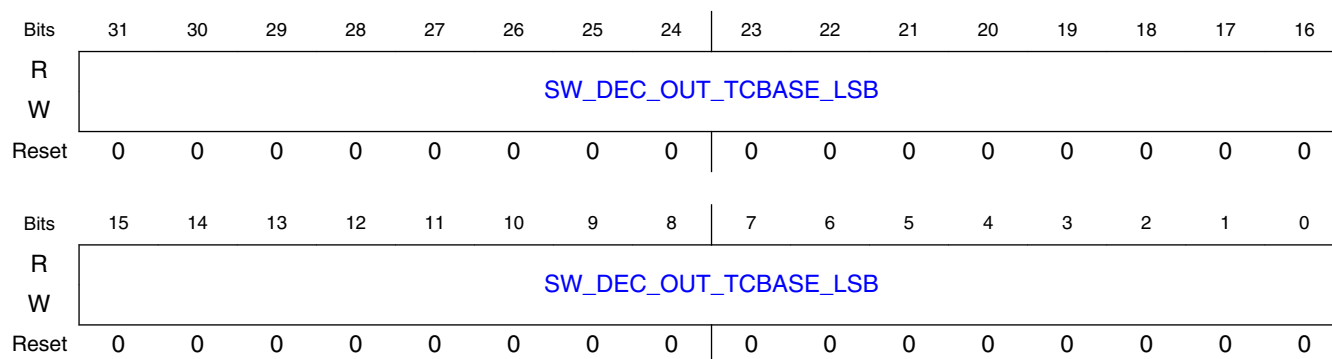
Field	Function
31-0 SW_DEC_OUT_ TCBASE_MSB	Base address MSB (bits 63:32) for decoder output compress chrominance table

14.2.5.1.226 Base address LSB (bits 31:0) for decoder output compress chrominance table (SWREG224)

14.2.5.1.226.1 Offset

Register	Offset
SWREG224	380h

14.2.5.1.226.2 Diagram



14.2.5.1.226.3 Fields

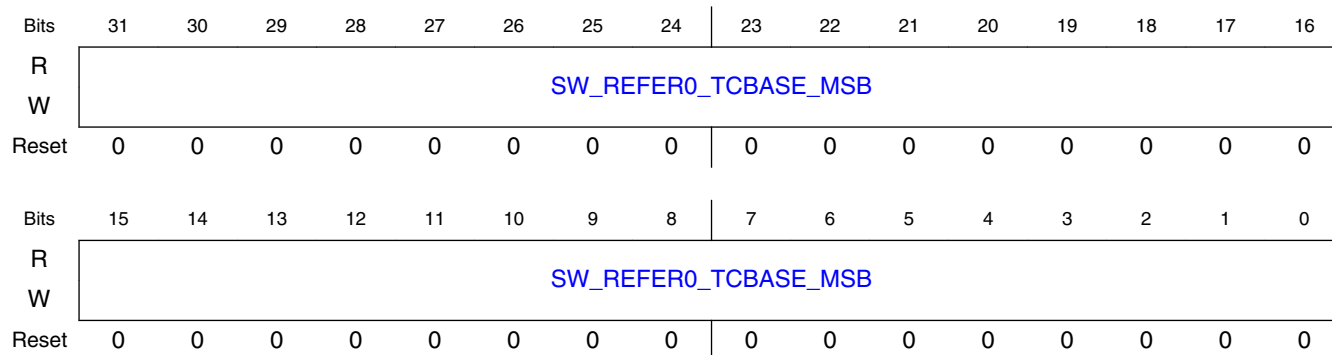
Field	Function
31-0 SW_DEC_OUT_TCBASE_LSB	Base address LSB (bits 31:0) for decoder output compress chrominance table

14.2.5.1.227 Base address MSB (bits 63:32) for reference compress chrominance table index 0 (SWREG225)

14.2.5.1.227.1 Offset

Register	Offset
SWREG225	384h

14.2.5.1.227.2 Diagram



14.2.5.1.227.3 Fields

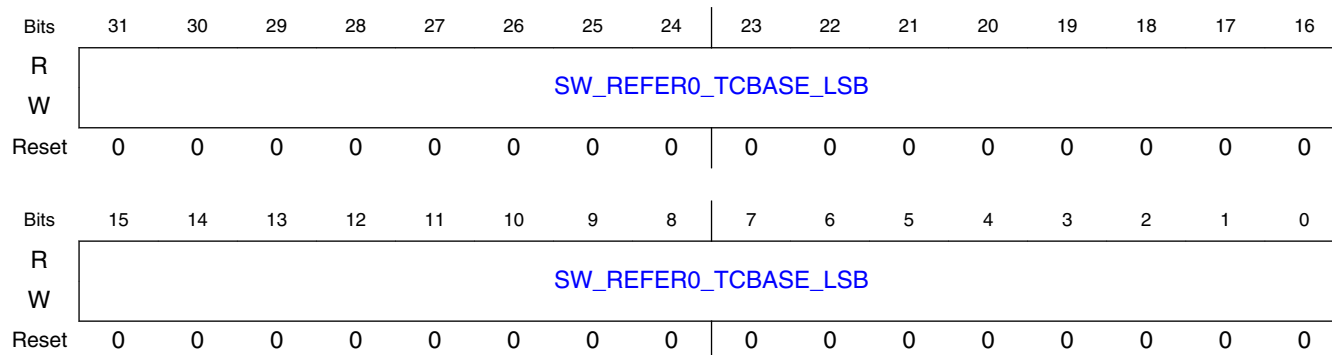
Field	Function
31-0 SW_REFER0_TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 0

14.2.5.1.228 Base address LSB (bits 31:0) for reference compress chrominance table index 0 (SWREG226)

14.2.5.1.228.1 Offset

Register	Offset
SWREG226	388h

14.2.5.1.228.2 Diagram



14.2.5.1.228.3 Fields

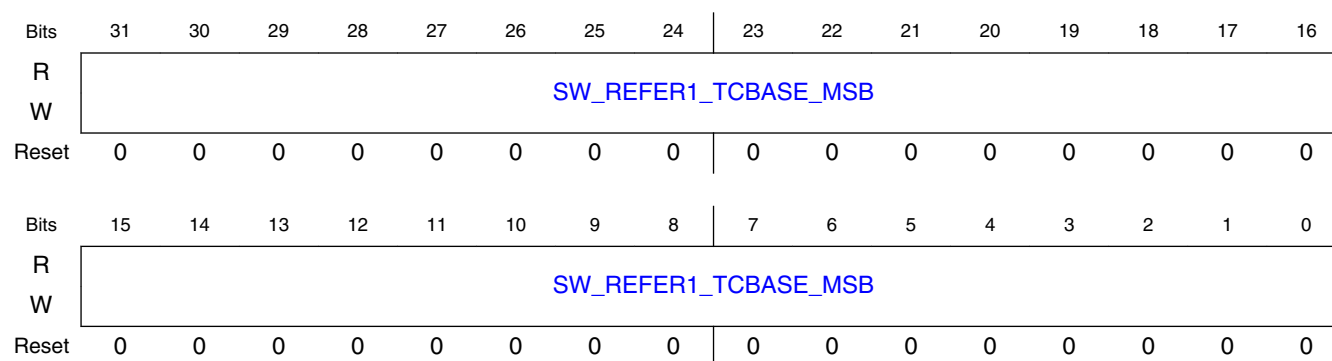
Field	Function
31-0 SW_REFER0_T CBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 0

14.2.5.1.229 Base address MSB (bits 63:32) for reference compress chrominance table index 1 (SWREG227)

14.2.5.1.229.1 Offset

Register	Offset
SWREG227	38Ch

14.2.5.1.229.2 Diagram



14.2.5.1.229.3 Fields

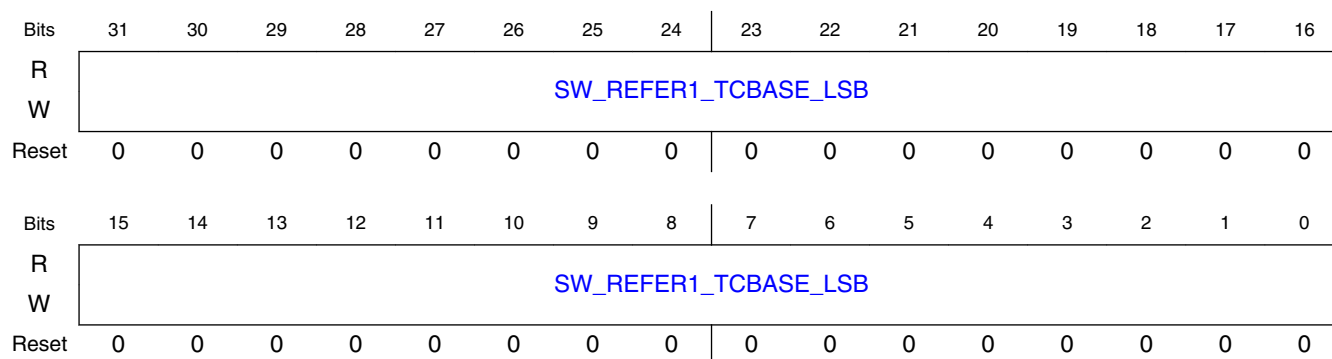
Field	Function
31-0 SW_REFER1_T CBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 1

14.2.5.1.230 Base address LSB (bits 31:0) for reference compress chrominance table index 1 (SWREG228)

14.2.5.1.230.1 Offset

Register	Offset
SWREG228	390h

14.2.5.1.230.2 Diagram



14.2.5.1.230.3 Fields

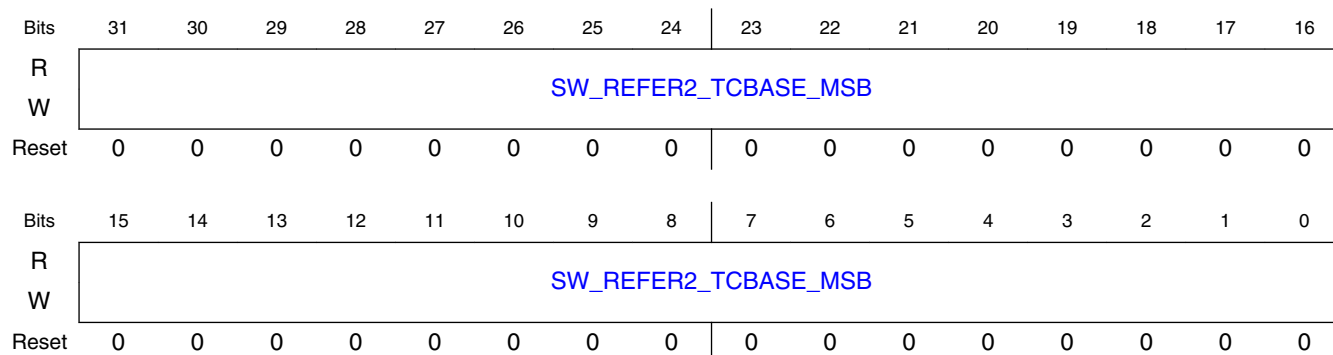
Field	Function
31-0 SW_REFER1_TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 1

14.2.5.1.231 Base address MSB (bits 63:32) for reference compress chrominance table index 2 (SWREG229)

14.2.5.1.231.1 Offset

Register	Offset
SWREG229	394h

14.2.5.1.231.2 Diagram



14.2.5.1.231.3 Fields

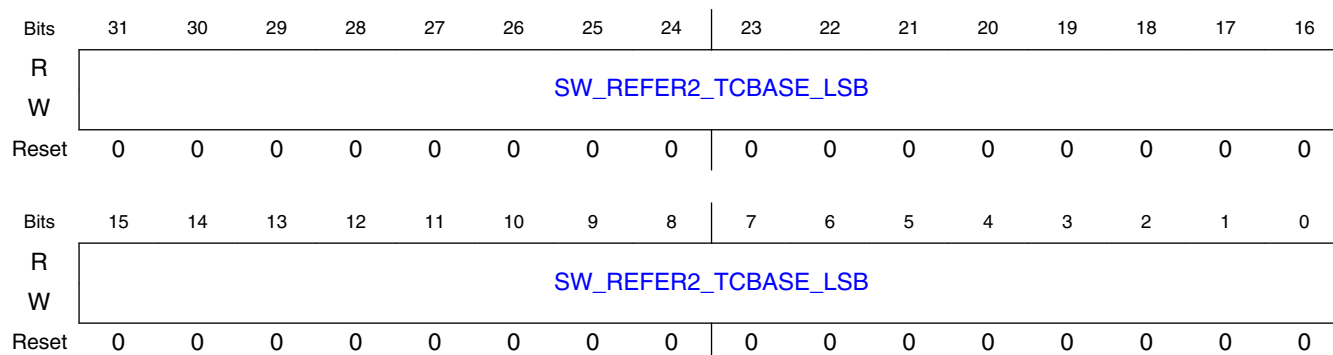
Field	Function
31-0 SW_REFER2_TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 2

14.2.5.1.232 Base address LSB (bits 31:0) for reference compress chrominance table index 2 (SWREG230)

14.2.5.1.232.1 Offset

Register	Offset
SWREG230	398h

14.2.5.1.232.2 Diagram



14.2.5.1.232.3 Fields

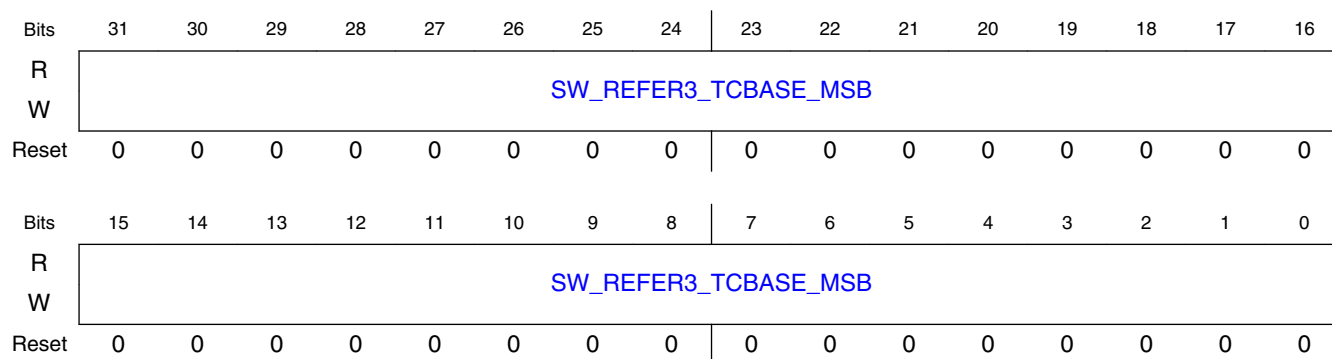
Field	Function
31-0 SW_REFER2_T CBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 2

14.2.5.1.233 Base address MSB (bits 63:32) for reference compress chrominance table index 3 (SWREG231)

14.2.5.1.233.1 Offset

Register	Offset
SWREG231	39Ch

14.2.5.1.233.2 Diagram



14.2.5.1.233.3 Fields

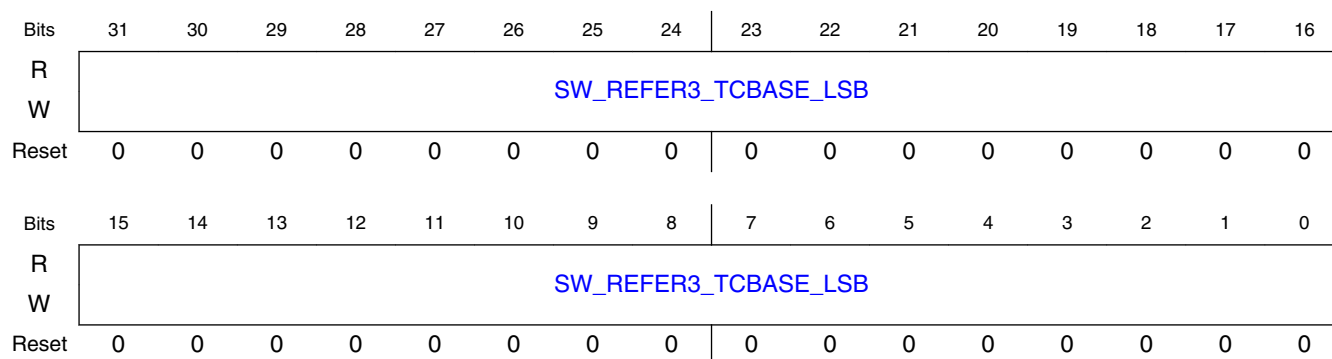
Field	Function
31-0 SW_REFER3_T CBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 3

14.2.5.1.234 Base address LSB (bits 31:0) for reference compress chrominance table index 3 (SWREG232)

14.2.5.1.234.1 Offset

Register	Offset
SWREG232	3A0h

14.2.5.1.234.2 Diagram



14.2.5.1.234.3 Fields

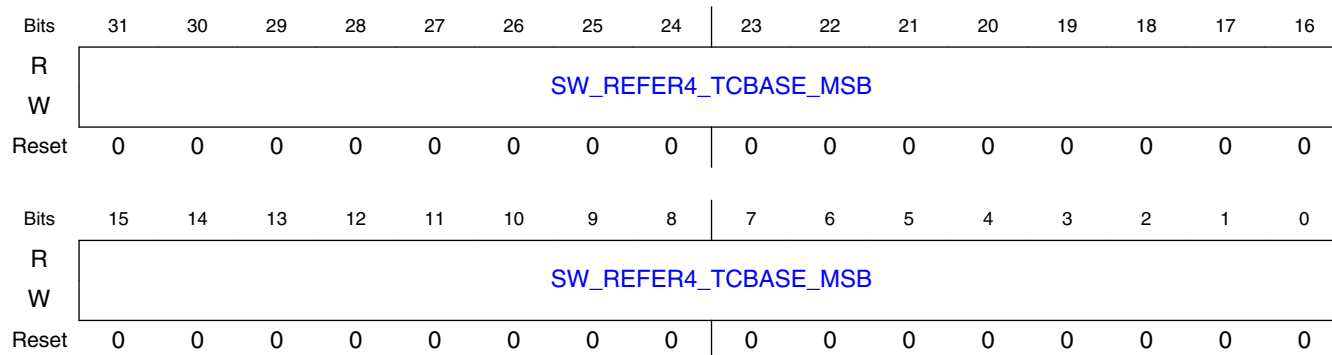
Field	Function
31-0 SW_REFER3_T CBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 3

14.2.5.1.235 Base address MSB (bits 63:32) for reference compress chrominance table index 4 (SWREG233)

14.2.5.1.235.1 Offset

Register	Offset
SWREG233	3A4h

14.2.5.1.235.2 Diagram



14.2.5.1.235.3 Fields

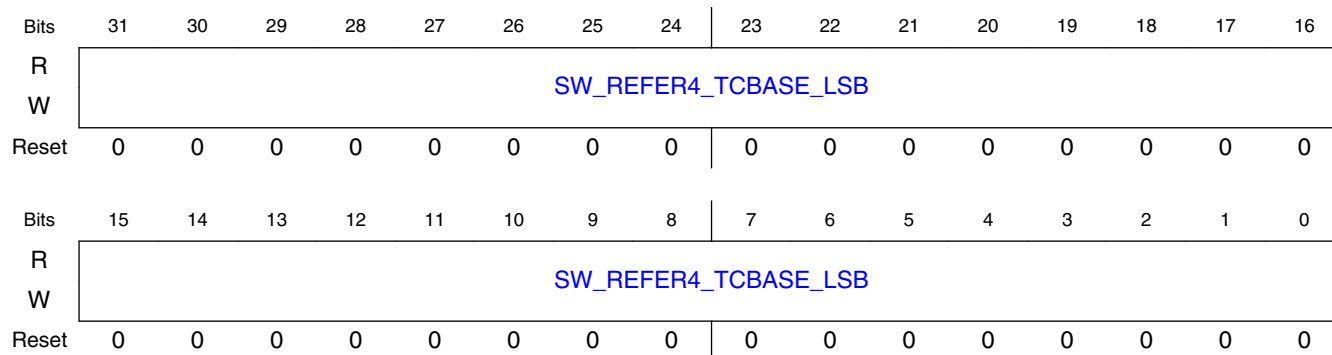
Field	Function
31-0 SW_REFER4_TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 4

14.2.5.1.236 Base address LSB (bits 31:0) for reference compress chrominance table index 4 (SWREG234)

14.2.5.1.236.1 Offset

Register	Offset
SWREG234	3A8h

14.2.5.1.236.2 Diagram



14.2.5.1.236.3 Fields

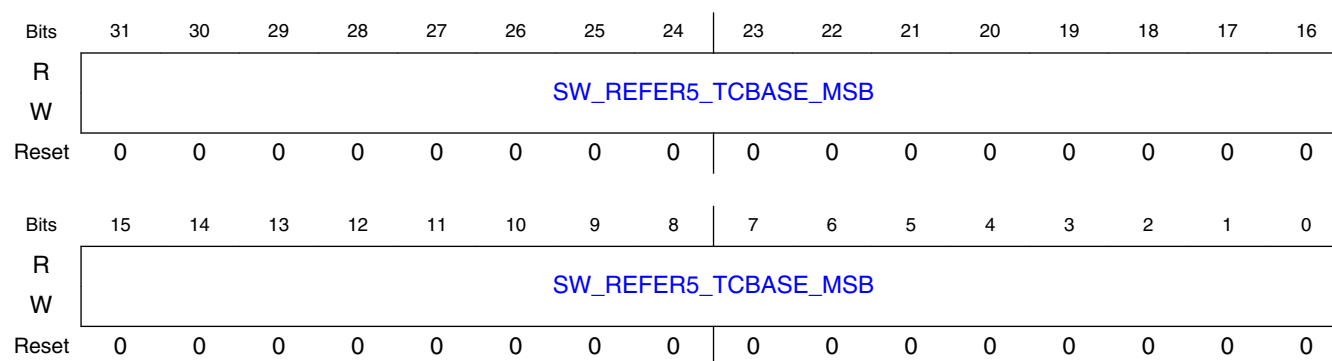
Field	Function
31-0 SW_REFER4_T CBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 4

14.2.5.1.237 Base address MSB (bits 63:32) for reference compress chrominance table index 5 (SWREG235)

14.2.5.1.237.1 Offset

Register	Offset
SWREG235	3ACh

14.2.5.1.237.2 Diagram



14.2.5.1.237.3 Fields

Field	Function
31-0 SW_REFER5_T CBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 5

14.2.5.1.238 Base address LSB (bits 31:0) for reference compress chrominance table index 5 (SWREG236)

14.2.5.1.238.1 Offset

Register	Offset
SWREG236	3B0h

14.2.5.1.238.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER5_TCBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER5_TCBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.238.3 Fields

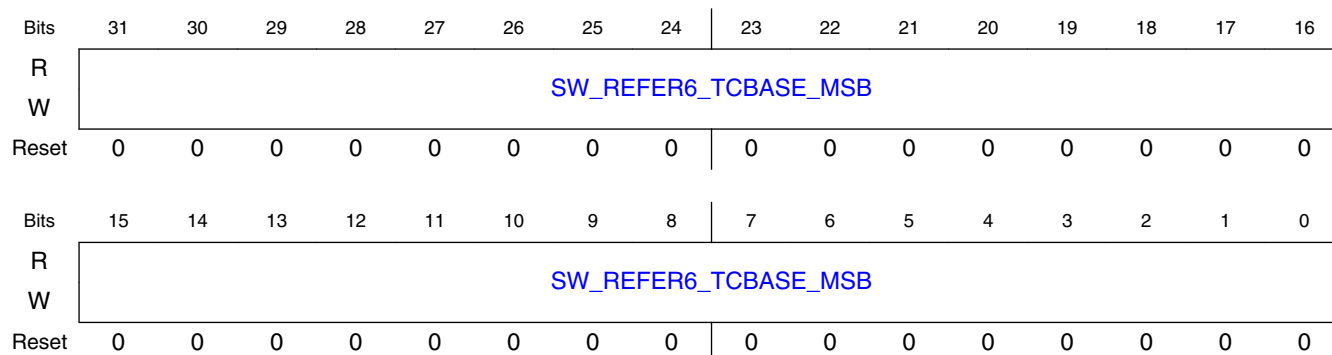
Field	Function
31-0 SW_REFER5_TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 5

14.2.5.1.239 Base address MSB (bits 63:32) for reference compress chrominance table index 6 (SWREG237)

14.2.5.1.239.1 Offset

Register	Offset
SWREG237	3B4h

14.2.5.1.239.2 Diagram



14.2.5.1.239.3 Fields

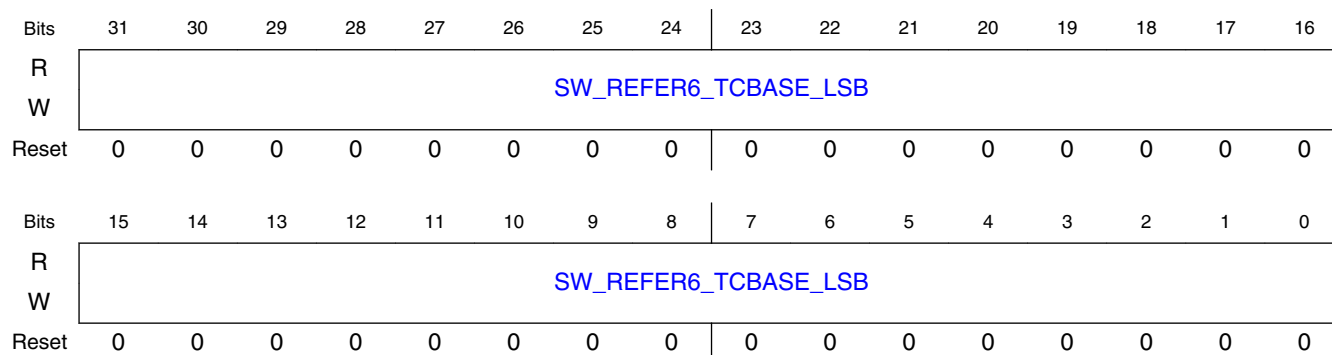
Field	Function
31-0 SW_REFER6_TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 6

14.2.5.1.240 Base address LSB (bits 31:0) for reference compress chrominance table index 6 (SWREG238)

14.2.5.1.240.1 Offset

Register	Offset
SWREG238	3B8h

14.2.5.1.240.2 Diagram



14.2.5.1.240.3 Fields

Field	Function
31-0 SW_REFER6_T CBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 6

14.2.5.1.241 Base address MSB (bits 63:32) for reference compress chrominance table index 7 (SWREG239)

14.2.5.1.241.1 Offset

Register	Offset
SWREG239	3BCh

14.2.5.1.241.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER7_TCBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER7_TCBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.241.3 Fields

Field	Function
31-0 SW_REFER7_T CBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 7

14.2.5.1.242 Base address LSB (bits 31:0) for reference compress chrominance table index 7 (SWREG240)

14.2.5.1.242.1 Offset

Register	Offset
SWREG240	3C0h

14.2.5.1.242.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER7_TCBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER7_TCBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.242.3 Fields

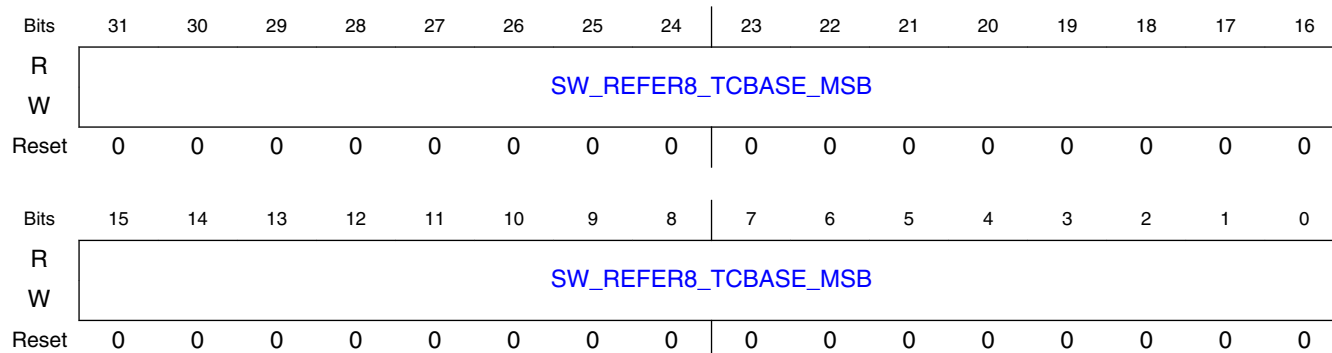
Field	Function
31-0 SW_REFER7_TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 7

14.2.5.1.243 Base address MSB (bits 63:32) for reference compress chrominance table index 8 (SWREG241)

14.2.5.1.243.1 Offset

Register	Offset
SWREG241	3C4h

14.2.5.1.243.2 Diagram



14.2.5.1.243.3 Fields

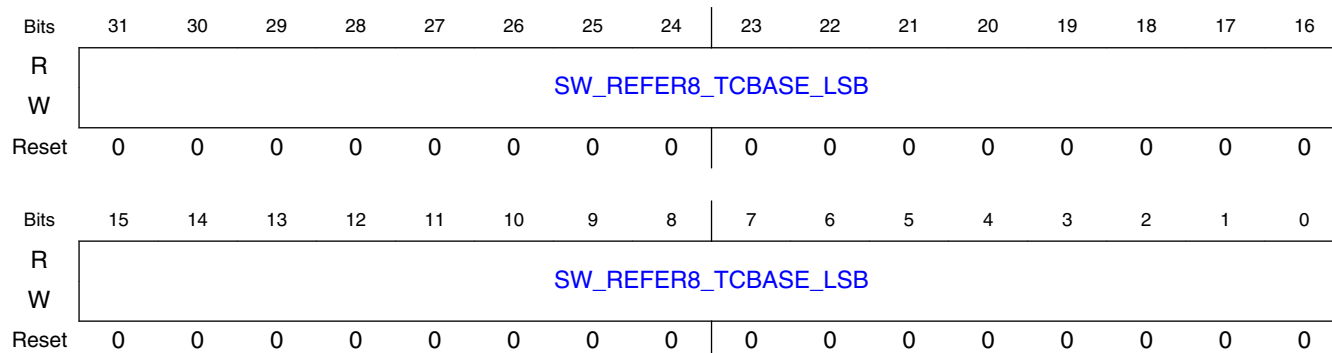
Field	Function
31-0 SW_REFER8_TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 8

14.2.5.1.244 Base address LSB (bits 31:0) for reference compress chrominance table index 8 (SWREG242)

14.2.5.1.244.1 Offset

Register	Offset
SWREG242	3C8h

14.2.5.1.244.2 Diagram



14.2.5.1.244.3 Fields

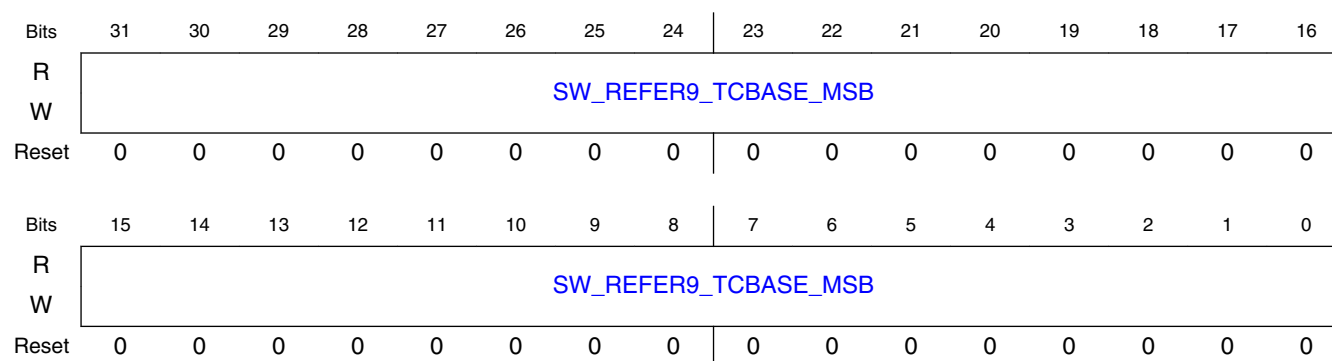
Field	Function
31-0 SW_REFER8_T CBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 8

14.2.5.1.245 Base address MSB (bits 63:32) for reference compress chrominance table index 9 (SWREG243)

14.2.5.1.245.1 Offset

Register	Offset
SWREG243	3CCh

14.2.5.1.245.2 Diagram



14.2.5.1.245.3 Fields

Field	Function
31-0 SW_REFER9_T CBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 9

14.2.5.1.246 Base address LSB (bits 31:0) for reference compress chrominance table index 9 (SWREG244)

14.2.5.1.246.1 Offset

Register	Offset
SWREG244	3D0h

14.2.5.1.246.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER9_TCBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER9_TCBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.246.3 Fields

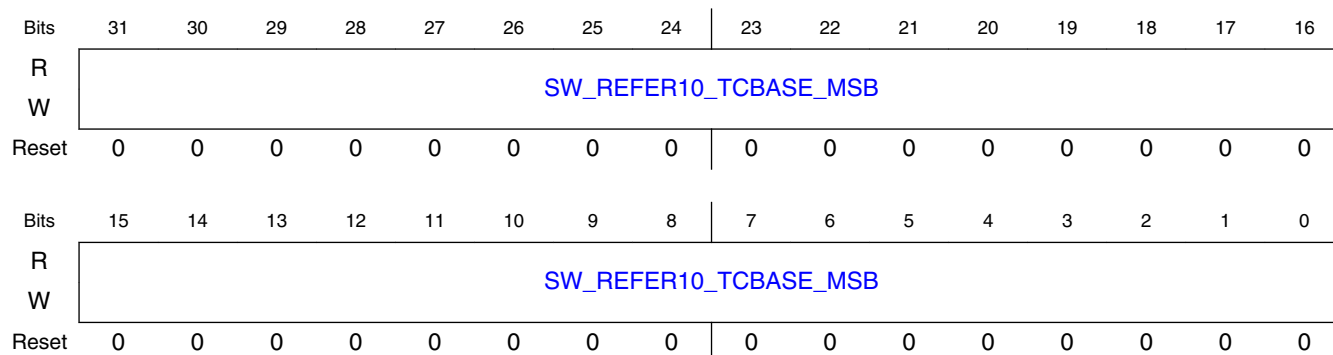
Field	Function
31-0 SW_REFER9_TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 9

14.2.5.1.247 Base address MSB (bits 63:32) for reference compress chrominance table index 10 (SWREG245)

14.2.5.1.247.1 Offset

Register	Offset
SWREG245	3D4h

14.2.5.1.247.2 Diagram



14.2.5.1.247.3 Fields

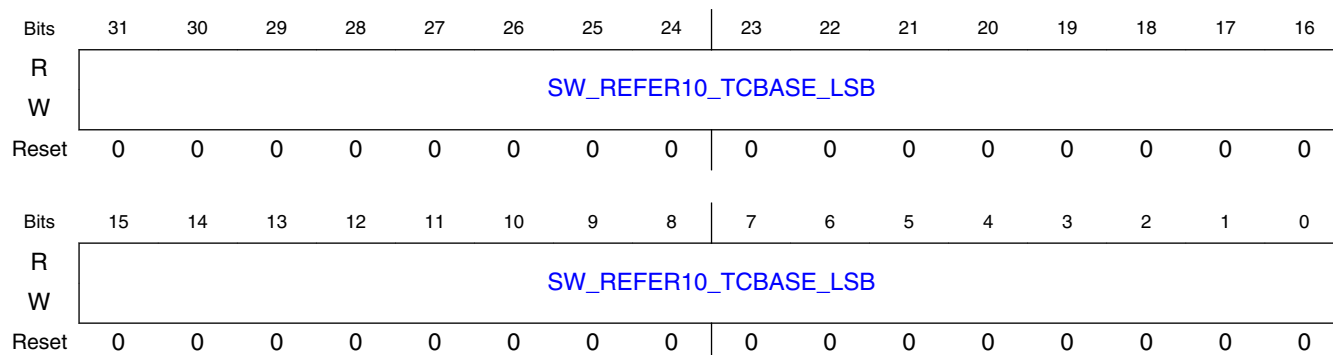
Field	Function
31-0 SW_REFER10_TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 10

14.2.5.1.248 Base address LSB (bits 31:0) for reference compress chrominance table index 10 (SWREG246)

14.2.5.1.248.1 Offset

Register	Offset
SWREG246	3D8h

14.2.5.1.248.2 Diagram

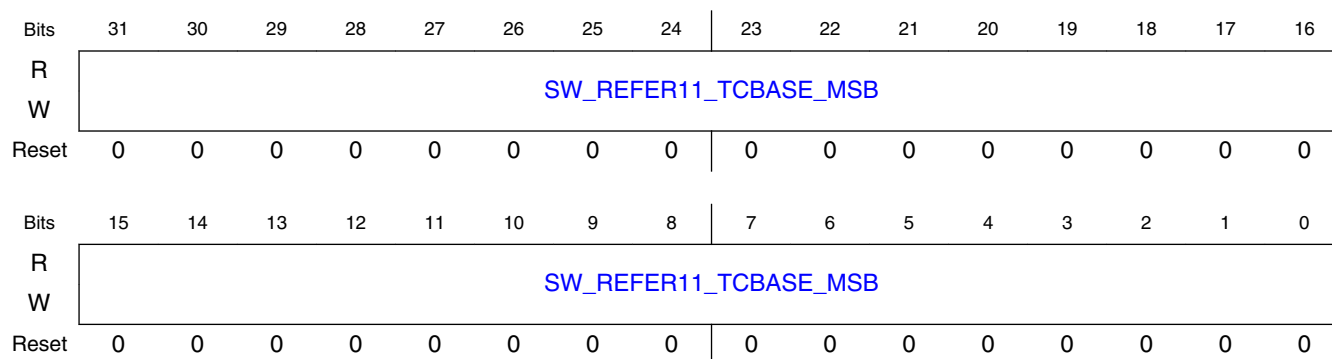


14.2.5.1.248.3 Fields

Field	Function
31-0 SW_REFER10_ TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 10

14.2.5.1.249 Base address MSB (bits 63:32) for reference compress chrominance table index 11 (SWREG247)**14.2.5.1.249.1 Offset**

Register	Offset
SWREG247	3DCh

14.2.5.1.249.2 Diagram**14.2.5.1.249.3 Fields**

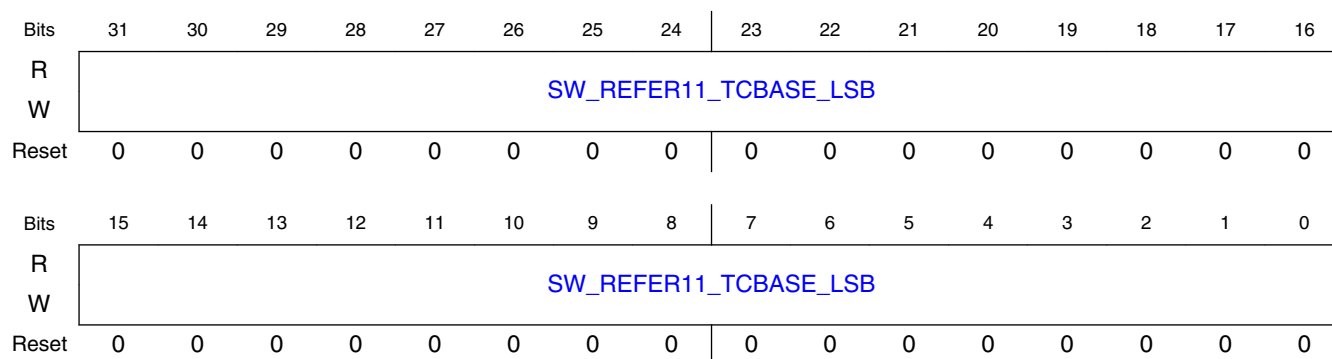
Field	Function
31-0 SW_REFER11_ TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 11

14.2.5.1.250 Base address LSB (bits 31:0) for reference compress chrominance table index 11 (SWREG248)

14.2.5.1.250.1 Offset

Register	Offset
SWREG248	3E0h

14.2.5.1.250.2 Diagram



14.2.5.1.250.3 Fields

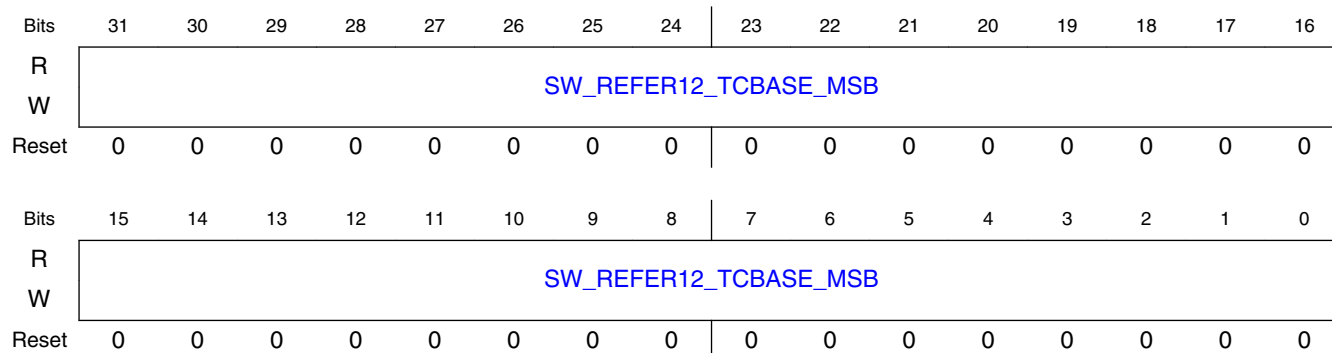
Field	Function
31-0 SW_REFER11_TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 11

14.2.5.1.251 Base address MSB (bits 63:32) for reference compress chrominance table index 12 (SWREG249)

14.2.5.1.251.1 Offset

Register	Offset
SWREG249	3E4h

14.2.5.1.251.2 Diagram



14.2.5.1.251.3 Fields

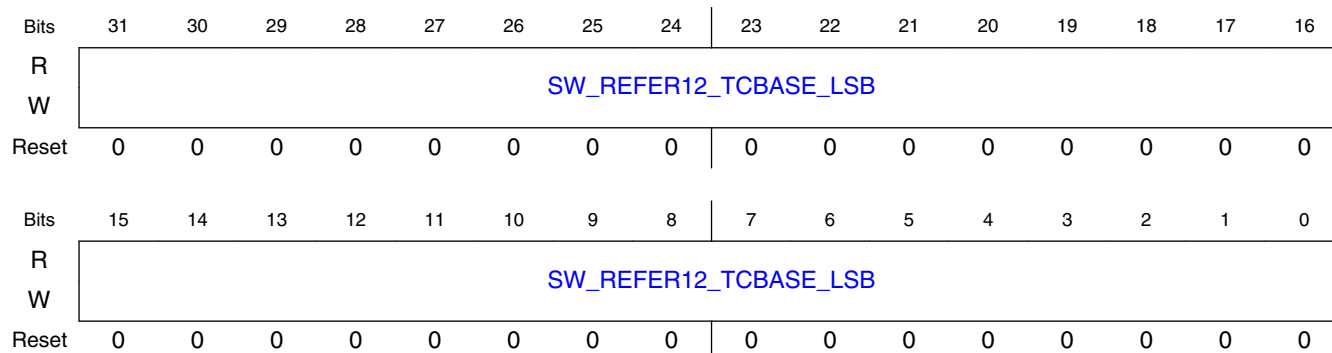
Field	Function
31-0 SW_REFER12_TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 12

14.2.5.1.252 Base address LSB (bits 31:0) for reference compress chrominance table index 12 (SWREG250)

14.2.5.1.252.1 Offset

Register	Offset
SWREG250	3E8h

14.2.5.1.252.2 Diagram



14.2.5.1.252.3 Fields

Field	Function
31-0 SW_REFER12_ TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 12

14.2.5.1.253 Base address MSB (bits 63:32) for reference compress chrominance table index 13 (SWREG251)

14.2.5.1.253.1 Offset

Register	Offset
SWREG251	3ECh

14.2.5.1.253.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER13_TCBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER13_TCBASE_MSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.253.3 Fields

Field	Function
31-0 SW_REFER13_ TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 13

14.2.5.1.254 Base address LSB (bits 31:0) for reference compress chrominance table index 13 (SWREG252)

14.2.5.1.254.1 Offset

Register	Offset
SWREG252	3F0h

14.2.5.1.254.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_REFER13_TCBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_REFER13_TCBASE_LSB															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.254.3 Fields

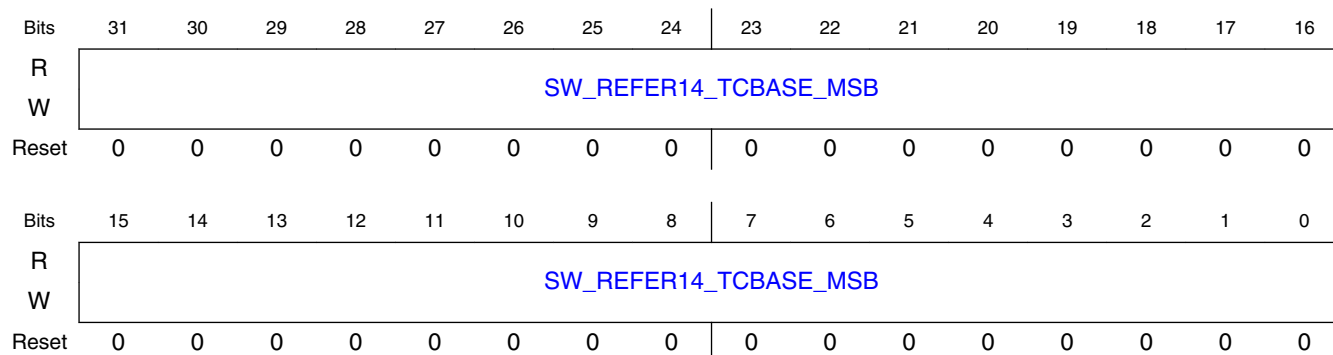
Field	Function
31-0 SW_REFER13_TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 13

14.2.5.1.255 Base address MSB (bits 63:32) for reference compress chrominance table index 14 (SWREG253)

14.2.5.1.255.1 Offset

Register	Offset
SWREG253	3F4h

14.2.5.1.255.2 Diagram



14.2.5.1.255.3 Fields

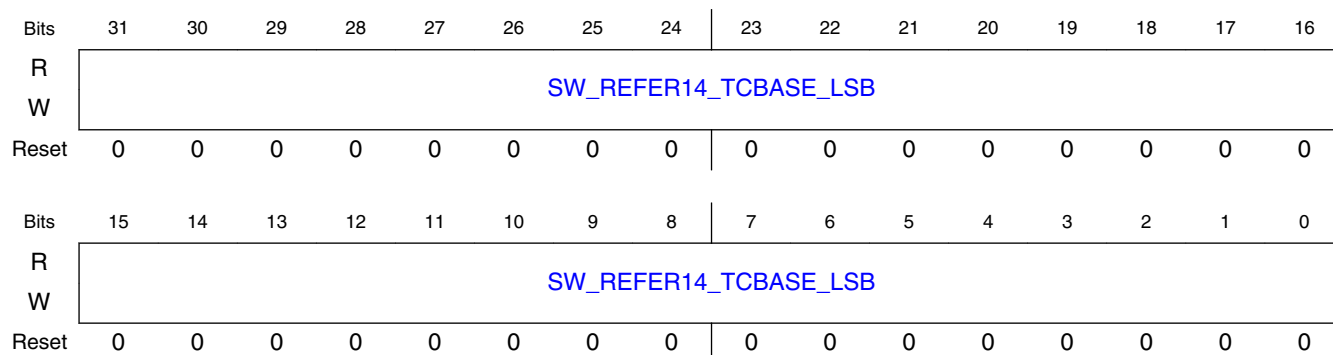
Field	Function
31-0 SW_REFER14_TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 14

14.2.5.1.256 Base address LSB (bits 31:0) for reference compress chrominance table index 14 (SWREG254)

14.2.5.1.256.1 Offset

Register	Offset
SWREG254	3F8h

14.2.5.1.256.2 Diagram



14.2.5.1.256.3 Fields

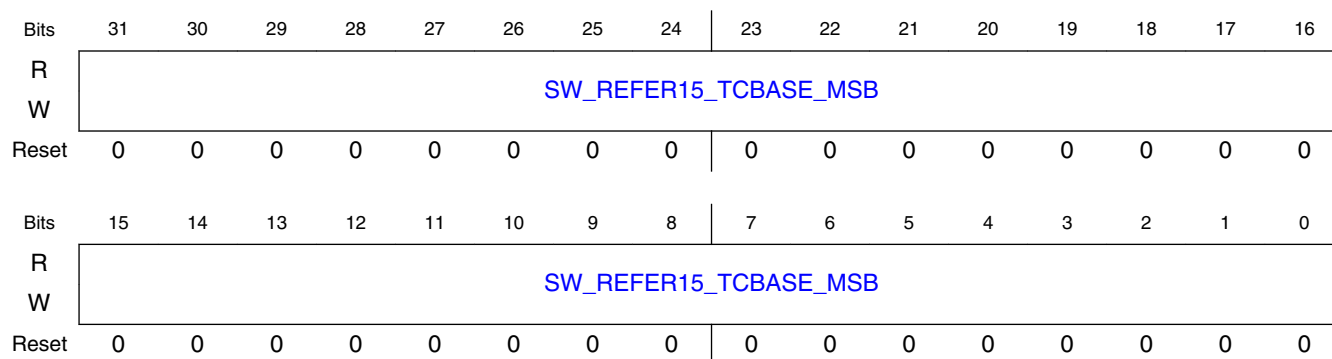
Field	Function
31-0 SW_REFER14_ TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 14

14.2.5.1.257 Base address MSB (bits 63:32) for reference compress chrominance table index 15 (SWREG255)

14.2.5.1.257.1 Offset

Register	Offset
SWREG255	3FCh

14.2.5.1.257.2 Diagram



14.2.5.1.257.3 Fields

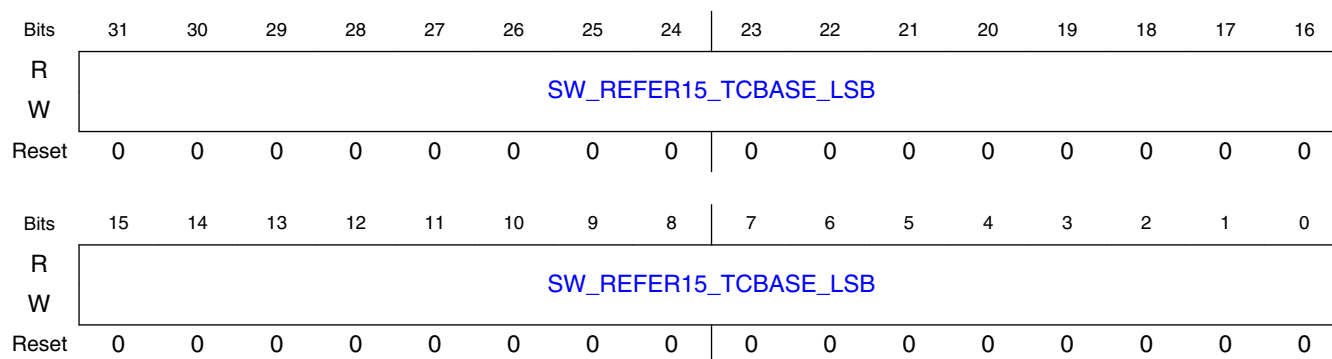
Field	Function
31-0 SW_REFER15_ TCBASE_MSB	Base address MSB (bits 63:32) for reference compress chrominance table index 15

14.2.5.1.258 Base address LSB (bits 31:0) for reference compress chrominance table index 15 (SWREG256)

14.2.5.1.258.1 Offset

Register	Offset
SWREG256	400h

14.2.5.1.258.2 Diagram



14.2.5.1.258.3 Fields

Field	Function
31-0 SW_REFER15_TCBASE_LSB	Base address LSB (bits 31:0) for reference compress chrominance table index 15

14.2.5.1.259 Not used (SWREG257)

14.2.5.1.259.1 Offset

Register	Offset
SWREG257	404h

14.2.5.1.259.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.259.3 Fields

Field	Function
31-0	Reserved.
—	

14.2.5.1.260 input stream buffer length (SWREG258)**14.2.5.1.260.1 Offset**

Register	Offset
SWREG258	408h

14.2.5.1.260.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SW_STRM_BUFFER_LEN															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SW_STRM_BUFFER_LEN															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

14.2.5.1.260.3 Fields

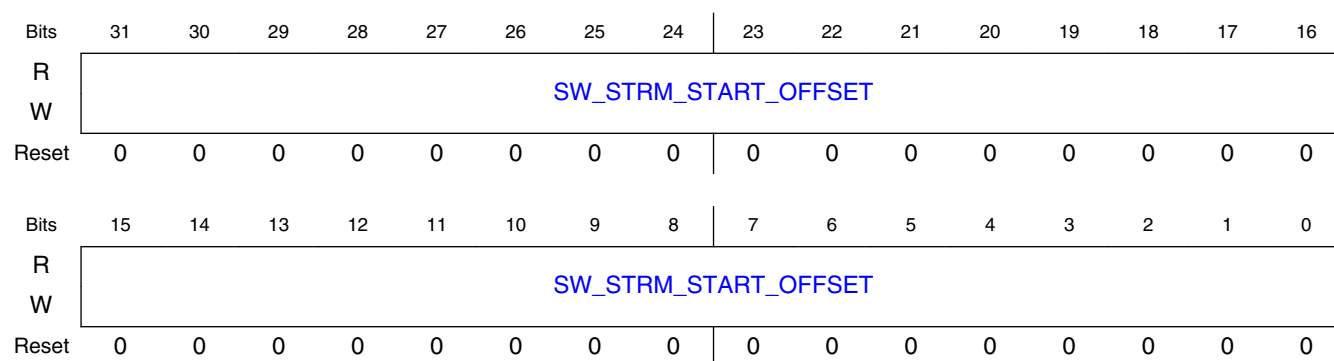
Field	Function
31-0 SW_STRM_BUFFER_LEN	input stream buffer length

14.2.5.1.261 input stream buffer start offset (SWREG259)

14.2.5.1.261.1 Offset

Register	Offset
SWREG259	40Ch

14.2.5.1.261.2 Diagram



14.2.5.1.261.3 Fields

Field	Function
31-0 SW_STRM_START_OFFSET	input stream buffer start offset

Chapter 15

Display Controller Subsystem (DCSS)

15.1 Display Controller Subsystem

15.1.1 Overview

The Display Controller Subsystem (DCSS) is used to source up to three display buffers, compose them, and drive a display using HDMI 2.0a with HDCP 2.2. The DCSS is intended to support up to 4kp60 displays. HDR10 image processing capabilities are included to provide a solution capable of driving next generation high dynamic range displays.

15.1.2 Features

The DCSS supports two major modes: HDR10 and Dolby Vision. These modes are mutually exclusive modes of operation. In HDR10 mode, there are 3 identical pipelines for video or graphics. In Dolby Vision mode, there are two pipelines, the Video pipeline (consists of 2 streams, base and enhanced layers) and the Overlay pipeline. The features of the DCSS include support for the following capabilities.

- Up to 4Kp60 video/graphics display over HDMI 2.0a with HDCP 2.2 encryption and audio formats including Dolby Digital, DTS, TrueHD, LPCM
- On-the-fly image composition of up to 3 source display buffers.
 - PIG (picture in GFX) overlay on video
 - PIP (picture in picture) scaled video on video with graphics
- On-the-fly decompression of compressed video or graphics frame buffers to optimize DRAM bandwidth.
- On-the-fly conversion of tile frame buffers to raster for display refresh.
- Integrated memory to store pre-fetched pixel data to manage DRAM access latencies.
- 3 independent 7x7 tap scaling to process 3 display buffers.

- Supports x7 up and x1/7 down scaling ratios, UHD/HD/SD/PIP
- PIP (picture in picture) scaled video on video with graphics
- Dolby or HDR10 image processing (see below for detailed feature list). These modes are mutually exclusive.
- Scaled output to DDR write back for 1 of three scaler engines (programmable). Used for large ratio downscaled images to manage dataflow from DDR and real time display processing.
- Read bus master to source raster frames from memory with direct input into Dolby and HDR10 pixel processing data pipes (bypass buffer conversion and scaling HW).
- CEA861 compatible display timing generator.
- Display state controller to manage display context changes.
- HDMI supporting HDCP output interface.
- YUV420/422/444 and ARGB with 8/10-bit per component source buffer formats

15.1.2.1 HDR10 Pixel Processing

In HDR10 mode, all three pixel processing pipelines are identical.

Each HDR10 pixel processing input pipeline supports

- 10-bit per component input
- Color space conversion
- LUT gamma correction per component

HDR10 Output Pipeline The HDR10 pixel processing output pipeline supports

- Linear to non-linear conversion per component
- Color space conversion
- 10-bit per component RGB/YUV444

15.1.2.2 Dolby Vision Pixel Processing

When the DCSS is configured to operate in Dolby Vision Mode, the follow capabilities are supported.

Dolby Vision Video pixel processing features supported.

- Algorithms
 - Dolby Vision base and enhanced layer processing and composition
 - Dolby Vision color space conversion
 - Dolby Vision color volume mapping
- Metadata and Tone Curve information processing.

- 422 10-bit pixel input
- Bypass supported for non-Dolby content

Dolby Overlay(Graphics) pixel processing features supported.

- Algorithms
 - Dolby Vision Input color volume conversion for overlay
 - Color space conversion
- Bypass supported for non-Dolby content
- Support for RGB and YUV422/444 8-bit pixel formats
- Tone Curve information processing

The overlay pipeline supports graphics and/or video overlay pixel processing. The video and graphics overlay pixels are composited at the output of these two Dolby pixel processing pipes. The overlay buffer can contain any content preprocessed by the SoC graphics accelerator or general purpose microprocessor. This can include preprocessed video and graphics composited buffers.

Dolby Output pixel processing features supported.

- Algorithms
 - Dolby Vision output color space mapping
 - Video+metadata scrambling
 - Dithering for 8/10 bit YUV output
- Output formats
 - Bypassed 8-bit RGB
 - 8/10-bit UV444

15.1.3 Use Cases

Some of the use cases the DCSS supports are described here:

- Video only – This can be optimized for power by turning off the CPUs and graphics and unused VPU decoders
- Graphics only - Supports GUI or gaming. This can be optimized for power by turning off the VPU
- Video with graphics overlay – Per pixel alpha blending supported
- PIG (picture in graphics) - Graphics with PIP video in corner
- PIP (picture in picture) – Full screen video with PIP video in corner
- Video with PIP and graphics – Full screen video with PIP video in corner and graphics overlay

Since the DCSS is a programmable display engine, the use cases are not limited to the cases listed above.

15.1.4 DCSS Block Diagram

The first diagram below contains the front end of the DCSS datapath, including all the busmaster components, decompression blocks, resolve memory, and scaler. All components feed the pixel processing components later in the DCSS pipeline.

The second diagram below contains the display pipeline, including the mutually exclusive Dolby and HDR10 processing pixel pipes, alpha blender, color subsampler, and HDMI controller and PHY.

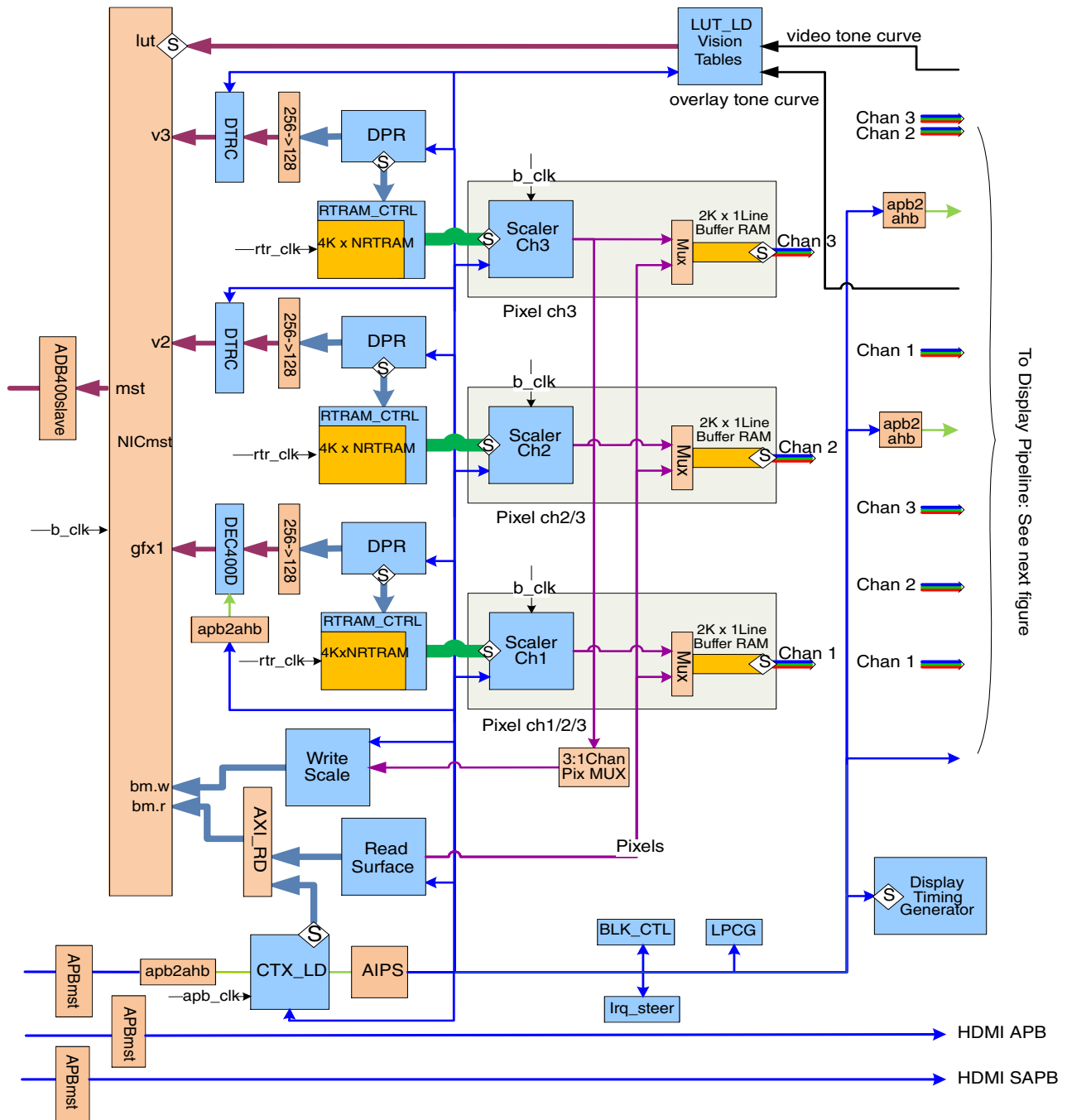


Figure 15-1. DCSS Front End Block Diagram

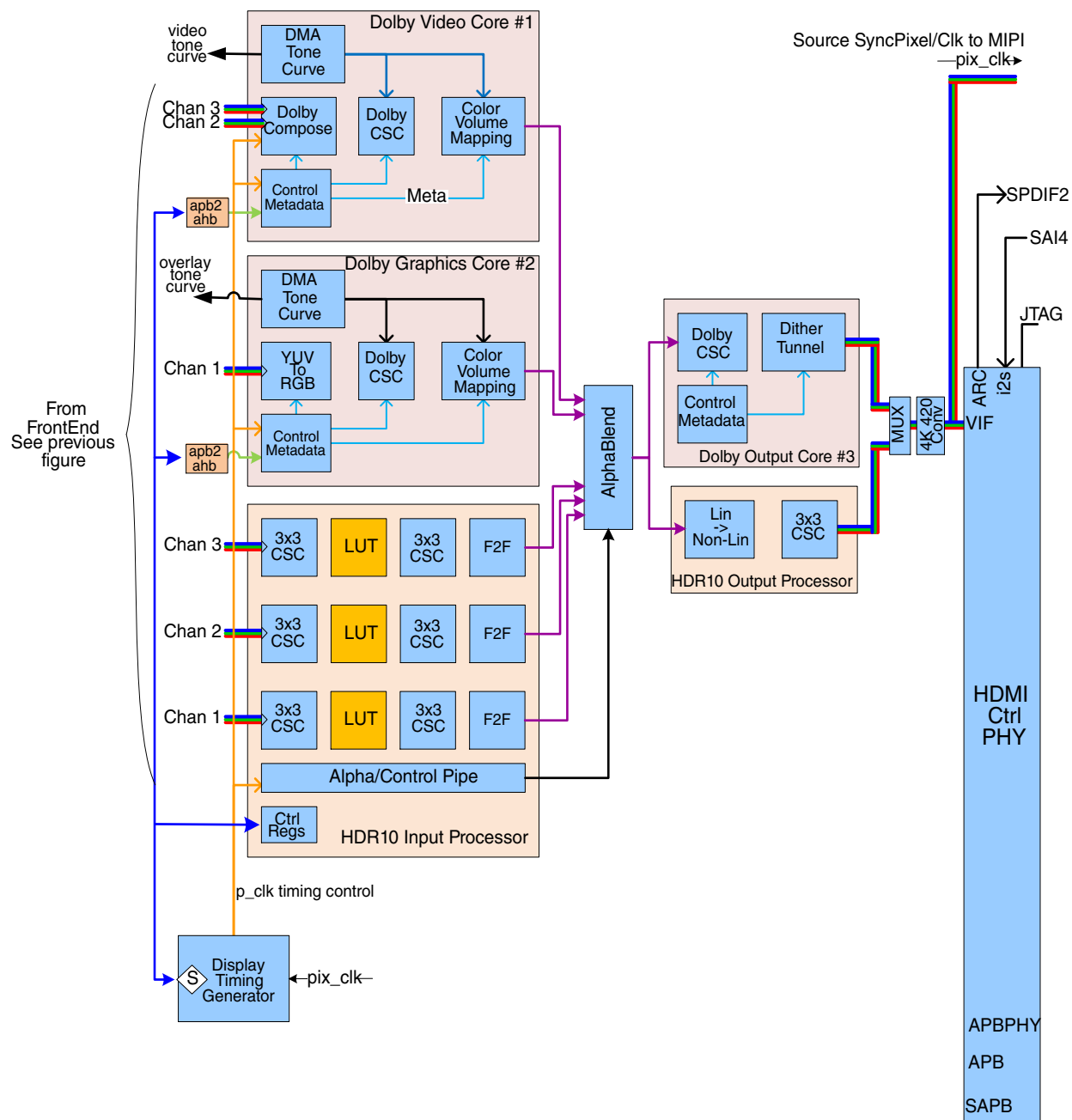


Figure 15-2. DCSS Pixel Processing Block Diagram

15.1.5 Memory Map and register definition

The HDMI controller has independent APB interfaces that are separate from the DCSS APB input bus. There is one HDMI APB for general purpose APB access, and another for secure SAPB access. Since they are separate DCSS input busses, their base address in the system is dependent on the address decoding outside of the DCSS. Refer to the top level memory map definition for the system.

The HDMI.APB interface uses 256KB of address space.

The HDMI.SAPB interface uses 64KB of address space.

The DCSS address space is configured as shown. All addresses are relative to the base address of the DCSS in the system. This memory map defines the offsets for all components from the base address which is defined outside of the DCSS.

Table 15-1. DCSS Memory Map

Start Address	End Address	Size	Allocation
0002_F000	0002_FFFF	4KB	BLK_CTL
0002_E000	0002_EFFF	4KB	LPCG See Note 1.
0002_D000	0002_DFFF	4KB	IRQ_STEER
0002_C000	0002_CFFF	4KB	HDMI_PHY registers
		28KB	Reserved
0002_4000	0002_4FFF	4KB	LUT_LD
0002_3000	0002_3FFF	4KB	CTX_LD
0002_2000	0002_2FFF	4KB	RD_SRC
0002_1000	0002_1FFF	4KB	WR_SCL
0002_0000	0002_0FFF	4KB	DTG
		12KB	Reserved
0001_C000	0001_CFFF	4KB	Scaler (All Channels 3 in 1)
0001_B000	0001_BFFF	4KB	SUBSAM
0001_A000	0001_AFFF	4KB	DPR-chan3
0001_9000	0001_9FFF	4KB	DPR-chan2
0001_8000	0001_8FFF	4KB	DPR-chan1
0001_7000	0001_7FFF	4KB	DTRC-chan3
0001_6000	0001_7FFF	4KB	DTRC-chan2
0001_5000	0001_7FFF	4KB	DEC400D-chan1
		8KB	Reserved
0001_2000	0001_2FFF	4KB	Dolby Vision Output (Core3)
0001_1000	0001_1FFF	4KB	Dolby Vision Graphics (Core2)
0001_0000	0001_0FFF	4KB	Dolby Vision Video (Core1)
0000_C000	0000_FFFF	16KB	HDR10 Tables and Registers
0000_8000	0000_BFFF	16KB	HDR10 Chan3 LUT
0000_4000	0000_7FFF	16KB	HDR10 Chan2 LUT
0000_0000	0000_3FFF	16KB	HDR10 Chan1 LUT

Note 1: The LPCG registers are at the base address in the table above. Each LPCG(i) register is located at an offset of $i*4$ bytes from the LPCG base address.

15.1.6 Functional description

At a high level, the DCSS consists of either three separate pipelines in HDR10 mode or two separate pipelines in Dolby Vision mode. These sources are composed to drive a single output display image.

The front end of the DCSS connects to DDR memory through an ADB400 slave interface with three decompression, DPR DMA channels, and three scalers. These component groups complete three separate channels for sourcing independent display buffers for display. Accesses for frame buffer data and control information is arbitrated by the bus interconnect component. There is also one write back channel and one read back channel, all clocked with the 800 MHz system bus clock. The display timing generator and the context loader work congruently to provide the overall DCSS control for updating the display between frame sequences. The pixel processing components (Dolby or HDR10) of the DCSS runs with a single display resolution and a single pixel clock. An HDMI 2.0 compliant controller and PHY complete the source solution for driving next generation HDR UHD displays with compatibility for current non-HDR HD displays.

The two operation modes, Dolby and HDR10, are mutually exclusive.

15.1.6.1 Operating Modes

15.1.6.1.1 Dolby Vision Mode Operation

In Dolby Vision mode, the video pipeline and the overlay pipeline get blended together, post processed and sent to the HDMI protocol controller. The HDMI output consists of a display window containing a video window and an overlay window. Both the video and overlay windows can be driven by the read back channel. Both the video and overlay windows can be full screen or smaller and offset inside the display window with the rest of the display window filled with the background color. The compositing of the video and overlay windows is controlled by the per pixel alpha channel. The overlay path creates an alpha based on the window priority and the graphics per pixel alpha. The diagram below shows a possible window configuration.

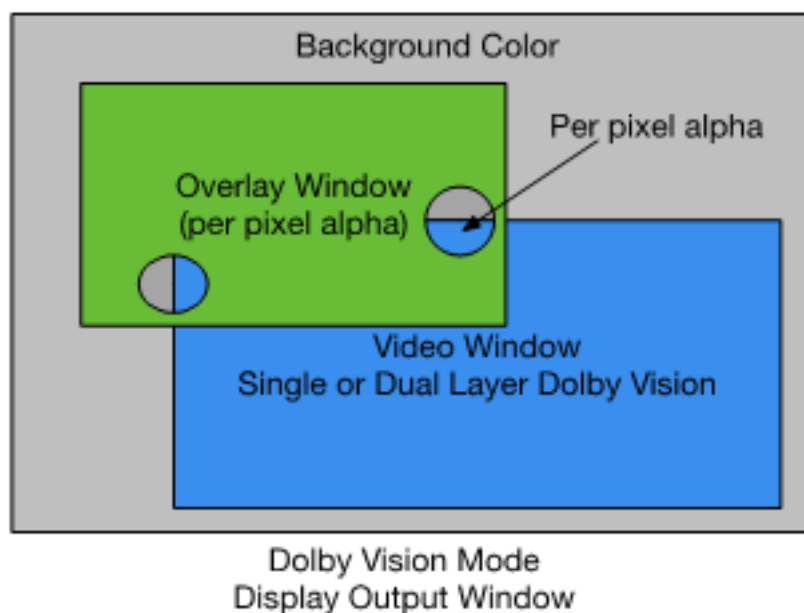


Figure 15-3. Dolby Processed Display

15.1.6.1.2 HDR10 Mode Operation

In HDR10 mode, the three inputs get merged or blended together, post processed and sent to the HDMI protocol controller. The HDMI output consists of a display window containing up to three input windows. Any of the three windows can be driven by the Readback channel. All three windows can be full screen or smaller and offset inside the display window with the rest of the display window filled with the background color. The compositing of the three windows is controlled by the per pixel alpha channel from channel 1. The alpha logic creates an alpha based on the window priority and the graphics per pixel alpha. The diagram below shows a possible window configuration. Note that window priority is fixed with channel 1 on top of channel 2 on top of channel 3. Only channel 1 can have a per pixel alpha.

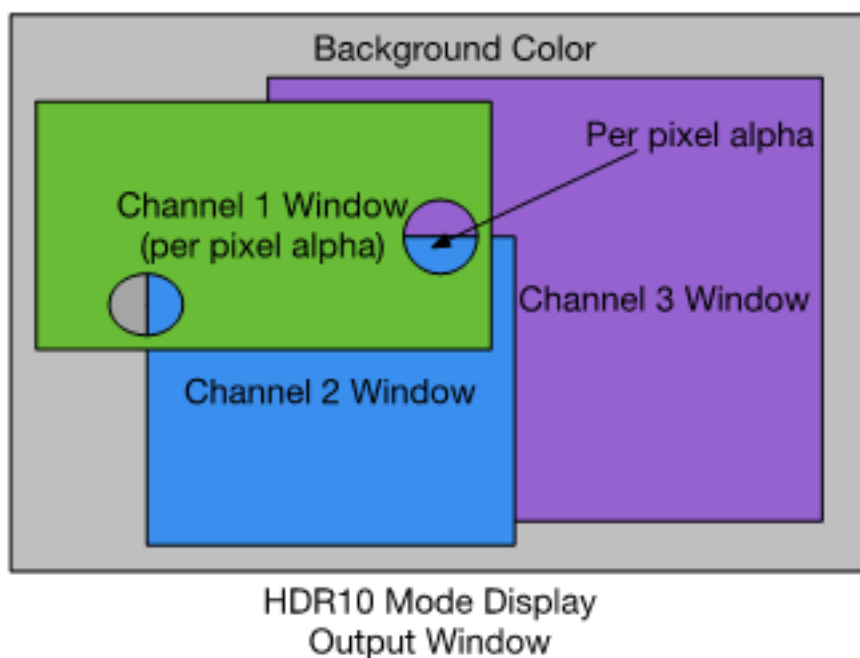


Figure 15-4. HDR10 Processed Display

15.1.6.2 Clocks

Clocks used in the DCSS are either sourced from the top level (see CCM) or the integrated HDMI PHY. The fundamental clock frequencies required to clock the DCSS are listed in the table below.

Table 15-2. DCSS Interrupt Mapping Table

Clock Domain	DCSS Clocked Components	Max Frequency
b_clk	NIC, DPR, RTRAM_CTRL, Scaler, etc	800MHz
rtr_clk	RTRAM_CTRL, DPR, Scaler. NOTE: the following condition must be met: $b_clk/2 \geq rtr_clk$	400MHz
apb_clk	All peripheral PIO access interfaces. All APB and core clocks for the HDMI controller.	133MHz
ss_refclock	Reference clock for HDMI PLL. This reference source clock is optional since the refclk_pln pins are the primary source for the reference clock for the PHY.	27MHz
p_clk	This is driven by the phy_pma_pixel_clk_out port of the PHY	max 594MHz

Table 15-2. DCSS Interrupt Mapping Table

Clock Domain	DCSS Clocked Components	Max Frequency
	(in HDMI mode), or the top level pixel clock input (in MIPI mode). Dolby Vision, HDR10, HDMI_CTRL, SUBSAM	This clock is programmed to the correct frequency to drive the desired display resolution and refresh rate. Refer to the CEA861 display timing specification for HDMI compliant pixel clock frequencies.

15.1.6.3 Interrupts

There are many sources for interrupts within the DCSS. The IRQ_STEER module is used to collect interrupts in the DC. SW can read the IRQ_STEER status bits to determine which DC interrupt sources are active. The IRQ_STEER provides the first level resource to determine the interrupt source. Refer to the chapter related to the IRQ_STEER module for detailed operation related to interrupt steering.

15.1.6.3.1 DCSS Interrupt Outputs

The DCSS dcss_int_out[0] is driven by internal_irq logically ANDed with IRQSTEER_CH0_MINTDIS[0] register. The DCSS dcss_int_out[1] is driven by the internal_irq directly. This bit is logically equal to dcss_int_out[0], with the exception that it is not gated with the IRQSTEER_CH0_MINTDIS[0] register. The DCSS dcss_int_out[2] is not routed through the IRQ_STEER module, but is driven directly from the HDMI PHY's source_hpd_out signal. This is the HDMI hot plug detect signal.

15.1.6.3.2 DTG General Purpose Interrupt Allocation

There are 8 general purpose interrupt collected by the Display Timing Generator (DTG). Refer to the DTG Interrupt Generation section for a description of the general purpose interrupt generator. The table below indicates the DCSS level allocation of the 8 interrupts implemented in the DTG.

Table 15-3. DTG Interrupt Map

DTG IRQ Input	Signal Source to generate interrupt
0	Panic Channel 0 – This is the panic signal output from the RTRAM_CTRL for channel 0.
1	Panic Channel 1 – This is the panic signal output from the RTRAM_CTRL for channel 1.
2	Panic Channel 2 – This is the panic signal output from the RTRAM_CTRL for channel 2.
3-7	reserved

15.1.6.3.3 DCSS Interrupt Mapping

The mapping of interrupts into the IRQ_STEER are defined by the following table. Details for the interrupt functionality including status and mask can be obtained from description of the respective interrupt registers within the source modules.

Table 15-4. DCSS Interrupt Mapping Table

IRQ Steer Interrupt Number	Interrupt Source
0	dolby_video_core
1	dolby_output_core
2	dolby_graphics_core
3	dpr_dc_ch1
4	dpr_dc_ch2
5	dpr_dc_ch3
6	ctx_ld
7	rd_src
8	dtg_programmable_1
9	dtg_programmable_2
10	dtg_programmable_3
11	dtg_programmable_4
12	dtg_dcsc_misc_8to1
13	hdmi_sirq
14	hdmi_pirq
15	dec400d_ch1
16	dtrc_ch2
17	dtrc_ch3
18	lut_ld
19	wr_scl
20	scaler_fifo_ch1_high_threshold_detect
21	scaler_fifo_ch2_high_threshold_detect
22	scaler_fifo_ch3_high_threshold_detect
23	scaler_fifo_ch1_low_threshold_detect
24	scaler_fifo_ch2_low_threshold_detect
25	scaler_fifo_ch3_low_threshold_detect
26	scaler_fifo_ch1_fifo_underrun_detect
27	scaler_fifo_ch2_fifo_underrun_detect
28	scaler_fifo_ch3_fifo_underrun_detect

15.2 DCSS Block Control (BLK_CTL)

15.2.1 Overview

The DC Block Control is the Display Controller Subsystem's general purpose control register block. The control registers contain various functionality including but not limited to:

- Async reset per clock domain
- Clock muxing control
- Async reset per clock domain
- HDMI source secure

15.2.2 Memory Map and Registers

15.2.2.1 register descriptions

15.2.2.1.1 MED_DCSS_BLK_CTL Memory map

MED_DCSS_BLK_CTL base address: 2_F000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Reset Control (RESET_CTRL)	32	RW	0000_0000h
4h	Reset Control (RESET_CTRL_SET)	32	RW	0000_0000h
8h	Reset Control (RESET_CTRL_CLR)	32	RW	0000_0000h
Ch	Reset Control (RESET_CTRL_TOG)	32	RW	0000_0000h
10h	Control (CONTROL0)	32	RW	0000_0000h
14h	Control (CONTROL0_SET)	32	RW	0000_0000h
18h	Control (CONTROL0_CLR)	32	RW	0000_0000h
1Ch	Control (CONTROL0_TOG)	32	RW	0000_0000h
20h	Spare Control0 (SPARE_CTRL0)	32	RW	0000_0000h
24h	Spare Control0 (SPARE_CTRL0_SET)	32	RW	0000_0000h
28h	Spare Control0 (SPARE_CTRL0_CLR)	32	RW	0000_0000h
2Ch	Spare Control0 (SPARE_CTRL0_TOG)	32	RW	0000_0000h
30h	Spare Control1 (SPARE_CTRL1)	32	RW	0000_0000h
34h	Spare Control1 (SPARE_CTRL1_SET)	32	RW	0000_0000h

Table continues on the next page...

Memory Map and Registers

Offset	Register	Width (In bits)	Access	Reset value
38h	Spare Control1 (SPARE_CTRL1_CLR)	32	RW	0000_0000h
3Ch	Spare Control1 (SPARE_CTRL1_TOG)	32	RW	0000_0000h
40h	Spare Status0 (SPARE_STATUS0)	32	RO	0000_0000h
44h	Spare Status0 (SPARE_STATUS0_SET)	32	RO	0000_0000h
48h	Spare Status0 (SPARE_STATUS0_CLR)	32	RO	0000_0000h
4Ch	Spare Status0 (SPARE_STATUS0_TOG)	32	RO	0000_0000h

15.2.2.1.2 Reset Control (RESET_CTRL)

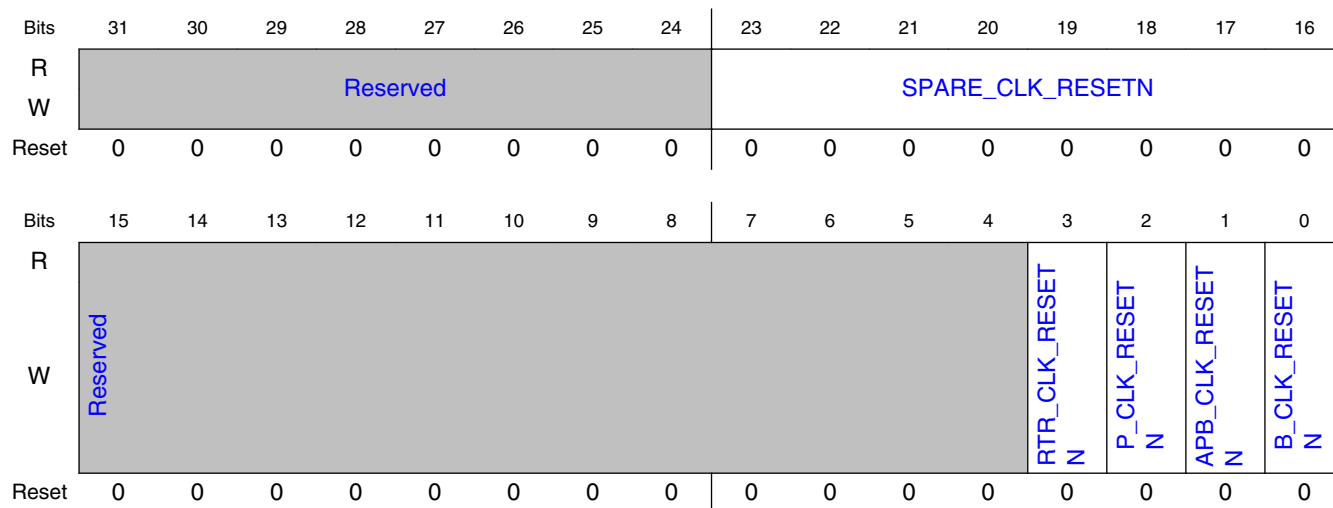
15.2.2.1.2.1 Offset

Register	Offset
RESET_CTRL	0h

15.2.2.1.2.2 Function

Asynchronous reset register control. There is one reset bit per clock domain. Each reset will be asynchronously asserted, and synchronized to synchronously deassert.

15.2.2.1.2.3 Diagram



15.2.2.1.2.4 Fields

Field	Function
31-24 —	Reserved.
23-16 SPARE_CLK_R ESETN	spare clk domain resetn Spare clock domain resets, implemented for potential ECO purposes. 0x0 = assert reset 0x1 = deassert reset
15-4 —	Reserved.
3 RTR_CLK_RES ETN	rtr_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
2 P_CLK_RESET N	p_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
1 APB_CLK_RES ETN	apb_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
0 B_CLK_RESET N	b_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset

15.2.2.1.3 Reset Control (RESET_CTRL_SET)

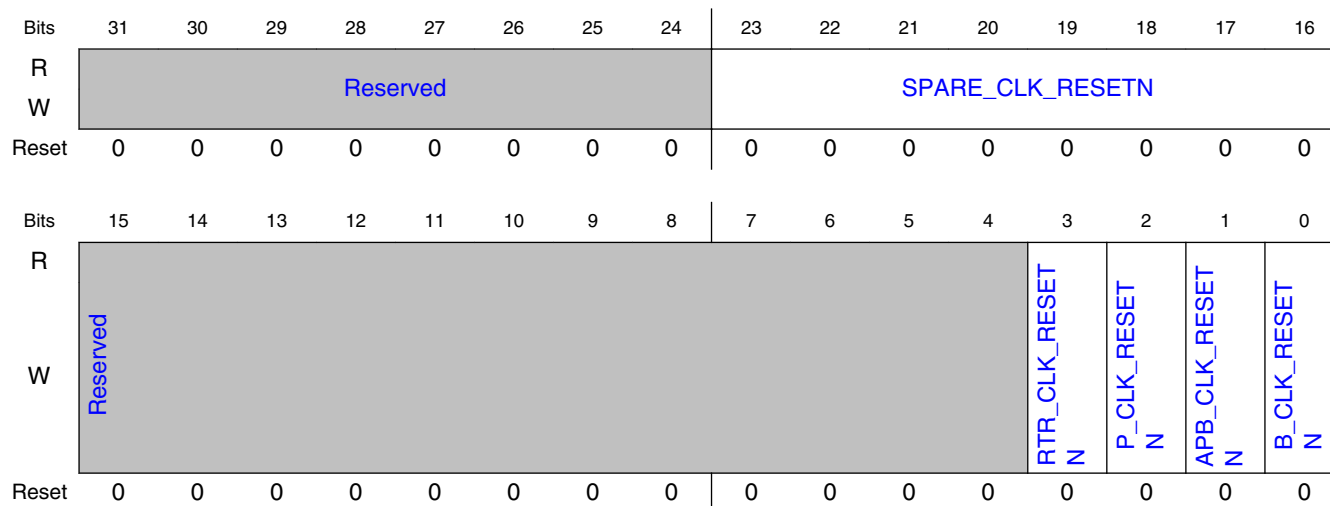
15.2.2.1.3.1 Offset

Register	Offset
RESET_CTRL_SET	4h

15.2.2.1.3.2 Function

Asynchronous reset register control. There is one reset bit per clock domain. Each reset will be asynchronously asserted, and synchronized to synchronously deassert.

15.2.2.1.3.3 Diagram



15.2.2.1.3.4 Fields

Field	Function
31-24 —	Reserved.
23-16 SPARE_CLK_RESETN	spare clk domain resetn Spare clock domain resets, implemented for potential ECO purposes. 0x0 = assert reset 0x1 = deassert reset
15-4 —	Reserved.
3 RTR_CLK_RESETN	rtr_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
2 P_CLK_RESETN	p_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
1 APB_CLK_RESETN	apb_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
0 B_CLK_RESETN	b_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset

15.2.2.1.4 Reset Control (RESET_CTRL_CLR)

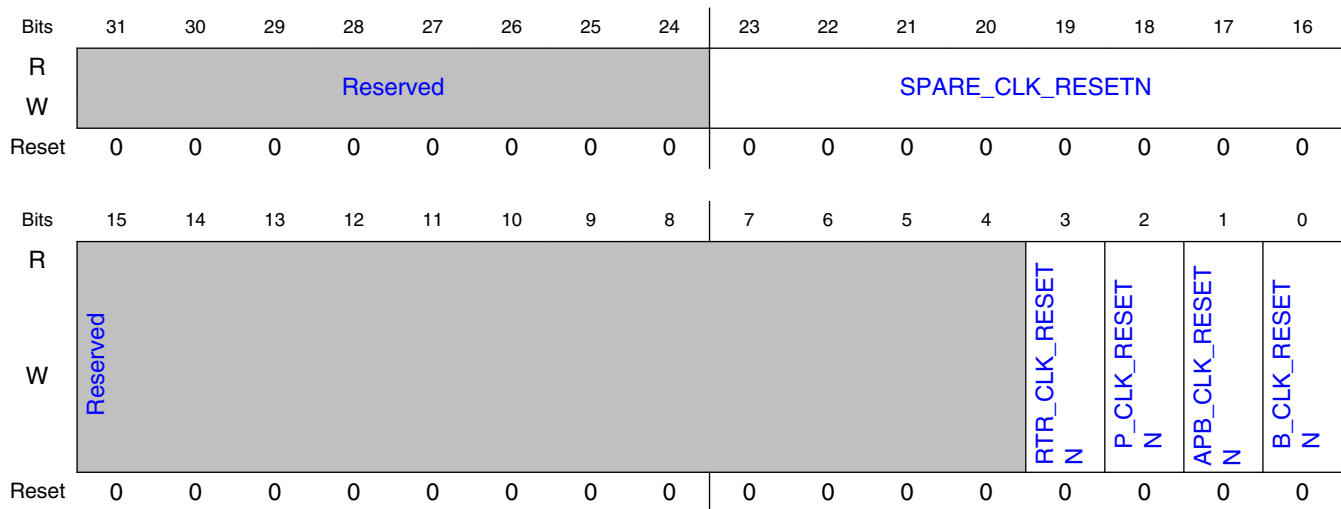
15.2.2.1.4.1 Offset

Register	Offset
RESET_CTRL_CLR	8h

15.2.2.1.4.2 Function

Asynchronous reset register control. There is one reset bit per clock domain. Each reset will be asynchronously asserted, and synchronized to synchronously deassert.

15.2.2.1.4.3 Diagram



15.2.2.1.4.4 Fields

Field	Function
31-24 —	Reserved.
23-16 SPARE_CLK_RESETN	spare clk domain reseth Spare clock domain resets, implemented for potential ECO purposes. 0x0 = assert reset 0x1 = deassert reset
15-4 —	Reserved.
3 RTR_CLK_RESETN	rtr_clk domain modules reseth 0x0 = assert reset 0x1 = deassert reset
2	p_clk domain modules reseth

Table continues on the next page...

Memory Map and Registers

Field	Function
P_CLK_RESET N	0x0 = assert reset 0x1 = deassert reset
1 APB_CLK_RESETN	apb_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
0 B_CLK_RESET N	b_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset

15.2.2.1.5 Reset Control (RESET_CTRL_TOG)

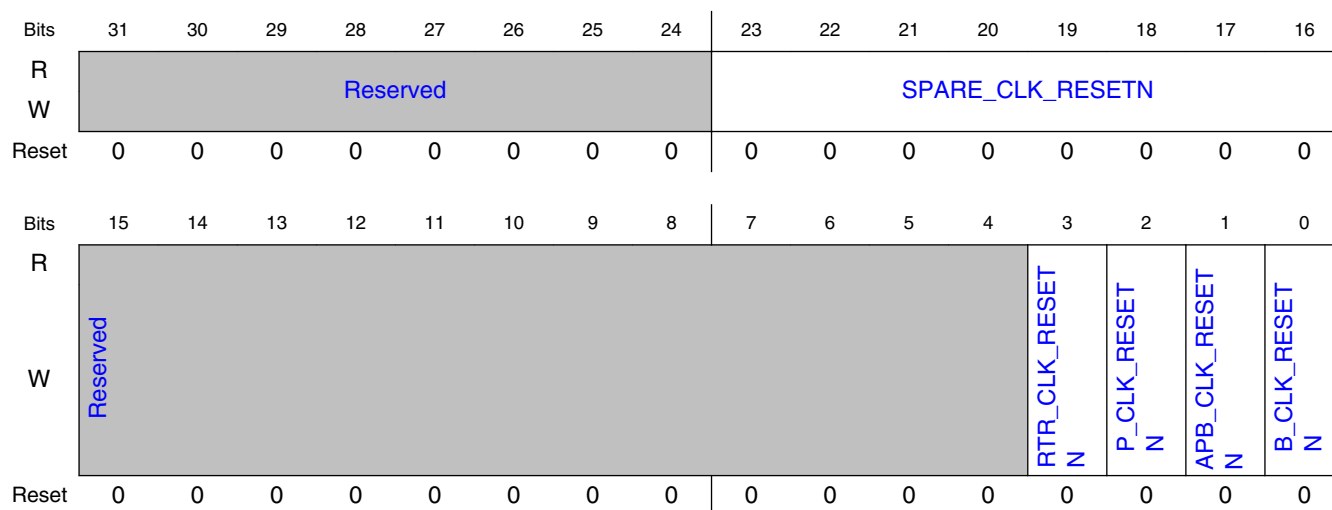
15.2.2.1.5.1 Offset

Register	Offset
RESET_CTRL_TOG	Ch

15.2.2.1.5.2 Function

Asynchronous reset register control. There is one reset bit per clock domain. Each reset will be asynchronously asserted, and synchronized to synchronously deassert.

15.2.2.1.5.3 Diagram



15.2.2.1.5.4 Fields

Field	Function
31-24 —	Reserved.
23-16 SPARE_CLK_R ESETN	spare clk domain resetn Spare clock domain resets, implemented for potential ECO purposes. 0x0 = assert reset 0x1 = deassert reset
15-4 —	Reserved.
3 RTR_CLK_RES ETN	rtr_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
2 P_CLK_RESET N	p_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
1 APB_CLK_RES ETN	apb_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset
0 B_CLK_RESET N	b_clk domain modules resetn 0x0 = assert reset 0x1 = deassert reset

15.2.2.1.6 Control (CONTROL0)

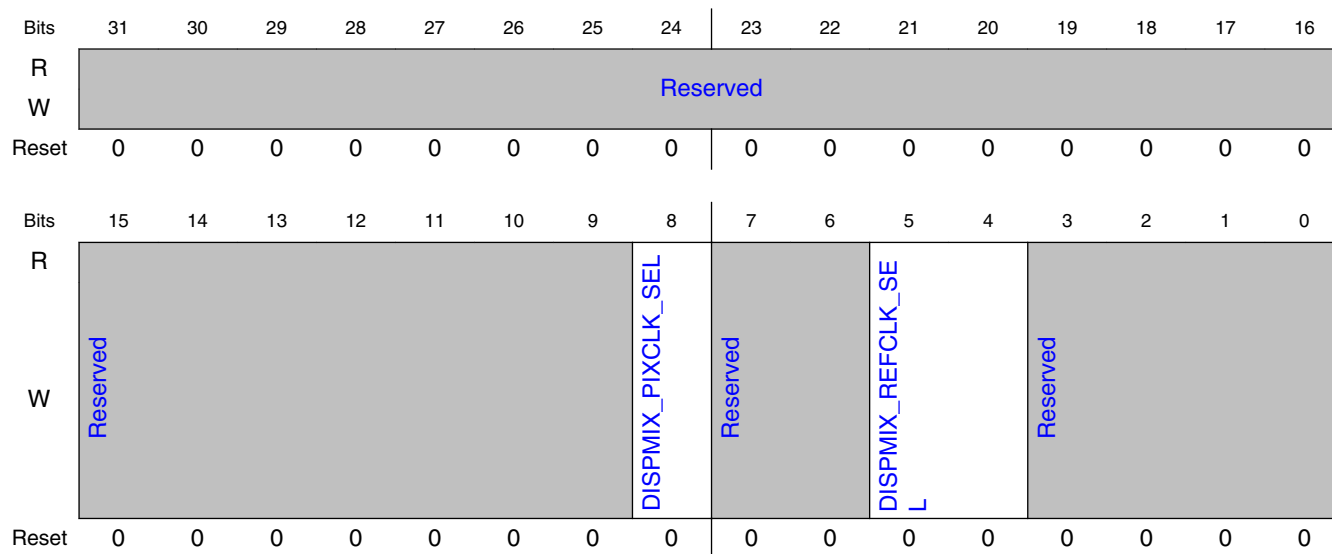
15.2.2.1.6.1 Offset

Register	Offset
CONTROL0	10h

15.2.2.1.6.2 Function

Misc. Control registers

15.2.2.1.6.3 Diagram



15.2.2.1.6.4 Fields

Field	Function
31-9 —	Reserved.
8 DISPMIX_PIXCLK_SEL	Display Subsystem Pixel Clock Select Pixel Clock source selection. 0x0 = Video PLL2 Clock 0x1 = CCM DC Pixel Clock
7-6 —	Reserved.
5-4 DISPMIX_REFCLK_SEL	Display Subsystem Reference Clock Select Reference clock source selection. 0x0 = 27 MHz Oscillator Reference Clock 0x1 = Video PLL2 Clock 0x2 = CCM DC Pixel Clock
3-0 —	Reserved.

15.2.2.1.7 Control (CONTROL0_SET)

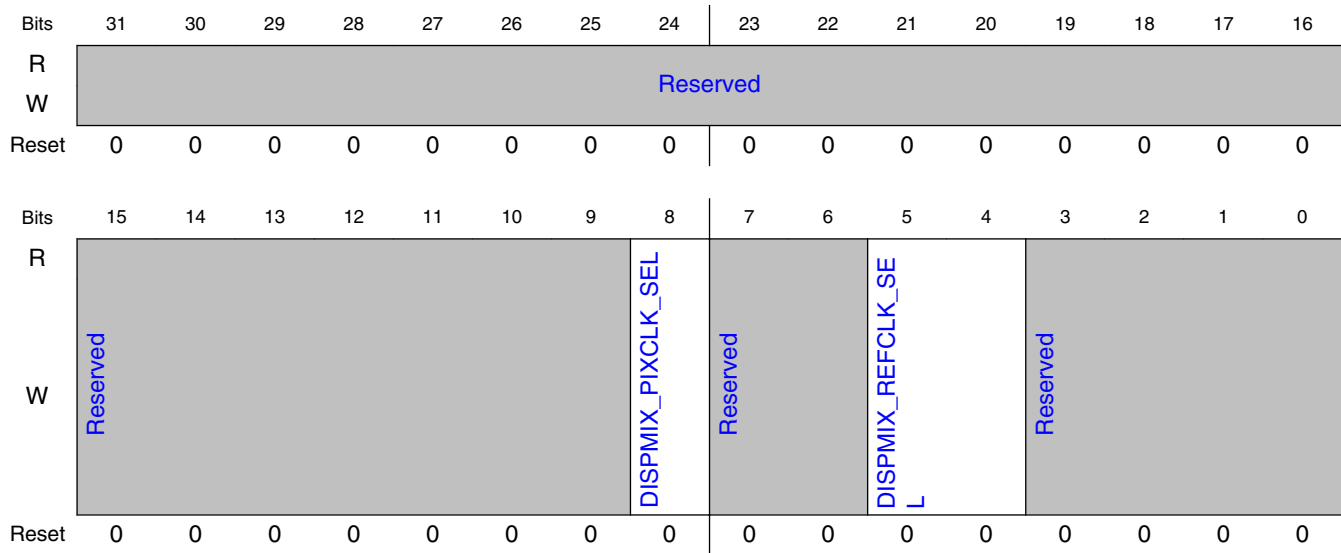
15.2.2.1.7.1 Offset

Register	Offset
CONTROL0_SET	14h

15.2.2.1.7.2 Function

Misc. Control registers

15.2.2.1.7.3 Diagram



15.2.2.1.7.4 Fields

Field	Function
31-9 —	Reserved.
8 DISPMIX_PIXCLK_SEL	Display Subsystem Pixel Clock Select Pixel Clock source selection. 0x0 = Video PLL2 Clock 0x1 = CCM DC Pixel Clock
7-6 —	Reserved.
5-4 DISPMIX_REFCLK_SEL	Display Subsystem Reference Clock Select Reference clock source selection. 0x0 = 27 MHz Oscillator Reference Clock 0x1 = Video PLL2 Clock 0x2 = CCM DC Pixel Clock

Table continues on the next page...

Memory Map and Registers

Field	Function
3-0	Reserved.
—	

15.2.2.1.8 Control (CONTROL0_CLR)

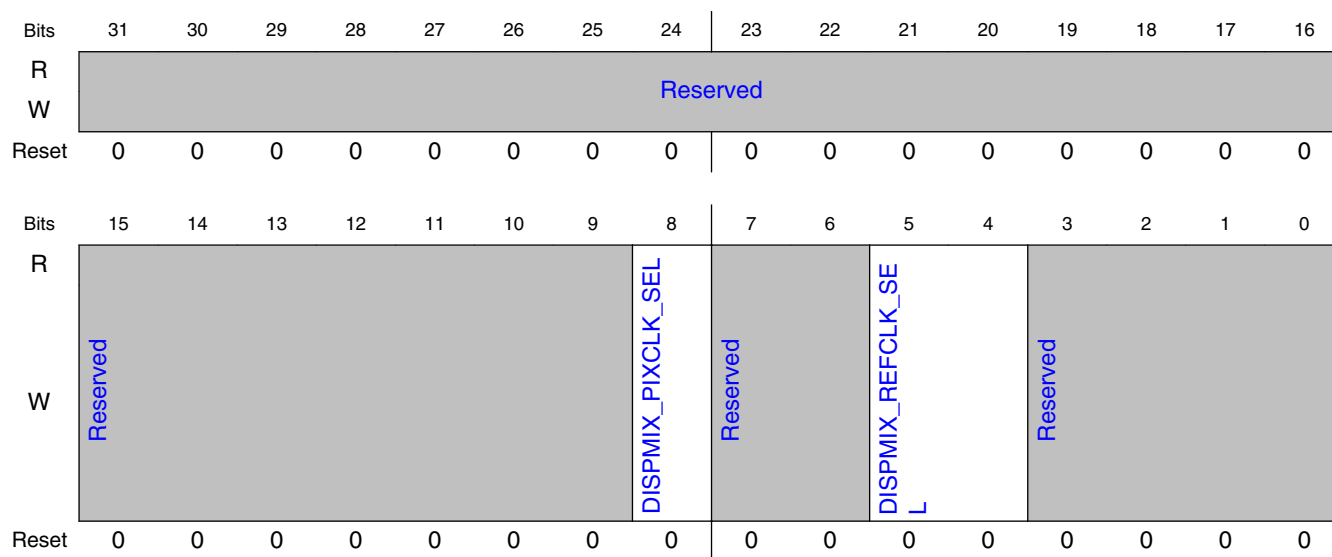
15.2.2.1.8.1 Offset

Register	Offset
CONTROL0_CLR	18h

15.2.2.1.8.2 Function

Misc. Control registers

15.2.2.1.8.3 Diagram



15.2.2.1.8.4 Fields

Field	Function
31-9	Reserved.
—	
8	Display Subsystem Pixel Clock Select

Table continues on the next page...

Field	Function
DISPMIX_PIXC LK_SEL	Pixel Clock source selection. 0x0 = Video PLL2 Clock 0x1 = CCM DC Pixel Clock
7-6 —	Reserved.
5-4 DISPMIX_REFC LK_SEL	Display Subsystem Reference Clock Select Reference clock source selection. 0x0 = 27 MHz Oscillator Reference Clock 0x1 = Video PLL2 Clock 0x2 = CCM DC Pixel Clock
3-0 —	Reserved.

15.2.2.1.9 Control (CONTROL0_TOG)

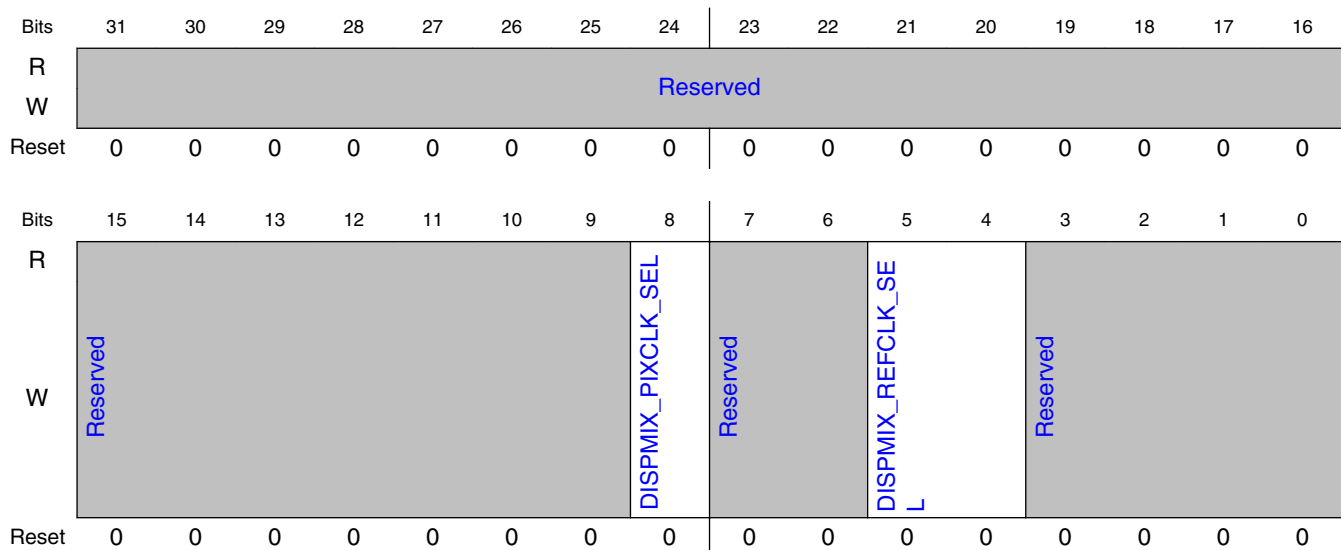
15.2.2.1.9.1 Offset

Register	Offset
CONTROL0_TOG	1Ch

15.2.2.1.9.2 Function

Misc. Control registers

15.2.2.1.9.3 Diagram



15.2.2.1.9.4 Fields

Field	Function
31-9 —	Reserved.
8 DISPMIX_PIXC LK_SEL	Display Subsystem Pixel Clock Select Pixel Clock source selection. 0x0 = Video PLL2 Clock 0x1 = CCM DC Pixel Clock
7-6 —	Reserved.
5-4 DISPMIX_REFC LK_SEL	Display Subsystem Reference Clock Select Reference clock source selection. 0x0 = 27 MHz Oscillator Reference Clock 0x1 = Video PLL2 Clock 0x2 = CCM DC Pixel Clock
3-0 —	Reserved.

15.2.2.1.10 Spare Control0 (SPARE_CTRL0)

15.2.2.1.10.1 Offset

Register	Offset
SPARE_CTRL0	20h

15.2.2.1.10.2 Function

Spare ECO Control Registers

15.2.2.1.10.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SPARE_CTRL															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SPARE_CTRL															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.2.2.1.10.4 Fields

Field	Function
31-0	SPARE Control
SPARE_CTRL	Spare control registers intended for ECO purposes.

15.2.2.1.11 Spare Control0 (SPARE_CTRL0_SET)

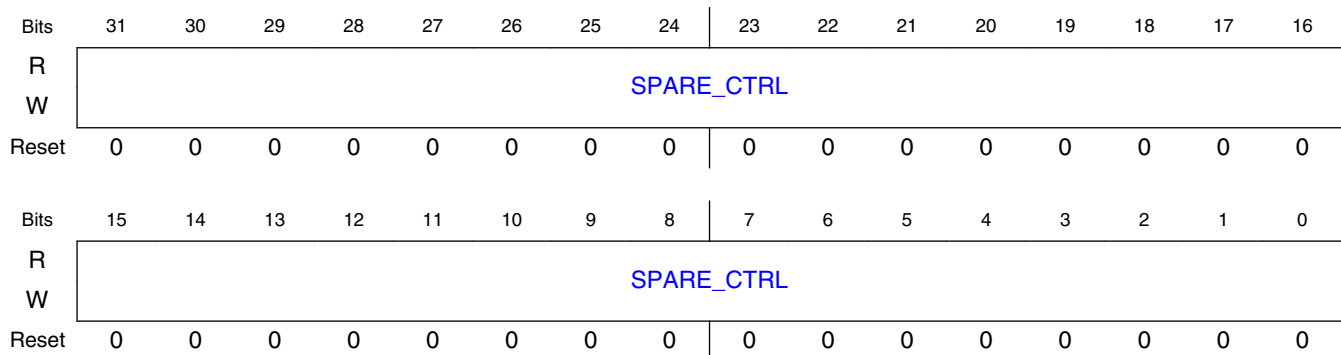
15.2.2.1.11.1 Offset

Register	Offset
SPARE_CTRL0_SET	24h

15.2.2.1.11.2 Function

Spare ECO Control Registers

15.2.2.1.11.3 Diagram



15.2.2.1.11.4 Fields

Field	Function
31-0	SPARE Control
SPARE_CTRL	Spare control registers intended for ECO purposes.

15.2.2.1.12 Spare Control0 (SPARE_CTRL0_CLR)

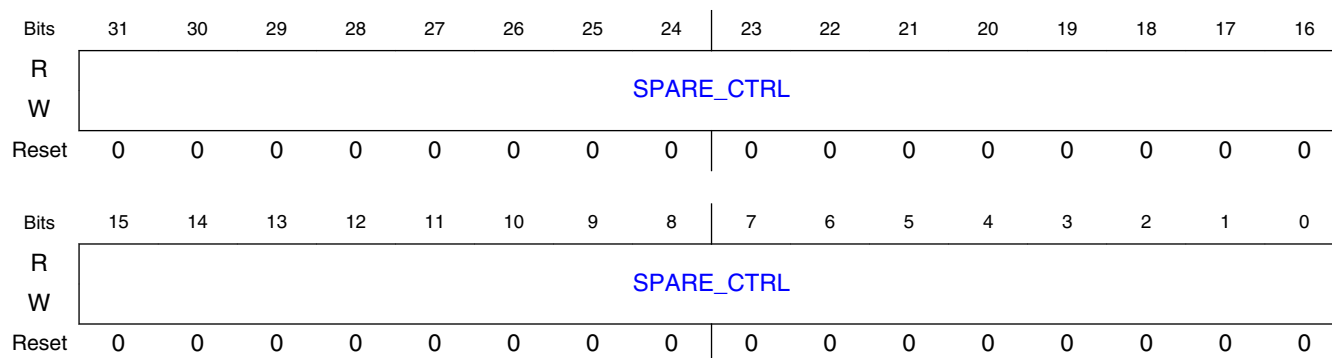
15.2.2.1.12.1 Offset

Register	Offset
SPARE_CTRL0_CLR	28h

15.2.2.1.12.2 Function

Spare ECO Control Registers

15.2.2.1.12.3 Diagram



15.2.2.1.12.4 Fields

Field	Function
31-0	SPARE Control
SPARE_CTRL	Spare control registers intended for ECO purposes.

15.2.2.1.13 Spare Control0 (SPARE_CTRL0_TOG)

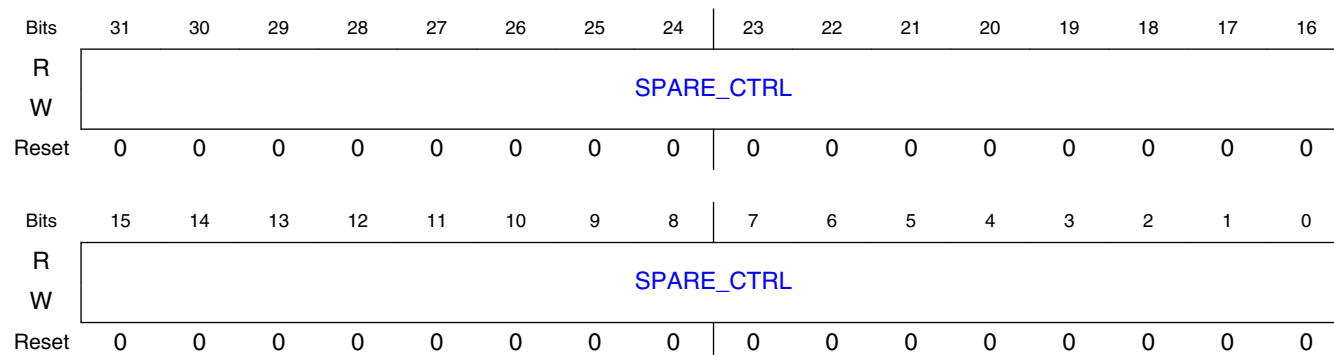
15.2.2.1.13.1 Offset

Register	Offset
SPARE_CTRL0_TOG	2Ch

15.2.2.1.13.2 Function

Spare ECO Control Registers

15.2.2.1.13.3 Diagram



15.2.2.1.13.4 Fields

Field	Function
31-0	SPARE Control
SPARE_CTRL	Spare control registers intended for ECO purposes.

15.2.2.1.14 Spare Control1 (SPARE_CTRL1)

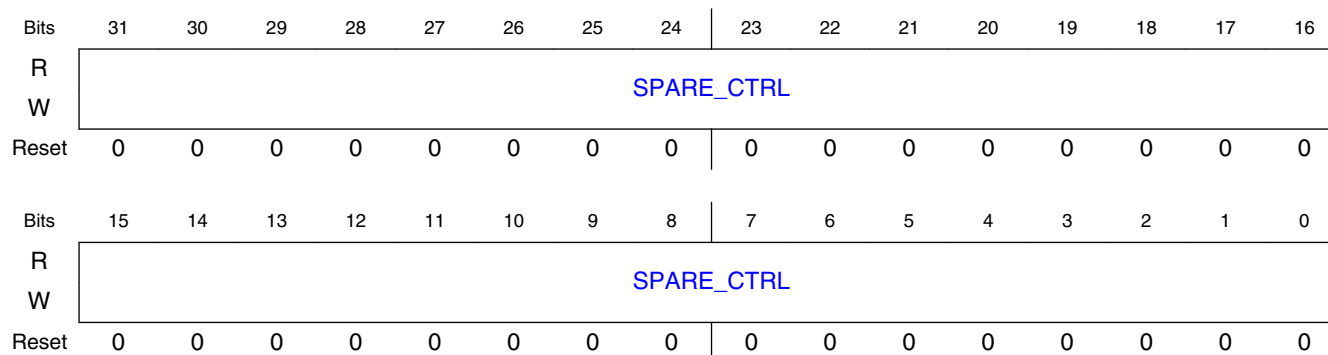
15.2.2.1.14.1 Offset

Register	Offset
SPARE_CTRL1	30h

15.2.2.1.14.2 Function

Spare ECO Control Registers

15.2.2.1.14.3 Diagram



15.2.2.1.14.4 Fields

Field	Function
31-0	SPARE Control
SPARE_CTRL	Spare control registers intended for ECO purposes.

15.2.2.1.15 Spare Control1 (SPARE_CTRL1_SET)

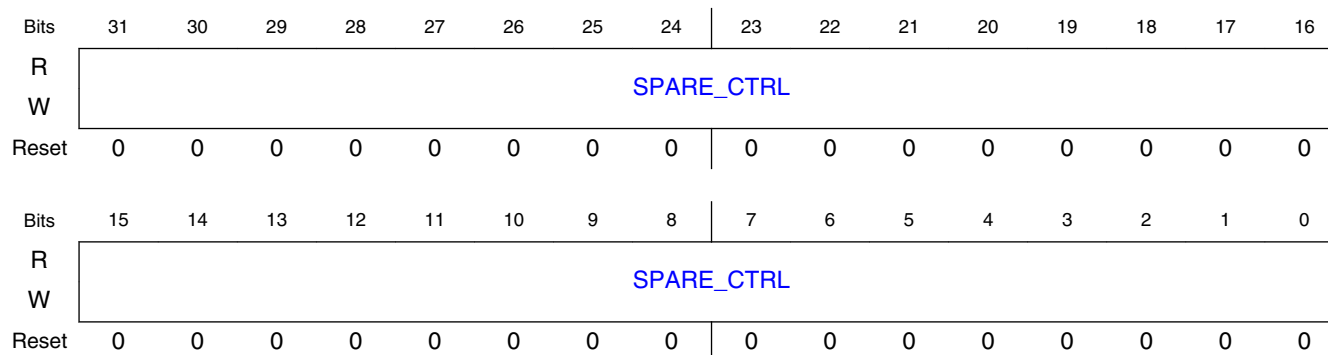
15.2.2.1.15.1 Offset

Register	Offset
SPARE_CTRL1_SET	34h

15.2.2.1.15.2 Function

Spare ECO Control Registers

15.2.2.1.15.3 Diagram



15.2.2.1.15.4 Fields

Field	Function
31-0	SPARE Control
SPARE_CTRL	Spare control registers intended for ECO purposes.

15.2.2.1.16 Spare Control1 (SPARE_CTRL1_CLR)

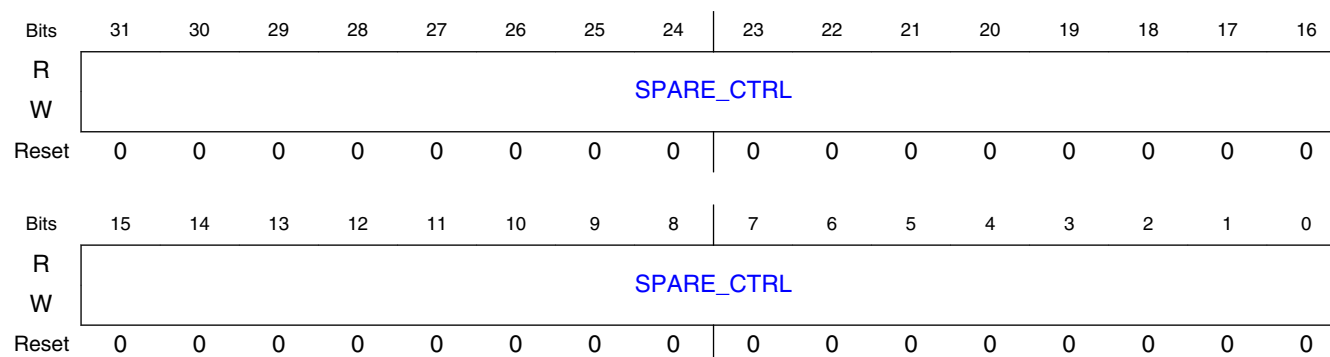
15.2.2.1.16.1 Offset

Register	Offset
SPARE_CTRL1_CLR	38h

15.2.2.1.16.2 Function

Spare ECO Control Registers

15.2.2.1.16.3 Diagram



15.2.2.1.16.4 Fields

Field	Function
31-0	SPARE Control
SPARE_CTRL	Spare control registers intended for ECO purposes.

15.2.2.1.17 Spare Control1 (SPARE_CTRL1_TOG)

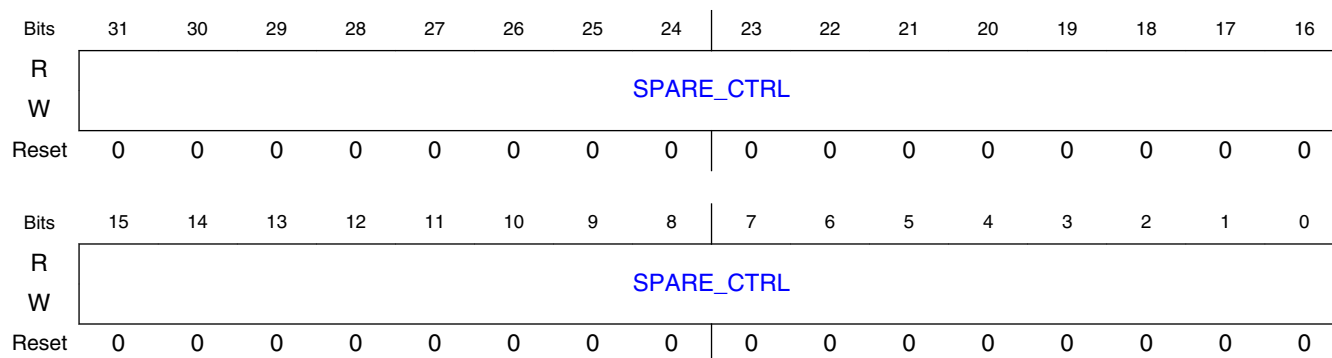
15.2.2.1.17.1 Offset

Register	Offset
SPARE_CTRL1_TOG	3Ch

15.2.2.1.17.2 Function

Spare ECO Control Registers

15.2.2.1.17.3 Diagram



15.2.2.1.17.4 Fields

Field	Function
31-0	SPARE Control
SPARE_CTRL	Spare control registers intended for ECO purposes.

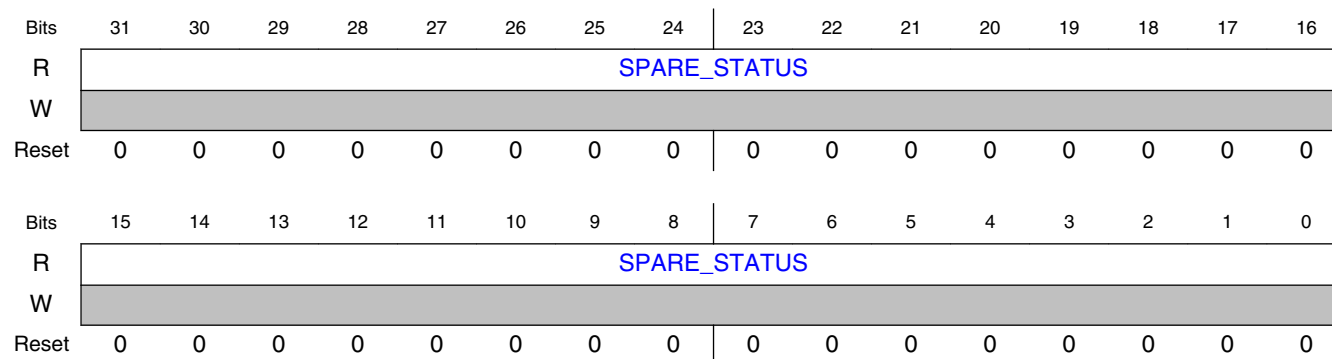
15.2.2.1.18 Spare Status0 (SPARE_STATUS0)

15.2.2.1.18.1 Offset

Register	Offset
SPARE_STATUS0	40h

15.2.2.1.18.2 Function

Spare ECO Control Status Registers

15.2.2.1.18.3 Diagram**15.2.2.1.18.4 Fields**

Field	Function
31-0	SPARE Control
SPARE_STATUS	Spare control status registers intended for ECO purposes.

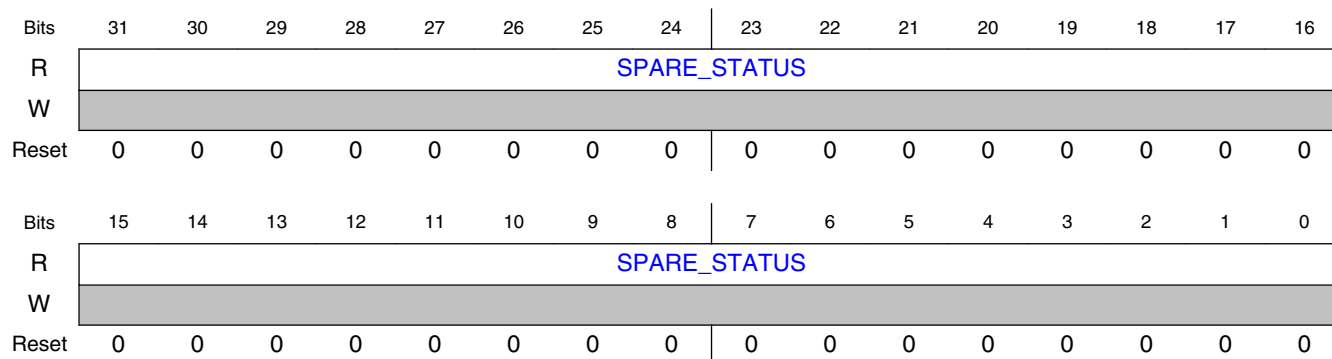
15.2.2.1.19 Spare Status0 (SPARE_STATUS0_SET)**15.2.2.1.19.1 Offset**

Register	Offset
SPARE_STATUS0_SET	44h

15.2.2.1.19.2 Function

Spare ECO Control Status Registers

15.2.2.1.19.3 Diagram



15.2.2.1.19.4 Fields

Field	Function
31-0	SPARE Control
SPARE_STATUS	Spare control status registers intended for ECO purposes.

15.2.2.1.20 Spare Status0 (SPARE_STATUS0_CLR)

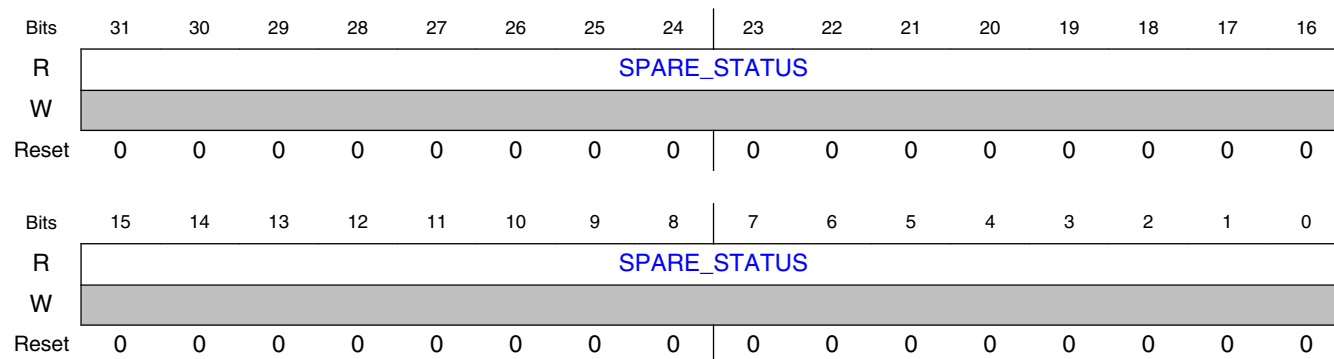
15.2.2.1.20.1 Offset

Register	Offset
SPARE_STATUS0_CLR	48h

15.2.2.1.20.2 Function

Spare ECO Control Status Registers

15.2.2.1.20.3 Diagram



15.2.2.1.20.4 Fields

Field	Function
31-0	SPARE Control
SPARE_STATUS	Spare control status registers intended for ECO purposes.

15.2.2.1.21 Spare Status0 (SPARE_STATUS0_TOG)

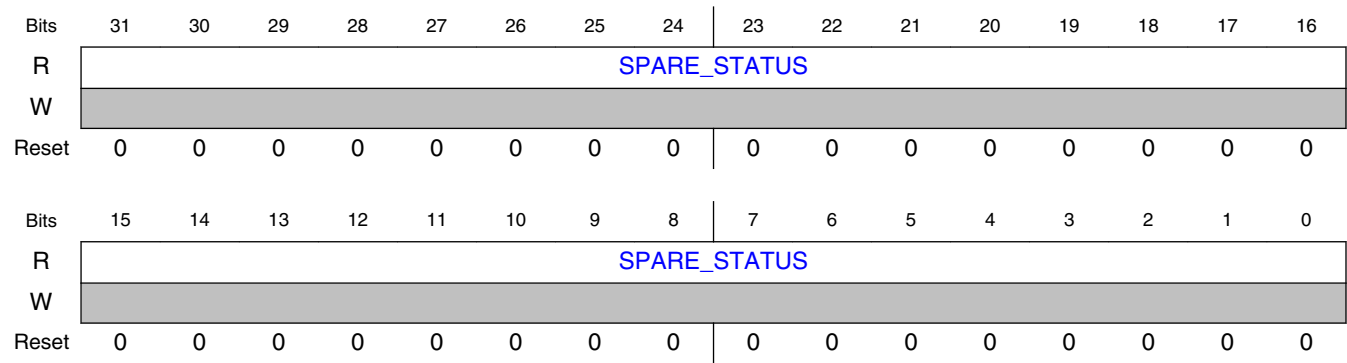
15.2.2.1.21.1 Offset

Register	Offset
SPARE_STATUS0_TOG	4Ch

15.2.2.1.21.2 Function

Spare ECO Control Status Registers

15.2.2.1.21.3 Diagram



15.2.2.1.21.4 Fields

Field	Function
31-0	SPARE Control
SPARE_STATUS	Spare control status registers intended for ECO purposes.

15.3 Display Timing Generator (DTG)

15.3.1 Overview

The purpose of the Display Timing Generator (DTG) is to generate the display timing control signals for the display and the pixel data path in the pixel clk domain. The outputs from the DTG controls the pixel data path and front end components (scaler, DPR, CTX_LD).

15.3.1.1 Block Diagram

The following diagram gives a high-level overview of the configuration of the DTG.

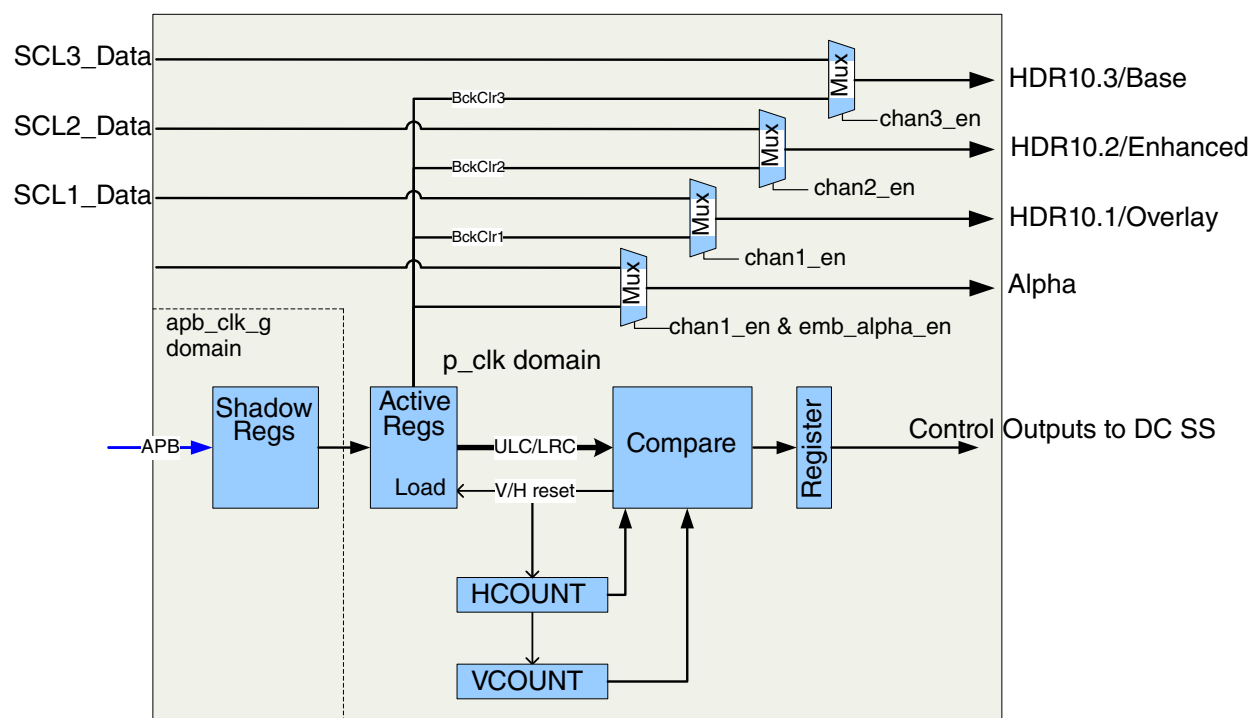


Figure 15-5. DTG block diagram

15.3.1.2 Features

The DTG contains the following components:

- **Shadow Registers** – These registers are loaded in advance of the next vertical retrace by system SW and used to load the active registers to setup for the next display trace at the vertical retrace time.
- **Active Registers** – These registers are loaded from the shadow registers in one clock in parallel. This occurs at the single p_clk edge where hcount = csr_dtg_lrc_x and vcount = csr_dtg_lrc_y.
- **Hcount, vcount** – Counters that determine the position in the display trace.
- **Compare** – Comparators that create active control signals for all DCSS components.
- **Register** – All control outputs are registered.
- **Datapath** – Muxes that select the scaler output data, or programmed background color based on whether the current H/Vcount are in the respective active regions.

Examples of control outputs from the DTG are:

- **VActive_n, HActive, HSYNC** – Note, these signals map to Dolby and HDR10 specific timing inputs. They are NOT equal to the traditional definition of display VSYNC/HSYNC/DE.

```

VActive_n = dolby_vsync_n
HActive = dolby_hsync
DE = dolby_de

```

- PS_EN[3,2,1] – Pixel Stream Enable, also referred to as VIDEO_ENABLE and OVERLAY_ENABLE. These active times are defined by the parameters *_ULC[X,Y]/*_LRC[X,Y], or the upper left and lower right display coordinates for the three pixel streams. These signals drive the bl_de, el_de, and de_in of the Dolby video and graphics pixel cores and the read_enable into the three scaler output FIFOs.
- CTX_LD__[DB|SB]_EN – Issue a control signal that indicates when a new display state can be loaded by the display context loader. There is one event for components with doubled buffered shadow registers, another with single buffered CSRs, since the timing events occur at different times in a single display trace.
- CONTROL[31:0] – Additional control signals that are used as the design evolves. These can be reduced at the final design release to match that actual implementation. Individual CSRs are used to determine when these signals are active within the display trace, and if they are horizontal or vertical control signals. Vertical means they are qualified by the horizontal count resetting to zero.

15.3.2 Functional description

15.3.2.1 DTG Timing Generator

The purpose of the DTG is to generate the display timing control signals for the display and the pixel data path in the p_clk domain. The DTG is fully synchronous to the p_clk domain except for the first set of double buffered CSR registers, also referred to as “shadow” registers. The shadow registers are synchronous to the apb_clk_s domain. The shadow registers are loaded into the active registers at the p_clk that defines the “origin” of the display. The active registers are clocked on p_clk and loaded when the shadow registers are not actively being updated. The outputs from the DTG control the pixel data path and front end components (scaler, DPR, CTX_LD). The “origin” of the display is defined as the p_clk cycle that causes the hcount and vcount to be set to zero. This occurs when the respective counters, hcount/vcount, reach their maximum programmed values and the p_clk results in hcount = vcount = 0x0. The origin of the display control is not arbitrarily defined, but must be the point at which the display enters the vertical front porch. This is the p_clk cycle that is just after the last pixel is sent to the display, or the lower right most pixel on the display. This origin is required to meet the proper

initialization of the display IP (Dolby and HDR10 components) AND to achieve synchronization with the output timing generator, or SUBSAM, that produces CEA861 HDMI compliant timing.

The Display Controller Subsystem (DCSS) does not support on-the-fly display resolution changes. Any display timing registers that effect the display resolution or refresh rate require the DCSS (and all components integrated) to be stopped, reset, and restarted with the correct pixel clock frequency and display parameters to drive the new resolution. One reason is that the SUBSAM (see subsequent descriptions) module doesn't implement double buffered display timing registers. Also, for a resolution update, the PLL, PHY, and clocking needs to be reinitialized. The double buffered registers that could be modified in the case of a display context change, and reloaded at retrace back to the origin, would be those that locate the three channels of display data within the active display region. In this case, the display resolution and timing is not changing, but the timing of the channels 1, 2, or 3, within the current display configuration, can change on-the-fly. This is one example of DTG registers that could change on-the-fly. Considering the transfer of the registers from shadows to active registers at retrace back to the origin, it is up to the application to determine which registers can be loaded using the double buffered scheme on-the-fly.

The CEA861 timing is generated at the end of the pixel data path in the SUBSAM module. The CEA861 timing is required to comply with the HDMI interface specification and provide the correct interface timing to the HDMI controller. The DTG timing is slightly different than the CEA861 timing. DTG produces the timing needed by Dolby and HDR10 pixel data path components, and the SUBSAM produces the CEA861 timing for HDMI controller and spec compliance. Refer to section Display Timing and Window Overlay for more information on display timing. A timing diagram that indicates the origin of the display is provided below. ALL timing parameters need to be programmed with the origin defined as the start of the vertical front porch AND the start of the horizontal front porch. The pixel data path IP (Dolby and HDR10) doesn't consider porches. There is only a blanking region, and a vertical active region (VActive). The vertical active region, VActive, is initiated when the VActive signal goes low. The vertical blanking region is initiated when the VActive signal goes high. The DTG and pixel data path logic needs to see the rising edge of the VActive signal as the first transition when the DTG is enabled. The reset state of the VActive signal is low. The low to high transition on VActive is the transition from the active display into the vertical front porch, and this event marks the origin (start) of the display trace.

In addition to timing and control, the DTG also implements the data path muxes for background color and alpha values.

NOTE

The timing of the VActive signal needs to be precise. No additional timing pulses during start up can exist since other components depend on a clean startup sequence.

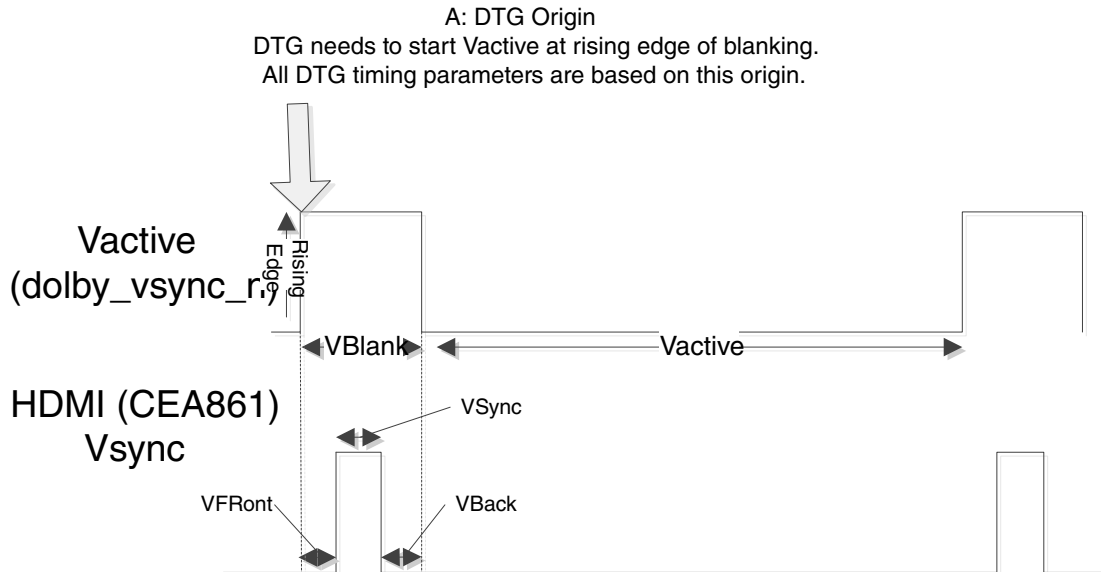


Figure 15-6. DTG Timing Diagram

15.3.2.2 DTG Window Overlay

The DTG generates the timing and control for the display and the pixel processing pipelines (Dolby/HDR10). Also, it generates timing for the CTX_LD to control when next state information is loaded to all DCSS components. An example of a display configuration is indicated as in the below diagram.

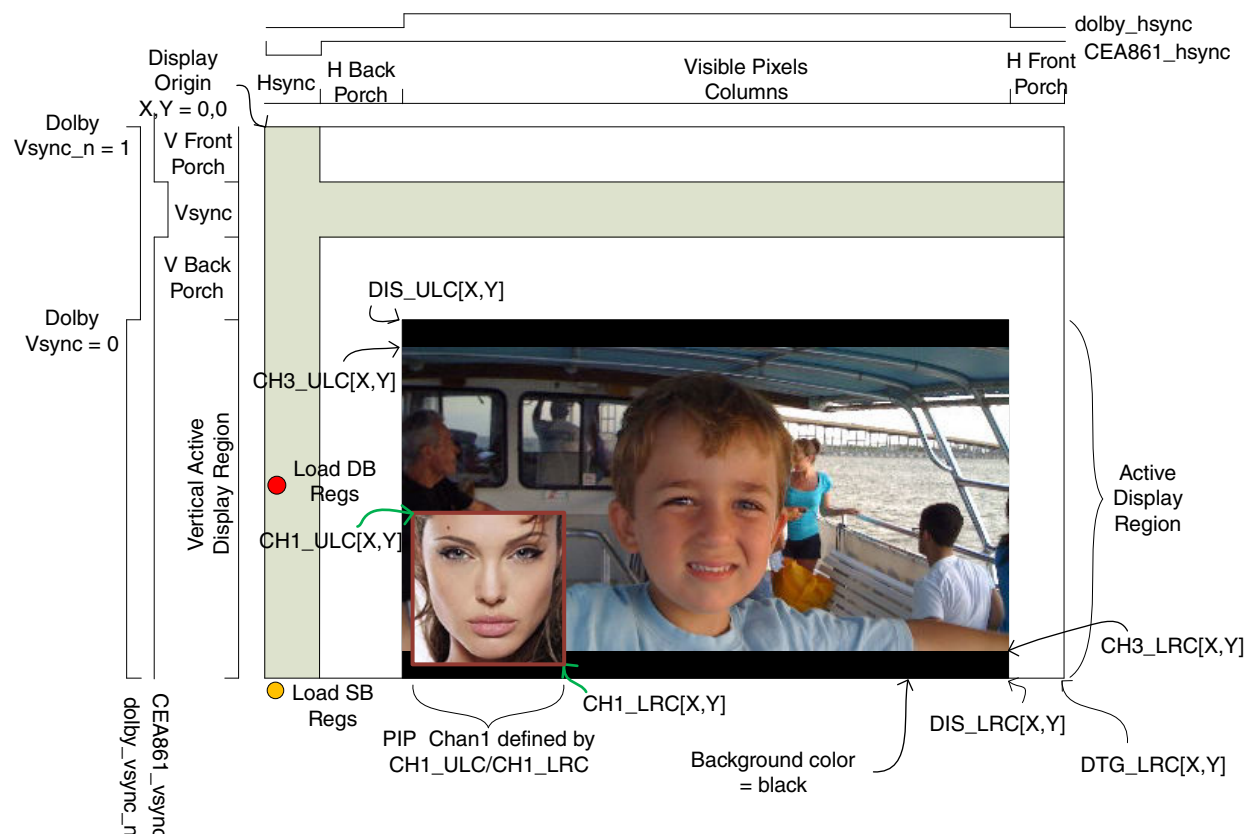


Figure 15-7. DTG Display Configuration

The DTG determines timing for the following events:

- When hcount and vcount counters are reset to zero, defined by $\text{TC_DTG_LOWER_RIGHT_}[X,Y]$, respectfully.
- When the active display region starts (Upper Left Coordinate or $\text{TC_DISPLAY_UPPER_LEFT_}[X,Y]$) and ends (Lower Right Coordinate or $\text{TC_DISPLAY_LOWER_RIGHT_}[X,Y]$).
- When Ch3/BL, starts/ends within the active display region defined by $\text{TC_CHANNEL_3_UPPER_LEFT_}[X,Y]$ /
 $\text{TC_CHANNEL_3_LOWER_RIGHT_}[X,Y]$. It is the same as the active display region in this example.
- When CH2/EL starts/ends indicated by $\text{TC_CHANNEL_2_UPPER_LEFT_}[X,Y]$ /
 $\text{TC_CHANNEL_2_LOWER_RIGHT_}[X,Y]$. This is the PIP window in this example.
- When the CH1/Overlay start/ends indicated by $\text{TC_CHANNEL_1_UPPER_LEFT_}[X,Y]$ /
 $\text{TC_CHANNEL_1_LOWER_RIGHT_}[X,Y]$.
- When the context loader fetches register contents for double buffered (Dolby/DTG), indicated by the red dot, and single buffered (DPR/bus masters) components, indicated by the orange dot, in the DCSS.

The DTG output signals drive the Dolby Vision or HDR10 pixel processing IP and also drive the pixel_enables of the scaler output buffers. Also, the DTG drives the context loader enable with several event triggers during a single frame trace. This is required since the Dolby IP and DTG have double buffered registers that need to be preloaded during the active display time. Other processing components may not have double buffered registers (NXP IP), so these parameters need to be loaded during the vertical blanking time (not during active display time). The context loader bus masters the display state for the next display trace for the Dolby IP and NXP IP at programmable times determined by the DTG.

The display state sequence controlled by the DTG is:

1. At the Red Dot, Context Load DB Regs – Context load engine bus masters the double buffered registers in the Dolby IP and DTG. Must follow the sequence using the Metadata Copy, Start Bit, and Finish Bit (see Dolby IP specs).
2. VSYNC Active – Active CSR registers automatically loaded from shadow registers.
3. At the Orange Dot, Context Load SB Regs - Context load engine bus masters the register contents from DRAM to the single buffered IP control registers.
4. Prefetch/Process – the DPRs prefetch data and store rows of blocks into the RTRAM. The RTRAM_CTRL indicates when data is available to the scalers to begin processing. The rd_src engine also prefetches a buffer if enabled.
5. Display – When in the active display region, the PS_EN signals modulate when pixel data is transferred from the scaler output buffers and consumed by the Dolby pixel data path cores. HSYNC indicates an increment in each line.
6. Display Front Porch – goto #1.

15.3.2.2.1 HCount

The DTG uses a horizontal and vertical counter and the necessary parameters to maintain the sequence of the display. The horizontal counter (hcount) increments once every p_clk and reset back to zero when the hcount counter matches the TC_DTG_LOWER_RIGHT_[X] value in the active register set. The follow pseudo code is an example of when hcount resets to zero.

```
if (hcount == dtg_lrc_x) hcount <= 0;
else                    hcount <= hcount + 1;
```


15.3.2.2 VCount

The vertical counter (vcount) increments when the hcount counter resets to zero (hcount == TC_DTG_LOWER_RIGHT_[X]). The vertical counter resets to zero in the p_clk domain when hcount resets to zero and vcount matches the TC_DTG_LOWER_RIGHT_[Y] value in the active register set. The following pseudo code is an example of when the vcount resets to zero.

```
if ((hcount == TC_DTG_LOWER_RIGHT_X) &&
    (vcount == TC_DTG_LOWER_RIGHT_Y)) vcount <= 0;
else if (hcount == TC_DTG_LOWER_RIGHT_X) vcount <= vcount + 1;
```

A region in the display sequence that would actively process pixels for a channel is defined by the upper left and lower right coordinates, i.e. ULC/LRC. As an example, the following code would determine when to process pixels in the overlay channel.

```
if ((TC_CHANNEL_1_UPPER_LEFT_X <= hcount) &&
    (TC_CHANNEL_1_UPPER_LEFT_Y <= vcount) &&
    (TC_CHANNEL_1_LOWER_RIGHT_X >= hcount) &&
    (TC_CHANNEL_1_LOWER_RIGHT_Y >= vcount)) chan1_active <= 1'b1;
else chan1_active <= 1'b0;
```

The chan1_active (overlay is active in Dolby mode) signal would be used to read signals from scaler #1 output FIFO. It is typically active within the region defined by TC_DISPLAY_UPPER_LEFT_/TC_DISPLAY_LOWER_RIGHT_. Similar logic is used to create the timing for active processing of the other channels #2/3 , or video channels.

15.3.2.3 DTG Datapath

The DTG also implement the three pixel data path muxes for pixel data and back ground color. With the respective regions defined by ULC/LRC, pixel data is used to drive the pixel data path. Outside of these regions, the background color is forwarded to the pixel data path. The alpha MUX is also implemented in the DTG. The per pixel alpha value embedded in channel 1 is forwarded to the data path when channel 1 is active (or the TC_DEFAULT_OVERLAY_ALPHA is used when TC_CH1_PER_PEL_ALPHA_SEL = 0). The blender alpha is based on the per pixel graphics alpha:

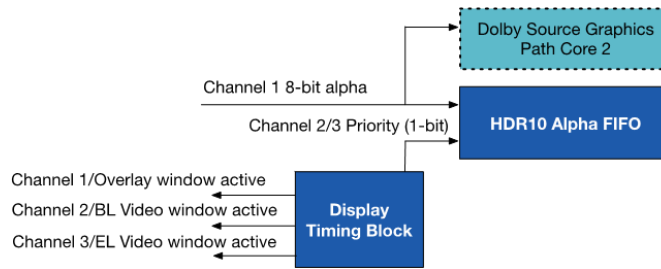


Figure 15-8. DTG Datapath Diagram

The Channel 2/3 Priority is 1 when Channel 2 is active and 0 when Channel 2 is inactive.

15.3.2.4 DTG Programming Example

The table below indicates the timing parameters for the Video Information Code (97, 107) defined for a progressive 4K TV at 60Hz.

Table 15-5. 4k TV at 60Hz Timing Parameters

Hactive	Vactive	Htotal	Hblank	Vtotal	Vblank	Pixel Freq
3840	2160	4400	560	2250	90	594
Hfront	Hsync	Hback	Vfront	Vsync	Vback	
176	88	296	8	10	72	

The DTG parameters for this display format are:

- $TC_DTG_LOWER_RIGHT_X = 4400 - 1 = 4399$; this is the Htotal from the table. Note the MINUS 1 value.
- $TC_DTG_LOWER_RIGHT_Y = 2250 - 1 = 2249$; this is the Vtotal from the table.
- $TC_DISPLAY_UPPER_LEFT_X = 4400 - 3840 - 176 - 1 = 383$; this is when the horizontal blanking ends, and active display region begins. dolby_hsync goes high at the p_clk when hcount == TC_DISPLAY_UPPER_LEFT_X
- $TC_DISPLAY_UPPER_LEFT_Y = 89$; this is when dolby_vsync_n goes low, and enters the active display region
- $TC_DISPLAY_LOWER_RIGHT_X = 4400 - 176 - 1 = 4223$; dolby_hsync goes low
- $TC_DISPLAY_LOWER_RIGHT_Y = 2249$; this is when dolby_vsync_n goes high, and enters vertical blanking

To extend the example to add a 64x64 pixel CH1 overlay with a 10 pixel offset into the active display area in both X and Y direction, the following CH1 parameters would be set:

- $TC_CHANNEL_1_UPPER_LEFT_X = 4400 - 3840 - 176 - 1 + 10 = 559 + 10 = 393$, or start of active display region ($TC_CHANNEL_1_UPPER_LEFT_X$) + 10

- $TC_CHANNEL_1_UPPER_LEFT_Y = 90 - 1 + 10 = 89 + 10 = 99$
- $TC_CHANNEL_1_LOWER_RIGHT_X = 4400 - 3840 - 176 - 1 + 10 + 64 = 393 + 64 + 10 = 457$ ($TC_DISPLAY_LOWER_RIGHT_X + 64$)
- $TC_CHANNEL_1_LOWER_RIGHT_Y = 89 + 64 + 10 = 163$

The timing to load SB registers starting with the second line (line #0, then line #1) in the vertical blanking region:

- $TC_CNTXT_SB_Y = 0$

And, to load double buffered registers in the active display region.

- $TC_CNTXT_DB_Y = 199$ // 110 lines after the active display area begins

NOTE

The ATG in the SUBSAM must be programmed with compatible values that are consistent with the DTG values.

15.3.2.5 DTG Interlaced Video Timing Generator

The DTG supports interlaced timing generation. Interlaced timings comply to the definition of “General Interlaced” and “Special Interlaced” timing in the CEA861 specification. The timing diagram for the General Interlaced timing from the CEA861 spec is shown below.

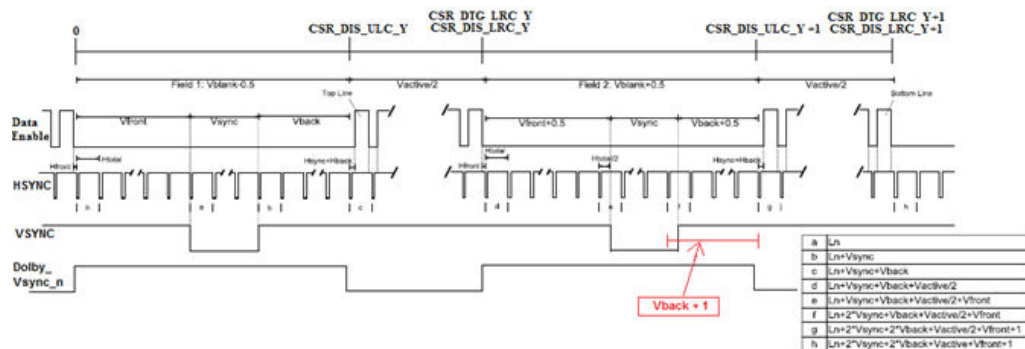


Figure 15-9. General Interlaced CEA861 Timing

The Dolby and HDR10 pixel processing engines do not use vertical front/back porches. The entire vertical blanking region is considered as one region.

$$V_{blank} = V_{front} + V_{sync} + V_{back}$$

In interlaced mode, “Field 1” is the first field displayed, and “Field 2” is the second displayed, when the DTG is enabled. The DTG alternates between these fields indefinitely after the GO bit is set. The DTG doesn’t have separate timing parameters for the two fields, only one set of programming parameters is provided. There is an additional control bit to enable interlaced timing.

For Special Interlaced mode, there is no difference for the DTG between the fields.

For General Interlaced mode, Field 2 has an additional line added to the vertical blanking region (see Vback + 1 in the diagram above). The addition of the extra line is done automatically when the DTG interlaced mode is selected. The ATG in the SUBSAM module also needs to be programmed with the same General Interlaced mode when the extra line is needed. This is required to be sure the two timing generators are synced for both the odd and even fields in interlaced mode.

NOTE

Only the General Interlaced mode requires the DTG interlaced enable bit to be set. Special Interlaced mode does NOT need the interlaced mode to be set. Simply, the timing parameters need to be programmed considering there are half as many lines in the active display region, i.e. 540 lines for 1080i.

15.3.2.6 DTG Interrupt Generation

The DTG module can generate interrupts under the following conditions:

- There are four interrupts that are programmable which can generate an interrupt at a programmable XY coordinate
- Each of these are identical and independently programmable
- When the programmed coordinate (XY) is reached within the sweep of the display, the interrupt goes active. The X value is compared to the HCOUNT, and the Y value is compared to the VCOUNT. When both compares are equal, the interrupt status bit is set
- These timing based interrupts can be used for any purpose. Examples of interrupts include:
 - Notify when the display enters/exits the active display region
 - Notify when the Vsync goes active

The DTG also collects events, or signals, from other components in the system. The DTG supports up to 8 asynchronous event inputs that can be used as general DCSS level interrupts. The status and mask bits associated with these event sources are implemented within the DTG to generate DCSS interrupts. These 8 events, and using the status and mask registers, are collected and multiplexed down to a single interrupt output from the DTG module.

15.3.2.7 DTG Timing Standards

Refer to the CEA861-F for timing standards to drive displays at different resolutions. A summary of timings is in the following table.

Table 15-6. CEA861-F Timing Summary

Name	Resolution	Refresh	Pixel Clock	Hblank	Vblank	Aspect Ratio
2160p60	3840x2160	60.00	594.00	560	90	16:09
2160p59.94	3840x2160	59.94	593.41	560	90	16:09
2160p50	3840x2160	50.00	594.00	1440	90	16:09
2160p30	3840x2160	30.00	297.00	560	90	16:09
2160p29.97	3840x2160	29.97	296.70	560	90	16:09
2160p25	3840x2160	25.00	297.00	1440	90	16:09
2160p24	3840x2160	24.00	297.00	1660	90	16:09
2160p23.98	3840x2160	23.98	296.70	1660	90	16:09
1080p60	1920x1080	60.00	148.50	280	45	16:09
1080p59.94	1920x1080	59.94	148.35	280	45	16:09
1080p50	1920x1080	50.00	148.50	720	45	16:09
1080p30	1920x1080	30.00	74.25	280	45	16:09
1080p29.97	1920x1080	29.97	74.18	280	45	16:09
1080p25	1920x1080	25.00	74.25	720	45	16:09
1080p24	1920x1080	24.00	74.25	830	45	16:09
1080p23.98	1920x1080	23.98	74.18	830	45	16:09
720p60	1280x720	60.00	74.25	370	30	16:09
720p59.94	1280x720	59.94	74.18	370	30	16:09
720p50	1280x720	50.00	74.25	700	30	16:09
720p30	1280x720	30.00	74.25	2020	30	16:09
720p29.97	1280x720	29.97	74.18	2020	30	16:09
720p25	1280x720	25.00	74.25	2680	30	16:09
720p24	1280x720	24.00	74.25	2845	30	16:09
720p23.98	1280x720	23.98	74.18	2845	30	16:09
576p50	720x576	50.00	27.00	144	49	4:03
576p25	720x576	25.00	13.50	144	49	4:03
480p59.94	720x480	59.94	27.00	138	45	4:03
480p29.97	720x480	29.97	13.50	138	45	4:03

The display timing creates all control signals for the entire pixel data path and the output FIFOs for the scalers and the readback FIFOs. Some of the control signals that are used by the pixel data path are listed below:

- **CHAN3_ACTIVE, CHAN2_ACTIVE** – Indicates when pixels are fetched from the video scaler output FIFO. This signal, for either the base layer (bl_*) or enhancement layer (el_*), is logically equal to ($\sim$ [bllel]_vsync_n && [bllel]_hsync && [bllel]_de).
- **CHAN1_ACTIVE** – Indicates when pixels are fetched from the overlay scaler output FIFO. This signal is logically equal to ($\sim$ vsync_n_in && hsync_in && de_in).

15.3.2.8 DTG Background Color

The YUV 3x10 bit and RGB 3x8 bit background colors are inserted at the inputs to the channel 1/2/3, or video and overlay, pipelines when CHAN1/2/3_ACTIVE are not asserted.

15.3.3 Memory Map and Registers

15.3.3.1 register descriptions

15.3.3.1.1 TIMING_CONTROL_AND_BLENDER Memory map

DTG base address: 2_0000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Timing Controller Control Register (TC_CONTROL_STATUS)	32	RW	0000_0000h
4h	DTG lower right corner locations (TC_DTG_REG1)	32	RW	0000_0000h
8h	Display Register: TOP Window Coordinates for Active display area (TC_DISPLAY_REG2)	32	RW	0000_0000h
Ch	Display Register: BOTTOM Window Coordinates for Active display area (TC_DISPLAY_REG3)	32	RW	0000_0000h
10h	Channel 1 window Register: TOP Window Coordinates for channel1 (TC_CH1_REG4)	32	RW	0000_0000h
14h	Channel_1 window Register: BOTTOM Window Coordinates for channel_1 window (TC_CH1_REG5)	32	RW	0000_0000h
18h	Channel 2 window Register: TOP Window Coordinates for channel_2 (TC_CH2_REG6)	32	RW	0000_0000h
1Ch	Channel_2 window Register: BOTTOM Window Coordinates for channel_2 pixel window (TC_CH2_REG7)	32	RW	0000_0000h
20h	Channel 3 window Register: TOP Window Coordinates for channel_3 (TC_CH3_REG8)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
24h	Channel_3 window Register: BOTTOM Window Coordinates for channel_3 pixel window (TC_CH3_REG9)	32	RW	0000_0000h
28h	Context Loader Register: Coordinates in the raster table where the context loader is started. (TC_CTX_LD_REG10)	32	RW	0000_0000h
2Ch	Channel_1 background pixel color. (TC_CH1_BKRND_REG11)	32	RW	0000_0000h
30h	Channel_2 background pixel color. (TC_CH2_BKRND_REG12)	32	RW	0000_0000h
38h	DBY MODE Blender control. (BLENDER_DBY_EOTF_RANGEINV)	32	RW	0000_0000h
3Ch	DBY MODE Blender control. (BLENDER_DBY_EOTF_RANGEMIN)	32	RW	0000_0000h
40h	DBY MODE blender control. (BLENDER_DBY_BDP)	32	RW	0000_0000h
44h	Background pixel color component sent to blender. Used when no valid pixels (BLENDER_BKRND_I_GRAPHICS)	32	RW	0000_0000h
48h	Background pixel color component sent to blender. Used when no valid pixels (BLENDER_BKRND_P_GRAPHICS)	32	RW	0000_0000h
4Ch	Background pixel color component sent to blender. Used when no valid pixels (BLENDER_BKRND_T_GRAPHICS)	32	RW	0000_0000h
50h	LINE1 interrupt control: Coordinate where line1 interrupt is asserted (TC_LINE1_INT_REG13)	32	RW	0000_0000h
54h	LINE2 interrupt control: Coordinate where line2 interrupt is asserted (TC_LINE2_INT_REG14)	32	RW	0000_0000h
58h	default alpha (TC_ALPHA_DEFAULT_REG15)	32	RW	0000_0000h
5Ch	Timing Controller interrupt status (TC_INTERRUPT_STATUS)	32	RO	0000_0000h
60h	Timing Controller interrupt control. (TC_INTRERRUPT_CONTROL_REG17)	32	RW	0000_0000h
64h	Channel_3 background pixel color. (TC_CH3_BKRND_REG18)	32	RW	0000_0000h
68h	Timing Controller interrupt masks (TC_INTRERRUPT_MASK)	32	RW	0000_0000h
6Ch	LINE3 interrupt control: Coordinate where line3 interrupt is asserted (TC_LINE3_INT_REG)	32	RW	0000_0000h
70h	LINE4 interrupt control: Coordinate where line4 interrupt is asserted (TC_LINE4_INT_REG)	32	RW	0000_0000h
74h	For DBY Mode: Controls the assertion and de-assertion DE signal (Overlay channel). (TC_OL_DE_CONTROL_REG)	32	RW	0000_0000h
78h	For DBY Mode: Controls the assertion and de-assertion DE signal (Base layer (BL) channel). (TC_BL_DE_CONTROL_REG)	32	RW	0000_0000h
7Ch	For DBY Mode: Controls the assertion and de-assertion DE signal (Enhancement layer (EL) channel). (TC_EL_DE_CONTROL_REG)	32	RW	0000_0000h

15.3.3.1.2 Timing Controller Control Register (TC_CONTROL_STATUS)

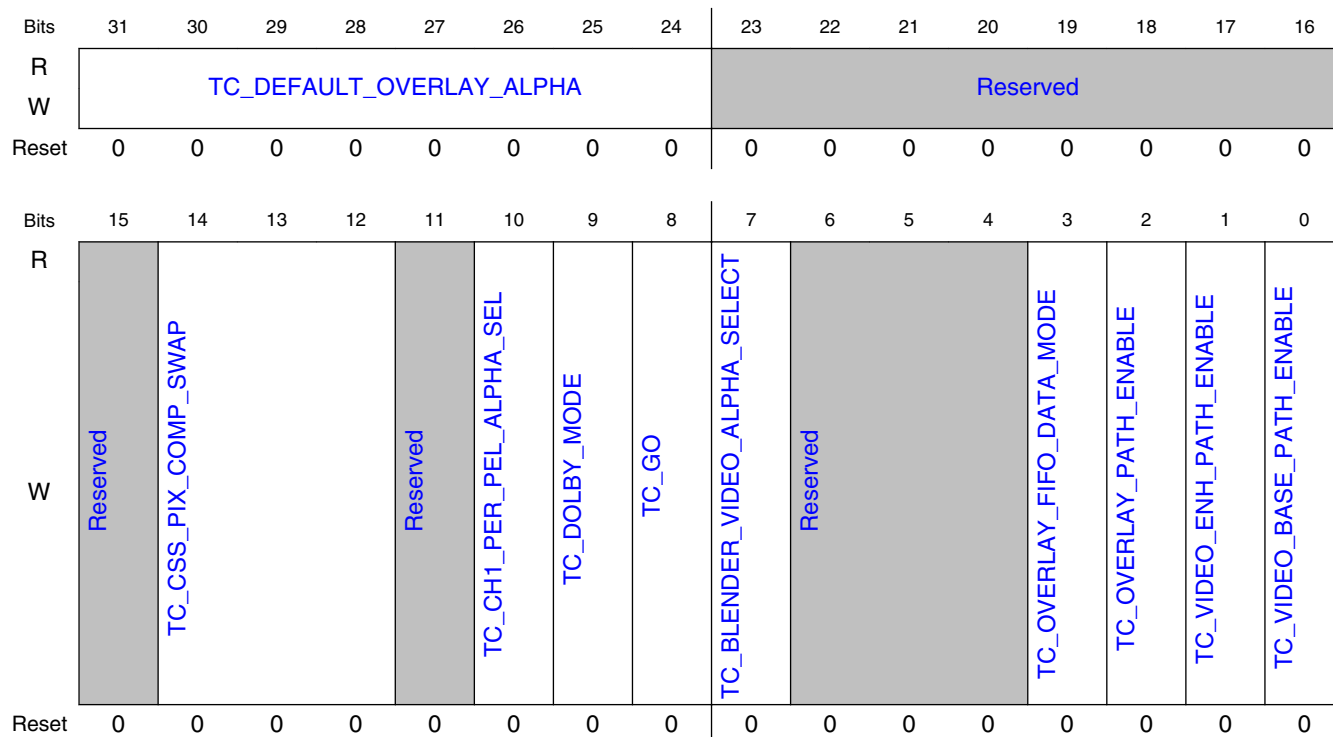
15.3.3.1.2.1 Offset

Register	Offset
TC_CONTROL_STATUS	0h

15.3.3.1.2.2 Function

Enables and controls the behaviour of timing controller and the three pixel channels in the display controller

15.3.3.1.2.3 Diagram



15.3.3.1.2.4 Fields

Field	Function
31-24 TC_DEFAULT_OVERLAY_ALPHA	Default alpha foreground used for the active regions where graphics channel does not provide an alpha value This is used when Graphics channel is ONLY RGB
23-15 —	Reserved.
14-12 TC_CSS_PIX_COMP_SWAP	Permutes the pixel component ordering into the chroma subsampler (CSS) block. 0: I P T 1: I T P 2: T I P 3: T P I 4: P I T 5: P T I

Table continues on the next page...

Field	Function
11 —	Reserved.
10 TC_CH1_PER_PEL_ALPHA_SELECT	Enables per pixel alpha for channel1 (Overlay or Graphics) 0: use alpha from TC_DEFAULT_OVERLAY_ALPHA or foreground alpha 1: use alpha from channel1 Data stream and apply blending per pixel
9 TC_DOLBY_MODE	Enables Dolby mode. 0: HDR10 pixel processing is performed. Dolby processing disabled 1: Dolby Processing Enabled. Dolby timing control features enabled. Dolby Blending is enabled
8 TC_GO	Used to start and stop timing controller 0: timing controller inactive. No display control signal assertions (HSYNC = 0) 1: timing controller enabled. Display control processing is per the programming of the timing controller. HSYNC and VSYNC DE are running
7 TC_BLENDER_VIDEO_ALPHA_SELECT	alpha_valid 0: No alpha in the ch1 pixel so use default foreground alpha TC_DEFAULT_OVERLAY_ALPHA 1: there is an alpha in ch1 pixel
6-4 —	Reserved.
3 TC_OVERLAY_FIFO_DATA_MODE	Overlay fifo output data in yuv or RGB mode 0: YUV 1: RGB
2 TC_OVERLAY_PATH_ENABLE	Enable channel_1 (Dolby_mode:Overlay /HDR10: (GFX)) processing. 0: channel_1 disabled. Pixels cannot pass thru channel_1 1: channel_1 enabled
1 TC_VIDEO_ENH_PATH_ENABLE	Enable channel_2 (Dolby_mode:Enhancement channel) processing. 0: channel_2 disabled. Pixels cannot pass thru channel_2 1: channel_2 enabled
0 TC_VIDEO_BASE_PATH_ENABLE	Enable channel3 (Dolby_mode:Base Layer channel) processing. 0: channel_3 disabled. Pixels cannot pass thru channel_3 1: channel_3 enabled

15.3.3.1.3 DTG lower right corner locations (TC_DTG_REG1)

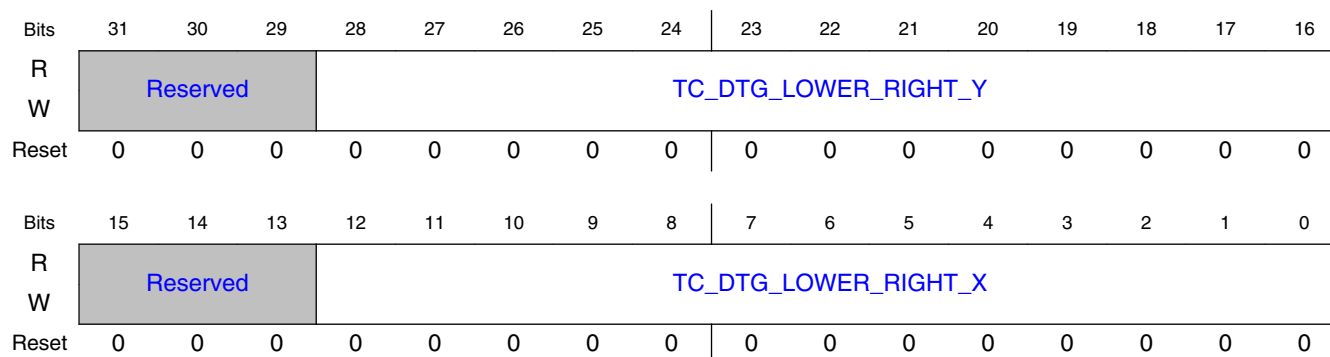
15.3.3.1.3.1 Offset

Register	Offset
TC_DTG_REG1	4h

15.3.3.1.3.2 Function

The X,Y coordinates in this register specify the size of the raster table including blanking regions. The upper left coordinate of this is (0,0)

15.3.3.1.3.3 Diagram



15.3.3.1.3.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_DTG_LOWER_RIGHT_Y	lower right corner Y (vertical) coordinate of the raster table
15-13 —	Reserved.
12-0 TC_DTG_LOWER_RIGHT_X	lower right corner X (horizontal) coordinate of the raster table

15.3.3.1.4 Display Register: TOP Window Coordinates for Active display area (TC_DISPLAY_REG2)

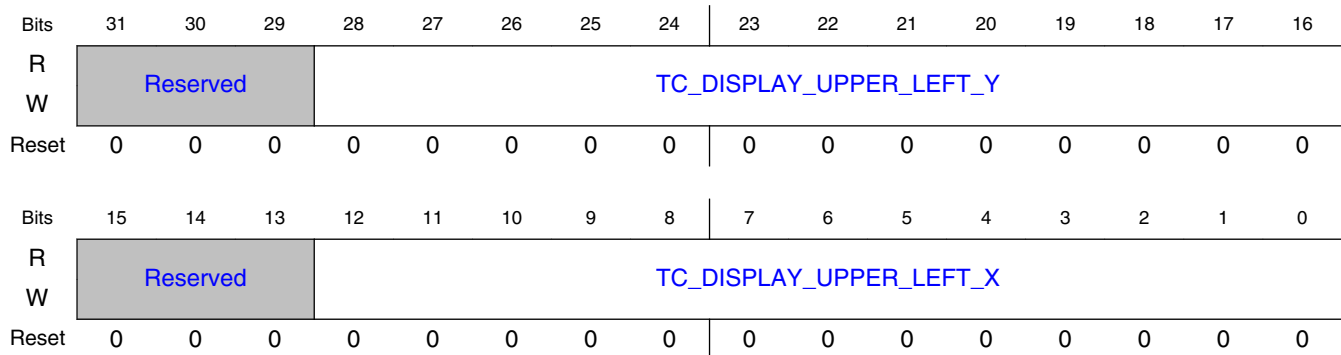
15.3.3.1.4.1 Offset

Register	Offset
TC_DISPLAY_REG2	8h

15.3.3.1.4.2 Function

The X,Y coordinates in this register specify the size of the active display. The upper left coordinate of the display window to specify the horizontal and vertical blankin region as an offset from (0,0). hactive and vactive are represented by the two values Any area in the raster table which is not covered by the active display area has a background color inserted.

15.3.3.1.4.3 Diagram



15.3.3.1.4.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_DISPLAY_UPPER_LEFT_Y	upper left corner Y (vertical) coordinate of the active display region
15-13 —	Reserved.
12-0 TC_DISPLAY_UPPER_LEFT_X	upper left corner X (horizontal) coordinate of the active display region

15.3.3.1.5 Display Register: BOTTOM Window Coordinates for Active display area (TC_DISPLAY_REG3)

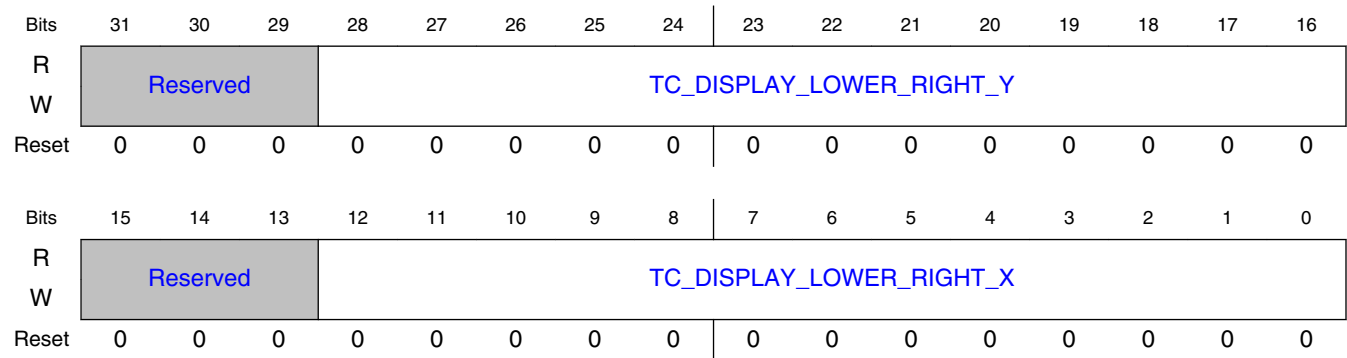
15.3.3.1.5.1 Offset

Register	Offset
TC_DISPLAY_REG3	Ch

15.3.3.1.5.2 Function

The X,Y coordinates in this register specify the size of the active display, Bottom Right Corner with respect to the Upper Left coordinates. Any area in the raster table which is not covered by the active display area has a background color inserted.

15.3.3.1.5.3 Diagram



15.3.3.1.5.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_DISPLAY_LOWER_RIGHT_Y	lower right corner Y (vertical) coordinate of the active display region
15-13 —	Reserved.
12-0 TC_DISPLAY_LOWER_RIGHT_X	lower right corner X (horizontal) coordinate of the active display region

15.3.3.1.6 Channel 1 window Register: TOP Window Coordinates for channel1 (TC_CH1_REG4)

15.3.3.1.6.1 Offset

Register	Offset
TC_CH1_REG4	10h

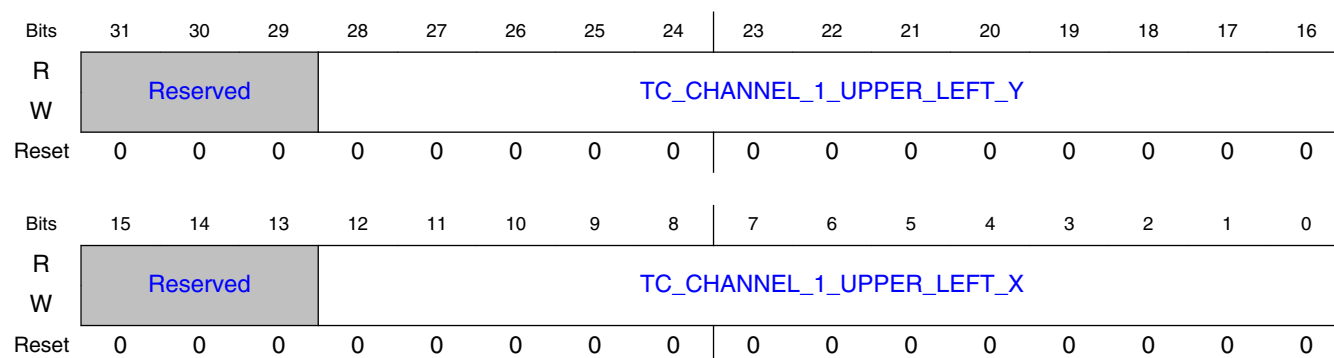
15.3.3.1.6.2 Function

The X,Y coordinates in this register specify the upper left coordinate of channel_1.

In HDR10 mode this may be the graphics channel or video channel.

In DBY mode this is the Overlay Channel

15.3.3.1.6.3 Diagram



15.3.3.1.6.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_CHANNEL_1_UPPER_LEFT_Y	upper left corner Y (vertical) coordinate of the channel_1 window
15-13 —	Reserved.
12-0 TC_CHANNEL_1_UPPER_LEFT_X	upper left corner X (horizontal) coordinate of the channel_1 window

15.3.3.1.7 Channel_1 window Register: BOTTOM Window Coordinates for channel_1 window (TC_CH1_REG5)

15.3.3.1.7.1 Offset

Register	Offset
TC_CH1_REG5	14h

15.3.3.1.7.2 Function

The X,Y coordinates in this register specify the Bottom Right coordinate of the channel_1 pixel window

15.3.3.1.7.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved			TC_CHANNEL_1_LOWER_RIGHT_Y												
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved			TC_CHANNEL_1_LOWER_RIGHT_X												
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.3.3.1.7.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_CHANNEL_1_LOWER_RIGHT_Y	lower right corner Y (vertical) coordinate of the channel_1 window
15-13 —	Reserved.
12-0 TC_CHANNEL_1_LOWER_RIGHT_X	lower right corner X (horizontal) coordinate of the channel_1 window

15.3.3.1.8 Channel 2 window Register: TOP Window Coordinates for channel_2 (TC_CH2_REG6)

15.3.3.1.8.1 Offset

Register	Offset
TC_CH2_REG6	18h

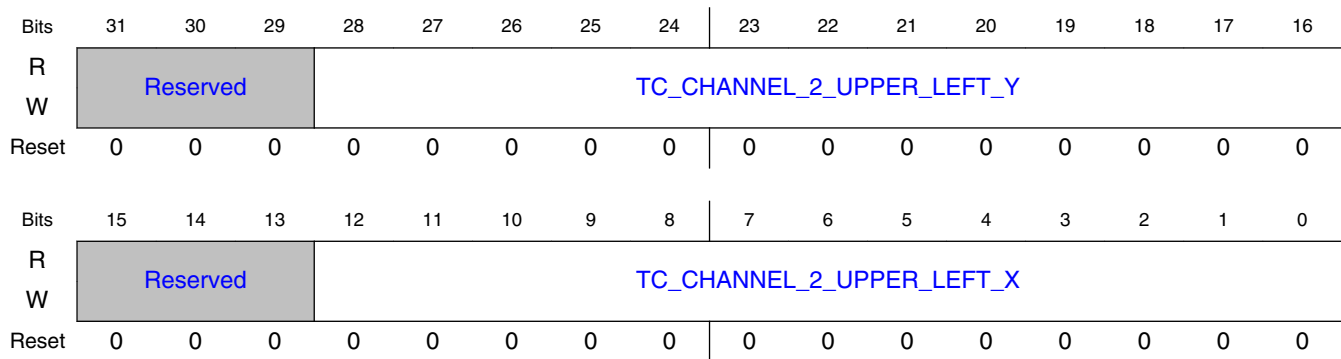
15.3.3.1.8.2 Function

The X,Y coordinates in this register specify the size of the channel_2 window. The upper left coordinate of channel_2.

In HDR10 mode this is a video channel.

In DBY mode this is the Enhancement Layer Channel

15.3.3.1.8.3 Diagram



15.3.3.1.8.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_CHANNEL_2_UPPER_LEFT_Y	upper left corner Y (vertical) coordinate of the channel_2 window
15-13 —	Reserved.
12-0	upper left corner X (horizontal) coordinate of the channel_2 window

Field	Function
TC_CHANNEL_2_UPPER_LEFT_X	

15.3.3.1.9 Channel_2 window Register: BOTTOM Window Coordinates for channel_2 pixel window (TC_CH2_REG7)

15.3.3.1.9.1 Offset

Register	Offset
TC_CH2_REG7	1Ch

15.3.3.1.9.2 Function

The X,Y coordinates in this register specify the Bottom Right coordinate of the channel_2 pixel window.

15.3.3.1.9.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved			TC_CHANNEL_2_LOWER_RIGHT_Y												
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved			TC_CHANNEL_2_LOWER_RIGHT_X												
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.3.3.1.9.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_CHANNEL_2_LOWER_RIGHT_Y	lower right corner Y (vertical) coordinate of the channel_2 window

Table continues on the next page...

Field	Function
15-13 —	Reserved.
12-0 TC_CHANNEL_2_LOWER_RIGHT_X	lower right corner X (horizontal) coordinate of the channel_2 window

15.3.3.1.10 Channel 3 window Register: TOP Window Coordinates for channel_3 (TC_CH3_REG8)

15.3.3.1.10.1 Offset

Register	Offset
TC_CH3_REG8	20h

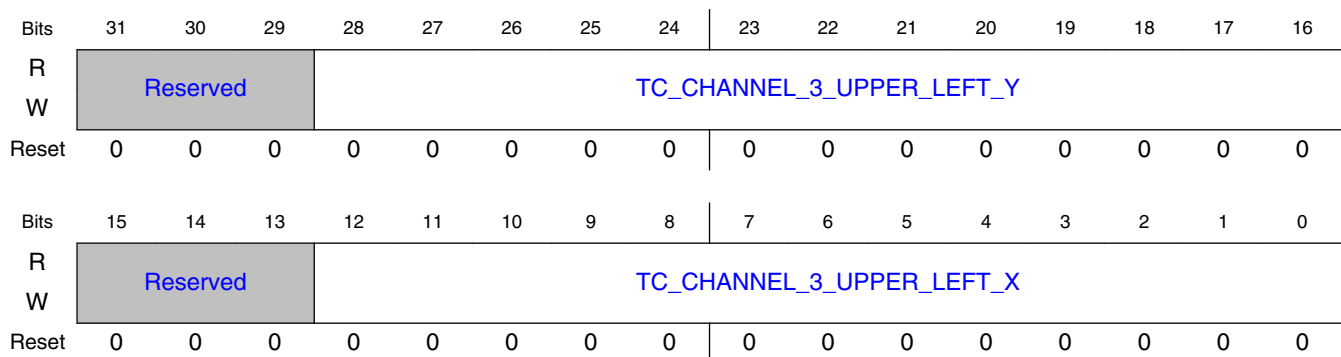
15.3.3.1.10.2 Function

The X,Y coordinates in this register specify the size of the channel_3 window. The upper left coordinate of channel_3.

In HDR10 mode this is a video channel.

In DBY mode this is the Base Layer (BL)

15.3.3.1.10.3 Diagram



15.3.3.1.10.4 Fields

Field	Function
31-29	Reserved.

Table continues on the next page...

Memory Map and Registers

Field	Function
—	
28-16 TC_CHANNEL_3_UPPER_LEF T_Y	upper left corner Y (vertical) coordinate of the channel_3 window
15-13 —	Reserved.
12-0 TC_CHANNEL_3_UPPER_LEF T_X	upper left corner X (horizontal) coordinate of the channel_3 window

15.3.3.1.11 Channel_3 window Register: BOTTOM Window Coordinates for channel_3 pixel window (TC_CH3_REG9)

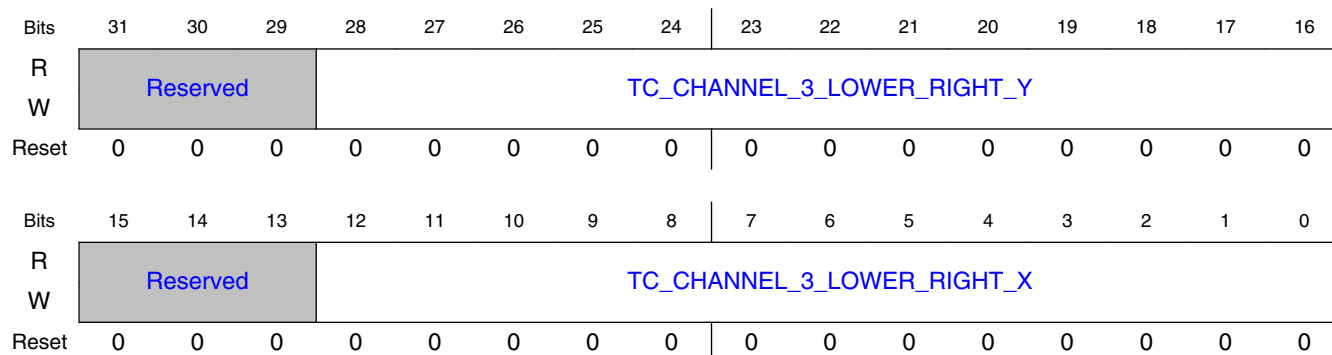
15.3.3.1.11.1 Offset

Register	Offset
TC_CH3_REG9	24h

15.3.3.1.11.2 Function

The X,Y coordinates in this register specify the size of the channel_3 pixel window.
Bottom Right

15.3.3.1.11.3 Diagram



15.3.3.1.11.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_CHANNEL_3_LOWER_RIGHT_Y	lower right corner Y (vertical) coordinate of the channel_3 window
15-13 —	Reserved.
12-0 TC_CHANNEL_3_LOWER_RIGHT_X	lower right corner X (horizontal) coordinate of the channel_3 window

15.3.3.1.12 Context Loader Register: Coordinates in the raster table where the context loader is started. (TC_CTX_LD_REG10)

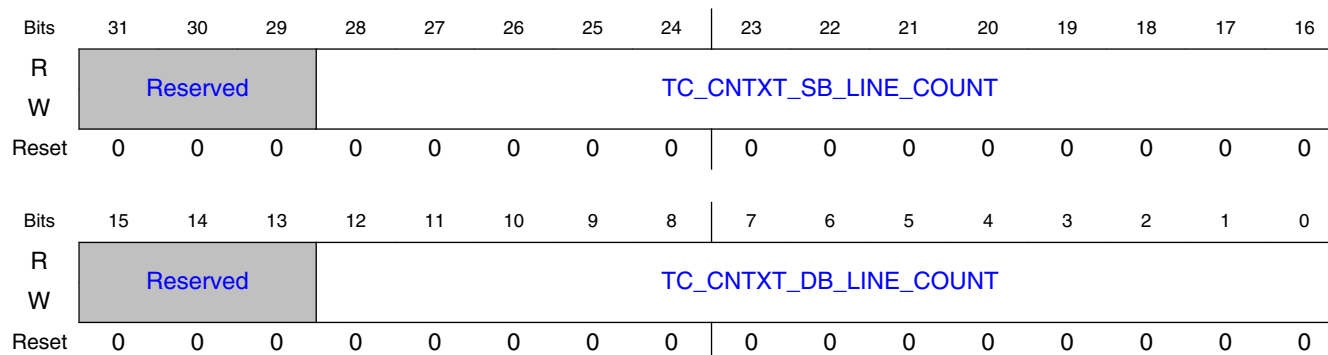
15.3.3.1.12.1 Offset

Register	Offset
TC_CTX_LD_REG10	28h

15.3.3.1.12.2 Function

Line Count in the raster table where the CTX_LD is triggered for loading context (some double buffered). The register field provides the line count that is compared with the VCOUNT counter value to generate the event for loading of Single Buffered or Double Buffered Context Timing Controller and Dolby context are double buffered. Rest of the blocks are single buffered

15.3.3.1.12.3 Diagram



15.3.3.1.12.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_CNTXT_SB_LINE_COUNT	Line Count for Single Bufferend (SB) Context Loading
15-13 —	Reserved.
12-0 TC_CNTXT_DB_LINE_COUNT	Line Count for Double Buffered (DB) Context Loading

15.3.3.1.13 Channel_1 background pixel color. (TC_CH1_BKRND_REG11)

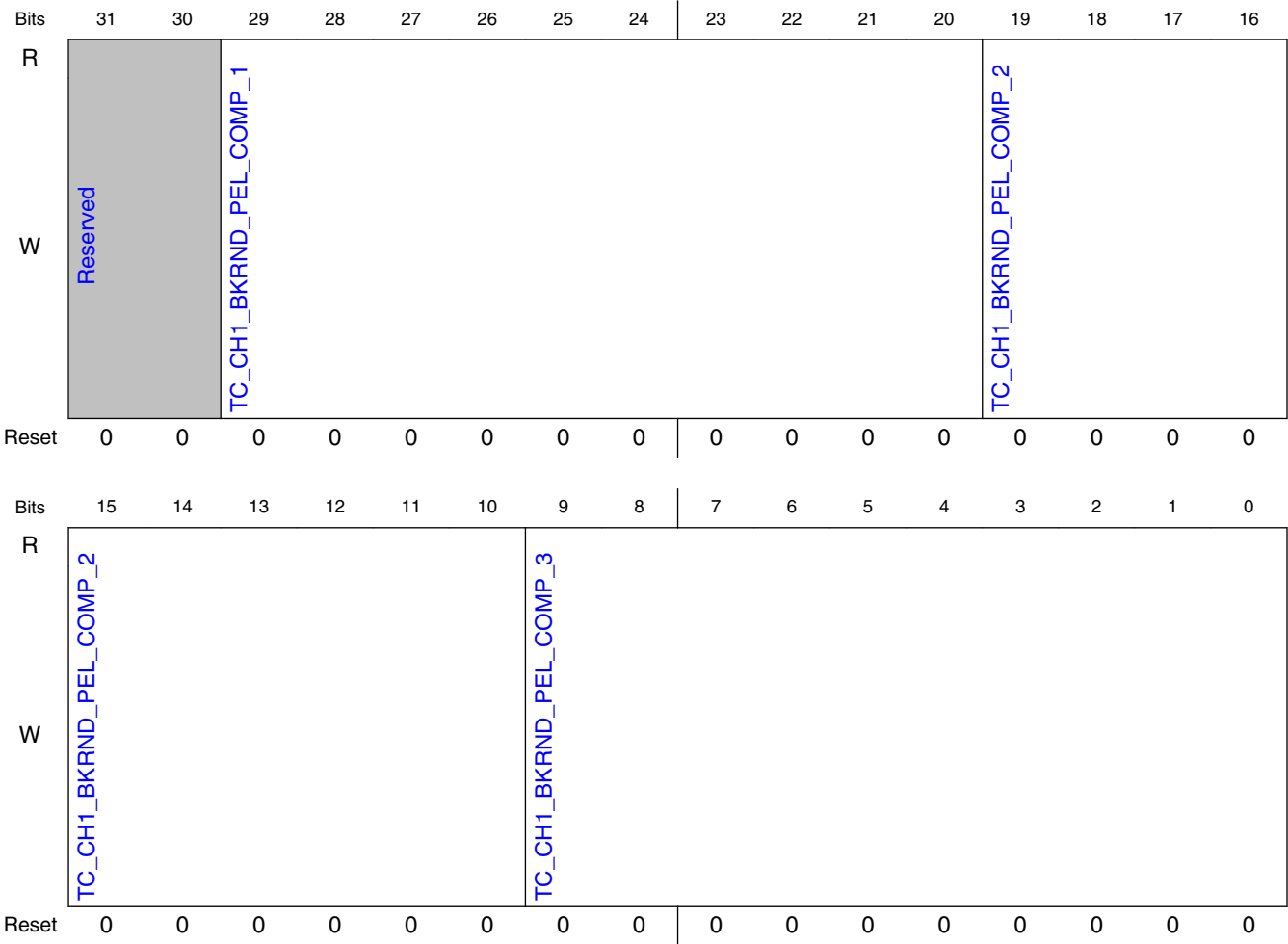
15.3.3.1.13.1 Offset

Register	Offset
TC_CH1_BKRND_REG 11	2Ch

15.3.3.1.13.2 Function

Outside of the Channel_1 pixel window, This background color is put on channel_1. 10 bits per component. Color space depending on programming

15.3.3.1.13.3 Diagram



15.3.3.1.13.4 Fields

Field	Function
31-30 —	Reserved.
29-20 TC_CH1_BKRND_PEL_COMP_1	10-bit component (R or Y)
19-10 TC_CH1_BKRND_PEL_COMP_2	10-bit component (G or Cb)
9-0	10-bit component (B or Cr)

Memory Map and Registers

Field	Function
TC_CH1_BKRN D_PEL_COMP_ 3	

15.3.3.1.14 Channel_2 background pixel color. (TC_CH2_BKRND_REG12)

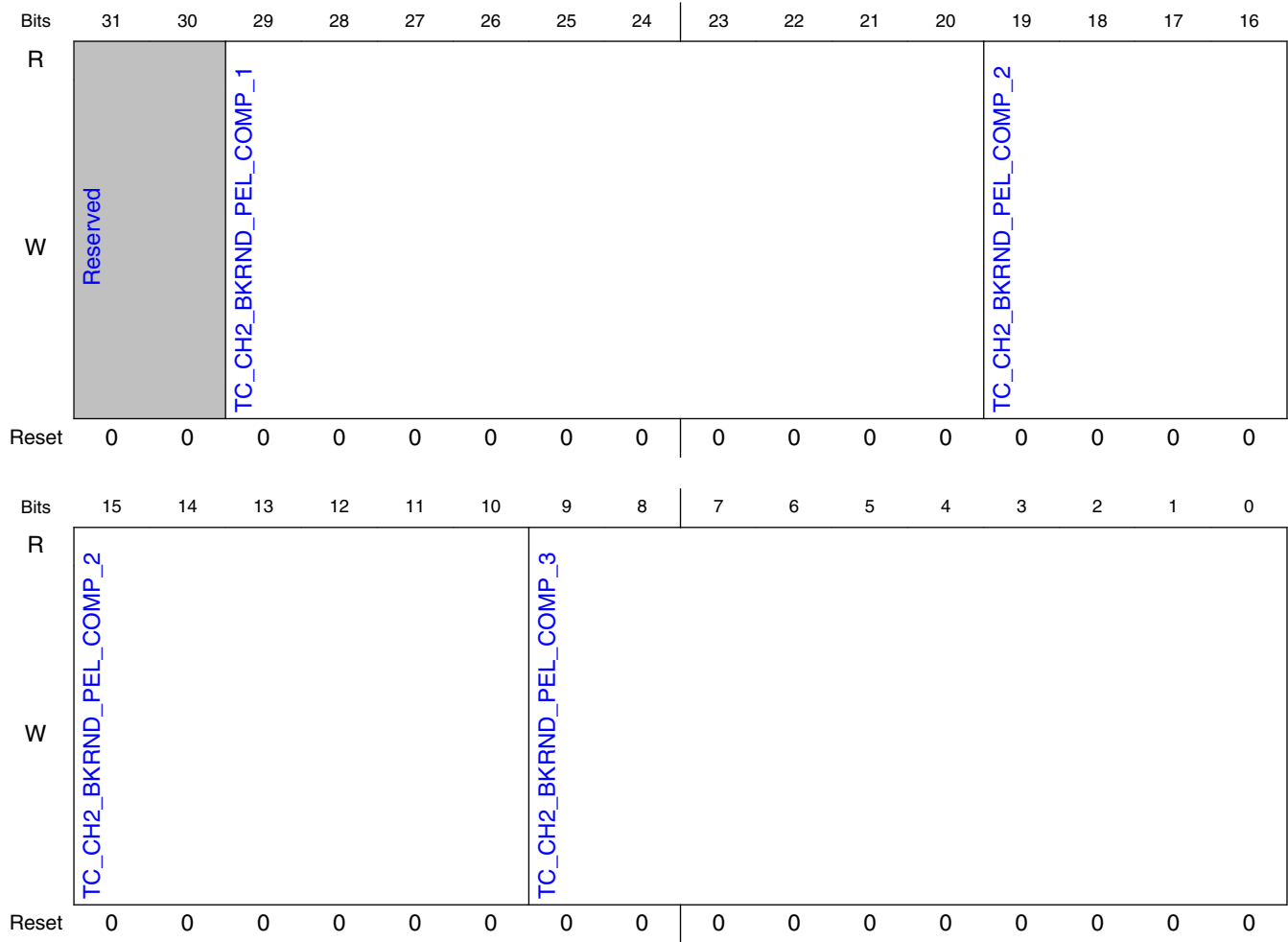
15.3.3.1.14.1 Offset

Register	Offset
TC_CH2_BKRND_REG 12	30h

15.3.3.1.14.2 Function

Outside of the Channel_2 pixel window, This background color is put on channel_2. 10 bits per component. Color space depending on programming

15.3.3.1.14.3 Diagram



15.3.3.1.14.4 Fields

Field	Function
31-30 —	Reserved.
29-20 TC_CH2_BKRND_PEL_COMP_1	10-bit component (R or Y)
19-10 TC_CH2_BKRND_PEL_COMP_2	10-bit component (G or Cb)
9-0	10-bit component (B or Cr)

Memory Map and Registers

Field	Function
TC_CH2_BKRN D_PEL_COMP_ 3	

15.3.3.1.15 DBY MODE Blender control. (BLENDER_DBY_EOTF_RANGEIN V)

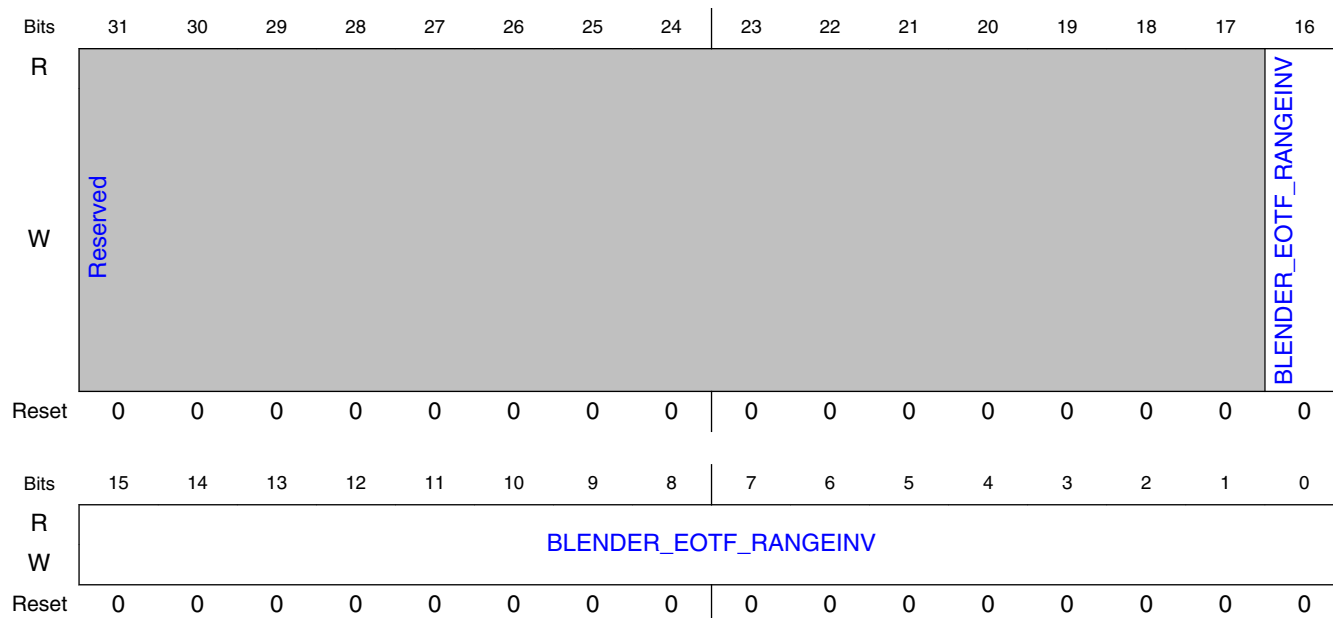
15.3.3.1.15.1 Offset

Register	Offset
BLENDER_DBY_EOTF_ RANGEINV	38h

15.3.3.1.15.2 Function

eotf_rangeInv in DBY Blender EQN.

15.3.3.1.15.3 Diagram



15.3.3.1.15.4 Fields

Field	Function
31-17 —	Reserved.
16-0 BLENDER_EOTF_RANGEINV	eotf_rangeInv parameter for DBY blender

15.3.3.1.16 DBY MODE Blender control. (BLENDER_DBY_EOTF_RANGEMIN)

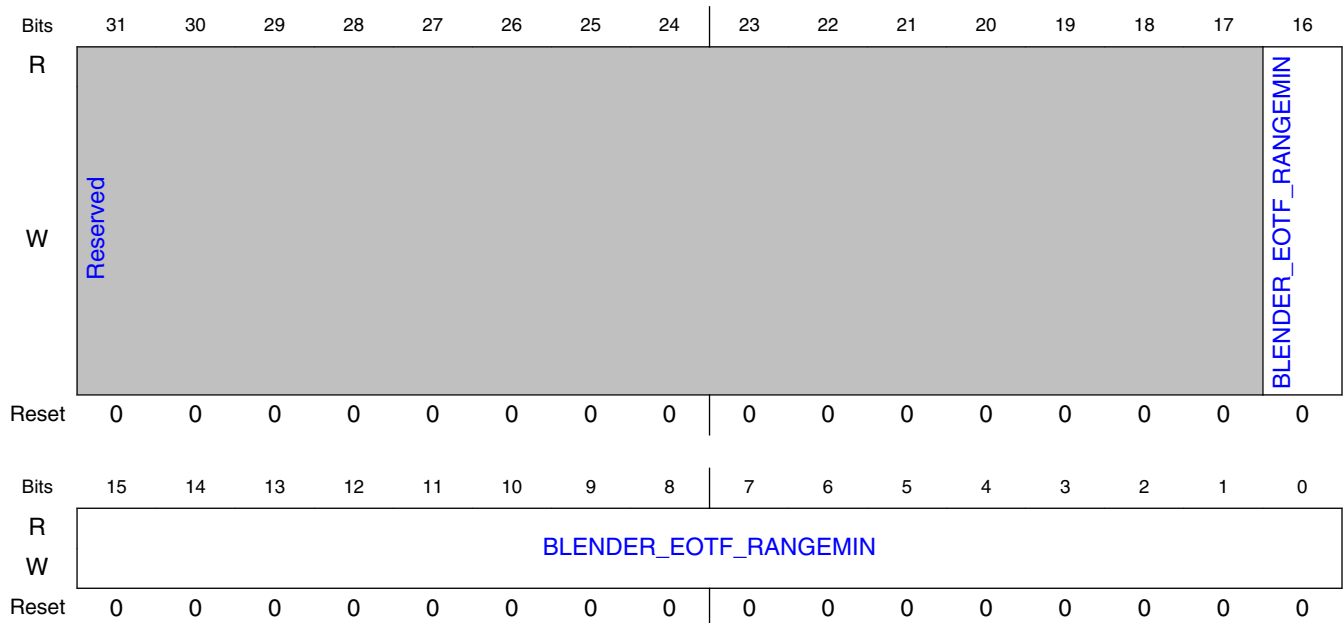
15.3.3.1.16.1 Offset

Register	Offset
BLENDER_DBY_EOTF_RANGEMIN	3Ch

15.3.3.1.16.2 Function

eotf_rangeMin in DBY blender EQN.

15.3.3.1.16.3 Diagram



15.3.3.1.16.4 Fields

Field	Function
31-17 —	Reserved.
16-0 BLENDER_EOT F_RANGEMIN	eotf_rangeMin parameter for DBY blender

15.3.3.1.17 DBY MODE blender control. (BLENDER_DBY_BDP)

15.3.3.1.17.1 Offset

Register	Offset
BLENDER_DBY_BDP	40h

15.3.3.1.17.2 Function

bitDepth parameter for DBY blender equation

15.3.3.1.17.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved												BLENDER_BDP			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.3.3.1.17.4 Fields

Field	Function
31-5 —	Reserved.

Table continues on the next page...

Field	Function
4-0 BLENDER_BDP	bitDepth parameter in DBY blender

15.3.3.1.18 Background pixel color component sent to blender. Used when no valid pixels (BLENDER_BKRND_I_GRAPHICS)

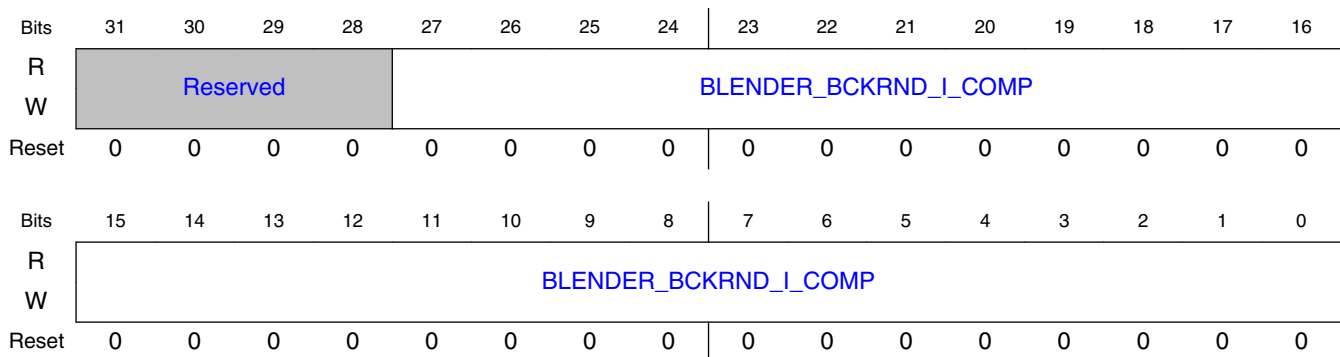
15.3.3.1.18.1 Offset

Register	Offset
BLENDER_BKRND_I_GRAPHICS	44h

15.3.3.1.18.2 Function

Background pixel color sent to blender. Used when no valid pixels 28 bits per component. Color space depending on programming

15.3.3.1.18.3 Diagram



15.3.3.1.18.4 Fields

Field	Function
31-28 —	Reserved.
27-0 BLENDER_BCKRND_I_COMP	28-bit component I component in DBY mode R/Y component in HDR10 MODE

15.3.3.1.19 Background pixel color component sent to blender. Used when no valid pixels (BLENDER_BKRND_P_GRAPHICS)

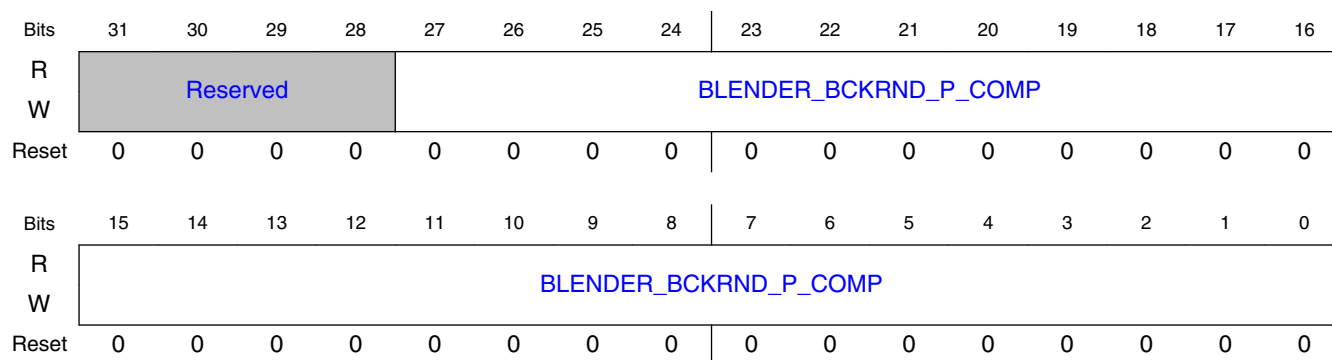
15.3.3.1.19.1 Offset

Register	Offset
BLENDER_BKRND_P_GRAPHICS	48h

15.3.3.1.19.2 Function

Background pixel color sent to blender. Used when no valid pixels 28 bits per component. Color space depending on programming

15.3.3.1.19.3 Diagram



15.3.3.1.19.4 Fields

Field	Function
31-28 —	Reserved.
27-0 BLENDER_BKRND_P_COMP	28-bit component P component in DBY mode G/Cb component in HDR10 MODE

15.3.3.1.20 Background pixel color component sent to blender. Used when no valid pixels (BLENDER_BKRND_T_GRAPHICS)

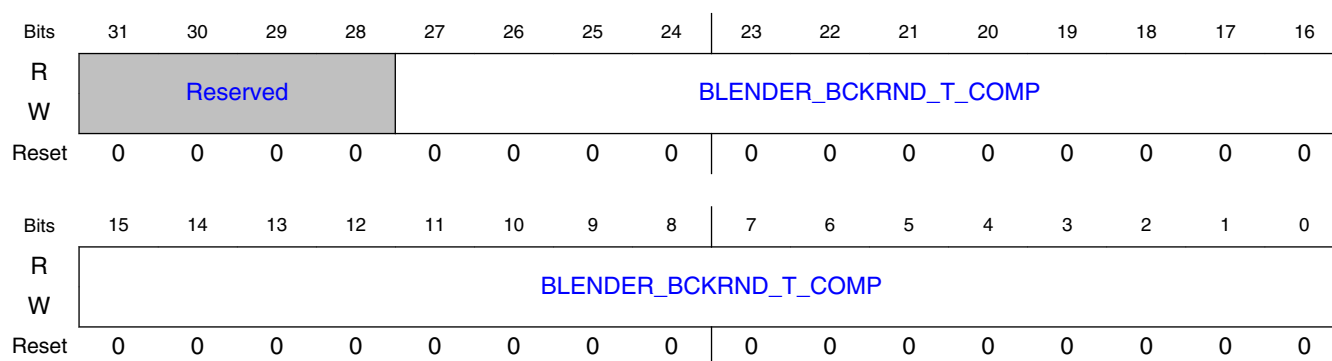
15.3.3.1.20.1 Offset

Register	Offset
BLENDER_BKRND_T_GRAPHICS	4Ch

15.3.3.1.20.2 Function

Background pixel color sent to blender. Used when no valid pixels 28 bits per component. Color space depending on programming

15.3.3.1.20.3 Diagram



15.3.3.1.20.4 Fields

Field	Function
31-28 —	Reserved.
27-0 BLENDER_BCKRND_T_COMP	28-bit component T component in DBY mode B/Cr component in HDR10 MODE

15.3.3.1.21 LINE1 interrupt control: Coordinate where line1 interrupt is asserted (TC_LINE1_INT_REG13)

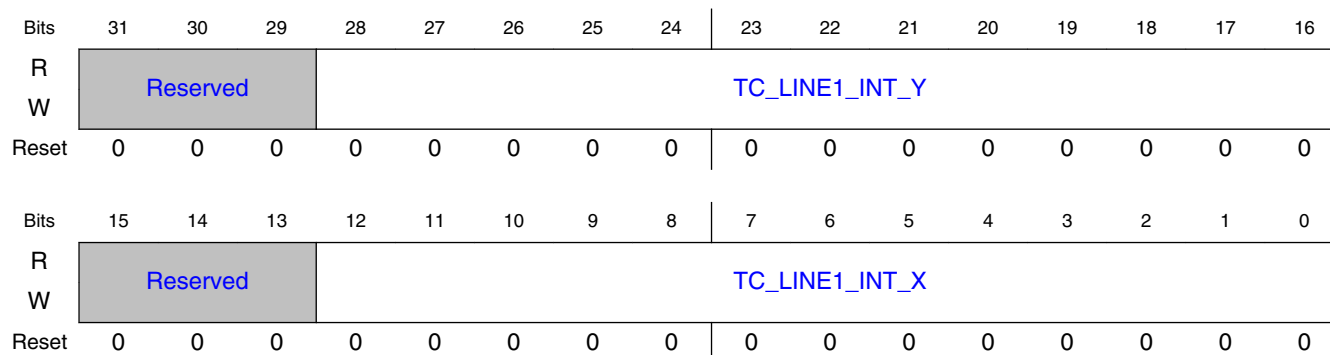
15.3.3.1.21.1 Offset

Register	Offset
TC_LINE1_INT_REG13	50h

15.3.3.1.21.2 Function

The X,Y coordinates in this register specify where line1 interrupt is asserted as timing controller encounters these coordinates. Can be anywhere on raster table. For historical reasons this is called LINE1 interrupt in RTL

15.3.3.1.21.3 Diagram



15.3.3.1.21.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_LINE1_INT_Y	Y (vertical) coordinate for line1 interrupt
15-13 —	Reserved.
12-0 TC_LINE1_INT_X	X (horizontal) coordinate for line1 interrupt

15.3.3.1.22 LINE2 interrupt control: Coordinate where line2 interrupt is asserted (TC_LINE2_INT_REG14)

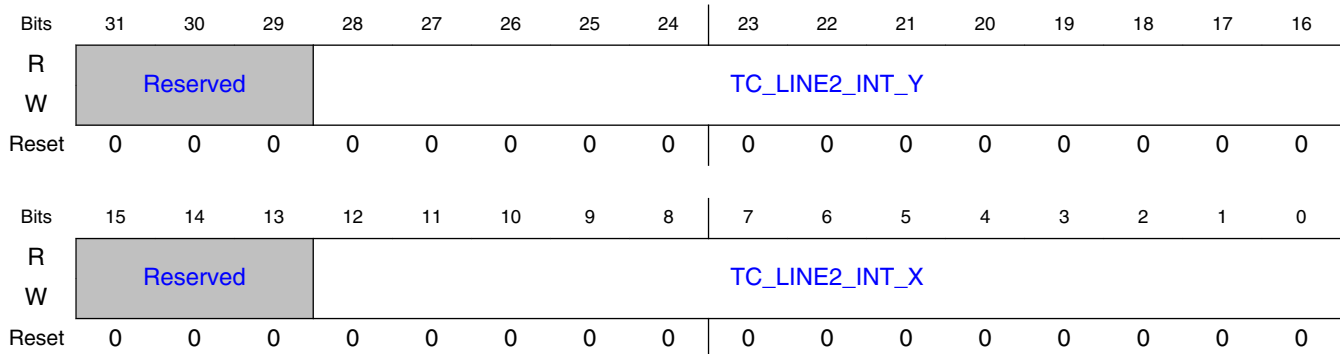
15.3.3.1.22.1 Offset

Register	Offset
TC_LINE2_INT_REG14	54h

15.3.3.1.22.2 Function

The X,Y coordinates in this register specify where line2 interrupt is asserted as timing controller encounters these coordinates. Can be anywhere on raster table. For historical reasons this is called LINE2 interrupt in RTL

15.3.3.1.22.3 Diagram



15.3.3.1.22.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_LINE2_INT_Y	Y (vertical) coordinate for line2 interrupt
15-13 —	Reserved.
12-0 TC_LINE2_INT_X	X (horizontal) coordinate for line2 interrupt

15.3.3.1.23 default alpha (TC_ALPHA_DEFAULT_REG15)

15.3.3.1.23.1 Offset

Register	Offset
TC_ALPHA_DEFAULT_REG15	58h

15.3.3.1.23.2 Function

This background alpha is applied when the per pixel alpha is not enabled

15.3.3.1.23.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								TC_ALPHA_DEF							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.3.3.1.23.4 Fields

Field	Function
31-8 —	Reserved.
7-0 TC_ALPHA_DEF	default background alpha value

15.3.3.1.24 Timing Controller interrupt status (TC_INTERRUPT_STATUS)

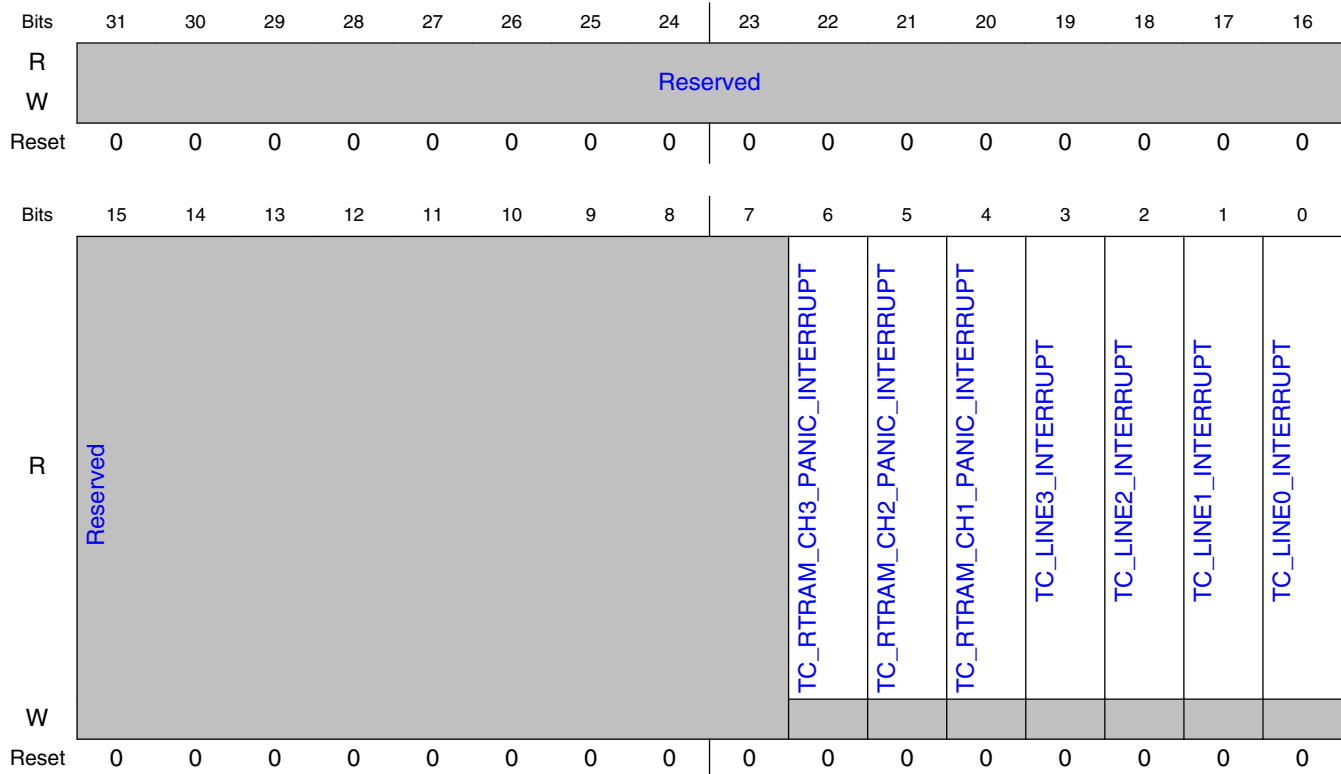
15.3.3.1.24.1 Offset

Register	Offset
TC_INTERRUPT_STATUS	5Ch

15.3.3.1.24.2 Function

This register has per interrupt status bits to indicate which interrupt(s) has occurred

15.3.3.1.24.3 Diagram



15.3.3.1.24.4 Fields

Field	Function
31-7 —	Reserved.
6 TC_RTRAM_CH3_PANIC_INTERRUPT	Panic interrupt is asserted by the scaler for channel3. Note: scaler doc may call this channel2 0: no interrupt 1: panic interrupt has been asserted
5 TC_RTRAM_CH2_PANIC_INTERRUPT	Panic interrupt is asserted by the scaler for channel2. Note: scaler doc may call this channel1 0: no interrupt 1: panic interrupt has been asserted
4 TC_RTRAM_CH1_PANIC_INTERRUPT	Panic interrupt is asserted by the scaler for channel1. Note: scaler doc may call this channel0 0: no interrupt 1: panic interrupt has been asserted
3 TC_LINE3_INTERRUPT	LINE3 interrupt status. LINE3 interrupt can be programmed to occur at any coordinate. 0: no interrupt 1: LINE3 interrupt has been asserted
2	LINE2 interrupt status. LINE2 interrupt can be programmed to occur at any coordinate.

Table continues on the next page...

Memory Map and Registers

Field	Function
TC_LINE2_INT ERRUPT	0: no interrupt 1: LINE2 interrupt has been asserted
1 TC_LINE1_INT ERRUPT	LINE1 interrupt status. LINE1 interrupt can be programmed to occur at any coordinate. 0: no interrupt 1: LINE1 interrupt has been asserted
0 TC_LINE0_INT ERRUPT	LINE0 interrupt status. LINE0 interrupt can be programmed to occur at any coordinate. 0: no interrupt 1: LINE0 interrupt has been asserted

15.3.3.1.25 Timing Controller interrupt control. (TC_INTRERRUPT_CONTROL_REG17)

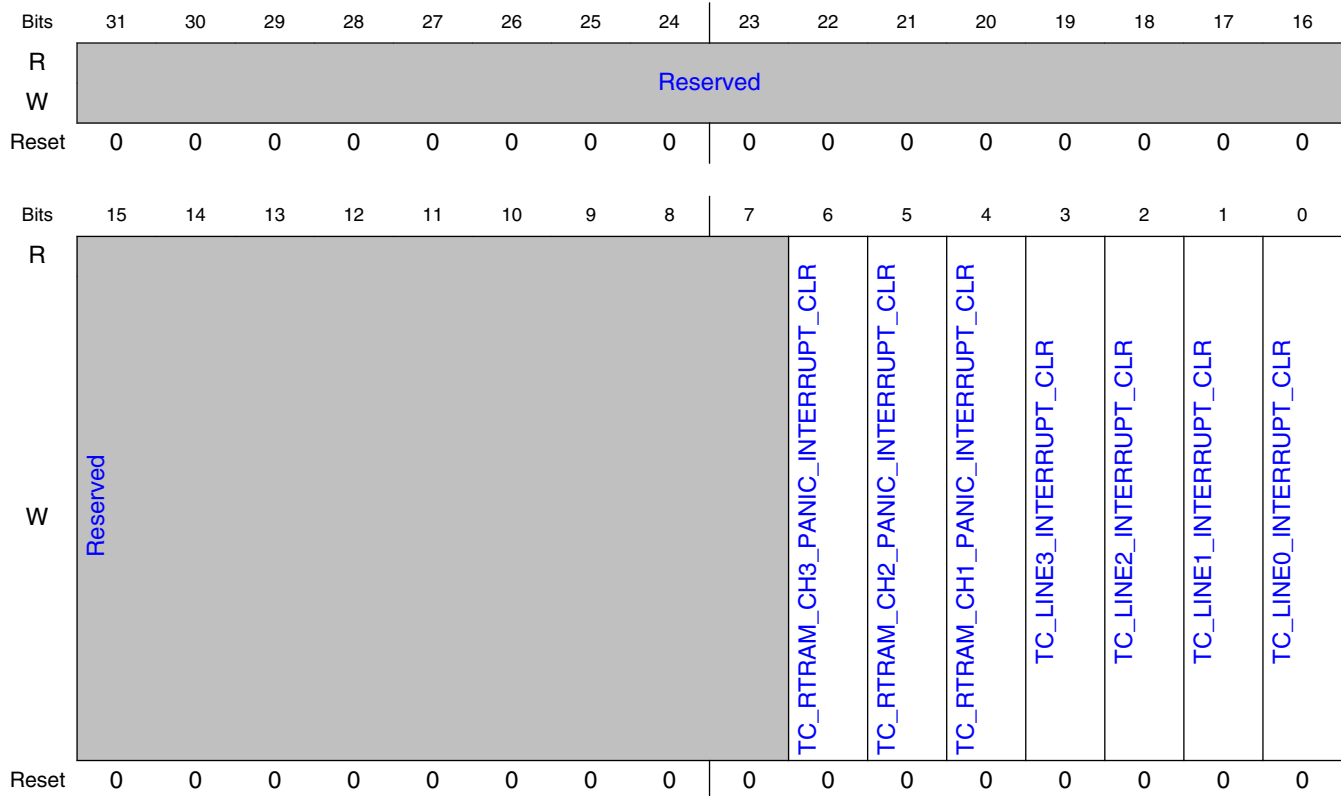
15.3.3.1.25.1 Offset

Register	Offset
TC_INTRERRUPT_CONTROL_REG17	60h

15.3.3.1.25.2 Function

This register has per interrupt control bits to clear interrupts in the TC_INTERRUPT_STATUS register

15.3.3.1.25.3 Diagram



15.3.3.1.25.4 Fields

Field	Function
31-7 —	Reserved.
6 TC_RTRAM_CH3_PANIC_INTERRUPT_CLR	Writing a 1 to this bit clears the respective interrupt in the TC_INTERRUPT_STATUS register. Channel 3 panic interrupt
5 TC_RTRAM_CH2_PANIC_INTERRUPT_CLR	Writing a 1 to this bit clears the respective interrupt in the TC_INTERRUPT_STATUS register. Channel 2 panic interrupt
4 TC_RTRAM_CH1_PANIC_INTERRUPT_CLR	Writing a 1 to this bit clears the respective interrupt in the TC_INTERRUPT_STATUS register. Channel 1 panic interrupt
3 TC_LINE3_INTERRUPT_CLR	Writing a 1 to this bit clears the respective interrupt in the TC_INTERRUPT_STATUS register.

Table continues on the next page...

Memory Map and Registers

Field	Function
2 TC_LINE2_INT ERRUPT_CLR	Writing a 1 to this bit clears the respective interrupt in the TC_INTERRUPT_STATUS register.
1 TC_LINE1_INT ERRUPT_CLR	Writing a 1 to this bit clears the respective interrupt in the TC_INTERRUPT_STATUS register.
0 TC_LINE0_INT ERRUPT_CLR	Writing a 1 to this bit clears the respective interrupt in the TC_INTERRUPT_STATUS register.

15.3.3.1.26 Channel_3 background pixel color. (TC_CH3_BKRND_REG18)

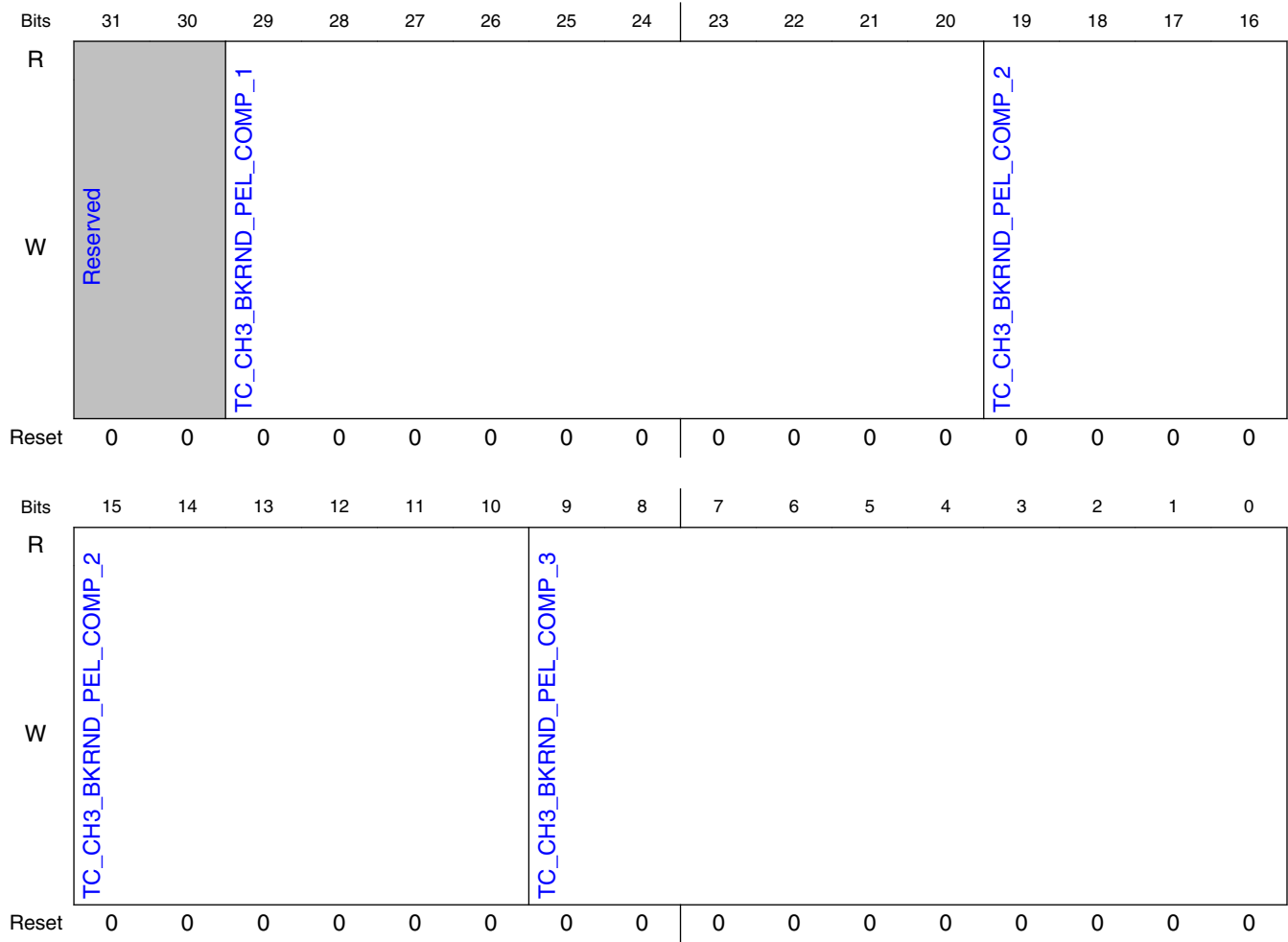
15.3.3.1.26.1 Offset

Register	Offset
TC_CH3_BKRND_REG 18	64h

15.3.3.1.26.2 Function

Outside of the Channel_3 pixel window, This background color is put on channel_3. 10 bits per component. Color space depending on programming

15.3.3.1.26.3 Diagram



15.3.3.1.26.4 Fields

Field	Function
31-30 —	Reserved.
29-20 TC_CH3_BKRND_PEL_COMP_1	10-bit component (R or Y)
19-10 TC_CH3_BKRND_PEL_COMP_2	10-bit component (G or Cb)
9-0	10-bit component (B or Cr)

Memory Map and Registers

Field	Function
TC_CH3_BKRN D_PEL_COMP_ 3	

15.3.3.1.27 Timing Controller interrupt masks (TC_INTRERRUPT_MASK)

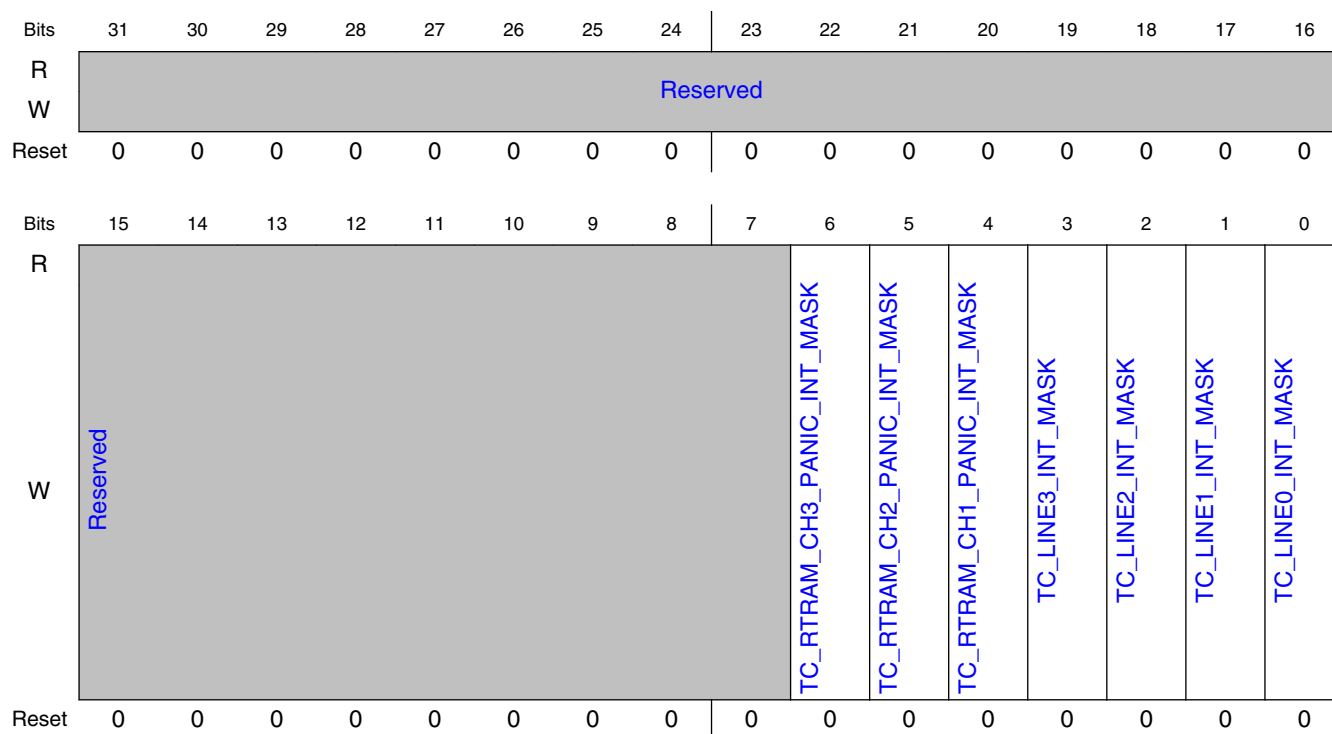
15.3.3.1.27.1 Offset

Register	Offset
TC_INTRERRUPT_MA SK	68h

15.3.3.1.27.2 Function

This register has per interrupt mask bits to allow interrupts to be disabled

15.3.3.1.27.3 Diagram



15.3.3.1.27.4 Fields

Field	Function
31-7 —	Reserved.
6 TC_RTRAM_CH 3_PANIC_INT_ MASK	Mask bit for Channel 3 Panic interrupt asserted by the scaler 0: interrupt is disabled (masked) 1: panic interrupt is enabled
5 TC_RTRAM_CH 2_PANIC_INT_ MASK	Mask bit for Channel 2 Panic interrupt asserted by the scaler 0: interrupt is disabled (masked) 1: panic interrupt is enabled
4 TC_RTRAM_CH 1_PANIC_INT_ MASK	Mask bit for Channel 1 Panic interrupt asserted by the scaler 0: interrupt is disabled (masked) 1: panic interrupt is enabled
3 TC_LINE3_INT_ MASK	LINE3 interrupt status. LINE3 interrupt can be programmed to occur at any coordinate. 0: interrupt is disabled (masked) 1: LINE3 interrupt has been enabled
2 TC_LINE2_INT_ MASK	LINE2 interrupt status. LINE2 interrupt can be programmed to occur at any coordinate. 0: interrupt is disabled (masked) 1: LINE2 interrupt has been enabled
1 TC_LINE1_INT_ MASK	LINE1 interrupt status. LINE1 interrupt can be programmed to occur at any coordinate. 0: interrupt is disabled (masked) 1: LINE1 interrupt has been enabled
0 TC_LINE0_INT_ MASK	LINE0 interrupt status. LINE0 interrupt can be programmed to occur at any coordinate. 0: interrupt is disabled (masked) 1: LINE0 interrupt has been enabled

15.3.3.1.28 LINE3 interrupt control: Coordinate where line3 interrupt is asserted (TC_LINE3_INT_REG)

15.3.3.1.28.1 Offset

Register	Offset
TC_LINE3_INT_REG	6Ch

15.3.3.1.28.2 Function

The X,Y coordinates in this register specify where line3 interrupt is asserted as timing controller encounters these coordinates. Can be anywhere on raster table. For historical reasons this is called LINE3 interrupt in RTL

15.3.3.1.28.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved			TC_LINE3_INT_Y												
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved			TC_LINE3_INT_X												
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.3.3.1.28.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_LINE3_INT_Y	Y (vertical) coordinate for line3 interrupt
15-13 —	Reserved.
12-0 TC_LINE3_INT_X	X (horizontal) coordinate for line3 interrupt

15.3.3.1.29 LINE4 interrupt control: Coordinate where line4 interrupt is asserted (TC_LINE4_INT_REG)

15.3.3.1.29.1 Offset

Register	Offset
TC_LINE4_INT_REG	70h

15.3.3.1.29.2 Function

The X,Y coordinates in this register specify where line4 interrupt is asserted as timing controller encounters these coordinates. Can be anywhere on raster table. For historical reasons this is called LINE4 interrupt in RTL.

15.3.3.1.29.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								TC_LINE4_INT_Y							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								TC_LINE4_INT_X							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.3.3.1.29.4 Fields

Field	Function
31-29 —	Reserved.
28-16 TC_LINE4_INT_Y	Y (vertical) coordinate for line4 interrupt
15-13 —	Reserved.
12-0 TC_LINE4_INT_X	X (horizontal) coordinate for line4 interrupt

15.3.3.1.30 For DBY Mode: Controls the assertion and de-assertion DE signal (Overlay channel). (TC_OL_DE_CONTROL_REG)

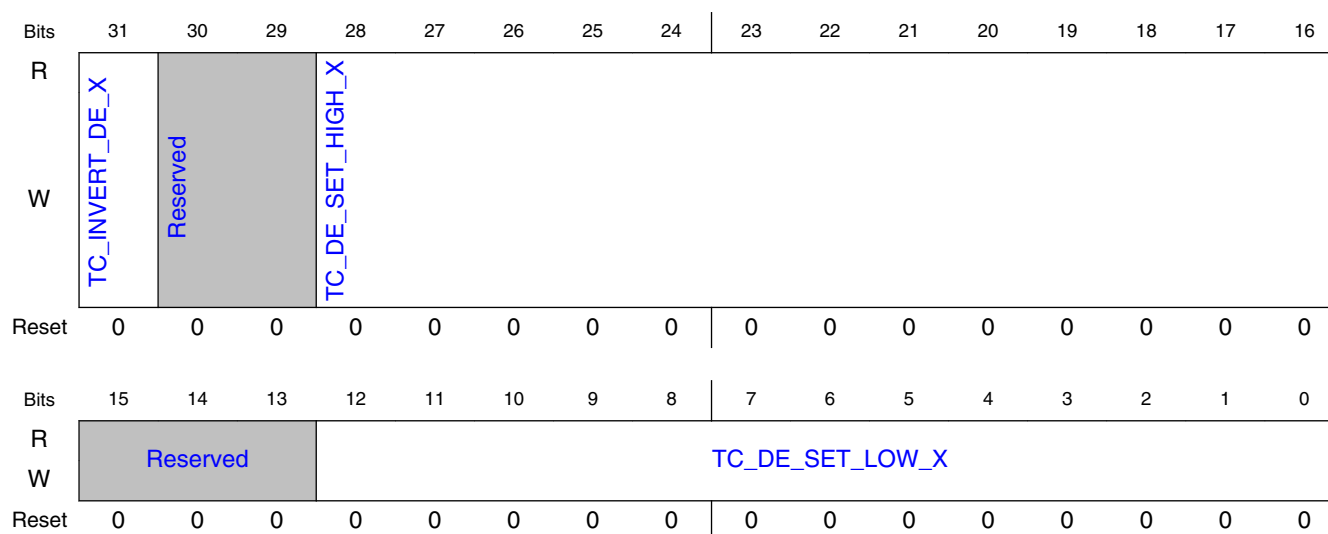
15.3.3.1.30.1 Offset

Register	Offset
TC_OL_DE_CONTROL_REG	74h

15.3.3.1.30.2 Function

This control register pertains to the DBY Overlay (OL) channel. The X coordinates in this register specify where the DE signal is made 0 or 1 in a line.

15.3.3.1.30.3 Diagram



15.3.3.1.30.4 Fields

Field	Function
31 TC_INVERT_DE_X	0: DE signal is not inverted 1: DE signal is INVERTED
30-29 —	Reserved.
28-16 TC_DE_SET_HIGH_X	X (horizontal) coordinate Where DE control signal is set to 1 in a line.
15-13 —	Reserved.
12-0 TC_DE_SET_LOW_X	X (horizontal) coordinate Where DE control signal is set to 0 in a line.

15.3.3.1.31 For DBY Mode: Controls the assertion and de-assertion DE signal (Base layer (BL) channel). (TC_BL_DE_CONTROL_REG)

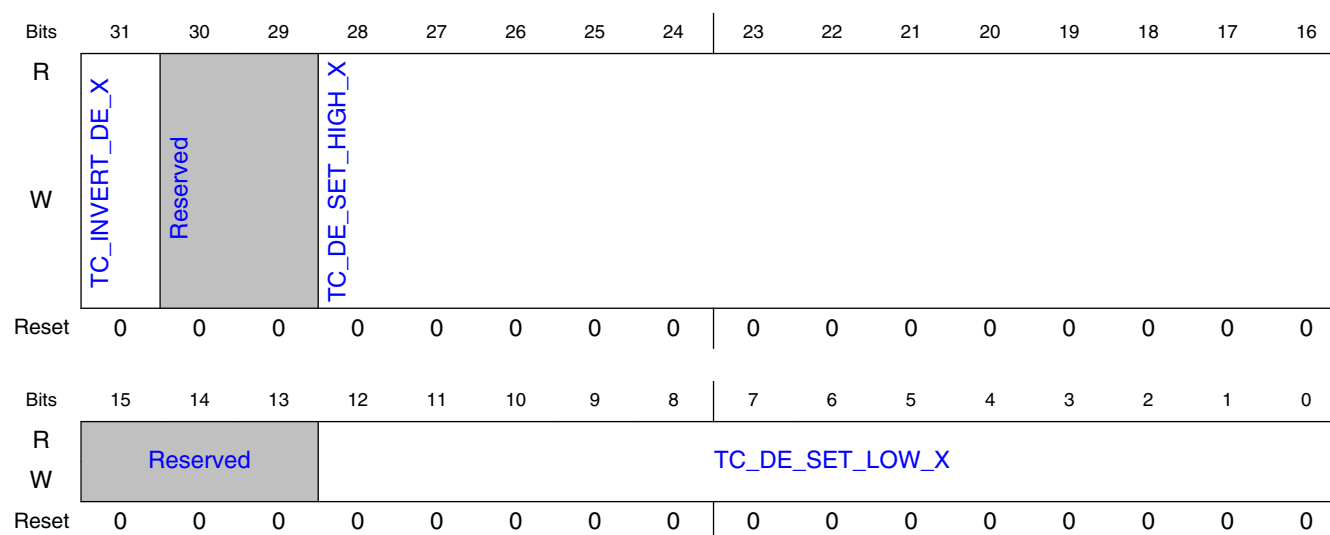
15.3.3.1.31.1 Offset

Register	Offset
TC_BL_DE_CONTROL_REG	78h

15.3.3.1.31.2 Function

This control register pertains to the DBY Base layer (BL) channel. The X coordinates in this register specify where the DE signal is made 0 or 1 in a line.

15.3.3.1.31.3 Diagram



15.3.3.1.31.4 Fields

Field	Function
31 TC_INVERT_DE_X	0: DE signal is not inverted 1: DE signal is INVERTED
30-29 —	Reserved.
28-16 TC_DE_SET_HIGH_X	X (horizontal) coordinate Where DE control signal is set to 1 in a line.
15-13 —	Reserved.
12-0	X (horizontal) coordinate Where DE control signal is set to 0 in a line.

Memory Map and Registers

Field	Function
TC_DE_SET_LOW_X	

15.3.3.1.32 For DBY Mode: Controls the assertion and de-assertion DE signal (Enhancement layer (EL) channel). (TC_EL_DE_CONTROL_REG)

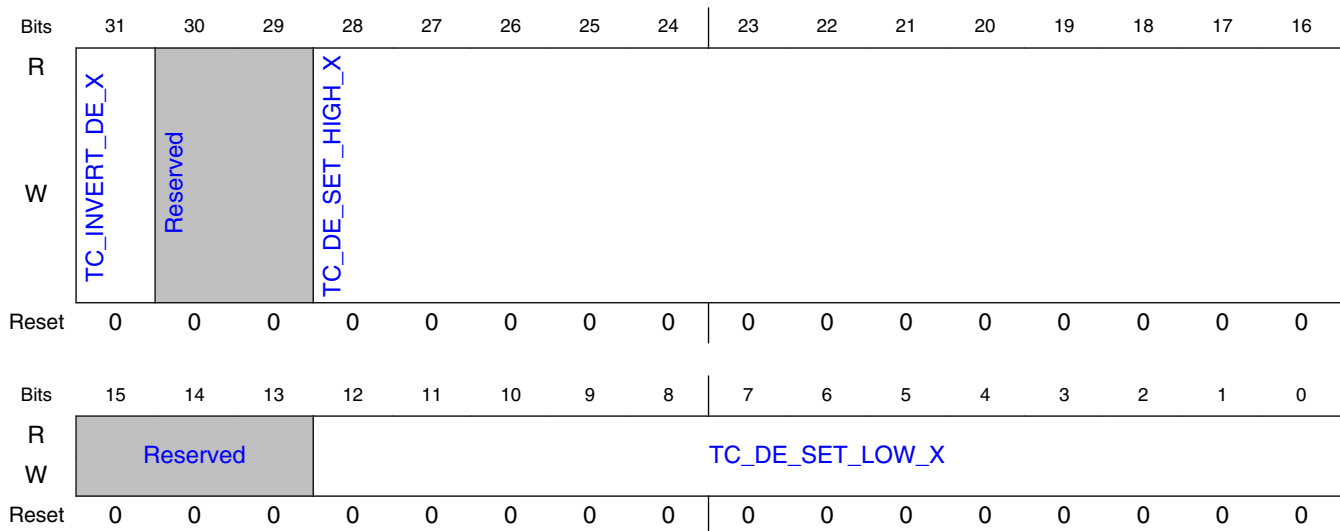
15.3.3.1.32.1 Offset

Register	Offset
TC_EL_DE_CONTROL_REG	7Ch

15.3.3.1.32.2 Function

This control register pertains to the DBY Enhancement layer (EL) channel. The X coordinates in this register specify where the DE signal is made 0 or 1 in a line.

15.3.3.1.32.3 Diagram



15.3.3.1.32.4 Fields

Field	Function
31	0: DE signal is not inverted

Table continues on the next page...

Field	Function
TC_INVERT_DE_X	1: DE signal is INVERTED
30-29 —	Reserved.
28-16 TC_DE_SET_HIGH_X	X (horizontal) coordinate Where DE control signal is set to 1 in a line.
15-13 —	Reserved.
12-0 TC_DE_SET_LOW_X	X (horizontal) coordinate Where DE control signal is set to 0 in a line.

15.4 Context Load (CTX_LD)

15.4.1 Overview

The Context Loader (CTX_LD) module fetches and loads the next state of the display from system memory.

15.4.1.1 Block Diagram

The diagram below indicates the functional components and data flow in the CTX_LD module.

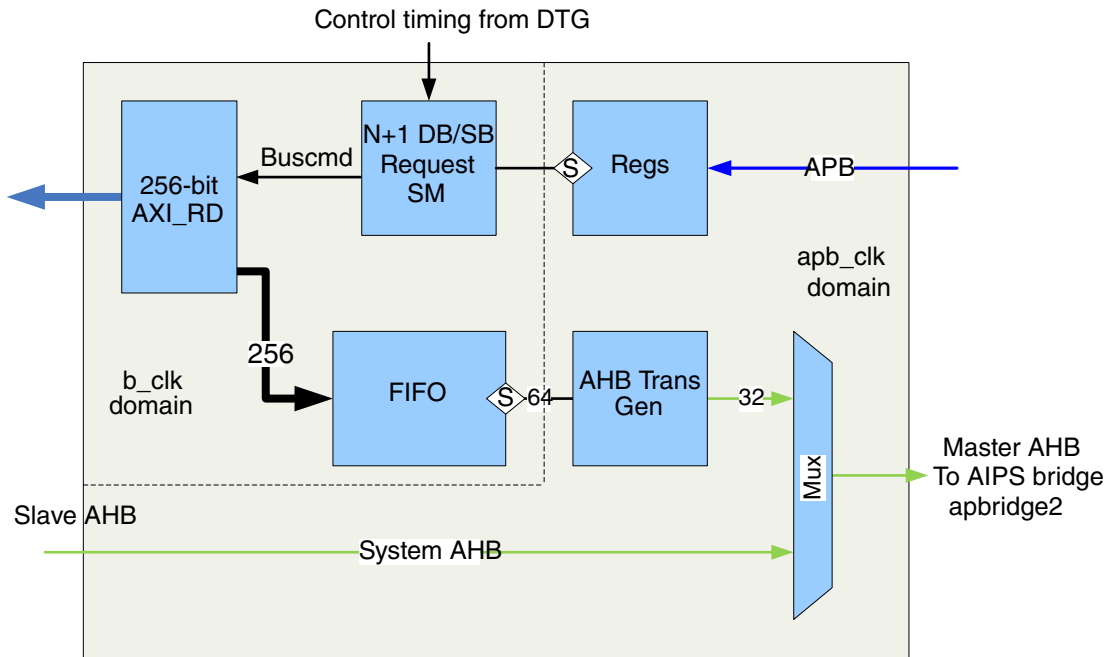


Figure 15-10. CTX_LD Block Diagram

- The request state machine issues transactions to fetch the register addr/data pairs from system memory (see [System Memory Display State Format](#)). It manages the FIFO to keep the register loading operation advancing until all the DB and SB registers are loaded.
- The loading sequence is described below in section [Display State Loading Sequence](#). A likely size for the FIFO is 512 bytes. Each request to system memory by the state machine is half the size of the FIFO. When the FIFO is half full/empty, the SM requests another half FIFO size of addr/data pairs from system memory. This process proceeds until all the DB and SB registers are loaded.
- The AHB transaction generator creates the transactions needed to empty the FIFO and load DC registers contents.

15.4.2 Functional Description

Two events indicate when the next display context is loaded. The timing of the CTX_LD is controlled by the Display Timing Generator (DTG).

The first event, marked by the red dot in the DTG chapter's timing example (Display Timing and Window Overlay section), and also in the timing diagram in the section [Timing Sequence](#), is for components that implement double buffered registers and require their next state to be loaded into shadow registers during the active display time. This interval is defined as the db_region, or double buffered region.

The second event, marked by the orange dot, occurs after the active display region for components that don't support double buffered registers and require the next state to be loaded during the vertical blanking time. This interval is defined as the `sb_region`, or single buffered region. The second region is subdivided into two regions. The first is high priority and critical from a timing perspective. The second includes all single buffered components that aren't critical from a timing perspective. These two sub regions are defined as `sb_region.high_prioity` and `sb_region.low_priority`.

The critical region should include loading component parameters that need to be issued first. As an example, the pixel prefetch components (DPR) would need to be programmed first and kicked to start their prefetch operation before the first scan line in the active display region. The DPR would not be enabled to start after the HDR10 LUTs are loaded, as an example. The idea is that low priority registers are being loaded in parallel with the DPR's prefetch operation. This provides the best opportunity to prefetch data BEFORE the first pixel is needed to drive the display when the active region begins

As stated before, the loading sequence has the following regions.

1. `db_region`; Region where all double buffered registers are loaded.
2. `sb_region`; Region where all single buffered registers are loaded.
 - a. `sb_region.high_priority` - critical time to load single buffered components.
 - b. `sb_region.low_priority` - time to load components not critical on timing.

The timing of db/sb regions are controlled and timed by the DTG, or respective display timing generator. These event trigger times are programmable based on vertical timing parameters in the DTG that go high for one scan line interval. The rising edge of these events cause the CTX_LD engine to fetch the next state parameters from DRAM.

The components that implement double buffered registers should transfer the shadow registers into the active registers when `ctx_ld_sb_en` goes active. This is the trigger used to start loading single buffered registers, and is used to allow the double buffered register components to start processing for the next frame. The idea is that the double buffered components should be enabled shortly after the active display region is complete, and the display enters the vertical front porch region. When the trigger is received by the double buffered component, then the component begins processing for trace N+1. The earliest possible event to enable the DB components is necessary to allow the most time to complete processing for trace N+1 before the active display area becomes active.

Components that implement single buffer registers receives their new values from the `ctx_ld` module after the `ctx_ld_sb_en` signal goes high. This is the event that indicates the CTX_LD fetches and loads the single buffered components in the display SS. Single buffered components starts processing using their last register write, or can also use the `ctx_ld_sb_hp_kick` output signal to "kick" their processing for trace N+1.

NOTE

It could be valid that the CTX_LD is NOT enabled to fetch the next state for N+1, i.e. there is no state change and the display system repeats the frame N+1 with the existing parameters already programmed. In this case, the CTX_LD still receives the `ctx_ld_sbldb_en` signal inputs and the CTX_LD does not issue DRAM accesses to load the parameters. The `ctx_ld_sb_hp_kick_en` and the `ctx_ld_sb_lp_kick_en` output strobes cycle low/high/low to indicate all modules should start processing using their existing parameters. Or, 0 registers were fetched in 0 time, and the process completes by issuing both `ctx_ld_sb_hp_kick` and `ctx_ld_sb_lp_kick` to “kick” all the SB components so processing can begin. The `ctx_ld_sb_[hpl lp]_kick` signal is asserted just after it receives the `ctx_ld_sb_en` input strobe if the CTX_LD register loading sequence was not enabled during trace N. This allows to restart the components with the previous state used in N.

Double buffered components use either vsync or the `ctx_ld_sb_hp_kick` signal to start processing for trace N+1.

Once the display state is completely loaded, both those that are doubled buffered and those that use a single buffering scheme, all components commence operation to refresh the display. Some of the next state parameters that are loaded by the CTX_LD engine are defined as, but not limited to, the following:

- The source buffer addresses, formats, and rotation parameters fetched by each DPR.
- The scale width, height, X/Y scaling factors and initial offsets, and coeffs for the scaling operation
- X,Y min/max extents for the three surfaces in the output display.
- WR_SCL and RD_SRC buffer addresses, pitch, and formats, if needed.
- Data flow configuration options (which scaler goes to DRAM if any).

The Context Loader Engine is also capable of loading the Dolby Vision tables or the HDR10 LUT entries from DDR. When the CTX_LD is done with the next state loading sequence, it returns to its idle state by auto clearing the ENABLE bit in the CTRL_STATUS register.

15.4.2.1 Timing Sequence

The timing diagram below indicates the sequence over the trace time N, in preparation for trace N+1. The DB load strobe starts the CTX_LD sequence. The first `ctx_ld_db_en` event that occurs AFTER the PIO write to set `CTX_LD_CTRL[ENABLE]` commences the register loading sequence for display trace N+1.

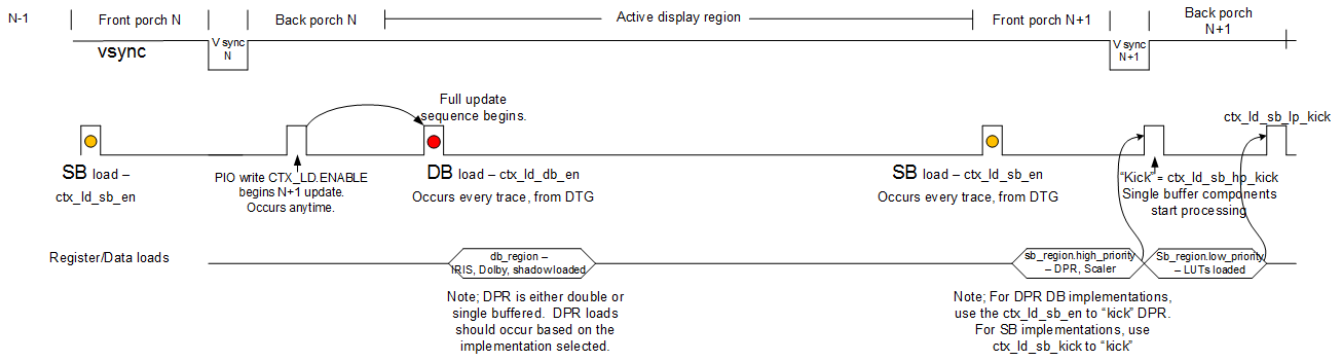


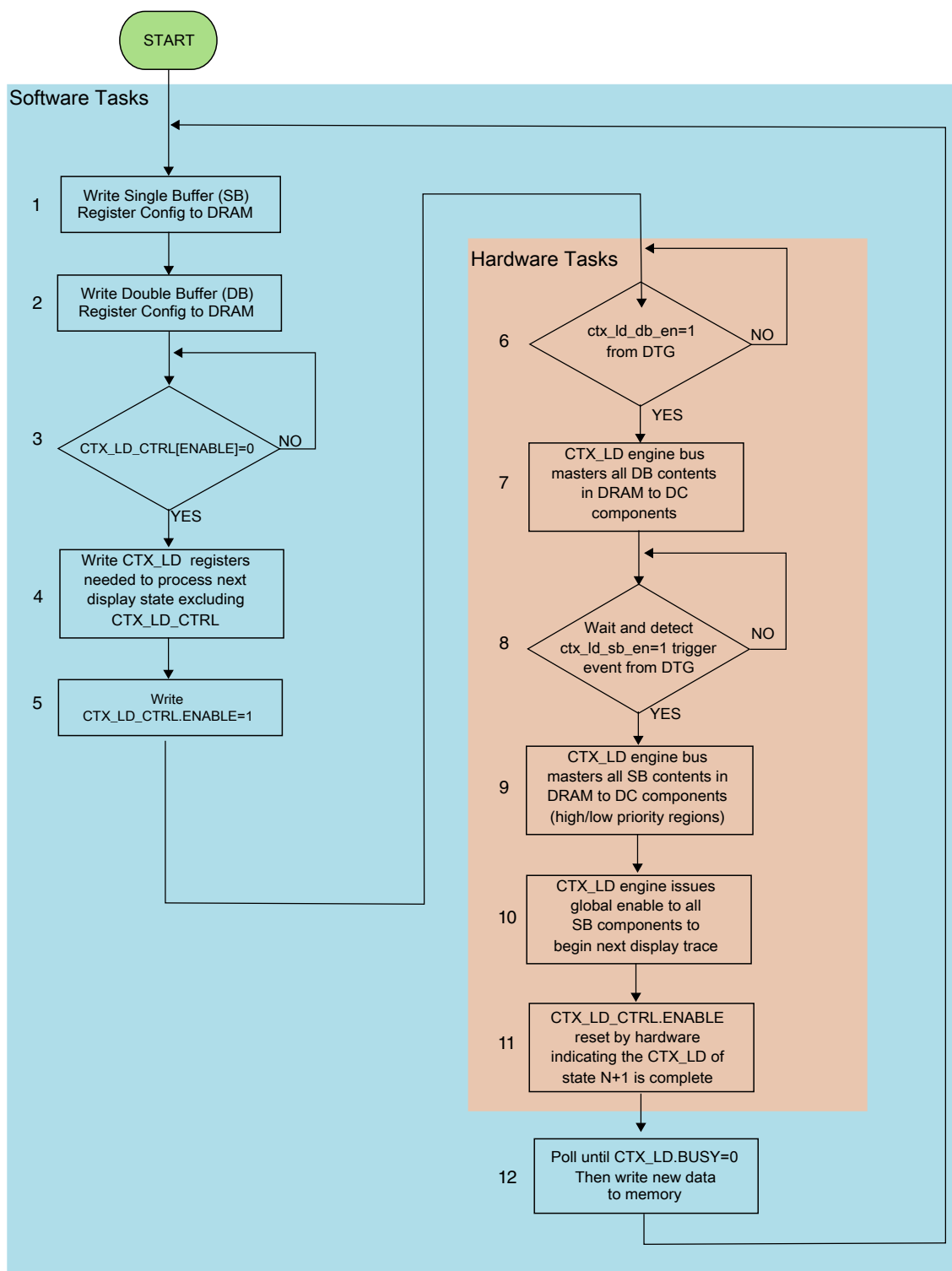
Figure 15-11. CTX_LD Timing Sequence

NOTE

If the PIO write to set the `CTX_LD_CTRL[ENABLE]` comes after the `ctx_ld_db_en` event AND before the `ctx_ld_sb_en` event, nothing occurs. The process to load DB and SB registers always starts at the `ctx_ld_db_en` event. So, DB registers are loaded first, and then the SB registers are loaded second to complete the full sequence.

15.4.2.2 Display State Loading Sequence

The sequence to properly load all registers for the next display state N+1 is shown below.



See detailed descriptions of these steps below for more information.

Figure 15-12. Display State Loading Sequence

NOTE: “SW” indicates an action taken by the software, and “HW” indicates action on behalf of the hardware.

1. SW: Write all single buffered register configurations in DRAM at address SB_BASE_ADDR. This should be a new address not used in the previous CTX_LD sequence.
2. SW: Write all double buffered register configurations in DRAM at address DB_BASE_ADDR. This should be a new address not used in the previous CTX_LD sequence.
3. SW: Read the CTX_LD_CTRL[ENABLE] = 0. This indicates the CTX_LD is idle and ready for the next state update.
 - a. CTX_LD_CTRL[ENABLE] = 1, wait, goto #3
 - b. CTX_LD_CTRL[ENABLE] = 0, goto #4
4. SW: Write all CTX_LD registers needed to process the next display state excluding CTX_LD_CTRL. CTX_LD_SB_HP_CNT cannot be 0 if CTX_LD_SB_LP_CNT != 0.
5. SW: Write the CTX_LD_CTRL[ENABLE] = 1. This bit can only be set by SW. Note, it is reset by HW when the CTX_LD engine is complete loading the entire next state. When the ENABLE bit is high, it indicates that
 - a. The CTX_LD is ready to start the loading sequence when the ctx_ld_db_en input goes high.
 - b. or CTX_LD is currently busy loading the state N+1.
6. HW: Wait and detect the ctx_ld_db_en trigger input event from the DTG. This occurs around the timing indicated by the red dot in the display timing diagram.
7. HW: The CTX_LD engine bus masters all double buffered register contents from DRAM to DC components.
8. HW: Wait and detect the ctx_ld_sb_en trigger event input from the DTG. Generally, this timing is indicated by the orange dot and is just after the completion of the active display region (start of vertical front porch).
9. HW: The ctx_ld engine bus masters all single buffered register contents from DRAM to DC components. Waits for the last transaction response via AHB. a. NOTE: During this interval, there are two regions to load registers divided into the high/low priority regions.
10. HW: The ctx_ld engine issues a global enable control signal to all single buffered components to begin processing for the next display trace after each of the high/low priority regions are complete. These control signals are the ctx_ld_sb_hp_kick and ctx_ld_sb_lp_kick. DC components that are considered high priority should use ctx_ld_sb_hp_kick signal to indicate when they should start processing. Likewise for low priority DC components.
11. The CTX_LD_CTRL[ENABLE] bit is reset by HW indicating the CTX_LD of state N+1 is complete.

12. Goto #1. When CTX_LD_CTRL[ENABLE] set by SW, it should not update the DRAM contents the CTX_LD uses to set the state for display trace N+1. SW should also refrain from writing CTX_LD registers that control the next context load for display trace N+1. If alternate buffers in DRAM are used for another display state, then they can be preloaded at a different address since the CTX_LD engine only uses the buffers pointed to by the base address registers. When the CTX_LD_CTRL[ENABLE] bit is low, SW can program CTX_LD registers and set CTX_LD_CTRL[ENABLE] to initiate the process again.

15.4.2.3 System Memory Display State Format

The format of the entries in DRAM used to load a register is indicated in the table below:

Table 15-7. 64-bit Address Offset and Data Pair in Memory

Address in DRAM	63:32 DC register address offset	31:0 DC register value
ctx_[dblsb]_base_addr	First reg offset to be loaded in DC	Respective first reg data value.
ctx_[dblsb]_base_addr + 8	Second reg offset to loaded in DC	Respective second reg value.
...	...	...
ctx_[dblsb]_base_addr + 8*count	CTX_[DBISB]_COUNT reg offset in DC	Respective last reg value.

15.4.2.4 Arbitration

The MUX arbitrates between transactions from the AHB transaction generator and the system AHB bus. When the CTX_LD state machine is actively fetching register contents from system memory and loading registers in the DC, accesses from the slave AHB bus (the host) are arbitrated. This occurs during the time when the CTX_LD is loading DB or SB registers. So, if concurrent access is required by the CTX_LD engine AND the host (via AHB), then the accesses are arbitrated in an alternating sequence. If the host is attempting a burst, a single burst is consider as a single access. This access completes, and then the CTX_LD engine wins the right to initiate a single burst.

As a programmable option (NOT the default), the host AHB bus can be blocked while the CTX_LD engine is actively processing transactions. This is intended to allow the CTX_LD to completely load the register state without interruption. It is expected that blocking the host is not permitted for long intervals. This option is an intent to avoid programming issues while the CTX_LD and host are concurrently accessing registers. Another way to view this is that the CTX_LD always has priority over the slave AHB accesses via the host. The following CTX_LD operational states would stall the slave AHB bus.

- CTX_LD request state machine is active and generating requests.
- CTX_LD request state machine is done, but the transaction FIFO is not empty and still has transactions to complete to components within the DC.

NOTE

System software is responsible for guaranteeing the system AHB transactions and the CTX_LD transactions do not violate the programming and operation model of the DC SS. This is required regardless if the arbitration scheme, or the round robin scheme, is used for processing transactions from the two sources.

15.4.3 Memory Map and Registers

15.4.3.1 register descriptions

15.4.3.1.1 CTX_LD Memory map

ctx_ld base address: 2_3000h

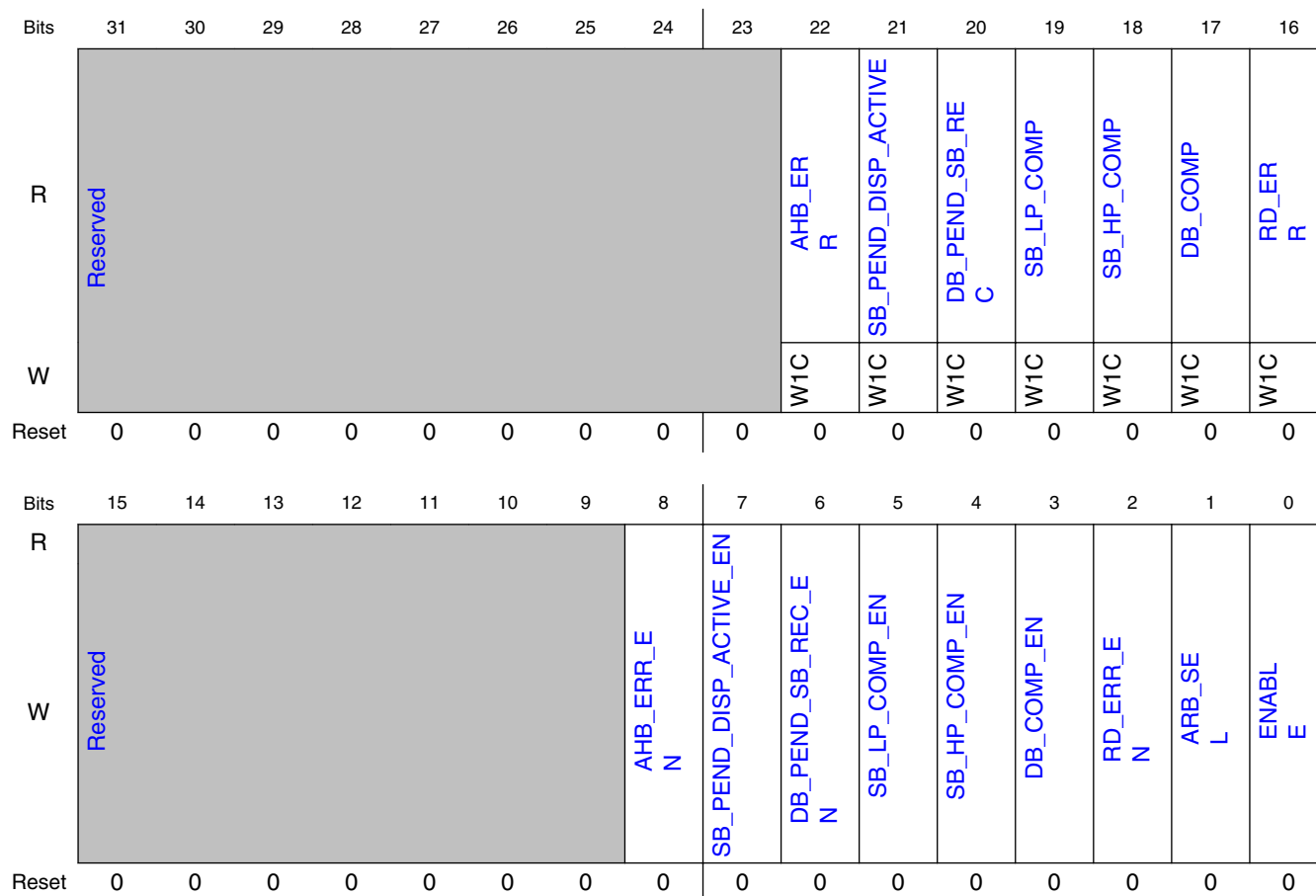
Offset	Register	Width (In bits)	Access	Reset value
0h	Control status register for Context Loader. (CTRL_STATUS)	32	RW	0000_0000h
4h	Control status register for Context Loader. (CTRL_STATUS_SET)	32	RW	0000_0000h
8h	Control status register for Context Loader. (CTRL_STATUS_CLR)	32	RW	0000_0000h
Ch	Control status register for Context Loader. (CTRL_STATUS_TOG)	32	RW	0000_0000h
10h	DRAM addr for double buffered register fetch. (DB_BASE_ADDR)	32	RW	0000_0000h
14h	Double buffer register count (DB_COUNT)	32	RW	0000_0000h
18h	DRAM addr for single buffered registers. (SB_BASE_ADDR)	32	RW	0000_0000h
1Ch	Single buffer register count (SB_COUNT)	32	RW	0000_0000h
20h	AHB address with error response. (AHB_ERR_ADDR)	32	RO	0000_0000h

15.4.3.1.2 Control status register for Context Loader. (CTRL_STATUS)

15.4.3.1.2.1 Offset

Register	Offset
CTRL_STATUS	0h

15.4.3.1.2.2 Diagram



15.4.3.1.2.3 Fields

Field	Function
31-23 —	Reserved
22 AHB_ERR	AHB error Status bit to indicate when the internal AHB bus returns an error response. This reflects an error on the AHB bus output from the CTX_LD to the DCSS memory mapped components. This could occur when accessing a region that is not mapped. This status bit can be set ONLY for CTX_LD initiated transactions

Table continues on the next page...

Field	Function
	that receive an AHB error. System transactions that respond with an error are passed through, and the initiating master manages the error response at the source.
21 SB_PEND_DIS P_ACTIVE	Single/active region overlap Status bit to indicate when the SB region is still actively loading registers to the display subsystem AND the display enters the active display time. This would indicate the SB registers are not loaded in time and the display trace could be corrupt. Unpredictable behaviour could result if this occurs.
20 DB_PEND_SB_ REC	Double/single region overlap Status bit to indicate when the DB region is still actively loading registers to the display subsystem AND a kick to load the SB region is received. This could indicate a timing or BW issue with loading the DB region in time before the SB region register loading should begin. Unpredictable behaviour could result if this occurs.
19 SB_LP_COMP	Single buffer low priority region loading complete IRQ enable Status bit to indicate when the low priority SB region loading sequence is complete.
18 SB_HP_COMP	Single buffer high priority region loading complete Status bit to indicate when the high priority SB region loading sequence is complete.
17 DB_COMP	Double buffer region loading complete Status bit to indicate when the DB (double buffer) region loading sequence is complete.
16 RD_ERR	AXI read error Status bit to indicate when the AXI read bus returns a SLV_ERR or DEC_ERR. This could result in an illegal or unmapped address programmed into the base address registers.
15-9 —	Reserved
8 AHB_ERR_EN	AHB error IRQ enable Enable and interrupt when the internal AHB bus returns an error response. This reflects an error on the AHB bus output from the CTX_LD to the DCSS memory mapped components. This could occur when accessing a region that is not mapped.
7 SB_PEND_DIS P_ACTIVE_EN	Single/active region overlap interrupt enable. Enable an interrupt when the SB region is still actively loading registers to the display subsystem AND the display enters the active display time. This would indicate the SB registers are not loaded in time and the display trace could be corrupt. Unpredictable behaviour could result if this occurs.
6 DB_PEND_SB_ REC_EN	Double/single region overlap interrupt enable Enable an interrupt when the DB region is still actively loading registers to the display subsystem AND a kick to load the SB region is received. This could indicate a timing or BW issue with loading the DB region in time before the SB region register loading should begin. Unpredictable behaviour could result if this occurs.
5 SB_LP_COMP_ EN	Single buffer low priority region loading complete IRQ enable Enable an interrupt when the low priority SB region loading sequence is complete.
4 SB_HP_COMP_ EN	Single buffer high priority region loading complete IRQ enable Enable an interrupt when the high priority SB region loading sequence is complete.
3 DB_COMP_EN	Double buffer region loading complete IRQ enable Enable an interrupt when the DB (double buffer) region loading sequence is complete.
2 RD_ERR_EN	AXI read error IRQ enable

Table continues on the next page...

Memory Map and Registers

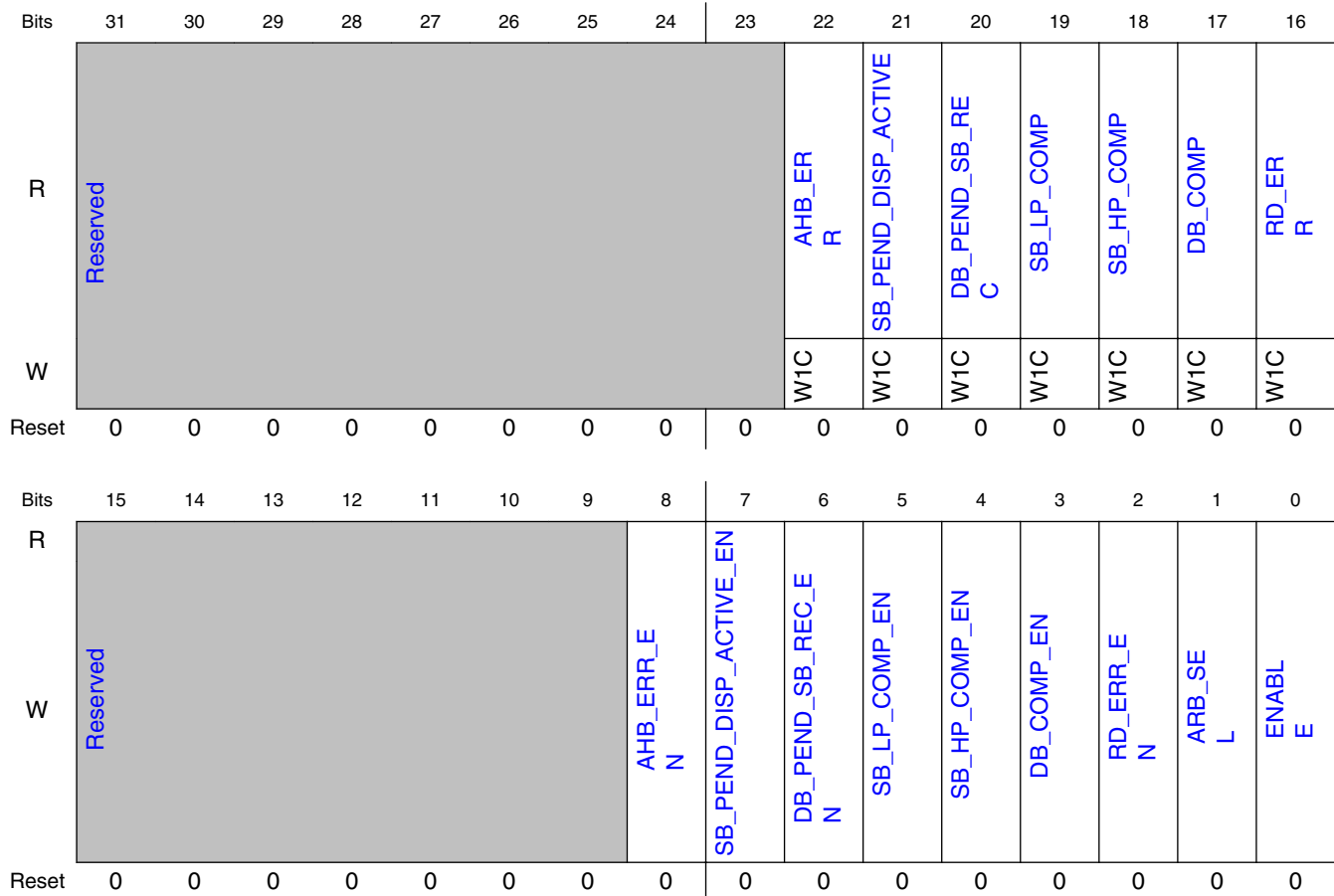
Field	Function
	Enable an interrupt when the AXI read bus returns a SLV_ERR or DEC_ERR. This could result in an illegal or unmapped address programmed into the base address registers.
1 ARB_SEL	Arbitration select This selects between always choosing the CTX_LD while it's active verses a round robin arbitration scheme of equal weight between the CTX_LD and the system register access bus. • 0 - Round robin arbitration at all times. • 1 - CTX_LD gets priority while it's active.
0 ENABLE	Enable/Busy Enable bit for the context loader. The context loader does not start the register loading sequence if this bit is not set. SW can set this bit only when it is logic 0 (never set this bit if it is already logic 1). The CTX_LD hardware resets this bit when the entire register loading sequence is complete. It is complete after the DB and SB regions are loaded and the CTX_LD state machine returns to the idle state to allow the next context loading sequence to be enabled.

15.4.3.1.3 Control status register for Context Loader. (CTRL_STATUS_SET)

15.4.3.1.3.1 Offset

Register	Offset
CTRL_STATUS_SET	4h

15.4.3.1.3.2 Diagram



15.4.3.1.3.3 Fields

Field	Function
31-23 —	Reserved
22 AHB_ERR	AHB error Status bit to indicate when the internal AHB bus returns an error response. This reflects an error on the AHB bus output from the CTX_LD to the DCSS memory mapped components. This could occur when accessing a region that is not mapped. This status bit can be set ONLY for CTX_LD initiated transactions that receive an AHB error. System transactions that respond with an error are passed through, and the initiating master manages the error response at the source.
21 SB_PEND_DISP_ACTIVE	Single/active region overlap Status bit to indicate when the SB region is still actively loading registers to the display subsystem AND the display enters the active display time. This would indicate the SB registers are not loaded in time and the display trace could be corrupt. Unpredictable behaviour could result if this occurs.
20 DB_PEND_SB_REC	Double/single region overlap Status bit to indicate when the DB region is still actively loading registers to the display subsystem AND a kick to load the SB region is received. This could indicate a timing or BW issue with loading the DB region

Table continues on the next page...

Memory Map and Registers

Field	Function
	in time before the SB region register loading should begin. Unpredictable behaviour could result if this occurs.
19 SB_LP_COMP	Single buffer low priority region loading complete IRQ enable Status bit to indicate when the low priority SB region loading sequence is complete.
18 SB_HP_COMP	Single buffer high priority region loading complete Status bit to indicate when the high priority SB region loading sequence is complete.
17 DB_COMP	Double buffer region loading complete Status bit to indicate when the DB (double buffer) region loading sequence is complete.
16 RD_ERR	AXI read error Status bit to indicate when the AXI read bus returns a SLV_ERR or DEC_ERR. This could result in an illegal or unmapped address programmed into the base address registers.
15-9 —	Reserved
8 AHB_ERR_EN	AHB error IRQ enable Enable and interrupt when the internal AHB bus returns an error response. This reflects an error on the AHB bus output from the CTX_LD to the DCSS memory mapped components. This could occur when accessing a region that is not mapped.
7 SB_PEND_DIS P_ACTIVE_EN	Single/active region overlap interrupt enable. Enable an interrupt when the SB region is still actively loading registers to the display subsystem AND the display enters the active display time. This would indicate the SB registers are not loaded in time and the display trace could be corrupt. Unpredictable behaviour could result if this occurs.
6 DB_PEND_SB_ REC_EN	Double/single region overlap interrupt enable Enable an interrupt when the DB region is still actively loading registers to the display subsystem AND a kick to load the SB region is received. This could indicate a timing or BW issue with loading the DB region in time before the SB region register loading should begin. Unpredictable behaviour could result if this occurs.
5 SB_LP_COMP_ EN	Single buffer low priority region loading complete IRQ enable Enable an interrupt when the low priority SB region loading sequence is complete.
4 SB_HP_COMP_ EN	Single buffer high priority region loading complete IRQ enable Enable an interrupt when the high priority SB region loading sequence is complete.
3 DB_COMP_EN	Double buffer region loading complete IRQ enable Enable an interrupt when the DB (double buffer) region loading sequence is complete.
2 RD_ERR_EN	AXI read error IRQ enable Enable an interrupt when the AXI read bus returns a SLV_ERR or DEC_ERR. This could result in an illegal or unmapped address programmed into the base address registers.
1 ARB_SEL	Arbitration select This selects between always choosing the CTX_LD while it's active verses a round robin arbitration scheme of equal weight between the CTX_LD and the system register access bus. • 0 - Round robin arbitration at all times. • 1 - CTX_LD gets priority while it's active.
0 ENABLE	Enable/Busy Enable bit for the context loader. The context loader does not start the register loading sequence if this bit is not set. SW can set this bit only when it is logic 0 (never set this bit if it is already logic 1). The CTX_LD

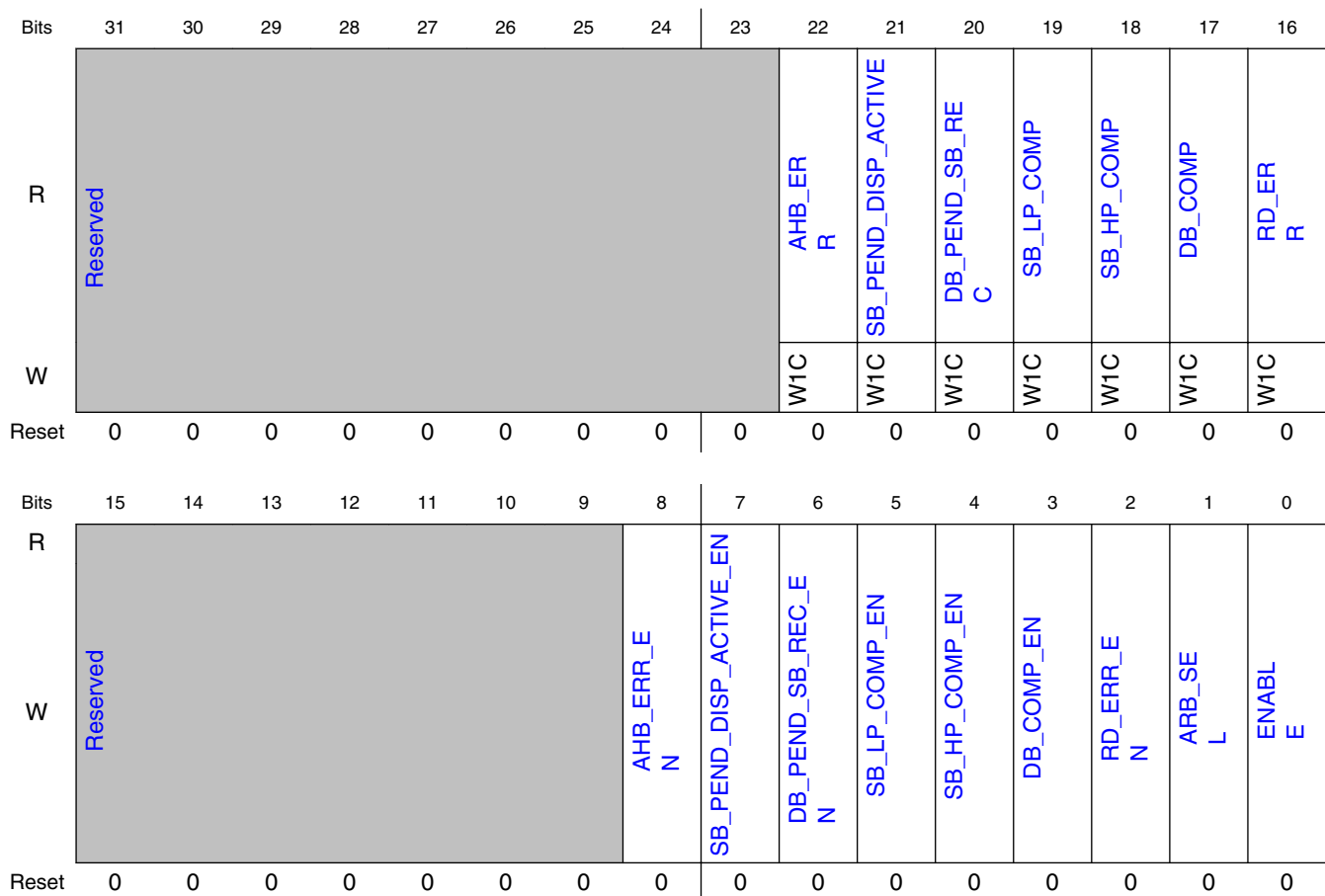
Field	Function
	hardware resets this bit when the entire register loading sequence is complete. It is complete after the DB and SB regions are loaded and the CTX_LD state machine returns to the idle state to allow the next context loading sequence to be enabled.

15.4.3.1.4 Control status register for Context Loader. (CTRL_STATUS_CLR)

15.4.3.1.4.1 Offset

Register	Offset
CTRL_STATUS_CLR	8h

15.4.3.1.4.2 Diagram



15.4.3.1.4.3 Fields

Field	Function
31-23 —	Reserved
22 AHB_ERR	AHB error Status bit to indicate when the internal AHB bus returns an error response. This reflects an error on the AHB bus output from the CTX_LD to the DCSS memory mapped components. This could occur when accessing a region that is not mapped. This status bit can be set ONLY for CTX_LD initiated transactions that receive an AHB error. System transactions that respond with an error are passed through, and the initiating master manages the error response at the source.
21 SB_PEND_DISP_ACTIVE	Single/active region overlap Status bit to indicate when the SB region is still actively loading registers to the display subsystem AND the display enters the active display time. This would indicate the SB registers are not loaded in time and the display trace could be corrupt. Unpredictable behaviour could result if this occurs.
20 DB_PEND_SB_REC	Double/single region overlap Status bit to indicate when the DB region is still actively loading registers to the display subsystem AND a kick to load the SB region is received. This could indicate a timing or BW issue with loading the DB region in time before the SB region register loading should begin. Unpredictable behaviour could result if this occurs.
19 SB_LP_COMP	Single buffer low priority region loading complete IRQ enable Status bit to indicate when the low priority SB region loading sequence is complete.
18 SB_HP_COMP	Single buffer high priority region loading complete Status bit to indicate when the high priority SB region loading sequence is complete.
17 DB_COMP	Double buffer region loading complete Status bit to indicate when the DB (double buffer) region loading sequence is complete.
16 RD_ERR	AXI read error Status bit to indicate when the AXI read bus returns a SLV_ERR or DEC_ERR. This could result in an illegal or unmapped address programmed into the base address registers.
15-9 —	Reserved
8 AHB_ERR_EN	AHB error IRQ enable Enable and interrupt when the internal AHB bus returns an error response. This reflects an error on the AHB bus output from the CTX_LD to the DCSS memory mapped components. This could occur when accessing a region that is not mapped.
7 SB_PEND_DISP_ACTIVE_EN	Single/active region overlap interrupt enable. Enable an interrupt when the SB region is still actively loading registers to the display subsystem AND the display enters the active display time. This would indicate the SB registers are not loaded in time and the display trace could be corrupt. Unpredictable behaviour could result if this occurs.
6 DB_PEND_SB_REC_EN	Double/single region overlap interrupt enable Enable an interrupt when the DB region is still actively loading registers to the display subsystem AND a kick to load the SB region is received. This could indicate a timing or BW issue with loading the DB region in time before the SB region register loading should begin. Unpredictable behaviour could result if this occurs.
5	Single buffer low priority region loading complete IRQ enable Enable an interrupt when the low priority SB region loading sequence is complete.

Table continues on the next page...

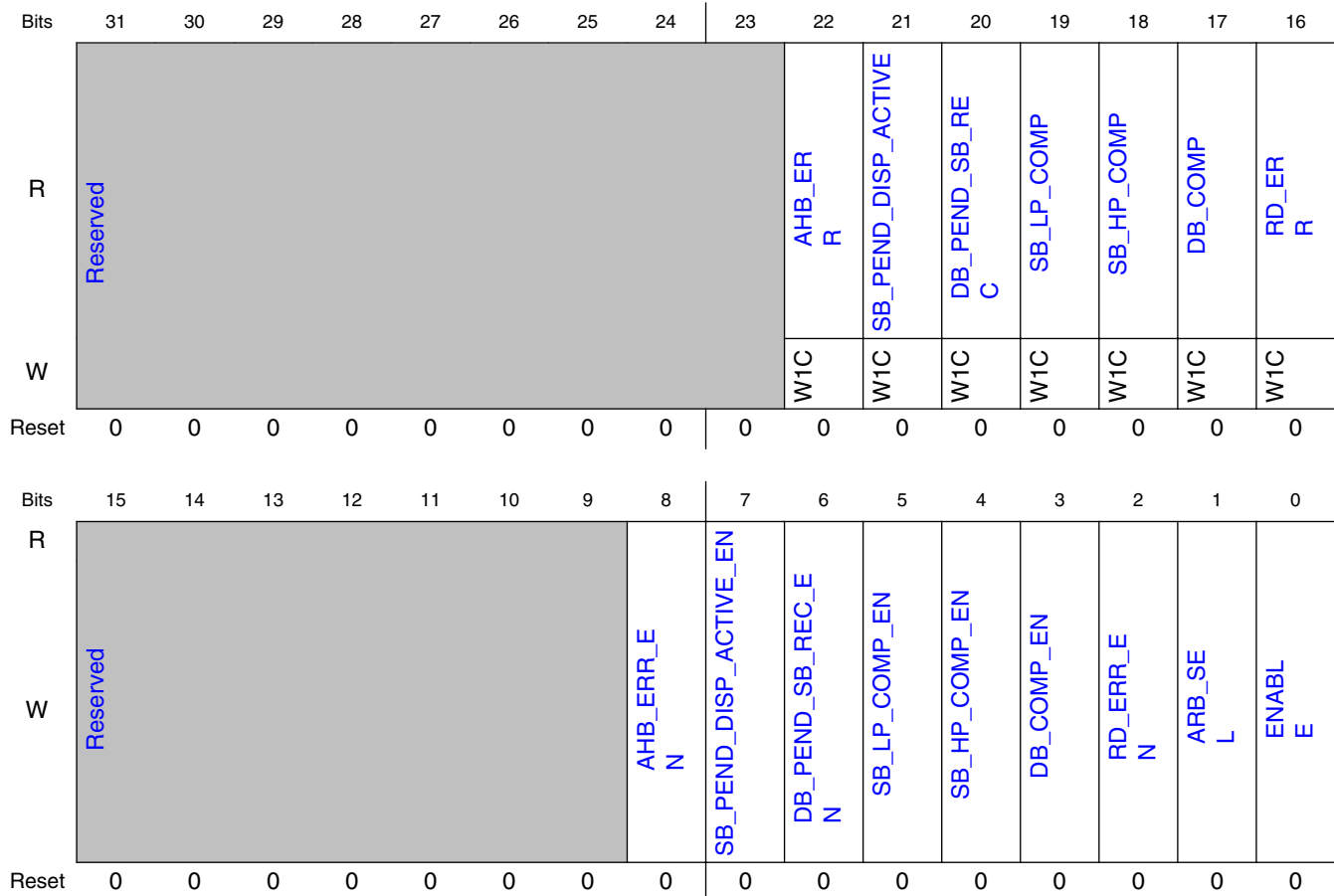
Field	Function
SB_LP_COMP_EN	
4	Single buffer high priority region loading complete IRQ enable
SB_HP_COMP_EN	Enable an interrupt when the high priority SB region loading sequence is complete.
3	Double buffer region loading complete IRQ enable
DB_COMP_EN	Enable an interrupt when the DB (double buffer) region loading sequence is complete.
2	AXI read error IRQ enable
RD_ERR_EN	Enable an interrupt when the AXI read bus returns a SLV_ERR or DEC_ERR. This could result in an illegal or unmapped address programmed into the base address registers.
1	Arbitration select
ARB_SEL	This selects between always choosing the CTX_LD while it's active verses a round robin arbitration scheme of equal weight between the CTX_LD and the system register access bus. • 0 - Round robin arbitration at all times. • 1 - CTX_LD gets priority while it's active.
0	Enable/Busy
ENABLE	Enable bit for the context loader. The context loader does not start the register loading sequence if this bit is not set. SW can set this bit only when it is logic 0 (never set this bit if it is already logic 1). The CTX_LD hardware resets this bit when the entire register loading sequence is complete. It is complete after the DB and SB regions are loaded and the CTX_LD state machine returns to the idle state to allow the next context loading sequence to be enabled.

15.4.3.1.5 Control status register for Context Loader. (CTRL_STATUS_TOG)

15.4.3.1.5.1 Offset

Register	Offset
CTRL_STATUS_TOG	Ch

15.4.3.1.5.2 Diagram



15.4.3.1.5.3 Fields

Field	Function
31-23 —	Reserved
22 AHB_ERR	AHB error Status bit to indicate when the internal AHB bus returns an error response. This reflects an error on the AHB bus output from the CTX_LD to the DCSS memory mapped components. This could occur when accessing a region that is not mapped. This status bit can be set ONLY for CTX_LD initiated transactions that receive an AHB error. System transactions that respond with an error are passed through, and the initiating master manages the error response at the source.
21 SB_PEND_DISP_ACTIVE	Single/active region overlap Status bit to indicate when the SB region is still actively loading registers to the display subsystem AND the display enters the active display time. This would indicate the SB registers are not loaded in time and the display trace could be corrupt. Unpredictable behaviour could result if this occurs.
20 DB_PEND_SB_REC	Double/single region overlap Status bit to indicate when the DB region is still actively loading registers to the display subsystem AND a kick to load the SB region is received. This could indicate a timing or BW issue with loading the DB region

Table continues on the next page...

Field	Function
	in time before the SB region register loading should begin. Unpredictable behaviour could result if this occurs.
19 SB_LP_COMP	Single buffer low priority region loading complete IRQ enable Status bit to indicate when the low priority SB region loading sequence is complete.
18 SB_HP_COMP	Single buffer high priority region loading complete Status bit to indicate when the high priority SB region loading sequence is complete.
17 DB_COMP	Double buffer region loading complete Status bit to indicate when the DB (double buffer) region loading sequence is complete.
16 RD_ERR	AXI read error Status bit to indicate when the AXI read bus returns a SLV_ERR or DEC_ERR. This could result in an illegal or unmapped address programmed into the base address registers.
15-9 —	Reserved
8 AHB_ERR_EN	AHB error IRQ enable Enable and interrupt when the internal AHB bus returns an error response. This reflects an error on the AHB bus output from the CTX_LD to the DCSS memory mapped components. This could occur when accessing a region that is not mapped.
7 SB_PEND_DIS P_ACTIVE_EN	Single/active region overlap interrupt enable. Enable an interrupt when the SB region is still actively loading registers to the display subsystem AND the display enters the active display time. This would indicate the SB registers are not loaded in time and the display trace could be corrupt. Unpredictable behaviour could result if this occurs.
6 DB_PEND_SB_ REC_EN	Double/single region overlap interrupt enable Enable an interrupt when the DB region is still actively loading registers to the display subsystem AND a kick to load the SB region is received. This could indicate a timing or BW issue with loading the DB region in time before the SB region register loading should begin. Unpredictable behaviour could result if this occurs.
5 SB_LP_COMP_ EN	Single buffer low priority region loading complete IRQ enable Enable an interrupt when the low priority SB region loading sequence is complete.
4 SB_HP_COMP_ EN	Single buffer high priority region loading complete IRQ enable Enable an interrupt when the high priority SB region loading sequence is complete.
3 DB_COMP_EN	Double buffer region loading complete IRQ enable Enable an interrupt when the DB (double buffer) region loading sequence is complete.
2 RD_ERR_EN	AXI read error IRQ enable Enable an interrupt when the AXI read bus returns a SLV_ERR or DEC_ERR. This could result in an illegal or unmapped address programmed into the base address registers.
1 ARB_SEL	Arbitration select This selects between always choosing the CTX_LD while it's active verses a round robin arbitration scheme of equal weight between the CTX_LD and the system register access bus. • 0 - Round robin arbitration at all times. • 1 - CTX_LD gets priority while it's active.
0 ENABLE	Enable/Busy Enable bit for the context loader. The context loader does not start the register loading sequence if this bit is not set. SW can set this bit only when it is logic 0 (never set this bit if it is already logic 1). The CTX_LD

Memory Map and Registers

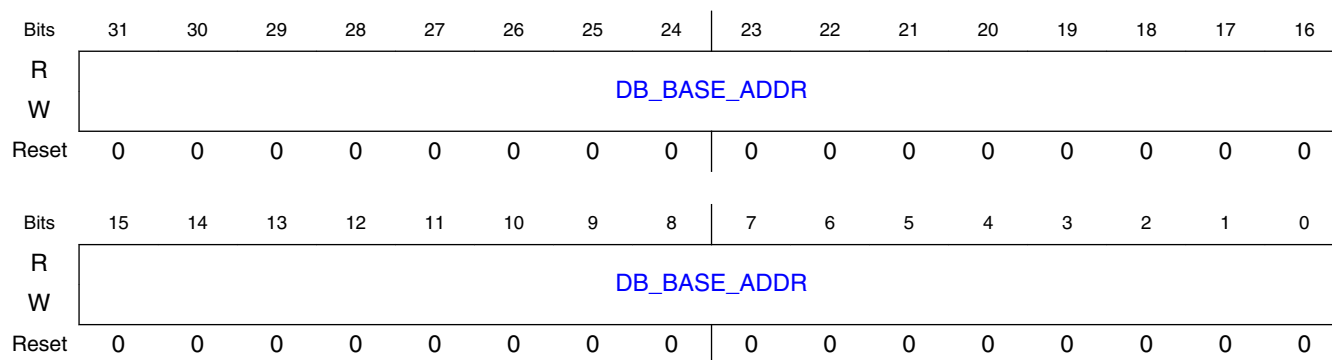
Field	Function
	hardware resets this bit when the entire register loading sequence is complete. It is complete after the DB and SB regions are loaded and the CTX_LD state machine returns to the idle state to allow the next context loading sequence to be enabled.

15.4.3.1.6 DRAM addr for double buffered register fetch. (DB_BASE_ADDR)

15.4.3.1.6.1 Offset

Register	Offset
DB_BASE_ADDR	10h

15.4.3.1.6.2 Diagram



15.4.3.1.6.3 Fields

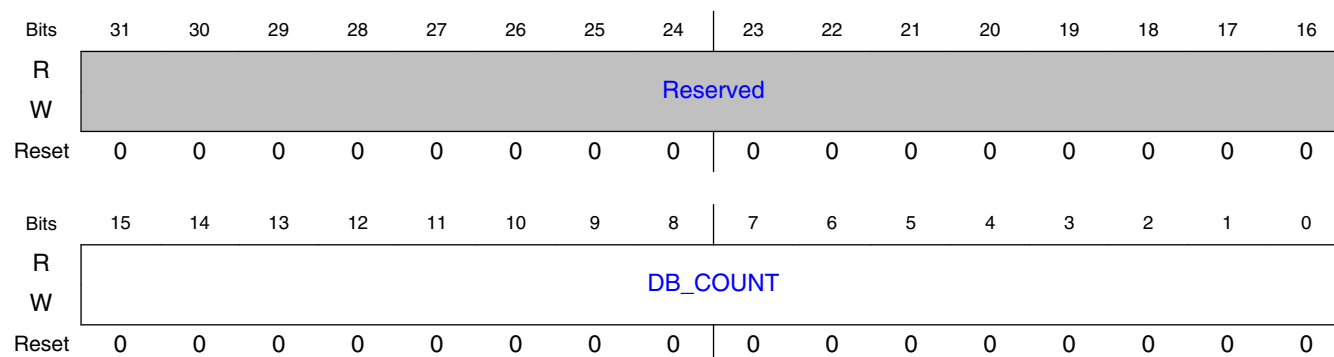
Field	Function
31-0	Double buffered registers base address.
DB_BASE_ADDR	Stores the system start address(DRAM) of double buffered registers.

15.4.3.1.7 Double buffer register count (DB_COUNT)

15.4.3.1.7.1 Offset

Register	Offset
DB_COUNT	14h

15.4.3.1.7.2 Diagram



15.4.3.1.7.3 Fields

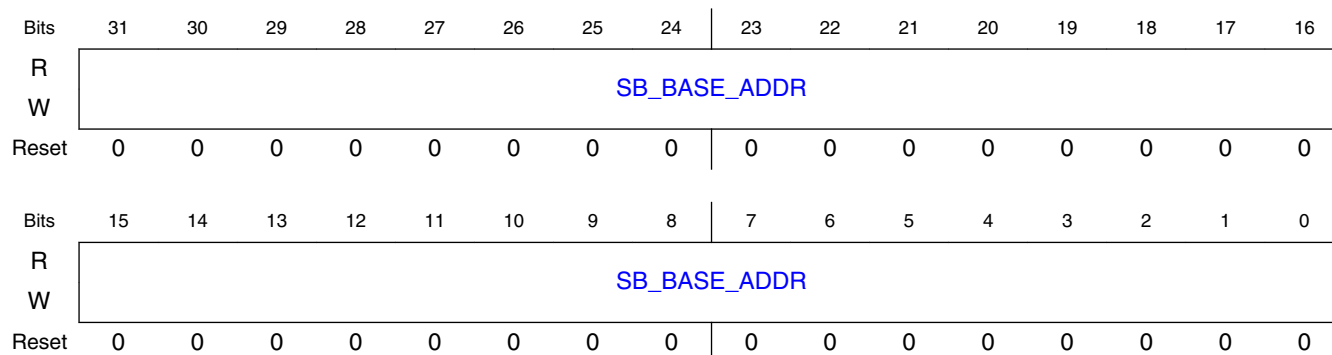
Field	Function
31-16 —	Reserved
15-0 DB_COUNT	Double buffered region fetch count Number of 64 bit entries of double buffered registers to fetch and load.

15.4.3.1.8 DRAM addr for single buffered registers. (SB_BASE_ADDR)

15.4.3.1.8.1 Offset

Register	Offset
SB_BASE_ADDR	18h

15.4.3.1.8.2 Diagram



15.4.3.1.8.3 Fields

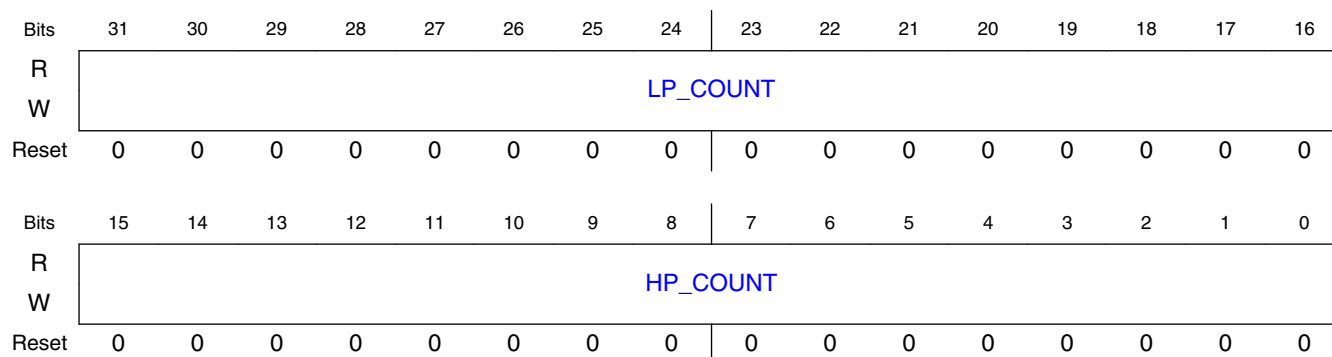
Field	Function
31-0	Single buffered register base address.
SB_BASE_ADDR R	Stores the system start address(DRAM) of single buffered registers.

15.4.3.1.9 Single buffer register count (SB_COUNT)

15.4.3.1.9.1 Offset

Register	Offset
SB_COUNT	1Ch

15.4.3.1.9.2 Diagram



15.4.3.1.9.3 Fields

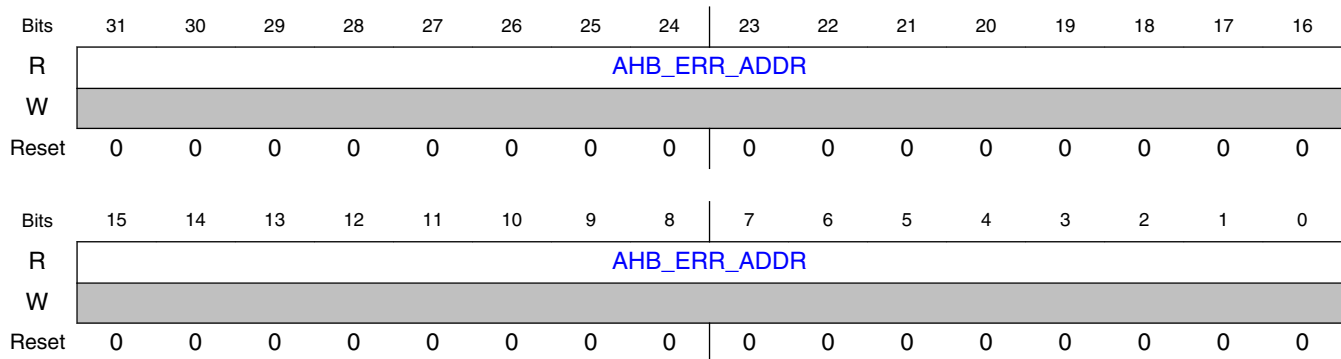
Field	Function
31-16 LP_COUNT	Single buffered low priority region fetch count Number of 64 bit low priority entries of single buffered registers to fetch and load.
15-0 HP_COUNT	Single buffered high priority region fetch count Number of 64 bit high priority entries of single buffer registers to fetch and load.

15.4.3.1.10 AHB address with error response. (AHB_ERR_ADDR)

15.4.3.1.10.1 Offset

Register	Offset
AHB_ERR_ADDR	20h

15.4.3.1.10.2 Diagram



15.4.3.1.10.3 Fields

Field	Function
31-0 AHB_ERR_ADDR	AHB error address.
R	Stores the address of the CTX_LD transaction that responded with an AHB error. A new address is stored only if ahb_error bit in ctrl_status is 0. If AHB_ERR is set, new address is not logged until the bit is cleared. This address is trapped only for CTX_LD initiated transactions when it is fetching the display state from memory. Error response for system transactions are not trapped and does NOT effect this register.

15.5 Graphics Decompression (DEC400D)

15.5.1 Overview

The DEC400D decompresses graphics data. The DEC400D decompresses the data on the fly, and the compressed format in DRAM is transparent to the Display Prefetch Resolve (DPR) block. The DPR is programmed as if the frame buffer is not compressed.

15.5.1.1 Block diagram

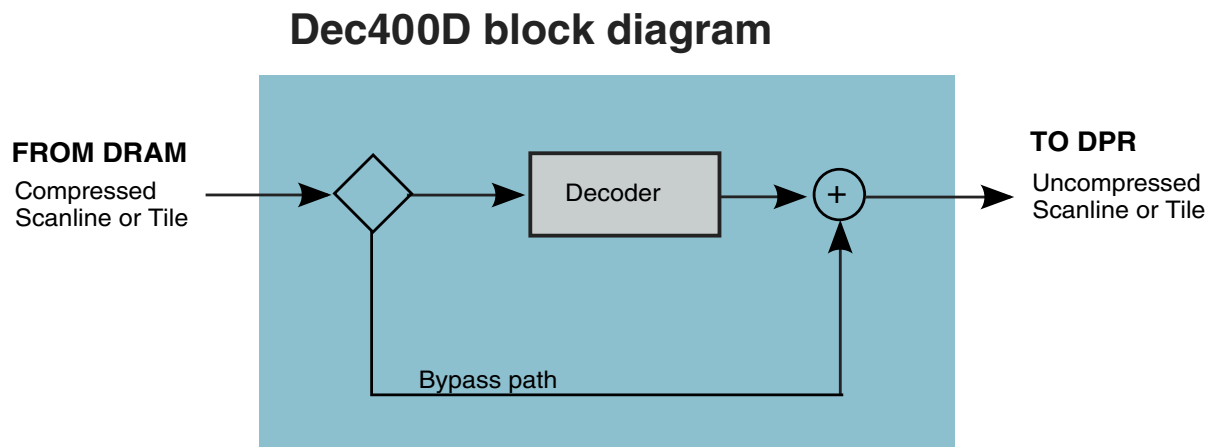


Figure 15-13. DEC400D block diagram

15.5.1.2 Features

The DEC400D includes the following host and memory features:

- 32-bit AHB Interface
- 128-bit width AXI data channel with dedicated pins for Pixel Engine and Memory
- 8-bit AXI ID, up to 16 read transactions supported for decompression
- 16-256 Byte AXI burst data range
- Supports interleaved return data
- Output arbitration, required to insure data output matches request
- Resource locks with CPU (Semaphore lock)

The DEC400D includes the following decode features:

- Up to 8/16 pixels per cycle decode for 32bpp with 128 bit AXI

- 2:1 to 3:1 average decompression ratio
- Pixel data organization support for Linear (scanline) or Tiled (4x4, 4x8, 8x4 and 8x8-XMajor)
- Bypass option controlled via software
- Full frame or partial screen update supported

15.5.2 Functional description

The following sections describe functional details of the DEC400D module.

15.5.2.1 Graphics compression/decompression

The DEC400D decompresses graphics data. The DPR issues AXI requests at a pixel size of 8x4 at 32bpp, or 128B per request, as an uncompressed AXI transaction. The DEC400D decompresses on the fly, and the compressed format in DRAM is transparent to the DPR. The DPR is programmed as if the frame buffer is not compressed.

Graphics buffer formats use the Z order super tile frame buffer format. A diagram depicting the tile format is shown below.

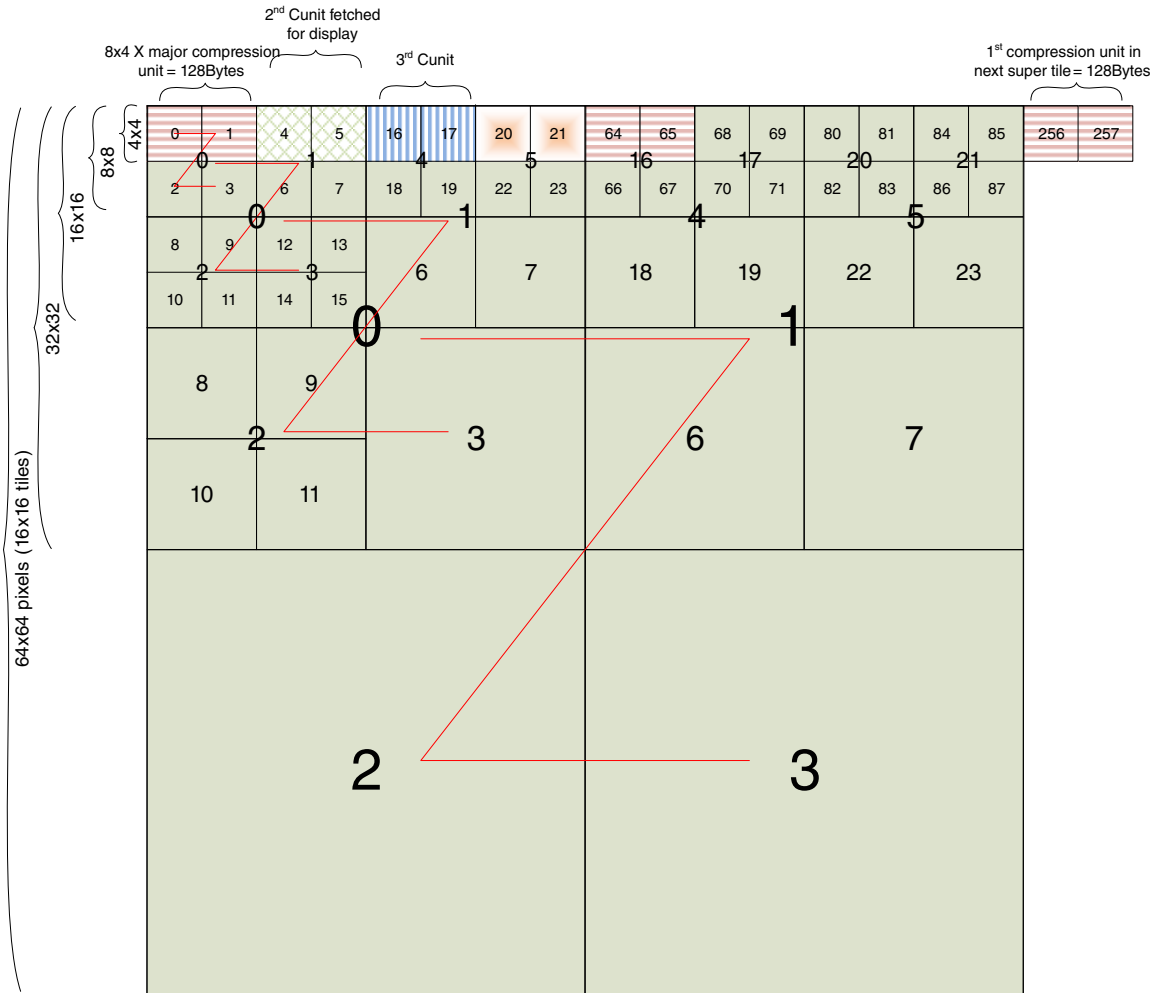


Figure 15-14. Graphics Tile Format

The DPR, or bus master that fetches data from DRAM uncompressed, is programmed to request 128B for each request. Each compression unit is 128B in the X major organization. The first compression unit in the diagram above is two 4x4 tiles labeled as 0,1, the second compression unit the DPR fetches contains the tiles labeled as 4,5. This fetch pattern continues left to right across the supertiled source buffer. The GPU compression scheme must compliment the DC's decompression scheme in that compressed units must be 8x4 pixel groups, which is 128B for two 4x4 tiles. The complete order of compression units the DPR fetches for GPU frames is:

- Bank 1 = cu(0,1), cu(4,5), cu(16,17), cu(20,21), cu(64,65), cu(68,69), cu(80,81), cu(84,85), cu(256,257), ... continues to the programmed width of the buffer.
- Bank 2 = cu(2,3), cu(6,7), cu(18,19), ...
- Bank 3 = cu(8,9), cu(12,13), cu(24,25), ...

The equivalent address fetch pattern for the first row of CUs is:

- 0x0, 0x100, 0x400, 0x500, 0x1000, 0x1100, 0x1400, 0x1500, 0x4000, ...

Each bank above is 4 lines high across the display. This data is stored locally in the RTRAM memory for subsequent use by the display controller for screen refresh (see DPR/RTRAM sections).

Note the Z order of 4x4 tiles, which extends to the next size of 8x8, 16x16, etc., to the super tile size that is 64x64 pixels. At a resolution above 64x64, the ordering of super tiles is raster ordered which extends for the entire frame size.

15.5.2.2 CU tile status

Each compression unit (CU) is two consecutive 4x4 tiles. The memory use before/after compression is depicted in the diagram below.

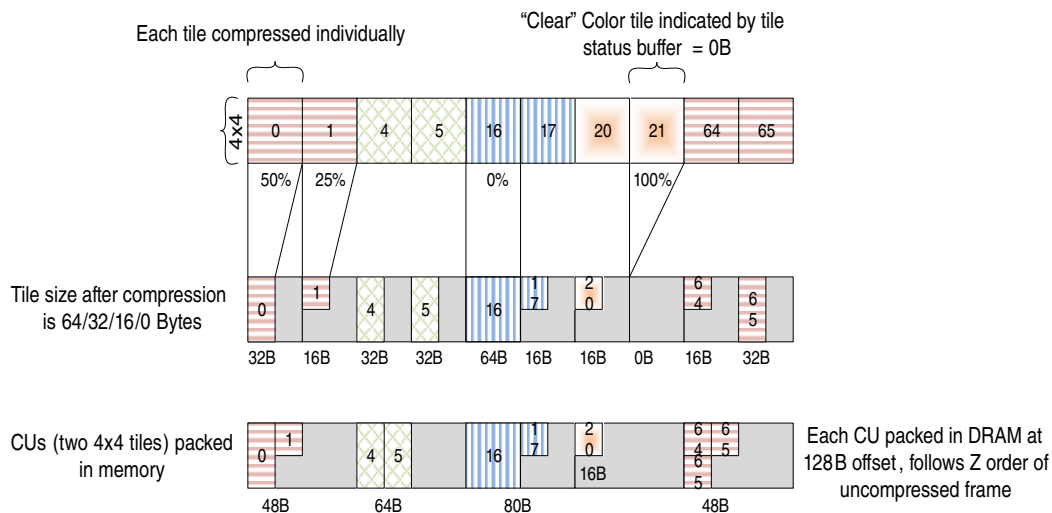


Figure 15-15. Memory use before/after compression

The uncompressed frame buffer is on the first row of tiles above, with each tile arranged as 4x4 pixels. The tiles are compressed to a reduced number of bytes, as in the second row. The compressed tiles are then packed into memory on a per CU basis (note the address order of CUs is still Z which is not depicted in the diagram). DRAM access to compressed tiles is on a per CU basis, and the AXI transaction size is determined by the tile status buffer entry for each CU. The CU data request could be from 16B – 128B on 16B increments. From an LPDDR4 32-bit implementation, 16/32/48/64B accesses are effectively 64B accesses. 80/96/112/128B accesses are effectively 128B accesses from the DRAM perspective.

15.5.3 Memory Map and register definition

This section includes the DEC400D module memory map and detailed descriptions of all registers.

15.5.3.1 DEC400D register descriptions

15.5.3.1.1 DEC400D Memory map

DEC400D_CHAN1 base address: 1_5000h

Offset	Register	Width (In bits)	Access	Reset value
24h	Revision ID (GCCHIPREV)	32	RO	See description.
28h	Release Date (GCCHIPDATE)	32	RO	See description.
98h	Patch Revision (GCREGHICHIPPATCHREV)	32	RO	See description.
A8h	Product ID (GCPRODUCTID)	32	RO	See description.
800h - 87Ch	Decode Read Configuration (GCREGAHBDECREADCONFIG0 - GCREGAHBDECREADCONFIG31)	32	RW	0000_0000h
900h - 97Ch	Decode Read Buffer Base (GCREGAHBDECREADBUFFERBASE0 - GCREGAHBDECREADBUFFERBASE31)	32	RW	0000_0000h
980h - 9FCh	Decode Read Cache Base (GCREGAHBDECREADCACHEBASE0 - GCREGAHBDECREADCACHEBASE31)	32	RW	0000_0000h
B00h	Dec400D Control (GCREGAHBDECCONTROL)	32	RW	0000_0000h
B04h	Interrupt Acknowledge (GCREGAHBDECINTRACKNOWLEDGE)	32	RO	0000_0000h
B08h	Interrupt Enable (GCREGAHBDECINTRENBL)	32	RW	0000_0000h
B0Ch	Tile Status Module Debug (GCREGAHBDECTILESTATUSDEBUG)	32	RO	0000_0000h
B14h	Decompression Module Debug (GCREGAHBDECDECODERDEBUG)	32	RO	0000_0000h
B18h	Total Reads In (GCREGAHBDECTOTALREADSIN)	32	RO	0000_0000h
B20h	Total Read Data Count (GCREGAHBDECTOTALREADBURSTSIN)	32	RO	0000_0000h
B28h	Total Read Request In (GCREGAHBDECTOTALREADREQIN)	32	RO	0000_0000h
B30h	Total Input Read Last Number (GCREGAHBDECTOTALREADLAST SIN)	32	RO	0000_0000h
B38h	Total Reads Out (GCREGAHBDECTOTALREADSOUT)	32	RO	0000_0000h
B40h	Total Read Bursts Out (GCREGAHBDECTOTALREADBURSTSOUT)	32	RO	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
B48h	Total Read Request Out (GCREGAHBDECTOTALREADREQOUT)	32	RO	0000_0000h
B50h	Total Read Last Out (GCREGAHBDECTOTALREADLASTSOUT)	32	RO	0000_0000h
B58h	Debug Register 0 (GCREGAHBDECDEBUG0)	32	RO	0000_0000h
B5Ch	Debug Register 1 (GCREGAHBDECDEBUG1)	32	RO	0000_0000h
B60h	Debug register 2 (GCREGAHBDECDEBUG2)	32	RO	0000_0000h
B64h	Debug Register 3 (GCREGAHBDECDEBUG3)	32	RO	0000_0000h
B68h	GCREGAHBDECCONTROLEX (GCREGAHBDECCONTROLEX)	32	RW	0000_0000h
B6Ch	GCREGAHBDECSTATECOMMIT (GCREGAHBDECSTATECOMMIT)	32	RW	0000_0000h
B70h	GCREGAHBDECSTATELOCK (GCREGAHBDECSTATELOCK)	32	RO	0000_0000h
C00h - C7Ch	Decode Read Extra Configuration (GCREGAHBDECREADSEXCONF0 - GCREGAHBDECREADSEXCONF31)	32	RW	0000_0000h
C80h - CFCh	Decoder Read Stride (GCREGAHBDECREADSTRIDE0 - GCREGAHBDECREADSTRIDE31)	32	RW	0000_0000h
E00h - E7Ch	Decoder Read Buffer End (GCREGAHBDECREADBUFFEREND0 - GCREGAHBDECREADBUFFEREND31)	32	RW	0000_0000h

15.5.3.1.2 Revision ID (GCCHIPREV)

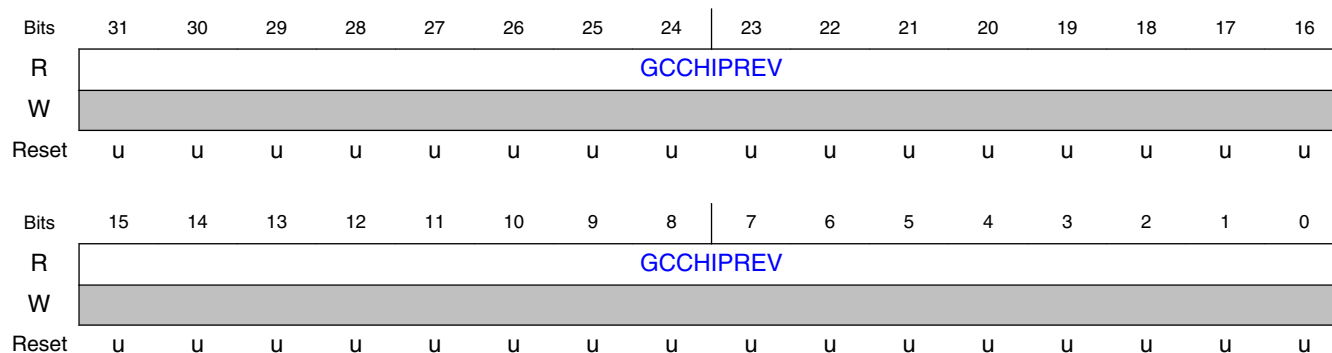
15.5.3.1.2.1 Offset

Register	Offset
GCCHIPREV	24h

15.5.3.1.2.2 Function

This read-only register shows the revision for the chip in BCD. This register has no set reset value. It varies with the implementation.

15.5.3.1.2.3 Diagram



15.5.3.1.2.4 Fields

Field	Function
31-0	Revision ID
GCCHIPREV	This read only field returns the revision number.

15.5.3.1.3 Release Date (GCCHIPDATE)

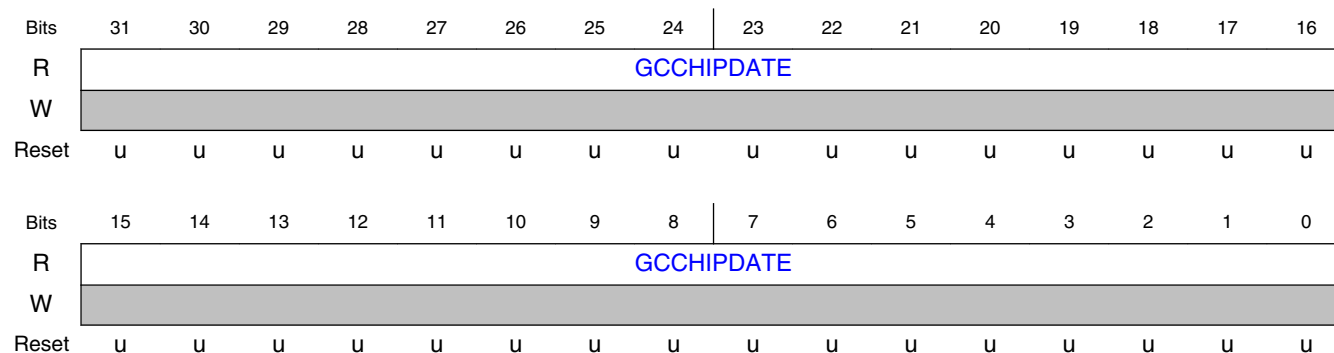
15.5.3.1.3.1 Offset

Register	Offset
GCCHIPDATE	28h

15.5.3.1.3.2 Function

Shows the release date for the IP in YYYYMMDD (year/month/day) format. This register has no set reset value. It varies with the implementation.

15.5.3.1.3.3 Diagram



15.5.3.1.3.4 Fields

Field	Function
31-0	Date
GCCHIPDATE	This read only field returns the date of design release.

15.5.3.1.4 Patch Revision (GCREGHICHIPPATCHREV)

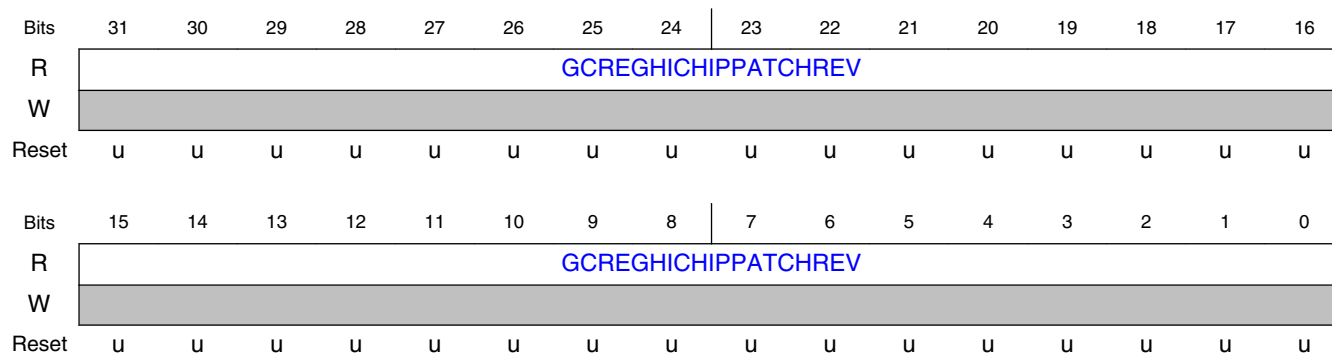
15.5.3.1.4.1 Offset

Register	Offset
GCREGHICHIPPATCHREV	98h

15.5.3.1.4.2 Function

Patch revision level for the chip.

15.5.3.1.4.3 Diagram



15.5.3.1.4.4 Fields

Field	Function
31-0	Product ID
GCREGHICHIP PATCHREV	This read only field returns patch revision level.

15.5.3.1.5 Product ID (GCPRODUCTID)

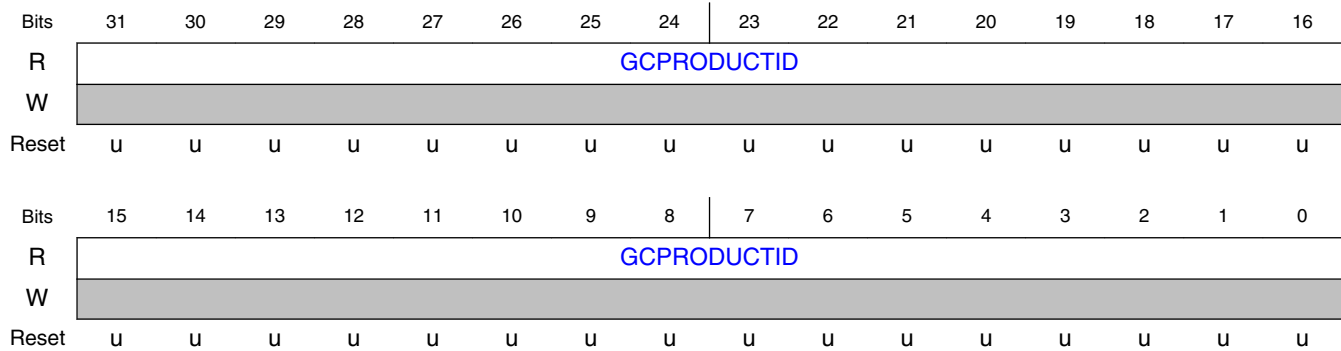
15.5.3.1.5.1 Offset

Register	Offset
GCPRODUCTID	A8h

15.5.3.1.5.2 Function

This read only register shows the Dec400D Product ID

15.5.3.1.5.3 Diagram



15.5.3.1.5.4 Fields

Field	Function
31-0	Product ID
GCPRODUCTID	This read only field returns the Dec400D Product ID for this implementation.

15.5.3.1.6 Decode Read Configuration (GCREGAHBDECREADCONFIG0 - GCREGAHBDECREADCONFIG31)

15.5.3.1.6.1 Offset

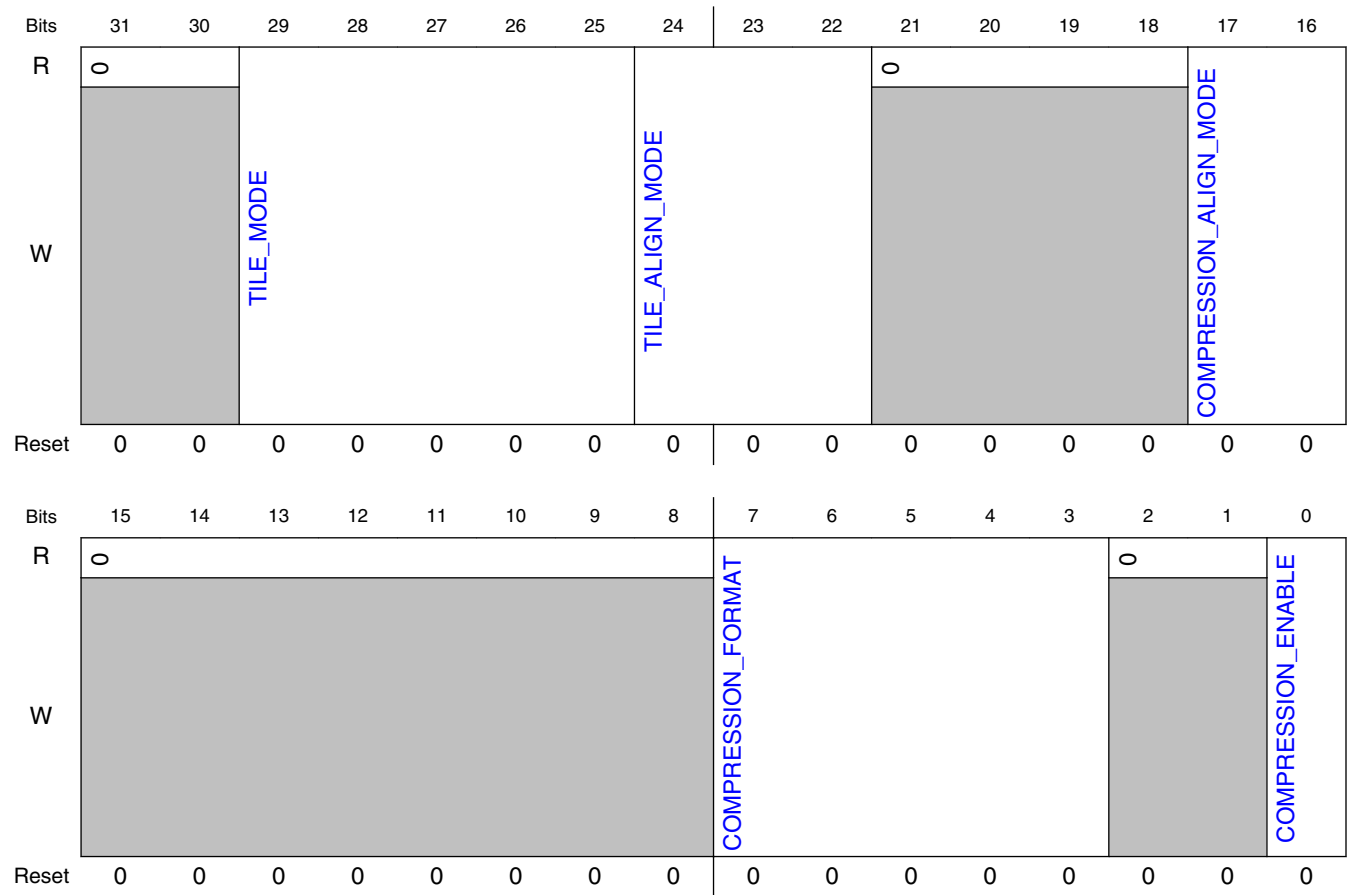
For n = 0 to 31:

Register	Offset
GCREGAHBDECREADC ONFIGn	800h + (n × 4h)

15.5.3.1.6.2 Function

Dec400D configuration register for read ID.

15.5.3.1.6.3 Diagram



15.5.3.1.6.4 Fields

Field	Function
31-30 —	Reserved
29-25 TILE_MODE	Tile Mode How many pixels in the tile and the walking direction in the tile. 00000b - TILE8X8_XMAJOR 00001b - TILE8X8_YMAJOR 00010b - TILE16X4 00011b - TILE8X4 00100b - TILE4X8 00101b - TILE4X4 00110b - RASTER16X4 00111b - TILE64X4 01000b - TILE32X4 01001b - RASTER256X1 01010b - RASTER128X1 01011b - RASTER64X4 01100b - RASTER256X2

Table continues on the next page...

Field	Function
	01101b - RASTER128X2 01110b - RASTER128X4 01111b - RASTER64X1
24-22 TILE_ALIGN_MODE	Tile Align Mode The tile alignment mode. 000b - TILE1_ALIGN 001b - TILE2_ALIGN 010b - TILE4_ALIGN 011b - CBSR_ALIGN
21-18 —	Reserved
17-16 COMPRESSION_ALIGN_MODE	Compression Align Mode Compression result size alignment mode. 00b - ALIGN1_BYTE 01b - ALIGN16_BYTE 10b - ALIGN32_BYTE 11b - ALIGN64_BYTE
15-8 —	Reserved
7-3 COMPRESSION_FORMAT	Compression Format Compression color format. 00000b - ARGB8
2-1 —	Reserved
0 COMPRESSION_ENABLE	Compression Enable Compression enable for individual streams. 0b - Disable 1b - Enable

15.5.3.1.7 Decode Read Buffer Base (GCREGAHBDECREADBUBUFFERBASE0 - GCREGAHBDECREADBUBUFFERBASE31)

15.5.3.1.7.1 Offset

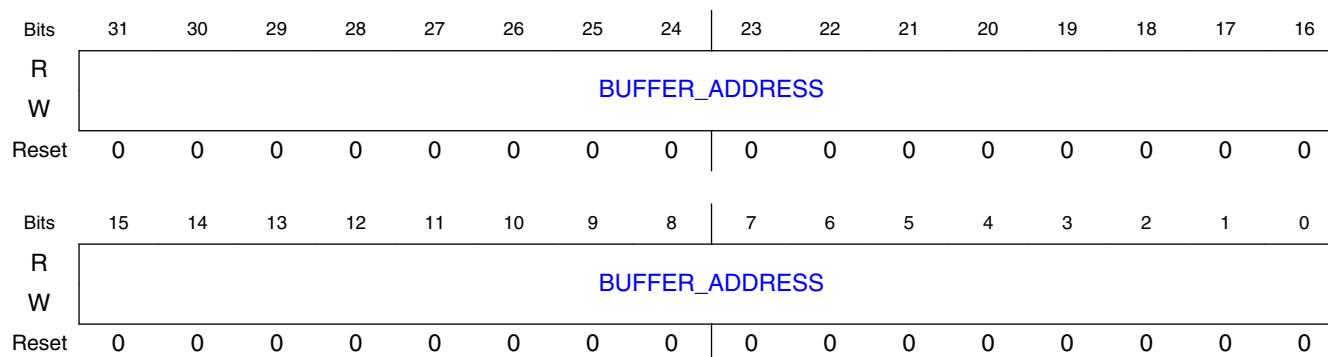
For n = 0 to 31:

Register	Offset
GCREGAHBDECREADBUBUFFERBASEn	900h + (n × 4h)

15.5.3.1.7.2 Function

Base address for pixel buffer for read ID.

15.5.3.1.7.3 Diagram



15.5.3.1.7.4 Fields

Field	Function
31-0 BUFFER_ADDRESS	Base address for pixel buffer for read ID.

15.5.3.1.8 Decode Read Cache Base (GCREGAHBDECREADCACHEBASE0 - GCREGAHBDECREADCACHEBASE31)

15.5.3.1.8.1 Offset

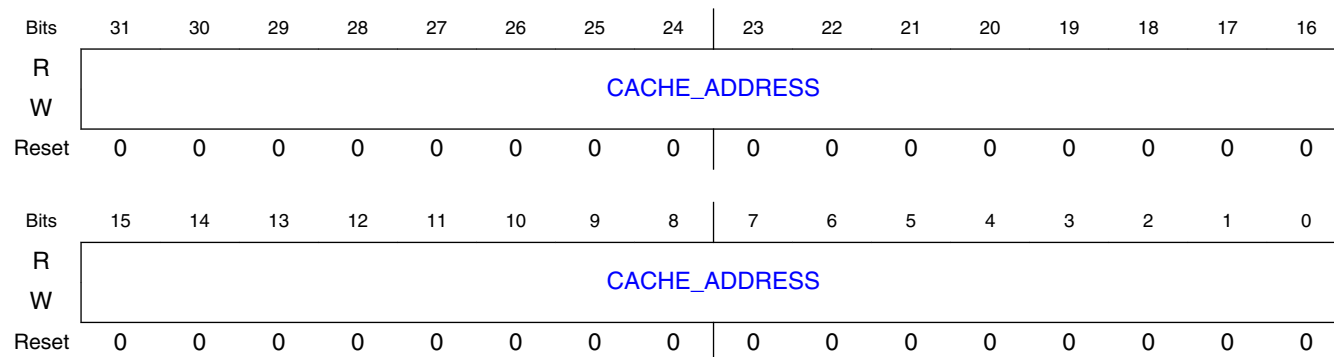
For $n = 0$ to 31:

Register	Offset
GCREGAHBDECREADCACHEBASE n	$980h + (n \times 4h)$

15.5.3.1.8.2 Function

Base address for tile status buffer for read ID.

15.5.3.1.8.3 Diagram



15.5.3.1.8.4 Fields

Field	Function
31-0 CACHE_ADDRESS	Base address for tile status buffer for read ID.

15.5.3.1.9 Dec400D Control (GCREGAHBDECCONTROL)

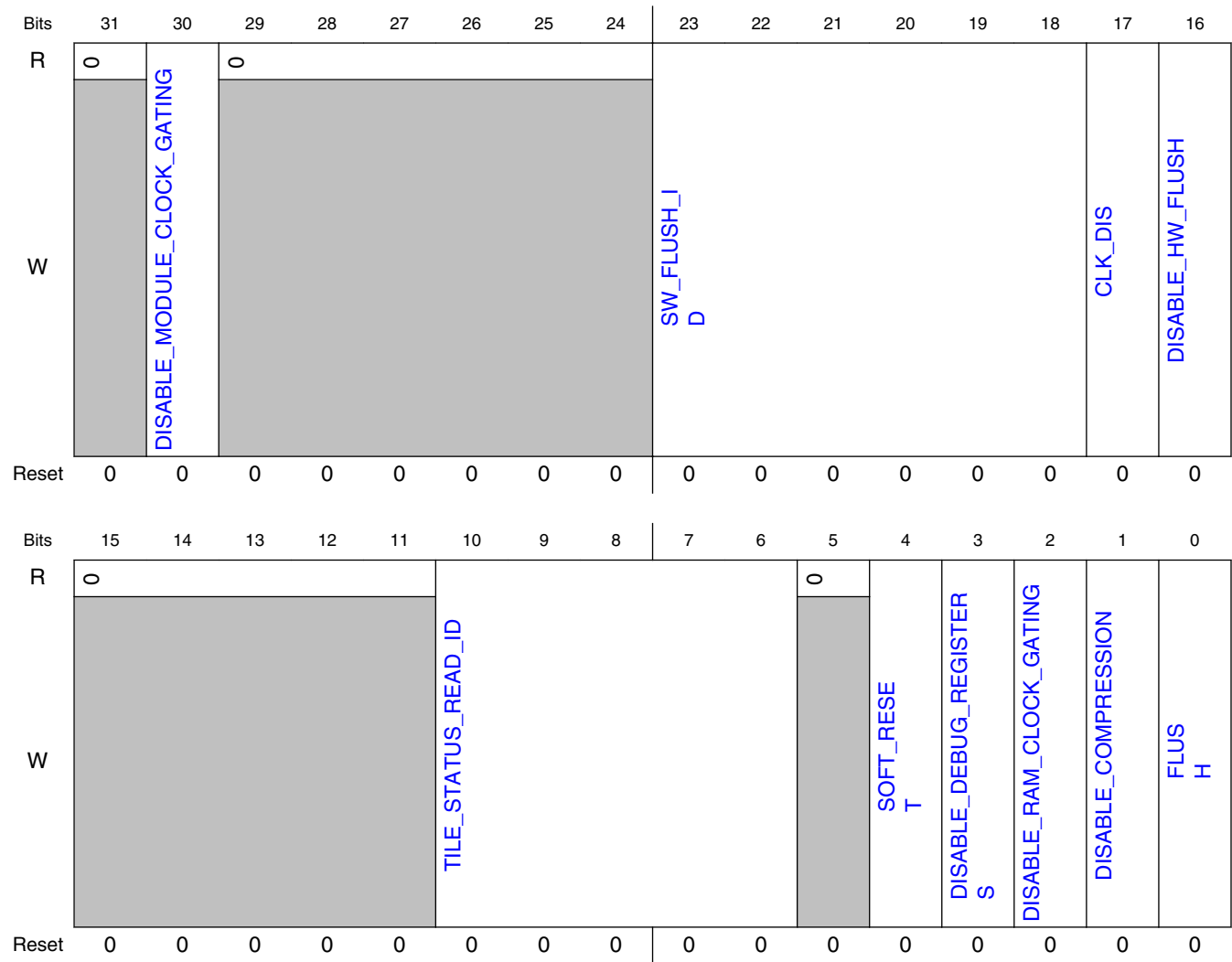
15.5.3.1.9.1 Offset

Register	Offset
GCREGAHBDECCONTROL	B00h

15.5.3.1.9.2 Function

Global Dec400D control register.

15.5.3.1.9.3 Diagram



15.5.3.1.9.4 Fields

Field	Function
31 —	Reserved
30 DISABLE_MODULE_CLOCK_GATING	Disable clock gating for sub modules 0b - Disable 1b - Enable
29-24 —	Reserved
23-18 SW_FLUSH_ID	ID of tile status flush. Bit[23] represents the request type, 0 is read, 1 is write. Bit[22:18] is the stream id

Table continues on the next page...

Field	Function
17 CLK_DIS	Disable clock. 0b - Disable 1b - Enable
16 DISABLE_HW_ FLUSH	Tile status cache flush through frame end pin is disabled. 0b - Disable 1b - Enable
15-11 —	Reserved
10-6 TILE_STATUS_ READ_ID	Tile status cache's AXI read ID.
5 —	Reserved
4 SOFT_RESET	Soft reset the Dec400D. 0b - Disable 1b - Enable
3 DISABLE_DEB UG_REGISTER S	Disable debug registers. If this bit is 1, debug registers are clock gated. 0b - Disable 1b - Enable
2 DISABLE_RAM _CLOCK_GATI NG	Disable clock gating for RAMs. 0b - Disable 1b - Enable
1 DISABLE_COM PRESSION	Bypass compression for all streams. 0b - Disable 1b - Enable
0 FLUSH	Flush tile status cache. 0b - Disable 1b - Enable

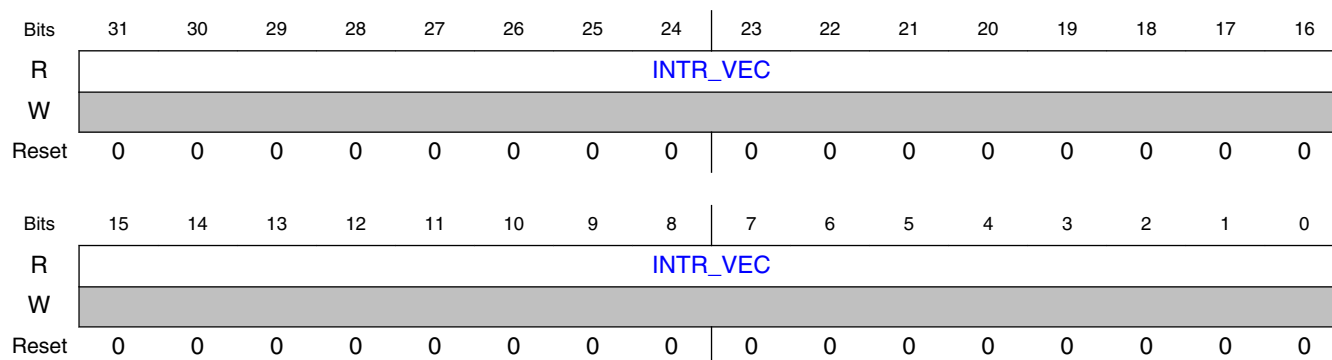
15.5.3.1.10 Interrupt Acknowledge (GCREGAHBDECINTRACKNOWLEDGE)

15.5.3.1.10.1 Offset

Register	Offset
GCREGAHBDECINTRACKNOWLEDGE	B04h

15.5.3.1.10.2 Function

Each bit represents a corresponding event being triggered. Reading from this register clears the outstanding interrupt

15.5.3.1.10.3 Diagram**15.5.3.1.10.4 Fields**

Field	Function
31-0	Interrupt vector
INTR_VEC	For each interrupt event, 0=Clear, 1=Interrupt Active Bit 31 is AXI bus error interrupt Bits 30:0 is tile status flush done interrupt for write stream

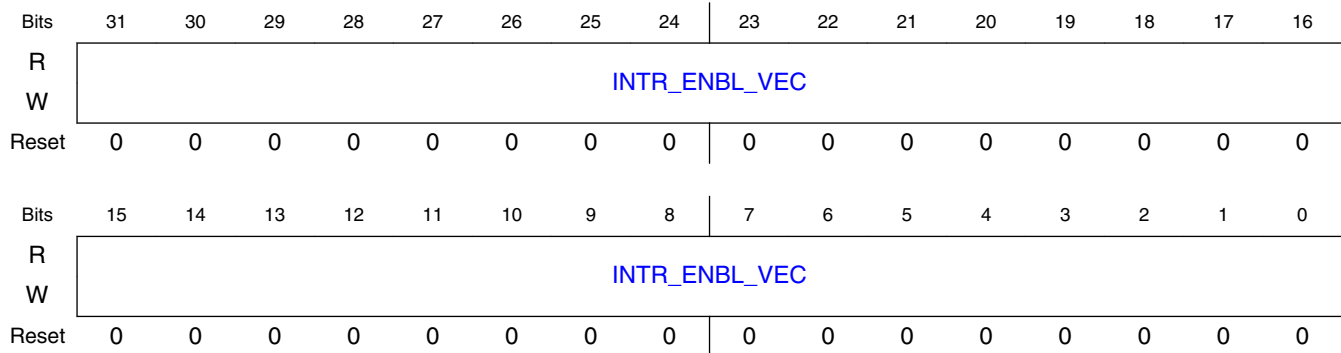
15.5.3.1.11 Interrupt Enable (GCREGAHBDECINTRENBL)**15.5.3.1.11.1 Offset**

Register	Offset
GCREGAHBDECINTRENBL	B08h

15.5.3.1.11.2 Function

Each bit enables a corresponding event. Bit 31 is AXI bus error interrupt Bit 30:0 is tile status flush done interrupt for write stream.

15.5.3.1.11.3 Diagram



15.5.3.1.11.4 Fields

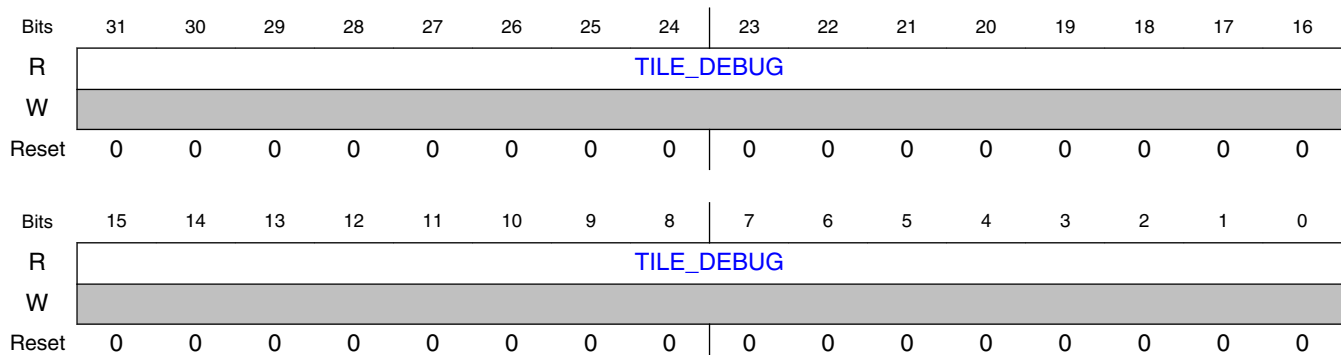
Field	Function
31-0	Interrupt enable vector
INTR_ENBL_VEC	For each bit, 0: Disable interrupt, 1: Enable interrupt. Bit 31 is AXI bus error interrupt. Bit 30:0 is tile status flush done interrupt for write stream.

15.5.3.1.12 Tile Status Module Debug (GCREGAHBDECTILESTATUSDEBUG)

15.5.3.1.12.1 Offset

Register	Offset
GCREGAHBDECTILESTATUSDEBUG	B0Ch

15.5.3.1.12.2 Diagram



15.5.3.1.12.3 Fields

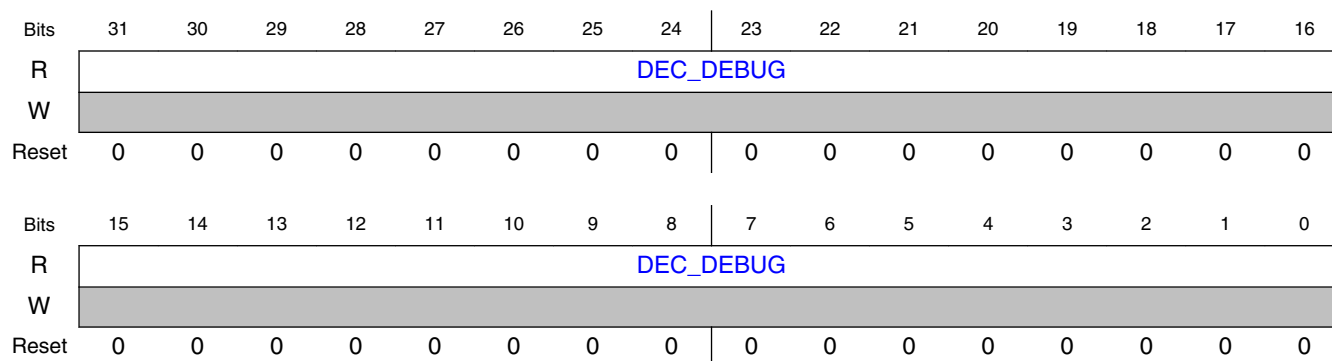
Field	Function
31-0 TILE_DEBUG	Debug

15.5.3.1.13 Decompression Module Debug (GCREGAHBDECDECODERDEBUG)

15.5.3.1.13.1 Offset

Register	Offset
GCREGAHBDECDECODERDEBUG	B14h

15.5.3.1.13.2 Diagram



15.5.3.1.13.3 Fields

Field	Function
31-0 DEC_DEBUG	Debug

15.5.3.1.14 Total Reads In (GCREGAHBDECTOTALREADSIN)

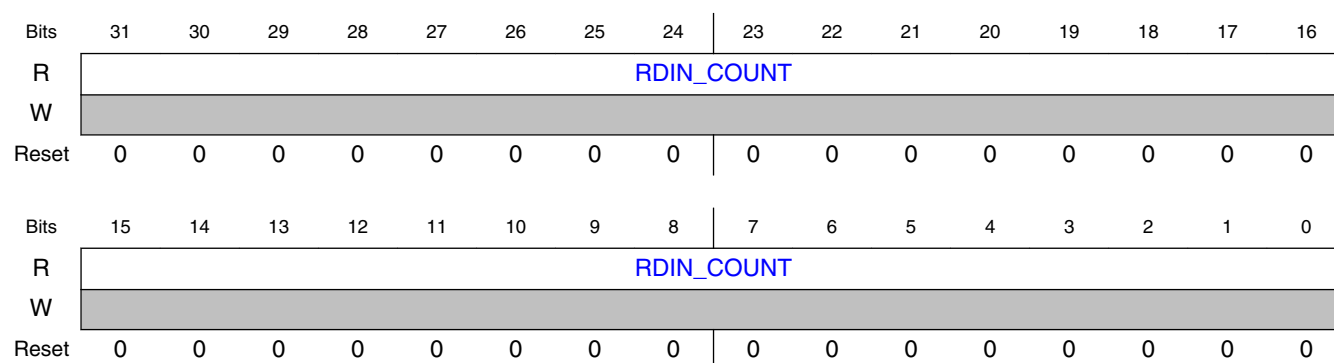
15.5.3.1.14.1 Offset

Register	Offset
GCREGAHBDECTOTAL READSIN	B18h

15.5.3.1.14.2 Function

Total reads in terms of dataWidth in input (pixel engine) side. Sum of ARLEN+1 of all requests sent to DEC.

15.5.3.1.14.3 Diagram



15.5.3.1.14.4 Fields

Field	Function
31-0 RDIN_COUNT	Count

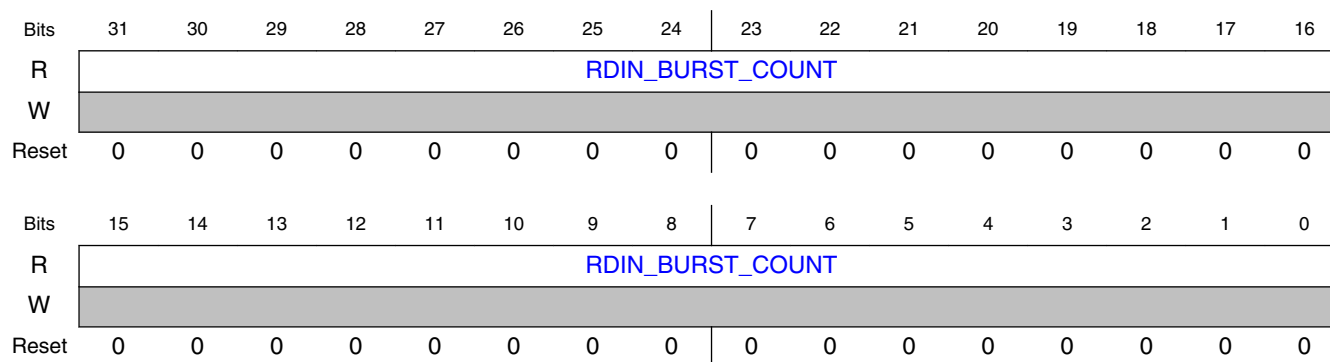
15.5.3.1.15 Total Read Data Count (GCREGAHBDECTOTALREADBURSTSIN)

15.5.3.1.15.1 Offset

Register	Offset
GCREGAHBDECTOTAL READBURSTSIN	B20h

15.5.3.1.15.2 Function

Total read data count in terms of dataWidth in input (pixel engine) side. Sum of returned data beats count sent back to pixel engine. Equal to gcregAHBDECTotalReadsIn.

15.5.3.1.15.3 Diagram**15.5.3.1.15.4 Fields**

Field	Function
31-0 RDIN_BURST_COUNT	Count

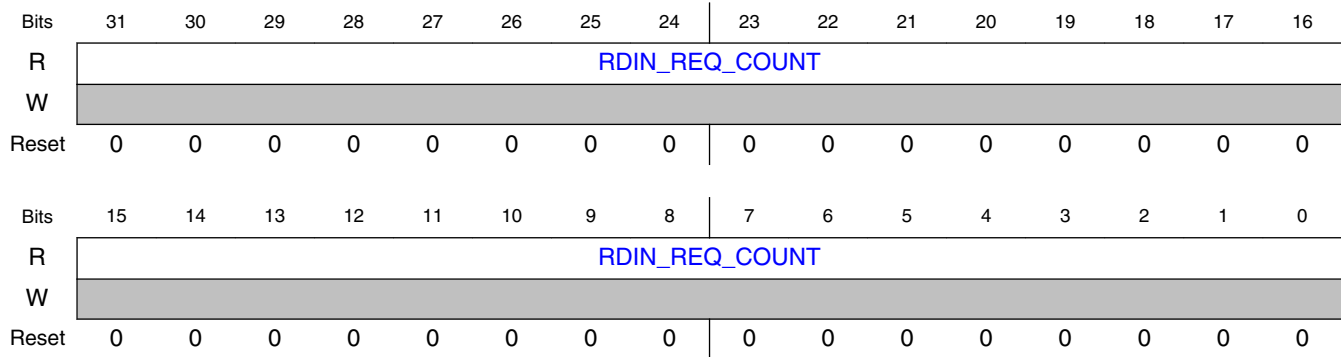
15.5.3.1.16 Total Read Request In (GCREGAHBDECTOTALREADREQIN)**15.5.3.1.16.1 Offset**

Register	Offset
GCREGAHBDECTOTALREADREQIN	B28h

15.5.3.1.16.2 Function

Total reads request number in input (pixel engine) side.

15.5.3.1.16.3 Diagram



15.5.3.1.16.4 Fields

Field	Function
31-0 RDIN_REQ_COUNT	Count

15.5.3.1.17 Total Input Read Last Number (GCREGAHBDECTOTALREADLASTSIN)

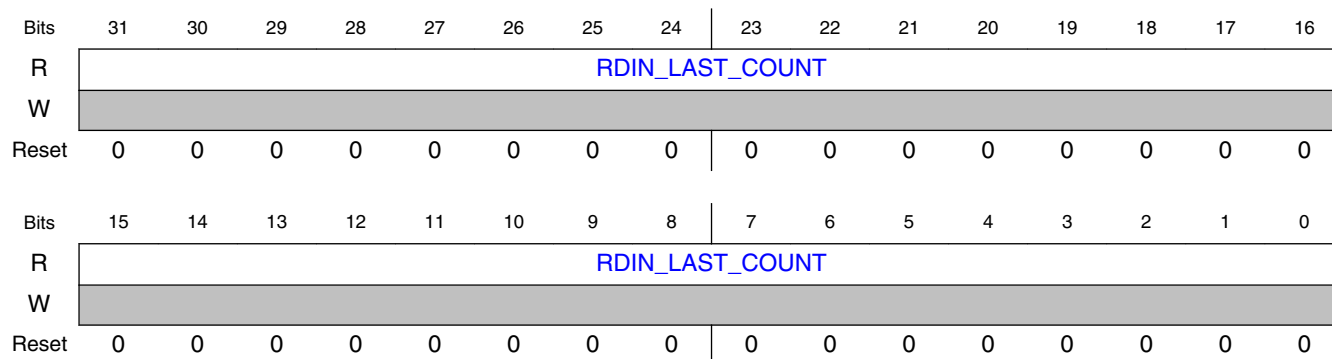
15.5.3.1.17.1 Offset

Register	Offset
GCREGAHBDECTOTALREADLASTSIN	B30h

15.5.3.1.17.2 Function

Total input read last number in input (pixel engine) side. Equal to gcregAHBDECTotalReadsReqIn.

15.5.3.1.17.3 Diagram



15.5.3.1.17.4 Fields

Field	Function
31-0 RDIN_LAST_COUNT	Count

15.5.3.1.18 Total Reads Out (GCREGAHBDECTOTALREADSOUT)

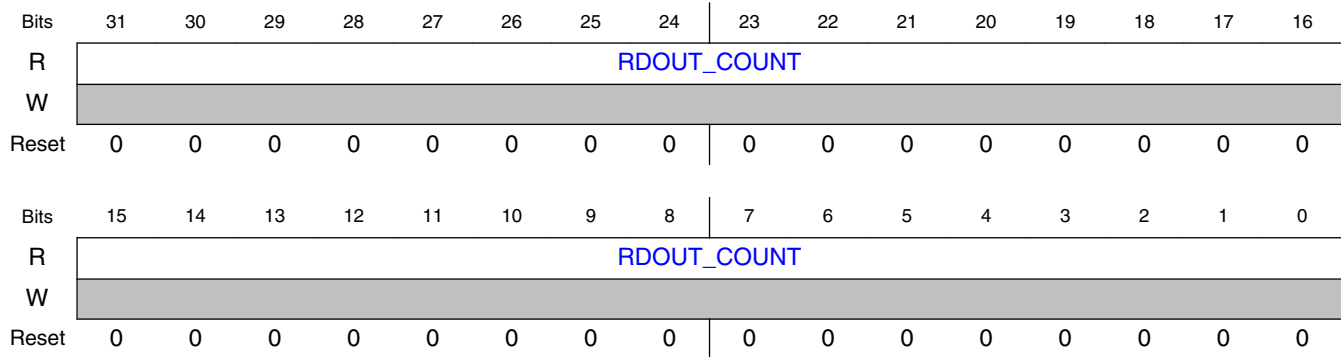
15.5.3.1.18.1 Offset

Register	Offset
GCREGAHBDECTOTALREADSOUT	B38h

15.5.3.1.18.2 Function

Total reads request in terms of dataWidth in output (DDR) side. Sum of ARLEN+1 of all the requests the Dec400D sends out.

15.5.3.1.18.3 Diagram



15.5.3.1.18.4 Fields

Field	Function
31-0 RDOUT_COUNT	Count

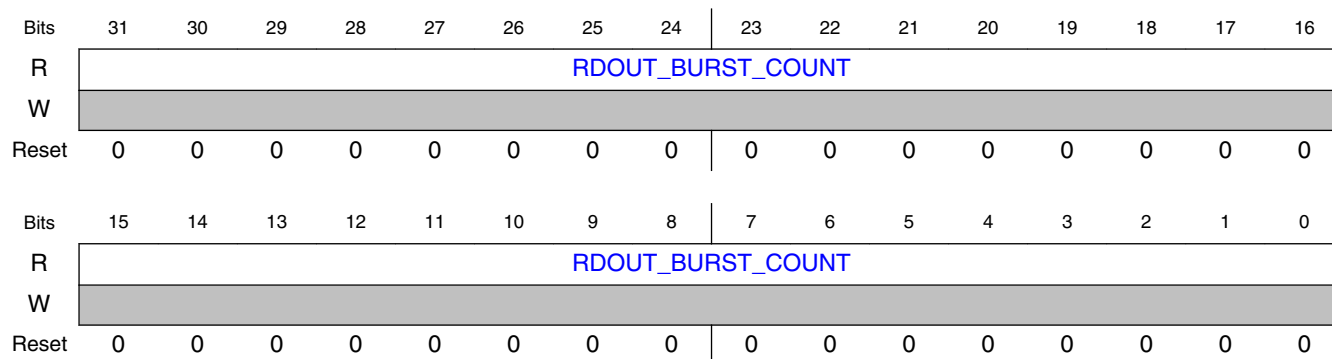
15.5.3.1.19 Total Read Bursts Out (GCREGAHBDECTOTALREADBURSTSOUT)

15.5.3.1.19.1 Offset

Register	Offset
GCREGAHBDECTOTALREADBURSTSOUT	B40h

15.5.3.1.19.2 Function

Total read data count in terms of dataWidth in output (DDR) side. Should be equal to gcregAHBDECTotalReadsOUT.

15.5.3.1.19.3 Diagram**15.5.3.1.19.4 Fields**

Field	Function
31-0 RDOUT_BURST_COUNT	Count

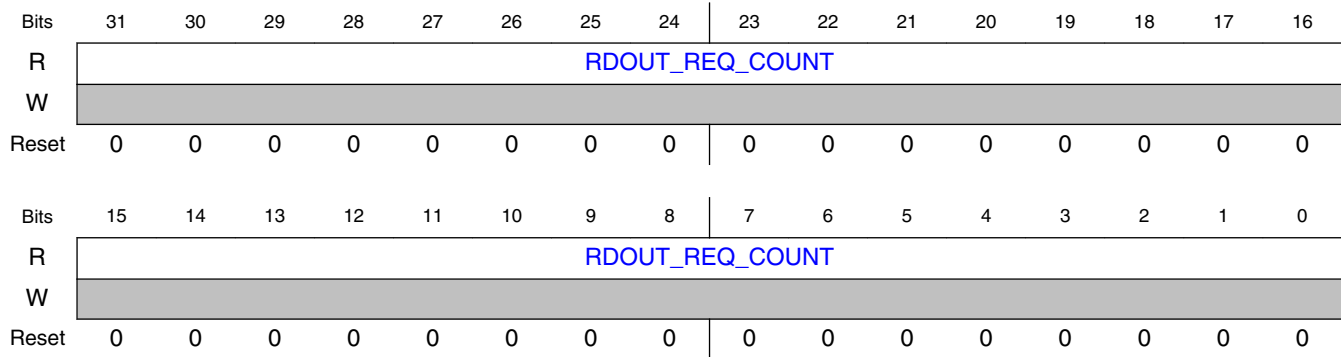
15.5.3.1.20 Total Read Request Out (GCREGAHBDECTOTALREADREQOUT)**15.5.3.1.20.1 Offset**

Register	Offset
GCREGAHBDECTOTALREADREQOUT	B48h

15.5.3.1.20.2 Function

Total reads request number in output (DDR) side..

15.5.3.1.20.3 Diagram



15.5.3.1.20.4 Fields

Field	Function
31-0 RDOUT_REQ_COUNT	Count

15.5.3.1.21 Total Read Last Out (GCREGAHBDECTOTALREADLASTSOUT)

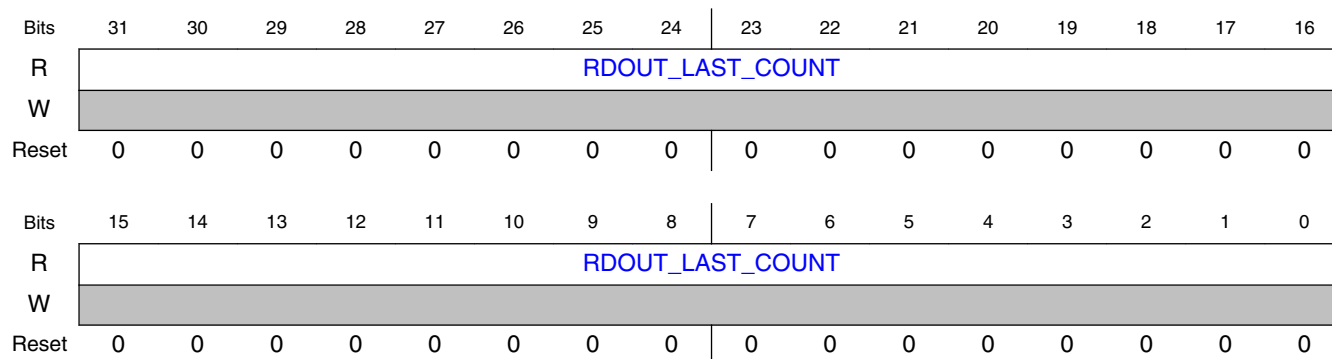
15.5.3.1.21.1 Offset

Register	Offset
GCREGAHBDECTOTALREADLASTSOUT	B50h

15.5.3.1.21.2 Function

Total read last number in output (DDR) side. Equal to gcregAHBDECTotalReadsReqOUT.

15.5.3.1.21.3 Diagram



15.5.3.1.21.4 Fields

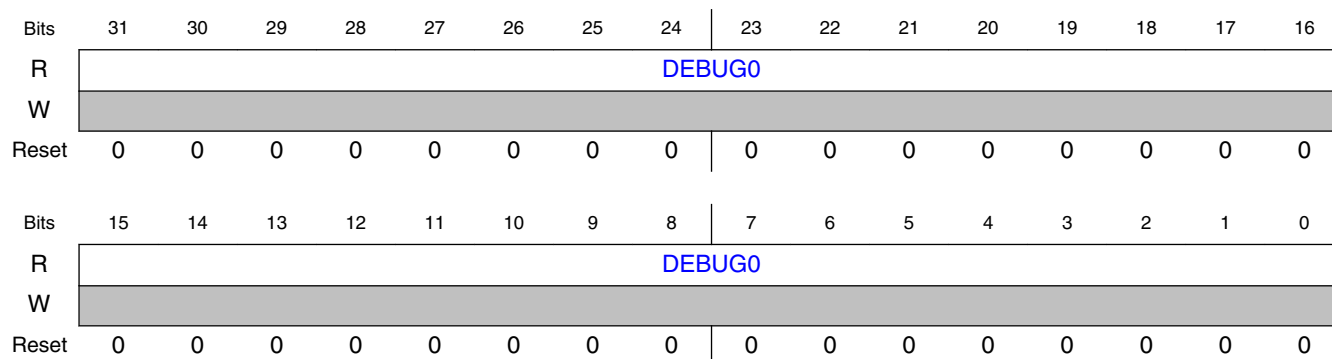
Field	Function
31-0 RDOUT_LAST_COUNT	Count

15.5.3.1.22 Debug Register 0 (GCREGAHBDECDEBUG0)

15.5.3.1.22.1 Offset

Register	Offset
GCREGAHBDECDEBUG0	B58h

15.5.3.1.22.2 Diagram



15.5.3.1.22.3 Fields

Field	Function
31-0 DEBUG0	Debug register 0

15.5.3.1.23 Debug Register 1 (GCREGAHBDECDEBUG1)

15.5.3.1.23.1 Offset

Register	Offset
GCREGAHBDECDEBUG1	B5Ch

15.5.3.1.23.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DEBUG1															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBUG1															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

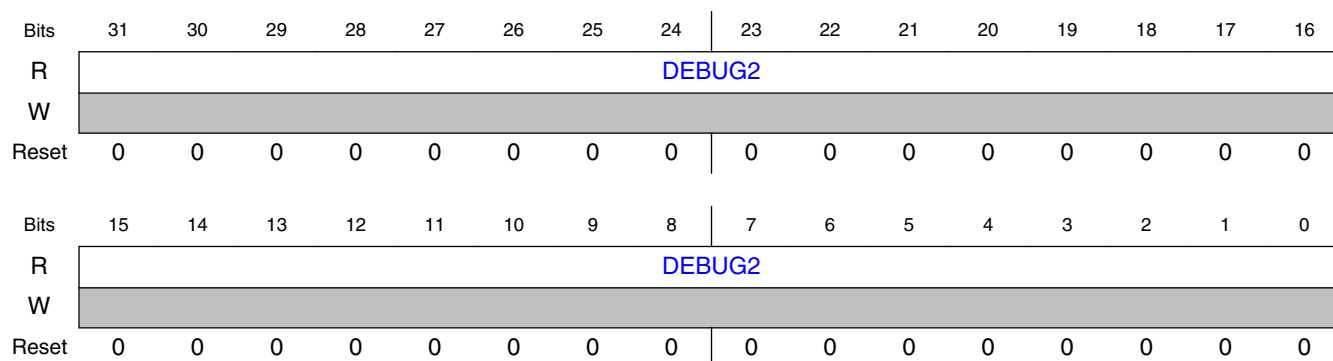
15.5.3.1.23.3 Fields

Field	Function
31-0 DEBUG1	Debug 1

15.5.3.1.24 Debug register 2 (GCREGAHBDECDEBUG2)

15.5.3.1.24.1 Offset

Register	Offset
GCREGAHBDECDEBUG 2	B60h

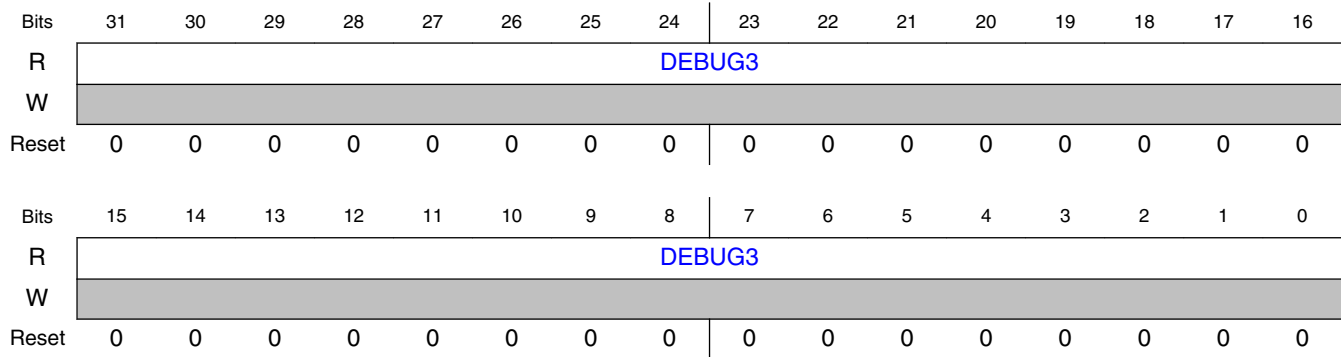
15.5.3.1.24.2 Diagram**15.5.3.1.24.3 Fields**

Field	Function
31-0 DEBUG2	Debug 2

15.5.3.1.25 Debug Register 3 (GCREGAHBDECDEBUG3)**15.5.3.1.25.1 Offset**

Register	Offset
GCREGAHBDECDEBUG 3	B64h

15.5.3.1.25.2 Diagram



15.5.3.1.25.3 Fields

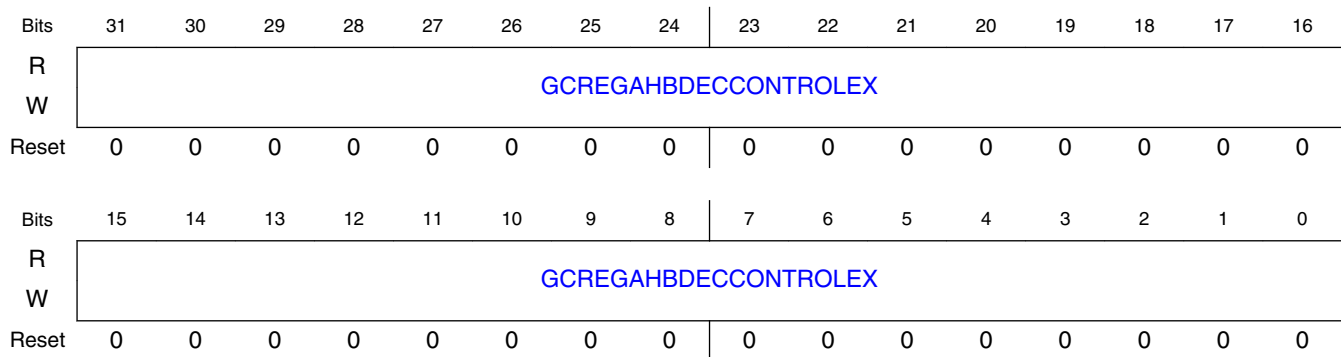
Field	Function
31-0 DEBUG3	Debug 3

15.5.3.1.26 GCREGAHBDECCONTROLEX (GCREGAHBDECCONTROLEX)

15.5.3.1.26.1 Offset

Register	Offset
GCREGAHBDECCONTROLEX	B68h

15.5.3.1.26.2 Diagram

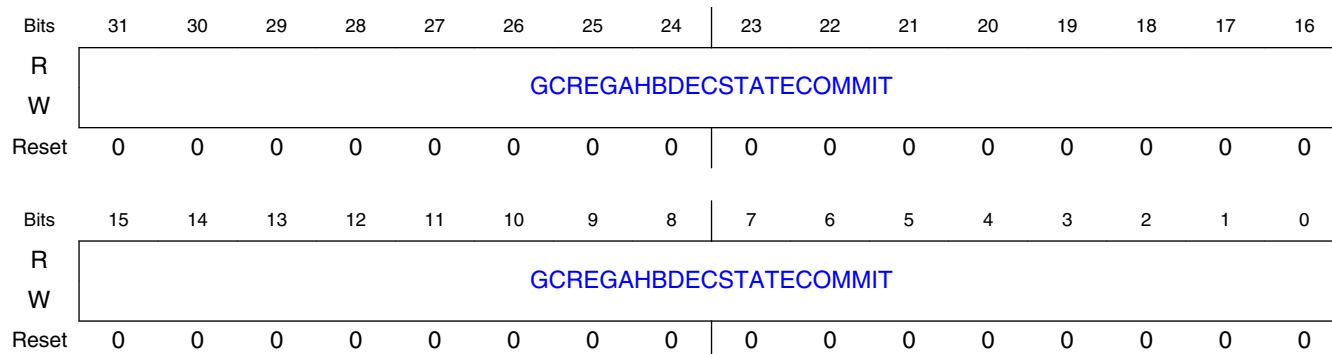


15.5.3.1.26.3 Fields

Field	Function
31-0 GCREGAHBDE CCONTROLEX	GCREGAHBDECCONTROLEX

15.5.3.1.27 GCREGAHBDECSTATECOMMIT (GCREGAHBDECSTATE COMMIT)**15.5.3.1.27.1 Offset**

Register	Offset
GCREGAHBDECSTATE COMMIT	B6Ch

15.5.3.1.27.2 Diagram**15.5.3.1.27.3 Fields**

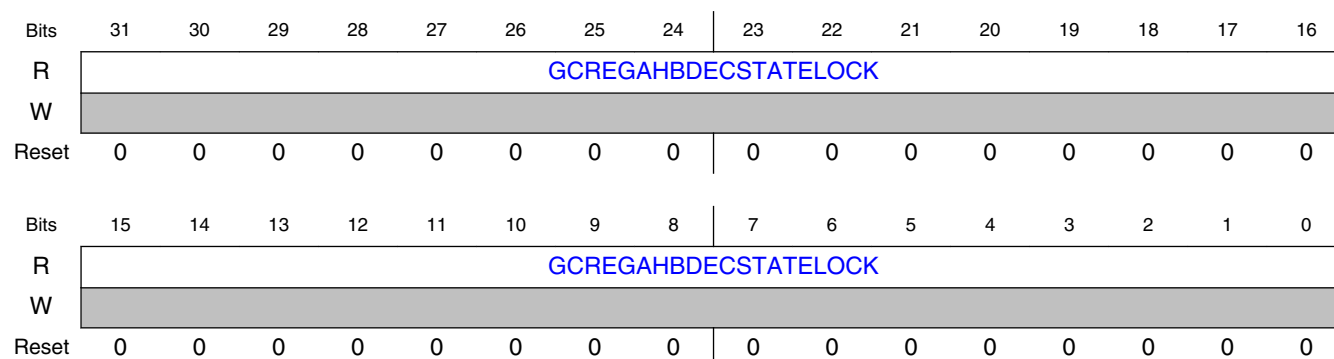
Field	Function
31-0 GCREGAHBDE CSTATECOMMI T	GCREGAHBDECSTATECOMMIT

15.5.3.1.28 GCREGAHBDECSTATELOCK (GCREGAHBDECSTATELOCK)

15.5.3.1.28.1 Offset

Register	Offset
GCREGAHBDECSTATE LOCK	B70h

15.5.3.1.28.2 Diagram



15.5.3.1.28.3 Fields

Field	Function
31-0 GCREGAHBDE CSTATELOCK	GCREGAHBDECSTATELOCK

15.5.3.1.29 Decode Read Extra Configuration (GCREGAHBDECREADEXCONFIG0 - GCREGAHBDECREADEXCONFIG31)

15.5.3.1.29.1 Offset

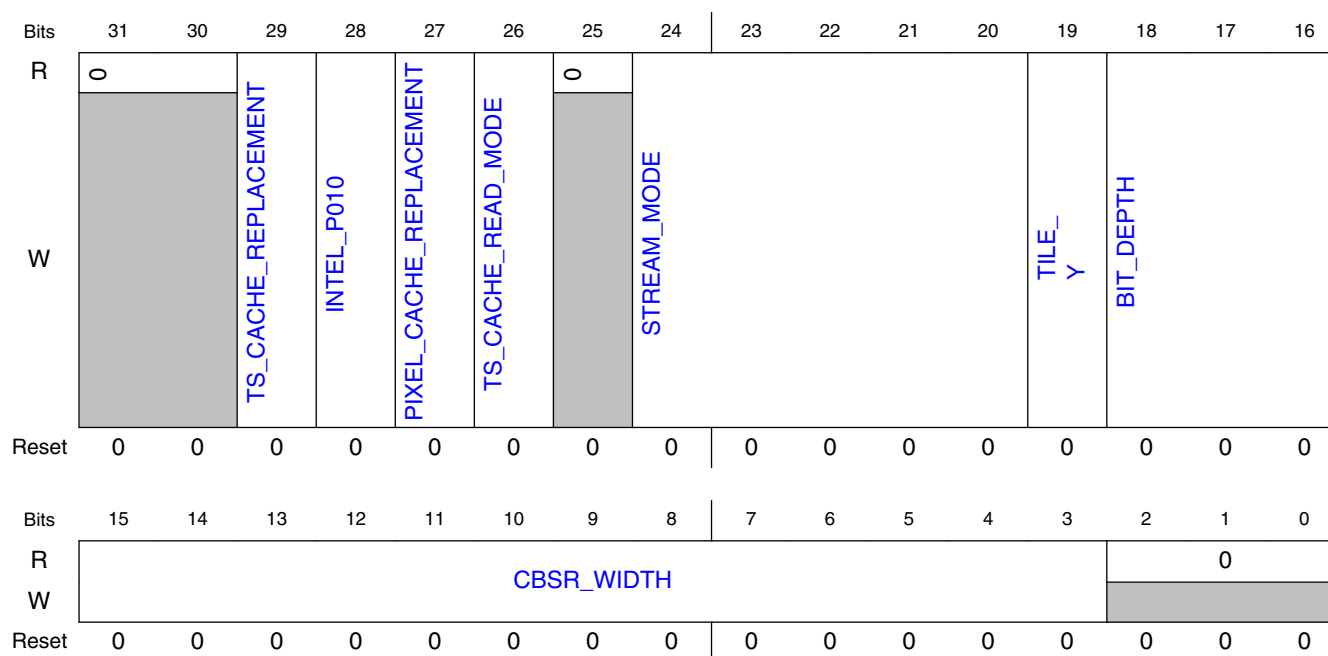
For $n = 0$ to 31:

Register	Offset
GCREGAHBDECREADE XCONFIG n	C00h + ($n \times 4h$)

15.5.3.1.29.2 Function

Dec400D extra configuration register for read ID.

15.5.3.1.29.3 Diagram



15.5.3.1.29.4 Fields

Field	Function
31-30 —	Reserved
29 TS_CACHE_REPLACEMENT	TS cache replacement Replacement method support for tile status cache. 0b - LRU 1b - FIFO
28 INTEL_P010	Intel's P010 format Intel's P010 format for 10 bit. 0b - Disable 1b - Enable
27 PIXEL_CACHE_REPLACEMENT	Pixel cache replacement Replacement method support for pixel overfetch cache. 0b - LRU 1b - FIFO
26 TS_CACHE_READ_MODE	TS cache read mode By default, the TS cache for write client is write only cache. If this bit is set, the cache reads back the cache line first, then modify it not valid for the read ID. 0b - Disable 1b - Enable
25	Reserved

Table continues on the next page...

Field	Function
—	
24-20 STREAM_MODE	Stream mode - Mode 1-9 only valid for DEC_IPU - Mode 10-13 only valid for DEC_VPU - Mode 14-19 only valid for DEC_XYZ Stream mode enumerates the possible streams sent by clients. In addition to the basic format configured for compression, streams have additional characteristics, such as reading or writing walking pattern, which distinguish one type from another 00000b - Default 00001b - ISA_STREAM0 00010b - ISA_STREAM1 00011b - ISA_STREAM2 00100b - ISA_STREAM3 00101b - TNR_STREAM_Y 00110b - TNR_STREAM_UV 00111b - GDC_STREAM_Y 01000b - GDC_STREAM_U 01001b - GDC_STREAM_V 01010b - VPU_SRC_Y 01011b - VPR_SRC_UV 01100b - VPU_REF_Y 01101b - VPU_REF_UV 01110b - XYZ_STREAM_AY 01111b - XYZ_STREAM_AU 10000b - XYZ_STREAM_AV 10001b - XYZ_STREAM_BY 10010b - XYZ_STREAM_BU 10011b - XYZ_STREAM_BV
19 TILE_Y	Tile Y Memory layout TileY 0b - Disable 1b - Enable
18-16 BIT_DEPTH	Bit depth Bit depth for Y/UV/Bayer format. 000b - 8 bit 001b - 10 bit 010b - 12 bit 011b - 16 bit
15-3 CBSR_WIDTH	CBSR width For CBSR Align mode, this bitfield indicates Y CBS number included in one CBSR since CBSG_SADDR field in CBSG table entry is based from each CBSR start address. Also used to judge whether CbCr needs decompression if Y is in odd CBS number.
2-0 —	Reserved

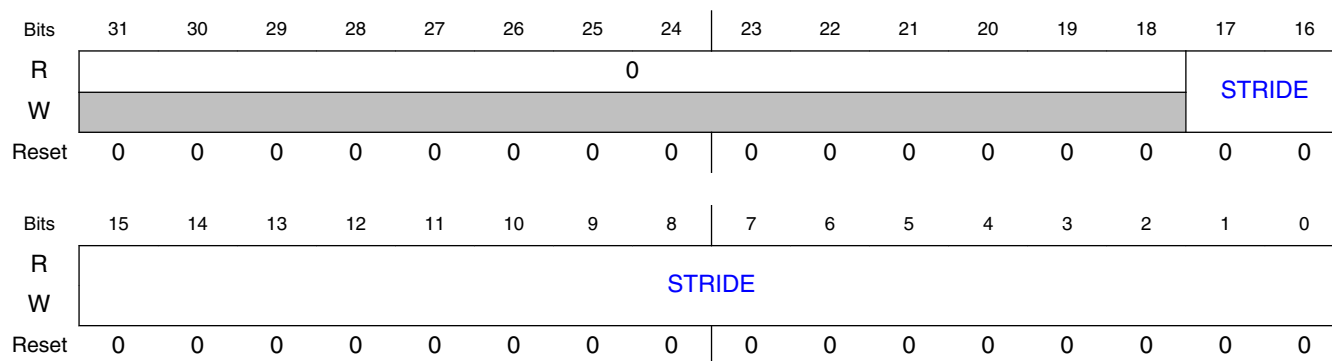
15.5.3.1.30 Decoder Read Stride (GCREGAHBDECREADSTRIDE0 - GCREGAHBDECREADSTRIDE31)

15.5.3.1.30.1 OffsetFor $n = 0$ to 31:

Register	Offset
GCREGAHBDECREADS TRIDEn	$C80h + (n \times 4h)$

15.5.3.1.30.2 Function

Dec400D surface stride in bytes for read ID.

15.5.3.1.30.3 Diagram**15.5.3.1.30.4 Fields**

Field	Function
31-18 —	Reserved
17-0 STRIDE	Surface stride

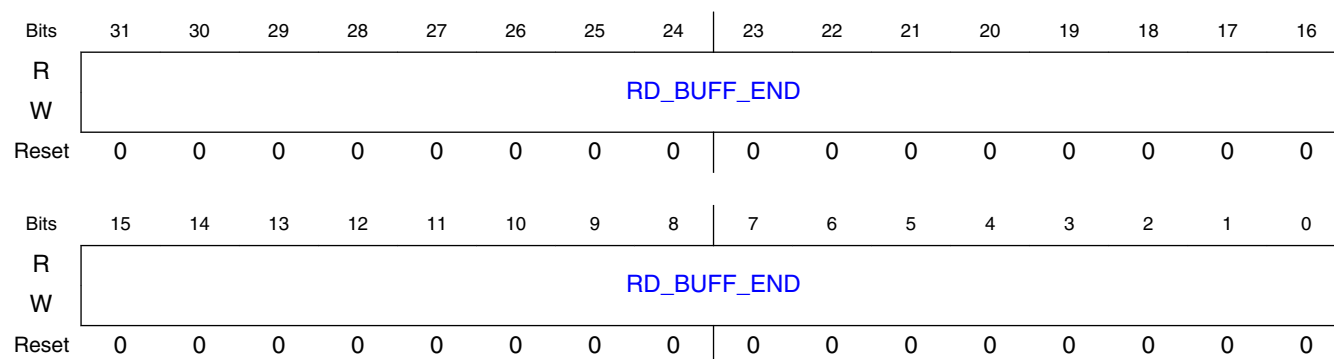
15.5.3.1.31 Decoder Read Buffer End (GCREGAHBDECREADBUFFEREND 0 - GCREGAHBDECREADBUFFEREND31)**15.5.3.1.31.1 Offset**For $n = 0$ to 31:

Register	Offset
GCREGAHBDECREADB UFFERENDn	E00h + (n × 4h)

15.5.3.1.31.2 Function

End address for pixel buffer for read ID. Pixel and tile status cache invalidation is triggered automatically when address of a request matches this address.

15.5.3.1.31.3 Diagram



15.5.3.1.31.4 Fields

Field	Function
31-0 RD_BUFF_END	Address

15.6 Decompression and Tile to Raster Conversion (DTRC)

15.6.1 Overview

The Decompression and Tile to Raster Conversion (DTRC) module decompresses content for video frames. It decompresses and resolves the tile data into linear scan lines.

The DTRC buffers a full 4K of raster lines in its local memory. This is required since the DPR fetches scan lines from the resolve memory attached to the DTRC.

15.6.2 Features

The DTRC includes the following functionality features:

- Input data format can be one of the following:
 - Compressed Hantro G2 reference frame buffer format
 - Uncompressed tile Hantro G2 reference frame buffer format
 - Uncompressed tile Hantro G1 reference frame buffer format
- Output data format is uncompressed YCbCr pixels on AXI bus
- Decompression for G2:
 - DTRC decompresses data in units of Coding Block Set Groups (CBSG)
 - DTRC decompression can be bypassed
- The DTRC converts G2/G1 tile format pixels into raster scan order pixels
- Picture size:
 - Minimum width of 144
 - Minimum height of 144
 - Maximum width of 4096
 - Maximum width of 4096
 - The width and height must be integer multiples of 8
- Maximum pixel sample rate is 4096 x 2304 at 60fps
- Output pixel scan order is normal whole picture raster scan
- Output picture crop:
 - Cropped picture origin (x,y) points to the top left corner of the cropped picture
 - Both x and y are even, and the unit is pixel
 - Cropped width is even and unit is pixel
 - Cropped height is even and unit is pixel
 - Minimum cropped picture size is 144 x 2
- Non-G2/G1 data read transactions directly pass through the DTRC
 - The transactions are distinguished by ARID or address
- Run-time hot reset triggered by software
 - G2/G1 data processing path is reset
 - Non-G2/G1 data path continues working
- Data prefetch and pre-decompression supported:
 - Prefetch between frames
 - Prefetch along with pixel scan order

15.6.3 Block diagram

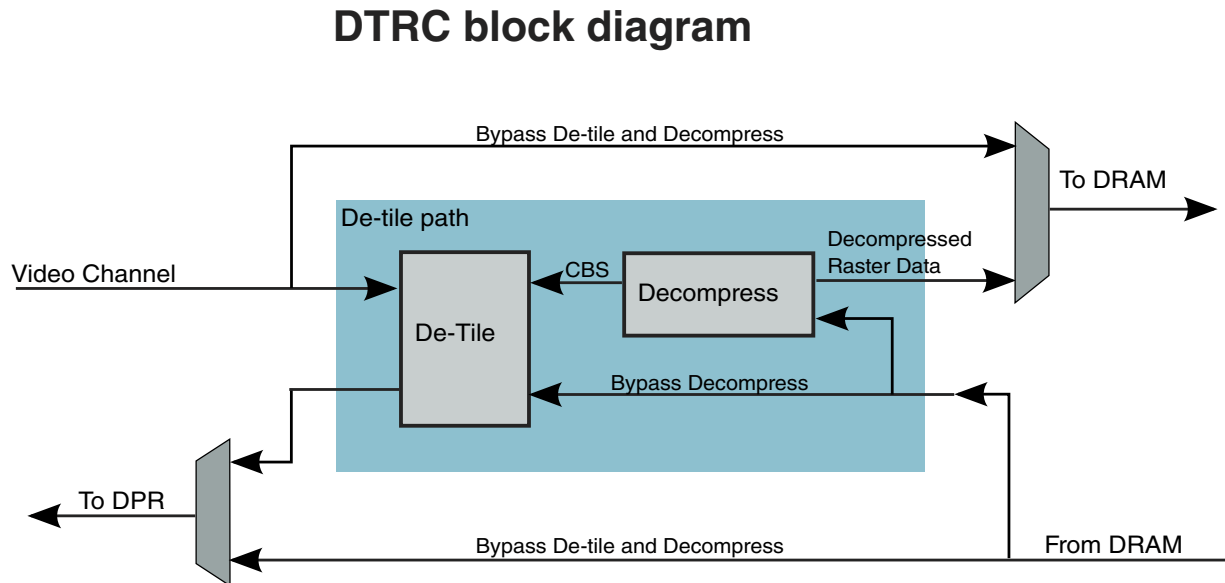


Figure 15-16. DTRC block diagram

15.6.4 Functional description

The following sections describe functional details of the DTRC module.

15.6.4.1 Video compression/decompression

Video decompression uses the DTRC module. This module decompresses CBSGs (Coding Block Set Groups) and store the resolved data in raster order in a local buffer exclusive to each DTRC instance. The DPR is programmed to fetch raster data, and the DTRC returns the scan line data to the DPRs.

The compression scheme used by the video engine is not Z ordered like in GPU frames. Instead, video frames are compressed using 8x8 pixel tiles (for Luma) as each compressed unit, and the compressed results are packed in memory. The 8x8 tiles are compressed in raster scan order from left to right, then top to bottom. The tile order of video frames is depicted in the following diagram.

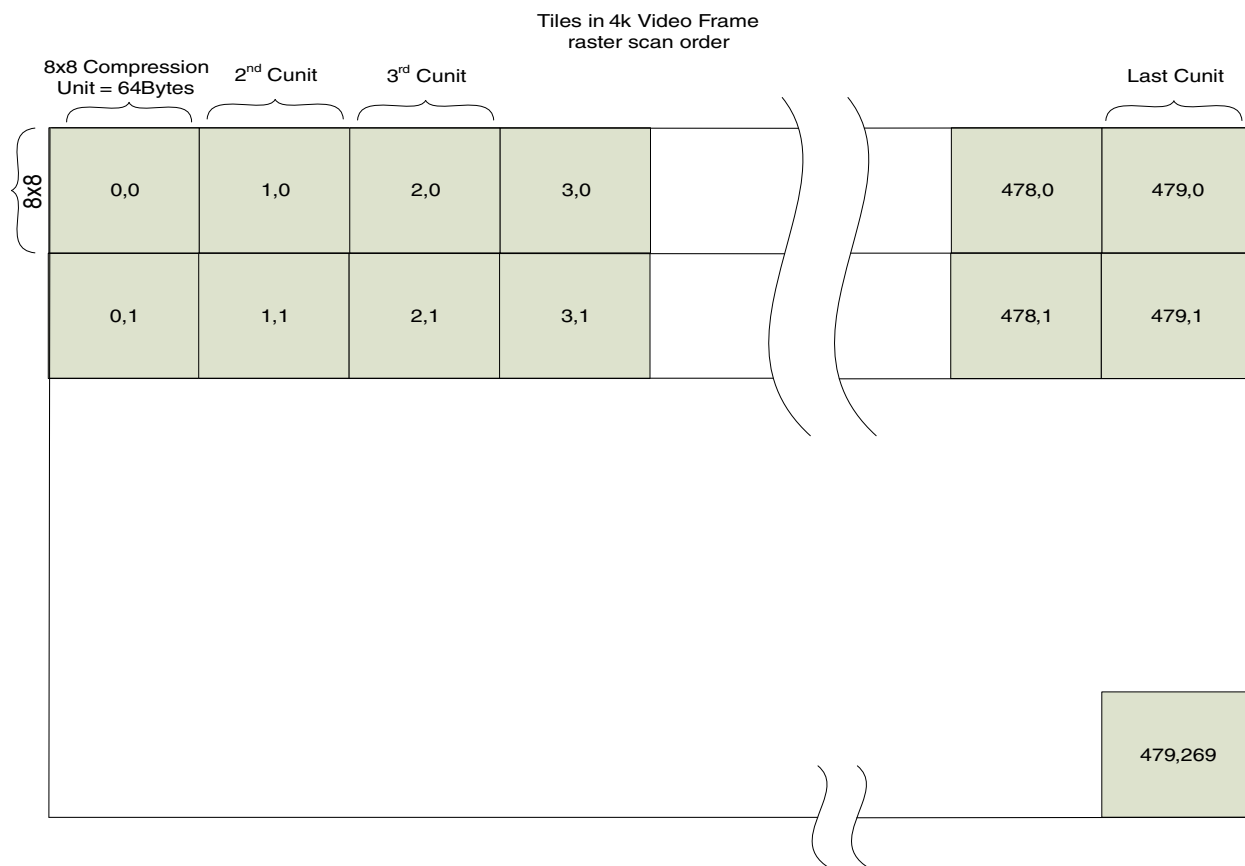


Figure 15-17. Video Frame Tile Order

NOTE

CbCr frame buffers are 8x4 compressed units, which are also 64B per compressed unit.

The video compression scheme is depicted in the diagram below.

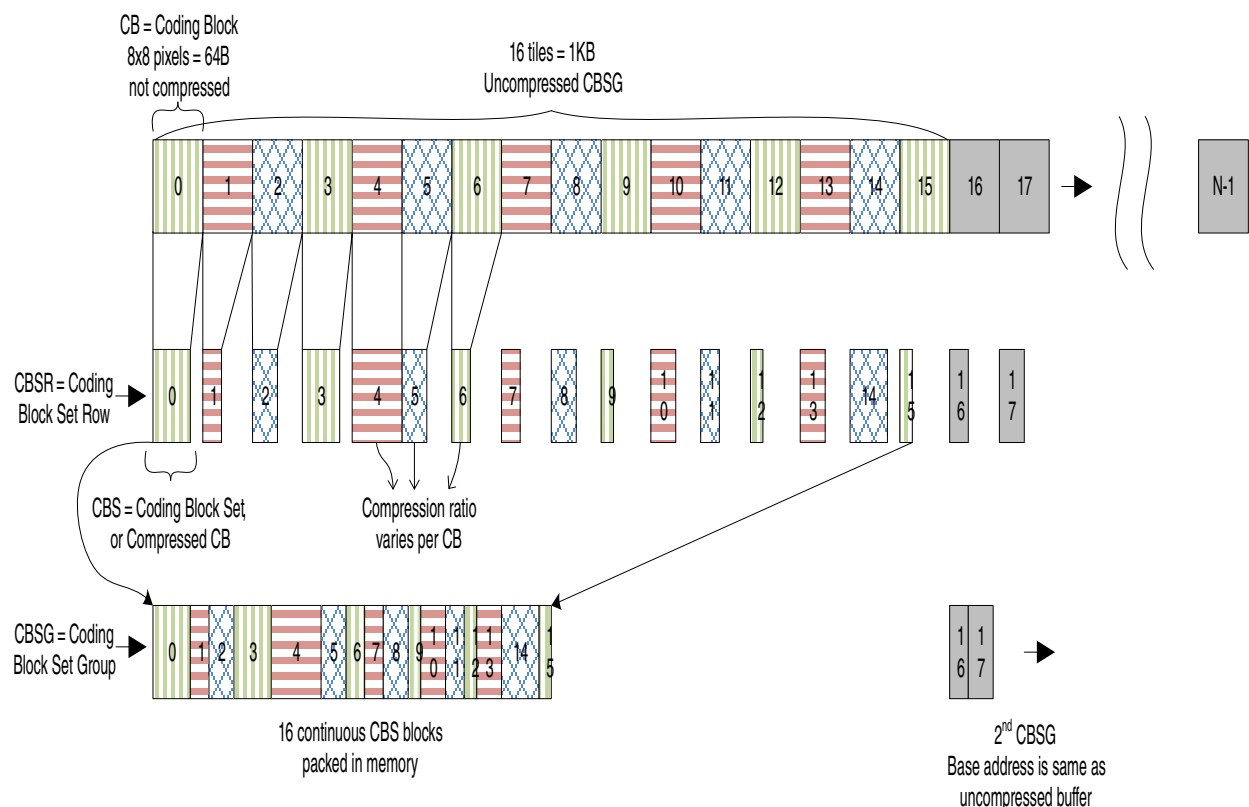


Figure 15-18. Video Compression Scheme

Each 8x8 tile (for Luma) is compressed separately as a single compressed unit, called a CBS unit. The compression ratio can be different for each unit. Then, 16 CBSs are packed in memory to form a CBSG. These 16 uncompressed tiles are 1KB, and when compressed and packed requires less than 1KB. In the example above, the compressed result for the 1st CBSG is about 50%, so the CBSG requires 512B in memory.

NOTE

The DTRC fetches compressed CBSGs, uncompress the units, and store them as raster scan lines in local memory attached to the DTRC. The DTRC then returns the data to the DPR in raster scan order. The DPR is programmed in raster scan mode, with a burst length of 4KB. This forces the DPR to fetch an entire scan line, and then move to the next scan line in the vertical direction. In this case, the DPR does not resolve tiles, but just process raster ordered scan lines.

15.6.4.2 DTRC scan line memory size

The maximum display is 3840 pixels. The scan line structure is 8 lines high for luma, and 4 lines high for chroma. Each pixel is 10-bit per component. The memory size is:

- Luma buffer = $3840 * 8 * 10/8 = 38400B$
- Chroma buffer = $3840 * 4 * 10/8 = 19200B$

This implementation supports 40960B for the luma buffer and 20480B for the chroma buffer to support 4096 horizontal displays. The total memory size is 61440B.

15.6.5 Programming considerations

The following sections describe software usage for the DTRC module.

15.6.5.1 Initial configuration

At the beginning to start the decompression of a video sequence, software needs to configure the following:

- Set DTRCINTEN register to enable/disable interrupt sources
- Set DTID2DDR register to configure AXI ID of DTRC generated transaction to fetch G2/G1 video data. If the default, AXI ID won't be used.
- Configure the mechanism to determine between G2/G1 and non-G2/G1 data
 - If ARID used to determine:
 - Set DTCTRL[30] to 0
 - Set DTCTRL[1:0] to select a mode of how to use IDs in ARIDR register
 - According to DTCTRL[1:0], set all of the four IDs in ARIDR. Duplicate if ID to be set is less than 4
 - If address used to determine:
 - Set DTCTRL[30] to 1
 - If all transactions are non-G2/G1 transactions, so that no meaningful address segment corresponding to G2/G1 video data, set all bit 0 of F0SYSSA, F0SYSEA, F0SUVSSA, F0SUVSEA, F1SYSSA, F1SYSEA, F1SUVSSA, F1SUVSEA to 1.

15.6.5.2 Picture-level configuration

There are two sets of picture level configuration registers corresponding to two back-to-back pictures, which allow seamless display between two pictures.

Software first looks at DTCTRL[31] to know which set of registers to program. After that, it sets addresses, picture size, vertical scan direction etc. into corresponding registers.

If ARADDR_S is used to tell G2/G1 and non-G2/G1 transactions, set luma and chroma video data address segment into F0SYSSA, F0SYSEA, F0SUVSSA, F0SUVSEA, or F1SYSSA, F1SYSEA, F1SUVSSA, F1SUVSEA.

After the whole set of configuration is done, set the F0DCTL[0] or F1DCTL[0] to 1.

In scenario that there is no G2/G1 transactions, set F0DCTL[0] or F1DCTL[0] to 0 and do not need to do any picture level configuration.

The DTRC internal state machine periodically performs the following action sequence:

1. (After an asynchronous reset) wait frame 0 bit 0 == 1'b1
2. Decompress frame
3. Wait frame 1 bit 0 == 1'b1
4. Decompress frame
5. Wait frame 0 bit 0 == 1'b1
6. Decompress frame
7. Wait frame 1 bit 0 == 1'b1
8. Decompress frame
9. ...

Although the switch of the two sets of picture configuration is predictable, it is recommended software always check the DTCTRL[31] to program picture settings.

15.6.5.3 Interrupts

If interrupt is enabled, DTRC raises the interrupt signal to high and sets the interrupt cause bit in FDINTR. Software should clean the interrupt after it finished the handling (such as program the next picture configurations).

15.6.5.4 Time out

To enable time out interrupt, software not only enables the interrupt but also enables the performance counter. Software calculates a reasonable maximum cycle count according to DTRC frequency and picture frame rate to set TOCR.

15.6.5.5 Hot reset

The video path of decompression and detile can be hot reset and do not influence the bypass non-G2/G1 transactions. To do the reset, the user needs to guarantee the following requirements:

NOTE

Requirement: The hot reset must be triggered when there is no outstanding G2/G1 video transactions remained at the DTRC AXI slave port.

The hot reset details are described below:

1. Software can write DTCTRL[2] by 1 to start a hot reset
2. A hot reset can start at any time in case the requirement described above is met
3. The reset action is hold a while after DTRC received the reset
4. When G2/G1 data related BUS transactions are clean, DTRC performs the reset
 - a. The following external visible register bits are reset, and others are not reset
 1. F0DCTL[0]
 2. F1DCTL[1]
 3. DTCTRL[31]
 4. PFCR[31:0]
 - b. Decompression and detile status is reset so that DTRC restarts from a new frame configured by the frame setting register group 0
 - c. If DTRCINTEN[4] is 1, DTRC raises hot reset finish interrupt as well as set the interrupt bit DTRCINTR[4]
 - d. DTRC clears DTCTRL[2] to 0
5. During the hot reset procedure, only bus error interrupt can be asserted, except for hot reset finish interrupt
6. During the hot reset procedure, the whole non-G2/G1 data path keeps working

15.6.6 Memory Map and register definition

This section includes the DTRC module memory map and detailed descriptions of all registers.

15.6.6.1 DTRC register descriptions

15.6.6.1.1 DTRC Memory map

DTRC_CHAN2 base address: 1_6000h

DTRC_CHAN3 base address: 1_7000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Luma video data start address (F0DYDSADDR)	32	RW	0000_0000h
4h	Chroma video data start address (F0DCDSADDR)	32	RW	0000_0000h
8h	Luma table data start address (F0DYTSADDR)	32	RW	0000_0000h
Ch	Chroma table data start address (F0DCTSADDR)	32	RW	0000_0000h
10h	Frame size (F0SIZE)	32	RW	0000_0000h
14h	Luma data slave start address (F0SYSSA)	32	RW	0000_0000h
18h	Luma data slave end address (F0SYSEA)	32	RW	0000_0000h
1Ch	Chroma data slave start address (F0SUVSSA)	32	RW	0000_0000h
20h	Chroma data slave end address (F0SUVSEA)	32	RW	0000_0000h
24h	Cropped picture origin (F0CROPORIG)	32	RW	0000_0000h
28h	Cropped picture size (F0CROPSIZE)	32	RW	0000_0000h
2Ch	Frame data control (F0DCTL)	32	RW	0000_0000h
60h	Luma video data start address (F1DYDSADDR)	32	RW	0000_0000h
64h	Chroma video data start address (F1DCDSADDR)	32	RW	0000_0000h
68h	Luma table data start address (F1DYTSADDR)	32	RW	0000_0000h
6Ch	Chroma table data start address (F1DCTSADDR)	32	RW	0000_0000h
70h	Frame size (F1SIZE)	32	RW	0000_0000h
74h	Luma data slave start address (F1SYSSA)	32	RW	0000_0000h
78h	Luma data slave end address (F1SYSEA)	32	RW	0000_0000h
7Ch	Chroma data slave start address (F1SUVSSA)	32	RW	0000_0000h
80h	Chroma data slave end address (F1SUVSEA)	32	RW	0000_0000h
84h	Cropped picture origin (F1CROPORIG)	32	RW	0000_0000h
88h	Cropped picture size (F1CROPSIZE)	32	RW	0000_0000h
8Ch	Frame data control (F1DCTL)	32	RW	0000_0000h
C0h	DTRC Interrupt enables (DTRCINTEN)	32	RW	0000_0000h
C4h	DTRC Interrupt Requests (FDINTR)	32	RW	0000_0000h
C8h	DTRC Control (DTCTRL)	32	RW	0000_0000h
CCh	ARIDR (ARIDR)	32	RW	0000_0000h
D0h	DTID2DDR (DTID2DDR)	32	RW	0000_0F0Eh
D4h	DTRCCONFIG (DTRCCONFIG)	32	RO	See description.
D8h	DTRC Version (DTRCVERSION)	32	RO	See description.
F0h	Performance counter control (PFCTRL)	32	RW	0000_0000h
F4h	Performance counter (PFCR)	32	RW	0000_0000h
F8h	Time Out Cycles (TOCR)	32	RW	0000_0000h

15.6.6.1.2 Luma video data start address (F0DYDSADDR)

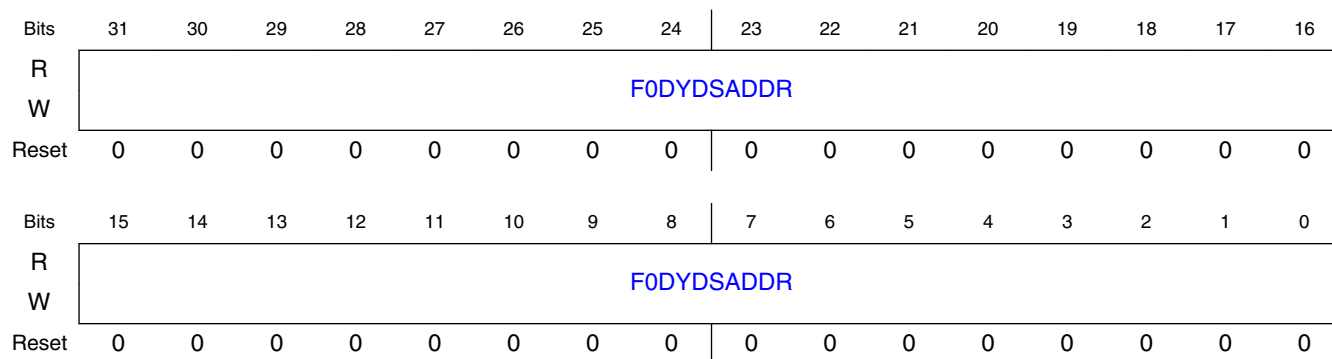
15.6.6.1.2.1 Offset

Register	Offset
F0DYDSADDR	0h

15.6.6.1.2.2 Function

This register specifies the start address for luma video data saved in external memory. This data is compressed or decompressed data.

15.6.6.1.2.3 Diagram



15.6.6.1.2.4 Fields

Field	Function
31-0	Luma video data start address
F0DYDSADDR	Start address for luma data saved in external memory. Unit is byte, but should be 16-byte aligned.

15.6.6.1.3 Chroma video data start address (F0DCDSADDR)

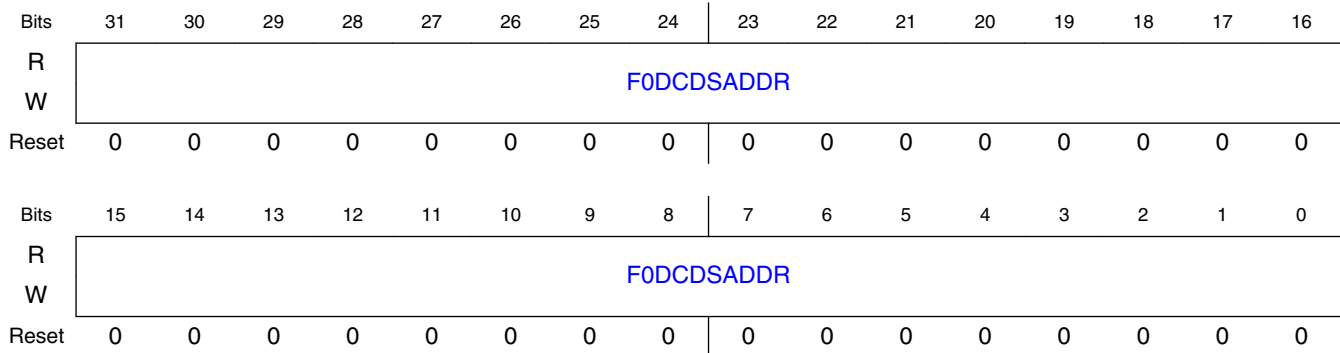
15.6.6.1.3.1 Offset

Register	Offset
F0DCDSADDR	4h

15.6.6.1.3.2 Function

This register specifies the start address for chroma video data saved in external memory.

15.6.6.1.3.3 Diagram



15.6.6.1.3.4 Fields

Field	Function
31-0	Chroma video data start address
F0DCDSADDR	Start address for chroma decompression data saved in external memory. Unit is byte, but should be 16-byte aligned.

15.6.6.1.4 Luma table data start address (F0DYTSADDR)

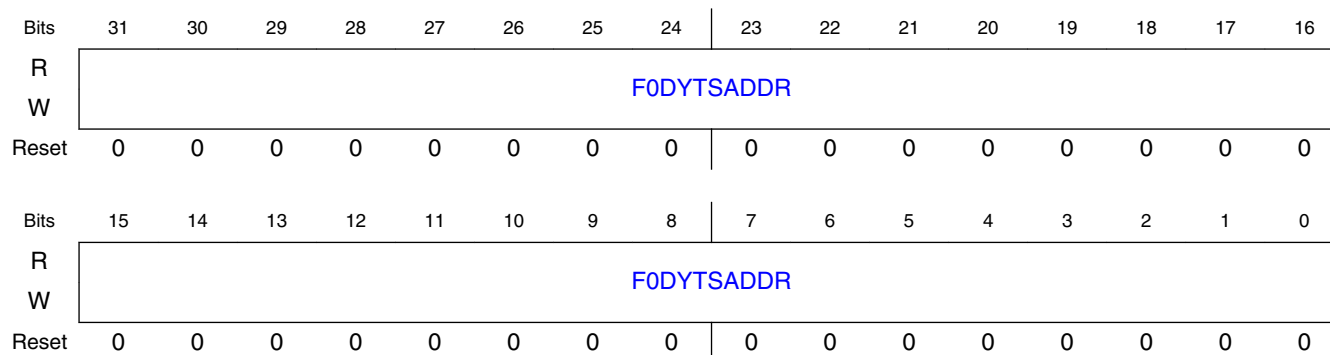
15.6.6.1.4.1 Offset

Register	Offset
F0DYTSADDR	8h

15.6.6.1.4.2 Function

This register specifies the start address for luma table data saved in external memory. Not used when compression is disabled in G2.

15.6.6.1.4.3 Diagram



15.6.6.1.4.4 Fields

Field	Function
31-0	Luma table data start address
F0DYTSADDR	Start address for luma table data saved in external memory. Unit is byte and 16-byte aligned.

15.6.6.1.5 Chroma table data start address (F0DCTSADDR)

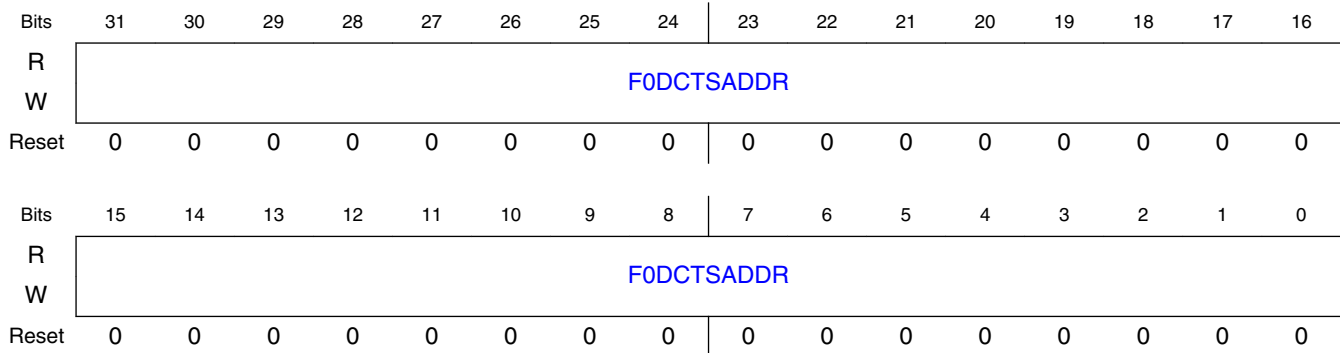
15.6.6.1.5.1 Offset

Register	Offset
F0DCTSADDR	Ch

15.6.6.1.5.2 Function

This register specifies the start address for chroma table data saved in external memory. Not used when compression is disabled in G2.

15.6.6.1.5.3 Diagram



15.6.6.1.5.4 Fields

Field	Function
31-0	Chroma table data start address
F0DCTSADDR	Start address for chroma table data saved in external memory. Unit is byte and 16-byte aligned.

15.6.6.1.6 Frame size (F0SIZE)

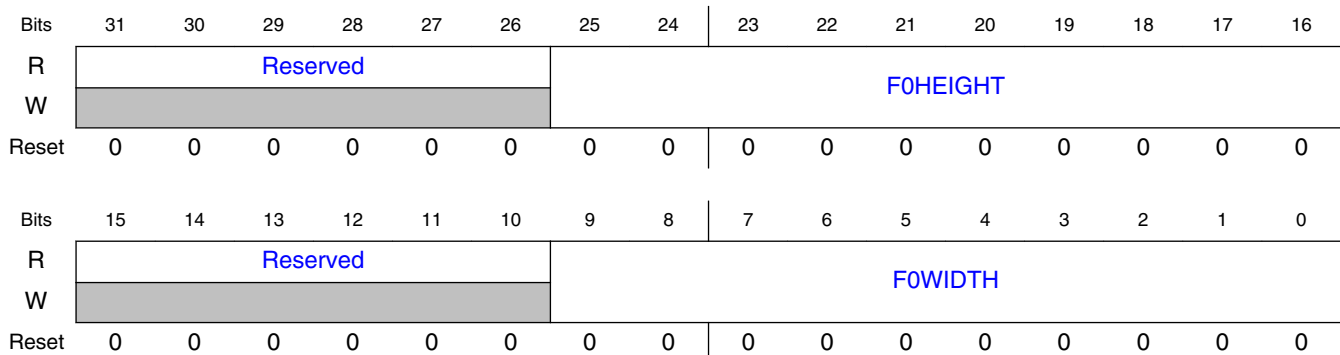
15.6.6.1.6.1 Offset

Register	Offset
F0SIZE	10h

15.6.6.1.6.2 Function

This register specifies frame width and frame height in units of 8.

15.6.6.1.6.3 Diagram



15.6.6.1.6.4 Fields

Field	Function
31-26 —	Reserved.
25-16 F0HEIGHT	Frame height Indicates frame height in units of 8 lines.
15-10 —	Reserved.
9-0 F0WIDTH	Frame width Indicates frame width in units of 8 pixels.

15.6.6.1.7 Luma data slave start address (F0SYSSA)

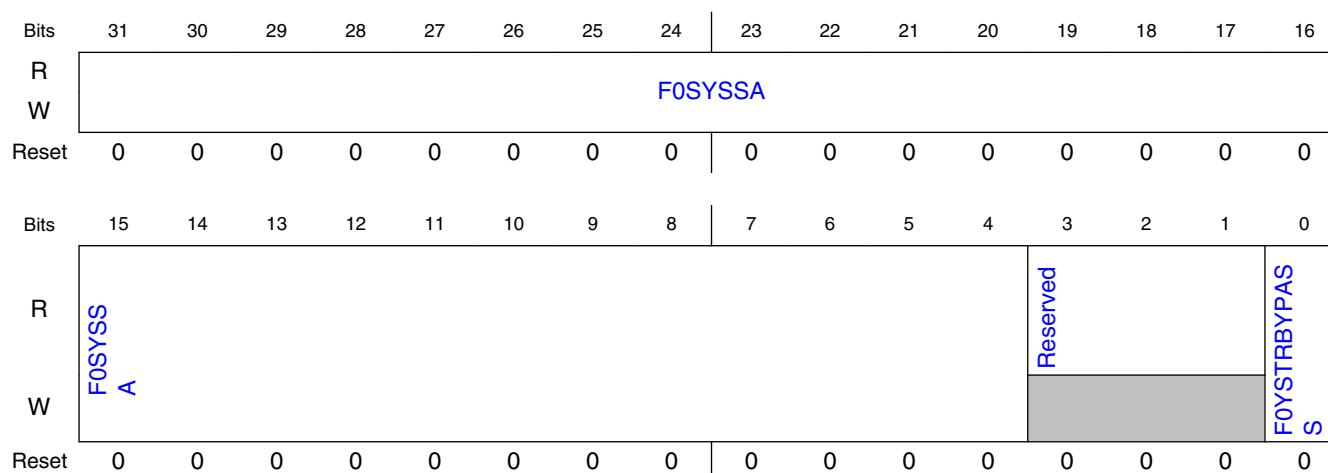
15.6.6.1.7.1 Offset

Register	Offset
F0SYSSA	14h

15.6.6.1.7.2 Function

This register specifies the start address range of the luma data associated to frame 0 and appears at ARADDR_S. Only used when G1/G2 transactions are told by ARADDR_S. It has nothing to do with F0DYDSADDR.

15.6.6.1.7.3 Diagram



15.6.6.1.7.4 Fields

Field	Function
31-4 F0SYSSA	Luma data slave start address Start of address space of slave interface luma data. This address should be 16-byte aligned and unit is 16 bytes, inclusive.
3-1 —	Reserved.
0 F0YSTRBYPASS	Luma Start Tile to Raster scan Bypass 0b - All ARADDR does NOT bypass the tile-to-rasterscan logic. 1b - All ARADDR bypasses the tile-to-rasterscan logic.

15.6.6.1.8 Luma data slave end address (F0SYSEA)

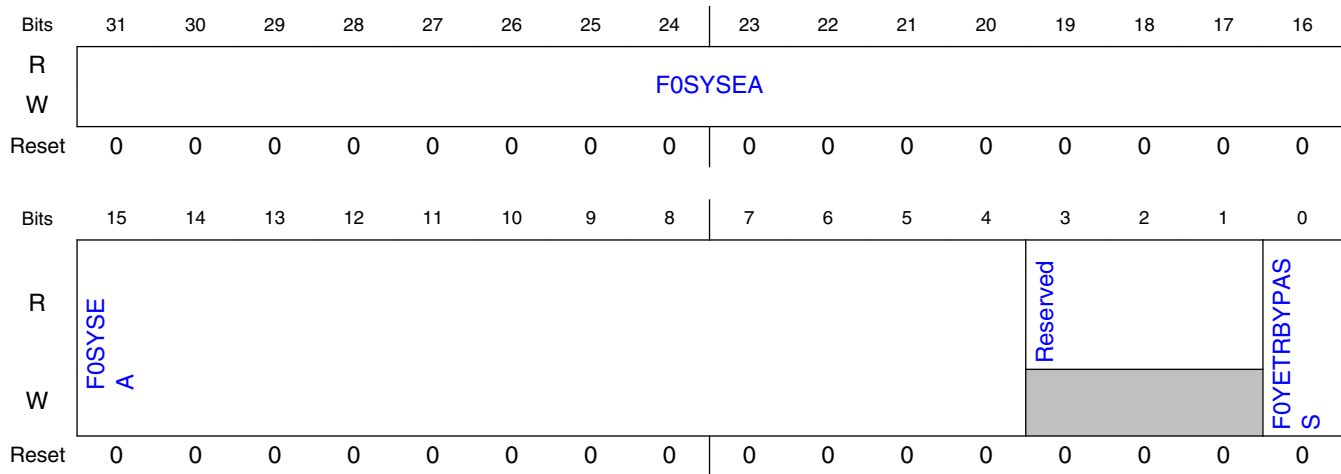
15.6.6.1.8.1 Offset

Register	Offset
F0SYSEA	18h

15.6.6.1.8.2 Function

This register specifies the end address space of luma data.

15.6.6.1.8.3 Diagram



15.6.6.1.8.4 Fields

Field	Function
31-4 F0SYSEA	Luma data slave end address End of address space of slave interface luma data. This address should be 16-byte aligned and unit is 16 bytes, exclusive.
3-1 —	Reserved.
0 F0YETRBYPASS	End Tile to Raster scan Bypass 0b - All ARADDR does NOT bypass the tile-to-rasterscan logic. 1b - All ARADDR bypasses the tile-to-rasterscan logic.

15.6.6.1.9 Chroma data slave start address (F0SUVSSA)

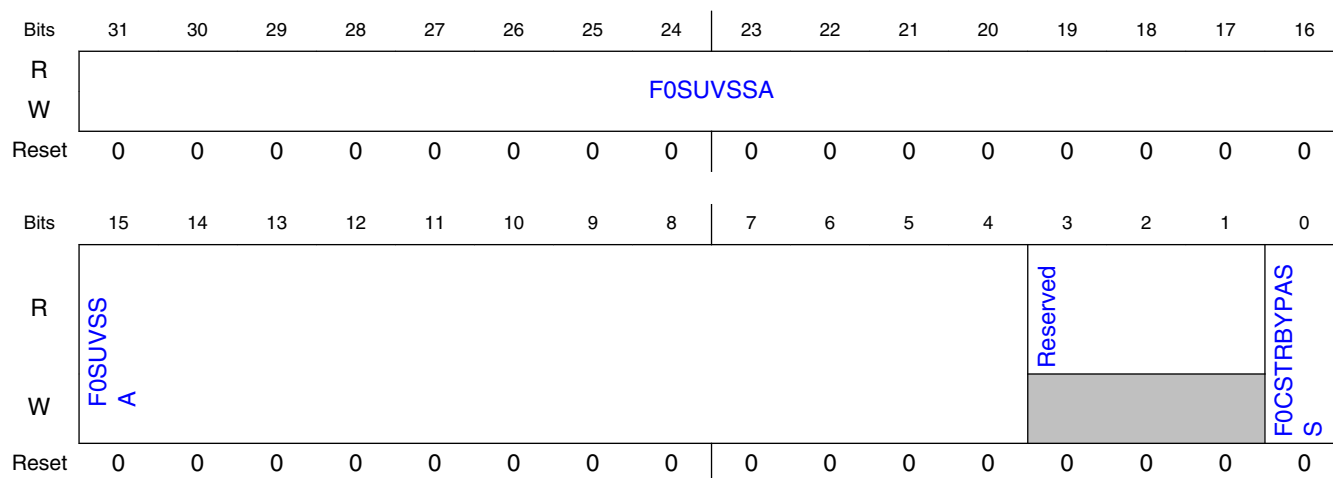
15.6.6.1.9.1 Offset

Register	Offset
F0SUVSSA	1Ch

15.6.6.1.9.2 Function

This register specifies the start address of the address range of the chroma data associated to frame 0 and appears at ARADDR_S. Only used when G2/G1 and non G2/G1 transactions are told by ARADDR_S. It has nothing to do with F0DCDSADDR.

15.6.6.1.9.3 Diagram



15.6.6.1.9.4 Fields

Field	Function
31-4 F0SUVSSA	Chroma data slave start address Start of address space of slave interface chroma data. This address should be 16-byte aligned and unit is 16 bytes, inclusive.
3-1 —	Reserved.
0 F0CSTRBYPAS S	Chroma Start Tile to Raster scan Bypass 0b - All ARADDR does NOT bypass the tile-to-rasterscan logic. 1b - All ARADDR bypasses the tile-to-rasterscan logic.

15.6.6.1.10 Chroma data slave end address (F0SUVSEA)

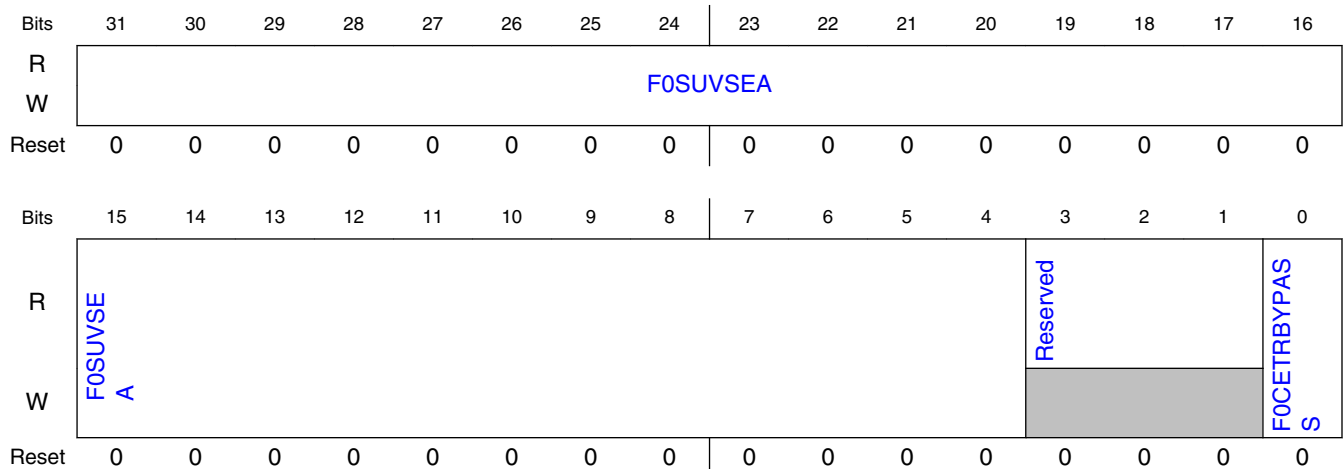
15.6.6.1.10.1 Offset

Register	Offset
F0SUVSEA	20h

15.6.6.1.10.2 Function

This register specifies the end address space of slave interface chroma data.

15.6.6.1.10.3 Diagram



15.6.6.1.10.4 Fields

Field	Function
31-4 F0SUVSEA	Chroma data slave end address End of address space of slave interface chroma data. This address should be 16-byte aligned and unit is 16 bytes, exclusive.
3-1 —	Reserved.
0 F0CETRBYPASS	End Tile to Raster scan Bypass 0b - All ARADDR does NOT bypass the tile-to-rasterscan logic. 1b - All ARADDR bypasses the tile-to-rasterscan logic.

15.6.6.1.11 Cropped picture origin (F0CROPORIG)

15.6.6.1.11.1 Offset

Register	Offset
F0CROPORIG	24h

15.6.6.1.11.2 Function

This register specifies the origin of the cropped picture.

15.6.6.1.11.3 Diagram

Bits	31	30	29	28	27	26	25	24		23	22	21	20	19	18	17	16
R	Reserved			F0CROPORIGY													
W																	
Reset	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8		7	6	5	4	3	2	1	0
R	Reserved			F0CROPORIGX													
W																	
Reset	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0

15.6.6.1.11.4 Fields

Field	Function
31-29	Reserved.

Table continues on the next page...

Field	Function
—	
28-16 F0CROPORIGY	Cropped picture y origin Indicates the y origin of the cropped picture in units of pixels. This number must be even. 0 points to the top left corner.
15-13 —	Reserved.
12-0 F0CROPORIGX	Cropped picture x origin Indicates the x origin of the cropped picture in units of pixels. This number must be even. 0 points to the top left corner.

15.6.6.1.12 Cropped picture size (F0CROPSIZE)

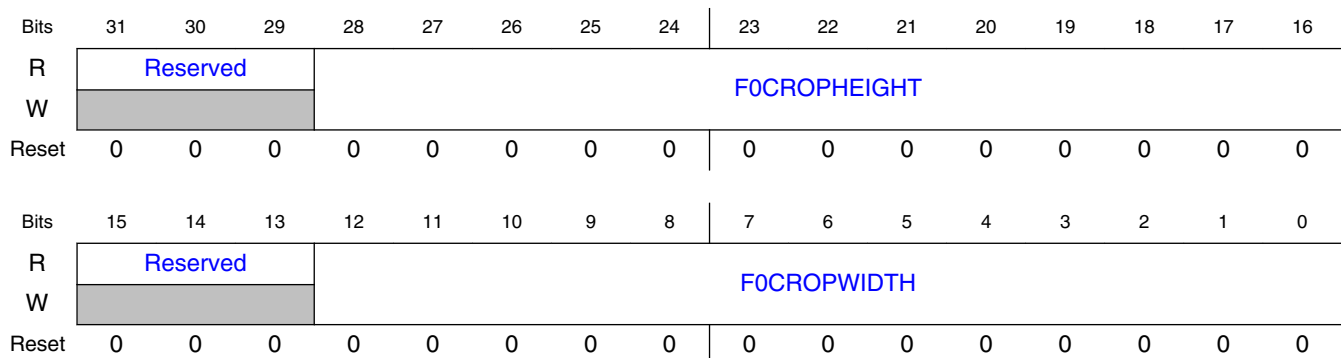
15.6.6.1.12.1 Offset

Register	Offset
F0CROPSIZE	28h

15.6.6.1.12.2 Function

This register specifies the size of the cropped picture.

15.6.6.1.12.3 Diagram



15.6.6.1.12.4 Fields

Field	Function
31-29	Reserved.

Table continues on the next page...

Decompression and Tile to Raster Conversion (DTRC)

Field	Function
—	
28-16 F0CROPHEIGHT	Cropped picture height Indicates the height of the cropped picture in units of pixels. This number must be even.
15-13 —	Reserved.
12-0 F0CROPWIDTH	Cropped picture width Indicates the width of the cropped picture in units of pixels. This number must be even.

15.6.6.1.13 Frame data control (F0DCTL)

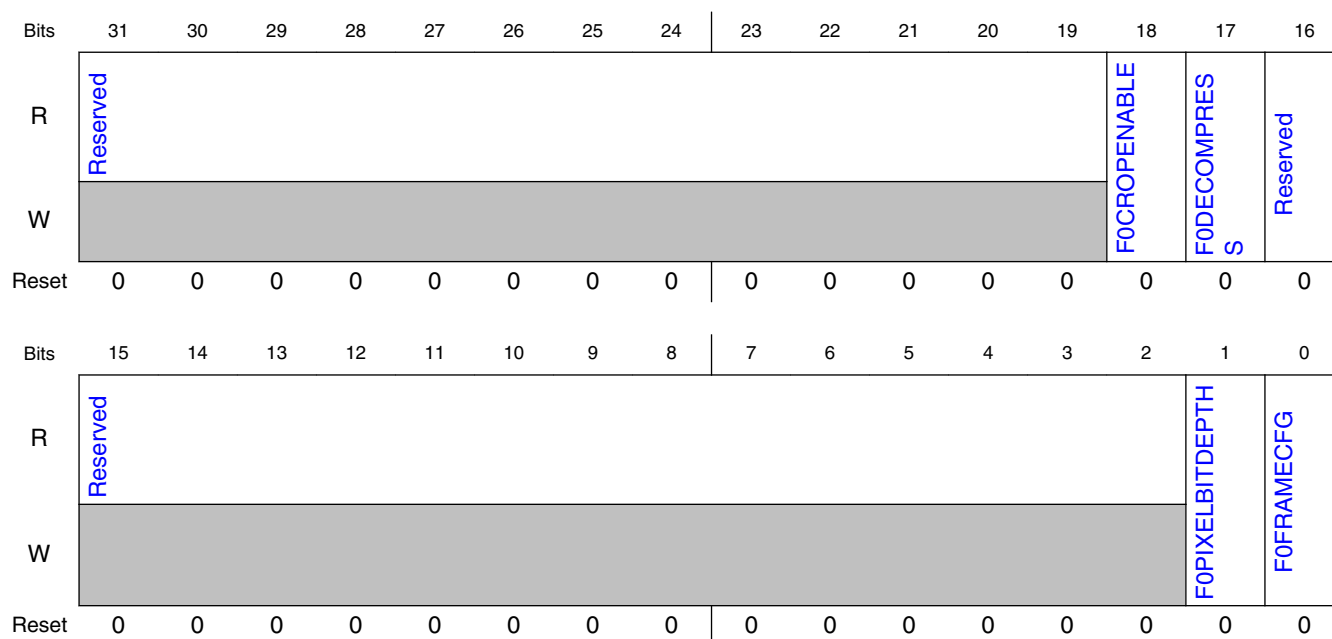
15.6.6.1.13.1 Offset

Register	Offset
F0DCTL	2Ch

15.6.6.1.13.2 Function

Hardware fetches data of Frame - first after reset.

15.6.6.1.13.3 Diagram



15.6.6.1.13.4 Fields

Field	Function
31-19 —	Reserved.
18 F0CROPENABLE	Cropped enable DTRC outputs the cropped picture data according to the settings of Frame0 Cropping Origin Register (F0CROPORIG) and Frame0 Cropping Size Register (F0CROPSIZE).
17 F0DECOMPRESS	Decompress bypass Indicates whether or not decompress is bypassed. For G1, this should always be set to 1. 0b - G2 reference frame is compressed. 1b - G2/G1 reference frame is not compressed.
16 —	Reserved.
15-2 —	Reserved.
1 F0PIXELBITDEPTH	Pixel bit depth Indicates pixel bit depth 0b - 10-bit pixel depth 1b - 8-bit pixel depth
0 F0FRAMECFG	Frame configuration ready Indicates whether or not the frame configuration is ready 0b - Frame 0 configuration is not ready. 1b - Frame 0 configuration is ready and decompress can start for the frame. All other configuration, such as main8/main10 are updated before setting this bit to 1. If there is no G1/G2 video transaction, set this bit to 0.

15.6.6.1.14 Luma video data start address (F1DYDSADDR)

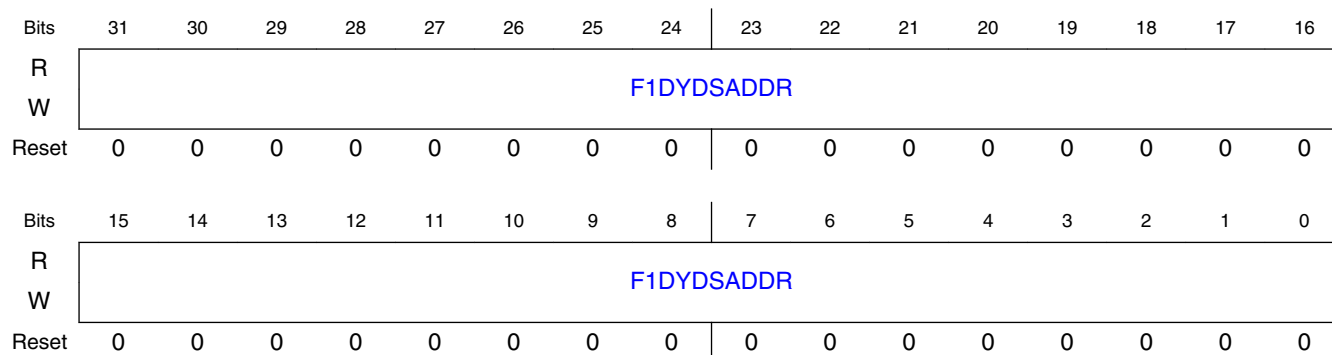
15.6.6.1.14.1 Offset

Register	Offset
F1DYDSADDR	60h

15.6.6.1.14.2 Function

This register specifies the start address for luma video data saved in external memory. This data is compressed or decompressed data.

15.6.6.1.14.3 Diagram



15.6.6.1.14.4 Fields

Field	Function
31-0	Luma video data start address
F1DYDSADDR	Start address for luma data saved in external memory. Unit is byte, but should be 16-byte aligned.

15.6.6.1.15 Chroma video data start address (F1DCDSADDR)

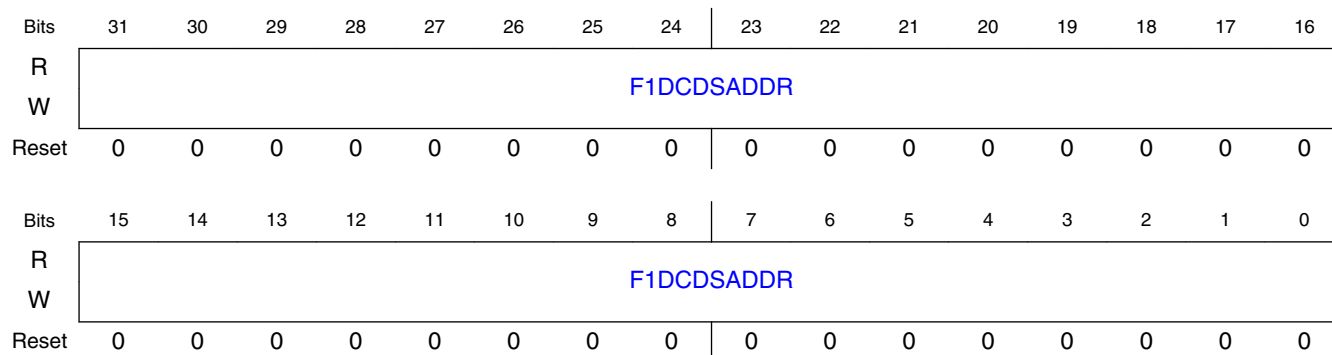
15.6.6.1.15.1 Offset

Register	Offset
F1DCDSADDR	64h

15.6.6.1.15.2 Function

This register specifies the start address for chroma video data saved in external memory.

15.6.6.1.15.3 Diagram



15.6.6.1.15.4 Fields

Field	Function
31-0 F1DCDSADDR	Chroma video data start address Start address for chroma decompression data saved in external memory. Unit is byte, but should be 16-byte aligned.

15.6.6.1.16 Luma table data start address (F1DYTSADDR)

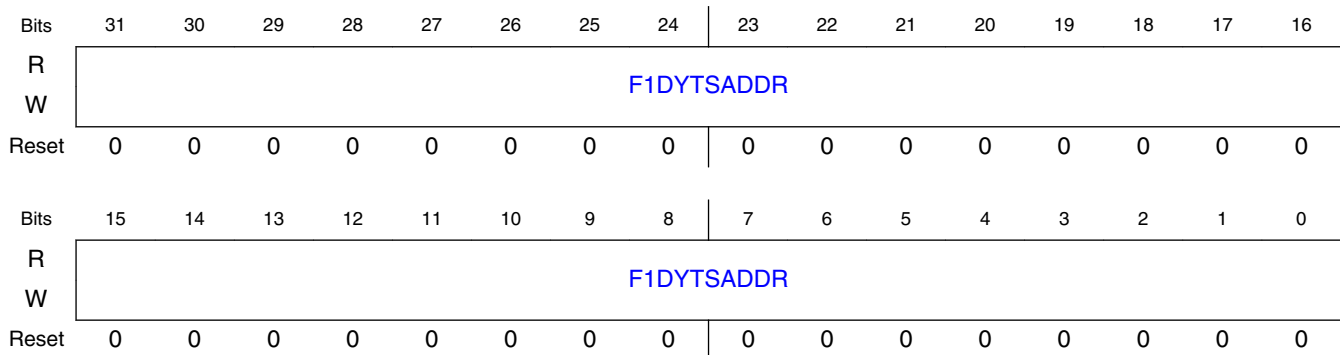
15.6.6.1.16.1 Offset

Register	Offset
F1DYTSADDR	68h

15.6.6.1.16.2 Function

This register specifies the start address for luma table data saved in external memory. Not used when compression is disabled in G2.

15.6.6.1.16.3 Diagram



15.6.6.1.16.4 Fields

Field	Function
31-0 F1DYTSADDR	Luma table data start address Start address for luma table data saved in external memory. Unit is byte and 16-byte aligned.

15.6.6.1.17 Chroma table data start address (F1DCTSADDR)

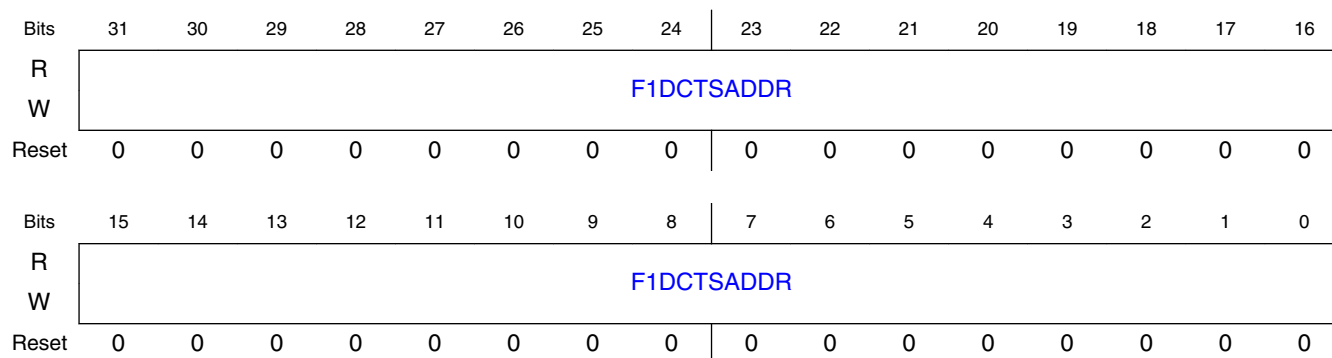
15.6.6.1.17.1 Offset

Register	Offset
F1DCTSADDR	6Ch

15.6.6.1.17.2 Function

This register specifies the start address for chroma table data saved in external memory. Not used when compression is disabled in G2.

15.6.6.1.17.3 Diagram



15.6.6.1.17.4 Fields

Field	Function
31-0	Chroma table data start address
F1DCTSADDR	Start address for chroma table data saved in external memory. Unit is byte and 16-byte aligned.

15.6.6.1.18 Frame size (F1SIZE)

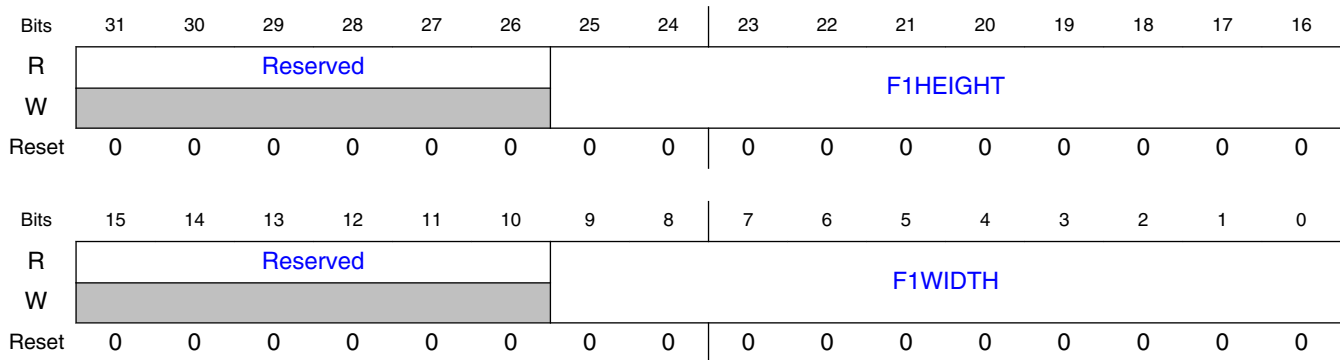
15.6.6.1.18.1 Offset

Register	Offset
F1SIZE	70h

15.6.6.1.18.2 Function

This register specifies frame width and frame height in units of 8.

15.6.6.1.18.3 Diagram



15.6.6.1.18.4 Fields

Field	Function
31-26 —	Reserved.
25-16 F1HEIGHT	Frame height Indicates frame height in units of 8 lines.
15-10 —	Reserved.
9-0 F1WIDTH	Frame width Indicates frame width in units of 8 pixels.

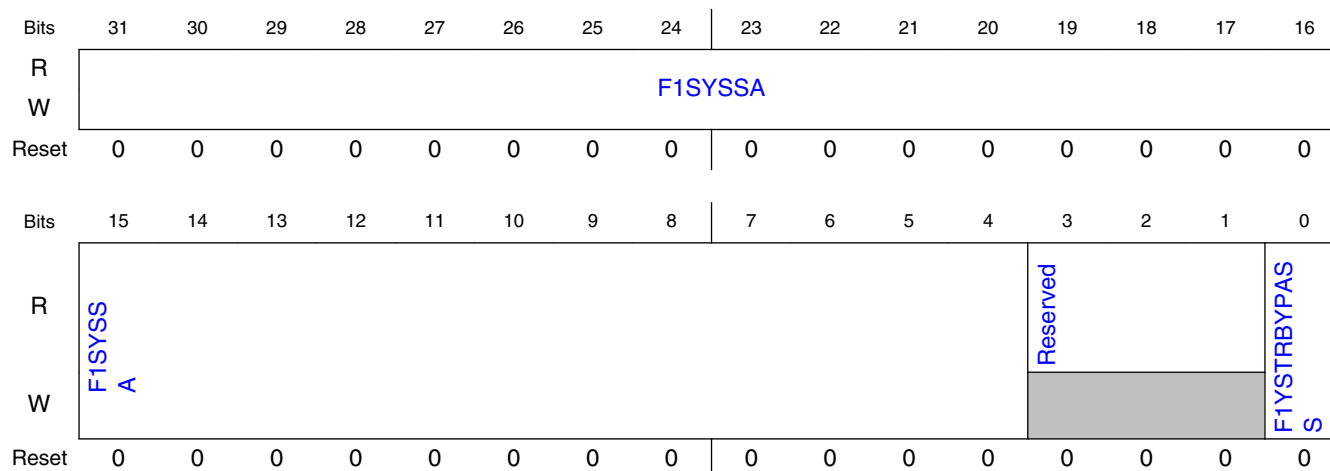
15.6.6.1.19 Luma data slave start address (F1SYSSA)

15.6.6.1.19.1 Offset

Register	Offset
F1SYSSA	74h

15.6.6.1.19.2 Function

This register specifies the start address range of the luma data associated to frame 0 and appears at ARADDR_S. Only used when G1/G2 transactions are told by ARADDR_S. It has nothing to do with F1DYDSADDR.

15.6.6.1.19.3 Diagram**15.6.6.1.19.4 Fields**

Field	Function
31-4 F1SYSSA	Luma data slave start address Start of address space of slave interface luma data. This address should be 16-byte aligned and unit is 16 bytes, inclusive.
3-1 —	Reserved.
0 F1YSTRBYPAS S	Luma Start Tile to Raster scan Bypass 0b - All ARADDR does NOT bypass the tile-to-rasterscan logic. 1b - All ARADDR bypasses the tile-to-rasterscan logic.

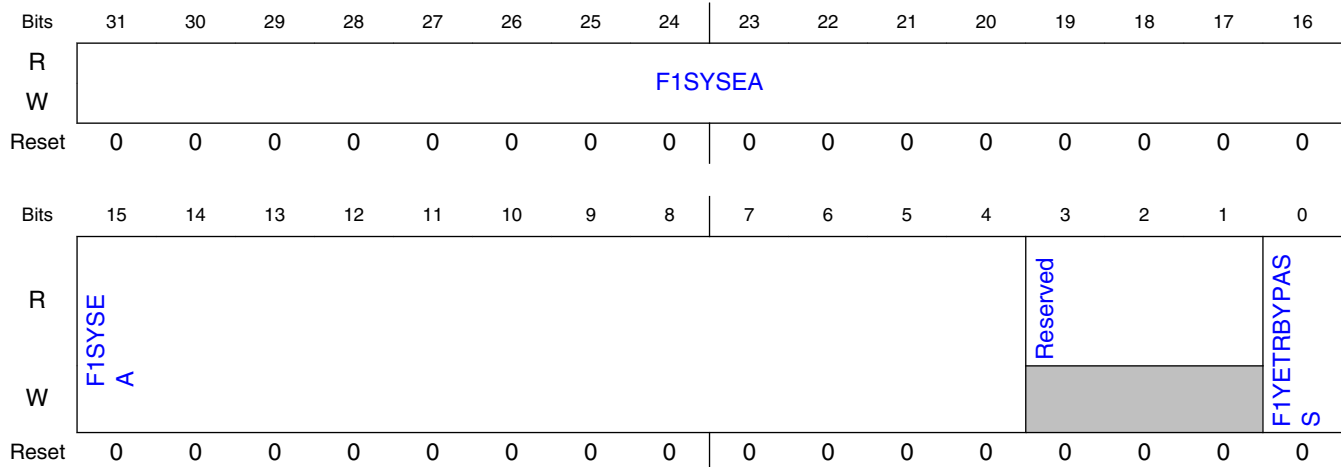
15.6.6.1.20 Luma data slave end address (F1SYSEA)**15.6.6.1.20.1 Offset**

Register	Offset
F1SYSEA	78h

15.6.6.1.20.2 Function

This register specifies the end address space of luma data.

15.6.6.1.20.3 Diagram



15.6.6.1.20.4 Fields

Field	Function
31-4 F1SYSEA	Luma data slave end address End of address space of slave interface luma data. This address should be 16-byte aligned and unit is 16 bytes, exclusive.
3-1 —	Reserved.
0 F1YETRBYPAS S	End Tile to Raster scan Bypass 0b - All ARADDR does NOT bypass the tile-to-rasterscan logic. 1b - All ARADDR bypasses the tile-to-rasterscan logic.

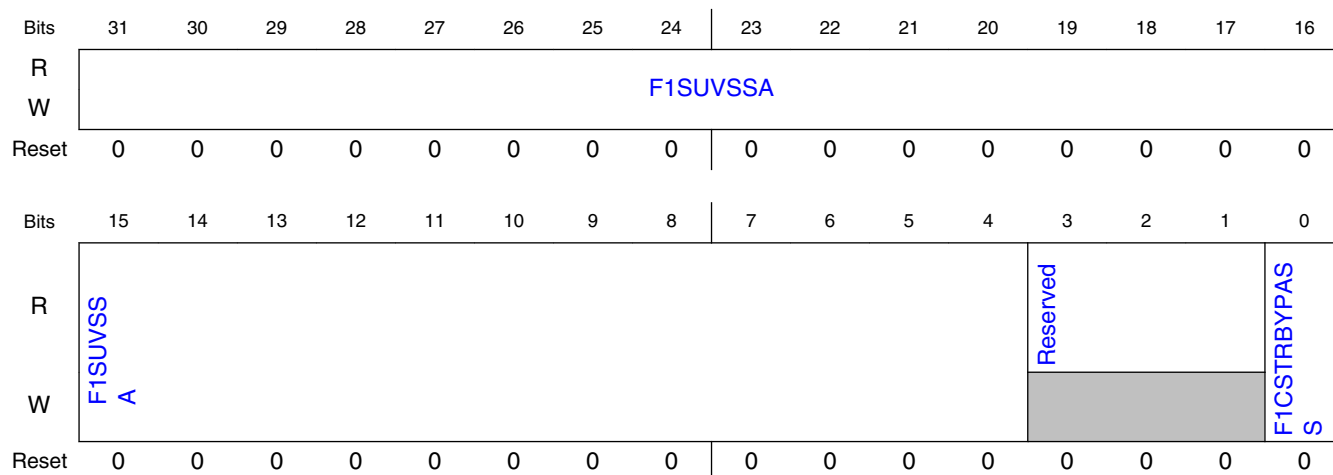
15.6.6.1.21 Chroma data slave start address (F1SUVSSA)

15.6.6.1.21.1 Offset

Register	Offset
F1SUVSSA	7Ch

15.6.6.1.21.2 Function

This register specifies the start address of the address range of the chroma data associated to frame 0 and appears at ARADDR_S. Only used when G2/G1 and non G2/G1 transactions are told by ARADDR_S. It has nothing to do with F1DCDSADDR.

15.6.6.1.21.3 Diagram**15.6.6.1.21.4 Fields**

Field	Function
31-4 F1SUVSSA	Chroma data slave start address Start of address space of slave interface chroma data. This address should be 16-byte aligned and unit is 16 bytes, inclusive.
3-1 —	Reserved.
0 F1CSTRBYPAS S	Chroma Start Tile to Raster scan Bypass 0b - All ARADDR does NOT bypass the tile-to-rasterscan logic. 1b - All ARADDR bypasses the tile-to-rasterscan logic.

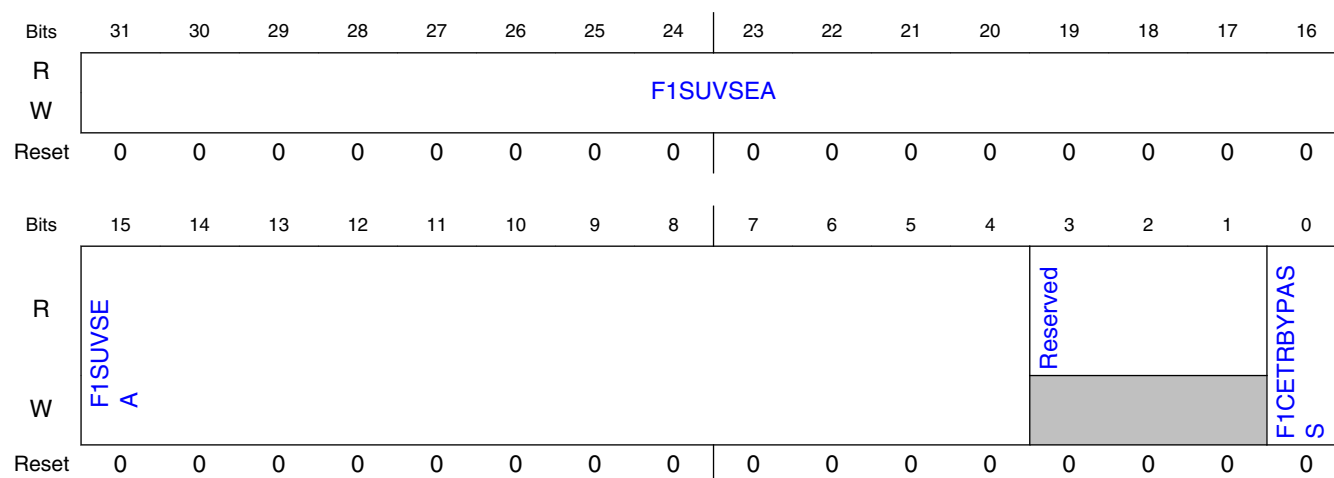
15.6.6.1.22 Chroma data slave end address (F1SUVSEA)**15.6.6.1.22.1 Offset**

Register	Offset
F1SUVSEA	80h

15.6.6.1.22.2 Function

This register specifies the end address space of slave interface chroma data.

15.6.6.1.22.3 Diagram



15.6.6.1.22.4 Fields

Field	Function
31-4 F1SUVSEA	Chroma data slave end address End of address space of slave interface chroma data. This address should be 16-byte aligned and unit is 16 bytes, exclusive.
3-1 —	Reserved.
0 F1CETRBYPASS	End Tile to Raster scan Bypass 0b - All ARADDR does NOT bypass the tile-to-rasterscan logic. 1b - All ARADDR bypasses the tile-to-rasterscan logic.

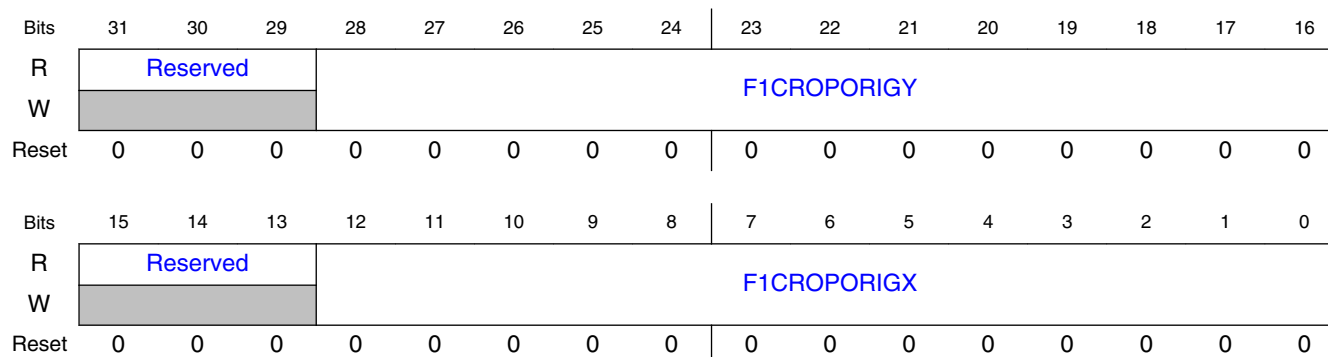
15.6.6.1.23 Cropped picture origin (F1CROPORIG)

15.6.6.1.23.1 Offset

Register	Offset
F1CROPORIG	84h

15.6.6.1.23.2 Function

This register specifies the origin of the cropped picture.

15.6.6.1.23.3 Diagram**15.6.6.1.23.4 Fields**

Field	Function
31-29 —	Reserved.
28-16 F1CROPORIGY	Cropped picture y origin Indicates the y origin of the cropped picture in units of pixels. This number must be even. 0 points to the top left corner.
15-13 —	Reserved.
12-0 F1CROPORIGX	Cropped picture x origin Indicates the x origin of the cropped picture in units of pixels. This number must be even. 0 points to the top left corner.

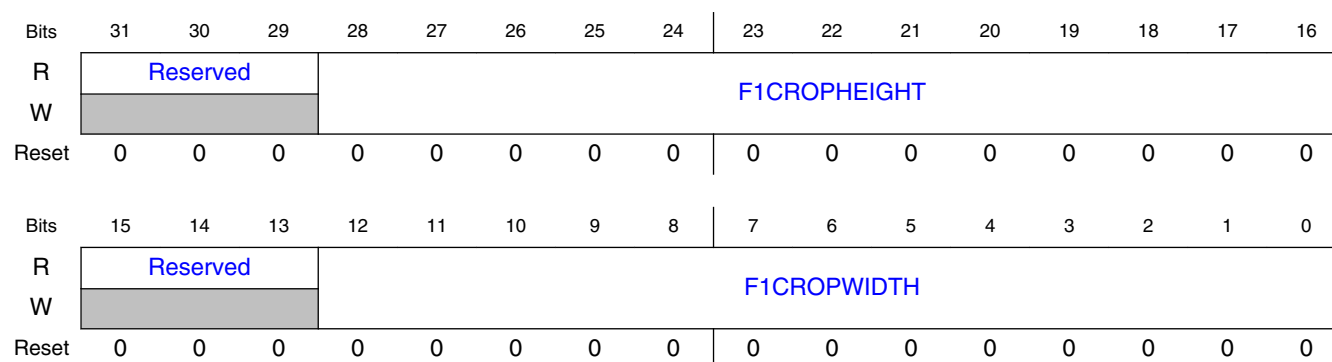
15.6.6.1.24 Cropped picture size (F1CROPSIZE)**15.6.6.1.24.1 Offset**

Register	Offset
F1CROPSIZE	88h

15.6.6.1.24.2 Function

This register specifies the size of the cropped picture.

15.6.6.1.24.3 Diagram



15.6.6.1.24.4 Fields

Field	Function
31-29 —	Reserved.
28-16 F1CROPHEIGHT	Cropped picture height Indicates the height of the cropped picture in units of pixels. This number must be even.
15-13 —	Reserved.
12-0 F1CROPWIDTH	Cropped picture width Indicates the width of the cropped picture in units of pixels. This number must be even.

15.6.6.1.25 Frame data control (F1DCTL)

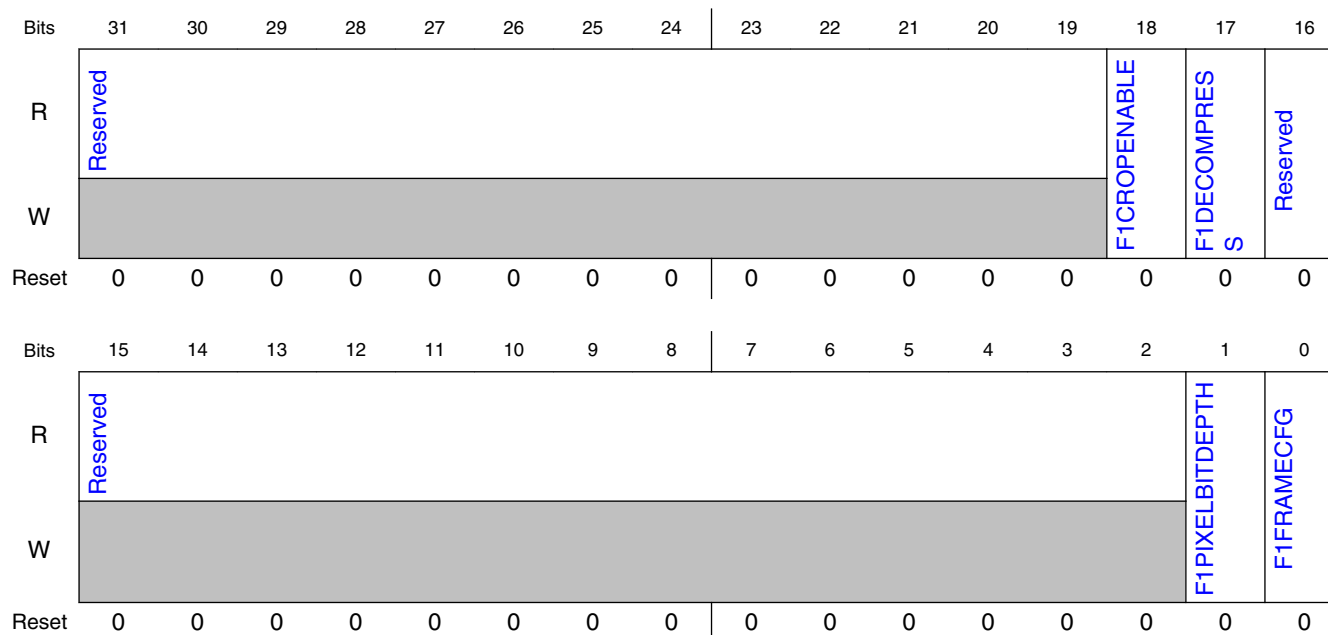
15.6.6.1.25.1 Offset

Register	Offset
F1DCTL	8Ch

15.6.6.1.25.2 Function

Hardware fetches data of Frame - first after reset.

15.6.6.1.25.3 Diagram



15.6.6.1.25.4 Fields

Field	Function
31-19 —	Reserved.
18 F1CROPENABLE	Cropped enable DTRC outputs the cropped picture data according to the settings of Frame0 Cropping Origin Register (F1CROPORIG) and Frame0 Cropping Size Register (F1CROPSIZE).
17 F1DECOMPRESS	Decompress bypass Indicates whether or not decompress is bypassed. For G1, this should always be set to 1. 0b - G2 reference frame is compressed. 1b - G2/G1 reference frame is not compressed.
16 —	Reserved.
15-2 —	Reserved.
1 F1PIXELBITDEPTH	Pixel bit depth Indicates pixel bit depth 0b - 10-bit pixel depth 1b - 8-bit pixel depth
0 F1FRAMECFG	Frame configuration ready Indicates whether or not the frame configuration is ready 0b - Frame 0 configuration is not ready.

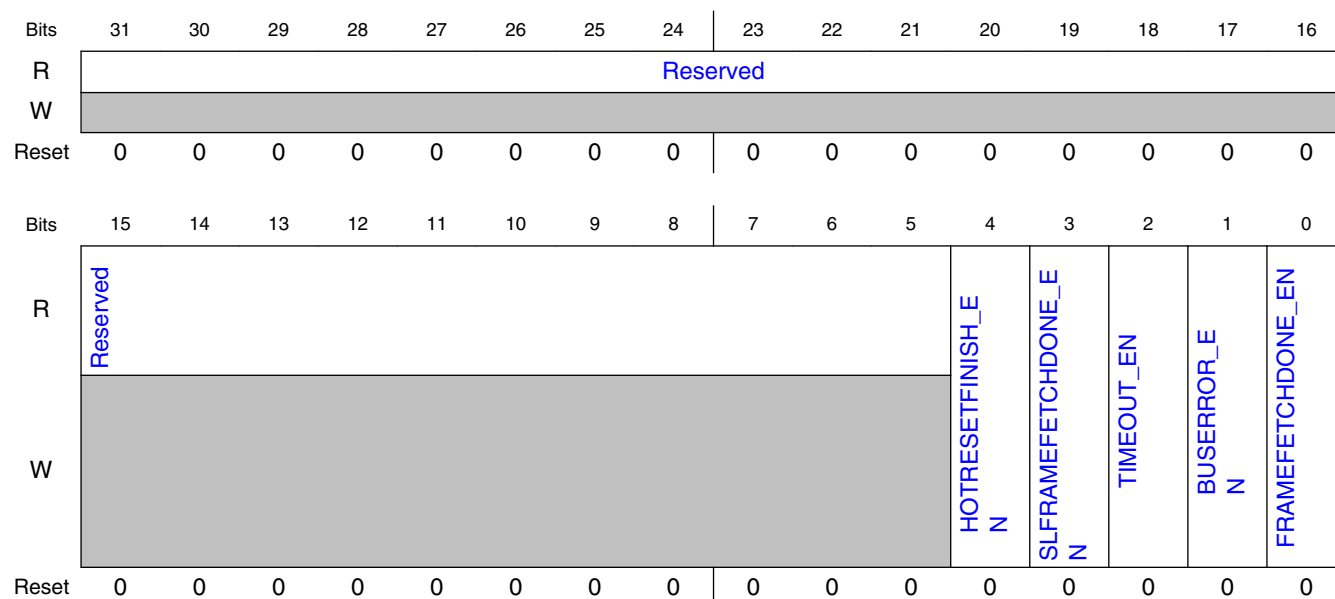
Field	Function
	1b - Frame 0 configuration is ready and decompress can start for the frame. All other configuration, such as main8/main10 are updated before setting this bit to 1. If there is no G1/G2 video transaction, set this bit to 0.

15.6.6.1.26 DTRC Interrupt enables (DTRCINTEN)

15.6.6.1.26.1 Offset

Register	Offset
DTRCINTEN	C0h

15.6.6.1.26.2 Diagram



15.6.6.1.26.3 Fields

Field	Function
31-5 —	Reserved.
4 HOTRESETFINISH_EN	Hot reset finish 0b - Hot reset finish disabled. 1b - Hot reset finish enabled.

Table continues on the next page...

Decompression and Tile to Raster Conversion (DTRC)

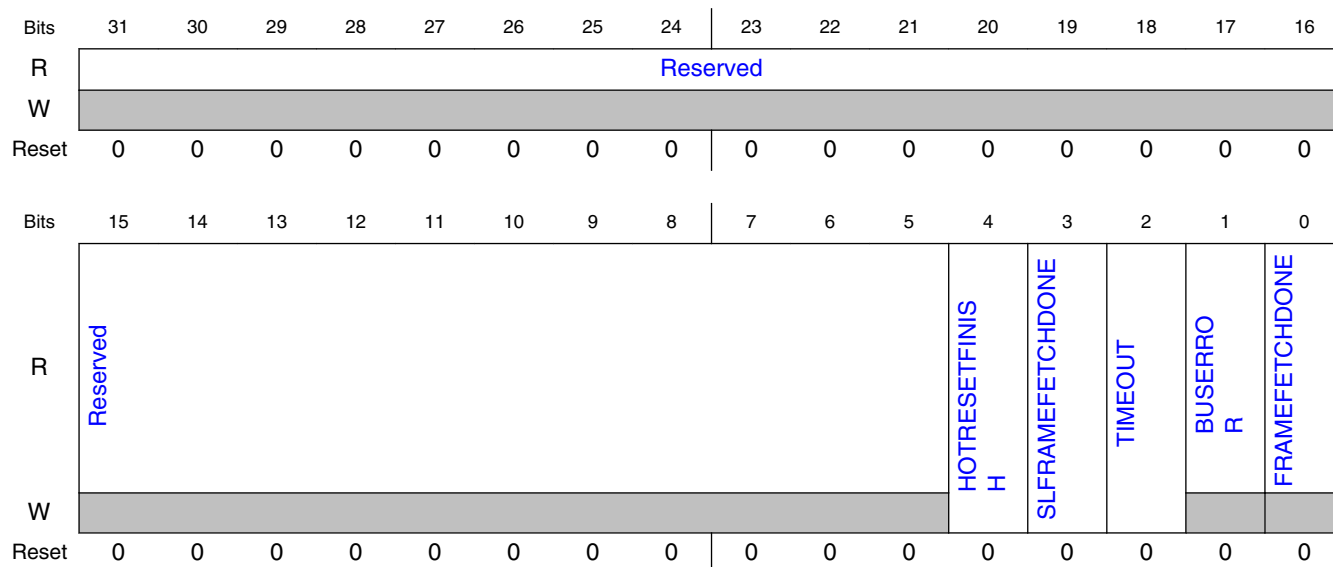
Field	Function
3 SLFRAMEFETCH HDONE_EN	Slave frame fetch done 0b - Slave frame fetch done disabled. 1b - Slave frame fetch done enabled.
2 TIMEOUT_EN	Time out enable 0b - Time out disabled. 1b - Time out enabled.
1 BUSERROR_EN	Bus error interrupt enable 0b - Bus error interrupt disabled. 1b - Bus error interrupt enabled.
0 FRAMEFETCH DONE_EN	Frame fetch done interrupt enable 0b - Frame fetch done interrupt disabled. 1b - Frame fetch done interrupt enabled.

15.6.6.1.27 DTRC Interrupt Requests (FDINTR)

15.6.6.1.27.1 Offset

Register	Offset
FDINTR	C4h

15.6.6.1.27.2 Diagram



15.6.6.1.27.3 Fields

Field	Function
31-5 —	Reserved.
4 HOTRESETFINI SH	Hot reset finish interrupt Write 1 clear.
3 SLFRAMEFETC HDONE	Slave frame fetch done interrupt Set by hardware when the down-stream IP finishes fetching the data of one frame from DTRC. Software usually ignore this interrupt. Write 1 clear.
2 TIMEOUT	Time out interrupt Set by hardware if the value of PFCR is larger than TOCR. It is cleared after software writes 1'b1.
1 BUSERROR	Bus error interrupt Set by hardware, and cleared after software writes 1'b1
0 FRAMEFETCH DONE	Frame fetch done interrupt Set by hardware and cleared after software writes 1'b1. Software watches this interrupt to configure frame parameters to DTRC.

15.6.6.1.28 DTRC Control (DTCTRL)

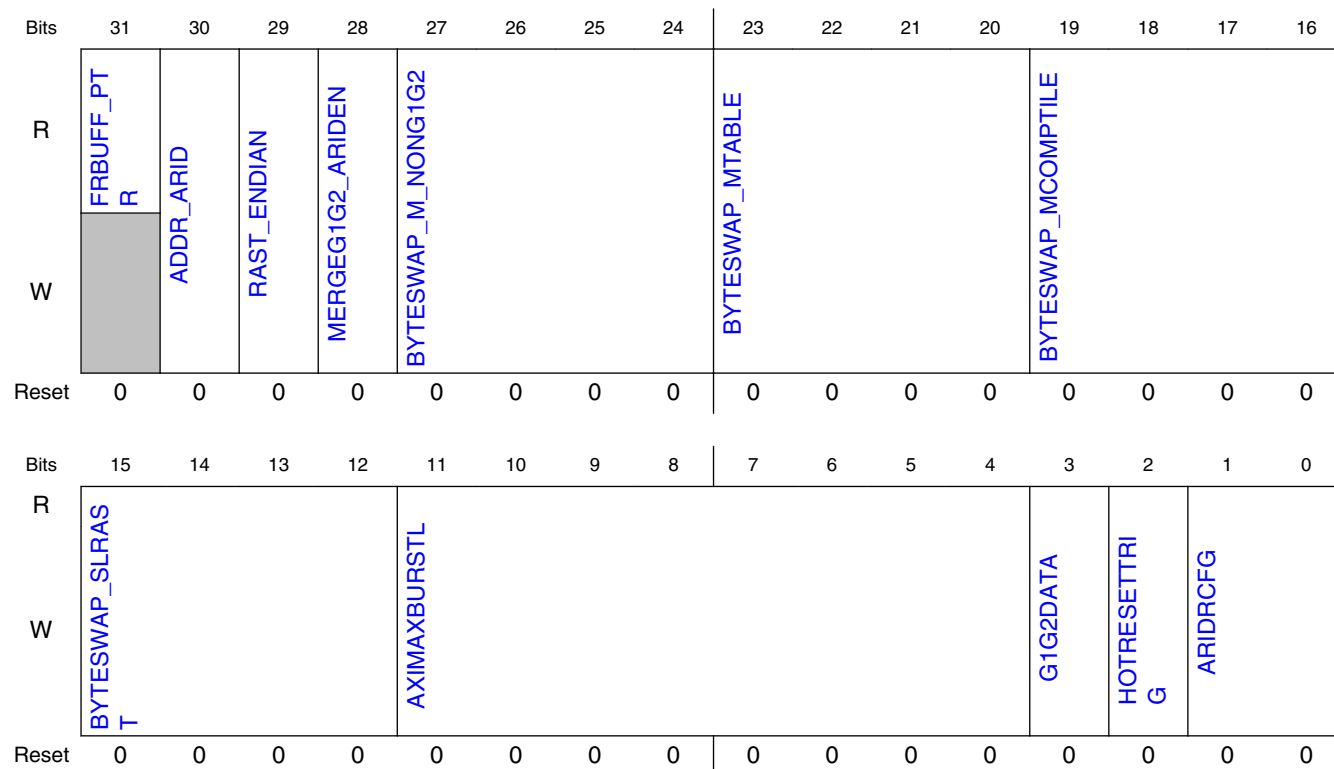
15.6.6.1.28.1 Offset

Register	Offset
DTCTRL	C8h

15.6.6.1.28.2 Function

This register specifies some global control information.

15.6.6.1.28.3 Diagram



15.6.6.1.28.4 Fields

Field	Function
31 FRBUFF_PTR	FRBUFF_PTR Frame buffer configure pointer, to indicate current pointer (Frme0 or Frame1) need to configure. 0b - Configure frame 0 registers. 1b - Configure frame 1 registers.
30 ADDR_ARID	ADDR_ARID Bypass tile-to-rasterscan by Address or AXI ID 0b - By ARID (See bit[1:0] of this register). 1b - By ARADDR
29 RAST_ENDIAN	Raster endian mode This bit indicates the endian mode of DTRC 10-bit raster scan output data. 0b - 10-bit output format is little-endian. Byte swap setting of DTCTRL[15:12] is used. 1b - 10-bit output format is big-endian. Byte swap setting of DTCTRL[15:12] is ignored.
28 MERGE1G2_ARIDEN	Merge G2/G1 ARID enable 0b - G2/G1 transactions at AXI master interface use different id for table/chroma and data/luma according to DTID2DDR definition. 1b - All G2/G1 transactions at AXI master interface use the same id configured in DTID2DDR[15:8] Please note that DTID2DDR[15:8] and DTID2DDR[7:0] still need to be set the same way as when DTCTRL[28] is 0.
27-24	Byte swap mode for master interface non-G1/G2 data

Table continues on the next page...

Field	Function
BYTESWAP_M_NONG1G2	• bit 4: 8 bit swap • bit 5: 16 bit swap • bit 6: 32 bit swap • bit 7: 64 bit swap • 4'b0000: 0-1-2-3-4-5-6-7-8-9-10-11-12-13-14-15 (the original bit sequence) • 4'b0001: 1-0-3-2-5-4-7-6... • 4'b0010: 2-3-0-1-6-7-4-5... • 4'b0011: 3-2-1-0-7-6-5-4... • 4'b0100: 4-5-6-7-0-1-2-3... • ... • 4'b0111: 7-6-5-4-3-2-1-0... • 4'b1000: 8-9-10-11-12-13-14-15-0-1-2-3-4-5-6-7... • ... • 4'b1111: 15-14-13-12-11-10-9-8-7-6-5-4-3-2-1-0
23-20 BYTESWAP_M TABLE	Byte swap mode for master interface table data • bit 4: 8 bit swap • bit 5: 16 bit swap • bit 6: 32 bit swap • bit 7: 64 bit swap • 4'b0000: 0-1-2-3-4-5-6-7-8-9-10-11-12-13-14-15 (the original bit sequence) • 4'b0001: 1-0-3-2-5-4-7-6... • 4'b0010: 2-3-0-1-6-7-4-5... • 4'b0011: 3-2-1-0-7-6-5-4... • 4'b0100: 4-5-6-7-0-1-2-3... • ... • 4'b0111: 7-6-5-4-3-2-1-0... • 4'b1000: 8-9-10-11-12-13-14-15-0-1-2-3-4-5-6-7... • ... • 4'b1111: 15-14-13-12-11-10-9-8-7-6-5-4-3-2-1-0
19-16 BYTESWAP_M COMPTILE	Byte swap mode for master interface compressed data and tiled data • bit 4: 8 bit swap • bit 5: 16 bit swap • bit 6: 32 bit swap • bit 7: 64 bit swap • 4'b0000: 0-1-2-3-4-5-6-7-8-9-10-11-12-13-14-15 (the original bit sequence) • 4'b0001: 1-0-3-2-5-4-7-6... • 4'b0010: 2-3-0-1-6-7-4-5... • 4'b0011: 3-2-1-0-7-6-5-4... • 4'b0100: 4-5-6-7-0-1-2-3... • ... • 4'b0111: 7-6-5-4-3-2-1-0... • 4'b1000: 8-9-10-11-12-13-14-15-0-1-2-3-4-5-6-7... • ... • 4'b1111: 15-14-13-12-11-10-9-8-7-6-5-4-3-2-1-0
15-12 BYTESWAP_SL RAST	Byte swap mode for slave interface raster scan data • bit 4: 8 bit swap • bit 5: 16 bit swap • bit 6: 32 bit swap • bit 7: 64 bit swap • 4'b0000: 0-1-2-3-4-5-6-7-8-9-10-11-12-13-14-15 (the original bit sequence) • 4'b0001: 1-0-3-2-5-4-7-6...

Table continues on the next page...

Decompression and Tile to Raster Conversion (DTRC)

Field	Function
	• 4'b0010: 2-3-0-1-6-7-4-5... • 4'b0011: 3-2-1-0-7-6-5-4... • 4'b0100: 4-5-6-7-0-1-2-3... • ... • 4'b0111: 7-6-5-4-3-2-1-0... • 4'b1000: 8-9-10-11-12-13-14-15-0-1-2-3-4-5-6-7... • ... • 4'b1111: 15-14-13-12-11-10-9-8-7-6-5-4-3-2-1-0
11-4 AXIMAXBURST L	Maximum burst length of AXI master port If maximum burst length feature is supported (see DTRCONFIG[0]), DTRC sends the G1/G2 relating requests with ARLEN less than or equal to (the maximum burst length - 1).
3 G1G2DATA	G2 or G1 source data When this bit is set, F0DCTL[17] and F1DCTL[17] must be 1'b1. DTRC converts G1 tile data to raster scan data. 0b - The source data is G2 data. 1b - The source data is G1 tile data
2 HOTRESETTRIG	Hot reset trigger Software writes 1 to initiate a hot reset of video decompression and detile path. Hardware clears the bit to 0 after the reset finished.
1-0 ARIDRCFG	ARIDR configuration Specify ARIDR mode. 00b - All ARID is de-tiled. 01b - ARID in ARIDR is de-tiled, and others are bypass. NOTE: ARID[0] specify decode luma or chroma, so 4 ARIDs in ARIDR should include both of ARID[0] ==1 and ARID[0] ==0. 10b - ARID in ARIDR is bypass_de-tile, and others are de-tiled. 11b - Same as 2'b01.

15.6.6.1.29 ARIDR (ARIDR)

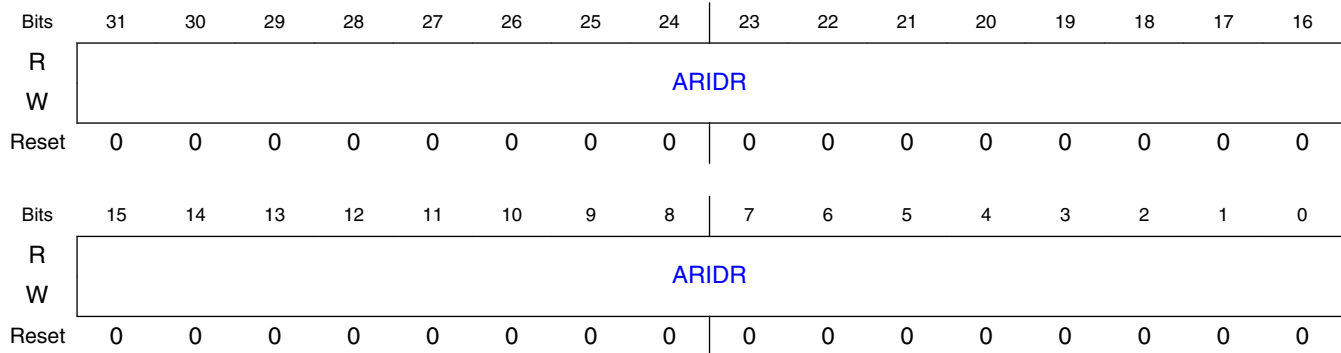
15.6.6.1.29.1 Offset

Register	Offset
ARIDR	CCh

15.6.6.1.29.2 Function

This register specifies ARID to use to fetch de-tile data or bypass de-tile data.

15.6.6.1.29.3 Diagram



15.6.6.1.29.4 Fields

Field	Function
31-0	ARIDR
ARIDR	4 ARID, each is 8 bits.

15.6.6.1.30 DTID2DDR (DTID2DDR)

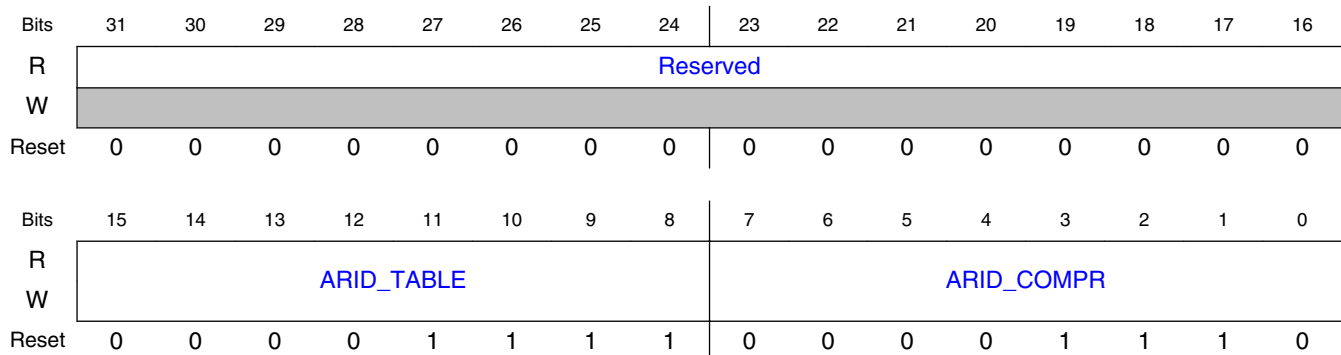
15.6.6.1.30.1 Offset

Register	Offset
DTID2DDR	D0h

15.6.6.1.30.2 Function

This register specifies ARID to use to fetch video data from DTRC to external memory.

15.6.6.1.30.3 Diagram



15.6.6.1.30.4 Fields

Field	Function
31-16 —	Reserved.
15-8 ARID_TABLE	ARID_TABLE If G2 frame is compressed, this is ARID of table fetching, from DTRC to external memory. If G2 frame is not compressed or G1 tile is the case, this is ARID of chroma data The value should be different from non-G2/G1 transaction ARID received from DTRC AXI slave port, and different from bits 7:0 of this register.
7-0 ARID_COMPR	ARID_COMPR If G2 frame is compressed, this is ARID of compressed data fetching, from DTRC to external memory. If G2 frame is not compressed or G1 tile is the case, this is ARID of luma data. The value should be different from non-G2/G1 transaction ARID received from DTRC AXI slave port, and different from bits 15:8 of this register.

15.6.6.1.31 DTRCCONFIG (DTRCCONFIG)

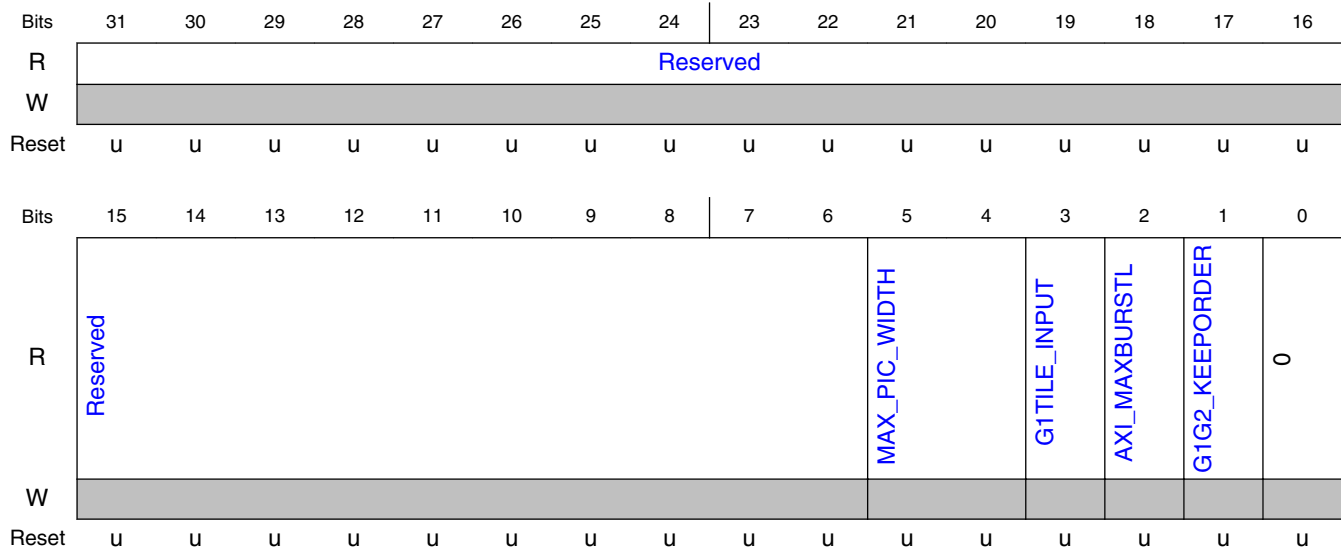
15.6.6.1.31.1 Offset

Register	Offset
DTRCCONFIG	D4h

15.6.6.1.31.2 Function

This register specifies ARID to use to fetch video data from DTRC to external memory.

15.6.6.1.31.3 Diagram



15.6.6.1.31.4 Fields

Field	Function
31-6 —	Reserved.
5-4 MAX_PIC_WIDTH	MAX_PIC_WIDTH Maximum supported picture width 00b - 4096 01b - 1920 10b - Reserved. 11b - Reserved.
3 G1TILE_INPUT	G1TILE_INPUT G1 tile input 0b - Not supported. 1b - Supported.
2 AXI_MAXBURSTL	AXI_MAXBURSTL Maximum burst length of AXI master port. 0b - Not supported. 1b - Supported. DTRC sends the G1/G2 relating requests with ARLEN <= (the maximum burst length - 1). The maximum burst length is set by DTCTRL[16:8].
1 G1G2_KEEPPORDER	G1G2_KEEPPORDER Keep the order between g1/g2 transactions and non-g1/g2 transactions at AXI slave interface. 0b - Not supported. DTRC sends out data transactions as soon as they are ready regardless of the address transactions order, The master connected to AXI slave interface must recognize the data transactions by the RID. 1b - Supported, DTRC ensure return read data from a sequence of read transactions in the same order in which it received the address.
0 —	Reserved.

15.6.6.1.32 DTRC Version (DTRCVERSION)

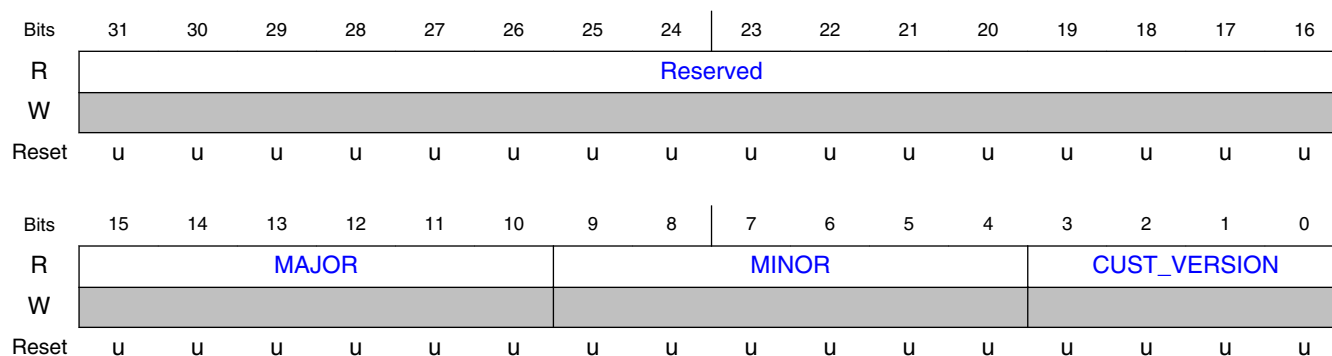
15.6.6.1.32.1 Offset

Register	Offset
DTRCVERSION	D8h

15.6.6.1.32.2 Function

This is a constant read-only register that contains encoded information about the hardware version.

15.6.6.1.32.3 Diagram



15.6.6.1.32.4 Fields

Field	Function
31-16 —	Reserved.
15-10 MAJOR	MAJOR Major version. 0x00: Version before 0831, including 0831. Support G2 tile + G2 compressed + raster scan output 0x01: Support G1 tile + G2 tile + G2 compressed + raster scan output. - All other values: Reserved.
9-4 MINOR	MINOR Minor version. Simply an update count (update at a release time) of a certain Major version. Count from 0.
3-0 CUST_VERSION	CUST_VERSION Customer Version.

15.6.6.1.33 Performance counter control (PFCTRL)

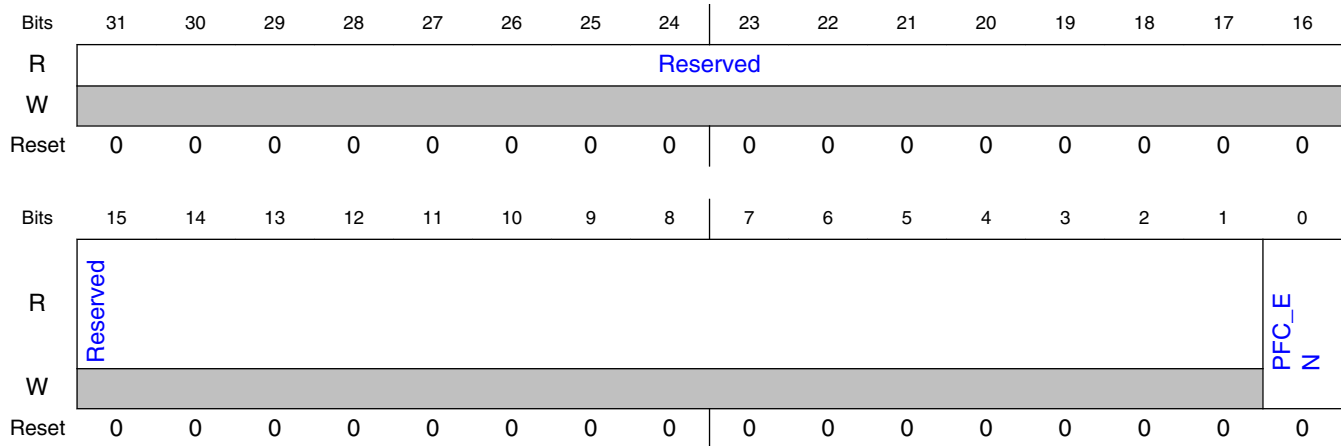
15.6.6.1.33.1 Offset

Register	Offset
PFCTRL	F0h

15.6.6.1.33.2 Function

This register enables or disables performance counter.

15.6.6.1.33.3 Diagram



15.6.6.1.33.4 Fields

Field	Function
31-1 —	Reserved.
0 PFC_EN	PFC_EN Performance Counter Enable. 0b - Performance Counter disabled. 1b - Performance Counter enabled.

15.6.6.1.34 Performance counter (PFCR)

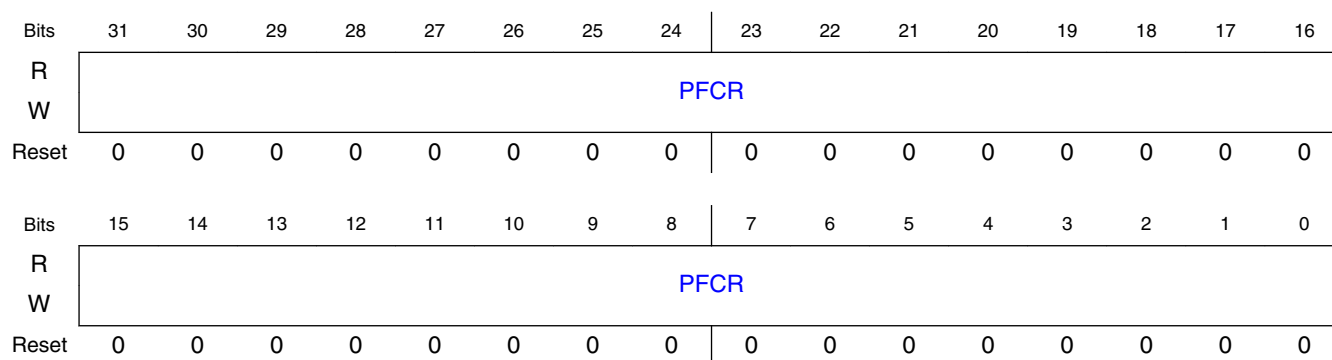
15.6.6.1.34.1 Offset

Register	Offset
PFCR	F4h

15.6.6.1.34.2 Function

This is performance counter to count total clock cycle from DTRC begins to send out a frame and finishes sending out the frame.

This counter belongs to decompression clock domain, not APB clock domain. The software can read this counter after received slave frame fetch done interrupt.

15.6.6.1.34.3 Diagram**15.6.6.1.34.4 Fields**

Field	Function
31-0 PFCR	Performance Counter

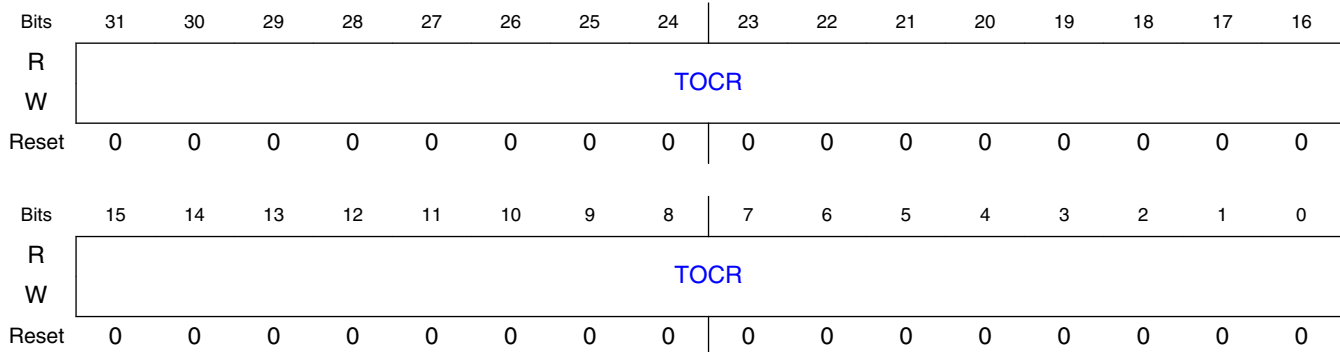
15.6.6.1.35 Time Out Cycles (TOCR)**15.6.6.1.35.1 Offset**

Register	Offset
TOCR	F8h

15.6.6.1.35.2 Function

This register sets the clock cycles (decompression clock domain) to wait before launching the time-out interrupt. When the value of performance counter register is larger than the value of this register, the time-out interrupt is set.

15.6.6.1.35.3 Diagram



15.6.6.1.35.4 Fields

Field	Function
31-0 TOCR	Time Out Cycles

15.7 Display, Prefetch and Resolve (DPR)

15.7.1 Overview

Display prefetch resolve, or DPR, is the process of fetching display data before the display pipeline needs the data to drive pixels in the active display region. This data is transformed, or resolved, from a variety of tiled buffer formats into linear format.

In a system with many bus masters, arbitration and transaction reordering, significant delays in the response time of returned read data can occur. Therefore, the prefetch process is required to issue transaction sequences which are processed with a high level of DRAM efficiency.

The resolve process is required to transform tile formatted frame buffers into linear buffers. Displays process data in linear scan line order. In the resolve process, local memory is used to store a small number of scan lines which the local display controller will source to refresh the panel. The DPR works with a double/triple bank memory structure. This memory structure is implemented in the RTRAM_CTRL and the banks are referred to as A, B, and optionally C. The DPR is configurable to work with either two or three banks, and each bank is either 4 or 8 lines high depending on the source frame buffer format. Zenverge VPU 2-Plane 420 source buffers use RTRAM banks that are 8 lines high. All other GPU, VP9 and VPU 1-Plane buffers use RTRAM banks that are 4 lines high.

15.7.1.1 Block Diagram

The following diagram gives a high-level overview of the configuration of the DPR.

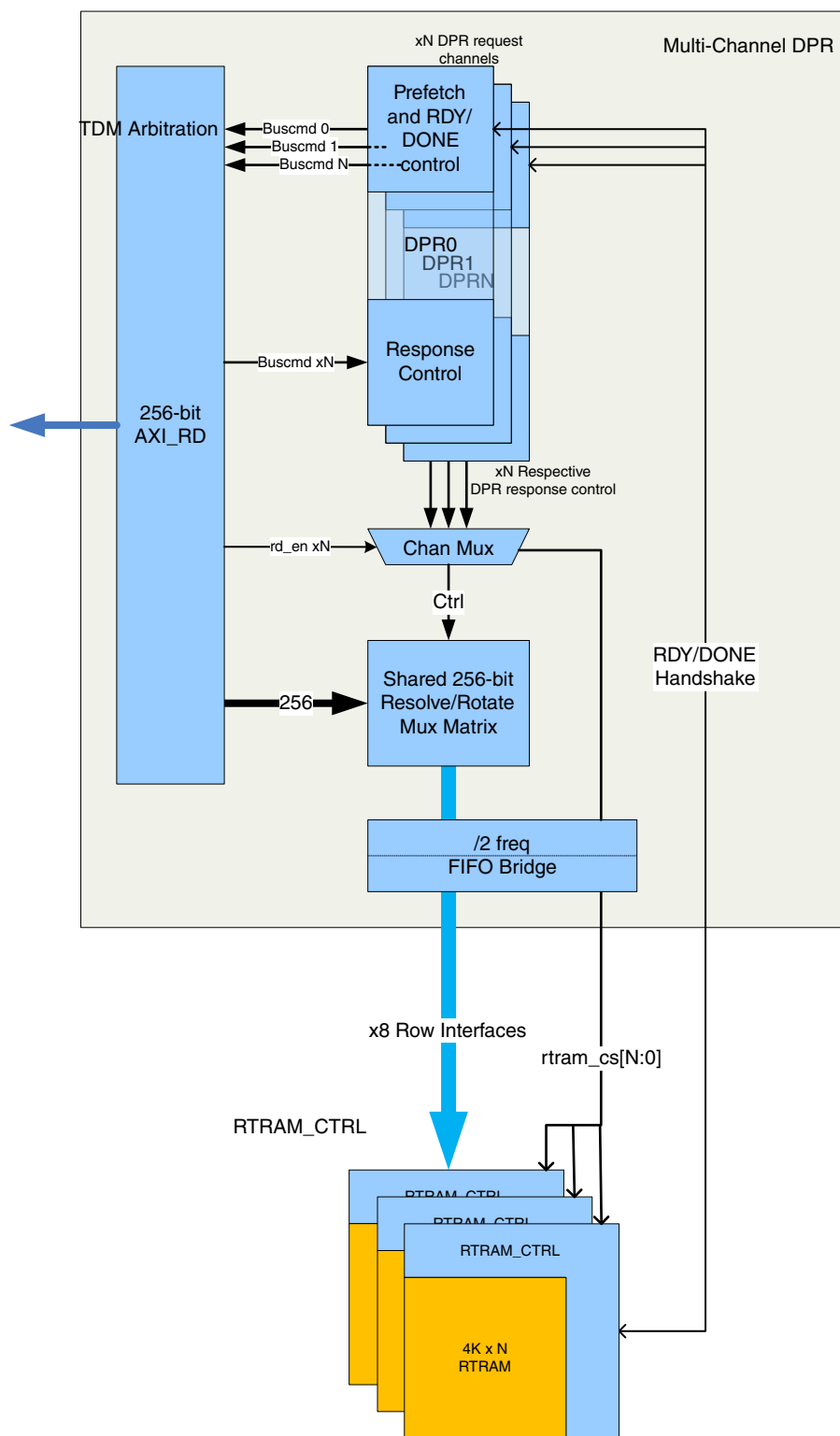


Figure 15-19. DPR block diagram

15.7.1.2 Features

The DPR module is capable of supporting the following features:

- Double/Triple (A,B and opt C) RTRAM buffers.
- Resolves and rotates at the bus width, 256-bit, without stalling the input bus.
- 32bpp ARGB pixel swapping
- 1-Plane YUV422 YUV pixel swapping
- 2-Plane 8bpp YUV420 UV pixel swapping
- RTRAM panic and stall underrun detection.

The supported prefetch and resolve modes are listed in the table below:

Mode	Linear or Tile	HFLIP	VFLIP	Rotation (0, 90, 180 or 270)	Lines per RTRAM Bank
GPU 32bpp xRGB	Linear	Yes	Yes	0 and 180 only	4
GPU 32bpp xRGB	X-Major Supertile and Standard tile	Yes	Yes	0, 90, 180 or 270	4
GPU 16bpp xRGB, 1-Plane YUV422	Linear	Yes	Yes	0 and 180 only	4
GPU 16bpp xRGB, 1-Plane YUV422	X-Major Supertile and Standard tile	Yes	Yes	0 and 180 only	4
VPU 8bpp 2-Plane YUV420	Linear	Yes	Yes	0 and 180 only	8
VPU 8bpp 2-Plane YUV420	Zenverge tile	Yes	Yes	0, 90, 180 or 270	8
VPU 10bpp 2-Plane YUV420	Linear	No	Yes	0 only	8
VPU 10bpp 2-Plane YUV420	Zenverge tile	No	Yes	0 only	8
VP9 8bpp 2-Plane YUV420	Linear	Yes	Yes	0 and 180 only	4
VP9 8bpp 2-Plane YUV420	Google tile	Yes	Yes	0 and 180 only	4
VP9 10bpp 2-Plane YUV420	Linear	No	Yes	0 only	4
VP9 10bpp 2-Plane YUV420	Google tile	No	Yes	0 only	4

For detailed GPU and VPU tile specifications, refer to the following application documents:

- Vivante.ArchNote.GPU.TileSpec-v2.0-20141003.pdf
- VPU_memory_tiling_v0.4.09092015.docx
- G-Series_2_v3_HW_Integration_Guide-2015-06-29.pdf

The following features are not supported:

- Progressive to interlaced conversion by dropping lines
- 3-plane YUV source buffers
- 2-plane to 1-plane conversion
- 90/270 rotation for all linear, all 16bpp and 10bpp 2-Plane YUV422 sources
- 10-bit packed input will be maintained as 10-bit packed output. The output will not unpack the components to 32/24/16-bit alignment.

15.7.1.3 DPR Signal Descriptions

Signal	Description	I/O
rtr_m_clk	RTMEM memory clock	I
rtr_aresetn	RTMEM async reset	I
dpr_apb_clk	APB Control register clock	I
dpr_b_clk	APB Control register gated clock	I
dpr_apb_aresetn	APB async reset	I
dpr_b_clk	DPR system clock	I
dpr_b_clkg	DPR system gated clock	I
dpr_chip_resetn	DPR system async reset	I
dpr_b_clkg_en	DPR system gated clock enable	O
dpr_irq[3-1:0]	DPR_IRQ per dpr_ctrl block	O
tg2dpr_start_en[3-1:0]	Logic high enable from timing generator (during VSYNC), to restart DPR with current CSR settings	I
tg2dpr_shadow_load_en[3-1:0]	Logic high strobe from timing generator indicating load shadow registers to active registers on next start.	I
dpr_apb_sel[3-1:0]	APB Select	I
dpr_apb_enable	APB Enable	I
dpr_apb_addr[16-1:0]	APB Address	I
dpr_apb_write	APB Write=1, Read=0	I
dpr_apb_wdata[31:0]	APB Write Data	I
dpr_apb_rdata[31:0]	APB Read Data	O
dpr_apb_ready	APB Ready	O
dpr_apb_slvrr	APB Slave Error	O
dpr_axi_ar	AXI Read Command Channel Bus	O
dpr_axi_arready	AXI Read Command Channel Ready	I
dpr_axi_r	AXI Read Data Channel Bus	I
dpr_axi_rready	AXI Read Data Channel Ready	O
rtr2dpr_abort_yrgb[(3*3)-1:0]	1=abort YRGB buffer being written to RTMEM bank	I
rtr2dpr_abort_uv[(3*3)-1:0]	1=abort UV buffer being written to RTMEM bank	I

Table continues on the next page...

Display, Prefetch and Resolve (DPR)

Signal	Description	I/O
rtr2dpr_buf_done_yrgb[(3*3)-1:0]	YRGB buffer becomes available to prefetch	I
rtr2dpr_buf_done_uv[(3*3)-1:0]	UV buffer becomes available to prefetch	I
dpr2rtr_ch_done_yrgb[3-1:0]	DPR completed prefetch/resolve of the YRGB channel	O
dpr2rtr_ch_done_uv[3-1:0]	DPR completed prefetch/resolve of the UV channel	O
dpr2rtr_buf_rdy_yrgb[(3*3)-1:0]	DPR has completed filling current YRGB row of blocks in the RTRAM buffer	O
dpr2rtr_buf_rdy_uv[(3*3)-1:0]	DPR has completed filling current UV row of blocks in the RTRAM buffer	O
dpr2rtr_cs[3-1:0]	Chip select which RTMEM is being written	O
dpr2rtr_row_data[(64*8)-1:0]	RTMEM row data	O
dpr2rtr_row_uv_sel	2-Plane UV memory select	O
dpr2rtr_bank_addr[1:0]	memory bank address	O
dpr2rtr_row_addr[13:3]	memory row address	O
dpr2rtr_row_sel[2:0]	memory row select	O
dpr2rtr_csr_mode[(3*3)-1:0]	RTRAM Mode Control	O
dpr2rtr_csr_num_rows[3-1:0]	Number of RTRAM lines/buffer: 0=8 lines, 1=4 lines	O
dpr2rtr_csr_num_planes[3-1:0]	0=Enable 1-Plane, 1=Enable 2-Plane	O
dpr2rtr_csr_num_rows_active[3-1:0]	0=Row0-4 enabled. 1=Row0-6, all rows, enabled	O
dpr2rtr_csr_thres_high[(3*3)-1:0]	panic high water mark threshold	O
dpr2rtr_csr_thres_low[(3*3)-1:0]	panic low water mark threshold	O
dpr2rtr_csr_abort_sel[3-1:0]	0=Panic Select. 1=Abort Select	O
dpr2rtr_csr_vflip_en[3-1:0]	1=vflip enable, 0=not vertical flipped	O
ipt_scan_mode	scan mode enable	I
ipt_se_gatedclk	ICG gated clock test enable	I

15.7.2 Functional description

15.7.2.1 Clocking and Resets

15.7.2.1.1 Functional clocks

The DPR b_clk clock, RTRAM rtr_m_clk and APB apb_clk clock domains are all asynchronous. The following clocking restriction exists in the DPR: m_clk >= b_clk/2

15.7.2.1.2 Chip resets

There is a separate asynchronous chip reset for each clock domain. The DPR implements reset synchronizers for each asynchronous reset, to deassert each reset synchronously.

15.7.2.2 DPR Sub-blocks

15.7.2.2.1 DPR Prefetch Control

This control engine is responsible for prefetching the source frame data. Request sizes are software programmable in a range from 64 bytes to 4k bytes. The prefetch engine is designed so the RTRAM buffer will always fill from left to right, top to bottom across the line regardless of the output image orientation.

15.7.2.2.2 DPR Response Control

The response control block is responsible for managing the DPR datapath resolve mux, horizontal flip, vertical flip and image rotation.

15.7.2.2.3 DPR Datapath

The DPR datapath block is shared across all DPR channel response blocks because on any given clock cycle, pixels for only 1 DPR response channel will be returned by the system. The datapath implements pixel cross bar logic for tile resolve, horizontal flip, vertical flip and image rotation by 0, 90, 180 or 270 degrees.

15.7.3 Memory Map and Registers

15.7.3.1 register descriptions

15.7.3.1.1 DPR Memory map

DPR1 base address: 1_8000h

DPR2 base address: 1_9000h

DPR3 base address: 1_A000h

Memory Map and Registers

Offset	Register	Width (In bits)	Access	Reset value
0h	System Control 0 (SYSTEM_CTRL0)	32	RW	0000_0010h
4h	System Control 0 (SYSTEM_CTRL0_SET)	32	RW	0000_0010h
8h	System Control 0 (SYSTEM_CTRL0_CLR)	32	RW	0000_0010h
Ch	System Control 0 (SYSTEM_CTRL0_TOG)	32	RW	0000_0010h
20h	Interrupt Mask (IRQ_MASK)	32	RW	0000_0000h
24h	Interrupt Mask (IRQ_MASK_SET)	32	RW	0000_0000h
28h	Interrupt Mask (IRQ_MASK_CLR)	32	RW	0000_0000h
2Ch	Interrupt Mask (IRQ_MASK_TOG)	32	RW	0000_0000h
30h	Status Register of Masked IRQ (IRQ_MASK_STATUS)	32	RO	0000_0000h
34h	Status Register of Masked IRQ (IRQ_MASK_STATUS_SET)	32	RO	0000_0000h
38h	Status Register of Masked IRQ (IRQ_MASK_STATUS_CLR)	32	RO	0000_0000h
3Ch	Status Register of Masked IRQ (IRQ_MASK_STATUS_TOG)	32	RO	0000_0000h
40h	Status of Non-Masked IRQ (IRQ_NONMASK_STATUS)	32	W1C	0000_0000h
44h	Status of Non-Masked IRQ (IRQ_NONMASK_STATUS_SET)	32	W1C	0000_0000h
48h	Status of Non-Masked IRQ (IRQ_NONMASK_STATUS_CLR)	32	W1C	0000_0000h
4Ch	Status of Non-Masked IRQ (IRQ_NONMASK_STATUS_TOG)	32	W1C	0000_0000h
50h	Mode Control 0 (MODE_CTRL0)	32	RW	0000_0000h
54h	Mode Control 0 (MODE_CTRL0_SET)	32	RW	0000_0000h
58h	Mode Control 0 (MODE_CTRL0_CLR)	32	RW	0000_0000h
5Ch	Mode Control 0 (MODE_CTRL0_TOG)	32	RW	0000_0000h
70h	Frame Control 0 (FRAME_CTRL0)	32	RW	0000_0000h
74h	Frame Control 0 (FRAME_CTRL0_SET)	32	RW	0000_0000h
78h	Frame Control 0 (FRAME_CTRL0_CLR)	32	RW	0000_0000h
7Ch	Frame Control 0 (FRAME_CTRL0_TOG)	32	RW	0000_0000h
90h	Frame 1-Plane Control 0 (FRAME_1P_CTRL0)	32	RW	0000_0000h
94h	Frame 1-Plane Control 0 (FRAME_1P_CTRL0_SET)	32	RW	0000_0000h
98h	Frame 1-Plane Control 0 (FRAME_1P_CTRL0_CLR)	32	RW	0000_0000h
9Ch	Frame 1-Plane Control 0 (FRAME_1P_CTRL0_TOG)	32	RW	0000_0000h
A0h	Frame 1-Plane Pix X Control (FRAME_1P_PIX_X_CTRL)	32	RW	0000_0000h
A4h	Frame 1-Plane Pix X Control (FRAME_1P_PIX_X_CTRL_SET)	32	RW	0000_0000h
A8h	Frame 1-Plane Pix X Control (FRAME_1P_PIX_X_CTRL_CLR)	32	RW	0000_0000h
ACh	Frame 1-Plane Pix X Control (FRAME_1P_PIX_X_CTRL_TOG)	32	RW	0000_0000h
B0h	Frame 1-Plane Pix Y Control (FRAME_1P_PIX_Y_CTRL)	32	RW	0000_0000h
B4h	Frame 1-Plane Pix Y Control (FRAME_1P_PIX_Y_CTRL_SET)	32	RW	0000_0000h
B8h	Frame 1-Plane Pix Y Control (FRAME_1P_PIX_Y_CTRL_CLR)	32	RW	0000_0000h
BCh	Frame 1-Plane Pix Y Control (FRAME_1P_PIX_Y_CTRL_TOG)	32	RW	0000_0000h
C0h	Frame 1-Plane Base Address Control 0 (FRAME_1P_BASE_ADDR_CTRL0)	32	RW	0000_0000h
C4h	Frame 1-Plane Base Address Control 0 (FRAME_1P_BASE_ADDR_CTRL0_SET)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
C8h	Frame 1-Plane Base Address Control 0 (FRAME_1P_BASE_ADDR_CTRL0_CLR)	32	RW	0000_0000h
CCh	Frame 1-Plane Base Address Control 0 (FRAME_1P_BASE_ADDR_CTRL0_TOG)	32	RW	0000_0000h
E0h	Frame 2-Plane Control 0 (FRAME_2P_CTRL0)	32	RW	0000_0000h
E4h	Frame 2-Plane Control 0 (FRAME_2P_CTRL0_SET)	32	RW	0000_0000h
E8h	Frame 2-Plane Control 0 (FRAME_2P_CTRL0_CLR)	32	RW	0000_0000h
ECh	Frame 2-Plane Control 0 (FRAME_2P_CTRL0_TOG)	32	RW	0000_0000h
F0h	Frame 2-Plane Pix X Control (FRAME_2P_PIX_X_CTRL)	32	RW	0000_0000h
F4h	Frame 2-Plane Pix X Control (FRAME_2P_PIX_X_CTRL_SET)	32	RW	0000_0000h
F8h	Frame 2-Plane Pix X Control (FRAME_2P_PIX_X_CTRL_CLR)	32	RW	0000_0000h
FCh	Frame 2-Plane Pix X Control (FRAME_2P_PIX_X_CTRL_TOG)	32	RW	0000_0000h
100h	Frame 2-Plane Pix Y Control (FRAME_2P_PIX_Y_CTRL)	32	RW	0000_0000h
104h	Frame 2-Plane Pix Y Control (FRAME_2P_PIX_Y_CTRL_SET)	32	RW	0000_0000h
108h	Frame 2-Plane Pix Y Control (FRAME_2P_PIX_Y_CTRL_CLR)	32	RW	0000_0000h
10Ch	Frame 2-Plane Pix Y Control (FRAME_2P_PIX_Y_CTRL_TOG)	32	RW	0000_0000h
110h	Frame 2-Plane Base Address Control 0 (FRAME_2P_BASE_ADDR_CTRL0)	32	RW	0000_0000h
114h	Frame 2-Plane Base Address Control 0 (FRAME_2P_BASE_ADDR_CTRL0_SET)	32	RW	0000_0000h
118h	Frame 2-Plane Base Address Control 0 (FRAME_2P_BASE_ADDR_CTRL0_CLR)	32	RW	0000_0000h
11Ch	Frame 2-Plane Base Address Control 0 (FRAME_2P_BASE_ADDR_CTRL0_TOG)	32	RW	0000_0000h
200h	RTRAM Control 0 (RTRAM_CTRL0)	32	RW	0000_0000h
204h	RTRAM Control 0 (RTRAM_CTRL0_SET)	32	RW	0000_0000h
208h	RTRAM Control 0 (RTRAM_CTRL0_CLR)	32	RW	0000_0000h
20Ch	RTRAM Control 0 (RTRAM_CTRL0_TOG)	32	RW	0000_0000h

15.7.3.1.2 System Control 0 (SYSTEM_CTRL0)

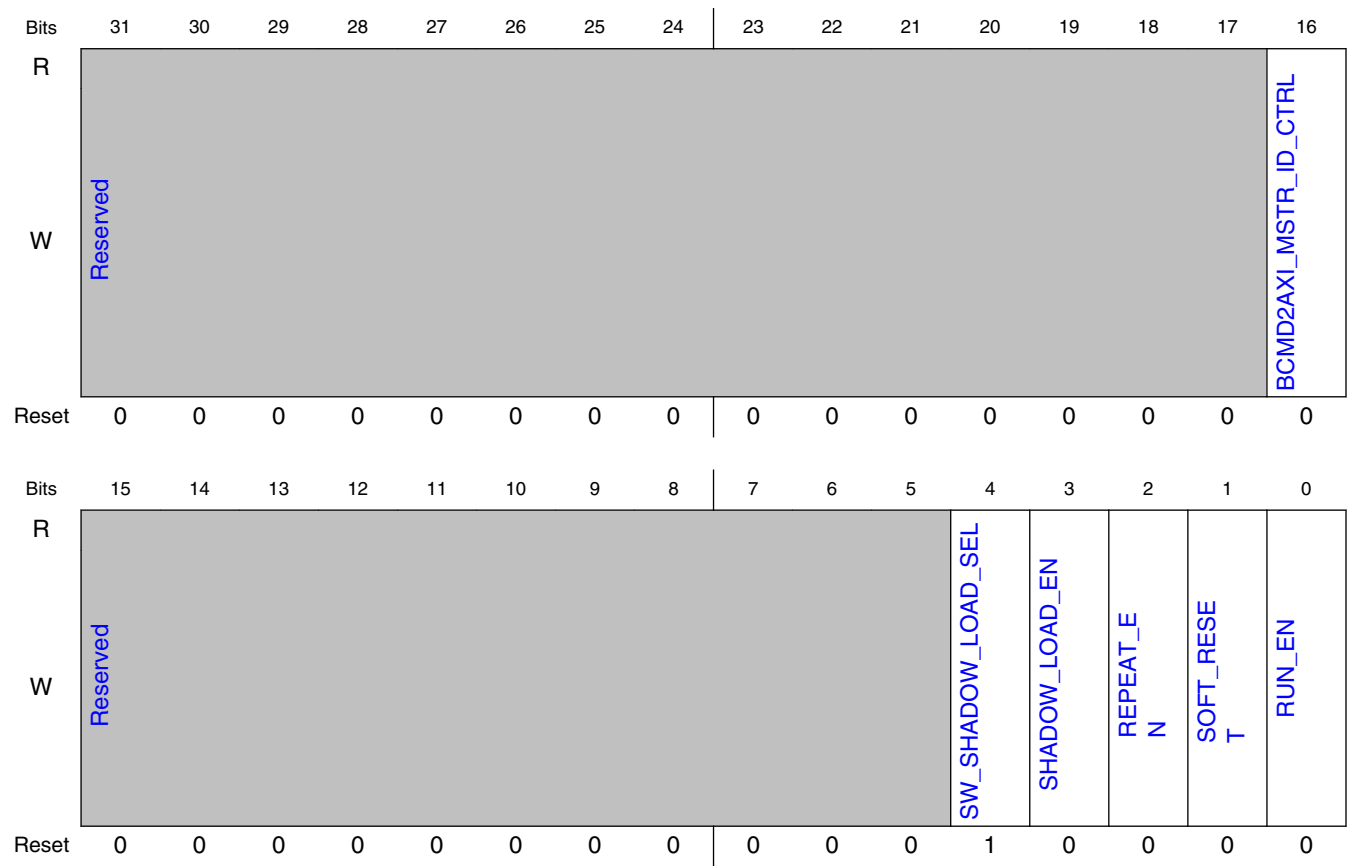
15.7.3.1.2.1 Offset

Register	Offset
SYSTEM_CTRL0	0h

15.7.3.1.2.2 Function

The System Control0 register contains register for DPR block level control functions.

15.7.3.1.2.3 Diagram



15.7.3.1.2.4 Fields

Field	Function
31-17 —	Reserved.
16 BCMD2AXI_MSTR_ID_CTRL	Buscmd To AXI Master ID Control 0x0 = All DPR buscmd master channels assigned a unique AXI ID 0x1 = All DPR buscmd master channels assigned the same AXI ID Note: If the DPR is configured to have > 1 DPR channel, then DPR channel 0, will control the BCMD2AXI_MSTR_ID_CTRL for all DPR Channels. DPR Channel 1-N's BCMD2AXI_MSTR_ID_CTRL will be reserved.
15-5 —	Reserved.
4 SW_SHADOW_LOAD_SEL	Software Shadow Load Select The SW_SHADOW_LOAD_SEL is used to select the shadow_load_en source.

Table continues on the next page...

Field	Function
	If set to 1, SYSTEM_CTRL0: SHADOW_LOAD_EN control register is used to allow update of the shadow control registers to the active control registers. If set to 0, Subsystem hardware signal is used to allow update of the shadow control registers to the active control registers. SW_SHADOW_LOAD_SEL should be set to 1, for initial DPR control register load and RUN_EN enablement. SW_SHADOW_LOAD_SEL should only be set to 0, after SYSTEM_CTRL0: REPEAT_EN is enabled.
3 SHADOW_LOAD_EN	Shadow Load Enable The SHADOW_LOAD_EN bit is written to 1 by software after all control registers are written. If set to 1, shadowed control registers are updated to the active control registers on next REPEAT_EN or RUN_EN to start DPR processing of the next frame. If set to 0, shadowed control registers are not loaded into the active control registers. The previous active control register settings will be used to process the next frame. Hardware will automatically clear this bit, when the shadow registers are loaded to the active control registers.
2 REPEAT_EN	Repeat Enable Writing a 1 to this register will enable the timing generator to write the RUN_EN register on subsequent restarts using the current control register settings. The REPEAT_EN will not take effect until after the 1st frame has been processed with RUN_EN issued by software. The REPEAT_EN may remain enabled for as many concurrent frames as desired to be automatically restarted. When automatic restart should be disabled, software must clear the REPEAT_EN before the next available hardware restart command.
1 SOFT_RESET	Soft Reset This bit is not self-clearing software will need to write a: 0x0 = deassert soft reset 0x1 = assert soft reset Note: Channel 0 soft reset will reset the SYSTEM_CTRL0:BCMD2AXI_MSTR_ID_CTRL. This should only be done if all DPR Channels are done processing.
0 RUN_EN	Run Enable Writing a 1 to this register will start processing the next frame image. This register is self-clearing.

15.7.3.1.3 System Control 0 (SYSTEM_CTRL0_SET)

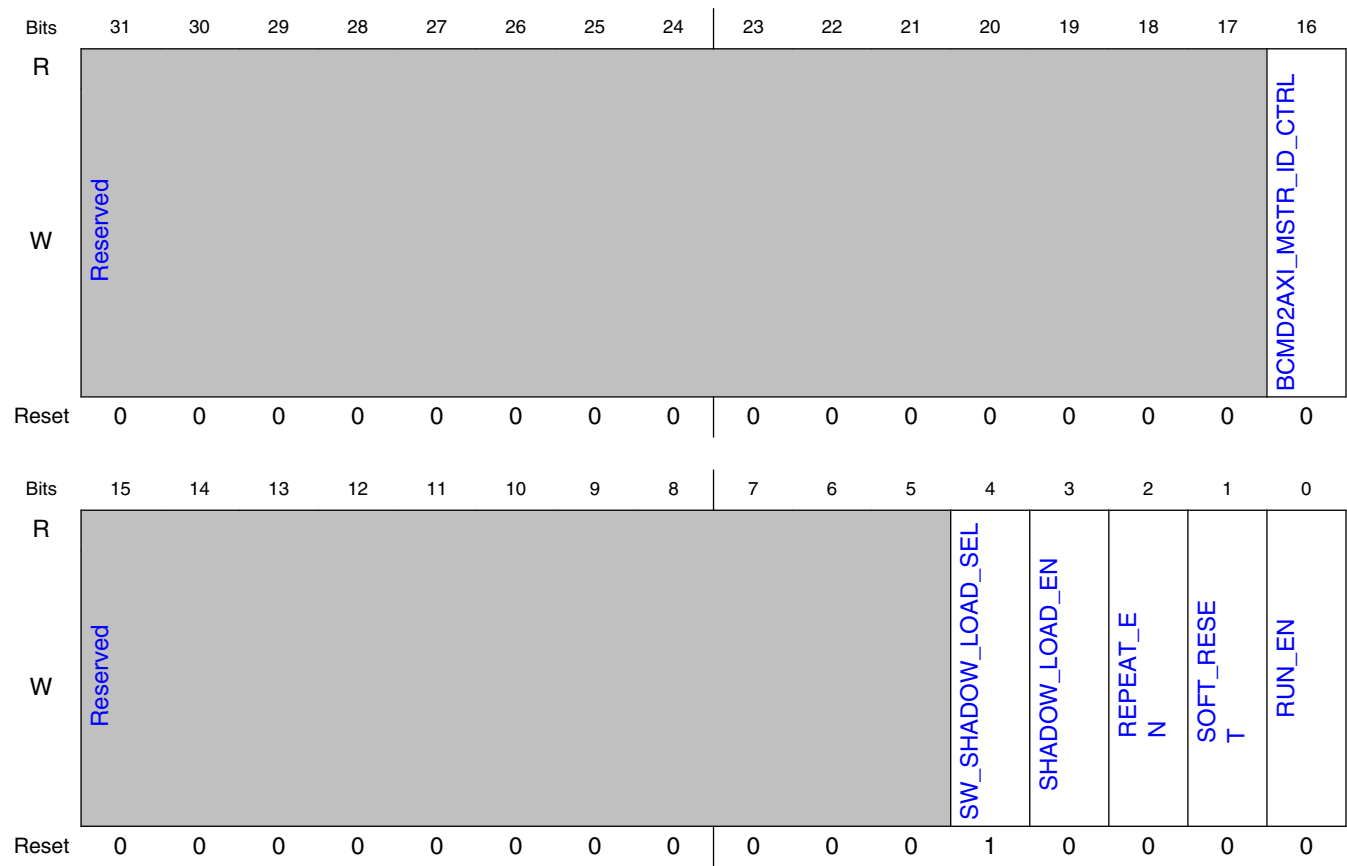
15.7.3.1.3.1 Offset

Register	Offset
SYSTEM_CTRL0_SET	4h

15.7.3.1.3.2 Function

The System Control0 register contains register for DPR block level control functions.

15.7.3.1.3.3 Diagram



15.7.3.1.3.4 Fields

Field	Function
31-17 —	Reserved.
16 BCMD2AXI_MSTR_ID_CTRL	Buscmd To AXI Master ID Control 0x0 = All DPR buscmd master channels assigned a unique AXI ID 0x1 = All DPR buscmd master channels assigned the same AXI ID Note: If the DPR is configured to have > 1 DPR channel, then DPR channel 0, will control the BCMD2AXI_MSTR_ID_CTRL for all DPR Channels. DPR Channel 1-N's BCMD2AXI_MSTR_ID_CTRL will be reserved.
15-5	Reserved.

Table continues on the next page...

Field	Function
—	
4 SW_SHADOW_LOAD_SEL	Software Shadow Load Select The SW_SHADOW_LOAD_SEL is used to select the shadow_load_en source. If set to 1, SYSTEM_CTRL0: SHADOW_LOAD_EN control register is used to allow update of the shadow control registers to the active control registers. If set to 0, Subsystem hardware signal is used to allow update of the shadow control registers to the active control registers. SW_SHADOW_LOAD_SEL should be set to 1, for initial DPR control register load and RUN_EN enablement. SW_SHADOW_LOAD_SEL should only be set to 0, after SYSTEM_CTRL0: REPEAT_EN is enabled.
3 SHADOW_LOAD_EN	Shadow Load Enable The SHADOW_LOAD_EN bit is written to 1 by software after all control registers are written. If set to 1, shadowed control registers are updated to the active control registers on next REPEAT_EN or RUN_EN to start DPR processing of the next frame. If set to 0, shadowed control registers are not loaded into the active control registers. The previous active control register settings will be used to process the next frame. Hardware will automatically clear this bit, when the shadow registers are loaded to the active control registers.
2 REPEAT_EN	Repeat Enable Writing a 1 to this register will enable the timing generator to write the RUN_EN register on subsequent restarts using the current control register settings. The REPEAT_EN will not take effect until after the 1st frame has been processed with RUN_EN issued by software. The REPEAT_EN may remain enabled for as many concurrent frames as desired to be automatically restarted. When automatic restart should be disabled, software must clear the REPEAT_EN before the next available hardware restart command.
1 SOFT_RESET	Soft Reset This bit is not self-clearing software will need to write a: 0x0 = deassert soft reset 0x1 = assert soft reset Note: Channel 0 soft reset will reset the SYSTEM_CTRL0:BCMD2AXI_MSTR_ID_CTRL. This should only be done if all DPR Channels are done processing.
0 RUN_EN	Run Enable Writing a 1 to this register will start processing the next frame image. This register is self-clearing.

15.7.3.1.4 System Control 0 (SYSTEM_CTRL0_CLR)

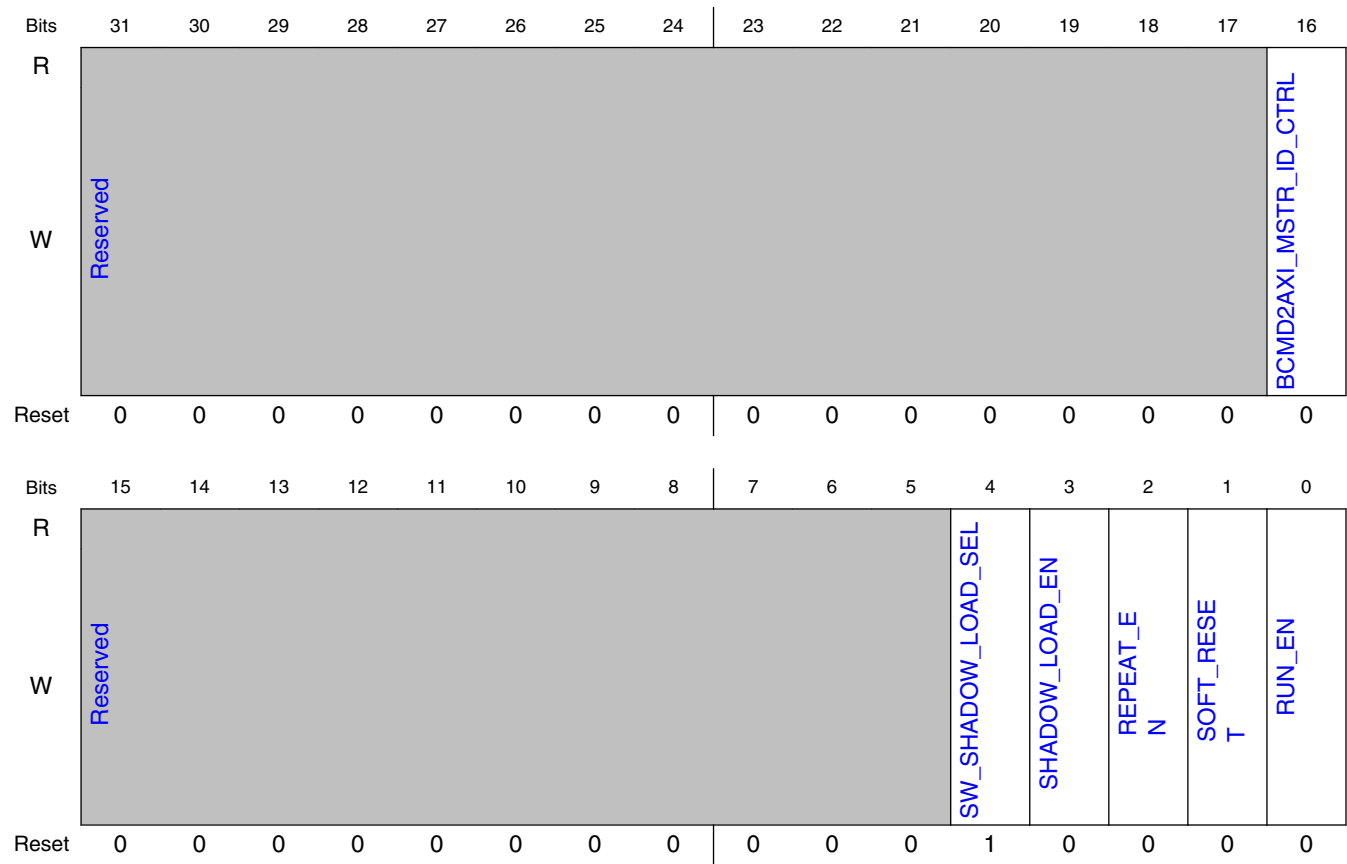
15.7.3.1.4.1 Offset

Register	Offset
SYSTEM_CTRL0_CLR	8h

15.7.3.1.4.2 Function

The System Control0 register contains register for DPR block level control functions.

15.7.3.1.4.3 Diagram



15.7.3.1.4.4 Fields

Field	Function
31-17	Reserved.
—	
16	Buscmd To AXI Master ID Control 0x0 = All DPR buscmd master channels assigned a unique AXI ID

Table continues on the next page...

Field	Function
BCMD2AXI_MSTR_ID_CTRL	0x1 = All DPR buscmd master channels assigned the same AXI ID Note: If the DPR is configured to have > 1 DPR channel, then DPR channel 0, will control the BCMD2AXI_MSTR_ID_CTRL for all DPR Channels. DPR Channel 1-N's BCMD2AXI_MSTR_ID_CTRL will be reserved.
15-5 —	Reserved.
4 SW_SHADOW_LOAD_SEL	Software Shadow Load Select The SW_SHADOW_LOAD_SEL is used to select the shadow_load_en source. If set to 1, SYSTEM_CTRL0: SHADOW_LOAD_EN control register is used to allow update of the shadow control registers to the active control registers. If set to 0, Subsystem hardware signal is used to allow update of the shadow control registers to the active control registers. SW_SHADOW_LOAD_SEL should be set to 1, for initial DPR control register load and RUN_EN enablement. SW_SHADOW_LOAD_SEL should only be set to 0, after SYSTEM_CTRL0: REPEAT_EN is enabled.
3 SHADOW_LOAD_EN	Shadow Load Enable The SHADOW_LOAD_EN bit is written to 1 by software after all control registers are written. If set to 1, shadowed control registers are updated to the active control registers on next REPEAT_EN or RUN_EN to start DPR processing of the next frame. If set to 0, shadowed control registers are not loaded into the active control registers. The previous active control register settings will be used to process the next frame. Hardware will automatically clear this bit, when the shadow registers are loaded to the active control registers.
2 REPEAT_EN	Repeat Enable Writing a 1 to this register will enable the timing generator to write the RUN_EN register on subsequent restarts using the current control register settings. The REPEAT_EN will not take effect until after the 1st frame has been processed with RUN_EN issued by software. The REPEAT_EN may remain enabled for as many concurrent frames as desired to be automatically restarted. When automatic restart should be disabled, software must clear the REPEAT_EN before the next available hardware restart command.
1 SOFT_RESET	Soft Reset This bit is not self-clearing software will need to write a: 0x0 = deassert soft reset 0x1 = assert soft reset Note: Channel 0 soft reset will reset the SYSTEM_CTRL0:BCMD2AXI_MSTR_ID_CTRL. This should only be done if all DPR Channels are done processing.
0 RUN_EN	Run Enable Writing a 1 to this register will start processing the next frame image. This register is self-clearing.

15.7.3.1.5 System Control 0 (SYSTEM_CTRL0_TOG)

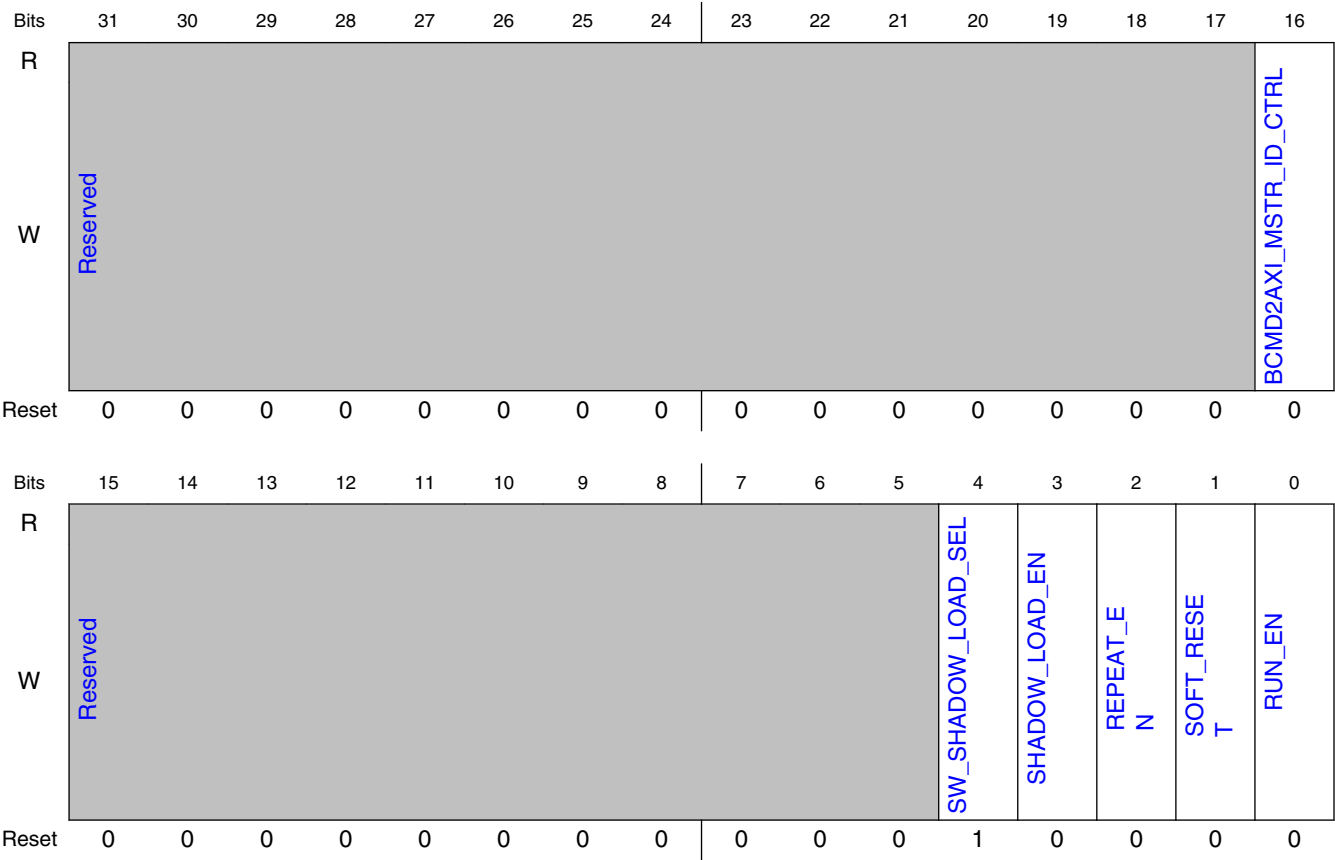
15.7.3.1.5.1 Offset

Register	Offset
SYSTEM_CTRL0_TOG	Ch

15.7.3.1.5.2 Function

The System Control0 register contains register for DPR block level control functions.

15.7.3.1.5.3 Diagram



15.7.3.1.5.4 Fields

Field	Function
31-17	Reserved.

Table continues on the next page...

Field	Function
—	
16 BCMD2AXI_MSTR_ID_CTRL	Buscmd To AXI Master ID Control 0x0 = All DPR buscmd master channels assigned a unique AXI ID 0x1 = All DPR buscmd master channels assigned the same AXI ID Note: If the DPR is configured to have > 1 DPR channel, then DPR channel 0, will control the BCMD2AXI_MSTR_ID_CTRL for all DPR Channels. DPR Channel 1-N's BCMD2AXI_MSTR_ID_CTRL will be reserved.
15-5 —	Reserved.
4 SW_SHADOW_LOAD_SEL	Software Shadow Load Select The SW_SHADOW_LOAD_SEL is used to select the shadow_load_en source. If set to 1, SYSTEM_CTRL0: SHADOW_LOAD_EN control register is used to allow update of the shadow control registers to the active control registers. If set to 0, Subsystem hardware signal is used to allow update of the shadow control registers to the active control registers. SW_SHADOW_LOAD_SEL should be set to 1, for initial DPR control register load and RUN_EN enablement. SW_SHADOW_LOAD_SEL should only be set to 0, after SYSTEM_CTRL0: REPEAT_EN is enabled.
3 SHADOW_LOAD_EN	Shadow Load Enable The SHADOW_LOAD_EN bit is written to 1 by software after all control registers are written. If set to 1, shadowed control registers are updated to the active control registers on next REPEAT_EN or RUN_EN to start DPR processing of the next frame. If set to 0, shadowed control registers are not loaded into the active control registers. The previous active control register settings will be used to process the next frame. Hardware will automatically clear this bit, when the shadow registers are loaded to the active control registers.
2 REPEAT_EN	Repeat Enable Writing a 1 to this register will enable the timing generator to write the RUN_EN register on subsequent restarts using the current control register settings. The REPEAT_EN will not take effect until after the 1st frame has been processed with RUN_EN issued by software. The REPEAT_EN may remain enabled for as many concurrent frames as desired to be automatically restarted. When automatic restart should be disabled, software must clear the REPEAT_EN before the next available hardware restart command.
1 SOFT_RESET	Soft Reset This bit is not self-clearing software will need to write a: 0x0 = deassert soft reset 0x1 = assert soft reset Note: Channel 0 soft reset will reset the SYSTEM_CTRL0:BCMD2AXI_MSTR_ID_CTRL. This should only be done if all DPR Channels are done processing.
0 RUN_EN	Run Enable

Memory Map and Registers

Field	Function
	Writing a 1 to this register will start processing the next frame image. This register is self-clearing.

15.7.3.1.6 Interrupt Mask (IRQ_MASK)

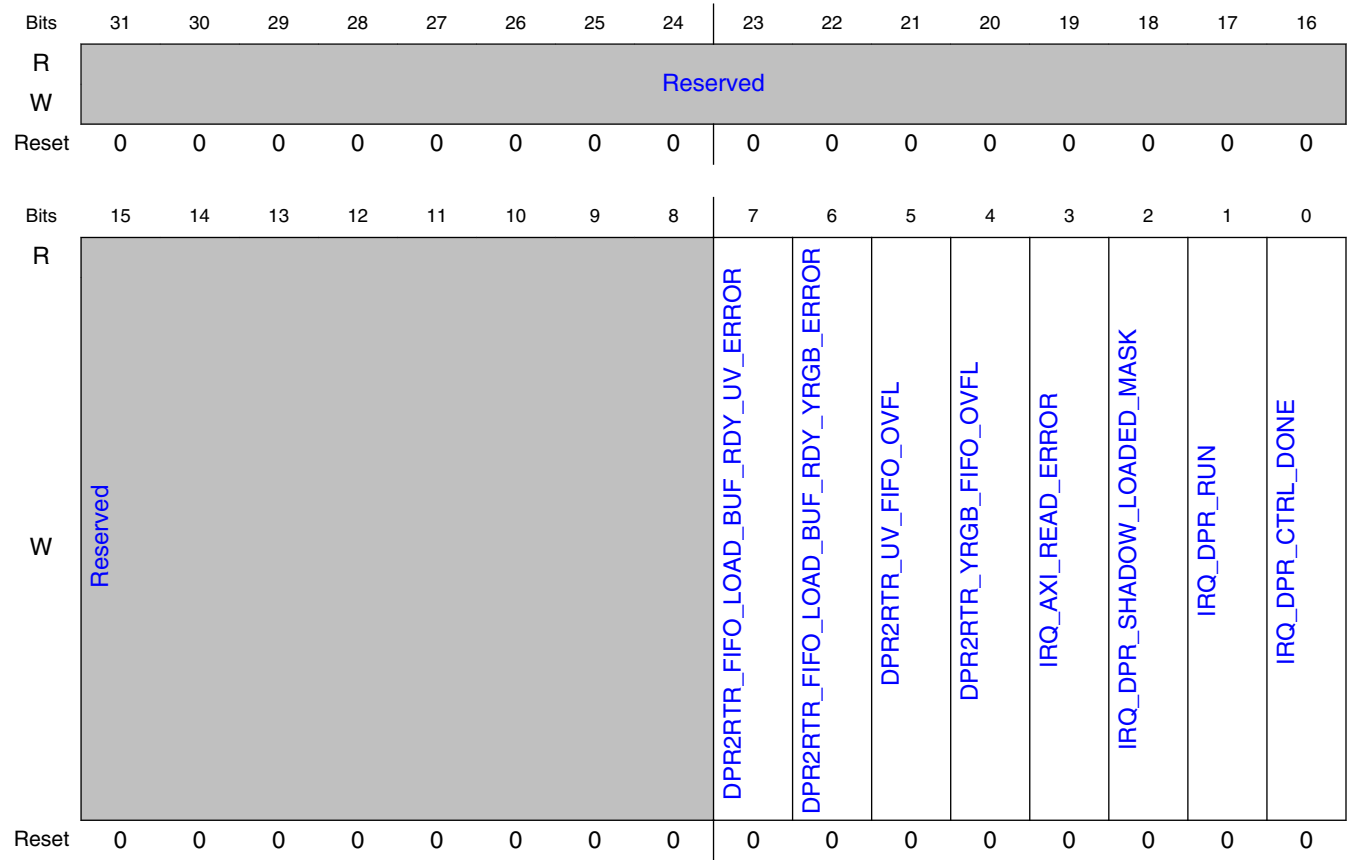
15.7.3.1.6.1 Offset

Register	Offset
IRQ_MASK	20h

15.7.3.1.6.2 Function

The IRQ_MASK control register contains interrupt mask control register bits. Writing a 1 to the IRQ_MASK control register will mask off the interrupt.

15.7.3.1.6.3 Diagram



15.7.3.1.6.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR	DPR to RTRAM Fifo load UV buffer ready IRQ error Mask Mask off the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.
6 DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	DPR to RTRAM Fifo load YRGB buffer ready IRQ error Mask Mask off the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow IRQ Mask Mask off the DPR2RTR_UV Fifo Overflow interrupt.
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow IRQ Mask Mask off the DPR2RTR_YRGB Fifo Overflow interrupt.
3 IRQ_AXI_READ_ERROR	AXI Read Error IRQ Mask Mask off the AXI Read Error interrupt.
2 IRQ_DPR_SHADOW_LOADED_MASK	DPR Shadow Loaded IRQ Mask Mask off the DPR_SHADOW_LOADED interrupt.
1 IRQ_DPR_RUN	DPR Run IRQ Mask Mask off the DPR_RUN interrupt.
0 IRQ_DPR_CTRL_DONE	DPR Control Done IRQ Mask Mask off the DPR_CTRL_DONE interrupt.

15.7.3.1.7 Interrupt Mask (IRQ_MASK_SET)

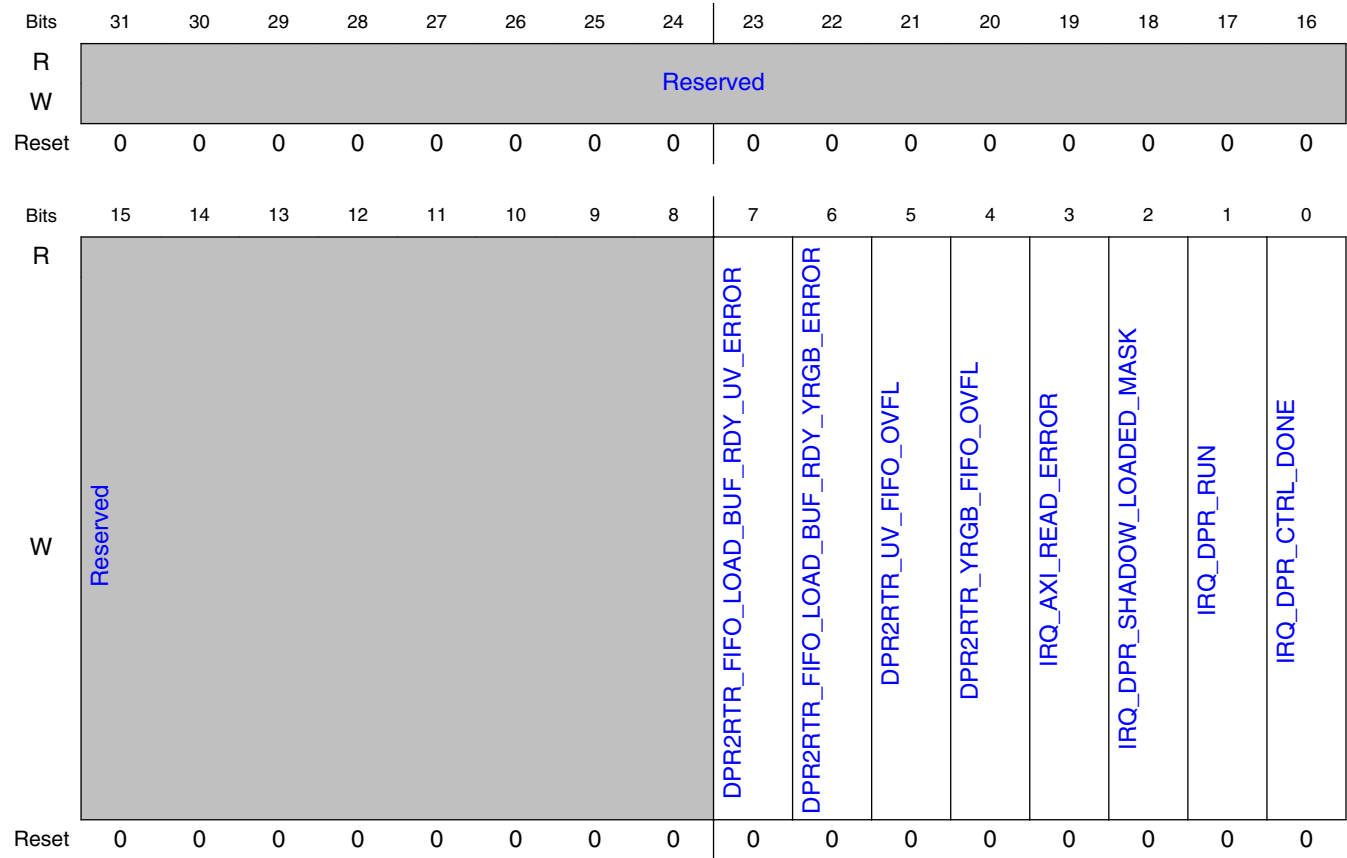
15.7.3.1.7.1 Offset

Register	Offset
IRQ_MASK_SET	24h

15.7.3.1.7.2 Function

The IRQ_MASK control register contains interrupt mask control register bits. Writing a 1 to the IRQ_MASK control register will mask off the interrupt.

15.7.3.1.7.3 Diagram



15.7.3.1.7.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR	DPR to RTRAM Fifo load UV buffer ready IRQ error Mask Mask off the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.
6	DPR to RTRAM Fifo load YRGB buffer ready IRQ error Mask Mask off the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt.

Table continues on the next page...

Field	Function
DPR2RTR_FIFO_LOAD_BUFFER_READY_YRGB_ERROR	
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow IRQ Mask Mask off the DPR2RTR_UV Fifo Overflow interrupt.
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow IRQ Mask Mask off the DPR2RTR_YRGB Fifo Overflow interrupt.
3 IRQ_AXI_READ_ERROR	AXI Read Error IRQ Mask Mask off the AXI Read Error interrupt.
2 IRQ_DPR_SHADOW_LOADED_MASK	DPR Shadow Loaded IRQ Mask Mask off the DPR_SHADOW_LOADED interrupt.
1 IRQ_DPR_RUN	DPR Run IRQ Mask Mask off the DPR_RUN interrupt.
0 IRQ_DPR_CTRL_DONE	DPR Control Done IRQ Mask Mask off the DPR_CTRL_DONE interrupt.

15.7.3.1.8 Interrupt Mask (IRQ_MASK_CLR)

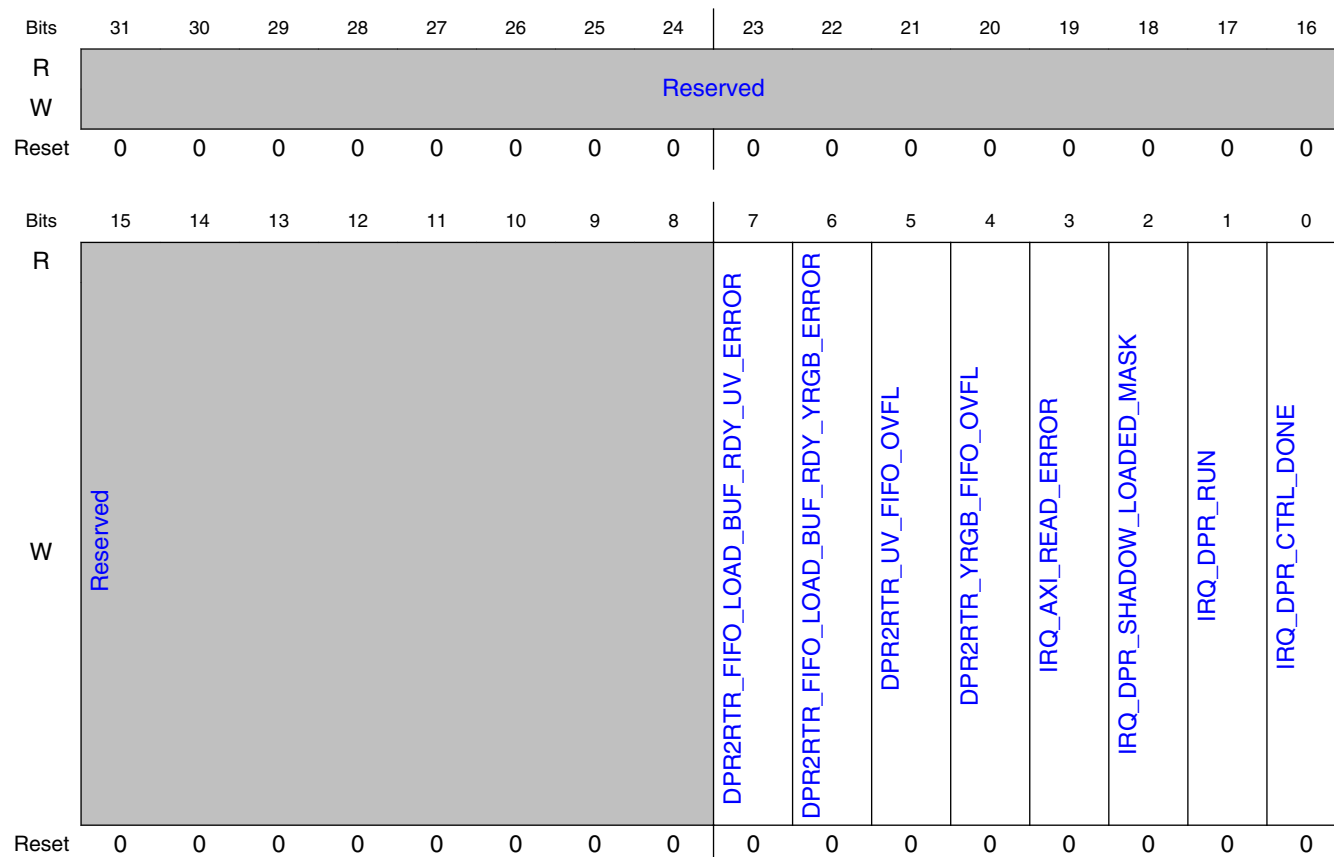
15.7.3.1.8.1 Offset

Register	Offset
IRQ_MASK_CLR	28h

15.7.3.1.8.2 Function

The IRQ_MASK control register contains interrupt mask control register bits. Writing a 1 to the IRQ_MASK control register will mask off the interrupt.

15.7.3.1.8.3 Diagram



15.7.3.1.8.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR	DPR to RTRAM Fifo load UV buffer ready IRQ error Mask Mask off the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.
6 DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	DPR to RTRAM Fifo load YRGB buffer ready IRQ error Mask Mask off the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow IRQ Mask Mask off the DPR2RTR_UV Fifo Overflow interrupt.
4	DPR to RTRAM YRGB Fifo Overflow IRQ Mask

Table continues on the next page...

Field	Function
DPR2RTR_YRGB_FIFO_OVFL	Mask off the DPR2RTR_YRGB Fifo Overflow interrupt.
3 IRQ_AXI_READ_ERROR	AXI Read Error IRQ Mask Mask off the AXI Read Error interrupt.
2 IRQ_DPR_SHADOW_LOADED_MASK	DPR Shadow Loaded IRQ Mask Mask off the DPR_SHADOW_LOADED interrupt.
1 IRQ_DPR_RUN	DPR Run IRQ Mask Mask off the DPR_RUN interrupt.
0 IRQ_DPR_CTRL_DONE	DPR Control Done IRQ Mask Mask off the DPR_CTRL_DONE interrupt.

15.7.3.1.9 Interrupt Mask (IRQ_MASK_TOG)

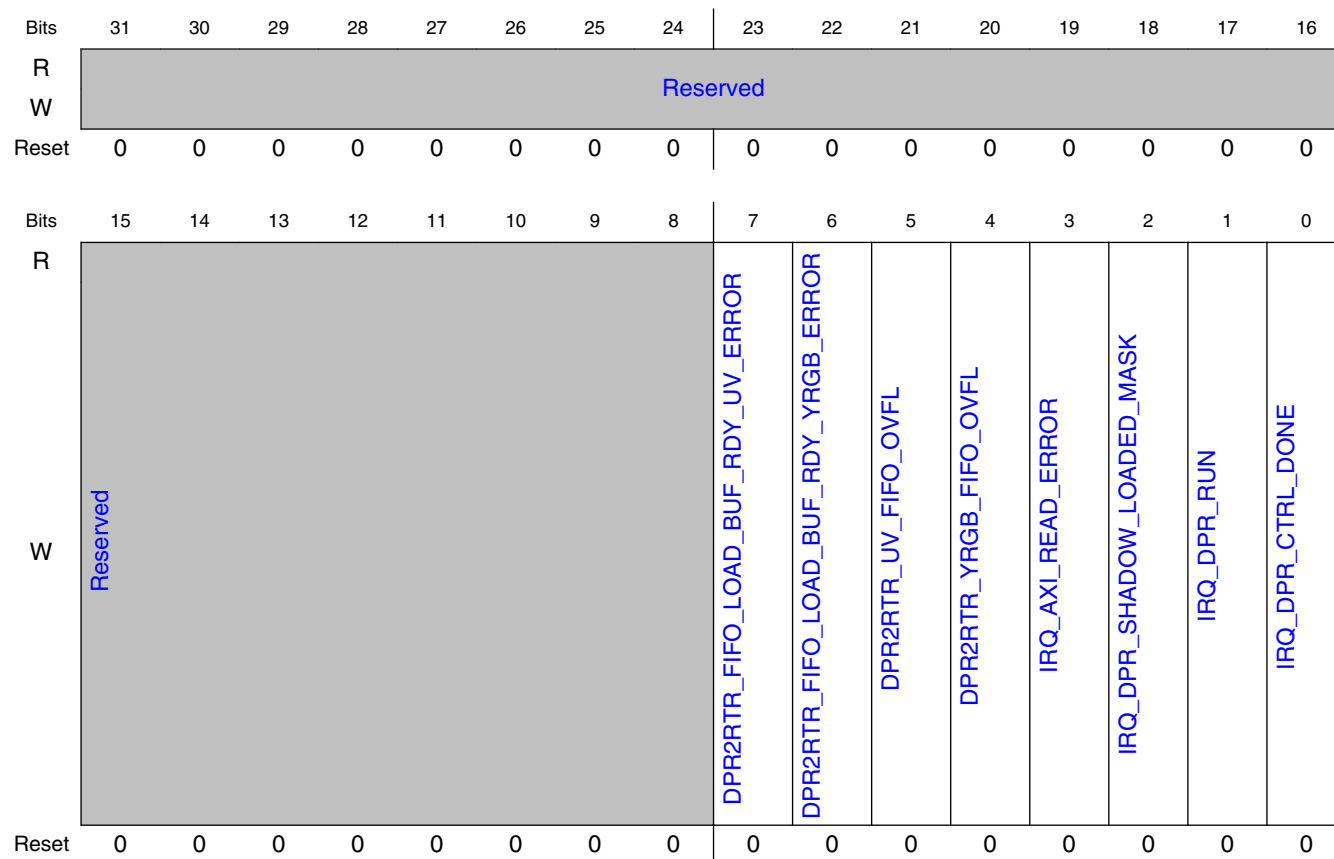
15.7.3.1.9.1 Offset

Register	Offset
IRQ_MASK_TOG	2Ch

15.7.3.1.9.2 Function

The IRQ_MASK control register contains interrupt mask control register bits. Writing a 1 to the IRQ_MASK control register will mask off the interrupt.

15.7.3.1.9.3 Diagram



15.7.3.1.9.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR	DPR to RTRAM Fifo load UV buffer ready IRQ error Mask Mask off the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.
6 DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	DPR to RTRAM Fifo load YRGB buffer ready IRQ error Mask Mask off the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow IRQ Mask Mask off the DPR2RTR_UV Fifo Overflow interrupt.
4	DPR to RTRAM YRGB Fifo Overflow IRQ Mask

Table continues on the next page...

Field	Function
DPR2RTR_YRGB_FIFO_OVFL	Mask off the DPR2RTR_YRGB Fifo Overflow interrupt.
3 IRQ_AXI_READ_ERROR	AXI Read Error IRQ Mask Mask off the AXI Read Error interrupt.
2 IRQ_DPR_SHADOW_LOADED_MASK	DPR Shadow Loaded IRQ Mask Mask off the DPR_SHADOW_LOADED interrupt.
1 IRQ_DPR_RUN	DPR Run IRQ Mask Mask off the DPR_RUN interrupt.
0 IRQ_DPR_CTRL_DONE	DPR Control Done IRQ Mask Mask off the DPR_CTRL_DONE interrupt.

15.7.3.1.10 Status Register of Masked IRQ (IRQ_MASK_STATUS)

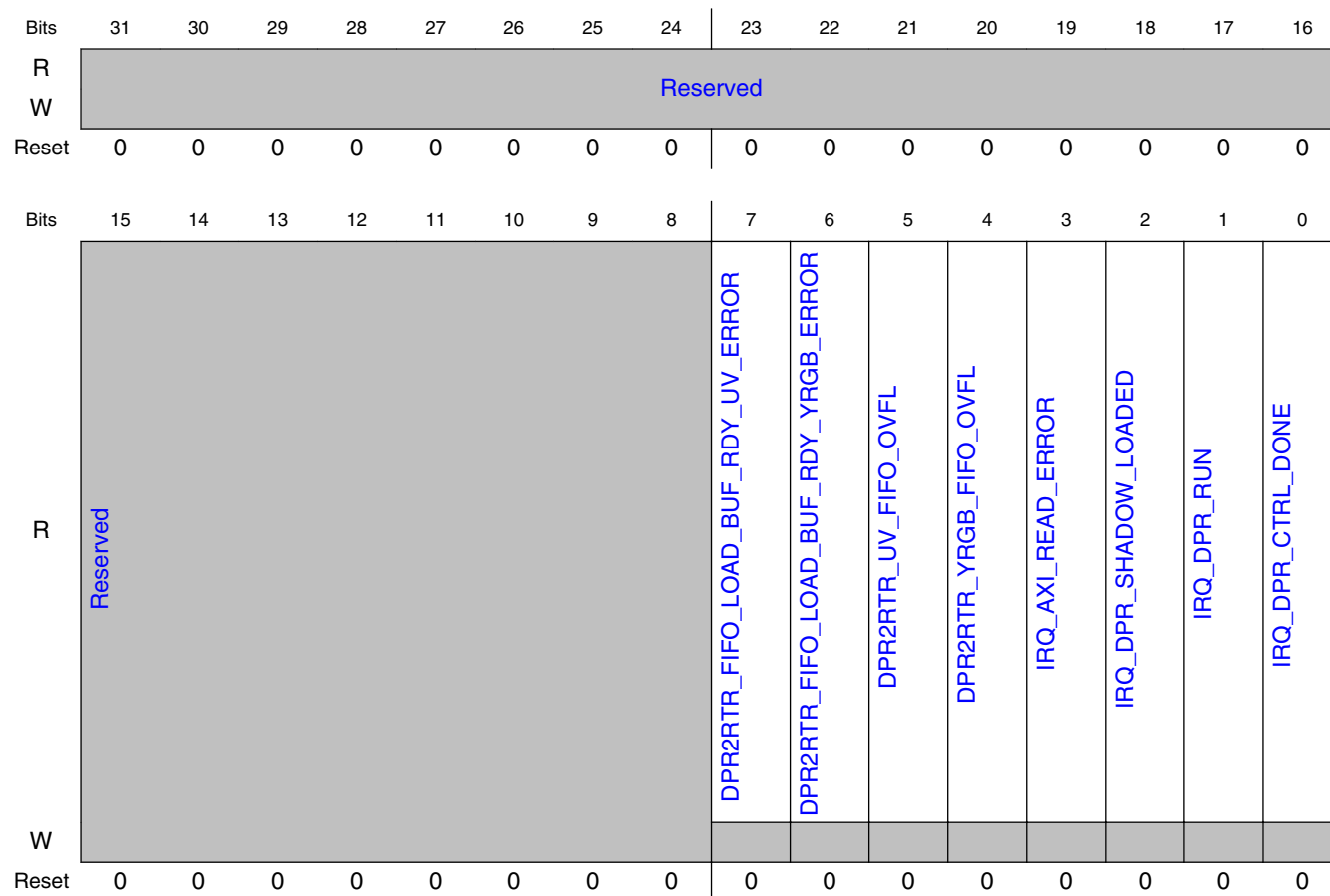
15.7.3.1.10.1 Offset

Register	Offset
IRQ_MASK_STATUS	30h

15.7.3.1.10.2 Function

The IRQ_MASK_STATUS control register contains the masked interrupt status.

15.7.3.1.10.3 Diagram



15.7.3.1.10.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR	DPR to RTRAM Fifo load UV buffer error Masked IRQ Masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.
6 DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	DPR to RTRAM Fifo load YRGB buffer error Masked IRQ Masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow Masked IRQ Masked interrupt status of the DPR2RTR_UV Fifo Overflow interrupt.

Table continues on the next page...

Field	Function
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow Masked IRQ Masked interrupt status of the DPR2RTR_YRGB Fifo Overflow interrupt.
3 IRQ_AXI_READ_ERROR	AXI Read Error Masked IRQ Masked interrupt status of the AXI Read Error interrupt.
2 IRQ_DPR_SHADOW_LOADED	DPR Shadow Loaded Masked IRQ Masked interrupt status of the DPR_SHADOW_LOADED interrupt.
1 IRQ_DPR_RUN	DPR Run Masked IRQ Masked interrupt status of the DPR_RUN interrupt.
0 IRQ_DPR_CTRL_DONE	DPR Control Done Masked IRQ Masked interrupt status of the DPR_CTRL_DONE interrupt.

15.7.3.1.11 Status Register of Masked IRQ (IRQ_MASK_STATUS_SET)

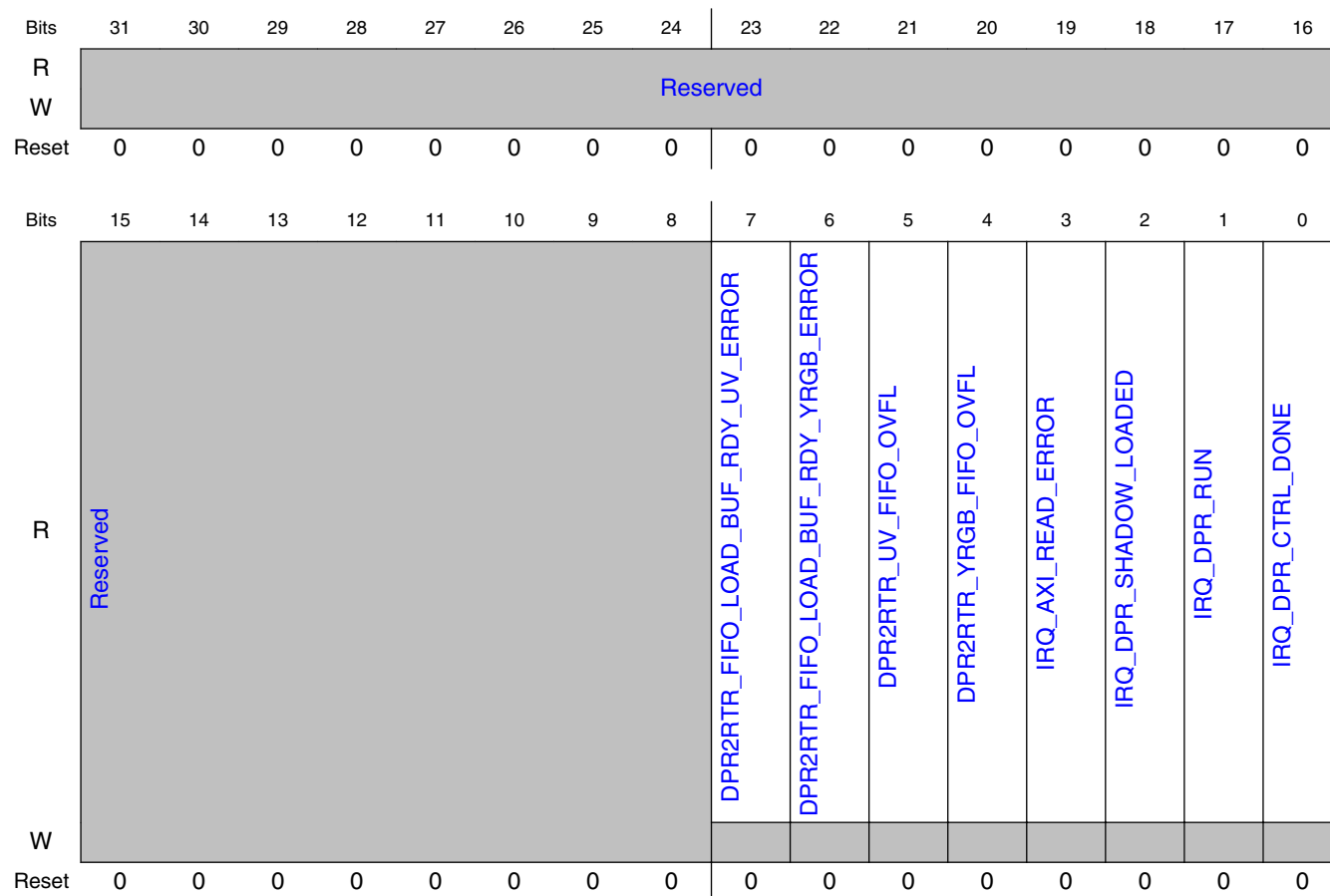
15.7.3.1.11.1 Offset

Register	Offset
IRQ_MASK_STATUS_SET	34h

15.7.3.1.11.2 Function

The IRQ_MASK_STATUS control register contains the masked interrupt status.

15.7.3.1.11.3 Diagram



15.7.3.1.11.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR	DPR to RTRAM Fifo load UV buffer error Masked IRQ Masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.
6 DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	DPR to RTRAM Fifo load YRGB buffer error Masked IRQ Masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow Masked IRQ Masked interrupt status of the DPR2RTR_UV Fifo Overflow interrupt.

Table continues on the next page...

Field	Function
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow Masked IRQ Masked interrupt status of the DPR2RTR_YRGB Fifo Overflow interrupt.
3 IRQ_AXI_READ_ERROR	AXI Read Error Masked IRQ Masked interrupt status of the AXI Read Error interrupt.
2 IRQ_DPR_SHADOW_LOADED	DPR Shadow Loaded Masked IRQ Masked interrupt status of the DPR_SHADOW_LOADED interrupt.
1 IRQ_DPR_RUN	DPR Run Masked IRQ Masked interrupt status of the DPR_RUN interrupt.
0 IRQ_DPR_CTRL_DONE	DPR Control Done Masked IRQ Masked interrupt status of the DPR_CTRL_DONE interrupt.

15.7.3.1.12 Status Register of Masked IRQ (IRQ_MASK_STATUS_CLR)

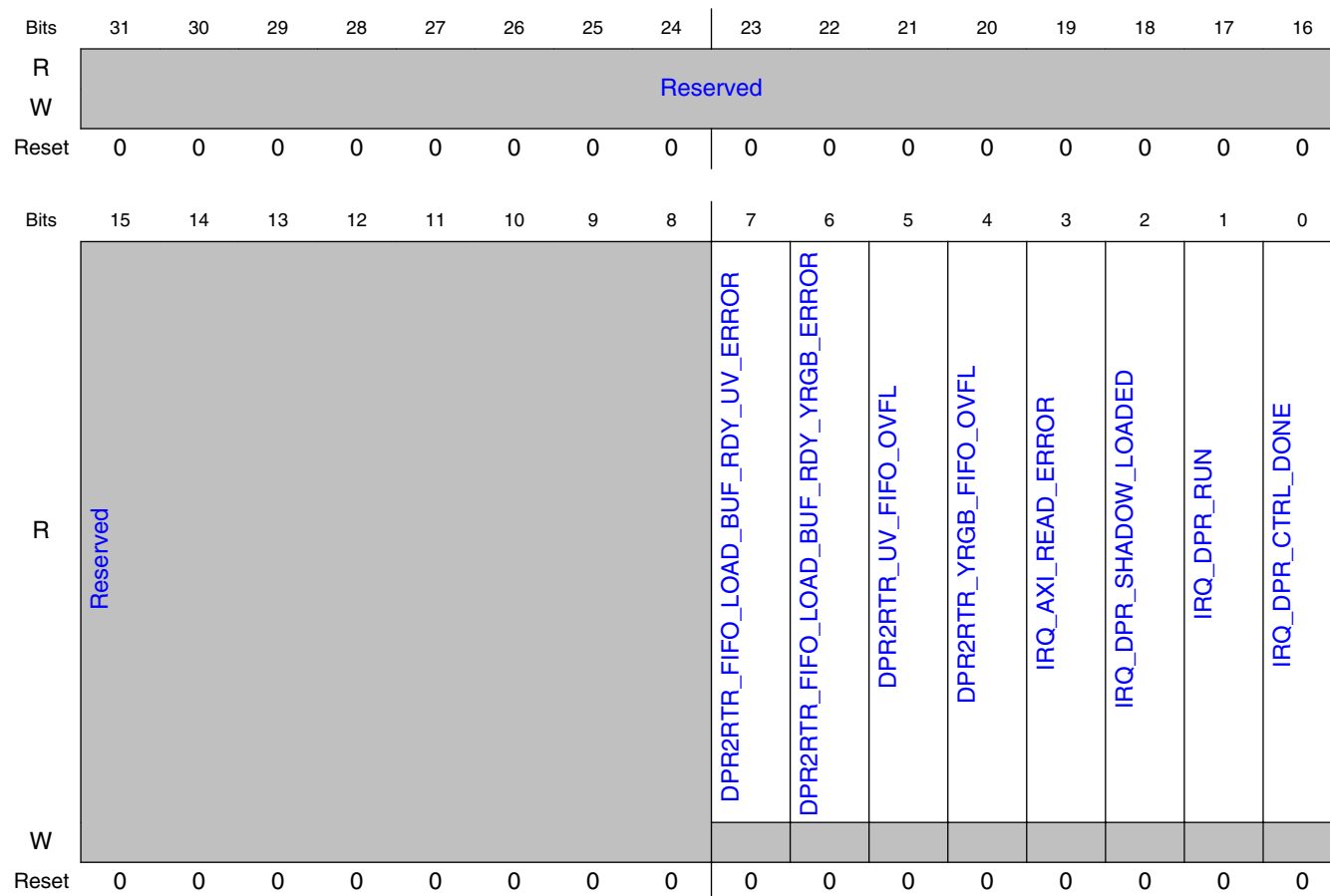
15.7.3.1.12.1 Offset

Register	Offset
IRQ_MASK_STATUS_CLR	38h

15.7.3.1.12.2 Function

The IRQ_MASK_STATUS control register contains the masked interrupt status.

15.7.3.1.12.3 Diagram



15.7.3.1.12.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR	DPR to RTRAM Fifo load UV buffer error Masked IRQ Masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.
6 DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	DPR to RTRAM Fifo load YRGB buffer error Masked IRQ Masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow Masked IRQ Masked interrupt status of the DPR2RTR_UV Fifo Overflow interrupt.

Table continues on the next page...

Field	Function
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow Masked IRQ Masked interrupt status of the DPR2RTR_YRGB Fifo Overflow interrupt.
3 IRQ_AXI_READ_ERROR	AXI Read Error Masked IRQ Masked interrupt status of the AXI Read Error interrupt.
2 IRQ_DPR_SHADOW_LOADED	DPR Shadow Loaded Masked IRQ Masked interrupt status of the DPR_SHADOW_LOADED interrupt.
1 IRQ_DPR_RUN	DPR Run Masked IRQ Masked interrupt status of the DPR_RUN interrupt.
0 IRQ_DPR_CTRL_DONE	DPR Control Done Masked IRQ Masked interrupt status of the DPR_CTRL_DONE interrupt.

15.7.3.1.13 Status Register of Masked IRQ (IRQ_MASK_STATUS_TOG)

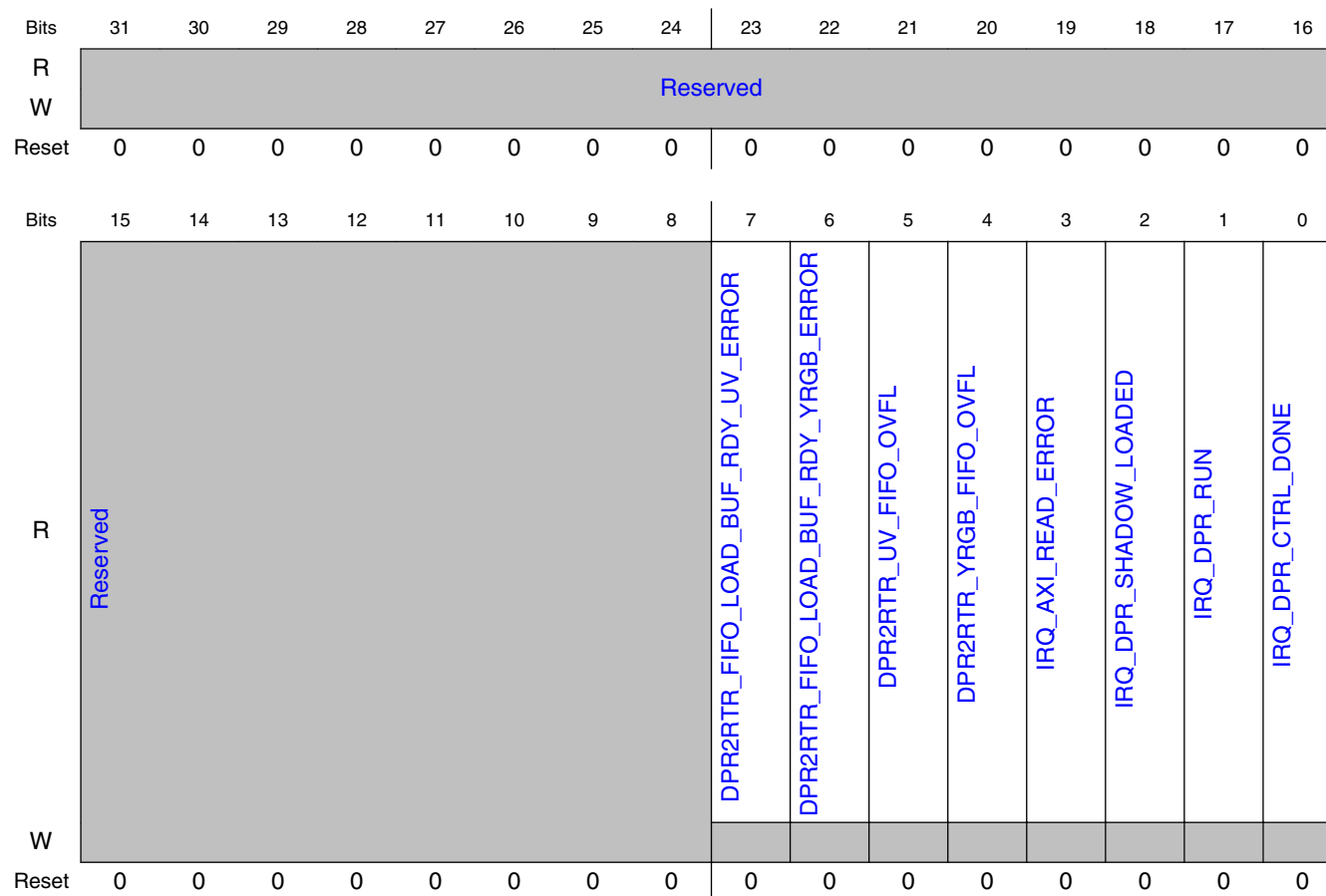
15.7.3.1.13.1 Offset

Register	Offset
IRQ_MASK_STATUS_TOG	3Ch

15.7.3.1.13.2 Function

The IRQ_MASK_STATUS control register contains the masked interrupt status.

15.7.3.1.13.3 Diagram



15.7.3.1.13.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR	DPR to RTRAM Fifo load UV buffer error Masked IRQ Masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.
6 DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	DPR to RTRAM Fifo load YRGB buffer error Masked IRQ Masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow Masked IRQ Masked interrupt status of the DPR2RTR_UV Fifo Overflow interrupt.

Table continues on the next page...

Field	Function
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow Masked IRQ Masked interrupt status of the DPR2RTR_YRGB Fifo Overflow interrupt.
3 IRQ_AXI_READ_ERROR	AXI Read Error Masked IRQ Masked interrupt status of the AXI Read Error interrupt.
2 IRQ_DPR_SHADOW_LOADED	DPR Shadow Loaded Masked IRQ Masked interrupt status of the DPR_SHADOW_LOADED interrupt.
1 IRQ_DPR_RUN	DPR Run Masked IRQ Masked interrupt status of the DPR_RUN interrupt.
0 IRQ_DPR_CTRL_DONE	DPR Control Done Masked IRQ Masked interrupt status of the DPR_CTRL_DONE interrupt.

15.7.3.1.14 Status of Non-Masked IRQ (IRQ_NONMASK_STATUS)

15.7.3.1.14.1 Offset

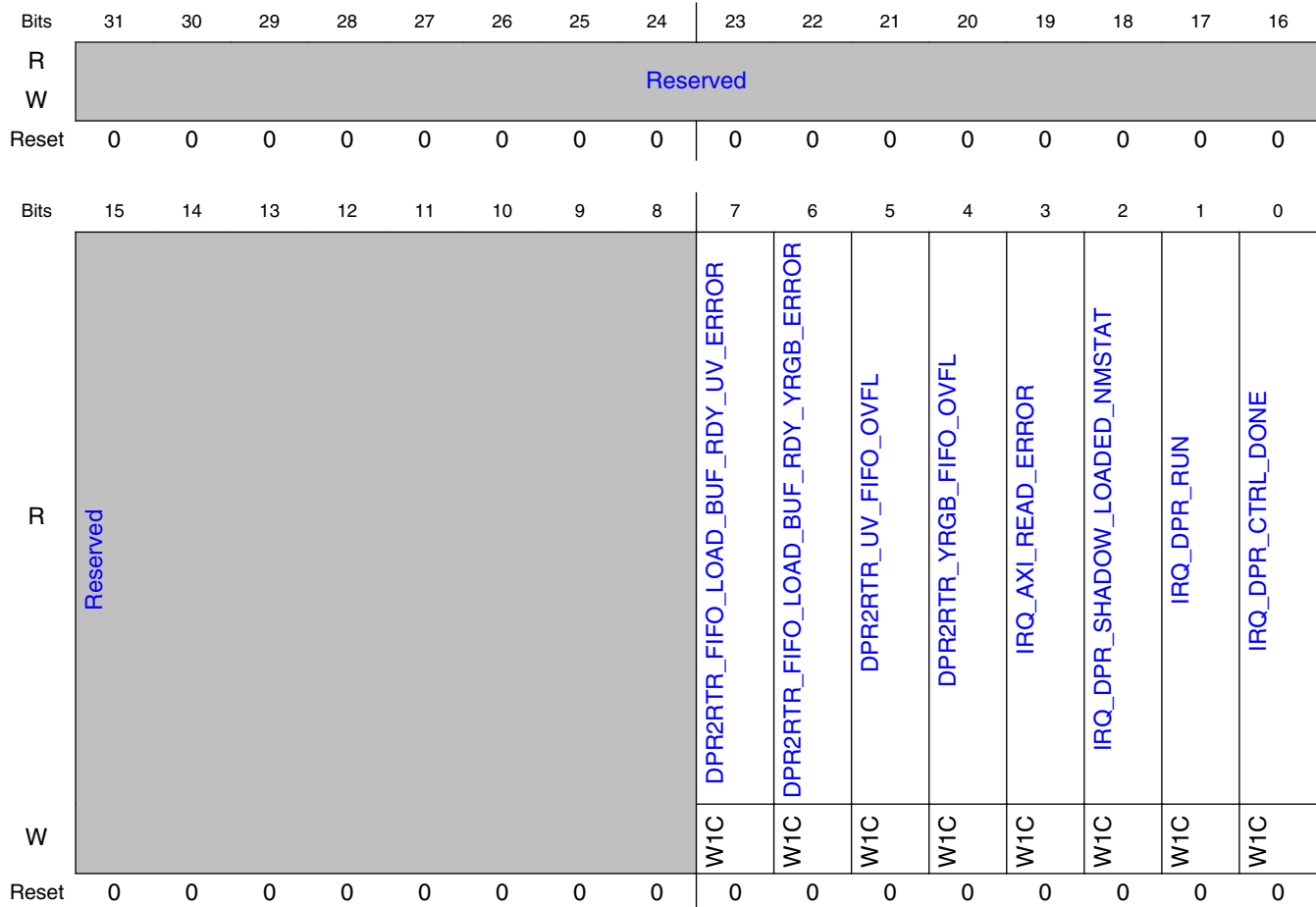
Register	Offset
IRQ_NONMASK_STATUS	40h

15.7.3.1.14.2 Function

The IRQ_NONMASK_STATUS control register contains the raw interrupt status before applying the mask.

Writing a 1 to the SCT's CLEAR mask bit, will clear the associated interrupt.

15.7.3.1.14.3 Diagram



15.7.3.1.14.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR OR	DPR to RTRAM Fifo load UV buffer ready error Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt. When this IRQ is set, the DPR UV buffer ready was set during the 1st 32-bytes instead of the 2nd 32-bytes of the 64-byte payload. This error indicates the following: • pixel data will be loaded incorrectly into the RTRAM • The error is (most likely) due to incorrect programming of the NUM_X_PIX_WIDE value not being evenly divisible by 64-bytes.
6	DPR to RTRAM Fifo load YRGB buffer ready error Non-Masked IRQ

Table continues on the next page...

Field	Function
DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	Non-masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt. When this IRQ is set, the DPR YRGB buffer ready was set during the 1st 32-bytes instead of the 2nd 32-bytes of the 64-byte payload. This error indicates the following: - pixel data will be loaded incorrectly into the RTRAM - The error is (most likely) due to incorrect programming of the NUM_X_PIX_WIDE value not being evenly divisible by 64-bytes.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_UV Fifo Overflow interrupt. When this IRQ is set, the DPR2RTR Fifo Overflowed writing 2PYUV420 UV data.
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_YRGB Fifo Overflow interrupt. When this is set, the DPR2RTR Fifo Overflowed writing YRGB data.
3 IRQ_AXI_READ_ERROR	AXI Read Error Non-Masked IRQ Non-masked interrupt status of the AXI Read Error interrupt. When this IRQ is set, the AXI Read Master received a bus transaction error code.
2 IRQ_DPR_SHADOW_LOADED_NMSTAT	DPR Shadow Loaded Non-Masked IRQ Non-masked interrupt status of the DPR_SHADOW_LOADED interrupt. When this IRQ IS set, the dpr_ctrl shadow control registers have been loaded into the active control registers.
1 IRQ_DPR_RUN	DPR Run Non-Masked IRQ Non-masked interrupt status of the DPR_RUN interrupt. When this IRQ IS set, the dpr_ctrl RUN_EN has been asserted in the control register clock domain. All control registers are considered active at this time.
0 IRQ_DPR_CTRL_DONE	DPR Control Done Non-Masked IRQ Non-masked interrupt status of the DPR_CTRL_DONE interrupt. When this IRQ is set, the dpr_ctrl_prefetch and dpr_ctrl_response blocks are done processing the current frame.

15.7.3.1.15 Status of Non-Masked IRQ (IRQ_NONMASK_STATUS_SET)

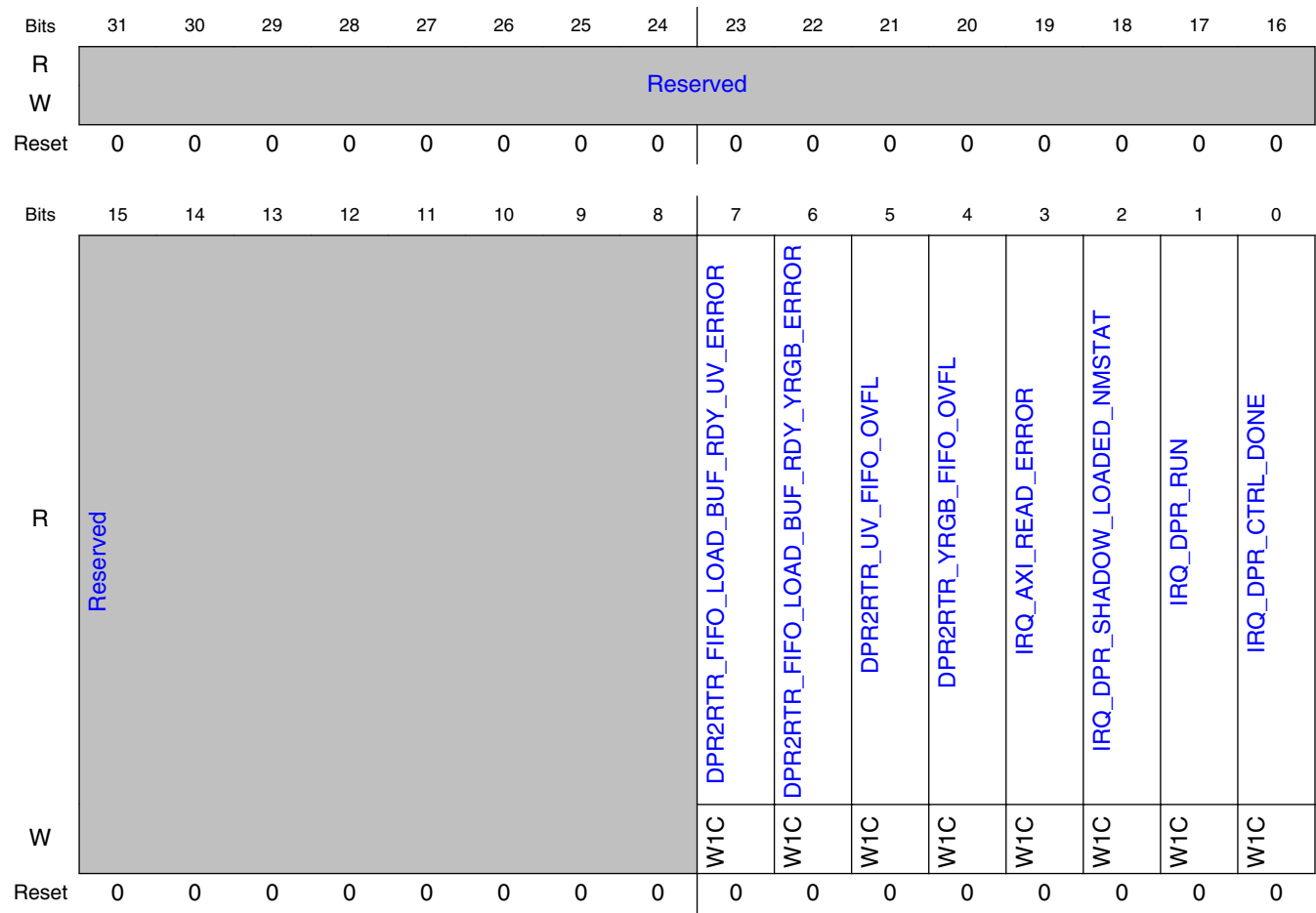
15.7.3.1.15.1 Offset

Register	Offset
IRQ_NONMASK_STATUS_SET	44h

15.7.3.1.15.2 Function

The IRQ_NONMASK_STATUS control register contains the raw interrupt status before applying the mask.
Writing a 1 to the SCT's CLEAR mask bit, will clear the associated interrupt.

15.7.3.1.15.3 Diagram



15.7.3.1.15.4 Fields

Field	Function
31-8 —	Reserved.
7	DPR to RTRAM Fifo load UV buffer ready error Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt.

Table continues on the next page...

Field	Function
DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR OR	When this IRQ is set, the DPR UV buffer ready was set during the 1st 32-bytes instead of the 2nd 32-bytes of the 64-byte payload. This error indicates the following: • pixel data will be loaded incorrectly into the RTRAM • The error is (most likely) due to incorrect programming of the NUM_X_PIX_WIDE value not being evenly divisible by 64-bytes.
6 DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	DPR to RTRAM Fifo load YRGB buffer ready error Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt. When this IRQ is set, the DPR YRGB buffer ready was set during the 1st 32-bytes instead of the 2nd 32-bytes of the 64-byte payload. This error indicates the following: • pixel data will be loaded incorrectly into the RTRAM • The error is (most likely) due to incorrect programming of the NUM_X_PIX_WIDE value not being evenly divisible by 64-bytes.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_UV Fifo Overflow interrupt. When this IRQ is set, the DPR2RTR Fifo Overflowed writing 2PYUV420 UV data.
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_YRGB Fifo Overflow interrupt. When this is set, the DPR2RTR Fifo Overflowed writing YRGB data.
3 IRQ_AXI_READ_ERROR	AXI Read Error Non-Masked IRQ Non-masked interrupt status of the AXI Read Error interrupt. When this IRQ is set, the AXI Read Master received a bus transaction error code.
2 IRQ_DPR_SHADOW_LOADED_NMSTAT	DPR Shadow Loaded Non-Masked IRQ Non-masked interrupt status of the DPR_SHADOW_LOADED interrupt. When this IRQ IS set, the dpr_ctrl shadow control registers have been loaded into the active control registers.
1 IRQ_DPR_RUN	DPR Run Non-Masked IRQ Non-masked interrupt status of the DPR_RUN interrupt. When this IRQ IS set, the dpr_ctrl RUN_EN has been asserted in the control register clock domain. All control registers are considered active at this time.
0 IRQ_DPR_CTRL_DONE	DPR Control Done Non-Masked IRQ Non-masked interrupt status of the DPR_CTRL_DONE interrupt. When this IRQ is set, the dpr_ctrl_prefetch and dpr_ctrl_response blocks are done processing the current frame.

15.7.3.1.16 Status of Non-Masked IRQ (IRQ_NONMASK_STATUS_CLR)

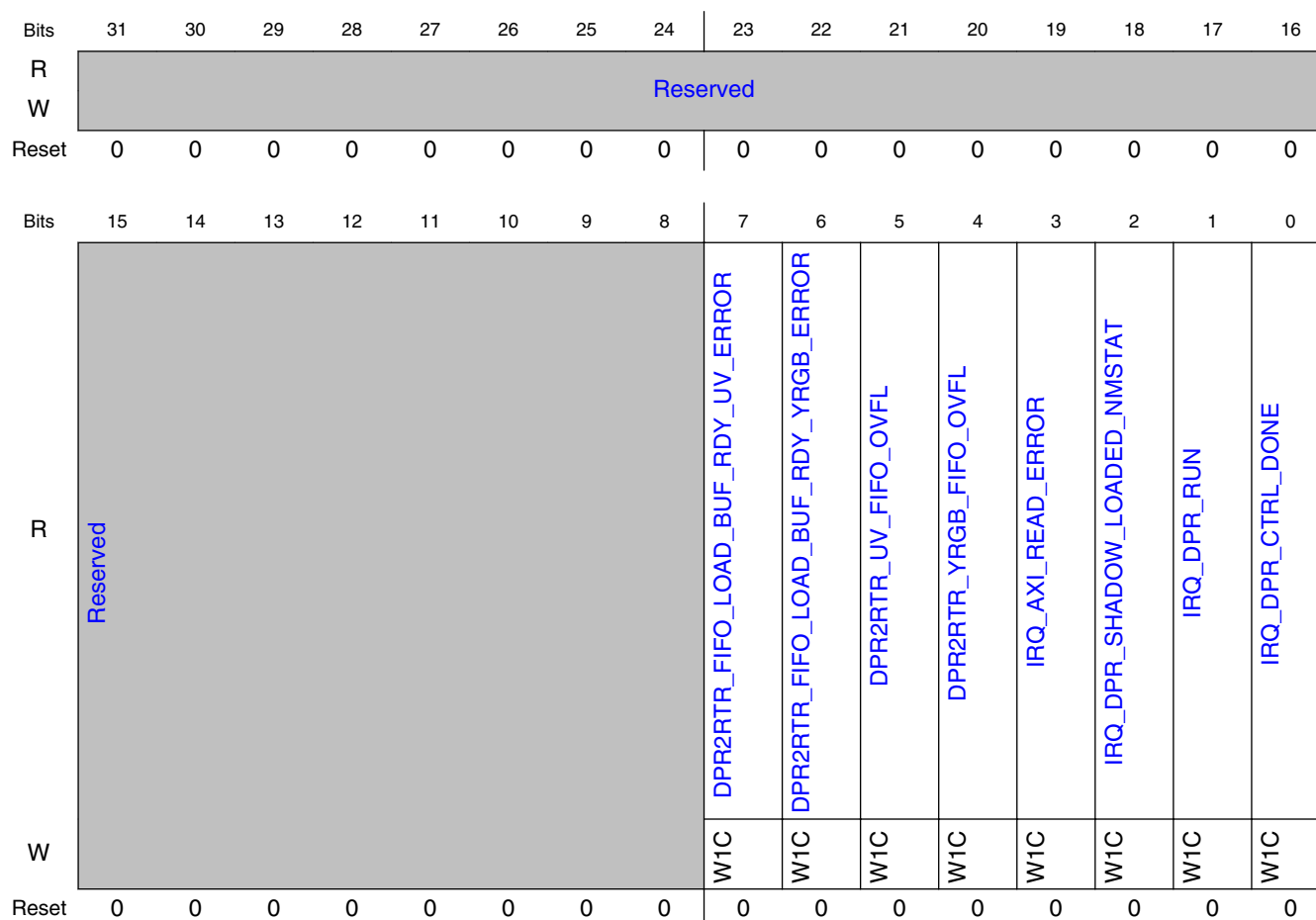
15.7.3.1.16.1 Offset

Register	Offset
IRQ_NONMASK_STAT US_CLR	48h

15.7.3.1.16.2 Function

The IRQ_NONMASK_STATUS control register contains the raw interrupt status before applying the mask.
Writing a 1 to the SCT's CLEAR mask bit, will clear the associated interrupt.

15.7.3.1.16.3 Diagram



15.7.3.1.16.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIF O_LOAD_BUF_ RDY_UV_ERR OR	DPR to RTRAM Fifo load UV buffer ready error Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt. When this IRQ is set, the DPR UV buffer ready was set during the 1st 32-bytes instead of the 2nd 32-bytes of the 64-byte payload. This error indicates the following: • pixel data will be loaded incorrectly into the RTRAM • The error is (most likely) due to incorrect programming of the NUM_X_PIX_WIDE value not being evenly divisible by 64-bytes.
6 DPR2RTR_FIF O_LOAD_BUF_ RDY_YRGB_ER ROR	DPR to RTRAM Fifo load YRGB buffer ready error Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt. When this IRQ is set, the DPR YRGB buffer ready was set during the 1st 32-bytes instead of the 2nd 32-bytes of the 64-byte payload. This error indicates the following: • pixel data will be loaded incorrectly into the RTRAM • The error is (most likely) due to incorrect programming of the NUM_X_PIX_WIDE value not being evenly divisible by 64-bytes.
5 DPR2RTR_UV_ FIFO_OVFL	DPR to RTRAM UV Fifo Overflow Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_UV Fifo Overflow interrupt. When this IRQ is set, the DPR2RTR Fifo Overflowed writing 2PYUV420 UV data.
4 DPR2RTR_YRG B_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_YRGB Fifo Overflow interrupt. When this is set, the DPR2RTR Fifo Overflowed writing YRGB data.
3 IRQ_AXI_READ _ERROR	AXI Read Error Non-Masked IRQ Non-masked interrupt status of the AXI Read Error interrupt. When this IRQ is set, the AXI Read Master received a bus transaction error code.
2 IRQ_DPR_SHA DOW_LOADED _NMSTAT	DPR Shadow Loaded Non-Masked IRQ Non-masked interrupt status of the DPR_SHADOW_LOADED interrupt. When this IRQ IS set, the dpr_ctrl shadow control registers have been loaded into the active control registers.
1 IRQ_DPR_RUN	DPR Run Non-Masked IRQ Non-masked interrupt status of the DPR_RUN interrupt. When this IRQ IS set, the dpr_ctrl RUN_EN has been asserted in the control registrer clock domain. All control registers are considered active at this time.
0	DPR Control Done Non-Masked IRQ

Memory Map and Registers

Field	Function
IRQ_DPR_CTRL_DONE	Non-masked interrupt status of the DPR_CTRL_DONE interrupt. When this IRQ is set, the dpr_ctrl_prefetch and dpr_ctrl_response blocks are done processing the current frame.

15.7.3.1.17 Status of Non-Masked IRQ (IRQ_NONMASK_STATUS_TOG)

15.7.3.1.17.1 Offset

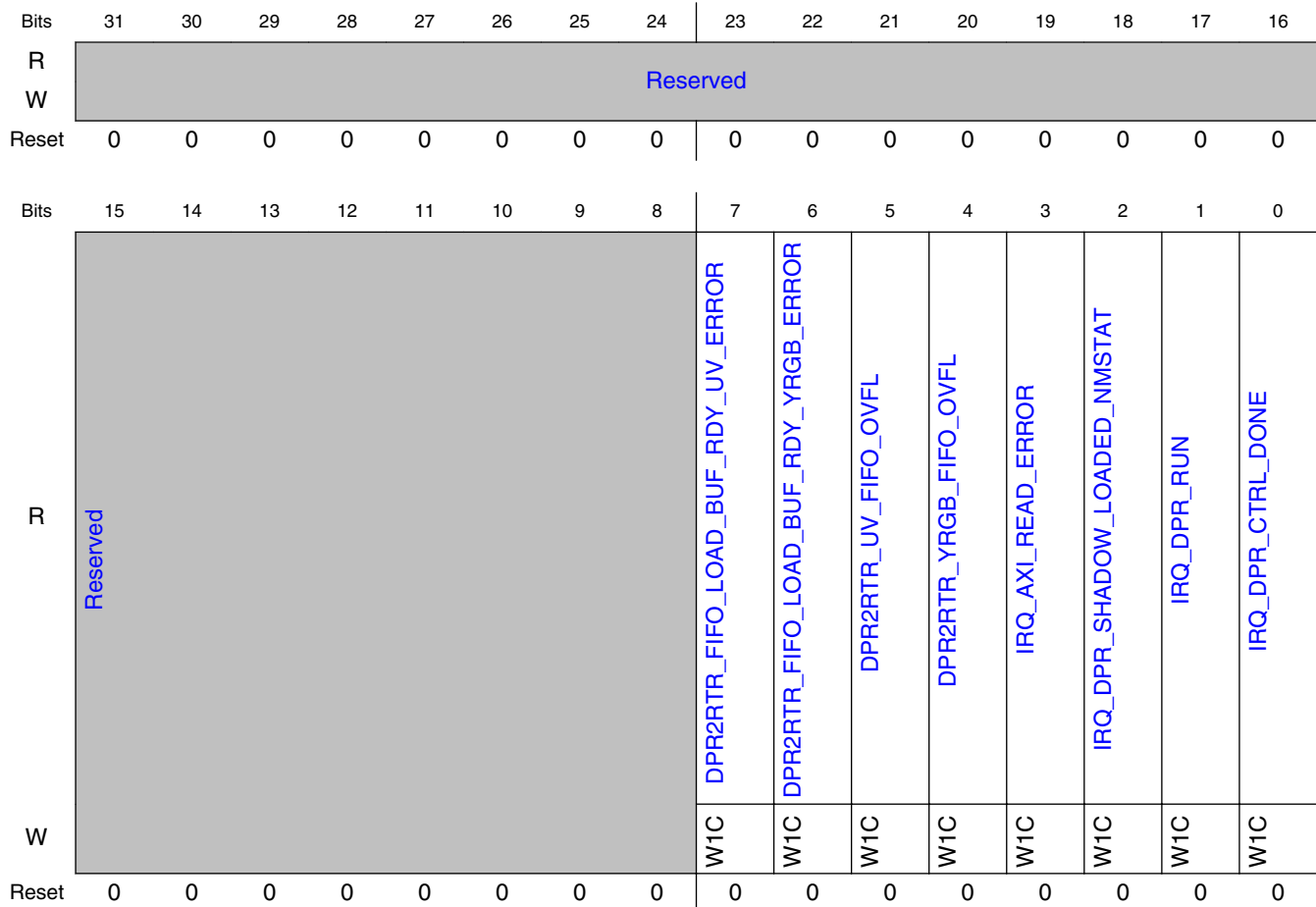
Register	Offset
IRQ_NONMASK_STATUS_TOG	4Ch

15.7.3.1.17.2 Function

The IRQ_NONMASK_STATUS control register contains the raw interrupt status before applying the mask.

Writing a 1 to the SCT's CLEAR mask bit, will clear the associated interrupt.

15.7.3.1.17.3 Diagram



15.7.3.1.17.4 Fields

Field	Function
31-8 —	Reserved.
7 DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR OR	DPR to RTRAM Fifo load UV buffer ready error Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_UV_ERROR interrupt. When this IRQ is set, the DPR UV buffer ready was set during the 1st 32-bytes instead of the 2nd 32-bytes of the 64-byte payload. This error indicates the following: - pixel data will be loaded incorrectly into the RTRAM - The error is (most likely) due to incorrect programming of the NUM_X_PIX_WIDE value not being evenly divisible by 64-bytes.
6	DPR to RTRAM Fifo load YRGB buffer ready error Non-Masked IRQ

Table continues on the next page...

Memory Map and Registers

Field	Function
DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR	Non-masked interrupt status of the DPR2RTR_FIFO_LOAD_BUF_RDY_YRGB_ERROR interrupt. When this IRQ is set, the DPR YRGB buffer ready was set during the 1st 32-bytes instead of the 2nd 32-bytes of the 64-byte payload. This error indicates the following: - pixel data will be loaded incorrectly into the RTRAM - The error is (most likely) due to incorrect programming of the NUM_X_PIX_WIDE value not being evenly divisible by 64-bytes.
5 DPR2RTR_UV_FIFO_OVFL	DPR to RTRAM UV Fifo Overflow Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_UV Fifo Overflow interrupt. When this IRQ is set, the DPR2RTR Fifo Overflowed writing 2PYUV420 UV data.
4 DPR2RTR_YRGB_FIFO_OVFL	DPR to RTRAM YRGB Fifo Overflow Non-Masked IRQ Non-masked interrupt status of the DPR2RTR_YRGB Fifo Overflow interrupt. When this is set, the DPR2RTR Fifo Overflowed writing YRGB data.
3 IRQ_AXI_READ_ERROR	AXI Read Error Non-Masked IRQ Non-masked interrupt status of the AXI Read Error interrupt. When this IRQ is set, the AXI Read Master received a bus transaction error code.
2 IRQ_DPR_SHADOW_LOADED_NMSTAT	DPR Shadow Loaded Non-Masked IRQ Non-masked interrupt status of the DPR_SHADOW_LOADED interrupt. When this IRQ IS set, the dpr_ctrl shadow control registers have been loaded into the active control registers.
1 IRQ_DPR_RUN	DPR Run Non-Masked IRQ Non-masked interrupt status of the DPR_RUN interrupt. When this IRQ IS set, the dpr_ctrl RUN_EN has been asserted in the control register clock domain. All control registers are considered active at this time.
0 IRQ_DPR_CTRL_DONE	DPR Control Done Non-Masked IRQ Non-masked interrupt status of the DPR_CTRL_DONE interrupt. When this IRQ is set, the dpr_ctrl_prefetch and dpr_ctrl_response blocks are done processing the current frame.

15.7.3.1.18 Mode Control 0 (MODE_CTRL0)

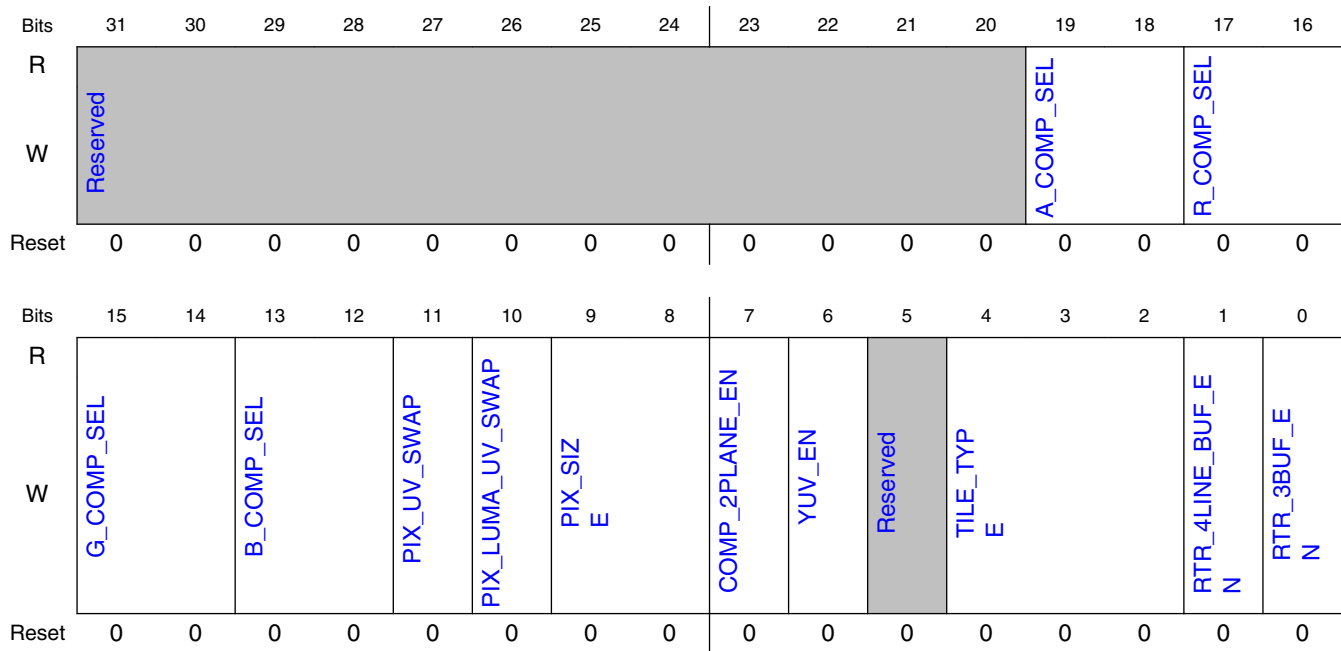
15.7.3.1.18.1 Offset

Register	Offset
MODE_CTRL0	50h

15.7.3.1.18.2 Function

The Mode Control0 register controls mode of operation configuration.

15.7.3.1.18.3 Diagram



15.7.3.1.18.4 Fields

Field	Function
31-20 —	Reserved.
19-18 A_COMP_SEL	A Component Select 32 bit per pixel, such as ARGB888, Alpha component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
17-16 R_COMP_SEL	R Component Select 32 bit per pixel, such as ARGB888, Red component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
15-14 G_COMP_SEL	G Component Select 32 bit per pixel, such as ARGB888, Green component byte select 0x0 = byte 0, bits[7:0]

Table continues on the next page...

Memory Map and Registers

Field	Function
	0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
13-12 B_COMP_SEL	B Component Select When processing 32 bit per pixel, such as ARGB888, Blue component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
11 PIX_UV_SWAP	Pixel UV Swap If processing 1PYUV422 or 2PYUV420 pixels, swap position of Chroma pixel component position. 0x0 = UV 0x1 = VU If processing 1PYUV422, the PIX_UV_SWAP and PIX_LUMA_UV_SWAP combinations will decode over 2 pixels. Default pixel orientation: Pixel_1 = Y0_U0, and Pixel_2 = Y1_V0: (PIX_LUMA_UV_SWAP, PIX_UV_SWAP) 0,0 = Y0_U0 Y1_V0 0,1 = Y0_V0 Y1_U0 1,0 = U0_Y0 V0_Y1 1,1 = V0_Y0 U0_Y1
10 PIX_LUMA_UV_SWAP	Pixel luma/UV position Swap If processing 1PYUV422 pixels, swap position of Luma and Chroma pixel component position. 0x0 = YUYV 0x1 = UYVY
9-8 PIX_SIZE	Pixel Size 0x0 = 8 bits per pixel 0x1 = 16 bits per pixel 0x2 = 32 bits per pixel 0x3 = reserved Note: The source frame buffer for 2PYUV420 10 bit pixels are packed horizontally for every row of pixels. The DPR should be programmed in 8 bits per pixel mode. The source frame buffer for VP9-10 10 bit pixels are unpacked horizontally, 16bpp for every 10 bit pixel. The DPR should be programmed in 16 bits per pixel mode.
7 COMP_2PLANE_EN	Component 2-Plane Enable 0x0 = 1-Plane operation. i.e. YUV422 (1PYUV422) or RGB operation. 0x1 = 2-Plane operation. i.e. YUV420 (2PYUV420), 2PVP9-8 or 2pVP9-10 operation.
6 YUV_EN	YUV Enable 0x0 = xRGB modes of operation. 0x1 = 2-Plane VP9-8 (2PVP9-8), 2-Plane VP9-10 (2PVP9-10), 2-Plane YUV420 (2PYUV420) or 1-Plane YUV422 (1PYUV422) mode of operation.
5 —	Reserved.
4-2 TILE_TYPE	Tile Type 0x0 = Linear (raster) tile. 0x1 = GPU Standard Tile.

Table continues on the next page...

Field	Function
	0x2 = GPU Super Tile. 0x3 = VPU 2PYUV420 Tile. 0x4 = VPU 2-Plane VP9 Tile.
1 RTR_4LINE_BUF_EN	RTRAM Lines Per Buffer 0x0 = 8 RTRAM lines per buffer. This setting would be applied for non-linear VPU tiles. 0x1 = 4 RTRAM lines per buffer. This setting would be applied for non-linear GPU tiles. For linear tile_type, the number of RTRAM lines per buffer depends on the Display Controller Scaler or PRG configuration.
0 RTR_3BUF_EN	RTRAM Buffer Implementation 0x0 = Process 2 RTRAM buffers. 0x1 = Process 3 RTRAM buffers.

15.7.3.1.19 Mode Control 0 (MODE_CTRL0_SET)

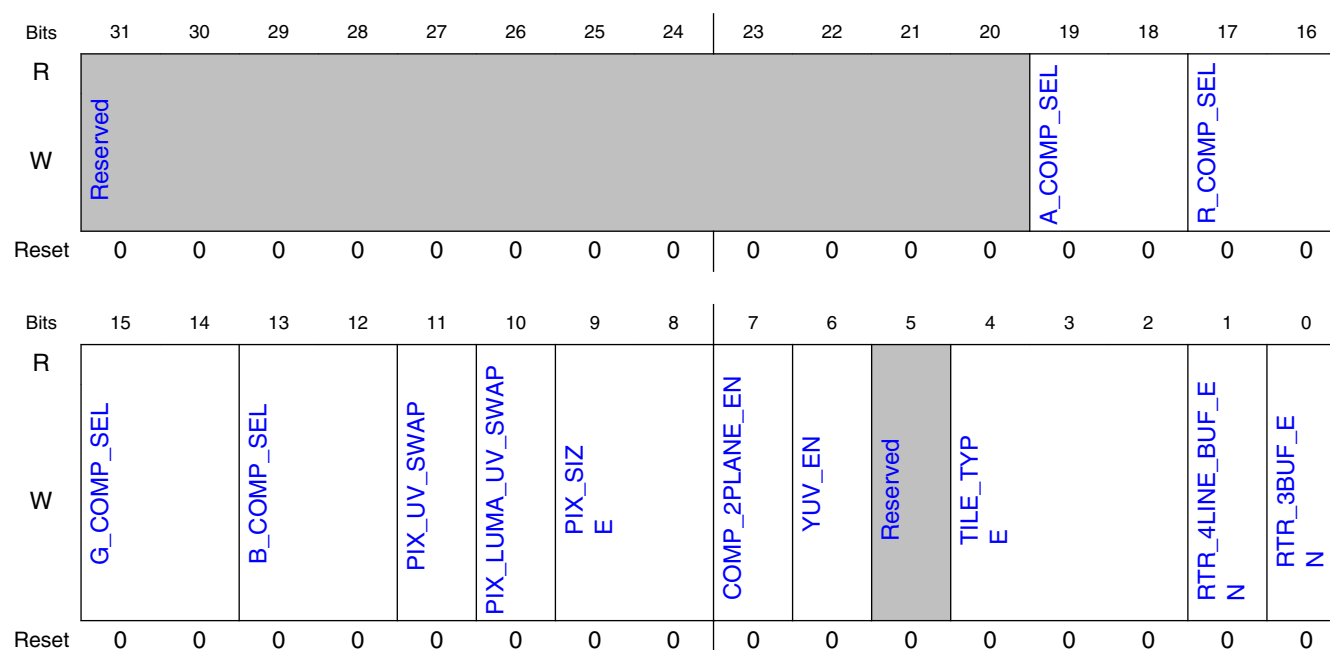
15.7.3.1.19.1 Offset

Register	Offset
MODE_CTRL0_SET	54h

15.7.3.1.19.2 Function

The Mode Control0 register controls mode of operation configuration.

15.7.3.1.19.3 Diagram



15.7.3.1.19.4 Fields

Field	Function
31-20 —	Reserved.
19-18 A_COMP_SEL	A Component Select 32 bit per pixel, such as ARGB888, Alpha component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
17-16 R_COMP_SEL	R Component Select 32 bit per pixel, such as ARGB888, Red component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
15-14 G_COMP_SEL	G Component Select 32 bit per pixel, such as ARGB888, Green component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
13-12 B_COMP_SEL	B Component Select When processing 32 bit per pixel, such as ARGB888, Blue component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
11 PIX_UV_SWAP	Pixel UV Swap If processing 1PYUV422 or 2PYUV420 pixels, swap position of Chroma pixel component position. 0x0 = UV 0x1 = VU If processing 1PYUV422, the PIX_UV_SWAP and PIX_LUMA_UV_SWAP combinations will decode over 2 pixels. Default pixel orientation: Pixel_1 = Y0_U0, and Pixel_2 = Y1_V0: (PIX_LUMA_UV_SWAP, PIX_UV_SWAP) 0,0 = Y0_U0 Y1_V0 0,1 = Y0_V0 Y1_U0 1,0 = U0_Y0 V0_Y1 1,1 = V0_Y0 U0_Y1
10 PIX_LUMA_UV_SWAP	Pixel luma/UV position Swap If processing 1PYUV422 pixels, swap position of Luma and Chroma pixel component position. 0x0 = YUYV 0x1 = UYVY
9-8	Pixel Size

Table continues on the next page...

Field	Function
PIX_SIZE	0x0 = 8 bits per pixel 0x1 = 16 bits per pixel 0x2 = 32 bits per pixel 0x3 = reserved Note: The source frame buffer for 2PYUV420 10 bit pixels are packed horizontally for every row of pixels. The DPR should be programmed in 8 bits per pixel mode. The source frame buffer for VP9-10 10 bit pixels are unpacked horizontally, 16bpp for every 10 bit pixel. The DPR should be programmed in 16 bits per pixel mode.
7 COMP_2PLANE_EN	Component 2-Plane Enable 0x0 = 1-Plane operation. i.e. YUV422 (1PYUV422) or RGB operation. 0x1 = 2-Plane operation. i.e. YUV420 (2PYUV420), 2PVP9-8 or 2pVP9-10 operation.
6 YUV_EN	YUV Enable 0x0 = xRGB modes of operation. 0x1 = 2-Plane VP9-8 (2PVP9-8), 2-Plane VP9-10 (2PVP9-10), 2-Plane YUV420 (2PYUV420) or 1-Plane YUV422 (1PYUV422) mode of operation.
5 —	Reserved.
4-2 TILE_TYPE	Tile Type 0x0 = Linear (raster) tile. 0x1 = GPU Standard Tile. 0x2 = GPU Super Tile. 0x3 = VPU 2PYUV420 Tile. 0x4 = VPU 2-Plane VP9 Tile.
1 RTR_4LINE_BUFFER_EN	RTRAM Lines Per Buffer 0x0 = 8 RTRAM lines per buffer. This setting would be applied for non-linear VPU tiles. 0x1 = 4 RTRAM lines per buffer. This setting would be applied for non-linear GPU tiles. For linear tile_type, the number of RTRAM lines per buffer depends on the Display Controller Scaler or PRG configuration.
0 RTR_3BUF_EN	RTRAM Buffer Implementation 0x0 = Process 2 RTRAM buffers. 0x1 = Process 3 RTRAM buffers.

15.7.3.1.20 Mode Control 0 (MODE_CTRL0_CLR)

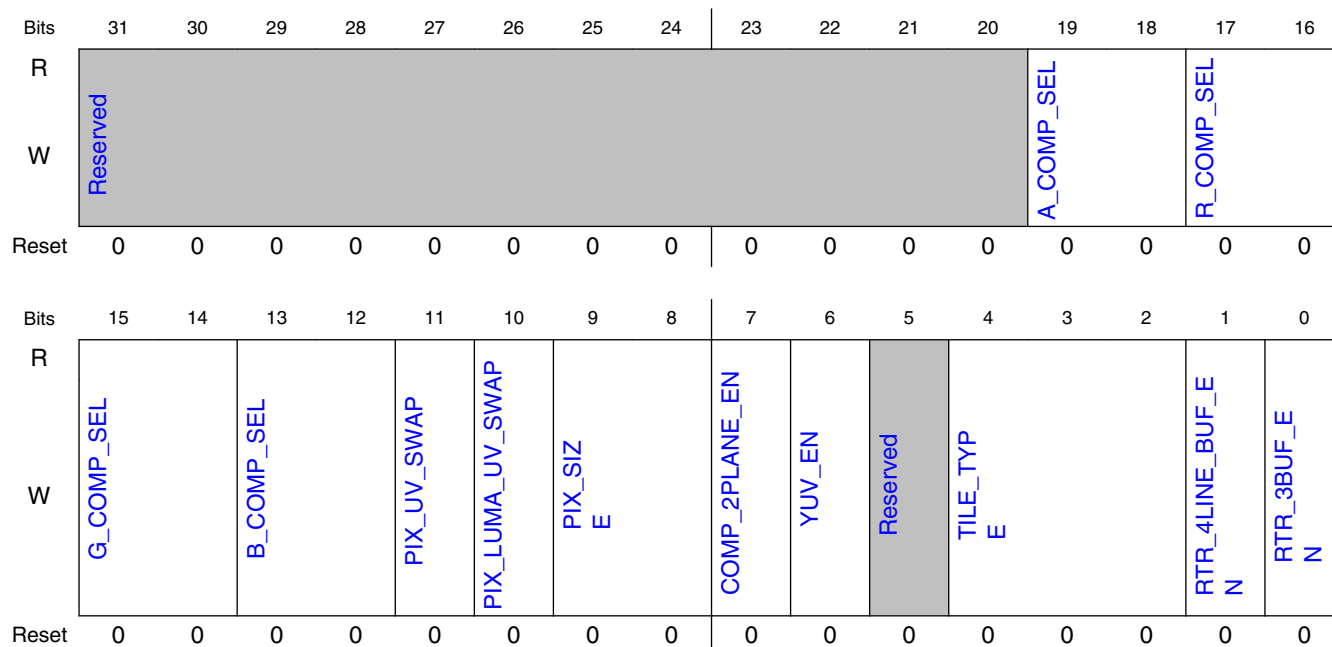
15.7.3.1.20.1 Offset

Register	Offset
MODE_CTRL0_CLR	58h

15.7.3.1.20.2 Function

The Mode Control0 register controls mode of operation configuration.

15.7.3.1.20.3 Diagram



15.7.3.1.20.4 Fields

Field	Function
31-20 —	Reserved.
19-18 A_COMP_SEL	A Component Select 32 bit per pixel, such as ARGB888, Alpha component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
17-16 R_COMP_SEL	R Component Select 32 bit per pixel, such as ARGB888, Red component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
15-14 G_COMP_SEL	G Component Select 32 bit per pixel, such as ARGB888, Green component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
13-12 B_COMP_SEL	B Component Select When processing 32 bit per pixel, such as ARGB888, Blue component byte select 0x0 = byte 0, bits[7:0]

Table continues on the next page...

Field	Function
	0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
11 PIX_UV_SWAP	Pixel UV Swap If processing 1PYUV422 or 2PYUV420 pixels, swap position of Chroma pixel component position. 0x0 = UV 0x1 = VU If processing 1PYUV422, the PIX_UV_SWAP and PIX_LUMA_UV_SWAP combinations will decode over 2 pixels. Default pixel orientation: Pixel_1 = Y0_U0, and Pixel_2 = Y1_V0: (PIX_LUMA_UV_SWAP, PIX_UV_SWAP) 0,0 = Y0_U0 Y1_V0 0,1 = Y0_V0 Y1_U0 1,0 = U0_Y0 V0_Y1 1,1 = V0_Y0 U0_Y1
10 PIX_LUMA_UV_SWAP	Pixel luma/UV position Swap If processing 1PYUV422 pixels, swap position of Luma and Chroma pixel component position. 0x0 = YUYV 0x1 = UYVY
9-8 PIX_SIZE	Pixel Size 0x0 = 8 bits per pixel 0x1 = 16 bits per pixel 0x2 = 32 bits per pixel 0x3 = reserved Note: The source frame buffer for 2PYUV420 10 bit pixels are packed horizontally for every row of pixels. The DPR should be programmed in 8 bits per pixel mode. The source frame buffer for VP9-10 10 bit pixels are unpacked horizontally, 16bpp for every 10 bit pixel. The DPR should be programmed in 16 bits per pixel mode.
7 COMP_2PLANE_EN	Component 2-Plane Enable 0x0 = 1-Plane operation. i.e. YUV422 (1PYUV422) or RGB operation. 0x1 = 2-Plane operation. i.e. YUV420 (2PYUV420), 2PVP9-8 or 2pVP9-10 operation.
6 YUV_EN	YUV Enable 0x0 = xRGB modes of operation. 0x1 = 2-Plane VP9-8 (2PVP9-8), 2-Plane VP9-10 (2PVP9-10), 2-Plane YUV420 (2PYUV420) or 1-Plane YUV422 (1PYUV422) mode of operation.
5 —	Reserved.
4-2 TILE_TYPE	Tile Type 0x0 = Linear (raster) tile. 0x1 = GPU Standard Tile. 0x2 = GPU Super Tile. 0x3 = VPU 2PYUV420 Tile. 0x4 = VPU 2-Plane VP9 Tile.
1 RTR_4LINE_BUFFER_EN	RTRAM Lines Per Buffer 0x0 = 8 RTRAM lines per buffer. This setting would be applied for non-linear VPU tiles. 0x1 = 4 RTRAM lines per buffer. This setting would be applied for non-linear GPU tiles.

Table continues on the next page...

Memory Map and Registers

Field	Function
	For linear tile_type, the number of RTRAM lines per buffer depends on the Display Controller Scaler or PRG configuration.
0 RTR_3BUF_EN	RTRAM Buffer Implementation 0x0 = Process 2 RTRAM buffers. 0x1 = Process 3 RTRAM buffers.

15.7.3.1.21 Mode Control 0 (MODE_CTRL0_TOG)

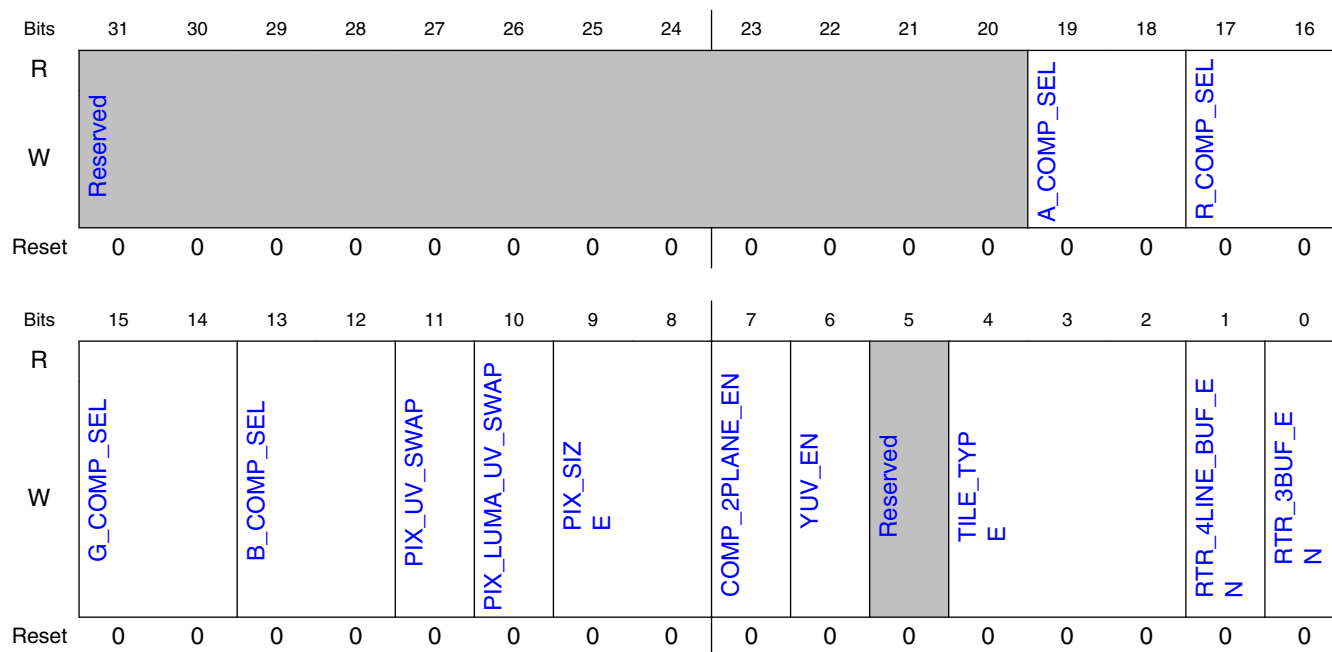
15.7.3.1.21.1 Offset

Register	Offset
MODE_CTRL0_TOG	5Ch

15.7.3.1.21.2 Function

The Mode Control0 register controls mode of operation configuration.

15.7.3.1.21.3 Diagram



15.7.3.1.21.4 Fields

Field	Function
31-20 —	Reserved.
19-18 A_COMP_SEL	A Component Select 32 bit per pixel, such as ARGB888, Alpha component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
17-16 R_COMP_SEL	R Component Select 32 bit per pixel, such as ARGB888, Red component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
15-14 G_COMP_SEL	G Component Select 32 bit per pixel, such as ARGB888, Green component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
13-12 B_COMP_SEL	B Component Select When processing 32 bit per pixel, such as ARGB888, Blue component byte select 0x0 = byte 0, bits[7:0] 0x1 = byte 1, bits[15:8] 0x2 = byte 2, bits[23:16] 0x3 = byte 3, bits[31:24]
11 PIX_UV_SWAP	Pixel UV Swap If processing 1PYUV422 or 2PYUV420 pixels, swap position of Chroma pixel component position. 0x0 = UV 0x1 = VU If processing 1PYUV422, the PIX_UV_SWAP and PIX_LUMA_UV_SWAP combinations will decode over 2 pixels. Default pixel orientation: Pixel_1 = Y0_U0, and Pixel_2 = Y1_V0: (PIX_LUMA_UV_SWAP, PIX_UV_SWAP) 0,0 = Y0_U0 Y1_V0 0,1 = Y0_V0 Y1_U0 1,0 = U0_Y0 V0_Y1 1,1 = V0_Y0 U0_Y1
10 PIX_LUMA_UV_SWAP	Pixel luma/UV position Swap If processing 1PYUV422 pixels, swap position of Luma and Chroma pixel component position. 0x0 = YUYV 0x1 = UYVY
9-8 PIX_SIZE	Pixel Size 0x0 = 8 bits per pixel 0x1 = 16 bits per pixel

Table continues on the next page...

Memory Map and Registers

Field	Function
	0x2 = 32 bits per pixel 0x3 = reserved Note: The source frame buffer for 2PYUV420 10 bit pixels are packed horizontally for every row of pixels. The DPR should be programmed in 8 bits per pixel mode. The source frame buffer for VP9-10 10 bit pixels are unpacked horizontally, 16bpp for every 10 bit pixel. The DPR should be programmed in 16 bits per pixel mode.
7 COMP_2PLANE_EN	Component 2-Plane Enable 0x0 = 1-Plane operation. i.e. YUV422 (1PYUV422) or RGB operation. 0x1 = 2-Plane operation. i.e. YUV420 (2PYUV420), 2PVP9-8 or 2pVP9-10 operation.
6 YUV_EN	YUV Enable 0x0 = xRGB modes of operation. 0x1 = 2-Plane VP9-8 (2PVP9-8), 2-Plane VP9-10 (2PVP9-10), 2-Plane YUV420 (2PYUV420) or 1-Plane YUV422 (1PYUV422) mode of operation.
5 —	Reserved.
4-2 TILE_TYPE	Tile Type 0x0 = Linear (raster) tile. 0x1 = GPU Standard Tile. 0x2 = GPU Super Tile. 0x3 = VPU 2PYUV420 Tile. 0x4 = VPU 2-Plane VP9 Tile.
1 RTR_4LINE_BUFFER_EN	RTRAM Lines Per Buffer 0x0 = 8 RTRAM lines per buffer. This setting would be applied for non-linear VPU tiles. 0x1 = 4 RTRAM lines per buffer. This setting would be applied for non-linear GPU tiles. For linear tile_type, the number of RTRAM lines per buffer depends on the Display Controller Scaler or PRG configuration.
0 RTR_3BUF_EN	RTRAM Buffer Implementation 0x0 = Process 2 RTRAM buffers. 0x1 = Process 3 RTRAM buffers.

15.7.3.1.22 Frame Control 0 (FRAME_CTRL0)

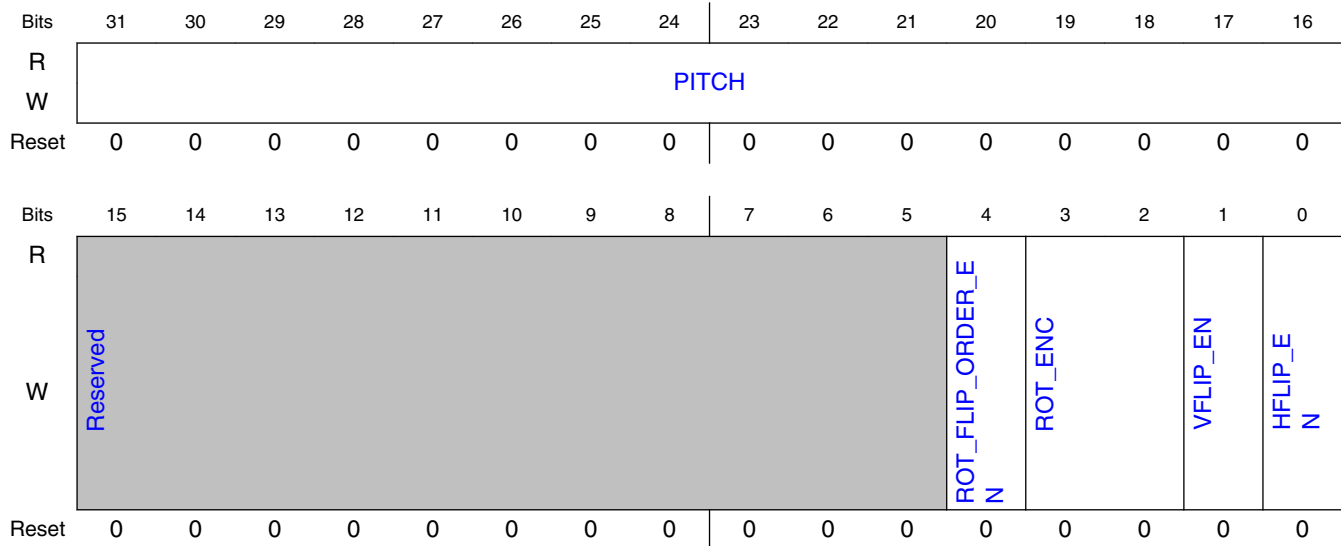
15.7.3.1.22.1 Offset

Register	Offset
FRAME_CTRL0	70h

15.7.3.1.22.2 Function

The Frame Control0 register controls fields that are common to processing both 1-Plane and 2-Plane image processing.

15.7.3.1.22.3 Diagram



15.7.3.1.22.4 Fields

Field	Function
31-16 PITCH	Image Pitch The number of bytes to next horizontal line. First calculate pitch as usual for a raster image <pre> bpp = (PIX_SIZE==2) ? 4 // 32bpp (PIX_SIZE==1) ? 2 // 16bpp ? 1; // 8bpp image_pitch = x_wide * bpp; </pre> Second calculate offset for GPU tile types // Calculate pixel offset for GPU Super Tile to prevent address alising of pixels // The Super Tile is 64x64 pixel block, so width and height must be divisible by 64 to prevent alising. // decode number of pixels in a 64byte boundary <pre> num_pix_x_dir = ((PIX_SIZE==2) && (TILE_TYPE==2)) ? 64 : // GPU ((PIX_SIZE==1) && (TILE_TYPE==2)) ? 64 : // GPU 32; // 8bpp super tile must be divisible by 64 pixels in x-direction to prevent alising </pre> 16bpp super tile must be divisible by 64 pixels in x-direction to prevent alising <pre> image_pitch; // </pre> // modulus of pix_x_width by number of pixels in a 64byte boundary. <pre> pix_x_mod = image_width % num_pix_x_dir; // number of pixels to offset in x-direction. pix_x_offset = (pix_x_mod == 0) ? 0 : (num_pix_x_dir - pix_x_mod); </pre> Third calculate final pitch value <pre> PITCH = (TILE_TYPE==2) pix_x_offset * bpp </pre>

Table continues on the next page...

Memory Map and Registers

Field	Function
	<pre> (TILE_TYPE==1) && (PIX_SIZE==2) && ((x_wide%4)>0) ? ({x_wide[15:2],2'h0} + 4) * bpp : (TILE_TYPE==1) && (PIX_SIZE==1) && ((x_wide%8)>0) ? ({x_wide[15:3],3'h0} + 8) * bpp : ((TILE_TYPE==3) (TILE_TYPE==4)) && ((x_wide%8)>0) ? ({x_wide[15:3],3'h0} + 8) * bpp : image_pitch; </pre>
15-5 —	Reserved.
4 ROT_FLIP_ORDER_EN	Rotation Flip Order The order of operation for rotate and/or horizontal/vertical flip of the output image 0x0 = Rotate followed by horizontal/vertical flip 0x1 = horizontal/vertical flip followed by rotate
3-2 ROT_ENC	Encoded Rotation Rotate the output image 0x0 = Rotate by 0 degrees 0x1 = Rotate by 90 degrees counter clock wise 0x2 = Rotate by 180 degrees counter clock wise 0x3 = Rotate by 270 degrees counter clock wise Rotation by 90 and 270 degrees is not supported for the following: • All linear (raster) tile modes of operation • 16bpp GPU Standard Tile mode of operation • 16bpp GPU Super Tile mode of operation • VPU 2PYUV420 10bpp mode • VPU 2PVP9 8bpp mode • VPU 2PVP9 10bpp mode
1 VFLIP_EN	Vertical Flip Enable Vertical Flip the output image. 0x0 = Disable vertical flip 0x1 = Enable Vertical flip
0 HFLIP_EN	Horizontal Flip Enable Horizontal Flip the output image. 0x0 = Disable horizontal flip 0x1 = Enable Horizontal flip Note: HFLIP is not supported for the following: • VPU 2PYUV420 10bpp mode • VPU 2-Plane VP9 10bpp mode

15.7.3.1.23 Frame Control 0 (FRAME_CTRL0_SET)

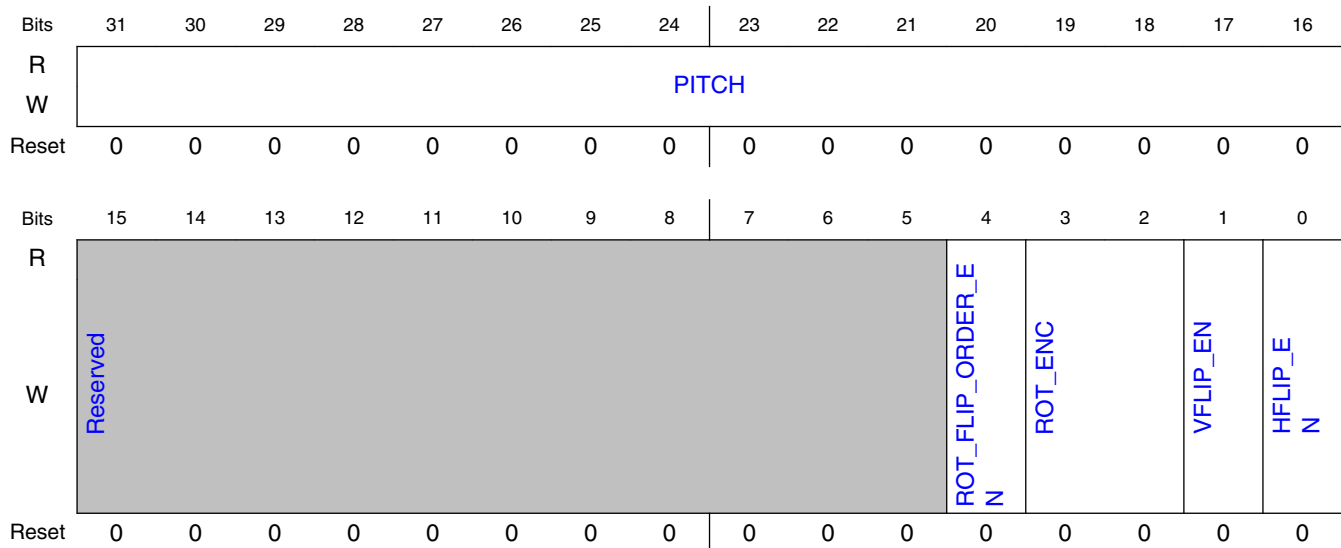
15.7.3.1.23.1 Offset

Register	Offset
FRAME_CTRL0_SET	74h

15.7.3.1.23.2 Function

The Frame Control0 register controls fields that are common to processing both 1-Plane and 2-Plane image processing.

15.7.3.1.23.3 Diagram



15.7.3.1.23.4 Fields

Field	Function
31-16 PITCH	Image Pitch The number of bytes to next horizontal line. First calculate pitch as usual for a raster image <pre> bpp = (PIX_SIZE==2) ? 4 // 32bpp (PIX_SIZE==1) ? 2 // 16bpp ? 1; // 8bpp image_pitch = x_wide * bpp; Second calculate offset for GPU tile types // Calculate pixel offset for GPU Super Tile to prevent address aliasing of pixels // The Super Tile is 64x64 pixel block, so width and height must be divisible by 64 to prevent aliasing. // decode number of pixels in a 64byte boundary num_pix_x_dir = ((PIX_SIZE==2) && (TILE_TYPE==2)) ? 64 : // GPU 32bpp super tile must be divisible by 64 pixels in x-direction to prevent aliasing ((PIX_SIZE==1) && (TILE_TYPE==2)) ? 64 : // GPU 16bpp super tile must be divisible by 64 pixels in x-direction to prevent aliasing image_pitch; // </pre>

Table continues on the next page...

Memory Map and Registers

Field	Function
	<pre> // modulus of pix_x_width by number of pixels in a 64byte boundary. pix_x_mod = image_width % num_pix_x_dir; // number of pixels to offset in x-direction. pix_x_offset = (pix_x_mod == 0) ? 0 : (num_pix_x_dir - pix_x_mod); Third calculate final pitch value PITCH = (TILE_TYPE==2) pix_x_offset * bpp : (TILE_TYPE==1) && (PIX_SIZE==2) && ((x_wide%4)>0) ? ({x_wide[15:2],2'h0} + 4) * bpp : (TILE_TYPE==1) && (PIX_SIZE==1) && ((x_wide%8)>0) ? ({x_wide[15:3],3'h0} + 8) * bpp : ((TILE_TYPE==3) (TILE_TYPE==4)) && ((x_wide%8)>0) ({x_wide[15:3],3'h0} + 8) * bpp : image_pitch; </pre>
15-5 —	Reserved.
4 ROT_FLIP_ORDER_EN	Rotation Flip Order The order of operation for rotate and/or horizontal/vertical flip of the output image 0x0 = Rotate followed by horizontal/vertical flip 0x1 = horizontal/vertical flip followed by rotate
3-2 ROT_ENC	Encoded Rotation Rotate the output image 0x0 = Rotate by 0 degrees 0x1 = Rotate by 90 degrees counter clock wise 0x2 = Rotate by 180 degrees counter clock wise 0x3 = Rotate by 270 degrees counter clock wise Rotation by 90 and 270 degrees is not supported for the following: • All linear (raster) tile modes of operation • 16bpp GPU Standard Tile mode of operation • 16bpp GPU Super Tile mode of operation • VPU 2PYUV420 10bpp mode • VPU 2PVP9 8bpp mode • VPU 2PVP9 10bpp mode
1 VFLIP_EN	Vertical Flip Enable Vertical Flip the output image. 0x0 = Disable vertical flip 0x1 = Enable Vertical flip
0 HFLIP_EN	Horizontal Flip Enable Horizontal Flip the output image. 0x0 = Disable horizontal flip 0x1 = Enable Horizontal flip Note: HFLIP is not supported for the following: • VPU 2PYUV420 10bpp mode • VPU 2-Plane VP9 10bpp mode

15.7.3.1.24 Frame Control 0 (FRAME_CTRL0_CLR)

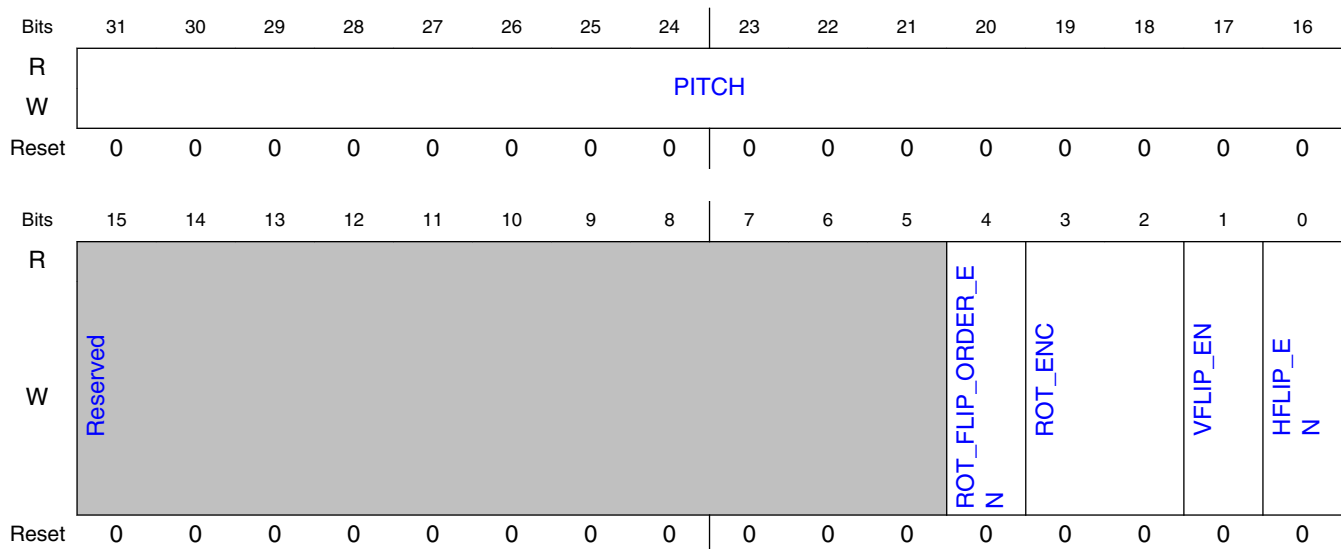
15.7.3.1.24.1 Offset

Register	Offset
FRAME_CTRL0_CLR	78h

15.7.3.1.24.2 Function

The Frame Control0 register controls fields that are common to processing both 1-Plane and 2-Plane image processing.

15.7.3.1.24.3 Diagram



15.7.3.1.24.4 Fields

Field	Function
31-16 PITCH	Image Pitch The number of bytes to next horizontal line. First calculate pitch as usual for a raster image <pre> bpp = (PIX_SIZE==2) ? 4 // 32bpp (PIX_SIZE==1) ? 2 // 16bpp ? 1; // 8bpp image_pitch = x_wide * bpp; </pre> Second calculate offset for GPU tile types // Calculate pixel offset for GPU Super Tile to prevent address alising of pixels

Table continues on the next page...

Memory Map and Registers

Field	Function
	<pre> // The Super Tile is 64x64 pixel block, so width and height must be divisible by 64 to prevent alising. // decode number of pixels in a 64byte boundary num_pix_x_dir = ((PIX_SIZE==2) && (TILE_TYPE==2)) ? 64 : // GPU 32bpp super tile must be divisible by 64 pixels in x-direction to prevent aliasing ((PIX_SIZE==1) && (TILE_TYPE==2)) ? 64 : // GPU 16bpp super tile must be divisible by 64 pixels in x-direction to prevent aliasing image_pitch; // // modulus of pix_x_width by number of pixels in a 64byte boundary. pix_x_mod = image_width % num_pix_x_dir; // number of pixels to offset in x-direction. pix_x_offset = (pix_x_mod == 0) ? 0 : (num_pix_x_dir - pix_x_mod); Third calculate final pitch value PITCH = (TILE_TYPE==2) pix_x_offset * bpp : (TILE_TYPE==1) && (PIX_SIZE==2) && ((x_wide%4)>0) ? ({x_wide[15:2],2'h0} + 4) * bpp : (TILE_TYPE==1) && (PIX_SIZE==1) && ((x_wide%8)>0) ? ({x_wide[15:3],3'h0} + 8) * bpp : ((TILE_TYPE==3) (TILE_TYPE==4)) && ((x_wide%8)>0) ({x_wide[15:3],3'h0} + 8) * bpp : ? image_pitch; </pre>
15-5 —	Reserved.
4 ROT_FLIP_OR DER_EN	Rotation Flip Order The order of operation for rotate and/or horizontal/vertical flip of the output image 0x0 = Rotate followed by horizontal/vertical flip 0x1 = horizontal/vertical flip followed by rotate
3-2 ROT_ENC	Encoded Rotation Rotate the output image 0x0 = Rotate by 0 degrees 0x1 = Rotate by 90 degrees counter clock wise 0x2 = Rotate by 180 degrees counter clock wise 0x3 = Rotate by 270 degrees counter clock wise Rotation by 90 and 270 degrees is not supported for the following: • All linear (raster) tile modes of operation • 16bpp GPU Standard Tile mode of operation • 16bpp GPU Super Tile mode of operation • VPU 2PYUV420 10bpp mode • VPU 2PVP9 8bpp mode • VPU 2PVP9 10bpp mode
1 VFLIP_EN	Vertical Flip Enable Vertical Flip the output image. 0x0 = Disable vertical flip 0x1 = Enable Vertical flip
0	Horizontal Flip Enable

Field	Function
HFLIP_EN	Horizontal Flip the output image. 0x0 = Disable horizontal flip 0x1 = Enable Horizontal flip Note: HFLIP is not supported for the following: • VPU 2PYUV420 10bpp mode • VPU 2-Plane VP9 10bpp mode

15.7.3.1.25 Frame Control 0 (FRAME_CTRL0_TOG)

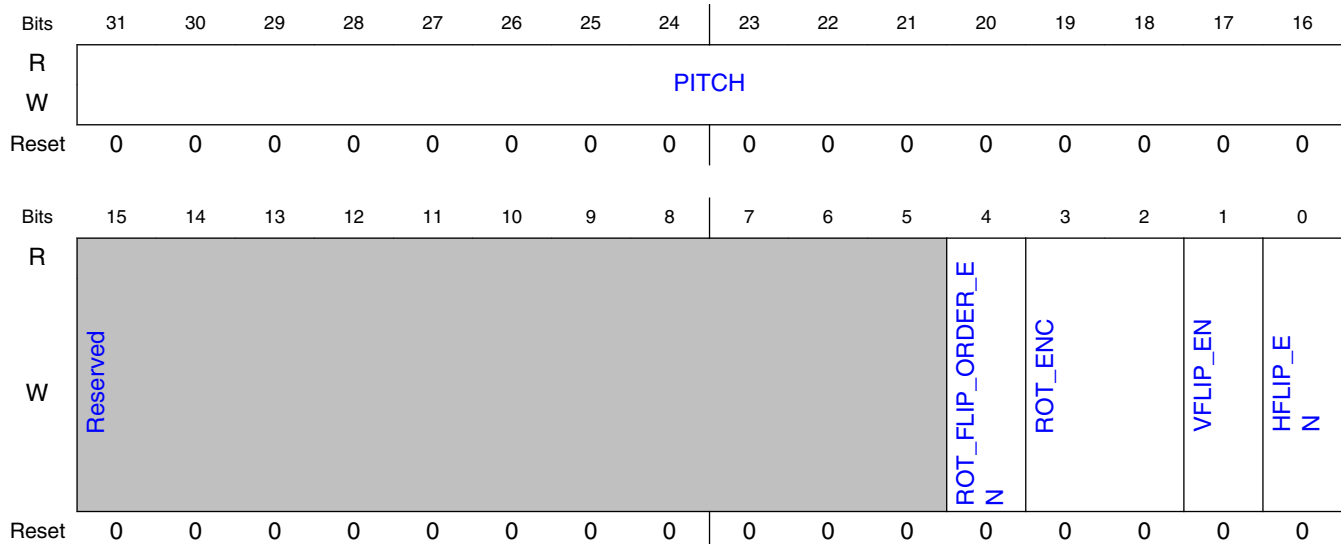
15.7.3.1.25.1 Offset

Register	Offset
FRAME_CTRL0_TOG	7Ch

15.7.3.1.25.2 Function

The Frame Control0 register controls fields that are common to processing both 1-Plane and 2-Plane image processing.

15.7.3.1.25.3 Diagram



15.7.3.1.25.4 Fields

Field	Function
31-16 PITCH	Image Pitch The number of bytes to next horizontal line. First calculate pitch as usual for a raster image <pre> bpp = (PIX_SIZE==2) ? 4 // 32bpp (PIX_SIZE==1) ? 2 // 16bpp ? 1; // 8bpp image_pitch = x_wide * bpp; </pre> Second calculate offset for GPU tile types // Calculate pixel offset for GPU Super Tile to prevent address alising of pixels // The Super Tile is 64x64 pixel block, so width and height must be divisible by 64 to prevent alising. // decode number of pixels in a 64byte boundary <pre> num_pix_x_dir = ((PIX_SIZE==2) && (TILE_TYPE==2)) ? 64 : // GPU 32bpp super tile must be divisible by 64 pixels in x-direction to prevent aliasing ((PIX_SIZE==1) && (TILE_TYPE==2)) ? 64 : // GPU 16bpp super tile must be divisible by 64 pixels in x-direction to prevent aliasing </pre> image_pitch; // // modulus of pix_x_width by number of pixels in a 64byte boundary. <pre> pix_x_mod = image_width % num_pix_x_dir; // number of pixels to offset in x-direction. pix_x_offset = (pix_x_mod == 0) ? 0 : (num_pix_x_dir - pix_x_mod); </pre> Third calculate final pitch value <pre> PITCH = (TILE_TYPE==2) pix_x_offset * bpp : (TILE_TYPE==1) && (PIX_SIZE==2) && ((x_wide%4)>0) ? ({x_wide[15:2],2'h0} + 4) * bpp : (TILE_TYPE==1) && (PIX_SIZE==1) && ((x_wide%8)>0) ? ({x_wide[15:3],3'h0} + 8) * bpp : ((TILE_TYPE==3) (TILE_TYPE==4)) && ((x_wide%8)>0) ({x_wide[15:3],3'h0} + 8) * bpp : </pre> image_pitch;
15-5 —	Reserved.
4 ROT_FLIP_OR DER_EN	Rotation Flip Order The order of operation for rotate and/or horizontal/vertical flip of the output image 0x0 = Rotate followed by horizontal/vertical flip 0x1 = horizontal/vertical flip followed by rotate
3-2 ROT_ENC	Encoded Rotation Rotate the output image 0x0 = Rotate by 0 degrees 0x1 = Rotate by 90 degrees counter clock wise 0x2 = Rotate by 180 degrees counter clock wise 0x3 = Rotate by 270 degrees counter clock wise

Table continues on the next page...

Field	Function
	Rotation by 90 and 270 degrees is not supported for the following: • All linear (raster) tile modes of operation • 16bpp GPU Standard Tile mode of operation • 16bpp GPU Super Tile mode of operation • VPU 2PYUV420 10bpp mode • VPU 2PVP9 8bpp mode • VPU 2PVP9 10bpp mode
1 VFLIP_EN	Vertical Flip Enable Vertical Flip the output image. 0x0 = Disable vertical flip 0x1 = Enable Vertical flip
0 HFLIP_EN	Horizontal Flip Enable Horizontal Flip the output image. 0x0 = Disable horizontal flip 0x1 = Enable Horizontal flip Note: HFLIP is not supported for the following: • VPU 2PYUV420 10bpp mode • VPU 2-Plane VP9 10bpp mode

15.7.3.1.26 Frame 1-Plane Control 0 (FRAME_1P_CTRL0)

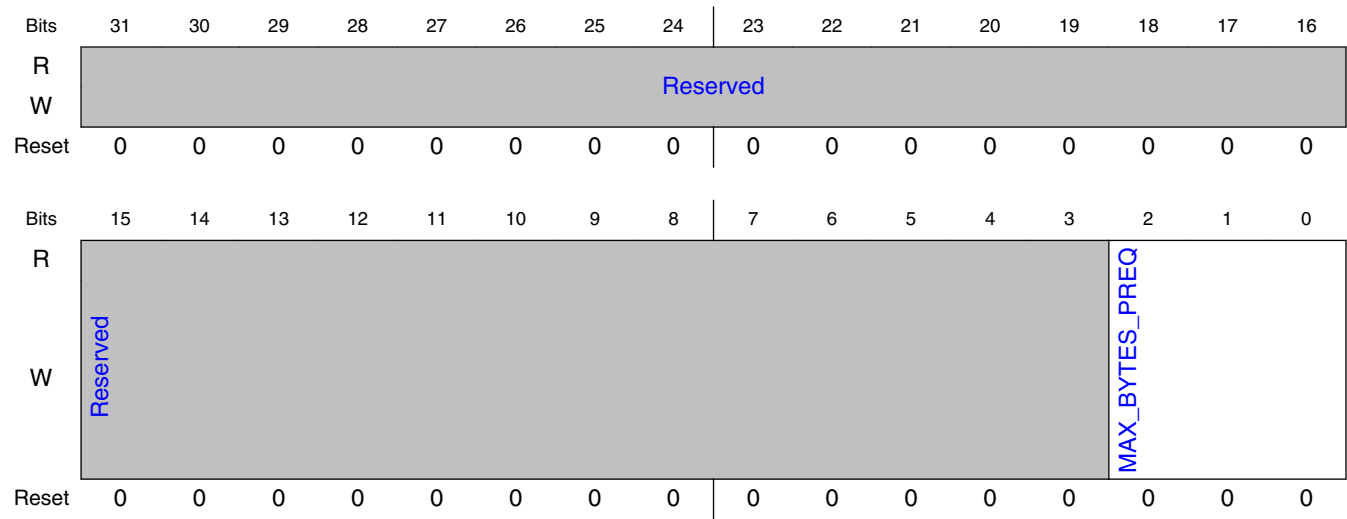
15.7.3.1.26.1 Offset

Register	Offset
FRAME_1P_CTRL0	90h

15.7.3.1.26.2 Function

The Frame 1P Control0 register controls fields only for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.26.3 Diagram



15.7.3.1.26.4 Fields

Field	Function
31-3	Reserved.
—	
2-0	Max Bytes Per Request
MAX_BYTES_P REQ	Maximum number of bytes to prefetch per request. 0x0 = 64 bytes 0x1 = 128 bytes 0x2 = 256 bytes 0x3 = 512 bytes 0x4 = 1K bytes 0x5 = 2K bytes 0x6 = 4K bytes 0x7 = 8K bytes Restrictions based on MODE_CTRL0:TILE_TYPE or FRAME_CTRL0:ROT_ENC - Rotation by 90 or 270: limit to 64 bytes - VPU Tile: limit to 64 bytes - GPU 16bpp super tile: limit to 64 bytes - GPU 32bpp super tile: limit to 128 bytes

15.7.3.1.27 Frame 1-Plane Control 0 (FRAME_1P_CTRL0_SET)

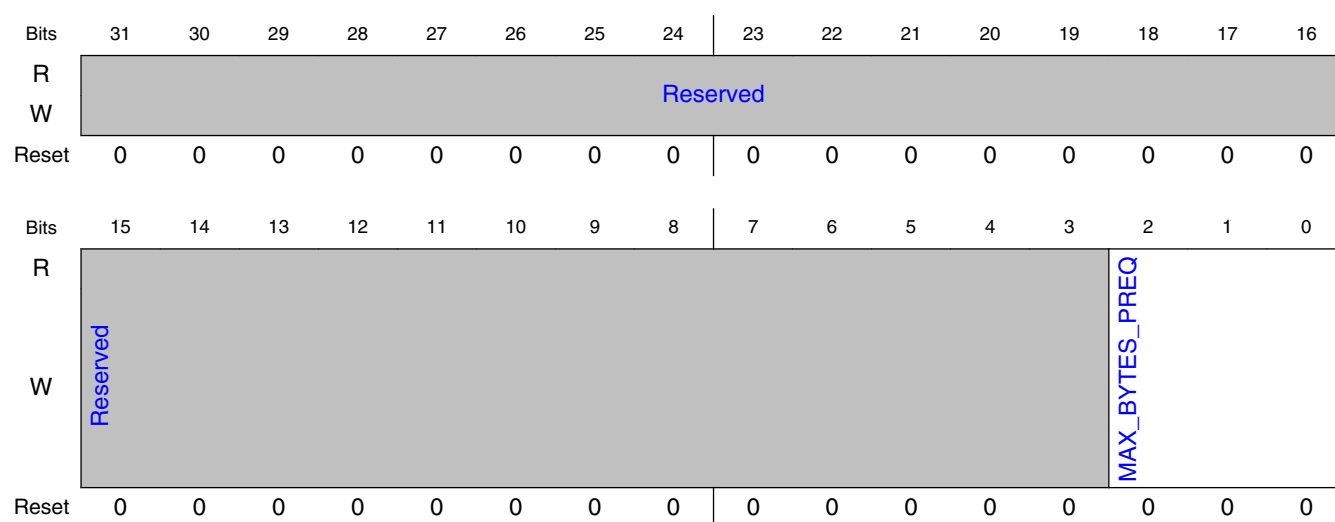
15.7.3.1.27.1 Offset

Register	Offset
FRAME_1P_CTRL0_SET	94h

15.7.3.1.27.2 Function

The Frame 1P Control0 register controls fields only for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.27.3 Diagram



15.7.3.1.27.4 Fields

Field	Function
31-3 —	Reserved.
2-0 MAX_BYTES_P REQ	Max Bytes Per Request Maximum number of bytes to prefetch per request. 0x0 = 64 bytes 0x1 = 128 bytes 0x2 = 256 bytes 0x3 = 512 bytes 0x4 = 1K bytes 0x5 = 2K bytes 0x6 = 4K bytes 0x7 = 8K bytes

Memory Map and Registers

Field	Function
	Restrictions based on MODE_CTRL0:TILE_TYPE or FRAME_CTRL0:ROT_ENC • Rotation by 90 or 270: limit to 64 bytes • VPU Tile: limit to 64 bytes • GPU 16bpp super tile: limit to 64 bytes • GPU 32bpp super tile: limit to 128 bytes

15.7.3.1.28 Frame 1-Plane Control 0 (FRAME_1P_CTRL0_CLR)

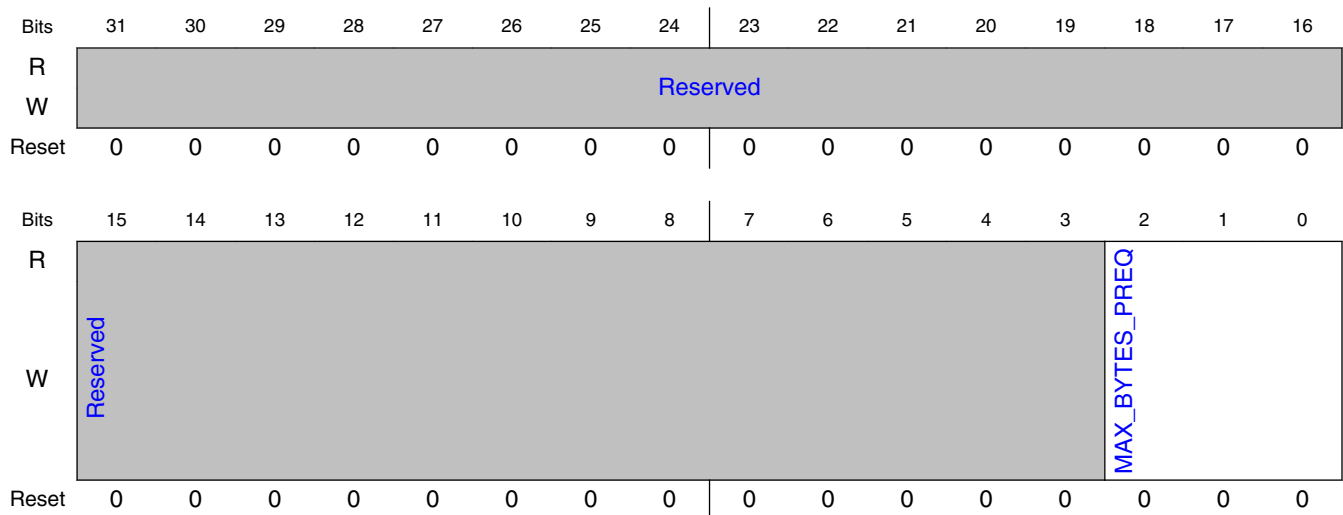
15.7.3.1.28.1 Offset

Register	Offset
FRAME_1P_CTRL0_CLR	98h

15.7.3.1.28.2 Function

The Frame 1P Control0 register controls fields only for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.28.3 Diagram



15.7.3.1.28.4 Fields

Field	Function
31-3	Reserved.

Table continues on the next page...

Field	Function
—	
2-0 MAX_BYTES_P REQ	Max Bytes Per Request Maximum number of bytes to prefetch per request. 0x0 = 64 bytes 0x1 = 128 bytes 0x2 = 256 bytes 0x3 = 512 bytes 0x4 = 1K bytes 0x5 = 2K bytes 0x6 = 4K bytes 0x7 = 8K bytes Restrictions based on MODE_CTRL0:TILE_TYPE or FRAME_CTRL0:ROT_ENC • Rotation by 90 or 270: limit to 64 bytes • VPU Tile: limit to 64 bytes • GPU 16bpp super tile: limit to 64 bytes • GPU 32bpp super tile: limit to 128 bytes

15.7.3.1.29 Frame 1-Plane Control 0 (FRAME_1P_CTRL0_TOG)

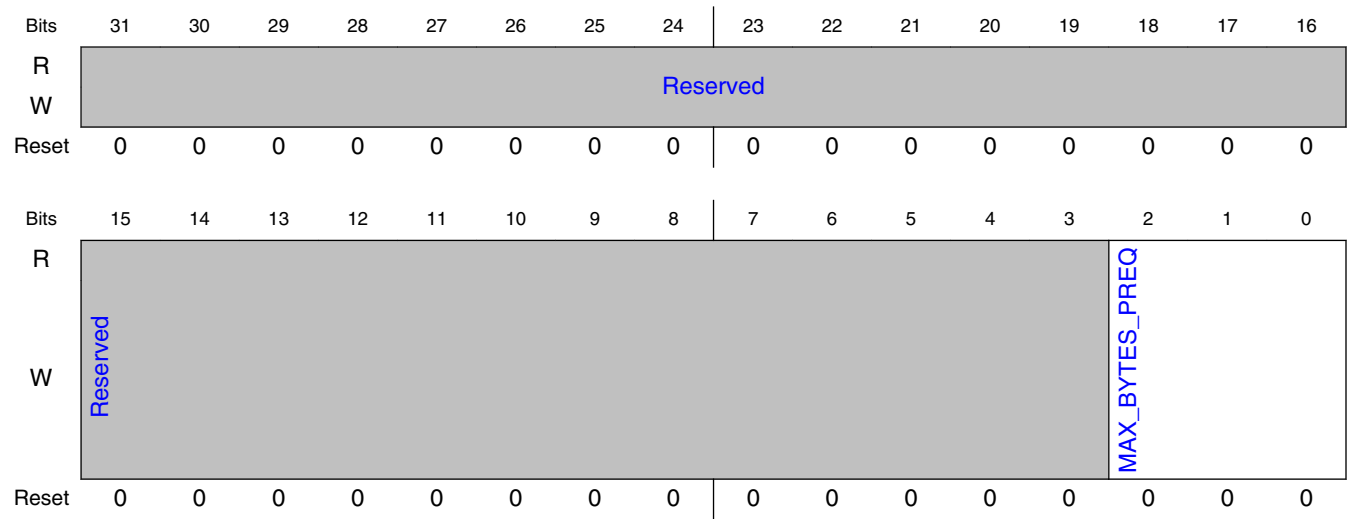
15.7.3.1.29.1 Offset

Register	Offset
FRAME_1P_CTRL0_TOG	9Ch

15.7.3.1.29.2 Function

The Frame 1P Control0 register controls fields only for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.29.3 Diagram



15.7.3.1.29.4 Fields

Field	Function
31-3 —	Reserved.
2-0 MAX_BYTES_P REQ	Max Bytes Per Request Maximum number of bytes to prefetch per request. 0x0 = 64 bytes 0x1 = 128 bytes 0x2 = 256 bytes 0x3 = 512 bytes 0x4 = 1K bytes 0x5 = 2K bytes 0x6 = 4K bytes 0x7 = 8K bytes Restrictions based on MODE_CTRL0:TILE_TYPE or FRAME_CTRL0:ROT_ENC • Rotation by 90 or 270: limit to 64 bytes • VPU Tile: limit to 64 bytes • GPU 16bpp super tile: limit to 64 bytes • GPU 32bpp super tile: limit to 128 bytes

15.7.3.1.30 Frame 1-Plane Pix X Control (FRAME_1P_PIX_X_CTRL)

15.7.3.1.30.1 Offset

Register	Offset
FRAME_1P_PIX_X_CTRL	A0h

15.7.3.1.30.2 Function

The Frame 1P PIX_X Control register controls x-axis pixel control for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.30.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_X															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_X_PIX_WIDE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.30.4 Fields

Field	Function																		
31-16 CROP_ULC_X	Starting Coordinate of Cropped Image X (1-Plane or 2-Plane Luma) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB, 1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB, 1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB, 1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	

Table continues on the next page...

Field	Function
15-0 NUM_X_PIX_WIDE	Number of Pixels Wide in X-direction The number of input frame pixels to fetch in the x-axis direction. In non-linear (raster) tile mode, the starting pixel must be at the start of a tile boundary. Partial tile fetch is not allowed. The NUM_X_PIX_WIDE must be sized so the prefetch is evenly divisible by 64 bytes. If the NUM_X_PIX_WIDE is not evenly divisible by 64 bytes, then an offset must be applied. To calculate if NUM_X_PIX_WIDE is divisible by 64 bytes: First determine the number of pix_x pixels divisible in 64 bytes, based on MODE_CTRL0: PIX_SIZE and TILE_TYPE settings: <pre> num_pix_x_in_64byte = (((PIX_SIZE==2) && (TILE_TYPE==0)) 16 : // GPU 32bpp linear tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==2) && ((TILE_TYPE==1) (TILE_TYPE==2))) ? 4 : // GPU 32bpp standard and/or super tile must be divisible by 4 pixels in x-direction ((PIX_SIZE==1) && (TILE_TYPE==0)) 32 : // GPU 16bpp linear tile must be divisible by 32 pixels in x- direction. VPU 10bpp VP9-10 UV linear tile must be divisible by 32 pixels in x-direction ((PIX_SIZE==1) && ((TILE_TYPE==1) (TILE_TYPE==2) (TILE_TYPE==4))) ? 8 : // GPU 16bpp standard and/or super tile must be divisible by 8 pixels in x-direction. VPU 10bpp VP9-10 UV tile must be divisible by 8 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==0)) 64 : // VPU 8bpp VP9 UV linear tile must be divisible by 64 pixels in x- direction ((PIX_SIZE==0) && (TILE_TYPE==4)) 16 : // VPU 8bpp VP9 UV tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==3)) 8 : // VPU 8bpp Luma zenverge tile must be divisible by 8 pixels in x- direction 8); // default case shouldn't be hit </pre> Second calculate the modulus of the NUM_X_PIX_WIDE by dpr_num_pix_x_in_64byte Note: for 10bpp source frame, NUM_X_PIX_WIDE must first be multiplied by 1.25 to account for MODE_CTRL0: PIX_SIZE=8bpp operation. <pre> pix_x_div_64byte_mod = NUM_X_PIX_WIDE % num_pix_x_in_64byte </pre> Third calculate the number of offset pixels <pre> pix_x_offset = (pix_x_div_64byte_mod == 0) ? 0 : (num_pix_x_in_64byte - pix_x_div_64byte_mod) </pre> Fourth add pix_x_offset to NUM_X_PIX_WIDE. This is the value written to the NUM_X_PIX_WIDE control registers. <pre> NUM_X_PIX_WIDE = NUM_X_PIX_WIDE + pix_x_offset </pre> Note1: If FRAME_1P_CTRL0:MAX_BYTES_PREQ > 64 bytes AND FRAME_CTRL0:HFLIP_EN=1, then the prefetch calculation above must be evenly divisible by 128 bytes instead of 64 bytes.

15.7.3.1.31 Frame 1-Plane Pix X Control (FRAME_1P_PIX_X_CTRL_SET)

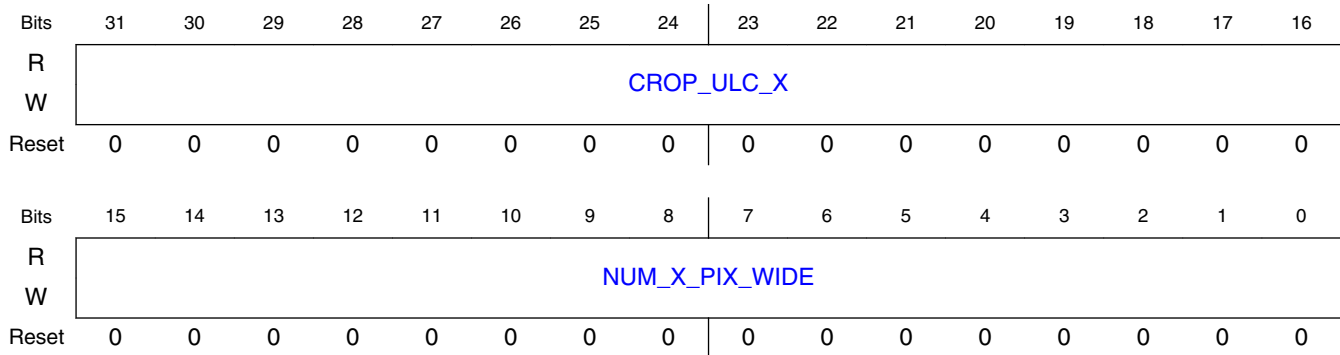
15.7.3.1.31.1 Offset

Register	Offset
FRAME_1P_PIX_X_CTRL_SET	A4h

15.7.3.1.31.2 Function

The Frame 1P PIX_X Control register controls x-axis pixel control for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.31.3 Diagram



15.7.3.1.31.4 Fields

Field	Function												
31-16 CROP_ULC_X	Starting Coordinate of Cropped Image X (1-Plane or 2-Plane Luma) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0
Tile Type	X Alignment	Y Alignment											
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0											
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0											
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0											

Table continues on the next page...

Memory Map and Registers

Field	Function
	VP9 8bpp 2-Plane YUV420 CROP_ULC_X[3:0] = 0 CROP_ULC_Y[3:0] = 0 VP9 10bpp 2-Plane YUV420 CROP_ULC_X[2:0] = 0 CROP_ULC_Y[2:0] = 0 -----
15-0 NUM_X_PIX_WIDE	Number of Pixels Wide in X-direction The number of input frame pixels to fetch in the x-axis direction. In non-linear (raster) tile mode, the starting pixel must be at the start of a tile boundary. Partial tile fetch is not allowed. The NUM_X_PIX_WIDE must be sized so the prefetch is evenly divisible by 64 bytes. If the NUM_X_PIX_WIDE is not evenly divisible by 64 bytes, then an offset must be applied. To calculate if NUM_X_PIX_WIDE is divisible by 64 bytes: First determine the number of pix_x pixels divisible in 64 bytes, based on MODE_CTRL0: PIX_SIZE and TILE_TYPE settings: <pre> num_pix_x_in_64byte = (((PIX_SIZE==2) && (TILE_TYPE==0)) 16 : // GPU 32bpp linear tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==2) && ((TILE_TYPE==1) (TILE_TYPE==2))) ? 4 : // GPU 32bpp standard and/or super tile must be divisible by 4 pixels in x-direction ((PIX_SIZE==1) && (TILE_TYPE==0)) 32 : // GPU 16bpp linear tile must be divisible by 32 pixels in x- direction. VPU 10bpp VP9-10 UV linear tile must be divisible by 32 pixels in x-direction ((PIX_SIZE==1) && ((TILE_TYPE==1) (TILE_TYPE==2) (TILE_TYPE==4))) ? 8 : // GPU 16bpp standard and/or super tile must be divisible by 8 pixels in x-direction. VPU 10bpp VP9-10 UV tile must be divisible by 8 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==0)) 64 : // VPU 8bpp VP9 UV linear tile must be divisible by 64 pixels in x- direction ((PIX_SIZE==0) && (TILE_TYPE==4)) 16 : // VPU 8bpp VP9 UV tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==3)) 8 : // VPU 8bpp Luma zenverge tile must be divisible by 8 pixels in x- direction 8); // default case shouldn't be hit </pre> Second calculate the modulus of the NUM_X_PIX_WIDE by dpr_num_pix_x_in_64byte Note: for 10bpp source frame, NUM_X_PIX_WIDE must first be multiplied by 1.25 to account for MODE_CTRL0: PIX_SIZE=8bpp operation. <pre> pix_x_div_64byte_mod = NUM_X_PIX_WIDE % num_pix_x_in_64byte </pre> Third calculate the number of offset pixels <pre> pix_x_offset = (pix_x_div_64byte_mod == 0) ? 0 : (num_pix_x_in_64byte - pix_x_div_64byte_mod) </pre> Fourth add pix_x_offset to NUM_X_PIX_WIDE. This is the value written to the NUM_X_PIX_WIDE control registers. <pre> NUM_X_PIX_WIDE = NUM_X_PIX_WIDE + pix_x_offset </pre> Note1: If FRAME_1P_CTRL0:MAX_BYTES_PREQ > 64 bytes AND FRAME_CTRL0:HFLIP_EN=1, then the prefetch calculation above must be evenly divisible by 128 bytes instead of 64 bytes.

15.7.3.1.32 Frame 1-Plane Pix X Control (FRAME_1P_PIX_X_CTRL_CLR)

15.7.3.1.32.1 Offset

Register	Offset
FRAME_1P_PIX_X_CTRL_CLR	A8h

15.7.3.1.32.2 Function

The Frame 1P PIX_X Control register controls x-axis pixel control for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.32.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_X															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_X_PIX_WIDE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.32.4 Fields

Field	Function												
31-16 CROP_ULC_X	Starting Coordinate of Cropped Image X (1-Plane or 2-Plane Luma) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>Tile Type</td><td>X Alignment</td><td>Y Alignment</td></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr></table>	-----	-----	-----	Tile Type	X Alignment	Y Alignment	-----	-----	-----	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0
-----	-----	-----											
Tile Type	X Alignment	Y Alignment											
-----	-----	-----											
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0											

Table continues on the next page...

Memory Map and Registers

Field	Function															
	<table><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr></table>	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	-----	-----	-----
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0														
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0														
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0														
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0														
-----	-----	-----														
15-0 NUM_X_PIX_WIDE	Number of Pixels Wide in X-direction The number of input frame pixels to fetch in the x-axis direction. In non-linear (raster) tile mode, the starting pixel must be at the start of a tile boundary. Partial tile fetch is not allowed. The NUM_X_PIX_WIDE must be sized so the prefetch is evenly divisible by 64 bytes. If the NUM_X_PIX_WIDE is not evenly divisible by 64 bytes, then an offset must be applied. To calculate if NUM_X_PIX_WIDE is divisible by 64 bytes: First determine the number of pix_x pixels divisible in 64 bytes, based on MODE_CTRL0: PIX_SIZE and TILE_TYPE settings: <pre>num_pix_x_in_64byte = (((PIX_SIZE==2) && (TILE_TYPE==0)) ? 16 : // GPU 32bpp linear tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==2) && ((TILE_TYPE==1) (TILE_TYPE==2))) ? 4 : // GPU 32bpp standard and/or super tile must be divisible by 4 pixels in x-direction ((PIX_SIZE==1) && (TILE_TYPE==0)) ? 32 : // GPU 16bpp linear tile must be divisible by 32 pixels in x-direction. VPU 10bpp VP9-10 UV linear tile must be divisible by 32 pixels in x-direction ((PIX_SIZE==1) && ((TILE_TYPE==1) (TILE_TYPE==2) (TILE_TYPE==4))) ? 8 : // GPU 16bpp standard and/or super tile must be divisible by 8 pixels in x-direction. VPU 10bpp VP9-10 UV tile must be divisible by 8 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==0)) ? 64 : // VPU 8bpp VP9 UV linear tile must be divisible by 64 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==4)) ? 16 : // VPU 8bpp VP9 UV tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==3)) ? 8 : // VPU 8bpp Luma zenverge tile must be divisible by 8 pixels in x-direction 8); // default case</pre> shouldn't be hit Second calculate the modulus of the NUM_X_PIX_WIDE by dpr_num_pix_x_in_64byte Note: for 10bpp source frame, NUM_X_PIX_WIDE must first be multiplied by 1.25 to account for MODE_CTRL0: PIX_SIZE=8bpp operation. <pre>pix_x_div_64byte_mod = NUM_X_PIX_WIDE % num_pix_x_in_64byte</pre> Third calculate the number of offset pixels <pre>pix_x_offset = (pix_x_div_64byte_mod == 0) ? 0 : (num_pix_x_in_64byte - pix_x_div_64byte_mod)</pre> Fourth add pix_x_offset to NUM_X_PIX_WIDE. This is the value written to the NUM_X_PIX_WIDE control registers. <pre>NUM_X_PIX_WIDE = NUM_X_PIX_WIDE + pix_x_offset</pre> Note1: If FRAME_1P_CTRL0:MAX_BYTES_PREQ > 64 bytes AND															

Field	Function
	FRAME_CTRL0:HFLIP_EN=1, then the prefetch calculation above must be evenly divisible by 128 bytes instead of 64 bytes.

15.7.3.1.33 Frame 1-Plane Pix X Control (FRAME_1P_PIX_X_CTRL_TOG)

15.7.3.1.33.1 Offset

Register	Offset
FRAME_1P_PIX_X_CTRL_TOG	ACh

15.7.3.1.33.2 Function

The Frame 1P PIX_X Control register controls x-axis pixel control for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.33.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_X															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_X_PIX_WIDE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.33.4 Fields

Field	Function
31-16 CROP_ULC_X	Starting Coordinate of Cropped Image X (1-Plane or 2-Plane Luma) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions

Table continues on the next page...

Memory Map and Registers

Field	Function																		
	apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><td>Tile Type</td><td>X Alignment</td><td>Y Alignment</td></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
15-0 NUM_X_PIX_WIDE	Number of Pixels Wide in X-direction The number of input frame pixels to fetch in the x-axis direction. In non-linear (raster) tile mode, the starting pixel must be at the start of a tile boundary. Partial tile fetch is not allowed. The NUM_X_PIX_WIDE must be sized so the prefetch is evenly divisible by 64 bytes. If the NUM_X_PIX_WIDE is not evenly divisible by 64 bytes, then an offset must be applied. To calculate if NUM_X_PIX_WIDE is divisible by 64 bytes: First determine the number of pix_x pixels divisible in 64 bytes, based on MODE_CTRL0: PIX_SIZE and TILE_TYPE settings: num_pix_x_in_64byte = (((PIX_SIZE==2) && (TILE_TYPE==0)) ? 16 : ((PIX_SIZE==2) && ((TILE_TYPE==1) (TILE_TYPE==2))) ? 4 : ((PIX_SIZE==1) && (TILE_TYPE==0)) ? 32 : ((PIX_SIZE==1) && ((TILE_TYPE==1) (TILE_TYPE==2) (TILE_TYPE==4))) ? 8 : ((PIX_SIZE==0) && (TILE_TYPE==0)) ? 64 : ((PIX_SIZE==0) && (TILE_TYPE==4)) ? 16 : ((PIX_SIZE==0) && (TILE_TYPE==3)) ? 8 : 8); // default case shouldn't be hit Second calculate the modulus of the NUM_X_PIX_WIDE by dpr_num_pix_x_in_64byte Note: for 10bpp source frame, NUM_X_PIX_WIDE must first be multiplied by 1.25 to account for MODE_CTRL0: PIX_SIZE=8bpp operation. pix_x_div_64byte_mod = NUM_X_PIX_WIDE % num_pix_x_in_64byte Third calculate the number of offset pixels pix_x_offset = (pix_x_div_64byte_mod == 0) ? 0 : (num_pix_x_in_64byte - pix_x_div_64byte_mod)																		

Field	Function
	Fourth add pix_x_offset to NUM_X_PIX_WIDE. This is the value written to the NUM_X_PIX_WIDE control registers. $\text{NUM_X_PIX_WIDE} = \text{NUM_X_PIX_WIDE} + \text{pix_x_offset}$ Note1: If FRAME_1P_CTRL0:MAX_BYTES_PREQ > 64 bytes AND FRAME_CTRL0:HFLIP_EN=1, then the prefetch calculation above must be evenly divisible by 128 bytes instead of 64 bytes.

15.7.3.1.34 Frame 1-Plane Pix Y Control (FRAME_1P_PIX_Y_CTRL)

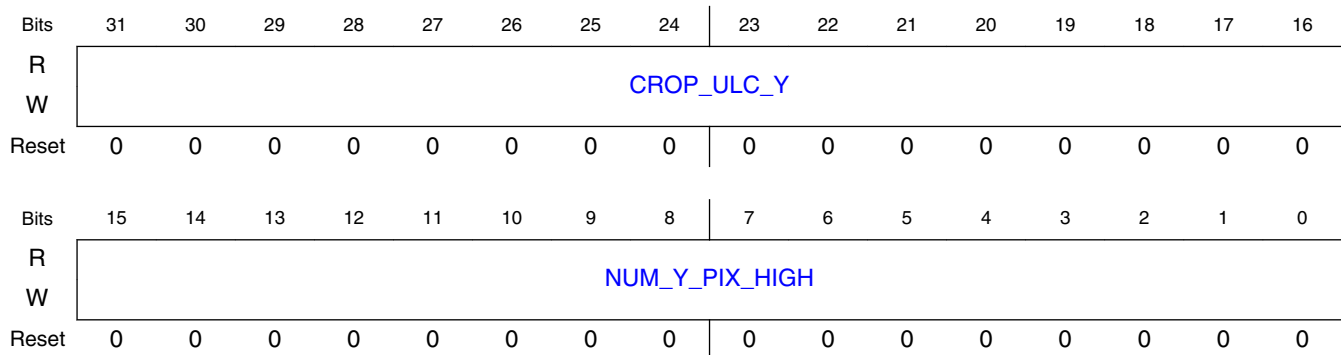
15.7.3.1.34.1 Offset

Register	Offset
FRAME_1P_PIX_Y_CTRL	B0h

15.7.3.1.34.2 Function

The Frame 1P PIX_Y Control register controls y-axis pixel control for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.34.3 Diagram



15.7.3.1.34.4 Fields

Field	Function
31-16 CROP_ULC_Y	Starting Coordinate of Cropped Image Y (1-Plane or 2-Plane Luma) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the

Table continues on the next page...

Field	Function																		
	NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
15-0 NUM_Y_PIX_HIGH	Number of Pixels High in Y-direction The number of input frame pixels to fetch in the y-axis direction. In non-linear (raster) t mode, the number of pixels must be sized for a full tile block boundary. Partial tile fetch is not allowed. The NUM_Y_PIX_HIGH value must be evenly divisible by the number of rows programmed in MODE_CTRL0: RTR_4LINE_BUF_EN. First determine the number of rows in the buffer num_rows_buf = (RTR_4LINE_BUF_EN==0) ? 8 : 4; Second calculate the modulus of NUM_Y_PIX_HIGH by the num_rows_buf pix_y_mod = NUM_Y_PIX_HIGH % num_rows_buf; Third calculate the number of rows to offset pix_y_offset = (pix_y_mod==0) ? 0 : (num_rows_buf - pix_y_mod); Fourth add pix_y_offset to NUM_Y_PIX_HIGH. This is the value written to the NUM_Y_PIX_HIGH control registers. NUM_Y_PIX_HIGH = NUM_Y_PIX_HIGH + pix_y_offset;																		

15.7.3.1.35 Frame 1-Plane Pix Y Control (FRAME_1P_PIX_Y_CTRL_SET)

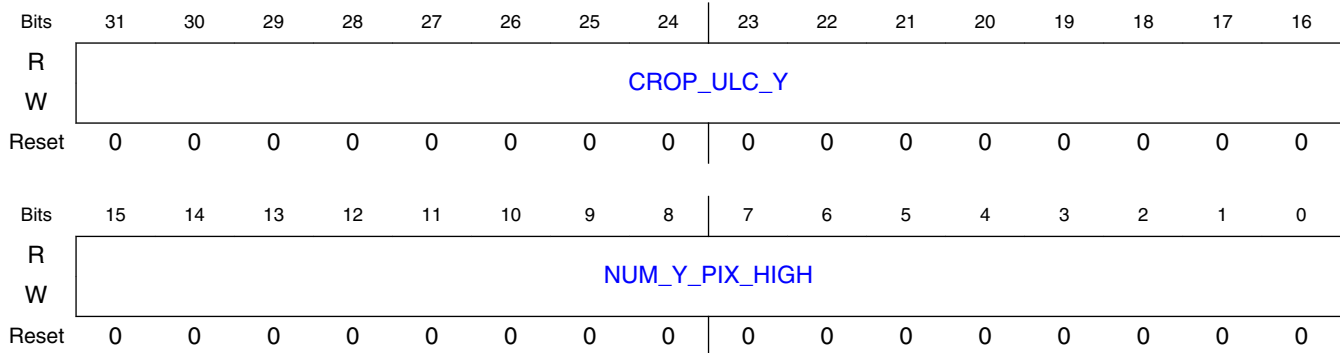
15.7.3.1.35.1 Offset

Register	Offset
FRAME_1P_PIX_Y_CTRL_SET	B4h

15.7.3.1.35.2 Function

The Frame 1P PIX_Y Control register controls y-axis pixel control for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.35.3 Diagram



15.7.3.1.35.4 Fields

Field	Function																		
31-16 CROP_ULC_Y	Starting Coordinate of Cropped Image Y (1-Plane or 2-Plane Luma) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
15-0 NUM_Y_PIX_HI GH	Number of Pixels High in Y-direction The number of input frame pixels to fetch in the y-axis direction. In non-linear (raster) t mode, the number of pixels must be sized for a full tile block boundary. Partial tile fetch is not allowed. The NUM_Y_PIX_HIGH value must be evenly divisible by the number of rows programmed in MODE_CTRL0: RTR_4LINE_BUF_EN. First determine the number of rows in the buffer num_rows_buf = (RTR_4LINE_BUF_EN==0) ? 8 : 4; Second calculate the modulus of NUM_Y_PIX_HIGH by the num_rows_buf pix_y_mod = NUM_Y_PIX_HIGH % num_rows_buf; Third calculate the number of rows to offset pix_y_offset = (pix_y_mod==0) ? 0 : (num_rows_buf - pix_y_mod);																		

Field	Function
	Fourth add pix_y_offset to NUM_Y_PIX_HIGH. This is the value written to the NUM_Y_PIX_HIGH control registers. $\text{NUM_Y_PIX_HIGH} = \text{NUM_Y_PIX_HIGH} + \text{pix_y_offset};$

15.7.3.1.36 Frame 1-Plane Pix Y Control (FRAME_1P_PIX_Y_CTRL_CLR)

15.7.3.1.36.1 Offset

Register	Offset
FRAME_1P_PIX_Y_CTRL_CLR	B8h

15.7.3.1.36.2 Function

The Frame 1P PIX_Y Control register controls y-axis pixel control for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.36.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_Y															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_Y_PIX_HIGH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.36.4 Fields

Field	Function
31-16 CROP_ULC_Y	Starting Coordinate of Cropped Image Y (1-Plane or 2-Plane Luma) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for

Table continues on the next page...

Field	Function																		
	proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE, NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB, 1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB, 1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB, 1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
15-0 NUM_Y_PIX_HIGH	Number of Pixels High in Y-direction The number of input frame pixels to fetch in the y-axis direction. In non-linear (raster) t mode, the number of pixels must be sized for a full tile block boundary. Partial tile fetch is not allowed. The NUM_Y_PIX_HIGH value must be evenly divisible by the number of rows programmed in MODE_CTRL0: RTR_4LINE_BUF_EN. First determine the number of rows in the buffer num_rows_buf = (RTR_4LINE_BUF_EN==0) ? 8 : 4; Second calculate the modulus of NUM_Y_PIX_HIGH by the num_rows_buf pix_y_mod = NUM_Y_PIX_HIGH % num_rows_buf; Third calculate the number of rows to offset pix_y_offset = (pix_y_mod==0) ? 0 : (num_rows_buf - pix_y_mod); Fourth add pix_y_offset to NUM_Y_PIX_HIGH. This is the value written to the NUM_Y_PIX_HIGH control registers. NUM_Y_PIX_HIGH = NUM_Y_PIX_HIGH + pix_y_offset;																		

15.7.3.1.37 Frame 1-Plane Pix Y Control (FRAME_1P_PIX_Y_CTRL_TOG)

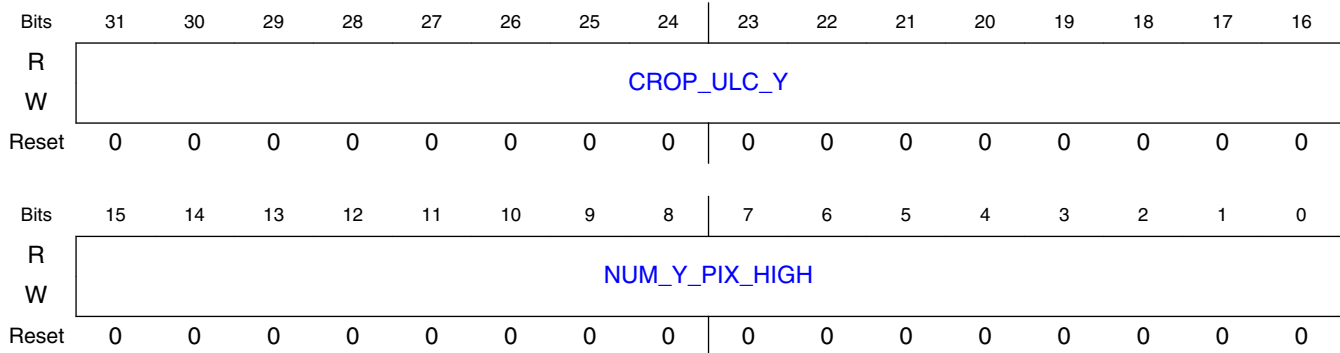
15.7.3.1.37.1 Offset

Register	Offset
FRAME_1P_PIX_Y_CTRL_TOG	BCh

15.7.3.1.37.2 Function

The Frame 1P PIX_Y Control register controls y-axis pixel control for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.37.3 Diagram



15.7.3.1.37.4 Fields

Field	Function																		
31-16 CROP_ULC_Y	Starting Coordinate of Cropped Image Y (1-Plane or 2-Plane Luma) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
15-0 NUM_Y_PIX_HI GH	Number of Pixels High in Y-direction The number of input frame pixels to fetch in the y-axis direction. In non-linear (raster) t mode, the number of pixels must be sized for a full tile block boundary. Partial tile fetch is not allowed. The NUM_Y_PIX_HIGH value must be evenly divisible by the number of rows programmed in MODE_CTRL0: RTR_4LINE_BUF_EN. First determine the number of rows in the buffer num_rows_buf = (RTR_4LINE_BUF_EN==0) ? 8 : 4; Second calculate the modulus of NUM_Y_PIX_HIGH by the num_rows_buf pix_y_mod = NUM_Y_PIX_HIGH % num_rows_buf; Third calculate the number of rows to offset pix_y_offset = (pix_y_mod==0) ? 0 : (num_rows_buf - pix_y_mod);																		

Field	Function
	Fourth add pix_y_offset to NUM_Y_PIX_HIGH. This is the value written to the NUM_Y_PIX_HIGH control registers. $\text{NUM_Y_PIX_HIGH} = \text{NUM_Y_PIX_HIGH} + \text{pix_y_offset};$

15.7.3.1.38 Frame 1-Plane Base Address Control 0 (FRAME_1P_BASE_ADDR_CTRL0)

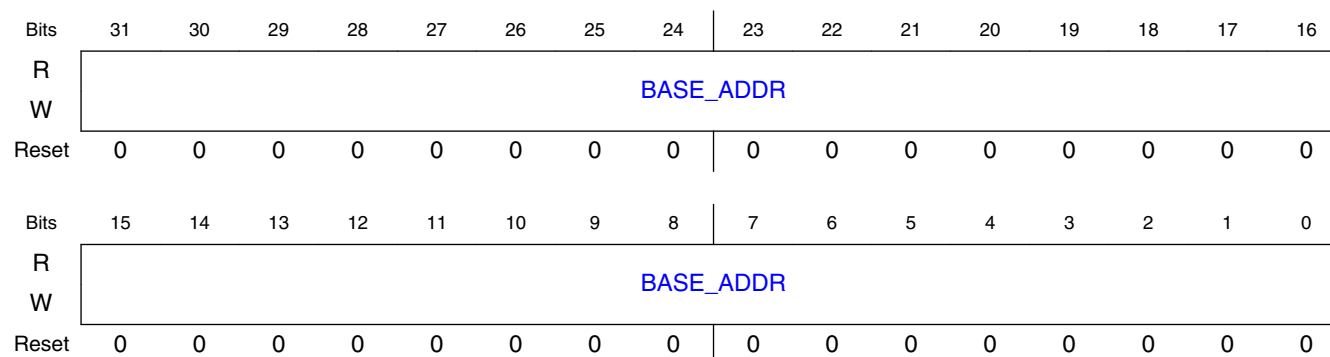
15.7.3.1.38.1 Offset

Register	Offset
FRAME_1P_BASE_ADDR_CTRL0	C0h

15.7.3.1.38.2 Function

The Frame 1P Base Address Control0 register is the input frame base address for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.38.3 Diagram



15.7.3.1.38.4 Fields

Field	Function
31-0 BASE_ADDR	Base Address The base address for 1-Plane and 2-Plane Luma. The base address is a source frame buffer offset to the non-rotated, non-hflip, non-vflip output image. This will correspond to the output image's pix_x=0, pix_y=0 position. If any non-linear GPU tile or VPU tile is being processed, then the base

Field	Function
	address must start at pixel 0 of the tile boundary. Under certain conditions, a offset must be applied to the BASE_ADDR pix_x=0, pix_y=0 position. 1. The DPR will perform an internal hflip AND the pix_x_offset calculation doesn't equal 0 as described in FRAME_1P_PIX_X_CTRL: NUM_X_PIX_WIDE. 2. The DPR will perform an internal vflip AND the pix_y_offset calculation doesn't equal 0 as described in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. Steps to apply an offset to the BASE_ADDR: First determine if the DPR will perform an internal hflip or vflip base on FRAME_CTRL0: ROT_FLIP_ORDER_EN, ROT_ENC, VFLIP_EN and HFLIP_EN. <pre> int_hflip = (((ROT_ENC==0) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en int_vflip = (((ROT_ENC==0) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> Second calculate the number of pix_x_byte_offset based on the pix_x_offset calculation described in FRAME_1P_PIX_X_CTRL: NUM_X_PIX_WIDE, and the MODE_CTRL0: PIX_SIZE. <pre> pix_x_byte_offset = int_hflip_en ? (pix_x_offset << PIX_SIZE) : 0; </pre> Third calculate the number of pix_y_byte_offset based on the pix_y_offset calculation described in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH, and the FRAME_CTRL0: PITCH. <pre> pix_y_byte_offset = int_vflip_en ? (PITCH * pix_y_offset) : 0; </pre> Fourth apply the following pix_x_byte_offset and pix_y_byte_offset to the BASE_ADDR. This is the value which should be written to BASE_ADDR. <pre> BASE_ADDR = (BASE_ADDR - pix_y_byte_offset) - pix_x_byte_offset; </pre>

15.7.3.1.39 Frame 1-Plane Base Address Control 0 (FRAME_1P_BASE_ADDR_CTRL0_SET)

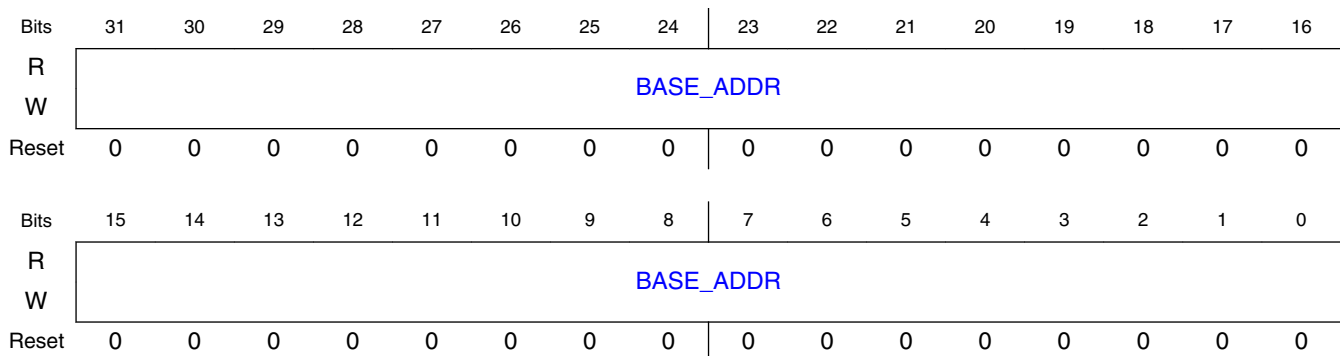
15.7.3.1.39.1 Offset

Register	Offset
FRAME_1P_BASE_ADDR_CTRL0_SET	C4h

15.7.3.1.39.2 Function

The Frame 1P Base Address Control0 register is the input frame base address for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.39.3 Diagram



15.7.3.1.39.4 Fields

Field	Function
31-0 BASE_ADDR	Base Address The base address for 1-Plane and 2-Plane Luma. The base address is a source frame buffer offset to the non-rotated, non-hflip, non-vflip output image. This will correspond to the output image's pix_x=0, pix_y=0 position. If any non-linear GPU tile or VPU tile is being processed, then the base address must start at pixel 0 of the tile boundary. Under certain conditions, a offset must be applied to the BASE_ADDR pix_x=0, pix_y=0 position. 1. The DPR will perform an internal hflip AND the pix_x_offset calculation doesn't equal 0 as described in FRAME_1P_PIX_X_CTRL: NUM_X_PIX_WIDE.

Field	Function
	2. The DPR will perform an internal vflip AND the pix_y_offset calculation doesn't equal 0 as described in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. Steps to apply an offset to the BASE_ADDR: First determine if the DPR will perform an internal hflip or vflip base on FRAME_CTRL0: ROT_FLIP_ORDER_EN, ROT_ENC, VFLIP_EN and HFLIP_EN. <pre> int_hflip = (((ROT_ENC==0) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> <pre> int_vflip = (((ROT_ENC==0) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> Second calculate the number of pix_x_byte_offset based on the pix_x_offset calculation described in FRAME_1P_PIX_X_CTRL: NUM_X_PIX_WIDE, and the MODE_CTRL0: PIX_SIZE. <pre> pix_x_byte_offset = int_hflip_en ? (pix_x_offset << PIX_SIZE) : 0; </pre> Third calculate the number of pix_y_byte_offset based on the pix_y_offset calculation described in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH, and the FRAME_CTRL0: PITCH. <pre> pix_y_byte_offset = int_vflip_en ? (PITCH * pix_y_offset) : 0; </pre> Fourth apply the following pix_x_byte_offset and pix_y_byte_offset to the BASE_ADDR. This is the value which should be written to BASE_ADDR. <pre> BASE_ADDR = (BASE_ADDR - pix_y_byte_offset) - pix_x_byte_offset; </pre>

15.7.3.1.40 Frame 1-Plane Base Address Control 0 (FRAME_1P_BASE_ADDR_CTRL0_CLR)

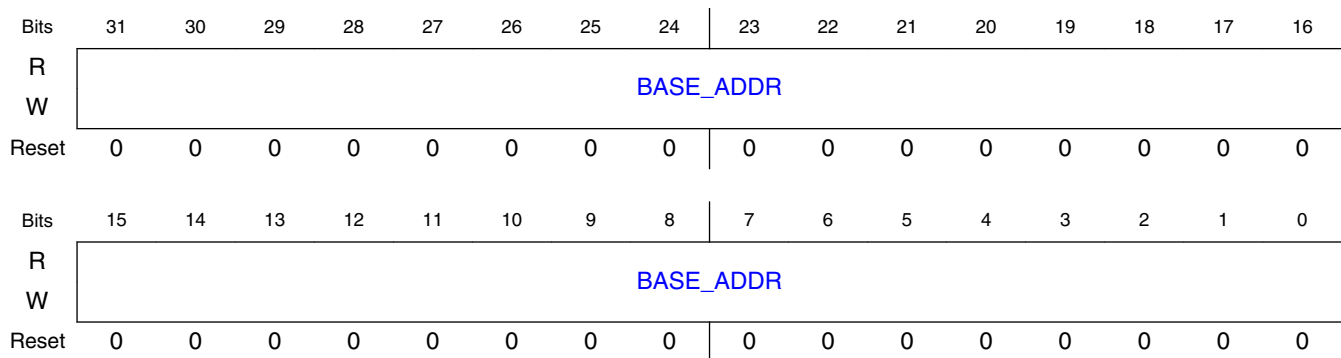
15.7.3.1.40.1 Offset

Register	Offset
FRAME_1P_BASE_ADDR_CTRL0_CLR	C8h

15.7.3.1.40.2 Function

The Frame 1P Base Address Control0 register is the input frame base address for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.40.3 Diagram



15.7.3.1.40.4 Fields

Field	Function
31-0 BASE_ADDR	Base Address The base address for 1-Plane and 2-Plane Luma. The base address is a source frame buffer offset to the non-rotated, non-hflip, non-vflip output image. This will correspond to the output image's pix_x=0, pix_y=0 position. If any non-linear GPU tile or VPU tile is being processed, then the base address must start at pixel 0 of the tile boundary. Under certain conditions, a offset must be applied to the BASE_ADDR pix_x=0, pix_y=0 position. 1. The DPR will perform an internal hflip AND the pix_x_offset calculation doesn't equal 0 as described in FRAME_1P_PIX_X_CTRL: NUM_X_PIX_WIDE. 2. The DPR will perform an internal vflip AND the pix_y_offset calculation doesn't equal 0 as described in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH.

Field	Function
	Steps to apply an offset to the BASE_ADDR: First determine if the DPR will perform an internal hflip or vflip base on FRAME_CTRL0: ROT_FLIP_ORDER_EN, ROT_ENC, VFLIP_EN and HFLIP_EN. <pre> int_hflip = (((ROT_ENC==0) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> <pre> int_vflip = (((ROT_ENC==0) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> Second calculate the number of pix_x_byte_offset based on the pix_x_offset calculation described in FRAME_1P_PIX_X_CTRL: NUM_X_PIX_WIDE, and the MODE_CTRL0: PIX_SIZE. <pre> pix_x_byte_offset = int_hflip_en ? (pix_x_offset << PIX_SIZE) : 0; </pre> Third calculate the number of pix_y_byte_offset based on the pix_y_offset calculation described in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH, and the FRAME_CTRL0: PITCH. <pre> pix_y_byte_offset = int_vflip_en ? (PITCH * pix_y_offset) : 0; </pre> Fourth apply the following pix_x_byte_offset and pix_y_byte_offset to the BASE_ADDR. This is the value which should be written to BASE_ADDR. <pre> BASE_ADDR = (BASE_ADDR - pix_y_byte_offset) - pix_x_byte_offset; </pre>

15.7.3.1.41 Frame 1-Plane Base Address Control 0 (FRAME_1P_BASE_ADDR_CTRL0_TOG)

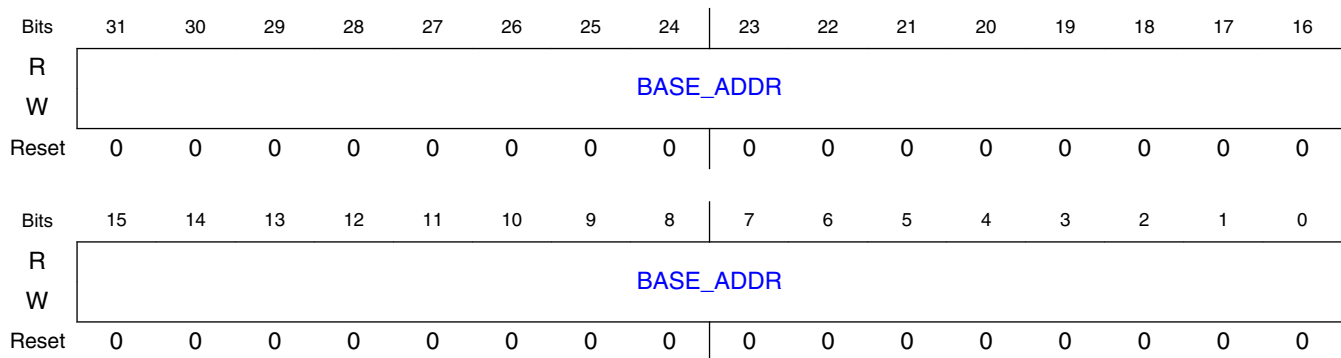
15.7.3.1.41.1 Offset

Register	Offset
FRAME_1P_BASE_AD DR_CTRL0_TOG	CCh

15.7.3.1.41.2 Function

The Frame 1P Base Address Control0 register is the input frame base address for 1-Plane or 2-Plane Luma image processing.

15.7.3.1.41.3 Diagram



15.7.3.1.41.4 Fields

Field	Function
31-0 BASE_ADDR	Base Address The base address for 1-Plane and 2-Plane Luma. The base address is a source frame buffer offset to the non-rotated, non-hflip, non-vflip output image. This will correspond to the output image's <code>pix_x=0</code>, <code>pix_y=0</code> position. If any non-linear GPU tile or VPU tile is being processed, then the base address must start at pixel 0 of the tile boundary. Under certain conditions, a offset must be applied to the BASE_ADDR <code>pix_x=0</code>, <code>pix_y=0</code> position. 1. The DPR will perform an internal hflip AND the <code>pix_x_offset</code> calculation doesn't equal 0 as described in <code>FRAME_1P_PIX_X_CTRL: NUM_X_PIX_WIDE</code>. 2. The DPR will perform an internal vflip AND the <code>pix_y_offset</code> calculation doesn't equal 0 as described in <code>FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH</code>. Steps to apply an offset to the BASE_ADDR: First determine if the DPR will perform an internal hflip or vflip base on <code>FRAME_CTRL0: ROT_FLIP_ORDER_EN</code>, <code>ROT_ENC</code>, <code>VFLIP_EN</code> and <code>HFLIP_EN</code>. <pre>int_hflip = (((ROT_ENC==0) && (~VFLIP_EN && HFLIP_EN)) (VFLIP_EN</pre>

Field	Function
	<pre> && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en int_vflip = (((ROT_ENC==0) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en Second calculate the number of pix_x_byte_offset based on the pix_x_offset calculation described in FRAME_1P_PIX_X_CTRL: NUM_X_PIX_WIDE, and the MODE_CTRL0: PIX_SIZE. pix_x_byte_offset = int_hflip_en ? (pix_x_offset << PIX_SIZE) : 0; Third calculate the number of pix_y_byte_offset based on the pix_y_offset calculation described in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH, and the FRAME_CTRL0: PITCH. pix_y_byte_offset = int_vflip_en ? (PITCH * pix_y_offset) : 0; Fourth apply the following pix_x_byte_offset and pix_y_byte_offset to the BASE_ADDR. This is the value which should be written to BASE_ADDR. BASE_ADDR = (BASE_ADDR - pix_y_byte_offset) - pix_x_byte_offset; </pre>

15.7.3.1.42 Frame 2-Plane Control 0 (FRAME_2P_CTRL0)

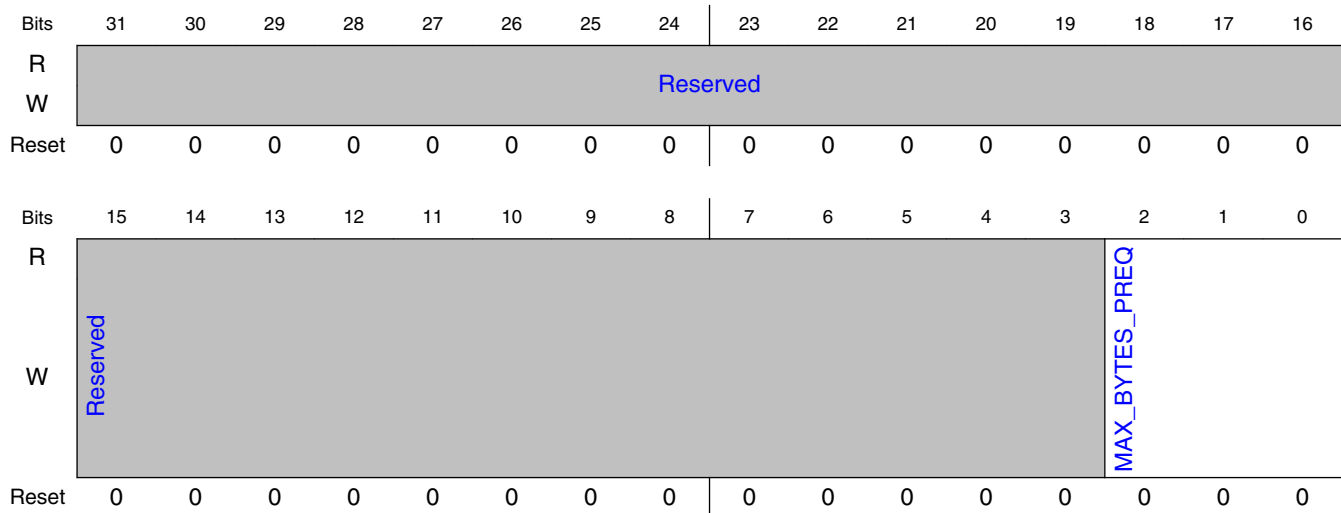
15.7.3.1.42.1 Offset

Register	Offset
FRAME_2P_CTRL0	E0h

15.7.3.1.42.2 Function

The Frame 2P Control0 register controls fields only for 2-Plane Chroma image processing.

15.7.3.1.42.3 Diagram



15.7.3.1.42.4 Fields

Field	Function
31-3 —	Reserved.
2-0 MAX_BYTES_P REQ	Max Bytes Per Request Maximum number of bytes to prefetch per request. 0x0 = 64 bytes 0x1 = 128 bytes 0x2 = 256 bytes 0x3 = 512 bytes 0x4 = 1K bytes 0x5 = 2K bytes 0x6 = 4K bytes 0x7 = 8K bytes Restrictions based on MODE_CTRL0:TILE_TYPE or FRAME_CTRL0:ROT_ENC • Rotation by 90 or 270: limit to 64 bytes • VPU Tile: limit to 64 bytes • GPU 16bpp super tile: limit to 64 bytes • GPU 32bpp super tile: limit to 128 bytes

15.7.3.1.43 Frame 2-Plane Control 0 (FRAME_2P_CTRL0_SET)

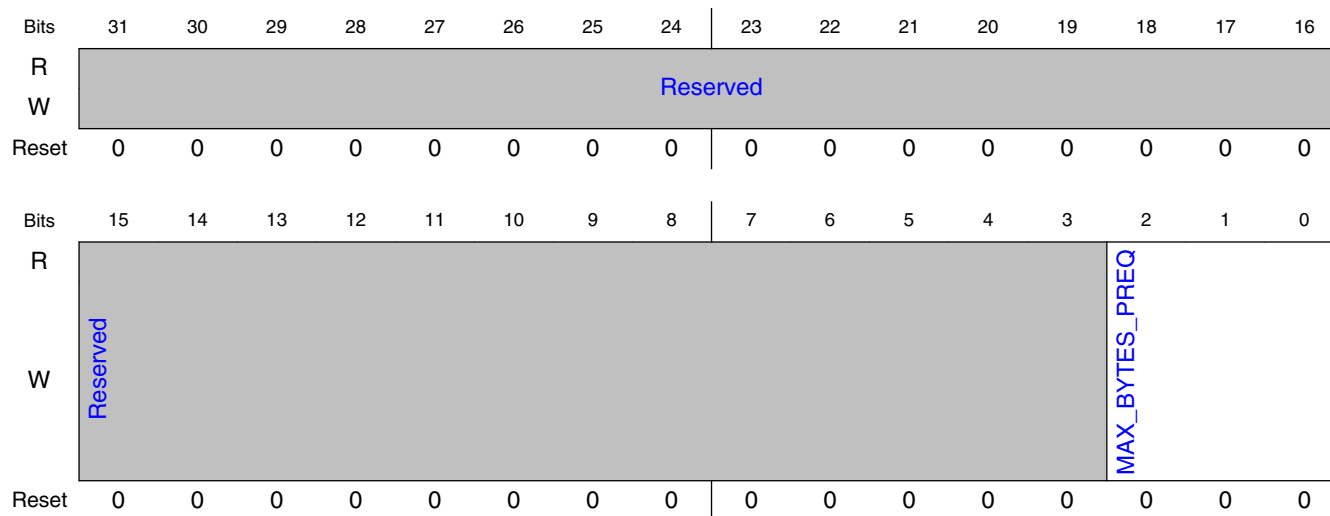
15.7.3.1.43.1 Offset

Register	Offset
FRAME_2P_CTRL0_SET	E4h

15.7.3.1.43.2 Function

The Frame 2P Control0 register controls fields only for 2-Plane Chroma image processing.

15.7.3.1.43.3 Diagram



15.7.3.1.43.4 Fields

Field	Function
31-3 —	Reserved.
2-0 MAX_BYTES_P REQ	Max Bytes Per Request Maximum number of bytes to prefetch per request. 0x0 = 64 bytes 0x1 = 128 bytes 0x2 = 256 bytes 0x3 = 512 bytes 0x4 = 1K bytes 0x5 = 2K bytes

Field	Function
	0x6 = 4K bytes 0x7 = 8K bytes Restrictions based on MODE_CTRL0:TILE_TYPE or FRAME_CTRL0:ROT_ENC • Rotation by 90 or 270: limit to 64 bytes • VPU Tile: limit to 64 bytes • GPU 16bpp super tile: limit to 64 bytes • GPU 32bpp super tile: limit to 128 bytes

15.7.3.1.44 Frame 2-Plane Control 0 (FRAME_2P_CTRL0_CLR)

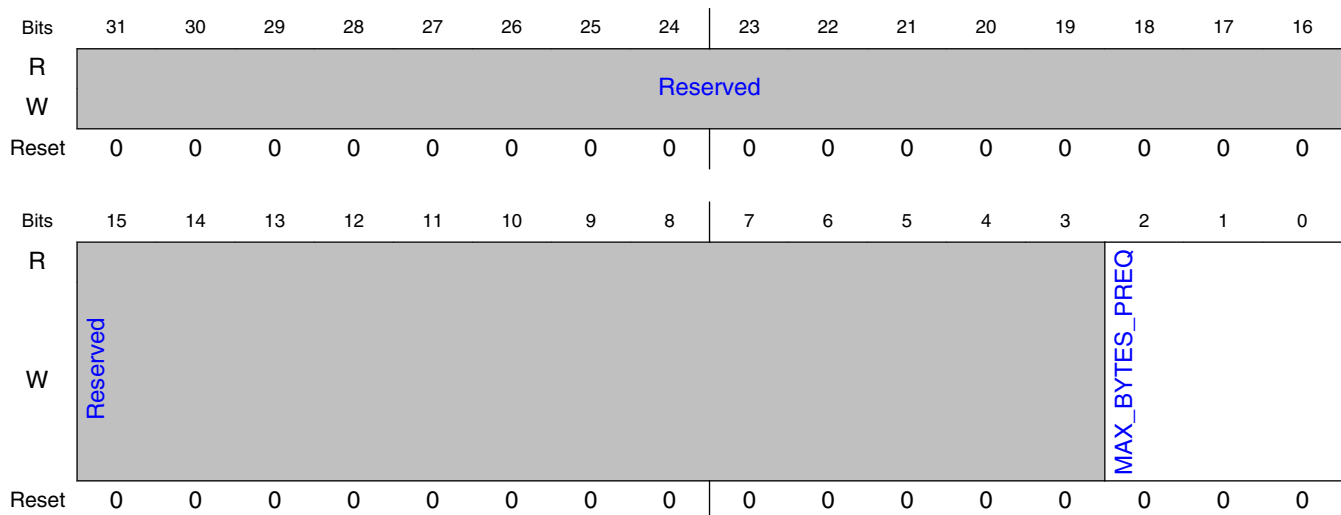
15.7.3.1.44.1 Offset

Register	Offset
FRAME_2P_CTRL0_CLR	E8h

15.7.3.1.44.2 Function

The Frame 2P Control0 register controls fields only for 2-Plane Chroma image processing.

15.7.3.1.44.3 Diagram



15.7.3.1.44.4 Fields

Field	Function
31-3 —	Reserved.
2-0 MAX_BYTES_P REQ	Max Bytes Per Request Maximum number of bytes to prefetch per request. 0x0 = 64 bytes 0x1 = 128 bytes 0x2 = 256 bytes 0x3 = 512 bytes 0x4 = 1K bytes 0x5 = 2K bytes 0x6 = 4K bytes 0x7 = 8K bytes Restrictions based on MODE_CTRL0:TILE_TYPE or FRAME_CTRL0:ROT_ENC • Rotation by 90 or 270: limit to 64 bytes • VPU Tile: limit to 64 bytes • GPU 16bpp super tile: limit to 64 bytes • GPU 32bpp super tile: limit to 128 bytes

15.7.3.1.45 Frame 2-Plane Control 0 (FRAME_2P_CTRL0_TOG)

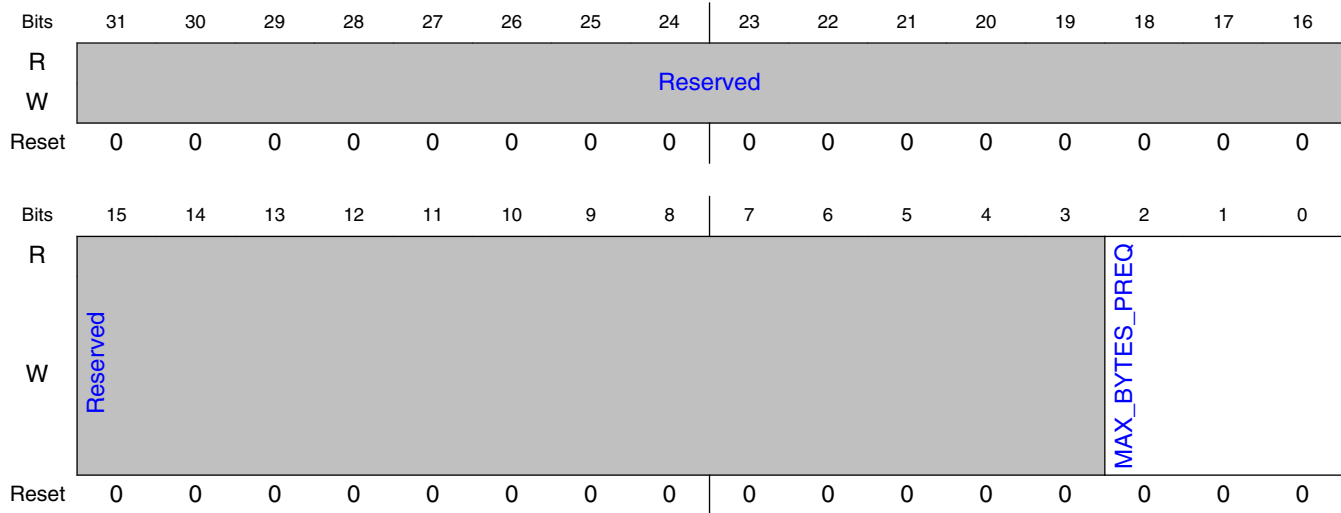
15.7.3.1.45.1 Offset

Register	Offset
FRAME_2P_CTRL0_TOG	ECh

15.7.3.1.45.2 Function

The Frame 2P Control0 register controls fields only for 2-Plane Chroma image processing.

15.7.3.1.45.3 Diagram



15.7.3.1.45.4 Fields

Field	Function
31-3 —	Reserved.
2-0 MAX_BYTES_P REQ	Max Bytes Per Request Maximum number of bytes to prefetch per request. 0x0 = 64 bytes 0x1 = 128 bytes 0x2 = 256 bytes 0x3 = 512 bytes 0x4 = 1K bytes 0x5 = 2K bytes 0x6 = 4K bytes 0x7 = 8K bytes Restrictions based on MODE_CTRL0:TILE_TYPE or FRAME_CTRL0:ROT_ENC - Rotation by 90 or 270: limit to 64 bytes - VPU Tile: limit to 64 bytes - GPU 16bpp super tile: limit to 64 bytes - GPU 32bpp super tile: limit to 128 bytes

15.7.3.1.46 Frame 2-Plane Pix X Control (FRAME_2P_PIX_X_CTRL)

15.7.3.1.46.1 Offset

Register	Offset
FRAME_2P_PIX_X_CTRL	F0h

15.7.3.1.46.2 Function

The Frame 2P PIX_X Control register controls x-axis pixel control for 2-Plane UV image processing.

15.7.3.1.46.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_X															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_X_PIX_WIDE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.46.4 Fields

Field	Function																		
31-16 CROP_ULC_X	Starting Coordinate of Cropped Image X (2-Plane UV) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB, 1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB, 1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB, 1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	

Table continues on the next page...

Field	Function
15-0 NUM_X_PIX_WIDE	Number of Pixels Wide in X-direction The number of input frame pixels to fetch in the x-axis direction. Note: To simplify the logic conversion from pixels to bytes, the U and V components will be calculated as separate pixels instead of UV as a combined pixel. As an example, in 2PYUV420, the U component will be treated as a 8-bit pixel and the V component as a 2nd 8-bit pixel instead of a single 16-bit pixel. In non-linear (raster) tile mode, the starting pixel must be at the start of a tile boundary. Partial tile fetch is not allowed. The NUM_X_PIX_WIDE must be sized so the prefetch is evenly divisible by 64 bytes. If the NUM_X_PIX_WIDE is not evenly divisible by 64 bytes, then an offset must be applied. To calculate if NUM_X_PIX_WIDE is divisible by 64 bytes: First determine the number of pix_x pixels divisible in 64 bytes, based on MODE_CTRL0: PIX_SIZE and TILE_TYPE settings: <pre> num_pix_x_in_64byte = (((PIX_SIZE==1) && (TILE_TYPE==0)) 32 : // VPU 10bpp VP9-10 UV linear tile must be divisible by 32 pixels in x- direction ((PIX_SIZE==1) && (TILE_TYPE==4)) 8 : // VPU 10bpp VP9-10 UV tile must be divisible by 8 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==0)) 64 : // VPU 8bpp VP9 UV linear tile must be divisible by 64 pixels in x- direction ((PIX_SIZE==0) && (TILE_TYPE==4)) 16 : // VPU 8bpp VP9 UV tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==3)) 8 : // VPU 8bpp 2PYUV420 UV zenverge tile must be divisible by 8 pixels in x-direction 8); // default case </pre> shouldn't be hit Second calculate the modulus of the NUM_X_PIX_WIDE by dpr_num_pix_x_in_64byte Note: for 2PYUV420 10bpp source frame, NUM_X_PIX_WIDE must first be multiplied by 1.25 to account for MODE_CTRL0: PIX_SIZE=8bpp operation. <pre> pix_x_div_64byte_mod = NUM_X_PIX_WIDE % num_pix_x_in_64byte </pre> Third calculate the number of offset pixels <pre> pix_x_offset = (pix_x_div_64byte_mod == 0) ? 0 : (num_pix_x_in_64byte - pix_x_div_64byte_mod) </pre> Fourth add pix_x_offset to NUM_X_PIX_WIDE. This is the value written to the NUM_X_PIX_WIDE control registers. <pre> NUM_X_PIX_WIDE = NUM_X_PIX_WIDE + pix_x_offset </pre> Note1: If FRAME_2P_CTRL0:MAX_BYTES_PREQ > 64 bytes AND FRAME_CTRL0:HFLIP_EN=1, then the prefetch calculation above must be evenly divisible by 128 bytes instead of 64 bytes.

15.7.3.1.47 Frame 2-Plane Pix X Control (FRAME_2P_PIX_X_CTRL_SET)

15.7.3.1.47.1 Offset

Register	Offset
FRAME_2P_PIX_X_CTRL_SET	F4h

15.7.3.1.47.2 Function

The Frame 2P PIX_X Control register controls x-axis pixel control for 2-Plane UV image processing.

15.7.3.1.47.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_X															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_X_PIX_WIDE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.47.4 Fields

Field	Function												
31-16 CROP_ULC_X	Starting Coordinate of Cropped Image X (2-Plane UV) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0
Tile Type	X Alignment	Y Alignment											
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0											
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0											
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0											

Table continues on the next page...

Field	Function
	VP9 8bpp 2-Plane YUV420 CROP_ULC_X[3:0] = 0 CROP_ULC_Y[3:0] = 0 VP9 10bpp 2-Plane YUV420 CROP_ULC_X[2:0] = 0 CROP_ULC_Y[2:0] = 0 -----
15-0 NUM_X_PIX_WIDE	Number of Pixels Wide in X-direction The number of input frame pixels to fetch in the x-axis direction. Note: To simplify the logic conversion from pixels to bytes, the U and V components will be calculated as separate pixels instead of UV as a combined pixel. As an example, in 2PYUV420, the U component will be treated as a 8-bit pixel and the V component as a 2nd 8-bit pixel instead of a single 16-bit pixel. In non-linear (raster) tile mode, the starting pixel must be at the start of a tile boundary. Partial tile fetch is not allowed. The NUM_X_PIX_WIDE must be sized so the prefetch is evenly divisible by 64 bytes. If the NUM_X_PIX_WIDE is not evenly divisible by 64 bytes, then an offset must be applied. To calculate if NUM_X_PIX_WIDE is divisible by 64 bytes: First determine the number of pix_x pixels divisible in 64 bytes, based on MODE_CTRL0: PIX_SIZE and TILE_TYPE settings: <pre> num_pix_x_in_64byte = (((PIX_SIZE==1) && (TILE_TYPE==0)) 32 : // VPU 10bpp VP9-10 UV linear tile must be divisible by 32 pixels in x- direction ((PIX_SIZE==1) && (TILE_TYPE==4)) 8 : // VPU 10bpp VP9-10 UV tile must be divisible by 8 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==0)) 64 : // VPU 8bpp VP9 UV linear tile must be divisible by 64 pixels in x- direction ((PIX_SIZE==0) && (TILE_TYPE==4)) 16 : // VPU 8bpp VP9 UV tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==3)) 8 : // VPU 8bpp 2PYUV420 UV zenverge tile must be divisible by 8 pixels in x-direction 8); // default case </pre> shouldn't be hit Second calculate the modulus of the NUM_X_PIX_WIDE by dpr_num_pix_x_in_64byte Note: for 2PYUV420 10bpp source frame, NUM_X_PIX_WIDE must first be multiplied by 1.25 to account for MODE_CTRL0: PIX_SIZE=8bpp operation. <pre> pix_x_div_64byte_mod = NUM_X_PIX_WIDE % num_pix_x_in_64byte </pre> Third calculate the number of offset pixels <pre> pix_x_offset = (pix_x_div_64byte_mod == 0) ? 0 : (num_pix_x_in_64byte - pix_x_div_64byte_mod) </pre> Fourth add pix_x_offset to NUM_X_PIX_WIDE. This is the value written to the NUM_X_PIX_WIDE control registers. <pre> NUM_X_PIX_WIDE = NUM_X_PIX_WIDE + pix_x_offset </pre> Note1: If FRAME_2P_CTRL0:MAX_BYTES_PREQ > 64 bytes AND FRAME_CTRL0:HFLIP_EN=1, then the prefetch calculation above must be evenly divisible by 128 bytes instead of 64 bytes.

15.7.3.1.48 Frame 2-Plane Pix X Control (FRAME_2P_PIX_X_CTRL_CLR)

15.7.3.1.48.1 Offset

Register	Offset
FRAME_2P_PIX_X_CTRL_CLR	F8h

15.7.3.1.48.2 Function

The Frame 2P PIX_X Control register controls x-axis pixel control for 2-Plane UV image processing.

15.7.3.1.48.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_X															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_X_PIX_WIDE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.48.4 Fields

Field	Function												
31-16 CROP_ULC_X	Starting Coordinate of Cropped Image X (2-Plane UV) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0
Tile Type	X Alignment	Y Alignment											
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0											
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0											
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0											

Table continues on the next page...

Field	Function
	VP9 8bpp 2-Plane YUV420 CROP_ULC_X[3:0] = 0 CROP_ULC_Y[3:0] = 0 VP9 10bpp 2-Plane YUV420 CROP_ULC_X[2:0] = 0 CROP_ULC_Y[2:0] = 0 -----
15-0 NUM_X_PIX_WIDE	Number of Pixels Wide in X-direction The number of input frame pixels to fetch in the x-axis direction. Note: To simplify the logic conversion from pixels to bytes, the U and V components will be calculated as separate pixels instead of UV as a combined pixel. As an example, in 2PYUV420, the U component will be treated as a 8-bit pixel and the V component as a 2nd 8-bit pixel instead of a single 16-bit pixel. In non-linear (raster) tile mode, the starting pixel must be at the start of a tile boundary. Partial tile fetch is not allowed. The NUM_X_PIX_WIDE must be sized so the prefetch is evenly divisible by 64 bytes. If the NUM_X_PIX_WIDE is not evenly divisible by 64 bytes, then an offset must be applied. To calculate if NUM_X_PIX_WIDE is divisible by 64 bytes: First determine the number of pix_x pixels divisible in 64 bytes, based on MODE_CTRL0: PIX_SIZE and TILE_TYPE settings: <pre> num_pix_x_in_64byte = (((PIX_SIZE==1) && (TILE_TYPE==0)) 32 : // VPU 10bpp VP9-10 UV linear tile must be divisible by 32 pixels in x- direction ((PIX_SIZE==1) && (TILE_TYPE==4)) 8 : // VPU 10bpp VP9-10 UV tile must be divisible by 8 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==0)) 64 : // VPU 8bpp VP9 UV linear tile must be divisible by 64 pixels in x- direction ((PIX_SIZE==0) && (TILE_TYPE==4)) 16 : // VPU 8bpp VP9 UV tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==3)) 8 : // VPU 8bpp 2PYUV420 UV zenverge tile must be divisible by 8 pixels in x-direction 8); // default case </pre> shouldn't be hit Second calculate the modulus of the NUM_X_PIX_WIDE by dpr_num_pix_x_in_64byte Note: for 2PYUV420 10bpp source frame, NUM_X_PIX_WIDE must first be multiplied by 1.25 to account for MODE_CTRL0: PIX_SIZE=8bpp operation. <pre> pix_x_div_64byte_mod = NUM_X_PIX_WIDE % num_pix_x_in_64byte </pre> Third calculate the number of offset pixels <pre> pix_x_offset = (pix_x_div_64byte_mod == 0) ? 0 : (num_pix_x_in_64byte - pix_x_div_64byte_mod) </pre> Fourth add pix_x_offset to NUM_X_PIX_WIDE. This is the value written to the NUM_X_PIX_WIDE control registers. <pre> NUM_X_PIX_WIDE = NUM_X_PIX_WIDE + pix_x_offset </pre> Note1: If FRAME_2P_CTRL0:MAX_BYTES_PREQ > 64 bytes AND FRAME_CTRL0:HFLIP_EN=1, then the prefetch calculation above must be evenly divisible by 128 bytes instead of 64 bytes.

15.7.3.1.49 Frame 2-Plane Pix X Control (FRAME_2P_PIX_X_CTRL_TOG)

15.7.3.1.49.1 Offset

Register	Offset
FRAME_2P_PIX_X_CTRL_TOG	FCh

15.7.3.1.49.2 Function

The Frame 2P PIX_X Control register controls x-axis pixel control for 2-Plane UV image processing.

15.7.3.1.49.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_X															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_X_PIX_WIDE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.49.4 Fields

Field	Function												
31-16 CROP_ULC_X	Starting Coordinate of Cropped Image X (2-Plane UV) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0
Tile Type	X Alignment	Y Alignment											
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0											
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0											
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0											

Table continues on the next page...

Field	Function
	VP9 8bpp 2-Plane YUV420 CROP_ULC_X[3:0] = 0 CROP_ULC_Y[3:0] = 0 VP9 10bpp 2-Plane YUV420 CROP_ULC_X[2:0] = 0 CROP_ULC_Y[2:0] = 0 -----
15-0 NUM_X_PIX_WIDE	Number of Pixels Wide in X-direction The number of input frame pixels to fetch in the x-axis direction. Note: To simplify the logic conversion from pixels to bytes, the U and V components will be calculated as separate pixels instead of UV as a combined pixel. As an example, in 2PYUV420, the U component will be treated as a 8-bit pixel and the V component as a 2nd 8-bit pixel instead of a single 16-bit pixel. In non-linear (raster) tile mode, the starting pixel must be at the start of a tile boundary. Partial tile fetch is not allowed. The NUM_X_PIX_WIDE must be sized so the prefetch is evenly divisible by 64 bytes. If the NUM_X_PIX_WIDE is not evenly divisible by 64 bytes, then an offset must be applied. To calculate if NUM_X_PIX_WIDE is divisible by 64 bytes: First determine the number of pix_x pixels divisible in 64 bytes, based on MODE_CTRL0: PIX_SIZE and TILE_TYPE settings: <pre> num_pix_x_in_64byte = (((PIX_SIZE==1) && (TILE_TYPE==0)) 32 : // VPU 10bpp VP9-10 UV linear tile must be divisible by 32 pixels in x- direction ((PIX_SIZE==1) && (TILE_TYPE==4)) 8 : // VPU 10bpp VP9-10 UV tile must be divisible by 8 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==0)) 64 : // VPU 8bpp VP9 UV linear tile must be divisible by 64 pixels in x- direction ((PIX_SIZE==0) && (TILE_TYPE==4)) 16 : // VPU 8bpp VP9 UV tile must be divisible by 16 pixels in x-direction ((PIX_SIZE==0) && (TILE_TYPE==3)) 8 : // VPU 8bpp 2PYUV420 UV zenverge tile must be divisible by 8 pixels in x-direction 8); // default case </pre> shouldn't be hit Second calculate the modulus of the NUM_X_PIX_WIDE by dpr_num_pix_x_in_64byte Note: for 2PYUV420 10bpp source frame, NUM_X_PIX_WIDE must first be multiplied by 1.25 to account for MODE_CTRL0: PIX_SIZE=8bpp operation. <pre> pix_x_div_64byte_mod = NUM_X_PIX_WIDE % num_pix_x_in_64byte </pre> Third calculate the number of offset pixels <pre> pix_x_offset = (pix_x_div_64byte_mod == 0) ? 0 : (num_pix_x_in_64byte - pix_x_div_64byte_mod) </pre> Fourth add pix_x_offset to NUM_X_PIX_WIDE. This is the value written to the NUM_X_PIX_WIDE control registers. <pre> NUM_X_PIX_WIDE = NUM_X_PIX_WIDE + pix_x_offset </pre> Note1: If FRAME_2P_CTRL0:MAX_BYTES_PREQ > 64 bytes AND FRAME_CTRL0:HFLIP_EN=1, then the prefetch calculation above must be evenly divisible by 128 bytes instead of 64 bytes.

15.7.3.1.50 Frame 2-Plane Pix Y Control (FRAME_2P_PIX_Y_CTRL)

15.7.3.1.50.1 Offset

Register	Offset
FRAME_2P_PIX_Y_CTRL	100h

15.7.3.1.50.2 Function

The Frame 2P PIX_Y Control register controls y-axis pixel control for 2-Plane UV image processing.

15.7.3.1.50.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CROP_ULC_Y															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NUM_Y_PIX_HIGH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.50.4 Fields

Field	Function												
31-16 CROP_ULC_Y	Starting Coordinate of Cropped Image Y (2-Plane UV) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0
Tile Type	X Alignment	Y Alignment											
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0											
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0											
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0											

Table continues on the next page...

Field	Function									
	<table><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr></table>	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	-----	-----	-----
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0								
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0								
-----	-----	-----								
15-0 NUM_Y_PIX_HI GH	Number of Pixels High in Y-direction The number of input frame pixels to fetch in the y-axis direction. In non-linear (raster) t mode, the number of pixels must be sized for a full tile block boundary. Partial tile fetch is not allowed. The NUM_Y_PIX_HIGH value must be evenly divisible by the number of rows programmed in MODE_CTRL0: RTR_4LINE_BUF_EN. First determine the number of rows in the buffer num_rows_buf = (RTR_4LINE_BUF_EN==0) ? 8 : 4; Second calculate the modulus of NUM_Y_PIX_HIGH by the num_rows_buf Note: The NUM_Y_PIX_HIGH UV height is 1/2 height of Luma from FRAME_1P_PIX_Y_CTRL:NUM_Y_PIX_HIGH before luma height offset was calculated pix_y_mod = NUM_Y_PIX_HIGH % num_rows_buf; Third calculate the number of rows to offset pix_y_offset = (pix_y_mod==0) ? 0 : (num_rows_buf - pix_y_mod); Fourth add pix_y_offset to NUM_Y_PIX_HIGH. This is the value written to the NUM_Y_PIX_HIGH control registers. NUM_Y_PIX_HIGH = NUM_Y_PIX_HIGH + pix_y_offset;									

15.7.3.1.51 Frame 2-Plane Pix Y Control (FRAME_2P_PIX_Y_CTRL_SET)

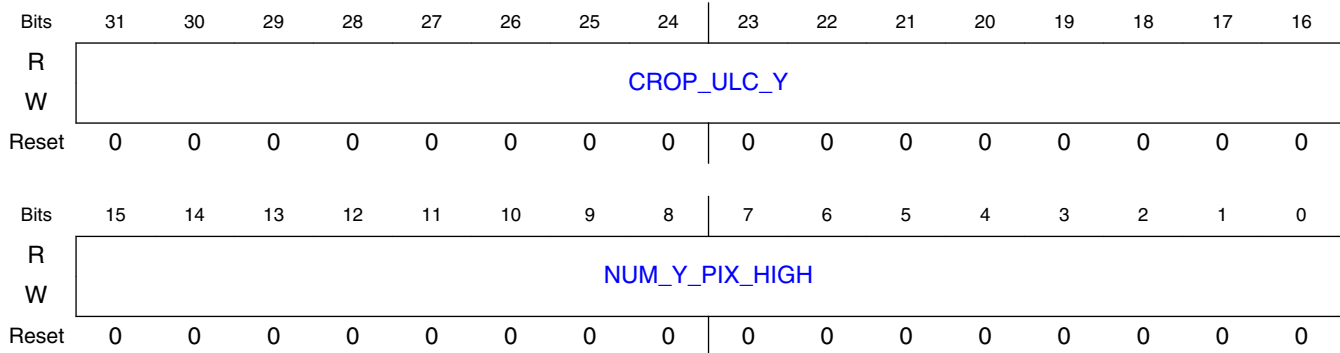
15.7.3.1.51.1 Offset

Register	Offset
FRAME_2P_PIX_Y_CTRL_SET	104h

15.7.3.1.51.2 Function

The Frame 2P PIX_Y Control register controls y-axis pixel control for 2-Plane UV image processing.

15.7.3.1.51.3 Diagram



15.7.3.1.51.4 Fields

Field	Function																		
31-16 CROP_ULC_Y	Starting Coordinate of Cropped Image Y (2-Plane UV) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
15-0 NUM_Y_PIX_HI GH	Number of Pixels High in Y-direction The number of input frame pixels to fetch in the y-axis direction. In non-linear (raster) t mode, the number of pixels must be sized for a full tile block boundary. Partial tile fetch is not allowed. The NUM_Y_PIX_HIGH value must be evenly divisible by the number of rows programmed in MODE_CTRL0: RTR_4LINE_BUF_EN. First determine the number of rows in the buffer num_rows_buf = (RTR_4LINE_BUF_EN==0) ? 8 : 4; Second calculate the modulus of NUM_Y_PIX_HIGH by the num_rows_buf Note: The NUM_Y_PIX_HIGH UV height is 1/2 height of Luma from FRAME_1P_PIX_Y_CTRL:NUM_Y_PIX_HIGH before luma height offset was calculated pix_y_mod = NUM_Y_PIX_HIGH % num_rows_buf;																		

Field	Function
	Third calculate the number of rows to offset $\text{pix_y_offset} = (\text{pix_y_mod} == 0) ? 0 : (\text{num_rows_buf} - \text{pix_y_mod});$ Fourth add <code>pix_y_offset</code> to <code>NUM_Y_PIX_HIGH</code>. This is the value written to the <code>NUM_Y_PIX_HIGH</code> control registers. $\text{NUM_Y_PIX_HIGH} = \text{NUM_Y_PIX_HIGH} + \text{pix_y_offset};$

15.7.3.1.52 Frame 2-Plane Pix Y Control (FRAME_2P_PIX_Y_CTRL_CLR)

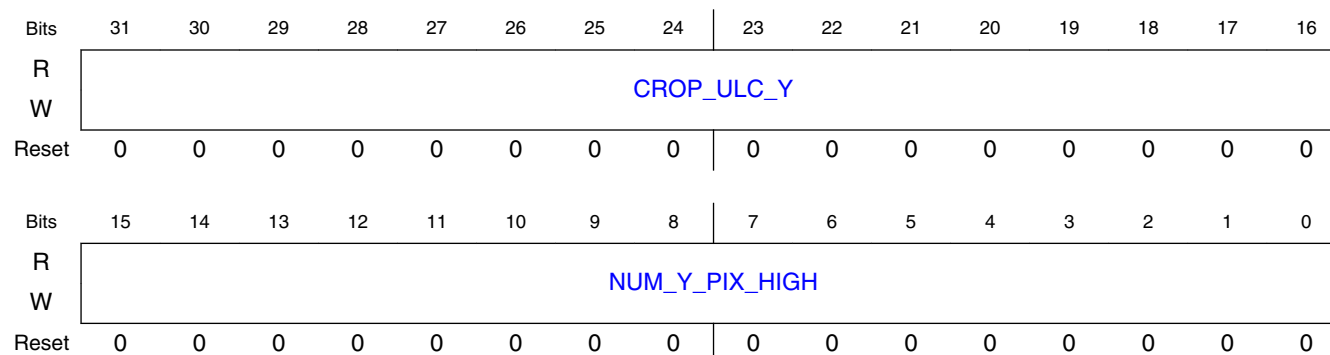
15.7.3.1.52.1 Offset

Register	Offset
FRAME_2P_PIX_Y_CTRL_CLR	108h

15.7.3.1.52.2 Function

The Frame 2P PIX_Y Control register controls y-axis pixel control for 2-Plane UV image processing.

15.7.3.1.52.3 Diagram



15.7.3.1.52.4 Fields

Field	Function
31-16 CROP_ULC_Y	Starting Coordinate of Cropped Image Y (2-Plane UV) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the <code>CROP_ULC_X</code> and <code>CROP_ULC_Y</code> coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the <code>NUM_X_PIX_WIDE</code> and <code>NUM_Y_PIX_HIGH</code> will indicate when to stop fetching

Table continues on the next page...

Field	Function																		
	pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><th>Tile Type</th><th>X Alignment</th><th>Y Alignment</th></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0																	
GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
15-0 NUM_Y_PIX_HI GH	Number of Pixels High in Y-direction The number of input frame pixels to fetch in the y-axis direction. In non-linear (raster) t mode, the number of pixels must be sized for a full tile block boundary. Partial tile fetch is not allowed. The NUM_Y_PIX_HIGH value must be evenly divisible by the number of rows programmed in MODE_CTRL0: RTR_4LINE_BUF_EN. First determine the number of rows in the buffer num_rows_buf = (RTR_4LINE_BUF_EN==0) ? 8 : 4; Second calculate the modulus of NUM_Y_PIX_HIGH by the num_rows_buf Note: The NUM_Y_PIX_HIGH UV height is 1/2 height of Luma from FRAME_1P_PIX_Y_CTRL:NUM_Y_PIX_HIGH before luma height offset was calculated pix_y_mod = NUM_Y_PIX_HIGH % num_rows_buf; Third calculate the number of rows to offset pix_y_offset = (pix_y_mod==0) ? 0 : (num_rows_buf - pix_y_mod); Fourth add pix_y_offset to NUM_Y_PIX_HIGH. This is the value written to the NUM_Y_PIX_HIGH control registers. NUM_Y_PIX_HIGH = NUM_Y_PIX_HIGH + pix_y_offset;																		

15.7.3.1.53 Frame 2-Plane Pix Y Control (FRAME_2P_PIX_Y_CTRL_TOG)

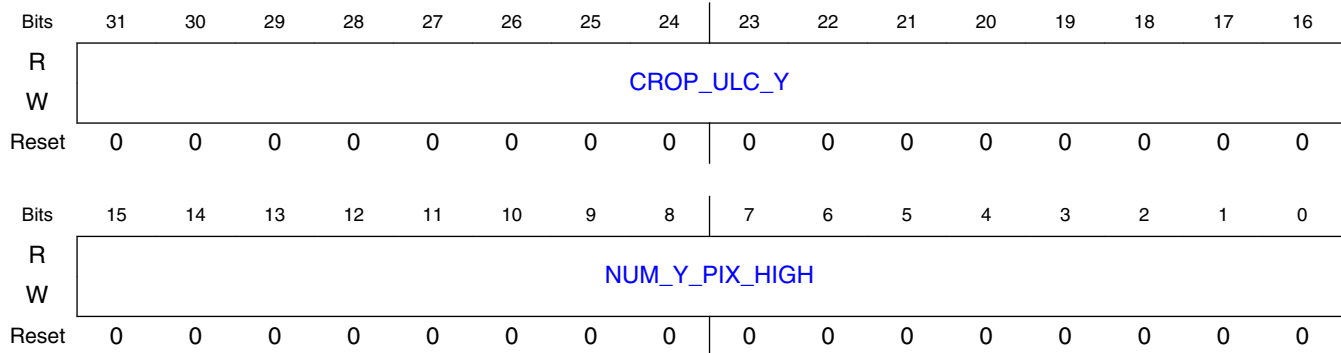
15.7.3.1.53.1 Offset

Register	Offset
FRAME_2P_PIX_Y_CTRL_TOG	10Ch

15.7.3.1.53.2 Function

The Frame 2P PIX_Y Control register controls y-axis pixel control for 2-Plane UV image processing.

15.7.3.1.53.3 Diagram



15.7.3.1.53.4 Fields

Field	Function																		
31-16 CROP_ULC_Y	Starting Coordinate of Cropped Image Y (2-Plane UV) Due to the tile geometry in VPU and Super-Tile formats, a simple address offset cannot be used to crop an image (in all cases). In these modes, the CROP_ULC_X and CROP_ULC_Y coordinate pair will be used to specify an offset to begin fetching pixels within an image. In addition, the NUM_X_PIX_WIDE and NUM_Y_PIX_HIGH will indicate when to stop fetching pixels. Note, these coordinate pairs (CROP_ULC_X,CROP_ULC_Y) and (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH) must be aligned to block boundaries for proper functionality. The following table illustrates coordinate restrictions for the tile types of interest (Note, the same restrictions apply to (NUM_X_PIX_WIDE,NUM_Y_PIX_HIGH)): <table><tr><td>Tile Type</td><td>X Alignment</td><td>Y Alignment</td></tr><tr><td>GPU 32bpp (X-Major Super-tile)</td><td>CROP_ULC_X[1:0] = 0</td><td>CROP_ULC_Y[1:0] = 0</td></tr><tr><td>GPU 16bpp xRGB,1-Plane YUV422</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr><tr><td>VPU 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 8bpp 2-Plane YUV420</td><td>CROP_ULC_X[3:0] = 0</td><td>CROP_ULC_Y[3:0] = 0</td></tr><tr><td>VP9 10bpp 2-Plane YUV420</td><td>CROP_ULC_X[2:0] = 0</td><td>CROP_ULC_Y[2:0] = 0</td></tr></table>	Tile Type	X Alignment	Y Alignment	GPU 32bpp (X-Major Super-tile)	CROP_ULC_X[1:0] = 0	CROP_ULC_Y[1:0] = 0	GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0	VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0	VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0
Tile Type	X Alignment	Y Alignment																	
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GPU 16bpp xRGB,1-Plane YUV422	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
VPU 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 8bpp 2-Plane YUV420	CROP_ULC_X[3:0] = 0	CROP_ULC_Y[3:0] = 0																	
VP9 10bpp 2-Plane YUV420	CROP_ULC_X[2:0] = 0	CROP_ULC_Y[2:0] = 0																	
15-0 NUM_Y_PIX_HI GH	Number of Pixels High in Y-direction The number of input frame pixels to fetch in the y-axis direction. In non-linear (raster) t mode, the number of pixels must be sized for a full tile block boundary. Partial tile fetch is not allowed. The NUM_Y_PIX_HIGH value must be evenly divisible by the number of rows programmed in MODE_CTRL0: RTR_4LINE_BUF_EN. First determine the number of rows in the buffer num_rows_buf = (RTR_4LINE_BUF_EN==0) ? 8 : 4; Second calculate the modulus of NUM_Y_PIX_HIGH by the num_rows_buf Note: The NUM_Y_PIX_HIGH UV height is 1/2 height of Luma from FRAME_1P_PIX_Y_CTRL:NUM_Y_PIX_HIGH before luma height offset was calculated pix_y_mod = NUM_Y_PIX_HIGH % num_rows_buf;																		

Memory Map and Registers

Field	Function
	Third calculate the number of rows to offset $\text{pix_y_offset} = (\text{pix_y_mod} == 0) ? 0 : (\text{num_rows_buf} - \text{pix_y_mod});$ Fourth add <code>pix_y_offset</code> to <code>NUM_Y_PIX_HIGH</code>. This is the value written to the <code>NUM_Y_PIX_HIGH</code> control registers. $\text{NUM_Y_PIX_HIGH} = \text{NUM_Y_PIX_HIGH} + \text{pix_y_offset};$

15.7.3.1.54 Frame 2-Plane Base Address Control 0 (FRAME_2P_BASE_AD DR_CTRL0)

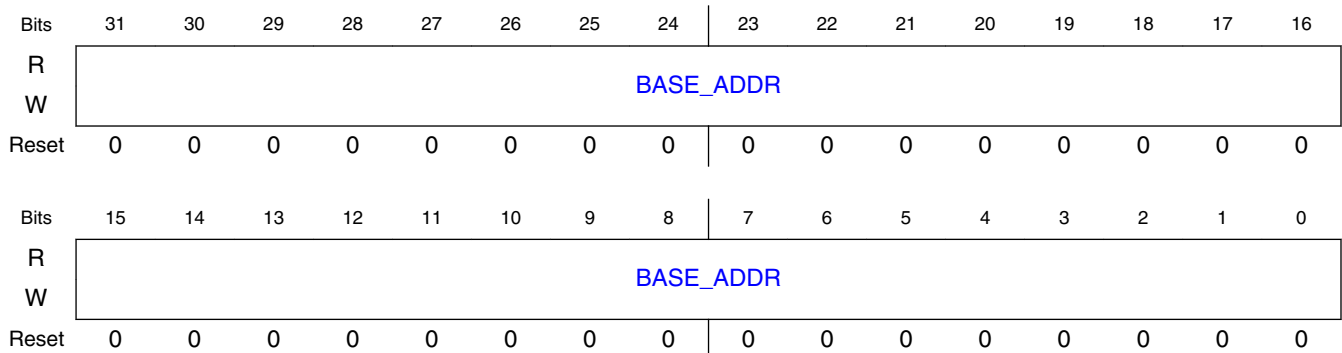
15.7.3.1.54.1 Offset

Register	Offset
FRAME_2P_BASE_AD DR_CTRL0	110h

15.7.3.1.54.2 Function

The Frame 2P Base Address Control0 register is the input frame base address for 2-Plane Chroma image processing.

15.7.3.1.54.3 Diagram



15.7.3.1.54.4 Fields

Field	Function
31-0 BASE_ADDR	Base Address The base address for 2-Plane Chroma. The base address is an input frame offset to the non-rotated, non-hflip, non-vflip output image. This will correspond to the output image's <code>pix_x=0</code>,

Field	Function
	pix_y=0 position. If any non-linear GPU tile or VPU tile is being processed, then the base address must start at pixel 0 of the tile boundary. Under certain conditions, a offset must be applied to the BASE_ADDR pix_x=0, pix_y=0 position. 1. The DPR will perform an internal hflip AND the pix_x_offset calculation doesn't equal 0 as described in FRAME_2P_PIX_X_CTRL: NUM_X_PIX_WIDE. 2. The DPR will perform an internal vflip AND the pix_y_offset calculation doesn't equal 0 as described in FRAME_2P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. Steps to apply an offset to the BASE_ADDR: First determine if the DPR will perform an internal hflip or vflip base on FRAME_CTRL0: ROT_FLIP_ORDER_EN, ROT_ENC, VFLIP_EN and HFLIP_EN. <pre> int_hflip = (((ROT_ENC==0) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> <pre> int_vflip = (((ROT_ENC==0) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> Second calculate the number of pix_x_byte_offset based on the pix_x_offset calculation described in FRAME_2P_PIX_X_CTRL: NUM_X_PIX_WIDE, and the MODE_CTRL0: PIX_SIZE. <pre> pix_x_byte_offset = int_hflip_en ? (pix_x_offset << PIX_SIZE) : 0; </pre> Third calculate the number of pix_y_byte_offset based on the pix_y_offset calculation described in FRAME_2P_PIX_Y_CTRL: NUM_Y_PIX_HIGH, and the FRAME_CTRL0: PITCH. <pre> pix_y_byte_offset = int_vflip_en ? (PITCH * pix_y_offset) : 0; </pre> Fourth apply the following pix_x_byte_offset and pix_y_byte_offset to the

Field	Function
	BASE_ADDR. This is the value which should be written to BASE_ADDR. Note: the pix_y_byte_offset is divided by 2 because the 2-Plane YUV420 UV height is 1/2 the pix_y_byte_offset calculated for the 2-Plane YUV420 Luma in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. BASE_ADDR = (BASE_ADDR - (pix_y_byte_offset >> 1)) - pix_x_byte_offset;

15.7.3.1.55 Frame 2-Plane Base Address Control 0 (FRAME_2P_BASE_ADDR_CTRL0_SET)

15.7.3.1.55.1 Offset

Register	Offset
FRAME_2P_BASE_ADDR_CTRL0_SET	114h

15.7.3.1.55.2 Function

The Frame 2P Base Address Control0 register is the input frame base address for 2-Plane Chroma image processing.

15.7.3.1.55.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	BASE_ADDR															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	BASE_ADDR															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.7.3.1.55.4 Fields

Field	Function
31-0 BASE_ADDR	Base Address The base address for 2-Plane Chroma. The base address is an input frame offset to the non-rotated, non-hflip, non-vflip output image. This will correspond to the output image's pix_x=0,

Field	Function
	pix_y=0 position. If any non-linear GPU tile or VPU tile is being processed, then the base address must start at pixel 0 of the tile boundary. Under certain conditions, a offset must be applied to the BASE_ADDR pix_x=0, pix_y=0 position. 1. The DPR will perform an internal hflip AND the pix_x_offset calculation doesn't equal 0 as described in FRAME_2P_PIX_X_CTRL: NUM_X_PIX_WIDE. 2. The DPR will perform an internal vflip AND the pix_y_offset calculation doesn't equal 0 as described in FRAME_2P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. Steps to apply an offset to the BASE_ADDR: First determine if the DPR will perform an internal hflip or vflip base on FRAME_CTRL0: ROT_FLIP_ORDER_EN, ROT_ENC, VFLIP_EN and HFLIP_EN. <pre> int_hflip = (((ROT_ENC==0) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> <pre> int_vflip = (((ROT_ENC==0) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> Second calculate the number of pix_x_byte_offset based on the pix_x_offset calculation described in FRAME_2P_PIX_X_CTRL: NUM_X_PIX_WIDE, and the MODE_CTRL0: PIX_SIZE. <pre> pix_x_byte_offset = int_hflip_en ? (pix_x_offset << PIX_SIZE) : 0; </pre> Third calculate the number of pix_y_byte_offset based on the pix_y_offset calculation described in FRAME_2P_PIX_Y_CTRL: NUM_Y_PIX_HIGH, and the FRAME_CTRL0: PITCH. <pre> pix_y_byte_offset = int_vflip_en ? (PITCH * pix_y_offset) : 0; </pre> Fourth apply the following pix_x_byte_offset and pix_y_byte_offset to the

Memory Map and Registers

Field	Function
	BASE_ADDR. This is the value which should be written to BASE_ADDR. Note: the pix_y_byte_offset is divided by 2 because the 2-Plane YUV420 UV height is 1/2 the pix_y_byte_offset calculated for the 2-Plane YUV420 Luma in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. BASE_ADDR = (BASE_ADDR - (pix_y_byte_offset >> 1)) - pix_x_byte_offset;

15.7.3.1.56 Frame 2-Plane Base Address Control 0 (FRAME_2P_BASE_ADDR_CTRL0_CLR)

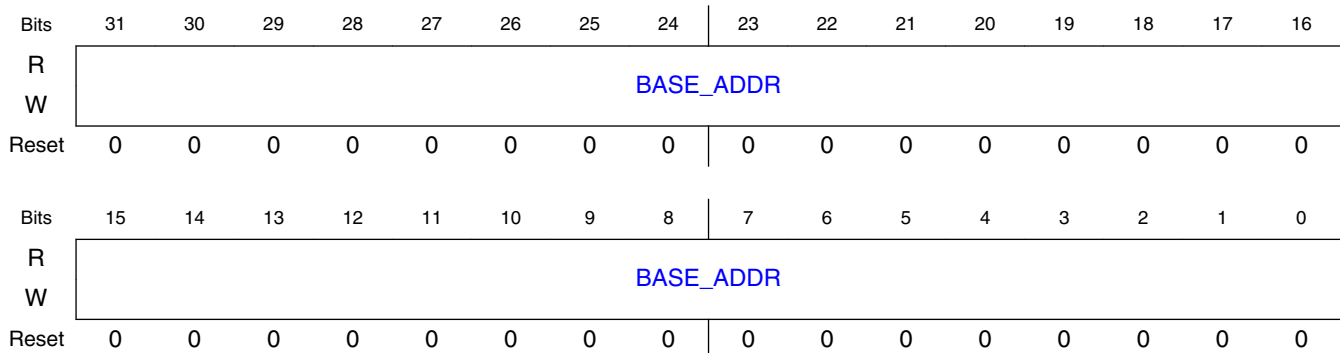
15.7.3.1.56.1 Offset

Register	Offset
FRAME_2P_BASE_ADDR_CTRL0_CLR	118h

15.7.3.1.56.2 Function

The Frame 2P Base Address Control0 register is the input frame base address for 2-Plane Chroma image processing.

15.7.3.1.56.3 Diagram



15.7.3.1.56.4 Fields

Field	Function
31-0 BASE_ADDR	Base Address The base address for 2-Plane Chroma. The base address is an input frame offset to the non-rotated, non-hflip, non-vflip output image. This will correspond to the output image's pix_x=0,

Field	Function
	pix_y=0 position. If any non-linear GPU tile or VPU tile is being processed, then the base address must start at pixel 0 of the tile boundary. Under certain conditions, a offset must be applied to the BASE_ADDR pix_x=0, pix_y=0 position. 1. The DPR will perform an internal hflip AND the pix_x_offset calculation doesn't equal 0 as described in FRAME_2P_PIX_X_CTRL: NUM_X_PIX_WIDE. 2. The DPR will perform an internal vflip AND the pix_y_offset calculation doesn't equal 0 as described in FRAME_2P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. Steps to apply an offset to the BASE_ADDR: First determine if the DPR will perform an internal hflip or vflip base on FRAME_CTRL0: ROT_FLIP_ORDER_EN, ROT_ENC, VFLIP_EN and HFLIP_EN. <pre> int_hflip = (((ROT_ENC==0) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> <pre> int_vflip = (((ROT_ENC==0) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> Second calculate the number of pix_x_byte_offset based on the pix_x_offset calculation described in FRAME_2P_PIX_X_CTRL: NUM_X_PIX_WIDE, and the MODE_CTRL0: PIX_SIZE. <pre> pix_x_byte_offset = int_hflip_en ? (pix_x_offset << PIX_SIZE) : 0; </pre> Third calculate the number of pix_y_byte_offset based on the pix_y_offset calculation described in FRAME_2P_PIX_Y_CTRL: NUM_Y_PIX_HIGH, and the FRAME_CTRL0: PITCH. <pre> pix_y_byte_offset = int_vflip_en ? (PITCH * pix_y_offset) : 0; </pre> Fourth apply the following pix_x_byte_offset and pix_y_byte_offset to the

Memory Map and Registers

Field	Function
	BASE_ADDR. This is the value which should be written to BASE_ADDR. Note: the pix_y_byte_offset is divided by 2 because the 2-Plane YUV420 UV height is 1/2 the pix_y_byte_offset calculated for the 2-Plane YUV420 Luma in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. BASE_ADDR = (BASE_ADDR - (pix_y_byte_offset >> 1)) - pix_x_byte_offset;

15.7.3.1.57 Frame 2-Plane Base Address Control 0 (FRAME_2P_BASE_ADDR_CTRL0_T0G)

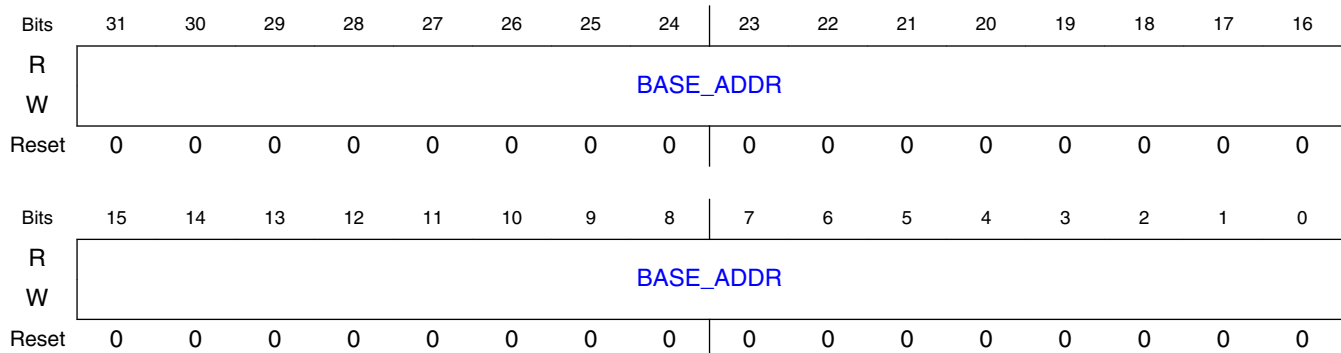
15.7.3.1.57.1 Offset

Register	Offset
FRAME_2P_BASE_ADDR_CTRL0_T0G	11Ch

15.7.3.1.57.2 Function

The Frame 2P Base Address Control0 register is the input frame base address for 2-Plane Chroma image processing.

15.7.3.1.57.3 Diagram



15.7.3.1.57.4 Fields

Field	Function
31-0 BASE_ADDR	Base Address The base address for 2-Plane Chroma. The base address is an input frame offset to the non-rotated, non-hflip, non-vflip output image. This will correspond to the output image's pix_x=0,

Field	Function
	pix_y=0 position. If any non-linear GPU tile or VPU tile is being processed, then the base address must start at pixel 0 of the tile boundary. Under certain conditions, a offset must be applied to the BASE_ADDR pix_x=0, pix_y=0 position. 1. The DPR will perform an internal hflip AND the pix_x_offset calculation doesn't equal 0 as described in FRAME_2P_PIX_X_CTRL: NUM_X_PIX_WIDE. 2. The DPR will perform an internal vflip AND the pix_y_offset calculation doesn't equal 0 as described in FRAME_2P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. Steps to apply an offset to the BASE_ADDR: First determine if the DPR will perform an internal hflip or vflip base on FRAME_CTRL0: ROT_FLIP_ORDER_EN, ROT_ENC, VFLIP_EN and HFLIP_EN. <pre> int_hflip = (((ROT_ENC==0) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> <pre> int_vflip = (((ROT_ENC==0) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN))) // rot_enc = 0 deg cases. ((ROT_ENC==2) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN))) // rot_enc = 180 deg cases. ((ROT_ENC==1) && ((VFLIP_EN && ~HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 90 deg cases if rotation before v/hflip_en ((ROT_ENC==1) && ((~VFLIP_EN && HFLIP_EN) (VFLIP_EN && HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1)) // rot_enc = 90 deg cases if rotation after v/hflip_en ((ROT_ENC==3) && ((~VFLIP_EN && HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==0)) // rot_enc = 270 deg cases if rotation before v/hflip_en ((ROT_ENC==3) && ((VFLIP_EN && ~HFLIP_EN) (~VFLIP_EN && ~HFLIP_EN)) && (ROT_FLIP_ORDER_EN==1))); // rot_enc = 270 deg cases if rotation after v/hflip_en </pre> Second calculate the number of pix_x_byte_offset based on the pix_x_offset calculation described in FRAME_2P_PIX_X_CTRL: NUM_X_PIX_WIDE, and the MODE_CTRL0: PIX_SIZE. <pre> pix_x_byte_offset = int_hflip_en ? (pix_x_offset << PIX_SIZE) : 0; </pre> Third calculate the number of pix_y_byte_offset based on the pix_y_offset calculation described in FRAME_2P_PIX_Y_CTRL: NUM_Y_PIX_HIGH, and the FRAME_CTRL0: PITCH. <pre> pix_y_byte_offset = int_vflip_en ? (PITCH * pix_y_offset) : 0; </pre> Fourth apply the following pix_x_byte_offset and pix_y_byte_offset to the

Memory Map and Registers

Field	Function
	BASE_ADDR. This is the value which should be written to BASE_ADDR. Note: the pix_y_byte_offset is divided by 2 because the 2-Plane YUV420 UV height is 1/2 the pix_y_byte_offset calculated for the 2-Plane YUV420 Luma in FRAME_1P_PIX_Y_CTRL: NUM_Y_PIX_HIGH. BASE_ADDR = (BASE_ADDR - (pix_y_byte_offset >> 1)) - pix_x_byte_offset;

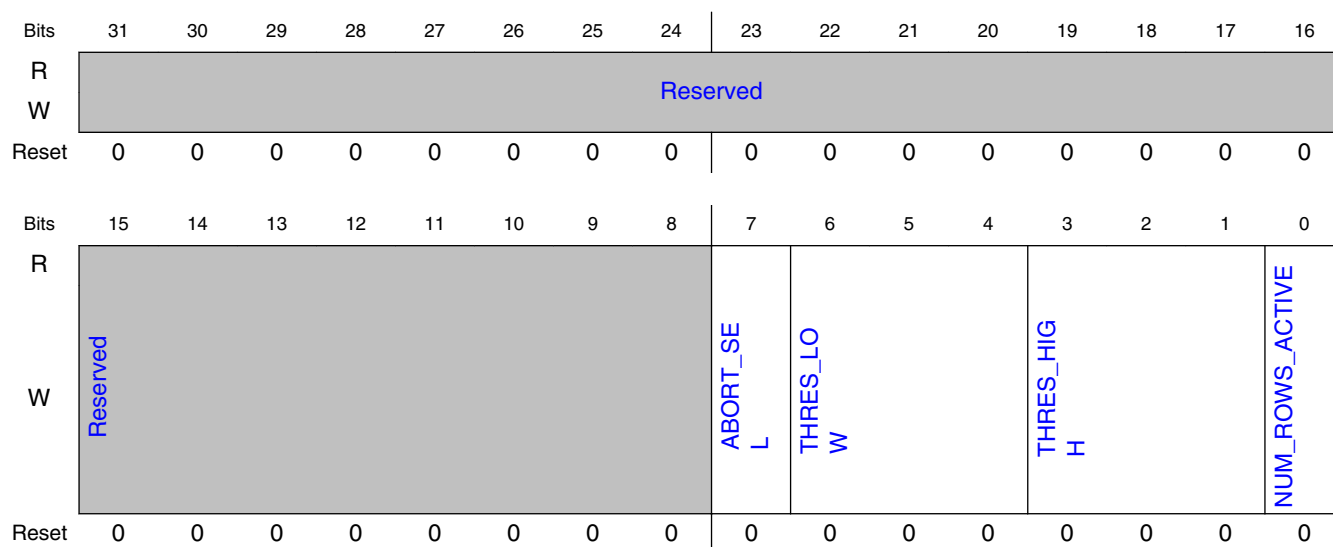
15.7.3.1.58 RTRAM Control 0 (RTRAM_CTRL0)

15.7.3.1.58.1 Offset

Register	Offset
RTRAM_CTRL0	200h

15.7.3.1.58.2 Function

15.7.3.1.58.3 Diagram



15.7.3.1.58.4 Fields

Field	Function
31-8	Reserved.
—	

Table continues on the next page...

Field	Function
7 ABORT_SEL	Abort Select Configuration to manage the bank abort or stall logic. 0x0 = STALL Select 0x1 = ABORT Select
6-4 THRES_LOW	Threshold Low Configuration to manage the panic state threshold, or hysteresis, detection logic for each channel. The THRES_LOW is the low water mark.
3-1 THRES_HIGH	Threshold High Configuration to manage the panic state threshold, or hysteresis, detection logic for each channel. The THRES_HIGH is the high water mark.
0 NUM_ROWS_ACTIVE	Number of Rows Active Configuration to manage the number of rows that are enabled to read. 0x0 = Row 0-4 are enabled. 0x1 = Row 0-6, all rows, enabled. NOTE: Configuring MODE_CTRL0: COMP_2PLANE_EN, will enable the UV rows when in 2-plane modes.

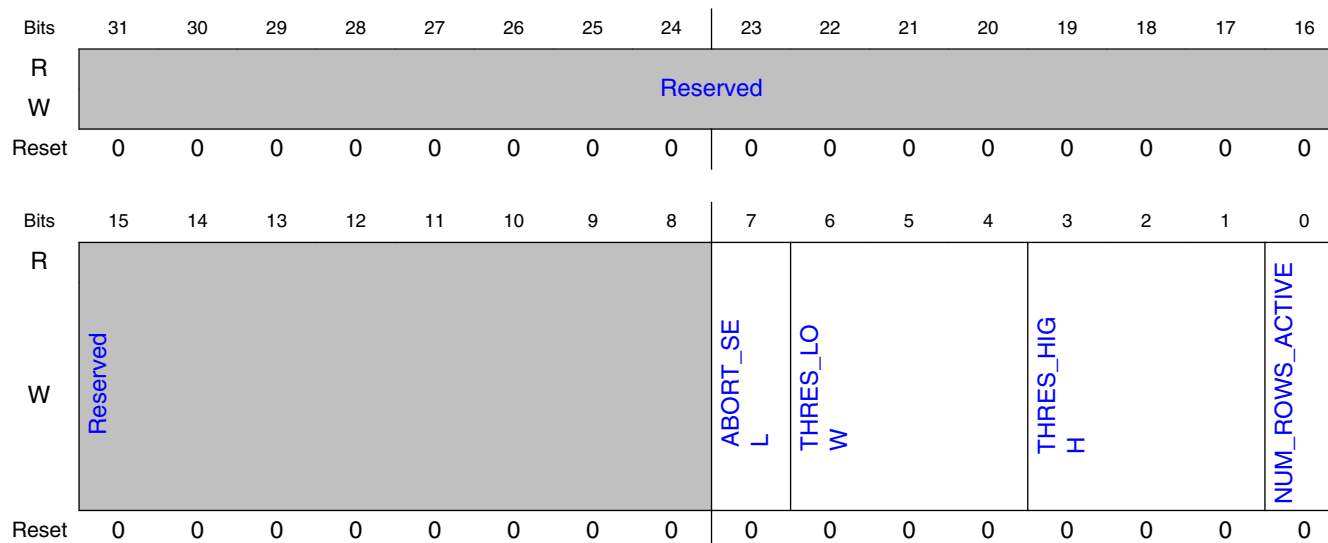
15.7.3.1.59 RTRAM Control 0 (RTRAM_CTRL0_SET)

15.7.3.1.59.1 Offset

Register	Offset
RTRAM_CTRL0_SET	204h

15.7.3.1.59.2 Function

15.7.3.1.59.3 Diagram



15.7.3.1.59.4 Fields

Field	Function
31-8 —	Reserved.
7 ABORT_SEL	Abort Select Configuration to manage the bank abort or stall logic. 0x0 = STALL Select 0x1 = ABORT Select
6-4 THRES_LOW	Threshold Low Configuration to manage the panic state threshold, or hysteresis, detection logic for each channel. The THRES_LOW is the low water mark.
3-1 THRES_HIGH	Threshold High Configuration to manage the panic state threshold, or hysteresis, detection logic for each channel. The THRES_HIGH is the high water mark.
0 NUM_ROWS_ACTIVE	Number of Rows Active Configuration to manage the number of rows that are enabled to read. 0x0 = Row 0-4 are enabled. 0x1 = Row 0-6, all rows, enabled. NOTE: Configuring MODE_CTRL0: COMP_2PLANE_EN, will enable the UV rows when in 2-plane modes.

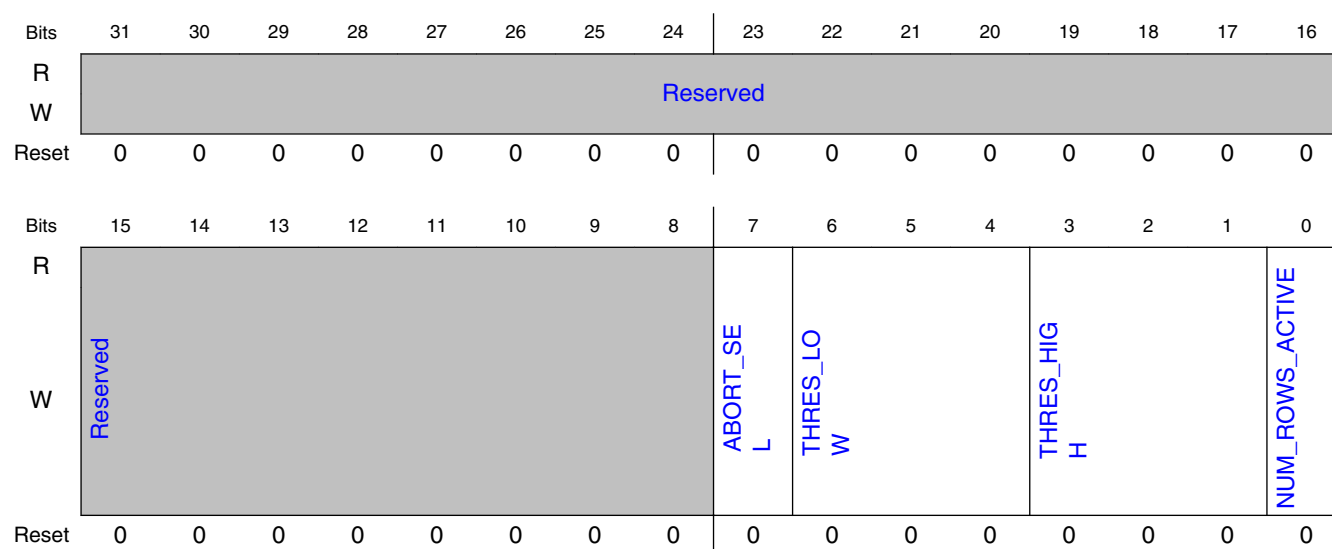
15.7.3.1.60 RTRAM Control 0 (RTRAM_CTRL0_CLR)

15.7.3.1.60.1 Offset

Register	Offset
RTRAM_CTRL0_CLR	208h

15.7.3.1.60.2 Function

15.7.3.1.60.3 Diagram



15.7.3.1.60.4 Fields

Field	Function
31-8 —	Reserved.
7 ABORT_SEL	Abort Select Configuration to manage the bank abort or stall logic. 0x0 = STALL Select 0x1 = ABORT Select
6-4 THRES_LOW	Threshold Low Configuration to manage the panic state threshold, or hysteresis, detection logic for each channel. The THRES_LOW is the low water mark.

Table continues on the next page...

Memory Map and Registers

Field	Function
3-1 THRES_HIGH	Threshold High Configuration to manage the panic state threshold, or hysteresis, detection logic for each channel. The THRES_HIGH is the high water mark.
0 NUM_ROWS_ACTIVE	Number of Rows Active Configuration to manage the number of rows that are enabled to read. 0x0 = Row 0-4 are enabled. 0x1 = Row 0-6, all rows, enabled. NOTE: Configuring MODE_CTRL0: COMP_2PLANE_EN, will enable the UV rows when in 2-plane modes.

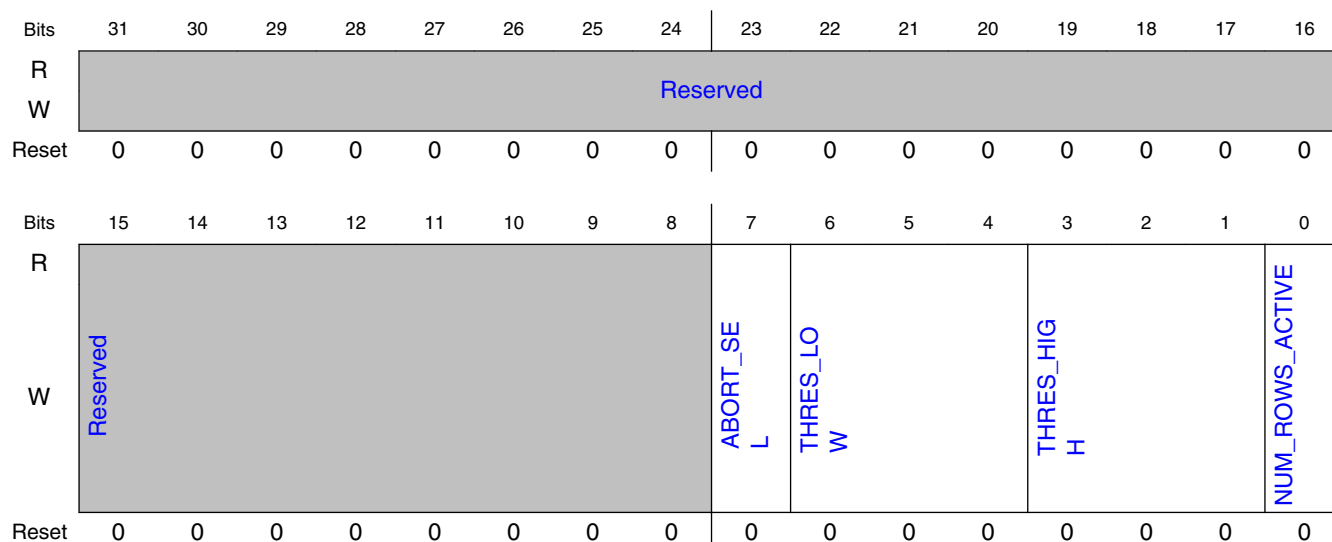
15.7.3.1.61 RTRAM Control 0 (RTRAM_CTRL0_TOG)

15.7.3.1.61.1 Offset

Register	Offset
RTRAM_CTRL0_TOG	20Ch

15.7.3.1.61.2 Function

15.7.3.1.61.3 Diagram



15.7.3.1.61.4 Fields

Field	Function
31-8 —	Reserved.
7 ABORT_SEL	Abort Select Configuration to manage the bank abort or stall logic. 0x0 = STALL Select 0x1 = ABORT Select
6-4 THRES_LOW	Threshold Low Configuration to manage the panic state threshold, or hysteresis, detection logic for each channel. The THRES_LOW is the low water mark.
3-1 THRES_HIGH	Threshold High Configuration to manage the panic state threshold, or hysteresis, detection logic for each channel. The THRES_HIGH is the high water mark.
0 NUM_ROWS_ACTIVE	Number of Rows Active Configuration to manage the number of rows that are enabled to read. 0x0 = Row 0-4 are enabled. 0x1 = Row 0-6, all rows, enabled. NOTE: Configuring MODE_CTRL0: COMP_2PLANE_EN, will enable the UV rows when in 2-plane modes.

15.8 Scaler (MED_DC_SCALE)

15.8.1 Overview

The Scaler's purpose is to resize images for display. Content can be delivered to the application at a resolution that different than the display resolution. Or, images may be reduced in size for PIP viewing. The scaler can resize the images to the required resolution for display. There are three scalers within the DCSS design, one dedicated to each of the three display channels.

15.8.1.1 Block Diagram

The diagram below illustrates the micro architecture of the RTRAM_CTRL, Scaling engine, and scaler output buffer. The example depicts an RTRAM_CTRL 12 row GPU configuration. The VPU configuration uses the same RAMs logically configured as 24 rows with 8 rows per buffer. It is intended that the Scaler, in conjunction with the RTRAM_CTRL, is a cohesive functional unit to resize images on the fly to the display.

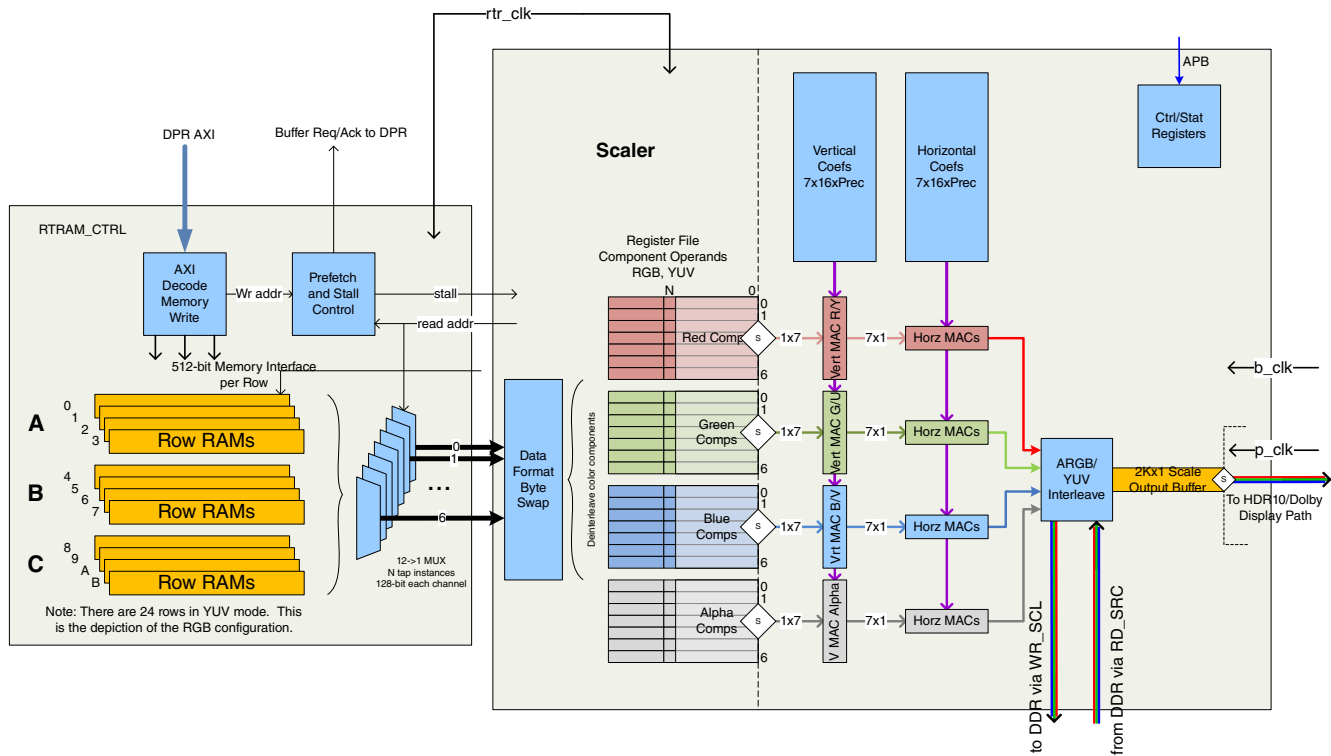


Figure 15-20. Block Diagram

The fundamental components within the Scaler design are as follows:

- Separate vertical and horizontal 7x1 MACs for independent image resizing.
- Separate vertical and horizontal coefficient memories for independent luma/chroma scaling.
- Data formatting and byte swapping to support pixel formats and data endianness.
- Color component FIFOs that contain pixel data used in the resizing of images. These FIFOs bridge between the `rtr_clk` and `b_clk` domains.
- Configuration registers to control the scaling configuration.
- Scaler output pixel FIFO to provide elasticity to meet the peak consumption rates of the pixel data path and display. This FIFO bridges between the `b_clk` and `p_clk` domains.

The Scaler takes in a block of 7 rows for video and 5 rows for graphics pipe and 128-bit wide data to process the current row. The input pixels go through vertical filtering first and horizontal filtering next to arrive at 1 output pixel. The input is fetched based on the size of the tap filter used. For the edges where the neighbor pixels aren't present, the boundary pixels are replicated. The scaling operation takes place in the raster format i.e. top-left to bottom right till the desired output size is obtained.

Based on the tap filter used (7 or 5), the input rows are filtered vertically giving out 1 pixel per cycle. These vertically filtered pixels are passed on to the horizontal pixels. Once the first 4 vertically filtered pixels are available, the horizontal filter kicks in and generates a scaled pixel output. Subsequently the horizontal tap filter is applied in sliding window manner as the vertical filter keeps giving out vertical filtered pixels till the end of the row. This gives out a row of scaled data and the same is continued till the end of the picture (based on the destination height).

For upscaling ≥ 1 , this results in the output performance of one pixel per clock since there are fewer pixels used at the source to scale images up. For downscaling, the scaling ratio limits the output pixel rate. With a down scale ratio $= 1/R$, the scaler produces an output pixel per R clock cycles.

The output pixels generated are either sent to the pixel pipe directly or to the DDR through Write Scale interface. If the channel is programmed to send out the data to the memory using the write scale path, it should be programmed to use the data from the DDR via the RD_SRC block to send the data to the pixel pipe. The output format can be either YUV444, YUV422 or ARGB based on the configuration. Only the first channel has the capacity to send out 38 bit output (including 8 bit Alpha component). The output of the scaler is always 10bit. In case of YUV422 each cycle contains only 2 components, i.e. Y0U0, Y1V0 etc.

15.8.1.2 Features

- Upscaling (up to 1:8).
- Downscaling (up to 7:1 for video channels, up to 5:1 for graphics channels). The down scale ratio, $1/R$, determines the rate at which the scaler produces output pixels. One pixel for every R cycles determines the down scaler performance.
- Alternate option for the Scaler to write data directly to the DRAM via the WR_SCL bus master interface.
- Alternate option for the Scaler Output FIFO to receive data directly from the DRAM via the RD_SRC bus master interface.
- 13-bit fractional accumulator to approximate fractional scaling ratios.
- Support 8/10-bits per component.
- Inputs formats supported, 8-bit components converted to 10-bit for processing.

- 1; A8R8G8B8
- 2; A8Y8U8V8
- 3; A2R10G10B10
- 4; A2Y10U10V10
- 5; Y8U8Y8V8
- 6; U8Y8V8Y8
- 7; Y8_U8V8, 2-plane
- 8; Y10_U10V10, 2-plane
- Output pixel format is 10-bit YUV422/444 (video) and ARGB-8:10:10:10 (graphics)
- 7-tap(video) and 5-tap(graphics) 16-phase FIR filtering
- Software programmable 12 bit filter coefficients in 2s complement
- Independent X/Y cropping

15.8.1.3 Scaler Use Cases

The horizontal and vertical scalers are used to resize source buffers for the following cases:

1. Downscaling for a lower resolution display or for PIP video with a maximum downscale of 7 to 1. The largest allowed 4K PIP window width is 2048 (determined by the scaler output FIFO size in pixels).
2. Upscaling to match the resolution of the display including upscaling HD graphics to 4K graphics with a maximum upscale of 1 to 8.

When the output resolution is UHD, and a UHD source is scaled down to cover less than the entire display, the largest sized PIP within the UHD output window is a HD PIP within the UHD display. This is limited by two factors:

- The Scaler Output FIFO is 2K pixels
- The rate of input into the scaling engine is 1 pixel per clock. So, it takes two clocks to derive one output pixel when scaling down by 2.

The FIFO allows the scaler to pre-compute pixels and store them for when the PIP window is active. This removes any issues with the scaler's performance to produce 1/R (R=scale ratio) pixels per clock. Or, the 2K pixel FIFO has a 2K PIP line worth of data for immediate consumption by the pixel pipeline for PIP display.

Cropping on the Left, Top, Right or Bottom boundaries to project only the area of interest in the incoming video

15.8.2 Functional Description

15.8.2.1 RTRAM Scaler Interface

RTRAM Interface

Scaler gets pixel data from RTRAM in raster format. The request is based on row number and address and RTRAM returns 7 rows of 128bits of data from the requested address and requested data. The row address is always a modulus of the total rows in the bank. If RTRAM operates in 4 lines per bank mode, the total rows is 12 (3 banks in RTRAM), if RTRAM operates in 8 lines per bank, the total rows is 24. For example, a request from ROW 38 and address 0 translates to row_address of 14 in 8 lines per bank mode and row_address of 2 in 4 lines per bank mode. RTRAM responds to the request from Scaler through ack signal. Scaler extends the request till the ack is received. The data from RTRAM is valid 1 cycle after the ack is received. The interface operates at 400MHz clock. In case of 2 plane mode operation i.e YUV420 semi planar, RTRAM has separate request bus to request for Y plane and UV plane. But the data bus is shared. There can be only 1 request at a time.

RTRAM Pixel Data

The data from the RTRAM is arranged in the following format:

1. 32 bpp A8R8G8B8/A8Y8U8V8 - 8bit input
2. 32 bpp A2R10G10B10/A2Y10U10V10 - 10 bit input
3. 16 bpp YUV 422 8 bit
4. 8 bpp Y/UV (2 planes) 420 - 8 bit input
5. 10 bpp Y/UV (2 planes) 420 - 10 bit input

Scaler supports swapping of the components based on the input pixel arrangement. In case of 32bpp, the pixels can be in the format of ARGB, ARBG, AGBR, RGBA, BRGA etc. The input data is modified to match the scaler processing to form ARGB as the pixel arrangement. In case of 16bpp, the data can be of the format Y0U0,Y1V0 or U0Y0,V0Y1 etc. Scaler swaps the data internally. All the pixels are in the little endian format from the RTRAM i.e. Pixel15..Pixel0. Scaler swaps it internally to convert to big endian format (pixel0..pixel15)

In case of 32bpp (ARGB or AYUV), a 5 tap filter is used. RTRAM returns only 5 rows of valid data per request from Scaler. The rows6 and rows7 are not used by Scaler.

Once Scaler is done processing the rows of a bank, bank_done signal is asserted so that RTRAM can fill the next rows in the bank. rtram_vsync_reset is asserted after all the rows of the picture are processed.

15.8.2.2 Scaler Horizontal and Vertical Operation

The scaler, based on the current X,Y position within the source image, operates on the 7x7 matrix of source pixels to produce a single output pixel. Note that for ARGB scaling, a 5x5 matrix of source pixels is used to produce a single output pixel.

For example, for 1:1 scaling, each source pixel is mapped to a destination pixel. For 7:1 downscale, only 1 out of 7 source pixels is mapped to a destination pixel. For 1:4 upscale, the same source group of 7 source pixels is used 4 times at the output but with different filter coefficients. Note that an output buffer is required to support the case of downscaling to manage the rate of BW at the source. In effect, the output buffer allows for the scaling process to spread the input peak BW over the width of the entire display, not just the region that we are displaying, which may not be at the resolution of the display. Of concern are the downscaling cases, fetching large amounts of data, and displaying that data in small regions of the display.

The scaling hardware relies on software to configure the following parameters for the horizontal and vertical scalers:

- 26 bit (13 int, 13 frac) Fractional pixel start
- 17 bit (4 int, 13 frac) Fractional pixel increment (max value is 8)
- 12 bit (12 int) Filter coefficients

The fractional pixel start corresponds to the starting bias, or the first source pixel position to be selected. The start location acts as a top and left crop value since all output pixels before the start location are dropped. The fractional pixel increment gives the scale ratio, or the number of source pixels to ‘skip’ for each output pixel. The fractional pixel start location consists of an integer and fractional part. The integer component is used to skip a selected number of pixels, usually the desired offset into a row of tiles in the X or Y direction. For a 4K image and a 13-bit fractional component, a maximum error of .5 pixel is achieved with a ratio of 4095/4094. The pixel increment is 17-bit in width made up of 4 integer and 13 fractional bits. This is limited to a maximum downscale value of 7 to 1 for video, or 5 to 1 for graphics that matches the filter width so that no pixels are dropped (except pixels outside the crop region). To limit the size of the polyphase filter table, 16 phases are used between source pixels. The phase accumulator keeps accumulating the pixel position based on the pixel_increment. Based on the phase index which is bit [12:9] of the phase accumulated, the pixel position to apply the 7 tap filter is arrived at.

The scaling method is output centric. The input pixels are filtered vertically first and then horizontally. A running output pixel location is maintained using the phase accumulator. Based on the location of the pixel, the corresponding pixels are chosen from the vertically filtered samples. The filter window slides across the vertically filtered samples to generate the row of pixels, and then the process is repeated until the end of the picture.

15.8.2.3 Scaler Filter

For each color component (R/Y, G/U, B/V, A), the scaler computes an output pixel value as a weighted average of 7 (5 for RGBA) source pixel values. The weights, or coefficients, are programmed by software. To produce output pixels of acceptable quality, the selected source pixel's neighbors must figure in the output pixel generation, instead of simply dropping them. It is the job of the filter to include the neighbors in generating the final output pixel value. For each component (R/Y, G/U, B/V, A), the design implements a 7-tap filter (5-tap filter for RGBA), i.e. the selected source pixel, plus its 3 neighboring pixels to either side. Each of the 7 pixels is given a different weight with the coefficients programmed by software. The output pixel is the sum-of-products of the 7 pixels multiplied by the corresponding coefficients. Scaling is performed in the vertical orientation first, followed by the horizontal orientation to determine each final output pixel value.

Polyphase Filter

The source pixel is selected by the accumulated phase from the scaler, rounded up to the next integer. In this way, there is no difference between 2 'real' source pixels with the same integer but different fractional components. To account for the fractional phase difference, a polyphase filter is implemented. The 4-bit fraction allows for 16 phases between 2 integer positions. Instead of using the same set of coefficients for all, each phase is configured with its own coefficient set of 7 coefficients for each component (YUV or 5-tap RGBA), one for each tap. While the current phase's integer component selects the source pixel (and its neighbors), the fractional component selects the coefficient set to operate over the source pixels. As shown in the diagram above, the index into the coefficient table is the accumulated phase's fractional component. The index into the source pixel array is the accumulated phase's integer component.

Boundary Conditions

If a source pixel is less than 3 pixels from the edge, there are an insufficient number of neighboring pixels. In this case, the value of the edge pixel is simply repeated, i.e. the 'missing' neighbors assume the value of the edge pixels.

Filter Operation

The output buffer is sized to handle the worst case downscale case allowed which is the largest width PIP or graphics window of 2048 from 4096 (2 to 1) requiring a buffer that is 2Kx32 bits to support the following output pixel formats:

1. 38 bit RGBA 8 bit Alpha, 10bit pixels per component for RGB.
2. 20 bit YUV 422 10 bit
3. 30 bit YUV 444 10 bit

The Alpha output from the scaler is always 8-bit. For the color components its always 10bit output.

Note that for downscaling 3 to 1, the peak bandwidth goes up by a factor of 3 (but not 9 due to the output buffer).

The scaler output buffer is the data structure used to cross the asynchronous boundary between the b_clk and p_clk domains. This allows the scaler to run at the rate of the system bus interface with no asynchronous boundary crossing.

Scaler Output Buffer with Status Interrupts

The scaler output buffer has a mechanism to detect when underflow could occur. There are three levels of state that determine how full or empty the fifo is at any time. These states are listed and ordered in levels of severity if these states are reached.

- Underflow - a buffer underflow has occurred. This is the most severe state, as the pointers into the write and read interfaces of the buffer may not recover.
- Near Empty - Also known as low threshold. The number of valid pixels in the buffer is less than or equal to this programmable threshold.
- Almost Full – Also known as high threshold. The number of valid pixels in the buffer is less than or equal to this programmable threshold.

The states are described here.

Underflow – The buffer was read while empty. An interrupt is generated and SW restarts the display.

Low Threshold – This is a SW programmable threshold, between 1 and 2048, of pixels that are valid in the FIFO. This value of valid pixels can be reached when the scaler is not able to produce pixels at the rate of consumption by the pixel data path. The SW programmable register that sets this value should be less than or equal to the high threshold.

High Threshold – This is a SW programmable threshold, between 1 and 2048, of pixels that are valid in the FIFO. This is a high threshold of pixels in the buffer and should be set to a value equal or greater than the low threshold.

All states reached can generate an interrupt so that SW can respond by changing system priorities and restarting the display if needed.

15.8.3 Memory Map and Registers

15.8.3.1 register descriptions

15.8.3.1.1 SCALE Memory map

med_dc_scaler base address: 1_C000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Scale Control Register (SCALE_CTRL)	32	RW	0000_0000h
4h	Scale Output FIFO Control Register (SCALE_OFIFO_CTRL)	32	RW	0000_0000h
8h	Scale Source Data Control Register (SCALE_SRC_DATA_CTRL)	32	RW	0000_0000h
Ch	Scale Bit Depth Control Register (SCALE_BIT_DEPTH)	32	RW	0000_0000h
10h	Scale Source Format Control Register (SCALE_SRC_FORMAT)	32	RW	0000_0000h
14h	Scale Destination Format Control Register (SCALE_DST_FORMAT)	32	RW	0000_0000h
18h	Scale Source Luma Resolution Register (SCALE_SRC_LUMA_RES)	32	RW	0000_0000h
1Ch	Scale Source Chroma Resolution Register (SCALE_SRC_CHROMA_RES)	32	RW	0000_0000h
20h	Scale Destination Luma Resolution Register (SCALE_DST_LUMA_RES)	32	RW	0000_0000h
24h	Scale Destination Chroma Resolution Register (SCALE_DST_CHROMA_RES)	32	RW	0000_0000h
48h	Scale Vertical Luma Start Register (SCALE_V_LUMA_START)	32	RW	0000_0000h
4Ch	Scale Vertical Luma Increment Register (SCALE_V_LUMA_INC)	32	RW	0000_0000h
50h	Scale Horizontal Luma Start Register (SCALE_H_LUMA_START)	32	RW	0000_0000h
54h	Scale Horizontal Luma Increment Register (SCALE_H_LUMA_INC)	32	RW	0000_0000h
58h	Scale Vertical Chroma Start Register (SCALE_V_CHROMA_START)	32	RW	0000_0000h
5Ch	Scale Vertical Chroma Increment Register (SCALE_V_CHROMA_INC)	32	RW	0000_0000h
60h	Scale Horizontal Chroma Start Register (SCALE_H_CHROMA_START)	32	RW	0000_0000h
64h	Scale Horizontal Chroma Increment Register (SCALE_H_CHROMA_INC)	32	RW	0000_0000h
80h	Scale Coefficient Memory Array (SCALE_COEF_ARRAY)	32	RW	0000_0000h

15.8.3.1.2 Scale Control Register (SCALE_CTRL)

15.8.3.1.2.1 Offset

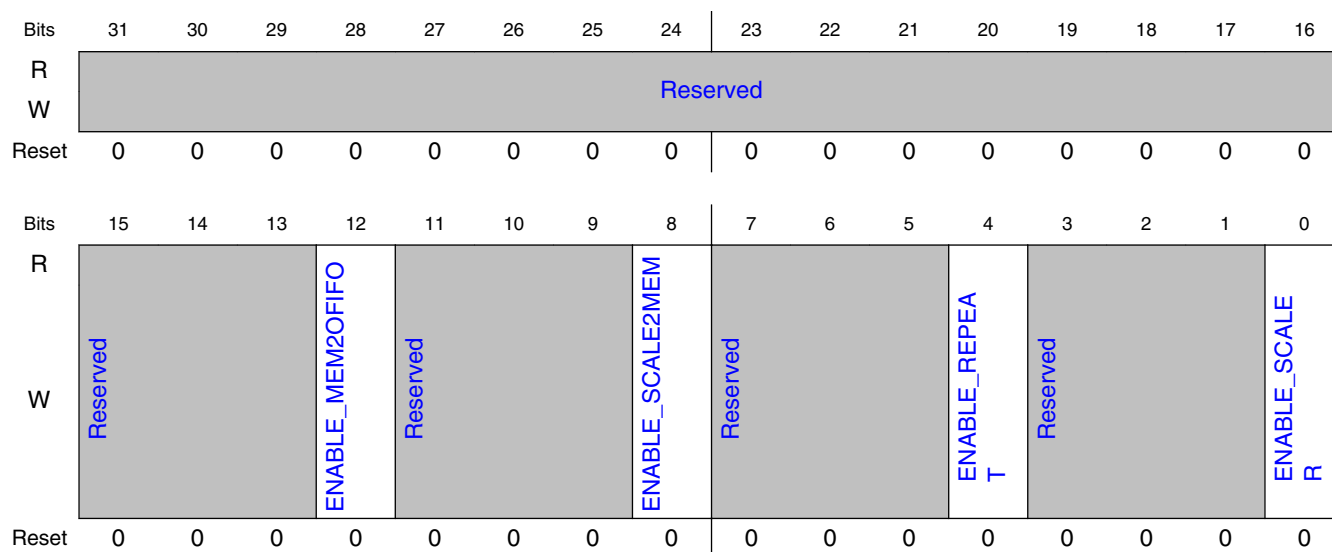
Register	Offset
SCALE_CTRL	0h

15.8.3.1.2.2 Function

This register is the main control register for the scaler.

Note: the bit fields ENABLE_MEM2OFIFO and ENABLE_SCALE2MEM MUST be programmed to the same state, both 1'b0 or both 1'b1.

15.8.3.1.2.3 Diagram



15.8.3.1.2.4 Fields

Field	Function
31-13 —	Reserved.
12 ENABLE_MEM2OFIFO	This bit enables the path from external memory to drive the data into the scale output FIFO. This option is used when the RD_SRC module is enabled to fetch data directly from the memory and bypassing decompression, DPR, and scaling is desired. 0: disabled, scaler drives the data directly into the scale output FIFO. 1: enabled, external pixel interface input bus is used to drive data into the scale output FIFO.
11-9 —	Reserved.

Table continues on the next page...

Field	Function
8 ENABLE_SCAL E2MEM	This bit enables the path from the scaler output back to the system memory. This option is used when the WR_SCL module is enabled to write data out to memory. It is intended this path is used when significant down scaling is required to manage peak bandwidth requirements. 0: disabled, the scaler drives data directly into the scale output FIFO. 1: enabled, the scaler drives data out on the pixel data bus to the WR_SCL module.
7-5 —	Reserved.
4 ENABLE_REPE AT	This bit enables the scaler to restart processing a frame buffer without receiving a SW "kick" event using the ENABLE_SCALER control bit. There is a HW event that can start the scaler to process a frame buffer when the ENABLE_REPEAT is high. This event usually occurs after the current active display region is complete. 0: disabled, the scaler will not repeat the processing when the HW repeat event occurs. 1: enabled, the scaler will repeat with the existing programming when the HW "kick" occurs.
3-1 —	Reserved.
0 ENABLE_SCAL ER	This bit, when set to logic 1, will cause the scaler to begin processing a frame buffer based on all current programmable settings. This bit is cleared after a few cycles, once the start is latched by the HW. If ENABLE_REPEAT is not set to 1, after every frame, this bit has to be programmed to restart the scaler for the next frame. SW can set this bit during the processing of the current frame, but the other parameters shouldn't be changed during frame processing. For predictable results, set ENABLE_SCALER (along with all other programmable parameters) after the scaler is done with processing the current frame. The CTX_LD engine is the best way to determine when to load and enable the scaler considering the tight timing relationship with display retrace. 0: disabled, 1: enabled, the scaler will start processing the frame based on the programmable settings.

15.8.3.1.3 Scale Output FIFO Control Register (SCALE_OFIFO_CTRL)

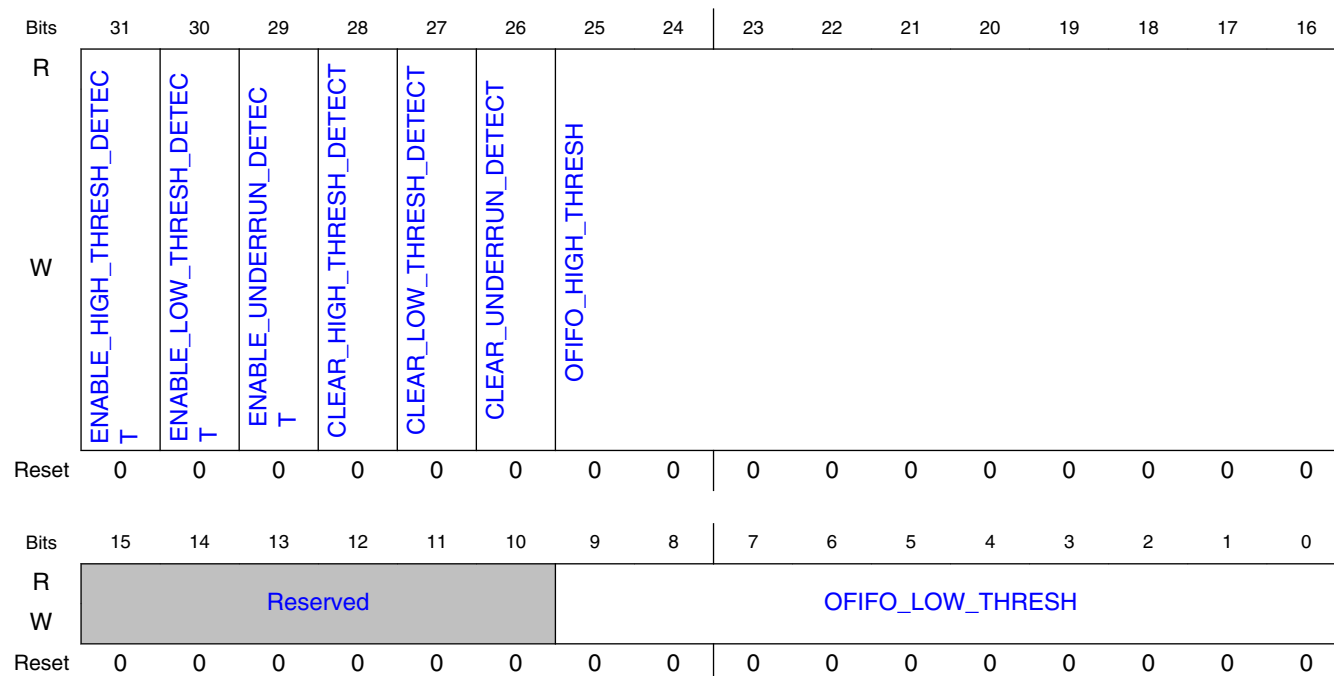
15.8.3.1.3.1 Offset

Register	Offset
SCALE_OFIFO_CTRL	4h

15.8.3.1.3.2 Function

This register controls when the output FIFO will generate an interrupt based on how many pixels are valid within the FIFO and the programmable thresholds.

15.8.3.1.3.3 Diagram



15.8.3.1.3.4 Fields

Field	Function
31 ENABLE_HIGH_THRESH_DETECT	This bit will control if the high threshold detection of the output FIFO can generate an interrupt. 0: disabled, no interrupt will be generated. 1: enabled, interrupt will occur if the high threshold is reached.
30 ENABLE_LOW_THRESH_DETECT	This bit will control if the low threshold detection of the output FIFO can generate an interrupt. 0: disabled, no interrupt will be generated. 1: enabled, interrupt will occur if the low threshold is reached.
29 ENABLE_UNDEERRUN_DETECT	This bit will control if the underrun on the output FIFO can generate an interrupt. 0: disabled, no interrupt will be generated. 1: enabled, interrupt will occur if the low threshold is reached.
28 CLEAR_HIGH_THRESH_DETECT	This bit clears the high threshold detect interrupt. 0: disabled, no action. 1: enabled, clear the high threshold reached interrupt.
27 CLEAR_LOW_THRESH_DETECT	This bit clears the low threshold detect interrupt 0: disabled, no action. 1: enabled, clear the low threshold detect interrupt.
26	This bit clears the underrun detected interrupt. 0: disabled, no action.

Table continues on the next page...

Field	Function
CLEAR_UNDER RUN_DETECT	1: enabled, underrun detected interrupt is cleared.
25-16 OFIFO_HIGH_T HRESH	This value is used to detect when the output FIFO has a number of pixels that are valid that first exceeds this value, and then decrements to a value equal to the value programmed in this register. Note: This value should be greater than the value programmed into OFIFO_LOW_THRESH.
15-10 —	Reserved.
9-0 OFIFO_LOW_T HRESH	This value is used to detect when the output FIFO has a number of pixels that are valid that first exceeds this value, and then decrements to a value equal to the value programmed in this register. Note: This value should be less than the value programmed into OFIFO_HIGH_THRESH.

15.8.3.1.4 Scale Source Data Control Register (SCALE_SRC_DATA_CTRL)

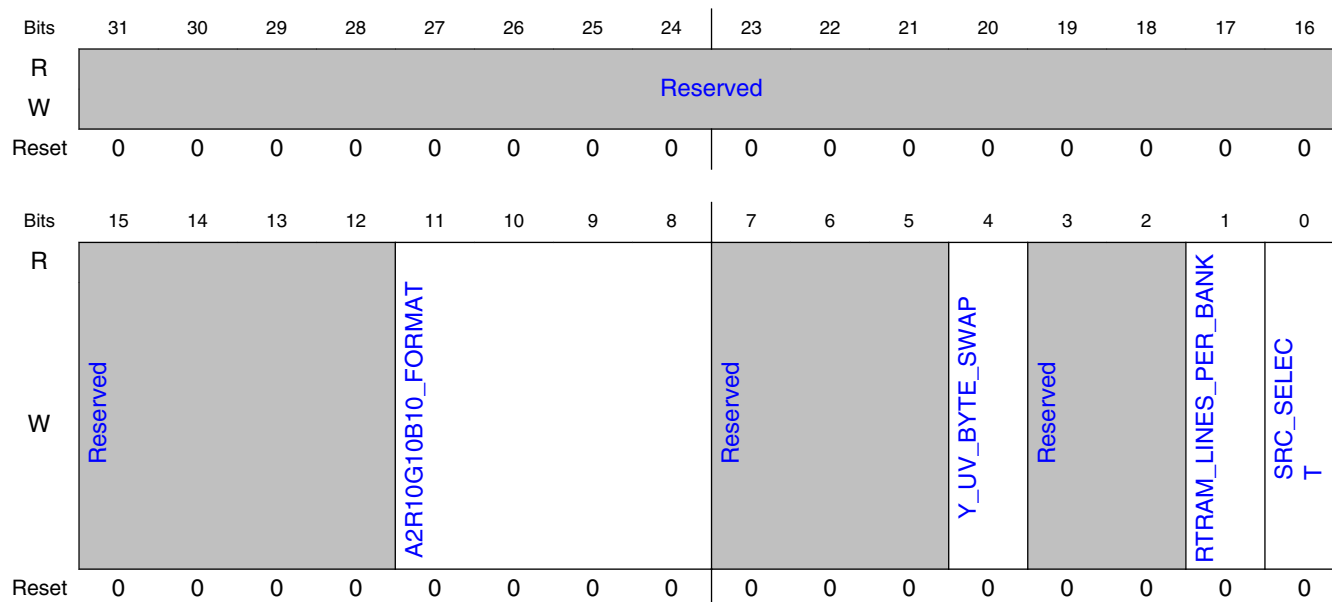
15.8.3.1.4.1 Offset

Register	Offset
SCALE_SRC_DATA_C TRL	8h

15.8.3.1.4.2 Function

This register control aspects related to data formats in the scaler. It also controls HW resources that are used based on the different modes that are programmed.

15.8.3.1.4.3 Diagram



15.8.3.1.4.4 Fields

Field	Function
31-12 —	Reserved.
11-8 A2R10G10B10_ FORMAT	This field represents the data arrangement in the memory in case 10 RGB mode. 0: ARGB is the format in 32bit data. A[31:30], R[29:20], G[19:10], B[9:0] 1: ARGB is the format in 32bit data. A[31:30], R[29:20], B[19:10], G[9:0] 2: AGRB is the format in 32bit data. A[31:30], G[29:20], R[19:10], B[9:0] 3: AGRB is the format in 32bit data. A[31:30], R[29:20], B[19:10], R[9:0] 4: ABRG is the format in 32bit data. A[31:30], B[29:20], R[19:10], G[9:0] 5: ABGR is the format in 32bit data. A[31:30], B[29:20], G[19:10], R[9:0] 6: RGBA is the format in 32bit data. R[31:22], G[21:12], B[11: 2], A[1:0] 7: RBGA is the format in 32bit data. R[31:22], B[21:12], G[11: 2], A[1:0] 8: GRBA is the format in 32bit data. G[31:22], R[21:12], B[11: 2], A[1:0] 9: GBRA is the format in 32bit data. G[31:22], B[21:12], R[11: 2], A[1:0] 10: BRGA is the format in 32bit data. B[31:22], R[21:12], G[11: 2], A[1:0] 11: BGRA is the format in 32bit data. B[31:22], G[21:12], R[11: 2], A[1:0]
7-5 —	Reserved.
4 Y_UV_BYTE_S WAP	This bit will control the swapping of alternate bytes in the incoming data word. 0: disabled, no byte steering. 1: enabled, each pair of bytes is swapped. This option can be used to achieve UYVY -> YUYV swapping of bytes.
3-2 —	Reserved.

Table continues on the next page...

Field	Function
1 RTRAM_LINES_PER_BANK	This field determines the number of lines that are used in each bank of the RTRAM_CTRL module. This should match the respective setting of lines per bank in the DPR. 0: RTRAM has 4 lines per bank. 1: RTRAM has 8 lines per bank.
0 SRC_SELECT	This bit controls whether the source is Video format(YUV) or Graphics(ARB). AYUV444 operates similar to ARGB, hence the SRC_SELECT bit has to be set to 0. However to work around a HW bug in YUV444 to YUV422, the bit has to be programmed to 0. The below are the scenarios and the corresponding programming ARGB input: SRC_SELECT should be set to 0 AYUV444 input to YUV444 output: SRC_SELECT should be set to 1 AYUV444 input to YUV422 output: SRC_SELECT should be set to 0 AYUV422 input to Any output: SRC_SELECT should be set to 1 AYUV420 input to Any output: SRC_SELECT should be set to 1 In particular, it determines if the chroma coefficients, start value, and increment value in HW are used. If the input source is an RGB format, the Luma coefficients, start, and increment values are used, and the chroma parameters are ignored (programming not needed). If the input source is any YUV format, both the Luma and Chroma parameters need to be programmed. Programming of the Luma/Chroma parameters start/inc need to be congruent and consistent with the input format. For example: YUV444: requires Luma/Chroma start/inc values to be identical. YUV422/YUV420: horizontal/vertical Chroma start/inc values are independent of Luma values The values in this register define the source format as: 0: Input source has ARGB or AYUV444 components. 1: Input source has YUV components.

15.8.3.1.5 Scale Bit Depth Control Register (SCALE_BIT_DEPTH)

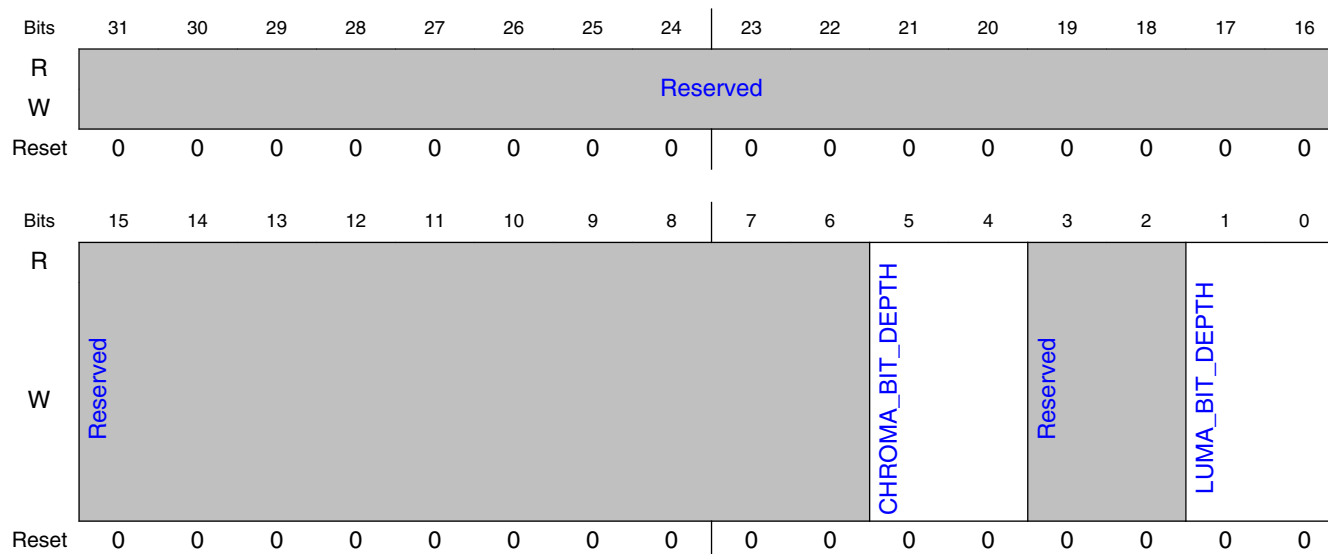
15.8.3.1.5.1 Offset

Register	Offset
SCALE_BIT_DEPTH	Ch

15.8.3.1.5.2 Function

This register determines the component bit depth of the source frame buffer.

15.8.3.1.5.3 Diagram



15.8.3.1.5.4 Fields

Field	Function
31-6 —	Reserved.
5-4 CHROMA_BIT_DEPTH	These bits determine the bit depth of the alternate component (UV 2-plane) processing. 0: 8-bit. 2: 10-bit. other: reserved Both Chroma and Luma bit depths MUST be programmed with the same value.
3-2 —	Reserved.
1-0 LUMA_BIT_DEPTH	These bits determine the bit depth of the primary component (Y, RGB) processing. 0: 8-bit. 2: 10-bit. other: reserved Both Chroma and Luma bit depths MUST be programmed with the same value.

15.8.3.1.6 Scale Source Format Control Register (SCALE_SRC_FORMAT)

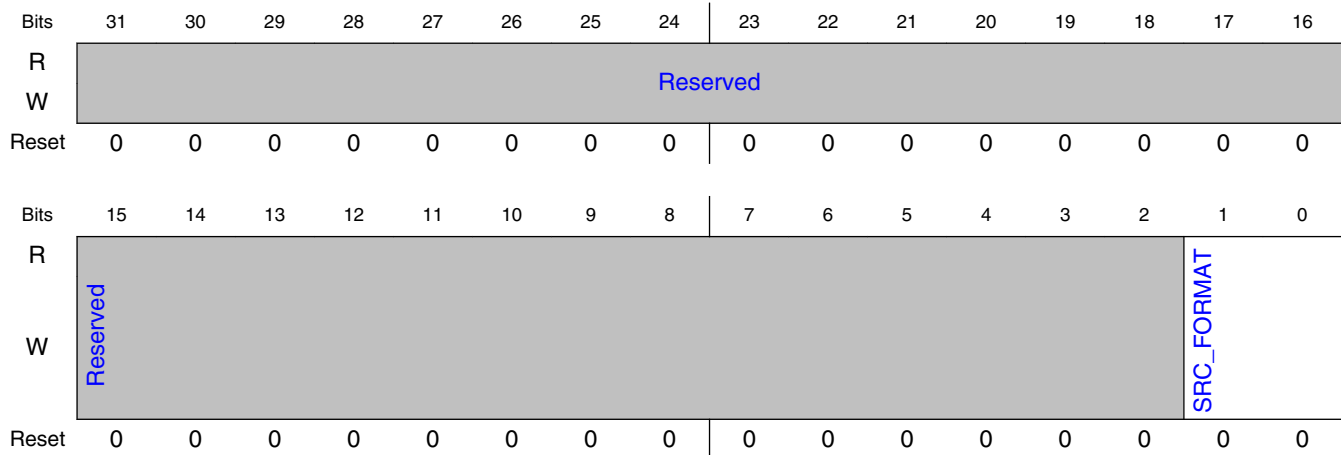
15.8.3.1.6.1 Offset

Register	Offset
SCALE_SRC_FORMAT	10h

15.8.3.1.6.2 Function

This register controls the format of the source frame buffer.

15.8.3.1.6.3 Diagram



15.8.3.1.6.4 Fields

Field	Function
31-2 —	Reserved.
1-0 SRC_FORMAT	These bits define the input buffer format. 0: YUV420 (2-plane chroma sub-sampled V/H). 1: YUV422 (1-plane chroma sub-sampled H). 2: ARGB8888/YUV444 (1-plane full color). Other: Reserved. This parameter needs to be congruent with the SRC_SELECT programming.

15.8.3.1.7 Scale Destination Format Control Register (SCALE_DST_FORMAT)

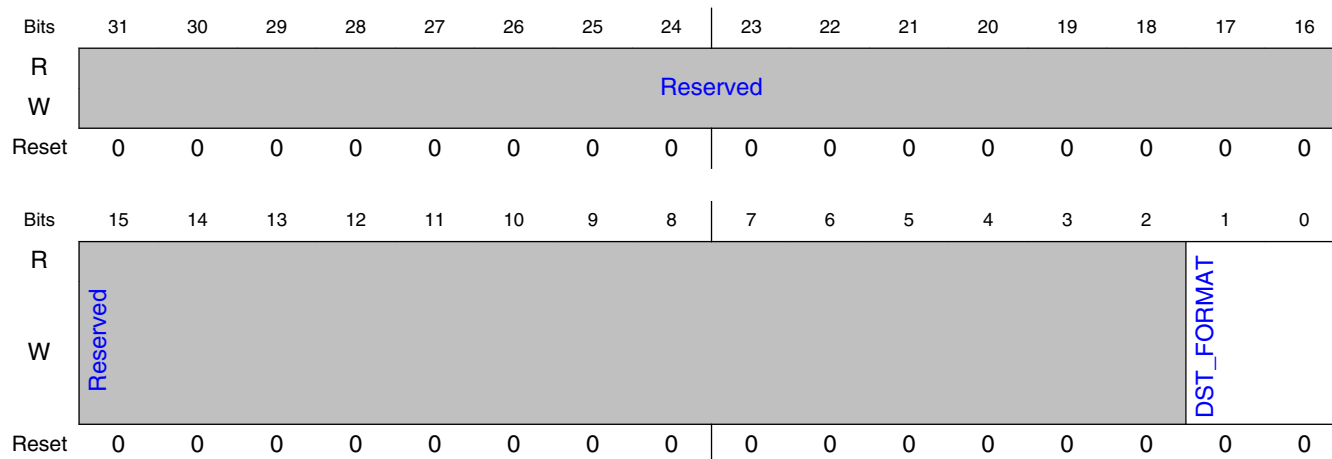
15.8.3.1.7.1 Offset

Register	Offset
SCALE_DST_FORMAT	14h

15.8.3.1.7.2 Function

This register controls the format of the scaler output.

15.8.3.1.7.3 Diagram



15.8.3.1.7.4 Fields

Field	Function
31-2	Reserved.
—	
1-0 DST_FORMAT	These bits define the scaler output format. 1: YUV422. 2: RGB888/YUV444. Other: Reserved.

15.8.3.1.8 Scale Source Luma Resolution Register (SCALE_SRC_LUMA_RES)

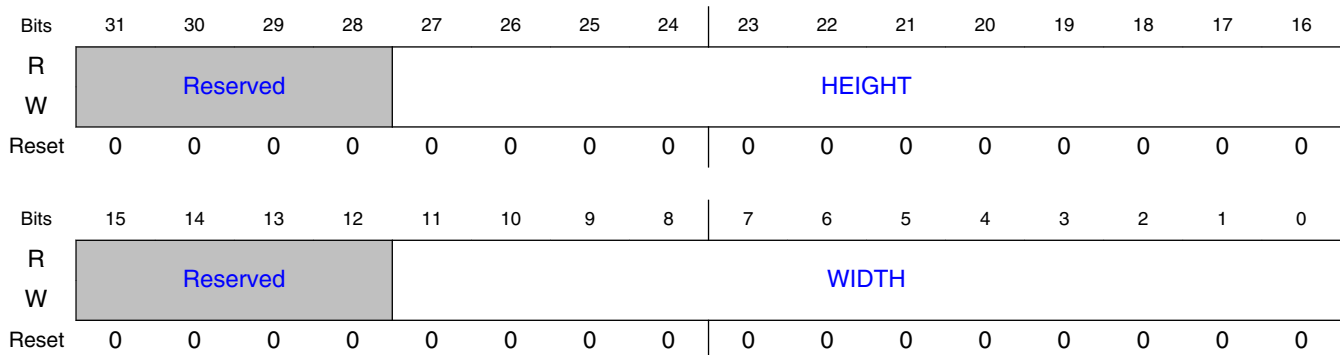
15.8.3.1.8.1 Offset

Register	Offset
SCALE_SRC_LUMA_RES	18h

15.8.3.1.8.2 Function

This register defines the resolution of the source image. The source image resolution needs to be congruent with the resolution programmed in the DPR to guarantee that all data prefetched by the DPR is consumed by the scaler engine. If the resolution programmed into the DPR is not congruent, scaler and DCSS operation will be unpredictable. This restriction pertains to all source buffers formats, either raster or tiled, and with any RTRAM bank size configuration, either 4 or 8 lines per bank.

15.8.3.1.8.3 Diagram



15.8.3.1.8.4 Fields

Field	Function
31-28 —	Reserved.
27-16 HEIGHT	This field defines the height of the source image in pixels minus 1. Incase of cropping, the height should be programmed removing the bottom lines to be cropped. The top lines cropped will be conveyed through vs_luma_start programming. Programmed height = Actual_height - bottom_crop - 1.
15-12 —	Reserved.
11-0 WIDTH	This field defines the width of the source image in pixels minus 1. Incase of cropping, the width should be programmed removing the right pixels to be cropped. The left pixels to be cropped will be conveyed through hs_luma_start programming. programmed width = actual_width - right_crop - 1.

15.8.3.1.9 Scale Source Chroma Resolution Register (SCALE_SRC_CH ROMA_RES)

15.8.3.1.9.1 Offset

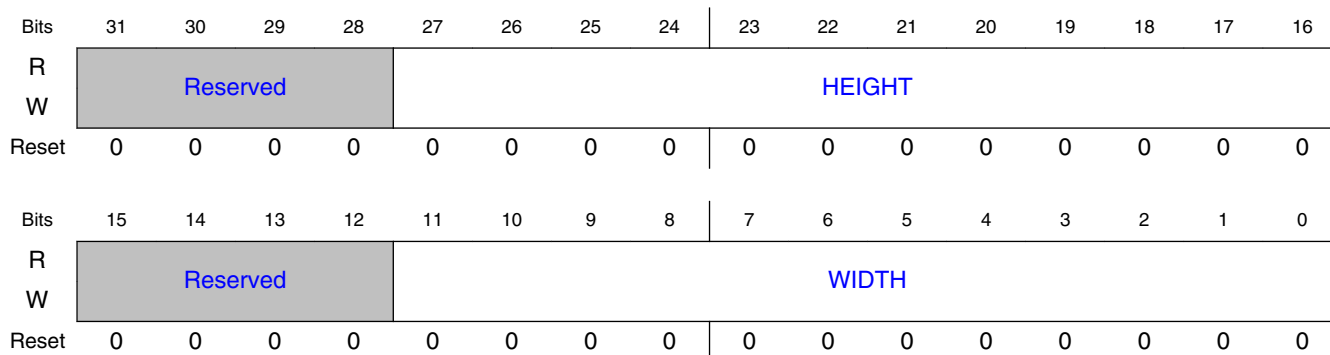
Register	Offset
SCALE_SRC_CHROMA_RES	1Ch

15.8.3.1.9.2 Function

This register defines the resolution of the source image. The value of the chroma resolution is dependent on the resolution of the source image programmed in the SCALE_SRC_LUMA_RES register, since the luma and chroma scaling channels are separate. The programming restrictions are described below.

ARGB8888/YUV444: SCALE_SRC_CHROMA_RES =
 SCALE_SRC_LUMA_RES, SCALE_SRC_FORMAT.SRC_FORMAT = 444
 YUV422: SCALE_SRC_CHROMA_RES.BUFFER_WIDTH =
 SCALE_SRC_LUMA_RES.BUFFER_WIDTH/2,
 SCALE_SRC_CHROMA_RES.BUFFER_HEIGHT =
 SCALE_SRC_LUMA_RES.BUFFER_HEIGHT,
 SCALE_SRC_FORMAT.SRC_FORMAT = 422
 YUV420: SCALE_SRC_CHROMA_RES.BUFFER_WIDTH =
 SCALE_SRC_LUMA_RES.BUFFER_WIDTH/2,
 SCALE_SRC_CHROMA_RES.BUFFER_HEIGHT =
 SCALE_SRC_LUMA_RES.BUFFER_HEIGHT/2,
 SCALE_SRC_FORMAT.SRC_FORMAT = 420

15.8.3.1.9.3 Diagram



15.8.3.1.9.4 Fields

Field	Function
31-28 —	Reserved.
27-16 HEIGHT	This field defines the height of the source image in pixels minus 1. In case of cropping, the height should be programmed removing the bottom lines to be cropped. The top lines cropped will be conveyed through vs_chroma_start programming. Programmed height = Actual_height - bottom_crop - 1.
15-12 —	Reserved.
11-0 WIDTH	This field defines the width of the source image in pixels minus 1. In case of cropping, the width should be programmed removing the right pixels to be cropped. The left pixels to be cropped will be conveyed through hs_chroma_start programming. programmed width = actual_width - right_crop - 1.

15.8.3.1.10 Scale Destination Luma Resolution Register (SCALE_DST_LUMA_RES)

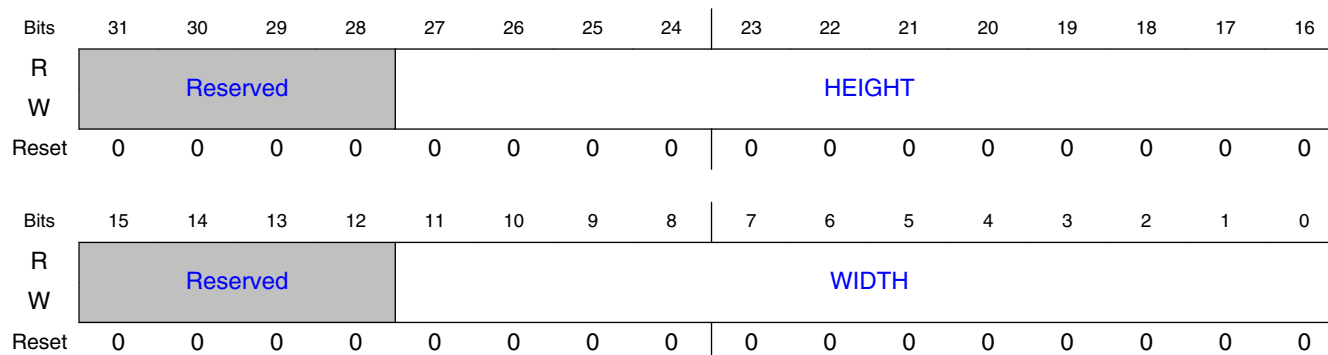
15.8.3.1.10.1 Offset

Register	Offset
SCALE_DST_LUMA_RES	20h

15.8.3.1.10.2 Function

This register defines the resolution of the scaled result, or destination image. The destination image resolution needs to be congruent with the source image resolution and the scaling factors applied. For example, if the source image is scaled down by a factor of 2, the destination resolution needs to be set to exactly half of the resolution of the source image. Fractional scaling ratios, cropping, and initial offsets into the source image all need to be congruent with the source and destination resolution values.

15.8.3.1.10.3 Diagram



15.8.3.1.10.4 Fields

Field	Function
31-28 —	Reserved.
27-16 HEIGHT	This field defines the height of the destination image in pixels minus 1.
15-12 —	Reserved.
11-0 WIDTH	This field defines the width of the destination image in pixels minus 1.

15.8.3.1.11 Scale Destination Chroma Resolution Register (SCALE_DST_CHROMA_RES)

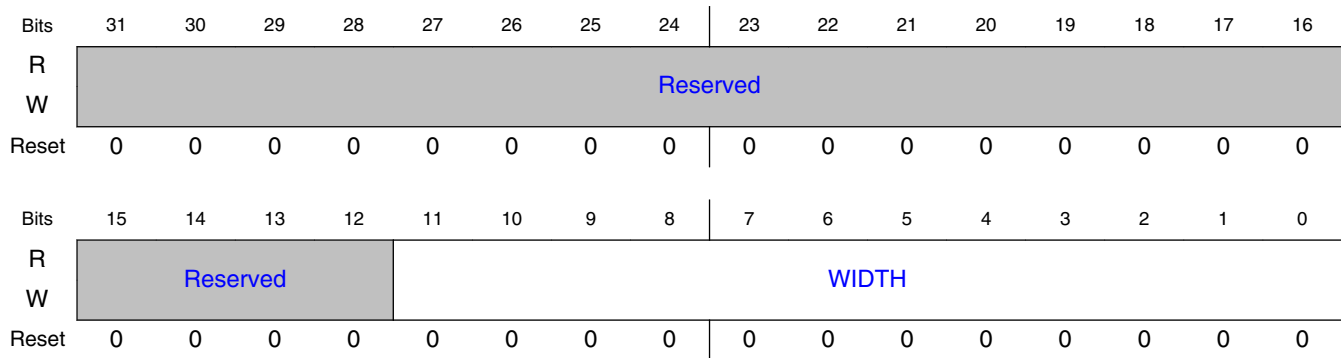
15.8.3.1.11.1 Offset

Register	Offset
SCALE_DST_CHROMA_RES	24h

15.8.3.1.11.2 Function

This register defines the resolution of the scaled result, or destination image. The height field is not required for this register. Similarly to the destination luma register, this register needs to be congruent with all other scaler register settings.

15.8.3.1.11.3 Diagram



15.8.3.1.11.4 Fields

Field	Function
31-12 —	Reserved.
11-0 WIDTH	This field defines the width of the destination image in pixels minus 1.

15.8.3.1.12 Scale Vertical Luma Start Register (SCALE_V_LUMA_START)

15.8.3.1.12.1 Offset

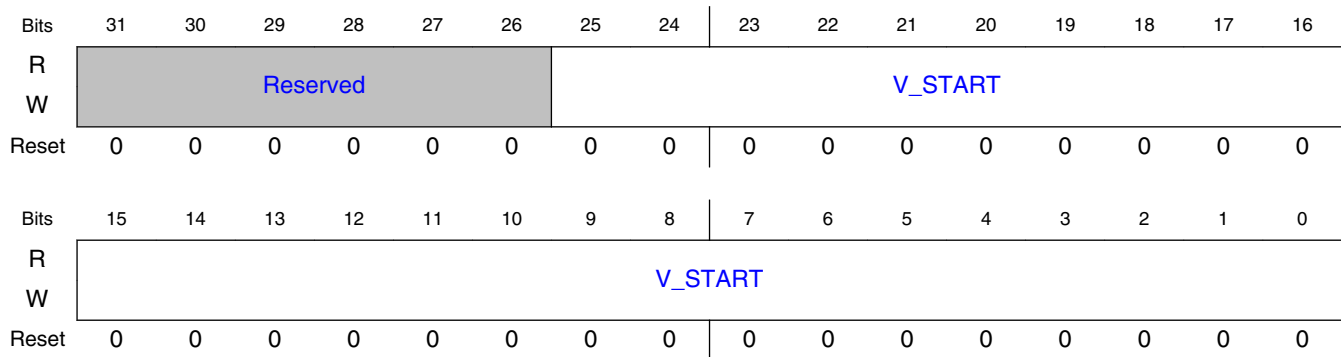
Register	Offset
SCALE_V_LUMA_START	48h

15.8.3.1.12.2 Function

This register contains the initial vertical pixel start, or offset, into the image that is fetched by the prefetch engine. Essentially it is used to discard pixels (integer component) and provide an initial phase (fractional component) into the image in RTRAM. Fundamentally, the images loaded into the RTRAM from the DPR are aligned to 64 byte addresses. This register is used for cropping the top rows from the source image. To achieve better DRAM efficiency, DPR should be used to not fetch the unwanted lines in the top of the picture. The remaining rows i.e. the rows that cannot be dropped by DPR can be cropped in Scaler. There is no limit on the number of lines that can be cropped in

the Scaler. If it crosses the bank boundary, Scaler gives out bank done without reading the bank data and this continues till the actual row is needed. If a few lines from the top are cropped, the source image pixels beyond the crop boundary aren't considered for scaling. Its treated as if they are outside the picture

15.8.3.1.12.3 Diagram



15.8.3.1.12.4 Fields

Field	Function
31-26 —	Reserved.
25-0 V_START	This register contains 13 integer and 13 fractional bits to define the vertical offset into the start of the prefetched image. It is a continuous 16 bit value with the following format ####_####_####.#_####_####_####. The start can be negative but greater than -1

15.8.3.1.13 Scale Vertical Luma Increment Register (SCALE_V_LUMA_INC)

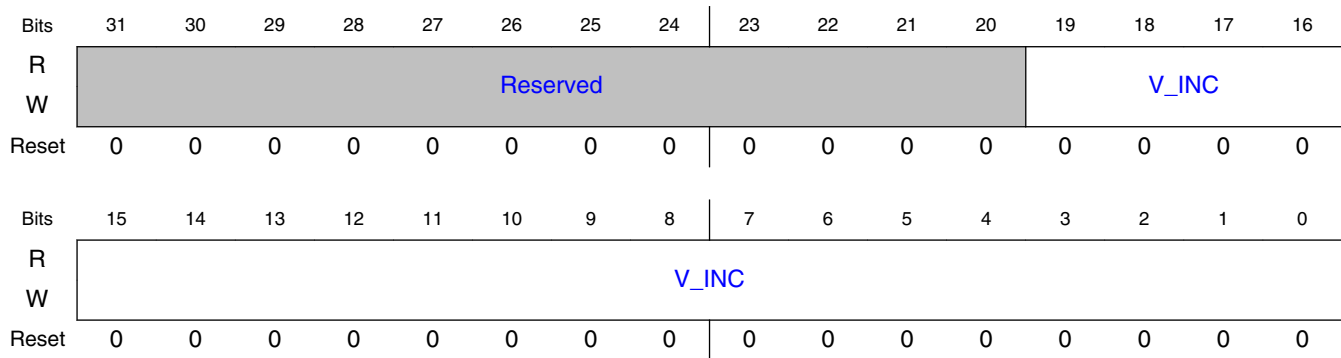
15.8.3.1.13.1 Offset

Register	Offset
SCALE_V_LUMA_INC	4Ch

15.8.3.1.13.2 Function

This register contains the increment which is used to define the scaling ratio. The upper 4 bits is the integer component of the step value. The lowest 13 bits is the fractional component.

15.8.3.1.13.3 Diagram



15.8.3.1.13.4 Fields

Field	Function
31-20 —	Reserved.
19-0 V_INC	Vertical increment value used for scaling the image. This register is programmed with the following format: #####.#_#####_#####_#####. The maximum value that can be programmed into this register is 0xE000, for a maximum scale down ratio of 1/7. Other programming examples could be: 0x1000, scale up by 2. 0x0800, scale up by 4. 0x6000, scale down by 3. 0x0AAB, scale up by 3, digital approximation of 1/3.

15.8.3.1.14 Scale Horizontal Luma Start Register (SCALE_H_LUMA_STA RT)

15.8.3.1.14.1 Offset

Register	Offset
SCALE_H_LUMA_STA RT	50h

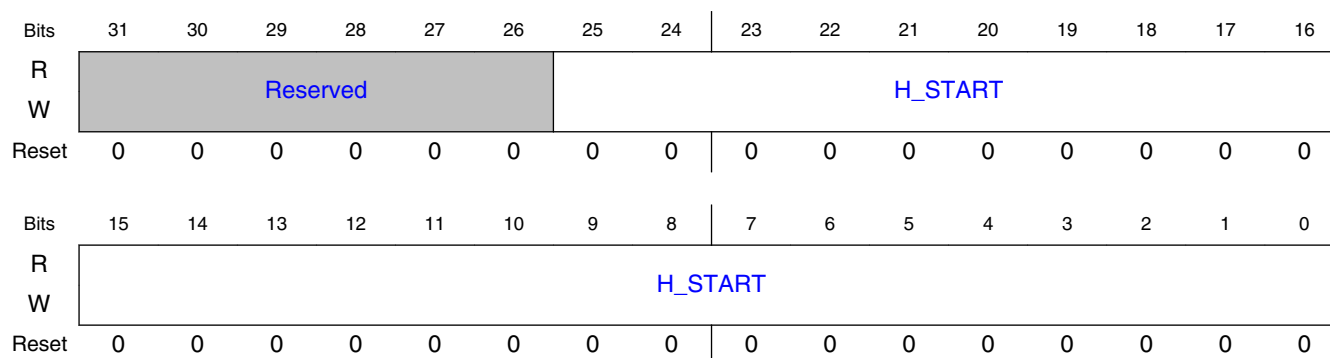
15.8.3.1.14.2 Function

This register contains the initial horizontal pixel start, or offset, into the image that is fetched by the prefetch engine. Essentially it is used to discard pixels (integer component) and provide an initial phase (fractional component) into the image in RTRAM. Fundamentally, the images loaded into the RTRAM from the DPR are aligned to 64 byte addresses. This is essential to achieve DRAM efficiency.

The integer component of this register is used to skip input pixels in the horizontal direction.

This register may use a fractional component to establish an initial horizontal phase offset.

15.8.3.1.14.3 Diagram



15.8.3.1.14.4 Fields

Field	Function
31-26 —	Reserved.
25-0 H_START	This register contains 13 integer and 13 fractional bits to define the horizontal offset into the start of the prefetched image. It is a continuous 19 bit value with the following format #####_####_####.#_####_####_####. The start can be negative but greater than -1

15.8.3.1.15 Scale Horizontal Luma Increment Register (SCALE_H_LUMA_INC)

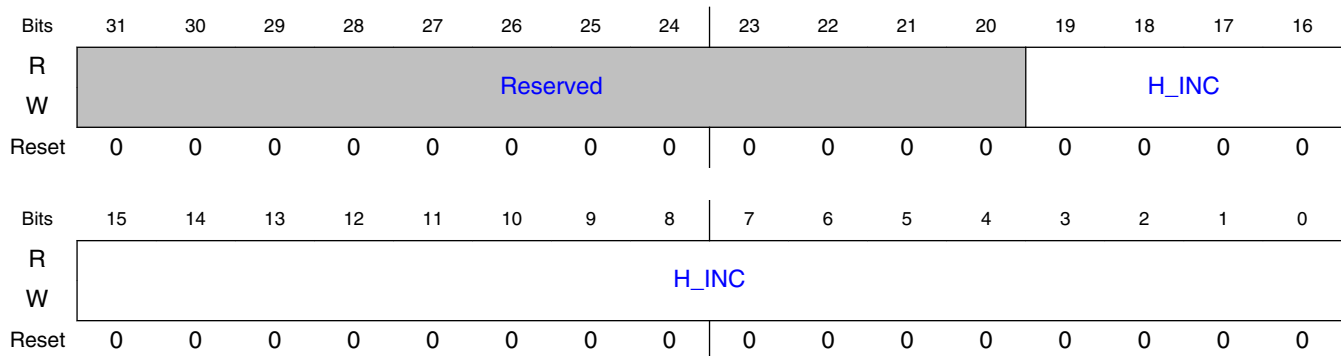
15.8.3.1.15.1 Offset

Register	Offset
SCALE_H_LUMA_INC	54h

15.8.3.1.15.2 Function

This register contains the increment which is used to define the scaling ratio. The upper 4 bits is the integer component of the step value. The lowest 13 bits is the fractional component.

15.8.3.1.15.3 Diagram



15.8.3.1.15.4 Fields

Field	Function
31-20 —	Reserved.
19-0 H_INC	Horizontal increment value used for scaling the image. This register is programmed with the following format: ###.#_####_####_####. The maximum value that can be programmed into this register is 0xE000, for a maximum scale down ratio of 1/7. Other programming examples could be: 0x1000, scale up by 2. 0x0800, scale up by 4. 0x6000, scale down by 3. 0x0AAB, scale up by 3, digital approximation of 1/3.

15.8.3.1.16 Scale Vertical Chroma Start Register (SCALE_V_CHROMA_START)

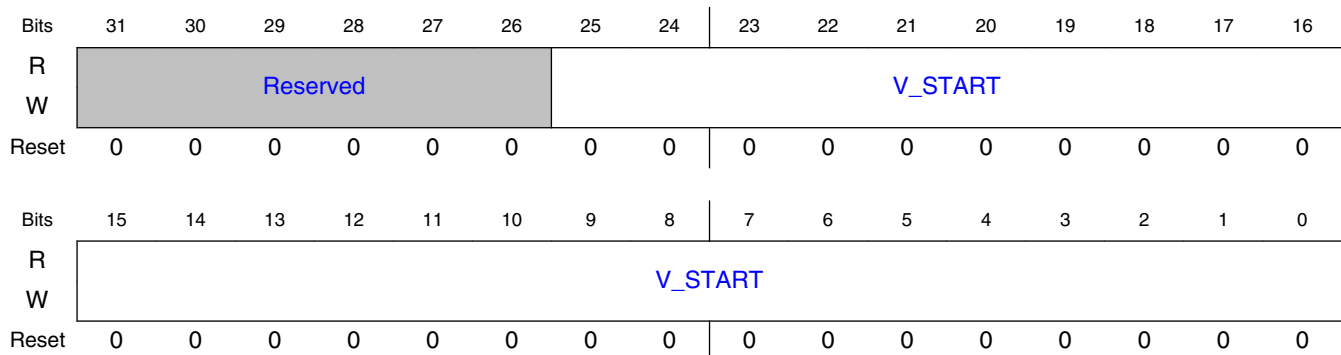
15.8.3.1.16.1 Offset

Register	Offset
SCALE_V_CHROMA_START	58h

15.8.3.1.16.2 Function

This register contains the initial vertical pixel start, or offset, into the image that is fetched by the prefetch engine. Essentially it is used to discard pixels (integer component) and provide an initial phase (fractional component) into the image in RTRAM. Fundamentally, the images loaded into the RTRAM from the DPR are aligned to 64 byte addresses. This register is used for cropping the top rows from the source image. To achieve better DRAM efficiency, DPR should be used to not fetch the unwanted lines in the top of the picture. The remaining rows i.e. the rows that cannot be dropped by DPR can be cropped in Scaler. There is no limit on the number of lines that can be cropped in the Scaler. If it crosses the bank boundary, Scaler gives out bank done without reading the bank data and this continues till the actual row is needed. If a few lines from the top are cropped, the source image pixels beyond the crop boundary aren't considered for scaling. Its treated as if they are outside the picture

15.8.3.1.16.3 Diagram



15.8.3.1.16.4 Fields

Field	Function
31-26 —	Reserved.
25-0 V_START	This register contains 13 integer and 13 fractional bits to define the vertical offset into the start of the prefetched image. It is a continuous 16 bit value with the following format ###.#_####_####_####.

15.8.3.1.17 Scale Vertical Chroma Increment Register (SCALE_V_CHROMA_INC)

15.8.3.1.17.1 Offset

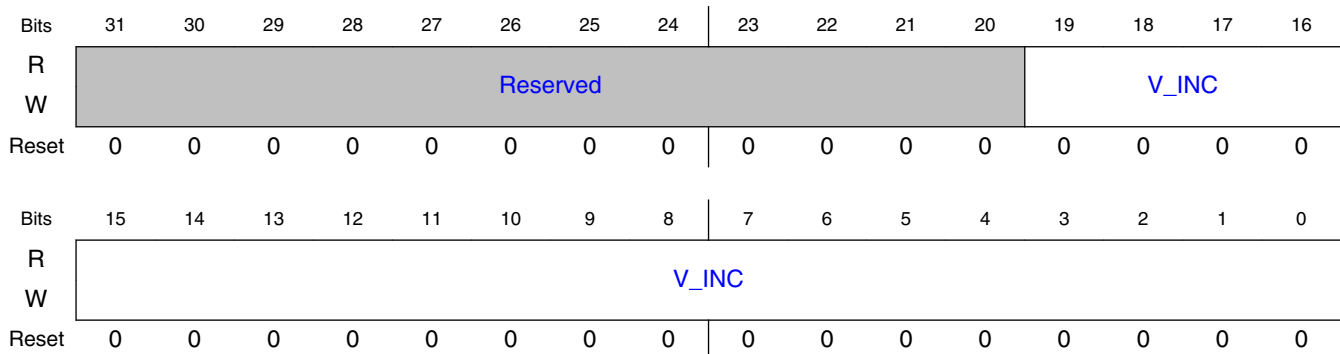
Register	Offset
SCALE_V_CHROMA_INC	5Ch

15.8.3.1.17.2 Function

This register contains the increment which is used to define the scaling ratio. The upper 4 bits is the integer component of the step value. The lowest 13 bits is the fractional component. The value programmed in this register is dependent on the value in the SCALE_V_LUMA_INC register AND the buffer format.

ARGB8888, YUV 1-plane: SCALE_V_CHROMA_INC = SCALE_V_LUMA_INC

15.8.3.1.17.3 Diagram



15.8.3.1.17.4 Fields

Field	Function
31-20 —	Reserved.
19-0 V_INC	Vertical increment value used for scaling the image. This register is programmed with the following format: ###.#_####_####_####. The maximum value that can be programmed into this register is 0xE000, for a maximum scale down ratio of 1/7. Other programming examples could be:

Field	Function
	0x1000, scale up by 2. 0x0800, scale up by 4. 0x6000, scale down by 3. 0x0AAB, scale up by 3, digital approximation of 1/3.

15.8.3.1.18 Scale Horizontal Chroma Start Register (SCALE_H_CHROMA_START)

15.8.3.1.18.1 Offset

Register	Offset
SCALE_H_CHROMA_START	60h

15.8.3.1.18.2 Function

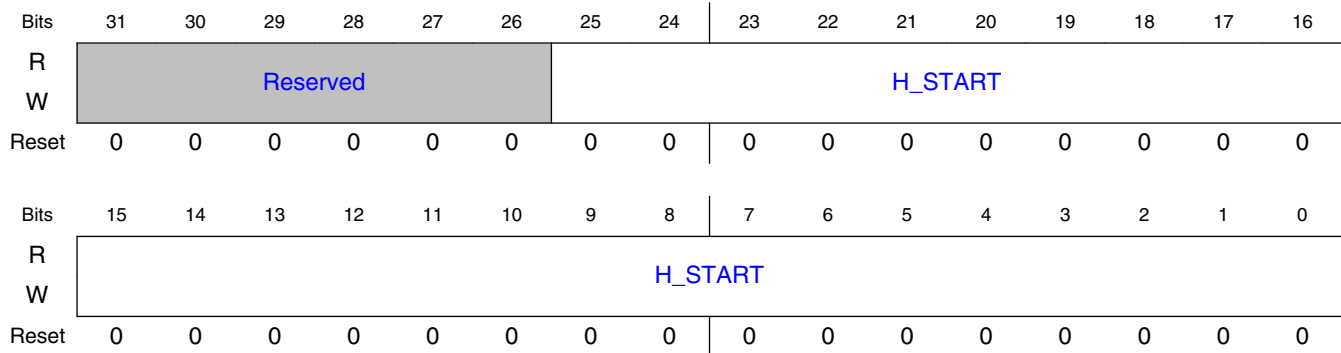
This register contains the initial horizontal pixel start, or offset, into the image that is fetched by the prefetch engine. Essentially it is used to discard pixels (integer component) and provide an initial phase (fractional component) into the image in RTRAM. Fundamentally, the images loaded into the RTRAM from the DPR are aligned to 64 byte addresses. This is essential to achieve DRAM efficiency.

The integer component of this register is used to skip input pixels in the horizontal direction.

If SW requires a high integer start value, change the base address of the source image in the DPR so that the undesired data is NOT prefetched.

This register may use a fractional component to establish an initial horizontal phase offset.

15.8.3.1.18.3 Diagram



15.8.3.1.18.4 Fields

Field	Function
31-26 —	Reserved.
25-0 H_START	This register contains 13 integer and 13 fractional bits to define the horizontal offset into the start of the prefetched image. It is a continuous 19 bit value with the following format ##_####.#_####_####_####. The start can be negative but greater than -1

15.8.3.1.19 Scale Horizontal Chroma Increment Register (SCALE_H_CHROMA_INC)

15.8.3.1.19.1 Offset

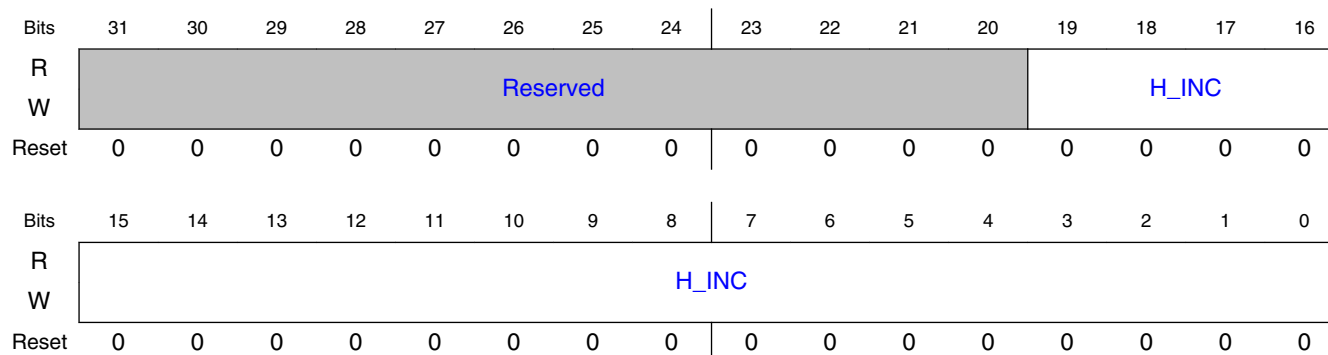
Register	Offset
SCALE_H_CHROMA_INC	64h

15.8.3.1.19.2 Function

This register contains the increment which is used to define the scaling ratio. The upper 4 bits is the integer component of the step value. The lowest 13 bits is the fractional component. The value programmed in this register is dependent on the value in the SCALE_H_LUMA_INC register AND the buffer format.

ARGB8888, YUV444: SCALE_H_CHROMA_INC = SCALE_H_LUMA_INC

15.8.3.1.19.3 Diagram



15.8.3.1.19.4 Fields

Field	Function
31-20 —	Reserved.
19-0 H_INC	Horizontal increment value used for scaling the image. This register is programmed with the following format: ###.#_####_####_####. The maximum value that can be programmed into this register is 0xE000, for a maximum scale down ratio of 1/7. Other programming examples could be: 0x1000, scale up by 2. 0x0800, scale up by 4. 0x6000, scale down by 3. 0x0AAB, scale up by 3, digital approximation of 1/3.

15.8.3.1.20 Scale Coefficient Memory Array (SCALE_COEF_ARRAY)

15.8.3.1.20.1 Offset

Register	Offset
SCALE_COEF_ARRAY	80h

15.8.3.1.20.2 Function

This register is not implemented in HW. This is place holder to document the address range used to program the scaling coefficients. The coefficients are defined as $c(P,T)$ where $P(0-15)$ are the possible 16 phases, and $T(0-6)$ are the possible 7 coefficient taps per phase. The base address of all coefficient RAMs starts at 0x80. The entire range of addresses for coefficient access ends with address 0x3BF. Each coefficient is 12-bits.

Note, there are 12-bits per coefficient, and 7 taps, for a total of $12 \times 7 = 84$ -bits. There are 3 memories to store the coefficients, and these memories have a word width of 28-bits. So, the coefficients are loaded 28-bits per PIO write. Since the coefficient bit width and memory word width are not evenly divisible, some coefficients will cross a memory word boundary. This is described below.

Vertical/Horizontal coefficients are accessed according to the following memory mapped layout.

ADDRESS	DESCRIPTION
0x80-0xBC	Vert/Luma, increment through 16-phases, tap 0, 1, and the high 4-bits of tap 2
0xC0-0xFF	Vert/Luma, increment through 16-phases, tap 2 low 8-bits, tap 3, and the high 8-bits of tap 4
0x100-0x13F	Vert/Luma, increment through 16-phases, tap 4 low 4-bits, tap 5, 6
0x140-0x17F	Horz/Luma, increment through 16-phases, taps 0, 1, and the high 4-bits of tap 2
0x180-0x1BF	Horz/Luma, increment through 16-phases, taps 2, 3, 4, as above
0x1C0-0x1FF	Horz/Luma, increment through 16-phases, taps 4, 5, 6, as above
0x200-0x23F	Vert/Chroma, increment through 16-phases, taps 0, 1, 2, as above
0x240-0x27F	Vert/Chroma, increment through 16-phases, taps 2, 3, 4, as above
0x280-0x2BF	Vert/Chroma, increment through 16-phases, taps 4, 5, 6, as above
0x300-0x33F	Horz/Chroma, increment through 16-phases, taps 0, 1, 2, as above
0x340-0x37F	Horz/Chroma, increment through 16-phases, taps 2, 3, 4, as above
0x380-0x3BF	Horz/Chroma, increment through 16-phases, taps 4, 5, 6, as above

```

As an example, the first few writes starting at address 0x80, 0x84, ..., would be:
# Adr = 0x80; Write phase 0, taps 0, 1, and the high 4-bits of tap 2.
# Adr = 0x84; Write phase 1, taps 0, 1, and the high 4-bits of tap 2.
...
# Adr = 0xBC; Write phase 15, taps 0, 1, and the high 4-bits of tap 2.
# Adr = 0xC0; Write phase 0, the low 8-bits of tap 2, tap 3, and the high 8-bits of tap 4.
WRITE(0x80, {4'b0, c(0,0),      c(0,1),  c(0,2)[11:8]})
WRITE(0x84, {4'b0, c(1,0),      c(1,1),  c(1,2)[11:8]})
...
WRITE(0xBC, {4'b0, c(15,0),     c(15,1), c(15,2)[11:8]})
WRITE(0xC0, {4'b0, c(0,2)[7:0], c(0,3),  c(0,4)[11:4]})
...

```

Using two dimensional arrays for vertical/horizontal and luma/chroma "c[v|h][l|c][P,T]" to store 12-bit coefficients, defined as c[15:0][6:0][11:0], the following code could be used to load the coefficient memories with the desired coefficient values.

```

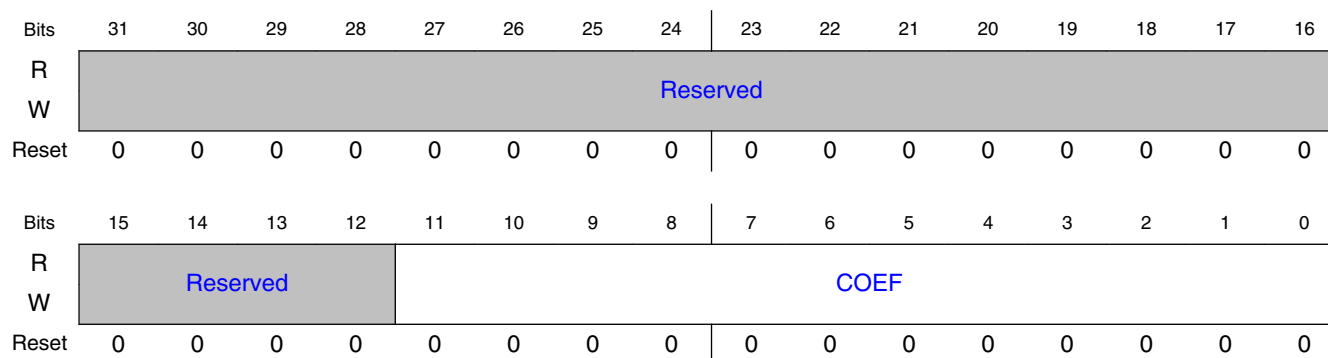
for (i=0, i.lt.16, i=i+1) (
  WRT(0x80+(4*i), {cvl(i,0)[11:0], cvl(i,1)[11:0], cvl(i,2)[11:8]});
  WRT(0xC0+(4*i), {cvl(i,2)[7:0],  cvl(i,3)[11:0], cvl(i,4)[11:4]});
  WRT(0x100+(4*i), {cvl(i,4)[3:0],  cvl(i,5)[11:0], cvl(i,6)[11:0]});
  WRT(0x140+(4*i), {chl(i,0)[11:0], chl(i,1)[11:0], chl(i,2)[11:8]});
  WRT(0x180+(4*i), {chl(i,2)[7:0],  chl(i,3)[11:0], chl(i,4)[11:4]});
  WRT(0x1C0+(4*i), {chl(i,4)[3:0],  chl(i,5)[11:0], chl(i,6)[11:0]});
  WRT(0x200+(4*i), {cvc(i,0)[11:0], cvc(i,1)[11:0], cvc(i,2)[11:8]});
  WRT(0x240+(4*i), {cvc(i,2)[7:0],  cvc(i,3)[11:0], cvc(i,4)[11:4]});
  WRT(0x280+(4*i), {cvc(i,4)[3:0],  cvc(i,5)[11:0], cvc(i,6)[11:0]});

```

Look Up Table Load (LUT_LD)

```
WRT(0x300+(4*i),{chc(i,0)[11:0], chc(i,1)[11:0], chc(i,2)[11:8]});
WRT(0x340+(4*i),{chc(i,2)[7:0], chc(i,3)[11:0], chc(i,4)[11:4]});
WRT(0x380+(4*i),{chc(i,4)[3:0], chc(i,5)[11:0], chc(i,6)[11:0]});
)
```

15.8.3.1.20.3 Diagram



15.8.3.1.20.4 Fields

Field	Function
31-12 —	Reserved.
11-0 COEF	The coefficients have 1-bit sign, 1-bit integer, and 10-bits fraction. A total of bits [11:0], or {sign,integer,frac[9:0]}

15.9 Look Up Table Load (LUT_LD)

15.9.1 Overview

The Look Up Table Load module (LUT_LD) provides the interface between the DMA interface of the Dolby IP and the AXI bus transactions to system RAM. This interface orchestrates the fetch of the tone curve LUT and the G2L LUT data from system RAM. Both the Dolby video and graphics pixel data pipelines require this LUT data. The LUT_LD engine will serve as the interface for both the video and graphics IP. Essentially, the LUT_LD will fetch the LUT data for the video interface first, followed by the LUT data for the graphics pipeline second. This module will not provide an asynchronous domain crossing between clock domains.

NOTE: More information on the DMA interface and LUT data processing can be obtained from the Dolby IP specs.

15.9.1.1 Block Diagram

The following diagram indicates the microarchitecture of the LUT_LD module.

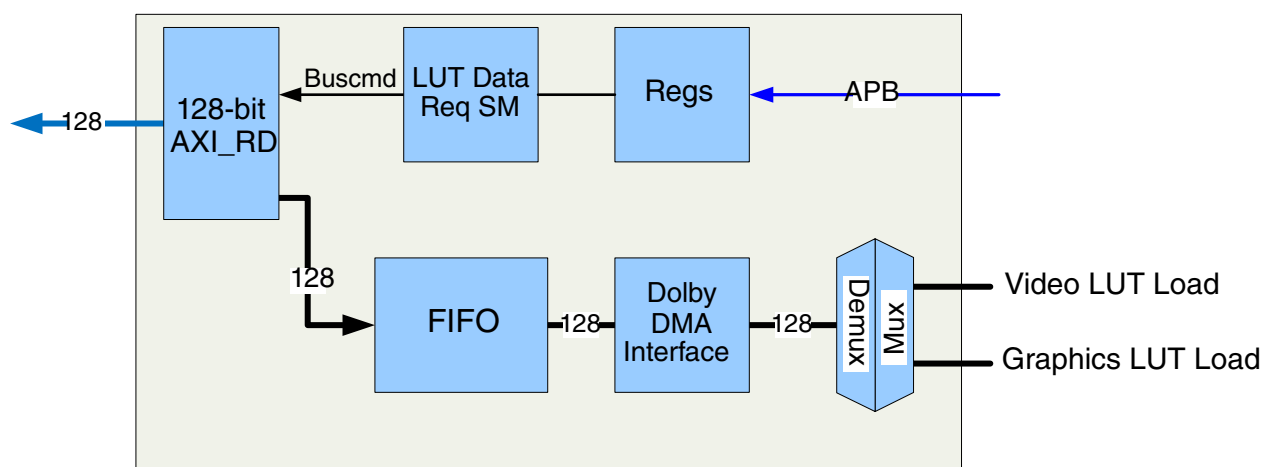


Figure 15-21. LUT_LD Block Diagram

15.9.1.2 Features

The LUT_LD integrates the following functional components:

- AXI_RD buscmd/AXI interface module at 128-bit bus width.
- LUT Data Request State Machine
- CSR registers to define the base address of the 10KB of LUT data.
- 512B FIFO.
- Dolby DMA interface SM

The amount of data to be fetched is defined in the Dolby IP specs. The video tone curve = 4KB and the G2L = 1KB.

Since both the video and graphics pixel pipelines require the same amount of LUT data (5KB), the LUT_LD engine serves the video pipeline first (5KB), followed by the graphics pipeline (5KB). Each Dolby pixel processing component needs 5KB of data to complete the LUT loading process. In all, 10KB is transferred over the two Dolby IP DMA interfaces (5KB each).

The FIFO is implemented to manage system memory latency and throughput.

The LUT_LD engine begins fetching LUT data from system memory when the CNTL_STATUS.ENABLE bit is set. The load sequence runs once and does not repeat every display trace. If the previous data needs to be fetched in subsequent display traces, the LUT_LD engine needs to be enabled for the loading sequence to commence.

15.9.2 Memory Map and Registers

15.9.2.1 register descriptions

15.9.2.1.1 LUT_LD Memory map

lut_ld base address: 2_4000h

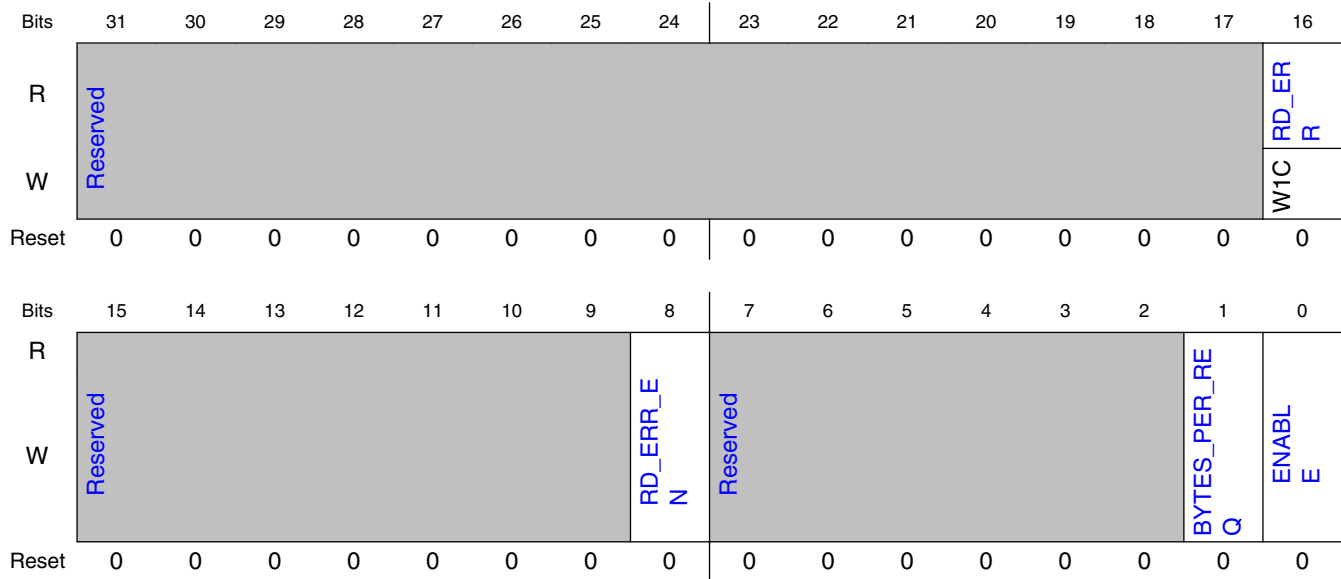
Offset	Register	Width (In bits)	Access	Reset value
0h	Control/Status register for LUT Loader. (CTRL_STATUS)	32	RW	0000_0000h
4h	Control/Status register for LUT Loader. (CTRL_STATUS_SET)	32	RW	0000_0000h
8h	Control/Status register for LUT Loader. (CTRL_STATUS_CLR)	32	RW	0000_0000h
Ch	Control/Status register for LUT Loader. (CTRL_STATUS_TOG)	32	RW	0000_0000h
10h	Address for data fetch. (BASE_ADDR)	32	RW	0000_0000h

15.9.2.1.2 Control/Status register for LUT Loader. (CTRL_STATUS)

15.9.2.1.2.1 Offset

Register	Offset
CTRL_STATUS	0h

15.9.2.1.2.2 Diagram



15.9.2.1.2.3 Fields

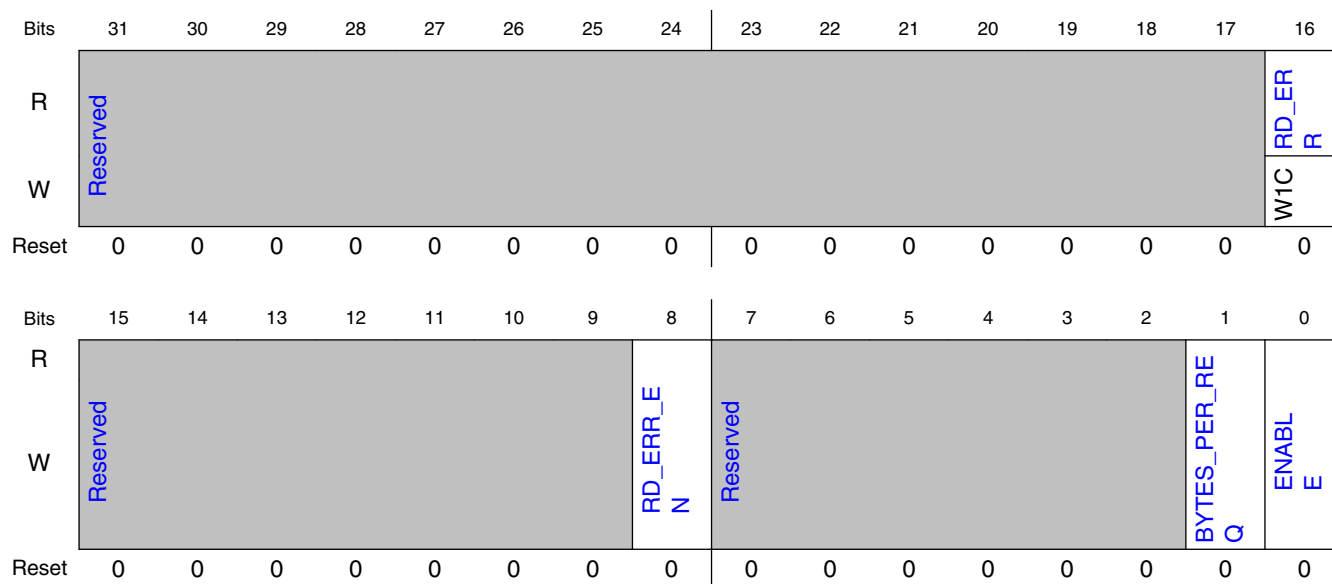
Field	Function
31-17 —	Reserved
16 RD_ERR	AXI Read Error. Status bit that indicates an AXI read response terminates with a SLV_ERR or DEC_ERR.
15-9 —	Reserved
8 RD_ERR_EN	AXI Read Error IRQ enable. Enable an interrupt when the AXI read response completes with a SLV_ERR or DEC_ERR.
7-2 —	Reserved
1 BYTES_PER_REQ	bytes per request control This field defines the number of bytes fetched for each AXI system memory request. 0 - 256 bytes of data for each AXI transaction. 1 - 128 bytes of data for each AXI transaction
0 ENABLE	Enable LUT_LD Enable bit for the lut loader. When set, the LUT_LD begins to fetch data from system memory. The data (Tone Curve, G2L) is transferred to the Dolby pixel engine via the DMA interface. A total of 10KB is fetched from system memory. 5 KB for video and 5 KB for graphics. Hardware automatically clears this bit when all 10KB of data is transferred to the Dolby pixel engine..

15.9.2.1.3 Control/Status register for LUT Loader. (CTRL_STATUS_SET)

15.9.2.1.3.1 Offset

Register	Offset
CTRL_STATUS_SET	4h

15.9.2.1.3.2 Diagram



15.9.2.1.3.3 Fields

Field	Function
31-17 —	Reserved
16 RD_ERR	AXI Read Error. Status bit that indicates an AXI read response terminates with a SLV_ERR or DEC_ERR.
15-9 —	Reserved
8 RD_ERR_EN	AXI Read Error IRQ enable. Enable an interrupt when the AXI read response completes with a SLV_ERR or DEC_ERR.
7-2 —	Reserved
1	bytes per request control

Table continues on the next page...

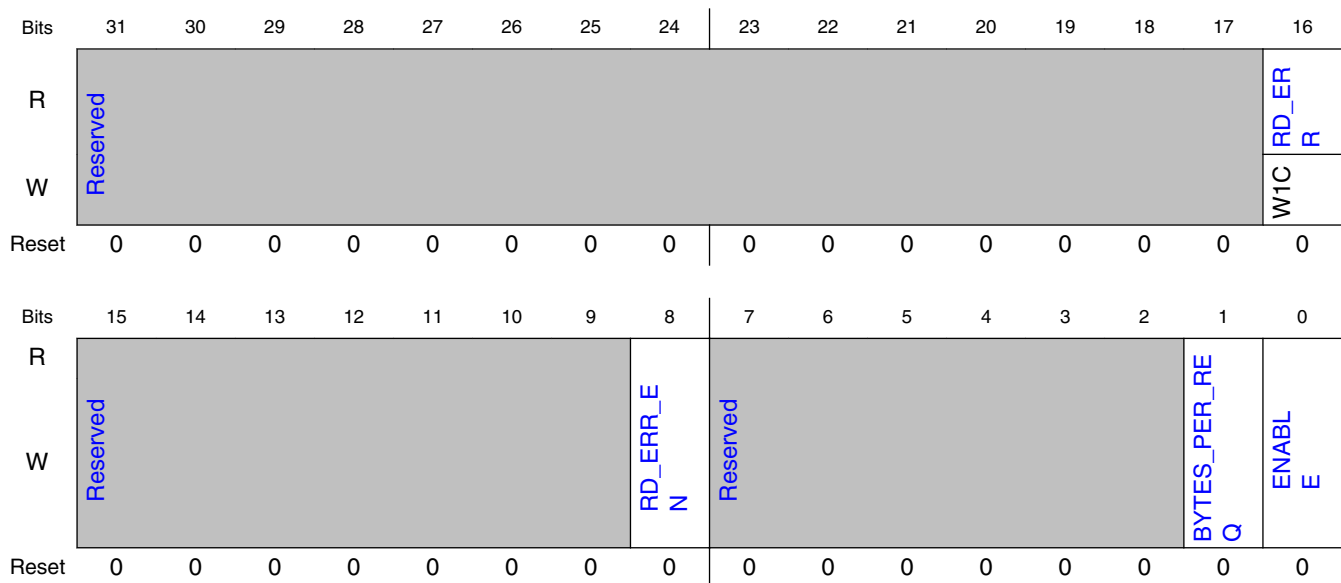
Field	Function
BYTES_PER_REQ	This field defines the number of bytes fetched for each AXI system memory request. 0 - 256 bytes of data for each AXI transaction. 1 - 128 bytes of data for each AXI transaction
0 ENABLE	Enable LUT_LD Enable bit for the lut loader. When set, the LUT_LD begins to fetch data from system memory. The data (Tone Curve, G2L) is transferred to the Dolby pixel engine via the DMA interface. A total of 10KB is fetched from system memory. 5 KB for video and 5 KB for graphics. Hardware automatically clears this bit when all 10KB of data is transferred to the Dolby pixel engine..

15.9.2.1.4 Control/Status register for LUT Loader. (CTRL_STATUS_CLR)

15.9.2.1.4.1 Offset

Register	Offset
CTRL_STATUS_CLR	8h

15.9.2.1.4.2 Diagram



15.9.2.1.4.3 Fields

Field	Function
31-17	Reserved

Table continues on the next page...

Memory Map and Registers

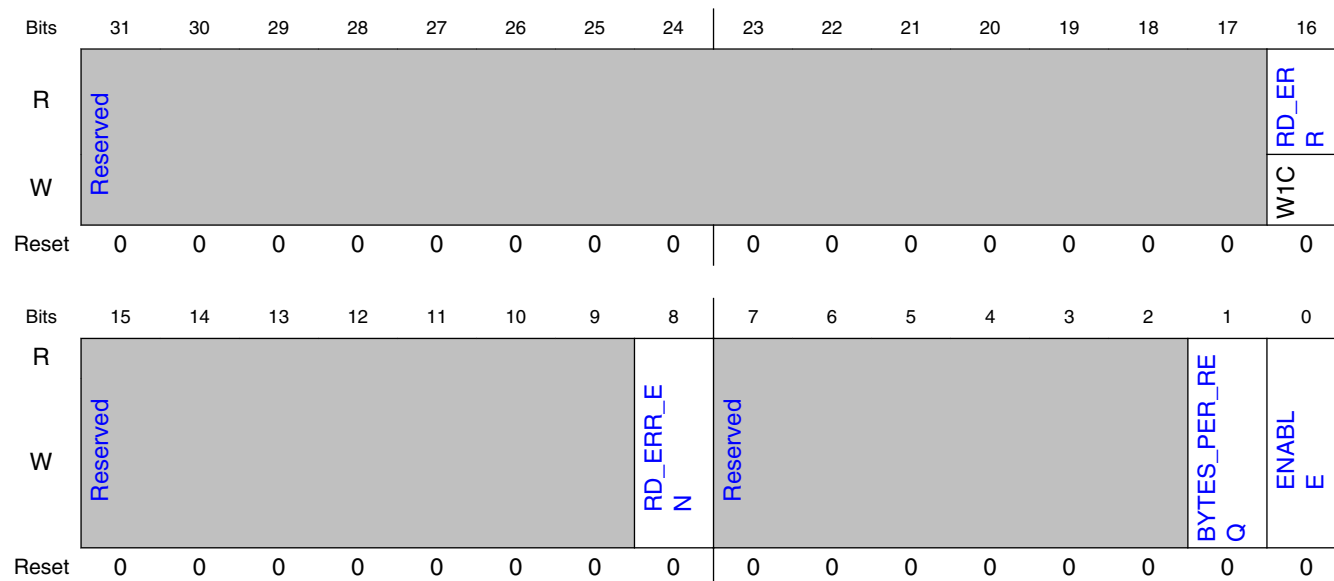
Field	Function
—	
16 RD_ERR	AXI Read Error. Status bit that indicates an AXI read response terminates with a SLV_ERR or DEC_ERR.
15-9 —	Reserved
8 RD_ERR_EN	AXI Read Error IRQ enable. Enable an interrupt when the AXI read response completes with a SLV_ERR or DEC_ERR.
7-2 —	Reserved
1 BYTES_PER_REQUEST	bytes per request control This field defines the number of bytes fetched for each AXI system memory request. • 0 - 256 bytes of data for each AXI transaction. • 1 - 128 bytes of data for each AXI transaction
0 ENABLE	Enable LUT_LD Enable bit for the lut loader. When set, the LUT_LD begins to fetch data from system memory. The data (Tone Curve, G2L) is transferred to the Dolby pixel engine via the DMA interface. A total of 10KB is fetched from system memory. 5 KB for video and 5 KB for graphics. Hardware automatically clears this bit when all 10KB of data is transferred to the Dolby pixel engine..

15.9.2.1.5 Control/Status register for LUT Loader. (CTRL_STATUS_TOG)

15.9.2.1.5.1 Offset

Register	Offset
CTRL_STATUS_TOG	Ch

15.9.2.1.5.2 Diagram



15.9.2.1.5.3 Fields

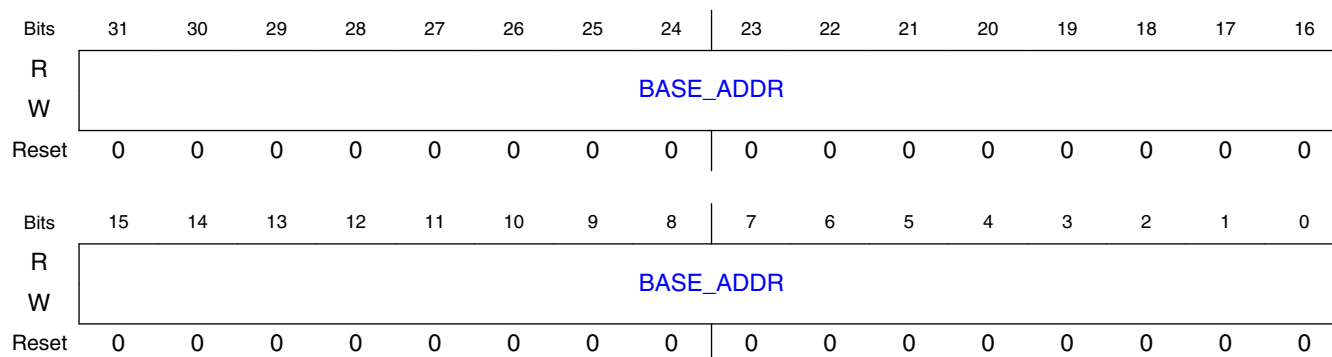
Field	Function
31-17 —	Reserved
16 RD_ERR	AXI Read Error. Status bit that indicates an AXI read response terminates with a SLV_ERR or DEC_ERR.
15-9 —	Reserved
8 RD_ERR_EN	AXI Read Error IRQ enable. Enable an interrupt when the AXI read response completes with a SLV_ERR or DEC_ERR.
7-2 —	Reserved
1 BYTES_PER_REQ	bytes per request control This field defines the number of bytes fetched for each AXI system memory request. 0 - 256 bytes of data for each AXI transaction. 1 - 128 bytes of data for each AXI transaction
0 ENABLE	Enable LUT_LD Enable bit for the lut loader. When set, the LUT_LD begins to fetch data from system memory. The data (Tone Curve, G2L) is transferred to the Dolby pixel engine via the DMA interface. A total of 10KB is fetched from system memory. 5 KB for video and 5 KB for graphics. Hardware automatically clears this bit when all 10KB of data is transferred to the Dolby pixel engine..

15.9.2.1.6 Address for data fetch. (BASE_ADDR)

15.9.2.1.6.1 Offset

Register	Offset
BASE_ADDR	10h

15.9.2.1.6.2 Diagram



15.9.2.1.6.3 Fields

Field	Function
31-0	Base Address
BASE_ADDR	Base address of continuous 5KB video and 5KB graphics LUT data. This address should be aligned to a 64B boundary in system memory.

15.10 HDR10 Image Processing (MED_HDR10)

15.10.1 Overview

In HDR10 mode, the three inputs get merged or blended together, post processed and sent to the HDMI protocol controller. The HDMI output consists of a display window containing up to three input windows. Any of the three windows can be driven by the Readback channel. All three windows can be full screen or smaller and offset inside the display window with the rest of the display window filled with the background color. The compositing of the three windows is controlled by the per pixel alpha channel from channel 1. The alpha logic creates an alpha based on the window priority and the graphics

per pixel alpha. The diagram below shows a possible window configuration. Note that window priority is fixed with channel 1 on top of channel 2 on top of channel 3. Only channel 1 can have a per pixel alpha.

Figure 15-22. HDR10 Processed Display

15.10.1.1 Features

In HDR10 mode, all three pixel processing pipelines are identical.

Each HDR10 pixel processing input pipeline supports:

- 10-bit per component input
- Color space conversion
- LUT gamma correction per component

HDR10 Output Pipeline The HDR10 pixel processing output pipeline supports:

- Linear to non-linear conversion per component
- Color space conversion
- 10-bit per component RGB/YUV444

15.10.2 Functional description

15.10.2.1 HDR10 Input Path

Below is a table showing the input to output flow depending on the source of HDR10 video, SDR video, or graphics and the output TV being HDR10 or SDR:

INPUT	Pre LUT	YCbCr CSC	EOTF LUT	2020 CSC	Output	TV
Graphics ARGB 8888 8b Rec. 709	Bypass	Bypass	Gamma EOTF (Non-linear RGB to Linear RGB) 3x1024 14b	RGB Rec.709 to RGB Rec.2020 Scale NIT 3x3 CSC 14b to 28b (1.14 coeff)	ARGB 8888 28b	HDR10
				Bypass		SDR
SDR Video YUV 444 8/10b Rec. 709	Remove offset 3x1024 10b	YCbCr to RGB 3x3 CSC 10b to 10b (1.14 coeff)		RGB Rec.709 to RGB Rec.2020 Scale NIT 3x3 CSC 14b to 28b (1.14 coeff)		HDR10
				Bypass		SDR

Table continues on the next page...

Memory Map and Registers

INPUT	Pre LUT	YCbCr CSC	EOTF LUT	2020 CSC	Output	TV
HDR10 Video YUV 444 10b Rec.2020			ST2084 EOTF (Non-linear RGB to Linear RGB) 3x1024 28b			HDR10

15.10.2.2 HDR10 Output Path

Below is a chart showing the input to output flow depending on the HDMI output format. The HDMI types include:

- HDMI 8-bit or 10-bit RGB
- HDMI 8-bit or 10-bit YCbCr 4:4:4
- HDMI 8-bit or 10-bit YCbCr 4:2:2
- HDMI 8-bit or 10-bit YCbCr 4:2:0

Output Path						
INPUT	Piece-wise linear interpolation	CSC	Post LUT	Chroma Sample Decimation	Output	HDMI Format
Linear RGB 444 28b	Reverse EOTF (Convert Linear RGB to Non-linear RGB OR Gamma) Piece-wise linear conversion 28b to 10b	Bypass RGB to YCbCr 3x3 CSC 10b to 10b	Add offset for limited range 3x1024 10b	Bypass	RGB 8/10b	HDMI RGB
				Horizontal Horizontal and Vertical	HDMI YCbCr 444	HDMI YCbCr 444
						HDMI YCbCr 422
						HDMI YCbCr 420

15.10.3 Memory Map and Registers

15.10.3.1 register descriptions

15.10.3.1.1 MED_HDR10 Memory map

med_hdr10 base address: C000h

Offset	Register	Width (In bits)	Access	Reset value
0h	A0 component Look-Up-Table. (LUT) (PIPE1_A0_LUT)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
1000h	A1 component Look-Up-Table. (LUT) (PIPE1_A1_LUT)	32	RW	0000_0000h
2000h	A2 component Look-Up-Table. (LUT) (PIPE1_A2_LUT)	32	RW	0000_0000h
3000h	Pipe1 Colorspace Converter A control. (HDR_PIPE1_CSCA_CONTR_OL_REG)	32	RW	0000_0000h
3004h	Pipe1 Colorspace Converter A (CSCA) h(0,0) matrix coefficient (HDR_PIPE1_CSCA_H00)	32	RW	0000_0000h
3008h	Pipe1 Colorspace Converter A (CSCA) h(1,0) matrix coefficient (HDR_PIPE1_CSCA_H10)	32	RW	0000_0000h
300Ch	Pipe1 Colorspace Converter A (CSCA) h(2,0) matrix coefficient (HDR_PIPE1_CSCA_H20)	32	RW	0000_0000h
3010h	Pipe1 Colorspace Converter A (CSCA) h(0,1) matrix coefficient (HDR_PIPE1_CSCA_H01)	32	RW	0000_0000h
3014h	Pipe1 Colorspace Converter A (CSCA) h(1,1) matrix coefficient (HDR_PIPE1_CSCA_H11)	32	RW	0000_0000h
3018h	Pipe1 Colorspace Converter A (CSCA) h(2,1) matrix coefficient (HDR_PIPE1_CSCA_H21)	32	RW	0000_0000h
301Ch	Pipe1 Colorspace Converter A (CSCA) h(0,2) matrix coefficient (HDR_PIPE1_CSCA_H02)	32	RW	0000_0000h
3020h	Pipe1 Colorspace Converter A (CSCA) h(1,2) matrix coefficient (HDR_PIPE1_CSCA_H12)	32	RW	0000_0000h
3024h	Pipe1 Colorspace Converter A (CSCA) h(2,2) matrix coefficient (HDR_PIPE1_CSCA_H22)	32	RW	0000_0000h
3028h	Pipe1 Colorspace Converter A (CSCA) component 0 pre-offset (HDR_PIPE1_CSCA_IO_0)	32	RW	0000_0000h
302Ch	Pipe1 Colorspace Converter A (CSCA) component 1 pre-offset (HDR_PIPE1_CSCA_IO_1)	32	RW	0000_0000h
3030h	Pipe1 Colorspace Converter A (CSCA) component 2 pre-offset (HDR_PIPE1_CSCA_IO_2)	32	RW	0000_0000h
3034h	Pipe1 Colorspace Converter A (CSCA) component 0 clip min. (HDR_PIPE1_CSCA_IO_MIN_0)	32	RW	0000_0000h
3038h	Pipe1 Colorspace Converter A (CSCA) component 1 clip min. (HDR_PIPE1_CSCA_IO_MIN_1)	32	RW	0000_0000h
303Ch	Pipe1 Colorspace Converter A (CSCA) component 2 clip min. (HDR_PIPE1_CSCA_IO_MIN_2)	32	RW	0000_0000h
3040h	Pipe1 Colorspace Converter A (CSCA) component 0 clip max value. (HDR_PIPE1_CSCA_IO_MAX_0)	32	RW	0000_0000h
3044h	Pipe1 Colorspace Converter A (CSCA) component 1 clip max value. (HDR_PIPE1_CSCA_IO_MAX_1)	32	RW	0000_0000h
3048h	Pipe1 Colorspace Converter A (CSCA) component 2 clip max value. (HDR_PIPE1_CSCA_IO_MAX_2)	32	RW	0000_0000h
304Ch	Pipe1 Colorspace Converter A (CSCA) normalization factor (HDR_PIPE1_CSCA_NORM)	32	RW	0000_0000h
3050h	Pipe1 Colorspace Converter A (CSCA): Post offset component 0 (HDR_PIPE1_CSCA_OO_0)	32	RW	0000_0000h
3054h	Pipe1 Colorspace Converter A (CSCA): Post offset component 1 (HDR_PIPE1_CSCA_OO_1)	32	RW	0000_0000h

Table continues on the next page...

Memory Map and Registers

Offset	Register	Width (In bits)	Access	Reset value
3058h	Pipe1 Colorspace Converter A (CSCA): Post offset component 2 (HDR_PIPE1_CSCA_OO_2)	32	RW	0000_0000h
305Ch	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 0 (HDR_PIPE1_CSCA_OMIN_0)	32	RW	0000_0000h
3060h	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 1 (HDR_PIPE1_CSCA_OMIN_1)	32	RW	0000_0000h
3064h	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 2 (HDR_PIPE1_CSCA_OMIN_2)	32	RW	0000_0000h
3068h	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 0 (HDR_PIPE1_CSCA_OMAX_0)	32	RW	0000_0000h
306Ch	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 1 (HDR_PIPE1_CSCA_OMAX_1)	32	RW	0000_0000h
3070h	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 2 (HDR_PIPE1_CSCA_OMAX_2)	32	RW	0000_0000h
3074h	PIPE1: NOT USED (HDR_PIPE1_ENTRY_29)	32	RU	0000_0000h
3080h	Pipe1 LUT control register (HDR_PIPE1_LUT_CONTROL_REG)	32	RW	0000_0000h
3800h	Pipe1 Colorspace Converter B control. (HDR_PIPE1_CSCB_CONTROL_REG)	32	RW	0000_0000h
3804h	Pipe1 Colorspace Converter A (CSCB) h(0,0) matrix coefficient (HDR_PIPE1_CSCB_H00)	32	RW	0000_0000h
3808h	Pipe1 Colorspace Converter B (CSCB) h(1,0) matrix coefficient (HDR_PIPE1_CSCB_H10)	32	RW	0000_0000h
380Ch	Pipe1 Colorspace Converter B (CSCB) h(2,0) matrix coefficient (HDR_PIPE1_CSCB_H20)	32	RW	0000_0000h
3810h	Pipe1 Colorspace Converter B (CSCB) h(0,1) matrix coefficient (HDR_PIPE1_CSCB_H01)	32	RW	0000_0000h
3814h	Pipe1 Colorspace Converter B (CSCB) h(1,1) matrix coefficient (HDR_PIPE1_CSCB_H11)	32	RW	0000_0000h
3818h	Pipe1 Colorspace Converter B (CSCB) h(2,1) matrix coefficient (HDR_PIPE1_CSCB_H21)	32	RW	0000_0000h
381Ch	Pipe1 Colorspace Converter B (CSCB) h(0,2) matrix coefficient (HDR_PIPE1_CSCB_H02)	32	RW	0000_0000h
3820h	Pipe1 Colorspace Converter B (CSCB) h(1,2) matrix coefficient (HDR_PIPE1_CSCB_H12)	32	RW	0000_0000h
3824h	Pipe1 Colorspace Converter B (CSCB) h(2,2) matrix coefficient (HDR_PIPE1_CSCB_H22)	32	RW	0000_0000h
3828h	Pipe1 Colorspace Converter B (CSCB) component 0 pre-offset (HDR_PIPE1_CSCB_IO_0)	32	RW	0000_0000h
382Ch	Pipe1 Colorspace Converter B (CSCB) component 1 pre-offset (HDR_PIPE1_CSCB_IO_1)	32	RW	0000_0000h
3830h	Pipe1 Colorspace Converter B (CSCB) component 2 pre-offset (HDR_PIPE1_CSCB_IO_2)	32	RW	0000_0000h
3834h	Pipe1 Colorspace Converter B (CSCB) component 0 clip min. (HDR_PIPE1_CSCB_IO_MIN_0)	32	RW	0000_0000h
3838h	Pipe1 Colorspace Converter B (CSCB) component 1 clip min. (HDR_PIPE1_CSCB_IO_MIN_1)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
383Ch	Pipe1 Colorspace Converter B (CSCB) component 2 clip min. (HDR_PIPE1_CSCB_IO_MIN_2)	32	RW	0000_0000h
3840h	Pipe1 Colorspace Converter B (CSCB) component 0 clip max value. (HDR_PIPE1_CSCB_IO_MAX_0)	32	RW	0000_0000h
3844h	Pipe1 Colorspace Converter B (CSCB) component 1 clip max value. (HDR_PIPE1_CSCB_IO_MAX_1)	32	RW	0000_0000h
3848h	Pipe1 Colorspace Converter B (CSCB) component 2 clip max value. (HDR_PIPE1_CSCB_IO_MAX_2)	32	RW	0000_0000h
384Ch	Pipe1 Colorspace Converter B (CSCB) normalization factor (HDR_PIPE1_CSCB_NORM)	32	RW	0000_0000h
3850h	Pipe1 Colorspace Converter B (CSCB): Post offset component 0 (HDR_PIPE1_CSCB_OO_0)	32	RW	0000_0000h
3854h	Pipe1 Colorspace Converter B (CSCB): Post offset component 1 (HDR_PIPE1_CSCB_OO_1)	32	RW	0000_0000h
3858h	Pipe1 Colorspace Converter B (CSCB): Post offset component 2 (HDR_PIPE1_CSCB_OO_2)	32	RW	0000_0000h
385Ch	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 0 (HDR_PIPE1_CSCB_OMIN_0)	32	RW	0000_0000h
3860h	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 1 (HDR_PIPE1_CSCB_OMIN_1)	32	RW	0000_0000h
3864h	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 2 (HDR_PIPE1_CSCB_OMIN_2)	32	RW	0000_0000h
3868h	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 0 (HDR_PIPE1_CSCB_OMAX_0)	32	RW	0000_0000h
386Ch	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 1 (HDR_PIPE1_CSCB_OMAX_1)	32	RW	0000_0000h
3870h	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 2 (HDR_PIPE1_CSCB_OMAX_2)	32	RW	0000_0000h
3874h	Pipe1 floating point to fixed point control (HDR_PIPE1_FL2FX)	32	RW	0000_0000h
3878h	PIPE1: NOT USED (HDR_PIPE1_ENTRY_30)	32	RU	0000_0000h
4000h	A0 component Look-Up-Table. (LUT) (PIPE2_A0_LUT)	32	RW	0000_0000h
5000h	A1 component Look-Up-Table. (LUT) (PIPE2_A1_LUT)	32	RW	0000_0000h
6000h	A2 component Look-Up-Table. (LUT) (PIPE2_A2_LUT)	32	RW	0000_0000h
7000h	Pipe1 Colorspace Converter A control. (HDR_PIPE2_CSCA_CONTR_OL_REG)	32	RW	0000_0000h
7004h	Pipe1 Colorspace Converter A (CSCA) h(0,0) matrix coefficient (HDR_PIPE2_CSCA_H00)	32	RW	0000_0000h
7008h	Pipe1 Colorspace Converter A (CSCA) h(1,0) matrix coefficient (HDR_PIPE2_CSCA_H10)	32	RW	0000_0000h
700Ch	Pipe1 Colorspace Converter A (CSCA) h(2,0) matrix coefficient (HDR_PIPE2_CSCA_H20)	32	RW	0000_0000h
7010h	Pipe1 Colorspace Converter A (CSCA) h(0,1) matrix coefficient (HDR_PIPE2_CSCA_H01)	32	RW	0000_0000h
7014h	Pipe1 Colorspace Converter A (CSCA) h(1,1) matrix coefficient (HDR_PIPE2_CSCA_H11)	32	RW	0000_0000h

Table continues on the next page...

Memory Map and Registers

Offset	Register	Width (In bits)	Access	Reset value
7018h	Pipe1 Colorspace Converter A (CSCA) h(2,1) matrix coefficient (HDR_PIPE2_CSCA_H21)	32	RW	0000_0000h
701Ch	Pipe1 Colorspace Converter A (CSCA) h(0,2) matrix coefficient (HDR_PIPE2_CSCA_H02)	32	RW	0000_0000h
7020h	Pipe1 Colorspace Converter A (CSCA) h(1,2) matrix coefficient (HDR_PIPE2_CSCA_H12)	32	RW	0000_0000h
7024h	Pipe1 Colorspace Converter A (CSCA) h(2,2) matrix coefficient (HDR_PIPE2_CSCA_H22)	32	RW	0000_0000h
7028h	Pipe1 Colorspace Converter A (CSCA) component 0 pre-offset (HDR_PIPE2_CSCA_IO_0)	32	RW	0000_0000h
702Ch	Pipe1 Colorspace Converter A (CSCA) component 1 pre-offset (HDR_PIPE2_CSCA_IO_1)	32	RW	0000_0000h
7030h	Pipe1 Colorspace Converter A (CSCA) component 2 pre-offset (HDR_PIPE2_CSCA_IO_2)	32	RW	0000_0000h
7034h	Pipe1 Colorspace Converter A (CSCA) component 0 clip min. (HDR_PIPE2_CSCA_IO_MIN_0)	32	RW	0000_0000h
7038h	Pipe1 Colorspace Converter A (CSCA) component 1 clip min. (HDR_PIPE2_CSCA_IO_MIN_1)	32	RW	0000_0000h
703Ch	Pipe1 Colorspace Converter A (CSCA) component 2 clip min. (HDR_PIPE2_CSCA_IO_MIN_2)	32	RW	0000_0000h
7040h	Pipe1 Colorspace Converter A (CSCA) component 0 clip max value. (HDR_PIPE2_CSCA_IO_MAX_0)	32	RW	0000_0000h
7044h	Pipe1 Colorspace Converter A (CSCA) component 1 clip max value. (HDR_PIPE2_CSCA_IO_MAX_1)	32	RW	0000_0000h
7048h	Pipe1 Colorspace Converter A (CSCA) component 2 clip max value. (HDR_PIPE2_CSCA_IO_MAX_2)	32	RW	0000_0000h
704Ch	Pipe1 Colorspace Converter A (CSCA) normalization factor (HDR_PIPE2_CSCA_NORM)	32	RW	0000_0000h
7050h	Pipe1 Colorspace Converter A (CSCA): Post offset component 0 (HDR_PIPE2_CSCA_OO_0)	32	RW	0000_0000h
7054h	Pipe1 Colorspace Converter A (CSCA): Post offset component 1 (HDR_PIPE2_CSCA_OO_1)	32	RW	0000_0000h
7058h	Pipe1 Colorspace Converter A (CSCA): Post offset component 2 (HDR_PIPE2_CSCA_OO_2)	32	RW	0000_0000h
705Ch	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 0 (HDR_PIPE2_CSCA_OMIN_0)	32	RW	0000_0000h
7060h	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 1 (HDR_PIPE2_CSCA_OMIN_1)	32	RW	0000_0000h
7064h	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 2 (HDR_PIPE2_CSCA_OMIN_2)	32	RW	0000_0000h
7068h	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 0 (HDR_PIPE2_CSCA_OMAX_0)	32	RW	0000_0000h
706Ch	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 1 (HDR_PIPE2_CSCA_OMAX_1)	32	RW	0000_0000h
7070h	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 2 (HDR_PIPE2_CSCA_OMAX_2)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
7074h	PIPE2: NOT USED (HDR_PIPE2_ENTRY_29)	32	RU	0000_0000h
7080h	Pipe1 LUT control register (HDR_PIPE2_LUT_CONTROL_REG)	32	RW	0000_0000h
7800h	Pipe1 Colorspace Converter B control. (HDR_PIPE2_CSCB_CONTR OL_REG)	32	RW	0000_0000h
7804h	Pipe1 Colorspace Converter A (CSCB) h(0,0) matrix coefficient (HDR_PIPE2_CSCB_H00)	32	RW	0000_0000h
7808h	Pipe1 Colorspace Converter B (CSCB) h(1,0) matrix coefficient (HDR_PIPE2_CSCB_H10)	32	RW	0000_0000h
780Ch	Pipe1 Colorspace Converter B (CSCB) h(2,0) matrix coefficient (HDR_PIPE2_CSCB_H20)	32	RW	0000_0000h
7810h	Pipe1 Colorspace Converter B (CSCB) h(0,1) matrix coefficient (HDR_PIPE2_CSCB_H01)	32	RW	0000_0000h
7814h	Pipe1 Colorspace Converter B (CSCB) h(1,1) matrix coefficient (HDR_PIPE2_CSCB_H11)	32	RW	0000_0000h
7818h	Pipe1 Colorspace Converter B (CSCB) h(2,1) matrix coefficient (HDR_PIPE2_CSCB_H21)	32	RW	0000_0000h
781Ch	Pipe1 Colorspace Converter B (CSCB) h(0,2) matrix coefficient (HDR_PIPE2_CSCB_H02)	32	RW	0000_0000h
7820h	Pipe1 Colorspace Converter B (CSCB) h(1,2) matrix coefficient (HDR_PIPE2_CSCB_H12)	32	RW	0000_0000h
7824h	Pipe1 Colorspace Converter B (CSCB) h(2,2) matrix coefficient (HDR_PIPE2_CSCB_H22)	32	RW	0000_0000h
7828h	Pipe1 Colorspace Converter B (CSCB) component 0 pre-offset (HDR_PIPE2_CSCB_IO_0)	32	RW	0000_0000h
782Ch	Pipe1 Colorspace Converter B (CSCB) component 1 pre-offset (HDR_PIPE2_CSCB_IO_1)	32	RW	0000_0000h
7830h	Pipe1 Colorspace Converter B (CSCB) component 2 pre-offset (HDR_PIPE2_CSCB_IO_2)	32	RW	0000_0000h
7834h	Pipe1 Colorspace Converter B (CSCB) component 0 clip min. (HDR_ PIPE2_CSCB_IO_MIN_0)	32	RW	0000_0000h
7838h	Pipe1 Colorspace Converter B (CSCB) component 1 clip min. (HDR_ PIPE2_CSCB_IO_MIN_1)	32	RW	0000_0000h
783Ch	Pipe1 Colorspace Converter B (CSCB) component 2 clip min. (HDR_ PIPE2_CSCB_IO_MIN_2)	32	RW	0000_0000h
7840h	Pipe1 Colorspace Converter B (CSCB) component 0 clip max value. (HDR_PIPE2_CSCB_IO_MAX_0)	32	RW	0000_0000h
7844h	Pipe1 Colorspace Converter B (CSCB) component 1 clip max value. (HDR_PIPE2_CSCB_IO_MAX_1)	32	RW	0000_0000h
7848h	Pipe1 Colorspace Converter B (CSCB) component 2 clip max value. (HDR_PIPE2_CSCB_IO_MAX_2)	32	RW	0000_0000h
784Ch	Pipe1 Colorspace Converter B (CSCB) normalization factor (HDR_ PIPE2_CSCB_NORM)	32	RW	0000_0000h
7850h	Pipe1 Colorspace Converter B (CSCB): Post offset component 0 (HDR_PIPE2_CSCB_OO_0)	32	RW	0000_0000h
7854h	Pipe1 Colorspace Converter B (CSCB): Post offset component 1 (HDR_PIPE2_CSCB_OO_1)	32	RW	0000_0000h

Table continues on the next page...

Memory Map and Registers

Offset	Register	Width (In bits)	Access	Reset value
7858h	Pipe1 Colorspace Converter B (CSCB): Post offset component 2 (HDR_PIPE2_CSCB_OO_2)	32	RW	0000_0000h
785Ch	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 0 (HDR_PIPE2_CSCB_OMIN_0)	32	RW	0000_0000h
7860h	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 1 (HDR_PIPE2_CSCB_OMIN_1)	32	RW	0000_0000h
7864h	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 2 (HDR_PIPE2_CSCB_OMIN_2)	32	RW	0000_0000h
7868h	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 0 (HDR_PIPE2_CSCB_OMAX_0)	32	RW	0000_0000h
786Ch	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 1 (HDR_PIPE2_CSCB_OMAX_1)	32	RW	0000_0000h
7870h	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 2 (HDR_PIPE2_CSCB_OMAX_2)	32	RW	0000_0000h
7874h	Pipe1 floating point to fixed point control (HDR_PIPE2_FL2FX)	32	RW	0000_0000h
7878h	PIPE2: NOT USED (HDR_PIPE2_ENTRY_30)	32	RU	0000_0000h
8000h	A0 component Look-Up-Table. (LUT) (PIPE3_A0_LUT)	32	RW	0000_0000h
9000h	A1 component Look-Up-Table. (LUT) (PIPE3_A1_LUT)	32	RW	0000_0000h
A000h	A2 component Look-Up-Table. (LUT) (PIPE3_A2_LUT)	32	RW	0000_0000h
B000h	Pipe1 Colorspace Converter A control. (HDR_PIPE3_CSCA_CONTR_OL_REG)	32	RW	0000_0000h
B004h	Pipe1 Colorspace Converter A (CSCA) h(0,0) matrix coefficient (HDR_PIPE3_CSCA_H00)	32	RW	0000_0000h
B008h	Pipe1 Colorspace Converter A (CSCA) h(1,0) matrix coefficient (HDR_PIPE3_CSCA_H10)	32	RW	0000_0000h
B00Ch	Pipe1 Colorspace Converter A (CSCA) h(2,0) matrix coefficient (HDR_PIPE3_CSCA_H20)	32	RW	0000_0000h
B010h	Pipe1 Colorspace Converter A (CSCA) h(0,1) matrix coefficient (HDR_PIPE3_CSCA_H01)	32	RW	0000_0000h
B014h	Pipe1 Colorspace Converter A (CSCA) h(1,1) matrix coefficient (HDR_PIPE3_CSCA_H11)	32	RW	0000_0000h
B018h	Pipe1 Colorspace Converter A (CSCA) h(2,1) matrix coefficient (HDR_PIPE3_CSCA_H21)	32	RW	0000_0000h
B01Ch	Pipe1 Colorspace Converter A (CSCA) h(0,2) matrix coefficient (HDR_PIPE3_CSCA_H02)	32	RW	0000_0000h
B020h	Pipe1 Colorspace Converter A (CSCA) h(1,2) matrix coefficient (HDR_PIPE3_CSCA_H12)	32	RW	0000_0000h
B024h	Pipe1 Colorspace Converter A (CSCA) h(2,2) matrix coefficient (HDR_PIPE3_CSCA_H22)	32	RW	0000_0000h
B028h	Pipe1 Colorspace Converter A (CSCA) component 0 pre-offset (HDR_PIPE3_CSCA_IO_0)	32	RW	0000_0000h
B02Ch	Pipe1 Colorspace Converter A (CSCA) component 1 pre-offset (HDR_PIPE3_CSCA_IO_1)	32	RW	0000_0000h
B030h	Pipe1 Colorspace Converter A (CSCA) component 2 pre-offset (HDR_PIPE3_CSCA_IO_2)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
B034h	Pipe1 Colorspace Converter A (CSCA) component 0 clip min. (HDR_PIPE3_CSCA_IO_MIN_0)	32	RW	0000_0000h
B038h	Pipe1 Colorspace Converter A (CSCA) component 1 clip min. (HDR_PIPE3_CSCA_IO_MIN_1)	32	RW	0000_0000h
B03Ch	Pipe1 Colorspace Converter A (CSCA) component 2 clip min. (HDR_PIPE3_CSCA_IO_MIN_2)	32	RW	0000_0000h
B040h	Pipe1 Colorspace Converter A (CSCA) component 0 clip max value. (HDR_PIPE3_CSCA_IO_MAX_0)	32	RW	0000_0000h
B044h	Pipe1 Colorspace Converter A (CSCA) component 1 clip max value. (HDR_PIPE3_CSCA_IO_MAX_1)	32	RW	0000_0000h
B048h	Pipe1 Colorspace Converter A (CSCA) component 2 clip max value. (HDR_PIPE3_CSCA_IO_MAX_2)	32	RW	0000_0000h
B04Ch	Pipe1 Colorspace Converter A (CSCA) normalization factor (HDR_PIPE3_CSCA_NORM)	32	RW	0000_0000h
B050h	Pipe1 Colorspace Converter A (CSCA): Post offset component 0 (HDR_PIPE3_CSCA_OO_0)	32	RW	0000_0000h
B054h	Pipe1 Colorspace Converter A (CSCA): Post offset component 1 (HDR_PIPE3_CSCA_OO_1)	32	RW	0000_0000h
B058h	Pipe1 Colorspace Converter A (CSCA): Post offset component 2 (HDR_PIPE3_CSCA_OO_2)	32	RW	0000_0000h
B05Ch	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 0 (HDR_PIPE3_CSCA_OMIN_0)	32	RW	0000_0000h
B060h	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 1 (HDR_PIPE3_CSCA_OMIN_1)	32	RW	0000_0000h
B064h	Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 2 (HDR_PIPE3_CSCA_OMIN_2)	32	RW	0000_0000h
B068h	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 0 (HDR_PIPE3_CSCA_OMAX_0)	32	RW	0000_0000h
B06Ch	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 1 (HDR_PIPE3_CSCA_OMAX_1)	32	RW	0000_0000h
B070h	Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 2 (HDR_PIPE3_CSCA_OMAX_2)	32	RW	0000_0000h
B074h	PIPE3: NOT USED (HDR_PIPE3_ENTRY_29)	32	RU	0000_0000h
B080h	Pipe1 LUT control register (HDR_PIPE3_LUT_CONTROL_REG)	32	RW	0000_0000h
B800h	Pipe1 Colorspace Converter B control. (HDR_PIPE3_CSCB_CONTROL_REG)	32	RW	0000_0000h
B804h	Pipe1 Colorspace Converter A (CSCB) h(0,0) matrix coefficient (HDR_PIPE3_CSCB_H00)	32	RW	0000_0000h
B808h	Pipe1 Colorspace Converter B (CSCB) h(1,0) matrix coefficient (HDR_PIPE3_CSCB_H10)	32	RW	0000_0000h
B80Ch	Pipe1 Colorspace Converter B (CSCB) h(2,0) matrix coefficient (HDR_PIPE3_CSCB_H20)	32	RW	0000_0000h
B810h	Pipe1 Colorspace Converter B (CSCB) h(0,1) matrix coefficient (HDR_PIPE3_CSCB_H01)	32	RW	0000_0000h
B814h	Pipe1 Colorspace Converter B (CSCB) h(1,1) matrix coefficient (HDR_PIPE3_CSCB_H11)	32	RW	0000_0000h

Table continues on the next page...

Memory Map and Registers

Offset	Register	Width (In bits)	Access	Reset value
B818h	Pipe1 Colorspace Converter B (CSCB) h(2,1) matrix coefficient (HDR_PIPE3_CSCB_H21)	32	RW	0000_0000h
B81Ch	Pipe1 Colorspace Converter B (CSCB) h(0,2) matrix coefficient (HDR_PIPE3_CSCB_H02)	32	RW	0000_0000h
B820h	Pipe1 Colorspace Converter B (CSCB) h(1,2) matrix coefficient (HDR_PIPE3_CSCB_H12)	32	RW	0000_0000h
B824h	Pipe1 Colorspace Converter B (CSCB) h(2,2) matrix coefficient (HDR_PIPE3_CSCB_H22)	32	RW	0000_0000h
B828h	Pipe1 Colorspace Converter B (CSCB) component 0 pre-offset (HDR_PIPE3_CSCB_IO_0)	32	RW	0000_0000h
B82Ch	Pipe1 Colorspace Converter B (CSCB) component 1 pre-offset (HDR_PIPE3_CSCB_IO_1)	32	RW	0000_0000h
B830h	Pipe1 Colorspace Converter B (CSCB) component 2 pre-offset (HDR_PIPE3_CSCB_IO_2)	32	RW	0000_0000h
B834h	Pipe1 Colorspace Converter B (CSCB) component 0 clip min. (HDR_PIPE3_CSCB_IO_MIN_0)	32	RW	0000_0000h
B838h	Pipe1 Colorspace Converter B (CSCB) component 1 clip min. (HDR_PIPE3_CSCB_IO_MIN_1)	32	RW	0000_0000h
B83Ch	Pipe1 Colorspace Converter B (CSCB) component 2 clip min. (HDR_PIPE3_CSCB_IO_MIN_2)	32	RW	0000_0000h
B840h	Pipe1 Colorspace Converter B (CSCB) component 0 clip max value. (HDR_PIPE3_CSCB_IO_MAX_0)	32	RW	0000_0000h
B844h	Pipe1 Colorspace Converter B (CSCB) component 1 clip max value. (HDR_PIPE3_CSCB_IO_MAX_1)	32	RW	0000_0000h
B848h	Pipe1 Colorspace Converter B (CSCB) component 2 clip max value. (HDR_PIPE3_CSCB_IO_MAX_2)	32	RW	0000_0000h
B84Ch	Pipe1 Colorspace Converter B (CSCB) normalization factor (HDR_PIPE3_CSCB_NORM)	32	RW	0000_0000h
B850h	Pipe1 Colorspace Converter B (CSCB): Post offset component 0 (HDR_PIPE3_CSCB_OO_0)	32	RW	0000_0000h
B854h	Pipe1 Colorspace Converter B (CSCB): Post offset component 1 (HDR_PIPE3_CSCB_OO_1)	32	RW	0000_0000h
B858h	Pipe1 Colorspace Converter B (CSCB): Post offset component 2 (HDR_PIPE3_CSCB_OO_2)	32	RW	0000_0000h
B85Ch	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 0 (HDR_PIPE3_CSCB_OMIN_0)	32	RW	0000_0000h
B860h	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 1 (HDR_PIPE3_CSCB_OMIN_1)	32	RW	0000_0000h
B864h	Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 2 (HDR_PIPE3_CSCB_OMIN_2)	32	RW	0000_0000h
B868h	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 0 (HDR_PIPE3_CSCB_OMAX_0)	32	RW	0000_0000h
B86Ch	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 1 (HDR_PIPE3_CSCB_OMAX_1)	32	RW	0000_0000h
B870h	Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 2 (HDR_PIPE3_CSCB_OMAX_2)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
B874h	Pipe1 floating point to fixed point control (HDR_PIPE3_FL2FX)	32	RW	0000_0000h
B878h	PIPE3: NOT USED (HDR_PIPE3_ENTRY_30)	32	RU	0000_0000h
C000h	A0 component Linear-to-Non-linear conversion table (OPIPE_A0_TABLE)	32	RW	0000_0000h
D000h	A1 component Linear-to-Non-linear conversion table (OPIPE_A1_TABLE)	32	RW	0000_0000h
E000h	A2 component Linear-to-Non-linear conversion table (OPIPE_A2_TABLE)	32	RW	0000_0000h
F000h	HDR output stage Colorspace Converter (CSCO) control. (HDR_OPIPE_CSC_CONTROL_REG)	32	RW	0000_0000h
F004h	Pipe1 Colorspace Converter (CSC) h(0,0) matrix coefficient (HDR_OPIPE_CSC_H00)	32	RW	0000_0000h
F008h	Pipe1 Colorspace Converter (CSC) h(1,0) matrix coefficient (HDR_OPIPE_CSC_H10)	32	RW	0000_0000h
F00Ch	HDR OUTPUT Colorspace Converter (CSCO) h(2,0) matrix coefficient (HDR_OPIPE_CSC_H20)	32	RW	0000_0000h
F010h	HDR OUTPUT pipe Colorspace Converter (CSCO) h(0,1) matrix coefficient (HDR_OPIPE_CSC_H01)	32	RW	0000_0000h
F014h	HDR OUTPUT pipe Colorspace Converter (CSCO) h(1,1) matrix coefficient (HDR_OPIPE_CSC_H11)	32	RW	0000_0000h
F018h	HDR_output pipe Colorspace Converter (CSCO) h(2,1) matrix coefficient (HDR_OPIPE_CSC_H21)	32	RW	0000_0000h
F01Ch	HDR OUTPUT pipe Colorspace Converter (CSCO) h(0,2) matrix coefficient (HDR_OPIPE_CSC_H02)	32	RW	0000_0000h
F020h	HDR OUPUT pipe Colorspace Converter (CSCO) h(1,2) matrix coefficient (HDR_OPIPE_CSC_H12)	32	RW	0000_0000h
F024h	HDR OUPUT pipe Colorspace Converter (CSCO) h(2,2) matrix coefficient (HDR_)	32	RW	0000_0000h
F028h	HDR OUTPUT pipe Colorspace Converter (CSCO) component 0 pre-offset (HDR_OPIPE_CSC_IO_0)	32	RW	0000_0000h
F02Ch	HDR OUPUT pipe Colorspace Converter (CSCO) component 1 pre-offset (HDR_OPIPE_CSC_IO_1)	32	RW	0000_0000h
F030h	HDR OUPUT pipe: Colorspace Converter (CSCO) component 2 pre-offset (HDR_OPIPE_CSC_IO_2)	32	RW	0000_0000h
F034h	HDR OUPUT pipe Colorspace Converter (CSCO) component 0 clip min. (HDR_OPIPE_CSC_MIN_0)	32	RW	0000_0000h
F038h	HDR OUPUT pipe Colorspace Converter (CSCO) component 1 clip min. (HDR_OPIPE_CSC_MIN_1)	32	RW	0000_0000h
F03Ch	HDR OUPUT pipe Colorspace Converter (CSCO) component 2 clip min. (HDR_OPIPE_CSC_MIN_2)	32	RW	0000_0000h
F040h	HDR OUPUT pipe Colorspace Converter O (CSC) component 0 clip max value. (HDR_OPIPE_CSC_MAX_0)	32	RW	0000_0000h
F044h	HDR OUTPUT pipe Colorspace Converter (CSCO) component 1 clip max value. (HDR_OPIPE_CSC_MAX_1)	32	RW	0000_0000h
F048h	HDR OUTPUT pipe Colorspace Converter (CSCO) component 2 clip max value. (HDR_OPIPE_CSC_MAX_2)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
F04Ch	HDR OUPUT pipe Colorspace Converter (CSCO) normalization factor (HDR_OPIPE_CSC_NORM)	32	RW	0000_0000h
F050h	HDR OUPUT pipe Colorspace Converter (CSC): Post offset component 0 (HDR_OPIPE_CSC_OO_0)	32	RW	0000_0000h
F054h	HDR OUTPUT pipe Colorspace Converter (CSC): Post offset component 1 (HDR_OPIPE_CSC_OO_1)	32	RW	0000_0000h
F058h	HDR OUPUT pipe Colorspace Converter (CSC): Post offset component 2 (HDR_OPIPE_CSC_OO_2)	32	RW	0000_0000h
F05Ch	HDR OUTPUT pipe Colorspace Converter (CSC): Post offset min clip value for component 0 (HDR_OPIPE_CSC_OMIN_0)	32	RW	0000_0000h
F060h	HDR OUTPUT pipe Colorspace Converter (CSC): Post offset min clip value for component 1 (HDR_OPIPE_CSC_OMIN_1)	32	RW	0000_0000h
F064h	HDR OUTPUT pipe Colorspace Converter (CSC): Post offset min clip value for component 2 (HDR_OPIPE_CSC_OMIN_2)	32	RW	0000_0000h
F068h	HDR OUPUT pipe Colorspace Converter (CSC): Post offset max clip value for component 0 (HDR_OPIPE_CSC_OMAX_0)	32	RW	0000_0000h
F06Ch	HDR OUTPUT pipe Colorspace Converter (CSC): Post offset max clip value for component 1 (HDR_OPIPE_CSC_OMAX_1)	32	RW	0000_0000h
F070h	HDR OUTPUT pipe Colorspace Converter (CSC): Post offset max clip value for component 2 (HDR_OPIPE_CSC_OMAX_2)	32	RW	0000_0000h
F874h	HDR OUTPUT -TO NON LINEAR pipeline control (HDR_OPIPE_2NL_CONTROL_REG)	32	RW	0000_0000h

15.10.3.1.2 A0 component Look-Up-Table. (LUT) (PIPE1_A0_LUT)

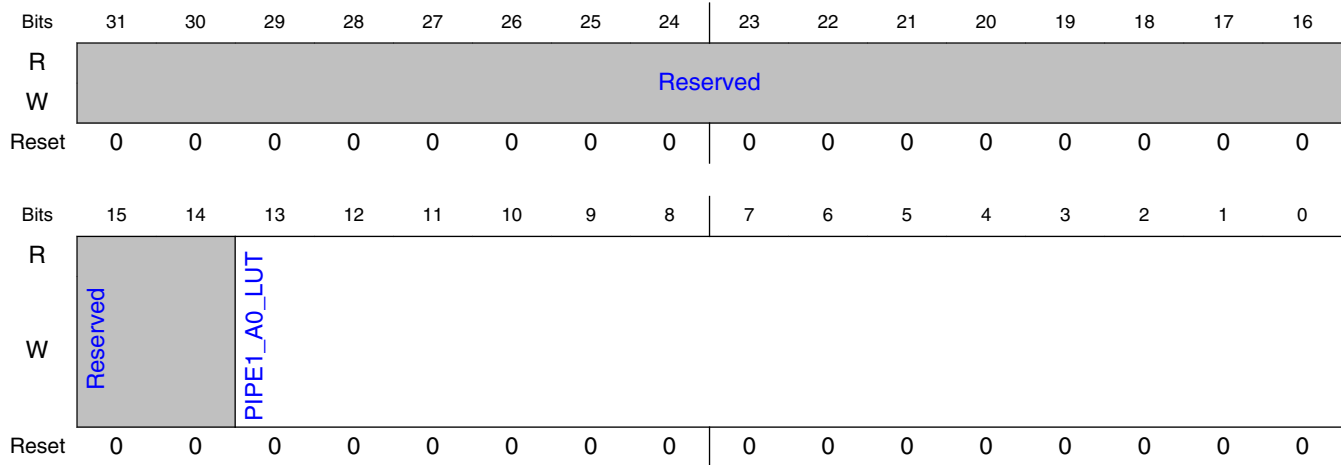
15.10.3.1.2.1 Offset

Register	Offset
PIPE1_A0_LUT	0h

15.10.3.1.2.2 Function

The LUT table has 1024 entries. A0 component may be R or Y or other

15.10.3.1.2.3 Diagram



15.10.3.1.2.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE1_A0_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.3 A1 component Look-Up-Table. (LUT) (PIPE1_A1_LUT)

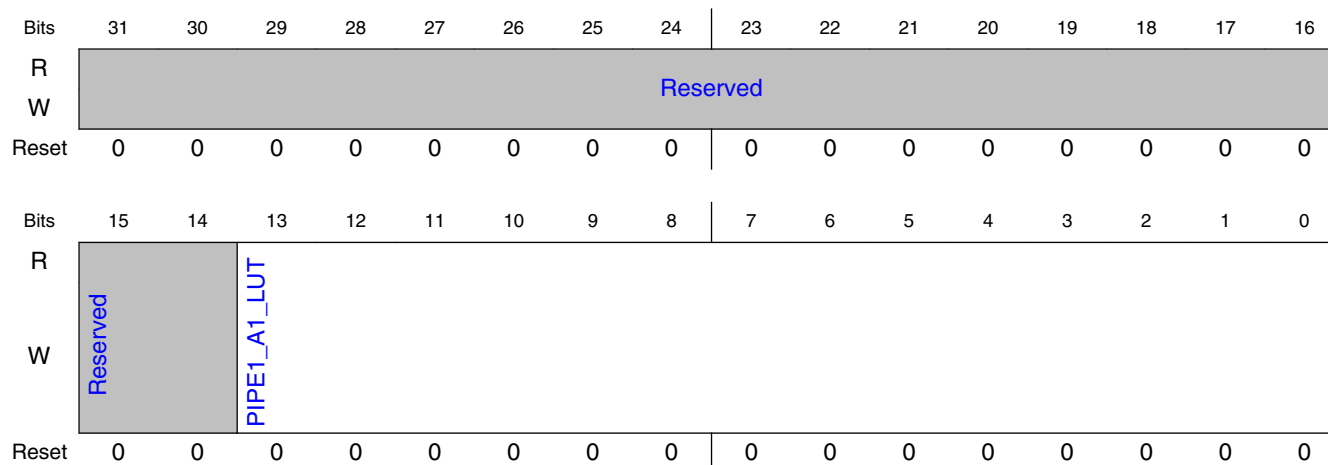
15.10.3.1.3.1 Offset

Register	Offset
PIPE1_A1_LUT	1000h

15.10.3.1.3.2 Function

The LUT table has 1024 entries. A1 component may be G or Cb or other

15.10.3.1.3.3 Diagram



15.10.3.1.3.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE1_A1_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.4 A2 component Look-Up-Table. (LUT) (PIPE1_A2_LUT)

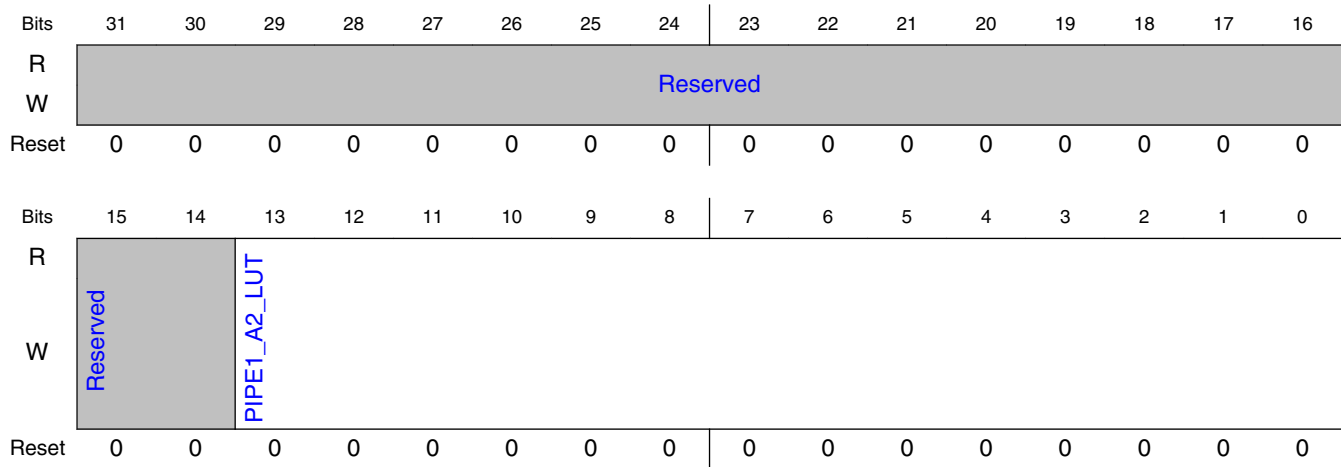
15.10.3.1.4.1 Offset

Register	Offset
PIPE1_A2_LUT	2000h

15.10.3.1.4.2 Function

The LUT table has 1024 entries. A2 component may be B or Cr or other

15.10.3.1.4.3 Diagram



15.10.3.1.4.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE1_A2_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.5 Pipe1 Colorspace Converter A control. (HDR_PIPE1_CSCA_CONTROL_REG)

15.10.3.1.5.1 Offset

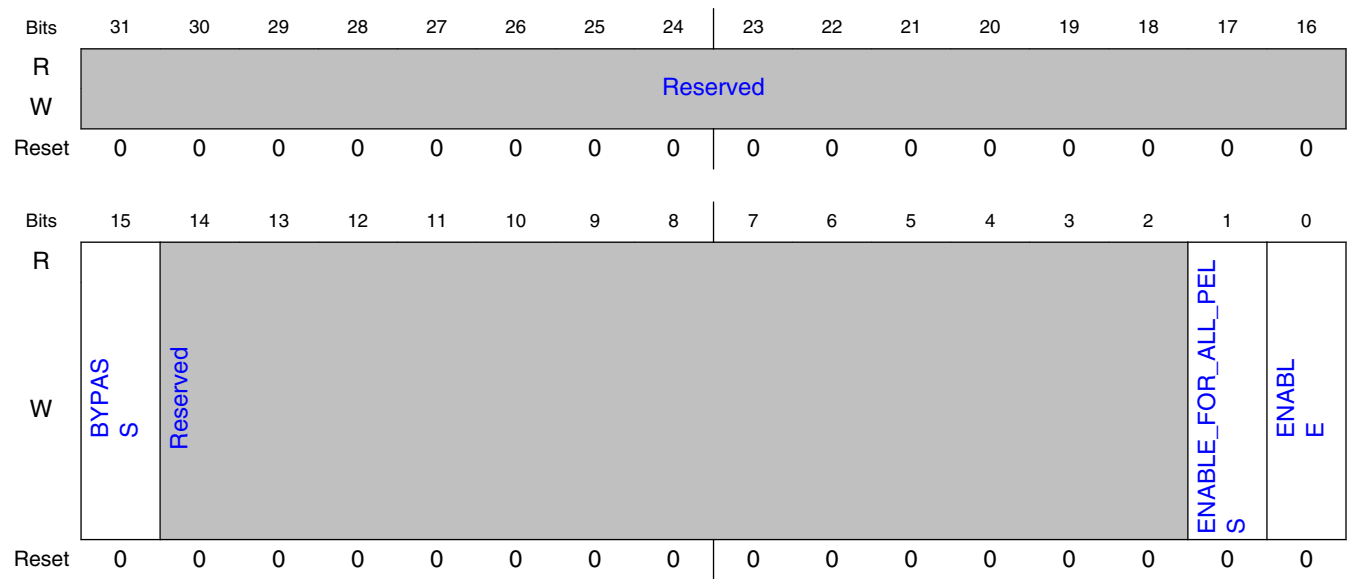
Register	Offset
HDR_PIPE1_CSCA_CONTROL_REG	3000h

15.10.3.1.5.2 Function

Controls the Color-Space-Converter-A (CSCA) in pipe1. This CSCA takes in 10-bit pixel components (component 0, component 1, component 2). A per-component offset (called pre-offset) is added to the pixel. After the pre-offset operation the pixel components are clipped. MAX_RANGE of PRE_CLIP is [-512, 1023]. A 3x3 matrix multiply (constant coefficients H(x,y)) is performed on the pixel to take to a different color space. After this matrix multiply, a 28 bit signed offset (post-offset) is added to the normalized result of

the matrix multiply. After the post-offset operation the results are clipped.
MAX_RANGE of POST_CLIP is [0,1023] Output of this CSCA are 10 bit per component pixels. After CSCA the pixels are fed into a LUT. Pipe1 is also called channel1.

15.10.3.1.5.3 Diagram



15.10.3.1.5.4 Fields

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this CSC 1: Pixels pass thru this CSC unmodified
14-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This CSC is enabled only for blened pixels 1: This CSC is enabled all pixels
0 ENABLE	0: Don't enable this CSC: Pixels pass thru the CSC unmodified 1: This CSC is enabled for current picture

15.10.3.1.6 Pipe1 Colorspace Converter A (CSCA) h(0,0) matrix coefficient (HDR_PIPE1_CSCA_H00)

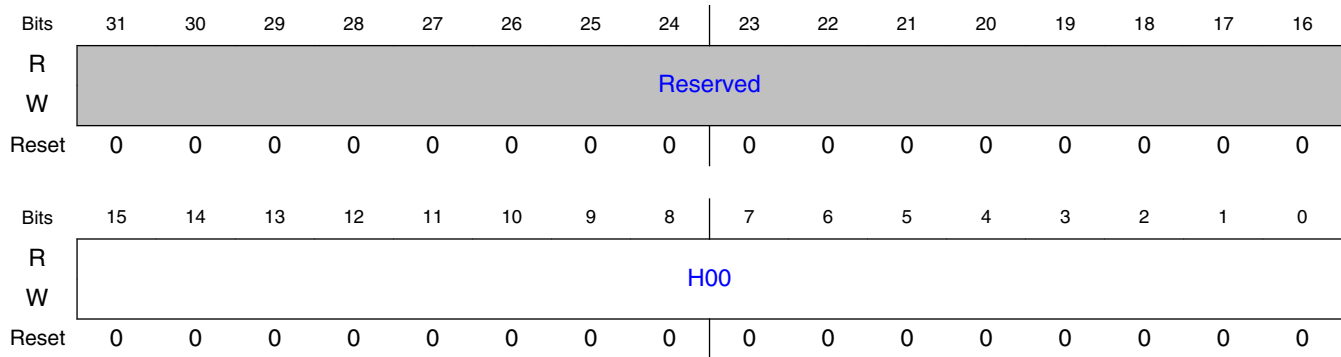
15.10.3.1.6.1 Offset

Register	Offset
HDR_PIPE1_CSCA_H00	3004h

15.10.3.1.6.2 Function

h(0,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.6.3 Diagram



15.10.3.1.6.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H00	h(0,0) 16 bit signed coefficient

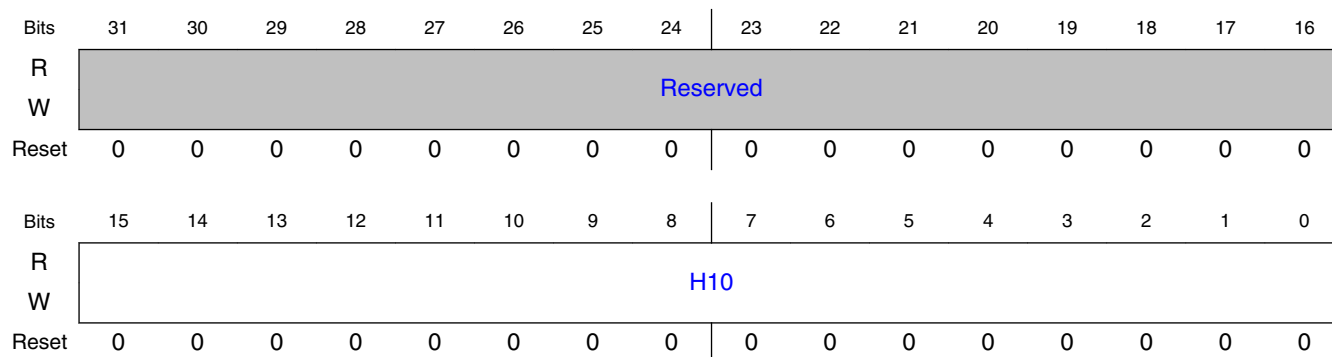
15.10.3.1.7 Pipe1 Colorspace Converter A (CSCA) h(1,0) matrix coefficient (HDR_PIPE1_CSCA_H10)

15.10.3.1.7.1 Offset

Register	Offset
HDR_PIPE1_CSCA_H10	3008h

15.10.3.1.7.2 Function

h(1,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.7.3 Diagram**15.10.3.1.7.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H10	h(1,0) 16 bit signed coefficient

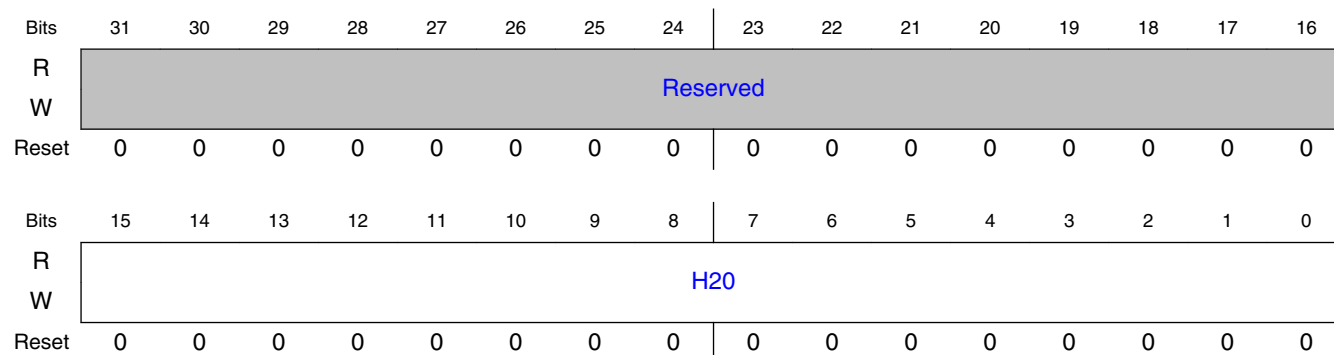
15.10.3.1.8 Pipe1 Colorspace Converter A (CSCA) h(2,0) matrix coefficient (HDR_PIPE1_CSCA_H20)**15.10.3.1.8.1 Offset**

Register	Offset
HDR_PIPE1_CSCA_H20	300Ch

15.10.3.1.8.2 Function

$h(2,0)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.8.3 Diagram



15.10.3.1.8.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H20	$h(2,0)$ 16 bit signed coefficient

15.10.3.1.9 Pipe1 Colorspace Converter A (CSCA) $h(0,1)$ matrix coefficient (HDR_PIPE1_CSCA_H01)

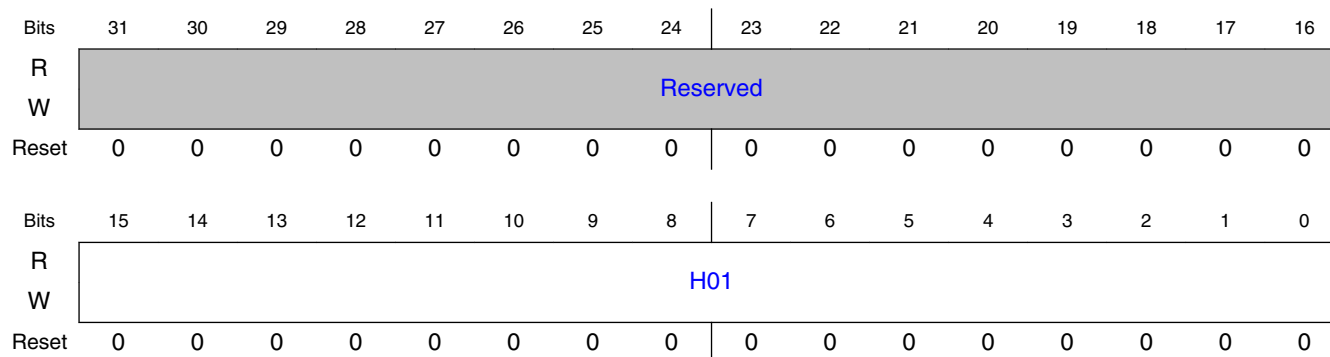
15.10.3.1.9.1 Offset

Register	Offset
HDR_PIPE1_CSCA_H01	3010h

15.10.3.1.9.2 Function

$h(0,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.9.3 Diagram



15.10.3.1.9.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H01	h(0,1) 16 bit signed coefficient

15.10.3.1.10 Pipe1 Colorspace Converter A (CSCA) h(1,1) matrix coefficient (HDR_PIPE1_CSCA_H11)

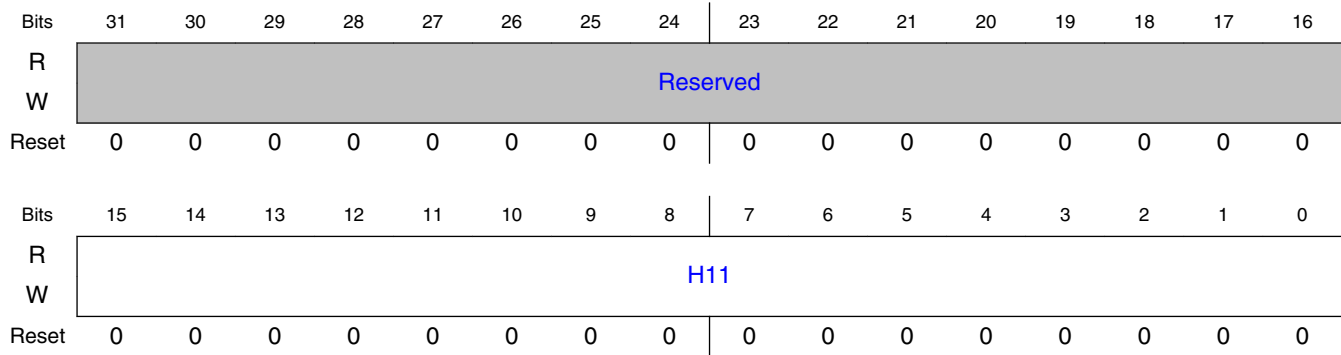
15.10.3.1.10.1 Offset

Register	Offset
HDR_PIPE1_CSCA_H11	3014h

15.10.3.1.10.2 Function

h(1,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.10.3 Diagram



15.10.3.1.10.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H11	h(1,1) 16 bit signed coefficient

15.10.3.1.11 Pipe1 Colorspace Converter A (CSCA) h(2,1) matrix coefficient (HDR_PIPE1_CSCA_H21)

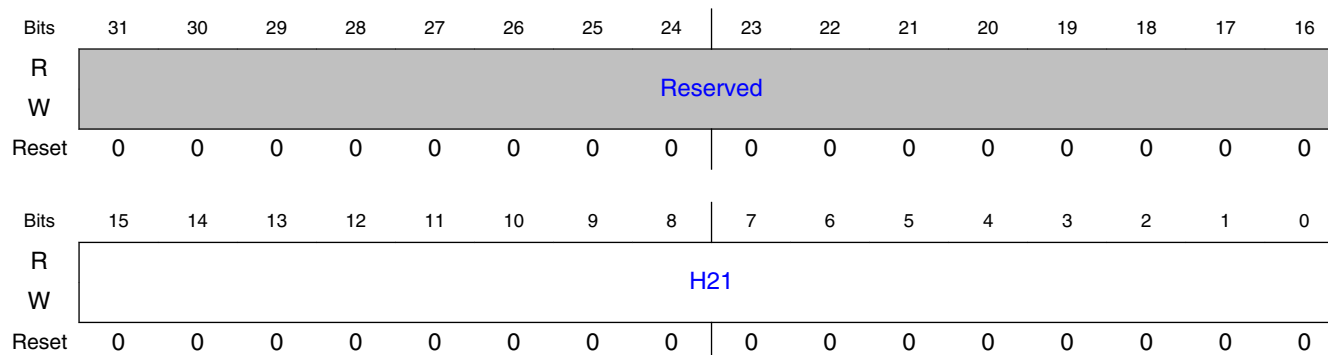
15.10.3.1.11.1 Offset

Register	Offset
HDR_PIPE1_CSCA_H21	3018h

15.10.3.1.11.2 Function

h(2,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.11.3 Diagram



15.10.3.1.11.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H21	h(2,1) 16 bit signed coefficient

15.10.3.1.12 Pipe1 Colorspace Converter A (CSCA) h(0,2) matrix coefficient (HDR_PIPE1_CSCA_H02)

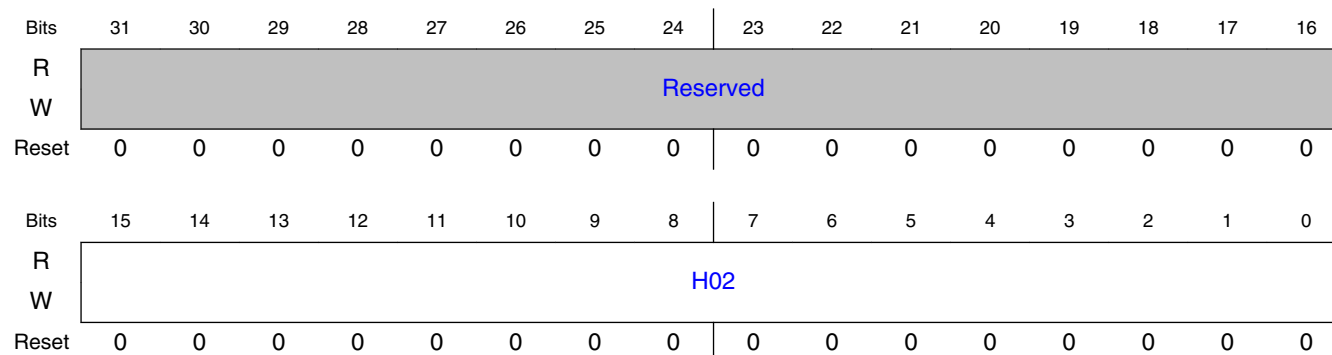
15.10.3.1.12.1 Offset

Register	Offset
HDR_PIPE1_CSCA_H02	301Ch

15.10.3.1.12.2 Function

h(0,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.12.3 Diagram



15.10.3.1.12.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H02	h(0,2) 16 bit signed coefficient

15.10.3.1.13 Pipe1 Colorspace Converter A (CSCA) h(1,2) matrix coefficient (HDR_PIPE1_CSCA_H12)

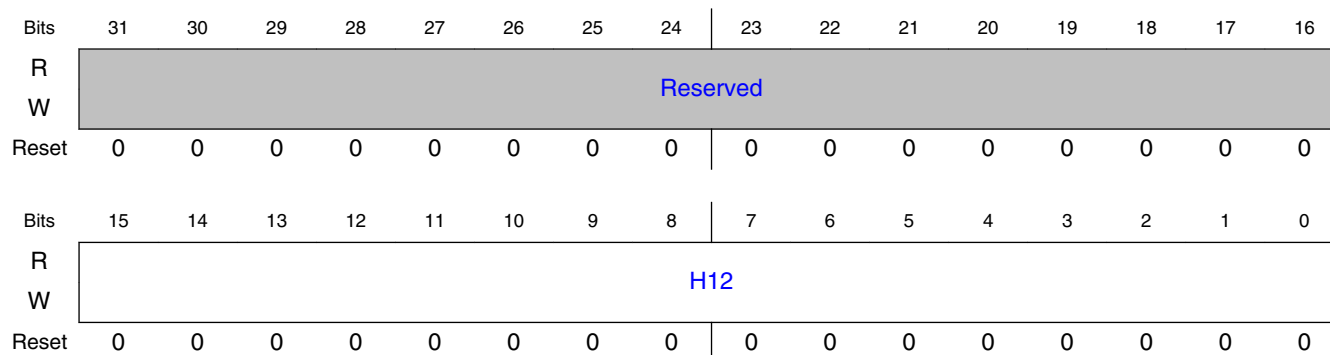
15.10.3.1.13.1 Offset

Register	Offset
HDR_PIPE1_CSCA_H12	3020h

15.10.3.1.13.2 Function

h(1,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.13.3 Diagram



15.10.3.1.13.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H12	h(1,2) 16 bit signed coefficient

15.10.3.1.14 Pipe1 Colorspace Converter A (CSCA) h(2,2) matrix coefficient (HDR_PIPE1_CSCA_H22)

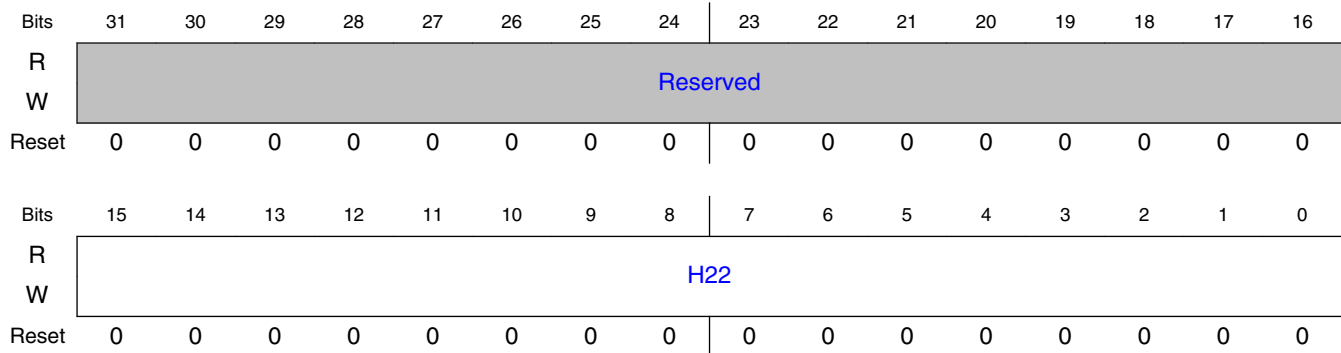
15.10.3.1.14.1 Offset

Register	Offset
HDR_PIPE1_CSCA_H22	3024h

15.10.3.1.14.2 Function

h(2,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.14.3 Diagram



15.10.3.1.14.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H22	h(2,2) 16 bit signed coefficient

15.10.3.1.15 Pipe1 Colorspace Converter A (CSCA) component 0 pre-offset (HDR_PIPE1_CSCA_IO_0)

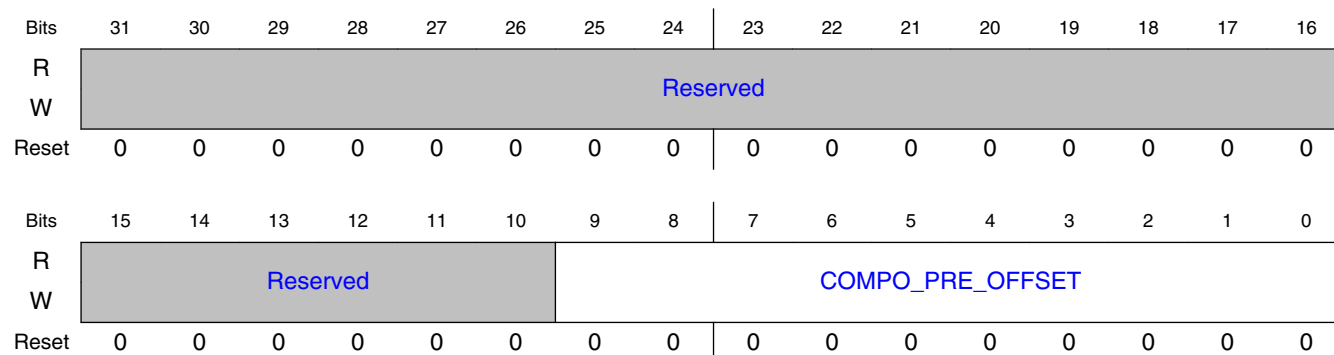
15.10.3.1.15.1 Offset

Register	Offset
HDR_PIPE1_CSCA_IO_0	3028h

15.10.3.1.15.2 Function

An signed 10-bit offset is added to component 0 (R,Y) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.15.3 Diagram



15.10.3.1.15.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMPO_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 0 of the pixel

15.10.3.1.16 Pipe1 Colorspace Converter A (CSCA) component 1 pre-offset (HDR_PIPE1_CSCA_IO_1)

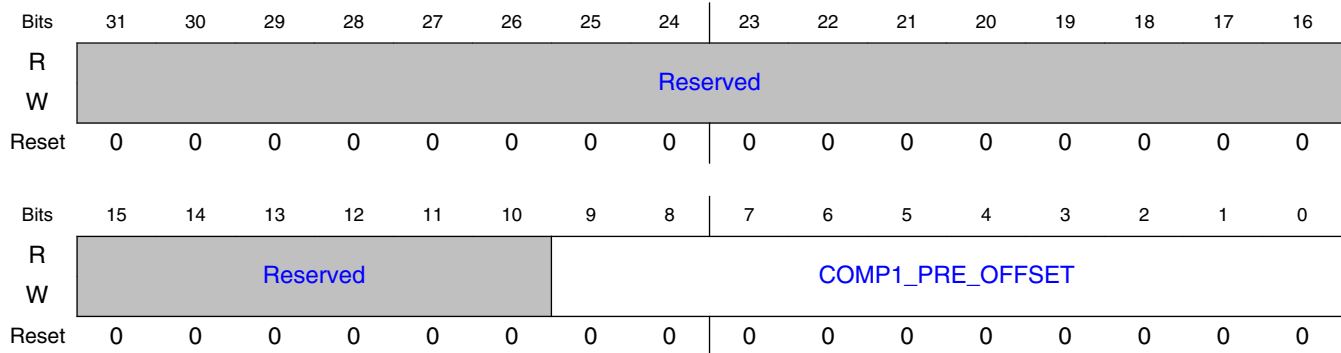
15.10.3.1.16.1 Offset

Register	Offset
HDR_PIPE1_CSCA_IO_1	302Ch

15.10.3.1.16.2 Function

An signed 10-bit offset is added to component 1 (G,Cb) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.16.3 Diagram



15.10.3.1.16.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP1_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 1 of the pixel

15.10.3.1.17 Pipe1 Colorspace Converter A (CSCA) component 2 pre-offset (HDR_PIPE1_CSCA_IO_2)

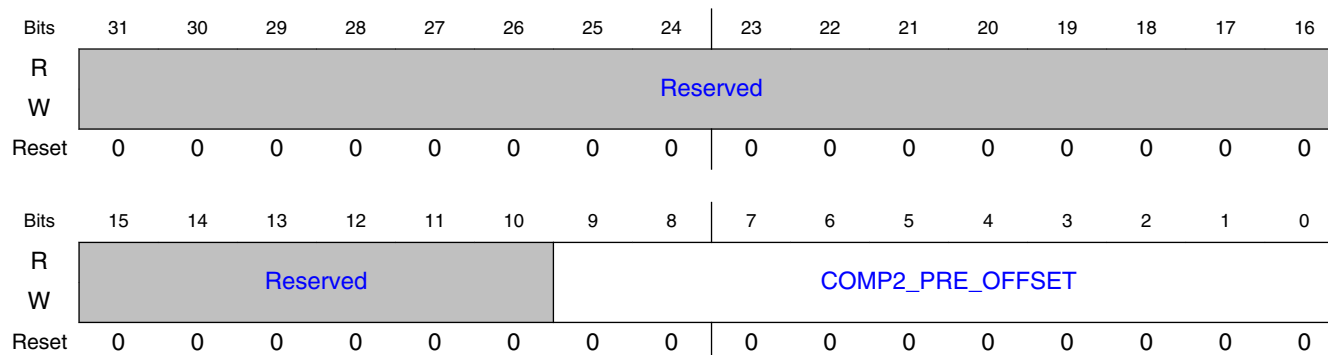
15.10.3.1.17.1 Offset

Register	Offset
HDR_PIPE1_CSCA_IO_2	3030h

15.10.3.1.17.2 Function

An signed 10-bit offset is added to component 2 (B,Cr) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.17.3 Diagram



15.10.3.1.17.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 2 of the pixel

15.10.3.1.18 Pipe1 Colorspace Converter A (CSCA) component 0 clip min. (HDR_PIPE1_CSCA_IO_MIN_0)

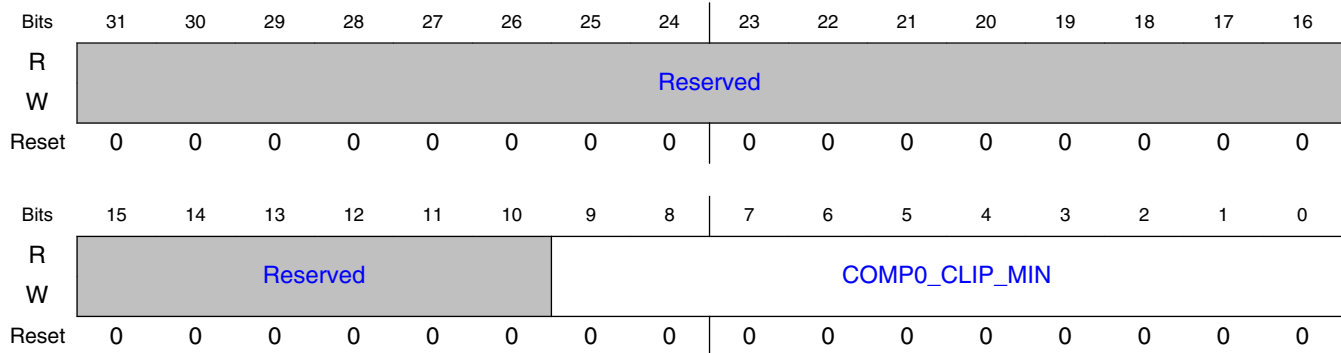
15.10.3.1.18.1 Offset

Register	Offset
HDR_PIPE1_CSCA_IO_MIN_0	3034h

15.10.3.1.18.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.18.3 Diagram



15.10.3.1.18.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP0_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

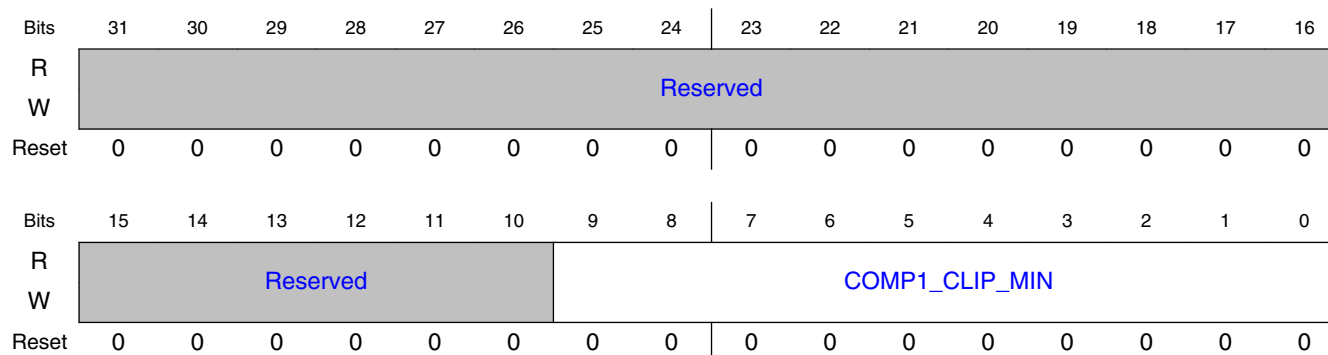
15.10.3.1.19 Pipe1 Colorspace Converter A (CSCA) component 1 clip min. (HDR_PIPE1_CSCA_IO_MIN_1)

15.10.3.1.19.1 Offset

Register	Offset
HDR_PIPE1_CSCA_IO_MIN_1	3038h

15.10.3.1.19.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.19.3 Diagram**15.10.3.1.19.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP1_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

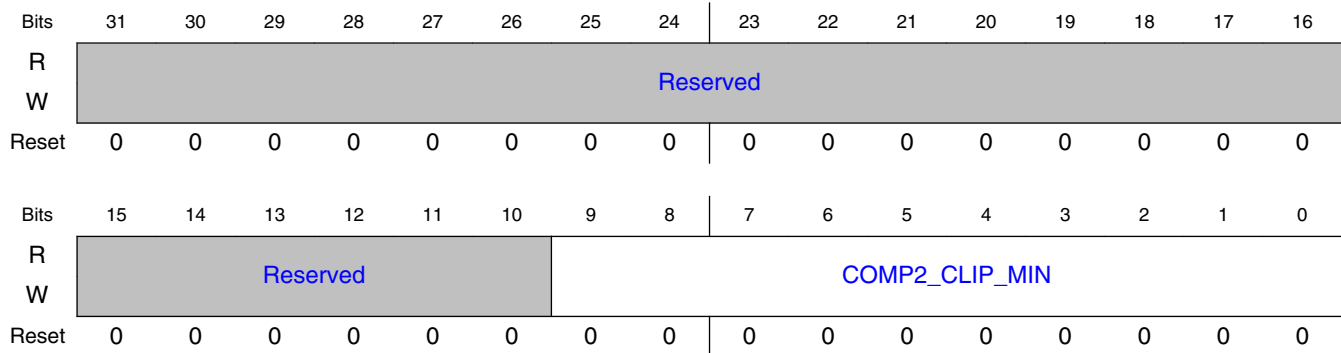
15.10.3.1.20 Pipe1 Colorspace Converter A (CSCA) component 2 clip min. (HDR_PIPE1_CSCA_IO_MIN_2)**15.10.3.1.20.1 Offset**

Register	Offset
HDR_PIPE1_CSCA_IO_MIN_2	303Ch

15.10.3.1.20.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.20.3 Diagram



15.10.3.1.20.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.21 Pipe1 Colorspace Converter A (CSCA) component 0 clip max value. (HDR_PIPE1_CSCA_IO_MAX_0)

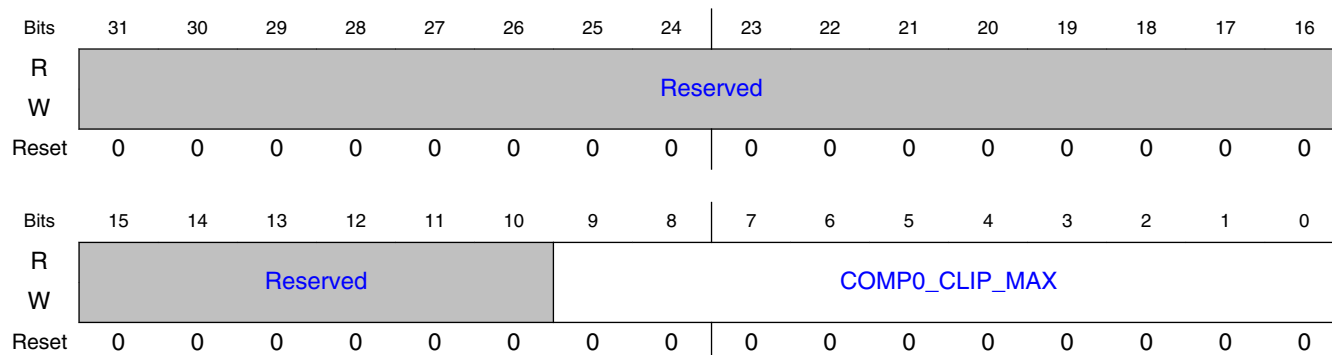
15.10.3.1.21.1 Offset

Register	Offset
HDR_PIPE1_CSCA_IO_MAX_0	3040h

15.10.3.1.21.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.21.3 Diagram



15.10.3.1.21.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP0_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.22 Pipe1 Colorspace Converter A (CSCA) component 1 clip max value. (HDR_PIPE1_CSCA_IO_MAX_1)

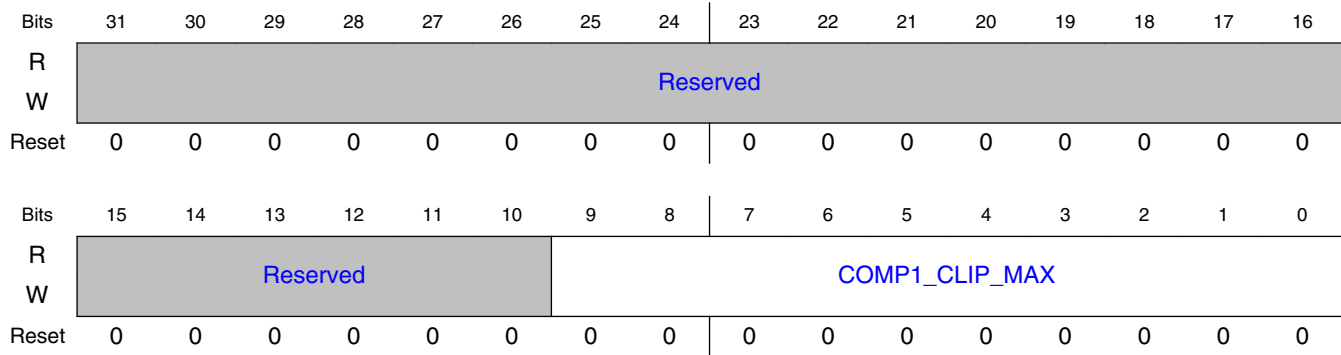
15.10.3.1.22.1 Offset

Register	Offset
HDR_PIPE1_CSCA_IO_MAX_1	3044h

15.10.3.1.22.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.22.3 Diagram



15.10.3.1.22.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP1_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.23 Pipe1 Colorspace Converter A (CSCA) component 2 clip max value. (HDR_PIPE1_CSCA_IO_MAX_2)

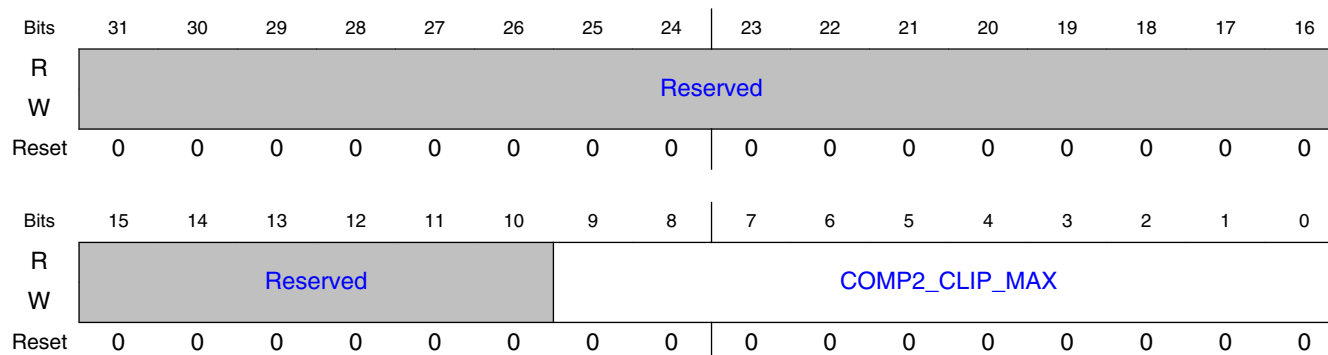
15.10.3.1.23.1 Offset

Register	Offset
HDR_PIPE1_CSCA_IO_MAX_2	3048h

15.10.3.1.23.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.23.3 Diagram



15.10.3.1.23.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.24 Pipe1 Colorspace Converter A (CSCA) normalization factor (HDR_PIPE1_CSCA_NORM)

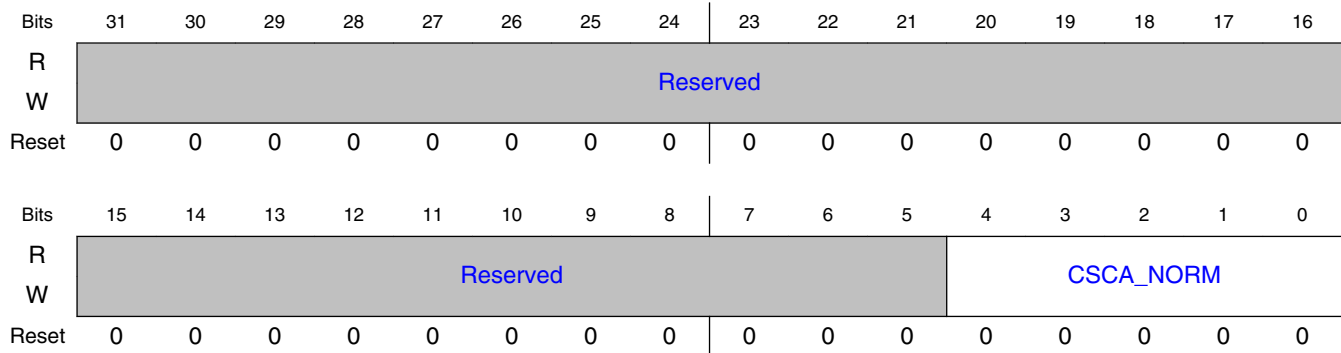
15.10.3.1.24.1 Offset

Register	Offset
HDR_PIPE1_CSCA_NORM	304Ch

15.10.3.1.24.2 Function

After the CSC matrix multiply, all components of the result are shifted right by the amount in this register. This is a signed right shift. Pipe1 is also called channel1.

15.10.3.1.24.3 Diagram



15.10.3.1.24.4 Fields

Field	Function
31-5 —	Reserved.
4-0 CSCA_NORM	This 5-bit unsigned value is size if arithmetic shift after matrix multiply.

15.10.3.1.25 Pipe1 Colorspace Converter A (CSCA): Post offset component 0 (HDR_PIPE1_CSCA_OO_0)

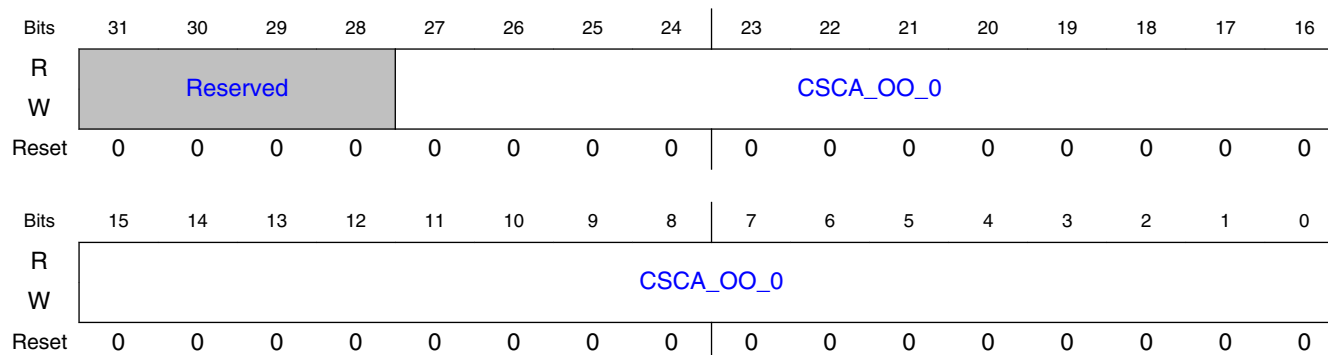
15.10.3.1.25.1 Offset

Register	Offset
HDR_PIPE1_CSCA_OO_0	3050h

15.10.3.1.25.2 Function

After the CSC normalization, this offset value is added to component 0. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.25.3 Diagram



15.10.3.1.25.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_0	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.26 Pipe1 Colorspace Converter A (CSCA): Post offset component 1 (HDR_PIPE1_CSCA_OO_1)

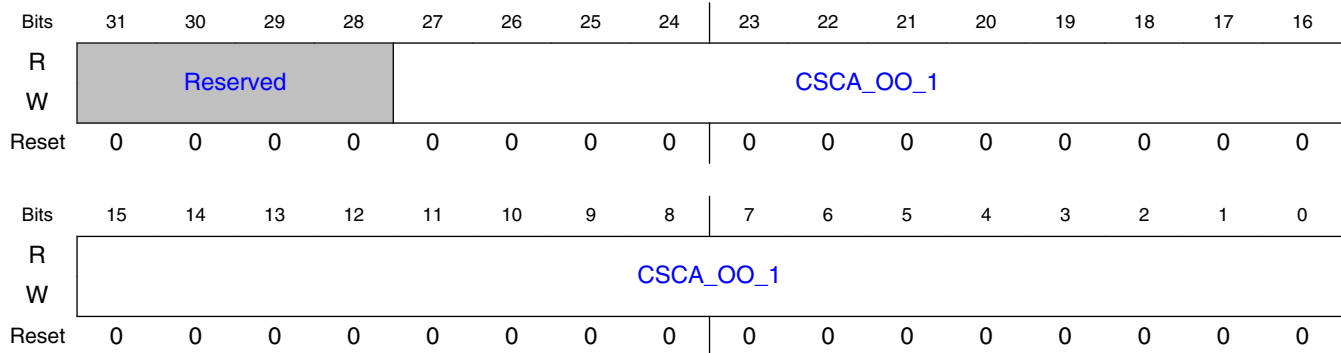
15.10.3.1.26.1 Offset

Register	Offset
HDR_PIPE1_CSCA_OO_1	3054h

15.10.3.1.26.2 Function

After the CSC normalization, this offset value is added to component 1. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.26.3 Diagram



15.10.3.1.26.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_1	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.27 Pipe1 Colorspace Converter A (CSCA): Post offset component 2 (HDR_PIPE1_CSCA_OO_2)

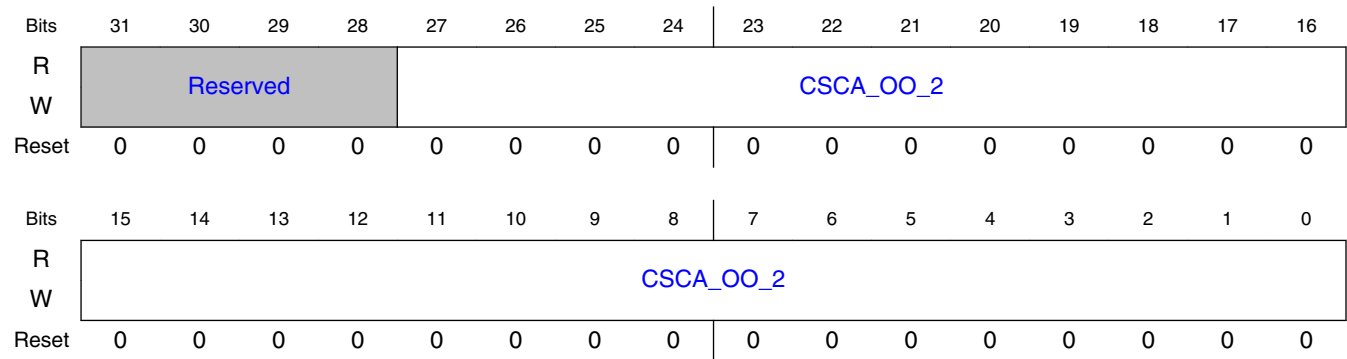
15.10.3.1.27.1 Offset

Register	Offset
HDR_PIPE1_CSCA_OO_2	3058h

15.10.3.1.27.2 Function

After the CSC normalization, this offset value is added to component 2. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.27.3 Diagram



15.10.3.1.27.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_2	Ouput Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.28 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 0 (HDR_PIPE1_CSCA_OMIN_0)

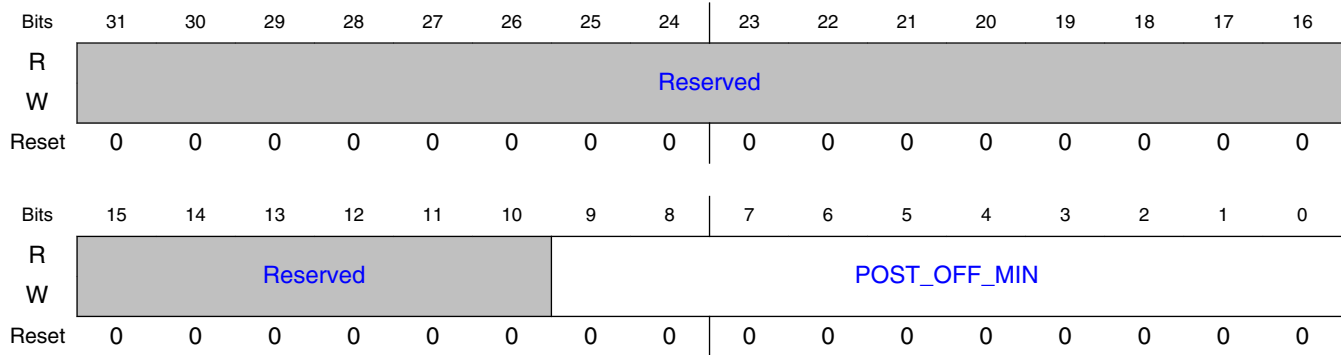
15.10.3.1.28.1 Offset

Register	Offset
HDR_PIPE1_CSCA_OMIN_0	305Ch

15.10.3.1.28.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.28.3 Diagram



15.10.3.1.28.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.29 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 1 (HDR_PIPE1_CSCA_OMIN_1)

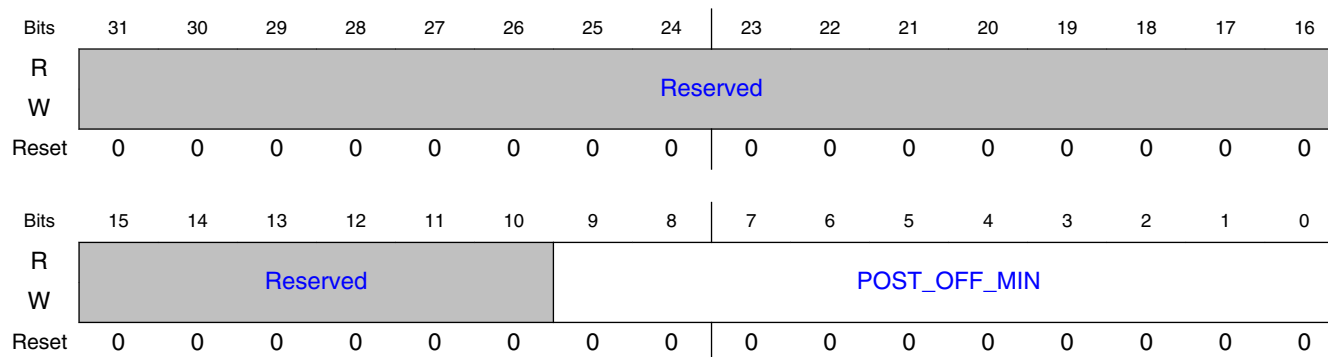
15.10.3.1.29.1 Offset

Register	Offset
HDR_PIPE1_CSCA_OMIN_1	3060h

15.10.3.1.29.2 Function

After the post offset is added this component is clipped. This is the minimum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.29.3 Diagram



15.10.3.1.29.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.30 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 2 (HDR_PIPE1_CSCA_OMIN_2)

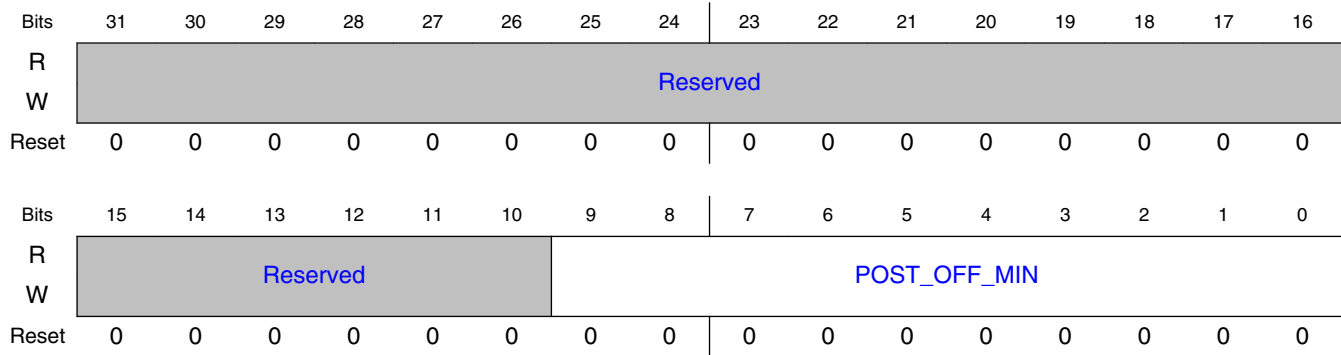
15.10.3.1.30.1 Offset

Register	Offset
HDR_PIPE1_CSCA_OMIN_2	3064h

15.10.3.1.30.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.30.3 Diagram



15.10.3.1.30.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

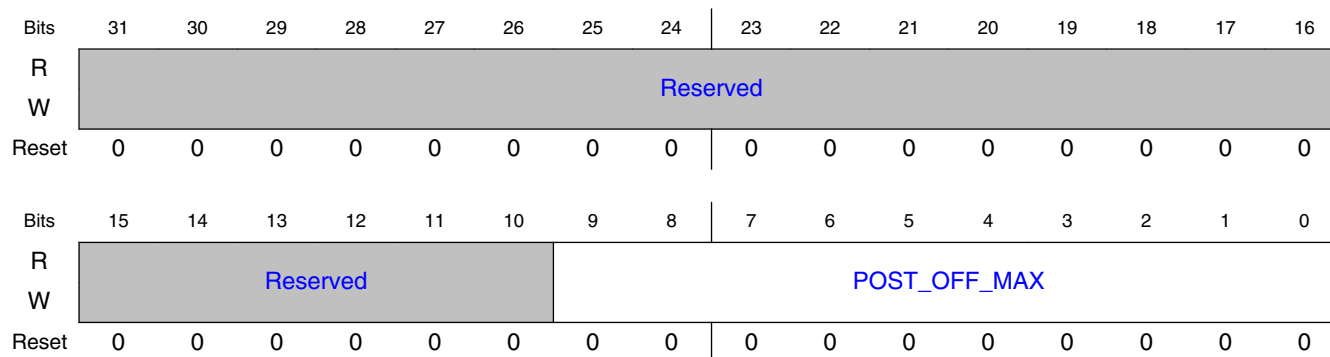
15.10.3.1.31 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 0 (HDR_PIPE1_CSCA_OMAX_0)

15.10.3.1.31.1 Offset

Register	Offset
HDR_PIPE1_CSCA_OMAX_0	3068h

15.10.3.1.31.2 Function

After the post offset is added thie component is clipped. This is the maximum clip value This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.31.3 Diagram**15.10.3.1.31.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

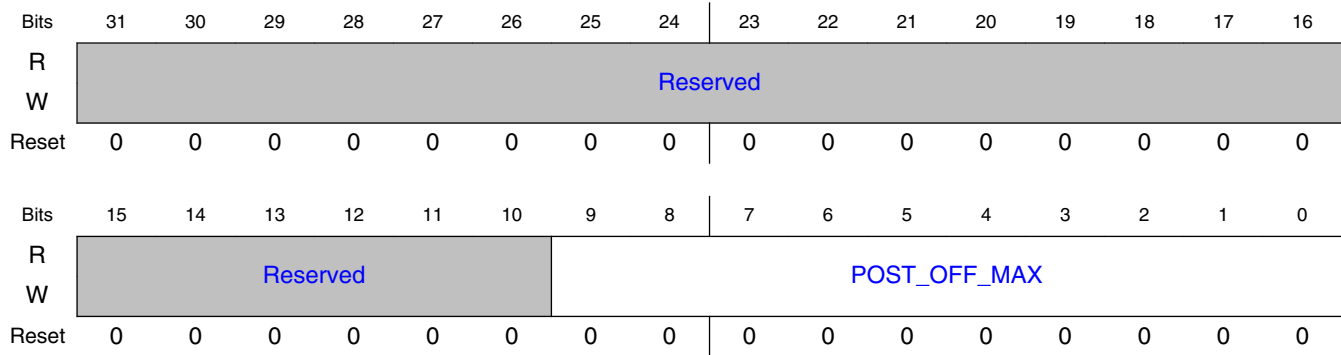
15.10.3.1.32 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 1 (HDR_PIPE1_CSCA_OMAX_1)**15.10.3.1.32.1 Offset**

Register	Offset
HDR_PIPE1_CSCA_OMAX_1	306Ch

15.10.3.1.32.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.32.3 Diagram



15.10.3.1.32.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

15.10.3.1.33 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 2 (HDR_PIPE1_CSCA_OMAX_2)

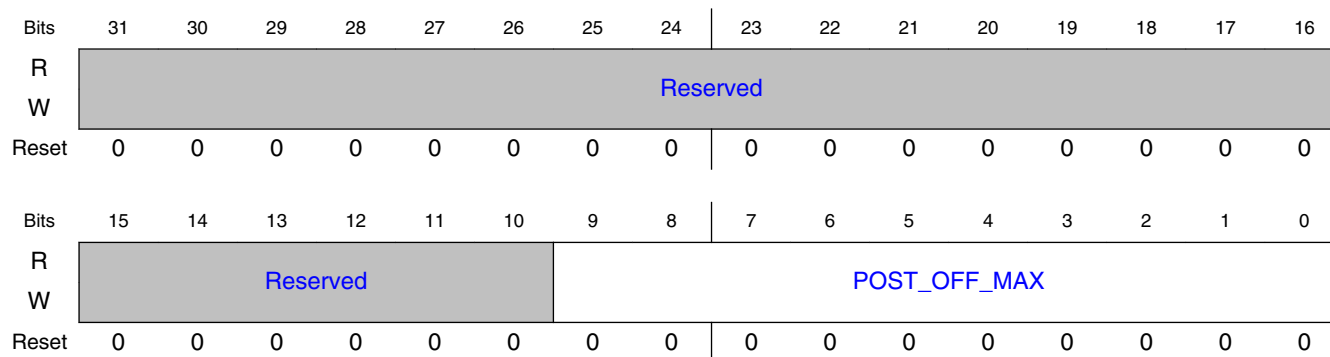
15.10.3.1.33.1 Offset

Register	Offset
HDR_PIPE1_CSCA_OMAX_2	3070h

15.10.3.1.33.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.33.3 Diagram



15.10.3.1.33.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

15.10.3.1.34 PIPE1: NOT USED (HDR_PIPE1_ENTRY_29)

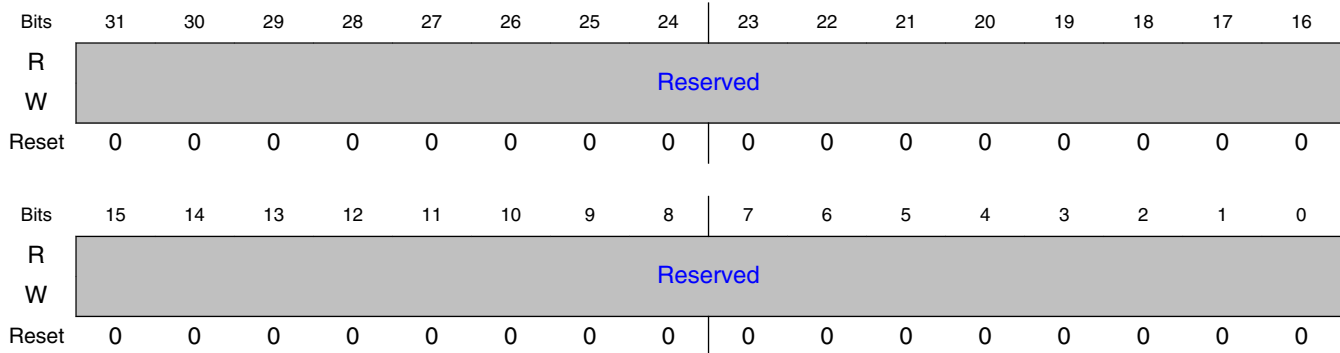
15.10.3.1.34.1 Offset

Register	Offset
HDR_PIPE1_ENTRY_29	3074h

15.10.3.1.34.2 Function

PIPE1: NOT USED

15.10.3.1.34.3 Diagram



15.10.3.1.34.4 Fields

Field	Function
31-0	Reserved.
—	

15.10.3.1.35 Pipe1 LUT control register (HDR_PIPE1_LUT_CONTROL_REG)

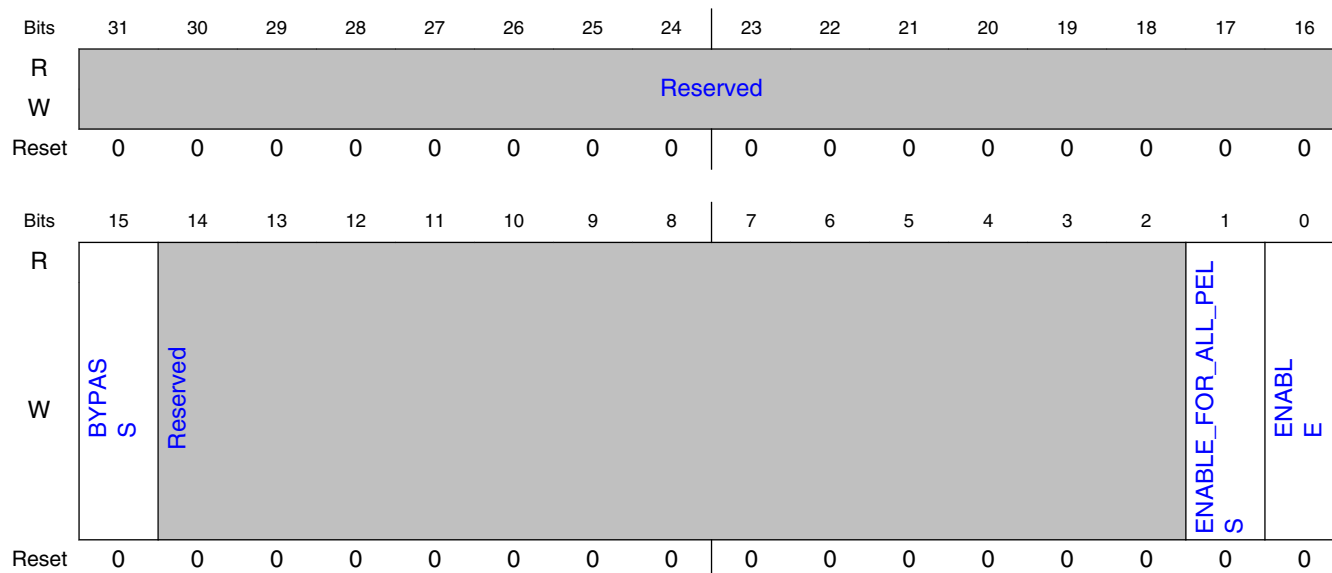
15.10.3.1.35.1 Offset

Register	Offset
HDR_PIPE1_LUT_CONTROL_REG	3080h

15.10.3.1.35.2 Function

Controls the Look-Up-Table in pipe1. Control of LUT is PER PIXEL not per component. LUT has 1024 14-bit, per component, entries. LUT maybe used for conversion from non-linear (gamma corrected) to linear pixels. LUT maybe used for conversion from linear to non linear pixels (gamma corrected). LUT can contain 14-bit fixed point or 14-bit floating point pixel value. Floating point format has 9 bit unsigned mantissa with hidden bit and 5 bit, unsigned, biased exponent. {exp[4:0], mantissa[8:0]} => 1.mantissa[8:0] * 2^(exp-1) When using floating piont numbers, in this LUT, these are used in HDR OUTPUT PIPE for HDR non-linear operation and must bypass Color Space Converter B (CSCB). Pipe1 is also called channel1.

15.10.3.1.35.3 Diagram



15.10.3.1.35.4 Fields

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this LUT 1: Pixels pass thru this LUT unmodified
14-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This LUT is enabled only for blended pixels 1: This LUT is enabled all pixels
0 ENABLE	0: Don't enable this LUT: Pixels pass thru the LUT unmodified 1: This LUT is enabled for current picture

15.10.3.1.36 Pipe1 Colorspace Converter B control. (HDR_PIPE1_CSCB_CONTROL_REG)

15.10.3.1.36.1 Offset

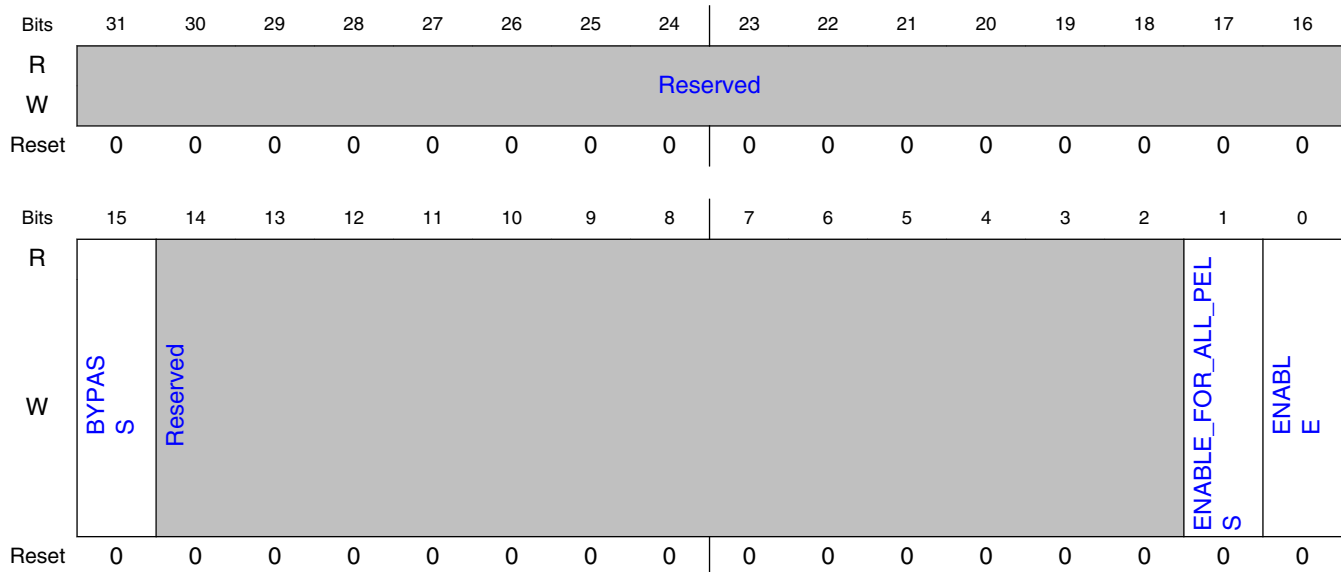
Register	Offset
HDR_PIPE1_CSCB_CONTROL_REG	3800h

15.10.3.1.36.2 Function

Controls the Color-Space-Converter-B (CSCB) in pipe1. This CSCB takes in 14-bit pixel components (component 0, component 1, component 2). A per-component offset (called pre-offset) is added to the pixel. After the pre-offset operation the pixel components are clipped. MAX_RANGE of PRE_CLIP is [-512, 1023]. A 3x3 matrix multiply (constant coefficients H(x,y)) is performed on the pixel to take to a different color space. After this matrix multiply, a 28 bit signed offset (post-offset) is added to the normalized result of the matrix multiply. After the post-offset operation the results are clipped.

MAX_RANGE of POST_CLIP is [0,1023] Output of this CSCA are 28 bit per component pixels. After CSCB the pixels are fed into the Blender. Pipe1 is also called channel1.

15.10.3.1.36.3 Diagram



15.10.3.1.36.4 Fields

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this CSC 1: Pixels pass thru this CSC unmodified
14-2 —	Reserved.

Table continues on the next page...

Memory Map and Registers

Field	Function
1 ENABLE_FOR_ ALL_PELS	0: This CSC is enabled only for blended pixels 1: This CSC is enabled all pixels
0 ENABLE	0: Don't enable this CSC: Pixels pass thru the CSC unmodified 1: This CSC is enabled for current picture

15.10.3.1.37 Pipe1 Colorspace Converter A (CSCB) h(0,0) matrix coefficient (HDR_PIPE1_CSCB_H00)

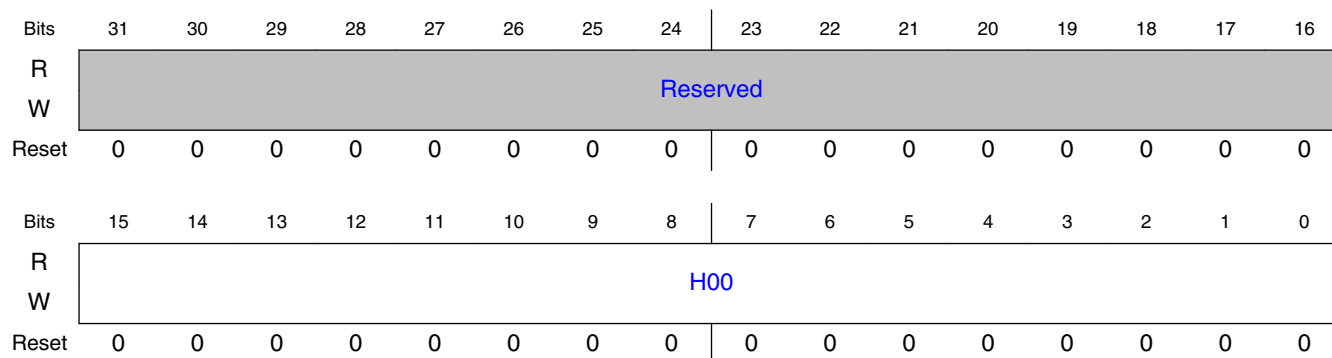
15.10.3.1.37.1 Offset

Register	Offset
HDR_PIPE1_CSCB_H00	3804h

15.10.3.1.37.2 Function

h(0,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.37.3 Diagram



15.10.3.1.37.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H00	h(0,0) 16 bit signed coefficient

15.10.3.1.38 Pipe1 Colorspace Converter B (CSCB) h(1,0) matrix coefficient (HDR_PIPE1_CSCB_H10)

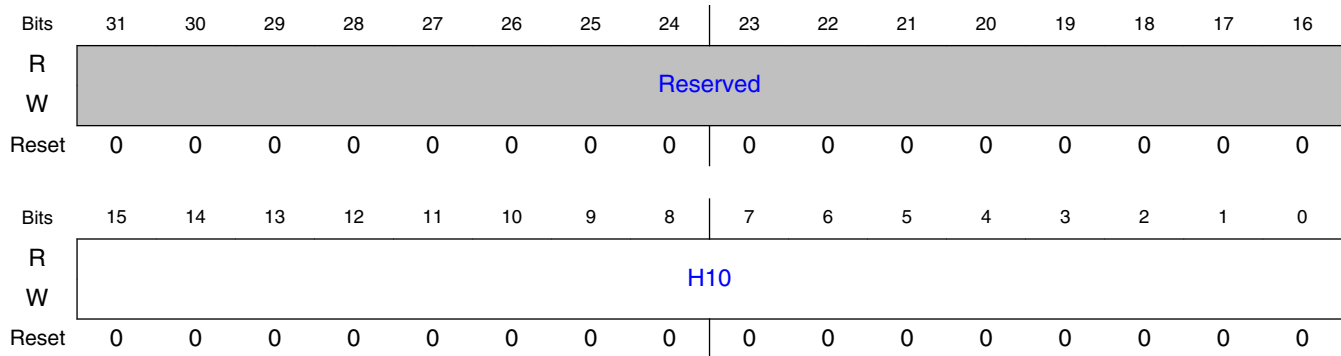
15.10.3.1.38.1 Offset

Register	Offset
HDR_PIPE1_CSCB_H10	3808h

15.10.3.1.38.2 Function

h(1,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.38.3 Diagram



15.10.3.1.38.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H10	h(1,0) 16 bit signed coefficient

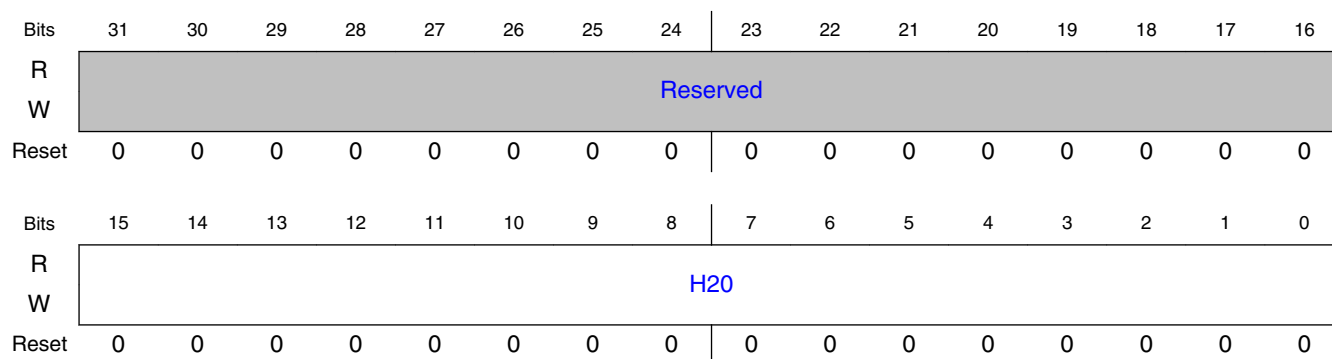
15.10.3.1.39 Pipe1 Colorspace Converter B (CSCB) h(2,0) matrix coefficient (HDR_PIPE1_CSCB_H20)

15.10.3.1.39.1 Offset

Register	Offset
HDR_PIPE1_CSCB_H20	380Ch

15.10.3.1.39.2 Function

h(2,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.39.3 Diagram**15.10.3.1.39.4 Fields**

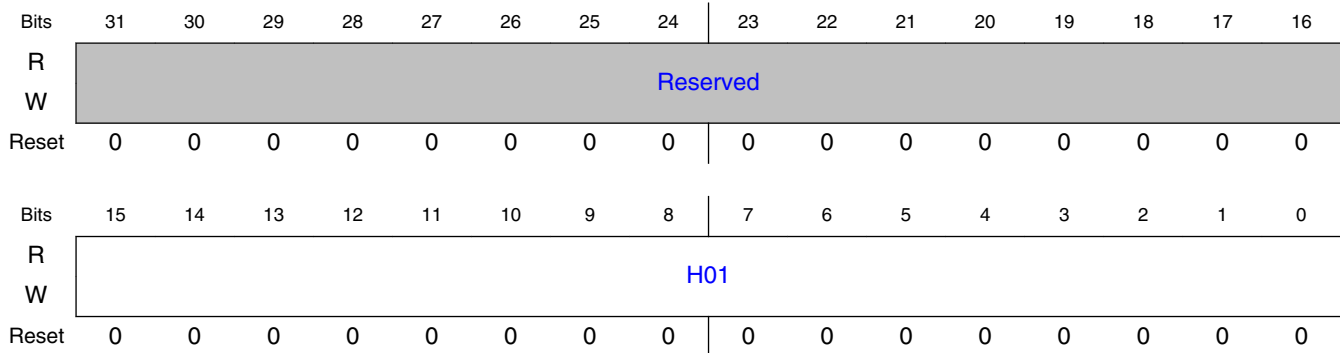
Field	Function
31-16 —	Reserved.
15-0 H20	h(2,0) 16 bit signed coefficient

15.10.3.1.40 Pipe1 Colorspace Converter B (CSCB) h(0,1) matrix coefficient (HDR_PIPE1_CSCB_H01)**15.10.3.1.40.1 Offset**

Register	Offset
HDR_PIPE1_CSCB_H01	3810h

15.10.3.1.40.2 Function

$h(0,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.40.3 Diagram**15.10.3.1.40.4 Fields**

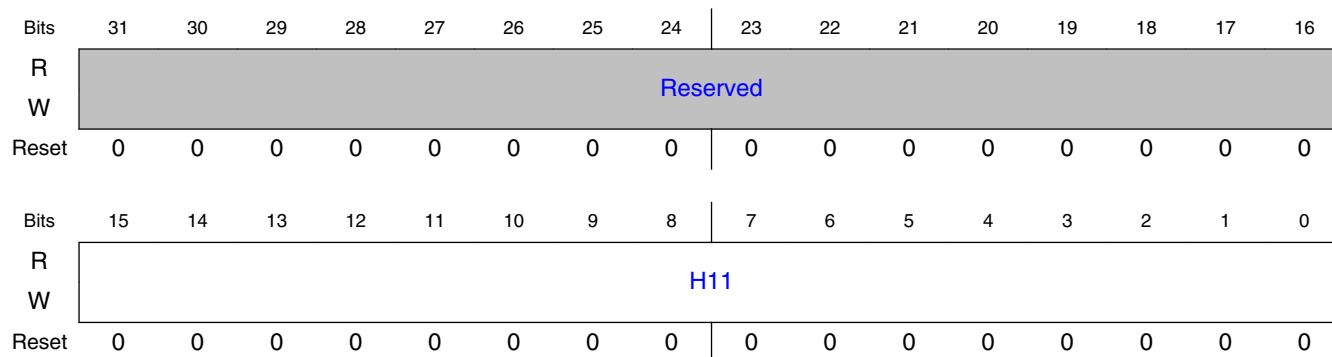
Field	Function
31-16 —	Reserved.
15-0 H01	$h(0,1)$ 16 bit signed coefficient

15.10.3.1.41 Pipe1 Colorspace Converter B (CSCB) $h(1,1)$ matrix coefficient (HDR_PIPE1_CSCB_H11)**15.10.3.1.41.1 Offset**

Register	Offset
HDR_PIPE1_CSCB_H11	3814h

15.10.3.1.41.2 Function

$h(1,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.41.3 Diagram**15.10.3.1.41.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H11	h(1,1) 16 bit signed coefficient

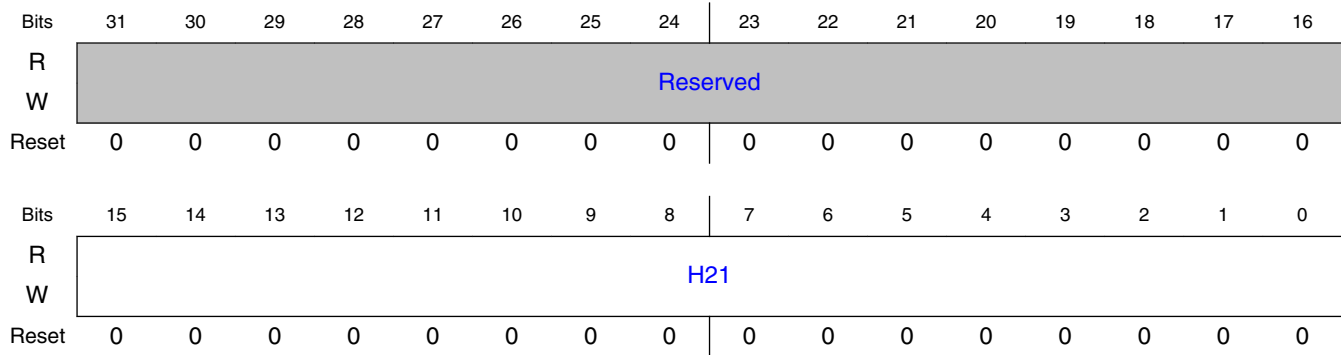
15.10.3.1.42 Pipe1 Colorspace Converter B (CSCB) h(2,1) matrix coefficient (HDR_PIPE1_CSCB_H21)**15.10.3.1.42.1 Offset**

Register	Offset
HDR_PIPE1_CSCB_H21	3818h

15.10.3.1.42.2 Function

h(2,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.42.3 Diagram



15.10.3.1.42.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H21	h(2,1) 16 bit signed coefficient

15.10.3.1.43 Pipe1 Colorspace Converter B (CSCB) h(0,2) matrix coefficient (HDR_PIPE1_CSCB_H02)

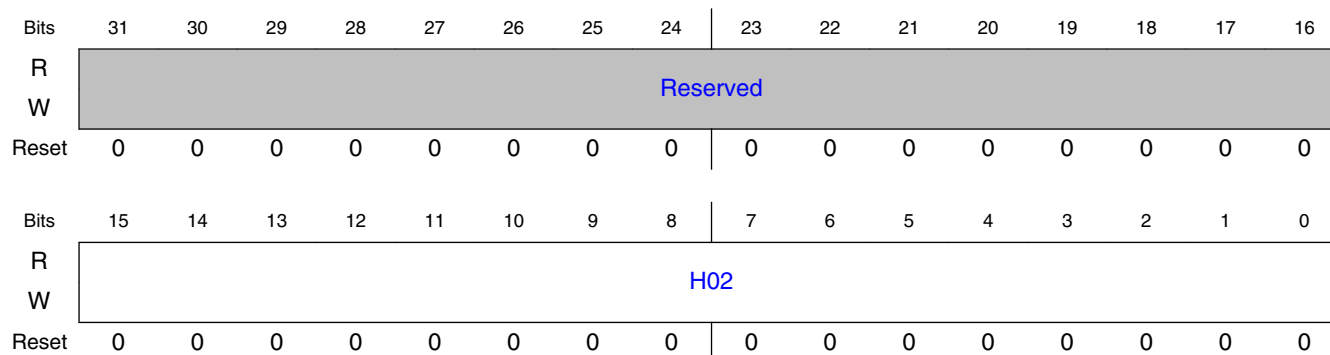
15.10.3.1.43.1 Offset

Register	Offset
HDR_PIPE1_CSCB_H02	381Ch

15.10.3.1.43.2 Function

h(0,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.43.3 Diagram



15.10.3.1.43.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H02	h(0,2) 16 bit signed coefficient

15.10.3.1.44 Pipe1 Colorspace Converter B (CSCB) h(1,2) matrix coefficient (HDR_PIPE1_CSCB_H12)

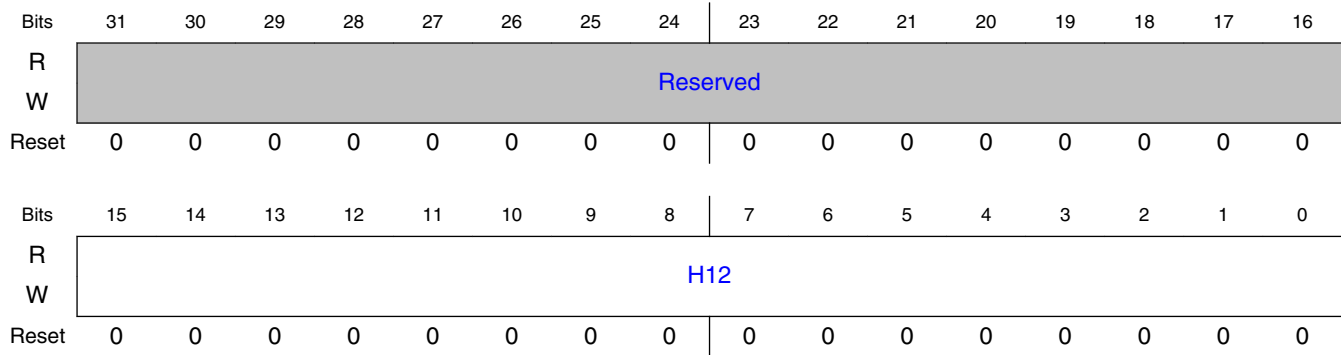
15.10.3.1.44.1 Offset

Register	Offset
HDR_PIPE1_CSCB_H12	3820h

15.10.3.1.44.2 Function

h(1,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.44.3 Diagram



15.10.3.1.44.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H12	h(1,2) 16 bit signed coefficient

15.10.3.1.45 Pipe1 Colorspace Converter B (CSCB) h(2,2) matrix coefficient (HDR_PIPE1_CSCB_H22)

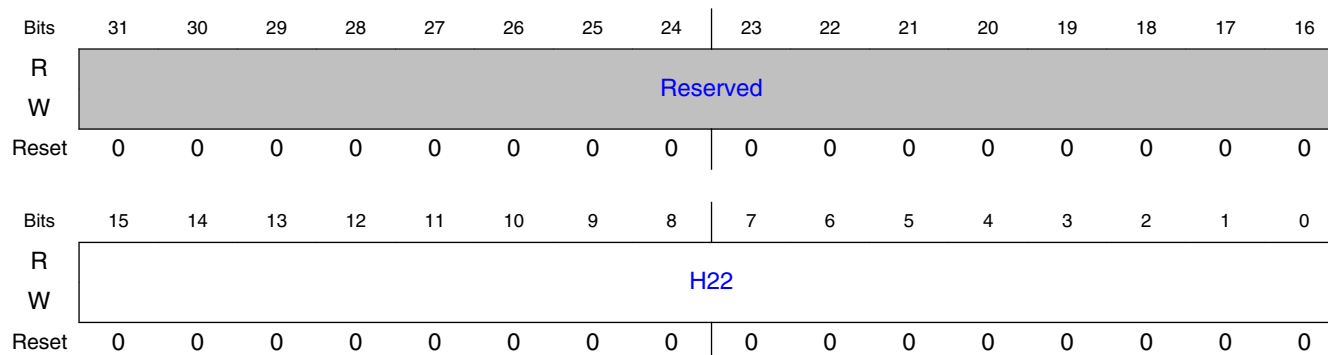
15.10.3.1.45.1 Offset

Register	Offset
HDR_PIPE1_CSCB_H22	3824h

15.10.3.1.45.2 Function

h(2,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.45.3 Diagram



15.10.3.1.45.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H22	h(2,2) 16 bit signed coefficient

15.10.3.1.46 Pipe1 Colorspace Converter B (CSCB) component 0 pre-offset (HDR_PIPE1_CSCB_IO_0)

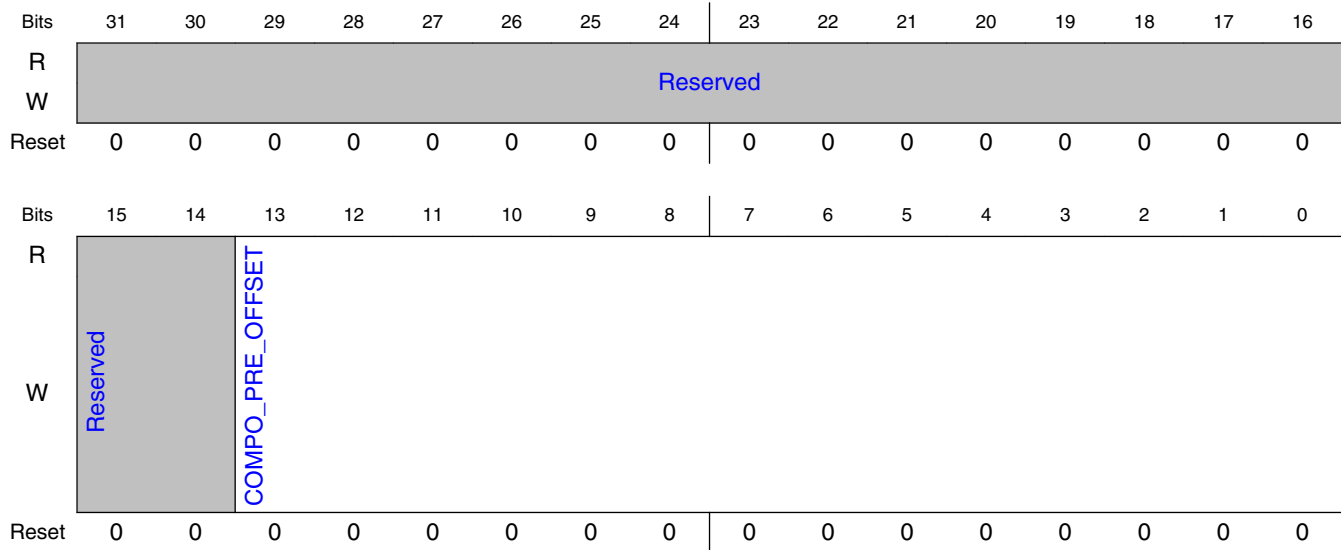
15.10.3.1.46.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_0	3828h

15.10.3.1.46.2 Function

A signed 14-bit offset is added to component 0 (R,Y) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.46.3 Diagram



15.10.3.1.46.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMPO_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 0 of the pixel

15.10.3.1.47 Pipe1 Colorspace Converter B (CSCB) component 1 pre-offset (HDR_PIPE1_CSCB_IO_1)

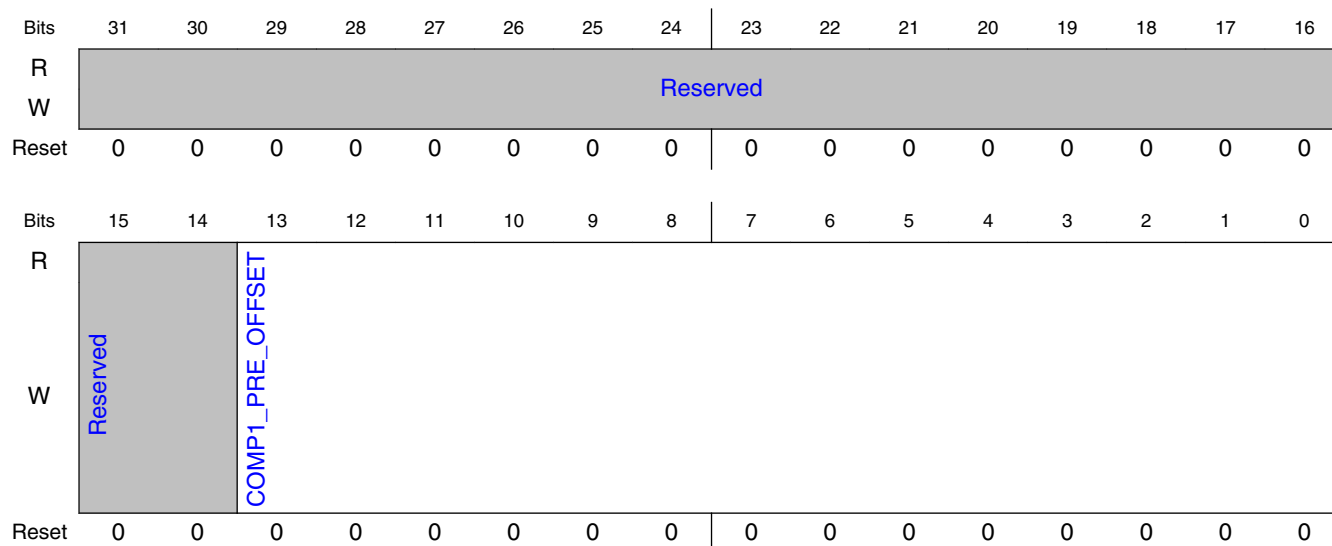
15.10.3.1.47.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_1	382Ch

15.10.3.1.47.2 Function

A signed 14-bit offset is added to component 1 (G,Cb) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.47.3 Diagram



15.10.3.1.47.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 1 of the pixel

15.10.3.1.48 Pipe1 Colorspace Converter B (CSCB) component 2 pre-offset (HDR_PIPE1_CSCB_IO_2)

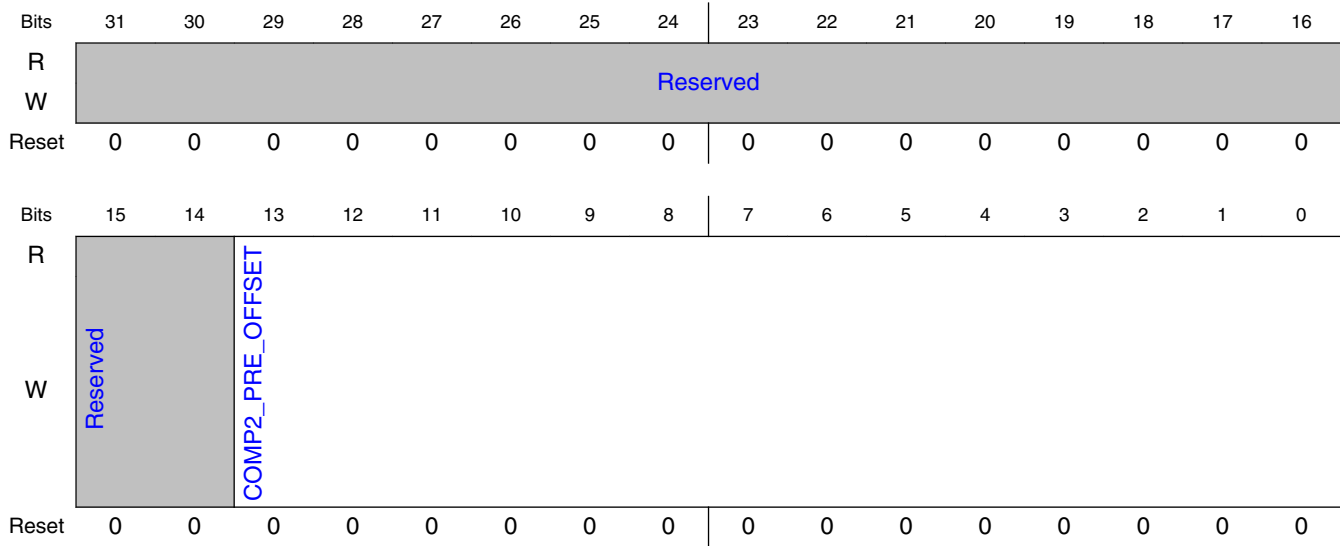
15.10.3.1.48.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_2	3830h

15.10.3.1.48.2 Function

A signed 14-bit offset is added to component 2 (B,Cr) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.48.3 Diagram



15.10.3.1.48.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 2 of the pixel

15.10.3.1.49 Pipe1 Colorspace Converter B (CSCB) component 0 clip min. (HDR_PIPE1_CSCB_IO_MIN_0)

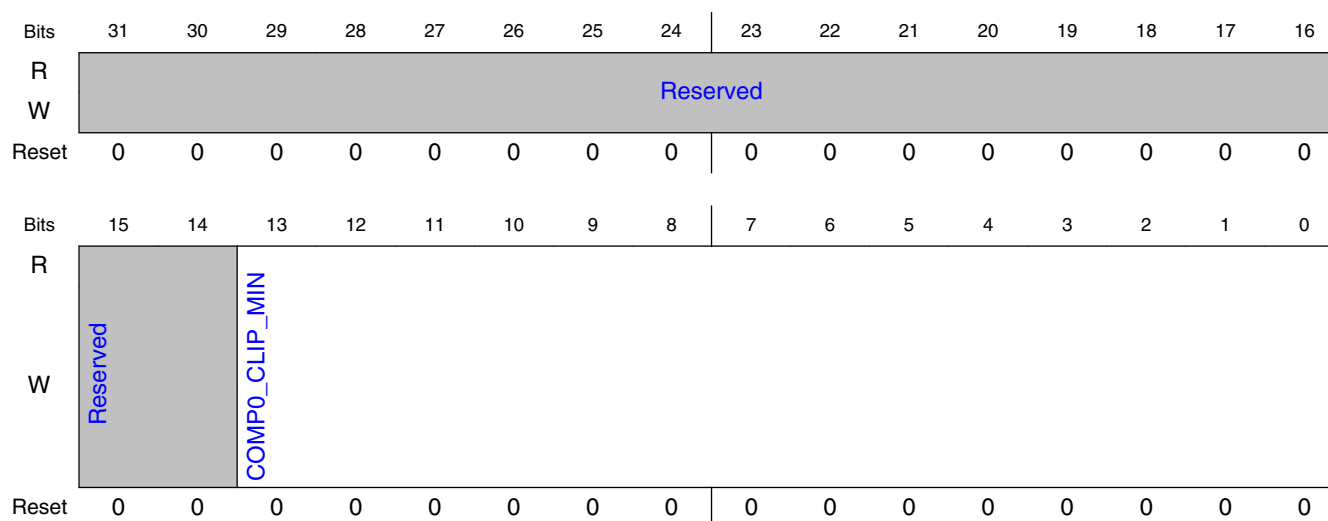
15.10.3.1.49.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_MIN_0	3834h

15.10.3.1.49.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192, 8191]. Pipe1 is also called channel1.

15.10.3.1.49.3 Diagram



15.10.3.1.49.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP0_CLIP_MIN	This 14-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.50 Pipe1 Colorspace Converter B (CSCB) component 1 clip min. (HDR_PIPE1_CSCB_IO_MIN_1)

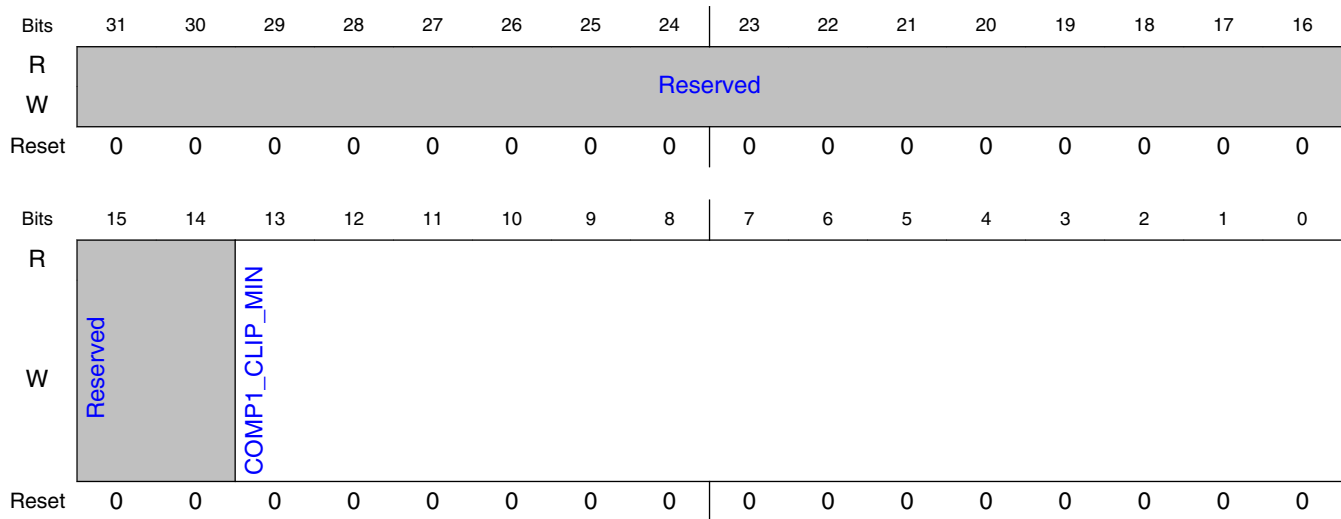
15.10.3.1.50.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_MIN_1	3838h

15.10.3.1.50.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192 , 8191]. Pipe1 is also called channel1.

15.10.3.1.50.3 Diagram



15.10.3.1.50.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_CLIP_MIN	This 14-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.51 Pipe1 Colorspace Converter B (CSCB) component 2 clip min. (HDR_PIPE1_CSCB_IO_MIN_2)

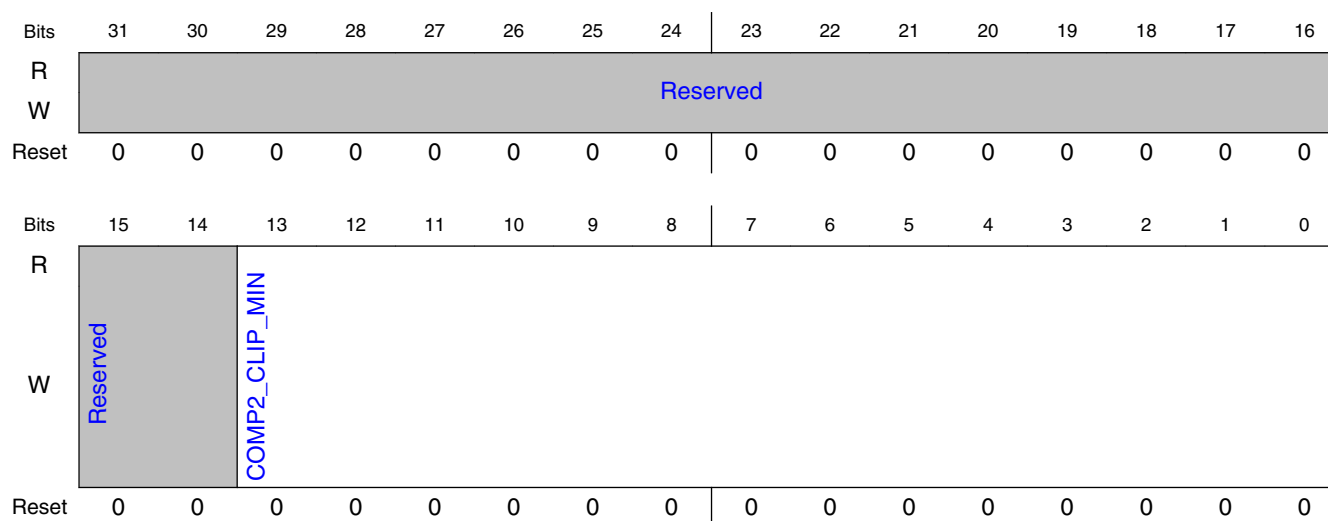
15.10.3.1.51.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_MIN_2	383Ch

15.10.3.1.51.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192 , 1891]. Pipe1 is also called channel1.

15.10.3.1.51.3 Diagram



15.10.3.1.51.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.52 Pipe1 Colorspace Converter B (CSCB) component 0 clip max value. (HDR_PIPE1_CSCB_IO_MAX_0)

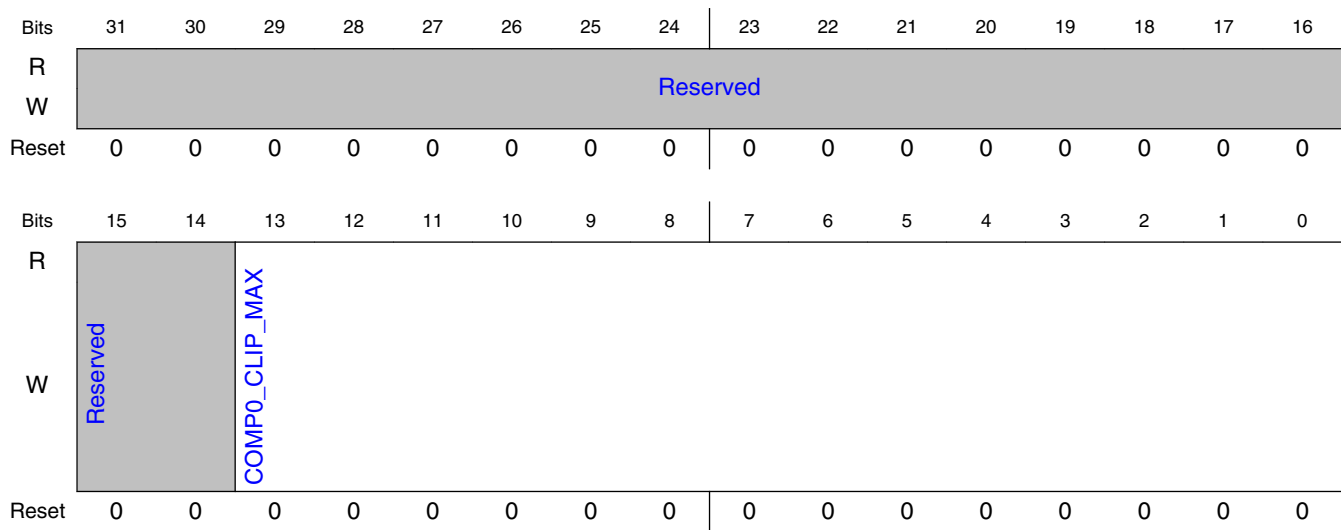
15.10.3.1.52.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_MAX_0	3840h

15.10.3.1.52.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0, 16383]. Pipe1 is also called channel1.

15.10.3.1.52.3 Diagram



15.10.3.1.52.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMPO_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.53 Pipe1 Colorspace Converter B (CSCB) component 1 clip max value. (HDR_PIPE1_CSCB_IO_MAX_1)

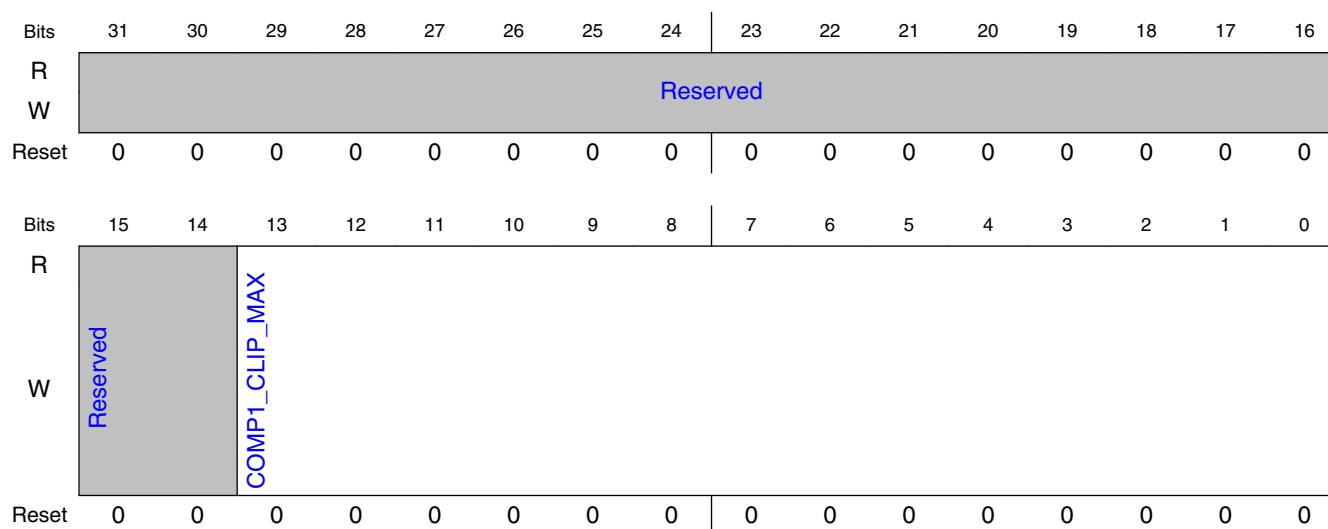
15.10.3.1.53.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_MAX_1	3844h

15.10.3.1.53.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,16383]. Pipe1 is also called channel1.

15.10.3.1.53.3 Diagram



15.10.3.1.53.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.54 Pipe1 Colorspace Converter B (CSCB) component 2 clip max value. (HDR_PIPE1_CSCB_IO_MAX_2)

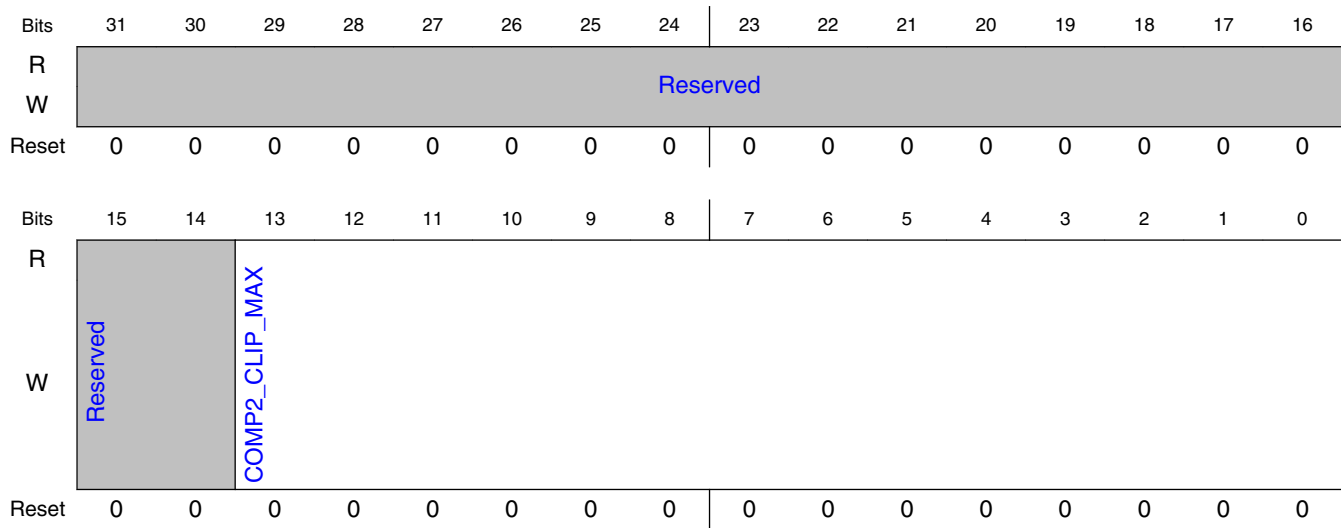
15.10.3.1.54.1 Offset

Register	Offset
HDR_PIPE1_CSCB_IO_MAX_2	3848h

15.10.3.1.54.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,16383]. Pipe1 is also called channel1.

15.10.3.1.54.3 Diagram



15.10.3.1.54.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.55 Pipe1 Colorspace Converter B (CSCB) normalization factor (HDR_PIPE1_CSCB_NORM)

15.10.3.1.55.1 Offset

Register	Offset
HDR_PIPE1_CSCB_NORM	384Ch

15.10.3.1.55.2 Function

After the CSC matrix multiply, all components of the result are shifted right by the amount in this register. This is a signed right shift. Pipe1 is also called channel1.

15.10.3.1.55.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved											CSCB_NORM				
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.10.3.1.55.4 Fields

Field	Function
31-5 —	Reserved.
4-0 CSCB_NORM	This 5-bit unsigned value is size if arithmetic shift after matrix multiply.

15.10.3.1.56 Pipe1 Colorspace Converter B (CSCB): Post offset component 0 (HDR_PIPE1_CSCB_OO_0)

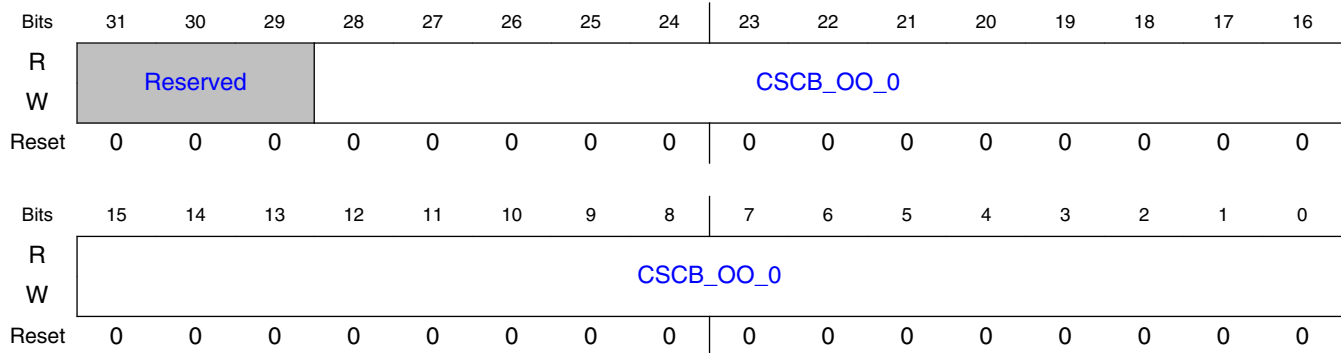
15.10.3.1.56.1 Offset

Register	Offset
HDR_PIPE1_CSCB_OO_0	3850h

15.10.3.1.56.2 Function

After the CSC normalization, this offset value is added to component 0. This is a signed 29-bit number. Pipe1 is also called channel1.

15.10.3.1.56.3 Diagram



15.10.3.1.56.4 Fields

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_0	Output Offset (OO) This is a signed 29-bit number. Per component

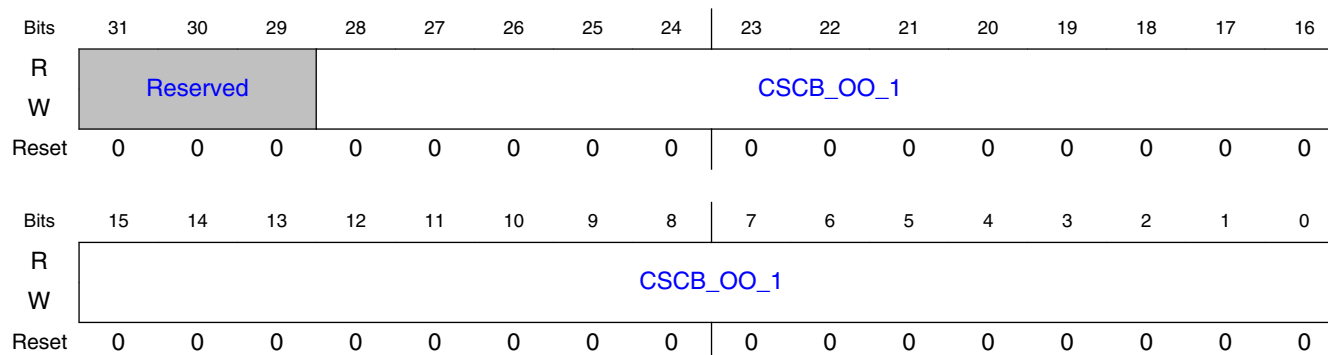
15.10.3.1.57 Pipe1 Colorspace Converter B (CSCB): Post offset component 1 (HDR_PIPE1_CSCB_OO_1)

15.10.3.1.57.1 Offset

Register	Offset
HDR_PIPE1_CSCB_OO_1	3854h

15.10.3.1.57.2 Function

After the CSC normalization, this offset value is added to component 1. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.57.3 Diagram**15.10.3.1.57.4 Fields**

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_1	Output Offset (OO) This is a signed 29-bit number. Per component

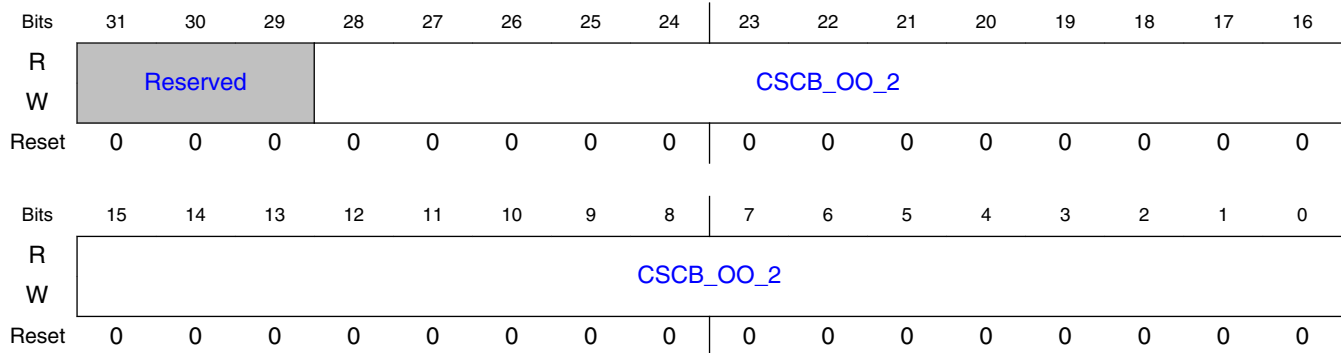
15.10.3.1.58 Pipe1 Colorspace Converter B (CSCB): Post offset component 2 (HDR_PIPE1_CSCB_OO_2)**15.10.3.1.58.1 Offset**

Register	Offset
HDR_PIPE1_CSCB_OO_2	3858h

15.10.3.1.58.2 Function

After the CSC normalization, this offset value is added to component 2. This is a signed 29-bit number. Pipe1 is also called channel1.

15.10.3.1.58.3 Diagram



15.10.3.1.58.4 Fields

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_2	Output Offset (OO) This is a signed 29-bit number. Per component

15.10.3.1.59 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 0 (HDR_PIPE1_CSCB_OMIN_0)

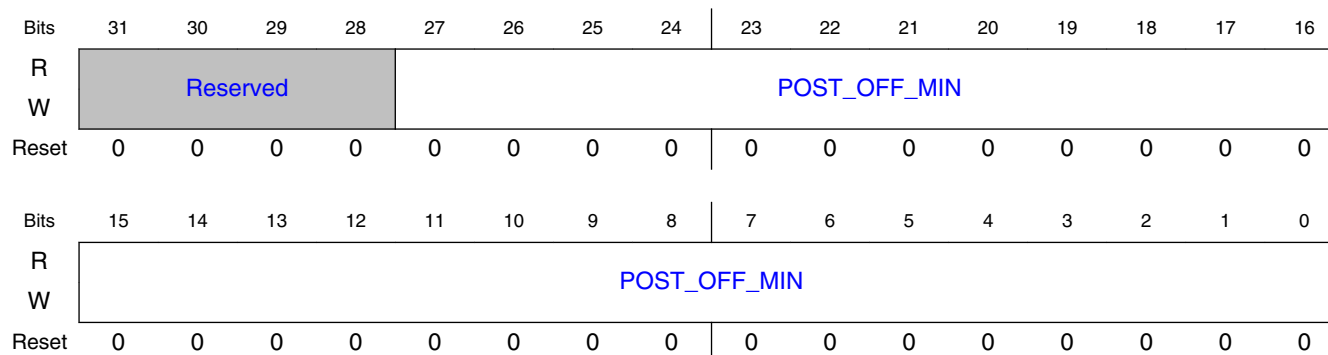
15.10.3.1.59.1 Offset

Register	Offset
HDR_PIPE1_CSCB_OMIN_0	385Ch

15.10.3.1.59.2 Function

After the post offset is added this component is clipped. This is the minimum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.59.3 Diagram



15.10.3.1.59.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minum clipped pixel component.

15.10.3.1.60 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 1 (HDR_PIPE1_CSCB_OMIN_1)

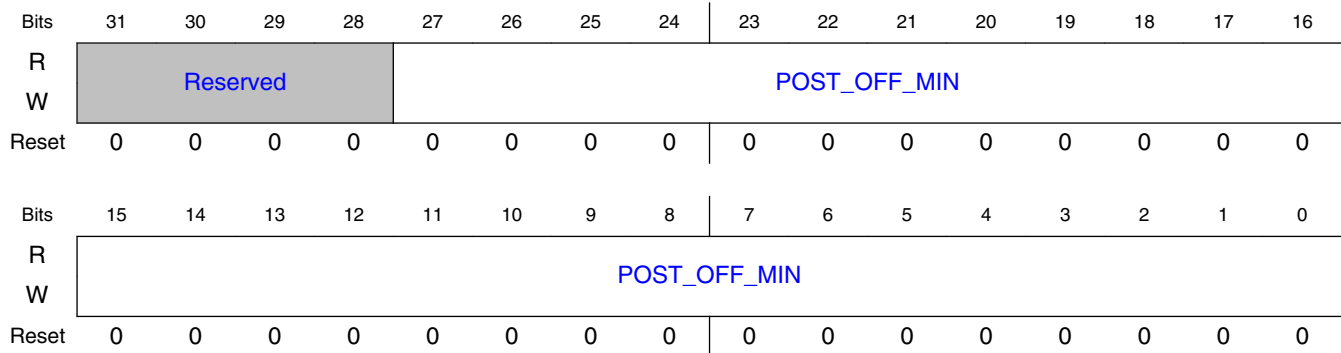
15.10.3.1.60.1 Offset

Register	Offset
HDR_PIPE1_CSCB_OMIN_1	3860h

15.10.3.1.60.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.60.3 Diagram



15.10.3.1.60.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minum clipped pixel component.

15.10.3.1.61 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 2 (HDR_PIPE1_CSCB_OMIN_2)

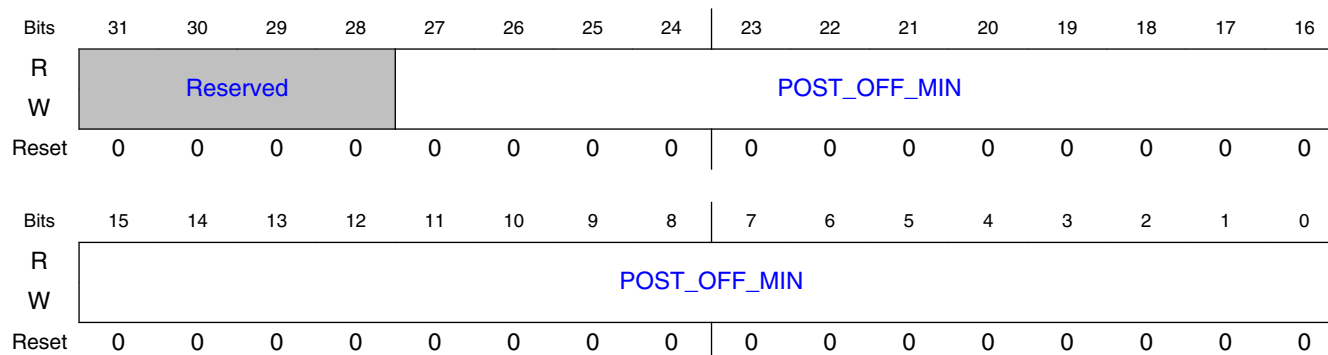
15.10.3.1.61.1 Offset

Register	Offset
HDR_PIPE1_CSCB_OMIN_2	3864h

15.10.3.1.61.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.61.3 Diagram



15.10.3.1.61.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minimum clipped pixel component.

15.10.3.1.62 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 0 (HDR_PIPE1_CSCB_OMAX_0)

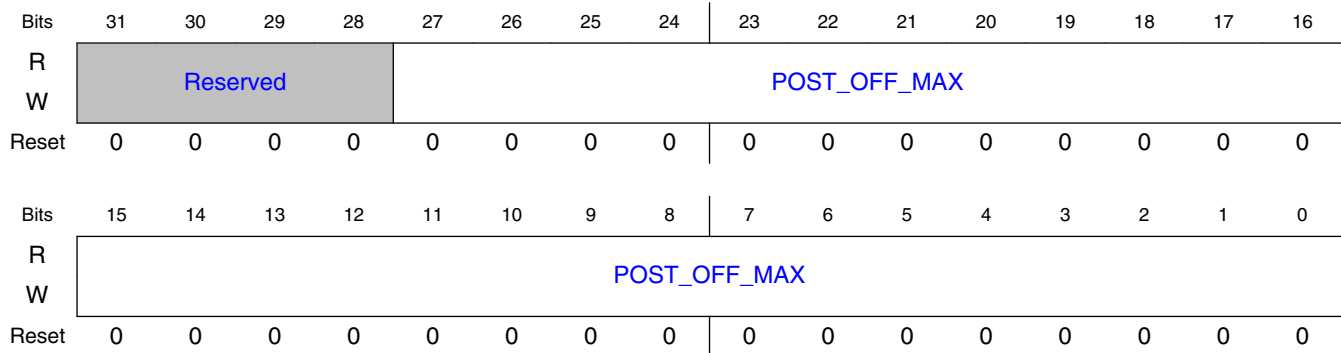
15.10.3.1.62.1 Offset

Register	Offset
HDR_PIPE1_CSCB_OMAX_0	3868h

15.10.3.1.62.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.62.3 Diagram



15.10.3.1.62.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MAX	Maximum clipped pixel component.

15.10.3.1.63 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 1 (HDR_PIPE1_CSCB_OMAX_1)

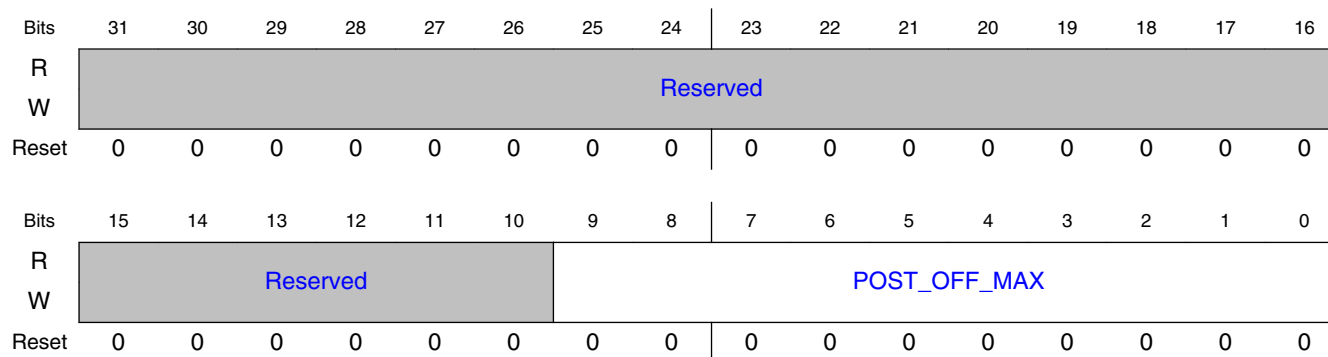
15.10.3.1.63.1 Offset

Register	Offset
HDR_PIPE1_CSCB_OMAX_1	386Ch

15.10.3.1.63.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.63.3 Diagram



15.10.3.1.63.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component.

15.10.3.1.64 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 2 (HDR_PIPE1_CSCB_OMAX_2)

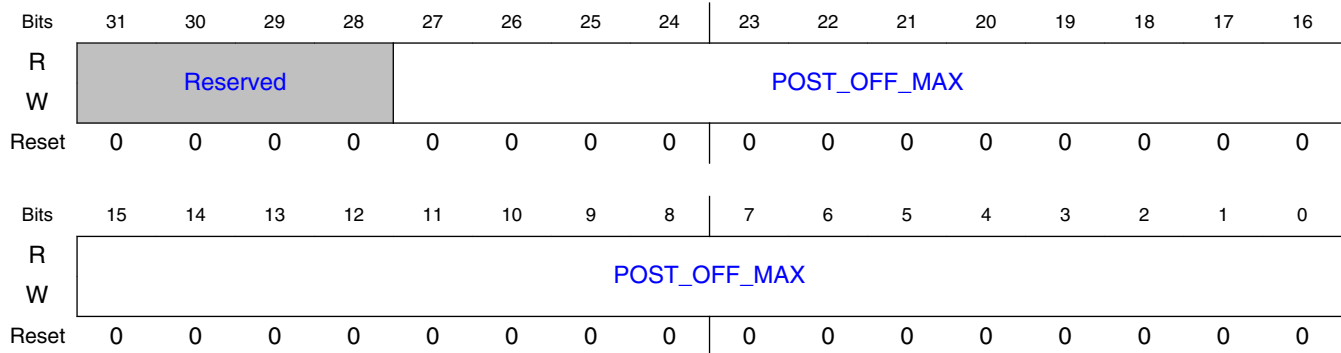
15.10.3.1.64.1 Offset

Register	Offset
HDR_PIPE1_CSCB_OMAX_2	3870h

15.10.3.1.64.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.64.3 Diagram



15.10.3.1.64.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MAX	Maximum clipped pixel component.

15.10.3.1.65 Pipe1 floating point to fixed point control (HDR_PIPE1_FL2FX)

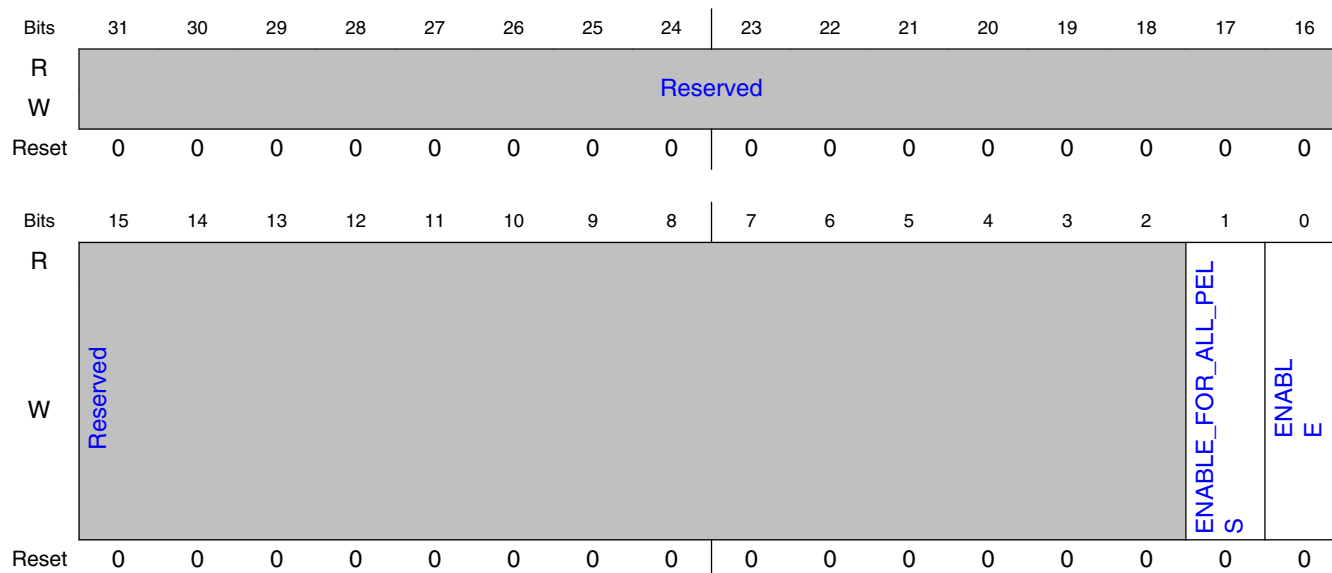
15.10.3.1.65.1 Offset

Register	Offset
HDR_PIPE1_FL2FX	3874h

15.10.3.1.65.2 Function

Pipe1 floating point to fixed point control Pipe1 is also called channel1.

15.10.3.1.65.3 Diagram



15.10.3.1.65.4 Fields

Field	Function
31-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This Float to Fixed operation is enabled only for blended pixels 1: This Float to Fixed operation is enabled all pixels
0 ENABLE	0: Don't enable this Float-to-Fixed converter: Pixels pass thru the Float-to-Fixed unmodified 1: This Float-to-Fixed converter is enabled for current picture

15.10.3.1.66 PIPE1: NOT USED (HDR_PIPE1_ENTRY_30)

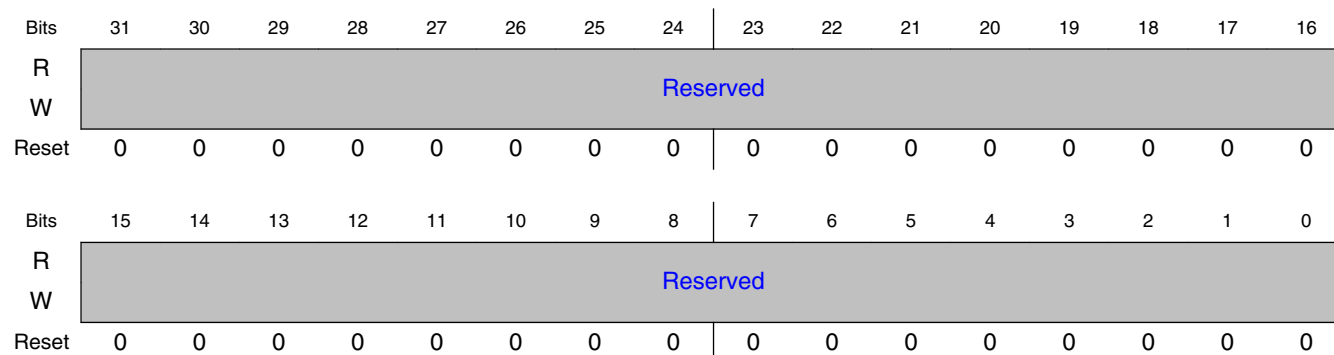
15.10.3.1.66.1 Offset

Register	Offset
HDR_PIPE1_ENTRY_30	3878h

15.10.3.1.66.2 Function

PIPE1: NOT USED

15.10.3.1.66.3 Diagram



15.10.3.1.66.4 Fields

Field	Function
31-0	Reserved.
—	

15.10.3.1.67 A0 component Look-Up-Table. (LUT) (PIPE2_A0_LUT)

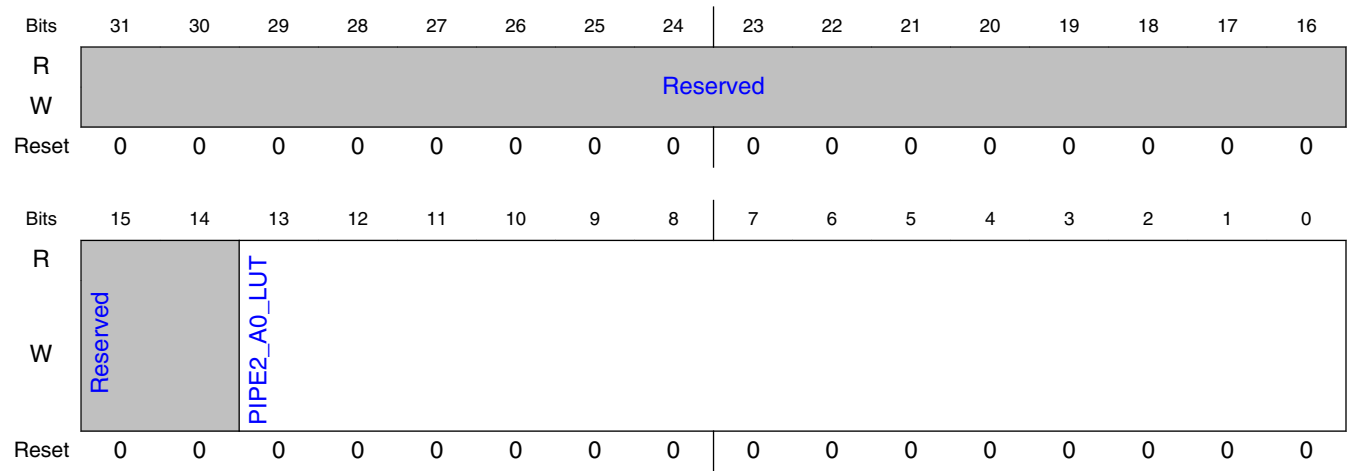
15.10.3.1.67.1 Offset

Register	Offset
PIPE2_A0_LUT	4000h

15.10.3.1.67.2 Function

The LUT table has 1024 entries. A0 component may be R or Y or other

15.10.3.1.67.3 Diagram



15.10.3.1.67.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE2_A0_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.68 A1 component Look-Up-Table. (LUT) (PIPE2_A1_LUT)

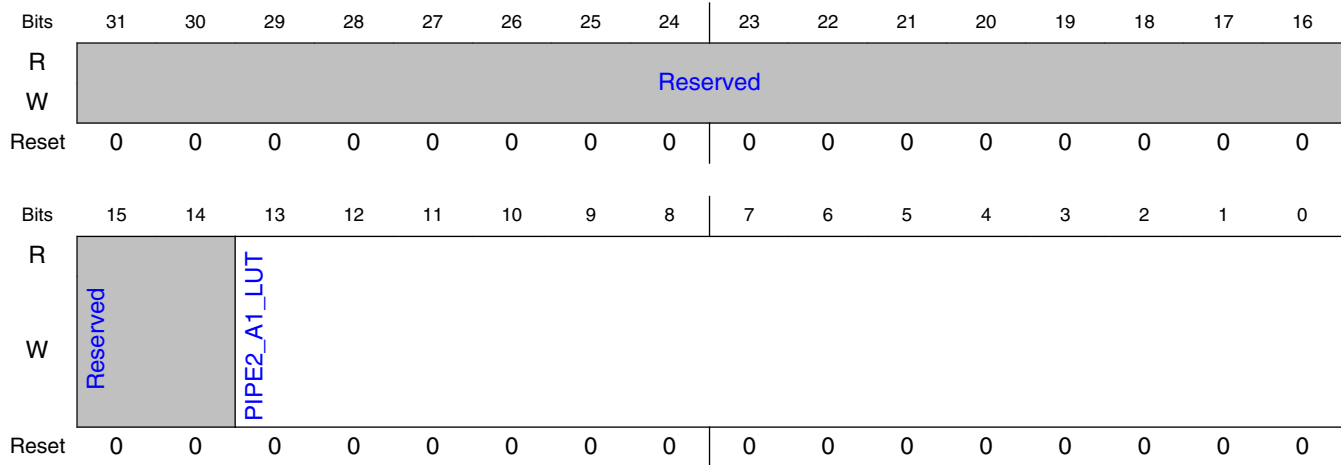
15.10.3.1.68.1 Offset

Register	Offset
PIPE2_A1_LUT	5000h

15.10.3.1.68.2 Function

The LUT table has 1024 entries. A1 component may be G or Cb or other

15.10.3.1.68.3 Diagram



15.10.3.1.68.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE2_A1_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.69 A2 component Look-Up-Table. (LUT) (PIPE2_A2_LUT)

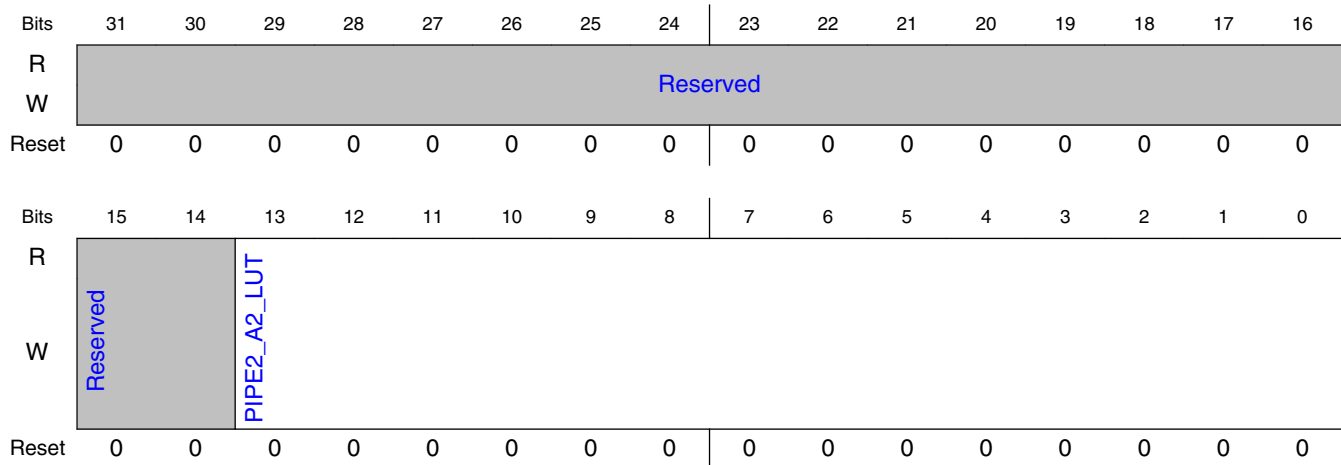
15.10.3.1.69.1 Offset

Register	Offset
PIPE2_A2_LUT	6000h

15.10.3.1.69.2 Function

The LUT table has 1024 entries. A2 component may be B or Cr or other

15.10.3.1.69.3 Diagram



15.10.3.1.69.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE2_A2_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.70 Pipe1 Colorspace Converter A control. (HDR_PIPE2_CSCA_CONTROL_REG)

15.10.3.1.70.1 Offset

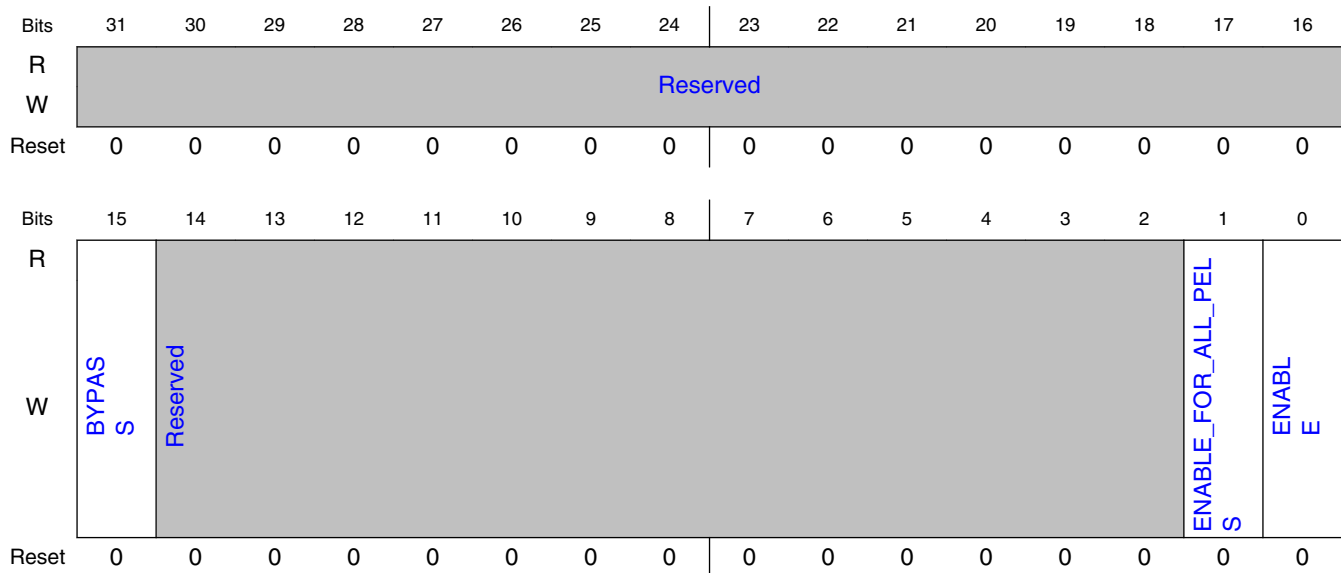
Register	Offset
HDR_PIPE2_CSCA_CONTROL_REG	7000h

15.10.3.1.70.2 Function

Controls the Color-Space-Converter-A (CSCA) in pipe1. This CSCA takes in 10-bit pixel components (component 0, component 1, component 2). A per-component offset (called pre-offset) is added to the pixel. After the pre-offset operation the pixel components are clipped. MAX_RANGE of PRE_CLIP is [-512, 1023]. A 3x3 matrix multiply (constant coefficients H(x,y)) is performed on the pixel to take to a different color space. After this matrix multiply, a 28 bit signed offset (post-offset) is added to the normalized result of

the matrix multiply. After the post-offset operation the results are clipped. MAX_RANGE of POST_CLIP is [0,1023] Output of this CSCA are 10 bit per component pixels. After CSCA the pixels are fed into a LUT. Pipe1 is also called channel1.

15.10.3.1.70.3 Diagram



15.10.3.1.70.4 Fields

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this CSC 1: Pixels pass thru this CSC unmodified
14-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This CSC is enabled only for blended pixels 1: This CSC is enabled all pixels
0 ENABLE	0: Don't enable this CSC: Pixels pass thru the CSC unmodified 1: This CSC is enabled for current picture

15.10.3.1.71 Pipe1 Colorspace Converter A (CSCA) h(0,0) matrix coefficient (HDR_PIPE2_CSCA_H00)

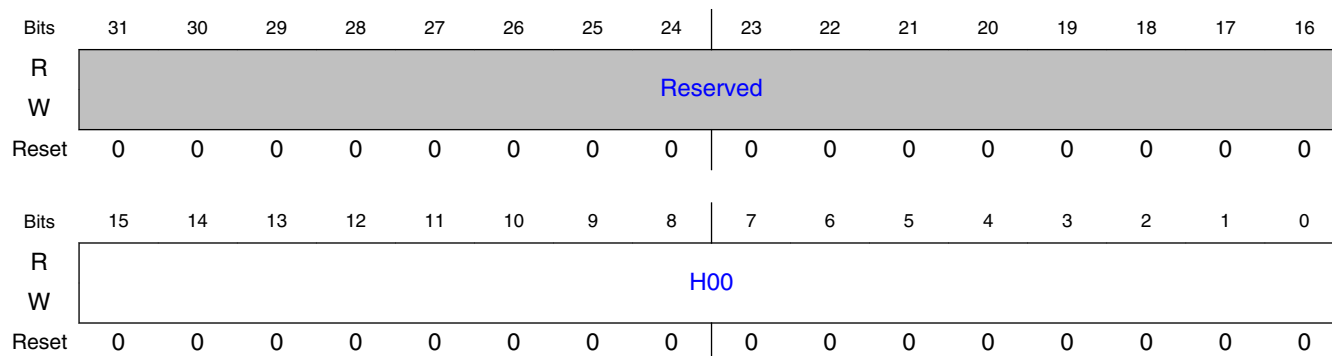
15.10.3.1.71.1 Offset

Register	Offset
HDR_PIPE2_CSCA_H00	7004h

15.10.3.1.71.2 Function

h(0,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.71.3 Diagram



15.10.3.1.71.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H00	h(0,0) 16 bit signed coefficient

15.10.3.1.72 Pipe1 Colorspace Converter A (CSCA) h(1,0) matrix coefficient (HDR_PIPE2_CSCA_H10)

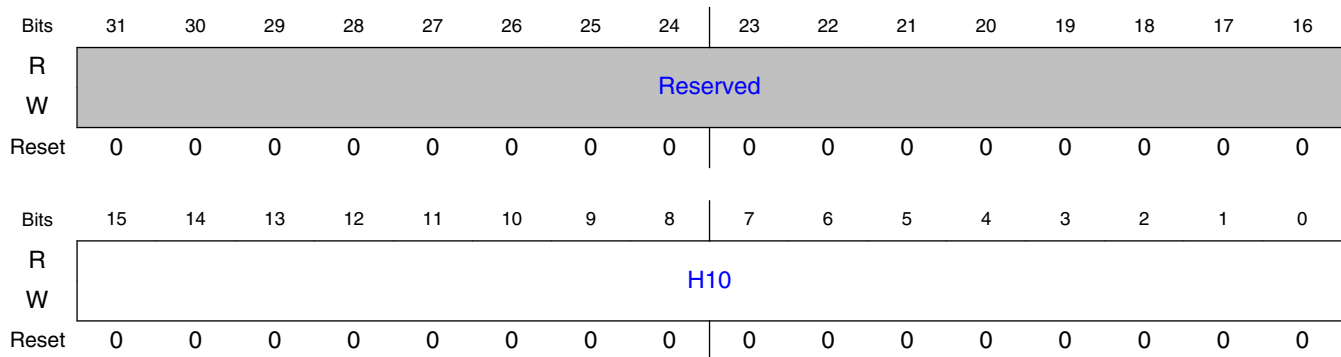
15.10.3.1.72.1 Offset

Register	Offset
HDR_PIPE2_CSCA_H10	7008h

15.10.3.1.72.2 Function

h(1,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.72.3 Diagram



15.10.3.1.72.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H10	h(1,0) 16 bit signed coefficient

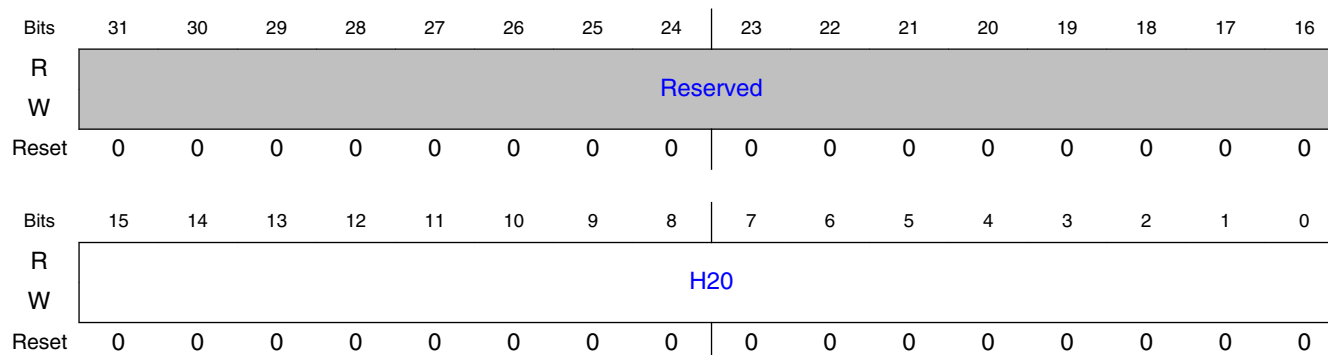
15.10.3.1.73 Pipe1 Colorspace Converter A (CSCA) h(2,0) matrix coefficient (HDR_PIPE2_CSCA_H20)

15.10.3.1.73.1 Offset

Register	Offset
HDR_PIPE2_CSCA_H20	700Ch

15.10.3.1.73.2 Function

$h(2,0)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.73.3 Diagram**15.10.3.1.73.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H20	$h(2,0)$ 16 bit signed coefficient

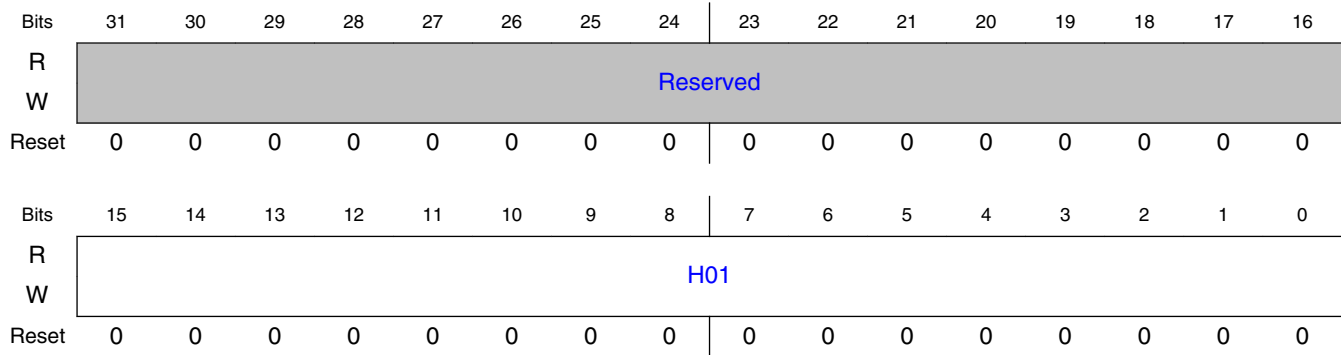
15.10.3.1.74 Pipe1 Colorspace Converter A (CSCA) $h(0,1)$ matrix coefficient (HDR_PIPE2_CSCA_H01)**15.10.3.1.74.1 Offset**

Register	Offset
HDR_PIPE2_CSCA_H01	7010h

15.10.3.1.74.2 Function

$h(0,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.74.3 Diagram



15.10.3.1.74.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H01	h(0,1) 16 bit signed coefficient

15.10.3.1.75 Pipe1 Colorspace Converter A (CSCA) h(1,1) matrix coefficient (HDR_PIPE2_CSCA_H11)

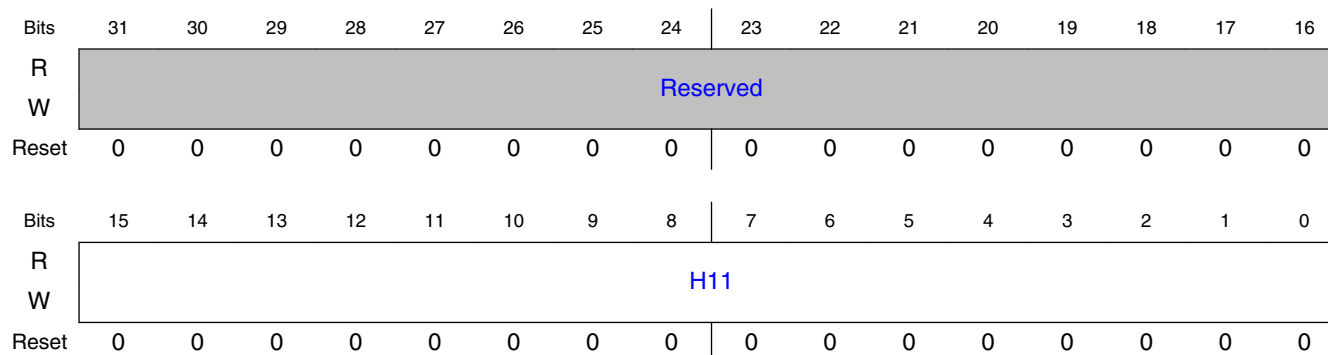
15.10.3.1.75.1 Offset

Register	Offset
HDR_PIPE2_CSCA_H11	7014h

15.10.3.1.75.2 Function

h(1,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.75.3 Diagram



15.10.3.1.75.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H11	h(1,1) 16 bit signed coefficient

15.10.3.1.76 Pipe1 Colorspace Converter A (CSCA) h(2,1) matrix coefficient (HDR_PIPE2_CSCA_H21)

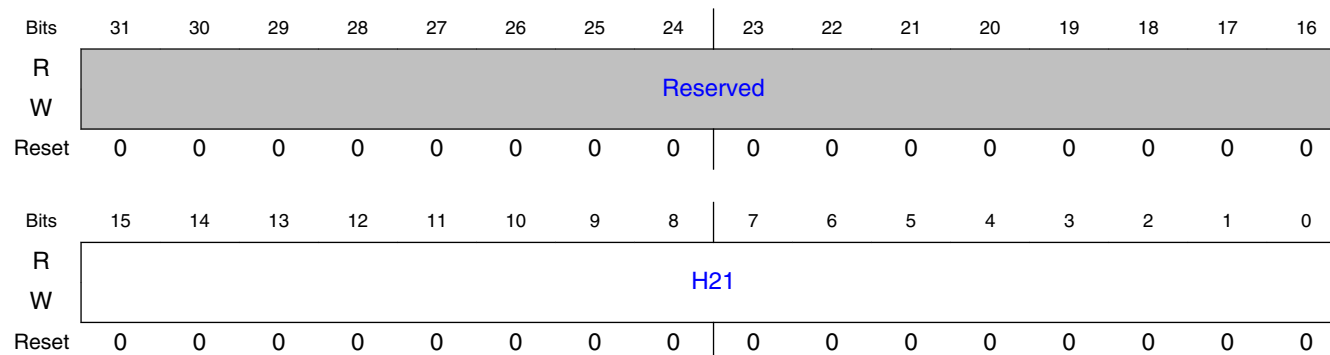
15.10.3.1.76.1 Offset

Register	Offset
HDR_PIPE2_CSCA_H21	7018h

15.10.3.1.76.2 Function

h(2,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.76.3 Diagram



15.10.3.1.76.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H21	h(2,1) 16 bit signed coefficient

15.10.3.1.77 Pipe1 Colorspace Converter A (CSCA) h(0,2) matrix coefficient (HDR_PIPE2_CSCA_H02)

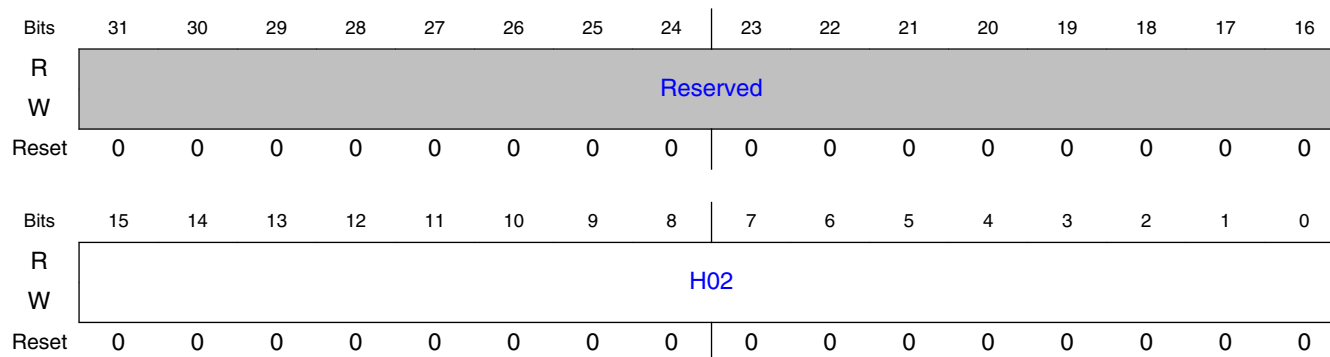
15.10.3.1.77.1 Offset

Register	Offset
HDR_PIPE2_CSCA_H02	701Ch

15.10.3.1.77.2 Function

h(0,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.77.3 Diagram



15.10.3.1.77.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H02	h(0,2) 16 bit signed coefficient

15.10.3.1.78 Pipe1 Colorspace Converter A (CSCA) h(1,2) matrix coefficient (HDR_PIPE2_CSCA_H12)

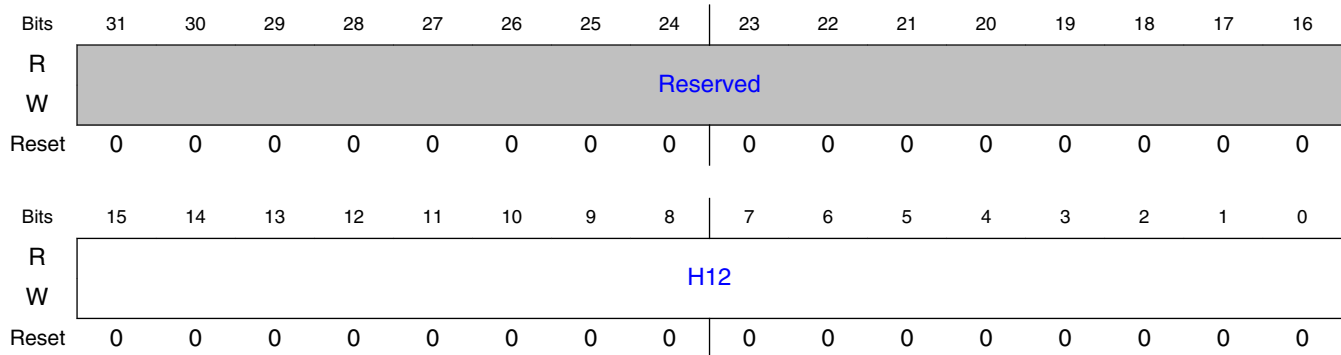
15.10.3.1.78.1 Offset

Register	Offset
HDR_PIPE2_CSCA_H12	7020h

15.10.3.1.78.2 Function

h(1,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.78.3 Diagram



15.10.3.1.78.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H12	h(1,2) 16 bit signed coefficient

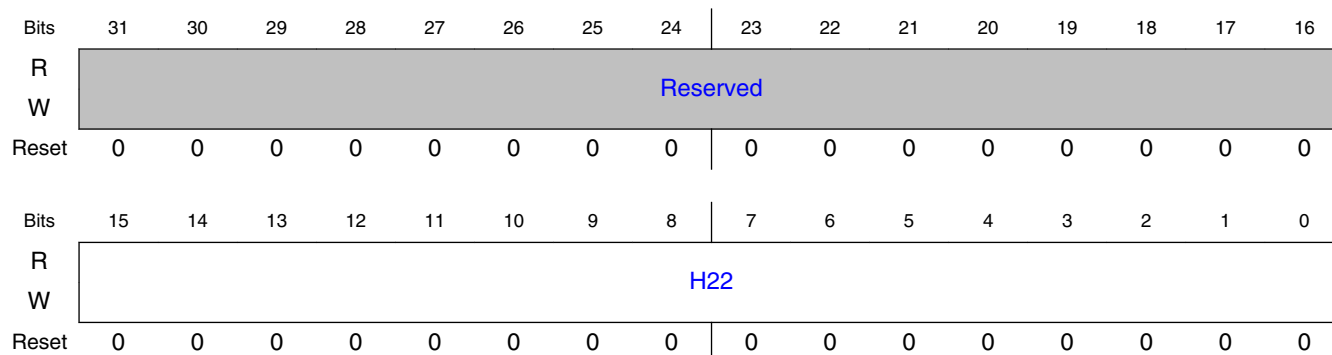
15.10.3.1.79 Pipe1 Colorspace Converter A (CSCA) h(2,2) matrix coefficient (HDR_PIPE2_CSCA_H22)

15.10.3.1.79.1 Offset

Register	Offset
HDR_PIPE2_CSCA_H22	7024h

15.10.3.1.79.2 Function

h(2,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.79.3 Diagram**15.10.3.1.79.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H22	h(2,2) 16 bit signed coefficient

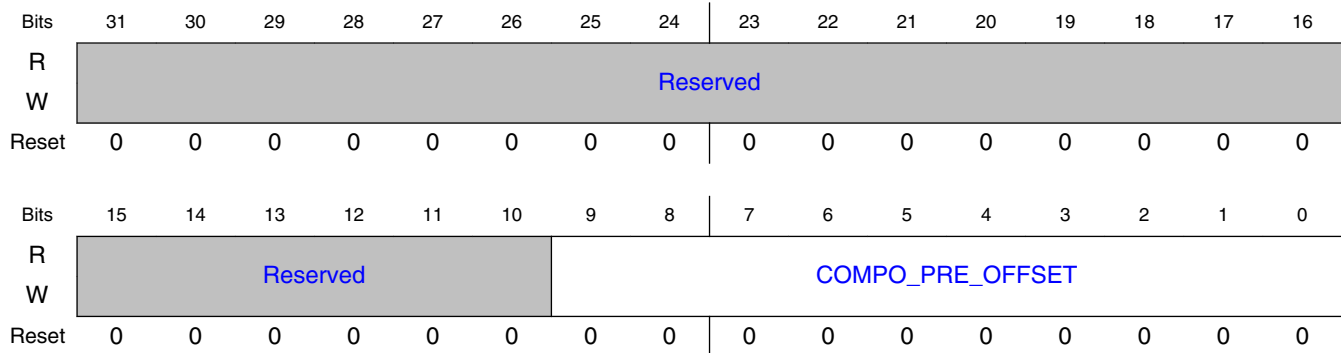
15.10.3.1.80 Pipe1 Colorspace Converter A (CSCA) component 0 pre-offset (HDR_PIPE2_CSCA_IO_0)**15.10.3.1.80.1 Offset**

Register	Offset
HDR_PIPE2_CSCA_IO_0	7028h

15.10.3.1.80.2 Function

An signed 10-bit offset is added to component 0 (R,Y) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.80.3 Diagram



15.10.3.1.80.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMPO_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 0 of the pixel

15.10.3.1.81 Pipe1 Colorspace Converter A (CSCA) component 1 pre-offset (HDR_PIPE2_CSCA_IO_1)

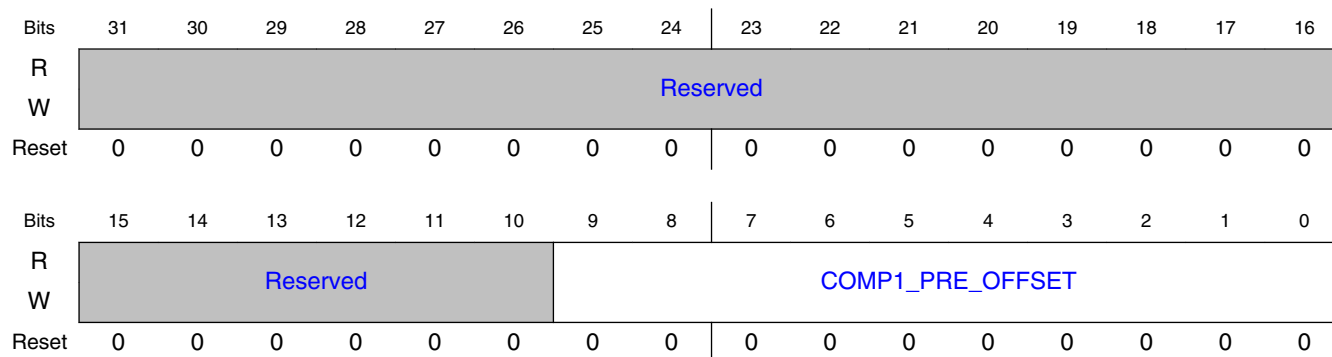
15.10.3.1.81.1 Offset

Register	Offset
HDR_PIPE2_CSCA_IO_1	702Ch

15.10.3.1.81.2 Function

An signed 10-bit offset is added to component 1 (G,Cb) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.81.3 Diagram



15.10.3.1.81.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP1_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 1 of the pixel

15.10.3.1.82 Pipe1 Colorspace Converter A (CSCA) component 2 pre-offset (HDR_PIPE2_CSCA_IO_2)

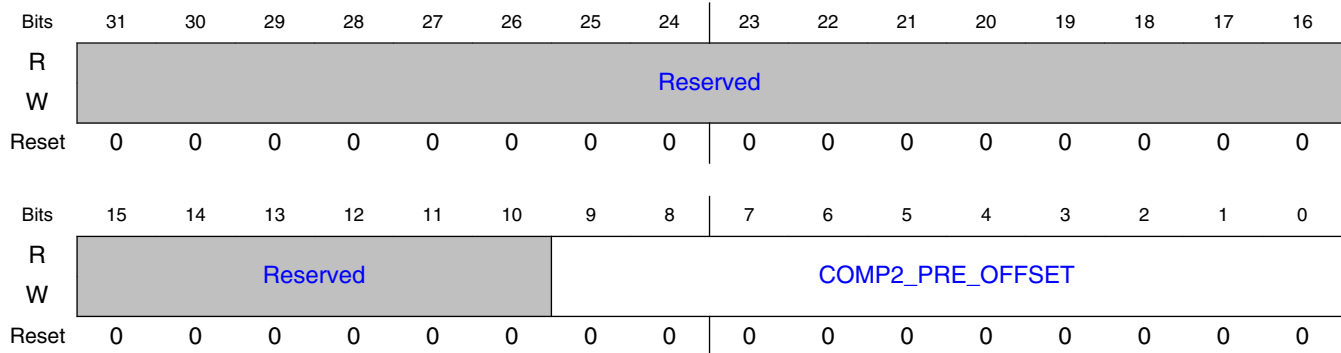
15.10.3.1.82.1 Offset

Register	Offset
HDR_PIPE2_CSCA_IO_2	7030h

15.10.3.1.82.2 Function

An signed 10-bit offset is added to component 2 (B,Cr) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.82.3 Diagram



15.10.3.1.82.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 2 of the pixel

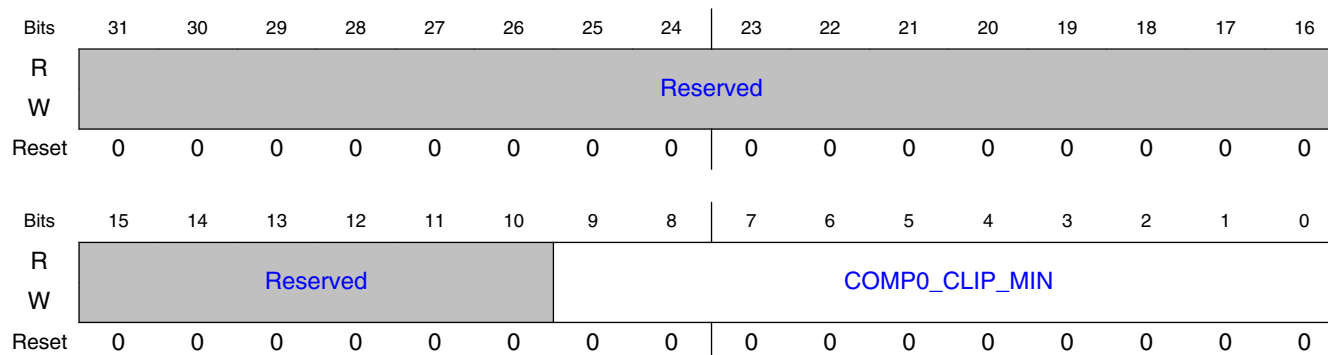
15.10.3.1.83 Pipe1 Colorspace Converter A (CSCA) component 0 clip min. (HDR_PIPE2_CSCA_IO_MIN_0)

15.10.3.1.83.1 Offset

Register	Offset
HDR_PIPE2_CSCA_IO_MIN_0	7034h

15.10.3.1.83.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.83.3 Diagram**15.10.3.1.83.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP0_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

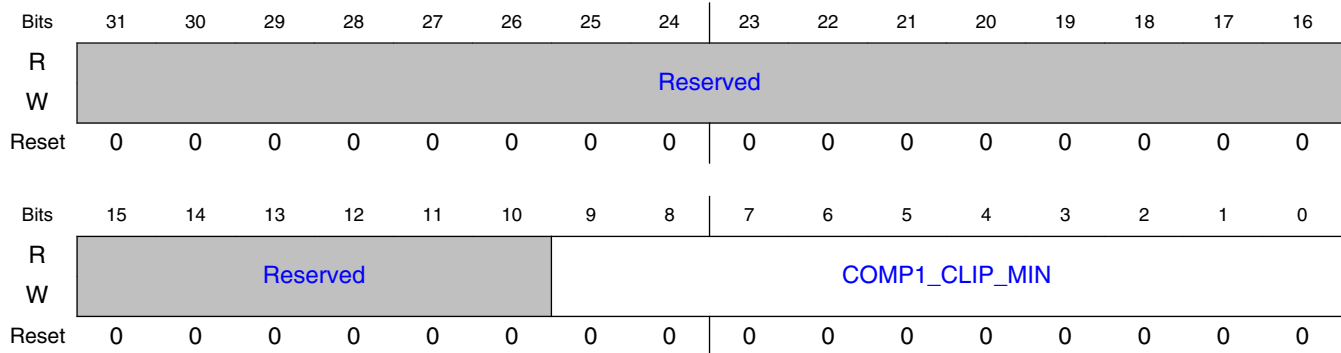
15.10.3.1.84 Pipe1 Colorspace Converter A (CSCA) component 1 clip min. (HDR_PIPE2_CSCA_IO_MIN_1)**15.10.3.1.84.1 Offset**

Register	Offset
HDR_PIPE2_CSCA_IO_MIN_1	7038h

15.10.3.1.84.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.84.3 Diagram



15.10.3.1.84.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP1_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.85 Pipe1 Colorspace Converter A (CSCA) component 2 clip min. (HDR_PIPE2_CSCA_IO_MIN_2)

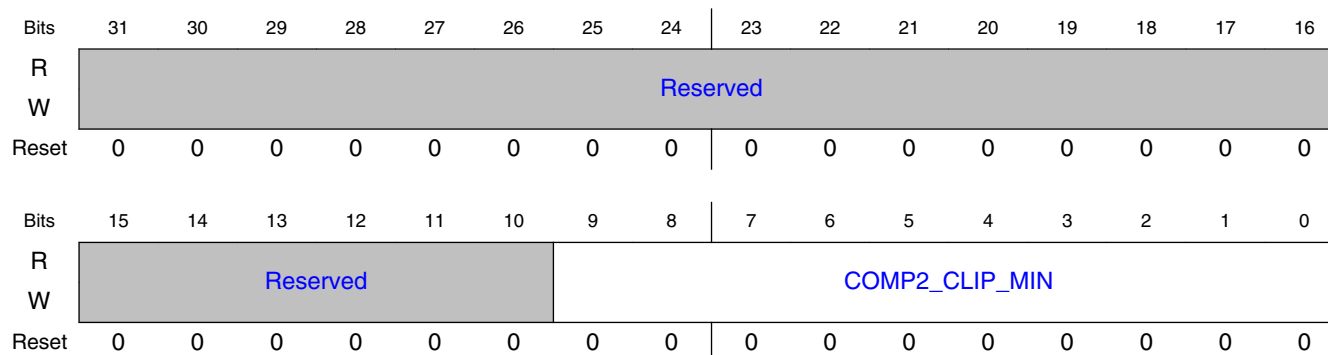
15.10.3.1.85.1 Offset

Register	Offset
HDR_PIPE2_CSCA_IO_MIN_2	703Ch

15.10.3.1.85.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.85.3 Diagram



15.10.3.1.85.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.86 Pipe1 Colorspace Converter A (CSCA) component 0 clip max value. (HDR_PIPE2_CSCA_IO_MAX_0)

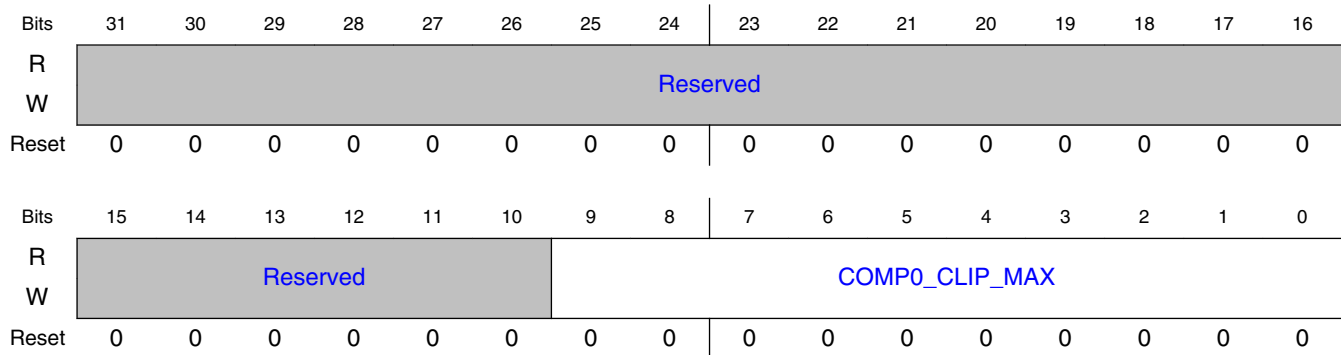
15.10.3.1.86.1 Offset

Register	Offset
HDR_PIPE2_CSCA_IO_MAX_0	7040h

15.10.3.1.86.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.86.3 Diagram



15.10.3.1.86.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP0_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

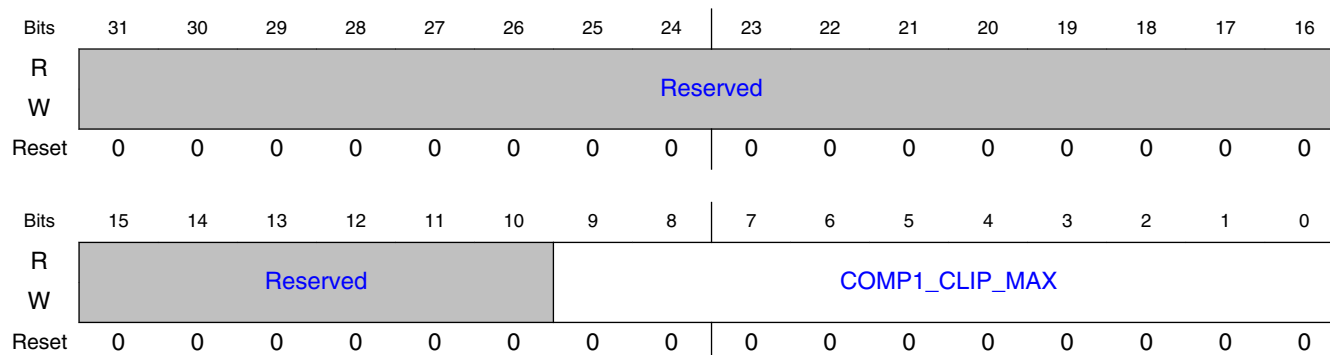
15.10.3.1.87 Pipe1 Colorspace Converter A (CSCA) component 1 clip max value. (HDR_PIPE2_CSCA_IO_MAX_1)

15.10.3.1.87.1 Offset

Register	Offset
HDR_PIPE2_CSCA_IO_MAX_1	7044h

15.10.3.1.87.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.87.3 Diagram**15.10.3.1.87.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP1_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

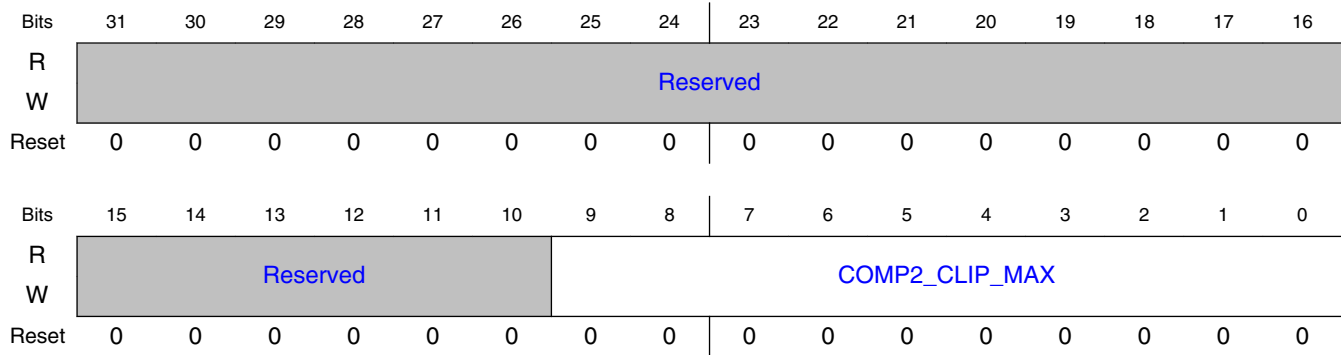
15.10.3.1.88 Pipe1 Colorspace Converter A (CSCA) component 2 clip max value. (HDR_PIPE2_CSCA_IO_MAX_2)**15.10.3.1.88.1 Offset**

Register	Offset
HDR_PIPE2_CSCA_IO_MAX_2	7048h

15.10.3.1.88.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.88.3 Diagram



15.10.3.1.88.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.89 Pipe1 Colorspace Converter A (CSCA) normalization factor (HDR_PIPE2_CSCA_NORM)

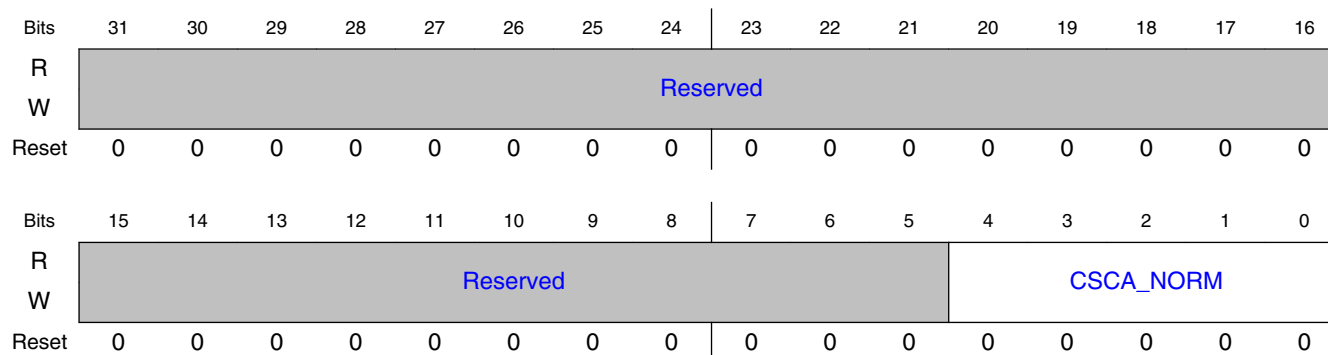
15.10.3.1.89.1 Offset

Register	Offset
HDR_PIPE2_CSCA_NORM	704Ch

15.10.3.1.89.2 Function

After the CSC matrix multiply, all components of the result are shifted right by the amount in this register. This is a signed right shift. Pipe1 is also called channel1.

15.10.3.1.89.3 Diagram



15.10.3.1.89.4 Fields

Field	Function
31-5 —	Reserved.
4-0 CSCA_NORM	This 5-bit unsigned value is size if arithmetic shift after matrix multiply.

15.10.3.1.90 Pipe1 Colorspace Converter A (CSCA): Post offset component 0 (HDR_PIPE2_CSCA_OO_0)

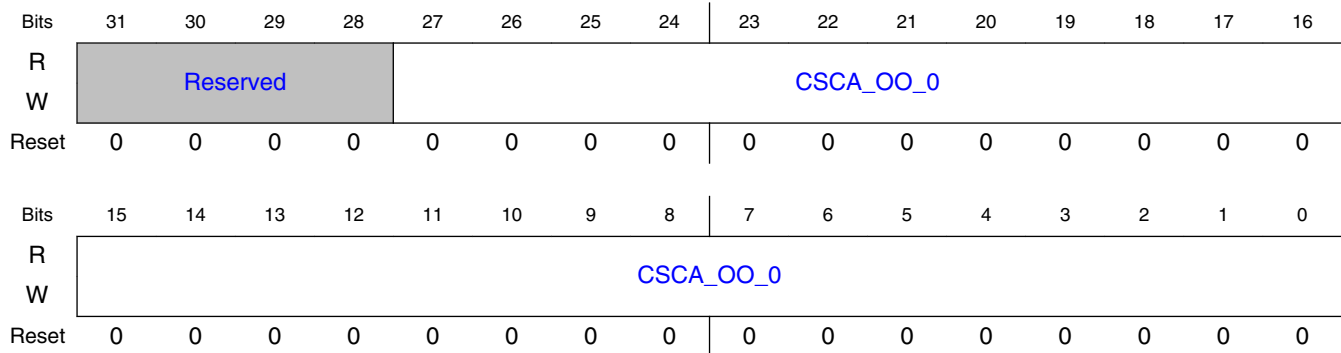
15.10.3.1.90.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OO_0	7050h

15.10.3.1.90.2 Function

After the CSC normalization, this offset value is added to component 0. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.90.3 Diagram



15.10.3.1.90.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_0	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.91 Pipe1 Colorspace Converter A (CSCA): Post offset component 1 (HDR_PIPE2_CSCA_OO_1)

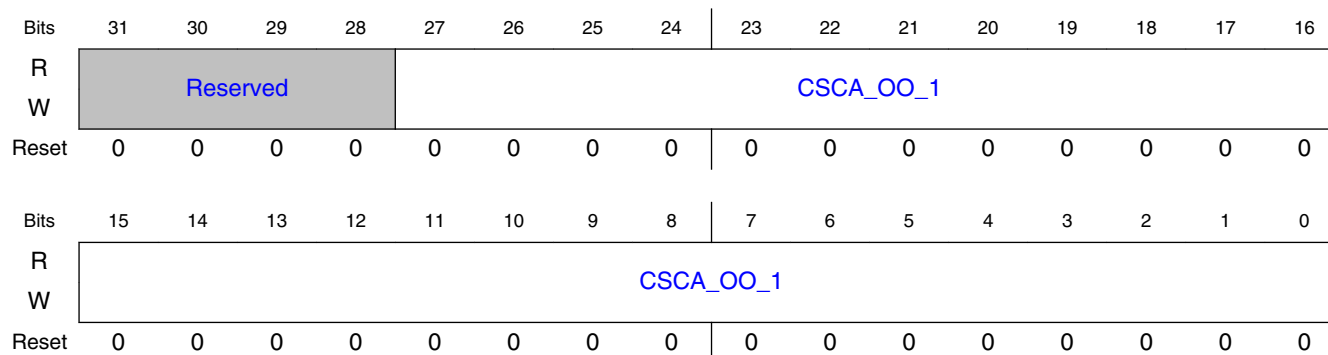
15.10.3.1.91.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OO_1	7054h

15.10.3.1.91.2 Function

After the CSC normalization, this offset value is added to component 1. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.91.3 Diagram



15.10.3.1.91.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_1	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.92 Pipe1 Colorspace Converter A (CSCA): Post offset component 2 (HDR_PIPE2_CSCA_OO_2)

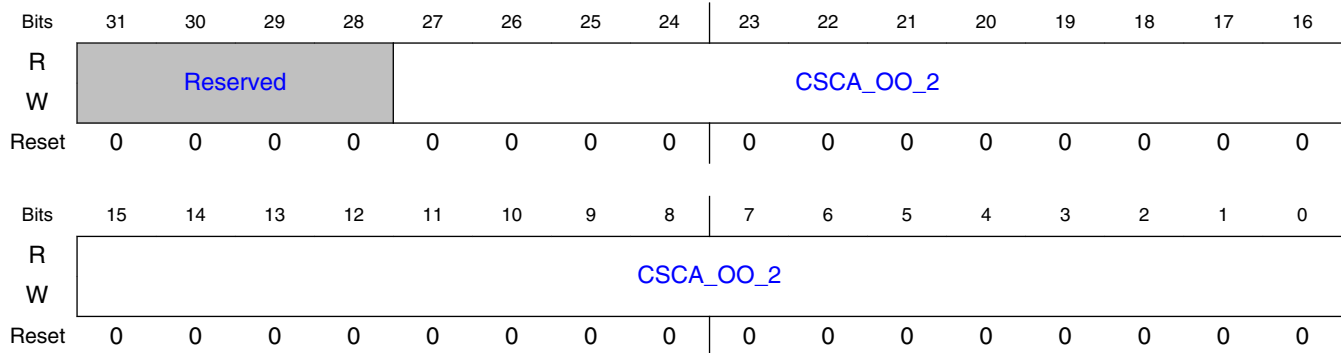
15.10.3.1.92.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OO_2	7058h

15.10.3.1.92.2 Function

After the CSC normalization, this offset value is added to component 2. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.92.3 Diagram



15.10.3.1.92.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_2	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.93 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 0 (HDR_PIPE2_CSCA_OMIN_0)

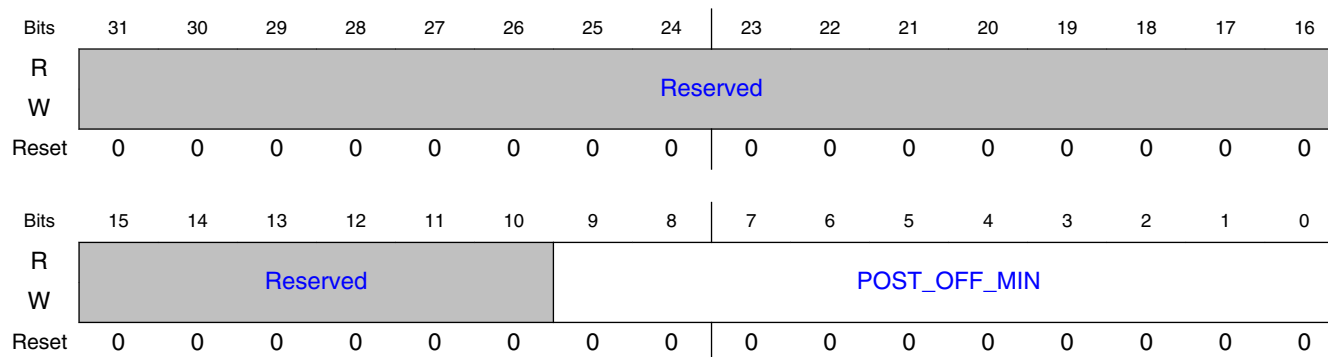
15.10.3.1.93.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OMIN_0	705Ch

15.10.3.1.93.2 Function

After the post offset is added this component is clipped. This is the minimum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.93.3 Diagram



15.10.3.1.93.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.94 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 1 (HDR_PIPE2_CSCA_OMIN_1)

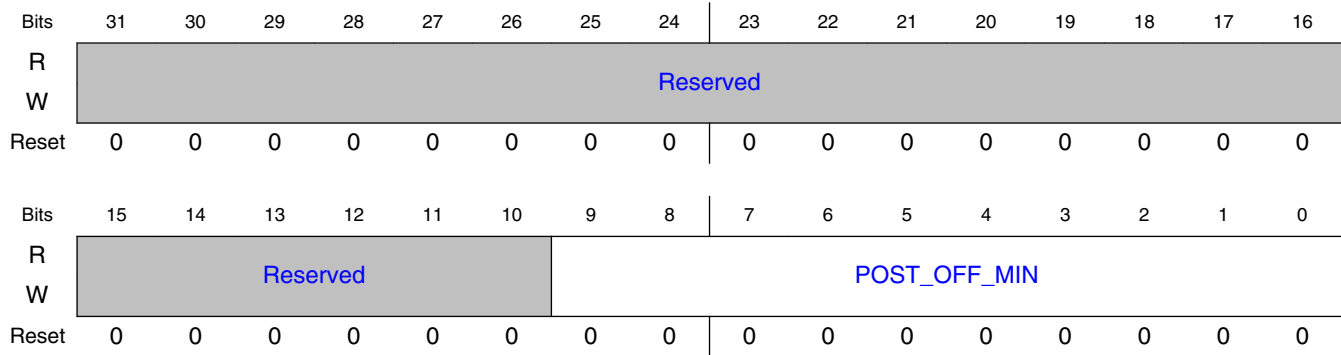
15.10.3.1.94.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OMIN_1	7060h

15.10.3.1.94.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.94.3 Diagram



15.10.3.1.94.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.95 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 2 (HDR_PIPE2_CSCA_OMIN_2)

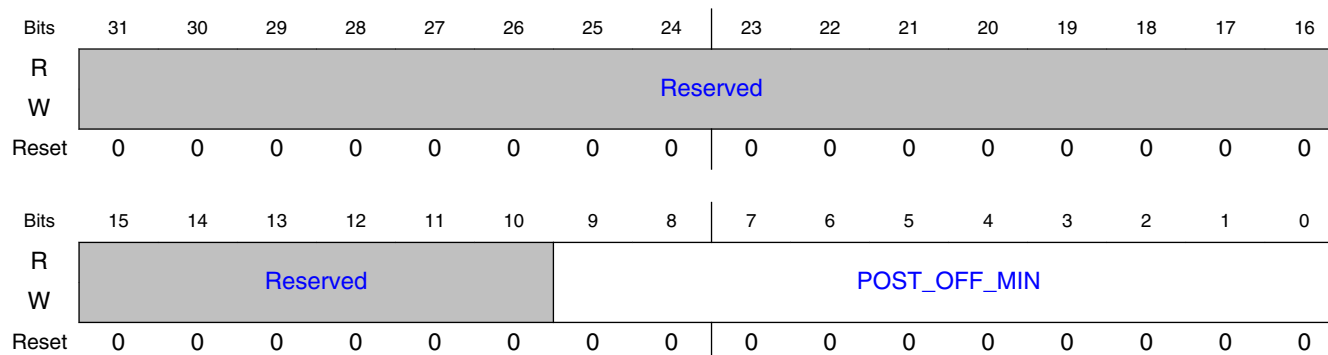
15.10.3.1.95.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OMIN_2	7064h

15.10.3.1.95.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.95.3 Diagram



15.10.3.1.95.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.96 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 0 (HDR_PIPE2_CSCA_OMAX_0)

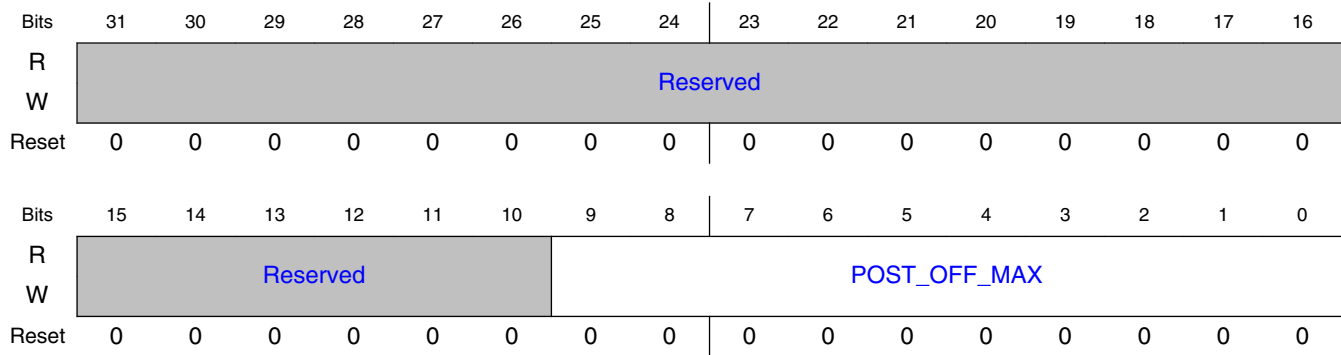
15.10.3.1.96.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OMAX_0	7068h

15.10.3.1.96.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.96.3 Diagram



15.10.3.1.96.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

15.10.3.1.97 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 1 (HDR_PIPE2_CSCA_OMAX_1)

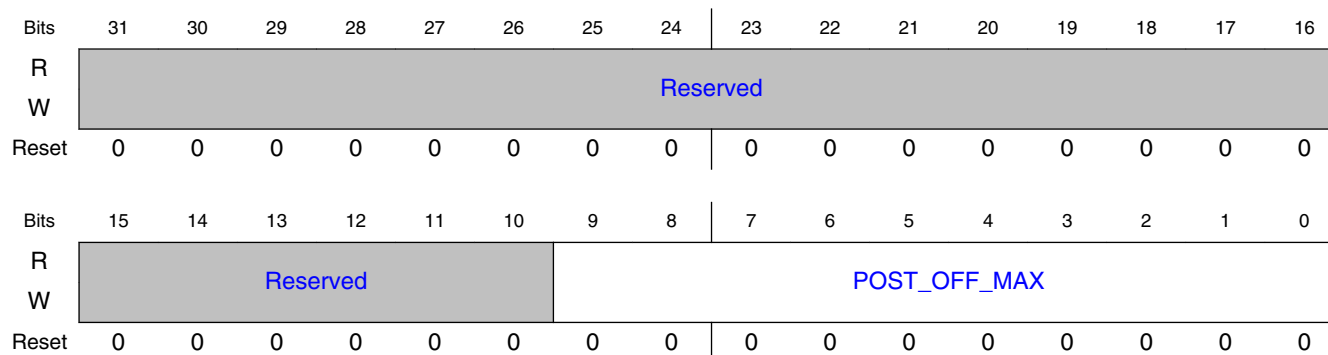
15.10.3.1.97.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OMAX_1	706Ch

15.10.3.1.97.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.97.3 Diagram



15.10.3.1.97.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

15.10.3.1.98 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 2 (HDR_PIPE2_CSCA_OMAX_2)

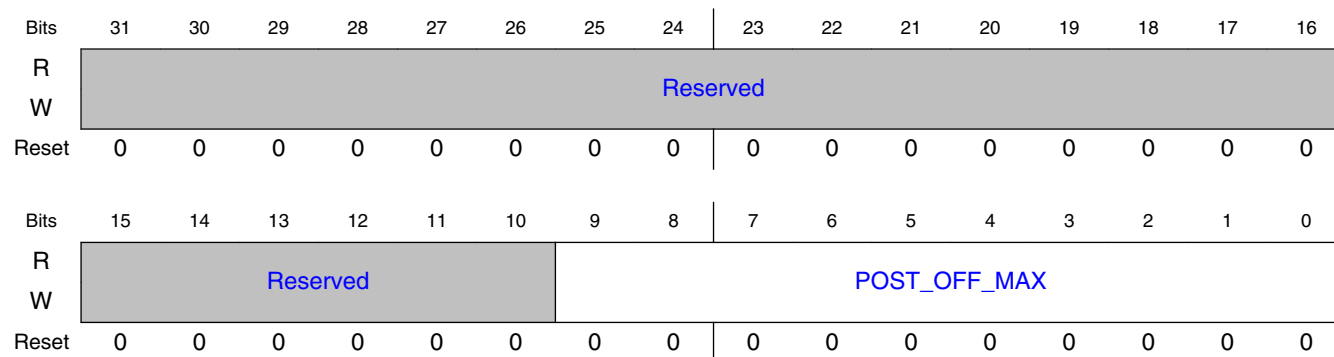
15.10.3.1.98.1 Offset

Register	Offset
HDR_PIPE2_CSCA_OMAX_2	7070h

15.10.3.1.98.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.98.3 Diagram



15.10.3.1.98.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

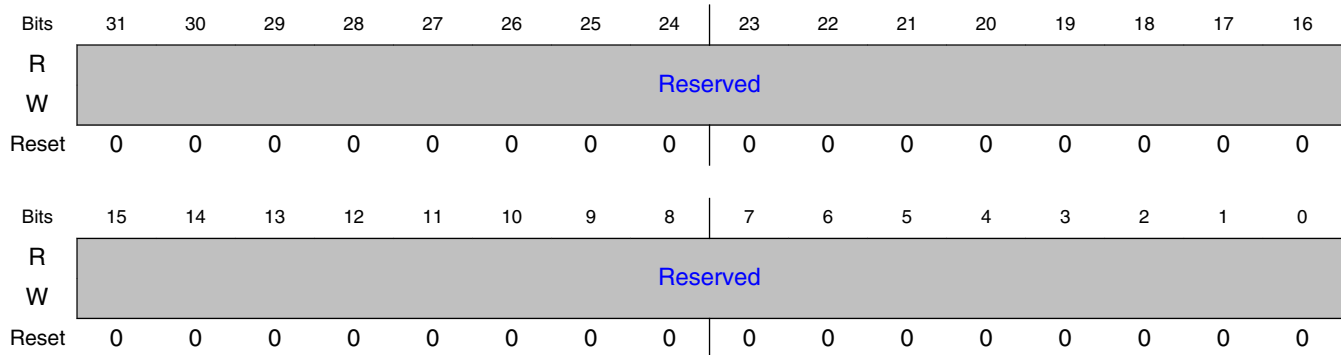
15.10.3.1.99 PIPE2: NOT USED (HDR_PIPE2_ENTRY_29)

15.10.3.1.99.1 Offset

Register	Offset
HDR_PIPE2_ENTRY_29	7074h

15.10.3.1.99.2 Function

PIPE2: NOT USED

15.10.3.1.99.3 Diagram**15.10.3.1.99.4 Fields**

Field	Function
31-0	Reserved.
—	

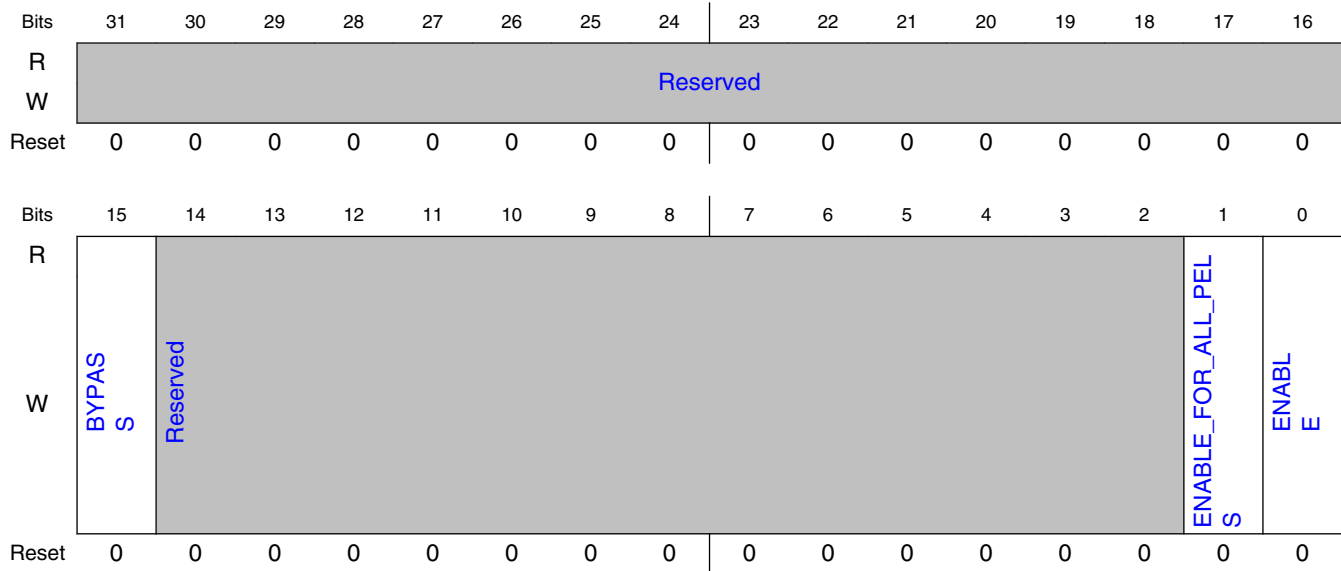
15.10.3.1.100 Pipe1 LUT control register (HDR_PIPE2_LUT_CONTROL_REG)**15.10.3.1.100.1 Offset**

Register	Offset
HDR_PIPE2_LUT_CONTROL_REG	7080h

15.10.3.1.100.2 Function

Controls the Look-Up-Table in pipe1. Control of LUT is PER PIXEL not per component. LUT has 1024 14-bit, per component, entries. LUT maybe used for conversion from non-linear (gamma corrected) to linear pixels. LUT maybe used for conversion from linear to non linear pixels (gamma corrected). LUT can contain 14-bit fixed point or 14-bit floating point pixel value. Floating point format has 9 bit unsigned mantissa with hidden bit and 5 bit, unsigned, biased exponent. {exp[4:0], mantissa[8:0]} => 1.mantissa[8:0] * 2^(exp-1) When using floating piont numbers, in this LUT, these are used in HDR OUTPUT PIPE for HDR non-linear operation and must bypass Color Space Converter B (CSCB). Pipe1 is also called channel1.

15.10.3.1.100.3 Diagram



15.10.3.1.100.4 Fields

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this LUT 1: Pixels pass thru this LUT unmodified
14-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This LUT is enabled only for blended pixels 1: This LUT is enabled all pixels
0 ENABLE	0: Don't enable this LUT: Pixels pass thru the LUT unmodified 1: This LUT is enabled for current picture

15.10.3.1.101 Pipe1 Colorspace Converter B control. (HDR_PIPE2_CSCB_CONTROL_REG)

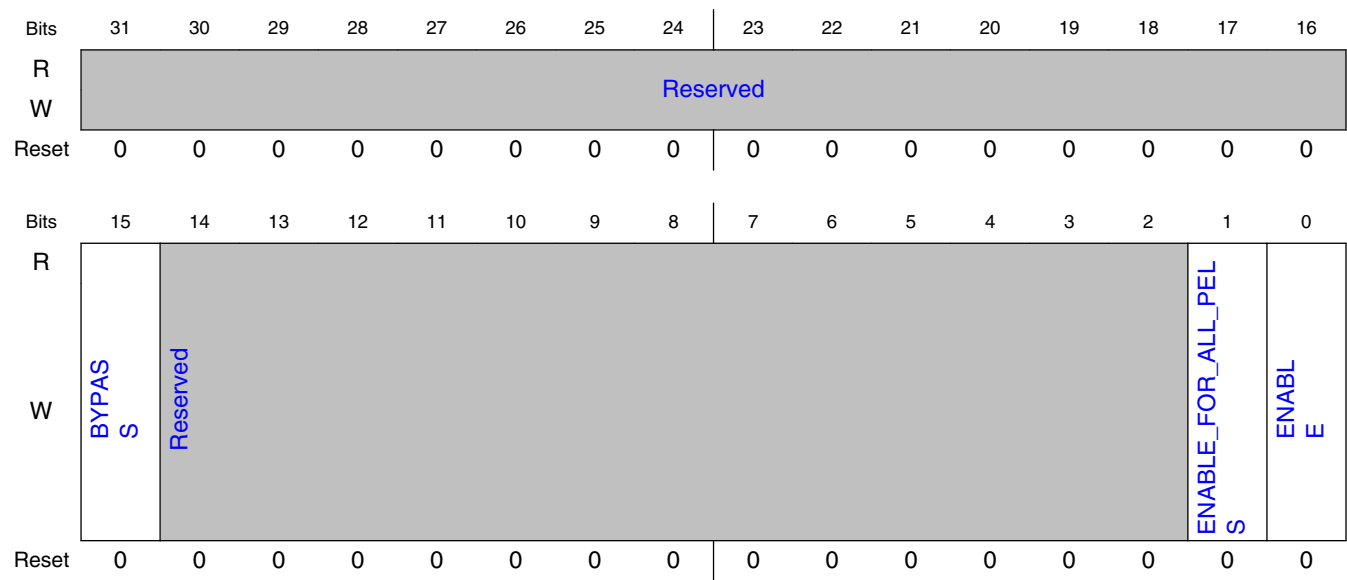
15.10.3.1.101.1 Offset

Register	Offset
HDR_PIPE2_CSCB_CONTROL_REG	7800h

15.10.3.1.101.2 **Function**

Controls the Color-Space-Converter-B (CSCB) in pipe1. This CSCB takes in 14-bit pixel components (component 0, component 1, component 2). A per-component offset (called pre-offset) is added to the pixel. After the pre-offset operation the pixel components are clipped. MAX_RANGE of PRE_CLIP is [-512, 1023]. A 3x3 matrix multiply (constant coefficients H(x,y)) is performed on the pixel to take to a different color space. After this matrix multiply, a 28 bit signed offset (post-offset) is added to the normalized result of the matrix multiply. After the post-offset operation the results are clipped. MAX_RANGE of POST_CLIP is [0,1023] Output of this CSCA are 28 bit per component pixels. After CSCB the pixels are fed into the Blender. Pipe1 is also called channel1.

15.10.3.1.101.3 **Diagram**



15.10.3.1.101.4 **Fields**

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this CSC 1: Pixels pass thru this CSC unmodified
14-2 —	Reserved.

Table continues on the next page...

Field	Function
1 ENABLE_FOR_ ALL_PELS	0: This CSC is enabled only for blended pixels 1: This CSC is enabled all pixels
0 ENABLE	0: Don't enable this CSC: Pixels pass thru the CSC unmodified 1: This CSC is enabled for current picture

15.10.3.1.102 Pipe1 Colorspace Converter A (CSCB) h(0,0) matrix coefficient (HDR_PIPE2_CSCB_H00)

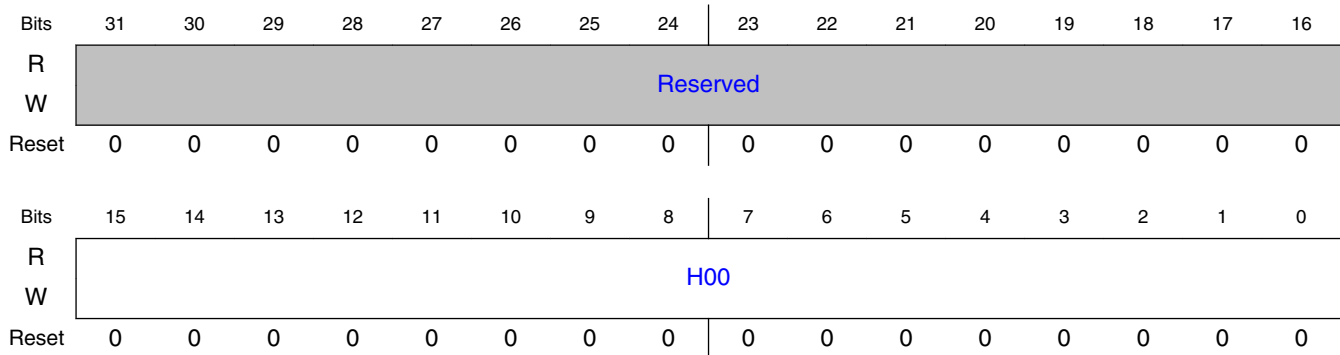
15.10.3.1.102.1 Offset

Register	Offset
HDR_PIPE2_CSCB_H00	7804h

15.10.3.1.102.2 Function

h(0,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.102.3 Diagram



15.10.3.1.102.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H00	h(0,0) 16 bit signed coefficient

15.10.3.1.103 Pipe1 Colorspace Converter B (CSCB) h(1,0) matrix coefficient (HDR_PIPE2_CSCB_H10)

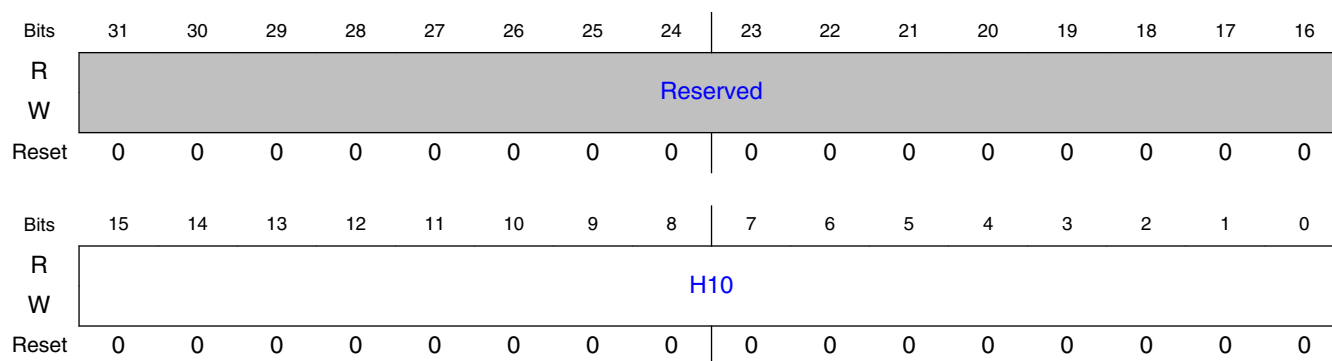
15.10.3.1.103.1 Offset

Register	Offset
HDR_PIPE2_CSCB_H10	7808h

15.10.3.1.103.2 Function

h(1,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.103.3 Diagram



15.10.3.1.103.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H10	h(1,0) 16 bit signed coefficient

15.10.3.1.104 Pipe1 Colorspace Converter B (CSCB) h(2,0) matrix coefficient (HDR_PIPE2_CSCB_H20)

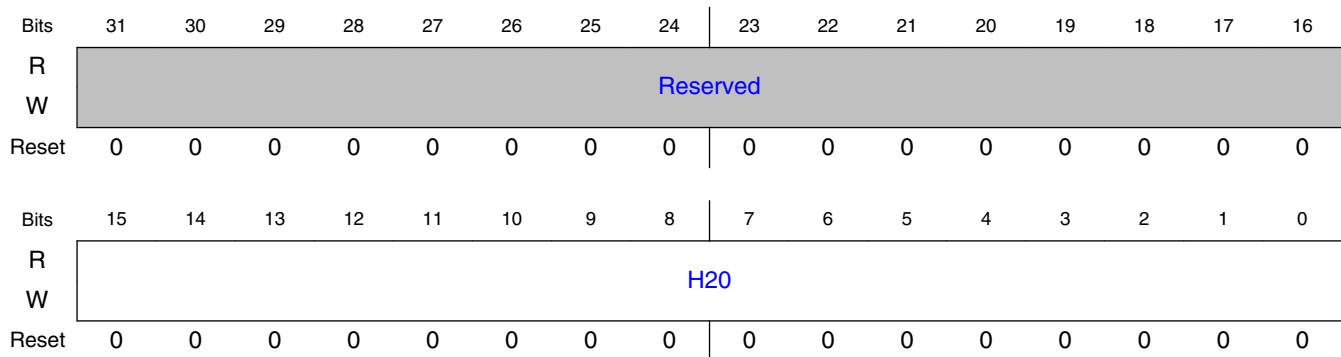
15.10.3.1.104.1 Offset

Register	Offset
HDR_PIPE2_CSCB_H20	780Ch

15.10.3.1.104.2 Function

h(2,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.104.3 Diagram



15.10.3.1.104.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H20	h(2,0) 16 bit signed coefficient

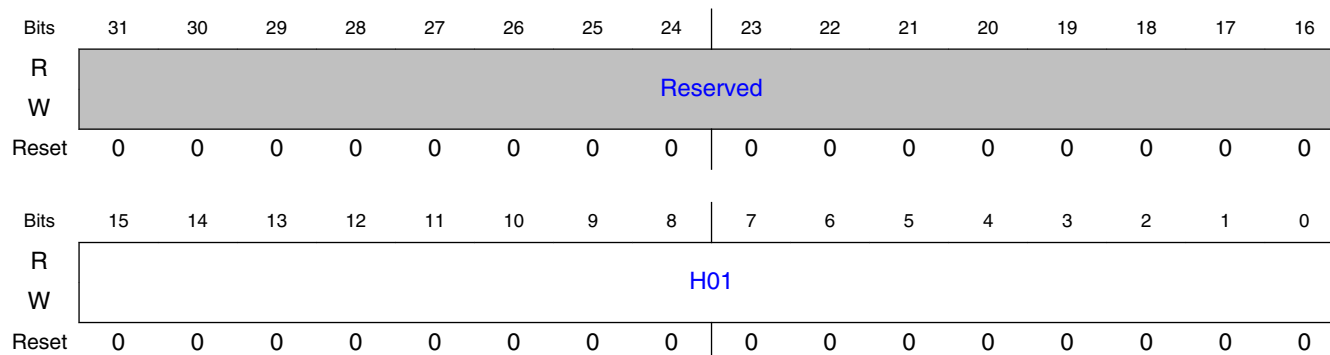
15.10.3.1.105 Pipe1 Colorspace Converter B (CSCB) h(0,1) matrix coefficient (HDR_PIPE2_CSCB_H01)

15.10.3.1.105.1 Offset

Register	Offset
HDR_PIPE2_CSCB_H01	7810h

15.10.3.1.105.2 Function

$h(0,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.105.3 Diagram**15.10.3.1.105.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H01	$h(0,1)$ 16 bit signed coefficient

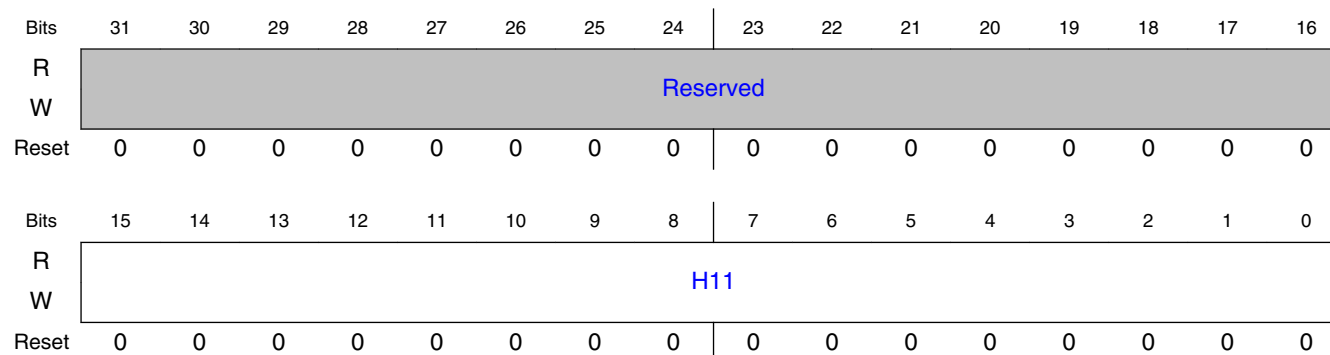
15.10.3.1.106 Pipe1 Colorspace Converter B (CSCB) $h(1,1)$ matrix coefficient (HDR_PIPE2_CSCB_H11)**15.10.3.1.106.1 Offset**

Register	Offset
HDR_PIPE2_CSCB_H11	7814h

15.10.3.1.106.2 Function

$h(1,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.106.3 Diagram



15.10.3.1.106.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H11	h(1,1) 16 bit signed coefficient

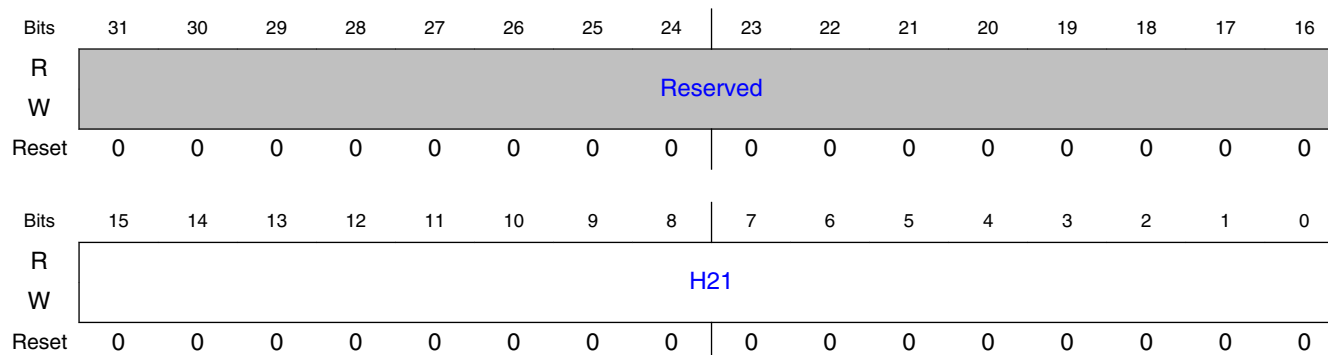
15.10.3.1.107 Pipe1 Colorspace Converter B (CSCB) h(2,1) matrix coefficient (HDR_PIPE2_CSCB_H21)

15.10.3.1.107.1 Offset

Register	Offset
HDR_PIPE2_CSCB_H21	7818h

15.10.3.1.107.2 Function

h(2,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.107.3 Diagram**15.10.3.1.107.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H21	h(2,1) 16 bit signed coefficient

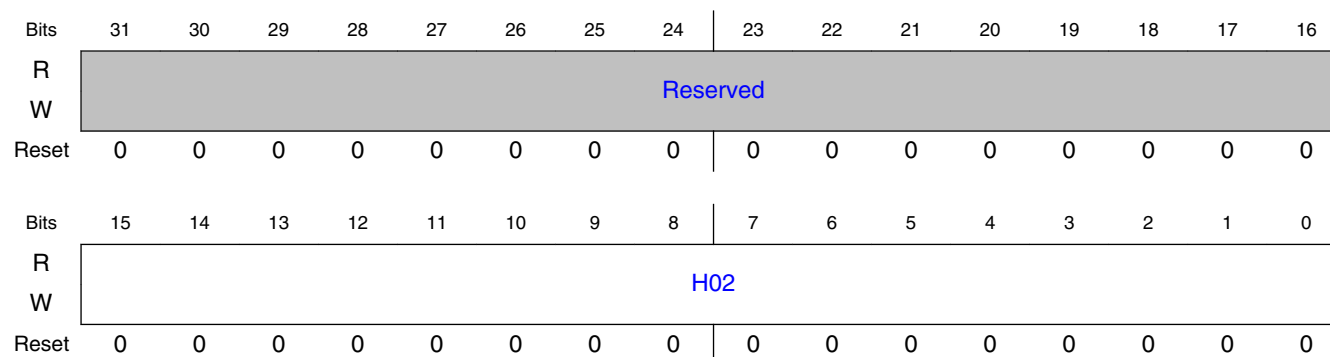
15.10.3.1.108 Pipe1 Colorspace Converter B (CSCB) h(0,2) matrix coefficient (HDR_PIPE2_CSCB_H02)**15.10.3.1.108.1 Offset**

Register	Offset
HDR_PIPE2_CSCB_H02	781Ch

15.10.3.1.108.2 Function

h(0,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.108.3 Diagram



15.10.3.1.108.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H02	h(0,2) 16 bit signed coefficient

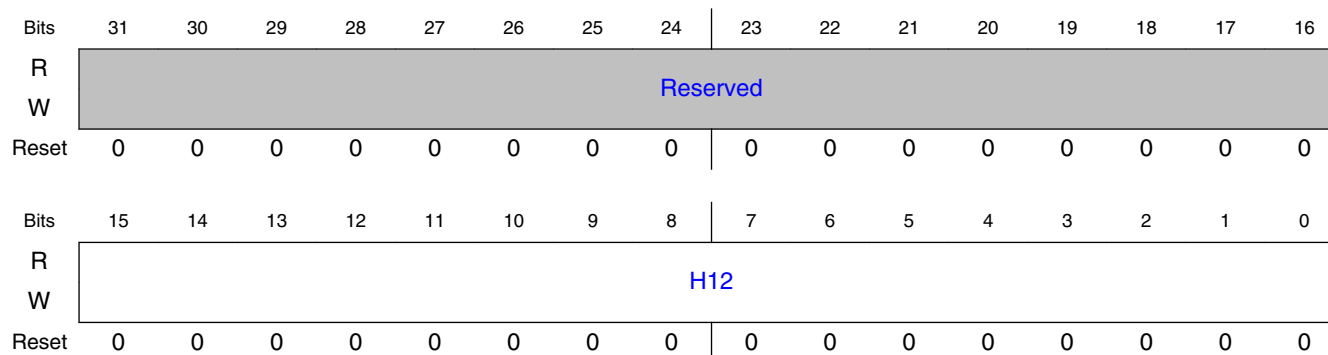
15.10.3.1.109 Pipe1 Colorspace Converter B (CSCB) h(1,2) matrix coefficient (HDR_PIPE2_CSCB_H12)

15.10.3.1.109.1 Offset

Register	Offset
HDR_PIPE2_CSCB_H12	7820h

15.10.3.1.109.2 Function

h(1,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.109.3 Diagram**15.10.3.1.109.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H12	h(1,2) 16 bit signed coefficient

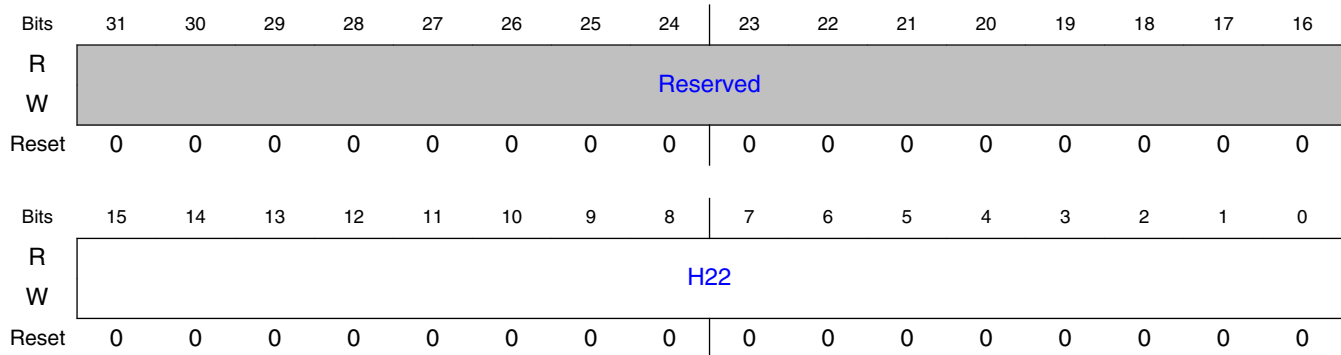
15.10.3.1.110 Pipe1 Colorspace Converter B (CSCB) h(2,2) matrix coefficient (HDR_PIPE2_CSCB_H22)**15.10.3.1.110.1 Offset**

Register	Offset
HDR_PIPE2_CSCB_H22	7824h

15.10.3.1.110.2 Function

h(2,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.110.3 Diagram



15.10.3.1.110.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H22	h(2,2) 16 bit signed coefficient

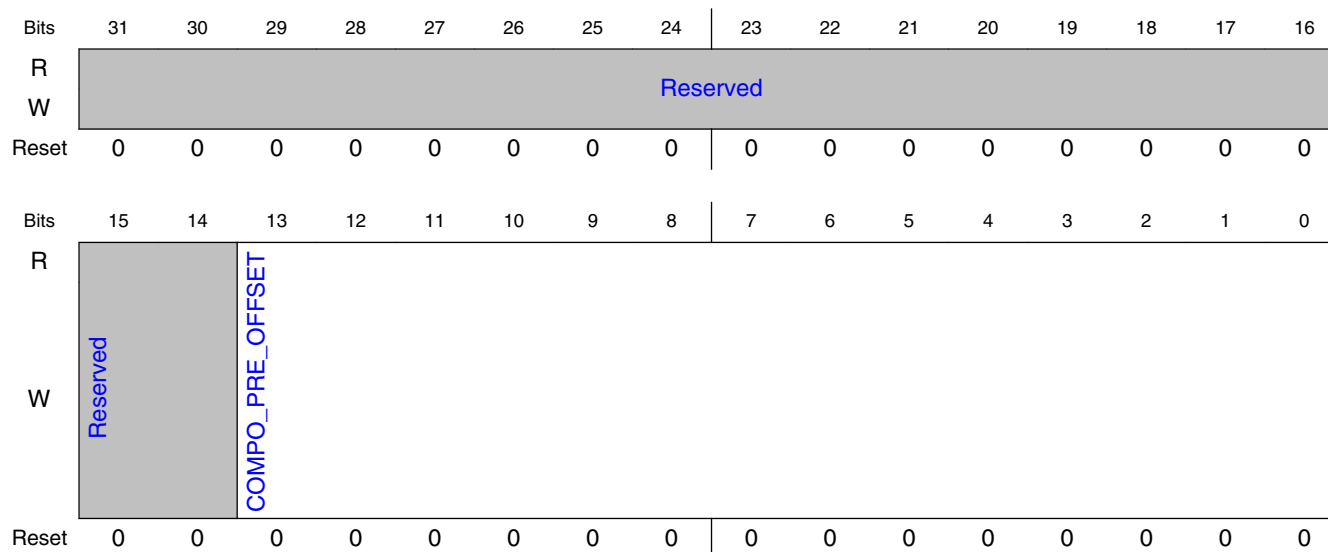
15.10.3.1.111 Pipe1 Colorspace Converter B (CSCB) component 0 pre-offset (HDR_PIPE2_CSCB_IO_0)

15.10.3.1.111.1 Offset

Register	Offset
HDR_PIPE2_CSCB_IO_0	7828h

15.10.3.1.111.2 Function

A signed 14-bit offset is added to component 0 (R,Y) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.111.3 Diagram**15.10.3.1.111.4 Fields**

Field	Function
31-14 —	Reserved.
13-0 COMPO_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 0 of the pixel

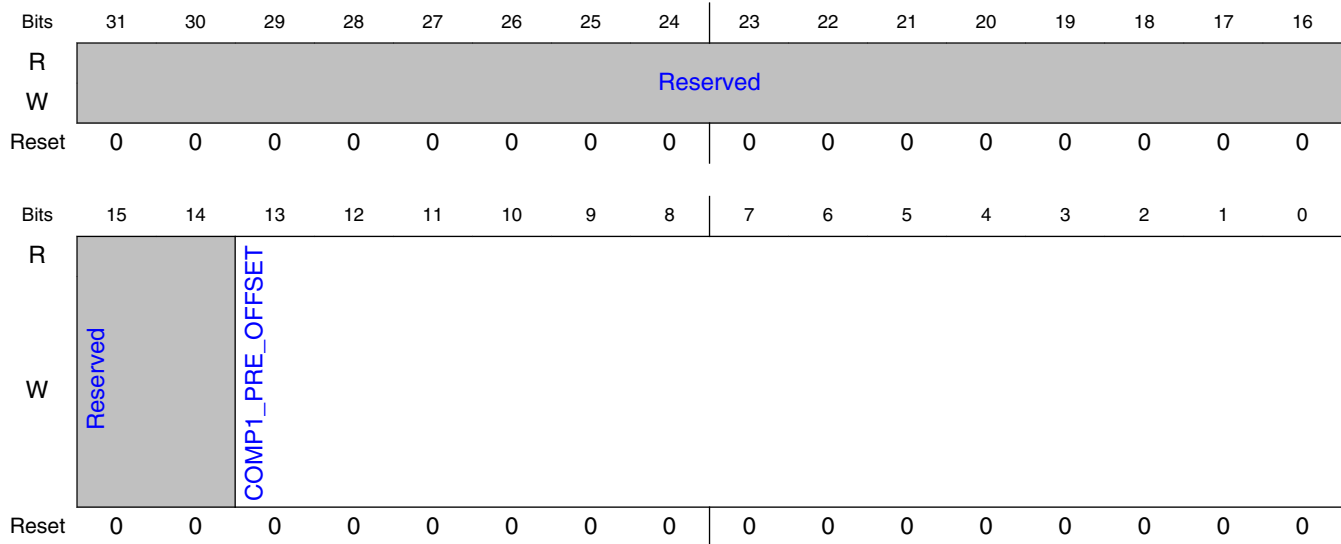
15.10.3.1.112 Pipe1 Colorspace Converter B (CSCB) component 1 pre-offset (HDR_PIPE2_CSCB_IO_1)**15.10.3.1.112.1 Offset**

Register	Offset
HDR_PIPE2_CSCB_IO_1	782Ch

15.10.3.1.112.2 Function

A signed 14-bit offset is added to component 1 (G,Cb) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.112.3 Diagram



15.10.3.1.112.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 1 of the pixel

15.10.3.1.113 Pipe1 Colorspace Converter B (CSCB) component 2 pre-offset (HDR_PIPE2_CSCB_IO_2)

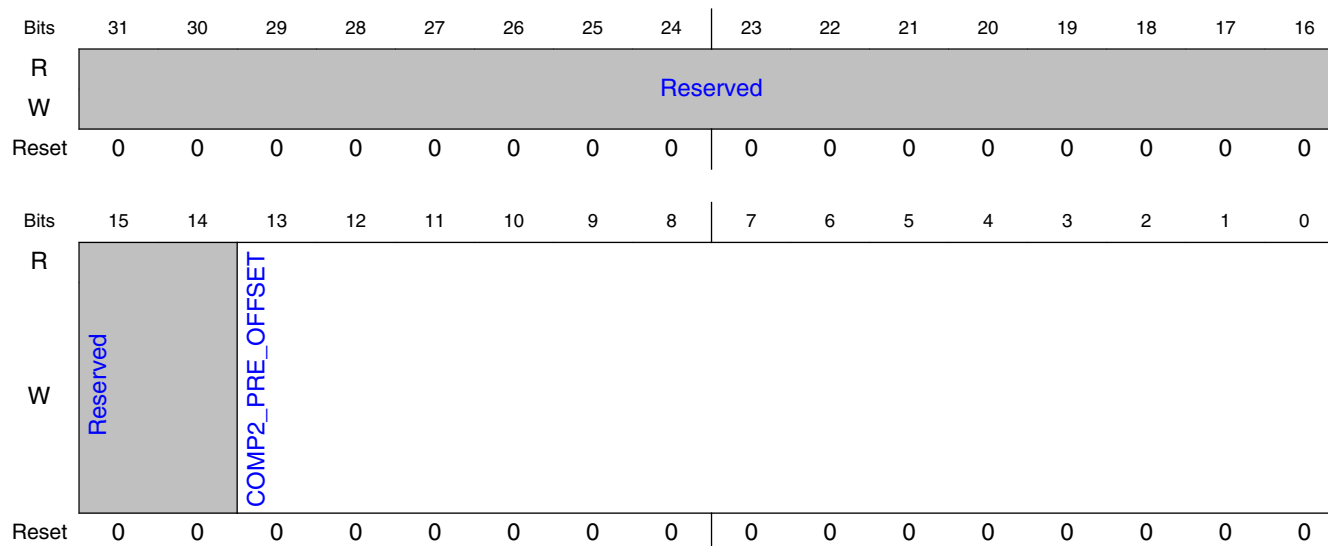
15.10.3.1.113.1 Offset

Register	Offset
HDR_PIPE2_CSCB_IO_2	7830h

15.10.3.1.113.2 Function

A signed 14-bit offset is added to component 2 (B,Cr) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.113.3 Diagram



15.10.3.1.113.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 2 of the pixel

15.10.3.1.114 Pipe1 Colorspace Converter B (CSCB) component 0 clip min. (HDR_PIPE2_CSCB_IO_MIN_0)

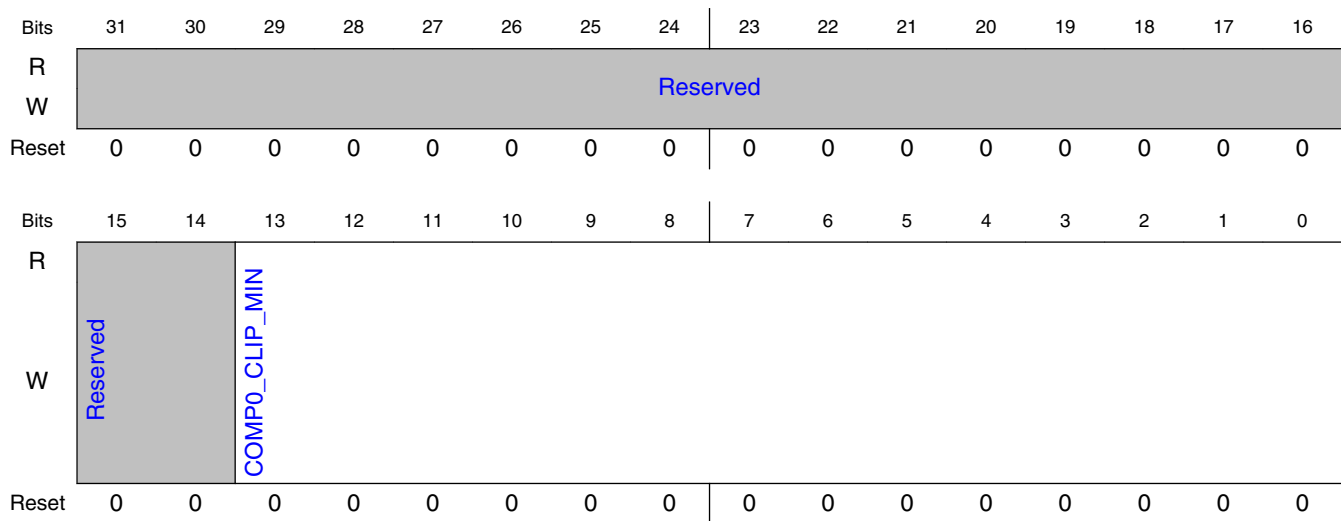
15.10.3.1.114.1 Offset

Register	Offset
HDR_PIPE2_CSCB_IO_MIN_0	7834h

15.10.3.1.114.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192, 8191]. Pipe1 is also called channel1.

15.10.3.1.114.3 Diagram



15.10.3.1.114.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP0_CLIP_MIN	This 14-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.115 Pipe1 Colorspace Converter B (CSCB) component 1 clip min. (HDR_PIPE2_CSCB_IO_MIN_1)

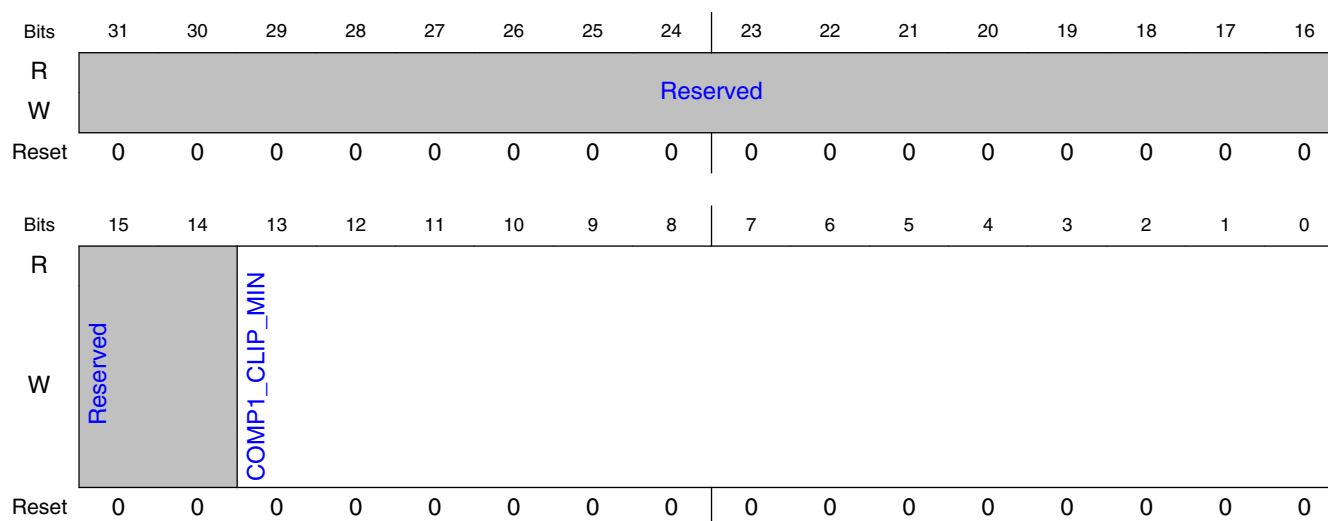
15.10.3.1.115.1 Offset

Register	Offset
HDR_PIPE2_CSCB_IO_MIN_1	7838h

15.10.3.1.115.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192 , 8191]. Pipe1 is also called channel1.

15.10.3.1.115.3 Diagram



15.10.3.1.115.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_CLIP_MIN	This 14-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.116 Pipe1 Colorspace Converter B (CSCB) component 2 clip min. (HDR_PIPE2_CSCB_IO_MIN_2)

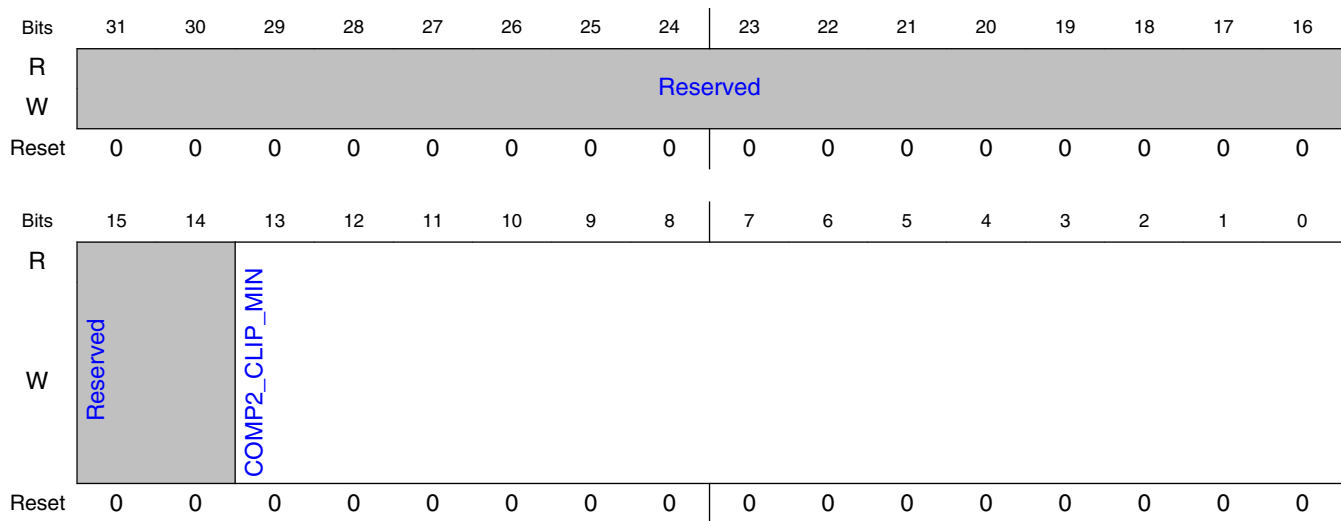
15.10.3.1.116.1 Offset

Register	Offset
HDR_PIPE2_CSCB_IO_MIN_2	783Ch

15.10.3.1.116.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192 , 1891]. Pipe1 is also called channel1.

15.10.3.1.116.3 Diagram



15.10.3.1.116.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.117 Pipe1 Colorspace Converter B (CSCB) component 0 clip max value. (HDR_PIPE2_CSCB_IO_MAX_0)

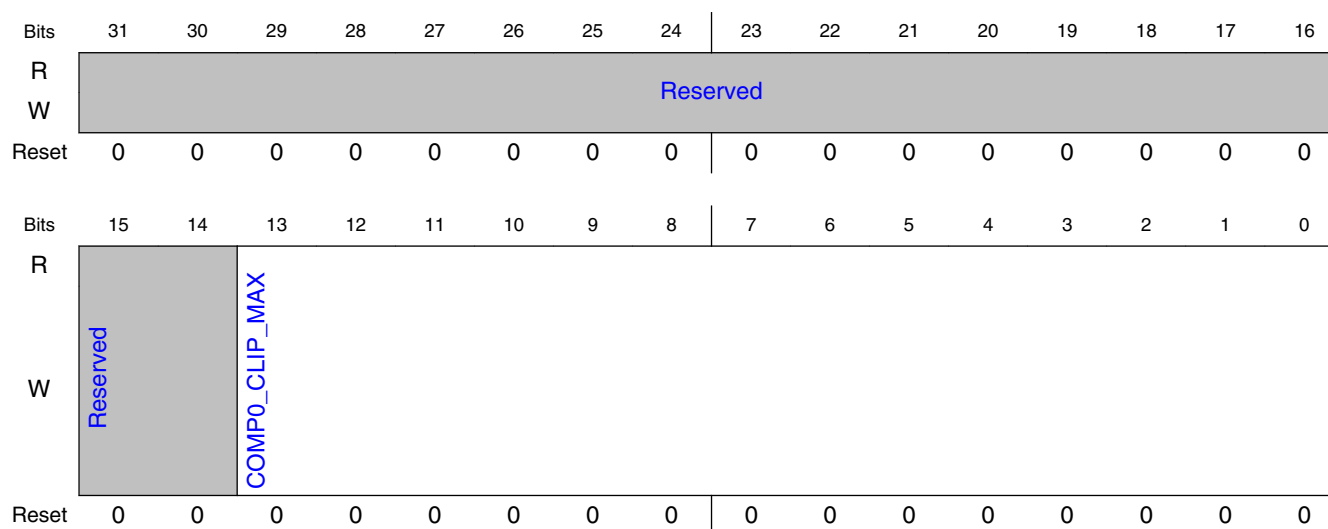
15.10.3.1.117.1 Offset

Register	Offset
HDR_PIPE2_CSCB_IO_MAX_0	7840h

15.10.3.1.117.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0, 16383]. Pipe1 is also called channel1.

15.10.3.1.117.3 Diagram



15.10.3.1.117.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP0_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.118 Pipe1 Colorspace Converter B (CSCB) component 1 clip max value. (HDR_PIPE2_CSCB_IO_MAX_1)

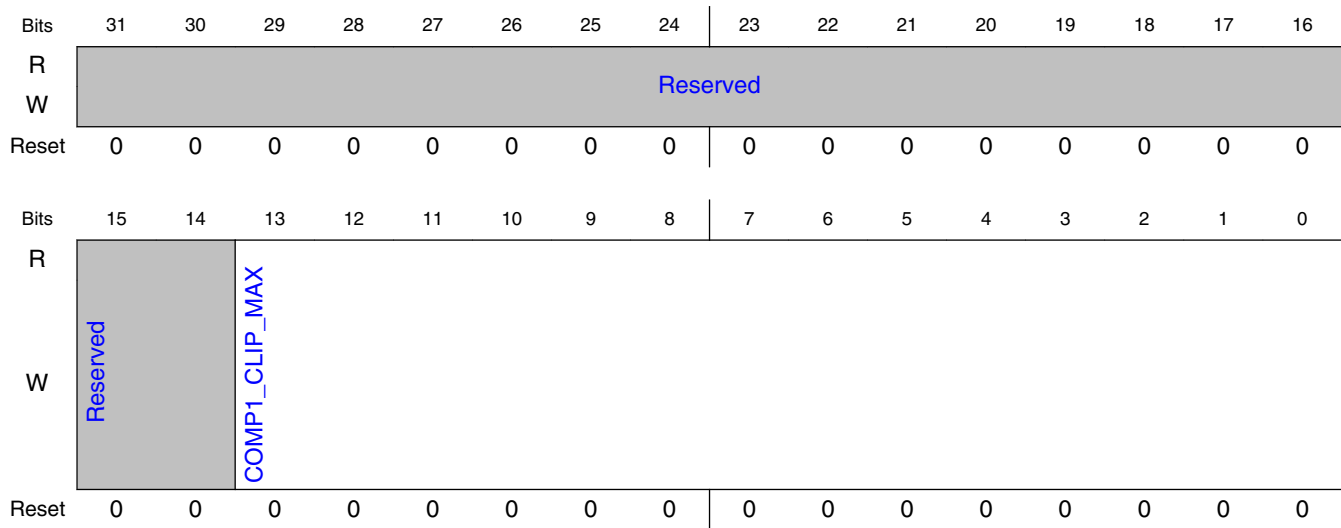
15.10.3.1.118.1 Offset

Register	Offset
HDR_PIPE2_CSCB_IO_MAX_1	7844h

15.10.3.1.118.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,16383]. Pipe1 is also called channel1.

15.10.3.1.118.3 Diagram



15.10.3.1.118.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.119 Pipe1 Colorspace Converter B (CSCB) component 2 clip max value. (HDR_PIPE2_CSCB_IO_MAX_2)

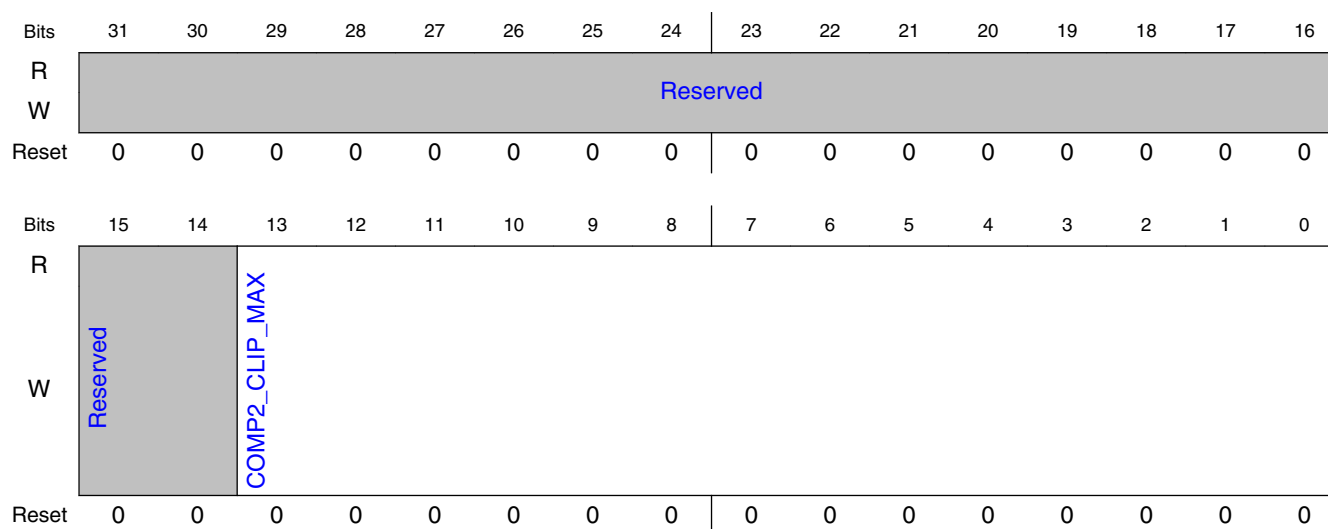
15.10.3.1.119.1 Offset

Register	Offset
HDR_PIPE2_CSCB_IO_MAX_2	7848h

15.10.3.1.119.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,16383]. Pipe1 is also called channel1.

15.10.3.1.119.3 Diagram



15.10.3.1.119.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.120 Pipe1 Colorspace Converter B (CSCB) normalization factor (HDR_PIPE2_CSCB_NORM)

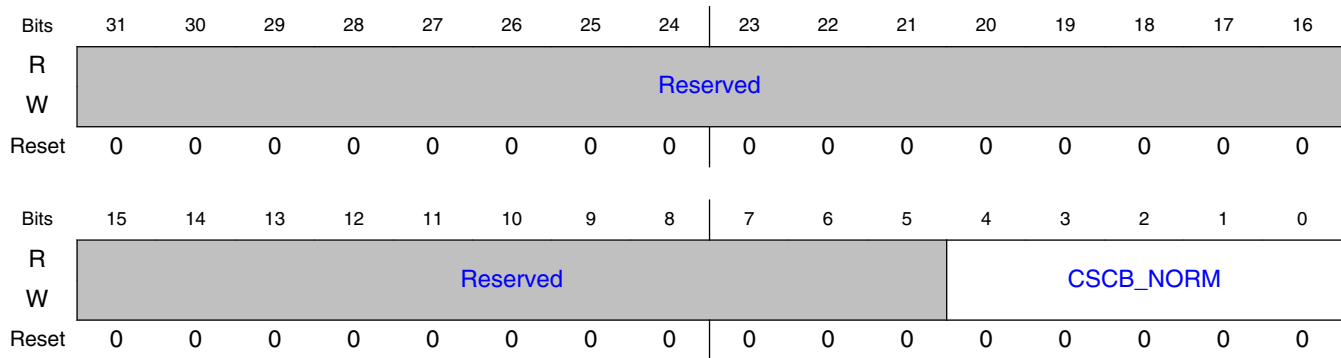
15.10.3.1.120.1 Offset

Register	Offset
HDR_PIPE2_CSCB_NORM	784Ch

15.10.3.1.120.2 Function

After the CSC matrix multiply, all components of the result are shifted right by the amount in this register. This is a signed right shift. Pipe1 is also called channel1.

15.10.3.1.120.3 Diagram



15.10.3.1.120.4 Fields

Field	Function
31-5 —	Reserved.
4-0 CSCB_NORM	This 5-bit unsigned value is size if arithmetic shift after matrix multiply.

15.10.3.1.121 Pipe1 Colorspace Converter B (CSCB): Post offset component 0 (HDR_PIPE2_CSCB_OO_0)

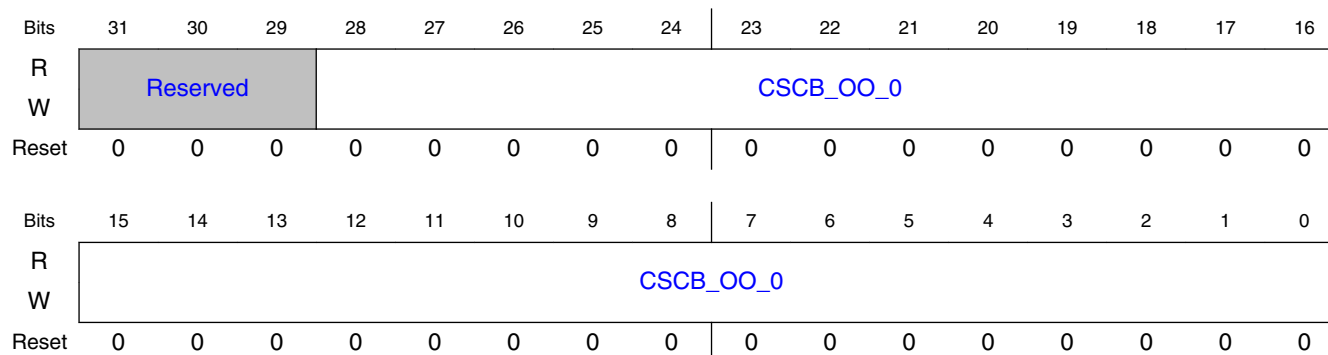
15.10.3.1.121.1 Offset

Register	Offset
HDR_PIPE2_CSCB_OO_0	7850h

15.10.3.1.121.2 Function

After the CSC normalization, this offset value is added to component 0. This is a signed 29-bit number. Pipe1 is also called channel1.

15.10.3.1.121.3 Diagram



15.10.3.1.121.4 Fields

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_0	Output Offset (OO) This is a signed 29-bit number. Per component

15.10.3.1.122 Pipe1 Colorspace Converter B (CSCB): Post offset component 1 (HDR_PIPE2_CSCB_OO_1)

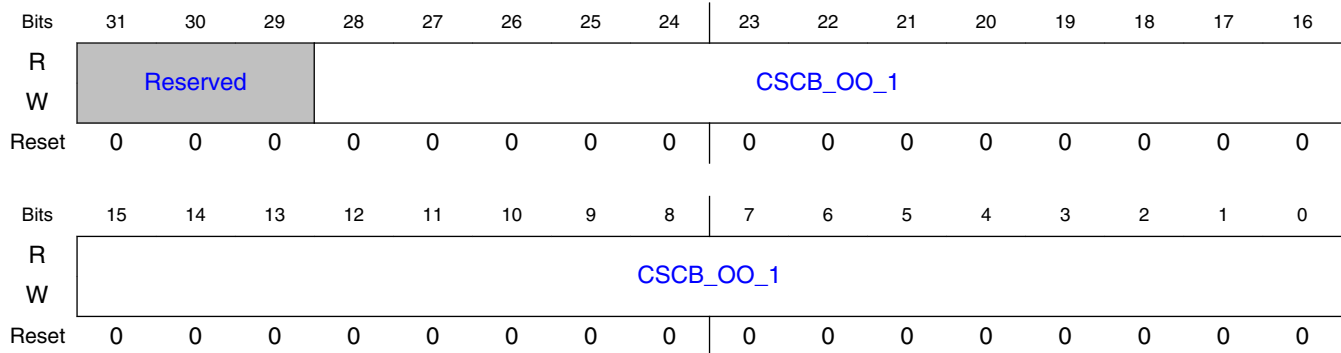
15.10.3.1.122.1 Offset

Register	Offset
HDR_PIPE2_CSCB_OO_1	7854h

15.10.3.1.122.2 Function

After the CSC normalization, this offset value is added to component 1. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.122.3 Diagram



15.10.3.1.122.4 Fields

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_1	Output Offset (OO) This is a signed 29-bit number. Per component

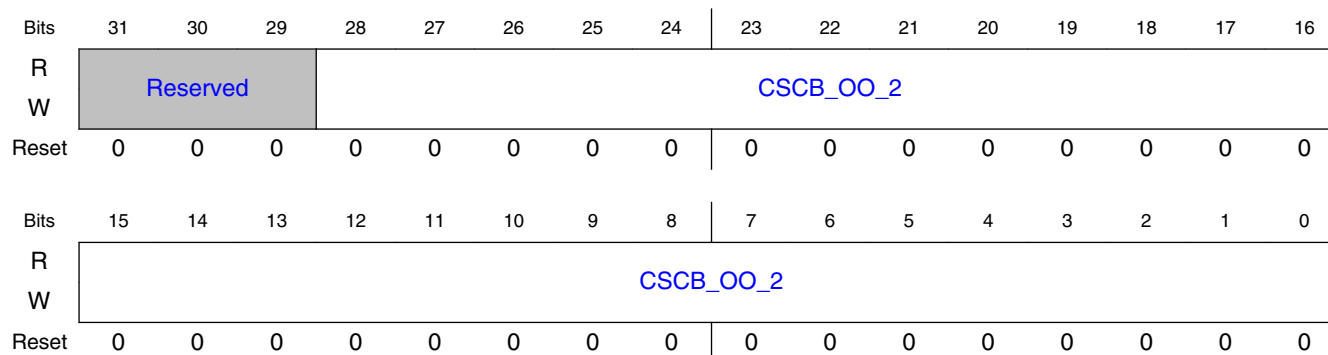
15.10.3.1.123 Pipe1 Colorspace Converter B (CSCB): Post offset component 2 (HDR_PIPE2_CSCB_OO_2)

15.10.3.1.123.1 Offset

Register	Offset
HDR_PIPE2_CSCB_OO_2	7858h

15.10.3.1.123.2 Function

After the CSC normalization, this offset value is added to component 2. This is a signed 29-bit number. Pipe1 is also called channel1.

15.10.3.1.123.3 Diagram**15.10.3.1.123.4 Fields**

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_2	Output Offset (OO) This is a signed 29-bit number. Per component

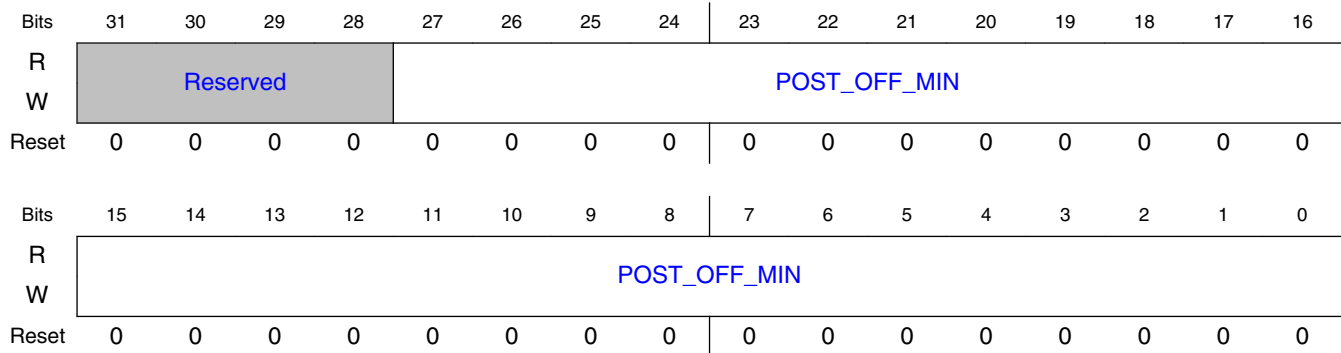
15.10.3.1.124 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 0 (HDR_PIPE2_CSCB_OMIN_0)**15.10.3.1.124.1 Offset**

Register	Offset
HDR_PIPE2_CSCB_OMIN_0	785Ch

15.10.3.1.124.2 Function

After the post offset is added this component is clipped. This is the minimum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.124.3 Diagram



15.10.3.1.124.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minum clipped pixel component.

15.10.3.1.125 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 1 (HDR_PIPE2_CSCB_OMIN_1)

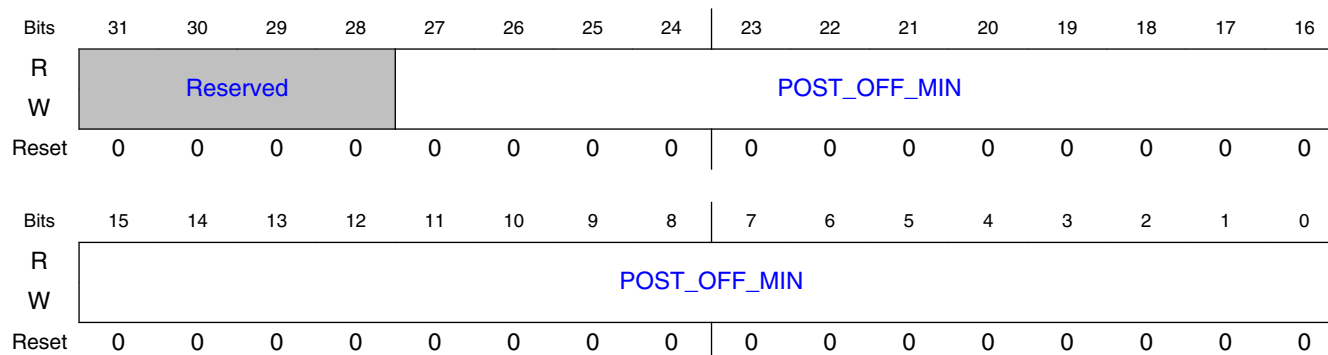
15.10.3.1.125.1 Offset

Register	Offset
HDR_PIPE2_CSCB_OMIN_1	7860h

15.10.3.1.125.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.125.3 Diagram



15.10.3.1.125.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minum clipped pixel component.

15.10.3.1.126 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 2 (HDR_PIPE2_CSCB_OMIN_2)

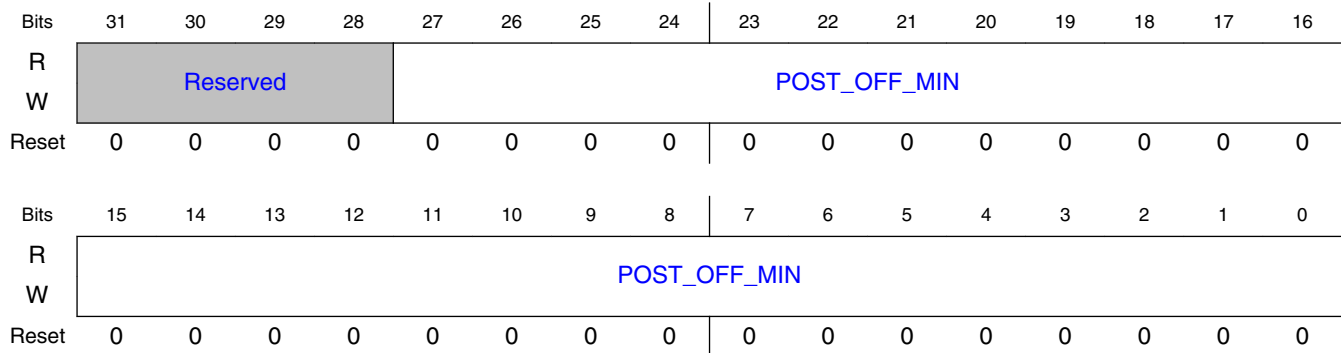
15.10.3.1.126.1 Offset

Register	Offset
HDR_PIPE2_CSCB_OMIN_2	7864h

15.10.3.1.126.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.126.3 Diagram



15.10.3.1.126.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minimum clipped pixel component.

15.10.3.1.127 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 0 (HDR_PIPE2_CSCB_OMAX_0)

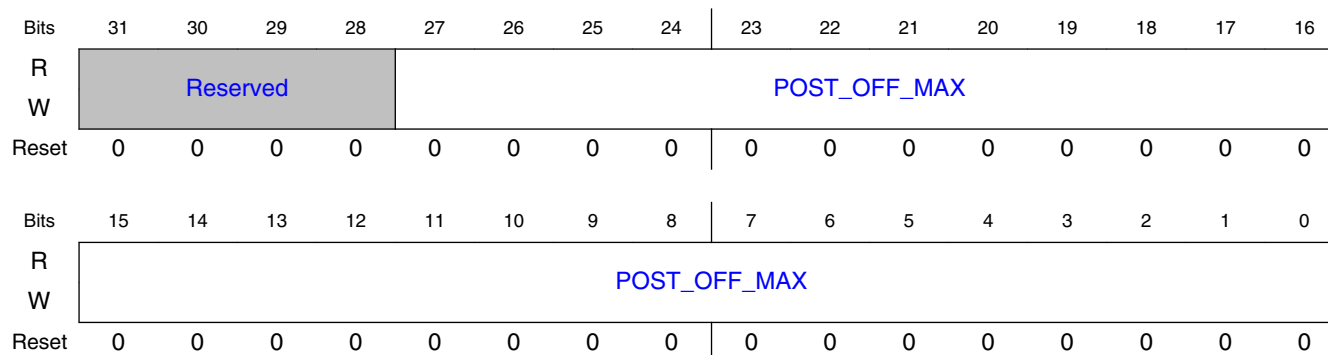
15.10.3.1.127.1 Offset

Register	Offset
HDR_PIPE2_CSCB_OMAX_0	7868h

15.10.3.1.127.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.127.3 Diagram



15.10.3.1.127.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MAX	Maximum clipped pixel component.

15.10.3.1.128 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 1 (HDR_PIPE2_CSCB_OMAX_1)

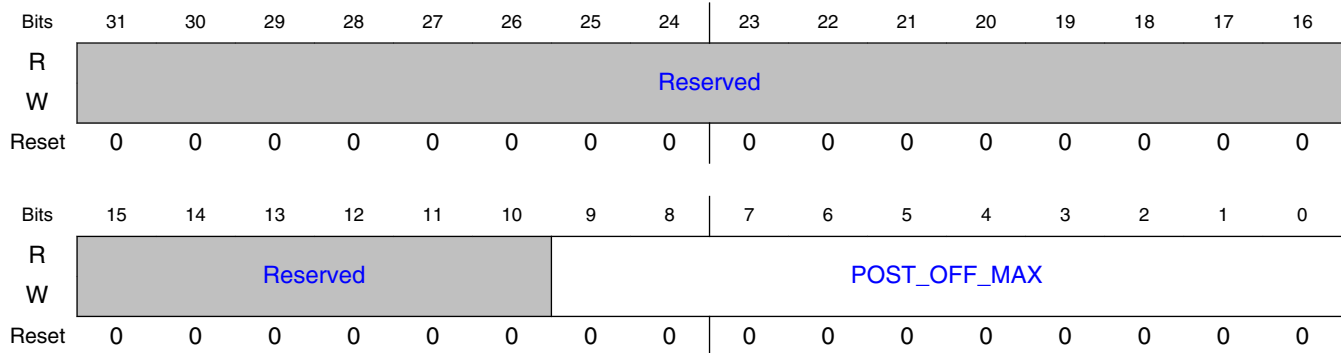
15.10.3.1.128.1 Offset

Register	Offset
HDR_PIPE2_CSCB_OMAX_1	786Ch

15.10.3.1.128.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.128.3 Diagram



15.10.3.1.128.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component.

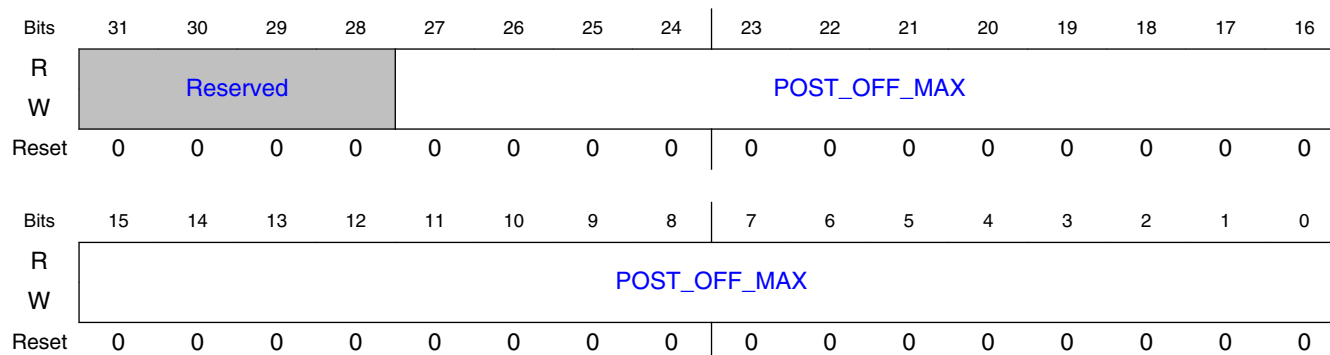
15.10.3.1.129 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 2 (HDR_PIPE2_CSCB_OMAX_2)

15.10.3.1.129.1 Offset

Register	Offset
HDR_PIPE2_CSCB_OMAX_2	7870h

15.10.3.1.129.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.129.3 Diagram**15.10.3.1.129.4 Fields**

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MAX	Maximun clipped pixel component.

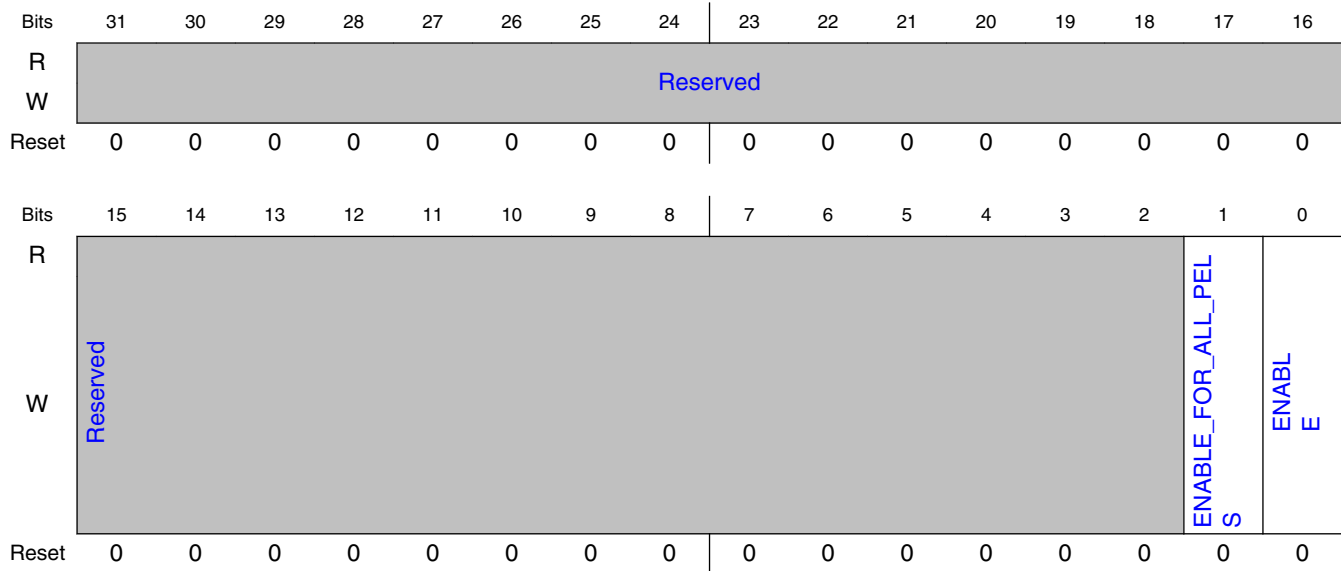
15.10.3.1.130 Pipe1 floating point to fixed point control (HDR_PIPE2_FL2FX)**15.10.3.1.130.1 Offset**

Register	Offset
HDR_PIPE2_FL2FX	7874h

15.10.3.1.130.2 Function

Pipe1 floating point to fixed point control Pipe1 is also called channel1.

15.10.3.1.130.3 Diagram



15.10.3.1.130.4 Fields

Field	Function
31-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This Float to Fixed operation is enabled only for blended pixels 1: This Float to Fixed operation is enabled all pixels
0 ENABLE	0: Don't enable this Float-to-Fixed converter: Pixels pass thru the Float-to-Fixed unmodified 1: This Float-to-Fixed converter is enabled for current picture

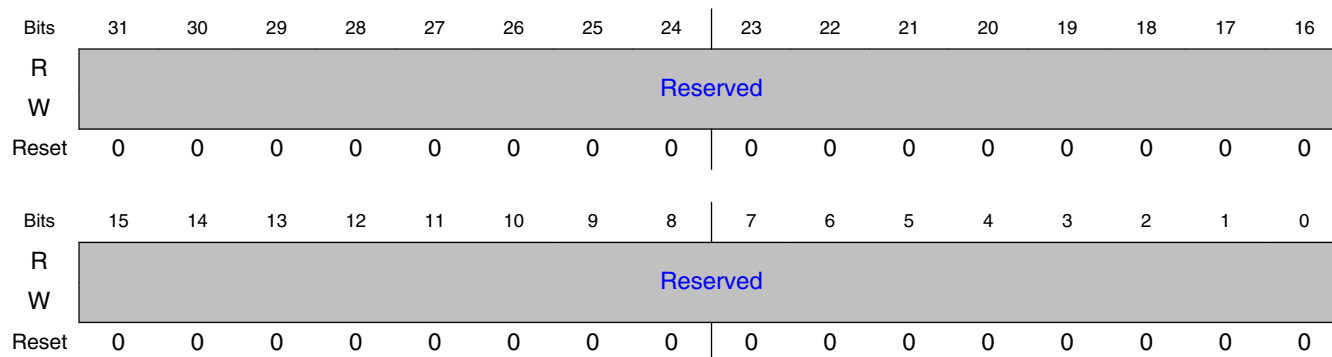
15.10.3.1.131 PIPE2: NOT USED (HDR_PIPE2_ENTRY_30)

15.10.3.1.131.1 Offset

Register	Offset
HDR_PIPE2_ENTRY_30	7878h

15.10.3.1.131.2 Function

PIPE2: NOT USED

15.10.3.1.131.3 Diagram**15.10.3.1.131.4 Fields**

Field	Function
31-0	Reserved.
—	

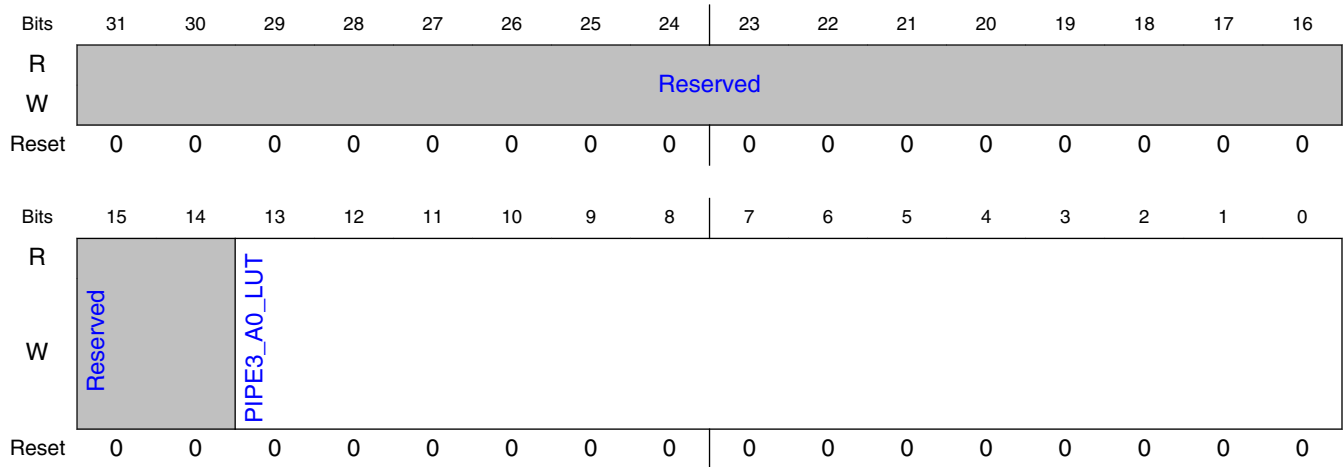
15.10.3.1.132 A0 component Look-Up-Table. (LUT) (PIPE3_A0_LUT)**15.10.3.1.132.1 Offset**

Register	Offset
PIPE3_A0_LUT	8000h

15.10.3.1.132.2 Function

The LUT table has 1024 entries. A0 component may be R or Y or other

15.10.3.1.132.3 Diagram



15.10.3.1.132.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE3_A0_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.133 A1 component Look-Up-Table. (LUT) (PIPE3_A1_LUT)

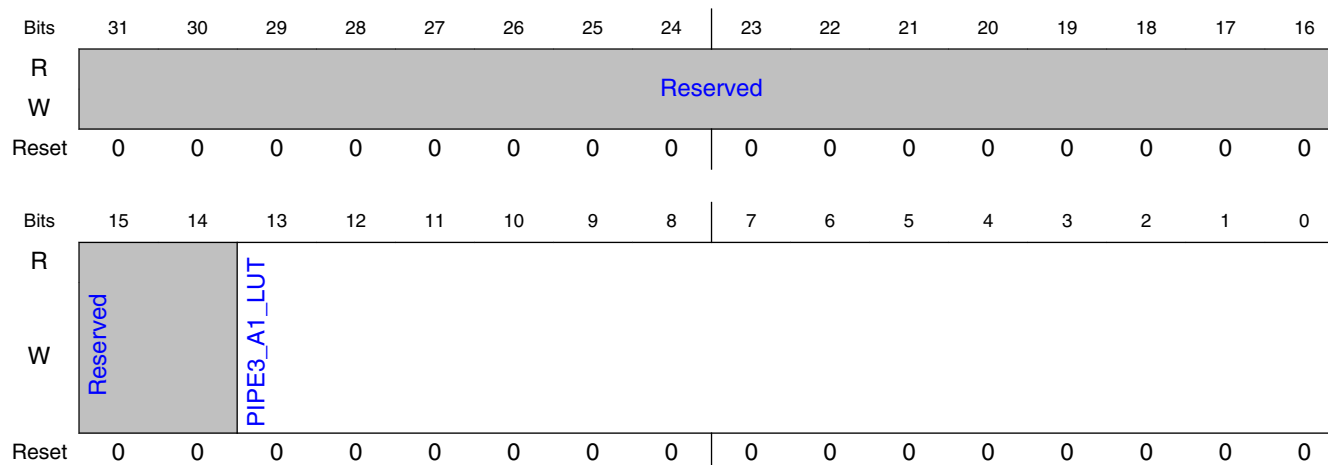
15.10.3.1.133.1 Offset

Register	Offset
PIPE3_A1_LUT	9000h

15.10.3.1.133.2 Function

The LUT table has 1024 entries. A1 component may be G or Cb or other

15.10.3.1.133.3 Diagram



15.10.3.1.133.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE3_A1_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.134 A2 component Look-Up-Table. (LUT) (PIPE3_A2_LUT)

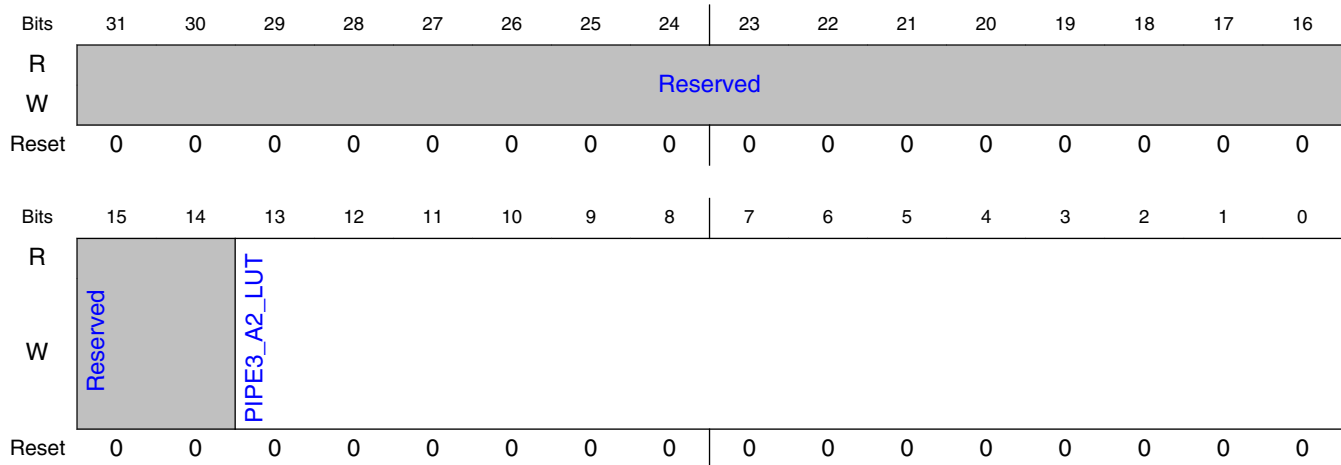
15.10.3.1.134.1 Offset

Register	Offset
PIPE3_A2_LUT	A000h

15.10.3.1.134.2 Function

The LUT table has 1024 entries. A2 component may be B or Cr or other

15.10.3.1.134.3 Diagram



15.10.3.1.134.4 Fields

Field	Function
31-14 —	Reserved.
13-0 PIPE3_A2_LUT	The LUT entries are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.135 Pipe1 Colorspace Converter A control. (HDR_PIPE3_CSCA_CONTROL_REG)

15.10.3.1.135.1 Offset

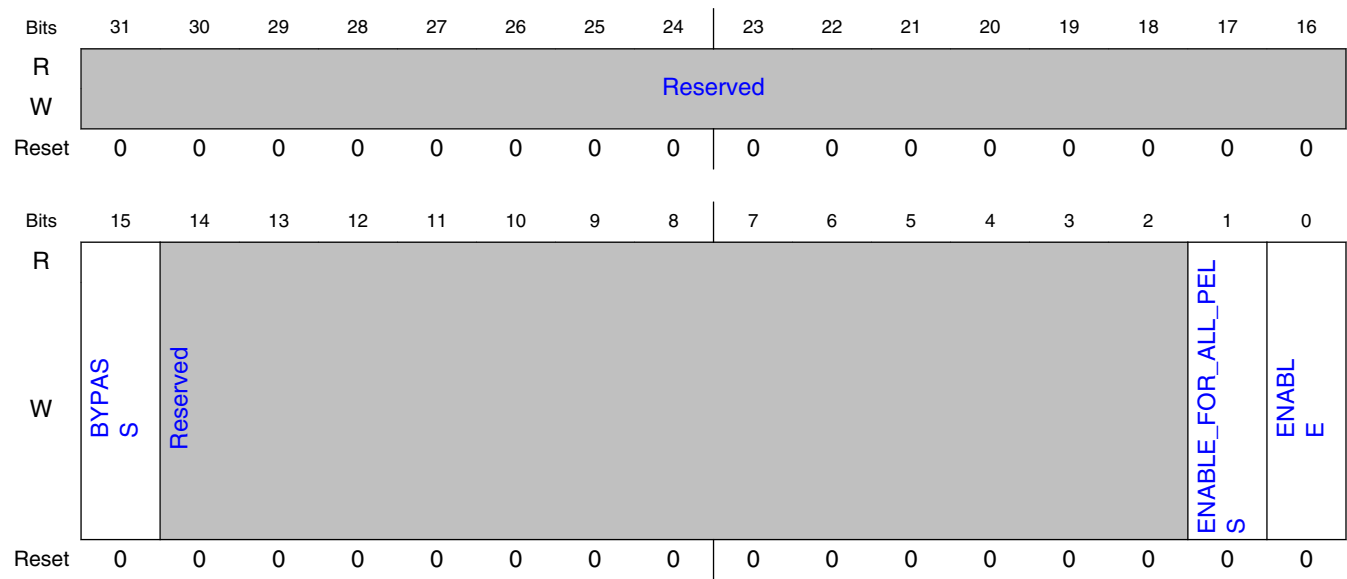
Register	Offset
HDR_PIPE3_CSCA_CONTROL_REG	B000h

15.10.3.1.135.2 Function

Controls the Color-Space-Converter-A (CSCA) in pipe1. This CSCA takes in 10-bit pixel components (component 0, component 1, component 2). A per-component offset (called pre-offset) is added to the pixel. After the pre-offset operation the pixel components are clipped. MAX_RANGE of PRE_CLIP is [-512, 1023]. A 3x3 matrix multiply (constant coefficients H(x,y)) is performed on the pixel to take to a different color space. After this matrix multiply, a 28 bit signed offset (post-offset) is added to the normalized result of

the matrix multiply. After the post-offset operation the results are clipped.
MAX_RANGE of POST_CLIP is [0,1023] Output of this CSCA are 10 bit per component pixels. After CSCA the pixels are fed into a LUT. Pipe1 is also called channel1.

15.10.3.1.135.3 Diagram



15.10.3.1.135.4 Fields

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this CSC 1: Pixels pass thru this CSC unmodified
14-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This CSC is enabled only for blended pixels 1: This CSC is enabled all pixels
0 ENABLE	0: Don't enable this CSC: Pixels pass thru the CSC unmodified 1: This CSC is enabled for current picture

15.10.3.1.136 Pipe1 Colorspace Converter A (CSCA) h(0,0) matrix coefficient (HDR_PIPE3_CSCA_H00)

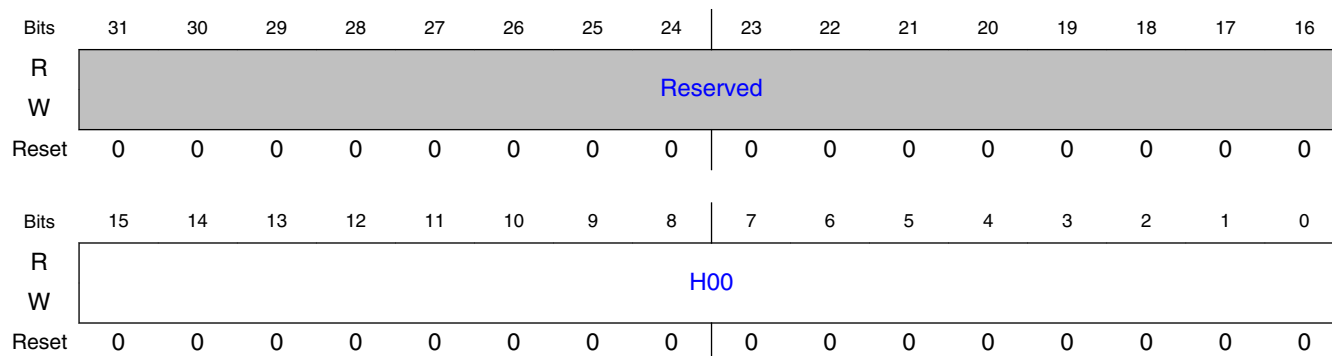
15.10.3.1.136.1 Offset

Register	Offset
HDR_PIPE3_CSCA_H00	B004h

15.10.3.1.136.2 Function

h(0,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.136.3 Diagram



15.10.3.1.136.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H00	h(0,0) 16 bit signed coefficient

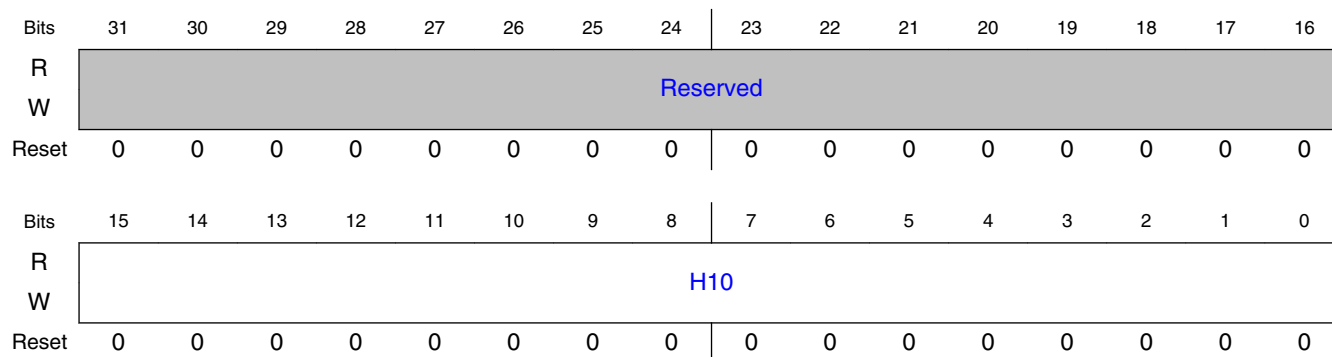
15.10.3.1.137 Pipe1 Colorspace Converter A (CSCA) h(1,0) matrix coefficient (HDR_PIPE3_CSCA_H10)

15.10.3.1.137.1 Offset

Register	Offset
HDR_PIPE3_CSCA_H10	B008h

15.10.3.1.137.2 Function

h(1,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.137.3 Diagram**15.10.3.1.137.4 Fields**

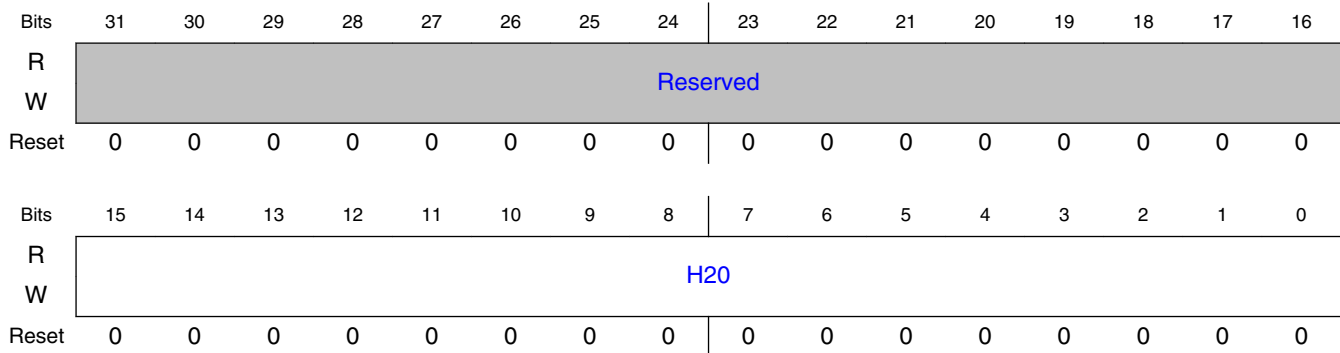
Field	Function
31-16 —	Reserved.
15-0 H10	h(1,0) 16 bit signed coefficient

15.10.3.1.138 Pipe1 Colorspace Converter A (CSCA) h(2,0) matrix coefficient (HDR_PIPE3_CSCA_H20)**15.10.3.1.138.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_H20	B00Ch

15.10.3.1.138.2 Function

$h(2,0)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.138.3 Diagram**15.10.3.1.138.4 Fields**

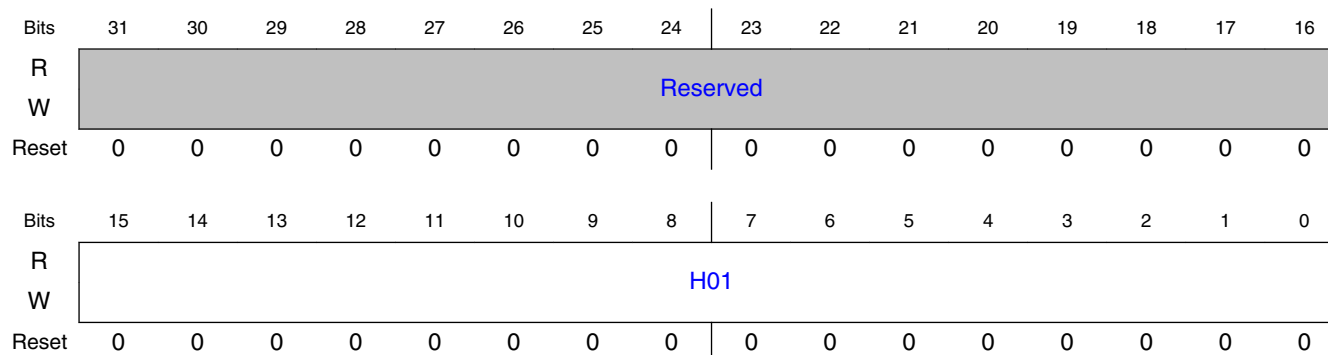
Field	Function
31-16 —	Reserved.
15-0 H20	$h(2,0)$ 16 bit signed coefficient

15.10.3.1.139 Pipe1 Colorspace Converter A (CSCA) $h(0,1)$ matrix coefficient (HDR_PIPE3_CSCA_H01)**15.10.3.1.139.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_H01	B010h

15.10.3.1.139.2 Function

$h(0,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.139.3 Diagram**15.10.3.1.139.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H01	h(0,1) 16 bit signed coefficient

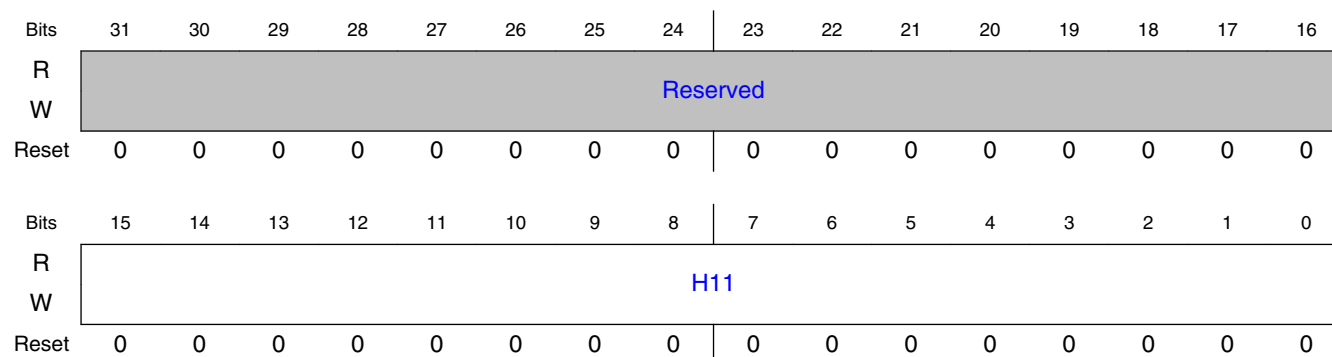
15.10.3.1.140 Pipe1 Colorspace Converter A (CSCA) h(1,1) matrix coefficient (HDR_PIPE3_CSCA_H11)**15.10.3.1.140.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_H11	B014h

15.10.3.1.140.2 Function

h(1,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.140.3 Diagram



15.10.3.1.140.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H11	h(1,1) 16 bit signed coefficient

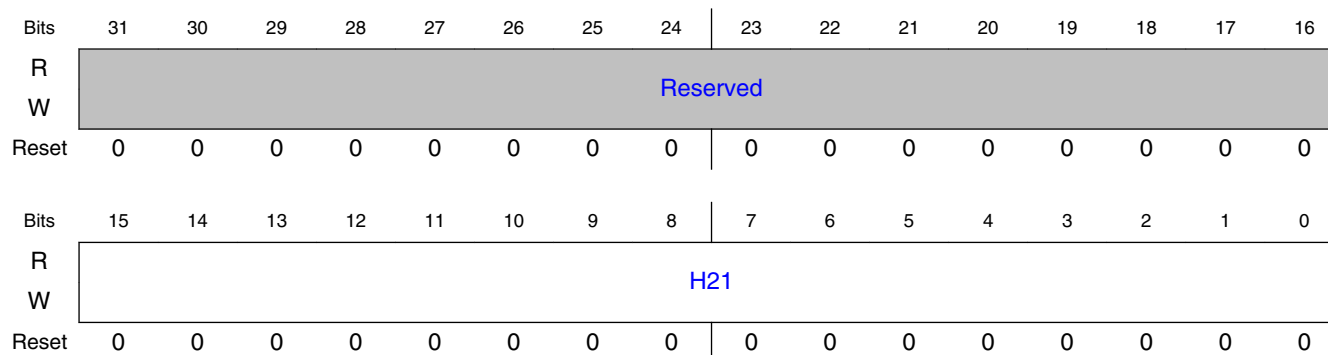
15.10.3.1.141 Pipe1 Colorspace Converter A (CSCA) h(2,1) matrix coefficient (HDR_PIPE3_CSCA_H21)

15.10.3.1.141.1 Offset

Register	Offset
HDR_PIPE3_CSCA_H21	B018h

15.10.3.1.141.2 Function

h(2,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.141.3 Diagram**15.10.3.1.141.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H21	h(2,1) 16 bit signed coefficient

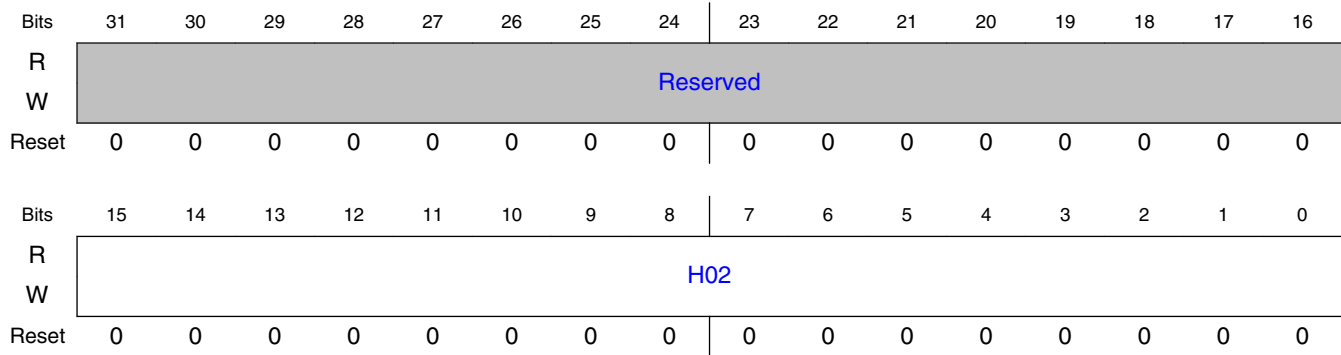
15.10.3.1.142 Pipe1 Colorspace Converter A (CSCA) h(0,2) matrix coefficient (HDR_PIPE3_CSCA_H02)**15.10.3.1.142.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_H02	B01Ch

15.10.3.1.142.2 Function

h(0,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.142.3 Diagram



15.10.3.1.142.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H02	h(0,2) 16 bit signed coefficient

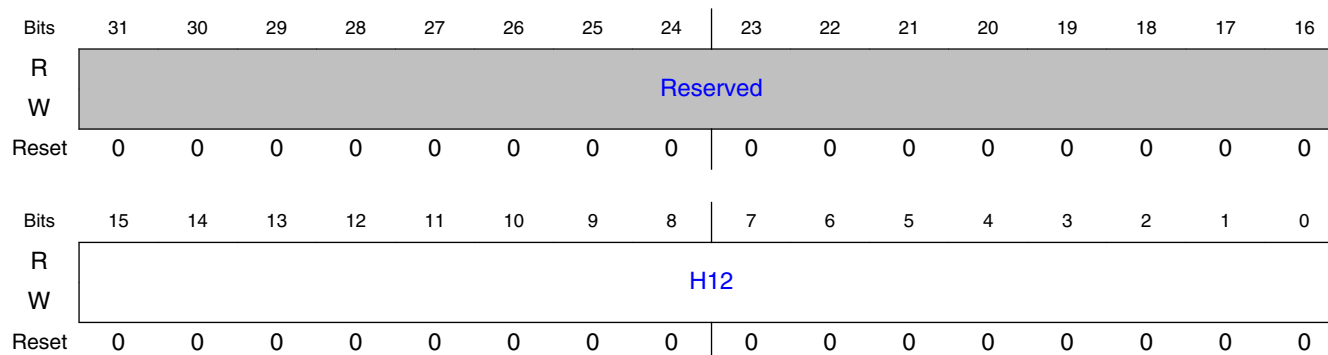
15.10.3.1.143 Pipe1 Colorspace Converter A (CSCA) h(1,2) matrix coefficient (HDR_PIPE3_CSCA_H12)

15.10.3.1.143.1 Offset

Register	Offset
HDR_PIPE3_CSCA_H12	B020h

15.10.3.1.143.2 Function

h(1,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.143.3 Diagram**15.10.3.1.143.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H12	h(1,2) 16 bit signed coefficient

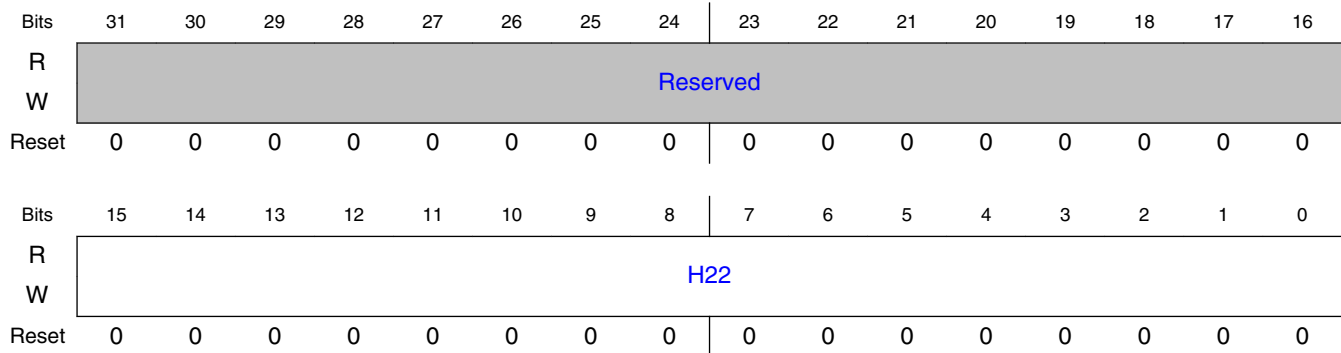
15.10.3.1.144 Pipe1 Colorspace Converter A (CSCA) h(2,2) matrix coefficient (HDR_PIPE3_CSCA_H22)**15.10.3.1.144.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_H22	B024h

15.10.3.1.144.2 Function

h(2,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.144.3 Diagram



15.10.3.1.144.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H22	h(2,2) 16 bit signed coefficient

15.10.3.1.145 Pipe1 Colorspace Converter A (CSCA) component 0 pre-offset (HDR_PIPE3_CSCA_IO_0)

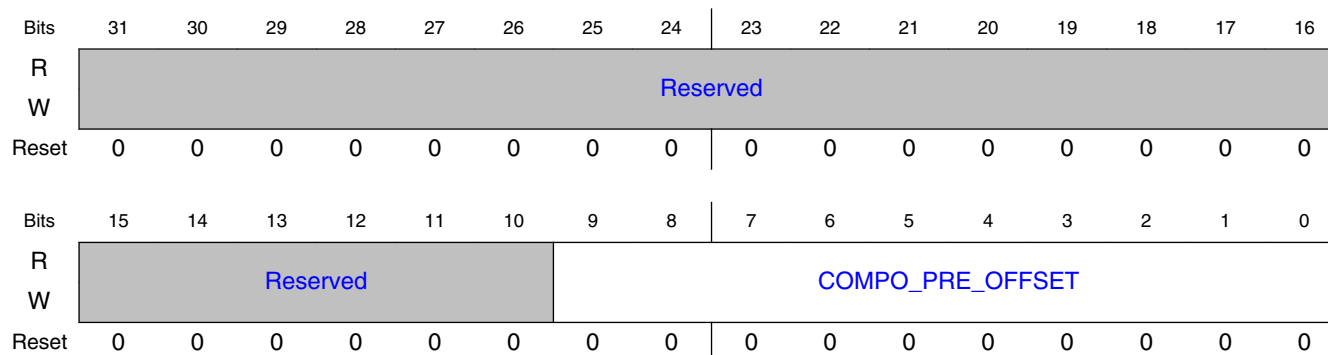
15.10.3.1.145.1 Offset

Register	Offset
HDR_PIPE3_CSCA_IO_0	B028h

15.10.3.1.145.2 Function

An signed 10-bit offset is added to component 0 (R,Y) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.145.3 Diagram



15.10.3.1.145.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMPO_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 0 of the pixel

15.10.3.1.146 Pipe1 Colorspace Converter A (CSCA) component 1 pre-offset (HDR_PIPE3_CSCA_IO_1)

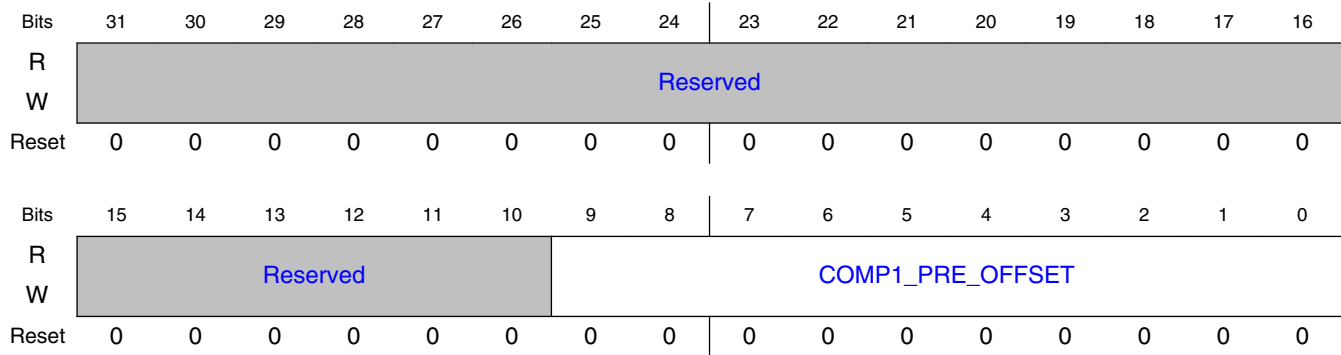
15.10.3.1.146.1 Offset

Register	Offset
HDR_PIPE3_CSCA_IO_1	B02Ch

15.10.3.1.146.2 Function

An signed 10-bit offset is added to component 1 (G,Cb) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.146.3 Diagram



15.10.3.1.146.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP1_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 1 of the pixel

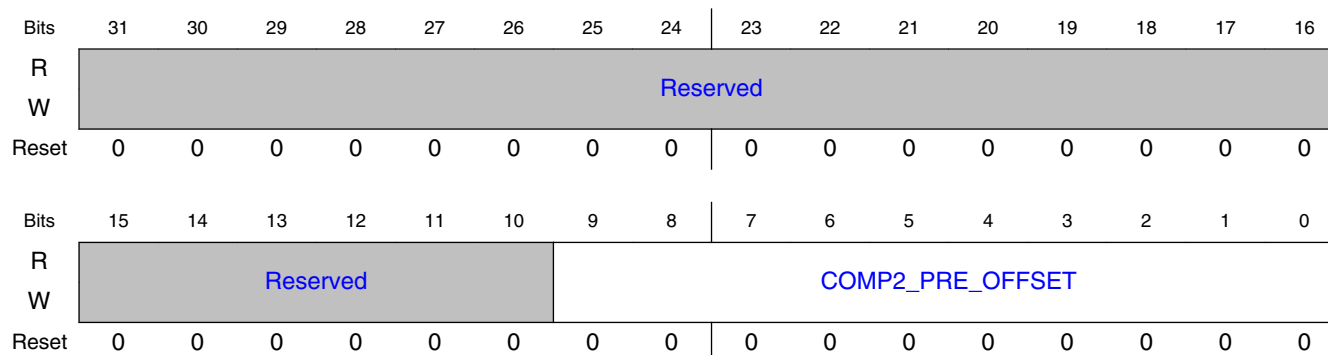
15.10.3.1.147 Pipe1 Colorspace Converter A (CSCA) component 2 pre-offset (HDR_PIPE3_CSCA_IO_2)

15.10.3.1.147.1 Offset

Register	Offset
HDR_PIPE3_CSCA_IO_2	B030h

15.10.3.1.147.2 Function

An signed 10-bit offset is added to component 2 (B,Cr) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.147.3 Diagram**15.10.3.1.147.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP2_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 2 of the pixel

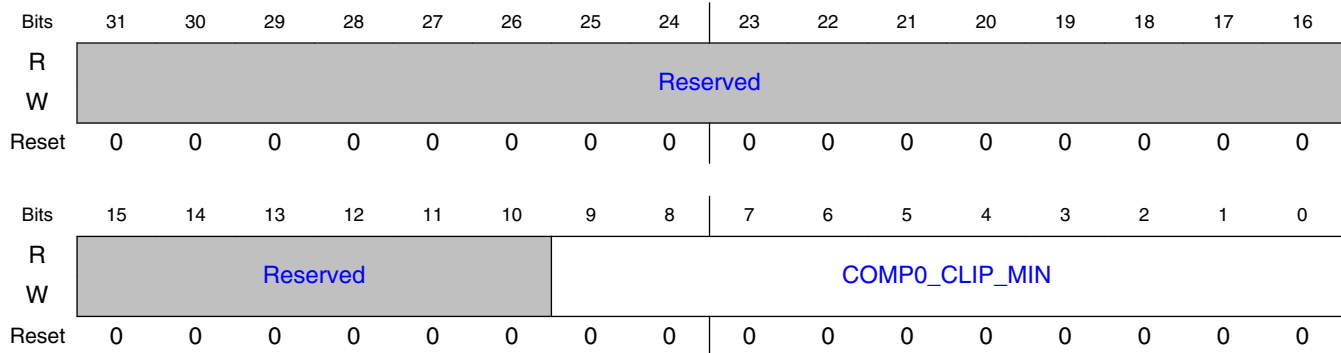
15.10.3.1.148 Pipe1 Colorspace Converter A (CSCA) component 0 clip min. (HDR_PIPE3_CSCA_IO_MIN_0)**15.10.3.1.148.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_IO_MIN_0	B034h

15.10.3.1.148.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.148.3 Diagram



15.10.3.1.148.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP0_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

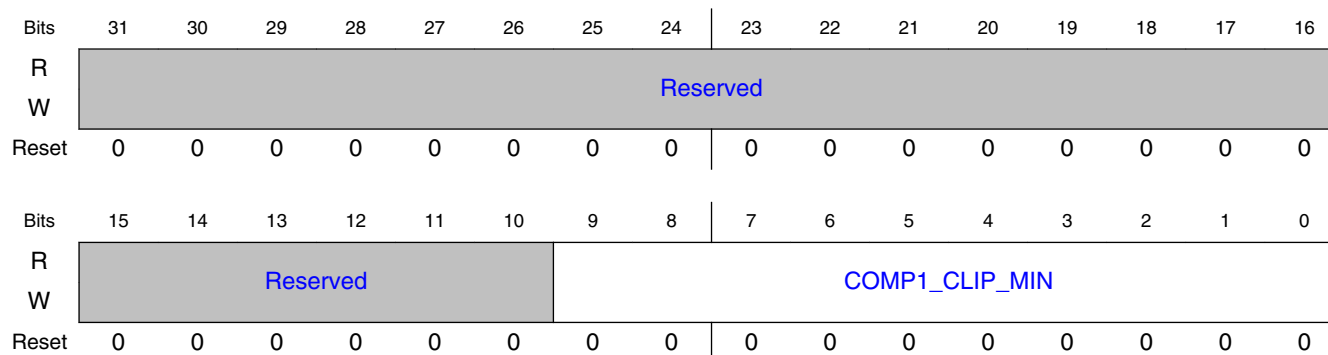
15.10.3.1.149 Pipe1 Colorspace Converter A (CSCA) component 1 clip min. (HDR_PIPE3_CSCA_IO_MIN_1)

15.10.3.1.149.1 Offset

Register	Offset
HDR_PIPE3_CSCA_IO_MIN_1	B038h

15.10.3.1.149.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.149.3 Diagram**15.10.3.1.149.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP1_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

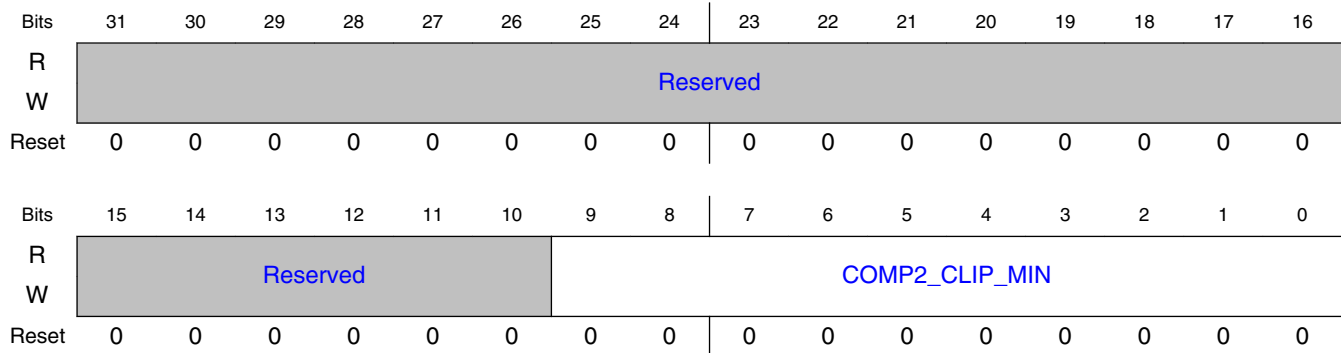
15.10.3.1.150 Pipe1 Colorspace Converter A (CSCA) component 2 clip min. (HDR_PIPE3_CSCA_IO_MIN_2)**15.10.3.1.150.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_IO_MIN_2	B03Ch

15.10.3.1.150.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511]. Pipe1 is also called channel1.

15.10.3.1.150.3 Diagram



15.10.3.1.150.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

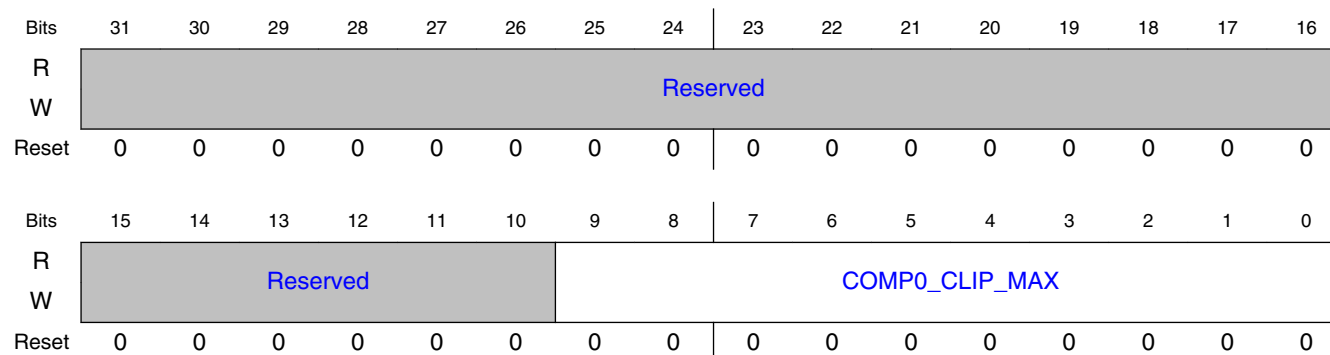
15.10.3.1.151 Pipe1 Colorspace Converter A (CSCA) component 0 clip max value. (HDR_PIPE3_CSCA_IO_MAX_0)

15.10.3.1.151.1 Offset

Register	Offset
HDR_PIPE3_CSCA_IO_MAX_0	B040h

15.10.3.1.151.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.151.3 Diagram**15.10.3.1.151.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP0_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

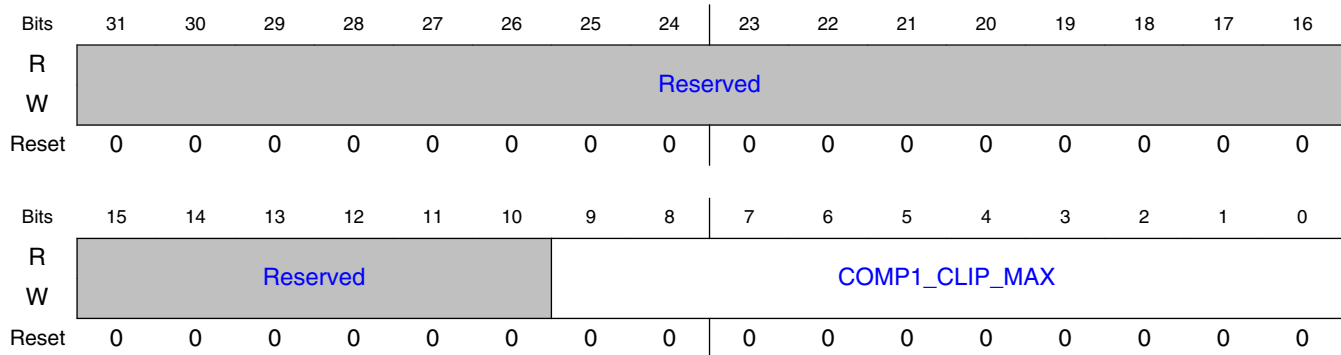
15.10.3.1.152 Pipe1 Colorspace Converter A (CSCA) component 1 clip max value. (HDR_PIPE3_CSCA_IO_MAX_1)**15.10.3.1.152.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_IO_MAX_1	B044h

15.10.3.1.152.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.152.3 Diagram



15.10.3.1.152.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP1_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.153 Pipe1 Colorspace Converter A (CSCA) component 2 clip max value. (HDR_PIPE3_CSCA_IO_MAX_2)

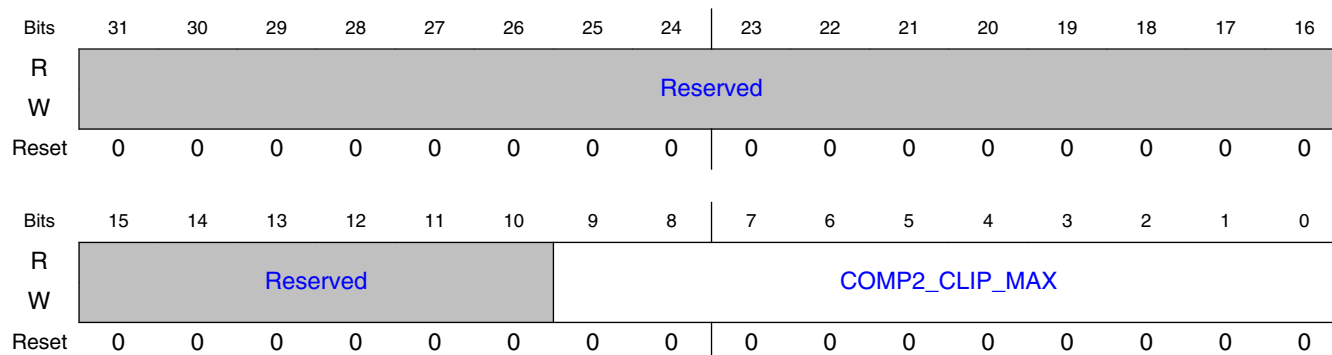
15.10.3.1.153.1 Offset

Register	Offset
HDR_PIPE3_CSCA_IO_MAX_2	B048h

15.10.3.1.153.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023]. Pipe1 is also called channel1.

15.10.3.1.153.3 Diagram



15.10.3.1.153.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.154 Pipe1 Colorspace Converter A (CSCA) normalization factor (HDR_PIPE3_CSCA_NORM)

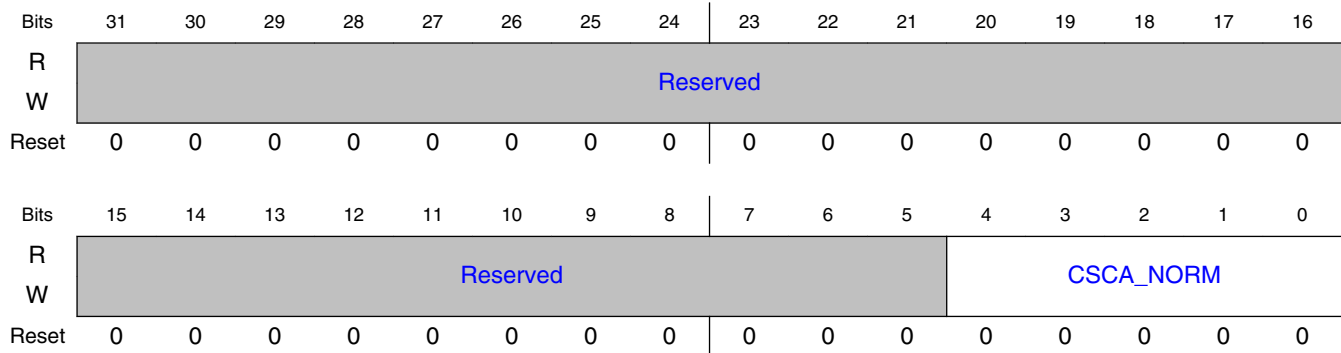
15.10.3.1.154.1 Offset

Register	Offset
HDR_PIPE3_CSCA_NORM	B04Ch

15.10.3.1.154.2 Function

After the CSC matrix multiply, all components of the result are shifted right by the amount in this register. This is a signed right shift. Pipe1 is also called channel1.

15.10.3.1.154.3 Diagram



15.10.3.1.154.4 Fields

Field	Function
31-5 —	Reserved.
4-0 CSCA_NORM	This 5-bit unsigned value is size if arithmetic shift after matrix multiply.

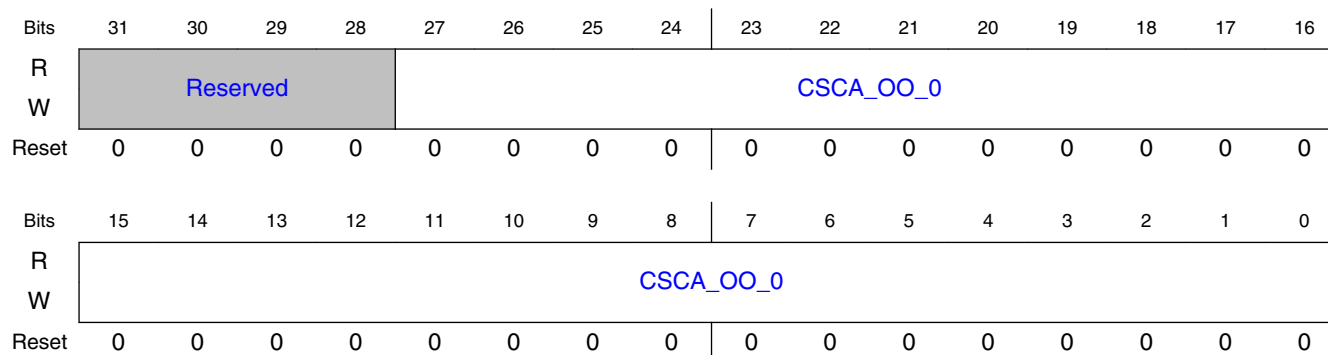
15.10.3.1.155 Pipe1 Colorspace Converter A (CSCA): Post offset component 0 (HDR_PIPE3_CSCA_OO_0)

15.10.3.1.155.1 Offset

Register	Offset
HDR_PIPE3_CSCA_OO_0	B050h

15.10.3.1.155.2 Function

After the CSC normalization, this offset value is added to component 0. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.155.3 Diagram**15.10.3.1.155.4 Fields**

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_0	Output Offset (OO) This is a signed 28-bit number. Per component

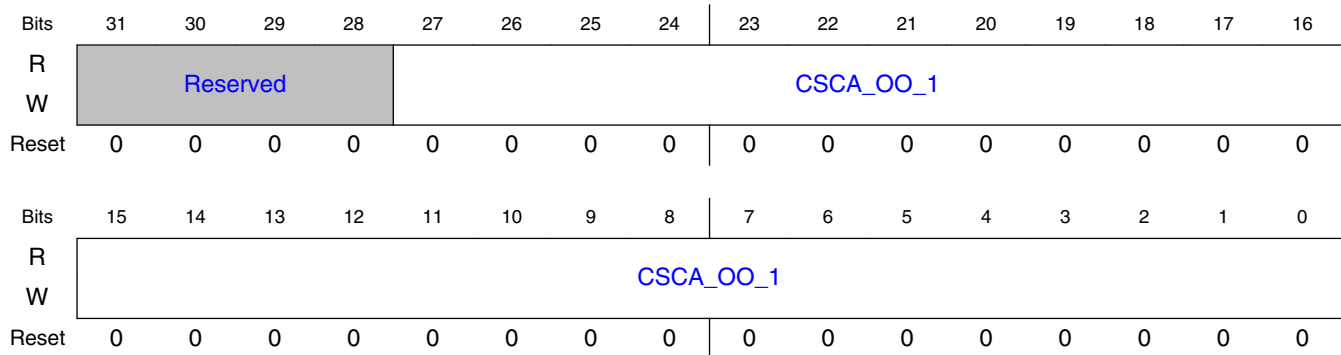
15.10.3.1.156 Pipe1 Colorspace Converter A (CSCA): Post offset component 1 (HDR_PIPE3_CSCA_OO_1)**15.10.3.1.156.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_OO_1	B054h

15.10.3.1.156.2 Function

After the CSC normalization, this offset value is added to component 1. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.156.3 Diagram



15.10.3.1.156.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_1	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.157 Pipe1 Colorspace Converter A (CSCA): Post offset component 2 (HDR_PIPE3_CSCA_OO_2)

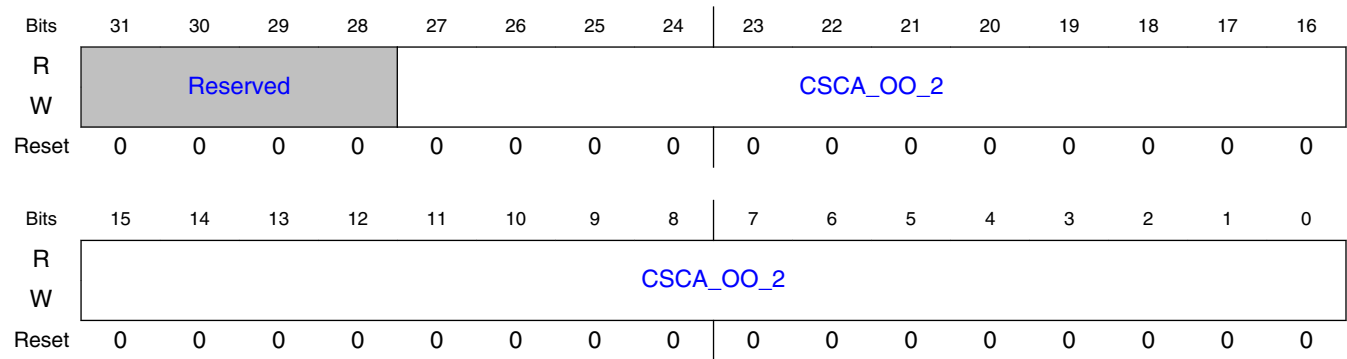
15.10.3.1.157.1 Offset

Register	Offset
HDR_PIPE3_CSCA_OO_2	B058h

15.10.3.1.157.2 Function

After the CSC normalization, this offset value is added to component 2. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.157.3 Diagram



15.10.3.1.157.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSCA_OO_2	Ouput Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.158 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 0 (HDR_PIPE3_CSCA_OMIN_0)

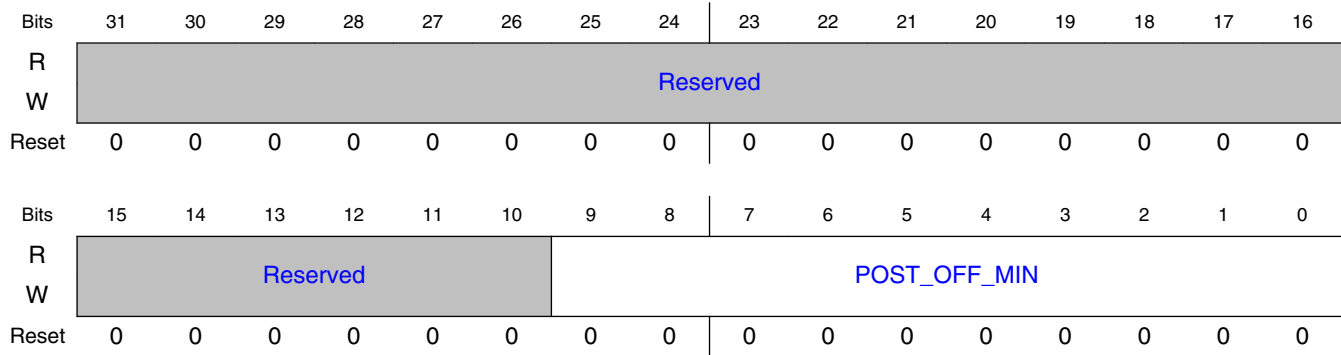
15.10.3.1.158.1 Offset

Register	Offset
HDR_PIPE3_CSCA_OMIN_0	B05Ch

15.10.3.1.158.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.158.3 Diagram



15.10.3.1.158.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

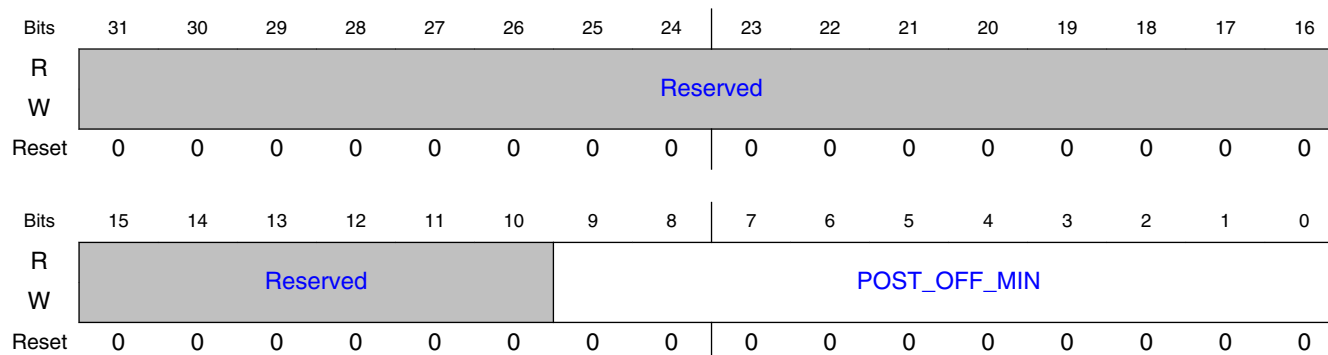
15.10.3.1.159 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 1 (HDR_PIPE3_CSCA_OMIN_1)

15.10.3.1.159.1 Offset

Register	Offset
HDR_PIPE3_CSCA_OMIN_1	B060h

15.10.3.1.159.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.159.3 Diagram**15.10.3.1.159.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

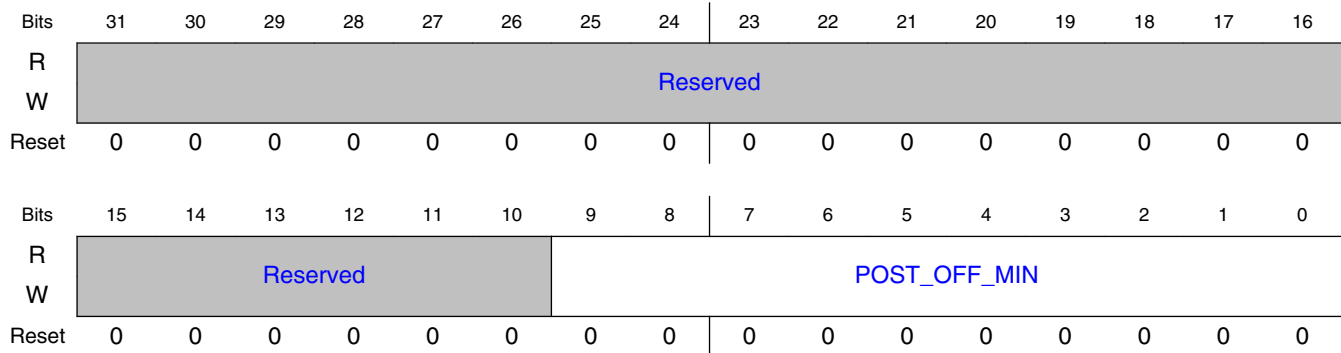
15.10.3.1.160 Pipe1 Colorspace Converter A (CSCA): Post offset min clip value for component 2 (HDR_PIPE3_CSCA_OMIN_2)**15.10.3.1.160.1 Offset**

Register	Offset
HDR_PIPE3_CSCA_OMIN_2	B064h

15.10.3.1.160.2 Function

After the post offset is added this component is clipped. This is the minimum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.160.3 Diagram



15.10.3.1.160.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.161 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 0 (HDR_PIPE3_CSCA_OMAX_0)

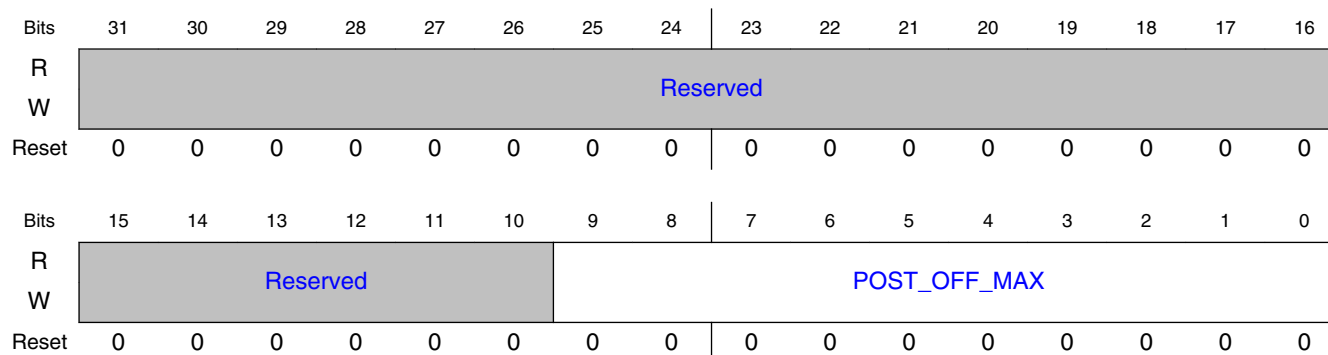
15.10.3.1.161.1 Offset

Register	Offset
HDR_PIPE3_CSCA_OMAX_0	B068h

15.10.3.1.161.2 Function

After the post offset is added thie component is clipped. This is the maximum clip value This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.161.3 Diagram



15.10.3.1.161.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

15.10.3.1.162 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 1 (HDR_PIPE3_CSCA_OMAX_1)

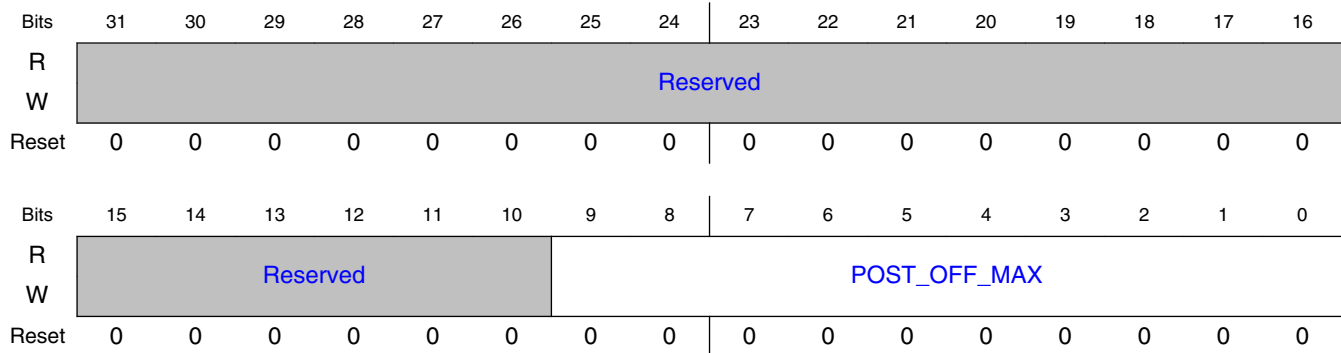
15.10.3.1.162.1 Offset

Register	Offset
HDR_PIPE3_CSCA_OMAX_1	B06Ch

15.10.3.1.162.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.162.3 Diagram



15.10.3.1.162.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

15.10.3.1.163 Pipe1 Colorspace Converter A (CSCA): Post offset max clip value for component 2 (HDR_PIPE3_CSCA_OMAX_2)

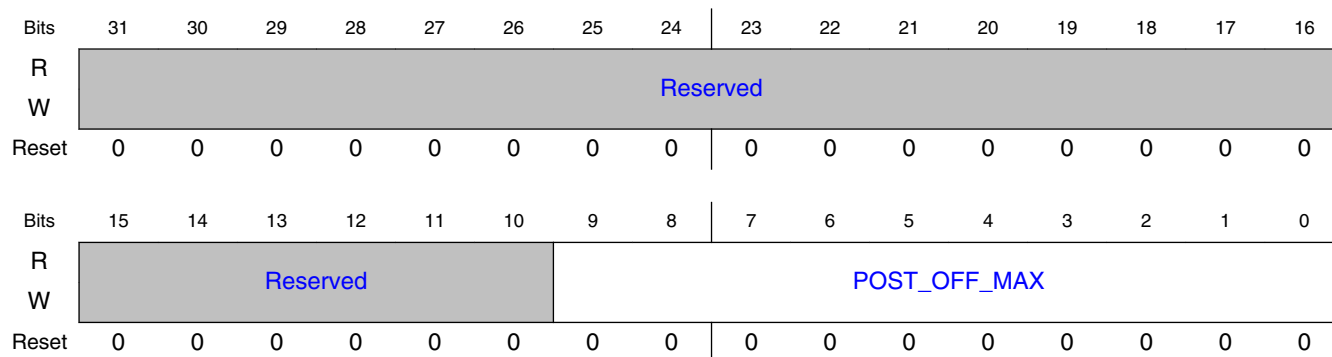
15.10.3.1.163.1 Offset

Register	Offset
HDR_PIPE3_CSCA_OMAX_2	B070h

15.10.3.1.163.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.163.3 Diagram



15.10.3.1.163.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

15.10.3.1.164 PIPE3: NOT USED (HDR_PIPE3_ENTRY_29)

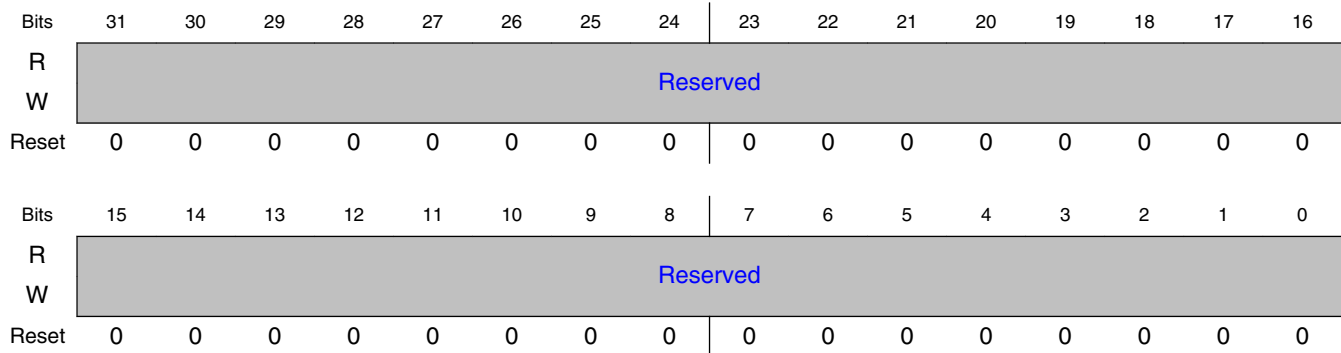
15.10.3.1.164.1 Offset

Register	Offset
HDR_PIPE3_ENTRY_29	B074h

15.10.3.1.164.2 Function

PIPE3: NOT USED

15.10.3.1.164.3 Diagram



15.10.3.1.164.4 Fields

Field	Function
31-0	Reserved.
—	

15.10.3.1.165 Pipe1 LUT control register (HDR_PIPE3_LUT_CONTROL_REG)

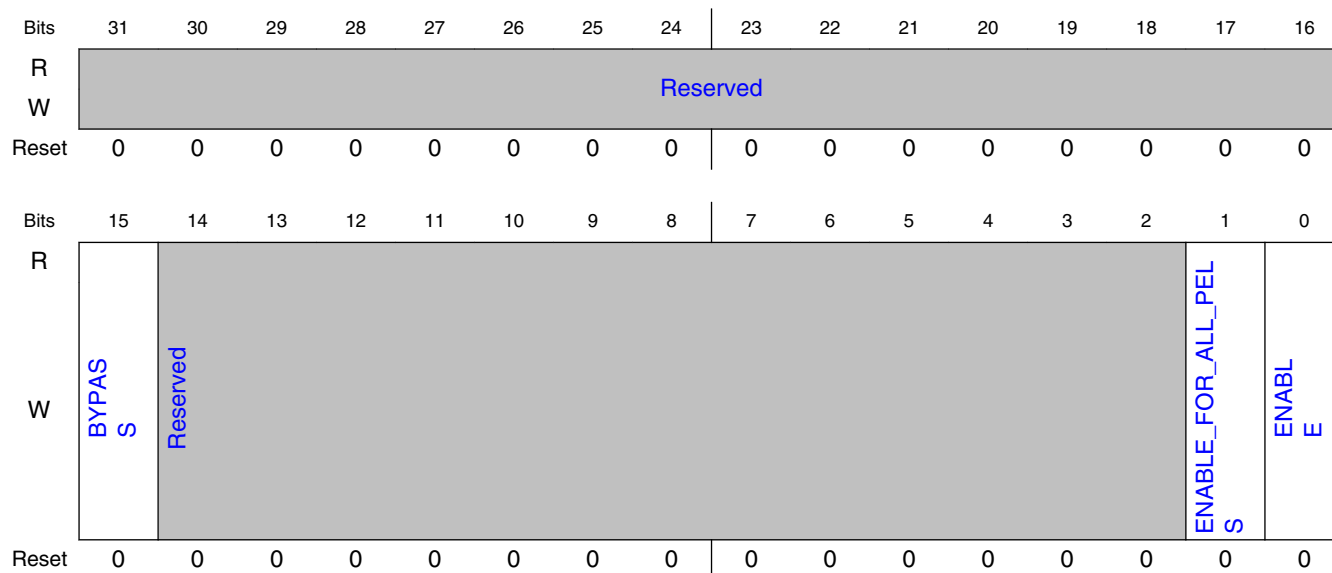
15.10.3.1.165.1 Offset

Register	Offset
HDR_PIPE3_LUT_CONTROL_REG	B080h

15.10.3.1.165.2 Function

Controls the Look-Up-Table in pipe1. Control of LUT is PER PIXEL not per component. LUT has 1024 14-bit, per component, entries. LUT maybe used for conversion from non-linear (gamma corrected) to linear pixels. LUT maybe used for conversion from linear to non linear pixels (gamma corrected). LUT can contain 14-bit fixed point or 14-bit floating point pixel value. Floating point format has 9 bit unsigned mantissa with hidden bit and 5 bit, unsigned, biased exponent. {exp[4:0], mantissa[8:0]} => 1.mantissa[8:0] * 2^(exp-1) When using floating piont numbers, in this LUT, these are used in HDR OUTPUT PIPE for HDR non-linear operation and must bypass Color Space Converter B (CSCB). Pipe1 is also called channel1.

15.10.3.1.165.3 Diagram



15.10.3.1.165.4 Fields

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this LUT 1: Pixels pass thru this LUT unmodified
14-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This LUT is enabled only for blended pixels 1: This LUT is enabled all pixels
0 ENABLE	0: Don't enable this LUT: Pixels pass thru the LUT unmodified 1: This LUT is enabled for current picture

15.10.3.1.166 Pipe1 Colorspace Converter B control. (HDR_PIPE3_CSCB_CONTROL_REG)

15.10.3.1.166.1 Offset

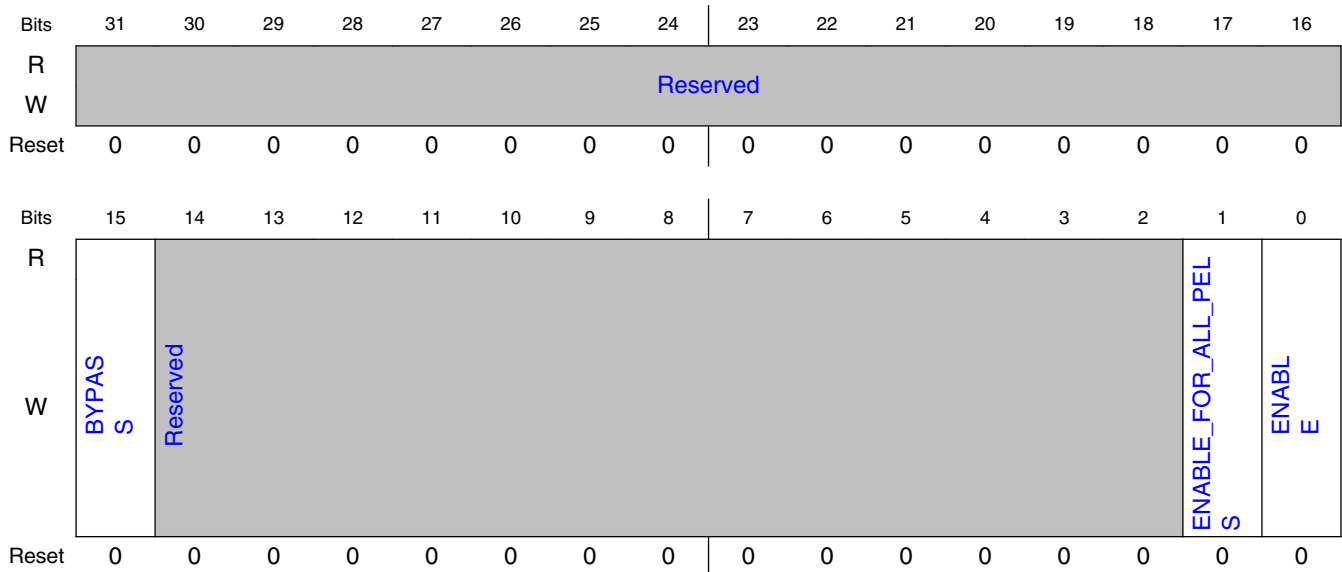
Register	Offset
HDR_PIPE3_CSCB_CONTROL_REG	B800h

15.10.3.1.166.2 Function

Controls the Color-Space-Converter-B (CSCB) in pipe1. This CSCB takes in 14-bit pixel components (component 0, component 1, component 2). A per-component offset (called pre-offset) is added to the pixel. After the pre-offset operation the pixel components are clipped. MAX_RANGE of PRE_CLIP is [-512, 1023]. A 3x3 matrix multiply (constant coefficients H(x,y)) is performed on the pixel to take to a different color space. After this matrix multiply, a 28 bit signed offset (post-offset) is added to the normalized result of the matrix multiply. After the post-offset operation the results are clipped.

MAX_RANGE of POST_CLIP is [0,1023] Output of this CSCA are 28 bit per component pixels. After CSCB the pixels are fed into the Blender. Pipe1 is also called channel1.

15.10.3.1.166.3 Diagram



15.10.3.1.166.4 Fields

Field	Function
31-16 —	Reserved.
15 BYPASS	0: Don't bypass this CSC 1: Pixels pass thru this CSC unmodified
14-2 —	Reserved.

Table continues on the next page...

Memory Map and Registers

Field	Function
1 ENABLE_FOR_ ALL_PELS	0: This CSC is enabled only for blended pixels 1: This CSC is enabled all pixels
0 ENABLE	0: Don't enable this CSC: Pixels pass thru the CSC unmodified 1: This CSC is enabled for current picture

15.10.3.1.167 Pipe1 Colorspace Converter A (CSCB) h(0,0) matrix coefficient (HDR_PIPE3_CSCB_H00)

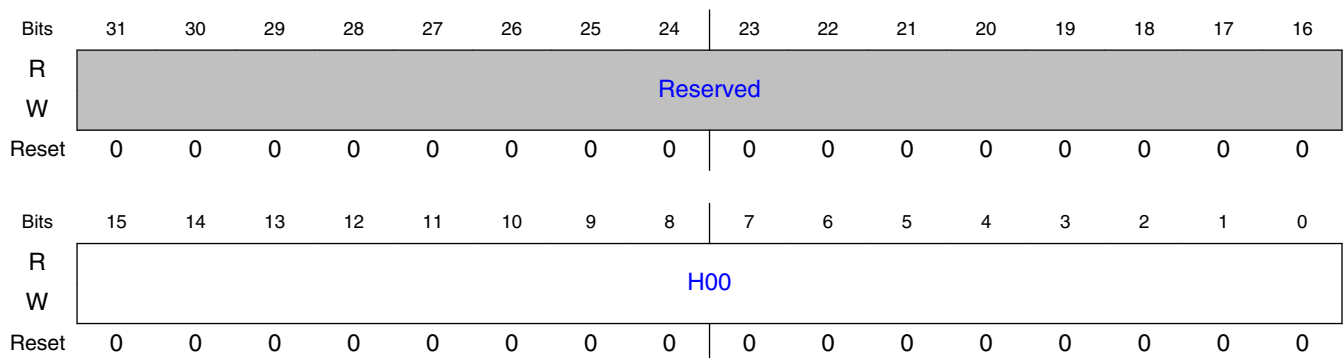
15.10.3.1.167.1 Offset

Register	Offset
HDR_PIPE3_CSCB_H00	B804h

15.10.3.1.167.2 Function

h(0,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.167.3 Diagram



15.10.3.1.167.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H00	h(0,0) 16 bit signed coefficient

15.10.3.1.168 Pipe1 Colorspace Converter B (CSCB) h(1,0) matrix coefficient (HDR_PIPE3_CSCB_H10)

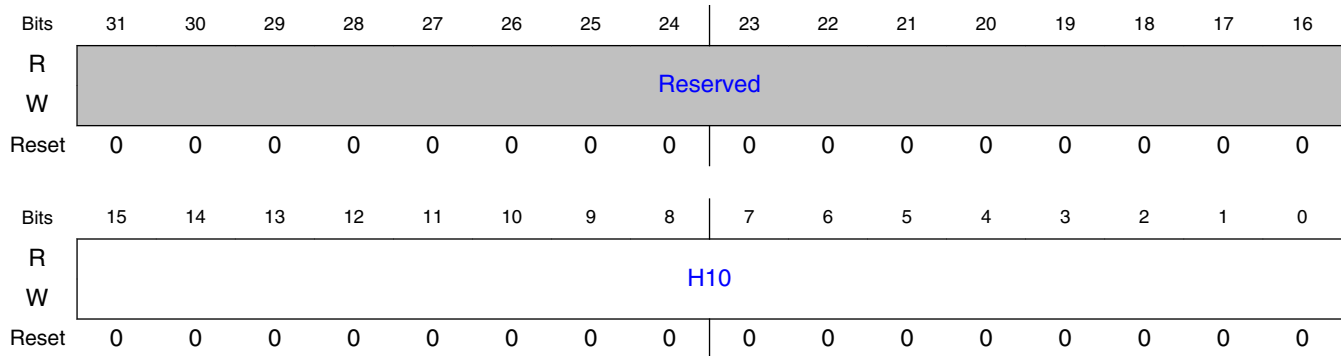
15.10.3.1.168.1 Offset

Register	Offset
HDR_PIPE3_CSCB_H10	B808h

15.10.3.1.168.2 Function

h(1,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.168.3 Diagram



15.10.3.1.168.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H10	h(1,0) 16 bit signed coefficient

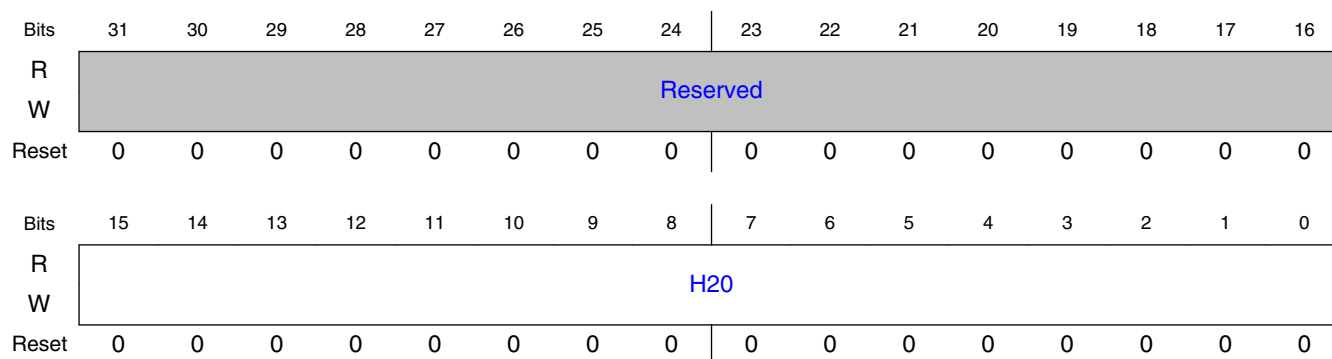
15.10.3.1.169 Pipe1 Colorspace Converter B (CSCB) h(2,0) matrix coefficient (HDR_PIPE3_CSCB_H20)

15.10.3.1.169.1 Offset

Register	Offset
HDR_PIPE3_CSCB_H20	B80Ch

15.10.3.1.169.2 Function

h(2,0) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.169.3 Diagram**15.10.3.1.169.4 Fields**

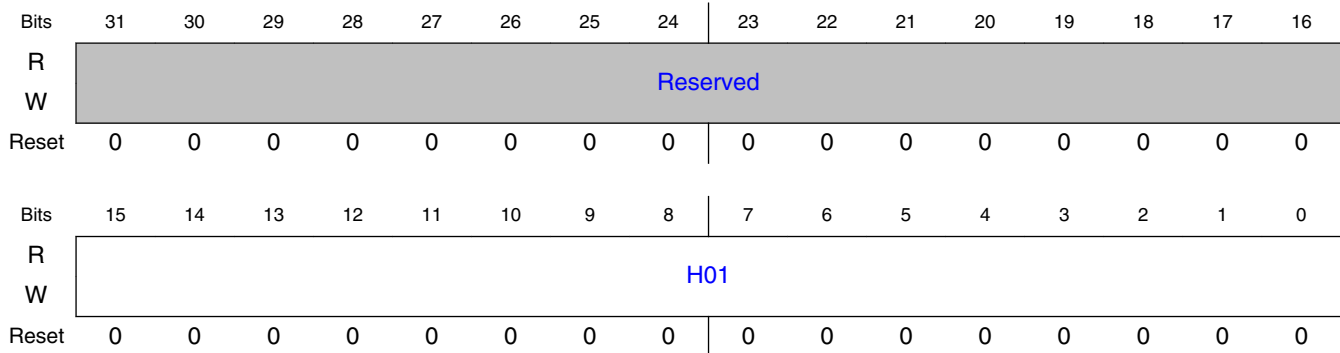
Field	Function
31-16 —	Reserved.
15-0 H20	h(2,0) 16 bit signed coefficient

15.10.3.1.170 Pipe1 Colorspace Converter B (CSCB) h(0,1) matrix coefficient (HDR_PIPE3_CSCB_H01)**15.10.3.1.170.1 Offset**

Register	Offset
HDR_PIPE3_CSCB_H01	B810h

15.10.3.1.170.2 Function

$h(0,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.170.3 Diagram**15.10.3.1.170.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H01	$h(0,1)$ 16 bit signed coefficient

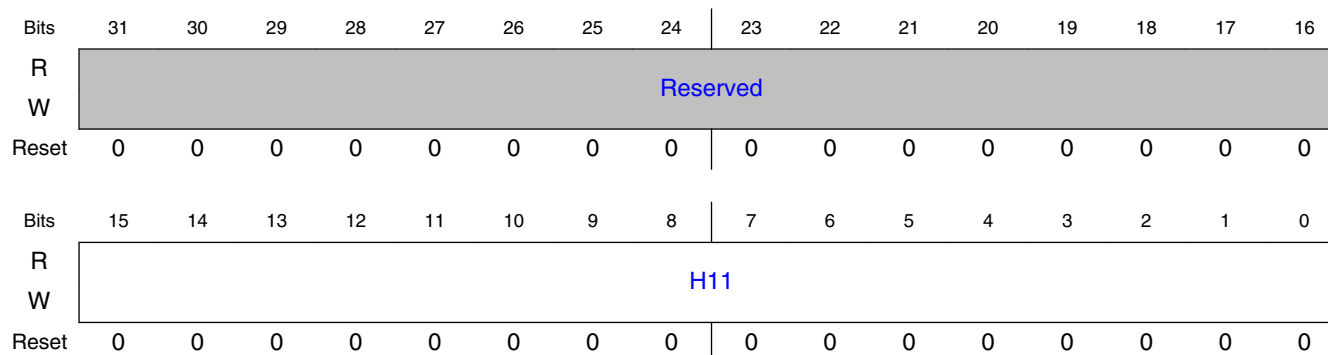
15.10.3.1.171 Pipe1 Colorspace Converter B (CSCB) $h(1,1)$ matrix coefficient (HDR_PIPE3_CSCB_H11)**15.10.3.1.171.1 Offset**

Register	Offset
HDR_PIPE3_CSCB_H11	B814h

15.10.3.1.171.2 Function

$h(1,1)$ coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.171.3 Diagram



15.10.3.1.171.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H11	h(1,1) 16 bit signed coefficient

15.10.3.1.172 Pipe1 Colorspace Converter B (CSCB) h(2,1) matrix coefficient (HDR_PIPE3_CSCB_H21)

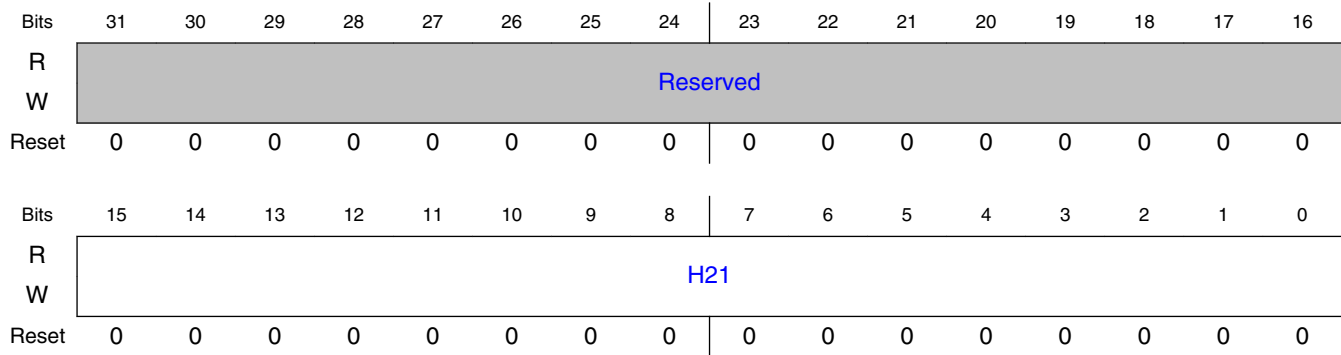
15.10.3.1.172.1 Offset

Register	Offset
HDR_PIPE3_CSCB_H21	B818h

15.10.3.1.172.2 Function

h(2,1) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.172.3 Diagram



15.10.3.1.172.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H21	h(2,1) 16 bit signed coefficient

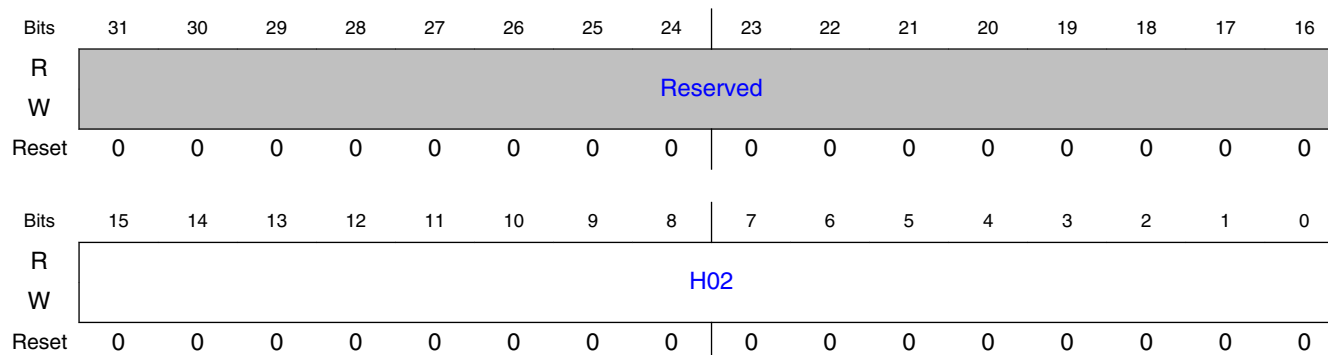
15.10.3.1.173 Pipe1 Colorspace Converter B (CSCB) h(0,2) matrix coefficient (HDR_PIPE3_CSCB_H02)

15.10.3.1.173.1 Offset

Register	Offset
HDR_PIPE3_CSCB_H02	B81Ch

15.10.3.1.173.2 Function

h(0,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.173.3 Diagram**15.10.3.1.173.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H02	h(0,2) 16 bit signed coefficient

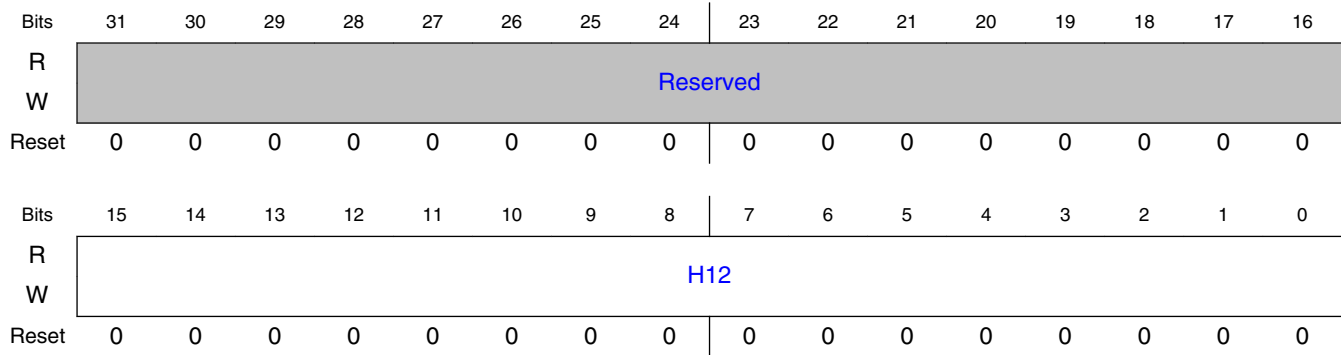
15.10.3.1.174 Pipe1 Colorspace Converter B (CSCB) h(1,2) matrix coefficient (HDR_PIPE3_CSCB_H12)**15.10.3.1.174.1 Offset**

Register	Offset
HDR_PIPE3_CSCB_H12	B820h

15.10.3.1.174.2 Function

h(1,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.174.3 Diagram



15.10.3.1.174.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H12	h(1,2) 16 bit signed coefficient

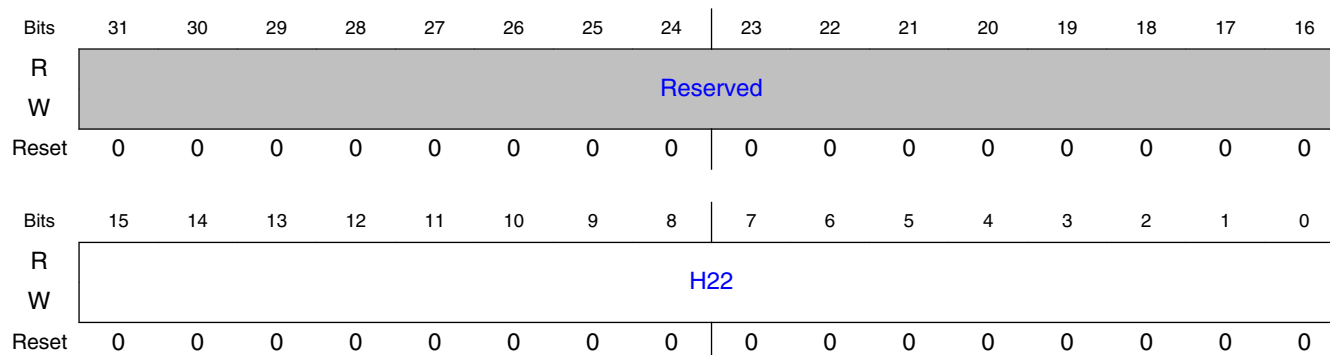
15.10.3.1.175 Pipe1 Colorspace Converter B (CSCB) h(2,2) matrix coefficient (HDR_PIPE3_CSCB_H22)

15.10.3.1.175.1 Offset

Register	Offset
HDR_PIPE3_CSCB_H22	B824h

15.10.3.1.175.2 Function

h(2,2) coefficient in color space matrix multiply Pipe1 is also called channel1.

15.10.3.1.175.3 Diagram**15.10.3.1.175.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H22	h(2,2) 16 bit signed coefficient

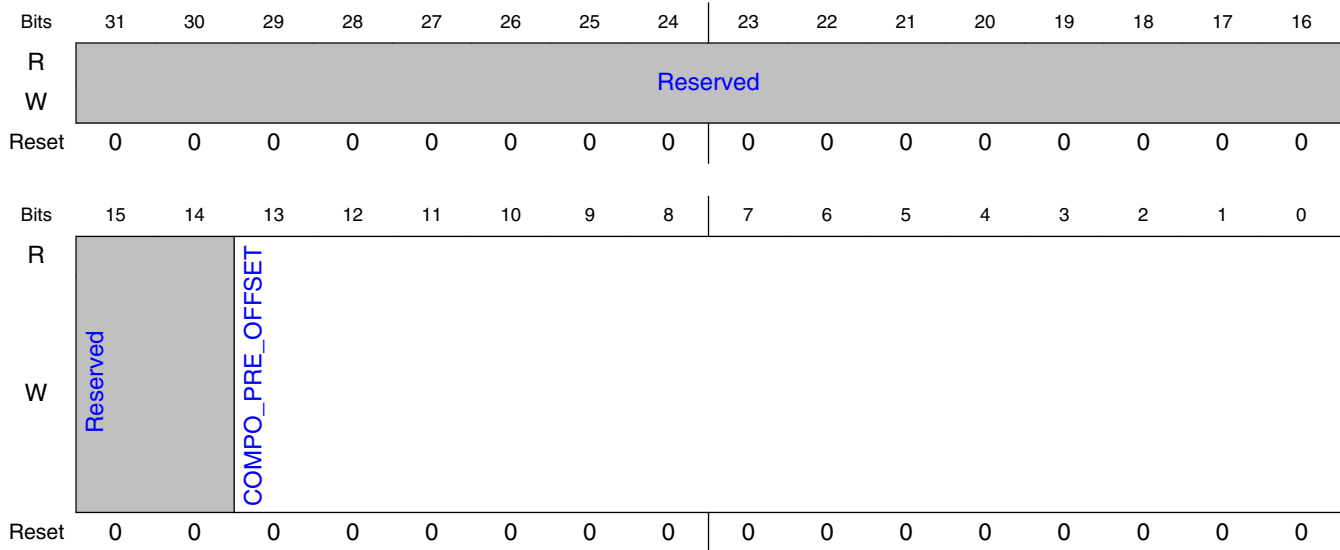
15.10.3.1.176 Pipe1 Colorspace Converter B (CSCB) component 0 pre-offset (HDR_PIPE3_CSCB_IO_0)**15.10.3.1.176.1 Offset**

Register	Offset
HDR_PIPE3_CSCB_IO_0	B828h

15.10.3.1.176.2 Function

A signed 14-bit offset is added to component 0 (R,Y) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.176.3 Diagram



15.10.3.1.176.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMPO_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 0 of the pixel

15.10.3.1.177 Pipe1 Colorspace Converter B (CSCB) component 1 pre-offset (HDR_PIPE3_CSCB_IO_1)

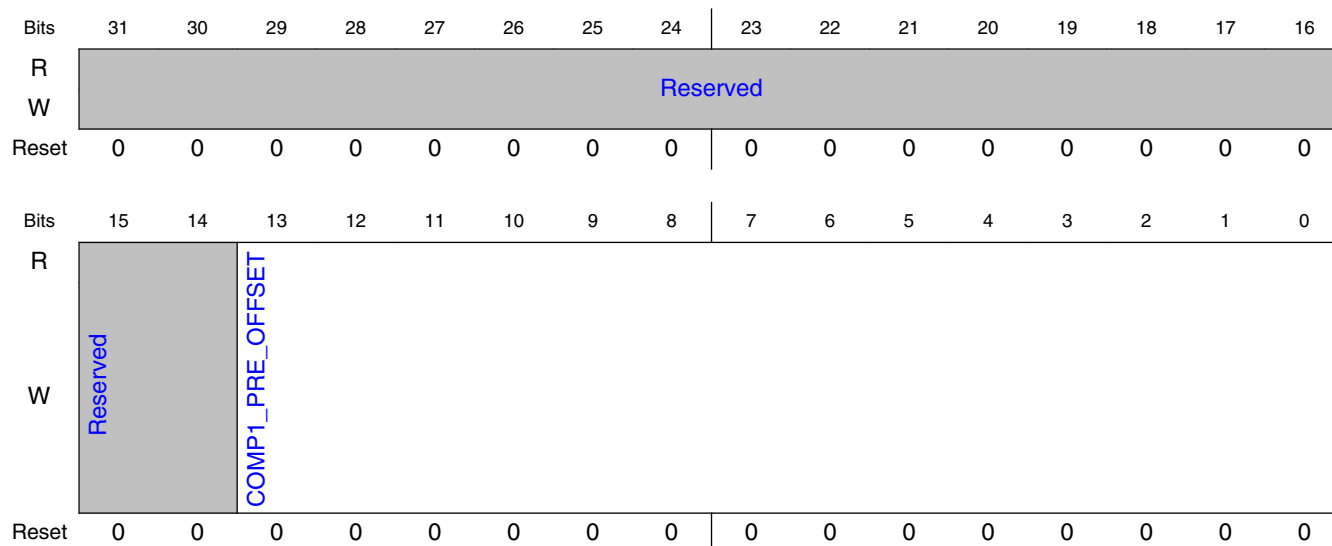
15.10.3.1.177.1 Offset

Register	Offset
HDR_PIPE3_CSCB_IO_1	B82Ch

15.10.3.1.177.2 Function

A signed 14-bit offset is added to component 1 (G,Cb) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.177.3 Diagram



15.10.3.1.177.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 1 of the pixel

15.10.3.1.178 Pipe1 Colorspace Converter B (CSCB) component 2 pre-offset (HDR_PIPE3_CSCB_IO_2)

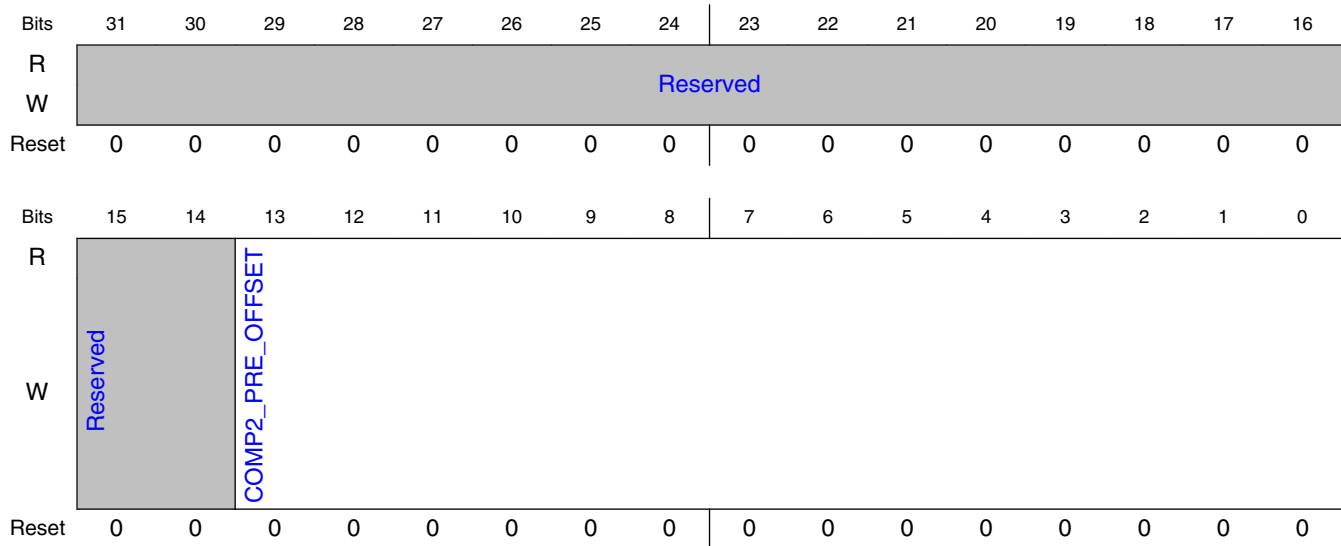
15.10.3.1.178.1 Offset

Register	Offset
HDR_PIPE3_CSCB_IO_2	B830h

15.10.3.1.178.2 Function

A signed 14-bit offset is added to component 2 (B,Cr) before matrix multiply Pipe1 is also called channel1.

15.10.3.1.178.3 Diagram



15.10.3.1.178.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 2 of the pixel

15.10.3.1.179 Pipe1 Colorspace Converter B (CSCB) component 0 clip min. (HDR_PIPE3_CSCB_IO_MIN_0)

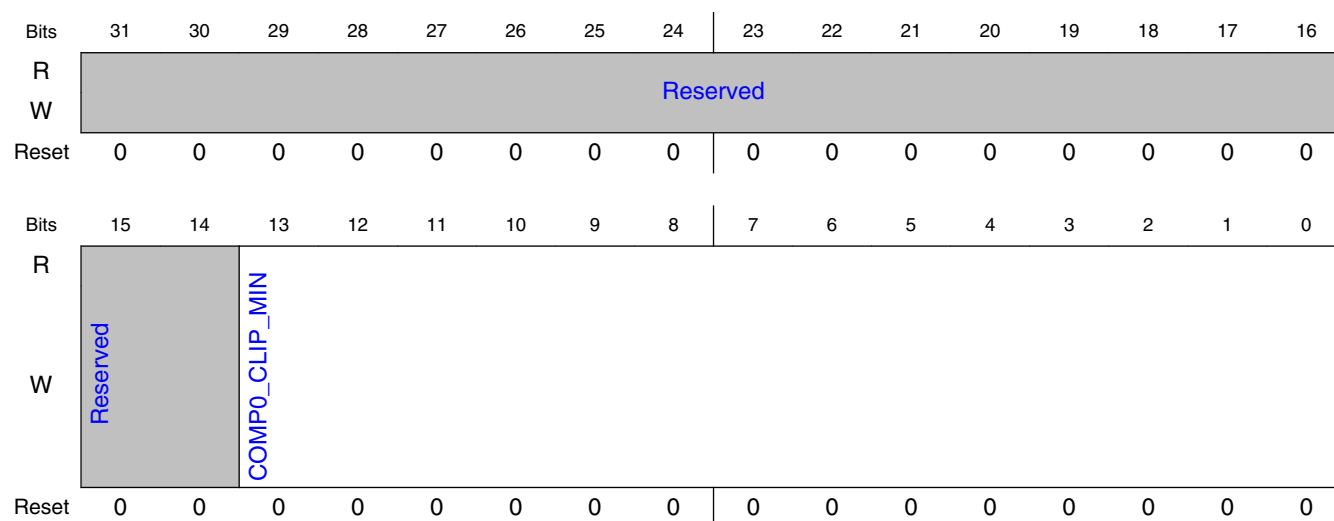
15.10.3.1.179.1 Offset

Register	Offset
HDR_PIPE3_CSCB_IO_MIN_0	B834h

15.10.3.1.179.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192, 8191]. Pipe1 is also called channel1.

15.10.3.1.179.3 Diagram



15.10.3.1.179.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP0_CLIP_MIN	This 14-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.180 Pipe1 Colorspace Converter B (CSCB) component 1 clip min. (HDR_PIPE3_CSCB_IO_MIN_1)

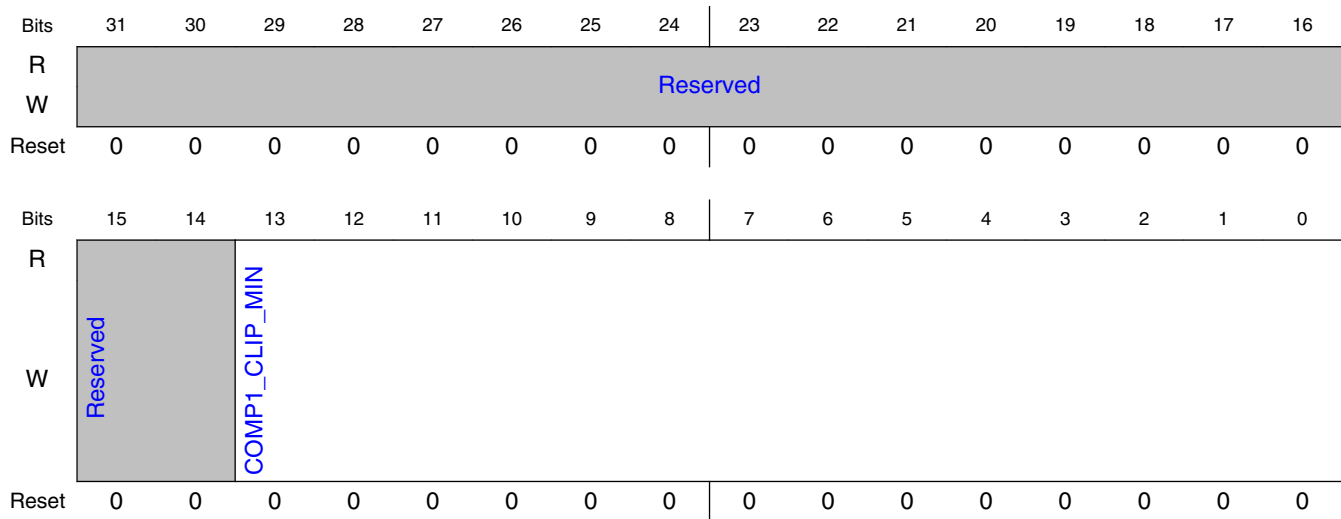
15.10.3.1.180.1 Offset

Register	Offset
HDR_PIPE3_CSCB_IO_MIN_1	B838h

15.10.3.1.180.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192 , 8191]. Pipe1 is also called channel1.

15.10.3.1.180.3 Diagram



15.10.3.1.180.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_CLIP_MIN	This 14-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.181 Pipe1 Colorspace Converter B (CSCB) component 2 clip min. (HDR_PIPE3_CSCB_IO_MIN_2)

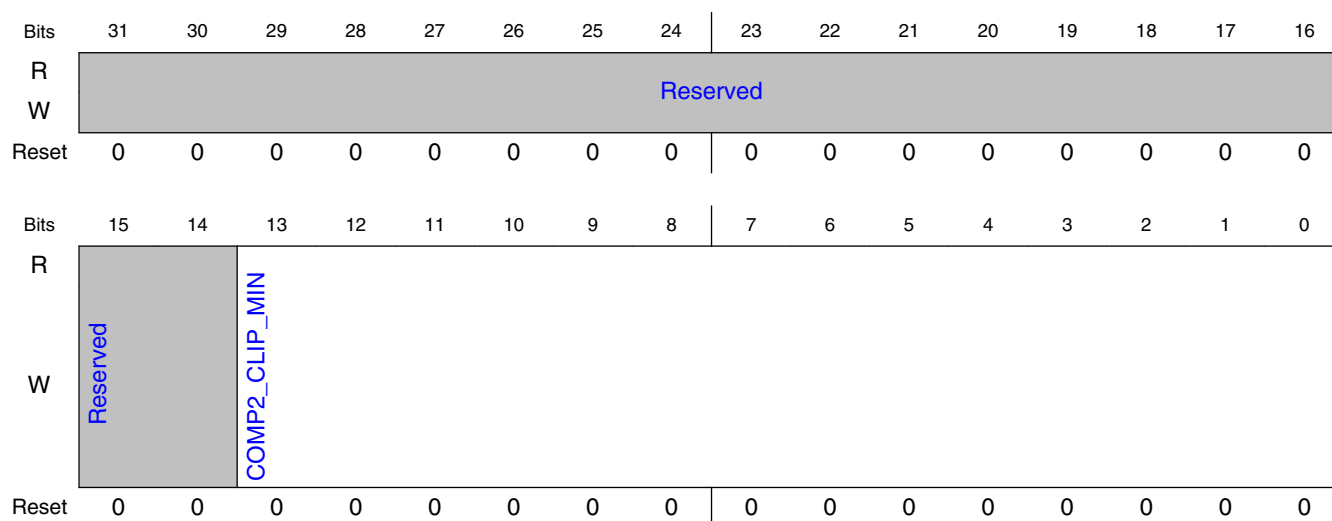
15.10.3.1.181.1 Offset

Register	Offset
HDR_PIPE3_CSCB_IO_MIN_2	B83Ch

15.10.3.1.181.2 Function

After the pre-increment, The result is clipped. This 14-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-8192 , 1891]. Pipe1 is also called channel1.

15.10.3.1.181.3 Diagram



15.10.3.1.181.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

15.10.3.1.182 Pipe1 Colorspace Converter B (CSCB) component 0 clip max value. (HDR_PIPE3_CSCB_IO_MAX_0)

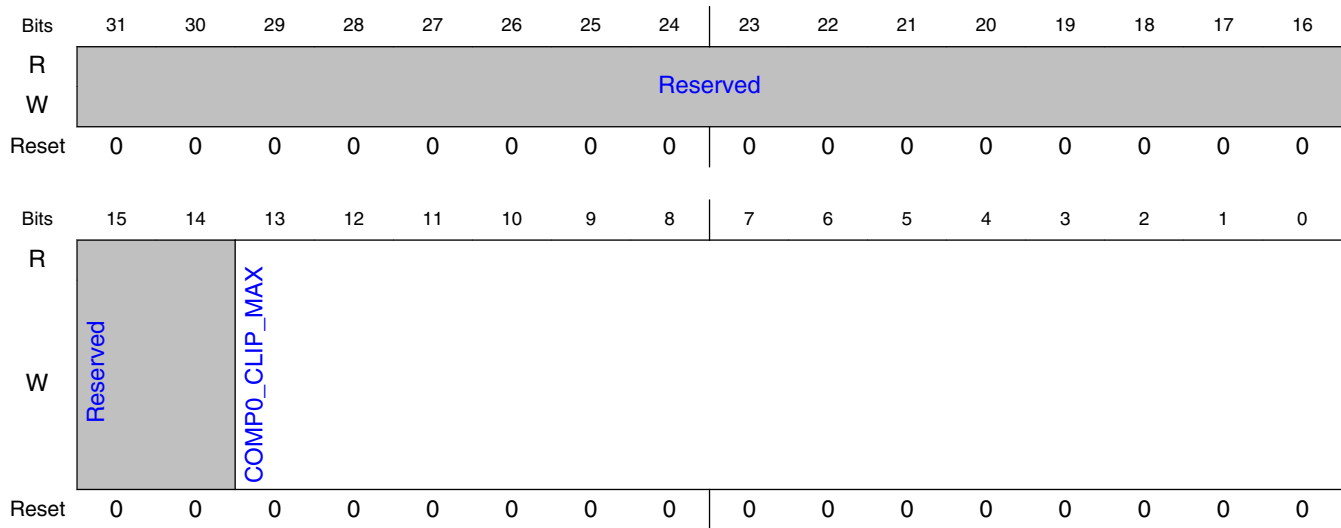
15.10.3.1.182.1 Offset

Register	Offset
HDR_PIPE3_CSCB_IO_MAX_0	B840h

15.10.3.1.182.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0, 16383]. Pipe1 is also called channel1.

15.10.3.1.182.3 Diagram



15.10.3.1.182.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMPO_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.183 Pipe1 Colorspace Converter B (CSCB) component 1 clip max value. (HDR_PIPE3_CSCB_IO_MAX_1)

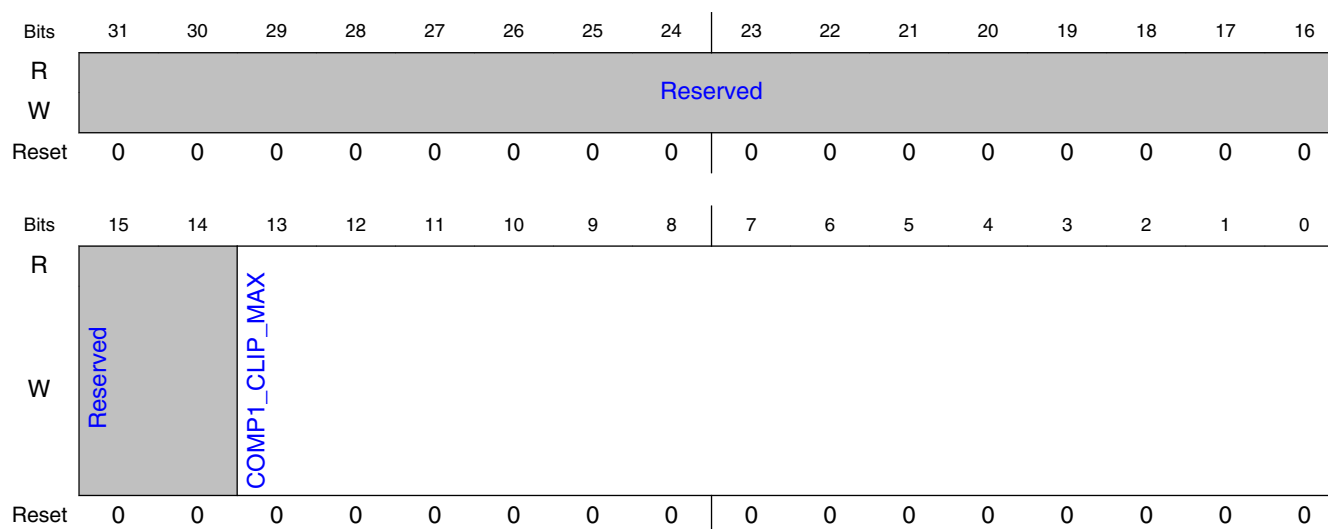
15.10.3.1.183.1 Offset

Register	Offset
HDR_PIPE3_CSCB_IO_MAX_1	B844h

15.10.3.1.183.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,16383]. Pipe1 is also called channel1.

15.10.3.1.183.3 Diagram



15.10.3.1.183.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP1_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.184 Pipe1 Colorspace Converter B (CSCB) component 2 clip max value. (HDR_PIPE3_CSCB_IO_MAX_2)

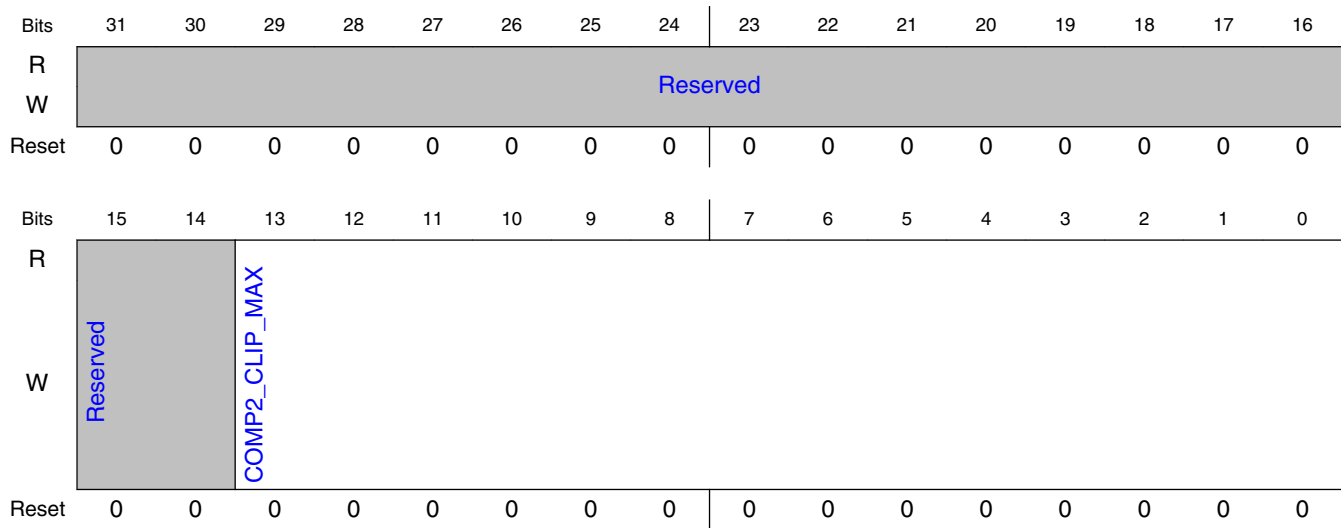
15.10.3.1.184.1 Offset

Register	Offset
HDR_PIPE3_CSCB_IO_MAX_2	B848h

15.10.3.1.184.2 Function

After the pre-increment, The result is clipped. This 14-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,16383]. Pipe1 is also called channel1.

15.10.3.1.184.3 Diagram



15.10.3.1.184.4 Fields

Field	Function
31-14 —	Reserved.
13-0 COMP2_CLIP_MAX	This 14-bit unsigned value is the maximum value of pixel component after the pre-increment.

15.10.3.1.185 Pipe1 Colorspace Converter B (CSCB) normalization factor (HDR_PIPE3_CSCB_NORM)

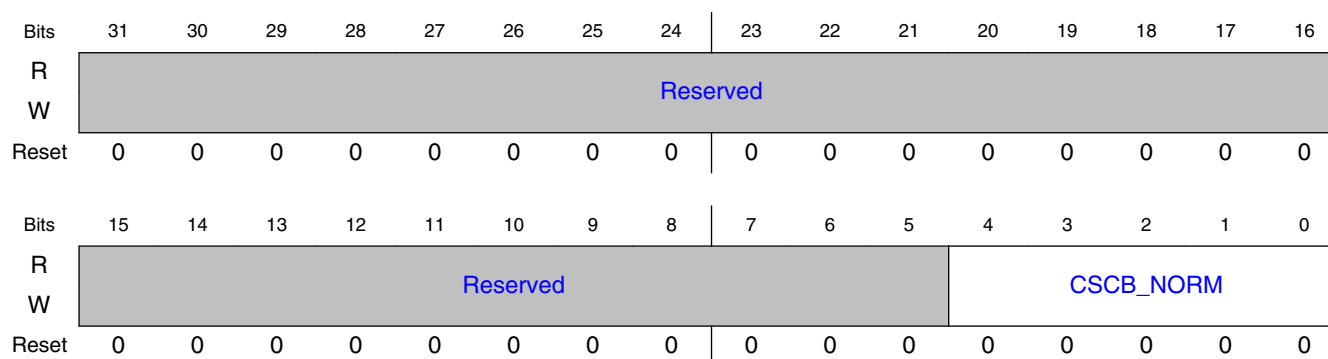
15.10.3.1.185.1 Offset

Register	Offset
HDR_PIPE3_CSCB_NORM	B84Ch

15.10.3.1.185.2 Function

After the CSC matrix multiply, all components of the result are shifted right by the amount in this register. This is a signed right shift. Pipe1 is also called channel1.

15.10.3.1.185.3 Diagram



15.10.3.1.185.4 Fields

Field	Function
31-5 —	Reserved.
4-0 CSCB_NORM	This 5-bit unsigned value is size if arithmetic shift after matrix multiply.

15.10.3.1.186 Pipe1 Colorspace Converter B (CSCB): Post offset component 0 (HDR_PIPE3_CSCB_OO_0)

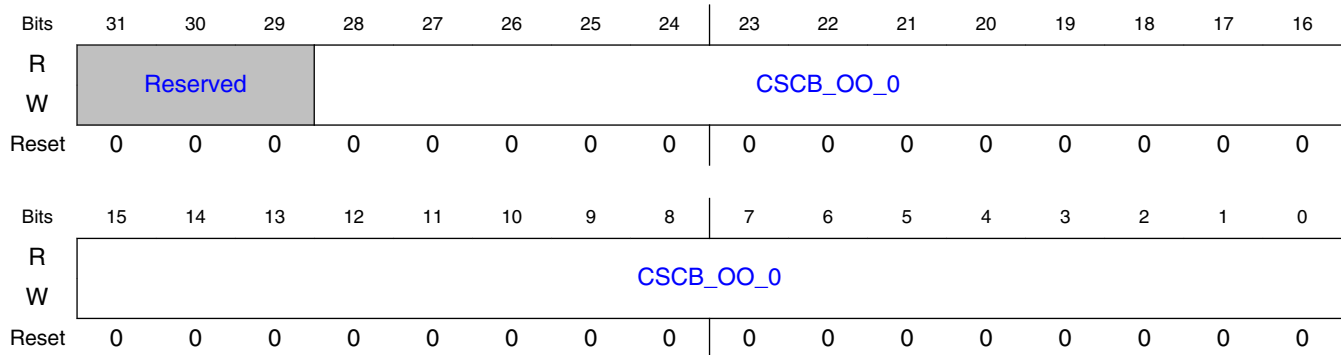
15.10.3.1.186.1 Offset

Register	Offset
HDR_PIPE3_CSCB_OO_0	B850h

15.10.3.1.186.2 Function

After the CSC normalization, this offset value is added to component 0. This is a signed 29-bit number. Pipe1 is also called channel1.

15.10.3.1.186.3 Diagram



15.10.3.1.186.4 Fields

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_0	Output Offset (OO) This is a signed 29-bit number. Per component

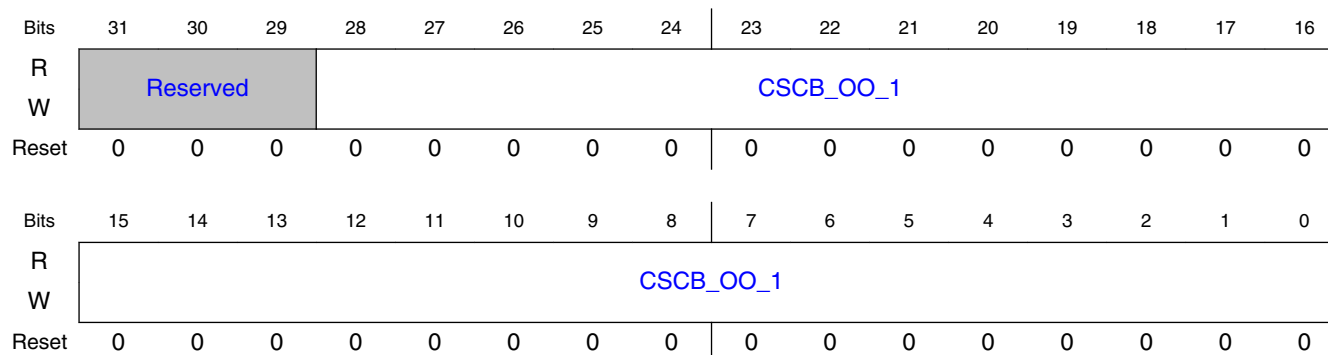
15.10.3.1.187 Pipe1 Colorspace Converter B (CSCB): Post offset component 1 (HDR_PIPE3_CSCB_OO_1)

15.10.3.1.187.1 Offset

Register	Offset
HDR_PIPE3_CSCB_OO_1	B854h

15.10.3.1.187.2 Function

After the CSC normalization, this offset value is added to component 1. This is a signed 28-bit number. Pipe1 is also called channel1.

15.10.3.1.187.3 Diagram**15.10.3.1.187.4 Fields**

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_1	Output Offset (OO) This is a signed 29-bit number. Per component

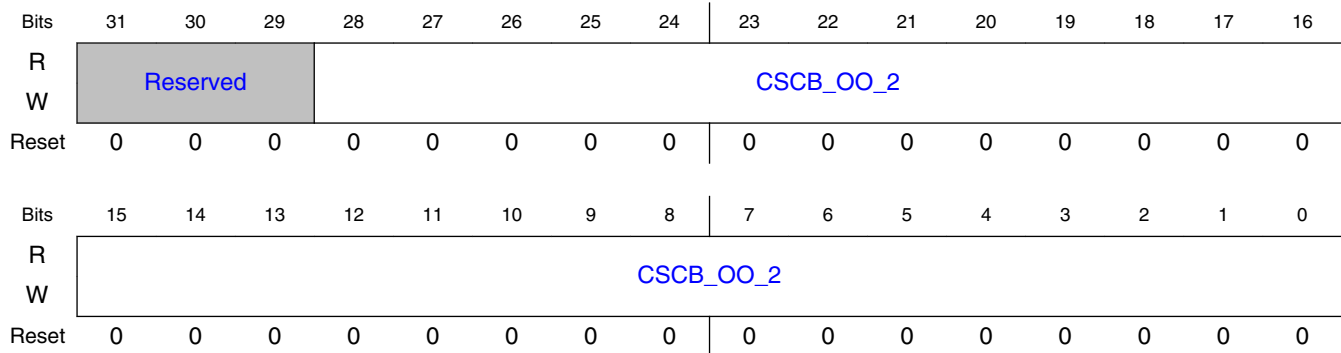
15.10.3.1.188 Pipe1 Colorspace Converter B (CSCB): Post offset component 2 (HDR_PIPE3_CSCB_OO_2)**15.10.3.1.188.1 Offset**

Register	Offset
HDR_PIPE3_CSCB_OO_2	B858h

15.10.3.1.188.2 Function

After the CSC normalization, this offset value is added to component 2. This is a signed 29-bit number. Pipe1 is also called channel1.

15.10.3.1.188.3 Diagram



15.10.3.1.188.4 Fields

Field	Function
31-29 —	Reserved.
28-0 CSCB_OO_2	Output Offset (OO) This is a signed 29-bit number. Per component

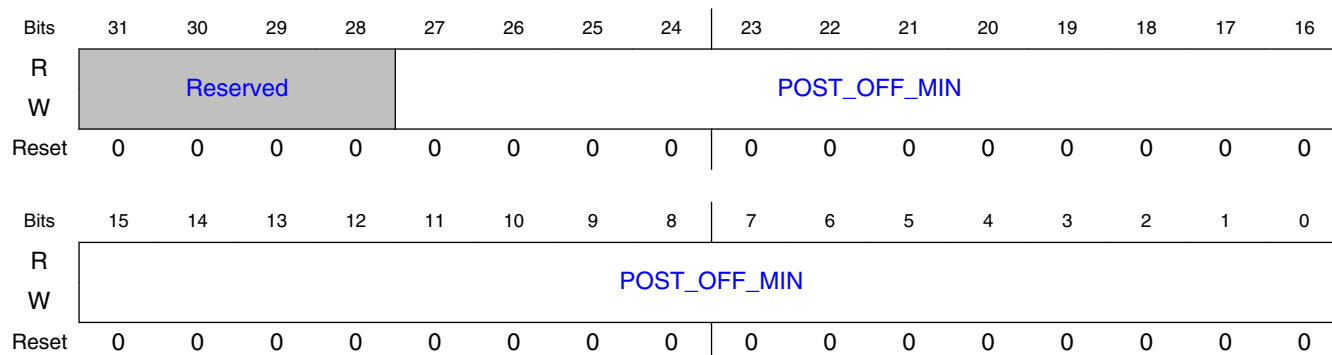
15.10.3.1.189 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 0 (HDR_PIPE3_CSCB_OMIN_0)

15.10.3.1.189.1 Offset

Register	Offset
HDR_PIPE3_CSCB_OMIN_0	B85Ch

15.10.3.1.189.2 Function

After the post offset is added this component is clipped. This is the minimum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.189.3 Diagram**15.10.3.1.189.4 Fields**

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minum clipped pixel component.

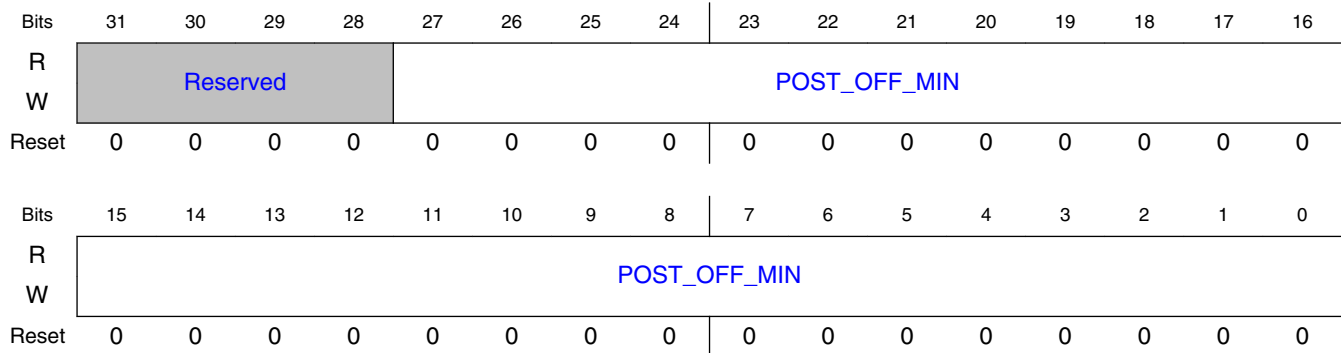
15.10.3.1.190 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 1 (HDR_PIPE3_CSCB_OMIN_1)**15.10.3.1.190.1 Offset**

Register	Offset
HDR_PIPE3_CSCB_OMIN_1	B860h

15.10.3.1.190.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.190.3 Diagram



15.10.3.1.190.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minum clipped pixel component.

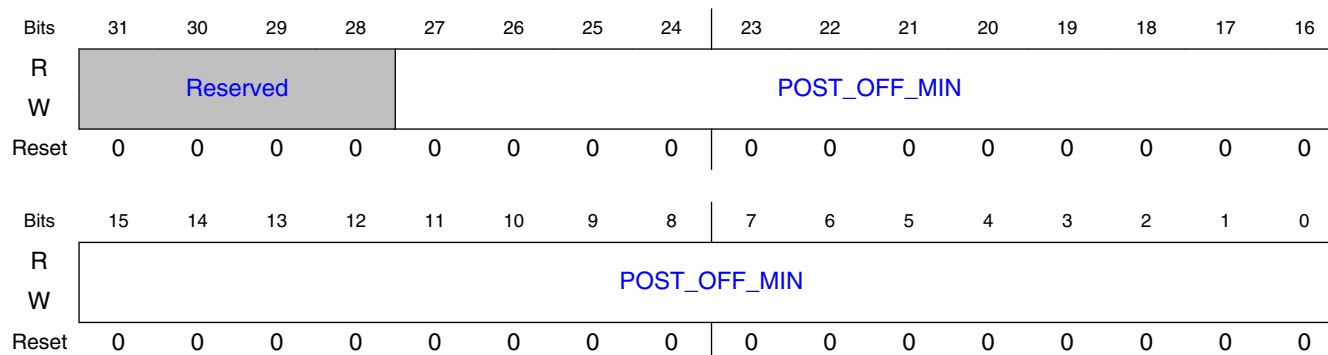
15.10.3.1.191 Pipe1 Colorspace Converter B (CSCB): Post offset min clip value for component 2 (HDR_PIPE3_CSCB_OMIN_2)

15.10.3.1.191.1 Offset

Register	Offset
HDR_PIPE3_CSCB_OMIN_2	B864h

15.10.3.1.191.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.191.3 Diagram**15.10.3.1.191.4 Fields**

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MIN	Minimum clipped pixel component.

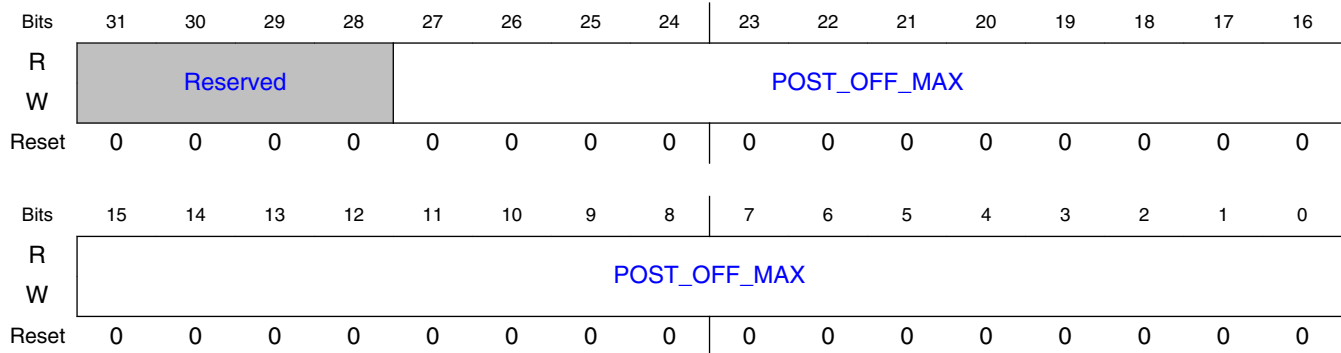
15.10.3.1.192 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 0 (HDR_PIPE3_CSCB_OMAX_0)**15.10.3.1.192.1 Offset**

Register	Offset
HDR_PIPE3_CSCB_OMAX_0	B868h

15.10.3.1.192.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.192.3 Diagram



15.10.3.1.192.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MAX	Maximum clipped pixel component.

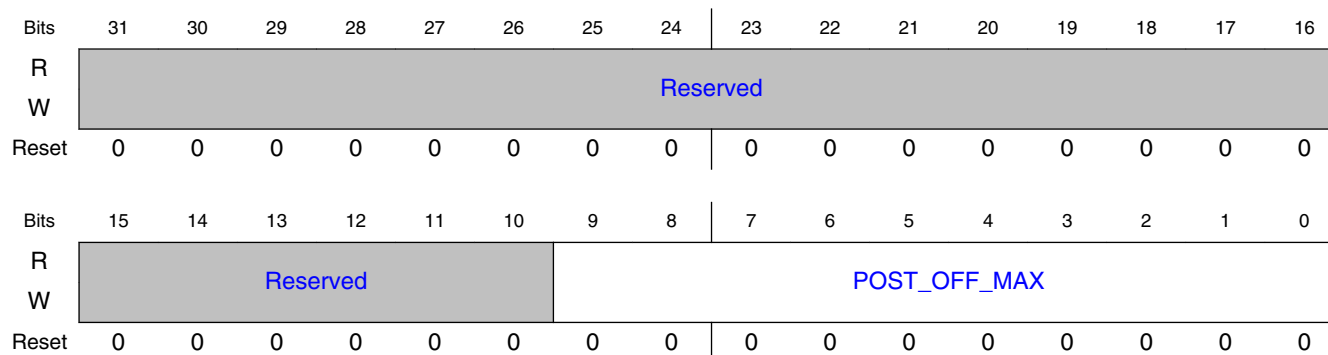
15.10.3.1.193 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 1 (HDR_PIPE3_CSCB_OMAX_1)

15.10.3.1.193.1 Offset

Register	Offset
HDR_PIPE3_CSCB_OMAX_1	B86Ch

15.10.3.1.193.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number. Pipe1 is also called channel1.

15.10.3.1.193.3 Diagram**15.10.3.1.193.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component.

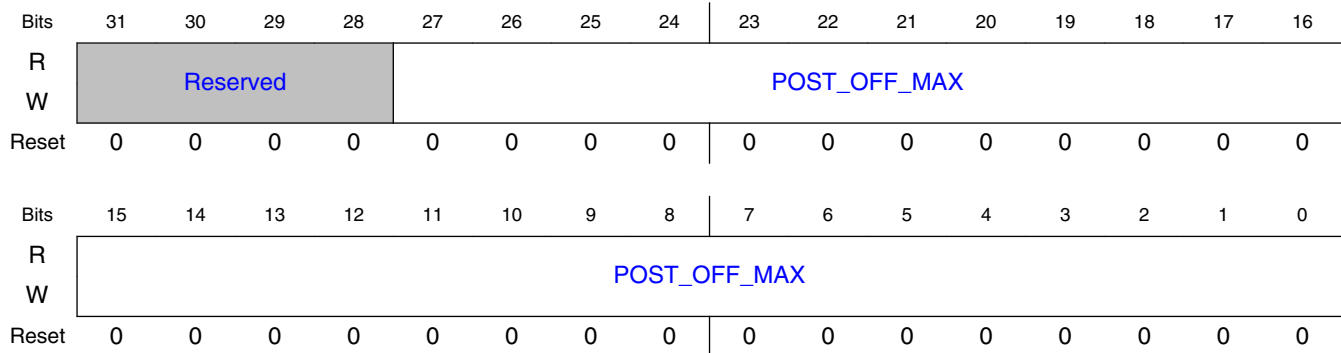
15.10.3.1.194 Pipe1 Colorspace Converter B (CSCB): Post offset max clip value for component 2 (HDR_PIPE3_CSCB_OMAX_2)**15.10.3.1.194.1 Offset**

Register	Offset
HDR_PIPE3_CSCB_OMAX_2	B870h

15.10.3.1.194.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 28-bit number. Pipe1 is also called channel1.

15.10.3.1.194.3 Diagram



15.10.3.1.194.4 Fields

Field	Function
31-28 —	Reserved.
27-0 POST_OFF_MAX	Maximum clipped pixel component.

15.10.3.1.195 Pipe1 floating point to fixed point control (HDR_PIPE3_FL2FX)

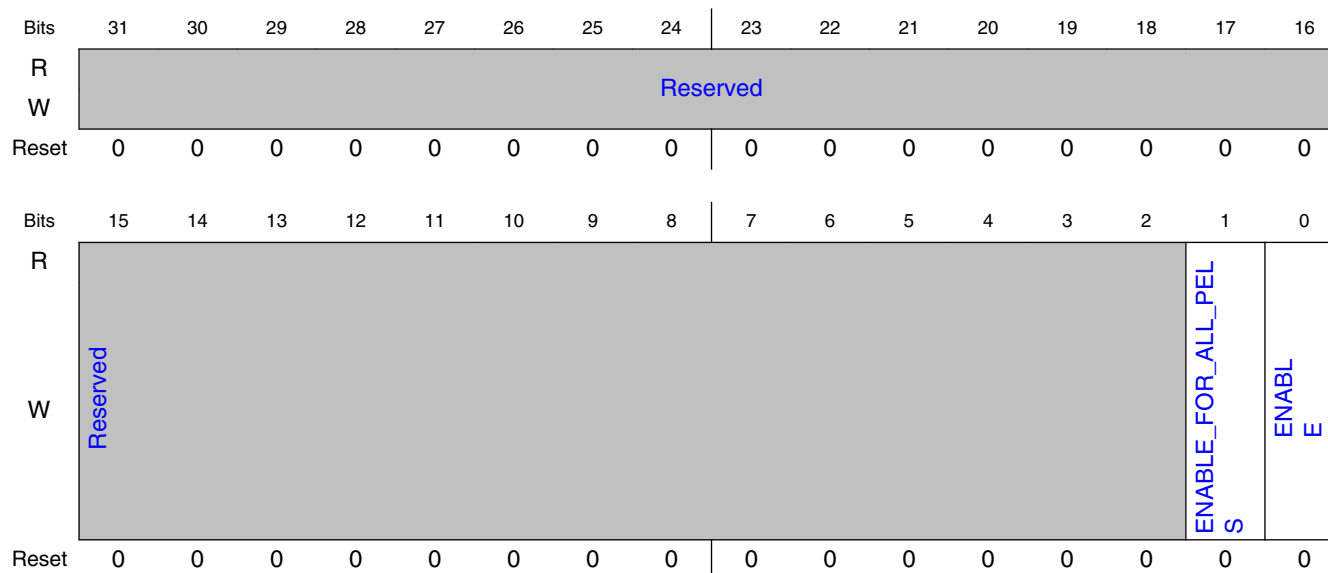
15.10.3.1.195.1 Offset

Register	Offset
HDR_PIPE3_FL2FX	B874h

15.10.3.1.195.2 Function

Pipe1 floating point to fixed point control Pipe1 is also called channel1.

15.10.3.1.195.3 Diagram



15.10.3.1.195.4 Fields

Field	Function
31-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This Float to Fixed operation is enabled only for blended pixels 1: This Float to Fixed operation is enabled all pixels
0 ENABLE	0: Don't enable this Float-to-Fixed converter: Pixels pass thru the Float-to-Fixed unmodified 1: This Float-to-Fixed converter is enabled for current picture

15.10.3.1.196 PIPE3: NOT USED (HDR_PIPE3_ENTRY_30)

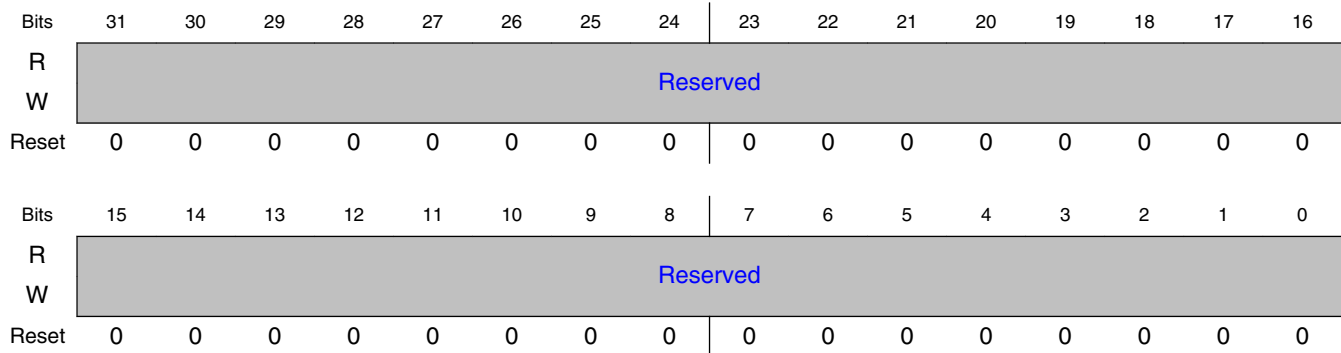
15.10.3.1.196.1 Offset

Register	Offset
HDR_PIPE3_ENTRY_30	B878h

15.10.3.1.196.2 Function

PIPE3: NOT USED

15.10.3.1.196.3 Diagram



15.10.3.1.196.4 Fields

Field	Function
31-0	Reserved.
—	

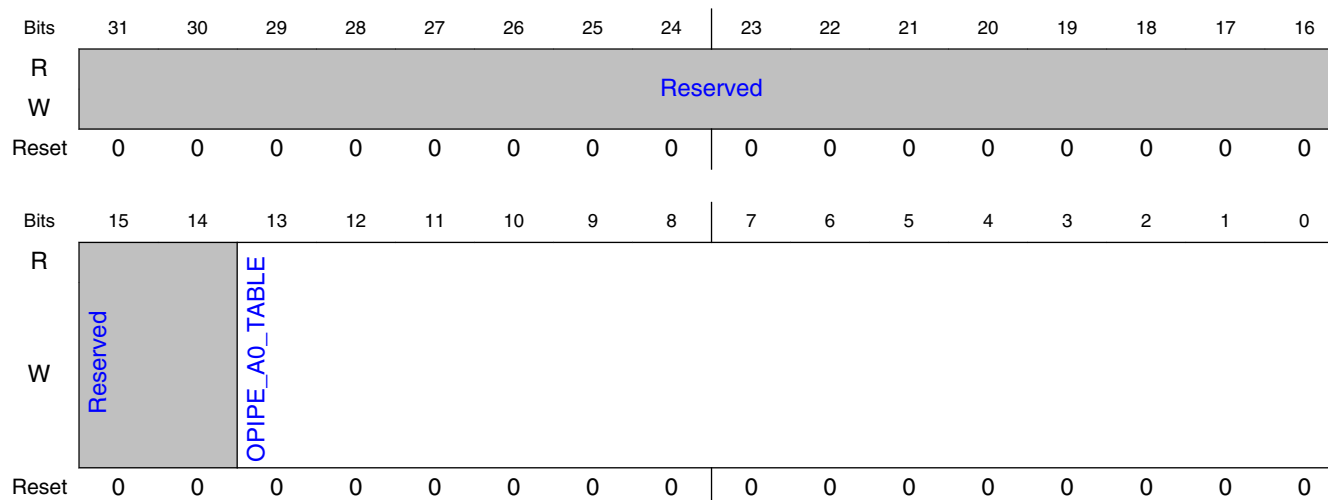
15.10.3.1.197 A0 component Linear-to-Non-linear conversion table (OPIPE_A0_TABLE)

15.10.3.1.197.1 Offset

Register	Offset
OPIPE_A0_TABLE	C000h

15.10.3.1.197.2 Function

The Linear to Non Linear Conversion is done using a binary search algorithm. 0th stage Node is at: address 0 or 1; 1st stage Nodes are at: addresses 2-3; 2nd stage Nodes are at addresses 4-7; 3rd stage Nodes are at addresses 8-15; 4th stage Nodes are at addresses 16-31; 5th stage Nodes are at addresses 32-63; 6th stage Nodes are at addresses 64-127; 7th stage Nodes are at addresses 128-255; 8th stage Nodes are at addresses 256-511; The binary search algorithm traces the Nodes starting at 0th stage, then proceeding to the next stage. Inside a stage, the search comparisons proceed "left-to-right" toward higher node address.

15.10.3.1.197.3 Diagram**15.10.3.1.197.4 Fields**

Field	Function
31-14 —	Reserved.
13-0 OPIPE_A0_TABLE	The TABLE Nodes are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.198 A1 component Linear-to-Non-linear conversion table (OPIPE_A1_TABLE)**15.10.3.1.198.1 Offset**

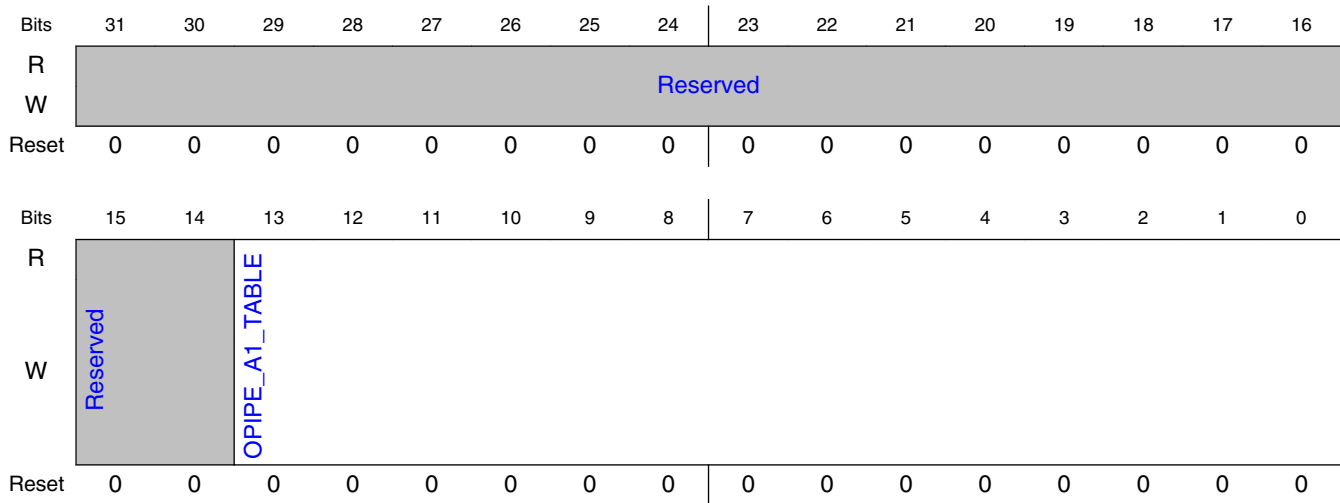
Register	Offset
OPIPE_A1_TABLE	D000h

15.10.3.1.198.2 Function

The Linear to Non Linear Conversion is done using a binary search algorithm. 0th stage Node is at: address 0 or 1; 1st stage Nodes are at: addresses 2-3; 2nd stage Nodes are at addresses 4-7; 3rd stage Nodes are at addresses 8-15; 4th stage Nodes are at addresses 16-31; 5th stage Nodes are at addresses 32-63; 6th stage Nodes are at addresses 64-127; 7th stage Nodes are at addresses 128-255; 8th stage Nodes are at addresses 256-511; The

binary search algorithm traces the Nodes starting at 0th stage, then proceeding to the next stage. Inside a stage, the search comparisons proceed "left-to-right" toward higher node address.

15.10.3.1.198.3 Diagram



15.10.3.1.198.4 Fields

Field	Function
31-14 —	Reserved.
13-0 OPIPE_A1_TABLE	The TABLE Nodes are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

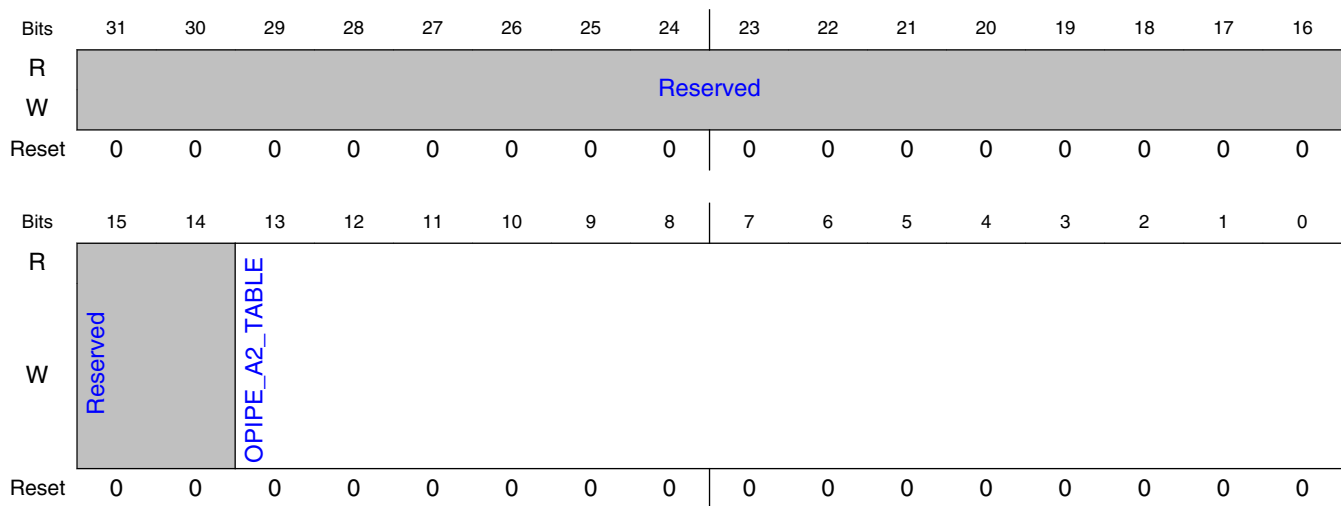
15.10.3.1.199 A2 component Linear-to-Non-linear conversion table (OPIPE_A2_TABLE)

15.10.3.1.199.1 Offset

Register	Offset
OPIPE_A2_TABLE	E000h

15.10.3.1.199.2 Function

The Linear to Non Linear Conversion is done using a binary search algorithm. 0th stage Node is at: address 0 or 1; 1st stage Nodes are at: addresses 2-3; 2nd stage Nodes are at addresses 4-7; 3rd stage Nodes are at addresses 8-15; 4th stage Nodes are at addresses 16-31; 5th stage Nodes are at addresses 32-63; 6th stage Nodes are at addresses 64-127; 7th stage Nodes are at addresses 128-255; 8th stage Nodes are at addresses 256-511; The binary search algorithm traces the Nodes starting at 0th stage, then proceeding to the next stage. Inside a stage, the search comparisons proceed "left-to-right" toward higher node address.

15.10.3.1.199.3 Diagram**15.10.3.1.199.4 Fields**

Field	Function
31-14 —	Reserved.
13-0 OPIPE_A2_TABLE	The TABLE Nodes are 14 bits wide. They may be unsigned integers OR 14-bit floating point numbers

15.10.3.1.200 HDR output stage Colorspace Converter (CSCO) control. (HDR_OPIPE_CSC_CONTROL_REG)

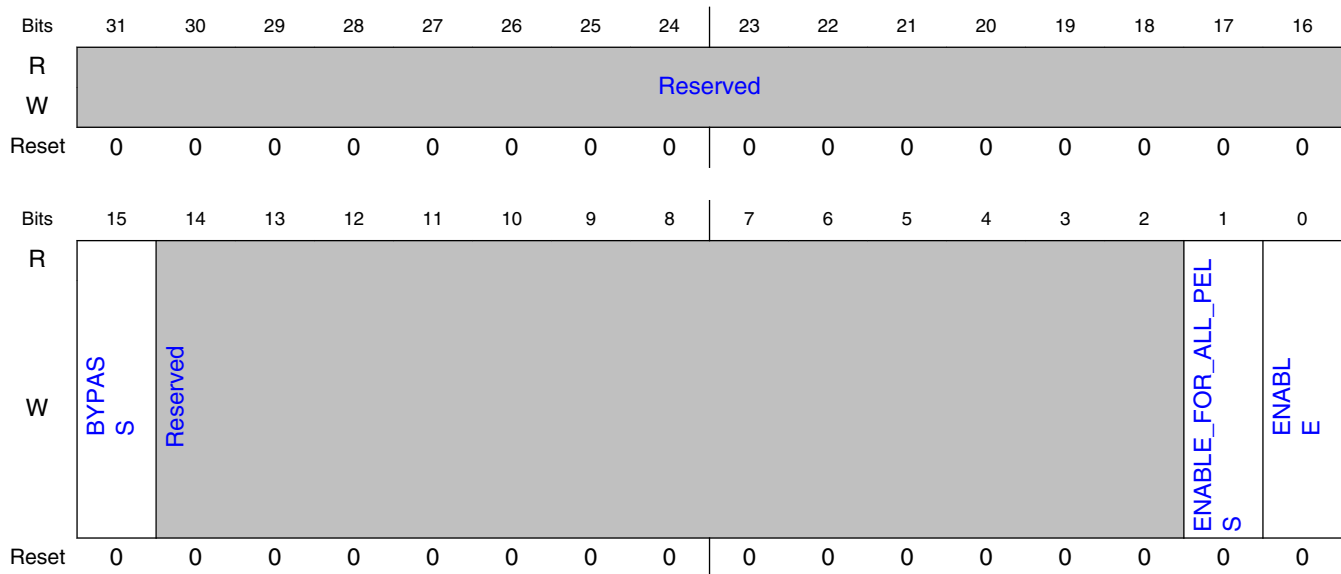
15.10.3.1.200.1 Offset

Register	Offset
HDR_OPIPE_CSC_CONTROL_REG	F000h

15.10.3.1.200.2 Function

Controls the Color-Space-Converter (CSC) in HDR output stage. This CSC takes in 10-bit pixel components (component 0, component 1, component 2). A per-component offset (called pre-offset) is added to the pixel. After the pre-offset operation the pixel components are clipped. MAX_RANGE of PRE_CLIP is [-512, 1023]. A 3x3 matrix multiply (constant coefficients H(x,y)) is performed on the pixel to take to a different color space. After this matrix multiply, a 28 bit signed offset (post-offset) is added to the normalized result of the matrix multiply. After the post-offset operation the results are clipped. MAX_RANGE of POST_CLIP is [0,1023] Output of this CSC are 10 bit per component pixels. After CSC the pixels are fed to the Chroma Subsampler.

15.10.3.1.200.3 Diagram



15.10.3.1.200.4 Fields

Field	Function
31-16	Reserved.
—	

Table continues on the next page...

Memory Map and Registers

Field	Function
15 BYPASS	0: Don't bypass this CSC 1: Pixels pass thru this CSC unmodified
14-2 —	Reserved.
1 ENABLE_FOR_ALL_PELS	0: This CSC is enabled only for blened pixels 1: This CSC is enabled all pixels
0 ENABLE	0: Don't enable this CSC: Pixels pass thru the CSC unmodified 1: This CSC is enabled for current picture

15.10.3.1.201 Pipe1 Colorspace Converter (CSC) h(0,0) matrix coefficient (HDR_OPIPE_CSC_H00)

15.10.3.1.201.1 Offset

Register	Offset
HDR_OPIPE_CSC_H00	F004h

15.10.3.1.201.2 Function

h(0,0) coefficient in color space matrix multiply

15.10.3.1.201.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	H00															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.10.3.1.201.4 Fields

Field	Function
31-16	Reserved.

Table continues on the next page...

Field	Function
—	
15-0 H00	h(0,0) 16 bit signed coefficient

15.10.3.1.202 Pipe1 Colorspace Converter (CSC) h(1,0) matrix coefficient (HDR_OPIPE_CSC_H10)

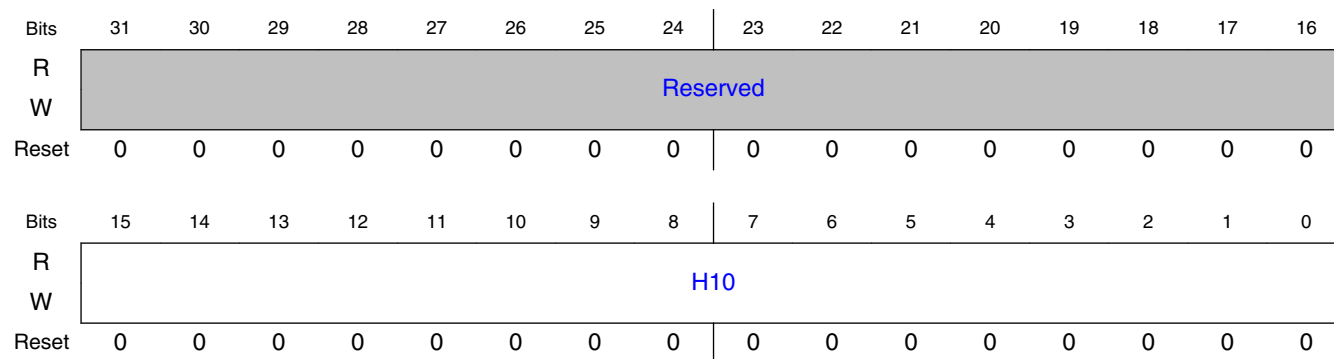
15.10.3.1.202.1 Offset

Register	Offset
HDR_OPIPE_CSC_H10	F008h

15.10.3.1.202.2 Function

h(1,0) coefficient in color space matrix multiply

15.10.3.1.202.3 Diagram



15.10.3.1.202.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H10	h(1,0) 16 bit signed coefficient

15.10.3.1.203 HDR OUTPUT Colorspace Converter (CSCO) h(2,0) matrix coefficient (HDR_OPIPE_CSC_H20)

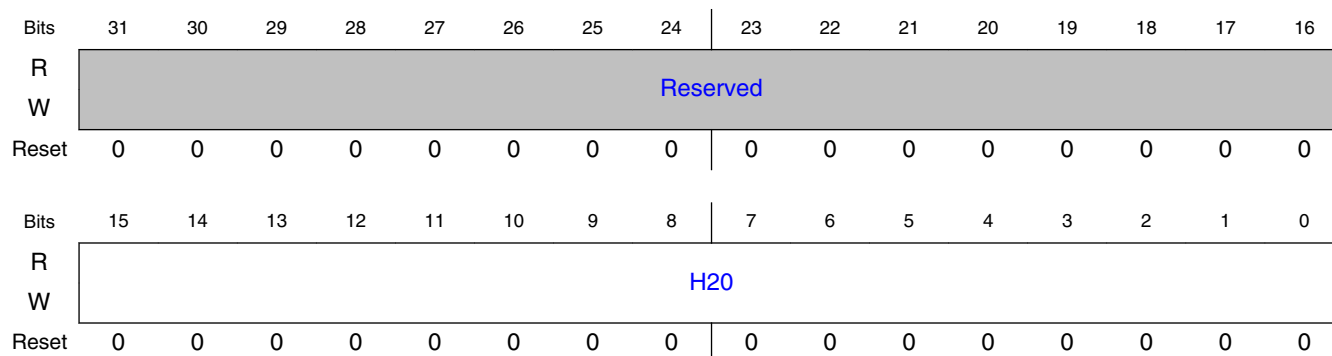
15.10.3.1.203.1 Offset

Register	Offset
HDR_OPIPE_CSC_H20	F00Ch

15.10.3.1.203.2 Function

h(2,0) coefficient in color space matrix multiply

15.10.3.1.203.3 Diagram



15.10.3.1.203.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H20	h(2,0) 16 bit signed coefficient

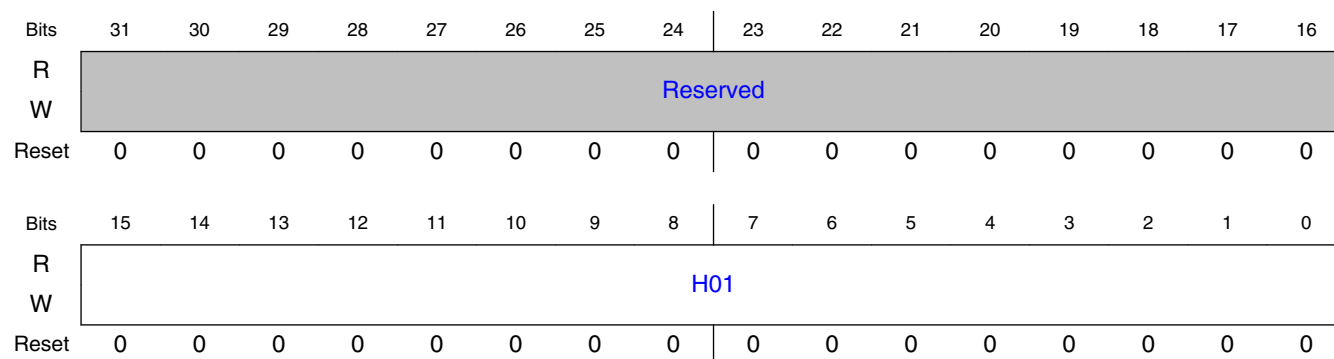
15.10.3.1.204 HDR OUTPUT pipe Colorspace Converter (CSCO) h(0,1) matrix coefficient (HDR_OPIPE_CSC_H01)

15.10.3.1.204.1 Offset

Register	Offset
HDR_OPIPE_CSC_H01	F010h

15.10.3.1.204.2 Function

h(0,1) coefficient in color space matrix multiply

15.10.3.1.204.3 Diagram**15.10.3.1.204.4 Fields**

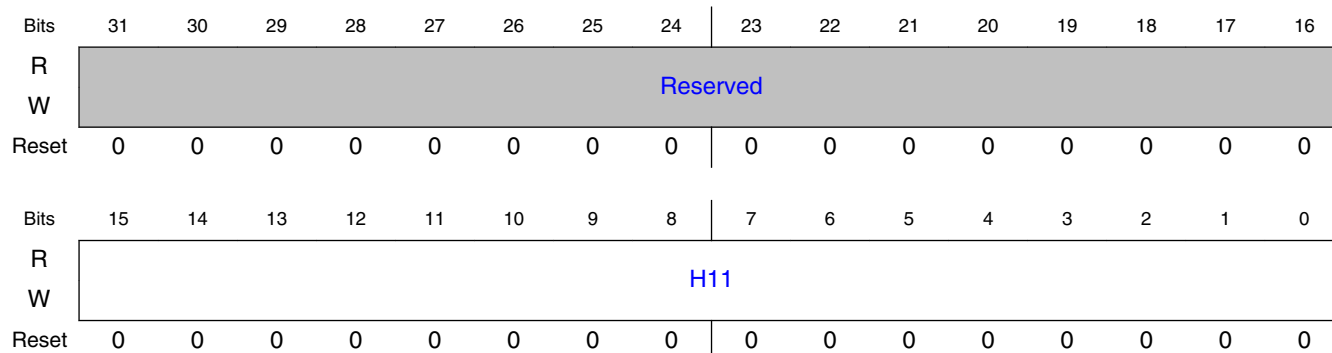
Field	Function
31-16 —	Reserved.
15-0 H01	h(0,1) 16 bit signed coefficient

15.10.3.1.205 HDR OUTPUT pipe Colorspace Converter (CSCO) h(1,1) matrix coefficient (HDR_OPIPE_CSC_H11)**15.10.3.1.205.1 Offset**

Register	Offset
HDR_OPIPE_CSC_H11	F014h

15.10.3.1.205.2 Function

h(1,1) coefficient in color space matrix multiply

15.10.3.1.205.3 Diagram**15.10.3.1.205.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H11	h(1,1) 16 bit signed coefficient

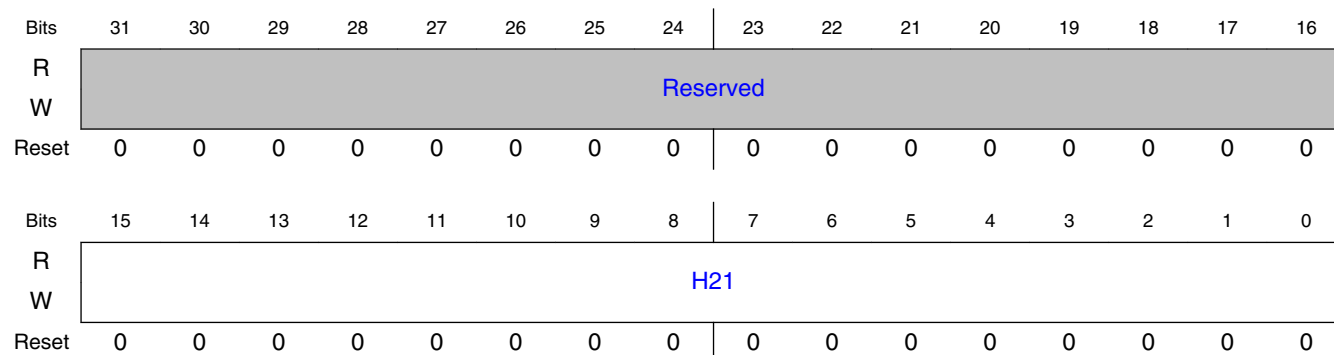
15.10.3.1.206 HDR_output pipe Colorspace Converter (CSCO) h(2,1) matrix coefficient (HDR_OPIPE_CSC_H21)**15.10.3.1.206.1 Offset**

Register	Offset
HDR_OPIPE_CSC_H21	F018h

15.10.3.1.206.2 Function

h(2,1) coefficient in color space matrix multiply

15.10.3.1.206.3 Diagram



15.10.3.1.206.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H21	h(2,1) 16 bit signed coefficient

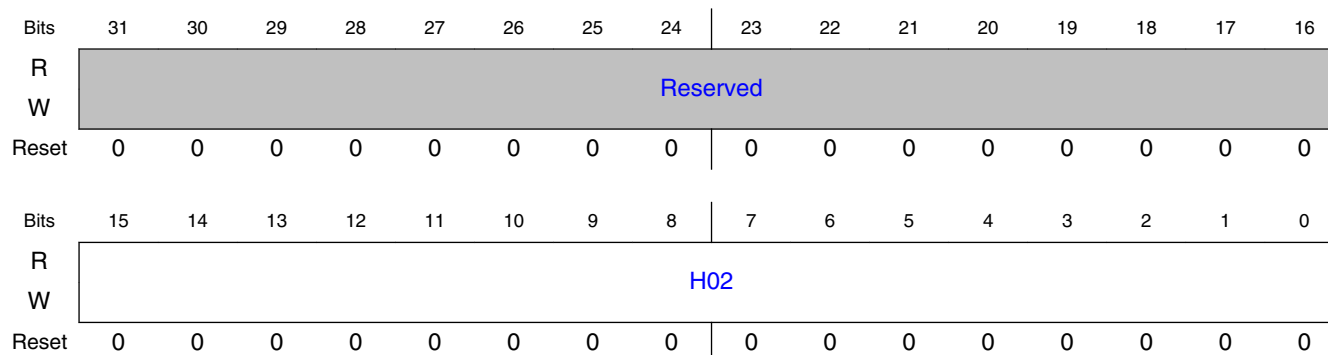
15.10.3.1.207 HDR OUTPUT pipe Colorspace Converter (CSCO) h(0,2) matrix coefficient (HDR_OPIPE_CSC_H02)

15.10.3.1.207.1 Offset

Register	Offset
HDR_OPIPE_CSC_H02	F01Ch

15.10.3.1.207.2 Function

h(0,2) coefficient in color space matrix multiply

15.10.3.1.207.3 Diagram**15.10.3.1.207.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H02	h(0,2) 16 bit signed coefficient

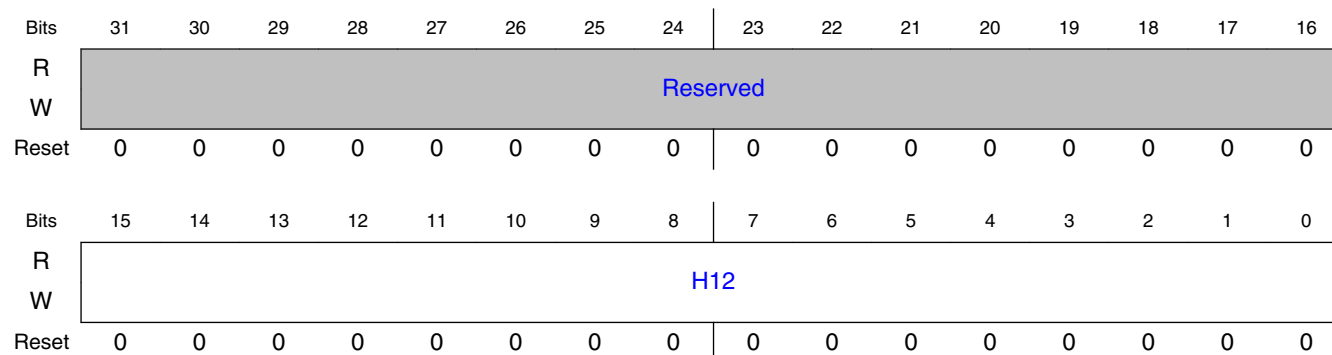
15.10.3.1.208 HDR OUPUT pipe Colorspace Converter (CSCO) h(1,2) matrix coefficient (HDR_OPIPE_CSC_H12)**15.10.3.1.208.1 Offset**

Register	Offset
HDR_OPIPE_CSC_H12	F020h

15.10.3.1.208.2 Function

h(1,2) coefficient in color space matrix multiply

15.10.3.1.208.3 Diagram



15.10.3.1.208.4 Fields

Field	Function
31-16 —	Reserved.
15-0 H12	h(1,2) 16 bit signed coefficient

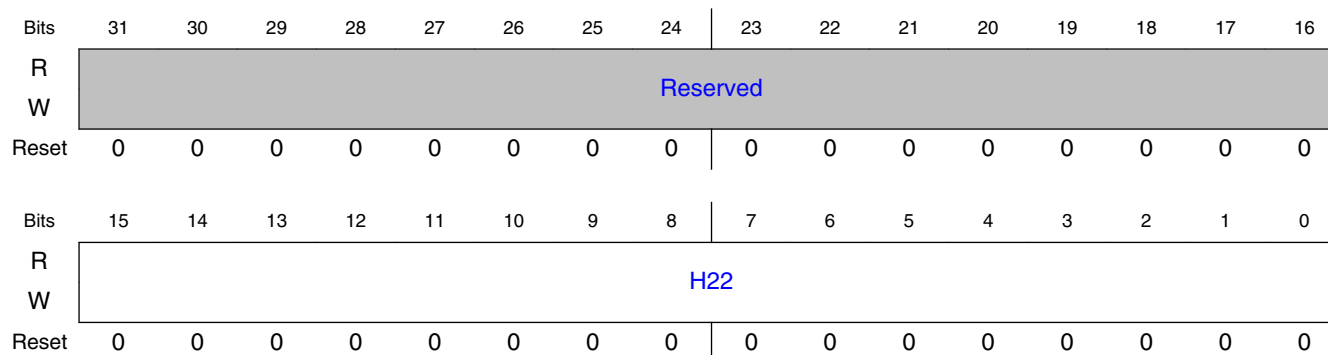
15.10.3.1.209 HDR OUPUT pipe Colorspace Converter (CSCO) h(2,2) matrix coefficient (HDR_)

15.10.3.1.209.1 Offset

Register	Offset
HDR_	F024h

15.10.3.1.209.2 Function

h(2,2) coefficient in color space matrix multiply

15.10.3.1.209.3 Diagram**15.10.3.1.209.4 Fields**

Field	Function
31-16 —	Reserved.
15-0 H22	h(2,2) 16 bit signed coefficient

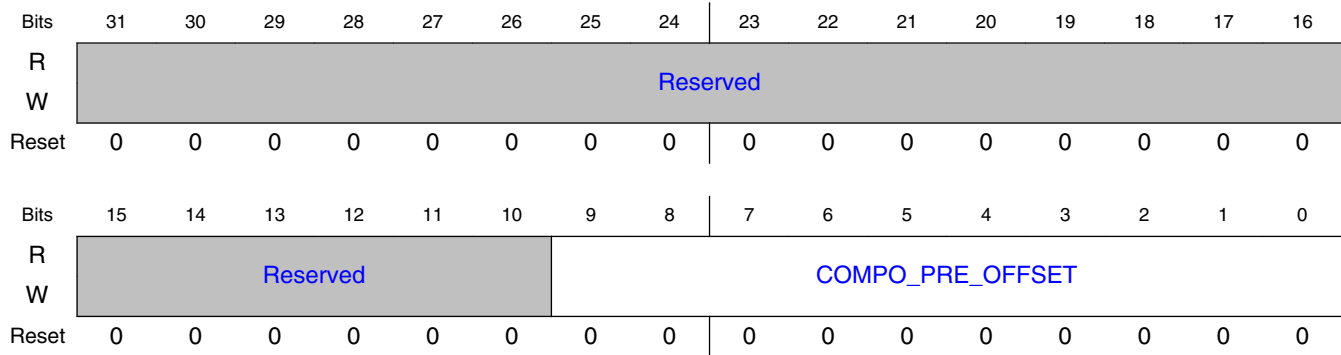
15.10.3.1.210 HDR OUTPUT pipe Colorspace Converter (CSCO) component 0 pre-offset (HDR_OPIPE_CSC_IO_0)**15.10.3.1.210.1 Offset**

Register	Offset
HDR_OPIPE_CSC_IO_0	F028h

15.10.3.1.210.2 Function

An signed 10-bit offset is added to component 0 (R,Y) before matrix multiply

15.10.3.1.210.3 Diagram



15.10.3.1.210.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMPO_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 0 of the pixel

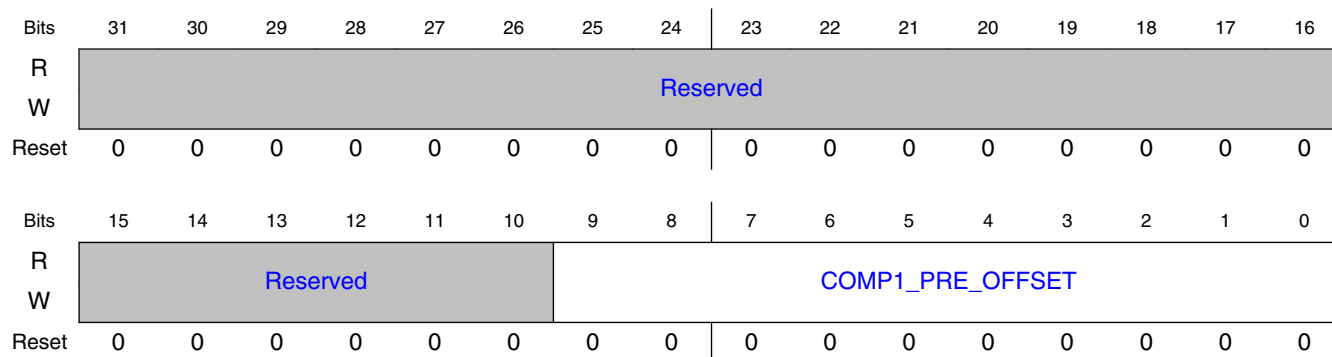
15.10.3.1.211 HDR OUPUT pipe Colorspace Converter (CSCO) component 1 pre-offset (HDR_OPIPE_CSC_IO_1)

15.10.3.1.211.1 Offset

Register	Offset
HDR_OPIPE_CSC_IO_1	F02Ch

15.10.3.1.211.2 Function

An signed 10-bit offset is added to component 1 (G,Cb) before matrix multiply

15.10.3.1.211.3 Diagram**15.10.3.1.211.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP1_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 1 of the pixel

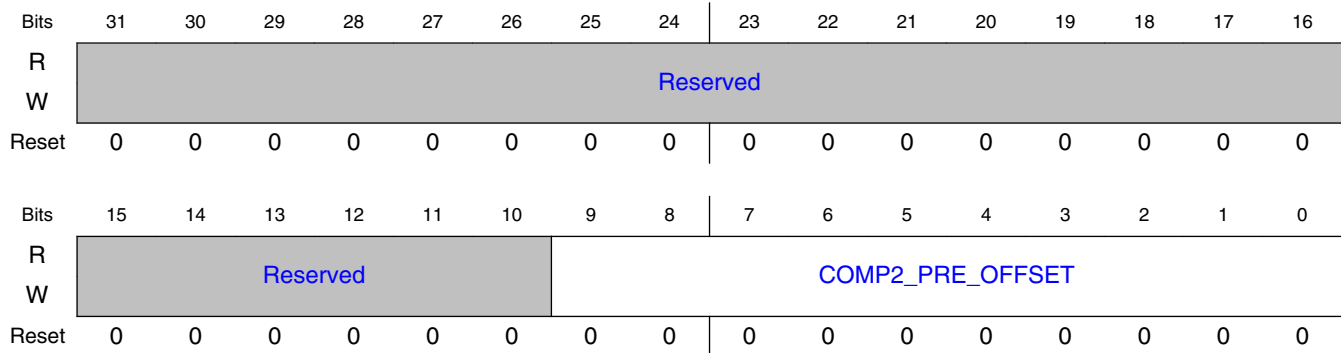
15.10.3.1.212 HDR OUPUT pipe: Colorspace Converter (CSCO) component 2 pre-offset (HDR_OPIPE_CSC_IO_2)**15.10.3.1.212.1 Offset**

Register	Offset
HDR_OPIPE_CSC_IO_2	F030h

15.10.3.1.212.2 Function

An signed 10-bit offset is added to component 2 (B,Cr) before matrix multiply

15.10.3.1.212.3 Diagram



15.10.3.1.212.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_PRE_OFFSET	Before the color space conversion matrix multiply, This offset is added to component 2 of the pixel

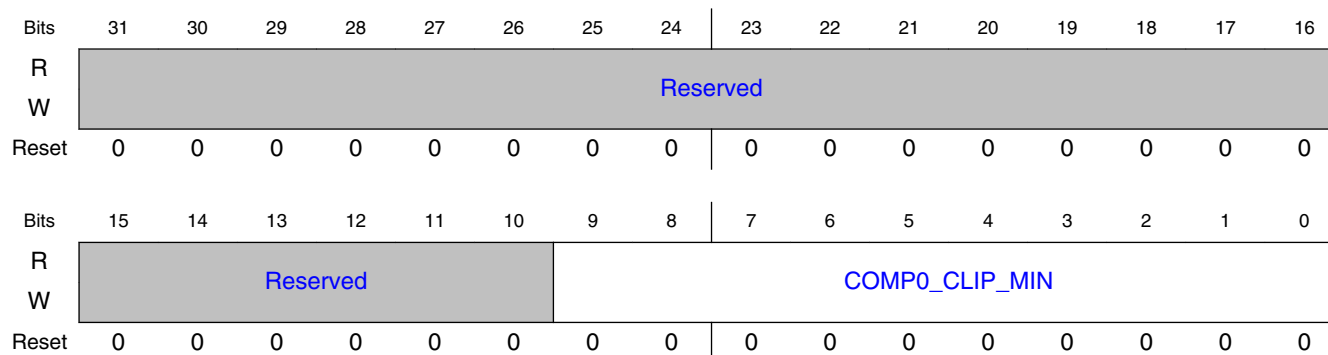
15.10.3.1.213 HDR OUPUTU pipe Colorspace Converter (CSCO) component 0 clip min. (HDR_OPIPE_CSC_MIN_0)

15.10.3.1.213.1 Offset

Register	Offset
HDR_OPIPE_CSC_MIN_0	F034h

15.10.3.1.213.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511].

15.10.3.1.213.3 Diagram**15.10.3.1.213.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP0_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

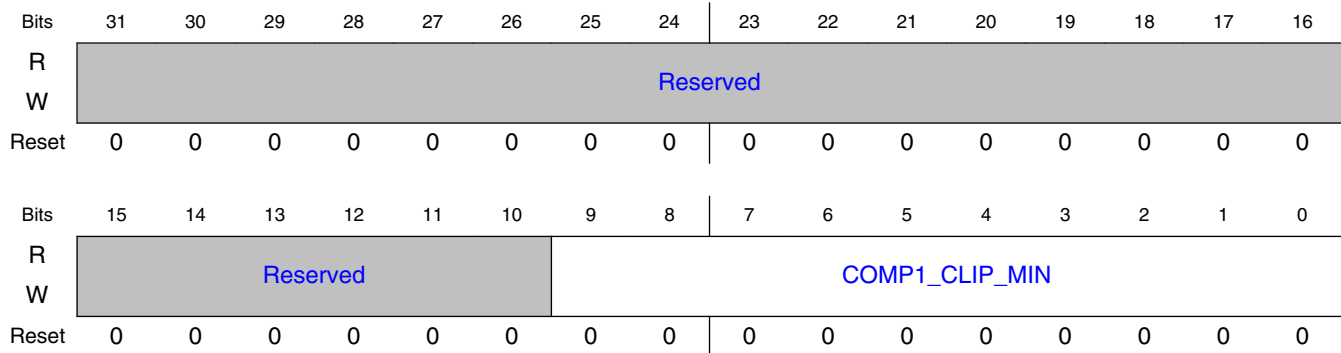
15.10.3.1.214 HDR OUPUT pipe Colorspace Converter (CSCO) component 1 clip min. (HDR_OPIPE_CSC_MIN_1)**15.10.3.1.214.1 Offset**

Register	Offset
HDR_OPIPE_CSC_MIN_1	F038h

15.10.3.1.214.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511].

15.10.3.1.214.3 Diagram



15.10.3.1.214.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP1_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

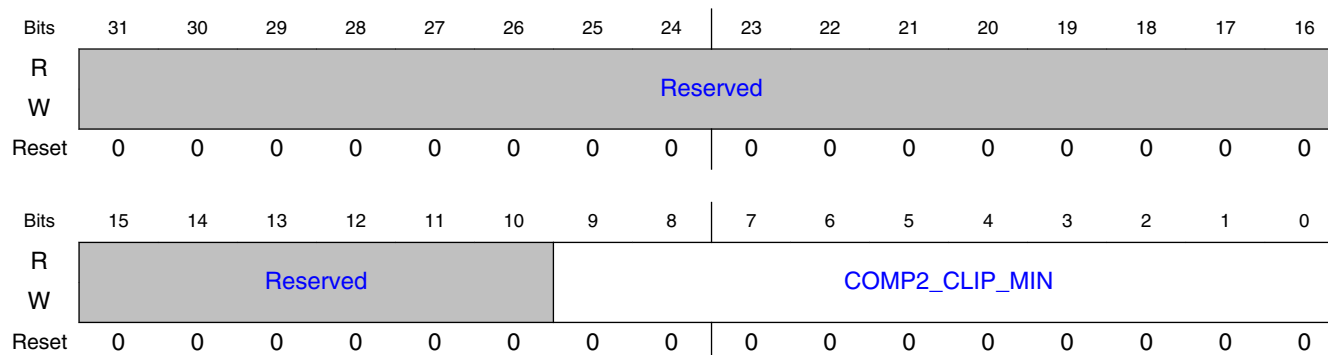
15.10.3.1.215 HDR OUPUTU pipe Colorspace Converter (CSCO) component 2 clip min. (HDR_OPIPE_CSC_MIN_2)

15.10.3.1.215.1 Offset

Register	Offset
HDR_OPIPE_CSC_MIN_2	F03Ch

15.10.3.1.215.2 Function

After the pre-increment, The result is clipped. This 10-bit signed value is the minimum value of pixel component after the pre-increment. So the minimum range is [-512,511].

15.10.3.1.215.3 Diagram**15.10.3.1.215.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP2_CLIP_MIN	This 10-bit signed value is the minimum value of pixel component after the pre-increment.

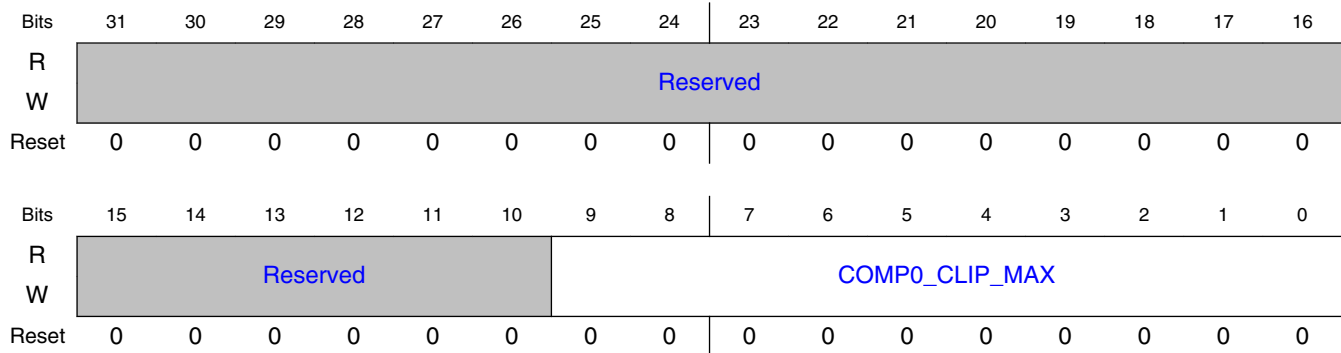
15.10.3.1.216 HDR OUPUT pipe Colorspace Converter O (CSC) component 0 clip max value. (HDR_OPIPE_CSC_MAX_0)**15.10.3.1.216.1 Offset**

Register	Offset
HDR_OPIPE_CSC_MAX_0	F040h

15.10.3.1.216.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023].

15.10.3.1.216.3 Diagram



15.10.3.1.216.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP0_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

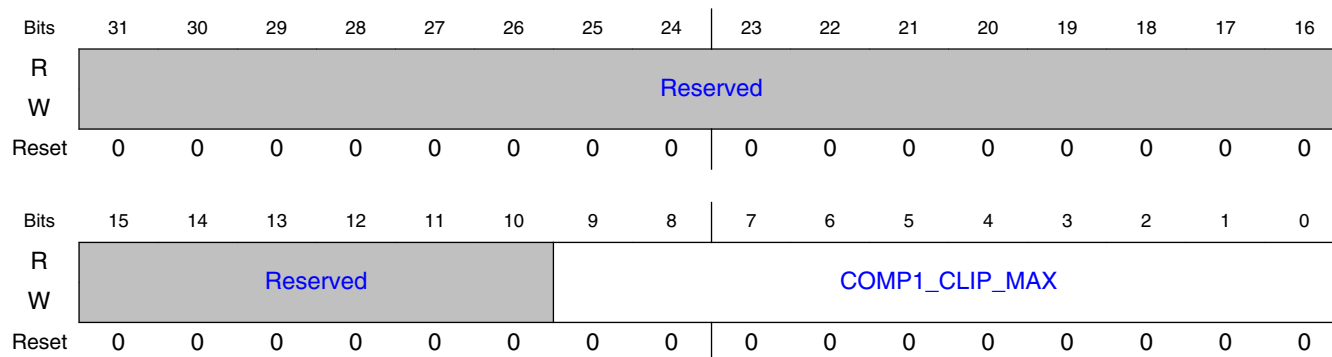
15.10.3.1.217 HDR OUTPUT pipe Colorspace Converter (CSCO) component 1 clip max value. (HDR_OPIPE_CSC_MAX_1)

15.10.3.1.217.1 Offset

Register	Offset
HDR_OPIPE_CSC_MAX_1	F044h

15.10.3.1.217.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023].

15.10.3.1.217.3 Diagram**15.10.3.1.217.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 COMP1_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

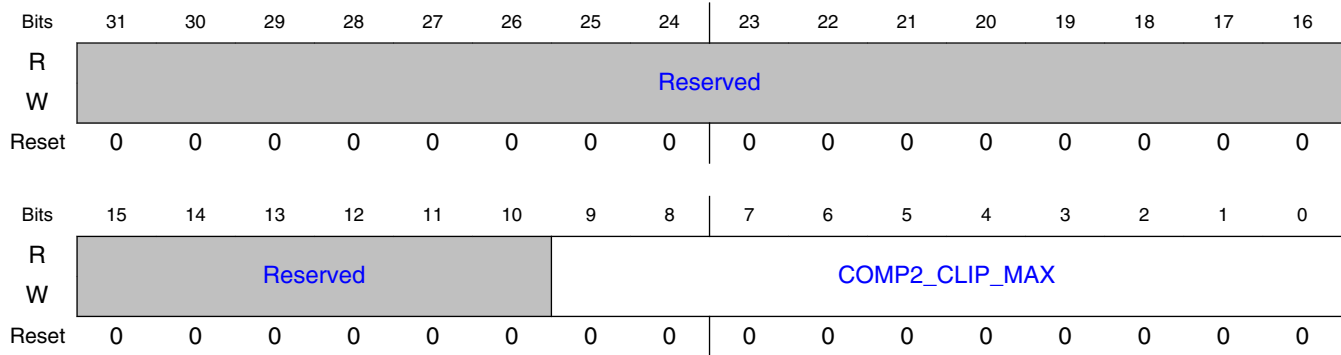
15.10.3.1.218 HDR OUTPUT pipe Colorspace Converter (CSCO) component 2 clip max value. (HDR_OPIPE_CSC_MAX_2)**15.10.3.1.218.1 Offset**

Register	Offset
HDR_OPIPE_CSC_MAX_2	F048h

15.10.3.1.218.2 Function

After the pre-increment, The result is clipped. This 10-bit unsigned value is the maximum value of pixel component after the pre-increment. So the maximum range is [0,1023].

15.10.3.1.218.3 Diagram



15.10.3.1.218.4 Fields

Field	Function
31-10 —	Reserved.
9-0 COMP2_CLIP_MAX	This 10-bit unsigned value is the maximum value of pixel component after the pre-increment.

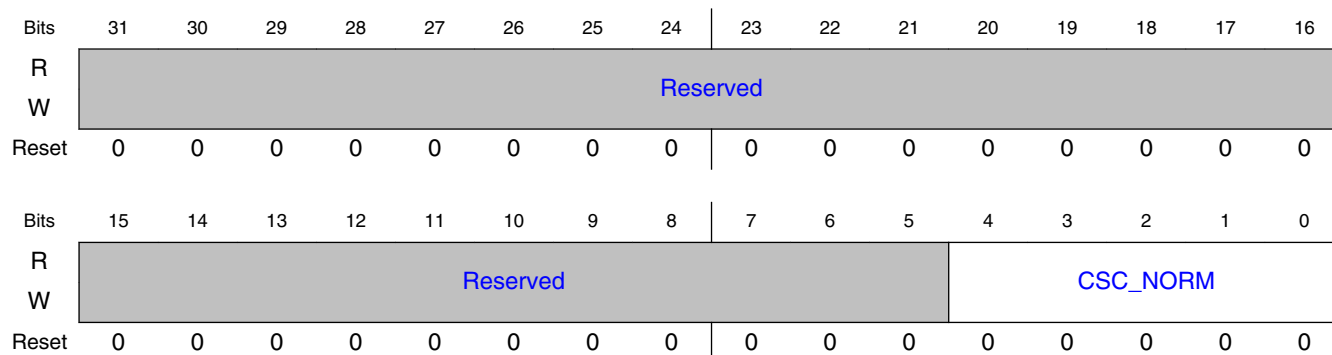
15.10.3.1.219 HDR OUPUT pipe Colorspace Converter (CSCO) normalization factor (HDR_OPIPE_CSC_NORM)

15.10.3.1.219.1 Offset

Register	Offset
HDR_OPIPE_CSC_NORM	F04Ch

15.10.3.1.219.2 Function

After the CSC matrix multiply, all components of the result are shifted right by the amount in this register. This is a signed right shift.

15.10.3.1.219.3 Diagram**15.10.3.1.219.4 Fields**

Field	Function
31-5 —	Reserved.
4-0 CSC_NORM	This 5-bit unsigned value is size if arithmetic shift after matrix multiply.

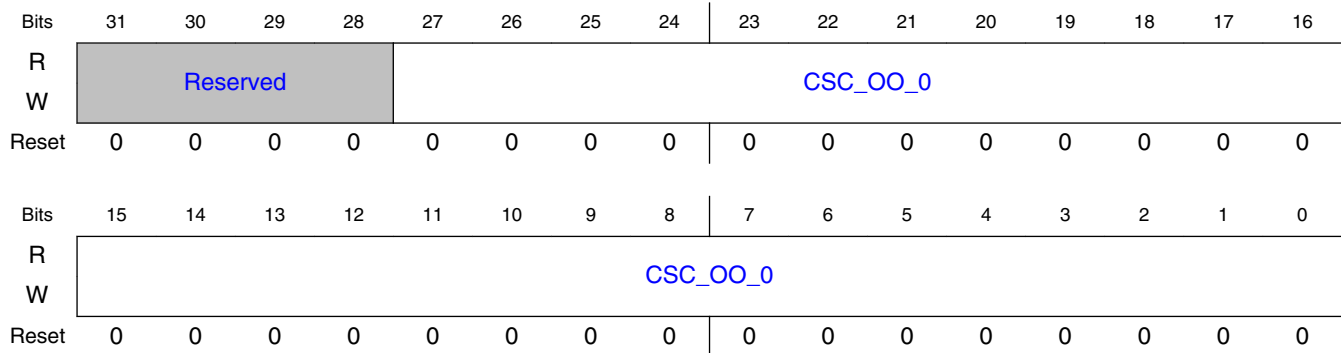
15.10.3.1.220 HDR OUPUT pipe Colorspace Converter (CSC): Post offset component 0 (HDR_OPIPE_CSC_OO_0)**15.10.3.1.220.1 Offset**

Register	Offset
HDR_OPIPE_CSC_OO_0	F050h

15.10.3.1.220.2 Function

After the CSC normalization, this offset value is added to component 0. This is a signed 28-bit number.

15.10.3.1.220.3 Diagram



15.10.3.1.220.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSC_OO_0	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.221 HDR OUTPUT pipe Colorspace Converter (CSC): Post offset component 1 (HDR_OPIPE_CSC_OO_1)

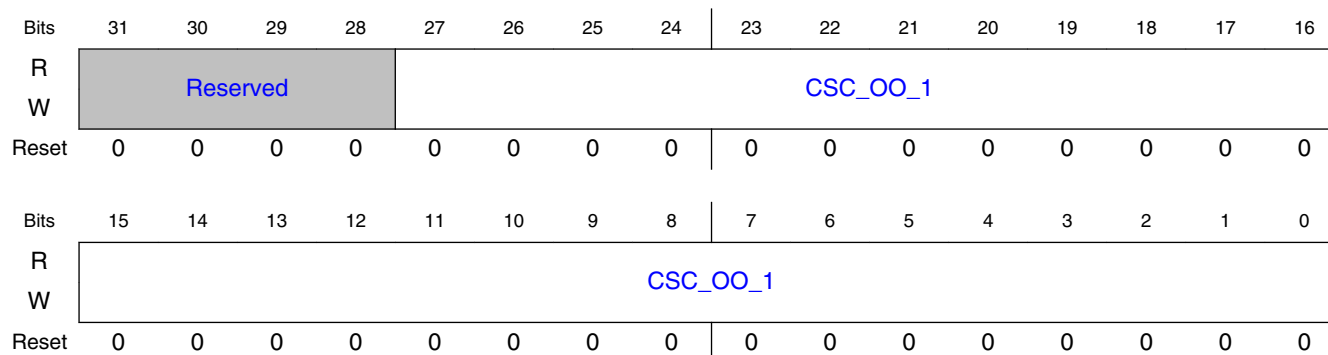
15.10.3.1.221.1 Offset

Register	Offset
HDR_OPIPE_CSC_OO_1	F054h

15.10.3.1.221.2 Function

After the CSC normalization, this offset value is added to component 1. This is a signed 28-bit number.

15.10.3.1.221.3 Diagram



15.10.3.1.221.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSC_OO_1	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.222 HDR OUPUT pipe Colorspace Converter (CSC): Post offset component 2 (HDR_OPIPE_CSC_OO_2)

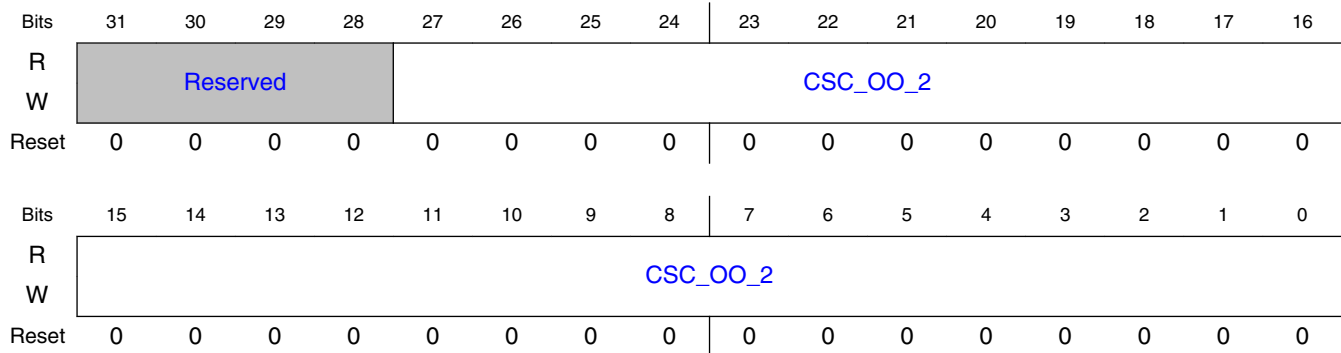
15.10.3.1.222.1 Offset

Register	Offset
HDR_OPIPE_CSC_OO_2	F058h

15.10.3.1.222.2 Function

After the CSC normalization, this offset value is added to component 2. This is a signed 28-bit number.

15.10.3.1.222.3 Diagram



15.10.3.1.222.4 Fields

Field	Function
31-28 —	Reserved.
27-0 CSC_OO_2	Output Offset (OO) This is a signed 28-bit number. Per component

15.10.3.1.223 HDR OUTPUT pipe Colorspace Converter (CSC): Post offset min clip value for component 0 (HDR_OPIPE_CSC_OMIN_0)

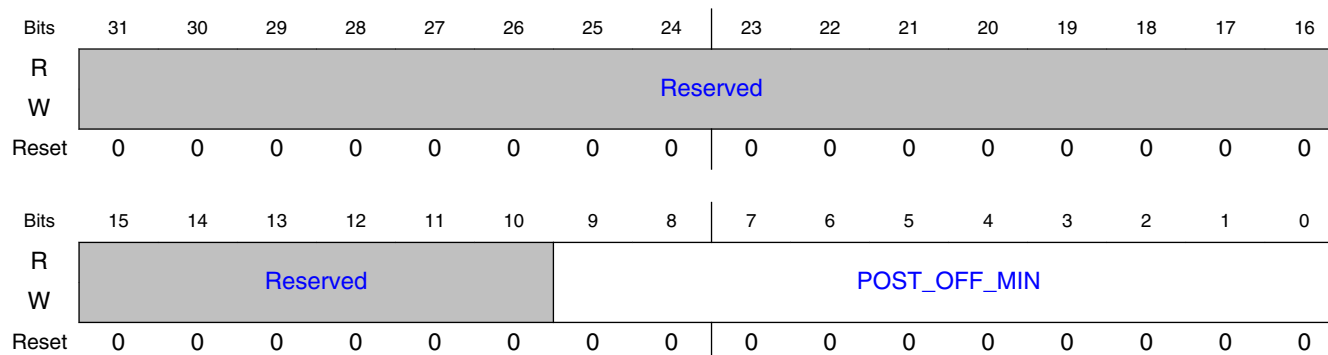
15.10.3.1.223.1 Offset

Register	Offset
HDR_OPIPE_CSC_OMIN_0	F05Ch

15.10.3.1.223.2 Function

After the post offset is added this component is clipped. This is the minimum clip value. This is an unsigned 10-bit number.

15.10.3.1.223.3 Diagram



15.10.3.1.223.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.224 HDR OUTPUT pipe Colorspace Converter (CSC): Post offset min clip value for component 1 (HDR_OPIPE_CSC_OMIN_1)

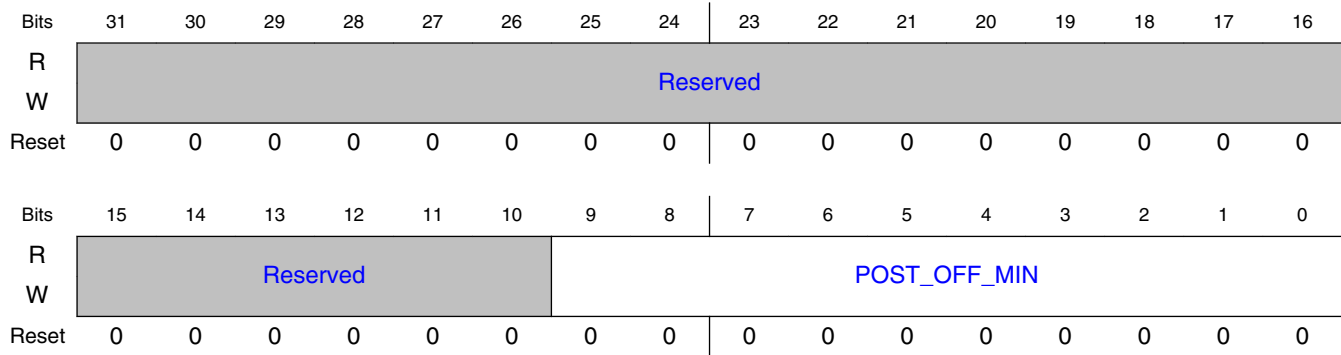
15.10.3.1.224.1 Offset

Register	Offset
HDR_OPIPE_CSC_OMIN_1	F060h

15.10.3.1.224.2 Function

After the post offset is added this component is clipped. This is the minimum clip value. This is an unsigned 10-bit number.

15.10.3.1.224.3 Diagram



15.10.3.1.224.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.225 HDR OUTPUT pipe Colorspace Converter (CSC): Post offset min clip value for component 2 (HDR_OPIPE_CSC_OMIN_2)

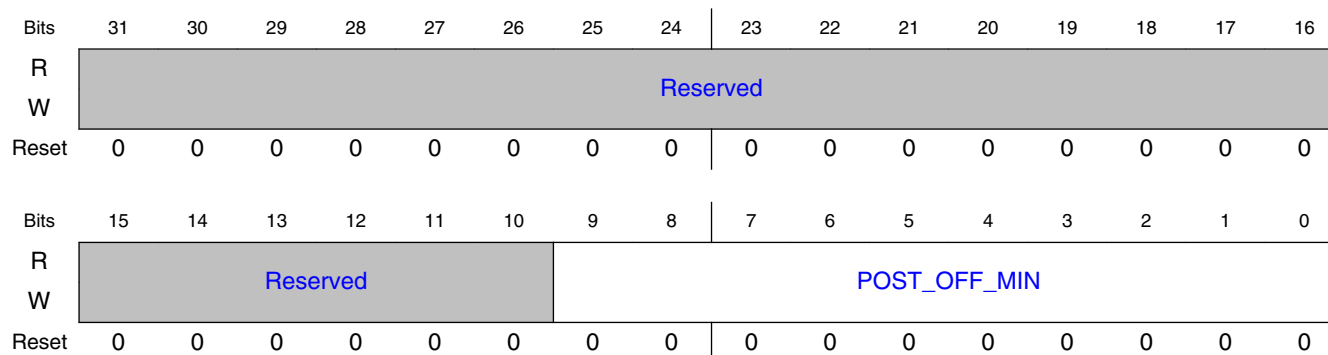
15.10.3.1.225.1 Offset

Register	Offset
HDR_OPIPE_CSC_OMIN_2	F064h

15.10.3.1.225.2 Function

After the post offset is added thie component is clipped. This is the minimum clip value
This is an unsigned 10-bit number.

15.10.3.1.225.3 Diagram



15.10.3.1.225.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MIN	Minum clipped pixel component. [0-1023]

15.10.3.1.226 HDR OUPUT pipe Colorspace Converter (CSC): Post offset max clip value for component 0 (HDR_OPIPE_CSC_OMAX_0)

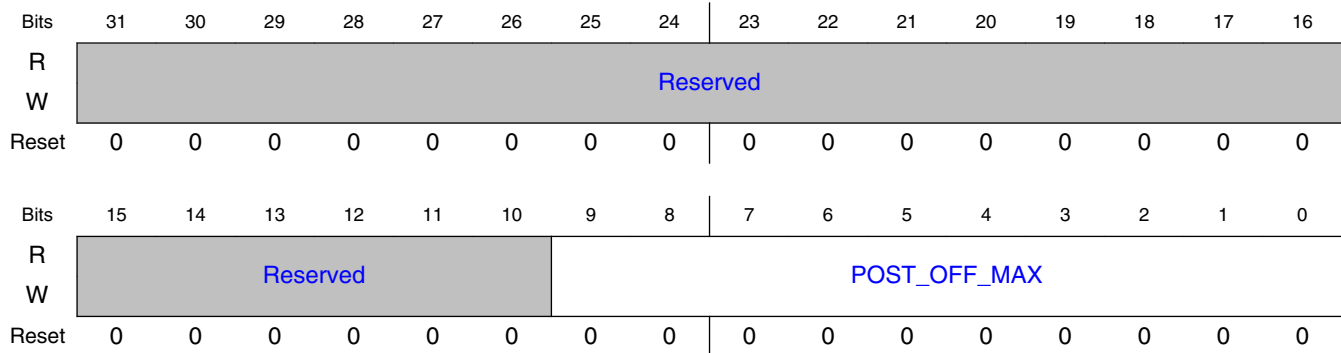
15.10.3.1.226.1 Offset

Register	Offset
HDR_OPIPE_CSC_OMAX_0	F068h

15.10.3.1.226.2 Function

After the post offset is added thie component is clipped. This is the maximum clip value
This is an unsigned 10-bit number.

15.10.3.1.226.3 Diagram



15.10.3.1.226.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

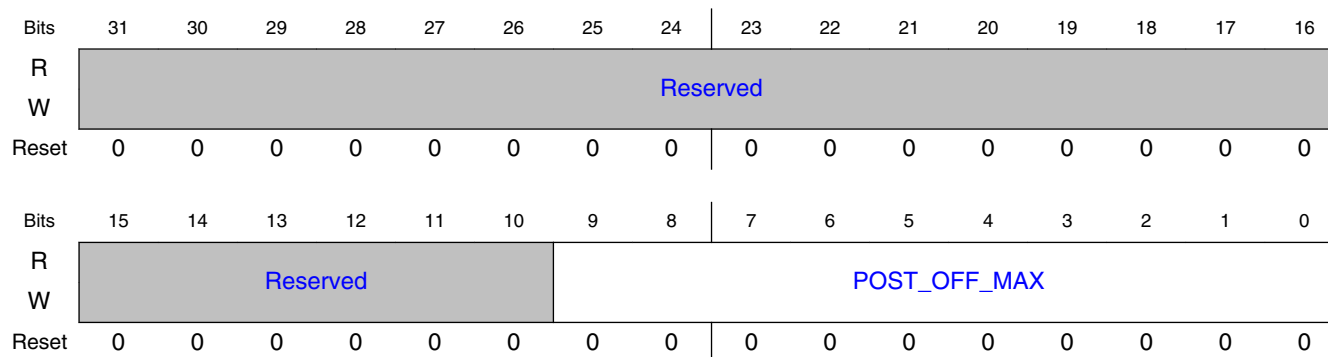
15.10.3.1.227 HDR OUTPUT pipe Colorspace Converter (CSC): Post offset max clip value for component 1 (HDR_OPIPE_CSC_OMAX_1)

15.10.3.1.227.1 Offset

Register	Offset
HDR_OPIPE_CSC_OMAX_1	F06Ch

15.10.3.1.227.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number.

15.10.3.1.227.3 Diagram**15.10.3.1.227.4 Fields**

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

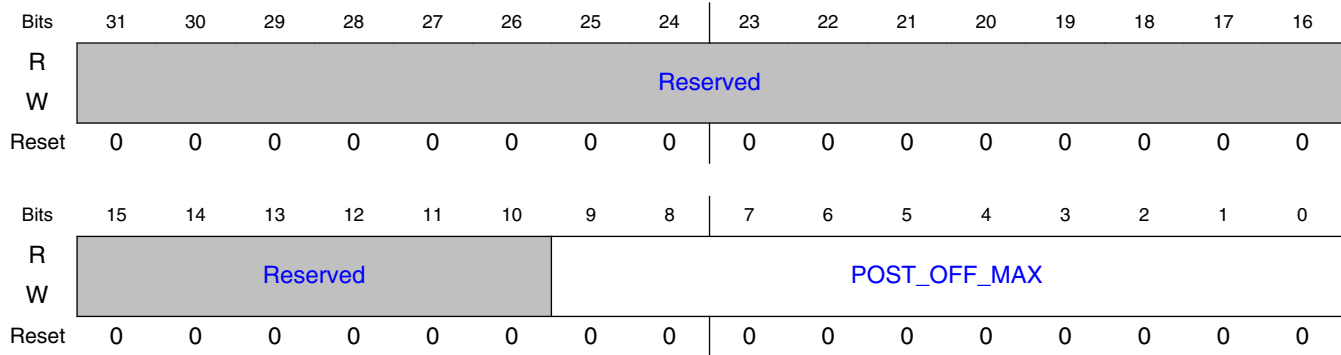
15.10.3.1.228 HDR OUTPUT pipe Colorspace Converter (CSC): Post offset max clip value for component 2 (HDR_OPIPE_CSC_OMAX_2)**15.10.3.1.228.1 Offset**

Register	Offset
HDR_OPIPE_CSC_OMAX_2	F070h

15.10.3.1.228.2 Function

After the post offset is added this component is clipped. This is the maximum clip value. This is an unsigned 10-bit number.

15.10.3.1.228.3 Diagram



15.10.3.1.228.4 Fields

Field	Function
31-10 —	Reserved.
9-0 POST_OFF_MAX	Maximum clipped pixel component. [0-1023]

15.10.3.1.229 HDR OUTPUT -TO NON LINEAR pipeline control (HDR_OPIPE_2NL_CONTROL_REG)

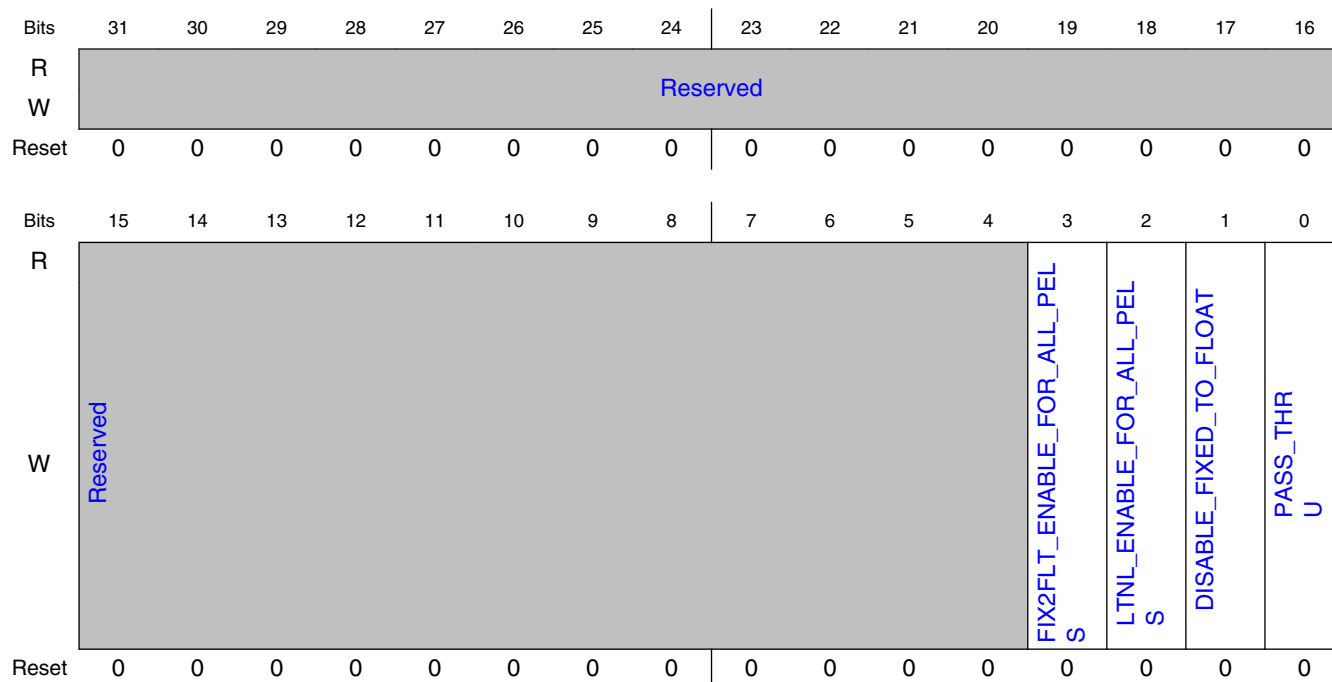
15.10.3.1.229.1 Offset

Register	Offset
HDR_OPIPE_2NL_CONTROL_REG	F874h

15.10.3.1.229.2 Function

HDR OUTPUT -TO NON LINEAR pipeline control and fixed_to_float control

15.10.3.1.229.3 Diagram



15.10.3.1.229.4 Fields

Field	Function
31-4 —	Reserved.
3 FIX2FLT_ENABLE_FOR_ALL_PELS	0: fixed-to_float is enabled for blended pels 1: fixed-to_float is enabled for all pels
2 LTNL_ENABLE_FOR_ALL_PELS	0: When 0 linear_to_non linear conversion is enabled for blended pels 1: When 1 linear_to_non linear conversion is enabled for all pels
1 DISABLE_FIXED_TO_FLOAT	0: The 28 bit input component is converted to floating point format to be processed by the linear to non-linear converter 1: The 28 bit fixed point number is truncated to 14 bits and passed the passed to the linear to non linear pipe
0 PASS_THRU	0: Process the pixels with linear to non-linear pipeline 1: Pass the data through the linear to non-linear pipeline unmodified

15.11 Color Sub-Sampler (SUBSAM)

15.11.1 Overview

The Color Sub-Sampler, or SUBSAM, is used to reduce the resolution of chroma samples for display that require YCbCr422/420 formats. Based on the HDMI2.0 specification, YCbCr420 is required for 4K displays at a 60Hz rate. There are three primary modes of operation: YCbCr444, the SUBSAM is in bypass. YCbCr422 conversions, a 3-tap horizontal linear filter is applied to the incoming pixel stream. YCbCr420, a 3x3 tap filter is used for sub-sampling. One Y line buffer and two of each Cb and Cr line buffers are used to facilitate the vertical scaling function. The chroma line buffers store the horizontally scaled result.

15.11.1.1 Block Diagram

The following diagram gives a high-level overview of the configuration of the SUBSAM.

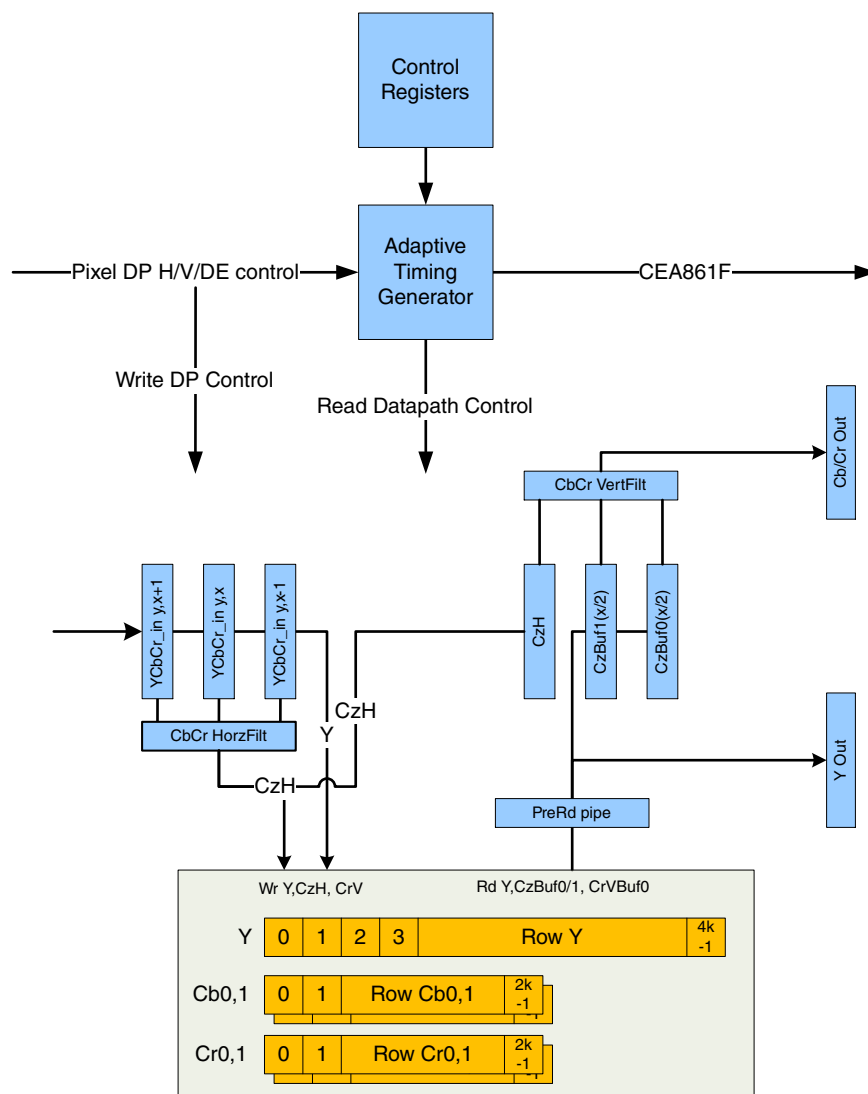


Figure 15-23. Color Subsampler Block Diagram

15.11.1.2 Features

From the block diagram above, the operands for the horizontal chroma subsampler, or CzH, are stored in shift registers. These registers are CbCr_in[x+1,x,x-1]. The results of the CzH are used to feed the vertical 3 tap filter and stored in line memories for subsequent vertical scaling. Line storage is used to store horizontally scaled chroma values and to store luma components. These components are sourced with one line of delay to feed the vertical scaling function. Luma values are stored unscaled, since luma is not subsampled.

These memories are single ported. The design pipelines the write and read data so that the same location is not accessed on the same clock. The preread function reads data out of the row rams so that reads lead locations that are written. Only 420 subsample mode uses

the memories due to the vertical scale requirement. 422 does not use the memories and the CzH is fed directly out of the subsample module aligned with the respective Y components.

15.11.2 Functional description

15.11.2.1 Sub-Sample Adaptive Timing Generator

The adaptive timing generator (ATG) uses the timing of the pixel data path signals to generate the timing of the CEA861F timing that is output to the HDMI controller. The timing is regenerated in the SUBSAM module, and the timing is completely programmable. Only the input `pixdp_vsync` is used to initiate the timing sequence at the SUBSAM output. When `pixdp_vsync` goes active (low to high transition), the SUBSAM resets its own H/V counters (`h_count` and `v_count`). The SUBSAM timing generator then walks across the entire display based on the programmable timing parameters defined in the SUBSAM PIO registers. It is expected the SUBSAM ATG timing parameters are consistent with the DTG timing parameters. As an example, the same `LRC_X` and `LRC_Y` values are programmed so that the display cycle time generated in the DTG is equivalent to the timing in the SUBSAM.

The display control is sequenced using the `h_count/v_count` counters. `h_count` is incremented on every clock, and is reset when it is equal to `CSR_SS_DISPLAY.LRC_X`. `v_count` is incremented, or reset, on the clock when `h_count` is reset to zero. The `h_count` value and respective comparators walks the display from left to right. The `v_count` value and vertical comparators walks the display timing from top to bottom. When the lower right corner (LRC) is reach, the timing generator resets back to the upper left corner (ULC) and begins to refresh the display again. This cycle continues indefinitely until the SUBSAM is disabled.

15.11.2.2 ATG Timing Details

Out of reset, the ATG timing sequence begins after receiving an initial positive edge of `pixdp_vsync`. This initial positive edge indicates the DTG is entering the vertical front porch. In order to synchronize the ATG HDMI output timing with the DTG pixel data path input timing, the ATG must also begin timing to enter the vertical front porch. This concept can be illustrated in the image below, and the timing begins when `pixdp_vsync` (also named `dolby_vsync_n`) transitions from low to high. Note the timing of `pixdp_vsync` is not the same as `hdmi_vsync`.

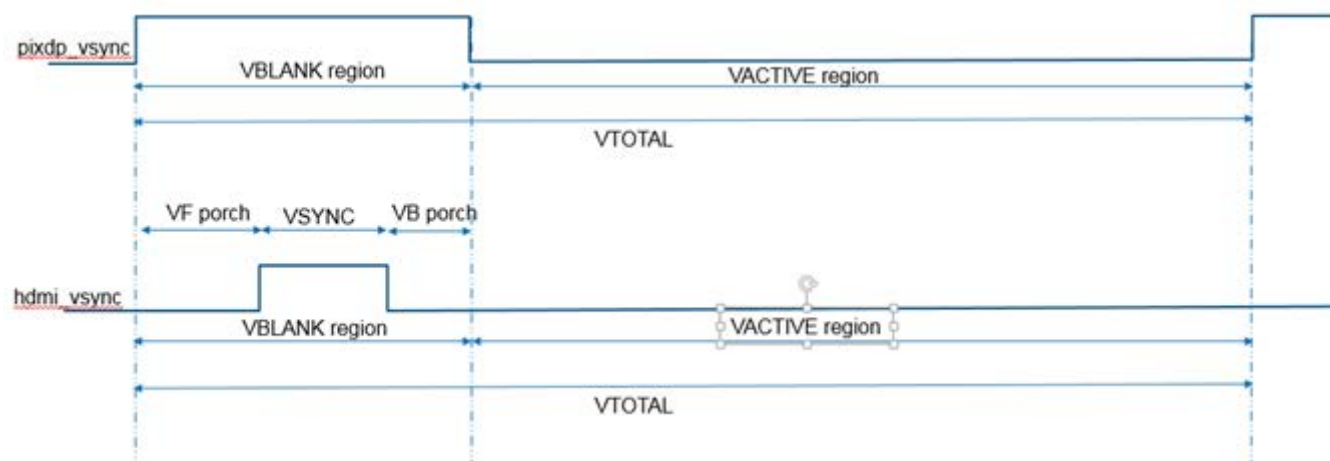


Figure 15-24. Initial VSYNC Timing

The following image illustrates the addition of pixdp_hsync and hdmi_de timing. Note the HDMI DE timing is based on the pixdp_hsync timing. This is based on how the pixel data path defines the timing of HSYNC and DE, which is not standard, and different then HDMI CEA861 timing standard.

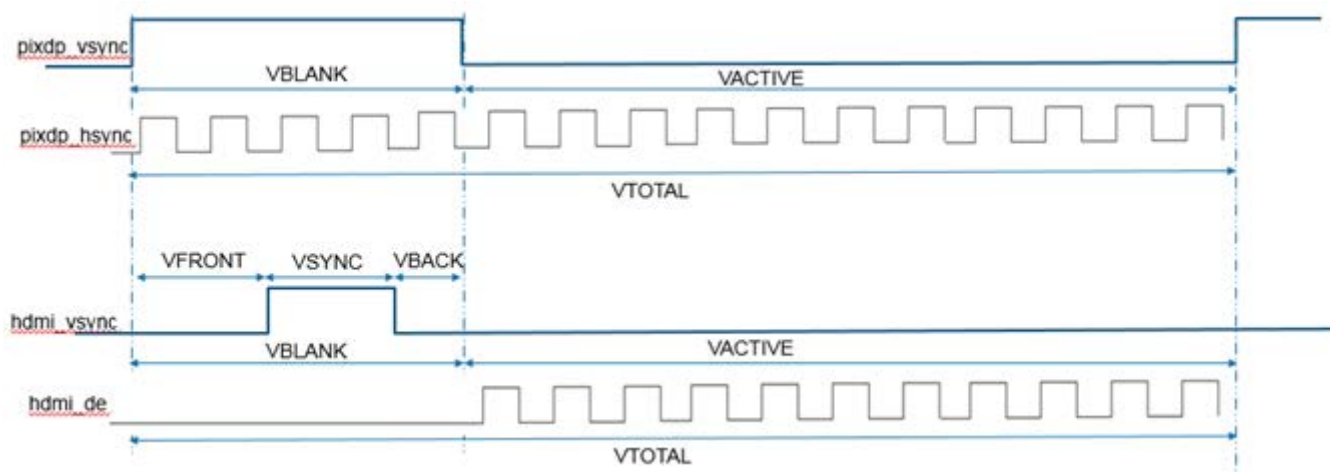


Figure 15-25. Initial VSYNC Timing with HSYNC and DE

The following provides an equivalence between standard HDMI timing and the ATG programming model. A waveform representation of CEA-861F timing is provided to illustrate which waveform sections correspond to which programming parameters. Typical HDMI timing parameters can be extracted from the following image:

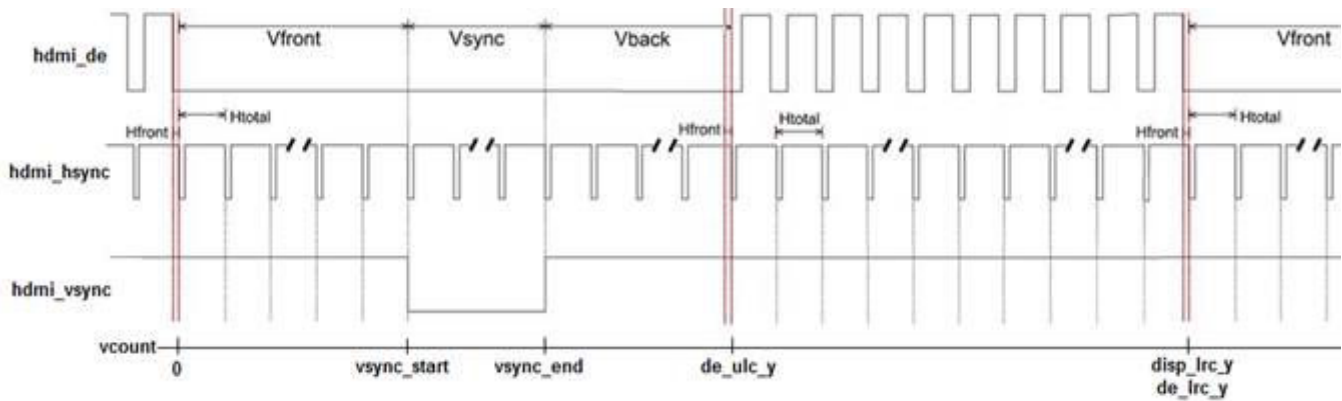


Figure 15-26. Typical HDMI Timing Parameters

- $V_{Front} = vsync_start + 1$ (duration of VFront porch)
- $V_{Sync} = vsync_end - vsync_start$
- $V_{Back} = de_ulc_y - 1 - vsync_end$
- $V_{Active} = disp_lrc_y - de_ulc_y - 1$
- $V_{Total} = disp_lrc_y + 1 = de_lrc_y + 1$
- $disp_lrc_y = de_lrc_y$

In addition to the vertical programming guidelines shown in the image above, a set of horizontal programming guidelines should also be followed.

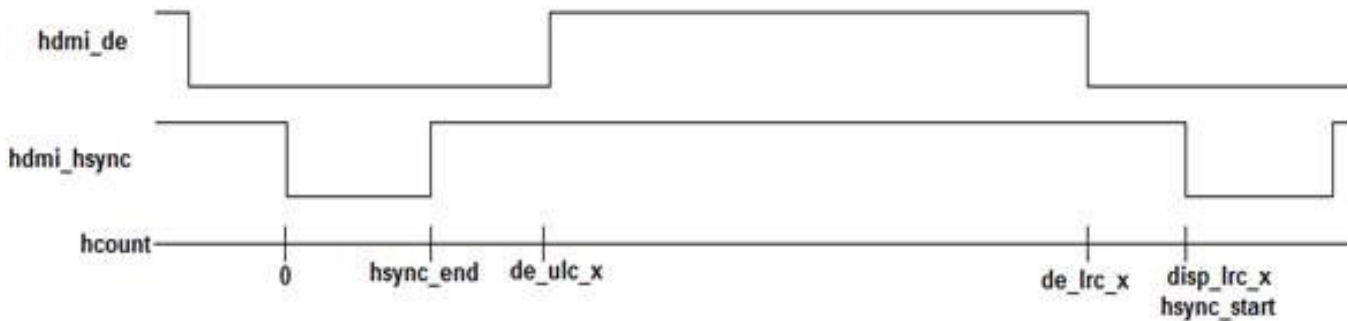


Figure 15-27. Typical HDMI horizontal timing parameters

Typical HDMI timing parameters can be extracted from the horizontal image above:

- $H_{Sync} = hsync_end + 1$
- $H_{Back} = de_ulc_x - hsync_end$
- $H_{Active} = de_lrc_x - de_ulc_x$
- $H_{Front} = disp_lrc_x - de_lrc_x$
- $H_{Total} = disp_lrc_x + 1 = hsync_start + 1$
- $disp_lrc_x = hsync_start$

A table representation of the programming example above is provided here:

Table 15-8. Subsampler Timing Registers

Subsampler Register	Register Setting
SS_DISPLAY[LRC_X]	Htotal - 1
SS_DISPLAY[LRC_Y]	Vtotal - 1
SS_HSYNC[START]	SS_DISPLAY[LRC_X]
SS_HSYNC[END]	Hsync - 1
SS_VSYNC[START]	Vfront - 1
SS_VSYNC[END]	Vsync + SS_VSYNC[START]
SS_DE_ULC[ULC_X]	Hback + SS_HSYNC[END]
SS_DE_ULC[ULC_Y]	Vback + SS_VSYNC[END] + 1
SS_DE_LRC[LRC_X]	Hactive + SS_DE_ULC[ULC_X]
SS_DE_LRC[LRC_Y]	SS_DISPLAY[LRC_Y]

15.11.2.3 SUBSAM Programming Example

The table below indicates the timing parameters for the Video Information Code (97, 107) defined for a progressive 4K TV at 60Hz.

Table 15-9. Video Information Code (97,107)

Hactive	Vactive	Htotal	Hblank	Vtotal	Vblank
3840	2160	4400	560	2250	90
Hfront	Hsync	Hback	Vfront	Vsync	Vback
176	88	296	8	10	72

The following programming example illustrates the practical use of the programming parameters derived in the ATG Timing Details section above. The SUBSAM parameters for this display format are:

- $SS_DISPLAY[LRC_X] = Htotal - 1 = 4400 - 1 = 4399$
- $SS_DISPLAY[LRC_Y] = Vtotal - 1 = 2250 - 1 = 2249$
- $SS_HSYNC[START] = SS_DISPLAY[LRC_X] = 4399$
- $SS_HSYNC[END] = Hsync - 1 = 88 - 1 = 87$
- $SS_DE_ULC[ULC_X] = Hback + SS_HSYNC[END] = 296 + 87 = 383$
- $SS_DE_LRC[LRC_X] = Hactive + SS_DE_ULC[ULC_X] = 3840 + 383 = 4223$
- $SS_VSYNC[START] = Vfront - 1 = 8 - 1 = 7$
- $SS_VSYNC[END] = Vsync + SS_VSYNC[START] = 10 + 7 = 17$
- $SS_DE_ULC[ULC_Y] = Vback + SS_VSYNC[END] + 1 = 72 + 17 + 1 = 90$
- $SS_DE_LRC[LRC_Y] = SS_DISPLAY[LRC_Y] = 2249$

Note that this example works for all subsampling modes (444, 422, and 420). To select between modes, the register field `SS_MODE[PIPE_MODE]` should be programmed. Each mode requires a different amount of processing latency. This latency is abstracted from the programming model and is accounted for in hardware based on the selection of `SS_MODE[PIPE_MODE]`.

15.11.2.4 Sub-Sample Frame Sequence

The following table indicates the sequence of 420 subsampling over the first 6 lines of output. There is a nested looping structure iterating over Y and X pixel positions. The inner loop is iterating over the possible X values of the display.

The table represents what is done on each pixel clock cycle. Basically, $x=0,1,\dots,4K-1$, then x resets to zero and the line increments. The first 7 lines of input processing is display, not the entire display of course. For 420, the Y loop repeats every two lines. The sequence reads as follows:

First horizontal iteration:

1. Pixel 0,0 is received, $x=0$
2. Hflit, $CzH = H((x-1,x,x_1)$, or $H(-1,0,1)$ on the first line. Both CbH and CrH are computed.
3. Vflit, ignored, since there is no vertical scaling when the first scan line is received.
4. No output, there is no output when the first input line is received, as we don't have enough chroma data for the vertical filtering operation.
5. Store the Luma value in `Ybuf(0)` at address 0.
6. Store the CbH value in `CbBuf[0](0)`
7. No storage to `CbBuf[1]`
8. Store `CrBuf[0](0)`
9. No storage to `CrBuf[1]`

Second horizontal iteration. Note for entires where $x/2$ are not integers, then the operation is not performed.

1. At this point, x is incremented to 1 and the operation in the first row repeat. The pixel position is now (1,0). Note all operations from 1-9 occur in the same clock cycle in a pipelined fashion in HW.
2. Hfilt is ignored, since x is odd, and the horizontal filter is valid only when x is even.
3. Again, ignored for the first input line.
4.

This loop continues until we reach the end of the scan line. This is determined when the logical DE goes low. The logical DE is a logic combination of Vsync, Hsync, and DE inputs from the pixel data path. Logical DE is active as follows:

Color Sub-Sampler (SUBSAM)

- $\text{logical_de} = \sim \text{pixdp_vsync} \ \&\& \ \text{pixdp_hsync} \ \&\& \ \text{pixdp_de}$

When all of the pixel are processed in the horizontal direction, processing increments to the next line, or all possible values of x for (x,1) for line 1.

Input Pixel YCbCr444(x,y) x = 0,1,... 3839	Hfilt (CzH) z = {b,r} x = even	Vfilt (CzV) z = {b,r}	Output Pixel 420	Ybuf(x)	CbBuf(0) (x/2)	CbBuf(1) (x/2)	CrBuf(0) (x/2)	CrBuf(1) (x/2)
1	2	3	4	5	6	7	8	9
(x, 0)	CzH(x, 0) = H(Cz(x-1, 0), Cz(x, 0), Cz(x +1,0))		none	Yin(x, 0)	CbH(x, 0)		CrH(x, 0)	
(x, 1)	CzH(x, 1) = H(Cz(x-1, 1), Cz(x, 1), Cz(x +1,1))	CzV = V(bg_z, CzBuf(0) (x/2), CzH(x, 1))	Ybuf(x), CbV	Yin(x, 1)		CbH(x, 1)	CrV(x, 0)	CrH(x, 1)
(x, 2)	CzH(x, 2) = H(Cz(x-1, 2), Cz(x, 2), Cz(x +1,2))	none	Ybuf(x), CrBuf(0) (x/2)	Yin(x, 2)	CbH(x, 2)		CrH(x, 2)	
(x, 3)	CzH(x, 3) = H(Cz(x-1, 3), CzBuf(0)	CzV = V(CzBuf(1) (x/2), CzBuf(0)	Ybuf(x), CbV	Yin(x, 3)		CbH(x, 3)	CrV(x, 2)	CrH(x, 3)

Table continues on the next page...

Input Pixel YCbCr444(x,y) x = 0,1,... 3839	Hfilt (CzH) z = {b,r} x = even	Vfilt (CzV) z = {b,r}	Output Pixel 420	Ybuf(x)	CbBuf(0) (x/2)	CbBuf(1) (x/2)	CrBuf(0) (x/2)	CrBuf(1) (x/2)
	Cz(x, 3), Cz(x +1,3))	(x/2), CzH(x,3))						
(x,4)	CzH(x,4) = H(Cz(x-1, 4), Cz(x, 4), Cz(x +1,4))	none	Ybuf(x), CrBuf(0) (x/2)	Yin(x,4)	CbH(x,4)		CrH(x,4)	
(x,5)	CzH(x,5) = H(Cz(x-1, 5), Cz(x, 5), Cz(x +1,5))	CzV = V(CzBuf(1) (x/2), CzBuf(0) (x/2), CzH(x,5))	Ybuf(x), CbV	Yin(x,5)		CbH(x,5)	CrV(x,4)	CrH(x,5)
(x,6)	CzH(x,6) = H(Cz(x-1, 6), Cz(x, 6), Cz(x +1,6))	none	Ybuf(x), CrBuf(0) (x/2)	Yin(x,6)	CbH(x,6)		CrH(x,6)	

15.11.2.5 Sub-Sample Filter

For 422 conversions, the memory is bypassed, since the scaled result is a horizontal linear filter. Line storage is not needed in this case. For 420, line memory is needed to perform the 3 tap vertical filter.

The conversion for 422 is as follows:

$$C_x = (Hcoef0 * C_x(x-1) + Hcoef1 * C_x(x) + Hcoef2 * C_x(x+1) + RoundX) / NormX \quad // \text{ Where } C_x = C_b | C_r$$

Note: Output chroma values are computed at pixel positions where x is even. So, 0, 2, 4, etc are positions where pixel values are generated and output to the pixel data path.

The conversion for 420 is as follows:

$$C_z = (Vcoef0 * (Hcoef0 * C_z(x-1)(y-1) + Hcoef1 * C_z(x)(y-1) + Hcoef2 * C_z(x+1)(y-1)) + Vcoef1 * (Hcoef0 * C_z(x-1)(y) + Hcoef1 * C_z(x)(y) + Hcoef2 * C_z(x+1)(y)) + Vcoef2 * (Hcoef0 * C_z(x-1)(y+1) + Hcoef1 * C_z(x)(y+1) + Hcoef2 * C_z(x+1)(y+1)) + RoundY) / NormY$$

Note: x,y => can only take even values in the calculation.

There is also a programmable clipping function applied in the sub_sample module. The final Cz output value is:

$$Cz_out = clip(Cz, minCzvalue, maxCzvalue)$$

The filter equations above use variable substitutions above for readability. To equate the variables to register settings, the following conversions should be used:

$$Hcoef0 = SS_COEFF[HORIZ_C]$$

$$Hcoef1 = SS_COEFF[HORIZ_B]$$

$$Hcoef2 = SS_COEFF[HORIZ_A]$$

$$\text{If } (SS_COEFF[HORIZ_NORM] = 0) \text{ RoundX} = 0; \text{ NormX} = 1;$$

$$\text{else if } (SS_COEFF[HORIZ_NORM] < 5) \text{ RoundX} = 2^{(NORM)}; \text{ NormX} = 2^{(NORM + 1)};$$

$$\text{else RoundX} = 2^{(5)}; \text{ NormX} = 2^{(5+1)};$$

$$Vcoef0 = SS_COEFF[VERT_C]$$

$$Vcoef1 = SS_COEFF[VERT_B]$$

$V_{coef2} = SS_COEFF[VERT_A]$

If $(SS_COEFF[VERT_NORM] = 0)$ RoundY = 0; NormY = 1;

else if $(SS_COEFF[VERT_NORM] < 5)$ RoundY = $2^{(SS_COEFF[VERT_NORM])}$;
NormY = $2^{(SS_COEFF[VERT_NORM]+1)}$;

else RoundY = 2^5 ; NormY = $2^{(5+1)}$;

15.11.3 Memory Map and Registers

15.11.3.1 register descriptions

15.11.3.1.1 SUBSAM Memory map

SUBSAM base address: 1_B000h

Offset	Register	Width (In bits)	Access	Reset value
0h	(SS_SYS_CTRL)	32	RW	0000_0000h
4h	(SS_SYS_CTRL_SET)	32	RW	0000_0000h
8h	(SS_SYS_CTRL_CLR)	32	RW	0000_0000h
Ch	(SS_SYS_CTRL_TOG)	32	RW	0000_0000h
10h	(SS_DISPLAY)	32	RW	0000_0000h
14h	(SS_DISPLAY_SET)	32	RW	0000_0000h
18h	(SS_DISPLAY_CLR)	32	RW	0000_0000h
1Ch	(SS_DISPLAY_TOG)	32	RW	0000_0000h
20h	(SS_HSYNC)	32	RW	0000_0000h
24h	(SS_HSYNC_SET)	32	RW	0000_0000h
28h	(SS_HSYNC_CLR)	32	RW	0000_0000h
2Ch	(SS_HSYNC_TOG)	32	RW	0000_0000h
30h	(SS_VSYNC)	32	RW	0000_0000h
34h	(SS_VSYNC_SET)	32	RW	0000_0000h
38h	(SS_VSYNC_CLR)	32	RW	0000_0000h
3Ch	(SS_VSYNC_TOG)	32	RW	0000_0000h
40h	(SS_DE_ULC)	32	RW	0000_0000h
44h	(SS_DE_ULC_SET)	32	RW	0000_0000h
48h	(SS_DE_ULC_CLR)	32	RW	0000_0000h
4Ch	(SS_DE_ULC_TOG)	32	RW	0000_0000h

Table continues on the next page...

Memory Map and Registers

Offset	Register	Width (In bits)	Access	Reset value
50h	(SS_DE_LRC)	32	RW	0000_0000h
54h	(SS_DE_LRC_SET)	32	RW	0000_0000h
58h	(SS_DE_LRC_CLR)	32	RW	0000_0000h
5Ch	(SS_DE_LRC_TOG)	32	RW	0000_0000h
60h	(SS_MODE)	32	RW	0000_0000h
64h	(SS_MODE_SET)	32	RW	0000_0000h
68h	(SS_MODE_CLR)	32	RW	0000_0000h
6Ch	(SS_MODE_TOG)	32	RW	0000_0000h
70h	(SS_COEFF)	32	RW	0000_0000h
74h	(SS_COEFF_SET)	32	RW	0000_0000h
78h	(SS_COEFF_CLR)	32	RW	0000_0000h
7Ch	(SS_COEFF_TOG)	32	RW	0000_0000h
80h	(SS_CLIP_CB)	32	RW	0000_0000h
84h	(SS_CLIP_CB_SET)	32	RW	0000_0000h
88h	(SS_CLIP_CB_CLR)	32	RW	0000_0000h
8Ch	(SS_CLIP_CB_TOG)	32	RW	0000_0000h
90h	(SS_CLIP_CR)	32	RW	0000_0000h
94h	(SS_CLIP_CR_SET)	32	RW	0000_0000h
98h	(SS_CLIP_CR_CLR)	32	RW	0000_0000h
9Ch	(SS_CLIP_CR_TOG)	32	RW	0000_0000h
A0h	(SS_INTER_MODE)	32	RW	0000_0000h
A4h	(SS_INTER_MODE_SET)	32	RW	0000_0000h
A8h	(SS_INTER_MODE_CLR)	32	RW	0000_0000h
ACh	(SS_INTER_MODE_TOG)	32	RW	0000_0000h

15.11.3.1.2 (SS_SYS_CTRL)

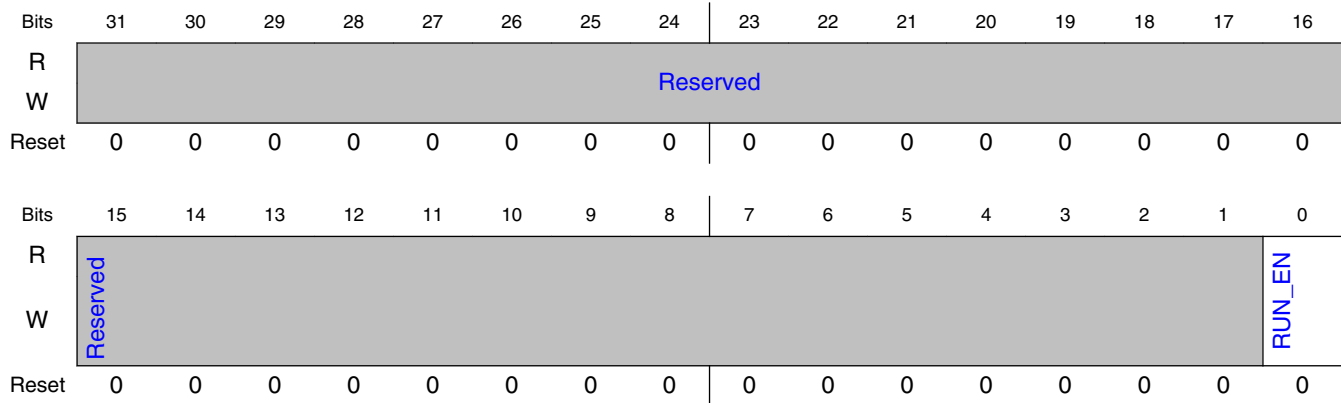
15.11.3.1.2.1 Offset

Register	Offset
SS_SYS_CTRL	0h

15.11.3.1.2.2 Function

The ss_sys_ctrl register contains all top level system control fields.

15.11.3.1.2.3 Diagram



15.11.3.1.2.4 Fields

Field	Function
31-1 —	Reserved
0 RUN_EN	Software sets this bit to indicate when the subsam module is ready to be enabled. Setting this bit from 1 to 0 causes a software reset.

15.11.3.1.3 (SS_SYS_CTRL_SET)

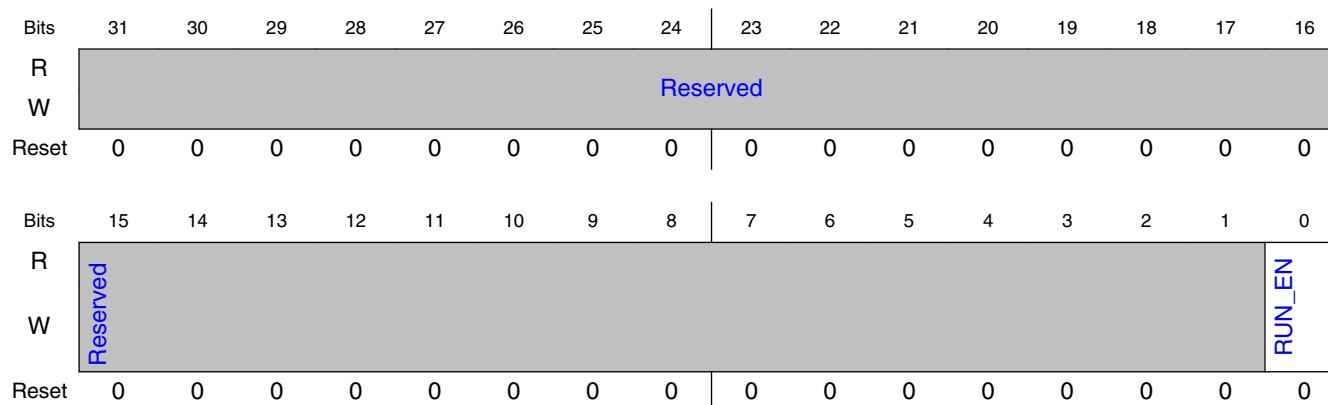
15.11.3.1.3.1 Offset

Register	Offset
SS_SYS_CTRL_SET	4h

15.11.3.1.3.2 Function

The `ss_sys_ctrl` register contains all top level system control fields.

15.11.3.1.3.3 Diagram



15.11.3.1.3.4 Fields

Field	Function
31-1 —	Reserved
0 RUN_EN	Software sets this bit to indicate when the subsam module is ready to be enabled. Setting this bit from 1 to 0 causes a software reset.

15.11.3.1.4 (SS_SYS_CTRL_CLR)

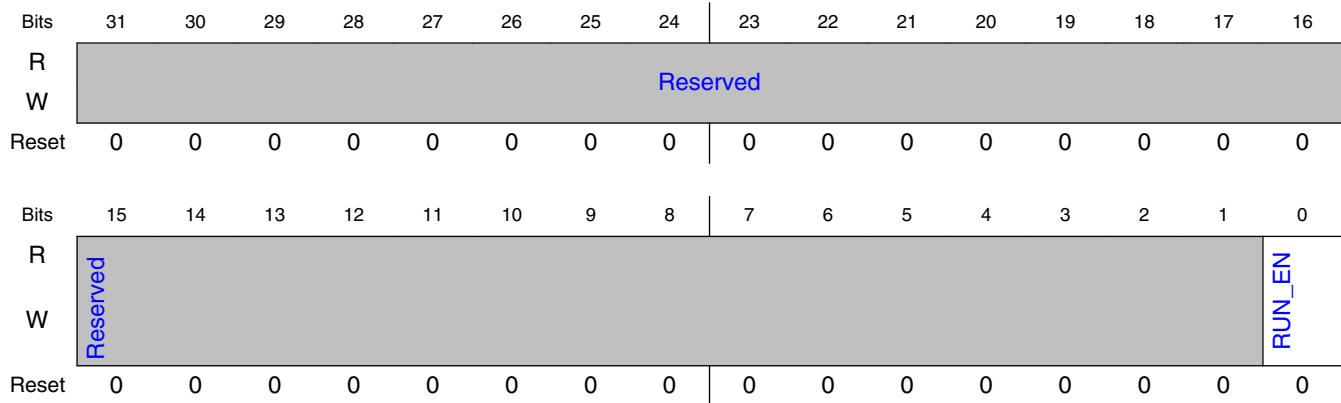
15.11.3.1.4.1 Offset

Register	Offset
SS_SYS_CTRL_CLR	8h

15.11.3.1.4.2 Function

The ss_sys_ctrl register contains all top level system control fields.

15.11.3.1.4.3 Diagram



15.11.3.1.4.4 Fields

Field	Function
31-1 —	Reserved
0 RUN_EN	Software sets this bit to indicate when the subsam module is ready to be enabled. Setting this bit from 1 to 0 causes a software reset.

15.11.3.1.5 (SS_SYS_CTRL_TOG)

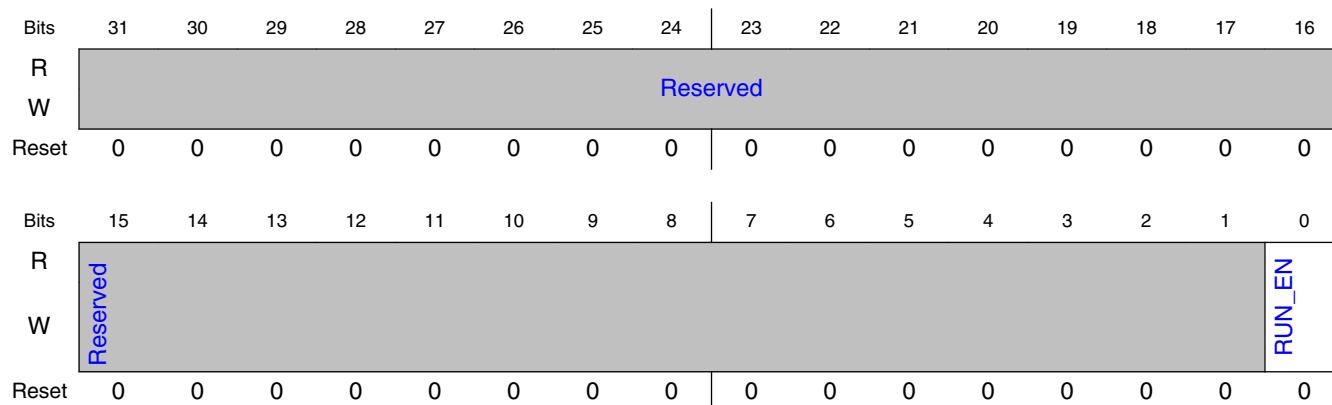
15.11.3.1.5.1 Offset

Register	Offset
SS_SYS_CTRL_TOG	Ch

15.11.3.1.5.2 Function

The `ss_sys_ctrl` register contains all top level system control fields.

15.11.3.1.5.3 Diagram



15.11.3.1.5.4 Fields

Field	Function
31-1 —	Reserved
0 RUN_EN	Software sets this bit to indicate when the subsam module is ready to be enabled. Setting this bit from 1 to 0 causes a software reset.

15.11.3.1.6 (SS_DISPLAY)

15.11.3.1.6.1 Offset

Register	Offset
SS_DISPLAY	10h

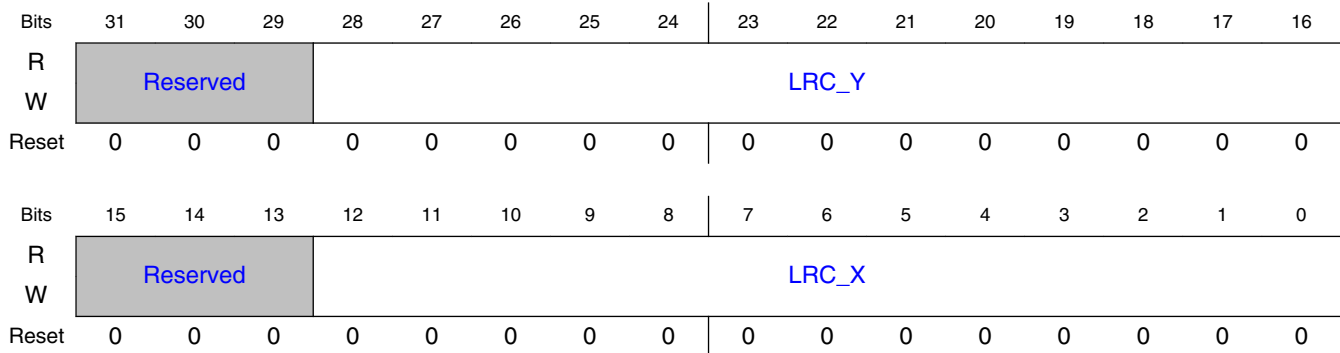
15.11.3.1.6.2 Function

The `csr_ss_display` register defines the entire size of the display timing window.

`SS_DISPLAY_LRX_Y = VACTIVE+VBLANK-1`

`SS_DISPLAY_LRC_X = HACTIVE+HBLANK-1`

15.11.3.1.6.3 Diagram



15.11.3.1.6.4 Fields

Field	Function
31-29 —	Reserved
28-16 LRC_Y	Holds the lower right corner y coordinate. The vertical count reg should conditionally reset to zero when it reaches this value and v_count = DTG_LRC_Y
15-13 —	Reserved
12-0 LRC_X	Holds the lower right corner x coordinate. The horizontal count reg should reset to zero when it reaches this value.

15.11.3.1.7 (SS_DISPLAY_SET)

15.11.3.1.7.1 Offset

Register	Offset
SS_DISPLAY_SET	14h

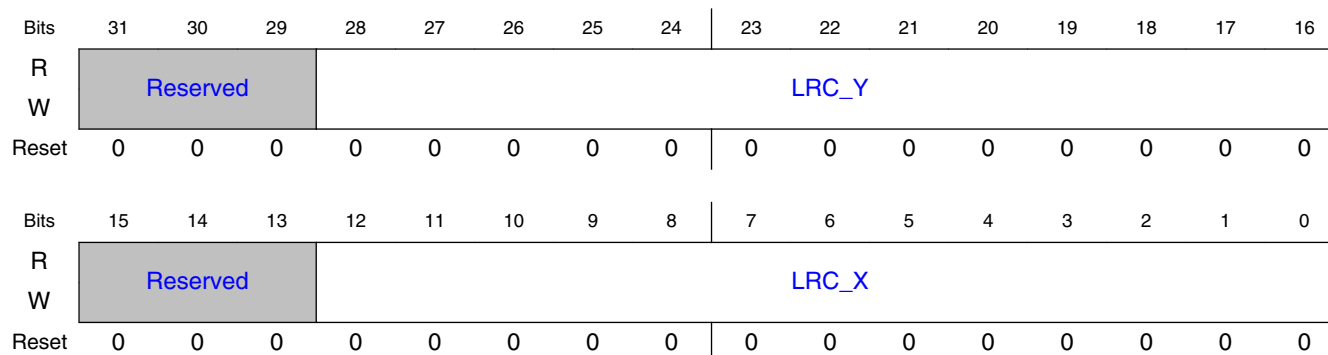
15.11.3.1.7.2 Function

The `csr_ss_display` register defines the entire size of the display timing window.

`SS_DISPLAY_LRX_Y = VACTIVE+VBLANK-1`

`SS_DISPLAY_LRC_X = HACTIVE+HBLANK-1`

15.11.3.1.7.3 Diagram



15.11.3.1.7.4 Fields

Field	Function
31-29 —	Reserved
28-16 LRC_Y	Holds the lower right corner y coordinate. The vertical count reg should conditionally reset to zero when it reaches this value and v_count = DTG_LRC_Y
15-13 —	Reserved
12-0 LRC_X	Holds the lower right corner x coordinate. The horizontal count reg should reset to zero when it reaches this value.

15.11.3.1.8 (SS_DISPLAY_CLR)

15.11.3.1.8.1 Offset

Register	Offset
SS_DISPLAY_CLR	18h

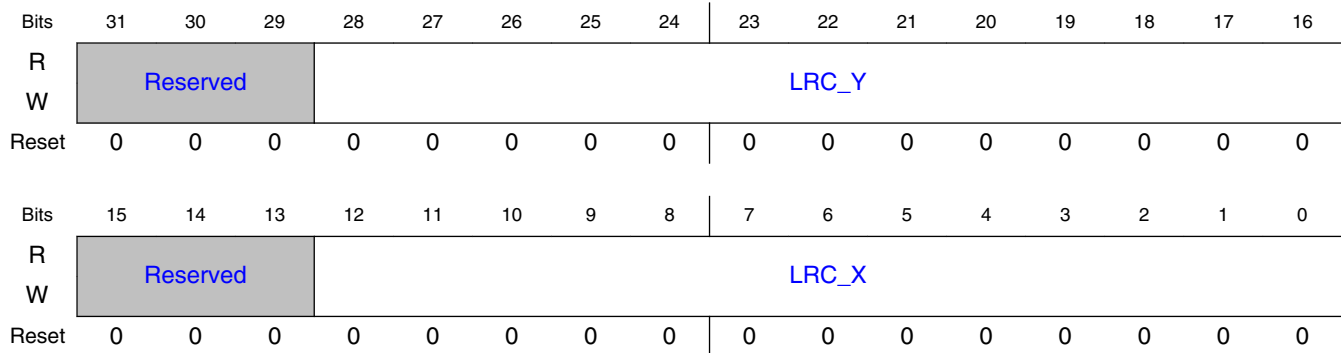
15.11.3.1.8.2 Function

The `csr_ss_display` register defines the entire size of the display timing window.

`SS_DISPLAY_LRX_Y = VACTIVE+VBLANK-1`

`SS_DISPLAY_LRC_X = HACTIVE+HBLANK-1`

15.11.3.1.8.3 Diagram



15.11.3.1.8.4 Fields

Field	Function
31-29 —	Reserved
28-16 LRC_Y	Holds the lower right corner y coordinate. The vertical count reg should conditionally reset to zero when it reaches this value and $v_count = DTG_LRC_Y$
15-13 —	Reserved
12-0 LRC_X	Holds the lower right corner x coordinate. The horizontal count reg should reset to zero when it reaches this value.

15.11.3.1.9 (SS_DISPLAY_TOG)

15.11.3.1.9.1 Offset

Register	Offset
SS_DISPLAY_TOG	1Ch

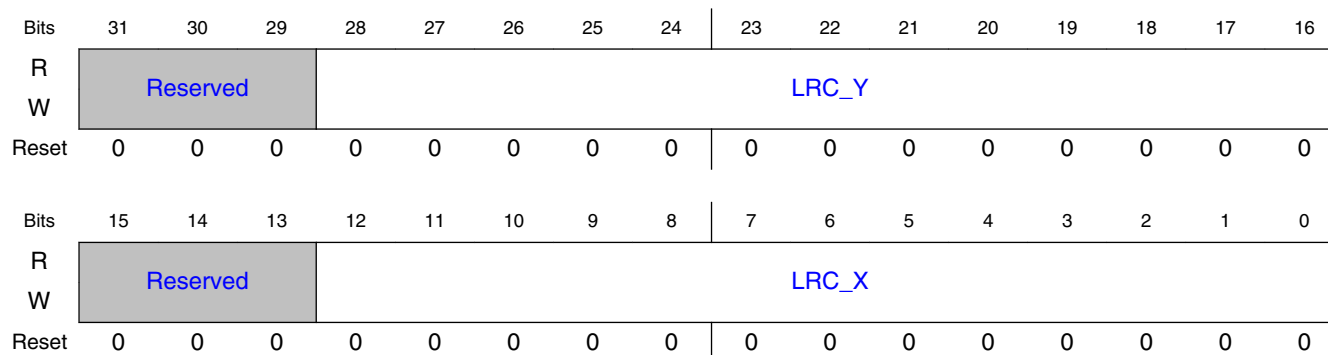
15.11.3.1.9.2 Function

The `csr_ss_display` register defines the entire size of the display timing window.

`SS_DISPLAY_LRX_Y = VACTIVE+VBLANK-1`

`SS_DISPLAY_LRC_X = HACTIVE+HBLANK-1`

15.11.3.1.9.3 Diagram



15.11.3.1.9.4 Fields

Field	Function
31-29 —	Reserved
28-16 LRC_Y	Holds the lower right corner y coordinate. The vertical count reg should conditionally reset to zero when it reaches this value and v_count = DTG_LRC_Y
15-13 —	Reserved
12-0 LRC_X	Holds the lower right corner x coordinate. The horizontal count reg should reset to zero when it reaches this value.

15.11.3.1.10 (SS_HSYNC)

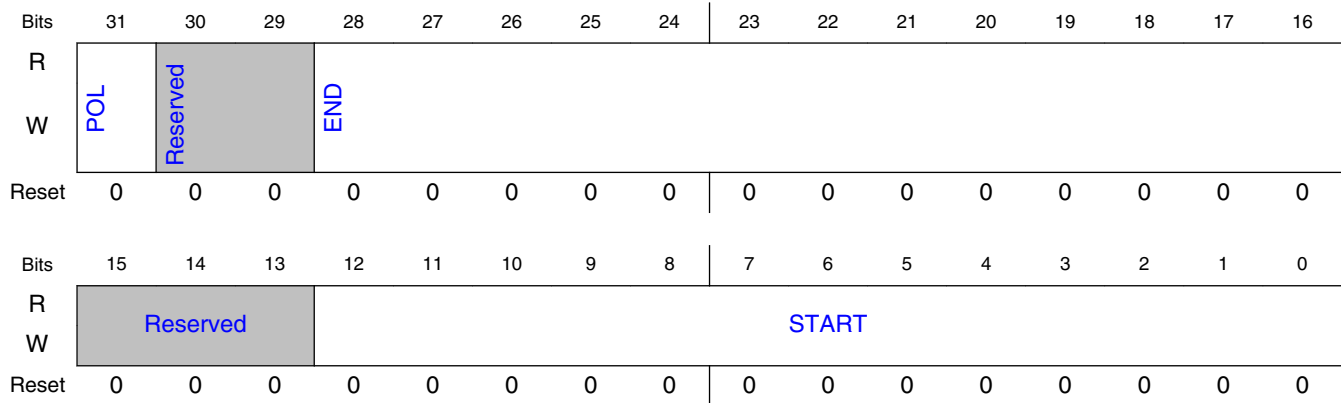
15.11.3.1.10.1 Offset

Register	Offset
SS_HSYNC	20h

15.11.3.1.10.2 Function

Defines when horizontal sync signal goes high/low.
There are two X coordinates defined in this register.
Horizontal sync width determined by END - START.

15.11.3.1.10.3 Diagram



15.11.3.1.10.4 Fields

Field	Function
31 POL	Polarity of sync signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 END	horizontal sync is de-asserted when horizontal count == END
15-13 —	Reserved
12-0 START	horizontal sync is asserted when horizontal count == START

15.11.3.1.11 (SS_HSYNC_SET)

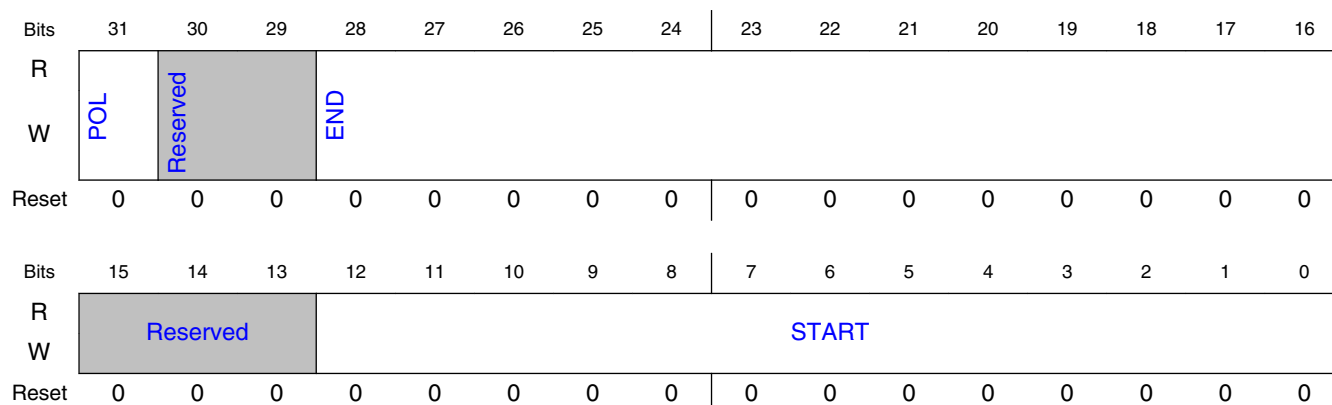
15.11.3.1.11.1 Offset

Register	Offset
SS_HSYNC_SET	24h

15.11.3.1.11.2 Function

Defines when horizontal sync signal goes high/low.
 There are two X coordinates defined in this register.
 Horizontal sync width determined by END - START.

15.11.3.1.11.3 Diagram



15.11.3.1.11.4 Fields

Field	Function
31 POL	Polarity of sync signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 END	horizontal sync is de-asserted when horizontal count == END
15-13 —	Reserved
12-0 START	horizontal sync is asserted when horizontal count == START

15.11.3.1.12 (SS_HSYNC_CLR)

15.11.3.1.12.1 Offset

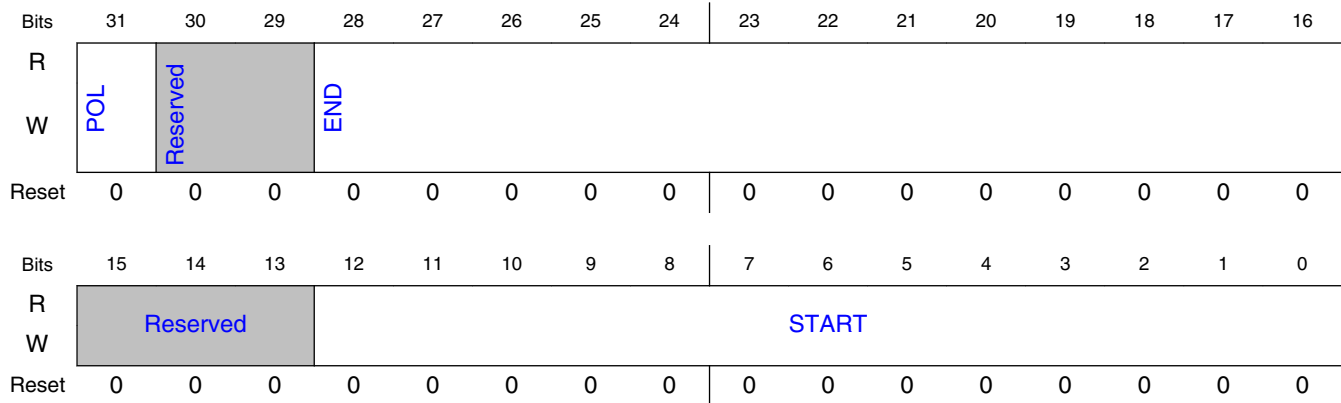
Register	Offset
SS_HSYNC_CLR	28h

15.11.3.1.12.2 Function

Defines when horizontal sync signal goes high/low.

There are two X coordinates defined in this register.
Horizontal sync width determined by END - START.

15.11.3.1.12.3 Diagram



15.11.3.1.12.4 Fields

Field	Function
31 POL	Polarity of sync signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 END	horizontal sync is de-asserted when horizontal count == END
15-13 —	Reserved
12-0 START	horizontal sync is asserted when horizontal count == START

15.11.3.1.13 (SS_HSYNC_TOG)

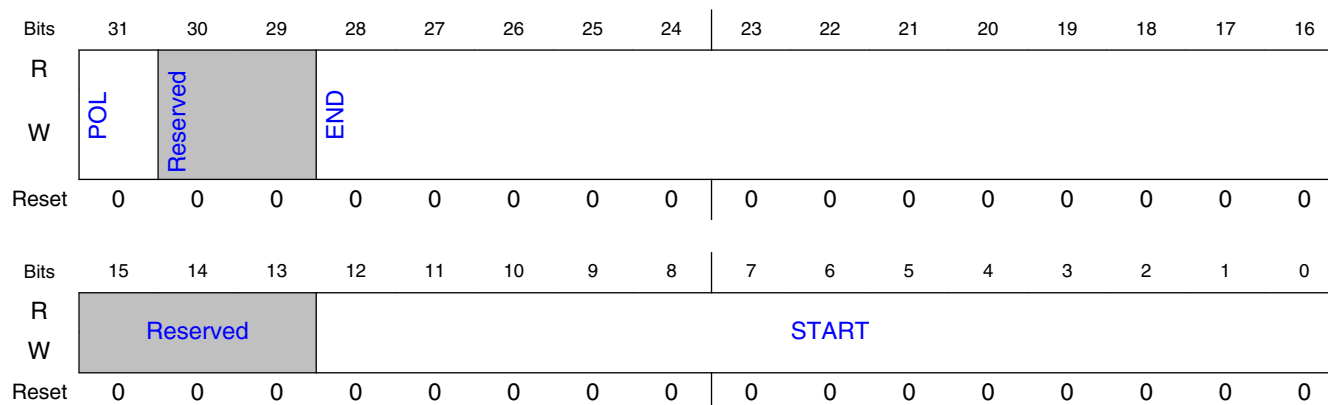
15.11.3.1.13.1 Offset

Register	Offset
SS_HSYNC_TOG	2Ch

15.11.3.1.13.2 Function

Defines when horizontal sync signal goes high/low.
There are two X coordinates defined in this register.
Horizontal sync width determined by END - START.

15.11.3.1.13.3 Diagram



15.11.3.1.13.4 Fields

Field	Function
31 POL	Polarity of sync signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 END	horizontal sync is de-asserted when horizontal count == END
15-13 —	Reserved
12-0 START	horizontal sync is asserted when horizontal count == START

15.11.3.1.14 (SS_VSYNC)

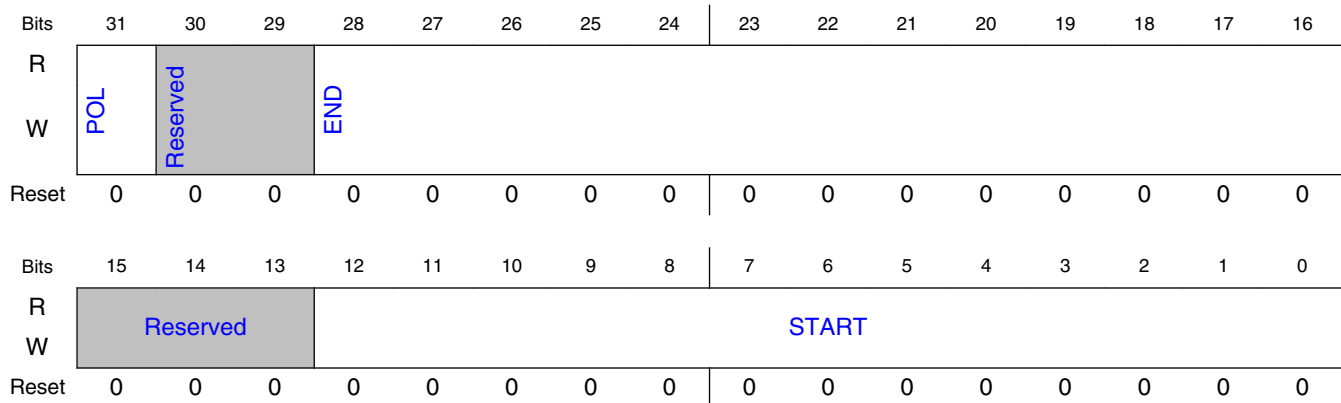
15.11.3.1.14.1 Offset

Register	Offset
SS_VSYNC	30h

15.11.3.1.14.2 Function

Defines Y coordinate when vertical sync signal goes high/low.
Horizontal sync width determined by END - START.

15.11.3.1.14.3 Diagram



15.11.3.1.14.4 Fields

Field	Function
31 POL	Polarity of sync signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 END	When v_count = END h_count = CSR_SS_LRC_X vsync goes inactive
15-13 —	Reserved
12-0 START	When v_count = START and h_count = CSR_SS_LRC_X then vsync goes active

15.11.3.1.15 (SS_VSYNC_SET)

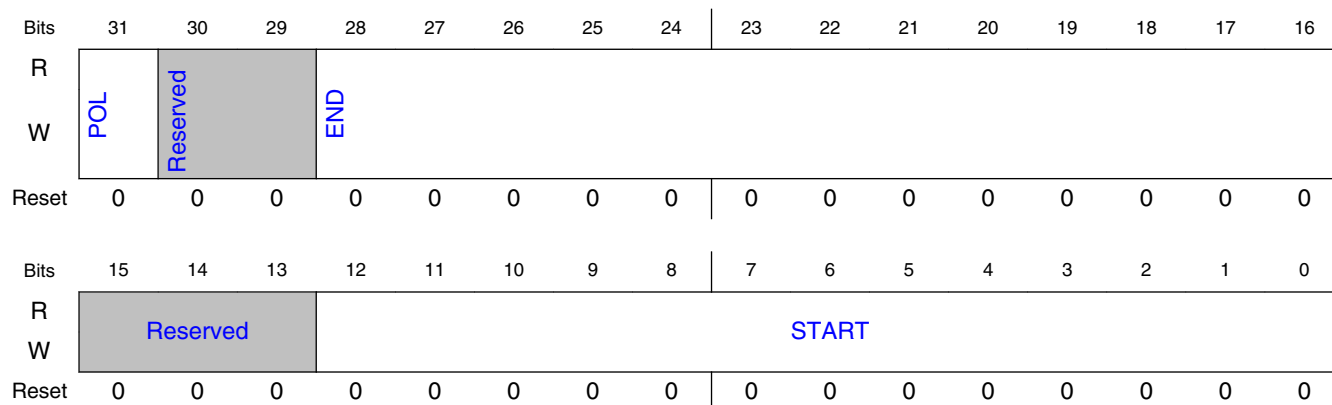
15.11.3.1.15.1 Offset

Register	Offset
SS_VSYNC_SET	34h

15.11.3.1.15.2 Function

Defines Y coordinate when vertical sync signal goes high/low.
Horizontal sync width determined by END - START.

15.11.3.1.15.3 Diagram



15.11.3.1.15.4 Fields

Field	Function
31 POL	Polarity of sync signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 END	When v_count = END h_count = CSR_SS_LRC_X vsync goes inactive
15-13 —	Reserved
12-0 START	When v_count = START and h_count = CSR_SS_LRC_X then vsync goes active

15.11.3.1.16 (SS_VSYNC_CLR)

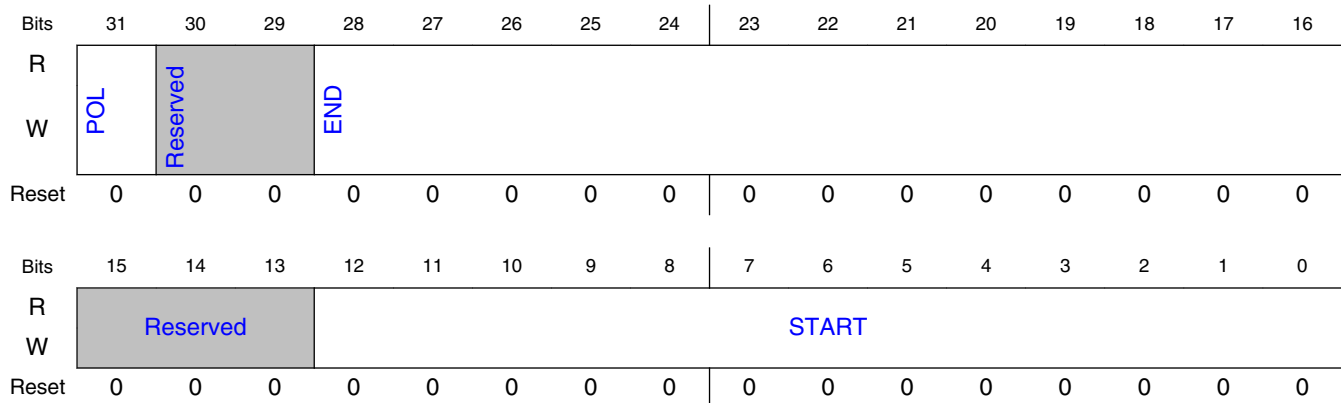
15.11.3.1.16.1 Offset

Register	Offset
SS_VSYNC_CLR	38h

15.11.3.1.16.2 Function

Defines Y coordinate when vertical sync signal goes high/low.
Horizontal sync width determined by END - START.

15.11.3.1.16.3 Diagram



15.11.3.1.16.4 Fields

Field	Function
31 POL	Polarity of sync signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 END	When v_count = END h_count = CSR_SS_LRC_X vsync goes inactive
15-13 —	Reserved
12-0 START	When v_count = START and h_count = CSR_SS_LRC_X then vsync goes active

15.11.3.1.17 (SS_VSYNC_TOG)

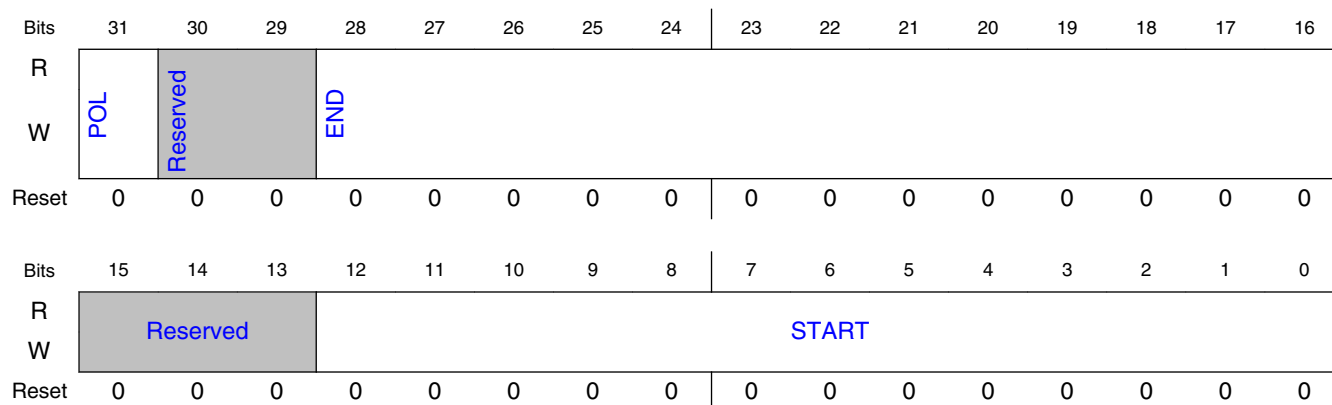
15.11.3.1.17.1 Offset

Register	Offset
SS_VSYNC_TOG	3Ch

15.11.3.1.17.2 Function

Defines Y coordinate when vertical sync signal goes high/low.
Horizontal sync width determined by END - START.

15.11.3.1.17.3 Diagram



15.11.3.1.17.4 Fields

Field	Function
31 POL	Polarity of sync signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 END	When v_count = END h_count = CSR_SS_LRC_X vsync goes inactive
15-13 —	Reserved
12-0 START	When v_count = START and h_count = CSR_SS_LRC_X then vsync goes active

15.11.3.1.18 (SS_DE_ULC)

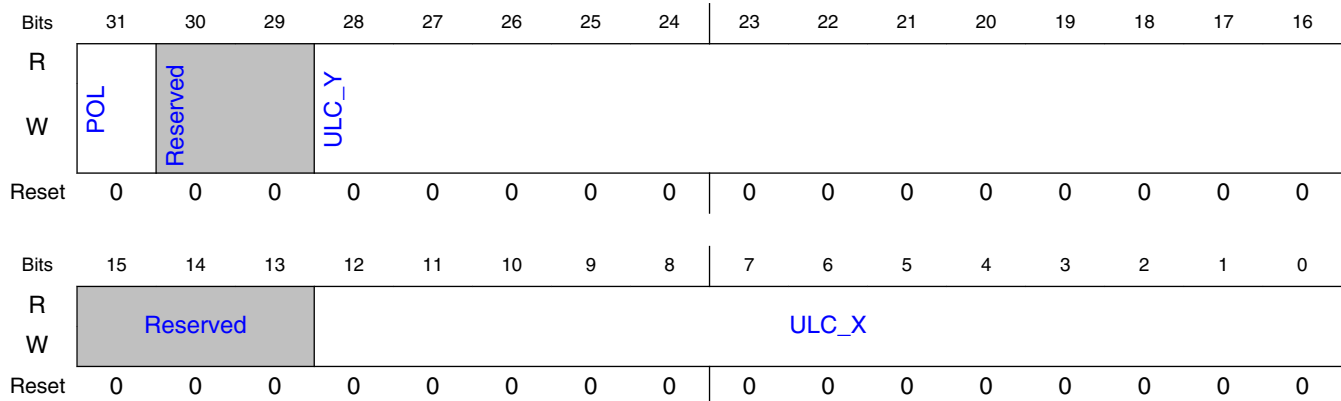
15.11.3.1.18.1 Offset

Register	Offset
SS_DE_ULC	40h

15.11.3.1.18.2 Function

Upper left coordinate of frame that defines when DE goes active.
Both X and Y coordinates are programmable as ULC_X and ULC_Y respectively.
Signal transitions occur when count == boundary.

15.11.3.1.18.3 Diagram



15.11.3.1.18.4 Fields

Field	Function
31 POL	Polarity of DE signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 ULC_Y	Vertical position in the display that determines when the DE goes active.
15-13 —	Reserved
12-0 ULC_X	Horizontal position in the scan line that determines when DE goes active.

15.11.3.1.19 (SS_DE_ULC_SET)

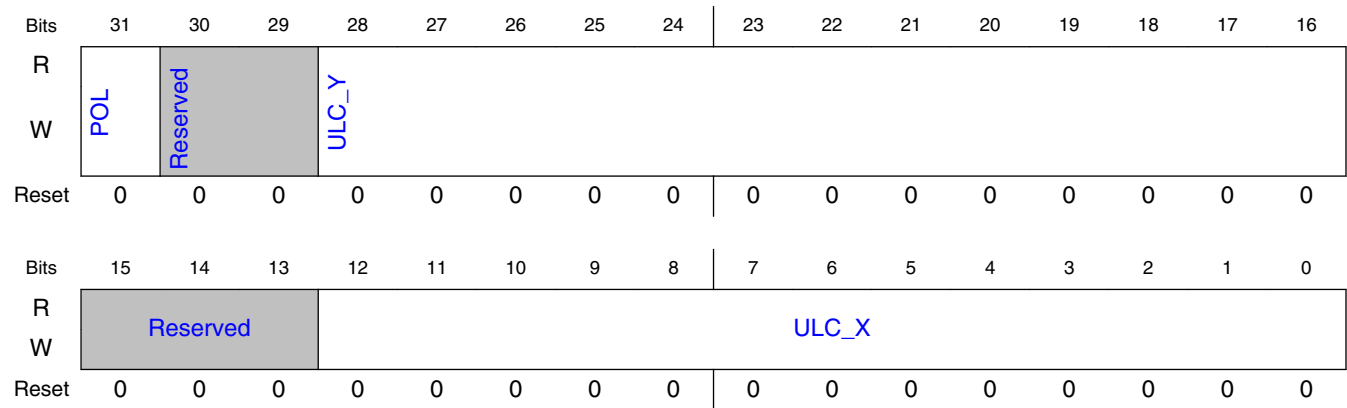
15.11.3.1.19.1 Offset

Register	Offset
SS_DE_ULC_SET	44h

15.11.3.1.19.2 Function

Upper left coordinate of frame that defines when DE goes active.
Both X and Y coordinates are programmable as ULC_X and ULC_Y respectively.
Signal transitions occur when count == boundary.

15.11.3.1.19.3 Diagram



15.11.3.1.19.4 Fields

Field	Function
31 POL	Polarity of DE signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 ULC_Y	Vertical position in the display that determines when the DE goes active.
15-13 —	Reserved
12-0 ULC_X	Horizontal position in the scan line that determines when DE goes active.

15.11.3.1.20 (SS_DE_ULC_CLR)

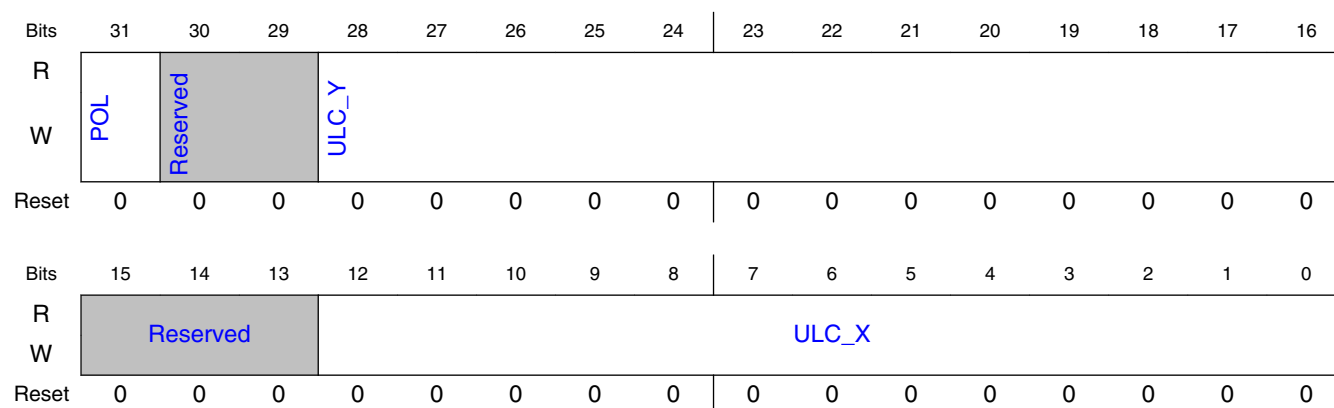
15.11.3.1.20.1 Offset

Register	Offset
SS_DE_ULC_CLR	48h

15.11.3.1.20.2 Function

Upper left coordinate of frame that defines when DE goes active.
Both X and Y coordinates are programmable as ULC_X and ULC_Y respectively.
Signal transitions occur when count == boundary.

15.11.3.1.20.3 Diagram



15.11.3.1.20.4 Fields

Field	Function
31 POL	Polarity of DE signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 ULC_Y	Vertical position in the display that determines when the DE goes active.
15-13 —	Reserved
12-0 ULC_X	Horizontal position in the scan line that determines when DE goes active.

15.11.3.1.21 (SS_DE_ULC_TOG)

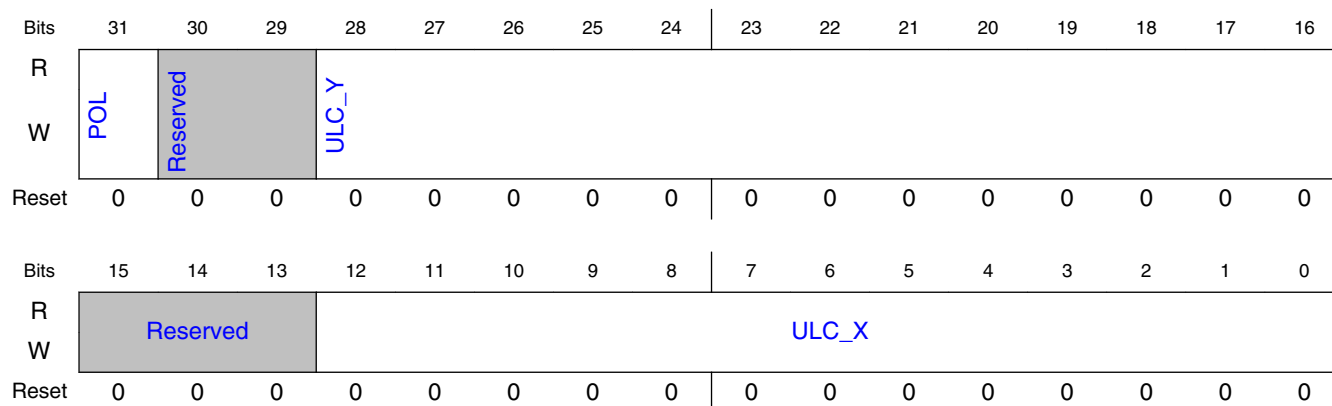
15.11.3.1.21.1 Offset

Register	Offset
SS_DE_ULC_TOG	4Ch

15.11.3.1.21.2 Function

Upper left coordinate of frame that defines when DE goes active.
Both X and Y coordinates are programmable as ULC_X and ULC_Y respectively.
Signal transitions occur when count == boundary.

15.11.3.1.21.3 Diagram



15.11.3.1.21.4 Fields

Field	Function
31 POL	Polarity of DE signal. 0 == active_low ; 1 == active_high
30-29 —	Reserved
28-16 ULC_Y	Vertical position in the display that determines when the DE goes active.
15-13 —	Reserved
12-0 ULC_X	Horizontal position in the scan line that determines when DE goes active.

15.11.3.1.22 (SS_DE_LRC)

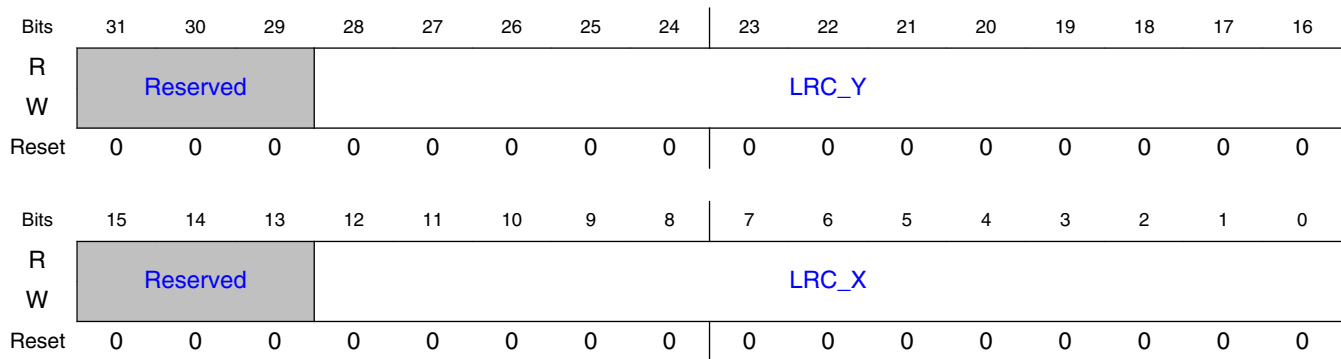
15.11.3.1.22.1 Offset

Register	Offset
SS_DE_LRC	50h

15.11.3.1.22.2 Function

Lower right coordinate that defines when DE goes NOT active.
Both X and Y coordinates are programmable as LRC_X and LRC_Y respectively.

15.11.3.1.22.3 Diagram



15.11.3.1.22.4 Fields

Field	Function
31-29 —	Reserved
28-16 LRC_Y	Vertical position in the display that determines when the DE goes inactive.
15-13 —	Reserved
12-0 LRC_X	Horizontal position in the scan line that determines when DE goes inactive.

15.11.3.1.23 (SS_DE_LRC_SET)

15.11.3.1.23.1 Offset

Register	Offset
SS_DE_LRC_SET	54h

15.11.3.1.23.2 Function

Lower right coordinate that defines when DE goes NOT active.
Both X and Y coordinates are programmable as LRC_X and LRC_Y respectively.

15.11.3.1.23.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								LRC_Y							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								LRC_X							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.11.3.1.23.4 Fields

Field	Function
31-29 —	Reserved
28-16 LRC_Y	Vertical position in the display that determines when the DE goes inactive.
15-13 —	Reserved
12-0 LRC_X	Horizontal position in the scan line that determines when DE goes inactive.

15.11.3.1.24 (SS_DE_LRC_CLR)

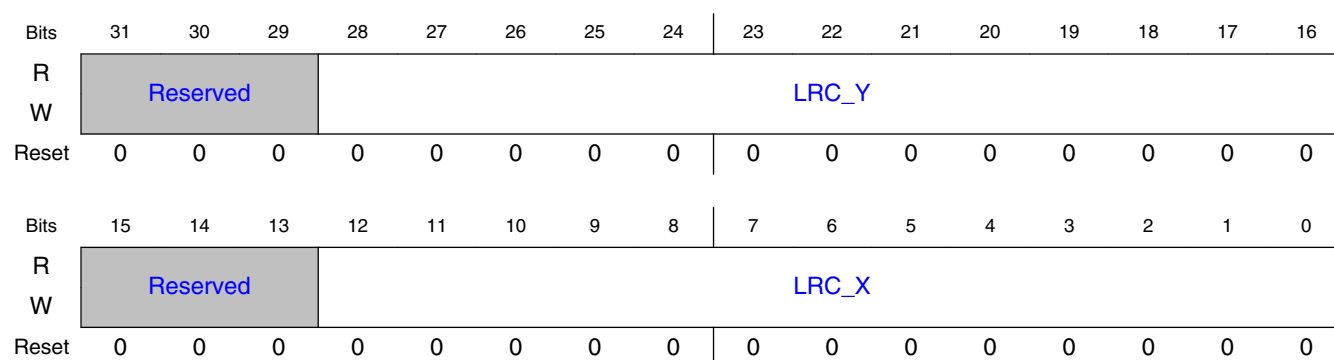
15.11.3.1.24.1 Offset

Register	Offset
SS_DE_LRC_CLR	58h

15.11.3.1.24.2 Function

Lower right coordinate that defines when DE goes NOT active.
Both X and Y coordinates are programmable as LRC_X and LRC_Y respectively.

15.11.3.1.24.3 Diagram



15.11.3.1.24.4 Fields

Field	Function
31-29 —	Reserved
28-16 LRC_Y	Vertical position in the display that determines when the DE goes inactive.
15-13 —	Reserved
12-0 LRC_X	Horizontal position in the scan line that determines when DE goes inactive.

15.11.3.1.25 (SS_DE_LRC_TOG)

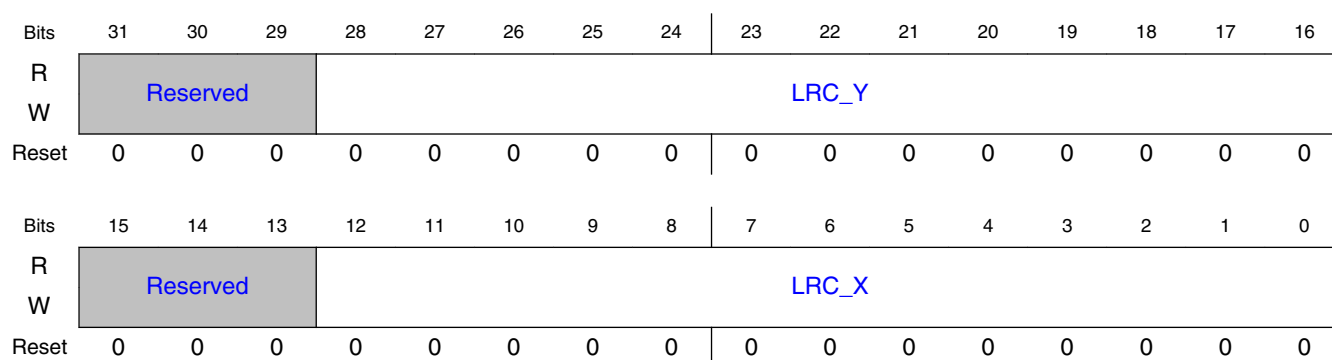
15.11.3.1.25.1 Offset

Register	Offset
SS_DE_LRC_TOG	5Ch

15.11.3.1.25.2 Function

Lower right coordinate that defines when DE goes NOT active.
Both X and Y coordinates are programmable as LRC_X and LRC_Y respectively.

15.11.3.1.25.3 Diagram



15.11.3.1.25.4 Fields

Field	Function
31-29 —	Reserved
28-16 LRC_Y	Vertical position in the display that determines when the DE goes inactive.
15-13 —	Reserved
12-0 LRC_X	Horizontal position in the scan line that determines when DE goes inactive.

15.11.3.1.26 (SS_MODE)

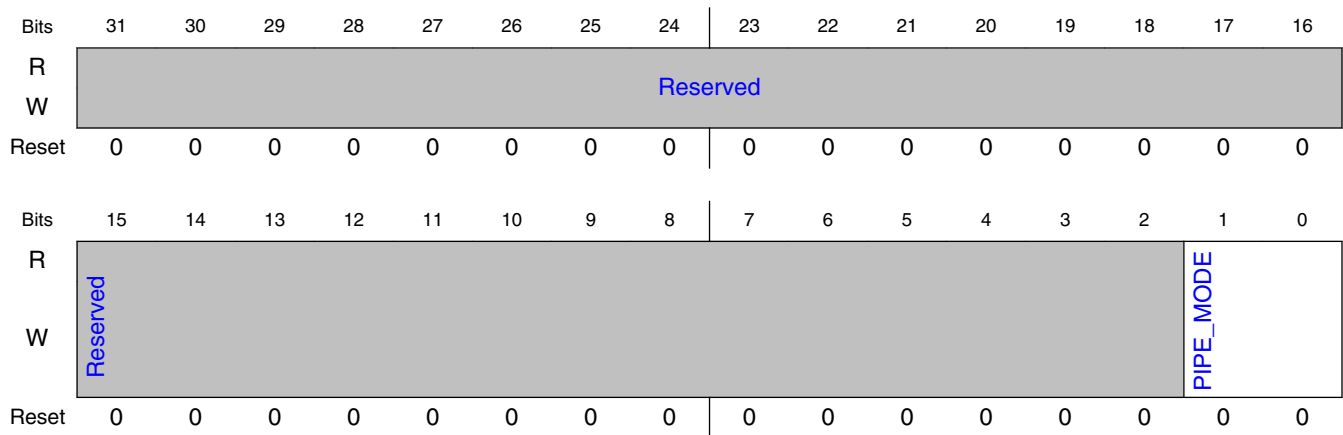
15.11.3.1.26.1 Offset

Register	Offset
SS_MODE	60h

15.11.3.1.26.2 Function

This register is used to select the type of chroma sub-sampling used. Three options are available: bypass(444), 422, and 420.

15.11.3.1.26.3 Diagram



15.11.3.1.26.4 Fields

Field	Function
31-2 —	Reserved
1-0 PIPE_MODE	Sets the sub-sampler mode. 00: bypass 01: 422 subsample 10: 420 subsample 11: bypass

15.11.3.1.27 (SS_MODE_SET)

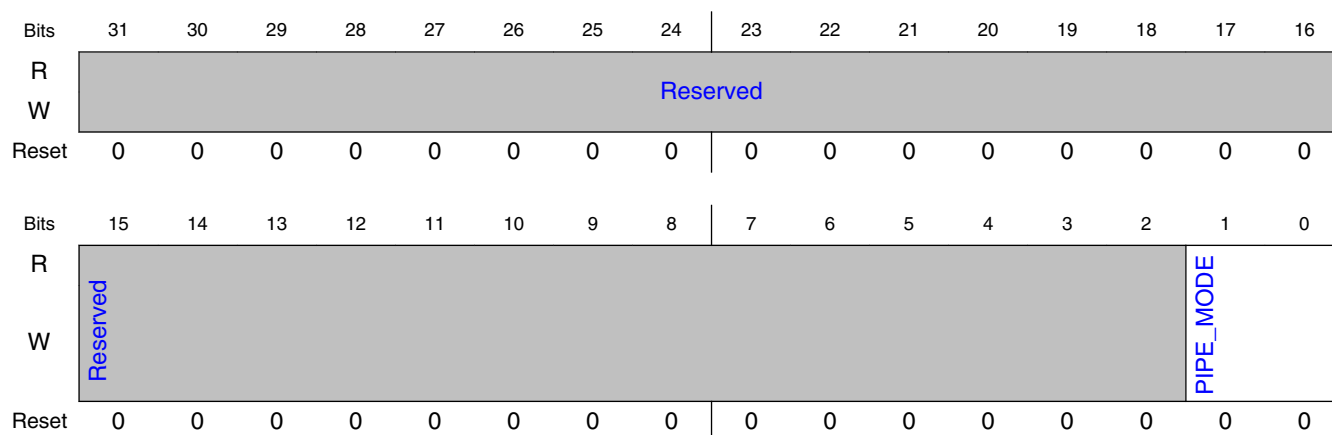
15.11.3.1.27.1 Offset

Register	Offset
SS_MODE_SET	64h

15.11.3.1.27.2 Function

This register is used to select the type of chroma sub-sampling used. Three options are available: bypass(444), 422, and 420.

15.11.3.1.27.3 Diagram



15.11.3.1.27.4 Fields

Field	Function
31-2 —	Reserved
1-0 PIPE_MODE	Sets the sub-sampler mode. 00: bypass 01: 422 subsample 10: 420 subsample 11: bypass

15.11.3.1.28 (SS_MODE_CLR)

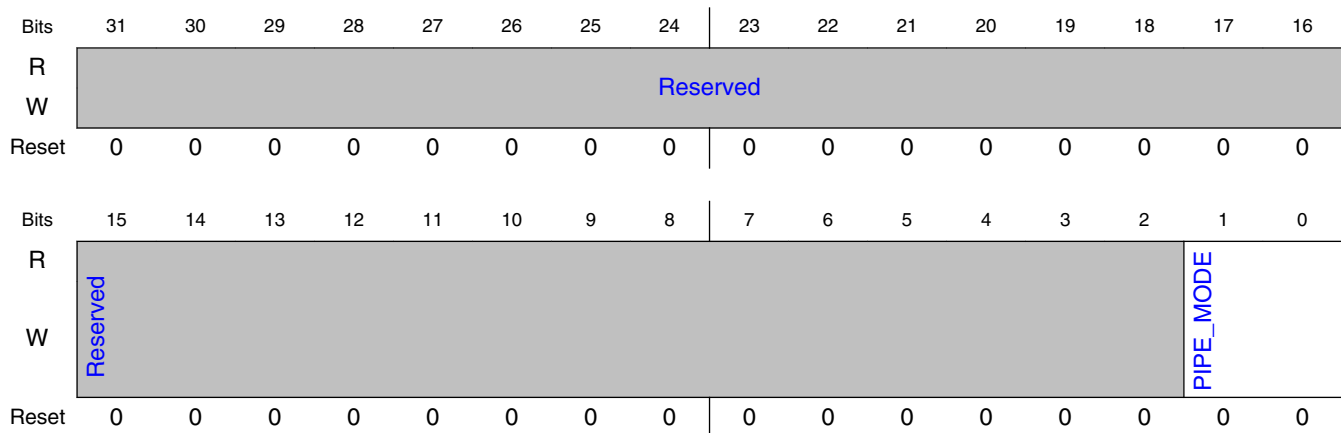
15.11.3.1.28.1 Offset

Register	Offset
SS_MODE_CLR	68h

15.11.3.1.28.2 Function

This register is used to select the type of chroma sub-sampling used. Three options are available: bypass(444), 422, and 420.

15.11.3.1.28.3 Diagram



15.11.3.1.28.4 Fields

Field	Function
31-2 —	Reserved
1-0 PIPE_MODE	Sets the sub-sampler mode. 00: bypass 01: 422 subsample 10: 420 subsample 11: bypass

15.11.3.1.29 (SS_MODE_TOG)

15.11.3.1.29.1 Offset

Register	Offset
SS_MODE_TOG	6Ch

15.11.3.1.29.2 Function

This register is used to select the type of chroma sub-sampling used. Three options are available: bypass(444), 422, and 420.

15.11.3.1.29.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved														PIPE_MODE	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15.11.3.1.29.4 Fields

Field	Function
31-2 —	Reserved
1-0 PIPE_MODE	Sets the sub-sampler mode. 00: bypass 01: 422 subsample 10: 420 subsample 11: bypass

15.11.3.1.30 (SS_COEFF)

15.11.3.1.30.1 Offset

Register	Offset
SS_COEFF	70h

15.11.3.1.30.2 Function

Holds coefficients for both vertical + horizontal filters.

Three coefficients used for 3 tap fir filter.

HORIZ_A, HORIZ_B, HORIZ_C is used for horizontal filter coefficients.

VERT_A, VERT_B, VERT_C is used for vertical filter coefficients.

In addition, the horizontal and vertical filters have a normalization factor called HORIZ_NORM and VERT_NORM respectively.

A general formula for these filters is:

$$\text{pixel_out} = (\text{A} * \text{pixel}[\text{x}+1] + \text{B} * \text{pixel}[\text{x}] + \text{C} * \text{pixel}[\text{x}-1] + \text{Round}) / \text{Normalize}$$

The values Round and Normalize are selected based on the values of HORIZ_NORM and VERT_NORM.

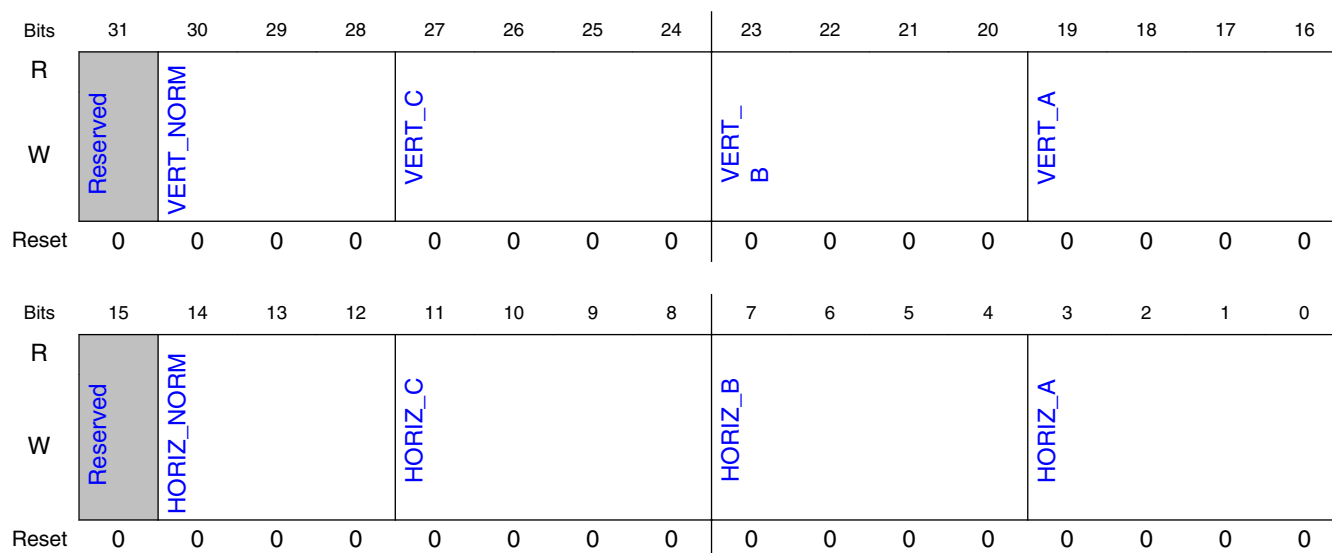
The mapping can be described by the following equations:

```

if (NORM = 0)      Round = 0; Normalize = 1;
else if (NORM < 5) Round = 2^(NORM); Normalize = 2^(NORM+1);
else              Round = 2^(5); Normalize = 2^(5+1);

```

15.11.3.1.30.3 Diagram



15.11.3.1.30.4 Fields

Field	Function
31	Reserved
—	
30-28	Sets the 3 bit vertical filter normalization factor.

Table continues on the next page...

Memory Map and Registers

Field	Function
VERT_NORM	
27-24 VERT_C	Sets the 4 bit vertical C coefficient.
23-20 VERT_B	Sets the 4 bit vertical B coefficient.
19-16 VERT_A	Sets the 4 bit vertical A coefficient.
15 —	Reserved
14-12 HORIZ_NORM	Sets the 3 bit horizontal filter normalization factor.
11-8 HORIZ_C	Sets the 4 bit horizontal C coefficient.
7-4 HORIZ_B	Sets the 4 bit horizontal B coefficient.
3-0 HORIZ_A	Sets the 4 bit horizontal A coefficient.

15.11.3.1.31 (SS_COEFF_SET)

15.11.3.1.31.1 Offset

Register	Offset
SS_COEFF_SET	74h

15.11.3.1.31.2 Function

Holds coefficients for both vertical + horizontal filters.
 Three coefficients used for 3 tap fir filter.
 HORIZ_A, HORIZ_B, HORIZ_C is used for horizontal filter coefficients.
 VERT_A, VERT_B, VERT_C is used for vertical filter coefficients.
 In addition, the horizontal and vertical filters have a normalization factor called HORIZ_NORM and VERT_NORM respectively.
 A general formula for these filters is:

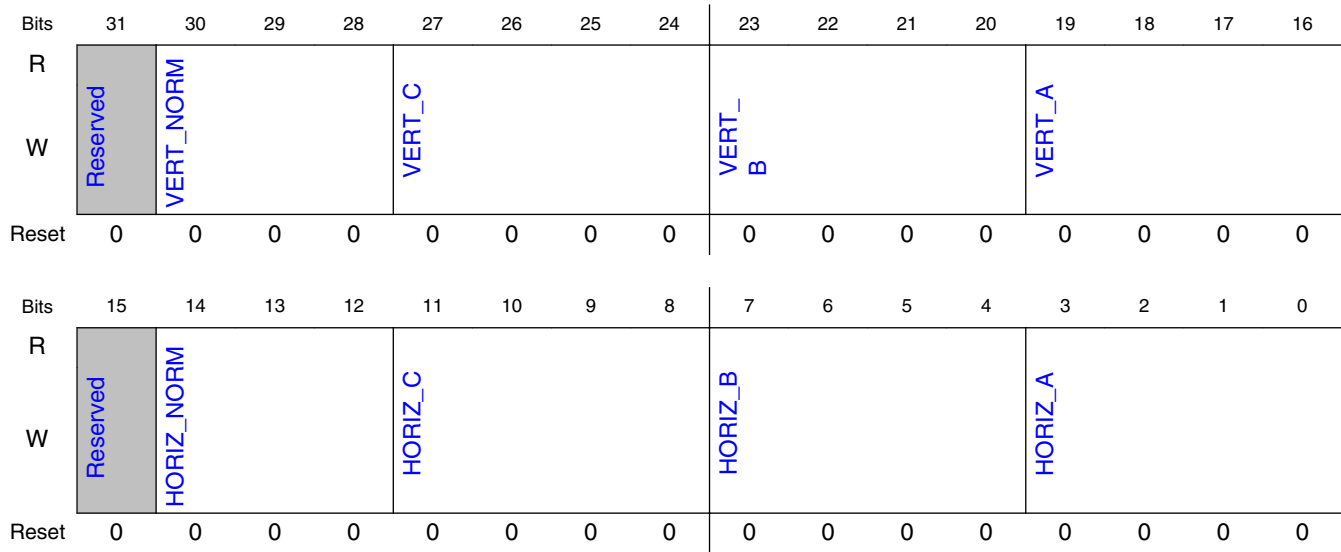
$$\text{pixel_out} = (\text{A} * \text{pixel}[\text{x}+1] + \text{B} * \text{pixel}[\text{x}] + \text{C} * \text{pixel}[\text{x}-1] + \text{Round}) / \text{Normalize}$$

 The values Round and Normalize are selected based on the values of HORIZ_NORM and VERT_NORM.
 The mapping can be described by the following equations:

```

if (NORM = 0)      Round = 0; Normalize = 1;
else if (NORM < 5) Round = 2^(NORM); Normalize = 2^(NORM+1);
else               Round = 2^5; Normalize = 2^(5+1);
  
```

15.11.3.1.31.3 Diagram



15.11.3.1.31.4 Fields

Field	Function
31 —	Reserved
30-28 VERT_NORM	Sets the 3 bit vertical filter normalization factor.
27-24 VERT_C	Sets the 4 bit vertical C coefficient.
23-20 VERT_B	Sets the 4 bit vertical B coefficient.
19-16 VERT_A	Sets the 4 bit vertical A coefficient.
15 —	Reserved
14-12 HORIZ_NORM	Sets the 3 bit horizontal filter normalization factor.
11-8 HORIZ_C	Sets the 4 bit horizontal C coefficient.
7-4 HORIZ_B	Sets the 4 bit horizontal B coefficient.
3-0 HORIZ_A	Sets the 4 bit horizontal A coefficient.

15.11.3.1.32 (SS_COEFF_CLR)

15.11.3.1.32.1 Offset

Register	Offset
SS_COEFF_CLR	78h

15.11.3.1.32.2 Function

Holds coefficients for both vertical + horizontal filters.

Three coefficients used for 3 tap fir filter.

HORIZ_A, HORIZ_B, HORIZ_C is used for horizontal filter coefficients.

VERT_A, VERT_B, VERT_C is used for vertical filter coefficients.

In addition, the horizontal and vertical filters have a normalization factor called HORIZ_NORM and VERT_NORM respectively.

A general formula for these filters is:

$$\text{pixel_out} = (\text{A} * \text{pixel}[\text{x}+1] + \text{B} * \text{pixel}[\text{x}] + \text{C} * \text{pixel}[\text{x}-1] + \text{Round}) / \text{Normalize}$$

The values Round and Normalize are selected based on the values of HORIZ_NORM and VERT_NORM.

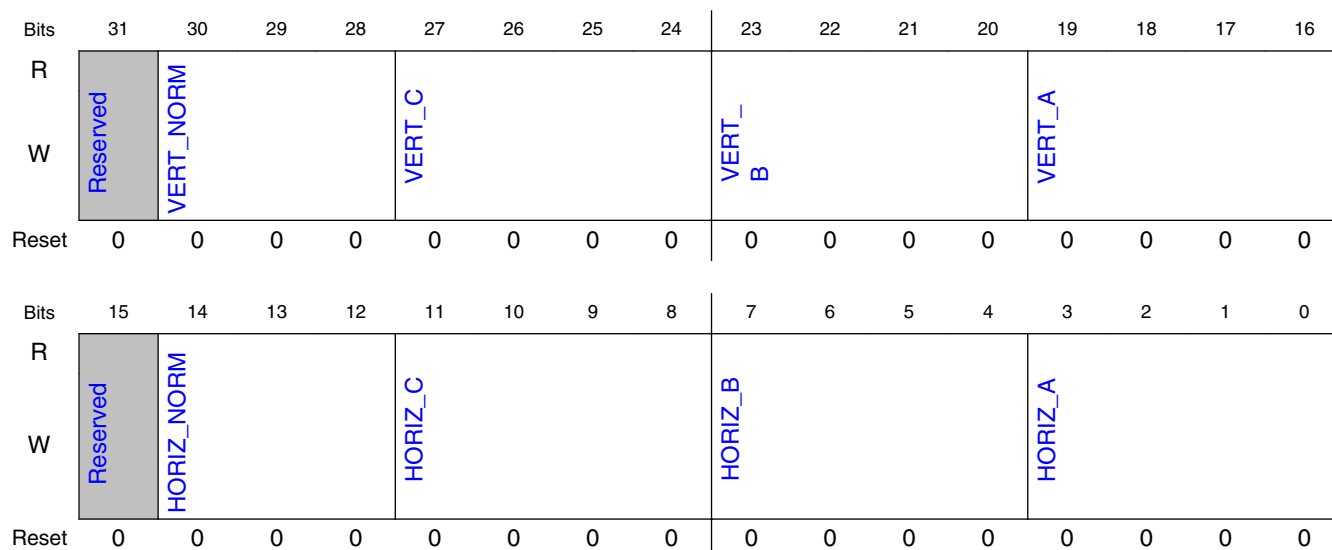
The mapping can be described by the following equations:

```

if (NORM = 0)      Round = 0; Normalize = 1;
else if (NORM < 5) Round = 2^(NORM); Normalize = 2^(NORM+1);
else              Round = 2^5; Normalize = 2^(5+1);

```

15.11.3.1.32.3 Diagram



15.11.3.1.32.4 Fields

Field	Function
31 —	Reserved
30-28 VERT_NORM	Sets the 3 bit vertical filter normalization factor.
27-24 VERT_C	Sets the 4 bit vertical C coefficient.
23-20 VERT_B	Sets the 4 bit vertical B coefficient.
19-16 VERT_A	Sets the 4 bit vertical A coefficient.
15 —	Reserved
14-12 HORIZ_NORM	Sets the 3 bit horizontal filter normalization factor.
11-8 HORIZ_C	Sets the 4 bit horizontal C coefficient.
7-4 HORIZ_B	Sets the 4 bit horizontal B coefficient.
3-0 HORIZ_A	Sets the 4 bit horizontal A coefficient.

15.11.3.1.33 (SS_COEFF_TOG)

15.11.3.1.33.1 Offset

Register	Offset
SS_COEFF_TOG	7Ch

15.11.3.1.33.2 Function

Holds coefficients for both vertical + horizontal filters.

Three coefficients used for 3 tap fir filter.

HORIZ_A, HORIZ_B, HORIZ_C is used for horizontal filter coefficients.

VERT_A, VERT_B, VERT_C is used for vertical filter coefficients.

In addition, the horizontal and vertical filters have a normalization factor called HORIZ_NORM and VERT_NORM respectively.

A general formula for these filters is:

$$\text{pixel_out} = (A * \text{pixel}[x+1] + B * \text{pixel}[x] + C * \text{pixel}[x-1] + \text{Round}) / \text{Normalize}$$

The values Round and Normalize are selected based on the values of HORIZ_NORM and VERT_NORM.

Memory Map and Registers

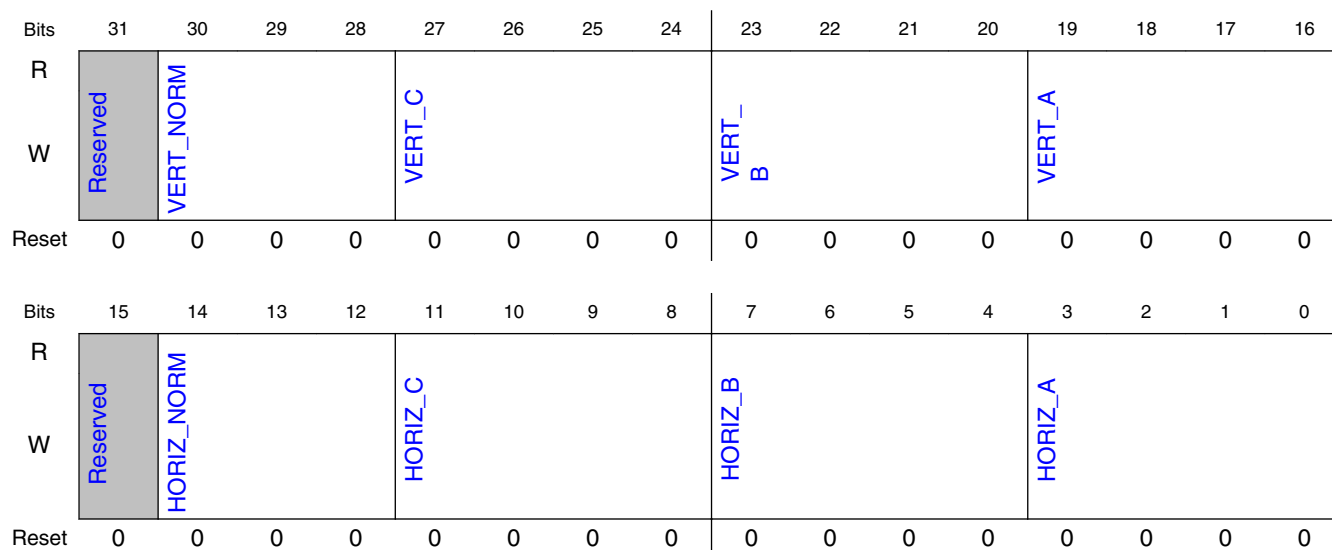
The mapping can be described by the following equations:

```

if (NORM = 0)      Round = 0; Normalize = 1;
else if (NORM < 5) Round = 2^(NORM); Normalize = 2^(NORM+1);
else               Round = 2^5; Normalize = 2^(5+1);

```

15.11.3.1.33.3 Diagram



15.11.3.1.33.4 Fields

Field	Function
31 —	Reserved
30-28 VERT_NORM	Sets the 3 bit vertical filter normalization factor.
27-24 VERT_C	Sets the 4 bit vertical C coefficient.
23-20 VERT_B	Sets the 4 bit vertical B coefficient.
19-16 VERT_A	Sets the 4 bit vertical A coefficient.
15 —	Reserved
14-12 HORIZ_NORM	Sets the 3 bit horizontal filter normalization factor.
11-8 HORIZ_C	Sets the 4 bit horizontal C coefficient.
7-4	Sets the 4 bit horizontal B coefficient.

Table continues on the next page...

Field	Function
HORIZ_B	
3-0 HORIZ_A	Sets the 4 bit horizontal A coefficient.

15.11.3.1.34 (SS_CLIP_CB)

15.11.3.1.34.1 Offset

Register	Offset
SS_CLIP_CB	80h

15.11.3.1.34.2 Function

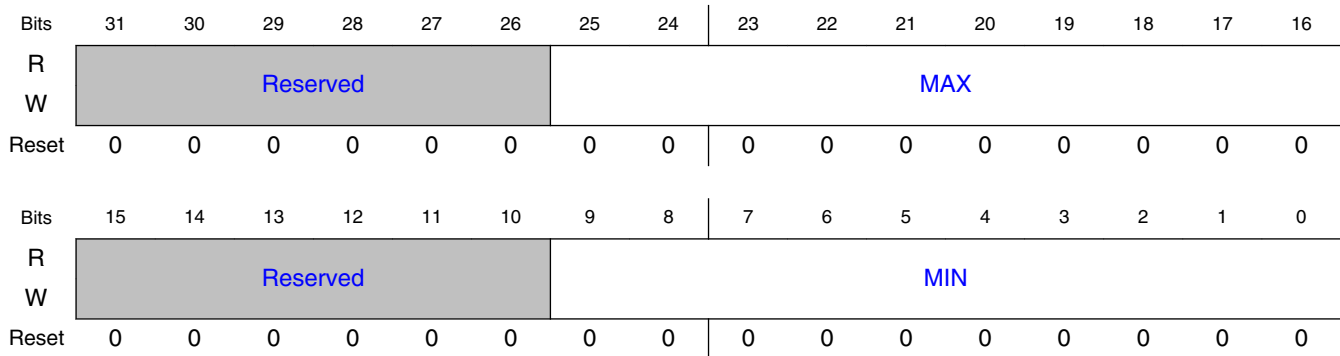
Contains clipping fields for Cb chroma components.

For data bypass, set MIN to 0x0 and MAX to 0x3FF.

Note, in 444 mode with 12 bit wide chroma components, clipping is applied to upper 10 bits only.

The lower 2 bits of 12 bit data in 444 mode passes through the subsampler untouched.

15.11.3.1.34.3 Diagram



15.11.3.1.34.4 Fields

Field	Function
31-26 —	Reserved
25-16	Set maximum value for Cb clipping function.

Table continues on the next page...

Memory Map and Registers

Field	Function
MAX	
15-10 —	Reserved
9-0 MIN	Set minimum value for Cb clipping function.

15.11.3.1.35 (SS_CLIP_CB_SET)

15.11.3.1.35.1 Offset

Register	Offset
SS_CLIP_CB_SET	84h

15.11.3.1.35.2 Function

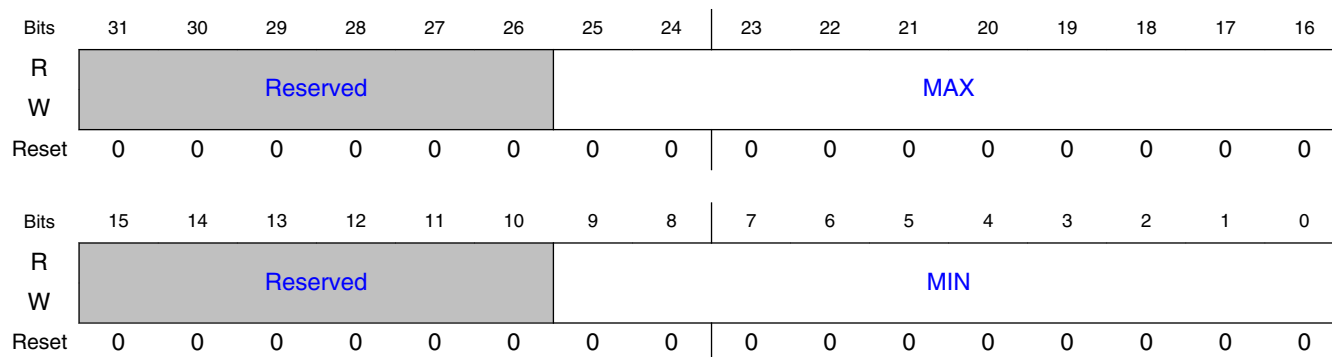
Contains clipping fields for Cb chroma components.

For data bypass, set MIN to 0x0 and MAX to 0x3FF.

Note, in 444 mode with 12 bit wide chroma components, clipping is applied to upper 10 bits only.

The lower 2 bits of 12 bit data in 444 mode passes through the subsampler untouched.

15.11.3.1.35.3 Diagram



15.11.3.1.35.4 Fields

Field	Function
31-26	Reserved

Table continues on the next page...

Field	Function
—	
25-16 MAX	Set maximum value for Cb clipping function.
15-10 —	Reserved
9-0 MIN	Set minimum value for Cb clipping function.

15.11.3.1.36 (SS_CLIP_CB_CLR)

15.11.3.1.36.1 Offset

Register	Offset
SS_CLIP_CB_CLR	88h

15.11.3.1.36.2 Function

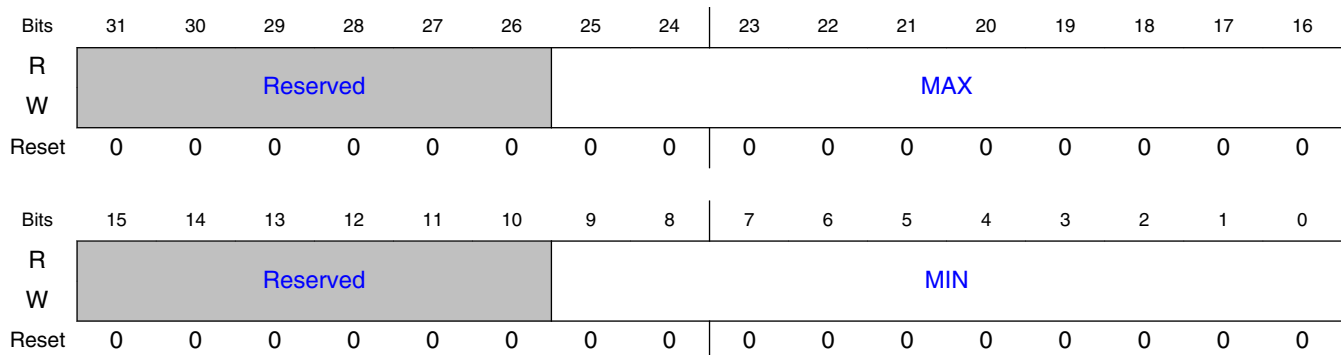
Contains clipping fields for Cb chroma components.

For data bypass, set MIN to 0x0 and MAX to 0x3FF.

Note, in 444 mode with 12 bit wide chroma components, clipping is applied to upper 10 bits only.

The lower 2 bits of 12 bit data in 444 mode passes through the subsampler untouched.

15.11.3.1.36.3 Diagram



15.11.3.1.36.4 Fields

Field	Function
31-26 —	Reserved
25-16 MAX	Set maximum value for Cb clipping function.
15-10 —	Reserved
9-0 MIN	Set minimum value for Cb clipping function.

15.11.3.1.37 (SS_CLIP_CB_TOG)

15.11.3.1.37.1 Offset

Register	Offset
SS_CLIP_CB_TOG	8Ch

15.11.3.1.37.2 Function

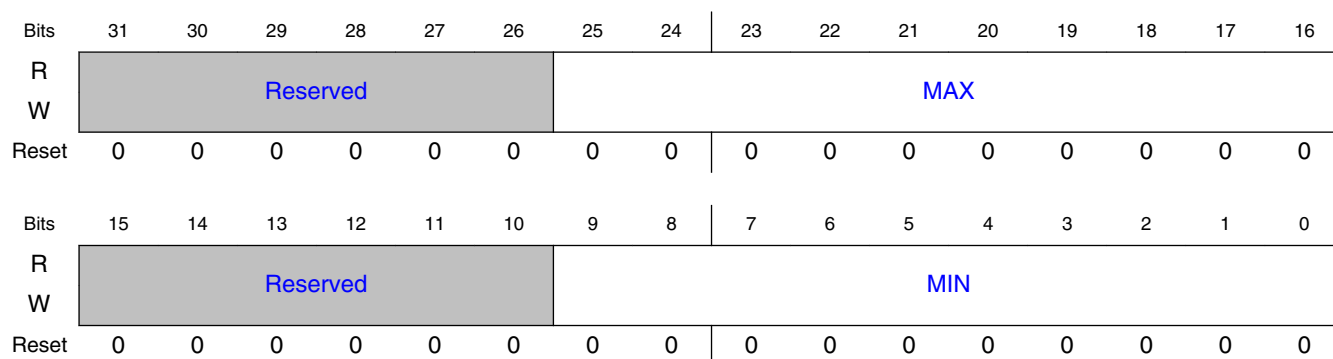
Contains clipping fields for Cb chroma components.

For data bypass, set MIN to 0x0 and MAX to 0x3FF.

Note, in 444 mode with 12 bit wide chroma components, clipping is applied to upper 10 bits only.

The lower 2 bits of 12 bit data in 444 mode passes through the subsampler untouched.

15.11.3.1.37.3 Diagram



15.11.3.1.37.4 Fields

Field	Function
31-26 —	Reserved
25-16 MAX	Set maximum value for Cb clipping function.
15-10 —	Reserved
9-0 MIN	Set minimum value for Cb clipping function.

15.11.3.1.38 (SS_CLIP_CR)

15.11.3.1.38.1 Offset

Register	Offset
SS_CLIP_CR	90h

15.11.3.1.38.2 Function

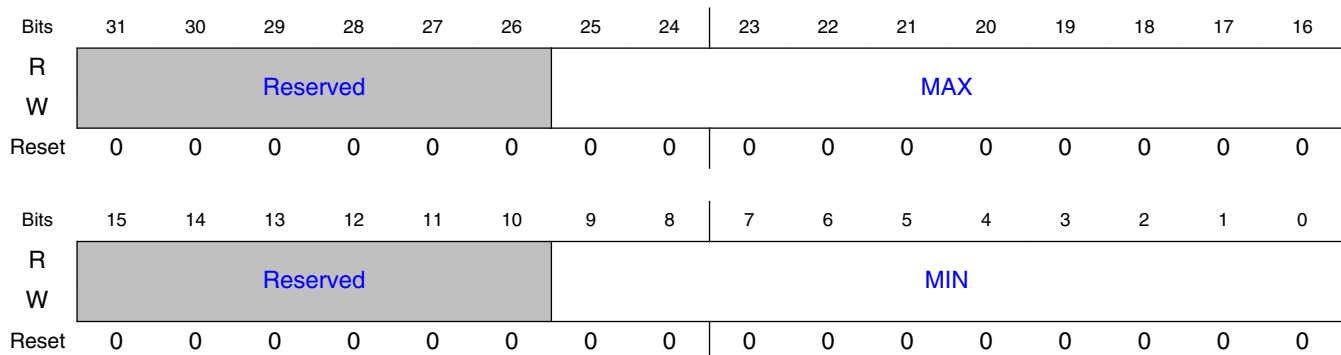
Contains clipping fields for Cr chroma components.

For data bypass, set MIN to 0x000 and MAX to 0x3FF.

Note, in 444 mode with 12 bit wide chroma components, clipping is applied to upper 10 bits only.

The lower 2 bits of 12 bit data in 444 mode passes through the subsampler untouched.

15.11.3.1.38.3 Diagram



15.11.3.1.38.4 Fields

Field	Function
31-26 —	Reserved
25-16 MAX	Set maximum value for Cr clipping function. If the Cr output data is greater than this value it is clipped to MAX.
15-10 —	Reserved
9-0 MIN	Set minimum value for Cr clipping function. If the Cr data is less than this value it is clipped to MIN.

15.11.3.1.39 (SS_CLIP_CR_SET)

15.11.3.1.39.1 Offset

Register	Offset
SS_CLIP_CR_SET	94h

15.11.3.1.39.2 Function

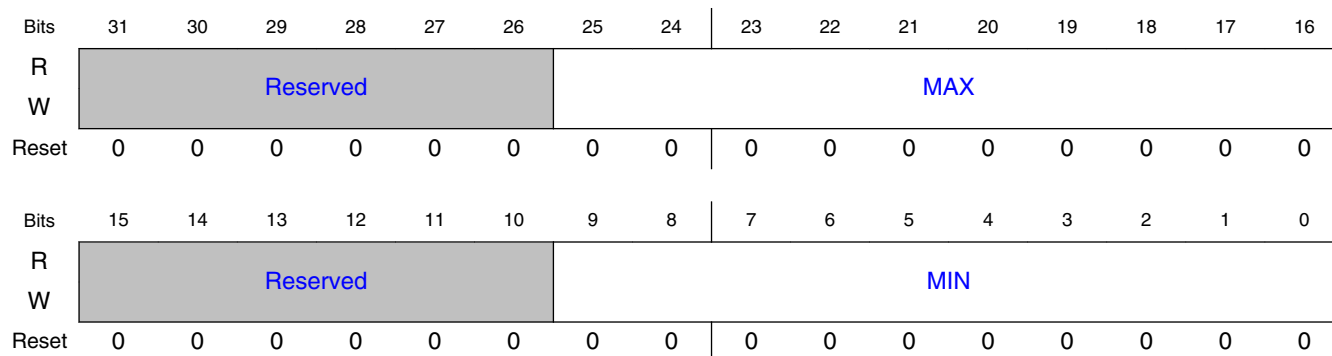
Contains clipping fields for Cr chroma components.

For data bypass, set MIN to 0x000 and MAX to 0x3FF.

Note, in 444 mode with 12 bit wide chroma components, clipping is applied to upper 10 bits only.

The lower 2 bits of 12 bit data in 444 mode passes through the subsampler untouched.

15.11.3.1.39.3 Diagram



15.11.3.1.39.4 Fields

Field	Function
31-26 —	Reserved
25-16 MAX	Set maximum value for Cr clipping function. If the Cr output data is greater than this value it is clipped to MAX.
15-10 —	Reserved
9-0 MIN	Set minimum value for Cr clipping function. If the Cr data is less than this value it is clipped to MIN.

15.11.3.1.40 (SS_CLIP_CR_CLR)

15.11.3.1.40.1 Offset

Register	Offset
SS_CLIP_CR_CLR	98h

15.11.3.1.40.2 Function

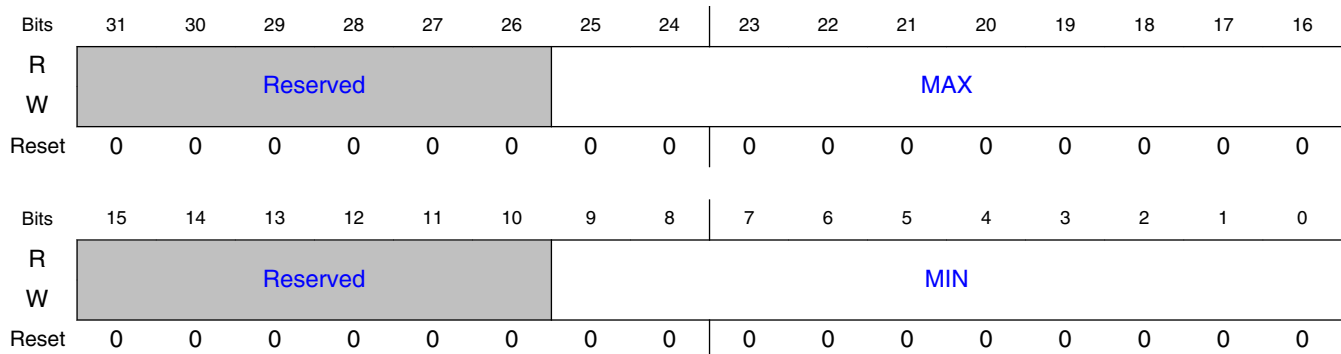
Contains clipping fields for Cr chroma components.

For data bypass, set MIN to 0x000 and MAX to 0x3FF.

Note, in 444 mode with 12 bit wide chroma components, clipping is applied to upper 10 bits only.

The lower 2 bits of 12 bit data in 444 mode passes through the subsampler untouched.

15.11.3.1.40.3 Diagram



15.11.3.1.40.4 Fields

Field	Function
31-26 —	Reserved
25-16 MAX	Set maximum value for Cr clipping function. If the Cr output data is greater than this value it is clipped to MAX.
15-10 —	Reserved
9-0 MIN	Set minimum value for Cr clipping function. If the Cr data is less than this value it is clipped to MIN.

15.11.3.1.41 (SS_CLIP_CR_TOG)

15.11.3.1.41.1 Offset

Register	Offset
SS_CLIP_CR_TOG	9Ch

15.11.3.1.41.2 Function

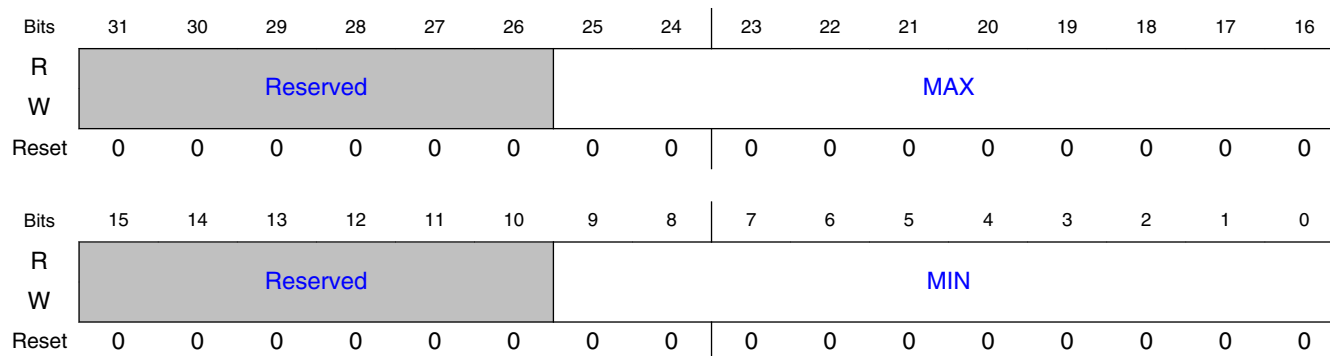
Contains clipping fields for Cr chroma components.

For data bypass, set MIN to 0x000 and MAX to 0x3FF.

Note, in 444 mode with 12 bit wide chroma components, clipping is applied to upper 10 bits only.

The lower 2 bits of 12 bit data in 444 mode passes through the subsampler untouched.

15.11.3.1.41.3 Diagram



15.11.3.1.41.4 Fields

Field	Function
31-26 —	Reserved
25-16 MAX	Set maximum value for Cr clipping function. If the Cr output data is greater than this value it is clipped to MAX.
15-10 —	Reserved
9-0 MIN	Set minimum value for Cr clipping function. If the Cr data is less than this value it is clipped to MIN.

15.11.3.1.42 (SS_INTER_MODE)

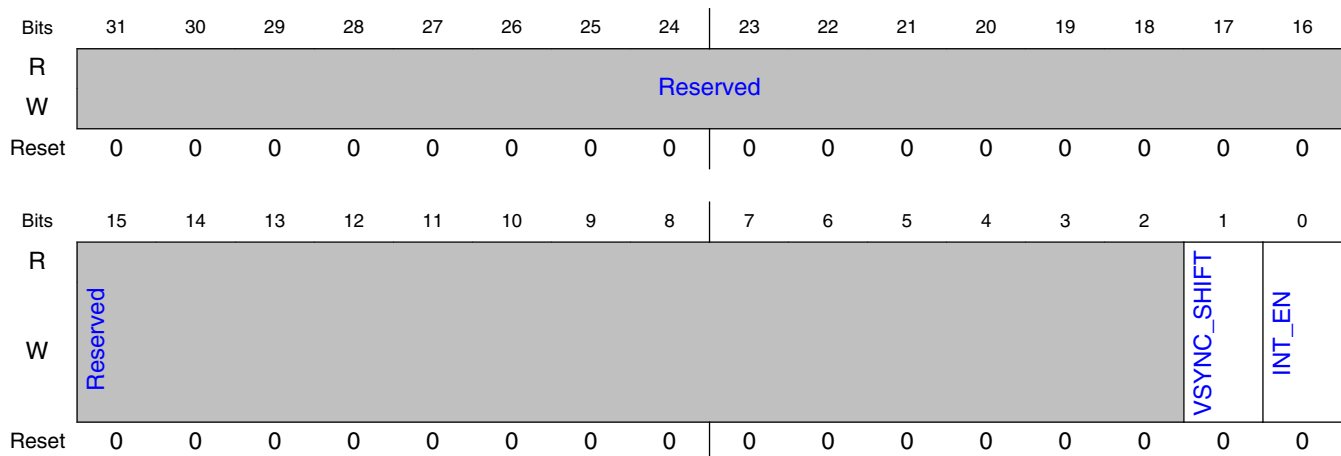
15.11.3.1.42.1 Offset

Register	Offset
SS_INTER_MODE	A0h

15.11.3.1.42.2 Function

Contains bits to enable HDMI timing in interlaced mode.
There are two interlaced formats supported in this module.
General + Special interlaced formats. General occurs when Vtotal is odd.

15.11.3.1.42.3 Diagram



15.11.3.1.42.4 Fields

Field	Function
31-2 —	Reserved
1 VSYNC_SHIFT	Set 0 to run in general interlaced mode. Every other field the vsync is shifted forward 1/2 scan lines. In addition, the horizontal back porch is extended 1 scan line. Set 1 run in special interlaced format. This shifts vsync backwards 1/2 scan line every other field. Remaining timing won't change.
0 INT_EN	Set this bit to 1 to enable interlaced HDMI timing.

15.11.3.1.43 (SS_INTER_MODE_SET)

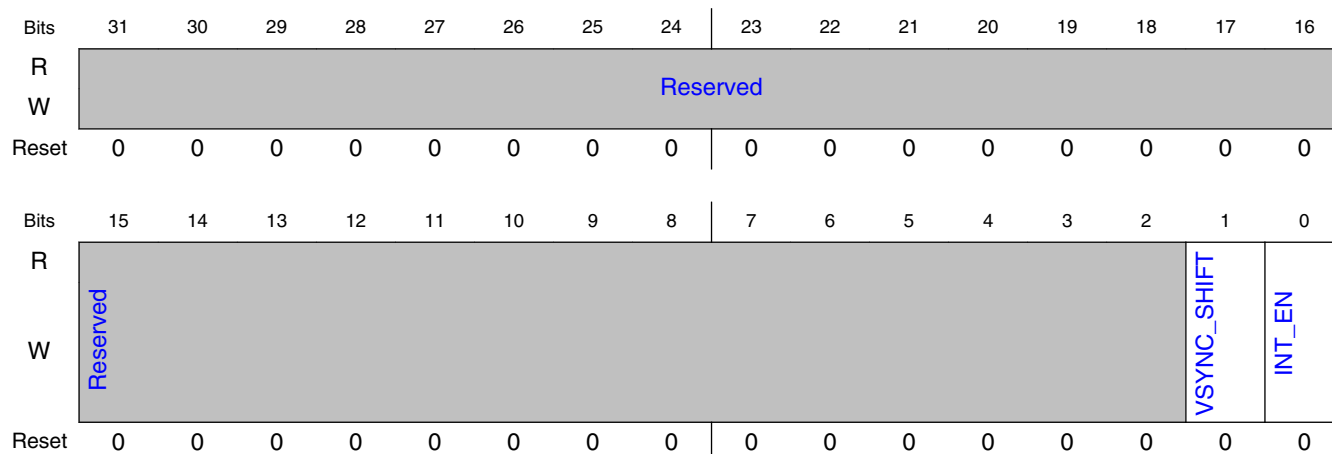
15.11.3.1.43.1 Offset

Register	Offset
SS_INTER_MODE_SET	A4h

15.11.3.1.43.2 Function

Contains bits to enable HDMI timing in interlaced mode.
There are two interlaced formats supported in this module.
General + Special interlaced formats. General occurs when Vtotal is odd.

15.11.3.1.43.3 Diagram



15.11.3.1.43.4 Fields

Field	Function
31-2 —	Reserved
1 VSYNC_SHIFT	Set 0 to run in general interlaced mode. Every other field the vsync is shifted forward 1/2 scan lines. In addition, the horizontal back porch is extended 1 scan line. Set 1 run in special interlaced format. This shifts vsync backwards 1/2 scan line every other field. Remaining timing won't change.
0 INT_EN	Set this bit to 1 to enable interlaced HDMI timing.

15.11.3.1.44 (SS_INTER_MODE_CLR)

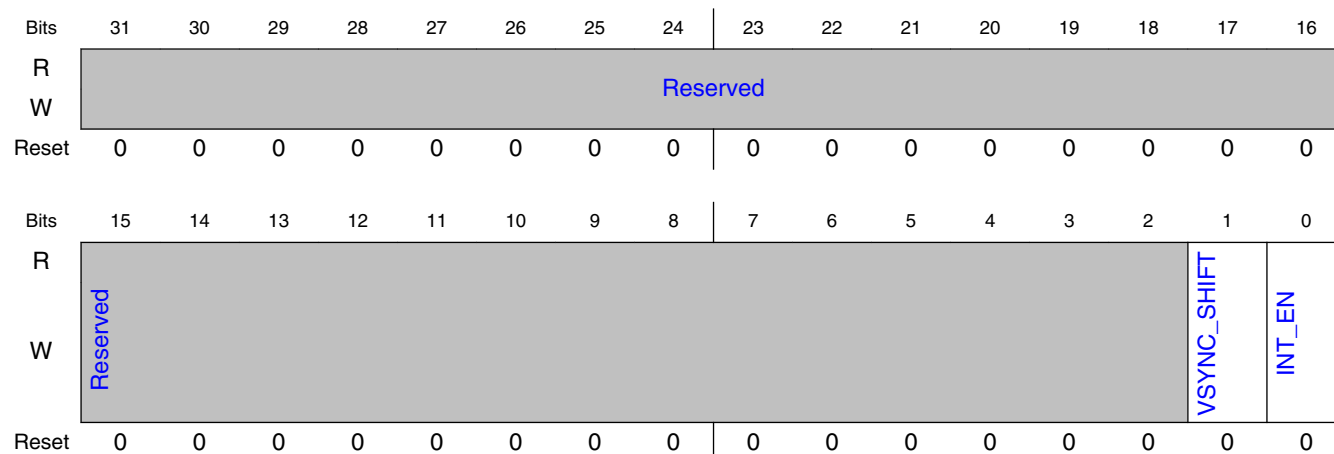
15.11.3.1.44.1 Offset

Register	Offset
SS_INTER_MODE_CLR	A8h

15.11.3.1.44.2 Function

Contains bits to enable HDMI timing in interlaced mode.
There are two interlaced formats supported in this module.
General + Special interlaced formats. General occurs when Vtotal is odd.

15.11.3.1.44.3 Diagram



15.11.3.1.44.4 Fields

Field	Function
31-2 —	Reserved
1 VSYNC_SHIFT	Set 0 to run in general interlaced mode. Every other field the vsync is shifted forward 1/2 scan lines. In addition, the horizontal back porch is extended 1 scan line. Set 1 run in special interlaced format. This shifts vsync backwards 1/2 scan line every other field. Remaining timing won't change.
0 INT_EN	Set this bit to 1 to enable interlaced HDMI timing.

15.11.3.1.45 (SS_INTER_MODE_TOG)

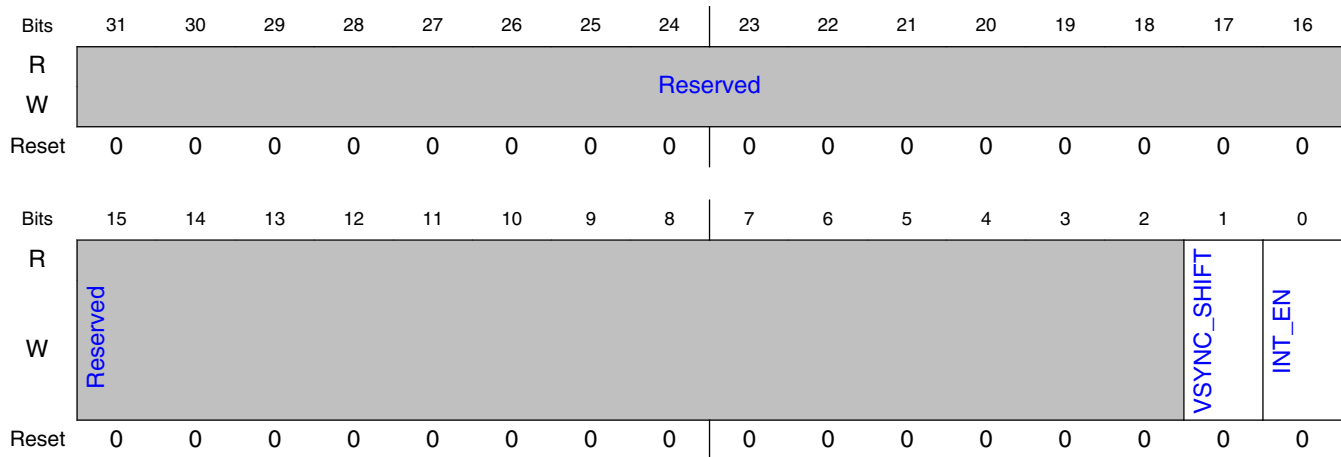
15.11.3.1.45.1 Offset

Register	Offset
SS_INTER_MODE_TOG	ACh

15.11.3.1.45.2 Function

Contains bits to enable HDMI timing in interlaced mode. There are two interlaced formats supported in this module. General + Special interlaced formats. General occurs when Vtotal is odd.

15.11.3.1.45.3 Diagram



15.11.3.1.45.4 Fields

Field	Function
31-2 —	Reserved
1 VSYNC_SHIFT	Set 0 to run in general interlaced mode. Every other field the vsync is shifted forward 1/2 scan lines. In addition, the horizontal back porch is extended 1 scan line. Set 1 run in special interlaced format. This shifts vsync backwards 1/2 scan line every other field. Remaining timing won't change.
0 INT_EN	Set this bit to 1 to enable interlaced HDMI timing.

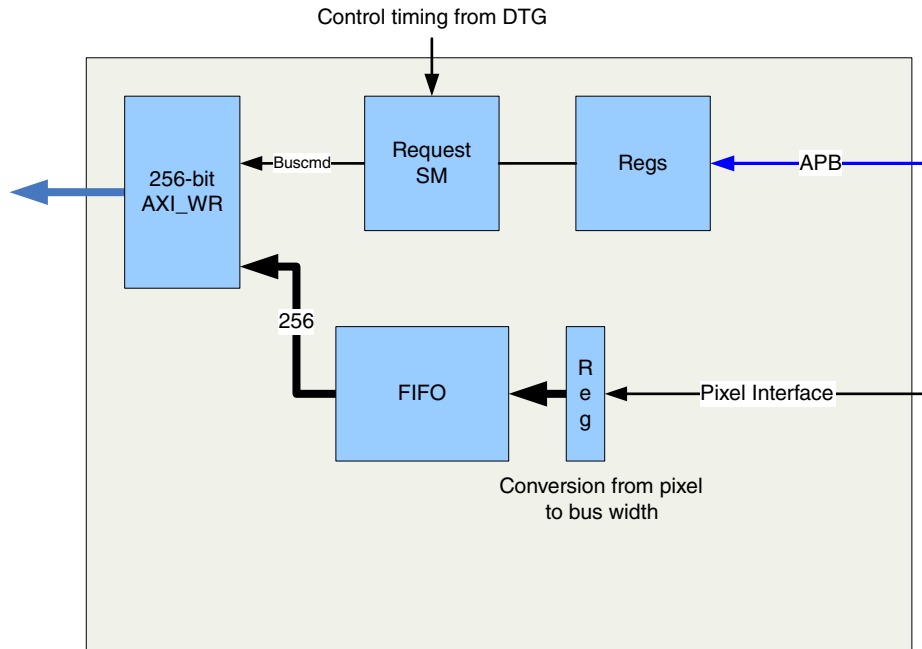
15.12 Write Scale (WR_SCL)

15.12.1 Overview

The Write Scale engine (WR_SCL) can be used to send pixel data from the scaler to DDR. This path to external memory would be used when significant downscaling factors are selected due to large bandwidth requirements on the source image within smaller regions of the display output. The WR_SCL module accepts valid pixels from the scaling engine and packs them into an output FIFO to manage bursts of data out onto the AXI interface to external memory.

15.12.1.1 Block Diagram

The diagram indicating the functional components in the WR_SCL module is below.



15.12.1.2 Features

The WR_SCL module is a bus master engine that accepts pixels from the scaler and generates packets of requests over the AXI interface.

The components in the WR_SCL are highlighted here:

- **Req_SM:** State machine to generate the sequence of transactions to write a buffer to system memory.
- **Regs:** All configuration registers used to store the WR_SCL configuration.
- **AXI_WR:** The buscmd to AXI interface convertor used in many AXI based designs.
- **FIFO:** used to store a sample set of data to be written to system memory.
- **Up sample Reg:** Register used to convert from pixel bus width to bus width.

15.12.2 Functional Description

It is expected that several parameters provides the configuration options to achieve buffer sizes and access patterns that maximizes DDR efficiency and design size. There are hardware and software parameters for the design as defined below.

WR_SCL SW parameters are:

- AXI transaction size (64B up to 512B per burst) is programmable.
- Buffer threshold: defines how much data needs to be in the FIFO before issuing a burst of system memory requests. When the number of bytes is greater than this value, the necessary AXI requests to transfer the data are issued
- Burst frequency: how often the wr_scale engine issues a sequence of AXI requests
- Buffer base address
- Line stride in bytes

15.12.2.1 WR_SCL Output Pixel Formatting

There are a variety of input to output format conversions that are available in the WR_SCL. The input is always 10-bit per component with an 8-bit alpha. Consumers of the data could be 8-bit or 10-bit, so the option to convert the output to 8-bit can be selected. Also, the consumer may not use an alpha value, so it can be truncated. The output pixel format should be influenced by the final consumer of pixels. It is up to SW to determine which to use. The idea is that data that is not needed at the consumer can be removed to save memory bandwidth.

The input pixel_data[37:0] to output conversions are programmed in the BPP field of the CSR_WR_SCL_CTRL register. The input to output conversion of the pixels is defined for each mode as below. Note that P0 is the first pixel received on the pixel_data[37:0] input bus, P1 is the second, etc.

BPP = 0x0: 40bpp, Full pass through of all 38-bits appended with 2'b00 for each pixel. Appending 2'b00 is intended to created byte alignment. The pixels in bold wrap on the 128-bit boundary.

This pattern repeats.

```
1st pixel_out[127:0] = {P3[7:0],
                       2'b00, P2[37:0],
                       2'b00, P1[37:0],
                       2'b00, P0[37:0]};
```

```
2nd pixel_out[127:0] = {P6[15:0],
                       2'b00, P5[37:0],
```

Write Scale (WR_SCL)

```
2'b00, P4[37:0],  
2'b00, P3[37:8]]};
```

```
3rd pixel_out[127:0] = {P9[23:0],  
2'b00, P8[37:0],  
2'b00, P7[37:0],  
2'b00, P6[37:16]]};
```

```
4th pixel_out[127:0] = {P12[31:0],  
2'b00, P11[37:0],  
2'b00, P10[37:0],  
2'b00, P9[37:24]]};
```

BPP = 0x1: 32bpp, ARGB8888/AYUV8888

The lowest two bits for each color component are dropped. Alpha is on the high order byte.

```
Pixel_out[127:0] = { P3[37:30, 29:22, 19:12, 9:2],  
P2[37:30, 29:22, 19:12, 9:2],  
P1[37:30, 29:22, 19:12, 9:2],  
P0[37:30, 29:22, 19:12, 9:2]}
```

BPP = 0x2: 32bpp, RGB/YUV 10-bit per component

All 10-bits of each component are used. The 2 highest bits of alpha are passed.

```
Pixel_out[127:0] = { P3[37:36, 29:20, 19:10, 9:0],  
P2[37:36, 29:20, 19:10, 9:0],  
P1[37:36, 29:20, 19:10, 9:0],  
P0[37:36, 29:20, 19:10, 9:0]}
```

BPP = 0x3: 24bpp, YUV422 10-bit per component

Y (pixel_data[29:20]) with alternate U/V (pixel_data[19:10]). The 20-bit pixel is concatenated with 4-bit Alpha.

```
1st pixel_out[127:0] = { P5[17:10],  
P4[37:34], P4[29:10],  
P3[37:34], P3[29:10],  
P2[37:34], P2[29:10],  
P1[37:34], P1[29:10],  
P0[37:34], P0[29:10] }
```

```
2nd pixel_out[127:0] = { P10[25:10],  
P9[37:34], P9[29:10],  
P8[37:34], P8[29:10],  
P7[37:34], P7[29:10],  
P6[37:34], P6[29:10],  
P5[37:34], P5[29:18] }
```

```
3rd pixel_out[127:0] = { P15[37:34], P15[27:10],  
P14[37:34], P14[29:10],  
P13[37:34], P13[29:10],  
P12[37:34], P12[29:10],  
P11[37:34], P11[29:10],  
P10[37:34], P10[29:26] }
```

This pattern repeats.

BPP = 0x4: 16bpp, YUV422 8-bit per component

The lowest two bits for each color component are dropped. Pixel_data[9:0] is dropped and not used in YUV422 mode. No alpha is used.

Pixel_out[127:0] = { P7[29:22,19:12], P6, ..., P3, P2[29:22,19:12], P1[29:22,19:12], P0[29:22,19:12] }

15.12.2.2 WR_SCL Interrupt

The WR_SCL module can generate interrupts under the following conditions.

- AXI SLVERR or DECERR is received.
- Frame processing complete. The last pixel coincident with pixel_v_sync is written to system memory.

15.12.3 Memory Map and Registers

15.12.3.1 register descriptions

15.12.3.1.1 WR_SCL Memory map

wr_scl base address: 2_1000h

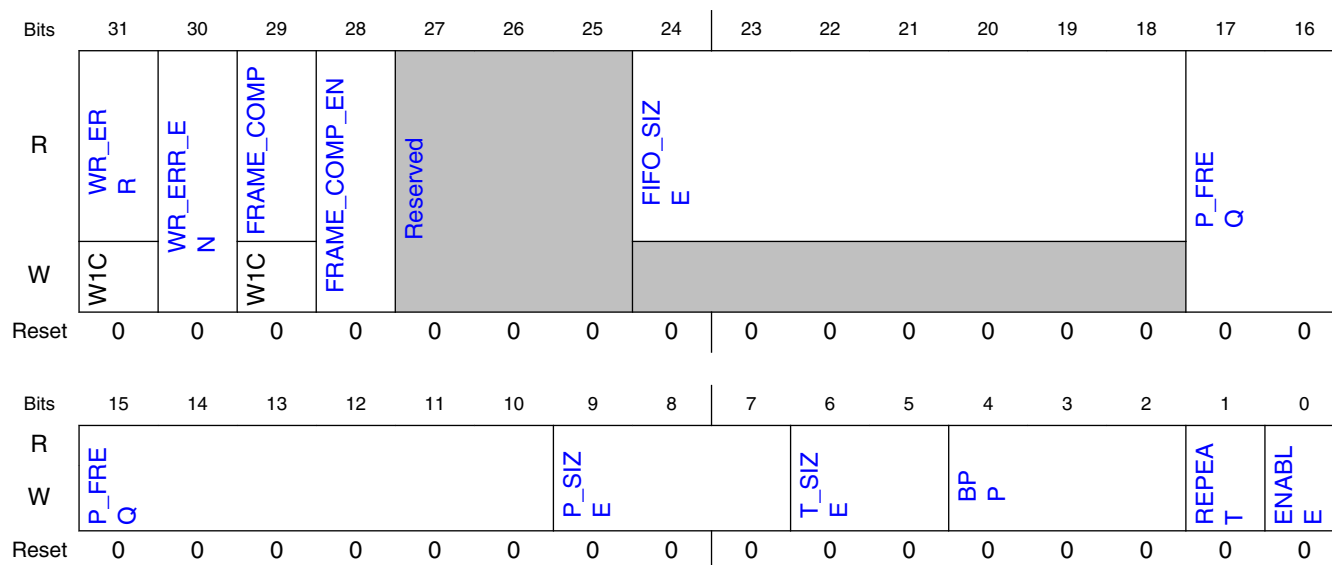
Offset	Register	Width (In bits)	Access	Reset value
0h	Control register for Context Loader. (CTRL_STATUS)	32	RW	0000_0000h
4h	Control register for Context Loader. (CTRL_STATUS_SET)	32	RW	0000_0000h
8h	Control register for Context Loader. (CTRL_STATUS_CLR)	32	RW	0000_0000h
Ch	Control register for Context Loader. (CTRL_STATUS_TOG)	32	RW	0000_0000h
10h	Holds the base address (BASE_ADDR)	32	RW	0000_0000h
14h	Pitch (PITCH)	32	RW	0000_0000h

15.12.3.1.2 Control register for Context Loader. (CTRL_STATUS)

15.12.3.1.2.1 Offset

Register	Offset
CTRL_STATUS	0h

15.12.3.1.2.2 Diagram



15.12.3.1.2.3 Fields

Field	Function
31 WR_ERR	Indicates a write error on the axi interface.
30 WR_ERR_EN	Write error IRQ enable If set, it allows WR_ERR to create an irq whenever there is an error response to a write request.
29 FRAME_COMP	Indicates the current frame being processed has finished.
28 FRAME_COMP_EN	Write error IRQ enable If set, it allows FRAME_COMP to create an irq whenever processing a frame finishes.
27-25 —	Reserved
24-18 FIFO_SIZE	Size of FIFO in design This is the fifo size in the HW. This can be used to program the appropriate P_SIZE. P_SIZE should ideally be 1/2 the FIFO size. 7'h0F - 512B 7'h1F - 1KB 7'h3F - 2KB 7'h7F - 4KB
17-10 P_FREQ	Payload frequency P_FREQ*8 is the number of clocks between payload requests. A value of 0 does not limit the frequency of the payload requests. Instead, the payload is written as frequently as data is available.
9-7	Payload Size

Table continues on the next page...

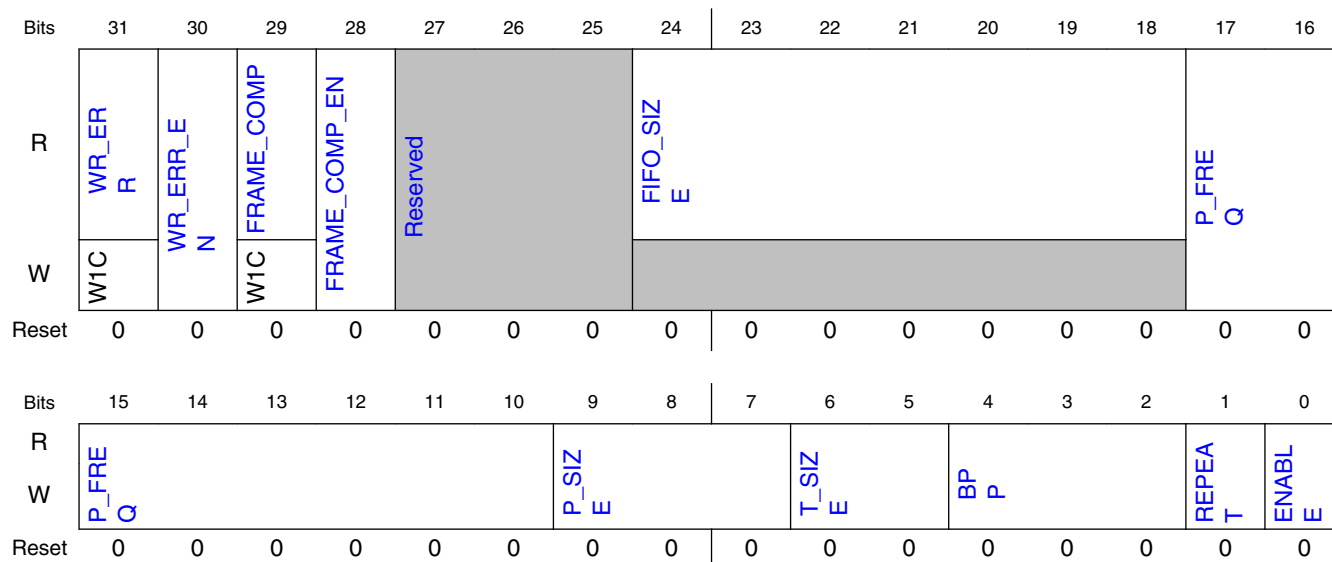
Field	Function
P_SIZE	Holds the payload size. This is used with the T_SIZE to calculate the number of transactions required per request. P_SIZE and T_SIZE are co-related and should be programmed as P_SIZE >= T_SIZE. • 000 -> 64B • 001 -> 128B • 010 -> 256B • 011 -> 512B • 100 -> 1KB • 101 -> 2KB • 110 -> 4KB • 111 -> Reserved
6-5 T_SIZE	Transaction Size Holds the transaction size / request. Transaction size determines how many AXI transactions a payload should be divided into. T_SIZE is related to P_SIZE and needs to be programmed as P_SIZE >= T_SIZE • 00 -> 64B • 01 -> 128B • 10 -> 256B • 11 -> 512B
4-2 BPP	Bits per pixel Holds the number of bits / pixel format interface on the output. • 0 -> 38 bpp, 38 bit components are unpacked • 1 -> 32 bpp, 8-bit to 10-bit upconversion • 2 -> 32 bpp, 10-bit output, alpha(37:30) tied to 0x0 • 3 -> 20 bpp, 10-bit YUV422. pixel_data[37:30]/pixel_data[9:0] tied to 0x0 • 4 -> 16 bpp, 8-bit component, YUV422
1 REPEAT	Repeat feature Repeat bit for the write_scalar. When set, write scalar does not lower the BUSY bit after each frame and processes the new frame of data with the same address, bpp, p_size and t_size values as the previous frame. If you do not expect a new frame, reset this signal before the current frame finishes. Once the frame finishes, and the design sees this bit set, it resets address to the programmed value and wait for new pixels. If you do not expect any more pixels, the only way to recover from this is through a reset.
0 ENABLE	Enable / Busy Enable bit for the write_scalar. When set, the wr_scalar waits for pixels and write them to memory as per the address and transfer sizes programmed in the control register. This bit is reset by HW when it has completed the frame write. If the REPEAT bit is set, this bit is not cleared as the WR_SCL resets it's pointers and waits for new pixels to be written to the same address in memory.

15.12.3.1.3 Control register for Context Loader. (CTRL_STATUS_SET)

15.12.3.1.3.1 Offset

Register	Offset
CTRL_STATUS_SET	4h

15.12.3.1.3.2 Diagram



15.12.3.1.3.3 Fields

Field	Function
31 WR_ERR	Indicates a write error on the axi interface.
30 WR_ERR_EN	Write error IRQ enable If set, it allows WR_ERR to create an irq whenever there is an error response to a write request.
29 FRAME_COMP	Indicates the current frame being processed has finished.
28 FRAME_COMP_EN	Write error IRQ enable If set, it allows FRAME_COMP to create an irq whenever processing a frame finishes.
27-25 —	Reserved
24-18 FIFO_SIZE	Size of FIFO in design This is the fifo size in the HW. This can be used to program the appropriate P_SIZE. P_SIZE should ideally be 1/2 the FIFO size. 7'h0F - 512B 7'h1F - 1KB 7'h3F - 2KB 7'h7F - 4KB
17-10 P_FREQ	Payload frequency P_FREQ*8 is the number of clocks between payload requests. A value of 0 does not limit the frequency of the payload requests. Instead, the payload is written as frequently as data is available.
9-7 P_SIZE	Payload Size

Table continues on the next page...

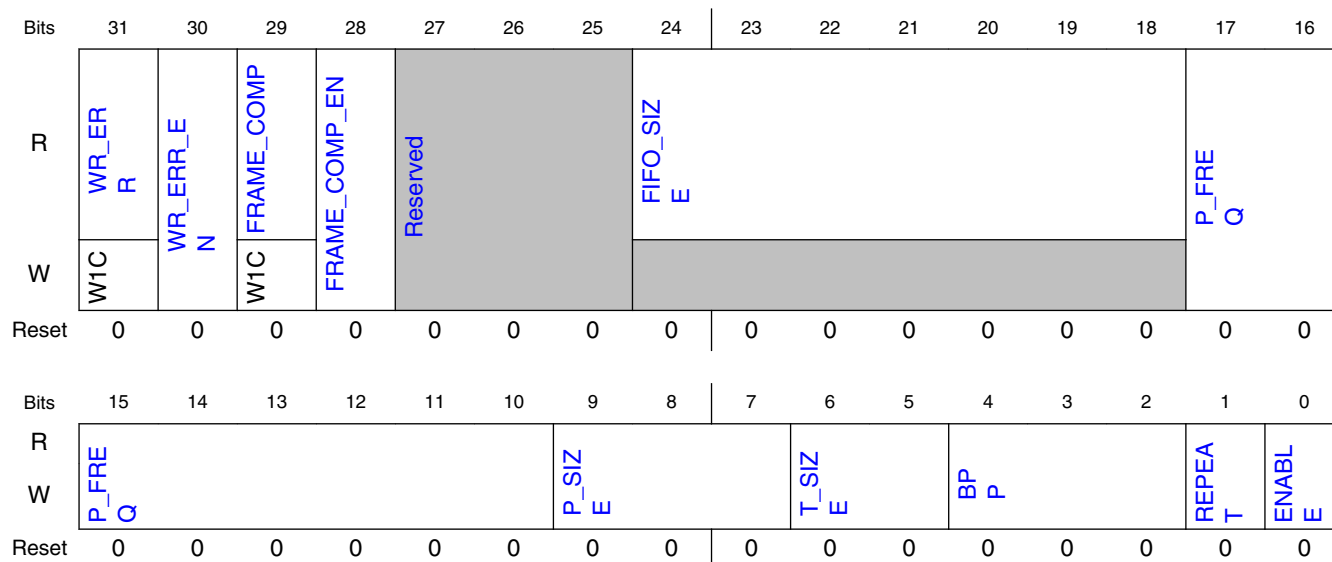
Field	Function
	Holds the payload size. This is used with the T_SIZE to calculate the number of transactions required per request. P_SIZE and T_SIZE are co-related and should be programmed as P_SIZE >= T_SIZE. • 000 -> 64B • 001 -> 128B • 010 -> 256B • 011 -> 512B • 100 -> 1KB • 101 -> 2KB • 110 -> 4KB • 111 -> Reserved
6-5 T_SIZE	Transaction Size Holds the transaction size / request. Transaction size determines how many AXI transactions a payload should be divided into. T_SIZE is related to P_SIZE and needs to be programmed as P_SIZE >= T_SIZE • 00 -> 64B • 01 -> 128B • 10 -> 256B • 11 -> 512B
4-2 BPP	Bits per pixel Holds the number of bits / pixel format interface on the output. • 0 -> 38 bpp, 38 bit components are unpacked • 1 -> 32 bpp, 8-bit to 10-bit upconversion • 2 -> 32 bpp, 10-bit output, alpha(37:30) tied to 0x0 • 3 -> 20 bpp, 10-bit YUV422. pixel_data[37:30]/pixel_data[9:0] tied to 0x0 • 4 -> 16 bpp, 8-bit component, YUV422
1 REPEAT	Repeat feature Repeat bit for the write_scalar. When set, write scalar does not lower the BUSY bit after each frame and processes the new frame of data with the same address, bpp, p_size and t_size values as the previous frame. If you do not expect a new frame, reset this signal before the current frame finishes. Once the frame finishes, and the design sees this bit set, it resets address to the programmed value and wait for new pixels. If you do not expect any more pixels, the only way to recover from this is through a reset.
0 ENABLE	Enable / Busy Enable bit for the write_scalar. When set, the wr_scalar waits for pixels and write them to memory as per the address and transfer sizes programmed in the control register. This bit is reset by HW when it has completed the frame write. If the REPEAT bit is set, this bit is not cleared as the WR_SCL resets it's pointers and waits for new pixels to be written to the same address in memory.

15.12.3.1.4 Control register for Context Loader. (CTRL_STATUS_CLR)

15.12.3.1.4.1 Offset

Register	Offset
CTRL_STATUS_CLR	8h

15.12.3.1.4.2 Diagram



15.12.3.1.4.3 Fields

Field	Function
31 WR_ERR	Indicates a write error on the axi interface.
30 WR_ERR_EN	Write error IRQ enable If set, it allows WR_ERR to create an irq whenever there is an error response to a write request.
29 FRAME_COMP	Indicates the current frame being processed has finished.
28 FRAME_COMP_EN	Write error IRQ enable If set, it allows FRAME_COMP to create an irq whenever processing a frame finishes.
27-25 —	Reserved
24-18 FIFO_SIZE	Size of FIFO in design This is the fifo size in the HW. This can be used to program the appropriate P_SIZE. P_SIZE should ideally be 1/2 the FIFO size. 7'h0F - 512B 7'h1F - 1KB 7'h3F - 2KB 7'h7F - 4KB
17-10 P_FREQ	Payload frequency P_FREQ*8 is the number of clocks between payload requests. A value of 0 does not limit the frequency of the payload requests. Instead, the payload is written as frequently as data is available.
9-7 P_SIZE	Payload Size

Table continues on the next page...

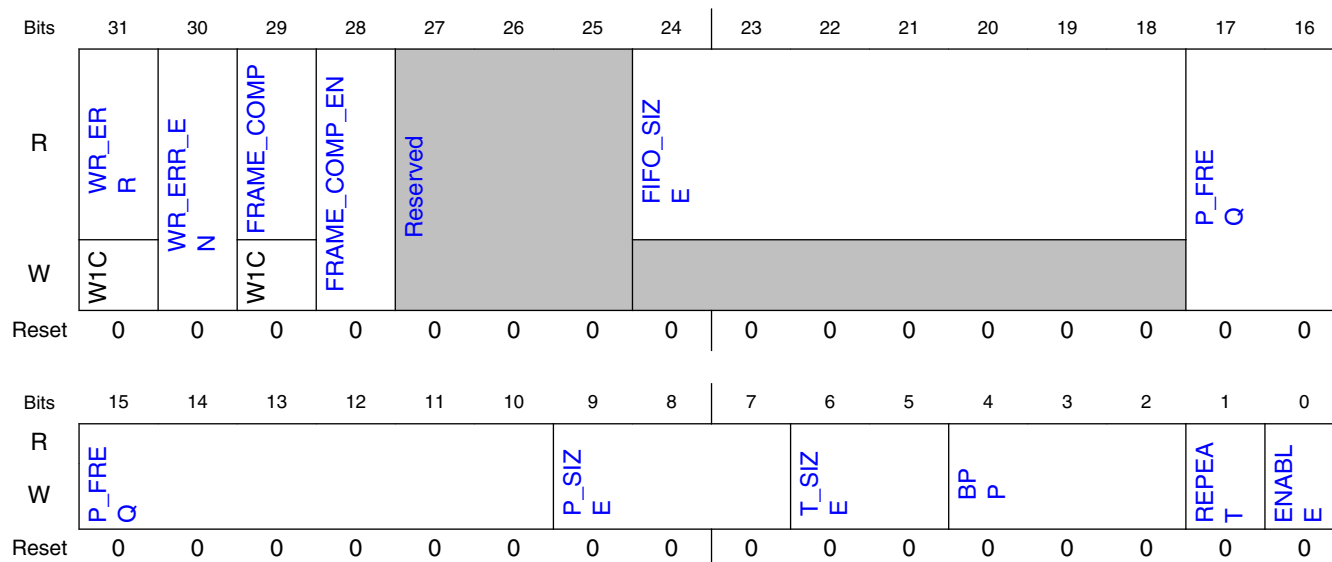
Field	Function
	Holds the payload size. This is used with the T_SIZE to calculate the number of transactions required per request. PSIZE and TSIZE are co-related and should be programmed as PSIZE >= TSIZE. • 000 -> 64B • 001 -> 128B • 010 -> 256B • 011 -> 512B • 100 -> 1KB • 101 -> 2KB • 110 -> 4KB • 111 -> Reserved
6-5 T_SIZE	Transaction Size Holds the transaction size / request. Transaction size determines how many AXI transactions a payload should be divided into. T_SIZE is related to P_SIZE and needs to be programmed as PSIZE >= TSIZE • 00 -> 64B • 01 -> 128B • 10 -> 256B • 11 -> 512B
4-2 BPP	Bits per pixel Holds the number of bits / pixel format interface on the output. • 0 -> 38 bpp, 38 bit components are unpacked • 1 -> 32 bpp, 8-bit to 10-bit upconversion • 2 -> 32 bpp, 10-bit output, alpha(37:30) tied to 0x0 • 3 -> 20 bpp, 10-bit YUV422. pixel_data[37:30]/pixel_data[9:0] tied to 0x0 • 4 -> 16 bpp, 8-bit component, YUV422
1 REPEAT	Repeat feature Repeat bit for the write_scalar. When set, write scalar does not lower the BUSY bit after each frame and processes the new frame of data with the same address, bpp, p_size and t_size values as the previous frame. If you do not expect a new frame, reset this signal before the current frame finishes. Once the frame finishes, and the design sees this bit set, it resets address to the programmed value and wait for new pixels. If you do not expect any more pixels, the only way to recover from this is through a reset.
0 ENABLE	Enable / Busy Enable bit for the write_scalar. When set, the wr_scalar waits for pixels and write them to memory as per the address and transfer sizes programmed in the control register. This bit is reset by HW when it has completed the frame write. If the REPEAT bit is set, this bit is not cleared as the WR_SCL resets it's pointers and waits for new pixels to be written to the same address in memory.

15.12.3.1.5 Control register for Context Loader. (CTRL_STATUS_TOG)

15.12.3.1.5.1 Offset

Register	Offset
CTRL_STATUS_TOG	Ch

15.12.3.1.5.2 Diagram



15.12.3.1.5.3 Fields

Field	Function
31 WR_ERR	Indicates a write error on the axi interface.
30 WR_ERR_EN	Write error IRQ enable If set, it allows WR_ERR to create an irq whenever there is an error response to a write request.
29 FRAME_COMP	Indicates the current frame being processed has finished.
28 FRAME_COMP_EN	Write error IRQ enable If set, it allows FRAME_COMP to create an irq whenever processing a frame finishes.
27-25 —	Reserved
24-18 FIFO_SIZE	Size of FIFO in design This is the fifo size in the HW. This can be used to program the appropriate P_SIZE. P_SIZE should ideally be 1/2 the FIFO size. 7'h0F - 512B 7'h1F - 1KB 7'h3F - 2KB 7'h7F - 4KB
17-10 P_FREQ	Payload frequency P_FREQ*8 is the number of clocks between payload requests. A value of 0 does not limit the frequency of the payload requests. Instead, the payload is written as frequently as data is available.
9-7 P_SIZE	Payload Size

Table continues on the next page...

Field	Function
	Holds the payload size. This is used with the T_SIZE to calculate the number of transactions required per request. PSIZE and TSIZE are co-related and should be programmed as PSIZE >= TSIZE. • 000 -> 64B • 001 -> 128B • 010 -> 256B • 011 -> 512B • 100 -> 1KB • 101 -> 2KB • 110 -> 4KB • 111 -> Reserved
6-5 T_SIZE	Transaction Size Holds the transaction size / request. Transaction size determines how many AXI transactions a payload should be divided into. T_SIZE is related to P_SIZE and needs to be programmed as PSIZE >= TSIZE • 00 -> 64B • 01 -> 128B • 10 -> 256B • 11 -> 512B
4-2 BPP	Bits per pixel Holds the number of bits / pixel format interface on the output. • 0 -> 38 bpp, 38 bit components are unpacked • 1 -> 32 bpp, 8-bit to 10-bit upconversion • 2 -> 32 bpp, 10-bit output, alpha(37:30) tied to 0x0 • 3 -> 20 bpp, 10-bit YUV422. pixel_data[37:30]/pixel_data[9:0] tied to 0x0 • 4 -> 16 bpp, 8-bit component, YUV422
1 REPEAT	Repeat feature Repeat bit for the write_scalar. When set, write scalar does not lower the BUSY bit after each frame and processes the new frame of data with the same address, bpp, p_size and t_size values as the previous frame. If you do not expect a new frame, reset this signal before the current frame finishes. Once the frame finishes, and the design sees this bit set, it resets address to the programmed value and wait for new pixels. If you do not expect any more pixels, the only way to recover from this is through a reset.
0 ENABLE	Enable / Busy Enable bit for the write_scalar. When set, the wr_scalar waits for pixels and write them to memory as per the address and transfer sizes programmed in the control register. This bit is reset by HW when it has completed the frame write. If the REPEAT bit is set, this bit is not cleared as the WR_SCL resets it's pointers and waits for new pixels to be written to the same address in memory.

15.12.3.1.6 Holds the base address (BASE_ADDR)

15.12.3.1.6.1 Offset

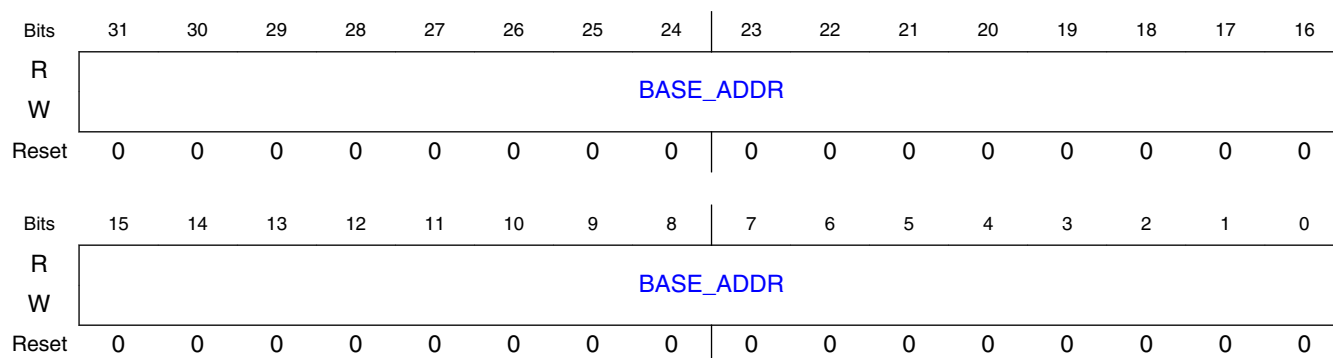
Register	Offset
BASE_ADDR	10h

15.12.3.1.6.2 Function

First comment.

secone third

15.12.3.1.6.3 Diagram



15.12.3.1.6.4 Fields

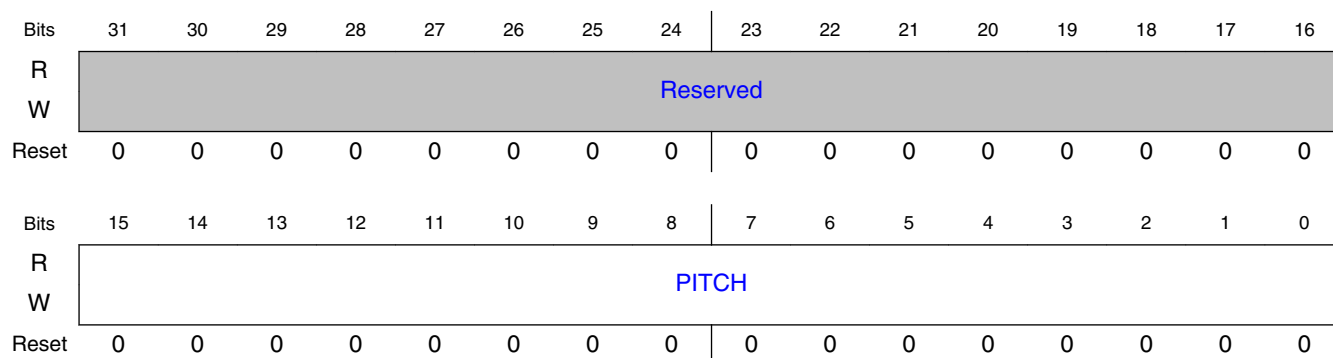
Field	Function
31-0	Base address
BASE_ADDR	System address of source buffer. Although byte granularity is supported, SW should align it to 64B boundry

15.12.3.1.7 Pitch (PITCH)

15.12.3.1.7.1 Offset

Register	Offset
PITCH	14h

15.12.3.1.7.2 Diagram



15.12.3.1.7.3 Fields

Field	Function
31-16 —	Reserved
15-0 PITCH	Vertical pitch for memory address calculation. Number of bytes between 2 vertically adjacent pixels in system memory. Byte granularity is supported, but SW should align to 64B boundary.

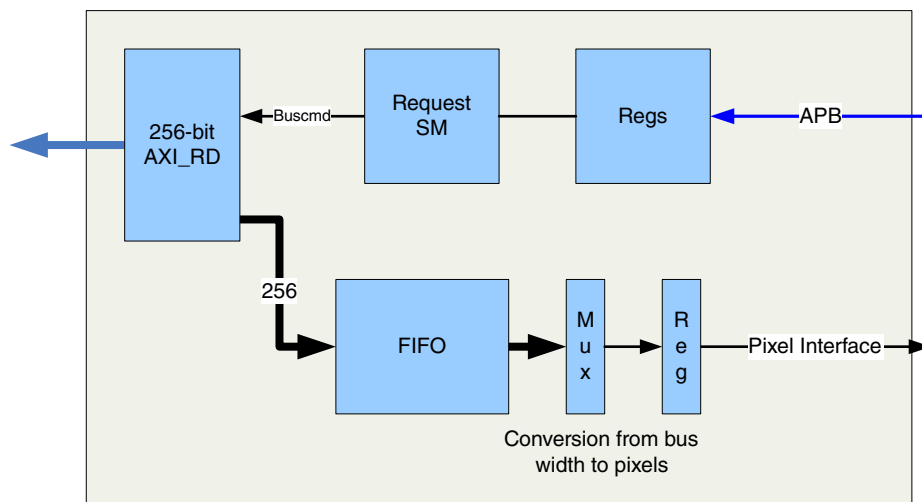
15.13 Read Surface (RD_SRC)

15.13.1 Overview

The Read Surface (RD_SRC) engine is a bus master that reads pixels from system memory and drive pixels out to the scale output buffer. The scale output buffer (1 of 3 not used for storing the scaled result) is reused as a read latency elasticity buffer. Minimal storage is needed in the RD_SRC module to store returned read data since the scale output buffer is repurposed as a pixel elasticity buffer. The RD_SRC engine supports raster scan source buffers. There is no support for tiled based buffers.

15.13.1.1 Block Diagram

The diagram indicating the functional components in the RD_SRC module is below.



The components are listed here:

- Req_SM: State machine to generate the sequence of transactions to read a buffer from system memory.
- Regs: All configuration registers used to store the RD_SRC configuration.
- AXI_RD: The buscmd to AXI interface convertor used in many AXI based designs. This instance is shared with the CTX_LD module.
- FIFO: used to store a sample set of data to be read from system memory.
- Down Sample Mux/Reg: Multiplexor used to down size from the bus width to a pixel interface to the scale output buffer.

The software parameters are:

- AXI transaction size (64B up to 512B per request) is programmable.
- Buffer threshold determines how empty the input data FIFO should be to trigger a sequence of read transactions from the system. When the number of available bytes is greater than this value, the necessary AXI requests to fill this buffer are issued.
- Buffer base address.
- Line stride in bytes.
- Source buffer format, either ARGB8888 (32bpp), or YUV422 1-plane (16bpp).
- Width, or number of pixels to fetch and write to the scale output buffer in a horizontal scan line.
- Height, number of lines in the source buffer.

15.13.2 Functional Description

The RD_SRC engine fetches and forwards only the pixels needed to drive the display. The granularity of the source buffer, in X and Y, forwarded to the display pipeline has no limits. Any width and height can be fetched and forwarded to the scale output buffer. Also, in YUV422 mode, the programmed width should be divisible by 2.

The RD_SRC has two events that can enable it to issue requests to fetch data for the next display trace.

1. SW can write the BUSY bit to 1'b1.
2. The rd_src_kick_enable signal input can go low to high.

The RD_SRC is disabled when the `csr_rd_src_width AND csr_rd_src_height = 0x0`. When these two registers are 0x0, both events that enable the RD_SRC are ignored. Also, if the RD_SRC is already enabled and processing a frame, any of the two events that are used to start processing would be ignored.

15.13.2.1 Input to Output Pixel Formatting

The pixel_data[37:0] output bus to the Scaler engine is normalized as 10-bit per color component, with an 8-bit alpha. Refer to section 4.21.1 Pixel Component Order for the standard on how to convert 8-bit to 10-bit color components when required depending on the format selected.

Input pixels from the system can be in 8/10-bit format, may contain an alpha value, or may have sub-sampled chroma components. The BPP field of the CSR_RD_SRC_CTRL register defines what the input format is. The definition of this field matches the respective control field in the WR_SCL module. Essentially, the RD_SRC reverses the pixel formatting process and send a full 38-bits back to the Scaler engine. The definition of this data format is defined as

```
Pixel_data[37:30] = Alpha
Pixel_data[29:20] = R/Y
Pixel_data[19:10] = G/U, or UV alternating in YUV422 modes
Pixel_data[9:0] = B/V
```

The output pixel modes are defined as:

- **BPP = 0x0: 40bpp**, Full pass through of all 38-bits with the appended 2'b00 bits dropped.
 - 1st pixel_data = [37:0],
 - 2nd pixel_data = [77:40], etc.
- **BPP = 0x1: 32bpp**, ARGB8888/AYUV8888
- 32bpp expanded to the full 38-bit vector. Refer to the standards section 4.21.2 Pixel Component Width Conversion to determine how to expand 8-bit components up to 10-bit.
- 1st pixel_data = {[31:24],[23:16],[23:22],[15:8],[15:14],[7:0],[7:6]} //ARGB: 8:10:10:10
- **BPP = 0x2: 32bpp**, RGB/YUV 10-bit per component
- All 10-bits of each component are passed to the output. The alpha is expanded from 2-bits to 8-bit.
- 1st pixel_data = {[4{[31:30]}],[29:20],[19:10],[9:0]} //Alpha expanded, component pass-through
- **BPP = 0x3: 24bpp**, YUV422 10-bit per component
- The full Y and alternating UV components are passed through as 10-bit. Bits 9:0 are driven as 10'b0. The 4-bit alpha is converted based on the standard conversion described in section 4.21.2 Pixel Component Width Conversion.
- 1st pixel_data = {[2{[23:20]}],[19:10],[9:0],10'b0} //Alpha expanded 4 to 8, Y and UV pass-through
- **BPP = 0x4: 16bpp**, YUV422 8-bit per component
- The Y and alternating UV components are expanded from 8 to 10 bits. Bits [9:0] and [37:30] are driven as 0x0.
- 1st pixel_data = {8'b0,[15:7],[15:14],[7:0],[7:6],10'b0} //Alpha = 0,

15.13.2.2 Interrupt

The RD_SRC module can generate interrupts under the following conditions.

- AXI SLVERR or DECERR is received.
- Frame processing complete. The last pixel coincident with pixel_v_sync is written to the scaler output FIFO.

15.13.3 Memory Map and Registers

15.13.3.1 register descriptions

15.13.3.1.1 RD_SRC Memory map

rd_src base address: 2_2000h

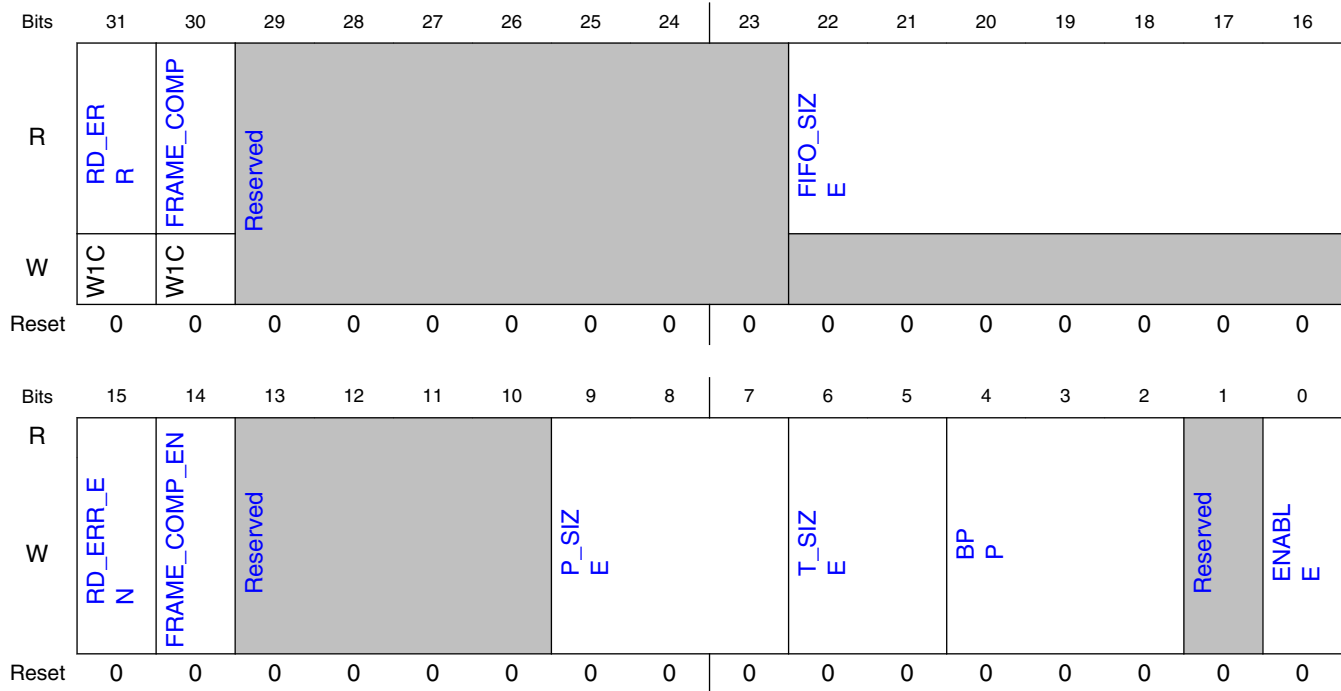
Offset	Register	Width (In bits)	Access	Reset value
0h	Control register for Read surface. (CTRL_STATUS)	32	RW	0000_0000h
4h	Control register for Read surface. (CTRL_STATUS_SET)	32	RW	0000_0000h
8h	Control register for Read surface. (CTRL_STATUS_CLR)	32	RW	0000_0000h
Ch	Control register for Read surface. (CTRL_STATUS_TOG)	32	RW	0000_0000h
10h	Read Surface Base address (BASE_ADDR)	32	RW	0000_0000h
14h	Read surface vertical pitch (PITCH)	32	RW	0000_0000h
18h	Source frame buffer width (WIDTH)	32	RW	0000_0000h
1Ch	Height of frame to be read (HEIGHT)	32	RW	0000_0000h

15.13.3.1.2 Control register for Read surface. (CTRL_STATUS)

15.13.3.1.2.1 Offset

Register	Offset
CTRL_STATUS	0h

15.13.3.1.2.2 Diagram



15.13.3.1.2.3 Fields

Field	Function
31 RD_ERR	AXI Read Error Status bit that indicates an AXI read response terminates with a SLV_ERR or DEC_ERR.
30 FRAME_COMP	Frame processing complete Status bit that indicates the current frame processing is complete. This is set even if the height and width are 0. If the respective interrupt enable bit is set, an interrupt is generated.
29-23 —	Reserved
22-16 FIFO_SIZE	FIFO size This is the fifo size implemented in the HW. The P_SIZE should never be set to a value higher than this value in bytes. 7'h0F - 512B 7'h1F - 1KB 7'h3F - 2KB 7'h7F - 4KB
15 RD_ERR_EN	AXI Read Error IRQ enable Enable an interrupt when the AXI read response completes with a SLV_ERR or DEC_ERR.
14 FRAME_COMP_EN	Frame complete IRQ enable Enable an interrupt when the read surface engine is finished processing a frame.
13-10	Reserved

Table continues on the next page...

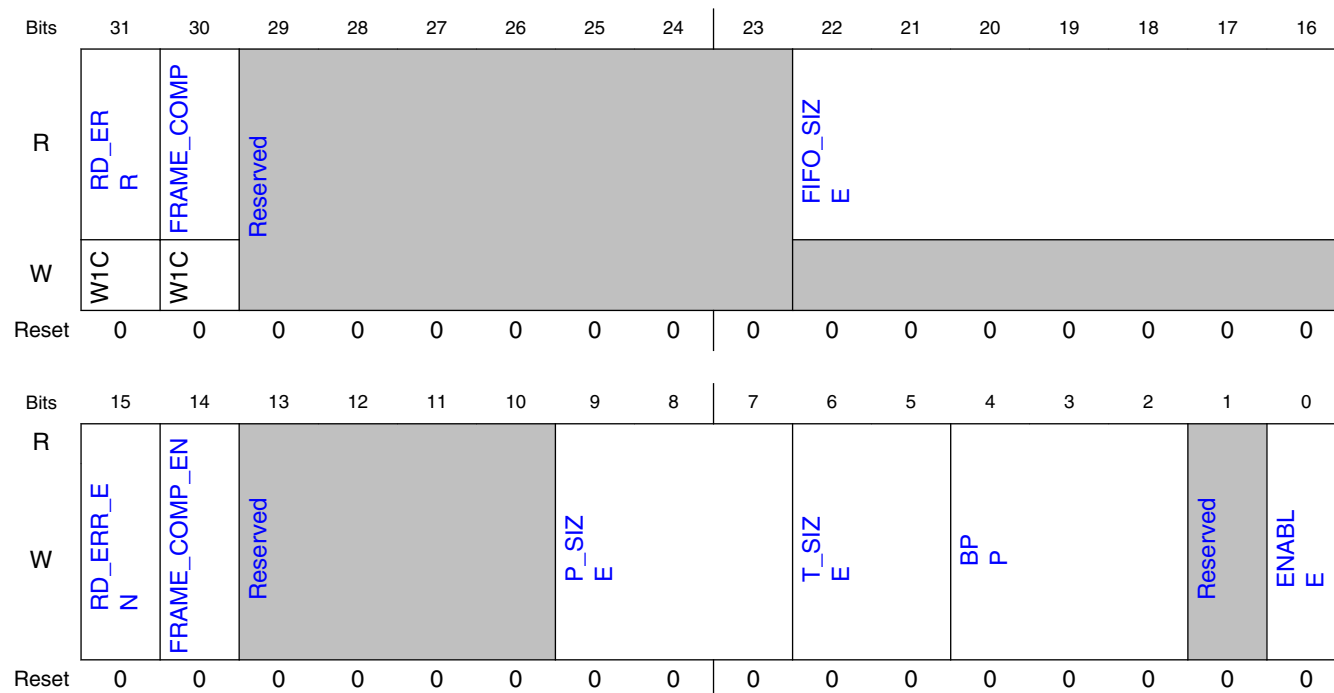
Field	Function
—	
9-7 P_SIZE	Payload size (P_SIZE) Holds the payload size per AXI burst sequence. The P_SIZE should never be programmed to be less than T_SIZE. If the T_SIZE and P_SIZE are both set to 64B, there is a high frequency of 64B requests issued to DRAM. If the P_SIZE is 256B, the requests are issued when there is 256B of space empty in the RD_SRC FIFO. The requests are issued 1/4 as frequently as when the P_SIZE is 64B. With a P_SIZE=256 and T_SIZE=128, two 128B transactions are issued to fulfill the 256B P_SIZE selection. The P_SIZE and T_SIZE should be set to optimize for DRAM efficiency. P_SIZE = T_SIZE = 256B is a good choice to start, but other settings could provide more optimal performance. • 000 -> 64B • 001 -> 128B • 010 -> 256B • 011 -> 512B • others -> Reserved FIXME this should be based on the BUFF_SZ parameter setting.
6-5 T_SIZE	Transaction Size (T_SIZE) Holds the AXI system bus transaction size per request in bytes. This should always be programmed in conjunction with P_SIZE where P_SIZE >= T_SIZE. • 00 -> 64B • 01 -> 128B • 10 -> 256B • 11 -> 512B
4-2 BPP	Bits per pixel Holds the number of bits/pixel format interface on the input. These settings are intended to reverse the operation of the respective settings in the WR_SCL engine. • 0 -> 40 bpp, 38 bit components are unpacked • 1 -> 32 bpp, 8-bit to 10-bit upconversion • 2 -> 32 bpp, 10-bit output, alpha(37:30) tied to 0x0 • 3 -> 24 bpp, 10-bit YUV422. pixel_data[37:30]/pixel_data[9:0] tied to 0x0 • 4 -> 16 bpp, 8-bit component, YUV422
1 —	Reserved
0 ENABLE	Read surface enable. Enable bit for read surface bus master. When this bit is set, AND the width and height are both not zero, the RD_SRC engine begins to fetch buffer based on existing programming.

15.13.3.1.3 Control register for Read surface. (CTRL_STATUS_SET)

15.13.3.1.3.1 Offset

Register	Offset
CTRL_STATUS_SET	4h

15.13.3.1.3.2 Diagram



15.13.3.1.3.3 Fields

Field	Function
31 RD_ERR	AXI Read Error Status bit that indicates an AXI read response terminates with a SLV_ERR or DEC_ERR.
30 FRAME_COMP	Frame processing complete Status bit that indicates the current frame processing is complete. This is set even if the height and width are 0. If the respective interrupt enable bit is set, an interrupt is generated.
29-23 —	Reserved
22-16 FIFO_SIZE	FIFO size This is the fifo size implemented in the HW. The P_SIZE should never be set to a value higher than this value in bytes. 7'h0F - 512B 7'h1F - 1KB 7'h3F - 2KB 7'h7F - 4KB
15 RD_ERR_EN	AXI Read Error IRQ enable Enable an interrupt when the AXI read response completes with a SLV_ERR or DEC_ERR.
14 FRAME_COMP_EN	Frame complete IRQ enable Enable an interrupt when the read surface engine is finished processing a frame.
13-10	Reserved

Table continues on the next page...

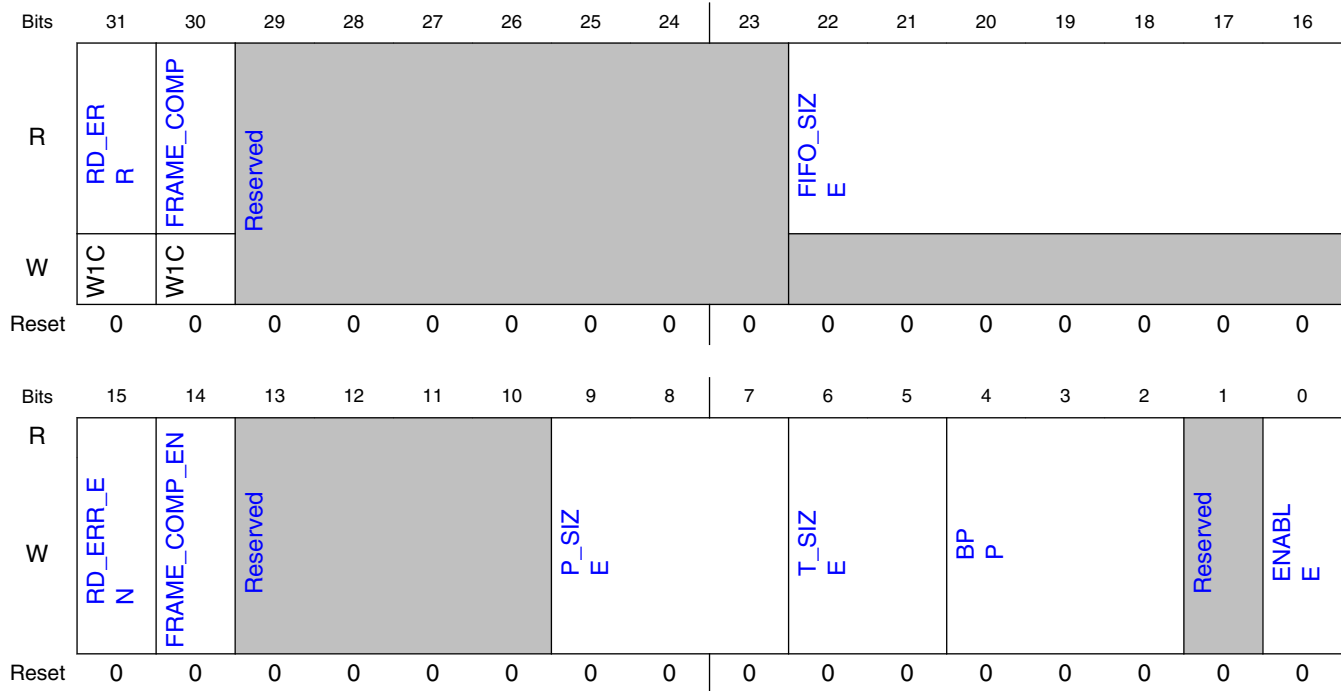
Field	Function
—	
9-7 P_SIZE	Payload size (P_SIZE) Holds the payload size per AXI burst sequence. The P_SIZE should never be programmed to be less than T_SIZE. If the T_SIZE and P_SIZE are both set to 64B, there is a high frequency of 64B requests issued to DRAM. If the P_SIZE is 256B, the requests are issued when there is 256B of space empty in the RD_SRC FIFO. The requests are issued 1/4 as frequently as when the P_SIZE is 64B. With a P_SIZE=256 and T_SIZE=128, two 128B transactions are issued to fulfill the 256B P_SIZE selection. The P_SIZE and T_SIZE should be set to optimize for DRAM efficiency. P_SIZE = T_SIZE = 256B is a good choice to start, but other settings could provide more optimal performance. • 000 -> 64B • 001 -> 128B • 010 -> 256B • 011 -> 512B • others -> Reserved FIXME this should be based on the BUFF_SZ parameter setting.
6-5 T_SIZE	Transaction Size (T_SIZE) Holds the AXI system bus transaction size per request in bytes. This should always be programmed in conjunction with P_SIZE where P_SIZE >= T_SIZE. • 00 -> 64B • 01 -> 128B • 10 -> 256B • 11 -> 512B
4-2 BPP	Bits per pixel Holds the number of bits/pixel format interface on the input. These settings are intended to reverse the operation of the respective settings in the WR_SCL engine. • 0 -> 40 bpp, 38 bit components are unpacked • 1 -> 32 bpp, 8-bit to 10-bit upconversion • 2 -> 32 bpp, 10-bit output, alpha(37:30) tied to 0x0 • 3 -> 24 bpp, 10-bit YUV422. pixel_data[37:30]/pixel_data[9:0] tied to 0x0 • 4 -> 16 bpp, 8-bit component, YUV422
1 —	Reserved
0 ENABLE	Read surface enable. Enable bit for read surface bus master. When this bit is set, AND the width and height are both not zero, the RD_SRC engine begins to fetch buffer based on existing programming.

15.13.3.1.4 Control register for Read surface. (CTRL_STATUS_CLR)

15.13.3.1.4.1 Offset

Register	Offset
CTRL_STATUS_CLR	8h

15.13.3.1.4.2 Diagram



15.13.3.1.4.3 Fields

Field	Function
31 RD_ERR	AXI Read Error Status bit that indicates an AXI read response terminates with a SLV_ERR or DEC_ERR.
30 FRAME_COMP	Frame processing complete Status bit that indicates the current frame processing is complete. This is set even if the height and width are 0. If the respective interrupt enable bit is set, an interrupt is generated.
29-23 —	Reserved
22-16 FIFO_SIZE	FIFO size This is the fifo size implemented in the HW. The P_SIZE should never be set to a value higher than this value in bytes. • 7'h0F - 512B • 7'h1F - 1KB • 7'h3F - 2KB • 7'h7F - 4KB
15 RD_ERR_EN	AXI Read Error IRQ enable Enable an interrupt when the AXI read response completes with a SLV_ERR or DEC_ERR.
14 FRAME_COMP_EN	Frame complete IRQ enable Enable an interrupt when the read surface engine is finished processing a frame.
13-10	Reserved

Table continues on the next page...

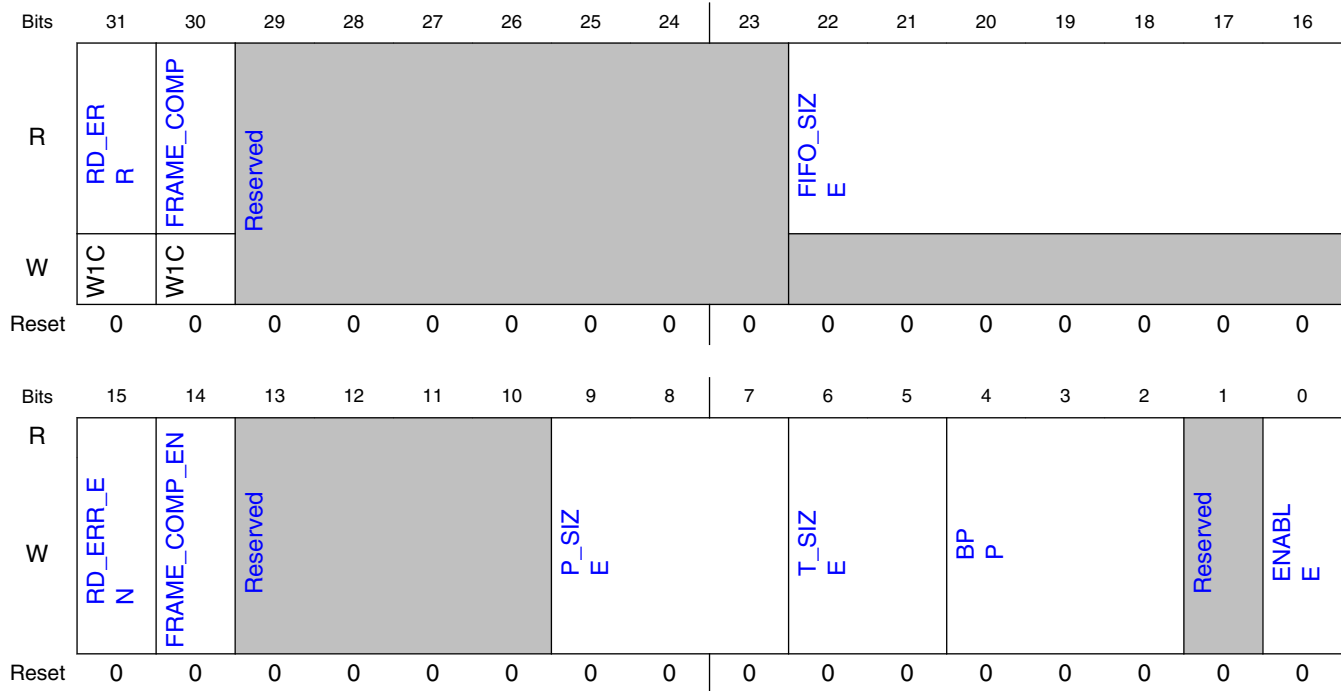
Field	Function
—	
9-7 P_SIZE	Payload size (P_SIZE) Holds the payload size per AXI burst sequence. The P_SIZE should never be programmed to be less than T_SIZE. If the T_SIZE and P_SIZE are both set to 64B, there is a high frequency of 64B requests issued to DRAM. If the P_SIZE is 256B, the requests are issued when there is 256B of space empty in the RD_SRC FIFO. The requests are issued 1/4 as frequently as when the P_SIZE is 64B. With a P_SIZE=256 and T_SIZE=128, two 128B transactions are issued to fulfill the 256B P_SIZE selection. The P_SIZE and T_SIZE should be set to optimize for DRAM efficiency. P_SIZE = T_SIZE = 256B is a good choice to start, but other settings could provide more optimal performance. • 000 -> 64B • 001 -> 128B • 010 -> 256B • 011 -> 512B • others -> Reserved FIXME this should be based on the BUFF_SZ parameter setting.
6-5 T_SIZE	Transaction Size (T_SIZE) Holds the AXI system bus transaction size per request in bytes. This should always be programmed in conjunction with P_SIZE where P_SIZE >= T_SIZE. • 00 -> 64B • 01 -> 128B • 10 -> 256B • 11 -> 512B
4-2 BPP	Bits per pixel Holds the number of bits/pixel format interface on the input. These settings are intended to reverse the operation of the respective settings in the WR_SCL engine. • 0 -> 40 bpp, 38 bit components are unpacked • 1 -> 32 bpp, 8-bit to 10-bit upconversion • 2 -> 32 bpp, 10-bit output, alpha(37:30) tied to 0x0 • 3 -> 24 bpp, 10-bit YUV422. pixel_data[37:30]/pixel_data[9:0] tied to 0x0 • 4 -> 16 bpp, 8-bit component, YUV422
1 —	Reserved
0 ENABLE	Read surface enable. Enable bit for read surface bus master. When this bit is set, AND the width and height are both not zero, the RD_SRC engine begins to fetch buffer based on existing programming.

15.13.3.1.5 Control register for Read surface. (CTRL_STATUS_TOG)

15.13.3.1.5.1 Offset

Register	Offset
CTRL_STATUS_TOG	Ch

15.13.3.1.5.2 Diagram



15.13.3.1.5.3 Fields

Field	Function
31 RD_ERR	AXI Read Error Status bit that indicates an AXI read response terminates with a SLV_ERR or DEC_ERR.
30 FRAME_COMP	Frame processing complete Status bit that indicates the current frame processing is complete. This is set even if the height and width are 0. If the respective interrupt enable bit is set, an interrupt is generated.
29-23 —	Reserved
22-16 FIFO_SIZE	FIFO size This is the fifo size implemented in the HW. The P_SIZE should never be set to a value higher than this value in bytes. 7'h0F - 512B 7'h1F - 1KB 7'h3F - 2KB 7'h7F - 4KB
15 RD_ERR_EN	AXI Read Error IRQ enable Enable an interrupt when the AXI read response completes with a SLV_ERR or DEC_ERR.
14 FRAME_COMP_EN	Frame complete IRQ enable Enable an interrupt when the read surface engine is finished processing a frame.
13-10	Reserved

Table continues on the next page...

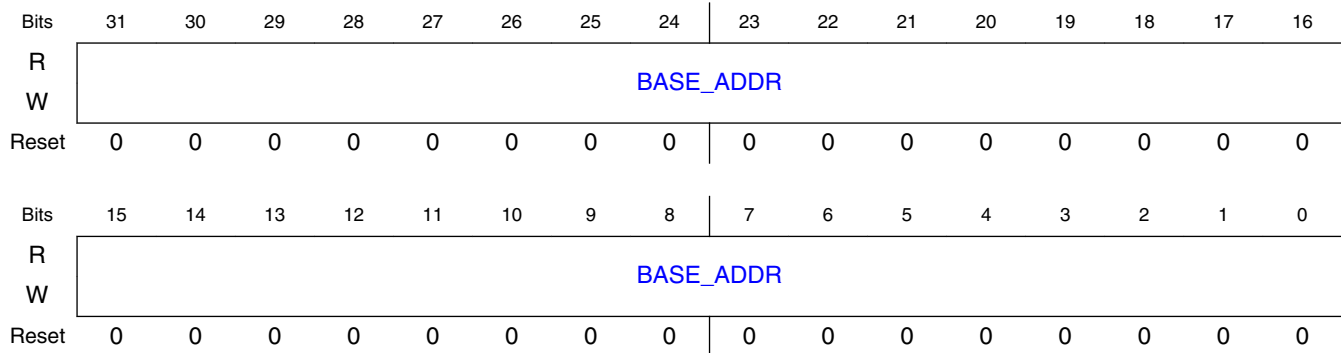
Field	Function
—	
9-7 P_SIZE	Payload size (P_SIZE) Holds the payload size per AXI burst sequence. The P_SIZE should never be programmed to be less than T_SIZE. If the T_SIZE and P_SIZE are both set to 64B, there is a high frequency of 64B requests issued to DRAM. If the P_SIZE is 256B, the requests are issued when there is 256B of space empty in the RD_SRC FIFO. The requests are issued 1/4 as frequently as when the P_SIZE is 64B. With a P_SIZE=256 and T_SIZE=128, two 128B transactions are issued to fulfill the 256B P_SIZE selection. The P_SIZE and T_SIZE should be set to optimize for DRAM efficiency. P_SIZE = T_SIZE = 256B is a good choice to start, but other settings could provide more optimal performance. • 000 -> 64B • 001 -> 128B • 010 -> 256B • 011 -> 512B • others -> Reserved FIXME this should be based on the BUFF_SZ parameter setting.
6-5 T_SIZE	Transaction Size (T_SIZE) Holds the AXI system bus transaction size per request in bytes. This should always be programmed in conjunction with P_SIZE where P_SIZE >= T_SIZE. • 00 -> 64B • 01 -> 128B • 10 -> 256B • 11 -> 512B
4-2 BPP	Bits per pixel Holds the number of bits/pixel format interface on the input. These settings are intended to reverse the operation of the respective settings in the WR_SCL engine. • 0 -> 40 bpp, 38 bit components are unpacked • 1 -> 32 bpp, 8-bit to 10-bit upconversion • 2 -> 32 bpp, 10-bit output, alpha(37:30) tied to 0x0 • 3 -> 24 bpp, 10-bit YUV422. pixel_data[37:30]/pixel_data[9:0] tied to 0x0 • 4 -> 16 bpp, 8-bit component, YUV422
1 —	Reserved
0 ENABLE	Read surface enable. Enable bit for read surface bus master. When this bit is set, AND the width and height are both not zero, the RD_SRC engine begins to fetch buffer based on existing programming.

15.13.3.1.6 Read Surface Base address (BASE_ADDR)

15.13.3.1.6.1 Offset

Register	Offset
BASE_ADDR	10h

15.13.3.1.6.2 Diagram



15.13.3.1.6.3 Fields

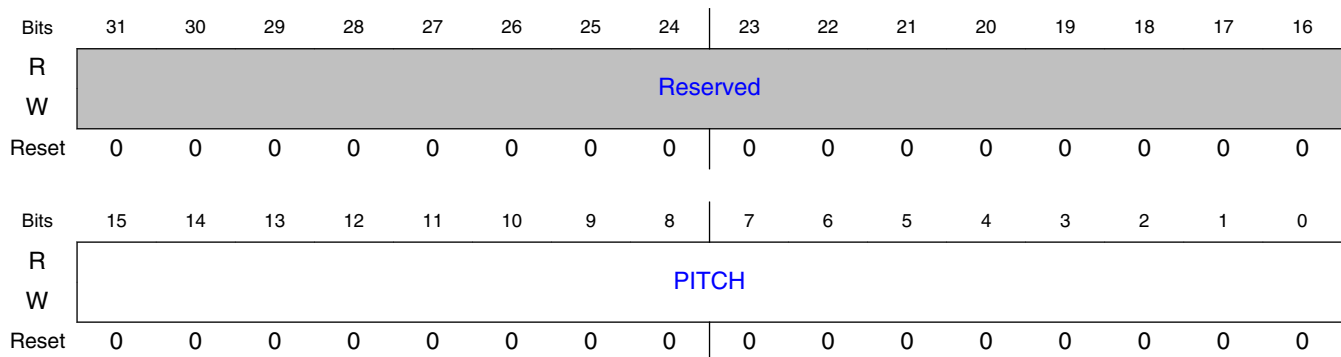
Field	Function
31-0	Base address
BASE_ADDR	System address of source buffer. Although byte granularity is supported, SW should align it to a 64B boundry to optimize system memory performance.

15.13.3.1.7 Read surface vertical pitch (PITCH)

15.13.3.1.7.1 Offset

Register	Offset
PITCH	14h

15.13.3.1.7.2 Diagram



15.13.3.1.7.3 Fields

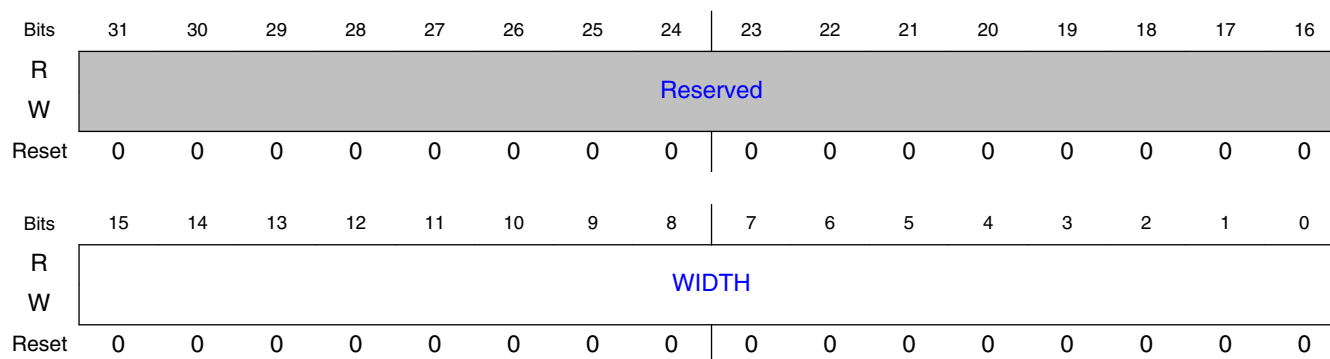
Field	Function
31-16 —	Reserved
15-0 PITCH	Pitch Number of bytes between 2 vertically adjacent pixels in system memory. Byte granularity is supported, but SW should align to 64B boundry.

15.13.3.1.8 Source frame buffer width (WIDTH)

15.13.3.1.8.1 Offset

Register	Offset
WIDTH	18h

15.13.3.1.8.2 Diagram



15.13.3.1.8.3 Fields

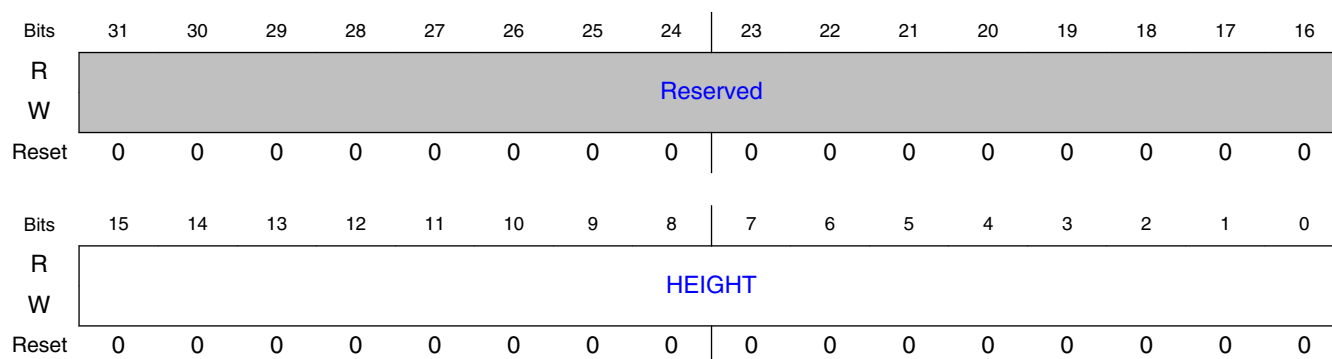
Field	Function
31-16 —	Reserved
15-0 WIDTH	Width Width, in pixels, of the source frame buffer. This is the number of pixels written to the output buffer pixel interface per scan line.

15.13.3.1.9 Height of frame to be read (HEIGHT)

15.13.3.1.9.1 Offset

Register	Offset
HEIGHT	1Ch

15.13.3.1.9.2 Diagram



15.13.3.1.9.3 Fields

Field	Function
31-16 —	Reserved
15-0 HEIGHT	Height Height, in pixels, of the source frame buffer. This is the number of scan lines written to the output buffer pixel interface per frame.

15.14 Interrupt Request Steering (IRQ_STEER)

15.14.1 Introduction

The irq_steer module redirects/steers the incoming interrupts to output interrupts of a selected/designated channel as specified by a set of configuration registers.

15.14.1.1 Block diagram

The figure below shows a high-level diagram of the irq_steer module.

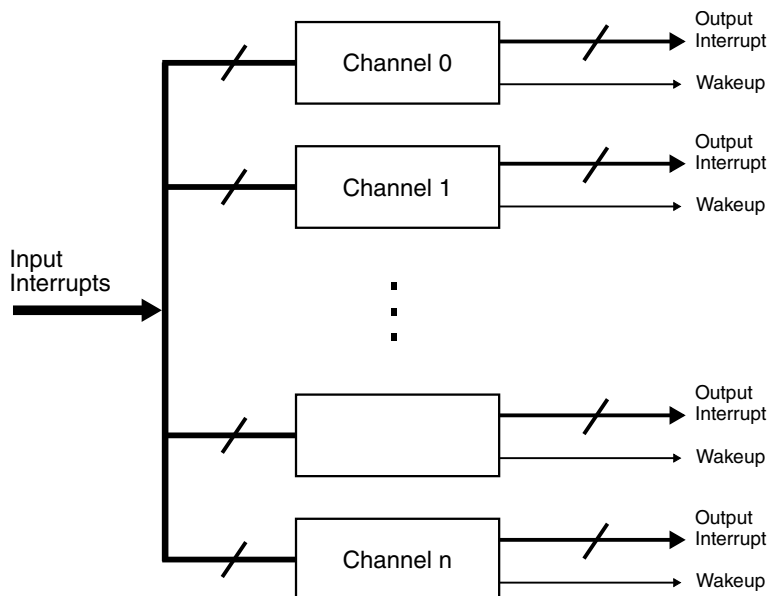


Figure 15-28. irq_steer block diagram

The figure below shows the high-level operation of one channel of the irq_steer module.

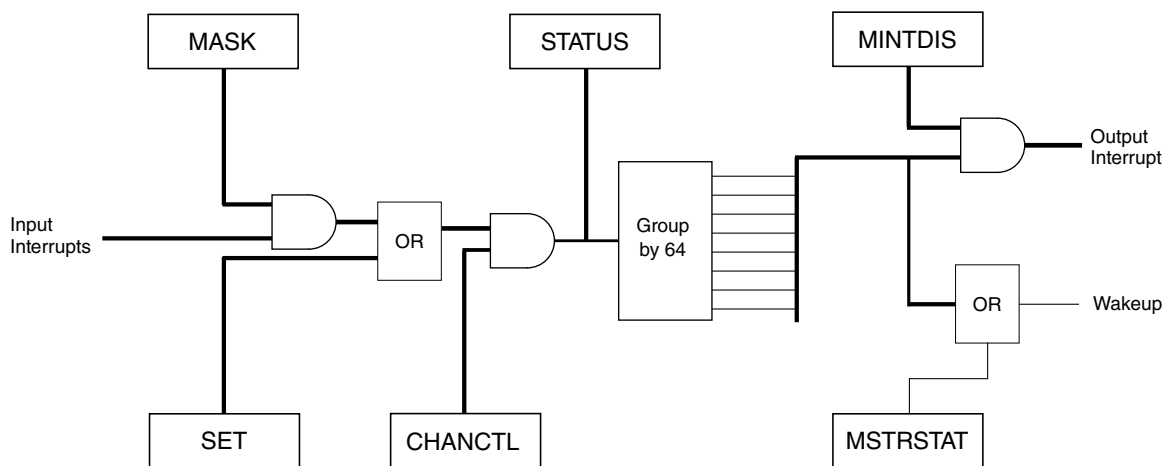


Figure 15-29. irq_steer channel block diagram

15.14.1.2 Features

This irq_steer module supports:

- 5 IRQ channels
- 512 interrupts per channel

15.14.2 Functional description

The irq_steer module allows the user to steer 512 interrupts into one of 5 channels.

15.14.3 Memory Map / Register Definition

This section describes the registers that are implemented for one channel. One channel in this module supports up to 512 interrupts per channel. Each channel has its own set of registers consisting of:

- One CHANnCTL register
- 16 CHn_MASK registers
- 16 CHn_SET registers
- 16 CHn_STATUS registers
- One CHn_MINTDIS register
- One CHn_MSTRSTAT register

This module supports 5 channels. Therefore, there are sets of 5 these registers. Each set of registers will have its own base offset, which must be added to the relative offset shown here.

The following table shows the correlation between an interrupt and its register and bit offset for the CHn_MASK, CHn_SET, and CHn_STATUS registers.

Table 15-10. Interrupt Mapping (incomplete)

n		Bit offset																															
		31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	511	510	509	508	507	506	505	504	503	502	501	500	499	498	497	496	495	494	493	492	491	490	489	488	487	486	485	484	483	482	481	480	
1	479	478	477	476	475	474	473	472	471	470	469	468	467	466	465	464	463	462	461	460	459	458	457	456	455	454	453	452	451	450	449	448	
2	447																															416	
3	415																															384	
4	383																															384	
5	351																															320	
6	319																															288	
7	287																															256	
8	255																															224	
9	223																															192	
10	191																															160	
11	159																															128	
12	127																															96	
13	95																															64	
14	63																															32	
15	31																															0	

15.14.3.1 IRQSTEER register descriptions

15.14.3.1.1 IRQSTEER Memory map

irq_steer base address: 0h

Offset	Register	Width (In bits)	Access	Reset value
0h	Channel n Control Register (CHANnCTL)	32	RW	0000_0000h
4h - 40h	Channel n Interrupt Mask Register (CHn_MASK0 - CHn_MASK15)	32	RW	0000_0000h
44h - 80h	Channel n Interrupt Set Register (CHn_SET0 - CHn_SET15)	32	RW	0000_0000h
84h - C0h	Channel n Interrupt Status Register (CHn_STATUS0 - CHn_STAT US15)	32	RO	0000_0000h
C4h	Channel n Master Interrupt Disable Register (CHn_MINTDIS)	32	RW	0000_0000h
C8h	Channel n Master Status Register (CHn_MSTRSTAT)	32	RO	0000_0000h

15.14.3.1.2 Channel n Control Register (CHANnCTL)

15.14.3.1.2.1 Offset

Register	Offset
CHANnCTL	0h

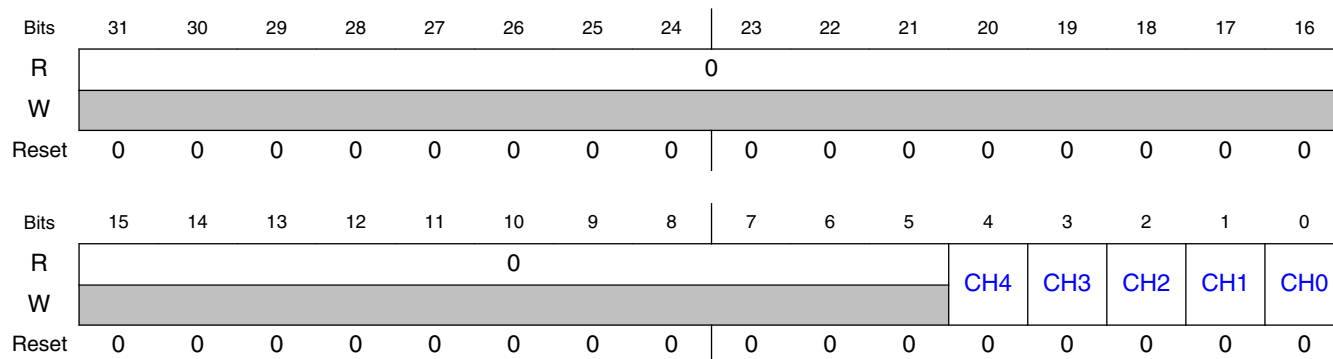
15.14.3.1.2.2 Function

This register is used to disable or "zero-out" outputs on a per-channel basis.

NOTE

Only one channel (CHn field) may be set at any given time.

15.14.3.1.2.3 Diagram



15.14.3.1.2.4 Fields

Field	Function
31-5 —	Reserved
4 CH4	Channel 4 control 0b - Disable channel 4 1b - Enable channel 4
3 CH3	Channel 3 control 0b - Disable channel 3 1b - Enable channel 3
2 CH2	Channel 2 control 0b - Disable channel 2 1b - Enable channel 2
1 CH1	Channel 1 control 0b - Disable channel 1 1b - Enable channel 1
0 CH0	Channel 0 control 0b - Disable channel 0 1b - Enable channel 0

15.14.3.1.3 Channel n Interrupt Mask Register (CHn_MASK0 - CHn_MASK 15)

15.14.3.1.3.1 Offset

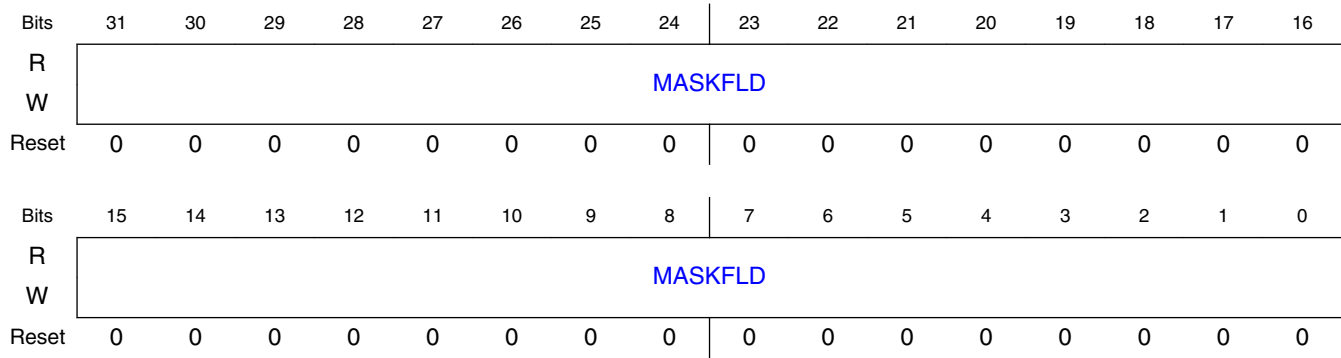
For a = 0 to 15:

Register	Offset
CHn_MASKa	4h + (a × 4h)

15.14.3.1.3.2 Function

The MASK registers are used to mask any of the 512 individual interrupts.

15.14.3.1.3.3 Diagram



15.14.3.1.3.4 Fields

Field	Function
31-0 MASKFLD	Mask bits See Table 15-10 for the correlation between an interrupt and its register and bit offset. 00000000000000000000000000000000b - Mask interrupt 00000000000000000000000000000001b - Do not mask interrupt

15.14.3.1.4 Channel n Interrupt Set Register (CHn_SET0 - CHn_SET15)

15.14.3.1.4.1 Offset

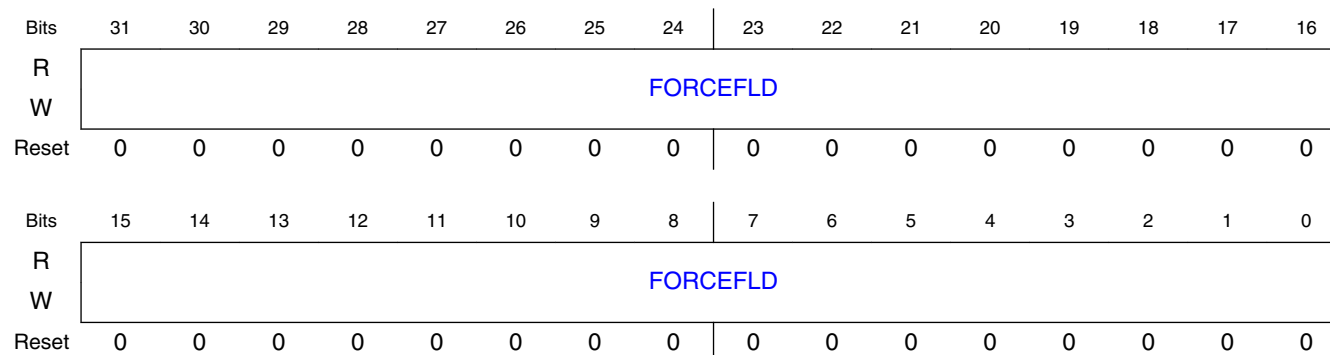
For a = 0 to 15:

Register	Offset
CHn_SETa	44h + (a × 4h)

15.14.3.1.4.2 Function

This register is used to force an interrupt.

15.14.3.1.4.3 Diagram



15.14.3.1.4.4 Fields

Field	Function
31-0	Brief bitfield description.
FORCEFLD	See Table 15-10 for the correlation between an interrupt and its register and bit offset. 00000000000000000000000000000000b - Normal operation 00000000000000000000000000000001b - Force interrupt

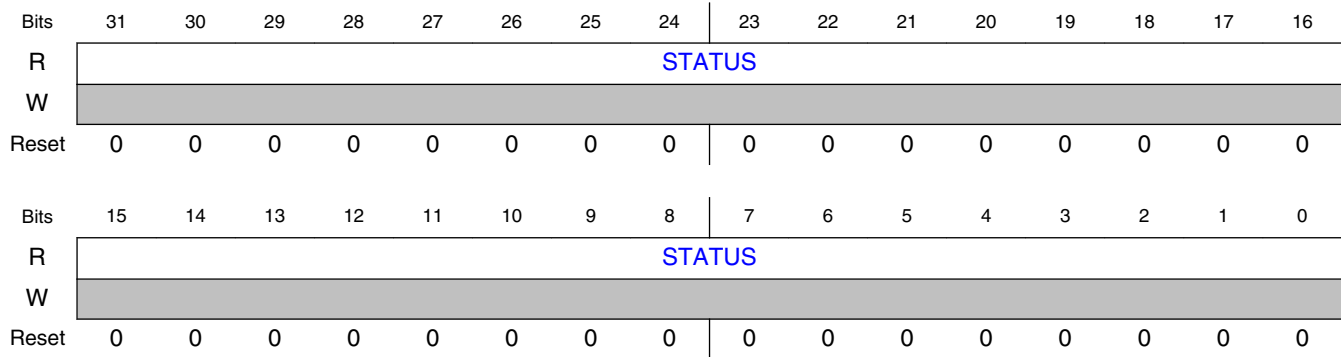
15.14.3.1.5 Channel n Interrupt Status Register (CHn_STATUS0 - CHn_STATUS15)

15.14.3.1.5.1 Offset

For a = 0 to 15:

Register	Offset
CHn_STATUSa	84h + (a × 4h)

15.14.3.1.5.2 Diagram



15.14.3.1.5.3 Fields

Field	Function
31-0 STATUS	Status of an interrupt See Table 15-10 for the correlation between an interrupt and its register and bit offset. 00000000000000000000000000000000b - Interrupt is not set. 00000000000000000000000000000001b - Interrupt is set.

15.14.3.1.6 Channel n Master Interrupt Disable Register (CHn_MINTDIS)

15.14.3.1.6.1 Offset

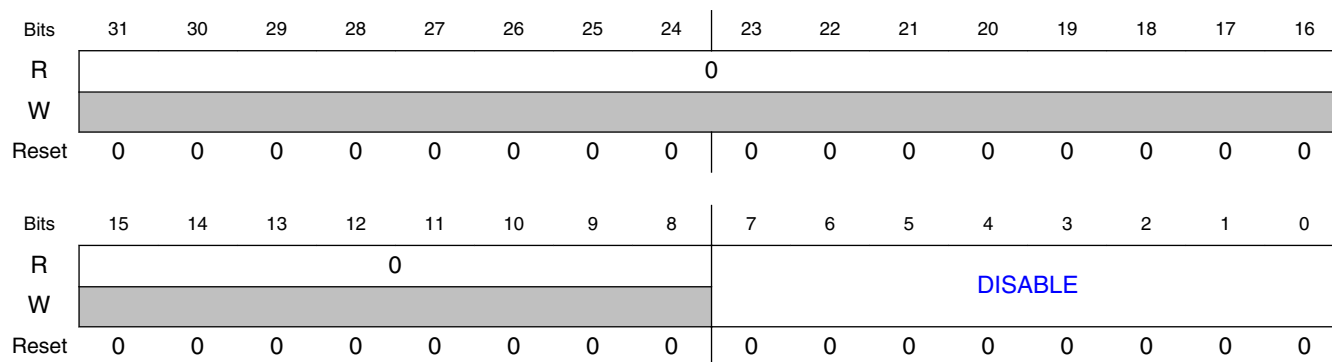
Register	Offset
CHn_MINTDIS	C4h

15.14.3.1.6.2 Function

Table 15-11. Interrupt disable mapping

STATUS bit	Interrupts disabled
7	511 - 448
6	447 - 384
5	383 - 320
4	319 - 256
3	255 - 192
2	191 - 128
1	127 - 64
0	63 - 0

15.14.3.1.6.3 Diagram



15.14.3.1.6.4 Fields

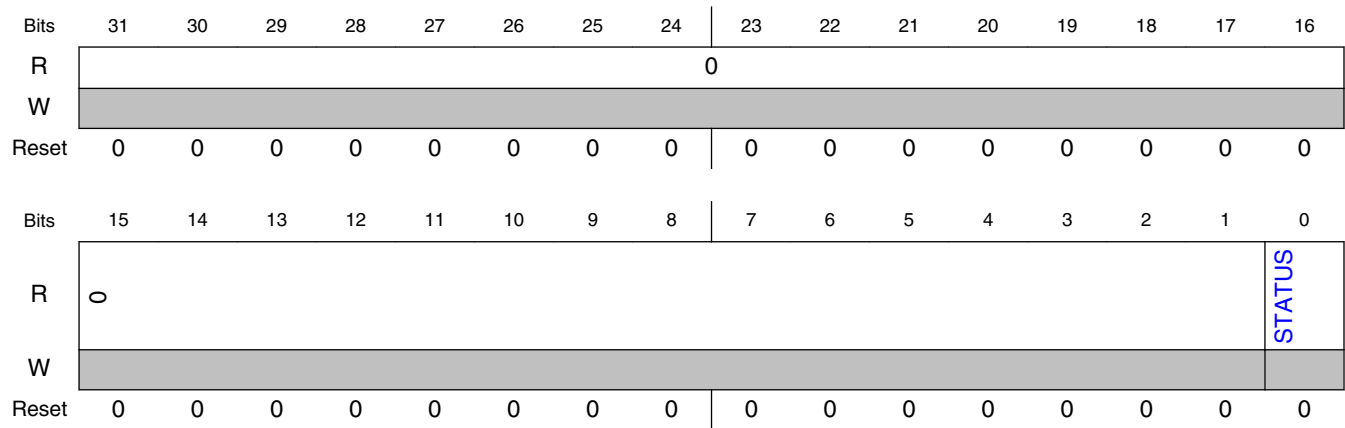
Field	Function
31-8 —	Reserved
7-0 DISABLE	Each bit of this field disables the corresponding interrupts in table above. 00000000b - Enable interrupts 00000001b - Disable interrupts

15.14.3.1.7 Channel n Master Status Register (CHn_MSTRSTAT)

15.14.3.1.7.1 Offset

Register	Offset
CHn_MSTRSTAT	C8h

15.14.3.1.7.2 Diagram



15.14.3.1.7.3 Fields

Field	Function
31-1 —	Reserved
0 STATUS	Status of all interrupts 0b - No interrupts are asserted. 1b - At least one interrupt is asserted.

Chapter 16

Low Speed Communication and Interconnects

16.1 I2C Controller (I2C)

16.1.1 Overview

This chapter describes block-level operation and programming of I2C. The chapter is intended for a block-driver software developer. To understand how the block is integrated at the SoC level, a system software developer should see discussions of the block in the appropriate SoC-level chapter(s).

References: This document assumes an understanding of the following document:

- *The I2C Bus Specification*, Version 2.1, by Philips Semiconductor

The Inter IC (I2C) provides functionality of a standard I2C slave and master. The I2C is designed to be compatible with the standard NXP I2C bus protocol.

NOTE

Four independent I2C channels are available.

I2C is a two-wire, bidirectional serial bus that provides a simple, efficient method of data exchange, minimizing the interconnection between devices. This bus is suitable for applications requiring occasional communications over a short distance between many devices. The flexible I2C standard allows additional devices to be connected to the bus for expansion and system development. See the connection diagram in the figure below.

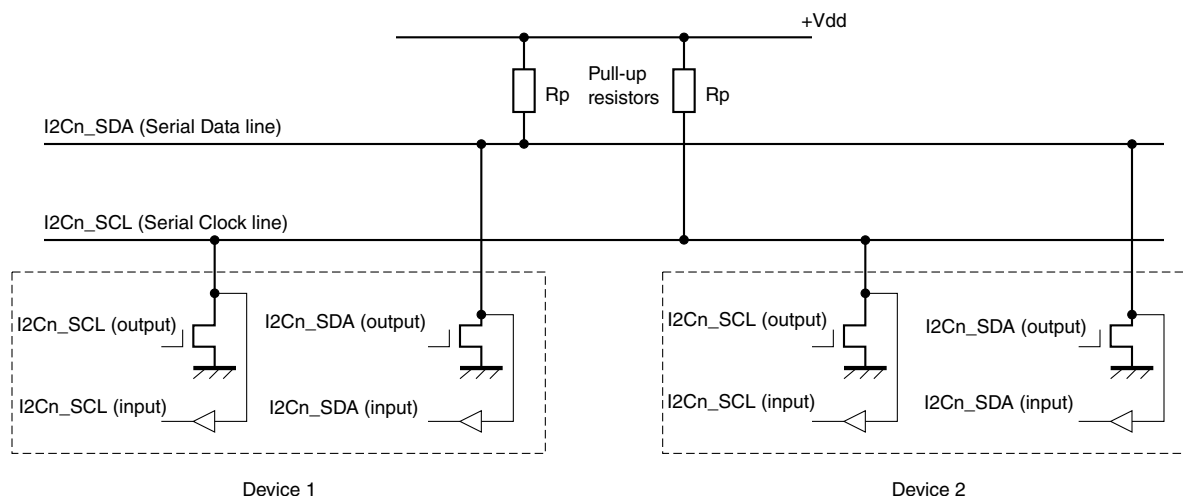


Figure 16-1. Connection of devices to I2C bus

The I2C interface speed is dependent on the I2C bus loading and timing characteristics. For pin requirement details, see *The I2C Bus Specification*. The I2C system is a true multimaster bus including arbitration and collision detection that prevents data corruption if multiple devices attempt to control the bus simultaneously. This feature supports complex applications with multiprocessor control and can be used for rapid testing and alignment of end products through external connections to an assembly-line computer. The figure below shows the block diagram of I2C.

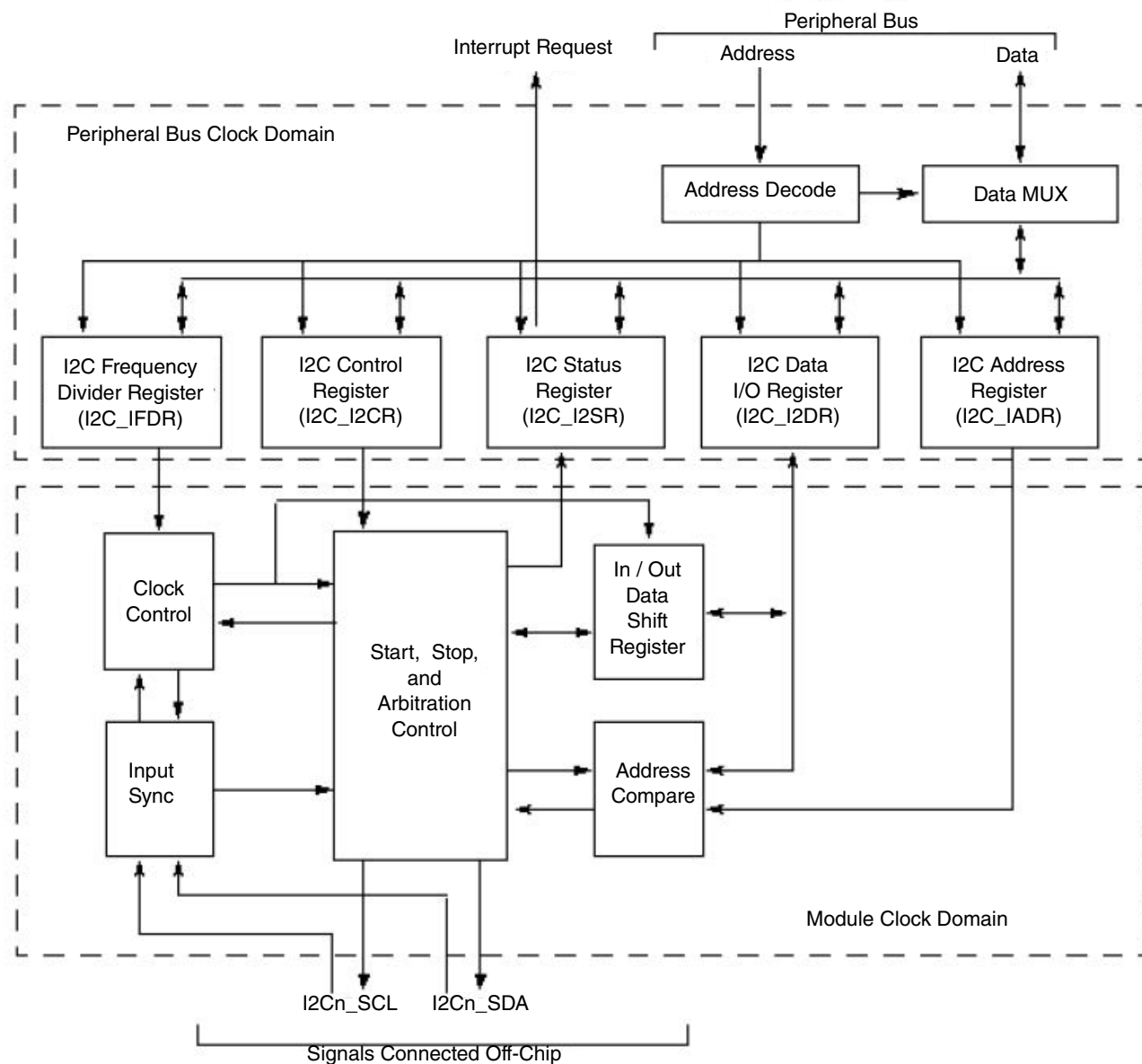


Figure 16-2. I2C block diagram

16.1.1.1 Features

The I2C has the following key features:

- Compatibility with I2C bus standard
- Multimaster operation
- Software programmability for one of 64 different serial clock frequencies
- Software-selectable acknowledge bit
- Interrupt-driven, byte-by-byte data transfer
- Arbitration-lost interrupt with automatic mode switching from master to slave

- Calling address identification interrupt
- Start and stop signal generation/detection
- Repeated Start signal generation
- Acknowledge bit generation/detection
- Bus-busy detection

16.1.1.2 Modes and operations

The I2C operates primarily in two functional modes: Standard mode and Fast mode.

- In Standard mode, I2C supports the data transfer rates up to 100 kbits/s.
- In Fast mode, data transfer rates up to 400 kbits/s can be achieved. Per block operation, there is no special configuration required for Fast or Standard mode. It is the data transfer rate that distinguishes Standard and Fast mode.

16.1.2 External Signals

This section discusses I2C signals that connect off-chip.

For I2C compliance, all devices connected to the I2Cn_SCL and I2Cn_SDA signals must have open-drain or open-collector outputs. The logic AND function is implemented on both lines with external pull-up resistors.

Inputs of I2Cn_SCL and I2Cn_SDA also need to be manually enabled by setting the SION bit in the IOMUX after the corresponding PADS are selected as I2C function.

16.1.3 Clocks

There are two input clocks for I2C.

The following table describes the clock sources for I2C. Please see [Clock Controller Module \(CCM\)](#) for clock setting, configuration and gating information.

Table 16-1. I2C Clocks

Clock name	Clock Root	Description
ipg_clk_patref		Module clock
ipg_clk_s		Peripheral access clock

- Peripheral clock: This clock is used for peripheral bus register read/writes.
- Module clock: This is the functional clock of the I2C. The serial bit clock frequency is derived from the module clock. The module clock and peripheral clocks are synchronous with each other. The minimum frequency of the module clock should be 12.8 MHz for Fast mode to achieve 400-kbps operation.

16.1.4 Functional description

This section provides a complete functional description of the block.

16.1.4.1 I2C system configuration

After a reset, the I2C defaults to Slave Receive operations. Thus, when not operating as a master or responding to a slave transmit address, the I2C defaults to the Slave Receive state.

For exceptions, see [Initialization sequence](#).

NOTE

The I2C is designed to be compatible with the PhilipsTM I2C bus protocol. For information on system configuration, protocol, and restrictions, see the *I2C Bus Specification*, version 2.1, by Philips Semiconductors. The I2C supports Standard and Fast modes only.

16.1.4.2 Arbitration procedure

If multiple devices simultaneously request the bus, the bus clock is determined by a synchronization procedure in which the low period equals the longest clock-low period among the devices, and the high period equals the shortest. A data arbitration procedure determines the relative priority of competing devices.

A device loses arbitration if it sends logic high while another sends logic low; it immediately switches to Slave Receive mode and stops driving I2Cn_SDA. In this case, the transition from master to Slave mode does not generate a Stop condition. Meanwhile, hardware sets the arbitration lost bit in the I2C Status register (I2C_I2SR[IAL] to indicate loss of arbitration).

16.1.4.3 Clock synchronization

Because wire-AND logic is used, a high-to-low transition on SCL affects devices connected to the bus. Devices start counting their low period when the master drives SCL low. When a device clock goes low, it holds SCL low until the Clock High state is reached. However, the low-to-high change in this device clock may not change the state of SCL if another device clock is still in its low period. Therefore, the device with the longest low period holds the synchronized clock SCL low.

Devices with shorter low periods enter a High Wait state during this time (see [Figure 16-3](#)). When all devices involved have counted off their low periods, the synchronized clock SCL is released and pulled high. There is then no difference between device clocks and the state of SCL, so all of the devices start counting their high periods. The first device to complete its high period pulls SCL low again.

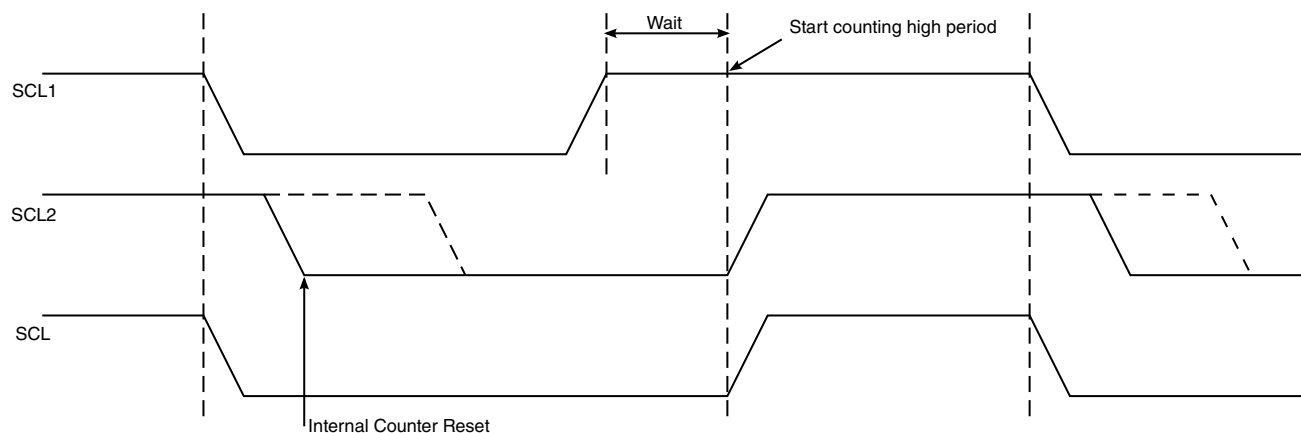


Figure 16-3. Synchronized clock SCL

16.1.4.4 Handshaking

The clock synchronization mechanism can be used as a handshake in data transfers. Slave devices can hold SCL low after completing one byte transfer (9 bits). In such a case, the clock mechanism halts the bus clock and forces the master clock into a Wait state until the slave releases SCL.

16.1.4.5 Clock stretching

Slaves can use the clock synchronization mechanism to slow down the transfer bit rate. After the master has driven SCL low, the slave can drive SCL low for the required period and then release it. If the slave SCL low period is longer than the master SCL low period, the resulting SCL bus signal low period is stretched.

16.1.4.6 Peripheral bus accesses

I2C is a 16-bit block. Only half-word accesses should be performed to the block.

16.1.4.7 Generation of transfer error on IP bus

If an address is received on the peripheral slave bus interface but it is not implemented, an access error is generated.

16.1.4.8 Reset

The I2C can be reset in the following ways:

- Global reset: A hard asynchronous reset of the whole I2C
- Software reset: An internal reset for the whole I2C (except for I2C_IADR and I2C_IFDR registers) initiated by deasserting the I2C_I2CR[IEN] bit

16.1.4.9 Interrupts

There is only one interrupt from the block, which is enabled by setting the I2C_I2CR[IIEN] bit.

The interrupt is generated in any one of the following conditions:

- One byte transfer is completed (the interrupt is set at the falling edge of the ninth clock).
- An address is received that matches its own specific address in Slave Receive mode.
- Arbitration is lost.

16.1.4.10 Byte order

The block only supports the Little-Endian mode.

16.1.5 Initialization

NOTE

Ensure the input select pins for IOMUXC are configured correctly for I2C.

16.1.5.1 Initialization sequence

Before the interface can transfer serial data, registers must be initialized, as listed here.

1. Set the data sampling rate (I2C_IFDR[IC]) to obtain SCL frequency from the system bus clock.
2. Update the address in the (I2C_IADR) to define its slave address (address can range from 0 to 0x7f).
3. Set the I2C enable bit (I2C_I2CR[IEN]) to enable the I2C bus interface system.
4. Modify the bits in the I2C_I2CR to select Master/Slave mode, Transmit/Receive mode, and Interrupt-Enable or not.

16.1.5.2 Generation of Start

After completion of the initialization procedure, serial data can be transmitted by selecting the Master Transmit mode. On a multimaster bus system, the busy bus (I2C_I2SR[IBB]) must be tested to determine whether the serial bus is free. If the bus is free (IBB = 0), the Start signal and the first byte (the slave address) can be sent. The data written to the data register comprises the address of the desired slave and the LSB indicates the transfer direction.

The free time between a Stop and the next Start condition is built into the hardware that generates the Start cycle. Depending on the relative frequencies of the system clock and the SCL period, it may be necessary to wait until the I2C is not busy after writing the calling address to the data register (I2C_I2DR), before proceeding to load data into the data register (I2C_I2DR).

16.1.5.3 Post-transfer software response

Sending or receiving a byte sets the data transferring bit (I2C_I2SR[ICF]), which indicates one byte of communication is finished. Upon completion, the interrupt status (I2C_I2SR[IIF]) is also set. An external interrupt is generated if the interrupt enable (I2C_I2CR[IEN]) is set. The software must first clear the interrupt status (I2C_I2SR[IIF]) in the interrupt routine.

See the flow chart in [Figure 16-5](#).

The data transferring bit (I2C_I2SR[ICF]) is cleared either by reading from I2C_I2DR in Receive mode or by writing to this register in Transmit mode.

The software can service the I2C I/O in the main program by monitoring the interrupt status (I2C_I2SR[IIF]) if the interrupt enable is deasserted. In this case, the interrupt status should be polled in the data transferring bit (I2C_I2SR[ICF]) because the operation is different when arbitration is lost.

When an interrupt occurs at the end of the address cycle, the master is always in Transmit mode; that is, the address is sent. If Master Receive mode is required, then I2C_I2CR[MTX] should be toggled and a dummy read of the I2C_I2DR register must be executed to trigger receive data.

During Slave-mode address cycles (I2C_I2SR[IAAS] = 1), the slave read/write bit I2C_I2SR[SRW] is read to determine the direction of the next transfer. The transmit/receive bit (I2C_I2CR[MTX]) should also be programmed accordingly. For Slave-mode data cycles (IAAS = 0), SRW is invalid. MTX should be read to determine the current transfer direction.

16.1.5.4 Generation of Stop

A data transfer ends when the master signals a Stop, which can occur after all data is sent.

For a master receiver to terminate a data transfer, it must inform the slave transmitter by not acknowledging the last data byte. This is done by setting the transmit acknowledge bit (I2C_I2CR[TXAK]) before reading the next-to-last byte. Before the last byte is read, a Stop signal must be generated.

16.1.5.5 Generation of Repeated Start

After the data transfer, if the master still requires the bus, it can signal another Start followed by another slave address without signaling a Stop.

16.1.5.6 Slave mode

In the slave interrupt service routine (see [Figure 16-5](#)), the block addressed as slave bit (IAAS) should be tested to check if a calling of its own address has just been received. If IAAS is set, software should set the Transmit/Receive mode select bit (I2C_I2CR[MTX]) according to the I2C_I2SR[SRW]. Writing to the I2C_I2CR clears the IAAS automatically. The only time IAAS is read as set is from the interrupt at the end of the address cycle where an address match occurred; interrupts resulting from subsequent data transfers will have IAAS cleared. A data transfer can now be initiated by writing information to I2C_I2DR for slave transmits, or read from I2C_I2DR in Slave Receive mode. A dummy read of I2C_I2DR in Slave Receive mode releases SCL, allowing the master to send data.

In the slave transmitter routine, the receive acknowledge bit (I2C_I2SR[RXAK]) must be tested before sending the next byte of data. Setting RXAK means an end-of-data signal from the master receiver, after which the software must switch it from Transmit to Receiver mode. Reading the data register (I2C_I2DR) then releases SCL so the master can generate a Stop signal.

16.1.5.7 Arbitration lost

If several devices try to engage the bus at the same time, one becomes master. Hardware immediately switches devices that lose arbitration to Slave Receive mode. Data output to 12Cn_SDA stops, but 12Cn_SCL is still generated until the end of the byte during which arbitration is lost. An interrupt occurs at the falling edge of the ninth clock of this transfer if the arbitration is lost (I2C_I2SR[IAL] = 1), and the Slave mode is selected (I2C_I2CR[MSTA] = 0).

See the flow chart in [Figure 16-5](#).

If a device that is not a master tries to transmit or do a Start, hardware inhibits the transmission, clears MSTA without signaling a Stop, generates an interrupt to the Arm platform, and sets I2C_I2SR[IAL] to indicate a failed attempt to engage the bus. When considering these cases, the slave service routine should first test I2C_I2SR[IAL], and the software should clear it if it is set.

For Multimaster mode, when an I2C is enabled when the bus is busy and asserts Start, the I2C_I2SR[IAL] bit gets set only for 12Cn_SDA=0, 12Cn_SCL=0/1, 12Cn_SDA=1, and 12Cn_SCL=0; but not for 12Cn_SDA=1 and I2Cn_SCL=1, which is the equivalent of Bus Idle state.

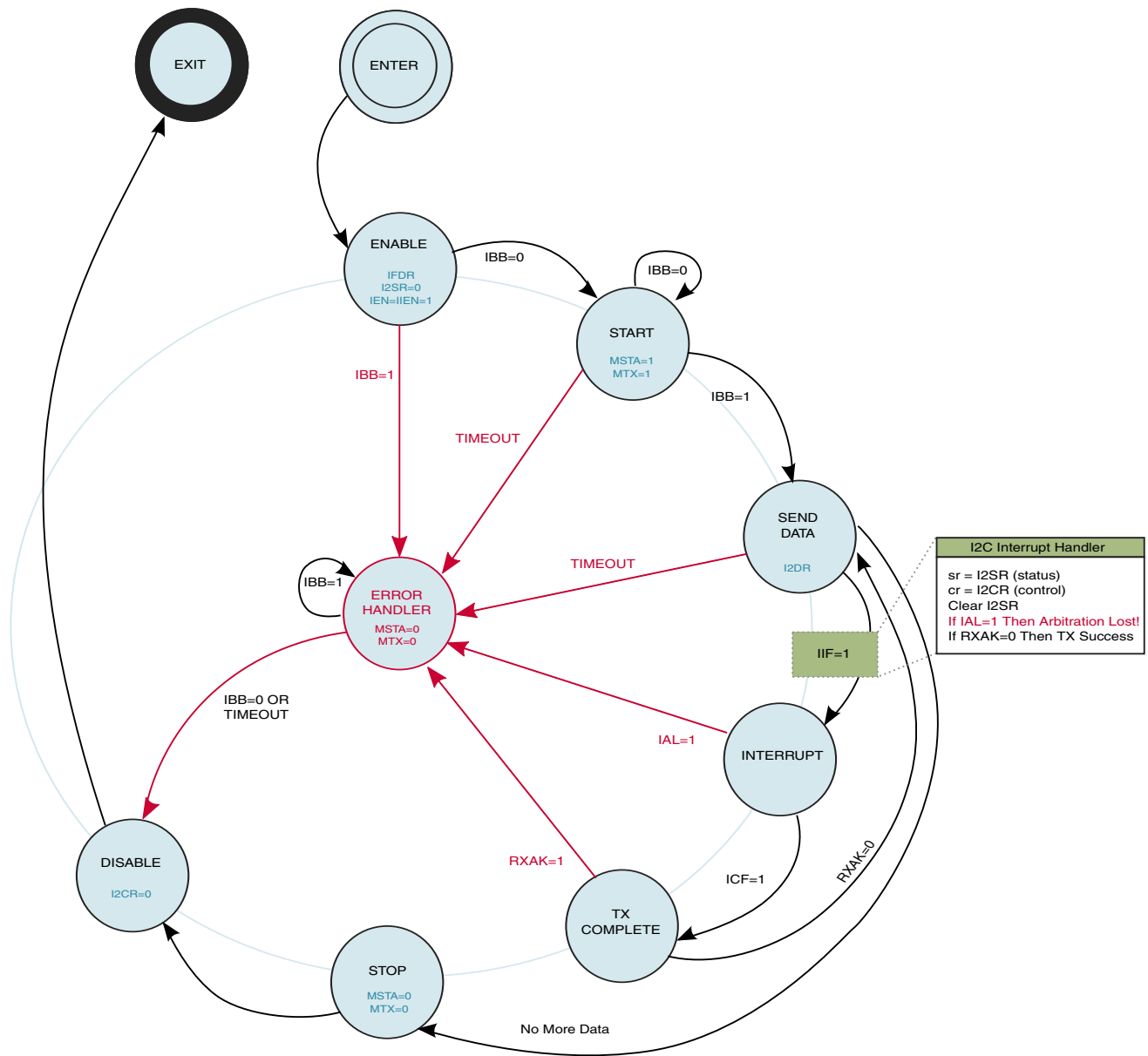


Figure 16-4. I2C Programming state diagram

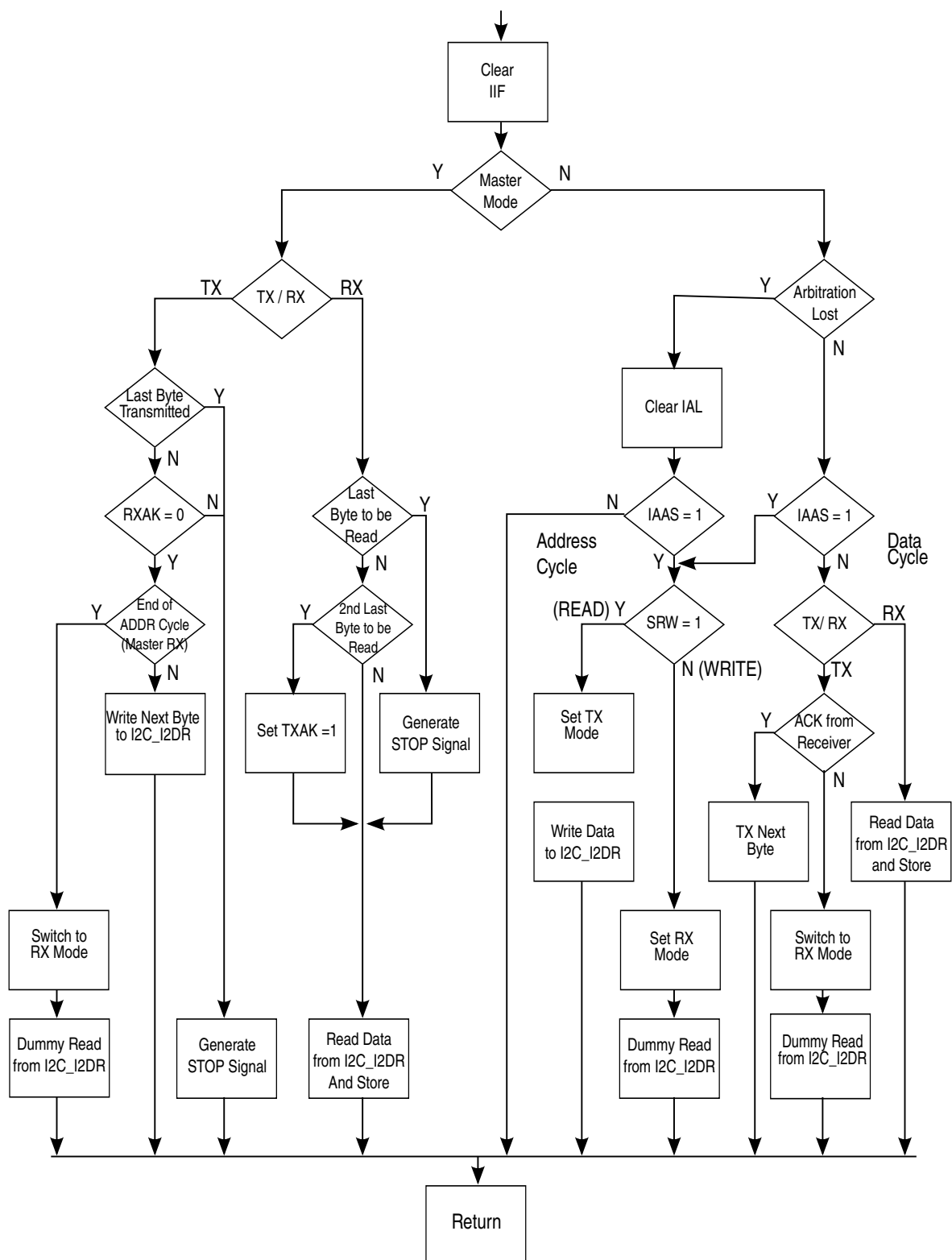


Figure 16-5. Flowchart of typical I2C interrupt routine

NOTE

For a Repeated Start only, the Stop-generation stage does not occur in Master mode. A loop repeats itself without stopping for the next start.

For Master Receive mode, I2C is programmed as Master Transmit during Address mode and after slave address transfer; the MTX bit should be cleared and a dummy read on the I2C_I2DR register should be performed so I2C can read the next receive data.

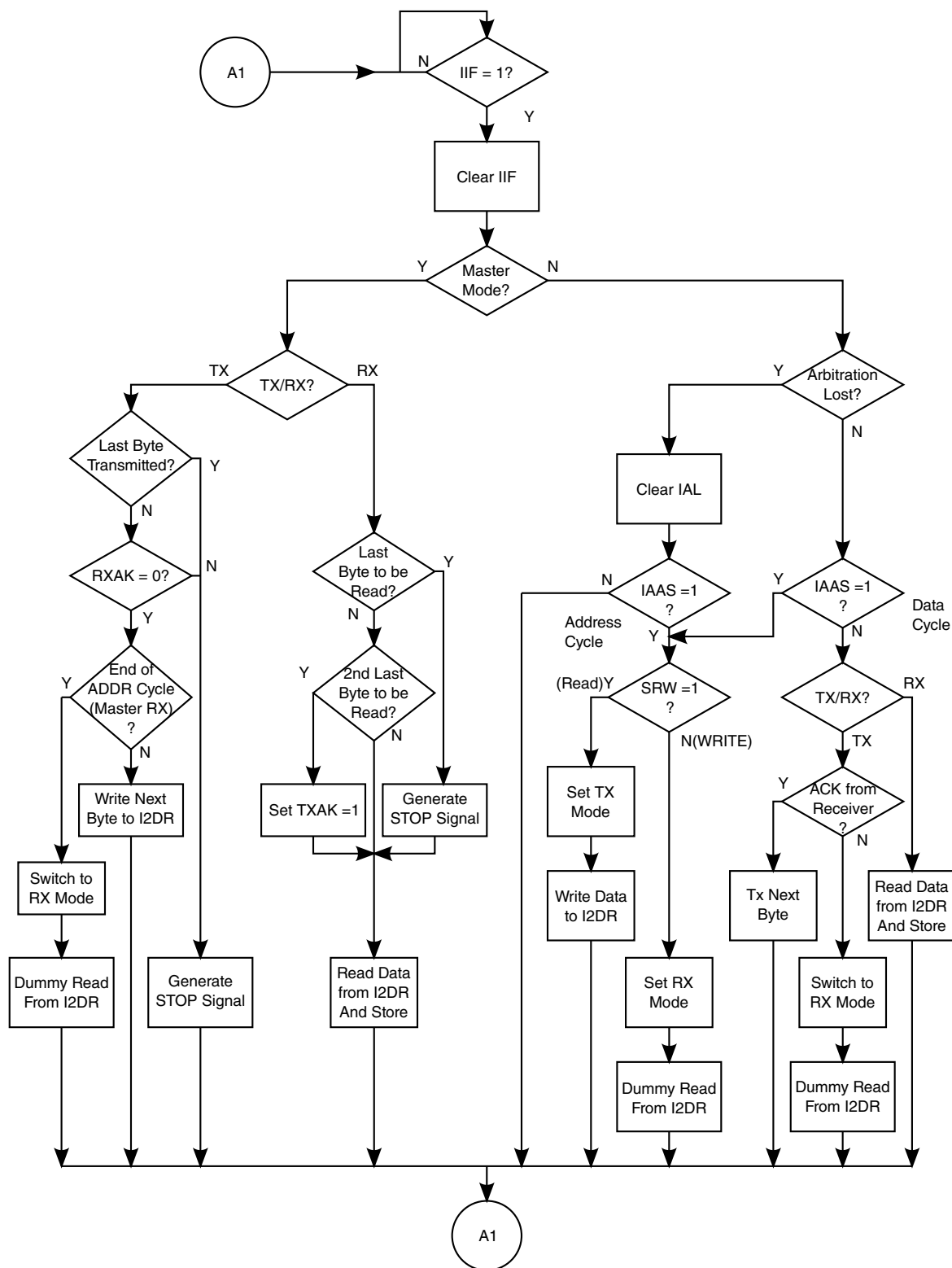


Figure 16-6. Flowchart for typical I2C polling routine

NOTE

The timeout value depends on the bus frequency at which I2C is operating. The minimum timeout for polling the IIF bit at a maximum I2C bus frequency of 400 kHz is $T_{\min} = 25 \mu\text{s}$ ($=2.5 \times 10 \mu\text{s}$). This value can be calculated for any bus frequency. The formula is $T_{\min} = 10/F_{\text{SCL}}$, where F_{SCL} is the frequency of the I2C clock (SCL).

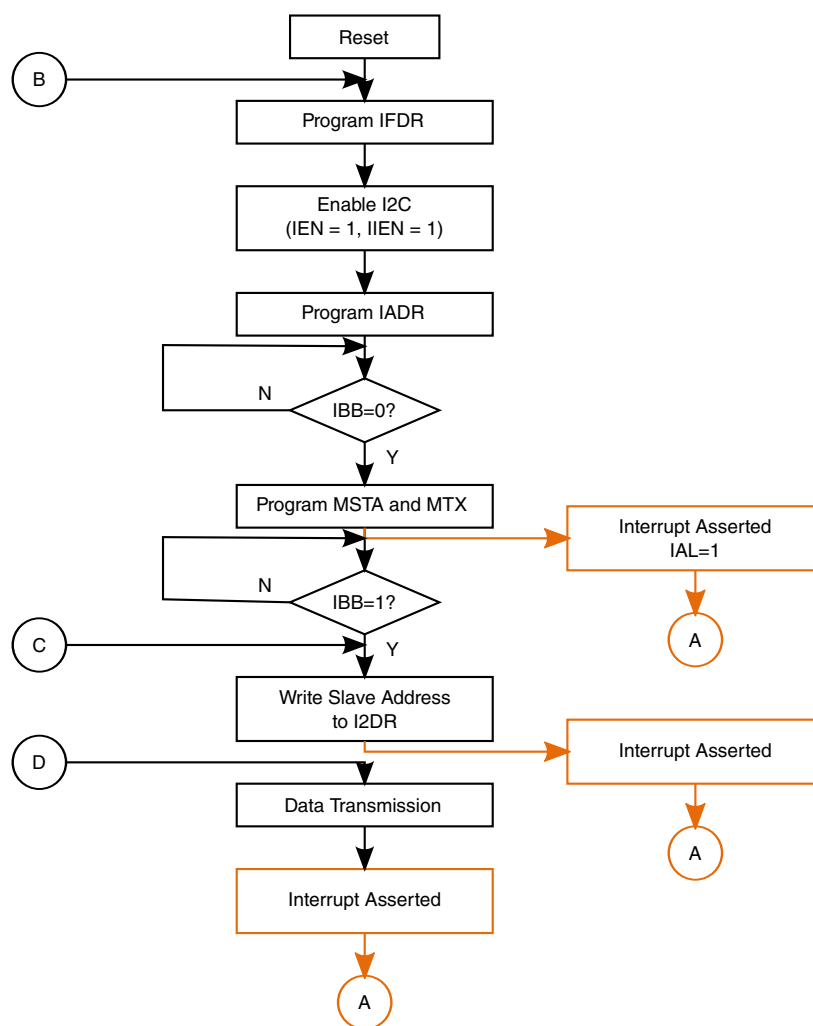


Figure 16-7. Detailed flowchart of a typical I2C Master Transmit mode, part 1

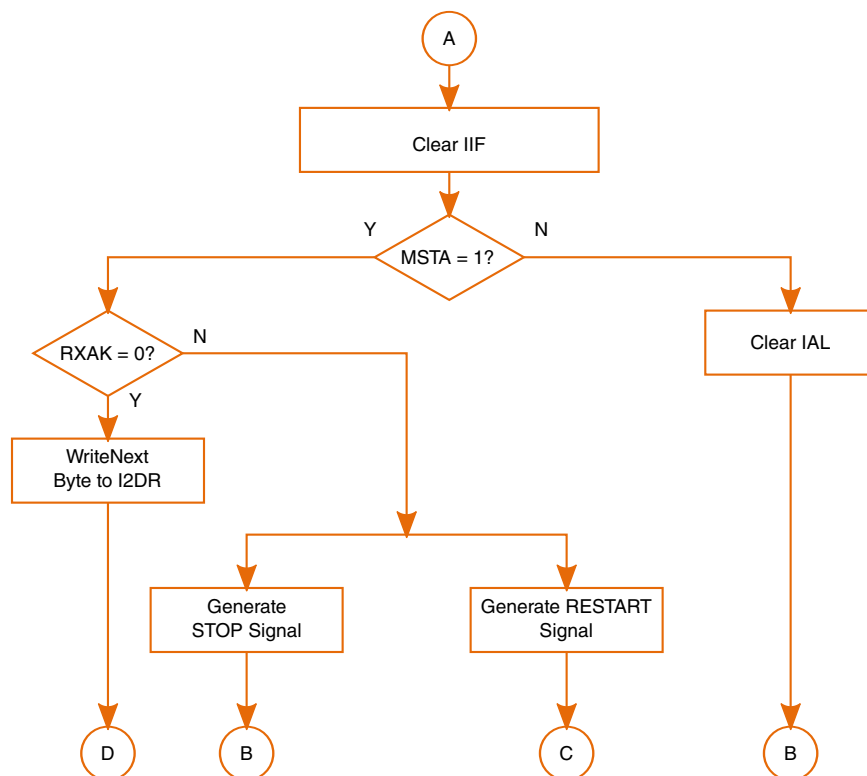


Figure 16-8. Detailed flowchart of a typical I2C Master Transmit mode, part 2

Figure 16-7 and Figure 16-8 show the Master Transmit mode operation with interrupt subroutine. If an interrupt is generated and the MSTA bit is 0, then bus arbitration is lost and IAL is set. Software can clear the IAL bit and reprogram I2C. If the MSTA bit is 1, then it is a transfer-generated interrupt. In this case, software can check the RXAK bit for a data receive acknowledgement by the slave and, accordingly, decide to do one of the following:

- Generate a STOP
- Generate a REPEATED START by writing to the I2C_I2CR register
- Perform the next data transfer by writing to the I2C_I2DR register

NOTE

The IBB bit is asserted by a Start condition on the bus, and it is deasserted by a Stop condition on the bus. Therefore, if arbitration is lost due to an unexpected Stop condition during transfer, then IBB is cleared. If arbitration is lost due to a data mismatch, then it is not cleared. Software should always clear the IEN bit and then set it if arbitration is lost.

16.1.6 Software restriction

Software should ensure that there is a delay of at least two module clock cycles after it sets the I2C_I2CR[RSTA] bit and before writing to the I2C_I2DR register. The maximum possible clock period of the module clock is 78 ns.

16.1.7 I2C Memory Map/Register Definition

The I2C contains five 16-bit registers.

NOTE

Registers at offsets 0x0002, 0x0006, 0x000A, and 0x000E are reserved for future additions.

I2C memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30A2_0000	I2C Address Register (I2C1_IADR)	16	R/W	0000h	16.1.7.1/6466
30A2_0004	I2C Frequency Divider Register (I2C1_IFDR)	16	R/W	0000h	16.1.7.2/6466
30A2_0008	I2C Control Register (I2C1_I2CR)	16	R/W	0000h	16.1.7.3/6468
30A2_000C	I2C Status Register (I2C1_I2SR)	16	R/W	0081h	16.1.7.4/6469
30A2_0010	I2C Data I/O Register (I2C1_I2DR)	16	R/W	0000h	16.1.7.5/6471
30A3_0000	I2C Address Register (I2C2_IADR)	16	R/W	0000h	16.1.7.1/6466
30A3_0004	I2C Frequency Divider Register (I2C2_IFDR)	16	R/W	0000h	16.1.7.2/6466
30A3_0008	I2C Control Register (I2C2_I2CR)	16	R/W	0000h	16.1.7.3/6468
30A3_000C	I2C Status Register (I2C2_I2SR)	16	R/W	0081h	16.1.7.4/6469
30A3_0010	I2C Data I/O Register (I2C2_I2DR)	16	R/W	0000h	16.1.7.5/6471
30A4_0000	I2C Address Register (I2C3_IADR)	16	R/W	0000h	16.1.7.1/6466
30A4_0004	I2C Frequency Divider Register (I2C3_IFDR)	16	R/W	0000h	16.1.7.2/6466

Table continues on the next page...

I2C memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30A4_0008	I2C Control Register (I2C3_I2CR)	16	R/W	0000h	16.1.7.3/6468
30A4_000C	I2C Status Register (I2C3_I2SR)	16	R/W	0081h	16.1.7.4/6469
30A4_0010	I2C Data I/O Register (I2C3_I2DR)	16	R/W	0000h	16.1.7.5/6471
30A5_0000	I2C Address Register (I2C4_IADR)	16	R/W	0000h	16.1.7.1/6466
30A5_0004	I2C Frequency Divider Register (I2C4_IFDR)	16	R/W	0000h	16.1.7.2/6466
30A5_0008	I2C Control Register (I2C4_I2CR)	16	R/W	0000h	16.1.7.3/6468
30A5_000C	I2C Status Register (I2C4_I2SR)	16	R/W	0081h	16.1.7.4/6469
30A5_0010	I2C Data I/O Register (I2C4_I2DR)	16	R/W	0000h	16.1.7.5/6471

16.1.7.1 I2C Address Register (I2Cx_IADR)

Address: Base address + 0h offset

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Read	0								ADR							0
Write																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

I2Cx_IADR field descriptions

Field	Description
15–8 Reserved	This read-only field is reserved and always has the value 0.
7–1 ADR	Slave address. Contains the specific slave address to be used by the I2C. Slave mode is the default I2C mode for an address match on the bus. NOTE: The I2C_IADR holds the address to which the I2C responds when addressed as a slave. The slave address is not the address sent on the bus during the address transfer. The register is not reset by a software reset.
0 Reserved	This read-only field is reserved and always has the value 0.

16.1.7.2 I2C Frequency Divider Register (I2Cx_IFDR)

The I2C_IFDR provides a programmable prescaler to configure the clock for bit-rate selection. The register does not get reset by a software reset.

The following table describes the divider and register values for the register field "IC."

Table 16-2. I2C_IFDR Register Field Values

IC	Divider		IC	Divider		IC	Divider		IC	Divider
0x00	30		0x10	288		0x20	22		0x30	160
0x01	32		0x11	320		0x21	24		0x31	192
0x02	36		0x12	384		0x22	26		0x32	224
0x03	42		0x13	480		0x23	28		0x33	256
0x04	48		0x14	576		0x24	32		0x34	320
0x05	52		0x15	640		0x25	36		0x35	384
0x06	60		0x16	768		0x26	40		0x36	448
0x07	72		0x17	960		0x27	44		0x37	512
0x08	80		0x18	1152		0x28	48		0x38	640
0x09	88		0x19	1280		0x29	56		0x39	768
0x0A	104		0x1A	1536		0x2A	64		0x3A	896
0x0B	128		0x1B	1920		0x2B	72		0x3B	1024
0x0C	144		0x1C	2304		0x2C	80		0x3C	1280
0x0D	160		0x1D	2560		0x2D	96		0x3D	1536
0x0E	192		0x1E	3072		0x2E	112		0x3E	1792
0x0F	240		0x1F	3840		0x2F	128		0x3F	2048

Address: Base address + 4h offset

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Read	0										IC					
Write																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

I2Cx_IFDR field descriptions

Field	Description
15–6 Reserved	This read-only field is reserved and always has the value 0.
IC	I2C clock rate. Prescales the clock for bit-rate selection. Due to potentially slow I2Cn_SCL and I2Cn_SDA rise and fall times, bus signals are sampled at the prescaler frequency. The serial bit clock frequency may be lower than IPG_CLK_ROOT divided by the divider shown in the I2C Data I/O Register. NOTE: The IC value should not be changed during the data transfer, however, it can be changed before a Repeat Start or Start programming sequence in I2C. The I2C protocol supports bit rates of up to 400 kbps. The IC bits need to be programmed in accordance with this constraint.

16.1.7.3 I2C Control Register (I2Cx_I2CR)

The I2C_I2CR is used to enable the I2C and the I2C interrupt. It also contains bits that govern operation as a slave or a master.

Address: Base address + 8h offset

Bit	15	14	13	12	11	10	9	8
Read	0							
Write								
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
Read	IEN	IEN	MSTA	MTX	TXAK	0	0	
Write						RSTA		
Reset	0	0	0	0	0	0	0	0

I2Cx_I2CR field descriptions

Field	Description
15–8 Reserved	This read-only field is reserved and always has the value 0.
7 IEN	I2C enable. Also controls the software reset of the entire I2C. Resetting the bit generates an internal reset to the block. If the block is enabled in the middle of a byte transfer, Slave mode ignores the current bus transfer and starts operating when the next Start condition is detected. Master mode is not aware that the bus is busy, so initiating a start cycle may corrupt the current bus cycle, ultimately causing either the current master or the I2C to lose arbitration. Subsequently, bus operation returns to normal. 0 The block is disabled, but registers can still be accessed. 1 The I2C is enabled. This bit must be set before any other I2C_I2CR bits have an effect.
6 I2EN	I2C interrupt enable. NOTE: If data is written during the Start condition, that is, just after setting the I2C_I2CR[MSTA] and I2C_I2CR[MTX] bits, then the ICF bit is cleared at the falling edge of SCLK after Start. If data is written after the Start condition and falling edge of SCLK, then the ICF bit is cleared as soon as data is written. 0 I2C interrupts are disabled, but the status flag I2C_I2SR[IIF] continues to be set when an Interrupt condition occurs. 1 I2C interrupts are enabled. An I2C interrupt occurs if I2C_I2SR[IIF] is also set.
5 MSTA	Master/Slave mode select bit. If the master loses arbitration, MSTA is cleared without generating a Stop signal. NOTE: The module clock should be on for writing to the MSTA bit. NOTE: The MSTA bit is cleared by software to generate a Stop condition; it can also be cleared by hardware when the I2C loses the bus arbitration.

Table continues on the next page...

I2Cx_I2CR field descriptions (continued)

Field	Description
	0 Slave mode. Changing MSTA from 1 to 0 generates a Stop and selects Slave mode. 1 Master mode. Changing MSTA from 0 to 1 signals a Start on the bus and selects Master mode.
4 MTX	Transmit/Receive mode select bit. Selects the direction of master and slave transfers. 0 Receive. When a slave is addressed, the software should set MTX according to the slave read/write bit in the I2C status register (I2C_I2SR[SRW]). 1 Transmit. In Master mode, MTX should be set according to the type of transfer required. Therefore, for address cycles, MTX is always 1.
3 TXAK	Transmit acknowledge enable. Specifies the value driven onto I2Cn_SDA during acknowledge cycles for both master and slave receivers. NOTE: Writing TXAK applies only when the I2C bus is a receiver. 0 An acknowledge signal is sent to the bus at the ninth clock bit after receiving one byte of data. 1 No acknowledge signal response is sent (that is, the acknowledge bit = 1).
2 RSTA	Repeat start. Always reads as 0. Attempting a repeat start without bus mastership causes loss of arbitration. 0 No repeat start 1 Generates a Repeated Start condition
Reserved	This read-only field is reserved and always has the value 0.

16.1.7.4 I2C Status Register (I2Cx_I2SR)

The I2C_I2SR contains bits that indicate transaction direction and status.

Address: Base address + Ch offset

Bit	15	14	13	12	11	10	9	8
Read	0							
Write								
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
Read	ICF	IAAS	IBB	IAL	0	SRW	IIF	RXAK
Write								
Reset	1	0	0	0	0	0	0	1

I2Cx_I2SR field descriptions

Field	Description
15–8 Reserved	This read-only field is reserved and always has the value 0.
7 ICF	Data transferring bit. While one byte of data is transferred, ICF is cleared. 0 Transfer is in progress. 1 Transfer is complete. This bit is set by the falling edge of the ninth clock of the last byte transfer.
6 IAAS	I2C addressed as a slave bit. The Arm platform is interrupted if the interrupt enable (I2C_I2CR[IEN]) is set. The Arm platform must check the slave read/write bit (SRW) and set its Transfer/Receive mode accordingly. Writing to I2C_I2CR clears this bit. 0 Not addressed 1 Addressed as a slave. Set when its own address (I2C_IADR) matches the calling address.
5 IBB	I2C bus busy bit. Indicates the status of the bus. NOTE: When I2C is enabled (I2C_I2CR[IEN] = 1), it continuously polls the bus data (SDA) and clock (SCL) signals to determine a Start or Stop condition. 0 Bus is idle. If a Stop signal is detected, IBB is cleared. 1 Bus is busy. When Start is detected, IBB is set.
4 IAL	Arbitration lost. Set by hardware in the following circumstances (IAL must be cleared by software by writing a "0" to it at the start of the interrupt service routine): - I2Cn_SDA input samples low when the master drives high during an address or data-transmit cycle. - I2Cn_SDA input samples low when the master drives high during the acknowledge bit of a data-receive cycle. For the above two cases, the bit is set at the falling edge of the ninth I2Cn_SCL clock during the ACK cycle. - A Start cycle is attempted when the bus is busy. - A Repeated Start cycle is requested in Slave mode. - A Stop condition is detected when the master did not request it. NOTE: Software cannot set the bit. 0 No arbitration lost. 1 Arbitration is lost.
3 Reserved	This read-only field is reserved and always has the value 0.
2 SRW	Slave read/write. When the I2C is addressed as a slave, IAAS is set, and the slave read/write bit (SRW) indicates the value of the R/W command bit of the calling address sent from the master. SRW is valid only when a complete transfer has occurred, no other transfers have been initiated, and the I2C is a slave and has an address match. 0 Slave receive, master writing to slave 1 Slave transmit, master reading from slave
1 IIF	I2C interrupt. Must be cleared by the software by writing a "0" to it in the interrupt routine. NOTE: The software cannot set the bit. 0 No I2C interrupt pending. 1 An interrupt is pending.

Table continues on the next page...

I2Cx_I2SR field descriptions (continued)

Field	Description
	This causes a processor interrupt request (if the interrupt enable is asserted [I1EN = 1]). The interrupt is set when one of the following occurs: • One byte transfer is completed (the interrupt is set at the falling edge of the ninth clock). • An address is received that matches its own specific address in Slave Receive mode. • Arbitration is lost.
0 RXAK	Received acknowledge. This is the value received from the I2Cn_SDA input for the acknowledge bit during a bus cycle. 0 An "acknowledge" signal was received after the completion of an 8-bit data transmission on the bus. 1 A "No acknowledge" signal was detected at the ninth clock.

16.1.7.5 I2C Data I/O Register (I2Cx_I2DR)

In Master Receive mode, reading the data register allows a read to occur and initiates the next byte to be received. In Slave mode, the same function is available after it is addressed.

Address: Base address + 10h offset

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Read	0								DATA							
Write																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

I2Cx_I2DR field descriptions

Field	Description
15–8 Reserved	This read-only field is reserved and always has the value 0.
DATA	Data Byte. Holds the last data byte received or the next data byte to be transferred. Software writes the next data byte to be transmitted or reads the data byte received. NOTE: The core-written value in I2C_I2DR cannot be read back by the core. Only data written by the I2C bus side can be read.

16.2 Universal Asynchronous Receiver/Transmitter(UART)

16.2.1 Overview

Universal Asynchronous Receiver/Transmitter (UART) provides serial communication capability with external devices through a level converter and an RS-232 cable or through use of external circuitry that converts infrared signals to electrical signals (for reception) or transforms electrical signals to signals that drive an infrared LED (for transmission) to provide low speed IrDA compatibility.

UART supports NRZ encoding format , RS485 compatible 9 bit data format and IrDA-compatible infrared slow data rate (SIR) format.

[Figure 16-9](#) is the UART block diagram.

The "Module Clock" is the UART_CLK which comes from CCM. The "Peripheral Clock" is the IPG_CLK which comes from CCM.

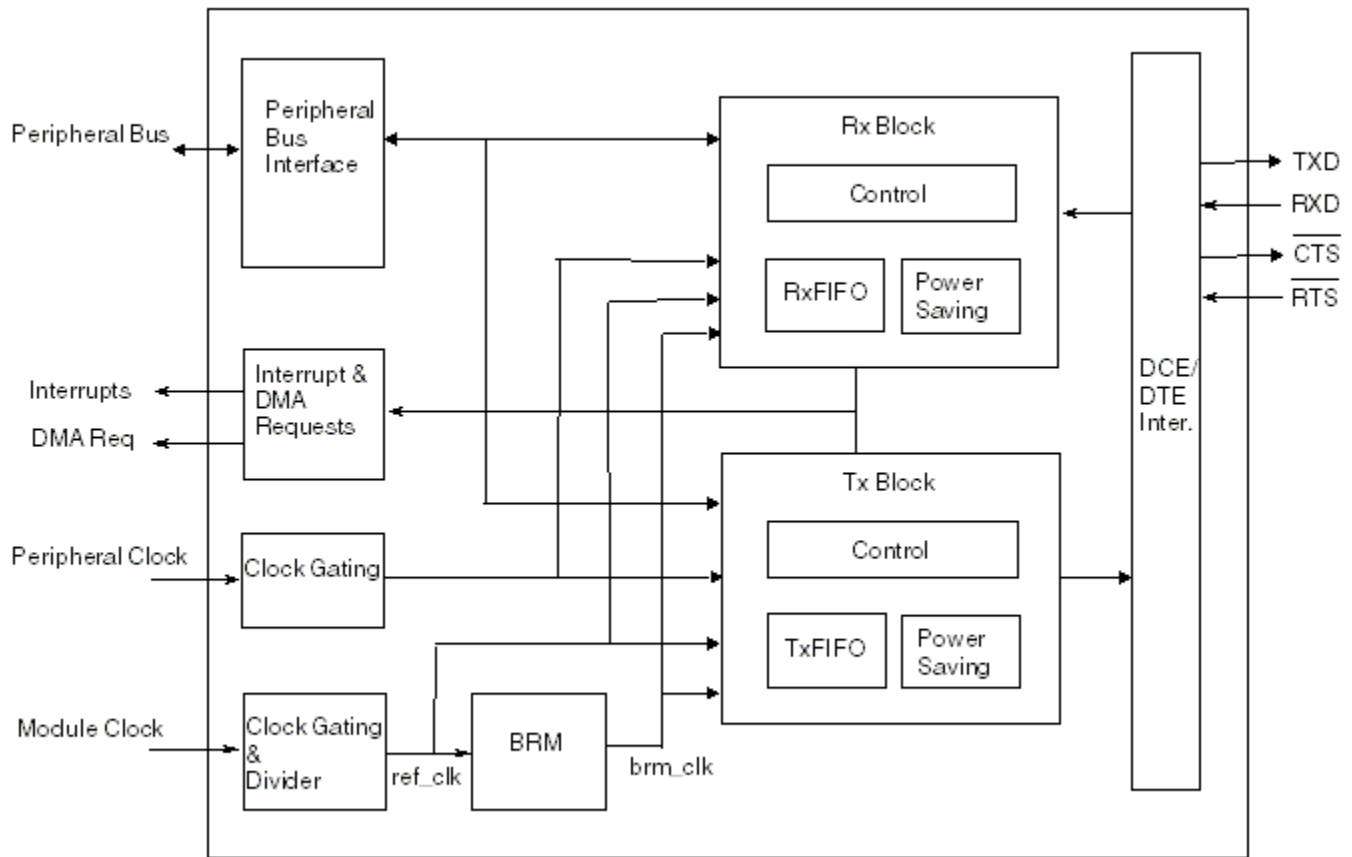


Figure 16-9. UART Block Diagram

16.2.1.1 Features

The UART includes the following features:

- High-speed TIA/EIA-232-F compatible, up to Mbit/s
- Serial IR interface low-speed, IrDA-compatible (up to 115.2 Kbit/s)
- 9-bit or Multidrop mode (RS-485) support (automatic slave address detection)
- 7 or 8 data bits for RS-232 characters, or 9 bit RS-485 format
- 1 or 2 stop bits
- Programmable parity (even, odd, and no parity)
- Hardware flow control support for request to send (RTS_B) and clear to send (CTS_B) signals
- RS-485 driver direction control via CTS_B signal
- Edge-selectable RTS_B and edge-detect interrupts
- Status flags for various flow control and FIFO states
- Voting logic for improved noise immunity (16x oversampling)
- Transmitter FIFO empty interrupt suppression

- UART internal clocks enable/disable
- Auto baud rate detection (up to 115.2 Kbit/s)
- Receiver and transmitter enable/disable for power saving
- RX_DATA input and TX_DATA output can be inverted respectively in RS-232/RS-485 mode
- DCE/DTE capability
- RTS_B, IrDA asynchronous wake (AIRINT), receive asynchronous wake (AWAKE) interrupts wake the processor from STOP mode
- Maskable interrupts
- Two DMA Requests (TxFIFO DMA Request and RxFIFO DMA Request)
- Escape character sequence detection
- Software reset (SRST_B)
- Two independent, 32-entry FIFOs for transmit and receive
- The peripheral clock can be totally asynchronous with the module clock. The module clock determines baud rate. This allows frequency scaling on peripheral clock (such as during DVFS mode) while remaining the module clock frequency and baud rate.

16.2.1.2 Modes of operation

- Serial RS-232NRZ mode
- 9-bit RS-485 mode
- IrDA mode

To set UART in different modes, see the table below.

Table 16-3. UART mode definition

MDEN (UMCR[0])	IREN (UCR1[7])	UART Mode	Description
0	0	RS-232	RXD/TXD data is serial RS-232 NRZ format
0	1	IrDA (Interface)	RXD/TXD data is IrDA-compatible infrared slow data rate (SIR) format
1	0	RS-485	RXD/TXD data is RS485 compatible 9 bit data format
1	1	Undefined	Undefined

16.2.2 External Signals

The chip-level IOMUX modifies the direction and routing of the UART signals based on whether the UART is operating in DCE mode (UARTn_UFCR[DCEDTE]=0) or DTE mode (UARTn_UFCR[DCEDTE]=1). The routing of the external signals to the UART module is shown in the figure below.

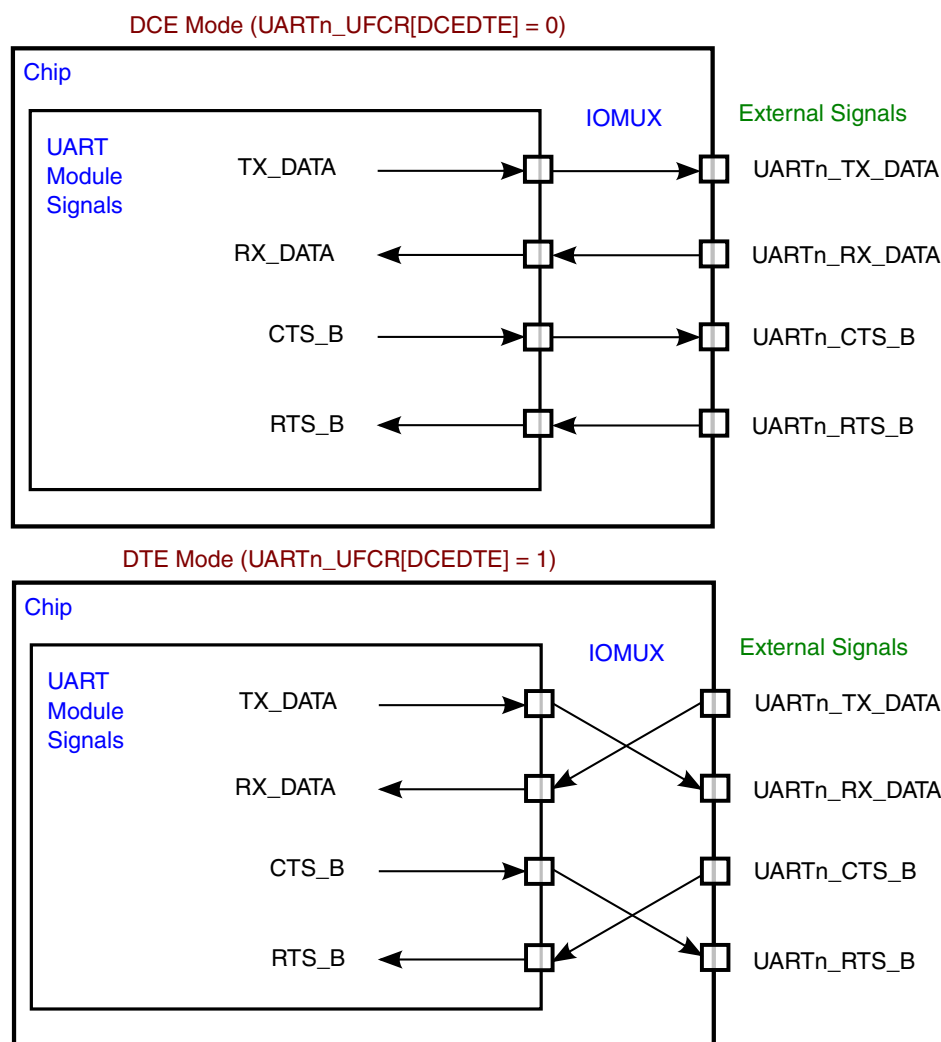


Figure 16-10. UART external signals to module signals routing with respect to DCE/DTE mode

The user must configure the input path to the UART by properly configuring the DAISY bits in the IOMUXC_UARTn_RX_DATA_INPUT and the IOMUXC_UARTn_UART_RTS_B_SELECT_INPUT registers.

For IOMUXC_UARTn_UART_RTS_B_SELECT_INPUT[DAISY]:

- Configurations that select UARTn_RTS_B for the pad are only valid when UARTn_UFCR[DCEDTE]=0 (DCE mode)
- Configurations that select UARTn_CTS_B for the pad are only valid when UARTn_UFCR[DCEDTE]=1 (DTE mode)

For IOMUXC_UARTn_UART_RX_DATA_B_SELECT_INPUT[DAISY]:

- Configurations that select UARTn_RX_DATA for the pad are only valid when UARTn_UFCR[DCEDTE]=0 (DCE mode)
- Configurations that select UARTn_TX_DATA for the pad are only valid when UARTn_UFCR[DCEDTE]=1 (DTE mode)

16.2.2.1 Detailed Signal Descriptions

16.2.2.1.1 Interrupt Signals

16.2.2.1.1.1 *interrupt_uart* - UART Interrupt

Output interrupt request.

16.2.2.1.2 DMA Request Signals

16.2.2.1.2.1 *dma_req_rx* - Receiver DMA Request

Output DMA Request signal for receiver interface.

16.2.2.1.2.2 *dma_req_tx* - Transmitter DMA Request

Output DMA Request signal for transmitter interface. Set at 0 when TXDMAEN (UCR1[3]) is at 1 and TRDY (USR1[13]) is also at 1.

16.2.2.1.3 Special Signals

16.2.2.1.3.1 *stop_req* - Stop Mode

Input stop mode. Indicates to UART that Arm platform is going to enter in Stop Mode and clocks are going to stop running.

See [Low Power Modes](#) for more information about Stop Mode.

16.2.2.1.3.2 *doze_req* - Doze Mode

Input doze mode. Arm platform requests UART to switch in doze mode (power saving mode).

See [Low Power Modes](#) for more information about Doze Mode.

16.2.2.1.3.3 *debug_req* - Debug Mode

Input debug mode. Indicates UART it has to enter in debug mode.

See [UART Operation in System Debug State](#), for more information about Debug Mode.

16.2.3 Clocks

The table found here describes the clock sources for UART.

Please see [Clock Controller Module \(CCM\)](#) for clock setting, configuration and gating information.

Table 16-4. UART Clocks

Clock name	Clock Root	Description
ipg_clk	ipg_clk_root	Peripheral clock
ipg_clk_s	ipg_clk_root	Peripheral access clock
ipg_perclk	uart_clk_root	Module clock

16.2.4 Functional Description

This section provides a complete functional description of the block.

16.2.4.1 Interrupts and DMA Requests

See the following table for the lists of all interrupt and DMA signals and associated interrupt and DMA sources of the UART. See register description section for explanation of interrupt/DMA enable and status.

Table 16-5. Interrupts and DMA

Interrupt/DMA Output	Interrupt/DMA Enable	Enable Register Location	Interrupt/DMA Flag	Flag Register Location
<i>interrupt_uart</i>	RRDYEN	UCR1 (bit 9)	RRDY	USR1 (bit 9)
	IDEN	UCR1 (bit 12)	IDLE	USR2 (bit 12)
	DREN	UCR4 (bit 0)	RDR	USR2 (bit 0)
	RXDSEN	UCR3 (bit 6)	RXDS	USR1 (bit 6)
	ATEN	UCR2 (bit 3)	AGTIM	USR1 (bit 8)
<i>interrupt_uart</i>	TXMPTYEN	UCR1 (bit 6)	TXFE	USR2 (bit 14)

Table continues on the next page...

Table 16-5. Interrupts and DMA (continued)

Interrupt/DMA Output	Interrupt/DMA Enable	Enable Register Location	Interrupt/DMA Flag	Flag Register Location
<i>interrupt_uart</i>	TRDYEN	UCR1 (bit 13)	TRDY	USR1 (bit 13)
	TCEN	UCR4 (bit 3)	TXDC	USR2 (bit 3)
	OREN	UCR4 (bit 1)	ORE	USR2 (bit 1)
	BKEN	UCR4 (bit 2)	BRCD	USR2 (bit 2)
	WKEN	UCR4 (bit 7)	WAKE	USR2 (bit 7)
	ADEN	UCR1 (bit 15)	ADET	USR2 (bit 15)
	ACIEN	UCR3 (bit 0)	ACST	USR2 (bit 11)
	ESCI	UCR2 (bit 15)	ESCF	USR1 (bit 11)
	ENIRI	UCR4 (bit 8)	IRINT	USR2 (bit 8)
	AIRINTEN	UCR3 (bit 5)	AIRINT	USR1 (bit 5)
	AWAKEN	UCR3 (bit 4)	AWAKE	USR1 (bit 4)
	FRAERREN	UCR3 (bit 11)	FRAERR	USR1 (bit 10)
	PARERREN	UCR3 (bit 12)	PARITYERR	USR1 (bit 15)
	RTSDEN	UCR1 (bit 5)	RTSD	USR1 (bit 12)
	RTSEN	UCR2 (bit 4)	RTSF	USR2 (bit 4)
	DTREN (DCE)	UCR3 (bit 13)	DTRF	USR2 (bit 13)
	RI (DTE)	UCR3 (bit 8)	RIDELT	USR2 (bit 10)
	DCD (DTE)	UCR3 (bit 9)	DCDDELT	USR2 (bit 6)
	DTRDEN	UCR3 (bit 3)	DTRD	USR1 (bit 7)
	SADEN	UMCR (bit 3)	SAD	USR1 (bit 3)
<i>dma_req_rx</i>	RXDMAEN	UCR1 (bit 8)	RRDY	USR1 (bit 9)
	ATDMAEN	UCR1 (bit 2)	AGTIM	USR1 (bit 8)
	IDDMAEN	UCR4 (bit 6)	IDLE	USR2 (bit 12)
<i>dma_req_tx</i>	TXDMAEN	UCR1 (bit 3)	TRDY	USR1 (bit 13)

16.2.4.2 Clocks

This section describes clocks and special clocking requirements of the UART.

16.2.4.2.1 Clock requirements

UART module receives 2 clocks, *peripheral_clock* and *module_clock*. The *peripheral_clock* is used as write clock of the TxFIFO, read clock of the RxFIFO and synchronization of the modem control input pins. It must always be running when UART is enabled. There is an exception in stop mode (see [Clocking in Low-Power Modes](#)).

The *module_clock* is for all the state machines, writing RxFIFO, reading TxFIFO, etc. It must always be running when UART is sending or receiving characters. This clock is used in order to allow frequency scaling on *peripheral_clock* without changing configuration of baud rate (*module_clock* staying at a fixed frequency).

The constraints on *peripheral_clock* and *module_clock* are as follows:

- *peripheral_clock* and *module_clock* can totally be asynchronous. They can also be synchronous.
- Due to the 16x oversampling of the incoming characters, *module_clock* frequency must always be greater or equal to 16x the maximum baud rate. For example, if max baud rate is 4 Mbit/s, *module_clock* must be greater or equal to $4 \text{ M} \times 16 = 64 \text{ MHz}$.

NOTE

The restriction that *peripheral_clock* frequency must be higher or equal to 16x baud rate has been removed. There is no limitation on *peripheral_clock* frequency to baud rate.

16.2.4.2.2 Maximum Baud Rate

The max baud rate the UART can support is determined by the max frequency of the *module_clock*.

For example, if the SoC can provide the fastest *module_clock* 66.5 MHz, the UART can transmit and receive serial data with the maximum baud rate $66.5\text{M}/16 = 4.15 \text{ Mbit/s}$.

The UART supports serial IR interface low speed. In the low speed IrDA mode, the max baud rate is 115.2 Kbit/s. To support the 115.2 Kbit/s, *module_clock* frequency must be higher or equal to 1.8432 MHz.

16.2.4.2.3 Clocking in Low-Power Modes

The UART supports 2 low-power modes: DOZE and STOP.

In STOP mode (input pin *stop_req* is at '1'), the UART doesn't need any clock. In this mode the UART can wake-up the Arm platform with the asynchronous interrupts (see [Low Power Modes](#)).

- If before entering in STOP mode the software has enabled RTSDEN interrupt, when RTS will change state (put at '0' by external device started to send), the asynchronous interrupt will wake-up the system, *peripheral_clock* and *module_clock* will be provided to the UART before first start bit, so that no data will be lost.
- If RTS doesn't change state (already at '0' before entering in STOP mode), then wake-up interrupt (AWAKE) will be sent at the arrival of first Start bit (on falling

edge). In this case, the UART must receive the *peripheral_clock* and *module_clock* during the first half of start bit to correctly receive this character (for example, at 115.2 Kbit/s, UART must receive *peripheral_clock* and *module_clock* at maximum 4.3 microseconds after falling edge of Start bit). If the UART receives *peripheral_clock* and *module_clock* too late, first character will be lost, and so should be dropped. Also, if autobaud detection is enabled, the first character won't be correctly received and another autobaud detection will need to be initiated.

In Doze mode, UART behavior is programmable through DOZE bit (UCR1[1]). If DOZE bit is set to '1', then UART is disabled in Doze mode, and in consequence, UART clocks can be switched-off (after being sure UART is not transmitting nor receiving). On the contrary, if DOZE bit is set to '0', UART is enabled and it must receive *peripheral_clock* and *module_clock*.

16.2.4.3 General UART Definitions

Definitions of terms that occurs the following discussions are given in this section.

- Bit Time-The period of time required to serially transmit or receive 1 bit of data (1 cycle of the baud rate frequency).
- Start bit-The bit time of a logic 0 that indicates the beginning of a data frame. A start bit begins with a 1-to-0 transition, and is preceded by at least 1 bit time of logic 1.
- Stop bit-1 bit time of logic 1 that indicates the end of a data frame.
- BREAK-A frame in which all of the data bits, including the stop bit, are logic 0. This type of frame is usually sent to signal the end of a message or the beginning of a new message.
- Mark - When no data is being sent, the serial port's transmit pin's voltage is 1 and is said to be in a MARK state.
- Space - The serial port can also be forced to keep the transmit pin at a 0 and is said to be the SPACE or BREAK state.
- Frame-A start bit followed by a specified number of data or information bits and terminated by a stop bit. The number of data or information bits depends on the format specified and must be the same for the transmitting device and the receiving device. The most common frame format is 1 start bit followed by 8 data bits (least significant bit first) and terminated by 1 stop bit. An additional stop bit and a parity bit also can be included.
- Framing Error-An error condition that occurs when the stop bit of a received frame is missing, usually when the frame boundaries in the received bit stream are not synchronized with the receiver bit counter. Framing errors can go undetected if a data bit in the expected stop bit time happens to be a logic 1. A framing error is always present on the receiver side when the transmitter is sending BREAKs. However,

when the UART is programmed to expect 2 stop bits and only the first stop bit is received, this is not a framing error by definition.

- **Parity Error**-An error condition that occurs when the calculated parity of the received data bits in a frame does not match the parity bit received on the RX_DATA input. Parity error is calculated only after an entire frame is received.
- **Idle-One** in NRZ encoding format and selectable polarity in IrDA mode.
- **Overrun Error**-An error condition that occurs when the latest character received is ignored to prevent overwriting a character already present in the UART receive buffer (RxFIFO). An overrun error indicates that the software reading the buffer (RxFIFO) is not keeping up with the actual reception of characters on the RX_DATA input.

16.2.4.3.1 RTS_B - UART Request To Send

The UART Request To Send input controls the transmitter. The modem or other terminal equipment signals the UART when it is ready to receive by setting '0' on the RTS_B pin.

Normally, the transmitter waits until this signal is active (low) before transmitting a character, however when the Ignore RTS (IRTS) bit is set, the transmitter sends a character as soon as it is ready to transmit. An interrupt (RTSD) can be posted on any transition of this pin and can wake the Arm platform from STOP mode on its assertion. When RTS_B is set to '1' during a transmission, the UART transmitter finishes transmitting the current character and shuts off. The contents of the TxFIFO (characters to be transmitted) remain undisturbed. The operation of this input is the same regardless of whether the UART is in DTE or DCE mode.

16.2.4.3.2 RTS Edge Triggered Interrupt

The input to the RTS_B pin can be programmed to generate an interrupt on a selectable edge.

See the table below for summary of the operation of the RTS edge triggered interrupt (RTSF).

To enable the RTS_B pin to generate an interrupt, set the request to send interrupt enable (RTSEN) bit (UCR2[4]) to 1. Writing 1 to the RTS_B edge triggered interrupt flag (RTSF) bit (USR2[4]) clears the interrupt flag. The interrupt can occur on the rising edge, falling edge, or either edge of the RTS_B input. The request to send edge control (RTEC) field (UCR2[10:9]) programs the edge that generates the interrupt. When RTEC is set to 0x00 and RTSEN = 1, the interrupt occurs on the rising edge (default). When RTEC is set to 0x01 and RTSEN = 1, the interrupt occurs on the falling edge. When RTEC is set to 0x1X and RTSEN = 1, the interrupt occurs on either edge. This is a synchronous interrupt. The RTSF bit is cleared by writing 1 to it. Writing 0 to RTSF has no effect.

Table 16-6. RTS_B Edge Triggered Interrupt Truth Table

RTS_B	RTSEN	RTEC [1]	RTEC [0]	RTSF	Interrupt Occurs On...	interrupt_uart
X	0	X	X	0	Interrupt disabled	1
1->0	1	0	0	0	Rising edge	1
0->1	1	0	0	1	Rising edge	0
1->0	1	0	1	1	Falling edge	0
0->1	1	0	1	0	Falling edge	1
1->0	1	1	X	1	Either edge	0
0->1	1	1	X	1	Either edge	0

There is another RTS_B interrupt that is not programmable. The status bit RTSD asserts the *interrupt_uart* interrupt when the RTS_B delta interrupt enable = 1. This is an asynchronous interrupt. The RTSD bit is cleared by writing 1 to it. Writing 0 to the RTSD bit has no effect.

16.2.4.3.3 CTS_B - Clear To Send

This output pin serves two purposes. Normally, the receiver indicates that it is ready to receive data by asserting this pin (low). When the CTS_B trigger level is programmed to trigger at 32 characters received and the receiver detects the valid start bit of the 33 character, it de-asserts this pin. The operation of this output is the same regardless of whether the UART is in DTE or DCE mode.

16.2.4.3.4 Programmable CTS_B Deassertion

The CTS_B output can also be programmed to deassert when the Rx FIFO reaches a certain level. Setting the CTS trigger level (UCR4[15:10]) at any value less than 32 deasserts the CTS_B pin on detection of the valid start bit of the N + 1 character (where N is the trigger level setting). However, the receiver continues to receive characters until the Rx FIFO is full.

16.2.4.3.5 TX_DATA - UART Transmit

This is the transmitter serial output. When operating in RS-232/RS-485 mode, NRZ encoded data is transmitted, and the data can be inverted (controlled by INVT (UCR3[1])) before transmitted. When operating in infrared mode, a 3/16 bit-period pulse is output for each 0 bit transmitted, and no pulse is output for each 1 bit transmitted.

For RS-232/RS-485 applications, this pin must be connected to an RS-232/RS-485 transmitter. The operation of this output is the same regardless of whether the UART is in DTE or DCE mode. See [Figure 16-11](#).

16.2.4.3.6 RX_DATA - UART Receive

This is the receiver serial input. When operating in RS-232/RS-485 mode, NRZ encoded data is expected, and the data can be inverted (controlled by INVR (UCR4[9])) before sampled. When operating in infrared mode, a narrow pulse is expected for each 0 bit received and no pulse is expected for each 1 bit received.

External circuitry must convert the IR signal to an electrical signal. RS-232/RS-485 applications require an external RS-232/RS-485 receiver to convert voltage levels. The operation of this input is the same regardless of whether the UART is in DTE or DCE mode. See the figure below.

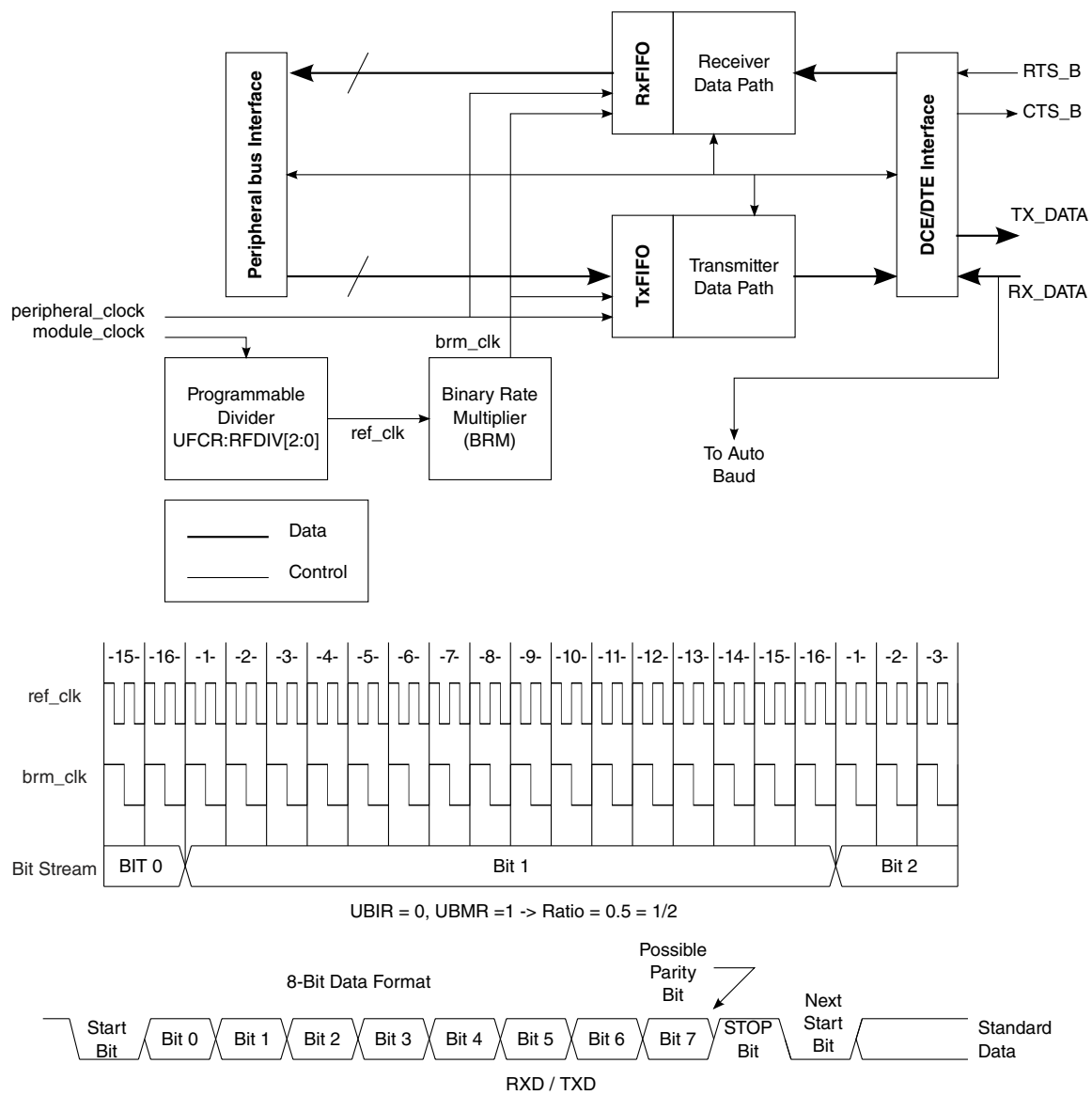


Figure 16-11. UART Simplified Block and Clock Generation Diagrams

16.2.4.4 Transmitter

The transmitter accepts a parallel character from the Arm platform and transmits it serially. The start, stop, and parity (when enabled) bits are added to the character.

When the ignore RTS bit (IRTS) is set, the transmitter sends a character as soon as it is ready to transmit. RTS_B can be used to provide flow-control of the serial data. When RTS_B is set to '1', the transmitter finishes sending the character in progress (if any), stops, and waits for RTS_B to be set to '0' again. Generation of BREAK characters and parity errors (for debugging purposes) is supported. The transmitter operates from the clock provided by the Binary Rate Multiplier(BRM). Normal NRZ encoded data is transmitted when the IR interface is disabled.

The transmitter FIFO (TxFIFO) contains 32 bytes. The data is written to TxFIFO by writing to the UTXD register with the byte data to the [7:0] bits. The data is written consecutively if the TxFIFO is not full. It is read (internally) consecutively if the TxFIFO is not empty. TXFULL bit (UTS[4]) can be used to control whether TXFIFO is full or not. The TxFIFO can be written regardless of the transmitter is disabled or enabled. If the UART is disabled, user can still write data into the TxFIFO correctly. But in this case the write access will yield to a transfer error.

16.2.4.4.1 Transmitter FIFO Empty Interrupt Suppression

The transmitter FIFO empty interrupt suppression logic suppresses the TXFE interrupt between writes to the TxFIFO.

When TxFIFO is empty, the software can either send one or several characters. If the software sends one character, it would write the character into the UTXD register, then that character is immediately transferred to the transmitter shift register, assuming the transmitter is already enabled. Without interrupt suppression logic, the TXFE interrupt flag would be set immediately. But, with this logic, the interrupt flag is set when the last bit of the character has been transmitted, for example, before the transmission of the parity bit (if exists) and the stop bit(s).

So, the suppression logic doesn't immediately send the TXFE interrupt flag. It allows the software to write another character to the TxFIFO before the interrupt flag is asserted.

When the transmitter shift register empties before another character is written to the TxFIFO, the interrupt flag is asserted. Writing data to the TxFIFO would release the interrupt flag. The interrupt flag is asserted on the following conditions:

- System Reset
- UART software reset

- When a single character has been written to Transmitter FIFO and then the Transmitter FIFO and the Transmitter Shift Register become empty until another character is written to the Transmitter FIFO
- The last character in the TxFIFO is transferred to the shift register, when TxFIFO contains two or more characters. See the figure below.

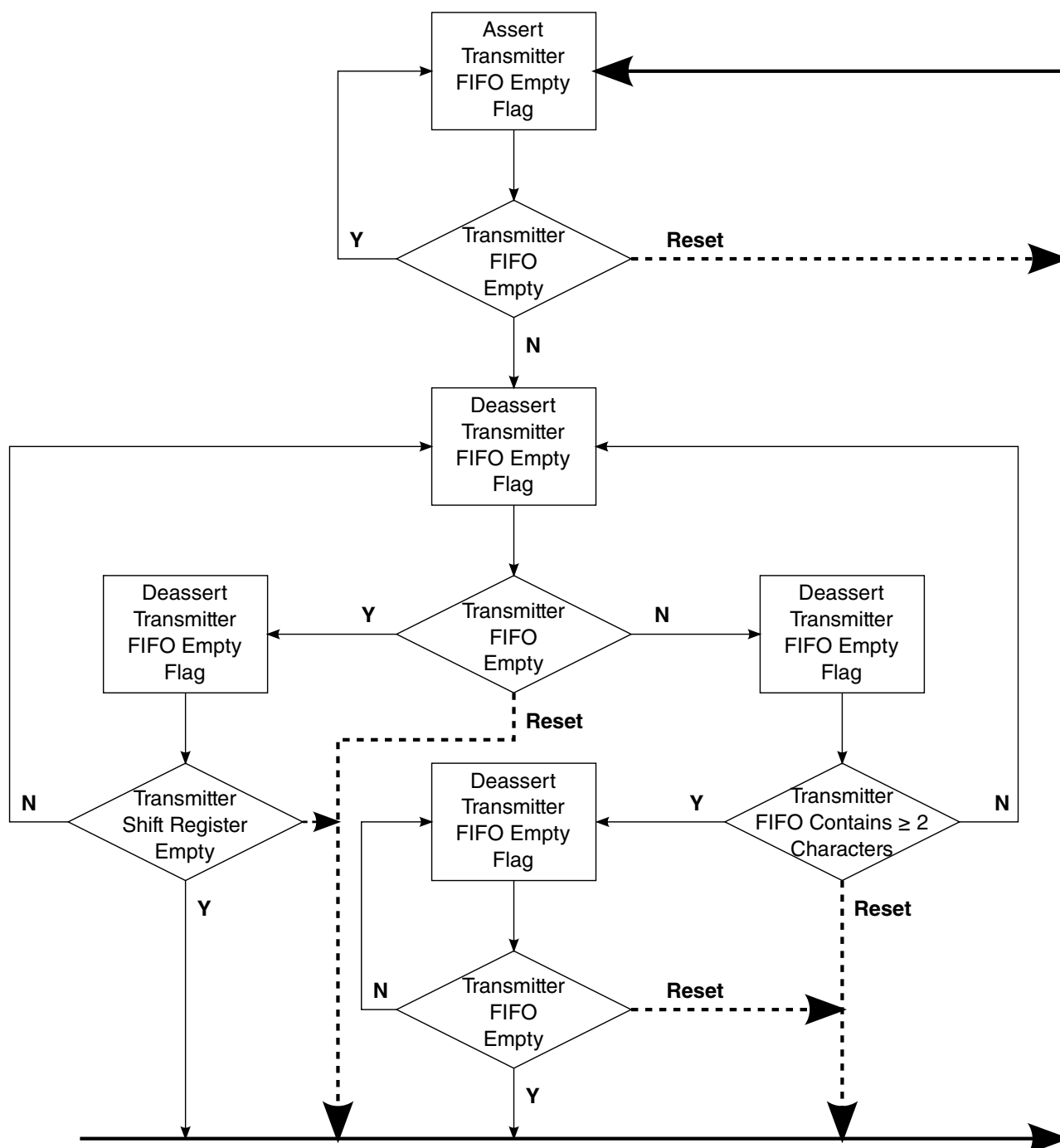


Figure 16-12. Transmitter FIFO Empty Interrupt Suppression Flow Chart

16.2.4.4.2 Transmitting a Break Condition

Asserting SNDBRK bit of the UCR1 Register forces the transmitter to send a break character (continuous zeros). The transmitter will finish sending the character in progress (if any) before sending break until this bit is reset.

The user is responsible to ensure that this bit is high for long enough to generate a valid BREAK. The transmitter samples SNDBRK after every bit is transmitted. Following completion of the BREAK transmission, the UART will transmit two mark bits. The user can continue to fill the FIFO and any character remaining will be transmitted when the break is terminated.

16.2.4.5 Receiver

See the figure below for the receiver flow chart.

The receiver accepts a serial data stream and converts it into parallel characters. When enabled, it searches for a start bit, qualifies it, and samples the following data bits at the bit-center.

Jitter tolerance and noise immunity are provided by sampling at a 16x rate and using voting techniques to clean up the samples. Once the start bit is found, the data bits, parity bit (if enabled), and stop bits (either 1 or 2 depending on user selection) are shifted in. Parity is checked and its status reported in the URXD register when parity is enabled. Frame errors and BREAKs are also checked and reported. When a new character is ready to be read by the Arm platform from the Rx FIFO, the receive data ready (RDR = $USR2[0]$) bit is asserted and an interrupt is posted (if $DREN = UCR4[0] = 1$). If the receiver trigger level is set to 2 ($RXTL[5:0] = UFCR[5:0] = 2$), and 2 chars have been received into Rx FIFO, the receiver ready interrupt flag ($RRDY = USR1[9]$) is asserted and an interrupt is posted if the receiver ready interrupt enable bit is set ($RRDYEN = UCR1[9] = 1$). If the UART Receiver Register (URXD) is read once, and in consequence there is only 1 character in the Rx FIFO, the interrupt generated by the RDR bit is automatically cleared. The RRDY bit is cleared when the data in the Rx FIFO falls below the programmed trigger level.

Normal NRZ encoded data is expected when the IR interface is disabled. The Rx FIFO contains 32 half-word entries. Characters received are written consecutively into this FIFO. If the FIFO is full and a 33rd character is received, this character will be ignored and the $USR2[ORE]$ bit will be set.

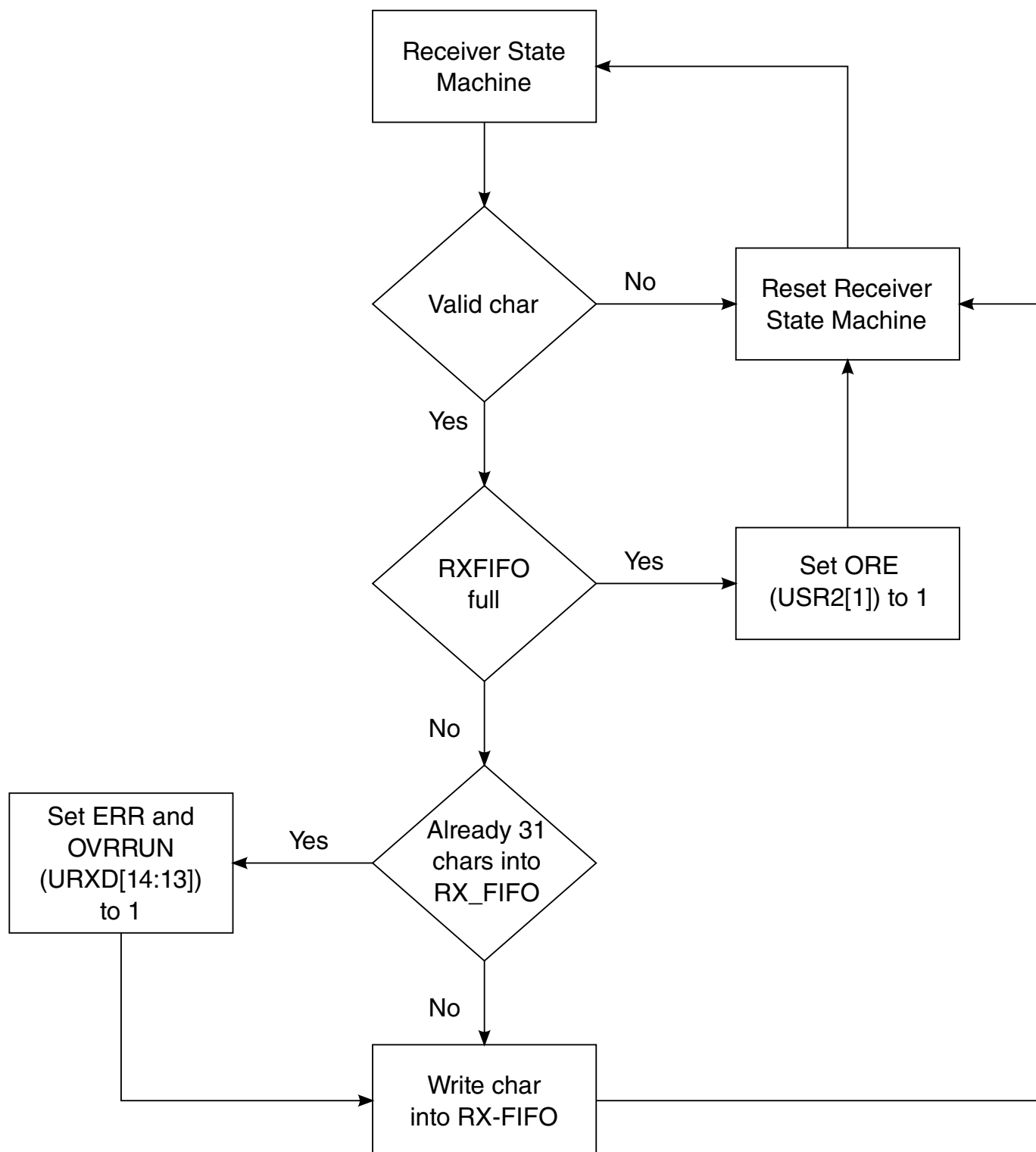


Figure 16-13. Receiver Flow Chart

16.2.4.5.1 Idle Line Detect

The receiver logic block includes the ability to detect an idle line. Idle lines indicate the end or the beginning of a message.

For an idle condition to occur:

- RxFIFO must be empty and
- RX_DATA pin must be idle for more than a configured number of frames (ICD[1:0] = UCR1[11:10]).

When the idle condition detected interrupt enable (IDEN = UCR1[12]) is set and the line is idle for 4 (default), 8, 16, or 32 (maximum) frames, the detection of an idle condition flags an interrupt (see the table below). When an idle condition is detected, the IDLE (USR2[12]) bit is set. Clear the IDLE bit by writing 1 to it. Writing 0 to the IDLE bit has no effect.

Table 16-7. Detection Truth Table

IDEN	ICD [1]	ICD [0]	IDLE	<i>interrupt_uart</i>
0	X	X	0	1
1	0	0	asserted after 4 idle frames	asserted after 4 idle frames
1	0	1	asserted after 8 idle frames	asserted after 8 idle frames
1	1	0	asserted after 16 idle frames	asserted after 16 idle frames
1	1	1	asserted after 32 idle frames	asserted after 32 idle frames
NOTE: This table assumes that no other interrupt is set at the same time this interrupt is set for the <i>interrupt_uart</i> signal. This table shows how this interrupt affects the <i>interrupt_uart</i> signal.				

During a normal message there is no idle time between frames. When all of the information bits in a frame are logic 1s, the start bit ensures that at least one logic 0 bit time occurs for each frame so that the IDLE bit is not asserted.

16.2.4.5.2 Aging Character Detect

The receiver block also includes the possibility to detect when at least one character has been sitting into the RxFIFO for a time corresponding to 8 characters. This aging character capability allows the UART to inform the Arm platform that there is less character into the RxFIFO than the Rx trigger and, no new character has been detected on the RXD line.

The aging capability is a timer which starts to count as soon as the RxFIFO is not empty and its trigger level is not reached (RRDY=0). This counter is reset when either a RxFIFO read is performed or another character starts to present on the RXD line. If none of those two events occurs, the bit AGTIM (USR1[8]) is set when the counter has

measured a time corresponding to 8 characters. AGTIM is cleared by writing a 1 to it. AGTIM can flag an interrupt to Arm platform on *interrupt_uart* if ATEN (UCR2[3]) has been set.

To summarize, AGTIM is set when:

- There is at least one character into RxFIFO.
- No read has occurred on RxFIFO and RXD line has stayed high, for a time corresponding to 8 characters.
- The RxFIFO trigger is not reached (RRDY=0)

16.2.4.5.3 Receiver Wake

The WAKE bit (USR2[7]) is set when the receiver detects a qualified Start bit. For this, two conditions must be fulfilled, firstly a falling edge on RX_DATA line must be detected and secondly the RX_DATA line must stay at low level for more than a half-bit duration.

When the wake interrupt enable WKEN (UCR4[7]) bit is enabled, the receiver flags an interrupt (*interrupt_uart*) if the WAKE status bit is set. The WAKE bit is cleared by writing 1 to it. Writing 0 to the WAKE bit has no effect. The WAKE status bit can be asserted in either serial RS-232 mode or IR mode. The generation of the WAKE interrupt needs the clock *module_clock* .

When the asynchronous wake interrupt (AWAKE) is enabled (AWAKEN = UCR3[4] = 1), and the Arm platform is in STOP mode, and UART clocks have been shut-off, then a falling edge detected on the receive pin (RX_DATA) asserts the AWAKE bit (USR1[4]) and the *interrupt_uart* interrupt to wake the Arm platform from STOP mode. Re-enable UART clocks and clear the AWAKE bit by writing 1 to it. Writing 0 to the AWAKE bit has no effect. When IR interface is enabled (UCR1[7]=1), the AWAKE bit is always not asserted. The generation of the asynchronous AWAKE interrupt does not need any clocks.

In IR mode, if the asynchronous IR WAKE interrupt is enabled (AIRINTEN = UCR3[5] = 1), and if the Arm platform is in STOP mode (UART clocks are off when Arm platform in STOP mode), then the detection of a falling edge on the receive pin (RXD_IR), asserts the AIRINT bit (USR1[5]), and the *interrupt_uart* interrupt. This interrupt wakes the Arm platform from STOP mode. Software re-enables UART clocks and clear the AIRINT bit by writing 1 to it. Writing 0 to the AIRINT bit has no effect. When IR interface is disabled (UCR1[7]=0), the AIRINT bit is always not asserted. The generation of the asynchronous AIRINT interrupt does not need any clocks.

Recommended procedure for programming the asynchronous interrupts is to first clear them by writing 1 to the appropriate bit in the UART Status Register 1 (USR1). Poll or enable the interrupt for the Receiver IDLE Interrupt Flag (RXDS) in the USR1. When asserted, the RXDS bit indicates to the software that the receiver state machine is in the idle state, the next state is idle, and the RX_DATA pin is idle (high). After following this procedure, enable the asynchronous interrupt and enter STOP mode.

16.2.4.5.4 Receiving a BREAK Condition

A BREAK condition is received when the receiver detects all 0s (including a 0 during the bit time of the stop bit) in a frame. The BREAK condition asserts the BRCD bit (USR2[2]) and writes only the first BREAK character to the RxFIFO. Clear the BRCD bit by writing 1 to it. Writing 0 to the BRCD bit has no effect.

Asserting BRCD would generate an interrupt on *interrupt_uart*. The interrupt generation can be masked using the control bit BKEN (UCR4[2]). Receiving a break condition will also effect the following bits in the receiver register URXD:

URXD(11) = BRK. While high this bit indicates that the current char was detected as a break.

URXD(12) = FRMERR. The frame error bit will always be set when BRK is set.

URXD(10) = PRERR. If odd parity was selected the parity error bit will also be set when BRK is set.

URXD(14) = ERR. The error detect bit indicates that the character present in the rx data field has an error status. This can be asserted by a break.

16.2.4.5.5 Vote Logic

The vote logic block provides jitter tolerance and noise immunity by sampling with respect to a 16x clock (*brm_clk*) and using voting techniques to clean up the samples. The voting is implemented by sampling the incoming signal constantly on the rising edge of the *brm_clk*.

See [Figure 16-14](#). The receiver is provided with the majority vote value, which is 2 out of the 3 samples. For examples of the majority vote results of the vote logic, see the following table.

Table 16-8. Majority Vote Results

Samples	Vote
000	0

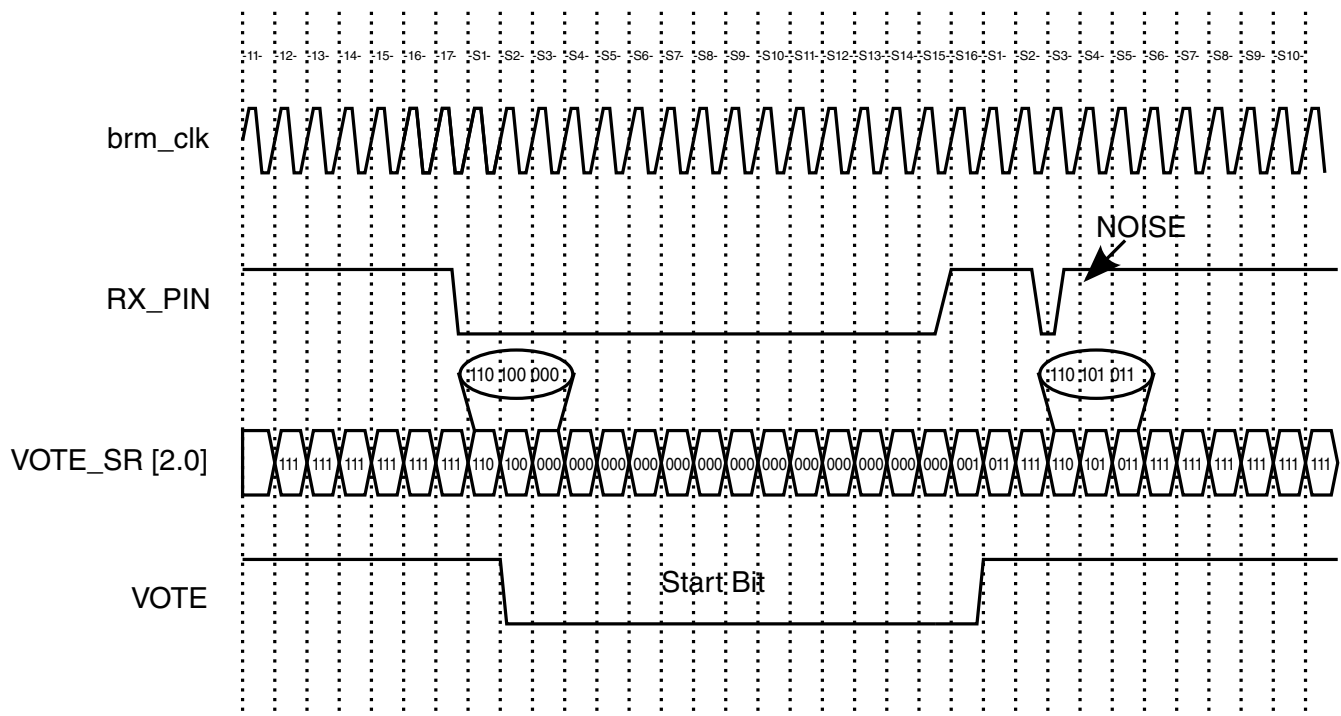
Table continues on the next page...

Table 16-8. Majority Vote Results (continued)

Samples	Vote
101	1
001	0
111	1

The vote logic captures a sample on every rising edge of *brm_clk*, however the receiver uses 16x oversampling to take its value in the middle of the sample character.

The receiver starts to count when the Start bit is set however it does not capture the contents of the RxFIFO at the time the Start bit is set. The start bit is validated when 0s are received for 7 consecutive 1/16 of bit times following the 1-to-0 transition. Once the counter reaches 0xF, it starts counting on the next bit and captures it in the middle of the sampling frame (see [Table 16-8](#)). All data bits are captured in the same manner. Once the stop bit is detected, the receiver shift register (SIPO_OUT) data is parallel shifted to the RxFIFO.

**Figure 16-14. Majority Vote Results**

A new feature has been recently implemented, it allows to re-synchronize the counter on each edge of RX_DATA line. This is automatic and allows to improve the immunity of UART against signal distortion.

There is a special case when the *brm_clk* frequency is too low and is unable to capture a 0 pulse in IrDA. In this case, the software must set the IRSC (UCR4[5]) bit so that the reference clock (after internal divider) is used for the voting logic. The pulse is validated by counting the length of the pulse.

Refer to [Infrared Interface](#) for more details.

16.2.4.5.6 Baud Rate Automatic Detection Logic

When the baud rate automatic detection logic is enabled, the UART locks onto the incoming baud rate. To enable this feature, set the automatic detection of baud rate bit (ADBR = UCR1[14] = 1) and write 1 to the ADET bit (USR2[15]) to clear it.

When ADET=0 and ADBR =1, the detection starts. Then, once the beginning of start bit (transition from 1-to-0 of RX_DATA) has been detected, UART starts a counter (UBRC) working at reference frequency. Once the end of start bit is detected (transition from 0-to-1 of RX_DATA), the value of UBRC - 1 is directly copied into UBMIR register. UBIR register is filled with 0x000F.

So, at the end of start bit, registers gets following values:

```
UBRC = number of reference clock periods (after divider) during Start bit.
UBIR = 0x000F
UBMR = UBRC - 1
```

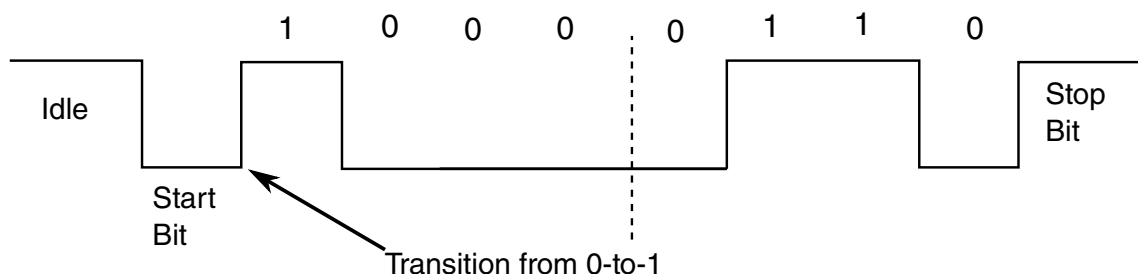
The updated values of the 3 registers can be read.

See [Table 16-9](#) for list of parameters for baud rate detection and [Figure 16-15](#) for baud rate detection protocol diagram.

If any of the UART BRM registers are simultaneously written by the baud rate automatic detection logic and by the peripheral data bus, the peripheral data bus would have lower priority.

Table 16-9. Baud Rate Automatic Detection

ADBR	ADET	Baud Rate Detection	<i>interrupt_uart</i>
0	X	Manual Configuration	1
1	0	Auto Detection Started	1
1	1	Auto Detection Complete	0
NOTE: This table assumes that no other interrupt is set at the same time this interrupt is set for the <i>interrupt_uart</i> signal.			



Note: LSB Transmitted first.

Figure 16-15. Baud Rate Detection Protocol Diagram

16.2.4.5.6.1 Baud Rate Automatic Detection Protocol

The receiver must receive an ASCII character "A" or "a" to verify proper detection of the incoming baud rate. When an ASCII character "A" (0x41) or "a" (0x61) is received and no error occurs, the Automatic Detect baud rate bit is set (ADET=1) and if the interrupt is enabled (ADEN=UCR1[15]=1), an interrupt *interrupt_uart* is generated.

When an ASCII character "A" or "a" is not received (because of a bit error or the reception of another character), the auto detection sequence restarts and waits for another 1-to-0 transition.

As long as ADET = 0 and ADBR = 1, the UART continues to try to lock onto the incoming baud rate. Once the ASCII character "A" or "a" is detected and the ADET bit is set, the receiver ignores the ADBR bit and continues normal operation with the calculated baud rate.

The UART interrupt is active (*interrupt_uart* = 0) as long as ADET = 1 and ADBR = 1. This can be disabled by clearing the automatic baud rate detection interrupt enable bit (ADEN = 0). Before starting an automatic baud rate detection sequence, set ADET = 0 and ADBR = 1.

The RxFIFO must contain the ASCII character "A" or "a" following the automatic baud rate detection interrupt.

The 16-bit UART Baud Rate Count Register (UBRC) is reset to 4 and stays at 0xFFFF when an overflow occurs. The UBRC register counts (measures) the duration of start bit. When the start bit is detected and counted, the UART Baud Rate Count Register retains its value until the next automatic baud rate detection sequence is initiated.

The Baud Rate Count Register counts only when auto detection is enabled.

16.2.4.5.6.2 New Baud Rate Determination

In order to fight against the problems caused by the distortion and the noise on the RX_DATA line, the duration of the baud rate measurement has been extended.

Previously, as described above, this determination was based on the measurement of the START bit duration. Now, this measurement is based on the duration of START bit + bit0. Bit0 is the first bit following the START bit. In fact, the counter which is started at the falling edge of START bit is no longer stopped at next rising edge (end of START bit), but it is stopped at the next falling edge (end of bit0). As the character sent is always a "A" (41h) or a "a" (61h), this second falling edge will always be present and it will indicate the end of bit0. Once this counter is stopped, the result is divided by 2 and used by the BRM to determine the incoming baud rate.

NOTE

UBRC register contains the result of this division by two, in consequence it reflects the measurement of the duration of one bit.

16.2.4.5.6.2.1 New Autobaud Counter Stopped bit and Interrupt

A new bit has been added in USR2 register: ACST (USR2[11]). This bit is set immediately after the determination of the baud rate.

So,

- if ADNIMP is not set (default), ACST is set to 1 after the end of bit0,
- If ADNIMP is set to 1, ACST is set to 1 at the end of START bit.

If ACIEN (UCR3[0]) is set to 1, ACST will flag an interrupt on *interrupt_uart* signal. This interrupt informs the Arm platform that the BRM has just been set with the result of the bit length measurement. If needed, the Arm platform can perform a read of UBMR (or UBRC) register and determine by itself the baud rate measured. Then the Arm platform has the possibility to correct the BRM registers with the nearest standardized baud rate.

NOTE

ACST is set only if ADBR is set to 1, for example, the UART is autobauding.

Clear the ACST bit by writing 1 to it. Writing 0 to the ACST bit has no effect.

16.2.4.6 Escape Sequence Detection

An escape sequence typically consists of 3 characters entered in rapid succession (such as +++). Because these are valid characters by themselves, the time between characters determines if it is a valid escape sequence.

Too much time between two of the "+" characters is interpreted as two "+" characters, and not part of an escape sequence.

The software chooses the escape character and writes its value to the UART Escape Character Register (UESC). The software must also enable escape detection feature by setting ESCEN (UCR2[11]) to 1. The hardware compares this value to incoming characters in the RxFIFO. When an escape character is detected, the internal escape timer starts to count. The software specifies a time-out value for the maximum allowable time between 2 successive escape characters (see the table below). The escape timer is programmable in intervals of 2 ms to a maximum interval of 8.192 seconds.

Table 16-10. Escape Timer Scaling

UTIM Register	Maximum Time Between Specified Escape Characters
0x000	2 ms
0x001	4 ms
0x002	6 ms
0x003	8 ms
0x004	10 ms
...	...
0F8	498 ms
0F9	500 ms
...	...
9C3	5 s
...	...
FFD	8.188 s
FFE	8.190 s
FFF	8.192 s
NOTE: To calculate the time interval: $(\text{UTIM_Value} + 1) \times 0.002 = \text{Time_Interval}$ Example: $(09C3 + 1) \times 0.002 = 5 \text{ s.}$	

The escape sequence detection feature is available for all the reference frequencies. Before using Escape Sequence Detection, the user must fill the ONEMS register. This 24-bit register must contain the value of the UART internal frequency divided by 1000. The internal frequency is obtained after the UART internal divider which is applied on *module_clock* clock.

Example I:

- If the input clock *module_clock* frequency is 66.5 MHz.
- And if the input clock *module_clock* is divided by 2 with the internal divider:
UFCR[9:7] = 3'b100

$$\text{ONEMS} = \frac{66.5 \times 10^6}{2 \times 1000} = 33250 = 81\text{E}2\text{h}$$

Figure 16-16. Calculation of Frequency for ONEMS Register

Example II:

- If the input clock *module_clock* frequency is 66.5 MHz.
- And if the input clock *module_clock* is divided by 1 with the internal divider:
UFCR[9:7] = 3'b101

$$\text{ONEMS} = \frac{66.5 \times 10^6}{1000} = 66500 = 103\text{C}4\text{h}$$

Figure 16-17. Calculation of Frequency for ONEMS Register

The escape sequence detection interrupt is asserted when the escape sequence interrupt enable (ESCI) bit is set and an escape sequence is detected (ESCF set). Clear the ESCF bit by writing 1 to it. Writing 0 to the ESCF bit has no effect.

16.2.5 Binary Rate Multiplier (BRM)

The BRM sub-block receives *ref_clk* (*module_clock* clock after divider). From this clock, and with integer and non-integer division, BRM generates a 16x baud rate clock .

The UART transmitter will shift data out based on this 16x baud rate clock. The UART receiver will sample the serial data line based on this 16x baud rate clock. The input and output frequency ratio is programmed in the UART BRM Incremental Register (UBIR) and UART BRM MOD Register (UBMR). The output frequency is divided by the input frequency to produce this ratio. For integer division, set the UBIR = 0x000F and write the divisor to the UBMR register. All values written to these registers must be one less than the actual value to eliminate division by 0 (undefined), and to increase the maximum range of the registers.

Updating the BRM registers requires writing to both registers. The UBIR register must be written before writing to the UBMR register. If only one register is written to by the software, the BRM continues to use the previous values.

The following examples show how to determine what values are to be programmed into UBIR and UBMR for a given reference frequency and desired baud rate. The following equation can be used to help determine these values:

$$\text{BaudRate} = \frac{\text{Ref Freq}}{\left(16 \times \frac{\text{UBMR} + 1}{\text{UBIR} + 1} \right)}$$

Figure 16-18. Frequency and Baud Rate for UBIR and UBMR

With:

Reference Frequency (Hz): UART Reference Frequency (*module_clock* after RFDIV divider)

Baud Rate (bit/s): Desired baud rate.

Integer Division ÷ 21

Reference Frequency = 19.44 MHz

UBIR = 0x000F

UBMR = 0x0014

Baud Rate = 925.7 kbit/s

NOTE

Observe that each value written to the registers is one less than the actual value.

Non-Integer Division

Universal Asynchronous Receiver/Transmitter(UART)

Reference Frequency = 16 MHz
Desired Baud Rate = 920 Kbits/s

$$\frac{UBMR + 1}{UBIR + 1} = \frac{RefFreq}{16 \times BaudRate} = \frac{16 \times 10^6}{16 \times 920 \times 10^3} = 1.087$$

Ratio = 1.087 = 1087 / 1000
UBIR = 999 (decimal)= 0x3E7
UBMR = 1086 (decimal)= 0x43E
Non-Integer Division
Reference Frequency = 25 MHz
Desired Baud Rate = 920 kbit/s
Ratio = 1.69837 = 625 / 368
UBIR = 367 (decimal)= 0x16F
UBMR = 624 (decimal)= 0x270

Non-Integer Division

Reference Frequency: 30 MHz
Desired Baud Rate = 115.2 kbit/s
Ratio = 16.276043 = 65153 / 4003
UBIR = 4002 (decimal) = 0x0FA2
UBMR = 65152 (decimal) = 0xFE80

16.2.6 Infrared Interface

16.2.6.1 Generalities-Infrared

The Infrared interface is selected when IREN (UCR1[7]) is set to 1.

The Infrared Interface is compatible with IrDA Serial Infrared Physical Layer Specification. In this specification, a "zero" is represented by a positive pulse, and a "one" is represented by no pulse (line remains low).

In the UART:

In TX: For each "zero" to be transmitted, a narrow positive pulse which is 3/16 of a bit time is generated. For each "one" to be transmitted no pulse is generated (output is low). External circuitry has to be provided to drive an Infrared LED.

In RX: When receiving, a narrow negative pulse is expected for each "zero" transmitted while no pulse is expected for each "one" transmitted (input is high).

NOTE

Rx part of IR block expects to receive an inverted signal compared to IrDA specification. Circuitry external to the IC transforms the Infrared signal to an electrical signal.

The IR interface has an edge triggered interrupt (IRINT). This interrupt validates a zero bit being received. This interrupt is enabled by writing a "one" to ENIRI bit.

The behavior of Infrared Interface is determined by 3 bits INVT (UCR3[1]), INVR (UCR4[9]) and IRSC (UCR4[5]).

16.2.6.2 Inverted Transmission and Reception bits (INVT & INVR)

The values of INVT and INVR depend of the IrDA transceiver connected on the TXD_IR and RXD_IR pins of the UART. If this transceiver is not inverting on both paths Tx and Rx, a Zero is represented by a positive pulse and a One is represented by no pulse (line remains low). In this case, the bit INVT must be set to 0 and the bit INVR must be set to 1 (because Rx IR block expects an inverted signal).

On the contrary user must set INVT=1 and INVR=0 if both paths of the transceiver are inverting, that is, a Zero is represented as a negative pulse and a One is represented by no pulse (line remains high). The transceiver can also be inverting on only one path (Tx or Rx), in this case INVT and INVR must be together equal to 1 or to 0, depending on which path is inverted.

16.2.6.3 InfraRed Special Case (IRSC) Bit

The value to apply to IRSC bit is based on 2 parameters: the baud rate and the Minimum Pulse Duration (MPD) of the transceiver.

According to IrDA Standard Specification, for SIR (Serial IR) baud rates from 2.4 Kbit/s to 115.2 Kbit/s this nominal pulse duration is equal to 3/16 of a bit duration (at the selected baud rate). But, for all the baud rates a Minimum Pulse Duration is also specified. According to IrDA Standard, a Zero is represented by a light pulse, so the IrDA transceiver can't emit a light pulse shorter than the MPD. For SIR, the MPD is constant and equal to 1.41 μ s.

But user must take into account the electrical MPD associated with the transceiver on the receiver path. Typically this value is 2.0 μ s, but for some manufacturers MPD can go down to 1.0 μ s.

In order to understand the meaning of IRSC bit, one must understand how the RX path works in IrDA mode.

When the UART is in IrDA mode, a Zero is not only detected by the state of the RXD_IR line, but also with the duration of the pulse. This pulse duration can be measured with 2 different clocks. In this case, clock is selected with the IRSC bit.

- If IRSC = 0, the clock used is the BRM clock.
- If IRSC = 1, the clock used is the UART internal clock (UART clock after the divider (RFDIV)).

In normal operation, IRSC=0. This means that at any time, the user must ensure that the frequency of BRM_clock is high enough to measure the pulse. The pulse must last at least 2 BRM clock cycles. If this condition is not fulfilled, IRSC must be set to 1.

Let's examine two examples, for a Minimum Pulse Duration equal to the MPD from the IrDA SIR specification (i.e., 1.41 μ s).

1: Calculation of BRM Clock Period (Clock Period < 1.41 μ s)

The user wants to receive IrDA data at 115.2 Kbit/s. The UBIR and UBMR registers are set in order to create the BRM_clock with a frequency of $16 \times \text{baud rate} = 16 \times 115.2\text{K} = 1.843\text{ MHz}$. But at the same time, in order to correctly detect the pulse, the user must be sure that $2 \times \text{BRM_clock period}$ is lower than 1.41 μ s. Let's check:

$$\text{BRM_clock period} = 1/1843000 = 542\text{ ns}$$

So $2 \times \text{BRM_clock period} = 1.09\text{ }\mu\text{s} < 1.41\text{ }\mu\text{s}$. It is fine.

2: Calculation of BRM Clock Period (Clock Period > 1.41 μ s)

This time the user wants to receive at 19.2 Kbit/s. So, the BRM_clock is set to $16 \times 19200 = 307.2\text{ kHz}$. Let's check if $2 \times \text{BRM_clock period} < 1.41\text{ }\mu\text{s}$:

$$1. \text{ BRM_clock period} = 1/307200 = 3.25\text{ }\mu\text{s}$$

So $2 \times \text{BRM_clock period} = 6.50\text{ }\mu\text{s} >> 1.41\text{ }\mu\text{s}$. It doesn't work.

So, in this case, the BRM clock can't be used to measure the pulse duration and the user must select the UART internal clock by setting IRSC = 1.

NOTE

Like for Escape character detection, when IR Special Case is enabled (IRSC=1), the UART must measure a duration. In order to do that, the user must fill the ONEMS register. Refer to [Escape Sequence Detection](#).

16.2.6.4 IrDA interrupt

Serial infrared mode (SIR) uses an edge triggered interrupt flag IRINT (USR2[8]). When INVR = 0, detection of a falling edge on the RXD pin asserts the IRINT bit. When INVR = 1, detection of a rising edge on the RXD pin asserts the IRINT bit. When IRINT and ENIRI bits are both asserted, the *interrupt_uart* interrupt is asserted. Clear the IRINT bit by writing 1 to it. Writing 0 to the IRINT bit has no effect.

16.2.6.5 Conclusion about IrDA

Before using the UART in IrDA, the baud rate limit must be calculated. This baud rate limit will inform the user if IRSC bit has to be set or not.

Let's determine this limit:

As already described, if IRSC = 0, the following condition must always be fulfilled

$$2 \times \text{BRMClockPeriod} < \text{MinPulseDuration}$$

Figure 16-19. Calculation of Baud Rate

So,

$$\text{BRMClockFrequency} > \frac{2}{\text{MPD}}$$

So, knowing BRM_clock frequency = 16 * Baud Rate, we get:

$$\text{BaudRate} > \frac{1}{8 \times \text{MinPulseDuration}}$$

So, the user needs to set IRSC = 0 when:

- If Minimum Pulse Duration = 2.5 us and Baud Rate > 50 Kbit/s.
- If Minimum Pulse Duration = 2.0 us and Baud Rate > 62.5 Kbit/s.
- If Minimum Pulse Duration = 1.41 us and Baud Rate > 88.6 Kbit/s.

NOTE

For baud rates lower than the limit, IRSC must be set to 1.

16.2.6.6 Programming IrDA Interface

16.2.6.6.1 High Speed

As an example, the following sequence can be used to program the IrDA interface in order to send and receive characters at 115.2 Kbit/s.

Assumptions:

- Input UART clock = 90 MHz
- Internal clock divider = 3 (divide Input UART clock by 3)
- Baud rate = 115.2 Kbit/s
- IrDA transceiver is not inverting on both channels: for Tx and Rx, a Zero is represented by a positive pulse, and a One is represented by no pulse (line stays low).
- Interrupt: Sent to Arm platform when 1 char is received into the Rx FIFO (RDR)

Registers values and Programming orders:

```

UCR1 = 0x0085
UCR1[7] = IREN = 1: Enable IR interface
UCR1[0] = UARTEN = 1: Enable UART
UTS = 0x0000
UFCR = 0x0981
TXTL[5:0] = 0x02: Default value
RFDIV[2:0] = 0x3: Divide Input UART clock by 3 (resulting internal clock is 30 MHz)
RXTL[5:0] = 0x01: Default value
UBIR = 0x0202
UBMR = 0x20BE Baud rate = 115.2 kbit/s with internal clock = 30 MHz
UCR2 = 0x4027
UCR2[14] = IRTS = 1: Ignore level of RTS input signal
UCR2[5] = WS = 1: Characters are 8-bit length
UCR2[2] = TXEN = 1: Enable Rx path
UCR2[1] = RXEN = 1: Enable Tx path
UCR2[0] = SRST_B = 1: No software reset
UCR3 = 0x0000

UCR4 = 0x8201
CTSTL[5:0] = 0x20: Default value
UCR4[9] = INVR = 1: Inverted Infrared Reception (because IrDA transceiver is not inverting)
UCR4[1] = DREN = 1: To enable RDR interrupt (sent when one char is received)

```

The UART is ready to send a character as soon as there is a write into UTXD register. And an interrupt will be sent to Arm platform when a character is received.

16.2.6.6.2 Low Speed

This time, we keep the same assumptions but the speed is now 9.6 Kbit/s. So, this baud rate is below the limit (even with a Min. Pulse Duration of 2.5 us) and thus IRSC must be set to 1.

Assumptions:

- Input UART clock = 90 MHz
- Internal clock divider = 3 (divide Input UART clock by 3)
- Baud rate = 9.6 Kbit/s
- IrDA transceiver is not inverting on both channels: for Tx and Rx, a Zero is represented by a positive pulse, and a One is represented by no pulse (line stays low).
- Interrupt: Sent to Arm platform when 1 char is received into the Rx FIFO (RDR).

Registers values and Programming orders:

```

UCR1 = 0x0085
UCR1[7] = IREN = 1: Enable IR interface
UCR1[0] = UARTEN = 1: Enable UART
UFCR = 0x0981
UFCR[15:10] = TXTL[5:0] = 0x02: Default value
RFDIV[2:0] = 0x3: Divide Input UART clock by 3 (resulting internal clock is 30 MHz)
UFCR[5:0] = RXTL[5:0] = 0x01: Default value
UBIR = 0x00FF
UBMR = 0xC354 Baud rate = 9.6 kbit/s with internal clock = 30 MHz
UCR2 = 0x4027
UCR2[14] = IRTS = 1: Ignore level of RTS input signal
UCR2[5] = WS = 1: Characters are 8-bit length
UCR2[2] = TXEN = 1: Enable Rx path
UCR2[1] = RXEN = 1: Enable Tx path
UCR2[0] = SRST_B = 1: No software reset
UCR3 = 0x0000
UCR3[1] = INVT = 0: Positive pulse represents 0.
UCR4 = 0x8221
UCR4[15:10] = CTSTL[5:0] = 0x20: Default value
UCR4[9] = INVR = 1: Inverted Infrared Reception (because IrDA transceiver is not inverting)
UCR4[5] = IRSC = 1: Because data rate is below the limit and thus the UART internal clock is
used to measure the pulse duration.
UCR4[1] = DREN = 1: To enable RDR interrupt (sent when one char is received)

```

The UART is now ready to send a character as soon as there is a write into UTXD register. An interrupt will be sent to Arm platform when a character is received.

16.2.7 9-bit RS-485 Mode

16.2.7.1 Generalities

The UART provides a 9-bit mode to facilitate multidrop (RS-485) network communication. To enable this mode, set MDEN bit in the UMCR register to 1. When 9-bit RS-485 mode is enabled, UART transmitter can transmit the ninth bit (9th bit) set by TXB8, and UART receiver can differentiate between data frames (9th bit = 0) and address frames (9th bit = 1).

The CTS_B pin can be used to control RS-485 output driver outside the chip.

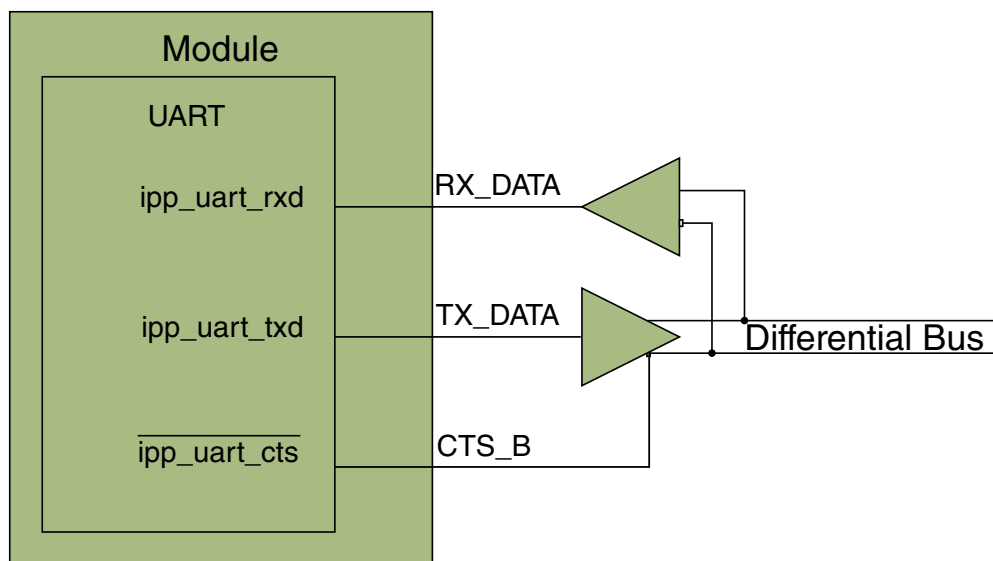


Figure 16-20. RS-485 driver connection (UART in DCE mode)

16.2.7.2 Transmit 9-bit RS-485 frames

To transmit 9-bit RS-485 frames, user need to enable parity (PREN=1) to enable trasmitting the ninth data bit, set 8-bit data word size (WS=1), and write TXB8 (UMCR[2]) as the 9th bit (bit [8]) to be transmitted (write '0' to TXB8 to transmit a data frame, write '1' to transmit a address frame). The other data bit [7:0] is written to TxFIFO by writing to the UTXD same as normal RS-232 operation.

16.2.7.3 Receive 9-bit RS-485 frames

To receive 9-bit RS-485 frames, user need to enable parity (PREN=1) to enable receiving the ninth data bit, set 8-bit data word size (WS=1). The receiver will save the 9-bit data to RxFIFO, and user should read the 9th databit (bit [8]) by reading the PRERR (URXD[10]) bit, and read data bit [7:0] by reading the RX_DATA (URXD[7:0]).

There are two slave address detect modes, normal detect mode and automatic detect mode, and can be selected by SLAM (UMCR[1]).

16.2.7.3.1 RS-485 Slave Address Normal Detect Mode

To enable Normal Detect mode, clear SLAM (UMCR[1] to 0). The receiver ignores all data frames (9th bit = 0) until an address frame is received (9th bit = 1). At that time, the slave address detected (SAD = USR1[3]) bit is asserted and the *interrupt_uart* interrupt is

generated (if SADEN = UMC[3] = 1). The address byte and subsequent bytes are all put into RxFIFO along with their 9th bit. The UART will also generate DMA request *dma_req_rx* when the RxFIFO reaches the selected threshold (controlled by RXTL) if receive ready DMA (RXDMAEN = UCR1[8]) request is enabled.

User should read the 9th databit (bit [8]) by reading the PRERR (URXD[10]) bit, and read data bit [7:0] by reading the RX_DATA (URXD[7:0]).

In this mode, once the UART has detected a 9th bit is equal to '1', it will always save the subsequent frames to RxFIFO. So the software must decide whether the address and data in RxFIFO are needed or not.

16.2.7.3.2 RS-485 Slave Address Automatic Detect Mode

To enable Automatic Detect Mode, set SLAM (UMCR[1]) to 1. The receiver tries to detect an address byte (frame 9th bit = 1) that matches the programmed SLADDR (UMCR[15:8]) character. If the received byte is a data or an address byte that does not match the programmed SLADDR character, the receiver will discard these data.

Once the UART receives a matching address byte, it will assert the slave address detected (SAD = USR1[3]) bit and the *interrupt_uart* interrupt will be generated (if SADEN = UMC[3] = 1). The address byte and subsequent bytes are all put into RxFIFO along with their 9th bit. If receive ready DMA (RXDMAEN = UCR1[8]) request is enabled, the UART will also generate DMA request *dma_req_rx* when the RxFIFO reaches the selected threshold (controlled by RXTL).

If another address byte is received and this address byte does not match SLADDR character, the receiver will discard the address byte and subsequent data byte. If the address byte again matches SLADDR character, the receiver will put this address byte and subsequent data byte in the RxFIFO along with their 9th bit.

User should read the 9th databit (bit [8]) by reading the PRERR (URXD[10]) bit, and read data bit [7:0] by reading the RX_DATA (URXD[7:0]).

See [Initialization](#) for 9-bit RS-485 programming guide.

16.2.8 Low Power Modes

These modes are controlled by the signals *doze_req* and *stop_req*. The control/status/data registers won't change when getting in/out of low power modes.

Table 16-11. UART Low Power State Operation

	Normal State (<i>doze_req</i> = 1'b0 & <i>stop_req</i> = 1'b0)	Doze State (<i>doze_req</i> = 1'b1)		Stop State (<i>stop_req</i> = 1'b1)
		DOZE bit = 0	DOZE bit = 1	
UART-Clock	ON	ON	ON	OFF
UART Serial / IrDA	ON	ON	OFF	OFF

16.2.8.1 UART Operation in System Doze Mode

While in Doze State (when *doze_req* input pin is set to 1'b1), the UART behavior depends on the DOZE (UCR1[1]) control bit.

While the DOZE bit is negated, the UART serial interface is enabled. While the system is in the Doze State, and the DOZE bit is asserted, the UART is disabled. If the Doze State is entered with the DOZE bit asserted while the UART serial interface was receiving or transmitting data, it will complete the receive/transmit of the current character and signal to the far-end transmitter/receiver to stop sending/receiving.

16.2.8.2 UART Operation in System Stop Mode

The internal baud rate clocks of the transmitter and receiver are gated off if the *stop_req* signal to UART is asserted. Even though the clocks at the input of the UART continue to run during system Stop mode, the UART will not do any transmission or reception.

The following UART interrupts wake the Arm platform processor from STOP mode:

- RTS (RTSD)
- IrDA Asynchronous WAKE (AIRINT)
- Asynchronous WAKE (AWAKE)
- RI (RIDELT in DTE mode only)
- DCD (DCDDELT in DTE mode only)
- DTR (DTRD in DCE mode only)
- DSR (DTRD in DTE mode only)

When an asynchronous WAKE (awake) interrupt exits the Arm platform from STOP mode, make sure that a dummy character is sent first because the first character may not be received correctly.

16.2.8.3 Power Saving Method in UART

The RXEN (UCR2[1]), TXEN (UCR2[2]) and UARTEN (UCR1[0]) bits are set by the user and provide software control of low-power modes.

Setting the UARTEN (UCR1[0]) bit to 0 shuts off the receiver and transmitter logic and the associated clocks.

If the UART is used only in transmit mode, UARTEN and TXEN must be set to 1. If the UART is used only in receive mode, UARTEN and RXEN must be set to 1. Setting TXEN or RXEN to 0 allows to save a lot of power.

16.2.9 UART Operation in System Debug State

The bit UTS [11] controls whether the UART will respond to the input signal *debug_req*, or whether it will continue to run as normal.

If the UART is programmed to respond to *debug_req*:

1. The UART will halt all operations upon detecting the *debug_req* input.
2. A transfer in progress, either to/from a core (using the IP Bus interface) or to/from an external device, will be completed before halting. This means a single byte/word transfer, not an entire FIFO. Reception of any further data from an external device will be disabled.
3. Internal registers will continue to be writable and readable using the IP Bus interface. A read will leave the contents unaffected.
4. The RX FIFO is affected in debug mode in the following way:
 - All writes into the RX FIFO are prevented.
 - The bit RXDBG (UTS[9]) is used to select the readability of the RX FIFO during debug mode:

RXDBG = 0: hold the read pointer at the location it had upon entering debug mode, and URXD register returns only the data value at that location, no matter how many reads attempted.

RXDBG = 1, selectable at any time: Allow to read the characters received in Rx FIFO. It will not be possible to re-read previously read locations, nor will it be possible to readjust the read pointer to the value it had prior to entering debug mode.

16.2.10 Reset

This section describes how to reset the block and explains special requirements related to reset.

16.2.10.1 Hardware reset

All of registers, FIFOs, state machines and sequential elements can be reset to their initial values by hardware reset or power on reset.

16.2.10.2 Software reset

The status registers USR1 and USR2, BRM registers UBIR and UBMIR, TxFIFO and RxFIFO, and transmitter and receiver state machines can be reset by software reset. Internal logic will keep the software reset asserted for about 4 *module_clock* cycles.

Programmer can follow the following software reset sequence:

1. Clear the SRST_B bit (UCR2[0])
2. Wait for software reset complete: poll SOFTRST bit (UTS[0]) until it is 0.
3. Re-program baud rate registers: Re-write UBIR and UBMIR.

16.2.11 Transfer Error

The UART can generate a transfer error on the peripheral bus in the following cases:

- Core is writing into a read-only register.
- Core is accessing (read or write) an unused location within the assigned address space reserved to UART.
- Core is writing into UTXD register with transmit interface disabled (TXEN=0 or UARTEN=0)
- Core is reading URXD register with receive interface disabled (RXEN=0 or UARTEN=0)

16.2.12 Functional Timing

This section includes timing diagrams for functional signaling.

16.2.12.1 IrDA Mode

According to IrDA specification, the low speed (115.2Kbit/s and below) IR frame format is compatible with UART frame.

In this figure, an example data 0x65 is used.

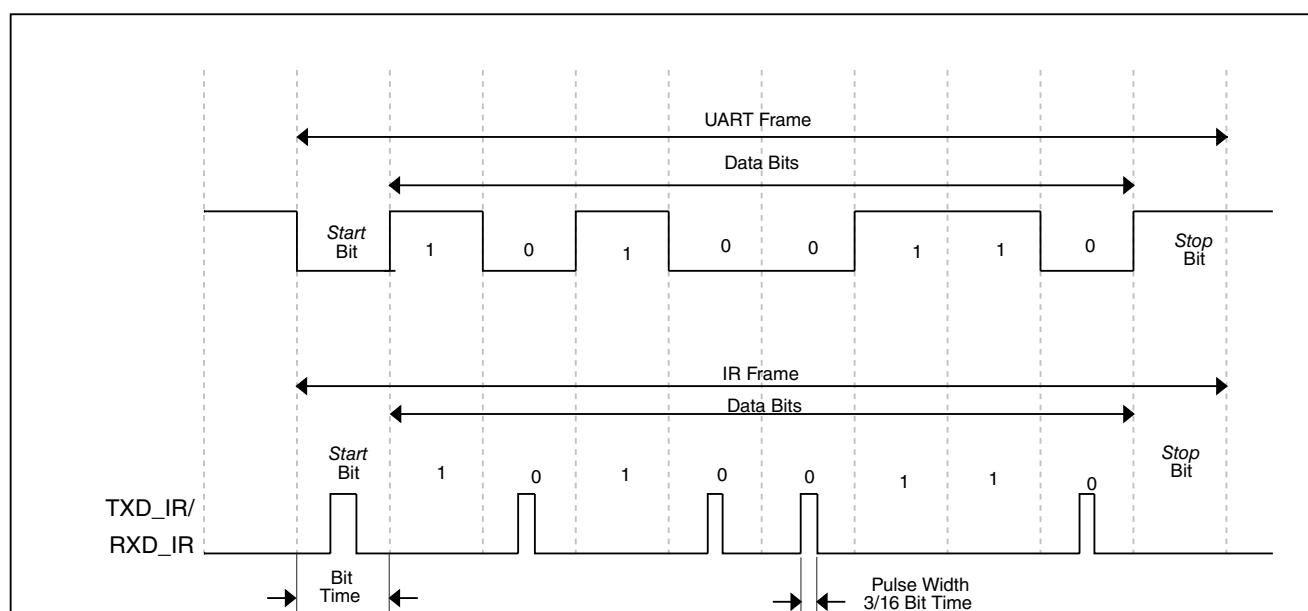


Figure 16-21. Timing diagram of Low Speed IR (<=115.2 Kbit/s) Data Line

16.2.13 Initialization

16.2.13.1 Programming the UART in RS-232 mode

As an example, the following sequence can be used to program the UART in order to send and receive characters in RS-232 mode.

Assumptions:

- Input uart clock = 100 MHz
- Baud rate = 921.6 Kbps

- Data bits = 8 bits
- Parity = Even
- Stop bits = 1 bit
- Flow control = Hardware

Main program:

1. UCR1 = 0x0001

Enable the UART.

2. UCR2 = 0x2127

Set hardware flow control, data format and enable transmitter and receiver.

3. UCR3 = 0x0704

Set UCR3[RXDMUXSEL] = 1.

4. UCR4 = 0x7C00

Set CTS trigger level to 31,

5. UFCR = 0x089E

Set internal clock divider = 5 (divide input uart clock by 5). So the reference clock is $100\text{ MHz}/5 = 20\text{ MHz}$.

Set TXTL = 2 and RXTL = 30.

6. UBIR = 0x08FF

7. UBMIR = 0x0C34

In the above two steps, set baud rate to 921.6Kbps based on the 20MHz reference clock.

8. UCR1 = 0x2201

Enable the TRDY and RRDY interrupts.

9. UMCR = 0x0000

UMCR stay at default value 0x0000

Interrupt service routine for the transmitter:

- Write characters into UTXD

The TRDY interrupt will be automatically de-asserted when the data level of the TxFIFO exceeds the TXTL=2. Note: For the first time the interrupt may be de-asserted after 4 characters are written into the TxFIFO because of the shift register.

Interrupt service routine for the receiver:

- Read characters from URXD

The RRDY interrupt will be automatically de-asserted when the data level of the RxFIFO is below the RXCTL=30.

16.2.13.2 Programming the UART in 9-bit RS-485 mode

As an example, the following sequence can be used to program the UART in order to send and receive frames in RS-485 mode.

Assumptions:

- Input uart clock = 100 MHz
- Baud rate = 5 Mbps

Main program:

1. UCR1 = 0x0001

Enable the UART.

2. UCR2 = 0x4127

Set software flow control ($\overline{\text{CTS}}$ pin is controlled by UCR2[12]), enable parity(enable 9th bit rxd/txd), 8-bit word size , and enable transmitter and receiver.

3. UCR4 = 0x7C00

Set CTS trigger level to 31,

4. UFCR = 0x0A9E

Set RFDIV = 5 (divide input uart clock by 1), so the reference clock is 100 MHz. Set UART in DCE mode (RS-485 driver connection outside the chip is the same as [Figure 16-20](#))

Set TXCTL = 2 and RXCTL = 30.

5. UBIR = 0x0003

6. UBMIR = 0x0004

In the above two steps, set baud rate to 5 Mbps based on the 100 MHz reference clock.

7. UCR1 = 0x2001 when UART as a master ,

or UCR1 = 0x0201 (or 0x0101) when UART as a slave.

Enable TRDY interrupt when UART as a master, enable RRDY interrupt or DMA request when UART as a slave.

8. UMCR = 0xA50B

Enable 9-bit RS-485 mode, enable SAD interrupt, set automatic slave address detect mode, set slave address is 0xA5.

Interrupt service routine for the transmitter:

- Transmit data: write its ninth bit (bit[8]) to UMCR[2], write its bit [7:0] into UTXD[7:0]

The TRDY interrupt will be automatically de-asserted when the data level of the TxFIFO exceeds the TXTL=2.

Note: For the first time the interrupt may be de-asserted after 4 characters are written into the TxFIFO because of the shift register.

Interrupt service routine for the receiver:

- Receive data: read its ninth bit (bit[8]) from URXD[10] , read its bit [7:0] from URXD[7:0].

Note: in RS-485 mode, URXD[10] bit is not the parity error, instead it holds the ninth bit (bit[8]) of the received data.

The SAD interrupt can not de-assert automatically, it needs MCU write 1 to USR1[3] to clear it . The RRDY interrupt or DMA request will be automatically de-asserted when the data level of the Rx FIFO is below the RXTL=30.

16.2.14 References

- EIA/TIA-232-F Interface Standard

<http://www.eia.org>, <http://www.tiaonline.org/standards>

- IrDA Standard

<http://www.irda.org>

16.2.15 UART Memory Map/Register Definition

UART supports 8-bit, 16-bit and 32-bit accesses to 32-bit memory-mapped addresses. Any access to unmapped memory location will yield a transfer error.

All registers except the ONEMS described in this section are 16-bit registers. The ONEMS register is a 24-bit register.

- For 32-bit write accesses, the upper two bytes will not be taken into account.
- For 32-bit read accesses the upper two bytes will return 0.

The ONEMS register is expanded from 16 bits to 24 bits in order to support the high frequency of the BRM internal clock *ref_clk* (*module_clock* after divider). The ONEMS register can be accessed as 8 bits, 16 bits or 32 bits.

- For 32-bit write accesses, the most significant byte of the ONEMS will be discarded.
- For 32-bit read accesses, the most significant byte of the ONEMS will be read as 0.

UART memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
3086_0000	UART Receiver Register (UART1_URXD)	32	R	0000_0000h	16.2.15.1/ 6517
3086_0040	UART Transmitter Register (UART1_UTXD)	32	W	0000_0000h	16.2.15.2/ 6519
3086_0080	UART Control Register 1 (UART1_UCR1)	32	R/W	0000_0000h	16.2.15.3/ 6520
3086_0084	UART Control Register 2 (UART1_UCR2)	32	R/W	0000_0001h	16.2.15.4/ 6522
3086_0088	UART Control Register 3 (UART1_UCR3)	32	R/W	0000_0700h	16.2.15.5/ 6525
3086_008C	UART Control Register 4 (UART1_UCR4)	32	R/W	0000_8000h	16.2.15.6/ 6527
3086_0090	UART FIFO Control Register (UART1_UFCR)	32	R/W	0000_0801h	16.2.15.7/ 6529
3086_0094	UART Status Register 1 (UART1_USR1)	32	R/W	0000_2040h	16.2.15.8/ 6531
3086_0098	UART Status Register 2 (UART1_USR2)	32	R/W	0000_4028h	16.2.15.9/ 6534
3086_009C	UART Escape Character Register (UART1_UESC)	32	R/W	0000_002Bh	16.2.15.10/ 6536
3086_00A0	UART Escape Timer Register (UART1_UTIM)	32	R/W	0000_0000h	16.2.15.11/ 6536
3086_00A4	UART BRM Incremental Register (UART1_UBIR)	32	R/W	0000_0000h	16.2.15.12/ 6537
3086_00A8	UART BRM Modulator Register (UART1_UBMR)	32	R/W	0000_0000h	16.2.15.13/ 6537
3086_00AC	UART Baud Rate Count Register (UART1_UBRC)	32	R	0000_0004h	16.2.15.14/ 6538

Table continues on the next page...

UART memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
3086_00B0	UART One Millisecond Register (UART1_ONEMS)	32	R/W	0000_0000h	16.2.15.15/ 6539
3086_00B4	UART Test Register (UART1_UTS)	32	R/W	0000_0060h	16.2.15.16/ 6540
3086_00B8	UART RS-485 Mode Control Register (UART1_UMCR)	32	R/W	0000_0000h	16.2.15.17/ 6541
3088_0000	UART Receiver Register (UART3_URXD)	32	R	0000_0000h	16.2.15.1/ 6517
3088_0040	UART Transmitter Register (UART3_UTXD)	32	W	0000_0000h	16.2.15.2/ 6519
3088_0080	UART Control Register 1 (UART3_UCR1)	32	R/W	0000_0000h	16.2.15.3/ 6520
3088_0084	UART Control Register 2 (UART3_UCR2)	32	R/W	0000_0001h	16.2.15.4/ 6522
3088_0088	UART Control Register 3 (UART3_UCR3)	32	R/W	0000_0700h	16.2.15.5/ 6525
3088_008C	UART Control Register 4 (UART3_UCR4)	32	R/W	0000_8000h	16.2.15.6/ 6527
3088_0090	UART FIFO Control Register (UART3_UFCR)	32	R/W	0000_0801h	16.2.15.7/ 6529
3088_0094	UART Status Register 1 (UART3_USR1)	32	R/W	0000_2040h	16.2.15.8/ 6531
3088_0098	UART Status Register 2 (UART3_USR2)	32	R/W	0000_4028h	16.2.15.9/ 6534
3088_009C	UART Escape Character Register (UART3_UESC)	32	R/W	0000_002Bh	16.2.15.10/ 6536
3088_00A0	UART Escape Timer Register (UART3_UTIM)	32	R/W	0000_0000h	16.2.15.11/ 6536
3088_00A4	UART BRM Incremental Register (UART3_UBIR)	32	R/W	0000_0000h	16.2.15.12/ 6537
3088_00A8	UART BRM Modulator Register (UART3_UBMR)	32	R/W	0000_0000h	16.2.15.13/ 6537
3088_00AC	UART Baud Rate Count Register (UART3_UBRC)	32	R	0000_0004h	16.2.15.14/ 6538
3088_00B0	UART One Millisecond Register (UART3_ONEMS)	32	R/W	0000_0000h	16.2.15.15/ 6539
3088_00B4	UART Test Register (UART3_UTS)	32	R/W	0000_0060h	16.2.15.16/ 6540
3088_00B8	UART RS-485 Mode Control Register (UART3_UMCR)	32	R/W	0000_0000h	16.2.15.17/ 6541
3089_0000	UART Receiver Register (UART2_URXD)	32	R	0000_0000h	16.2.15.1/ 6517
3089_0040	UART Transmitter Register (UART2_UTXD)	32	W	0000_0000h	16.2.15.2/ 6519

Table continues on the next page...

UART memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
3089_0080	UART Control Register 1 (UART2_UCR1)	32	R/W	0000_0000h	16.2.15.3/ 6520
3089_0084	UART Control Register 2 (UART2_UCR2)	32	R/W	0000_0001h	16.2.15.4/ 6522
3089_0088	UART Control Register 3 (UART2_UCR3)	32	R/W	0000_0700h	16.2.15.5/ 6525
3089_008C	UART Control Register 4 (UART2_UCR4)	32	R/W	0000_8000h	16.2.15.6/ 6527
3089_0090	UART FIFO Control Register (UART2_UFCR)	32	R/W	0000_0801h	16.2.15.7/ 6529
3089_0094	UART Status Register 1 (UART2_USR1)	32	R/W	0000_2040h	16.2.15.8/ 6531
3089_0098	UART Status Register 2 (UART2_USR2)	32	R/W	0000_4028h	16.2.15.9/ 6534
3089_009C	UART Escape Character Register (UART2_UESC)	32	R/W	0000_002Bh	16.2.15.10/ 6536
3089_00A0	UART Escape Timer Register (UART2_UTIM)	32	R/W	0000_0000h	16.2.15.11/ 6536
3089_00A4	UART BRM Incremental Register (UART2_UBIR)	32	R/W	0000_0000h	16.2.15.12/ 6537
3089_00A8	UART BRM Modulator Register (UART2_UBMR)	32	R/W	0000_0000h	16.2.15.13/ 6537
3089_00AC	UART Baud Rate Count Register (UART2_UBRC)	32	R	0000_0004h	16.2.15.14/ 6538
3089_00B0	UART One Millisecond Register (UART2_ONEMS)	32	R/W	0000_0000h	16.2.15.15/ 6539
3089_00B4	UART Test Register (UART2_UTS)	32	R/W	0000_0060h	16.2.15.16/ 6540
3089_00B8	UART RS-485 Mode Control Register (UART2_UMCR)	32	R/W	0000_0000h	16.2.15.17/ 6541
30A6_0000	UART Receiver Register (UART4_URXD)	32	R	0000_0000h	16.2.15.1/ 6517
30A6_0040	UART Transmitter Register (UART4_UTXD)	32	W	0000_0000h	16.2.15.2/ 6519
30A6_0080	UART Control Register 1 (UART4_UCR1)	32	R/W	0000_0000h	16.2.15.3/ 6520
30A6_0084	UART Control Register 2 (UART4_UCR2)	32	R/W	0000_0001h	16.2.15.4/ 6522
30A6_0088	UART Control Register 3 (UART4_UCR3)	32	R/W	0000_0700h	16.2.15.5/ 6525
30A6_008C	UART Control Register 4 (UART4_UCR4)	32	R/W	0000_8000h	16.2.15.6/ 6527
30A6_0090	UART FIFO Control Register (UART4_UFCR)	32	R/W	0000_0801h	16.2.15.7/ 6529

Table continues on the next page...

UART memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
30A6_0094	UART Status Register 1 (UART4_USR1)	32	R/W	0000_2040h	16.2.15.8/6531
30A6_0098	UART Status Register 2 (UART4_USR2)	32	R/W	0000_4028h	16.2.15.9/6534
30A6_009C	UART Escape Character Register (UART4_UESC)	32	R/W	0000_002Bh	16.2.15.10/6536
30A6_00A0	UART Escape Timer Register (UART4_UTIM)	32	R/W	0000_0000h	16.2.15.11/6536
30A6_00A4	UART BRM Incremental Register (UART4_UBIR)	32	R/W	0000_0000h	16.2.15.12/6537
30A6_00A8	UART BRM Modulator Register (UART4_UBMR)	32	R/W	0000_0000h	16.2.15.13/6537
30A6_00AC	UART Baud Rate Count Register (UART4_UBRC)	32	R	0000_0004h	16.2.15.14/6538
30A6_00B0	UART One Millisecond Register (UART4_ONEMS)	32	R/W	0000_0000h	16.2.15.15/6539
30A6_00B4	UART Test Register (UART4_UTS)	32	R/W	0000_0060h	16.2.15.16/6540
30A6_00B8	UART RS-485 Mode Control Register (UART4_UMCR)	32	R/W	0000_0000h	16.2.15.17/6541

16.2.15.1 UART Receiver Register (UARTx_URXD)

NOTE

The UART will yield a transfer error on the peripheral bus when core is reading URXD register with receive interface disabled (RXEN=0 or UARTEN=0).

Address: Base address + 0h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CHARRDY	ERR	OVRUN	FRMERR	BRK	PRERR	Reserved		RX_DATA							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UARTx_URXD field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15 CHARRDY	Character Ready. This read-only bit indicates an invalid read when the FIFO becomes empty and software tries to read the same old data. This bit should not be used for polling for data written to the RX FIFO. 0 Character in RX_DATA field and associated flags are invalid. 1 Character in RX_DATA field and associated flags valid and ready for reading.

Table continues on the next page...

UARTx_URXD field descriptions (continued)

Field	Description
14 ERR	Error Detect. Indicates whether the character present in the RX_DATA field has an error (OVRUN, FRMERR, BRK or PRERR) status. The ERR bit is updated and valid for each received character. 0 No error status was detected 1 An error status was detected
13 OVRUN	Receiver Overrun. This read-only bit, when HIGH, indicates that the corresponding character was stored in the last position (32nd) of the Rx FIFO. Even if a 33rd character has not been detected, this bit will be set to '1' for the 32nd character. 0 No RxFIFO overrun was detected 1 A RxFIFO overrun was detected
12 FRMERR	Frame Error. Indicates whether the current character had a framing error (a missing stop bit) and is possibly corrupted. FRMERR is updated for each character read from the RxFIFO. 0 The current character has no framing error 1 The current character has a framing error
11 BRK	BREAK Detect. Indicates whether the current character was detected as a BREAK character. The data bits and the stop bit are all 0. The FRMERR bit is set when BRK is set. When odd parity is selected, PRERR is also set when BRK is set. BRK is valid for each character read from the RxFIFO. 0 The current character is not a BREAK character 1 The current character is a BREAK character
10 PRERR	In RS-485 mode, it holds the ninth data bit (bit [8]) of received 9-bit RS-485 data In RS232/IrDA mode, it is the Parity Error flag. Indicates whether the current character was detected with a parity error and is possibly corrupted. PRERR is updated for each character read from the RxFIFO. When parity is disabled, PRERR always reads as 0. 0 = No parity error was detected for data in the RX_DATA field 1 = A parity error was detected for data in the RX_DATA field
9–8 -	This field is reserved. Reserved
RX_DATA	Received Data. Holds the received character. In 7-bit mode, the most significant bit (MSB) is forced to 0. In 8-bit mode, all bits are active.

16.2.15.2 UART Transmitter Register (UARTx_UTXD)

NOTE

The UART will yield a transfer error on the peripheral bus when core is writing into UART_URXD register with transmit interface disabled (TXEN=0 or UARTEN=0).

Memory space between UART_URXD and UART_UTXD registers is reserved. Any read or write access to this space will be considered as an invalid access and yield a transfer error.

Address: Base address + 40h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																0															
W																									TX_DATA							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UARTx_UTXD field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15–8 Reserved	This read-only field is reserved and always has the value 0.
TX_DATA	Transmit Data. Holds the parallel transmit data inputs. In 7-bit mode, D7 is ignored. In 8-bit mode, all bits are used. Data is transmitted least significant bit (LSB) first. A new character is transmitted when the TX_DATA field is written. The TX_DATA field must be written only when the TRDY bit is high to ensure that corrupted data is not sent.

16.2.15.3 UART Control Register 1 (UARTx_UCR1)

Address: Base address + 80h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ADEN	ADBR	TRDYEN	IDEN	ICD	RRDYEN	RXDMAEN	IREN	TXEMPTYEN	RTSDEN	SNDBRK	TXDMAEN	ATDMAEN	DOZE	UARTEN	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UARTx_UCR1 field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15 ADEN	Automatic Baud Rate Detection Interrupt Enable. Enables/Disables the automatic baud rate detect complete (ADET) bit to generate an interrupt (<i>interrupt_uart</i> = 0). 0 Disable the automatic baud rate detection interrupt 1 Enable the automatic baud rate detection interrupt
14 ADBR	Automatic Detection of Baud Rate. Enables/Disables automatic baud rate detection. When the ADBR bit is set and the ADET bit is cleared, the receiver detects the incoming baud rate automatically. The ADET flag is set when the receiver verifies that the incoming baud rate is detected properly by detecting an ASCII character "A" or "a" (0x41 or 0x61). 0 Disable automatic detection of baud rate 1 Enable automatic detection of baud rate
13 TRDYEN	Transmitter Ready Interrupt Enable. Enables/Disables the transmitter Ready Interrupt (TRDY) when the transmitter has one or more slots available in the TxFIFO. The fill level in the TXFIFO at which an interrupt is generated is controlled by TxTL bits. When TRDYEN is negated, the transmitter ready interrupt is disabled. NOTE: An interrupt will be issued as long as TRDYEN and TRDY are high even if the transmitter is not enabled. In general, user should enable the transmitter before enabling the TRDY interrupt. 0 Disable the transmitter ready interrupt 1 Enable the transmitter ready interrupt
12 IDEN	Idle Condition Detected Interrupt Enable. Enables/Disables the IDLE bit to generate an interrupt (<i>interrupt_uart</i> = 0). 0 Disable the IDLE interrupt 1 Enable the IDLE interrupt

Table continues on the next page...

UARTx_UCR1 field descriptions (continued)

Field	Description
11–10 ICD	Idle Condition Detect. Controls the number of frames RXD is allowed to be idle before an idle condition is reported. 00 Idle for more than 4 frames 01 Idle for more than 8 frames 10 Idle for more than 16 frames 11 Idle for more than 32 frames
9 RRDYEN	Receiver Ready Interrupt Enable. Enables/Disables the RRDY interrupt when the RxFIFO contains data. The fill level in the RxFIFO at which an interrupt is generated is controlled by the RXTL bits. When RRDYEN is negated, the receiver ready interrupt is disabled. 0 Disables the RRDY interrupt 1 Enables the RRDY interrupt
8 RXDMAEN	Receive Ready DMA Enable. Enables/Disables the receive DMA request <i>dma_req_rx</i> when the receiver has data in the RxFIFO. The fill level in the RxFIFO at which a DMA request is generated is controlled by the RXTL bits. When negated, the receive DMA request is disabled. 0 Disable DMA request 1 Enable DMA request
7 IREN	Infrared Interface Enable. Enables/Disables the IR interface. See the IR interface description in Infrared Interface , for more information. Note: MDEN(UMCR[0]) must be cleared to 0 when using IrDA interface. See Table 16-3 0 Disable the IR interface 1 Enable the IR interface
6 TXMPTYEN	Transmitter Empty Interrupt Enable. Enables/Disables the transmitter FIFO empty (TXFE) interrupt. <i>interrupt_uart</i> . When negated, the TXFE interrupt is disabled. NOTE: An interrupt will be issued as long as TXMPTYEN and TXFE are high even if the transmitter is not enabled. In general, user should enable the transmitter before enabling the TXFE interrupt. 0 Disable the transmitter FIFO empty interrupt 1 Enable the transmitter FIFO empty interrupt
5 RTSDEN	RTS Delta Interrupt Enable. Enables/Disables the RTSD interrupt. The current status of the RTS_B pin is read in the RTSS bit. 0 Disable RTSD interrupt 1 Enable RTSD interrupt
4 SNDBRK	Send BREAK. Forces the transmitter to send a BREAK character. The transmitter finishes sending the character in progress (if any) and sends BREAK characters until SNDBRK is reset. Because the transmitter samples SNDBRK after every bit is transmitted, it is important that SNDBRK is asserted high for a sufficient period of time to generate a valid BREAK. After the BREAK transmission completes, the UART transmits 2 mark bits. The user can continue to fill the TxFIFO and any characters remaining are transmitted when the BREAK is terminated. 0 Do not send a BREAK character 1 Send a BREAK character (continuous 0s)
3 TXDMAEN	Transmitter Ready DMA Enable. Enables/Disables the transmit DMA request <i>dma_req_tx</i> when the transmitter has one or more slots available in the TxFIFO. The fill level in the TxFIFO that generates the <i>dma_req_tx</i> is controlled by the TXTL bits.

Table continues on the next page...

UARTx_UCR1 field descriptions (continued)

Field	Description
	NOTE: A DMA request will be issued as long as TXDMAEN and TRDY are high even if the transmitter is not enabled. In general, user should enable the transmitter before enabling the transmit DMA request. 0 Disable transmit DMA request 1 Enable transmit DMA request
2 ATDMAEN	Aging DMA Timer Enable. Enables/Disables the receive DMA request <i>dma_req_rx</i> for the aging timer interrupt (triggered with AGTIM flag in USR1[8]). 0 Disable AGTIM DMA request 1 Enable AGTIM DMA request
1 DOZE	DOZE. Determines the UART enable condition in the DOZE state. When <i>doze_req</i> input pin is at '1', (the Arm Platform executes a doze instruction and the system is placed in the Doze State), the DOZE bit affects operation of the UART. While in the Doze State, if this bit is asserted, the UART is disabled. See the description in Low Power Modes. 0 The UART is enabled when in DOZE state 1 The UART is disabled when in DOZE state
0 UARTEN	UART Enable. Enables/Disables the UART. If UARTEN is negated in the middle of a transmission, the transmitter stops and pulls the TXD line to a logic 1. UARTEN must be set to 1 before any access to UTXD and URXD registers, otherwise a transfer error is returned. This bit can be set to 1 along with other bits in this register. There is no restriction to the sequence of programing this bit and other control registers. 0 Disable the UART 1 Enable the UART

16.2.15.4 UART Control Register 2 (UARTx_UCR2)

Address: Base address + 84h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ESCI	IRTS	CTSC	CTS	ESSEN	RTEC	PREN	PRO E	STPB	WS	RTSEN	ATEN	TXEN	RXEN	SRST	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

UARTx_UCR2 field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15 ESCI	Escape Sequence Interrupt Enable. Enables/Disables the ESCF bit to generate an interrupt. 0 Disable the escape sequence interrupt 1 Enable the escape sequence interrupt
14 IRTS	Ignore RTS Pin. Forces the RTS input signal presented to the transmitter to always be asserted (set to low), effectively ignoring the external pin. When in this mode, the RTS pin serves as a general purpose input. 0 Transmit only when the RTS pin is asserted 1 Ignore the RTS pin
13 CTSC	CTS Pin Control. Controls the operation of the CTS_B module output. When CTSC is asserted, the CTS_B module output is controlled by the receiver. When the RxFIFO is filled to the level of the programmed trigger level and the start bit of the overflowing character (TRIGGER LEVEL + 1) is validated, the CTS_B module output is negated to indicate to the far-end transmitter to stop transmitting. When the trigger level is programmed for less than 32, the receiver continues to receive data until the RxFIFO is full. When the CTSC bit is negated, the CTS_B module output is controlled by the CTS bit. On reset, because CTSC is cleared to 0, the CTS_B pin is controlled by the CTS bit, which again is cleared to 0 on reset. This means that on reset the CTS_B signal is negated. 0 The CTS_B pin is controlled by the CTS bit 1 The CTS_B pin is controlled by the receiver
12 CTS	Clear to Send. Controls the CTS_B pin when the CTSC bit is negated. CTS has no function when CTSC is asserted. 0 The CTS_B pin is high (inactive) 1 The CTS_B pin is low (active)
11 ESCEN	Escape Enable. Enables/Disables the escape sequence detection logic. 0 Disable escape sequence detection 1 Enable escape sequence detection
10–9 RTEC	Request to Send Edge Control. Selects the edge that triggers the RTS interrupt. This has no effect on the RTS delta interrupt. RTEC has an effect only when RTSEN = 1 (see Table 16-6). 00 Trigger interrupt on a rising edge 01 Trigger interrupt on a falling edge 1X Trigger interrupt on any edge
8 PREN	Parity Enable. Enables/Disables the parity generator in the transmitter and parity checker in the receiver. When PREN is asserted, the parity generator and checker are enabled, and disabled when PREN is negated. 0 Disable parity generator and checker 1 Enable parity generator and checker
7 PROE	Parity Odd/Even. Controls the sense of the parity generator and checker. When PROE is high, odd parity is generated and expected. When PROE is low, even parity is generated and expected. PROE has no function if PREN is low. 0 Even parity 1 Odd parity

Table continues on the next page...

UARTx_UCR2 field descriptions (continued)

Field	Description
6 STPB	Stop. Controls the number of stop bits after a character. When STPB is low, 1 stop bit is sent. When STPB is high, 2 stop bits are sent. STPB also affects the receiver. 0 The transmitter sends 1 stop bit. The receiver expects 1 or more stop bits. 1 The transmitter sends 2 stop bits. The receiver expects 2 or more stop bits.
5 WS	Word Size. Controls the character length. When WS is high, the transmitter and receiver are in 8-bit mode. When WS is low, they are in 7-bit mode. The transmitter ignores bit 7 and the receiver sets bit 7 to 0. WS can be changed in-between transmission (reception) of characters, however not when a transmission (reception) is in progress, in which case the length of the current character being transmitted (received) is unpredictable. 0 7-bit transmit and receive character length (not including START, STOP or PARITY bits) 1 8-bit transmit and receive character length (not including START, STOP or PARITY bits)
4 RTSEN	Request to Send Interrupt Enable. Controls the RTS edge sensitive interrupt. When RTSEN is asserted and the programmed edge is detected on the RTS_B pin (the RTSF bit is asserted), an interrupt will be generated on the <i>interrupt_uart</i> pin. (See Table 16-6.) 0 Disable request to send interrupt 1 Enable request to send interrupt
3 ATEN	Aging Timer Enable. This bit is used to enable the aging timer interrupt (triggered with AGTIM) 0 AGTIM interrupt disabled 1 AGTIM interrupt enabled
2 TXEN	Transmitter Enable. Enables/Disables the transmitter. When TXEN is negated the transmitter is disabled and idle. When the UARTEN and TXEN bits are set the transmitter is enabled. If TXEN is negated in the middle of a transmission, the UART disables the transmitter immediately, and starts marking 1s. The transmitter FIFO cannot be written when this bit is cleared. 0 Disable the transmitter 1 Enable the transmitter
1 RXEN	Receiver Enable. Enables/Disables the receiver. When the receiver is enabled, if the RXD input is already low, the receiver does not recognize BREAK characters, because it requires a valid 1-to-0 transition before it can accept any character. 0 Disable the receiver 1 Enable the receiver
0 SRST	Software Reset. Once the software writes 0 to SRST_B, the software reset remains active for 4 <i>module_clock</i> cycles before the hardware deasserts SRST_B. The software can only write 0 to SRST_B. Writing 1 to SRST_B is ignored. 0 Reset the transmit and receive state machines, all FIFOs and register USR1, USR2, UBIR, UBMR, UBRC, URXD, UTXD and UTS[6-3]. 1 No reset

16.2.15.5 UART Control Register 3 (UARTx_UCR3)

Address: Base address + 88h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DPEC		DTREN	PARERREN	FRAERREN	DSR	DCD	RI	ADNIMP	RXDSEN	AIRINTEN	AWAKEN	DTRDEN	RXDMUXSEL	INVT	ACIEN
W																
Reset	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0

UARTx_UCR3 field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15–14 DPEC	This bit is not used in this chip.
13 DTREN	This bit is not used in this chip.
12 PARERREN	Parity Error Interrupt Enable. Enables/Disables the interrupt. When asserted, PARERREN causes the PARITYERR bit to generate an interrupt. 0 Disable the parity error interrupt 1 Enable the parity error interrupt
11 FRAERREN	Frame Error Interrupt Enable. Enables/Disables the interrupt. When asserted, FRAERREN causes the FRAMERR bit to generate an interrupt. 0 Disable the frame error interrupt 1 Enable the frame error interrupt
10 DSR	This bit is not used in this chip.
9 DCD	This bit is not used in this chip.
8 RI	This bit is not used in this chip.
7 ADNIMP	Autobaud Detection Not Improved-. Disables new features of autobaud detection (See Baud Rate Automatic Detection Protocol, for more details). 0 Autobaud detection new features selected 1 Keep old autobaud detection mechanism

Table continues on the next page...

UARTx_UCR3 field descriptions (continued)

Field	Description
6 RXDSEN	Receive Status Interrupt Enable. Controls the receive status interrupt (<i>interrupt_uart</i>). When this bit is enabled and RXDS status bit is set, the interrupt <i>interrupt_uart</i> will be generated. 0 Disable the RXDS interrupt 1 Enable the RXDS interrupt
5 AIRINTEN	Asynchronous IR WAKE Interrupt Enable. Controls the asynchronous IR WAKE interrupt. An interrupt is generated when AIRINTEN is asserted and a pulse is detected on the RXD pin. 0 Disable the AIRINT interrupt 1 Enable the AIRINT interrupt
4 AWAKEN	Asynchronous WAKE Interrupt Enable. Controls the asynchronous WAKE interrupt. An interrupt is generated when AWAKEN is asserted and a falling edge is detected on the RXD pin. 0 Disable the AWAKE interrupt 1 Enable the AWAKE interrupt
3 DTRDEN	This bit is not used in this chip.
2 RXDMUXSEL	RXD Muxed Input Selected. Selects proper input pins for serial and Infrared input signal. NOTE: In this chip, UARTs are used in MUXED mode, so that this bit should always be set.
1 INVT	Invert TXD output in RS-232/RS-485 mode, set TXD active level in IrDA mode. In RS232/RS-485 mode(UMCR[0] = 1), if this bit is set to 1, the TXD output is inverted before transmitted. In IrDA mode , when INVT is cleared, the infrared logic block transmits a positive IR 3/16 pulse for all 0s and 0s are transmitted for 1s. When INVT is set (INVT = 1), the infrared logic block transmits an active low or negative infrared 3/16 pulse for all 0s and 1s are transmitted for 1s. 0 TXD is not inverted 1 TXD is inverted 0 TXD Active low transmission 1 TXD Active high transmission
0 ACIEN	Autobaud Counter Interrupt Enable. This bit is used to enable the autobaud counter stopped interrupt (triggered with ACST (USR2[11]). 0 ACST interrupt disabled 1 ACST interrupt enabled

16.2.15.6 UART Control Register 4 (UARTx_UCR4)

Address: Base address + 8Ch offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CTSTL						INVR	ENIRI	WKEN	IDDMAEN	IRSC	LPBYP	TCEN	BKEN	OREN	DREN
W																
Reset	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UARTx_UCR4 field descriptions

Field	Description										
31–16 Reserved	This read-only field is reserved and always has the value 0.										
15–10 CTSTL	CTS Trigger Level. Controls the threshold at which the CTS_B pin is deasserted by the RxFIFO. After the trigger level is reached and the CTS_B pin is deasserted, the RxFIFO continues to receive data until it is full. The CTSTL bits are encoded as shown in the Settings column. Settings 0 to 32 are in use. All other settings are Reserved. <table> <tr> <td>000000</td><td>0 characters received</td></tr> <tr> <td>000001</td><td>1 characters in the RxFIFO</td></tr> <tr> <td>...</td><td>—</td></tr> <tr> <td>...</td><td>—</td></tr> <tr> <td>100000</td><td>32 characters in the RxFIFO (maximum)</td></tr> </table>	000000	0 characters received	000001	1 characters in the RxFIFO	...	—	...	—	100000	32 characters in the RxFIFO (maximum)
000000	0 characters received										
000001	1 characters in the RxFIFO										
...	—										
...	—										
100000	32 characters in the RxFIFO (maximum)										
9 INVR	Invert RXD input in RS-232/RS-485 Mode, determine RXD input logic level being sampled in In IrDA mode. In RS232/RS-485 Mode(UMCR[0] = 1), if this bit is set to 1, the RXD input is inverted before sampled. In IrDA mode,when cleared, the infrared logic block expects an active low or negative IR 3/16 pulse for 0s and 1s are expected for 1s. When INVR is set (INVR 1), the infrared logic block expects an active high or positive IR 3/16 pulse for 0s and 0s are expected for 1s. <table> <tr> <td>0</td><td>RXD input is not inverted</td></tr> <tr> <td>1</td><td>RXD input is inverted</td></tr> <tr> <td>0</td><td>RXD active low detection</td></tr> <tr> <td>1</td><td>RXD active high detection</td></tr> </table>	0	RXD input is not inverted	1	RXD input is inverted	0	RXD active low detection	1	RXD active high detection		
0	RXD input is not inverted										
1	RXD input is inverted										
0	RXD active low detection										
1	RXD active high detection										
8 ENIRI	Serial Infrared Interrupt Enable. Enables/Disables the serial infrared interrupt. <table> <tr> <td>0</td><td>Serial infrared Interrupt disabled</td></tr> <tr> <td>1</td><td>Serial infrared Interrupt enabled</td></tr> </table>	0	Serial infrared Interrupt disabled	1	Serial infrared Interrupt enabled						
0	Serial infrared Interrupt disabled										
1	Serial infrared Interrupt enabled										

Table continues on the next page...

UARTx_UCR4 field descriptions (continued)

Field	Description
7 WKEN	WAKE Interrupt Enable. Enables/Disables the WAKE bit to generate an interrupt. The WAKE bit is set at the detection of a start bit by the receiver. 0 Disable the WAKE interrupt 1 Enable the WAKE interrupt
6 IDDMAEN	DMA IDLE Condition Detected Interrupt Enable Enables/Disables the receive DMA request <i>dma_req_rx</i> for the IDLE interrupt (triggered with IDLE flag in USR2[12]). 0 DMA IDLE interrupt disabled 1 DMA IDLE interrupt enabled
5 IRSC	IR Special Case. Selects the clock for the vote logic. When set, IRSC switches the vote logic clock from the sampling clock to the UART reference clock. The IR pulses are counted a predetermined amount of time depending on the reference frequency. See InfraRed Special Case (IRSC) Bit . 0 The vote logic uses the sampling clock (16x baud rate) for normal operation 1 The vote logic uses the UART reference clock
4 LPBYP	Low Power Bypass. Allows to bypass the low power new features in UART. To use during debug phase. 0 Low power features enabled 1 Low power features disabled
3 TCEN	TransmitComplete Interrupt Enable. Enables/Disables the TXDC bit to generate an interrupt (<i>interrupt_uart</i> = 0) NOTE: An interrupt will be issued as long as TCEN and TXDC are high even if the transmitter is not enabled. In general, user should enable the transmitter before enabling the TXDC interrupt. 0 Disable TXDC interrupt 1 Enable TXDC interrupt
2 BKEN	BREAK Condition Detected Interrupt Enable. Enables/Disables the BRCD bit to generate an interrupt. 0 Disable the BRCD interrupt 1 Enable the BRCD interrupt
1 OREN	Receiver Overrun Interrupt Enable. Enables/Disables the ORE bit to generate an interrupt. 0 Disable ORE interrupt 1 Enable ORE interrupt
0 DREN	Receive Data Ready Interrupt Enable. Enables/Disables the RDR bit to generate an interrupt. 0 Disable RDR interrupt 1 Enable RDR interrupt

16.2.15.7 UART FIFO Control Register (UARTx_UFCR)

Address: Base address + 90h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TXTL							RFDIV			DCEDTE	RXTL				
W																
Reset	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1

UARTx_UFCR field descriptions

Field	Description																
31–16 Reserved	This read-only field is reserved and always has the value 0.																
15–10 TXTL	Transmitter Trigger Level. Controls the threshold at which a maskable interrupt is generated by the TxFIFO. A maskable interrupt is generated whenever the data level in the TxFIFO falls below the selected threshold. The bits are encoded as shown in the Settings column. Settings 0 to 32 are in use. All other settings are Reserved. <table> <tr><td>000000</td><td>Reserved</td></tr> <tr><td>000001</td><td>Reserved</td></tr> <tr><td>000010</td><td>TxFIFO has 2 or fewer characters</td></tr> <tr><td>...</td><td>—</td></tr> <tr><td>...</td><td>—</td></tr> <tr><td>011111</td><td>TxFIFO has 31 or fewer characters</td></tr> <tr><td>100000</td><td>TxFIFO has 32 characters (maximum)</td></tr> </table>	000000	Reserved	000001	Reserved	000010	TxFIFO has 2 or fewer characters	...	—	...	—	011111	TxFIFO has 31 or fewer characters	100000	TxFIFO has 32 characters (maximum)		
000000	Reserved																
000001	Reserved																
000010	TxFIFO has 2 or fewer characters																
...	—																
...	—																
011111	TxFIFO has 31 or fewer characters																
100000	TxFIFO has 32 characters (maximum)																
9–7 RFDIV	Reference Frequency Divider. Controls the divide ratio for the reference clock. The input clock is <i>module_clock</i>. The output from the divider is <i>ref_clk</i> which is used by BRM to create the 16x baud rate oversampling clock (<i>brm_clk</i>). <table> <tr><td>000</td><td>Divide input clock by 6</td></tr> <tr><td>001</td><td>Divide input clock by 5</td></tr> <tr><td>010</td><td>Divide input clock by 4</td></tr> <tr><td>011</td><td>Divide input clock by 3</td></tr> <tr><td>100</td><td>Divide input clock by 2</td></tr> <tr><td>101</td><td>Divide input clock by 1</td></tr> <tr><td>110</td><td>Divide input clock by 7</td></tr> <tr><td>111</td><td>Reserved</td></tr> </table>	000	Divide input clock by 6	001	Divide input clock by 5	010	Divide input clock by 4	011	Divide input clock by 3	100	Divide input clock by 2	101	Divide input clock by 1	110	Divide input clock by 7	111	Reserved
000	Divide input clock by 6																
001	Divide input clock by 5																
010	Divide input clock by 4																
011	Divide input clock by 3																
100	Divide input clock by 2																
101	Divide input clock by 1																
110	Divide input clock by 7																
111	Reserved																
6 DCEDTE	DCE/DTE mode select. Select UART as data communication equipment (DCE mode) or as data terminal equipment (DTE mode).																

Table continues on the next page...

UARTx_UFCR field descriptions (continued)

Field	Description
	0 DCE mode selected 1 DTE mode selected
RXTL	Receiver Trigger Level. Controls the threshold at which a maskable interrupt is generated by the RxFIFO. A maskable interrupt is generated whenever the data level in the RxFIFO reaches the selected threshold. The RXTL bits are encoded as shown in the Settings column. Setting 0 to 32 are in use. All other settings are Reserved. 000000 0 characters received 000001 RxFIFO has 1 character ... — ... — 011111 RxFIFO has 31 characters 100000 RxFIFO has 32 characters (maximum)

16.2.15.8 UART Status Register 1 (UARTx_USR1)

Address: Base address + 94h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PARITYERR	RTSS	TRDY	RTSD	ESCF	FRAMERR	RRDY	AGTIM	DTRD	RXDS	AIRINT	AWAKE	SAD	Reserved		
W	w1c			w1c	w1c	w1c		w1c	w1c		w1c	w1c	w1c			
Reset	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0

UARTx_USR1 field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15 PARITYERR	Parity Error Interrupt Flag. Indicates a parity error is detected. PARITYERR is cleared by writing 1 to it. Writing 0 to PARITYERR has no effect. When parity is disabled, PARITYERR always reads 0. At reset, PARITYERR is set to 0. 0 No parity error detected 1 Parity error detected (write 1 to clear)
14 RTSS	RTS_B Pin Status. Indicates the current status of the RTS_B pin. A "snapshot" of RTS_B is taken immediately before RTSS is presented to the data bus. RTSS cannot be cleared because all writes to RTSS are ignored. At reset, RTSS is set to 0.

Table continues on the next page...

UARTx_USR1 field descriptions (continued)

Field	Description
	0 The RTS_B module input is high (inactive) 1 The RTS_B module input is low (active)
13 TRDY	Transmitter Ready Interrupt / DMA Flag. Indicates that the TxFIFO emptied below its target threshold and requires data. TRDY is automatically cleared when the data level in the TxFIFO exceeds the threshold set by TXTL bits. At reset, TRDY is set to 1. 0 The transmitter does not require data 1 The transmitter requires data (interrupt posted)
12 RTSD	RTS Delta. Indicates whether the RTS_B pin changed state. It (RTSD) generates a maskable interrupt. When in STOP mode, RTS assertion sets RTSD and can be used to wake the processor. The current state of the RTS_B pin is available on the RTSS bit. Clear RTSD by writing 1 to it. Writing 0 to RTSD has no effect. At reset, RTSD is set to 0. 0 RTS_B pin did not change state since last cleared 1 RTS_B pin changed state (write 1 to clear)
11 ESCF	Escape Sequence Interrupt Flag. Indicates if an escape sequence was detected. ESCF is asserted when the ESCEN bit is set and an escape sequence is detected in the RxFIFO. Clear ESCF by writing 1 to it. Writing 0 to ESCF has no effect. 0 No escape sequence detected 1 Escape sequence detected (write 1 to clear).
10 FRAMERR	Frame Error Interrupt Flag. Indicates that a frame error is detected. The <i>interrupt_uart</i> interrupt will be generated if a frame error is detected and the interrupt is enabled. Clear FRAMERR by writing 1 to it. Writing 0 to FRAMERR has no effect. 0 No frame error detected 1 Frame error detected (write 1 to clear)
9 RRDY	Receiver Ready Interrupt / DMA Flag. Indicates that the RxFIFO data level is above the threshold set by the RXTL bits. (See the RXTL bits description in UART FIFO Control Register (UART_UFCR) for setting the interrupt threshold.) When asserted, RRDY generates a maskable interrupt or DMA request. RRDY is automatically cleared when data level in the RxFIFO goes below the set threshold level. At reset, RRDY is set to 0. 0 No character ready 1 Character(s) ready (interrupt posted)
8 AGTIM	Ageing Timer Interrupt Flag. Indicates that data in the RxFIFO has been idle for a time of 8 character lengths (where a character length consists of 7 or 8 bits, depending on the setting of the WS bit in UCR2, with the bit time corresponding to the baud rate setting) and FIFO data level is less than RxFIFO threshold level (RXTL in the UFCR). Clear by writing a 1 to it. 0 AGTIM is not active 1 AGTIM is active (write 1 to clear)
7 DTRD	This bit is not used in this chip.
6 RXDS	Receiver IDLE Interrupt Flag. Indicates that the receiver state machine is in an IDLE state, the next state is IDLE, and the receive pin is high. RXDS is automatically cleared when a character is received. RXDS is active only when the receiver is enabled. 0 Receive in progress 1 Receiver is IDLE

Table continues on the next page...

UARTx_USR1 field descriptions (continued)

Field	Description
5 AIRINT	Asynchronous IR WAKE Interrupt Flag. Indicates that the IR WAKE pulse was detected on the RXD pin. Clear AIRINT by writing 1 to it. Writing 0 to AIRINT has no effect. 0 No pulse was detected on the RXD IrDA pin 1 A pulse was detected on the RXD IrDA pin
4 AWAKE	Asynchronous WAKE Interrupt Flag. Indicates that a falling edge was detected on the RXD pin. Clear AWAKE by writing 1 to it. Writing 0 to AWAKE has no effect. 0 No falling edge was detected on the RXD Serial pin 1 A falling edge was detected on the RXD Serial pin
3 SAD	RS-485 Slave Address Detected Interrupt Flag. Indicates if RS-485 Slave Address was detected . SAD was asserted in RS-485 mode when the SADEN bit is set and Slave Address is detected in RxFIFO (in Nomal Address Detect Mode, the 9 th data bit = 1; in Automatic Address Detect Mode, the received charater matches the programmed SLADDR). 0 No slave address detected 1 Slave address detected
-	This field is reserved. Reserved

16.2.15.9 UART Status Register 2 (UARTx_USR2)

Address: Base address + 98h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ADET	TXFE	DTRF	IDLE	ACST	RIDELT	RIIN	IRINT	WAKE	DCDELT	DCDIN	RTSF	TXDC	BRCD	ORE	RDR
W	w1c		w1c	w1c	w1c	w1c		w1c	w1c	w1c		w1c		w1c	w1c	
Reset	0	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0

UARTx_USR2 field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15 ADET	Automatic Baud Rate Detect Complete. Indicates that an "A" or "a" was received and that the receiver detected and verified the incoming baud rate. Clear ADET by writing 1 to it. Writing 0 to ADET has no effect. 0 ASCII "A" or "a" was not received 1 ASCII "A" or "a" was received (write 1 to clear)
14 TXFE	Transmit Buffer FIFO Empty. Indicates that the transmit buffer (TxFIFO) is empty. TXFE is cleared automatically when data is written to the TxFIFO. Even though TXFE is high, the transmission might still be in progress. 0 The transmit buffer (TxFIFO) is not empty 1 The transmit buffer (TxFIFO) is empty
13 DTRF	This bit is not used in this chip.

Table continues on the next page...

UARTx_USR2 field descriptions (continued)

Field	Description
12 IDLE	Idle Condition. Indicates that an idle condition has existed for more than a programmed amount frame (see Idle Line Detect). An interrupt can be generated by this IDLE bit if IDEN (UCR1[12]) is enabled. IDLE is cleared by writing 1 to it. Writing 0 to IDLE has no effect. 0 No idle condition detected 1 Idle condition detected (write 1 to clear)
11 ACST	Autobaud Counter Stopped. In autobaud detection (ADBR=1), indicates the counter which determines the baud rate was running and is now stopped. This means either START bit is finished (if ADNIMP=1), or Bit 0 is finished (if ADNIMP=0). See New Autobaud Counter Stopped bit and Interrupt, for more details. An interrupt can be flagged on <i>interrupt_uart</i> if ACIEN=1. 0 Measurement of bit length not finished (in autobaud) 1 Measurement of bit length finished (in autobaud). (write 1 to clear)
10 RIDELT	This bit is not used in this chip.
9 RIIN	This bit is not used in this chip.
8 IRINT	Serial Infrared Interrupt Flag. When an edge is detected on the RXD pin during SIR Mode, this flag will be asserted. This flag can cause an interrupt which can be masked using the control bit ENIRI: UCR4 [8]. 0 no edge detected 1 valid edge detected (write 1 to clear)
7 WAKE	Wake. Indicates the start bit is detected. WAKE can generate an interrupt that can be masked using the WKEN bit. Clear WAKE by writing 1 to it. Writing 0 to WAKE has no effect. 0 start bit not detected 1 start bit detected (write 1 to clear)
6 DCDELDT	This bit is not used in this chip.
5 DCDIN	This bit is not used in this chip.
4 RTSF	RTS Edge Triggered Interrupt Flag. Indicates if a programmed edge is detected on the RTS_B pin. The RTEC bits select the edge that generates an interrupt (see Table 16-6). RTSF can generate an interrupt that can be masked using the RTSEN bit. Clear RTSF by writing 1 to it. Writing 0 to RTSF has no effect. 0 Programmed edge not detected on RTS_B 1 Programmed edge detected on RTS_B (write 1 to clear)
3 TXDC	Transmitter Complete. Indicates that the transmit buffer (TxFIFO) and Shift Register is empty; therefore the transmission is complete. TXDC is cleared automatically when data is written to the TxFIFO. 0 Transmit is incomplete 1 Transmit is complete
2 BRCD	BREAK Condition Detected. Indicates that a BREAK condition was detected by the receiver. Clear BRCD by writing 1 to it. Writing 0 to BRCD has no effect. 0 No BREAK condition was detected 1 A BREAK condition was detected (write 1 to clear)
1 ORE	Overrun Error. When set to 1, ORE indicates that the receive buffer (Rx FIFO) was full (32 chars inside), and a 33rd character has been fully received. This 33rd character has been discarded. Clear ORE by writing 1 to it. Writing 0 to ORE has no effect.

Table continues on the next page...

UARTx_USR2 field descriptions (continued)

Field	Description
	0 No overrun error 1 Overrun error (write 1 to clear)
0 RDR	Receive Data Ready -Indicates that at least 1 character is received and written to the RxFIFO. If the URXD register is read and there is only 1 character in the RxFIFO, RDR is automatically cleared. 0 No receive data ready 1 Receive data ready

16.2.15.10 UART Escape Character Register (UARTx_UESC)

Address: Base address + 9Ch offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																ESC_CHAR															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1

UARTx_UESC field descriptions

Field	Description
31–8 Reserved	This read-only field is reserved and always has the value 0.
ESC_CHAR	UART Escape Character. Holds the selected escape character that all received characters are compared against to detect an escape sequence.

16.2.15.11 UART Escape Timer Register (UARTx_UTIM)

Address: Base address + A0h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TIM															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UARTx_UTIM field descriptions

Field	Description
31–12 Reserved	This read-only field is reserved and always has the value 0.
TIM	UART Escape Timer. Holds the maximum time interval (in ms) allowed between escape characters. The escape timer register is programmable in intervals of 2 ms. See Escape Sequence Detection and Table 16-10 for more information on the UART escape sequence detection. Reset value 0x000 = 2 ms up to 0xFFFF = 8.192 s.

16.2.15.12 UART BRM Incremental Register (UARTx_UBIR)

This register can be written by both software and hardware. When enabling the automatic baud rate detection feature hardware can write 0x000F value into the UBIR after finishing detecting baud rate. Hardware has higher priority when both software and hardware try to write it at the same cycle³.

Please note software reset will reset the register to its reset value.

Address: Base address + A4h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																INC															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UARTx_UBIR field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
INC	Incremental Numerator. Holds the numerator value minus one of the BRM ratio (see Binary Rate Multiplier (BRM)). The UBIR register MUST be updated before the UBMR register for the baud rate to be updated correctly. If only one register is written to by software, the BRM will ignore this data until the other register is written to by software. Updating this field using byte accesses is not recommended and is undefined.

16.2.15.13 UART BRM Modulator Register (UARTx_UBMR)

This register can be written by both software and hardware. When enabling the automatic baud rate detection feature hardware can write a proper value into the UBMR based on detected baud rate. Hardware has higher priority when both software and hardware try to write it at the same cycle⁴.

Please note software reset will reset the register to its reset value.

Address: Base address + A8h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																MOD															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

3. Note: The write priority in the new design is not same as the original UART. In the original design, software has higher priority than hardware when writing this register at the same time.

4. Note: The write priority in the new design is not same as the original UART. In the original design, software has higher priority than hardware when writing this register at the same time.

UARTx_UBMR field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
MOD	Modulator Denominator. Holds the value of the denominator minus one of the BRM ratio (see Binary Rate Multiplier (BRM)). The UBIR register MUST be updated before the UBMR register for the baud rate to be updated correctly. If only one register is written to by software, the BRM will ignore this data until the other register is written to by software. Updating this register using byte accesses is not recommended and undefined.

16.2.15.14 UART Baud Rate Count Register (UARTx_UBRC)

Address: Base address + ACh offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																BCNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

UARTx_UBRC field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
BCNT	Baud Rate Count Register. This read only register is used to count the start bit of the incoming baud rate (if ADNIMP=1), or start bit + bit0 (if ADNIMP=0). When the measurement is done, the Baud Rate Count Register contains the number of UART internal clock cycles (clock after divider) present in an incoming bit. BCNT retains its value until the next Automatic Baud Rate Detection sequence has been initiated. The 16 bit Baud Rate Count register is reset to 4 and stays at hex FFFF in the case of an overflow.

16.2.15.15 UART One Millisecond Register (UARTx_ONEMS)

NOTE

This register has been expanded from 16 bits to 24 bits. In previous versions, the 16-bit ONEMS can only support the maximum 65.535MHz (0xFFFFx1000) *ref_clk*. To support 4Mbps Bluetooth application with 66.5MHz *module_clock*, the value 0x103C4 (66.5M/1000) should be written into this register. In this case, the 16 bits are not enough to contain the 0x103C4. So this register was expanded to 24 bits to support high frequency of the *ref_clk*.

Address: Base address + B0h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0								ONEMS																							
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UARTx_ONEMS field descriptions

Field	Description
31–24 Reserved	This read-only field is reserved and always has the value 0.
ONEMS	One Millisecond Register. This 24-bit register must contain the value of the UART internal frequency (<i>ref_clk</i> in Figure 16-9) divided by 1000. The internal frequency is obtained after the UART BRM internal divider ($F(\text{ref_clk}) = F(\text{module_clock}) / \text{RFDIV}$). In fact this register contains the value corresponding to the number of UART BRM internal clock cycles present in one millisecond. The ONEMS (and UTIM) registers value are used in the escape character detection feature (Escape Sequence Detection) to count the number of clock cycles left between two escape characters. The ONEMS register is also used in infrared special case mode (IRSC = UCR4[5] = 1'b1), see InfraRed Special Case (IRSC) Bit.

16.2.15.16 UART Test Register (UARTx_UTS)

Address: Base address + B4h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0		FRCPERR	LOOP	DBGEN	LOOPIR	RXDBG	0	TXEMPTY	RXEMPTY	TXFULL	RXFULL	0			
W																
Reset	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0

UARTx_UTS field descriptions

Field	Description
31–14 Reserved	This read-only field is reserved and always has the value 0.
13 FRCPERR	Force Parity Error. Forces the transmitter to generate a parity error if parity is enabled. FRCPERR is provided for system debugging. 0 Generate normal parity 1 Generate inverted parity (error)
12 LOOP	Loop TX and RX for Test. Controls loopback for test purposes. When LOOP is high, the receiver input is internally connected to the transmitter and ignores the RXD pin. The transmitter is unaffected by LOOP. If RXDMUXSEL (UCR3[2]) is set to 1, the loopback is applied on serial and IrDA signals. If RXDMUXSEL is set to 0, the loopback is only applied on serial signals. 0 Normal receiver operation 1 Internally connect the transmitter output to the receiver input
11 DBGEN	This bit is not used in this chip. debug_enable_B. This bit controls whether to respond to the <i>debug_req</i> input signal. 0 UART will go into debug mode when <i>debug_req</i> is HIGH 1 UART will not go into debug mode even if <i>debug_req</i> is HIGH
10 LOOPIR	Loop TX and RX for IR Test (LOOPIR). This bit controls loopback from transmitter to receiver in the InfraRed interface. 0 No IR loop 1 Connect IR transmitter to IR receiver
9 RXDBG	This bit is not used in this chip. RX_fifo_debug_mode. This bit controls the operation of the RX fifo read counter when in debug mode. 0 rx fifo read pointer does not increment 1 rx_fifo read pointer increments as normal

Table continues on the next page...

UARTx_UTS field descriptions (continued)

Field	Description
8–7 Reserved	This read-only field is reserved and always has the value 0.
6 TXEMPTY	TxFIFO Empty. Indicates that the TxFIFO is empty. 0 The TxFIFO is not empty 1 The TxFIFO is empty
5 RXEMPTY	RxFIFO Empty. Indicates the RxFIFO is empty. 0 The RxFIFO is not empty 1 The RxFIFO is empty
4 TXFULL	TxFIFO FULL. Indicates the TxFIFO is full. 0 The TxFIFO is not full 1 The TxFIFO is full
3 RXFULL	RxFIFO FULL. Indicates the RxFIFO is full. 0 The RxFIFO is not full 1 The RxFIFO is full
2–1 Reserved	This read-only field is reserved and always has the value 0.
0 SOFTTRST	Software Reset. Indicates the status of the software reset (SRST_B bit of UCR2). 0 Software reset inactive 1 Software reset active

16.2.15.17 UART RS-485 Mode Control Register (UARTx_UMCR)

Address: Base address + B8h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SLADDR								0				SADEN	TXB8	SLAM	MDEN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UARTx_UMCR field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15–8 SLADDR	RS-485 Slave Address Character. Holds the selected slave address character that the receiver will try to detect.
7–4 Reserved	This read-only field is reserved and always has the value 0.
3 SADEN	RS-485 Slave Address Detected Interrupt Enable. 0 Disable RS-485 Slave Address Detected Interrupt 1 Enable RS-485 Slave Address Detected Interrupt
2 TXB8	Transmit RS-485 bit 8 (the ninth bit or 9 th bit). In RS-485 mode, software writes TXB8 bit as the 9 th data bit to be transmitted. 0 0 will be transmitted as the RS485 9 th data bit 1 1 will be transmitted as the RS485 9 th data bit
1 SLAM	RS-485 Slave Address Detect Mode Selection. 0 Select Normal Address Detect mode 1 Select Automatic Address Detect mode
0 MDEN	9-bit data or Multidrop Mode (RS-485) Enable. 0 Normal RS-232 or IrDA mode, see Table 16-3 for detail. 1 Enable RS-485 mode, see Table 16-3 for detail

Appendix A

Revision History

A.1 Substantive changes from revision 3 to revision 3.1

Substantive changes from revision 3 to revision 3.1 are as follows:

A.1.1 Reference Manual Revision History

Topic Cross-Reference	Change Description
-	New PCIe PHY chapter replaces previous version.
IOMUXC GPR Memory Map/Register Definition	Updated bit field descriptions and values for IOMUXC_GPR_GPR14[GPR_PCIE1_VREG_BYPASS] and IOMUXC_GPR_GPR16[GPR_PCIE2_VREG_BYPASS].

A.2 Substantive changes from revision 2 to revision 3

Substantive changes from revision 2 to revision 3 are as follows:

A.2.1 Reference Manual Revision History

Topic Cross-Reference	Change Description
-	MIPI CSI2 support for 4Kp30 Camera Input added
-	Added System Counter (SYS_CTR) chapter

A.2.2 Introduction Revision History

No substantive changes

A.2.3 Memory Maps Revision History

Topic Cross-Reference	Change Description
AIPS Memory Maps	Updated MIPI DSI and DPHY Addresses
AIPS Memory Maps	Added DCSS Addresses

A.2.4 System Security Revision History

No substantive changes

A.2.5 RDC Revision History

No substantive changes

A.2.6 A53 Revision History

No substantive changes

A.2.7 CM4 Revision History

No substantive changes

A.2.8 MU Revision History

No substantive changes

A.2.9 SEMA4 Revision History

No substantive changes

A.2.10 OCRAM Revision History

No substantive changes

A.2.11 NIC Revision History

No substantive changes

A.2.12 AIPSTZ Revision History

No substantive changes

A.2.13 SPBA Revision History

No substantive changes

A.2.14 ROMCP Revision History

No substantive changes

A.2.15 ROMCP Revision History

No substantive changes

A.2.16 TZASC Revision History

No substantive changes

A.2.17 System Debug Revision History

No substantive changes

A.2.18 SJC Revision History

No substantive changes

A.2.19 CCM Revision History

Topic Cross-Reference	Change Description
Clock Tree	Updated clock tree diagram
Clock Root Selects	Changed Max freq of MIPI_CSI_PHY_REF_CLK_ROOT to 333MHz

A.2.20 GPC Revision History

No substantive changes

A.2.21 XTALOSC Revision History

No substantive changes

A.2.22 TMU Revision History

Topic Cross-Reference	Change Description
Features	Changed Maximum temperature to 125C

A.2.23 System Boot Revision History

Topic Cross-Reference	Change Description
Low-power boot	Renamed BT_MMU_DISABLE to BT_DCACHE_DISABLE
Boot eFUSE descriptions	Updated ARM frequency for various low power boot modes
Clocks at boot time	Updated ARM_PLL and ARM_A53_ROOT frequencies

A.2.24 Fusemap Revision History

No substantive changes

A.2.25 OCOTP Revision History

No substantive changes

A.2.26 SNVS Revision History

No substantive changes

A.2.27 SRC Revision History

No substantive changes

A.2.28 Interrupts Revision History

No substantive changes

A.2.29 SDMA Revision History

No substantive changes

A.2.30 External Signals Revision History

No substantive changes

A.2.31 IOMUXC Revision History

Topic Cross-Reference	Change Description
-	Replaced ARM with Arm where appropriate. IO names remain all capital letters, but references to Arm replaced.

A.2.32 GPIO Revision History

No substantive changes

A.2.33 External Memory Overview Revision History

No substantive changes

A.2.34 DDR Performance Monitor Revision History

No substantive changes

A.2.35 DDR Controller Revision History

No substantive changes

A.2.36 DDR PHY Revision History

No substantive changes

A.2.37 APBH Revision History

No substantive changes

A.2.38 BCH Revision History

No substantive changes

A.2.39 GPMI Revision History

No substantive changes

A.2.40 ECSPI Revision History

No substantive changes

A.2.41 QSPI Revision History

No substantive changes

A.2.42 USDHC Revision History

No substantive changes

A.2.43 USB 3.0 Revision History

No substantive changes

A.2.44 Peripheral Component Interconnect express (PCIe) Revision History

No substantive changes

A.2.45 PCIe PHY Revision History

No substantive changes

A.2.46 ENET Revision History

No substantive changes

A.2.47 GPT Revision History

No substantive changes

A.2.48 PWM Revision History

No substantive changes

A.2.49 Multimedia Overview Revision History

No substantive changes

A.2.50 eLCDIF Revision History

Topic Cross-Reference	Customer-facing Description
Functional Description	Added Top-level Block Diagram
-	Removed pinout references, eLCDIF is not pinned out

A.2.51 Graphics Processing Unit (GPU) Revision History

No substantive changes

A.2.52 HDMI TX Revision History

No substantive changes

A.2.53 HDMI TX PHY Revision History

No substantive changes

A.2.54 MIPI DSI Revision History

No substantive changes

A.2.55 CSI Revision History

No substantive changes

A.2.56 MIPI CSI2 Revision History

No substantive changes

A.2.57 SPDIF Revision History

No substantive changes

A.2.58 SAI Revision History

No substantive changes

A.2.59 VPU G1 Revision History

No substantive changes

A.2.60 VPU G2 Revision History

No substantive changes

A.2.61 Display Controller Subsystem (DCSS) Revision History

No substantive changes

A.2.62 DCSS Block Control (BLK_CTL) Revision History

No substantive changes

A.2.63 Display Timing Generator (DTG) Revision History

No substantive changes

A.2.64 Context Load (CTX_LD) Revision History

No substantive changes

A.2.65 Graphics Decompression (DEC400D) Revision History

No substantive changes

A.2.66 Decompression and Tile to Raster Conversion (DTRC) Revision History

No substantive changes

A.2.67 Display, Prefetch and Resolve (DPR) Revision History

No substantive changes

A.2.68 Scaler (MED_DC_SCALER) Revision History

No substantive changes

A.2.69 Look Up Table Load (LUT_LD) Revision History

No substantive changes

A.2.70 HDR10 Image Processing (MED_HDR10) Revision History

No substantive changes

A.2.71 Color Sub-Sampler (SUBSAM) Revision History

No substantive changes

A.2.72 Write Scale (WR_SCL) Revision History

No substantive changes

A.2.73 Read Surface (RD_SRC) Revision History

No substantive changes

A.2.74 Interrupt Request Steering (IrqSteer) Revision History

No substantive changes

A.2.75 I2C Revision History

No substantive changes

A.2.76 UART Revision History

No substantive changes

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